Passage — Product Requirements Document

Status	Draft v0.2 — for review by Ben Russell and Zach Demanche
Date	2026-10-21
Owners	@BenjaminSRussell (product), @zdemanche (engineering/security)
Method	Moonaan Scribe requirements workflow (understand → research → draft → validate → publish). Jira/Confluence research was unavailable this session; research covered public AX/SAP documentation and both owners' GitHub accounts (no prior ERP work found).
Companion	roadmap/waves.json is the machine-readable source for the 50 waves; roadmap/README.md is generated from it.
Thesis	Emulate the top-tier manufacturing ERPs (S/4HANA, D365 F&SCM, Infor LN, Epicor Kinetic, IFS, Plex) feature-for-feature, then be **10× easier, faster, more accurate and more open** — see Part 0A.

Document map

This master document (Part 0) holds the problem, goals, scope, delivery model, phase-level requirements and NFRs. The detailed parts hold wave-level requirements and are assembled into `docs/PRD-full.md` by `scripts/build_prd.py`.

Part	File	Content
0	`docs/PRD.md` (this file)	Problem, goals, personas, principles, scope map, delivery model, phase-level FRs, NFRs, risks
0A	prd/part-00-thesis.md	The 10× thesis: pain-point ledger (usability, speed, accuracy, openness), interoperability contract, what we keep from AX and SAP
1	prd/phase-0-foundation.md	Waves 1–5: pipeline, identity/tenancy, metadata kernel, events/workflow, shell/audit/observability
2	prd/phase-1-master-data.md	Waves 6–10: enterprise structure, party model, products, engineering data, migration framework
3	prd/phase-2-financial-core.md	Waves 11–15: universal journal, currencies/ledgers, tax, close, reporting
4	prd/phase-3-procure-to-pay.md	Waves 16–20: inventory ledger, purchasing, receiving/matching, AP/payments, costing/supplier portal
5	prd/phase-4-manufacturing.md	Waves 21–25: production orders, MRP, scheduling, process/lean, shop floor/IoT
6	prd/phase-5-order-to-cash.md	Waves 26–30: pricing, orders/ATP, fulfillment, billing/rev rec, AR/collections — go-live candidate
7	prd/phase-6-logistics.md	Waves 31–35: WMS, mobile, quality, transportation, EDI/e-invoicing/integration hub
8	prd/phase-7-treasury-controlling.md	Waves 36–40: cash/bank, fixed assets, budgeting/cost accounting, product costing, intercompany/consolidation
9	prd/phase-8-projects-people-service.md	Waves 41–45: projects, HR, payroll/expenses, EAM, field service
10	prd/phase-9-intelligence-golive.md	Waves 46–50: data platform, AI agents, localization/compliance, migration factory, scale/go-live
11	prd/appendices.md	Glossary, cross-module data model, event/API conventions, security model, NFR detail, numbering, localization, testing standards, traceability, references, competitive benchmark
12	prd/coverage-ax.md	Completeness oracle: every AX 2012 R3 / D365 F&SCM module → forms, inquiries/reports, periodic jobs, setup forms, each mapped to a wave or flagged GAP; gap register
13	prd/coverage-sap.md	Completeness oracle: every SAP ECC / S/4 component → transactions, reports, periodic processing, IMG customizing, each mapped to a wave or flagged GAP; gap register

Each wave section follows the fixed template in prd/TEMPLATE.md: scope and pain points solved → AX/SAP parity map → functional requirements → data model → APIs/events → UX → business rules → acceptance criteria → test plan → deployment → migration → dependencies/risks/open questions → parameters and setup → inquiries and reports → printed/electronic documents → periodic jobs → duties and privileges → message catalog → performance budgets → localization hooks.

Completeness rule. A wave is not specified until every row of Parts 12 and 13 that maps to it is either covered by a numbered FR in that wave's section or explicitly marked SUPERSEDED/OUT-OF-SCOPE with a reason. The gap registers (12.99, 13.99) must be empty before Phase 0 starts.

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1. Problem statement

Microsoft Dynamics AX (Axapta → AX 2012 → D365 Finance & Supply Chain) and SAP (ECC → S/4HANA) are the systems of record for most mid-to-large manufacturers, distributors and services firms. Both are:

- **Closed and expensive to change** — customization lives in proprietary languages (X++, ABAP) and licensing/upgrade costs dominate TCO.
- **Architecturally dated** — monolithic application servers, synchronous batch frameworks, and vendor-locked cloud operation (LCS, RISE).
- **Hard to integrate and extend with AI** — bolted-on copilots that can't act safely across the system.

Passage is a ground-up rebuild of the complete feature surface of AX/SAP as an open-architecture, cloud-agnostic platform: Kubernetes-native, Postgres-per-domain, Kafka-eventing, metadata-driven, with AI agents as first-class, governed actors.

2. Goals and non-goals

Goals

1. **Feature parity**: every AX 2012 / D365 F&SCM and SAP ECC / S/4 module in scope (§5) has an equivalent in Passage, mapped wave-by-wave.
2. **Provable delivery**: every one of the 50 waves ends with an artifact that is (a) **provable** — automated tests and fixtures that demonstrate correctness, (b) **testable** — a stakeholder can exercise it in a running environment, and (c) **deployable** — it ships to a Kubernetes environment through the GitOps pipeline.
3. **Sequential, top-down**: waves are strictly ordered; each wave depends only on earlier waves (dependency graph is a chain with stated back-references). A manufacturer can go live at the end of wave 30; full AX/SAP parity at wave 50.
4. **Cloud-agnostic and self-hostable**: runs on any conformant Kubernetes (EKS/GKE/AKS/bare-metal/k3d) with no provider-specific managed services in the critical path.
5. **Extensible without forking**: customers extend entities, workflows, forms and reports through metadata and sanctioned extension points, never by overlayering code.

6. **Financially rigorous**: a single append-only universal journal is the only source of truth; every subledger reconciles to it by automated check.
7. **10× on four measured axes** (defined in Part 0A): easier (≤ 1/10 the clicks per task; new user productive in < 30 min), faster (MRP incremental < 2 min, continuous close, real-time statements), more accurate (zero unexplained subledger-to-GL breaks, < 0.5% duplicate master data), open (any system on the customer's network connects in < 1 day through REST/GraphQL/Kafka/webhooks/MCP/EDI/OPC-UA/SFTP with one identity and one policy engine).
8. **Best-of-breed manufacturing**: planning, scheduling, shop floor, quality, costing and warehouse are benchmarked against S/4HANA, D365 SCM, Infor LN, Epicor Kinetic, IFS Cloud and Plex (Appendix L), not only against AX 2012.

Non-goals (v1)

- Replacing CRM/marketing automation suites (we integrate with Salesforce/HubSpot; a native CRM is v2).
- Retail POS and e-commerce storefront (we integrate with Shopify/Medusa; native POS is v2).
- Country packs beyond the first set (US, UK, DE, FR, IN, MX, CA) — framework ships in wave 48; more packs are v1.x.
- Supporting SAP's or Microsoft's proprietary extension languages. Migration tooling moves **data**, not X++/ABAP code.

3. Users and personas

Persona	Primary jobs	Waves that serve them first
Platform engineer / SRE	Deploy, observe, upgrade, scale, secure	1–5, 48, 50
System administrator	Tenants, users, roles, extensions, number sequences, audit	2, 3, 5, 48
Master-data steward	Parties, products, BOMs, org structure, migration	6–10
Controller / accountant	Journal, close, statements, tax, FX, subledgers	11–15, 19, 30, 36–40
Buyer / AP clerk	Requisition to payment	17–20
Production planner / supervisor	Orders, MRP, scheduling, shop floor	21–25
Sales rep / order desk / AR clerk	Quote to cash	26–30
Warehouse manager / operator	WMS, mobile, shipping, quality, transport	28, 31–35
Project manager / HR / payroll admin	Projects, people, payroll, assets, service	41–45
Analyst / executive	Dashboards, KPIs, copilots	46–47
Implementation partner	Migration, localization, cutover	10, 48–50

4. Product principles

1. **Journal first.** Nothing is financially true until it is in the universal journal. Subledgers are projections; they reconcile automatically every day.
2. **Metadata over code.** Entities, forms, workflows, reports and extensions are data. Code generators produce schema, API, clients and UI from it.
3. **Events are the integration surface.** Every state change is an immutable CloudEvent on Kafka via transactional outbox — no dual writes, no polling.
4. **Everything is tenant-scoped and policy-authorized.** Postgres RLS + OPA on every request; agents inherit (never exceed) their user's permissions.
5. **Idempotent, replayable, auditable.** Posting APIs take idempotency keys; audit is derived from CDC, not from application code that can forget.
6. **Ship the thin vertical slice.** A wave is done when a stakeholder can run the business process on a deployed environment with reconciled numbers — not when the code is written.

5. Scope map — AX/SAP modules → Passage phases

AX / D365 module	SAP module	Passage phase (waves)
AOT, security, workflow, batch, LCS, number sequences, DB log	DDIC, PFCG, Business Workflow, SM36, STMS, SNRO, CDHDR	**0 Foundation (1–5)**
Org model, global address book, products, BOM/route, DIXF	Enterprise structure, Business Partner, material master, CS/CA, LSMW	**1 Master data (6–10)**
General ledger, dimensions, currencies, tax, period close, financial reporting	FI-GL/ACDOCA, ledgers, FTXP, OB52, FSV	**2 Financial core (11–15)**
Inventory management, procurement & sourcing, AP, inventory costing	MM-IM, MM-PUR, LIV, FI-AP, F110, material ledger	**3 Inventory & P2P (16–20)**
Production control, master planning, scheduling, process/lean, MES	PP, PP-PI, REM, KANBAN, PP/DS, MII	**4 Manufacturing & planning (21–25)**
Trade agreements, sales, shipping, invoicing, rev rec, AR, collections	SD pricing, SD orders, LE-SHP, billing, RAR, FI-AR, FSCM	**5 Order-to-cash (26–30)**
Warehouse management, mobile app, quality, TMS, AIF/ER/EDI	EWM, RF, QM, TM, IDoc/PI, DRC	**6 Warehouse, quality & logistics (31–35)**
Cash & bank, fixed assets, budgeting/cost accounting, costing sheets, intercompany/consolidation	Cash mgmt, FI-AA, CO, product costing, Group Reporting	**7 Treasury, assets & controlling (36–40)**
Projects, HR, payroll, expense, asset management, service	PS, HCM/SF, PY, Concur, PM, CS/FSM	**8 Projects, people & service (41–45)**
Entity store/Power BI, Copilot, localizations/SoD, upgrade tooling, LCS prod	BW/SAC, Joule, country versions/GRC, conversion tools, Basis	**9 Intelligence, compliance & go-live (46–50)**

Full per-wave mapping with references is in roadmap/waves.json (`replaces` field).

6. Delivery model — what a "wave" is

A wave is a 4–8 week increment with four mandatory outputs, enforced by the wave issue template and the Definition of Done:

Output	Definition	Evidence required to close the wave
Deliverable	The capabilities listed in the wave's `deliverable` array, built on the agreed stack.	Merged PRs linked from the wave issue.
Proof	The automated tests, fixtures, property tests and performance targets in `proof`.	CI run link; fixture datasets committed; perf report.
Deploy	What is running where, as stated in `deploy`.	Argo CD app healthy in `stage`; demo URL; recorded walkthrough for phase-exit waves.
References	The AX/SAP documentation and open-source/standards we built against.	Linked in the wave issue; ADRs for any deviation.

Sequencing rules

- Wave *n* may start when wave *n−1* has passed Proof and Deploy; overlapping *build* is allowed, overlapping *close* is not.
- Every 5th wave is a **phase-exit wave** with a stakeholder sign-off (demo + reconciliation evidence).
- Wave 30 is the **first go-live candidate** (manufacturer buy → make → sell → collect). Wave 50 is **production go-live + v1.0**.

7. Functional requirements (phase level)

Phase-level requirements below are the contract; wave-level detail (deliverables, proofs, references) is normative in `roadmap/waves.json` and rendered in `roadmap/README.md`.

FR-0 Foundation (waves 1–5)

- **FR-0.1** The platform deploys to any conformant Kubernetes via GitOps with signed images, SBOMs and digest-pinned rollback.
- **FR-0.2** Identity is federated (OIDC/SAML/SCIM); authorization is AX-style roles/duties/privileges compiled to OPA policies; tenant isolation is enforced by Postgres RLS.
- **FR-0.3** Business entities are declared in metadata that generates schema, API, clients, validation and UI; customers extend entities without code.
- **FR-0.4** All state changes publish CloudEvents via transactional outbox; approvals and batch jobs run on a durable workflow engine with timers, retries and human tasks.
- **FR-0.5** A single web shell renders any entity as accessible (WCAG 2.1 AA) list/detail/grid; number sequences support gap-free mode; audit is field-level and immutable; every request is traceable end-to-end.

FR-1 Master data (waves 6–10)

- **FR-1.1** Legal entities, operating units, sites, warehouses and effective-dated multi-purpose hierarchies; fiscal calendars; ISO reference data; UoM conversions.
- **FR-1.2** One party per real-world organization/person with layered roles, duplicate detection and merge, tax-id validation, governance workflow.
- **FR-1.3** Products with product/storage/tracking dimension groups, released products per entity, attributes/classification, GS1 identifiers, configurator.
- **FR-1.4** Multi-level versioned BOMs and formulas, routings/operations/resources/calendars, engineering change management with where-used.
- **FR-1.5** A migration framework with AX and SAP connectors, declarative mapping, validation, reconciliation and idempotent re-runs.

FR-2 Financial core (waves 11–15)

- **FR-2.1** Append-only double-entry universal journal with dimensions, account structures, idempotent atomic posting, balance projections rebuilt from the journal.
- **FR-2.2** Three-currency amounts per line, rate types/providers, parallel ledgers, FX revaluation and translation.
- **FR-2.3** Tax determination/calculation/posting for sales & use, VAT/GST and withholding; pluggable external providers; settlement and statutory extracts (SAF-T).
- **FR-2.4** Period control per ledger × module, close checklists, allocations, accrual schemes, year-end close.
- **FR-2.5** Statement designer and standard statements with drill-down to source document; audit trail reporting.

FR-3 Inventory & P2P (waves 16–20)

- **FR-3.1** Immutable inventory transactions with physical/financial status and dimensions; journals; reservations; cycle counting; GL integration.
- **FR-3.2** Requisition → RFQ → agreement → PO with policy-driven approvals, versioned change management and vendor pricing.
- **FR-3.3** Receiving with GR/IR accrual; vendor invoice capture; 2/3-way matching with tolerances and exception workflow; ERS.
- **FR-3.4** AP open-item settlement, payment proposals, ISO 20022/NACHA/check payment media, positive pay, prepayments.
- **FR-3.5** FIFO/LIFO/weighted-average/standard costing with inventory close and GL reconciliation; supplier portal.

FR-4 Manufacturing & planning (waves 21–25)

- **FR-4.1** Production order lifecycle with picking, route consumption, RAF, WIP and variance settlement; subcontracting.
- **FR-4.2** MRP/MPS with coverage groups, planned orders, action/delay messages, pegging, statistical forecasting.
- **FR-4.3** Finite/infinite capacity scheduling with Gantt and optional CP-SAT optimization.
- **FR-4.4** Process manufacturing (formulas, potency, co/by-products, catch weight, batch attributes) and lean (production flows, kanban).
- **FR-4.5** Shop-floor kiosk with time registration, IoT ingestion (OPC-UA/MQTT), OEE, offline tolerance.

FR-5 Order-to-cash (waves 26–30)

- **FR-5.1** Deterministic pricing engine with condition tables/access sequences, discounts, promotions, rebates, explain.
- **FR-5.2** Quotations, sales orders, ATP/CTP, credit management, holds, multi-channel intake.
- **FR-5.3** Release, pick/pack/ship, packing slips with COGS, carrier integration, drop-ship.
- **FR-5.4** Invoicing (one-off, consolidated, subscription), ASC 606/IFRS 15 revenue recognition.
- **FR-5.5** Cash application, collections/dunning/disputes, RMA, rebate settlement, customer portal with payments. **Go-live candidate.**

FR-6 Warehouse, quality & logistics (waves 31–35)

- **FR-6.1** Location directives, work templates, waves, license plates, replenishment, load planning.
- **FR-6.2** Mobile scanner app with configurable flows, GS1 parsing, offline queue, label printing.
- **FR-6.3** Quality orders by rule, sampling, usage decisions, inventory blocking, NCR/CAPA, CoA, supplier scorecards.
- **FR-6.4** Load/route planning, rating, tendering, tracking, freight reconciliation, landed cost.
- **FR-6.5** EDI (X12/EDIFACT), Peppol e-invoicing, integration hub (APIs, webhooks, catalog, developer portal).

FR-7 Treasury, assets & controlling (waves 36–40)

- **FR-7.1** Bank accounts, statement import and auto-reconciliation, cash positioning/forecasting, bank connectivity with vaulted keys.
- **FR-7.2** Fixed assets with multi-book depreciation, full transaction set, CIP from projects, IFRS 16 leases.
- **FR-7.3** Budgeting with workflow and budget control on commitments; cost accounting ledger with allocations and ABC.
- **FR-7.4** Costing sheets, BOM cost roll-up, standard cost activation and revaluation, variance decomposition.
- **FR-7.5** Intercompany trade/accounting with mirror documents and reconciliation; group consolidation with eliminations and translation. **Finance parity.**

FR-8 Projects, people & service (waves 41–45)

- **FR-8.1** WBS, resources, hour/expense/item journals, T&M/fixed-price/milestone billing, WIP and POC accounting, EVM.
- **FR-8.2** Workers/positions/jobs with effective dating, absence, benefits, self-service, talent basics, PII controls.
- **FR-8.3** Country-pluggable gross-to-net payroll (US + one EU reference) and expense management with policies and card feeds.
- **FR-8.4** Functional locations/assets, preventive and condition-based plans, work orders with costs to asset/GL, IoT triggers.
- **FR-8.5** Installed base, agreements/SLAs, cases, field work orders, dispatch optimization, technician mobile app.

FR-9 Intelligence, compliance & go-live (waves 46–50)

- **FR-9.1** CDC lakehouse (Iceberg) + semantic layer with governed metrics, embedded dashboards, document designer, catalog/lineage.
- **FR-9.2** MCP-exposed APIs; copilot UI; governed agents with approval gates, evals in CI, cost budgets, model-agnostic provider layer.
- **FR-9.3** Localization framework and first country packs; electronic reporting designer; SoD engine; SOC 2 control evidence; privacy requests.
- **FR-9.4** Transactional migration packs for AX and SAP, parallel-run diff harness, cutover runbooks, coexistence adapters.
- **FR-9.5** Scale/perf targets met, multi-region DR, operations and documentation complete, first customer in production, v1.0 released.

8. Non-functional requirements

Area	Requirement
Availability	99.9% per tenant (v1), 99.95% target with multi-region (wave 50). RPO 5 min, RTO 1 h.
Performance	Journal posting 5k lines/s sustained; generated entity reads p95 < 50 ms; pricing p95 < 20 ms; order intake p99 < 500 ms; dashboard lag p95 < 60 s.
Scale	10k concurrent users; 1M journal lines/h; 100k order lines/h; 50k-item MRP in < 10 min.
Security	OIDC/SAML/SCIM; OPA authorization on every request; Postgres RLS; signed images/SBOM; secrets in Vault; SoD engine; pen-test clean at high/critical before wave 48 exit.
Compliance	SOC 2 TSC control mapping, GDPR/CCPA subject requests, SAF-T, Peppol, ISO 20022, GS1.
Accessibility	WCAG 2.1 AA for every generated form and every hand-built workspace (axe in CI).
Observability	OpenTelemetry traces/metrics/logs for 100% of services; SLO dashboards; alerting.
Portability	No provider-managed service in the critical path; Helm + Terraform/OpenTofu modules per provider; k3d local parity.
Extensibility	Entity/enum/field/workflow/form/report extensions as metadata; versioned public APIs with additive-change guarantees.
Data integrity	Every subledger reconciles to the universal journal daily by automated check; no negative inventory unless policy allows; gap-free fiscal sequences.
Auditability	Field-level immutable audit from CDC; every agent action logged with inputs/outputs/approver.

9. Technical architecture (summary — see ADR-0001)

- **Runtime**: Kubernetes (any conformant distribution), Argo CD GitOps, Envoy Gateway, cert-manager.
- **Services**: Go for ledgers/engines (journal, inventory, pricing, MRP, scheduling), TypeScript (Node) for workflow/integration/UI-adjacent services, Python for forecasting/ML/agent evals.
- **Data**: Postgres per domain (CloudNativePG), RLS, Atlas migrations generated from metadata; Apache Iceberg lakehouse with Trino/ClickHouse for analytics.
- **Eventing**: Kafka (Strimzi), Debezium CDC + transactional outbox, CloudEvents + AsyncAPI, schema registry.
- **Workflow**: Temporal for approvals, batch, agents.
- **Identity/authz**: Keycloak, OPA/Rego.
- **Frontend**: React + module federation shell; PWA for shop floor, warehouse and field mobile.
- **AI**: MCP server over Passage APIs; Claude as default model via a provider-agnostic layer; eval harness in CI.
- **Observability**: OpenTelemetry → Prometheus/Tempo/Loki/Grafana.

10. Success metrics

Milestone	Metric
Wave 5	Platform demo: deploy → login → extend entity → approve workflow → trace a request, all live.
Wave 15	A period closed in Passage with statements that tie to the trial balance to the cent.
Wave 30	Demo manufacturer runs one simulated month buy → make → sell → collect; all subledgers reconcile; stakeholder sign-off.
Wave 40	Group consolidation produced; finance parity sign-off by a practicing controller.
Wave 50	First customer month-end in production; load/DR targets met; v1.0 released.
Ongoing	Every wave: CI green, proofs committed, Argo app healthy in stage, references linked.

11. Risks and mitigations

Risk	Mitigation
Scope is enormous (50 waves)	Strict sequencing; each wave independently demoable; phase-exit sign-offs; `waves.json` is the single plan of record and is re-estimated each phase exit.
Accounting correctness	Fixture sets reviewed by a practicing accountant per finance wave; daily automated subledger-to-GL reconciliation from wave 16 onward.
Postgres-per-domain vs cross-domain transactions	Sagas over Temporal + outbox; the journal posting API is the only cross-domain financial truth; no distributed 2PC.
Cloud-agnostic ambition increases ops burden	Reference stack is entirely CNCF/open-source with Helm charts; provider-specific Terraform modules are thin and optional.
Payroll/tax/localization legal exposure	Ship provider adapters (Avalara/ADP-style) alongside internal engines; country packs gated by regulator-schema fixtures.
AI agent misuse	Agents inherit user permissions via OPA, approval gates, evals in CI, kill switch, cost budgets.
Two-person team	GitHub Projects + wave issues as the operating system; AI-assisted generation (Moonaan sdlc `generate-architecture` → `generate-backlog` → `generate-code`) per wave.

12. Open questions

1. Licensing model for v1.0 (open-core vs. source-available vs. commercial) — decide before wave 35 (integration hub/public APIs).
2. First real customer: identify by wave 20 to shape go-live scope at wave 30.
3. Which EU country is the payroll/localization reference (DE vs. UK)?
4. Hosting of the demo environments (which provider first, even though the stack is agnostic)?

13. Next steps in the Moonaan SDLC pipeline

1. **This PRD** is the requirements source.
2. Run `/sdlc:generate-architecture docs/PRD.md docs/architecture/` per phase to produce the architecture document and compiled fact sheets.
3. Run `/sdlc:generate-backlog` per wave to turn `waves.json` entries into epics/stories with dependency ordering (GitHub Issues are the backlog store for now; Jira when the Moonaan proxy is reachable).
4. Run `/sdlc:generate-code` per story.

Validation checklist (Scribe step 4)

- [x] Every functional requirement is traceable to a wave with a provable, testable, deployable output.
- [x] Every wave names the AX and SAP features it replaces (grounding).
- [x] Every wave links external references (standards, vendor docs, open-source implementations).
- [x] Non-functional requirements are measurable.
- [x] Sequencing and go-live milestones are explicit.
- [ ] Reviewed by @zdemanche.
- [ ] Accountant review of finance-wave fixtures (owner TBD).

Part 0A — The 10× Thesis: What Top-Tier Manufacturing ERPs Get Wrong and How Passage Fixes It

Passage is not a cheaper AX or a friendlier SAP. It emulates the full capability surface of the best manufacturing ERPs — SAP S/4HANA, Dynamics 365 Finance & Supply Chain, Infor LN / CloudSuite Industrial, Epicor Kinetic, IFS Cloud and Plex — and then removes the structural reasons those systems are slow, error-prone, expensive to use and closed. "10×" is measured, per wave, on four axes:

Axis	What we measure	Baseline (typical AX 2012 / D365 / S/4 deployment)	Passage target
Easier	Clicks and screens per task; training hours to competence; share of tasks completed without a super-user	Sales order with 5 lines: 40–60 clicks across 4 forms; AP clerk training 3–5 days	≤ 1/10 of clicks; a new user completes a guided task in < 30 min
Faster	Latency of the business answer (not the screen): MRP, inventory close, ATP, period close, report	MRP regeneration 2–8 h nightly; inventory close 1–6 h; month-end close 5–10 business days	MRP incremental < 2 min, full < 10 min; close continuous; statements real-time
More accurate	Reconciliation breaks per month; silent rounding/FX errors; duplicate master data; manual journal share	Subledger-to-GL breaks every close; 3–8% duplicate customers/vendors; 20–40% of journals are corrections	Zero unexplained breaks (checked daily by machine); < 0.5% duplicates; corrections < 5%
Open	Days to connect a new system; share of data reachable by API/event; integration projects per year	Weeks per interface (AIF/IDoc mapping projects); large share of logic reachable only from the UI	Any system on the network connects in < 1 day with no middleware; 100% of entities and events reachable

0A.1 The structural pain-point ledger

Each pain point below names the AX/SAP mechanism that causes it, the root cause, the Passage design that removes it, and the wave where it ships. Wave files repeat this at wave granularity under "Pain points solved (10× lens)".

A. Usability

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
U1	Dozens of forms to complete one business task; fields scattered across tabs and FastTabs	AX `SalesTable` form with 20+ tabs; SAP `VA01` header/item/schedule-line screens	Forms mirror tables, not tasks	Task-oriented screens generated from metadata with the *task* as the unit (order intake, receipt, close task); line grids with inline edit; the full document on one scrollable page	5, 27
U2	Lookups require knowing the code (vendor account, main account, item number)	`MainAccount` lookup in `LedgerJournalTrans`; SAP GL account search help	Keys are the primary identifier; search is secondary	Global typeahead on name/description/any attribute, ranked by recency and role; keys shown but never typed	5
U3	Personalization is per-user, fragile, lost on upgrade	AX usage data (`SysLastValue`); SAP GUI variants	Personalization stored outside metadata	Saved views are first-class, versioned, shareable, role-defaulted; survive upgrades because they target metadata ids	5
U4	Batch jobs hide status; users poll "Batch jobs" form to see if posting finished	`BatchJob`/`BatchJobHistory`; SAP `SM37`	Fire-and-forget batch framework with no UI contract	Every long-running action returns a tracked workflow with progress, ETA and a notification; the originating screen updates live	4, 5
U5	Error messages name internal objects ("Account structure does not allow this combination")	AX infolog; SAP message classes	Validation written for developers	Validation messages are metadata with user text, remediation link and the exact field; errors shown at the field as you type	3, 5
U6	Shop-floor and warehouse UIs are RF text menus or require a PC	AX `JmgRegistration` terminal; SAP ITSmobile/RF	UI frameworks predate touch and offline	Touch-first PWA kiosk and scanner apps with offline queues; one scan completes a step	25, 32
U7	Excel round-trips to do bulk edits	AX Excel add-in; SAP GUI copy/paste	No bulk-edit grid	Every list page supports multi-row inline edit, paste from spreadsheet with validation preview, and undo	5
U8	Approval requires logging in to the ERP	AX workflow work items; SAP Business Workplace (`SBWP`)	Workflow inbox is inside the client	Approvals in email/Slack/Teams/mobile with one-tap approve, full context and the same audit trail	4
U9	Copilot features are bolted on and read-only	D365 Copilot summaries; SAP Joule Q&A	AI layer has no safe action surface	Every API is an MCP tool under the user's permissions; agents draft, users approve; explain-this-record everywhere	47

B. Speed

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
S1	MRP runs overnight; planners work on stale data; a changed order waits until tomorrow	AX `ReqCalc` regeneration; SAP `MD01` total planning	Full regeneration over all items; pegging recomputed from scratch	Net-change planning engine keyed by events (order/receipt/BOM change) with incremental re-plan of affected low-level codes; full regen < 10 min for 50k items	22
S2	Inventory close/recalculation blocks posting for hours	AX `InventClosing`; SAP material ledger `CKMLCP`	Settlement of issues to receipts is a batch over all transactions	Running costing (continuous settlement on each issue for FIFO/avg) so close is a verification pass, not a computation; partitioned by item	20
S3	ATP is computed per line with locks; order entry slows under load	AX `InventOnHand`/CTP; SAP ATP with `VBBE` requirements	Row-level locking on on-hand and requirement tables	Cumulative ATP projection maintained from the inventory event stream; reads lock-free; promise computed in memory	27
S4	Month-end close takes 5–10 days of manual steps	AX period close checklist is a list; SAP closing cockpit sequences manual transactions	Postings are batch; reconciliations are manual	Continuous close: every subledger reconciles to the universal journal daily by machine; close checklist tasks auto-complete when their check passes	14, 16+
S5	Reports run against OLTP and time out, or against a cube that is a day old	AX SSRS on `GeneralJournalAccountEntry`; SAP BW loads	No streaming path from transaction to analytics	CDC → Iceberg lakehouse with p95 < 60 s lag; statements served from pre-aggregated balances rebuilt from the journal	11, 46
S6	Posting a 10k-line journal takes minutes and locks the period	`LedgerJournalCheckPost`; SAP `FB50` with `BSEG` inserts	Row-by-row posting through class hierarchies; synchronous number sequences	Append-only journal with batch insert, idempotency keys and asynchronous balance projection: 5,000 lines/s sustained	11
S7	Deploying a customization takes a release window	AX model store deploy with AOS restart; SAP transports with downtime	Code and metadata are compiled into the runtime	Metadata extensions apply live; code ships via GitOps with digest rollback in < 15 min	1, 3

C. Accuracy

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
A1	Subledgers drift from the GL (inventory value ≠ inventory account; AR aging ≠ AR control)	`InventSum` vs `GeneralJournalAccountEntry`; SAP `BSID` vs `SKB1` balances	Subledgers post to the GL through separate code paths with their own rounding	One posting API; subledgers are projections of the same event stream; daily machine reconciliation with zero unexplained breaks as a release gate	11, 16, 19, 30
A2	Rounding and FX differences appear as "penny" entries	AX `LedgerJournalTrans` rounding to accounting currency; SAP `BSEG` rounding differences	Rounding at line level then summing	Three-currency amounts stored per line; rounding applied once at document level with explicit rounding lines; ISO 4217 minor units enforced	12
A3	Duplicate customers/vendors/items	`CustTable` created freely; SAP `KNA1` duplicates across sales orgs	No dedup at creation; roles not layered on one party	One party per real-world entity with fuzzy-match blocking at creation (precision ≥ 0.95) and governed merge	7
A4	Wrong tax because tax group ↔ item tax group intersections are configured by hand	`TaxGroup` / `TaxItemGroup` intersection; SAP `FTXP` condition records	Determination is a configuration lattice with no test harness	Determination engine with a 500-scenario fixture pack reviewed by accountants; explain-this-tax on every line	13
A5	Negative inventory and phantom on-hand	`InventSum` allows negative physical when `InventModelGroup` permits; SAP `MMBE` vs `MSEG` timing	Projection updated by application code that can skip	On-hand is a projection of immutable inventory events; policy enforced at the event write; projection rebuilt from events and compared nightly	16
A6	Costing variances nobody can explain	`InventCostTrans` variances; SAP `KKS1`	Variance types computed after the fact with no linkage	Variance decomposition attached to each ended order with lot-size, quantity, price and substitution components that sum exactly	21, 39
A7	Master data changed without approval or trace	Direct edit of `VendTable` bank account; SAP `XK02`	Audit log optional (`SysDatabaseLog`) and slow	Field-level immutable audit from CDC for every entity; sensitive fields (bank accounts, credit limits) require workflow	5, 7

D. Openness and interoperability

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
O1	Each interface is a project (AIF service, IDoc mapping, custom data entity)	AX AIF/`DMF` entities; SAP `WE20` partner profiles, PI/PO	Integration surface is bespoke per object	Every entity and event is published automatically (OpenAPI, AsyncAPI, MCP) from metadata; new entity ⇒ new API with no work	3, 4, 35
O2	External systems poll for changes	`DMF` change tracking; SAP change pointers (`BDCP2`)	No event stream	Kafka topics per entity with CloudEvents, replay from any offset, webhooks for partners	4
O3	MES/PLM/SCADA need a separate middleware product	SAP MII, PI/PO; AX via BizTalk/Logic Apps	ERP speaks only its own protocols	Native OPC-UA/MQTT ingestion, EDI engine, Peppol, SFTP, MCP — eight integration surfaces with one auth model	25, 35
O4	Vendor cloud lock-in (LCS, RISE)	D365 only on Microsoft-managed Azure; S/4 Cloud on RISE	Platform tied to the vendor's operations	Any conformant Kubernetes; Helm charts; self-host or managed	1
O5	Customizations block upgrades	X++ overlayering; ABAP modifications	Extension model is code in the vendor's layer	Extensions are metadata and sanctioned hooks; additive API guarantees; upgrades are image promotions	3
O6	Analytics requires exporting to BYOD/BW	Entity store; BW extractors	No open analytical surface	Iceberg tables readable by any engine (Trino, Spark, DuckDB, Power BI, Tableau) with governed metrics	46

0A.2 Interoperability contract ("all systems on the network can use it")

Passage treats the ERP as a service on the customer's network, not a destination. Every capability is reachable through the same eight surfaces with one identity and one policy engine:

Surface	Used by	Auth	Guarantee
REST (OpenAPI 3.1, generated per entity)	Apps, scripts, partners	OIDC bearer token or API key, OPA-authorized	p95 < 50 ms simple reads; additive versioning
GraphQL (read model)	BI tools, portals	Same	Cross-entity reads without N+1
Kafka topics (CloudEvents, AsyncAPI)	MES, WMS, data platforms, other ERPs	mTLS + ACLs per tenant	At-least-once, ordered per key, 7-day replay
Webhooks	SaaS partners	Signed (HMAC) deliveries	Retries with backoff, dead-letter, replay
MCP	AI agents (Claude and others)	Delegated user token; agent ≤ user permissions	Every tool call audited
EDI (X12/EDIFACT via AS2/SFTP) and Peppol	Trading partners, tax authorities	Partner certificates	Acknowledgments, SLA monitor
OPC-UA / MQTT	Machines, PLCs, SCADA	Device certificates	10k events/min per plant, no loss on broker restart
File / SFTP	Banks, legacy systems	Keys in Vault	ISO 20022, BAI2, NACHA, CSV with schema validation

Rule: **if it is not reachable through these surfaces, it does not ship.** Each wave's API/event section is reviewed against this rule before close.

0A.3 How "deep knowledge of AX and SAP" is applied

We deliberately keep what those systems got right and name it in every wave's parity map:

- **From AX:** the role/duty/privilege security model; inventory dimensions (product/storage/tracking) as a first-class concept; `InventTrans` status semantics; trade-agreement search order; number sequences per legal entity; financial dimensions with account structures; coverage groups; lean production flows.
- **From SAP:** the universal journal (ACDOCA) as single source of truth; parallel ledgers and currency types; condition technique for pricing and tax; posting periods per module; EWM's location-directive rigor; QM inspection lots and usage decisions; results analysis for projects; Group Reporting eliminations.

And we remove the parts that exist only because of their era: batch-first processing, table-shaped forms, code-based extensions, closed integration layers, vendor-bound operations.

Part 0 — Phase 0: Foundation & Platform Kernel (waves 1–5)

0.0 Phase overview

Business outcome at phase exit. At the end of wave 5 nothing business-facing exists, but everything a business module needs does: a change merged to `main` reaches a Kubernetes environment with digest-pinned rollback in under 15 minutes; a user signs in once through a federated IdP and lands in a legal entity; every API call is authorized by OPA against AX-style roles/duties/privileges and fenced by Postgres row-level security; an entity declared in YAML becomes a table, an API, a client and a form without hand-written code; every state change is a CloudEvent on Kafka via a transactional outbox; approvals and batch jobs run on Temporal; number sequences support gap-free mode; every field change is in an immutable, hash-chained audit log; and a click in the shell can be followed through gateway, service, database and Kafka consumer in one trace. This layer is what AX calls the AOT, SysSecurity, batch and workflow frameworks, number sequences and SysDatabaseLog, and what SAP calls DDIC, PFCG, SM36/SM37, Business Workflow, SNRO and CDHDR/CDPOS — rebuilt as separately deployable services on the ADR-0001 stack.

Phase-exit sign-off criteria

- `make deploy-local` on a clean laptop produces a working k3d environment with all Phase 0 services in under 10 minutes; the same Helm/Argo manifests are healthy in the cloud `stage` cluster.
- Stakeholder demo executes, live, the wave-5 success metric from PRD §10: deploy a change → log in via a federated IdP → extend the Vendor-placeholder entity with a field and an enum value → run a 3-level approval → open the audit entry → open the trace in Tempo.
- Table-driven authorization suite (≥ 20 duties × allow/deny) and RLS cross-tenant suite are green; effective-permission report equals OPA decision for 100 sampled resources.
- Golden-generation suite for 10 reference entities is deterministic across 3 consecutive CI runs; additive v2 change passes, breaking change fails.
- Outbox chaos test shows zero lost and zero duplicated events over 10 kill cycles; 1M-row batch resumes after pod kill with 0 duplicates.
- Gap-free number sequence Jepsen-style test (10k concurrent requests) shows 0 gaps, 0 duplicates.
- 100% of generated forms pass axe WCAG 2.1 AA; audit entry visible within 2 s (p95); every service exports OTel traces, metrics and logs.
- SLO dashboards exist for every Phase 0 service with alert routes to the PagerDuty-compatible webhook.

Cross-wave data model summary

Entity	Owning service	Key relationships
Release, Environment	platform (wave 1, GitOps repo + registry DB)	Release → Environment (promotion history)
Tenant, LegalEntity	identity (wave 2; regenerated from metadata in wave 3)	Tenant 1→N LegalEntity; every row in every domain carries tenant_id
User, Role, Duty, Privilege, RoleAssignment	identity / authz (wave 2)	User N↔N Role via RoleAssignment scoped to LegalEntity; Role 1→N Duty 1→N Privilege
EntityDefinition, FieldDefinition, EnumDefinition, TypeDefinition, ExtensionPackage	metadata-registry (wave 3)	Entity 1→N Field; Field → Type or Enum; ExtensionPackage → Entity (additive)
OutboxEvent, TopicDefinition, SchemaVersion	every service / events (wave 4)	Outbox row → CloudEvent; TopicDefinition → SchemaVersion (from EntityDefinition)
WorkflowDefinition, WorkflowInstance, WorkItem, BatchJob, BatchRun, BatchTask	workflow (wave 4)	Instance → Definition version; WorkItem → Instance + assignee User; BatchRun 1→N BatchTask
NumberSequence, NumberSequenceAllocation	numseq (wave 5)	Sequence scoped to Tenant/LegalEntity; Allocation → Sequence (gap-free audit)
AuditEntry, AuditField	audit (wave 5)	Entry → any entity/record; hash chain per tenant
SavedView, Personalization, ImportJob	shell-bff / importexport (wave 5)	SavedView → EntityDefinition + User; ImportJob → EntityDefinition
RuleSet, DecisionTable, DecisionRow, UsageSnapshot	metadata-registry (wave 3)	RuleSet 1→N DecisionTable 1→N DecisionRow; rows may come from an ExtensionPackage
OutboundMessage, MessageTemplate, NotificationPreference	notify (wave 4)	Message → Template version + recipient User; Preference per User × event class
Attachment, AttachmentVersion, AttachmentLink, DocumentType, FileType	docs (wave 5)	Attachment 1→N Version; Attachment N↔N any record via Link; Attachment → DocumentType
AlertRule, AlertEvent, StandardText	shell-bff (wave 5)	AlertRule → EntityDefinition; AlertEvent → Rule + record + recipient; StandardText referenced by MessageTemplate

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
dev.passage.identity.user.created.v1 / .deactivated.v1	identity	audit, workflow (assignee cache), shell	user_id, tenant_id, email, status
dev.passage.authz.role-assignment.granted.v1 / .revoked.v1	authz	audit, OPA bundle builder	user_id, role_id, legal_entity_ids, effective_from/to
dev.passage.authz.policy.compiled.v1	authz	OPA sidecars (bundle pull), audit	bundle_version, sha256, role_count
dev.passage.metadata.entity.published.v1	metadata-registry	generator, events (topic/schema), shell (form cache)	entity_name, version, diff summary
dev.passage.metadata.extension.applied.v1	metadata-registry	generator, audit, shell	package_id, entity_name, fields_added, enum_values_added
dev.passage.workflow.instance.submitted.v1 / .completed.v1	workflow	owning entity service, audit, shell inbox	instance_id, entity, record_id, outcome
dev.passage.workflow.workitem.assigned.v1 / .escalated.v1	workflow	shell inbox, notifications	workitem_id, assignee_id, due_at
dev.passage.batch.run.completed.v1 / .failed.v1	workflow (batch)	shell job monitor, alerting	job_id, run_id, rows_processed, error
dev.passage.audit.entry.recorded.v1	audit	shell audit viewer (live), analytics (wave 46)	entry_id, entity, record_id, operation, hash
dev.passage.rules.ruleset.published.v1	metadata-registry	every service holding compiled rule trees, audit	code, version, tables, row_count
dev.passage.notify.message.sent.v1 / .failed.v1 / .bounced.v1	notify	audit, alerting, identity	message_id, channel, template_code, recipient_user_id
dev.passage.docs.attachment.available.v1 / .purged.v1	docs	shell, audit, OCR capture (wave 18)	attachment_id, version_no, document_type, links
dev.passage.alerts.alert.fired.v1	shell-bff (alert evaluator)	notify, shell	rule_id, entity, record_id, recipient_id

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Metadata kernel under-specified before business waves need it	Medium	High	10 reference entities in wave 3 include the shapes of Party, Product, JournalLine; golden tests frozen at phase exit	Product
Two-person team cannot operate Kafka + Temporal + Keycloak + CNPG	Medium	High	All operators installed via Helm umbrella; runbooks per component; k3d parity so every incident reproduces locally	Engineering
OPA policy compile from roles produces unexpected denies	Medium	Medium	Effective-permission viewer and `decision` debug endpoint; deny-reason strings in every 403	Engineering
RLS misconfiguration leaks tenant data	Low	Critical	`FORCE ROW LEVEL SECURITY` on every table; app role NOBYPASSRLS; CI test queries every table cross-tenant	Security
Gap-free sequence becomes a throughput bottleneck	Medium	Medium	Continuous mode only where fiscal law requires it; non-continuous is the default with block pre-allocation	Engineering
Audit volume from CDC overwhelms Postgres	Medium	Medium	Monthly partitions; audit flag per field; PII masking at capture; 10-year retention tiered to object storage from wave 46	Engineering

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0.1 Wave 1: Delivery Pipeline & Cluster Baseline

0.1.1 Scope and business context

AX ships code as LCS deployable packages built on a build VM and applied with AxUpdateInstaller; model store exports (`AxUtil export`) and database restores are the rollback. SAP moves ABAP through STMS transport requests (SE09/SE10 workbench requests, DEV → QAS → PRD routes, `tp import`), with back-out by transport of copies. Both couple release to a vendor tool and give no software bill of materials, no cryptographic signature and no sub-hour rollback. Wave 1 replaces this with a GitOps pipeline on any conformant Kubernetes. At the end of the wave an engineer can merge a pull request and watch the change reach `dev`, promote the same image digest to `stage`, and roll back to the previous digest, each in under 15 minutes. Out of scope: multi-region, production-grade HA Postgres, secrets rotation policy, cost reporting, and any business service other than the `hello-ledger` placeholder.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Applying an LCS deployable package to a D365 sandbox takes 2–4 h of downtime and a service request for production; a failed apply means a database restore	Package = whole-system binary; environment managed by vendor tooling	Per-service OCI image + Helm chart, promoted by digest through Argo CD; rolling update with no downtime	Merge → running in env: 4 h → < 15 min; downtime per release: hours → 0
STMS transport import to PRD (STMS_IMPORT) is a manual queue with no automatic back-out; a bad transport is reversed by hand-building a transport of copies	Transports carry object deltas, not immutable artifacts	Immutable digests; rollback is a one-line overlay revert	Rollback: 1–3 h and a Basis ticket → < 5 min, self-service
No SBOM or signature for an AX model or SAP transport; security teams cannot answer "which CVE is in production"	Build tooling predates supply-chain attestation	Syft SBOM, Grype/Trivy gate, cosign signature verified at admission	CVE-to-affected-environment lookup: days of manual inventory → one query; unsigned artifacts in prod: 0
OneBox/VHD developer environments (AX) or NetWeaver dev editions (SAP) take days to provision and drift from production	Monolithic app server image per developer	`make deploy-local` on k3d with the same charts as cloud	New engineer to a running environment: 2–5 days → < 10 min
Customers cannot host on their own infrastructure without LCS (D365) or RISE (S/4)	Vendor-managed cloud is the only supported topology	One OpenTofu interface across k3d/EKS/GKE/AKS/bare-metal; nothing provider-specific in the critical path	Supported deployment targets: 1 → 5 with identical manifests

0.1.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Package a release	LCS deployable package, `AxUtil export` model store, AXModel files	STMS transport request (E070/E071), `tp export`	OCI image (digest) + Helm chart version in `deploy/`	improved
Build and validate	Build VM, `AxBuild.exe`, Best Practice checks	SE80 activation, ATC (code inspector SCI), ABAP unit	GitHub Actions: lint, unit, container build, SBOM, scan	improved
Environment tiers	LCS dev/test/UAT/prod topology	STMS transport route DEV→QAS→PRD, STMS_QA	Argo CD app-of-apps per environment: dev, stage, prod	full
Promote between tiers	LCS "apply package" to the next environment	STMS import queue, `tp import`, STMS_IMPORT	`passage promote --to stage --digest sha256:…` opens a PR against the env overlay	full
Rollback	Restore DB + model store, "AOT revert"	Transport of copies back-out, SNOTE de-implement	Argo CD rollback to prior digest, `passage rollback --env stage`	improved
Code signing / trust	Strong-name model signing (`AxUtil … /key`)	Digitally signed SAP Notes, TMS trusted systems	cosign keyless signatures, policy-controller admission denies unsigned images	improved
Hotfix path	KB hotfix via LCS, binary update	SNOTE note implementation, SPAU	Hotfix branch → cherry-pick → same pipeline, same gates	full
Deployment log / history	LCS environment history, AxUpdateInstaller logs	STMS import log, SLOG/ALOG	Argo CD history + GitHub Deployments API + `platform.release` table	improved
Local dev environment	OneBox VM / VHD from LCS	SAP NetWeaver developer edition, mini-basis	`make deploy-local` on k3d, same charts as cloud	improved

0.1.3 Functional requirements

- **FR-1.1 Monorepo layout** — Must. The system shall keep all Passage source in one repository with the layout `services/go/<svc>`, `services/ts/<svc>`, `services/py/<svc>`, `web/shell`, `web/remotes/<name>`, `deploy/charts/<svc>`, `deploy/envs/{dev,stage,prod}`, `infra/modules/<provider>`, `metadata/entities`, `docs/`.
  - Notes: each service owns a `Dockerfile`, `Makefile`, `CODEOWNERS` entry and a Helm chart; CI path filters build only changed services; a `go.work` and `pnpm` workspace root pin toolchain versions (Go 1.23, Node 22, Python 3.12).
- **FR-1.2 Cluster bootstrap module** — Must. The system shall provide an OpenTofu module `infra/modules/cluster` with one variable interface (`provider ∈ {k3d, eks, gke, aks, baremetal}`, `name`, `node_pools[]`, `kubernetes_version`) that outputs a kubeconfig and installs the base add-ons.

    - Notes: base add-ons are Argo CD, Envoy Gateway, cert-manager, External Secrets Operator, CloudNativePG operator, sigstore policy-controller, metrics-server; provider modules contain only networking, node pools and storage classes; k3d module runs with 3 agent nodes and 8 GB RAM.
- **FR-1.3 Argo CD app-of-apps** — Must. The system shall define a root Application per environment that reconciles child Applications from `deploy/envs/<env>/apps/*.yaml`, each pinning a chart version and an image digest.
    - Notes: sync policy automated for dev, manual-approve PR for stage and prod; prune and self-heal enabled; sync waves order operators (–2), databases (–1), services (0), gateway routes (1).
- **FR-1.4 Digest-only promotion** — Must. The system shall promote images by immutable digest; tags are informational only.
    - Notes: `passage promote` rewrites `image.digest` in the target overlay and opens a PR with the SBOM diff and scan summary; promotion to prod requires the digest to have run in stage for ≥ 60 minutes with no SLO burn alert.
- **FR-1.5 CI pipeline** — Must. The system shall run on every PR and on `main`: lint (golangci-lint, eslint, ruff), unit tests with coverage ≥ 70% per service, container build (BuildKit, reproducible), Syft SBOM (SPDX 2.3), Grype and Trivy scan, cosign sign and attest.
    - Notes: required checks block merge; `main` builds push to the registry under `ghcr.io/passage/<svc>@sha256:…`; median PR pipeline ≤ 8 minutes; cache keyed by lockfile hash.
- **FR-1.6 Vulnerability gate** — Must. The system shall fail the build on any CRITICAL finding, or any HIGH finding older than 14 days with a fix available, unless an exception file `.vuln-exceptions.yaml` lists the CVE with an expiry ≤ 30 days.
- **FR-1.7 Signature verification admission** — Must. The system shall reject any pod whose image lacks a valid cosign signature from the repository identity (`https://github.com/passage/*` OIDC issuer) in namespaces labelled `passage.dev/verified=true`.
    - Notes: SLSA provenance attestation level 2 at wave 1, level 3 target by wave 48; a negative test deploys an unsigned image and asserts the admission denial message.
- **FR-1.8 `hello-ledger` placeholder** — Must. The system shall deploy a Go service with one Postgres table that proves the path: migration on start, REST endpoints, health probes, TLS ingress, metrics endpoint.
- **FR-1.9 Local parity** — Must. `make deploy-local` shall stand up k3d, Argo CD and `hello-ledger` behind `https://app.local.passage.dev` with a locally trusted cert-manager CA in under 10 minutes on a 16 GB laptop.
- **FR-1.10 Rollback** — Must. The system shall roll back any environment to the previous release by digest in ≤ 5 minutes via `passage rollback --env <env>` (which reverts the overlay commit) or the Argo CD history UI.
    - Notes: database migrations in Phase 0 are additive-only so rollback never requires a down migration; a rollback records a `Release` row with `kind=rollback`.
- **FR-1.11 Secrets and config** — Must. The system shall source all secrets from Vault through External Secrets Operator; no secret value appears in Git, Helm values or CI logs.
    - Notes: config via ConfigMap generated from `deploy/envs/<env>/values.yaml`; secret names follow `passage/<env>/<svc>/<key>`.

0.1.4 Data model

Release (`platform.release`, registry DB, also materialized in Git history)

Field	Type	Required	Notes
id	uuid	yes	PK
service	text	yes	matches chart name
image_digest	text	yes	`sha256:` + 64 hex; unique with service
git_sha	char(40)	yes	commit that produced the build
chart_version	text	yes	semver
sbom_url	text	yes	SPDX artifact in registry
signed	bool	yes	cosign verification result
status	enum	yes	built, scanned, signed, deployed_dev, deployed_stage, deployed_prod, superseded, rolled_back
created_at	timestamptz	yes	

Environment (`platform.environment`)

Field	Type	Required	Notes
name	text	yes	PK; dev, stage, prod, local-
cluster_provider	enum	yes	k3d, eks, gke, aks, baremetal
argo_root_app	text	yes	app-of-apps name
promotion_policy	enum	yes	auto, pr_approval
soak_minutes	int	yes	default 0 dev, 60 stage→prod
base_url	text	yes	`https://app.<env>.passage.dev`

HelloLedgerEntry (`hello_ledger.entry`)

Field	Type	Required	Notes
id	uuid	yes	PK
memo	varchar(200)	yes	
amount_minor	bigint	yes	integer minor units, USD/EUR examples: 12345 = $123.45
currency	char(3)	yes	ISO 4217
created_at	timestamptz	yes	default now()
created_by	text	yes	`anonymous` until wave 2

Invariants: one `Release` row per (service, image_digest); at most one `deployed_<env>` release per service per environment; `Environment.name` is immutable. Indexing: `release(service, created_at desc)`; `entry(created_at)`.

0.1.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	/healthz, /readyz	Liveness / readiness	n/a	none

Method	Path	Purpose	Idempotency	Auth duty
GET	/v1/version	git sha, digest, chart version	n/a	none
GET	/v1/hello-ledger/entries?limit=&cursor=	List entries	n/a	none (token required from wave 2)
POST	/v1/hello-ledger/entries	Create entry	`Idempotency-Key` header, 24 h window	Maintain hello ledger
GET	/v1/hello-ledger/entries/{id}	Fetch one	n/a	none
GET	/v1/platform/releases?service=&env=	Release history	n/a	View platform releases
POST	/v1/platform/releases/{id}/promote	Open promotion PR	release id + env	Maintain platform releases

CLI commands `passage promote`, `passage rollback`, `passage status` call these endpoints or edit the overlay directly.

Event	Trigger	Payload fields	Consumers
dev.passage.platform.release.built.v1	CI publishes signed image	release_id, service, image_digest, git_sha, sbom_url	registry DB writer
dev.passage.platform.release.promoted.v1	Argo sync Healthy in target env	release_id, environment, actor, previous_digest	audit (wave 5), shell notifications
dev.passage.platform.release.rolledback.v1	Rollback sync Healthy	release_id, environment, actor, reason	audit, alerting
dev.passage.hello.entry.created.v1	Entry insert	entry_id, amount_minor, currency	none (proves outbox path in wave 4)

In wave 1 events are written as CloudEvents JSON to stdout and to the registry DB; Kafka transport replaces the sink in wave 4 without changing types.

0.1.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Argo CD UI (per env)	Sync, view history, rollback	app, health, sync status, revision, digest	Platform engineer (Argo RBAC via GitHub team)
GitHub PR checks	View gate results, SBOM diff	lint, unit, build, scan, sign	repository write
hello-ledger page `/`	Create entry, list entries	memo, amount, currency, created_at	none

Interaction rules: promotion PRs are templated with a checklist (soak time met, scan clean, rollback target named); CI annotates failing lines inline; `make deploy-local` prints a single URL table at the end.

0.1.7 Business rules and edge cases

- **BR-1.1** Release status machine: built → scanned → signed → deployed_dev → deployed_stage → deployed_prod → superseded; any deployed_* may go to rolled_back; no transition skips `signed`.
- **BR-1.2** A tag may be rewritten; a digest may not. Any overlay referencing a tag fails the `deploy-lint` CI check.
- **BR-1.3** Promotion to prod requires soak ≥ `Environment.soak_minutes` in stage with zero `SLOBurnRateHigh` alerts; the check is computed in UTC with ISO 8601 timestamps.
- **BR-1.4** Two concurrent promotions of the same service to the same environment: the second PR is auto-closed with a comment pointing at the first.
- **BR-1.5** A vulnerability exception past its expiry date fails the build as if no exception existed.
- **BR-1.6** Rollback target is always the last `deployed_<env>` release with a different digest; if none exists the command refuses.
- **BR-1.7** Images are retained for 180 days or while referenced by any environment, whichever is longer; SBOMs retained 7 years.
- **BR-1.8** A hotfix branch must be cut from the prod digest's `git_sha`; CI refuses `hotfix/*` branches whose merge base is not a deployed_prod release.

0.1.8 Acceptance criteria

- **AC-1.1** Given a PR with an injected lint error, When CI runs, Then the `lint` check fails and merge is blocked. (FR-1.5)
- **AC-1.2** Given a PR with a failing unit test, When CI runs, Then `unit` fails and the failing assertion is annotated inline. (FR-1.5)
- **AC-1.3** Given a clean macOS or Linux laptop with Docker, When `make deploy-local` runs, Then Argo CD and hello-ledger are reachable over TLS within 10 minutes. (FR-1.9)
- **AC-1.4** Given a signed image in dev, When `passage promote --to stage` is merged, Then Argo shows Healthy with that digest within 15 minutes of merge. (FR-1.3, FR-1.4)
- **AC-1.5** Given stage running digest B after A, When `passage rollback --env stage` runs, Then Argo history shows A active within 5 minutes and a `rolled_back` Release row exists. (FR-1.10)
- **AC-1.6** Given an unsigned image applied to a verified namespace, When the pod is created, Then admission denies it with a message naming the missing signature. (FR-1.7)
- **AC-1.7** Given an image with one CRITICAL CVE, When CI runs, Then the scan gate fails; adding a valid exception with expiry in 10 days passes it. (FR-1.6)
- **AC-1.8** Given a secret in Vault, When hello-ledger starts, Then it reads `DATABASE_URL` via ESO and the value is absent from Git, values files and CI logs (grep check). (FR-1.11)

0.1.9 Test plan and fixtures

Fixtures: `fixtures/pipeline/negative-lint.go` (committed lint failure behind a build tag), `fixtures/pipeline/unsigned-image.yaml`, `fixtures/pipeline/cve-exception.yaml` (expired and valid), `fixtures/hello-ledger/entries-100.json` (100 entries in USD and EUR).

Test	Type	Target	Pass criterion
Lint/unit negative	CI contract	pipeline	both injected failures block merge
Local bootstrap	E2E	`make deploy-local`	< 10 min, all pods Ready, TLS valid
Promote and rollback	E2E	Argo CD	both < 15 min, history correct
Admission unsigned	Negative	policy-controller	pod denied
SBOM presence	Contract	registry	SPDX attached to every digest
hello-ledger API	Unit + contract	handlers	100% of OpenAPI paths covered
Build reproducibility	Property	BuildKit	same source → same digest twice

Reconciliation: `Release` rows with status `deployed_<env>` equal the digests in each env overlay and the running pods' image IDs (nightly check).

0.1.10 Deployment and operations

Charts: `deploy/charts/platform-base` (umbrella for operators), `deploy/charts/hello-ledger`. Config: `values.yaml` per env; env vars `PASSAGE_ENV`, `OTEL_EXPORTER_OTLP_ENDPOINT` (used from wave 5), `DATABASE_URL` (ESO). Migrations: Atlas, additive-only, run as an init job. Feature flags: none. Scheduled jobs: nightly release reconciliation. SLOs: pipeline p50 ≤ 8 min, promotion merge→Healthy p95 ≤ 15 min, hello-ledger availability 99.9%. Alerts: `ArgoAppDegraded` > 10 min, `PipelineFailureRate` > 20% over 1 h, `AdmissionDeniedSpike`. Runbook entries: Argo out of sync, failed migration job, expired local CA. Rollback: `passage rollback`, then verify `/v1/version`.

0.1.11 Migration from AX / SAP

Nothing is migrated. LCS packages, AX model store layers and SAP transport requests contain code (X++/ABAP), which PRD §2 excludes. The only carry-over is organizational: the STMS DEV→QAS→PRD route maps to dev→stage→prod and LCS environment names are recorded in `Environment.description` for operator familiarity.

0.1.12 Dependencies, risks, open questions

Depends on: nothing (first wave).

Risk	Likelihood	Impact	Mitigation
k3d cannot run the full stack in 16 GB by wave 5	Medium	Medium	Profiles: `make deploy-local PROFILE=min` omits observability
Keyless cosign requires outbound access to Fulcio/Rekor	Medium	Low	Private sigstore instance documented for air-gapped installs
GitHub Actions minutes cost	Low	Low	Self-hosted runner module in `infra/`

Open questions: (1) First cloud provider for `dev/stage` (PRD §12 Q4) — default EKS; decide by wave 2 start. (2) Private registry vs ghcr.io for customers — default ghcr.io with mirror instructions; decide by wave 5. (3) Mesh (none vs Linkerd) for mTLS — default none, gateway-to-service mTLS only; decide by wave 4.

0.1.13 Parameters and setup

Platform parameters live in Git (`deploy/envs/<env>/values.yaml`, `infra/modules/cluster/variables.tf`, repository settings) rather than in a database form; "scope" below names the file or object that owns the value.

Parameter	Scope	Type	Default	Allowed values	Effect	AX
`cluster.provider`	cluster (OpenTofu variable)	enum	`k3d`	k3d, eks, gke, aks, baremetal	selects the provider sub-module for networking, node pools, storage classes	LCS topo host pren
`cluster.kubernetes_version`	cluster	semver	1.31	n–2 … n of upstream	pins the control plane and node pools	LCS upd
`cluster.node_pools[]`	cluster	list of {name, size, count, labels}	3 × 8 GB (k3d)	provider-specific	sizes compute; `passage.dev/workload` label routes stateful sets	LCS sizi
`environment.promotion_policy`	environment (`platform.environment`)	enum	auto (dev), pr_approval (stage, prod)	auto, pr_approval	auto-sync vs. PR with required reviewers	LCS pack appr
`environment.soak_minutes`	environment	int	0 dev, 60 stage	0–1,440	minimum time a digest must run in the prior tier before promotion	non
`argo.sync.self_heal`, `argo.sync.prune`	environment overlay	bool	true, true	true/false	drift corrected automatically; removed manifests deleted	non
`argo.sync_waves`	overlay annotation	ordered ints	operators –2, databases –1, services 0, routes 1	–10 … 10	start order per sync	AxU step
`ci.unit_coverage_min_pct`	repository (`.passage-ci.yaml`)	int	70	0–100	fails the `unit` check below threshold	Best chec
`ci.vuln.fail_on`	repository	enum set	critical, high_with_fix_older_than_14d	critical, high, high_with_fix_older_than_14d, medium	scan gate policy (FR-1.6)	non
`ci.vuln.exception_max_days`	repository	int	30	1–90	maximum validity of a `.vuln-exceptions.yaml` entry	non
`ci.sbom.format`	repository	enum	spdx-2.3	spdx-2.3, cyclonedx-1.6	SBOM attestation format	non
`admission.verified_namespace_label`	cluster (policy-controller `ClusterImagePolicy`)	string	`passage.dev/verified=true`	label selector	namespaces where unsigned images are denied	stro sign requ mod /ke
`admission.oidc_identity`	cluster	regex	`https://github.com/passage/.*`	issuer + subject regex	identity accepted by cosign keyless verification	non
`registry.image_retention_days`	registry	int	180	30–3,650	untagged, unreferenced digests deleted after this	LCS rete
`registry.sbom_retention_years`	registry	int	7	1–10	SBOM and attestation retention	non
`secrets.path_template`	cluster (ESO `ClusterSecretStore`)	string	`passage/<env>/<svc>/<key>`	Vault KV v2 path	where every service secret is read from	non / AO

Parameter	Scope	Type	Default	Allowed values	Effect	AX
`local.profile`	developer (`make deploy-local PROFILE=`)	enum	full	min, full	`min` omits observability and Temporal UI to fit 16 GB	One size
`hello_ledger.idempotency_window_hours`	service values	int	24	1–168	replay window for `Idempotency-Key`	none

0.1.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Release history	list inquiry	service, environment, status, date range	service, digest (short), git sha, chart version, status, created_at, signed, sbom link	`platform.release`	LCS Environment details → History / Applied updates	STMS import history (STMS_IMPORT → Goto → Import history), SE10 request log
Environment status	drill-down	environment	app, health, sync status, revision, digest, last sync, soak elapsed	Argo CD Application CRs + `platform.environment`	LCS environment monitoring page	STMS system overview, SM51
Deployment reconciliation	operational report (nightly)	environment	service, overlay digest, running digest, release status, mismatch flag	overlay files vs. pod image IDs vs. `platform.release`	LCS "Environment details → Version" check	STMS import queue vs. SE03 object directory comparison
CVE exposure by environment	operational report	severity, environment, fixed-available flag	CVE id, package, version, fixed version, digest, environment, first seen	Grype/Trivy results attached to each digest	none	Solution Manager System Recommendations (SysRec) for security notes
Vulnerability exceptions	list inquiry	expiry window	CVE, service, justification, approver, expires_at, days left	`.vuln-exceptions.yaml` (parsed in CI)	none	SNOTE note status / RSECNOTE
Pipeline statistics	KPI	period (day/week), branch	runs, median duration, p95 duration, failure rate by check	GitHub Actions API	Azure DevOps build pipeline analytics (D365)	STMS import log statistics, ChaRM cycle reports
Admission denials	operational report	namespace, period	pod, image, reason (unsigned / wrong issuer), time	policy-controller Kubernetes events	none	TP system log (SLOG/ALOG) rejected imports
Service version page	drill-down	service	git sha, digest, chart version, build time, Go/Node version	`GET /v1/version` of each service	Help → About (kernel and application build numbers)	System → Status (component versions, SPAM/SAINT level)
Hotfix lineage	list inquiry	service	hotfix branch, base prod digest, cherry-picked commits, status	Git refs + `platform.release`	LCS hotfix KB list, "Applied hotfixes"	SNOTE implemented notes, SPAU
Image retention candidates	operational report	registry	digest, service, last referenced env, age, SBOM present	registry API + `platform.release`	LCS asset library	none

0.1.15 Printed and electronic documents

No printed documents. Electronic artifacts produced by the wave:

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
SBOM	every `main` build	SPDX document: creator, package list (name, version, license, PURL), relationships, digest it describes	SPDX 2.3 JSON, attached to the image as a cosign attestation	identified by image digest, no sequence	retained 7 years (BR-1.7); evidence for wave 48 SOC 2 CC8.1
Provenance attestation	every signed build	in-toto statement: builder id, source repo + sha, materials, build config	in-toto JSON (SLSA v1), Rekor transparency log entry	Rekor log index	retained with the image; immutable in Rekor
Promotion PR summary	`passage promote`	title (service, from → to env, digest), checklist (soak met, scan clean, rollback target), SBOM diff table, scan summary	GitHub PR body (Markdown)	PR number	retained with repository history
Release notes (per promotion to prod)	PR merge to prod overlay	commits since previous prod digest grouped by Conventional Commit type, linked issues	Markdown in GitHub Deployment description	release id	none

0.1.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Release reconciliation (overlay vs. running vs. DB)	nightly 02:00 UTC	environment	re-runnable; writes a report row per (env, service) keyed by date	< 2 min	report row `mismatch=true`; no automatic correction	`ReleaseDriftDetected` (ticket)
Image retention sweep	weekly Sunday 03:00 UTC	service	deletes only digests not referenced by any `platform.environment`; dry-run list stored first	< 10 min	abort on registry 5xx; rerun next week	`RegistryRetentionFailed` (notify)
Deployed-digest re-scan	daily 04:00 UTC	environment × service	scan result keyed by (digest, scanner db version)	< 15 min for 30 images	partial results kept; missing scans listed	`NewCriticalCVEInProd` (page during business hours)
Vulnerability exception expiry check	daily 05:00 UTC	repository	stateless	< 1 min	opens a GitHub issue per expiring exception (≤ 7 days)	`VulnExceptionExpiring` (notify)
Local CA renewal (k3d)	on `make deploy-local`, and every 300 days	developer machine	reissues only when < 30 days left	< 30 s	prints instruction to re-trust the CA	n/a
Rekor inclusion verification	weekly	service	verifies each prod digest's signature entry is still present	< 5 min	alert; no deployment change	`SignatureVerificationFailed` (page)

0.1.17 Security duties and privileges

Wave 1 has no OPA-enforced API yet (wave 2); the duties below are enforced by GitHub branch protection, CODEOWNERS, Argo CD RBAC (mapped from GitHub teams) and Vault policies, and are re-declared as Passage duties in wave 2 so the same names hold later.

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
View platform releases	`GET /v1/platform/releases`, Argo CD read-only UI, `passage status`	none	Platform engineer, Release manager, Auditor
Maintain platform releases	`POST /v1/platform/releases/{id}/promote` (dev → stage), `passage promote --to stage`, merge to `deploy/envs/dev`, `deploy/envs/stage`	with Approve production promotion (same person cannot author and sole-approve the prod PR; enforced by required reviewers ≠ author)	Platform engineer
Approve production promotion	review + merge of PRs touching `deploy/envs/prod` (CODEOWNERS `@passage/release-managers`)	with Maintain vulnerability exceptions	Release manager
Rollback environment	`passage rollback --env <env>`, Argo CD rollback action	none (rollback is always allowed to anyone who can promote to that env)	Platform engineer (dev, stage), Release manager (prod)
Maintain vulnerability exceptions	edit `.vuln-exceptions.yaml` (CODEOWNERS `@passage/security`)	with Approve production promotion	Security engineer
Maintain cluster configuration	`tofu apply` on `infra/`, edit `deploy/charts/platform-base`	with Maintain secrets (one person should not hold both cluster admin and Vault root)	Platform engineer
Maintain secrets	Vault policy `passage/<env>/*` write; ESO `ClusterSecretStore` edit	with Maintain cluster configuration	Security engineer
Maintain hello ledger	`POST /v1/hello-ledger/entries` (token-gated from wave 2)	none	Platform engineer (demo only)

0.1.18 Message catalog

Codes `PLT-01xx` are emitted by CI checks, the `passage` CLI, the admission controller and the hello-ledger service; CI messages are also written as GitHub check annotations.

Code	Severity	User text	Cause	Remediation	Raised by
PLT-0101	error	Overlay references image by tag `{tag}`; digests are required	`image.tag` or `image:tag` found in `deploy/envs/**`	replace with `image.digest: sha256:…` from the release history	BR-1.2, FR-1.4
PLT-0102	error	Promotion to prod blocked: digest ran {minutes} of {required} soak minutes in stage	soak not met	wait, or lower `soak_minutes` with Release manager approval	BR-1.3
PLT-0103	error	Promotion to prod blocked: SLO burn alert `{alert}` fired during soak	`SLOBurnRateHigh` within the window	investigate the alert; re-soak after fix	BR-1.3
PLT-0104	warning	A promotion of {service} to {env} is already open (#{pr}); this PR was closed	concurrent promotion	wait for the first PR	BR-1.4
PLT-0105	error	No rollback target: no earlier `deployed_{env}` release with a different digest	first release in the environment	deploy an earlier digest explicitly with `passage promote --digest`	BR-1.6
PLT-0106	error	Vulnerability exception for {cve} expired on {date}	exception past expiry	renew with new justification (≤ 30 days) or fix	BR-1.5, FR-1.6
PLT-0107	error	Scan gate failed: {n} CRITICAL / {m} actionable HIGH findings	FR-1.6 policy	upgrade the package or add a justified exception	FR-1.6
PLT-0108	error	Pod denied: image {image} has no valid cosign signature from `{identity}`	admission policy	build through CI; never push hand-built images to verified namespaces	FR-1.7
PLT-0109	error	Hotfix branch `{branch}` is not based on a deployed_prod release	wrong merge base	`git rebase --onto <prod sha>`	BR-1.8
PLT-0110	error	Unit coverage {pct}% below minimum {min}% for {service}	`ci.unit_coverage_min_pct`	add tests	FR-1.5
PLT-0111	error	Possible secret committed: {rule} at {path}:{line}	gitleaks rule hit	remove, rotate the secret in Vault, force-push is not allowed; add to history rewrite ticket	FR-1.11
PLT-0112	error	Migration job for {service} failed: {atlas error}	Atlas apply error	fix migration; additive-only rule means no down migration is needed	FR-1.10
PLT-0113	warning	Local bootstrap needs 16 GB RAM; {gb} GB detected — using PROFILE=min	insufficient memory	accept reduced profile or free memory	FR-1.9
PLT-0114	info	Idempotent replay: returning the entry created at {time} for this Idempotency-Key	duplicate POST within 24 h	none	FR-1.8
PLT-0115	error	Build not reproducible: digest {a} ≠ {b} for the same source	non-deterministic build input	pin the input (timestamp, dependency)	0.1.9 reproducibility test

0.1.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
PR pipeline (lint → sign)	40 runs/day, peak 10 concurrent	6 min / 8 min / 12 min	20 concurrent runners	CI cache ≤ 20 GB (LRU)	path-filtered per service; cache keyed by lockfile hash	queued runs wait; no timeouts under 60 min
Promotion merge → Argo Healthy	10/day across envs	5 min / 15 min / 20 min	n/a	`platform.release` ≈ 10 k rows/year (≈ 5 MB)	`release(service, created_at desc)`	Argo reconciles in order; sync waves prevent partial start
Rollback	1/week	2 min / 5 min / 8 min	n/a	n/a	n/a	same as promotion
Admission webhook (signature verify)	500 pod creations/day, 50/min at rollout	50 ms / 200 ms / 500 ms	20/s	none	verification result cached per digest for 1 h	failurePolicy=Fail in verified namespaces; cached digests still admitted
hello-ledger `POST /entries`	10 k/day (demo)	10 ms / 40 ms / 80 ms	200 rps	≈ 1 GB/year at 10 M rows	`entry(created_at)`	Postgres connection pool 20; 503 beyond queue depth 100
Registry pulls at rollout	30 images × 3 nodes per env	5 s / 20 s / 40 s per image	100 MB/s	images ≈ 200 GB/year before retention	retention sweep (0.1.16)	node image cache; rollouts slower, not failed
`make deploy-local`	5/day per engineer	7 min / 10 min / 12 min	n/a	15 GB disk per environment	n/a	PROFILE=min

0.1.20 Localization and compliance hooks

- Nothing in this wave varies by country or language; all timestamps are UTC ISO 8601 and all CLI/CI messages are English only (label packs for CLI output are not planned).
- Data residency: a customer runs the `infra/modules/cluster` module in the region of their choice; nothing in `deploy/` assumes a region. Wave 48 country packs do not touch this wave; wave 50 (multi-region) adds region-aware overlays under `deploy/envs/<env>-<region>`.
- Air-gapped and export-controlled sites: registry mirror instructions (open question 2) and a private sigstore (Fulcio/Rekor) profile in `infra/modules/sigstore-private`; `admission.oidc_identity` is the extension point.
- Supply-chain regulation (EU Cyber Resilience Act, US EO 14028): SBOM + provenance per digest (0.1.15) are the evidence artifacts; retention 7 years.
- Privacy: no personal data is processed by this wave except Git author identities in commit history (retained as part of source control; excluded from erasure requests under wave 48 privacy tooling, documented).

0.1.21 Coverage closure

No `GAP` or `PARTIAL` row in coverage-ax.md §12.99 or coverage-sap.md §13.99 is assigned to wave 1 (both registers parsed on 2026-10-21). Inventory rows tagged `COVERED-1` (SAP §13.30: SE09/SE10/SE01/SE03/STMS/STMS_IMPORT/SE06/SE07 transport organizer and management; STRUST/SSFA/SSO2/SAML2/SNCWIZARD trust manager; STMS import history and SE10 request logs; STMS import jobs / transport of copies / ChaRM cycles) are met by FR-1.1–FR-1.11 and 0.1.14.

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
none	—	—	Wave 1 has no register rows; see paragraph above for the `COVERED-1` confirmation

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0.2 Wave 2: Identity, Tenancy & Authorization

0.2.1 Scope and business context

AX 2012 authorizes through `SecurityRole` → `SecurityDuty` → `SecurityPrivilege` → `SecurityEntryPoint` with access levels (NoAccess, Read, Update, Create, Correct, Delete), table permissions in the AOT, extensible data security policies (XDS) and `UserInfo` rows tied to Active Directory; companies are isolated by `DataAreaId` on every per-company table. SAP uses `SU01` user masters (USR02), `PFCG` roles (AGR_DEFINE, AGR_1251 field values, AGR_USERS), authorization objects such as `S_TCODE`, `F_BKPF_BUK` (company code) and `M_MATE_WRK` (plant), `SU24` proposals, `SU53` failed-check traces, and `MANDT` for client isolation. Wave 2 gives Passage one identity provider (Keycloak) federated to Entra ID, Okta or Google, SCIM provisioning, an AX-shaped role model compiled to Rego, OPA as a sidecar on every service, Envoy Gateway JWT validation with per-tenant rate limits, and Postgres RLS on every table. At the end of the wave a user logs in once, lands in a legal entity, and every API call is allowed or denied by policy, not code. Out of scope: segregation-of-duties conflict analysis (wave 48), position hierarchies (wave 42), record-level XDS-style data policies beyond tenant and legal entity (wave 15 adds dimension security), customer-facing portals (waves 20, 30).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Diagnosing "no authorization" in SAP needs SU53 on the user's session, then SUIM, then PFCG edit, role generation and user comparison (PFCG → Authorizations → Generate → User tab → User comparison): 6 screens and often a transport	Authorization data lives in profiles generated from roles; failures are only visible in the user's own buffer	Every 403 carries `duty_missing`; the effective-permission viewer shows the decision trace for any user × entity × action; assignment takes effect in ≤ 30 s without a transport	Access-fix cycle: 1–3 days (ticket + transport window) → < 5 min; screens: 6 → 1
AX security development: a new entry point needs a privilege, duty and role in the AOT, synchronization and an AOS restart; D365 needs a build and package	Security objects are AOT metadata compiled into the model store	Role/duty/privilege are data compiled to a Rego bundle on save	Role change to live: 1–2 h (AX) / 4 h (D365 package) → ≤ 5 s compile + 10 s propagation
Company isolation relies on every X++ query honoring `DataAreaId` (and `crossCompany` misuse) or every ABAP select carrying MANDT; leaks are code bugs	Isolation enforced in application code, not the database	Postgres RLS with `FORCE ROW LEVEL SECURITY`; the app role cannot bypass it	Code paths that can leak a tenant: every query → 0; cross-tenant CI test on 100% of tables
Joiners/leavers: SU10 mass maintenance or D365 user import run manually; leavers keep access for days	No standard provisioning protocol in ECC; D365 ties to Entra only	SCIM 2.0 server with deprovision revoking sessions in ≤ 60 s	Deprovision latency: days → < 60 s; manual user admin steps per joiner: ~10 → 0
MFA and federation to Okta/Google are unsupported on AX 2012; S/4 needs SAP IAS/SNC projects	IdP coupling to AD/Entra or SAP's own stack	Keycloak brokering any OIDC/SAML IdP with TOTP/WebAuthn	Federation to a new IdP: a multi-week project → < 1 h of configuration

0.2.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
User master	`UserInfo`, SysUserInfoPage form, AD import wizard	SU01, USR02/USR21, SU10 mass maintenance	identity `User` entity, Admin UI Users, SCIM /Users	full
Role	`SecurityRole`, SysSecRoles form	PFCG single/composite/derived roles, AGR_DEFINE	authz `Role` (+ `derived_from` for org-level variants)	full
Duty / privilege	`SecurityDuty`, `SecurityPrivilege`, `SecurityRoleTaskGrant`, `SecurityTaskEntryPoint`	Authorization objects/fields (TOBJ, AGR_1251), SU24 defaults	`Duty`, `Privilege`, `Permission(entity, action, scope)`	full
Access levels	NoAccess/Read/Update/Create/Correct/Delete on entry points and tables	ACTVT 01/02/03/06 etc. per object	action enum: read, create, update, correct, delete, post, approve	full
Company isolation	`DataAreaId`, `CompanyInfo`, `changecompany()`	MANDT client, BUKRS org level in roles	tenant_id RLS + legal_entity_id scope on RoleAssignment	improved
Org-scoped roles	`OMUserRoleOrganization` (role valid for selected legal entities)	Derived roles with org-level values (BUKRS, WERKS)	`RoleAssignment.legal_entity_ids[]` or `all`	full
Effective permissions	Security development tool, "View permissions"	SUIM, SU56 user buffer, SU53 trace	`GET /v1/authz/effective-permissions`, decision debug endpoint	improved
Identity federation / MFA	Entra ID (AAD) only on D365; AD on AX 2012	SAML via Identity Authentication Service, SNC	Keycloak OIDC/SAML brokering, TOTP/WebAuthn MFA	improved
Provisioning	AD sync, D365 user import	SAP IPS SCIM, CUA	SCIM 2.0 server (RFC 7644)	full

0.2.3 Functional requirements

- **FR-2.1 Keycloak as IdP** — Must. The system shall deploy Keycloak with one realm per tenant, brokering OIDC and SAML 2.0 to external IdPs, issuing RS256 JWTs with claims `sub`, `tid` (tenant), `email`, `preferred_username`, `pv` (policy bundle version), `amr`.
  - Notes: access token TTL 10 minutes, refresh 8 hours, absolute session 12 hours; MFA (TOTP or WebAuthn) required for roles flagged `mfa_required`; realm export committed as Helm values for reproducibility.
- **FR-2.2 SCIM provisioning** — Must. The system shall expose `/scim/v2/Users` and `/scim/v2/Groups` per tenant with filter, PATCH and bulk (≤ 1,000 operations) per RFC 7644; group membership maps to role assignment via a configurable mapping table.
  - Notes: deprovision sets `status=deprovisioned`, revokes sessions within 60 s, never deletes the row (audit needs it).
- **FR-2.3 Tenant and legal entity model** — Must. The system shall model `Tenant` → `LegalEntity` → `Site` (shell only; detailed in wave 6) and stamp `tenant_id` on every table in every domain database.
  - Notes: `legal_entity_id` is stamped on per-company tables (AX `SaveDataPerCompany=Yes` equivalent, declared in metadata in wave 3); a user's current legal entity is a session attribute, not a token claim, so switching does not require re-login.
- **FR-2.4 Row-level security** — Must. Every table shall have `ENABLE ROW LEVEL SECURITY` and `FORCE ROW LEVEL SECURITY` with policy `tenant_id = current_setting('app.tenant_id')::uuid`; services connect as role `passage_app` with `NOBYPASSRLS` and set the variable with `SET LOCAL` inside every transaction.
  - Notes: a missing `app.tenant_id` yields zero rows, not an error; migrations run as `passage_migrate`, which is also NOBYPASSRLS; a CI check fails if any table lacks the policy.
- **FR-2.5 Role/duty/privilege model** — Must. The system shall store Role → Duty → Privilege → Permission where Permission = (entity, action, field_mask?, condition?), and compile the closure into one Rego bundle per tenant.
  - Notes: actions are read, create, update, correct, delete, post, approve, export; `correct` retains AX semantics (edit date-effective rows); 20 seeded duties ship (Maintain users, Maintain security roles, Maintain vendors, View vendors, Maintain entity metadata, Apply extensions, Maintain workflows, Approve workflow, Maintain batch jobs, View batch history, Maintain number sequences, View audit log, Import data, Export data, Maintain platform releases, View platform releases, Maintain tenants, Maintain legal entities, Maintain saved views, Maintain hello ledger).
- **FR-2.6 Policy compilation** — Must. The system shall compile a bundle within 5 s of any role/duty/privilege/assignment change and distribute it to OPA sidecars via bundle polling (interval 10 s) with the version in `pv`.
  - Notes: a token whose `pv` is older than the active bundle by more than 2 versions is still honored (bundles are backward compatible); deny reasons are returned as `{"code":"AUTHZ_DENIED","duty_missing":"Maintain vendors"}`.
- **FR-2.7 OPA decision point** — Must. Every service shall call its OPA sidecar with `input = {subject, resource, action, context}` and enforce the decision before any database call; p95 decision latency ≤ 2 ms, p99 ≤ 5 ms.
  - Notes: `resource` carries entity, id, tenant_id, legal_entity_id and the field list for field-level masks; the gateway performs coarse checks (token validity, tenant match), services perform fine checks.
- **FR-2.8 Gateway** — Must. Envoy Gateway shall validate JWT signature, issuer and audience against the tenant realm's JWKS (cached 5 min), reject tokens with `tid` not matching the host's tenant, and enforce per-tenant rate limits (default 1,000 rps sustained, 2,000 burst; 429 with `Retry-After`).
- **FR-2.9 Role assignment with scope and effective dates** — Must. The system shall assign roles to users with `legal_entity_ids` (or `all`) and `effective_from`/`effective_to` (ISO 8601 dates, tenant time zone, inclusive).
  - Notes: expired assignments are excluded at compile time; a nightly job recompiles bundles for tenants with assignments crossing midnight.
- **FR-2.10 Admin UI** — Must. The system shall provide screens for users, roles, duties, privileges, assignments and an effective-permission viewer that renders the OPA decision for a chosen user × entity × action × legal entity.
- **FR-2.11 Service accounts and agents** — Should. The system shall support non-human principals (`kind=service|agent`) with client-credentials tokens and role assignments; an agent principal may hold only roles that are a subset of its sponsoring user's roles.

0.2.4 Data model

User (`identity.user`)

Field	Type	Required	Notes
id	uuid	yes	PK; equals Keycloak `sub`
tenant_id	uuid	yes	FK tenant; RLS
username	varchar(100)	yes	unique per tenant, case-insensitive
email	varchar(254)	yes	unique per tenant; PII
display_name	varchar(200)	yes	PII
kind	enum	yes	human, service, agent
status	enum	yes	invited, active, locked, disabled, deprovisioned
default_legal_entity_id	uuid	no	FK legal_entity
external_id	varchar(200)	no	SCIM externalId
mfa_enrolled	bool	yes	default false
last_login_at	timestamptz	no	
created_at, updated_at, version	timestamptz, timestamptz, int	yes	optimistic lock

Role / Duty / Privilege / Permission (`authz.*`)

Field	Type	Required	Notes
id	uuid	yes	PK (each table)
tenant_id	uuid	yes	`sys` rows have the reserved system tenant and are read-only
code	varchar(64)	yes	unique per tenant; e.g. `VendMaintain`
name, description	varchar(120), text	yes/no	localized labels in wave 48
layer	enum	yes	sys, partner, customer
parent_id	uuid	no	Role→Duty, Duty→Privilege via link tables `role_duty`, `duty_privilege`, `privilege_permission`
entity, action	varchar(64), enum	yes (Permission)	action ∈ read, create, update, correct, delete, post, approve, export
field_mask	text[]	no (Permission)	fields hidden when present
condition	jsonb	no (Permission)	Rego-compatible expression, e.g. `{"legal_entity_in":"$subject.legal_entities"}`
mfa_required	bool	yes (Role)	

RoleAssignment (`authz.role_assignment`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS
user_id	uuid	yes	FK user
role_id	uuid	yes	FK role
legal_entity_ids	uuid[]	yes	empty array means all
effective_from	date	yes	default today
effective_to	date	no	null = open-ended
granted_by	uuid	yes	FK user
reason	text	no	required when `role.break_glass=true`

Tenant / **LegalEntity**: id, code (varchar 10, unique), name, status (active, suspended), default_currency (char 3), time_zone (IANA), created_at; LegalEntity adds tenant_id and `country` (ISO 3166-1 alpha-2).

Invariants: a user has at most one active assignment per (role, legal entity set) overlapping in time; `sys` layer rows are immutable in customer tenants; every Permission entity name must exist in the metadata registry from wave 3. Indexes: `user(tenant_id, lower(email))` unique; `role_assignment(tenant_id, user_id)`; `role_assignment(tenant_id, effective_to)` for expiry sweeps. Keycloak keeps its own Postgres (CNPG) per environment.

0.2.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET/POST	/v1/identity/users	List / create users	POST: Idempotency-Key	Maintain users
PATCH	/v1/identity/users/{id}	Update status, default legal entity	version check (If-Match)	Maintain users
GET/POST/PUT	/scim/v2/Users, /scim/v2/Groups	SCIM provisioning	SCIM semantics	SCIM client credential
GET/POST/PUT	/v1/authz/roles, /duties, /privileges	Maintain the model	PUT by code	Maintain security roles
GET/POST/DELETE	/v1/authz/assignments	Grant / revoke	POST: natural key	Maintain security roles
GET	/v1/authz/effective-permissions?user_id=&legal_entity_id=	Flattened permissions	n/a	Maintain security roles or self
POST	/v1/authz/decision	Dry-run an OPA decision (debug)	n/a	Maintain security roles
POST	/v1/authz/policies/compile	Force recompile	bundle version	Maintain security roles
GET/POST	/v1/tenants, /v1/legal-entities	Tenancy shells	POST: code	Maintain tenants / Maintain legal entities
POST	/v1/session/legal-entity	Switch current legal entity	n/a	any active user

Commands validate: email RFC 5322; role code `^[A-Za-z][A-Za-z0-9]{2,63}$`; assignment dates `effective_from ≤ effective_to`; a revoke of a `sys`-layer role from the last `Tenant owner` is rejected (409).

Event	Trigger	Payload fields	Consumers
dev.passage.identity.user.created.v1	User insert (UI, SCIM, federation first login)	user_id, tenant_id, email, kind, source	workflow, audit, shell
dev.passage.identity.user.status-changed.v1	status transition	user_id, from, to, actor, reason	audit, session revoker
dev.passage.authz.role-assignment.granted.v1 / revoked.v1	assignment insert/delete	assignment_id, user_id, role_id, legal_entity_ids, effective_from, effective_to	policy compiler, audit
dev.passage.authz.policy.compiled.v1	bundle built	tenant_id, bundle_version, sha256	OPA sidecars, audit
dev.passage.identity.login.succeeded.v1 / failed.v1	Keycloak event listener	user_id, ip, idp, amr	audit, anomaly alert

0.2.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Users list/detail	Invite, lock, disable, assign roles, set default legal entity	username, email, status, kind, last login, MFA	Maintain users
Roles	Create, add duties, mark MFA/break-glass	code, name, layer, duty count, user count	Maintain security roles
Duties / Privileges	Create, add privileges / permissions	code, entity, action, field mask	Maintain security roles
Role assignment	Grant with legal entity picker and dates	user, role, legal entities, from, to, reason	Maintain security roles
Effective permissions	Pick user × legal entity, inspect decision trace	entity, action, allow/deny, via role → duty → privilege	Maintain security roles
Legal entity switcher (shell header)	Switch current legal entity	code, name	any

Rules: all grids are keyboard navigable (wave 5 shell completes this); role edits are saved explicitly, not on blur; a decision trace shows the first allowing path and all denying reasons; the switcher remembers the last legal entity per user in `Personalization` (wave 5).

0.2.7 Business rules and edge cases

- **BR-2.1** User status machine: invited → active; active ↔ locked (5 failed logins in 15 min locks for 30 min, admin may unlock); active → disabled → active; any → deprovisioned (terminal).
- **BR-2.2** Deny by default: no matching Permission means deny; explicit deny permissions do not exist (AX semantics), so removing access means removing the grant.
- **BR-2.3** Access level ordering for merge: delete > correct > update > create > read; the highest level across all roles wins per entity (AX `SecurityEntryPoint` union behavior).
- **BR-2.4** Legal-entity scoping: a permission is effective only in legal entities listed on the assignment; `all` includes legal entities created later.
- **BR-2.5** Effective dates are evaluated in the tenant time zone at compile time and at 00:05 nightly; a grant with `effective_from` in the future compiles into the bundle inactive.
- **BR-2.6** Revoking a session: status change to locked/disabled/deprovisioned revokes refresh tokens; access tokens remain valid at most 10 minutes (TTL), which is the documented worst case.

- **BR-2.7** Cross-tenant request (token `tid` ≠ host tenant) returns 404 rather than 403 to avoid tenant enumeration.
- **BR-2.8** Rate limit is per tenant per gateway replica, aggregated through the gateway's shared rate-limit service; service accounts count against the same bucket.
- **BR-2.9** Concurrency: two admins editing the same role use optimistic locking (`version`); the loser gets 409 with the current version.
- **BR-2.10** Agent principals cannot be granted `Tenant owner`, `Maintain security roles` or break-glass roles.

0.2.8 Acceptance criteria

- **AC-2.1** Given a user with duty "Maintain vendors", When they `PUT /vendors/{id}` (stub entity), Then 200; Given a user without it, Then 403 with `duty_missing`. Table-driven across 20 duties. (FR-2.5, FR-2.7)
- **AC-2.2** Given tenant A's app session, When a query filters `tenant_id = <tenant B>`, Then zero rows are returned under `passage_app`. (FR-2.4)
- **AC-2.3** Given two Keycloak realms federated to a stand-in Keycloak IdP in CI, When a user logs in through each, Then both sessions carry the correct `tid` and land in the user's default legal entity. (FR-2.1, FR-2.3)
- **AC-2.4** Given 100 random (user, entity, action, legal entity) tuples, When effective-permission report and `POST /v1/authz/decision` are compared, Then they agree 100%. (FR-2.6, FR-2.10)
- **AC-2.5** Given a SCIM PATCH deactivating a user, When processed, Then `status=deprovisioned`, refresh tokens are revoked within 60 s and the row persists. (FR-2.2)
- **AC-2.6** Given an assignment with `effective_to` yesterday, When the nightly recompile runs, Then the user's requests for that role's permissions are denied. (FR-2.9)
- **AC-2.7** Given 1,500 rps from one tenant for 10 s, When routed through the gateway, Then requests above the limit receive 429 with `Retry-After`. (FR-2.8)
- **AC-2.8** Given a role flagged `mfa_required`, When a user without MFA logs in, Then Keycloak forces enrollment before issuing a token. (FR-2.1)
- **AC-2.9** Given an agent principal sponsored by user U, When an assignment outside U's roles is attempted, Then 422 with `AGENT_EXCEEDS_SPONSOR`. (FR-2.11)

0.2.9 Test plan and fixtures

Fixtures: `fixtures/authz/duties-20.yaml` (20 duties with allow/deny expectations, 240 rows), `fixtures/identity/users-50.json` (two tenants, three legal entities), `fixtures/identity/keycloak-realm-ci.json` (two realms + broker), `fixtures/authz/decision-sample-100.json`.

Test	Type	Target	Pass criterion
Duty matrix	Contract (table-driven)	gateway + OPA + stub service	240/240 expected codes
RLS cross-tenant	Integration	every table (generated list)	0 rows for foreign tenant; CI fails on any table without policy
Federation login	E2E (Playwright)	Keycloak broker	2 IdPs, correct claims
Effective vs decision	Property	compiler	100/100 agree
Bundle compile latency	Performance	compiler	≤ 5 s for 500 roles / 2,000 duties
OPA decision latency	Performance	sidecar	p95 ≤ 2 ms at 2,000 rps
SCIM conformance	Contract	SCIM server	RFC 7644 test suite subset (filter, PATCH, bulk)

Reconciliation: count of Keycloak users per realm equals `identity.user` rows with status ≠ deprovisioned; sum of permissions in the compiled bundle equals the flattened SQL closure.

0.2.10 Deployment and operations

Charts: `keycloak` (with CNPG database, 2 replicas), `identity`, `authz` (compiler + bundle server), `opa` as a sidecar template in `deploy/charts/_lib`, `envoy-gateway` policies (`SecurityPolicy` for JWT, `BackendTrafficPolicy` for rate limits). Config: `KEYCLOAK_ISSUER_TEMPLATE`, `OPA_BUNDLE_URL`, `AUTHZ_BUNDLE_POLL_SECONDS=10`. Migrations: additive Atlas. Feature flags: `authz.enforce` (log-only mode for the first week in dev). Scheduled jobs: nightly assignment expiry recompile 00:05 tenant time; JWKS refresh. SLOs: login success ≥ 99.9%, token issuance p95 ≤ 300 ms, OPA p95 ≤ 2 ms, bundle propagation ≤ 30 s. Alerts: `AuthzDenySpike` (> 5× baseline), `BundleStale` (> 5 min), `KeycloakDown`. Runbook: stale bundle, locked-out tenant owner (break-glass via Keycloak admin console), JWKS rotation. Rollback: digest rollback; bundles are versioned so a rollback of authz re-serves the prior bundle.

0.2.11 Migration from AX / SAP

Source	Passage entity	Notes
AX `UserInfo` (id, name, networkAlias, enable)	User	`networkAlias` → username; `enable=0` → disabled
AX `SecurityUserRole` (user, securityRole, validFrom/validTo)	RoleAssignment	dates carried; `OMUserRoleOrganization` rows → `legal_entity_ids`
AX `SecurityRole` / `SecurityDuty` / `SecurityPrivilege` + `SecurityRoleTaskGrant`, `SecurityTaskEntryPoint`	Role / Duty / Privilege	custom (CUS/VAR layer) roles only; SYS roles are mapped by name to Passage seeds
SAP USR02 (BNAME, UFLAG, GLTGV/GLTGB)	User	UFLAG 64/128 → locked; GLTGV/GLTGB → status dates
SAP AGR_USERS (AGR_NAME, UNAME, FROM_DAT, TO_DAT)	RoleAssignment	dates carried
SAP AGR_DEFINE / AGR_1251 (OBJECT, FIELD, LOW/HIGH)	Role / Duty / Privilege	authorization objects map via a crosswalk file to Passage permissions; BUKRS values → legal_entity_ids

Transformations run through the wave 10 migration framework; this wave ships only the crosswalk format and a dry-run report. Not migrated: passwords (federated), SU24 proposals, AX entry-point grants on forms that have no Passage equivalent (reported as "unmapped" with counts).

0.2.12 Dependencies, risks, open questions

Depends on: wave 1 (cluster, Argo, Vault/ESO, gateway).

Risk	Likelihood	Impact	Mitigation
Keycloak realm-per-tenant does not scale past ~500 tenants	Low (v1)	Medium	Organizations feature evaluated by wave 48
Rego closure bundle size for large role models	Medium	Medium	Partial evaluation; bundle per tenant; limit 50k permissions
SAML IdPs with non-standard attribute names	Medium	Low	Mapper templates for Entra ID, Okta, ADFS

Open questions: (1) Field-level masks in P0 or deferred to wave 15 — default: masks compiled but enforced only for PII-flagged fields; decide by wave 3. (2) Session legal entity in a cookie vs header — default header `X-Passage-Legal-Entity`; decide by wave 5. (3) Tenant realm per tenant vs shared realm with groups — default per

tenant; decide by wave 2 mid-point.

0.2.13 Parameters and setup

Tenant-scoped values are rows of `identity.tenant_settings` (edited on the Admin workspace → Tenant settings form, the AX "System parameters → Security" and SAP RZ10 `login/*` equivalent); realm values are Keycloak realm settings exported to Helm values.

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`auth.access_token_ttl_minutes`	tenant (realm)	int	10	5–60	JWT lifetime; worst-case access after revoke (BR-2.6)	none (AD Kerberos ticket)	`rdisp/gui_auto_logout`, `login/ticket_expiration_tim`
`auth.refresh_token_ttl_hours`	tenant (realm)	int	8	1–24	silent re-login window	none	`login/ticket_expiration_tim`
`auth.session_max_hours`	tenant (realm)	int	12	1–24	absolute session cap	System parameters → Security: session timeout (D365 `SysUserLogTimeout`)	`rdisp/max_wprun_time`, SICF session timeout
`auth.lockout.failures`, `auth.lockout.window_minutes`, `auth.lockout.duration_minutes`	tenant (realm brute-force detection)	int, int, int	5, 15, 30	3–10, 5–60, 5–1,440	BR-2.1 lock behavior	none (AD policy)	`login/fails_to_user_lock`, `login/failed_user_auto_unlo`
`auth.mfa.required_roles`	tenant	role code list	`TenantOwner`, `SecurityAdmin`	any role codes	roles whose holders must enroll TOTP/WebAuthn	none	SNC / SAML2 enforced for user grou
`auth.mfa.allowed_methods`	tenant	enum set	totp, webauthn	totp, webauthn, recovery_code	enrollment options	none	none
`auth.local_password_policy`	tenant	{min_length, history, max_age_days}	14, 5, 0 (no expiry)	min_length 12–64	applies only to break-glass local accounts (federated users have none)	none	`login/min_password_lng`, `login/password_history_size` `login/password_expiration_t` USR40
`auth.idp_brokers[]`	tenant	list {alias, protocol oidc/saml, issuer/metadata URL, attribute mappers, first-login flow}	empty	—	federation targets; first login auto-links by verified email	Users → Active Directory import wizard, Claims users / Identity providers (D365 Entra tenant)	SAML2 trusted providers, SNC configuration, SM30 `VUSREXTID`
`scim.enabled`, `scim.group_role_mapping`	tenant	bool, map {group → role code}	false, {}	—	SCIM endpoints on/off; group → role	AD group import	SAP IPS mapping, CUA
`scim.bulk_max_operations`	tenant	int	1,000	100–1,000	RFC 7644 bulk limit	none	none
`authz.enforce`	environment / tenant feature flag	enum	enforce (log_only in dev first week)	enforce, log_only	log_only records would-be denies without blocking	none	`auth/no_check_in_some_cases` (SU24 proposals)
`authz.bundle_poll_seconds`	environment	int	10	5–60	OPA bundle polling	none (AOS cache refresh)	PFCG user comparison (PFUD)
`authz.bundle_max_permissions`	tenant	int	50,000	≤ 100,000	compile guard (risk table)	none	none
`authz.recompile_time`	tenant	HH:MM tenant time zone	00:05	any	nightly effective-date recompile (BR-2.5)	none	PFUD / PFCG_TIME_DEPENDENCY
`gateway.rate_limit.sustained_rps`, `gateway.rate_limit.burst`	tenant	int, int	1,000, 2,000	100–50,000	FR-2.8 limits; 429 with `Retry-After`	none	ICM `icm/max_conn`, `rdisp/tm_max_no`
`gateway.jwks_cache_minutes`	environment	int	5	1–60	JWKS refresh	none	none
`session.legal_entity_header`	environment	string	`X-Passage-Legal-Entity`	header name	carries the current legal entity (open question 2)	`changecompany()` / company switch	SU3 default parameter BUK
`tenant.time_zone`, `tenant.default_currency`	tenant	IANA tz, ISO 4217	`UTC`, `USD`	valid values	effective-date evaluation, defaults for later waves	System parameters → General (time zone), CompanyInfo currency	SCC4 client, OY01 country, OB22 company code currency
`legal_entity.country`	legal entity	ISO 3166-1 alpha-2	required	valid code	country-pack hook (0.2.20)	Legal entities form → Addresses (country/region)	OX02 company code country
`agent.max_sponsor_subset`	tenant	bool	true	true	agents may hold only a subset of the sponsor's roles (FR-2.11)	none	SU01 reference user
`identity.deprovision_revoke_seconds`	tenant	int	60	10–300	session revocation deadline after SCIM deactivation	none	SU10 mass lock

0.2.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Users	list inquiry	status, kind, role, legal entity, last login before	username, email, display name, kind, status, MFA, last login, default legal entity	`identity.user`	Users list (SysUserInfoPage)	SU01 / SU10 list, RSUSR002 users by complex criteria
Users by role	list inquiry	role, legal entity, as-of date	user, role, legal entities, effective from/to, granted by	`authz.role_assignment` ⋈ user	Security → Users by role (SSRS "User role assignments")	SUIM → User → by roles; AGR_USERS
Roles by user	drill-down	user	role, duties, legal entities, effective dates, break-glass flag	same	User details → Roles tab	SU01 → Roles tab, SUIM roles by user
Role permissions (closure)	drill-down	role	duty → privilege → permission (entity, action, field mask, condition)	compiled closure	Security roles → View permissions; SSRS "Security role report"	PFCG → Authorizations (display), SUIM → Roles → by authorization object
Effective permissions	drill-down	user × legal entity (× entity, action)	allow/deny, first allowing path, all denying reasons, bundle version	`GET /v1/authz/effective-permissions`, `POST /v1/authz/decision`	Security development tool → View permissions	SU53, SU56, STAUTHTRACE
Login history	list inquiry	user, outcome, IdP, date range, IP	time, user, outcome, IdP alias, `amr`, IP, user agent	Keycloak event listener → `identity.login_event`	Users → User log (SysUserLog)	SM20 security audit log (logon events), RSUSR200 logon dates
Online sessions	list inquiry	tenant, user	user, session start, last activity, IdP, client; action: revoke	Keycloak sessions API	Online users (SysUsersOnline)	SM04, AL08
Expiring role assignments	operational report	horizon (default 30 days)	user, role, effective_to, granted by	`authz.role_assignment`	none	PFCG_TIME_DEPENDENCY log
Authorization denials	KPI / operational	period, duty, entity, user	denials per duty per hour, top users, top entities	OPA decision logs (Loki) → `authz.decision_stat`	none	SU53 (per user), SM20 RFC/auth failures
SCIM reconciliation	operational report	last run	users in IdP not in Passage, deprovisioned in IdP still active, group → role mismatches	SCIM sync job output	AD import wizard log	SAP IPS job log, CUA distribution log (SCUL)
Policy bundle versions	list inquiry	tenant	version, sha256, compiled at, role count, permission count, trigger	`authz.bundle`	none	PFCG profile generation log
Tenant and legal entity list	list inquiry	status	code, name, country, currency, time zone, user count	`identity.tenant`, `identity.legal_entity`	Legal entities (OMLegalEntity)	OX02 company codes, SCC4 clients

0.2.15 Printed and electronic documents

No printed documents. Electronic documents:

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
User invitation email	user created with `status=invited` (UI or SCIM without federation link)	tenant name, inviter, link (24 h validity), IdP instructions	HTML + text email sent by Keycloak "execute actions" mail (tenant SMTP settings in the realm until wave 4 moves outbound email to FR-4.13)	none	link token single-use; email retained in Keycloak event log 30 days
MFA reset / enrollment email	admin resets MFA; `mfa_required` role granted to an unenrolled user	user, action, link	same	none	same
Effective-permission export	administrator action on the viewer	user, legal entity, bundle version, rows (entity, action, decision, path)	CSV, JSON	none	kept by the requester; wave 48 access reviews re-generate on demand
Security role report	administrator action	role, duties, privileges, permissions, assigned users as of date	CSV, PDF (wave 15 report designer; CSV only in wave 2)	none	SOC 2 evidence (wave 48) retained 7 years

0.2.16 Periodic and batch jobs

Jobs run as Kubernetes CronJobs in wave 2 and are re-registered on the wave 4 batch framework without behavior change.

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Effective-date recompile	daily 00:05 tenant time (BR-2.5)	tenant	compile is deterministic; skipped when no assignment crosses midnight	< 5 s per tenant	retry 3×; last good bundle keeps serving	`BundleStale` (> 5 min)
SCIM reconciliation	daily 01:00 UTC	tenant	compares IdP export to `identity.user`; writes report only	< 2 min per 10 k users	report with `errors[]`; no automatic deprovision	`ScimDriftDetected` (ticket)
Session cleanup	hourly	realm	Keycloak offline-session purge	< 30 s	n/a	none
Login anomaly aggregation	every 15 min	tenant	window keyed by (tenant, bucket)	< 10 s	skipped bucket recomputed next run	`LoginFailureSpike` (> 50 failures / 15 min / tenant)
Bundle garbage collection	daily 03:00 UTC	tenant	keeps last 10 versions + any referenced by a live `pv`	< 10 s	none	none
JWKS refresh	every 5 min (gateway)	realm	cache keyed by `kid`	< 200 ms	previous keys kept until rotation grace (24 h)	`JwksFetchFailed`
Keycloak realm export (backup)	daily 02:30 UTC	realm	export to object storage keyed by date	< 1 min	alert	`RealmBackupFailed` (ticket)

0.2.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
Maintain users	`GET/POST /v1/identity/users`, `PATCH /v1/identity/users/{id}`, Users screen (invite, lock, disable), revoke session	with Maintain security roles (a holder of both can grant anything to anyone; self-grant is rejected with IAM-0416 regardless)	Security administrator
View users	`GET /v1/identity/users`, Users screen read-only, Login history, Online sessions	none	Security administrator, Auditor, Help desk
Maintain security roles	`/v1/authz/roles`, `/duties`, `/privileges`, `/assignments` (POST/DELETE), `/policies/compile`, Roles/Duties/Privileges/Assignment screens	with Maintain users; with Approve workflow for the same entity (wave 4)	Security administrator
View security reports	`GET /v1/authz/effective-permissions`, `POST /v1/authz/decision`, Role permissions, Users by role, Authorization denials	none	Security administrator, Auditor
Maintain identity providers	realm broker settings (`auth.idp_brokers[]`), mappers, MFA policy	with Maintain users	Tenant owner
Maintain SCIM clients	create/rotate SCIM client credentials; `scim.*` settings	none	Tenant owner
Maintain tenants	`GET/POST /v1/tenants`, tenant settings form	with Maintain legal entities (only in multi-tenant operator context)	Platform operator (system tenant)
Maintain legal entities	`GET/POST /v1/legal-entities`, Legal entities screen	none	Tenant owner, Organization administrator
Tenant owner (break-glass)	all of the above plus Keycloak admin console for the realm; every use requires `reason` (RoleAssignment.reason) and raises an audit event	with everything; limited to ≤ 2 human users per tenant, never agents (BR-2.10)	Tenant owner

0.2.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
IAM-0401	error	You do not have the duty "{duty}" required for this action	OPA deny; `duty_missing`	request the role containing the duty from a Security administrator	FR-2.6, FR-2.7
IAM-0402	error	Not found	token `tid` ≠ host tenant (returned as 404)	sign in to the correct tenant host	BR-2.7
IAM-0403	error	Too many requests; retry after {seconds} s	tenant rate limit	back off; service accounts share the bucket	FR-2.8, BR-2.8
IAM-0404	error	Session expired; sign in again	access and refresh tokens expired	sign in	FR-2.1
IAM-0405	warning	Your permissions were updated; refresh to apply	token `pv` more than 2 bundle versions behind	refresh the page (silent re-login)	FR-2.6
IAM-0406	info	Multi-factor enrollment is required for your role "{role}"	`mfa_required` role without enrolled factor	enroll TOTP or WebAuthn	FR-2.1, AC-2.8
IAM-0407	error	Account locked for {minutes} minutes after {n} failed sign-ins	lockout policy	wait or ask an administrator to unlock	BR-2.1
IAM-0408	error	Cannot revoke: {user} is the last Tenant owner	last-owner protection (409)	grant Tenant owner to another user first	0.2.5 validation
IAM-0409	error	Agent cannot hold "{role}": not within sponsor {user}'s roles	FR-2.11 subset rule (422 `AGENT_EXCEEDS_SPONSOR`)	grant the role to the sponsor first or choose another sponsor	FR-2.11, BR-2.10
IAM-0410	error	This role was changed by {user} at {time}; reload to see the current version	optimistic lock (409)	reload and reapply	BR-2.9
IAM-0411	error	Role code must match `^[A-Za-z][A-Za-z0-9]{2,63}$`	validation	rename	0.2.5 validation
IAM-0412	error	Effective from {from} is after effective to {to}	validation	correct the dates	FR-2.9
IAM-0413	error	SCIM bulk request exceeds {max} operations	RFC 7644 limit	split the request	FR-2.2
IAM-0414	error (alert)	Policy bundle compile failed for tenant {tenant}: {reason}	compiler error; previous bundle still serves	fix the offending permission condition	FR-2.6
IAM-0415	warning	Sign-in attempt by deprovisioned user {user} from {ip}	post-deprovision login	none; recorded in audit and login anomaly report	FR-2.2
IAM-0416	error	You cannot grant a role to yourself	self-grant attempt	ask another Security administrator	0.2.17 SoD
IAM-0417	error	Reason is required when granting a break-glass role	`RoleAssignment.reason` missing	provide a reason	0.2.4 RoleAssignment

0.2.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Interactive login (federated)	5,000 users/tenant, 2 logins/user/day; peak 300/min at 08:00	400 ms / 1.2 s / 2 s (excluding IdP)	50 logins/s per Keycloak replica	login events ≈ 2.5 M rows (≈ 500 MB) per tenant	`login_event(tenant_id, occurred_at)` monthly partitions	Keycloak HPA 2 → 6 replicas; IdP latency dominates
Token refresh	5,000 users × 6 refreshes/day	50 ms / 150 ms / 300 ms	500/s	none	n/a	same
OPA decision (sidecar)	2,000 decisions/s per service at peak	0.5 ms / 2 ms / 5 ms	5,000/s per sidecar	none	bundle in memory, partial evaluation per entity	CPU-bound; sidecar requests 100 m / limit 500 m; latency rises linearly, no errors
Gateway JWT validation	1,000 rps sustained per tenant	0.3 ms / 1 ms / 2 ms	20,000 rps per gateway replica	none	JWKS cache 5 min	rate limit 429 beyond `burst`

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Bundle compile	500 roles, 2,000 duties, 50 k permissions	1 s / 5 s / 8 s	10 compiles/min/tenant (debounced 2 s)	bundles ≈ 2 MB × 10 kept per tenant	one bundle per tenant	compile queue; stale bundle serves (`BundleStale` alert at 5 min)
Effective-permission report	100/day	200 ms / 800 ms / 1.5 s	10/s	none	closure computed from bundle, cached 60 s	cached result served
SCIM bulk (1,000 ops)	10/day	5 s / 15 s / 30 s	100 ops/s	`identity.user` ≈ 1 MB per 5 k users	`user(tenant_id, lower(email))`	queued; 429 to the SCIM client with `Retry-After`
Role assignment table	5,000 users × 4 roles	n/a	n/a	≈ 20 k rows/tenant (≈ 5 MB)	`role_assignment(tenant_id, user_id)`, `(tenant_id, effective_to)`	n/a

0.2.20 Localization and compliance hooks

- Privacy: `identity.user.email`, `display_name` and login IPs are personal data (GDPR Art. 4, CCPA). Erasure requests (wave 48 privacy tooling) pseudonymize these columns (`user-<id>@erased.invalid`) and purge `login_event` rows older than the tenant's retention (`identity.login_event_retention_days`, default 365; EU tenants typically 180); the row itself is never deleted (audit integrity). `GET /v1/identity/users/{id}/export` serves the data-subject access export (JSON).
- Residency: Keycloak and the identity database run in the customer's chosen region (0.1.20); realm per tenant lets a multi-tenant operator pin a tenant's realm database to a regional CNPG cluster from wave 50.
- Country and legal entity: `LegalEntity.country` is the key the wave 48 country packs use to attach statutory roles (for example, a German "Datenschutzbeauftragter" read-only role) and password/MFA minimums (for example, NIS2-driven MFA for all administrative roles). Extension point: `TenantSettings.overrides` jsonb keyed by country, consumed by the Keycloak values template and the compiler.
- Language and formats: the Keycloak login theme ships English only in wave 2; locale files are loaded from the wave 48 label packs; dates in effective-date fields are stored as dates (no time) and rendered per user preference from wave 5.
- Time zone: effective dates are evaluated in `tenant.time_zone` (BR-2.5), not per legal entity; multi-time-zone tenants are an accepted limitation until wave 6 calendars.
- Accessibility: Keycloak login pages are themed to meet WCAG 2.1 AA (checked by the wave 5 axe suite).
- Audit and evidence: role grants, revokes, break-glass use and IdP configuration changes are events consumed by wave 5 audit; wave 48 SOC 2 CC6.1–CC6.3 evidence is generated from them.

0.2.21 Coverage closure

No `GAP` or `PARTIAL` row in coverage-ax.md §12.99 or coverage-sap.md §13.99 is assigned to wave 2 (both registers parsed on 2026-10-21). Inventory rows tagged `COVERED-2` (AX §12.30 users, user groups, security roles/duties/privileges, automatic role assignment rules, online users, system parameters, user log, security inquiries, SSRS security reports, AD import / claims users / identity providers, password/MFA policies; AX §12.39 external user roles and provisioning; AX §12.7 cost accounting ledger security; AX §12.26 mobile device user authentication; SAP §13.30 SU01/SU10, SUGR, PFCG, SU53/SU56/ST01/STAUTHTRACE, SUIM, SECPOL/RZ10, SU01 user types, SAML/OIDC/SNC, UCON, profile parameters, PFCG org levels, RSUSR* reports, PFUD, SU10 mass lock; SAP §13.2 OBA4 tolerance groups; SAP §13.12 OMET buyer authorizations and authorization groups) are met by FR-2.1–FR-2.11, 0.2.14 and 0.2.17.

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
none	—	—	Wave 2 has no register rows; see paragraph above for the `COVERED-2` confirmation. The automatic role-assignment rule (`SysSecRoleAssignmentRule`, query-based) is met by `scim.group_role_mapping` plus the wave 4 batch job that evaluates a saved filter over users; the SAP OBA4/OMET amount limits are met by `Permission.condition`

0.3 Wave 3: Metadata Kernel (Entities, Fields, Extensions)

0.3.1 Scope and business context

In AX every business object is an AOT artifact: a Table with properties (`TableGroup`, `SaveDataPerCompany`, `PrimaryIndex`, `ClusteredIndex`, `TitleField1/2`, `CacheLookup`), Fields typed by Extended Data Types and Base Enums, Relations with `DeleteActions`, Indexes with `AlternateKey`, and table methods (`validateField`, `validateWrite`, `validateDelete`, `initValue`, `insert`, `update`) in X++; D365 adds data entities for OData and the extension model that replaced overlayering (SYS/GLS/FPK/SLN/ISV/VAR/CUS/USR layers). In SAP the equivalents are DDIC objects: tables (DD02L), fields (DD03L), domains (DD01L), data elements (DD04L), foreign keys (DD05S/DD08L), search helps (DD30L), append structures and `CI_*` customer includes, delivery classes and `SE11` activation, with BAPIs and CDS/OData exposing them. Wave 3 replaces both with a YAML entity definition language and a generator: declare an entity once and get the Postgres migration, Go model and repository, REST handlers, OpenAPI 3.1, TypeScript client and JSON Schema. Customers extend any entity with fields, enum values and declarative rules as metadata applied to a running environment. At wave end `Tenant`, `LegalEntity` and `User` from wave 2 are regenerated from metadata and the hand-written code is deleted. Out of scope: generated forms (wave 5), AsyncAPI topics (wave 4), scripted customer hooks (open question), views/maps/table collections (wave 46 analytics covers views).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Adding one field to VendTable in D365: table extension, EDT, form extension, data entity extension, label, build, deployable package, LCS apply — a 2–5 day developer task with an environment outage	Schema, API, form and entity are separate artifacts kept in sync by hand	One `x_` field in an `ExtensionPackage`: migration, API, client, JSON Schema and (wave 5) form generated and applied with zero downtime	Field addition: 2–5 days + outage → < 60 s, no outage; artifacts touched by hand: 6 → 1
SAP append structure on LFA1 needs SE11, activation (table conversion risk on large tables), CMOD/BAdI for validation, SE51 screen exit, and a transport through DEV→QAS→PRD	DDIC activation rewrites tables; UI exits are separate from DDIC	Additive `ADD COLUMN` with fast default, `CREATE INDEX CONCURRENTLY`, declarative validation in the same package	Lock time on a 100M-row table: minutes → < 2 s; transport cycle: 1–2 weeks → one PR
Integration partners (MES, PLM, WMS) must reverse-engineer OData entities or BAPI parameter lists; no machine-readable contract versioning	APIs are hand-exposed per entity (data entities, SEGW projects)	Every entity publishes OpenAPI 3.1 and JSON Schema with additive-change guarantees enforced in CI	Time for an external team to consume a new entity: weeks → the day it is published; breaking changes reaching partners: 0

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Silent type drift: AX `real` fields and SAP `CURR` with DECIMALS=2 round amounts differently in X++/ABAP arithmetic vs. the database	Type semantics defined in two places (EDT/domain and language runtime)	One type system (`decimal(p,s)`, `money` + currency, `half_even`) enforced at the API and column	Rounding discrepancies between API and storage: possible → 0 (property-tested)
Finding "where is this field used" requires cross-reference tools (AX cross-reference update takes hours; SAP SE11 where-used across packages)	Reflection is an offline build artifact	Registry `/v1/metadata/*` answers entity, field, relation and extension queries live	Where-used answer: hours (after a cross-reference build) → < 20 ms

0.3.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Table definition	AOT Tables node, `TableGroup`, `SaveDataPerCompany`, `RecId`, `RecVersion`, `DataAreaId`, `Partition`	DD02L, delivery class, MANDT, SE11	`EntityDefinition` YAML; system fields id, tenant_id, legal_entity_id, version, created_*, updated_*	full
Typed fields	Extended Data Types (StringSize, Label, HelpText), `SysDictField`	Domains (DD01L) + data elements (DD04L)	`TypeDefinition` (EDT) + `FieldDefinition`	full
Enumerations	Base Enums, `SysDictEnum`, extensible enums (D365)	Domain fixed values (DD07L), check tables	`EnumDefinition` / `EnumValue`, customer values ≥ 1000	full
Relations / FK	Relations node, `ForeignKey`, `DeleteActions` None/Cascade/Restricted	DD05S/DD08L foreign keys, check tables, cardinality	`RelationDefinition` with on_delete restrict/cascade/set_null	full
Indexes	Indexes node, `AllowDuplicates`, `AlternateKey`	DD12L secondary indexes, unique flag	`IndexDefinition` unique/non-unique, partial predicate	full
Validation hooks	`validateField`, `validateWrite`, `validateDelete`, `initValue`, `modifiedField`	Table maintenance events, BAdIs, check tables	lifecycle hooks: initValue, validateField, validateWrite, validateDelete, afterWrite, afterDelete	full (declarative + registered code)
API exposure	Data entities (IsPublic, PublicCollectionName), OData, custom services	BAPIs, RFC, OData via SEGW / RAP, CDS views	generated REST (OpenAPI 3.1) per entity; GraphQL deferred	partial (no OData $batch)
Extensions	Extensions (`.extension`), Chain of Command, extensible enums	Append structures, CI_* includes, Z/Y namespace	`ExtensionPackage` (customer layer), `x_` field prefix	improved (zero-downtime apply)
Versioning / diff	Model store versions, compare tool	Version management (SE09 versions), transport diffs	`MetadataVersion` + `/diff` endpoint, snapshot golden tests	improved
Business rules / decision tables	Policy rules framework (`SysPolicy`, `SysPolicyRuleType`), audit policy rules, X++ conditions in code	BRF+ (`FDT_WORKBENCH`): applications, functions, rulesets, decision tables, expressions; Decision Service Management	`RuleSet` / `DecisionTable` / `DecisionRow` metadata, `POST /v1/rules/{code}/evaluate`, `rules.eval()` in hooks and workflow conditions	full
Usage and license measurement	License inquiries, named user license counts (`SysUserLicenseCount`), D365 "License usage" report	USMM / SLAW system measurement, ST03N user profile, SUIM users by logon date	`metadata.usage_snapshot` (nightly), Usage report, CSV export	full

0.3.3 Functional requirements

- **FR-3.1 Entity definition language** — Must. The system shall accept entity definitions in YAML validated by a published JSON Schema (`metadata/schema/entity.v1.json`) with keys `name`, `domain`, `table_group` (main, group, transaction, worksheet_header, worksheet_line, parameter, reference), `per_legal_entity`, `title_fields[]`, `fields[]`, `enums[]`, `relations[]`, `indexes[]`, `hooks[]`, `audit`, `api`.
  - Notes: `name` is UpperCamelCase ≤ 48 chars; `domain` selects the owning Postgres database; one entity per file under `metadata/entities/<domain>/`.
- **FR-3.2 Type system** — Must. The system shall support field types string(n ≤ 4000), text, int32, int64, decimal(p ≤ 38, s ≤ 12), money (decimal(28,6) + currency reference), bool, date, datetime (timestamptz), time, uuid, enum, json, bytes, and named `TypeDefinition`s (EDT-equivalents) that bundle type, length, label, help, format and lookup entity.
  - Notes: `money` fields always pair with a `currency` char(3); `decimal` default scale 6 for quantities, 2 for displayed amounts, with rounding `half_even`.
- **FR-3.3 Field attributes** — Must. Each field shall declare `required`, `unique`, `default` (literal or `now()`, `current_user()`, `numseq(<code>)` from wave 5), `computed` (expression over sibling fields, evaluated server-side), `visibility` (always, advanced, hidden), `pii` (none, personal, sensitive), `audit` (bool, default inherits entity), `allow_edit`, `allow_edit_on_create`, `label`, `help`, `min`, `max`, `pattern`.
- **FR-3.4 System fields** — Must. Every generated table shall carry `id uuid`, `tenant_id uuid`, `legal_entity_id uuid` (when `per_legal_entity=true`), `version int` (RecVersion), `created_at`, `created_by`, `updated_at`, `updated_by`, and optional `deleted_at` for soft delete.
  - Notes: RLS policy generated per FR-2.4; `version` increments on every update and is required in `If-Match`.
- **FR-3.5 Migration generation** — Must. The generator shall emit Atlas migrations that are deterministic for a given definition version; additive changes (new nullable field, new enum value, new index) generate forward-only migrations; destructive changes require an explicit `breaking: true` flag and a new API major version.
  - Notes: `ADD COLUMN … DEFAULT` uses Postgres fast default; index creation uses `CONCURRENTLY`; migrations run as a pre-sync Argo hook job.
- **FR-3.6 Code generation** — Must. For each entity the generator shall emit a Go struct, repository (list with filter/sort/cursor pagination, get, create, update, patch, delete), HTTP handlers, OpenAPI 3.1 document, TypeScript client (fetch-based, typed), and JSON Schema for request validation.
  - Notes: list filter grammar `?filter=field op value` with ops eq, ne, gt, ge, lt, le, in, like, is_null; max page size 500, default 50; cursor is opaque base64.
- **FR-3.7 Lifecycle hooks** — Must. The system shall execute hooks in order initValue → validateField (per changed field) → validateWrite → write → afterWrite, and validateDelete → delete → afterDelete; declarative hooks are expressions returning `{ok, code, message, field}`; coded hooks are Go/TS functions registered by name in the owning service.
  - Notes: validation errors return 422 with `errors[]{field, code, message}`; a coded hook that panics rolls back the transaction and returns 500 with a trace id.
- **FR-3.8 Metadata registry** — Must. The system shall store every definition version, expose diff between versions, publish an immutable version on approval, and serve the current version to generators, the gateway and the shell with ETag caching.
  - Notes: registry DB is `metadata`; publishing emits `dev.passage.metadata.entity.published.v1`; registry read p95 ≤ 20 ms.
- **FR-3.9 Extension packages** — Must. Customers shall create an `ExtensionPackage` (layer customer or partner) that adds fields (prefixed `x_`), enum values (code ≥ 1000), indexes, declarative hooks and labels to any `sys` entity; applying a package generates and runs the migration, regenerates OpenAPI/JSON Schema and invalidates the shell form cache with zero downtime.
  - Notes: limits: 200 extension fields per entity, 50 enum values per enum per package; packages are versioned and reversible only while additive.
- **FR-3.10 Backward compatibility** — Must. CI shall run the previous published version's generated TypeScript client against the new server and fail on any breaking change (removed field, type change, required added, enum value removed); additive changes pass.
- **FR-3.11 Golden tests** — Must. Ten reference entities (Tenant, LegalEntity, User, Role, VendorStub, ProductStub, JournalLineStub, Address, Note, Attachment) shall have committed snapshots of migration, OpenAPI and client; CI fails on any byte-level drift.

- **FR-3.12 Performance** — Must. A generated entity API shall serve 1,000 rps of simple reads with p95 < 50 ms on the 3-node dev cluster and 200 rps of creates with p95 < 120 ms.
- **FR-3.13 Decision tables and rule sets** — Should. The system shall provide a `RuleSet` metadata object (same draft → review → published lifecycle, layers and extension packages as entities) containing one or more `DecisionTable`s with typed input columns (bound to a `TypeDefinition` or an entity field), condition cells (operators eq, ne, lt, le, gt, ge, between, in, like, any), typed result columns, a hit policy (`first`, `priority`, `all`, `collect_sum`), and per-row `effective_from/effective_to`; published tables shall be evaluable by `POST /v1/rules/{code}/evaluate`, from declarative hooks via `rules.eval('<code>', {inputs})` and from wave 4 workflow conditions, with an `explain` mode listing the rows that matched.
    - Notes: limits 10,000 rows and 20 columns per table, 50 tables per rule set; a published version is compiled to an in-memory decision tree per service (p95 ≤ 5 ms for a 1,000-row table); customers add or override rows through an `ExtensionPackage` (`layer=customer` rows evaluate before `sys` rows under `priority`); rows are imported and exported as CSV through wave 5 import/export; later consumers are tax determination (wave 13), output determination (wave 15), credit rules (wave 30), substitution and delegation rules (wave 4), audit policies (wave 48) — none of them may ship a private rules engine.
- **FR-3.14 Usage metering** — Should. The system shall compute, nightly per tenant, a `UsageSnapshot` with active named users (distinct humans with a login or token use in the last 30 and 90 days) per legal entity and per role, principals by kind (human, service, agent), API requests per entity per day (from gateway metrics), and row counts per entity; expose it as the Usage report (0.3.14) with CSV export; retain snapshots 3 years.
    - Notes: a user counted in several legal entities counts once at tenant level; service and agent principals are reported but never counted as named users (PRD §12 Q1: no per-seat charge for shop-floor, portal or agent users); the snapshot is the input to the licensing decision and to capacity planning, not an enforcement point — nothing is blocked by a count.

0.3.4 Data model

EntityDefinition (`metadata.entity`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	`sys` tenant for shipped entities
name	varchar(48)	yes	unique with domain and version
domain	varchar(32)	yes	Postgres database / service
version	int	yes	monotonically increasing
status	enum	yes	draft, review, published, deprecated
table_group	enum	yes	FR-3.1 values
per_legal_entity	bool	yes	default true for main/transaction
title_fields	text[]	no	≤ 2
audit	bool	yes	default true for main/transaction
api_major	int	yes	default 1
spec	jsonb	yes	canonical YAML as JSON
spec_sha256	char(64)	yes	drift detection
published_at, published_by	timestamptz, uuid	no	

FieldDefinition (`metadata.field`)

Field	Type	Required	Notes
id	uuid	yes	PK
entity_id	uuid	yes	FK entity version
name	varchar(48)	yes	snake_case; `x_` reserved for extensions
type	enum	yes	FR-3.2
type_def_id	uuid	no	FK TypeDefinition
enum_id	uuid	no	FK EnumDefinition when type=enum
length, precision, scale	int	no	per type
required, unique	bool	yes	
default_expr	text	no	
computed_expr	text	no	
visibility	enum	yes	always, advanced, hidden
pii	enum	yes	none, personal, sensitive
audit	bool	yes	
layer	enum	yes	sys, partner, customer
extension_package_id	uuid	no	FK when layer ≠ sys
ordinal	int	yes	form/column order default

ExtensionPackage (`metadata.extension_package`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	
code	varchar(32)	yes	unique per tenant
layer	enum	yes	partner, customer
version	int	yes	
status	enum	yes	draft, validated, applied, failed, retired
targets	jsonb	yes	entities, fields, enum values, hooks, labels
applied_at, applied_by	timestamptz, uuid	no	
migration_id	text	no	Atlas migration name

EnumDefinition / EnumValue / TypeDefinition / RelationDefinition / IndexDefinition follow the same shape: id, entity or enum FK, name/code, layer, ordinal; EnumValue has `code int` (sys < 1000, extensions ≥ 1000) and `label`; RelationDefinition has `from_field`, `to_entity`, `to_field`, `on_delete`; IndexDefinition has `fields[]`, `unique`, `where`.

RuleSet / DecisionTable / DecisionRow (`metadata.rule_set`, `metadata.decision_table`, `metadata.decision_row`)

Field	Type	Required	Notes
id	uuid	yes	PK (each table)
tenant_id	uuid	yes	`sys` tenant for shipped rule sets
code	varchar(48)	yes (RuleSet)	unique per tenant and version; e.g. `TaxGroupDetermination`
version, status	int, enum	yes (RuleSet)	draft, review, published, deprecated (BR-3.1 semantics)
rule_set_id, name	uuid, varchar(48)	yes (DecisionTable)	name unique within the set
hit_policy	enum	yes (DecisionTable)	first, priority, all, collect_sum
input_columns, result_columns	jsonb	yes (DecisionTable)	`[{name, type_def_id
table_id, ordinal, priority	uuid, int, int	yes (DecisionRow)	ordinal = evaluation order under `first`; priority used under `priority`
conditions	jsonb	yes (DecisionRow)	`{column: {op, value
results	jsonb	yes (DecisionRow)	typed result values
effective_from, effective_to	date	no (DecisionRow)	evaluated in the tenant time zone
layer, extension_package_id	enum, uuid	yes / no (DecisionRow)	sys, partner, customer

UsageSnapshot (`metadata.usage_snapshot`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	
snapshot_date	date	yes	unique with tenant_id
active_users_30d, active_users_90d	int	yes	distinct humans
by_legal_entity	jsonb	yes	`{legal_entity_id: {active_30d, active_90d}}`
by_role	jsonb	yes	`{role_code: active_30d}`
principals_by_kind	jsonb	yes	`{human, service, agent}`
api_requests_by_entity	jsonb	yes	`{entity: count}` for the day
row_counts	jsonb	yes	`{entity: rows}` from `pg_class.reltuples` (estimate)
computed_at	timestamptz	yes	

Invariants: a published version is immutable; field names unique per entity across layers; an extension cannot change a `sys` field's type or required flag; every `enum` field references an enum; `title_fields` must be string type. Indexing: `entity(tenant_id, domain, name, version)` unique; `field(entity_id, ordinal)`. Partitioning: none (metadata volume < 1M rows).

0.3.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	/v1/metadata/entities?domain=&status=	List entity definitions	n/a	any authenticated
GET	/v1/metadata/entities/{name}	Current published version (ETag)	n/a	any
PUT	/v1/metadata/entities/{name}/versions/{v}	Create/replace a draft version	by version	Maintain entity metadata
POST	/v1/metadata/entities/{name}/versions/{v}/publish	Publish	version	Maintain entity metadata
GET	/v1/metadata/entities/{name}/diff?from=&to=	Structured diff	n/a	any
POST	/v1/metadata/compat-check	Classify a diff as additive/breaking	n/a	Maintain entity metadata
GET/POST	/v1/metadata/extensions	List / create packages	POST: code+version	Apply extensions
POST	/v1/metadata/extensions/{id}/validate, /apply, /retire	Lifecycle	package version	Apply extensions
GET	/v1/metadata/openapi/{entity}	Generated OpenAPI 3.1	n/a	any
GET	/v1/metadata/types, /v1/metadata/enums	EDT and enum catalogs	n/a	any
GET/POST/PATCH/DELETE	/v1///{id}	Generated entity API	POST: Idempotency-Key; PATCH: If-Match	per Permission(entity, action)
GET/PUT/POST	/v1/rules/sets, /{code}/versions/{v}, /{code}/versions/{v}/publish	Maintain and publish rule sets	by version	Maintain business rules
POST	/v1/rules/{code}/evaluate?table=&explain=true	Evaluate a published table; `explain` returns matched rows	n/a (pure function)	any authenticated (results filtered by the calling service)
POST	/v1/rules/{code}/versions/{v}/rows:import	CSV row import (wave 5 import framework)	file sha256	Maintain business rules
GET	/v1/metadata/usage?from=&to=&legal_entity_id=	Usage snapshots	n/a	View usage reports

Event	Trigger	Payload fields	Consumers
dev.passage.rules.ruleset.published.v1	rule set publish	code, version, tables[], row_count	services holding compiled trees (reload), audit
dev.passage.metadata.usage.snapshot.v1	nightly snapshot	tenant_id, snapshot_date, active_users_30d, active_users_90d	analytics (wave 46), licensing report (PRD §12 Q1)
dev.passage.metadata.entity.published.v1	publish	entity, domain, version, api_major, diff_summary{added, changed, removed}	generator pipeline, events (wave 4), shell
dev.passage.metadata.entity.deprecated.v1	deprecate	entity, version, sunset_date	shell, API docs
dev.passage.metadata.extension.applied.v1	apply success	package_id, code, version, entities[], fields_added, enum_values_added, migration_id	audit, shell, generator
dev.passage.metadata.extension.failed.v1	apply failure	package_id, stage, error	alerting

Event	Trigger	Payload fields	Consumers
dev.passage.metadata.schema.migrated.v1	Atlas job completes	domain, migration_id, duration_ms	audit

0.3.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Entity catalog	Browse, open, view diff, publish	name, domain, version, status, field count, extensions	Maintain entity metadata
Entity designer (YAML + form view)	Add field/enum/relation/index/hook, validate, save draft	name, type, required, default, pii, visibility	Maintain entity metadata
Extension packages	Create, validate, apply, retire, view migration log	code, layer, version, status, targets	Apply extensions
Type and enum catalogs	Create EDT, add enum values	code, base type, length, label	Maintain entity metadata
Rule set designer	Add table, define columns, edit rows in a grid with typed cells, test panel (inputs → result + matched rows), publish	code, version, status, table, hit policy, rows, effective dates	Maintain business rules
Usage report	Pick date range and legal entity, export CSV	active users 30/90 d, by role, by legal entity, principals by kind, API requests by entity	View usage reports

Rules: the designer validates against the JSON Schema on every keystroke (debounced 300 ms) and against the registry on save; apply shows a dry-run migration SQL with an estimated lock time and refuses if > 2 s; keyboard: `Ctrl+S` save draft, `Ctrl+Shift+P` publish.

0.3.7 Business rules and edge cases

- **BR-3.1** Entity version status machine: draft → review → published → deprecated; published versions are immutable; a deprecated version serves for 180 days after `sunset_date` announcement.
- **BR-3.2** Breaking vs additive: removing a field, narrowing a type, adding `required` without default, removing an enum value, or renaming are breaking; everything else is additive. Breaking requires `api_major + 1` and a parallel `/v{n+1}` route for ≥ 1 release.
- **BR-3.3** Enum value codes are permanent; retiring a value sets `retired=true` and the value remains readable but not assignable.
- **BR-3.4** Extension apply is transactional per domain database: migration, registry update and cache invalidation succeed together or the package goes to `failed` with the SQL error.
- **BR-3.5** Concurrent publishes of the same entity: second publish gets 409 and must rebase on the new version.
- **BR-3.6** Decimal rounding at write uses the field's scale with `half_even`; values exceeding precision return 422 `VALUE_OUT_OF_RANGE`.
- **BR-3.7** Soft-deleted rows (`deleted_at` set) are excluded from list and get by default; `?include_deleted=true` requires the `delete` action.
- **BR-3.8** `on_delete: restrict` returns 409 `REFERENCED_BY` with the referencing entity and count; cascade is limited to 10,000 child rows per request.
- **BR-3.9** PII `sensitive` fields are masked (`***`) in responses unless the Permission grants the field explicitly; they are never included in list endpoints.
- **BR-3.10** Decision table hit policies: `first` returns the lowest-ordinal matching row; `priority` returns the highest `priority` (ties broken by layer customer > partner > sys, then ordinal); `all` returns every match in ordinal order; `collect_sum` sums numeric result columns across matches. A table with no matching row returns `{matched: false}` and the caller decides (hooks treat it as validation code `RULE_NO_MATCH`). `between` is inclusive on both ends; string comparison is case-insensitive; dates and `effective_*` are evaluated in the tenant time zone.
- **BR-3.11** A named user is "active" on a day when a human principal obtained a token or used one (gateway request) on that day; counts are distinct per tenant, never summed across legal entities; the snapshot for day D is computed at 01:30 tenant time on D+1 and is never recomputed except by an explicit rebuild with the reason recorded.

0.3.8 Acceptance criteria

- **AC-3.1** Given the 10 reference entities, When the generator runs 3 times, Then the migration, OpenAPI and client snapshots are byte-identical and match the committed golden files. (FR-3.5, FR-3.6, FR-3.11)
- **AC-3.2** Given a customer package adding `x_risk_rating` (enum) and enum value `PreferredPlus (1001)` to VendorStub, When applied in dev, Then the API accepts the field within 60 s, no request fails during apply, and the wave-5 form shows it. (FR-3.9)
- **AC-3.3** Given a published v1 client, When the server publishes an additive v2, Then the v1 contract suite passes; Given a breaking change without `breaking: true`, Then CI fails naming the change. (FR-3.10, BR-3.2)
- **AC-3.4** Given 1,000 rps of `GET /v1/identity/users/{id}` for 5 minutes on the dev cluster, When measured at the gateway, Then p95 < 50 ms and error rate < 0.1%. (FR-3.12)
- **AC-3.5** Given a `validateWrite` declarative hook `amount > 0`, When a client posts `amount = -1`, Then 422 with `field=amount`. (FR-3.7)
- **AC-3.6** Given Tenant, LegalEntity and User regenerated from metadata, When the wave-2 test suites run, Then they pass unchanged and the hand-written handlers are removed from the repo. (FR-3.6)
- **AC-3.7** Given a diff request between versions 3 and 5, When called, Then the response lists each added/changed/removed field with classification additive/breaking. (FR-3.8)
- **AC-3.8** Given a field with `pii=sensitive`, When listed by a user without the field permission, Then the field is absent; when fetched by id, Then masked. (FR-3.3, BR-3.9)
- **AC-3.9** Given a published rule set `VendorRiskBand` with a 1,000-row `priority` table (inputs country, spend_band, payment_terms), When evaluated 10,000 times with random inputs from the fixture, Then every result equals the reference evaluator's result, `explain` lists the matched rows, and p95 ≤ 5 ms. (FR-3.13, BR-3.10)
- **AC-3.10** Given a customer extension package adding 3 rows to `VendorRiskBand`, When applied, Then a hook calling `rules.eval` returns the customer row for the overlapping inputs and the sys row otherwise. (FR-3.9, FR-3.13)
- **AC-3.11** Given 50 users of whom 30 obtained tokens in the last 30 days (2 of them in two legal entities) plus 5 service accounts, When the nightly snapshot runs, Then `active_users_30d = 30`, the legal-entity breakdown sums to 32, and `principals_by_kind.service = 5`. (FR-3.14, BR-3.11)

0.3.9 Test plan and fixtures

Fixtures: `metadata/entities/_reference/*.yaml` (10 entities), `fixtures/metadata/golden/` (snapshots), `fixtures/metadata/ext-vendor-risk.yaml`, `fixtures/metadata/breaking-change-cases.yaml` (12 breaking, 12 additive cases), `fixtures/load/users-100k.sql` (100k users for the read benchmark).

Test	Type	Target	Pass criterion
Golden generation	Snapshot	generator	zero drift

Test	Type	Target	Pass criterion
Schema validation	Unit	JSON Schema	100% of invalid fixtures rejected with path
Compat classifier	Table-driven	compat-check	24/24
Extension apply	E2E	registry + Atlas + API	applied, 0 failed requests during apply (k6 background at 100 rps)
Hook ordering	Property	repository	hooks fire in FR-3.7 order for random mutations
Read benchmark	Performance (k6)	generated API	p95 < 50 ms @ 1,000 rps
RLS on generated tables	Contract	every generated table	policy present and effective

Reconciliation: registry field count per published entity equals `information_schema.columns` for the generated table minus system fields; OpenAPI property count equals field count minus hidden fields.

0.3.10 Deployment and operations

Charts: `metadata-registry` (Go, CNPG `metadata` database), `passage-gen` runs in CI and as an in-cluster Job for extension apply. Config: `REGISTRY_URL`, `GEN_OUTPUT_DIR`, `ATLAS_DEV_URL` (scratch database for migration planning). Migrations: generated; applied by Argo pre-sync hook with `atlas migrate apply --tx-mode all`. Feature flags: `metadata.extensions.enabled` per tenant. Scheduled jobs: nightly drift check (registry vs live schema). SLOs: registry availability 99.9%, read p95 ≤ 20 ms, extension apply ≤ 60 s. Alerts: `SchemaDrift`, `ExtensionApplyFailed`, `MigrationJobFailed`. Runbook: failed migration (mark package failed, run `atlas migrate down` only in dev), registry restore from CNPG backup. Rollback: registry and generated services roll back together by digest; because migrations are additive the previous digest runs against the newer schema.

0.3.11 Migration from AX / SAP

Source	Passage entity	Notes
AX AOT Tables (via `SysDictTable` export or AxModel XML) with `TableGroup`, `SaveDataPerCompany`	EntityDefinition	only customer-layer (CUS/VAR/USR) tables and fields are candidates; SYS tables map to Passage entities by crosswalk
AX EDTs, Base Enums	TypeDefinition, EnumDefinition	enum values keep their integer codes where < 1000
AX table extensions (`.extension` XML)	ExtensionPackage	field additions become `x_` fields
SAP DD02L/DD03L for Z*/Y* tables, append structures, CI_* includes	EntityDefinition / ExtensionPackage	DD04L data elements → TypeDefinition; DD07L fixed values → EnumValue

Transformation is a definition import (`passage-gen import-ax`, `import-sap`) producing draft YAML for review; nothing is auto-published. Not migrated: X++ table methods and ABAP user exits (code), SAP delivery-class semantics, AX `CacheLookup` and `Partition`.

0.3.12 Dependencies, risks, open questions

Depends on: wave 2 (tenant_id/RLS, duties `Maintain entity metadata`, `Apply extensions`; the three entities to regenerate).

Risk	Likelihood	Impact	Mitigation
Generator becomes a bottleneck for every later wave	High	High	Generator is itself golden-tested; business waves add entities by YAML only; escape hatch for hand-written endpoints alongside generated ones
Expression language for hooks grows into a scripting language	Medium	Medium	CEL-style expression subset frozen at wave 3; scripted hooks decided in OQ 2
Large-table `ADD COLUMN` with non-constant default locks	Low	Medium	Dry-run lock estimate; non-constant defaults applied by backfill job

Open questions: (1) GraphQL alongside REST — default no until wave 35; decide by wave 10. (2) Customer scripted hooks (sandboxed TS or WASM) — default declarative only through wave 30; decide by wave 20. (3) Soft delete default for main tables — default on for master data groups, off for transactions; decide by wave 6.

0.3.13 Parameters and setup

Registry parameters are rows of `metadata.parameter` edited on Admin workspace → Metadata → Parameters (the AX "System parameters" / SAP SE11 technical-settings equivalent for the kernel); entity-level attributes are part of the YAML and are listed in FR-3.1–FR-3.4.

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`metadata.extensions.enabled`	tenant (feature flag)	bool	true	true/false	allows `ExtensionPackage` apply in the tenant	Feature management (D365)	SCC4 client role "customizing allowed", SE06 system change option
`metadata.extension.max_fields_per_entity`	tenant	int	200	1–500	FR-3.9 limit	none	append structure field limit (practical)
`metadata.extension.max_enum_values_per_package`	tenant	int	50	1–200	FR-3.9 limit	none	DD07L fixed values
`metadata.extension.max_lock_seconds`	environment	decimal	2	0.5–10	apply refuses when the dry-run lock estimate exceeds this	none	SE14 table conversion (no guard)
`metadata.decimal.default_scale_qty`, `metadata.decimal.default_scale_amount`	tenant	int, int	6, 2	0–12	FR-3.2 defaults	EDT `NoOfDecimals`	Domain DECIMALS (CURR/QUAN)
`metadata.rounding_mode`	tenant	enum	half_even	half_even, half_away_from_zero	write-time rounding (BR-3.6); wave 11 ledger may override per legal entity	`round()` X++ semantics	`ROUND` ABAP semantics per currency (TCURX)
`metadata.api.page_size_default`, `metadata.api.page_size_max`	environment	int, int	50, 500	10–200, 100–2,000	FR-3.6 list paging	none (`maxbuffersize`)	OData `$top` limits (/IWFND/MAINT_SERVICE)
`metadata.api.deprecated_version_days`	tenant	int	180	30–365	BR-3.1 sunset window	none	none

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`metadata.soft_delete_default`	tenant	enum per table group	on for main/group, off for transaction (OQ 3)	on/off per table_group	default of `deleted_at` generation	none	deletion flag (LOEVM) convention
`metadata.pii.sensitive_mask`	tenant	string	`***`	≤ 8 chars	BR-3.9 mask text	none	none (UI masking via SU24 field auth)
`metadata.registry.cache_ttl_seconds`	environment	int	60	10–600	ETag cache in gateway/shell	AOS metadata cache	SE11 buffer settings (technical settings → buffering)
`metadata.golden.entities[]`	repository	list	10 reference entities (FR-3.11)	entity names	golden snapshot set	none	none
`metadata.hooks.expression_timeout_ms`	environment	int	50	10–500	declarative hook evaluation guard	none	none
`metadata.hooks.coded_timeout_ms`	environment	int	2,000	100–10,000	coded hook guard; exceeded → 500 with trace id	none	none
`rules.table.max_rows`, `rules.table.max_columns`	tenant	int, int	10,000, 20	≤ 50,000, ≤ 30	FR-3.13 limits	none	BRF+ decision table limits (none enforced)
`rules.compile_on_publish`	environment	bool	true	true	compiled tree built on `ruleset.published`	none	BRF+ generation (FDT_GENERATE)
`usage.snapshot_time`	tenant	HH:MM tenant time	01:30	any	FR-3.14 / BR-3.11	License count refresh batch	USMM schedule
`usage.retention_years`	tenant	int	3	1–10	snapshot retention	none	none
`usage.active_window_days[]`	tenant	int list	30, 90	7–365	windows reported	"Named user license count" window (D365: 30 days)	ST03N month aggregates

0.3.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Entity catalog	list inquiry	domain, status, layer, has extensions	name, domain, version, status, table group, field count, extension count, published at	`metadata.entity`	AOT Tables node, Data entities list (D365)	SE11 table list, SE16 DD02L
Field list	drill-down	entity, layer, pii, audit	field, type, length/precision, required, default, pii, audit, visibility, layer	`metadata.field`	Table → Fields (AOT), SysDictField	SE11 → Fields, DD03L
Where-used	drill-down	field, type definition, enum	referencing entities, relations, indexes, hooks, rule sets, saved views (wave 5)	registry cross-reference	Cross-reference (xRef)	SE11 → Where-used list (RSWBO052)
Entity diff	drill-down	entity, from version, to version	added / changed / removed items with classification additive/breaking	`/diff` endpoint	Model store compare	SE09 version management, SE39 split-screen
Extension package log	list inquiry	status, entity, date	package, layer, version, status, migration id, applied by/at, error	`metadata.extension_package`	Models form (installed models)	SE03 object directory, transport log
Schema drift report	operational report (nightly)	domain	entity, column, registry type, live type, drift kind	registry vs. `information_schema`	none	SE14 consistency check (database utility)
Type and enum catalogs	list inquiry	base type, layer	code, base type, length, label, usages	`metadata.type_definition`, `metadata.enum_definition`	EDTs, Base Enums (AOT)	SE11 domains / data elements, DD07L
API catalog	list inquiry	domain, api_major, status	entity, OpenAPI link, AsyncAPI link (wave 4), deprecated until	`/v1/metadata/openapi/{entity}`	Data entities (IsPublic)	/IWFND/MAINT_SERVICE (OData services), SEGW projects
Generated-API performance	KPI	entity, period	rps, p50/p95/p99, error rate	gateway metrics (Prometheus)	SQL insights (LCS)	ST03N transaction profile, STAD
Rule set catalog and row inquiry	list inquiry / drill-down	rule set, version, table, effective date, layer	code, version, status, tables, rows; per row: conditions, results, effective dates, layer	`metadata.rule_set`, `metadata.decision_row`	Policy rules list	FDT_WORKBENCH object list, decision table display
Rule evaluation trace	drill-down	rule set, inputs	matched rows, hit policy applied, result, duration	`evaluate?explain=true`	none	BRF+ simulation (FDT_WORKBENCH → Simulate), trace (FDT_TRACE)
Usage report	operational report	date range, legal entity	active users 30/90 d, by role, by legal entity, principals by kind, API requests by entity, row counts	`metadata.usage_snapshot`	License inquiries → Named user license counts, License usage by role; SSRS License report	USMM / SLAW, ST03N user profile, RSUSR200

0.3.15 Printed and electronic documents

No printed documents. Electronic documents:

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
OpenAPI 3.1 document per entity	publish	info (entity, version, api_major), paths, components (schemas from JSON Schema), security scheme	JSON, YAML	version = entity version	retained for every published version (immutable)

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
JSON Schema per entity	publish	`$id`, properties with types/formats, required, enum values	JSON Schema 2020-12	same	same
TypeScript client package	publish	typed fetch client, request/response types	npm tarball `@passage/client-<domain>@<version>`	semver from version	retained in registry 7 years (same as images)
Migration file	publish / extension apply	Atlas SQL with header (entity, version, package)	`.sql` under `migrations/<domain>/`	timestamped Atlas name	permanent in Git
Extension package manifest	package validated	code, layer, version, targets, dry-run SQL, lock estimate	YAML (`passage-gen export`)	package code + version	permanent in `metadata.extension_package`
Rule set export	administrator action	tables with columns, rows (conditions, results, effective dates, layer)	CSV per table, YAML for the whole set	rule set code + version	retained with the version
Usage report export	administrator action	snapshot rows	CSV, JSON	none	retained by requester; snapshots 3 years

0.3.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Schema drift check	nightly 01:00 UTC	domain	compares registry to `information_schema`; report rows keyed by (date, domain)	< 1 min per domain	report with drift rows; no automatic change	`SchemaDrift` (ticket)
Extension apply job	on demand (`/apply`)	package	transactional per domain (BR-3.4); package status is the checkpoint	≤ 60 s	package → `failed` with SQL error; rerun after fix	`ExtensionApplyFailed` (notify)
Non-constant default backfill	after an apply that needs it	entity, id range (10 k rows)	checkpoint per range in `metadata.backfill`	100 k rows/min	resume from checkpoint (wave 4 framework after wave 4; CronJob loop before)	`BackfillStalled` (> 30 min no progress)
Deprecated version sunset	daily 02:00 UTC	entity	removes `/v{n}` routes whose sunset passed; idempotent	< 10 s	none	none
Golden snapshot verification	on every CI run and nightly	entity	deterministic	< 2 min	CI failure	CI status
Rule set compile	on `ruleset.published`	rule set	compiled tree keyed by (code, version)	< 5 s for 10 k rows	publish rolled back with error	`RuleCompileFailed` (notify)
Usage snapshot	daily 01:30 tenant time	tenant	one row per (tenant, date); rebuild only with explicit reason (BR-3.11)	< 30 s per tenant	retry 3×; missing day flagged in the report	`UsageSnapshotMissing` (ticket)
Usage snapshot retention	monthly	tenant	deletes rows older than `usage.retention_years`	< 10 s	none	none

0.3.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
Maintain entity metadata	`PUT /v1/metadata/entities/{name}/versions/{v}`, `/publish`, `/compat-check`, Entity designer, Type and enum catalogs	with Apply extensions in production tenants (designer of a sys change should not also apply customer packages without review)	Platform engineer, Solution architect
View entity metadata	`GET /v1/metadata/*`, Entity catalog, Field list, Where-used, Entity diff, API catalog	none	all authenticated users (read of the published version is `any`), Auditor
Apply extensions	`POST /v1/metadata/extensions`, `/validate`, `/apply`, `/retire`, Extension packages screen	with Maintain entity metadata; with Approve workflow for the extended entity (wave 4)	Customer administrator, Partner consultant
Maintain business rules	`/v1/rules/sets`, versions, publish, `rows:import`, Rule set designer	with the business duties that consume a rule set (for example Maintain tax setup, wave 13): a rule author should not also post the documents the rule governs; enforced by wave 48 SoD rule `RULE_AUTHOR_VS_POSTER`	Solution architect, Business administrator
Evaluate business rules	`POST /v1/rules/{code}/evaluate`	none	all services (internal), any authenticated user for `explain` on rules they may read
View usage reports	`GET /v1/metadata/usage`, Usage report	none	Tenant owner, Auditor, Platform operator
Run metadata jobs	trigger drift check, backfill, snapshot rebuild	none	Platform engineer

0.3.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
MDK-0301	error	Entity definition invalid at `{json_path}`: {schema error}	JSON Schema validation	fix the YAML	FR-3.1
MDK-0302	error	Breaking change `{change}` on {entity} requires `breaking: true` and api_major {n+1}	compat classifier	mark breaking and bump, or make the change additive	FR-3.5, BR-3.2
MDK-0303	error	Version {v} of {entity} is published and immutable	PUT on a published version	create a new draft version	BR-3.1
MDK-0304	error	{entity} version {v} was published by {user} after your draft; rebase	concurrent publish (409)	reload and reapply	BR-3.5
MDK-0305	error	Extension field names must start with `x_`	naming rule	rename	FR-3.9
MDK-0306	error	Enum value code {code} must be ≥ 1000 for extensions	layer rule	choose a code ≥ 1000	FR-3.9

Code	Severity	User text	Cause	Remediation	Raised by
MDK-0307	error	Extension cannot change type or required flag of sys field {field}	invariant	add a new field instead	0.3.4 invariants
MDK-0308	error	Apply refused: estimated lock {s} s exceeds {max} s on {table} ({rows} rows)	dry-run lock estimate	schedule a backfill or split the change	0.3.6 rules
MDK-0309	error	Package {code} v{v} failed at {stage}: {sql error}; nothing was applied	transactional apply	fix and rerun	BR-3.4
MDK-0310	error	Value for {field} exceeds precision ({p},{s})	VALUE_OUT_OF_RANGE (422)	correct the value	BR-3.6
MDK-0311	error	Cannot delete {entity} {id}: referenced by {n} {child_entity} rows	REFERENCED_BY (409)	delete or reassign children	BR-3.8
MDK-0312	error	Cascade delete limit of 10,000 child rows exceeded	BR-3.8 limit	delete in batches (wave 4 job)	BR-3.8
MDK-0313	error	Validation failed: {message} (field {field})	declarative/coded hook returned ok=false	correct the field	FR-3.7
MDK-0314	error	Internal error in hook {name}; trace {trace_id}	coded hook panic / timeout	report with trace id	FR-3.7
MDK-0315	warning	Record changed by {user} at {time}; your version {v} is stale	If-Match mismatch (409)	reload	FR-3.4
MDK-0316	error	Decision table {table} has {n} rows; limit {max}	FR-3.13 limit	split the table or raise `rules.table.max_rows`	FR-3.13
MDK-0317	warning	No row matched in {table} for the given inputs	`{matched:false}`	add a catch-all row or handle `RULE_NO_MATCH`	BR-3.10
MDK-0318	error	Row {ordinal}: condition on column {column} uses operator {op} not allowed for type {type}	typed cell validation	choose an allowed operator	FR-3.13
MDK-0319	error	Rule set {code} v{v} is referenced by hook {hook} on {entity}; deprecate instead of retiring	reference check	deprecate with sunset date	FR-3.13
MDK-0320	info	Usage snapshot for {date} rebuilt by {user}: {reason}	explicit rebuild	none	BR-3.11

0.3.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Registry read (current version, ETag)	5,000/min at shell cold start; 200/min steady	5 ms / 20 ms / 40 ms	500 rps	`metadata.*` < 1 M rows total (< 1 GB)	`entity(tenant_id, domain, name, version)` unique; ETag served from cache	cache serves stale ≤ 60 s; registry HPA 2 → 4
Generated entity read (`GET /{id}`)	1,000 rps on dev cluster (FR-3.12)	10 ms / 50 ms / 90 ms	1,000 rps per 3 pods	per entity: rows × avg 1 KB	PK + declared indexes; `(tenant_id, …)` leading column everywhere	pool exhaustion → 503 with `Retry-After`; p95 may reach 150 ms
Generated entity create	200 rps	30 ms / 120 ms / 200 ms	200 rps	same	hooks inline; outbox row (wave 4)	same
List with filter/sort, page 50	300 rps	20 ms / 80 ms / 150 ms	300 rps	n/a	cursor pagination on (sort key, id)	large offsets refused (cursor only)
Publish (generate + migrate)	20/day	10 s / 30 s / 60 s	2 concurrent per domain	migrations ≈ 2 MB/year	one Atlas job per domain	queued; CI concurrency limit
Extension apply (100 M-row table)	5/month	15 s / 45 s / 60 s	1 at a time per domain	n/a	`ADD COLUMN` fast default; `CREATE INDEX CONCURRENTLY`	refused above lock estimate (MDK-0308)
Rule evaluation (1,000-row table)	2,000/s across services at peak	0.5 ms / 5 ms / 10 ms	5,000/s per service instance	rule tables ≤ 50 MB per tenant	compiled tree in memory per (code, version)	CPU-bound; no errors
Usage snapshot	1/day/tenant	5 s / 30 s / 60 s	n/a	365 rows × ≈ 20 KB per tenant per year	`(tenant_id, snapshot_date)` unique	runs later; `UsageSnapshotMissing` after 24 h

0.3.20 Localization and compliance hooks

- Labels and help text: `label` and `help` on entities, fields, enum values and rule columns are stored as keys in the registry; English is the only language in wave 3. Wave 48 label packs supply translations per country/language and the shell (wave 5) resolves them per user locale; the extension point is `metadata.label` keyed by (object, locale).
- Formats: dates, decimals and money are stored canonically (ISO 8601, `decimal`, `money` + ISO 4217) and formatted in the shell per user preference (wave 5); country packs never change storage types. Currency minor-unit scale comes from the wave 11 currency table; until then `money` uses scale 6 for storage and 2 for display.
- PII classification (`pii=personal|sensitive`) is the hook used by wave 48 privacy tooling (GDPR/CCPA erasure, data-subject export) and by wave 5 audit masking; country packs may raise a field's classification but never lower it.
- Statutory fields: country packs add fields through `ExtensionPackage` with `layer=partner` (for example Brazil `x_cnpj`, India `x_gstin`) — the same mechanism customers use; validation patterns go in `pattern` and in rule sets (FR-3.13).
- Decision tables are the extension point for country-dependent determinations (tax groups, document types, numbering formats): a country pack ships `sys`-layer rows with a `country` input column; customers override with `customer`-layer rows.
- Retention: metadata versions are never deleted (audit trail of schema), which satisfies SOX/ITGC change-management evidence (wave 48) together with the Git history of `metadata/entities`.
- Usage metering stores no personal data beyond counts; per-user activity lives in wave 2 login events and wave 5 audit.

0.3.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.30.2 System administration → Inquiries & Reports: License inquiries (Named user license counts, License usage by role) (register row 265)	PARTIAL-3	ADDED as FR-3.14	Usage report in 0.3.14; `UsageSnapshot` entity; named users distinct per tenant (BR-3.11); counts inform PRD §12 Q1, never block

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.30.2 System administration → SSRS: License report / Named user counts (register row 266)	PARTIAL-3	ADDED as FR-3.14	CSV/JSON export in 0.3.15; PDF layout arrives with the wave 15 report designer
AX §12.30.3 System administration → Periodic: Licensing → Named user license count refresh (register row 268)	PARTIAL-3	ADDED as FR-3.14	nightly Usage snapshot job (0.3.16) at `usage.snapshot_time`; rebuild with reason (MDK-0320)
SAP §13.31 Cross-Application → BRF+ (BRFPLUS / FDT_WORKBENCH) / Decision Service Management: business rules framework (register row 348)	PARTIAL-3	ADDED as FR-3.13	`RuleSet` / `DecisionTable` / `DecisionRow`; hit policies in BR-3.10; evaluate and explain endpoints; consumers in waves 4, 13, 15, 30, 48 must use it
SAP §13.31 Cross-Application → SAP Business Rules (BRF+ in GRC, tax, dispute, output) (register row 352)	PARTIAL-3	ADDED as FR-3.13	same mechanism; per-domain rule sets are authored in the consuming waves (tax 13, output 15, credit/dispute 30, GRC 48)
SAP §13.31 Customizing (SPRO) → BRF+ applications, functions, rulesets, decision tables, expressions (FDT_*) (register row 354)	PARTIAL-3	ADDED as FR-3.13	application = rule set, function = table evaluation, ruleset/decision table = `DecisionTable`, expression = condition cells and `computed_expr` from FR-3.3; formulas beyond the expression subset are deferred with open question 2

0.4 Wave 4: Event Backbone, Workflow & Batch

0.4.1 Scope and business context

AX runs approvals through the workflow framework (`WorkflowTable`, `WorkflowVersionTable`, `WorkflowWorkItemTable`, `WorkflowTrackingStatusTable`, `WorkflowTrackingTable`; approval, task and conditional-decision elements; work item statuses Pending/Delegated/Completed/Canceled; escalation and due dates) and background work through the batch framework (`BatchJob`, `Batch`, `BatchJobHistory`, `BatchGroup`, `RunBaseBatch`, `SysOperationServiceController` with `SysOperationExecutionMode` Synchronous/Asynchronous/ReliableAsynchronous/ScheduledBatch, job statuses Waiting/Executing/Ended/Error/Hold/Canceling/Canceled); D365 adds business events. SAP has Business Workflow (SWDD, work items SWWWIHEAD, SWIA/SWI1, SBWP inbox, event linkage SWE2, deadline monitoring), background jobs (SM36 definition, SM37 monitor, TBTCO/TBTCP, statuses Scheduled/Released/Ready/Active/Finished/Canceled, job classes A/B/C, SM62 events) and IDoc status processing (EDIDC/EDIDS status 03/51/53/64, BD87). Wave 4 replaces these with Strimzi Kafka carrying CloudEvents, a transactional outbox in every service via Debezium, a schema registry fed by entity metadata, Temporal for approvals and batch, a configuration-driven approval chain, a partitioned checkpointing batch framework and a job monitor. Out of scope: notification channels beyond in-app and email (wave 46 adds Teams/Slack), position-hierarchy assignee resolution (wave 42), line-item workflows (wave 17 first consumer), IDoc/EDI payload semantics (wave 35).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Downstream systems (MES, WMS, BI) poll `ModifiedDateTime` on AX tables or wait for the nightly IDoc/BD21 change-pointer job; IDocs stuck in status 51/64 are reprocessed by hand in BD87	No reliable push of changes; outbound integration is batch	Every committed change is a CloudEvent on Kafka via Debezium outbox within 500 ms; consumers subscribe, never poll	Change-to-consumer latency: 15 min–24 h → < 2 s p95; manual reprocessing runs per week: dozens → 0 (DLQ with reason, replayable)
AX workflow approvals: the workflow batch job (`WorkflowMessageProcessingTask`) runs every minute and the work-item list is refreshed manually; escalation "due dates" silently fail when the batch server is down	Workflow runtime is a polling batch job on the AOS	Temporal workflows with durable timers; work item pushed to the inbox by event	Submit-to-inbox latency: 1–5 min → < 2 s; missed escalations due to batch outage: possible → 0 (timers survive restarts)
SAP Business Workflow (SWDD) changes need an ABAP workflow developer, a transport and SWU3 checks; amount bands for release strategies live in customizing tables (CL20N/CL24N)	Chains are code and classification objects, not configuration	Designer with amount bands, conditions and SLAs as versioned data, activated per legal entity	Approval chain change: 1–3 weeks → < 30 min by an administrator; developer involvement: required → none
A failed AX batch job over 1M rows (e.g. inventory recalculation) restarts from zero; SM37 jobs show "Canceled" with a dump in ST22 and no restart point	Batch frameworks have no checkpoints or partitions	Partitioned tasks with checkpoints every 10k rows; resume from last checkpoint; rerun failed partitions only	Rework after a failure at 90%: 90% of the run → < 1%; mean time to resume: hours → < 1 min
Exactly-once delivery requires custom dedupe tables in X++/ABAP and still double-posts under AOS/work-process crashes	Dual writes (DB + queue) in application code	Outbox in the same transaction; consumer library dedupes on CloudEvents `id`	Duplicate side effects under chaos: observed → 0 over 10 kill cycles

0.4.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Change notification	Business events (D365 `BusinessEventsBase`), AIF outbound ports, alerts (`EventInbox`)	Event type linkage SWE2, change pointers BD21, IDoc outbound	CloudEvents on Kafka via outbox; AsyncAPI per topic	improved
Reliable publish	AIF queue (`AifGatewayQueue`), dual-write risk in X++	tRFC/qRFC (SM58), IDoc status 03	Debezium outbox event router; exactly-once effect	improved
Workflow definition	Workflow editor, `WorkflowTypeTable`, `WorkflowDocument` class, approval/task elements	SWDD workflow builder, tasks TS*, agent rules PFAC	`WorkflowDefinition` JSON (steps, conditions, SLAs), designer UI	full
Work items	`WorkflowWorkItemTable`, Work list, Delegate/Approve/Reject/Request change	SWWWIHEAD, SBWP inbox, forward/resubmit	`WorkItem` with approve/reject/delegate/request_change/recall	full
Escalation and deadlines	Time limits, escalation path, `WorkflowEscalation*`	Deadline monitoring (latest end), SWWDHEX	Temporal timers; SLA + escalation action	full
Workflow history	`WorkflowTrackingTable`, tracking status form	SWIA, workflow log (SWI1/SWEL)	Temporal history + `WorkflowInstance` timeline API	improved
Background jobs	`BatchJob`, `Batch`, `BatchGroup`, `SysOperation` data contracts, recurrence (`SysRecurrence`)	SM36/SM37, TBTCO/TBTCP, variants, job classes A/B/C	`BatchJob` definition, `BatchRun`, `BatchTask`, cron recurrence, priority	full
Job monitoring	Batch job history, `BatchJobHistory`, Infolog	SM37, job log (TBTC5), SM50	Job monitor UI, run log, Prometheus metrics	improved
Restart / reprocess	Rerun batch job, no checkpoints	BD87 IDoc reprocess, job copy in SM37	checkpoint per partition; resume from last checkpoint	improved
Outbound email	Email parameters (`SysEmailParameters`: SMTP, Exchange, mail sending mode batch/client), Email distributor batch (`SysEmailBatch`), Outbound email status (`SysOutgoingEmailTable`), Retry failed	SAPconnect (SCOT) nodes, send job RSCONN01, SOST send status, SOSV/SOSG	notify service: `OutboundMessage` queue, provider adapter (SMTP / HTTP API), distributor job, status inquiry, retry, bounce ingest	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Message templates and notification settings	Email templates (`SysEmailTable` / `SysEmailMessageTable`, per language), Workflow approval email templates, Workflow parameters → Notifications, User options → Workflow notifications (email, in-app, digest), Alert email templates	SO10 standard texts in workflow tasks, SWNCONFIG extended notifications (digest schedule, delivery), SBWP settings, SU3 user defaults	`MessageTemplate` (channel, language, placeholders, layout), sys templates per workflow event, `NotificationPreference` per user, per-definition notify flags	full

0.4.3 Functional requirements

- **FR-4.1 Kafka and topics** — Must. The system shall run Strimzi-managed Kafka (3 brokers in cloud, 1 in k3d, KRaft) with topics named `passage.<domain>.<aggregate>.v<n>`, partition key = aggregate id, 12 partitions default, retention 7 days (entity-state topics compacted), and CloudEvents 1.0 JSON envelopes with extensions `tenantid`, `legalentityid`, `traceparent`, `actor`.
  - Notes: no per-tenant topics; consumers filter by `tenantid` header; max message 1 MB, larger payloads use a claim-check reference (object storage URL).
- **FR-4.2 Transactional outbox** — Must. Every service shall write domain changes and an `outbox_event` row in the same transaction; Debezium (one connector per domain database, pgoutput) with the outbox event router publishes and never writes to Kafka from application code.
  - Notes: outbox columns id, aggregate_type, aggregate_id, event_type, payload (jsonb), headers (jsonb), created_at; rows deleted after 24 h by a sweep; publish lag p95 ≤ 500 ms.
- **FR-4.3 Schema registry** — Must. Event payload schemas shall be JSON Schema derived from the metadata registry (`dev.passage.<domain>.<entity>.<verb>.v<n>` ↔ entity version) registered with BACKWARD_TRANSITIVE compatibility; CI rejects an incompatible schema.
- **FR-4.4 AsyncAPI generation** — Must. The generator from wave 3 shall emit an AsyncAPI 3.0 document per domain listing topics, event types and schemas; published at `/v1/events/asyncapi/{domain}`.
- **FR-4.5 Consumer library** — Must. Go and TS libraries shall provide idempotent consumption keyed by CloudEvents `id` (dedupe table per consumer group, 7-day window), ordered processing per partition, retry with exponential backoff (1 s → 5 min, 10 attempts) and a per-topic dead-letter topic `…dlq`.
- **FR-4.6 Temporal cluster** — Must. The system shall run Temporal (CNPG persistence, 1 namespace per environment, `tenant_id` as a search attribute) with Go and TS SDK wrappers that inject tenant, actor and trace context into every workflow and activity.
  - Notes: workflow ids `wf-{tenant}-{entity}-{record_id}-{seq}`; history retention 90 days; activity timeouts default 30 s start-to-close, heartbeat for long activities.
- **FR-4.7 Approval workflow template** — Must. The system shall provide a configurable approval workflow with steps (assignee resolution: user, role, expression, `record.owner`, manager placeholder), conditions (expression over the document, e.g. `amount_minor > 1000000`), completion policy (single, all, majority, percentage n), SLA (ISO 8601 duration) and escalation action (delegate_to, auto_approve, auto_reject, notify_only), and actions submit, approve, reject, delegate, request_change, recall, cancel.
  - Notes: a step with zero resolved assignees escalates immediately to the workflow administrator; delegation may chain at most 3 levels; delegation records keep the original assignee for audit.
- **FR-4.8 Workflow designer** — Must. Administrators shall configure approval chains per entity with amount bands and conditions in a form (no code), version them, and activate one version per entity × legal entity.
- **FR-4.9 Batch job framework** — Must. The system shall define jobs (`BatchJob`: handler name, parameters JSON Schema, partition strategy rangeIhashIlist, partition count ≤ 256, concurrency ≤ 32, priority A/B/C, retry policy), schedule them by cron (tenant time zone) or on demand, execute each partition as a Temporal activity with checkpoints every N rows (default 10,000) and resume from the last checkpoint after failure.
  - Notes: statuses waiting, scheduled, executing, ended, error, hold, canceling, canceled; max run time default 6 h; a job may declare `exclusive=true` to prevent overlapping runs.
- **FR-4.10 Job monitor** — Must. The system shall show runs, tasks, progress (rows/sec, ETA), logs and errors; allow cancel, hold, resume and rerun-failed-partitions; retain history 90 days.
- **FR-4.11 Metadata-changed events** — Must. The metadata registry shall emit `dev.passage.metadata.entity.published.v1` through its own outbox (first production consumer of the backbone).
- **FR-4.12 Chaos and replay** — Should. The platform shall support topic replay from an offset or timestamp per consumer group and ship a chaos test that kills a service between commit and publish.
- **FR-4.13 Outbound email and message delivery** — Must. The system shall provide a `notify` service (TS) with a provider adapter interface (`Send`, `GetStatus`, `IngestBounce`) implemented for SMTP (STARTTLS or implicit TLS, credentials via ESO) and for an HTTP API provider (the SMTP relay is the P0 default per open question 2); a per-tenant sender identity (from address, display name, reply-to, envelope domain whose SPF/DKIM/DMARC are the customer's responsibility and are checked by a "verify sender" action); an `OutboundMessage` queue (channel email I in_app I webhook; status queued → sending → sent I failed I bounced; attempts; provider message id); a distributor job (every 60 s, batch 500, 5 attempts with backoff 1 min → 1 h); a mail sending mode per tenant (`immediate` for workflow and alerts, `batch` for digests and reports); attachments ≤ 10 MB total per message; an outbound status inquiry; manual retry; and a bounce webhook/IMAP ingest that marks the message `bounced` and, after 3 hard bounces, flags the recipient user (`identity.user.email_status=bouncing`).
  - Notes: in-app messages are rows in the same table read by the shell notification tray (wave 5) via SSE; webhook deliveries sign the body with HMAC-SHA256 (`X-Passage-Signature`) and retry on non-2xx; messages are retained 90 days (body) and 3 years (metadata); a tenant with no sender identity falls back to the platform default `no-reply@<env>.passage.dev` with a warning on the parameters form; the service never generates email in test environments unless `notify.allowlist_domains` matches the recipient.
- **FR-4.14 Message templates and notification settings** — Must. The system shall provide `MessageTemplate` (code, channel, language, subject, body in Mustache syntax with placeholders resolved from the entity record via metadata (`{{record.number}}`, `{{record.amount | money}}`), the workflow context (`{{workitem.due_at | date}}`, `{{instance.submitted_by.display_name}}`, `{{link}}` deep link to the work item or record) and standard texts from wave 5 (`{{text:CODE}}`); an HTML layout wrapper per tenant (logo, footer); versioning and `sys/customer` layers so customers override shipped templates per language); shipped `sys` templates for `workflow.workitem.assigned`, `.escalated`, `.delegated`, `instance.completed`, `instance.change_requested`, `batch.run.failed`, `identity.user.invited`; per-user `NotificationPreference` (per event class: in_app, email, none; digest none I daily at HH:MM I weekly on day; language; quiet hours); per-`WorkflowDefinition` flags (notify_on_assign, notify_on_escalate, notify_submitter_on_complete, comment_required); and a "preview with sample record" action.
  - Notes: rendering is fail-closed — a missing placeholder or render error marks the message `failed` with NTF-0407 instead of sending a blank; templates are rendered in the recipient's language, falling back to the tenant default then `en`; digests aggregate pending in-app items into one email at the chosen time (BR-4.12); user-level workflow delegation stays in `WorkItem` (FR-4.7) and is referenced, not duplicated, by preferences.

0.4.4 Data model

OutboxEvent (`<domain>.outbox_event`, every service)

Field	Type	Required	Notes
id	uuid	yes	PK; becomes CloudEvents `id`

Field	Type	Required	Notes
tenant_id	uuid	yes	RLS; copied to header `tenantid`
aggregate_type	varchar(64)	yes	entity name; routes to topic
aggregate_id	uuid	yes	partition key
event_type	varchar(128)	yes	`dev.passage.<domain>.<entity>.<verb>.v<n>`
payload	jsonb	yes	CloudEvents `data`
headers	jsonb	yes	traceparent, actor, legalentityid, schema_version
created_at	timestamptz	yes	CloudEvents `time`

WorkflowInstance (`workflow.instance`)

Field	Type	Required	Notes
id	uuid	yes	PK; equals Temporal workflow id suffix
tenant_id, legal_entity_id	uuid	yes	
definition_id, definition_version	uuid, int	yes	FK WorkflowDefinition
entity, record_id	varchar(48), uuid	yes	document under approval
status	enum	yes	submitted, in_review, change_requested, approved, rejected, recalled, canceled, failed
submitted_by, submitted_at	uuid, timestamptz	yes	
current_step	int	no	1-based
completed_at, outcome_reason	timestamptz, text	no	
temporal_run_id	varchar(64)	yes	

WorkItem (`workflow.work_item`)

Field	Type	Required	Notes
id	uuid	yes	PK
instance_id	uuid	yes	FK
step	int	yes	
assignee_id	uuid	yes	FK user
original_assignee_id	uuid	no	set when delegated
status	enum	yes	pending, claimed, approved, rejected, delegated, escalated, change_requested, canceled, expired
due_at	timestamptz	yes	SLA
acted_at, comment	timestamptz, text	no	

BatchJob / BatchRun / BatchTask (`workflow.batch_*`): BatchJob has id, tenant_id, code (unique), handler, params_schema jsonb, partition_strategy, partition_count, concurrency, priority, cron, time_zone, exclusive, retry_policy jsonb, status (active, hold); BatchRun has id, job_id, params jsonb, status (FR-4.9), scheduled_at, started_at, ended_at, rows_total, rows_done, error, triggered_by; BatchTask has id, run_id, partition_no, status, checkpoint jsonb, attempts, last_error, rows_done.

OutboundMessage (`notify.outbound_message`, partitioned monthly)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS
channel	enum	yes	email, in_app, webhook
recipient_user_id, recipient_address	uuid, varchar(254)	one of	user for in_app/email; address for external email; URL for webhook
template_code, template_version, language	varchar(48), int, char(5)	no	null when sent raw by a service
subject, body_html, body_text	varchar(255), text, text	yes (email)	rendered content; body retained 90 days
related_entity, related_record_id	varchar(48), uuid	no	deep link target
status	enum	yes	queued, sending, sent, failed, bounced, read (in_app only)
attempts, next_attempt_at, last_error	int, timestamptz, text	yes/no/no	BR-4.8-style backoff
provider_message_id	varchar(200)	no	for bounce correlation
priority	enum	yes	immediate, batch
created_at, sent_at	timestamptz	yes/no	

MessageTemplate (`notify.message_template`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	`sys` tenant for shipped templates
code	varchar(64)	yes	e.g. `workflow.workitem.assigned`; unique with language, layer, version
channel	enum	yes	email, in_app, webhook
language	char(5)	yes	BCP 47 (`en`, `de-DE`)
layer	enum	yes	sys, customer
version, status	int, enum	yes	draft, published, retired
subject, body	varchar(255), text	yes	Mustache; placeholders validated against the event's payload schema at publish
layout_id	uuid	no	FK tenant HTML layout
sample_payload	jsonb	no	for preview

NotificationPreference (`notify.preference`)

Field	Type	Required	Notes
user_id	uuid	yes	PK with event_class
tenant_id	uuid	yes	
event_class	enum	yes	workflow_assigned, workflow_escalated, workflow_completed, batch_failed, alert (wave 5), system
channels	enum set	yes	in_app, email; default in_app + email for workflow_assigned
digest	enum	yes	none, daily, weekly
digest_time, digest_day	time, int	no	tenant time zone; day 1–7
quiet_hours	int4range	no	hours suppressed for email (digest still sent)
language	char(5)	no	null = user locale

Invariants: exactly one active `WorkflowDefinition` version per (entity, legal_entity); a WorkItem is terminal once acted; `BatchRun` of an exclusive job cannot overlap another run of the same job; outbox rows are never updated. Indexing: `outbox_event(created_at)`; `work_item(tenant_id, assignee_id, status)`; `instance(tenant_id, entity, record_id)`; `batch_run(job_id, scheduled_at desc)`. Partitioning: `work_item` and `batch_task` by month from wave 30.

0.4.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	/v1/events/topics, /v1/events/asyncapi/{domain}	Topic catalog, AsyncAPI	n/a	any
GET/POST	/v1/events/schemas/{type}	Read / register schema	by type+version	Maintain entity metadata
POST	/v1/events/replay	Replay topic for consumer group from offset/time	request id	Maintain batch jobs
GET/POST/PUT	/v1/workflows/definitions, /{id}/versions/{v}, /{id}/activate	Definitions	by version	Maintain workflows
POST	/v1/workflows/instances	Submit a document	entity+record_id+seq	per entity `post/update`
GET	/v1/workflows/instances/{id}, /{id}/timeline	Status and history	n/a	submitter, assignee, Maintain workflows
POST	/v1/workflows/instances/{id}/actions	recall, cancel	action+instance version	submitter / Maintain workflows
GET	/v1/workitems?assignee=me&status=pending	Inbox	n/a	any
POST	/v1/workitems/{id}/actions	approve, reject, delegate, request_change	workitem id + action	assignee (Approve workflow)
GET/POST/PUT	/v1/batch/jobs, /{id}	Define jobs	by code	Maintain batch jobs
POST	/v1/batch/jobs/{id}/runs	Start run	Idempotency-Key	Maintain batch jobs
GET/POST	/v1/batch/runs/{id}, /{id}/cancel, /{id}/resume, /{id}/rerun-failed	Monitor and control	run id + action	View batch history / Maintain batch jobs
GET/POST/PUT	/v1/notify/templates, /{code}/versions/{v}, /{code}/versions/{v}/publish, /{code}/preview	Templates and preview	by version	Maintain message templates
POST	/v1/notify/messages	Enqueue a message (internal services; template or raw)	Idempotency-Key (24 h)	internal / Send notifications
GET/POST	/v1/notify/messages?status=&recipient=&from=&to=, /{id}/retry	Outbound status and retry	retry: message id	View outbound messages / Maintain notification settings
GET/PUT	/v1/notify/preferences/me, /v1/notify/preferences/{user_id}	Notification preferences	last write wins	self / Maintain users
GET/PUT/POST	/v1/notify/settings, /v1/notify/settings/verify-sender, /test-send	Tenant sender identity, provider, mode; test	n/a	Maintain notification settings
POST	/v1/notify/bounces	Provider bounce webhook (HMAC-verified)	provider message id	provider credential
GET	/v1/notify/inbox?unread=true	In-app messages for the shell tray (SSE at `/v1/notify/inbox/stream`)	n/a	self

Validation: approve/reject require a comment when the definition flags `comment_required`; delegate target must hold the step's role; a run request against a `hold` job returns 409; a template publish fails when a placeholder is not in the event's payload schema (NTF-0406); `test-send` is limited to 10 per hour per tenant.

Event	Trigger	Payload fields	Consumers
dev.passage.workflow.instance.submitted.v1	submit	instance_id, entity, record_id, definition_version, submitted_by	owning service (sets record status), audit
dev.passage.workflow.workitem.assigned.v1	step entered / delegated	workitem_id, instance_id, assignee_id, due_at, step	shell inbox, notifier
dev.passage.workflow.workitem.escalated.v1	SLA timer fired	workitem_id, from_assignee, to_assignee, action	notifier, audit
dev.passage.workflow.instance.completed.v1	terminal status	instance_id, outcome (approved/rejected/canceled/recalled), completed_at	owning service, audit, shell
dev.passage.batch.run.started.v1 / completed.v1 / failed.v1	run transitions	job_code, run_id, rows_total, rows_done, duration_ms, error	job monitor, alerting, audit
dev.passage.metadata.entity.published.v1	registry outbox	(see wave 3)	events service (topic/schema create)
dev.passage.notify.message.sent.v1 / failed.v1 / bounced.v1	distributor outcome	message_id, channel, template_code, recipient_user_id (no address in payload), attempts, error	audit, alerting (`NotifyFailureRate`), identity (bounce flag)
dev.passage.notify.template.published.v1	template publish	code, version, language, layer	notify workers (cache reload), audit

0.4.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Work items (inbox)	Approve, reject, delegate, request change, bulk approve	document, step, submitted by, due, age	Approve workflow
Workflow instance timeline	View steps, comments, timers; recall	status, step, assignee, acted at, SLA	submitter / Maintain workflows
Workflow designer	Add step, condition, SLA, escalation; version; activate	entity, legal entity, bands, assignees	Maintain workflows
Batch jobs	Define, schedule, hold, run now	code, handler, cron, last run, next run	Maintain batch jobs
Job monitor	Watch progress, cancel, resume, rerun failed, view log	run, status, rows/sec, ETA, errors by partition	View batch history
Topic catalog	Browse topics, schemas, consumer lag	topic, partitions, schema version, lag	Maintain batch jobs
Message templates	Create, edit (subject/body with placeholder picker), preview with sample record, publish per language	code, channel, language, layer, version, status	Maintain message templates

Screen	Primary actions	Key fields/columns	Duty required
Outbound messages	Filter by status/recipient/date, open rendered message, retry, cancel queued	sent at, channel, recipient, subject, status, attempts, last error, provider id	View outbound messages
Notification settings (tenant)	Sender identity, provider, mode, allowlist, verify sender, test send	from, reply-to, provider, mode, SPF/DKIM check result	Maintain notification settings
My notification preferences	Per event class choose channels, digest, quiet hours, language	event class, in-app, email, digest, time	self

Rules: bulk approve is limited to 100 items per action and requires a confirmation listing document totals; the inbox refreshes from events within 2 s; designer validates expressions against the entity's current metadata; keyboard `a` approve, `r` reject, `d` delegate on a focused row.

0.4.7 Business rules and edge cases

- **BR-4.1** Instance status machine: submitted → in_review → (approved | rejected | change_requested → submitted on resubmit); submitted/in_review → recalled (by submitter, before any approval) or canceled (by administrator); any → failed on unrecoverable activity error (alert raised).
- **BR-4.2** Work item status machine: pending → claimed → approved | rejected | change_requested; pending/claimed → delegated (new item created) | escalated (on SLA) | expired (SLA with `auto_reject`) | canceled (instance terminal).
- **BR-4.3** SLA timers use the tenant time zone and the tenant work calendar when one exists (wave 6); in P0 they are wall-clock; the timer survives Temporal worker restarts.
- **BR-4.4** Completion policy `majority` rounds up (3 of 4); `percentage 60` on 5 assignees needs 3; a single reject under `all` rejects the step immediately.
- **BR-4.5** Exactly-once effect: a consumer that sees a CloudEvents `id` already in its dedupe table acknowledges without side effects; the dedupe table is keyed (consumer_group, event_id) and swept after 7 days.
- **BR-4.6** Ordering is guaranteed per aggregate id only; consumers must not assume cross-aggregate order.
- **BR-4.7** Batch checkpoint is committed in the same transaction as the processed rows; after resume, rows with ids ≤ checkpoint are skipped, so handlers must process in key order within a partition.
- **BR-4.8** Retry policy: activity failures retry up to 5 times with backoff 2 s × 2^n capped at 5 min; after that the task is `error` and the run continues other partitions; run ends `error` if any task is `error`.
- **BR-4.9** A workflow definition change never alters running instances; they finish on their version.
- **BR-4.10** Poison events after 10 delivery attempts go to the DLQ with the last error in header `dlqreason`; a DLQ depth > 0 for 10 min alerts.
- **BR-4.11** Outbound message status machine: queued → sending → sent | failed; sent → bounced (provider bounce); queued → canceled (administrator); in_app: queued → sent → read. Rendering happens at enqueue time (the stored body is what was sent); a render error is `failed` with NTF-0407 and never a partially rendered send. Retries: attempts 1–5 at 1 min, 5 min, 15 min, 1 h, 1 h; after the fifth failure the message stays `failed` and only a manual retry resets `attempts`.
- **BR-4.12** Digest: for a user with `digest=daily 08:00`, every in-app item of the configured event classes created since the previous digest is listed in one email at 08:00 tenant time; items already read in the shell are omitted; an empty digest is not sent; `workflow_escalated` always sends immediately regardless of digest (escalations are time-critical); quiet hours delay immediate email to the end of the window but never delay in-app.
- **BR-4.13** Bounce handling: a hard bounce (5xx, unknown recipient) marks the message `bounced`; the third hard bounce for the same address within 30 days sets `identity.user.email_status=bouncing`, suppresses further email to that user (in-app continues) and notifies Security administrators; a soft bounce (4xx, mailbox full) is retried per BR-4.11.

0.4.8 Acceptance criteria

- **AC-4.1** Given a service killed by SIGKILL between DB commit and Debezium capture, 10 times in a loop, When it restarts, Then every committed outbox row appears on Kafka exactly once (consumer dedupe table shows 0 duplicates, 0 gaps). (FR-4.2, FR-4.5)
- **AC-4.2** Given a 3-level chain (role → user → expression) with 4-hour SLAs, When step 2's assignee delegates and step 3 times out with `delegate_to admin`, Then the instance completes approved after simulated clock advance and the timeline shows 5 work items. (FR-4.7)
- **AC-4.3** Given a batch job over 1,000,000 synthetic rows in 16 hash partitions, When the worker pod is killed at ~40%, Then the run resumes and ends with rows_done = 1,000,000 and no duplicate side effects. (FR-4.9)
- **AC-4.4** Given a schema change that removes a required field, When CI registers it, Then the registry rejects it with the incompatible path. (FR-4.3)
- **AC-4.5** Given an entity publish in the registry, When the outbox is captured, Then a topic `dev.passage.metadata.entity.v1` message with type `dev.passage.metadata.entity.published.v1` exists within 1 s. (FR-4.11)
- **AC-4.6** Given a submitter recalls after step 1 approved, When the action is sent, Then 409 `RECALL_NOT_ALLOWED`. (BR-4.1)
- **AC-4.7** Given a consumer group replay from `T-1h`, When executed, Then events are redelivered and the dedupe table suppresses side effects for already-processed ids. (FR-4.12)
- **AC-4.8** Given an exclusive job with a running instance, When a second run is requested, Then 409 `RUN_OVERLAP`. (FR-4.9)
- **AC-4.9** Given a step whose assignee expression resolves to nobody, When the step is entered, Then it escalates to the workflow administrator within 5 s and emits `workitem.escalated`. (FR-4.7)
- **AC-4.10** Given a tenant with a verified SMTP sender and a user whose `workflow_assigned` preference is in_app + email, When a work item is assigned, Then an in-app message appears in the tray within 2 s and an email rendered from the `sys` template (subject contains the document number, body contains the deep link) reaches the CI mail sink within 60 s. (FR-4.13, FR-4.14)
- **AC-4.11** Given the SMTP relay is down for 20 minutes, When 50 messages are enqueued, Then all 50 show `queued`/`sending` with attempts in the outbound inquiry, none is lost, and all are `sent` within 10 minutes of the relay returning. (FR-4.13, BR-4.11)
- **AC-4.12** Given a user with `digest=daily 08:00` and 7 unread assignments of which 2 were read in the shell, When 08:00 tenant time passes, Then exactly one email lists 5 items and no per-item emails were sent. (FR-4.14, BR-4.12)
- **AC-4.13** Given a customer template override for `workflow.workitem.assigned` in `de-DE` referencing placeholder `{{record.nonexistent}}`, When published, Then the publish is rejected with NTF-0406 naming the placeholder. (FR-4.14)
- **AC-4.14** Given the provider reports a hard bounce three times for the same address within 30 days, When the third bounce is ingested, Then the user's `email_status` is `bouncing`, subsequent messages to that user are in-app only, and Security administrators receive a notification. (BR-4.13)

0.4.9 Test plan and fixtures

Fixtures: `fixtures/events/chaos-outbox.yaml` (kill schedule), `fixtures/workflow/chain-3-level.json`, `fixtures/workflow/completion-policies.json` (12 cases), `fixtures/batch/rows-1m.sql` (synthetic), `fixtures/events/schema-breaking-cases/` (8 breaking, 8 compatible).

Test	Type	Target	Pass criterion
Outbox chaos	Chaos / integration	Debezium + consumer lib	0 dup, 0 loss over 10 kills

Test	Type	Target	Pass criterion
Approval chain	Temporal test server (time-skipping)	workflow	expected timeline, timers fire
Completion policy	Property	step evaluator	12/12 fixtures; random assignee sets obey rounding
Batch resume	Integration	batch framework	1M rows, resume, idempotent
Schema compatibility	Contract	registry	16/16 classified correctly
Publish lag	Performance	outbox → Kafka	p95 ≤ 500 ms at 2,000 events/s
DLQ routing	Negative	consumer lib	poison event lands in DLQ with reason

Reconciliation: count of outbox rows created in a window equals Kafka messages for that domain in the same window (nightly check, tolerance 0); sum of `BatchTask.rows_done` equals `BatchRun.rows_done`.

0.4.10 Deployment and operations

Charts: `strimzi-kafka` (cluster CR), `debezium-connect` (one connector per domain DB, secrets via ESO), `schema-registry`, `temporal` (frontend/history/matching/worker, CNPG persistence), `workflow` (TS workers + API), `events` (catalog API). Config: `KAFKA_BOOTSTRAP`, `SCHEMA_REGISTRY_URL`, `TEMPORAL_ADDRESS`, `TEMPORAL_NAMESPACE`, `OUTBOX_SWEEP_HOURS=24`. Migrations: outbox table added to every domain by generator; `logical_decoding` and `wal_level=logical` set in CNPG clusters. Feature flags: `workflow.enforce` per entity. Scheduled jobs: outbox sweep hourly, DLQ report daily, Temporal history archival. SLOs: publish lag p95 ≤ 500 ms, consumer lag ≤ 10 s p95, workflow action p95 ≤ 300 ms, batch scheduler drift ≤ 30 s. Alerts: `DebeziumConnectorFailed`, `ConsumerLagHigh`, `DLQNonEmpty`, `TemporalWorkflowFailedRate`, `BatchRunError`. Runbook: connector restart with slot reset, replay procedure, stuck workflow reset via Temporal CLI, DLQ reprocess. Rollback: digest rollback; schemas are backward compatible so older consumers keep working; Temporal workers are versioned with build ids.

0.4.11 Migration from AX / SAP

Source	Passage entity	Notes
AX `WorkflowTable` / `WorkflowVersionTable` (approval elements, conditions)	WorkflowDefinition	exported via workflow export XML; steps mapped where assignee type is user/role; hierarchy-based assignees flagged for wave 42
AX `BatchJob` / `Batch` / `SysRecurrence`	BatchJob	only jobs whose handler has a Passage equivalent (listed per later wave); recurrence → cron
SAP TBTCO/TBTCP (job name, program, variant, period)	BatchJob	same rule; variants → params
SAP SWDD workflow templates (WS*), PFAC rules	WorkflowDefinition	manual rebuild from a generated inventory report
In-flight work items (`WorkflowWorkItemTable`, SWWWIHEAD)	not migrated	cut over by completing or canceling in the source system (wave 49 runbook)

Not migrated: workflow history (kept read-only in the source for audit), AIF/IDoc queues, batch job logs.

0.4.12 Dependencies, risks, open questions

Depends on: wave 3 (entity metadata for schemas and outbox generation, registry as first producer), wave 2 (actor identity, duties `Approve workflow`, `Maintain workflows`, `Maintain batch jobs`, `View batch history`).

Risk	Likelihood	Impact	Mitigation
Debezium replication slot bloat when connector is down	Medium	High	Slot lag alert at 1 GB; max WAL retention 20 GB; runbook
Temporal operational complexity on k3d	Medium	Medium	Single-binary dev mode for `PROFILE=min`
Expression language in conditions diverges from metadata hooks	Medium	Medium	Same expression engine as wave 3

Open questions: (1) JSON vs Avro/Protobuf for payloads — default JSON Schema (matches metadata kernel); decide by wave 4 start. (2) Email provider in P0 — default SMTP relay via ESO secret; decide by wave 5. (3) Temporal namespace per tenant vs shared with search attribute — default shared; decide by wave 48.

0.4.13 Parameters and setup

Tenant values live on Admin workspace → Workflow parameters, Batch parameters and Notification settings (the AX "Workflow parameters", "Email parameters" and "System parameters → Alerts" forms; SAP SWU3, SWNCONFIG, SCOT and SM36 settings).

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`events.topic.partitions_default`	environment	int	12	1–96	FR-4.1 default partition count	none	qRFC queue count (SMQR)
`events.topic.retention_days`	environment	int	7	1–30 (compacted topics unlimited)	Kafka retention	AIF message retention	IDoc reorganization (RSETESTD / SARA IDOC)
`events.max_message_bytes`	environment	int	1,048,576	≤ 1 MB	larger payloads use claim-check	none	IDoc segment limits
`outbox.sweep_hours`	environment	int	24	1–168	outbox row deletion after capture	none	tRFC LUW deletion (SM58 reorg)
`consumer.dedupe_window_days`	environment	int	7	1–30	BR-4.5	none	none
`consumer.max_attempts`, `consumer.backoff_max_seconds`	environment	int, int	10, 300	1–50, 10–3,600	FR-4.5 retry policy before DLQ	Business events retry (D365)	tRFC retry (SM59 `Connection attempts`)
`workflow.enforce`	tenant × entity (flag)	bool	false until a definition is activated	true/false	submit required before posting	Workflow configuration → Activate	Flexible workflow scenario activation

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`workflow.admin_user_id`	tenant	uuid	Tenant owner	any active user	escalation target when nobody resolves (FR-4.7)	Workflow parameters → Workflow administrator	SWU3 workflow system administrator; SWIA
`workflow.delegation_max_depth`	tenant	int	3	1–5	FR-4.7 chain limit	Workflow delegation	HRUS_D2 substitution chain
`workflow.default_sla`	tenant	ISO 8601 duration	P2D	PT1H–P30D	default due date per step	Workflow → time limits (default)	Deadline monitoring latest end
`workflow.history_retention_days`	environment	int	90	30–3,650	Temporal history retention	Workflow cleanup batch (`WorkflowCleanup`)	SWWL / RSWWWIDE deletion
`batch.max_runtime_hours`	tenant	int	6	1–48	FR-4.9 run timeout	Batch job "Execution time" alerts	SM37 job runtime / `rdisp/max_wprun_time`
`batch.default_checkpoint_rows`	environment	int	10,000	1,000–1,000,000	FR-4.9	none	none
`batch.max_concurrency_per_tenant`	tenant	int	32	1–128	total concurrent tasks per tenant	Batch server groups / max batch threads	SM61 server groups, `rdisp/wp_no_btc`
`batch.history_retention_days`	tenant	int	90	30–730	FR-4.10	Batch job history cleanup	SM37 → delete old jobs (RSBTCDEL2)
`notify.provider`	tenant	enum	smtp	smtp, http_api	FR-4.13 adapter	Email parameters → SMTP / Exchange	SCOT SMTP node
`notify.smtp.host`, `.port`, `.tls`, `.credential_ref`	tenant	string, int, enum, secret ref	platform relay, 587, starttls, ESO path	—	outbound SMTP	Email parameters → SMTP server, port, SSL, user/password	SCOT node → host, port, TLS, logon data
`notify.sender.from`, `.display_name`, `.reply_to`	tenant (legal entity override)	email, string, email	`no-reply@<env>.passage.dev`	verified domain	sender identity	Email parameters → "Send from"	SCOT default domain, SO_USER_AUTOMATIC
`notify.mode`	tenant	enum per event class	immediate (workflow, alerts), batch (digests)	immediate, batch	when the distributor sends	Mail sending mode: batch / client	SCOT send job RSCONN01 period
`notify.distributor_interval_seconds`	environment	int	60	15–900	distributor polling	Email distributor batch recurrence	RSCONN01 job period
`notify.max_attempts`	tenant	int	5	1–10	BR-4.11	Email distributor "Retries"	SCOT retry settings
`notify.allowlist_domains`	environment	domain list	empty in prod, `passage.dev` in dev/stage	domains	non-prod recipient guard	none	SCOT address area restrictions
`notify.body_retention_days`, `notify.metadata_retention_years`	tenant	int, int	90, 3	7–365, 1–10	outbound history retention	Outbound email status cleanup	SOST reorg (RSBCS_REORG)
`notify.default_language`	tenant	BCP 47	en	any	template fallback	Email templates default language	SO10 language key fallback

0.4.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Work items (by user, queue, status, overdue)	list inquiry	assignee, status, entity, due before, legal entity	document, step, assignee, original assignee, due, age, SLA state	`workflow.work_item`	Work items assigned to me / Work item inquiries	SBWP, SWI5, SWI2_DEAD
Workflow history per document	drill-down	entity, record	steps, assignees, actions, comments, timers, durations	`workflow.instance` + Temporal history	Workflow history / tracking details	SWIA, SWI1, workflow log
Workflow performance	KPI	definition, period	submitted, approved, rejected, median cycle time, % escalated, % within SLA	`workflow.*` projections	SSRS Workflow performance	SWI2_DURA, SWI2_FREQ
Delegation inquiry	list inquiry	user, date range	delegator, delegate, scope, from, to	`WorkItem.original_assignee_id` + delegation settings	Delegation inquiry (WorkflowDelegation)	HRUS_D2 substitutes
Batch job list	list inquiry	status, priority, handler	code, handler, cron, next run, last run status, exclusive	`workflow.batch_job`	Batch jobs list	SM37 job overview
Batch run history and errors	drill-down	job, status, date range	run, status, rows, duration, failed partitions, error	`workflow.batch_run`, `batch_task`	Batch job history / Batch job errors; SSRS Batch job report	SM37 job log (TBTC5)
Topic and consumer lag	operational report	topic, consumer group	partitions, end offset, committed offset, lag, lag seconds	Kafka admin API	Business events catalog / AIF queue	SMQ1/SMQ2, SM58, WE05
DLQ report	operational report (daily)	topic, period	event type, count, first/last seen, top `dlqreason`	`…dlq` topics	AIF exceptions	BD87 IDoc status 51 list, SM58 errors
Outbound message status	list inquiry	status, channel, recipient, template, date	sent at, recipient, subject, status, attempts, last error, provider id	`notify.outbound_message`	Outbound email status (E-mail sending status)	SOST
Notification delivery KPI	KPI	period, template	sent, failed, bounced, median send latency	`notify.outbound_message`	none	SOSV send statistics
Template catalog	list inquiry	code, language, layer, status	code, channel, languages present, customer override flag, version	`notify.message_template`	Email templates (`SysEmailTable`)	SO10 text list, SE75 text objects

0.4.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Work item notification email	`workitem.assigned` / `.delegated` (preference allows email)	header: tenant layout logo; subject: `{entity} {record.number} awaits your approval`; body: submitter, amount, due, comment, deep link, approve/reject links (signed, single-use, 72 h); footer: preference link	email (HTML + text)	none (message id)	body 90 days, metadata 3 years
Escalation email	`workitem.escalated`	from/to assignee, overdue by, deep link	email	none	same
Completion email to submitter	`instance.completed` with `notify_submitter_on_complete`	outcome, approver, comment, link	email	none	same
Daily / weekly digest	digest schedule (BR-4.12)	list of pending items grouped by entity with due dates	email	none	same
Batch failure email	`batch.run.failed` to job owner	job, run, failed partitions, first error, link to monitor	email	none	same
Webhook delivery	subscribed event class	CloudEvents JSON body, `X-Passage-Signature`	HTTPS POST	CloudEvents `id`	delivery log 3 years
AsyncAPI document	metadata publish	channels, messages, schemas	AsyncAPI 3.0 YAML/JSON	entity version	retained per version

0.4.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Outbox sweep	hourly	domain database	deletes captured rows older than `outbox.sweep_hours` (by `created_at` and Debezium LSN)	< 1 min	retry next hour	`OutboxBacklog` (> 100 k rows)
Dedupe table sweep	daily 02:00 UTC	consumer group	deletes rows older than window	< 5 min	retry	none
DLQ report	daily 06:00 UTC	topic	report keyed by date	< 1 min	none	`DLQNonEmpty` (10 min)
Temporal history archival	daily	namespace	Temporal archival to object storage	< 30 min	retried by Temporal	`TemporalArchivalFailed`
Workflow SLA timers	event-driven (durable timers, not polled)	workflow id	Temporal timer	fires within 1 s of due	survives restarts	`TemporalWorkflowFailedRate`
Email distributor	every 60 s	tenant × priority	`SELECT … FOR UPDATE SKIP LOCKED` batch 500; status is the checkpoint	< 10 s per batch	BR-4.11 retries	`NotifyFailureRate` (> 5% over 15 min), `NotifyQueueAge` (oldest queued > 10 min)
Digest builder	every 15 min (sends users whose digest time fell in the window)	tenant	one digest per (user, period) keyed in `notify.digest_log`	< 1 min	missing digest sent next window	`DigestBacklog`
Bounce ingest (IMAP mode)	every 5 min	tenant mailbox	provider message id unique	< 30 s	retry	`BounceIngestFailed`
Outbound message retention	daily 03:30 UTC	month partition	nulls bodies after 90 days; detaches partitions after 3 years	< 5 min	retry	none
Batch history cleanup	daily 04:00 UTC	tenant	deletes runs older than `batch.history_retention_days`	< 5 min	retry	none

0.4.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
Maintain workflows	`/v1/workflows/definitions*`, activate, cancel instances, Workflow designer	with Approve workflow for the same entity (designing your own approval chain)	Workflow administrator
Approve workflow	`/v1/workitems/{id}/actions` (approve, reject, delegate, request_change), Inbox	with the entity's create/post duty on the same record (submitter cannot approve own submission; enforced at runtime, NTF-0410)	every business role with approval steps
Submit for approval	`POST /v1/workflows/instances`, recall	none	entity maintainers
Maintain batch jobs	`/v1/batch/jobs*`, run, cancel, hold, resume, rerun-failed, `/v1/events/replay`	none	Platform engineer, Batch operator
View batch history	`GET /v1/batch/runs*`, Job monitor, Topic catalog read	none	Batch operator, Auditor
Maintain message templates	`/v1/notify/templates*`, preview, publish	none	Workflow administrator, Communications administrator
Maintain notification settings	`/v1/notify/settings*`, verify sender, test send, retry, cancel queued, other users' preferences	with View outbound messages is acceptable; with Maintain secrets (SMTP credentials are ESO references, never visible)	Tenant owner, Workflow administrator
View outbound messages	`GET /v1/notify/messages*`, rendered body view	message bodies may contain document data — masked for `pii=sensitive` placeholders	Workflow administrator, Help desk, Auditor
Send notifications	`POST /v1/notify/messages`	none	internal service principals only

0.4.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
WFL-0401	error	Recall is not allowed after step {n} was approved	RECALL_NOT_ALLOWED (409)	ask the workflow administrator to cancel	BR-4.1, AC-4.6
WFL-0402	error	A comment is required to {action}	`comment_required`	add a comment	0.4.5 validation
WFL-0403	error	{user} cannot receive this item: missing role {role}	delegate target check	pick a user who holds the role	0.4.5 validation

Code	Severity	User text	Cause	Remediation	Raised by
WFL-0404	error	Delegation chain limit ({max}) reached	FR-4.7	approve or reject directly	FR-4.7
WFL-0405	warning	No approver resolved for step {n}; escalated to the workflow administrator	empty assignee set	fix the assignee rule	FR-4.7, AC-4.9
WFL-0406	error	This document is already in approval (instance {id})	duplicate submit	open the existing instance	FR-4.7
WFL-0407	error	Bulk approve is limited to 100 items	UX rule	approve in batches	0.4.6 rules
WFL-0408	error	Job {code} is on hold	run on hold job (409)	release the hold	0.4.5 validation
WFL-0409	error	Job {code} already has a running instance	RUN_OVERLAP (409)	wait or cancel the running run	FR-4.9, AC-4.8
WFL-0410	error	You cannot approve a document you submitted	runtime SoD	delegate to another approver	0.4.17
WFL-0411	error	Run {id} exceeded the maximum runtime of {h} h and was canceled	`batch.max_runtime_hours`	raise the limit or add partitions	FR-4.9
EVT-0401	error	Schema change for {type} is incompatible at `{path}`	BACKWARD_TRANSITIVE check	make the change additive or bump `v<n+1>`	FR-4.3, AC-4.4
EVT-0402	warning	Event {id} moved to DLQ after {n} attempts: {reason}	poison event	fix consumer, then replay	BR-4.10
EVT-0403	error	Payload exceeds 1 MB; use a claim-check reference	message size	store payload in object storage	FR-4.1
NTF-0401	error	Sender {address} is not verified for tenant {tenant}	SPF/DKIM check failed	publish DNS records and re-verify	FR-4.13
NTF-0402	warning	Delivery to {recipient} failed (attempt {n} of {max}): {provider error}; retrying at {time}	provider/transient error	none; watch the outbound inquiry	BR-4.11
NTF-0403	error	Delivery to {recipient} failed permanently after {max} attempts	retries exhausted	fix the cause and retry manually	BR-4.11
NTF-0404	warning	{user}'s email address is bouncing; email suppressed, in-app delivery continues	3 hard bounces	correct the address in the IdP	BR-4.13
NTF-0405	info	Recipient {address} is outside the non-production allowlist; message not sent	`notify.allowlist_domains`	add the domain in dev/stage only	FR-4.13
NTF-0406	error	Template {code} ({language}) uses unknown placeholder `{{placeholder}}`	publish-time validation	use a placeholder from the picker	FR-4.14, AC-4.13
NTF-0407	error	Message not sent: template render failed ({reason})	runtime render error	fix the template; retry	FR-4.14, BR-4.11
NTF-0408	error	Attachments exceed 10 MB for one message	size limit	send a link instead	FR-4.13

0.4.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Outbox publish (commit → Kafka)	2 M events/day/tenant, peak 2,000/s	100 ms / 500 ms / 1 s	6,000 events/s per Kafka cluster	Kafka 7-day retention ≈ 1.2 TB at 1 KB × 170 M; outbox rows transient	partition key = aggregate id; 12 partitions	lag grows; no loss; `ConsumerLagHigh` at 10 s
Consumer processing	same	5 ms / 50 ms / 200 ms per event	scale by partitions	dedupe table ≈ 14 M rows per group (7 days)	dedupe PK (consumer_group, event_id)	lag, not errors
Workflow action (approve)	20 k/day/tenant, peak 50/s	80 ms / 300 ms / 600 ms	200/s	`work_item` ≈ 7 M rows/year/tenant	`work_item(tenant_id, assignee_id, status)`; monthly partitions from wave 30	Temporal task queue backlog; UI shows "processing"
Inbox query	5,000 users × 20 views/day	20 ms / 80 ms / 150 ms	300 rps	n/a	same index	cached 2 s per user
Batch run (1 M rows, 16 partitions)	50 runs/day/tenant	n/a	≥ 20 k rows/s aggregate	`batch_task` ≈ 300 k rows/year	`batch_run(job_id, scheduled_at desc)`	concurrency capped per tenant; runs queue by priority A > B > C
Email enqueue + render	50 k messages/day/tenant, peak 100/s	10 ms / 40 ms / 100 ms	500/s	≈ 18 M rows/year/tenant; bodies ≈ 100 GB/year before 90-day nulling	monthly partitions; `(tenant_id, status, next_attempt_at)`	queue grows; immediate priority served first
Email send (enqueue → provider accepted)	same	5 s / 60 s / 3 min	provider limit (default 50/s per tenant)	n/a	n/a	throttled to provider limit; `NotifyQueueAge` alert
In-app delivery (event → tray)	same	300 ms / 2 s / 4 s	1,000/s	same table	SSE fan-out by user	SSE reconnect with `Last-Event-ID`

0.4.20 Localization and compliance hooks

- Language: templates exist per BCP 47 language (FR-4.14); recipient language → tenant default → `en`. Wave 48 label packs ship `sys` template translations for each supported language; formatting filters (`money`, `date`, `number`) use the recipient's locale and the legal entity currency.
- Calendars and time zones: SLA timers are wall-clock in P0 (BR-4.3); wave 6 adds work calendars per legal entity so `P2D` means two working days; digest times and cron schedules use the tenant time zone (DST transitions: a 02:30 job on a spring-forward day runs at 03:00; on fall-back it runs once).
- Email compliance: CAN-SPAM / CASL / EU ePrivacy do not apply to transactional workflow mail, but every email carries a preference link and the sender's postal address from the legal entity footer when `notify.include_legal_footer=true` (default true for marketing-like digests in EU legal entities). Wave 48 adds country-specific footer content (for example German Impressum fields) through the tenant HTML layout.
- Privacy: message bodies may contain personal data; bodies are nulled after 90 days (`notify.body_retention_days`), event payloads carry user ids not addresses, and erasure requests (wave 48) redact `recipient_address` and bodies for the subject.
- Workflow evidence: approval history (Temporal + `workflow.instance`) is the SOX approval-control evidence; retention follows the record's table group (wave 5 audit) because approval actions are also audited entries.
- Data residency: Kafka and Temporal run in the tenant's region; email providers are configurable so EU tenants can choose an EU-hosted relay.

0.4.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.29.4 Organization administration → Setup: Organization → Workflow approval email templates (register row 261)	PARTIAL-4	ADDED as FR-4.14	`MessageTemplate` with sys templates per workflow event and customer overrides per language
AX §12.30.1 System administration → Alerts → Alert parameters / Email templates / Email parameters (SMTP) / Outbound email status (`EventParameters`, `SysEmailTable`, `SysEmailParameters`) (register row 262)	PARTIAL-4	ADDED as FR-4.13	SMTP/provider parameters (0.4.13), templates (FR-4.14), outbound status inquiry (0.4.14); alert parameters themselves are in wave 5 (FR-5.17)
AX §12.30.1 System administration → Email → Email distributor batch (`SysEmailBatch`) (register row 263)	PARTIAL-4	ADDED as FR-4.13	distributor job every 60 s (0.4.16)
AX §12.30.3 System administration → Periodic: Email → Send email distributor batch / Retry failed (register row 267)	PARTIAL-4	ADDED as FR-4.13	distributor + BR-4.11 retries + manual `/retry`
AX §12.30.4 System administration → Setup: Alerts → Alert parameters (polling intervals, retention, drill-down links) / Email templates / Email parameters (SMTP, Exchange, mail sending mode: batch/client) / Email distributor (register row 269)	PARTIAL-4	ADDED as FR-4.13	`notify.mode` = mail sending mode; Exchange via SMTP submission or HTTP API adapter; alert polling/retention parameters in 0.5.13 (FR-5.17)
AX §12.31.4 Home / role centers / workflow → Setup: Workflow → Notification settings (email, in-app, digest) / Workflow email templates (register row 274)	PARTIAL-4	ADDED as FR-4.14	`NotificationPreference` (channels, digest, quiet hours) and per-definition notify flags; BR-4.12

0.5 Wave 5: UX Shell, Number Sequences, Audit & Observability

0.5.1 Scope and business context

AX users work in list pages and details forms generated from the AOT (form datasources, `ListPageInteraction`, action panes, FactBoxes, `SysQueryForm` advanced filters, `SysLastValue` personalization, `SysTableLookup` lookups, Infolog) and D365 workspaces, saved views and the Excel add-in; SAP users work in SAP GUI dynpros and ALV grids with layout variants, F4 search helps, SE16N, and the Fiori launchpad with catalogs and groups. Document numbering comes from AX `NumberSequenceTable` / `NumberSequenceReference` / `NumberSequenceScope` (shared, company, legal entity, fiscal calendar period, operating unit) with continuous and non-continuous modes and `NumberSequenceList` pre-allocation, or SAP `SNRO` number range objects with `NRIV` intervals, buffering (`NRIV_LOKAL`) and gap-free settings. Change tracking is AX `SysDatabaseLog` / `SysDatabaseLogSetup` (Insert/Update/Delete/RenameKey) or SAP change documents (`CDHDR`/`CDPOS`, `SCDO` objects, `DBTABLOG` table logging). Monitoring is LCS environment monitoring or Solution Manager / ST03N / SM21 / ST22. Wave 5 replaces all of this with a React module-federation shell, a metadata-driven form renderer, a number sequence service with a Jepsen-tested gap-free mode, an audit service that derives field-level history from CDC into a hash-chained append-only store, OpenTelemetry across every service into Prometheus/Tempo/Loki/Grafana, and CSV/XLSX import-export against any entity. At wave end the admin workspace runs at `app.<env>.passage.dev` and Phase 0 is demonstrated to stakeholders. Out of scope: full-text search across business documents and analytics content beyond attachment text (wave 46), report designer (wave 15), mobile PWA (wave 32), localization of labels (wave 48), fiscal-period scopes for sequences (wave 6 supplies calendars; scope type is reserved here).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Editing 50 vendor records in AX means 50 details-form round trips (open, edit, save) or an Excel add-in session that fails on the first validation error; SAP needs an LSMW recording or XK02 one by one	Forms are record-at-a-time; bulk edit is an external tool	Grid inline edit, bulk edit of a field across ≤ 1,000 selected rows, validated per row with per-row error reporting	Clicks for 50 edits: ~300 → ~6; failed bulk sessions due to one bad row: all-or-nothing → partial with error file
Continuous number sequences in AX serialize on `NumberSequenceTable` for the whole transaction, causing lock waits on sales-order entry; SAP gap-free NRIV without buffering shows as `NRIV` lock contention in SM12	Lock held for the caller's full transaction	Allocation row with reserved → used/released lifecycle; lock held only for the allocation statement	Gap-free allocation p99: 100s of ms under load → ≤ 30 ms; buffered p99 ≤ 2 ms; gaps and duplicates: 0 (Jepsen-tested)
AX `SysDatabaseLog` must be enabled per table/field in `SysDatabaseLogSetup` and slows every write via triggers in application code; SAP change documents exist only where the developer called `CHANGEDOCUMENT_*` function modules	Audit is opt-in application code that can be forgotten	Audit derived from CDC on every `audit=true` field; no code path can skip it; hash chain proves integrity	Fields with audit coverage: whatever developers remembered → 100% of flagged fields; write-path overhead: trigger cost → 0
Tracing a slow save: AX trace parser / SQL profiler sessions, SAP ST05/SAT/STAD per work process; no end-to-end id across UI, server, DB and interfaces	No distributed trace context	W3C `traceparent` from the shell through gateway, service, Postgres, outbox, Kafka consumer and Temporal	Time to locate the slow span of one user action: hours of expert time → < 1 min from the audit entry's trace link
SAP GUI and AX client forms fail basic accessibility; keyboard navigation differs per form; ALV variants and `SysLastValue` are per-user and non-shareable	Forms hand-built per screen; no accessibility gate	Generated forms with ARIA grid semantics, one keyboard model, axe in CI; saved views shareable by role	WCAG 2.1 AA violations on generated forms: unknown → 0 in CI; shared views: 0 → any role

0.5.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Navigation / workspaces	Navigation pane, area pages, D365 workspaces and tiles	Fiori launchpad (FLP), catalogs/groups, SAP menu (SE43)	Shell navigation by workspace, tiles bound to saved views	full
List page	List page framework, `ListPageInteraction`, action pane, quick filter	ALV grid (SALV), SE16N, layout variants	Generated list page with grid, filters, saved views	full
Details form / line grids	Details form, `FormRun`, datasources, line grid	Dynpro header/item screens, table controls	Generated detail page with header + line grids	full
Lookups	`SysTableLookup`, EDT relation lookups	F4 search helps (DD30L), personal value list	Generated lookup from `RelationDefinition` with recent values	full
Advanced filter / query	`SysQueryForm`, `Query` ranges, saved queries	Dynamic selections, SE16N selection screen	Filter builder over the list filter grammar; saved as view	full
Personalization	`SysLastValue`, usage data, D365 personalization	Layout variants, user parameters (SU3)	`SavedView`, `Personalization` entities per user and shared	full
Number sequences	`NumberSequenceTable`, `NumberSequenceReference`, `NumberSequenceScope`, `NumberSeq::newGetNum`, continuous flag	SNRO objects, NRIV intervals, buffering NRIV_LOKAL, gap-free flag	numseq service: sequence, scope, continuous/non-continuous, block pre-allocation	full
Database log / change documents	`SysDatabaseLog`, `SysDatabaseLogSetup`, database log viewer	CDHDR/CDPOS, SCDO, DBTABLOG (SCU3)	audit service from CDC; `AuditEntry`/`AuditField`; viewer per record	improved (hash chain)

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Excel integration	Excel add-in (OData data entities), Open in Excel	ALV export, Analysis for Office, LSMW upload	CSV/XLSX import-export per entity with staging and validation	partial (no live Excel binding)
Monitoring	LCS environment monitoring, Event Viewer, SQL insights	Solution Manager, ST03N, SM21, ST22, STAD	OpenTelemetry → Grafana/Tempo/Loki/Prometheus; SLO dashboards	improved
Document management	Document handling (`DocuRef`, `DocuValue`, `DocuView`), Document types (`DocuType`: file/note/URL/image/template, group, location archive/database/SharePoint), File types (`DocuFileTypes`), Active document tables, Document management parameters (`DocuParameters`: archive directory, max file size)	GOS attachments (SOOD/SOFFCONT1), ArchiveLink (OAC0 content repositories, OAC3 links, TOA01), DMS document info records (CV01N, DRAW) with originals check-in/check-out, DC30 workstation applications (TDWP), conversion runs	`Attachment` / `AttachmentVersion` on any record, `DocumentType`, `FileType`, S3-compatible storage locations, check-out/check-in, previews, retention and purge, content search	full (native CAD viewing dropped: PLM owns CAD)
Alerts	Alert rules (`EventRule`, `EventRuleField`), change-based and due-date alerts (`EventJobCUD`, `EventJobDueDate` batch), Notification list / My alerts (`EventInbox`), Alert parameters (`EventParameters`)	Alert Management (ALRTCATDEF, ALRTPROC), S/4 Situation Handling (SIT_TEMPLATE), SWE2 change-document event → workflow, Fiori My Notifications	`AlertRule` (change-based / due-date) evaluated from CDC events and a due-date job; alert inbox in the shell; delivery via wave 4 notify	improved (event-driven, no polling batch)
Standard texts	Notes with document type "Note" and restriction internal/external; sales/purchase "Form notes" (`FormLetterRemarks`)	SO10 standard texts (STXH/STXL), SE75 text objects and IDs, SO72	`StandardText` library with language versions and placeholders, referenced by templates and document layouts	full

0.5.3 Functional requirements

- **FR-5.1 Shell** — Must. The system shall ship a React host (module federation, `web/shell`) that loads remotes by manifest, renders workspace navigation, a global search box (title fields of any entity the user may read, p95 ≤ 300 ms), the legal entity switcher, the work item inbox badge, and a notification tray; first contentful paint ≤ 1.5 s on a 10 Mbps connection.
  - Notes: remotes are versioned by digest and pinned per environment; the shell degrades to an error tile when a remote fails to load.
- **FR-5.2 Metadata-driven forms** — Must. The renderer shall generate, from `EntityDefinition`, a list page (grid, quick filter, filter builder, saved views, column chooser), a detail page (sections by `visibility`, tabs when > 24 fields, line grids for child relations, lookups from relations, attachments and notes panels) and dialogs for create; no entity needs hand-written UI to be usable.
  - Notes: field order from `ordinal`; `advanced` visibility collapses into a "More" section; `hidden` never renders; `pii=sensitive` renders masked with a reveal action gated by permission.
- **FR-5.3 Grid** — Must. The grid shall support virtual scrolling to 100,000 rows, inline edit with per-cell validation, multi-row select, bulk edit of one field across selected rows (≤ 1,000), column sort/resize/reorder/pin, group by one column, and export of the current view.
- **FR-5.4 Keyboard-first** — Must. Every action shall be reachable by keyboard: `/` focus search, `n` new, `Ctrl+S` save, `Ctrl+Enter` save and close, `Esc` cancel, `Ctrl+Shift+F` filter builder, arrows and `Tab` for cell navigation, `F2` edit cell, `Enter` open row, `Ctrl+Shift+V` saved views; focus order follows reading order; shortcuts are listed in a `?` overlay.
- **FR-5.5 Accessibility** — Must. All shell screens and generated forms shall pass axe-core WCAG 2.1 AA rules with zero violations in CI (Playwright + axe), support 200% zoom without horizontal scroll, and expose ARIA grid semantics.
- **FR-5.6 Validation timing** — Must. Field-level validation runs on blur (client JSON Schema, then `validateField` server call when the field has a hook); write validation runs on save; server `422 errors[]` map to fields and unresolved errors show in a summary list with focus jump.
- **FR-5.7 Saved views and personalization** — Must. Users shall save filter + sort + columns + grouping as a `SavedView` (personal or shared with a role), set one as default per entity, and have column widths/order and last legal entity persisted as `Personalization`.
- **FR-5.8 Number sequences** — Must. The numseq service shall define sequences with `code`, `scope_type` (shared, legal_entity, legal_entity_fiscal_period (reserved), operating_unit (reserved)), `format` segments (constant, number with fixed width 1–18, year `YYYY`/`YY`, legal entity code, scope value), `smallest`, `largest`, `next`, `continuous` flag, `manual` flag, `allow_to_lower`, `block_size` (non-continuous, default 50, max 1,000), `wrap` flag, and `in_use` flag; total formatted length ≤ 40.
  - Notes: references bind (entity, field, legal entity) to a sequence, mirroring `NumberSequenceReference`; `numseq(<code>)` is a valid field default from wave 3.
- **FR-5.9 Gap-free and buffered allocation** — Must. Continuous sequences shall allocate inside the caller's transaction with a `SELECT … FOR UPDATE` on the sequence/scope row and record an `Allocation` row (status reserved → used | released); a rollback releases the number to be reused by the next caller; p99 ≤ 30 ms at 500 rps per sequence. Non-continuous sequences shall pre-allocate a block per service instance and serve from memory, p99 ≤ 2 ms; unused numbers in a block are lost on restart (documented, like `NRIV_LOKAL`).
- **FR-5.10 Audit capture** — Must. The audit service shall consume Debezium change events for every table whose entity or field has `audit=true`, and write `AuditEntry` (operation insert/update/delete, actor from `updated_by`, `trace_id` from the transaction's outbox header or `pg_current_xact_id` correlation, `occurred_at`) and `AuditField` rows for changed fields only, with `pii=sensitive` values stored as SHA-256 digests plus masked display.
  - Notes: entry visible in the viewer within 2 s (p95) of commit; the audit database is `audit`, separate from domain databases; role `passage_app` has INSERT and SELECT only.
- **FR-5.11 Immutable hash chain** — Must. Each `AuditEntry` shall store `prev_hash` (previous entry for the same tenant) and `hash = SHA-256(prev_hash ‖ canonical_json(entry + fields))`; a daily job writes the chain head per tenant to an anchor table and to object storage; `GET …/verify` recomputes a range.
- **FR-5.12 Audit viewer** — Must. From any record the user shall open "History" showing entries with user, time, trace id, operation and before/after per field, filterable by field and date, exportable; a global audit workspace searches by entity, record, user, date range.
- **FR-5.13 OpenTelemetry** — Must. Every service shall emit traces (W3C `traceparent` propagated gateway → service → Postgres spans → outbox header → Kafka consumer → Temporal activities), RED metrics (`http_server_requests`, `db_query_duration`, `kafka_consume_lag`, `temporal_activity_duration`) and structured JSON logs with `trace_id`, `tenant_id`, `actor`; the shell emits browser spans for navigation and API calls.
  - Notes: sampling 100% in dev, tail-based 10% + 100% of errors and > 1 s in stage/prod; PII never in span attributes (lint rule).
- **FR-5.14 Dashboards and alerting** — Must. Grafana shall ship per-service SLO dashboards (availability, latency, error budget burn), platform dashboards (Kafka lag, Temporal, Postgres, Argo), and Alertmanager routes to a PagerDuty-compatible webhook with severities page/ticket/notify.
- **FR-5.15 Import/export** — Must. Users shall import CSV or XLSX into any entity through staging (header mapping saved as a template per entity, type coercion, validation using the entity's JSON Schema and hooks, error file with row and field) and export any list view to CSV/XLSX respecting RLS, permissions and PII masks; 100,000 rows import ≤ 5 min; runs as a wave-4 batch job.
- **FR-5.16 Document management** — Must. The system shall attach documents to any record of any entity whose metadata sets `attachments: true` (default true for main and transaction table groups), and one document may be linked to many records. Each attachment has a `DocumentType` (code, kind file | note | url | image | template, group, storage location, allowed file types, max size, restriction internal | external, retention class, `active_entities[]`), versions (`AttachmentVersion`: file, sha256, size, mime, uploaded by/at), and an optional check-out lock. Storage locations are S3-compatible buckets (per tenant prefix; credentials via ESO) or `database` for notes; uploads use pre-signed URLs (≤ 100 MB default, `FileType.max_bytes` per type); every upload is malware-scanned before it becomes `available`; previews (PDF first page, image thumbnail 320 px, Office documents converted to PDF) are generated by a conversion job; text is extracted from PDF, Office and text files into a Postgres `tsvector` so global search (FR-5.1) and the documents inquiry find attachment content.

    - Notes: check-out sets `checked_out_by/at` and blocks new versions by others until check-in or an administrator force-release (lock expires after 7 days); `external` restriction marks documents that may print or send with external documents (wave 15/46 layouts); retention classes (`retain_years`, `on_expiry` purge | archive) drive the purge/archive job, which never purges a document under legal hold (flag set by wave 48) or linked to a record whose own retention has not expired; deleting a record leaves its attachments for the retention job; `active_entities[]` reproduces AX "Active document tables"; native CAD 2D/3D viewing is not provided (PLM systems keep CAD and link by URL document type).
- **FR-5.17 Alert rules** — Must. Users shall create `AlertRule`s on any entity the user may read: `change-based` (field changes to / from a value, any change of a field, record created, record deleted; optional filter in the list filter grammar) or `due-date` (a date field is within N days before/after today, evaluated daily); with subject and message from a wave 4 `MessageTemplate` or inline text with placeholders, recipients (self, users, role, record owner field), delivery (in-app, email, both), legal entity scope (current or all), active from/to, and "organization-wide" rules (maintained by an administrator for all users of a role). Fired alerts appear in the shell Alerts list with a drill-down link to the record and can be marked read or deleted.
    - Notes: change-based rules evaluate from the CDC stream already consumed by audit (no polling job; AX change-based batch is replaced); due-date rules evaluate in a daily job at `alerts.due_date_time` tenant time; the recipient must hold read on the entity at evaluation time, otherwise the alert is dropped and counted; a rule firing more than `alerts.max_per_rule_per_hour` (default 500) is suspended and its owner notified (ALR-0505); alerts are retained `alerts.retention_days` (default 90).
- **FR-5.18 Standard texts** — Should. The system shall provide a `StandardText` library (code, purpose: document header / document footer / line / email / workflow, language versions, body in plain or limited Markdown with placeholders `{{record.field}}`, legal entity scope, effective dates, layer sys | customer) that wave 4 message templates reference as `{{text:CODE}}`, users insert into notes on any record (expanded at insert time), and wave 15/46 document layouts reference by code with the recipient's language.
    - Notes: language fallback recipient → legal entity default → `en`; a text referenced by a published template or layout cannot be retired (STX-0503); edits create a new version, documents already issued keep the version they printed (layouts store the version id).

0.5.4 Data model

NumberSequence (`numseq.sequence`) and **NumberSequenceAllocation** (`numseq.allocation`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS
code	varchar(20)	yes	unique per tenant (`Vend_Acct`, `SO_Num`)
scope_type	enum	yes	shared, legal_entity, legal_entity_fiscal_period, operating_unit
scope_value	varchar(32)	no	legal entity code etc.; unique with code
format	jsonb	yes	segments; e.g. `[{"const":"SO-"},{"year":"YYYY"},{"const":"-"},{"number":6}]`
smallest, largest, next	bigint	yes	1 ≤ smallest ≤ next ≤ largest ≤ 10^18
continuous, manual, allow_to_lower, wrap, in_use	bool	yes	defaults false except in_use
block_size	int	yes	1 for continuous; 50 default
version	int	yes	optimistic lock
Allocation: id, sequence_id, number, status (reserved, used, released), tx_ref, requested_by, requested_at, used_at	—	—	only for continuous sequences

AuditEntry (`audit.entry`, partitioned monthly) and **AuditField** (`audit.field`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id, legal_entity_id	uuid	yes/no	
entity	varchar(48)	yes	
record_id	uuid	yes	
operation	enum	yes	insert, update, delete
actor_id	uuid	yes	from updated_by / created_by; `system` for jobs
occurred_at	timestamptz	yes	commit time
trace_id	char(32)	no	W3C trace id
tx_id	bigint	yes	Postgres xid for grouping
seq	bigint	yes	per-tenant monotonic
prev_hash, hash	char(64)	yes	SHA-256 hex
AuditField: entry_id, field, old_value jsonb, new_value jsonb, pii enum, digest char(64)	—	—	old/new null when masked

SavedView (`ui.saved_view`) and **Personalization** (`ui.personalization`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id, owner_id	uuid	yes	
entity	varchar(48)	yes	
name	varchar(60)	yes	unique per owner + entity
filter, sort, columns, group_by	jsonb	yes/no	list grammar from FR-3.6
shared_with_role_id	uuid	no	null = personal
is_default	bool	yes	one per owner + entity
version	int	yes	
Personalization: user_id, key (varchar 80), value jsonb, updated_at	—	—	e.g. `grid.Vendor.columns`, `session.legal_entity`

ImportJob (`io.import_job`): id, tenant_id, entity, template_id, file_ref, status (uploaded, validating, validated, loading, completed, failed, canceled), rows_total, rows_ok, rows_failed, error_file_ref, batch_run_id, created_by, created_at.

Attachment / AttachmentVersion / AttachmentLink (`docs.attachment`, `docs.attachment_version`, `docs.attachment_link`)

Field	Type	Required	Notes
id	uuid	yes	PK (each table)
tenant_id, legal_entity_id	uuid	yes / no	RLS
document_type_id	uuid	yes (Attachment)	FK DocumentType
name, description	varchar(255), text	yes / no	
restriction	enum	yes	internal, external
status	enum	yes	uploading, scanning, available, quarantined, archived, purged
current_version_id	uuid	no	FK AttachmentVersion
checked_out_by, checked_out_at	uuid, timestamptz	no	lock (FR-5.16)
legal_hold	bool	yes	default false; set by wave 48
retain_until	date	no	computed from retention class and linked record
note_body	text	no	kind=note (location database)
url	varchar(2,000)	no	kind=url
Version: attachment_id, version_no, storage_key, sha256, size_bytes, mime, preview_key, extracted_text tsvector, uploaded_by, uploaded_at	—	—	storage_key = `t/<tenant>/docs/<attachment>/<version>`
Link: attachment_id, entity, record_id, linked_by, linked_at	—	—	unique (attachment_id, entity, record_id)

DocumentType / FileType (`docs.document_type`, `docs.file_type`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	`sys` defaults: File, Note, URL, Image, Scanned invoice
code	varchar(20)	yes	unique per tenant
kind	enum	yes	file, note, url, image, template
group	enum	yes	file, note, url, image, other
storage_location	enum	yes	object_storage, database
bucket_ref	varchar(120)	no	ESO secret path for a customer bucket; default platform bucket
allowed_file_types	text[]	no	FileType codes; empty = all active
max_bytes	bigint	yes	default 104,857,600
retention_years, on_expiry	int, enum	yes	default 10 / archive; purge, archive
active_entities	text[]	no	empty = all entities with `attachments: true`
FileType: code (extension), mime, max_bytes, preview (pdf_first_page, thumbnail, office_to_pdf, none), extract_text bool, blocked bool	—	—	blocked defaults: exe, bat, js, msi, ps1, scr

AlertRule / AlertEvent (`ui.alert_rule`, `ui.alert_event`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	
owner_id	uuid	yes	creator
entity	varchar(48)	yes	
kind	enum	yes	change_based, due_date
event	enum	yes	field_changed, field_changed_to, field_changed_from, record_created, record_deleted, due_in, overdue_by
field, value	varchar(48), jsonb	no	per event
days	int	no	due-date offset 0–365
filter	jsonb	no	list grammar (FR-3.6)
recipients	jsonb	yes	`{self, user_ids[], role_ids[], owner_field}`
delivery	enum set	yes	in_app, email
template_code, subject, message	varchar(64), varchar(255), text	no	template or inline
legal_entity_scope	enum	yes	current, all
org_wide	bool	yes	administrator-maintained
active_from, active_to, status	date, date, enum	yes/no/yes	active, suspended, expired
Event: rule_id, entity, record_id, recipient_id, fired_at, subject, message, read_at, message_id (wave 4 OutboundMessage)	—	—	retained 90 days

StandardText (`ui.standard_text`): id, tenant_id, code (varchar 32, unique per tenant/version), purpose (enum), language (BCP 47), version, body (text ≤ 8,000), legal_entity_id (nullable = all), effective_from, effective_to, layer (sys, customer), status (draft, published, retired).

Invariants: `next ≤ largest` unless `wrap`; for continuous sequences the set of `used` allocation numbers has no gaps between `smallest` and `next−1`; `hash` of entry n equals SHA-256(prev_hash of n ‖ canonical(n)) and `prev_hash` of n+1 equals `hash` of n; one default SavedView per (owner, entity). Indexing: `sequence(tenant_id, code, scope_value)` unique; `allocation(sequence_id, number)` unique; `entry(tenant_id, entity, record_id, occurred_at desc)`; `entry(tenant_id, seq)` unique; `saved_view(tenant_id, entity, owner_id)`. Partitioning: `audit.entry` and `audit.field` by month; retention 10 years for `transaction` table-group entities, 3 years otherwise, enforced by partition detachment to object storage from wave 46.

0.5.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET/POST/PUT	/v1/numseq/sequences, /{id}	Maintain sequences and references	PUT: If-Match	Maintain number sequences
POST	/v1/numseq/sequences/{code}/next?scope=	Allocate (continuous: inside caller tx via `tx_ref`)	`Idempotency-Key` returns the same number	internal / per entity create
POST	/v1/numseq/allocations/{id}/use, /release	Finalize or release a continuous allocation	allocation id	internal
GET	/v1/audit/entries?entity=&record_id=&actor=&from=&to=	Query history	n/a	View audit log (or owner of record)
GET	/v1/audit/entries/{id}	Entry with fields	n/a	View audit log
POST	/v1/audit/verify?tenant=&from_seq=&to_seq=	Recompute chain	n/a	View audit log
GET/POST/PUT/DELETE	/v1/ui/views, /{id}	Saved views	PUT: If-Match	Maintain saved views (own)
GET/PUT	/v1/ui/personalization/{key}	Personalization	last write wins	self
GET	/v1/ui/forms/{entity}	Form definition (ETag)	n/a	any
GET	/v1/ui/search?q=&limit=	Global search	n/a	any (results filtered by read permission)
POST/GET	/v1/io/imports, /{id}, /{id}/validate, /{id}/load, /{id}/errors	Import lifecycle	POST: file sha256	Import data
POST	/v1/io/exports	Export a view (async, returns file ref)	request id	Export data
POST	/v1/docs/attachments (returns pre-signed upload URL), /{id}/versions, /{id}/links	Create attachment, add version, link to another record	Idempotency-Key; version: sha256	entity update + Maintain documents
GET	/v1/docs/attachments?entity=&record_id=&type=&q=	List by record, type, or content search	n/a	entity read
GET	/v1/docs/attachments/{id}/content?version=, /preview	Pre-signed download (5 min), preview	n/a	entity read; `external` restriction irrelevant for read
POST	/v1/docs/attachments/{id}/checkout, /checkin, /release	Lock management	attachment version	Maintain documents (release: Administer document management)
GET/POST/PUT	/v1/docs/document-types, /v1/docs/file-types, /v1/docs/parameters	Setup	PUT: If-Match	Administer document management
GET/POST/PUT/DELETE	/v1/alerts/rules, /{id}	Maintain alert rules (own; org-wide needs admin duty)	PUT: If-Match	Maintain alert rules / Administer alerts
GET/POST	/v1/alerts/events?unread=, /{id}/read, /{id}/delete	Alert inbox	event id	self
GET/PUT/POST	/v1/texts, /{code}/versions/{v}, /publish, /render?language=&record=	Standard texts	by version	Maintain standard texts (render: any)

Event	Trigger	Payload fields	Consumers
dev.passage.numseq.number.allocated.v1	continuous allocation `used` or `released`	sequence_code, scope_value, number, status, tx_ref, actor	audit
dev.passage.audit.entry.recorded.v1	entry committed	entry_id, tenant_id, entity, record_id, operation, hash, seq	shell live history, analytics (wave 46)
dev.passage.audit.chain.anchored.v1	daily anchor	tenant_id, seq, hash, anchor_ref	compliance evidence (wave 48)
dev.passage.io.import.completed.v1 / failed.v1	import terminal	import_id, entity, rows_ok, rows_failed, error_file_ref	shell inbox, audit
dev.passage.ui.view.shared.v1	saved view shared	view_id, entity, role_id, owner	shell cache
dev.passage.docs.attachment.available.v1 / quarantined.v1	scan outcome	attachment_id, version_no, document_type, links[], sha256, size	shell (refresh panel), audit, OCR capture (wave 18)
dev.passage.docs.attachment.purged.v1 / archived.v1	retention job	attachment_id, retention class, reason	audit, compliance evidence (wave 48)
dev.passage.alerts.alert.fired.v1	rule match	rule_id, entity, record_id, recipient_id, delivery	notify (wave 4), shell alert badge
dev.passage.texts.standard-text.published.v1	publish	code, language, version	notify template cache, layouts (wave 46)

0.5.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Admin workspace (home)	Tiles: users, roles, entities, extensions, workflows, jobs, sequences, audit, imports	counts, failed runs, pending work items	per tile duty
Generated list page (any entity)	New, open, inline edit, bulk edit, filter, save view, export, history	columns from metadata, status, updated	entity read/update
Generated detail page	Save, save and close, delete, open related, history, attachments	sections, tabs, line grids, lookups	entity actions
Number sequences	Create, set format, test next, mark in use	code, scope, format preview, next, continuous	Maintain number sequences
Audit viewer (record and global)	Filter, verify chain, export	time, actor, operation, field, old, new, trace link	View audit log
Import/export	Upload, map, validate, load, download errors	entity, template, rows, status	Import data / Export data
Observability (Grafana, linked)	Open trace from any record history; SLO dashboards	trace id, service, latency, error budget	Platform engineer
Attachments panel (on any detail page)	Upload (drag-drop), new version, check out/in, link existing, preview, download, delete link	name, type, version, size, uploaded by/at, checked out by, restriction	entity read/update; Maintain documents
Documents workspace	Search by type, record, content; bulk re-link; release locks; view quarantined	name, type, linked records, versions, status, retain until, legal hold	View documents / Administer document management
Document management parameters	Document types, file types, storage locations, max sizes, retention classes	code, kind, location, max size, retention	Administer document management
Alerts (shell bell)	Open record, mark read, delete, create rule from a field ("Notify me when…")	fired at, rule, record, message	self
Alert rules	Create, edit, suspend, test against a sample record; org-wide rules	entity, event, field, recipients, delivery, active dates, status, fires/24 h	Maintain alert rules / Administer alerts
Standard texts	Create, translate, preview with record, publish	code, purpose, language, version, used by	Maintain standard texts

Rules: inline edit commits per row on `Enter` or row blur with optimistic locking (409 shows a diff and lets the user reload or overwrite with permission); bulk edit previews affected rows and runs as a batch job above 200 rows; validation messages appear under the field and in a summary list; personalization never changes required-field visibility; the trace link from the audit viewer deep-links into Tempo with the `trace_id`.

0.5.7 Business rules and edge cases

- **BR-5.1** Continuous allocation lifecycle: reserved → used (on caller commit) | released (on caller rollback or 10-minute reservation timeout); a released number is the next number served, preserving gap-freedom; `next` only advances when no released numbers exist.
- **BR-5.2** Two concurrent continuous requests serialize on the row lock; the lock is held only for the allocation statement, not the caller's whole transaction, which is why `Allocation` rows exist.
- **BR-5.3** Non-continuous blocks are per pod; a pod restart loses at most `block_size − 1` numbers per sequence; `allow_to_lower=false` forbids editing `next` downward.
- **BR-5.4** Year segment is evaluated in the legal entity's time zone at allocation time; a sequence with a year segment and `continuous=true` resets `next` to `smallest` on 1 January only when `reset_on_year=true` (default false).
- **BR-5.5** `in_use=false` sequences reject allocation with 409 `SEQUENCE_NOT_IN_USE`; `manual=true` sequences require the caller to supply the number and enforce uniqueness per scope.
- **BR-5.6** Reaching `largest` with `wrap=false` returns 409 `SEQUENCE_EXHAUSTED` and raises an alert at 90% consumption.
- **BR-5.7** Audit entries are never updated or deleted by the application; a verification mismatch raises `AuditChainBroken` (page severity) and freezes further anchoring until acknowledged.
- **BR-5.8** Updates that change only `updated_at`/`version` produce no audit entry; updates to `audit=false` fields produce an entry with no `AuditField` rows only if another audited field changed.
- **BR-5.9** Deleting a record writes an entry with all audited fields as `old_value`; soft delete is an update with field `deleted_at`.
- **BR-5.10** Saved view sharing: a shared view is read-only to others; copying creates a personal view; deleting a shared default view reverts affected users to the system default.
- **BR-5.11** Import loads are all-or-nothing per file by default; `partial=true` loads valid rows and produces the error file for the rest; re-importing the same file sha256 is a no-op with 200 and the previous import id.
- **BR-5.12** Attachment lifecycle: uploading → scanning → available | quarantined; available → archived (retention `on_expiry=archive`: object moved to the archive storage class, metadata kept) → purged; available → purged (retention `purge`). A version upload while `checked_out_by ≠ caller` returns 423 DOC-0504. `retain_until` = max over linked records of (record retention per its table group, wave 5 audit retention) and the document type's `retention_years` from upload; legal hold blocks archive and purge regardless.
- **BR-5.13** Content dedupe: an upload whose sha256 equals an existing version in the same tenant stores no new object (the version points to the existing key); a purge deletes the object only when no version references it.
- **BR-5.14** Change-based alert evaluation: a rule fires once per (rule, record, committed transaction) when its event matches the before/after values in the CDC change and the record passes `filter` after the change; changes by the alert's own recipient fire too unless `exclude_own_changes=true` (default true); due-date rules fire once per (rule, record, due date value) so re-running the daily job never duplicates; a rule on a field the recipient cannot read (field mask) is rejected at save (ALR-0502).
- **BR-5.15** Standard text expansion: placeholders resolve against the record at insert or render time; an unknown placeholder fails publish (STX-0502); a text with `effective_to` in the past is not offered for insertion but still renders for documents that reference its version.

0.5.8 Acceptance criteria

- **AC-5.1** Given the 10 reference entities plus 5 extension cases, When Playwright + axe runs list, detail and create dialogs, Then 0 WCAG 2.1 AA violations and the keyboard suite completes every action without a mouse. (FR-5.2, FR-5.4, FR-5.5)
- **AC-5.2** Given a continuous sequence and 10,000 concurrent allocation requests with 10% random rollbacks (Jepsen-style with clock skew and partition injection), When the run ends, Then used numbers are contiguous, unique and `next − smallest = count(used)`. (FR-5.9, BR-5.1)
- **AC-5.3** Given a non-continuous sequence under 2,000 rps across 3 pods, When measured, Then p99 ≤ 2 ms and no duplicates. (FR-5.9)
- **AC-5.4** Given a field change on any audited entity, When committed, Then the audit viewer shows user, time, old/new and the trace id within 2 s (p95 over 1,000 changes). (FR-5.10, FR-5.12)
- **AC-5.5** Given a tampered `AuditField.new_value` (edited directly in the DB in test), When `verify` runs, Then it reports the first broken seq and `AuditChainBroken` fires. (FR-5.11, BR-5.7)
- **AC-5.6** Given a click "Save" on a vendor stub, When the request flows, Then one trace in Tempo contains spans for gateway, service, Postgres, outbox publish, Kafka consumer (audit) and the shell. (FR-5.13)
- **AC-5.7** Given a CSV of 100,000 vendor stubs with 1% invalid rows, When imported with `partial=true`, Then 99,000 rows load in ≤ 5 min and the error file lists 1,000 rows with field and code. (FR-5.15)
- **AC-5.8** Given a user saves a view with filter and columns and marks it default, When they reopen the list, Then the view applies; another user in the shared role sees it read-only. (FR-5.7, BR-5.10)
- **AC-5.9** Given the SLO dashboard for `identity`, When error rate is forced to 2% for 10 min in stage, Then `SLOBurnRateHigh` reaches the webhook with severity page. (FR-5.14)
- **AC-5.10** Given a PDF uploaded to a VendorStub record with document type "Scanned invoice", When the upload completes, Then the attachment is `available` within 30 s with a first-page preview, its text is found by global search, and linking the same attachment to a second record shows it on both. (FR-5.16)
- **AC-5.11** Given the EICAR test file uploaded with an allowed extension, When scanning completes, Then the attachment is `quarantined`, no download URL is issued, and DOC-0503 is shown. (FR-5.16, BR-5.12)
- **AC-5.12** Given user A has checked out an attachment, When user B uploads a new version, Then 423 DOC-0504 names A; after A checks in, B's upload succeeds as version n+1. (FR-5.16, BR-5.12)
- **AC-5.13** Given a document type with `retention_years=1, on_expiry=purge` and one attachment under legal hold, When the retention job runs past expiry, Then the unheld attachment's object is deleted and `attachment.purged` is emitted, while the held one remains `available`. (FR-5.16, BR-5.12)
- **AC-5.14** Given a change-based rule "VendorStub.status changes to Blocked, notify role AP clerk by in-app + email", When a vendor is blocked, Then each role member with read access receives one alert in the bell within 5 s and one email via wave 4 notify, and the user who made the change does not. (FR-5.17, BR-5.14)
- **AC-5.15** Given a due-date rule "due_date within 3 days" on 1,000 records, When the daily job runs twice on the same day, Then each qualifying record fires exactly once per recipient. (FR-5.17, BR-5.14)
- **AC-5.16** Given a standard text `PAY_TERMS_FOOTER` in `en` and `de-DE` referenced as `{{text:PAY_TERMS_FOOTER}}` in a workflow template, When rendered for a German-language recipient, Then the German version appears with placeholders resolved, and retiring the text is refused with STX-0503. (FR-5.18, BR-5.15)

0.5.9 Test plan and fixtures

Fixtures: `fixtures/numseq/jepsen-config.yaml` (10k clients, nemesis schedule), `fixtures/audit/tamper-cases.sql`, `fixtures/io/vendors-100k.csv` (1% invalid), `fixtures/ui/keyboard-script.json` (every shortcut), `fixtures/ui/entities-15.yaml` (10 reference + 5 extension variants), `fixtures/otel/trace-expectation.json` (span list for AC-5.6).

Test	Type	Target	Pass criterion
axe + keyboard	E2E (Playwright)	shell + renderer	0 violations, 100% shortcuts
Gap-free Jepsen	Property / chaos	numseq	0 gaps, 0 dups under partitions and skew
Buffered throughput	Performance	numseq	p99 ≤ 2 ms @ 2,000 rps
Audit latency	Integration	CDC → audit	p95 ≤ 2 s
Chain verify	Negative	audit	tamper detected at exact seq
Trace continuity	Integration	OTel pipeline	all expected spans share one trace id
Import 100k	Performance	importexport	≤ 5 min, error file correct
Form renderer snapshot	Snapshot	renderer	15 entities × 3 pages stable
Grid virtualization	Performance (browser)	grid	100k rows scroll at ≥ 50 fps on reference laptop

Reconciliation: for each continuous sequence, `count(used allocations) = next − smallest − count(released outstanding)`; audit entries per table per day equal Debezium change count for audited tables; import `rows_ok + rows_failed = rows_total`.

0.5.10 Deployment and operations

Charts: `shell-web` (static assets behind the gateway, remotes manifest in ConfigMap), `forms-renderer` (remote), `shell-bff` (TS: views, personalization, search, SSE), `numseq` (Go, CNPG `numseq`), `audit` (Go consumer + API, CNPG `audit`, monthly partition job), `importexport` (TS, object storage bucket via ESO credentials), `observability` umbrella (OTel Collector DaemonSet + gateway, Prometheus, Tempo, Loki, Grafana, Alertmanager). Config: `REMOTES_MANIFEST_URL`, `OTEL_TRACES_SAMPLER=parentbased_traceidratio`, `AUDIT_RETENTION_YEARS`, `NUMSEQ_RESERVATION_TIMEOUT_SECONDS=600`, `ALERT_WEBHOOK_URL` (ESO). Migrations: additive; `audit.entry` partitions created 3 months ahead by a scheduled job. Feature flags: `ui.inline_edit`, `ui.bulk_edit`, `audit.capture` per domain. Scheduled jobs: audit anchor 00:30 UTC, partition maintenance weekly, sequence exhaustion check hourly, import staging cleanup after 7 days. SLOs: shell availability 99.9%, form load p95 ≤ 800 ms, numseq continuous p99 ≤ 30 ms, audit visibility p95 ≤ 2 s, trace ingestion loss < 0.1%. Alerts: `SLOBurnRateHigh` (multi-window 1h/6h), `AuditLagHigh` (> 10 s), `AuditChainBroken`, `SequenceNearExhaustion`, `RemoteLoadFailure`, `OTelCollectorDropping`. Runbook: audit consumer rebuild from Kafka retention, sequence reservation stuck, remote rollback (manifest pin), Grafana datasource auth. Rollback: digest rollback per chart; shell remotes roll back by manifest pin without redeploying the host.

0.5.11 Migration from AX / SAP

Source	Passage entity	Notes
AX `NumberSequenceTable` (NumberSequence, Txt, Format, Lowest, Highest, NextRec, Continuous, Manual, InUse, Wrap) + `NumberSequenceScope`	NumberSequence	`Format ###-######` converted to segments; `NextRec` → `next` after cutover freeze; scope `DataAreaId` → legal entity code
AX `NumberSequenceReference` (DataTypeId → NumberSequenceId)	sequence reference	EDT name mapped to Passage entity/field by crosswalk
SAP NRIV (OBJECT, SUBOBJECT, NRRANGENR, FROMNUMBER, TONUMBER, NRLEVEL, EXTERNIND)	NumberSequence	SUBOBJECT (BUKRS) → scope value; EXTERNIND='X' → manual; buffering settings from TNRO → block_size
AX `SysDatabaseLog` (table, recId, logType, createdBy, createdDateTime, data container)	AuditEntry/AuditField (read-only import)	loaded as historical entries with `source=ax`, excluded from the hash chain (separate chain `legacy`)
SAP CDHDR/CDPOS (OBJECTCLAS, OBJECTID, CHANGENR, USERNAME, UDATE, UTIME, TABNAME, FNAME, VALUE_OLD, VALUE_NEW)	AuditEntry/AuditField	same treatment; OBJECTID resolved to Passage record ids via the migration run ledger (wave 10)
D365 saved views / AX `SysLastValue` usage data; SAP ALV layout variants (LTDX)	SavedView	optional; only named, shared variants are imported
AX `DocuRef` (RefTableId, RefRecId, TypeId, Name, Notes, Restriction) + `DocuValue` (FileName, FileType, Path / File container) + `DocuType`	Attachment / AttachmentVersion / AttachmentLink / DocumentType	file content copied from the archive directory, database container or SharePoint location to object storage; RefTableId/RefRecId resolved via the wave 10 run ledger; notes → kind=note
SAP GOS (SRGBTBREL, SOOD, SOFFCONT1), ArchiveLink (TOA01–TOA03, TOAAT), DMS (DRAW, DRAD object links, originals in KPro)	same	content fetched via `SO_DOCUMENT_READ_API1` / ArchiveLink HTTP or KPro export; DRAD links → AttachmentLink; DMS versions → AttachmentVersion
AX `EventRule` (alert rules)	AlertRule	change-based and due-date rules mapped when the table/field crosswalk resolves; others reported unmapped
SAP SO10 texts (STXH/STXL via `READ_TEXT`, TDOBJECT=TEXT)	StandardText	per language; SAPscript formatting reduced to plain text with line breaks

Not migrated: AX Infolog history, SAP SM21/ST22 logs, SAP DBTABLOG (table logging) beyond customizing tables, personal (unnamed) usage data. Number sequence `next` values are taken at cutover (wave 49) and never earlier.

0.5.12 Dependencies, risks, open questions

Depends on: wave 4 (CDC stream for audit, batch jobs for import/export, events for inbox), wave 3 (form definitions from metadata, `numseq()` defaults), wave 2 (duties `Maintain number sequences`, `View audit log`, `Import data`, `Export data`, `Maintain saved views`), wave 1 (gateway, TLS, charts).

Risk	Likelihood	Impact	Mitigation
Generated forms are usable but ugly; stakeholders judge the platform by them	High	Medium	Design tokens and a 15-entity visual review before the demo; hand-built workspaces for admin home
Audit consumer falls behind on bulk loads	Medium	Medium	Partitioned consumer group (12), batching 500 rows per commit, `AuditLagHigh` alert
Gap-free sequences used where not required (throughput)	Medium	Medium	Default non-continuous; continuous requires a documented legal reason in the sequence description
Tail sampling hides a rare slow path	Low	Medium	100% sampling of > 1 s and errors; exemplar links from metrics

Open questions: (1) Grid implementation (in-house vs. open-source data grid library) — default open-source MIT grid with virtualization; decide by wave 5 start. (2) Audit anchor target (object storage only vs. external timestamping) — default object storage with WORM policy; decide by wave 48. (3) SSE vs WebSocket for the inbox — default SSE; decide by wave 5 mid-point. (4) Whether `legal_entity_fiscal_period` scope ships in wave 5 or wave 6 — default wave 6 with the enum reserved now; decide by wave 6 start.

0.5.13 Parameters and setup

Values live on Admin workspace → Shell parameters, Number sequence parameters, Audit parameters, Document management parameters, Alert parameters and Import/export parameters (AX "Number sequence parameters", "Database log setup", "Document management parameters", "Alert parameters", "System parameters"; SAP SNRO/TNRO, SCDO/SCU3, OAC0/DC10, SM19 and SPRO → SAP NetWeaver → General settings).

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`ui.inline_edit`, `ui.bulk_edit`	tenant (feature flags)	bool	true, true	true/false	grid editing modes (FR-5.3)	Form personalization "Edit mode" (System parameters → default view/edit mode)	ALV edit mode (layout)
`ui.bulk_edit.max_rows`, `ui.bulk_edit.batch_threshold`	tenant	int, int	1,000, 200	10–10,000, 10–1,000	bulk edit cap; above threshold runs as batch job	Excel add-in row limits	MASS transaction selection limits
`ui.grid.page_fetch_rows`	environment	int	200	50–1,000	virtual-scroll fetch size	`FormDataSource` cache size	ALV `maxrows`
`ui.search.entities[]`, `ui.search.max_results`	tenant	entity list, int	all with title fields, 20	—	global search scope (FR-5.1)	Search configuration (SysSearchConfiguration)	ESH_COCKPIT search models
`ui.user_defaults` (language, date format, number format, time zone, default legal entity, start workspace)	user (tenant default)	jsonb	tenant defaults	locale codes, IANA tz	My user options / personalization	User options → Preferences (SysUserSetup)	SU3 defaults (date format, decimal notation, time zone)
`numseq.reservation_timeout_seconds`	environment	int	600	60–3,600	BR-5.1 release of unconfirmed numbers	Number sequence parameters → "Clean up" interval (continuous)	none (NRIV commit-bound)
`numseq.block_size_default`	tenant	int	50	1–1,000	non-continuous pre-allocation	Number sequence → Performance → Preallocation quantity	SNRO → number of numbers in buffer (TNRO-NOIVBUFFER)
`numseq.exhaustion_warn_pct`	tenant	int	90	50–99	`SequenceNearExhaustion` alert	Number sequence → "Warning at"	SNRO → warning % (TNRO-PERCENTAGE)
`numseq.reset_on_year`	sequence	bool	false	true/false	BR-5.4	Segment configuration with fiscal calendar period scope	year-dependent intervals (NRIV-TOYEAR), OBH1 copy to fiscal year
`audit.capture`	domain (flag)	bool	true	true/false	CDC audit capture on/off per domain	Database log setup (SysDatabaseLogSetup) per table	SCDO change document objects; table logging `rec/client`, SE13 log flag
`audit.retention_years.transaction`, `.other`	tenant	int, int	10, 3	1–30	partition retention (0.5.4)	Database log cleanup (SysDatabaseLogCleanup)	SCU3 / SARA CHANGEDOCU archiving
`audit.anchor_time_utc`	environment	HH:MM	00:30	any	daily hash anchor (FR-5.11)	none	none
`docs.max_upload_bytes`	tenant	bigint	104,857,600	≤ 2 GB	upload ceiling across types	Document management parameters → Maximum file size	OAC0 repository settings, `SO_DOCUMENT` size limits
`docs.default_bucket_ref`, `docs.archive_storage_class`	tenant	ESO path, enum	platform bucket, `GLACIER_IR`-equivalent cold class	provider classes	storage location and archive tier	Document management parameters → Archive directory; DocuType location (archive / database / SharePoint)	OAC0 content repositories, OACT categories
`docs.blocked_extensions`	tenant	text[]	exe, bat, cmd, js, msi, ps1, scr, vbs	any	uploads refused (DOC-0502)	File types (DocuFileTypes)	DC30 workstation applications (TDWP)
`docs.checkout_lock_days`	tenant	int	7	1–90	auto-release of stale check-outs	none	CV01N check-out status (no expiry)
`docs.preview.enabled`, `docs.extract_text.enabled`	tenant	bool, bool	true, true	true/false	conversion and content indexing	none (AX 2012); D365 Office preview	DMS conversion server, TREX/HANA full-text activation
`alerts.due_date_time`	tenant	HH:MM tenant tz	06:00	any	daily due-date evaluation	Alert parameters → Due-date alert batch recurrence (`EventJobDueDate`)	ALRTCATDEF scheduling, SWE2 deadline events
`alerts.max_per_rule_per_hour`	tenant	int	500	10–10,000	runaway-rule suspension (ALR-0505)	none	none
`alerts.retention_days`	tenant	int	90	7–365	alert inbox retention	Alert parameters → "Alerts retention" (EventParameters)	ALRTPROC cleanup (SALRT_DELETE)
`alerts.drilldown_base_url`	environment	URL	`https://app.<env>.passage.dev`	URL	deep-link target in alert email	Alert parameters → Drill-down target	Fiori launchpad intent base
`io.max_import_rows`, `io.staging_retention_days`	tenant	int, int	1,000,000, 7	≤ 5 M, 1–30	import size and staging cleanup	Data management parameters (DMFParameters) → staging cleanup	LSMW / Migration Cockpit project retention
`otel.sampling.ratio`, `otel.sampling.latency_threshold_ms`	environment	decimal, int	1.0 dev / 0.1 stage-prod, 1,000	0–1, 100–10,000	FR-5.13 tail sampling	LCS trace capture settings	ST05 / SAT trace settings
`alerting.webhook_url`, `alerting.routes`	environment	ESO ref, YAML	PagerDuty-compatible	—	Alertmanager routing (FR-5.14)	LCS monitoring alerts	Solution Manager alerting (CCMS RZ20 thresholds)

0.5.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Number sequence list and status	list inquiry	scope, continuous, in use	code, scope value, format preview, next, largest, % consumed, continuous, block size	`numseq.sequence`	Number sequences list page; SSRS Number sequence list / references	SNRO intervals, NRIV display
Continuous allocation status (gaps audit)	operational report	sequence, date range, status	number, status (reserved/used/released), requested by, tx_ref, age	`numseq.allocation`	Number sequence → Status list (`NumberSequenceList`)	S_ALR_87012342 gaps in document number assignment; S_ALR_87012341 numbers allocated twice
Record history	drill-down	entity, record, field, date	time, actor, operation, field, old, new, trace link	`audit.entry`, `audit.field`	Database log viewer (per record)	change documents per object (RSSCD100), S_ALR_87012308
Global audit search	list inquiry	entity, user, date range, operation	same plus record id	same	Database log inquiry; SSRS Database log report	SCU3, RSSCD100, S_ALR_87012293
Audit chain verification	operational report	tenant, seq range	range, entries verified, first broken seq, anchor ref	`POST /v1/audit/verify`	none	none (SM20 integrity is file-based)
Import/export job history	list inquiry	entity, status, user, date	job, entity, rows total/ok/failed, duration, error file	`io.import_job`	Data management → Job history (DMFDefinitionGroupExecution)	LSMW / Migration Cockpit logs
Documents by type / record	list inquiry	document type, entity, record, uploaded by, date, content text	name, type, linked records, version, size, status, restriction, retain until	`docs.attachment*`	Document management inquiries (DocuRef list)	GOS attachment list, CV04N document search, OAOR
Document retention preview	operational report	retention class, horizon (days)	attachment, type, retain until, action (archive/purge), legal hold	`docs.attachment`	none (archive purge not standard in AX)	ILM retention (IRMPOL) preview, SARA write variant test
Checked-out documents	list inquiry	user, age	attachment, checked out by, since, linked record	`docs.attachment`	none	CV04N with status "checked out"
My alerts / Alert inquiry by rule or user	list inquiry	rule, user, read status, date	fired at, rule, record, recipient, message, read at, delivery status	`ui.alert_event`	Notification list / My alerts (EventInbox); alert inquiry	ALRTINBOX, SALRT1 alert inbox
Alert rule statistics	KPI	rule, period	fires per day, suspended flag, recipients, email failures	`ui.alert_rule` + events	none	Situation Handling monitor
Standard text where-used	drill-down	text code	templates and layouts referencing it, versions	`ui.standard_text` + notify templates	none	SO10 where-used (none standard)
SLO and service dashboards	KPI	service, window	availability, p95 latency, error budget remaining, burn rate	Prometheus	LCS environment monitoring	Solution Manager technical monitoring, ST03N
Trace explorer	drill-down	trace id, service, duration	spans, attributes, errors	Tempo	Trace parser	ST05, SAT, STAD
Log explorer	list inquiry	service, level, trace id, tenant	time, level, message, trace id	Loki	Event Viewer / Infolog history	SM21 system log, SLG1 application log, ST22 dumps

0.5.15 Printed and electronic documents

No printed business documents (wave 15/46 layouts consume FR-5.18 texts and FR-5.16 `external` attachments). Electronic documents:

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
List export	user export of a view (FR-5.15)	header row (column labels), rows per view, footer row with export time, user, filter, legal entity	CSV (UTF-8, RFC 4180), XLSX	none	not retained by Passage after 7 days (download link expiry)
Import error file	import validation	original columns + `row`, `field`, `code`, `message`	CSV, XLSX	import id	7 days
Audit history export	audit viewer export	entity, record, entries, chain seq range, verification result	CSV, JSON (with hashes for third-party verification)	none	requester's custody; source entries retained per 0.5.4
Audit chain anchor	daily job	tenant, seq, hash, timestamp	JSON object in WORM bucket	per-tenant seq	10 years, object lock (compliance mode)
Alert email	alert fired with email delivery	subject (rule or template), record title, changed field old → new or due date, deep link, unsubscribe-from-rule link	email via wave 4 notify	none	wave 4 retention
Attachment preview	conversion job	first page PDF or 320 px thumbnail	PDF, PNG	attachment version	deleted with the version

0.5.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Audit chain anchor	daily 00:30 UTC	tenant	anchor row keyed by (tenant, date)	< 1 min for 1,000 tenants	retry 3×; freeze on mismatch (BR-5.7)	`AuditChainBroken` (page), `AuditAnchorMissing`
Audit partition maintenance	weekly	month	creates partitions 3 months ahead; idempotent	< 1 min	alert	`AuditPartitionMissing`
Audit retention detach	monthly	month partition	detaches partitions older than retention to object storage (from wave 46)	< 10 min	retry	none
Sequence exhaustion check	hourly	sequence	stateless	< 10 s	none	`SequenceNearExhaustion`
Stale reservation release	every minute	sequence	releases `reserved` older than timeout (BR-5.1)	< 1 s	retry	`ReservationStuck` (> 3 timeouts/h)
Import staging cleanup	daily 03:00 UTC	tenant	deletes staging and files older than 7 days	< 5 min	retry	none
Notification and alert cleanup	daily 03:15 UTC	tenant	deletes alert events older than retention	< 1 min	retry	none
Due-date alert evaluation	daily at `alerts.due_date_time`	tenant × rule	fired key (rule, record, due value) (BR-5.14)	< 2 min per 100 rules × 100 k records	resume per rule	`AlertJobFailed`

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Change-based alert evaluator	continuous (CDC consumer)	Kafka partition	consumer dedupe (BR-4.5)	≤ 5 s event → alert	DLQ	`AlertLagHigh` (> 30 s)
Attachment scan and preview	on upload (queue)	attachment version	status is the checkpoint	≤ 30 s for 20 MB	3 retries, then `quarantined` with reason `scan_failed`	`DocScanBacklog` (> 100 queued)
Attachment text extraction	on `available`	attachment version	writes `extracted_text` once per version	≤ 60 s for 200 pages	skip with warning DOC-0507	none
Document retention (archive / purge)	weekly Sunday 01:00 tenant time	tenant × document type	dry-run list stored first; each attachment processed once (status)	< 30 min per 1 M attachments	per-item errors logged, run continues	`DocRetentionErrors`
Stale check-out release	daily	tenant	releases locks older than `docs.checkout_lock_days`	< 1 min	none	none
Personalization / usage data reset	on demand (administrator)	user	deletes Personalization rows for selected users	< 10 s	none	none

0.5.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
Maintain number sequences	`/v1/numseq/sequences*`, Number sequences screen, test next	with posting duties for fiscal documents in production (lowering `next` on a gap-free sequence is a fraud vector; `allow_to_lower=false` by default)	Financial administrator, Platform engineer
View audit log	`/v1/audit/entries*`, `/verify`, audit workspace, history on any record the user can read	with Maintain users / Maintain security roles for the security auditor role (auditors should not administer what they audit)	Auditor, Security administrator (own scope)
Import data	`/v1/io/imports*`, Import screen	per entity: also requires `create` / `update` on the target entity	Data steward, Migration consultant
Export data	`/v1/io/exports`, export button on lists	export of `pii=sensitive` columns requires explicit field permission	most business roles
Maintain saved views	`/v1/ui/views*` own; share with role	none	all users (own); Business administrator (shared, publish to role)
Maintain documents	upload, version, check out/in, link on records the user may update	none	all entity maintainers
View documents	download, preview, documents workspace search (filtered by read on linked records)	none	all users with entity read
Administer document management	`/v1/docs/document-types`, `/file-types`, `/parameters`, force-release, view quarantined, retention preview	with legal-hold management (wave 48 duty)	Document administrator
Maintain alert rules	`/v1/alerts/rules*` own rules	none	all users
Administer alerts	org-wide rules, suspend/resume any rule, alert statistics	none	Business administrator
Maintain standard texts	`/v1/texts*`, Standard texts screen	none	Communications administrator, Business administrator
Platform observability	Grafana, Tempo, Loki (Keycloak SSO group)	log access may expose payload fragments; lint keeps PII out of spans	Platform engineer, SRE

0.5.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
NSQ-0501	error	Number sequence {code} is not in use	`in_use=false` (409 SEQUENCE_NOT_IN_USE)	mark in use or bind another sequence	BR-5.5
NSQ-0502	error	Number sequence {code} is exhausted at {largest}	SEQUENCE_EXHAUSTED (409)	raise `largest` or enable wrap where legal	BR-5.6
NSQ-0503	warning	Number sequence {code} is {pct}% consumed	exhaustion threshold	plan a new range	BR-5.6
NSQ-0504	error	{code} is manual: supply a number	`manual=true` without number	enter the number	BR-5.5
NSQ-0505	error	Number {n} already used in {code} for scope {scope}	manual duplicate	choose another number	BR-5.5
NSQ-0506	error	Next number cannot be lowered for {code}	`allow_to_lower=false`	none (gap-free protection)	BR-5.3
NSQ-0507	error	Formatted number exceeds 40 characters	format length	shorten segments	FR-5.8
AUD-0501	error (page)	Audit chain broken for tenant {tenant} at seq {seq}	hash mismatch	follow runbook; anchoring frozen	BR-5.7, FR-5.11
AUD-0502	warning	Audit is {s} s behind; recent changes may not show yet	consumer lag	none; wait	FR-5.10
UIX-0501	error	This record was changed by {user} at {time}; reload or overwrite	optimistic lock (409)	reload, or overwrite with permission	0.5.6 rules
UIX-0502	error	{n} fields need attention	client/server validation summary	jump to each field	FR-5.6
UIX-0503	error	Bulk edit is limited to {max} rows	`ui.bulk_edit.max_rows`	narrow the selection	FR-5.3
UIX-0504	info	Bulk edit of {n} rows started as job {run}; you will be notified	above batch threshold	none	0.5.6 rules
UIX-0505	warning	Module {remote} failed to load; other areas still work	remote load failure	retry; platform rolls back pin	FR-5.1
IOX-0501	error	Row {row}, column {column}: {message}	import validation	fix and re-import or use partial	FR-5.15
IOX-0502	info	This file was already imported (import {id})	same sha256	none	BR-5.11
IOX-0503	error	Import exceeds {max} rows	`io.max_import_rows`	split the file	FR-5.15
DOC-0501	error	File exceeds the {max} MB limit for document type {type}	size check	compress or split	FR-5.16
DOC-0502	error	File type .{ext} is not allowed	blocked or not in allowed list	convert to an allowed type	FR-5.16

Code	Severity	User text	Cause	Remediation	Raised by
DOC-0503	error	File quarantined: malware detected ({signature})	scan result	contact the sender; file is not downloadable	FR-5.16, BR-5.12
DOC-0504	error	Checked out by {user} since {time}; new versions are blocked	check-out lock (423)	ask {user} to check in, or request a release	BR-5.12
DOC-0505	error	Document is under legal hold and cannot be deleted or purged	legal hold	none	BR-5.12
DOC-0506	warning	Preview not available for this file type	FileType.preview=none	download to view	FR-5.16
DOC-0507	warning	Text could not be extracted; the document is not content-searchable	extraction failure (scanned image, encrypted PDF)	OCR arrives in wave 18 capture	FR-5.16
ALR-0501	info	Alert: {subject}	rule fired	open the record	FR-5.17
ALR-0502	error	You cannot create an alert on {field}: you do not have read access to it	field mask	choose another field	BR-5.14
ALR-0503	error	Due-date rules require a date or datetime field	wrong field type	choose a date field	FR-5.17
ALR-0504	warning	{n} recipients were skipped because they cannot read {entity}	recipient lacks read	adjust recipients	FR-5.17
ALR-0505	warning	Alert rule {name} was suspended after {n} alerts in one hour	runaway rule	narrow the filter and resume	FR-5.17
STX-0501	error	Standard text {code} has no version in {language} or the fallback language	missing translation	add the language version	FR-5.18
STX-0502	error	Unknown placeholder `{{placeholder}}` in {code}	publish validation	correct the placeholder	BR-5.15
STX-0503	error	{code} is used by {n} published templates or layouts and cannot be retired	where-used check	update the references first	FR-5.18

0.5.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Shell first contentful paint	5,000 users, 2 sessions/day	0.8 s / 1.5 s / 2.5 s (10 Mbps)	CDN-cached assets	n/a	remotes cached by digest	static assets unaffected
Form load (metadata + record)	200 k/day, peak 100/s	250 ms / 800 ms / 1.5 s	300/s	n/a	form definition ETag cache	p95 ≤ 2 s; skeleton rendering
Global search	50 k/day, peak 30/s	80 ms / 300 ms / 600 ms	100/s	title + content index ≈ 2 GB/tenant/year	GIN on `tsvector` per entity; attachment content GIN	results truncated to 20; timeouts at 1 s return partial
Gap-free allocation	50 k/day per sequence, peak 500/s	5 ms / 15 ms / 30 ms	500/s per sequence	`allocation` ≈ 18 M rows/year for a busy sequence	`(sequence_id, number)` unique; monthly cleanup of used rows > 13 months to archive	row-lock queueing; p99 rises to ≈ 90 ms, no gaps
Buffered allocation	2,000/s across pods	0.1 ms / 1 ms / 2 ms	2,000/s	none	in-memory block	block refill contention; still < 5 ms
Audit capture (commit → visible)	5 M changed rows/day/tenant, peak 2,000/s	500 ms / 2 s / 5 s	3,000 rows/s with 12 consumers	≈ 1.8 B entries/year at peak volume (≈ 900 GB)	monthly partitions; `(tenant_id, entity, record_id, occurred_at desc)`	lag grows; `AuditLagHigh` at 10 s; no loss (Kafka retention 7 days)
Record history query	20 k/day	30 ms / 120 ms / 300 ms	50/s	—	same index, partition pruning by date	paging at 100 entries
Import 100 k rows	50/day/tenant	2 min / 5 min / 8 min	400 rows/s per job	staging transient	batch insert 1,000 rows	jobs queue by priority
Attachment upload (20 MB)	20 k/day/tenant, peak 20/s	1 s / 4 s / 8 s to `scanning`; ≤ 30 s to `available`	100 MB/s per tenant to object storage	≈ 4 TB/year/tenant at 20 MB average (before dedupe)	object key per version; DB rows ≈ 7 M/year	scan queue grows; uploads still accepted
Change-based alert evaluation	500 rules/tenant over the audit stream	1 s / 5 s / 10 s event → alert	3,000 changes/s	`alert_event` ≈ 10 M rows/year	rules indexed by (entity, field) in memory; events by (tenant_id, recipient_id, read_at)	lag; suspension of runaway rules
Trace ingestion	2 k spans/s per service at 10% sampling	n/a	50 k spans/s per collector gateway	Tempo ≈ 5 TB/year (14-day retention by default)	object storage backend	collector drops lowest-priority spans; `OTelCollectorDropping`

0.5.20 Localization and compliance hooks

- Number sequences and fiscal law: gap-free continuous numbering is the hook for invoice-numbering rules (EU VAT Directive Art. 226(2) "sequential number", France and Italy chronological numbering, Mexico CFDI folio series, India GST invoice series ≤ 16 characters per financial year). Country packs (wave 48) ship sequence templates per document type (format, `reset_on_year`, scope `legal_entity_fiscal_period` from wave 6) and validate `continuous=true` for fiscal documents of their country; the extension point is the sequence template + decision table `NumberingPolicy` (FR-3.13).
- Audit retention: defaults 10/3 years; country packs raise minimums (Germany GoBD 10 years for booking records, France 6–10, US SOX 7); `audit.retention_years.*` can only be raised by a country pack, not lowered below its minimum.
- Document retention: retention classes per document type are the hook for GoBD/HGB §257, US IRS records retention and similar rules; legal hold (wave 48) overrides purge; archive storage class with object lock satisfies WORM requirements where mandated.
- Privacy: audit stores SHA-256 digests for `pii=sensitive` fields; erasure requests (wave 48) pseudonymize actor display names in the viewer but never rewrite audit rows (lawful basis: legal obligation, GDPR Art. 17(3)(b)); attachments linked to an erased subject are listed for the privacy officer; alert events and personalization are deleted on erasure.
- Language and formats: shell strings, form labels, alert subjects and standard texts are resolved per user locale; English ships in wave 5, wave 48 label packs add languages; right-to-left layout support is a wave 48 deliverable but the shell uses CSS logical properties from wave 5. Date, number and currency formatting follow `ui.user_defaults` via `Intl` APIs.
- Calendar: due-date alerts and year segments use the legal entity time zone; fiscal-period scopes arrive in wave 6.
- Accessibility law: WCAG 2.1 AA (FR-5.5) covers EN 301 549 (EU Accessibility Act, from 2025-06-28) and US Section 508 for the generated UI.

- E-invoicing: attachments with `restriction=external` and document type "E-invoice original" are the storage hook for wave 35 Peppol/Factur-X archives (original XML retained unchanged, sha256 recorded).

0.5.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.29.1 Organization administration → Alerts (org-level alert rules) (register row 82)	GAP	ADDED as FR-5.17	org-wide rules (`org_wide=true`) maintained by Administer alerts
AX §12.29.4 Organization administration → Setup: Alerts → Alert rules (change-based, due-date; event rule setup, alert email templates) (register row 88)	GAP	ADDED as FR-5.17	rule kinds per FR-5.17; email via wave 4 templates (FR-4.14)
AX §12.30.3 System administration → Periodic: Alerts → Change-based alerts / Due date alerts (batch processors) (register row 91)	GAP	ADDED as FR-5.17	change-based evaluated continuously from CDC; due-date daily job (0.5.16)
AX §12.31.1 Home / role centers / workflow → Alerts (My alerts / Notification list) (register row 94)	GAP	ADDED as FR-5.17	shell Alerts bell and list (0.5.6), `ui.alert_event`
AX §12.31.2 Home / role centers / workflow → Inquiries: Alert inquiry (alerts by rule/user) (register row 95)	GAP	ADDED as FR-5.17	My alerts / Alert inquiry and Alert rule statistics (0.5.14)
AX §12.31.3 Home / role centers / workflow → Periodic: Alerts → Process change-based alerts / Due-date alerts (register row 98)	GAP	ADDED as FR-5.17	same jobs as row 91
AX §12.31.4 Home / role centers / workflow → Setup: Alerts → Alert rules (field/record changes, due date; subject/message; recipients; organization-wide) / Alert rule templates (register row 100)	GAP	ADDED as FR-5.17	rule templates = `MessageTemplate` codes; "Notify me when…" creates a rule from a field
AX §12.29.1 Organization administration → Document management → Document handling (attachments on any record), Document types, File types (`DocuView`, `DocuType`) (register row 235)	PARTIAL-5	ADDED as FR-5.16	`Attachment`, `DocumentType`, `FileType` entities
AX §12.29.2 Organization administration → Document management inquiries (documents by type/record) (register row 243)	PARTIAL-5	ADDED as FR-5.16	Documents by type / record inquiry (0.5.14)
AX §12.29.3 Organization administration → Periodic: Document management → Archive documents / Purge by retention (register row 249)	PARTIAL-5	ADDED as FR-5.16	retention job (0.5.16), BR-5.12, legal hold
AX §12.29.4 Organization administration → Setup: Document management parameters (archive directory, SharePoint, file types, max size), Document types (file, note, URL, image, template; group; location), Active document tables, File types (register row 252)	PARTIAL-5	ADDED as FR-5.16	parameters in 0.5.13; `active_entities[]` = active document tables; location = object storage or database; SharePoint location is not a storage backend (see row 253)
AX §12.29.4 Organization administration → Setup: Document management → OneDrive/SharePoint integration (D365), Attachment migration (register row 253)	PARTIAL-5	ADDED as FR-5.16	S3-compatible storage with customer bucket (`bucket_ref`) and attachment migration from `DocuRef`/`DocuValue` and SharePoint paths (0.5.11); live SharePoint/OneDrive storage as a document location is not offered — files are copied into object storage at migration, and a `url` document type links to SharePoint items
AX §12.30.1 System administration → Document handling (link) (register row 264)	PARTIAL-5	ADDED as FR-5.16	same capability as row 235
AX §12.30.4 System administration → Setup: Document management (link) (register row 270)	PARTIAL-5	ADDED as FR-5.16	same as row 252
AX §12.31.1 Home / role centers / workflow → Document handling on any record (`DocuView`) (register row 273)	PARTIAL-5	ADDED as FR-5.16	attachments panel on every generated detail page (0.5.6)
SAP §13.31 Cross-Application → SO10 / SO72 / SE75 (text objects/IDs): standard texts, text management (register row 349)	PARTIAL-5	ADDED as FR-5.18	register proposed wave 46 text blocks; added here because wave 4 templates (FR-4.14) and wave 46 layouts both consume it; layouts reference by code
SAP §13.35 DMS → CV01N originals (check-in/check-out, KPro storage categories, viewer, conversion) (register row 377)	PARTIAL-5	ADDED as FR-5.16	check-out/check-in, multiple versions per attachment, storage locations, previews; native CAD 2D/3D viewing is dropped (PLM keeps CAD; wave 9 engineering change links PLM documents by URL type)
SAP §13.35 DMS → Periodic: CV conversion runs (originals → PDF/neutral formats) (register row 389)	PARTIAL-5	ADDED as FR-5.16	attachment scan and preview job (Office → PDF, PDF first page, image thumbnails); CAD neutral-format conversion dropped with the viewer
SAP §13.35 DMS → Customizing: DC30 workstation applications (file types, viewer/editor, conversion) (TDWP) (register row 394)	PARTIAL-5	ADDED as FR-5.16	`FileType` (mime, max size, preview method, extract_text, blocked)
SAP §13.35 DMS → Customizing: full-text search activation (TREX/HANA) for originals (register row 398)	PARTIAL-5	ADDED as FR-5.16	text extraction to Postgres `tsvector`, searchable from global search and the documents inquiry; OCR of scanned images is wave 18

Part 1 — Phase 1: Enterprise Structure, Master Data & Engineering Data (waves 6–10)

1.0 Phase overview

Business outcome at phase exit. A master-data steward can model the demo manufacturer end to end on Passage without touching AX or SAP: three legal entities, two plants, five warehouses and their hierarchies; one governed party record per customer, vendor and contact; a 500-item, 50-master product catalog released per legal entity; multi-level BOMs and routings under engineering change control; and a migration framework that has loaded all of it from a synthetic AX extract with 100% reconciliation. Every later phase (journal in wave 11, inventory in wave 16, orders in wave 26) filters by this enterprise structure, posts against these parties and products, and loads its own legacy data through this framework.

Phase-exit sign-off criteria

- `make demo-seed` loads the demo manufacturer (3 legal entities, 2 sites, 5 warehouses, 3 hierarchy purposes) in under 2 minutes and every list page re-scopes on legal-entity switch (property test over 20 entities, 0 leaks).
- Duplicate detection on the 10k-party fixture achieves precision ≥ 0.95 and recall ≥ 0.90; merge leaves zero dangling references (referential-integrity audit returns 0 rows).
- Variant explosion of a 3×10×10 master creates 1,000 variants in < 5 s; GS1 check-digit tests pass for all 40 published GS1 vectors.
- 7-level, 2,000-component BOM explosion completes in < 500 ms and equals the golden result byte for byte.
- Synthetic AX extract (50k customers, 20k vendors, 100k items, 10k BOMs) migrates in < 30 min; re-run produces 0 changes; reconciliation report shows 0 unexplained exceptions.
- Stakeholder walkthrough recorded: create party → release product → build BOM → approve ECO → run migration.

Cross-wave data model summary

Entity	Owning service	Key relationships
LegalEntity	org-service (Go)	tenant 1:N legal_entity; legal_entity 1:N fiscal_calendar_assignment
OperatingUnit	org-service	type ∈ {department, cost_center, business_unit, value_stream}; node of HierarchyVersion
Site / Warehouse / Location	org-service	legal_entity 1:N site 1:N warehouse 1:N location
HierarchyType / HierarchyVersion / HierarchyNode	org-service	purpose-bound, effective-dated tree over OrgUnit ids
PostalAddress / AddressPurpose	org-service	referenced by Party, Site, Warehouse, LegalEntity
FiscalCalendar / FiscalPeriod	org-service	consumed by ledger (wave 11) for period status
UnitOfMeasure / UomConversion	org-service	product-specific conversions live in product-service
Party (Person/Organization)	party-service (Go)	1:N PartyRole; 1:N PartyAddress; 1:N TaxRegistration
Customer / Vendor / Contact / Employee role	party-service	role rows keyed (party_id, legal_entity_id)
DuplicateCandidate / MergeCase	party-service	pairs of party_id with score; Temporal workflow id
Product / ProductMaster / ProductVariant	product-service (Go)	master 1:N variant; dimension_group FK
ReleasedProduct	product-service	(product_id, legal_entity_id) with site/warehouse defaults
ItemGroup / ItemModelGroup / Category	product-service	referenced from ReleasedProduct
ConfigurationModel	product-service (Python solver)	master 1:N model; constraints in JSON
BomHeader / BomVersion / BomLine	engineering-service (Go)	version → (product_id, site_id, effective range)
RouteHeader / RouteVersion / Operation / OperationRelation	engineering-service	operation N:1 resource requirement
Resource / ResourceGroup / Capability / WorkCalendar	engineering-service	calendar consumed by scheduling (wave 23)
EngineeringChangeRequest / EngineeringChangeOrder	engineering-service	ECO 1:N affected BomVersion/RouteVersion
MigrationProject / MigrationRun / MappingSpec / StagingRow / ReconciliationResult	migration-service (TS + Go loaders)	run N:1 project; row N:1 run

Cross-wave event catalog

Event name	Producer	Consumers	Payload summary
dev.passage.org.legalentity.created.v1	org-service	party, product, ledger (w11), shell	legal_entity_id, code, name, country, currency, fiscal_calendar_id
dev.passage.org.hierarchy.version.activated.v1	org-service	security (OPA bundle), reporting, procurement	hierarchy_type_id, version_id, effective_from, node_count
dev.passage.org.site.created.v1 / warehouse.created.v1	org-service	product (released defaults), inventory (w16)	id, legal_entity_id, parent ids, address_id
dev.passage.org.fiscalcalendar.generated.v1	org-service	ledger (w11), period close (w14)	calendar_id, pattern, period_count, year
dev.passage.party.created.v1 / updated.v1	party-service	search index, migration, audit	party_id, type, name, primary_address_id
dev.passage.party.customer.created.v1 / vendor.created.v1	party-service	AR (w30), AP (w19), sales (w26), purchasing (w17)	party_id, legal_entity_id, account_number, group, terms
dev.passage.party.merged.v1	party-service	every service holding party_id FKs	survivor_id, merged_ids[], reference_rewrite_count
dev.passage.party.change.approved.v1	party-service (Temporal)	audit, notification	party_id, field, old, new, approver
dev.passage.product.created.v1 / variant.created.v1	product-service	engineering, inventory, pricing (w26)	product_id, master_id, dimension_values
dev.passage.product.released.v1	product-service	inventory, purchasing, sales, costing	product_id, legal_entity_id, item_model_group_id, defaults
dev.passage.engineering.bom.version.activated.v1	engineering-service	planning (w22), production (w21), costing (w39)	bom_version_id, product_id, site_id, effective_from, line_count
dev.passage.engineering.route.version.activated.v1	engineering-service	planning, scheduling (w23), production	route_version_id, operations[]
dev.passage.engineering.eco.released.v1	engineering-service	notification, audit, PLM adapter	eco_id, affected_versions[], release_date

Event name	Producer	Consumers	Payload summary
dev.passage.migration.run.completed.v1	migration-service	notification, reconciliation UI, audit	run_id, project_id, counts{staged, loaded, rejected}, hash_match

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Party dedup thresholds tuned to synthetic data miss real-world noise	Medium	High	Splink model trained on 10k fixture plus 2k hand-labeled real-shaped records; thresholds adjustable per tenant	Product
Effective-dated hierarchies become slow under as-of queries at 50k nodes	Medium	Medium	Closure table per version, materialized on activation; p95 target 20 ms enforced in CI	Engineering
Product dimension explosion makes variant counts unbounded	Low	High	Hard cap 10,000 variants per master; explosion is a Temporal job with checkpointing	Engineering
BOM cycles or phantom recursion break explosion	Medium	High	Cycle check on every line insert (closure table), depth cap 25	Engineering
AX/SAP extracts arrive with codepage and enum drift	High	Medium	Mapping DSL with explicit enum tables and reject-with-reason; dry-run before load	Implementation partner
Wave 10 depends on three prior waves landing on time	Medium	Medium	Connectors for DirParty/CustTable built against wave 7 as soon as it passes Proof	Product

1.6 Wave 6: Enterprise Structure & Organization Model

1.6.1 Scope and business context

AX scopes everything by `DataAreaId` (CompanyInfo) and organizes units in `OMOperatingUnit` with hierarchies in `OMHierarchyType` / `OMHierarchyRelationship`; SAP scopes by `BUKRS` (T001) with plants (T001W), storage locations (T001L), sales organizations (TVKO) and purchasing organizations (T024E) wired together in SPRO. Wave 6 builds the single organization model every later service filters by. At the end of the wave an administrator can create a tenant's legal entities, operating units, sites, warehouses and locations; build effective-dated hierarchies for reporting, procurement and security purposes; attach validated postal addresses; generate fiscal calendars; and maintain countries, currencies, units of measure and languages. The shell's legal-entity switcher re-scopes every list page.

Out of scope: ledger period *statuses* (wave 14 owns open/closed per module), warehouse layout detail such as zones and location directives (wave 31), intercompany relationships (wave 40), and any transactional data.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX organization hierarchy designer (OMHierarchyRelationship) only shows one effective date at a time; previewing "what changes on 2026-07-01" means publishing and re-opening; a mistaken publish cannot be undone	Hierarchy relationships are stored as flat ValidFrom/ValidTo rows with no version concept	Versioned trees with draft/published states, as-of slider and a diff view before publish (FR-6.3, FR-6.11)	Preview clicks 9→1; wrong-date publishes recoverable in 100% of cases via superseding version
SAP enterprise structure (OX02/OX10/OX09/OX18/OB29) requires 6+ IMG transactions plus assignments to define one plant; a missing assignment surfaces only when the first goods movement fails	Configuration spread across customizing tables (T001, T001W, T001K, T001L, T024E) with assignment tables between them	One `Site` create carries its legal entity, calendar and default warehouse; invariants are checked at save, not at first transaction (FR-6.5)	Transactions 6→1; structural errors found at save: 100% vs ~0%
AX company switch (DataAreaId) reloads the whole client and loses open forms; cross-company inquiries need `crossCompany` X++	Company is a session property baked into every table	Header-scoped entity context; the shell switches without reload; read-only `ALL` scope behind a duty (FR-6.10)	Switch time 8–15 s→<300 ms; open views preserved 100%
Fiscal calendar period creation in AX (FiscalCalendarPeriod) and SAP (OB29 period tables) is manual per year; 4-4-5 week-53 years are keyed by hand and frequently wrong	No pattern generator	Pattern-driven generation with contiguity proof and golden fixtures (FR-6.7)	Setup effort 45 min/year→<10 s; boundary errors 0 against fixture
UoM conversion rounding silent in AX (UnitOfMeasureConversion stores Numerator/Denominator with decimal drift) and SAP (T006 integer ratios)	Ratio stored as scaled integers, rounded at storage	numeric(28,12) factor+offset, rounding only at display (FR-6.9, BR-6.8)	Round-trip error 1e-6→0 at 12 decimals across 50 pairs

1.6.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Legal entity	CompanyInfo, `OMLegalEntity` form, DataAreaId	T001 company code (OX02)	org-service `LegalEntity`; Legal entities screen	full
Operating units	OMOperatingUnit (dept, cost center, BU, value stream)	Cost center CSKS (KS01), profit center CEPC, controlling area TKA01	`OperatingUnit` with `type` enum	full
Org hierarchies	OMHierarchyType, OMHierarchyPurpose, OMHierarchyRelationship, Organization hierarchy designer	Standard hierarchy KSH1/KCH1, enterprise organization	`HierarchyType/HierarchyVersion/HierarchyNode`; Hierarchy designer	improved (multiple purposes with effective-dated versions and as-of API)
Sites	InventSite, Sites form	T001W plant (OX10)	`Site`	full
Warehouses / locations	InventLocation, WMSLocation	T001L storage location (OX09); LAGP bins	`Warehouse`, `Location` (bin-level detail deferred to wave 31)	partial
Addresses	LogisticsPostalAddress, LogisticsLocation, LogisticsLocationRole, DirPartyLocation	ADRC / ADRCT, address version	`PostalAddress`, `AddressPurpose`	full
Fiscal calendars	FiscalCalendar, FiscalCalendarYear, FiscalCalendarPeriod, Ledger calendar	T009 fiscal year variant (OB29), T009B periods	`FiscalCalendar`, `FiscalYear`, `FiscalPeriod`	full (status per ledger deferred to wave 14)
Currencies	Currency table, ISO 4217	TCURC, TCURX decimals	`Currency` reference with minor units	full
Units of measure	UnitOfMeasure, UnitOfMeasureConversion (standard + product-specific)	T006 (CUNI), T006A	`UnitOfMeasure`, `UomConversion`; product-specific in wave 8	full
Countries / regions / languages	LogisticsAddressCountryRegion, LogisticsAddressState, LanguageTable	T005, T005S, T002	`Country`, `Region`, `Language`	full
Company-level parameters	CompanyInfo fields, Ledger (currency, calendar)	T001 fields (WAERS, PERIV, KTOPL)	`LegalEntity` fields + ledger linkage in wave 11	partial (chart of accounts linkage arrives in wave 11)

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Teams and team types	OMTeam, OMTeamType, OMTeamMembershipCriterion (Organization administration → Teams)	HR OM work groups (PP01 organizational units), S/4 Responsibility Management teams (Fiori "Manage Teams and Responsibilities")	org-service `TeamType`, `Team`, `TeamMember`	full
Legal entity copy / template	Legal entity copy (SysDataExport/SysDataImport), D365 "Copy into legal entity" configuration project	EC01 copy company code, EC02 copy plant (organizational object copy)	`POST /org/v1/legal-entities/{id}:copy` (FR-6.15)	improved (invariants checked at copy)
Currency rounding rules	Currency form: general, price, purchase, sales rounding with round-off type normal/downward/rounding-up	TCURX decimals, OB90 rounding rules per company code and currency	`CurrencyRounding` per (currency, purpose, legal entity) (FR-6.16)	full
Organization name translations	Organization translations on OMOperatingUnit / CompanyInfo	T001 / CSKT / CEPCT texts per SPRAS	`org_unit_translation` (FR-6.17)	full

1.6.3 Functional requirements

- **FR-6.1 Tenant and legal-entity creation** — Must. The system shall create legal entities under a tenant with a unique 4–8 character code, legal name, country, accounting currency, default language and time zone.
  - Code is immutable after creation; names are effective-dated (name history table).
  - A legal entity cannot be deleted; it can be set `inactive`, which hides it from the switcher but keeps all references.
  - Maximum 500 legal entities per tenant in v1.
- **FR-6.2 Operating units** — Must. The system shall maintain operating units of type department, cost_center, business_unit and value_stream, each with a code unique within tenant and type.
  - Operating units are tenant-scoped, not legal-entity-scoped (mirrors AX, where OMOperatingUnit is shared), with an optional default legal entity.
  - A cost center may carry an optional `manager_party_id` resolved after wave 7.
- **FR-6.3 Multi-purpose effective-dated hierarchies** — Must. The system shall support hierarchy types with one or more purposes (reporting, procurement, expenditure, security, centralized_payments, custom) and versioned trees with `effective_from`/`effective_to` ranges that cannot overlap for the same type.
  - A node is any LegalEntity or OperatingUnit; one node appears at most once per version.
  - Draft versions may be edited freely; published versions are immutable; a new version is created by copying the current one.
  - Maximum depth 15, maximum 50,000 nodes per version.
- **FR-6.4 As-of hierarchy queries** — Must. The system shall answer parent, ancestors, descendants and path-to-root for any node as of any date with p95 < 20 ms at 50,000 nodes.
  - Implemented with a closure table materialized at publish time.
- **FR-6.5 Sites, warehouses and locations** — Must. The system shall maintain sites per legal entity, warehouses per site, and locations per warehouse, each with address, time zone and type (`standard`, `transit`, `quarantine`, `subcontractor` for warehouses).
  - Warehouse code unique per legal entity (as in AX InventLocationId per company); site code unique per legal entity.
  - A warehouse cannot move between sites once referenced by an inventory transaction (enforced from wave 16 via a reference-count check).
- **FR-6.6 Postal addresses with purposes** — Must. The system shall store structured postal addresses (ISO 3166-1 alpha-2 country, region, city, postal code, street lines, building, PO box) and attach them to any owner with purposes (business, delivery, invoice, remit_to, service, home, other) and a primary flag per purpose.
  - Country-specific formatting rules for the first seven countries (US, UK, DE, FR, IN, MX, CA); postal code regex per country.
  - Geocoding hook: an adapter interface returning lat/long and a confidence; default adapter is a no-op.
  - Addresses are effective-dated; an address edit creates a new version and keeps the old one for documents already printed.
- **FR-6.7 Fiscal calendars** — Must. The system shall generate fiscal years and periods from patterns: calendar-year (12 periods), 4-4-5 / 4-5-4 / 5-4-4 (12 periods + 53rd week handling), 13-period, and custom (user-entered period boundaries).
  - Each year has exactly one opening period and one closing period in addition to operating periods (AX convention), configurable off for SAP-style special periods 13–16.
  - Periods within a year are contiguous and non-overlapping; the generator proves this before saving.
- **FR-6.8 Reference data** — Must. The system shall seed and maintain ISO 3166 countries and regions, ISO 4217 currencies with minor units (JPY 0, KWD 3, USD/EUR 2), ISO 639-1 languages, and UCUM-aligned units of measure in classes (quantity, mass, length, volume, area, time, temperature, each).
  - Reference data is tenant-visible but platform-owned; tenants may add custom units and deactivate rows, not delete.
- **FR-6.9 UoM conversions** — Must. The system shall store conversions as `factor` and `offset` with numeric(28,12) precision and compute any-to-any within a class by path search of at most two hops.
  - Round-trip of 1 unit through A→B→A must return exactly 1 to 12 decimal places for the 50-pair fixture.
  - Temperature uses the offset (°C→°F: factor 1.8, offset 32).
- **FR-6.10 Legal-entity context switch** — Must. The system shall carry `X-Passage-LegalEntity` on every request; generated list pages filter by it, and the OPA policy denies requests for entities the user has no role assignment in.
- **FR-6.11 Hierarchy designer UI** — Should. The system shall provide a drag-and-drop tree editor with an effective-date slider that previews the tree as of a chosen date and highlights nodes that change between versions.
- **FR-6.12 Seed and export** — Should. The system shall export and import the complete organization model as a versioned JSON package (used by `make demo-seed` and by the migration framework in wave 10).
- **FR-6.13 Address validation adapter** — Could. The system shall expose an address-validation adapter interface (normalize, suggest) with a null default; third-party adapters are out of scope for this wave.
- **FR-6.14 Teams and team types** — Should. The system shall maintain team types (code, name, allowed member kinds `user` / `party`) and teams (code unique per tenant, name, description, team type, optional owning operating unit, `effective_from`/`effective_to`, status) with members (Keycloak user id or wave-7 party id, member role `member` / `lead`, `valid_from`/`valid_to`).
  - Teams are cross-organization groups, not hierarchy nodes; they are referenced by stewardship ownership (FR-7.11 `owner_team_id`), wave-4 work-item queues, address-book access (FR-7.18) and readiness-check assignment (FR-9.17).
  - Limits: 5,000 teams per tenant, 2,000 members per team; membership changes publish `dev.passage.org.team.membership.changed.v1` (team_id, added[], removed[], as_of).
  - Membership criteria (AX OMTeamMembershipCriterion) are an optional Rego filter over user attributes (e.g. `department == "D100"`), evaluated nightly by job ORG-J5; manual members are never removed by the rule.
- **FR-6.15 Legal-entity copy from template** — Should. The system shall create a legal entity by copying an existing legal entity or a template package: number sequences (reset to start values), fiscal calendar assignment, parameters (§1.6.13), address purposes, and optionally sites and warehouses; code, name, country and accounting currency are supplied by the user.
  - Runs as Temporal workflow `copy-le-{target_code}`; every copied object receives a new id; transactional data is never copied.

    - Copy of a 5-site, 20-warehouse entity completes in < 30 s; the result is validated against the same invariants as a manual create before commit.
- **FR-6.16 Currency rounding rules** — Must. The system shall store, per currency, purpose (general, price, purchase, sales, tax, cash) and optional legal entity, a rounding unit (0.000001–1,000) and method (half_even, half_up, down, up); the default is the ISO 4217 minor unit with half_even, and consumers call `GET /org/v1/currencies/{code}/round?amount=&purpose=`.
    - Cash rounding (CHF 0.05, SEK 1.00) uses the `cash` purpose; consumed by waves 26 and 30.
    - Legal-entity rows override tenant rows for the same (currency, purpose).
- **FR-6.17 Organization name translations** — Should. The system shall store translated names of legal entities, operating units, sites and warehouses per BCP 47 language, with fallback to the base name; printed documents (wave 46) render the name in the document language.

1.6.4 Data model

LegalEntity

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS key
code	text(8)	yes	unique per tenant, immutable, `^[A-Z0-9]{2,8}$`
name	text(100)	yes	current name; history in legal_entity_name
country_code	char(2)	yes	FK country
accounting_currency	char(3)	yes	FK currency; immutable after first posting (wave 11)
default_language	text(5)	yes	BCP 47, default `en-US`
time_zone	text	yes	IANA zone
primary_address_id	uuid	no	FK postal_address
registration_number	text(40)	no	company registry number
status	enum	yes	active, inactive
created_at / updated_at / version	timestamptz, int	yes	optimistic lock on `version`

HierarchyVersion / HierarchyNode

Field	Type	Required	Notes
id	uuid	yes	PK
hierarchy_type_id	uuid	yes	FK; type carries purposes[]
version_no	int	yes	monotonic per type
effective_from	date	yes	inclusive
effective_to	date	no	exclusive; null = open-ended
status	enum	yes	draft, published, superseded
node.id	uuid	yes	PK of hierarchy_node
node.version_id	uuid	yes	FK
node.org_unit_id	uuid	yes	LegalEntity or OperatingUnit id; unique per version
node.parent_node_id	uuid	no	null for root; exactly one root per version
node.sort_order	int	yes	sibling order
closure(ancestor_id, descendant_id, depth)	table	yes	rebuilt on publish

PostalAddress

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS
country_code	char(2)	yes	FK
region_code	text(10)	conditional	required for US, CA, IN, MX
city	text(60)	yes	
postal_code	text(12)	conditional	validated by country regex
street_1 / street_2 / street_3	text(100)	street_1 yes	
building / po_box	text(20)	no	
latitude / longitude	numeric(9,6)	no	from geocoding hook
valid_from / valid_to	date	yes/no	effective dating
formatted	text	computed	rendered per country template

FiscalCalendar / FiscalPeriod — calendar: id, tenant_id, code, name, pattern (enum), week_start; period: id, calendar_id, fiscal_year, period_no (0 = opening, 1–13 operating, 99 = closing), name, start_date, end_date, type (opening/operating/closing/adjustment).

UnitOfMeasure / UomConversion — uom: id, code (`^[A-Za-z0-9/%]{1,10}$`), name, class, decimal_precision (0–6), system_flag; conversion: from_uom_id, to_uom_id, factor numeric(28,12), offset numeric(28,12), product_id (null here, used in wave 8).

Invariants: exactly one root per published hierarchy version; no overlapping effective ranges per hierarchy type; fiscal periods of a year tile the year with no gap or overlap; a warehouse's legal entity equals its site's legal entity; every `primary` address purpose is unique per (owner, purpose). Indexing: `hierarchy_node(version_id, org_unit_id)` unique; closure table PK (ancestor_id, descendant_id); `postal_address` GIN trigram on `formatted` for search; all tables partitioned logically by tenant_id with RLS policies, no physical partitioning in this wave.

Team / TeamMember / CurrencyRounding (FR-6.14, FR-6.16) — team: id, tenant_id, code text(20) unique per tenant, name text(60), team_type_id FK, operating_unit_id FK null, effective_from date, effective_to date null, status (active, inactive), membership_rule text null (Rego), version; team_member: team_id, member_kind (user, party), member_id uuid, member_role (member, lead), valid_from date, valid_to date null, source (manual, rule), unique (team_id, member_kind,

member_id, valid_from); currency_rounding: currency_code char(3), purpose enum, legal_entity_id uuid null, unit numeric(16,6) > 0, method enum, PK (currency_code, purpose, coalesce(legal_entity_id, nil-uuid)). org_unit_translation: owner_type, owner_id, language text(5), name text(100), PK (owner_type, owner_id, language).

1.6.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/org/v1/legal-entities	Create legal entity	`Idempotency-Key` header, 24 h	Maintain legal entities
GET	/org/v1/legal-entities/{id}	Read	n/a	View organization
PATCH	/org/v1/legal-entities/{id}	Update name/addresses/status	version check (409 on mismatch)	Maintain legal entities
POST	/org/v1/operating-units	Create operating unit	key header	Maintain operating units
POST	/org/v1/hierarchy-types/{id}/versions	Create draft version (copy of current)	key header	Maintain organization hierarchies
PUT	/org/v1/hierarchy-versions/{id}/nodes	Replace the full node set of a draft	full replace, naturally idempotent	Maintain organization hierarchies
POST	/org/v1/hierarchy-versions/{id}:publish	Publish; sets effective_from; builds closure	key header	Publish organization hierarchies
GET	/org/v1/hierarchy-types/{id}/as-of?date=2026-06-01&node=...	Ancestors/descendants/path as of a date	n/a	View organization
POST	/org/v1/sites, /org/v1/warehouses, /org/v1/locations	Create physical structure	key header	Maintain sites and warehouses
POST	/org/v1/addresses	Create address; `owner_type`, `owner_id`, `purposes[]`	key header	Maintain addresses
POST	/org/v1/fiscal-calendars/{id}:generate	Generate years from pattern	idempotent per (calendar, year)	Maintain fiscal calendars
GET	/org/v1/uom/convert?from=KG&to=LB&qty=10	Convert quantities	n/a	any authenticated
POST	/org/v1/export, /org/v1/import	Org model package	import idempotent by code	Administer organization

Commands validate: codes against regex; address against country rules; hierarchy publish rejects cycles, duplicate nodes, multiple roots, overlapping ranges; fiscal generation rejects a year already generated unless `force=true` and the year has no postings (checked via a ledger callback that returns `true` until wave 11 exists).

Event (CloudEvents type)	Trigger	Payload fields	Consumers
dev.passage.org.legalentity.created.v1	commit of legal entity	id, code, name, country_code, accounting_currency, time_zone	party, product, ledger, shell switcher
dev.passage.org.legalentity.updated.v1	PATCH commit	id, changed_fields[], version	shell, search
dev.passage.org.operatingunit.created.v1	commit	id, type, code, name	reporting, ledger dimensions
dev.passage.org.hierarchy.version.activated.v1	publish	hierarchy_type_id, purposes[], version_id, effective_from, node_count	OPA bundle builder, reporting, procurement
dev.passage.org.site.created.v1 / warehouse.created.v1 / location.created.v1	commit	id, legal_entity_id, parent_id, type, address_id	product, inventory
dev.passage.org.address.changed.v1	new address version	address_id, owner_type, owner_id, purposes[], valid_from	document services
dev.passage.org.fiscalcalendar.generated.v1	generate	calendar_id, fiscal_year, periods[]	ledger, period close

1.6.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Organization administration workspace	Tiles: legal entities, operating units, hierarchies, sites/warehouses, calendars, reference data	counts, last-changed, pending drafts	View organization
Legal entities list/detail	New, edit, deactivate, addresses tab, name history	code, name, country, currency, status	Maintain legal entities
Operating units list	New, bulk import (CSV), type filter	type, code, name, default legal entity	Maintain operating units
Hierarchy designer	Drag-drop nodes, add from picker, effective-date slider, diff vs previous version, publish	tree, effective range, node count, change badges	Maintain / Publish organization hierarchies
Sites and warehouses	New site, new warehouse under site, locations grid	code, name, type, address, time zone	Maintain sites and warehouses
Fiscal calendars	New calendar, generate years, period grid	pattern, year, period no, start, end	Maintain fiscal calendars
Reference data (countries, currencies, UoM, languages)	Deactivate, add custom UoM, add conversion, test conversion	code, name, class, factor, offset	Maintain reference data

Interaction rules: all grids support inline edit with validation on blur and a single save per row; `Ctrl+S` saves, `Esc` cancels, `Alt+N` new; the hierarchy designer validates structurally on every drop and semantically (effective range) on publish only; personalization (column order, saved filters) is stored per user per screen by the wave-5 shell; the legal-entity switcher is in the top bar and persists the last choice.

1.6.7 Business rules and edge cases

- **BR-6.1** Legal entity codes are upper-cased on save and compared case-insensitively; `DAT`, `SYS` and `ALL` are reserved.
- **BR-6.2** Hierarchy version state machine: `draft → published → superseded`; `draft` may be deleted; `published` becomes `superseded` only when a later version publishes with `effective_from` greater than its own; a published version can be ended early by publishing a successor, never by editing.
- **BR-6.3** Publishing a version whose `effective_from` is earlier than today requires the duty "Backdate organization hierarchies" and is audited with a reason text (min 10 chars).
- **BR-6.4** Moving a node moves its subtree; dropping a node onto its own descendant is rejected with `ORG-HIER-CYCLE`.
- **BR-6.5** A security-purpose hierarchy triggers an OPA bundle rebuild within 60 s of activation; until the rebuild completes, the previous bundle stays in force (no window with no policy).
- **BR-6.6** Fiscal 4-4-5 years ending on a 53-week boundary add the extra week to the last period of the year; the generator records `week53 = true` on that period.
- **BR-6.7** A fiscal year may be regenerated only while no period of it has any posting; otherwise `ORG-CAL-LOCKED` (409).
- **BR-6.8** UoM conversions are stored once in a canonical direction; the inverse is computed as 1/factor with the offset negated and divided; results are rounded only at display, using the target unit's `decimal_precision`.

- **BR-6.9** Address edits are copy-on-write: documents hold the address version id, not the address id, so a printed invoice never changes retroactively.
- **BR-6.10** Concurrent edits use the `version` column; a stale PATCH returns 409 with the current representation; the UI offers merge or reload.
- **BR-6.11** Deactivating a legal entity is refused while any user has it as their default entity (`ORG-LE-DEFAULT-IN-USE`).
- **BR-6.12** A team member's validity must lie inside the team's effective range (`ORG-TEAM-MEMBER-RANGE`); a team whose `effective_to` is in the past accepts no new members (`ORG-TEAM-EXPIRED`); deactivating a team that owns parties reassigns ownership to the team lead or, with no lead, raises a stewardship item per party.
- **BR-6.13** Rounding: `round(a, u, m) = u × R_m(a / u)` computed in numeric(28,12); half_even sends exact ties to the even multiple; a unit that is not a power-of-ten multiple of the minor unit or 0.05/0.25/0.5 multiples thereof is rejected (`ORG-CUR-ROUNDING-UNIT`).
- **BR-6.14** Legal-entity copy keeps site and warehouse codes (unique per legal entity) and rewrites tenant-unique codes with the target code as prefix; a collision aborts the whole copy with no partial result.

1.6.8 Acceptance criteria

- **AC-6.1** (FR-6.3, FR-6.4) Given 1,000 effective-dated nodes across 5 purposes with 3 versions each, when the as-of API is called for 10,000 random (node, date) pairs, then every parent matches the fixture oracle and p95 latency < 20 ms.
- **AC-6.2** (FR-6.10) Given a user assigned to legal entities USMF and DEMF, when they switch from USMF to DEMF, then all 20 generated list pages show only DEMF rows and a direct API call with `X-Passage-LegalEntity: FRMF` returns 403.
- **AC-6.3** (FR-6.9) Given the 50-pair conversion fixture, when each quantity is converted A→B→A, then the result equals the input to 12 decimal places.
- **AC-6.4** (FR-6.7) Given calendar patterns 4-4-5, 13-period and calendar-year for fiscal 2026, when generated, then period boundaries equal the committed fixture files exactly, including the 53rd week in the 4-4-5 variant for 2028.
- **AC-6.5** (FR-6.3, BR-6.2) Given a published version effective 2026-01-01, when a draft with `effective_from` 2025-12-01 is published without the backdate duty, then the request is rejected with 403 and no version changes state.
- **AC-6.6** (FR-6.6) Given a US address without `region_code`, when saved, then validation fails with field-level error `ORG-ADDR-REGION-REQUIRED`; with `region_code = CA` and postal code `9410`, then error `ORG-ADDR-POSTAL-FORMAT`.
- **AC-6.7** (FR-6.5) Given warehouse WH1 under site S1 in USMF, when a PATCH attempts to set `site_id` to a site in DEMF, then 422 `ORG-WH-LE-MISMATCH`.
- **AC-6.8** (FR-6.12) Given the demo seed package, when imported twice, then the second import reports 0 creates, 0 updates.
- **AC-6.9** (FR-6.11) Given the designer with the slider at 2026-06-01, when a node moved only in the draft version is viewed, then it renders with a "changed" badge and the previous parent shown in a tooltip.
- **AC-6.10** (FR-6.14, BR-6.12) Given team "MD stewards" effective 2026-01-01..2026-12-31, when a member is added with `valid_from` 2027-01-15, then 422 `ORG-TEAM-MEMBER-RANGE`; given a party owned by the team, then it appears in each current member's stewardship queue.
- **AC-6.11** (FR-6.16, BR-6.13) Given CHF cash rounding 0.05 half_up, when 10.024 and 10.025 are rounded, then 10.00 and 10.05; given EUR general 0.01 half_even, then 2.345 → 2.34 and 2.355 → 2.36.
- **AC-6.12** (FR-6.15) Given USMF with 2 sites, 5 warehouses and 12 number sequences, when copied to CAMF (CA, CAD), then CAMF has 2 sites, 5 warehouses, 12 sequences at their start values, 0 transactional rows, and completes in < 30 s.

1.6.9 Test plan and fixtures

Fixtures: `org/demo-manufacturer.json` (3 legal entities USMF/DEMF/FRMF, 2 sites, 5 warehouses, 40 operating units, 3 hierarchy types, generated); `org/hierarchy-1k.json` (1,000 nodes, 5 purposes, 3 versions each, generated with seed 6); `org/uom-pairs.csv` (50 pairs, hand-checked); `org/fiscal-patterns/*.csv` (4-4-5, 4-5-4, 5-4-4, 13-period, calendar for 2025–2028, hand-checked against the NRF retail calendar); `org/addresses-7-countries.csv` (140 valid, 70 invalid).

Test	Type	Target	Pass criterion
Hierarchy as-of oracle	property	as-of API	10k samples, 0 mismatches, p95 < 20 ms
Closure rebuild	unit	publish	closure row count = Σ depth over nodes
Entity re-scope	property (Playwright)	20 list pages	0 cross-entity rows
UoM round-trip	unit	convert	12-decimal equality
Fiscal generation	golden	generator	byte-equal CSV
Address validation	table-driven	validator	210/210 expected outcomes
Publish under load	performance	50k-node publish	< 30 s, as-of p95 < 20 ms after
Event contract	contract (AsyncAPI)	7 events	schema registry accepts; consumer stubs decode

Reconciliation: node count per published version = closure rows with depth 0; count of warehouses per legal entity in export = count in database; every legal entity has exactly one `primary` business address or none.

1.6.10 Deployment and operations

Helm chart `passage-org` (Deployment ×2 replicas, CloudNativePG cluster `org-db` with 3 instances, Debezium connector CR for the outbox table). Config: `ORG_HIERARCHY_MAX_NODES=50000`, `ORG_GEOCODER=null`, `ORG_SEED_PACKAGE=demo-manufacturer`. Migrations via Atlas from entity metadata, applied by an Argo CD pre-sync Job. Feature flags: `org.hierarchy_designer`, `org.address_validation_adapter`. Scheduled job: nightly closure-table integrity check (Temporal cron `0 2 * * *`). SLOs: read p95 < 50 ms, as-of p95 < 20 ms, availability 99.9%. Alerts: outbox lag > 60 s, closure integrity failure, OPA bundle rebuild older than 5 min after a security hierarchy activation. Runbook: "Hierarchy publish stuck" (check Temporal workflow `publish-hierarchy-{id}`), "Rebuild closure table" (`passagectl org rebuild-closure --version`). Rollback: Argo rollback to the previous digest; migrations in this wave are additive only, so no down-migration is needed.

1.6.11 Migration from AX / SAP

Source	Passage entity	Notes
AX CompanyInfo / OMLegalEntity (DataAreaId, Name, LanguageId, CurrencyCode via Ledger)	LegalEntity	DataAreaId → code (upper-cased); Ledger.AccountingCurrency → accounting_currency
AX OMOperatingUnit (OMOperatingUnitNumber, OMOperatingUnitType)	OperatingUnit	type enum 1..4 → department/cost_center/business_unit/value_stream
AX OMHierarchyType, OMHierarchyPurpose, OMHierarchyRelationship (ValidFrom/ValidTo)	HierarchyType, HierarchyVersion, HierarchyNode	each distinct ValidFrom in AX becomes a version; ValidTo max date → null
AX InventSite, InventLocation	Site, Warehouse	InventLocation.InventSiteId → site FK; warehouse type from InventLocation.WMSLocationId usage
AX LogisticsPostalAddress, LogisticsLocation, LogisticsLocationRole, DirPartyLocation	PostalAddress, AddressPurpose	roles Business/Delivery/Invoice/RemitTo map 1:1; others → other

Source	Passage entity	Notes
AX FiscalCalendar, FiscalCalendarYear, FiscalCalendarPeriod	FiscalCalendar, FiscalPeriod	Type Opening/Operating/Closing → period type
AX UnitOfMeasure, UnitOfMeasureConversion (non product-specific)	UnitOfMeasure, UomConversion	Factor/Numerator/Denominator → factor; InnerOffset/OuterOffset → offset
SAP T001 (BUKRS, BUTXT, LAND1, WAERS, SPRAS, PERIV)	LegalEntity	PERIV → fiscal_calendar assignment
SAP T001W (WERKS, NAME1, ADRNR) / T001L (LGORT)	Site + Warehouse	plant → site; a default warehouse per plant is created when T001L is empty
SAP CSKS / CEPC / T001K	OperatingUnit	cost center, profit center; KOKRS stored as attribute
SAP ADRC (ADDRNUMBER, NAME1, STREET, POST_CODE1, CITY1, COUNTRY, REGION)	PostalAddress	DATE_FROM → valid_from
SAP T009 / T009B	FiscalCalendar, FiscalPeriod	special periods 13–16 → adjustment periods when `sap_special_periods=true`
SAP T006 / T006A / TCURC / TCURX / T005	reference data	matched to ISO seed by code; unmatched custom units created

Validation: every migrated warehouse resolves to a site; every address passes country rules or lands in the exception queue; hierarchy versions must tile without gaps. Not migrated: AX `OMInternalOrganization` team records (no Passage equivalent in v1), SAP sales/purchasing organizations (modeled in waves 17 and 26 as procurement and sales hierarchies), SAP client (MANDT) which maps to tenant by configuration.

1.6.12 Dependencies, risks, open questions

Depends on: wave 2 (tenant model, RLS, OPA duties), wave 3 (entity metadata and generator), wave 4 (outbox, Temporal for publish jobs), wave 5 (shell, generated forms, audit, number sequences for operating unit numbers).

Risk	Likelihood	Impact	Mitigation
Closure-table rebuild blocks publishing large versions	Medium	Medium	Rebuild runs as a Temporal activity; version stays `publishing` until done
Country address rules incomplete beyond 7 countries	High	Low	Generic fallback rule; country packs in wave 48
Shell switcher semantics conflict with cross-entity screens (consolidation)	Low	Medium	Switcher supports `ALL` for read-only cross-entity views behind a duty

Open questions: (1) Should operating units be legal-entity-scoped as an option for SAP-style cost centers tied to a controlling area? Proposed default: tenant-scoped with optional default legal entity; decide by wave 11. (2) Do we store SAP special periods as adjustment periods or as flags on period 12? Proposed: adjustment periods; decide by wave 14. (3) Is UCUM the canonical UoM code set or ISO 80000? Proposed: UCUM codes with ISO display names; decide by wave 8.

1.6.13 Parameters and setup

Organization parameters live in the wave-3 metadata registry as typed settings; each row below is one setting key in `org.*`.

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`org.le.code_pattern`	tenant	regex	`^[A-Z0-9]{2,8}$`	regex ≤ 40 chars	Validates legal entity codes (FR-6.1, BR-6.1)	DataAreaId (4 chars) on CompanyInfo	BUKRS (4 chars), OX02
`org.le.default_language`	legal entity	BCP 47	`en-US`	seeded languages	Default UI and document language	CompanyInfo.LanguageId	T001-SPRAS
`org.time_zone`	legal entity / site / warehouse	IANA zone	tenant default	IANA list	Date boundaries for postings, cut-offs, timers	Legal entities → Time zone (D365)	ADRC-TIME_ZONE on plant address, system time zone STZAC
`org.le.accounting_currency`	legal entity	ISO 4217	none (required)	active currencies	Ledger currency; immutable after first posting	Ledger form → Accounting currency	T001-WAERS (OB22 for parall currencies)
`org.le.fiscal_calendar`	legal entity	FK	none (required before wave 11 posting)	calendars with ≥ 1 generated year	Period resolution for every posting	Ledger form → Fiscal calendar	OB37 (T001-PERIV)
`org.cal.opening_closing_periods`	calendar	bool	true	true / false	Generates period 0 and 99 per year (FR-6.7)	FiscalCalendar → Create year (opening/closing periods)	— (SAP uses special periods)
`org.cal.special_periods`	calendar	int	0	0–4	Adds adjustment periods 13–16 (open question 2)	—	T009-ANZBP via OB29
`org.cal.week_start`	calendar	enum	Monday	Monday–Sunday	Anchors 4-4-5 / 13-period generation	FiscalCalendar week start	Week-based fiscal year variant (OB29)
`org.hier.max_depth`	tenant	int	15	5–25	Publish validation (FR-6.3)	—	Standard hierarchy levels (OKEON)
`org.hier.max_nodes`	tenant	int	50,000	1,000–200,000	Publish validation; `ORG_HIERARCHY_MAX_NODES`	—	—
`org.hier.backdate_allowed`	hierarchy type	bool	false	true / false	When false, BR-6.3 duty is not grantable for that type	Hierarchy designer publish date rule	—
`org.addr.format`	country	template	UPU-style template for US, UK, DE, FR, IN, MX, CA	template tokens `{street_1}` … `{postal_code}`	Renders `formatted` (FR-6.6)	Address setup → Address format (LogisticsAddressFormatHeading)	Address layout key T005-LAND (ADDRESS_INTO_PRINTFOR
`org.addr.postal_check`	country	enum	strict for 7 v1 countries, off otherwise	off / warn / strict	Postal code regex enforcement	Address parameters → ZIP/postal code validation	T005 postal code check rule (OY17)
`org.addr.geocoder`	tenant	adapter id	`null`	`null` / registered adapter	Geocoding hook (FR-6.6)	Address parameters → geocoding (Bing Maps, D365)	Geographical Enablement Framework / DQM

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`org.cur.rounding`	currency × purpose (× legal entity)	unit + method	ISO minor unit, half_even	BR-6.13	Monetary rounding service (FR-6.16)	Currency → Rounding rules	OB90, TCURX
`org.uom.decimal_precision`	unit	int	per seed (EA 0, KG 3)	0–6	Display rounding (BR-6.8)	Units → Decimal precision	T006-ANDEC
`org.ou.number_sequence`	tenant	sequence	`OU-{#####}`	wave-5 sequences	Operating unit numbers	Organization administration parameters → Number sequences	Cost center / profit center manu keys (KS01/KE51)
`org.team.rule_evaluation`	tenant	bool	false	true / false	Enables job ORG-J5	OMTeamMembershipCriterion	—
`org.switcher.all_scope`	tenant	bool	false	true / false	Allows read-only `ALL` legal-entity scope behind a duty (FR-6.10)	Cross-company queries	—

1.6.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Legal entities	list inquiry	status, country, currency	code, name, country, currency, calendar, time zone, status	`org.legal_entity`	OMLegalEntity list; SSRS Legal entities list	OX02 overview (T001)
Organization hierarchy as of date	drill-down	hierarchy type, purpose, as-of date, root node	tree, depth, node type, effective range, child count	closure table	Organization hierarchy inquiry; SSRS Organization chart	OKENN, KSH3, KCH3
Hierarchy version comparison	drill-down	type, version A, version B	added, removed, moved nodes with old/new parent	`hierarchy_node`	Hierarchy designer "view changes"	Change documents on SETHEADER/SETNODE (RSSCD100)
Operating units	list inquiry	type, default legal entity, status, hierarchy membership	number, name, type, manager, legal entity, hierarchies	`org.operating_unit`	SSRS Operating units list	KS13 cost center report, KE5X profit center list
Sites, warehouses and locations	list inquiry	legal entity, type, status	site, warehouse, type, address, time zone, location count	`org.site`, `org.warehouse`	InventSite / InventLocation lists	OX10 / OX09 (T001W, T001L)
Fiscal periods	list inquiry	calendar, year	period no, name, start, end, type, week53	`org.fiscal_period`	Fiscal calendars → periods	OB29 periods (T009B)
Unit conversions	operational report	class, unit, product-specific yes/no	from, to, factor, offset, inverse, round-trip check	`org.uom_conversion`	SSRS Unit conversions list	CUNI (T006)
Unassigned structure	KPI	legal entity	warehouses without address, legal entities without calendar, operating units in no hierarchy, sites without warehouse	integrity projection	—	1KE1 profit center assignment monitor
Address validation exceptions	operational report	country, owner type, rule	address, failed rule, owner, age	`org.postal_address`	Address validation (D365)	DQM address check results
Teams and members	list inquiry	team type, as-of date, member	team, member, role, source, valid from/to	`org.team_member`	Teams (OMTeam)	Responsibility Management teams
Organization change log	drill-down	entity, user, date range	field, old, new, user, timestamp	wave-5 audit	Database log (SysDatabaseLog)	Change documents (KS05, OX02 table logs)

1.6.15 Printed and electronic documents

Document	Trigger	Layout elements (header/line/footer fields)	Formats	Numbering sequence	Legal / retention requirement
Organization chart	on demand from hierarchy designer	header: hierarchy type, purpose, as-of date; body: node boxes (code, name, manager); footer: version no, printed by, timestamp	PDF, SVG	none	none
Fiscal period schedule	on calendar generation or on demand	header: calendar, year, pattern; lines: period no, name, start, end, type; footer: generated by	PDF, XLSX	none	retained with ledger configuration for 10 years (HGB §257, Code de commerce L123-22)
Organization model package	export (FR-6.12)	manifest (schema version, sha256, counts) + JSON bodies	JSON (electronic)	`ORGPKG-{yyyy}-{####}`	7 years as configuration evidence

1.6.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
ORG-J1 Closure-table integrity check	daily 02:00 tenant local	hierarchy_type_id	read-only; checkpoint per type	< 5 min for 10 types × 50k nodes	report only, never auto-repair; runbook "Rebuild closure table"	`org_closure_integrity_failures > 0`
ORG-J2 Future-dated hierarchy activation	Temporal timer at `effective_from` 00:00 legal-entity time	version_id	one timer per version; activation idempotent	< 30 s	5 retries with backoff, then alert; previous version stays active	activation lag > 5 min
ORG-J3 Fiscal year roll-forward	daily check; generates next year 90 days before the last generated year ends	calendar_id	idempotent per (calendar, year)	< 10 s per calendar	custom-pattern calendars raise a stewardship item instead	no next year 30 days before year end
ORG-J4 Address re-validation and geocode refresh	weekly Sunday 03:00	country_code	checkpoint per 1,000 addresses	< 1 h per 1M addresses	adapter outage leaves status unchanged	adapter error rate > 20%
ORG-J5 Team membership rule evaluation	nightly 01:30	team_id	diff-based; re-run yields no change	< 10 min for 5,000 teams	rule error disables the rule and notifies the team lead	rule failures > 0
ORG-J6 Reference data refresh (ISO 3166/4217/639 updates)	quarterly, applied after admin approval	dataset	versioned package; re-apply is a no-op	< 1 min	rejected package keeps prior version	package pending > 14 days

1.6.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
View organization	`GET /org/v1/**`; Organization administration workspace; all org list pages	none	every role
Maintain legal entities	`POST/PATCH /org/v1/legal-entities`, `:copy`; Legal entities screen	Administer security roles (wave 2)	Organization administrator
Maintain operating units and teams	`/org/v1/operating-units`, `/org/v1/teams`, `/org/v1/team-types`; job ORG-J5 config	none	Organization administrator, HR administrator
Maintain organization hierarchies	draft create, `PUT …/nodes`; Hierarchy designer	Publish organization hierarchies for `security` purpose (maker-checker)	Organization administrator
Publish organization hierarchies	`POST …:publish`; ORG-J2 timers	Maintain organization hierarchies on security-purpose types	Organization administrator lead, Security administrator
Backdate organization hierarchies	publish with `effective_from` < today	Maintain organization hierarchies	Controller
Maintain sites and warehouses	`/org/v1/sites`, `/warehouses`, `/locations`	none	Organization administrator, Warehouse manager
Maintain fiscal calendars	`/org/v1/fiscal-calendars`, `:generate`; ORG-J3	Close fiscal periods (wave 14)	Accounting manager
Maintain reference data	countries, currencies, rounding, UoM, languages; ORG-J6 approval	none	Master data steward
Administer organization	`/org/v1/export`, `/org/v1/import`	Approve migration sign-off (wave 10)	System administrator

1.6.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
ORG-LE-CODE-RESERVED	error	"Code {code} is reserved."	DAT, SYS or ALL used	Choose another code	BR-6.1
ORG-LE-DEFAULT-IN-USE	error	"{n} users have this legal entity as default."	Deactivation while users default to it	Change users' default entity	BR-6.11
ORG-HIER-CYCLE	error	"Moving {node} under {target} creates a cycle."	Drop onto own descendant	Drop on a non-descendant	BR-6.4
ORG-HIER-MULTI-ROOT	error	"A version must have exactly one root; found {n}."	Publish with 0 or ≥ 2 roots	Attach orphans to the root	FR-6.3
ORG-HIER-OVERLAP	error	"Effective range overlaps version {v}."	Overlapping ranges per type	Adjust `effective_from`	FR-6.3
ORG-HIER-BACKDATE-DUTY	error	"Backdated publish requires the Backdate duty and a reason."	Past `effective_from` without duty	Request duty or use today	BR-6.3
ORG-CAL-LOCKED	error	"Fiscal year {y} has postings and cannot be regenerated."	Regenerate with postings	Create adjustment periods instead	BR-6.7
ORG-CAL-GAP	error	"Periods leave a gap on {date}."	Custom boundaries not contiguous	Fix boundaries	FR-6.7
ORG-ADDR-REGION-REQUIRED	error	"Region is required for {country}."	Missing region for US/CA/IN/MX	Enter region	FR-6.6
ORG-ADDR-POSTAL-FORMAT	error	"Postal code {code} is not valid for {country}."	Regex failure in strict mode	Correct code or set mode warn	FR-6.6
ORG-WH-LE-MISMATCH	error	"Warehouse and site belong to different legal entities."	Site change across entities	Pick a site in the same entity	FR-6.5
ORG-UOM-NO-PATH	error	"No conversion from {from} to {to} within two steps."	Missing conversion	Add a conversion	FR-6.9
ORG-TEAM-MEMBER-RANGE	error	"Member validity must be inside the team's effective range."	Out-of-range dates	Adjust dates	BR-6.12
ORG-TEAM-EXPIRED	error	"Team {code} has ended."	Member added to ended team	Extend the team	BR-6.12
ORG-CUR-ROUNDING-UNIT	error	"Rounding unit {u} is not allowed for {currency}."	Unit not a permitted multiple	Use a permitted unit	BR-6.13
ORG-VERSION-CONFLICT	warning	"This record changed since you opened it."	Stale `version`	Merge or reload	BR-6.10

1.6.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
As-of hierarchy query	2M/day; peak 3,000/min	4 / 20 / 40 ms	500 req/s per replica	closure ≈ 400k rows per 50k-node version, ~30 MB/yr	PK (ancestor_id, descendant_id) per version; partial index on published	served from current closure; p95 ≤ 40 ms; exports queued first
Legal-entity switch (shell re-scope)	20k/day; peak 200/min	80 / 300 / 500 ms	50 req/s	none	OPA decision cache per (user, entity) 60 s	cache hit ratio keeps p95 ≤ 600 ms
Hierarchy publish (50k nodes)	20/day	8 / 30 / 45 s	1 concurrent per hierarchy type	~5 MB per version	Temporal activity; bulk insert closure	queued FIFO per type; no timeout
UoM conversion	5M/day; peak 10k/min	1 / 5 / 10 ms	2,000 req/s per replica	negligible	in-memory conversion graph, invalidated by event	unchanged (CPU-bound, cached)
Address create with validation	50k/day; peak 500/min	20 / 80 / 150 ms (adapter excluded)	100 req/s	~200 MB per 1M addresses	GIN trigram on `formatted`; btree (owner_type, owner_id)	adapter calls skipped, status `unverified`
Fiscal year generation	100/yr	0.5 / 2 / 5 s	n/a	< 1 MB	(calendar_id, fiscal_year, period_no) unique	unchanged
Organization export package	10/day	2 / 10 / 20 s	2 concurrent per tenant	1–20 MB per package	n/a	requests beyond 2 queued

1.6.20 Localization and compliance hooks

Item	What varies	Country packs touching it (wave 48)	Extension point
Address format and postal validation	Line order, region requirement, postal regex (US ZIP+4, UK postcode, CA A1A 1A1, IN 6-digit PIN, MX 5-digit CP)	US, UK, DE, FR, IN, MX, CA in this wave; others via pack	`AddressFormatProvider` registry keyed by country
Legal entity registration identifiers	DE Handelsregister (HRB + court), FR SIREN/SIRET + NAF, IN CIN + PAN + TAN, MX RFC + régimen fiscal, UK Companies House number, US EIN + state of incorporation, CA BN	all seven	typed `legal_entity_registration` list contributed by pack metadata
Fiscal year conventions	IN April–March statutory year; UK companies any year end; MX statutory calendar year (CFF Art. 11); US 52/53-week allowed (IRC §441(f))	IN, MX, US	calendar templates shipped by pack; FR-6.7 generator
Currency and cash rounding	CHF/SEK cash rounding; CAD penny elimination (cash to 0.05); MXN 2 decimals; INR 2 decimals, invoice totals often to 1.00	CA, IN	`org.cur.rounding` rows seeded by pack

Item	What varies	Country packs touching it (wave 48)	Extension point
Holidays and time zones	Bank holidays for due-date shifting (wave 19), DST rules	all	IANA zones; holiday calendars provided as pack data, consumed by FR-9.6 and wave 19
Language of documents	FR Loi Toubon (French on documents for French counterparties); CA Quebec Charter of the French language (Bill 96)	FR, CA	FR-6.17 translations; document language on legal entity
Privacy	Home addresses of persons are personal data (GDPR, UK GDPR, CCPA/CPRA, India DPDP Act 2023, Mexico LFPDPPP, PIPEDA)	all	`home` purpose flagged PII; masking per BR-7.9
Retention	Organization master and calendars kept for tenant life + 10 years (DE HGB §257 10 y, FR 10 y, US IRS 7 y, IN Companies Act §128 8 y)	all	retention class `org-config` in wave-5 audit store

1.6.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.12.1 Human resources, Common — Teams (Org admin: Teams, Team types), OMTeam; register §12.99.4 #107	PARTIAL-6	ADDED as FR-6.14	Team, team type, members, membership criteria; HR uses the same entity
AX §12.12.4 Human resources, Setup — Teams → Team types, Teams, OMTeam; register §12.99.4 #114	PARTIAL-6	ADDED as FR-6.14	Setup screen is the Teams list with type filter (§1.6.17 duty "Maintain operating units and teams")
AX §12.29.1 Organization administration, Common — Teams / Team types, OMTeam; register §12.99.4 #232	PARTIAL-6	ADDED as FR-6.14	Inquiry "Teams and members" in §1.6.14
AX §12.29.4 Organization administration, Setup — Organization → Teams / Team types; register §12.99.4 #251	PARTIAL-6	ADDED as FR-6.14	Rule evaluation job ORG-J5; parameter `org.team.rule_evaluation`

1.7 Wave 7: Global Party & Business Partner Model

1.7.1 Scope and business context

AX keeps one `DirPartyTable` (subtyped into `DirPerson` and `DirOrganization`) under the global address book, with `CustTable` and `VendTable` as company-specific roles pointing at the party; SAP ECC keeps separate `KNA1`/`KNB1` (customer general/company-code) and `LFA1`/`LFB1` (vendor) masters, and S/4 forces them into Business Partner (`BUT000`, roles in `BUT100`, addresses in `BUT020`/`ADRC`) through Customer-Vendor Integration. Wave 7 builds the party model once: one Party per real-world organization or person, with Customer, Vendor, Employee, Contact, Prospect and Bank roles layered per legal entity. At the end of the wave a steward can create and govern parties and roles, run duplicate detection and merge, validate tax registrations, and import a legacy CustTable/VendTable extract. Posting profiles, payment terms and credit limits are captured now as data; the GL hooks that use them arrive in waves 11, 19 and 30.

Out of scope: customer/vendor transactions and balances, bank account verification (wave 36), employee HR attributes beyond the role stub (wave 42), CRM pipeline (non-goal).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
In ECC the same supplier who is also a customer exists as LFA1 and KNA1 with two addresses that drift; S/4 CVI (`CVI_FS_CHECK_CUSTOMIZING`, `MDS_LOAD_COCKPIT`) is a months-long synchronization project	Two masters designed independently, linked by KNA1-LIFNR / LFA1-KUNNR cross references	Party is the only identity; roles reference it; addresses are shared by purpose (FR-7.1, FR-7.2)	Duplicate identities per real-world entity 2+→1; address drift incidents 0
AX duplicate check on DirPartyTable is exact-name only (`DirPartyTable.checkDuplicate`); near-duplicates ("ACME Corp." vs "Acme Corporation") pass and are found at year-end	No probabilistic matching	Fellegi-Sunter model (Splink) on name/tax id/address/phone with blocking (FR-7.5)	Planted near-duplicates caught: recall ≥ 0.90 vs ≈0.30 for exact match
Merging two vendors in AX is manual: re-point every open PO, settlement, and address, or leave one blocked forever; SAP has no merge, only block (XK05)	No merge primitive; FKs scattered across modules	Merge workflow rewrites references atomically through a consumer contract and keeps full role data (FR-7.6)	Merge effort 2–4 h→<1 min; dangling references 0
VAT number entry (CustTable.VATNum / KNA1-STCEG) is free text; wrong numbers surface as EU sales list rejections months later	No validation at entry	Country-specific format checks plus VIES/TIN adapters at save (FR-7.4)	Invalid tax ids at save-time rejected 100%; downstream filing rejections →0
Creating a customer in AX touches CustTable form, address tab, contact info, posting profile, payment terms: 12+ clicks across 4 tabs; SAP XD01 walks 3 views × 5 tabs	Role and party data mixed in one wide form	One create flow with inline role panels and defaults from customer group (FR-7.3, FR-7.7)	Clicks 12→3; time-to-create 4 min→30 s

1.7.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Party	DirPartyTable, DirPerson, DirOrganization, DirPartyTable form (Global address book)	BUT000 (BP), BUT100 roles; ECC KNA1/LFA1 as separate masters	party-service `Party` with `party_type` person/organization	full
Customer role	CustTable (AccountNum, CustGroup, PaymentTermId, CreditMax, InvoiceAccount), CustGroup, Customers list page	KNA1 (KUNNR), KNB1 (company code), KNVV (sales area), XD01/FD01	`CustomerRole` (party_id, legal_entity_id)	full (sales-area data KNVV arrives in wave 26)
Vendor role	VendTable (AccountNum, VendGroup, PaymentTermId), VendGroup, Vendors list page	LFA1, LFB1, LFM1 (purchasing org), XK01/FK01	`VendorRole`	full (LFM1 in wave 17)
Contacts / persons	ContactPerson, DirPersonName (first/middle/last), DirPersonUser	KNVK contact person, BP person category	`ContactRole`, `PersonName`	full
Employee / worker	HcmWorker, DirPerson link	PA0001/PA0002, BP employee role	`EmployeeRole` stub (personnel number, legal entity)	partial (HR data in wave 42)
Addresses and contact info	DirPartyLocation, LogisticsPostalAddress, LogisticsElectronicAddress (phone/email/URL)	ADRC, ADR2 (phone), ADR6 (email), BUT020	`PartyAddress` (wave-6 address + purposes), `ElectronicAddress`	full
Tax registrations	TaxRegistration, TaxRegistrationType, CustTable.VATNum, VendTable.VATNum	KNA1-STCEG/STCD1..5, LFA1-STCEG, DFKKBPTAXNUM (BP tax numbers)	`TaxRegistration` (type, country, number, validated_at, result)	improved (validation adapters)

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Duplicate detection / merge	DirPartyTable duplicate check (exact), Merge parties (D365 "Merge" on DirPartyTable, limited)	None in ECC; S/4 MDG-BP matching (separate license)	`DuplicateCandidate`, `MergeCase`, Temporal merge workflow	improved
Groups, posting profiles, terms	CustGroup/VendGroup, CustLedger/VendLedger posting profiles, PaymentTerm, CustPaymMode, DlvTerm, DlvMode	KNB1-AKONT reconciliation account, T052 payment terms (ZTERM), TINC Incoterms, KNVV-VSBED	`CustomerGroup`, `VendorGroup`, `PostingProfile` (account fields filled in wave 11), `PaymentTerm`, `DeliveryTerm`, `DeliveryMode`	full (GL accounts bound in wave 11)
Credit management	CustTable.CreditMax, CreditRating, blocked status (CustBlocked)	KNKK credit limit (FD32), UKM in S/4	`CreditProfile` (limit, currency, rating, hold)	partial (credit checks run in wave 27)
Bank role	BankAccountTable per customer/vendor (CustBankAccount/VendBankAccount)	BP bank details BUT0BK, LFBK/KNBK	`BankRole` plus `PartyBankAccount` (IBAN/BIC format check only)	partial (verification in wave 36)
Governance	Workflow on vendor changes (VendTable change proposals in D365), DirPartyTable data owner	MDG change requests, SAP workflow	Stewardship queue, change-approval workflow on protected fields	improved
Vendor certifications and diversity	VendCertification, VendCertificationType (Vendors → Certification types), Diversity classifications (D365 procurement)	No ECC master field; SAP SLC/Ariba SLP supplier certificates; BP attributes	`CertificationType`, `VendorCertification`, `DiversityClassification` (FR-7.14)	full
Installment / payment schedules	PaymSched, PaymSchedLine (Payment schedules form)	T052R installment terms (OBB9) linked to T052 ZTERM	`PaymentScheduleLine` on `PaymentTerm` (FR-7.15)	full
One-time accounts	One-time customer/supplier (AR/AP parameters → One-time customer account / One-time supplier account; CustTable.OneTimeCustomer)	Account groups CPD/CPDL (T077D/T077K XCPDK), document data in BSEC	`one_time` role flag + document-level address (FR-7.16)	full
Head office, alternate payer, permitted payee	CustTable/VendTable.InvoiceAccount, DirPartyRelationship	KNA1-KNRZE alternate payer, KNB1-KNRZB head office (branch/head office), LFA1-LNRZA, LFZA permitted payees	`PartyRelationship` (FR-7.17)	full
Address books	DirAddressBook, DirAddressBookParty, address book team security (Global address book parameters)	BP authorization group BU_BEGRU, BP grouping	`AddressBook` with team access (FR-7.18)	full
Customer hierarchies	TAMCustomerCategory hierarchy (D365 trade allowance)	KNVH customer hierarchy (VDH1N/VDH2N), hierarchy types OVH1	hierarchy purpose `customer_*` over Customer roles (FR-7.19)	improved (effective-dated versions)
Classification code lists and hold reasons	CustClassificationGroup, LineOfBusiness, smmSegment/smmSubsegment, SalesDistrict, Vendor/Customer hold reasons	KNA1-BRSCH industry, KNA1-KUKLA classification, KNVV-KDGRP, KNVV-BZIRK, block reasons	`ClassificationCode` lists (FR-7.20)	full

1.7.3 Functional requirements

- **FR-7.1 Party core** — Must. The system shall create parties of type organization (legal name, trade name, registration number, industry code NAICS/NACE) or person (given, middle, family name, name prefix/suffix, date of birth optional, gender optional) with a tenant-wide party number from a number sequence.
  - Name search uses trigram similarity on `search_name` (normalized: casefold, diacritics stripped, punctuation removed).
  - Party `status` ∈ {active, inactive, merged}; merged parties keep a `merged_into_id` and all reads redirect.
- **FR-7.2 Roles per legal entity** — Must. The system shall attach Customer, Vendor, Contact, Employee, Prospect and Bank roles to a party; Customer and Vendor roles are keyed by (party_id, legal_entity_id) and carry an account number from the legal entity's customer/vendor number sequence.
  - A party may hold Customer and Vendor roles in the same legal entity (the AX "one party, two accounts" case).
  - Role deactivation keeps the row and sets `blocked = all` with a reason.
- **FR-7.3 Group defaults** — Must. The system shall maintain customer groups and vendor groups carrying default payment term, delivery term, delivery mode, posting profile, price/discount group placeholders and tax group placeholders; creating a role copies group defaults into the role and records `defaulted_from_group`.
- **FR-7.4 Tax registrations with validation** — Must. The system shall store tax registration numbers by type (VAT, EIN, SSN-masked, GSTIN, ABN, CNPJ, RFC, UTR) with per-country format validation (regex plus check digit where defined: GSTIN mod-36, ABN mod-89, CNPJ/RFC algorithms, UK VAT mod-97) and optional online validation through adapters.
  - VIES adapter calls `checkVatService` with a 5 s timeout; result cached 30 days; outage yields `validation_status = unverified`, never a save failure.
  - IRS TIN matching adapter is an interface with a recorded fixture implementation in v1.
- **FR-7.5 Duplicate detection** — Must. The system shall score candidate pairs on every create/update and on demand, using blocking on (normalized name first 4 chars | tax id | postal code+street number | phone last 7 | email domain) and a Fellegi-Sunter model with comparison levels for name (Jaro-Winkler ≥ 0.92 / ≥ 0.85 / else), tax id (exact), address (street+postal exact / postal only / else), phone, email.
  - Match weight ≥ 12 (probability ≈ 0.99) auto-flags as `likely_duplicate`; 6–12 flags `possible`; below 6 ignored.
  - Create with a `likely_duplicate` candidate returns 409 with the candidates unless `X-Passage-Override-Duplicate: true` and the duty "Override duplicate check".
  - Scoring of a single party against 1M parties completes in < 300 ms using blocking indexes.
- **FR-7.6 Merge workflow and survivorship** — Must. The system shall merge N parties into one survivor through a Temporal workflow: pick survivor, resolve field conflicts by survivorship rules (most-recently-verified tax id, most-recently-updated address, non-empty wins, manual override), union roles (conflicting roles in the same legal entity require explicit choice), then rewrite references in every consuming service via `party.merged.v1` with acknowledgement tracking.
  - Merge is reversible for 30 days through a stored pre-merge snapshot.
  - The workflow does not complete until every registered consumer acknowledges rewrite; stragglers alert after 10 min.
- **FR-7.7 Electronic addresses** — Must. The system shall store phone (E.164 normalized), email (RFC 5322 syntax check), URL, fax, with purposes and a primary flag per type.
- **FR-7.8 Credit profile** — Should. The system shall store credit limit with currency, credit rating, hold flag, and hold reason on the Customer role; changes to limit are protected fields under FR-7.10.
- **FR-7.9 Payment, delivery terms and modes** — Must. The system shall maintain payment terms (net days, day-of-month, cash discount percent and days, up to 3 discount tiers), delivery terms (Incoterms 2020 list seeded), and delivery modes; terms compute due dates deterministically (`due = invoice_date + net_days`, or next day-of-month ≥ invoice_date + offset_months).
- **FR-7.10 Change-approval workflow on protected fields** — Must. The system shall route changes to protected fields (bank account, tax registration, credit limit, payment term on vendor, remit-to address) through an approval workflow configured per legal entity; the change is held in `pending_change` and not visible to consumers until approved.
  - SLA 2 business days then escalation; rejection records a reason.
- **FR-7.11 Stewardship queue and ownership** — Should. The system shall assign a data owner (user or team) per party and show stewards a queue of duplicates, unverified tax ids, pending changes and incomplete records (missing primary address or primary email), sorted by age.

- **FR-7.12 Legacy import** — Must. The system shall import CustTable/VendTable/DirPartyTable and KNA1/LFA1 extracts (CSV) through the wave-5 import tool with the wave-10 mapping specs pre-seeded; duplicates found during import go to the stewardship queue rather than blocking the load.
- **FR-7.13 Party search API** — Must. The system shall expose typeahead search returning up to 20 parties in < 100 ms p95 over 1M rows, filtered by role and legal entity.
- **FR-7.14 Vendor certifications and diversity classifications** — Should. The system shall maintain certification types (code, name, issuing body, `requires_expiry`, `reminder_days` default 30, `attachment_required`) and vendor certifications (vendor role, type, certificate number, issuer, `valid_from`/`valid_to`, document id from FR-8.20, status) plus diversity classifications per vendor (seeded: SB, WOSB, EDWOSB, MBE, VOSB, SDVOSB, HUBZone, 8(a), LGBTBE, DOBE) each optionally linked to the certification that evidences it.
  - Vendor groups may flag certification types as mandatory; an expired mandatory certification sets vendor hold `requisition` through job PTY-J6 and notifies the data owner.
  - Endpoints: `POST /party/v1/vendors/{roleId}/certifications`, `GET /party/v1/certifications?status=expiring&days=30`.
- **FR-7.15 Installment payment schedules** — Must. The system shall attach payment schedules to payment terms with 1–24 lines, each line carrying a percent or fixed amount, its own net-days or day-of-month rule and an optional cash discount; allocation modes mirror AX PaymSched: `percent`, `fixed_amount`, and `count` (N equal installments every M days or months).
  - `GET /party/v1/payment-terms/{id}/due-date` returns one (due_date, amount, discount_date, discount_amount) tuple per installment when `amount` is supplied.
  - SAP T052R rows (ZTERM, RATNR, RATPZ, RATZT) map one-to-one to schedule lines.
- **FR-7.16 One-time customers and vendors** — Must. The system shall provide one or more one-time Customer and Vendor roles per legal entity (role flag `one_time = true`, from a group with `one_time_default = true`) whose payee/sold-to name, address, tax id and bank details are captured on each document instead of on the role.
  - Consumers (purchasing 17, AP 19, sales 26, AR 30) store the captured data on the document header, the BSEC equivalent; `POST /party/v1/one-time-data:validate` runs FR-6.6 address and FR-7.4 tax-id validation and returns normalized values.
  - One-time roles are excluded from duplicate scoring; a document counterparty name/tax id used on ≥ 3 documents in 12 months raises a "convert to regular account" stewardship item (job PTY-J7).
- **FR-7.17 Party relationships: head office, alternate payer, permitted payee** — Must. The system shall maintain typed, effective-dated relationships between parties or roles: `invoice_account`, `head_office_of` / `branch_of`, `alternate_payer`, `permitted_payee`, `contact_for`, `parent_company` / `subsidiary_of`, plus tenant-defined types (AX DirRelationshipType).
  - `head_office_of` carries `decentralized_processing` (KNB1-XDEZV equivalent): when false, AR/AP open items of the branch post to the head-office account in waves 19 and 30.
  - Payment proposals (wave 19) pay only to the role's own bank accounts or to active permitted payees; adding a permitted payee is a protected-field change (FR-7.10).
  - `head_office_of` chains are acyclic and at most 5 levels deep (`PTY-REL-CYCLE`).
- **FR-7.18 Address books** — Should. The system shall maintain named address books (e.g. "US vendors", "EMEA customers") that group parties; a party may belong to many books; books carry team access lists (FR-6.14) enforced by OPA on party reads when the tenant enables `party.address_book_security`; each user has a default book applied to parties they create.
- **FR-7.19 Customer hierarchies** — Should. The system shall build effective-dated customer hierarchies with purposes `customer_pricing`, `customer_rebate` and `customer_reporting` whose nodes are Customer roles, reusing the wave-6 hierarchy engine (versions, closure table, as-of API); pricing (wave 26) and rebates (wave 29) resolve ancestors as of the document date in < 20 ms p95.
- **FR-7.20 Classification code lists and hold reasons** — Must. The system shall maintain tenant code lists for line of business, segment and subsegment, sales district, customer classification group, industry (NAICS 2022 / NACE Rev. 2.1 seeded), customer hold reasons and vendor hold reasons; role fields reference them, and every block or unblock records reason code, user, timestamp and free text.
  - Vendor `blocked` values are extended to the AX set: none, invoice, payment, requisition, all, never (never = AX "Never", exempt from inactivity blocking job PTY-J9).

1.7.4 Data model

Party

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS
party_number	text(20)	yes	number sequence `PARTY`, unique per tenant
party_type	enum	yes	organization, person
name	text(200)	yes	display name; computed for persons
search_name	text(200)	yes	normalized; trigram index
org.legal_name / org.trade_name / org.registration_number / org.industry_code	text	legal_name yes	organization subtype
person.given_name / middle_name / family_name / prefix / suffix / dob	text, date	given+family yes	person subtype; PII-flagged
status	enum	yes	active, inactive, merged
merged_into_id	uuid	no	set when status = merged
owner_user_id / owner_team_id	uuid	no	stewardship
language_code / time_zone	text	no	
version, created_at, updated_at	int, timestamptz	yes	

CustomerRole (VendorRole is symmetric with `vendor_group_id`, `vendor_hold`)

Field	Type	Required	Notes
id	uuid	yes	PK
party_id	uuid	yes	FK party
legal_entity_id	uuid	yes	FK; unique (party_id, legal_entity_id)
account_number	text(20)	yes	sequence `CUST-{LE}`; unique per legal entity
customer_group_id	uuid	yes	FK
payment_term_id / delivery_term_id / delivery_mode_id	uuid	payment yes	FK
posting_profile_id	uuid	yes	FK; GL accounts attached in wave 11

Field	Type	Required	Notes
invoice_account_party_id	uuid	no	bill-to party (AX InvoiceAccount)
currency_code	char(3)	yes	default accounting currency
credit_limit / credit_currency / credit_rating / credit_hold	numeric(18,2), char(3), text, bool	no	protected fields
blocked	enum	yes	none, invoice, all
sales_tax_group_id	uuid	no	bound in wave 13
version	int	yes	

TaxRegistration

Field	Type	Required	Notes
id	uuid	yes	PK
party_id	uuid	yes	FK
registration_type	enum	yes	VAT, EIN, GSTIN, ABN, CNPJ, RFC, UTR, OTHER
country_code	char(2)	yes	
number	text(30)	yes	stored normalized (no spaces); unique (type, country, number) per tenant
valid_from / valid_to	date	no	
validation_status	enum	yes	unverified, valid, invalid, format_only
validated_at / validation_source	timestamptz, text	no	VIES, IRS_TIN, manual
is_primary	bool	yes	one per (party, country)

DuplicateCandidate / MergeCase — candidate: id, party_a_id, party_b_id (a < b), match_weight numeric(8,4), probability, level_detail jsonb, status (open, dismissed, merged), created_at; merge_case: id, survivor_id, merged_ids uuid[], survivorship jsonb, snapshot jsonb, workflow_id, status (draft, running, awaiting_ack, completed, reverted, failed), consumer_acks jsonb.

Invariants: a party with status `merged` has no active roles; `account_number` never reused after merge (merged role numbers are retained as aliases in `customer_alias`); at most one primary tax registration per (party, country); every role has a posting profile. Indexing: `party` GIN trigram on `search_name`; btree on `(tenant_id, party_type, status)`; blocking indexes on `left(search_name,4)`, `tax_registration.number`, `electronic_address.normalized`; `duplicate_candidate(status, match_weight desc)`.

VendorCertification / PaymentScheduleLine / PartyRelationship / AddressBook (FR-7.14 – FR-7.18) — vendor_certification: id, vendor_role_id FK, certification_type_id FK, number text(60), issuer text(100), valid_from date, valid_to date (required when type requires_expiry), document_id uuid null, status (valid, expiring, expired, revoked), revoked_reason text; vendor_diversity: vendor_role_id, classification_code, certification_id null, valid_from, valid_to. payment_schedule_line: payment_term_id FK, line_no int (1–24), allocation (percent, fixed_amount, count), value numeric(18,6), net_days int, day_of_month int null, offset_months int, discount_pct numeric(5,2) null, discount_days int null. party_relationship: id, type_code FK relationship_type, from_party_id, from_role_id null, to_party_id, to_role_id null, legal_entity_id null, valid_from, valid_to, attributes jsonb (e.g. decentralized_processing); unique (type_code, from_role_id, to_role_id, valid_from). address_book: id, code text(20), name, team_ids uuid[]; address_book_party (address_book_id, party_id) PK.

1.7.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/party/v1/parties	Create party (runs dedup)	`Idempotency-Key`	Maintain parties
GET	/party/v1/parties/{id}	Read (follows merged_into)	n/a	View parties
PATCH	/party/v1/parties/{id}	Update; protected fields → pending change	version check	Maintain parties
GET	/party/v1/parties:search?q=&role=customer&legalEntity=	Typeahead	n/a	View parties
POST	/party/v1/parties/{id}/roles/customer	Add customer role in `X-Passage-LegalEntity`	key header	Maintain customers
POST	/party/v1/parties/{id}/roles/vendor	Add vendor role	key header	Maintain vendors
POST	/party/v1/parties/{id}/tax-registrations	Add and validate	key header	Maintain parties
POST	/party/v1/tax-registrations/{id}:validate	Re-run adapter	idempotent (cached 30 d)	Maintain parties
GET	/party/v1/duplicates?status=open&minWeight=6	Stewardship list	n/a	Steward master data
POST	/party/v1/merge-cases	Start merge workflow	key header	Merge parties
POST	/party/v1/merge-cases/{id}:revert	Revert within 30 days	key header	Merge parties
POST	/party/v1/pending-changes/{id}:approve / :reject	Workflow decisions	key header	Approve master data changes
GET	/party/v1/payment-terms/{id}/due-date?invoiceDate=2026-03-31	Compute due date	n/a	any authenticated

Validation: tax registration format per country before adapter call; role creation requires the legal entity header and an existing group; merge requires all parties active and in the same tenant; search `q` minimum 2 characters.

Event	Trigger	Payload fields	Consumers
dev.passage.party.created.v1	commit	party_id, party_number, party_type, name, primary_address_id	search index, migration, audit
dev.passage.party.updated.v1	commit of non-protected change	party_id, changed_fields[], version	search, consumers caching names
dev.passage.party.customer.created.v1	role commit	party_id, legal_entity_id, account_number, customer_group_id, payment_term_id, posting_profile_id, currency_code	AR, sales, pricing, credit
dev.passage.party.vendor.created.v1	role commit	party_id, legal_entity_id, account_number, vendor_group_id, payment_term_id	AP, purchasing
dev.passage.party.duplicate.flagged.v1	score ≥ 6	candidate_id, party_a_id, party_b_id, match_weight	stewardship queue, notification
dev.passage.party.merged.v1	merge workflow enters awaiting_ack	survivor_id, merged_ids[], role_map{old_account→new_account}, merge_case_id	every service with party FKs; each replies on `dev.passage.party.merge.ack.v1`
dev.passage.party.change.approved.v1 / rejected.v1	workflow decision	party_id, field, old, new, approver_id, reason	audit, notification
dev.passage.party.taxregistration.validated.v1	adapter result	registration_id, status, source, validated_at	tax (wave 13), audit

1.7.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Global address book (parties list)	New party, search, filter by role/entity, open duplicates	party number, name, type, roles, primary address, status	View parties
Party detail	Edit, add role panels (customer/vendor/contact), addresses, electronic addresses, tax registrations, relationships, audit tab	all party fields; role panels collapsed by default	Maintain parties
Customers workspace (per legal entity)	New customer (party picker or inline create), block, edit terms, credit tab	account, name, group, terms, credit limit, hold, blocked	Maintain customers
Vendors workspace	New vendor, block, bank accounts (pending-change aware)	account, name, group, terms, bank verified flag	Maintain vendors
Duplicate review	Side-by-side compare, dismiss, start merge, pick survivor per field	match weight, field-level comparison levels	Steward master data
Merge case monitor	Progress, consumer acks, revert	consumers acked / pending, elapsed	Merge parties
Stewardship queue	Claim, resolve, bulk dismiss	item type, party, age, owner	Steward master data
Pending changes (approvals)	Approve/reject with reason	field, old/new, requester, SLA remaining	Approve master data changes

Interaction rules: the create-party dialog runs dedup after name and country are entered (debounced 400 ms) and shows candidates inline before save; `Ctrl+Enter` saves and opens the role panel; bulk block/unblock from the list is limited to 500 rows per action; protected fields render with a lock badge and show "pending" with the proposed value until approval; personalization of role panels (which are expanded) is per user.

1.7.7 Business rules and edge cases

- **BR-7.1** Party status machine: `active ↔ inactive`; `active → merged` only through a merge case; `merged` is terminal except via revert within 30 days, which restores the snapshot and sets `active`.
- **BR-7.2** Merge case machine: `draft → running → awaiting_ack → completed`; `running` or `awaiting_ack → failed` on activity error after 5 retries; `completed → reverted` within 30 days only when no consumer reports a post-merge transaction referencing the survivor that cannot be split (consumer rejects revert with reason).
- **BR-7.3** Survivorship default order per field: manual override > most recently validated (tax ids) > most recently updated > non-empty > survivor's value.
- **BR-7.4** Account numbers of merged roles become aliases; lookups by alias resolve to the survivor role and are logged for 12 months to catch stale integrations.
- **BR-7.5** Dedup weights are recomputed for an updated party only against candidates from its blocking keys; a full re-score job runs weekly (Temporal cron, Sunday 01:00 tenant local).
- **BR-7.6** Tax registration numbers are normalized before uniqueness checks (strip spaces, hyphens, dots; uppercase); two parties with the same VAT number in the same country are auto-flagged with weight +15.
- **BR-7.7** Payment term due date: `net_days` adds calendar days; `day_of_month` moves to that day in the month `offset_months` ahead, clamped to the last day; cash discount date computed the same way per tier; if discount date > due date the term is invalid at save.
- **BR-7.8** Protected-field changes by a user holding "Approve master data changes" still require a second approver (no self-approval).
- **BR-7.9** A person party's `dob` and government ids are PII-flagged: masked in lists, visible in detail only to duties with `pii.read`, excluded from search index and from CloudEvent payloads.
- **BR-7.10** Deleting a party is never allowed; `inactive` parties cannot receive new roles; roles on an `inactive` party cannot be unblocked.
- **BR-7.11** Concurrent merges touching the same party are serialized by a Temporal workflow id `merge-{tenant}-{min(party_ids)}`; the second request receives 409 with the running case id.
- **BR-7.12** Import rows with a `likely_duplicate` candidate are loaded with status `active` and a candidate record; they are not rejected, because migrations must be complete and reviewed afterward.
- **BR-7.13** Installment amounts: `amount_i = round(total × pct_i / 100)` using the currency's general rounding (FR-6.16); the last installment absorbs the remainder (`total − Σ previous`); percent lines must sum to exactly 100.00 (`PTY-SCHED-PCT-SUM`); each due date applies BR-7.7 to the baseline (invoice) date.
- **BR-7.14** Certification status is computed daily: `valid` while today < `valid_to − reminder_days`; `expiring` inside the window; `expired` after `valid_to`; `revoked` only manually with reason; each change emits `dev.passage.party.certification.status.changed.v1`.
- **BR-7.15** One-time roles cannot carry credit limits, bank accounts, tax registrations or posting-profile overrides on the role (`PTY-ONETIME-FIELD`); they cannot be merged as survivors.
- **BR-7.16** `head_office_of` requires both ends to hold the same role type (Customer–Customer or Vendor–Vendor) in the same legal entity; changing the head office while the branch has open items is refused through a wave-30/19 callback (returns "no open items" until those waves exist).
- **BR-7.17** A permitted payee must be an active party with at least one verified bank account; removal takes effect immediately, addition only after approval.

1.7.8 Acceptance criteria

- **AC-7.1** (FR-7.5) Given 10,000 seeded parties with 500 planted near-duplicates (name typos, swapped address lines, formatted tax ids), when the full re-score job runs, then flagged pairs with weight ≥ 6 achieve precision ≥ 0.95 and recall ≥ 0.90 against the labels.
- **AC-7.2** (FR-7.6) Given parties A and B each with a Customer role in USMF and B with a Vendor role in DEMF, when merged into A, then A holds all three roles, B's account numbers are aliases, every registered consumer acks, and the referential-integrity audit reports 0 rows referencing B.
- **AC-7.3** (FR-7.4) Given the recorded VIES fixture of 200 VAT numbers, when validated, then results equal the fixture; given the adapter stub returns a timeout, then the registration saves with `unverified` and an alert is raised after 3 consecutive timeouts.
- **AC-7.4** (FR-7.10) Given a customer with credit limit USD 50,000, when a user raises it to USD 120,000, then the stored limit stays 50,000, a pending change exists, the Temporal workflow has an open approval task, and after approval the limit is 120,000 with an `approved` event.
- **AC-7.5** (FR-7.2) Given a party with a Customer role in USMF, when a Vendor role is created in USMF, then both exist with distinct account numbers from the `CUST-USMF` and `VEND-USMF` sequences.
- **AC-7.6** (FR-7.9) Given payment term "Net 30 EOM+1" (day_of_month 31, offset 1) and invoice date 2026-01-31, when due date is computed, then it equals 2026-02-28; with cash discount 2% at 10 days, discount date is 2026-02-10.
- **AC-7.7** (FR-7.5) Given a `likely_duplicate` candidate at create, when the client retries with the override header but without the override duty, then 403.
- **AC-7.8** (FR-7.12) Given a 1,000-row CustTable extract with 20 duplicates of existing parties, when imported, then 1,000 roles exist, 20 duplicate candidates are open, and 0 rows are rejected for duplication.
- **AC-7.9** (FR-7.13) Given 1M parties, when typeahead `q=acm` is called 1,000 times, then p95 < 100 ms and results are ordered by similarity then name.
- **AC-7.10** (BR-7.9) Given a person with `dob`, when the list page is rendered by a user without `pii.read`, then the column shows `••••` and the CloudEvent payload has no `dob` key.
- **AC-7.11** (FR-7.15, BR-7.13) Given term "3 installments 40/30/30 at net 30/60/90" and an invoice of USD 1,000.01 dated 2026-03-01, when due dates are computed, then the amounts are 400.00 / 300.00 / 300.01 and the due dates 2026-03-31 / 2026-04-30 / 2026-05-30.

- **AC-7.12** (FR-7.14, BR-7.14) Given a vendor in group "Regulated" whose mandatory ISO 13485 certificate expires 2026-11-15 with 30 reminder days, when PTY-J6 runs on 2026-10-16, then status is `expiring` and the owner is notified; on 2026-11-16 status is `expired` and the vendor hold is `requisition`.
- **AC-7.13** (FR-7.16, BR-7.15) Given one-time vendor OTV-USMF, when a party with an identical name is created, then no duplicate candidate references OTV-USMF; when a credit limit is set on OTV-USMF, then 422 `PTY-ONETIME-FIELD`.
- **AC-7.14** (FR-7.17) Given branch B whose head office is H, when H is made a branch of B, then 422 `PTY-REL-CYCLE`; given permitted payee P added to vendor V, then P is not payable until the pending change is approved.

1.7.9 Test plan and fixtures

Fixtures: `party/parties-10k.jsonl` (generated with Faker seed 7, 500 labeled duplicate pairs in `party/dup-labels.csv`); `party/vies-recorded.json` (200 numbers, 150 valid, 50 invalid, recorded 2026-09); `party/tax-format-vectors.csv` (300 valid/invalid across 8 types); `party/custtable-extract.csv` and `party/kna1-extract.csv` (1,000 rows each, AX/SAP column names verbatim); `party/payment-terms.csv` (25 terms with 100 invoice-date → due-date oracle rows).

Test	Type	Target	Pass criterion
Dedup precision/recall	property	Splink model	P ≥ 0.95, R ≥ 0.90
Single-party scoring latency	performance	scorer	< 300 ms at 1M
Merge integrity	E2E (Temporal test server)	merge workflow	0 dangling FKs, all acks
Merge revert	E2E	revert	snapshot restored byte-equal
Tax format vectors	table-driven	validators	300/300
VIES adapter	contract (recorded)	adapter	matches fixture; timeout path
Change approval	Temporal workflow test	protected fields	blocked until approve; SLA timer fires after 2 business days
Due-date oracle	unit	payment terms	100/100
PII masking	contract	API + events	no PII fields in list or events
Typeahead	performance	search	p95 < 100 ms

Reconciliation: count(roles) per legal entity in import = source row count; Σ open candidates + dismissed + merged = total pairs ever flagged; every merged party has exactly one merge case.

1.7.10 Deployment and operations

Helm chart `passage-party` (service ×2, CloudNativePG `party-db`, Debezium connector, Splink scoring worker as a Python Deployment with 2 replicas, Temporal namespace `party`). Config: `PARTY_DEDUP_LIKELY_WEIGHT=12`, `PARTY_DEDUP_POSSIBLE_WEIGHT=6`, `PARTY_VIES_TIMEOUT_MS=5000`, `PARTY_MERGE_ACK_TIMEOUT_MIN=10`, `PARTY_APPROVAL_SLA_BUSINESS_DAYS=2`. Migrations additive via Atlas. Feature flags: `party.online_tax_validation`, `party.merge_revert`, `party.pii_masking` (default on). Scheduled jobs: weekly full re-score; daily VIES re-validation of registrations older than 30 days (max 5,000 per run); hourly stale-merge sweeper. SLOs: create p95 < 400 ms including dedup; search p95 < 100 ms; merge completion p95 < 5 min. Alerts: merge awaiting ack > 10 min; VIES failure rate > 20% over 15 min; duplicate queue > 1,000 open. Runbook entries: "Consumer not acking merge" (list registered consumers, `passagectl party merge-status`), "Re-train Splink model" (export labeled pairs, run `party-train`, publish model version). Rollback: previous digest; the Splink model version is pinned in config so a rollback restores scoring behavior.

1.7.11 Migration from AX / SAP

Source	Passage entity	Notes
AX DirPartyTable (PartyNumber, Name, NameAlias, InstanceRelationType, LanguageId)	Party	InstanceRelationType DirPerson/DirOrganization → party_type
AX DirPerson (FirstName, MiddleName, LastName, Gender, BirthDate via DirPersonName)	Party.person	names from DirPersonName valid-time current row
AX DirOrganization (OrgNumber), DirOrganizationName	Party.org	OrgNumber → registration_number
AX CustTable (AccountNum, Party, CustGroup, PaymentTermId, DlvTerm, DlvMode, Currency, CreditMax, CreditRating, Blocked, InvoiceAccount, VATNum, DataAreaId)	CustomerRole, TaxRegistration	DataAreaId → legal_entity; Blocked enum No/Invoice/All → blocked; VATNum → VAT registration with country from primary address
AX VendTable (AccountNum, Party, VendGroup, PaymentTermId, Blocked, VATNum, DataAreaId)	VendorRole	same pattern
AX ContactPerson (ContactPersonId, Party, ContactForParty)	ContactRole	ContactForParty → relationship row
AX DirPartyLocation + LogisticsLocationRole	PartyAddress with purposes	reuses wave-6 address load; IsPrimary → primary flag
AX LogisticsElectronicAddress (Type, Locator)	ElectronicAddress	Type Phone/Email/URL/Fax; phones normalized E.164 using party country
AX CustGroup/VendGroup, PaymentTerm, DlvTerm, DlvMode	groups, terms, modes	PaymentTerm.NumOfDays/PaymMethod (Net/CurrentMonth/…) mapped to net_days/day_of_month
SAP KNA1 (KUNNR, NAME1, LAND1, STCEG, STCD1, SPERR, LOEVM, ADRNR) + KNB1 (BUKRS, AKONT, ZTERM, ZUAWA)	Party, CustomerRole, TaxRegistration	one Party per KUNNR unless matched to an LFA1 by STCEG or ADRNR; BUKRS → legal_entity; ZTERM → payment term by code
SAP LFA1 (LIFNR, NAME1, STCEG, SPERR, SPERM) + LFB1 (BUKRS, AKONT, ZTERM)	Party, VendorRole	KNA1-LIFNR / LFA1-KUNNR cross references pre-merge the party
SAP BUT000 (PARTNER, TYPE, NAME_ORG1, NAME_FIRST, NAME_LAST) + BUT100 (RLTYP FLCU01/FLVN01/BUP001) + DFKKBPTAXNUM	Party, roles, TaxRegistration	roles map FLCU01→customer, FLVN01→vendor, BUP001→contact
SAP ADRC / ADR2 / ADR6 / KNVK	PartyAddress, ElectronicAddress, ContactRole	ADR2 TEL_NUMBER + TELNR_CALL merged
SAP T052 (ZTERM, ZTAGG, ZTAG1..3, ZPRZ1..2), TINC, KNKK (KLIMK, CTLPC)	PaymentTerm, DeliveryTerm, CreditProfile	T052 day-limit variants produce one Passage term per variant

Transformation notes: AX `Party` RecId joins are resolved through DirPartyTable.PartyNumber; SAP customer and vendor numbers are kept as `account_number` when they fit 20 characters, else a new sequence number is assigned and the SAP number stored as alias. Validation: every role must resolve to a legal entity and a group; tax ids failing format go to the stewardship queue, not rejection. Not migrated: AX `DirPartyRelationship` types other than contact-for; SAP sales-area (KNVV) and purchasing-org (LFM1) data (waves 26 and 17); SAP dunning data KNB5 (wave 30); AX `CustTrans`/`VendTrans` open items (transactional packs in wave 49).

1.7.12 Dependencies, risks, open questions

Depends on: wave 6 (legal entities, addresses, countries, number sequences per entity), wave 4 (Temporal approval template, outbox), wave 5 (import tool, audit, PII attributes from wave-3 metadata).

Risk	Likelihood	Impact	Mitigation
Consumers added in later waves forget to implement merge ack	Medium	High	Consumer registry with contract test required for any service subscribing to party events
VIES rate limits or outages block daily re-validation	High	Low	Backoff and cap per run; unverified is a non-blocking state
Splink model drifts for non-Latin names	Medium	Medium	Name normalization with Unicode NFKD; per-tenant threshold override

Open questions: (1) Do Prospect roles need a number sequence or stay numberless until conversion to Customer? Proposed: numberless; decide by wave 26. (2) Is the employee role created here or only in wave 42 with HcmWorker migration? Proposed: stub here (personnel number only); decide by wave 10. (3) Which survivorship rules are tenant-configurable in v1? Proposed: order list editable, levels fixed; decide by wave 10.

1.7.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`party.dedup.mode`	tenant	enum	block_likely	off / warn / block_likely	Create behavior with candidates (FR-7.5)	Global address book parameters → Duplicate check (DirParameters)	BP duplicate check (Business Address Services / MDG matching)
`party.dedup.likely_weight`	tenant	decimal	12	8–20	Likely-duplicate threshold	—	MDG match threshold
`party.dedup.possible_weight`	tenant	decimal	6	3–12	Possible-duplicate threshold	—	—
`party.number_sequence`	tenant	sequence	`PARTY-{########}`	wave-5 sequences	Party numbers (FR-7.1)	Global address book parameters → Number sequences (Party ID)	BP number ranges (BUCF) and groupings (BUC2)
`party.customer.account_sequence`	legal entity	sequence	`CUST-{LE}-{######}`	sequences	Customer account numbers	AR parameters → Number sequences → Customer account	XDN1 number ranges per account group (OBD2)
`party.vendor.account_sequence`	legal entity	sequence	`VEND-{LE}-{######}`	sequences	Vendor account numbers	AP parameters → Number sequences → Vendor account	XKN1 number ranges per account group (OBD3)
`party.person.name_sequence`	tenant / country	enum	given_family	given_family / family_given / family_comma_given	Display name of persons	Name sequences (DirNameSequence)	BP name format rules per country (address layout key)
`party.tax.online_validation`	tenant	bool	true	true / false	Calls VIES/TIN adapters (FR-7.4)	EU VAT number validation (D365 sales tax parameters)	VIES check on BP tax numbers (DFKKBPTAXNUM)
`party.tax.cache_days`	tenant	int	30	1–365	Validation result reuse	—	—
`party.protected_fields`	legal entity	field set	bank account, tax registration, credit limit, vendor payment term, remit-to address, permitted payee	any role field	Fields routed to approval (FR-7.10)	Vendor approval workflow fields (D365 AP parameters → Vendor approval)	T055F sensitive fields (S_ALR_87003378), FK08/FD08
`party.approval.sla_days`	legal entity	int (business days)	2	1–10	Escalation timer	Workflow escalation	—
`party.group.mandatory_fields`	customer/vendor group	field set	primary address; tax id for organizations	role and party fields	Save-time required fields	Customer/vendor group mandatory fields (AR/AP parameters → mandatory fields)	Field status groups per account group (OBD2/OBD3, T077D/T077K)
`party.group.one_time_default`	group	bool	false	true / false	Roles in group are one-time (FR-7.16)	AR/AP parameters → One-time customer/supplier	Account group one-time indicator (XCPDK)
`party.customer.default_group`	legal entity	FK	none	customer groups	Default on create	AR parameters → General → Default customer group	Default account group
`party.credit.check_type`	legal entity	enum	balance	off / balance / balance_incl_open_orders	Credit check basis for wave 27	AR parameters → Credit rating → Credit limit type	OVA8 credit control (FSCM UKM in S/4)
`party.merge.revert_days`	tenant	int	30	0–90	Revert window (FR-7.6)	—	—
`party.cert.reminder_days`	certification type	int	30	0–180	Expiring window (BR-7.14)	Vendor certification types	—
`party.address_book_security`	tenant	bool	false	true / false	Enforce book team access (FR-7.18)	Global address book parameters → Use address book security	BP authorization groups (B_BUPA_GRP)
`party.inactivity_block_days`	legal entity	int	0 (off)	0–1,825	PTY-J9 blocks roles without transactions	Vendor inactivation (D365 AP periodic "Vendor inactivation")	—
`party.pii_masking`	tenant	bool	true	true / false	BR-7.9 masking	Personal data / privacy (D365 GDPR)	ILM blocking / end-of-purpose check

1.7.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Global address book	list inquiry	role, legal entity, type, address book, status	party number, name, type, roles, primary address, status	`party.party`	DirPartyTableListPage	BP search, S_ALR_87012180 address list
Customer list	list inquiry	legal entity, group, blocked, classification	account, name, group, terms, currency, credit limit, blocked	`party.customer_role`	CustTableListPage; SSRS Customer base data	S_ALR_87012179, XD03
Vendor list	list inquiry	legal entity, group, hold, diversity	account, name, group, terms, hold, certifications	`party.vendor_role`	VendTableListPage; SSRS Vendor base data	S_ALR_87012086, MKVZ
Duplicate candidates	operational report	weight range, status, age	pair, weight, probability, level detail, age	`party.duplicate_candidate`	Duplicate check results (DirDuplicateCheck)	MDG duplicate check results
Critical changes	operational report	field, legal entity, status, requester	field, old (masked), new (masked), requester, approver, SLA	`party.pending_change`	Vendor change proposals (D365)	S_ALR_87012090, S_ALR_87012183 (FK09/FD09)
Payment terms and schedules	list inquiry	term code	net rule, discount tiers, installments, due-date preview	`party.payment_term`	SSRS Terms of payment list; Cash discount list	OBB8 / OBB9 listing (T052, T052R)
Certification expiry	KPI	days to expiry, type, vendor group	expiring count, expired count, vendors on hold	`party.vendor_certification`	Vendor certifications (D365)	—
Diversity vendor base	KPI	classification, legal entity	vendor count per classification (spend added in wave 17)	`party.vendor_diversity`	Vendor diversity (D365)	—
Tax registration status	operational report	status, country, type	number, status, source, validated at	`party.tax_registration`	VAT number validation log	VIES check log
Hold history	drill-down	role, reason, date range	hold, reason, user, timestamp	wave-5 audit	Customer/Vendor hold history	XD04 / XK04 change documents
Relationships and head offices	list inquiry	type, party	from, to, type, validity, decentralized flag	`party.party_relationship`	Party relationships	KNA1-KNRZE / KNB1-KNRZB lists, VDH2N
Merge history	drill-down	date range, status	case, survivor, merged ids, ack count, revert	`party.merge_case`	—	—

1.7.15 Printed and electronic documents

Document	Trigger	Layout elements (header/line/footer fields)	Formats	Numbering sequence	Legal / retention requirement
Party master data sheet	on demand	header: party number, name, type, primary address; lines: roles per legal entity, terms, tax ids (masked); footer: printed by, timestamp	PDF	none	none
Bank-detail change confirmation	approval of a protected bank-account or permitted-payee change	header: party, role, change id; body: old/new IBAN masked to last 4, effective date, approvers; footer: callback-verification statement	PDF, email to the party's verified contact	`PCHG-{yyyy}-{#####}`	10 years as payment-fraud control evidence (SOX ITGC)
Data subject access export	privacy officer request for a person party	header: request id, subject, date; body: all personal fields, addresses, roles, processing purposes	JSON + PDF	`DSAR-{yyyy}-{####}`	respond within 1 month (GDPR Art. 12(3)), 45 days (CCPA); request log 3 years
Merge certificate	merge case completed	header: case id; lines: merged ids → survivor, consumer acks; footer: operator, timestamp	PDF	merge case number	10 years
Mailing labels	none	—	—	—	none: mailing labels are CRM scope (PRD non-goal)

1.7.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
PTY-J1 Full duplicate re-score	weekly Sunday 01:00 tenant local	blocking key hash (16 partitions)	candidates upserted by (a, b); checkpoint per partition	< 2 h for 1M parties on 8 workers	partition retried 3×; partial results kept	queue > 1,000 open
PTY-J2 VIES re-validation	daily 04:00	country	max 5,000 per run; cursor by `validated_at`	< 30 min	backoff on rate limit; leaves `unverified`	failure rate > 20% / 15 min
PTY-J3 Stale merge sweeper	hourly	merge_case_id	idempotent	< 1 min	alerts on cases awaiting ack > 10 min	stale cases > 0
PTY-J4 Approval SLA escalation	every 15 min	legal_entity_id	Temporal timers; idempotent per task	< 1 min	escalation to approver's manager	SLA breaches per day
PTY-J5 Alias usage log purge	monthly	tenant	deletes rows > 12 months	< 5 min	retried next run	none
PTY-J6 Certification status update	daily 00:30	vendor_role_id range	status recomputed; events only on change	< 10 min for 100k certificates	retried; holds applied only after status commit	expired-mandatory count
PTY-J7 One-time conversion candidates	weekly	legal_entity_id	candidate upsert by normalized name/tax id	< 15 min	none	candidates created
PTY-J8 Personal data end-of-purpose check	monthly	party_id range	flags candidates; never deletes without approval	< 1 h per 1M persons	report only	candidates > 0
PTY-J9 Role inactivity blocking	monthly (off by default)	legal_entity_id	idempotent; skips `never`	< 15 min	dry-run report first	blocked count

1.7.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
View parties	`GET /party/v1/**` (PII masked); Global address book	none	every business role
Maintain parties	`POST/PATCH /party/v1/parties`, tax registrations, relationships, address books	Approve master data changes	Master data steward, AR clerk, AP clerk
Maintain customers	customer role endpoints; Customers workspace	Post customer payments (wave 30)	AR clerk, Sales administrator
Maintain vendors	vendor role endpoints; certifications; Vendors workspace	Approve vendor invoices (wave 19), Release payments (wave 19)	AP clerk, Purchasing agent

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
Maintain payment and delivery terms	terms, schedules, delivery terms/modes	none	AR manager, AP manager
Steward master data	duplicates, stewardship queue, PTY-J1/J7 results	none	Master data steward
Merge parties	merge cases, `:revert`	Maintain vendors (same party), to prevent payee substitution	Master data steward lead
Approve master data changes	`pending-changes/{id}:approve/:reject`	Maintain vendors / Maintain customers (no self-approval, BR-7.8)	AP manager, Credit manager, Controller
Override duplicate check	`X-Passage-Override-Duplicate`	none	Master data steward lead
Read personal data	`pii.read`; DSAR export	Administer migration (bulk export risk)	Privacy officer, HR administrator

1.7.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
PTY-DUP-LIKELY	error	"{n} likely duplicates found."	Weight ≥ likely threshold on create	Review candidates or override with duty	FR-7.5
PTY-DUP-OVERRIDE-DUTY	error	"You are not allowed to override the duplicate check."	Override header without duty	Ask a steward	AC-7.7
PTY-TAX-FORMAT	error	"{type} {number} does not match the {country} format."	Regex failure	Correct the number	FR-7.4
PTY-TAX-CHECKDIGIT	error	"Check digit of {number} is wrong."	Algorithm failure (mod-97, mod-89, mod-36)	Correct the number	FR-7.4
PTY-TAX-UNVERIFIED	warning	"Online validation unavailable; saved as unverified."	VIES/TIN timeout	None; PTY-J2 retries	FR-7.4
PTY-ROLE-LE-HEADER	error	"Select a legal entity before adding a role."	Missing `X-Passage-LegalEntity`	Set the entity	FR-7.2
PTY-ROLE-GROUP-MISSING	error	"Group {code} does not exist in {entity}."	Unknown group	Create or pick a group	FR-7.3
PTY-MERGE-RUNNING	error	"Merge case {id} is already running for this party."	Concurrent merge	Wait for the running case	BR-7.11
PTY-MERGE-REVERT-BLOCKED	error	"{service} cannot split post-merge transactions."	Consumer refuses revert	Correct manually	BR-7.2
PTY-PENDING-SELF-APPROVAL	error	"You cannot approve your own change."	Requester = approver	Second approver	BR-7.8
PTY-TERM-DISCOUNT-AFTER-DUE	error	"Discount date is after the due date."	Invalid term	Fix days	BR-7.7
PTY-SCHED-PCT-SUM	error	"Installment percentages total {x}, not 100."	Percent lines ≠ 100	Adjust lines	BR-7.13
PTY-ONETIME-FIELD	error	"One-time accounts cannot hold {field}."	Forbidden field on one-time role	Capture on the document	BR-7.15
PTY-REL-CYCLE	error	"This relationship creates a cycle: {path}."	Head-office loop	Choose another head office	FR-7.17
PTY-CERT-EXPIRED	warning	"Mandatory certificate {type} expired on {date}; vendor on hold."	BR-7.14	Upload renewed certificate	FR-7.14
PTY-INACTIVE-ROLE	error	"Party is inactive; roles cannot be added or unblocked."	BR-7.10	Reactivate the party	BR-7.10
PTY-SEARCH-MIN-CHARS	info	"Type at least 2 characters."	Short query	Type more	FR-7.13

1.7.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Party create with dedup	5k/day; peak 200/min (imports via bulk path)	120 / 400 / 800 ms	50 req/s	~2 KB per party; 1M parties ≈ 2 GB incl. indexes	GIN trigram `search_name`; blocking indexes	scoring moves to async; create returns 202 with candidates later
Typeahead search	500k/day; peak 2,000/min	25 / 100 / 180 ms	300 req/s per replica	none	trigram + (tenant_id, status)	result cap drops 20 → 10
Single-party scoring	per create/update	60 / 300 / 500 ms at 1M	100/s on 2 workers	candidates ~5% of parties	blocking keys	queue grows; HPA to 6 workers
Weekly full re-score	1M parties	—	140 parties/s per worker	—	16 hash partitions	runtime extends; no user impact
Due-date / schedule computation	2M/day from AR/AP	1 / 3 / 6 ms	2,000 req/s	none	term cache	unchanged
Merge workflow	50/day	20 s / 5 min / 15 min (ack-bound)	10 concurrent	snapshot ~50 KB per case	workflow id per min(party)	queued; acks are the bottleneck
VIES validation	5k/day	300 ms / 2 s / 5 s (external)	20 concurrent	cache rows	—	`unverified` returned immediately

1.7.20 Localization and compliance hooks

Item	What varies	Country packs touching it (wave 48)	Extension point
Tax identifiers	US EIN/SSN (masked) and 1099 vendor flag (box type consumed by wave 19); UK VAT (HMRC VAT API after VIES exit) and UTR; DE USt-IdNr and Steuernummer; FR TVA intracommunautaire and SIREN/SIRET; IN GSTIN (state code + PAN) and PAN; MX RFC with CFDI 4.0 receptor data (régimen fiscal, código postal); CA BN with GST/HST and QST	all seven	`TaxIdValidator` registry; `TaxIdAdapter` (VIES, HMRC, IRS TIN Matching, GSTN, SAT LCO)
Name and address conventions	Person name order, Hindi/Devanagari names, Spanish double surnames (MX), Quebec bilingual	IN, MX, CA	`party.person.name_sequence`; transliteration attribute
Diversity classifications	US SBA programs (8(a), HUBZone, WOSB, SDVOSB); CA Indigenous business registry; UK none statutory	US, CA	classification code list seeded by pack
Payment term conventions	FR LME late-payment cap (60 days from invoice or 45 days end of month, Code de commerce L441-10); DE §286 BGB 30-day default; IN MSMED Act 45-day limit for MSE suppliers; UK Late Payment Act	FR, DE, IN, UK	term validation rule per country pack (warn on terms exceeding statutory maximum)
Privacy and retention	GDPR/UK GDPR erasure vs retention (HGB/AO 10 y for accounting-relevant data), CCPA/CPRA, DPDP Act 2023, LFPDPPP ARCO rights, PIPEDA	all	PII flags (BR-7.9), DSAR export, PTY-J8 end-of-purpose check

Item	What varies	Country packs touching it (wave 48)	Extension point
Sanctions screening	OFAC SDN (US), UK OFSI, EU consolidated list on vendor/customer create	all	screening adapter interface (wave 48 compliance) called at role creation; result stored on role

1.7.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.16.1 Procurement and sourcing, Common — Vendor certifications / Vendor diversity (D365), VendCertification; register §12.99.4 #158	PARTIAL-7	ADDED as FR-7.14	Expiry job PTY-J6, KPI "Certification expiry"
AX §12.16.4 Procurement and sourcing, Setup — Vendor certification types / Diversity classifications; register §12.99.4 #165	PARTIAL-7	ADDED as FR-7.14	Parameter `party.cert.reminder_days`; seeded diversity list
SAP §13.1 FI-GL Customizing — OBB8 / OBB9 Terms of payment / installment terms (T052, T052R); register §13.99.2 #17	PARTIAL-7	ADDED as FR-7.15	Installments via payment schedule lines; BR-7.13 rounding
SAP §13.2 FI-AP Master data — One-time vendor (account group CPD); register §13.99.2 #23	PARTIAL-7	ADDED as FR-7.16	Document-level capture stored by waves 17/19
SAP §13.2 FI-AP Customizing — OBB9 Installment terms (T052R); register §13.99.2 #35	PARTIAL-7	ADDED as FR-7.15	Same schedule model for vendor terms
SAP §13.2 FI-AP Customizing — OBD3 "one-time account" (T077K); register §13.99.2 #37	PARTIAL-7	ADDED as FR-7.16	Parameter `party.group.one_time_default`
SAP §13.3 FI-AR Customizing — OBB9 Installment terms (customers) (T052R); register §13.99.2 #47	PARTIAL-7	ADDED as FR-7.15	AR due dates per installment in wave 30
SAP §13.3 FI-AR Customizing — Define alternate payer / head office–branch (KNA1-KNRZE, KNB1); register §13.99.2 #50	PARTIAL-7	ADDED as FR-7.17	`head_office_of` with decentralized flag; `alternate_payer`; posting effect in waves 19/30

1.8 Wave 8: Product Master, Variants & Catalogs

1.8.1 Scope and business context

AX defines a shared `EcoResProduct` (subtyped `EcoResDistinctProduct` and `EcoResProductMaster`) with variants in `EcoResDistinctProductVariant`, dimension values in `EcoResSize/Color/Style/Configuration`, and releases it per company into `InventTable` with `InventDimGroup`-style product/storage/tracking dimension groups, `InventItemGroup` and `InventModelGroup`. SAP keeps `MARA` (client-level), `MARC` (plant), `MARD` (storage location), `MVKE` (sales), `MBEW` (valuation), material types (`T134`), variant configuration in LO-VC (`CUKB`, `CUOB`, `CABN/CAWN` characteristics) and the classification system (`KLAH/KSML`). Wave 8 defines the product once at tenant level, explodes variants from dimension groups, releases per legal entity with site/warehouse defaults, and adds attributes, categories, item groups, item model groups, GS1 identifiers and a constraint-based configurator foundation. At the end of the wave the product information management workspace holds the demo manufacturer's 500 items and 50 masters.

Out of scope: on-hand, costing values and inventory postings (waves 16, 20), sales/purchase prices (waves 17, 26), BOMs (wave 9), batch/serial *instances* (wave 16; this wave defines only whether an item is tracked), quality specs (wave 33).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Releasing an AX product to 10 companies means 10 passes through "Release products" plus editing InventTable defaults (item group, model group, site, warehouse) in each company; a missed default blocks the first PO with "Item model group not specified"	Release is a per-company copy with no template	Release profiles: one call releases to N entities with defaults derived from item group and site; missing-default check at release (FR-8.4)	Clicks per 10-entity release 60→3; first-transaction failures from missing defaults →0
SAP MM01 walks 15–20 views (basic, purchasing, MRP 1–4, accounting, costing); a material is "created" but unusable in a plant until every view is extended (MM01 extend)	View-based material master with per-plant/storage segments (MARC/MARD) each needing separate maintenance	One product record plus released-product per entity; plant-level data derives from site defaults (FR-8.4)	Time to a usable material in a new plant 25 min→<1 min
AX variant creation (EcoResProductVariant "Variant suggestions") is a modal that times out above ~500 combinations; D365 batch job needed	Synchronous form-driven explosion	Explosion is a Temporal job with checkpoints; 1,000 variants in <5 s (FR-8.2)	10k variants: 40 min batch→<60 s with progress
GTIN entry in AX (InventItemGTIN) and SAP (MARM-EAN11) accepts any 13 digits; bad check digits break EDI 856/ASN and retailer onboarding	No check-digit validation at entry	GS1 check-digit and prefix validation, generation from a GS1 company prefix (FR-8.9)	Invalid GTINs stored 0; retailer rejections for GTIN →0
Product attributes in AX (EcoResProductAttribute) require defining attribute types, attributes, categories and category attributes across four forms before a value can be entered; SAP class/characteristic (CL02/CT04) is similar	Classification metadata modeled as a separate hierarchy from products	Attribute sets attached to categories with inline attribute creation and typed validation (FR-8.6)	Setup of a 10-attribute class 30 min→3 min

1.8.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Shared product	EcoResProduct, EcoResDistinctProduct, EcoResProductMaster, Product details form, ProductNumber	MARA (MATNR, MTART, MEINS, MATKL), MM01/MM02/MM03	product-service `Product` (product_type item/service, subtype product/master)	full
Variants and dimensions	EcoResProductDimensionGroup, EcoResSize/Color/Style/Configuration, EcoResDistinctProductVariant, EcoResProductVariantDimensionValue	Generic material + variants (MARA-SATNR, KZKFG), characteristics CABN/CAWN	`DimensionGroup`, `DimensionValue`, `ProductVariant`	full
Storage / tracking dimension groups	EcoResStorageDimensionGroup, EcoResTrackingDimensionGroup (batch/serial active, physical/financial inventory flags)	MARA-XCHPF batch management, serial profile MARC-SERNP, valuation level	`StorageDimensionGroup`, `TrackingDimensionGroup`	full
Released product per company	InventTable (ItemId, ItemGroupId, ModelGroupId, DataAreaId), InventTableModule, InventItemSetupFilter, default order settings (InventItemInventSetup/PurchSetup/SalesSetup)	MARC (plant), MARD (storage location), MVKE (sales org), MBEW (valuation)	`ReleasedProduct` with `ReleasedProductSiteDefault`	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Item groups / model groups	InventItemGroup, InventModelGroup (InventModel FIFO/LIFO/WAvg/Std, PhysicalNegative, FinancialNegative, PostPhysical, PostFinancial)	Material type T134, material group T023, valuation class T025, price control MBEW-VPRSV	`ItemGroup`, `ItemModelGroup`	full
Units and conversions	UnitOfMeasureConversion product-specific, InventTable unit per module	MARA-MEINS, MARM alternative units	`ProductUomConversion`, unit per module (inventory/purchase/sales/BOM)	full
Attributes / classification	EcoResAttribute, EcoResAttributeType, EcoResCategory, EcoResCategoryHierarchy, EcoResProductCategory	Classification: KLAH, KSML, CABN, CAWN, AUSP (CL01/CL02/CT04)	`AttributeType`, `Attribute`, `AttributeSet`, `Category`, `CategoryHierarchy`, `ProductAttributeValue`	full
Configurator	Product configuration models (PCConfiguration, PCComponent, PCConstraint — D365 constraint-based configurator)	LO-VC: CU41 configuration profile, dependencies CUKB, object dependencies CUOB	`ConfigurationModel` solved by OR-Tools CP-SAT in a Python worker	partial (pricing and BOM/route generation from configuration in waves 26 and 9)
GS1 identifiers	InventItemGTIN, InventItemBarcode, BarcodeSetup	MARM-EAN11, EAN category T006M (NUMTP)	`ProductIdentifier` (GTIN-8/12/13/14, SSCC generation)	improved (validation and generation)
Images / documents	DocuRef attachments on EcoResProduct, product image	DMS (CV01N) links	`ProductDocument` (object storage key, type)	full
Product lifecycle	D365 product lifecycle state (EcoResProductLifecycleState)	MARA-MSTAE cross-plant status, MARC-MMSTA plant status	`lifecycle_state` on Product and ReleasedProduct	full
Related products	EcoResProductRelation, EcoResProductRelationType (Released product → Related products)	SD cross-selling via material determination (VB41/VB42/VB43), cross-selling profile	`ProductRelation`, `ProductRelationType` (FR-8.13)	full
Configurable lifecycle states	EcoResProductLifecycleState with enabled business processes (D365)	MARA-MSTAE / MARC-MMSTA with status control table T141 (OMS4)	`LifecycleState`, `LifecycleProcessPolicy` (FR-8.14)	full
Controlled group conversion	Change product dimension group / Change item model group (D365 periodic tasks), mass "Delete released products"	MM17 mass maintenance; MM06 flag for deletion; MMDE archive deletion	`group-change` and `obsolete` Temporal jobs (FR-8.15, FR-8.16)	improved (preconditions checked by callback)
Service master	EcoResProduct product type Service; procurement categories for services	AC01–AC06 service master (ASMD), service categories, standard service catalog	`Product` type service with `ServiceAttributes` (FR-8.17)	full
Variant tables and configuration templates	Table constraints (PCTableConstraint, user-defined/system-defined), configuration reuse	CU60–CU62 variant tables (CUVTAB/CUVTAB_VALC), CU7/CUCPT configuration templates	`DecisionTable`, `SavedConfiguration` (FR-8.18, FR-8.19)	full
Document info records	DocuRef, DocuType (Document management → Document types)	CV01N–CV04N document info record (DRAW, DRAT, DRAD), DC10 document types (TDWA)	`DocumentType`, `DocumentInfoRecord`, `DocumentOriginal`, `DocumentObjectLink` (FR-8.20, FR-8.21)	full

1.8.3 Functional requirements

- **FR-8.1 Product and product master** — Must. The system shall create tenant-level products (subtype `product`) and product masters (subtype `master`) with product number (sequence `PROD` or manual), name, description, search name, product type (item/service), base unit, and lifecycle state.
  - Product number regex `^[A-Z0-9][A-Z0-9._-]{0,39}$`; immutable after release.
  - Masters carry a `dimension_group_id` with 1–4 product dimensions chosen from size, color, style, configuration (named per tenant, e.g. "length").
- **FR-8.2 Variant explosion** — Must. The system shall generate variants for a master from the Cartesian product of selected dimension values, or from an explicit list, as a Temporal job with 500-row checkpoints.
  - Hard cap 10,000 variants per master; 1,000 variants in < 5 s.
  - Variant number is `{master}-{size}-{color}-{style}-{config}` with empty segments collapsed; override pattern per tenant.
  - Lookup by (master_id, dimension values) resolves with a unique composite index.
- **FR-8.3 Storage and tracking dimension groups** — Must. The system shall define storage dimension groups (site, warehouse, location, inventory status, license plate — each active, mandatory, primary stocking, physical/financial inventory flags) and tracking dimension groups (batch active/mandatory, serial active/mandatory, serial capture at sale/receipt), assigned to a product and immutable once a released product has inventory transactions (callback from wave 16).
- **FR-8.4 Released products with site/warehouse defaults** — Must. The system shall release a product to one or more legal entities, creating `ReleasedProduct` with item group, item model group, units per module, default site and warehouse, and per-site default warehouse and order settings (minimum/multiple/standard order quantities, lead time) for inventory, purchase and sales.
  - Release profiles: a named template of defaults per (legal entity, item group); release of 1,000 products to 3 entities completes in < 60 s.
  - Cascade: a per-site override wins over the entity default; absent both, the request is rejected with the missing field named.
- **FR-8.5 Item groups and item model groups** — Must. The system shall maintain item groups (posting hooks for GL arrive in wave 11) and item model groups with costing method (FIFO, LIFO, weighted_average, weighted_average_date, standard, moving_average), negative physical/financial inventory flags, post physical/financial inventory flags, stocked flag, and quarantine/registration/picking requirements.
- **FR-8.6 Attributes and categories** — Must. The system shall maintain attribute types (text, integer, decimal, boolean, date, enum, currency, with unit and min/max), attributes, attribute sets, and category hierarchies (procurement, sales, retail, reporting) where a category carries attribute sets; a product in a category inherits required attributes and is incomplete until they are filled.
  - Maximum 200 attributes per product; enum attributes have up to 500 values.
- **FR-8.7 Product-specific UoM conversions** — Must. The system shall store per-product conversions (e.g. 1 BOX = 12 EA for product X) that override class conversions; resolution order product → class → error.
- **FR-8.8 Configurator foundation** — Should. The system shall define configuration models per master: components, attribute domains, constraints (expression language with `==, !=, <, >, in, implies, allDifferent, sum`), default values, and a `solve` endpoint that returns a valid configuration or the minimal conflicting constraint set, using CP-SAT with a 2 s time limit.
  - A 50-constraint model solves in < 200 ms; the result is a variant or a configured-product instance with attribute values.
- **FR-8.9 GS1 identifiers** — Must. The system shall store GTIN-8/12/13/14 per (product/variant, unit) with check-digit validation (mod-10 weighting 3-1), allocate GTINs from a tenant GS1 company prefix with a next-number counter, generate SSCC-18 with extension digit, and render barcodes (EAN-13, GS1-128, DataMatrix) as SVG/PNG.
- **FR-8.10 Product lifecycle and blocking** — Must. The system shall enforce lifecycle states `draft → active → phase_out → obsolete` per product and per released product, with module-level blocks (purchase, sales, inventory, production) on released products.

- **FR-8.11 Images and documents** — Should. The system shall attach images (max 10 MB each, 20 per product) and documents to products in object storage with a primary image used in lists.
- **FR-8.12 Product search and catalogs** — Should. The system shall provide typeahead over product number, name, GTIN and attribute values, filtered by legal entity and category, p95 < 100 ms at 1M products, and allow catalogs as named category subtrees with a validity range.
- **FR-8.13 Product relations** — Should. The system shall maintain relation types (seeded: accessory, cross_sell, up_sell, substitute, spare_part, consumable, same_family; tenant-extensible) with a `symmetric` flag, and relations from a product or variant to another product or variant with sort order and `valid_from`/`valid_to`.
  - Maximum 200 relations per product; symmetric types create and delete the inverse in the same transaction.
  - Sales order entry (waves 26–27) reads `GET /product/v1/products/{id}/relations?type=` for suggestions; event `dev.passage.product.relation.changed.v1`.
- **FR-8.14 Configurable lifecycle states with process policies** — Must. The system shall let tenants define lifecycle states (code, name, category ∈ {draft, active, phase_out, obsolete} from FR-8.10, `default_for_new`, applies to product / released product / engineering version) and, per state, a policy of `allow`, `warn` or `block` for each business process: purchase_requisition, purchase_order, sales_quotation, sales_order, transfer_order, production_order, bom_line_use, master_planning, inventory_receipt, inventory_issue, counting, return_order.
  - `GET /product/v1/released-products/{id}/process-policy?process=` answers in < 10 ms p95; consumers cache the result and refresh on `releasedproduct.updated.v1`.
  - FR-8.10 boolean blocks stay as overrides (BR-8.12).
  - Seed mirrors D365 defaults: Prototype (draft: allows bom_line_use and planning only), Active, Phase out (blocks purchase_order and master_planning, warns on sales_order), Obsolete (blocks all except inventory_issue, counting and return_order).
  - Engineering versions (FR-9.14) carry their own state with the same policy model; the most restrictive of product, released-product and version policy applies.
- **FR-8.15 Mass obsolete and delete of released products** — Should. The system shall run a job that sets selected released products to an obsolete-category state (validated by BR-8.1) and deletes a released product only when every registered reference callback (inventory, purchasing, sales, production, engineering, pricing) reports zero references; otherwise the item is set obsolete and reported as "obsolete instead of deleted".
  - Max 10,000 items per run; a dry-run report precedes execution; deletion removes the released product only, never the shared product.
- **FR-8.16 Controlled change of dimension and model groups** — Should. The system shall change the item model group, storage dimension group or tracking dimension group of selected released products at an effective date through Temporal job `group-change-{id}`, after verifying through callbacks: on-hand = 0 (wave 16), no open purchase, sales, transfer or production orders (waves 16, 17, 21, 26), and inventory closed up to the effective date for costing-method changes (wave 20).
  - Before those waves exist the callbacks return "no references" and the change applies directly; each change writes an audit record and emits `releasedproduct.updated.v1` with `changed_fields`.
  - This job is the only path around BR-8.5 and FR-8.3 immutability.
- **FR-8.17 Service products** — Must. The system shall extend products of type `service` with: service category (code list with its own number sequence, e.g. `SVC-{cat}-{#####}`), service unit (hour, day, each, lump_sum, or any time-class unit), long purchasing text (≤ 4,000 characters, translatable), membership in standard service catalogs (category hierarchy role `service_catalog`), valuation class placeholder (bound in waves 11 and 20) and `requires_service_entry` flag read by purchasing (wave 17).
  - Service products never have storage or tracking dimension groups and never use a stocked item model group (BR-8.17).
- **FR-8.18 Configurator decision tables** — Should. The system shall maintain decision tables (columns = configuration attributes, rows = allowed value combinations; ≤ 50 columns, ≤ 100,000 rows) in a grid with CSV import/export, referenced from constraints as `table(name, attr1, …, attrN)` and pinned to the model version on activation.
  - LO-VC variant functions (ABAP function modules) are replaced by the sanctioned expression functions `round`, `min`, `max`, `abs`, `lookup`, `concat`, `substr`.
- **FR-8.19 Saved configurations and templates** — Should. The system shall save a solved configuration as a named `SavedConfiguration` (model id and version, selections, derived attribute values) at tenant scope or for a customer party, reuse it as the starting point of later solves, and re-validate saved configurations when a new model version activates; incompatible ones are flagged `stale` with the violated constraints.
- **FR-8.20 Document info records and document types** — Should. The system shall maintain document types (code, name, number sequence, allowed file types, max originals default 5 / max 20, version scheme numeric or alpha, part numbering on/off, `change_controlled` flag enforced by FR-9.25, status network id from FR-9.24, attribute set from FR-8.6, allowed object-link targets: product, released product, BOM version, route version, party, ECO) and document info records (DIR: document number, part `000`, version `00`, description, status, owner, originals with object-storage key, sha256 and size ≤ 200 MB, object links, attribute values).
  - Product images and documents from FR-8.11 become DIRs of system types `IMG` and `DOC`; the FR-8.11 endpoints are unchanged.
  - Endpoints: `POST /product/v1/documents`, `POST /product/v1/documents/{id}/versions`, `PUT /product/v1/documents/{id}/links`.
- **FR-8.21 Document search** — Should. The system shall search DIRs by number, description, type, status, attribute values, object link and full text of PDF, DOCX and TXT originals (text extracted at upload by a Python worker, ≤ 50 MB of text per original, Postgres `tsvector` with language-specific dictionaries), p95 < 300 ms at 2M DIRs.

1.8.4 Data model

Product

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS
product_number	text(40)	yes	unique per tenant
subtype	enum	yes	product, master, variant
master_id	uuid	conditional	FK for variants
product_type	enum	yes	item, service
name / description / search_name	text(60)/text(1000)/text(60)	name yes	translations in product_translation
base_uom_id	uuid	yes	FK
dimension_group_id	uuid	masters only	product dimensions
storage_dimension_group_id / tracking_dimension_group_id	uuid	yes at release	immutable after first transaction
lifecycle_state	enum	yes	draft, active, phase_out, obsolete
dimension_values	jsonb	variants	{size, color, style, config}; unique with master_id
primary_image_key	text	no	object storage
version	int	yes	

ReleasedProduct

Field	Type	Required	Notes
id	uuid	yes	PK
product_id	uuid	yes	FK; unique (product_id, legal_entity_id)
legal_entity_id	uuid	yes	
item_number	text(40)	yes	defaults to product_number; unique per legal entity
item_group_id / item_model_group_id	uuid	yes	FK
inventory_uom_id / purchase_uom_id / sales_uom_id / bom_uom_id	uuid	inventory yes	convertible to base
default_site_id / default_warehouse_id	uuid	yes for stocked	warehouse must belong to site
lifecycle_state	enum	yes	same enum as product
blocked_purchase / blocked_sales / blocked_inventory / blocked_production	bool	yes	default false
net_weight / gross_weight / volume + units	numeric(18,6)	no	
site_defaults	1:N		ReleasedProductSiteDefault(site_id, warehouse_id, min_qty, multiple_qty, std_qty, lead_time_days per module)
version	int	yes	

ItemModelGroup

Field	Type	Required	Notes
id / tenant_id / code / name	uuid, uuid, text(10), text(60)	yes	code unique per tenant
costing_method	enum	yes	fifo, lifo, weighted_average, weighted_average_date, standard, moving_average
stocked	bool	yes	default true
physical_negative / financial_negative	bool	yes	default false
post_physical / post_financial	bool	yes	default true
registration_required / receiving_required / picking_required / deduction_required	bool	yes	
quarantine_required	bool	yes	default false
include_physical_value	bool	yes	used by costing (wave 20)

ConfigurationModel — id, master_id, version, status (draft/active/retired), components jsonb, attributes jsonb (name, domain), constraints jsonb (expression list, ≤ 2,000), defaults jsonb. **ProductIdentifier** — id, product_id, uom_id, type (GTIN8/12/13/14, SSCC, UPC, custom), value text(18), is_primary, check_digit_valid bool.

Invariants: a variant's dimension_values keys equal its master's dimension group; every released stocked product has item model group and default site/warehouse; GTIN values pass check digit; `obsolete` products have no released product in `active`. Indexing: unique `(master_id, dimension_values)` via a generated hash column; GIN trigram on `search_name`; `product_identifier(value)` unique per tenant; `released_product(legal_entity_id, item_number)` unique; attribute values in `product_attribute_value(product_id, attribute_id)` with a GIN index on a jsonb projection for filtered search.

Addendum entities (FR-8.13 – FR-8.21) — product_relation: id, type_code FK, from_product_id, to_product_id, sort_order int, valid_from, valid_to, unique (type_code, from_product_id, to_product_id). lifecycle_state: id, code text(20), name, category enum, default_for_new bool, applies_to text[]; lifecycle_process_policy: state_id, process enum, policy (allow, warn, block), PK (state_id, process). service_attributes: product_id PK/FK, service_category_code, service_uom_id, purchasing_text_id (translation), requires_service_entry bool, valuation_class text(10) null. decision_table: id, model_id, name, columns jsonb, row_count int, version int, checksum; rows in decision_table_row (table_id, row_no, values jsonb) with a GIN index. saved_configuration: id, model_id, model_version, scope (tenant, customer), party_id null, name, selections jsonb, status (valid, stale). document_type: code PK per tenant, settings jsonb; document_info_record: id, type_code, doc_number text(25), part char(3), doc_version text(2), description text(200), status_code, owner_user_id, unique (type_code, doc_number, part, doc_version); document_original: dir_id, seq, storage_key, sha256 bytea, size_bytes, mime, text_tsv tsvector; document_object_link: dir_id, target_type, target_id. group_change_job: id, change jsonb, effective_date, status, items jsonb.

1.8.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/product/v1/products	Create product or master	`Idempotency-Key`	Maintain products
GET	/product/v1/products/{id}	Read with variants summary	n/a	View products
POST	/product/v1/products/{id}/variants:explode	Start explosion job; returns job id	key header; job id keyed by request hash	Maintain products
GET	/product/v1/products/{id}/variants:lookup?size=M&color=RED	Resolve variant	n/a	View products
POST	/product/v1/products/{id}:release	Release to legal entities with profile	idempotent per (product, entity)	Release products
PATCH	/product/v1/released-products/{id}	Defaults, blocks, lifecycle	version check	Maintain released products
PUT	/product/v1/released-products/{id}/site-defaults/{siteId}	Per-site defaults	full replace	Maintain released products
POST	/product/v1/products/{id}/attributes	Set attribute values (validated against type)	full replace of set	Maintain products
POST	/product/v1/products/{id}/identifiers	Add GTIN; `allocate=true` draws from prefix	key header	Maintain products
POST	/product/v1/configuration-models/{id}:solve	Solve with given selections	pure function	Configure products
GET	/product/v1/products:search?q=&legalEntity=&category=	Typeahead	n/a	View products
GET	/product/v1/uom/convert?product=&from=&to=&qty=	Product-aware conversion	n/a	any authenticated

Validation: dimension values must exist in the dimension group; release rejects a `draft` product; identifier type must match length; `solve` rejects selections outside domains with the offending attribute.

Event	Trigger	Payload fields	Consumers
dev.passage.product.created.v1	commit	product_id, product_number, subtype, product_type, base_uom	engineering, search
dev.passage.product.variant.created.v1	per explosion checkpoint (batched, ≤ 500 ids)	master_id, variant_ids[], dimension_values[]	engineering, pricing, inventory
dev.passage.product.released.v1	release commit	product_id, legal_entity_id, item_number, item_group_id, item_model_group_id, default_site_id, default_warehouse_id, uoms	inventory, purchasing, sales, costing, planning
dev.passage.product.releasedproduct.updated.v1	PATCH	released_product_id, changed_fields[], blocks, lifecycle_state	all transactional services
dev.passage.product.identifier.added.v1	commit	product_id, type, value, uom_id	WMS (wave 31), EDI (wave 35)
dev.passage.product.configuration.solved.v1	solve returning a variant	model_id, master_id, selections, result_product_id	sales (wave 27), engineering

1.8.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Product information management workspace	Tiles: products, masters, released per entity, incomplete attributes, pending releases	counts, incomplete badges	View products
All products and masters	New, release, explode variants, lifecycle change	number, name, subtype, type, dimension groups, state	Maintain products
Product detail	Tabs: general, dimensions/variants grid, attributes, identifiers, images, releases, translations		Maintain products
Variant explosion dialog	Select values per dimension, preview count, start job, progress bar	dimension values, count, job status	Maintain products
Released products (per legal entity)	Edit defaults grid (inline), block, site defaults sub-grid, bulk edit by item group	item number, item group, model group, units, site, warehouse, blocks	Maintain released products
Item groups / item model groups	Edit, usage count	code, name, costing method, flags	Maintain product setup
Categories and attributes	Hierarchy tree, attribute set editor, assign products	category path, attributes, required	Maintain product categories
Configurator model editor	Add components/attributes/constraints, test solve, activate version	constraints, domains, solve result	Configure products
GS1 identifiers	Allocate, validate, print barcode	type, value, uom, valid	Maintain products

Interaction rules: the released-products grid supports bulk edit of selected rows (max 2,000) with one save; attribute inputs validate on blur against type and range; the explosion dialog warns above 2,000 variants and requires confirmation; barcode print uses the browser print dialog with an SVG preview; the legal-entity switcher filters the released-product views only.

1.8.7 Business rules and edge cases

- **BR-8.1** Lifecycle machine for Product and ReleasedProduct: `draft → active`, `active → phase_out`, `phase_out → active | obsolete`, `obsolete` terminal; a released product cannot be more "alive" than its product (product `obsolete` forces released `obsolete`).
- **BR-8.2** A master cannot change its dimension group once any variant exists; a dimension value cannot be deleted while any variant uses it.
- **BR-8.3** Explosion is idempotent: existing variants for a combination are skipped and reported; the job result lists created/skipped counts.
- **BR-8.4** Release copies defaults at release time; later changes to the release profile do not propagate (explicit "re-apply profile" action does, with a diff preview).
- **BR-8.5** Changing `costing_method` on an item model group is refused once any released product in it has a financial inventory transaction (wave-16 callback; returns false before wave 16).
- **BR-8.6** UoM conversion quantities round half-even to the target unit's `decimal_precision`; conversions that would produce zero from a non-zero input are rejected (`PRD-UOM-UNDERFLOW`).
- **BR-8.7** GTIN check digit: weights 3,1,3,… from the right of the payload; GTIN-12 stored as GTIN-14 with leading zeros for uniqueness; the same GTIN may not belong to two products in one tenant.
- **BR-8.8** Configurator time limit 2 s; on timeout the response is 409 `PRD-CFG-TIMEOUT` with the partial assignment; infeasible selections return the irreducible infeasible subset (IIS) of at most 10 constraints.
- **BR-8.9** Required attributes from a category are enforced at `active` transition, not at create, so drafts can be saved incomplete.
- **BR-8.10** Translations fall back language → base language of the tenant → product name; never empty.
- **BR-8.11** Variant numbers longer than 40 characters are truncated to 32 plus an 8-character hash, and the full dimension values remain the lookup key.
- **BR-8.12** Process policy precedence: an FR-8.10 block flag set to true wins; otherwise the most restrictive of the released-product state, product state and (when versions are tracked) engineering-version state applies; `warn` returns success with `PRD-LC-WARN`, `block` returns `PRD-LC-BLOCKED` naming state and process.
- **BR-8.13** Each tenant state maps to exactly one category; transitions between states follow the BR-8.1 category machine; a state in use by any product or released product cannot be deleted; exactly one state per `applies_to` target is `default_for_new`.
- **BR-8.14** Symmetric relations are stored as two rows created and deleted together; self-relations are rejected (`PRD-REL-SELF`); a relation to an obsolete-category product is rejected for new relations and kept for existing ones.
- **BR-8.15** Group-change preconditions are evaluated at scheduling and again at the effective date; a product failing at the effective date is skipped with its reason code; no product is ever partially changed.
- **BR-8.16** A new DIR version copies object links and attribute values but not originals; at most one version per (type, number, part) may be in a status with the release flag at a time.
- **BR-8.17** Setting a storage or tracking dimension group, or a stocked item model group, on a service product is rejected (`PRD-SVC-STOCKED`).

1.8.8 Acceptance criteria

- **AC-8.1** (FR-8.2) Given a master with 3 dimensions × 10 values, when explosion runs, then 1,000 variants exist in < 5 s, lookup by any combination returns the right variant in < 10 ms, and a second run creates 0 and skips 1,000.
- **AC-8.2** (FR-8.8) Given the 50-constraint fixture model and 100 selection sets (60 valid, 40 invalid), when solved, then valid sets return a configuration consistent with every constraint and invalid sets return an IIS, each in < 200 ms.
- **AC-8.3** (FR-8.4) Given release profile P for item group "FG" in USMF with default site S1/warehouse W1 and a site default for S2/W3, when 1,000 products are released and 10,000 random (product, site) default lookups run, then each returns the site override if present, else the entity default (property test).
- **AC-8.4** (FR-8.9) Given the 40 GS1 published test vectors plus 40 mutated ones, when validated, then 40 accepted and 40 rejected; given a prefix 0614141 with next number 123457, when allocated, then GTIN-13 `0614141123457x` has the correct check digit and the counter increments.
- **AC-8.5** (FR-8.3, BR-8.5) Given a released product with a financial transaction (simulated callback true), when its tracking dimension group is changed, then 409 `PRD-DIMGRP-LOCKED`.
- **AC-8.6** (FR-8.10) Given a released product with `blocked_sales = true`, then `releasedproduct.updated.v1` carries `blocks.sales = true` and a later wave's contract stub refuses a sales line.

- **AC-8.7** (FR-8.6) Given category "Fasteners" requiring attributes thread_pitch (decimal, 0.2–6.0 mm) and material (enum), when a product in it is activated without `material`, then 422 lists the missing attribute; with `thread_pitch = 7`, then 422 names the range.
- **AC-8.8** (FR-8.7) Given product X with 1 BOX = 12 EA and class conversion 1 CASE = 24 EA, when converting 2 BOX → CASE, then result is 1 CASE; when converting for product Y with no product conversion, then the class conversion applies.
- **AC-8.9** (FR-8.12) Given 1M products, when searched by a GTIN fragment, then p95 < 100 ms.
- **AC-8.10** (FR-8.14, BR-8.12) Given state "Phase out" with purchase_order = block and sales_order = warn, when the process policy is queried for purchase_order, then `PRD-LC-BLOCKED`; for sales_order, then allowed with `PRD-LC-WARN`; with `blocked_sales = true`, then sales_order is blocked.
- **AC-8.11** (FR-8.13, BR-8.14) Given symmetric type `substitute`, when A→B is created, then B→A exists; when B→A is deleted, then A→B is gone; when A→A is created, then 422 `PRD-REL-SELF`.
- **AC-8.12** (FR-8.16, BR-8.15) Given 100 released products scheduled to move to model group STD effective 2026-12-01 and an on-hand callback returning 5 units for 3 of them on that date, when the job runs, then 97 are changed, 3 are skipped with `PRD-GRPCHG-ONHAND`, and 97 update events are published.
- **AC-8.13** (FR-8.20, FR-8.21) Given the 2M-DIR fixture, when searching attribute `drawing_size = A3` with text "torque", then p95 < 300 ms; when version 01 of DRW-0000123-000 is created, then it has the 3 object links of version 00 and 0 originals.
- **AC-8.14** (FR-8.18) Given a decision table of 10,000 allowed (frame, fork, wheel) rows, when a selection outside the table is solved, then the IIS names the table constraint and the response time is < 200 ms.
- **AC-8.15** (FR-8.17, BR-8.17) Given service product "Calibration – hourly" in category CAL, when a tracking dimension group is assigned, then 422 `PRD-SVC-STOCKED`; its number is `SVC-CAL-00001`.

1.8.9 Test plan and fixtures

Fixtures: `product/demo-catalog.json` (500 items, 50 masters, 12 item groups, 4 model groups, 3 category hierarchies); `product/master-3x10x10.json`; `product/configurator-50.yaml` (bicycle model: frame, wheel, gear, color with 50 constraints; 100 selection sets labeled); `product/gs1-vectors.csv` (40 valid from GS1 check-digit calculator documentation, 40 mutated); `product/uom-product.csv` (30 product conversions); `product/mara-extract.csv` and `product/inventtable-extract.csv` (10,000 rows each).

Test	Type	Target	Pass criterion
Explosion performance and idempotency	performance + unit	explode job	< 5 s for 1,000; second run 0 creates
Variant lookup	property	lookup index	100% hits, p95 < 10 ms
Configurator fixture	table-driven	CP-SAT worker	100/100 expected outcomes, each < 200 ms
Release cascade	property	defaults resolver	10k lookups, 0 mismatches
GTIN vectors	unit	validator/allocator	80/80
Lifecycle transitions	state-machine property	product/released	every illegal transition rejected
Attribute validation	table-driven	attribute types	120 cases
Search latency	performance	typeahead	p95 < 100 ms at 1M
Event contracts	contract	6 events	registry accepts; consumer stubs decode

Reconciliation: count(variants) = |size| × |color| × |style| × |config| for fully exploded masters; count(released products) per entity = count in demo catalog; count(GTINs) = count(products with identifiers) in fixtures.

1.8.10 Deployment and operations

Helm chart `passage-product` (Go service ×2, CloudNativePG `product-db`, Debezium connector, `product-configurator` Python worker with OR-Tools, 2 replicas, 1 CPU / 1 GiB each). Config: `PRODUCT_MAX_VARIANTS=10000`, `PRODUCT_EXPLODE_CHECKPOINT=500`, `PRODUCT_CFG_TIME_LIMIT_MS=2000`, `PRODUCT_GS1_PREFIX` per tenant in the tenant settings entity, object storage bucket `passage-product-media`. Feature flags: `product.configurator`, `product.gs1_allocation`, `product.catalogs`. Scheduled jobs: nightly incomplete-attribute report; weekly orphan media sweep. SLOs: read p95 < 50 ms; release of 1,000 products < 60 s; solve p95 < 500 ms. Alerts: explosion job failed; configurator worker queue depth > 100; media bucket errors. Runbook: "Explosion job stuck" (Temporal `explode-{master_id}` inspect, resume from checkpoint), "Re-apply release profile". Rollback: previous digest; additive migrations only.

1.8.11 Migration from AX / SAP

Source	Passage entity	Notes
AX EcoResProduct (DisplayProductNumber, ProductType, InstanceRelationType), EcoResProductTranslation (Name, Description)	Product, translations	InstanceRelationType EcoResProductMaster → subtype master
AX EcoResProductMaster + EcoResProductDimensionGroup, EcoResProductMasterDimensionValue (Size/Color/Style/Configuration tables)	DimensionGroup, DimensionValue	dimension names kept as AX names unless renamed in mapping
AX EcoResDistinctProductVariant + EcoResProductVariantDimensionValue	ProductVariant	dimension_values from the four value tables
AX EcoResStorageDimensionGroup / EcoResTrackingDimensionGroup + *DimensionGroupFldSetup	storage/tracking groups	field flags per dimension
AX InventTable (ItemId, Product, ItemGroupId, ModelGroupId, DataAreaId, NetWeight, UnitVolume), InventTableModule (UnitId per module), InventItemSetupFilter / InventItemInventSetup (default site/warehouse), InventItemPurchSetup, InventItemSalesSetup	ReleasedProduct, ReleasedProductSiteDefault	one row per DataAreaId; InventDim resolved for site/warehouse defaults
AX InventItemGroup, InventModelGroup (InventModel, PhysicalNegative, FinancialNegative, PostPhysical, PostFinancial, Stocked)	ItemGroup, ItemModelGroup	InventModel enum → costing_method
AX UnitOfMeasureConversion with Product ≠ 0	ProductUomConversion	
AX EcoResAttribute*, EcoResCategory*, EcoResProductCategory, EcoResProductInstanceValue	attributes, categories, values	typed values from EcoResTextValue/IntValue/FloatValue/BooleanValue/DateTimeValue
AX InventItemGTIN (GTIN, ItemId, UnitId)	ProductIdentifier	check digit validated; failures to exception queue
SAP MARA (MATNR, MTART, MATKL, MEINS, MSTAE, XCHPF, SATNR, KZKFG, BRGEW/NTGEW/VOLUM)	Product, dimension/tracking hints	MTART → item group suggestion; XCHPF → batch active; SATNR → master link
SAP MAKT (MAKTX by SPRAS)	translations	
SAP MARC (WERKS, DISMM, PLIFZ, BSTMI, BSTRF, MMSTA, SERNP), MARD (LGORT), MVKE (VKORG, VTWEG), MBEW (BWKEY, VPRSV, BKLAS)	ReleasedProduct per company code of the plant; site defaults per plant	PLIFZ → lead_time_days; BSTMI/BSTRF → min/multiple qty; VPRSV S/V → standard/moving_average
SAP MARM (MEINH, UMREZ, UMREN), MEAN/MARM-EAN11	ProductUomConversion, ProductIdentifier	factor = UMREZ/UMREN

Source	Passage entity	Notes
SAP KLAH/KSML/CABN/CAWN/AUSP (class type 001)	AttributeSet, Attribute, values	characteristic data types CHAR/NUM/DATE mapped; multi-value → enum set
SAP CUKB/CUOB/CUKN (object dependencies)	ConfigurationModel	only `precondition/selection condition` dependency types with simple syntax are translated; others logged to exception queue

Validation: every released product resolves an item group and model group; units per module convertible to base; variants resolve to a master. Not migrated: AX `InventSum`/on-hand and `InventTrans` (wave 16 opening balances); SAP `MBEW` prices (wave 20); SAP `MLAN` tax classification (wave 13); AX retail-specific product tables (non-goal); LO-VC procedures and functions (no equivalent; re-authored as constraints).

1.8.12 Dependencies, risks, open questions

Depends on: wave 6 (sites, warehouses, UoM classes, legal entities), wave 5 (number sequences, import tool, media attachments), wave 4 (Temporal jobs, outbox). Does not depend on wave 7 (parties) except for vendor-item references, deferred to wave 17.

Risk	Likelihood	Impact	Mitigation
Configurator expression language grows into LO-VC complexity	Medium	Medium	Freeze grammar at 8 operators in v1; extensions via sanctioned functions only
Customers with 5+ product dimensions (apparel) exceed the 4-dimension model	Medium	Medium	Extra dimensions modeled as required attributes; evaluate a 5th dimension slot by wave 26
Release profiles diverge from AX default order settings semantics	Low	Medium	Fixture comparison against InventItemInventSetup exports

Open questions: (1) Should `item_number` be allowed to differ from `product_number` per entity (AX allows it)? Proposed: allowed but defaults equal; decide by wave 10. (2) Is moving-average a separate costing method or a parameter of weighted average? Proposed: separate, matching AX; decide by wave 20. (3) Catalogs: product-level or released-product-level membership? Proposed: product-level with entity filter at read; decide by wave 26.

1.8.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`product.number_mode`	tenant	enum	sequence	sequence / manual / both	How product numbers are assigned (FR-8.1)	PIM parameters → Number sequences (Product number)	Material number ranges per material type (MMNR)
`product.number_pattern`	tenant	regex	`^[A-Z0-9][A-Z0-9._-]{0,39}$`	regex	Manual number validation	Product number format	OMSL material number format (TMCNM: length, leading zeros, lexicographic)
`product.variant_nomenclature`	tenant / master	template	`{master}-{size}-{color}-{style}-{config}`	tokens, separators, max 40 chars	Variant numbers (FR-8.2, BR-8.11)	Product nomenclature (EcoResProductNomenclature, D365)	Material variant number assignment
`product.max_variants`	tenant	int	10,000	100–10,000	Explosion cap	—	—
`product.explode_checkpoint`	tenant	int	500	100–2,000	Checkpoint size	—	—
`product.default_dimension_groups`	tenant / item group	FK × 3	none	groups	Pre-fill storage/tracking groups at create	PIM parameters → default dimension groups	Material type defaults (OMS2), batch management default
`product.release_profile`	legal entity × item group	FK	none	release profiles	Defaults at release (FR-8.4)	Default order settings templates / released product templates (D365)	Reference material on MM01 / material type views (T134)
`product.lifecycle.default_state`	tenant × target	FK	Prototype (product), Active (released)	lifecycle states	State on create (FR-8.14)	Product lifecycle states → Default for new products	MARA-MSTAE default
`product.lifecycle.process_policy`	state × process	enum	seeded per FR-8.14	allow / warn / block	Process enablement	Product lifecycle state → enabled business processes	OMS4 material status (T141: purchasing, BOM, routing, MRP, inventory messages)
`product.gs1.company_prefix`	tenant	digits	none	6–12 digits	GTIN allocation (FR-8.9)	GTIN setup / Bar code setup	EAN categories (T006M) and EAN number ranges (W4ES)
`product.barcode.symbologies`	tenant	set	EAN-13, GS1-128, DataMatrix	+ UPC-A, ITF-14, Code 128, Code 39, QR	Rendering choices	Bar code setup (BarcodeSetup)	—
`product.cfg.time_limit_ms`	tenant	int	2,000	200–10,000	Solver limit (BR-8.8)	Product configuration parameters (solver strategy)	Configuration profile settings (CUCO)
`product.media.limits`	tenant	size / count	10 MB, 20 per product	1–50 MB, 1–100	FR-8.11 limits	Product image settings	DC10 originals per document
`product.translation.fallback`	tenant	BCP 47	tenant base language	languages	BR-8.10	Product translations	MAKT fallback to logon language
`product.attr.required_enforcement`	tenant	enum	on_activate	on_create / on_activate / off	BR-8.9	Category attribute "Mandatory"	CT04 "entry required"
`product.doc.types`	tenant	records	IMG, DOC, DRW, SPC	FR-8.20 fields	Document behavior	Document types (DocuType)	DC10 (TDWA, TDWO, TDWP)
`product.service.category_sequence`	service category	sequence	`SVC-{cat}-{#####}`	sequences	Service numbers (FR-8.17)	—	Service master number ranges and service categories (IMG MM-SRV)
`product.relation.types`	tenant	list	7 seeded types	+ custom	FR-8.13	Product relation types (EcoResProductRelationType)	Cross-selling profiles (SD material determination)

1.8.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
All products and masters	list inquiry	subtype, type, state, category, attribute	number, name, subtype, dimension groups, state	`product.product`	EcoResProductListPage	MM60 material list, MM03
Released products per legal entity	list inquiry	legal entity, item group, model group, state, blocks	item number, groups, units, site, warehouse, state	`product.released_product`	Released products; SSRS Item base data	MM60 by plant
Variants of a master	list inquiry	master, dimension values	variant number, dimension values, released entities	`product.product` (variant)	EcoResProductVariantListPage	MM60 filtered by SATNR
Incomplete products	KPI	category, legal entity	missing attributes, missing defaults, counts by category	completeness projection	Incomplete attribute report (nightly)	MM50 list of extendable materials
Lifecycle and process policy	KPI	state, process	products per state, blocked processes	`released_product` × policy	Product lifecycle state inquiry (D365)	MM60 by MSTAE/MMSTA
GTIN register	list inquiry	prefix, type, unit	GTIN, product, unit, primary, check digit	`product.product_identifier`	Bar codes / GTIN codes	MEAN / MARM-EAN11 list
Attribute search	drill-down	attribute, value range, category	product, attribute values	attribute value projection	Product attributes inquiry	CL30N, CT06
Configuration models and solves	drill-down	model, version, date range	solves, feasible %, p95 time, IIS frequency	solve log	Configuration inquiries; SSRS Product configuration model	CU50 simulation, CUTRACE
Item model and dimension group usage	list inquiry	group	products per group, costing method, flags	`product.item_model_group`	SSRS Item model group / Dimension group listings	OMS2, T134 overview
Document search	list inquiry	type, status, attribute, link, text	number, part, version, description, status, links	`product.document_info_record`	Document handling (DocuView)	CV04N
Product relations	list inquiry	type, product	from, to, type, validity	`product.product_relation`	Related products	VB43 material determination display
Group change and obsolete job history	drill-down	job, date	items changed, skipped (reason), deleted, obsolete	job projection	Periodic task history	MM17 log

1.8.15 Printed and electronic documents

Document	Trigger	Layout elements (header/line/footer fields)	Formats	Numbering sequence	Legal / retention requirement
Product data sheet	on demand	header: product number, name, primary image; lines: attributes, units, conversions, GTINs; footer: legal entity, date, translation language	PDF, email	none	none
Product barcode label	on demand or job PRD-J4	GTIN, human-readable interpretation, description, unit, optional batch/expiry AIs for GS1-128	PDF, ZPL (printer routing in wave 31)	none	GS1 General Specifications: symbol X-dimension, quiet zones, HRI
Product catalog export	on demand per catalog	header: catalog, validity, supplier; lines: product, description, unit, GTIN, attributes, price placeholder	BMEcat 2005 XML, CSV	`CAT-{yyyy}-{####}`	none
Configuration summary	solve saved as configuration	header: model, version, customer; lines: attribute selections; footer: constraint check result	PDF	saved configuration id	retained with the sales document (wave 27)
GDSN item publication	none in this wave	—	—	—	GDSN data-pool synchronization deferred to wave 35 integration hub

1.8.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
PRD-J1 Variant explosion	on demand	master_id	500-row checkpoints; existing combinations skipped (BR-8.3)	1,000 variants < 5 s; 10,000 < 60 s	resume from checkpoint	job failed
PRD-J2 Incomplete-attribute report	nightly 03:00	category_id	read-only	< 10 min at 1M products	retried next night	none
PRD-J3 Orphan media sweep	weekly Saturday 02:00	storage prefix	marks then deletes after 7 days	< 30 min	stops on storage errors	bucket errors
PRD-J4 Barcode batch generation	on demand / nightly for new GTINs	product_id range	output keyed by (GTIN, symbology)	10,000 labels < 5 min	per-item error list	failures > 1%
PRD-J5 Group change conversion	at effective date (Temporal timer)	released_product_id	per-item idempotent; BR-8.15	1,000 items < 2 min	item skipped with reason	skipped > 0
PRD-J6 Mass obsolete / delete	on demand after dry run	legal_entity_id	per-item; delete only with zero references	10,000 items < 10 min	item set obsolete instead	none
PRD-J7 Document text extraction	on upload (queue)	original id	content hash	< 20 s per 10 MB PDF	retried 3×; DIR searchable by metadata	queue depth > 500
PRD-J8 Saved-configuration revalidation	on model activation	model_id	stale flag recomputed	10,000 configurations < 2 min	per-item error	stale count

1.8.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
View products	`GET /product/v1/**`; PIM workspace	none	every business role
Maintain products	product create/edit, variants, attributes, relations, identifiers	none	Product manager, Master data steward
Release products	`:release`, release profiles	none	Product manager, Master data steward
Maintain released products	released-product edits, site defaults, blocks	none	Inventory manager, Purchasing manager
Maintain product setup	item groups, model groups, dimension groups, lifecycle states and policies	Post inventory adjustments (wave 16)	Product manager lead, Cost accountant
Run product group conversions	PRD-J5, PRD-J6	Maintain product setup (maker-checker on costing-method change)	Cost accountant

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
Configure products	configuration models, decision tables, `:solve` test	none	Product configurator modeler
Maintain product categories	category hierarchies, attribute sets	none	Category manager
Maintain documents	DIR create, versions, originals, links	Release engineering documents (FR-9.24) on the same DIR	Engineer, Document controller
Allocate GTINs	`identifiers?allocate=true`, prefix settings	none	Master data steward

1.8.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
PRD-NUM-FORMAT	error	"Product number {n} does not match the pattern."	Regex failure	Correct number	FR-8.1
PRD-NUM-IMMUTABLE	error	"Product number cannot change after release."	Edit after release	Create a new product	FR-8.1
PRD-VAR-CAP	error	"Explosion would create {n} variants; limit is {max}."	Cartesian product above cap	Reduce values	FR-8.2
PRD-VAR-DIMVAL-UNKNOWN	error	"{value} is not a value of dimension {dim}."	Unknown dimension value	Add the value first	FR-8.2
PRD-RELEASE-DRAFT	error	"Draft products cannot be released."	Release of draft	Activate first	FR-8.4
PRD-RELEASE-DEFAULT-MISSING	error	"Missing {field} for {entity}/{site}."	No profile or site default	Add the default	FR-8.4
PRD-DIMGRP-LOCKED	error	"Dimension group is locked by inventory transactions."	Change after transactions	Use the group-change job	FR-8.3, BR-8.5
PRD-UOM-UNDERFLOW	error	"Conversion rounds {qty} to zero."	Precision too low	Increase precision	BR-8.6
PRD-GTIN-CHECKDIGIT	error	"GTIN {v} has an invalid check digit."	mod-10 failure	Correct GTIN	BR-8.7
PRD-GTIN-DUPLICATE	error	"GTIN {v} already belongs to {product}."	Duplicate	Use another GTIN	BR-8.7
PRD-CFG-TIMEOUT	error	"Configuration not solved within {ms} ms."	Solver limit	Simplify or retry	BR-8.8
PRD-CFG-INFEASIBLE	error	"Selections conflict: {constraints}."	IIS found	Change selections	FR-8.8
PRD-ATTR-REQUIRED	error	"Required attributes missing: {list}."	Activation incomplete	Fill attributes	BR-8.9
PRD-ATTR-RANGE	error	"{attr} must be between {min} and {max}."	Out of range	Correct value	FR-8.6
PRD-LC-BLOCKED	error	"{process} is blocked for state {state}."	Policy block	Change state or use another product	BR-8.12
PRD-LC-WARN	warning	"{process} is discouraged for state {state}."	Policy warn	Confirm	BR-8.12
PRD-REL-SELF	error	"A product cannot be related to itself."	Self-relation	Choose another product	BR-8.14
PRD-GRPCHG-ONHAND	warning	"Skipped: on-hand {qty} on {date}."	Precondition failed	Empty stock, reschedule	BR-8.15
PRD-SVC-STOCKED	error	"Service products cannot be stocked."	Stocking setup on service	Remove group	BR-8.17
PRD-DOC-ORIGINAL-LIMIT	error	"Document type {t} allows {n} originals."	Too many files	Remove a file or new version	FR-8.20

1.8.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Product read	3M/day; peak 5,000/min	8 / 50 / 90 ms	500 req/s per replica	~3 KB per product + attributes	PK; (tenant_id, product_number) unique	read replicas (CNPG) absorb load
Variant lookup	10M/day from order lines	2 / 10 / 20 ms	2,000 req/s	none	unique (master_id, dimension hash)	unchanged
Release 1,000 products × 3 entities	20/day	25 / 60 / 90 s	2 concurrent per tenant	~2 KB per released product	(legal_entity_id, item_number) unique	queued; no timeout
Process-policy query	20M/day	1 / 10 / 20 ms	5,000 req/s (cached)	none	in-memory policy cache	stale cache ≤ 60 s
Typeahead search	1M/day	25 / 100 / 180 ms at 1M	300 req/s	trigram ~1 GB at 1M	GIN trigram on search_name, GTIN	results capped at 10
Configurator solve	20k/day; peak 100/min	60 / 200 / 500 ms (50 constraints)	20/s per worker	solve log ~1 KB	worker queue	HPA to 6 workers; queue > 100 alerts
Document search	50k/day	80 / 300 / 600 ms at 2M DIRs	50 req/s	originals 2–5 TB/yr in object storage; tsvector ~5%	GIN tsvector; attribute jsonb GIN	full-text limited to 1,000 hits

1.8.20 Localization and compliance hooks

Item	What varies	Country packs touching it (wave 48)	Extension point
Commodity and tax classification	HS/CN8 for Intrastat (DE, FR); HSN/SAC codes mandatory on IN GST invoices (6 digits above INR 5 crore turnover); SAT c_ClaveProdServ and c_ClaveUnidad on MX CFDI 4.0; UNSPSC optional	DE, FR, IN, MX	attribute sets supplied by pack; `ProductTaxClassificationHook` read by wave 13
Units and labeling	US FPLA dual metric/US customary net quantity; CA Consumer Packaging and Labelling Act bilingual; EU Directive 76/211/EEC ℮ mark	US, CA, DE, FR	unit display rules; translation requirements per country
GS1 prefixes	Prefix issued by GS1 US, UK, Germany, France, India, Mexico, Canada	all	prefix per legal entity allowed
Product names and descriptions	FR/CA-QC French-language requirement; MX Spanish CFDI description	FR, CA, MX	FR-8.12 translations; BR-8.10 fallback
Regulated attributes	Hazardous substance data (REACH, GHS), export control classification (ECCN/EAR, EU 2021/821), RoHS	all	attribute sets contributed by wave 48 compliance packs
Document retention	Technical documentation kept 10 years after last unit placed on market (EU MDR Art. 10(8), Machinery Regulation 2023/1230)	DE, FR	retention class per document type (FR-8.20)

1.8.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.15.1 PIM, Common — Released product → Related products, EcoResProductRelation; register §12.99.4 #141	PARTIAL-8	ADDED as FR-8.13	Symmetric types, sales suggestions read in waves 26–27
AX §12.15.1 PIM, Common — Released product → Product lifecycle state (D365), EcoResProductLifecycleState; register §12.99.4 #143	PARTIAL-8	ADDED as FR-8.14	Per-process allow/warn/block
AX §12.15.3 PIM, Periodic — Change product dimension groups / Change item model group; register §12.99.4 #146	PARTIAL-8	ADDED as FR-8.16	Callback-checked conversion job PRD-J5
AX §12.15.3 PIM, Periodic — Delete released products / Mark obsolete; register §12.99.4 #147	PARTIAL-8	ADDED as FR-8.15	Job PRD-J6 with reference callbacks
AX §12.15.4 PIM, Setup — Product lifecycle states (D365); register §12.99.4 #148	PARTIAL-8	ADDED as FR-8.14	Parameters `product.lifecycle.*`
AX §12.16.4 Procurement, Setup — Categories → category vendor assignment, category questionnaires; register §12.99.4 #162	PARTIAL-8	DEFERRED to wave 17	Category hierarchies exist here (FR-8.6); vendor-per-category assignment and questionnaires belong to sourcing in wave 17, as the register proposes
AX §12.34.1 ECM, Common — Product lifecycle states (per version, controlling process enablement); register §12.99.4 #283	PARTIAL-8	ADDED as FR-8.14	Engineering-version states applied by FR-9.14
AX §12.34.4 ECM, Setup — Product lifecycle states (enabled business processes per state, default states); register §12.99.4 #297	PARTIAL-8	ADDED as FR-8.14	`default_for_new` per target
SAP §13.15 MM-SRV Master data — AC01/AC02/AC03/AC04/AC06 service master; register §13.99.2 #163	PARTIAL-8	ADDED as FR-8.17	Service categories, purchasing text, service catalog
SAP §13.15 MM-SRV Customizing — Service master number ranges, service categories, organizational status, field selection (ASMD); register §13.99.2 #181	PARTIAL-8	ADDED as FR-8.17	Number range per category; field selection via wave-3 metadata field rules
SAP §13.32 LO-VC Master data — CU60/CU61/CU62/CU65/CU66/CU67 variant tables and functions; register §13.99.2 #356	PARTIAL-8	ADDED as FR-8.18	Table grid added; ABAP variant functions replaced by expression functions
SAP §13.32 LO-VC Master data — CU7 / CUCPT configuration templates; register §13.99.2 #359	PARTIAL-8	ADDED as FR-8.19	Tenant and customer scope, stale detection
SAP §13.35 DMS Master data — CV01N/CV02N/CV03N document info record; register §13.99.2 #374	PARTIAL-8	ADDED as FR-8.20	Status network, ECO versioning and signatures in FR-9.24/FR-9.25
SAP §13.35 DMS Master data — CV04N find document; register §13.99.2 #375	PARTIAL-8	ADDED as FR-8.21	Attribute, link and full-text search
SAP §13.35 DMS Master data — Fiori Manage Documents / Manage Document Info Records; register §13.99.2 #384	PARTIAL-8	ADDED as FR-8.20	Document screens in the PIM and engineering workspaces
SAP §13.35 DMS Reports — CV04N document search; register §13.99.2 #385	PARTIAL-8	ADDED as FR-8.21	"Document search" in §1.8.14
SAP §13.35 DMS Customizing — DC10 document types (TDWA, TDWS, TDWO, TDWP); register §13.99.2 #392	PARTIAL-8	ADDED as FR-8.20	Parameter `product.doc.types`

1.9 Wave 9: Engineering Data: BOMs, Routings, Resources & Change Management

1.9.1 Scope and business context

AX stores bills of material as `BOMTable` (BOM id) with `BOMVersion` (which product/site/date range uses which BOM) and `BOM` lines; routes as `RouteTable`/`RouteVersion`/`RouteOpr` with `RouteOprTable` operations, `WrkCtrTable`/`WrkCtrResourceGroup` resources, `WrkCtrCapability` and `WorkCalendarTable`; D365 adds engineering change management (ECR/ECO, product versions). SAP keeps `MAST` (material-to-BOM link), `STKO` (BOM header), `STPO` (BOM items), `STAS` (alternatives), routings in `PLKO`/`PLPO`/`PLAS` with `MAPL` assignment, work centers in `CRHD`/`CRCA`/`KAKO` (capacity), and change numbers in `AENR` (CC01). Wave 9 delivers BOM headers with versions and lines (effective dates, scrap, phantoms, by-/co-products, process formulas), routing versions with operations, relations and resource requirements by capability, resources with calendars, an ECR/ECO workflow with where-used impact analysis, and an explosion/implosion API. The cost roll-up only records the structure the wave-39 roll-up needs.

Out of scope: production orders (wave 21), capacity scheduling and Gantt (wave 23), potency/catch weight execution (wave 24), costing sheets and standard-cost roll (wave 39), PLM adapters beyond the event contract.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX BOM designer (BOMDesigner form) loads a 7-level, 2,000-component BOM in 30–90 s and the "Calculate" explosion runs as a batch (BOMCalcJob) that users wait minutes for	Recursive X++ explosion row by row over BOMVersion/BOM	Closure-table-backed explosion service; 2,000 components in < 500 ms (FR-9.9)	Explosion 60 s–5 min→<0.5 s
SAP CS03 shows one BOM alternative at one date; "what is active on 2026-06-01" means CS03 with a key date, CS80 where-used, and CC03 change number, three transactions and still no diff	BOM validity scattered across STKO/STPO DATUV with AENNR change numbers	Versioned BOMs with as-of resolution, side-by-side version diff and effective-date preview (FR-9.1, FR-9.3)	Transactions to answer "what is active when" 3→1; clicks 15→2
Engineering change in AX 2012 is an unaudited edit of BOM lines; two engineers editing the same BOM overwrite each other; D365 ECM exists but is a separate product-version model many customers never turn on	No change object in the core BOM model	ECR→ECO workflow is the only path to change an active version; optimistic locking on drafts (FR-9.7, BR-9.3)	Unaudited changes to active BOMs 100%→0; lost-update incidents 0
Where-used (AX BOMWhereUsed, SAP CS15) is a flat list; an engineer cannot see which finished goods, orders, and sites a component change impacts before releasing	No impact analysis in the change flow	ECO impact analysis: multi-level where-used across sites with affected product count and open-order hooks (FR-9.8)	Impact assessment effort 2 h→<10 s
Work center capacity in SAP (CR02 capacities, KAKO) and AX (WrkCtrTable calendar + efficiency %) is keyed in several places; routing lead time calculated by hand in spreadsheets	Calendar, efficiency and operation times modeled separately	One calendar model with shifts and efficiency; routing lead-time calculation API (FR-9.6, FR-9.10)	Lead-time calculation manual→API, fixture-verified

1.9.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
BOM header / version	BOMTable (BOMId, Name, ItemGroupId, SiteId), BOMVersion (ItemId, BOMId, FromDate, ToDate, FromQty, ToQty, Active, Approved)	STKO (STLNR, STLAL, BMENG, DATUV), MAST (MATNR, WERKS, STLNR, STLAL), STAS, CS01/CS02/CS03	engineering-service `BomHeader`, `BomVersion`	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
BOM lines	BOM (LineNum, ItemId, BOMQty, BOMQtySerie, UnitId, FromDate, ToDate, ScrapConst, ScrapVar, BOMType Item/Phantom/Pegged/Vendor, Position, RoundUp, Formula)	STPO (POSNR, IDNRK, MENGE, MEINS, AUSCH, DUMPS phantom, ALPOS, POSTP item category, DATUV)	`BomLine`	full
Phantoms, by-/co-products	BOMType Phantom; co-products/by-products in PmfCoByProduct (process industries)	DUMPS, co-products MARC-KZKUP / STPO negative qty by-products	`BomLine.line_type` ∈ {item, phantom, pegged_supply, vendor, co_product, by_product}	full
Formulas (process)	PmfFormula, formula versions, scalable, yield percent	PP-PI master recipes (C201), STKO BMENG base qty	`BomHeader.kind = formula`, `yield_percent`, `scalable` flag	partial (batch execution in wave 24)
Routing header / version	RouteTable (RouteId), RouteVersion (ItemId, RouteId, FromDate, ToDate, Active, Approved)	PLKO (PLNNR, PLNAL, DATUV), MAPL (MATNR, WERKS, PLNNR), CA01/CA02/CA03	`RouteHeader`, `RouteVersion`	full
Operations and times	RouteOpr (OprNum, OprId, SetUpTime, ProcessTime, ProcessPerQty, TransitTime, QueueTimeBefore/After, OprPriority, NextOprNum, LinkType), RouteOprTable	PLPO (VORNR, ARBPL, VGW01..06 standard values, LAR01..06 activity types, PLIFZ), PLAS	`Operation` with times, `OperationRelation` (sequential/parallel via `next_operation_no` + priority)	full
Resources / work centers	WrkCtrTable (WrkCtrId, WrkCtrType Vendor/Machine/HumanResources/Tool/Location, Efficiency, CalendarId), WrkCtrResourceGroup, WrkCtrCapability, WrkCtrCapabilityResource	CRHD (ARBPL, VERWE category, KAPID), CRCA capacities, KAKO (capacity header, utilization), CRHD-STEUS control key	`Resource`, `ResourceGroup`, `Capability`, `ResourceCapability`	full
Calendars	WorkCalendarTable, WorkCalendarDate, WorkCalendarDateLine (shifts), WorkCalendarTemplate	Factory calendar TFACD/TFACS (SCAL), shift definitions	`WorkCalendar`, `CalendarDay`, `Shift`	full
Engineering change	D365 ECM: EngChgRequest, EngChgOrder, product versions, impact analysis, release to legal entities	ECM: AENR change master (CC01), ECR/ECO (CC31/CC32), effectivity	`EngineeringChangeRequest`, `EngineeringChangeOrder`, Temporal workflow	full
Where-used / explosion	BOMWhereUsed, BOM calculation (BOMCalcTable), BOM designer	CS11/CS12/CS13 (explosion), CS15 (where-used)	`/engineering/v1/bom:explode`, `:implode`, `:where-used`	improved (latency, as-of)
Cost roll-up	BOMCalcTable, costing versions	CK11N/CK40N	structure hooks only; roll-up in wave 39	partial by design
Engineering organization and release product structure	EngChgEngineeringOrganization, EngChgReleaseProductStructure, product release policies, Open product releases (acceptance)	ALE distribution of material and BOM (BD10, BD30, distribution model BD64, status BD87); MDG-M replication	`EngineeringOrganization` flag, `ReleasePolicy`, `ReleaseSession` / `ReleaseLine` (FR-9.14 – FR-9.16)	improved (one workflow with per-line status)
Product readiness	Product readiness policies and checks (D365 ECM)	No direct equivalent (MM50 extension lists, material status)	`ReadinessPolicy`, `ReadinessCheckResult` (FR-9.17)	improved
Engineering categories and version nomenclature	EngChgEngineeringProductCategory, version nomenclature	Revision levels (OS13 material revision levels, CC01 revision)	`EngineeringCategory`, `VersionNomenclature` (FR-9.18)	full
BOM usage and category	BOMTable (production BOM vs formula), no usage concept	STLAN usage (OS20), STLTY category M/D/E/T/K/P	`BomHeader.usage`, `BomHeader.category` (FR-9.19)	partial (order, project, equipment BOM instances in waves 27, 41, 44)
Alternative items and discontinuation	Plan material substitution (D365 planned-order substitution), BOM line substitution	STPO-ALPGR alternative item group with ALPST strategy / ALPRF priority / EWAHR probability; discontinuation STPO-KZAUS / NFMAT follow-up material	`BomLine.alt_*`, `BomLine.discontinuation_*` (FR-9.20)	full (MRP logic in wave 22)
Serial and parameter effectivity	Date effectivity only (BOMVersion/BOM FromDate/ToDate)	ECM effectivity types: date, serial number, parameter (IMG ECM → Effectivity)	`Effectivity` record on ECO, version and line (FR-9.21)	improved over AX, full vs SAP
Super BOM and super routing	Constraint-based configuration BOM line and route operation templates (PCBOMLineTemplate, PCRouteOperationTemplate)	CS40–CS42 super BOM with selection conditions (CU01), CA01 operation dependencies	`selection_condition` and `qty_expr` on BomLine / Operation (FR-9.22)	full
Engineering documents	DocuRef attached to engineering versions	DIR status network (DC20), document BOM item category D (CS01), document structures (CV11–CV15), ECM-controlled versions	DIR status network, signatures, ECO-controlled versions, document lines, structures (FR-9.23 – FR-9.26)	full
CAD / PLM engineering BOM synchronization	PLM integration via Dataverse (Teamcenter, Windchill partner connectors)	ECTR Engineering Control Center (CAD BOM transfer, CAD desktop CDESK)	eBOM import contract, comparison report, polling job (FR-9.27)	partial (vendor connectors in wave 35)

1.9.3 Functional requirements

- **FR-9.1 BOM headers and versions** — Must. The system shall create BOM headers (bom_id from sequence `BOM`, name, kind item/formula, site optional) and versions binding a header to a product (released product optional for site-specific), with `effective_from`/`effective_to`, `from_qty`/`to_qty` ranges, `approved` and `active` flags.
  - At most one active version per (product, site, overlapping date range, overlapping quantity range); activation checks the overlap.
  - Active versions are immutable; changes go through an ECO (FR-9.7) or by creating a new draft version.
- **FR-9.2 BOM lines** — Must. The system shall maintain lines with line number, component product/variant, quantity per `series` (per N parents), unit, line type (item, phantom, pegged_supply, vendor, co_product, by_product), position/reference designator, constant and variable scrap percent, `round_up` policy (none, quantity, measurement, consumption), effective dates per line, resource consumption flag and operation number.
  - Maximum 5,000 lines per version; quantity numeric(18,6); scrap 0–100% with 2 decimals.
  - Co-/by-product lines carry a cost allocation percent (sum ≤ 100 across co-products) used in wave 24.
- **FR-9.3 As-of resolution and diff** — Must. The system shall resolve the applicable BOM and route version for (product, site, date, quantity) and produce a line-level diff between any two versions.
- **FR-9.4 Routing headers, versions and operations** — Must. The system shall create routes (route_id from sequence `ROUTE`), versions bound to a product/site with effective dates, and operations with operation number, operation id (reusable definition), setup time, run time per `process_qty`, queue before/after, transit time, overlap quantity, `next_operation_no` and `priority` (primary, secondary1..3) for parallel branches, scrap percent, and link type (hard/soft).
  - Maximum 500 operations per version; times in hours with 4 decimals.

- **FR-9.5 Resource requirements by capability** — Must. The system shall express each operation's requirement as resource, resource group, capability (with minimum level), resource type, or skill, with quantity of resources (default 1); scheduling (wave 23) resolves requirements, this wave validates that at least one resource satisfies each requirement at activation.
- **FR-9.6 Resources, groups, capabilities, calendars** — Must. The system shall maintain resources (type machine, tool, labor, vendor, location), resource groups per site, capabilities with levels and effective dates, and work calendars with templates (7 day patterns), shifts (start, end, efficiency percent), closed days, and a generator for a date range.
  - Calendar generation for 3 years × 5 shifts completes in < 2 s; calendars are inherited group → resource unless overridden.
- **FR-9.7 ECR/ECO workflow** — Must. The system shall create engineering change requests (title, reason, priority, affected products, requester) that move through a Temporal workflow `draft → submitted → under_review → approved | rejected`, and change orders that bundle draft BOM/route versions, go through approval, and on release atomically expire old versions (`effective_to = release_date − 1`) and activate new ones with `effective_from = release_date`.
  - Release date may be future; the workflow holds a timer until then.
  - An ECO may affect up to 500 versions; release across them is one transaction per service and compensated on failure.
- **FR-9.8 Where-used and impact analysis** — Must. The system shall return single- and multi-level where-used for a component as of a date, across sites, with distinct top-level products; an ECO impact report lists affected products, sites, active versions, and counts from consumer hooks (open production and planned orders — stubs return 0 until waves 21–22).
- **FR-9.9 Explosion and implosion API** — Must. The system shall explode a product to N levels (default all, max 25) as of a date and quantity, resolving phantoms inline, applying series and scrap, returning a tree and a flattened summed-component list; and implode (where-used tree) in the reverse direction.
  - 7-level, 2,000-component explosion < 500 ms; results are cacheable by (version set hash, date, qty).
- **FR-9.10 Routing lead-time calculation** — Should. The system shall compute operation and total lead time for a quantity: setup + run × qty / process_qty, queue and transit times added, parallel branches taking the max, and efficiency applied from the calendar; results in hours and in calendar days against the site calendar.
- **FR-9.11 Cycle and integrity checks** — Must. The system shall reject a line whose component's own active BOM (recursively) contains the parent (`ENG-BOM-CYCLE`), using the product-level closure table, and reject activation when any component is `obsolete`.
- **FR-9.12 Copy and template** — Should. The system shall copy BOM and route versions to another product or site, with optional line filtering, and keep a `copied_from_version_id` lineage.
- **FR-9.13 PLM event contract** — Could. The system shall publish ECO lifecycle events in a shape that a PLM adapter can consume and accept inbound `eco.proposed` payloads through the import tool.
- **FR-9.14 Engineering organization and engineering products** — Must. The system shall let a legal entity be flagged `engineering_organization` (one or more per tenant); products whose engineering category (FR-9.18) has `track_versions = true` are engineering products owned by one engineering organization, and their engineering versions (version code from the nomenclature, effective range, lifecycle state from FR-8.14, attached BOM and route versions) are maintained only in that organization.
  - Operational legal entities receive products, BOMs and routes by release (FR-9.15) and cannot edit engineering-owned fields (`ENG-OWNED-FIELD`); local fields (site defaults, local BOM site) stay editable.
  - When the category sets `version_as_dimension = true`, the engineering version is an additional product dimension `version` so that inventory (wave 16) and orders track it (D365 "version" product dimension).
- **FR-9.15 Release policies and release product structure** — Must. The system shall release engineering products to operational legal entities through `ReleaseSession` workflows (Temporal `release-{id}`), created manually, by job ENG-J7, or automatically on ECO release, governed by a release policy that lists what is copied: product, released-product template (FR-8.4 profile), BOM versions (copy yes/no, activate yes/no), route versions (copy yes/no, activate yes/no), attributes, default order settings, document links (FR-8.20).
  - Each (product, entity) line moves through BR-9.13; resubmission creates a new line; a site-mapping table per receiving entity maps engineering sites to operational sites.
  - Releasing a 1,000-component structure to 3 entities completes in < 2 min.
- **FR-9.16 Product acceptance** — Should. The system shall let a receiving entity whose policy sets `require_acceptance = true` review release lines in "Open product releases", adjust site defaults and BOM/route site, and accept or reject with reason; accepted lines create or update the released product and activate copied versions; entities without acceptance accept automatically.
- **FR-9.17 Product readiness policies and checks** — Must. The system shall evaluate readiness policies (per engineering category, item group or release policy) made of automated checks — item model group set, storage/tracking groups set, default order settings per site, active BOM for manufactured items, active route, tax group (wave 13 hook), coverage group (wave 22 hook), purchase price agreement (wave 17 hook), sales price (wave 26 hook), warehouse setup (wave 31 hook) — and manual checks (text, assigned user or team from FR-6.14, due in N days, mandatory flag).
  - A released product stays in a state with category draft named `readiness_pending` (blocks transactional processes via FR-8.14) until every mandatory check is passed or not applicable, then moves to the policy's target state (default Active).
  - Hooks of waves not yet live return `not_applicable`; checks re-evaluate on relevant events and by job ENG-J9.
- **FR-9.18 Engineering product categories and version nomenclature** — Must. The system shall maintain engineering categories (code, name, product type, `track_versions`, `version_as_dimension`, nomenclature, default readiness policy, default release policy, attribute set, `change_controlled` = ECO mandatory) and version nomenclatures made of ordered segments: number (width, start), letter (A–Z, option to skip I and O), constant text, date (`yyyyMMdd` or `yyMM`), with separators; e.g. `V-{NN}` → V-01, V-02; `{A}` → A … Z, AA.
- **FR-9.19 BOM usage and category** — Must. The system shall carry `usage` ∈ {production, engineering, costing, maintenance, sales, universal} (SAP STLAN 1–6) and `category` ∈ {material, document, equipment, functional_location, sales_order, project} on BOM headers; this wave maintains material and document BOMs, while equipment/functional-location (wave 44), sales-order (wave 27) and project/WBS (wave 41) BOMs reuse the same structures with an owner reference and the same explosion API.
  - `bom:resolve` and `bom:explode` accept `usage` with a configurable fallback order (default production → universal).
- **FR-9.20 Alternative item groups and discontinuation** — Should. The system shall store on BOM lines an alternative group (`alt_group` 2 chars, `alt_priority` 1–99, `alt_strategy` priority or usage_probability, `usage_probability_pct`) and discontinuation data (`discontinuation_group`, `discontinue_type` single or simultaneous, `follow_up_group`, `follow_up_component_id`, `discontinue_on` date); explosion returns groups unresolved with this metadata and planning (wave 22) resolves them against availability.
- **FR-9.21 Serial and parameter effectivity** — Should. The system shall support effectivity type date (default), serial (end-item product plus serial range from–to) and parameter (tenant-defined parameters such as `model_year`, `customer_party_id`, `order_type`, with values or ranges) on ECOs and on the versions and lines they release; `bom:resolve` and `bom:explode` accept `effectivity_context` (e.g. `{"serial":"SN100045","parameters":{"model_year":2027}}`).
  - Without context, date effectivity applies and serial/parameter-conditioned lines are excluded unless `include_unconditional = true`.
  - Maximum 20 effectivity parameters per tenant; serial ranges compare lexicographically after zero-padding to the product's serial mask length.
- **FR-9.22 Configurable (super) BOM and routing** — Should. The system shall let BOM lines and route operations carry a `selection_condition` in the FR-8.8 expression grammar over the parent master's configuration attributes and an optional `qty_expr` (e.g. `length_m * 2`); exploding with a solved configuration includes only lines whose condition is true and evaluates quantity expressions; activation validates conditions against the active model version (`ENG-COND-UNKNOWN-ATTR`).
- **FR-9.23 Document and text BOM lines** — Should. The system shall add BOM line types `document` (references a DIR number/part/version, no consumption, printed on production papers in wave 21) and `text` (≤ 2,000 characters, translatable); planning and costing explosion modes skip both, `mode=full` includes

them.

- **FR-9.24 Document status network and release with electronic signature** — Should. The system shall let document types (FR-8.20) define a status network (statuses with release flag, allowed transitions, required duty, signature required, meaning text); default network `in_work → in_review → released → obsolete` plus `in_work → cancelled`; a signature transition requires Keycloak step-up re-authentication (password or WebAuthn) and records signer, printed name, timestamp, meaning and a hash of all originals, satisfying 21 CFR 11.50 (manifestation) and 11.70 (signature/record linking).
- **FR-9.25 Document versions under ECO** — Should. For document types with `change_controlled = true`, the system shall release a new DIR version only as an affected object of an ECO; ECO release sets the new version to the released status and the previous one to obsolete in the same compensated transaction as BOM/route versions (BR-9.9); ECO impact analysis lists documents linked to the affected products.
- **FR-9.26 Document structures and where-used** — Could. The system shall let a DIR carry a document structure (ordered child DIRs with optional item text, versioned with the parent) and answer where-used for a document across parent documents and BOM document lines (`GET /engineering/v1/documents/{id}:where-used`); cycles are rejected (`ENG-DOC-CYCLE`).
- **FR-9.27 CAD/PLM engineering BOM synchronization contract** — Should. The system shall accept engineering BOM payloads (`ebom.published`, JSON schema in the registry: header part number and revision; lines with find number, component part number, quantity, unit, reference designators, file URIs) through the import tool or `POST /engineering/v1/ebom-imports`, resolve part numbers via product number or `ProductIdentifier` type `cad_part`, create a draft BOM version plus an ECR, and produce an eBOM-versus-active-mBOM comparison (added, removed, changed lines).
  - Source configuration: application name (SolidWorks PDM, Inventor Vault, Teamcenter, Windchill, other), transport (SFTP or REST pull), mapping rules (unit codes, find number → line_no, phantom attribute), auto-ECR on/off; job ENG-J8 polls every 15 min.
  - Native CAD file management and 3D viewing are not provided; vendor-specific connectors ship in wave 35.

1.9.4 Data model

BomVersion

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS
bom_header_id	uuid	yes	FK bom_header(bom_id, name, kind, site_id)
product_id	uuid	yes	FK product (variant allowed)
site_id	uuid	no	null = all sites
effective_from / effective_to	date	from yes	inclusive
from_qty / to_qty	numeric(18,6)	no	quantity-dependent versions
status	enum	yes	draft, active, expired, superseded
approved_by / approved_at	uuid, timestamptz	no	
base_qty / yield_percent	numeric(18,6), numeric(5,2)	yes / formulas	base_qty default 1
eco_id	uuid	no	ECO that activated it
copied_from_version_id	uuid	no	lineage
version	int	yes	optimistic lock

BomLine

Field	Type	Required	Notes
id / bom_version_id	uuid	yes	FK
line_no	int	yes	unique per version, step 10
component_product_id	uuid	yes	FK product
quantity / series / uom_id	numeric(18,6), numeric(18,6), uuid	yes	quantity per `series` parents
line_type	enum	yes	item, phantom, pegged_supply, vendor, co_product, by_product
scrap_const / scrap_var_pct	numeric(18,6), numeric(5,2)	yes	default 0
round_up	enum	yes	none, quantity, measurement, consumption
position / reference_designators	text(20), text[]	no	
operation_no	int	no	consumption at operation
effective_from / effective_to	date	no	line-level
cost_allocation_pct	numeric(5,2)	co/by only	
warehouse_id / resource_consumption	uuid, bool	no	

Operation

Field	Type	Required	Notes
id / route_version_id	uuid	yes	FK route_version (same shape as BomVersion minus qty range)
operation_no	int	yes	unique per version
operation_id	text(20)	yes	reusable operation definition
next_operation_no / priority / link_type	int, enum, enum	no	priority primary/secondary1..3; link hard/soft
setup_time_h / run_time_h / process_qty	numeric(10,4), numeric(10,4), numeric(18,6)	yes	run time per process_qty
queue_before_h / queue_after_h / transit_h	numeric(10,4)	no	
overlap_qty / scrap_pct	numeric(18,6), numeric(5,2)	no	
requirements	1:N	yes	OperationRequirement(kind resource/group/capability/type/skill, ref_id, level, quantity)
cost_category_setup_id / run_id	uuid	no	bound in wave 39

Resource / WorkCalendar — resource: id, code, name, type, site_id, group_id, calendar_id, efficiency_pct (default 100), capacity_per_day_h, exclusive bool, vendor_party_id (type vendor); calendar: id, code, template jsonb (7 days × shifts), generated_from/to; calendar_day: calendar_id, date, open bool, shifts jsonb [{start, end, efficiency_pct}], total_h.

EngineeringChangeOrder — id, eco_number (sequence `ECO`), title, reason, priority, status (draft, submitted, approved, scheduled, released, rejected, cancelled), release_date, affected_versions uuid[] (draft versions to activate, with the versions they supersede), impact jsonb snapshot, workflow_id.

Invariants: no two active BOM versions overlap for the same (product, site, date, qty); every active version's lines reference non-obsolete components; product closure table has no cycles; a released ECO's versions are all `active` and their predecessors `expired`; operation numbers referenced by `next_operation_no` exist in the same version. Indexing: `bom_version(product_id, site_id, effective_from, status)`; `bom_line(component_product_id)` for where-used; product closure (parent_product_id, child_product_id, depth) maintained on activation; `calendar_day(calendar_id, date)` PK.

Addendum entities (FR-9.14 – FR-9.27) — engineering_category: id, code text(20), name, product_type, track_versions bool, version_as_dimension bool, nomenclature_id FK, readiness_policy_id FK null, release_policy_id FK null, attribute_set_id FK null, change_controlled bool. version_nomenclature: id, code, segments jsonb [{type: number|letter|constant|date, width, start, skip_io, format, text}], separator text(3). engineering_version: id, product_id, version_code text(20), effective_from, effective_to, lifecycle_state_id, engineering_le_id, unique (product_id, version_code). release_policy: id, code, copy_bom bool, activate_bom bool, copy_route bool, activate_route bool, copy_attributes bool, copy_order_settings bool, copy_documents bool, require_acceptance jsonb {legal_entity_id: bool}; release_session: id, number `REL-{yyyy}-{#####}`, created_by, source (manual, eco, job), workflow_id; release_line: session_id, product_id, engineering_version_id, legal_entity_id, status (pending, sent, accepted, rejected, failed), reason, accepted_by. readiness_policy: id, code, checks jsonb [{key, kind auto|manual, mandatory, assignee, due_days}], target_state_id, continuous bool; readiness_check_result: released_product_id, check_key, status (pending, passed, failed, not_applicable), evaluated_at, signed_by. effectivity: id, type (date, serial, parameter), product_id null, serial_from, serial_to, parameters jsonb; referenced by eco, bom_version and bom_line via effectivity_id. bom_header gains usage, category, owner_ref; bom_line gains alt_group, alt_priority, alt_strategy, usage_probability_pct, discontinuation_group, discontinue_type, follow_up_group, follow_up_component_id, discontinue_on, selection_condition text, qty_expr text, document_ref (dir_id), text_id. document_status_network: id, statuses jsonb, transitions jsonb; document_signature: dir_id, version, transition, signer_id, meaning, signed_at, originals_hash. ebom_import: id, source_code, part_number, revision, payload jsonb, status, bom_version_id, ecr_id, unique (source_code, part_number, revision).

1.9.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/engineering/v1/bom-headers	Create header	`Idempotency-Key`	Maintain BOMs
POST	/engineering/v1/bom-versions	Create draft version (optionally copy from)	key header	Maintain BOMs
PUT	/engineering/v1/bom-versions/{id}/lines	Replace line set of a draft	full replace	Maintain BOMs
POST	/engineering/v1/bom-versions/{id}:activate	Activate outside ECO (requires duty "Activate BOM without ECO")	key header	Activate BOMs
GET	/engineering/v1/bom:resolve?product=&site=&date=&qty=	As-of version	n/a	View engineering data
POST	/engineering/v1/bom:explode	Body: product, site, date, qty, levels	pure	View engineering data
GET	/engineering/v1/bom:where-used?component=&date=&levels=	Implosion	n/a	View engineering data
GET	/engineering/v1/bom-versions/{a}:diff?against={b}	Line diff	n/a	View engineering data
POST	/engineering/v1/route-versions, PUT …/{id}/operations	Routes	as BOM	Maintain routes
POST	/engineering/v1/route-versions/{id}:lead-time?qty=	Lead-time calc	pure	View engineering data
POST	/engineering/v1/resources, /resource-groups, /capabilities, /calendars/{id}:generate	Resources and calendars	key header; generate idempotent per range	Maintain resources
POST	/engineering/v1/ecrs, /ecos, /ecos/{id}:submit, :approve, :release	Change management	key header	Maintain / Approve / Release engineering changes
GET	/engineering/v1/ecos/{id}/impact	Impact analysis	n/a	View engineering data

Validation: line set replacement runs cycle, unit-convertibility and component-state checks; activation validates overlap and resource satisfiability; `:release` requires `approved` status and all affected drafts to pass activation checks in a dry run before the transaction.

Event	Trigger	Payload fields	Consumers
dev.passage.engineering.bom.version.activated.v1	activation (direct or ECO)	bom_version_id, product_id, site_id, effective_from, effective_to, line_count, supersedes_version_id	planning, production, costing
dev.passage.engineering.bom.version.expired.v1	expiry by ECO	bom_version_id, effective_to, eco_id	planning, production
dev.passage.engineering.route.version.activated.v1	activation	route_version_id, product_id, site_id, operations[{no, resource requirements}]	planning, scheduling, production
dev.passage.engineering.resource.changed.v1	resource/calendar commit	resource_id, site_id, calendar_id, efficiency_pct, changed_fields[]	scheduling, shop floor
dev.passage.engineering.eco.submitted.v1 / approved.v1 / released.v1	workflow transitions	eco_id, eco_number, affected_versions[], release_date, approver_id	notification, audit, PLM adapter
dev.passage.engineering.calendar.generated.v1	generate	calendar_id, from, to, open_days	scheduling

1.9.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Engineering workspace	Tiles: BOMs, routes, resources, open ECRs/ECOs, versions expiring in 30 days	counts	View engineering data
BOM designer	Tree with inline line grid, add/remove, drag to reorder, as-of date picker, explode/implode toggle, version diff	line no, component, qty, series, unit, type, scrap, position, operation	Maintain BOMs
BOM versions	List per product, activate (with ECO or direct), copy	version, site, dates, qty range, status, ECO	Maintain BOMs
Route editor	Operation grid, network view (parallel branches), lead-time panel	op no, op id, next, priority, setup, run, process qty, requirements	Maintain routes
Resources and groups	New, capabilities tab, calendar assignment, utilization preview	code, type, site, group, calendar, efficiency	Maintain resources
Work calendars	Template editor, generate range, closed days	day, shifts, hours	Maintain resources
ECR / ECO	Create, add affected products/versions, impact report, submit/approve/release, schedule release date	number, title, status, affected, release date	Maintain / Approve / Release engineering changes
Where-used	Component picker, as-of date, levels, export CSV	parent, qty, level, site, version status	View engineering data

Interaction rules: BOM designer supports keyboard line entry (`Tab` through component/qty/unit, `Enter` adds next line at +10); component lookup searches wave-8 products filtered by lifecycle `active`; validation of cycles runs on line commit, overlap on activation; designer shows an "as of" ribbon with a date slider; network view in the route editor validates `next_operation_no` references live; personalization persists tree expansion depth.

1.9.7 Business rules and edge cases

- **BR-9.1** BOM/route version machine: `draft → active` (direct activation or ECO release), `active → expired` (effective_to passes or ECO), `active → superseded` (newer version activated for the same range by ECO), `draft → cancelled`; `active` is immutable.
- **BR-9.2** ECO machine: `draft → submitted → approved → scheduled → released`; `submitted | approved → rejected`; `draft | submitted | approved | scheduled → cancelled`; `released` terminal. `scheduled` waits on a Temporal timer for `release_date`.
- **BR-9.3** Draft versions use `version` optimistic locking; two engineers editing the same draft get 409 on the second save and a line-level merge view.
- **BR-9.4** Scrap: consumed quantity = `ceil_policy((qty_parent / series) × quantity × (1 + scrap_var_pct/100) + scrap_const)`; `round_up = quantity` rounds to the next whole unit, `measurement` to the unit's `decimal_precision`, `consumption` rounds the total across the order, not per line.
- **BR-9.5** Phantom lines are replaced by their own active BOM at explosion time, as of the same date; a phantom with no active BOM is reported as an explosion error, not silently skipped.
- **BR-9.6** Explosion depth cap is 25; exceeding it returns 422 `ENG-BOM-DEPTH`; cycle detection on insert makes this reachable only for legitimately deep structures.
- **BR-9.7** Parallel operations: an operation with `priority = secondary1` runs alongside the primary with the same `next_operation_no`; lead time takes the max of the branches; a secondary without a primary at the same join is rejected.
- **BR-9.8** Lead time with efficiency: run hours ÷ (resource efficiency × calendar shift efficiency); setup is not scaled by quantity; queue and transit are calendar-elapsed, not capacity-consuming.
- **BR-9.9** ECO release is all-or-nothing within engineering-service; failure after partial activation triggers Temporal compensation that reverts activated versions to `draft` and restores expired ones to `active`, then marks the ECO `approved` again with an error note.
- **BR-9.10** A release date earlier than today requires the duty "Backdate engineering changes" and is audited; a release date more than 365 days ahead is rejected.
- **BR-9.11** Deleting a resource is refused while any active route version requires it directly; capability-based requirements are unaffected.
- **BR-9.12** Co-product allocation percents on a version must sum to ≤ 100; by-products carry no allocation and may have negative cost in wave 24.
- **BR-9.13** Release line machine: `pending → sent → accepted | rejected`; `sent → failed` after 5 retries; `rejected` and `failed` are terminal for the line and resubmission creates a new line; an ECO release triggers release lines only for entities that previously accepted the product.
- **BR-9.14** A released product is ready when every mandatory check is `passed` or `not_applicable`; a passed automated check that later fails (e.g. the BOM expired) reverts the product to `readiness_pending` only when the policy sets `continuous = true` (default false); manual checks cannot be signed by the user who created the release session.
- **BR-9.15** Nomenclature increments: a number segment that overflows its width is rejected (`ENG-NOMEN-OVERFLOW`); letters roll Z → AA; date segments use the release date in the engineering organization's time zone; a version code is never reused for the same product.
- **BR-9.16** Alternative groups: with `usage_probability`, probabilities in a group sum to exactly 100; with `priority`, priorities are unique in the group; a line may not be both an alternative-group member and a follow-up component in the same group (`ENG-ALT-CONFLICT`).
- **BR-9.17** Effectivity precedence at resolution is serial > parameter > date; two serial-effective versions with overlapping ranges for the same (product, site) are rejected at ECO approval (`ENG-EFF-OVERLAP`).
- **BR-9.18** A signature binds DIR id, version, transition and the sha256 of all originals; changing originals after signature is refused (`ENG-DOC-SIGNED`); where the type requires independent review the signer cannot be the last editor.
- **BR-9.19** eBOM import is idempotent on (source, part number, revision): re-import updates the draft while it is still draft; once released, the same revision is rejected (`ENG-EBOM-REVISION-RELEASED`).

1.9.8 Acceptance criteria

- **AC-9.1** (FR-9.9) Given the 7-level, 2,000-component golden product, when exploded for qty 10 as of 2026-03-01, then the flattened component list equals `engineering/golden-explosion.csv` and p95 over 100 runs < 500 ms.
- **AC-9.2** (FR-9.3) Given BOM versions V1 (2025-01-01..2026-05-31) and V2 (2026-06-01..), when resolved as of 2026-01-01, then V1; as of 2026-06-01, then V2; as of 2026-05-31 with qty 1,000 where V3 covers qty ≥ 500, then V3.
- **AC-9.3** (FR-9.7, BR-9.2) Given an approved ECO replacing V1 with draft V2 at release date 2026-07-01, when the Temporal test clock passes 2026-07-01, then V1 is `expired` with effective_to 2026-06-30, V2 is `active`, both events are published, and the ECO is `released`.
- **AC-9.4** (FR-9.10) Given the 12 routing fixtures (sequential, two-branch parallel, overlap), when lead time is computed for qty 100, then results match fixture hours to 4 decimals.
- **AC-9.5** (FR-9.11) Given product A whose active BOM contains B, when a line adding A to B's draft is saved, then 422 `ENG-BOM-CYCLE` naming the path A→B→A.
- **AC-9.6** (FR-9.8) Given component C used in 30 products across 2 sites, when where-used is run as of today with levels = all, then 30 distinct top-level products and the ECO impact report shows the same set.
- **AC-9.7** (FR-9.5) Given an operation requiring capability "welding ≥ level 3" and no resource with it, when the route version is activated, then 422 `ENG-ROUTE-UNSATISFIED` names the operation.
- **AC-9.8** (BR-9.9) Given an ECO with 3 versions where the third fails activation, when released, then none are active, the ECO returns to `approved` with the error, and the audit shows the compensation.
- **AC-9.9** (FR-9.6) Given a 3-year calendar with 2 shifts and 10 closed days, when generated, then open-day count equals the fixture and generation < 2 s.
- **AC-9.10** (BR-9.4) Given a line qty 3 per series 2 with scrap_var 5% and scrap_const 1, round_up quantity, when the parent qty is 7, then consumption = ceil(7/2×3×1.05 + 1) = 13.
- **AC-9.11** (FR-9.15, FR-9.16, BR-9.13) Given engineering organization ENG1 and a policy that copies and activates BOM and route, requiring acceptance in USMF and not in DEMF, when product P version V-02 is released to both, then the DEMF line is `accepted` with active BOM and route on the mapped DEMF site and the USMF line waits in Open product releases; after acceptance, USMF has the released product and both events are published.
- **AC-9.12** (FR-9.17, BR-9.14) Given a readiness policy with 3 automated checks and 1 mandatory manual check, when P is released to USMF, then the released product is `readiness_pending` and a sales-order policy query returns `PRD-LC-BLOCKED`; after the manual check is signed and the BOM is active, the state becomes Active.
- **AC-9.13** (FR-9.18, BR-9.15) Given nomenclature `V-{NN}` with last version V-99, when the next version is requested, then `ENG-NOMEN-OVERFLOW`; given `{A}` with last Z, then AA.
- **AC-9.14** (FR-9.21, BR-9.17) Given an ECO with serial effectivity SN1000–SN1999 replacing line 30, when exploding with serial SN1500, then the new component is returned; with SN2000, the old component.
- **AC-9.15** (FR-9.22) Given a super BOM line with `frame == "carbon"` and a wheel line with `qty_expr = wheel_count`, when exploded with frame = carbon and wheel_count = 2, then only the carbon frame line appears and wheel quantity is 2.
- **AC-9.16** (FR-9.24, FR-9.25, BR-9.18) Given change-controlled type DRW with signature on release, when version 02 is released outside an ECO, then 409 `ENG-DOC-ECO-REQUIRED`; through an ECO, a signature record is stored, version 02 is released and version 01 obsolete.

- **AC-9.17** (FR-9.27, BR-9.19) Given an eBOM of 120 lines with 5 unknown part numbers, when imported, then a draft BOM version with 115 lines and an ECR exist, 5 lines are exceptions `ENG-EBOM-PART-UNKNOWN`, and the comparison counts equal the fixture.
- **AC-9.18** (FR-9.20, BR-9.16) Given alternative group A1 with probabilities 60/40, when exploded, then both lines return with group metadata; when probabilities are saved as 60/30, then 422 `ENG-ALT-CONFLICT`.

1.9.9 Test plan and fixtures

Fixtures: `engineering/golden-product.json` (7 levels, 2,000 components, 40 phantoms, 3 sites) with `golden-explosion.csv`; `engineering/version-dates.json` (12 overlapping-version scenarios); `engineering/routes-12.yaml` with lead-time oracle; `engineering/calendars.csv`; `engineering/eco-scenarios.yaml` (8 workflows including compensation); `engineering/bom-extract-ax.csv` (BOMVersion/BOM 10k rows) and `engineering/stko-stpo-extract.csv`.

Test	Type	Target	Pass criterion
Golden explosion	golden + performance	explode	byte-equal, p95 < 500 ms
As-of resolution	table-driven	resolve	12/12
Cycle detection	property (random DAGs + injected cycles)	line insert	0 cycles accepted, 0 false rejections
ECO workflow	Temporal test server	ECO	8/8 scenarios incl. compensation
Lead time	unit	calc	12/12 to 4 decimals
Scrap rounding	table-driven	consumption	40 cases
Calendar generation	unit + performance	generate	fixture equal, < 2 s
Where-used scale	performance	implode	100k lines, < 300 ms
Event contracts	contract	7 events	registry accepts

Reconciliation: Σ flattened explosion quantities = Σ over tree leaves; active version count per (product, site, date) ≤ 1 (nightly check); ECO affected versions = activated + expired events emitted.

1.9.10 Deployment and operations

Helm chart `passage-engineering` (Go service ×2, CloudNativePG `engineering-db`, Debezium connector, Temporal namespace `engineering`). Config: `ENG_BOM_MAX_LINES=5000`, `ENG_BOM_MAX_DEPTH=25`, `ENG_ROUTE_MAX_OPS=500`, `ENG_ECO_MAX_VERSIONS=500`, `ENG_EXPLODE_CACHE_TTL_S=300`. Feature flags: `engineering.direct_activation` (default off in production tenants), `engineering.formulas`, `engineering.plm_inbound`. Scheduled jobs: nightly overlap check; nightly expiry job setting `expired` on versions whose `effective_to` passed; 30-day expiry warning digest. SLOs: explode p95 < 500 ms; resolve p95 < 30 ms; ECO release p95 < 60 s. Alerts: overlap check finds > 0 rows; ECO stuck in `scheduled` past release_date + 10 min; explosion errors > 1% of calls. Runbook: "ECO compensation ran" (review error note, re-validate drafts, resubmit), "Rebuild product closure" (`passagectl eng rebuild-closure`). Rollback: previous digest; migrations additive.

1.9.11 Migration from AX / SAP

Source	Passage entity	Notes
AX BOMTable (BOMId, Name, SiteId, ItemGroupId)	BomHeader	
AX BOMVersion (ItemId, BOMId, FromDate, ToDate, FromQty, ToQty, Active, Approved, ApprovedBy, InventDimId→site)	BomVersion	Active=Yes → active; Approved without Active → draft with approved_by; date 1900-01-01 → null
AX BOM (BOMId, LineNum, ItemId, BOMQty, BOMQtySerie, UnitId, BOMType, ScrapConst, ScrapVar, Position, RoundUp, FromDate, ToDate, OprNum, InventDimId)	BomLine	BOMType enum Item/Phantom/Pegged/Vendor; co/by-products from PmfCoByProduct
AX RouteTable, RouteVersion, RouteOpr (OprNum, OprId, SetUpTime, ProcessTime, ProcessPerQty, TransitTime, QueueTimeBefore/After, OprPriority, NextOprNum, LinkType, Overlap), RouteOprTable (WrkCtrIdCost, requirements)	RouteHeader, RouteVersion, Operation, OperationRequirement	OprPriority Primary/Secondary1..3; requirements from RouteOprTable WrkCtrId / WrkCtrGroupId / WrkCtrCapability
AX WrkCtrTable (WrkCtrId, WrkCtrType, Efficiency, CalendarId, WrkCtrGroupId, VendAccount), WrkCtrResourceGroup, WrkCtrCapability, WrkCtrCapabilityResource (Level, FromDate, ToDate)	Resource, ResourceGroup, Capability, ResourceCapability	VendAccount → vendor_party_id via wave-7 alias
AX WorkCalendarTable, WorkCalendarDate, WorkCalendarDateLine (FromTime, ToTime, Efficiency), WorkCalendarTemplate	WorkCalendar, CalendarDay, Shift	times in seconds → hh:mm
D365 EngChgRequest / EngChgOrder (if present)	ECR / ECO	historical ECOs loaded as `released` with their versions
SAP MAST (MATNR, WERKS, STLNR, STLAL, STLAN usage) + STKO (STLAL, BMENG, BMEIN, DATUV, STLST status, AENNR) + STAS	BomHeader, BomVersion	one header per STLNR; one version per STLAL × DATUV; STLAN 1 (production) only; base_qty from BMENG
SAP STPO (POSNR, IDNRK, MENGE, MEINS, AUSCH scrap, DUMPS phantom, ALPOS, POSTP L/N/R, DATUV, DATUB, SANKA costing relevance)	BomLine	POSTP N (non-stock) → vendor; negative MENGE → by_product
SAP PLKO (PLNTY N, PLNNR, PLNAL, DATUV, VERWE, STATU) + MAPL + PLAS (sequence) + PLPO (VORNR, ARBPL, STEUS, VGW01..06, VGE0x units, PLIFZ, ZMERH, UEMUS overlap)	RouteHeader, RouteVersion, Operation	VGW01 setup, VGW02 machine, VGW03 labor → setup_time_h / run_time_h (labor kept as attribute); PLAS alternative sequences → parallel branches
SAP CRHD (OBJID, ARBPL, WERKS, VERWE, KAPID), CRCA, KAKO (KAPAZ, NGRAD utilization), CRTX	Resource, ResourceGroup, capacity_per_day_h	NGRAD → efficiency_pct
SAP TFACD/TFACS factory calendar, shift definitions	WorkCalendar	holiday calendars expanded for the migration range
SAP AENR (AENNR, DATUV, AETXT, AESTA)	ECO (historical, released)	effectivity dates recreated on versions

Validation: every line's component exists as a product in wave 8 and is not obsolete; every version's product is released at the version's site; no overlapping active versions after load (overlaps go to the exception queue with both versions listed); route requirements satisfiable. Not migrated: AX `BOMCalcTable` results (recomputed in wave 39), SAP `PLPO` activity-type costing fields `LAR0x` (wave 39), SAP document BOMs (STLTY D) and sales-order BOMs (wave 27 configured products), classification-based variant BOMs that depend on LO-VC procedures.

1.9.12 Dependencies, risks, open questions

Depends on: wave 8 (products, variants, UoM conversions, lifecycle), wave 6 (sites, calendars reference data), wave 7 (vendor parties for vendor-type resources), wave 4 (Temporal, outbox), wave 5 (number sequences `BOM`/`ROUTE`/`ECO`, audit).

Risk	Likelihood	Impact	Mitigation
Customers with un-versioned AX BOMs rely on in-place edits; forcing ECOs slows them	High	Medium	`engineering.direct_activation` flag per tenant with audit; ECO required only for products flagged `controlled`
SAP alternative sequences and parallel sequences map imperfectly to priority branches	Medium	Medium	Fixture comparison; unmapped PLAS structures reported, not silently flattened
Explosion cache serves stale results after ECO release	Low	High	Cache key includes the set of active version ids; release invalidates by product

Open questions: (1) Should BOM lines reference product variants directly or master plus dimension expressions (AX "dimension-based" lines)? Proposed: direct variant with an optional "inherit parent dimension" flag; decide by wave 21. (2) Formula yield at header or per line? Proposed: header yield plus line scrap; decide by wave 24. (3) Do ECOs cover released-product changes (item model group) too? Proposed: no in v1; decide by wave 10.

1.9.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`eng.bom.approval_required`	tenant / site	bool	true	true / false	Activation requires approval (BR-9.1)	Inventory and warehouse management parameters → Bills of materials → BOM approval check	OS28 BOM modification parameters
`eng.direct_activation`	tenant	bool	false in production tenants	true / false	Activation outside ECO (feature flag)	— (in-place BOM edits)	OS28 "ECM mandatory" setting
`eng.bom.max_depth`	tenant	int	25	5–25	BR-9.6	BOM levels setting	CS12 explosion level limit
`eng.bom.max_lines`	tenant	int	5,000	100–10,000	FR-9.2	—	—
`eng.bom.usage_fallback`	tenant	list	production → universal	ordered usages	FR-9.19 resolution	—	OPJI/OPJM selection ID, OS20 usages
`eng.bom.scrap_default`	item group	percent	0	0–100	Default variable scrap on new lines	BOM line ScrapVar default	OS27 default values, MARC-AUSSS assembly scrap
`eng.route.time_unit`	tenant	enum	hours (4 dec)	hours / minutes	Operation time storage and display	Production control parameters → time unit	Standard value keys (OP19)
`eng.route.default_link_type`	tenant	enum	soft	hard / soft	Default operation relation	RouteOpr.LinkType	PLAS relationship type
`eng.cal.horizon_months`	resource group	int	36	1–60	Calendar generation window (FR-9.6)	Compose working times period	Factory calendar validity (SCAL)
`eng.cal.expiry_warning_days`	tenant	int	60	7–365	Job ENG-J4 warning	Update calendars	SCAL valid-to
`eng.eco.number_sequence`	tenant	sequence	`ECO-{######}`	sequences	ECO numbers	ECM parameters → Number sequences	Change number number-range object (CC01)
`eng.eco.max_horizon_days`	tenant	int	365	30–1,095	BR-9.10	—	—
`eng.effectivity.types`	tenant	set	{date}	date, serial, parameter	FR-9.21	—	IMG ECM → Effectivity types and parameters
`eng.org.engineering`	legal entity	bool	false	true / false	FR-9.14	ECM parameters → Engineering organization	—
`eng.release.default_policy`	engineering category	FK	none	release policies	FR-9.15	Product release policies	ALE distribution model (BD64)
`eng.release.require_acceptance`	release policy × legal entity	bool	false	true / false	FR-9.16	Release policy "require acceptance"	—
`eng.readiness.policy`	engineering category / item group	FK	none	readiness policies	FR-9.17	Product readiness policies	—
`eng.nomenclature`	engineering category	FK	`V-{NN}`	nomenclatures	FR-9.18	Version nomenclature	Revision levels (OS13)
`eng.doc.signature_meanings`	document status	text	"Approved for release"	≤ 100 chars	FR-9.24	Electronic signature requirements (D365)	Digital signature strategy on DC20 status
`eng.ebom.sources`	tenant	records	none	SFTP / REST source definitions	FR-9.27	PLM integration (Dataverse)	ECTR configuration
`eng.explode.cache_ttl_s`	tenant	int	300	0–3,600	Explosion cache	—	—

1.9.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
BOM versions per product	list inquiry	product, site, status, usage, date	version, site, dates, qty range, status, ECO, approved by	`engineering.bom_version`	BOM versions; SSRS BOM versions	CS03, CS05
Multi-level explosion	drill-down	product, site, date, qty, usage, effectivity context, configuration	level, line, component, qty, scrap, alt group, effectivity	explode API	BOM designer; SSRS BOM lines	CS11 / CS12 / CS13
Where-used	drill-down	component, date, levels, site	parent, level, qty, version status, top-level count	product closure	BOMWhereUsed; SSRS BOM where-used	CS15
BOM version comparison	drill-down	version A, version B	added, removed, changed fields per line	`bom_line`	Engineering version comparison (D365)	CS14
Route operations and lead time	operational report	route version, qty	op, resource req., setup, run, queue, transit, lead time	route projection	SSRS Routes / Operations listings	CA03, CA97
Resource where-used	list inquiry	resource, group, capability	route versions, operations, site	operation requirements	Resource → Route versions	CA80, CR15
Work calendar	list inquiry	calendar, date range	day, open, shifts, hours, efficiency	`calendar_day`	SSRS Calendars / Working time	SCAL, CR03 capacity

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
ECR/ECO by status	KPI	status, priority, category, age	count, median cycle time, overdue	`engineering.eco`	Change request / change order inquiries	CC04 change master information system
ECO impact report	operational report	ECO	affected products, sites, versions, documents, open-order counts (hooks)	impact snapshot	Impact analysis report	CC04, CS15
Release status (product × legal entity)	list inquiry	product, entity, status, session	line status, version, accepted by, reason	`release_line`	Release status inquiry; SSRS Product release report	BD87 IDoc status for BD10/BD30
Readiness check status	operational report	entity, policy, check, status	pending checks, assignee, due date, age	`readiness_check_result`	Readiness check status inquiry	—
Engineering versions by category	list inquiry	category, product, state	version code, effective range, state, BOM/route	`engineering_version`	Engineering version inquiry	Revision level display (CC03)
eBOM vs mBOM comparison	drill-down	eBOM import, active BOM	added, removed, changed lines	`ebom_import`	—	ECTR CAD BOM comparison
Document structure and where-used	drill-down	DIR	children, parents, BOM document lines	document structure	—	CV13, CV15

1.9.15 Printed and electronic documents

Document	Trigger	Layout elements (header/line/footer fields)	Formats	Numbering sequence	Legal / retention requirement
Indented BOM report	on demand, attached to ECO	header: product, version, site, effective range, usage; lines: level, line no, component, qty/series, unit, scrap, position, operation; footer: approved by, ECO, page x/y	PDF, XLSX	none	kept with ECO record
Route sheet	on demand	header: product, route version, site; lines: op no, resource requirement, setup, run per qty, queue, transit; footer: lead time for reference qty	PDF	none	kept with ECO record
ECR / ECO document	submission and release	header: number, title, reason, priority, release date, effectivity; lines: affected versions with diff summary, affected documents, impact counts; footer: approvals with signature manifestation	PDF, email to approvers	`ECO-{######}`	10 years after product becomes obsolete (ISO 9001 §7.5.3; ISO 13485 §4.2.5 and 21 CFR 820 for medical-device tenants)
Product release report	release session completion	header: session number, policy; lines: product, version, entity, line status, reason	PDF	`REL-{yyyy}-{#####}`	7 years
Signed engineering document rendition	released status with signature	original plus signature page: printed name, date/time (UTC and local), meaning, hash	PDF/A-2b	DIR number/part/version	per document-type retention class; 21 CFR 11.10(c) record retention
Work calendar printout	on demand	calendar, month grid, shifts, closed days	PDF	none	none

1.9.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
ENG-J1 Active-version overlap check	nightly 01:00	product_id hash (8 partitions)	read-only	< 10 min at 1M versions	report only	rows > 0
ENG-J2 Version expiry	nightly 00:05 site time	site_id	sets `expired` where `effective_to` < today; idempotent	< 5 min	retried hourly	failures > 0
ENG-J3 Expiry warning digest	weekly Monday 06:00	owner team	read-only	< 2 min	none	none
ENG-J4 Calendar horizon extension	monthly 1st 02:00	calendar_id	generate idempotent per range	< 2 s per calendar	per-calendar retry	calendars within warning window
ENG-J5 Scheduled ECO release	Temporal timer at release_date 00:00	eco_id	workflow id per ECO	< 60 s p95	compensation BR-9.9	ECO in `scheduled` > release_date + 10 min
ENG-J6 Product closure integrity	nightly 03:30	tenant	read-only	< 15 min	runbook "Rebuild product closure"	mismatches > 0
ENG-J7 Release product structures	every 15 min for pending sessions; on ECO release	session_id	line-level idempotent	1,000 components × 3 entities < 2 min	line `failed` after 5 retries	failed lines > 0
ENG-J8 eBOM source poll	every 15 min	source_code	idempotent per (source, part, revision)	< 1 min per 100 payloads	payload to exceptions with reason	source unreachable 3 polls
ENG-J9 Readiness evaluation	hourly and on events	legal_entity_id	recompute per (product, check)	< 5 min for 100k released products	check left `pending`	overdue manual checks

1.9.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
View engineering data	`GET /engineering/v1/**`, explode, where-used, diff	none	every manufacturing role
Maintain BOMs	BOM headers, draft versions, lines; BOM designer	Approve engineering changes on the same ECO	Design engineer
Maintain routes and resources	route versions, operations, resources, calendars	none	Manufacturing engineer
Activate BOM without ECO	`bom-versions/{id}:activate`	Maintain BOMs (when `change_controlled`)	Engineering manager
Maintain engineering changes	ECR/ECO create, submit, eBOM import	Approve engineering changes	Design engineer, Manufacturing engineer
Approve engineering changes	`ecos/{id}:approve`	Maintain engineering changes on the same ECO (no self-approval)	Engineering manager, Quality manager
Release engineering changes	`ecos/{id}:release`, backdate with separate duty	none	Engineering manager
Release product structures	release sessions, ENG-J7, release policies	Accept product releases in the same receiving entity	Engineering administrator
Accept product releases	Open product releases accept/reject	Release product structures	Plant master data steward

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
Maintain readiness policies and sign checks	readiness policies; manual check sign-off	sign-off conflicts with creating the release session (BR-9.14)	Plant master data steward, Quality manager
Sign engineering documents	signature transitions (FR-9.24)	Maintain documents on the same DIR when independent review is required	Quality manager, Engineering manager
Administer eBOM sources	`eng.ebom.sources`, ENG-J8	none	Engineering administrator

1.9.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
ENG-BOM-CYCLE	error	"Adding {c} creates a cycle: {path}."	Recursive structure	Remove the line	FR-9.11
ENG-BOM-DEPTH	error	"Structure exceeds {max} levels."	Depth cap	Flatten or raise limit	BR-9.6
ENG-BOM-OVERLAP	error	"Version overlaps active version {v} for this site, date and quantity."	Overlap on activation	Adjust ranges	FR-9.1
ENG-BOM-COMP-OBSOLETE	error	"Component {c} is obsolete."	Obsolete component	Replace component	FR-9.11
ENG-BOM-PHANTOM-NOBOM	error	"Phantom {c} has no active BOM on {date}."	Phantom without BOM	Activate its BOM	BR-9.5
ENG-ROUTE-UNSATISFIED	error	"No resource satisfies operation {op} requirement {req}."	Requirement unmet	Add resource or capability	FR-9.5
ENG-ROUTE-NEXT-MISSING	error	"Next operation {n} does not exist."	Bad reference	Fix operation number	FR-9.4
ENG-ROUTE-SECONDARY-NOPRIMARY	error	"Secondary operation {op} has no primary at its join."	BR-9.7	Add primary	BR-9.7
ENG-ECO-NOT-APPROVED	error	"ECO {n} must be approved before release."	Wrong status	Approve first	FR-9.7
ENG-ECO-BACKDATE-DUTY	error	"Backdated release requires the Backdate duty."	Past release date	Use today or request duty	BR-9.10
ENG-ECO-HORIZON	error	"Release date is more than {d} days ahead."	Horizon exceeded	Choose an earlier date	BR-9.10
ENG-RES-IN-USE	error	"Resource {r} is required by active route {v}."	Delete in use	Change routes first	BR-9.11
ENG-OWNED-FIELD	error	"{field} is owned by engineering organization {eo}."	Local edit of engineering data	Raise an ECR	FR-9.14
ENG-READINESS-PENDING	warning	"{n} readiness checks are pending."	FR-9.17	Complete checks	FR-9.17
ENG-NOMEN-OVERFLOW	error	"Version numbering exhausted for {product}."	BR-9.15	Change nomenclature width	BR-9.15
ENG-ALT-CONFLICT	error	"Alternative group {g} is inconsistent: {reason}."	BR-9.16	Correct priorities/probabilities	BR-9.16
ENG-EFF-OVERLAP	error	"Serial ranges overlap with ECO {n}."	BR-9.17	Adjust ranges	BR-9.17
ENG-COND-UNKNOWN-ATTR	error	"Condition references unknown attribute {a}."	FR-9.22	Fix condition or model	FR-9.22
ENG-DOC-ECO-REQUIRED	error	"Document type {t} is change-controlled; release through an ECO."	FR-9.25	Add to an ECO	FR-9.25
ENG-DOC-SIGNED	error	"Originals are locked by signature on {date}."	BR-9.18	Create a new version	BR-9.18
ENG-EBOM-PART-UNKNOWN	warning	"Part {p} is not a known product."	FR-9.27	Create product or map identifier	FR-9.27
ENG-EBOM-REVISION-RELEASED	error	"Revision {r} of {p} is already released."	BR-9.19	Send a new revision	BR-9.19

1.9.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Explosion (7 levels, 2,000 components)	200k/day from planning and costing; peak 1,000/min	120 / 500 / 900 ms	50 req/s per replica	cache only	product closure; cache keyed by active version set	cache hit ratio rises; uncached p95 ≤ 1.2 s
As-of resolve	5M/day	5 / 30 / 60 ms	1,000 req/s	none	(product_id, site_id, effective_from, status)	unchanged
Where-used (100k lines)	20k/day	60 / 300 / 600 ms	50 req/s	none	btree bom_line(component_product_id)	levels capped at request value
BOM line save with cycle check	50k/day	20 / 80 / 150 ms	100 req/s	~200 B per line; 10M lines ≈ 3 GB	closure lookup	unchanged
ECO release (500 versions)	100/day	10 / 60 / 120 s	5 concurrent	ECO snapshots ~100 KB	Temporal workflow	queued; timers unaffected
Release session (1,000 components × 3 entities)	50/day	40 s / 2 min / 4 min	4 concurrent sessions	release lines ~1 KB	session_id partition of work	lines queued, no failure
eBOM import (500 lines)	500/day	2 / 8 / 15 s	10 concurrent	payloads ~200 KB	unique (source, part, revision)	ENG-J8 extends poll interval to 30 min

1.9.20 Localization and compliance hooks

Item	What varies	Country packs touching it (wave 48)	Extension point
Public holidays in work calendars	US federal; UK bank holidays per nation; DE holidays per Land; FR 11 jours fériés plus Alsace-Moselle; IN national and state holidays; MX mandatory rest days (Ley Federal del Trabajo Art. 74); CA federal and provincial	all seven	`HolidayCalendarProvider` feeding FR-9.6 closed days
Electronic records and signatures	US 21 CFR Part 11; EU GMP Annex 11; EU eIDAS for qualified signatures (optional)	US, DE, FR	signature policy per document type (FR-9.24)
Design and change control	ISO 9001 §8.3/§8.5.6; ISO 13485 §7.3.9; 21 CFR 820 (QMSR effective 2026-02-02); EU MDR Annex II technical documentation	US, DE, FR	`change_controlled` engineering categories; ECO record retention
Export control on technical data	US EAR/ITAR technical data, EU dual-use Regulation 2021/821	US, DE, FR	export classification attribute on engineering category; access restricted through OPA attributes (wave 48)
Units on drawings and BOMs	US customary vs metric	US	unit conversion (FR-6.9, FR-8.7) at display
Working time	Shift patterns must respect local limits (EU Working Time Directive 2003/88/EC, IN Factories Act 1948 §51 48 h/week); enforced for workers in wave 42, warned here when calendar shifts exceed 48 h/week	DE, FR, IN	calendar validation hook (warning only)

1.9.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.15.1 PIM, Common — Product readiness checks (D365), EcoResProductReadiness; register §12.99.4 #144	PARTIAL-9	ADDED as FR-9.17	Automated and manual checks; blocks via FR-8.14
AX §12.15.4 PIM, Setup — Product readiness policies (D365); register §12.99.4 #151	PARTIAL-9	ADDED as FR-9.17	Parameter `eng.readiness.policy`
AX §12.34.1 ECM, Common — Release product structure, EngChgReleaseProductStructure; register §12.99.4 #280	PARTIAL-9	ADDED as FR-9.15	Engineering vs operational split in FR-9.14
AX §12.34.1 ECM, Common — Product readiness checks (per LE checklist), EngChgProductReadiness; register §12.99.4 #281	PARTIAL-9	ADDED as FR-9.17	Per receiving legal entity
AX §12.34.1 ECM, Common — Product acceptance, EngChgProductAcceptance; register §12.99.4 #282	PARTIAL-9	ADDED as FR-9.16	Open product releases screen
AX §12.34.1 ECM, Common — Engineering product categories, EngChgEngineeringProductCategory; register §12.99.4 #284	PARTIAL-9	ADDED as FR-9.18	Category carries nomenclature, readiness and release policies
AX §12.34.2 ECM, Inquiries — Release status inquiry (product × legal entity); register §12.99.4 #286	PARTIAL-9	ADDED as FR-9.15	§1.9.14 "Release status"
AX §12.34.2 ECM, Inquiries — Readiness check status inquiry; register §12.99.4 #287	PARTIAL-9	ADDED as FR-9.17	§1.9.14 "Readiness check status"
AX §12.34.2 ECM, Reports — SSRS Product release report; register §12.99.4 #288	PARTIAL-9	ADDED as FR-9.15	§1.9.15 "Product release report", rendered by wave 46
AX §12.34.3 ECM, Periodic — Release product structures (batch); register §12.99.4 #289	PARTIAL-9	ADDED as FR-9.15	Job ENG-J7
AX §12.34.3 ECM, Periodic — Readiness check evaluation (batch); register §12.99.4 #290	PARTIAL-9	ADDED as FR-9.17	Job ENG-J9
AX §12.34.4 ECM, Setup — Engineering organizations; register §12.99.4 #292	PARTIAL-9	ADDED as FR-9.14	Parameter `eng.org.engineering`
AX §12.34.4 ECM, Setup — Engineering product categories (version nomenclature, attributes, readiness policy, release policy); register §12.99.4 #293	PARTIAL-9	ADDED as FR-9.18	—
AX §12.34.4 ECM, Setup — Version nomenclature (number/letter/date segments); register §12.99.4 #294	PARTIAL-9	ADDED as FR-9.18	BR-9.15 increment rules
AX §12.34.4 ECM, Setup — Product readiness policies / readiness checks; register §12.99.4 #295	PARTIAL-9	ADDED as FR-9.17	Hooks to waves 13, 17, 22, 26, 31 return not_applicable until live
AX §12.34.4 ECM, Setup — Product release policies; register §12.99.4 #296	PARTIAL-9	ADDED as FR-9.15	Parameter `eng.release.default_policy`
SAP §13.21 PP Master data — CS51/CS52/CS53/CS61/CS62/CS63/CS71 BOM categories; register §13.99.2 #238	PARTIAL-9	ADDED as FR-9.19	Usage and category on header; sales-order BOM instances in wave 27, WBS BOMs in wave 41, equipment/FL BOMs in wave 44 reuse the model
SAP §13.21 PP Master data — alternative items, discontinuation / follow-up material; register §13.99.2 #243	PARTIAL-9	ADDED as FR-9.20	Resolution logic in MRP wave 22
SAP §13.21 PP Customizing — OS40/OS41/OS44 ECM effectivity types (date, serial, parameter); register §13.99.2 #250	PARTIAL-9	ADDED as FR-9.21	Parameter `eng.effectivity.types`
SAP §13.32 LO-VC Master data — PSM / iPPE product variant structures; register §13.99.2 #360	PARTIAL-9	SUPERSEDED by FR-9.22 super BOM/routing with selection conditions	iPPE-specific objects (PVS nodes, access diagrams, CMP) have no equivalent; multi-level configurable structures are modeled as super BOMs over configuration attributes
SAP §13.32 LO-VC Master data — VC with batch/serial effectivity, parameter effectivity; register §13.99.2 #362	PARTIAL-9	ADDED as FR-9.21	Effectivity context combined with configuration in explode
SAP §13.35 DMS Master data — CV11/CV12/CV13/CV15 document structure; register §13.99.2 #376	GAP	ADDED as FR-9.26	Structures versioned with parent DIR
SAP §13.35 DMS Master data — CV01N status network with release workflow and digital signature; register §13.99.2 #378	PARTIAL-9	ADDED as FR-9.24	21 CFR 11 manifestation and linking
SAP §13.35 DMS Master data — CV01N version / part / change number (ECM); register §13.99.2 #379	PARTIAL-9	ADDED as FR-9.25	ECO-controlled DIR versions
SAP §13.35 DMS Master data — ECTR CAD integration, CAD BOM derivation; register §13.99.2 #381	GAP	ADDED as FR-9.27	Contract and import path here; native CAD file management out of scope; vendor connectors deferred to wave 35
SAP §13.35 DMS Master data — Document BOM items (CS01 item category D); register §13.99.2 #382	PARTIAL-9	ADDED as FR-9.23	Line types `document` and `text`
SAP §13.35 DMS Master data — PLM suite (Engineering Record, PSM, Visual Enterprise, Recipe Development, Specification management, Labeling); register §13.99.2 #383	PARTIAL-9	DEFERRED to wave 24	Recipe development and specifications go to wave 24 as the register proposes; global label management to wave 48; Visual Enterprise 3D viewing is out of scope; Engineering Record is covered by ECR/ECO (FR-9.7)
SAP §13.35 DMS Reports — CV13 / CV15 document structure display / where-used; register §13.99.2 #386	GAP	ADDED as FR-9.26	§1.9.14 "Document structure and where-used"
SAP §13.35 DMS Reports — ECTR reports (CAD BOM comparison); register §13.99.2 #388	GAP	ADDED as FR-9.27	§1.9.14 "eBOM vs mBOM comparison"
SAP §13.35 DMS Periodic — ECTR sync jobs; register §13.99.2 #391	GAP	ADDED as FR-9.27	Job ENG-J8
SAP §13.35 DMS Customizing — DC20 document status definitions; register §13.99.2 #393	PARTIAL-9	ADDED as FR-9.24	Status network per document type
SAP §13.35 DMS Customizing — ECTR configuration (CAD applications, templates, BOM sync rules); register §13.99.2 #396	GAP	ADDED as FR-9.27	Parameter `eng.ebom.sources`

1.10 Wave 10: Data Migration Framework & Legacy Connectors

1.10.1 Scope and business context

AX ships the Data Import Export Framework (DIXF/DMF: `DMFDefinitionGroup`, `DMFEntity`, staging tables such as `DMFCustomerEntity`, `DMFVendorEntity`, `DMFProductEntity`, `DMFBOMEntity`) and D365 the data management framework with data entities (`CustCustomerV3Entity`, `VendVendorV2Entity`, `EcoResReleasedProductV2Entity`, `BOMEntity`), data projects and packages; SAP ships LSMW (recording, batch input, BAPI, IDoc), the S/4HANA Migration Cockpit

(LTMC/LTMOM with migration objects such as `Customer`, `Supplier`, `Material`, `Bill of material`, `Routing`, `Work center`) and BAPIs (`BAPI_MATERIAL_SAVEDATA`, `BAPI_CUSTOMER_CREATEFROMDATA1`). Wave 10 builds the pipeline every later wave reuses: extract from AX SQL (ODBC) or SAP (RFC/OData or flat files), stage into a Postgres staging schema, map with a declarative DSL, validate, load through the public APIs of waves 6–9, and reconcile with row counts, hash totals and sampled field compares. Dry-run and idempotent re-run, a run ledger with lineage, and an exception queue with re-run close the loop. Phase 1 exits when the demo manufacturer's master data is fully migrated from a synthetic AX extract.

Out of scope: transactional data packs (open items, on-hand, open orders — wave 49), parallel-run diff harness (wave 49), live coexistence adapters (wave 49), X++/ABAP code conversion (non-goal).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
DIXF staging-to-target fails at row 48,213 of 100,000 with "Function DMFEntityWriter... has been incorrectly called"; the partial load leaves half-created customers and the fix is manual SQL cleanup before re-run	Non-idempotent target writes; errors swallowed into DMFExecution log	Every load is idempotent by natural key and run id; failed rows go to the exception queue, successful rows stay; re-run touches only changed rows (FR-10.6, FR-10.7)	Re-run cost after failure: hours of cleanup→0; duplicate rows after re-run 0
LSMW requires recording a transaction (XD01) screen by screen; a screen layout change in a support pack breaks the recording silently	Batch-input emulation of GUI screens	Loads go through versioned public APIs with contract tests (FR-10.5)	Recording breakage per upgrade: recurring→0
Reconciliation in both tools is "count rows in source, count in target, eyeball"; field-level drift (truncated names, dropped leading zeros in account numbers) is discovered weeks later	No hash or sampled compare	Row counts + per-column hash totals + stratified sampled field compare with tolerance per type (FR-10.8)	Field-level drift found at load: 0%→100% of sampled columns; sign-off evidence auto-generated
Mapping lives in Excel and in the heads of consultants; the enum mapping for CustTable.Blocked or MARA-MTART is re-discovered on every project	Mappings not declarative or versioned	Mapping DSL in git with unit tests, enum tables and rejection reasons (FR-10.3)	Mapping reuse across projects: 0→100% of standard objects; mapping defects found by tests before load
A 100k-item load through DIXF takes 6–10 hours; SAP LTMC similar	Row-at-a-time batch processing	Partitioned parallel loaders (Go) against bulk endpoints with checkpoints (FR-10.6)	170k master records in < 30 min vs 6–10 h

1.10.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Migration project / definition	DMFDefinitionGroup, D365 data project, data package (manifest + package header)	LSMW project/subproject/object; LTMC migration project	migration-service `MigrationProject` with `MappingSpec` versions	full
Source extraction	DIXF ODBC/file source, AX SQL direct, D365 entity export	RFC (RFC_READ_TABLE, BAPIs), OData, SE16 extracts, IDoc outbound	`Connector` (ax-odbc, d365-odata, sap-rfc, sap-odata, csv/parquet)	full
Staging	DMF*Entity staging tables	LSMW read/convert data files; LTMC staging tables (DB staging)	Postgres schema `staging` with typed tables per source object + raw jsonb	full
Mapping / transformation	DMF mapping (source → staging → target), X++ transforms	LSMW field mapping and conversion rules (ABAP), LTMOM	Mapping DSL (YAML) with functions, lookups, enum tables	improved (tested, versioned)
Validation	Entity validation on target write	LTMC simulate; BAPI return messages	Validation stage with declarative expectations plus target-API dry run	improved
Load	DMFEntityWriter, data entity insert	Batch input / BAPI / IDoc posting	Parallel loaders against wave 6–9 bulk endpoints	full
Reconciliation	DMF execution log counts	LTMC monitoring counts	Row counts, hash totals, sampled compare, exception queue	improved
Run history / lineage	DMFExecution, job history	LSMW logs, LTMC run history	`MigrationRun` ledger with per-row lineage	improved
Standard objects	DMF customer, vendor, product, BOM, route entities	Migration Cockpit objects Customer, Supplier, Material, BOM, Routing, Work center	Connector packs for DirParty/CustTable/VendTable/InventTable/EcoResProduct/BOM/Route and KNA1/LFA1/BUT000/MARA/MAST-STKO-STPO/PLKO-PLPO/CRHD	full for Phase 1 scope
Scheduling / monitoring	Batch jobs, Data management workspace	SM37, LTMC monitor	Temporal workflows, migration workspace	full
Copy data between legal entities	DMFCopyEntity (Copy entity data between companies), D365 "Copy into legal entity"	EC01/EC02 organizational object copy; LSMW cross-company reload	`copy` project type (FR-10.14)	full
Configuration data packages	D365 configuration data projects, data templates, LCS asset library	Customizing transports (SE09/SE10, SCC1), BC Sets (SCPR20), cross-system viewer SCU0	signed configuration packages with diff (FR-10.15)	improved (diff before apply)
Recurring imports	D365 recurring data jobs (recurring integrations)	LSMW periodic runs, IDoc inbound processing (BD87)	scheduled incremental runs until cutover (FR-10.16)	full

1.10.3 Functional requirements

- **FR-10.1 Migration projects and runs** — Must. The system shall define projects (name, source system type ax2012/d365fo/sapecc/s4, target tenant, object list in dependency order, mapping spec versions) and execute runs with modes `dry_run`, `load`, and `reconcile_only`; each run records start/end, user, counts per stage, and the mapping and connector versions used.
  - Object dependency order for Phase 1 is fixed: reference data → org → parties → products → released products → resources/calendars → BOMs → routes → ECOs.
- **FR-10.2 Connectors** — Must. The system shall provide connectors: `ax-odbc` (read-only SQL over the AX 2012 database with DataAreaId and RecId-based paging), `d365-odata` (data entities with `$skiptoken` paging and change tracking), `sap-rfc` (RFC_READ_TABLE with 512-byte row limit handling, BAPI reads), `sap-odata` (API_BUSINESS_PARTNER, API_PRODUCT_SRV, API_BILL_OF_MATERIAL_SRV), and `file` (CSV, XLSX, Parquet with declared schema and codepage).
  - Each connector supports incremental extraction by `ModifiedDateTime/RecVersion` (AX) or `AEDAT/LAEDA` (SAP) when available; otherwise full extract.
  - Extraction throughput target 20,000 rows/s per connector worker on the dev cluster.
- **FR-10.3 Mapping DSL** — Must. The system shall express mappings as versioned YAML: source object, target entity, key definition (natural key columns), field rules (`copy`, `const`, `concat`, `lookup` against staged or target tables, `enum` with explicit table and default, `coerce` with type and format, `expr` in a restricted expression language, `default`), and `reject_when` conditions with reason codes.

    - Type coercions: AX `utcdatetime`/SAP `DATS`+`TIMS` to timestamptz with source zone; `NoYes` and `X`/blank to boolean; decimals with explicit scale; trim and codepage normalization.
    - Every mapping ships unit tests (input row → expected output or rejection) runnable in CI.
- **FR-10.4 Validation stage** — Must. The system shall validate staged, mapped rows with declarative expectations (not null, unique, FK-exists in target or staging, regex, range, set membership, row-count within ±N% of previous run) and a target-API dry run that calls the wave 6–9 endpoints with `X-Passage-Dry-Run: true` to collect business validation errors without writing.
    - Validation output: per-row status (ok, warn, reject) with reason codes; a run proceeds to load only when rejects ≤ the project's threshold (default 0.5%).
- **FR-10.5 Load through public APIs** — Must. The system shall load exclusively through versioned bulk endpoints (`POST /org/v1/import`, `/party/v1/parties:bulk`, `/product/v1/products:bulk`, `/engineering/v1/bom-versions:bulk`, …) with batches of 500 and `Idempotency-Key = sha256(run_id, object, natural_key)`, never by direct SQL into domain databases.
- **FR-10.6 Parallel, checkpointed, resumable loads** — Must. The system shall partition each object by hash of the natural key into N workers (default 8), checkpoint every batch in the run ledger, and resume from the last checkpoint after worker or pod failure without duplicates.
    - Target: 50k customers + 20k vendors + 100k items + 10k BOMs in < 30 min including reconciliation.
- **FR-10.7 Idempotent re-run** — Must. The system shall compute a content hash per mapped row; on re-run, rows with an unchanged hash and a successful prior load are skipped, changed rows are upserted by natural key, and rows absent from the source are reported (never deleted) unless the project enables `retire_missing` which sets the target to `inactive`.
- **FR-10.8 Reconciliation** — Must. The system shall produce, per object and run: source count, staged, mapped, rejected, loaded, skipped; per-column hash totals (sum of `hashtext` over normalized values for 1–20 chosen columns) on source-mapped vs target-read; and a stratified random sample (default 1%, min 100, max 5,000 rows) with field-by-field compare and per-type tolerance (decimals to scale, strings after trim/casefold, dates exact).
    - A run is `reconciled` only when counts tie, hash totals match, and sample mismatches = 0; otherwise `reconciled_with_exceptions` with every mismatch in the exception queue.
- **FR-10.9 Exception queue and re-run** — Must. The system shall list rejected and mismatched rows with reason, source key, mapped payload and target response; a user may edit the staged row, add a mapping override, or mark `accepted_difference` with a reason, then re-run only the exception set.
- **FR-10.10 Run ledger and lineage** — Must. The system shall keep an append-only ledger: for every loaded target record, the run id, source object, source key, content hash and mapped payload hash, queryable from the target record's audit tab ("migrated from AX CustTable US-00123 in run 42").
- **FR-10.11 Migration workspace** — Must. The system shall provide a workspace with project list, run history, per-stage progress, reconciliation report (downloadable as XLSX/PDF for sign-off), and exception queue.
- **FR-10.12 Packaging** — Should. The system shall package connectors and standard mapping packs as Helm sub-charts and OCI artifacts (`passage-migration-ax`, `passage-migration-sap`) versioned independently of the core service.
- **FR-10.13 PII handling in staging** — Must. The system shall encrypt PII-flagged staging columns at rest with a per-project key, purge staging data after a configurable retention (default 30 days after the run is signed off), and exclude PII from exception-queue exports unless the user holds `pii.read`.
- **FR-10.14 Copy entity data between legal entities** — Should. The system shall support a project type `copy` whose source is a tenant legal entity and whose target is another legal entity in the same tenant, reusing the mapping DSL with identity rules plus key-rewrite rules (source code → target code), dry run, idempotent load and reconciliation; supported objects: customer and vendor roles, released products with site defaults, BOM and route versions, resources and calendars.
- **FR-10.15 Configuration data packages** — Should. The system shall export setup data (the parameters of §1.6.13–§1.9.13, groups, terms, lifecycle states, document types, nomenclatures, readiness and release policies, mapping specs) as a versioned package (JSON bodies, manifest with schema version, dependency order, sha256 per file, signature by the tenant key) and import it into another tenant or environment after a dry-run diff showing create / update / unchanged per record; deletes are never applied.
- **FR-10.16 Recurring import jobs** — Should. The system shall let a project schedule incremental `load` runs (cron, minimum interval 15 min) from a connector or a file drop (SFTP or S3-compatible object storage path) during the coexistence and cutover window, using incremental extraction (FR-10.2); scheduling stops automatically at the project's `cutover_at`.
- **FR-10.17 Mapping packs for coverage-closure objects** — Must. The system shall extend the standard AX and SAP mapping packs, with tests, to the objects added by the phase-1 coverage closure: AX OMTeam/OMTeamMember → Team (FR-6.14); VendCertification/VendCertificationType → VendorCertification (FR-7.14); PaymSched/PaymSchedLine and SAP T052R → PaymentScheduleLine (FR-7.15); KNA1-KNRZE, KNB1-KNRZB/XDEZV, LFA1-LNRZA, LFZA and AX DirPartyRelationship → PartyRelationship (FR-7.17); DirAddressBook/DirAddressBookParty → AddressBook (FR-7.18); KNVH → customer hierarchy (FR-7.19); EcoResProductRelation → ProductRelation (FR-8.13); EcoResProductLifecycleState and MARA-MSTAE/MARC-MMSTA with T141 → LifecycleState (FR-8.14); ASMD/ASMDT → service products (FR-8.17); CUVTAB/CUVTAB_VALC → DecisionTable (FR-8.18); DRAW/DRAT/DRAD and DocuRef/DocuType → DocumentInfoRecord and links (FR-8.20); EngChgEngineeringProductCategory, version nomenclature, release and readiness policies (FR-9.15–FR-9.18); STPO ALPGR/ALPST/ALPRF/EWAHR/KZAUS/NFMAT (FR-9.20); AENR effectivity data (FR-9.21); STKO/STPO with STLTY = D document structures (FR-9.26).
    - The STLTY = D load supersedes the "not migrated: SAP document BOMs" note in §1.9.11.
- **FR-10.18 Data templates** — Could. The system shall let a project attach per-object default-value templates (D365 data templates / default templates per module) applied before mapping rules, so that fields absent in the source receive project-specific defaults recorded as `default_from_template` in lineage.

1.10.4 Data model

MigrationRun

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id / project_id	uuid	yes	FK project(name, source_type, objects[], thresholds)
run_no	int	yes	sequence per project
mode	enum	yes	dry_run, load, reconcile_only, exceptions_only
status	enum	yes	created, extracting, staging, mapping, validating, loading, reconciling, reconciled, reconciled_with_exceptions, failed, cancelled
mapping_versions / connector_versions	jsonb	yes	pinned at start
started_at / ended_at / started_by	timestamptz, uuid	yes	
counts	jsonb	yes	per object: source, staged, mapped, rejected, loaded, skipped, mismatched
parent_run_id	uuid	no	for exceptions_only re-runs
workflow_id	text	yes	Temporal
signed_off_by / signed_off_at	uuid, timestamptz	no	enables staging purge timer

StagingRow (one physical table per source object, common columns)

Field	Type	Required	Notes
run_id / object	uuid, text	yes	partition key
source_key	text(200)	yes	e.g. `USMF
raw	jsonb	yes	source columns verbatim (PII columns encrypted)
mapped	jsonb	no	output of mapping
content_hash	bytea(32)	no	sha256 of normalized mapped payload
status	enum	yes	staged, mapped, rejected, validated, loaded, skipped, failed, mismatched, accepted_difference
reason_code / reason_detail	text	no	e.g. `MAP-ENUM-UNMAPPED`, `VAL-FK-MISSING`, `API-422`
target_id	uuid	no	id returned by load
attempt	int	yes	increments per re-run

MappingSpec

Field	Type	Required	Notes
id / project_id	uuid	yes	
source_object / target_entity	text	yes	e.g. `CustTable` → `party.customer_role`
version	int	yes	immutable once a run pinned it
natural_key	text[]	yes	columns of the mapped payload
rules	jsonb	yes	ordered field rules
enum_tables	jsonb	no	{field: {source_value: target_value}, default}
reject_when	jsonb	no	[{expr, reason_code}]
tests	jsonb	yes	at least 1 case; CI fails on 0
checksum	text	yes	sha256 of the YAML

LineageEntry — run_id, object, source_key, target_entity, target_id, content_hash, loaded_at, idempotency_key (unique). **ReconciliationResult** — run_id, object, counts jsonb, hash_totals jsonb {column: {source, target, match}}, sample jsonb {size, mismatches[]}, status.

Invariants: a run's mapping versions never change after `extracting`; `loaded` rows have a lineage entry and a target_id; idempotency keys are unique across all runs; a `reconciled` status requires mismatches = 0 and rejects = 0 or all `accepted_difference`. Indexing: staging tables partitioned by `run_id` (list partitioning, dropped on purge); `(run_id, object, status)` btree; `lineage(target_entity, target_id)` for the audit tab; `lineage(idempotency_key)` unique.

Addendum entities (FR-10.14 – FR-10.18) — migration_project gains `project_type` (migrate, copy, config_package), `source_legal_entity_id`, `target_legal_entity_id`, `cutover_at`, `schedule_cron`; key_rewrite_rule: project_id, field, from_value, to_value; config_package: id, version text(20), schema_version int, manifest jsonb, signature bytea, created_by, created_at, applied jsonb [{tenant_id, applied_at, diff_counts}]; data_template: project_id, object, defaults jsonb.

1.10.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/migration/v1/projects	Create project	`Idempotency-Key`	Administer migration
POST	/migration/v1/projects/{id}/mapping-specs	Upload YAML; returns version and test results	checksum-idempotent	Maintain migration mappings
POST	/migration/v1/projects/{id}/runs	Start run (`mode`, `objects[]`, `from_checkpoint`)	key header; one active run per project	Run migrations
GET	/migration/v1/runs/{id}	Status and counts per stage	n/a	View migrations
POST	/migration/v1/runs/{id}:cancel	Cancel after current batch	key header	Run migrations
GET	/migration/v1/runs/{id}/reconciliation	Report (JSON; `Accept: application/vnd.openxmlformats…` for XLSX)	n/a	View migrations
GET	/migration/v1/runs/{id}/exceptions?object=&reason=	Exception queue	n/a	View migrations
PATCH	/migration/v1/exceptions/{rowId}	Edit staged row, set accepted_difference	version check	Resolve migration exceptions
POST	/migration/v1/runs/{id}:rerun-exceptions	Child run over exception set	key header	Run migrations
POST	/migration/v1/runs/{id}:sign-off	Sign-off; starts purge timer	key header	Approve migration sign-off
GET	/migration/v1/lineage?entity=party&id=	Lineage for a target record	n/a	View migrations
POST	/migration/v1/connectors/{type}:test	Connectivity and schema probe	pure	Administer migration

Validation: run start rejects if any object's mapping tests fail or a dependency object has no reconciled run in the project (override with `allow_unreconciled_deps=true` and the Administer duty); mapping upload rejects unknown functions, unmapped enum values without a default, and natural keys not produced by the rules.

Event	Trigger	Payload fields	Consumers
dev.passage.migration.run.started.v1	run start	run_id, project_id, mode, objects[], mapping_versions	notification, audit
dev.passage.migration.run.stage.completed.v1	each stage	run_id, object, stage, counts	workspace progress, audit
dev.passage.migration.run.completed.v1	terminal status	run_id, status, counts{staged, loaded, rejected, skipped, mismatched}, hash_match bool	notification, reconciliation UI, audit
dev.passage.migration.exception.raised.v1	reject or mismatch	run_id, object, source_key, reason_code	stewardship queue (wave 7 style)
dev.passage.migration.run.signedoff.v1	sign-off	run_id, signed_off_by, purge_at	audit, purge job

1.10.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Migration workspace	Tiles: projects, active runs, exceptions open, last reconciliation status	counts, status chips	View migrations
Project detail	Edit objects/order, upload mapping, view test results, start run (mode picker), connector test	object, mapping version, tests passed, last run	Administer migration

Screen	Primary actions	Key fields/columns	Duty required
Run monitor	Per-object stage pipeline with counts and throughput, cancel, logs link (Tempo trace)	stage, rows/s, ETA, errors	Run migrations
Reconciliation report	Counts table, hash totals per column, sample mismatches grid, download XLSX/PDF, sign-off	object, source/target counts, hash match, mismatch fields	View / Approve migration sign-off
Exception queue	Filter by reason, edit row inline, bulk accept difference with reason, re-run selection	source key, reason code, detail, mapped payload, target response	Resolve migration exceptions
Mapping editor (read-mostly)	View rules, enum tables, test cases; edit opens YAML with schema validation	rule, source, target, function	Maintain migration mappings
Lineage tab (on any migrated record)	Open run, open staged row	run no, source object, source key, loaded at	View migrations

Interaction rules: run monitor auto-refreshes every 2 s via SSE; exception grid supports keyboard navigation and inline JSON edit with schema hints; bulk accept limited to 1,000 rows per action and requires a reason of ≥ 10 characters; reconciliation report is read-only once signed off; PII columns render masked unless `pii.read`.

1.10.7 Business rules and edge cases

- **BR-10.1** Run state machine: `created → extracting → staging → mapping → validating → loading → reconciling → reconciled | reconciled_with_exceptions`; any stage `→ failed` after exhausting retries (5 per batch, exponential backoff 1–60 s); `→ cancelled` from any non-terminal state after the current batch commits; `dry_run` stops after `validating` with a simulated load report.
- **BR-10.2** Only one active run per project; a second start returns 409 with the active run id.
- **BR-10.3** Natural keys are normalized before hashing: trim, casefold for codes that the target treats case-insensitively, leading-zero preservation for SAP numbers (never cast to integer).
- **BR-10.4** Enum mapping with no match and no default rejects the row with `MAP-ENUM-UNMAPPED` and the value; mapping with default records a `warn` and the default used.
- **BR-10.5** Decimal coercion: AX `real` values are read as numeric(32,16) and rounded half-even to the target scale; a row whose rounding changes the value by more than 0.5 units of the last target decimal is flagged `warn` with code `MAP-PRECISION-LOSS`.
- **BR-10.6** Dates: AX `1900-01-01` and SAP `00000000` / `99991231` are nulls or open-ended markers by rule, never literal dates.
- **BR-10.7** Load batches are atomic per batch on the target side; a batch with one failing row is retried as singletons so one bad row rejects only itself.
- **BR-10.8** Re-run skip rule: skip when `content_hash` equals the last `loaded` lineage hash and the target record's version has not changed since (target `updated_at` ≤ lineage `loaded_at`); otherwise upsert; a target changed by a user after the load is reported as `TARGET-DRIFT` and not overwritten unless `overwrite_drift=true`.
- **BR-10.9** Hash totals use the same normalization on both sides; a hash mismatch with matching counts triggers an automatic widened sample (10%) before the run is marked with exceptions.
- **BR-10.10** Dependency between objects: a party role row whose party failed in the same run is rejected with `VAL-DEP-FAILED` referencing the parent source key, not retried blindly.
- **BR-10.11** Staging purge runs 30 days after sign-off (configurable 7–365); purge drops the run's partitions; lineage and reconciliation results are never purged.
- **BR-10.12** Concurrency with live users: loads run as a service account under OPA with the duties of the project owner; records created carry `created_by = migration:{run_id}` for audit.
- **BR-10.13** Copy projects rewrite every legal-entity-scoped natural key; a rewritten key that already exists in the target is rejected with `COPY-KEY-EXISTS` unless the project selects `upsert`.
- **BR-10.14** Configuration packages apply in manifest dependency order; a package whose schema version is newer than the target's is refused (`MIG-PKG-SCHEMA-NEWER`); a package whose signature does not verify is refused (`MIG-PKG-SIGNATURE`).
- **BR-10.15** Recurring runs never overlap (BR-10.2); a trigger that fires while a run is active is skipped and logged, never queued.

1.10.8 Acceptance criteria

- **AC-10.1** (FR-10.6, FR-10.8) Given the seeded AX-shaped dataset (50k CustTable, 20k VendTable, 100k InventTable/EcoResProduct, 10k BOMVersion/BOM), when a `load` run executes on the dev cluster, then it reaches `reconciled` in < 30 min with counts tied, all hash totals matching and 0 sample mismatches.
- **AC-10.2** (FR-10.7) Given the completed run, when the identical run executes again, then loaded = 0, skipped = 170,000, and no target `updated.v1` events are published.
- **AC-10.3** (FR-10.3) Given the mapping test suites for the 12 standard Phase 1 objects, when CI runs, then every test passes including type coercion (utcdatetime, NoYes, DATS), enum mapping (Blocked, MTART), default fill and rejection reasons.
- **AC-10.4** (FR-10.6) Given a load in progress, when a loader pod is killed at batch 137, then the resumed run continues from checkpoint 136, finishes, and the lineage table has no duplicate idempotency keys.
- **AC-10.5** (FR-10.4) Given a `dry_run` with 10 rows pointing at a non-existent customer group, when validated, then 10 rows are `reject` with `VAL-FK-MISSING` and nothing is written to party-db.
- **AC-10.6** (FR-10.9) Given 25 exceptions, when 20 are edited and 5 marked accepted_difference, then `rerun-exceptions` loads 20, skips 5, and the parent run moves to `reconciled`.
- **AC-10.7** (BR-10.8) Given a customer modified by a user after load, when the run is repeated with changed source data, then the row is `TARGET-DRIFT` and the user's change is preserved.
- **AC-10.8** (FR-10.2) Given the `sap-rfc` connector against the recorded RFC_READ_TABLE fixture with 600-byte rows, when extracting KNA1, then all columns are reassembled across two field groups with 0 truncation.
- **AC-10.9** (FR-10.13) Given a signed-off run, when 30 days pass on the Temporal test clock, then staging partitions are dropped and lineage rows remain.
- **AC-10.10** (FR-10.1) Phase 1 exit: given the synthetic AX extract of the demo manufacturer, when the full project runs, then every object is `reconciled`, the wave 6–9 workspaces show the migrated data, and the reconciliation XLSX is attached to the sign-off record.
- **AC-10.11** (FR-10.15, BR-10.14) Given a package of 240 setup records from tenant A, when it is dry-run against tenant B where 40 records are identical and 10 differ, then the diff shows 190 create, 10 update, 40 unchanged; after apply, a second apply reports 0 changes.
- **AC-10.12** (FR-10.14, BR-10.13) Given 500 released products in USMF, when a copy project loads them into CAMF, then the run is `reconciled`, every item number exists in CAMF, and a second run skips 500.
- **AC-10.13** (FR-10.17) Given the mapping test suites for the 15 added objects, when CI runs, then all pass; an AX PaymSched with 3 percent lines maps to 3 schedule lines summing 100.00.

1.10.9 Test plan and fixtures

Fixtures: `migration/ax-seed/` (generated AX-shaped CSVs with real column names: DirPartyTable, DirPerson, DirOrganization, LogisticsPostalAddress, CustTable 50k, VendTable 20k, EcoResProduct/InventTable 100k, EcoResProductMaster 2k, BOMVersion/BOM 10k, RouteVersion/RouteOpr 3k, WrkCtrTable 200; generator seed 10); `migration/sap-seed/` (KNA1/KNB1 10k, LFA1/LFB1 5k, BUT000/BUT100 15k, MARA/MARC/MARD/MARM 20k, MAST/STKO/STPO 5k, PLKO/PLPO/MAPL 2k, CRHD 100); `migration/mapping-tests/*.yaml` (≥ 10 cases per object); `migration/rfc-recorded.json`; `migration/demo-manufacturer-ax/` (the phase-exit extract).

Test	Type	Target	Pass criterion
Mapping DSL	unit	every rule function	100% branch coverage on functions; all object test cases pass
Full load performance	performance	end-to-end run	< 30 min, reconciled
Idempotent re-run	E2E	run twice	0 loaded second time
Pod-kill resume	chaos	loaders	no duplicate idempotency keys
Dry run isolation	contract	target APIs	0 writes on `X-Passage-Dry-Run`
Connector contracts	contract (recorded)	ax-odbc, d365-odata, sap-rfc, sap-odata, file	schema probes and paging match fixtures
Reconciliation math	property	hash totals and sampling	injected drift in 1 of 10k rows detected ≥ 99% with widened sample
Exception re-run	E2E	exceptions_only	parent reaches reconciled
PII at rest	security	staging	encrypted columns unreadable with DB role; purge verified
Event contracts	contract	5 events	registry accepts

Reconciliation checks: source count = staged; staged = mapped + rejected; mapped = loaded + skipped + failed; Σ per-object loaded = lineage rows for the run; target row count delta (before/after run) = loaded − upserts of existing keys.

1.10.10 Deployment and operations

Helm chart `passage-migration` (TS orchestration service ×2, Go loader workers as a scalable Deployment 2–16 replicas via HPA on queue depth, CloudNativePG `migration-db` with staging partitions on a separate tablespace, Temporal namespace `migration`); sub-charts `passage-migration-ax` (ODBC driver image with FreeTDS, mapping packs) and `passage-migration-sap` (RFC SDK not redistributable: image expects the customer-mounted NW RFC library; OData path works without it). Config: `MIG_WORKERS_DEFAULT=8`, `MIG_BATCH_SIZE=500`, `MIG_REJECT_THRESHOLD_PCT=0.5`, `MIG_SAMPLE_PCT=1`, `MIG_STAGING_RETENTION_DAYS=30`, `MIG_PII_KEY_RING` (Vault path), connector secrets in Vault. Feature flags: `migration.retire_missing`, `migration.overwrite_drift`, `migration.sap_rfc`. Scheduled jobs: hourly purge timer check; nightly lineage integrity check. SLOs: loader throughput ≥ 100 rows/s per worker sustained; run monitor update lag < 2 s; availability 99.5% (non-critical path). Alerts: run `failed`; batch retry rate > 5%; staging tablespace > 80%; connector auth failure. Runbook: "Resume a failed run" (`passagectl mig resume --run`), "Rotate PII key" (re-encrypt staging for active runs), "Widen sample manually". Rollback: previous digest; staging schema per run is unaffected; mapping packs are OCI artifacts pinned by digest in the project.

1.10.11 Migration from AX / SAP

This wave is the migration tool; its own "migration" is the catalog of standard objects it ships with, each a mapping pack with tests:

Source object (connector)	Target entity (wave)	Pack notes
AX CompanyInfo/OMLegalEntity, OMOperatingUnit, OMHierarchy*, InventSite, InventLocation, LogisticsPostalAddress, FiscalCalendar*, UnitOfMeasure* (ax-odbc)	wave 6 entities	natural keys DataAreaId, OMOperatingUnitNumber, SiteId, InventLocationId
AX DirPartyTable, DirPerson, DirOrganization, DirPartyLocation, LogisticsElectronicAddress, CustTable, VendTable, ContactPerson, CustGroup, VendGroup, PaymentTerm, DlvTerm, DlvMode (ax-odbc)	wave 7 entities	natural keys PartyNumber, `DataAreaId
AX EcoResProduct*, InventTable, InventTableModule, InventItemSetupFilter, InventItemGroup, InventModelGroup, UnitOfMeasureConversion, EcoResAttribute*/Category*, InventItemGTIN (ax-odbc)	wave 8 entities	natural keys DisplayProductNumber, `DataAreaId
AX BOMTable, BOMVersion, BOM, RouteTable, RouteVersion, RouteOpr, RouteOprTable, WrkCtrTable, WrkCtrResourceGroup, WrkCtrCapability*, WorkCalendar* (ax-odbc)	wave 9 entities	natural keys BOMId, `BOMId
D365 F&O entities CustCustomerV3Entity, VendVendorV2Entity, EcoResReleasedProductV2Entity, BOMEntity, RouteEntity, WorkCenterEntity (d365-odata)	waves 7–9	same natural keys via entity key fields
SAP T001, T001W, T001L, CSKS, ADRC, T009/T009B, T006 (sap-rfc / file)	wave 6	keys BUKRS, WERKS, `WERKS
SAP KNA1/KNB1, LFA1/LFB1, BUT000/BUT100/BUT020, ADR2/ADR6, KNVK, T052, KNKK (sap-rfc / sap-odata API_BUSINESS_PARTNER)	wave 7	keys KUNNR, LIFNR, PARTNER; CVI links honored
SAP MARA, MAKT, MARC, MARD, MARM, MEAN, KLAH/KSML/CABN/CAWN/AUSP (sap-rfc / API_PRODUCT_SRV)	wave 8	keys MATNR, `MATNR
SAP MAST/STKO/STPO/STAS, PLKO/PLPO/PLAS/MAPL, CRHD/CRCA/KAKO, AENR (sap-rfc / API_BILL_OF_MATERIAL_SRV)	wave 9	keys `STLNR

Transformation notes are owned by the waves above; this wave contributes the shared functions (`ax_datetime`, `ax_noyes`, `sap_dats`, `sap_lead_zero`, `codepage_normalize`, `lookup_party_alias`). Validation rules are the union of each target wave's rules plus cross-object dependency checks. Not migrated by this wave: any transactional table (CustTrans, VendTrans, InventTrans, InventSum, BKPF/BSEG, MSEG, VBAK/VBAP, EKKO/EKPO — wave 49 packs), AX security roles (re-modeled in wave 2's admin UI), DMF/LSMW definitions themselves (replaced by mapping packs), X++/ABAP customizations (non-goal).

1.10.12 Dependencies, risks, open questions

Depends on: waves 6, 7, 8 and 9 (bulk endpoints with `X-Passage-Dry-Run` support and natural-key upsert semantics), wave 4 (Temporal, outbox), wave 5 (audit tab integration for lineage, import/export tool), wave 2 (service-account duties, Vault secrets).

Risk	Likelihood	Impact	Mitigation
Bulk endpoints in waves 6–9 lack dry-run or upsert semantics when wave 10 starts	Medium	High	Contract requirement added to each wave's Definition of Done; wave 10 contract tests run against waves 6–9 in CI
SAP RFC SDK licensing blocks a self-contained connector image	High	Medium	OData and flat-file paths are first-class; RFC is a customer-mounted library
Synthetic AX fixture does not resemble real data quality	Medium	Medium	Fixture generator injects 2% noise (codepage, trailing spaces, invalid enums); first real customer extract (PRD open question 2) replaces it by wave 20
Reconciliation sampling misses rare drift	Low	Medium	Hash totals per column catch systematic drift; widened sample on mismatch

Open questions: (1) Is the mapping DSL YAML plus a restricted expression language enough, or do packs need a plug-in hook in TypeScript? Proposed: YAML only in v1, hooks behind a flag by wave 49; decide by wave 16. (2) Should lineage be stored in each domain database (next to the record) or only in migration-db? Proposed: migration-db with the audit tab querying it; decide by wave 11. (3) Who owns sign-off on reconciliation for the phase-exit demo — product or the implementation partner persona? Proposed: both sign; decide at the phase-1 exit review.

1.10.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`mig.workers`	project	int	8	1–16	Loader partitions (FR-10.6)	Data management parameters → entity execution threads	Migration Cockpit number of background jobs
`mig.batch_size`	project	int	500	50–5,000	Rows per API call	Set-based processing / bundle size	LTMC transfer package size
`mig.reject_threshold_pct`	project	decimal	0.5	0–10	Load gate after validation (FR-10.4)	—	—
`mig.sample_pct`	project	decimal	1 (min 100, max 5,000 rows)	0.1–100	Reconciliation sample (FR-10.8)	Compare data	—
`mig.widened_sample_pct`	project	decimal	10	1–100	BR-10.9	—	—
`mig.staging_retention_days`	project	int	30	7–365	Purge after sign-off (BR-10.11)	Staging cleanup (job history cleanup)	Migration Cockpit staging table deletion
`mig.source.codepage`	connector	enum	UTF-8	UTF-8, Windows-1252, ISO-8859-1, UTF-16LE, SAP 1100, SAP 1160	Decode source bytes	Source data formats → code page	LSMW file code page, LTMC file encoding
`mig.source.timezone`	connector	IANA	source server zone	IANA	`utcdatetime` / DATS+TIMS conversion	Source data formats → time zone	User / system time zone of extract
`mig.retire_missing`	project	bool	false	true / false	FR-10.7	—	—
`mig.overwrite_drift`	project	bool	false	true / false	BR-10.8	—	—
`mig.allow_unreconciled_deps`	project	bool	false	true / false	Run-start dependency check	Entity sequencing	Migration object dependencies
`mig.recurring.min_interval_min`	tenant	int	15	15–1,440	FR-10.16	Recurring data jobs → recurrence	Job scheduling (SM36)
`mig.cutover_at`	project	timestamptz	none	future timestamp	Stops recurring runs	—	—
`mig.signoff.approvers`	project	int	2	1–3	Required signatures (§1.10.15)	—	—
`mig.connector.paging`	connector	record	ax-odbc 10,000 RecIds; d365 `$top` 10,000; RFC_READ_TABLE ROWCOUNT 50,000	per connector	Extraction page size	Entity paging / skip staging	RFC_READ_TABLE ROWCOUNT / ROWSKIPS
`mig.pii_key_ring`	project	Vault path	`passage/mig/{project}`	Vault paths	FR-10.13	—	—
`mig.package.signing_key`	tenant	Vault path	`passage/config-signing`	Vault paths	FR-10.15	—	—

1.10.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Run history	list inquiry	project, mode, status, date	run no, mode, status, start, duration, counts	`migration_run`	Execution history (DMFExecution); job history	Migration Cockpit monitor; SM37
Run stage progress	drill-down	run, object	stage, rows, rows/s, ETA, errors	stage events	Execution details	Migration Cockpit "Monitoring"
Reconciliation report	operational report (sign-off evidence)	run, object	counts per stage, hash totals per column, sample mismatches	`reconciliation_result`	Compare data results	Migration object statistics
Exception queue	operational report	run, object, reason code	source key, reason, mapped payload, target response	`staging_row` (rejected, mismatched)	Staging data with errors	Migration Cockpit messages; LSMW error log
Mapping catalog	list inquiry	project, object, version	rules, enum tables, tests passed, checksum	`mapping_spec`	Entity mapping inquiry	LTMOM migration object modeler
Record lineage	drill-down	target entity and id	run, source object, source key, loaded at, hashes	`lineage_entry`	—	—
Throughput KPI	KPI	project, object, worker	rows/s, retries, batch failures	run metrics	—	—
Configuration package diff	drill-down	package, target tenant	create / update / unchanged per record	`config_package`	Configuration data project comparison	SCU0 customizing cross-system viewer
Recurring schedule	list inquiry	project	next run, last result, skipped triggers	schedule projection	Recurring data jobs	SM37
Copy project summary	operational report	copy project	objects, rewritten keys, collisions	copy run	Copy entity data between companies log	EC01 log

1.10.15 Printed and electronic documents

Document	Trigger	Layout elements (header/line/footer fields)	Formats	Numbering sequence	Legal / retention requirement
Reconciliation sign-off report	`:sign-off`	header: project, run no, mode, mapping and connector versions, signers; lines: per object counts, hash totals, sample size and mismatches, accepted differences with reasons; footer: signatures, sha256 of the report	PDF, XLSX	`{project}-RUN-{run_no}`	data-conversion control evidence for SOX ITGC; retain 7 years minimum (10 years where HGB/AO applies)

Document	Trigger	Layout elements (header/line/footer fields)	Formats	Numbering sequence	Legal / retention requirement
Exception export	on demand	run, object, reason, source key, payload (PII masked)	XLSX, CSV	none	follows staging retention
Configuration package	export (FR-10.15)	manifest, dependency order, signatures	JSON (zip)	package version	7 years as change evidence
Migration evidence bundle	after sign-off	zip: sign-off report, mapping checksums, lineage summary, connector probe results	ZIP	`{project}-EVID-{run_no}`	7–10 years with go-live audit file

1.10.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
MIG-J1 Staging purge timer	hourly	run_id	drops partitions once; re-run no-op	< 1 min per run	retried next hour	purge overdue > 24 h
MIG-J2 Lineage integrity	nightly 02:30	target_entity	read-only	< 20 min for 10M lineage rows	report only	orphans > 0
MIG-J3 Recurring incremental runs	per project cron (min 15 min)	project_id	BR-10.15; incremental watermark per object	< 5 min for 10k changed rows	run `failed`, next trigger proceeds	consecutive failures ≥ 3
MIG-J4 Staging tablespace monitor	every 5 min	tablespace	read-only	< 5 s	none	usage > 80%
MIG-J5 Connector credential expiry check	daily 06:00	connector	read-only	< 1 min	none	expiry within 14 days
MIG-J6 Orphaned staging partition sweep	weekly Sunday 04:00	partition	drops partitions with no run row	< 10 min	none	orphans found

1.10.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
View migrations	`GET /migration/v1/**` (PII masked)	none	Implementation consultant, Data migration lead, Auditor
Administer migration	projects, connectors, `:test`, schedules, copy projects	Approve migration sign-off	Data migration lead
Maintain migration mappings	mapping-spec upload, data templates	Approve migration sign-off	Implementation consultant
Run migrations	start, cancel, rerun-exceptions	Approve migration sign-off	Data migration lead, Implementation consultant
Resolve migration exceptions	`PATCH /exceptions/{rowId}`, bulk accept difference	Approve migration sign-off	Master data steward
Approve migration sign-off	`:sign-off`	Run migrations, Resolve migration exceptions, Maintain migration mappings on the same project	Business process owner, Controller
Apply configuration packages	config package import and apply	Maintain migration mappings (package author cannot apply to production)	System administrator
Read staged personal data	`pii.read` on staging and exceptions	Administer migration	Privacy officer

1.10.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
MIG-RUN-ACTIVE	error	"Run {n} is already active for this project."	Second start	Wait or cancel	BR-10.2
MIG-MAP-TEST-FAIL	error	"{n} mapping tests failed for {object}."	CI or upload test failure	Fix mapping	FR-10.3
MIG-MAP-UNKNOWN-FN	error	"Unknown mapping function {fn}."	Upload validation	Use sanctioned function	FR-10.3
MAP-ENUM-UNMAPPED	error	"Value {v} of {field} has no mapping."	Enum without default	Add enum row or default	BR-10.4
MAP-PRECISION-LOSS	warning	"Rounding changed {field} by {delta}."	Scale reduction	Review target scale	BR-10.5
VAL-FK-MISSING	error	"{field} references {key}, which does not exist."	FK expectation	Load parent or fix value	FR-10.4
VAL-DEP-FAILED	error	"Parent {key} failed in this run."	Dependency failure	Fix parent row	BR-10.10
API-422	error	"Target rejected the row: {detail}."	Business validation	Edit row	FR-10.5
TARGET-DRIFT	warning	"Target changed after load; not overwritten."	User edit after load	Set overwrite_drift or accept	BR-10.8
MIG-THRESHOLD-EXCEEDED	error	"Rejects {x}% exceed threshold {t}%."	Validation gate	Resolve rejects	FR-10.4
MIG-DEP-UNRECONCILED	error	"Dependency {object} has no reconciled run."	Run-start check	Reconcile dependency	§1.10.5
MIG-HASH-MISMATCH	warning	"Hash total differs for {column}; sample widened."	BR-10.9	Review mismatches	BR-10.9
MIG-PKG-SCHEMA-NEWER	error	"Package schema {v} is newer than this environment."	Version skew	Upgrade target	BR-10.14
MIG-PKG-SIGNATURE	error	"Package signature is invalid."	Tampered or wrong key	Re-export	BR-10.14
COPY-KEY-EXISTS	error	"{key} already exists in {target}."	Copy collision	Choose upsert or rewrite rule	BR-10.13
MIG-CONN-AUTH	error	"Connector {c} authentication failed."	Bad credentials	Update Vault secret	FR-10.2
MIG-RFC-ROWWIDTH	info	"Rows of {table} exceed 512 bytes; read in {n} field groups."	RFC_READ_TABLE limit	none	FR-10.2

1.10.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Full phase-1 load (170k master records)	1–5 runs per project	end-to-end < 30 min	≥ 100 rows/s per worker; 8 workers	staging ~2 KB per row, dropped after purge	staging list-partitioned by run_id	HPA to 16 workers; runtime grows linearly

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Extraction per connector worker	20k rows/s target	—	20,000 rows/s	none	source paging	backpressure on staging writes
Re-run with no changes	per run	< 5 min for 170k rows	hash compare 5,000 rows/s	none	(run_id, object, status)	unchanged
Run monitor update	SSE every 2 s	lag 0.5 / 2 / 4 s	200 concurrent viewers	none	event fan-out	refresh drops to 5 s
Reconciliation sample compare	1% sample, min 100, max 5,000 per object	< 60 s per object	500 target reads/s	results ~1 MB per run	—	sample compare serialized per object
Lineage lookup (audit tab)	50k/day	5 / 30 / 60 ms	200 req/s	~150 B per loaded row; 10M rows ≈ 2 GB	(target_entity, target_id)	unchanged
Configuration package apply (5,000 records)	10/day	20 / 60 / 120 s	1 concurrent per tenant	packages 1–10 MB	dependency order	queued

1.10.20 Localization and compliance hooks

Item	What varies	Country packs touching it (wave 48)	Extension point
Code pages and scripts	AX 2012 non-Unicode collations (Windows-1252 for US/UK/DE/FR/MX/CA data), SAP non-Unicode code pages 1100/1160, Devanagari and other Indic scripts in IN data	IN	`mig.source.codepage`; `codepage_normalize` function
Date and number formats in files	DD.MM.YYYY and decimal comma (DE, FR), DD/MM/YYYY (UK, IN, FR), MM/DD/YYYY (US), YYYY-MM-DD (CA ISO)	all	`coerce` rule with explicit format per source column
Country-specific source fields	SAP KNA1-STCD1..5 meaning per country (MX RFC, IN PAN/GSTIN in J_1I fields, FR SIRET in STCD2), AX country-specific tables (e.g. Mexico CFDI fields, India tax registration)	IN, MX, FR	mapping pack overlays keyed `country: XX` layered over the base pack
Data residency and transfer	GDPR Chapter V transfers, India DPDP Act 2023 cross-border rules, Mexico LFPDPPP transfer notice; staging stays in the tenant's region	all	per-tenant region pinning of migration-db; no cross-region connectors
Retention and erasure	staging purge (BR-10.11); sign-off evidence kept 7–10 years; personal data in exceptions masked	all	retention classes `mig-staging`, `mig-evidence`
Statutory master-data checks on load	MX RFC structure and SAT list checks, IN GSTIN state code vs address state, EU VAT format	MX, IN, DE, FR	target-API dry run (FR-10.4) runs the wave-7 validators

1.10.21 Coverage closure

No row in the AX gap register (§12.99) or the SAP gap register (§13.99) carries wave 10 as its current or proposed wave. Every AX and SAP inventory row tagged for wave 10 is `COVERED-10`. Three of those rows are covered in the inventory but were not named in this section until now: DMFCopyEntity, D365 configuration data projects and data templates, and recurring data jobs. They are now FR-10.14, FR-10.15, FR-10.18 and FR-10.16. The table below lists register-derived consequences for this wave: migration packs needed by the capabilities that waves 6–9 added to close their register rows.

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
(none assigned to wave 10)	—	—	0 register rows
Consequence of §1.6.21 (AX OMTeam rows #107, #114, #232, #251)	n/a (consequential)	ADDED as FR-10.17	OMTeam / membership pack
Consequence of §1.7.21 (AX #158, #165; SAP #17, #23, #35, #37, #47, #50)	n/a (consequential)	ADDED as FR-10.17	VendCertification, PaymSched/T052R, KNRZE/KNRZB/LFZA, CPD account groups flagged one-time
Consequence of §1.8.21 (AX #141, #143, #146, #147, #148, #283, #297; SAP #163, #181, #356, #359, #374, #375, #384, #385, #392)	n/a (consequential)	ADDED as FR-10.17	EcoResProductRelation, lifecycle states/T141, ASMD, CUVTAB, DRAW/DRAT/DRAD
Consequence of §1.9.21 (AX #144–#296 ECM rows; SAP #238, #243, #250, #362, #376, #378, #379, #381, #382, #386, #388, #391, #393, #396)	n/a (consequential)	ADDED as FR-10.17	Engineering categories, policies, STPO alternative/discontinuation fields, effectivity, STLTY = D structures

Part 2 — Phase 2: Financial Core (waves 11–15)

2.0 Phase overview

Business outcome at phase exit. At the end of wave 15 a controller closes a fiscal period for the demo manufacturer's three legal entities entirely inside Passage: every financial fact lives in one append-only universal journal (`ledger` service, Go + Postgres) that balances by construction; each line carries transaction, accounting and reporting currency amounts and is posted simultaneously to the leading, local-GAAP and tax ledgers; tax is determined and settled to the cent; periods are controlled per ledger × module; allocations, accruals and year-end carry-forward are automated; and the balance sheet, income statement and cash flow statement are generated from the journal with drill-down to the originating document. Every later module (inventory, AP, AR, assets, projects) posts only through the wave 11 posting API — there is no second path to the general ledger.

Phase-exit sign-off criteria

- Trial balance for each demo entity and each ledger sums to 0.00 in all three currencies after 1M synthetic lines; balance projection equals a full re-aggregation of the journal (zero rows differ).
- Posting API sustains 5,000 lines/s for 30 min on the dev cluster with p99 < 200 ms; duplicate `Idempotency-Key` yields a byte-identical 200 response.
- Parallel-ledger fixture (20 documents) posts per the matrix in §2.12.7 and each ledger's trial balance is zero.
- 500-scenario tax matrix (US/EU/IN/CA) matches expected results; settlement run ties to GL tax accounts to the cent.
- One simulated period closed across 3 entities with a 40-task checklist, dependencies honored; year-end carry-forward produces opening balances equal to closing balances per dimension combination.
- Statement pack (TB, BS, P&L, CF) ties to the trial balance to the cent; cash flow reconciles to the change in cash; finance stakeholders record sign-off in the wave 15 issue.

Cross-wave data model summary

Entity	Owning service	Key relationships
Ledger, LedgerGroup	ledger	LegalEntity (wave 6) 1:N Ledger; Ledger → FiscalCalendar, AccountingCurrency, ReportingCurrency
MainAccount, AccountStructure, DimensionAttribute, DimensionValue, DimensionCombination	ledger	ChartOfAccounts 1:N MainAccount; AccountStructure → ordered DimensionAttributes; DimensionCombination = hashed tuple of DimensionValues
JournalEntry, JournalLine, JournalBatch, JournalBatchLine	ledger	JournalEntry 1:N JournalLine; JournalEntry N:1 Ledger; JournalBatch (unposted) → JournalEntry (posted); SourceDocument reference (URI)
Balance (projection)	ledger	Keyed by ledger × account × DimensionCombination × fiscal period × currency type
ExchangeRateType, ExchangeRate, CurrencyPolicy	fx	ExchangeRate → (type, from, to, valid_from); Ledger → ExchangeRateType default
RevaluationRun, RevaluationLine	ledger	Produces JournalEntry pairs (posting + reversal)
TaxCode, TaxRate, TaxGroup, ItemTaxGroup, TaxJurisdiction, ExemptionCertificate, WithholdingTaxCode	tax	TaxGroup N:M TaxCode; ItemTaxGroup N:M TaxCode; TaxRate effective-dated per TaxCode × Jurisdiction
TaxTransaction, TaxSettlementRun	tax	TaxTransaction → JournalLine (posted tax line); SettlementRun → JournalEntry
PeriodStatus, CloseTemplate, CloseTask, CloseRun	close (TS/Temporal) + ledger (PeriodStatus)	PeriodStatus keyed ledger × period × module; CloseRun → CloseTasks → assignees (Party/Employee role)
AllocationRule, AllocationRun, AccrualScheme, RecurringJournal	ledger	Runs produce JournalEntry; rules reference AccountStructure filters
ReportDefinition, ReportRow, ReportColumn, ReportTree, StatementRun, StatementCell	reporting	ReportDefinition → MainAccount ranges + DimensionValue filters; StatementCell → Balance keys → JournalLines

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
dev.passage.ledger.journal.posted.v1	ledger	reporting, tax, audit, lakehouse, all subledgers (waves 16+)	entry id, ledger id, document id, posting date, period, line count, totals per currency type, source ref, trace id
dev.passage.ledger.journal.reversed.v1	ledger	reporting, tax, subledgers	original entry id, reversing entry id, reason code
dev.passage.ledger.balance.rebuilt.v1	ledger	reporting, ops alerting	ledger id, high-water sequence, rows, diffs
dev.passage.fx.rate.published.v1	fx	ledger, reporting, AP/AR (later)	rate type, pair, valid_from, rate, source, previous rate
dev.passage.fx.provider.degraded.v1	fx	ops alerting	provider, last success, fallback mode
dev.passage.ledger.revaluation.completed.v1	ledger	reporting, close	run id, ledger, period, unrealized gain/loss totals, entry ids
dev.passage.tax.calculated.v1	tax	document services (waves 17+, 26+)	document ref, lines, tax amounts, provider used, fallback flag
dev.passage.tax.settled.v1	tax	reporting, ledger	settlement period, payable/receivable totals, entry id
dev.passage.close.period.status-changed.v1	ledger	all posting services, close, reporting	ledger, period, module, old → new status, actor, approvals
dev.passage.close.task.completed.v1	close	close board, notifications	run id, task id, assignee, evidence link
dev.passage.ledger.allocation.completed.v1	ledger	reporting, close	run id, rule ids, iterations, entry ids
dev.passage.ledger.year-end.closed.v1	ledger	reporting, close	ledger, fiscal year, opening entry ids, retained earnings per dimension
dev.passage.reporting.statement.generated.v1	reporting	distribution, audit	run id, definition id, period, hash, output URIs

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Balance projection drifts from journal under concurrent posting	Medium	High	Projection updated in the same Postgres transaction as the entry; nightly rebuild-and-compare job; drift alert at 1 row	ledger lead
Rounding differences across three currencies and N ledgers accumulate	Medium	High	ISO 4217 minor-unit rules in one shared Go package `money`; property tests over JPY/KWD/USD/EUR; rounding-difference account with tolerance	ledger lead
Tax engine correctness outside fixture coverage	High	High	Provider adapters (Avalara/Vertex) as production default for US; internal engine for VAT with accountant-reviewed fixtures	tax lead

Risk	Likelihood	Impact	Mitigation	Owner
Period reopened silently	Low	High	Reopen requires two distinct approvers (Temporal human tasks) and emits an audit event; no API bypass	close lead
Reporting projection lags and statements are stale at close	Medium	Medium	Statement run records the journal high-water sequence it covered; close task blocks until lag = 0	reporting lead
Accountant review capacity (two-person team)	High	Medium	Fixtures reviewed per wave by an external practicing accountant; sign-off recorded in the wave issue	product

2.11 Wave 11: Universal Journal & Chart of Accounts

2.11.1 Scope and business context

AX stores financial truth across `GeneralJournalEntry`/`GeneralJournalAccountEntry` with `DimensionAttributeValueCombination` as the account key, and SAP consolidated FI/CO into `ACDOCA`. Both rely on a posting layer that every module calls. Wave 11 builds that layer first: one append-only, double-entry journal per ledger, financial dimensions validated by account structures, a posting API with idempotency keys, a balance projection, and a journal workspace for manual, recurring and template journals with approval and attachments.

At the end of the wave a controller can define a chart of accounts and dimensions, post a manual journal (with approval), reverse it, and see the trial balance update in real time. Subledgers (waves 16+) will call `POST /api/v1/ledger/journal-entries` and nothing else.

Out of scope: multiple currencies beyond a single accounting currency (wave 12), tax lines (wave 13), period status enforcement beyond open/closed defaults (wave 14), statements (wave 15), intercompany (wave 40), budget control (wave 38).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX `LedgerJournalTable` form: posting a 2,000-line journal runs `LedgerJournalCheckPost` synchronously; validation errors surface one at a time in the Infolog after minutes of processing, forcing repeated post attempts	Validation and posting interleaved in X++ with per-line dimension lookups against `DimensionAttributeValueCombination`	`POST /ledger/validate` returns every violation with `line_index` in one pass; posting is a single atomic transaction	Validation of 2,000 lines < 1 s (vs 2–5 min); all errors in one response (vs 1 per attempt)
SAP FB50/F-02: a document that fails mid-posting (dump, lock on `NRIV` number range) can consume a document number or leave a parked `VBKPF` that is re-keyed	Number assignment and insert into `BKPF`/`BSEG` are not one atomic unit with the application-level checks	Gap-free `posting_sequence` assigned inside the same Postgres transaction; idempotency key makes retries safe	0 lost numbers; 0 duplicate postings under crash test; retry needs 0 user action
Integrations double-post when a timeout hides a success (AIF/IDoc retries, Data management framework re-runs)	No idempotency contract on `LedgerJournalTrans` import or BAPI_ACC_DOCUMENT_POST	`Idempotency-Key` header with 400-day retention; same key + same payload → replay, different payload → 409	Duplicate-post incidents 1–3/month typical → 0
AX trial balance list page / SAP FAGLB03 are read from totals tables (`FAGLFLEXT`) that drift from line items after failed batches; reconciliation jobs (RFINDEX) run for hours	Totals maintained by separate update paths from line inserts	`Balance` upserted in the posting transaction; rebuild-and-compare job emits a diff count	Balance vs journal drift = 0 rows; rebuild of 10M lines < 15 min (vs multi-hour RFINDEX)
Reaching the source document from a GL line takes 6–10 clicks (AX voucher transactions → origin → subledger; SAP FB03 → environment → document flow)	Linkage stored in `SubledgerVoucherGeneralJournalEntry` / `BKPF-AWKEY` with per-module resolution logic	Every entry carries `source.uri`; UI resolves it as one link; `trace_id` on every entry	Clicks to source 8 → 1; any posting traceable through Tempo end-to-end
External systems (MES, BI, PLM) need custom X++/ABAP or middleware to learn that a posting happened	No event stream from the ledger; polling on `GeneralJournalEntry`/`ACDOCA` timestamps	`dev.passage.ledger.journal.posted.v1` on Kafka via outbox, AsyncAPI-described, plus REST/MCP	Integration lead time weeks → same day; event lag p95 < 2 s (vs 15-min+ polling)

2.11.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Chart of accounts and main accounts	`MainAccount`, `LedgerChartOfAccounts`, form MainAccountListPage	`SKA1` (chart), `SKB1` (company-code view), FS00, OB13	`ledger.MainAccount`, `ledger.ChartOfAccounts`; screen Chart of accounts	full
Financial dimensions	`DimensionAttribute`, `DimensionAttributeValue`, `DimensionFinancialTag`	CO objects (cost center `CSKS`, profit center `CEPC`), ACDOCA fields `RCNTR`, `PRCTR`, `RBUSA`	`ledger.DimensionAttribute`, `ledger.DimensionValue` (up to 20 per ledger)	full
Account structures / dimension validation	`DimensionHierarchy` (account structure), advanced rules	Field status groups (OBC4), validation/substitution (OB28/GGB0)	`ledger.AccountStructure` with ordered constraint rows	full
Ledger account combination	`DimensionAttributeValueCombination` (`RecId` as LedgerDimension)	ACDOCA field combination (no single key)	`ledger.DimensionCombination` (SHA-256 of ordered tuple)	improved: content-addressed, immutable
Journal entry / lines	`GeneralJournalEntry`, `GeneralJournalAccountEntry`, `SubledgerVoucherGeneralJournalEntry`	`BKPF`/`BSEG`, `ACDOCA`	`ledger.JournalEntry`, `ledger.JournalLine`	full
Manual journals (unposted)	`LedgerJournalTable`/`LedgerJournalTrans`, LedgerJournalTable form, journal names `LedgerJournalName`	FB50/F-02 park (`VBKPF`/`VBSEG`), FV50	`ledger.JournalBatch`, `ledger.JournalBatchLine`; screen Journals	full
Recurring / periodic journals	Periodic journals (`LedgerJournalTable.JournalType = Periodic`)	FBD1 recurring entries (`BKDF`)	`ledger.RecurringJournal` (template + schedule)	full
Journal approval	Ledger journal workflow `LedgerJournalApproval`	Parked document workflow (WS10000051)	Temporal approval workflow template (wave 4) bound to JournalBatch	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Reversal	`LedgerJournalTrans` reverse transaction, `TransactionReversalTrans`	FB08, FBRA (reset and reverse)	`POST /journal-entries/{id}/reverse` creating negating entry	full
Voucher numbering	`NumberSequenceTable` per journal name	Document number ranges per document type (FBN1, `NRIV`)	Number sequence service (wave 5), gap-free per ledger × document type	full
Trial balance / balances	`GeneralJournalAccountEntry` aggregation, Trial balance list page, `LedgerBalance...`	`FAGLFLEXT` (totals), `GLT0`, FAGLB03	`ledger.Balance` projection; Trial balance screen	full
Posting definitions / posting profiles	Posting definitions, `LedgerPostingSetup`	Account determination (OBYC, T030)	Deferred to the owning subledger waves; wave 11 exposes only account + dimension resolution hooks	partial (by design)
Financial reason codes	`ReasonTable`, `ReasonTableRef`, Reason codes form (Organization administration > Setup > Financial reasons); reversal reason prompt in `LedgerTransReverse`	`T041C` reasons for reversal (OB_REVERSE), negative-posting flag, alternative posting date	`ledger.ReasonCode` with `applies_to[]` modules; required on reversal when the journal name demands it (FR-11.15)	full
Advanced rule structures	`DimensionHierarchy` (type Advanced rule structure), `DimensionAdvancedRule`, `DimensionRuleCriteria`, form "Advanced rules"	Field status groups (OBC4) and validations only; no conditional segment model	`ledger.AdvancedRule` (filter → appended structure segments) (FR-11.16)	improved: versioned with the structure
Default descriptions (transaction text)	Default descriptions form (`TransactionTxt`), placeholders `%1`–`%3` per transaction type and language	Header text `BKPF-BKTXT` / line text `BSEG-SGTXT` set by substitution exits	`ledger.DescriptionTemplate` per source_type × language (FR-11.17)	full
Open-item managed G/L accounts and clearing	Not available for ledger accounts (only customer/vendor settlement and bank reconciliation)	`SKB1-XOPVW` open-item management, F-03/F-04 manual clearing, F.13/F13E automatic clearing, FBRA reset, FAGL_ACTIVATE_OP / FAGL_SWITCH_TO_OPEN_ITEM	`ledger.OpenItem`, `ledger.ClearingDocument`, clearing rules, tolerance groups (FR-11.19, FR-11.20); reused by waves 18/19/30	full
User-defined validation rules	Journal controls (`LedgerJournalControl`), main account validation (currency, user, posting type)	OB28 / GGB0 validations (`GB90*`, `T001Q`) with message classes at header/line/document	`ledger.ValidationRule` (prerequisite → check → message, CEL) (FR-11.21)	full
Dimension derivation / substitution	Default dimensions on master data; D365 default dimension templates; no rule engine	OBBH / GGB1 substitutions (`GB91*`, `T001R`), KEDR derivation strategy (`TKEDR*`, `K9R*`), 0KEM/0KEP PCA substitution (`T8A*`), FMDERIVE strategy	`ledger.DerivationStrategy` with ordered steps (table lookup, move, clear, expression, constant) applied before structure validation (FR-11.22)	full
Functional area / cost-of-sales accounting	Custom dimension attribute	OKBD functional areas (`TFKB`), FAGL_FUNC_AREA activation, OBBZ/OBBW derivation, `ACDOCA-RFAREA`	Standard dimension attribute `FUNC_AREA` with a shipped lookup-table derivation step (FR-11.22)	full
Document splitting	Not available (account structures require dimensions per line; no inheritance to AP/AR/tax lines)	FAGL_SPLIT*: item categories, splitting method 0000000012, zero-balance clearing account, constants (`T8G01`–`T8G40`)	`ledger.SplittingRule` per source_type: proportional inheritance of splitting dimensions to balance-sheet and tax lines plus zero-balance clearing lines (FR-11.23)	full
Dimension realignment	Not available (manual reclassification journals)	KEND / KEDR realignment run (CO-PA), re-derivation of characteristics on history	`ledger.RealignmentRun` posting reclassification entries, append-only (FR-11.24)	improved: no in-place update of history
Attachments with document types	`DocuRef`, `DocuType`, `DocuValue`, Document handling form	GOS attachments (`SRGBTBREL`), ArchiveLink (OAC0/OAC3) document types	Wave 5 document framework: `document_type`, retention class, store reference on every attachment (FR-11.18)	full

2.11.3 Functional requirements

- **FR-11.1 Ledger and chart of accounts** — Must. The system shall maintain one or more ledgers per legal entity, each bound to a chart of accounts, a fiscal calendar (wave 6), and an accounting currency.
  - A chart of accounts may be shared across legal entities (SAP `SKA1` semantics); company-specific attributes (`SKB1` semantics: suspended, posting allowed, default dimensions) live on `MainAccountLedgerSetting`.
  - Main account id is 1–20 characters, alphanumeric, unique per chart; type ∈ {asset, liability, equity, revenue, expense, profit_and_loss_total, balance_sheet_total, reporting}; totals and reporting types are non-postable.
  - Maximum 50,000 main accounts per chart; 500 charts per tenant.
- **FR-11.2 Financial dimensions** — Must. The system shall support up to 20 dimension attributes per ledger, each backed either by a Passage entity (department, cost center, site, project, customer, vendor, item) or by a custom value list.
  - Dimension values are effective-dated (`active_from`, `active_to`) and may be suspended; values from backing entities are synchronized by consuming that entity's CloudEvents.
  - Default dimensions on master data (party, product, site) are stored as a `DimensionCombination` id and merged at posting time by precedence: explicit line > source document header > master data default > account default.
- **FR-11.3 Account structures and validation** — Must. The system shall validate every line's (main account, dimensions) tuple against the ledger's active account structures before posting.
  - An account structure is an ordered list of segments (main account first) with per-segment constraint rows: value ranges, lists, wildcards (`*`), and "blank allowed". Advanced rules add extra segments when a filter matches (AX advanced rule semantics).
  - Validation is deterministic: the first structure whose main-account filter matches is applied; no match → `LEDGER_STRUCTURE_NOT_FOUND`.
  - Structures are versioned with `effective_from`; a posting uses the version effective on the posting date.
- **FR-11.4 Dimension combinations** — Must. The system shall intern each validated tuple as an immutable `DimensionCombination` whose id is the SHA-256 of the canonical tuple `(structure_version_id, account_id, dim1_value_id, …, dim20_value_id)`.
  - Combinations are never deleted; display name is derived (`1010-DEPT01-CC200`).
  - Lookup by id is a primary-key read; lookup by tuple is a hash computation plus one read.
- **FR-11.5 Posting API** — Must. The system shall expose `POST /api/v1/ledger/journal-entries` that posts a balanced entry atomically or rejects it entirely, with an idempotency key.
  - Request: `Idempotency-Key` header (1–128 chars, scoped to tenant × legal entity), body per §2.11.5. Maximum 10,000 lines per entry; larger documents must be split by the caller with a shared `document_id`.
  - Response 201 with the posted entry; 200 on replay of the identical payload; 409 `LEDGER_IDEMPOTENCY_CONFLICT` if the same key is reused with a different payload hash; 409 `LEDGER_IDEMPOTENCY_IN_PROGRESS` with `Retry-After: 1` while the first request is still in flight.
  - 422 with a `violations[]` array, each `{code, line_index, field, message}`; codes in BR-11.9. Posting never partially succeeds.
  - Idempotency records are retained 400 days; replays after expiry are treated as new requests (documented contract).

- **FR-11.6 Append-only journal** — Must. The system shall never update or delete a posted `JournalEntry` or `JournalLine`; corrections are new entries.
  - Enforced by Postgres: the application role has INSERT and SELECT only on `journal_entry/journal_line`; a trigger raises on UPDATE/DELETE for all roles.
  - Every entry receives a gap-free `posting_sequence` (bigint) per ledger from the number sequence service in gap-free mode, assigned inside the posting transaction.
- **FR-11.7 Balance projection** — Must. The system shall maintain `ledger.Balance` rows keyed by (ledger, account, dimension combination, fiscal period, currency type) and update them in the same database transaction as the posting.
  - Fields: `debit`, `credit`, `net`, `line_count`, `last_posting_sequence`; period-to-date only; year-to-date and opening balances are computed at query time by summing periods (opening balances live in period 0, see wave 14).
  - A rebuild job (§2.11.7 BR-11.7) reconstructs the table from the journal and reports differences.
- **FR-11.8 Reversal** — Must. The system shall create an exact negating entry for any posted entry via `POST /journal-entries/{id}/reverse`.
  - Reversal date defaults to the original posting date; may be any open date ≥ original. Each reversing line has `debit/credit` swapped and `reversal_of_line_id` set.
  - An entry can be reversed once; reversing a reversal is permitted (re-instating), each linked to its predecessor.
- **FR-11.9 Journal workspace (manual journals)** — Must. The system shall provide journal batches (AX `LedgerJournalTable` semantics) with lines, totals, validation on save, submit-for-approval, and post.
  - Journal names carry: voucher number sequence, default offset account, approval workflow id, `posting_restriction` (own journals only / by user group), and `lines_limit` (default 5,000).
  - Batch statuses: draft → validated → in_approval → approved → posting → posted | rejected | cancelled (state machine in §2.11.7).
- **FR-11.10 Recurring journals and templates** — Should. The system shall allow a posted or draft batch to be saved as a template (amounts or percentages) and a template to be scheduled (cron expression, fiscal-period-relative dates) to generate batches.
  - Generation runs on Temporal at the scheduled time; generated batches enter the approval workflow unless the template is flagged `auto_post` and the user who scheduled it holds duty `Post ledger journals without approval`.
- **FR-11.11 Attachments and notes** — Should. The system shall attach files (max 25 MB each, 50 per batch) and notes to batches and carry references to the posted entry.
- **FR-11.12 Source document linkage** — Must. The system shall record on each entry a `source` object `{system, type, id, uri}` and on each line an optional `source_line_ref`, enabling drill-down from statements (wave 15) to the originating document.
- **FR-11.13 Opening balances load** — Must. The system shall accept opening balances through the migration framework (wave 10) as a journal entry in period 0 of the first fiscal year, balanced against a conversion-suspense account that must be zero after load.
- **FR-11.14 Query API** — Must. The system shall expose line and balance queries with filters on ledger, account ranges, dimension values, posting date and period, paginated by keyset cursor (max 1,000 rows per page).
- **FR-11.15 Financial reason codes** — Must. The system shall maintain one shared `ReasonCode` list per tenant (code ≤ 10 characters, name, `applies_to[]` ⊆ {ledger, bank, customer, vendor, fixed_asset, inventory_counting, tax, project}, `requires_comment`, `negative_posting_allowed`, `alternative_posting_date_allowed`, `active`) that later modules reuse, and shall require a reason code on journal reversal when the journal name or the ledger parameter `reversal_reason_required` is set.
  - The reverse request carries `reason_code` and optional `reason_comment` (≤ 255 characters); both are stored on the reversing entry and emitted in `journal.reversed.reason_code`.
  - `negative_posting_allowed` (SAP `T041C` semantics, AX "Correction" parameter): the reversing lines are flagged `negative_posting = true` and the `Balance` projection reduces the original side (debit −100) instead of adding to the opposite side (credit +100); trial balance totals of debits and credits therefore return to their pre-posting values.
  - `alternative_posting_date_allowed = false` forces the reversal date to equal the original posting date when that period is open.
  - Maximum 500 codes per tenant; a referenced code cannot be deleted, only deactivated.
- **FR-11.16 Advanced rule structures** — Must. Beyond FR-11.3, the system shall maintain advanced rules: a filter over (main account ranges, dimension values, wildcards) that, when it matches a line, appends an advanced rule structure (additional segments with their own constraint rows) to the account structure applied to that line.
  - Several advanced rules may match; their structures are appended in `priority` order; the resulting segment list is capped at 20 attributes (`LEDGER_ADVANCED_RULE_SEGMENT_LIMIT`).
  - Advanced rules and advanced rule structures share the account structure's version and activation step (AX "Activate" on the structure); activation re-validates the 1,000 most recently used combinations under the new version and lists those that would become invalid before the user confirms.
- **FR-11.17 Default descriptions** — Should. The system shall maintain description templates keyed by (source_type, legal entity or tenant, language) with placeholders `%1` … `%9` bound to named fields (`{voucher}`, `{party_name}`, `{document_date}`, `{amount_txn}`, `{currency_txn}`, `{source_id}`, `{period}`), applied when a posting request omits `description` or a line omits `text`.
  - Substituted text is truncated to 255 characters with a trailing ellipsis; language resolves from the legal entity default language (wave 6), or from the user's UI language for manual journals when the journal name sets `description_language = user`.
  - Legal-entity templates override tenant templates (AX Default descriptions per company semantics).
- **FR-11.18 Attachment document types and retention** — Should. Attachments (FR-11.11) shall carry a `document_type` from the wave 5 document framework (shipped types `invoice_image`, `approval_evidence`, `supporting_calculation`, `correspondence`, `other`), a retention class (default `financial_record_10y`), and a store reference (object-storage key or external DMS URI), and shall remain readable from the posted entry after the batch is posted.
  - Attachments become immutable when the batch reaches `posted`; deleting an attachment before its retention class expires is rejected with `LEDGER_ATTACHMENT_RETAINED`.
  - Maximum 25 MB per file, 50 per batch, as in FR-11.11; virus scanning and content-type allow-list are inherited from wave 5.
- **FR-11.19 Open-item managed G/L accounts and clearing** — Must. The system shall allow `MainAccountLedgerSetting.open_item_managed = true` on balance-sheet accounts (clearing accounts, bank sub-accounts, GR/IR, suspense, payroll clearing); every posted line on such an account creates an `OpenItem`; open items are cleared by a `ClearingDocument` grouping items whose Σ(debit − credit) in the clearing currency is 0 within the applicable tolerance.
  - Manual clearing (SAP F-03 semantics): the user selects items (filters: assignment, `source_line_ref`, document id, date range, amount, dimension values); the system shows the residual; a residual within the user's and account's tolerance group posts a difference line to the tolerance group's difference account; a residual outside tolerance either creates a residual item (new open item for the remainder, `residual_of_item_id`) or is rejected with `LEDGER_CLEARING_UNBALANCED`.
  - Post with clearing (F-04 semantics): a manual journal line on an open-item account may list `clear_item_ids[]`; clearing happens in the same posting transaction and fails the whole post on mismatch.
  - Reset clearing (FBRA semantics): reopens the items, marks the clearing document `reset` with a reason code (FR-11.15) and actor; if the clearing posted difference or FX lines, a negating entry is posted. Posted entries are never modified.
  - Activation on an account with history (FAGL_ACTIVATE_OP / FAGL_SWITCH_TO_OPEN_ITEM semantics): permitted only through an activation run that creates one opening open item per non-zero (dimension combination, transaction currency) balance dated the activation date and records the run id on the account setting; deactivation requires zero open items.

    - Clearing currency per account: `accounting` (default) or `transaction` (foreign-currency sub-accounts); FX differences on clearing post to the realized gain/loss accounts of the wave 12 `CurrencyPolicy` and are excluded from wave 12 revaluation (`revalue_level = open_item`).
    - AP and AR settlement (waves 19/30) and GR/IR clearing (wave 18) reuse this engine through the open-item contract in §2.11.5 with their own item sources; this wave ships the GL account source only.
- **FR-11.20 Automatic clearing rules and tolerance groups** — Must. The system shall define automatic clearing rules per (ledger, account range) as an ordered list of up to 5 criteria fields (`assignment`, `source_line_ref`, `document_id`, `dim.<attr>`, `amount_txn`, `posting_date` bucket) that group open items (SAP OB74 semantics), and tolerance groups (SAP OBA0 / AX "Maximum penny difference" semantics) with absolute and percentage limits per currency, assigned to accounts and to users.
    - The scheduled job `gl-auto-clear` (§2.11.16) evaluates each rule, clears every group whose Σ = 0 (or within the account tolerance when the rule sets `allow_tolerance`) as one `ClearingDocument` per group, and never creates residual items; a simulation mode returns the proposed groups.
    - The effective tolerance is the minimum of the user's and the account's group; service-account runs use the account group only.
- **FR-11.21 User-defined validation rules** — Must. The system shall evaluate ordered validation rules per ledger at header, line and document level, each with `prerequisite` (CEL boolean over header fields, line fields, dimension values, main account attributes, source, user groups), `check` (CEL boolean) and `message` (message-catalog code, severity error | warning, parameter bindings).
    - A failed error rule produces a 422 violation `{code, rule_id, line_index}`; a failed warning rule produces a `warnings[]` entry and, for manual batches, a confirmation step before submit.
    - Rules are effective-dated and versioned; evaluation adds ≤ 2 ms per 100 lines; the rule editor runs a rule against saved sample entries and shows the result per line.
    - AX journal controls and main-account validations (currency, user, posting type) and SAP OB28/GGB0 validations are shipped as rule templates.
- **FR-11.22 Dimension derivation and substitution rules** — Must. The system shall apply a per-ledger `DerivationStrategy` before account-structure validation: ordered steps of type `table_lookup` (source fields → target dimension through a validity-dated lookup table, e.g., (account category, cost center) → functional area), `move` (copy a field or dimension to another), `clear`, `expression` (CEL) and `constant`, each with an optional condition and an `overwrite` flag.
    - Precedence extends FR-11.2: explicit line value > step with `overwrite` > source document header > master data default > account default > step without `overwrite`.
    - Every derived value is recorded per line as `derived_by_step_ids[]`, shown in the journal line derivation trace and returned by `POST /ledger/validate`.
    - Functional area for cost-of-sales accounting is a standard dimension attribute `FUNC_AREA` with a shipped lookup-table step; the P&L-by-function statement in wave 15 reads it. Segment derivation from profit center (SAP FAGL_SEGM) is a shipped `table_lookup` step.
    - Lookup tables are maintained in the UI and by CSV import (≤ 100,000 rows per table); SAP KEDR / 0KEM / FMDERIVE strategies and OBBH substitutions migrate to steps (§2.11.11).
- **FR-11.23 Document splitting: dimension inheritance and zero balance** — Must. The system shall, per source_type, inherit a configured set of splitting dimensions (for example profit center, segment, business unit) from the lines that carry them to the lines that do not (payable, receivable, tax, cash discount, bank, rounding) in proportion to the carrying lines' amounts, and post zero-balance clearing lines so that every entry balances per splitting dimension value.
    - Item categories per account range (expense_revenue, payable, receivable, tax, cash, balance_sheet_other) and the method (`active` = proportional inheritance; `passive` = inherit from the open items being cleared, FR-11.19) are configuration; each ledger names a `zero_balance_clearing_account` and a `default_splitting_value` constant used when no carrying line exists.
    - Splitting runs after derivation (FR-11.22) and before structure validation; produced lines carry `split_of_line_id` and `inherited = true`; proportional rounding uses the BR-12.4 residual rule; a line that cannot be split and has no default raises `LEDGER_SPLIT_NO_SOURCE`.
    - A balance sheet by profit center or segment (wave 15 `BS-STD` with a tree over the splitting dimension) therefore balances per unit.
- **FR-11.24 Dimension realignment** — Should. The system shall provide a controlled restatement that, for a filter (ledger, account ranges, period range, dimension values), re-derives selected dimensions with the current derivation strategy or an explicit old → new value mapping and posts reclassification entries per (account, period, old combination, new combination) flagged `is_realignment = true`, without modifying any posted line.
    - Simulation lists affected combinations, line counts and amounts; posting requires duty `Run dimension realignment`, a reason code and an open or on-hold GL period with the runner in the allowed group; a run is reversible as a unit.
    - Realignment entries are ordinary entries for balances and statements and appear in the wave 15 audit trail as `kind = realignment_run`.

2.11.4 Data model

MainAccount

Field	Type	Required	Notes
id	uuid	yes	PK
chart_of_accounts_id	uuid	yes	FK ChartOfAccounts
account_number	text(20)	yes	unique per chart; `^[A-Za-z0-9._-]{1,20}$`
name	text(120)	yes	translatable via wave 3 metadata
account_type	enum	yes	asset, liability, equity, revenue, expense, profit_and_loss_total, balance_sheet_total, reporting
debit_credit_default	enum	no	debit/credit; UI default only
is_postable	bool	yes	false for totals/reporting
suspended_from	date	no	posting rejected on/after this date
default_dimension_combination_id	bytea(32)	no	FK DimensionCombination
reconciliation_account_type	enum	no	none, customer, vendor, bank, inventory, fixed_asset — subledger-only posting (SAP recon account semantics)
created_at / created_by	timestamptz / uuid	yes	audit from CDC

AccountStructure (header) and **AccountStructureConstraint** (rows)

Field	Type	Required	Notes
id	uuid	yes	PK
ledger_id	uuid	yes	FK Ledger
version	int	yes	unique (ledger_id, name, version)
name	text(60)	yes	
segments	jsonb	yes	ordered array of dimension_attribute_id; first is MAIN_ACCOUNT
effective_from	date	yes	
status	enum	yes	draft, active, retired
constraint rows: segment_index	int	yes	0 = main account

Field	Type	Required	Notes
constraint rows: parent_path	jsonb	yes	the matched values of previous segments this row applies under
constraint rows: allowed	jsonb	yes	list of `{from,to}` ranges, exact values, `*`, `blank`

JournalEntry and **JournalLine**

Field	Type	Required	Notes
id	uuid	yes	PK
ledger_id	uuid	yes	FK Ledger
legal_entity_id	uuid	yes	denormalized for RLS
document_id	uuid	yes	groups entries across ledgers and split batches
voucher	text(30)	yes	from number sequence; unique (ledger_id, voucher)
posting_sequence	bigint	yes	gap-free per ledger; unique (ledger_id, posting_sequence)
posting_date	date	yes	
fiscal_period_id	uuid	yes	FK FiscalPeriod (wave 6) resolved from posting_date
document_date	date	no	
description	text(255)	no	
source_system / source_type / source_id / source_uri	text	yes	e.g. `ledger`, `manual_journal`, batch id, `passage://ledger/journal-batches/{id}`
idempotency_key	text(128)	yes	unique (tenant_id, legal_entity_id, idempotency_key)
payload_hash	bytea(32)	yes	SHA-256 of canonical request
reversal_of_entry_id	uuid	no	FK JournalEntry
posted_at / posted_by	timestamptz / uuid	yes	
trace_id	text(32)	yes	OpenTelemetry trace
line: id	uuid	yes	PK
line: entry_id	uuid	yes	FK
line: line_number	int	yes	unique per entry, 1..10000
line: dimension_combination_id	bytea(32)	yes	FK DimensionCombination (includes account)
line: main_account_id	uuid	yes	denormalized for indexing
line: debit_credit	enum	yes	D or C; amounts always positive
line: amount_txn / currency_txn	numeric(23,6) / char(3)	yes	wave 11: equals accounting currency
line: amount_acct / amount_rpt	numeric(23,6)	yes	rounded to minor units (BR-11.3)
line: quantity / unit	numeric(18,6) / text	no	statistical quantity
line: text	text(255)	no	
line: source_line_ref	text(200)	no	
line: reversal_of_line_id	uuid	no	

Balance (projection): `ledger_id`, `main_account_id`, `dimension_combination_id`, `fiscal_period_id`, `currency_type` (txn/acct/rpt), `currency` char(3), `debit`, `credit`, `net`, `line_count`, `last_posting_sequence`; PK over the first five.

ReasonCode (FR-11.15)

Field	Type	Required	Notes
id	uuid	yes	PK
code	text(10)	yes	unique per tenant; `^[A-Z0-9_]{1,10}$`
name	text(60)	yes	translatable
applies_to	enum[]	yes	ledger, bank, customer, vendor, fixed_asset, inventory_counting, tax, project
requires_comment	bool	yes	default false
negative_posting_allowed	bool	yes	default false; SAP `T041C-XNEGP`
alternative_posting_date_allowed	bool	yes	default true
active	bool	yes	

OpenItem and **ClearingDocument** (FR-11.19, FR-11.20)

Field	Type	Required	Notes
item: id	uuid	yes	PK
item: ledger_id / main_account_id / dimension_combination_id	uuid / uuid / bytea(32)	yes	account must be `open_item_managed`
item: journal_line_id	uuid	yes	unique; the line that created the item (or activation run)
item: source_type	enum	yes	gl_line, activation, residual; waves 18/19/30 add gr_ir, vendor, customer
item: amount_clearing / currency_clearing	numeric(23,6) / char(3)	yes	signed (debit positive) in the account's clearing currency
item: amount_acct	numeric(23,6)	yes	
item: assignment	text(50)	no	free matching key (SAP `ZUONR` semantics)
item: posting_date / due_date	date	yes / no	
item: status	enum	yes	open, cleared, reset_open
item: clearing_document_id	uuid	no	set when cleared
item: residual_of_item_id	uuid	no	FK OpenItem
doc: id	uuid	yes	PK
doc: ledger_id / main_account_id	uuid	yes	
doc: clearing_date	date	yes	posting date of difference/FX lines
doc: mode	enum	yes	manual, post_with_clearing, automatic, activation

Field	Type	Required	Notes
doc: rule_id	uuid	no	FK AutoClearingRule when automatic
doc: item_ids	uuid[]	yes	2..10,000
doc: difference_amount / difference_entry_id	numeric(23,6) / uuid	no	tolerance posting
doc: fx_difference_entry_id	uuid	no	wave 12 realized gain/loss
doc: status	enum	yes	cleared, reset
doc: reset_reason_code / reset_by / reset_at	text / uuid / timestamptz	conditional	

AutoClearingRule: `id`, `ledger_id`, `account_from`, `account_to`, `criteria` (jsonb ordered list ≤ 5 of field names), `allow_tolerance` bool, `active`. **ToleranceGroup**: `id`, `legal_entity_id`, `code`, `scope` (account | user), `limits` (jsonb per currency: `{abs_debit, abs_credit, pct}`), `difference_account_id`; assignment tables `tolerance_group_account` (account range) and `tolerance_group_user`.

ValidationRule and **DerivationStrategy** (FR-11.21, FR-11.22)

Field	Type	Required	Notes
rule: id / ledger_id / sequence	uuid / uuid / int	yes	evaluated in sequence
rule: level	enum	yes	header, line, document
rule: prerequisite	text (CEL)	no	empty = always
rule: check	text (CEL)	yes	must evaluate true, else message
rule: message_code / severity / params	text / enum / jsonb	yes	catalog code (§2.11.18), error or warning
rule: effective_from / effective_to / version / status	date / date / int / enum	yes	draft, active, retired
strategy: id / ledger_id / version / status		yes	one active version per ledger
step: id / strategy_id / sequence		yes	≤ 50 steps
step: kind	enum	yes	table_lookup, move, clear, expression, constant
step: condition	text (CEL)	no	
step: source_fields / target	jsonb / text	yes	target is a dimension attribute or line field
step: lookup_table_id	uuid	conditional	FK LookupTable (rows: source tuple, target value, valid_from, valid_to)
step: overwrite	bool	yes	default false

SplittingRule (FR-11.23): `id`, `ledger_id`, `source_type`, `splitting_dimensions` (attribute ids), `method` (active | passive), `item_categories` (jsonb account range → category), `zero_balance_clearing_account_id`, `default_splitting_value` (jsonb attribute → value), `active`. **RealignmentRun** (FR-11.24): `id`, `ledger_id`, `filter` (jsonb), `mapping` (jsonb old → new per attribute) or `strategy_version`, `mode` (simulate | post), `status` (proposed, posted, reversed, cancelled), `reason_code`, `entry_ids[]`, `reversal_entry_ids[]`, `combinations_affected`, `created_by`. **DescriptionTemplate** (FR-11.17): `id`, `scope` (tenant | legal_entity), `legal_entity_id`, `source_type`, `language`, `template` text(255), `field_bindings` (jsonb `%n` → field path).

Invariants

- For every JournalEntry: Σ debit lines = Σ credit lines in each of amount_txn (per currency), amount_acct and amount_rpt, exactly, after rounding.
- Every JournalLine's `dimension_combination_id` was valid under the account structure version effective on `posting_date`.
- `posting_sequence` is dense (no gaps) per ledger; `Balance.last_posting_sequence` ≤ max posted sequence.
- Σ Balance.net over all accounts for a (ledger, period, currency_type) = 0.

Indexing / partitioning: `journal_line` is range-partitioned by `fiscal_year` (one partition per year per tenant shard); indexes on (ledger_id, main_account_id, posting_date), (dimension_combination_id, posting_date), (entry_id); `journal_entry` indexed on (ledger_id, posting_sequence), (document_id), (source_system, source_type, source_id). `balance` is unpartitioned, PK is the clustered access path. All tables carry `tenant_id` with RLS.

2.11.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/ledger/journal-entries	Post a balanced entry	`Idempotency-Key` required	Post journal entries (service accounts) / Post ledger journals
POST	/api/v1/ledger/journal-entries/{id}/reverse	Create negating entry	`Idempotency-Key` required	Reverse journal entries
GET	/api/v1/ledger/journal-entries/{id}	Entry with lines	n/a	View ledger transactions
GET	/api/v1/ledger/journal-lines?ledger=&account_from=&account_to=&dim.{attr}=&date_from=&date_to=&cursor=	Line query	n/a	View ledger transactions
GET	/api/v1/ledger/balances?ledger=&period_from=&period_to=&account_from=&account_to=&dim.{attr}=&group_by=	Balance query	n/a	View ledger balances
POST	/api/v1/ledger/validate	Dry-run validation of an entry (no post)	none (stateless)	Post ledger journals
GET/PUT	/api/v1/ledger/charts/{chart}/accounts[/{number}]	Main account CRUD	PUT is naturally idempotent	Maintain chart of accounts
GET/PUT	/api/v1/ledger/ledgers/{ledger}/account-structures[/{id}]	Structure CRUD and activation	PUT idempotent	Maintain account structures
POST	/api/v1/ledger/dimension-combinations/resolve	Tuple → combination id	stateless	View ledger transactions
POST/GET/PATCH	/api/v1/ledger/journal-batches[/{id}] and /{id}:submit, :approve, :post	Manual journal lifecycle	`Idempotency-Key` on :post	Maintain / Approve / Post ledger journals
POST	/api/v1/ledger/balances:rebuild	Trigger rebuild (admin)	run id returned	Administer ledger
GET/PUT	/api/v1/ledger/reason-codes[/{code}]	Reason code CRUD (shared)	PUT idempotent	Maintain reason codes
GET/PUT	/api/v1/ledger/ledgers/{ledger}/advanced-rules[/{id}]	Advanced rules and advanced rule structures	PUT idempotent	Maintain account structures
GET/PUT	/api/v1/ledger/description-templates[/{id}]	Default description templates	PUT idempotent	Maintain ledger setup

Method	Path	Purpose	Idempotency	Auth duty
GET	/api/v1/ledger/open-items?ledger=&account=&status=&assignment=&dim.{attr}=&cursor=	Open items on OI-managed accounts	n/a	View ledger transactions
POST	/api/v1/ledger/clearings	Manual clearing of selected items (with tolerance / residual options)	`Idempotency-Key` required	Clear open items
POST	/api/v1/ledger/clearings:simulate	Residual and tolerance preview	stateless	Clear open items
POST	/api/v1/ledger/clearings/{id}:reset	Reset clearing (reason code required)	`Idempotency-Key` required	Reset clearings
POST	/api/v1/ledger/accounts/{account}:activate-open-items	Activation run for an account with history	`Idempotency-Key` required	Administer ledger
GET/PUT	/api/v1/ledger/ledgers/{ledger}/auto-clearing-rules[/{id}] and /tolerance-groups[/{code}]	Automatic clearing rules, tolerance groups and assignments	PUT idempotent	Maintain clearing rules
POST	/api/v1/ledger/auto-clearing:run	Run automatic clearing now (simulate or clear)	run id	Run automatic clearing
GET/PUT	/api/v1/ledger/ledgers/{ledger}/validation-rules[/{id}] and POST /validation-rules/{id}:test	Validation rules CRUD and test against samples	PUT idempotent	Maintain validation and derivation rules
GET/PUT	/api/v1/ledger/ledgers/{ledger}/derivation-strategies[/{version}] and /lookup-tables[/{id}] (CSV import)	Derivation strategy, steps, lookup tables	PUT idempotent; import `Idempotency-Key`	Maintain validation and derivation rules
GET/PUT	/api/v1/ledger/ledgers/{ledger}/splitting-rules[/{source_type}]	Document splitting configuration	PUT idempotent	Maintain ledger setup
POST	/api/v1/ledger/realignments and /{id}:post, :reverse	Dimension realignment simulate / post / reverse	`Idempotency-Key` required	Run dimension realignment

Posting request (canonical JSON)

```
POST /api/v1/ledger/journal-entries
Idempotency-Key: ap-invoice-7F3A-post
{ "ledger_group": "ALL" | "ledger_id": "...",
  "document_id": "uuid", "posting_date": "2026-03-31", "document_date": "2026-03-28",
  "description": "Vendor invoice 7F3A",
  "source": {"system":"ap","type":"vendor_invoice","id":"7F3A","uri":"passage://ap/invoices/7F3A"},
  "lines": [
    {"line_number":1,"account":"6010","dimensions":{"DEPT":"D100","CC":"C200"},
     "debit_credit":"D","amount":{"currency":"USD","value":"1250.00"},"text":"Freight"},
    {"line_number":2,"account":"2100","dimensions":{},"debit_credit":"C",
     "amount":{"currency":"USD","value":"1250.00"}} ] }
```

Response 201: `{ "entries":[{"id","ledger_id","voucher","posting_sequence","fiscal_period_id","totals":{"txn":{...},"acct":"1250.00","rpt":"1150.75"},"lines":[...]}], "idempotency":{"key","replayed":false}, "trace_id" }`. Amounts are strings with exactly the currency's minor-unit scale.

Commands / validation: `PostJournalEntry` (validation order: schema → ledger/period resolution → account existence and postability → structure validation → combination interning → currency/rounding (wave 12) → period status (wave 14) → balance check → sequence assignment → insert → balance upsert → outbox). `ReverseJournalEntry` (original exists, not already reversed unless `allow_reinstate`, target date open). `SubmitJournalBatch` (every line valid, totals balanced, user owns batch or holds duty). `ApproveJournalBatch` (approver ≠ submitter unless journal name allows self-approval).

Event	Trigger	Payload fields	Consumers
dev.passage.ledger.journal.posted.v1	entry committed	entry_id, ledger_id, document_id, voucher, posting_sequence, posting_date, fiscal_period_id, line_count, totals{txn[],acct,rpt}, source, trace_id	reporting, tax, audit, lakehouse, subledgers
dev.passage.ledger.journal.reversed.v1	reversal committed	original_entry_id, reversing_entry_id, reason_code, posting_date	reporting, subledgers
dev.passage.ledger.batch.status-changed.v1	batch transition	batch_id, journal_name, from, to, actor, workflow_run_id	notifications, audit
dev.passage.ledger.balance.rebuilt.v1	rebuild finished	ledger_id, high_water_sequence, rows_written, rows_differing, duration_ms	ops alerting
dev.passage.ledger.account.changed.v1	account/structure change	chart_id, account_number, change_type, effective_from	reporting, dimension sync
dev.passage.ledger.open-item.cleared.v1	clearing document committed or reset	clearing_document_id, ledger_id, account, mode, item_ids, difference_amount, fx_difference_entry_id, status (cleared/reset), reason_code	bank reconciliation (wave 36), AP/AR (waves 19/30), reporting, audit
dev.passage.ledger.realignment.completed.v1	realignment posted or reversed	run_id, ledger_id, period_range, combinations_affected, entry_ids, reversal	reporting, audit
dev.passage.ledger.rules.changed.v1	validation rule, derivation strategy or splitting rule activated	ledger_id, kind, id, version, actor	posting cache, audit

Open-item contract for subledgers (used by waves 18/19/30): a service registers an item source `{source_type, resolver_uri}`; the clearing engine calls `resolver_uri` to fetch candidate items and posts the clearing through the same `ClearingDocument` path; the source service consumes `open-item.cleared` to update its own status. The posting request accepts per line `"clear_item_ids": [...]` (FR-11.19) and the derivation/splitting/validation stages run in the order shown in the command pipeline below.

Posting pipeline (revised with FR-11.21–FR-11.23): schema → ledger/period resolution → default descriptions (FR-11.17) → derivation strategy (FR-11.22) → document splitting (FR-11.23) → account existence and postability → structure and advanced rule validation (FR-11.3, FR-11.16) → validation rules (FR-11.21) → combination interning → currency/rounding (wave 12) → period status (wave 14) → balance check → open-item clearing (FR-11.19) → sequence assignment → insert → balance upsert → outbox. `POST /ledger/validate` runs the same stages up to the balance check and returns the derivation trace and splitting result.

2.11.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Chart of accounts	Create, edit, suspend, import CSV, view balances	number, name, type, postable, suspended_from, default dimensions	Maintain chart of accounts
Financial dimensions	Add attribute, bind to entity, maintain custom values	attribute, backing entity, active_from/to, suspended	Maintain financial dimensions
Account structures	Design segments, constraints grid, validate sample, activate version	segment, allowed ranges, effective_from, status	Maintain account structures

Screen	Primary actions	Key fields/columns	Duty required
Journals (list)	New batch, filter by status/name/owner, bulk submit	voucher, journal name, status, total debit/credit, owner, posting date	Maintain ledger journals
Journal batch (detail)	Add/edit lines inline, validate, submit, approve, post, attach, save as template	line grid: account, dimensions (segmented entry), D/C, amount, text; totals bar	Maintain / Approve / Post ledger journals
Trial balance	Filter ledger/period/dimension, drill to lines, export	account, opening, debit, credit, closing, net	View ledger balances
Ledger transactions	Filter, drill to source document, trace id link	voucher, date, account, dims, D/C, amount, source	View ledger transactions
Reason codes	Create, deactivate, set modules and flags	code, name, applies_to, requires_comment, negative_posting_allowed	Maintain reason codes
Open items and clearing	Filter items, select, see residual, clear, create residual item, reset clearing	account, assignment, date, amount (clearing / acct), status, clearing document	Clear open items / Reset clearings
Clearing rules and tolerances	Define auto-clearing criteria, tolerance groups, assign to accounts and users, simulate run	account range, criteria order, allow_tolerance, limits per currency, difference account	Maintain clearing rules
Validation and derivation rules	Rule editor with CEL assist, lookup table grid and CSV import, test against sample entries, activate version	sequence, level, prerequisite, check, message; step kind, source, target, overwrite	Maintain validation and derivation rules
Document splitting setup	Item categories by account range, splitting dimensions, method, zero-balance account, test with a sample document	source_type, dimensions, method, categories, clearing account	Maintain ledger setup
Dimension realignment	Define filter and mapping, simulate, review affected combinations, post, reverse	account range, periods, old value, new value, lines, amount	Run dimension realignment

Interaction rules: segmented account entry accepts `1010-D100-C200` typed as one string with per-segment lookup (Tab moves segment); validation on blur per line and on save for the batch; grid supports paste from Excel (up to 5,000 rows) and bulk fill-down of dimensions; keyboard: Ctrl+Enter = save line, Alt+P = post (confirm dialog), Esc = cancel edit; saved views and column personalization persist per user (wave 5 shell).

2.11.7 Business rules and edge cases

- **BR-11.1 Balance by construction.** An entry is accepted only if Σ debits = Σ credits in every amount column; the check is performed on rounded amounts and repeated inside the database transaction (CHECK via deferred constraint trigger).
- **BR-11.2 Zero-amount lines** are allowed only when `quantity` ≠ 0 (statistical) or the line carries `text`; otherwise rejected with `LEDGER_LINE_EMPTY`.
- **BR-11.3 Rounding.** Amounts are stored with scale equal to the currency's ISO 4217 minor units (USD/EUR 2, JPY 0, KWD/BHD 3); requests with excess precision are rejected with `LEDGER_AMOUNT_SCALE` (no silent rounding of transaction amounts). Derived amounts (wave 12) are rounded half-away-from-zero.
- **BR-11.4 Posting date.** Must fall in a fiscal period of the ledger's calendar; date before the ledger's `first_open_date` or in a closed period → `LEDGER_PERIOD_CLOSED`. Future dates are allowed up to 366 days.
- **BR-11.5 Concurrency.** Posting takes a row lock on `ledger` only for sequence assignment (sub-millisecond); balance upserts use `INSERT … ON CONFLICT DO UPDATE` ordered by PK to prevent deadlocks; the service retries serialization failures up to 3 times with jitter.
- **BR-11.6 Idempotency key semantics.** Key is compared after trimming; scope is (tenant, legal entity). A replay returns the original response including the original `trace_id` and `replayed: true`. Keys are persisted before the entry insert in the same transaction, so a crash leaves either both or neither.
- **BR-11.7 Balance rebuild algorithm.** (1) Record `hwm = max(posting_sequence)` per ledger. (2) In a shadow table, `INSERT … SELECT ledger_id, main_account_id, dimension_combination_id, fiscal_period_id, currency_type, SUM(debit), SUM(credit), COUNT(*), MAX(posting_sequence) FROM journal_line JOIN journal_entry WHERE posting_sequence ≤ hwm GROUP BY …`, chunked by fiscal period (each chunk ≤ 5M lines, committed separately). (3) Catch-up loop: apply lines with sequence in (hwm, new_hwm] as deltas; repeat until the delta is < 1,000 lines. (4) Take a 2-second advisory lock on posting for the ledger, apply the final delta, compare shadow to live row by row (FULL OUTER JOIN on PK, any non-zero difference recorded), swap tables atomically, release. (5) Emit `balance.rebuilt` with `rows_differing`; > 0 raises a P1 alert and the live table is kept as the rebuilt copy (the journal is truth).
- **BR-11.8 Reconciliation account protection.** Accounts with `reconciliation_account_type ≠ none` reject manual batch lines (`LEDGER_RECON_ACCOUNT_MANUAL`) unless the journal name is flagged `allow_recon_accounts` (migration and correction use only).
- **BR-11.9 Error codes (422 unless noted).** `LEDGER_UNBALANCED`, `LEDGER_LINE_LIMIT`, `LEDGER_LINE_EMPTY`, `LEDGER_AMOUNT_SCALE`, `LEDGER_ACCOUNT_NOT_FOUND`, `LEDGER_ACCOUNT_NOT_POSTABLE`, `LEDGER_ACCOUNT_SUSPENDED`, `LEDGER_RECON_ACCOUNT_MANUAL`, `LEDGER_STRUCTURE_NOT_FOUND`, `LEDGER_DIMENSION_REQUIRED`, `LEDGER_DIMENSION_INVALID`, `LEDGER_DIMENSION_INACTIVE`, `LEDGER_COMBINATION_INVALID`, `LEDGER_PERIOD_NOT_FOUND`, `LEDGER_PERIOD_CLOSED`, `LEDGER_PERIOD_ON_HOLD` (403), `LEDGER_DATE_INVALID`, `LEDGER_CURRENCY_MISMATCH`, `LEDGER_RATE_MISSING`, `LEDGER_ROUNDING_EXCEEDED`, `LEDGER_REVERSAL_ALREADY_REVERSED`, `LEDGER_IDEMPOTENCY_CONFLICT` (409), `LEDGER_IDEMPOTENCY_IN_PROGRESS` (409), `LEDGER_FORBIDDEN` (403).
- **BR-11.10 Journal batch state machine.** `draft → validated` (all lines valid) → `in_approval` (submit; skipped if journal name has no workflow) → `approved | rejected` → `posting` (lock) → `posted` (entry ids recorded) or back to `approved` with error on failure; `draft|validated|rejected → cancelled`; any line edit in `validated|approved` returns the batch to `draft`. Posted and cancelled are terminal.
- **BR-11.11 Document split.** Entries sharing a `document_id` are independent for balancing; callers that split a document must balance each part (use a clearing account) — the API does not defer balancing across requests.
- **BR-11.12 Clearing tolerance.** For a selected item set, `residual = Σ amount_clearing`. Let `T_user` and `T_account` be the applicable tolerance groups (absent = 0). The posting limit is `min(T_user, T_account)` per side (`abs_debit` for a debit residual, `abs_credit` for a credit residual) and additionally `pct × Σ|amount|` when `pct` is set. `|residual| = 0` → clear; `0 < |residual| ≤ limit` → clear and post the residual to the tolerance group's `difference_account_id` with the dimensions of the largest item; `|residual| > limit` → `residual_item = true` creates a residual item for `residual` on the same account and combination, else `LEDGER_CLEARING_UNBALANCED`.
- **BR-11.13 Open-item lifecycle.** `open → cleared` (exactly once per clearing document) → `reset_open` (reset) → `cleared` (again, by a new document); an item in `cleared` cannot be in a second active clearing document (`LEDGER_OPEN_ITEM_ALREADY_CLEARED`); a residual item's `residual_of_item_id` chain is at most 20 deep; clearing dates may not precede the latest posting date of the items cleared; a clearing document is reset as a whole, never partially.
- **BR-11.14 Derivation evaluation.** Steps run strictly in `sequence`; a step whose target was set by an earlier step in the same run is skipped unless `overwrite = true`; a step may not read a field it also writes; lookup tables are matched on the full source tuple with `*` wildcards and the most specific row wins (fewest wildcards, then latest `valid_from`); a strategy with > 50 steps or a step with a CEL parse error cannot be activated (`LEDGER_DERIVATION_INVALID`).
- **BR-11.15 Splitting proportions.** For an entry with carrying lines (amounts `a_i`, splitting values `v_i`) and a non-carrying line amount `b`: produce lines `b × a_i / Σa` per distinct `v`, rounded per BR-12.4 with the residual on the largest share; when Σa = 0 or no carrying line exists use `default_splitting_value` or raise `LEDGER_SPLIT_NO_SOURCE`; after inheritance, for each splitting value with Σ(debit – credit) ≠ 0, add a pair of zero-balance lines (clearing account) so each value balances; zero-balance lines are excluded from P&L statements and flagged `is_zero_balance = true`.
- **BR-11.16 Negative posting.** When a reversal uses a reason code with `negative_posting_allowed`, the reversing lines keep the original side with the opposite sign in the `Balance` projection (`debit −x`), while `JournalLine.debit_credit` stays D/C with positive amounts and `negative_posting = true`; reports that sum debits and credits (trial balance, document journal) apply the flag so turnover totals equal the pre-posting values.
- **BR-11.17 Validation rule evaluation.** All rules are evaluated (no short-circuit) so the response lists every violation; a rule referencing a dimension attribute not in the ledger is skipped and logged; warnings do not block API callers (service accounts) but are returned in `warnings[]`; the same rule set applies to reversals, allocations, accruals and year-end entries unless the rule's prerequisite excludes the `source_type`.

- **BR-11.18 Open-item account constraints.** `open_item_managed` is allowed only for balance-sheet accounts that are not reconciliation accounts of type customer/vendor (those are managed by waves 19/30) and not `is_postable = false`; the flag change is effective-dated and requires an activation run when the account has non-zero balances; open items are revalued (wave 12) at item level, so the account is excluded from balance-level revaluation.

2.11.8 Acceptance criteria

- **AC-11.1** (FR-11.5, FR-11.6) Given a balanced two-line USD entry in an open period, when posted with a new key, then 201 is returned within 200 ms p99, a `posting_sequence` is assigned, and `journal.posted` is observed on Kafka within 2 s.
- **AC-11.2** (FR-11.5) Given the same request repeated with the same key, when posted, then 200 is returned with `replayed: true` and no new entry exists.
- **AC-11.3** (FR-11.5) Given the same key with a changed amount, when posted, then 409 `LEDGER_IDEMPOTENCY_CONFLICT`.
- **AC-11.4** (FR-11.3) Given account 6010 requires DEPT under the active structure, when a line omits DEPT, then 422 `LEDGER_DIMENSION_REQUIRED` with `line_index`.
- **AC-11.5** (BR-11.1) Given lines summing to debit 100.00 / credit 99.99, when posted, then 422 `LEDGER_UNBALANCED` and no rows are written.
- **AC-11.6** (FR-11.7) Given 1M random balanced entries posted by 64 concurrent workers, when complete, then Σ Balance.net = 0 for every period and a rebuild reports `rows_differing = 0`.
- **AC-11.7** (FR-11.8) Given a posted entry, when reversed, then a new entry exists whose lines equal the original with D/C swapped and `reversal_of_line_id` set, and a second reversal returns `LEDGER_REVERSAL_ALREADY_REVERSED`.
- **AC-11.8** (FR-11.9, BR-11.10) Given a journal name with a 2-level approval, when a clerk submits a batch, then it is `in_approval`, the approver sees a Temporal task, and after approval the clerk can post.
- **AC-11.9** (FR-11.13) Given the demo opening balance file, when loaded through the migration framework, then the conversion-suspense account balance is 0.00 and the trial balance equals the AX extract totals.
- **AC-11.10** (FR-11.14) Given 10M lines, when querying lines for one account and month, then the first page returns in < 300 ms p95 using the keyset cursor.
- **AC-11.11** (FR-11.15) Given a journal name with `reversal_reason_required = true`, when a reversal is requested without `reason_code`, then 422 `LEDGER_REASON_REQUIRED`; when requested with a code flagged `negative_posting_allowed`, then the trial balance debit and credit turnover equal their values before the original posting.
- **AC-11.12** (FR-11.19, BR-11.12) Given open items of USD 100.00 (D) and USD 99.98 (C) on a clearing account and a user tolerance of USD 0.05, when cleared manually, then one clearing document exists, a USD 0.02 difference line is posted to the tolerance account, and both items are `cleared`; with tolerance USD 0.01 and `residual_item = true`, then a residual item of USD 0.02 is created instead.
- **AC-11.13** (FR-11.20) Given 1,000 open items on GR/IR-style accounts with matching `assignment` values summing to zero in 400 groups and 50 unmatched items, when `gl-auto-clear` runs, then 400 clearing documents exist, 50 items remain open, and no residual items were created.
- **AC-11.14** (FR-11.21) Given an active rule "lines on accounts 6000–6999 require DEPT when amount_acct > 1,000", when a 1,500.00 line without DEPT is validated, then 422 with the rule id and `line_index`; a 500.00 line without DEPT passes.
- **AC-11.15** (FR-11.22) Given a lookup step (account category = COGS, cost center = C200) → FUNC_AREA = PROD, when a line on a COGS account with C200 and no FUNC_AREA is posted, then the stored combination contains FUNC_AREA = PROD and `derived_by_step_ids` names the step.
- **AC-11.16** (FR-11.23, BR-11.15) Given a vendor invoice with expense lines 600.00 (PC1) and 400.00 (PC2), a tax line 190.00 and a payable line 1,190.00, when posted with an active splitting rule on PC, then the payable is split 714.00 / 476.00 and the tax 114.00 / 76.00, and Σ(debit – credit) per profit center is 0.00.
- **AC-11.17** (FR-11.24) Given 2,000 lines on cost center C100 across 3 periods, when a realignment C100 → C150 is posted and then reversed, then balances per combination equal the pre-run values and both runs appear in the audit trail.
- **AC-11.18** (FR-11.16) Given an advanced rule "account 5000–5999 and DEPT = D100 adds segment PROJECT (required)", when a line on 5100/D100 omits PROJECT, then 422 `LEDGER_DIMENSION_REQUIRED` for PROJECT; a line on 5100/D200 posts without PROJECT.
- **AC-11.19** (FR-11.17) Given a tenant template for `manual_journal` "Journal %1 of %2" bound to `{voucher}` and `{document_date}`, when a batch without description is posted, then the entry description is "Journal GJ-000123 of 2026-03-31".

2.11.9 Test plan and fixtures

- Fixtures: `coa-demo` (1,200 accounts, 3 charts, from the demo manufacturer's AX `MainAccount` extract); `dims-demo` (8 attributes, 2,400 values); `structures-demo` (4 structures, 2 advanced rules); `entries-random-1m` (generator: seeded, 1M entries, 2–50 lines each, balanced by construction); `entries-negative-120` (120 hand-written rejection cases, one per error code × variant); `opening-balances-demo` (3 entities, CSV from AX `GeneralJournalAccountEntry` aggregation).

Test	Type	Target	Pass criterion
Balanced-entry property test	property (Go rapid)	posting service	any generated balanced entry posts; any mutated unbalanced entry is rejected
Error-code table	unit	validator	120 cases map to expected code and line_index
Idempotency contract	contract (OpenAPI + Pact)	posting API	replay/conflict/in-progress semantics as specified
Concurrent posting	integration	Postgres + service	64 workers × 1M entries: zero sequence gaps/duplicates, TB = 0
Rebuild equivalence	integration	rebuild job	rows_differing = 0 after concurrent load
Throughput	performance (k6)	dev cluster 3 nodes	5,000 lines/s for 30 min, p99 < 200 ms, error rate 0
Journal workspace E2E	Playwright + axe	React shell	create → submit → approve → post; WCAG 2.1 AA pass
Append-only enforcement	unit (SQL)	triggers/grants	UPDATE/DELETE on journal tables raise for app role and superuser

Reconciliation checks: Σ Balance per (ledger, period) = 0; Σ Balance per account = Σ journal lines per account; count of `journal.posted` events = count of entries (outbox audit); opening-balance conversion suspense = 0.

2.11.10 Deployment and operations

Helm chart `passage-ledger` (deployment, HPA 3–12 pods on CPU 60%, PodDisruptionBudget 2, CloudNativePG cluster `ledger-db` 3 instances, Kafka topics `ledger.journal`, `ledger.batch`, Debezium outbox connector). Config: `LEDGER_MAX_LINES=10000`, `LEDGER_IDEMPOTENCY_TTL_DAYS=400`, `LEDGER_REBUILD_CHUNK_LINES=5000000`, `LEDGER_FUTURE_DATE_DAYS=366`. Migrations via Atlas generated from entity metadata; partitions for the next fiscal year created by a monthly job. Feature flags: `ledger.recurring_journals`, `ledger.advanced_rules`. Scheduled jobs (Temporal): nightly `balance-verify` (rebuild into shadow, compare, no swap unless diff), monthly `create-partitions`, recurring journal generator. SLOs: posting availability 99.9%; posting p99 < 200 ms; `journal.posted` publish lag p95 < 2 s. Alerts: balance verify diff > 0 (P1), sequence gap detected (P1), outbox lag > 60 s (P2), posting error rate > 1% over 5 min (P2). Runbook: rebuild procedure, sequence gap investigation, idempotency table growth. Rollback: Argo CD previous digest; schema migrations are additive-only in this wave; no data rollback required because the journal is append-only.

2.11.11 Migration from AX / SAP

Source	Passage entity	Notes
AX `MainAccount`, `LedgerChartOfAccounts`, `MainAccountLegalEntity`	MainAccount, ChartOfAccounts, MainAccountLedgerSetting	type mapping from `MainAccountType`; suspended → `suspended_from`
AX `DimensionAttribute`, `DimensionAttributeValue`, `DimensionFinancialTag`	DimensionAttribute, DimensionValue	entity-backed attributes re-bound to Passage parties/sites; custom lists copied
AX `DimensionHierarchy` (account structures), `DimensionConstraintTree/Node`	AccountStructure(+Constraint)	advanced rules flattened into constraint rows; activated version 1
AX `DimensionAttributeValueCombination`	DimensionCombination	recomputed from tuples; `RecId` kept in `legacy_ref` for cross-reference
AX `GeneralJournalEntry/GeneralJournalAccountEntry` (closing balances only)	JournalEntry (period 0)	opening balances per combination; detail history not migrated in Phase 2 (wave 49 transactional packs)
SAP `SKA1/SKAT/SKB1`	ChartOfAccounts, MainAccount, MainAccountLedgerSetting	`XSPEB` (blocked) → suspended; `MITKZ` → reconciliation_account_type
SAP `CSKS`, `CEPC`, `TGSB`	DimensionValue (CC, PC, BA attributes)	validity dates preserved
SAP `FAGLFLEXT/ACDOCA` closing balances by period	JournalEntry (period 0)	per ledger (`RLDNR`) → Passage ledger
AX `ReasonTable`, `ReasonTableRef`	ReasonCode	`applies_to` from the Ledger/Bank/Cust/Vend/Asset flags
AX `DimensionHierarchy` (advanced rule structures), `DimensionAdvancedRule`, `DimensionRuleCriteria`	AdvancedRule (+ structure)	criteria → filter; structure segments → appended segments
AX `TransactionTxt` (default descriptions)	DescriptionTemplate	`%1`–`%3` mapped to field bindings per transaction type
AX `DocuRef/DocuValue` on `LedgerJournalTable`	Attachment (wave 5 framework)	`DocuType` → document_type; files copied to object storage
SAP `T041C`	ReasonCode	`XNEGP` → negative_posting_allowed; `XABWD` → alternative_posting_date_allowed
SAP `SKB1–XOPVW`, open items on OI-managed G/L accounts (`BSIS` with `AUGBL` blank), `T043S`, `TF123`	MainAccountLedgerSetting.open_item_managed, OpenItem (activation run), ToleranceGroup, AutoClearingRule	one open item per uncleared `BSIS` row dated cutover; cleared history (`BSAS`) not migrated
SAP `GB90*/T001Q` validations, `GB91*/T001R` substitutions, `TKEDR*/K9R*` derivation, `T8A*` PCA substitution, `FMDERIVE*`, `TFKB` functional areas	ValidationRule, DerivationStrategy steps, LookupTable, dimension values for FUNC_AREA	Boolean rules converted to CEL where the exit is table-based; user exits (ZGGB*) listed in a conversion report for re-authoring
SAP `T8G01`–`T8G40` document splitting	SplittingRule	item categories → categories; splitting method → active; zero-balance account → clearing account

Validation: account count match; every migrated combination validates under the migrated structure (exceptions queued); opening TB per entity equals source to the cent. Not migrated: AX `LedgerJournalTable` unposted batches (users re-key), SAP parked documents, posting definitions (re-implemented per subledger wave).

2.11.12 Dependencies, risks, open questions

Depends on: wave 3 (metadata entity generation), wave 4 (outbox, Temporal approval template), wave 5 (gap-free number sequences, audit), wave 6 (legal entities, fiscal calendars, currencies), wave 7 (party dimension backing), wave 10 (opening balance load).

Risk	Likelihood	Impact	Mitigation
Gap-free sequence becomes a throughput bottleneck	Medium	Medium	Sequence lock held < 1 ms; batch allocation for multi-entry documents; measured in the 5k lines/s test
Dimension combination table growth (millions of tuples)	Medium	Low	32-byte hash key; combinations are reused; monitored
Structure change invalidates existing combinations	Medium	Medium	Combinations are tied to structure version; old versions stay queryable

Open questions: (1) Should `amount_txn` be stored as integer minor units instead of numeric(23,6)? Default: numeric with scale check; decide by wave 12 start. (2) Idempotency scope per legal entity vs per tenant? Default: per legal entity; decide by wave 11 mid-point. (3) Maximum dimensions 20 vs AX's practical 10? Default 20; decide by wave 11 exit.

2.11.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`max_lines_per_entry`	tenant	int	10,000	100–50,000	Posting request line limit (FR-11.5)	— (journal `LinesLimit` on journal name)	999 line limit per `BSEG` document (hard)
`idempotency_ttl_days`	tenant	int	400	30–3,650	Retention of idempotency records (FR-11.5)	—	—
`future_posting_days`	legal entity	int	366	0–730	Maximum days ahead for `posting_date` (BR-11.4)	General ledger parameters > Ledger > "Allow posting to future periods" (implicit)	OB52 open-period range
`reversal_reason_required`	legal entity	bool	true	true / false	Reason code mandatory on reversal (FR-11.15)	General ledger parameters > Ledger > Reason code requirements (reversal)	T041C reason required on FB08
`negative_posting_enabled`	ledger	bool	false	true / false	Permits reason codes with `negative_posting_allowed` (BR-11.16)	General ledger parameters > Ledger > Correction (storno)	OBY6 "Negative postings permitted"

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`check_voucher_used`	ledger	enum	reject_duplicates	reject_duplicates / warn / accept	Duplicate voucher detection across journal names	General ledger parameters > Ledger > Check for voucher used	FBN1 number range object `RF_BELEG` (unique by design)
`use_document_date_for_voucher`	ledger	bool	false	true / false	Voucher sequence drawn from document date year	General ledger parameters > Ledger > Use document date for vouchers	OBA7 document type "Doc. date = posting date"
`journal_name.posting_restriction`	journal name	enum	by_user_group	none / own_journals_only / by_user_group	Who may post batches of this name (FR-11.9)	Journal names > Posting restrictions	OBA7 authorization group
`journal_name.lines_limit`	journal name	int	5,000	1–50,000	Lines per batch (FR-11.9)	Journal names > Lines limit	—
`journal_name.allow_recon_accounts`	journal name	bool	false	true / false	Manual lines on reconciliation accounts (BR-11.8)	Journal names > Journal control (account types allowed)	OBA7 "Account types allowed" (D/K)
`journal_name.self_approval_allowed`	journal name	bool	false	true / false	Approver may equal submitter	Workflow configuration > automatic actions	WS10000051 agent rule
`journal_name.voucher_sequence_id`	journal name	ref	required	number sequence (wave 5), gap-free	Voucher numbering per journal name	Journal names > Voucher series	OBA7 number range per document type (FBN1)
`journal_name.description_language`	journal name	enum	legal_entity	legal_entity / user	Language for default descriptions (FR-11.17)	Default descriptions > Language	—
`max_dimension_attributes`	ledger	int	20	1–20	Attributes per ledger (FR-11.2)	Financial dimensions (practical limit ~10)	ACDOCA fixed fields + coding block extension (OXK3)
`account.open_item_managed`	main account × ledger	bool	false	true / false	Creates open items per line (FR-11.19)	—	FS00 company-code data "Open item management" (`SKB1-XOPVW`)
`account.clearing_currency`	main account × ledger	enum	accounting	accounting / transaction	Currency in which items must balance to clear	—	FS00 "Only balances in local currency"
`tolerance_group.limits`	legal entity (per group)	jsonb	`{USD:{abs_debit:0.05, abs_credit:0.05, pct:0}}`	per currency	Clearing difference posting limits (BR-11.12)	General ledger parameters > Ledger > Maximum penny difference	OBA0 tolerance groups for G/L accounts; OB57 user assignment
`auto_clearing.schedule`	ledger	cron	`0 2 * * *`	cron	Schedule of `gl-auto-clear` (FR-11.20)	—	F.13 batch variant (SM36)
`validation_rules.active_version`	ledger	int	latest active	activated version	Rule set in force (FR-11.21)	Journal controls	OB28 activation level 1/2 per company code
`derivation_strategy.active_version`	ledger	int	latest active	activated version	Strategy in force (FR-11.22)	—	KEDR strategy active; OBBH activation
`splitting.method` per source_type	ledger	enum	off	off / active / passive	Document splitting (FR-11.23)	—	FAGL_SPLIT activation per company code; splitting method
`splitting.zero_balance_clearing_account_id`	ledger	ref	none	balance-sheet account	Zero-balance lines (BR-11.15)	—	Zero-balance clearing account (T8G30)
`realignment.enabled`	ledger	bool	false	true / false	Enables FR-11.24 runs	—	KEND authorization
`balance_rebuild_chunk_lines`	tenant	int	5,000,000	1M–20M	Chunk size in BR-11.7	Financial dimension sets > Rebuild balances (no tuning)	RFINDEX variant
`attachment.default_retention_class`	legal entity	ref	`financial_record_10y`	wave 5 retention classes	Retention applied to voucher attachments (FR-11.18)	Document management parameters > Document types	OAC3 ArchiveLink document type / retention

2.11.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Trial balance	list inquiry	ledger, period range, dimension set, currency type, account range	account, opening, debit, credit, closing, net, line count	`Balance`	Trial balance list page (`LedgerTrialBalanceListPage`); SSRS Trial balance	FAGLB03, S_ALR_87012277, F.08
Ledger transactions (voucher transactions)	drill-down	ledger, voucher, date range, account, dimension values, source system	voucher, posting date, account, dimensions, D/C, txn/acct/rpt amounts, text, source link, trace id	`JournalLine` ⋈ `JournalEntry`	Voucher transactions / Ledger transactions inquiry; SSRS Ledger transaction list	FBL3N, FAGLL03, S_ALR_87012282, Fiori Display Line Items in General Ledger
Dimension statement	list inquiry	ledger, dimension attribute + value, period range	account, net per period, YTD	`Balance`	Dimension balance inquiry (`DimensionFocus` balances)	KE5Z (profit center), S_ALR_87013611 (cost center)
Chart of accounts listing	operational report	chart, account type, postable, suspended	number, name, type, category, default dimensions, OI flag, recon type	`MainAccount` ⋈ `MainAccountLedgerSetting`	SSRS Chart of accounts	S_ALR_87012326, S_ALR_87012328
Journal batches (unposted)	list inquiry	journal name, status, owner, date, amount	voucher, name, status, Σ debit, Σ credit, owner, approver, posting date	`JournalBatch`	General journal list page; SSRS Journal (LedgerJournal)	FBV0 / FBV3 parked documents list
Posted journals	operational report	journal name, posted date range, posted by	batch, vouchers, entry ids, totals, posted by/at	`JournalBatch` (posted) ⋈ `JournalEntry`	Posted journals list; SSRS Posted journals	S_ALR_87012287 document journal
Voucher (document) view	drill-down	voucher or entry id	header, all lines with dimensions and three currencies, reversal links, attachments	`JournalEntry`	SSRS Voucher (LedgerTransVoucher); Voucher transactions	FB03, S_ALR_87012289 compact document journal
Open items by account	list inquiry	ledger, account range, status, assignment, date, dimension	item, posting date, assignment, amount clearing/acct, days open, clearing document	`OpenItem`	— (customer/vendor open transactions only)	FBL3N open items, FAGLL03 (OI status)
Clearing documents	drill-down	account, clearing date range, mode, status	document, mode, items, difference, FX difference, reset reason/by	`ClearingDocument`	—	FB03 clearing document, FBRA history
Automatic clearing proposal / log	operational report	run id, ledger, rule	groups proposed, cleared, skipped (reason), items per group	`gl-auto-clear` run log	—	F.13 log (SE38 `SAPF124` output)
Validation rule test result	drill-down	rule id, sample entry	per line: prerequisite result, check result, message	`ValidationRule` test harness	—	GGB0 simulation
Derivation trace	drill-down	batch line or `validate` response	step sequence, kind, matched lookup row, value before/after	posting pipeline	—	KEDR "Analyze derivation" (KE4ST)
Document splitting simulation	drill-down	sample document	lines before/after split, inherited flags, zero-balance lines	posting pipeline	—	FB50 "Simulate general ledger" (split view)
Reason code usage	operational report	reason code, module, period	count of reversals, clearings resets, amounts by code	`JournalEntry` (reversal) ⋈ `ReasonCode`	Reason codes form (no report)	T041C usage via FB03 reversal reason
Balance projection drift KPI	KPI	ledger	rows_differing from last `balance-verify`, duration, HWM	`balance.rebuilt` events	Financial dimension sets rebuild log	RFINDEX / SAPF190 comparison report
Idempotency replays	operational report	legal entity, date range, source system	key, first seen, replays, payload hash, conflicts (409)	idempotency table	—	—

2.11.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Journal batch printout (unposted)	user action on Journal batch; optional on submit for approval	header: journal name, batch id, owner, status, totals; lines: line no., account, dimensions, D/C, currency, amount, text; footer: validation messages, approval chain	PDF, XLSX	batch id (not a legal number)	none; internal working paper, retained with batch 90 days after posting
Posted voucher	user action on entry; attached automatically to approval evidence when the journal name sets `print_on_post`	header: ledger, voucher, posting date, document date, description, source, trace id; lines: account, dimensions, D/C, txn/acct/rpt amounts, text, derived/inherited flags; footer: reversal links, attachments list, posted by/at, result hash	PDF, XLSX, CSV	voucher (gap-free, FR-11.6)	financial record; retention per `attachment.default_retention_class` (default 10 years)
Clearing statement	on clearing document creation or reset	header: account, clearing date, mode, document id; lines: cleared items with assignment, dates, amounts, residual/difference lines; footer: tolerance applied, actor	PDF	clearing document id	financial record; 10 years
Chart of accounts export	user action	account attributes incl. default dimensions, OI flag, recon type, suspended_from	XLSX, CSV, JSON	—	none
Account structure / advanced rule definition export	user action or version activation	segments, constraint rows, advanced rules and structures, version, effective_from	JSON, XLSX	structure version	configuration record; kept with the version indefinitely
Approval evidence pack	batch posted after approval	batch printout + workflow history (submitter, approvers, timestamps, comments)	PDF	batch id	audit evidence; 10 years

No e-invoice or EDI documents originate in this wave.

2.11.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`balance-verify` (rebuild into shadow, compare, swap only on diff)	daily 01:00 tenant time zone	ledger	run id per (ledger, date); chunk checkpoints per fiscal period (BR-11.7)	10M lines < 15 min	retry 1× after 10 min; partial chunks rolled back per chunk	`balance.rebuilt.rows_differing > 0` → P1; duration > 60 min → P3
`create-partitions`	monthly, 1st 00:30	tenant shard	idempotent `CREATE TABLE IF NOT EXISTS` for next fiscal year	< 1 min	alert, retry daily	missing partition for next year within 60 days → P2
`recurring-journal-generate`	per template cron; default last business day of period 06:00	template	one batch per (template, scheduled date) key; replay returns existing batch	1,000 templates < 5 min	failed template marked; others continue	failed templates > 0 → P3; batch not generated by due date → P3
`gl-auto-clear`	daily 02:00	ledger × rule	run id per (ledger, rule, date); groups cleared individually with idempotency key `auto:{rule}:{group_hash}`	1M open items < 10 min	group-level failures logged, run continues	cleared count = 0 for 3 consecutive runs on a rule with open items → P3
`idempotency-purge`	daily 03:00	tenant	deletes records older than `idempotency_ttl_days`; checkpoint by created_at	< 5 min per 10M keys	retry next day	table growth > 2× expected → P3
`dimension-value-reconcile` (compare backing entities with `DimensionValue`)	daily 04:00	dimension attribute	idempotent upsert; diff report stored	< 2 min	alert	diff count > 0 → P3
`attachment-retention-sweep` (wave 5 framework, ledger hook)	weekly Sunday 05:00	legal entity	retention class evaluation; deletes only past expiry	< 10 min	skip and retry	failures → P3
`open-item-aging-snapshot`	daily 05:30	ledger	one row per (account, date)	< 2 min	retry	—

2.11.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
Maintain chart of accounts	PUT /ledger/charts/*/accounts; screen Chart of accounts; import CSV	Post ledger journals; Maintain ledger journals	Financial controller, Accounting manager
Maintain financial dimensions	PUT dimension attributes/values; screen Financial dimensions	Post ledger journals	Accounting manager
Maintain account structures	PUT account-structures, advanced-rules; activation; screen Account structures	Post ledger journals	Accounting manager
Maintain ledger setup	PUT description-templates, splitting-rules, ledgers; screen Ledger setup, Document splitting setup	Post ledger journals; Maintain ledger journals	Accounting manager, System administrator (finance)
Maintain reason codes	PUT /ledger/reason-codes; screen Reason codes	—	Accounting manager
Maintain ledger journals	POST/PATCH journal-batches, :submit; screen Journals, Journal batch; attachments	Approve ledger journals (same batch); Maintain chart of accounts	Accountant, Accounting clerk
Approve ledger journals	:approve, :reject; Temporal task UI	Maintain ledger journals (same batch); Post ledger journals without approval	Accounting supervisor, Financial controller
Post ledger journals	:post on batches; POST /ledger/validate	Maintain chart of accounts; Maintain validation and derivation rules	Accountant (senior), Accounting supervisor
Post ledger journals without approval	`auto_post` templates, direct post when journal name has no workflow	Maintain ledger journals; Approve ledger journals	Financial controller (limited)
Post journal entries (service accounts)	POST /ledger/journal-entries with `source.system ≠ ledger`	not assignable to human users	Subledger service accounts (waves 16+)
Reverse journal entries	POST /journal-entries/{id}/reverse; screen Ledger transactions > Reverse	Approve ledger journals (same entry)	Accounting supervisor, Financial controller
View ledger transactions	GET journal-entries, journal-lines, open-items; drill-downs	—	All finance roles, Internal auditor
View ledger balances	GET balances; Trial balance; Dimension statement	—	All finance roles, Internal auditor, Department manager (own dimensions via OPA)
Clear open items	POST /ledger/clearings, :simulate; screen Open items and clearing	Reset clearings; Maintain clearing rules; Maintain bank reconciliation (wave 36)	Accountant (senior), Treasury accountant
Reset clearings	POST /clearings/{id}:reset	Clear open items	Accounting supervisor
Maintain clearing rules	PUT auto-clearing-rules, tolerance-groups and assignments	Clear open items; Run automatic clearing	Accounting manager
Run automatic clearing	POST /ledger/auto-clearing:run; job `gl-auto-clear`	Maintain clearing rules	Accounting supervisor, Finance batch operator
Maintain validation and derivation rules	PUT validation-rules, derivation-strategies, lookup-tables; :test	Post ledger journals; Run dimension realignment	Accounting manager, Finance systems analyst
Run dimension realignment	POST /ledger/realignments, :post, :reverse	Maintain validation and derivation rules; Approve ledger journals	Financial controller
Administer ledger	balances:rebuild, accounts:activate-open-items, config, partitions	Post ledger journals	Finance platform administrator

2.11.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
LEDGER-0001 `LEDGER_UNBALANCED`	error	Entry does not balance in {currency}: debits {debits}, credits {credits}.	Σ D ≠ Σ C in a currency column	Correct the line amounts or add the clearing line	BR-11.1, BR-12.5
LEDGER-0002 `LEDGER_LINE_LIMIT`	error	Entry has {count} lines; the limit is {max}.	More than `max_lines_per_entry`	Split the document by `document_id`	FR-11.5

Code	Severity	User text	Cause	Remediation	Raised by
LEDGER-0003 `LEDGER_LINE_EMPTY`	error	Line {line} has no amount, quantity or text.	Zero line without statistical content	Remove the line or add quantity/text	BR-11.2
LEDGER-0004 `LEDGER_AMOUNT_SCALE`	error	Amount {amount} has more decimals than {currency} allows ({scale}).	Excess precision	Round in the caller	BR-11.3
LEDGER-0005 `LEDGER_ACCOUNT_NOT_FOUND`	error	Account {account} does not exist in chart {chart}.	Unknown account	Create the account or correct the number	FR-11.1
LEDGER-0006 `LEDGER_ACCOUNT_NOT_POSTABLE`	error	Account {account} is a total or reporting account.	`is_postable = false`	Use a postable account	FR-11.1
LEDGER-0007 `LEDGER_ACCOUNT_SUSPENDED`	error	Account {account} is suspended from {date}.	`suspended_from ≤ posting_date`	Use another account or lift the suspension	FR-11.1
LEDGER-0008 `LEDGER_RECON_ACCOUNT_MANUAL`	error	Account {account} accepts subledger postings only.	Manual line on a reconciliation account	Use a journal name with `allow_recon_accounts` or post through the subledger	BR-11.8
LEDGER-0009 `LEDGER_STRUCTURE_NOT_FOUND`	error	No active account structure covers account {account} on {date}.	Structure filter gap	Extend a structure's main-account filter	FR-11.3
LEDGER-0010 `LEDGER_DIMENSION_REQUIRED`	error	Dimension {attribute} is required for account {account}.	Blank not allowed by the structure or advanced rule	Supply the dimension value	FR-11.3, FR-11.16
LEDGER-0011 `LEDGER_DIMENSION_INVALID`	error	Value {value} is not allowed for {attribute} under structure {structure}.	Constraint row mismatch	Choose an allowed value	FR-11.3
LEDGER-0012 `LEDGER_DIMENSION_INACTIVE`	error	Value {value} of {attribute} is not active on {date}.	Effective dates / suspended	Use an active value or adjust dates	FR-11.2
LEDGER-0013 `LEDGER_PERIOD_CLOSED`	error	Period {period} is closed for module {module}.	Period status	Post in an open period or request reopen (wave 14)	BR-11.4, BR-14.1
LEDGER-0014 `LEDGER_PERIOD_ON_HOLD`	error (403)	Period {period} is on hold; your user group may not post to it.	On-hold without membership	Ask the controller group to post or reopen	BR-14.1
LEDGER-0015 `LEDGER_IDEMPOTENCY_CONFLICT`	error (409)	Idempotency key {key} was already used with a different payload.	Key reuse	Use a new key for a new request	BR-11.6
LEDGER-0016 `LEDGER_IDEMPOTENCY_IN_PROGRESS`	error (409)	A request with key {key} is still being processed.	Concurrent duplicate	Retry after 1 s	FR-11.5
LEDGER-0017 `LEDGER_REVERSAL_ALREADY_REVERSED`	error	Entry {voucher} was already reversed by {reversing_voucher}.	Second reversal	Reverse the reversing entry to reinstate	FR-11.8
LEDGER-0018 `LEDGER_REASON_REQUIRED`	error	A reason code is required for this reversal.	`reversal_reason_required`	Select a reason code (and comment if required)	FR-11.15
LEDGER-0019 `LEDGER_REASON_INVALID_MODULE`	error	Reason code {code} does not apply to {module}.	`applies_to` mismatch	Choose a code valid for the module	FR-11.15
LEDGER-0020 `LEDGER_CLEARING_UNBALANCED`	error	Selected items leave a residual of {residual} {currency}, above the tolerance {limit}.	BR-11.12	Add items, create a residual item, or adjust the selection	FR-11.19
LEDGER-0021 `LEDGER_OPEN_ITEM_ALREADY_CLEARED`	error	Item {item} is already cleared by document {document}.	BR-11.13	Reset the earlier clearing first	FR-11.19
LEDGER-0022 `LEDGER_OPEN_ITEM_ACCOUNT_REQUIRED`	error	Account {account} is not open-item managed.	Clearing on a non-OI account	Activate open-item management	BR-11.18
LEDGER-0023 `LEDGER_OI_ACTIVATION_BALANCE`	error	Open-item activation needs an activation run because account {account} has a balance.	Flag changed on an account with history	Run accounts:activate-open-items	FR-11.19
LEDGER-0024 `LEDGER_VALIDATION_RULE_FAILED`	error / warning	{rule_message} (rule {rule_id}, line {line}).	User-defined rule	Follow the rule's remediation text	FR-11.21
LEDGER-0025 `LEDGER_DERIVATION_INVALID`	error	Derivation strategy version {version} cannot be activated: {detail}.	Parse error, > 50 steps, cyclic read/write	Fix the step	BR-11.14
LEDGER-0026 `LEDGER_SPLIT_NO_SOURCE`	error	Line {line} cannot inherit {attribute}: no carrying line and no default value.	FR-11.23	Add a value on the line or set `default_splitting_value`	BR-11.15
LEDGER-0027 `LEDGER_SPLIT_CLEARING_ACCOUNT_MISSING`	error	Document splitting is active but no zero-balance clearing account is set for ledger {ledger}.	Configuration	Set the account in Document splitting setup	FR-11.23
LEDGER-0028 `LEDGER_ADVANCED_RULE_SEGMENT_LIMIT`	error	Advanced rules add more than 20 segments for account {account}.	FR-11.16	Reduce the matching rules' segments	FR-11.16
LEDGER-0029 `LEDGER_ATTACHMENT_RETAINED`	error	Attachment {name} is retained until {date} and cannot be deleted.	Retention class	Wait for expiry or request a legal-hold exception (wave 5)	FR-11.18
LEDGER-0030 `LEDGER_REALIGNMENT_PERIOD_CLOSED`	error	Realignment touches period {period}, which is closed for GL.	Period control	Exclude the period or reopen it	FR-11.24
LEDGER-0031 `LEDGER_DESCRIPTION_TRUNCATED`	info	Description was truncated to 255 characters.	Template expansion	Shorten the template	FR-11.17
LEDGER-0032 `LEDGER_NEGATIVE_POSTING_DISABLED`	error	Reason code {code} allows negative posting but the ledger does not.	`negative_posting_enabled = false`	Use another reason code or enable the parameter	BR-11.16

2.11.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Post journal entry (2–50 lines, incl. derivation, splitting, validation rules)	200k lines/day per tenant; peak 5,000 lines/s at month-end	40 / 120 / 200 ms	5,000 lines/s sustained 30 min	400 B/line × 50M lines = 20 GB + 60% indexes ≈ 32 GB	`journal_line` range-partitioned by fiscal year; indexes §2.11.4	HPA to 12 pods; sequence lock serializes per ledger; p99 may rise to 600 ms; no rejection below 3×

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Validate (dry run)	50k calls/day	20 / 60 / 100 ms	1,000 req/s	none	stateless	same as posting
Balance query (one account, 12 periods, group by dimension)	100k/day	30 / 150 / 300 ms	500 req/s	—	PK clustered on `balance`	read replica; cached dimension sets 60 s
Line query page (keyset, 1,000 rows)	50k/day	50 / 200 / 300 ms	300 req/s	—	(ledger, account, posting_date) partition pruning	replica; page size reduced to 500 under pressure
Balance rebuild	nightly, 10M lines	—	10M lines < 15 min	shadow table transient	chunk per fiscal period ≤ 5M lines	runs on replica snapshot; swap skipped if lock not acquired in 2 s
Manual clearing (≤ 10,000 items)	5k clearings/day	100 / 400 / 800 ms	50/s	150 B/open item; 2M items/year ≈ 300 MB	`open_item` index (ledger, account, status, assignment)	item selection capped at 10,000; larger via auto-clear
`gl-auto-clear`	1M open items nightly	—	1M items < 10 min	—	grouping in SQL per rule with hash on criteria	per-rule timeout 20 min; remaining rules next run
Derivation strategy evaluation	every posted line	≤ 0.2 ms per line (50 steps)	—	lookup tables ≤ 100k rows each, memory-resident per pod	lookup tables cached, invalidated by `rules.changed`	strategy cache miss adds one query per pod
Document splitting	30% of entries	≤ 1 ms per entry of 50 lines	—	+25% lines on split entries	—	—
Realignment run	10/month; ≤ 5M lines	—	1M lines < 5 min (simulate), < 10 min (post)	reclassification entries ≈ 2 lines per combination × period	chunked by period	run queued; one active run per ledger
Idempotency table	200k keys/day × 400 days ≈ 80M rows	lookup ≤ 1 ms	—	120 B/key ≈ 10 GB	PK (tenant, legal_entity, key); purged daily	—

2.11.20 Localization and compliance hooks

- Chart of accounts templates by country (German SKR03/SKR04, French Plan Comptable Général with mandatory classes 1–7, Indian Schedule III grouping, Spanish PGC, US/UK free-form): wave 48 country packs ship `ChartTemplate` JSON imported through PUT /ledger/charts; account number validation regex per template.
- Account name translations: `name` is translatable through wave 3 metadata; statutory languages per legal entity come from wave 6.
- Reversal practice: negative posting (storno) is the default expectation in DE/AT/CH/IT; country packs set `negative_posting_enabled = true` and ship reason codes flagged `negative_posting_allowed`; US/UK packs keep opposite-side reversal.
- Gap-free voucher numbering (FR-11.6) satisfies IT, FR, PT, ES and PL sequential-numbering rules; country packs may require a separate sequence per document class (e.g., IT sectional numbering) through the wave 5 number sequence scopes.
- Posting date vs document date: FR-11.5 accepts both; country packs add validation rules (FR-11.21) such as "document date ≤ posting date" (DE GoBD) or "posting within 30 days of document date".
- Retention: `attachment.default_retention_class` and journal retention map to statutory minimums (DE HGB §257 / AO §147: 10 years; US IRS: 7 years; UK: 6 years; FR: 10 years); the journal itself is never purged in Phase 2; archival export hooks are provided to wave 48 audit files (SAF-T via wave 13, GDPdU/GoBD export, FEC for France) through the line query API with stable ordering.
- Audit-file fields: `JournalLine` carries every field FEC (FR) and GoBD (DE) require (voucher, journal code via journal name, posting date, account, text, debit, credit, document date, validation date = posted_at); SAF-T `GeneralLedgerEntries` is produced by wave 13 from the same API.
- Privacy: `text`, `description` and attachment contents may hold personal data; they are tagged `pii_possible` for wave 7 data-subject request searches; no deletion from the journal — masking is applied on export only when legally required.
- Calendar and currency: fiscal calendars (wave 6) and ISO 4217 minor units (wave 12) are the only currency/calendar dependencies; rounding mode per ledger (BR-12.2) is set by country packs where commercial rounding differs.
- Extension point: `ledger.country_pack` hook set = chart template import, validation rule pack, description template language pack, reason code pack, and audit-file exporter registration; packs are declarative (JSON) and versioned with the wave 48 pack registry.

2.11.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.1.1 Common / Journals — Voucher attachments / document handling (`DocuView`) (AX register row 1)	PARTIAL-11	ADDED as FR-11.18	document types, retention class, store reference from wave 5 framework; attachment immutability after posting
AX §12.1.3 Periodic — Reason code assignment (reversal reasons) (AX register row 2)	PARTIAL-11	ADDED as FR-11.15	reason code required on reversal; negative posting semantics BR-11.16
AX §12.1.4 Setup — Chart of accounts → Advanced rule structures / Advanced rules (AX register row 3)	PARTIAL-11	ADDED as FR-11.16	advanced rules versioned with account structures; AC-11.18
AX §12.1.4 Setup — Journals → Default descriptions (Transaction text) (AX register row 5)	PARTIAL-11	ADDED as FR-11.17	templates per source_type × language; AC-11.19
AX §12.1.4 Setup — Posting → Reason codes (financial reasons) (AX register row 6)	PARTIAL-11	ADDED as FR-11.15	shared `ReasonCode` with `applies_to[]`
AX §12.3.4 Setup — Sales → Sales reasons / Reason codes (customer) (AX register row 32)	PARTIAL-11	ADDED as FR-11.15	`applies_to` includes customer; AR (wave 30) consumes
AX §12.6.4 Setup — Fixed asset → Reason codes (fixed assets) (AX register row 44)	PARTIAL-11	ADDED as FR-11.15	`applies_to` includes fixed_asset; wave 37 consumes
AX §12.14.4 Setup — Inventory → Counting → Counting reason code policies (AX register row 137)	PARTIAL-11	ADDED as FR-11.15 (codes); policies DEFERRED to wave 16	code list here; the counting policy (which codes are mandatory per warehouse) belongs to the inventory ledger wave
AX §12.29.1 Common / Journals — Reason codes (shared financial reasons across modules) (`ReasonTable`) (AX register row 239)	PARTIAL-11	ADDED as FR-11.15	
AX §12.29.4 Setup — Reason codes (Financial reasons: ledger, bank, customer, vendor, fixed asset) (AX register row 258)	PARTIAL-11	ADDED as FR-11.15	screen Reason codes; API /ledger/reason-codes
SAP §13.1 FI-GL Master data & documents — F-03 / F-04 Clear G/L account / post with clearing (SAP register row 1)	PARTIAL-11	ADDED as FR-11.19	manual clearing, post-with-clearing, reset (FBRA); AC-11.12

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP §13.1 FI-GL Master data & documents — FAGL_ACTIVATE_OP (SAP register row 4)	PARTIAL-11	ADDED as FR-11.19	activation run creating opening open items
SAP §13.1 FI-GL Master data & documents — FAGL_SWITCH_TO_OPEN_ITEM (SAP register row 5)	PARTIAL-11	ADDED as FR-11.19	same activation run; deactivation requires zero open items
SAP §13.1 FI-GL Periodic processing — F.13 / F13E Automatic clearing (SAP register row 7)	PARTIAL-11	ADDED as FR-11.20	job `gl-auto-clear`; AC-11.13
SAP §13.1 FI-GL Customizing — OB28 / GGB0 Validations (SAP register row 12)	PARTIAL-11	ADDED as FR-11.21	CEL prerequisite/check/message; AC-11.14
SAP §13.1 FI-GL Customizing — OBBH / GGB1 Substitutions (SAP register row 13)	PARTIAL-11	ADDED as FR-11.22	derivation strategy steps
SAP §13.1 FI-GL Customizing — OBBZ / OBBW Substitution/validation for cost of sales accounting (SAP register row 14)	PARTIAL-11	ADDED as FR-11.22	FUNC_AREA attribute + shipped lookup step; AC-11.15
SAP §13.1 FI-GL Customizing — Document splitting (FAGL_SPLIT*) (SAP register row 15)	PARTIAL-11	ADDED as FR-11.23	active/passive inheritance, zero-balance lines; AC-11.16
SAP §13.1 FI-GL Customizing — OBA0 Tolerance groups for G/L accounts (SAP register row 18)	PARTIAL-11	ADDED as FR-11.20	`ToleranceGroup` per account and user; BR-11.12
SAP §13.1 FI-GL Customizing — OB74 Additional rules for automatic clearing (SAP register row 20)	PARTIAL-11	ADDED as FR-11.20	`AutoClearingRule.criteria`
SAP §13.1 FI-GL Customizing — Cost of sales accounting: functional areas (OKBD), FAGL_FUNC_AREA (SAP register row 22)	PARTIAL-11	ADDED as FR-11.22	functional area is a standard dimension attribute; P&L by function in wave 15
SAP §13.9 CO-PA Master data & documents — KEDR / KEDE derivation strategy (SAP register row 102)	PARTIAL-11	ADDED as FR-11.22	table_lookup steps with validity dates
SAP §13.9 CO-PA Periodic processing — KEND / KEDR realignment run (SAP register row 105)	PARTIAL-11	ADDED as FR-11.24	append-only reclassification; AC-11.17
SAP §13.9 CO-PA Customizing — KEDR derivation rules (table lookup, rule, move, clear, enhancement) (SAP register row 106)	PARTIAL-11	ADDED as FR-11.22	enhancement exits → `expression` (CEL) steps
SAP §13.9 CO-PA Customizing — 0KEM / 0KEP PCA substitution / derivation of PC (SAP register row 107)	PARTIAL-11	ADDED as FR-11.22	profit center derived from (sales org, material group) lookup tables once waves 8/27 supply the fields
SAP §13.11 FM / PSM Master data & documents — FMDERIVE (SAP register row 121)	PARTIAL-11	ADDED as FR-11.22	FM account assignment (fund, funds center, commitment item) modeled as dimension attributes; budget control in wave 38
SAP §13.11 FM / PSM Customizing — FMDERIVE strategy (SAP register row 122)	PARTIAL-11	ADDED as FR-11.22	strategy versions per ledger

2.12 Wave 12: Multi-Currency, Parallel Ledgers & Revaluation

2.12.1 Scope and business context

AX carries transaction, accounting and reporting currency on `GeneralJournalAccountEntry` and uses posting layers (current/operations/tax); SAP New GL carries currency types (00 document, 10 company code, 30 group) on `ACDOCA` and posts to leading and non-leading ledgers through ledger groups. Wave 12 adds to the journal: an exchange rate service (`fx`, Go) with rate types, provider adapters, triangulation and audit; three-currency amounts with ISO 4217 rounding; ledger groups so one document posts to N ledgers with per-ledger accounting differences; FX revaluation of GL balances (open-item revaluation for AP/AR uses the same engine when those waves arrive); and currency translation for reporting-currency restatement.

After this wave a user can post a EUR invoice into a USD-accounting entity with EUR reporting currency, see the amounts in all three currencies, run the month-end revaluation and see it auto-reverse. Out of scope: realized gain/loss on settlement (waves 19 and 30), group-currency consolidation translation (wave 40), hedge accounting.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
SAP OB08 / `TCURR`: rates entered per 100 units with `TCURF` factors; a wrong factor silently converts amounts 100x; no deviation check	Factor table separate from rate table; no validation against the previous rate	Rates stored as plain `numeric(20,10)` "1 from = rate to"; > 10% deviation requires confirmation and emits a warning event	Factor-related misposting incidents → 0; deviation flagged in < 100 ms at entry
AX `ExchangeRate` import batch (`ExchangeRateProviderImport`) runs on a batch group; a failed run leaves no rate and postings fail hours later with "no exchange rate"	Batch framework with no fallback or alerting	Temporal schedule with retries; last-known rate stays valid; `fx.provider.degraded` alerts immediately	Posting failures from missing rates → 0; outage detected < 1 min (vs discovered at posting)
Rounding differences between `AccountingCurrencyAmount` and `ReportingCurrencyAmount` land in penny-difference accounts with no trace to the line (AX `LedgerParameters`); SAP rounding to OB00 account per document	Each line rounded independently; residual posted to a separate account without lineage	Residual allocated to the largest line and recorded as `rounding_adjustment` with the tolerance rule in BR-12.4	Unexplained rounding postings → 0; every residual traceable to its line
Parallel valuation needs SAP ledger groups + FB50L and separate runs per ledger, or AX posting layers that most reports ignore	Ledger-specific posting bolted onto a single-ledger document model	One request to `ledger_group: ALL` yields N balanced entries sharing `document_id`, atomically	Per-ledger reconciliation effort 2 days/close → automated check; ledger-specific adjustments 1 call
FAGL_FC_VAL / AX revaluation runs take hours on large `BSEG` sets and users rerun after rate corrections, producing duplicate adjustments	Open-item scan over unindexed valuation sets; no run idempotency	Revaluation selects from `Balance` keyed by currency; runs are idempotent per (ledger, period, rate type) and auto-reverse	Revaluation of 100k combinations < 5 min (vs 1–4 h); duplicate adjustment runs → 0

2.12.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Exchange rate types	`ExchangeRateType`, Currency exchange rate types form	`TCURV` (rate types M, EURX, P), OB07	`fx.ExchangeRateType`	full
Exchange rates	`ExchangeRate`, `ExchangeRateCurrencyPair`, OB08-style form	`TCURR`, OB08, `TCURF` (factors)	`fx.ExchangeRate` with `from`, `to`, `valid_from`, `rate` numeric(20,10); factor handled by rate precision	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Rate providers	Exchange rate providers (OANDA, Central Bank of Europe)	Market data interface (datafeed)	`fx` provider adapters: ECB, OpenExchangeRates, manual, CSV	full
Triangulation	Euro triangulation	`TCURR` via reference currency (`TCURV-BWAER`)	`fx` triangulation through ledger reference currency	full
Three currencies per line	`TransactionCurrencyAmount`, `AccountingCurrencyAmount`, `ReportingCurrencyAmount`	`WSL` (document), `HSL` (local), `KSL` (group) in ACDOCA	`amount_txn`, `amount_acct`, `amount_rpt` on JournalLine	full
Parallel ledgers / posting layers	Posting layers Current/Operations/Tax (`LedgerPostingLayer`)	Leading ledger 0L, non-leading ledgers, ledger groups, `RLDNR`	`ledger.Ledger` × `LedgerGroup`; entries per ledger share `document_id`	improved: unlimited ledgers, per-ledger calendar and currency
FX revaluation (GL)	Foreign currency revaluation (General ledger), `LedgerExchAdj`	FAGL_FC_VAL (balances), valuation methods OB59, valuation areas	`ledger.RevaluationRun` with method per ledger	full
Unrealized gain/loss accounts	Currency revaluation accounts per currency / ledger	OBA1 (KDB/KDF), OB09 per recon account	`CurrencyPolicy` per ledger × currency: unrealized_gain, unrealized_loss, adjustment accounts	full
Currency translation (reporting currency)	Reporting currency on ledger, revaluation of reporting currency	Currency translation (FAGL_FCV / consolidation)	`ledger.TranslationRun` (monthly, average/closing/historical by account type)	partial: group consolidation in wave 40
Rounding differences	Penny difference accounts (`LedgerParameters`)	Rounding difference account (OB00), tolerance	`CurrencyPolicy.rounding_difference_account`, tolerance per BR-12.4	full
Rate audit / history	Database log on `ExchangeRate`	`TCURR` change documents	`fx.ExchangeRate` is append-only with `superseded_by`	improved
Market data beyond FX (interest rates, indices, prices)	Not available (exchange rates only)	TBDM market data import, market data classes (`VTB_MARKET`, `JBD*`), rate types / yield curves / volatility customizing	`fx.MarketDataSeries` (kind ∈ {fx_rate, interest_rate, index, price}) with the same providers, audit and lookup (FR-12.13); curves and security valuation in wave 36	partial: series storage and import here, valuation in wave 36

2.12.3 Functional requirements

- **FR-12.1 Currency reference data and policy** — Must. The system shall use ISO 4217 currencies from wave 6 (code, numeric code, minor units) and maintain a `CurrencyPolicy` per ledger × currency with gain/loss/adjustment/rounding-difference accounts.
  - Minor units default from ISO 4217 (JPY 0, USD/EUR/GBP/INR 2, KWD/BHD/OMR 3, CLF 4); a tenant may not override them.
  - Non-ISO currencies (e.g., XTS test, loyalty points) can be defined with explicit minor units and flagged `is_statistical`.
- **FR-12.2 Exchange rate types and rates** — Must. The system shall maintain rate types (default, average, closing, budget, custom) and rates by (type, from, to, valid_from) with precision numeric(20,10).
  - Rate direction is `from → to` as "1 from = rate to"; the reciprocal is computed only if no explicit inverse rate exists and `allow_inverse` is set on the type.
  - Rates are append-only; a corrected rate for the same key supersedes the previous row and both are kept with `superseded_by`.
  - Maximum 10,000 pairs per type; history retained indefinitely.
- **FR-12.3 Rate lookup and triangulation** — Must. The system shall resolve a rate for (type, from, to, date) by: exact pair valid on date → inverse pair (if allowed) → triangulation through the ledger's reference currency (two hops, both must resolve) → `LEDGER_RATE_MISSING`.
  - Lookup is served from an in-memory cache per pod refreshed from `fx.rate.published` events; cold lookup p99 < 5 ms.
  - Triangulated amounts are computed in one step `amount × r1 × r2` at full precision, then rounded once.
- **FR-12.4 Provider adapters** — Must. The system shall import rates from ECB (daily reference rates), OpenExchangeRates and CSV through a Temporal schedule per rate type.
  - Provider timeout 10 s, 3 retries with exponential back-off; on failure the last-known rate remains valid and `fx.provider.degraded` is emitted; rates older than `max_rate_age_days` (default 5 business days) cause posting warnings, not rejections, unless the ledger sets `reject_stale_rates`.
  - Imported rates are stored with `source`, `provider_timestamp` and raw payload hash for audit.
- **FR-12.5 Three-currency amounts on every line** — Must. The posting API shall accept `amount` in any currency, convert to accounting and reporting currency using the ledger's default rate type (or an explicit `rate_override` with rate type and value), and store all three amounts rounded per currency.
  - If a line's currency equals the accounting currency, `amount_acct = amount_txn` exactly (no rate applied); likewise for reporting.
  - Explicit `rate_override` requires duty `Override exchange rates` and is recorded on the line.
  - An entry may contain lines in multiple transaction currencies; balancing is checked per transaction currency as well as in accounting and reporting currency.
- **FR-12.6 Rounding and residual allocation** — Must. The system shall apply ISO 4217 minor-unit rounding (half away from zero) per derived amount, then allocate any residual so that each derived column balances.
  - Algorithm in BR-12.4; the residual goes to the line with the largest absolute amount on the side that needs it; if the residual exceeds tolerance the post is rejected with `LEDGER_ROUNDING_EXCEEDED`.
- **FR-12.7 Ledger groups and parallel posting** — Must. The system shall define ledger groups (e.g., `ALL`, `LEADING`, `LOCAL`, `TAX`) per legal entity; a posting request addressed to a group produces one `JournalEntry` per member ledger, all committed in one transaction and sharing `document_id`.
  - Each ledger has its own accounting principle code (IFRS, US-GAAP, HGB, TAX), fiscal calendar, accounting currency, reporting currency and default rate type; conversions are performed per ledger.
  - Per-ledger differences are expressed by the caller (subledgers send ledger-specific lines, e.g., different depreciation) or by `LedgerAccountMapping` rows that remap an account per ledger (e.g., IFRS 16 lease accounts to operating lease expense in local GAAP).
  - A request addressed to a single non-leading ledger (ledger-specific adjustment, SAP FB50L semantics) is allowed when the user holds duty `Post ledger-specific adjustments`.
- **FR-12.8 FX revaluation of GL balances** — Must. The system shall run a revaluation per ledger × period × rate type over balance-sheet accounts flagged `revalue = true`, computing unrealized gain/loss per account × dimension combination × transaction currency, posting it on the period-end date and auto-reversing it on the first day of the next period.
  - Method per ledger: `closing_rate` (default, FAGL_FC_VAL semantics with reversal) or `cumulative_no_reversal` (delta against prior revaluation, AX semantics); both are supported, the ledger chooses.
  - Simulation mode produces the proposal without posting; the proposal is retained 90 days.
- **FR-12.9 Currency translation for reporting currency** — Should. The system shall restate reporting-currency balances monthly: monetary balance-sheet accounts at closing rate, non-monetary at historical (the posted `amount_rpt`), P&L at average rate, with the difference posted to a cumulative translation adjustment (CTA) equity account.

- **FR-12.10 Exchange rates workspace** — Must. The system shall provide screens for rate types, rate entry/import, provider schedules, lookup tester and rate history.
- **FR-12.11 Rate audit** — Must. The system shall record who/what created each rate, the provider payload hash, and expose a history view by pair; a rate change that moves more than 10% from the previous value requires confirmation in the UI and emits a warning event.
- **FR-12.12 Posting-time currency validation** — Must. The system shall reject a transaction currency that is suspended or not enabled for the legal entity with `LEDGER_CURRENCY_MISMATCH`, and a missing rate with `LEDGER_RATE_MISSING` including the (type, pair, date) in the violation.
- **FR-12.13 Market data series** — Should. The `fx` service shall store, besides exchange rates, generic market data series of kind `interest_rate` (reference rates such as SOFR, €STR, EURIBOR 3M, SONIA by tenor), `index` (CPI, commodity indices) and `price` (security or commodity prices by identifier), each a series of (series_code, valid_from, value numeric(20,10), source, provider_timestamp, payload_hash), append-only with `superseded_by`, imported by the same provider schedule framework as FR-12.4 (ECB SDW, CSV, manual) and queryable by `GET /fx/series/{code}?date=`.
  - Series carry `unit` (percent per annum, index points, currency per unit), `day_count` for rates (ACT/360, ACT/365, 30/360) and `tenor` (ON, 1W, 1M, 3M, 6M, 12M) as metadata; no interpolation or curve bootstrapping in this wave.
  - Deviation warning (FR-12.11) applies with a series-specific threshold (default 25% for rates, 10% for prices).
  - Consumers: wave 36 (interest calculation, yield curves, security valuation), wave 38 (budget rate assumptions), wave 29 (index-linked subscription pricing).

2.12.4 Data model

ExchangeRate (fx)

Field	Type	Required	Notes
id	uuid	yes	PK
rate_type_id	uuid	yes	FK ExchangeRateType
from_currency / to_currency	char(3)	yes	ISO 4217; unique (type, from, to, valid_from, version)
valid_from	date	yes	effective until next valid_from for the same key
rate	numeric(20,10)	yes	> 0
source	enum	yes	manual, ecb, openexchangerates, csv, migration
provider_timestamp	timestamptz	no	
payload_hash	bytea(32)	no	SHA-256 of provider record
superseded_by	uuid	no	FK ExchangeRate; non-null means not current
created_at / created_by	timestamptz / uuid	yes	

Ledger (extended) and **LedgerGroup**

Field	Type	Required	Notes
id	uuid	yes	PK
legal_entity_id	uuid	yes	FK
code	text(10)	yes	e.g. `0L`, `IFRS`, `HGB`, `TAX`; unique per legal entity
accounting_principle	enum	yes	IFRS, US_GAAP, LOCAL_GAAP, TAX, MANAGEMENT
is_leading	bool	yes	exactly one leading ledger per legal entity
accounting_currency / reporting_currency	char(3)	yes	
reference_currency	char(3)	no	triangulation pivot (default accounting currency)
default_rate_type_id / closing_rate_type_id / average_rate_type_id	uuid	yes	FK ExchangeRateType
fiscal_calendar_id	uuid	yes	FK (wave 6)
rounding_mode	enum	yes	half_away_from_zero (default), half_even
revaluation_method	enum	yes	closing_rate, cumulative_no_reversal
group: id / legal_entity_id / code / member_ledger_ids	uuid / uuid / text / uuid[]	yes	`ALL` auto-maintained

RevaluationRun and **RevaluationLine**

Field	Type	Required	Notes
id	uuid	yes	PK
ledger_id	uuid	yes	
fiscal_period_id	uuid	yes	
rate_type_id	uuid	yes	typically closing
valuation_date	date	yes	period end
mode	enum	yes	simulate, post
status	enum	yes	proposed, posted, reversed, cancelled
posting_entry_id / reversal_entry_id	uuid	no	FK JournalEntry
totals_gain / totals_loss	numeric(23,6)	yes	accounting currency
line: account_id, dimension_combination_id, currency_txn		yes	key
line: balance_txn, balance_acct_posted, rate_used, balance_acct_revalued, adjustment	numeric	yes	adjustment = revalued – posted

MarketDataSeries (fx, FR-12.13): `id`, `code` text(30) unique per tenant (e.g., `EURIBOR_3M`, `SOFR_ON`, `CPI_US`), `kind` enum (interest_rate, index, price), `unit` text, `currency` char(3) optional, `tenor` enum optional, `day_count` enum optional, `provider_id` optional, `active` bool; **MarketDataPoint**: `series_id`, `valid_from` date, `value` numeric(20,10), `source` enum, `provider_timestamp`, `payload_hash` bytea(32), `superseded_by` uuid, `created_at/by`; unique (series_id, valid_from, version); index (series_id, valid_from DESC) WHERE superseded_by IS NULL.

Invariants: `amount_acct = round(amount_txn × rate, minor_units(acct))` per line before residual allocation; after allocation Σ D = Σ C in every column; exactly one leading ledger per legal entity; every member of a ledger group belongs to the same legal entity; a revaluation run that is `posted` with method `closing_rate` has a reversal entry dated period_end + 1 day.

Indexing: `exchange_rate` index on (rate_type_id, from_currency, to_currency, valid_from DESC) WHERE superseded_by IS NULL; `journal_line` adds index on (ledger_id, currency_txn, posting_date) for revaluation selection; `revaluation_line` PK (run_id, account_id, dimension_combination_id, currency_txn).

2.12.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET/PUT	/api/v1/fx/rate-types[/{code}]	Rate type CRUD	PUT idempotent	Maintain exchange rates
POST	/api/v1/fx/rates	Create rate(s), bulk up to 5,000	`Idempotency-Key`	Maintain exchange rates
GET	/api/v1/fx/rates?type=&from=&to=&date=	Resolve a rate (with triangulation path)	n/a	View exchange rates
GET	/api/v1/fx/rates/history?type=&from=&to=	Rate history incl. superseded	n/a	View exchange rates
POST	/api/v1/fx/providers/{provider}:import	Trigger import now	run id	Administer exchange rates
POST	/api/v1/fx/convert	Convert amount(s) with rounding (stateless)	none	View exchange rates
GET/PUT	/api/v1/ledger/ledgers/{id}/currency-policies/{currency}	Gain/loss/rounding accounts	PUT idempotent	Maintain ledger setup
GET/PUT	/api/v1/ledger/legal-entities/{le}/ledger-groups[/{code}]	Ledger group membership	PUT idempotent	Maintain ledger setup
POST	/api/v1/ledger/revaluations	Create run (simulate or post)	`Idempotency-Key`	Run foreign currency revaluation
GET	/api/v1/ledger/revaluations/{id}[/lines]	Inspect proposal/results	n/a	View ledger balances
POST	/api/v1/ledger/revaluations/{id}:post	Post a simulated proposal	`Idempotency-Key`	Run foreign currency revaluation
POST	/api/v1/ledger/translations	Reporting-currency translation run	`Idempotency-Key`	Run currency translation
GET/PUT	/api/v1/fx/series[/{code}]	Market data series definitions (FR-12.13)	PUT idempotent	Maintain exchange rates
POST	/api/v1/fx/series/{code}/points	Append points, bulk up to 5,000	`Idempotency-Key`	Maintain exchange rates
GET	/api/v1/fx/series/{code}?date= and /history	Resolve a value on a date; full history	n/a	View exchange rates

Posting request extensions (wave 11 contract, additive): `"ledger_group": "ALL"`, per line `"amount": {"currency":"EUR","value":"1000.00"}`, optional `"rate_override": {"rate_type":"M","rate":"1.0850000000"}`, optional `"ledger_lines": {"HGB": [ … ]}` for ledger-specific substitutions. Response contains `entries[]` one per ledger with `amounts: {txn, acct, rpt}` and `rates_applied[]`.

Commands: `PublishRate` (rate > 0, currencies enabled, 10% deviation warning), `RunRevaluation` (period open for module GL, accounts flagged, closing rate exists for every currency with balance else `LEDGER_RATE_MISSING`), `RunTranslation`.

Event	Trigger	Payload fields	Consumers
dev.passage.fx.rate.published.v1	rate committed	rate_type, from, to, valid_from, rate, previous_rate, source, deviation_pct	ledger cache, reporting, AP/AR
dev.passage.fx.provider.degraded.v1	import failure after retries	provider, last_success_at, rate_type, error	ops alerting
dev.passage.ledger.revaluation.completed.v1	run posted	run_id, ledger_id, period, rate_type, gain_total, loss_total, posting_entry_id, reversal_entry_id	reporting, close
dev.passage.ledger.translation.completed.v1	translation posted	run_id, ledger_id, period, cta_amount, entry_id	reporting, close
dev.passage.fx.series.published.v1	market data point committed	series_code, kind, valid_from, value, previous_value, source, deviation_pct	treasury (wave 36), budgeting (wave 38), subscription pricing (wave 29)

2.12.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Exchange rate types	Create, set defaults per ledger	code, name, allow_inverse, provider	Maintain exchange rates
Exchange rates	Enter, import CSV, view history, compare to provider	type, from, to, valid_from, rate, source, deviation vs previous	Maintain exchange rates
Provider schedules	Configure provider, cron, test connection, run now	provider, rate type, pairs, last run, status	Administer exchange rates
Rate lookup tester	Enter type/pair/date, see resolution path	path (direct/inverse/triangulated), rate, valid_from	View exchange rates
Ledger setup	Ledgers, groups, currency policies, account mapping	code, principle, currencies, rate types, revaluation method, accounts	Maintain ledger setup
Foreign currency revaluation	Simulate, review proposal grid, post, view reversal	account, dims, currency, balance txn, posted acct, rate, revalued, adjustment	Run foreign currency revaluation
Journal line (extended)	Enter amount in any currency; see acct/rpt amounts live	currency, amount, rate (editable with duty), acct amount, rpt amount	Maintain ledger journals

Interaction rules: the line grid shows derived amounts as read-only cells that recompute on currency/amount/date change within 100 ms; a rate override cell is editable only with the duty and is highlighted; the revaluation proposal grid supports excluding rows (recorded as exclusions on the run) and exporting to XLSX; a rate deviation > 10% opens a confirmation dialog with the previous rate.

2.12.7 Business rules and edge cases

- **BR-12.1 Minor units.** `scale(currency)` is the ISO 4217 minor unit; a stored amount must satisfy `amount = round(amount, scale)`; CHECK constraint `scale(amount) ≤ scale(currency)` is validated in the service and by a trigger.
- **BR-12.2 Rounding mode.** Default half-away-from-zero (matches AX `round()` and SAP commercial rounding); `half_even` is selectable per ledger for statistical ledgers only. Rounding occurs once per derived amount, after full-precision multiplication (numeric(38,16) intermediate).
- **BR-12.3 Rate date.** The rate for conversion is the one valid on `posting_date` unless the request supplies `rate_date` (allowed for AP/AR documents using document date); rate_date must be ≤ posting_date + 0 days (no future rates).
- **BR-12.4 Residual allocation.** For each derived column (acct, rpt): compute `rounded_i` per line; `residual = Σ rounded_debits − Σ rounded_credits`. If `|residual| ≤ tolerance = 0.5 × unit × line_count` (unit = 10^−scale), add `−residual` to the rounded amount of the line with the largest `|amount|` on the side that reduces the residual (ties: lowest line_number); the adjusted line records `rounding_adjustment`. If `|residual| > tolerance` reject with `LEDGER_ROUNDING_EXCEEDED`. When the entry has only one line per side and the residual is non-zero, post the residual to `rounding_difference_account` instead of adjusting the line, provided `|residual| ≤ 1 unit`.
- **BR-12.5 Multi-currency balancing.** Balancing is checked per transaction currency; an entry with EUR debit and USD credit must include explicit cross-currency lines (clearing) or is rejected with `LEDGER_UNBALANCED` listing the unbalanced currency.
- **BR-12.6 Parallel-ledger posting matrix.** For a request to ledger group `ALL` with leading `0L` (IFRS), `HGB` (local GAAP) and `TAX`:

Source document type	0L (IFRS)	HGB (local)	TAX	Rule
Manual journal, ledger_group ALL	post	post	post	identical lines, converted per ledger
Ledger-specific adjustment (single ledger)	post only if addressed	post only if addressed	post only if addressed	duty-gated
FX revaluation	post	post	skip if ledger.revalue = false	run per ledger
Account-mapped lines (`LedgerAccountMapping`)	original account	mapped account	mapped account	mapping keyed (ledger, account, effective_from)
Subledger with principle-specific lines (`ledger_lines`)	common lines + 0L lines	common lines + HGB lines	common lines + TAX lines	each ledger must balance independently
Year-end close (wave 14)	per ledger calendar	per ledger calendar	per ledger calendar	independent runs

- **BR-12.7 Stale rates.** A rate older than `max_rate_age_days` yields a `warnings[]` entry `FX_RATE_STALE` in the response; with `reject_stale_rates` it is a 422 `LEDGER_RATE_MISSING`.
- **BR-12.8 Revaluation state machine.** `proposed → posted → reversed` (automatic at period+1 for closing_rate method) | `proposed → cancelled`; a second run for the same (ledger, period, rate type) while one is `posted` is rejected unless the first is reversed; cumulative method runs post the delta versus the last posted run.
- **BR-12.9 Revaluation selection.** Only balance-sheet accounts with `revalue = true` and non-zero `balance_txn` in a currency ≠ accounting currency; reconciliation accounts for AP/AR are revalued at open-item level in waves 19/30 and are excluded here (`revalue_level = open_item`).
- **BR-12.10 Provider precedence.** If two providers deliver the same (type, pair, date), the one with higher `priority` wins and the other is stored as `superseded`.
- **BR-12.11 Rate precision in display.** Rates display 6 decimals by default with full 10 on hover/export; amounts never display more than the currency scale.
- **BR-12.12 Market data series resolution.** A value for (series, date) is the point with the greatest `valid_from ≤ date` that is not superseded; no point → `FX_SERIES_VALUE_MISSING` (422); points older than `max_series_age_days` (default 10 business days for rates, 45 for indices) return `FX_SERIES_STALE` in `warnings[]`; the `fx` service never interpolates between points.

2.12.8 Acceptance criteria

- **AC-12.1** (FR-12.5, FR-12.6) Given a 3-line EUR entry posted to a USD ledger with rate 1.0850000000, when posted, then each `amount_acct` equals the rounded product, Σ D = Σ C in USD, and at most one line carries a `rounding_adjustment` of ≤ 0.01.
- **AC-12.2** (BR-12.1) Given 10,000 random amounts in JPY, KWD, USD and EUR converted in all directions, when summed per entry, then every entry balances to the minor unit and no stored amount has scale beyond the currency's.
- **AC-12.3** (FR-12.3) Given no GBP→CHF rate but GBP→EUR and EUR→CHF, when converting, then the response path is `triangulated` and the amount equals one-step rounding of `a × r1 × r2`.
- **AC-12.4** (FR-12.7, BR-12.6) Given the 20-document parallel-ledger fixture, when posted to `ALL`, then 0L, HGB and TAX each receive the expected lines (mapped accounts and ledger-specific lines) and each ledger's TB is 0.
- **AC-12.5** (FR-12.8) Given the revaluation fixture (12 accounts × 4 currencies), when run with closing rates, then unrealized gain/loss per line equals the workbook to the cent and a reversal entry exists dated period end + 1.
- **AC-12.6** (FR-12.4) Given the ECB adapter times out 3 times, when the schedule runs, then no rates change, `fx.provider.degraded` is emitted and a P2 alert fires.
- **AC-12.7** (FR-12.11) Given a manual rate 12% above the previous, when saved, then the UI requires confirmation and the event carries `deviation_pct = 12.0`.
- **AC-12.8** (FR-12.12) Given a line in a currency not enabled for the entity, when posted, then 422 `LEDGER_CURRENCY_MISMATCH`.
- **AC-12.9** (FR-12.9) Given the translation fixture, when run, then CTA equals total assets – liabilities – equity restated at the specified rates, to the cent.
- **AC-12.10** (FR-12.13, BR-12.12) Given series `EURIBOR_3M` with points on 2026-03-02 (2.4100) and 2026-03-03 (2.4300), when `GET /fx/series/EURIBOR_3M?date=2026-03-02` is called, then value 2.4100000000 with `valid_from = 2026-03-02`; when a point of 3.1000 is appended for 2026-03-04, then `fx.series.published` carries `deviation_pct = 27.57` and the UI requires confirmation.

2.12.9 Test plan and fixtures

- Fixtures: `iso4217-2026` (all active currencies with minor units); `rates-ecb-2025` (recorded ECB daily reference rates, 12 months); `amounts-random-10k` (seeded amounts across JPY/KWD/USD/EUR incl. extremes 0.001 and 9,999,999,999.999); `parallel-ledger-20` (20 documents with expected per-ledger lines, accountant-reviewed); `reval-workbook` (12 accounts × 4 currencies with XLSX expected values, FAGL_FC_VAL semantics); `translation-workbook`.

Test	Type	Target	Pass criterion
Rounding property test	property	`money` package	balance after allocation for all generated entries; residual ≤ tolerance
Rate resolution unit tests	unit	fx resolver	direct/inverse/triangulated/missing per table
Provider adapter contract	contract (recorded HTTP)	ECB, OXR adapters	parse recorded payloads; failure modes
Parallel-ledger fixture	integration	posting API	per-ledger lines and TB = 0
Revaluation workbook	integration	revaluation job	adjustment per line equals workbook
Reversal timing	integration (Temporal time-skip)	revaluation workflow	reversal posted on period+1
Rate cache consistency	integration	ledger pods	a published rate is used by all pods within 2 s
Conversion latency	performance	fx + posting	posting p99 stays < 200 ms with 3-currency conversion

Reconciliation: per ledger per currency type TB = 0; Σ revaluation adjustments = posted gain – loss; `fx.rate.published` count = rate rows.

2.12.10 Deployment and operations

Helm chart `passage-fx` (deployment 2–6 pods, CloudNativePG `fx-db`, topics `fx.rates`, `fx.providers`); `passage-ledger` upgraded with currency policy tables. Config: `FX_MAX_RATE_AGE_DAYS=5`, `FX_PROVIDER_TIMEOUT_S=10`, `FX_DEVIATION_WARN_PCT=10`, `LEDGER_ROUNDING_MODE=half_away_from_zero`. Migrations: add `amount_rpt/currency_txn` backfill for wave 11 entries (= accounting amounts), add `ledger_group`, `currency_policy`, `revaluation_*`. Feature flags: `fx.triangulation`, `ledger.parallel_ledgers`, `ledger.translation`. Scheduled jobs: ECB import 16:30 CET business days; OXR hourly (if licensed); month-end revaluation on the close checklist (wave 14). SLOs: rate lookup p99 < 5 ms; import success ≥ 99% of business days. Alerts: provider degraded (P2), stale rate used in posting > 100/h (P3), revaluation run failure (P2). Runbook: manual rate entry during provider outage; re-run revaluation after rate correction (reverse then re-run). Rollback: digest rollback; feature flags off disable parallel posting (entries then go only to the leading ledger — documented as a degraded mode, not for production).

2.12.11 Migration from AX / SAP

Source	Passage entity	Notes
AX `ExchangeRateType`, `ExchangeRateCurrencyPair`, `ExchangeRate`	ExchangeRateType, ExchangeRate	AX stores rate per 100 units by default → divided; `ValidFrom` preserved
AX `Ledger` (accounting/reporting currency, posting layers)	Ledger, LedgerGroup	Current → leading; Operations/Tax layers → additional ledgers only if the customer used them
AX `LedgerExchAdj` history	not migrated	prior revaluations are embedded in migrated closing balances
SAP `TCURV`, `TCURR`, `TCURF`, `TCURX`	ExchangeRateType, ExchangeRate	rate = `UKURS × TFACT/FFACT`; `TCURX` decimals cross-checked against ISO 4217, exceptions reported
SAP `T882` (ledger assignment), `T881` (ledgers), ledger groups	Ledger, LedgerGroup	leading `0L` → is_leading; currency types 10/30 → accounting/reporting
SAP OBA1/OB09 gain/loss accounts	CurrencyPolicy	per ledger × currency
SAP TBDM market data (`VTB_MARKET`, reference interest rates `T056R`/`T056P`, index tables)	MarketDataSeries, MarketDataPoint	reference rate ids → series codes; yield curve definitions not migrated (wave 36 re-defines curves over migrated series)

Validation: rate counts per type; sampled 1,000 conversions compared with source values to the cent; opening balances in all three currencies tie per ledger. Not migrated: provider credentials (re-entered in Vault), valuation run history.

2.12.12 Dependencies, risks, open questions

Depends on: wave 11 (journal, posting API), wave 6 (currencies), wave 4 (Temporal schedules), wave 5 (audit).

Risk	Likelihood	Impact	Mitigation
SAP `TCURX` non-ISO decimals conflict with ISO 4217	Medium	Medium	Migration report flags conflicts; ISO wins; amounts rescaled with sign-off
Residual allocation changes a user-visible line amount	Medium	Low	Adjustment recorded on the line and shown in UI; alternative rounding-difference account configurable
Provider licensing (OXR)	Low	Low	ECB and manual are license-free defaults

Open questions: (1) Should reporting currency be per ledger or per legal entity? Default per ledger; decide by wave 12 mid-point. (2) Historical rate for non-monetary items in translation: store rate on line at posting (default) or look up by date? Default store; decide by wave 12 exit. (3) Support SAP-style "second local currency" (currency type 40) as a fourth amount? Default no (use a parallel ledger); revisit at wave 40.

2.12.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent
`ledger.accounting_currency`	ledger	char(3)	required	enabled ISO 4217	Currency of `amount_acct`	Ledger form > Accounting currency
`ledger.reporting_currency`	ledger	char(3)	= accounting	enabled ISO 4217	Currency of `amount_rpt`	Ledger form > Reporting currency
`ledger.reference_currency`	ledger	char(3)	accounting currency	ISO 4217	Triangulation pivot (FR-12.3)	Currency exchange rate types > Euro triangulation
`ledger.default_rate_type_id` / `closing_rate_type_id` / `average_rate_type_id`	ledger	ref	`DEFAULT` / `CLOSING` / `AVERAGE`	rate types	Conversion at posting, revaluation, translation	Ledger form > Exchange rate type, budget exchange rate type
`rate_type.allow_inverse`	rate type	bool	true	true / false	Reciprocal lookup when no explicit inverse (FR-12.2)	— (AX stores both directions)
`rate_type.provider_id` and schedule	rate type	ref + cron	ECB, `30 16 * * 1–5` CET	providers: ecb, openexchangerates, csv, manual	Import schedule (FR-12.4)	Currency exchange rate providers; Import currency exchange rates batch
`fx.max_rate_age_days`	tenant	int	5	1–60 business days	Stale rate warning threshold (BR-12.7)	—
`ledger.reject_stale_rates`	ledger	bool	false	true / false	Stale rate becomes a 422	—
`fx.deviation_warn_pct`	tenant	numeric	10	1–100	Confirmation threshold on rate entry (FR-12.11)	—
`fx.provider_timeout_s` / `fx.provider_retries`	tenant	int	10 / 3	1–60 / 0–5	Provider call behavior (FR-12.4)	Batch retry on `ExchangeRateProviderImport`
`ledger.rounding_mode`	ledger	enum	half_away_from_zero	half_away_from_zero / half_even	BR-12.2	— (fixed)
`currency_policy.unrealized_gain_account_id` / `unrealized_loss_account_id` / `realized_gain_account_id` / `realized_loss_account_id`	ledger × currency	ref	required when currency enabled	P&L accounts	Revaluation and clearing FX postings	Currency revaluation accounts (General ledger > Setup > Currency)
`currency_policy.rounding_difference_account_id` and `rounding_tolerance_units`	ledger × currency	ref + int	required; 1	balance-sheet or P&L account; 0–5 units	BR-12.4 single-line residual	General ledger parameters > Ledger > Penny difference accounts (`LedgerParameters`)
`ledger.revaluation_method`	ledger	enum	closing_rate	closing_rate / cumulative_no_reversal	FR-12.8	Foreign currency revaluation > "Financial dimensions / Reversal" behavior
`account.revalue` and `account.revalue_level`	main account × ledger	bool + enum	false; balance	balance / open_item	Revaluation selection (BR-12.9)	Main account > Foreign currency revaluation checkbox
`ledger_group.member_ledger_ids`	legal entity	uuid[]	`ALL` auto	ledgers of the entity	Parallel posting target (FR-12.7)	Posting layers (Current/Operations/Tax)
`ledger.accounting_principle`	ledger	enum	IFRS	IFRS / US_GAAP / LOCAL_GAAP / TAX / MANAGEMENT	Drives `LedgerAccountMapping` and subledger ledger-specific lines	—

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent
`translation.rate_by_account_type`	ledger	jsonb	monetary: closing; non_monetary: historical; pnl: average; equity: historical	rate kinds	FR-12.9	Reporting currency adjustment settings (D365)
`translation.cta_account_id`	ledger	ref	required for translation	equity account	CTA posting (FR-12.9)	—
`fx.provider_priority`	tenant	jsonb	ecb 10, openexchangerates 20, csv 30, manual 40	ordered	BR-12.10 precedence	—
`series.max_series_age_days`	series kind	int	rates 10, indices 45, prices 5	1–120	BR-12.12 staleness warning	—
`series.deviation_warn_pct`	series kind	numeric	rates 25, prices 10, indices 10	1–100	FR-12.13 confirmation	—

2.12.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Exchange rates by pair	list inquiry	rate type, from, to, date range, source	valid_from, rate, inverse, source, provider timestamp, superseded, deviation vs previous	`ExchangeRate`	Currency exchange rates form; SSRS Exchange rate (Currency exchange rates)	OB08 display, S_ALR_87012XXX via `TCURR` table view, Fiori Manage Exchange Rates
Rate lookup tester	drill-down	type, pair, date	resolution path (direct / inverse / triangulated), hops, rates used, result	fx resolver	—	—
Rate history / audit	drill-down	pair, type	version chain with `superseded_by`, created by, payload hash, provider	`ExchangeRate`	Database log on `ExchangeRate`	`TCURR` change documents (SCU3)
Provider import log	operational report	provider, rate type, date range	run, status, pairs imported, pairs unchanged, errors, duration	import run log	Batch job history for `ExchangeRateProviderImport`	TBDM import log
Ledger trial balance per ledger and currency type	list inquiry	ledger, period range, currency type (txn/acct/rpt), currency filter	account, opening, debit, credit, closing in selected currency type	`Balance`	Trial balance with posting layer filter	FAGLB03 per ledger (`RLDNR`), S_PL0_86000030
Ledger comparison	operational report	two ledgers, period, account range	balance ledger A, ledger B, difference, mapped account	`Balance` × `LedgerAccountMapping`	—	GCAC ledger comparison
Multi-currency line inquiry	drill-down	ledger, currency_txn, date range, account	line, txn amount/currency, rate applied, rate type, override flag, acct, rpt, rounding adjustment	`JournalLine`	Voucher transactions (three currency columns)	FAGLL03 with document/local/group currency columns
Foreign currency revaluation proposal / results	drill-down	run id	account, dimensions, currency, balance txn, posted acct, rate used, revalued, adjustment, exclusion flag	`RevaluationLine`	Foreign currency revaluation form (simulation/results); SSRS Exchange adjustment	FAGL_FC_VAL log / posting list, S_ALR_87012XXX valuation list
Revaluation history	list inquiry	ledger, period range, rate type	run, mode, status, gain, loss, posting entry, reversal entry	`RevaluationRun`	Foreign currency revaluation history (`LedgerExchAdj`)	FAGL_FC_VAL run history (`FAGL_BSBW_HISTRY`)
Translation (CTA) report	operational report	ledger, period	account group, historical/closing/average rate, restated amount, CTA contribution	translation run	Reporting currency adjustment report (D365)	FAGL_FC_TRANS log
Stale rate usage	KPI	ledger, period	postings with `FX_RATE_STALE` warnings per day, pairs affected	posting warnings log	—	—
Market data series inquiry	list inquiry	series code, kind, date range	valid_from, value, unit, tenor, source, deviation	`MarketDataPoint`	—	TBDM market data display, `T056P` reference rates
Rate overrides audit	operational report	ledger, period, user	entry, line, rate type, override rate, market rate, deviation, user	`JournalLine.rate_override`	—	Document header exchange rate ≠ `TCURR` (FB03 rate field)

2.12.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Exchange rate table printout	user action; scheduled monthly	header: rate type, date range, ledger reference currency; lines: pair, valid_from, rate, source; footer: generated at, provider payload hashes	PDF, XLSX, CSV	—	supporting record for audit of conversions; 10 years when attached to a close run
Foreign currency revaluation posting list	run posted	header: ledger, period, rate type, valuation date, method; lines: account, dimensions, currency, balances, rate, adjustment; footer: totals gain/loss, posting and reversal vouchers, result hash	PDF, XLSX	revaluation run id; vouchers from the ledger sequence	financial record; 10 years
Currency translation statement	translation run posted	header: ledger, period, rates applied per class; lines: account group, historical/closing/average amounts; footer: CTA movement, voucher	PDF, XLSX	run id	financial record; 10 years
Rate deviation confirmation record	rate saved above `fx.deviation_warn_pct`	pair, previous rate, new rate, deviation, user, timestamp, confirmation text	PDF (audit evidence), JSON	—	audit evidence; 10 years

No external e-documents; rate files received from providers are stored as raw payloads (hash in `ExchangeRate.payload_hash`) for 13 months.

2.12.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`fx-import-ecb`	business days 16:30 CET	rate type	run id per (provider, rate type, date); points upserted by key, unchanged values skipped	< 30 s	3 retries, exponential back-off; `fx.provider.degraded` after final failure; last-known rates stay valid	provider degraded → P2; no import in 2 business days → P2
`fx-import-openexchangerates` (optional)	hourly	rate type	same	< 30 s	same	same
`fx-import-csv-inbox`	every 15 min	file	file hash as idempotency key	< 1 min	file quarantined with error report	parse errors → P3
`fx-series-import` (interest/index/price series)	business days 18:00	series	run id per (series, date)	< 1 min	same as rates	stale series beyond `max_series_age_days` → P3
`ledger-fx-revaluation` (automation hook on close checklist, wave 14)	period end + 1 business day 06:00, per ledger	ledger × rate type	idempotent per (ledger, period, rate type); second run rejected while one is posted (BR-12.8)	100k combinations < 5 min	run marked failed; close task blocked	failure → P2
`ledger-fx-revaluation-reverse`	first day of next period 00:10	run	reversal idempotent (`reversal_entry_id` set once)	< 1 min	retry hourly up to 24 h	reversal missing by day 1 12:00 → P2
`ledger-translation` (close hook)	period end + 2 business days	ledger	idempotent per (ledger, period)	200k balances < 3 min	failed → close task blocked	failure → P2
`fx-rate-cache-refresh-audit`	every 5 min	pod	compares pod cache version with latest published rate	< 1 s	pod restarts cache	pod lag > 2 s → P3
`fx-stale-rate-report`	daily 07:00	ledger	daily snapshot	< 1 min	—	stale postings > 100/h → P3

2.12.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
View exchange rates	GET /fx/rates, /rates/history, /convert, /series; screens Exchange rates (read), Rate lookup tester	—	All finance roles
Maintain exchange rates	PUT /fx/rate-types, POST /fx/rates, POST /fx/series/*/points; screens Exchange rate types, Exchange rates, Market data series	Override exchange rates; Run foreign currency revaluation	Treasury accountant, Accounting manager
Administer exchange rates	POST /fx/providers/{p}:import; provider schedules and credentials (Vault); screen Provider schedules	Maintain exchange rates	Finance platform administrator
Override exchange rates	`rate_override` on posting lines; journal line rate cell	Maintain exchange rates; Approve ledger journals	Accounting supervisor (limited), Treasury accountant
Maintain ledger setup	PUT currency-policies, ledger-groups, ledgers, account mapping; screen Ledger setup	Post ledger journals; Run foreign currency revaluation	Accounting manager
Post ledger-specific adjustments	POST /ledger/journal-entries addressed to a single non-leading ledger	Approve ledger journals (same entry)	Financial controller, IFRS/local GAAP specialist
Run foreign currency revaluation	POST /ledger/revaluations, :post; screen Foreign currency revaluation	Maintain exchange rates; Maintain ledger setup	Accounting supervisor, Financial controller
Run currency translation	POST /ledger/translations	Maintain ledger setup	Financial controller, Group accountant
View ledger balances (per ledger)	GET balances with `ledger` filter; Ledger comparison	—	All finance roles

2.12.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
FX-0001 `LEDGER_RATE_MISSING`	error	No {rate_type} rate from {from} to {to} valid on {date} (direct, inverse or triangulated).	FR-12.3 lookup exhausted	Enter the rate or import; check `allow_inverse` and reference currency	FR-12.3, FR-12.12
FX-0002 `LEDGER_CURRENCY_MISMATCH`	error	Currency {currency} is not enabled or is suspended for {legal_entity}.	Currency not enabled	Enable the currency (wave 6)	FR-12.12
FX-0003 `LEDGER_ROUNDING_EXCEEDED`	error	Rounding residual {residual} {currency} exceeds the tolerance {tolerance}.	BR-12.4	Review amounts; a line split may be needed	FR-12.6
FX-0004 `FX_RATE_STALE`	warning	Rate {rate_type} {from}/{to} is {age} days old (limit {max}).	BR-12.7	Import or enter a current rate	FR-12.4
FX-0005 `FX_RATE_DEVIATION`	warning	New rate {rate} deviates {pct}% from the previous {previous}.	FR-12.11	Confirm or correct the rate	FR-12.11
FX-0006 `FX_RATE_INVALID`	error	Rate must be greater than 0 with at most 10 decimals.	Validation	Correct the value	FR-12.2
FX-0007 `FX_PROVIDER_DEGRADED`	warning (ops)	Provider {provider} failed after {retries} retries; last success {last_success}.	FR-12.4	Check provider status; enter rates manually	FR-12.4
FX-0008 `FX_RATE_OVERRIDE_FORBIDDEN`	error (403)	You do not hold the duty to override exchange rates.	FR-12.5	Remove the override or request the duty	FR-12.5
FX-0009 `LEDGER_GROUP_UNKNOWN`	error	Ledger group {code} does not exist for {legal_entity}.	FR-12.7	Create the group or use a ledger id	FR-12.7
FX-0010 `LEDGER_SPECIFIC_FORBIDDEN`	error (403)	Posting to a single non-leading ledger requires the duty "Post ledger-specific adjustments".	FR-12.7	Address the ledger group or request the duty	FR-12.7
FX-0011 `LEDGER_LEDGER_LINES_UNBALANCED`	error	Ledger-specific lines for {ledger} do not balance ({debits} / {credits}).	BR-12.6	Balance each ledger's lines	FR-12.7
FX-0012 `REVAL_RUN_EXISTS`	error	A posted revaluation already exists for {ledger} {period} {rate_type}.	BR-12.8	Reverse it before re-running	FR-12.8
FX-0013 `REVAL_RATE_MISSING`	error	No closing rate for {currency} on {valuation_date}; {count} balances cannot be revalued.	FR-12.8	Enter the closing rate	FR-12.8
FX-0014 `REVAL_PERIOD_CLOSED`	error	GL is closed for {period}; revaluation cannot post.	BR-14.1	Reopen or choose another period	FR-12.8
FX-0015 `TRANSLATION_CTA_ACCOUNT_MISSING`	error	No CTA account configured for ledger {ledger}.	FR-12.9	Set `translation.cta_account_id`	FR-12.9

Code	Severity	User text	Cause	Remediation	Raised by
FX-0016 `FX_SERIES_VALUE_MISSING`	error	Series {code} has no value on or before {date}.	BR-12.12	Import or enter a point	FR-12.13
FX-0017 `FX_SERIES_STALE`	warning	Series {code} value is {age} days old.	BR-12.12	Import a current point	FR-12.13
FX-0018 `FX_RATE_DATE_FUTURE`	error	Rate date {rate_date} is after the posting date.	BR-12.3	Use a rate date ≤ posting date	FR-12.5

2.12.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Rate lookup (cached)	every posted line in foreign currency: 60k/day; peak 5,000/s	0.05 / 1 / 5 ms	20,000/s per pod	—	in-memory map (type, from, to) → sorted valid_from	cache miss → one indexed query; posting p99 budget unchanged
Rate publish (manual or import)	2,000 points/day	20 / 80 / 150 ms	500/s bulk	200 B × 500k rows ≈ 100 MB	index §2.12.4	import queued
`POST /fx/convert` (stateless)	100k/day	5 / 20 / 40 ms	2,000/s	—	—	—
Parallel posting to 3 ledgers	30% of entries	+15 / +40 / +60 ms over single ledger	5,000 lines/s aggregate	3× journal rows for parallel entries	shared `document_id` index	HPA; entries stay atomic
Revaluation run	monthly per ledger; 100k combinations × 4 currencies	—	100k combinations < 5 min	revaluation lines 120 B × 1.2M/year ≈ 150 MB	`journal_line` (ledger, currency_txn, posting_date); `revaluation_line` PK	run time scales linearly; proposal retained 90 days
Translation run	monthly; 200k balances	—	< 3 min	—	balance PK	—
Rate history query	5k/day	20 / 100 / 200 ms	—	—	(type, from, to, valid_from DESC)	—
Market data series storage	50 series × 260 points/year	lookup 1 ms	—	< 5 MB	(series_id, valid_from DESC)	—

2.12.20 Localization and compliance hooks

- Statutory rate sources: country packs (wave 48) register the mandatory provider per legal entity — ECB reference rates (EU), Bundesbank/ÖNB monthly VAT rates (DE/AT, `Umsatzsteuer-Umrechnungskurse`), HMRC monthly rates (UK), Banco de México FIX (MX), RBI reference rates (IN), Bank of Canada (CA) — as additional `fx` provider adapters with their own rate types (e.g., `TAX_MONTHLY`); the tax engine (wave 13) selects the statutory rate type per country for VAT reporting.
- Rate direction and quotation conventions (direct vs indirect quotation, per-100 units as in AX and `TCURF` factors) are normalized at import; display conventions per locale (decimal separator, 6 decimals) come from wave 5 formatting.
- Rounding: ISO 4217 minor units are not overridable; Swiss 0.05 cash rounding and similar are invoice-level concerns handled by waves 29/30 pricing, not by the ledger.
- Parallel ledgers for local GAAP: country packs ship ledger templates (`HGB`, `PCG`, `Ind AS`, `JGAAP`) with accounting principle, calendar (e.g., Japan April–March), and `LedgerAccountMapping` starter sets.
- Hyperinflation (IAS 29) restatement and functional-currency changes are out of scope for Phase 2; the `MarketDataSeries` of kind `index` (CPI) is the hook wave 40 uses.
- Privacy: none (no personal data in rates); provider credentials are secrets in Vault.
- Extension point: `fx.provider` adapter interface (`Fetch(rateType, date) → []RatePoint`), registered by country packs; `ledger.account_mapping_pack` for principle-specific account remaps.

2.12.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP §13.6 FSCM Master data & documents — TBDM market data import (rates, prices) (SAP register row 74)	PARTIAL-12	ADDED as FR-12.13 (series storage and import); valuation DEFERRED to wave 36	exchange-rate import was already FR-12.4; interest, index and price series now stored with the same provider and audit model
SAP §13.6 FSCM Periodic processing — TBDM import run (SAP register row 82)	PARTIAL-12	ADDED as FR-12.13	job `fx-series-import` (§2.12.16)
SAP §13.6 FSCM Customizing — Market data: rate types, yield curves, volatility (`VTB*`, `TCURR`) (SAP register row 85)	PARTIAL-12	ADDED as FR-12.13 for rate types and series metadata; yield curves and volatility DEFERRED to wave 36	curve bootstrapping, interpolation and volatility surfaces belong to treasury (wave 36), which consumes `fx.series.published`

2.13 Wave 13: Tax Engine (Sales/Use, VAT/GST, Withholding)

2.13.1 Scope and business context

AX determines tax by intersecting a sales tax group (party side, `TaxGroupHeading`/`TaxGroupData`) with an item sales tax group (`TaxItemGroupHeading`/`TaxItemGroupData`) to find tax codes (`TaxTable`) with rates (`TaxData`), posting results to `TaxTrans`. SAP uses tax procedures with condition types, tax codes per country (`T007A`, FTXP), jurisdiction codes for US, and withholding tax types/codes. Wave 13 builds the `tax` service (Go) that any document line calls for determination, calculation and posting lines, with an internal rules engine as default and an Avalara/Vertex-compatible provider adapter interface, plus settlement to the GL and statutory extracts (VAT return, EU sales list, Intrastat, SAF-T).

After this wave a manual journal or any later document gets correct tax lines for US sales/use tax, EU VAT (standard, reduced, reverse charge, intra-community), Indian GST (CGST/SGST/IGST) and Canadian GST/HST/PST, and the tax settlement run posts the net payable and reconciles to the tax accounts. Out of scope: document-level invoicing (waves 18, 29), 1099/26AS statutory filing formats (hooks only; forms in wave 48 country packs), customs duty.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
SAP FTXP: a tax code is a single rate per procedure step with no validity dates; a rate change (e.g., DE 19% → 16% in 2020) forces new codes and mass master-data updates in `KNVI`/`MLAN`	`T007A` has no effective dating	`TaxRate` rows are effective-dated under a stable `TaxCode`; groups and parties keep their assignments	Rate change rollout 2–5 days of config → 1 rate row; zero master-data touches
AX `TaxTrans` and SAP `BSET` show the result but not why: users cannot see which group intersection, exemption or jurisdiction produced the amount; support tickets replay the document in a test client	Determination logic in X++ `Tax` class / condition technique with no trace output	`POST /tax/calculate:explain` returns `determination_trace[]` step by step; tester screen renders it	"Why this tax?" investigation 30–60 min → < 1 min
US multi-jurisdiction via SAP external tax interface or AX ISV connectors: provider outage blocks invoicing (RFC timeout, document cannot save)	Synchronous dependency with no fallback	Provider adapter with 2 s timeout, internal-engine fallback, `tax_fallback` flag and recommit queue	Invoicing downtime during provider outage → 0; fallback decision < 2.5 s
Sales tax settlement (AX "Settle and post sales tax", SAP F.12 / S_ALR_87012357) posts a total that does not match the GL tax accounts; controllers reconcile by hand in Excel	Tax transactions and GL postings written by different code paths with separate rounding	`TaxTransaction` linked to `JournalLine` by id; settlement blocks on mismatch (`TAX_SETTLEMENT_MISMATCH`)	Settlement-to-GL reconciliation 4 h/period → automated, 0.00 variance enforced
Price-inclusive rounding differs across SAP condition steps and AX "per line / per document" settings, producing 1-cent mismatches with customer invoices and credit notes	Multiple rounding points without a distribution rule	One algorithm (BR-13.3) with largest-remainder distribution; credit notes use the original bracket	1-cent invoice disputes → 0 in the 500-scenario matrix

2.13.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Tax codes and rates	`TaxTable`, `TaxData` (effective-dated values, limits)	`T007A` (tax codes), FTXP, condition records `KONP`/`A003`	`tax.TaxCode`, `tax.TaxRate` (effective-dated, bracketed)	full
Party-side tax group	`TaxGroupHeading`, `TaxGroupData`, Sales tax groups form	Customer/vendor tax classification (`KNVI`/`LFA1`–`STKZU`), tax category on BP	`tax.TaxGroup` (N:M TaxCode)	full
Item-side tax group	`TaxItemGroupHeading`, `TaxItemGroupData`	Material tax classification (`MLAN`)	`tax.ItemTaxGroup` (N:M TaxCode)	full
Determination	Intersection of group and item group	Tax procedure (TAXUSJ/TAXEUR), access sequences, condition technique	Intersection + jurisdiction resolution + rule overrides (§2.13.7 BR-13.1)	improved: explainable trace
Jurisdictions	`TaxJurisdiction` (US), sales tax authorities `TaxAuthorityAddress`	Jurisdiction codes (`TTXJ`), external tax interface (Vertex/Sabrix)	`tax.TaxJurisdiction` hierarchy (country > state > county > city > district)	full
Exemptions	Tax exempt numbers, `TaxExempt` codes	Exemption licenses (withholding), tax code 0%	`tax.ExemptionCertificate` (party × jurisdiction × validity × reason)	full
Calculation modes	Price includes tax, calculation method (whole amount/interval), tax on tax	Condition calculation types, `KSCHL` steps, from/to step references	Inclusive/exclusive, compound sequence, bracketed rates	full
Reverse charge / use tax	Reverse charge item groups, Use tax checkbox on tax code	Acquisition tax codes (EU), use tax (US) via tax procedure	`TaxCode.mechanism` ∈ {standard, reverse_charge, use_tax, withholding}	full
Partial recoverability	Non-deductible percentage on tax code	Non-deductible tax (NAV/NVV condition types)	`TaxCode.recoverable_pct` with expense distribution	full
Withholding tax	Withholding tax codes/groups (`TaxWithholdTable`), 1099 fields	Extended withholding tax types/codes (`T059P`, `T059Z`), `WITH_ITEM`	`tax.WithholdingTaxCode`, `tax.WithholdingTransaction`; trigger at invoice or payment	full
Posted tax	`TaxTrans`	`BSET`, ACDOCA tax fields	`tax.TaxTransaction` linked to JournalLine	full
Settlement and reporting	Settlement periods, "Settle and post sales tax", EU sales list, Intrastat	Tax payable posting (F.12/S_ALR_87012357), ACR (Advanced Compliance Reporting), Intrastat (VE01)	`tax.TaxSettlementRun`, VAT return, ESL, Intrastat, SAF-T extracts	full (formats in wave 48)
External provider	Avalara/Vertex ISV connectors	External tax interface (RFC_CALCULATE_TAXES_DOC)	Provider adapter interface (`TaxProvider`) with failover	full
Intrastat master data	Foreign trade parameters, Commodity codes (`IntrastatCommodity`), Transaction codes (`IntrastatTransactionCode`), Transport methods, Ports, Statistics procedures, Country/region properties (EU), Intrastat compression / check setup	Foreign trade customizing (VEB/VE0: `T604`, `T605`, `T606`, `T616` modes of transport, `T618` customs offices), commodity codes (`T604`), Intrastat settings per country	`tax.CommodityCode`, `tax.IntrastatTransactionCode`, `tax.TransportMode`, `tax.Port`, `tax.StatisticsProcedure`, `tax.CountryTradeProperty` (FR-13.15)	full
Intrastat journal and declaration	Intrastat journal (`IntrastatArchiveGeneral`, `Intrastat` form): transfer from invoices, validate, compress, export file	VE01 / MEIS / VI99 declaration runs (`VEIAV`), Intrastat checks, file export per country	`tax.IntrastatLine` with transfer, validation, compression, correction and export (FR-13.16); country formats via wave 48	full
Commodity code on products	Released product > Foreign trade > Commodity code	Material master Foreign trade export view (`MARC-STAWN`)	`product.commodity_code_id` (wave 8 field, FK to `tax.CommodityCode`), default per category	full
VAT recovery on expenses	Expense management > Setup > Tax configuration (recoverable VAT per country), Recover VAT (ER-based claim), SSRS VAT recovery report	Non-deductible tax (NAV/NVV), ESC claim via DRC / country-specific reports; no native EU refund directive flow	`tax.VatRecoveryPolicy` per (country, expense category), foreign VAT reclaim extract (FR-13.17)	full
Withholding remittance and certificates	Withholding tax payment (settle), 1099 summary; India TDS challan via GL/Indian localization	J1INCHLN (TDS challan), J1INBANK (bank challan update), J1INCERT (Form 16A), generic WHT reporting RFIDYYWT	`tax.WithholdingRemittanceRun`, certificate numbering and hooks (FR-13.18); country forms in wave 48	partial: forms in wave 48

2.13.3 Functional requirements

- **FR-13.1 Tax codes and effective-dated rates** — Must. The system shall define tax codes per legal entity with mechanism (standard, reverse_charge, use_tax, withholding), rate rows effective-dated with optional amount brackets (`from_amount`, `to_amount`), recoverable percentage, jurisdiction, and posting accounts (payable, receivable, non-recoverable expense, use-tax expense).
  - Rate precision numeric(9,6) in percent (e.g., 19.000000); brackets evaluated on the line base or document base per code setting (AX "calculation method": whole amount vs interval).
  - Maximum 2,000 codes per legal entity; rate history retained indefinitely.
- **FR-13.2 Tax groups and item tax groups** — Must. The system shall maintain party-side tax groups and item-side item tax groups, each a set of tax codes; defaults come from party roles (customer/vendor), delivery address jurisdiction, and product/category.
- **FR-13.3 Jurisdiction hierarchy** — Must. The system shall model jurisdictions as a tree (country > state/province > county > city > special district) with address-to-jurisdiction resolution by postal code tables or provider lookup, and compute stacked rates (e.g., CA state 7.25% + county + district).

- **FR-13.4 Determination and calculation API** — Must. The system shall expose `POST /api/v1/tax/calculate` taking a document (header: legal entity, document type, dates, party, addresses, currency; lines: amount, quantity, product, tax group, item tax group, overrides) and returning per-line and per-document tax lines with an explain trace.
  - Stateless, p99 < 50 ms for 100-line documents with the internal engine; provider calls add their latency (timeout 2 s).
  - Order of operations in BR-13.1. Results carry `engine: internal | avalara | vertex`, `fallback: bool`, and `determination_trace[]`.
- **FR-13.5 Inclusive and exclusive prices** — Must. The system shall support price-inclusive lines by backing out tax: `base = gross / (1 + Σ non-compound rates)`, iteratively resolving compound codes; rounding per BR-13.3.
- **FR-13.6 Compound tax, reverse charge, use tax, partial recoverability** — Must. The system shall calculate tax-on-tax by code sequence, generate equal input and output lines for reverse charge, accrue use tax as a liability with the expense to the line's account, and split recoverable vs non-recoverable amounts.
- **FR-13.7 Exemption certificates** — Must. The system shall apply certificates by (party, jurisdiction, validity, reason code, optional product category), zeroing the applicable codes and recording the certificate id on the tax transaction; expired certificates produce `TAX_EXEMPTION_EXPIRED` warnings and full tax.
- **FR-13.8 Withholding tax** — Must. The system shall calculate withholding (vendor and customer) with code, base (gross, net, tax-inclusive), rate, threshold and annual accumulation, triggered at invoice or payment per code; results are stored as `WithholdingTransaction` and exposed to AP/AR (waves 19/30) and to 1099/26AS reporting hooks.
- **FR-13.9 Tax posting** — Must. The system shall convert a calculated document into journal lines (posting API, wave 11) with the tax account per code, dimension inheritance from the taxed line, and create `TaxTransaction` rows linked to the posted `JournalLine` ids, within the caller's document transaction via a prepared-lines contract (`POST /tax/calculate` → caller includes returned lines in its posting request → tax consumes `journal.posted` to link).
- **FR-13.10 Provider adapters and failover** — Must. The system shall implement a `TaxProvider` interface (`Calculate`, `Commit`, `Void`, `ValidateAddress`) with Avalara AvaTax REST v2 and Vertex O-Series adapters, selectable per legal entity × document type; on timeout or 5xx the internal engine is used and the document is flagged `tax_fallback = true` for later recommit.
  - Provider credentials in Vault; requests and responses hashed and stored 7 years for audit (payload itself retained 13 months).
- **FR-13.11 Settlement periods and settlement run** — Must. The system shall define settlement periods per tax authority (monthly, quarterly, annual) and run a settlement that sums tax transactions by code and period, posts the net payable/receivable to the authority's vendor/customer (party role) or a settlement account, and marks transactions settled.
  - Re-runs for the same period post only the delta (late-posted transactions) as a new settlement with `version` incremented.
  - Settlement totals must equal the GL tax account movements for the period; a difference > 0.00 blocks posting with `TAX_SETTLEMENT_MISMATCH`.
- **FR-13.12 Statutory extracts** — Should. The system shall produce VAT return boxes (configurable box mapping per country), EU sales list (ESL, by customer VAT id and transaction type), Intrastat lines (commodity code, weight, value, partner country) and SAF-T (OECD v2.0 XSD-valid XML) for a period.
- **FR-13.13 Tax configuration workspace** — Must. Screens for codes, rates, groups, jurisdictions, certificates, withholding, settlement periods, provider settings and a calculation tester.
- **FR-13.14 Tax matrix test harness** — Must. The system shall ship a fixture runner that executes the 500-scenario matrix against the engine in CI and reports diffs per scenario.
- **FR-13.15 Intrastat master data** — Must. The system shall maintain, per tenant with legal-entity overrides: commodity codes (Combined Nomenclature CN8, with validity year, supplementary unit, description, hierarchy by HS6), nature-of-transaction codes (two-digit EU list plus national second digit), transport modes (1–9 per EU regulation 2019/2152), ports and airports (UN/LOCODE), statistics procedures, delivery terms (Incoterms 2020) mapping, country/region trade properties (EU member, Intrastat threshold arrival/dispatch, reporting currency, arrival/dispatch statistical value rules), and the legal entity's Intrastat registration (VAT id, third-party declarant, agent).
  - CN8 code lists are imported yearly from the EU RAMON/Eurostat CSV through a CSV importer (≤ 20,000 rows); expired codes stay for historic lines.
  - Commodity code defaults on products via category (wave 8 exposes `commodity_code_id` and `country_of_origin`); a line without a commodity code at Intrastat transfer time is reported as an exception, not dropped.
- **FR-13.16 Intrastat journal, validation, compression and export** — Must. The system shall build `IntrastatLine` rows from posted documents (wave 13: manual journals flagged with trade data; waves 18/27/28/29 register vendor invoices, sales invoices, shipments and receipts as sources through the `IntrastatSource` contract) with direction (arrival/dispatch), commodity code, partner country and partner VAT id (dispatches, EU 2022 requirement), country of origin, nature of transaction, transport mode, delivery terms, net mass, supplementary units, invoice value, statistical value, and period.
  - Validation rules per country (mandatory fields by direction, mass > 0 unless supplementary unit, partner VAT id format via wave 7) produce `TAX_INTRASTAT_INVALID` exceptions on a worklist; lines can be corrected in the journal with an audit trail (field, old, new, user).
  - Compression merges lines with identical (direction, commodity, partner country, origin, transaction nature, transport mode, delivery terms, partner VAT id) per period into one declaration line (summed mass, units, values) per the country's compression setup.
  - Export produces the declaration dataset for the period (JSON canonical) and country formats registered by wave 48 (DE XML (Destatis), FR DEB/EMEBI, NL CBS IDEP, IT Elenchi; wave 13 ships the canonical format and a generic CSV); thresholds per country decide whether a declaration is required and are reported on the period summary.
  - Corrections after export create delta lines flagged `correction_of_line_id` and a correction declaration for the original period.
- **FR-13.17 VAT recovery on expenses** — Should. The system shall maintain `VatRecoveryPolicy` rows per (expense category, country of expense, legal entity country) with `recoverable_pct` (domestic input VAT per category, e.g., DE entertainment 70%, FR passenger cars 0%, BE 50%), `foreign_reclaim_eligible` (EU Directive 2008/9/EC or 13th Directive) and the minimum claim thresholds; the calculation API applies the policy when a document line carries an `expense_category` (expense reports in wave 43, vendor invoices in wave 19) by setting the code's recoverable percentage per line.
  - A foreign VAT reclaim extract per (legal entity, refund country, calendar quarter or year) lists eligible transactions with invoice reference, vendor VAT id, amounts, and expense codes per Annex of the directive (codes 1–10), and a VAT recovery report shows recoverable, non-recoverable and reclaimable amounts per period and country.
  - Non-recoverable amounts post to the line's expense account (FR-13.6) and reclaimable foreign VAT posts to a `foreign_vat_receivable_account_id` per policy when `book_foreign_vat_as_receivable = true`, otherwise to expense.
- **FR-13.18 Withholding remittance and certificate hooks** — Should. The system shall group posted `WithholdingTransaction` rows by (legal entity, authority, withholding code section, period) into a `WithholdingRemittanceRun` that posts the liability transfer to the authority party and records the remittance reference (bank challan number, payment date, bank branch code) and, per payee and period, assigns certificate numbers from a number sequence and exposes the data set (payee, tax id, section, base, rate, tax, remittance references) that country packs render as Form 16A / 26Q (IN), 1099-NEC/MISC and 1042-S (US), CIS statements (UK) and `Lohnsteuerbescheinigung`-style certificates (wave 48).
  - A remittance run may be `proposed → posted → cancelled` and its transactions are flagged `remitted`; quarterly returns reference remittance runs; a certificate is immutable once issued and can be reissued with a new number and `supersedes_certificate_id`.

2.13.4 Data model

TaxCode and **TaxRate**

Field	Type	Required	Notes
id	uuid	yes	PK
legal_entity_id	uuid	yes	

Field	Type	Required	Notes
code	text(20)	yes	unique per legal entity
name	text(120)	yes	
mechanism	enum	yes	standard, reverse_charge, use_tax, withholding
jurisdiction_id	uuid	no	FK TaxJurisdiction
authority_party_id	uuid	yes	FK Party (role tax authority)
settlement_period_id	uuid	yes	FK TaxSettlementPeriod
calc_basis	enum	yes	line, document (bracket evaluation)
compound_sequence	int	no	null = not compound; higher sequences apply on base + earlier tax
recoverable_pct	numeric(9,6)	yes	default 100
payable_account_id / receivable_account_id / nonrecoverable_account_id / use_tax_expense_account_id	uuid	conditional	FK MainAccount
rounding	enum	yes	per_line, per_document (default per_document for VAT, per_line for US)
rate: tax_code_id, valid_from, valid_to		yes	non-overlapping intervals
rate: from_amount / to_amount	numeric(23,6)	no	bracket
rate: rate_pct	numeric(9,6)	yes	
rate: min_tax / max_tax	numeric(23,6)	no	caps

TaxTransaction

Field	Type	Required	Notes
id	uuid	yes	PK
legal_entity_id / ledger_id	uuid	yes	leading ledger
tax_code_id	uuid	yes	
jurisdiction_id	uuid	no	
source (system, type, id, line_ref)	text	yes	document line
journal_line_id	uuid	no	set when `journal.posted` consumed
tax_direction	enum	yes	output (sales), input (purchase), use, withholding
base_amount_txn / tax_amount_txn / currency_txn	numeric / char(3)	yes	
base_amount_acct / tax_amount_acct	numeric(23,6)	yes	
recoverable_amount_acct / nonrecoverable_amount_acct	numeric(23,6)	yes	sum = tax_amount_acct
rate_pct_applied	numeric(9,6)	yes	
exemption_certificate_id	uuid	no	
counterparty_party_id / counterparty_tax_id	uuid / text	no	for ESL
engine / fallback / provider_ref	enum / bool / text	yes	
settlement_run_id	uuid	no	null = unsettled
posting_date / fiscal_period_id	date / uuid	yes	

TaxSettlementRun: `id`, `legal_entity_id`, `authority_party_id`, `settlement_period_id`, `period_from`, `period_to`, `version`, `status` (proposed, posted, cancelled), `output_tax_total`, `input_tax_total`, `net_amount`, `entry_id`, `transaction_count`, `created_by`, `posted_at`.

ExemptionCertificate: `id`, `party_id`, `jurisdiction_id`, `certificate_number`, `reason_code`, `valid_from`, `valid_to`, `product_category_id` (optional), `document_uri`, `status` (active, expired, revoked).

IntrastatLine (FR-13.16) and Intrastat master data (FR-13.15)

Field	Type	Required	Notes
id	uuid	yes	PK
legal_entity_id / reporting_period	uuid / text(7)	yes	`YYYY-MM`
direction	enum	yes	arrival, dispatch
source (system, type, id, line_ref)	text	yes	originating document line
commodity_code_id	uuid	conditional	FK CommodityCode (CN8, `validity_year`); missing → exception
partner_country / origin_country	char(2)	yes / conditional	ISO 3166-1; origin mandatory for dispatches since 2022
partner_vat_id	text(20)	conditional	dispatches; format-checked (wave 7)
transaction_nature_code	text(2)	yes	FK IntrastatTransactionCode
transport_mode	text(1)	yes	FK TransportMode (1–9)
delivery_terms	text(3)	conditional	Incoterms 2020
port_id	uuid	no	FK Port (UN/LOCODE)
statistics_procedure_id	uuid	no	FK StatisticsProcedure
net_mass_kg / supplementary_units	numeric(18,3) / numeric(18,3)	conditional	mass required unless CN8 has supplementary unit only
invoice_value / statistical_value / currency	numeric(23,6) / numeric(23,6) / char(3)	yes	statistical value per country rule (FR-13.15)
status	enum	yes	transferred, validated, exception, compressed, exported, corrected
compressed_into_id / correction_of_line_id / export_run_id	uuid	no	
corrections	jsonb	no	[{field, old, new, user, at}]

`CommodityCode`: `id`, `code` text(8), `validity_year`, `description`, `supplementary_unit`, `parent_hs6`; `IntrastatTransactionCode`: `code` text(2), `description`, `country` (null = EU list); `TransportMode`: `code` text(1), `description`; `Port`: `locode` text(5), `name`, `country`; `StatisticsProcedure`: `code`, `description`, `country`; `CountryTradeProperty`: `country`, `is_eu_member`, `threshold_arrival`, `threshold_dispatch`, `reporting_currency`, `statistical_value_rule` (invoice_value |

invoice_plus_freight_to_border I custom), `compression_fields` jsonb, `export_format_id` (wave 48). `IntrastatExportRun`: `id`, `legal_entity_id`, `period`, `direction`, `line_count`, `total_value`, `format`, `output_uri`, `hash`, `status`, `declared_at`.

VatRecoveryPolicy (FR-13.17): `id`, `legal_entity_id`, `expense_category_id` (wave 43/19 category list; `*` allowed), `expense_country` char(2), `recoverable_pct` numeric(9,6), `foreign_reclaim_eligible` bool, `directive` enum (2008_9, 13th, none), `expense_code_annex` text(2), `book_foreign_vat_as_receivable` bool, `foreign_vat_receivable_account_id`, `min_claim_amount` numeric(23,6), `valid_from`, `valid_to`. `VatReclaimExtract`: `id`, `legal_entity_id`, `refund_country`, `period_from`, `period_to`, `transaction_ids[]`, `total_reclaimable`, `output_uri`, `status`.

WithholdingRemittanceRun (FR-13.18): `id`, `legal_entity_id`, `authority_party_id`, `section_code`, `period_from`, `period_to`, `status` (proposed, posted, cancelled), `transaction_ids[]`, `total_withheld`, `entry_id`, `remittance_reference` jsonb (`challan_no`, `bank_code`, `paid_on`, `bsr_code`), `posted_at`. `WithholdingCertificate`: `id`, `payee_party_id`, `period`, `certificate_number` (sequence per legal entity × section), `transaction_ids[]`, `issued_at`, `supersedes_certificate_id`, `output_uri`.

Invariants: Σ TaxTransaction.tax_amount_acct per (code, period) = movement on the code's payable/receivable account for that period (checked by settlement); recoverable + nonrecoverable = tax; rate intervals per code do not overlap; a settled transaction cannot change `settlement_run_id` except to null via a cancelled run. Indexing: `tax_transaction` on (legal_entity_id, tax_code_id, fiscal_period_id, settlement_run_id), (source_system, source_type, source_id), (journal_line_id); partitioned by fiscal year.

2.13.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/tax/calculate	Determine and calculate for a document	stateless (optional `Idempotency-Key` for provider commit)	Calculate tax (service accounts)
POST	/api/v1/tax/calculate:explain	Same with full determination trace	stateless	View tax configuration
POST	/api/v1/tax/commit	Commit provider transaction after document posting	`Idempotency-Key` required	Calculate tax
GET/PUT	/api/v1/tax/legal-entities/{le}/codes[/{code}] and /codes/{code}/rates	Codes and rates	PUT idempotent	Maintain tax codes
GET/PUT	/api/v1/tax/legal-entities/{le}/groups[/{id}] and /item-groups[/{id}]	Groups	PUT idempotent	Maintain tax groups
GET/PUT	/api/v1/tax/jurisdictions[/{id}] and POST /jurisdictions:resolve	Jurisdiction tree and address resolution	PUT idempotent	Maintain tax jurisdictions
GET/POST/PATCH	/api/v1/tax/exemption-certificates[/{id}]	Certificates	`Idempotency-Key` on POST	Maintain tax exemptions
GET/PUT	/api/v1/tax/legal-entities/{le}/withholding-codes[/{code}]	Withholding	PUT idempotent	Maintain withholding tax
GET	/api/v1/tax/transactions?le=&code=&period=&settled=	Posted tax transactions	n/a	View tax transactions
POST	/api/v1/tax/settlements	Propose or post a settlement run	`Idempotency-Key`	Settle tax
POST	/api/v1/tax/settlements/{id}:post	Post a proposal	`Idempotency-Key`	Settle tax
POST	/api/v1/tax/extracts/{vat-return	esl	intrastat	saf-t}
GET/PUT	/api/v1/tax/intrastat/commodity-codes[/{code}], /transaction-codes, /transport-modes, /ports, /statistics-procedures, /country-properties[/{country}] and POST /commodity-codes:import	Intrastat master data (FR-13.15)	PUT idempotent; import `Idempotency-Key`	Maintain foreign trade setup
POST	/api/v1/tax/intrastat/lines:transfer	Transfer lines from registered sources for a period	`Idempotency-Key` (period)	Maintain Intrastat journal
GET/PATCH	/api/v1/tax/intrastat/lines[/{id}]?le=&period=&direction=&status=	Intrastat journal and corrections	PATCH audited	Maintain Intrastat journal
POST	/api/v1/tax/intrastat/lines:validate, :compress and /exports	Validate, compress, export declaration	run id	Produce tax reports
GET/PUT	/api/v1/tax/legal-entities/{le}/vat-recovery-policies[/{id}]	VAT recovery policies (FR-13.17)	PUT idempotent	Maintain tax codes
POST	/api/v1/tax/vat-reclaims	Foreign VAT reclaim extract for (refund country, period)	run id	Produce tax reports
POST	/api/v1/tax/withholding/remittances and /{id}:post, :cancel	Withholding remittance runs (FR-13.18)	`Idempotency-Key`	Settle tax
POST	/api/v1/tax/withholding/certificates:issue	Issue certificates per payee and period	`Idempotency-Key`	Produce tax reports

Calculation request/response (abridged): request `{legal_entity, document_type, document_date, tax_date, currency, party:{id, role, tax_id}, ship_from, ship_to, lines:[{line_ref, amount, quantity, product_id, tax_group, item_tax_group, price_includes_tax, account, dimensions, override:{tax_code?, exempt_reason?}}]}`; response `{engine, fallback, lines:[{line_ref, base, taxes:[{code, jurisdiction, rate_pct, tax_amount, recoverable, nonrecoverable, direction}]}], document_taxes_by_code:[…], posting_lines:[… wave 11 line objects …], trace_id, determination_trace[]}`.

Validation: every line has resolvable groups or an override; dates within code validity; currency enabled; party tax id format valid for reverse charge; totals of `posting_lines` balance against the caller's lines.

Event	Trigger	Payload fields	Consumers
dev.passage.tax.calculated.v1	calculate with commit intent	document ref, engine, fallback, totals per code, trace_id	document services, audit
dev.passage.tax.transaction.recorded.v1	TaxTransaction linked to journal line	transaction id, code, period, base, tax, direction, journal_line_id	reporting, lakehouse
dev.passage.tax.settled.v1	settlement posted	run id, authority, period, output, input, net, entry_id, version	reporting, ledger
dev.passage.tax.provider.fallback.v1	provider failure	provider, document ref, error, retry_after	ops, AP/AR recommit job
dev.passage.tax.rate.changed.v1	rate row created	code, valid_from, rate_pct, previous	document services cache
dev.passage.tax.intrastat.exported.v1	declaration export completed	export_run_id, legal_entity, period, direction, line_count, total_value, format, hash	audit, wave 48 submission connectors

Event	Trigger	Payload fields	Consumers
dev.passage.tax.withholding.remitted.v1	remittance run posted	run_id, authority, section, period, total_withheld, entry_id, remittance_reference	AP (wave 19), reporting, wave 48 returns

`IntrastatSource` contract (used by waves 18/27/28/29): a document service registers `{source_type, resolver_uri}`; at transfer the tax service calls the resolver for posted documents in the period and receives lines with trade fields; documents reversed after transfer produce negative lines on the next transfer.

2.13.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Tax codes	Create, add rate rows, set accounts, mechanism	code, mechanism, jurisdiction, current rate, authority, settlement period	Maintain tax codes
Tax groups / item tax groups	Assign codes, set defaults for party/product	group, codes, exempt flags	Maintain tax groups
Jurisdictions	Tree view, import postal tables, resolve address tester	level, code, parent, rates stacked	Maintain tax jurisdictions
Exemption certificates	Register, upload document, expire	party, jurisdiction, number, validity, reason	Maintain tax exemptions
Withholding codes	Rates, thresholds, trigger point, reporting category	code, rate, base, threshold, 1099 box	Maintain withholding tax
Calculation tester	Enter a sample document, see tax lines and trace	line, code, rate, base, tax, explanation	View tax configuration
Tax transactions	Filter by code/period/document, drill to journal line and source	document, code, base, tax, direction, settled	View tax transactions
Settlement	Propose, review by code, post, view history	authority, period, output, input, net, status, version	Settle tax
Provider settings	Select provider per entity/document type, test connection	provider, environment, company code, last success	Administer tax
Foreign trade setup	Maintain commodity codes (import yearly CN8), transaction codes, transport modes, ports, statistics procedures, country properties and thresholds	code, description, validity year, supplementary unit; country, threshold arrival/dispatch, compression fields	Maintain foreign trade setup
Intrastat journal	Transfer period, work exceptions, correct fields, validate, compress, export, view declaration summary vs threshold	direction, commodity, partner country, VAT id, nature, transport, mass, units, invoice/statistical value, status	Maintain Intrastat journal / Produce tax reports
VAT recovery policies	Define recoverable % per category and country, reclaim eligibility, accounts; run reclaim extract; view recovery report	category, country, recoverable %, directive, annex code, receivable account	Maintain tax codes / Produce tax reports
Withholding remittance	Propose run per authority/section/period, enter remittance reference, post, issue certificates	payee, section, base, rate, withheld, challan no., certificate no.	Settle tax / Produce tax reports

Interaction rules: the tester renders the determination trace as a step list (groups → intersection → jurisdiction → exemptions → brackets → compounding → rounding); rate grids validate non-overlapping intervals on save; settlement proposals show a reconciliation panel (transactions vs GL) and block the post button on mismatch.

2.13.7 Business rules and edge cases

- **BR-13.1 Order of operations.** (1) Resolve `tax_date` (document date unless the document type specifies delivery date). (2) Resolve party tax group (override > party address jurisdiction default > party role default) and item tax group (override > product > category > legal entity default). (3) Candidate codes = TaxGroup.codes ∩ ItemTaxGroup.codes; if the ship-to jurisdiction requires stacked rates, expand jurisdiction codes. (4) Apply exemption certificates: remove exempted codes, record certificate. (5) Apply explicit line override (duty-gated). (6) Compute base per line: exclusive → amount; inclusive → back out (FR-13.5). (7) Evaluate brackets on line or document basis per code. (8) Calculate non-compound codes, then compound codes in `compound_sequence` order on `base + Σ earlier taxes`. (9) Apply min/max caps. (10) Reverse charge → emit input and output lines of equal amount; use tax → liability line with offset to the line's expense account. (11) Split recoverable/non-recoverable. (12) Round per BR-13.3. (13) Withholding is computed separately on the configured base and never alters the document tax. Every step writes to `determination_trace`.
- **BR-13.2 Provider precedence.** If a provider is configured for (legal entity, document type), steps 3–11 are delegated; the provider's results are normalized into the same structures; the internal engine still validates rounding and balancing of the returned lines.
- **BR-13.3 Rounding.** `per_document` (VAT default): sum unrounded tax per code across lines, round once to currency minor units, then distribute to lines proportionally with largest-remainder allocation so line taxes sum to the document tax. `per_line` (US default): round each line's tax per code. Rounding mode is the ledger's (wave 12).
- **BR-13.4 Inclusive back-out with compound taxes.** Solve `gross = base × Π(1 + r_i)` for compound stacks and `gross = base × (1 + Σ r_i)` for parallel codes; base rounded last; the sum of base + taxes must equal gross to the minor unit (residual to the largest tax line).
- **BR-13.5 Brackets.** Evaluated on the absolute base; credit notes (negative amounts) use the same bracket as the positive amount; `calc_basis = document` sums all lines of the same code before bracket lookup.
- **BR-13.6 Effective dates.** A code with no rate valid on `tax_date` → `TAX_RATE_NOT_FOUND` (422); a rate change mid-period is applied by date, never pro-rated.
- **BR-13.7 Reverse charge eligibility.** Requires a counterparty tax id passing the country format check (wave 7) and both parties in different EU member states (or domestic reverse-charge categories configured); otherwise falls back to standard rate with warning `TAX_REVERSE_CHARGE_INELIGIBLE`.
- **BR-13.8 Withholding thresholds.** Annual accumulation per (vendor, code, fiscal year); withholding applies to the amount exceeding the threshold in the document that crosses it; accumulation is updated only on posting, not calculation.
- **BR-13.9 Settlement state machine.** `proposed → posted | cancelled`; `posted` is terminal; a cancelled run releases its transactions; settlement for a period may not post while the period is `closed` for module TAX (wave 14); a posted run cannot include transactions dated after `period_to`.
- **BR-13.10 Provider commit idempotency.** `commit` with the same key is replayed; a provider `Void` is issued when a document is reversed (`journal.reversed` consumer) and failures queue for retry with alerting after 24 h.
- **BR-13.11 Negative and zero rates.** Rate 0 is valid (zero-rated vs exempt distinguished by `reason_code` on the code); negative rates are rejected (`TAX_RATE_INVALID`).
- **BR-13.12 Multi-currency.** Tax is calculated in transaction currency and converted with the document's rates (wave 12); settlement is in accounting currency; the ESL reports in EUR using the ECB rate on tax_date when the accounting currency is not EUR.
- **BR-13.13 Intrastat line lifecycle.** `transferred → validated | exception`; `exception → validated` after correction; `validated → compressed` (lines merged into one compressed line referencing members) → `exported`; an exported line is immutable; a later correction creates a new line with `correction_of_line_id` dated in the original period and marks the original `corrected`; a period can be exported more than once (each export run is a version; the latest is the declaration of record). Statistical value = invoice value × (1 + `freight_pct`) when the country rule is `invoice_plus_freight_to_border` and no explicit freight is available, with `freight_pct` on the country property (default 0).
- **BR-13.14 Intrastat thresholds.** The period summary compares cumulative calendar-year value per direction with the country threshold; below threshold the export is produced but flagged `declaration_not_required`; crossing the threshold mid-year sets `declaration_required_from` on the legal entity's country property and raises `TAX_INTRASTAT_THRESHOLD_CROSSED` (info).

- **BR-13.15 VAT recovery precedence.** Policy resolution: (expense category, expense country, legal entity) > (category, country, `*`) > (`*`, country, legal entity) > (`*`, country, `*`); the resolved `recoverable_pct` overrides the tax code's default for that line only; foreign reclaim applies only when expense country ≠ legal entity country, the vendor VAT id is valid for the expense country, and the invoice meets the country's minimum (EU: simplified invoice ≤ EUR 1,000 accepted; fuel ≤ EUR 250); amounts below `min_claim_amount` per quarter (EUR 400) or year (EUR 50) are carried to the annual claim.
- **BR-13.16 Withholding remittance.** A remittance run may include only `WithholdingTransaction` rows with `posting_date ≤ period_to`, not yet remitted, for one (authority, section); posting transfers the withholding payable to the authority party's open item (wave 19) or settlement account and sets `remitted = true`; cancelling a posted run is allowed only before the certificate issue and posts a negating entry; certificates are numbered per (legal entity, section, fiscal year) gap-free.

2.13.8 Acceptance criteria

- **AC-13.1** (FR-13.4, FR-13.14) Given the 500-scenario matrix, when run in CI, then 100% of scenarios match expected tax per line and per document to the minor unit.
- **AC-13.2** (FR-13.5, BR-13.4) Given a EUR 119.00 inclusive line at 19% VAT, when calculated, then base 100.00 and tax 19.00; given 3 inclusive lines of 33.33 each at 19%, then the document tax equals round(Σ unrounded) and line taxes sum to it.
- **AC-13.3** (FR-13.6) Given a Canadian QC line of CAD 100.00 with GST 5% and QST 9.975% (non-compound since 2013), when calculated, then taxes are 5.00 and 9.98; given a configured compound stack, then the second tax applies on base + first tax.
- **AC-13.4** (FR-13.6) Given an intra-EU purchase from a DE vendor into FR with a valid VAT id, when calculated, then an input and an output line of equal amount are produced and the net tax is 0.00.
- **AC-13.5** (FR-13.7) Given an expired exemption certificate, when calculated, then full tax applies and `TAX_EXEMPTION_EXPIRED` is in warnings.
- **AC-13.6** (FR-13.10) Given AvaTax times out, when calculating, then the internal engine responds within 2.5 s total, `fallback = true`, and `tax.provider.fallback` is emitted.
- **AC-13.7** (FR-13.11) Given the settlement fixture for March 2026, when the run is posted, then net payable equals output – input, the posting balances, and Σ settled transactions equals the GL tax account movement to the cent.
- **AC-13.8** (FR-13.12) Given the demo EU entity, when SAF-T is generated, then the XML validates against the OECD v2.0 XSD.
- **AC-13.9** (FR-13.8, BR-13.8) Given a vendor with USD 550 cumulative and a USD 600 threshold, when a USD 100 invoice is posted, then withholding applies to USD 50.
- **AC-13.10** (BR-13.9) Given the TAX module is closed for the period, when a settlement post is attempted, then 422 `LEDGER_PERIOD_CLOSED`.
- **AC-13.11** (FR-13.15) Given the Eurostat CN8 file for 2026 (≈ 9,500 codes), when imported, then every code has `validity_year = 2026`, the 2025 codes remain for historic lines, and a product category default resolves `commodity_code_id` on a sample product.
- **AC-13.12** (FR-13.16, BR-13.13) Given 1,200 dispatch lines for DE in March 2026 of which 30 lack net mass, when validated and compressed, then 30 exceptions are listed, the remaining lines compress to the expected 180 declaration lines (fixture), Σ invoice value equals the source documents, and the canonical export carries the partner VAT ids.
- **AC-13.13** (FR-13.16) Given an exported line whose commodity code is later corrected, when the journal is corrected, then a correction line with `correction_of_line_id` exists, the original is `corrected`, and a correction export run for the same period is produced.
- **AC-13.14** (FR-13.17, BR-13.15) Given a DE entity with an entertainment policy of 70% and an expense line of EUR 119.00 incl. 19% VAT, when calculated, then recoverable 13.30, non-recoverable 5.70 posted to the expense account; given a FR hotel invoice for the same DE entity, then the line is marked reclaimable under Directive 2008/9 with annex code 6 and appears in the Q1 reclaim extract for FR.
- **AC-13.15** (FR-13.18, BR-13.16) Given 40 Indian TDS transactions under section 194J for April 2026, when a remittance run is posted with challan reference, then the liability transfer balances, all 40 are `remitted`, and certificates issued for 12 payees are numbered gap-free and carry the challan reference.

2.13.9 Test plan and fixtures

- Fixtures: `tax-matrix-500` (US 200: multi-jurisdiction, exemptions, use tax; EU 180: standard/reduced/reverse charge/triangulation; IN 70: CGST/SGST/IGST, TDS; CA 50: GST/HST/PST/QST; each with expected results reviewed by an accountant); `jurisdictions-us-sample` (5 states' postal tables); `settlement-2026-03` (1,200 transactions with GL expectations); `saf-t-demo`; recorded AvaTax/Vertex HTTP fixtures.

Test	Type	Target	Pass criterion
Matrix runner	fixture (CI)	internal engine	500/500 match
Rounding distribution property test	property	per_document rounding	line taxes sum to document tax; each line within 1 unit of proportional
Inclusive back-out property	property	FR-13.5	base + taxes = gross exactly
Provider contract	contract (recorded)	AvaTax, Vertex adapters	request mapping and response normalization
Failover	integration (fault injection)	provider client	fallback within 2.5 s, event emitted
Settlement reconciliation	integration	settlement job	transactions = GL movement
SAF-T schema	unit	extract generator	XSD valid
Calculation latency	performance	tax API	100-line document p99 < 50 ms internal

Reconciliation: Σ TaxTransaction per code and period = GL tax account movement; settlement net = VAT return net box; ESL totals = Σ reverse-charge/intra-community output transactions.

2.13.10 Deployment and operations

Helm chart `passage-tax` (deployment 3–10 pods, CloudNativePG `tax-db`, topics `tax.calculations`, `tax.transactions`, `tax.settlements`; Kafka consumer of `ledger.journal`). Config: `TAX_PROVIDER_TIMEOUT_MS=2000`, `TAX_PROVIDER_RETRIES=1`, `TAX_PAYLOAD_RETENTION_DAYS=395`, `TAX_DEFAULT_ROUNDING=per_document`. Secrets: provider credentials in Vault paths `tax/providers/{le}`. Migrations: tax schema; partitions per fiscal year. Feature flags: `tax.provider.avalara`, `tax.provider.vertex`, `tax.extracts.saft`. Scheduled jobs: provider recommit retry (hourly), certificate expiry notifications (daily), settlement reminders per period. SLOs: calculate p99 < 50 ms (internal), availability 99.9%, provider fallback rate < 0.5%/day. Alerts: fallback rate > 2% over 1 h (P2), commit retry queue > 100 (P2), settlement mismatch (P2). Runbook: provider outage, re-commit queue drain, rate correction after posting (reverse documents, never edit transactions). Rollback: digest rollback; flags disable providers (internal engine only).

2.13.11 Migration from AX / SAP

Source	Passage entity	Notes
AX `TaxTable`, `TaxData`, `TaxAuthorityAddress`, `TaxPeriodHead`	TaxCode, TaxRate, authority Party, TaxSettlementPeriod	`TaxData` intervals → brackets; ledger posting group `TaxLedgerAccountGroup` → accounts

Source	Passage entity	Notes
AX `TaxGroupHeading/TaxGroupData`, `TaxItemGroupHeading/TaxItemGroupData`	TaxGroup, ItemTaxGroup	exempt flags preserved
AX `TaxExemptCodeTable`, customer `TaxExemptNumber`	ExemptionCertificate	validity defaulted to migration date → open; steward review queue
AX `TaxWithholdTable`, `TaxWithholdGroupHeading`, 1099 setup	WithholdingTaxCode	
AX `TaxTrans` (unsettled only)	TaxTransaction	linked to migrated opening balance lines by code
SAP `T007A/T007S`, `KONP/A003`, `T030K`	TaxCode, TaxRate, accounts	tax procedure steps → compound sequences
SAP `TTXJ/TTXD` jurisdictions	TaxJurisdiction	
SAP `T059P/T059Z`, `LFBW`	WithholdingTaxCode, party assignments	
SAP `BSET` (unsettled)	TaxTransaction	
AX `IntrastatCommodity`, `IntrastatTransactionCode`, `IntrastatTransportMode`, `IntrastatPort`, `IntrastatStatProc`, `IntrastatParameters`, `IntrastatCountryRegionParameters`	CommodityCode, IntrastatTransactionCode, TransportMode, Port, StatisticsProcedure, CountryTradeProperty	commodity codes re-keyed to CN8 validity year; `InventTable.IntrastatCommodity` → product `commodity_code_id` (wave 8)
AX `Intrastat` (journal, current open period only), `IntrastatArchiveGeneral`	IntrastatLine	open period lines migrated as `transferred`; archived declarations kept in the legacy archive
AX `TrvTaxConfiguration` / expense tax setup (recoverable VAT per country)	VatRecoveryPolicy	category mapping to wave 43 expense categories
SAP `T604`, `T605`, `T616`, `T618`, `MARC-STAWN`, `VEIAV` (open period)	CommodityCode, IntrastatTransactionCode, TransportMode, Port, product commodity code, IntrastatLine	`VEIAV` rows for the current period only
SAP `T059P/T059Z` sections, J1INCHLN challan history (open quarter)	WithholdingRemittanceRun (posted, historical reference only)	prior certificates not re-issued

Validation: code counts; a 200-document sample recalculated in Passage matches the source tax to the cent; unsettled totals per code match. Not migrated: settled history (kept in the legacy archive), provider transaction logs.

2.13.12 Dependencies, risks, open questions

Depends on: wave 11 (posting), wave 12 (currency conversion for tax amounts), wave 7 (party tax ids, VIES check), wave 8 (product categories), wave 6 (addresses).

Risk	Likelihood	Impact	Mitigation
Jurisdiction tables for US are vast and change monthly	High	Medium	Providers are the production default for US; internal engine ships sample states only
Country-specific VAT box mappings	High	Medium	Mapping is configuration (wave 48 packs); wave 13 ships DE/FR/UK/IN reference mappings
Rounding disputes with legacy figures	Medium	Low	Rounding mode configurable per code; migration validation sample

Open questions: (1) Settlement counterpart as tax authority Party (vendor role) vs settlement account? Default Party; decide by wave 13 mid-point. (2) Store provider raw payloads for 13 months or 7 years? Default 13 months payload + 7 years hash; decide by wave 13 exit. (3) Should `tax/calculate` be callable from the posting API automatically for manual journals with a tax group? Default yes behind flag `ledger.auto_tax`; decide by wave 14.

2.13.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`tax.provider` per document type	legal entity × document type	enum	internal	internal / avalara / vertex	Engine selection (FR-13.10)	Sales tax parameters > External tax (ISV) / D365 Tax calculation parameters	External tax interface activation (`TTXD`, RFC destination) per tax procedure
`tax.provider_timeout_ms` / `tax.provider_retries`	tenant	int	2,000 / 1	500–10,000 / 0–3	Failover timing (FR-13.10)	—	RFC timeout
`tax.default_rounding`	legal entity	enum	per_document (VAT), per_line (US)	per_line / per_document	BR-13.3	General ledger parameters > Sales tax > Rounding (per line / per document) "Calculation"	OBY6 "Tax rounding" / `T001-TXKRS`; condition rounding rule
`tax.price_includes_tax_default`	legal entity × document type	bool	false (US), true (B2C retail)	true / false	Default for `price_includes_tax` (FR-13.5)	Accounts receivable parameters > Prices include sales tax	Pricing procedure gross/net condition (`KZWI`)
`tax.tax_date_basis`	legal entity × document type	enum	document_date	document_date / delivery_date / posting_date	BR-13.1 step 1	General ledger parameters > Sales tax > "Date of sales tax calculation"	OBCK/`T001-XTXDT` tax reporting date
`tax.conditional_settlement` (cash basis)	legal entity	bool	false	true / false	Tax becomes due on payment (AX conditional sales tax)	General ledger parameters > Sales tax > Conditional sales tax	`T007A` "deferred tax" (target tax code, RFUMSV50)
`tax.tax_on_prepayment`	legal entity	bool	false	true / false	Tax lines on prepayment invoices (waves 19/30)	General ledger parameters > Sales tax > Sales tax on prepayment	Down payment tax (F-29 with tax code)

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`tax.cash_discount_base`	legal entity	enum	net_of_tax	net_of_tax / incl_tax	Cash discount tax adjustment base (waves 19/30)	General ledger parameters > Sales tax > Cash discount on invoice incl. sales tax	OBY6 "Discount base is net value"
`tax_code.settlement_period_id` and `authority_party_id`	tax code	ref	required	settlement periods, authority parties	Settlement grouping (FR-13.11)	Sales tax codes > Settlement period; Sales tax authorities	Tax payable posting F.12 by tax code; tax office vendor in `T001`
`tax_code.rounding`	tax code	enum	inherits entity	per_line / per_document	BR-13.3 per code	Sales tax authorities > Rounding	—
`tax.reporting_box_mapping`	legal entity (country)	jsonb	DE/FR/UK/IN reference mappings	box id → (code, direction, base/tax)	VAT return boxes (FR-13.12)	Sales tax reporting codes	ACR report definitions / `T007K` tax report box
`tax.saft_version`	legal entity	enum	OECD_2_0	OECD_2_0 / country variants (wave 48)	SAF-T schema	Electronic reporting SAF-T format configuration	ACR SAF-T report
`withholding.trigger`	withholding code	enum	payment	invoice / payment	FR-13.8	Withholding tax codes > "Withholding tax on invoice/payment"	`T059P` "accumulation / posting at payment"
`withholding.threshold` / `withholding.annual_accumulation`	withholding code	numeric / bool	0 / false	≥ 0	BR-13.8	Withholding tax codes > Limits	`T059Z` minimum amount, accumulation type
`intrastat.registration` (VAT id, declarant, agent)	legal entity	jsonb	required for EU entities	—	Export header (FR-13.15)	Foreign trade parameters > Intrastat	Foreign trade customizing: company code Intrastat data (VE0)
`intrastat.country_property.threshold_arrival` / `threshold_dispatch`	country	numeric	per country (DE EUR 3,000,000 arrival / 1,000,000 dispatch)	≥ 0	BR-13.14	Country/region properties (EU)	Country-specific threshold in VEB settings
`intrastat.compression_fields`	country	jsonb	EU default set (FR-13.16)	field list	Compression keys	Intrastat compression setup	VE01 aggregation settings
`intrastat.statistical_value_rule` / `freight_pct`	country	enum / numeric	invoice_value / 0	invoice_value / invoice_plus_freight_to_border / custom	BR-13.13	Foreign trade parameters > Statistical value	VEB statistical value calculation
`vat_recovery.min_claim_quarter` / `min_claim_year`	legal entity	numeric	EUR 400 / EUR 50	≥ 0	BR-13.15 carry-forward	—	—
`tax.payload_retention_days`	tenant	int	395	30–2,555	Provider payload retention (FR-13.10)	—	—
`tax.recommit_retry_interval`	tenant	duration	1 h	5 min–24 h	Provider recommit queue	—	—

2.13.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Posted tax transactions	list inquiry	legal entity, code, period, direction, settled, document	document, code, jurisdiction, base, tax, recoverable, non-recoverable, direction, settled run	`TaxTransaction`	Posted sales tax inquiry; SSRS Sales tax transactions — details	S_ALR_87012357 (line items), FBL3N on tax accounts, `BSET` view
Tax specification by voucher	drill-down	voucher / document	lines with codes, base, tax, determination trace link	`TaxTransaction` ⋈ `JournalLine`	SSRS Sales tax specification by ledger transaction	FB03 → Tax data
Sales tax settlement report	statutory report	authority, period, version	output, input, use, net, by code; GL reconciliation panel	`TaxSettlementRun`	SSRS Sales tax payments; Settled sales tax inquiry	S_ALR_87012357 advance return, F.12 (RFUMSV10)
VAT return (box report)	statutory report	legal entity, period	box id, base, tax per box mapping	`TaxTransaction` grouped by `reporting_box_mapping`	Sales tax report (by reporting codes)	RFUMSV00 / S_ALR_87099918, ACR VAT return
EU sales list	statutory report	legal entity, period	customer VAT id, country, value goods, services, triangulation	`TaxTransaction` (reverse charge / intra-community output)	SSRS EU sales list	RFASLM00 / S_ALR_87012400
Intrastat declaration summary	statutory report	legal entity, period, direction	lines, Σ invoice value, Σ statistical value, Σ mass, exceptions, threshold status	`IntrastatLine`	Intrastat report / Intrastat list	VE02 / VE03, MEIS output
Intrastat exceptions worklist	list inquiry	period, direction, rule	line, missing/invalid field, source document	`IntrastatLine` (status exception)	Intrastat validation result	VE01 error log
Withholding tax report	statutory report	legal entity, code / section, period, payee	payee, tax id, base, rate, withheld, remitted, certificate	`WithholdingTransaction`, `WithholdingRemittanceRun`	SSRS Withholding tax; Tax 1099 summary / detail	S_ALR_87012111, RFIDYYWT, S_P00_07000134

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
VAT recovery report	operational report	legal entity, period, country, category	recoverable, non-recoverable, reclaimable, carried forward	`TaxTransaction` ⋈ `VatRecoveryPolicy`	SSRS VAT recovery report (expense)	— (custom report over `BSET` NAV/NVV)
Foreign VAT reclaim extract	statutory report	legal entity, refund country, period	invoice ref, vendor VAT id, date, taxable amount, VAT, annex code	`VatReclaimExtract`	Recover VAT (ER-based claim)	—
SAF-T audit file	statutory report	legal entity, period	GL entries, master files, source documents (OECD 2.0)	ledger + tax + party APIs	Electronic reporting SAF-T	ACR SAF-T (FIUAMSAFT)
Calculation tester / explain	drill-down	sample document	determination trace steps	tax engine	Sales tax calculation (tax information dialog)	Tax procedure analysis (FB60 > tax simulation)
Provider fallback KPI	KPI	legal entity, day	fallback rate, recommit queue size, mean provider latency	`tax.provider.fallback` events	—	—
Exemption certificate expiry	operational report	party, jurisdiction, days to expiry	certificate, valid_to, status, document link	`ExemptionCertificate`	Tax exempt numbers listing	Exemption licenses list (J_1BLFZ-type / `LFBW` exemption dates)
Tax configuration listing	operational report	legal entity	codes, rates, groups, item groups, accounts	tax setup	SSRS Sales tax list / Sales tax codes	`T007A` / FTXP listing

2.13.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
VAT return	settlement period close / user	header: entity, VAT id, period, version; lines: boxes with base and tax; footer: net payable/refundable, preparer, settlement voucher	PDF, canonical JSON; country e-formats (ELSTER XML, UK MTD JSON, FR EDI-TVA) via wave 48	settlement run id + version	statutory; 10 years (DE/FR), 6 (UK), 7 (US)
EU sales list	monthly / quarterly	header: entity VAT id, period; lines: customer VAT id, country, goods, services, triangulation	PDF, CSV, canonical XML; country e-formats in wave 48	extract run id	statutory; 10 years
Intrastat declaration	monthly per direction	header: PSI VAT id, declarant, period, direction; lines: CN8, partner country, partner VAT id, origin, nature, transport, delivery terms, mass, units, values	canonical JSON, CSV; DE XML (Destatis IDEV), FR EMEBI, NL CBS via wave 48	export run id (versioned)	statutory; 10 years
SAF-T file	on request / monthly (PT, NO, PL mandates)	OECD 2.0 header, master files, GL entries, source documents	XML (XSD-validated)	extract run id	statutory; equals ledger retention
Tax settlement voucher	settlement posted	entity, authority, period, codes, output/input/net, voucher	PDF	ledger voucher	financial record; 10 years
Withholding certificate	certificate issue	payer, payee, tax ids, section, period, base, rate, withheld, remittance references, certificate no.	PDF; country forms (Form 16A, 1099-NEC/MISC, 1042-S) rendered by wave 48	per (legal entity, section, fiscal year), gap-free	statutory; 7–10 years per country
Withholding remittance statement	remittance run posted	authority, section, period, transactions, challan/bank reference	PDF, CSV	remittance run id	statutory; 10 years
Foreign VAT reclaim file	reclaim extract	applicant, refund country, period, invoices with annex codes	XML per refund-country portal (wave 48), CSV	extract run id	statutory; 10 years
Exemption certificate image	certificate registered	uploaded document	PDF / image	certificate number (external)	statutory; certificate validity + 4 years (US streamlined)

2.13.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`tax-provider-recommit`	hourly	legal entity × provider	commit with stored `Idempotency-Key`; queue item deleted on success	1,000 items < 2 min	exponential back-off; alert after 24 h	queue > 100 → P2
`tax-journal-link` (consume `journal.posted`, link TaxTransaction)	continuous	legal entity	upsert by (source, line_ref)	lag p95 < 5 s	dead-letter after 5 attempts	DLQ > 0 → P2
`tax-settlement-propose`	per settlement period end + 1 day	authority	run per (authority, period, version)	1M transactions < 3 min	proposal marked failed; close task blocked	mismatch → P2
`tax-certificate-expiry`	daily 06:00	legal entity	daily snapshot	< 1 min	—	certificates expiring ≤ 30 days → notification
`intrastat-transfer`	monthly, 3rd business day 05:00	legal entity × direction	transfer keyed by (source line, period); re-run adds only new and reversed lines	100k lines < 5 min	source resolver errors listed; run continues	exceptions > 0 → close task "Intrastat exceptions"
`intrastat-cn8-import`	yearly, 1 December	tenant	file hash	< 1 min	quarantine file	import missing by 31 December → P3
`withholding-accumulation-rebuild`	daily 02:30	vendor	recomputes annual accumulation from posted transactions; compare	100k vendors < 5 min	—	diff > 0 → P2
`vat-reclaim-quarterly`	quarter end + 10 days	legal entity × refund country	run per (country, quarter)	< 2 min	—	—
`saft-monthly` (where mandated)	5th day of month	legal entity	run per period	1M lines < 10 min	retry once	XSD failure → P2

2.13.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
Calculate tax (service accounts)	POST /tax/calculate, /tax/commit	not assignable to humans	Document service accounts
View tax configuration	/tax/calculate:explain; read screens	—	Tax accountant, Accountant, Internal auditor
Maintain tax codes	PUT codes, rates, VAT recovery policies; screen Tax codes, VAT recovery policies	Settle tax; Produce tax reports (same period sign-off)	Tax manager

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
Maintain tax groups	PUT groups / item-groups	Maintain tax codes (optional)	Tax manager, Tax accountant
Maintain tax jurisdictions	PUT jurisdictions, postal table import	—	Tax manager
Maintain tax exemptions	certificates CRUD	Maintain customer master (wave 7)	Tax accountant, AR credit specialist
Maintain withholding tax	withholding codes	Settle tax	Tax manager
View tax transactions	GET transactions, Intrastat lines (read)	—	Tax accountant, Accountant, Internal auditor
Settle tax	settlements, :post; withholding remittances	Maintain tax codes; Maintain withholding tax	Tax accountant (senior), Financial controller
Produce tax reports	extracts, Intrastat exports, VAT reclaims, certificates	Maintain tax codes	Tax accountant
Maintain foreign trade setup	Intrastat master data CRUD and CN8 import	Maintain Intrastat journal	Tax manager, Trade compliance officer
Maintain Intrastat journal	transfer, PATCH lines (corrections), validate, compress	Maintain foreign trade setup	Trade compliance clerk, Tax accountant
Administer tax	provider settings, credentials (Vault), recommit queue	Settle tax	Finance platform administrator

2.13.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
TAX-0001 `TAX_RATE_NOT_FOUND`	error	Tax code {code} has no rate valid on {tax_date}.	BR-13.6	Add a rate row	FR-13.1
TAX-0002 `TAX_RATE_INVALID`	error	Rate {rate} is not valid; rates must be ≥ 0 with intervals that do not overlap.	BR-13.11	Correct the rate rows	FR-13.1
TAX-0003 `TAX_GROUP_UNRESOLVED`	error	No tax group or item tax group could be resolved for line {line}.	BR-13.1 step 2	Set party/product defaults or override	FR-13.2
TAX-0004 `TAX_NO_INTERSECTION`	warning	Groups {group} and {item_group} share no tax code; line {line} is untaxed.	BR-13.1 step 3	Review group setup	FR-13.4
TAX-0005 `TAX_EXEMPTION_EXPIRED`	warning	Exemption certificate {number} expired on {valid_to}; full tax applied.	FR-13.7	Renew the certificate	FR-13.7
TAX-0006 `TAX_REVERSE_CHARGE_INELIGIBLE`	warning	Reverse charge not applicable: {reason}; standard rate applied.	BR-13.7	Check VAT id and countries	FR-13.6
TAX-0007 `TAX_PROVIDER_FALLBACK`	warning	Provider {provider} unavailable; internal engine used. Document flagged for recommit.	FR-13.10	None; recommit runs automatically	FR-13.10
TAX-0008 `TAX_SETTLEMENT_MISMATCH`	error	Settlement {total} differs from GL tax account movement {gl} by {diff}.	FR-13.11	Investigate unlinked tax lines or manual postings on tax accounts	FR-13.11
TAX-0009 `TAX_SETTLEMENT_PERIOD_CLOSED`	error	Module TAX is closed for {period}.	BR-13.9	Reopen TAX or settle next period	BR-13.9
TAX-0010 `TAX_OVERRIDE_FORBIDDEN`	error (403)	You may not override the tax code on this line.	BR-13.1 step 5	Remove the override or request the duty	FR-13.4
TAX-0011 `TAX_INCLUSIVE_RESIDUAL`	info	Inclusive back-out residual {residual} applied to the largest tax line.	BR-13.4	None	FR-13.5
TAX-0012 `TAX_INTRASTAT_INVALID`	error	Intrastat line {line}: {field} {reason}.	FR-13.16 validation	Correct the line or the source master data	FR-13.16
TAX-0013 `TAX_INTRASTAT_COMMODITY_MISSING`	error	No commodity code for product {product} on line {line}.	Product without CN8	Set the product or category commodity code	FR-13.15
TAX-0014 `TAX_INTRASTAT_LINE_EXPORTED`	error	Line {line} was exported in run {run} and cannot be edited; create a correction.	BR-13.13	Use the correction action	FR-13.16
TAX-0015 `TAX_INTRASTAT_THRESHOLD_CROSSED`	info	Cumulative {direction} value {value} exceeds the {country} threshold {threshold}; declarations required from {period}.	BR-13.14	Submit declarations from that period	FR-13.16
TAX-0016 `TAX_COMMODITY_CODE_EXPIRED`	warning	Commodity code {code} is not valid for {year}.	CN8 yearly change	Map to the successor code	FR-13.15
TAX-0017 `TAX_VAT_RECOVERY_POLICY_MISSING`	warning	No VAT recovery policy for category {category} in {country}; tax code default {pct}% used.	BR-13.15	Add a policy	FR-13.17
TAX-0018 `TAX_RECLAIM_BELOW_MINIMUM`	info	Reclaimable VAT {amount} for {country} is below the minimum; carried to the annual claim.	BR-13.15	None	FR-13.17
TAX-0019 `TAX_WHT_ALREADY_REMITTED`	error	Transaction {id} is already included in remittance {run}.	BR-13.16	Exclude it	FR-13.18
TAX-0020 `TAX_WHT_CERTIFICATE_ISSUED`	error	Remittance {run} has issued certificates and cannot be cancelled.	BR-13.16	Reissue certificates with corrections	FR-13.18
TAX-0021 `TAX_SAFT_SCHEMA_INVALID`	error	SAF-T file failed XSD validation: {detail}.	FR-13.12	Correct source master data (e.g., missing tax registration)	FR-13.12

2.13.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
`POST /tax/calculate` (internal, 100 lines)	300k documents/day; peak 200/s	8 / 25 / 50 ms	1,000 documents/s per 10 pods	none (stateless)	rate/group caches per pod	HPA to 10 pods; p99 ≤ 120 ms
`POST /tax/calculate` (provider)	30% of US documents	provider + 10 ms	per provider quota	payload 13 months ≈ 50 GB	—	timeout 2 s → internal fallback
TaxTransaction linking	1M tax lines/month	lag p95 < 5 s	2,000/s	350 B/row × 12M = 4 GB	partition by fiscal year; indexes §2.13.4	lag grows; settlement waits for lag = 0
Settlement proposal	monthly; 1M transactions	—	< 3 min	settlement rows small	(le, code, period, settlement_run_id)	runs on replica
Intrastat transfer + validate + compress	100k lines/month per entity	—	< 5 min	250 B × 1.2M/year = 300 MB	(le, period, direction, status)	chunked per source type
VAT return / ESL extract	monthly	—	< 1 min	outputs ≈ 1 MB	—	—
SAF-T export	monthly / on request; 1M lines	—	< 10 min	1 GB XML per month (object storage, compressed 10:1)	streamed generation	queued, one per entity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3x load
Calculation explain	5k/day	15 / 60 / 120 ms	100/s	—	—	—

2.13.20 Localization and compliance hooks

- Tax regimes vary by country in rates, mechanisms (VAT vs GST vs sales/use), reverse-charge categories (domestic reverse charge for construction, scrap, mobile phones in DE/UK), cash-basis VAT (FR services, PT), split payment (IT, PL), and tax point rules; the internal engine models them as configuration; wave 48 packs ship tax code sets, box mappings and settlement periods per country.
- Statutory filings: VAT return e-formats (ELSTER DE, MTD UK, EDI-TVA FR, SII ES real-time ledger, Fattura/LIPE IT), ESL formats, Intrastat national formats, SAF-T variants (PT, NO, PL JPK, RO D406), US 1099 IRIS/FIRE files, India GSTR-1/3B and TDS 26Q, Canada GST34 are produced by pack exporters registered on the `tax.extract_format` extension point over the canonical datasets of FR-13.12, FR-13.16, FR-13.18.
- E-invoicing clearance (SDI, CFDI, Peppol) is wave 35/48; the tax engine supplies tax breakdowns in the canonical form these require.
- Currency: VAT returns in national currency; ESL in EUR (BR-13.12); statutory rate type per country (wave 12 §2.12.20).
- Rounding: per-document vs per-line default per country (DE/FR per document, US per line, IN per line per GST rules).
- Retention: VAT records 10 years (DE, FR), 6 (UK), 7 (US), 8 (IN); tax transactions are part of the append-only ledger domain and never purged in Phase 2.
- Privacy: party tax ids and addresses are personal data for sole traders; Intrastat partner VAT id and withholding payee data are tagged `pii` for wave 7 data-subject requests; exports are access-logged.
- Foreign trade beyond Intrastat (export documents, customs declarations, export control screening) is a wave 48 foreign trade pack; wave 13 provides commodity codes, country of origin and trade fields reused by that pack.
- Extension points: `tax.provider` adapter, `tax.extract_format`, `tax.box_mapping_pack`, `tax.intrastat_validation_pack`, `tax.withholding_form` renderer.

2.13.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.1.4 Setup — Foreign trade → Intrastat parameters, commodity codes, transaction codes, transport methods, ports, statistics procedure (AX register row 7)	PARTIAL-13	ADDED as FR-13.15	master data entities and screen Foreign trade setup
AX §12.10.2 Inquiries & Reports — SSRS VAT recovery report (expense tax) (AX register row 78)	PARTIAL-13	ADDED as FR-13.17	VAT recovery report in §2.13.14
AX §12.10.3 Periodic — Expense → Recover VAT (ER-based claim) (AX register row 81)	PARTIAL-13	ADDED as FR-13.17	foreign VAT reclaim extract; portal file formats DEFERRED to wave 48
AX §12.10.4 Setup — Expense → Tax configuration (recoverable VAT per country, tax groups on expenses) (AX register row 86)	PARTIAL-13	ADDED as FR-13.17	`VatRecoveryPolicy`; expense categories come from wave 43
AX §12.14.1 Common — Inventory → Commodity codes (Intrastat on item) (AX register row 124)	PARTIAL-13	ADDED as FR-13.15	product `commodity_code_id` (wave 8 field) referencing `tax.CommodityCode`
AX §12.29.1 Common — Foreign trade → Intrastat (journal) / EU sales list (link) (AX register row 236)	PARTIAL-13	ADDED as FR-13.16	Intrastat journal with corrections and audit
AX §12.29.3 Periodic — Foreign trade → Intrastat transfer / validation / export (file) (AX register row 247)	PARTIAL-13	ADDED as FR-13.16	transfer, validate, compress, export; national file formats via wave 48
AX §12.29.4 Setup — Foreign trade parameters, Commodity codes, Transaction codes, Transport methods, Ports, Statistics procedures, Country/region properties, Intrastat compression, check setup, EU sales list reporting (AX register row 255)	PARTIAL-13	ADDED as FR-13.15 (setup), FR-13.16 (compression, checks)	ESL reporting was already FR-13.12
SAP §13.2 FI-AP — J1INCHLN / J1INBANK / J1INCERT (IN WHT) (SAP register row 25)	PARTIAL-13	ADDED as FR-13.18 (remittance, challan reference, certificate numbering); Form 16A / 26Q rendering DEFERRED to wave 48	IN country pack renders forms over the canonical dataset
SAP §13.17 LE-SHP — Foreign trade / export documents in delivery (customs, commercial invoice, EUR.1, Intrastat from delivery; GTS) (SAP register row 205)	PARTIAL-13	ADDED as FR-13.16 for Intrastat from deliveries (via `IntrastatSource`); export documents and customs DEFERRED to wave 48	register proposes a wave 48 foreign trade/customs pack; wave 28 shipments register as an Intrastat source
SAP §13.17 LE-SHP Customizing — Foreign trade: export control, preference, Intrastat settings (VEB, VE0; `T604`, `T605`) (SAP register row 208)	PARTIAL-13	ADDED as FR-13.15 for Intrastat settings; export control and preference DEFERRED to wave 48	denied-party screening and preference calculation are trade compliance, not tax
SAP §13.18 TM — Customs declarations via GTS from TM (SAP register row 212)	PARTIAL-13	DEFERRED to wave 48	customs declarations belong to the foreign trade/customs pack; wave 13 supplies commodity codes and origin data

2.14 Wave 14: Period Close, Year-End & Allocations

2.14.1 Scope and business context

AX controls posting through `LedgerPeriod` statuses (open, on hold, closed) per module (`LedgerPeriodModuleAccessControl`) and offers the financial period close workspace, allocation rules (`LedgerAllocation`), accrual schemes and year-end close; SAP uses OB52 posting period intervals per account type with authorization groups, the Financial Closing Cockpit, assessment/distribution cycles (KSV5/KSU5), the accrual engine and balance carry-forward (FAGLGVTR). Wave 14 adds to Passage: a period-control matrix enforced inside the posting API; a close checklist run on Temporal with templates, dependencies and a cross-entity board; an allocation engine (fixed, basis, iterative) with reversal; accrual schemes and recurring journal scheduling (completing FR-11.10); and year-end close with carry-forward per dimension.

After this wave a controller can close March 2026 for three entities from one board, with every subledger path rejected after close, allocations and accruals posted and reversed on schedule, and year-end producing opening balances. Out of scope: subledger-specific close tasks (inventory close wave 20, AP/AR aging waves 19/30), consolidation (wave 40), budget vs actual (wave 38).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
SAP OB52 (`T001B`): closing a period is an edit of interval rows per account type and company code, done by Basis/FI power users across many company codes; a typo opens the wrong range and postings land in a closed month	Period control is a maintenance view, not a controlled transition with validation	Period-control grid with state machine (BR-14.3), GL-closes-last rule, bulk close across ledgers with confirmation	Close of 3 entities × 10 modules: 30 OB52 edits → 1 bulk action; wrong-period postings → 0
Reopening a closed period in AX (Ledger calendar form) or SAP is a single-user action with no approval; auditors find silent reopenings in the database log weeks later	No workflow on period status	Reopen requires two approvers, a reason, an expiry timer and emits `close.period.status-changed`	Unapproved reopenings → 0; audit evidence available in < 1 s rather than at audit time
Close tracking lives in Excel or the AX financial period close workspace that does not block dependent tasks; SAP Closing Cockpit setup requires consultants	Task status is informational; automation hooks are limited to batch jobs	Temporal-driven CloseRun with DAG dependencies, automation hooks that complete on completion events, cross-entity board	Close coordination emails/period 50+ → 0; task status latency hours → seconds
AX allocation rules (`LedgerAllocation`) and SAP KSV5/KSU5 cycles post rounding residues to odd destinations and have no reciprocal (iterative) option without CO-OM workarounds; reruns double-post	Residual handling undefined; cycles not idempotent	Largest-share residual rule, iterative solver with convergence check, runs reversible as a unit and idempotent by key	Allocation rerun duplicates → 0; reciprocal allocations supported natively (vs manual spreadsheets)
FAGLGVTR / AX year-end close must be rerun after late postings; operators delete and regenerate opening entries, breaking the audit trail	Carry-forward implemented as mutable opening balances	Year-end run reverses its predecessor and regenerates (append-only), verifies opening = closing per combination before commit	Opening/closing mismatch after reruns → 0; audit trail unbroken

2.14.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Period status	`LedgerPeriod` (Open/On hold/Closed), Ledger calendar form	`T001B` intervals via OB52 (periods 1–12, 13–16 special)	`ledger.PeriodStatus` per ledger × period × module	full
Module access control	`LedgerPeriodModuleAccessControl`, user groups per module	OB52 authorization group (`BRGRU`), account type (+, A, D, K, M, S)	PeriodStatus.module × `allowed_group_id`	full
Special periods	Closing period type on fiscal calendar	Periods 13–16	Fiscal period `kind = adjustment` (wave 6) with own PeriodStatus	full
Close workspace	Financial period close workspace (templates, tasks, dependencies, calendar)	Financial Closing Cockpit / Advanced Financial Closing	`close` service: CloseTemplate, CloseRun, CloseTask (Temporal)	full
Allocations	`LedgerAllocation`, allocation rules (basis/fixed/equally/spread), process allocation request	KSV5 (distribution), KSU5 (assessment), cycles/segments, FAGLGA35	`ledger.AllocationRule`, `AllocationRun` with fixed, basis, equal, iterative methods	full
Accruals	Accrual schemes (`LedgerAccrualScheme`), ledger accruals	Accrual engine (ACE), FBS1 with reversal	`ledger.AccrualScheme`; reversal via wave 11 reversal	full
Recurring entries	Periodic journals	FBD1/F.14	`ledger.RecurringJournal` (wave 11) with calendar-relative schedules	full
Year-end close	General ledger > Year-end close (`LedgerFiscalYearClose...`), opening transactions, retained earnings per dimension	FAGLGVTR balance carry-forward, retained earnings account (OB53)	`ledger.YearEndCloseRun` producing period-0 opening entries	full
Reopen period	Change status back to Open	OB52 edit	Dual-approval reopen workflow	improved: four-eyes, audited
Close calendar / status board	Workspace tiles per entity, task list by due date	Cockpit task list, status overview	Close board across legal entities	full
Balance-sheet regrouping / reclassification	Not standard (manual reclassification journals; vendor debit balances via "Customer/vendor balance list" and manual journal)	FAGLF101 / F101 sort and reclassify by remaining term, FAGL_CL_REGROUP (S/4 closing regrouping), FAGL_CL_RGRP sort method and adjustment accounts (`T030`, `T043`), regroup vendors with debit balances and customers with credit balances	`ledger.RegroupingRule`, `ledger.RegroupingRun` with auto-reversal (FR-14.13); AP/AR item sources from waves 19/30	full

2.14.3 Functional requirements

- **FR-14.1 Period-control matrix** — Must. The system shall maintain a status per (ledger, fiscal period, module) where module ∈ {GL, AP, AR, INV, FA, PROJ, TAX, BANK, PAYROLL, ALL} and status ∈ {never_opened, open, on_hold, closed}.
  - `ALL` is the default when no module row exists; a module row overrides `ALL`. `on_hold` carries `allowed_group_id` (user group that may still post).
  - The posting API resolves module from `source.system` (mapping table, default GL for manual journals) and rejects per BR-14.1. The check is inside the posting transaction (status row read with `FOR SHARE`), so a close committing concurrently wins or loses atomically.
- **FR-14.2 Period status transitions** — Must. The system shall allow `never_opened → open`, `open ↔ on_hold`, `open|on_hold → closed`, and `closed → open` only through the reopen workflow (FR-14.3).
  - Closing a period for `ALL` requires every module row to be closed or absent; closing GL requires all other modules closed (GL closes last).
  - Period 13–16 adjustment periods may be open while 12 is closed.
- **FR-14.3 Reopen with dual approval** — Must. The system shall reopen a closed period only via a Temporal workflow requiring two distinct approvers holding duty `Approve period reopen`, a reason, and an expiry (reopened period auto-closes after N days, default 3).
  - Each reopen increments `reopen_count` and emits `close.period.status-changed` with both approvers.
- **FR-14.4 Close templates and runs** — Must. The system shall define close templates (tasks with module, description, assignee role, due offset in business days from period end, dependencies, evidence requirement, automation hook) and instantiate a `CloseRun` per (template, period, set of legal entities).
  - Maximum 500 tasks per template; dependencies form a DAG (cycle rejected on save).
  - Automation hooks call Passage jobs (revaluation, allocation, accrual, settlement, year-end, statement pack) and mark the task complete when the job emits its completion event.
- **FR-14.5 Close board** — Must. The system shall show tasks across entities by status (not started, in progress, blocked, completed, skipped) with due dates, owners, overdue counts, and allow bulk reassignment and evidence attachment.
- **FR-14.6 Allocation rules** — Must. The system shall define allocation rules with source (account/dimension filter, period, ledger), destination (account/dimension targets), method ∈ {fixed_percentage, fixed_weight, equally, basis}, basis source (balances of statistical or financial accounts by dimension, e.g., headcount), offset account, and `allocate_from_zero` behavior.
  - Percentages must sum to 100.000000 (tolerance 0); weights normalized; basis computed from the Balance projection for the period.
  - Rounding per BR-14.4; destination lines inherit dimensions from the basis.
- **FR-14.7 Iterative (reciprocal) allocations** — Should. The system shall solve mutually dependent rules (service departments allocating to each other) iteratively until the residual per source < 1 minor unit or 100 iterations, then post one entry per rule.

- **FR-14.8 Allocation runs and reversal** — Must. Runs support simulate and post modes, record the basis snapshot, and are reversible as a unit (reversing every produced entry); a rule may be flagged `auto_reverse_next_period`.
- **FR-14.9 Accrual schemes** — Must. The system shall define schemes (debit/credit accounts, frequency, number of occurrences, spread method ∈ {equal, by_days, custom}) that split a source line into future-dated entries, each auto-reversed per scheme (e.g., monthly accrual reversed on day 1).
  - Generated entries are scheduled on Temporal and posted on their dates if the period is open; otherwise the task appears as blocked on the close board.
- **FR-14.10 Year-end close** — Must. The system shall close a fiscal year per ledger: (a) for each P&L account × dimension combination, post the net to retained earnings (per dimension when `retain_dimensions = true`, else to the ledger's retained earnings combination) in the closing period, and (b) create opening entries in period 0 of the next year for each balance-sheet account × combination equal to its closing balance.
  - Re-runnable: a subsequent run reverses the prior run's entries and regenerates (append-only).
  - Requires all periods of the year closed for all modules except `GL` in `on_hold` for the closing user group; year-end entries are flagged `is_closing_entry` / `is_opening_entry` and excluded from P&L statements.
- **FR-14.11 Recurring journal scheduling** — Must. Completes FR-11.10: schedules by fiscal period offset (e.g., "last business day of period"), with pause/resume, next-run preview and a generated-batch history.
- **FR-14.12 Close reporting** — Should. The system shall report close duration per period and entity (period end to GL closed), tasks completed late, and reopen history.
- **FR-14.13 Balance-sheet regrouping (reclassification)** — Must. The system shall run, per ledger and key date, regrouping rules that reclassify balance-sheet positions for presentation and post adjustment entries to adjustment accounts, auto-reversed on the first day of the next period:
  - Sort method `remaining_term` (FAGLF101 semantics): open items of a source (GL open items from FR-11.19 in this wave; vendor and customer open items and receivable/payable reconciliation accounts when waves 19/30 register their `open_item_source`) are bucketed by remaining term from the key date to `due_date` (default buckets ≤ 1 year, 1–5 years, > 5 years) and each bucket's amount is reclassified from the source account to the bucket's adjustment account with the configured target (e.g., long-term portion of a loan or receivable).
  - Sort method `debit_credit_balance` per party: vendors whose net balance is a debit are reclassified from payables to "receivables – vendors with debit balance"; customers with a credit net balance from receivables to "payables – customers with credit balance"; the balance is computed per party × reconciliation account × splitting dimensions (FR-11.23) and currency.
  - Sort method `account_balance_sign` for GL accounts (bank accounts with overdraft reclassified to short-term borrowings; VAT receivable/payable offset positions): the whole account × combination balance is reclassified when its sign is opposite the account's `normal_balance`.
  - Each run supports simulate and post, records the snapshot it used, is idempotent per (ledger, key date, rule set), and is reversible as a unit; posted entries carry `is_regrouping = true` and keep the source line's dimensions; the close checklist (FR-14.4) offers a "Regrouping" automation hook.

2.14.4 Data model

PeriodStatus

Field	Type	Required	Notes
ledger_id	uuid	yes	PK part
fiscal_period_id	uuid	yes	PK part
module	enum	yes	PK part; ALL, GL, AP, AR, INV, FA, PROJ, TAX, BANK, PAYROLL
status	enum	yes	never_opened, open, on_hold, closed
allowed_group_id	uuid	no	required when on_hold
reopen_count	int	yes	default 0
reopen_expires_at	timestamptz	no	
changed_at / changed_by / approvals	timestamptz / uuid / jsonb	yes	approvals = [{user_id, at}] for reopen

CloseTask (within CloseRun)

Field	Type	Required	Notes
id	uuid	yes	PK
run_id	uuid	yes	FK CloseRun (template_id, fiscal_period_id, legal_entity_ids[], status)
legal_entity_id	uuid	yes	
template_task_id	uuid	yes	
module	enum	yes	
name / description	text	yes	
assignee_user_id / assignee_role	uuid / text	conditional	
due_at	timestamptz	yes	period_end + offset business days
depends_on	uuid[]	no	task ids
status	enum	yes	not_started, in_progress, blocked, completed, skipped
evidence_uri / evidence_required	text / bool	conditional	
automation	jsonb	no	{job, params, completion_event}
completed_at / completed_by	timestamptz / uuid	no	
temporal_workflow_id	text	yes	

AllocationRule and **AllocationRun**

Field	Type	Required	Notes
id	uuid	yes	PK
ledger_id	uuid	yes	
name	text(60)	yes	unique per ledger
source_filter	jsonb	yes	account ranges + dimension filters + `include_opening`
method	enum	yes	fixed_percentage, fixed_weight, equally, basis
destinations	jsonb	yes	[{account, dimensions, pct
offset_account_id / offset_dimension_mode	uuid / enum	yes	mode: source_dimensions, fixed

Field	Type	Required	Notes
iterative_group	text	no	rules in the same group are solved together
auto_reverse_next_period	bool	yes	
run: id, rule_ids[], fiscal_period_id, mode (simulate/post), status (proposed, posted, reversed, cancelled), iterations, basis_snapshot (jsonb), entry_ids[], reversal_entry_ids[]			

AccrualScheme: `id`, `ledger_id`, `code`, `debit_account_id`, `credit_account_id`, `frequency` (monthly, quarterly, custom), `occurrences`, `spread` (equal, by_days, custom[]), `start_rule` (document_date, next_period_start), `auto_reverse` bool, `reverse_day` int. **YearEndCloseRun**: `id`, `ledger_id`, `fiscal_year_id`, `retained_earnings_combination_id`, `retain_dimensions`, `status`, `closing_entry_ids[]`, `opening_entry_ids[]`, `previous_run_id`.

RegroupingRule and **RegroupingRun** (FR-14.13)

Field	Type	Required	Notes
rule: id / ledger_id / code	uuid / uuid / text(20)	yes	unique (ledger_id, code)
rule: sort_method	enum	yes	remaining_term, debit_credit_balance, account_balance_sign
rule: source	jsonb	yes	{source_type: gl_open_item / vendor / customer / account_balance, account ranges, dimension filter}
rule: buckets	jsonb	conditional	remaining_term: [{from_days, to_days, adjustment_account_id, target_account_id}]
rule: adjustment_account_id / target_account_id	uuid	conditional	debit/credit balance and sign methods
rule: group_by	text[]	yes	party, reconciliation account, splitting dimensions, currency
rule: active / sequence	bool / int	yes	
run: id / ledger_id / key_date / rule_ids	uuid / uuid / date / uuid[]	yes	
run: mode / status	enum / enum	yes	simulate, post; proposed, posted, reversed, cancelled
run: snapshot	jsonb	yes	per rule: groups with balances, buckets, items count (compressed)
run: entry_ids / reversal_entry_ids	uuid[]	no	reversal dated key_date + 1 day
run: totals	jsonb	yes	per adjustment account

Invariants: at most one PeriodStatus per (ledger, period, module); `closed` GL implies every other module closed; an AllocationRun that is `posted` has Σ destination amounts = source amount per rule (after rounding); YearEndCloseRun opening balances per combination = closing balances (verified by the run before commit). Indexing: `period_status` PK; `close_task` on (run_id, status), (assignee_user_id, due_at); `allocation_run` on (ledger_id, fiscal_period_id).

2.14.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	/api/v1/ledger/ledgers/{id}/periods?year=	Period-control matrix	n/a	View period status
PATCH	/api/v1/ledger/ledgers/{id}/periods/{period}/modules/{module}	Transition status (open/on_hold/closed)	naturally idempotent on target status	Maintain period status
POST	/api/v1/ledger/ledgers/{id}/periods/{period}:request-reopen	Start reopen workflow	`Idempotency-Key`	Request period reopen
POST	/api/v1/close/templates and GET/PUT /{id}	Templates	PUT idempotent	Maintain close templates
POST	/api/v1/close/runs	Instantiate run for period × entities	`Idempotency-Key`	Run period close
GET	/api/v1/close/runs/{id}/tasks?entity=&status=	Board data	n/a	View close board
POST	/api/v1/close/tasks/{id}:start, :complete, :skip, :reassign	Task actions	`Idempotency-Key`	Work close tasks
GET/PUT	/api/v1/ledger/ledgers/{id}/allocation-rules[/{rule}]	Rules	PUT idempotent	Maintain allocation rules
POST	/api/v1/ledger/allocations	Run (simulate/post)	`Idempotency-Key`	Run allocations
POST	/api/v1/ledger/allocations/{id}:post, :reverse	Post proposal / reverse run	`Idempotency-Key`	Run allocations
GET/PUT	/api/v1/ledger/ledgers/{id}/accrual-schemes[/{code}]	Schemes	PUT idempotent	Maintain accrual schemes
POST	/api/v1/ledger/year-end-closes	Run year-end close (simulate/post)	`Idempotency-Key`	Run year-end close
GET/PUT	/api/v1/ledger/recurring-journals[/{id}] and :pause, :resume	Schedules	PUT idempotent	Maintain ledger journals
GET/PUT	/api/v1/ledger/ledgers/{id}/regrouping-rules[/{code}]	Regrouping rules (FR-14.13)	PUT idempotent	Maintain regrouping rules
POST	/api/v1/ledger/regroupings and /{id}:post, :reverse	Regrouping run simulate / post / reverse	`Idempotency-Key`	Run regrouping

Commands: `TransitionPeriodStatus` (validates FR-14.2 ordering; closing with unposted approved batches dated in the period → `CLOSE_PENDING_BATCHES` unless `force` with duty); `RequestReopen` (reason ≥ 10 chars, two approvers distinct from requester); `RunAllocation` (period open for GL, basis non-zero or `allocate_from_zero`); `RunYearEndClose` (all periods closed per FR-14.10, retained earnings account exists and is equity).

Event	Trigger	Payload fields	Consumers
dev.passage.close.period.status-changed.v1	PeriodStatus transition	ledger_id, fiscal_period_id, module, from, to, actor, approvals[], reason	all posting services (cache), close, reporting, audit
dev.passage.close.run.created.v1	run instantiated	run_id, template_id, period, legal_entity_ids, task_count	notifications
dev.passage.close.task.completed.v1	task completed/skipped	run_id, task_id, status, actor, evidence_uri, late (bool)	board, notifications, reporting
dev.passage.ledger.allocation.completed.v1	run posted/reversed	run_id, rule_ids, period, iterations, entry_ids, reversal	reporting, close
dev.passage.ledger.accrual.scheduled.v1	scheme applied	source_entry_id, scheme, occurrences[] (dates, amounts)	close board
dev.passage.ledger.year-end.closed.v1	year-end posted	ledger_id, fiscal_year_id, closing_entry_ids, opening_entry_ids, retained_earnings_total	reporting, close
dev.passage.ledger.regrouping.completed.v1	regrouping posted or reversed	run_id, ledger_id, key_date, rule_ids, totals per adjustment account, entry_ids, reversal	reporting, close

2.14.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Ledger calendar / period control	Grid ledger × period × module; set status; bulk close; request reopen	period, module, status, allowed group, changed by, reopen count	Maintain period status
Close templates	Build task list, dependencies (DAG view), due offsets, automation	task, module, role, due offset, depends on, evidence, hook	Maintain close templates

Screen	Primary actions	Key fields/columns	Duty required
Close board	Instantiate run, kanban/list across entities, filter overdue, reassign, attach evidence, complete	entity, task, owner, due, status, blocked by, evidence	View / Work close tasks
Allocation rules	Define source, method, destinations, basis; simulate	name, method, source filter, destinations, offset, iterative group	Maintain allocation rules
Allocation runs	Simulate, review proposal (source → destinations), post, reverse	rule, source amount, destination, basis value, pct, amount	Run allocations
Accrual schemes	Define, preview schedule on a sample amount	code, accounts, frequency, occurrences, spread, reverse day	Maintain accrual schemes
Year-end close	Select year/ledger, simulate, review retained earnings by dimension, post	account, combination, closing balance, opening balance, retained earnings	Run year-end close
Regrouping	Define rules (method, source, buckets, accounts), simulate for a key date, review groups (party / bucket / amount), post, reverse	rule, source account, party, bucket, balance, adjustment account, target account, amount	Maintain regrouping rules / Run regrouping

Interaction rules: period grid supports keyboard range selection and bulk status change with a confirmation summarizing affected modules; close board tasks open in a side panel without losing filters; blocked tasks show the blocking task inline; allocation proposals support "what-if" edits to percentages that recompute live but require saving the rule to post.

2.14.7 Business rules and edge cases

- **BR-14.1 Period-control decision.** For a line with posting_date → period P, module M (from source mapping), user U: status = row(P, M) if present else row(P, ALL) if present else `never_opened`. `open` → allow; `on_hold` → allow iff U ∈ allowed_group else 403 `LEDGER_PERIOD_ON_HOLD`; `closed` or `never_opened` → 422 `LEDGER_PERIOD_CLOSED`. Service accounts for subledgers are members of no on-hold group by default.
- **BR-14.2 Period-control matrix (reference configuration for March close).**

Module	Day 1–3 after period end	Day 4–5	After GL close
INV, PROJ, PAYROLL	on_hold (ops group)	closed	closed
AP, AR, BANK	open	on_hold (AP/AR supervisors)	closed
TAX	open	open	closed after settlement
FA	open	closed	closed
GL	open	on_hold (controllers)	closed
ALL (default for unmapped)	open	on_hold (controllers)	closed

- **BR-14.3 Period status state machine.** `never_opened → open`; `open → on_hold`; `on_hold → open`; `open|on_hold → closed`; `closed → open` only via reopen workflow (two approvals) which sets `reopen_expires_at`; on expiry a Temporal timer sets `closed` again and emits the event. Transitions are rejected with `CLOSE_INVALID_TRANSITION`.
- **BR-14.4 Allocation rounding.** Destination amounts = round(source × share, scale); residual = source − Σ destinations; residual applied to the destination with the largest share (ties: first by definition order). Basis values of zero across all destinations → skip rule with `ALLOC_ZERO_BASIS` unless `allocate_from_zero` = equally.
- **BR-14.5 Iterative convergence.** Reciprocal groups solved by Gauss–Seidel iteration on the unrounded amounts; stop when max |Δ| < 10^−scale / 2 or at 100 iterations (then fail `ALLOC_NO_CONVERGENCE`); rounding applied once after convergence per rule.
- **BR-14.6 Allocation period lock.** A run posts only if GL is open or on_hold with the runner in the allowed group; the basis period may differ from the posting period (e.g., prior-period headcount) and is snapshotted on the run.
- **BR-14.7 Accrual spread by days.** Amount_i = round(total × days_i / total_days); residual to the last occurrence; occurrences dated the first day of each period (or scheme `post_day`) and reversed the first day of the following period.
- **BR-14.8 Year-end retained earnings per dimension.** When `retain_dimensions`, the retained earnings line keeps the P&L line's dimension combination (replacing only the main account); dimension values inactive on the opening date are remapped to the ledger's `default_closing_combination` and listed in the run report.
- **BR-14.9 Year-end re-run.** A new run for the same (ledger, year) first reverses all entries of `previous_run_id` (dated the same dates), then regenerates; opening period 0 is not subject to period control of the new year's period 1.
- **BR-14.10 Close task dependencies.** A task becomes `in_progress`-eligible only when all `depends_on` tasks are `completed` or `skipped`; skipping requires a reason and duty `Skip close tasks`; automation tasks cannot be completed manually unless marked `manual_override_allowed`.
- **BR-14.11 Concurrency on close.** Closing GL takes `FOR UPDATE` on the PeriodStatus row and waits for in-flight postings (which hold `FOR SHARE`) up to 5 s before failing with `CLOSE_BUSY`; retried by the UI.
- **BR-14.12 Calendar mismatch across ledgers.** Period-control is per ledger; a ledger group posting where any member ledger's period is closed is rejected as a whole with the offending ledger named.
- **BR-14.13 Regrouping computation.** Remaining term = `due_date − key_date` in days; items without `due_date` fall into the shortest bucket and are listed as `no_due_date`; overdue items (negative term) fall into the shortest bucket. For `debit_credit_balance`, the group balance is Σ open items per (party, reconciliation account, splitting dimension values, currency) at the key date; only groups whose sign is opposite the account's normal balance are reclassified, for their full balance. Entries: debit/credit the adjustment account and the target account in accounting and reporting currency using the posted amounts (revalued amounts when a wave 12 revaluation for the key date is posted, so reclassification follows valuation); one entry per rule per run; auto-reversal on key_date + 1 day. A second posted run for the same (ledger, key date, rule) is rejected (`CLOSE_REGROUPING_EXISTS`) until the first is reversed. Rules run in `sequence`; an item is used by at most one rule per run.

2.14.8 Acceptance criteria

- **AC-14.1** (FR-14.1, BR-14.1) Given AP is closed for March and GL open, when a vendor-invoice posting (source.system = ap) dated 2026-03-31 is attempted, then 422 `LEDGER_PERIOD_CLOSED`; a manual GL journal on the same date posts.
- **AC-14.2** (FR-14.1) Given every subledger posting path in the contract suite (manual, ap, ar, inv, fa, proj, tax, bank stubs), when the period is closed for ALL, then every path is rejected (table-driven test, 0 exceptions).
- **AC-14.3** (FR-14.2) Given AP is still open, when GL close is requested, then `CLOSE_INVALID_TRANSITION` naming AP.
- **AC-14.4** (FR-14.3) Given a closed period, when a reopen is requested and approved by two distinct users, then status is `open`, `reopen_count = 1`, and after the expiry timer the period is `closed` again (Temporal time-skip).
- **AC-14.5** (FR-14.6, BR-14.4) Given the allocation fixture (rent USD 10,000.01 by headcount 3/5/2), when run, then destinations are 3,000.00, 5,000.01, 2,000.00 and Σ = source.
- **AC-14.6** (FR-14.7) Given two service departments allocating 10% to each other, when run iteratively, then the result converges within 100 iterations and matches the reciprocal workbook to the cent.

- **AC-14.7** (FR-14.9) Given a USD 1,200.00 annual insurance invoice with a 12-occurrence equal scheme, when applied, then 12 entries of 100.00 are scheduled on period starts with reversals on the next period start.
- **AC-14.8** (FR-14.10) Given a 12-period dataset, when the year-end close is posted, then every balance-sheet combination's opening balance equals its closing balance, P&L accounts open at 0.00, and retained earnings change equals net income.
- **AC-14.9** (FR-14.4, FR-14.5) Given a 40-task template across 3 entities with dependencies, when the run completes, then no task was completed before its dependencies and the board shows 120 tasks completed.
- **AC-14.10** (BR-14.11) Given 50 concurrent postings in flight, when GL close is requested, then it either waits ≤ 5 s and succeeds with no posting committed after the close timestamp, or returns `CLOSE_BUSY`.
- **AC-14.11** (FR-14.13, BR-14.13) Given a loan clearing account with open items due 2026-09-30 (USD 40,000.00), 2028-03-31 (USD 120,000.00) and 2032-03-31 (USD 200,000.00) and key date 2026-03-31, when the `remaining_term` rule runs, then USD 40,000.00 stays short-term, USD 320,000.00 is reclassified to long-term in one entry, and the reversal is dated 2026-04-01.
- **AC-14.12** (FR-14.13) Given stub vendor open items where vendor V7 nets to a debit of USD 1,250.00 and customer C3 nets to a credit of USD 800.00, when the `debit_credit_balance` rules run, then USD 1,250.00 moves from payables to "receivables – vendors with debit balance" and USD 800.00 from receivables to "payables – customers with credit balance"; a second post for the same key date returns `CLOSE_REGROUPING_EXISTS`.

2.14.9 Test plan and fixtures

- Fixtures: `period-matrix-demo` (3 ledgers × 16 periods × 10 modules); `close-template-40` (40 tasks, 3 entities, 12 dependencies, 6 automation hooks); `alloc-fixtures-25` (fixed, weight, equal, basis, 3 reciprocal groups; XLSX expected); `accrual-fixtures-10`; `fy2025-12-periods` (12 posted periods, 400 combinations, expected opening balances).

Test	Type	Target	Pass criterion
Period decision table	unit	BR-14.1	60 cases (status × module × group)
Subledger path contract	contract	all posting clients' stubs	closed period rejected on every path
Close race	integration	ledger + close	no post commits after close timestamp
Reopen workflow	Temporal test	close service	two approvals, expiry auto-close
Allocation workbook	integration	allocation engine	totals equal, residual rule honored
Reciprocal convergence property	property	iterative solver	converges for random 2–6 department matrices with shares < 50%
Year-end equivalence	integration	year-end job	opening = closing per combination
Close board E2E	Playwright + axe	React	40-task run completes; WCAG AA

Reconciliation: Σ allocation destinations = Σ sources per run; Σ accrual occurrences = source amount; opening balances per combination = closing; retained earnings Δ = net income per ledger.

2.14.10 Deployment and operations

Helm charts: `passage-close` (TypeScript service, 2–4 pods, CloudNativePG `close-db`, Temporal namespace `close`), `passage-ledger` upgraded (period status, allocation, accrual, year-end). Config: `CLOSE_REOPEN_DEFAULT_DAYS=3`, `CLOSE_LOCK_WAIT_MS=5000`, `ALLOC_MAX_ITERATIONS=100`, `ALLOC_CONVERGENCE_FRACTION=0.5`. Migrations: `period_status` seeded from fiscal calendars (`never_opened`), existing demo periods set `open`. Feature flags: `close.reopen_workflow` (always on in prod), `ledger.iterative_allocations`. Scheduled jobs: accrual and recurring journal generators (Temporal schedules), reopen expiry timers, close reminder notifications (daily 08:00 entity time zone). SLOs: period decision adds < 2 ms to posting p99; close board p95 < 500 ms for 1,000 tasks. Alerts: reopen without expiry (P2), posting rejected rate for closed periods > 50/h (P3, indicates a subledger misconfiguration), allocation non-convergence (P3). Runbook: emergency reopen, stuck close lock, re-running year-end. Rollback: digest rollback; period status rows are retained; disabling the flag never bypasses period control.

2.14.11 Migration from AX / SAP

Source	Passage entity	Notes
AX `LedgerPeriod`, `LedgerPeriodModuleAccessControl`, `LedgerFiscalCalendarPeriod`	PeriodStatus	statuses copied; modules mapped (Ledger→GL, VendInvoice→AP, CustInvoice→AR, Inventory→INV, Asset→FA, Project→PROJ, Sales tax→TAX, Bank→BANK)
AX `LedgerAllocation`, `LedgerAllocationRule*` (basis/destination/source)	AllocationRule	basis "statistical" → basis spec on statistical accounts
AX `LedgerAccrualScheme`	AccrualScheme	
AX financial period close templates/tasks (`FinancialPeriodCloseTemplate*`)	CloseTemplate, template tasks	dependencies and due offsets preserved
SAP `T001B` (OB52)	PeriodStatus	account type + → ALL, D → AR, K → AP, A → FA, M → INV, S → GL; authorization group → allowed_group
SAP KSV5/KSU5 cycles (`T811C`, `T811S`, `T811K`)	AllocationRule	segments → rules; tracing factor → basis spec
SAP `T030` retained earnings (OB53)	Ledger.retained_earnings_combination_id	per ledger
SAP closing cockpit templates	CloseTemplate	
SAP FAGLF101 sort methods and adjustment accounts (`T030` transactions PYO, PYU, PYT; `T043`), FAGL_CL_RGRP settings	RegroupingRule	sort method buckets → buckets; adjustment/target accounts per reconciliation account; prior regrouping runs not migrated (they auto-reverse)

Validation: period statuses per module equal the source at cutover; allocation rules re-executed on the last closed period in simulate mode and compared to source postings (tolerance 0.01 per destination). Not migrated: completed close task history, prior allocation run history, SAP accrual engine objects (re-keyed as schemes).

2.14.12 Dependencies, risks, open questions

Depends on: wave 11 (posting API hook, reversal), wave 12 (revaluation hook on checklist), wave 13 (settlement hook), wave 4 (Temporal human tasks and timers), wave 6 (fiscal calendars with adjustment periods), wave 2 (user groups).

Risk	Likelihood	Impact	Mitigation
Subledger services (future waves) bypass module mapping	Medium	High	Source mapping is mandatory; unmapped `source.system` → GL and rejected when GL on hold; contract test suite grows per wave
Iterative allocation performance with many dimensions	Low	Medium	Solver limited to 50 rules per group; measured in fixtures

Risk	Likelihood	Impact	Mitigation
Year-end with millions of combinations	Medium	Medium	Chunked generation (50k lines per entry, multiple entries per run); run measured on 1M-combination synthetic set

Open questions: (1) Should on_hold allow service accounts of the owning module by default? Default no; decide by wave 14 mid-point. (2) Retained earnings per dimension as default? Default yes for the demo; decide with accountant review by wave 14 exit. (3) Do adjustment periods 13–16 get their own close checklist? Default: same run, tasks flagged `adjustment_only`.

2.14.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivale
`period_control.module_mapping`	tenant	jsonb	ledger→GL, ap→AP, ar→AR, inventory→INV, fa→FA, project→PROJ, tax→TAX, bank→BANK, payroll→PAYROLL	source.system → module	BR-14.1 module resolution	Ledger calendar > Module access (`LedgerPeriodModuleAccessControl`)	OB52 ac types (+, K, M, S,
`period_control.on_hold_group_id`	ledger × period × module	ref	none	user groups (wave 2)	Who may post while on hold	Ledger calendar > Access level / user group	OB52 authoriz group (`BRGRU`) second interval
`close.reopen_default_days`	tenant	int	3	1–30	Reopen expiry (FR-14.3)	—	—
`close.reopen_approvers`	tenant	int	2	2–3	Distinct approvers for reopen	—	—
`close.lock_wait_ms`	tenant	int	5,000	1,000–30,000	BR-14.11 wait	—	—
`close.gl_closes_last`	ledger	bool	true	true (fixed in prod)	FR-14.2 ordering	—	—
`close.block_on_pending_batches`	ledger	bool	true	true / false	`CLOSE_PENDING_BATCHES` on close	—	—
`close.template.due_offset` / `business_calendar_id`	template task	int / ref	0 / legal entity calendar	−10 to +30 business days	Due date computation	Financial period close > Templates > Due day	Closing cockpit t offset (CLOCO
`allocation.max_iterations` / `convergence_fraction`	tenant	int / numeric	100 / 0.5	10–1,000 / 0.1–1	BR-14.5	Allocation rules (no iterative)	KSV5/KS iterative flag
`allocation.allocate_from_zero`	rule	enum	skip	skip / equally	BR-14.4 zero basis	Allocation rule > "Allocate from zero"	Cycle segment "zero tra factors"
`allocation.auto_reverse_next_period`	rule	bool	false	true / false	FR-14.8	—	Cycle "reverse" KSV6/KS
`accrual.reverse_day` / `post_day`	scheme	int	1 / 1	1–28	BR-14.7	Accrual scheme > Reversal / Period key	FBS1 re date; AC reversal settings
`year_end.retained_earnings_combination_id`	ledger	ref	required	equity account combination	FR-14.10	Ledger > Year-end close > Retained earnings account	OB53 retained earnings account (`T030` E
`year_end.retain_dimensions`	ledger	bool	true	true / false	BR-14.8	Year-end close template > "Transfer dimensions" (D365)	FAGLGV carry-for of characte (`T881` fi
`year_end.default_closing_combination_id`	ledger	ref	required when retain_dimensions	combination	Remap of inactive values	Year-end close template > Financial dimension defaults	—
`year_end.chunk_lines`	tenant	int	50,000	10,000–100,000	Entry size in year-end runs	—	—
`regrouping.buckets` default	ledger	jsonb	≤ 365 d, 366–1,825 d, > 1,825 d	ordered day ranges	BR-14.13	—	FAGLF1 sort meth (`T030` PYO/PY
`regrouping.auto_reverse`	ledger	bool	true	true (fixed)	Reversal on key_date + 1	—	FAGLF1 reversal posting d
`close.reminder_time`	legal entity	time	08:00	local time	Close reminders	—	—

2.14.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Period-control matrix	list inquiry	ledger, fiscal year, module	period, module, status, on-hold group, changed by/at, reopen count	`PeriodStatus`	Ledger calendar form	OB52 view (`T001B`), OKP1 (CO period lock), MMPV log

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Reopen history	operational report	ledger, year, requester, approver	period, module, reason, requester, approvers, opened/closed at, expiry	`PeriodStatus` audit	Database log on `LedgerPeriod`	`T001B` change documents (SCU3)
Close board / task list	list inquiry	run, entity, status, owner, overdue	task, entity, owner, due, status, blocked by, evidence, late	`CloseTask`	Financial period close workspace (Tasks, Calendar)	CLOCOS monitor, Fiori Manage Closing Tasks
Close duration KPI	KPI	entity, period range	business days period end → GL closed, late tasks %, reopen count	`CloseRun`, `PeriodStatus`	Financial period close workspace tiles	Advanced Financial Closing analytics
Allocation run results	drill-down	run, rule	source amount, destination, basis value, share, amount, iteration count	`AllocationRun`	Allocation journal (review before post); SSRS Allocation rule report	KSV5/KSU5 detail list, KSB5
Allocation rule listing	operational report	ledger, method	rule, source filter, destinations, basis, offset, iterative group	`AllocationRule`	SSRS Allocation rule report	KSV3/KSU3 cycle display
Accrual schedule	list inquiry	scheme, source entry, date range	occurrence, date, amount, posted, reversal	`AccrualScheme` schedules	Ledger accruals inquiry	ACACPSITEMS, F.81 list
Recurring journal next runs	list inquiry	template, owner	next run, last run, generated batches, paused	`RecurringJournal`	Periodic journals list	F.15 list recurring entries
Year-end close results	operational report	ledger, fiscal year	retained earnings by dimension, opening per combination, remapped combinations	`YearEndCloseRun`	SSRS Fiscal year close report / Opening balances	FAGLGVTR log, 2KET, F.07 log
Regrouping proposal / posting list	drill-down	run, rule, key date	party / item, bucket, balance, adjustment account, target account, amount	`RegroupingRun.snapshot`	—	FAGLF101 sorted list and posting log
Rejected postings in closed periods	KPI	ledger, period, source system	count, amount, first/last attempt	posting rejection log	—	—

2.14.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Close pack (checklist evidence)	CloseRun completed	header: entities, period, template; lines: task, owner, completed at, evidence links, skipped reasons; footer: GL close time, approvers	PDF, XLSX	close run id	audit evidence; 10 years
Period reopen approval record	reopen approved	ledger, period, module, reason, requester, approvers, timestamps, expiry	PDF, JSON	—	audit evidence; 10 years
Allocation posting list	allocation run posted	rules, basis snapshot, destination amounts, vouchers	PDF, XLSX	run id; ledger vouchers	financial record; 10 years
Year-end closing report	year-end run posted	retained earnings per dimension, opening entries summary, remapped combinations, vouchers	PDF, XLSX	run id	financial record; permanent for the fiscal year file
Regrouping posting list	regrouping run posted	rule, buckets, groups, adjustment amounts, vouchers, reversal date	PDF, XLSX	run id	supporting record for balance sheet presentation; 10 years

No external electronic documents in this wave.

2.14.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`accrual-generator`	daily 00:30 entity time zone	scheme × occurrence	occurrence keyed (scheme, source entry, n)	10k occurrences < 2 min	occurrence blocked if period closed → close task	blocked > 0 at day 3 → P3
`recurring-journal-generate` (scheduling per FR-14.11)	per template	template	(template, scheduled date)	< 5 min for 1,000 templates	per template	—
`reopen-expiry` (Temporal timer)	at `reopen_expires_at`	period status row	transition idempotent	< 1 s	retry; alert	timer missing → P2
`close-reminders`	daily `close.reminder_time`	legal entity	one notification per (task, day)	< 1 min	—	—
`allocation-run` (close hook)	per checklist	ledger × rule group	(ledger, period, rule ids) key	1,000 rules < 5 min	run failed; task blocked	non-convergence → P3
`year-end-close`	on demand / checklist	ledger	(ledger, year, run id); chunks of 50k lines	1M combinations < 20 min	failed chunk rolls back run	failure → P2
`regrouping-run` (close hook)	key date = period end, day +2	ledger × rule	(ledger, key date, rule set)	1M open items < 10 min	run failed; task blocked	failure → P2
`regrouping-reverse`	key date + 1 day 00:15	run	reversal set once	< 1 min	retry hourly	missing by 12:00 → P2
`period-status-cache-audit`	every 5 min	posting service	compares cached status vs table	< 1 s	cache refresh	mismatch → P2

2.14.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
View period status	GET periods matrix	—	All finance roles
Maintain period status	PATCH periods/{p}/modules/{m}; bulk close	Approve period reopen; Post ledger journals in on-hold group	Financial controller
Request period reopen	POST :request-reopen	Approve period reopen (same request)	Accounting supervisor, Financial controller
Approve period reopen	Temporal approval task	Request period reopen (same request); Maintain period status	CFO, Group controller
Maintain close templates	close templates CRUD	—	Financial controller, Close manager
Run period close	POST /close/runs	—	Close manager
Work close tasks / View close board	task actions; board	Skip close tasks (own tasks)	All finance roles
Skip close tasks	:skip	Work close tasks (same task)	Close manager
Maintain allocation rules	allocation rules CRUD	Run allocations	Cost accountant (senior)

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
Run allocations	POST /ledger/allocations, :post, :reverse	Maintain allocation rules	Cost accountant
Maintain accrual schemes	accrual schemes CRUD	Approve ledger journals	Accounting manager
Run year-end close	POST /ledger/year-end-closes	Maintain period status	Financial controller
Maintain regrouping rules	regrouping rules CRUD	Run regrouping	Accounting manager
Run regrouping	POST /ledger/regroupings, :post, :reverse	Maintain regrouping rules	Financial controller, Accounting supervisor

2.14.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
CLOSE-0001 `CLOSE_INVALID_TRANSITION`	error	Cannot change {module} {period} from {from} to {to}: {reason}.	BR-14.3, FR-14.2	Close dependent modules first or use the reopen workflow	FR-14.2
CLOSE-0002 `CLOSE_PENDING_BATCHES`	error	{count} approved journal batches dated in {period} are not posted.	Close with unposted batches	Post or cancel them, or force with duty	FR-14.2
CLOSE-0003 `CLOSE_BUSY`	error	Postings are in progress for {ledger}; retry the close.	BR-14.11 lock wait	Retry	BR-14.11
CLOSE-0004 `CLOSE_REOPEN_REASON_REQUIRED`	error	A reason of at least 10 characters is required.	FR-14.3	Enter a reason	FR-14.3
CLOSE-0005 `CLOSE_REOPEN_APPROVER_INVALID`	error	Approver {user} is the requester or already approved.	FR-14.3	Choose a distinct approver	FR-14.3
CLOSE-0006 `CLOSE_REOPEN_EXPIRED`	info	Period {period} was reclosed automatically after the reopen window.	Expiry timer	Request a new reopen if needed	BR-14.3
CLOSE-0007 `CLOSE_TEMPLATE_CYCLE`	error	Task dependencies contain a cycle: {path}.	FR-14.4	Remove a dependency	FR-14.4
CLOSE-0008 `CLOSE_TASK_BLOCKED`	error	Task {task} is blocked by {blocking}.	BR-14.10	Complete or skip the blocking tasks	BR-14.10
CLOSE-0009 `CLOSE_TASK_AUTOMATED`	error	Task {task} completes automatically when {job} finishes.	BR-14.10	Run the job, or allow manual override	BR-14.10
CLOSE-0010 `ALLOC_PCT_NOT_100`	error	Destination percentages total {sum}; they must total 100.000000.	FR-14.6	Adjust percentages	FR-14.6
CLOSE-0011 `ALLOC_ZERO_BASIS`	warning	Rule {rule} skipped: basis is zero for all destinations.	BR-14.4	Check basis data or set allocate_from_zero	BR-14.4
CLOSE-0012 `ALLOC_NO_CONVERGENCE`	error	Iterative group {group} did not converge in {n} iterations.	BR-14.5	Review reciprocal shares (≥ 50% loops)	FR-14.7
CLOSE-0013 `YEAR_END_PERIODS_OPEN`	error	Year-end close needs all periods of {year} closed; open: {list}.	FR-14.10	Close the listed periods	FR-14.10
CLOSE-0014 `YEAR_END_RE_ACCOUNT_INVALID`	error	Retained earnings account {account} must be an equity account.	FR-14.10	Fix the ledger setting	FR-14.10
CLOSE-0015 `YEAR_END_DIMENSION_REMAPPED`	warning	{count} combinations had inactive values and were remapped to {default}.	BR-14.8	Review the run report	BR-14.8
CLOSE-0016 `CLOSE_REGROUPING_EXISTS`	error	A posted regrouping exists for {ledger} {key_date} rule {rule}.	BR-14.13	Reverse it first	FR-14.13
CLOSE-0017 `CLOSE_REGROUPING_SOURCE_UNAVAILABLE`	error	Open-item source {source} is not registered for {ledger}.	AP/AR wave not live	Use GL sources only or wait for the source	FR-14.13
CLOSE-0018 `CLOSE_REGROUPING_NO_DUE_DATE`	warning	{count} items without due date were placed in the shortest bucket.	BR-14.13	Maintain due dates on the source documents	BR-14.13

2.14.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Period decision in posting	every posting line batch	+0.3 / +1 / +2 ms	at 5,000 lines/s	—	`period_status` PK cached per pod, invalidated by event	cache miss → PK read
Period transition (close GL)	3 entities × 10 modules × 16 periods per year	50 / 500 / 5,000 ms (lock wait)	—	480 rows/year/ledger	PK	`CLOSE_BUSY` returned after 5 s
Close board load	1,000 tasks	80 / 300 / 500 ms	50 req/s	1,000 tasks × 12 runs × 1 KB = 12 MB	(run_id, status), (assignee, due_at)	page size reduced
Allocation run	1,000 rules, 50k destinations	—	< 5 min	basis snapshots 10 MB/run	balance PK queries per rule	runs queued per ledger
Iterative allocation group	≤ 50 rules	—	< 60 s	—	in-memory solver	fail at 100 iterations
Accrual generation	10k occurrences/day	—	< 2 min	—	(scheme, date)	—
Year-end close	1M combinations	—	< 20 min	2M lines/year/ledger	chunked entries of 50k lines	chunk size reduced, run longer
Regrouping	1M open items	—	< 10 min	2 entries per rule per period	`open_item` (account, status, due_date)	per-rule timeout 20 min

2.14.20 Localization and compliance hooks

- Fiscal calendars vary (JP/IN April–March, AU July–June, 4-4-5 retail calendars, 13-period calendars) — handled by wave 6 calendars; period control is per ledger so local-GAAP ledgers can follow a different year end.
- Year-end: some jurisdictions require closing to the profit-and-loss summary account then to retained earnings (FR, ES PGC class 129 "Resultado del ejercicio"; IT "Utile d'esercizio") and separate opening/closing balance sheets (DE "Eröffnungsbilanzkonto"/"Schlussbilanzkonto" 9000/9008); country packs configure an intermediate result account and closing/opening balance sheet accounts as year-end template steps.
- Regrouping: remaining-term disclosure buckets vary — HGB §268(4)/(5) "Restlaufzeit bis 1 Jahr / mehr als 5 Jahre", IFRS current/non-current (IAS 1.66–1.76: 12 months), US GAAP current maturities; country packs ship bucket sets and adjustment account sets; debit-balance vendor reclassification is mandatory under HGB and common under IFRS.

- Statutory locking: DE GoBD requires "Festschreibung" (posted entries immutable within the period of the VAT return) — satisfied by append-only journal plus period close; FR FEC closes the year with a "clôture" that is irreversible — country pack sets `reopen_approvers = 3` and disallows reopen after the statutory deadline.
- Retention of close evidence: 10 years (DE/FR), 7 (US SOX workpapers per PCAOB AS 1215).
- SOX: reopen dual approval, close pack evidence and SoD duties above are the SOX 404 control hooks consumed by wave 48 controls monitoring.
- Extension points: `close.template_pack` (country/industry close templates), `ledger.year_end_step` (intermediate result account, closing/opening BS accounts), `ledger.regrouping_bucket_pack`.

2.14.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP §13.1 FI-GL Periodic processing — FAGLF101 / F101 Regroup receivables/payables (sorted list, reclassification by remaining term) (SAP register row 8)	GAP (propose 14)	ADDED as FR-14.13	`remaining_term` and `debit_credit_balance` methods; AC-14.11
SAP §13.1 FI-GL Periodic processing — FAGL_CL_REGROUP / FAGL_CL_MIGRATION Closing: regrouping (S/4) (SAP register row 10)	GAP (propose 14)	ADDED as FR-14.13	same engine; migration of FAGL_CL settings in §2.14.11
SAP §13.1 FI-GL Customizing — FAGL_CL_RGRP / Define sort method & adjustment accounts for regrouping (`T043`, `T030`) (SAP register row 21)	GAP (propose 14)	ADDED as FR-14.13	`RegroupingRule` buckets and adjustment/target accounts; parameters in §2.14.13
SAP §13.2 FI-AP Periodic processing — FAGLF101 (AP) Regroup vendors with debit balances (SAP register row 30)	GAP (propose 14)	ADDED as FR-14.13	rule ships in this wave; vendor item source activates when wave 19 registers `open_item_source = vendor` (AC-14.12 uses stub items)
SAP §13.3 FI-AR Periodic processing — FAGLF101 (AR) Regroup customers with credit balances (SAP register row 42)	GAP (propose 14)	ADDED as FR-14.13	customer item source activates with wave 30

2.15 Wave 15: Financial Reporting, Statements & Audit Trail

2.15.1 Scope and business context

AX ships Management Reporter (row, column, reporting tree definitions over dimensions) plus trial balance and ledger transaction inquiries; SAP uses financial statement versions (OB58, `T011`/`T011T`, `FAGL_011*`), Fiori financial statement apps and document-flow drill-down. Wave 15 builds the `reporting` service (Go, with its own Postgres read model fed by ledger events) with a row/column/tree report designer, standard statements (trial balance, balance sheet, P&L, cash flow indirect, comparatives, by-entity consolidated view), drill-down to the journal line and source document, scheduled distribution with row-level security on dimensions, and the audit trail report. This is the Phase 2 exit wave: finance stakeholders sign off a closed period produced entirely from Passage.

Out of scope: group consolidation with eliminations and translation (wave 40), budgets (wave 38), lakehouse analytics and dashboards (wave 46), XBRL/iXBRL (wave 48).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Management Reporter reads from its own data mart refreshed by an integration service; statements lag the GL by 15–60 min and the mart silently breaks after AX upgrades or dimension changes	Separate ETL (MR integration) with no "as of" stamping	Read model fed by `journal.posted` events with per-ledger HWM; every statement shows the sequence it reflects; nightly zero-diff verify	Statement lag p95 < 10 s (vs 15–60 min); "mart out of sync" incidents → 0
SAP F.01/RFBILA00 and FSV (OB58) drill to FAGLL03 but lose the FSV context; reaching the source document from a balance-sheet line is 7+ screens and often impossible for migrated items	Hierarchy, totals and line items are separate programs without a shared key	Drill-down contract (BR-15.4): cell → balance keys → lines → `source_uri` → trace id, with sums that must equal the cell	Clicks balance → document 7 → 3; drill sum mismatches → 0 (property-tested)
Report designer in MR requires a Windows client, separate security (tree security) and definitions that cannot be versioned or promoted between environments except by manual export	Desktop tooling, no API, no versioning	Web designer, JSON export/import with schema version, immutable published versions, OPA dimension security shared with the rest of Passage	Definition promotion dev → prod 1 h manual → < 1 min API call; security rule sets 2 → 1
Cash flow statement is hand-built in Excel from the trial balance because MR/FSV cash flow setups are fragile and do not reconcile	No tagging model linking balance movements to cash flow categories with an enforced check	`cash_flow_tag` on rows, `CF-STD` indirect derivation with `CF_UNRECONCILED` failure and a "not classified" check row	Cash flow preparation 1–2 days/period → generated in < 20 s with 0.00 reconciliation enforced
Audit requests ("who posted, who approved, when was the period reopened") require pulling `SysDatabaseLog` / `CDHDR` extracts and workflow history separately, usually by IT	Audit data spread across logs with no finance-facing report	Audit trail report over CDC audit + ledger events with hashed export	Audit evidence turnaround days → minutes; controller self-service, 0 IT tickets

2.15.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Report definitions	Management Reporter row/column/tree definitions, report definitions	Financial statement version (OB58, `T011`, `FAGL_011PC/ZC`), Report Painter/Writer	`reporting.ReportDefinition` (rows, columns, tree)	full
Row definitions	MR row: description, link to financial dimensions, format codes (TOT, CAL), normal balance	FSV nodes with account intervals, debit/credit split, net result	`ReportRow` with kind ∈ {account_range, dimension_filter, total, calculation, header}	full
Column definitions	MR column: FD (period), CALC, comparative, YTD, budget	Period/comparison columns in F.01 / Fiori	`ReportColumn` with period expressions (`P`, `P-1`, `YTD`, `PY`, `PYTD`, `CALC`)	full (budget columns in wave 38)
Reporting tree	MR tree of reporting units (dimension values)	Profit center hierarchy (KE5X), FSV per company	`ReportTree` over dimension hierarchies (wave 6)	full
Trial balance	Trial balance list page, `LedgerTrialBalanceListPage`	FAGLB03, S_ALR_87012277, F.08	Trial balance statement (standard definition)	full
Balance sheet / P&L	MR default reports	F.01 (RFBILA00), Fiori Balance Sheet / P&L	Standard definitions `BS-STD`, `PL-STD`	full
Cash flow	MR cash flow (manual definitions)	Cash flow via FSV + S/4 Cash Flow app	`CF-STD` indirect with cash-account tagging and non-cash adjustments	full
Drill-down	MR drill to account/transactions; ledger transaction inquiry; voucher transactions	Document flow (FB03), FAGLL03 line items, drill from Fiori statements	Statement cell → balance keys → lines → source document URI → trace id	improved: trace id
Distribution	MR report library, scheduled generation, email	Report scheduling (SM36), SAP Analytics Cloud publication	`StatementRun` schedules with PDF/XLSX/CSV, email, link	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Security	MR tree security, AX dimension security	Authorization objects (F_BKPF_BUK, K_CSKS), FSV authorization	OPA dimension-value policies applied to every balance query	full
Audit trail	Database log, voucher audit, workflow history	Change documents (CDHDR/CDPOS), FB03 header changes, SE16 on BKPF	Audit trail report over CDC audit (wave 5) + ledger events	full
Main account categories and derived hierarchies	Main account categories (`MainAccountCategory`), D365 derived financial hierarchies (`DimensionHierarchy` type Derived), Financial dimension sets	Account groups / G/L account hierarchies (KDH1–KDH3, `SETHEADER`/`SETNODE`), FSV nodes, profit center groups (KCH1)	`reporting.DerivedHierarchy` over main account categories and dimension values, effective-dated versions, usable as row sets and trees (FR-15.13)	full
P&L by function of expense (cost of sales accounting)	Custom MR rows by dimension	FSV with functional area (`RFAREA`), S_ALR_87012284 with FSV for cost of sales	Standard definition `PL-FUNC` over `FUNC_AREA` (FR-15.14)	full

2.15.3 Functional requirements

- **FR-15.1 Reporting read model** — Must. The system shall maintain in `reporting-db` a `ReportingBalance` fact (ledger × account × dimension combination × period × currency type) built from `ledger.journal.posted/reversed/balance.rebuilt/year-end.closed` events, with the covered `posting_sequence` high-water mark per ledger.
  - Lag SLO: p95 < 10 s after posting; every statement records the HWM it reflects and shows "as of sequence N / time T".
  - A nightly comparison against `ledger.Balance` must show zero differences.
- **FR-15.2 Report designer: rows** — Must. Row definitions contain ordered rows of kind header, account_range (account from/to, optional dimension filter), dimension_filter (any dimension values with wildcards), total (sum of row references), calculation (expression over row references with + − × ÷ and constants), with format (bold, indent, sign flip for normal credit balance, print zero, page break).
  - Maximum 2,000 rows per definition; expressions validated on save (unknown references, division by zero handled → blank).
- **FR-15.3 Report designer: columns** — Must. Column definitions specify period expressions relative to the run period (`P`, `P-n`, `YTD`, `PY`, `PYTD`, `Q`, `FY`), currency type (acct/rpt/txn with currency filter), ledger, dimension filter, variance/percent calculations between columns, and formatting.
  - Maximum 60 columns; comparative column sets (actual vs prior year) are templates.
- **FR-15.4 Reporting trees** — Must. Trees list reporting units (dimension value or hierarchy node from wave 6 hierarchies) with roll-up; a statement run produces one report per unit plus the roll-up, honoring OPA dimension security per viewer.
- **FR-15.5 Standard statements** — Must. The system shall ship definitions `TB-STD` (trial balance: opening, debit, credit, closing, net), `BS-STD`, `PL-STD`, `CF-STD` (indirect method: net income + non-cash adjustments + working-capital changes derived from balance movements of tagged accounts + investing + financing = change in cash), `TB-DIM` (trial balance by dimension), each parameterized by ledger, period, currency type, and tree.
  - Statements mark `is_closing_entry` / `is_opening_entry` handling: P&L excludes closing entries; BS includes opening entries.
  - Consolidated-by-entity view: side-by-side columns per legal entity and a simple sum column in reporting currency (no eliminations; wave 40).
- **FR-15.6 Statement runs** — Must. A run evaluates a definition for parameters and stores `StatementCell` values with the balance keys used, the HWM, duration, and a SHA-256 of the result; outputs rendered as HTML (interactive), PDF, XLSX and CSV.
  - 2,000-row × 12-column statement over 200k balances completes in < 20 s.
- **FR-15.7 Drill-down** — Must. From any cell: (1) balances view listing the (account, combination, period) keys with amounts summing to the cell; (2) journal lines for a selected key or the whole cell, paginated, with `voucher`, date, D/C, amounts, source; (3) source document link (`source_uri` resolved to a Passage route or external URL) and `trace_id` link to Tempo.
  - Drill honors the viewer's dimension security; the line sum shown equals the cell value exactly or the UI displays a reconciliation warning with the filtered-out amount (never silently).
- **FR-15.8 Scheduling and distribution** — Must. Definitions can be scheduled (cron or close-checklist hook) to run per entity set and distribute via email (PDF/XLSX attachments ≤ 20 MB, else link), shared link with expiry, and a report library with retention (default 7 years for statements run against closed periods, 90 days for ad-hoc).
- **FR-15.9 Row-level security on dimensions** — Must. OPA policies grant dimension values (e.g., cost centers) per user; balance queries, statements, drill-down and exports are filtered server-side; a report run by a scheduler uses the recipient's permissions per rendered copy.
- **FR-15.10 Audit trail report** — Must. The system shall report for a period and ledger: entries posted (who, when, source, idempotency replay flag), batch approvals (submitter, approvers, timestamps), reversals, period status changes including reopen approvals, rate overrides, tax fallbacks, allocation and year-end runs; exportable to CSV/PDF with a hash.
- **FR-15.11 Definition import/export** — Must. Definitions round-trip as JSON (schema-versioned) for backup, environment promotion and Management Reporter conversion.
- **FR-15.12 Management Reporter and FSV converters** — Should. The migration framework shall convert MR row/column/tree definitions (from the MR database `ControlRowDetail`, `ControlColumnDetail`, `ControlTreeDetail`) and SAP FSV (`T011`, `FAGL_011PC`, `FAGL_011ZC`) into Passage definitions with a conversion report of unsupported constructs.
- **FR-15.13 Derived financial hierarchies** — Must. The system shall maintain derived hierarchies: trees whose leaf assignment is derived by rules rather than listed, over (a) main account categories (each main account carries `category_id`; categories nest under hierarchy nodes) and (b) dimension values (nodes with membership rules: value ranges, wildcards, attribute filters such as `cost_center.region = EMEA`), with effective-dated versions.
  - A derived hierarchy can be used as a row set (one row per node, totals rolled up, leaves expanded on demand) and as a reporting tree (FR-15.4); statements record the hierarchy version used.
  - Validation on publish: every postable account (or every active dimension value) maps to exactly one leaf node, or is listed in an "unassigned" check node that must be empty for `strict = true` hierarchies (`REPORT_HIERARCHY_UNASSIGNED`); overlaps are rejected (`REPORT_HIERARCHY_OVERLAP`).
  - Membership is re-evaluated when accounts or dimension values change (`ledger.account.changed`, dimension events); a new value that matches no rule appears in the unassigned node within 60 s and raises a notification to the hierarchy owner.
  - Up to 12 levels and 20,000 nodes per hierarchy; JSON import/export like FR-15.11.
- **FR-15.14 P&L by function of expense** — Should. The system shall ship `PL-FUNC` (revenue, cost of sales, distribution costs, administrative expenses, other operating income/expense, financial result) whose expense rows filter on the `FUNC_AREA` dimension (FR-11.22) instead of account ranges, with a check row that lists P&L lines without `FUNC_AREA` and must be 0.00.

2.15.4 Data model

ReportDefinition, **ReportRow**, **ReportColumn**

Field	Type	Required	Notes
definition: id	uuid	yes	PK
definition: code / name / version	text(30) / text(120) / int	yes	unique (tenant, code, version)

Field	Type	Required	Notes
definition: kind	enum	yes	trial_balance, balance_sheet, income_statement, cash_flow, custom
definition: row_set_id / column_set_id / tree_id	uuid	yes/yes/no	
definition: default_ledger_id / currency_type	uuid / enum	yes	
definition: status	enum	yes	draft, published, retired
definition: schema_version	text	yes	for JSON import/export
row: id, row_set_id, sequence	uuid, uuid, int	yes	
row: kind	enum	yes	header, account_range, dimension_filter, total, calculation
row: label	text(200)	yes	
row: account_from / account_to	text(20)	conditional	
row: dimension_filter	jsonb	no	{attr: [values or ranges or *]}
row: references / expression	uuid[] / text	conditional	for total / calculation
row: normal_balance	enum	yes	debit, credit (sign presentation)
row: format	jsonb	no	bold, indent, print_zero, page_break, underline
row: cash_flow_tag	enum	no	operating_adjustment, working_capital, investing, financing, cash
column: id, column_set_id, sequence, label		yes	
column: period_expr	text	yes	`P`, `P-1`, `YTD`, `PY`, `PYTD`, `Q`, `FY`, `CALC`
column: currency_type / currency	enum / char(3)	yes/no	
column: ledger_code_override	text	no	compare ledgers side by side
column: dimension_filter	jsonb	no	
column: calc_expression	text	no	`C2-C1`, `(C2-C1)/C1`
column: format	jsonb	no	decimals, percent, width

StatementRun and **StatementCell**

Field	Type	Required	Notes
run: id	uuid	yes	PK
run: definition_id / version	uuid / int	yes	
run: parameters	jsonb	yes	ledger(s), period, tree unit, currency type, entities
run: hwm_by_ledger	jsonb	yes	{ledger_id: posting_sequence}
run: status	enum	yes	queued, running, completed, failed
run: started_at / completed_at / duration_ms		yes	
run: result_hash	bytea(32)	yes	SHA-256 over cells
run: outputs	jsonb	yes	[{format, uri, size}]
run: requested_by / security_principal	uuid	yes	whose dimension security applied
cell: run_id, row_id, column_id		yes	PK
cell: value	numeric(23,6)	yes	
cell: balance_keys	jsonb	yes	[{ledger, account, combination, period, currency_type}] (compressed; up to 50k keys, overflow stored as a query spec)
cell: line_count	int	yes	

DerivedHierarchy and **HierarchyNode** (FR-15.13)

Field	Type	Required	Notes
hierarchy: id / code / name	uuid / text(30) / text(120)	yes	unique (tenant, code, version)
hierarchy: subject	enum	yes	main_account_category, main_account, dimension_value
hierarchy: dimension_attribute_id	uuid	conditional	when subject = dimension_value
hierarchy: version / effective_from / status	int / date / enum	yes	draft, published, retired
hierarchy: strict	bool	yes	unassigned must be empty to publish
hierarchy: owner_user_id	uuid	yes	notified on unassigned members
node: id / hierarchy_id / parent_id / sequence	uuid	yes	depth ≤ 12
node: label / code	text(120) / text(30)	yes	
node: membership_rules	jsonb	no	leaves only: ranges, lists, wildcards, attribute filters, category ids
node: normal_balance / cash_flow_tag	enum	no	inherited by generated rows
membership (materialized): hierarchy_version, node_id, member_id		yes	rebuilt on change; PK (hierarchy_version, member_id)

MainAccountCategory: `id`, `code` text(20), `name`, `account_type` (must match member accounts' type), `sequence`; `MainAccount.category_id` FK (wave 11 entity extended).

AuditTrailEntry (materialized view over audit store + ledger events): `occurred_at`, `ledger_id`, `kind` (posted, replayed, reversed, batch_submitted, batch_approved, batch_rejected, period_changed, reopen_approved, rate_override, tax_fallback, allocation_run, year_end_run, rebuild), `actor_id`, `object_ref`, `details` jsonb, `trace_id`.

Invariants: for every cell, Σ ReportingBalance over `balance_keys` = `value` (property-tested); total rows equal Σ referenced rows; `TB-STD` closing per account = opening + debit − credit; `CF-STD` change in cash = Σ cash-tagged account movements; `result_hash` is reproducible for the same HWM and definition version. Indexing: `reporting_balance` PK (ledger, period, account, combination, currency_type) plus (ledger, account, period); `statement_cell` PK (run_id, row_id, column_id); audit view indexed on (ledger_id, occurred_at).

2.15.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET/POST/PUT	/api/v1/reporting/definitions[/{code}] and /{code}:publish	Definitions CRUD and publish	PUT idempotent	Maintain report definitions
GET/PUT	/api/v1/reporting/row-sets[/{id}], /column-sets[/{id}], /trees[/{id}]	Designer parts	PUT idempotent	Maintain report definitions
POST	/api/v1/reporting/definitions/{code}:validate	Validate expressions, references	stateless	Maintain report definitions
GET/POST	/api/v1/reporting/definitions/{code}/export, /import	JSON round-trip	import `Idempotency-Key`	Maintain report definitions
POST	/api/v1/reporting/runs	Run a statement	`Idempotency-Key`	Run financial statements
GET	/api/v1/reporting/runs/{id}[/cells] and /outputs/{format}	Results and files	n/a	View financial statements
GET	/api/v1/reporting/runs/{id}/cells/{row}/{col}/balances	Drill level 1	n/a	View financial statements
GET	/api/v1/reporting/runs/{id}/cells/{row}/{col}/lines?cursor=	Drill level 2 (lines)	n/a	View ledger transactions
GET	/api/v1/reporting/balances?ledger=&period=&account_from=&… &group_by=	Ad-hoc balance query (secured)	n/a	View ledger balances
GET/PUT	/api/v1/reporting/schedules[/{id}]	Distribution schedules	PUT idempotent	Schedule financial statements
GET	/api/v1/reporting/audit-trail?ledger=&period=&kind= and /audit-trail:export	Audit trail report	export run id	View audit trail
GET	/api/v1/reporting/lag	Read-model lag per ledger	n/a	View financial statements
GET/PUT	/api/v1/reporting/hierarchies[/{code}] and /{code}:publish, /{code}/members?version=	Derived hierarchies, publish, materialized membership (FR-15.13)	PUT idempotent	Maintain report definitions
GET/PUT	/api/v1/ledger/main-account-categories[/{code}]	Main account categories	PUT idempotent	Maintain chart of accounts

Queries and validation: `RunStatement` requires a published definition, period exists for every selected ledger, viewer has at least one permitted dimension value when a tree is used, and read-model lag ≤ `max_lag_seconds` (default 30) unless `allow_stale = true` (result flagged). `ImportDefinition` validates schema version and rejects duplicate codes without `overwrite`.

Event	Trigger	Payload fields	Consumers
dev.passage.reporting.statement.generated.v1	run completed	run_id, definition code/version, parameters, hwm_by_ledger, result_hash, outputs[]	distribution, close (automation hook), audit
dev.passage.reporting.statement.failed.v1	run failed	run_id, error, stage	ops, requester notification
dev.passage.reporting.definition.published.v1	publish	code, version, actor, diff_summary	audit, cache
dev.passage.reporting.lag.exceeded.v1	lag > threshold for 60 s	ledger_id, lag_seconds, hwm_ledger, hwm_reporting	ops alerting, close board
dev.passage.reporting.distribution.sent.v1	email/link delivered	run_id, recipients (hashed), format, size	audit
dev.passage.reporting.hierarchy.unassigned.v1	new account or dimension value matches no leaf	hierarchy code, version, member_id, member kind	hierarchy owner notification, close board

2.15.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Report designer	Create definition, edit rows/columns/tree in split panes, preview with live data, validate, publish, export/import	row label, kind, accounts, dimension filter, expression, format; column label, period expr, currency type	Maintain report definitions
Financial statements	Run (ledger, period, tree, currency), view interactive statement, compare runs, export	row, columns, as-of HWM, run status	Run / View financial statements
Drill-down panel	Cell → balances → lines → source document, open trace	account, combination, period, amount; voucher, date, D/C, amounts, source link	View ledger transactions
Trial balance (standard)	Period range, dimension filters, drill, export	account, opening, debit, credit, closing	View ledger balances
Report library	Browse outputs by entity/period/definition, share link, retention	definition, period, entity, generated at, hash	View financial statements
Schedules	Define cron/close hook, recipients, formats	definition, parameters, cron, recipients, last run	Schedule financial statements
Audit trail	Filter period/ledger/kind/actor, export with hash	occurred at, kind, actor, object, details, trace id	View audit trail
Derived hierarchies	Build tree, define membership rules per leaf, view unassigned/overlaps, publish version, use as row set or tree	node, level, rules, members count, unassigned count, version	Maintain report definitions

Interaction rules: designer previews render within 3 s against the demo dataset and update on each save; row grid supports drag-reorder, multi-select indent and "insert total of selection"; statements are keyboard navigable cell-by-cell (arrow keys) with Enter to drill; drill panels stack as breadcrumbs (Statement › Cell › Balances › Lines › Document); exports preserve number formatting and include an "as of" footer with HWM, definition version and result hash; all screens pass axe WCAG 2.1 AA.

2.15.7 Business rules and edge cases

- **BR-15.1 Period expression evaluation.** `P` = run period; `P-n` = n periods earlier in the ledger calendar (crossing fiscal years, adjustment periods skipped unless `include_adjustment_periods`); `YTD` = periods from fiscal year start through P, including adjustment periods ≤ P's year when flagged; `PY` = same period prior year; `Q` = fiscal quarter containing P.
- **BR-15.2 Balance-sheet amounts** are cumulative: Σ ReportingBalance.net from period 0 (opening) through the column's period; P&L amounts are the sum over the column's period range only and exclude `is_closing_entry` lines.
- **BR-15.3 Sign presentation.** Stored net = debit – credit; rows with `normal_balance = credit` display `-net`; calculation rows use displayed values unless the expression is prefixed `raw:`.
- **BR-15.4 Statement drill-down contract.** Level 1 returns balance keys whose Σ equals the cell value exactly (including rounding of the cell to the currency scale — none is applied since balances are already at scale). Level 2 returns lines for one key or the whole cell with keyset pagination (page size 500, max

1,000) ordered by (posting_date, posting_sequence, line_number); a `sum_of_lines` field accompanies every page and the final page's cumulative sum must equal the cell value; when dimension security filters lines, the response includes `filtered_amount` and the UI shows a warning.

- **BR-15.5 Cash flow (indirect) derivation.** Net income = P&L total for the range; operating adjustments = movements of rows tagged `operating_adjustment` (depreciation, provisions); working capital = −Δ of asset rows and +Δ of liability rows tagged `working_capital`; investing and financing = movements of tagged rows; cash = Δ of rows tagged `cash`. The statement fails validation (`CF_UNRECONCILED`) if net cash flow ≠ Δ cash by more than 0.00; untagged balance-sheet accounts appear in a "not classified" check row that must be zero.
- **BR-15.6 Read-model consistency.** A run captures HWM per ledger at start and evaluates only lines with `posting_sequence ≤ HWM`; two runs with equal HWM and definition version produce equal `result_hash`.
- **BR-15.7 Security.** Dimension security is applied as a filter on `ReportingBalance` queries (OPA returns allowed value sets per attribute, cached 60 s); a scheduled run renders per recipient; totals never include values the viewer cannot see; an "unrestricted" copy is produced only for recipients holding `View all dimensions`.
- **BR-15.8 Definition versioning.** Publishing creates an immutable version; runs reference the version; editing a published definition creates a draft of version n+1; retiring prevents new runs but keeps history.
- **BR-15.9 Comparative columns across calendars.** If `PY` period does not exist (calendar changed), the column shows blank with a footnote, not zero.
- **BR-15.10 Multi-ledger columns.** Columns with `ledger_code_override` are allowed only when the ledgers share the chart of accounts; otherwise `REPORT_LEDGER_INCOMPATIBLE`.
- **BR-15.11 Large cells.** Cells with > 50k balance keys store a query spec instead of the key list; drill level 1 re-executes the query at the run's HWM.
- **BR-15.12 Audit trail completeness.** Every `journal.posted` event must have an audit row within 60 s; a daily job counts events vs rows and alerts on any difference.
- **BR-15.13 Derived hierarchy resolution.** A statement run uses the hierarchy version effective on the run period's end date (or an explicit version); membership is materialized per version so re-running at the same HWM and version reproduces `result_hash`; a member matching several leaves blocks publish; for non-strict hierarchies unassigned members roll into an "Unassigned" node shown last with a footnote; a member's category change after publish takes effect only in the next version (or immediately when the hierarchy sets `follow_master_data = true`, which then records the change time in the run).

2.15.8 Acceptance criteria

- **AC-15.1** (FR-15.5) Given the demo entities' closed March 2026, when `BS-STD` and `PL-STD` run, then total assets = liabilities + equity (including current-period net income) to the cent and the P&L net equals `TB-STD` net of P&L accounts.
- **AC-15.2** (FR-15.5, BR-15.5) Given `CF-STD` for the same period, when run, then net cash flow equals the change in cash-tagged accounts to the cent and the "not classified" row is 0.00.
- **AC-15.3** (FR-15.7, BR-15.4) Given any cell of a run, when drilled to lines across all pages, then the cumulative sum equals the cell value exactly (property test over 1,000 random cells).
- **AC-15.4** (FR-15.7) Given a line from an AP-sourced entry (stub), when the source link is opened, then it resolves to the document route and the trace id opens in Tempo.
- **AC-15.5** (FR-15.11) Given a 200-row, 12-column definition, when exported to JSON and re-imported under a new code, then the two definitions produce identical cell values for the same parameters.
- **AC-15.6** (FR-15.9) Given a user permitted only cost center C200, when they run `TB-DIM`, then only C200 rows appear and drill-down lines are limited to C200 with `filtered_amount` shown on mixed cells.
- **AC-15.7** (FR-15.1, BR-15.6) Given postings continue during a run, when the run completes, then its `result_hash` is reproducible by re-running at the same HWM.
- **AC-15.8** (FR-15.8) Given a nightly schedule for 3 entities, when it fires, then PDF and XLSX outputs are in the library and the email is sent with attachments < 20 MB.
- **AC-15.9** (FR-15.10) Given a reopened period with a posting and re-close, when the audit trail runs, then it lists the reopen with both approvers, the posting, and the re-close in order with trace ids.
- **AC-15.10** (FR-15.6) Given 200k balances, when a 2,000 × 12 statement runs, then it completes in < 20 s.
- **AC-15.11** (Phase exit) Given the finance stakeholders' review, when they compare the Passage statement pack to the AX/SAP-derived expectations for the demo period, then every line ties and sign-off is recorded in the wave 15 issue.
- **AC-15.12** (FR-15.13, BR-15.13) Given a derived hierarchy over main account categories with `strict = true`, when a new postable account without a category is created, then publish of the next version is rejected with `REPORT_HIERARCHY_UNASSIGNED` naming the account and `hierarchy.unassigned` is emitted; after assigning the category, `BS-STD` run with the hierarchy as row set ties to `TB-STD` to the cent.
- **AC-15.13** (FR-15.14) Given the demo period with every P&L line carrying `FUNC_AREA`, when `PL-FUNC` runs, then its net result equals `PL-STD` net result to the cent and the check row is 0.00.

2.15.9 Test plan and fixtures

- Fixtures: `demo-fy2026-closed` (the dataset produced by waves 11–14 for the demo entities, three ledgers, EUR reporting currency); `definitions-std` (TB-STD, BS-STD, PL-STD, CF-STD, TB-DIM); `definition-200x12` (designer round-trip fixture); `mr-sample-definitions` (Management Reporter export of 10 definitions); `fsv-sample` (SAP `T011/FAGL_011PC` extract for 2 FSVs); `security-cases` (8 users with distinct dimension grants).

Test	Type	Target	Pass criterion
Statement tie-out	integration	standard definitions	BS balances; P&L = TB; CF reconciles
Drill-down sum property	property	drill API	Σ lines = cell for 1,000 random cells
Period expression unit tests	unit	evaluator	40 cases incl. adjustment periods and calendar change
Designer round-trip	contract	export/import	equal cell values
MR/FSV converters	unit	migration converters	10 MR + 2 FSV definitions convert; unsupported constructs listed
Security filtering	integration	OPA + reporting	8 users see only permitted values; totals exclude
Read-model lag	integration	event consumer	p95 < 10 s under 5k lines/s
Statement performance	performance	run engine	2,000 × 12 over 200k balances < 20 s
Reporting UI	Playwright + axe	designer, statements, drill	WCAG AA; keyboard drill works

Reconciliation: `ReportingBalance` = `ledger.Balance` nightly (0 diffs); TB closing = opening + movements; BS = L + E; CF net = Δ cash; audit rows = posted events.

2.15.10 Deployment and operations

Helm chart `passage-reporting` (Go service 3–8 pods, CloudNativePG `reporting-db` 3 instances with 1 read replica for statement runs, Kafka consumer group `reporting-balances`, renderer sidecar for PDF (Chromium headless) with 2 CPU limit). Config: `REPORTING_MAX_LAG_S=30`, `REPORTING_PAGE_SIZE=500`, `REPORTING_CELL_KEY_LIMIT=50000`, `REPORTING_RETENTION_CLOSED_YEARS=7`, `REPORTING_RETENTION_ADHOC_DAYS=90`. Migrations: reporting schema; backfill of `ReportingBalance` from the ledger via a one-time replay of `ledger.journal` topic (compacted snapshot + replay). Feature flags: `reporting.cash_flow`, `reporting.multi_ledger_columns`, `reporting.pdf`. Scheduled jobs: nightly `reporting-balance-verify`, nightly statement pack for demo entities (23:00 UTC), library retention sweep (weekly), audit completeness check (daily). SLOs: read-model lag p95 < 10 s; statement run p95 < 20 s (reference size); drill level 2 first page p95 < 300 ms; availability 99.9%. Alerts: lag > 60 s (P2), verify diff > 0 (P1), audit completeness diff (P1), PDF renderer failures > 5% (P3). Runbook: rebuild read model from ledger replay (≈ 15 min per 10M lines), stuck run cancellation, retention restore from object storage. Rollback: digest rollback; read model can be rebuilt from the journal at any time, so no data rollback is needed.

2.15.11 Migration from AX / SAP

Source	Passage entity	Notes
Management Reporter `ControlRowDetail`, `ControlColumnDetail`, `ControlTreeDetail`, `ControlReport`	ReportRow, ReportColumn, ReportTree, ReportDefinition	row link types (FD, dimension sets) → account_range/dimension_filter; CALC with + −×÷ converted; unsupported: XBRL tags, cross-report links (listed)
AX trial balance / inquiry saved queries	not migrated	standard definitions replace them
SAP `T011`, `T011T`, `FAGL_011PC`, `FAGL_011ZC`, `FAGL_011QT` (FSV)	ReportDefinition (BS/P&L), ReportRow	account intervals → account_range; debit/credit split nodes → two rows with sign rules; net result node → calculation row
SAP Report Painter libraries	not migrated	re-authored in the designer; conversion report lists them
AX `MainAccountCategory`, `DimensionHierarchy` (derived hierarchies), MR reporting trees	MainAccountCategory, DerivedHierarchy, HierarchyNode	category → `MainAccount.category_id`; derived hierarchy rules copied as membership rules
SAP account groups / G/L hierarchies (`SETHEADER` / `SETNODE` / `SETLEAF` class 0109), profit center groups (KCH1)	DerivedHierarchy (subject main_account or dimension_value)	set intervals → range rules
AX database log / SAP CDHDR history	not migrated	legacy audit remains in the archive; Passage audit begins at cutover

Validation: converted definitions run against migrated opening balances and are compared to the legacy final statements for the cutover period (tolerance 0.00 after sign normalization); conversion report reviewed by the controller.

2.15.12 Dependencies, risks, open questions

Depends on: wave 11 (journal events, balances), wave 12 (reporting currency amounts), wave 13 (tax audit events), wave 14 (period events, closing/opening flags, close-checklist hooks), wave 6 (dimension hierarchies for trees), wave 5 (audit store, Tempo trace links), wave 2 (OPA).

Risk	Likelihood	Impact	Mitigation
Read model lag at month-end peak invalidates "as of" statements	Medium	Medium	HWM stamping; close task waits for lag = 0; replica for runs
MR definitions use constructs with no equivalent	High	Low	Conversion report; manual re-authoring budgeted in migration plan
PDF rendering resource spikes	Medium	Low	Renderer sidecar limits; queue with concurrency 4 per pod
Stakeholder sign-off depends on accountant availability	Medium	High	Sign-off session scheduled at wave start; fixtures reviewed incrementally

Open questions: (1) Should the reporting read model live in `reporting-db` (default) or query `ledger.Balance` directly with replicas? Default separate read model; decide by wave 15 start. (2) Cash flow direct method in v1? Default no (indirect only); revisit at wave 36 (cash and bank). (3) Retention of ad-hoc runs 90 days vs 1 year? Default 90 days; decide by wave 15 exit.

2.15.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`reporting.max_lag_seconds`	tenant	int	30	0–600	Run blocked above lag unless `allow_stale` (§2.15.5)	Financial reporting parameters > data mart refresh	—
`reporting.page_size`	tenant	int	500	100–1,000	Drill level 2 page size (BR-15.4)	—	ALV page size
`reporting.cell_key_limit`	tenant	int	50,000	1,000–200,000	BR-15.11 key list vs query spec	—	—
`reporting.retention_closed_years`	legal entity	int	7	1–30	Library retention for closed-period runs (FR-15.8)	Report library retention	Spool retention (SP01)
`reporting.retention_adhoc_days`	tenant	int	90	7–730	Ad-hoc run retention	—	Spool retention
`reporting.email_attachment_limit_mb`	tenant	int	20	1–25	Attach vs link (FR-15.8)	Report distribution email settings	SCOT / SOST size limits
`reporting.share_link_expiry_days`	tenant	int	14	1–90	Shared link lifetime	—	—
`reporting.include_adjustment_periods` default	definition	bool	false	true / false	BR-15.1	MR column "Include closing/adjustment periods"	F.01 "special periods" selection
`reporting.opa_cache_seconds`	tenant	int	60	5–600	BR-15.7 security cache	MR tree security sync	Authorization buffer
`reporting.pdf_concurrency_per_pod`	tenant	int	4	1–16	Renderer queue	—	—
`definition.currency_type` default	definition	enum	acct	txn / acct / rpt	Column currency	MR report > Currency	F.01 currency type selection
`definition.rounding_display`	definition	enum	units	units / thousands / millions	Display rounding (cells keep exact values; rounding difference row shown)	MR "Rounding" (thousands)	F.01 "Display in thousands" / RFBILA00 scale
`definition.zero_rows`	definition	enum	hide	hide / show	Print-zero default	MR row format "Print zero"	FSV "Display zero balances"
`hierarchy.strict`	hierarchy	bool	true for account hierarchies	true / false	FR-15.13 publish check	Derived hierarchies (D365) validation	FSV "Not assigned" node check (OB58)
`hierarchy.follow_master_data`	hierarchy	bool	false	true / false	BR-15.13	—	Set maintenance dynamic sets
`audit_trail.export_hash_algorithm`	tenant	enum	sha256	sha256	Export integrity (FR-15.10)	—	—

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`schedule.timezone`	schedule	tz	entity time zone	IANA	Cron evaluation	Batch job time zone	SM36 server time

2.15.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Trial balance `TB-STD` / `TB-DIM`	statutory report	ledger, period range, currency type, dimension filter, tree	opening, debit, credit, closing, net per account (per dimension)	`ReportingBalance`	Trial balance list page; SSRS Trial balance / Trial balance summary	FAGLB03, S_ALR_87012277, S_ALR_87012301, F.08, Fiori Trial Balance
Balance sheet `BS-STD`	statutory report	ledger, period, comparison (PY), tree, currency type	rows per FSV-like definition, comparison, variance	`ReportingBalance`	MR Balance sheet; SSRS Balance control	F.01 (RFBILA00), S_ALR_87012284, Fiori Display Financial Statement
Income statement `PL-STD` / `PL-FUNC`	statutory report	ledger, period / YTD, comparison, tree	rows, current, PY, variance, %	`ReportingBalance`	MR Income statement; SSRS Dimension statement	S_ALR_87012284, S_PL0_86000028/29
Cash flow `CF-STD`	statutory report	ledger, period range	operating, investing, financing, net change, check rows	`ReportingBalance`	MR cash flow definitions	S_ALR_87012382 (FSV-based), S/4 Cash Flow app
Consolidated-by-entity view	operational report	legal entities, period, reporting currency	per-entity columns, sum column	`ReportingBalance`	MR multi-company report	S_ALR_87012284 multiple company codes
G/L account statement	operational report	account range, period, dimension	opening, lines, running balance, closing	`ReportingBalance` + journal lines	SSRS Ledger account balance	S_ALR_87012332
Document journal / line item journal	operational report	ledger, posting date range, voucher range	voucher, date, lines with accounts and amounts, totals	journal line API	SSRS Posted journals / Voucher	S_ALR_87012287, S_ALR_87012289, S_ALR_87012291
Statement drill-down (cell → balances → lines → document)	drill-down	run, row, column	keys, amounts, lines, source link, trace id	`StatementCell` + ledger API	MR drill-down; Transaction origin	FS10NA, FAGLB03 → FAGLL03 → FB03
Audit trail	operational report	ledger, period, kind, actor	occurred at, kind, actor, object, details, trace id	`AuditTrailEntry`	SSRS Audit trail; Database log	Fiori Audit Journal, RSSCD100 (change documents), S_ALR_87012287
Receivables/payables by profit center	operational report	ledger, period, profit center tree	AR/AP reconciliation balances per PC (requires FR-11.23 splitting)	`ReportingBalance`	—	S_AC0_52000887 / 52000888
Balance sheet by profit center / segment	statutory report	ledger, period, segment tree	BS per segment (IFRS 8)	`ReportingBalance` with splitting dimensions	MR tree by dimension	S_ALR_87013336, FSV by segment
Derived hierarchy membership	list inquiry	hierarchy, version	node, members, unassigned, overlaps	`HierarchyNode` membership	Derived financial hierarchies form	KDH3 / GS03 set display
Read-model lag KPI	KPI	ledger	lag seconds, HWM ledger vs reporting	`/reporting/lag`	MR data mart integration status	—
Report library	list inquiry	definition, entity, period	run, generated at, formats, hash, retention until	`StatementRun`	MR report library	Spool / SAP Analytics Cloud stories

2.15.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Financial statement pack (TB, BS, P&L, CF)	close checklist hook, schedule or user	header: entity, ledger, period, currency, definition code/version, as-of HWM; rows/columns; footer: result hash, generated at/by, "as of sequence N"	PDF, XLSX, CSV, interactive HTML	statement run id	annual statements: statutory (DE HGB §257 10 years, US 7 years); monthly packs: internal, 7 years
Trial balance printout	user / schedule	accounts with opening, debit, credit, closing; totals; hash	PDF, XLSX, CSV	run id	supporting record; 10 years for year-end TB
G/L account statement	user	account header, lines with running balance, closing	PDF, XLSX	run id	supporting record
Audit trail export	user / auditor request	filters used, entries, hash of the file	CSV, PDF	export run id	audit evidence; 10 years
Statement distribution email	schedule	subject with entity/period, body with as-of and link; attachments ≤ 20 MB	email (SMTP via wave 4)	—	distribution log retained 7 years
Definition export	user	JSON definition with schema version	JSON	definition version	configuration record

XBRL / iXBRL filings are wave 48.

2.15.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`reporting-balance-consumer`	continuous	ledger (Kafka partition)	upsert by event id with HWM per ledger	lag p95 < 10 s	DLQ after 5 attempts; replay	lag > 60 s → P2
`reporting-balance-verify`	daily 01:30	ledger	compare with `ledger.Balance` at HWM	10M balances < 10 min	rebuild ledger from replay	diff > 0 → P1
`statement-schedules`	per schedule cron (demo pack 23:00 UTC)	schedule × entity	run per (schedule, fire time)	2,000 × 12 < 20 s each	retry once; notify owner	failures > 5% → P3
`library-retention-sweep`	weekly Sunday 04:00	legal entity	deletes outputs past retention; legal hold respected	< 10 min	skip and retry	—
`audit-completeness-check`	daily 02:00	ledger	count events vs audit rows per day	< 2 min	—	diff > 0 → P1
`hierarchy-membership-refresh`	on account/dimension events; full nightly 03:00	hierarchy	rebuild per version	20k nodes < 1 min	—	unassigned > 0 on strict → notification

2.15.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
Maintain report definitions	definitions, row/column sets, trees, hierarchies CRUD, publish, import/export	Post ledger journals (for published statutory definitions)	Financial reporting analyst, Financial controller
Run financial statements	POST /reporting/runs	—	Accountant, Financial controller, FP&A analyst
View financial statements	GET runs, cells, outputs, library, lag	—	Finance roles, Executives (via OPA dimension grants)
Schedule financial statements	schedules CRUD	—	Financial reporting analyst
View all dimensions	bypass of dimension filtering for unrestricted copies (BR-15.7)	Maintain dimension security policies	CFO, Group controller, External auditor (time-boxed)
View audit trail	audit-trail query and export	Post ledger journals; Maintain period status	Internal auditor, External auditor, Financial controller
View ledger balances / transactions (drill)	balances, drill level 2	—	as wave 11

2.15.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
REPORT-0001 `REPORT_DEFINITION_NOT_PUBLISHED`	error	Definition {code} has no published version.	RunStatement	Publish the definition	FR-15.6
REPORT-0002 `REPORT_EXPRESSION_INVALID`	error	Row {row}: {detail} in expression {expr}.	FR-15.2 validation	Fix references	FR-15.2
REPORT-0003 `REPORT_PERIOD_NOT_FOUND`	error	Period {period} does not exist in ledger {ledger}.	Parameters	Choose an existing period	FR-15.6
REPORT-0004 `REPORT_STALE`	warning	Read model is {lag} s behind ledger {ledger}; results flagged as stale.	lag > max	Wait or rerun	FR-15.1
REPORT-0005 `REPORT_LEDGER_INCOMPATIBLE`	error	Ledgers {a} and {b} do not share a chart of accounts.	BR-15.10	Remove the column override	BR-15.10
REPORT-0006 `REPORT_NO_PERMITTED_VALUES`	error (403)	You have no permitted values for {attribute} in tree {tree}.	BR-15.7	Request dimension access	FR-15.9
REPORT-0007 `REPORT_DRILL_FILTERED`	warning	{amount} of this cell is outside your dimension permissions.	BR-15.4	None	FR-15.7
REPORT-0008 `CF_UNRECONCILED`	error	Net cash flow {cf} differs from change in cash {delta}.	BR-15.5	Tag accounts; fix "not classified" row	FR-15.5
REPORT-0009 `CF_UNCLASSIFIED_ACCOUNTS`	error	{count} balance-sheet accounts have no cash flow tag.	BR-15.5	Tag the listed accounts	FR-15.5
REPORT-0010 `REPORT_PY_PERIOD_MISSING`	info	Prior-year period not found; column left blank.	BR-15.9	None	BR-15.9
REPORT-0011 `REPORT_IMPORT_DUPLICATE`	error	Definition code {code} exists; use overwrite to replace.	FR-15.11	Rename or overwrite	FR-15.11
REPORT-0012 `REPORT_IMPORT_SCHEMA`	error	Schema version {v} is not supported (supported: {list}).	FR-15.11	Export from a compatible version	FR-15.11
REPORT-0013 `REPORT_HIERARCHY_UNASSIGNED`	error	{count} members are not assigned to any leaf of {hierarchy}: {sample}.	FR-15.13 strict	Assign members or relax `strict`	FR-15.13
REPORT-0014 `REPORT_HIERARCHY_OVERLAP`	error	Member {member} matches leaves {a} and {b}.	BR-15.13	Narrow membership rules	FR-15.13
REPORT-0015 `REPORT_PLFUNC_MISSING_FUNC_AREA`	warning	P&L lines totaling {amount} have no functional area.	FR-15.14	Add derivation steps (FR-11.22)	FR-15.14
REPORT-0016 `REPORT_RENDER_FAILED`	error	{format} rendering failed for run {run}.	Renderer	Retry; other formats unaffected	FR-15.6
REPORT-0017 `REPORT_ATTACHMENT_TOO_LARGE`	info	Output exceeds {limit} MB; a link was sent instead.	FR-15.8	None	FR-15.8

2.15.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Read-model update	5,000 lines/s peak	lag 2 / 10 / 30 s	5,000 lines/s	`reporting_balance` ≈ ledger Balance size (≈ 5 GB per 10M combinations × periods)	PK (ledger, period, account, combination, currency_type)	lag grows; runs flagged stale or wait
Statement run 2,000 × 12	500 runs/day; 50 concurrent at close	6 / 20 / 35 s	50 concurrent on replica	cells 60 B × 24k × 500 × 365 ≈ 260 GB raw → kept per retention, compressed	replica; per-run query plans by row kind	queued with concurrency limit; p95 40 s
Drill level 1 / 2	20k/day	80 / 300 / 500 ms	200 req/s	—	cell keys / query spec; journal line indexes	page size 250
PDF render	500/day	2 / 6 / 12 s	4 per pod	outputs 1 MB avg × 180k/year = 180 GB object storage	—	queue
Audit trail query (one month)	2k/day	200 ms / 1 s / 2 s	—	audit view 1 KB/event × 50M = 50 GB	(ledger_id, occurred_at)	narrowed date range enforced
Hierarchy membership rebuild	1,000 events/day	< 5 s per event; nightly < 1 min	—	small	(hierarchy_version, member_id)	—

2.15.20 Localization and compliance hooks

- Statutory statement layouts by country: HGB §266/§275 (DE balance sheet and P&L by nature/function), French PCG "bilan / compte de résultat" formats, UK FRS 102 formats, Indian Schedule III, Spanish PGC models, US GAAP/SEC Regulation S-X ordering; wave 48 packs ship `ReportDefinition` JSON per format and language.
- Presentation: number formats, negative-number display (parentheses vs minus), thousands scaling and statement language come from wave 5 locale settings; definitions carry translatable labels.
- Segment reporting (IFRS 8 / ASC 280) relies on document splitting (FR-11.23) and segment trees.
- XBRL/iXBRL tagging (ESEF in the EU, SEC EDGAR, HMRC CT600 iXBRL, MCA India) is wave 48; `ReportRow` reserves `taxonomy_concept` for tagging.
- Audit-file exports (FEC FR, GoBD DE, SAF-T) use the ledger line API; wave 15's audit trail adds the who/when/approval layer auditors request (ISA 315/330 journal-entry testing).
- Retention: statements and audit exports per §2.15.15; legal hold suspends `library-retention-sweep` per entity.

- Privacy: audit trail contains user ids and names; exports to external auditors are scoped by time-boxed duty and logged; data-subject requests mask user display names in exports older than retention when required (actor ids remain).
- Extension points: `reporting.definition_pack`, `reporting.renderer` (statutory PDF templates), `reporting.taxonomy_mapping`.

2.15.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.1.4 Setup — Chart of accounts → Derived financial hierarchies (R3) (AX register row 4)	PARTIAL-15	ADDED as FR-15.13	derived hierarchies over main account categories and dimension values, versioned, usable as row sets and trees; AC-15.12

Part 3 — Phase 3: Inventory & Procure-to-Pay (waves 16–20)

3.0 Phase overview

Business outcome at phase exit. At the end of wave 20 the demo plant buys, receives, pays for and values physical goods on Passage with no AX or SAP system in the loop. Every quantity movement is an immutable inventory transaction carrying physical and financial status plus storage and tracking dimensions (the InventTrans / MSEG role). Buyers run requisition → RFQ → agreement → purchase order under policy-driven approval (PurchReqTable / EBAN through PurchTable / EKKO). Receiving posts inventory and a GR/IR accrual to the universal journal; vendor invoices match 2-way or 3-way inside tolerances or route to an exception queue (VendInvoiceInfoTable / RBKP-RSEG). Open vendor items settle against payment runs that emit ISO 20022 pain.001 and NACHA files (VendSettlement / F110-REGUH). Inventory is valued under FIFO, LIFO, weighted average or standard cost with a close/recalculation job whose value report ties to the GL inventory accounts (InventClosing / CKMLCP). Suppliers acknowledge POs, send ASNs and submit invoices through a portal.

Phase-exit sign-off criteria

- Inventory on-hand projection equals SUM(inventory transactions) for a 1M-transaction fixture after a full rebuild; zero rows differ.
- Each of the 11 journal/posting types in this phase (movement, adjustment, transfer, BOM, counting, product receipt, vendor invoice, payment, settlement, close adjustment, revaluation) produces a balanced universal-journal entry; the daily subledger-to-GL reconciliation for inventory, GR/IR and AP control accounts reports a difference of 0.00 in accounting currency for the demo entity.
- 30 requisitions in the approval matrix route to the expected chain; 200 matching scenarios yield the expected match status and postings; 300 settlement scenarios post the expected discount and FX gain/loss; 100 costing sequences per method tie to the reference workbook to the cent.
- pain.001.001.09 output validates against the ISO 20022 XSD and is accepted by the bank test harness; a payment run killed mid-flight and resumed pays each open item exactly once.
- Inventory close on 1M transactions completes in under 20 minutes on the stage cluster (4 vCPU / 16 GiB inventory pod, Postgres 8 vCPU).
- Recorded end-to-end walkthrough (buy → receive → match → pay → close) signed off by both owners and the reviewing accountant.

Cross-wave data model summary

Entity	Owning service	Key relationships
InventoryTransaction (InventTrans / MSEG)	inventory (Go, `inventory` Postgres)	→ InventoryTransactionOrigin, → Dimension set, → Product (wave 8), → Site/Warehouse (wave 6)
InventoryTransactionOrigin (InventTransOrigin / MKPF)	inventory	1:N InventoryTransaction; references source document id+type (journal line, PO line, receipt line, production order line)
OnHand (InventSum / MARD+MCHB)	inventory (projection)	Keyed by product × dimension hash; rebuilt from InventoryTransaction
InventoryJournal / JournalLine (InventJournalTable/Trans)	inventory	Lines → InventoryTransactionOrigin; posts via ledger `Post(JournalEntry)` (wave 11)
Reservation	inventory	→ InventoryTransaction (status change), → batch/serial
CountPlan / CountSheet	inventory	→ JournalLine (variance)
PurchaseRequisition / Line (PurchReqTable / EBAN)	procurement (Go, `procurement` Postgres)	→ PurchaseOrderLine on consolidation; → Temporal approval workflow
RFQ / RFQReply (PurchRFQTable / EKKO type AN)	procurement	→ PurchaseOrder on award
PurchaseAgreement (PurchAgreementHeader / EKKO type K/L)	procurement	→ PurchaseOrderLine releases
TradeAgreement (PriceDiscTable / A017-KONP)	procurement	Vendor pricing lookup
PurchaseOrder / Line / Version (PurchTable/PurchLine, EKKO/EKPO)	procurement	→ InventoryTransaction (OnOrder), → Receipt, → VendorInvoice
ProductReceipt / Line (VendPackingSlipJour/Trans, MKPF/MSEG 101)	receiving (Go, `procurement` Postgres)	→ PurchaseOrderLine, → InventoryTransaction (Received), → GR/IR journal entry
VendorInvoice / Line (VendInvoiceInfoTable/Line → VendInvoiceJour, RBKP/RSEG)	payables (Go, `payables` Postgres)	→ ReceiptLine (3-way), → PurchaseOrderLine (2-way), → VendorTransaction, → GR/IR clearing
MatchResult	payables	→ VendorInvoiceLine; stores price/qty/total variance and status
VendorTransaction (VendTrans / BSIK→BSAK)	payables	Open item; → Settlement
Settlement (VendSettlement / BSE_CLR)	payables	Links payment/credit/invoice transactions with discount and FX
PaymentProposal / Payment / PaymentFile (LedgerJournalTrans payment journal, REGUH/REGUP)	payables	→ VendorTransaction, → bank account (wave 36 placeholder: bank master lite)
CostTransaction (InventCostTrans / CKMLPP-CKMLCR)	costing (Go, `inventory` Postgres schema `costing`)	→ InventoryTransaction; produced by close/recalc
CostingVersion / ItemCost (CostingVersion/InventItemPrice, MBEW)	costing	Standard and planned costs per product × site
SupplierPortalUser / Acknowledgement / ASN	supplier-portal (TypeScript BFF + `supplier` Postgres)	→ PurchaseOrder, → ProductReceipt (expected)

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
`dev.passage.inventory.transaction.posted.v1`	inventory	costing, planning (w22), analytics	txn id, origin, product, dims, qty, issue/receipt status, physical/financial dates
`dev.passage.inventory.onhand.changed.v1`	inventory	procurement (ATP for PO), sales (w27), warehouse (w31)	product, dim hash, avail_physical, avail_ordered, delta
`dev.passage.inventory.journal.posted.v1`	inventory	ledger (reconciliation), costing	journal id, type, line count, ledger entry id
`dev.passage.procurement.requisition.approved.v1`	procurement	procurement consolidation, budget control (w38)	req id, lines, approver chain
`dev.passage.procurement.po.confirmed.v1`	procurement	inventory (OnOrder txns), supplier-portal, EDI (w35)	po id, version, lines, vendor
`dev.passage.procurement.po.version.created.v1`	procurement	supplier-portal, receiving	po id, version, diff summary
`dev.passage.receiving.receipt.posted.v1`	receiving	inventory, payables (match candidates), costing	receipt id, lines, po lines, qty, GR/IR amount

Event (CloudEvents type)	Producer	Consumers	Payload summary
`dev.passage.payables.invoice.posted.v1`	payables	ledger, costing (PPV/invoice-cost settle), tax (w13)	invoice id, vendor, lines, match status, amounts
`dev.passage.payables.invoice.match.exception.v1`	payables	workflow (Temporal), notifications	invoice id, line, variance type, amounts
`dev.passage.payables.settlement.posted.v1`	payables	ledger, analytics	settlement id, items, discount, fx gain/loss
`dev.passage.payables.payment.sent.v1`	payables	bank connectivity (w36), supplier-portal, cash (w36)	payment id, file id, amount, method, vendor
`dev.passage.payables.payment.status.changed.v1`	payables	notifications, analytics	payment id, old/new status, pain.002 reason code
`dev.passage.costing.close.completed.v1`	costing	ledger, analytics	close id, period, adjustments, duration
`dev.passage.costing.cost.changed.v1`	costing	inventory, planning, product costing (w39)	product, site, version, old/new cost, effective date
`dev.passage.supplier.asn.submitted.v1`	supplier-portal	receiving, warehouse (w31)	asn id, po lines, expected qty, carrier

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
On-hand projection drifts from transactions under concurrency	Medium	High	Single-writer per product × dimension partition (advisory lock), nightly rebuild-and-compare job, alert on any diff	Engineering
Matching tolerance semantics differ between AX (per item/vendor %) and SAP (tolerance keys) and users expect their old behavior	High	Medium	Tolerance model supports both percentage and absolute limits at vendor, item, vendor×item and entity levels; migration maps both	Product
FX gain/loss and discount postings fail accountant review	Medium	High	300-scenario fixture set reviewed by accountant before wave 19 close; property test that settlement always balances	Product + accountant
Inventory close performance on large transaction history	Medium	Medium	Settlement by product partition in Temporal child workflows; incremental close from last closed date; 1M-txn perf gate	Engineering
Bank file rejection in production (schema or bank-specific rules)	Medium	High	XSD validation in CI, bank test harness, per-bank profile overrides (SEPA, CH, UK)	Engineering
Supplier portal becomes an attack surface	Medium	High	Separate Keycloak realm, OPA policies scoped to vendor id, rate limits 60 req/min, no inbound write to GL	Security

3.16 Wave 16: Inventory Ledger, Dimensions & On-Hand

3.16.1 Scope and business context

Wave 16 creates the inventory ledger that every later quantity-bearing module (purchasing, receiving, production, sales, warehouse) writes to. In AX this is InventTrans with InventTransOrigin, InventSum and InventDim; in SAP it is the material document (MKPF/MSEG) with stock segments in MARD (storage location), MCHB (batch) and MSKA/MSPR (special stock). Before this wave Passage has products with dimension groups (wave 8) and sites/warehouses (wave 6) but no quantities.

At the end of the wave a user can: load opening balances through a movement journal; move, adjust, transfer and count stock; reserve stock manually or by FEFO/FIFO policy; block or quarantine a batch; run a cycle-count plan; and see on-hand by any combination of site, warehouse, location, batch, serial, inventory status. Every posting produces a balanced universal-journal entry through the wave 11 `Post(JournalEntry)` API.

Out of scope: purchase/sales-driven transactions (waves 17–18, 27), production consumption (wave 21), warehouse work and license plates (wave 31), quality orders (wave 33), costing and value adjustments beyond the running cost price captured on receipt (wave 20). Inventory in this wave values at the item's current cost price on receipt and issues at the running average; wave 20 replaces that with the full costing engine.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX On-hand form (InventOnhand) requires choosing a dimension display (InventDimParm) and re-querying per combination; SAP MMBE shows plant/storage location/batch only, MB52 is a batch report; users export to Excel to get location × batch × status	InventSum is pre-aggregated per InventDimId and queried through fixed display sets; MARD/MCHB are separate tables with no common query	Single `GET /onhand` with `groupBy` any dimension subset, served from one projection; shell grid pivots client-side	Clicks to answer "how much of X in batch B at location L": 9 → 2; query p95 < 50 ms vs 3–10 s
InventSum drift from InventTrans after failed batch or kernel patch; fix is the "Check inventory on-hand" consistency check run at night with users off the system	Dual-write of InventTrans and InventSum under X++ ttsbegin with retry-on-deadlock; no automated compare	Projection updated in the same Postgres transaction; weekly rebuild-and-compare; diff event alerts	Reconciliation effort: 2 h/month manual → 0 (automated, 0 diffs); detection latency: weeks → < 24 h
SAP MIGO posts synchronously but stock overview in BW/analytics lags the nightly extract; AX entity store refresh is hourly at best	Analytics copies instead of event stream	`onhand.changed` CloudEvent via outbox within 5 s; any consumer (BI, MES, WMS) subscribes without middleware	Data latency to external systems: 1–24 h → < 5 s
Negative inventory silently allowed per InventModelGroup; errors surface at inventory close as unsettled issues (InventClosing "Cannot settle") months later	Check done at posting with a flag, no alerting; financial negative evaluated only at close	Negative check atomic under lock; negative rows surfaced on workspace tile and in daily reconciliation	Time to detect unauthorized negative stock: 30–90 days → same day; close-time settlement errors from negatives: 0
Cycle counting in AX needs the Counting journal, InventCountGroup setup and the "Create lines" dialog with 14 fields; SAP MI01/MI04/MI07 is three transactions and a printed count list	Journal-centric design with no plan object; count execution not mobile	`CountPlan` generates sheets on a cron; PWA count screen with blind count and offline queue; one `:post`	Steps from plan to posted variance: 3 transactions / ~25 clicks → 1 plan + 1 tap per line; counts per hour per operator ×3

3.16.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Inventory transaction record	InventTrans, InventTransOrigin, InventTransPosting; form InventTrans	MSEG/MKPF (material document), MATDOC in S/4; MB03/MIGO display	`inventory.InventoryTransaction`, `InventoryTransactionOrigin`; screen Inventory transactions	full
On-hand by dimension	InventSum, InventOnhand form, InventDimParm	MARD, MCHB, MSKA; MMBE stock overview, MB52	`inventory.OnHand` projection; On-hand inquiry	improved (any dimension combination, real time)

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Dimensions	InventDim (InventSiteId, InventLocationId, wMSLocationId, InventBatchId, InventSerialId, InventStatusId, LicensePlateId, ConfigId/Size/Color/Style)	Plant (WERKS), storage location (LGORT), batch (CHARG), serial (SERNR via SERI), special stock indicator, valuation type	`DimensionSet` (site, warehouse, location, batch, serial, status, config/size/color/style, license_plate reserved for w31)	full
Movement / adjustment journal	InventJournalTable/Trans types Movement, LossProfit; InventJournalCheckPost	MIGO movement types 501/502 (receipt w/o PO), 551/552 (scrapping), 561/562 (initial stock)	`InventoryJournal` type MOVEMENT, ADJUSTMENT	full
Transfer journal	InventJournalTrans type Transfer	301/302 (plant to plant), 311/312 (storage loc), MB1B	`InventoryJournal` type TRANSFER (paired issue/receipt txns)	full
BOM journal	InventJournalTrans type BOM (InventBOMJournal)	261/262 consumption to order, 201 cost center	`InventoryJournal` type BOM (explode wave 9 BOM, issue components, receive parent)	partial (no route/overhead until w21)
Counting	InventJournalTrans type Count, InventCountJournal, counting groups (InventCountGroup)	Physical inventory MI01/MI04/MI07, IKPF/ISEG, cycle counting indicator ABC (MIBC)	`CountPlan`, `CountSheet`, journal type COUNT	full
Item model group	InventModelGroup (InventModel FIFO/LIFO/WeightedAvg/Std, PhysNegative, FinancialNegative, PostPhysicalInvent)	MBEW price control (S/V), negative stock allowed per plant/storage location (OMJ1)	`ItemModelGroup` (wave 8 entity, enforced here)	full
Reservation	InventTrans status ReservPhysical/ReservOrdered, InventMovement, reservation form, FEFO date-controlled (InventBatch expiry, PdsFEFO)	Stock reservations (MB21, RESB), batch determination (FEFO via strategy), batch status	`Reservation` with policy FEFO/FIFO/MANUAL	full
Blocking / quarantine	Inventory blocking (InventBlocking), quarantine order (InventQuarantineOrder), inventory status (D365)	Blocked stock (movement 344/343), quality inspection stock, batch restricted status	Inventory status `AVAILABLE`, `BLOCKED`, `QUARANTINE`; `Blocking` record	full
Unit conversion on movement	UnitOfMeasureConversion (intra/inter-class, product-specific)	MARM alternative UoM, conversion on MIGO	Wave 6/8 UoM service; quantity stored in inventory unit plus entered unit and qty	full
Inventory closing/recalc	InventClosing	CKMLCP	Deferred to wave 20	dropped-with-reason (wave 20)
Transfer orders	InventTransferTable/InventTransferLine, InventTransferJour (shipment, receipt), in-transit warehouse (InventLocation type Transit), Transfer order parameters, Transfer order picking release	Stock transport order (ME21N item category U, document type UB intra-company / NB cross-company), replenishment delivery NL/NLCC via VL10B, movement 351/641 to stock in transit, GR 101, MB5T	`TransferOrder`, `TransferOrderLine`, `TransferShipment`, `TransferReceipt`; in-transit warehouse dimension (FR-16.15, FR-16.16)	full (cross-company STO pricing and intercompany invoice in wave 40; picking work in wave 31)
Vendor consignment	D365 consignment replenishment orders (ConsignmentReplenishmentOrder), inventory ownership change journal (InventOwnershipChangeJournal), owner inventory dimension (InventDim.InventOwnerId), Consignment parameters	Item category K, special stock K (MKOL), MB54, PI for consignment MIK1, settlement MRKO, stock determination OSPX/OSPW, consignment info record price	`owner` dimension, `ConsignmentReplenishmentOrder`, journal type OWNERSHIP_CHANGE (FR-16.17); settlement invoice in wave 18 FR-18.16; portal view wave 20 FR-20.22	full
Customer consignment	Not standard in AX 2012 (warehouse per customer convention)	Special stock W (MSKU), MB58, fill-up 631 / issue 633	Warehouse type `CUSTOMER_CONSIGNMENT` with `customer_id` (FR-16.17); fill-up and consumption documents in wave 27	partial (documents in wave 27)
Sales-order / project stock	Not in AX 2012 (marking only)	Special stock E (MSKA) / Q (MSPR), valuated/non-valuated, MBBS	`segment` dimension (SALES_ORDER / PROJECT + reference) (FR-16.20)	full (value report by segment in wave 20)
Tag counting	InventJournalTable type TagCounting (InventJournalTrans with TagId), Create tag counting, Transfer to counting journal	No direct equivalent (MI09 count without reference, count tags printed from MI21)	Journal type TAG_COUNT with `CountTag` rows → COUNT journal (FR-16.18)	full
ABC classification	InventTable ABCRevenue/ABCValue/ABCMargin/ABCContribution, periodic ABC classification (InventABCUpdate), ABC classification report	MIBC ABC analysis for cycle counting (MARC-ABCIN), cycle counting indicators OMCO	`AbcClassification` per product × site with periodic job (FR-16.19); feeds CountPlan selection	full
Statistical inventory sampling	n/a	MIS1 / MIS2 inventory sampling	Not provided; CountPlan `sample_pct` counts a random sample without extrapolation	dropped-with-reason (see 3.16.21)

3.16.3 Functional requirements

- **FR-16.1 Immutable inventory transactions** — Must. The system shall record every quantity movement as an append-only `InventoryTransaction` row linked to an `InventoryTransactionOrigin`; updates only change status fields and dates, never quantity or identity.
    - Status changes are written as a new `InventoryTransactionStatusChange` audit row; the current status lives on the transaction for query speed.
    - A transaction has exactly one of `issue_status` or `receipt_status` set (sign determined by quantity).
    - Corrections are new transactions with opposite sign referencing `corrects_transaction_id`.
    - Partition key: tenant_id, product_id; up to 50M transactions per tenant supported before range partitioning by `physical_date` year is required.
- **FR-16.2 Transaction status model** — Must. The system shall implement the AX issue/receipt status model: issue statuses `ON_ORDER → RESERVED_ORDERED → RESERVED_PHYSICAL → PICKED → DEDUCTED → SOLD`; receipt statuses `ORDERED → ARRIVED → REGISTERED → RECEIVED → PURCHASED`.
    - Physical statuses (affect physical on-hand): PICKED, DEDUCTED, REGISTERED, RECEIVED. Financial statuses: SOLD, PURCHASED.
    - Transitions may skip forward (ORDERED → RECEIVED directly) but never go backward except via the explicit `Unpick`, `Unregister`, `CancelReservation` commands.
    - SAP mapping on migration: 101 → RECEIVED, 102 → reversal, 261 → DEDUCTED, 301 → DEDUCTED+RECEIVED pair, 561 → RECEIVED+PURCHASED opening.
- **FR-16.3 On-hand projection** — Must. The system shall maintain `OnHand` per product × dimension set with the columns `posted_qty, posted_value, received, registered, arrived, ordered, deducted, picked, reserved_physical, reserved_ordered, on_order` and computed `avail_physical`, `avail_ordered`,

`physical_invent`.

  - `physical_invent = posted_qty + received + registered − deducted − picked`.
  - `avail_physical = physical_invent − reserved_physical`.
  - `avail_ordered = avail_physical + ordered + arrived − on_order − reserved_ordered`.
  - Projection is updated in the same Postgres transaction as the inventory transaction (not via event) to guarantee the negative-stock check; the event is emitted from the outbox afterward.
  - Query at any dimension subset: aggregation over the finest-grain rows, p95 < 50 ms for ≤ 10k rows.

- **FR-16.4 Negative inventory policy** — Must. The system shall reject an issue that would make `avail_physical` (physical negative forbidden) or `avail_ordered` (financial negative forbidden) negative when the item model group forbids it.

  - Enforcement uses `SELECT … FOR UPDATE` on the OnHand row; the check and the write are atomic.
  - When allowed, negative on-hand is flagged in the On-hand inquiry and in the daily reconciliation.
  - Error code `INV-NEG-001` with product, dimension set and shortfall quantity.

- **FR-16.5 Inventory dimensions** — Must. The system shall store the dimension set as a normalized `DimensionSet` row keyed by a SHA-256 hash of its canonical values (AX InventDimId pattern), with active dimensions determined by the product's storage and tracking dimension groups.

  - Dimensions: site, warehouse, location, batch, serial, inventory_status, config, size, color, style, license_plate (reserved), owner (activated by FR-16.17 consignment).
  - Site is mandatory on every transaction; warehouse mandatory if the storage dimension group marks it mandatory.
  - A serial-tracked product requires quantity 1 per transaction and a unique serial per product in physical stock (duplicate serial → `INV-SER-001`).

- **FR-16.6 Inventory journals** — Must. The system shall support journal types MOVEMENT, ADJUSTMENT, TRANSFER, BOM, COUNT, OPENING_BALANCE with header/lines, validate, approval workflow (Temporal template from wave 4), and post.

  - Posting creates transactions, updates on-hand and calls `Post(JournalEntry)` with the journal id as idempotency key; a ledger failure rolls back the inventory posting (saga compensating step with explicit reversal if the ledger acknowledged).
  - Journal offset accounts: MOVEMENT requires a user-entered offset account; ADJUSTMENT uses the posting profile's inventory adjustment account; TRANSFER posts no GL unless cost differs by site (wave 20).
  - Maximum 10,000 lines per journal; posting throughput target 2,000 lines/s.

- **FR-16.7 Reservations** — Must. The system shall reserve quantity against on-hand (RESERVED_PHYSICAL) or expected receipts (RESERVED_ORDERED) by manual choice or by policy FIFO, FEFO, LIFO, with batch/serial selection.

  - FEFO orders candidate batches by `expiry_date ASC, receipt_date ASC`; batches with status BLOCKED or QUARANTINE or `expiry_date < required_date + shelf_life_margin_days` are excluded.
  - Reservation holds for `reservation_ttl_hours` (default 0 = until released or consumed).
  - Over-reservation is rejected; partial reservation returns the shortfall.

- **FR-16.8 Blocking and quarantine** — Must. The system shall allow blocking a quantity, batch or serial with reason code; blocked quantity is excluded from `avail_physical` and from policy-based reservation.

  - Blocking is implemented as inventory status change (status dimension) producing a status-transfer transaction pair.
  - Quarantine requires a disposition (release, scrap, return) to exit.

- **FR-16.9 Cycle counting** — Must. The system shall generate count sheets from a `CountPlan` (ABC class frequency, location range, last-count age, random sample) and post counted variances as COUNT journal lines.

  - Counting freezes on-hand for the counted dimension set (`count_lock`) while a sheet is open unless the plan allows movement with recount.
  - Variance tolerance per plan (qty % and value); above tolerance requires supervisor approval before posting.
  - Blind count option hides expected quantity.

- **FR-16.10 Unit conversions on movements** — Must. The system shall accept quantity in any unit with a conversion to the product's inventory unit, storing `qty` (inventory unit), `entered_qty`, `entered_unit`.

  - Rounding to the product's inventory unit decimals (default 2, max 6); residual beyond precision is rejected (`INV-UOM-002`).
  - Catch-weight is deferred to wave 24.

- **FR-16.11 Physical and financial dates** — Must. The system shall store `physical_date` (goods movement) and `financial_date` (value recognition) separately; the ledger entry uses `financial_date` and is validated against period control (wave 14) for module INVENTORY.

- **FR-16.12 Daily reconciliation** — Must. The system shall run a job that compares SUM(OnHand.posted_value) by inventory account and dimension to the ledger balance projection and raises `dev.passage.inventory.reconciliation.diff.v1` on any non-zero difference.

- **FR-16.13 Inventory transaction inquiry** — Should. The system shall provide a filterable, exportable inquiry over transactions with origin, status, dates and GL voucher drill-down.

- **FR-16.14 Opening balance load** — Should. The system shall load opening balances from AX InventSum/InventTrans or SAP MARD/MCHB/MBEW extracts through the wave 10 migration framework into an OPENING_BALANCE journal.

- **FR-16.15 Transfer orders with in-transit stock** — Must. The system shall provide a `TransferOrder` document (AX InventTransferTable; SAP stock transport order UB) with header `from_warehouse_id`, `to_warehouse_id`, `transit_warehouse_id`, `ship_date`, `receipt_date`, `delivery_terms`, `mode_of_delivery`, status `DRAFT, CREATED, SHIPPED, PARTIALLY_RECEIVED, RECEIVED, CANCELLED`, and lines with `qty`, `shipped_qty`, `received_qty`, `remaining_qty`, `underdelivery_pct`, `overdelivery_pct`, `close_short` and tracking dimensions.

  - Creation writes an ON_ORDER issue at the from-warehouse and an ORDERED receipt at the to-warehouse (supply and demand for wave 22 planning).
  - `:ship` (AX "Ship transfer order", SAP VL10B + PGI 641) posts in one database transaction a DEDUCTED issue at the from-dimension and a RECEIVED receipt at the transit dimension (`warehouse = transit_warehouse_id`, status `IN_TRANSIT`); cost of the transit receipt equals the cost of the issue (BR-16.3).
  - `:receive` (AX "Receive", SAP MIGO 101 against STO) posts a DEDUCTED issue at the transit dimension and a RECEIVED receipt at the to-dimension with the registered batch/serial/location; under-delivery within `underdelivery_pct` with `close_short=true` closes the line; over-receipt beyond

`overdelivery_pct` is rejected with `INV-TRF-002`.
  - `transit_warehouse_id` is mandatory when the legal-entity parameter `transfer_use_transit=true`; otherwise `:ship` performs ship-and-receive in one step (SAP one-step 301).
  - Transfer order picking release emits `dev.passage.inventory.transfer.release.v1` for wave 31 warehouse work (and wave 28 outbound picking); until those waves exist the shipment is posted from the transfer order screen or PWA.
  - Cancellation of unshipped remaining quantity cancels the ON_ORDER/ORDERED pair; delete of CREATED orders with no postings is allowed (AX "Delete transfer orders").
  - Documents: transfer shipping document and transfer receiving document (3.16.15).

- **FR-16.16 In-transit stock inquiry** — Should. The system shall list stock in transit by transfer order, product, from/to warehouse, ship date and days in transit (SAP MB5T), sourced from OnHand rows at transit warehouses, with drill-down to the transfer order.

- **FR-16.17 Ownership dimension, vendor consignment and customer consignment** — Must. The system shall activate the `owner` dimension (D365 InventOwnerId; SAP special stock indicator K) through the tracking dimension group, with `owner = legal entity` by default or a vendor id for vendor-owned stock.

  - `ConsignmentReplenishmentOrder` (vendor, lines product/qty/date/warehouse) receives vendor-owned quantity as RECEIVED transactions with `owner = vendor`, `posted_value = 0`, no ledger entry and no GR/IR (quantity-only stock).
  - Journal type OWNERSHIP_CHANGE moves quantity from `owner = vendor` to `owner = legal entity` as a paired issue/receipt at the consignment price (wave 17 trade agreement with `consignment=true`, else the replenishment order price), posting Dr inventory / Cr consignment accrual (posting profile account `consignment_accrual`; SAP OBYC KON) and emitting `dev.passage.inventory.ownership.changed.v1`; wave 18 FR-18.16 consumes it to create the consignment settlement invoice (SAP MRKO).
  - When the item model group sets `auto_ownership_change_on_issue=true`, an issue that would consume vendor-owned stock creates the ownership change automatically before the issue; stock determination order (`OWN_FIRST` / `CONSIGNMENT_FIRST`, SAP OSPX) is set per warehouse.
  - A periodic ownership-change batch (3.16.16) converts consumption-driven and age-driven (`max_consignment_days`) quantities.
  - Vendor-owned stock is counted in cycle counts (SAP MIK1) and is excluded from the inventory value report; consignment inventory reports (3.16.14) list vendor-owned quantity by vendor, warehouse and age.
  - Customer consignment: a warehouse of type `CUSTOMER_CONSIGNMENT` carries `customer_id`; stock there remains `owner = legal entity`, is reported by customer (SAP MB58), and is consumed by wave 27 consignment fill-up/issue documents.

- **FR-16.18 Tag counting** — Should. The system shall support two-step tag counting (AX tag counting journal): a `CountTag` range is generated per plan or warehouse with sequence `TAG-`, each tag printed with a barcode; a tag entry records product, dimensions and counted quantity per tag with status `USED`, `VOID`, `MISSING`; "transfer to counting journal" aggregates USED tags by product × dimension set into a COUNT journal where expected quantity is compared; missing and void tags are listed before transfer and each tag number is used at most once (`INV-TAG-001`).

- **FR-16.19 ABC classification** — Should. The system shall compute `AbcClassification` per product × site for criteria `VALUE` (on-hand value), `REVENUE`, `MARGIN`, `CARRYING_COST` over a lookback (default 365 days) with cumulative-percentage thresholds (default A 80 %, B 15 %, C 5 %, must sum to 100 — `INV-ABC-001`), store the class on the product (AX InventTable ABC fields; SAP MARC-ABCIN), run it as a monthly job and expose a report; CountPlan selection by `abc_class[]` uses the latest run.

- **FR-16.20 Segregated stock (sales-order and project stock)** — Should. The system shall support a `segment` dimension with `segment_type` (`NONE`, `SALES_ORDER`, `PROJECT`) and `segment_ref_id` (SAP special stock E / Q), activated per tracking dimension group, with an item model group flag `segment_valuated` (valuated vs non-valuated special stock); policy-based reservation for a demand carrying a segment considers only matching-segment stock first, then unsegmented stock when `segment_fallback=true`; the wave 20 value report groups by segment (SAP MBBS).

3.16.4 Data model

InventoryTransaction

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS
origin_id	uuid	yes	FK InventoryTransactionOrigin
product_id	uuid	yes	FK product (wave 8)
dimension_set_id	bytea(32)	yes	FK DimensionSet hash
qty	numeric(28,6)	yes	signed; inventory unit; ≠ 0
entered_qty / entered_unit	numeric(28,6) / text	yes	as keyed
issue_status	enum	cond.	ON_ORDER, RESERVED_ORDERED, RESERVED_PHYSICAL, PICKED, DEDUCTED, SOLD; null for receipts
receipt_status	enum	cond.	ORDERED, ARRIVED, REGISTERED, RECEIVED, PURCHASED; null for issues
physical_date	date	yes	goods movement date
financial_date	date	no	set when financial status reached
cost_amount_physical	numeric(19,4)	no	accounting currency, at physical posting
cost_amount_posted	numeric(19,4)	no	accounting currency, financial
cost_amount_adjustment	numeric(19,4)	no	written by wave 20 close
ledger_entry_id	uuid	no	voucher in universal journal
corrects_transaction_id	uuid	no	reversal/correction link
marked_transaction_id	uuid	no	explicit issue→receipt marking (wave 20)
created_at	timestamptz	yes	append-only

InventoryTransactionOrigin

Field	Type	Required	Notes
id	uuid	yes	PK
source_type	enum	yes	INVENT_JOURNAL_LINE, PURCH_LINE, RECEIPT_LINE, SALES_LINE, PROD_LINE, COUNT_LINE, TRANSFER_LINE, OPENING
source_id	uuid	yes	polymorphic FK; unique (source_type, source_id)

Field	Type	Required	Notes
product_id	uuid	yes	denormalized
reference_number	text	no	PO number, journal number
direction	enum	yes	ISSUE / RECEIPT
created_at	timestamptz	yes	

OnHand

Field	Type	Required	Notes
tenant_id, product_id, dimension_set_id	composite PK	yes	
posted_qty, posted_value	numeric	yes	financially posted
received, registered, arrived, ordered	numeric	yes	receipt-side buckets
deducted, picked, reserved_physical, reserved_ordered, on_order	numeric	yes	issue-side buckets
last_txn_seq	bigint	yes	monotonic per row for rebuild diff
updated_at	timestamptz	yes	

InventoryJournal / InventoryJournalLine

Field	Type	Required	Notes
id, journal_number	uuid, text	yes	number sequence `INVJ-` gap-free optional
journal_type	enum	yes	MOVEMENT, ADJUSTMENT, TRANSFER, BOM, COUNT, OPENING_BALANCE
status	enum	yes	DRAFT, SUBMITTED, APPROVED, POSTING, POSTED, REJECTED, CANCELLED
posting_date	date	yes	financial_date default
offset_account_id	uuid	cond.	MOVEMENT mandatory
line: line_no, product_id, dimension_set_id, qty, entered_unit, cost_price, counted_qty, expected_qty, to_dimension_set_id (TRANSFER), bom_id (BOM)			
ledger_entry_id	uuid	no	after posting

Reservation: id, product_id, dimension_set_id, qty, demand_type, demand_id, policy (MANUAL/FIFO/FEFO/LIFO), status (ACTIVE, RELEASED, CONSUMED), expires_at, created_by.

CountPlan / CountSheet: plan id, name, selection (abc_class[], warehouse_ids[], location_range, max_days_since_count, sample_pct), frequency (cron), tolerance_qty_pct, tolerance_value, blind (bool); sheet id, plan_id, status (OPEN, COUNTED, APPROVED, POSTED), lines (product, dims, expected_qty, counted_qty, variance_qty, counted_by, counted_at).

Invariants: Σ qty over transactions with physical status for a (product, dimension set) = OnHand.physical_invent; posted_value changes only through financial status transitions and wave 20 adjustments; a serial has at most one open physical receipt not matched by an issue; DimensionSet hash is deterministic across services (shared Go package `passage/invdim`). Indexing: InventoryTransaction (tenant, product, dimension_set, physical_date), (origin_id), partial index on open statuses; OnHand clustered by (tenant, product); monthly range partitions on InventoryTransaction by physical_date when a tenant exceeds 20M rows.

3.16.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/inventory/journals	Create journal header	client key	InventoryJournalMaintain
POST	/api/v1/inventory/journals/{id}/lines:bulk	Add up to 10,000 lines	per-line client key	InventoryJournalMaintain
POST	/api/v1/inventory/journals/{id}:validate	Dry-run validation	n/a	InventoryJournalMaintain
POST	/api/v1/inventory/journals/{id}:post	Post (starts Temporal `PostInventoryJournal`)	journal id	InventoryJournalPost
GET	/api/v1/inventory/onhand?product=&site=&warehouse=&batch=&groupBy=	On-hand inquiry	n/a	InventoryInquire
GET	/api/v1/inventory/transactions?product=&status=&from=&to=	Transaction inquiry	n/a	InventoryInquire
POST	/api/v1/inventory/reservations	Reserve (manual or policy)	client key	InventoryReserve
DELETE	/api/v1/inventory/reservations/{id}	Release	idempotent	InventoryReserve
POST	/api/v1/inventory/blockings	Block qty/batch/serial	client key	InventoryBlock
POST	/api/v1/inventory/count-plans/{id}:generate	Create count sheets	plan+date	CycleCountMaintain
POST	/api/v1/inventory/count-sheets/{id}:post	Post variance journal	sheet id	CycleCountPost
POST	/internal/v1/inventory/transactions:apply	Service-to-service status transition (used by waves 17–20)	origin+step	service account
POST	/internal/v1/inventory/onhand:rebuild	Rebuild projection from transactions	run id	InventoryAdmin

Commands: `CreateJournal`, `AddLines`, `PostJournal` (validates period control, negative policy, dimension mandatory flags, UoM conversion existence, offset account structure), `Reserve` (validates qty > 0, policy-eligible batches, status), `Block`, `ApplyTransactionStep` (validates monotonic status order, quantity ≤ remaining at previous status).

Event	Trigger	Payload fields	Consumers
`dev.passage.inventory.transaction.posted.v1`	any transaction insert or status change	transaction_id, origin, product_id, dimension_set, qty, issue_status, receipt_status, physical_date, financial_date, cost_amount_*	costing, analytics, planning
`dev.passage.inventory.onhand.changed.v1`	OnHand row update	product_id, dimension_set, avail_physical, avail_ordered, delta_bucket, delta_qty	procurement, sales, warehouse
`dev.passage.inventory.journal.posted.v1`	journal reaches POSTED	journal_id, type, lines, ledger_entry_id, posting_date	ledger reconciliation, costing
`dev.passage.inventory.reservation.created.v1` / `.released.v1`	reservation lifecycle	reservation_id, demand, product, dims, qty, policy	sales, production

Event	Trigger	Payload fields	Consumers
`dev.passage.inventory.reconciliation.diff.v1`	daily job finds diff ≠ 0	account, dimension, inventory_value, ledger_balance, diff	alerting
`dev.passage.inventory.transfer.shipped.v1` / `.received.v1` / `.release.v1`	transfer order ship / receive / picking release	transfer_order_id, lines (product, dims, qty), from/to/transit warehouse, ship_date, receipt_date	planning (w22), warehouse (w31), fulfillment (w28), analytics
`dev.passage.inventory.ownership.changed.v1`	OWNERSHIP_CHANGE journal posted	journal_id, vendor_id, lines (product, dims, qty, consignment_price, amount_acct), ledger_entry_id	payables (w18 consignment settlement), supplier-portal (w20), analytics

3.16.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Inventory management workspace	Tiles: journals awaiting approval, negative on-hand, open count sheets, reconciliation diffs	counts, last run	InventoryInquire
On-hand inquiry	Filter, group by any dimension, drill to transactions, export	product, site, warehouse, location, batch, serial, status, physical_invent, avail_physical, avail_ordered, posted_value	InventoryInquire
Inventory journal (header + grid)	New, add lines, validate, submit, post, copy, import CSV	journal_number, type, status, line grid: product, dims, qty, unit, cost_price, offset account	InventoryJournalMaintain
Transaction inquiry	Filter by status/date/origin, drill to voucher	reference, status, qty, dates, cost amounts	InventoryInquire
Reservation dialog	Reserve by policy or pick batches/serials	candidate batches with expiry, available qty	InventoryReserve
Count plan / count sheet	Generate, assign, count (mobile-friendly), approve, post	expected (hidden when blind), counted, variance	CycleCountMaintain
Blocking	Block / release with reason	product, dims, qty, reason, expected release date	InventoryBlock

Interaction rules: grid supports keyboard row entry (Tab across cells, Ctrl+Enter to add line), paste from spreadsheet with dimension column mapping, inline validation on blur for product/dimension and deferred validation (`:validate`) for on-hand and period checks; column sets and filters personalized per user (wave 5 shell). Count sheet works on the PWA shell with offline queue of counted quantities (synced on reconnect, conflict = latest count wins with audit).

3.16.7 Business rules and edge cases

- **BR-16.1** Status transitions are monotonic per transaction: issue `ON_ORDER(0) < RESERVED_ORDERED(1) < RESERVED_PHYSICAL(2) < PICKED(3) < DEDUCTED(4) < SOLD(5)`; receipt `ORDERED(0) < ARRIVED(1) < REGISTERED(2) < RECEIVED(3) < PURCHASED(4)`. Backward moves only through named commands that create a status-change audit row and an on-hand delta.
- **BR-16.2** Splitting: applying a step for a quantity less than the transaction quantity splits the transaction into two rows with the same origin (AX InventTrans split); sum of split quantities equals the original.
- **BR-16.3** Transfers create an issue (DEDUCTED) at the from-dimension and a receipt (RECEIVED) at the to-dimension in one database transaction; cost of the receipt equals the cost of the issue at the time of posting.
- **BR-16.4** Negative-policy check reads `avail_physical` under row lock; two concurrent issues against the last unit serialize, the second fails with `INV-NEG-001`. Deadlock avoidance: locks acquired in ascending (product_id, dimension_set_id) order within a journal.
- **BR-16.5** Rounding: quantities rounded half-up to the inventory unit's decimals; values to the accounting currency's minor unit (wave 12 rounding rules). Transfer of 1/3 of a 1.000-unit batch under 3 decimals yields 0.333 and leaves 0.667.
- **BR-16.6** Physical date may be in a closed inventory period only if financial date is in an open period and the item model group allows backdated physical posting; otherwise `INV-PER-001`.
- **BR-16.7** FEFO tie-break: equal expiry → earliest receipt_date → lowest batch_id (lexical). Batches with no expiry sort last.
- **BR-16.8** Count sheet posting posts only lines where counted_qty differs from expected_qty at the time of posting (not generation); movements after generation trigger a recount flag when the plan has `freeze=false`.
- **BR-16.9** Serial-tracked products reject quantities other than ±1 and reject a receipt of a serial already in physical stock for the same product (across all warehouses in the legal entity).
- **BR-16.10** Blocking never changes physical_invent; it moves quantity between inventory statuses so `physical_invent` is unchanged and `avail_physical` decreases.
- **BR-16.11** Journal state machine: `DRAFT → SUBMITTED → APPROVED → POSTING → POSTED`; `SUBMITTED → REJECTED → DRAFT`; `DRAFT|REJECTED → CANCELLED`. POSTING is a crash-safe intermediate state; the Temporal workflow owns it and is the only writer.
- **BR-16.12** Reversal of a posted journal creates a new journal of the same type with negated quantities, referencing the original; original rows are never modified.
- **BR-16.13** Transfer order state machine: `DRAFT → CREATED → SHIPPED → PARTIALLY_RECEIVED → RECEIVED`; `CREATED|SHIPPED → CANCELLED` only for unshipped remaining quantity (shipped quantity in transit must be received or returned by a reverse transfer); `received_qty ≤ shipped_qty` per line; the transit dimension never carries reservations.
- **BR-16.14** Vendor-owned stock (`owner ≠ legal entity`) never carries `posted_value` and never reaches a financial status; an OWNERSHIP_CHANGE line cannot exceed the vendor-owned physical quantity at the dimension set (`INV-OWN-002`); the ownership change is a paired issue/receipt whose only difference is the owner dimension, so `physical_invent` summed across owners is unchanged.
- **BR-16.15** Tag counting: a tag number is consumed once per count tag range; a USED tag with zero quantity is a valid count of zero; the transfer to the COUNT journal rejects when any tag is still OPEN unless the supervisor marks remaining tags MISSING.
- **BR-16.16** ABC thresholds are cumulative percentages of the ranked criterion; ties at a boundary go to the higher class; products with zero activity in the lookback are class `C` unless `abc_zero_class` is set to `N` (not classified).
- **BR-16.17** Segmented stock: an issue for demand with segment S may consume stock with segment S, or unsegmented stock when `segment_fallback=true`; it never consumes stock of a different segment; transfers between segments are TRANSFER journal lines with `to_dimension_set_id` differing only in segment.

3.16.8 Acceptance criteria

- **AC-16.1** (FR-16.4) Given item model group forbids physical negative and on-hand 1,000 units, when 10,000 concurrent issues of 1 unit are submitted, then exactly 1,000 succeed, 9,000 return `INV-NEG-001`, and on-hand is 0.
- **AC-16.2** (FR-16.3) Given 1,000,000 posted transactions, when the rebuild job runs, then the rebuilt OnHand table equals the live table row-for-row and the job completes in < 10 minutes.
- **AC-16.3** (FR-16.6) Given a MOVEMENT journal of 100 lines with offset account 500100, when posted, then one ledger entry exists with Σ debits = Σ credits, inventory account per posting profile, and `ledger_entry_id` set on all 100 transactions.
- **AC-16.4** (FR-16.7) Given 100 batches with distinct expiry dates and a FEFO reservation of 50 units, when reserved, then the batch with the earliest non-blocked expiry is selected first and the reservation spans batches in expiry order until 50 units are reached.
- **AC-16.5** (FR-16.5) Given a serial-tracked product with serial S-001 in stock, when a receipt of S-001 is posted, then it is rejected with `INV-SER-001`.

- **AC-16.6** (FR-16.9) Given a count plan with 5% qty tolerance, when a sheet line variance is 8%, then posting requires an approver with CycleCountApprove and the posted journal line equals counted − expected.
- **AC-16.7** (FR-16.11, FR-16.6) Given inventory period 2026-09 closed for module INVENTORY, when a journal with financial_date 2026-09-30 is posted, then it is rejected with the period-control error from wave 14 and no transaction is written.
- **AC-16.8** (FR-16.2) Given a receipt transaction at REGISTERED, when `Unregister` is called, then status returns to ARRIVED, registered bucket decreases, and a status-change audit row exists.
- **AC-16.9** (FR-16.12) Given a manual ledger journal posted to the inventory account without an inventory transaction, when the daily reconciliation runs, then a diff event is emitted with the exact amount.
- **AC-16.10** (FR-16.10) Given product unit EA with conversion 1 BOX = 12 EA, when a line of 2.5 BOX is posted, then qty = 30 EA, entered_qty = 2.5, entered_unit = BOX.
- **AC-16.11** (FR-16.15) Given a transfer order of 100 EA from warehouse W1 to W2 with transit warehouse T1 and `transfer_use_transit=true`, when shipped and then 95 EA received with `close_short=true` and `underdelivery_pct=10`, then on-hand is W1 −100, T1 +5 in transit (flagged for investigation), W2 +95, the line is closed, and both postings have `ledger_entry_id` null when the sites share a cost (no GL effect).
- **AC-16.12** (FR-16.17) Given 500 EA vendor-owned stock (owner = V1) and an OWNERSHIP_CHANGE journal of 120 EA at consignment price 4.00 USD, when posted, then on-hand shows 380 EA owner V1 (value 0) and 120 EA owner legal entity (value 480.00), the ledger entry is Dr inventory 480.00 / Cr consignment accrual 480.00, and `ownership.changed` is emitted with amount 480.00.
- **AC-16.13** (FR-16.18) Given 50 tags generated for warehouse W1, when 48 are entered as USED, 1 VOID and 1 left open, then transfer to the COUNT journal is rejected until the open tag is marked MISSING, after which the COUNT journal contains one line per product × dimension set with the summed tag quantities.
- **AC-16.14** (FR-16.19) Given 1,000 products with on-hand value following a Zipf distribution, when ABC classification by VALUE runs with 80/15/5, then the cumulative value of class A products is ≥ 80 % and the class is visible on the product and usable as a CountPlan filter.

3.16.9 Test plan and fixtures

Fixtures: `inv-demo-plant` (200 products, 3 sites, 6 warehouses, 400 locations, 5,000 opening lines; generated); `inv-1m-txn` (1M synthetic transactions over 12 months, Zipf-distributed across 2,000 products; generated by `scripts/fixtures/inv_txn_gen.py`); `inv-fefo-100` (100 batches with expiry grid, 10 blocked); `inv-ax-extract` (InventSum/InventTrans/InventDim CSV sample from AX 2012 demo company USMF, 50k rows); `inv-sap-extract` (MARD/MCHB/MBEW/MSEG sample, 50k rows).

Test	Type	Target	Pass criterion
Status transition monotonicity	property (Go rapid)	ApplyTransactionStep	no sequence of commands produces a backward step without a named reversal command
On-hand formula	unit	OnHand compute	formulas in FR-16.3 hold for 10k random bucket vectors
Concurrency negative	E2E (k6)	journals:post	AC-16.1
Rebuild equality	performance	onhand:rebuild on inv-1m-txn	zero diff, < 10 min
Journal GL balance	contract	ledger Post	every journal type → balanced entry
FEFO selection	unit + E2E	reservations	AC-16.4
Outbox event schema	contract (AsyncAPI)	all 5 events	schema registry compatibility BACKWARD
Period control	contract	journals:post	AC-16.7
PWA offline count	E2E (Playwright)	count sheet	counts queued offline sync without loss

Reconciliation checks: Σ OnHand.posted_value per inventory account = ledger balance; Σ transaction qty (physical) = OnHand.physical_invent; count sheet variance total = COUNT journal total; AX migration: InventSum.PostedQty + Received + Registered − Deducted − Picked = Passage physical_invent per InventDimId.

3.16.10 Deployment and operations

Helm chart `passage-inventory` (Deployment ×3, HPA 3–12 on CPU 70%, Postgres CloudNativePG cluster `inventory-db` 3 instances, outbox Debezium connector `inventory-outbox`). Config: `INV_NEGATIVE_DEFAULT=forbid`, `INV_RESERVATION_TTL_HOURS=0`, `INV_JOURNAL_MAX_LINES=10000`, `INV_RECON_CRON=0 2 * * *`. Migrations via Atlas from entity metadata (`inventory/v16.*`). Feature flags: `inv.fefo_policy`, `inv.count_pwa`, `inv.transfer_cost_diff` (off until w20). Scheduled jobs (Temporal cron): daily reconciliation 02:00 tenant local, count sheet generation per plan cron, weekly rebuild-and-compare Sunday 03:00. SLOs: journal post p95 < 2 s for 1,000 lines; on-hand query p95 < 50 ms; event lag p95 < 5 s. Alerts: reconciliation diff ≠ 0 (page), outbox lag > 60 s (ticket), negative on-hand where forbidden (page, indicates bug). Runbook: rebuild projection, replay outbox, reverse journal. Rollback: Argo CD digest rollback; migrations are additive in this wave (no destructive change), so downgrade is safe.

3.16.11 Migration from AX / SAP

Source	Passage entity	Notes
AX InventSum (per InventDimId)	OPENING_BALANCE journal lines → InventoryTransaction + OnHand	PostedQty+Received+Registered−Deducted−Picked → qty; PostedValue+PhysicalValue → cost; reserved/ordered buckets not migrated (open orders re-created in w17/w27)
AX InventDim	DimensionSet	map InventSiteId→site, InventLocationId→warehouse, wMSLocationId→location, InventBatchId→batch, InventSerialId→serial, InventStatusId→inventory_status
AX InventTrans (closed history)	InventoryTransaction (read-only history, flag `migrated=true`)	optional; last 24 months; StatusIssue/StatusReceipt mapped 1:1
AX InventBatch, InventSerial	Batch, Serial (wave 8 tracking entities)	expiry, shelf-life advice
AX InventModelGroup	ItemModelGroup (wave 8)	already migrated; verified here
SAP MARD (LABST unrestricted, INSME quality, SPEME blocked)	OpeningBalance per plant/storage location with inventory_status AVAILABLE / QUARANTINE / BLOCKED	
SAP MCHB (CLABS per batch)	OpeningBalance with batch dim	batch master from MCH1/MCHA
SAP MBEW (LBKUM, SALK3, VERPR/STPRS)	cost on opening lines	value = SALK3 split proportionally by qty
SAP MSEG/MKPF history	InventoryTransaction history	BWART → status mapping table (101, 102, 261, 262, 301, 311, 551, 561)

Transformation notes: dimension hash computed after mapping; quantities converted to Passage inventory unit via MARM/UnitOfMeasureConversion; opening journal posts with financial_date = cutover date − 1 and offset account = migration clearing account. Validation: Σ migrated qty per product/site equals source extract; Σ value equals source within 0.01 × product count; no serial duplicates. Not migrated: InventSum reservation buckets (recreated by open documents), AX quarantine orders

(re-entered manually; typically < 50), SAP special stock O (subcontracting; wave 21); special stock K migrates as `owner = vendor` stock per FR-16.17, InventClosing history (wave 20 decides).

3.16.12 Dependencies, risks, open questions

Depends on: wave 6 (sites, warehouses, UoM), wave 8 (products, dimension groups, item model groups, batch/serial masters), wave 11 (`Post(JournalEntry)`, posting profiles), wave 12 (currency rounding), wave 14 (period control module INVENTORY), wave 4 (Temporal, outbox), wave 10 (migration framework).

Risk	Likelihood	Impact	Mitigation
Row-lock contention on hot SKUs (one dimension set)	Medium	Medium	Lock per OnHand row only; batch issues in a journal sorted; measured by k6 at 10k concurrent
Dimension hash collision semantics across services	Low	High	Shared package with golden test vectors; hash includes dimension names
Count PWA offline conflicts	Medium	Low	Last-write-wins with audit; recount flag

Open questions:

1. Store license plate as a dimension now or add in wave 31? Default: reserve the column, null until wave 31. Decide by wave 17 close.
2. Should RESERVED_ORDERED be supported in wave 16 or only when sales/production demand exists? Default: implement now with manual demand type. Decide by wave 16 mid-point.
3. Monthly vs yearly partitioning threshold? Default: monthly when > 20M rows. Decide by wave 20.

3.16.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent
`negative_physical_allowed`	item model group	bool	false	true/false	FR-16.4 rejects issues below zero `avail_physical` when false	InventModelGroup.PhysNegative ("Physical negative inventory")
`negative_financial_allowed`	item model group	bool	false	true/false	FR-16.4 on `avail_ordered`	InventModelGroup.FinancialNegativ
`post_physical_inventory`	item model group	bool	true	true/false	Physical statuses post `cost_amount_physical` to the ledger (accrual accounts)	InventModelGroup.PostPhysicalInv ("Post physical inventory")
`backdated_physical_posting`	item model group	bool	false	true/false	BR-16.6 physical date in closed inventory period	n/a
`reservation_policy_default`	legal entity	enum	FIFO	MANUAL, FIFO, FEFO, LIFO	FR-16.7 default when the caller passes none	Inventory parameters → Date-contr reservation; Batch reservation parameters (FEFO date-controlled)
`reservation_ttl_hours`	tenant	int	0	0–720	FR-16.7 auto-release of stale reservations	n/a
`shelf_life_margin_days`	product tracking group	int	0	0–365	FEFO excludes batches expiring inside margin	PdsShelfAdvice / best-before margi
`journal_names`	legal entity	table	6 seeded (MOV, ADJ, TRF, BOM, CNT, OPN)	journal type, voucher sequence, posting restriction (USER/GROUP), approval workflow id, delete lines after posting	FR-16.6 header defaults and approval routing	Inventory → Journals → Journal na
`journal_max_lines`	tenant	int	10,000	1,000–50,000	FR-16.6 line cap (`INV-JNL-001`)	Inventory parameters → Journals – Journal size
`inventory_posting_profile`	legal entity	table	seeded from wave 11 chart	item code (table/group/all) × account code → inventory, adjustment, loss/profit, consignment accrual, transit accounts	FR-16.6 account resolution	Inventory → Posting → Inventory posting profiles (Inventory tab)

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent
`count_tolerance_qty_pct` / `count_tolerance_value`	count plan	numeric	5 / 100.00 acct ccy	0–100 / ≥ 0	FR-16.9 supervisor approval threshold	Counting groups (InventCountGrou counting reason codes
`count_freeze`	count plan	bool	true	true/false	BR-16.8 count lock vs recount flag	n/a (uses inventory blocking)
`blind_count`	count plan	bool	false	true/false	FR-16.9 hides expected quantity	n/a
`blocking_reason_codes`	legal entity	list	QUALITY, DAMAGE, RECALL, LEGAL, OTHER	user-defined	FR-16.8 mandatory reason	Inventory blocking reasons
`inventory_status_transitions`	legal entity	matrix	AVAILABLE↔BLOCKED, AVAILABLE→QUARANTINE, QUARANTINE→AVAILABLE/BLOCKED (disposition)	allowed pairs	FR-16.8 status change validation	Inventory statuses / inventory statu transitions (D365)
`serial_number_groups`	product	table	none	prefix, counter, length, auto-on-receipt	FR-16.5 auto serial assignment	Serial number groups
`recon_cron`	tenant	cron	`0 2 * * *` (tenant local)	valid cron	FR-16.12 schedule	n/a (Check inventory consistency, manual)
`transfer_use_transit`	legal entity	bool	true	true/false	FR-16.15 two-step vs one-step transfer	Transfer order parameters → In-tra warehouse required
`transit_warehouse_id`	warehouse (from)	reference	none	warehouse of type TRANSIT	FR-16.15 transit dimension per shipping warehouse	Warehouse → "Transit" warehouse assignment (InventLocation.InventLocationIdTra
`transfer_underdelivery_pct` / `transfer_overdelivery_pct`	product, overridable per line	numeric	0 / 0	0–100	FR-16.15 close-short and over-receipt check	Released product → Deliver → Underdelivery/Overdelivery
`owner_dimension_active`	tracking dimension group	bool	false	true/false	FR-16.17 enables `owner` on the hash	Tracking dimension group → Owne (D365)
`auto_ownership_change_on_issue` / `stock_determination`	item model group / warehouse	bool / enum	false / OWN_FIRST	OWN_FIRST, CONSIGNMENT_FIRST	FR-16.17 consumption-driven ownership change	Consignment parameters (D365)
`max_consignment_days`	vendor × product (trade agreement)	int	0 (off)	0–365	Ownership change batch converts stock older than N days	n/a
`abc_thresholds` / `abc_criterion` / `abc_lookback_days`	legal entity	numeric triple / enum / int	80-15-5 / VALUE / 365	sum = 100; VALUE, REVENUE, MARGIN, CARRYING_COST	FR-16.19	Inventory → ABC classification setu
`segment_dimension_active` / `segment_valuated` / `segment_fallback`	tracking dimension group / item model group / legal entity	bool	false / true / true	true/false	FR-16.20	n/a
`tag_number_sequence`	legal entity	number sequence	`TAG-`	wave 5 sequence	FR-16.18	Tag counting journal name

3.16.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
On-hand inquiry	list inquiry	product, site, warehouse, location, batch, serial, status, owner, segment, `groupBy`, as-of date	physical_invent, avail_physical, avail_ordered, ordered, on_order, reserved_*, posted_value	`inventory.OnHand` (as-of replays transactions)	On-hand inventory (InventOnhand), Physical inventory by inventory dimension	MMBE, MB52, MB53 plant availability, MB5B stocks on posting date

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Inventory transactions	drill-down	product, dims, status, origin type, physical/financial date range, voucher	reference, direction, qty, statuses, dates, cost amounts, ledger voucher	`InventoryTransaction`	Inventory transactions (InventTrans), Transaction origin	MB51 material document list, MB03 / MIGO display, MBGR by reason
Posted journals by type	list inquiry	journal type, status, date, posted by	journal number, lines, total qty, voucher	`InventoryJournal`	Journal inquiries (posted journals by type)	MB51 by movement type / document type
Count variance report	operational report	plan, sheet, warehouse, date, counter	expected, counted, variance qty, variance value, approval status	`CountSheet`	Counting history / Count variance inquiry	MI20 / MI24 difference list
Reservation inquiry	list inquiry	product, demand type/id, policy, status	reserved qty, batch/serial, expires_at	`Reservation`	Reservation form (InventTransReservation)	MB24 / MB25 reservations list
Blocking and quarantine inquiry	list inquiry	product, reason, status, expected release	blocked qty, reason, created by, release date, disposition	`Blocking` + OnHand status buckets	Inventory blocking inquiry, Quarantine order inquiry	MMBE blocked / quality stock columns
Negative on-hand list	KPI / list	site, warehouse	product, dims, shortfall qty, first negative date	`OnHand` where `physical_invent < 0`	Inventory management workspace tile	MB52 with negative filter
Inventory-to-ledger reconciliation	operational report (daily)	legal entity, account, dimension, date	inventory value, ledger balance, difference, sample vouchers	FR-16.12 job output	Ledger reconciliation inquiry (Inventory/ledger)	MB5L stock values vs G/L
Dimension combinations inquiry	list inquiry	product	dimension set hash, active dimensions, first/last use	`DimensionSet`	Inventory dimension inquiry	n/a
Batch / serial trace	drill-down	batch or serial, direction (forward/backward)	transactions, documents, locations through time	`InventoryTransaction` by tracking dimension	Batch/serial trace (Item tracing)	MB56 batch where-used list
Transfer order lines (shipped / received)	list inquiry	from/to warehouse, status, dates, product	ordered, shipped, received, remaining, days in transit	`TransferOrderLine`	Transfer order lines inquiry (InventTransferLine)	ME2N (document type UB), VL06O STO deliveries
Stock in transit	list inquiry	product, from/to warehouse, days in transit > N	qty, value at from-cost, transfer order, ship date	`OnHand` at TRANSIT warehouses	On-hand at transit warehouse	MB5T stock in transit
Consignment stock by vendor	operational report	vendor, warehouse, product, age	vendor-owned qty, consignment price, extended value (statistical), age buckets	`OnHand` where `owner ≠ legal entity`	Consignment inventory reports (D365)	MB54 consignment stocks
Customer consignment stock	operational report	customer, warehouse	qty, value, last movement	`OnHand` at CUSTOMER_CONSIGNMENT warehouses	n/a	MB58 customer consignment
Sales-order / project stock	list inquiry	segment type, segment reference	qty, value (if valuated), reservations	`OnHand` by `segment`	n/a	MBBS valuated special stock, MMBE special stock view
ABC classification report	operational report	site, criterion, run date	product, criterion value, cumulative %, class, previous class	`AbcClassification`	ABC classification report (SSRS)	MIBC result list
Tag status / missing tags	list inquiry	tag range, plan	tag number, status, product, qty, entered by	`CountTag`	Tag counting journal	n/a
Transfer order overview	operational report	legal entity, date range, status	orders by status, late shipments, late receipts	`TransferOrder`	SSRS Transfer order overview	ME2N + VL06I inbound delivery monitor

3.16.15 Printed and electronic documents

Document	Trigger	Layout elements (header / line / footer)	Formats	Numbering sequence	Legal / retention requirement
Inventory journal posting list	journal POSTED (on demand)	header: journal number, type, posting date, posted by, voucher; lines: product, dimensions, qty, unit, cost; footer: totals, offset account	PDF (A4/Letter), CSV	journal number `INVJ–`	Accounting record; retain with ledger vouchers (10 years DE/FR, 7 years US)
Count sheet / count list	count sheet generated	header: plan, sheet number, warehouse, counter, generated at; lines: location, product, batch/serial, expected (blank when blind), counted; footer: signature line	PDF, PWA screen	sheet number `CNT–`	Physical inventory evidence (HGB §240/§257, French CGI annual inventory); retain 10 years
Count tags	tag range generated	tag number barcode (Code 128), warehouse, range id, product/qty entry boxes	PDF (tag strips), ZPL	`TAG–`	Same as count sheet
Transfer shipping document (packing slip)	transfer order `:ship`	header: transfer number, from/to/transit warehouse, ship date, carrier, delivery terms; lines: product, qty, batch/serial, weight; footer: package count, signatures	PDF, email; EDI DESADV via wave 35 hook	shipment number `TRS–`	Transport document (CMR-like data for cross-border; Intrastat source, see 3.16.20)
Transfer receiving document	transfer order `:receive`	header: transfer number, receipt date, receiver; lines: shipped qty, received qty, variance, location	PDF	receipt number `TRR–`	Goods receipt evidence; retain 10 years
Consignment replenishment order	order confirmed	header: vendor, warehouse, requested date; lines: product, qty, consignment price	PDF, email, portal (wave 20)	`CRO–`	Commercial document; retain 10 years
Batch / serial / location label	registration or on demand	product, batch, expiry, serial, GS1-128 barcode (AI 01/10/17/21)	ZPL (4×6 in, 100×150 mm), PDF	none	GS1 compliance where required (FMD serial for pharma, wave 33)
Blocking / quarantine notice	block created	product, dims, qty, reason, expected release, approver	email (wave 4 notifications)	none	none

3.16.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Inventory-to-ledger reconciliation (FR-16.12)	daily 02:00 tenant local	legal entity × inventory account	run id = (entity, date); recomputes fully, upserts result rows	< 5 min per 1M transactions	retry 3× with 10 min backoff; then alert	diff ≠ 0 pages inventory accountant; missed run ticket

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
On-hand rebuild-and-compare	weekly Sunday 03:00	tenant × product range (1,000 products per child workflow)	checkpoint per child; compare table keyed by run id	< 10 min per 1M transactions	child retry; whole run resumable from last child	any row diff pages engineering (indicates bug)
Count sheet generation per plan	plan cron (e.g., `0 6 * * 1-5`)	count plan	key = (plan id, run date); re-run returns existing sheets	< 30 s per 10k lines	alert plan owner	sheets generated = 0 for an active plan → warning
Reservation TTL expiry	every 5 min	tenant	selects `expires_at < now()`; release is idempotent	< 10 s	retry next tick	expired-but-active reservations > 100 → ticket
ABC classification recalculation (FR-16.19)	monthly, 1st 04:00	legal entity × site	key = (entity, site, period); overwrites class with previous retained	< 2 min per 100k products	retry 3×	class churn > 30 % → warning to inventory manager
Ownership change batch (FR-16.17)	daily 23:00	vendor × warehouse	key = (vendor, warehouse, date); converts consumption-driven and aged quantities once	< 5 min	failed vendor skipped and listed	lines failed > 0 → ticket
Transfer order batch ship / receive (from confirmed shipments or ASNs)	hourly	legal entity	key = (transfer line, step)	< 1 min per 1k lines	line-level errors reported, rest posted	error lines > 0 → ticket
Journal and projection cleanup	weekly Sunday 05:00	tenant	deletes journal line drafts older than `journal_draft_retention_days` (90) and OnHand rows with all-zero buckets older than 30 days	< 2 min	skip and retry	none
Partition maintenance	monthly 1st 01:00	tenant	creates next-month partition on InventoryTransaction when over the row threshold; idempotent DDL	< 1 min	page (future inserts would fail)	partition missing for next month → page

3.16.17 Security duties and privileges

Duty	Privileges (entry points: API paths / screens / jobs)	Segregation-of-duties conflicts	Default roles that include it	AX duty / SAP authorization reference
InventoryInquire	GET /inventory/onhand, /transactions, /reservations; all inquiries in 3.16.14; workspace tiles	none	Inventory clerk, Warehouse supervisor, Inventory accountant, Buyer, Planner	AX "Inquire into inventory status" (InventOnhandInquire); SAP M_MSEG_LGO display, S_ALR stock reports
InventoryJournalMaintain	POST /inventory/journals, lines:bulk, :validate; Inventory journal screen (create/edit/submit)	with InventoryJournalPost when `journal_names.posting_restriction=SEGREGATED`	Inventory clerk	AX "Maintain inventory journals" (InventJournalMaintain); SAP M_MSEG_BWA activity 01
InventoryJournalPost	POST /inventory/journals/{id}:post; approval task	with InventoryJournalMaintain (same journal), with InventorySetupMaintain	Warehouse supervisor, Inventory accountant	AX "Post inventory journals"; SAP M_MSEG_BWA activity 02 per movement type
InventoryReserve	POST/DELETE /inventory/reservations; reservation dialog	none	Inventory clerk, Order processor (w27), Production planner (w21)	AX "Maintain inventory reservations"; SAP M_MRES_BWA
InventoryBlock	POST /inventory/blockings, status change; blocking screen; quarantine disposition	with CycleCountPost for the same dimension set (hides shrinkage)	Quality inspector (w33), Warehouse supervisor	AX "Maintain inventory blocking"; SAP M_MSEG_BWA for 343/344, QM stock
CycleCountMaintain	count-plans CRUD, :generate; count sheet entry (desktop/PWA); tag entry	with CycleCountApprove	Inventory clerk, Counter	AX "Maintain counting journals"; SAP M_ISEG_WIB (PI count)
CycleCountApprove	approve sheet lines above tolerance	with CycleCountMaintain, CycleCountPost (when policy `four_eyes_count=true`)	Warehouse supervisor	AX "Approve counting variances"; SAP M_ISEG_WDB (post differences)
CycleCountPost	POST /count-sheets/{id}:post; tag transfer to journal	with CycleCountMaintain under four-eyes	Warehouse supervisor, Inventory accountant	AX "Post counting journals"; SAP MI07 M_ISEG_WDB
TransferOrderMaintain	transfer orders CRUD, :cancel, delete; transfer order screen	none	Inventory clerk, Planner	AX "Maintain transfer orders" (InventTransferOrderMaintain); SAP M_BEST_BSA document type UB
TransferOrderPost	:ship, :receive; batch ship/receive job	with TransferOrderMaintain when `transfer_four_eyes=true`	Warehouse supervisor	AX "Ship and receive transfer orders"; SAP M_MSEG_BWA 641/101, V_VLIK
ConsignmentMaintain	consignment replenishment orders, OWNERSHIP_CHANGE journals, ownership change batch	with VendorInvoicePost (w18) for the same vendor	Inventory clerk, Buyer	D365 "Maintain consignment inventory"; SAP M_BEST_WRK item cat K, MRKO
InventoryAdmin	/internal/onhand:rebuild, partition maintenance, reconciliation re-run	with InventoryJournalPost	Inventory system administrator	AX "Maintain inventory parameters" + batch; SAP S_BTCH_ADM, MB5L
InventorySetupMaintain	parameters in 3.16.13, journal names, posting profiles, status transitions, ABC setup	with InventoryJournalPost, TransferOrderPost	Inventory system administrator, Inventory accountant (posting profiles only)	AX "Maintain inventory setup" (InventParameters); SAP SPRO MM-IM customizing (S_TABU_DIS)

3.16.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
INV-NEG-001	error	Issue of {qty} {unit} would make available physical stock negative for {product} at {dims} (shortfall {shortfall})	Item model group forbids physical negative	Receive stock, reduce quantity or change item model group policy	FR-16.4, BR-16.4
INV-NEG-002	error	Issue would make financial (ordered) stock negative for {product} at {dims}	Item model group forbids financial negative	Confirm the expected receipt first or allow financial negative	FR-16.4
INV-SER-001	error	Serial {serial} of {product} is already in physical stock at {warehouse}	Duplicate serial receipt	Correct the serial or issue the existing one first	FR-16.5, BR-16.9
INV-SER-002	error	Serial-tracked product {product} requires quantity ±1 per transaction	Quantity not ±1	Split the line into one row per serial	BR-16.9
INV-UOM-002	error	Quantity {entered_qty} {entered_unit} converts to {qty} with residual beyond {decimals} decimals	Conversion residual	Enter a quantity that converts exactly or adjust unit decimals	FR-16.10

Code	Severity	User text	Cause	Remediation	Raised by
INV-DIM-001	error	Dimension {dimension} is mandatory for {product}	Storage/tracking dimension group marks dimension mandatory	Fill the dimension	FR-16.5
INV-PER-001	error	Physical date {date} is in a closed inventory period and backdated posting is not allowed	BR-16.6	Change the physical date or allow backdated posting on the item model group	BR-16.6
INV-RES-001	error	Only {available} of {requested} {unit} available to reserve for {product}	Over-reservation	Reserve the partial quantity returned or wait for receipts	FR-16.7
INV-RES-002	warning	No eligible batch found for FEFO reservation (expiry margin {days} days)	All batches blocked, quarantined or expiring	Review blocked batches or reduce the shelf-life margin	FR-16.7
INV-BLK-001	error	Quarantined quantity requires a disposition (release, scrap, return) before exit	Missing disposition	Choose a disposition	FR-16.8
INV-JNL-001	error	Journal exceeds {max} lines	`journal_max_lines`	Split the journal	FR-16.6
INV-JNL-002	error	Movement journal line {line} requires an offset account	Missing offset	Enter an offset account valid for the account structure	FR-16.6
INV-JNL-003	error	Journal {journal} is already posting (workflow {id})	Concurrent post	Wait for the workflow to finish	BR-16.11
INV-CNT-001	warning	Variance {pct} % exceeds tolerance {tol} %; supervisor approval required	FR-16.9	Request approval from a user with CycleCountApprove	FR-16.9
INV-CNT-002	error	Dimension set {dims} is locked by open count sheet {sheet}	Count freeze	Post or cancel the sheet, or allow movement with recount on the plan	FR-16.9
INV-TRF-001	error	Shipping warehouse {warehouse} has no transit warehouse configured	`transit_warehouse_id` missing	Assign a transit warehouse or disable `transfer_use_transit`	FR-16.15
INV-TRF-002	error	Received quantity {qty} exceeds shipped quantity {shipped} plus tolerance {pct} %	Over-receipt on transfer	Correct the quantity or raise the tolerance on the line	FR-16.15
INV-TRF-003	error	Transfer order {number} cannot be cancelled: {qty} in transit	BR-16.13	Receive the in-transit quantity or create a reverse transfer	BR-16.13
INV-OWN-001	error	Vendor-owned stock cannot be financially posted or valued	Attempt to post value on `owner ≠ legal entity`	Post an ownership change journal first	BR-16.14
INV-OWN-002	error	Ownership change of {qty} exceeds vendor-owned quantity {available} at {dims}	BR-16.14	Reduce the quantity	FR-16.17
INV-TAG-001	error	Tag {tag} has already been used	Duplicate tag entry	Void the earlier entry or use a different tag	FR-16.18
INV-TAG-002	error	{n} tags are still open; mark them MISSING or enter them before transferring	BR-16.15	Resolve open tags	FR-16.18
INV-ABC-001	error	ABC thresholds must sum to 100 (got {sum})	Setup error	Correct the thresholds	FR-16.19
INV-SEG-001	error	Stock with segment {segment} cannot be issued to demand with segment {demand_segment}	BR-16.17	Transfer between segments or enable fallback	FR-16.20
INV-LCK-001	error	Could not lock on-hand row for {product} at {dims} within {ms} ms	Lock contention	Retry; the client should back off 200 ms × attempt	BR-16.4

3.16.19 Performance budgets and capacity

Operation	Volume assumption (per tenant per day / peak per minute)	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Journal post (1,000 lines)	500 journals / 20 per min	0.8 s / 2 s / 4 s	2,000 lines/s per pod	InventoryTransaction ≈ 420 B/row + 280 B indexes; 50M rows ≈ 35 GB	(tenant, product, dimension_set, physical_date) B-tree; partial index on open statuses; monthly range partitions over 20M rows	Temporal task queue absorbs backlog; HPA 3→12 pods; posting latency rises to ≤ 10 s p95, no rejections
Internal `transactions:apply` (waves 17–20)	200k steps / 1,500 per min	15 ms / 40 ms / 80 ms	1,500 rps sustained	counted above	OnHand PK (tenant, product, dimension_set); `FOR UPDATE` per row	429 with `Retry-After` beyond 2,500 rps; callers (receiving, PO confirm) retry with jitter
On-hand query (≤ 10k rows aggregated)	50k queries / 600 per min	12 ms / 50 ms / 120 ms	1,000 rps	OnHand ≈ 300 B/row; ≤ 2M rows per tenant	clustered by (tenant, product); covering index on (tenant, site, warehouse, product)	Reads shift to the Postgres read replica (`inventory-db-ro`); stale-by ≤ 2 s shown in UI
Reservation (policy FEFO over 100 batches)	30k / 300 per min	25 ms / 80 ms / 150 ms	500 rps	Reservation ≈ 200 B/row	index (product, dimension_set, status)	Serializes per product; p95 ≤ 300 ms
Count sheet generation (10k lines)	20 sheets	5 s / 30 s / 60 s	n/a	CountSheet lines ≈ 250 B/row	(plan_id, status)	Queued per plan; no user-facing path
Rebuild-and-compare (1M transactions)	weekly	6 min / 10 min / 15 min	2,000 rows/s	temp compare table dropped after run	parallel by product range, 32 children	Runs on replica snapshot; skipped if lag > 60 s and retried next hour
Reconciliation (per entity)	daily	1 min / 5 min / 8 min	n/a	result rows ≈ 1 KB × accounts × days	aggregate on (ledger_entry_id) index	Off-peak only; alert if > 15 min
Transfer order ship / receive (100 lines)	2,000 orders / 50 per min	0.4 s / 1.2 s / 2.5 s	200 orders/min	TransferOrder ≈ 1.5 KB + lines 300 B	(from_warehouse, status), (to_warehouse, status)	Same as journal post (paired transactions)
Outbox event publish	300k events / 3,000 per min	lag 1 s / 5 s / 10 s	5,000 events/s per Debezium connector	Kafka retention 7 days	partition by tenant_id + product_id	Lag alert at 60 s; consumers idempotent on transaction id

3.16.20 Localization and compliance hooks

Concern	Varies by	Country packs touching it (wave 48)	Extension point used
Statutory stock ledger / inventory book (Italy libro giornale di magazzino, Portugal/Spain stock register, India stock register, Russia stock card)	Country reporting rules	IT, ES, PT, IN, RU, TR	Report projection `inventory.transaction_ledger` (wave 15 report definitions) with country layout; SAP S_P00_07000139 / AX country-specific inventory reports
Annual physical inventory obligation and evidence retention	HGB §240/§257 (10 years), French CGI, US GAAP evidence	DE, FR, US	`CountSheet` retention policy from wave 5 audit retention; immutable PDF archive in object store
Electronic signature on count approval, blocking release and quarantine disposition	FDA 21 CFR Part 11, EU GMP Annex 11	US (pharma), EU (pharma)	Wave 4 approval step with `signature_required=true`; wave 33 extends to quality orders
Batch and serial regulatory tracing (EU FMD, DSCSA, food lot tracing FSMA 204)	Industry regulation	EU, US	GS1 label template (3.16.15) and batch trace inquiry (3.16.14); key data elements stored on batch (wave 8)
Intrastat and cross-border transfers (intra-EU transfer orders between warehouses in different member states, STO cross-border)	EU Intrastat thresholds	All EU packs	`transfer.received` event carries country of dispatch/arrival, commodity code (wave 8), net mass; wave 48 Intrastat consumer
Consignment stock VAT and call-off stock simplification (EU "quick fixes" 2020)	EU VAT directive Art. 17a	EU packs	`ownership.changed` event feeds wave 13 tax engine for supply point; call-off register report hook
Units of measure, decimals and quantity rounding	Metric vs imperial; product decimals	n/a (product-level)	Wave 6 UoM service; BR-16.5 rounding
Currency rounding of `posted_value`	Accounting currency minor unit (JPY 0, KWD 3)	n/a	Wave 12 rounding rules
Calendar and time zone for cron jobs and physical date default	Tenant/site local time, local holidays	All	Wave 6 site calendar; jobs scheduled in site time zone
Language of journal types, statuses, reason codes, PWA count screen	User language	n/a	Wave 3 metadata labels (en, de, fr, es in v1)
Privacy of counter identity (`counted_by`) and PWA device identifiers	GDPR, state privacy laws	EU, US-CA	Wave 2 pseudonymization hook on worker references after retention

3.16.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.2.1 — Vendor → Consignment (D365 consignment replenishment orders) [#2]	GAP	ADDED as FR-16.17	`ConsignmentReplenishmentOrder`; settlement invoice in wave 18 FR-18.16
AX 12.2.3 — Consignment → Consignment replenishment order processing [#5]	GAP	ADDED as FR-16.17	Replenishment order receive; ownership change batch (3.16.16)
AX 12.2.4 — Vendor → Vendor consignment setup (consignment parameters, ownership change journal names) [#7]	GAP	ADDED as FR-16.17	Parameters `owner_dimension_active`, `auto_ownership_change_on_issue`, journal name OWNERSHIP_CHANGE (3.16.13)
AX 12.14.1 — Journals → Inventory ownership change journal [#44]	GAP	ADDED as FR-16.17	Journal type OWNERSHIP_CHANGE
AX 12.14.2 — SSRS: Consignment inventory reports [#45]	GAP	ADDED as FR-16.17	"Consignment stock by vendor" report (3.16.14)
AX 12.14.3 — Inventory → Consignment: ownership change batch [#46]	GAP	ADDED as FR-16.17	Ownership change batch job (3.16.16)
AX 12.14.4 — Inventory → Consignment → Consignment parameters [#47]	GAP	ADDED as FR-16.17	3.16.13 rows
AX 12.16.1 — Consignment (link) [#57]	GAP	ADDED as FR-16.17	Procurement entry point to replenishment orders
AX 12.39.1 — Vendor → Consignment inventory view (on-hand, replenishment orders) [#108]	GAP	DEFERRED to wave 20	Portal consignment view is FR-20.22 (data from FR-16.17)
AX 12.14.1 — Transfer orders (list page) [#119]	PARTIAL	ADDED as FR-16.15	`TransferOrder` with in-transit warehouse
AX 12.14.1 — Transfer order details [#120]	PARTIAL	ADDED as FR-16.15	Header/lines, ship, receive, underdelivery
AX 12.14.1 — Journals → Tag counting journal [#121]	PARTIAL	ADDED as FR-16.18	Two-step tag counting
AX 12.14.1 — Inventory → ABC classification (on item) [#123]	PARTIAL	ADDED as FR-16.19	Class stored per product × site
AX 12.14.2 — Transfer order inquiries (lines, shipped/received) [#125]	PARTIAL	ADDED as FR-16.15	3.16.14 "Transfer order lines"
AX 12.14.2 — SSRS: Transfer order (shipping document, receiving document, overview) [#127]	PARTIAL	ADDED as FR-16.15	3.16.15 shipping/receiving documents; 3.16.14 overview
AX 12.14.2 — SSRS: ABC classification report [#129]	PARTIAL	ADDED as FR-16.19	3.16.14
AX 12.14.2 — SSRS: Packing slip (inventory transfer) [#130]	PARTIAL	ADDED as FR-16.15	Transfer shipping document (3.16.15)
AX 12.14.3 — Tag counting → Create tag counting / Transfer to counting journal [#131]	PARTIAL	ADDED as FR-16.18	Tag range generation and transfer
AX 12.14.3 — Transfer orders → Ship / Receive (batch) / Transfer order picking release [#132]	PARTIAL	ADDED as FR-16.15	Batch ship/receive job (3.16.16); picking release event for wave 31
AX 12.14.3 — Transfer orders → Delete transfer orders [#133]	PARTIAL	ADDED as FR-16.15	Delete of CREATED orders without postings
AX 12.14.3 — ABC classification (periodic calculation) [#134]	PARTIAL	ADDED as FR-16.19	Monthly job (3.16.16)
AX 12.14.4 — Inventory and warehouse management parameters — Transfer orders [#136]	PARTIAL	ADDED as FR-16.15	`transfer_use_transit`, tolerances (3.16.13)
AX 12.14.4 — Inventory → ABC classification setup [#138]	PARTIAL	ADDED as FR-16.19	`abc_thresholds`, criterion, lookback (3.16.13)

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.14.4 — Inventory → Transfer orders → Transfer order parameters, In-transit warehouses [#139]	PARTIAL	ADDED as FR-16.15	`transit_warehouse_id` per shipping warehouse
SAP 13.12 — ME21N item category K (consignment) [#126]	PARTIAL-16	ADDED as FR-16.17	Replenishment order replaces consignment PO; price from trade agreement `consignment=true` (wave 17)
SAP 13.12 — ME21N item category U / UB STO / NB cross-company STO [#127]	PARTIAL-16	ADDED as FR-16.15	Intra-company STO full; cross-company pricing and intercompany invoice in wave 40
SAP 13.12 — MRKO consignment settlement [#134]	PARTIAL-16	ADDED as FR-16.17	Ownership change emits event; invoice creation in wave 18 FR-18.16
SAP 13.12 — OMGN set up stock transport order [#136]	PARTIAL-16	ADDED as FR-16.15	`transfer_use_transit`, transit warehouse, tolerances
SAP 13.13 — MB53 / MB54 plant availability / consignment stock [#140]	PARTIAL-16	ADDED as FR-16.17	On-hand inquiry with `owner`; consignment report (3.16.14)
SAP 13.13 — MIK1 / MIQ1 / MIM1 PI for consignment / project stock / RTP [#142]	PARTIAL-16	ADDED as FR-16.17, FR-16.20	Count plans filter by owner and segment; returnable transport packaging OUT-OF-SCOPE (handled as ordinary product with owner dimension in v1)
SAP 13.13 — MIS1 / MIS2 inventory sampling [#144]	GAP (propose 16)	OUT-OF-SCOPE statistical sampling with extrapolation is not provided in v1; random-sample counting without extrapolation exists via CountPlan `sample_pct`	Revisit if a customer requires sampling-based book adjustment
SAP 13.13 — Special stock: sales order stock (E), project stock (Q) [#145]	PARTIAL-16	ADDED as FR-16.20	`segment` dimension, valuated flag
SAP 13.13 — Stock in transit (MB5T; S/4 stock in transit for STO) [#146]	PARTIAL-16	ADDED as FR-16.15, FR-16.16	Transit warehouse dimension and inquiry
SAP 13.13 — MB5T stock in transit report [#148]	PARTIAL-16	ADDED as FR-16.16	3.16.14
SAP 13.13 — MB54 / MB58 / MBLB consignment / customer consignment / subcontractor stock [#149]	PARTIAL-16	ADDED as FR-16.17	MB54 and MB58 reports added; subcontractor stock (MBLB) DEFERRED to wave 21 subcontracting
SAP 13.13 — MBBS valuated special stock [#150]	PARTIAL-16	ADDED as FR-16.20	Inquiry in 3.16.14; valued grouping in wave 20 value report
SAP 13.13 — MRKO consignment and pipeline settlement [#151]	PARTIAL-16	ADDED as FR-16.17	Pipeline material (utilities without stock) OUT-OF-SCOPE; settlement in wave 18
SAP 13.13 — Special stock settings (consignment price determination, pipeline) [#153]	PARTIAL-16	ADDED as FR-16.17	Consignment price via trade agreement flag; pipeline OUT-OF-SCOPE
SAP 13.13 — OSPX / OSPW stock determination strategies [#154]	PARTIAL-16	ADDED as FR-16.17	`stock_determination` OWN_FIRST / CONSIGNMENT_FIRST per warehouse
SAP 13.14 — MRKO consignment settlement (MM-IV) [#155]	PARTIAL-16	ADDED as FR-16.17	Invoice side in wave 18 FR-18.16
SAP 13.14 — MRKO scheduled [#158]	PARTIAL-16	ADDED as FR-16.17	Ownership change batch (3.16.16) + wave 18 settlement job
SAP 13.17 — VL10 → STO deliveries (NL/NLCC) [#203]	PARTIAL-16	ADDED as FR-16.15	`TransferShipment` with shipping document; carrier/freight in wave 34

3.17 Wave 17: Purchasing: Requisitions, RFQs, Agreements & Purchase Orders

3.17.1 Scope and business context

Wave 17 gives buyers the procurement document chain: purchase requisition (AX PurchReqTable/PurchReqLine, SAP EBAN), request for quotation with supplier replies (PurchRFQCaseTable/PurchRFQTable/PurchRFQReplyTable, SAP EKKO document type AN with quotation maintenance ME47), purchase agreements with release tracking (PurchAgreementHeader/PurchAgreementLine, SAP contracts type MK/WK and scheduling agreements LP via ME31K/ME31L), vendor trade agreements (PriceDiscTable journal type PriceDiscAdmTrans relation PurchPrice/PurchLineDisc, SAP condition records A017/KONP via MEK1), and purchase orders with versioned change management (PurchTable/PurchLine with PurchTableHistory and change-management parameters, SAP EKKO/EKPO with change documents CDHDR/CDPOS and version management in EREV). Purchase order confirmation writes ON_ORDER/ORDERED inventory transactions from wave 16.

At the end of the wave a requester can raise a requisition that routes by amount band and category to the right approvers, a buyer can consolidate approved requisition lines into POs, run an RFQ to several suppliers and award, create blanket agreements that POs release against, and send a PO as PDF/email with an EDI 850/ORDERS hook stubbed for wave 35. Out of scope: receiving (wave 18), invoice (wave 18), budget control on commitments (wave 38), subcontracting POs from production (wave 21), landed cost (wave 34), supplier portal acknowledgement (wave 20), catalog punch-out (v1.x).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX purchase order confirmation runs PurchFormLetter with totals recalculation and PDF through SSRS; 20–90 s per PO on a loaded AOS; ME21N saves fast but output (NEU message, ME9F) depends on a scheduled RSNAST00 job	Synchronous document class chain plus report server; SAP output via NAST condition technique and batch	Confirmation is a state transition that enqueues rendering (Temporal activity) and emits `po.confirmed`; PDF rendered by a stateless worker within 3 s	Confirm latency: 20–90 s → < 2 s UI, PDF < 5 s; no batch wait
Price lookup in AX (PriceDisc class, PriceDiscTable search over 3 account codes × 3 item codes) is opaque; buyers cannot see why a price was chosen; SAP condition analysis (ME23N > Conditions > Analysis) exists but is 6 clicks deep	No explain output; access sequence hidden	Every PO line carries `price_source` (agreement id, trade agreement id, last price, manual) with an "explain" popover; `GET /pricing/explain`	Clicks to see why a price applied: 6–8 → 1; price disputes per 1,000 POs: tracked, target –80%
Change management in AX creates a full PurchTableHistory copy only when the parameter is on; comparing versions is a manual form; SAP version management requires EREV config and a reason per change	Version stored as whole-document copies without diff	Versions are stored as JSON-patch diffs against the previous version with a rendered side-by-side diff; re-approval rule evaluated per diff	Time to identify what changed between PO versions: 5 min → 10 s
Requisition approval in AX workflow (PurchReqReview) evaluates in the batch workflow processor every 60 s; escalation and delegation are per-user configuration pages; SAP release strategy (CL20N/CL24N classification) is nearly impossible to change without a consultant	Batch-polled workflow runtime; classification-based release strategy	Temporal approval workflow started synchronously on submit; policy is a declarative matrix (amount band × category × org) editable by a procurement admin	Submit-to-first-approver latency: 60–120 s → < 2 s; policy change lead time: days (consultant) → minutes (admin)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
PO total in AX recomputed by PurchTotals class on demand; header/line totals can disagree until "Totals" is opened; SAP EKKO has no stored total	Totals derived lazily	Totals persisted on every line change with a database constraint test `header.total = Σ lines + charges + tax`	Total mismatches: non-zero → 0 across 1,000 random POs

3.17.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Purchase requisition	PurchReqTable, PurchReqLine; forms PurchReqTable, PurchReqReview workflow; policies (PurchReqControlRule)	EBAN/EBKN; ME51N/ME52N/ME53N; release strategy ME54N; ME57 assign source	`procurement.PurchaseRequisition`, `RequisitionLine`; Requisition workspace; policy matrix	full
Approval policy	Purchasing policies (PurchPolicy), amount bands via workflow conditions, category access policy	Release strategy (release groups/codes, classification characteristics CEBAN)	`ApprovalPolicy` (bands, category, org unit, requester role) compiled to Temporal workflow input	improved
Consolidation into PO	PurchReqConsolidation (consolidation opportunities), release approved lines (PurchReqPurchaseOrderCreate)	ME57/ME58 (assign and process), automatic PO creation ME59N	`POST /requisitions:consolidate` grouping by vendor × currency × site × delivery date window	full
RFQ	PurchRFQCaseTable, PurchRFQTable (per vendor), PurchRFQReplyTable, PurchRFQCompare; vendor collaboration portal reply (D365)	RFQ ME41/ME42/ME43 (EKKO BSTYP A), quotation ME47, price comparison ME49, rejection ME9A	`RFQ`, `RFQInvitation`, `RFQReply`; compare screen; award creates PO or agreement	full
Purchase agreement	PurchAgreementHeader, PurchAgreementLine, AgreementLineReleasedLine (release tracking); commitment types (product qty/value, category value, value)	Contracts (EKKO BSTYP K, ME31K/ME32K/ME33K), release orders, scheduling agreements (L, ME31L) with delivery schedules ME38	`PurchaseAgreement`, `AgreementLine` (commitment types QUANTITY, VALUE, CATEGORY_VALUE, TOTAL_VALUE); release tracking	full (scheduling agreement delivery schedules partial until w22 MRP)
Vendor pricing	Trade agreements PriceDiscTable (PurchPrice, PurchLineDisc, PurchMultiLnDisc, PurchEndDisc), account/item code hierarchy, date validity, quantity breaks	Conditions A017 (material info record), A018, A019, A025; KONP; info record EINA/EINE; MEK1	`TradeAgreement` (vendor > vendor group > all; product > product group > all; qty breaks; validity; currency) with explain	full
Purchase order	PurchTable, PurchLine, PurchTableHistory/PurchLineHistory (versions), PurchOrderStatus, DocumentStatus, PurchFormLetter_PurchOrder (confirm)	EKKO/EKPO/EKET (schedule lines)/EKKN (account assignment), ME21N/ME22N/ME23N, EREV versions	`PurchaseOrder`, `PurchaseOrderLine`, `DeliverySchedule`, `PurchaseOrderVersion`	full
Charges	MarkupTrans (misc. charges header/line), MarkupTable, auto charges (MarkupAutoTable)	Pricing conditions FRA1 (freight), ZOA1 (customs), header/item conditions	`Charge` (header/line; fixed, per unit, percent; debit/credit to item, vendor, ledger)	full
Delivery schedules	PurchLine delivery schedule (multiple delivery lines), confirmed delivery date	EKET schedule lines	`DeliverySchedule` rows per PO line	full
PO output	Print management (PrintMgmt), SSRS PurchPurchaseOrder report, email, AIF service PurchPurchaseOrderService	Message output NAST, forms MEDRUCK, IDoc ORDERS05	PDF via render worker, email via wave 4 notification service; EDI hook emits `po.confirmed` for wave 35	partial (EDI transport in w35)
Category hierarchy procurement	EcoResCategory procurement hierarchy, category policy	Material groups (T023), product categories	Wave 8 category hierarchy with procurement purpose	full
Vendor on-hold / status	VendTable Blocked (invoice/payment/all)	LFA1 SPERR/LOEVM, purchasing block LFM1 SPERM	Vendor role status (wave 7) checked on PO create/confirm	full
Scheduling agreements and releases	Purchase agreement with delivery schedule lines (partial); D365 no FRC/JIT releases	Scheduling agreement ME31L/ME38 delivery schedule, release creation profile (T163M), firm/trade-off zones, FRC/JIT releases ME84, release documentation ME85/ME86, Monitor SA Releases	`SchedulingAgreement` + `ReleaseSchedule` + `Release` (FRC/JIT) with cumulative quantities (FR-17.15)	full (EDI DELFOR/DELJIT transport in wave 35)
Policy framework	Policy framework (SysPolicy): Purchasing policies, Expense policies, Audit policies, Vendor invoice policies, Signing limit policies scoped by organization hierarchy purpose	Release strategy classification; no generic policy framework	`Policy` / `PolicyRule` engine scoped by org hierarchy purpose (FR-17.16); reused by waves 43, 48	improved
Signing limits	Signing limit policies, default signing limits by job/compensation level, worker signing limit requests, recertification	Release codes per user (CL24N classification by amount)	`SigningLimit` per worker with request workflow and recertification (FR-17.17)	full
Procurement catalog	Procurement catalogs (CatProcurementCatalog), catalog policy rule, catalog images, vendor catalog import (CMR XML), external catalogs (cXML/OCI punch-out)	Fiori self-service requisition with catalogs; OCI/cXML punch-out; SRM-MDM catalog	`ProcurementCatalog` with publication by org unit, images, vendor catalog import (FR-17.18); vendor upload via portal (wave 20 FR-20.19); punch-out in wave 35	partial (punch-out in wave 35)
RFQ bidding (sealed bids, scoring)	RFQ types, scoring criteria/groups, reply templates, sealed bids, amendments, solicitation types (D365 public sector)	ME47 quotation, ME49 price comparison; weighted scoring only in Ariba Sourcing	`RfqType`, `ScoringCriterion`, sealed reply storage, amendments (FR-17.19)	full (reverse auctions OUT-OF-SCOPE)
Category setup depth	Procurement categories, category policy rules, category managers, category vendor assignment, category questionnaires	Material groups T023, source list ME01, purchasing group	`CategoryVendorAssignment`, `CategoryManager`, `CategoryQuestionnaire` (FR-17.20)	full
Invoicing plans	n/a (prepayments only)	Invoicing plan on PO item (periodic/partial), MRIS settlement run, invoicing plan types T163*	`InvoicingPlan` on PO line (FR-17.21); run in wave 18 FR-18.17	full
Subsequent price change	n/a	MEI1/MEI2/MEI4 document index and automatic document adjustment; MRNB revaluation of invoiced items	`:reprice` run over open PO lines (FR-17.22); invoiced items via wave 18 FR-18.18 subsequent debit/credit	full
Service purchase orders and service specifications	Procurement category lines (no hierarchy)	ME21N item category D with service outline, value limits, contract limits; model service specifications ML10–ML12; standard service catalog; service conditions ML33–ML48, calculation schema MS0000	Service lines with `ServiceOutline` hierarchy, limits, `ModelServiceSpecification` templates, service prices via TradeAgreement (FR-17.23)	full
Consignment price agreements	Trade agreement with consignment flag (D365)	Consignment info record (ME11 info category K), price determination for consignment	TradeAgreement `consignment=true` consumed by wave 16 ownership change (FR-17.24)	full
Legal contract authoring	n/a	SAP CLM / Ariba Contracts clause library, obligations	Not provided; agreements carry attachments, validity and expiry alerts	dropped-with-reason (3.17.21)

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Configurable PO items	n/a	Configuration in purchasing (LO-VC configured items on PO)	Deferred to the wave that introduces the configurator (wave 24/27); PO line stores `configuration_id` placeholder	dropped-with-reason (3.17.21)

3.17.3 Functional requirements

- **FR-17.1 Purchase requisition** — Must. The system shall let any user with RequisitionCreate raise a requisition with lines for catalog products, non-catalog items (description + category) or services, with requested date, quantity, unit, estimated price, site/warehouse, financial dimensions and attachments.
    - Statuses: DRAFT, IN_REVIEW, APPROVED, REJECTED, CLOSED (all lines consolidated), CANCELLED.
    - Estimated price defaults from TradeAgreement lookup for the preferred vendor, else last PO price, else zero with warning.
    - Maximum 500 lines; account structure validated at submit using wave 11 rules.
- **FR-17.2 Approval policy matrix** — Must. The system shall evaluate an `ApprovalPolicy` with rules (amount band in accounting currency, category, operating unit, requester role) producing an ordered approver chain, executed by the wave 4 Temporal approval template with 48 h SLA timers, escalation and delegation.
    - Bands default: 0–1,000 USD auto-approve, 1,000–10,000 manager, 10,000–100,000 manager + category owner, > 100,000 + CFO.
    - Policy versions are effective-dated; the version at submit time governs the whole instance.
    - Any line edit after submit restarts the workflow.
- **FR-17.3 Consolidation into purchase orders** — Must. The system shall group approved requisition lines into POs by vendor × currency × site × delivery window (default 7 days) with buyer override of vendor and the ability to split lines.
    - Lines with no vendor route to a "Needs sourcing" queue from which an RFQ can be started.
    - Consolidated lines keep `requisition_line_id` on the PO line for commitment tracking.
- **FR-17.4 RFQ cycle** — Must. The system shall create an RFQ with lines, invite N vendors, capture replies (price, lead time, quantity breaks, validity, notes, attachments) manually or via the supplier portal (wave 20), compare, and award whole or per line to create a PO or a purchase agreement.
    - Statuses: DRAFT, SENT, REPLIES_OPEN, CLOSED, AWARDED, CANCELLED; reply deadline enforced (late replies flagged, accepted only with override).
    - Compare view ranks by total landed estimate = price × qty + charges; ranking rule configurable (lowest price, shortest lead time, weighted score).
- **FR-17.5 Purchase agreements** — Must. The system shall support blanket agreements with commitment types QUANTITY (product × qty), VALUE (product × amount), CATEGORY_VALUE and TOTAL_VALUE, validity dates, price/discount terms, max enforcement (hard/soft) and release tracking.
    - A PO line referencing an agreement line draws down `released_qty` / `released_amount`; exceeding a hard limit rejects with `PRC-AGR-003`.
    - Agreement prices override trade agreements when `price_override=true`.
- **FR-17.6 Vendor pricing (trade agreements)** — Must. The system shall resolve unit price and line discount from `TradeAgreement` rows by most-specific match: vendor > vendor group > all vendors, product > product group > all products, with currency, unit, quantity break (from_qty) and validity (from/to dates).
    - Deterministic ordering: specificity score (vendor 4/group 2/all 0 + product 4/group 2/all 0), then latest from_date, then lowest price when still tied (configurable `tie_rule`).
    - Returns `explain` listing considered rows and the winner.
    - Lookup p95 < 20 ms (aligns with wave 26 pricing SLO; wave 26 replaces this implementation with the condition engine while keeping the API).
- **FR-17.7 Purchase order lifecycle** — Must. The system shall manage POs with approval status (DRAFT, IN_REVIEW, APPROVED, REJECTED, CONFIRMED, FINALIZED, CANCELLED) and document status (NONE, RECEIVED_PARTIAL, RECEIVED, INVOICED_PARTIAL, INVOICED) derived from downstream documents.
    - Confirmation requires approval status APPROVED and vendor not blocked; on confirm the service calls inventory `transactions:apply` to create ORDERED receipts (one per delivery schedule line) and renders output.
    - Lines carry product or category, qty, unit, unit price, discount %, line amount, delivery date, confirmed delivery date, site, warehouse, financial dimensions, tax group/item tax group (wave 13), agreement line, requisition line.
- **FR-17.8 Versioned change management** — Must. The system shall snapshot a `PurchaseOrderVersion` at each confirmation and, when change management is enabled for the vendor or legal entity, require re-approval of changes after confirmation according to the policy's `reapproval_rules` (any change / price or qty increase > X % / new line / delivery date change).
    - Version numbers monotonic from 1; diff stored as JSON Patch (RFC 6902); version document rendered and sent to vendor marked "Change order v{n}".
    - Received or invoiced quantities cannot be reduced below received/invoiced totals (`PRC-PO-007`).
- **FR-17.9 Charges** — Must. The system shall support header and line charges with calculation (FIXED, PER_UNIT, PERCENT), debit/credit sides (ITEM = capitalize into inventory cost; LEDGER = expense account; VENDOR = payable to a different vendor), and auto-charges by vendor/product group.
    - Header charges allocate to lines at invoice time by quantity, amount or line count (wave 18 consumes).
- **FR-17.10 Delivery schedules** — Must. The system shall allow one PO line to have multiple delivery schedule rows (qty, date, confirmed date) summing to line qty, each producing its own ORDERED inventory transaction.
- **FR-17.11 Totals and tax** — Must. The system shall persist header totals (`subtotal`, `discount_total`, `charges_total`, `tax_total`, `total`) recalculated on every line/charge change, calling the wave 13 tax engine, with `total = subtotal − discount_total + charges_total + tax_total` to the cent.
- **FR-17.12 PO output** — Must. The system shall render the PO/change order to PDF (template per legal entity, wave 15 document designer basics), email it to the vendor's purchasing contact, and emit `po.confirmed` for EDI/portal consumers; rendering failures do not block confirmation (retry with backoff, alert after 3 failures).
- **FR-17.13 Purchasing policies** — Should. The system shall provide entity-level policy settings: change management on/off, requisition required for PO creation (yes/no per category), PO required before invoice (consumed by wave 18), reapproval rules, consolidation window.

- **FR-17.14 Buyer workspace** — Should. The system shall show the buyer the queue of approved requisition lines, RFQs awaiting replies, POs awaiting confirmation, agreements near expiry (< 30 days) or near limit (> 90 %).
- **FR-17.15 Scheduling agreements and releases** — Must. The system shall support a `SchedulingAgreement` (SAP ME31L type LP/LPA; AX agreement with delivery schedule) per vendor × product × site with target quantity, validity, price terms, `release_profile` (FRC forecast horizon days, JIT horizon days, firm zone days, trade-off zone days, release frequency), and a `ReleaseSchedule` of dated quantities maintained manually in this wave and by wave 22 MRP later.
    - A release run (manual `:release` or the job in 3.17.16; SAP ME84) snapshots the schedule into a numbered `Release` (kind FRC or JIT) carrying per-line cumulative released quantity, cumulative received quantity (from wave 18 receipts) and the last receipt reference; the release document is rendered (3.17.15) and emitted as `dev.passage.procurement.sa.release.created.v1` for wave 35 EDI DELFOR/DELJIT.
    - Schedule lines inside the firm zone cannot be reduced without a reason code; lines in the trade-off zone warn; beyond it they are forecast only.
    - Each schedule line writes an ORDERED inventory transaction (wave 16) like a PO delivery schedule; receipts reference the schedule line.
    - Inquiry "Monitor scheduling agreement releases" lists agreements with open cumulative quantity, last release date, next release due, and release documentation per release number (SAP ME85/ME86).
- **FR-17.16 Policy framework** — Must. The system shall implement a reusable `Policy` engine (AX SysPolicy framework): policy types `PURCHASING`, `SIGNING_LIMIT`, `VENDOR_INVOICE` (consumed by wave 18), `EXPENSE` (wave 43), `AUDIT` (wave 48); each policy has an organization hierarchy purpose (wave 6), effective dates, and ordered `PolicyRule` rows of a rule type (e.g., purchasing: catalog policy, category access, purchase agreement policy, PO creation and demand consolidation, requisition control, RFQ control) with a JSON body validated by a per-rule-type schema.
    - Resolution: the most specific organization node in the hierarchy wins; rule evaluation is exposed as `POST /policies:evaluate` (pure, used by FR-17.2 and the policy tester).
    - Policies are versioned; the version at document submit time governs the instance; a "re-evaluate policies on open requisitions" job exists (AX Procurement → Policy evaluation).
    - Reports: policy rules per hierarchy node (AX SSRS Policies report) in 3.17.14.
- **FR-17.17 Signing limits** — Must. The system shall maintain `SigningLimit` per worker (wave 7/42 worker) with document types REQUISITION, PURCHASE_ORDER, INVOICE (wave 18), EXPENSE (wave 43), amount per currency, `spending_limit` and `approval_limit`, derived from `DefaultSigningLimit` rows by job and compensation level (policy type SIGNING_LIMIT) with worker-level overrides approved through a `SigningLimitRequest` workflow (wave 4), annual recertification job, and a signing limits report; FR-17.2 approval routing uses `approval_limit` so an approver whose limit is below the document amount is skipped to the next level.
- **FR-17.18 Procurement catalog** — Must. The system shall provide `ProcurementCatalog` objects (name, org units published to, validity, category navigation tree from the wave 8 procurement hierarchy, product entries with images, vendor, price source, unit, lead time, `preferred` flag) with publication status DRAFT/PUBLISHED/RETIRED; the requisition line picker searches published catalogs for the requester's org unit (catalog policy rule in FR-17.16), shows images and prices, and marks non-catalog lines.
    - Vendor catalog import: `POST /catalogs/{id}/vendor-import` accepts CSV/JSON (columns vendor product number, name, category, unit, price, currency, validity, image URL, lead time) in a staging table with buyer review (accept/reject per row) before publication; the same endpoint is used by the wave 20 supplier portal upload (FR-20.19); AX CMR XML is accepted through a mapping in the wave 10 migration framework.
    - External catalog punch-out (cXML/OCI) is a wave 35 integration hub capability; the requisition line carries `external_catalog_ref` reserved for it.
- **FR-17.19 RFQ types, sealed bids and scoring** — Should. The system shall add `RfqType` (sealed yes/no, amendment allowed, default scoring group, reply template), sealed bidding (replies encrypted at rest with a per-RFQ key and unreadable to buyers until `reply_deadline`, opening logged with user and time), `ScoringCriterion` groups (criterion, weight, scale 1–10) with buyer scoring per reply and a weighted total used by the compare view's `WEIGHTED_SCORE` ranking rule, amendments/addenda (new RFQ version re-sent to invited vendors with acknowledgement), and reply templates (questions the vendor must answer, rendered in the wave 20 portal).
- **FR-17.20 Category vendor assignment, category managers and questionnaires** — Should. The system shall allow per procurement category: `CategoryManager` (worker), `CategoryVendorAssignment` (vendor, status APPROVED/PROVISIONAL/BLOCKED, validity) enforced by the category access rule (`PRC-CAT-002` when a requisition or PO line uses a non-assigned vendor and the rule is HARD), and `CategoryQuestionnaire` (questions attached to a category, answered on the requisition line and shown to approvers).
- **FR-17.21 Invoicing plans on purchase order lines** — Should. The system shall allow a PO line (service or category line) to carry an `InvoicingPlan` of type PERIODIC (start, end, frequency MONTHLY/QUARTERLY, amount per period, billing day) or PARTIAL (milestone rows with date, percent or amount), with `settle_without_receipt=true` so that wave 18 FR-18.17 generates the invoice (SAP MRIS) on each due date without a product receipt; Σ partial percents = 100 and the plan total equals the line amount.
- **FR-17.22 Subsequent price change on open purchase orders** — Should. The system shall provide `POST /purchase-orders:reprice` (SAP MEI1/MEI2 document index; AX has no equivalent) selecting open, un-invoiced PO lines whose `price_source` is a trade agreement or agreement line changed since confirmation, previewing old/new price and amount deltas, and applying the change as a PO version (FR-17.8) with reapproval per policy; lines already received or invoiced at the old price are passed to wave 18 FR-18.18 for a subsequent debit/credit when `reprice_invoiced=true`.
- **FR-17.23 Service purchase orders, service outlines and model service specifications** — Must. The system shall support PO lines of kind SERVICE (SAP item category D; AX procurement category line) with an optional `ServiceOutline` hierarchy (outline levels with description, and service items with service product or free text, quantity, unit, unit price), `value_limit` and `expected_value` for unplanned services (SAP limit items), an optional reference to an agreement line whose remaining value constrains the limit (`contract_limit`), and reusable `ModelServiceSpecification` templates (SAP ML10–ML12, standard service catalog) copied into a PO or RFQ line.
    - Service prices resolve through FR-17.6 with `item_code=PRODUCT` on service products (SAP service conditions ML33–ML48, calculation schema MS0000); source determination picks an agreement for the service product and site when one exists.
    - Wave 18 FR-18.14 receives against service lines through service entry sheets.
- **FR-17.24 Consignment price agreements** — Should. The system shall allow a TradeAgreement row with `consignment=true` (SAP consignment info record; D365 consignment trade agreement) used by wave 16 FR-16.17 ownership changes and ignored by PO pricing unless the PO line is a consignment replenishment order line.

3.17.4 Data model

PurchaseRequisition / RequisitionLine

Field	Type	Required	Notes
id, requisition_number	uuid, text	yes	sequence `PR–`
requester_id, on_behalf_of_id	uuid	yes/no	wave 7 worker/user
legal_entity_id, operating_unit_id	uuid	yes	
status	enum	yes	DRAFT, IN_REVIEW, APPROVED, REJECTED, CLOSED, CANCELLED
policy_version_id	uuid	cond.	set at submit
total_estimated (acct ccy)	numeric(19,4)	yes	recomputed
line: line_no, product_id or (description, category_id), qty, unit, estimated_unit_price, currency, requested_date, site_id, warehouse_id, preferred_vendor_id, dimension_default_id, agreement_line_id, status (OPEN, CONSOLIDATED, CANCELLED), po_line_id			

PurchaseOrder / PurchaseOrderLine / DeliverySchedule / PurchaseOrderVersion

Field	Type	Required	Notes
id, po_number	uuid, text	yes	sequence `PO–`, gap-free optional
vendor_id, invoice_vendor_id	uuid	yes	wave 7 vendor role; invoice vendor defaults to vendor
legal_entity_id, site_id, warehouse_id	uuid	yes	
currency, exchange_rate_type	text	yes	wave 12
approval_status	enum	yes	DRAFT, IN_REVIEW, APPROVED, REJECTED, CONFIRMED, FINALIZED, CANCELLED
document_status	enum	yes	derived (FR-17.7)
version_no	int	yes	current confirmed version; 0 before first confirmation
change_management	bool	yes	copied from policy at creation
payment_terms_id, delivery_terms_id, mode_of_delivery	uuid/text	yes/no	wave 7
subtotal, discount_total, charges_total, tax_total, total	numeric(19,4)	yes	transaction currency
line: line_no, product_id or category_id, description, qty, unit, unit_price, price_unit (per 1/100/1000), discount_pct, discount_amount, line_amount, delivery_date, confirmed_delivery_date, site_id, warehouse_id, dimension_set_defaults, tax_group_id, item_tax_group_id, agreement_line_id, requisition_line_id, price_source, received_qty, invoiced_qty, remaining_qty, status (OPEN, RECEIVED, INVOICED, CANCELLED)			
schedule: po_line_id, seq, qty, delivery_date, confirmed_date, inventory_transaction_id			Σ qty = line qty
version: po_id, version_no, confirmed_at, confirmed_by, snapshot (jsonb), patch (jsonb RFC 6902), reapproval_required (bool)			

PurchaseAgreement / AgreementLine: id, agreement_number, vendor_id, legal_entity_id, currency, valid_from, valid_to, status (DRAFT, EFFECTIVE, ON_HOLD, CLOSED), default_commitment_type; line: commitment_type, product_id/category_id, committed_qty/committed_amount, unit, unit_price, discount_pct, max_enforced (HARD/SOFT/NONE), price_override (bool), released_qty, released_amount, remaining.

TradeAgreement: id, relation (PRICE, LINE_DISC, MULTILINE_DISC, TOTAL_DISC), account_code (VENDOR/GROUP/ALL), account_id, item_code (PRODUCT/GROUP/ALL), item_id, currency, unit, from_qty, to_qty, amount or percent, price_unit, valid_from, valid_to, site_id (optional), lead_time_days, status (DRAFT, ACTIVE, EXPIRED).

RFQ / RFQInvitation / RFQReply: rfq id, number, status, reply_deadline, lines (product/category, qty, unit, required_date); invitation (vendor_id, sent_at, channel EMAIL/PORTAL, status INVITED, DECLINED, REPLIED); reply (invitation_id, line_id, unit_price, currency, lead_time_days, qty_breaks jsonb, valid_until, notes, attachments, awarded (bool)).

Charge: id, owner_type (PO_HEADER, PO_LINE, RECEIPT, INVOICE), owner_id, charge_code_id, calc (FIXED, PER_UNIT, PERCENT), value, amount, currency, debit_type (ITEM, LEDGER, VENDOR), credit_type, debit_account_id, credit_account_id, allocation_rule.

Invariants: `po.total = Σ line_amount − discount + charges + tax` enforced by a deferred constraint trigger; Σ schedule qty = line qty; `received_qty ≤ qty` and `invoiced_qty ≤ max(received_qty, qty)`; agreement `released ≤ committed` when HARD; exactly one ACTIVE TradeAgreement wins per lookup (ties resolved by `tie_rule`). Indexes: PO (tenant, vendor, approval_status), line (po_id, line_no), schedule (po_line_id), TradeAgreement (tenant, relation, account_code, account_id, item_code, item_id, valid_from) for lookup; requisitions by requester and status.

3.17.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/procurement/requisitions	Create requisition	client key	RequisitionCreate
POST	/api/v1/procurement/requisitions/{id}:submit	Submit for approval	req id + version	RequisitionCreate
POST	/api/v1/procurement/requisitions:consolidate	Create POs from approved lines	line id set hash	PurchaseOrderMaintain
POST	/api/v1/procurement/rfqs	Create RFQ	client key	RFQMaintain
POST	/api/v1/procurement/rfqs/{id}:send	Send invitations	rfq id	RFQMaintain
POST	/api/v1/procurement/rfqs/{id}/replies	Record a reply	invitation+line	RFQMaintain / portal vendor
POST	/api/v1/procurement/rfqs/{id}:award	Award to vendor(s)	rfq id	RFQAward
POST	/api/v1/procurement/agreements	Create agreement	client key	AgreementMaintain
GET	/api/v1/procurement/pricing/lookup?vendor=&product=&qty=&date=&currency=	Resolve price/discount with explain	n/a	PurchaseOrderMaintain
POST	/api/v1/procurement/purchase-orders	Create PO	client key	PurchaseOrderMaintain
PATCH	/api/v1/procurement/purchase-orders/{id}	Edit header/lines (creates pending change)	ETag	PurchaseOrderMaintain
POST	/api/v1/procurement/purchase-orders/{id}:confirm	Confirm (new version)	po id + version_no+1	PurchaseOrderConfirm
POST	/api/v1/procurement/purchase-orders/{id}:cancel	Cancel remaining	po id	PurchaseOrderMaintain
GET	/api/v1/procurement/purchase-orders/{id}/versions/{n}/diff	JSON Patch and rendered diff	n/a	PurchaseOrderInquire
GET	/api/v1/procurement/purchase-orders/{id}/document.pdf	Rendered PO	n/a	PurchaseOrderInquire

Commands: `SubmitRequisition` (account structure valid, lines ≥ 1, no closed category), `ConsolidateLines` (same vendor/currency/site/entity, vendor not blocked), `AwardRFQ` (reply not expired, award qty ≤ RFQ qty), `ConfirmPO` (approved, vendor active, agreement limits, delivery schedules sum, tax calculated, totals constraint), `ApplyChange` (post-confirmation changes evaluated against reapproval rules; sets approval_status back to IN_REVIEW if required).

Event	Trigger	Payload fields	Consumers
`dev.passage.procurement.requisition.submitted.v1` / `.approved.v1` / `.rejected.v1`	workflow transitions	requisition_id, lines (product, qty, amount), total, approver chain, policy_version	notifications, budget (w38), analytics
`dev.passage.procurement.rfq.sent.v1` / `.awarded.v1`	RFQ send/award	rfq_id, vendors, lines, awarded vendor per line	supplier-portal (w20), notifications
`dev.passage.procurement.po.confirmed.v1`	confirm	po_id, version_no, vendor_id, lines (product, qty, unit, price, delivery dates, site/warehouse), totals, output channel	inventory, receiving (w18), supplier-portal, EDI (w35), analytics
`dev.passage.procurement.po.version.created.v1`	any confirmed version > 1	po_id, version_no, patch, reapproval_required	supplier-portal, receiving
`dev.passage.procurement.po.cancelled.v1`	cancel	po_id, lines cancelled, remaining qty	inventory (cancel ORDERED), supplier-portal
`dev.passage.procurement.agreement.threshold.v1`	released ≥ 90 % or expiry < 30 d	agreement_id, line, pct, expiry	notifications
`dev.passage.procurement.sa.release.created.v1`	scheduling agreement release run	agreement_id, release_number, kind (FRC/JIT), vendor_id, lines (product, schedule dates, qty, cumulative released, cumulative received)	EDI (w35 DELFOR/DELJIT), supplier-portal (w20), planning (w22)
`dev.passage.procurement.policy.changed.v1` / `signinglimit.changed.v1`	policy version activated / signing limit approved	policy_id, type, version, hierarchy purpose; worker_id, document type, limits	procurement (re-evaluation), expense (w43), audit (w48)
`dev.passage.procurement.po.repriced.v1`	`:reprice` applied	po_id, version_no, lines (old price, new price, delta), invoiced lines passed to payables	payables (w18 subsequent debit/credit), analytics

3.17.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Procurement workspace (buyer)	Tiles and lists: lines to source, RFQs open, POs to confirm, agreements expiring, change orders pending re-approval	counts, aging	PurchaseOrderInquire
Requisition (requester, self-service)	Add product from catalog search, non-catalog line, submit, track approval	product, qty, price est., date, dimensions, approval timeline	RequisitionCreate
Approval task (shell inbox)	Approve, reject, delegate, request change, comment	requisition summary, lines, policy band hit, attachments	RequisitionApprove
Consolidation board	Group suggestions, drag lines between POs, change vendor, create POs	vendor, currency, site, lines, estimated total	PurchaseOrderMaintain
RFQ	Create, invite vendors, record replies, compare, award	lines, invitations, replies grid (price, lead time, validity), ranking	RFQMaintain
Purchase agreement	Create, add lines, view releases	commitment type, committed, released, remaining %, validity	AgreementMaintain
Purchase order	Create, add lines (price explain), charges, delivery schedules, confirm, view versions diff, print/email	header, lines grid, totals panel live, version timeline	PurchaseOrderMaintain
Trade agreement journal	Create rows in grid, validate overlap, post (activate)	relation, account/item codes, qty break, price, validity	TradeAgreementMaintain

Interaction rules: line grid keyboard entry with product search-as-you-type (p95 < 100 ms via wave 8 search index); price lookup fires on product/qty/vendor/date change and shows source badge; totals panel updates within 200 ms of any edit (server computed, optimistic client); bulk edit on selected lines (delivery date, site, dimensions); validation timing: inline for required/format, on `:confirm` for agreement limits and tax; approvers can act from email deep links with Keycloak session; personalization: saved views per buyer.

3.17.7 Business rules and edge cases

- **BR-17.1** Requisition state machine: `DRAFT → IN_REVIEW → APPROVED → CLOSED`; `IN_REVIEW → REJECTED → DRAFT`; `DRAFT|REJECTED → CANCELLED`; `APPROVED` lines may be individually CANCELLED by the requester before consolidation.
- **BR-17.2** PO approval state machine: `DRAFT → IN_REVIEW → APPROVED → CONFIRMED`; `CONFIRMED −(change requiring re-approval)→ IN_REVIEW`; `CONFIRMED −(change not requiring)→ CONFIRMED (version+1)`; `CONFIRMED → FINALIZED` when document_status = INVOICED and remaining = 0; any non-finalized → CANCELLED if received_qty = 0 and invoiced_qty = 0.
- **BR-17.3** Price resolution date is the PO line creation date by default, switchable to delivery date per policy; a trade agreement with `valid_to` null is open-ended; overlapping rows with equal specificity and dates are rejected at activation (`PRC-TA-002`).
- **BR-17.4** Quantity breaks use `from_qty ≤ qty < to_qty` in the agreement's unit after conversion; conversion failure falls back to the next less-specific row and records the fallback in explain.
- **BR-17.5** Rounding: unit price stored to 4 decimals of currency minor units (e.g., 12.3456 USD per price_unit); line_amount = round_half_even(qty × unit_price / price_unit × (1 − discount_pct/100), minor units); header totals are sums of rounded line amounts (never re-rounded), matching AX PurchTotals and SAP KOMP netwr behavior.
- **BR-17.6** Confirmation in a different fiscal period than creation is allowed; commitment date for budget (wave 38) is the confirmation date.
- **BR-17.7** Reducing PO line qty below `received_qty` is rejected; reducing below `invoiced_qty` but above `received_qty` is allowed only when the policy flag `allow_invoice_before_receipt` is true.
- **BR-17.8** Concurrency: PO edits use optimistic concurrency via ETag (`row_version`); a stale PATCH returns 412 with the current version and diff.
- **BR-17.9** Deleting a line on a confirmed PO cancels its remaining ORDERED inventory transactions; the line remains with status CANCELLED in versions.
- **BR-17.10** Agreement validity: a PO line may reference an agreement line only if the PO's requested delivery date falls within `valid_from..valid_to`; SOFT limit overrun emits a warning and `agreement.threshold`; HARD rejects.
- **BR-17.11** RFQ replies received after `reply_deadline` are stored with `late=true` and excluded from ranking unless the buyer overrides with a reason.
- **BR-17.12** Vendor currency differing from PO currency: PO currency defaults from the vendor's purchase currency; overriding requires an exchange rate available on the PO date (wave 12), else `PRC-FX-001`.
- **BR-17.13** Scheduling agreement zones: a schedule line with date ≤ today + `firm_zone_days` is FIRM (reduction requires reason code and emits a change release), ≤ today + `tradeoff_zone_days` is TRADEOFF (reduction warns), otherwise FORECAST; cumulative released quantity on a release never decreases unless a cumulative reset (new agreement year) is recorded explicitly.

- **BR-17.14** Policy resolution walks the organization hierarchy of the policy type's purpose from the document's operating unit upward; the first node with an active policy version wins; if no node has one, the legal-entity default policy applies; absence of a rule type in the winning policy means "no restriction".
- **BR-17.15** Signing limits are evaluated in the document currency converted to the worker's limit currency at the document date rate; an approver whose `approval_limit < amount` is skipped and the chain continues upward; a requester whose `spending_limit < amount` triggers an additional approval level rather than a rejection.
- **BR-17.16** Sealed RFQ replies cannot be read, scored or compared before `reply_deadline`; opening early requires two users with RFQAward and is logged as a sealed-bid exception; a late reply to a sealed RFQ is rejected (not merely flagged).
- **BR-17.17** Invoicing plan totals: PERIODIC plan total = periods × amount and must equal the line amount; PARTIAL plan percents sum to 100; changing the line amount after plan generation reproportions ungenerated plan rows only.
- **BR-17.18** Reprice selects only lines with `invoiced_qty = 0` by default; lines with receipts but no invoice reprice the GR/IR expectation (wave 18 GrirOpenItem keeps the receipt-side amount; difference flows at invoice); the run never changes manually priced lines (`price_source = MANUAL`).
- **BR-17.19** Service outline items sum to the outline parent; `value_limit` applies to the sum of unplanned service entry sheet amounts (wave 18) and `contract_limit` to the referenced agreement line's remaining value; exceeding either rejects the entry sheet with `PRC-SRV-003`.

3.17.8 Acceptance criteria

- **AC-17.1** (FR-17.2) Given the default band policy, when 30 requisitions spanning 6 amount bands × 5 categories are submitted, then each routes to the expected approver chain per the test matrix in `fixtures/procurement/approval-matrix.csv`.
- **AC-17.2** (FR-17.8) Given a confirmed PO with change management on and rule "qty increase > 5 % requires re-approval", when a line qty is raised from 100 to 110, then version 2 is created with a JSON Patch on `/lines/0/qty`, approval_status = IN_REVIEW, and the vendor is not notified until re-confirmation.
- **AC-17.3** (FR-17.8) Given the same PO, when the delivery date moves by one day and the rule excludes dates, then version 2 is created with approval_status CONFIRMED and the change order PDF is emailed.
- **AC-17.4** (FR-17.6) Given rows for vendor V (price 10), vendor group G containing V (price 9), and all vendors (price 8), all valid, when lookup for V runs, then price 10 is returned and explain lists all three rows (property test over random overlapping sets: most specific valid row always wins).
- **AC-17.5** (FR-17.11) Given 1,000 random POs with 1–50 lines, charges and tax, when totals are read, then `total = subtotal − discount_total + charges_total + tax_total` to the cent for all 1,000.
- **AC-17.6** (FR-17.7) Given an approved PO with 2 delivery schedule rows, when confirmed, then 2 inventory transactions at ORDERED exist with matching quantities and dates and on-hand `ordered` increases accordingly.
- **AC-17.7** (FR-17.5) Given a QUANTITY agreement line of 1,000 EA with HARD enforcement and 950 released, when a PO line of 100 EA references it, then confirmation fails with `PRC-AGR-003`.
- **AC-17.8** (FR-17.4) Given an RFQ sent to 3 vendors with 2 replies before deadline and 1 after, when compared, then the late reply is excluded and awarding to the lowest creates a PO with the reply price and lead time.
- **AC-17.9** (FR-17.3) Given 10 approved requisition lines across 2 vendors and 2 sites, when consolidated with the default window, then 4 POs are created and each requisition line references its PO line.
- **AC-17.10** (FR-17.12) Given the PDF renderer is down, when a PO is confirmed, then confirmation succeeds, `po.confirmed` is emitted, and the render activity retries 3 times before alerting.
- **AC-17.11** (FR-17.15) Given a scheduling agreement with schedule lines of 100 EA weekly for 12 weeks, firm zone 14 days, when a JIT release is created on day 0 and 100 EA are received against week 1, then release 1 shows cumulative released 1,200 and cumulative received 0, release 2 (run on day 8) shows cumulative received 100, and reducing week-2 quantity to 50 requires a reason code.
- **AC-17.12** (FR-17.16, FR-17.17) Given a purchasing policy on operating unit "Plant A" with category access rule blocking category "IT hardware" for requesters outside IT, and a requester with approval limit 5,000 USD, when a 7,000 USD requisition for IT hardware is submitted from Plant A by a non-IT requester, then it is rejected with `PRC-CAT-002`; submitted by an IT requester, it routes past the requester's manager (limit 5,000) to the next approver with limit ≥ 7,000.
- **AC-17.13** (FR-17.18) Given a published catalog for org unit "Finance" with 200 items and a vendor import file of 50 rows of which 3 lack a unit, when imported, then 47 rows stage as ACCEPTABLE and 3 as ERROR with reason, and after buyer acceptance the 47 appear in the catalog and in the requisition picker for Finance requesters only.
- **AC-17.14** (FR-17.19) Given a sealed RFQ with deadline 2026-11-01T17:00Z and 3 replies, when a buyer opens the compare view at 2026-11-01T16:59Z, then no reply prices are returned (403 `PRC-RFQ-004`); at 17:00:01Z all three are visible and the opening is logged.
- **AC-17.15** (FR-17.21) Given a service PO line of 12,000.00 EUR with a PERIODIC plan monthly 1,000.00 from 2027-01-01, when generated, then 12 plan rows exist with `due_date` on the billing day and Σ amount = 12,000.00; changing the line to 13,200.00 before any invoice reproportions all 12 rows to 1,100.00.
- **AC-17.16** (FR-17.22) Given 100 open PO lines priced from trade agreement T1 and T1's price changed from 10.00 to 9.50, when `:reprice` previews and applies, then 100 PO versions are created with patch `/lines/n/unit_price` 10.00 → 9.50, lines with invoices are excluded from the version and listed for wave 18 subsequent credit, and manually priced lines are untouched.
- **AC-17.17** (FR-17.23) Given a service PO line with outline "Maintenance" containing 3 service items and `value_limit` 2,000.00 for unplanned work, when wave 18 service entry sheets total 1,900.00 unplanned, then a further 200.00 unplanned entry is rejected with `PRC-SRV-003`.

3.17.9 Test plan and fixtures

Fixtures: `proc-demo-vendors` (40 vendors with groups, payment terms, currencies USD/EUR; 300 trade agreement rows); `proc-approval-matrix` (30 requisitions, expected chains); `proc-po-1000` (1,000 random POs generated with seed for totals test); `proc-rfq-cases` (12 RFQs); `proc-ax-extract` (PurchTable/PurchLine/PurchAgreementHeader/PriceDiscTable from USMF); `proc-sap-extract` (EKKO/EKPO/EKET/KONP/A017 samples).

Test	Type	Target	Pass criterion
Price specificity	property	pricing/lookup	most-specific valid row wins over 10k random sets
Totals identity	property + E2E	PO totals	AC-17.5
Approval routing	E2E (Temporal test server)	requisition workflow	AC-17.1
Version diff	unit	JSON Patch generation/apply	apply(patch, v_n) = v_{n+1} for all fixtures
Confirm → inventory	contract	inventory `transactions:apply`	AC-17.6
Agreement limits	unit	ConfirmPO	AC-17.7
PDF render	E2E	render worker	PDF/A-2b valid, < 5 s
Lookup latency	performance	pricing/lookup at 500 rps	p95 < 20 ms
Event schemas	contract	6 events	AsyncAPI BACKWARD compatible

Reconciliation checks: Σ requisition lines CONSOLIDATED = Σ PO lines with requisition_line_id; Σ agreement released = Σ confirmed PO line qty/amount referencing it; Σ inventory ORDERED qty per PO line = remaining open schedule qty.

3.17.10 Deployment and operations

Helm chart `passage-procurement` (API Deployment ×3, render worker ×2, Temporal worker ×2, CloudNativePG `procurement-db`, outbox connector). Config: `PRC_CONSOLIDATION_WINDOW_DAYS=7`, `PRC_CHANGE_MGMT_DEFAULT=true`, `PRC_RFQ_LATE_POLICY=flag`, `PRC_PDF_TIMEOUT_S=20`. Migrations `procurement/v17.*`. Feature flags: `prc.rfq_portal_replies` (on in w20), `prc.edi_output` (w35), `prc.delivery_date_pricing`. Jobs: agreement threshold scan daily 06:00; RFQ deadline closer hourly; approval SLA escalation (Temporal timers). SLOs: PO create/confirm API p95 < 500 ms; requisition submit to workflow start < 2 s; PDF < 5 s p95. Alerts: workflow task backlog > 100 for 10 min; render failures > 5/h; pricing lookup p95 > 50 ms. Runbook: re-render document, restart stuck approval (Temporal signal), re-emit `po.confirmed`. Rollback: additive migrations; flag off `prc.change_mgmt` if version logic misbehaves (POs remain confirmable without versions).

3.17.11 Migration from AX / SAP

Source	Passage entity	Notes
AX PurchTable/PurchLine (open: PurchStatus Backorder, open qty > 0)	PurchaseOrder/Line version 1, approval_status CONFIRMED	remaining qty = PurchQty – received; received history not migrated (opening on-hand covers stock)
AX PurchAgreementHeader/Line, AgreementLineReleasedLine	PurchaseAgreement/Line with released totals	released computed from source releases
AX PriceDiscTable (relation PurchPrice, PurchLineDisc)	TradeAgreement	AccountCode/ItemCode map 1:1; QuantityAmountFrom/To → from_qty/to_qty
AX PurchReqTable approved-not-consolidated	PurchaseRequisition APPROVED	in-review requisitions re-submitted after cutover
AX PurchRFQ*	not migrated	open RFQs re-issued
SAP EKKO/EKPO/EKET (BSTYP F, open per EKPO ELIKZ ≠ X)	PurchaseOrder/Line/DeliverySchedule	NETPR/PEINH → unit_price/price_unit; MENGE – delivered
SAP EKKO BSTYP K/L, EKAB release docs	PurchaseAgreement/Line	KTMNG target qty, KTWRT target value
SAP A017/A018 + KONP (PB00, RA01)	TradeAgreement	KBETR/KPEIN; KONM scales → qty breaks
SAP EBAN open (BANPR approved)	PurchaseRequisition	

Validation: count and Σ open amount per vendor equal source; sample 50 POs compared field-by-field; price lookup for 200 random vendor/product pairs matches source price. Not migrated: closed POs (reporting only, in lakehouse wave 46), PurchTableHistory versions, SAP change documents, NAST output logs.

3.17.12 Dependencies, risks, open questions

Depends on: wave 16 (ORDERED transactions, on-hand), wave 7 (vendors, groups, terms, blocking), wave 8 (products, categories, search), wave 11–13 (dimensions, currency, tax), wave 4 (approval workflow, notifications), wave 5 (shell inbox, document templates basics).

Risk	Likelihood	Impact	Mitigation
Wave 26 pricing engine replaces trade agreement lookup; behavior drift	Medium	Medium	Keep `pricing/lookup` contract and fixture set; wave 26 must pass wave 17 tests
Re-approval rule complexity grows	High	Low	Rules are a small DSL with fixtures; defaults cover 90 %
PDF template per legal entity before document designer (w15 basics only)	Medium	Low	One default template plus logo/address substitution

Open questions:

1. Should requisitions support punch-out catalogs in v1? Default: no, v1.x. Decide by wave 20.
2. Gap-free PO numbering per legal entity by default? Default: off (sequence with gaps allowed); on for countries requiring it. Decide by wave 18.
3. Scheduling-agreement delivery schedules from MRP: wave 22 or wave 17 manual only? Default: manual only now. Decide by wave 21.

3.17.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`change_management_enabled` / `change_management_vendor_override`	legal entity / vendor	bool	true / inherit	true/false/inherit	FR-17.8 versioning and re-approval	Procurement and sourcing parameters → Purchase order change management; Vendor → "Activate change management"	OMFI/OMFJ version management per document type (T160VS)
`reapproval_rules`	legal entity (policy)	rule set	any price or qty increase > 5 %, new line	DSL (field, operator, threshold)	FR-17.8	Purchase order approval workflow conditions	Release strategy re-triggering (CEKKO changes)
`requisition_required_categories`	legal entity	category list	none	procurement categories	FR-17.13 PO creation requires requisition	Purchasing policy rule "Purchase order creation and demand consolidation"	n/a (PR-to-PO link enforced by T161 "PR required")
`po_required_before_invoice`	legal entity	bool	false	true/false	consumed by wave 18 expense invoices	Accounts payable parameters → "Require purchase order"	n/a
`consolidation_window_days`	legal entity	int	7	1–60	FR-17.3 grouping	Purchasing policy → Demand consolidation	ME59N grouping by delivery date (variant)
`approval_bands`	policy (PURCHASING)	table	0–1k auto; 1k–10k manager; 10k–100k + category owner; >100k + CFO	amount band × category × org unit × role	FR-17.2	Purchase requisition workflow conditions; purchasing policy	Release strategy CL24N characteristics (amount, material group)
`price_resolution_date`	legal entity	enum	ORDER_DATE	ORDER_DATE, DELIVERY_DATE	BR-17.3	Procurement parameters → Prices → "Price date"	Pricing date control on vendor master (LFM1-MEPRF)
`tie_rule`	legal entity	enum	LOWEST_PRICE	LOWEST_PRICE, LATEST_FROM_DATE	FR-17.6	Trade agreement "Find next" / search precedence	Access sequence exclusion flags

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`rfq_late_policy` / `rfq_default_ranking`	legal entity	enum	FLAG / LOWEST_PRICE	FLAG, REJECT; LOWEST_PRICE, LEAD_TIME, WEIGHTED_SCORE	FR-17.4, BR-17.11	RFQ parameters, bidding parameters	n/a (ME49 ranking manual)
`rfq_types`	legal entity	table	STANDARD, SEALED	sealed flag, amendments allowed, scoring group, reply template	FR-17.19	RFQ types, solicitation types, scoring criteria/groups, reply templates	Document types for RFQ (OMEB)
`agreement_threshold_pct` / `agreement_expiry_days`	legal entity	numeric / int	90 / 30	50–100 / 1–180	FR-17.14 alerts	Purchase agreements → Agreement classifications (expiry)	ME3M validity monitoring (custom)
`charge_codes` / `auto_charges`	legal entity	table	seeded FREIGHT, HANDLING, CUSTOMS	code, calc, debit/credit types, accounts; auto charge by vendor/item group	FR-17.9	Charges codes, Auto charges, Charges groups	Condition types FRA1/ZOA1, calculation schema RM0000
`pdf_template_id` / `po_email_from`	legal entity	reference / email	default template / entity address	wave 15 templates	FR-17.12	Form setup (purchase documents), Print management	NACE EF output type NEU, MEDRUCK form
`po_number_gap_free`	legal entity	bool	false	true/false	number sequence continuity (open question 2)	Number sequence "Continuous"	Number range internal, no gaps (T161 ext. numbering)
`sa_release_profile`	scheduling agreement (default per legal entity)	record	FRC 90 d / JIT 14 d / firm 14 d / trade-off 28 d / weekly	horizons in days; frequency	FR-17.15	n/a	T163M release creation profile; firm zone / trade-off zone on SA
`policy_types` / `hierarchy_purpose_per_type`	tenant / legal entity	table	PURCHASING → "Procurement internal control"; SIGNING_LIMIT → "Signing limit"	wave 6 hierarchy purposes	FR-17.16	Policy parameters (organization hierarchy purpose per policy type)	n/a
`default_signing_limits`	policy (SIGNING_LIMIT)	table	none	job, compensation level, document type, currency, spending/approval limit	FR-17.17	Default signing limits (per job / compensation level), Signing limit parameters	Release codes per user
`signing_limit_recertification_months`	legal entity	int	12	3–24	FR-17.17 job	Signing limit request workflow (recertification)	n/a
`catalog_policy_rule`	policy (PURCHASING)	rule	catalog required for categories X	categories, org units, allow non-catalog	FR-17.18	Catalog policy rule	n/a
`vendor_catalog_import_params`	legal entity	record	max 50,000 rows; image max 2 MB; currencies allowed		FR-17.18	Vendor catalog import parameters, Catalog images	n/a
`category_access_mode`	policy (PURCHASING)	enum	SOFT	HARD, SOFT, OFF	FR-17.20 `PRC–CAT–002`	Category access policy rule, Category vendor assignment	Source list "fixed/blocked" (ME01)
`invoicing_plan_types`	legal entity	table	MONTHLY, QUARTERLY, MILESTONE	type, frequency, billing day, settle without receipt	FR-17.21	n/a	Invoicing plan types (T163*), OMEC
`reprice_invoiced`	legal entity	bool	false	true/false	FR-17.22 passes invoiced lines to wave 18	n/a	MRNB revaluation activation
`service_limit_defaults`	legal entity	record	expected value = 0, limit = 0 (must be set), overall limit check HARD		FR-17.23	n/a	Limit item defaults, MM-SRV screen layout, value limits
`model_service_specifications`	legal entity	table	none	outline template rows	FR-17.23	n/a	ML10–ML12 model service specs, standard service catalog

3.17.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Purchase orders (all / open / received not invoiced / backordered)	list inquiry	vendor, status, document status, site, buyer, date range	PO number, vendor, approval status, document status, total, version	`PurchaseOrder`	Purchase orders list page; Backordered purchase lines; Received but not invoiced	ME2L / ME2M / ME2N with selection parameters WE101 / RECHNUNG; ME80FN
Purchase order lines and history	drill-down	product, vendor, delivery date, status	line qty, received, invoiced, remaining, price, price source, versions	`PurchaseOrderLine`, `PurchaseOrderVersion`	Purchase order lines / Purchase order history / Versions	ME23N purchase order history (EKBE), ME2N line view
Requisitions by requester / status / consolidation	list inquiry	requester, status, org unit, category, date	requisition, lines, estimated total, approver chain position, PO link	`PurchaseRequisition`	Purchase requisition inquiries; My requisitions	ME5A / ME5K / ME5J, Monitor Purchase Requisitions
Approval status and SLA	drill-down	document, approver, overdue	step, approver, assigned at, due, decision	Temporal workflow projection	Workflow history	SWI1 / release log
RFQ status, replies and award history	list inquiry	RFQ, vendor, status, deadline	invitations, replies, ranking, scores, awarded	`RFQ`, `RFQReply`	RFQ inquiries; Compare replies	ME4L / ME4M / ME4N, ME49 price comparison, ME47

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Purchase agreement fulfillment	list inquiry	agreement, vendor, product, validity	committed, released, remaining %, expiry	`PurchaseAgreement`	Purchase agreement inquiries (fulfillment / released quantities)	ME3L / ME3M / ME3N release documentation, ME80RN
Pricing explain	drill-down	vendor, product, qty, date, currency	candidate rows, specificity, winner	FR-17.6 explain	Trade agreement "Find prices"	ME23N → Conditions → Analysis (condition analysis)
Trade agreement list by vendor / product	list inquiry	relation, account/item codes, validity	price, breaks, currency, validity	`TradeAgreement`	Trade agreements (PriceDiscTable) inquiry	MEKA / MEKB / MEKC / MEKD condition lists, ME1L / ME1M info records
Order price history	operational report	product, vendor, date range	PO date, vendor, price, qty, agreement	`PurchaseOrderLine`	n/a (PurchLine query)	ME1P order price history
Charges inquiry	list inquiry	PO, charge code, vendor	charge, calc, amount, debit/credit sides	`Charge`	Charges inquiry	Condition records on PO (KONV)
Vendor delivery performance (confirmed vs actual)	KPI	vendor, period	on-time %, average days late, lines	`DeliverySchedule` + receipts (w18)	Vendor delivery performance (D365)	ME2V goods receipt forecast; vendor evaluation ME6H
Buyer workspace queues	KPI	buyer	lines to source, RFQs open, POs to confirm, agreements expiring, change orders pending	FR-17.14	Procurement workspace	Fiori Procurement Overview
Scheduling agreement releases monitor	list inquiry	agreement, vendor, release kind, date	release number, cumulative released, cumulative received, next release due	`Release`	n/a	Monitor Scheduling Agreement Releases; ME85 / ME86 release documentation
Policy rules per hierarchy node	operational report	policy type, hierarchy, date	node, policy version, rule type, rule body	`Policy`	SSRS Policies report	n/a
Signing limits report	operational report	org unit, document type, as-of	worker, job, spending limit, approval limit, source (default/override), recertified at	`SigningLimit`	SSRS Signing limits report	Release codes assignment list
Procurement catalog report	operational report	catalog, category, org unit	items, vendor, price, publication status, missing images	`ProcurementCatalog`	SSRS Procurement catalog report	n/a
Services per purchase order	list inquiry	PO, vendor, service product	outline, planned value, entered (w18), invoiced, limit remaining	`ServiceOutline`	n/a	ME2S services per PO, MSRV1–MSRV6 service lists
Invoicing plan schedule	list inquiry	PO, vendor, due date range	plan row, due date, amount, status (OPEN/INVOICED)	`InvoicingPlan`	n/a	MRIS selection list

3.17.15 Printed and electronic documents

Document	Trigger	Layout elements (header / line / footer)	Formats	Numbering sequence	Legal / retention requirement
Purchase order / change order	`:confirm` (every version)	header: PO number, version, vendor name/address, ship-to, bill-to, buyer, currency, payment and delivery terms, dates; lines: line no, product/vendor item no, description, qty, unit, price, discount, amount, delivery date, agreement ref; footer: subtotal, charges, tax, total, terms text, signature	PDF/A-2b, email (wave 4), EDI 850/ORDERS via `po.confirmed` (wave 35), portal (wave 20)	PO number `PO-` + `v{n}`	Commercial document; retain 10 years (DE/FR), 7 years (US); e-signature optional
Request for quotation	`:send`	header: RFQ number, deadline, terms, contact; lines: product/description, qty, unit, required date; questions from reply template	PDF, email, portal	RFQ number `RFQ-`	Public-sector solicitation rules where applicable (wave 48)
RFQ amendment / addendum	amendment issued	as RFQ with change summary	PDF, email, portal	`RFQ-{n}-A{k}`	Same
Purchase agreement document	agreement EFFECTIVE	header: agreement number, vendor, validity, commitment type; lines: product/category, committed qty/amount, price terms	PDF, email	`PA-`	Contract; retain 10 years after expiry
Scheduling agreement release (FRC / JIT)	release run	header: agreement, release number, kind, vendor, cumulative received, last receipt; lines: schedule date, qty, cumulative qty	PDF, email; EDI DELFOR (FRC) / DELJIT (JIT) via wave 35	`REL-{agreement}-{n}`	Commercial; retain with agreement
Consolidated requisition approval summary	approval task	requisition header, lines, policy band, approver chain	HTML email with deep link	n/a	none
Requisition printout	on demand	header, lines, dimensions, approval history	PDF	`PR-`	none (internal)
Vendor catalog import result	import finished	accepted/rejected rows with reasons	CSV	n/a	none

3.17.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Agreement threshold and expiry scan	daily 06:00	legal entity	key = (agreement line, date); emits each threshold once	< 1 min per 10k lines	retry 3×	missed run ticket
RFQ deadline closer	hourly	legal entity	sets REPLIES_OPEN → CLOSED where deadline passed; idempotent	< 10 s	retry next hour	RFQs past deadline still open > 2 h → ticket
Approval SLA escalation	Temporal timers per instance	workflow id	timer fires once per step	n/a	workflow retry	task backlog > 100 → ticket
Release approved requisitions (auto-create POs)	daily 07:00 (opt-in)	legal entity × vendor	key = line id set hash; lines already consolidated skipped	< 2 min per 5k lines	lines with errors left in queue	errors > 0 → buyer notification
Scheduling agreement release run (FR-17.15)	per profile frequency (default weekly Monday 05:00; JIT daily)	agreement	key = (agreement, kind, run date); unchanged schedule produces no new release	< 1 min per 1k agreements	failed agreement listed, rest released	releases failed > 0 → buyer ticket
Policy re-evaluation on open requisitions (FR-17.16)	on policy version activation + nightly 01:00	legal entity	key = (policy version, requisition); re-routes only when chain differs	< 2 min per 10k requisitions	retry 3×	re-routed count reported

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Signing limit recertification (FR-17.17)	yearly per `signing_limit_recertification_months` + daily sweep	legal entity	key = (worker, cycle); creates one request per worker per cycle	< 1 min	retry	unrecertified limits past due → manager notification
Catalog publication / vendor import processing (FR-17.18)	on demand + nightly 02:00	catalog	key = (import file hash); rows staged once	< 5 min per 50k rows	row-level errors	errors > 5 % → import marked FAILED
Subsequent price change run (FR-17.22)	on demand; optional weekly Sunday 06:00	legal entity × trade agreement	key = (agreement version, run); lines already repriced skipped	< 5 min per 10k lines	per-PO failure isolated	versions created / failed reported
Delete and clean-up (cancelled POs older than N days, expired requisitions, update history)	monthly 1st 03:00	legal entity	idempotent delete by cut-off	< 5 min	skip	none

3.17.17 Security duties and privileges

Duty	Privileges (entry points: API paths / screens / jobs)	Segregation-of-duties conflicts	Default roles that include it	AX duty / SAP authorization reference
RequisitionCreate	POST /requisitions, :submit; requisition screen; catalog picker	none	Employee (all), Requester	AX "Maintain purchase requisitions" (PurchReqMaintain); SAP M_BANF_BSA, M_BANF_WRK
RequisitionApprove	approval tasks for requisitions; delegate	with RequisitionCreate on own requisitions (self-approval blocked by workflow)	Manager, Category owner, CFO	AX "Approve purchase requisitions"; SAP M_BANF_FRG release codes
PurchaseOrderInquire	GET purchase orders, versions, diff, document.pdf; buyer workspace; inquiries 3.17.14	none	Buyer, Receiving clerk, AP clerk, Controller	AX "Inquire into purchase orders"; SAP M_BEST_BSA activity 03
PurchaseOrderMaintain	POST/PATCH purchase orders, :cancel, requisitions:consolidate, pricing/lookup, charges	with PurchaseOrderConfirm when `po_four_eyes=true`; with ReceivingPost (w18) and VendorInvoicePost (w18)	Buyer	AX "Maintain purchase orders" (PurchTableMaintain); SAP M_BEST_BSA 01/02, M_BEST_EKG purchasing group
PurchaseOrderConfirm	:confirm; approval tasks for POs; :reprice apply	with PurchaseOrderMaintain under four-eyes; with VendorMaintain (w7)	Senior buyer, Purchasing manager	AX "Confirm purchase orders"; SAP M_EINK_FRG release PO (ME28/ME29N)
RFQMaintain	POST rfqs, :send, replies; RFQ screen; amendments	with RFQAward on same RFQ when sealed	Buyer	AX "Maintain requests for quotation"; SAP M_ANFR_BSA, M_ANGB_BSA
RFQAward	:award; sealed-bid opening; scoring	with RFQMaintain (sealed)	Purchasing manager	AX "Award RFQ"; SAP M_ANGB_FRG
AgreementMaintain	POST agreements, lines; scheduling agreements, release schedules	with PurchaseOrderConfirm when policy requires	Buyer, Contract manager	AX "Maintain purchase agreements"; SAP M_RAHM_BSA, M_RAHM_FRG
TradeAgreementMaintain	trade agreement journal create/post; consignment price rows	with PurchaseOrderConfirm (price manipulation)	Pricing analyst, Purchasing manager	AX "Maintain trade agreements" (PriceDiscAdm); SAP M_EINF_EKO conditions, MEK1
ProcurementPolicyMaintain	policies, policy rules, approval bands, category access, catalog policy, signing limit defaults	with RequisitionApprove, PurchaseOrderConfirm	Procurement administrator	AX "Maintain purchasing policies" (PurchPolicy); SAP release strategy customizing (S_TABU_DIS, CL24N)
SigningLimitApprove	signing limit requests approval; recertification	with own SigningLimitRequest	HR manager, Procurement administrator	AX "Approve signing limit requests"; n/a
CatalogMaintain	catalogs CRUD, vendor import review, publication	none	Catalog manager, Buyer	AX "Maintain procurement catalogs" (CatProcurementCatalogMaintain); SAP SRM catalog admin
CategoryMaintain	category managers, vendor assignments, questionnaires	with VendorMaintain (w7) when assignment approval required	Category manager, Procurement administrator	AX "Maintain procurement categories"; SAP source list ME01 (M_ORDR_WRK)
ProcurementSetupMaintain	parameters in 3.17.13, charge codes, number sequences, templates	with PurchaseOrderConfirm, TradeAgreementMaintain	Procurement administrator	AX "Maintain procurement parameters" (PurchParameters); SAP SPRO MM-PUR (S_TABU_DIS)

3.17.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
PRC-REQ-001	error	Requisition line {line}: account structure invalid for {dimensions}	Wave 11 account structure	Correct the financial dimensions	FR-17.1
PRC-REQ-002	error	Requisition has more than {max} lines	Line cap 500	Split the requisition	FR-17.1
PRC-REQ-003	warning	No price found for {product}; estimated price set to 0	No trade agreement or last price	Enter an estimated price	FR-17.1
PRC-POL-001	error	No approver with approval limit ≥ {amount} {currency} found above {requester}	Signing limits exhausted	Add a default signing limit for the top level or route to CFO	FR-17.2, FR-17.17
PRC-CAT-002	error	Vendor {vendor} is not assigned to category {category} (policy {policy})	Category access HARD	Choose an assigned vendor or request a provisional assignment	FR-17.20
PRC-CAT-003	error	Catalog item required for category {category} under policy {policy}	Catalog policy rule	Pick a catalog item or request a non-catalog exception	FR-17.18
PRC-AGR-003	error	Agreement line {line} would be exceeded: released {released} + {qty} > committed {committed} (HARD)	FR-17.5	Reduce quantity, extend the agreement or switch to SOFT	FR-17.5
PRC-AGR-004	warning	Agreement line {line} is at {pct} % of commitment	Threshold ≥ 90 %	Plan a renewal	FR-17.14
PRC-AGR-005	error	PO delivery date {date} is outside agreement validity {from}–{to}	BR-17.10	Change the date or the agreement reference	BR-17.10
PRC-TA-002	error	Overlapping trade agreement rows with equal specificity for {account}/{item} from {date}	Activation check	Adjust validity or quantity breaks	BR-17.3
PRC-PO-007	error	Line {line} quantity {qty} cannot be below received {received} / invoiced {invoiced}	FR-17.8, BR-17.7	Receive a correction or credit first	FR-17.8

Code	Severity	User text	Cause	Remediation	Raised by
PRC-PO-008	error	PO {po} must be APPROVED and the vendor active before confirmation (vendor status {status})	Confirm precondition	Complete approval or release the vendor hold	FR-17.7
PRC-PO-009	error	Concurrent update: PO {po} changed by {user} at {time}	ETag mismatch	Reload and reapply the change	BR-17.8
PRC-FX-001	error	No exchange rate {from}→{to} of type {type} on {date}	Wave 12 rates	Enter the rate or change the PO currency	BR-17.12
PRC-RFQ-003	warning	Reply from {vendor} received after deadline {deadline}; excluded from ranking	BR-17.11	Override with reason to include	FR-17.4
PRC-RFQ-004	error	Sealed replies for RFQ {rfq} cannot be viewed before {deadline}	BR-17.16	Wait for the deadline or perform a logged two-person opening	FR-17.19
PRC-SA-001	error	Schedule line {date} is in the firm zone; reduction requires a reason code	BR-17.13	Provide a reason code	FR-17.15
PRC-SA-002	error	Cumulative received {received} exceeds cumulative released {released} for {product}	Receipt beyond release	Create a release or check receipts	FR-17.15
PRC-IP-001	error	Invoicing plan total {plan_total} ≠ line amount {line_amount}	BR-17.17	Adjust plan rows	FR-17.21
PRC-RP-001	info	Reprice preview: {n} lines, total delta {delta} {currency}; {m} invoiced lines excluded	FR-17.22	Apply or cancel	FR-17.22
PRC-SRV-003	error	Unplanned services {amount} exceed limit {limit} on PO line {line}	BR-17.19	Raise the limit via a PO change or reject the entry	FR-17.23
PRC-SRV-004	error	Service outline items ({sum}) do not sum to outline total ({total})	Outline consistency	Correct item amounts	FR-17.23
PRC-CAT-005	error	Vendor catalog row {row}: {field} missing or invalid ({value})	Import validation	Fix the file and re-import	FR-17.18
PRC-SL-001	warning	Signing limit for {worker} expired on {date}; approvals routed to manager until recertified	Recertification overdue	Complete recertification	FR-17.17
PRC-OUT-001	warning	PO {po} confirmed but document rendering failed {n} times; vendor not yet notified	Render worker failure	Re-render from the PO screen	FR-17.12

3.17.19 Performance budgets and capacity

Operation	Volume assumption (per tenant per day / peak per minute)	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
PO create / line edit with totals and tax	5,000 POs, 40k line edits / 200 per min	120 ms / 500 ms / 900 ms	100 rps	PO ≈ 3 KB + lines 600 B + versions 2–20 KB (JSON patch); 5k POs/day ≈ 6 GB/year	PO (tenant, vendor, approval_status); line (po_id, line_no); versions by (po_id, version_no)	Totals recalculation batched per request; tax engine timeout 2 s → totals marked STALE and recomputed on confirm
PO confirm (incl. inventory apply + render enqueue)	4,000 / 100 per min	200 ms / 500 ms / 1.5 s (render async < 5 s)	60 rps	documents ≈ 150 KB PDF each in object store ≈ 220 GB/year	render worker queue	Render queue grows; confirmation unaffected; PDF SLO relaxed to 60 s with user notice
Pricing lookup / explain	300k / 2,000 per min	5 ms / 20 ms / 40 ms	500 rps	TradeAgreement ≤ 1M rows ≈ 400 MB	composite (tenant, relation, account_code, account_id, item_code, item_id, valid_from) + currency	Served from in-pod cache (5 s TTL) of agreement rows per vendor; p95 ≤ 50 ms
Requisition submit → workflow start	2,000 / 60 per min	300 ms / 2 s / 4 s	30 rps	requisition ≈ 2 KB + lines	by requester, status	Temporal queue; submit returns 202 with workflow id
Consolidation board (5k open lines)	50 runs	1 s / 4 s / 8 s	n/a	n/a	lines (status, vendor, site, delivery_date)	Paginated grouping; hard cap 20k lines per run
RFQ compare (50 replies × 200 lines)	100	300 ms / 1 s / 2 s	n/a	replies ≈ 1 KB/line	reply (rfq_id, line_id)	Server-side ranking cached per RFQ until a reply changes
Scheduling agreement release run (1k agreements)	weekly	20 s / 60 s / 120 s	n/a	release ≈ 5 KB	(agreement_id, release_number)	Child workflows per 100 agreements
Policy evaluate	10k / 300 per min	3 ms / 10 ms / 25 ms	1,000 rps	policies ≈ 50 KB each	hierarchy path materialized (wave 6)	Cached compiled policy per version in pod memory
Catalog search (requisition picker)	100k / 1,000 per min	30 ms / 100 ms / 200 ms	300 rps	catalog entries ≈ 1 KB + images in object store	wave 8 search index with catalog filter	Falls back to product search without images
Reprice run (10k lines)	monthly	1 min / 5 min / 10 min	n/a	versions as above	(price_source_id, status)	Chunked 500 lines per activity

3.17.20 Localization and compliance hooks

Concern	Varies by	Country packs touching it (wave 48)	Extension point used
PO document legal content (company registration numbers, VAT ids, mandatory clauses, language)	Country commercial law; vendor language	All packs	Wave 15 template per legal entity with country block; vendor language from wave 7 party preferences
Tax group defaulting and reverse charge on PO lines	VAT/GST regimes, US sales/use tax	EU, UK, CA, AU, IN, BR	Wave 13 tax engine via tax group / item tax group on line; `tax_total` in FR-17.11
Public procurement rules (sealed bids, solicitation types, bid protests, thresholds)	Public sector regulations (US FAR, EU directives 2014/24)	US-PS, EU-PS	FR-17.19 `RfqType` with pack-supplied solicitation types; audit log of sealed-bid opening
Gap-free document numbering for POs/RFQs	IT, ES, PT, TR, some LATAM	IT, ES, PT, TR, BR	`po_number_gap_free` per legal entity (wave 5 sequences)
Currency and rounding of unit price (4 decimals) and line amounts	Currency minor units	n/a	BR-17.5 using wave 12 rounding rules
Calendar: delivery dates and SLA timers on business days	Site calendar and holidays	All	Wave 6 calendars; approval SLA timers skip non-working days
Signing limits and approval evidence (SOX 404, internal control)	Audit regimes	US (SOX), JP (J-SOX)	FR-17.17 recertification report; wave 48 audit policy consumes `signinglimit.changed`

Concern	Varies by	Country packs touching it (wave 48)	Extension point used
Vendor sanctions / denied-party screening before PO confirmation	OFAC, EU sanctions lists	US, EU	Wave 7 party screening hook invoked in `ConfirmPO` (vendor status)
EDI/e-order formats (EDIFACT ORDERS, X12 850, Peppol Order 3.0)	Trading-partner region	n/a (wave 35)	`po.confirmed` event payload complete enough for each format (wave 35 mapping)
Data residency of PO documents and attachments	Tenant region	n/a	Object store bucket per region (wave 1)
Privacy of requester and approver identities in printed documents	GDPR	EU	Template fields default to role names, not personal names, in EU packs

3.17.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.16.1 — RFQ → Bidding (vendor portal RFQ reply, sealed bids, scoring criteria) [#152]	PARTIAL	ADDED as FR-17.19	Sealed bids, scoring groups, amendments; portal reply in wave 20 FR-20.13
AX 12.16.1 — Procurement catalogs [#153]	PARTIAL	ADDED as FR-17.18	Catalog with images, category navigation, publication by org unit
AX 12.16.1 — Procurement → Signing limits (link to Org admin) [#156]	PARTIAL	ADDED as FR-17.17	Signing limits with requests and recertification
AX 12.16.2 — SSRS: Procurement catalog report [#159]	PARTIAL	ADDED as FR-17.18	3.17.14
AX 12.16.3 — Catalog → Import vendor catalogs (CMR) / Publish procurement catalog [#160]	PARTIAL	ADDED as FR-17.18	CSV/JSON import with staging; CMR XML mapped by wave 10 framework; job in 3.17.16
AX 12.16.4 — Policies → Purchase requisition approval amount thresholds / Signing limit policies [#161]	PARTIAL	ADDED as FR-17.16, FR-17.17	Approval bands as policy rules; default signing limits
AX 12.16.4 — Categories → Procurement categories, Category policy rules, Category managers, Category vendor assignment, Category questionnaires [#162]	PARTIAL	ADDED as FR-17.20	Managers, vendor assignment with HARD/SOFT mode, questionnaires
AX 12.16.4 — Catalogs → Procurement catalogs, Catalog policies, External catalogs (cXML config), Vendor catalog import parameters, Catalog images [#163]	PARTIAL	ADDED as FR-17.18	Catalog policy rule and import parameters in 3.17.13; external catalog (cXML/OCI) configuration DEFERRED to wave 35
AX 12.16.4 — RFQ → RFQ types, RFQ scoring criteria/groups, Reply templates, Solicitation types, Bidding parameters [#166]	PARTIAL	ADDED as FR-17.19	`rfq_types` setup in 3.17.13
AX 12.29.1 — Policies → Signing limits (policies, worker requests, defaults by job/compensation) [#237]	PARTIAL	ADDED as FR-17.17	
AX 12.29.1 — Policies → Policy framework forms (Purchasing, Expense, Audit, Vendor invoice, Signing limit) [#238]	PARTIAL	ADDED as FR-17.16	Policy types PURCHASING and SIGNING_LIMIT implemented here; VENDOR_INVOICE used by wave 18; EXPENSE (wave 43) and AUDIT (wave 48) reuse the engine
AX 12.29.2 — SSRS: Signing limits report [#245]	PARTIAL	ADDED as FR-17.17	3.17.14
AX 12.29.2 — SSRS: Policies report (policy rules per hierarchy) [#246]	PARTIAL	ADDED as FR-17.16	3.17.14
AX 12.29.3 — Signing limits → Signing limit request processing (annual recertification) [#250]	PARTIAL	ADDED as FR-17.17	Recertification job (3.17.16)
AX 12.29.4 — Policies → Signing limit policies / parameters / default signing limits / request workflow [#256]	PARTIAL	ADDED as FR-17.17	3.17.13 `default_signing_limits`, `signing_limit_recertification_months`
AX 12.29.4 — Policies → Policy parameters (organization hierarchy purpose per policy type) [#257]	PARTIAL	ADDED as FR-17.16	3.17.13 `hierarchy_purpose_per_type`
SAP 13.12 — ME84 / ME85 / ME86 SA releases (FRC/JIT), renumber, release documentation [#124]	PARTIAL-17	ADDED as FR-17.15	Releases with cumulative quantities; DELFOR/DELJIT transport in wave 35
SAP 13.12 — Fiori: Create Purchase Requisition (self-service, catalog, free text) with OCI punch-out [#129]	PARTIAL-17	ADDED as FR-17.18	Catalog item selection added; OCI/cXML punch-out DEFERRED to wave 35
SAP 13.12 — Fiori: Monitor Supplier Confirmations / Monitor Scheduling Agreement Releases [#132]	PARTIAL-17	ADDED as FR-17.15	Releases monitor in 3.17.14; supplier confirmations monitor from FR-17.7 confirmed dates and wave 20 acknowledgements
SAP 13.12 — ME84 batch SA release creation run [#133]	PARTIAL-17	ADDED as FR-17.15	Job in 3.17.16
SAP 13.12 — MEI1 / MEI2 / MEI4 subsequent price change [#135]	PARTIAL-17	ADDED as FR-17.22	Open lines repriced via PO version; invoiced lines via wave 18 FR-18.18
SAP 13.12 — Scheduling agreement release creation profile, firm/trade-off zone, JIT indicator (T163M) [#138]	PARTIAL-17	ADDED as FR-17.15	`sa_release_profile` in 3.17.13
SAP 13.14 — MRIS invoicing plan settlement [#156]	PARTIAL-17	ADDED as FR-17.21	Plan on PO line; settlement run in wave 18 FR-18.17
SAP 13.14 — MRNB revaluation (retroactive price change on invoiced items) [#157]	PARTIAL-17	ADDED as FR-17.22	`reprice_invoiced` hands invoiced lines to wave 18 FR-18.18
SAP 13.14 — MRIS scheduled invoicing plan run [#159]	PARTIAL-17	ADDED as FR-17.21	Run defined in wave 18 (3.18.16)
SAP 13.14 — MRNB scheduled revaluation run [#160]	PARTIAL-17	ADDED as FR-17.22	Job in 3.17.16 (weekly, opt-in)
SAP 13.14 — Invoicing plan types (periodic/partial) [#162]	PARTIAL-17	ADDED as FR-17.21	`invoicing_plan_types` in 3.17.13

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP 13.15 — ML10 / ML11 / ML12 model service specifications [#164]	GAP (propose 17)	ADDED as FR-17.23	`ModelServiceSpecification` templates
SAP 13.15 — Standard service catalog (SSC) [#165]	GAP (propose 17)	ADDED as FR-17.23	Standard texts as service products (wave 8) referenced from templates
SAP 13.15 — ME21N item category D with service lines, value limits, contract limits [#166]	PARTIAL-17	ADDED as FR-17.23	Service outline, `value_limit`, `contract_limit`
SAP 13.15 — ML33 … ML48 service conditions (vendor / service / vendor+plant) [#168]	PARTIAL-17	ADDED as FR-17.23	Service prices via FR-17.6 trade agreements on service products with site
SAP 13.15 — Fiori: Manage Purchase Orders (service items), Manage Supplier Quotations for services [#173]	PARTIAL-17	ADDED as FR-17.23	PO and RFQ lines of kind SERVICE with outline
SAP 13.15 — Service conditions calculation schema MS0000, condition types PRS [#183]	PARTIAL-17	ADDED as FR-17.23	Trade agreement relation PRICE on service products; schema concept SUPERSEDED by wave 26 condition engine
SAP 13.15 — Source determination for services (contracts) [#185]	PARTIAL-17	ADDED as FR-17.23	Agreement lookup for service product × site
SAP 13.31 — Legal Content Management / contract management (CLM, Ariba Contracts) [#351]	PARTIAL-17	OUT-OF-SCOPE clause library, obligation tracking and contract authoring are not an ERP v1 capability; agreements keep attachments, validity and expiry alerts (FR-17.5, FR-17.14)	Candidate for v1.x or external CLM integration via wave 35
SAP 13.32 — Configuration in purchasing (configurable PO items) [#361]	PARTIAL-17	DEFERRED to wave 24	Depends on the product configurator (wave 24 process/variant configuration); PO line reserves `configuration_id`

3.18 Wave 18: Receiving, Product Receipts & Three-Way Match

3.18.1 Scope and business context

Wave 18 closes the physical and financial loop on a purchase order. Receiving moves a PO line's inventory transaction from ORDERED through ARRIVED (arrival journal, SAP inbound delivery VL31N), REGISTERED (registration, SAP GR to quality/blocked stock) to RECEIVED by posting a product receipt (AX VendPackingSlipJour/VendPackingSlipTrans, SAP MKPF/MSEG movement type 101 with EKBE history). The product receipt posts inventory value against the GR/IR accrual account (AX "Purchase, accrual" / "Accrued purchases" in the posting profile, SAP WRX GR/IR clearing via OBYC). A vendor invoice (AX VendInvoiceInfoTable/VendInvoiceInfoLine pending invoice → VendInvoiceJour/VendInvoiceTrans; SAP RBKP/RSEG via MIRO) is captured manually, from an OCR/AP-automation adapter, or (wave 35) as UBL/Peppol, and the matching engine compares invoice price, quantity and total to PO and receipt within tolerances (AX price tolerance per item/vendor, quantity tolerance, invoice totals tolerance; SAP tolerance keys PP, DQ, BD, AN/AP, DW, KW). Matched invoices post the vendor liability (VendTrans / BSIK) and clear GR/IR; exceptions route to a Temporal resolution workflow. Evaluated receipt settlement (ERS / MRRL) generates invoices from receipts for vendors flagged for self-billing.

Out of scope: payment (wave 19), warehouse-directed put-away and license plates (wave 31), quality orders on receipt (wave 33), landed cost (wave 34), Peppol transport (wave 35), invoice-cost adjustments to inventory beyond the direct posting (wave 20 settles invoice-receipt cost differences), non-PO expense invoices with GL distribution (included here as 0-way "invoice register" with coding, approval by wave 4 workflow).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX pending vendor invoice form (VendInvoiceInfoTable) runs "Match product receipts" as a dialog, then matching icons update only after "Update match status"; users re-key invoice lines from paper; SAP MIRO proposes lines but the tolerance log (RBKP block indicator, MRBR release) is a separate transaction	Matching is a form-triggered class (VendInvoiceMatch), not an engine; capture is manual	Matching engine runs on every invoice save and on every receipt posting (re-match candidates), status visible inline; OCR adapter pre-fills lines; `MatchResult` per line with variances	Invoice entry time: 4–6 min → < 60 s; manual re-match steps: 3 → 0
GR/IR reconciliation: AX "Accrued purchases" report and SAP MR11 (GR/IR maintenance) plus F.19 regrouping are month-end chores; differences hide for months	GR/IR balance is an account balance with no per-PO-line open-item ledger in AX; SAP EKBE requires MR11 to clear residuals	`GrirOpenItem` per PO line × receipt, cleared by invoice line; GR/IR report = Σ open items and must equal the account balance (daily reconciliation)	Month-end GR/IR effort: 4–8 h → < 15 min; unexplained residual: non-zero → 0
Receipt posting latency: AX product receipt via PurchFormLetter_PackingSlip takes 10–40 s per receipt; SAP MIGO posts in ~1 s but inventory valuation updates MBEW synchronously causing lock waits (table lock on MBEW/MARD during month-end)	Document class chain and serialized valuation updates	Receipt posting is one inventory `transactions:apply` + one ledger `Post`; value updated per transaction, no material-level lock	Receipt post p95: 10–40 s (AX) / lock waits at close (SAP) → < 1.5 s, no cross-document locks
Tolerance configuration: AX has three separate forms (price tolerance per item/vendor/company, quantity tolerance on item, totals tolerance in parameters) with "Allow price tolerance" flags scattered; SAP OMR6 tolerance keys use cryptic codes and a company-code-level default	Settings scattered without a single evaluation order	One `MatchingPolicy` entity with ordered scopes (vendor×item > vendor > item > entity) and explicit percent/absolute limits, previewable with "test an invoice"	Time to configure and verify a tolerance change: 1–2 h with testing → 5 min with the policy tester
ERS/self-billing in AX requires a batch "Generate invoices from product receipts"; SAP MRRL is a batch run; both mis-handle partial receipts with price changes	Batch design; invoice creation decoupled from the receipt event	ERS invoice created by the `receipt.posted` consumer within seconds using the price at receipt date and locked PO price	ERS invoice latency: nightly batch → < 10 s; ERS price mismatches: tracked, target 0

3.18.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Arrival / registration	Arrival overview (WMSArrivalOverview), arrival journal (WMSJournalTable type Reception), registration (InventTransRegister)	Inbound delivery VL31N, GR into blocked stock (103/105), quality inspection stock	`Arrival` (ARRIVED), `Registration` (REGISTERED) steps on ReceiptLine; arrival overview screen	full
Product receipt	PurchFormLetter_PackingSlip → VendPackingSlipJour/Trans, InventTrans Received, ledger posting PurchPackingSlip/accrual	MIGO 101 → MKPF/MSEG, EKBE (VGABE 1), GR/IR via WRX, inventory BSX	`ProductReceipt`/`ReceiptLine`; RECEIVED transactions; ledger entry Dr inventory / Cr GR/IR	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Receipt reversal / correction	Product receipt correction, cancel (VendPackingSlipJour cancel)	102 reversal, MBST cancel document	`:correct` (new receipt with delta qty), `:cancel` (full reversal)	full
Vendor invoice (PO)	VendInvoiceInfoTable/Line (pending), VendInvoiceJour/Trans (posted), invoice register/approval journals	MIRO → RBKP/RSEG, EKBE VGABE 2; invoice park MIR7	`VendorInvoice`/`VendorInvoiceLine` (DRAFT → POSTED)	full
Non-PO invoice	Invoice journal (LedgerJournalTrans type VendInvoiceRegister / Vend), invoice register + approval	FB60, MIRO without PO	`VendorInvoice` kind EXPENSE with GL distribution lines	full
Matching	VendInvoiceMatch, match status (VendInvoiceMatchingStatus), price/qty/totals tolerances (VendInvoicePriceTolerance, InventTable quantity tolerance, Accounts payable parameters)	LIV tolerance keys OMR6 (PP, DQ, BD, AN, AP, DW, KW, ST, VP), blocking indicator RBKP ZLSPR, MRBR release	`MatchingEngine`, `MatchingPolicy`, `MatchResult`; exception workflow	full
Accrued purchases / GR/IR	Accrued purchases report (PurchAccrual), posting types PurchaseAccrual/PurchasePackingSlip	GR/IR clearing WRX, MR11, F.19, MB5S	`GrirOpenItem`, GR/IR report	improved
ERS	Vendor self-billing: "Generate invoices from product receipts" (PurchFormLetter with VendInvoiceInfoTable auto)	MRRL evaluated receipt settlement, LFM1 XERSY	`ErsPolicy` on vendor; consumer creates invoice from receipt	full
Invoice capture adapters	Invoice capture (D365 add-in), AIF VendVendInvoiceService	OpenText VIM, SAP Invoice Management, IDoc INVOIC	`POST /invoices:capture` with OCR adapter interface (JSON), UBL mapping stub	partial (Peppol w35)
Charges allocation at invoice	MarkupTrans allocation on invoice	Unplanned delivery costs, planned conditions FRA1	`Charge` allocation rules applied to invoice lines	full
Period control AP	Ledger calendar module Accounts payable	OB52 posting periods account type K	wave 14 period control module AP at `:post`	full
Service entry sheets	n/a (service lines invoiced directly)	ML81N service entry sheet, ML82/ML83/ML84 display/list/accept, ML85 collective release, ESSR, lean services (S/4 Manage Service Entry Sheets), MSRV lists, ME2S, MRRL for services, MIRO with SES reference	`ServiceEntrySheet` with acceptance workflow, partial and unplanned entries within limits, collective acceptance, ERS for services, invoice match to SES (FR-18.14)	full
Stochastic invoice block	n/a	OMRX stochastic block (threshold amount, percentage), RBKP ZLSPR = S, MRBR release	`stochastic_block` policy (FR-18.15) producing status `AUDIT_SAMPLE`	full
Consignment settlement	D365 consignment: vendor invoice from ownership change	MRKO consignment and pipeline settlement	Consignment settlement run creating `VendorInvoice` kind CONSIGNMENT from wave 16 ownership changes (FR-18.16)	full (pipeline OUT-OF-SCOPE)
Invoicing plan settlement	n/a	MRIS invoicing plan settlement (periodic/partial)	Invoicing plan run creating invoices without receipt (FR-18.17)	full
Subsequent debit / credit	Invoice with reference to invoiced PO line (price adjustment)	MIRO subsequent debit/credit; MRNB revaluation	`VendorInvoice` kind SUBSEQUENT_DEBIT / SUBSEQUENT_CREDIT (FR-18.18)	full
One-time vendor invoice	VendTable one-time supplier (OneTimeVendor flag, VendParameters one-time vendor account)	Account group CPD, one-time account with address/bank on document (OBD3)	`VendorInvoice.one_time_payee` with document-level name, address, tax id and bank account (FR-18.19); payment in wave 19 FR-19.23	full

3.18.3 Functional requirements

- **FR-18.1 Arrival and registration** — Must. The system shall record arrival (ARRIVED) with carrier, packing slip number, pallet count and expected PO lines, then registration (REGISTERED) with quantity, batch, serial, location and inventory status, for any confirmed PO line with remaining quantity.
  - Over-receipt beyond `qty × (1 + over_delivery_pct)` (item default 0 %, policy per vendor×item) is rejected with `RCV-OVR-001`; under-delivery below `under_delivery_pct` closes the line only if `close_short=true`.
  - Serial-tracked lines require one serial per unit; duplicate serial rejected by inventory (`INV-SER-001`).
- **FR-18.2 Product receipt posting** — Must. The system shall post a `ProductReceipt` for registered or directly received quantities, moving inventory transactions to RECEIVED, writing `cost_amount_physical = qty × PO unit price (converted to accounting currency at the receipt date rate) + ITEM charges`, and posting Dr inventory (posting profile) / Cr GR/IR accrual through `Post(JournalEntry)` with idempotency key = receipt id.
  - For category (non-stock) lines the debit goes to the line's expense account; GR/IR credit is unchanged.
  - A `GrirOpenItem` is created per receipt line with amount in transaction and accounting currency.
  - Receipt date must be in an open INVENTORY period; packing slip number unique per vendor (`RCV-DUP-001`).
- **FR-18.3 Receipt correction and cancellation** — Must. The system shall support correction (delta quantity, positive or negative) and cancellation (full reversal) of a posted receipt while the receipt line has not been invoiced; invoiced lines require an invoice credit first.
- **FR-18.4 Vendor invoice capture** — Must. The system shall create a `VendorInvoice` in DRAFT from manual entry, from the capture adapter (`:capture` JSON with header fields, lines, confidence scores, page images) or from ERS, with vendor, invoice number, dates, currency, totals, lines linked to PO lines and/or receipt lines, charges, tax groups.
  - Duplicate check: same vendor + invoice number (case-insensitive, whitespace-stripped) within 24 months → `AP-DUP-001`.
  - Lines may be proposed from open receipts (3-way) or open PO lines (2-way); proposal fills qty = uninvoiced received qty.
  - Expense invoices (no PO) carry GL distribution lines validated by account structure.
- **FR-18.5 Matching policy** — Must. The system shall evaluate a `MatchingPolicy` resolved by scope precedence vendor×item > vendor > item > legal entity with fields: `match_level` (NONE, TWO_WAY, THREE_WAY), `price_tol_pct` (default 2.0), `price_tol_abs` (default 25.00 acct ccy per line), `qty_tol_pct` (default 5.0), `total_tol_pct` (default 1.0), `total_tol_abs` (default 50.00), `small_diff_abs` (default 0.50, auto-posted to small-differences account), `price_tol_mode` (UPPER_ONLY / BOTH), `date_tol_days` (0 = off).
  - A line passes price if `|invoice_unit_price − po_unit_price| ≤ max(po_price × pct, abs)` (lower prices always pass in UPPER_ONLY).
  - A line passes quantity (3-way) if `invoice_qty ≤ received_uninvoiced_qty × (1 + qty_tol_pct/100)`; 2-way compares to PO remaining qty.

- Total passes if `|invoice_total − Σ matched line totals − charges − tax| ≤ max(total_tol_pct, total_tol_abs)`.

- **FR-18.6 Matching engine and status** — Must. The system shall compute `MatchResult` per line and header with statuses `NOT_MATCHED`, `MATCHED`, `PRICE_VARIANCE`, `QTY_VARIANCE`, `TOTAL_VARIANCE`, `NO_RECEIPT`, `WAITING_RECEIPT`; header status is the worst line status; the engine re-runs automatically when a receipt posts for a referenced PO line.
  - Engine is pure (inputs → result) and exposed at `POST /matching:evaluate` for the policy tester.
  - Price variance within tolerance posts the difference to the purchase price variance account (standard-cost items, wave 20 PPV) or to inventory/expense (actual-cost items).
- **FR-18.7 Exception workflow** — Must. The system shall start a Temporal `InvoiceMatchException` workflow for any invoice with a variance status, assigning to the buyer of the PO (price) or receiving supervisor (quantity) with actions: accept variance (post with reason code), request credit note, adjust PO price (creates PO version via wave 17), wait for receipt (timer, re-evaluate), reject invoice.
  - SLA 3 business days then escalation to AP manager; all actions audited on the invoice.
- **FR-18.8 Invoice posting** — Must. The system shall post a matched or accepted invoice: Dr GR/IR (clearing `GrirOpenItem` to the invoiced amount) or Dr expense/inventory (2-way/expense), Dr tax recoverable (wave 13), Cr vendor control account (posting profile) creating a `VendorTransaction` open item with due date and cash discount date per payment terms, moving PO inventory transactions to PURCHASED for invoiced quantity and setting `cost_amount_posted` from the invoice.
  - Idempotency key = invoice id; AP period must be open; posting date ≠ invoice date allowed (both stored).
  - Differences between receipt value and invoice value for stock items post to inventory (actual cost methods) or PPV (standard), and are recorded on the transaction as `cost_amount_adjustment` for wave 20.
- **FR-18.9 GR/IR reconciliation** — Must. The system shall provide the GR/IR report listing open items (received not invoiced, invoiced not received) by vendor, PO, line, age, with totals that equal the GR/IR ledger balance; a daily job emits `reconciliation.diff` if they differ.
- **FR-18.10 Evaluated receipt settlement** — Must. The system shall, for vendors with `ers_enabled`, create and post an invoice from each receipt within 10 s using PO price and the vendor's ERS invoice number format (`ERS-{receipt_number}`), tax per PO line, and send the self-billing document to the vendor.
- **FR-18.11 Charges on receipt and invoice** — Must. The system shall allocate header charges to lines by the rule on the charge (QTY, AMOUNT, LINES) at invoice posting, with ITEM-type charges capitalized into `cost_amount_posted`.
- **FR-18.12 Invoice credit notes** — Should. The system shall create a credit note (negative invoice) referencing an invoice and optionally a receipt correction; it posts the reverse entries and creates a negative `VendorTransaction` for wave 19 settlement.
- **FR-18.13 Receiving workspace and mobile receipt** — Should. The system shall provide a receiving workspace (expected today, overdue, registered not posted) and a PWA receipt screen scanning PO number, product GTIN and batch/serial barcodes (GS1 parsing from wave 8).
- **FR-18.14 Service entry sheets (service receipts)** — Must. The system shall provide a `ServiceEntrySheet` (SAP ML81N / ESSR; S/4 lean service entry sheet) against a SERVICE PO line (wave 17 FR-17.23) with header (vendor, PO, period from/to, external reference, entered by), lines referencing outline service items (planned) or free-text unplanned services with quantity, unit, unit price and amount, status `DRAFT, SUBMITTED, ACCEPTED, REJECTED, REVOKED`, and an acceptance workflow (wave 4 template; approver = PO requester or cost owner from the policy type VENDOR_INVOICE of FR-17.16).
  - Acceptance posts the service receipt: Dr expense / cost object per line account assignment, Cr GR/IR accrual, creating a `GrirOpenItem` per sheet line (SAP "GR for SES"); the PO line's `received_qty` / `received_amount` and `document_status` update; for lines with `settle_without_receipt=false` the invoice (FR-18.4) proposes lines from accepted sheets and matching (FR-18.5) compares to the sheet amount (quantity tolerance applies to amount for services).
  - Unplanned services are limited by the PO line `value_limit` and `contract_limit` (BR-17.19, `PRC-SRV-003`); partial entries accumulate against planned quantities.
  - Collective acceptance (`POST /service-entry-sheets:accept-bulk`, SAP ML85) and revoke (reverses the posting if not invoiced).
  - ERS vendors (FR-18.10) get an invoice per accepted sheet (SAP MRRL for services) with number `ERS-{sheet_number}`.
  - Lists: by service, by PO, by sheet, by vendor, by agreement (SAP MSRV1–MSRV6), acceptance list (ML83/ML84); the wave 20 portal lets vendors enter sheets (FR-20.20).
- **FR-18.15 Stochastic (random-sample) invoice block** — Must. The system shall evaluate on every invoice reaching MATCHED a `stochastic_block` rule from the matching policy scope (SAP OMRX T169P): `threshold_amount` (acct ccy) and `percentage`; invoices with total ≥ threshold are blocked with probability `percentage` % (linear decrease below threshold proportional to total/threshold), using a seeded CSPRNG with the invoice id so the decision is reproducible; a blocked invoice gets match status `AUDIT_SAMPLE`, is routed to the AP auditor queue (duty InvoiceAuditRelease), and is released or rejected with reason; the sample rate actually achieved is reported monthly.
- **FR-18.16 Consignment settlement** — Must. The system shall consume `dev.passage.inventory.ownership.changed.v1` (wave 16 FR-16.17) and, per vendor and settlement frequency (`consignment_settlement_frequency` DAILY/WEEKLY/MONTHLY, SAP MRKO), create and post a `VendorInvoice` of kind CONSIGNMENT with one line per ownership change line (product, qty, consignment price, tax group from the trade agreement), clearing the consignment accrual (Dr consignment accrual / Cr vendor control), skipping matching (status MATCHED by construction), and sending a consignment settlement statement (3.18.15) to the vendor; when `consignment_self_billing=false` the vendor's own invoice is matched against the ownership change lines instead (2-way to accrual).
- **FR-18.17 Invoicing plan settlement run** — Should. The system shall run daily (3.18.16; SAP MRIS) over wave 17 invoicing plan rows with `due_date ≤ run date` and status OPEN, creating per vendor a `VendorInvoice` kind PLAN with lines for each due row (amount, tax group of the PO line, account assignment), posting it when `settle_without_receipt=true` and the plan is marked ERS-like (`auto_post=true`), otherwise leaving it in DRAFT for the vendor's invoice to be matched 2-way to the plan row; plan rows move to INVOICED with the invoice reference.
- **FR-18.18 Subsequent debit and credit** — Should. The system shall support `VendorInvoice` kinds SUBSEQUENT_DEBIT and SUBSEQUENT_CREDIT (SAP MIRO subsequent debit/credit; AX price-adjustment invoice) referencing invoiced PO lines with a value-only change (quantity 0): posting Dr/Cr inventory (actual-cost items, via `cost_amount_adjustment` for wave 20) or PPV (standard cost) against the vendor control account; wave 17 FR-17.22 `:reprice` with `reprice_invoiced=true` creates SUBSEQUENT_CREDIT drafts for repriced invoiced lines (SAP MRNB).
- **FR-18.19 One-time vendor invoices** — Should. The system shall allow an invoice (kind EXPENSE or PO with the legal entity's one-time vendor account; AX VendParameters one-time vendor; SAP CPD account group) to carry a document-level `one_time_payee` block (name, address, country, tax id, email, bank

account IBAN/BIC or routing/account) validated like vendor bank accounts (BR-19 IBAN mod-97 / ABA), stored encrypted, printed on the payment advice and used by wave 19 FR-19.23 for payment; duplicate check uses payee name + invoice number; one-time payees exceeding `one_time_payee_max_amount` (default 10,000 USD) or appearing more than `one_time_payee_max_count` times (default 3 per 12 months) are rejected with `AP-OTV-001` and must be set up as vendors (wave 7).

3.18.4 Data model

ProductReceipt / ReceiptLine

Field	Type	Required	Notes
id, receipt_number	uuid, text	yes	sequence `RCV-`
vendor_id, po_id	uuid	yes	one PO per receipt (multi-PO receipts split)
packing_slip_number	text	yes	unique per vendor
receipt_date, posting_date	date	yes	inventory physical/financial dates
status	enum	yes	ARRIVED, REGISTERED, POSTED, CORRECTED, CANCELLED
ledger_entry_id	uuid	no	
exchange_rate	numeric(18,8)	yes	PO currency → accounting at receipt_date
line: po_line_id, schedule_id, qty, unit, dimension_set_id, batch_id, serial_ids[], inventory_status, unit_price (from PO), amount_txn, amount_acct, charges_item_amount, inventory_transaction_id, invoiced_qty, grir_open_item_id			

VendorInvoice / VendorInvoiceLine / MatchResult

Field	Type	Required	Notes
id, internal_number	uuid, text	yes	sequence `VI-`
vendor_id, invoice_vendor_id	uuid	yes	
vendor_invoice_number	text	yes	duplicate check
kind	enum	yes	PO, EXPENSE, CREDIT_NOTE, ERS
invoice_date, posting_date, due_date, discount_date	date	yes/no	from payment terms
currency, exchange_rate	text, numeric	yes	
status	enum	yes	DRAFT, IN_MATCHING, EXCEPTION, APPROVED, POSTED, REJECTED, CANCELLED
match_status	enum	yes	header worst-of
subtotal, charges_total, tax_total, total, total_entered	numeric(19,4)	yes	`total_entered` keyed from paper; must equal computed total within small_diff
source	enum	yes	MANUAL, OCR, UBL, ERS, PORTAL
capture_confidence	numeric(5,2)	no	from adapter
ledger_entry_id, vendor_transaction_id	uuid	no	after post
line: po_line_id, receipt_line_id, product_id/description, qty, unit, unit_price, line_amount, tax_group_id, item_tax_group_id, gl_account_id (expense), dimension_default_id, charges_allocated			
match_result: line_id, policy_id, price_variance_amt, price_variance_pct, qty_variance, total_variance, status, evaluated_at, accepted_by, reason_code			

GrirOpenItem

Field	Type	Required	Notes
id	uuid	yes	
po_line_id, receipt_line_id	uuid	yes	
vendor_id	uuid	yes	
received_qty, invoiced_qty	numeric	yes	
received_amount_acct, invoiced_amount_acct	numeric(19,4)	yes	
status	enum	yes	OPEN, PARTIAL, CLEARED, WRITTEN_OFF
cleared_at	timestamptz	no	

MatchingPolicy: id, scope (ENTITY, ITEM, VENDOR, VENDOR_ITEM), vendor_id, product_id, match_level, price_tol_pct, price_tol_abs, qty_tol_pct, total_tol_pct, total_tol_abs, small_diff_abs, price_tol_mode, date_tol_days, valid_from, valid_to.

Invariants: Σ GrirOpenItem (received – invoiced) in accounting currency = GR/IR ledger balance; receipt line `invoiced_qty ≤ qty`; invoice POSTED implies every line has MATCHED or accepted status; VendorTransaction amount = invoice total in transaction currency. Indexes: receipts (tenant, vendor, packing_slip_number) unique; invoices (tenant, vendor_id, lower(vendor_invoice_number)) unique partial where status ≠ CANCELLED; GrirOpenItem (po_line_id, status).

3.18.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/receiving/arrivals	Record arrival	client key	ReceivingMaintain
POST	/api/v1/receiving/receipts	Create receipt (register lines)	client key	ReceivingMaintain
POST	/api/v1/receiving/receipts/{id}:post	Post product receipt	receipt id	ReceivingPost
POST	/api/v1/receiving/receipts/{id}:correct	Delta correction	receipt id + seq	ReceivingPost
POST	/api/v1/receiving/receipts/{id}:cancel	Reverse	receipt id	ReceivingPost
POST	/api/v1/payables/invoices	Create invoice (manual)	client key	VendorInvoiceMaintain
POST	/api/v1/payables/invoices:capture	Adapter/OCR intake	adapter doc id	service account / VendorInvoiceCapture
POST	/api/v1/payables/invoices/{id}:propose-lines	Propose from receipts/PO	n/a	VendorInvoiceMaintain

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/payables/invoices/{id}:match	Run matching now	n/a	VendorInvoiceMaintain
POST	/api/v1/payables/invoices/{id}:post	Post	invoice id	VendorInvoicePost
POST	/api/v1/payables/matching:evaluate	Policy tester (pure)	n/a	MatchingPolicyMaintain
GET	/api/v1/payables/grir?vendor=&age_gt=	GR/IR open items	n/a	APInquire
POST	/api/v1/payables/invoices/{id}/exceptions/{line}:accept	Accept variance	line+reason	InvoiceExceptionResolve

Commands: `PostReceipt` (open period, over-delivery, serial uniqueness, PO confirmed, exchange rate exists), `PostInvoice` (AP period open, match status acceptable, totals within small_diff, tax calculated, duplicate check), `AcceptVariance` (reason code required, amount ≤ approver limit from policy).

Event	Trigger	Payload fields	Consumers
`dev.passage.receiving.receipt.posted.v1`	receipt POSTED	receipt_id, po_id, lines (po_line, qty, dims, batch, serials, amount_acct), ledger_entry_id	inventory, payables (re-match, ERS), costing, supplier-portal
`dev.passage.receiving.receipt.corrected.v1`	correction/cancel	receipt_id, delta lines	payables, costing
`dev.passage.payables.invoice.captured.v1`	DRAFT created from adapter	invoice_id, vendor, source, confidence	notifications
`dev.passage.payables.invoice.match.exception.v1`	variance status	invoice_id, line_id, status, variances, assignee	workflow, notifications
`dev.passage.payables.invoice.posted.v1`	POSTED	invoice_id, vendor_transaction_id, lines, totals, grir cleared amounts, ledger_entry_id	ledger reconciliation, costing, tax, analytics
`dev.passage.payables.grir.cleared.v1`	GrirOpenItem CLEARED/WRITTEN_OFF	open_item_id, amounts	analytics
`dev.passage.receiving.service-entry-sheet.accepted.v1` / `.revoked.v1`	sheet accepted / revoked	sheet_id, po_id, po_line_id, vendor_id, lines (service item, qty, amount, account assignment), grir amounts, ledger_entry_id	payables (match, ERS), costing (expense), project accounting (w41), supplier-portal (w20)
`dev.passage.payables.invoice.audit-sampled.v1`	stochastic block applied	invoice_id, vendor_id, total, rule (threshold, percentage), assignee	workflow, notifications, audit (w48)
`dev.passage.payables.consignment.settled.v1`	consignment settlement invoice posted	invoice_id, vendor_id, ownership change journal ids, lines, total	supplier-portal (w20), analytics

3.18.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Receiving workspace	Expected today/overdue, arrivals, registered-not-posted, post	PO, vendor, line, expected qty, arrived qty	ReceivingMaintain
Receipt (desktop + PWA)	Scan PO/GTIN/batch/serial, register, post, print receipt label	lines with remaining qty, batch/serial entry, location, status	ReceivingMaintain
Vendor invoice	Enter/review captured header, propose lines, match status badges, resolve, post	header, lines with PO/receipt links, variance columns, totals vs entered	VendorInvoiceMaintain
Invoice exception task	Accept, request credit, adjust PO, wait, reject	variance detail, policy limits, history	InvoiceExceptionResolve
Matching policy + tester	Create policy rows, "test an invoice"	scopes, tolerances, sample invoice result	MatchingPolicyMaintain
GR/IR report	Filter by age/vendor, drill to PO/receipt/invoice, write-off	open items, totals vs ledger balance	APInquire

Interaction rules: captured fields with confidence < 0.85 are highlighted and require explicit confirmation; totals-entered vs computed difference shown live; keyboard: Ctrl+M match, Ctrl+P post; PWA receipt supports camera scan and offline registration queue; bulk post of registered receipts; personalization of invoice line columns.

3.18.7 Business rules and edge cases

- **BR-18.1** Receipt state machine: `ARRIVED → REGISTERED → POSTED`; `POSTED → CORRECTED` (new correction receipt, original unchanged); `POSTED → CANCELLED` only if invoiced_qty = 0 on all lines; `ARRIVED|REGISTERED → CANCELLED` releases inventory statuses back to ORDERED.
- **BR-18.2** Invoice state machine: `DRAFT → IN_MATCHING → (MATCHED→) APPROVED → POSTED`; `IN_MATCHING → EXCEPTION → APPROVED|REJECTED`; `DRAFT|EXCEPTION → CANCELLED`; POSTED is terminal (reverse via credit note).
- **BR-18.3** Price comparison uses PO unit price in PO currency normalized to invoice price unit; invoice in a different currency than the PO is compared after conversion at the invoice date rate and flagged `currency_mismatch=true` (always exception unless policy allows).
- **BR-18.4** Quantity tolerance applies to over-invoicing only; under-invoicing leaves the receipt line partially open (`PARTIAL`) for a later invoice.
- **BR-18.5** Small differences: if the header total variance ≤ `small_diff_abs` the difference posts to the small-differences account and the invoice is MATCHED; the amount is recorded on `MatchResult`.
- **BR-18.6** Rounding: variance percent computed on unrounded values and compared at 4 decimals; amounts rounded half-even to minor unit; the GR/IR clearing amount is the receipt-side accounting amount for the invoiced quantity (pro-rata, rounded), so GR/IR residual after full invoicing of a line is exactly 0.
- **BR-18.7** Receipt date rate vs invoice date rate: GR/IR is cleared at the receipt-side accounting amount; the difference between invoice accounting amount and that clearing amount posts to inventory/expense (not FX gain/loss), matching SAP KDM/AX exchange adjustment on purchase.
- **BR-18.8** Concurrency: posting a receipt and an invoice for the same PO line serializes on the PO line row lock (`SELECT … FOR UPDATE` on `purchase_order_line`); the invoice's matching always sees the committed receipt state.
- **BR-18.9** ERS invoices skip matching (status MATCHED by construction) and are blocked if the PO price has changed since receipt until a buyer confirms the price.
- **BR-18.10** Invoice posting date in a closed AP period is rejected (`PER-AP-001`) even when the invoice date is open; the user may change posting date to the first open day with an audit note.
- **BR-18.11** Invoice vendor ≠ ordering vendor (factoring) is allowed when the ordering vendor's `invoice_account` points to the invoice vendor; the VendorTransaction belongs to the invoice vendor.
- **BR-18.12** Credit note quantity cannot exceed invoiced quantity on the referenced line; posting a credit note re-opens the GR/IR item and reverts inventory transaction status from PURCHASED to RECEIVED for that quantity.
- **BR-18.13** Service entry sheet state machine: `DRAFT → SUBMITTED → ACCEPTED`; `SUBMITTED → REJECTED → DRAFT`; `ACCEPTED → REVOKED` only while `invoiced_amount = 0` on all its lines; acceptance is the posting event (SAP "GR for SES"); a sheet period may not overlap another ACCEPTED sheet of the same PO line for the same planned service item unless the item is quantity-based.
- **BR-18.14** Stochastic block decision = `hash(invoice_id ‖ rule_version) mod 10,000 < percentage × 100 × min(1, total / threshold)`; the decision is recorded with rule version so re-matching does not re-draw; an invoice already in a variance exception is not additionally sampled.
- **BR-18.15** Consignment settlement invoices are always posted in the vendor's currency at the trade-agreement price captured on the ownership change line (not re-priced at settlement); price changes after the ownership change are handled as SUBSEQUENT_DEBIT/CREDIT.

- **BR-18.16** Invoicing plan rows generate at most one invoice line each; a plan row whose PO line is cancelled is cancelled with it; plan invoices for lines with `settle_without_receipt=false` require an accepted service entry sheet or product receipt covering the period before posting.
- **BR-18.17** Subsequent debit/credit lines have quantity 0 and reference an invoiced PO line; the amount is capped at the invoiced amount for credits (`AP-SUB-001`); GR/IR is not touched; for actual-cost items the adjustment is written to the receipt transaction's `cost_amount_adjustment` and flows to issues at the next wave 20 close (BR-20.6).
- **BR-18.18** One-time payee data is stored encrypted (wave 2 field-level encryption) and never copied to the party model; the payee block is immutable after posting; payment to a one-time payee requires the bank account to pass the same validation as vendor bank accounts and the invoice to have passed a four-eyes approval regardless of amount band.

3.18.8 Acceptance criteria

- **AC-18.1** (FR-18.5, FR-18.6) Given the 200-scenario matching fixture, when each invoice is evaluated and posted, then every match status, variance amount and ledger posting equals the expected row (0 mismatches).
- **AC-18.2** (FR-18.2, FR-18.8) Given 1,000 random PO lines received then fully invoiced at PO price, when GR/IR open items and the ledger balance are compared, then both are 0.00.
- **AC-18.3** (FR-18.1) Given a serial-tracked PO line of 3 units, when serials S1, S2, S2 are registered, then registration fails on the duplicate and records S1, S2 only after correction.
- **AC-18.4** (FR-18.8) Given AP period 2026-08 closed, when an invoice with posting date 2026-08-15 is posted, then it is rejected with `PER-AP-001` and no VendorTransaction or ledger entry exists.
- **AC-18.5** (FR-18.5) Given policy price_tol_pct 2 and price_tol_abs 25 and a PO price 100.00 USD, when an invoice line at 101.50 is matched, then status MATCHED with price variance 1.50 posted to PPV/inventory; at 103.00 status PRICE_VARIANCE and an exception workflow starts.
- **AC-18.6** (FR-18.7) Given a PRICE_VARIANCE exception, when the buyer accepts with reason "AGREED", then the invoice posts, the variance posts to the variance account, and the audit shows actor, reason and amount.
- **AC-18.7** (FR-18.10) Given an ERS vendor, when a receipt of 50 units at 10.00 EUR posts, then an invoice `ERS-{receipt}` of 500.00 EUR plus tax is posted within 10 s and a self-billing PDF is emailed.
- **AC-18.8** (FR-18.3) Given a posted receipt of 100 with 40 invoiced, when a correction of −70 is attempted, then it is rejected; −60 succeeds and GR/IR open item received_qty becomes 40.
- **AC-18.9** (FR-18.4) Given an invoice captured with confidence 0.6 on the total field, when opened, then the field is flagged and posting is blocked until confirmed.
- **AC-18.10** (FR-18.9) Given a manual journal posted to the GR/IR account, when the daily GR/IR reconciliation runs, then a diff event with the journal amount is emitted.
- **AC-18.11** (FR-18.14) Given a service PO line with planned items A (10 h × 100.00 EUR) and B (1 × 500.00 EUR) and unplanned limit 300.00, when a sheet with A 6 h, B 1 and unplanned 250.00 is submitted and accepted, then a ledger entry Dr expense 1,350.00 / Cr GR/IR 1,350.00 posts, three GrirOpenItem rows exist, and an invoice proposed from the sheet matches with status MATCHED; a second sheet with unplanned 100.00 is rejected with `PRC-SRV-003`.
- **AC-18.12** (FR-18.15) Given a stochastic rule threshold 10,000.00 USD and percentage 10, when 10,000 matched invoices of 20,000.00 USD are evaluated, then between 900 and 1,100 receive `AUDIT_SAMPLE`, re-matching any invoice yields the same decision, and released invoices post normally.
- **AC-18.13** (FR-18.16) Given 3 ownership changes for vendor V1 in a week (120 EA × 4.00, 50 EA × 4.00, 30 EA × 4.20 USD) and weekly settlement, when the settlement run executes, then one CONSIGNMENT invoice of 806.00 USD plus tax posts, the consignment accrual account nets to 0.00 for those lines, and the settlement statement is emailed.
- **AC-18.14** (FR-18.17) Given a PERIODIC plan of 1,000.00 EUR monthly with `auto_post=true`, when the run executes on the due date twice (rerun), then exactly one invoice per due row exists and the plan row is INVOICED.
- **AC-18.15** (FR-18.18) Given an invoiced line of 100 EA at 10.00 USD for a FIFO item with 60 EA still on hand, when a SUBSEQUENT_CREDIT of 50.00 USD posts, then AP is debited 50.00, inventory credited 50.00, the receipt transaction carries `cost_amount_adjustment` −50.00, and GR/IR is unchanged.
- **AC-18.16** (FR-18.19) Given a one-time payee with an invalid IBAN checksum, when the invoice is saved, then it is rejected with `AP-BNK-001`; with a valid IBAN and amount 12,000.00 USD (> 10,000.00 limit), it is rejected with `AP-OTV-001`.

3.18.9 Test plan and fixtures

Fixtures: `match-200` (200 invoice scenarios CSV with expected status/postings, authored with the reviewing accountant; covers over/under-delivery, price up/down, multi-receipt, partial, currency mismatch, small difference, ERS, credit note); `grir-1000` (1,000 PO lines generated); `rcv-serial-batch` (tracked items); `ap-period-closed`; `ocr-samples` (25 invoice JSON payloads with confidence scores).

Test	Type	Target	Pass criterion
Matching engine	unit (table-driven)	matching:evaluate	match-200 all pass
Tolerance monotonicity	property	engine	increasing invoice price never moves status from variance to matched
GR/IR zero	E2E	receipt+invoice flow	AC-18.2
Receipt → inventory	contract	inventory apply	RECEIVED qty and cost equal receipt
Invoice → ledger	contract	Post	balanced entry; accounts per posting profile
Period control	contract	invoices:post	AC-18.4
ERS latency	E2E	receipt.posted consumer	< 10 s p95
Receipt post latency	performance (k6, 200 receipts/min)	receipts:post	p95 < 1.5 s
PWA scan	E2E (Playwright)	receipt screen	GS1 barcode → line, batch, serial filled

Reconciliation checks: Σ GrirOpenItem = GR/IR ledger balance; Σ VendorTransaction created in period = Σ invoice totals posted; Σ receipt amount_acct = Δ inventory account from receipts; invoiced_qty on receipt lines = Σ invoice line qty referencing them.

3.18.10 Deployment and operations

Helm charts: `passage-procurement` gains the receiving module (same Postgres, separate Deployment `receiving-api` ×2); `passage-payables` (API ×3, matching worker ×2, Temporal worker ×2, CloudNativePG `payables-db`, outbox connector). Config: `AP_SMALL_DIFF_DEFAULT=0.50`, `AP_PRICE_TOL_PCT=2.0`, `AP_QTY_TOL_PCT=5.0`, `AP_DUP_WINDOW_MONTHS=24`, `AP_EXCEPTION_SLA_DAYS=3`, `RCV_OVER_DELIVERY_PCT=0`. Migrations `procurement/v18.*`, `payables/v18.*`. Flags: `ap.ocr_adapter`, `ap.ers`, `ap.ubl_intake` (w35), `rcv.pwa`. Jobs: GR/IR reconciliation daily 02:30; exception SLA timers (Temporal); ERS consumer (Kafka consumer group `payables-ers`). SLOs: receipt post p95 < 1.5 s; invoice post p95 < 2 s; matching evaluate p95 < 100 ms; event lag < 5 s. Alerts: GR/IR diff ≠ 0 (page); exception backlog > 50 (ticket); ERS consumer lag > 100 messages. Runbook: re-run match for invoice set, reprocess failed capture payloads from DLQ topic `payables.capture.dlq`, write off GR/IR residual with approval. Rollback: additive migrations; disable `ap.ers` consumer; invoices in EXCEPTION are unaffected by rollback of the matching worker.

3.18.11 Migration from AX / SAP

Source	Passage entity	Notes
AX VendPackingSlipJour/Trans for open PO lines (received not invoiced)	ProductReceipt POSTED + GrirOpenItem	amount from VendPackingSlipTrans value; creates GR/IR opening balance matched to GL migration
AX VendInvoiceInfoTable (pending invoices)	VendorInvoice DRAFT	re-matched on load
AX VendInvoicePriceTolerance, InventTable tolerance fields, AP parameters	MatchingPolicy rows	scope mapping: item+vendor → VENDOR_ITEM etc.
AX VendTable self-billing flags	Vendor `ers_enabled`	
SAP EKBE VGABE 1 without matching VGABE 2 (GR not IR)	ProductReceipt + GrirOpenItem	WRX balance per PO item
SAP RBKP/RSEG parked (RBSTAT A/B) or blocked (ZLSPR)	VendorInvoice DRAFT / EXCEPTION	
SAP OMR6 tolerance keys (T169G) PP, DQ, BD, AN/AP	MatchingPolicy ENTITY scope	PP → price_tol; DQ → qty_tol; BD → small_diff; AN/AP → total_tol
SAP LFM1 XERSY	ers_enabled	

Validation: GR/IR opening Σ equals GL migrated balance per legal entity; count of open receipt lines equals source; parked invoices re-matched produce the same block/no-block result as SAP for a 50-invoice sample. Not migrated: posted invoice history (becomes VendorTransaction open items in wave 19 migration; line detail stays in lakehouse), MRBR release logs, AX invoice register journals already approved (post before cutover).

3.18.12 Dependencies, risks, open questions

Depends on: wave 17 (PO lines, versions, charges), wave 16 (transaction steps, serial/batch), wave 11–14 (posting profiles, currency, tax, period control), wave 4 (Temporal workflows), wave 8 (GS1 parsing).

Risk	Likelihood	Impact	Mitigation
Accountant disagrees with GR/IR FX treatment (BR-18.7)	Medium	Medium	Confirm with accountant in match-200 review before posting logic is frozen
OCR adapter quality varies; low-confidence fields	High	Low	Confidence threshold gating; no auto-post below 0.95
Multi-PO invoices	Medium	Low	Supported by line-level PO links; receipts stay one PO per document

Open questions:

1. Auto-post invoices with 100 % match and confidence ≥ 0.95 without human review? Default: configurable per vendor, off by default. Decide by wave 19.
2. Should quantity tolerance also apply to under-invoicing close-out (auto-close residual GR/IR)? Default: no; manual write-off. Decide by wave 20.
3. Non-PO invoice approval: route by GL account owner or by amount only? Default: amount bands reuse of the wave 17 policy. Decide by wave 18 mid-point.

3.18.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent
`match_level`	matching policy scope	enum	THREE_WAY	NONE, TWO_WAY, THREE_WAY	FR-18.5	Accounts payable parameters → Invoice validation → "Enable invoice matching validation", line matching policy
`price_tol_pct` / `price_tol_abs` / `price_tol_mode`	matching policy scope	numeric / numeric / enum	2.0 / 25.00 / UPPER_ONLY	≥ 0; UPPER_ONLY, BOTH	FR-18.5 price check	Price tolerances (VendInvoicePriceTolerance) per item/vendor/group
`qty_tol_pct`	matching policy scope	numeric	5.0	0–100	FR-18.5 quantity check	Released product → quantity tolerance
`total_tol_pct` / `total_tol_abs`	matching policy scope	numeric	1.0 / 50.00	≥ 0	FR-18.5 total check	Invoice totals tolerances (D365 invoice totals matching)
`small_diff_abs`	matching policy scope	numeric	0.50	≥ 0	BR-18.5 auto-posting to small-differences account	Accounts payable parameters → "Maximum penny difference"
`date_tol_days`	matching policy scope	int	0 (off)	0–90	invoice before planned delivery date	n/a
`charges_tol_pct`	matching policy scope	numeric	5.0	≥ 0	FR-18.11 charge variance	Invoice matching → Charges tolerances
`stochastic_block`	matching policy scope	record	off (threshold 0, percentage 0)	threshold_amount ≥ 0, percentage 0–100	FR-18.15	n/a
`over_delivery_pct` / `under_delivery_pct` / `close_short`	product, vendor × product	numeric / numeric / bool	0 / 0 / false	0–100	FR-18.1 receiving tolerance	Released product → Over/underdelivery

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent
`receipt_requirements`	item model group	enum	RECEIPT	NONE, REGISTRATION, RECEIPT	FR-18.1 registration mandatory before receipt	Item model group → Registration requirements / Receiving requirements
`default_receipt_status`	warehouse	enum	AVAILABLE	AVAILABLE, QUARANTINE, BLOCKED	inventory status on registration	Default inventory status on warehouse
`duplicate_check_window_months` / `duplicate_check_fields`	legal entity	int / set	24 / vendor + invoice number	+ invoice date, + amount	FR-18.4 `AP-DUP-001`	Accounts payable parameters → "Check for invoice numbers used"
`capture_confidence_threshold` / `auto_post_confidence`	legal entity	numeric	0.85 / off (0.95 when enabled)	0–1	UX highlight and auto-post gate	Invoice capture parameters (D365)
`ers_enabled` / `ers_invoice_number_format`	vendor	bool / pattern	false / `ERS-{receipt_number}`		FR-18.10	Vendor → "Self-billing"; Generate invoices from product receipts
`exception_sla_days` / `exception_assignee_rule`	legal entity	int / enum	3 / BUYER_FOR_PRICE_RECEIVER_FOR_QTY	1–30	FR-18.7	Vendor invoice workflow escalation
`grir_writeoff_limit` / `grir_writeoff_age_days`	legal entity	numeric / int	5.00 / 180	≥ 0	GR/IR maintenance proposal (3.18.16)	Accrued purchases write-off (manual)
`posting_profile.grir_account` / `ppv_account` / `small_diff_account` / `consignment_accrual`	posting profile	account	seeded	ledger accounts	FR-18.2, FR-18.6, FR-18.16	Posting profiles: Purchase expenditure un-invoiced, Price variance, Penny difference
`ses_acceptance_workflow` / `ses_approver_rule`	policy (VENDOR_INVOICE)	workflow id / enum	wave 4 default / PO_REQUESTER	PO_REQUESTER, COST_CENTER_OWNER, PROJECT_MANAGER	FR-18.14	n/a
`ses_number_sequence`	legal entity	sequence	`SES-`	wave 5 sequences	FR-18.14	n/a
`consignment_settlement_frequency` / `consignment_self_billing`	vendor	enum / bool	WEEKLY / true	DAILY, WEEKLY, MONTHLY	FR-18.16	Consignment parameters (D365)
`invoicing_plan_run_time`	legal entity	time	04:00		FR-18.17	n/a
`one_time_vendor_account` / `one_time_payee_max_amount` / `one_time_payee_max_count`	legal entity	reference / numeric / int	none / 10,000.00 USD / 3 per 12 months		FR-18.19	Accounts payable parameters → One-time supplier account
`non_po_invoice_approval`	policy (VENDOR_INVOICE)	enum	AMOUNT_BANDS	AMOUNT_BANDS, GL_ACCOUNT_OWNER	open question 3	Invoice register approval journal

3.18.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Expected receipts (arrival overview)	list inquiry	site, warehouse, vendor, date range, ASN	PO, line, expected qty, arrived, registered, ASN	PO schedules + `ProductReceipt`	Arrival overview (WMSArrivalOverview)	ME2V goods receipt forecast, VL06I inbound delivery monitor
Product receipt journal	list inquiry	vendor, PO, packing slip, date	receipt number, lines, qty, amount, voucher	`ProductReceipt`	Product receipt journal inquiry (VendPackingSlipJour)	MB51 movement 101, ME23N PO history
Vendor invoice list / status	list inquiry	vendor, status, match status, source, date	invoice, vendor invoice number, total, match status, assignee	`VendorInvoice`	Pending vendor invoices; Invoice journal inquiry	MIR5 / MIR6 invoice overview (parked, held, blocked, posted)
Match details	drill-down	invoice	per line PO price, invoice price, qty received/invoiced, variances, policy applied	`MatchResult`	Invoice history and matching details	MIRO tolerance log, MRBR block reasons
Exception queue	operational report	assignee, variance type, age	invoice, line, variance, SLA due	Temporal workflow projection	Vendor invoice workflow history	MRBR blocked invoices list
GR/IR open items (accrued purchases)	operational report	vendor, PO, age, as-of	received not invoiced, invoiced not received, totals vs ledger	`GrirOpenItem`	Accrued purchases report (Received not invoiced)	MB5S GR/IR balances, F.19 GR/IR analysis
Received not invoiced by PO	list inquiry	vendor, buyer	PO line, received qty/amount, invoiced, days open	`GrirOpenItem` + PO	Purchase orders → Received but not invoiced	ME2L with RECHNUNG, ME80FN
Duplicate invoice suspects	KPI / list	vendor, window	pairs with same normalized number or same amount/date	`VendorInvoice`	n/a	OMRDC report (custom)
Invoice capture quality	KPI	adapter, vendor, period	invoices captured, fields corrected %, avg confidence, touchless %	`VendorInvoice.source`, audit	Invoice capture dashboard (D365)	VIM analytics
Service entry sheets (by service / PO / sheet / vendor / contract)	list inquiry	vendor, PO, status, period	sheet, lines planned/unplanned, amount, accepted by, invoiced	`ServiceEntrySheet`	n/a	MSRV1–MSRV6, MSRV3 entry sheets, ME2S
SES acceptance list	list inquiry	approver, status	sheet, amount, submitted at, SLA	`ServiceEntrySheet`	n/a	ML83 / ML84
Audit-sampled invoices	operational report	period, vendor, result	invoice, total, sampled at, released/rejected, reason	`MatchResult` status AUDIT_SAMPLE	n/a	MRBR filtered on stochastic block (ZLSPR S)

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Consignment settlement history	list inquiry	vendor, period	invoice, ownership change journals, qty, amount	`VendorInvoice` kind CONSIGNMENT	Consignment vendor invoices (D365)	MRKO settlement list
ERS / self-billing log	list inquiry	vendor, date	receipt or sheet, ERS invoice, status	`VendorInvoice` kind ERS	Self-billing invoices	MR90 ERS output, MRRL log

3.18.15 Printed and electronic documents

Document	Trigger	Layout elements (header / line / footer)	Formats	Numbering sequence	Legal / retention requirement
Goods receipt slip / product receipt	receipt POSTED	header: receipt number, PO, vendor, packing slip, receipt date, receiver; lines: product, qty, unit, batch/serial, location, status; footer: signature	PDF, PWA print	`RCV-`	Goods receipt evidence; retain 10 years
Receipt label	registration	product, batch, expiry, serial, PO, GS1-128	ZPL, PDF	none	GS1 where required
Self-billing (ERS) invoice	ERS invoice POSTED	header: "Self-billed invoice" legend, vendor and buyer VAT ids, invoice number, date, receipt reference; lines: product, qty, price, tax; footer: totals per tax rate, self-billing agreement reference	PDF/A-3, email; Peppol self-billing (wave 35)	`ERS-{receipt}` / `ERS-{sheet}`	EU VAT Directive Art. 224 (prior self-billing agreement, acceptance procedure); retain 10 years
Consignment settlement statement / self-billing invoice	consignment invoice POSTED	header: vendor, period, settlement number; lines: ownership change date, product, qty, price, amount; footer: tax, total	PDF/A-3, email, portal (wave 20)	`CSS-`	Same as ERS (self-billing rules)
Service entry sheet	sheet SUBMITTED / ACCEPTED	header: sheet number, PO line, period, vendor, accepted by/at; lines: service item, qty, unit, price, amount, planned/unplanned; footer: totals, limit remaining	PDF, email, portal	`SES-`	Service acceptance evidence; retain 10 years
Invoice reduction / debit note to vendor	exception resolved with "request credit note"	header: invoice reference, reason; lines: disputed lines, expected vs invoiced	PDF, email	`DN-`	Commercial correspondence
Invoice approval cover (non-PO)	approval task	invoice image link, coding lines, approver chain	HTML email	n/a	none
GR/IR reconciliation report	month end (on demand)	open items with totals vs ledger	PDF, XLSX	n/a	Audit workpaper; retain with period close

3.18.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
GR/IR reconciliation (FR-18.9)	daily 02:30	legal entity × GR/IR account	run id (entity, date); full recompute	< 3 min per 1M open items	retry 3×	diff ≠ 0 pages AP lead
Post pending matched invoices (batch, opt-in)	every 15 min	legal entity	key = invoice id; POSTED skipped	< 1 min per 1k	invoice-level errors left in queue	errors > 20 → ticket
Automatic release of blocked invoices	hourly	legal entity	re-evaluates EXCEPTION invoices whose cause cleared (receipt arrived, PO price changed); idempotent	< 2 min	retry	none
ERS run (receipts and sheets missed by the consumer)	daily 05:00	vendor	key = receipt line / sheet line; existing ERS invoice skipped	< 5 min	vendor skipped	ERS invoices created by batch > 0 indicates consumer lag → ticket
Consignment settlement run (FR-18.16)	per vendor frequency (default Monday 04:30)	vendor	key = (vendor, period); ownership change lines settled once	< 2 min per 1k lines	vendor skipped and listed	failures > 0 → ticket
Invoicing plan run (FR-18.17)	daily 04:00	legal entity	key = plan row id	< 1 min per 5k rows	row-level error	errors > 0 → ticket
GR/IR maintenance proposal (write-off of aged residuals)	monthly, day 1 06:00	legal entity	key = (open item, period); proposal reviewed before posting	< 2 min	n/a	proposal ready notification
Capture DLQ reprocessing	every 30 min	adapter	key = adapter document id	< 1 min	after 5 attempts park	DLQ depth > 50 → ticket
Stochastic sample rate report	monthly day 2	legal entity	key = period	< 30 s	retry	achieved rate deviates > 2 points from configured → warning
SES acceptance reminders and escalation	Temporal timers	sheet	timer per step	n/a	workflow retry	sheets pending > SLA → manager notification

3.18.17 Security duties and privileges

Duty	Privileges (entry points: API paths / screens / jobs)	Segregation-of-duties conflicts	Default roles that include it	AX duty / SAP authorization reference
ReceivingMaintain	arrivals, receipts create/register; receiving workspace; PWA	none	Receiving clerk	AX "Maintain product receipts" (VendPackingSlipMaintain); SAP M_MSEG_BWE 101 activity 01
ReceivingPost	receipts :post, :correct, :cancel	with PurchaseOrderMaintain (w17), VendorInvoicePost	Receiving supervisor	AX "Post product receipts"; SAP M_MSEG_BWE post
ServiceEntryMaintain	service entry sheets create/submit; portal entry (w20)	with ServiceEntryAccept on same sheet	Requester, Vendor user (portal), Project manager	n/a; SAP ML81N (M_SRV_LS activity 01)
ServiceEntryAccept	accept, revoke, accept-bulk	with ServiceEntryMaintain (same sheet), PurchaseOrderMaintain (same PO)	Cost center owner, Project manager	SAP ML85 / M_SRV_LS release (activity 43)
VendorInvoiceMaintain	invoices create, :propose-lines, :match; invoice screen	with VendorMaintain (w7), PaymentProposalMaintain (w19)	AP clerk	AX "Maintain vendor invoices" (VendInvoiceInfoTableMaintain); SAP M_RECH_WRK, MIRO/MIR7
VendorInvoiceCapture	invoices:capture (adapter/service), capture review	none	AP clerk, service account	AX "Process vendor invoice capture" (D365); SAP VIM roles

Duty	Privileges (entry points: API paths / screens / jobs)	Segregation-of-duties conflicts	Default roles that include it	AX duty / SAP authorization reference
VendorInvoicePost	invoices :post; credit notes; subsequent debit/credit; ERS, consignment, plan runs	with VendorInvoiceMaintain when `ap_four_eyes=true`, ReceivingPost, PaymentRunExecute (w19)	AP supervisor	AX "Post vendor invoices"; SAP M_RECH_WRK post, FB60
InvoiceExceptionResolve	exception tasks :accept, request credit, adjust PO, reject	with VendorInvoicePost under four-eyes; amount ≤ approver limit (FR-17.17)	Buyer, Receiving supervisor, AP manager	AX "Approve vendor invoice variances"; SAP MRBR (M_RECH_SPG release blocked invoices)
InvoiceAuditRelease	audit-sampled invoices release/reject	with VendorInvoicePost (same invoice)	Internal auditor, AP manager	SAP MRBR stochastic block release (M_RECH_SPG)
MatchingPolicyMaintain	matching policies, stochastic rule, matching:evaluate tester	with InvoiceExceptionResolve	AP administrator	AX "Maintain invoice matching setup"; SAP OMR6/OMRX (S_TABU_DIS)
APInquire	GR/IR report, invoice lists, match details, SES lists	none	AP clerk, Controller, Auditor	AX "Inquire into vendor invoices"; SAP F_BKPF_BUK display, MB5S
GrirWriteOff	GR/IR maintenance proposal post	with VendorInvoicePost	AP manager, Controller	SAP MR11 (M_RECH_WRK + F_BKPF_BUK)

3.18.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
RCV-OVR-001	error	Receipt of {qty} exceeds ordered {ordered} plus over-delivery tolerance {pct} %	FR-18.1	Reduce quantity or raise tolerance on the PO line	FR-18.1
RCV-UND-001	warning	Line {line} received {qty} below under-delivery tolerance; line stays open	FR-18.1	Set close short or await remaining delivery	FR-18.1
RCV-DUP-001	error	Packing slip {number} already used for vendor {vendor} on receipt {receipt}	FR-18.2	Correct the packing slip number	FR-18.2
RCV-PER-001	error	Receipt date {date} is in a closed INVENTORY period	Period control	Use the first open date	FR-18.2
RCV-COR-001	error	Correction {delta} would reduce received quantity below invoiced {invoiced}	FR-18.3	Post a credit note first	FR-18.3, BR-18.12
RCV-REG-001	error	Registration is required before receipt for {product}	`receipt_requirements`	Register the line first	FR-18.1
AP-DUP-001	error	Invoice {number} from {vendor} already exists ({existing}, {date})	Duplicate check	Verify with vendor; cancel the duplicate	FR-18.4
AP-CAP-001	warning	{n} captured fields below confidence {threshold}; confirm before posting	OCR confidence	Review highlighted fields	FR-18.4
AP-TOT-001	error	Entered total {entered} differs from computed total {computed} by more than {small_diff}	Keying error	Correct the total or lines	FR-18.4, BR-18.5
AP-MAT-001	warning	Line {line}: price variance {amount} ({pct} %) exceeds tolerance	FR-18.5	Accept, adjust PO or request credit	FR-18.5, FR-18.7
AP-MAT-002	warning	Line {line}: invoiced quantity {qty} exceeds received uninvoiced {received}	FR-18.5	Wait for receipt or request credit	FR-18.5
AP-MAT-003	warning	Invoice total variance {amount} exceeds tolerance	FR-18.5	Review charges and tax	FR-18.5
AP-MAT-004	warning	Invoice currency {inv_ccy} differs from PO currency {po_ccy}	BR-18.3	Confirm or request corrected invoice	BR-18.3
AP-MAT-005	info	Invoice selected for audit sample (rule {rule}); posting held for release	FR-18.15	Internal auditor releases or rejects	FR-18.15
AP-ACC-001	error	Variance {amount} exceeds your approval limit {limit}	AcceptVariance limit	Escalate to an approver with a higher limit	FR-18.7
PER-AP-001	error	Posting date {date} is in a closed AP period	BR-18.10	Change posting date to first open day	BR-18.10
AP-ERS-001	error	ERS invoice blocked: PO price changed since receipt ({old} → {new})	BR-18.9	Buyer confirms price	BR-18.9
AP-SES-001	error	Service entry sheet {sheet} overlaps accepted sheet {other} for item {item}	BR-18.13	Adjust period or item	BR-18.13
AP-SES-002	error	Sheet {sheet} cannot be revoked: {amount} already invoiced	BR-18.13	Credit the invoice first	FR-18.14
AP-CON-001	error	Consignment settlement for {vendor}: no trade agreement price for {product} on {date}	Missing consignment price	Create consignment trade agreement row	FR-18.16
AP-IP-001	error	Invoicing plan row {row} requires an accepted receipt or service entry sheet for the period	BR-18.16	Accept the sheet or set settle without receipt	FR-18.17
AP-SUB-001	error	Subsequent credit {amount} exceeds invoiced amount {invoiced} on line {line}	BR-18.17	Reduce the amount	FR-18.18
AP-OTV-001	error	One-time payee {name} exceeds limits ({amount} / {count} uses); create a vendor	FR-18.19	Create a vendor in the party model	FR-18.19
AP-BNK-001	error	Bank account {account} fails validation ({rule})	IBAN mod-97 / ABA checksum	Correct the bank account	FR-18.19
AP-GRIR-001	error	GR/IR report total {report} differs from ledger balance {ledger} by {diff}	Manual journal or defect	Investigate vouchers listed in the diff event	FR-18.9

3.18.19 Performance budgets and capacity

Operation	Volume assumption (per tenant per day / peak per minute)	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Receipt post	8,000 receipts / 200 per min	400 ms / 1.5 s / 3 s	200 receipts/min	receipt ≈ 1.5 KB + lines 500 B; GrirOpenItem 300 B	(tenant, vendor, packing_slip_number) unique; GrirOpenItem (po_line_id, status)	Ledger `Post` queued in Temporal; receipt returns POSTING then POSTED ≤ 10 s
Invoice create + match	10,000 invoices / 150 per min	80 ms / 300 ms / 700 ms	150 rps	invoice ≈ 3 KB + lines 600 B + page images 300 KB in object store (≈ 1 TB/year at 10k/day)	invoices (tenant, vendor_id, lower(vendor_invoice_number)) partial unique	Matching moved to async worker; UI shows MATCHING spinner ≤ 5 s
Matching evaluate (pure)	50k / 1,000 per min	10 ms / 100 ms / 200 ms	500 rps	none	in-memory policy cache by scope	Policy cache warm; p95 ≤ 250 ms
Invoice post	10,000 / 150 per min	500 ms / 2 s / 4 s	100 rps	VendorTransaction (w19) 400 B	n/a	Ledger post backpressure via Temporal
ERS consumer	3,000 receipts / 100 per min	lag 2 s / 10 s / 30 s	50 invoices/s	as invoices	consumer group `payables-ers` partitions = 12	lag alert at 100 messages; batch job catches up

Operation	Volume assumption (per tenant per day / peak per minute)	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Service entry sheet accept	2,000 / 50 per min	300 ms / 1 s / 2 s	50 rps	sheet ≈ 2 KB	(po_line_id, status)	same as receipt post
GR/IR report (1M open items)	50 runs	1 s / 4 s / 8 s	n/a	n/a	aggregate on (vendor_id, status)	Served from materialized daily snapshot + intraday delta
Capture intake (OCR adapter)	5,000 / 100 per min	200 ms / 800 ms / 2 s	100 rps	payload JSON 20 KB + images	DLQ topic `payables.capture.dlq`	429 to adapter with `Retry-After`

3.18.20 Localization and compliance hooks

Concern	Varies by	Country packs touching it (wave 48)	Extension point used
E-invoice intake formats (Peppol BIS 3.0, XRechnung, FatturaPA/SDI, Factur-X, CFDI, NF-e) and mandatory clearance	Country e-invoicing mandates	DE, IT, FR, ES, PL, MX, BR, IN	`:capture` source UBL/structured (wave 35 adapters); invoice keeps original XML as legal archive
Self-billing rules (prior agreement, "self-billing" legend, vendor acceptance)	EU VAT Art. 224, UK HMRC self-billing	EU, UK	ERS/consignment templates (3.18.15) with country legend; vendor `self_billing_agreement_ref`
Invoice archiving (original format, integrity, 8–10 year retention; GoBD, French PAF)	Country retention law	DE, FR, IT, ES	Object store write-once bucket with retention lock; hash on VendorInvoice
Tax determination on receipt and invoice (reverse charge, use tax accrual, import VAT)	VAT/GST/sales tax regimes	all	Wave 13 tax engine call in `PostInvoice`; use-tax accrual on 2-way invoices
GR/IR and accrual presentation (balance sheet regrouping for GR/IR debit balances)	Local GAAP	DE (HGB), FR (PCG)	F.19-style regrouping job hook in wave 14 close; GrirOpenItem sign split
Withholding on services (e.g., IN TDS, BR ISS/IRRF, US backup withholding) at invoice	Country withholding	IN, BR, MX, US	Wave 13 withholding engine invoked on invoice lines with service products; payment-time withholding in wave 19
Invoice numbering of self-billed documents (gap-free)	IT, ES, PT	IT, ES, PT	Number sequence per legal entity with `gap_free=true`
Language of vendor-facing documents	Vendor language	all	Wave 15 templates per language
Date and amount formats in capture parsing	Locale	all	Capture adapter locale hint per vendor
Privacy of one-time payee data	GDPR, CCPA	EU, US-CA	Field-level encryption, retention purge after legal retention (wave 2)

3.18.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP 13.12 — ME21N service item (D) services PO [#128]	PARTIAL-18	ADDED as FR-17.23 (order) and FR-18.14 (receipt)	Service PO defined in wave 17; acceptance and posting here
SAP 13.12 — MSRV1 / MSRV3 service reports [#131]	PARTIAL-18	ADDED as FR-18.14	3.18.14 service entry sheet lists
SAP 13.14 — OMRX stochastic block (T169P) [#161]	GAP (propose 18)	ADDED as FR-18.15	Deterministic seeded sampling, auditor release
SAP 13.15 — ML81N / ML82 / ML83 / ML84 / ML85 service entry sheet create, display, list, accept/revoke, collective release [#167]	PARTIAL-18	ADDED as FR-18.14	Acceptance workflow, revoke, accept-bulk
SAP 13.15 — MSRV1 … MSRV6 service lists (master data) [#169]	PARTIAL-18	ADDED as FR-18.14	3.18.14
SAP 13.15 — MIRO with SES reference [#170]	PARTIAL-18	ADDED as FR-18.14	Invoice proposes lines from accepted sheets; match to sheet amount
SAP 13.15 — ML91 / ML92 / ML93 SES archiving / lists [#171]	PARTIAL-18	SUPERSEDED by wave 46 archiving policy	Lists covered in 3.18.14; archiving uses the platform retention/archival service rather than a per-object archive run
SAP 13.15 — Fiori: Manage Service Entry Sheets – Lean Services / Create Service Entry Sheet [#172]	PARTIAL-18	ADDED as FR-18.14	Desktop and portal (FR-20.20) entry
SAP 13.15 — MSRV1 … MSRV6 service lists (reports) [#175]	PARTIAL-18	ADDED as FR-18.14	3.18.14
SAP 13.15 — ME2S services per purchase order [#176]	PARTIAL-18	ADDED as FR-18.14	3.17.14 "Services per purchase order" and 3.18.14
SAP 13.15 — MSRV3 entry sheets list [#177]	PARTIAL-18	ADDED as FR-18.14	3.18.14
SAP 13.15 — ML83 / ML84 SES acceptance list [#178]	PARTIAL-18	ADDED as FR-18.14	3.18.14 "SES acceptance list"
SAP 13.15 — ML85 collective release of SES [#179]	PARTIAL-18	ADDED as FR-18.14	`:accept-bulk`
SAP 13.15 — MRRL for services (ERS for SES) [#180]	PARTIAL-18	ADDED as FR-18.14	ERS invoice per accepted sheet
SAP 13.15 — Service specification screen layout, SES number ranges, SES release procedure (ESSR) [#182]	PARTIAL-18	ADDED as FR-18.14	`ses_number_sequence`, `ses_acceptance_workflow` (3.18.13); screen layout SUPERSEDED by wave 3 metadata form layouts
SAP 13.15 — Default values for SES acceptance (immediate GR, account assignment) [#184]	PARTIAL-18	ADDED as FR-18.14	Acceptance is the posting; account assignment defaults from the PO line
SAP 13.15 — Lean services activation, product type service, lean SES settings [#186]	PARTIAL-18	ADDED as FR-18.14	Service products (wave 8 product type SERVICE) and SES settings in 3.18.13
SAP 13.2 — One-time vendor (account group CPD) [#23, register wave 7]	PARTIAL-7	ADDED as FR-18.19 (document-level payee on invoice)	Register assigns the party role to wave 7; the document-level capture and payment use are closed here and in FR-19.23
SAP 13.2 — OBD3 one-time account (T077K) [#37, register wave 7]	PARTIAL-7	ADDED as FR-18.19	`one_time_vendor_account` parameter (3.18.13)

3.19 Wave 19: Accounts Payable, Settlement & Payments

3.19.1 Scope and business context

Wave 19 turns posted vendor invoices into paid liabilities. Invoices from wave 18 are open items (AX VendTrans with VendTransOpen; SAP BSIK) that settle against payments, credit notes and prepayments (AX VendSettlement / SpecTrans marking; SAP clearing to BSAK with BSE_CLR, F-53/F-44). A payment proposal (AX LedgerJournalTrans payment journal with VendPaymProposal; SAP F110 proposal run producing REGUH/REGUP) selects items by due date and cash discount, applies holds, approves, generates payment media (ISO 20022 pain.001 for SEPA/ACH-via-ISO, NACHA for US ACH, check print, positive pay file; SAP DMEE /

PMW, AX payment formats via ER), posts the payment to the bank clearing account and vendor control account, and ingests bank status (pain.002) to update payment status. Prepayments, credit notes, reversals and vendor aging complete the AP subledger.

Out of scope: bank statement reconciliation and cash positioning (wave 36, which consumes this wave's payment events), bank connectivity transport (wave 36; this wave writes files to an object store path and exposes them for download), customer-side settlement (wave 30), 1099/withholding statutory reporting beyond withholding calculation (wave 13 provides calculation; reporting in wave 48), intercompany payments (wave 40).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX payment proposal (VendPaymProposal) runs as a modal dialog from the payment journal, locks the journal while running, and must be re-run from scratch to change selection; SAP F110 parameters → proposal → edit proposal → payment run → print is five screens, with proposal logs (REGUP exceptions) read in a separate list	Proposal is a transient batch producing journal lines; no proposal entity	`PaymentProposal` is a persisted entity with selection criteria, line-level include/exclude, holds, and re-selection diff; approve → run is one action	Screens per payment run: 5–7 → 2; time from "select" to "approved file": 45 min → < 5 min
Settlement in AX via "Settle open transactions" (CustVendOpenTransManager) silently computes discount and FX differences; users discover realized gain/loss or discount-lost postings only in the ledger; SAP F-53 shows the clearing document but FX difference accounts (OB09) are hidden until posted	No preview of settlement postings	`POST /settlements:preview` returns the exact journal (discount, FX gain/loss, write-off) before commit; UI shows it inline	Unexpected settlement postings found at month-end: typical 5–20/month → 0; review time per settlement: 2 min → 10 s
Payment formats: AX requires Electronic Reporting configurations per bank (hundreds of nodes), SAP DMEE trees; a bank-specific tag change takes a consultant day	Format defined in a proprietary tree editor	Canonical pain.001.001.09 generator with per-bank profile overrides as JSON (≤ 20 keys), validated against the XSD in CI and a bank harness	Time to onboard a new bank profile: 2–5 days → < 2 h; file rejections at bank: tracked, target < 0.1 %
Payment run idempotency: AX journal posting interrupted mid-batch leaves half-posted lines and a generated file; re-posting can pay twice; SAP F110 "restart" needs SM37 analysis	No durable workflow; file generation and posting are separate steps without a shared idempotent record	Temporal `PaymentRun` workflow with idempotency keys per open item; file is derived from posted payments, never the reverse	Duplicate payments from crash/retry: non-zero historical → 0 by test
Vendor aging in AX (VendAgingReport) is an SSRS batch over VendTrans; SAP S_ALR_87012078 runs minutes for large vendors; neither reconciles to the control account automatically	Report computed from open items with no tie to GL	Aging served from the `VendorTransaction` projection with bucket config; daily job asserts Σ open = AP control balance per vendor group	Aging report time: 2–10 min → < 2 s; monthly AP-to-GL reconciliation: 2 h → automated, 0 diff

3.19.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Vendor open items	VendTrans, VendTransOpen; Vendor transactions form; VendTable balance	BSIK (open), BSAK (cleared), FBL1N	`VendorTransaction` with `open_amount`; vendor transactions screen	full
Settlement	VendSettlement, CustVendOpenTransManager (Settle open transactions), marking (SpecTrans), partial settlement, cross-rate	Clearing F-44/F-53/F-51, residual/partial payment, BSE_CLR, AUGBL	`Settlement`, `SettlementLine`; manual, during payment, automatic	full
Cash discount	CashDisc (terms), VendTrans CashDiscDate, discount taken/lost accounts, "use cash discount" normal/always/never	Payment terms T052 ZBD1T/ZBD1P, SKTO/SKE accounts, OBXU/OBXV	`PaymentTerms` (wave 7) extended with discount tiers; settlement algorithm	full
FX on settlement	Exchange adjustment realized gain/loss (LedgerParameters, CurrencyExchangeAdjustment)	OB09 realized gain/loss accounts, KDF	realized gain/loss in settlement journal (wave 12 accounts)	full
Payment journal / proposal	LedgerJournalTable type Payment, LedgerJournalTrans, VendPaymProposal (by due date / cash discount / both), VendPaymModeTable (method of payment), payment specification, CustVendPaymJournal	F110 (FBZP config), REGUH/REGUP, payment methods T042Z, house banks FI12	`PaymentProposal`, `PaymentMethod`, `Payment`, `PaymentRun` (Temporal)	full
Payment media	ER formats (ISO 20022 credit transfer, NACHA CCD/CTX, check layout), VendOutPaym classes	DMEE/PMW formats, F110 print program RFFOUS_T, check lot FCHI	`PaymentFile` generators: pain.001.001.09, NACHA (PPD/CCD/CTX), check PDF + MICR, positive pay CSV	full
Bank status ingestion	Return file import (ER pain.002), payment status	pain.002 via bank communication management (BCM)	`POST /payments/status:ingest` pain.002 parser	full
Positive pay	Positive pay format (BankPositivePayExport)	Not standard (custom/BCM)	positive pay CSV per bank profile	full
Prepayments	Prepayment invoice on PO (PurchPrepayTable), prepayment journal with posting profile	Down payment request F-47, down payment F-48 (special G/L A), clearing F-54	`Prepayment` (PO-linked or free) with separate posting profile, applied at invoice posting or settlement	full
Credit notes and reversal	Negative invoice, reversal (TransactionReversal), void check	Credit memo MIRO/FB65, reverse FB08, void FCH8/FCH9	wave 18 credit note; `Payment:reverse`, `Payment:void`	full
Vendor aging	Vendor aging report (VendAgingReport), aging period definitions	Aging S_ALR_87012078, due date analysis	`GET /aging` with bucket definitions; ties to control account	improved
Withholding tax at payment	Withholding tax (TaxWithhold*) calculated at payment	Extended withholding at payment (T059P)	wave 13 withholding engine invoked in settlement; posts withholding payable	full
Payment advice	Vendor payment advice (print management)	Payment advice RFFOAVIS	email/portal advice PDF with settled invoices	full
ACH prenotes	Payments → Prenotes (ACH), vendor bank account prenote status, "Update vendor bank accounts status", Bank account → Prenote setup	Not standard (DMEE custom zero-dollar entry)	`Prenote` (NACHA transaction code 23/33, zero amount) with bank account status lifecycle (FR-19.15)	full
Payment calendar and bank holidays	Payment calendar (D365), Payment days, bank holidays / calendar per bank	Factory calendar on house bank / payment method (FBZP "bank calendar"), T052 payment terms "next business day"	`PaymentCalendar` rules shifting due dates and execution dates by bank calendar (FR-19.16)	full
Balance confirmations	n/a (vendor statement only)	F.18 / F.98 balance confirmation (vendor), reply processing F.19-style	`BalanceConfirmation` runs with reply capture (FR-19.17)	full
AP correspondence	Vendor account statement (VendAccountStatement), payment advice	Correspondence F.64, types OB77/OB78/OB79 (account statement, payment notice, open items list)	`CorrespondenceType` (statement, payment notice, open items, balance confirmation) with template and trigger (FR-19.18)	full (conditional print routing in wave 46)
Special G/L transactions (vendors)	Prepayment posting profiles; no guarantees / noted items	OBYR special G/L indicators A (down payment), F (down payment request), G (guarantee), noted items; bill of exchange W	`VendorTransaction` types NOTED_ITEM (statistical) and GUARANTEE with separate reconciliation accounts (FR-19.19); bills of exchange in wave 36	partial (bills of exchange in wave 36)

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Installment payment terms	Payment schedules (PaymSched, multiple due dates per invoice)	OBB9 installment terms (T052R)	Installment split of vendor open items (FR-19.20)	full
Vendor-initiated direct debits (SEPA B2B)	Method of payment with "vendor collects"; no mandate register	Payment method for direct debit at vendor; mandate reference stored as text	`VendorDebitMandate` and payment method `DEBIT_BY_VENDOR` (FR-19.21)	full
Bank account change control	D365 vendor bank account change workflow / approval	Dual control on sensitive fields (FK08 confirm, S_ALR_87010039 list, FI12)	Dual approval and payment cooling-off on vendor bank account changes (FR-19.22)	improved
One-time payee payment	One-time supplier payment via payment journal	CPD account paid via F110 with document-level bank details	Payment to `one_time_payee` (FR-19.23)	full

3.19.3 Functional requirements

- **FR-19.1 Vendor open items** — Must. The system shall maintain a `VendorTransaction` for every posted invoice, credit note, payment and prepayment with `amount_txn`, `amount_acct`, `open_amount_txn`, `open_amount_acct`, `due_date`, `discount_date`, `discount_amount`, `status` (OPEN, PARTIALLY_SETTLED, SETTLED), and a `hold` flag with reason.
  - Open amount is updated only by settlement rows; `Σ open_amount_acct` per posting profile account equals the AP control balance (daily assertion).
  - Vendor balance query p95 < 50 ms for vendors with ≤ 100k transactions.
- **FR-19.2 Settlement algorithm** — Must. The system shall settle a set of credit-side items (payment, credit note, prepayment) against debit-side items (invoices) per vendor with the following algorithm, previewable before commit:
  1. Order target invoices by `settlement_priority` (user marks first, then due_date ASC, then amount DESC).
  2. For each invoice, `discount = round(invoice_amount_txn × discount_pct, minor)` if `settlement_date ≤ discount_date` (or terms are `ALWAYS`), else 0; for partial settlement, discount is pro-rated by `settled/invoice_amount` unless policy `discount_on_partial=false`.
  3. `applied_txn = min(available_credit, open_amount_txn − discount)`; invoice open amount reduces by `applied_txn + discount` (discount closes the item only when `applied_txn = open − discount`).
  4. Accounting amounts: `applied_acct_invoice = applied_txn × invoice_rate`, `applied_acct_credit = applied_txn × credit_rate`; `fx_diff = applied_acct_credit − applied_acct_invoice`; positive = realized loss for a vendor payable, negative = realized gain (wave 12 accounts).
  5. If a residual `|open_amount_txn| ≤ write_off_limit` (default 1.00 in transaction currency) and `auto_write_off=true`, post the residual to the write-off account and close the item.
  6. Cross-currency settlement (credit currency ≠ invoice currency) converts both to accounting currency at their own rates; the settlement unit is accounting currency; residuals follow step 5.
  7. Withholding (wave 13) is computed on `applied_txn` and reduces the payment, posting withholding payable.
  - Worked example: invoice 1,000.00 EUR posted at 1.1000 USD/EUR (1,100.00 USD), terms 2 % 10 net 30, paid on day 8 at rate 1.0800. Discount = 20.00 EUR. Applied = 980.00 EUR. Invoice-side acct = 980 × 1.10 = 1,078.00 USD; discount acct = 20 × 1.10 = 22.00; payment acct = 980 × 1.08 = 1,058.40. Journal: Dr AP control 1,100.00; Cr bank clearing 1,058.40; Cr discount taken 22.00; Cr realized FX gain 19.60. Balanced to the cent.
- **FR-19.3 Payment terms and discounts** — Must. The system shall compute `due_date` and up to two discount tiers (e.g., 2/10, 1/20, net 30) from the wave 7 payment terms, with `discount_basis` (gross, net of tax) and `use_cash_discount` (NORMAL, ALWAYS, NEVER) per vendor.
- **FR-19.4 Payment proposal** — Must. The system shall create a `PaymentProposal` from criteria (legal entity, vendor range/group, method of payment, due date to, discount date to, selection rule BY_DUE_DATE / BY_DISCOUNT / BOTH, minimum/maximum payment, currency, bank account), listing candidate open items with proposed amount and discount, excluding held vendors/items, and allowing line-level include/exclude, amount change and reselection with diff.
  - Discount optimization: when `BY_DISCOUNT`, items whose discount date is within `discount_lookahead_days` (default 7) are included; annualized discount value vs `cost_of_capital_pct` (default 8 %) decides early payment when `optimize=true`.
  - Proposal approval uses the wave 4 workflow with amount bands (default: > 100,000 USD two approvers).
- **FR-19.5 Payment run** — Must. The system shall execute an approved proposal as a Temporal `PaymentRun` workflow: for each vendor group create `Payment` records (one per vendor × currency × bank account × method), settle items (FR-19.2), post the payment journal (Dr AP control, Cr bank clearing per method), generate the `PaymentFile`, and mark payments SENT.
  - Each activity is idempotent on `(proposal_id, vendor_transaction_id)`; a resumed run skips already-settled items and already-generated files; the file is regenerated from posted payments if missing.
  - Throughput target: 10,000 open items settled and 2,000 payments generated in < 5 minutes.
- **FR-19.6 ISO 20022 pain.001** — Must. The system shall generate pain.001.001.09 files validated against the XSD with the following mapping:
  - `GrpHdr/MsgId` = `PaymentFile.file_number` (≤ 35 chars); `CreDtTm` = generation timestamp; `NbOfTxs` = count of CdtTrfTxInf; `CtrlSum` = Σ InstdAmt; `InitgPty/Nm` = legal entity name; `InitgPty/Id/OrgId/Othr/Id` = bank-assigned customer id from bank profile.
  - `PmtInf` per (debtor account × execution date × method): `PmtInfId` = `{file_number}-{seq}`; `PmtMtd` = `TRF`; `BtchBookg` = profile `batch_booking` (default true); `PmtTpInf/SvcLvl/Cd` = `SEPA` for SEPA profiles else `NURG/URGP`; `ReqdExctnDt/Dt` = payment date; `Dbtr/Nm`, `Dbtr/PstlAdr` = legal entity; `DbtrAcct/Id/IBAN` (or `Othr/Id` for non-IBAN) = bank account; `DbtrAgt/FinInstnId/BICFI` = bank BIC; `ChrgBr` = `SLEV` (SEPA) or profile default `SHAR`.
  - `CdtTrfTxInf` per Payment: `PmtId/InstrId` = payment number; `PmtId/EndToEndId` = payment number (≤ 35 chars, alphanumeric); `Amt/InstdAmt @Ccy` = amount and currency; `CdtrAgt/FinInstnId/BICFI` = vendor bank BIC (omitted when IBAN-only SEPA and profile allows); `Cdtr/Nm` = vendor name (≤ 140 chars, transliterated to the Latin character set required by the bank profile); `Cdtr/PstlAdr/Ctry` plus structured address when profile requires; `CdtrAcct/Id/IBAN` or `Othr/Id` + `CdtrAgt/.../ClrSysMmbId` for routing-number countries; `RmtInf/Ustrd` = concatenated invoice numbers (≤ 140 chars) or `RmtInf/Strd/RfrdDocInf/Nb` + `RfrdDocAmt` per settled invoice when `structured_remittance=true`; `Purp/Cd` = `SUPP`.
  - AX/SAP lineage: REGUH ZBNKN/ZBNKL → CdtrAcct, REGUH VBLNR → EndToEndId, REGUP BELNR → RfrdDocInf/Nb; AX LedgerJournalTrans PaymReference → EndToEndId.
- **FR-19.7 NACHA, checks, positive pay** — Must. The system shall generate NACHA files (file header 1, batch header 5 with SEC code PPD/CCD/CTX, entry detail 6, addenda 7 for CTX remittance, batch control 8, file control 9, blocking factor 10, 94-char records, hash of routing numbers), check PDFs with MICR line and check number from a check lot, and positive pay CSV (bank account, check number, amount, payee, issue date, void flag) per bank profile.
- **FR-19.8 Payment status and pain.002** — Must. The system shall ingest pain.002.001.10 status reports and map `GrpSts`/`TxSts` (ACCP, ACSC, ACSP, PDNG, RJCT) with `StsRsnInf/Rsn/Cd` (AC01, AC04, AM04, …) to payment statuses, moving RJCT payments to REJECTED, reversing settlement and re-opening items; NACHA returns (R01–R85) map likewise.

- **FR-19.9 Payment reversal and void** — Must. The system shall reverse a posted payment (new journal, settlement undone, items re-opened, file not regenerated) and void a check (check status VOID, positive pay void record), with reason codes and audit.
- **FR-19.10 Prepayments** — Must. The system shall post prepayment invoices (PO-linked, with prepayment posting profile and tax) and prepayment journals; at vendor invoice posting or manual settlement the prepayment applies against the invoice, reclassifying from the prepayment account to AP control.
- **FR-19.11 Vendor aging** — Must. The system shall produce aging by vendor/group with configurable buckets (default 0–30, 31–60, 61–90, 91–180, > 180 by due date or document date), as of any date, in transaction and accounting currency; totals tie to the AP control account per vendor group (daily job + report footer).
- **FR-19.12 Payment advice** — Should. The system shall email (or publish to the portal, wave 20) a payment advice PDF per payment listing settled invoices, discounts and withholding.
- **FR-19.13 Holds** — Should. The system shall support vendor-level holds (INVOICE, PAYMENT, ALL, from wave 7 status) and item-level holds with release date; held items are excluded from proposals and listed on the AP workspace.
- **FR-19.14 Automatic settlement** — Could. The system shall optionally settle credit notes against the oldest open invoices of the same vendor at posting (`auto_settle_credit_notes`).
- **FR-19.15 ACH prenotes** — Should. The system shall generate prenote entries (NACHA entry detail with transaction code 23 checking / 33 savings, amount 0.00, SEC code per method) for new or changed vendor bank accounts whose payment method requires a prenote, set the bank account `prenote_status` to `PENDING` with `prenote_sent_at`, and move it to `CONFIRMED` automatically after `prenote_wait_days` (default 6 banking days, per NACHA) unless a return (R03, R04, C01–C13 NOC) is ingested via FR-19.8, which sets `REJECTED` or applies the notification of change; payment proposals exclude vendors whose required prenote is not CONFIRMED (`AP-PRN-001`) and a periodic job updates statuses (AX "Update vendor bank accounts status").
- **FR-19.16 Payment calendar and bank-holiday shifting** — Must. The system shall maintain `PaymentCalendar` rules per legal entity, payment method and bank account (AX payment calendar / payment days; SAP bank calendar on house bank) referencing wave 6 calendars with bank holidays (TARGET2, US Federal Reserve, UK BACS etc.) and apply them: (a) invoice due dates computed from terms that fall on a non-banking day shift per rule `NEXT`, `PREVIOUS` or `NEXT_UNLESS_MONTH_CHANGE`; (b) proposal `payment_date` and pain.001 `ReqdExctnDt` are validated to be banking days of the debtor bank (`AP-CAL-001`), with `cutoff_time` per method; (c) payment days (e.g., pay only on Tuesdays) restrict proposal payment dates. The same rule set is used by AR (wave 30) for customer due dates.
- **FR-19.17 Vendor balance confirmations** — Should. The system shall run a balance confirmation (SAP F.18 / F.98) for selected vendors as of a key date, rendering a confirmation letter per vendor (open items and balance, reply slip), sending it by email or portal (wave 20), capturing replies (AGREED, DISAGREED with vendor balance and comment, NO_REPLY) with a reconciliation list of differences, and storing the run as audit evidence; the same framework serves AR in wave 30.
- **FR-19.18 AP correspondence types** — Should. The system shall define `CorrespondenceType` records (VENDOR_STATEMENT, PAYMENT_NOTICE, OPEN_ITEMS_LIST, BALANCE_CONFIRMATION, PAYMENT_ADVICE) each with a wave 15 template, channel (email, portal, print), trigger (on demand, on event such as `payment.posted`, or scheduled), and a correspondence log per vendor (SAP F.64 maintain/print, OB77–OB79); conditional routing per recipient is extended in wave 46.
- **FR-19.19 Noted items and guarantees** — Could. The system shall support vendor transaction types NOTED_ITEM (statistical, no ledger posting, visible in open items and proposals as reminders, e.g., expected down payment requests; SAP special G/L F) and GUARANTEE (guarantee received from vendor, posted to a statistical guarantee account pair with expiry date and alert; SAP special G/L G), both excluded from AP control reconciliation and shown separately in aging; bills of exchange and promissory notes (SAP W, AX promissory note journals) are wave 36.
- **FR-19.20 Installment payment terms** — Should. The system shall split a posted invoice's vendor transaction into installment open items when its payment terms (wave 7) define installments (AX payment schedule; SAP OBB9 T052R: installment n, percent, own terms per installment), each installment carrying its own due date, discount date and amount, with rounding residual on the last installment; settlement, proposals and aging operate on installments; the invoice header shows total and installment status.
- **FR-19.21 Vendor-initiated direct debits** — Could. The system shall record `VendorDebitMandate` (vendor, creditor identifier, mandate reference, scheme SEPA CORE/B2B or US ACH debit authorization, max amount, valid from/to, status) for vendors that collect by direct debit, a payment method of type `DEBIT_BY_VENDOR` whose proposals create Payments without a file (status SENT on the pre-notified collection date), and expected-debit lines for bank reconciliation in wave 36; a debit without an active mandate or above `max_amount` is flagged on the AP workspace.
- **FR-19.22 Vendor bank account change control** — Must. The system shall require dual approval (approver ≠ requester) for any create/change of a vendor bank account (wave 7 data) used by payment, log old/new values, notify the vendor's primary contact through a channel other than the one used for the change request, and block payment to the changed account for `bank_change_cooling_days` (default 3) after approval unless an AP manager overrides with reason (`AP-BNK-003`); the sensitive-field change report lists changes per period (SAP S_ALR_87010039 / FK08).
- **FR-19.23 Payment to one-time payees** — Should. The system shall pay invoices carrying a `one_time_payee` (wave 18 FR-18.19) as one Payment per invoice using the document-level bank account, include them in proposals only after four-eyes approval of the invoice, emit them in pain.001/NACHA with the payee name and address from the document, and print the payee on the advice and positive-pay record.

3.19.4 Data model

VendorTransaction

Field	Type	Required	Notes
id, voucher	uuid, text	yes	ledger entry voucher
vendor_id, legal_entity_id	uuid	yes	
type	enum	yes	INVOICE, CREDIT_NOTE, PAYMENT, PREPAYMENT, PREPAYMENT_INVOICE, WRITE_OFF, FX_ADJ
source_id	uuid	yes	invoice/payment id
document_date, posting_date, due_date, discount_date, discount_date_2	date	yes/no	
currency, exchange_rate	text, numeric(18,8)	yes	

Field	Type	Required	Notes
amount_txn, amount_acct	numeric(19,4)	yes	signed (invoice negative = credit to vendor)
open_amount_txn, open_amount_acct	numeric(19,4)	yes	
discount_amount, discount_amount_2	numeric(19,4)	no	
status	enum	yes	OPEN, PARTIALLY_SETTLED, SETTLED
hold_reason, hold_release_date	text, date	no	
posting_profile_id, control_account_id	uuid	yes	
payment_method_id, bank_account_id (vendor)	uuid	no	default from vendor

Settlement / SettlementLine

Field	Type	Required	Notes
id, settlement_number	uuid, text	yes	
vendor_id, settlement_date	uuid, date	yes	
kind	enum	yes	MANUAL, PAYMENT_RUN, AUTO, REVERSAL
ledger_entry_id	uuid	yes	discount/FX/write-off/withholding postings
reversed_by_id	uuid	no	
line: credit_transaction_id, debit_transaction_id, applied_txn, applied_acct_debit, applied_acct_credit, discount_txn, discount_acct, fx_diff_acct, write_off_txn, withholding_txn, closes_debit (bool), closes_credit (bool)			

PaymentProposal / Payment / PaymentFile

Field	Type	Required	Notes
proposal: id, number, criteria (jsonb), status (DRAFT, SELECTED, IN_APPROVAL, APPROVED, RUNNING, COMPLETED, CANCELLED), payment_date, bank_account_id, method_id, total_by_currency (jsonb), approved_by[]			
proposal_line: proposal_id, vendor_transaction_id, proposed_txn, discount_txn, included (bool), hold (bool), exclusion_reason			
payment: id, payment_number, proposal_id, vendor_id, vendor_bank_account_id, method_id, currency, amount_txn, amount_acct, payment_date, status (CREATED, POSTED, SENT, ACCEPTED, SETTLED_AT_BANK, REJECTED, REVERSED, VOID), end_to_end_id, check_number, settlement_id, ledger_entry_id, file_id, status_reason_code			
file: id, file_number, format (PAIN001, NACHA, CHECK_PDF, POSITIVE_PAY), bank_profile_id, generated_at, payment_count, control_sum, sha256, storage_uri, status (GENERATED, DOWNLOADED, TRANSMITTED, ACKNOWLEDGED, REJECTED)			

PaymentMethod / BankProfile: method (id, code, type TRANSFER/ACH/CHECK/CARD/WIRE, format, bank_account_id, clearing_account_id, check_lot_id, grace_days, cutoff_time); bank profile (id, bank, format variant SEPA/CH/UK/US, overrides jsonb ≤ 20 keys: charge bearer, batch booking, BIC required, structured remittance, char set, max Ustrd length).

Invariants: for every settlement line `applied_acct_credit − applied_acct_debit − fx_diff_acct = 0`; `Σ settlement lines applied on a transaction ≤ |amount|`; transaction SETTLED ⇔ open_amount = 0; `PaymentFile.control_sum = Σ payments in file`; one Payment per (proposal, vendor, currency, bank account, method). Indexes: VendorTransaction (tenant, vendor_id, status, due_date), (control_account_id, status); payments (proposal_id), (end_to_end_id) unique; settlements by transaction ids.

3.19.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	/api/v1/payables/vendors/{id}/transactions?status=open	Open items	n/a	APInquire
POST	/api/v1/payables/settlements:preview	Compute settlement journal	n/a	APSettle
POST	/api/v1/payables/settlements	Commit manual settlement	client key	APSettle
POST	/api/v1/payables/settlements/{id}:reverse	Reverse	settlement id	APSettle
POST	/api/v1/payables/payment-proposals	Create and select	client key	PaymentProposalMaintain
POST	/api/v1/payables/payment-proposals/{id}:reselect	Re-run selection with diff	proposal id + criteria hash	PaymentProposalMaintain
PATCH	/api/v1/payables/payment-proposals/{id}/lines/{lineId}	Include/exclude/amount	ETag	PaymentProposalMaintain
POST	/api/v1/payables/payment-proposals/{id}:submit	Start approval	proposal id	PaymentProposalMaintain
POST	/api/v1/payables/payment-proposals/{id}:run	Execute PaymentRun	proposal id	PaymentRunExecute
GET	/api/v1/payables/payment-files/{id}/content	Download file	n/a	PaymentFileDownload
POST	/api/v1/payables/payments/status:ingest	pain.002 / NACHA return intake	message id	service account / PaymentStatusIngest
POST	/api/v1/payables/payments/{id}:reverse	Reverse payment	payment id	APSettle
POST	/api/v1/payables/payments/{id}:void	Void check	payment id	APSettle
POST	/api/v1/payables/prepayments	Prepayment invoice/journal	client key	VendorInvoicePost
GET	/api/v1/payables/aging?asOf=&by=due	document&buckets=	Aging	n/a

Commands: `CommitSettlement` (same vendor, no held items, rates available, AP period open, Σ applied ≤ available credit), `RunPayment` (proposal APPROVED, bank account active, cutoff time, vendor bank account validated — IBAN mod-97, ABA checksum), `IngestStatus` (file matches known MsgId/OrgnlEndToEndId; unknown ids → DLQ).

Event	Trigger	Payload fields	Consumers
`dev.passage.payables.settlement.posted.v1` / `.reversed.v1`	settlement commit/reverse	settlement_id, vendor_id, lines (ids, applied, discount, fx_diff), ledger_entry_id	ledger reconciliation, analytics
`dev.passage.payables.proposal.approved.v1`	approval complete	proposal_id, totals by currency, payment_date	notifications
`dev.passage.payables.payment.posted.v1`	payment journal posted	payment_id, vendor, amount, method, bank account, settled items	cash (w36), analytics

Event	Trigger	Payload fields	Consumers
`dev.passage.payables.payment.sent.v1`	file generated & marked sent	payment_id, file_id, format, end_to_end_id	bank connectivity (w36), supplier-portal (w20)
`dev.passage.payables.payment.status.changed.v1`	pain.002/NACHA return	payment_id, old_status, new_status, reason_code	notifications, supplier-portal
`dev.passage.payables.aging.reconciled.v1`	daily job	as_of, vendor_group, open_total, control_balance, diff	alerting
`dev.passage.payables.prenote.status.changed.v1`	prenote sent / confirmed / rejected / NOC applied	vendor_id, bank_account_id, old_status, new_status, return_code	notifications, vendor master (w7)
`dev.passage.payables.vendor-bank-account.changed.v1`	bank account change approved	vendor_id, bank_account_id, changed fields (masked), approved_by, cooling_until	notifications (vendor contact), audit (w48), security monitoring
`dev.passage.payables.balance-confirmation.replied.v1`	vendor reply captured	run_id, vendor_id, our_balance, vendor_balance, status	notifications, audit (w48)

3.19.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
AP workspace	Due this week, discounts expiring, held items, proposals awaiting approval, rejected payments	counts, amounts by currency	APInquire
Vendor transactions	Filter open/all, settle (opens preview), hold/release, view voucher	document, dates, amounts, open, discount available, status	APInquire / APSettle
Settlement dialog	Mark items, set date, preview journal, commit	credit/debit items, applied, discount, FX diff, write-off, resulting journal	APSettle
Payment proposal	Criteria form, selection grid, include/exclude, reselect diff, submit, run	vendor, item, due, discount, proposed amount, hold reason	PaymentProposalMaintain
Payment run monitor	Progress, failures, retry, download files, advices	payments by status, file list, control sums	PaymentRunExecute
Payment files / bank profiles	Download, mark transmitted, configure profile overrides, test generate	format, bank, count, control sum, status	PaymentFileDownload / BankProfileMaintain
Aging	Bucket view, drill to items, export	vendor group, buckets, total, control balance, diff	APInquire

Interaction rules: settlement preview updates within 300 ms of marking an item; proposal grid supports multi-select include/exclude and keyboard navigation; amounts always shown with currency code; approvals from shell inbox; bank profile editor validates with a sample file generation and XSD check before save; personalization of buckets and columns.

3.19.7 Business rules and edge cases

- **BR-19.1** Payment state machine: `CREATED → POSTED → SENT → ACCEPTED → SETTLED_AT_BANK`; `SENT|ACCEPTED → REJECTED` (settlement reversed, items re-opened, payment stays for audit); `POSTED|SENT → REVERSED` (manual); `SENT (check) → VOID`. Proposal: `DRAFT → SELECTED → IN_APPROVAL → APPROVED → RUNNING → COMPLETED`; `IN_APPROVAL → SELECTED` on rejection; `DRAFT|SELECTED → CANCELLED`.
- **BR-19.2** Discount eligibility uses `settlement_date` (payment date), not run date; a proposal with payment_date ≤ discount_date takes the discount even if generated after it; `use_cash_discount=ALWAYS` takes it regardless of date, `NEVER` never.
- **BR-19.3** Partial payment discount: default `discount_on_partial=true` pro-rata; residual item keeps the remaining discount entitlement with the same discount date.
- **BR-19.4** Rounding: discount and FX amounts rounded half-even to the currency minor unit; the FX difference is computed as the balancing figure so the settlement journal always balances by construction; any rounding residual ≤ 0.01 in accounting currency posts to the rounding account (wave 12).
- **BR-19.5** Cross-currency: when credit and debit currencies differ, `applied` is expressed in accounting currency; the transaction-currency open amounts are reduced by `applied_acct / own_rate`; residual below `write_off_limit` closes the item.
- **BR-19.6** Settlement in a closed AP period is rejected; reversal posts in the current open period with a reversal voucher referencing the original.
- **BR-19.7** Concurrency: settlement locks VendorTransaction rows in ascending id order; a payment run and a manual settlement on the same item serialize; the loser sees `AP-SET-009 item no longer open` and the proposal line is marked excluded with reason.
- **BR-19.8** Idempotency: `PaymentRun` activities key on `(proposal_id, vendor_transaction_id)`; the file activity keys on `(proposal_id, bank_account_id, method_id, payment_date)` and is deterministic (same payments → same bytes except `CreDtTm`), so a regenerated file is byte-identical in content fields and carries the same `MsgId`.
- **BR-19.9** `EndToEndId` and `MsgId` are restricted to `[A-Za-z0-9/-?:().,'+ ]` ≤ 35 chars; vendor names are transliterated per bank profile (e.g., `ß → ss`); failure to transliterate rejects the payment line before file generation.
- **BR-19.10** Minimum payment amount (default 1.00) and negative vendor balance: a vendor whose included items net to ≤ 0 is excluded from the proposal with reason `NET_CREDIT`; credit notes are settled against invoices inside the proposal before amounts are computed.
- **BR-19.11** Prepayment applied at invoice posting cannot exceed the invoice total; the remainder stays open as a prepayment transaction.
- **BR-19.12** Rejected payment (pain.002 RJCT AC01 invalid account) re-opens items with a `hold_reason=BANK_DETAILS` on the vendor's bank account until corrected.
- **BR-19.13** Prenote status machine: `NONE → PENDING → CONFIRMED`; `PENDING → REJECTED` on return R02/R03/R04; NOC (C01–C13) updates the bank account fields and restarts at PENDING when the account number or routing changed; any change to account or routing number of a CONFIRMED account resets to NONE and requires a new prenote.
- **BR-19.14** Due-date shifting applies to the due date and to discount dates only when `shift_discount_dates=true` (default false, matching SAP behavior of not shifting discount days); a shifted due date never crosses the next due-date of another installment of the same invoice.
- **BR-19.15** Installments: amount of installment k = round(total × pct_k, minor) for k < n; the last installment absorbs the residual so Σ installments = invoice total exactly; cash discount applies per installment from its own terms.
- **BR-19.16** Bank account change cooling-off is measured in banking days of the paying bank calendar; payments already in a SENT file are unaffected; an override records approver, reason and the out-of-band confirmation reference.
- **BR-19.17** Noted items and guarantees never affect `open_amount_acct` totals reconciled to the AP control account; they cannot be settled against invoices and are closed by explicit release or expiry.

3.19.8 Acceptance criteria

- **AC-19.1** (FR-19.2) Given the 300-scenario settlement fixture, when each scenario is previewed and committed, then discount, FX gain/loss, write-off and withholding amounts and accounts equal expected values for all 300, and every settlement journal balances.

- **AC-19.2** (FR-19.2) Given the worked example in FR-19.2, when settled, then the journal lines are exactly Dr AP 1,100.00; Cr bank clearing 1,058.40; Cr discount 22.00; Cr FX gain 19.60.
- **AC-19.3** (FR-19.6) Given a proposal of 50 SEPA payments, when the run completes, then the pain.001.001.09 file validates against the XSD, `CtrlSum` equals Σ amounts, and the bank test harness returns ACCP for all 50.
- **AC-19.4** (FR-19.5) Given a run of 2,000 payments killed after 1,200 payments are posted, when the workflow resumes, then exactly 2,000 payments exist, no item is settled twice, and the file contains 2,000 transactions.
- **AC-19.5** (FR-19.11) Given open items across 3 vendor groups, when aging is run as of today, then Σ buckets per group equals the AP control account balance per posting profile and the daily job emits diff = 0.
- **AC-19.6** (FR-19.4) Given an invoice with 2 % discount expiring in 5 days and cost of capital 8 %, when a BY_DISCOUNT proposal with optimize=true runs, then the item is included with the discount (annualized 2 % × 365/20 ≈ 36.5 % > 8 %).
- **AC-19.7** (FR-19.8) Given a SENT payment, when a pain.002 with TxSts RJCT and Rsn AC01 is ingested, then the payment is REJECTED, its settlement reversed, the invoice re-opened with hold BANK_DETAILS, and `payment.status.changed` emitted.
- **AC-19.8** (FR-19.7) Given 10 US vendors paid by ACH CCD, when the NACHA file is generated, then record counts, block count (multiple of 10), entry hash and total debit/credit match the file control record and a NACHA validator passes.
- **AC-19.9** (FR-19.10) Given a PO prepayment of 300.00 USD and a later invoice of 1,000.00 USD, when the invoice posts, then 300.00 is applied, AP open = 700.00, and the prepayment account is reduced by 300.00.
- **AC-19.10** (FR-19.9) Given a posted check payment, when voided, then the check status is VOID, the positive pay file contains a void record, items are re-opened, and the payment journal is reversed.
- **AC-19.11** (FR-19.15) Given a new vendor bank account with ACH method requiring a prenote, when the prenote file is generated and no return arrives for 6 banking days, then status becomes CONFIRMED and the vendor is included in the next proposal; when an R03 return is ingested on day 2, status is REJECTED and the proposal excludes the vendor with `AP-PRN-001`.
- **AC-19.12** (FR-19.16) Given terms net 30 producing due date Saturday 2026-12-26 and rule NEXT with the TARGET2 calendar (2026-12-28 is a banking day), when the invoice posts, then the due date is 2026-12-28; a proposal with payment date 2026-12-25 is rejected with `AP-CAL-001`.
- **AC-19.13** (FR-19.20) Given an invoice of 1,000.00 USD with installment terms 3 × 33.33 %, when posted, then three open items of 333.33, 333.33 and 333.34 exist with due dates 30/60/90 days, and aging shows each in its bucket.
- **AC-19.14** (FR-19.22) Given a vendor bank account change requested by user A, when user A tries to approve it, then it is rejected; approved by user B, a payment proposal on the next banking day excludes the vendor with `AP-BNK-003` until 3 banking days pass, and the change report lists the change with old/new masked values.
- **AC-19.15** (FR-19.17) Given a balance confirmation run for 20 vendors, when 15 reply AGREED, 3 DISAGREED with balances and 2 do not reply, then the reconciliation list shows 3 differences with amounts and 2 NO_REPLY after the reply deadline.

3.19.9 Test plan and fixtures

Fixtures: `ap-settle-300` (300 scenarios authored with the accountant: full/partial, discount taken/lost/always, cross-currency, write-off, withholding, prepayment, credit note netting); `ap-pain001-bank-profiles` (SEPA, UK, CH, US-ISO) with expected XML; `ap-nacha-10`; `ap-aging-demo` (3 vendor groups, 5,000 items); `ap-pain002-samples` (ACCP, RJCT codes); `ap-crash-run` (2,000 payments with injected failure).

Test	Type	Target	Pass criterion
Settlement scenarios	unit (table-driven)	settlement engine	AC-19.1
Settlement balances	property	engine	random items/rates/dates: journal always balances; Σ applied ≤ credit
pain.001 XSD	contract	file generator	valid against `pain.001.001.09.xsd` for all profiles
Bank harness	E2E	sandbox bank	ACCP for fixture file
NACHA validator	unit	NACHA generator	AC-19.8
Run idempotency	E2E (Temporal test server with activity failure injection)	PaymentRun	AC-19.4
Aging ties	E2E	aging job	AC-19.5
Status ingest	unit	pain.002 parser	all codes map; unknown → DLQ
Proposal throughput	performance	10k items	< 5 min run, p95 settle activity < 50 ms

Reconciliation checks: Σ VendorTransaction.open_amount_acct per control account = ledger balance; Σ payments POSTED per bank clearing account = ledger clearing balance; Σ file control sums = Σ payments SENT; settlement discount Σ = discount account movement for the period.

3.19.10 Deployment and operations

Helm chart `passage-payables` extended: payment-run Temporal worker ×2, file generator worker ×2, object store bucket `passage-payment-files` (S3-compatible, server-side encryption, 7-year retention, write-once). Config: `AP_WRITE_OFF_LIMIT=1.00`, `AP_DISCOUNT_LOOKAHEAD_DAYS=7`, `AP_COST_OF_CAPITAL_PCT=8`, `AP_MIN_PAYMENT=1.00`, `AP_FILE_RETENTION_DAYS=2555`. Migrations `payables/v19.*`. Flags: `ap.nacha`, `ap.checks`, `ap.positive_pay`, `ap.auto_settle_credit_notes`, `ap.structured_remittance`. Jobs: aging reconciliation daily 03:00; discount-expiry digest daily 07:00; pain.002 poller (wave 36 replaces with bank connectivity). SLOs: settlement preview p95 < 300 ms; proposal selection of 10k items < 10 s; file generation < 30 s for 2,000 payments. Alerts: aging diff ≠ 0 (page); payment run failed activity > 3 retries (page); files generated but not downloaded/transmitted within 4 h (ticket); rejected payments > 1 % of a run (ticket). Runbook: resume stuck run (Temporal), regenerate file, reverse payment, correct vendor bank account and re-propose. Rollback: additive migrations; payment files already transmitted are not affected; disable `ap.*` format flags to stop generation.

3.19.11 Migration from AX / SAP

Source	Passage entity	Notes
AX VendTrans + VendTransOpen (open items)	VendorTransaction OPEN/PARTIALLY_SETTLED	AmountCur/AmountMST → amount_txn/acct; VendTransOpen.AmountCur → open; CashDiscDate, DueDate; vendor balance Σ must equal migrated AP control
AX VendSettlement history	not migrated	closed history in lakehouse
AX VendPaymModeTable, VendPaymModeSpec	PaymentMethod	ER format name → format enum
AX BankAccountTable (vendor bank accounts, VendBankAccount)	vendor bank accounts (wave 7 role data)	IBAN/BIC/ABA validated on load
AX CashDisc, PaymTerm	PaymentTerms discount tiers	
AX LedgerJournalTrans open payment journals	not migrated	post or delete before cutover
SAP BSIK (open vendor items)	VendorTransaction	DMBTR/WRBTR, ZFBDT+ZBD1T → due/discount dates, ZLSPR → hold; UMSKZ A → PREPAYMENT

Source	Passage entity	Notes
SAP T042Z/T042E (payment methods), FBZP house banks	PaymentMethod, bank accounts	
SAP LFBK (vendor bank details)	vendor bank accounts	
SAP REGUH/REGUP of unexecuted F110 proposals	not migrated	complete or delete before cutover
SAP T052 payment terms	PaymentTerms	

Validation: Σ open items per vendor and per control account equals the source trial balance at cutover (0.00 difference); count of open items equals source; 100 random vendors' bank details re-validated. Not migrated: cleared items (BSAK, settled VendTrans), check registers (void history), positive pay history, payment files.

3.19.12 Dependencies, risks, open questions

Depends on: wave 18 (VendorTransaction creation), wave 12 (rates, realized gain/loss accounts, rounding), wave 13 (withholding), wave 14 (AP period control), wave 7 (vendor bank accounts, terms, holds), wave 4 (Temporal, approvals), wave 36 for bank transport (forward dependency; this wave uses file download).

Risk	Likelihood	Impact	Mitigation
Bank-specific pain.001 variants exceed the 20-key profile model	Medium	Medium	Profile model reviewed against 4 banks' implementation guides; escape hatch: XSLT post-processor per profile
Accountant rejects discount-on-partial default	Low	Low	Policy flag; fixture covers both
Payment file leakage (contains bank details)	Low	High	Object store with per-tenant prefix, OPA-gated download, audit on every download, no email attachment of files

Open questions:

1. pain.001 version: .09 (2019) as default or .03 for banks still on the older schema? Default: .09 with profile override to .03. Decide by wave 19 mid-point.
2. Should the proposal auto-net credit notes across currencies? Default: same currency only. Decide by wave 19 close.
3. Card payments (virtual cards) in v1? Default: method type exists, no provider adapter until v1.x. Decide by wave 35.

3.19.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`write_off_limit` / `auto_write_off`	legal entity (per currency)	numeric / bool	1.00 / true	≥ 0	FR-19.2 step 5	Accounts payable parameters → Settlement → "Maximum penny difference" / over-under payment	OBA3 / OBXT tolerance groups for vendors and payment differences (T043G)
`discount_on_partial`	legal entity	bool	true	true/false	BR-19.3	Settlement → "Calculate cash discount for partial payments"	OBA3 cash discount adjustment for partial payments
`use_cash_discount`	vendor	enum	NORMAL	NORMAL, ALWAYS, NEVER	FR-19.3	Vendor → "Use cash discount"	n/a (always per terms; ZTERM override)
`discount_lookahead_days` / `cost_of_capital_pct` / `optimize`	legal entity	int / numeric / bool	7 / 8 / false		FR-19.4	Payment proposal → "Cash discount" selection	F110 parameter "next payment run date" logic
`min_payment` / `max_payment`	payment method	numeric	1.00 / none	≥ 0	BR-19.10	Method of payment → minimum/maximum	FBZP payment method per company code: minimum/maximum amount
`payment_methods`	legal entity	table	TRANSFER (pain.001), ACH (NACHA), CHECK, WIRE	type, format, bank account, clearing account, check lot, grace days, cutoff time	FR-19.5	Methods of payment (VendPaymModeTable), payment specifications, payment fees	T042Z payment methods per country, T042E per company code, FBZP
`bank_profiles`	bank account	record	SEPA / US / UK / CH variants	≤ 20 override keys	FR-19.6	ER format configurations per bank	DMEE tree / PMW format (OBPM1–OBPM4)
`pain001_version`	bank profile	enum	001.001.09	.03, .09	open question 1	ER format version	DMEE format version
`check_lots` / `check_layout`	bank account	table / template	none / default MICR layout	lot ranges, layout fields	FR-19.7	Check layout (per bank account), check number sequences	FCHI check lots, FBZ5 print program RFFOUS_C
`positive_pay_format`	bank account	enum	CSV	CSV, BAI-like fixed, bank XML	FR-19.7	Positive pay formats	FCHX check extract
`proposal_approval_bands`	legal entity	table	> 100,000 USD two approvers	bands	FR-19.4	Payment journal approval workflow	F110 release via BCM approval (FBPM rules)
`aging_buckets` / `aging_basis`	legal entity, user override	list / enum	0–30, 31–60, 61–90, 91–180, >180 / DUE_DATE	DUE_DATE, DOCUMENT_DATE, POSTING_DATE	FR-19.11	Aging period definitions	S_ALR_87012078 variant (days in arrears)
`prenote_required` / `prenote_wait_days`	payment method / legal entity	bool / int	true for ACH / 6	3–10 banking days	FR-19.15	Bank account → Prenote setup; Method of payment "Prenote required"	n/a
`payment_calendar_rules`	legal entity × method × bank account	table	NEXT, bank calendar from wave 6	NEXT, PREVIOUS, NEXT_UNLESS_MONTH_CHANGE; payment days	FR-19.16	Payment calendar (D365), Payment days, Bank holidays	House bank calendar, FBZP "bank calendar", T052 "next business day"

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`shift_discount_dates`	legal entity	bool	false	true/false	BR-19.14	Payment calendar "Cash discount date" shift option	n/a
`correspondence_types`	legal entity	table	VENDOR_STATEMENT, PAYMENT_NOTICE, OPEN_ITEMS_LIST, BALANCE_CONFIRMATION, PAYMENT_ADVICE	template, channel, trigger	FR-19.18	Print management for vendor documents	OB77 / OB78 / OB79 correspondence types and print programs (T048, T048B)
`balance_confirmation_reply_days`	legal entity	int	21	7–90	FR-19.17	n/a	F.18 reply deadline (letter text)
`special_gl_types`	legal entity	table	NOTED_ITEM, GUARANTEE with accounts		FR-19.19	Prepayment posting profiles	OBYR special G/L indicators (T074)
`bank_change_cooling_days` / `bank_change_dual_approval`	legal entity	int / bool	3 / true	0–15	FR-19.22	Vendor bank account change workflow (D365)	Sensitive fields for dual control (V_T055F), FK08
`auto_settle_credit_notes`	legal entity	bool	false	true/false	FR-19.14	Settlement → "Automatic settlement"	F.13 automatic clearing
`withholding_at_payment`	legal entity	bool	per wave 13 country pack		FR-19.2 step 7	Withholding tax parameters	Extended withholding at payment (T059P)

3.19.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Vendor transactions (open / all)	list inquiry	vendor, status, type, date range, currency	voucher, document, dates, amount, open, discount available, hold	`VendorTransaction`	Vendor transactions (VendTrans), Open transactions	FBL1N / FBL1H vendor line items
Vendor balance	list inquiry	vendor group, currency, as-of	balance txn/acct, last payment	`VendorTransaction` aggregate	Vendor balance list	FK10N / S_ALR_87012082 vendor balances
Vendor aging	operational report	as-of, basis, buckets, group	buckets, total, control balance, diff	FR-19.11 projection	Vendor aging report (VendAgingReport)	S_ALR_87012078 due date analysis, RFKOPR00 aging
Cash requirements / due forecast	operational report	due date range, method, bank account	amount due by day and currency, discounts available	`VendorTransaction` open	Cash flow forecast (AP part)	S_ALR_87012083 open items for printing; FF7A cash position
Settlement history	drill-down	vendor, date, kind	settlement, items, discount, FX, write-off	`Settlement`	Closed transaction editing / Settlement history	FBL1N cleared items, FB03 clearing document
Cash discount inquiry	list inquiry	date range, taken/lost	invoice, discount available, taken, lost	`SettlementLine`	Cash discount inquiry	S_ALR_87012085 payment history with discount
Payment proposal log	drill-down	proposal	included/excluded items with reasons	`PaymentProposal`	Payment proposal (VendPaymProposal)	SAPF110V proposal log, FBZ0
Payment history / payment list	list inquiry	method, status, date, bank account	payment number, vendor, amount, file, status, reason code	`Payment`	Payment history	S_P99_41000099 payment list, FBL1N payments
Payment files	list inquiry	format, bank, status	file number, count, control sum, sha256, downloaded by	`PaymentFile`	Generated payment files	FDTA DME administration
Positive pay inquiry	list inquiry	bank account, date	check number, amount, payee, void	`Payment` CHECK	Positive pay inquiry	FCHN check register
Check register	operational report	check lot, date	check number, payee, amount, status (issued/void/cleared)	`Payment` CHECK	Check journal (BankChequeTable)	FCHN, FCH6
Prenote status	list inquiry	status, vendor	bank account (masked), status, sent at, return code	vendor bank accounts	Prenote inquiry	n/a
Down payments / prepayments open at key date	list inquiry	as-of, vendor	prepayment, applied, open	`VendorTransaction` PREPAYMENT	Prepayment inquiry	S_ALR_87012105 down payments at key date
Balance confirmation reconciliation	operational report	run	vendor, our balance, vendor balance, status, difference	`BalanceConfirmation`	n/a	F.18/F.98 reply list
Vendor bank account changes (sensitive fields)	operational report (audit)	period, vendor	field, old/new (masked), requested by, approved by, cooling until	audit log	Vendor bank account change tracking (D365)	S_ALR_87010039 sensitive field changes, FK08
AP control reconciliation	operational report	entity, account, date	open items total, ledger balance, diff	aging job output	Ledger reconciliation (AP)	FAGLF03 / F.03 comparative analysis

3.19.15 Printed and electronic documents

Document	Trigger	Layout elements (header / line / footer)	Formats	Numbering sequence	Legal / retention requirement
ISO 20022 credit transfer	payment run	per FR-19.6 mapping	XML pain.001.001.09 (or .03)	file number `PF–`	Bank instruction; retain 7–10 years (write-once bucket, 2,555 days default)
NACHA file	payment run (ACH) / prenote run	records 1/5/6/7/8/9 per FR-19.7	fixed-width 94-char ASCII	file id modifier A–Z/0–9 per day	NACHA Operating Rules record retention (2 years min); retained 7 years
Check with remittance stub	payment run (CHECK)	check face: date, payee, amount in words and digits, MICR (E-13B) line with check number, routing, account; stub: invoices settled, discounts	PDF on check stock (MICR toner)	check lot number	Escheat/unclaimed property follow-up records; 7 years
Positive pay file	after check run / void	bank account, check number, amount, payee, issue date, void flag	CSV / bank format	per file	Bank agreement

Document	Trigger	Layout elements (header / line / footer)	Formats	Numbering sequence	Legal / retention requirement
Payment advice / remittance advice	payment posted / sent	header: payer, payee, payment number, date, method, amount; lines: invoice number, date, gross, discount, withholding, paid	PDF, email, portal (wave 20), EDI 820/REMADV (wave 35)	payment number	none beyond accounting retention
Vendor account statement	on demand / scheduled	header: vendor, period; lines: opening, documents, payments, closing balance	PDF, email, portal	n/a	none
Balance confirmation letter	confirmation run	header: key date, vendor, our balance; lines: open items; reply slip	PDF, email, portal	run number BC–	Audit evidence (ISA 505 external confirmations); retain with audit file 10 years
Withholding certificate	payment with withholding	payer, payee tax ids, gross, withheld, rate, period	PDF (country form in wave 48)	per country	Country tax law (e.g., IN Form 16A, MX constancia)
Prenote notification	prenote sent	vendor, account (masked), expected confirmation date	email	n/a	none

3.19.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Aging reconciliation to AP control	daily 03:00	legal entity × control account	run id (entity, date)	< 2 min per 1M open items	retry 3×	diff ≠ 0 pages AP lead
Discount-expiry digest	daily 07:00	legal entity	key = (entity, date)	< 30 s	retry	none
Scheduled payment proposal (opt-in, SAP F110S)	per schedule (e.g., Tue/Thu 06:00)	legal entity × method	key = (schedule, run date); creates proposal in SELECTED only	< 10 s per 10k items	alert treasurer	proposal not created → ticket
pain.002 / NACHA return poller	every 15 min (until wave 36)	bank account	key = message id / return trace number	< 30 s	DLQ unknown ids	DLQ depth > 0 → ticket
Prenote status update (FR-19.15)	daily 05:00	legal entity	key = (bank account, date)	< 1 min	retry	PENDING > 10 banking days → ticket
Automatic settlement of credit notes (FR-19.14)	every 30 min	vendor	key = credit note id	< 1 min	item-level error	errors > 0 → ticket
Recalculate cash discount and due dates after calendar change (FR-19.16)	on calendar change + nightly 01:30	legal entity	key = (calendar version, item)	< 2 min per 100k items	retry	changed items reported
Balance confirmation run (FR-19.17)	annually / on demand	vendor range	key = run id; letters generated once	< 5 min per 5k vendors	vendor-level errors	NO_REPLY after deadline → list to auditor
Correspondence scheduled sends (statements)	monthly day 3	vendor	key = (vendor, period, type)	< 10 min per 10k vendors	retry 3×	bounce rate > 5 % → ticket
Guarantee expiry alert (FR-19.19)	daily 06:00	legal entity	key = (guarantee, date)	< 10 s	retry	guarantees expiring < 30 days → notification
Payment file retention sweep	monthly	bucket prefix	object lock ensures no early delete	< 5 min	none	n/a

3.19.17 Security duties and privileges

Duty	Privileges (entry points: API paths / screens / jobs)	Segregation-of-duties conflicts	Default roles that include it	AX duty / SAP authorization reference
APInquire	vendor transactions, balance, aging, settlement history, payment history	none	AP clerk, Controller, Auditor	AX "Inquire into vendor transactions"; SAP F_LFA1_BUK display, FBL1N
APSettle	settlements preview/commit/reverse; payments :reverse, :void; holds	with PaymentRunExecute when four-eyes; with VendorMaintain (w7)	AP supervisor	AX "Settle vendor transactions" (VendSettle); SAP F_BKPF_KOA K clearing F-44
PaymentProposalMaintain	payment proposals create/reselect/edit lines/submit	with PaymentProposalApprove; with VendorMaintain and VendorBankAccountMaintain	AP clerk, Treasury analyst	AX "Generate payment proposals" (VendPaymProposal); SAP F_REGU_BUK activity 02/11 (F110 parameters/proposal)
PaymentProposalApprove	proposal approval tasks	with PaymentProposalMaintain, PaymentRunExecute (same proposal)	Treasurer, Controller	AX payment journal approval workflow; SAP BCM release (F_PAYRQ, FBPM approval)
PaymentRunExecute	:run; run monitor; retry	with PaymentProposalMaintain, VendorBankAccountMaintain, VendorInvoicePost (w18)	Treasury operator	AX "Generate payments"; SAP F_REGU_BUK activity 21/25 (F110 payment run, print)
PaymentFileDownload	payment-files content download, mark transmitted	with VendorBankAccountMaintain	Treasury operator	AX "Export payment files"; SAP FDTA (F_REGU_BUK 25)
PaymentStatusIngest	payments/status:ingest (service account), manual return entry	none	service account, Treasury operator	AX "Import payment return files"; SAP FEBAN/BCM status
BankProfileMaintain	bank profiles, payment methods, check lots, payment calendars	with PaymentRunExecute	Treasury administrator	AX "Maintain methods of payment"; SAP FBZP (S_TABU_DIS T042*)
VendorBankAccountMaintain	request vendor bank account create/change	with VendorBankAccountApprove (same change), PaymentRunExecute, PaymentProposalMaintain	Vendor master clerk	AX "Maintain vendor bank accounts"; SAP F_LFA1_AEN sensitive fields
VendorBankAccountApprove	approve bank account changes; cooling-off override	with VendorBankAccountMaintain (same change)	AP manager, Treasurer	SAP FK08 confirm change (F_LFA1_AEN activity 08)
CorrespondenceSend	statements, balance confirmations, payment notices; reply capture	none	AP clerk	SAP F.64 / F.18 (F_BKPF_BUK display + correspondence)
PrepaymentMaintain	prepayments, noted items, guarantees	with PaymentRunExecute	AP supervisor	AX prepayment journal; SAP F-47/F-48 (F_BKPF_BUK, special G/L)

3.19.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
AP-SET-001	error	Items belong to different vendors ({v1}, {v2})	CommitSettlement	Settle per vendor	FR-19.2
AP-SET-002	error	Applied amount {applied} exceeds available credit {available}	CommitSettlement	Reduce applied amounts	FR-19.2

Code	Severity	User text	Cause	Remediation	Raised by
AP-SET-003	error	No exchange rate {ccy}→{acct_ccy} on {date}	Missing rate	Enter the rate (wave 12)	FR-19.2
AP-SET-004	error	Settlement date {date} is in a closed AP period	BR-19.6	Use an open date	BR-19.6
AP-SET-009	warning	Item {item} is no longer open; excluded from proposal	BR-19.7 race	None (informational)	BR-19.7
AP-PRP-001	warning	Vendor {vendor} excluded: net credit balance ({amount})	BR-19.10	Review credit notes	BR-19.10
AP-PRP-002	warning	Item {item} on hold ({reason}) excluded	FR-19.13	Release hold	FR-19.13
AP-PRP-003	error	Proposal total {amount} requires approval by {n} approvers	approval bands	Submit for approval	FR-19.4
AP-RUN-001	error	Bank account {account} inactive or method cutoff {time} passed for {date}	RunPayment	Change payment date or bank account	FR-19.5
AP-RUN-002	error	Vendor bank account {account} failed {rule} validation	IBAN mod-97 / ABA	Correct bank details	FR-19.5
AP-FIL-001	error	Payment {payment}: creditor name cannot be transliterated to {charset}	BR-19.9	Edit vendor name or profile charset	BR-19.9
AP-FIL-002	error	Generated file failed XSD validation: {xpath} {reason}	Generator / profile defect	Fix bank profile override; regenerate	FR-19.6
AP-STS-001	warning	Payment {payment} rejected by bank ({code}: {text}); items re-opened	pain.002 RJCT / NACHA return	Correct bank details and re-propose	FR-19.8, BR-19.12
AP-STS-002	error	Status message references unknown MsgId/EndToEndId {id}	IngestStatus	Investigate DLQ entry	FR-19.8
AP-VOI-001	error	Check {check} cannot be voided: status {status}	FR-19.9	Reverse instead	FR-19.9
AP-PRE-001	error	Prepayment application {amount} exceeds invoice total {total}	BR-19.11	Reduce application	BR-19.11
AP-PRN-001	warning	Vendor {vendor} excluded: prenote for account {account} is {status}	FR-19.15	Wait for confirmation or change method	FR-19.15
AP-CAL-001	error	Payment date {date} is not a banking day for {bank} (calendar {calendar})	FR-19.16	Choose the next banking day {next}	FR-19.16
AP-INS-001	error	Installment percentages sum to {sum} %, expected 100 %	Terms setup (wave 7)	Correct payment terms	FR-19.20
AP-BNK-002	error	Bank account change must be approved by a different user	FR-19.22	Ask another approver	FR-19.22
AP-BNK-003	warning	Vendor {vendor} excluded: bank account changed on {date}, cooling-off until {until}	FR-19.22, BR-19.16	Wait or obtain manager override with confirmation reference	FR-19.22
AP-MND-001	warning	Direct debit by {vendor} of {amount} has no active mandate or exceeds max {max}	FR-19.21	Review with vendor; dispute debit with bank (wave 36)	FR-19.21
AP-OTV-002	error	One-time payee invoice {invoice} requires four-eyes approval before payment	FR-19.23	Complete approval	FR-19.23
AP-AGE-001	error	Aging total {aging} differs from AP control balance {ledger} by {diff}	Daily reconciliation	Investigate listed vouchers	FR-19.11

3.19.19 Performance budgets and capacity

Operation	Volume assumption (per tenant per day / peak per minute)	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Settlement preview	5,000 / 100 per min	60 ms / 300 ms / 600 ms	100 rps	none	VendorTransaction (tenant, vendor_id, status, due_date)	p95 ≤ 1 s; preview limited to 500 items per request
Settlement commit	20,000 (incl. runs) / 2,000 per min in runs	30 ms / 50 ms / 120 ms per item activity	200 items/s per worker	Settlement ≈ 500 B + lines 300 B	lock order by id (BR-19.7)	Run extends; deadline-based alert if ETA past cutoff
Proposal selection (10k open items)	20 runs	3 s / 10 s / 20 s	n/a	proposal lines ≈ 200 B	partial index on status = OPEN	Selection paginated by vendor range
Payment run (2,000 payments, 10k items)	5 runs / day	2 min / 5 min / 8 min	40 items/s end-to-end	Payment ≈ 600 B; files 1–5 MB	payments (proposal_id), (end_to_end_id) unique	Worker HPA 2→6; cutoff-time ETA alert
File generation (2,000 payments)	5 / day	8 s / 30 s / 60 s	n/a	object store 7-year retention ≈ 20 GB/year	per-tenant prefix	Streaming writer; memory bounded at 256 MiB
Status ingest (pain.002)	50 files / day	1 s / 5 s / 10 s per file	10k transactions per file	raw messages retained 7 years	(end_to_end_id) lookup	Queue; status lag ≤ 15 min
Aging (5k vendors, 1M items)	200 queries	300 ms / 2 s / 4 s	n/a	daily snapshot ≈ 50 MB	snapshot table by (as_of, vendor_group)	Served from snapshot + intraday delta; exact recompute async
Vendor balance	50k / 500 per min	10 ms / 50 ms / 100 ms	500 rps	n/a	running balance table per vendor × currency	Read replica

3.19.20 Localization and compliance hooks

Concern	Varies by	Country packs touching it (wave 48)	Extension point used
Payment formats beyond the canonical pain.001/NACHA (UK BACS Standard 18, CH pain.001 SPS, DE DTAZV, JP Zengin, MX SPEI, BR CNAB 240, AU ABA, CA CPA 005)	Local clearing systems	UK, CH, DE, JP, MX, BR, AU, CA	`PaymentFile` generator plug-in interface (format enum + bank profile); XSLT post-processor escape hatch
Payment purpose codes and regulatory reporting codes (e.g., CN, IN, AE purpose codes, CBR reporting)	Central bank rules	CN, IN, AE, RU	`Purp/Cd` and `RgltryRptg` from vendor/payment purpose code table (AX "Payment purpose codes")
Bills of exchange, promissory notes, postdated checks	IT, ES, FR, TR, JP, BR, APAC practice	IT, ES, FR, TR, JP, BR, MY, SG	Wave 36 instrument model on top of `VendorTransaction` type extension; DEFERRED (see 3.19.21)

Concern	Varies by	Country packs touching it (wave 48)	Extension point used
Withholding at payment and certificates (US 1099/backup withholding, IN TDS, MX ISR, BR IRRF)	Country tax law	US, IN, MX, BR, PH	Wave 13 withholding engine called in settlement step 7; certificate templates per pack; 1099 reporting in wave 48
Cash discount VAT adjustment (discount reduces taxable base in some countries)	VAT regimes	DE, AT, NL, BE	Settlement journal hook `tax.adjust_on_discount` invoking wave 13
Payment calendars and bank holidays	Clearing system calendars (TARGET2, Fedwire, BACS)	all	Wave 6 calendars referenced by FR-19.16
Unclaimed property / escheatment of uncashed checks	US state law	US	Check register aging report + `escheat_due_date` on CHECK payments (wave 48 report)
Strong customer authentication / payment approval evidence	PSD2 (bank side), SOX controls	EU, US	Proposal approval workflow evidence, file hash, download audit
Vendor bank detail change fraud controls	Internal control frameworks	all	FR-19.22 dual approval and cooling-off
Data protection of bank data in files and advices	GDPR, PCI (card)	EU	Masked display, OPA-gated download, encryption at rest
Language and format of advices, statements and confirmation letters	Vendor language	all	Wave 15 templates

3.19.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.2.1 — Payments → Prenotes (ACH) [#10]	PARTIAL	ADDED as FR-19.15	Zero-dollar NACHA entries with status lifecycle
AX 12.2.3 — Vendor → Update vendor bank accounts status (prenote results) [#13]	PARTIAL	ADDED as FR-19.15	Job in 3.19.16
AX 12.2.4 — Payment → Payment calendar (D365) [#15]	PARTIAL	ADDED as FR-19.16	Due-date and execution-date shifting
AX 12.3.4 — Payment → Payment calendar (AR) [#26]	PARTIAL	ADDED as FR-19.16	Same rule set reused by wave 30 for customer due dates
AX 12.5.4 — Bank account → Prenote setup [#38]	PARTIAL	ADDED as FR-19.15	`prenote_required`, `prenote_wait_days` (3.19.13)
AX 12.5.4 — Bank → Bank holidays / Payment calendar [#39]	PARTIAL	ADDED as FR-19.16	Bank calendars from wave 6
SAP 13.1 — F.97 / F.98 / F.99 balance confirmations (G/L, AP, AR) [#11]	GAP (propose 19)	ADDED as FR-19.17	Vendor side here; customer side reuses the framework in wave 30; G/L confirmations (F.97) SUPERSEDED by wave 15 audit trail exports
SAP 13.1 — OBXR / OBYR special G/L indicators (customers / vendors) [#16]	PARTIAL-19	ADDED as FR-19.19	Noted items and guarantees added; bills of exchange DEFERRED to wave 36; customer side wave 30
SAP 13.2 — F.98 vendor balance confirmation [#31]	GAP (propose 19)	ADDED as FR-19.17	
SAP 13.2 — F.64 AP correspondence maintain / print [#33]	PARTIAL-19	ADDED as FR-19.18	Correspondence types with log; conditional routing in wave 46
SAP 13.2 — OBYR special G/L indicators vendors (down payments, guarantees, BoE) [#34]	PARTIAL-19	ADDED as FR-19.19	Down payments already FR-19.10; guarantees added; BoE DEFERRED to wave 36
SAP 13.2 — Define correspondence types / forms OB77, OB78, OB79 [#36]	PARTIAL-19	ADDED as FR-19.18	`correspondence_types` (3.19.13)
SAP 13.20 — Supplier condition contracts (purchasing rebates, chargebacks) [#229]	PARTIAL-19	DEFERRED to wave 26	Rebate accrual and claim engine belongs with the wave 26 condition engine; AP settlement of the resulting vendor credit notes already uses FR-18.12 and FR-19.2
AX 12.2.1 / 12.2.2 / 12.2.3 / 12.2.4 — Promissory note journals, documents, draw/remit/settle batches, setup [#1, #3, #4, #6; register wave 36]	GAP	DEFERRED to wave 36	Register assigns treasury instruments to wave 36; noted here because the instrument settles vendor open items; `VendorTransaction.type` is extensible for it
AX 12.5.1 / 12.5.4 — Bills of exchange / promissory notes and postdated checks (bank side) [#16, #17, #18, #19; register wave 36]	GAP	DEFERRED to wave 36	Same
SAP 13.2 — F-40 / F-36 / FBW6 bill of exchange payment (AP) [#24; register wave 36]	GAP	DEFERRED to wave 36	Same
SAP 13.1 / 13.2 — OBB8 / OBB9 installment terms [#17, #35; register wave 7]	PARTIAL-7	ADDED as FR-19.20 (open-item split)	Terms definition stays in wave 7; splitting vendor open items into installments is closed here
AX 12.3.1 — Direct debit mandates (SEPA) [#18; register wave 30] / SAP 13.3 SEPA mandate management [#44; register wave 30]	PARTIAL-30	ADDED as FR-19.21 (vendor-initiated debits only)	Customer-side creditor mandates (pain.008) remain wave 30
SAP 13.2 — One-time vendor (CPD) payment [#23; register wave 7]	PARTIAL-7	ADDED as FR-19.23	Invoice capture in FR-18.19

3.20 Wave 20: Inventory Costing & Supplier Collaboration

3.20.1 Scope and business context

Wave 20 makes inventory value correct. Waves 16–18 wrote `cost_amount_physical` and `cost_amount_posted` on every inventory transaction at the running estimate (receipt price, running average on issues). This wave adds the costing engine that settles issues against receipts under the item model group's method — FIFO, LIFO, weighted average (running or periodic) or standard cost — and the inventory close/recalculation job (AX InventClosing with InventSettlement and InventCostTrans; SAP material ledger period-end closing CKMLCP with CKMLPP/CKMLCR and MBEW price control S/V, MR21 price change, MR22 debit/credit) that posts `cost_amount_adjustment` rows and GL adjustments so that the inventory value report ties to the GL inventory accounts by site, warehouse, item group and financial dimension. Standard cost captures purchase price variance (PPV) on receipt and invoice. Costing versions (AX CostingVersion/InventItemPrice; SAP CK11N/CK24 marking and release for material-level prices) hold planned and standard costs by product × site with activation and revaluation of on-hand.

The supplier portal (D365 vendor collaboration; SAP Ariba Supplier Network / SNC) at `supplier.<env>.passage.dev` lets vendor users acknowledge POs (accept/reject/propose changes), send ASNs, reply to RFQs (wave 17 hook), submit invoices against POs (wave 18 `:capture` source PORTAL), view payment status and advices (wave 19), and exchange documents.

Out of scope: BOM cost roll-up, costing sheets and production variance decomposition (wave 39), production order WIP settlement (wave 21 uses this engine for its receipts/issues), landed cost (wave 34), cost accounting ledger (wave 38), portal e-invoicing via Peppol (wave 35), supplier onboarding/qualification workflow (v1.x).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX inventory close (InventClosing) takes 4–12 h for large companies, blocks posting in the period, fails on unsettled negative issues with cryptic "cannot settle" messages, and must be cancelled and re-run end to end	Single-threaded settlement per item in X++ with helper threads; whole-period scope; no incremental	Close runs as a Temporal workflow with per-product child workflows (parallelism 32), incremental from the last closed date, resumable, with a pre-check report listing items that cannot settle and why	Close duration on 1M transactions: 4–12 h → < 20 min; re-run after failure: full → incremental from checkpoint
Inventory value does not tie to the GL: AX "Inventory value report" (InventValueReport) requires report configuration (InventValueReportSetup) and still differs from the GL because of manual journals and timing; SAP MB5L vs FAGLB03 differences investigated monthly	Value report computed from InventTrans; GL has postings from other sources; no per-dimension open-item relationship	Value report is a projection from `CostTransaction` rows with the GL voucher on each; daily reconciliation per inventory account × dimension; every difference row drills to the causing voucher	Monthly inventory-to-GL reconciliation: 1–3 days → < 1 h; unexplained differences: non-zero → 0
Weighted average in AX during the period uses running average per receipt, then close recalculates to periodic average, changing COGS retroactively with surprises at close; SAP MAP (VERPR) can go negative or spike on small stock with large invoice differences	Two-phase (running vs periodic) design; MAP recalculated on invoice with stock near zero	Method choice `WEIGHTED_AVG_RUNNING` (what you see during the period is final) or `WEIGHTED_AVG_PERIODIC` (explicit), with a preview of close adjustments by product before posting; guard that invoice differences on zero stock post to price difference, not to a residual unit	Unexpected COGS adjustments at close: typical 1–3 % of COGS → < 0.1 %; MAP spike incidents: tracked, target 0
Standard cost activation in AX requires costing version setup, pending prices, "Activate pending prices", then a revaluation posting that is hard to reconcile; SAP CK24 marking/release with CKM3 analysis is a multi-transaction month-end routine	Multi-step manual procedure	`CostingVersion` with pending/active state; `:activate` posts the revaluation of on-hand in one voucher and shows the per-site delta before activation	Steps to activate standard costs for a plant: 6–9 transactions → 2 (review, activate); activation latency: hours → < 2 min for 10k products
Vendors call or email for PO status, ASN entry is EDI-only or re-keyed by the warehouse, invoices arrive by email PDF; D365 vendor collaboration needs Azure AD B2B guest setup per vendor user; Ariba requires a separate subscription	Portal is an add-on with separate identity	Portal built on the same APIs/events with its own Keycloak realm; vendor admin invites users self-service; ASN pre-fills receiving (wave 18)	Vendor onboarding to portal: days → < 15 min; PO acknowledgement latency: 2–5 days → < 1 day median; receipts pre-filled from ASN: 0 % → > 80 % of portal vendors' receipts

3.20.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Costing method per item	InventModelGroup.InventModel (FIFO, LIFO, LIFO date, Weighted avg, Weighted avg date, Standard cost, Moving average)	MBEW VPRSV (S standard / V moving average), material ledger actual costing for periodic	`ItemModelGroup.costing_method` (FIFO, LIFO, LIFO_DATE, WEIGHTED_AVG_RUNNING, WEIGHTED_AVG_PERIODIC, STANDARD, MOVING_AVG)	full
Inventory close	InventClosing, InventSettlement, InventCostTrans, InventClosingRecalculation; Close and adjustment form	CKMLCP (material ledger cockpit: selection, determine sequence, single/multi-level price determination, post closing), CKMLPP/CKMLCR	`InventoryClose` workflow, `CostSettlement`, `CostTransaction`	full
Recalculation	Recalculation (without closing), cancel recalculation	Material ledger price determination preview; CKM3 analysis	`:recalculate` (preview/post without closing period)	full
Adjustment	Adjustment of on-hand / transactions (InventAdjOnhand, InventAdjTrans)	MR22 debit/credit material, MR21 price change	`CostAdjustment` (on-hand or transaction level)	full
Standard cost / costing versions	CostingVersion, InventItemPrice (pending/active), activation, revaluation posting	Standard price STPRS, CK11N estimate, CK24 mark/release, price change document MR21, revaluation	`CostingVersion`, `ItemCost` (PENDING/ACTIVE), `:activate` with revaluation voucher	full (roll-up in w39)
Purchase price variance	Standard cost variance posting types PurchPriceVariance at receipt/invoice	PRD price difference account (OBYC PRD) on GR/IR	PPV posting at receipt and invoice (wave 18 hooks) with `CostTransaction` kind PPV	full
Inventory value report	InventValueReport, InventValueReportSetup, Inventory aging	MB5L (stock value), MC.9 / MB52, CKM3	`GET /costing/value-report` with dimension grouping and GL tie-out column	improved
Marking	InventTrans marking (MarkingRefInventTransOrigin)	Not standard (batch-specific valuation partial)	`marked_transaction_id` on transactions (wave 16 field) honored by settlement	full
Moving average	Moving average (InventModel MovingAverage) with product cost adjustment at invoice	MAP VERPR update on GR/IR	`MOVING_AVG` with invoice-difference settlement to stock proportionally	full
Vendor collaboration	D365 Vendor collaboration workspace (PO confirmation, consignment, invoices, RFQ replies)	Ariba Supplier Network, SNC/Supplier Portal	Supplier portal (PO ack, ASN, invoice, RFQ reply, payments, documents)	full
ASN	D365 vendor collaboration ASN; AIF ASN inbound	ASN via EDI DESADV, SNC	`AdvanceShipNotice` → receiving expected lines	full
Parallel inventory valuation	D365 Global inventory accounting add-in (multiple cost ledgers, accounting conventions, accounting policies per ledger)	Material ledger with multiple currencies/valuations (group valuation, profit-center valuation, parallel currencies in CKMLCR), transfer prices	`ValuationLedger` per wave 12 ledger group with its own costing method and currency (FR-20.15)	full
Split valuation / batch valuation	Inventory by dimension cost price (site/warehouse/batch "Financial inventory" flag on dimension group)	Split valuation OMWC/OMWD (valuation categories and types, e.g., domestic/import, new/refurbished), batch valuation (valuation type = batch), T149	`valuation_type` dimension with `ValuationCategory` (FR-20.16)	full
Inventory aging	Inventory aging report (D365, InventAgingReport) with aging period definitions, inventory aging report storage	MC.A / MC50 dead stock / MC46 slow-moving, inventory turnover MC.9	Inventory aging report by receipt-date layers (FR-20.17)	full
Vendor requests and onboarding	Vendor requests (R3: VendRequest*), vendor request types, D365 prospective vendor registration wizard and questionnaires	Ariba SLP registration and qualification; MDG-S supplier requests	`VendorRequest` workflow and portal self-registration with questionnaires (FR-20.18)	full
Vendor catalog upload	Vendor collaboration → vendor catalogs (CMR upload)	Ariba catalog (CIF) upload	Portal catalog upload into wave 17 staging (FR-20.19)	full
Supplier service entry	n/a	SAP Business Network / Fieldglass supplier SES and timesheets	Portal service entry sheet and timesheet entry (FR-20.20)	full
Document distribution to suppliers	Document handling + vendor collaboration documents	DMS document distribution CVI1/CVI2/CVI8, recipient lists, distribution log	`DocumentDistribution` with recipient lists, acknowledgement and log (FR-20.21)	full
Portal consignment view	Vendor collaboration consignment inventory (on-hand, replenishment orders, product receipts)	SNC consignment monitor	Portal consignment pages (FR-20.22)	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Alternative valuation / usage variance distribution / WIP at actual / CO-PA revaluation	n/a	CKMLCPAVR, CKMDUVACT/CKMDUVMAT, WIP at actual, KE27	Not in this wave; production actual costing in waves 21/39	dropped-with-reason (3.20.21)

3.20.3 Functional requirements

- **FR-20.1 Costing methods** — Must. The system shall value issues against receipts by the item model group's method:
  - **FIFO**: issues settle against the oldest unsettled receipts by `financial_date`, then `physical_date`, then id. Example: receipts 100 @ 10.00 (Jan 5), 100 @ 12.00 (Jan 20); issue 150 on Jan 25 → cost = 100 × 10.00 + 50 × 12.00 = 1,600.00; remaining on-hand 50 @ 12.00 = 600.00.
  - **LIFO**: issues settle against the newest receipts preceding the issue date (LIFO_DATE) or the newest in the period (LIFO). Same data, LIFO_DATE: 150 issue = 100 × 12.00 + 50 × 10.00 = 1,700.00; remaining 50 @ 10.00 = 500.00.
  - **WEIGHTED_AVG_PERIODIC**: all issues in the period settle at `(opening value + Σ receipts value) / (opening qty + Σ receipts qty)`. Same data: average = 2,200 / 200 = 11.00; issue cost 1,650.00; remaining 550.00.
  - **WEIGHTED_AVG_RUNNING**: each issue uses the average at its moment (in date order): issue on Jan 25 after both receipts = 11.00 → 1,650.00; an issue of 50 on Jan 10 would be at 10.00.
  - **MOVING_AVG**: like running average, but invoice-price differences on already-issued quantity post to price difference (not to stock); differences on remaining stock adjust its value.
  - **STANDARD**: all transactions at the active `ItemCost` for product × site; receipt at PO price 10.50 against standard 10.00 for 100 units posts PPV 50.00 Dr; invoice at 10.80 posts additional PPV 30.00; inventory stays 1,000.00.
  - Settlement rows (`CostSettlement`) link issue to receipt with qty and value; `cost_amount_adjustment` on the issue = settled value – previously posted value.
- **FR-20.2 Inventory close** — Must. The system shall run `InventoryClose` for a legal entity up to a close date as a Temporal workflow: pre-check (unsettled negative on-hand, missing invoices for RECEIVED-not-PURCHASED when `close_requires_invoice=true`, open count sheets), settlement per product (child workflows, parallelism configurable, default 32), adjustment postings to the ledger (one voucher per product group × account, idempotent on close id), and lock of transactions with `financial_date ≤ close_date` against cost changes.
  - Performance: 1M transactions, 2,000 products in < 20 min on the stage sizing.
  - Cancel close reverses adjustment vouchers and unlocks; a close cannot be cancelled if a later close exists.
  - Close requires the INVENTORY period for the close date to be OPEN; after close completes it may be set CLOSED (wave 14).
- **FR-20.3 Recalculation** — Must. The system shall run the same settlement as close without locking, posting adjustments with `kind=RECALC` reversible by the next recalculation or close; preview mode returns per-product adjustment totals without posting.
- **FR-20.4 Cost transactions and value report** — Must. The system shall write a `CostTransaction` for every value-affecting event (physical posting, financial posting, invoice difference, settlement adjustment, PPV, revaluation, manual adjustment) with the ledger voucher, and serve the inventory value report (quantity and value by product, site, warehouse, item group, dimension; posted vs physical value; opening, receipts, issues, adjustments, closing) whose totals equal the ledger inventory account balances per dimension on the report date.
  - Daily reconciliation job `costing.reconcile` emits `reconciliation.diff` on any difference.
  - Value report for 10k products p95 < 5 s.
- **FR-20.5 Costing versions and standard cost activation** — Must. The system shall maintain `CostingVersion` (type STANDARD, PLANNED; blocking flags; fallback version) with `ItemCost` rows (product, site, unit cost, cost breakdown placeholder for wave 39, effective_from, status PENDING/ACTIVE/SUPERSEDED). Activation sets ACTIVE, supersedes the previous, posts revaluation of on-hand `(new − old) × physical qty` per product × site to inventory / revaluation account, and emits `cost.changed`.
  - Pending prices can be imported (CSV/API) and copied from the previous version with a percent change.
  - Activation is atomic per product and idempotent per version id.
- **FR-20.6 Cost adjustments** — Must. The system shall allow manual adjustments of on-hand value (per product × dimension: new unit cost or amount) and of specific receipt transactions (amount), posting to inventory / adjustment account and flowing to issues on next close.
- **FR-20.7 Purchase price variance** — Must. The system shall, for STANDARD items, post PPV at receipt `(PO price − standard) × qty` and at invoice `(invoice price − PO price) × qty` (wave 18 hook), to the PPV account from the posting profile; for actual-cost methods the invoice difference adjusts the receipt's cost (`cost_amount_posted`) and flows to issues through settlement.
- **FR-20.8 Marking** — Should. The system shall honor explicit marking between an issue and a receipt (set by users or by wave 21/27 processes) so settlement pairs marked transactions first regardless of method.
- **FR-20.9 Supplier portal: PO acknowledgement** — Must. The portal shall show confirmed POs and versions to the vendor's users; they shall accept, reject (reason) or propose changes (date, qty, price per line) which create a `PurchaseOrderAcknowledgement`; proposed changes become a pending change on the PO for the buyer to accept (creating a new version via wave 17) or decline.
  - SLA tracking: acknowledgement due within `ack_due_days` (default 2); overdue listed on the buyer workspace.
- **FR-20.10 Supplier portal: ASN** — Must. The portal shall create an `AdvanceShipNotice` (carrier, tracking, ship date, expected arrival, packages with SSCC optional, lines with PO line, qty, batch, serials, expiry) that pre-creates ARRIVED expected lines in receiving; receipt registration from an ASN pre-fills quantities and tracking dimensions.
- **FR-20.11 Supplier portal: invoices and payments** — Must. The portal shall let vendors submit invoices against received PO lines (source PORTAL into wave 18 `:capture`, structured, no OCR), view invoice match/approval status (without internal variance amounts), and see payment status, advices and remittance details from wave 19.
- **FR-20.12 Supplier portal: identity and authorization** — Must. The portal shall use a separate Keycloak realm `suppliers` with vendor organizations as groups, vendor-admin self-service invitation of users (email verification, MFA optional per vendor), OPA policies scoping every query to the vendor ids the user belongs to, and rate limits (60 requests/min per user, 1,000/min per vendor).

- **FR-20.13 Supplier portal: RFQ replies and documents** — Should. The portal shall accept RFQ replies (wave 17 FR-17.4) and exchange documents (certificates, contracts) with virus scanning and 25 MB size limit.
- **FR-20.14 Phase-exit walkthrough** — Must. The system shall support the recorded end-to-end demo: requisition → PO → portal acknowledgement → ASN → receipt → invoice (portal) → match → payment run → close → value report tie-out, on the stage environment with the demo plant.
- **FR-20.15 Parallel inventory valuation ledgers** — Should. The system shall allow a legal entity to define additional `ValuationLedger` rows bound to wave 12 ledger groups (e.g., IFRS primary in EUR with WEIGHTED_AVG_PERIODIC, US GAAP group ledger in USD with FIFO, tax ledger with LIFO_DATE), each with `costing_method_override` per item model group, `currency`, and `exchange_rate_type`; every value-bearing inventory event writes one `CostTransaction` per valuation ledger (column `valuation_ledger_id`, default ledger = existing behavior), close and recalculation run per valuation ledger (child workflows keyed by ledger × product), and adjustments post only to the ledgers of that ledger group via `Post(JournalEntry)` with ledger-group scope; the value report takes a `valuationLedger` parameter. Cost lookup for waves 21/27 returns the primary ledger unless asked otherwise.
- **FR-20.16 Split valuation and batch valuation** — Should. The system shall support a `valuation_type` storage/tracking dimension governed by a `ValuationCategory` on the product (e.g., ORIGIN with types DOMESTIC/IMPORT; CONDITION with NEW/REFURBISHED; BATCH where each batch is its own valuation type), so that cost layers, running/periodic averages, standard costs and the value report are kept per product × site × valuation type (SAP OMWC/OMWD split valuation, valuation type = batch); the item posting profile can resolve accounts per valuation type (valuation class equivalent); a transfer between valuation types is a TRANSFER journal posting the value difference to revaluation; the category is set at product creation and cannot change while stock exists (`CST-VAL-002`).
- **FR-20.17 Inventory aging report** — Must. The system shall produce an inventory aging report as of a date by product, site, warehouse, item group and valuation type with configurable buckets (default 0–30, 31–90, 91–180, 181–365, > 365 days) based on the receipt date of the remaining cost layers (FIFO layers after settlement; for average methods, oldest-unconsumed approximation using FIFO layer order), showing quantity and value per bucket and totals that tie to the value report; report setups (bucket sets, grouping) are saved per user or legal entity (AX Inventory aging report setup) and the report can be stored as a snapshot for month-end evidence.
- **FR-20.18 Vendor requests and prospective vendor onboarding** — Should. The system shall support `VendorRequest` (types ADD_VENDOR, ADD_TO_CATEGORY, CHANGE_STATUS, per AX vendor request types) raised internally or via a public portal registration page (D365 prospective vendor registration): the prospect enters company data, contacts, bank details (not activated), tax ids and answers to the `OnboardingQuestionnaire` assigned by category or request type (file uploads for certificates); the request runs a wave 4 approval workflow (category manager, compliance, AP master data) with duplicate check against wave 7 parties (tax id, name similarity ≥ 0.9), sanctions screening hook, and on approval creates the wave 7 vendor role (status ON_HOLD until bank account approval per FR-19.22) and invites the contact as VENDOR_ADMIN; rejected requests keep data 90 days then purge.
- **FR-20.19 Vendor catalog upload via the portal** — Should. The portal shall let vendor users upload catalog files (CSV/JSON per FR-17.18, max 50,000 rows, images ≤ 2 MB each, 25 MB total) into the wave 17 staging area for a catalog the buyer has opened to that vendor, show row-level validation results, and notify the buyer for review; vendors see only their own staged and published rows.
- **FR-20.20 Supplier service entry and timesheets** — Should. The portal shall let vendor users create service entry sheets (wave 18 FR-18.14) against their open SERVICE PO lines, including timesheet-style entry (worker name or contractor id, date, hours, service item) aggregated into sheet lines, submit them for acceptance, and view acceptance status and rejection reasons; internal limits and expected values are not displayed, only remaining limit percentage when `portal_show_limit=true`.
- **FR-20.21 Controlled document distribution to suppliers** — Could. The system shall provide `DocumentDistribution` (SAP CVI1/CVI2): a document (drawing, specification, quality agreement, policy) from the wave 5 attachment store is distributed to a `RecipientList` (vendors, vendor users, categories) via the portal, each recipient must acknowledge (read and accept) the specific version by a due date, reminders follow wave 4 timers, and the distribution log (SAP CVI8) shows sent/viewed/acknowledged per recipient; a new document version supersedes open distributions and re-requests acknowledgement.
- **FR-20.22 Portal consignment view** — Should. The portal shall show vendor users their vendor-owned stock (wave 16 FR-16.17) by warehouse and product with age, open consignment replenishment orders, receipts against them, ownership changes (consumption) and consignment settlement invoices (wave 18 FR-18.16), with export.

3.20.4 Data model

CostTransaction

Field	Type	Required	Notes
id	uuid	yes	
inventory_transaction_id	uuid	yes	FK
product_id, site_id, dimension_set_id	uuid/bytea	yes	denormalized for the report
kind	enum	yes	PHYSICAL, FINANCIAL, INVOICE_DIFF, SETTLEMENT_ADJ, PPV, REVALUATION, MANUAL_ADJ, RECALC
qty	numeric(28,6)	yes	0 for value-only
amount_acct	numeric(19,4)	yes	signed
close_id / recalc_id	uuid	no	
ledger_entry_id	uuid	yes	voucher
financial_date	date	yes	
created_at	timestamptz	yes	append-only

CostSettlement

Field	Type	Required	Notes
id	uuid	yes	
close_id	uuid	yes	or recalc id
issue_transaction_id, receipt_transaction_id	uuid	yes	
qty_settled	numeric(28,6)	yes	> 0
value_settled_acct	numeric(19,4)	yes	
method	enum	yes	method used

Field	Type	Required	Notes
sequence	int	yes	order of settlement for determinism
cancelled (bool), cancelled_by_close_id		no	

InventoryClose

Field	Type	Required	Notes
id, close_number	uuid, text	yes	
legal_entity_id, close_date	uuid, date	yes	
kind	enum	yes	CLOSE, RECALC, PREVIEW
status	enum	yes	PRECHECK, PRECHECK_FAILED, SETTLING, POSTING, COMPLETED, CANCELLING, CANCELLED, FAILED
parallelism	int	yes	default 32
products_total, products_done, settlements_count, adjustment_total_acct	int/numeric	yes	progress
started_at, completed_at, duration_s	timestamptz/int		
temporal_workflow_id	text	yes	

CostingVersion / ItemCost: version (id, code, type STANDARD/PLANNED, status DRAFT/ACTIVE/ARCHIVED, fallback_version_id, block_activation (bool), site_scope[]); item cost (version_id, product_id, site_id, unit, unit_cost_acct, effective_from, status PENDING/ACTIVE/SUPERSEDED, activated_at, revaluation_ledger_entry_id, source MANUAL/IMPORT/COPY/ROLLUP (w39)).

PurchaseOrderAcknowledgement / AdvanceShipNotice / SupplierUser: ack (id, po_id, version_no, vendor_user_id, response ACCEPT/REJECT/PROPOSE, reason, proposed_lines jsonb, status PENDING_BUYER/ACCEPTED/DECLINED, created_at); asn (id, asn_number, vendor_id, po_id, carrier, tracking_number, ship_date, expected_arrival, packages jsonb (sscc, weight), lines (po_line_id, qty, batch, serials[], expiry), status SUBMITTED/ARRIVED/RECEIVED/CANCELLED); supplier user (keycloak_subject, vendor_ids[], role VENDOR_ADMIN/VENDOR_USER, invited_by, mfa_enabled, last_login).

Invariants: for each issue transaction Σ qty_settled over non-cancelled settlements ≤ |qty|; after a COMPLETED close every issue with `financial_date ≤ close_date` is fully settled or listed in the close exception report; Σ CostTransaction.amount_acct per inventory account = ledger balance; exactly one ACTIVE ItemCost per (STANDARD version chain, product, site); a vendor user can read only rows where vendor_id ∈ their vendor_ids (OPA + RLS policy on the `supplier` database). Indexes: CostTransaction (tenant, product_id, financial_date), (ledger_entry_id); CostSettlement (issue_transaction_id), (receipt_transaction_id); ItemCost (product_id, site_id, status).

3.20.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/costing/closes	Start close (kind CLOSE/RECALC/PREVIEW)	entity+close_date+kind	InventoryCloseRun
GET	/api/v1/costing/closes/{id}	Progress and exceptions	n/a	CostingInquire
POST	/api/v1/costing/closes/{id}:cancel	Cancel/reverse	close id	InventoryCloseRun
GET	/api/v1/costing/value-report?asOf=&groupBy=&site=	Inventory value with GL tie-out	n/a	CostingInquire
POST	/api/v1/costing/adjustments	On-hand or transaction adjustment	client key	CostAdjust
POST	/api/v1/costing/versions	Create costing version	client key	CostingVersionMaintain
POST	/api/v1/costing/versions/{id}/item-costs:import	Import pending costs (CSV/JSON)	file hash	CostingVersionMaintain
POST	/api/v1/costing/versions/{id}:activate	Activate pending costs with revaluation	version id	CostingVersionActivate
GET	/api/v1/costing/products/{id}/cost?site=&date=	Current cost (used by waves 21, 27)	n/a	CostingInquire
POST	/internal/v1/costing/ppv	Hook from receiving/payables for STANDARD items	source doc id	service account
GET	/portal/v1/purchase-orders	Vendor's POs	n/a	portal: VendorUser
POST	/portal/v1/purchase-orders/{id}:acknowledge	Accept/reject/propose	po+version	portal: VendorUser
POST	/portal/v1/asns	Submit ASN	client key	portal: VendorUser
POST	/portal/v1/invoices	Submit invoice (→ payables capture)	vendor invoice number	portal: VendorUser
GET	/portal/v1/payments	Payment status and advices	n/a	portal: VendorUser
POST	/portal/v1/users:invite	Vendor admin invites user	email	portal: VendorAdmin

Commands: `StartClose` (period open, no running close for entity, close_date ≥ last close), `ActivateVersion` (no block flag, all pending costs > 0 unless `allow_zero_cost`, site in scope), `Acknowledge` (latest version only; proposals require at least one changed line), `SubmitASN` (PO confirmed, qty ≤ remaining × (1 + over_delivery_pct), serial count = qty for serial items).

Event	Trigger	Payload fields	Consumers
`dev.passage.costing.close.completed.v1` / `.cancelled.v1` / `.failed.v1`	close lifecycle	close_id, entity, close_date, kind, products, settlements, adjustment_total, duration_s, exceptions_count	ledger reconciliation, notifications, analytics
`dev.passage.costing.cost.changed.v1`	activation, manual adjustment affecting unit cost	product_id, site_id, version, old_cost, new_cost, effective_from, revaluation_amount	inventory (cost price cache), planning (w22), product costing (w39)
`dev.passage.costing.reconciliation.diff.v1`	daily tie-out diff ≠ 0	account, dimension, inventory_value, ledger_balance, diff, sample vouchers	alerting
`dev.passage.supplier.po.acknowledged.v1`	portal ack	po_id, version, response, proposed changes	procurement (pending change), notifications
`dev.passage.supplier.asn.submitted.v1`	ASN submit	asn_id, po_id, lines, expected_arrival, tracking	receiving (expected lines), warehouse (w31), notifications
`dev.passage.supplier.invoice.submitted.v1`	portal invoice	invoice_id (payables), vendor_id, po_id, total	payables (already captured), notifications
`dev.passage.supplier.vendor-request.submitted.v1` / `.approved.v1` / `.rejected.v1`	vendor request lifecycle	request_id, type, prospect (name, country, tax id masked), category, questionnaire score	party master (w7 vendor creation), notifications, compliance screening

Event	Trigger	Payload fields	Consumers
`dev.passage.supplier.document.acknowledged.v1`	recipient acknowledges distributed document	distribution_id, document_id, version, vendor_id, user_id, acknowledged_at	quality (w33), notifications
`dev.passage.costing.aging.snapshot.created.v1`	aging snapshot stored	snapshot_id, entity, as_of, valuation_ledger, totals per bucket	analytics, close checklist (w14)

3.20.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Costing workspace	Last close status, reconciliation diffs, pending costing versions, products with negative value or unsettled issues	counts, amounts, durations	CostingInquire
Inventory close	Pre-check, start, monitor (progress bar per product), exceptions list, cancel	close date, kind, parallelism, exceptions (product, reason, qty)	InventoryCloseRun
Inventory value report	Group by, as-of date, GL tie-out column, drill to cost transactions and vouchers, export	product, site, qty, physical value, financial value, GL balance, diff	CostingInquire
Costing version / item costs	Create, import, copy with %, compare to active, activate with revaluation preview	product, site, pending cost, active cost, delta, on-hand qty, revaluation amount	CostingVersionMaintain
Cost adjustment	Adjust on-hand or receipt, preview voucher, post	product, dims, qty, current value, new unit cost, delta	CostAdjust
Supplier portal: home	Open POs needing ack, ASNs, invoices in progress, payments	counts, due dates	portal VendorUser
Supplier portal: PO	View versions, accept/reject/propose per line, print	lines, dates, prices, version diff	portal VendorUser
Supplier portal: ASN	Create from PO lines, add packages, batches/serials, submit	PO lines, qty, batch, expiry, serials, tracking	portal VendorUser
Supplier portal: invoices / payments	Submit invoice from receipts, status timeline, payment advices	invoice no, amount, status, payment date, reference	portal VendorUser
Supplier portal: users	Invite, deactivate, MFA	email, role, status	portal VendorAdmin

Interaction rules: the close screen refreshes progress via server-sent events every 2 s; value report defaults to the last close date; activation shows a mandatory revaluation preview with total and top-20 deltas; portal is WCAG 2.1 AA, mobile-friendly, English + German + French + Spanish strings in v1, no internal data (variance amounts, standard costs, approvals) exposed; portal file uploads scanned before storage.

3.20.7 Business rules and edge cases

- **BR-20.1** Close state machine: `PRECHECK → SETTLING → POSTING → COMPLETED`; `PRECHECK → PRECHECK_FAILED`; `SETTLING|POSTING → FAILED` (resumable); `COMPLETED → CANCELLING → CANCELLED` only when it is the latest close for the entity.
- **BR-20.2** Settlement determinism: receipts are ordered by (financial_date, physical_date, id) and issues by (financial_date, physical_date, id); the same inputs always produce the same `CostSettlement` sequence (property-tested), so re-runs after crashes are idempotent.
- **BR-20.3** Unsettled issues (negative on-hand at settlement time) are valued at the running average of the product × site (fallback: active standard, then last receipt cost) and listed as exceptions; a later receipt settles them at the next close with an adjustment.
- **BR-20.4** Rounding: settlement values are computed in full precision and rounded half-even to the minor unit per settlement row; the last settlement row of a receipt absorbs the rounding residual so Σ settled value = receipt value exactly.
- **BR-20.5** Periodic weighted average with transfers between sites: the average is per product × site (`costing_level=SITE`, default) or per product (`costing_level=ENTITY`) as configured on the item model group; transfers at site level settle like issues/receipts.
- **BR-20.6** Invoice difference after close: an invoice posted after the receipt's period was closed posts the difference to the invoice's period as `INVOICE_DIFF` and flows to issues in the next close (never reopens a closed period).
- **BR-20.7** Standard cost revaluation uses the physical on-hand quantity at activation time; transactions already financially posted are not restated; activation during an in-flight close is rejected (`CST-ACT-004`).
- **BR-20.8** Close date must be ≤ today and ≥ the last completed close; transactions with `financial_date ≤ close_date` are locked against new cost adjustments except through `:cancel`.
- **BR-20.9** Portal acknowledgement applies to the latest PO version only; a new version invalidates pending acknowledgements (status SUPERSEDED) and re-requests one.
- **BR-20.10** ASN quantity may exceed PO remaining only within the over-delivery tolerance (wave 18); duplicate tracking number per vendor is rejected; an ASN can be cancelled until any of its lines is RECEIVED.
- **BR-20.11** Portal invoice submission is limited to lines with received, uninvoiced quantity; the vendor cannot submit a price different from the PO price unless the buyer enabled `portal_price_override` for the vendor (creating a price-variance exception in wave 18).
- **BR-20.12** Concurrency: close child workflows take an advisory lock per product; wave 16 journal posting for a product whose close child is settling waits up to 5 s then fails with `CST-LOCK-001` (retry); costs read by waves 21/27 are the last committed values.
- **BR-20.13** Valuation ledgers share quantities and settlement order inputs but not values: each ledger's settlement is computed independently from the same transaction sequence using its own method and currency; the primary ledger's results drive `cost_amount_adjustment` on the inventory transaction, the others are stored only in `CostTransaction`; a close is COMPLETED for the entity only when all valuation ledgers completed.
- **BR-20.14** Split valuation: issues must name a valuation type when more than one has stock (`CST-VAL-001`) unless the product's default valuation type is set; batch valuation creates the valuation type automatically at batch receipt with the receipt cost.
- **BR-20.15** Aging buckets use calendar days between layer receipt `financial_date` (physical date when not yet invoiced) and the as-of date; layers from migration opening balances carry the source receipt date when available, else the cutover date (flagged).
- **BR-20.16** A vendor request cannot create a vendor when a duplicate party match ≥ 0.9 exists unless a master data steward links or overrides with reason; bank details captured in onboarding are always subject to FR-19.22 dual approval.
- **BR-20.17** Portal-entered service entry sheets enter status SUBMITTED directly; vendors may withdraw only while not ACCEPTED; timesheet hours per worker per day are capped at 24 and per sheet at the period length.

3.20.8 Acceptance criteria

- **AC-20.1** (FR-20.1) Given the 100 receipt/issue sequences per method fixture, when settled by close, then every issue cost, remaining on-hand value and adjustment equals the reference workbook to the cent for FIFO, LIFO, LIFO_DATE, WEIGHTED_AVG_PERIODIC, WEIGHTED_AVG_RUNNING, MOVING_AVG and STANDARD.
- **AC-20.2** (FR-20.1) Given the FR-20.1 example data, when closed under FIFO, LIFO_DATE and WEIGHTED_AVG_PERIODIC, then issue cost is 1,600.00, 1,700.00 and 1,650.00 and remaining value 600.00, 500.00 and 550.00.

- **AC-20.3** (FR-20.2) Given the 1M-transaction fixture, when a close runs on stage, then it completes in < 20 min, and the value report at close date equals the GL inventory balances per site and item group with diff 0.00.
- **AC-20.4** (FR-20.7) Given a STANDARD item at 10.00 with a PO at 10.50 and invoice at 10.80 for 100 units, when received and invoiced, then PPV postings are 50.00 and 30.00 Dr, inventory value is 1,000.00, GR/IR nets to zero.
- **AC-20.5** (FR-20.5) Given 10,000 pending item costs, when the version is activated, then all are ACTIVE within 2 min, one revaluation voucher per site balances to `Σ (new − old) × on-hand`, and `cost.changed` is emitted per product.
- **AC-20.6** (FR-20.2) Given a close killed during SETTLING at 40 %, when resumed, then it completes with the same settlement rows as an uninterrupted run (byte-equal export).
- **AC-20.7** (FR-20.9) Given a confirmed PO visible in the portal, when the vendor proposes a delivery date change, then a pending change appears on the buyer's PO and accepting it creates version 2 and emits `po.version.created`.
- **AC-20.8** (FR-20.10) Given an ASN for 2 batch-tracked lines, when the goods arrive and the receiver opens the receipt from the ASN, then quantities, batches and expiry dates are pre-filled and registration takes one confirmation.
- **AC-20.9** (FR-20.12) Given a user of vendor A, when they request vendor B's PO by id, then the portal returns 404 and the OPA decision log records the denial.
- **AC-20.10** (FR-20.14) Given the demo plant, when the phase-exit walkthrough is executed, then every step completes on stage and the final value report, GR/IR report and AP aging each reconcile to the GL with 0.00 difference.
- **AC-20.11** (FR-20.15) Given the FR-20.1 example data, a primary ledger (EUR, WEIGHTED_AVG_PERIODIC) and a group ledger (USD at 1.10, FIFO), when the close runs, then the primary issue cost is 1,650.00 EUR and the group ledger issue cost is 1,760.00 USD (1,600.00 × 1.10), each ledger's value report ties to its own ledger balances, and the group ledger adjustment voucher posts only to the group ledger.
- **AC-20.12** (FR-20.16) Given a product with valuation category ORIGIN and receipts 100 DOMESTIC @ 10.00 and 100 IMPORT @ 14.00 under WEIGHTED_AVG_RUNNING, when 50 IMPORT are issued, then issue cost is 700.00 (not 600.00) and the value report shows DOMESTIC 1,000.00 and IMPORT 700.00.
- **AC-20.13** (FR-20.17) Given FIFO layers received 400, 100 and 20 days before the as-of date with remaining 10, 50 and 100 units at 5.00, when the aging report runs with default buckets, then buckets 0–30 = 500.00, 91–180 = 250.00, 181–365 = 50.00 and the total equals the value report for the product.
- **AC-20.14** (FR-20.18) Given a prospect registering with a tax id equal to an existing vendor's, when the request is submitted, then it is flagged as duplicate and cannot create a vendor until a steward links it; a non-duplicate approved request creates a vendor ON_HOLD and emails a VENDOR_ADMIN invitation.
- **AC-20.15** (FR-20.21) Given a specification distributed to 10 vendors with due date 5 days, when 7 acknowledge and a new version is released on day 3, then the 7 acknowledgements are marked SUPERSEDED, all 10 receive the new version, and the log shows per-recipient status for both versions.

3.20.9 Test plan and fixtures

Fixtures: `cost-methods-700` (100 sequences × 7 methods with reference workbook `cost_reference.xlsx` maintained by the accountant; includes negative on-hand, transfers, invoice differences, returns); `cost-1m` (reuses `inv-1m-txn` with receipts/invoices from waves 17–18 generators); `cost-std-activation-10k`; `portal-demo-vendor` (vendor with 3 users, 20 POs, 5 ASNs); `cost-ax-extract` (InventClosing, InventSettlement, InventItemPrice from USMF); `cost-sap-extract` (MBEW, CKMLHD/CKMLPP/CKMLCR samples).

Test	Type	Target	Pass criterion
Method correctness	unit (table-driven)	settlement engine	AC-20.1
Determinism/idempotency	property	engine	identical inputs → identical settlements; resume = uninterrupted
Close performance	performance (stage)	InventoryClose	< 20 min on cost-1m
Value report tie-out	E2E	value-report vs ledger	diff 0.00 after close
PPV	contract	receiving/payables → costing/ppv	AC-20.4
Activation	E2E	versions:activate	AC-20.5
Portal authz	security (OPA unit + E2E)	portal APIs	AC-20.9; no cross-vendor leakage across 1,000 random requests
Portal accessibility	E2E (axe)	portal screens	0 serious/critical violations
Phase-exit walkthrough	E2E (Playwright, recorded)	full flow	AC-20.10

Reconciliation checks: Σ CostTransaction per inventory account = ledger balance; Σ settlement value per receipt = receipt value; Σ PPV cost transactions = PPV account movement; revaluation voucher = Σ item deltas × on-hand; portal invoice totals = payables captured totals.

3.20.10 Deployment and operations

Helm charts: `passage-costing` (API ×2, Temporal close workers ×4 with CPU 2 each, schema `costing` in `inventory-db`), `passage-supplier-portal` (React app served by the shell at `supplier.<env>.passage.dev`, BFF ×2 in TypeScript, CloudNativePG `supplier-db`, Keycloak realm `suppliers` provisioned by Helm hook, Envoy Gateway route with rate limiting and WAF rules). Config: `CST_CLOSE_PARALLELISM=32`, `CST_LOCK_WAIT_S=5`, `CST_LEVEL_DEFAULT=SITE`, `CST_ALLOW_ZERO_COST=false`, `PORTAL_RATE_LIMIT_USER=60`, `PORTAL_UPLOAD_MAX_MB=25`, `PORTAL_ACK_DUE_DAYS=2`. Migrations `inventory/v20.*` (costing schema), `supplier/v20.*`. Flags: `cst.moving_avg`, `cst.marking`, `portal.rfq_replies`, `portal.invoice_submit`, `portal.mfa_required`. Jobs: reconciliation daily 02:15; ack-overdue scan daily 08:00; portal inactive-user cleanup weekly. SLOs: value report p95 < 5 s; cost lookup p95 < 20 ms; close < 20 min per 1M txns; portal page load p95 < 1.5 s; portal availability 99.9 %. Alerts: reconciliation diff ≠ 0 (page); close FAILED (page); close running > 60 min (ticket); portal auth failures > 100/min from one IP (security ticket, auto-block). Runbook: resume close, cancel close, re-run reconciliation, rotate portal realm client secret, revoke vendor user. Rollback: additive migrations; costing schema can be disabled (flag `cst.enabled=false` reverts to wave 16 running estimate without data loss); portal can be taken down independently (DNS) without affecting internal flows.

3.20.11 Migration from AX / SAP

Source	Passage entity	Notes
AX InventModelGroup.InventModel	ItemModelGroup.costing_method	FIFO→FIFO, LIFO→LIFO, LIFODate→LIFO_DATE, WeightedAvg→WEIGHTED_AVG_PERIODIC, WeightedAvgDate→WEIGHTED_AVG_RUNNING, StdCost→STANDARD, MovingAverage→MOVING_AVG
AX InventItemPrice (active, PriceType Cost) per CostingVersion	ItemCost ACTIVE in version `MIG-STD`	per site; pending prices migrated as PENDING
AX InventSum PostedValue / wave 16 opening	opening cost layers: one receipt layer per product × dimension at average cost	FIFO/LIFO layer detail from last InventClosing unsettled receipts (InventTrans with open qty) when available, else collapsed to one layer
AX InventClosing history	not migrated	reporting only

Source	Passage entity	Notes
SAP MBEW (VPRSV, STPRS, VERPR, LBKUM, SALK3, PEINH)	costing_method (S→STANDARD, V→WEIGHTED_AVG_RUNNING or MOVING_AVG per decision), ItemCost (STPRS/PEINH), opening layer value SALK3	per plant (BWKEY) → site
SAP CKMLPP/CKMLCR (material ledger period values)	opening layer value when ML active	actual price of last closed period
SAP MLCD / CKMLHD history	not migrated	
D365 vendor collaboration users / Ariba supplier accounts	SupplierUser invitations (not credentials)	vendors re-register via invitation email

Validation: Σ opening layer value per product × site equals source value (0.01 tolerance per product, 0.00 total per account); standard costs for 200 random products equal source; costing method distribution equals source counts. Not migrated: settled cost history, SAP ML price analysis, AX recalculation logs, portal documents.

3.20.12 Dependencies, risks, open questions

Depends on: wave 16 (transactions, cost fields, marking field), wave 17 (PO prices, versions, pending changes), wave 18 (receipts, invoice differences, capture), wave 19 (payment status), wave 11–14 (posting, period control, vouchers), wave 4 (Temporal), wave 2–3 (Keycloak realms, OPA, metadata), wave 7 (vendor contacts).

Risk	Likelihood	Impact	Mitigation
Opening cost layers from AX/SAP are collapsed, so first FIFO/LIFO close differs from legacy	High	Medium	Document in cutover; parallel-run compare in wave 49; collapsed layer flagged in value report
Close parallelism contends with daytime posting	Medium	Low	Run off-hours by default; lock wait 5 s with retry; advisory locks per product only
Portal security incident	Low	High	Separate realm/DB, OPA deny-by-default, pen-test before phase exit, WAF, rate limits, audit
Accountant requires LIFO by period layers (US tax) beyond LIFO_DATE	Medium	Low	Periodic LIFO pools deferred; documented as v1.x

Open questions:

1. SAP MAP (V) maps to WEIGHTED_AVG_RUNNING or MOVING_AVG by default? Default: MOVING_AVG (closest to VERPR behavior on invoice differences). Decide by wave 20 mid-point.
2. Should the portal expose expected payment dates (from proposals) or only executed payments? Default: executed only. Decide by wave 20 close.
3. Costing level ENTITY vs SITE default for weighted average? Default: SITE. Decide by wave 20 mid-point with the accountant.
4. Dollar-value LIFO pools for US tax reporting: v1.x or wave 39? Default: v1.x. Decide by wave 39 planning.

3.20.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`costing_method`	item model group	enum	WEIGHTED_AVG_RUNNING	FIFO, LIFO, LIFO_DATE, WEIGHTED_AVG_RUNNING, WEIGHTED_AVG_PERIODIC, STANDARD, MOVING_AVG	FR-20.1	InventModelGroup.InventModel	MBEW-VPRSV price control S/V (OMW0/OMW1)
`include_physical_value`	item model group	bool	true	true/false	physically posted (un-invoiced) receipts participate in running average	Item model group → "Include physical value"	n/a (MAP updated at GR)
`costing_level`	item model group	enum	SITE	SITE, ENTITY	BR-20.5	Dimension-based cost price (financial inventory flag on site/warehouse)	Valuation level plant vs company code (OX14)
`close_parallelism`	legal entity	int	32	1–128	FR-20.2	Inventory close helper threads	CKMLCP parallel processing settings
`close_requires_invoice`	legal entity	bool	false	true/false	FR-20.2 pre-check	n/a	CKMLCP "materials not invoiced" check
`lock_wait_s`	tenant	int	5	1–30	BR-20.12	n/a	n/a
`unsettled_issue_fallback`	legal entity	enum	RUNNING_AVG	RUNNING_AVG, STANDARD, LAST_RECEIPT	BR-20.3	Inventory parameters → "Use latest cost price" / break-even	ML price determination fallback (CKMLCP single-level)
`costing_versions`	legal entity	table	`STD-{year}`	type STANDARD/PLANNED, block activation, fallback version, site scope	FR-20.5	Costing versions	CK11N costing variant, CK24 marking allowance (CK24 "allow marking")
`allow_zero_cost`	legal entity	bool	false	true/false	FR-20.5 activation check	n/a	n/a
`posting_profile.revaluation_account` / `ppv_account` / `cost_adjustment_account`	posting profile	account	seeded	ledger accounts	FR-20.5–20.7	Posting profiles: Standard cost revaluation, Price variance, Inventory adjustment	OBYC UMB (revaluation), PRD (price difference), BSX

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`valuation_ledgers`	legal entity	table	primary only	ledger group (w12), currency, rate type, method override per item model group	FR-20.15	Global inventory accounting: ledgers, accounting policies, conventions	ML currency types (OMX2/OMX3), group/profit-center valuation (8KEM)
`valuation_categories`	tenant	table	none	category code, types, account mapping per type, BATCH flag	FR-20.16	Financial inventory dimension (batch/warehouse)	OMWC split valuation (T149C/T149D), OMWD valuation grouping code
`aging_bucket_sets`	legal entity, user	table	0–30, 31–90, 91–180, 181–365, > 365	bucket boundaries	FR-20.17	Inventory aging report setup	MC50 / MC46 day thresholds
`value_report_setup`	legal entity	table	default columns	columns, dimensions, resource type, GL tie-out	FR-20.4	Inventory value report configuration	MB5L variant
`portal_rate_limit_user` / `portal_rate_limit_vendor`	tenant	int	60 / 1,000 per min		FR-20.12	n/a	n/a
`portal_mfa_required`	vendor	bool	false	true/false	FR-20.12	Vendor collaboration security (Azure AD B2B)	Ariba MFA setting
`ack_due_days`	legal entity, vendor	int	2	0–30	FR-20.9	Vendor collaboration → confirmation response time	Confirmation control keys (T163D)
`portal_price_override`	vendor	bool	false	true/false	BR-20.11	Vendor collaboration "Allow price change on invoice"	n/a
`portal_upload_max_mb` / `portal_allowed_file_types`	tenant	int / list	25 / pdf, png, jpg, csv, xlsx, xml		FR-20.13	Document management file types	DMS document type file extensions
`vendor_request_types` / `onboarding_questionnaires`	legal entity	table	ADD_VENDOR with default questionnaire	type, workflow id, questionnaire per category	FR-20.18	Vendor request types; Vendor onboarding questionnaires; Registration wizard configuration	Ariba SLP registration questionnaires
`prospect_data_retention_days`	tenant	int	90	30–365	FR-20.18 purge	n/a	n/a
`portal_show_limit`	vendor	bool	false	true/false	FR-20.20	n/a	n/a
`document_recipient_lists` / `distribution_types`	legal entity	table	none	recipients, acknowledgement required, due days	FR-20.21	n/a	DMS distribution types and recipient lists (CVI customizing)

3.20.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Inventory value report	statutory / operational report	as-of, group by, site, valuation ledger, valuation type, segment	qty, physical value, financial value, GL balance, diff	`CostTransaction`	Inventory value report (InventValueReport)	MB5L, MB5W, MC.9
Inventory close list and exceptions	list inquiry	entity, close date, kind	status, duration, products, settlements, exceptions	`InventoryClose`	Closing and adjustment list	CKMLCP cockpit status, CKMLPP log
Settlement trace (cost explorer)	drill-down	issue or receipt transaction	settled pairs, qty, value, method, sequence	`CostSettlement`	Inventory value / Cost explorer, Settlements	CKM3N material price analysis
Adjustment history	list inquiry	product, period	adjustment kind, amount, voucher	`CostTransaction` kinds SETTLEMENT_ADJ/RECALC/MANUAL_ADJ	Adjustment history	MR22 documents, ML price change documents
Costing version comparison	operational report	version A vs B, site	product, cost A, cost B, delta, on-hand, revaluation	`ItemCost`	Costing version comparison	CK13N / CK40N comparison, MR21 preview
Price change history	list inquiry	product, site	effective date, old/new cost, source, voucher	`cost.changed` projection	Item price history (InventItemPrice)	MR21 price change documents, S_P99_41000062 price history
PPV analysis	operational report	vendor, product, period	PO price, standard, invoice price, PPV at receipt, PPV at invoice	`CostTransaction` PPV	Standard cost variance report	PRD account analysis, CKM3
Inventory-to-GL reconciliation	operational report (daily)	account, dimension	inventory value, ledger, diff, sample vouchers	`costing.reconcile`	Ledger reconciliation inquiry	MB5L vs FAGLB03
Inventory aging	operational report	as-of, bucket set, grouping, valuation ledger	qty and value per bucket, total, % > 180 days	cost layers after settlement	Inventory aging report (D365)	MC50 dead stock, MC46 slow-moving, MC.A
Inventory turnover / days on hand	KPI	period, item group, site	COGS, average inventory, turns, days on hand	`CostTransaction`	Inventory turnover (D365 PBI)	MC.9 / MC44 inventory turnover
Portal acknowledgement monitor	list inquiry	vendor, due, status	PO, version, sent, due, response	`PurchaseOrderAcknowledgement`	Vendor collaboration → PO confirmation requests	ME92F/ME92K monitor order acknowledgment, ME2A
ASN list	list inquiry	vendor, expected arrival, status	ASN, PO, lines, tracking	`AdvanceShipNotice`	Vendor collaboration ASN	VL06I inbound delivery monitor
Portal user activity and provisioning	operational report	vendor, last login, status	user, role, MFA, invited by, last login	`SupplierUser`	Portal user activity / provisioning status	Ariba user report

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Vendor requests pipeline	list inquiry	type, status, category, age	request, prospect, step, assignee, duplicate flag	`VendorRequest`	Vendor requests list	MDG-S change requests (USMD)
Document distribution log	list inquiry	document, recipient list, status	recipient, version, sent, viewed, acknowledged	`DocumentDistribution`	n/a	CVI8 distribution log
Consignment portal activity	list inquiry	vendor	replenishment orders, receipts, ownership changes, settlements	waves 16/18 projections	Vendor collaboration consignment	SNC consignment monitor

3.20.15 Printed and electronic documents

Document	Trigger	Layout elements (header / line / footer)	Formats	Numbering sequence	Legal / retention requirement
Inventory value report (period-end)	close COMPLETED / on demand	header: entity, as-of, valuation ledger, method; lines: grouping, qty, value, GL balance, diff; footer: totals per account	PDF/A, XLSX	close number `CLS–`	Year-end inventory valuation evidence (IAS 2, HGB §253, US ASC 330); retain 10 years
Inventory aging snapshot	month end	buckets per grouping, totals	PDF, XLSX	snapshot id	Audit workpaper (obsolescence reserve support)
Standard cost revaluation voucher report	version activation	per site: products, old/new cost, on-hand, revaluation amount	PDF, XLSX	version code	Audit evidence for cost changes
Cost adjustment voucher report	manual adjustment	product, dims, old/new value, reason, approver	PDF	adjustment number `CAD–`	Audit evidence
PO acknowledgement confirmation	vendor acknowledges	PO, version, response, proposed changes	email to buyer and vendor	n/a	Commercial correspondence
ASN / delivery note (vendor-generated)	ASN submitted	ASN number, PO, packages with SSCC, lines, batches	PDF (vendor print), SSCC label ZPL	vendor ASN number	GS1 logistics label where required
Vendor onboarding confirmation	request approved / rejected	prospect, decision, next steps	email	request number `VR–`	Privacy notice reference (GDPR Art. 13)
Distributed document acknowledgement receipt	acknowledgement	document, version, user, timestamp	PDF in log, email	n/a	Quality agreement evidence (ISO 9001 §7.5, IATF 16949)

3.20.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Inventory close / recalculation (FR-20.2, FR-20.3)	monthly after period end (manual start or schedule day 2 22:00)	legal entity × valuation ledger × product	close id; settlement deterministic; child checkpoints	< 20 min per 1M transactions per ledger	FAILED resumable; cancel reverses	close FAILED page; > 60 min ticket
Inventory-to-GL reconciliation (`costing.reconcile`)	daily 02:15	legal entity × account	run id	< 5 min	retry 3×	diff ≠ 0 page
Running average / on-hand cost refresh	every 10 min	product	idempotent recompute from cost layers	< 1 min	retry	lag > 30 min ticket
Pending cost activation (scheduled versions)	daily 00:30	version × site	version id idempotent	< 2 min per 10k products	per-product failure listed	failures > 0 ticket
Moving-average revaluation (invoice differences on remaining stock)	hourly	product	key = invoice diff cost transaction	< 1 min	retry	none
Inventory value report and aging snapshots	month end day 1 03:00	legal entity × valuation ledger	key = (entity, ledger, as-of)	< 5 min per 10k products	retry	missing snapshot → close checklist blocker
PO acknowledgement overdue scan	daily 08:00	vendor	key = (ack, date)	< 30 s	retry	overdue list on buyer workspace
Portal inactive-user cleanup	weekly Sunday 04:00	vendor	deactivates users with no login > 180 days	< 1 min	retry	deactivations reported to vendor admin
Vendor request SLA and prospect data purge	daily 01:00	legal entity	purge key = request id after retention	< 1 min	retry	none
Document distribution reminders	daily 07:30	distribution	timer per recipient	< 1 min	retry	overdue acknowledgements → owner notification

3.20.17 Security duties and privileges

Duty	Privileges (entry points: API paths / screens / jobs)	Segregation-of-duties conflicts	Default roles that include it	AX duty / SAP authorization reference
CostingInquire	closes GET, value report, aging, cost lookup, settlement trace	none	Cost accountant, Controller, Auditor	AX "Inquire into inventory value"; SAP CKM3 (K_ML_VA display)
InventoryCloseRun	closes POST (CLOSE/RECALC/PREVIEW), :cancel	with CostAdjust, CostingVersionActivate	Cost accountant	AX "Close inventory" (InventClosing); SAP CKMLCP (K_ML_VA activity 'post closing')
CostAdjust	adjustments POST	with InventoryCloseRun, InventoryJournalPost (w16)	Senior cost accountant	AX "Adjust inventory" (InventAdjustment); SAP MR22 (M_MSEG_WMB / K_ML_VA)
CostingVersionMaintain	versions create, item-costs:import, copy	with CostingVersionActivate	Cost accountant	AX "Maintain costing versions"; SAP CK11N / CK24 mark (K_KEKO)
CostingVersionActivate	versions :activate (revaluation)	with CostingVersionMaintain, CostAdjust	Controller	AX "Activate pending prices"; SAP CK24 release, MR21 (M_MATE_WRK + F_BKPF_BUK)
ValuationSetupMaintain	valuation ledgers, valuation categories, item model group costing fields, posting profile accounts	with InventoryCloseRun	Finance system administrator	AX "Maintain cost management setup"; SAP OMWC/OMX2 (S_TABU_DIS)

Duty	Privileges (entry points: API paths / screens / jobs)	Segregation-of-duties conflicts	Default roles that include it	AX duty / SAP authorization reference
SupplierPortalAdmin (internal)	invite vendor admins, deactivate users, vendor request approval queue, distribution lists	with VendorBankAccountApprove (w19)	Supplier relationship manager	D365 "Provision vendor collaboration users"; Ariba administrator
VendorRequestApprove	approve/reject vendor requests, duplicate override	with VendorMaintain (w7) when the same person entered prospect data	Category manager, Compliance officer, Vendor master steward	AX "Approve vendor requests"; SAP MDG-S approval
DocumentDistributionMaintain	distributions create, recipient lists	none	Quality engineer, Buyer	SAP CVI1/CVI2 (C_DRAD_OBJ)
portal: VendorUser	/portal/v1 purchase orders, ASNs, invoices, payments, SES/timesheets, catalog upload (when granted), consignment view, documents	scoped by OPA to own vendor ids	external vendor staff	D365 "Vendor collaboration user" role; Ariba supplier user
portal: VendorAdmin	users:invite, deactivate, MFA settings, contact data	none (external)	external vendor administrator	D365 "Vendor admin (external)"; Ariba supplier administrator
portal: ProspectiveVendor	public registration endpoint, own request status	rate limited, CAPTCHA	anonymous prospect	D365 "Prospective vendor" role

3.20.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
CST-CLS-001	error	A close is already running for {entity} ({close})	StartClose	Wait for completion	FR-20.2
CST-CLS-002	error	Close date {date} is before the last completed close {last}	BR-20.8	Cancel later closes first	BR-20.8
CST-CLS-003	error	INVENTORY period for {date} is not open	FR-20.2	Open the period (wave 14)	FR-20.2
CST-CLS-004	warning	Pre-check: {n} products with unsettled negative on-hand; valued at fallback	BR-20.3	Post missing receipts or accept fallback	FR-20.2
CST-CLS-005	warning	Pre-check: {n} receipts not invoiced and `close_requires_invoice=true`	FR-20.2	Post invoices or disable the check	FR-20.2
CST-CLS-006	error	Close {close} cannot be cancelled: later close {later} exists	BR-20.1	Cancel the later close first	BR-20.1
CST-LOCK-001	error	Product {product} is being settled by close {close}; retry	BR-20.12	Retry after a few seconds	BR-20.12
CST-ACT-001	error	Version {version} is blocked for activation	ActivateVersion	Remove the block flag	FR-20.5
CST-ACT-002	error	{n} pending costs are zero and zero cost is not allowed	ActivateVersion	Enter costs or allow zero cost	FR-20.5
CST-ACT-004	error	Cannot activate during an in-flight close	BR-20.7	Wait for the close	BR-20.7
CST-ADJ-001	error	Transaction {txn} is in a closed inventory period (close {close})	BR-20.8	Adjust in an open period	FR-20.6
CST-REC-001	error	Inventory value {inv} differs from ledger {gl} for {account}/{dimension} by {diff}	Daily reconciliation	Drill to sample vouchers	FR-20.4
CST-VLG-001	error	Valuation ledger {ledger} has no exchange rate {from}→{to} on {date}	FR-20.15	Enter the rate	FR-20.15
CST-VAL-001	error	Valuation type required: {product} has stock in {types}	BR-20.14	Specify the valuation type	FR-20.16
CST-VAL-002	error	Valuation category of {product} cannot change while stock exists	FR-20.16	Issue or transfer all stock first	FR-20.16
CST-AGE-001	info	{pct} % of layers originate from migration opening balances (dated at cutover)	BR-20.15	None; flagged in report	FR-20.17
SUP-ACK-001	error	Only the latest PO version {version} can be acknowledged	BR-20.9	Open the latest version	FR-20.9
SUP-ASN-001	error	ASN quantity {qty} exceeds remaining {remaining} plus tolerance	BR-20.10	Correct quantity	FR-20.10
SUP-ASN-002	error	Tracking number {tracking} already used on ASN {asn}	BR-20.10	Use the correct tracking number	FR-20.10
SUP-INV-001	error	Invoice price {price} differs from PO price {po_price}; price override not enabled	BR-20.11	Contact the buyer	FR-20.11
SUP-AUTH-001	error	Not found	Cross-vendor access attempt (404 by design)	None	FR-20.12
SUP-RATE-001	error	Too many requests; retry after {seconds} s	Rate limit	Retry later	FR-20.12
SUP-UPL-001	error	File {name} rejected: {reason} (max {mb} MB, types {types}, malware scan)	Upload validation	Upload an allowed clean file	FR-20.13
SUP-REQ-001	warning	A party with matching tax id or name already exists; steward review required	BR-20.16	Steward links or overrides	FR-20.18
SUP-SES-001	error	Hours for {worker} on {date} exceed 24	BR-20.17	Correct timesheet	FR-20.20
SUP-DOC-001	warning	Acknowledgement of {document} v{version} overdue since {date}	FR-20.21	Vendor acknowledges	FR-20.21

3.20.19 Performance budgets and capacity

Operation	Volume assumption (per tenant per day / peak per minute)	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Inventory close (1M transactions, 2,000 products, per valuation ledger)	monthly	12 min / 20 min / 30 min	1,000 settlements/s across 32 children	CostSettlement ≈ 200 B/row (~1.2M rows per close ≈ 240 MB) ; CostTransaction ≈ 250 B/row	CostTransaction (tenant, product_id, financial_date), (ledger_entry_id), (valuation_ledger_id); monthly partitions	Parallelism capped by DB CPU; additional ledgers run sequentially if CPU > 80 %
Value report (10k products)	200 runs	1 s / 5 s / 10 s	n/a	n/a	materialized balance per (product, site, account, valuation ledger) refreshed on each cost transaction	Served from last snapshot with "as of HH:MM" banner
Cost lookup (waves 21/27)	500k / 5,000 per min	3 ms / 20 ms / 40 ms	2,000 rps	n/a	in-pod cache keyed (product, site, valuation type) invalidated by `cost.changed`	Cache-only reads, stale ≤ 60 s
Standard cost activation (10k products)	12 / year	40 s / 2 min / 4 min	n/a	ItemCost ≈ 200 B/row	(product_id, site_id, status)	Chunked by 500 products
Aging report (10k products, 200k layers)	50 runs	2 s / 8 s / 15 s	n/a	snapshots ≈ 20 MB/month	layers (product, site, receipt_date)	Snapshot fallback

Operation	Volume assumption (per tenant per day / peak per minute)	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Portal page load / API	2,000 vendor users, 50k requests / 500 per min	300 ms / 1.5 s / 3 s (page) ; 50 ms / 200 ms / 400 ms (API)	200 rps	supplier-db ≈ 5 GB/year + documents 50 GB/year	RLS by vendor_id; indexes (vendor_id, status)	Rate limits per user/vendor; WAF; static assets cached at edge
ASN submit / receipt pre-fill	1,500 / 50 per min	150 ms / 500 ms / 1 s	50 rps	ASN ≈ 2 KB + lines	(vendor_id, tracking_number) unique	Queue to receiving consumer
Vendor registration (public)	50 / 5 per min	300 ms / 1 s / 2 s	10 rps	prospect ≈ 5 KB + uploads	n/a	CAPTCHA + 5 req/min/IP limit

3.20.20 Localization and compliance hooks

Concern	Varies by	Country packs touching it (wave 48)	Extension point used
Permitted costing methods (LIFO prohibited under IFRS/IAS 2; allowed under US GAAP and US tax with LIFO conformity)	Accounting framework, tax	US (LIFO, dollar-value LIFO in v1.x), IFRS countries	Item model group validation per ledger framework; FR-20.15 lets a tax/US GAAP ledger use LIFO while IFRS uses FIFO/average
Lower of cost and net realizable value / write-downs	IAS 2, ASC 330, HGB §253	all	Manual cost adjustment (FR-20.6) + aging report (FR-20.17) as evidence; NRV calculation hook for wave 39
Statutory inventory valuation reports (e.g., FR état des stocks, IT LIFO a scatti, BR Bloco K / SPED, MX inventory reports)	Country	FR, IT, BR, MX, IN	Value report projection exposed to wave 15 report definitions; SPED Bloco K mapping in wave 48
Currency of valuation ledgers and hyperinflation (IAS 29 restatement)	Functional currency	AR, TR, VE	Valuation ledger with index-adjusted revaluation hook (wave 12 revaluation)
Transfer pricing for intercompany stock (group valuation)	OECD guidelines	all multi-entity	Group valuation ledger in FR-20.15; intercompany profit elimination in wave 40
Supplier portal accessibility and language	WCAG 2.1 AA, EU EAA 2025	EU	Shell i18n (en, de, fr, es); axe tests
Supplier data privacy (prospect and user personal data, retention, consent)	GDPR, CCPA, LGPD	EU, US-CA, BR	Privacy notice at registration; purge job; DSAR export of SupplierUser data
Supplier due diligence and sanctions screening (EU CSDDD, German LkSG supply-chain act, UFLPA)	Country law	DE, EU, US	FR-20.18 questionnaire + screening hook; questionnaire answers stored as evidence
E-invoice submission from portal must respect clearance mandates	IT SDI, MX CFDI, IN IRN	IT, MX, IN, PL (KSeF)	Portal invoice converted via wave 35 country adapter before posting
Document distribution evidence for quality agreements	ISO 9001, IATF 16949, GMP	all (regulated industries)	FR-20.21 acknowledgement log retained 10 years

3.20.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.8.1 — Global inventory accounting (multiple cost ledgers / accounting conventions) [#20]	GAP	ADDED as FR-20.15	Valuation ledgers bound to wave 12 ledger groups
AX 12.8.4 — Global inventory accounting setup (ledgers, accounting policies, conventions) [#21]	GAP	ADDED as FR-20.15	`valuation_ledgers` in 3.20.13
AX 12.2.1 — Vendor requests (R3) [#9]	PARTIAL	ADDED as FR-20.18	Internal and prospect-initiated requests
AX 12.2.4 — Vendors → Vendor request types [#14]	PARTIAL	ADDED as FR-20.18	`vendor_request_types`
AX 12.8.2 — Inventory aging report (D365) [#48]	PARTIAL	ADDED as FR-20.17	Layer-based buckets tying to value report
AX 12.8.4 — Inventory aging report setup [#52]	PARTIAL	ADDED as FR-20.17	`aging_bucket_sets`
AX 12.14.2 — SSRS: Inventory aging (link) [#128]	PARTIAL	ADDED as FR-20.17	3.20.14 and snapshot document (3.20.15)
AX 12.16.1 — Vendor catalogs (vendor catalog import, CMR files) [#154]	PARTIAL	ADDED as FR-20.19	Portal upload into FR-17.18 staging
AX 12.16.1 — Vendor requests (link) [#155]	PARTIAL	ADDED as FR-20.18	
AX 12.16.1 — Procurement → Vendor onboarding (prospective vendor registration) [#157]	PARTIAL	ADDED as FR-20.18	Public registration with questionnaires
AX 12.16.4 — Vendors → Vendor request types / Vendor onboarding questionnaires [#164]	PARTIAL	ADDED as FR-20.18	`onboarding_questionnaires`
AX 12.39.1 — Vendor → Prospective vendor registration (wizard, questionnaires) [#316]	PARTIAL	ADDED as FR-20.18	
AX 12.39.4 — Vendor collaboration → Onboarding: registration wizard configuration, questionnaires [#317]	PARTIAL	ADDED as FR-20.18	
AX 12.39.1 — Vendor → Consignment inventory view [#108, deferred from wave 16]	GAP	ADDED as FR-20.22	Data from FR-16.17 and FR-18.16
SAP 13.8 — CKMLCPAVR alternative valuation run (cumulative periods) [#93]	PARTIAL-20	DEFERRED to wave 39	Multi-period cumulative actual costing belongs with product costing and multi-level actual cost roll-up; the wave 20 engine provides per-period settlement only
SAP 13.8 — CKMDUVACT / CKMDUVMAT distribution of usage variances [#94]	PARTIAL-20	DEFERRED to wave 39	Requires production consumption variances (waves 21/39)
SAP 13.8 — WIP at actual cost (ML) [#95]	PARTIAL-20	DEFERRED to wave 39	WIP valuation from wave 21 production orders; revaluation at actual in wave 39
SAP 13.9 — KE27 periodic valuation (revaluate profitability with actual costs) [#104]	PARTIAL-20	DEFERRED to wave 39	COGS revaluation to profitability segments uses `close.completed` adjustments; profitability analysis lands with wave 38/39 cost accounting
SAP 13.12 — SAP Ariba Buying / Sourcing / Contracts / SLP [#130]	PARTIAL-20	ADDED as FR-20.18 (supplier onboarding/qualification), FR-17.19 (weighted RFP scoring)	Reverse auctions and contract authoring OUT-OF-SCOPE for v1

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP 13.13 — OMWD / OMWC / OMSK split valuation, valuation types, valuation classes [#152]	PARTIAL-20	ADDED as FR-20.16	`valuation_type` dimension and per-type account mapping
SAP 13.15 — Supplier self-service SES (Business Network / Fieldglass) [#174]	PARTIAL-20	ADDED as FR-20.20	Portal sheets and timesheets; contingent-labor management (Fieldglass worker onboarding) OUT-OF-SCOPE
SAP 13.33 — Batch valuation (split valuation per batch) [#366]	PARTIAL-20	ADDED as FR-20.16	Valuation category BATCH
SAP 13.33 — Batch valuation activation (T001K, T149) [#372]	PARTIAL-20	ADDED as FR-20.16	`valuation_categories` with BATCH flag
SAP 13.35 — CV01N document distribution (recipients, CVI1/CVI2/CVI8) [#380]	PARTIAL-20	ADDED as FR-20.21	
SAP 13.35 — CVI8 distribution log [#387]	PARTIAL-20	ADDED as FR-20.21	3.20.14
SAP 13.35 — Document distribution runs (CVI) [#390]	PARTIAL-20	ADDED as FR-20.21	Reminder job (3.20.16)
SAP 13.35 — Document distribution customizing (recipient lists, distribution types) [#395]	PARTIAL-20	ADDED as FR-20.21	`document_recipient_lists`

Part 4 — Phase 4: Manufacturing & Planning (waves 21–25)

4.0 Phase overview

Business outcome at phase exit. A discrete or process manufacturer can run its plant on Passage: master planning nets demand against supply across a multi-level BOM and generates planned purchase, production and transfer orders; planners firm those into production or batch orders, schedule them against finite resource capacity on a Gantt, release them to the floor, and operators execute them from a kiosk while machines report counts over OPC-UA/MQTT. Every material issue, route consumption and report-as-finished posts WIP through the universal journal (wave 11) and the inventory ledger (wave 16); order end settles actual against planned cost and posts variances. Lean plants run kanban replenishment and production flows with periodic backflush costing. The phase-exit demo (wave 25 proof) manufactures a configured product from planned order to finished goods with full costing, and the WIP, inventory and variance accounts reconcile to the journal.

Phase-exit sign-off criteria

- 100% of production order status transitions in the wave-21 state machine are covered by a property test; every illegal transition is rejected and every legal one posts the fixture-defined inventory and GL entries.
- WIP control account balance = Σ(open production orders' posted consumption – posted RAF value) to the cent, verified by the nightly subledger reconciliation job for 30 consecutive days in `stage`.
- The 40 APICS-style MRP fixtures (wave 22) pass; a 50k-item × 12-period × 5-level plan completes in < 10 min on the 3-node reference cluster and produces byte-identical planned orders on re-run.
- Scheduler produces feasible, non-overlapping schedules on all 30 OR-Library job-shop instances; 5k operations scheduled in < 60 s; CP-SAT mode within 5% of known optimum on instances ≤ 10×10.
- Process fixtures: potency consumption, co/by-product allocation (sum equals batch cost exactly), catch-weight dual quantities, and the event-kanban simulation pass.
- Kiosk E2E: an 8-hour shift of 6 operators clocks jobs offline for 30 min, syncs without duplicates, and the production order's labor cost equals Σ(hours × pay rate) from the approved registrations.
- IoT: a simulated machine publishes 10,000 events/min for 60 min with a broker restart at minute 30; zero events lost (sequence check); OEE tile lag p95 < 5 s.
- A practicing cost accountant signs off the WIP/variance fixture set (50 orders) against a reference workbook.

Cross-wave data model summary

Entity	Owning service (Postgres DB)	Key relationships
ProductionOrder	`production` (production_db)	→ ReleasedProduct (w8), BomVersion/RouteVersion (w9), Site/Warehouse (w6), PlannedOrder (w22), SalesOrderLine (w27, pegged)
ProductionBomLine, ProductionOperation, ProductionJob	`production`	child of ProductionOrder; ProductionJob → Resource (w9)
ProductionJournal (+ lines)	`production`	→ ProductionOrder; posts to InventoryTransaction (w16) and JournalEntry (w11)
CostEstimate, WipPosting, Variance	`production`	→ ProductionOrder; → JournalEntry
BatchOrder, Formula, CoProductLine, BatchAttribute, CatchWeightQty	`production` (process extension)	BatchOrder is a ProductionOrder subtype; → InventoryBatch (w16)
PlanVersion, Requirement, PlannedOrder, Pegging, ActionMessage, CoverageGroup, ForecastLine	`planning` (planning_db)	Requirement ↔ Pegging ↔ Requirement; PlannedOrder → ProductionOrder/PurchaseOrder/TransferOrder on firming
ResourceCapacity, ScheduledJob, ScheduleRun	`scheduling` (scheduling_db)	ScheduledJob → ProductionJob; ResourceCapacity → Resource + WorkCalendar (w9)
ValueStream, ProductionFlow, FlowActivity, KanbanRule, Kanban, KanbanCard	`lean` (lean_db)	Kanban → ProductionOrder (manufacturing kanban) or TransferOrder (withdrawal)
Worker, TimeRegistration, PayLine, ProfileDay	`shopfloor` (shopfloor_db)	TimeRegistration → ProductionJob; PayLine → payroll export (w43)
MachineEvent, OeeBucket, DowntimeRecord	`shopfloor`	MachineEvent → Resource; OeeBucket aggregated per resource × 15 min

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
`passage.production.order.status-changed.v1`	production	planning, scheduling, shopfloor, lean, analytics	orderId, from, to, qty, dates, actor
`dev.passage.production.journal.posted.v1`	production	inventory (w16), ledger (w11), planning	journalId, type, lines[item, qty, cost], journalEntryId
`dev.passage.production.order.settled.v1`	production	ledger, costing (w20), analytics	orderId, actualCost, plannedCost, variances[]
`dev.passage.planning.plan.completed.v1`	planning	production, purchasing (w17), scheduling	planVersionId, counts, duration, messages
`passage.planning.planned-order.firmed.v1`	planning	production, purchasing, inventory	plannedOrderId, targetType, targetId
`dev.passage.scheduling.job.scheduled.v1`	scheduling	production, shopfloor, planning	jobId, resourceId, start, end, runId
`passage.lean.kanban.state-changed.v1`	lean	production, inventory, shopfloor	kanbanId, from, to, handlingUnits
`dev.passage.shopfloor.registration.posted.v1`	shopfloor	production, payroll (w43)	registrationId, workerId, jobId, hours, payType
`dev.passage.shopfloor.machine.event.v1`	iot-edge → shopfloor	shopfloor (OEE), production, analytics	machineId, resourceId, kind, counts, state, ts

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
WIP postings drift from the journal under concurrent journal posting on one order	Medium	High	Per-order advisory lock; idempotency key per journal; nightly WIP reconciliation job with alert at any non-zero delta	Engineering
MRP non-determinism from map iteration / parallelism	Medium	High	Deterministic sort keys (item, date, source id) at every stage; plan hash in `plan.completed` event; CI compares hashes across 3 runs	Engineering
Scheduling heuristic produces infeasible schedules on calendar edge cases (DST, zero-capacity days)	Medium	Medium	Calendar materialized in UTC minutes per resource; property test "no job outside open capacity" over 10k jobs	Engineering
Process-industry semantics (potency, co-products) misunderstood	Medium	High	Fixtures reviewed by a process-industry consultant before wave 24 close	Product

Risk	Likelihood	Impact	Mitigation	Owner
IoT connector drops events on broker restart	Low	High	Edge buffer (SQLite ring, 24 h) + Kafka idempotent producer + per-machine sequence numbers	Engineering
Two-person team on five engine-heavy waves	High	High	Reuse OR-Tools, statsforecast, frePPLe comparison harness; cut scope per the open questions rather than slip proofs	Product

4.21 Wave 21: Production Orders, WIP & Order-Based Costing

4.21.1 Scope and business context

Wave 21 is where Passage first consumes components and produces finished goods. Before it, the demo plant buys, receives, pays and values inventory (phase 3) but cannot transform it. After it, a production supervisor can create a production order for a BOM/route item, estimate its planned cost, release it, start it (auto-consuming materials by backflush or manually via a picking list), report route time on route and job cards, report quantities as finished including by-/co-products and scrap, and end the order so that actual cost settles against planned cost with material, labor and overhead variances posted to the GL. Subcontracted operations create a linked purchase order to the vendor.

Out of scope: scheduling against capacity (wave 23; wave 21 schedules with infinite capacity only), MRP-generated orders (wave 22), formula/batch orders, co-product allocation methods beyond the basic by-quantity method (wave 24), operator kiosk and time registration (wave 25), standard-cost roll-up and activation (wave 39; wave 21 reads an active standard or weighted-average cost from wave 20).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX ProdTable status update dialogs (Estimate → Schedule → Release → Start) each open a modal with 8–15 parameters (ProdParmStartUp, ProdParmRelease); a supervisor clicks ~40 times to take one order to Started; SAP CO02 needs CO01 → release → CO11N → MIGO 261 across four transactions	Each status is a separate parameter table and batch class (ProdMultiStartUp); no sensible defaults persisted per site	One `:start` command with site-level defaults; parameter overrides only in an "advanced" drawer; bulk start on the list page	Clicks per order to Started 40 → 3; bulk 500 orders in < 5 min
WIP vs GL drift is found at month-end: AX ProdCalcTrans and the ledger disagree after a failed ProdUpd posting left half a journal; SAP KKAX WIP calculation runs as a batch (CO88 settlement) hours before close	Non-atomic multi-table posting (ProdJournal → InventTrans → LedgerTrans) without idempotency; WIP computed periodically rather than per posting	Journal post is one transaction with idempotency key; WipPosting rows per journal; nightly reconciliation job, delta alerts	WIP reconciliation effort 2 days/month → 0; drift detected < 24 h instead of at close
Backflush quantities silently rounded: AX BOM lines with `Round-up` = No and InventTable unit precision truncate 0.3333 to 0.33, accumulating variance; SAP RESB rounding profile (MARC.RDPRF) applied inconsistently to backflush	Rounding applied at explosion, not at posting; precision taken from the unit not the line	BR-21.2 rounds at posting with 6-decimal internal precision and explicit per-line round-up flag; variance of rounding reported	Rounding-induced variance < 0.01% of material cost (today 0.3–1%)
Cost analysis requires re-keying into Excel: AX "Price calculation" form (ProdCalcTrans) and SAP CO03 cost itemization have no drill to the GL line; auditors re-key order costs into spreadsheets	Cost itemization stored separately from the ledger posting; no shared trace id	Costs tab drills from variance → WipPosting → JournalEntry (w15 drill) with one trace id; `/costs` API for BI	Time to explain one order's variance 30 min → < 1 min; 100% of cost lines link to a journal entry
Epicor Kinetic and Infor LN require a separate MES (Plex, Infor MES) to see live order progress; AX users run the ProdTable list with manual refresh	Status and quantities updated only by batch posting classes	`order.status-changed` and `journal.posted` CloudEvents on Kafka within 1 s; any MES/BI consumes them without middleware	Order-status latency to external systems 15 min polling → < 2 s event

4.21.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Production order header	ProdTable, ProdStatus enum, form ProdTable, menu item ProdTable	AUFK/AFKO/AFPO, CO01/CO02/CO03	`production` service, entity ProductionOrder, screen Production orders	full
Order BOM and route snapshot	ProdBOM, ProdRoute, ProdRouteJob (ProdUpdCreate explosion)	RESB (components), AFVC/AFVV (operations), order read of BOM/routing	ProductionBomLine, ProductionOperation, ProductionJob	full
Estimation (planned cost)	ProdParmCostEstimation, ProdCalcTrans, status Estimated	Preliminary costing (CO01 → cost), KKF6N	CostEstimate entity, `:estimate` action	full
Picking list	ProdJournalTable (JournalType=Bom), ProdJournalBOM, backflush via BOM line flushing principle	Goods issue MIGO 261, backflush at CO11N	ProductionJournal type `picking-list`	full
Route card / job card	ProdJournalRoute (route card), ProdJournalRoute by job (job card), ProdParmReportFinished	CO11N confirmation (AFRU), CO15 order confirmation	ProductionJournal types `route-card`, `job-card`	full
Report as finished	ProdJournalProd, ProdParmReportFinished, accept error quantity	Goods receipt MIGO 101 for order, CO15 final confirmation	ProductionJournal type `report-as-finished`	full
WIP posting	Production posting profiles (ProdPosting), ledger accounts WIP issue/receipt, Picking list issue/offset	WIP calculation KKAX/KKAO, results analysis (RA key)	WipPosting rows, posting profile per item group × site	full
Order end and settlement	ProdParmCostUpdate (End), ProdCalcTrans actual, variance accounts	Settlement KO88/CO88, variance calculation KKS2	`:end` action, Variance entity, settlement to journal	full
Subcontracting	Vendor resource type, BOM line type Vendor, auto PO	External processing operation (PP02 control key), PO item category L	ProductionOperation.subcontract → PurchaseOrderLine (w17)	full
Production order inquiry	ProdTable listpage, ProdCalcTrans inquiry, ProdJournal inquiries	COOIS, CO03 cost analysis	Production order detail, Cost tab, Status log	improved (single trace id per transaction)
Scrap and error quantity	ProdJournalProd.QtyError, BOM line scrap %, route scrap %	Assembly scrap (MARC), component scrap (STPO), operation scrap	Scrap fields on BOM line, operation and RAF line	full
Reset status	ProdTable reset status (ProdUpdStatusDown)	Not native (reversal via CO13 cancel confirmation)	`:reset-status` with reversal journals	full
Production pools / groups	ProdPool, ProdGroup	Production scheduler profile, order type	OrderGroup (posting profile key)	partial — pools dropped; replaced by saved views
Order split / group	ProdSplit (Split production order), ProdGroup	CO02 → Functions → Split order (AFKO.SPLIT), order group	`:split` and `:merge` commands (FR-21.15)	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Order change management	ECO re-explosion on ProdTable before Started (BOMCalc), no change-cost tracking	OCM (COCM / COCM1), change profiles, change costs (COCM*)	OrderChangeRecord, `:apply-change` with cost preview (FR-21.16)	full
Trigger points / rework operation insertion	n/a (manual route edit)	Trigger points (CO31/CO33, OPMP/OPM, TTRIG), rework order CO07	TriggerPoint rules on operation feedback (FR-21.17); rework order (FR-21.20)	full
Order-specific BOM / route (ETO)	ProdBOM / ProdRoute line edits on the order (ProdBOM form)	Order BOM CU51/CU52/CU55/CU56, order routing	Order BOM editor with audit (FR-21.18)	full
As-built genealogy (serial → component serial)	InventTrans tracing by serial (InventDim), no as-built record	Serial numbers at component level (IQ09, as-built configuration)	AsBuiltRecord captured at picking/RAF (FR-21.19)	full

4.21.3 Functional requirements

- **FR-21.1 Create production order** — Must. The system shall create a ProductionOrder for a released product with a BOM version and route version resolved as of the order's delivery date and site, and explode them into ProductionBomLine, ProductionOperation and ProductionJob rows at creation.
  - Notes: Order number from the `production-order` number sequence (continuous not required). Default quantity from released-product default order settings; multiple/min/max enforced. Manual BOM/route version override allowed only before Estimated. Phantom BOM lines are exploded in place; line numbering keeps the parent's line number with a suffix (`30.1`, `30.2`). Explosion of a 7-level, 2,000-component product completes in < 1 s (wave 9 explosion API).
- **FR-21.2 Status lifecycle** — Must. The system shall enforce the state machine in BR-21.1 with statuses Created, Estimated, Scheduled, Released, Started, ReportedAsFinished, Ended and Cancelled.
  - Notes: Transitions are commands (`:estimate`, `:schedule`, `:release`, `:start`, `:report-finished`, `:end`, `:cancel`, `:reset-status`); each is idempotent by `Idempotency-Key`. Each transition writes a StatusLog row with actor, timestamp, trace id. SAP mapping on export: CRTD→Created, REL→Released, PCNF→Started, CNF/DLV→ReportedAsFinished, TECO/CLSD→Ended.
- **FR-21.3 Estimation** — Must. The system shall compute planned cost per order: material (Σ BOM line qty × (1 + scrap%) × unit cost from the active costing version), labor (Σ operation setup + run time × resource cost category rate), overhead (costing-sheet surcharges from wave 20; full costing sheets in wave 39), and subcontract (PO price).
  - Notes: Creates inventory transactions: receipt `ordered` for the finished item, issues `on-order` for BOM lines (w16 statuses). Re-estimate is allowed until Started and replaces the estimate atomically. Rounding: cost amounts to 4 decimals internally, 2 on posting.
- **FR-21.4 Release and printing** — Must. The system shall release an order, generate the picking list and job card documents (PDF), and set BOM-line reservations according to the order's reservation policy (none, estimated, released, started).
  - Notes: Release is blocked if any BOM line has an unreleased product in this legal entity. Release with shortages raises a warning listing short lines; a `strict` flag turns it into an error.
- **FR-21.5 Picking list journal** — Must. The system shall post picking-list journals that issue BOM-line quantities from inventory to WIP, with proposal modes: remaining quantity, consumption proportional to RAF quantity, or manual.
  - Notes: Flushing principle per BOM line: `manual`, `start` (consumed at Start), `finish` (consumed at RAF), `available-at-location` (consumed when job reports). Over-consumption beyond 100% + line tolerance (default 10%) requires duty *Post production over-consumption*. Posting creates one InventoryTransaction per line with status `deducted` and the GL entry Dr WIP / Cr Inventory.
- **FR-21.6 Route card and job card journals** — Must. The system shall post route consumption (setup hours, run hours, quantity good, quantity error) per operation (route card) or per job (job card), valued at the resource's cost category rate, posting Dr WIP / Cr Labor absorption (and Cr Overhead absorption for indirect costs).
  - Notes: Hours in decimals to 2 places; minimum postable 0.01 h. Reporting on an operation of a non-Started order auto-starts the order if the order's parameter `autoStartOnFeedback` is true (default true, AX parity). Operation `finished` flag closes it to further feedback unless reopened.
- **FR-21.7 Report as finished** — Must. The system shall post RAF journals that receive the finished item (and by-/co-product lines) into inventory at the estimated unit cost (standard cost if the item's costing method is standard), consume `finish`-flushed BOM lines, and post Dr Finished goods / Cr WIP.
  - Notes: `acceptError` flag posts error quantity as scrap without inventory receipt. RAF quantity > remaining + tolerance (default 0%) is rejected. The last RAF sets `endJob = true` on all jobs when the `endJobs` parameter is set. Tracking dimensions (batch/serial) required per the item's tracking dimension group; serials must be unique across the tenant (w16).
- **FR-21.8 End and settle** — Must. The system shall end an order by computing actual cost from posted journals, reversing WIP, posting actual cost to the consumption and inventory accounts, and posting variances per BR-21.6.
  - Notes: End date must be ≥ last journal date and in an open inventory and GL period (w14 period control). Remaining `on-order` and `ordered` inventory transactions are cancelled. Orders with RAF quantity 0 and posted consumption are ended with 100% of cost to variance (AX "end with zero quantity" behavior) and require duty *End production order with zero output*.
- **FR-21.9 Reset status** — Should. The system shall reset an order to an earlier status, reversing postings made by later statuses (reversal journals with negative quantities, linked to originals).
  - Notes: Reset from Ended is not allowed (AX parity); reset from ReportedAsFinished to Started reverses all RAF journals. Reset is blocked if any reversed inventory transaction has been consumed by a later issue (on-hand would go negative).
- **FR-21.10 Subcontracting** — Should. The system shall create a purchase order line (w17) for an operation whose resource is of type Vendor, linked to the operation, and shall post the vendor invoice amount as the operation's actual cost on product receipt.
  - Notes: PO line quantity = operation quantity in the service item's UoM; delivery date = operation end. Changes to order quantity after PO confirmation raise a PO change (w17 versioning).
- **FR-21.11 Order inquiry and cost view** — Must. The system shall show, per order, planned vs actual cost by cost group (material, labor, overhead, subcontract), the journal list, inventory transactions, GL entries with drill-down to the journal entry (w15 drill-down) and the status log.
- **FR-21.12 Bulk status update** — Should. The system shall apply a status transition to up to 500 selected orders in one batch job (Temporal), reporting per-order success/failure.
- **FR-21.13 Posting profiles** — Must. The system shall resolve ledger accounts for each posting type (picking list issue, picking list offset, WIP issue, WIP account, RAF receipt, RAF offset, variance accounts per variance type) by item group × site, falling back to All.
- **FR-21.14 Co-product RAF (basic)** — Could. The system shall accept RAF lines for co-/by-products declared on the BOM (w9), allocating cost by quantity only; allocation methods arrive in wave 24.
- **FR-21.15 Order split and merge** — Should. The system shall split a production order in status Created…Started into two orders (original keeps the remaining quantity; the new order receives `splitQty`), carrying BOM lines, operations and jobs proportionally, and shall merge orders of the same product, BOM/route version and site that are in status ≤ Released into one.
  - Notes: Split of a Started order distributes posted consumption and route hours proportionally to the split ratio unless `allocation=keep-on-original`; WIP postings move via an internal transfer journal (Dr WIP new / Cr WIP original) so each order's WIP stays reconcilable. Split quantity must be ≥ the item's minimum order quantity and a multiple of the item's multiple. Scheduled jobs of the new order are rescheduled (w23) with the original's dates as the preferred window. Merge is refused if either order has posted journals. AX ProdSplit and SAP CO02 split both stop at Released; Passage allows Started with the allocation rules above.

- **FR-21.16 Order change management** — Should. The system shall detect, for open orders (status < Started, or Started with `ocm.allowStarted`), that the BOM or route version they snapshot has been superseded by an ECO (w9 `bom.version.activated` / `route.version.activated` events) and shall offer `:apply-change` that re-explodes the order and reports the delta (added/removed/changed lines and operations) with a change cost = Σ (new planned cost – old planned cost) plus a configurable change-cost surcharge per OCM profile.
    - Notes: OCM profile per item group × site: `automatic` (apply at next status transition), `manual` (planner confirms), `none` (ignore). Lines already consumed are never removed; they become `substituted` and feed the production substitution variance (BR-21.6). Each application writes an OrderChangeRecord (ECO id, before/after snapshot hash, change cost, actor). Configured products (w8) re-explode through the configurator with the stored configuration; a changed model that no longer yields the same configuration raises `OCM_CONFIG_DRIFT` and requires manual review.
- **FR-21.17 Trigger points** — Should. The system shall evaluate TriggerPoint rules on operation feedback events (route card / job card post, operation status change, qty_error > 0, quality check fail from w25) and perform the configured function: insert a rework operation sequence after the current operation (standard operation set from w9), create a rework order (FR-21.20) for the error quantity, release the next operation, start a Temporal workflow, or raise a notification.
    - Notes: Trigger points are defined on route operations (w9 route version) and copied to ProductionOperation at explosion; conditions use the same expression language as alert rules (w5) over the feedback payload. Inserted operations get oper_number = current + 5 (renumbered if collision) and are scheduled forward (w23). One trigger point fires at most once per operation unless `repeatable`.
- **FR-21.18 Order-specific BOM and route** — Should. The system shall allow editing BOM lines and operations on an order after explosion and before Started (add, remove, change quantity/item/resource requirement), marking the order `orderBomModified = true`, keeping an audit diff against the source versions, and optionally saving the result as a new BOM/route version for the product (ETO reuse).
    - Notes: Edits after Released require duty *Maintain order BOM after release* and re-run the reservation policy. Planned cost is re-estimated after each edit. Edited orders are excluded from automatic OCM `automatic` profile and flagged for manual review instead.
- **FR-21.19 As-built genealogy** — Should. For serial- or batch-tracked finished items the system shall record, per finished serial/batch, which component serials/batches were consumed (AsBuiltRecord), captured on picking-list posting (component serials against the order) and resolved per finished serial at RAF (operator assignment on the kiosk in w25, or automatic when the order quantity is 1).
    - Notes: Genealogy is queryable in both directions (where-used of a component serial; as-built of a finished serial) and feeds the w16 trace inquiry and w33 nonconformance. Unresolved component serials at End raise a warning, not an error, unless the item's tracking policy sets `asBuiltRequired`.
- **FR-21.20 Rework production orders** — Should. The system shall create a rework order for a finished item quantity (optionally referencing the original order, batch or serial): no BOM explosion by default (components added manually), a route of rework operations, and a `reworkReference` so that the rework cost is posted to the item's rework cost group and reported against the original order.
    - Notes: Rework output is received at the original item's current cost; rework cost is posted as a variance of type `rework` (added to the Variance enum) and not capitalized into inventory unless the parameter `rework.capitalize` is set. SAP CO07 (order without material) semantics; AX equivalent is a production order with an empty BOM and the "rework" posting profile.

4.21.4 Data model

ProductionOrder

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id, legal_entity_id	uuid	yes	RLS
order_number	text	yes	unique per legal entity; number sequence
product_id, variant_id	uuid	yes/no	FK ReleasedProduct (w8)
site_id, warehouse_id	uuid	yes	FK (w6)
quantity	numeric(18,6)	yes	> 0, in inventory UoM
qty_started, qty_raf, qty_error	numeric(18,6)	yes	default 0
bom_version_id, route_version_id	uuid	yes	FK (w9), snapshot taken at creation
status	enum	yes	Created…Cancelled
delivery_date, start_date, end_date	date	yes/no/no	start/end set by scheduling
scheduling_method	enum	yes	`operations` or `jobs`
reservation_policy	enum	yes	none/estimated/released/started
posting_profile_key	text	yes	item group × site resolved key
origin_type, origin_id	enum, uuid	no	PlannedOrder, SalesOrderLine, Manual, Kanban
version	int	yes	optimistic concurrency

ProductionBomLine: id, order_id (FK), line_number (text), item_id, variant_id, bom_qty_per (numeric 18,6), bom_series_size (numeric), scrap_pct (numeric 7,4, 0–100), qty_required (numeric 18,6, computed), qty_consumed (numeric 18,6), flushing_principle (enum), line_type (enum: item, phantom, pegged-supply, vendor), warehouse_id, location_id, reservation_status (enum), unit_cost_est (numeric 18,6).

ProductionOperation: id, order_id, oper_number (int), next_oper_number (int, 0 = last), priority (enum: primary, secondary1–4), resource_group_id, resource_id (nullable until scheduled), setup_time_h, run_time_h_per (numeric 10,4), process_qty (numeric), queue_before_h, queue_after_h, transit_h, scrap_pct, cost_category_setup_id, cost_category_run_id, subcontract (bool), purchase_line_id (nullable), status (enum: open, started, finished), scheduled_start, scheduled_end (timestamptz).

ProductionJob: id, operation_id, job_type (enum: queue-before, setup, process, overlap, transport, queue-after), resource_id, planned_start, planned_end, planned_hours, reported_hours, qty_good, qty_error, status.

ProductionJournal: id, order_id, journal_type (enum: picking-list, route-card, job-card, report-as-finished), journal_number, posting_date, posted (bool), posted_at, posted_by, journal_entry_id (w11 FK), reversal_of (nullable self FK), idempotency_key (unique).

CostEstimate / WipPosting / Variance: CostEstimate(order_id, cost_group, planned_qty, planned_unit_cost, planned_amount, costing_version_id); WipPosting(order_id, journal_id, posting_type, amount_dr, amount_cr, ledger_account, journal_entry_id); Variance(order_id, variance_type enum {lot-size, production-quantity, production-price, production-substitution}, cost_group, amount, journal_entry_id).

Invariants:

- Σ WipPosting.amount_dr – Σ amount_cr for an order = 0 once status = Ended.
- qty_raf + qty_error ≤ quantity × (1 + raf_tolerance_pct/100).
- Every posted ProductionJournal has exactly one journal_entry_id and its inventory transactions sum to the journal's quantities.

- A ProductionOperation with subcontract = true has exactly one purchase_line_id once the order is Released.

Indexing: ProductionOrder (tenant_id, legal_entity_id, status, delivery_date); (tenant_id, product_id, status); ProductionJournal unique (tenant_id, idempotency_key); WipPosting (order_id); partition ProductionJournal and WipPosting by legal_entity_id hash (16) once rows > 50M.

OrderChangeRecord (FR-21.16): id, order_id (FK), eco_id (w9 FK), source_version_old_id, source_version_new_id, delta (jsonb: added/removed/changed lines and operations), change_cost (numeric 18,4), surcharge (numeric 18,4), profile (enum automatic/manual/none), status (enum detected/previewed/applied/dismissed), applied_by, applied_at, snapshot_hash_before, snapshot_hash_after (bytea).

TriggerPoint (FR-21.17): id, order_id, oper_number, sequence (int), condition (text, w5 expression), function (enum: insert-operations, create-rework-order, release-next, start-workflow, notify), standard_operation_set_id (nullable, w9), workflow_key (nullable), repeatable (bool), fired_count (int), last_fired_at.

AsBuiltRecord (FR-21.19): id, order_id, finished_item_id, finished_serial_or_batch (text), component_item_id, component_serial_or_batch (text), qty (numeric 18,6), journal_id (picking list), raf_journal_id, assigned_by, assigned_at; unique (tenant_id, component_item_id, component_serial_or_batch) for serialized components.

ProductionOrder additions: split_from_order_id (nullable self FK), merged_into_order_id (nullable), order_bom_modified (bool, default false), rework_reference_type (enum: order/batch/serial, nullable), rework_reference_id; ProductionOrder.origin_type gains `Split` and `Rework`; Variance.variance_type gains `rework`.

Invariants (additions): Σ quantity of an original order and orders split from it = original quantity at split time; an order with merged_into_order_id set has status Cancelled and no posted journals; each serialized component serial appears in at most one AsBuiltRecord. Indexing: AsBuiltRecord (finished_serial_or_batch), (component_serial_or_batch); OrderChangeRecord (status, order_id).

4.21.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/production/orders	Create order (explodes BOM/route)	Idempotency-Key header	Maintain production orders
GET	/api/v1/production/orders/{id}	Read order with lines, ops, jobs, cost summary	n/a	View production orders
PATCH	/api/v1/production/orders/{id}	Edit header fields allowed for current status	ETag/version	Maintain production orders
POST	/api/v1/production/orders/{id}:estimate	Estimate	Idempotency-Key	Maintain production orders
POST	/api/v1/production/orders/{id}:release	Release (and :schedule, :start, :end, :cancel, :reset-status)	Idempotency-Key	Release production orders / End production orders
POST	/api/v1/production/orders/{id}/journals	Create journal with proposal (`proposal=remaining	proportional	none`)
POST	/api/v1/production/journals/{id}:post	Post journal	Idempotency-Key	Post production journals
GET	/api/v1/production/orders/{id}/costs	Planned vs actual by cost group	n/a	View production costs
POST	/api/v1/production/orders:bulk-status	Batch transition up to 500 ids	Idempotency-Key	Release production orders
GET	/api/v1/production/posting-profiles	Resolve accounts for item group × site	n/a	View production setup

Commands: `CreateOrder` (validates product released, BOM/route active as of delivery date, qty multiples); `PostJournal` (validates order status allows journal type, period open, quantities within tolerance, tracking dims present); `EndOrder` (validates no unposted journals, all subcontract POs invoiced or explicitly accrued).

Event	Trigger	Payload fields	Consumers
`dev.passage.production.order.created.v1`	order committed	orderId, orderNumber, productId, qty, siteId, deliveryDate, originType, originId	planning, analytics
`passage.production.order.status-changed.v1`	any transition	orderId, from, to, qty, startDate, endDate, actor, occurredAt	planning, scheduling, shopfloor, lean
`dev.passage.production.journal.posted.v1`	journal posted	journalId, orderId, type, postingDate, lines[{itemId, qty, unitCost, operNumber, hours}], journalEntryId	inventory, ledger, planning
`dev.passage.production.order.settled.v1`	End posted	orderId, plannedCost{material,labor,overhead,subcontract}, actualCost{…}, variances[{type,costGroup,amount}]	ledger, costing, analytics
`passage.production.subcontract.po-requested.v1`	release of subcontract op	orderId, operNumber, vendorId, serviceItemId, qty, dueDate	purchasing

4.21.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Production orders (list)	Create, Estimate, Release, Start, RAF, End, bulk status, saved views by status/site	Order, Item, Qty, Remaining, Status, Delivery date, Site, Planned cost, Actual cost	View production orders
Production order detail	Tabs: Lines (BOM), Route, Jobs, Journals, Costs, Inventory transactions, Status log	Header fields; per-tab grids	Maintain production orders
Journal editor	Add lines, proposal, validate, post, print	Item/operation, qty, hours, error qty, reason, batch/serial	Post production journals
Cost analysis	Compare planned vs actual, drill to journal entry	Cost group, planned, actual, variance, variance %	View production costs
Posting profile setup	Maintain accounts by item group × site	Posting type, item group, site, account, dimensions	Maintain production setup

Interaction rules: `Ctrl+Shift+S` opens the status action menu; list page supports multi-select with inline status badge; validation runs on blur per field and fully on post; journal editor keeps unsaved lines in `localStorage` and warns on navigation; the Costs tab is read-only and recomputes on each journal post event (live update via SSE from the BFF).

4.21.7 Business rules and edge cases

- **BR-21.1 State machine.** Allowed transitions and their posting effects:

From → To	Preconditions	Posting
Created → Estimated	BOM/route exploded	Inventory transactions `ordered` / `on-order`; CostEstimate rows
Estimated → Scheduled	estimate exists	Operation/job dates (infinite capacity in w21)
Scheduled → Released	no unreleased components	Reservations per policy; documents
Released → Started	period open	Auto-post picking list for `start`-flushed lines; route card for ops with `autoStart`; Dr WIP / Cr Inventory

From → To	Preconditions	Posting
Started → ReportedAsFinished	qty_raf > 0	Dr FG inventory (est. cost) / Cr WIP; consume `finish`-flushed lines
ReportedAsFinished → Ended	no open journals; period open	Reverse WIP; post actual; post variances
Started → Ended	qty_raf = 0 and duty held	Same as above with 100% variance
Created/Estimated/Scheduled/Released → Cancelled	no posted journals	Cancel `ordered`/`on-order` transactions
Any non-Ended → earlier status (`:reset-status`)	reversal possible (BR-21.5)	Reversal journals

Any other transition returns HTTP 409 `PRODUCTION_ILLEGAL_TRANSITION`.

- **BR-21.2 Quantity rounding.** Required quantity = ceil-to-UoM-precision(qty_per × order_qty / series_size × (1 + scrap/100)); UoM precision from the unit definition (w6), default 6 decimals; "round up to whole unit" flag on the BOM line forces integer.
- **BR-21.3 Posting date.** A journal's posting date must be ≥ order creation date and within an open inventory period and GL period for the Production module; otherwise 422 `PERIOD_CLOSED`.
- **BR-21.4 Concurrency.** Journal posting takes `pg_advisory_xact_lock(hash(order_id))`; concurrent posts serialize; a stale `version` on PATCH returns 412.
- **BR-21.5 Reversal safety.** Reset status is refused when any inventory transaction created by a to-be-reversed journal has a downstream issue with status `deducted` or `picked`.
- **BR-21.6 Variance decomposition at End** (standard-cost items; weighted-average items post only a total variance to the item's inventory adjustment account):
  - Production quantity variance = Σ (actual qty – planned qty for actual output) × standard price.
  - Production price variance = Σ actual qty × (actual price – standard price).
  - Lot-size variance = planned fixed cost (setup, constant overhead) × (1 – actual output / planned output).
  - Production substitution variance = cost of consumed items/operations not on the estimate minus estimate of items not consumed.
  - Sum of the four equals actual cost – (qty_raf × standard unit cost); property-tested.
- **BR-21.7 Worked settlement example.** Item FG-100, standard cost USD 50.00/unit = material 30.00 + labor 15.00 + overhead 5.00. Order 100 units. Estimate: 300 kg RM @ 10.00 = 3,000.00; 50 h @ 30.00 = 1,500.00; overhead 10.00/h × 50 = 500.00; total 5,000.00. Actual: picking list 305 kg at weighted-average 10.20 = 3,111.00; route cards 46 h = 1,380.00; overhead 46 × 10.00 = 460.00; actual total 4,951.00. RAF 100 units posts Dr Finished goods 5,000.00 / Cr WIP 5,000.00. WIP before End: Dr 4,951.00, Cr 5,000.00, balance Cr 49.00. End posts: material quantity variance Dr 50.00 ((305 – 300) × 10.00), material price variance Dr 61.00 (305 × 0.20), labor quantity variance Cr 120.00 ((46 – 50) × 30.00), overhead quantity variance Cr 40.00; WIP cleared to 0.00; net variance Cr 49.00 (favorable). EUR equivalents are carried on each line at the posting-date rate (w12).
- **BR-21.8 Scrap.** Operation scrap compounds backward along the route: quantity to start at operation n = order qty / Π(1 – scrap_k) for k ≥ n, rounded up at each operation.
- **BR-21.9 Negative inventory.** Picking list posting that would drive on-hand negative is rejected unless the item model group allows physical negative inventory (w16 policy).
- **BR-21.10 Serial/batch at RAF.** A RAF line for a serial-tracked item must carry exactly qty_good serial numbers; duplicates across the tenant are rejected at 422 `SERIAL_DUPLICATE`.
- **BR-21.11 Split allocation.** With split ratio r = splitQty / order qty: posted consumption qty, route hours and WIP amounts move to the new order at round(x × r, 6) (quantities) and round(x × r, 2) (amounts); the rounding residual stays on the original. Each order's Σ WipPosting must equal its moved consumption after the transfer journal; property-tested.
- **BR-21.12 OCM change cost.** change_cost = Σ_new planned – Σ_old planned + surcharge(profile), where surcharge is a fixed amount or % of the delta; a negative delta still carries the fixed surcharge. Applying a change to an order whose scheduled jobs are firmed (w23) requires un-firming or the `ocm.forceReschedule` option.
- **BR-21.13 Trigger point evaluation order.** Trigger points on one operation evaluate in ascending sequence number; a trigger that inserts operations re-evaluates nothing on the inserted operations in the same feedback transaction (prevents loops). A trigger's function failing (e.g., no standard operation set) raises `TRIGGER_FUNCTION_FAILED` and does not block the journal post.
- **BR-21.14 As-built resolution.** Σ component serials assigned across finished serials ≤ component serials consumed on the order; one component serial belongs to exactly one finished serial; reversal of a RAF journal detaches the finished serial's record.

4.21.8 Acceptance criteria

- **AC-21.1** (FR-21.1) Given a released product with a 3-level BOM and a 4-operation route, when I create an order for 100 units, then the order has the exploded lines with phantoms flattened and quantities per BR-21.2.
- **AC-21.2** (FR-21.2) Given an order in Created, when I call `:start`, then I get 409 `PRODUCTION_ILLEGAL_TRANSITION`.
- **AC-21.3** (FR-21.3) Given an estimated order, when I read `/costs`, then planned material equals Σ line qty × unit cost from the active costing version to 4 decimals.
- **AC-21.4** (FR-21.5) Given a Started order with `start`-flushed lines, when I RAF 100 units with proportional consumption, then consumed quantities equal BOM qty × 100 × (1 + scrap%) exactly (wave proof).
- **AC-21.5** (FR-21.7) Given a batch-tracked finished item, when I post RAF without a batch, then the post fails with 422 and no inventory transaction is created.
- **AC-21.6** (FR-21.8) Given the BR-21.7 fixture, when I end the order, then the four variance rows total –49.00 and the WIP account balance for the order is 0.00.
- **AC-21.7** (FR-21.8) Given a partially RAF'd order (60 of 100) ended with duty, then remaining WIP equals posted consumption – 60 × standard cost and reconciles to the GL WIP account.
- **AC-21.8** (FR-21.9) Given an order in ReportedAsFinished whose finished goods were already shipped, when I reset to Started, then the reset is refused with `REVERSAL_BLOCKED_DOWNSTREAM`.
- **AC-21.9** (FR-21.10) Given an operation on a Vendor resource, when I release the order, then a PO line exists for the vendor with the operation quantity and due date.
- **AC-21.10** (FR-21.12) Given 500 Released orders, when I bulk-start them, then the batch completes in < 5 min and the report lists per-order results.
- **AC-21.11** (FR-21.15, BR-21.11) Given a Started order of 100 with 305 kg consumed and 20.00 h posted, when I split 40 units with proportional allocation, then the new order shows 122.000000 kg and 8.00 h, the original 183.000000 kg and 12.00 h, and both WIP balances reconcile to the GL.
- **AC-21.12** (FR-21.16, BR-21.12) Given a Released order whose BOM version was superseded by an ECO adding one line, when I preview `:apply-change`, then the delta lists the added line and a change cost equal to the line's planned cost plus the profile surcharge; applying it re-explodes without touching consumed lines.
- **AC-21.13** (FR-21.17) Given a trigger point "qty_error > 0 → insert rework operations R10, R20", when a job card posts 3 error units, then operations 25 and 26 (renumbered) exist after the current operation and are scheduled forward.
- **AC-21.14** (FR-21.19, BR-21.14) Given an order for 2 serialized units consuming 2 serialized motors, when RAF assigns motor M-1 to unit S-1 and M-2 to S-2, then the as-built inquiry for S-2 returns M-2 and the where-used of M-1 returns S-1.
- **AC-21.15** (FR-21.20) Given a rework order for 5 units of FG-100 with 2.00 h of rework at 30.00/h, when ended, then a variance of type `rework` of 60.00 is posted and the 5 units are received at their existing unit cost.

4.21.9 Test plan and fixtures

Fixtures: `prod-orders-50` (50 orders with known consumptions, standard and weighted-average items, generated from the demo plant BOMs; expected WIP and variance postings in a reviewed XLSX); `prod-bom-7level` (from wave 9); `prod-state-machine` (table of all 9×9 transition pairs with expected result); `prod-subcontract-5`.

Test	Type	Target	Pass criterion
State machine	property (Go rapid)	transitions	every illegal transition rejected, every legal one posts expected entries
Settlement fixtures	unit	`settle()`	50/50 orders match workbook to the cent
Backflush	unit	RAF proportional	consumed = BOM × qty × (1+scrap)
Posting contract	contract	ledger/inventory APIs	journals produce balanced entries, idempotent on replay
Period control	contract	journal post	closed period rejected on every journal type
Order E2E	Playwright	create→end	demo order ends with 0 WIP
Journal post latency	performance	200-line picking list	p95 < 2 s

Reconciliation: Σ open-order WIP = GL WIP control account; Σ RAF value = inventory receipts from production (w16) for the period; Σ variances = GL variance accounts.

4.21.10 Deployment and operations

Helm chart `production` (Go service, 3 replicas, HPA on CPU 70%), `production-db` (CloudNativePG, 2 instances, 100 GB). Config: `PRODUCTION_RAF_TOLERANCE_PCT` (0), `PRODUCTION_OVERCONSUMPTION_PCT` (10), `PRODUCTION_AUTOSTART_ON_FEEDBACK` (true). Migrations via Atlas from entity metadata. Feature flags: `prod.subcontracting`, `prod.coproduct-basic`. Scheduled jobs: nightly `wip-reconcile` (Temporal, 02:00 tenant local), `cancel-stale-created-orders` (weekly). SLOs: create p95 < 300 ms; transition p95 < 500 ms; journal post p95 < 2 s; availability 99.9%. Alerts: WIP reconciliation delta ≠ 0 (page); journal post error rate > 1% 5 min (warn). Runbook: "WIP delta" — list orders with unbalanced WipPosting, check outbox lag, replay. Rollback: Argo CD to prior digest; migrations are additive within the wave.

4.21.11 Migration from AX / SAP

Source	Passage entity	Notes
ProdTable (AX) / AUFK+AFKO+AFPO (SAP)	ProductionOrder	status map per FR-21.2; only open orders (status < Ended / not TECO)
ProdBOM / RESB	ProductionBomLine	qty_required recomputed; consumed qty from InventTrans / MSEG 261
ProdRoute, ProdRouteJob / AFVC, AFVV	ProductionOperation, ProductionJob	reported hours from ProdRouteTrans / AFRU
ProdJournalTable+Bom/Route/Prod / AFRU, MSEG	ProductionJournal (history)	posted journals loaded read-only, no re-posting
ProdCalcTrans / CO cost itemization	CostEstimate (planned), WipPosting (actual, as opening balance)	WIP opening = Σ actual − Σ RAF value, posted as one opening journal per order

Validation: per-order WIP opening = legacy WIP report (AX ProdCalcTrans actual, SAP KKAX) within 0.01; counts of open orders by status match. Not migrated: ended/TECO orders (go to the lakehouse via w46 history load), production pools, legacy posting-profile accounts (re-mapped to the new chart). Recommended cutover: end all orders in legacy that are ≥ 95% complete.

4.21.12 Dependencies, risks, open questions

Depends on: w9 BOM/route explosion and resources; w16 inventory transactions and statuses; w11 posting API; w14 period control; w17 PO creation; w20 costing versions and standard cost.

Risk	Likelihood	Impact	Mitigation
Variance semantics differ between AX (4 types) and SAP (price/quantity/resource-usage/lot-size/…)	High	Medium	Ship AX's four; document SAP mapping; w39 adds decomposition
Backflush under weighted-average produces cost at post time not at End	Medium	Low	Revalue at End using inventory close (w20) settlement

Open questions: (1) Should Cancelled be exposed for Released orders with reservations? Default yes with automatic un-reserve; decide by wave 21 mid-point. (2) Co-product basic allocation in w21 or defer entirely to w24? Default defer; decide at w21 kickoff.

4.21.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Production order number sequence	legal entity	sequence ref	`production-order`	any w5 sequence	Numbers new orders	Production control parameters → Number sequences	CORU/OPJH number ranges per order type (CO82)
Default reservation policy	site	enum	`released`	none / estimated / released / started	When BOM-line reservations are created	Production control parameters → General → Reservation	Order type-dependent parameters OPL8 (reservation/purchase req. creation)
RAF tolerance %	site	numeric(5,2)	0	0–100	Max over-delivery on RAF	Production parameters → Report as finished → Accept error / over-delivery	Overdelivery tolerance in material master (MARC.UEETO) / order
Over-consumption tolerance %	site	numeric(5,2)	10	0–1000	Picking beyond required qty before duty is needed	Production parameters → Journals → Over-consumption	None native (user status / exit)
Auto-start on feedback	site	bool	true	true/false	Route card on a non-Started order starts it	Production parameters → Status update → Auto-start	Confirmation parameters OPK4 → "Set order status PCNF/CNF automatically"
End jobs on last RAF	site	bool	true	true/false	Sets endJob on all jobs at final RAF	ProdParmReportFinished.EndJob	Final confirmation flag (AFRU.AUERU)
Estimate on create	site	bool	false	true/false	Runs `:estimate` immediately after creation	Production parameters → Estimation on creation	Preliminary costing at order creation (OPL8 costing variant)

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Picking-list proposal default	site	enum	`remaining`	remaining / proportional / none	Default journal proposal mode	ProdParmBOMConsumption → Quantity	Backflush indicator (MARC.RGEKZ / work center)
Route-card proposal basis	site	enum	`remaining`	remaining / proportional	Default hours proposal	ProdParmRouteConsumption	Confirmation proposal (OPK4 → propose quantities/activities)
Posting profile	legal entity	table	per item group × site	item group, site, All	Ledger accounts per posting type	Production posting profiles (ProdPosting)	OBYC keys GBB-VBR/AUF, WIP via results analysis OKG1–OKG4
Variance accounts per type	legal entity	table	per posting profile	account per variance type	GL account for lot-size/qty/price/substitution/rework	ProdPosting variance accounts	Variance categories OKV1, settlement profile
Period control for Production module	legal entity	ref	w14 module `PROD`	open/closed per period	Blocks journal posting in closed periods	Ledger calendar module access	Posting period OB52 (MM/CO)
OCM profile	item group × site	enum	`manual`	automatic / manual / none	FR-21.16 behavior	n/a	OCM change profile (COCM profiles)
OCM change-cost surcharge	tenant	amount or %	0	≥ 0	Added to change cost	n/a	Change cost in OCM profile
Trigger points enabled	site	bool	false	true/false	FR-21.17 evaluation	n/a	OPMP standard trigger points
Split allocation default	site	enum	`proportional`	proportional / keep-on-original	FR-21.15 allocation	ProdSplit dialog	CO02 split dialog
Rework capitalize	legal entity	bool	false	true/false	FR-21.20 cost treatment	n/a	Settlement rule of rework order (CO07)
As-built required	item tracking policy	bool	false	true/false	FR-21.19 End enforcement	n/a	Serial number profile (OIS2) procedure PPAU/PPRL
Status transition duties	tenant	duty map	per 4.21.17	duties	Which duty each `:command` needs	Security roles on ProdTable menu items	Authorization object C_AFKO_AWK per status/activity

4.21.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Production orders list	list inquiry	status, site, product, delivery date range, origin, saved view	order, item, qty, remaining, status, dates, planned/actual cost	`production.order_summary`	ProdTable list page	COOIS (order headers)
Production order detail / status log	drill-down	order id	header, lines, operations, jobs, journals, status log with actor/trace id	order aggregate	ProdTable form, ProdTable status log	CO03, order status log (AFKO status, JEST)
Cost analysis per order	drill-down	order, cost group	planned, actual, variance, variance %, drill to journal entry	`production.cost_view`	ProdCalcTrans "Price calculation" inquiry	CO03 → Cost analysis (KKBC_ORD)
Production journals	list inquiry	journal type, date range, posted/unposted, order	journal number, type, posting date, lines, journal entry id	ProductionJournal	ProdJournalTable inquiries (Picking list, Route card, Job card, RAF)	CO14 confirmation display, MB51 (movement types 261/101)
Operations overview	list inquiry	site, resource, status, date range	order, oper, resource, planned/reported hours, qty good/error, status	ProductionOperation	ProdRoute inquiry "Operations"	COOIS (operations), CO26
WIP balance by order	operational report	site, as-of date, status	order, posted consumption, posted RAF value, WIP balance, GL WIP account delta	WipPosting	SSRS ProdCalcTrans (WIP), "Production orders in process"	KKAO/KKAX WIP calculation list, S_ALR_87013127
Variance report	operational report	period, site, item group, variance type	order, variance type, cost group, amount, % of planned	Variance	SSRS "Production variance" (ProdVariance)	KKS1/KKS2 variance report, S_ALR_87013139
Material shortage / release check	list inquiry	order selection	line, item, required, available, short qty	ProductionBomLine × on-hand (w16)	Release warning list, "Missing materials"	CO24 missing parts list, CO09 availability
Subcontract operations	list inquiry	vendor, status, due date	order, oper, vendor, PO line, qty, price, receipt status	ProductionOperation.subcontract	Vendor resource operations (ProdRoute)	ME2O subcontracting stock monitor, COOIS (external ops)
Order change management worklist	list inquiry	site, OCM status	order, ECO, delta count, change cost, profile	OrderChangeRecord	n/a	COCM / COCM1 worklist
As-built / where-used by serial	drill-down	finished serial or component serial	serial tree, order, posting date	AsBuiltRecord	InventTrans serial trace	IQ09 / serial number history
Production order throughput KPI	KPI	site, period	orders ended, on-time %, average cycle days, scrap %	`production.order_kpi` (w46)	n/a (BI)	SAP MC.* PP information system (MCP1/MCP3)

4.21.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Picking list	Release, journal create, manual print	Header: order, item, qty, site/warehouse, dates; line: line no., item, variant, qty required/issued, location, batch/serial, barcode; footer: signatures, page x/y	PDF (w5 document service), email, print queue	`prod-doc` per legal entity	None statutory; keep with order 7 years where production records are GxP (w24 EBR)
Job card	Release, re-print	Header: order, operation, resource, scheduled start/end; line: job type, planned hours, qty; barcode (GS1 `(90)jobId`) for kiosk scan	PDF, print queue	`prod-doc`	None
Route card	Release	Header: order, route version; line per operation: resource, setup/run time, scrap %, instructions	PDF	`prod-doc`	None

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Production order summary (shop packet)	On demand	Header, BOM lines, operations, documents (w5 attachments list), QR code to order	PDF	`prod-doc`	None
Rework instruction	Rework order release (FR-21.20)	Reference order/batch/serial, defect reason, rework operations	PDF	`prod-doc`	Kept with original order record
Order change notice	OCM apply (FR-21.16)	ECO id, before/after lines and operations, change cost, approver	PDF, email to planner	`prod-doc`	None
Subcontract dispatch note	Release of subcontract operation	Vendor, PO line, components shipped, qty, due date	PDF, email, EDI DESADV via w35	`prod-doc`	Customs / transit documents in w34 where goods cross borders
Electronic order events	Each status change / journal post	CloudEvents on Kafka (4.21.5)	JSON	n/a	Retained 7 years in Iceberg (w46)

4.21.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`wip-reconcile`	daily 02:00 tenant local	legal entity	Compares Σ WipPosting per order vs GL WIP account; writes ReconcileRun row; re-runnable	< 5 min per 50k open orders	Per-entity failure isolates; retries 3×	Delta ≠ 0 pages on-call
`bulk-status-update` (Temporal)	on demand	order id batches of 50	Idempotency-Key per batch; per-order result rows	500 orders < 5 min	Continue-on-error, per-order status	Error rate > 5% warns
`cancel-stale-created-orders`	weekly Sunday 03:00	site	Orders Created > N days (param, 90) without estimate → Cancelled; checkpoint per 1,000	< 2 min	Skip orders with origin PlannedOrder	Count cancelled reported
`ocm-detect`	on ECO event + daily 04:00	site	Marks orders whose snapshot version is superseded; idempotent by (order, ECO)	< 1 min per 10k orders	Logs and continues	Worklist size > 500 warns
`subcontract-accrual`	month-end (w14 close task)	legal entity	Accrues received-not-invoiced subcontract ops; reversal next period	< 3 min	Blocks close task on failure	Close checklist
`order-cost-snapshot`	daily 23:30	legal entity	Snapshots planned vs actual per open order for trend KPI; upsert by (order, date)	< 5 min	Non-blocking	Missing snapshot warns
`auto-end-fully-reported`	daily 22:00 (opt-in)	site	Ends orders with qty_raf = qty and no open journals; param `autoEnd.enabled`	< 5 min	Per-order errors listed	Report emailed to supervisor

4.21.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
View production orders	GET /production/orders*, order list and detail screens, status log	none	Production supervisor, Planner, Cost accountant, Shop floor supervisor
Maintain production orders	POST/PATCH /production/orders, `:estimate`, `:schedule`, order BOM editor (FR-21.18 pre-release)	with *Post production journals* when the same user also approves cost (w39 cost approval)	Production supervisor, Planner
Release production orders	`:release`, `:start`, `:cancel`, `:reset-status`, `:bulk-status`, `:split`, `:merge`	with *End production orders* for the same order (release and settle) — flagged, not blocked	Production supervisor
Post production journals	POST /production/orders/{id}/journals, `:post`, journal editor	with *Maintain production setup* (posting profiles)	Production supervisor, Shop floor supervisor, Kiosk service account (w25)
Post production over-consumption	journal post beyond tolerance	—	Production supervisor
End production orders	`:end`, `end with zero output` (separate privilege), settlement	with *Maintain production setup*; with *Post production journals* flagged	Production supervisor, Cost accountant
End production order with zero output	`:end` when qty_raf = 0	—	Cost accountant
View production costs	GET /production/orders/{id}/costs, cost analysis, variance report	none	Cost accountant, Controller, Production supervisor
Maintain production setup	posting profiles, parameters (4.21.13), OCM profiles, trigger points	with *Post production journals*, *End production orders*	ERP administrator, Cost accountant (posting profiles only)
Maintain order BOM after release	FR-21.18 edits on Released orders	with *Release production orders* flagged	Production engineer
Apply order change	`:apply-change` (FR-21.16)	none	Planner, Production engineer
Create rework orders	FR-21.20 create/release	none	Production supervisor, Quality engineer (w33)

4.21.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
PROD-0001	error	Transition {from} → {to} is not allowed for order {order}.	Illegal state-machine transition	Use an allowed command; see status log	BR-21.1 (`PRODUCTION_ILLEGAL_TRANSITION`)
PROD-0002	error	Posting date {date} is in a closed period for Production.	Period closed in w14 period control	Change posting date or reopen period	BR-21.3 (`PERIOD_CLOSED`)
PROD-0003	error	Order {order} was changed by another user. Reload and retry.	Stale version on PATCH	Reload	BR-21.4
PROD-0004	error	Reset is blocked: inventory from journal {journal} has been consumed downstream.	Reversal would drive on-hand negative	Reverse the downstream issue first	BR-21.5 (`REVERSAL_BLOCKED_DOWNSTREAM`)
PROD-0005	warning	Release with shortages: {n} lines short (first: {item} short {qty}).	On-hand < required at release	Release anyway, or wait/expedite supply	FR-21.4
PROD-0006	error	Release blocked: line {line} item {item} is not released in {legalEntity}.	Unreleased product	Release the product (w8)	FR-21.4
PROD-0007	error	Consumption {qty} exceeds required {required} plus tolerance {pct}%.	Over-consumption without duty	Reduce qty or obtain duty	FR-21.5

Code	Severity	User text	Cause	Remediation	Raised by
PROD-0008	error	RAF quantity {qty} exceeds remaining {remaining} plus tolerance.	Over-delivery	Reduce qty or raise tolerance	FR-21.7
PROD-0009	error	Serial {serial} already exists in this tenant.	Duplicate serial at RAF	Use a unique serial	BR-21.10 (`SERIAL_DUPLICATE`)
PROD-0010	error	Tracking dimension {dim} is required for item {item}.	Missing batch/serial	Enter batch/serial	FR-21.7
PROD-0011	error	Picking would drive on-hand of {item} at {location} negative ({qty}).	Negative inventory not allowed	Receive stock or change location	BR-21.9
PROD-0012	error	Cannot end: {n} journals are unposted.	Open journals at End	Post or delete the journals	FR-21.8
PROD-0013	error	Cannot end with zero output without the duty "End production order with zero output".	qty_raf = 0	Obtain duty or RAF	FR-21.8
PROD-0014	warning	{n} subcontract operations have no vendor invoice; cost will be accrued at PO price.	Uninvoiced subcontract ops at End	Accept accrual or wait for invoice	FR-21.10
PROD-0015	error	Split quantity {qty} is below the minimum order quantity {min} or not a multiple of {multiple}.	Invalid split	Adjust split quantity	FR-21.15
PROD-0016	error	Merge refused: order {order} has posted journals.	Merge precondition	Only merge unposted orders	FR-21.15
PROD-0017	info	A newer BOM/route version exists (ECO {eco}); change cost {amount}.	OCM detection	Preview and apply the change	FR-21.16
PROD-0018	error	Configuration drift: the model change no longer yields this order's configuration.	OCM on configured product	Review configuration manually	FR-21.16 (`OCM_CONFIG_DRIFT`)
PROD-0019	warning	Trigger point {tp} on operation {oper} failed: {reason}. Journal was posted.	Trigger function failure	Fix the trigger setup and re-run	BR-21.13 (`TRIGGER_FUNCTION_FAILED`)
PROD-0020	warning	{n} component serials are not assigned to a finished serial.	Unresolved as-built at End	Assign on the order's as-built tab	FR-21.19
PROD-0021	error	As-built assignment is required for item {item} before End.	`asBuiltRequired` policy	Assign component serials	FR-21.19
PROD-0022	error	Bulk status update: {succeeded} succeeded, {failed} failed; see result list.	Partial batch failure	Open result list and retry failed orders	FR-21.12

4.21.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation under 3× load
Create order (explode 7-level, 2,000 components)	2,000 orders/tenant/day; peak 50/min	150 / 300 / 800 ms	100 creates/s per 3 replicas	2,000 × 365 × ~40 rows × 0.5 KB ≈ 15 GB	ProductionOrder (tenant, legal_entity, status, delivery_date); lines by order_id	Queue creates behind HPA scale-out; p95 ≤ 2 s
Status transition	10,000/day; peak 500/min (bulk)	100 / 500 / 1,000 ms	200/s	StatusLog 10k × 365 × 0.3 KB ≈ 1 GB	advisory lock per order	Bulk batches throttle to 50 concurrent
Journal post (200 lines)	5,000 journals/day; peak 100/min	600 ms / 2 s / 4 s	50 posts/s	ProductionJournal + lines + WipPosting ≈ 20 GB	unique (tenant, idempotency_key); hash partition by legal entity at > 50M rows	Serialize per order; shed with 429 above 3× for > 60 s
Cost view read	20,000/day	80 / 300 / 600 ms	300/s	n/a (projection)	materialized `cost_view` refreshed on journal.posted	Serve last snapshot with staleness header
Order list (saved view, 50 rows)	100,000/day	100 / 300 / 600 ms	500/s	n/a	covering index + keyset pagination	Reduce page size to 25
wip-reconcile	nightly over 50k open orders	n/a	50k orders in < 5 min	ReconcileRun 365 rows/entity	batch by legal entity	Runs serially per entity; alerts if > 15 min
Event publishing (outbox)	50,000 events/day	lag p95 < 1 s	1,000 events/s	Kafka retention 7 days; Iceberg forever	Debezium outbox per DB	Lag alert at 30 s; consumers idempotent

4.21.20 Localization and compliance hooks

- **Period control and fiscal calendars**: journal posting dates obey the legal entity's fiscal calendar (w14) including 4-4-5 and non-January year starts; no production-specific statutory calendars.
- **Currency rounding**: WIP and variance amounts post in the legal entity's accounting currency with the w12 rounding rule (0.01 USD/EUR; 1 JPY; 0.05 CHF cash rounding does not apply to GL postings); EUR reporting-currency amounts carried per line.
- **Statutory production reporting**: none in v1. Country packs (wave 48) may add production statistics extracts (e.g., EU PRODCOM, Brazil Bloco K of SPED Fiscal, which needs consumption per order by NCM code) through the `production.statutory-extract` extension point on the `journal.posted` projection; Bloco K is registered as a wave 48 BR country-pack item.
- **Retention**: production journals and WIP postings are financial records — 7 years (configurable per legal entity, w48 retention policy); GxP plants extend to batch expiry + 1 year via the w24 EBR profile.
- **Language**: document templates (4.21.15) are per-language in the w5 document service; item descriptions print in the site's language with fallback to the tenant default.
- **Units**: metric/imperial dual display from w6 unit conversions; quantities stored in inventory UoM.
- **Privacy**: actor ids on StatusLog are Keycloak subject ids; worker names resolve at read time so GDPR erasure (w48) pseudonymizes without rewriting history.
- **Extension points**: `production.posting-profile.resolve` (OPA policy hook for country-specific account derivation), `production.document.render` (template override per country), `production.statutory-extract`.

4.21.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.22.1 Production control — Production order → Split / Group (ProdSplit) [AX #195]	PARTIAL-21	ADDED as FR-21.15	Split and merge with proportional WIP allocation (BR-21.11); AX stops at Released, Passage allows Started
SAP 13.21 PP — Rework (CO07 / trigger points CO31 / CO33) [SAP #245]	PARTIAL-21	ADDED as FR-21.17, FR-21.20	Trigger points insert rework operations or create rework orders; rework order semantics of CO07

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP 13.21 PP — Order split (CO02 → split), order change management (OCM: COCM / COCM1) [#246]	PARTIAL-21	ADDED as FR-21.15, FR-21.16	Split per FR-21.15; OCM detection, delta preview and change cost per FR-21.16 / BR-21.12
SAP 13.21 PP Customizing — OPMP / OPM trigger points (TTRIG) [#249]	PARTIAL-21	ADDED as FR-21.17	Trigger point definitions on w9 route operations; parameter "Trigger points enabled" in 4.21.13
SAP 13.21 PP Customizing — OCM profiles (change profiles, change cost) [#252]	PARTIAL-21	ADDED as FR-21.16	OCM profile and surcharge parameters in 4.21.13; worklist in 4.21.14
SAP 13.32 LO-VC — CU51 / CU52 / CU55 / CU56 order BOM [#357]	PARTIAL-21	ADDED as FR-21.18	Order-specific BOM/route edits with audit diff and save-as-version; sales-order-level order BOM before order creation stays with wave 27
SAP 13.32 LO-VC Periodic — Re-explosion of configured orders after model change (OCM) [#363]	PARTIAL-21	ADDED as FR-21.16	Re-explosion through the configurator with `OCM_CONFIG_DRIFT` guard; `ocm-detect` job in 4.21.16
SAP 13.32 LO-VC Customizing — Order BOM settings (CU51 processing) [#364]	PARTIAL-21	ADDED as FR-21.18	Duty *Maintain order BOM after release* and OCM exclusion flag replace CU51 processing settings
SAP 13.33 Batch/Serial — Serial numbers at component level (as-built genealogy) [#368]	PARTIAL-21	ADDED as FR-21.19	AsBuiltRecord captured at picking/RAF; kiosk assignment in w25; inquiry in 4.21.14

4.22 Wave 22: Master Planning: MRP, MPS & Forecasting

4.22.1 Scope and business context

Wave 22 gives planners the engine that decides what to make, buy and move. Before it, orders are created by hand. After it, a nightly (or on-demand) plan nets all demand (sales orders, forecasts, safety stock, dependent demand, transfer demand) against all supply (on-hand, purchase orders, production orders, transfers, planned orders) level by level through the BOM, applying coverage groups and lead times, and produces planned purchase, production and transfer orders with action and futures/delay messages, each fully pegged to the demand it serves. Planners firm planned orders into real orders, and a statistical forecast (ETS/Croston) seeds demand with manual override and a consensus workflow. Out of scope: finite capacity (wave 23; wave 22 plans infinite), DDMRP buffers, multi-site network optimization, and ATP (wave 27 reuses the engine's supply picture).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Nightly MRP regeneration takes hours: AX ReqCalcTask on 50k items runs 3–6 h with helper threads contending on ReqTrans; SAP MD01 NEUPL regenerative run 2–4 h; Epicor Kinetic "MRP Full Regen" and Infor LN "Order Planning" routinely 4+ h, so planners work on yesterday's plan	Row-at-a-time netting with database round trips per item; planned orders deleted and recreated in place	In-memory level-by-level engine with sparse buckets, snapshot per PlanVersion, pointer swap; dynamic net-change within 60 s of an order event	50k items 4 h → < 10 min; net change 4 h → 60 s
No live pegging: AX ReqTransCov is only populated during the run and lost on re-run; SAP MD04 pegged requirements (MD09) require multi-level order report and are unavailable for planned orders mid-run; Plex "Where Used/Demand" pegging is single-level	Pegging stored as a transient by-product of the run	Pegging rows persisted per version; `/pegging` explain tree to any depth; "Explain this plan" in the UI	Time to answer "why is this planned order here" 20 min → < 5 s; 100% of qty explained
Exception noise: SAP MD06 exception groups and AX action messages generate tens of thousands of rows planners ignore; Infor LN exception messages cannot be filtered by cause	Messages generated for every supply without fences or root-cause grouping; no deduplication	Action/futures fences per coverage group; messages grouped by root cause (one delayed PO → one message with propagated effects)	Messages per planner per run 5,000 → < 300; 90% of messages actionable (survey)
Non-deterministic results: AX plans differ run to run under helper threads; SAP parallel MRP (OMIQ) ordering varies, so planners cannot diff two runs	Parallel processing without stable ordering	Deterministic sort keys and plan hash; version-to-version diff API	Identical input → identical hash 100% of runs; diff of two versions in < 2 s
Forecast re-keying: AX ForecastSales grid vs D365 demand forecasting Excel round-trips; SAP MD61 PIR upload by LSMW; Epicor forecast import by CSV	Forecasting done outside the ERP and pasted back	Statistical baseline in the `forecast` service, in-grid override, consensus workflow, published as a version	Forecast cycle 2 days → 2 h; zero CSV round-trips

4.22.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Plan definition	ReqPlan (static/dynamic), ReqPlanVersion	MRP area, planning run parameters (MD01/MD02 processing keys NETCH/NEUPL)	Plan, PlanVersion	full
Coverage group	ReqGroup (ReqPOType: Requirement/Period/MinMax/Manual)	MRP type (PD/VB/ND), lot-sizing procedure (T438A: EX, TB, FX, WB), MRP group	CoverageGroup	full
Item coverage overrides	ReqItemTable	MARC MRP views 1–4 (DISMM, DISLS, EISBE)	ItemCoverage	full
Requirement transactions	ReqTrans (ReqRefType), ReqPO (planned orders)	MDTB (MRP list), PLAF (planned orders), MD04 stock/requirements list	Requirement, PlannedOrder	full
Pegging	ReqTransCov	Pegged requirements (MD09), order report	Pegging	improved (explain tree API)
Action messages	ReqTrans action (Advance/Postpone/Decrease/Increase/Derived)	Exception messages (MD06, groups 1–8)	ActionMessage	full
Futures/delay messages	ReqTrans futures date	Exception 10/15 (reschedule in/out), 30 (plan process according to schedule)	DelayMessage	full
Low-level codes	BOMCalcLevel / ReqCalcTask level loop	Low-level code (MARA.DISST)	Item.low_level_code	full
Forecast	ForecastSales, ForecastPurch, reduction keys, D365 demand forecasting (Azure ML)	PIR (MD61), consumption mode (MARC.VRMOD), flexible planning/SOP	ForecastLine, forecast service	full
MPS	Master scheduling via item coverage + planning time fence	MPS (MD40), MPS items (MRP type M0–M4)	plan scope `mps` with time fence	full
Firming	ReqTransPO firming to ProdTable/PurchTable	MD14/MD15 convert planned order	`:firm`	full
Intercompany planned orders	Intercompany master planning	Special procurement key 40/70	planned transfer orders only; intercompany in w40	partial

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Simulation / long-term planning	ReqPlan static plans as copies (ReqPlanVersion), forecast plan, D365 planning optimization "simulation"	LTP MS01–MS08, MS31–MS70 planning scenarios, simulative MRP, PURCHIS transfer; S/4 pMRP; APO /SAPAPO/MC01 planning versions	Plan kind `simulation` with ScenarioOverride (FR-22.15)	full
DDMRP buffers / priority-based planning	D365 priority-based planning (planning priority on coverage, ReqPlanningPriority)	S/4 DDMRP: buffer positioning, buffer profiles (PPH_DD_*), buffer level calculation, replenishment planning Fiori apps	Coverage code `ddmrp`, DdmrpBuffer, BufferProfile (FR-22.16)	full
Quota arrangements	Item coverage vendor + "intercompany/transfer" source; no percentage split	MEQ1/MEQ3/MEQ4/MEQ6/MEQ8, quota arrangement (EQUK/EQUP), source list rules (TMQ1)	QuotaArrangement (FR-22.17)	full
MRP areas (sub-site planning)	Warehouse-level coverage (ReqItemTable by InventDim warehouse)	MRP areas OMIZ (MDLV, MDLG, MDLW), storage location and subcontractor MRP areas	Planning scope `warehouse` and `subcontractor` (FR-22.18)	full
Planning strategies	Coverage code + forecast reduction keys (ReqReduceKey) + pegged supply BOM lines	Strategy groups OPPS/OPPT (T461S, T461P), requirements types/classes (OVZG/OVZH), consumption mode MARC.VRMOD	PlanningStrategy (FR-22.19)	full
Shelf-life planning	Shelf-life on InventBatch, master planning "Use shelf life" (PdsShelfLife on coverage)	PP/DS shelf-life planning (SAP_PP_SL001)	Shelf-life aware pegging (FR-22.20)	full
Supply schedule workbench	ReqSupplySchedule (Supply schedule form, R3), SSRS supply schedule report	MD04 period totals, MD07 collective list	Supply schedule screen (FR-22.21)	full
Inventory forecast	ForecastInvent (Inventory forecast inquiry)	MD04 projected available stock, MS04 simulation	InventoryForecast projection (FR-22.22)	full

4.22.3 Functional requirements

- **FR-22.1 Plan versions** — Must. The system shall run a plan as a batch job (Temporal) that snapshots inputs into a PlanVersion and writes outputs there, swapping the "current" pointer atomically on success.
  - Notes: Two plan kinds, `static` (nightly, full regeneration) and `dynamic` (net-change, triggered by order events within 60 s, limited to affected items). Regeneration deletes unfirmed planned orders of the version; firmed ones are supply. Max 20 retained versions per plan; older ones purged by nightly job.
- **FR-22.2 Coverage groups** — Must. The system shall support coverage codes Requirement (lot-for-lot), Period (coverage time fence in days, orders grouped per period), Min/Max (reorder to max when PAB < min) and Manual (no planned orders, only messages), with safety stock, positive/negative days, coverage time fence, firming time fence, freeze time fence, action and futures message settings.
  - Notes: Defaults: coverage time fence 100 days, negative days 1, positive days 0 (ignore existing supply earlier than requirement when 0 means unlimited; AX semantics documented in BR-22.4), firming fence 0, freeze fence 0. SAP mapping: PD→Requirement/Period; VB→Min/Max with reorder point = min; ND→Manual.
- **FR-22.3 Netting algorithm** — Must. The system shall compute net requirements per BR-22.1 and produce planned orders with order type by the item's default order settings (purchase, production, transfer, kanban).
- **FR-22.4 Lot sizing** — Must. The system shall apply minimum, maximum, multiple and standard order quantities from default order settings per site, and period grouping.
  - Notes: After rounding up to multiple, if the result exceeds maximum, split into ⌈qty/max⌉ orders. Max 1,000 splits per requirement.
- **FR-22.5 Lead time and calendars** — Must. The system shall offset dates by purchase lead time (vendor-specific from trade agreements, else item), production lead time (route-based: Σ operation times at infinite capacity, else item lead time), transfer lead time, plus safety margins (receipt, issue, reorder margins) using the site's working calendar.
- **FR-22.6 Pegging** — Must. The system shall record Pegging rows linking each unit of supply to the demand it covers, FIFO by requirement date then by priority (sales order > forecast > safety stock), and expose an explain tree.
- **FR-22.7 Action and delay messages** — Must. The system shall generate Advance, Postpone, Decrease, Increase, Cancel and Derived actions for existing supply, and Futures/Delay messages when a planned date falls before today (plus the delay propagates upward to the demand).
- **FR-22.8 Forecast and reduction** — Must. The system shall accept forecast lines by item/site/period (and allocation keys) and reduce them by sales orders using reduction principles: none, percent per period, transactions (forecast minus open orders in the period), transactions-dynamic period.
- **FR-22.9 Statistical forecast** — Should. The system shall generate a baseline forecast per item/site from 24+ months of history using ETS (seasonal/non-seasonal selected by AICc) or Croston/TSB for intermittent items (ADI > 1.32), via the Python `forecast` service, with MAPE/wMAPE reported per item.
  - Notes: Batch for 50k items < 15 min. Manual override per cell; consensus workflow (Temporal approval) locks the forecast version.
- **FR-22.10 Firming** — Must. The system shall firm a planned order into a ProductionOrder (`:estimate` optional), PurchaseOrder line (grouped by vendor/date per setting) or TransferOrder, preserving pegging to the firmed supply.
- **FR-22.11 Planned order maintenance** — Must. The system shall allow planners to change quantity/date/vendor on a planned order, approve it (becomes firmed on next run), or delete it.
- **FR-22.12 Net requirements view** — Must. The system shall show per item/site the MD04-style time-phased list: period, gross requirements, scheduled receipts, planned receipts, PAB, net requirement, with drill to pegging.
- **FR-22.13 Plan scope** — Should. The system shall support plan filters by site, item group, planner, low-level-code range, and `mps` scope (MPS items only; MRP uses MPS output as fixed supply).
- **FR-22.14 Determinism** — Must. The system shall produce byte-identical outputs for identical inputs; the plan hash is emitted in `plan.completed`.
- **FR-22.15 Simulation plans and planning scenarios** — Should. The system shall run plans of kind `simulation` that read the same snapshot inputs as an operational plan but apply a ScenarioOverride (demand source = a named forecast version or demand program, demand multipliers by item group, added/removed sales demand, BOM/route version effective date, lead-time and capacity overrides, starting inventory as of a date), write to their own PlanVersion, never firm or emit action messages to operational consumers, and offer evaluations: rough-cut capacity load per resource group and period (using route-based times, or w23 finite load with flag), purchase volume by vendor × period (value at trade-agreement price), and inventory value projection.
  - Notes: Up to 10 active scenarios per plan; each scenario may run as many times as needed (versions retained per FR-22.1). Simulation results can be copied into a forecast version (LTP "transfer to PURCHIS"/vendor information) and published to suppliers via the w20 supplier portal as a non-binding forecast. pMRP-style rough simulation is the scenario run with `bucket=week` and `bomDepth=1..n` limit for speed (≤ 2 min for 50k items at depth 2). Planned orders in a simulation version carry `simulation = true` and are rejected by `:firm` with 409 `PLAN_SIMULATION_NOT_FIRMABLE`. Version-to-version diff (FR-22.14 hash, diff API) works across a scenario and the current operational version.
- **FR-22.16 DDMRP buffers and priority-based planning** — Should. The system shall support coverage code `ddmrp` for items marked as decoupling points (strategic buffers), computing per item/site: average daily usage (ADU, rolling past and/or forward window), decoupled lead time (DLT), buffer zones per BufferProfile (lead-time factor, variability factor, minimum order cycle, minimum order qty) — red base = ADU × DLT × LTF, red safety = red base × VF, yellow = ADU × DLT, green = max(ADU × DLT × LTF, MOQ, ADU × order cycle) — and planned orders by the net-flow equation (BR-22.11), with planning priority = net-flow position / top of green (TOG) used to sort planned orders and the planner worklist.

    - Notes: Buffer positioning is set per item coverage (`decouplingPoint = true`); DLT is the longest unprotected lead-time path through the BOM to the next decoupling point or purchase (computed from w9 BOM and lead times). Order-spike threshold (default 50% of red zone) and horizon (default DLT) qualify spikes into qualified demand. Dynamic adjustment factors (planned adjustment factor per period, e.g., seasonal ramp) multiply ADU. Non-buffered items below a decoupling point are planned by standard MRP. Buffer recalculation runs nightly (4.22.16). Priority-based planning (D365) uses the same priority field: coverage groups with `priorityBased = true` replace date-ordering with priority ordering for planned-order generation within the horizon.
- **FR-22.17 Quota arrangements** — Should. The system shall split net requirements of an item/site across sources defined in a QuotaArrangement (valid from/to; sources: vendor, own production with production version, transfer from site/warehouse; quota per source; quota base qty; maximum quantity per period; minimum lot per source) by quota rating = (allocated qty + base qty) / quota, assigning each planned order to the source with the lowest rating, optionally splitting a single requirement across sources (`split = true`, SAP quota "splitting quota").
    - Notes: Allocated qty accumulates from firmed and open orders created against the arrangement (purchase orders, production orders, transfer orders carry `quota_arrangement_id`). Sources with quota 0 are used only when all others are blocked (e.g., trade agreement expired). Quota arrangement analysis shows planned vs actual split by source for the validity period. Takes precedence over the item coverage default vendor; trade-agreement lead times apply per chosen source.
- **FR-22.18 Planning scope by warehouse and subcontractor (MRP areas)** — Should. The system shall plan items at site level (default) or at warehouse level when item coverage exists for that warehouse (MRP area), with warehouse-level demand and supply netted separately and replenished from the site by planned transfers or by direct purchase/production per the warehouse's coverage; and shall plan components provided to a subcontractor (w21 FR-21.10 vendor operations) in a subcontractor planning scope per vendor, creating planned transfer (provision) orders to the vendor location.
    - Notes: A warehouse in its own planning scope is excluded from site-level netting. Subcontractor scope requires the vendor to have a w16 vendor-location warehouse (consignment-out type). Planning scope is a field on ItemCoverage (`site`, `warehouse`, `subcontractor`).
- **FR-22.19 Planning strategies** — Should. The system shall assign a PlanningStrategy to item coverage that controls how forecasts and sales demand interact: `mts-net` (SAP 10: forecast is demand, sales orders reduce forecast per reduction principle), `mts-gross` (SAP 11: forecast planned regardless of on-hand), `mto` (SAP 20: no forecast; supply created per sales line and marked/pegged 1:1, segregated as order-specific stock when w16 sales-order stock dimension is active), `assemble-to-order-final` (SAP 40: forecast at the finished item consumed by sales orders with consumption mode backward/forward/backward-then-forward and consumption periods in days), `plan-without-final-assembly` (SAP 50/52: forecast plans components; finished item supply created only on sales demand), `subassembly` (SAP 70: forecast on the subassembly consumed by dependent demand).
    - Notes: Consumption window: backward days (default 30) and forward days (default 0); unconsumed sales demand beyond the window is added on top of forecast. `mto` and `plan-without-final-assembly` set `is_marking = true` on Pegging so the supply cannot be re-pegged by later runs. Strategies map from SAP strategy groups (T461P) on migration; AX customers map ReqReduceKey + coverage code combinations to the nearest strategy (mapping table in 4.22.11 addendum below).
- **FR-22.20 Shelf-life aware pegging** — Should. For items with shelf-life tracking (w24 FR-24.6) the planning engine shall treat batch on-hand and supply as usable for a demand only if expiry (or best-before, per item setting) – demand date ≥ the demand's required remaining shelf life (customer sellable days from w24 FR-24.19, else item default), creating planned supply for the shortfall and a `Expiring supply` message for batches that will expire unpegged.
- **FR-22.21 Supply schedule workbench** — Should. The system shall provide a period-bucketed (day/week/month) grid per item × site for a selection of items showing demand, supply, planned supply, PAB and safety stock per period, with inline creation of planned orders in a period cell, quantity edits, firming, and a printable supply schedule report.
- **FR-22.22 Inventory forecast** — Could. The system shall expose an InventoryForecast projection per item/site/period: projected on-hand = PAB at period end of the current plan version (or a chosen scenario), with value at current cost, and days-of-supply = projected on-hand / average daily demand of the next 30 days.

4.22.4 Data model

CoverageGroup: id, tenant_id, code (text, unique), coverage_code (enum), coverage_time_fence_days (int, default 100), negative_days, positive_days (int), safety_stock_policy (enum: today, first-issue), firming_fence_days, freeze_fence_days, action_fence_days, futures_fence_days, period_days (int, Period only), rescheduling_margin_days, generate_actions (bool), generate_futures (bool).

ItemCoverage: id, item_id, site_id, warehouse_id (nullable), coverage_group_id, min_qty, max_qty, safety_margin_receipt_days, lead_time_days (override), vendor_id (default), planned_order_type (enum).

PlanVersion: id, plan_id, number (int), kind (static/dynamic), snapshot_at (timestamptz), status (running/complete/failed/current), items_planned, planned_orders_count, messages_count, duration_ms, plan_hash (bytea, sha256).

Requirement (ReqTrans): id, plan_version_id, item_id, variant_id, site_id, warehouse_id, ref_type (enum: sales-line, forecast, safety-stock, bom-line, transfer-issue, on-hand, purchase-line, production-order, transfer-receipt, planned-order), ref_id, direction (demand/supply), qty (numeric 18,6, +supply/–demand), requirement_date (date), futures_date (date, nullable), low_level_code (int), priority (int).

PlannedOrder (ReqPO): id, plan_version_id, number, item_id, site_id, warehouse_id, order_type (purchase/production/transfer/kanban), qty, order_date, delivery_date, vendor_id, from_warehouse_id, status (enum: unprocessed, approved, firmed), firmed_target_type, firmed_target_id, bom_version_id, route_version_id, lot_rule_applied (text).

Pegging (ReqTransCov): id, supply_requirement_id, demand_requirement_id, qty, is_marking (bool).

ActionMessage: id, requirement_id, action (enum), suggested_qty, suggested_date, reason, delay_days.

ForecastLine: id, model_id, item_id, site_id, period_start, period_end, qty, source (statistical/manual/consensus), reduced_qty, version.

Invariants: Σ Pegging.qty per supply ≤ supply qty; Σ Pegging.qty per demand = demand qty when fully covered (else a `Cancel`-free shortage message exists); PAB(t) = PAB(t–1) + Σ supply(t) – Σ demand(t) for every period; only one PlanVersion per plan has status `current`.

Indexing: Requirement (plan_version_id, item_id, site_id, requirement_date); Pegging (demand_requirement_id), (supply_requirement_id); PlanVersion rows partitioned by plan_version_id (one partition per version, dropped on purge — avoids 50M-row deletes).

ScenarioOverride (FR-22.15): id, plan_id, name, demand_source (enum: operational, forecast-version, demand-program), forecast_version_id, demand_multipliers (jsonb: [{itemGroupId, factor}]), added_demand (jsonb lines), bom_effective_date (date), lead_time_overrides (jsonb), capacity_mode (enum: none, rough-cut, finite), start_inventory_as_of (date), bucket (enum day/week/month), bom_depth_limit (int, nullable), active (bool).

BufferProfile / DdmrpBuffer (FR-22.16): BufferProfile(id, code, item_type (manufactured/purchased/distributed), lead_time_factor numeric(5,3), variability_factor numeric(5,3), order_cycle_days int, spike_threshold_pct numeric(5,2) default 50, spike_horizon_days int nullable = DLT); DdmrpBuffer(id, item_id, site_id, warehouse_id, profile_id, adu numeric(18,6), adu_window_past_days int default 90, adu_window_future_days int default 0, dlt_days numeric(8,2), daf numeric(6,3) default 1, red_base, red_safety, yellow, green, top_of_red, top_of_yellow, top_of_green (numeric 18,6), calculated_at, manual_override (bool)).

QuotaArrangement / QuotaSource (FR-22.17): QuotaArrangement(id, item_id, site_id, valid_from, valid_to, split (bool), min_split_qty); QuotaSource(id, arrangement_id, source_type (vendor/production/transfer), vendor_id, production_version (bom_version_id, route_version_id), from_warehouse_id, quota numeric(9,4), base_qty numeric(18,6), allocated_qty numeric(18,6), max_qty_per_period, period_days, priority int).

PlanningStrategy (FR-22.19): code (enum per FR-22.19), consumption_mode (backward/forward/backward-forward), backward_days int default 30, forward_days int default 0, uses_marking (bool), segregated_stock (bool); ItemCoverage gains planning_strategy_code, planning_scope (site/warehouse/subcontractor), decoupling_point (bool), buffer_profile_id, required_shelf_life_days, priority_based (bool). PlannedOrder gains simulation (bool), planning_priority numeric(7,4), quota_source_id.

Invariants (additions): for a DdmrpBuffer, top_of_red = red_base + red_safety < top_of_yellow < top_of_green; for an active QuotaArrangement Σ quota of non-zero sources > 0 and validity periods of arrangements for the same item/site do not overlap; no PlannedOrder with simulation = true has status firmed.

4.22.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/planning/plans/{planId}:run	Start plan (static/dynamic, scope filter)	Idempotency-Key; one running version per plan	Run master planning
GET	/api/v1/planning/plans/{planId}/versions/{v}	Version status, counts, hash	n/a	View planning
GET	/api/v1/planning/planned-orders?planId&status&site&itemGroup	List/filter planned orders	n/a	View planning
PATCH	/api/v1/planning/planned-orders/{id}	Change qty/date/vendor; approve	ETag	Maintain planned orders
POST	/api/v1/planning/planned-orders:firm	Firm up to 1,000 ids with grouping options	Idempotency-Key	Firm planned orders
GET	/api/v1/planning/net-requirements?itemId&siteId&planId	MD04-style time-phased view	n/a	View planning
GET	/api/v1/planning/planned-orders/{id}/pegging	Explain tree (depth ≤ 20)	n/a	View planning
GET	/api/v1/planning/messages?planId&type	Action/delay messages	n/a	View planning
POST	/api/v1/planning/forecasts/{modelId}:generate	Statistical forecast batch	Idempotency-Key	Maintain forecasts
PUT	/api/v1/planning/forecasts/{modelId}/lines	Bulk upsert manual lines	full replace per item/site/period	Maintain forecasts
GET	/api/v1/planning/coverage-groups	Setup	n/a	View planning setup

Event	Trigger	Payload fields	Consumers
`dev.passage.planning.plan.started.v1`	run begins	planId, versionId, kind, scope	analytics
`dev.passage.planning.plan.completed.v1`	version becomes current	planId, versionId, itemsPlanned, plannedOrders{purchase,production,transfer}, messages, durationMs, planHash	production, purchasing, scheduling
`passage.planning.planned-order.firmed.v1`	firm	plannedOrderId, orderType, targetId, qty, dates	production, purchasing, inventory
`dev.passage.planning.message.raised.v1`	message created	requirementId, refType, refId, action, delayDays	purchasing, sales (w27), notifications
`dev.passage.planning.forecast.published.v1`	consensus approved	modelId, version, items, horizonPeriods	planning run

4.22.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Master planning workspace	Run plan, view last run, tiles: delayed orders, actions, planned orders by type	Plan, version, status, duration, counts	View planning
Planned orders	Filter, approve, firm (grouped), change, delete, bulk	Number, item, site, type, qty, order/delivery date, vendor, status, delay	Maintain planned orders
Net requirements (MD04)	Time-phased grid, pegging drill, simulate (what-if qty change)	Period, GR, SR, PR, PAB, net req, messages	View planning
Action messages	Apply action to the source order (advance/postpone PO date), dismiss	Action, order, qty, date, suggested date, delay	Maintain planned orders
Forecast grid	Edit cells, generate statistical, submit consensus	Item, site, period columns, source badge, MAPE	Maintain forecasts
Coverage groups / item coverage	Setup	per data model	Maintain planning setup

Interaction rules: planned-orders grid supports keyboard range selection and `F` to firm; cell edits in the forecast grid validate ≥ 0 immediately; the net-requirements grid is virtualized to 365 periods × 20 columns; pegging tree lazy-loads children.

4.22.7 Business rules and edge cases

- **BR-22.1 Netting algorithm.** Per plan run:
  1. Snapshot inputs (on-hand by dimension at physical-available, open orders, forecasts after reduction, safety stock, item coverage, BOM/route versions as of each requirement date, calendars) into the PlanVersion.
  2. Assign low-level codes: LLC(item) = max over all active BOMs of the depth at which the item appears (finished items 0). Recomputed when any BOM changes (w9 event). Cycles are rejected at BOM activation.
  3. For LLC = 0 … max: for each item/site in that level, sort requirements by date then priority then ref id (determinism).
  4. Build time buckets by day; PAB_0 = on-hand – safety stock (policy `today`) or on-hand (policy `first-issue`, safety stock added as a demand on the first issue date).
  5. For each day t: PAB_t = PAB_{t–1} + SR_t + PR_t – GR_t. If PAB_t < 0 (Requirement/Period) or PAB_t < min (Min/Max), net requirement N = –PAB_t (or max – PAB_t).
  6. Lot-size N: Requirement → N; Period → aggregate N over the next `period_days`; Min/Max → max – PAB; then apply min/multiple/max/standard qty.
  7. Create planned order with delivery date t – receipt margin; order date = delivery date – lead time (working days); if order date < today → futures_date = today + lead time; raise Delay message with delay_days and propagate the futures date to pegged demand.
  8. Peg the planned order to the demand(s) that caused it, FIFO.
  9. Explode planned production orders: for each BOM line, create dependent demand Requirement (ref_type bom-line) on the item at date = order start (or operation start when route-based), qty per BR-21.2; these appear at a higher LLC and are processed in step 3 of that level.
  10. For existing supply earlier than its earliest pegged demand by more than positive_days (0 = unlimited) raise Postpone; later than demand by more than negative_days raise Advance; unpegged supply beyond the fence raises Cancel/Decrease.

11. Write outputs, compute plan hash over (planned orders, pegging, messages) sorted, mark version current.

- **BR-22.2 Worked example (3-level BOM, weekly buckets, today = Monday of week 1).** FG-100 (LLC 0) uses 2 × SA-200 (LLC 1) and 1 × RM-300; SA-200 uses 3 × RM-300, so RM-300 has LLC 2. Coverage: all Requirement; FG-100 lead time 1 wk, safety stock 10, on-hand 20; SA-200 lead time 1 wk, multiple 100, on-hand 50; RM-300 purchase lead time 2 wk, minimum 500, on-hand 200, open PO 300 due week 2. Demand: sales order 120 FG week 4; forecast 80 FG week 5.
  - FG-100: PAB_0 = 20 − 10 = 10. Week 4: 10 − 120 = −110 → planned production PP-1 110, due wk 4, start wk 3. Week 5: 0 − 80 = −80 → PP-2 80, due wk 5, start wk 4. Dependent demand: SA-200 220 wk 3, 160 wk 4; RM-300 110 wk 3, 80 wk 4.
  - SA-200: PAB_0 = 50. Week 3: 50 − 220 = −170 → multiple 100 → PP-3 200, due wk 3, start wk 2; PAB 30. Week 4: 30 − 160 = −130 → PP-4 200, due wk 4, start wk 3; PAB 70. Dependent RM-300: 600 wk 2, 600 wk 3.
  - RM-300: PAB_0 = 200. Week 2: 200 + 300 (PO) − 600 = −100 → min 500 → planned purchase PO-1 500 due wk 2, order date wk 0 < today → futures date wk 3, Delay message 7 days; PAB 400. Week 3: 400 − 600 − 110 = −310 → PO-2 500 due wk 3, order date wk 1 = today (no delay); PAB 190. Week 4: 190 − 80 = 110.
  - Propagation: PP-3 pegged to PO-1 inherits futures wk 3 → PP-3 cannot finish before wk 4 → PP-1 futures wk 5 → Delay message on sales order line (requested wk 4, expected wk 5, delay 7 days). Pegging explains 100% of each planned order: PP-1 → SO 110 of 120 (10 from on-hand after safety), PO-1 → PP-3 (100 of 500; 400 surplus unpegged, no Cancel because within coverage time fence).
- **BR-22.3 Forecast reduction.** `transactions` principle: reduced forecast for a period = max(0, forecast − Σ sales order qty in the period); periods are forecast-line periods, not calendar months.
- **BR-22.4 Fences.** Requirements inside the freeze fence generate no planned orders; inside the firming fence, planned orders are auto-firmed; action messages are generated only inside the action fence; futures messages only inside the futures fence.
- **BR-22.5 Safety stock replenishment date.** With policy `today`, the safety-stock planned order is dated today + lead time; with `first-issue`, the first demand date.
- **BR-22.6 Dynamic plan scope.** A dynamic run replans the changed item and all items whose LLC is greater and that are reachable through its BOM; it never changes planned orders outside that set.
- **BR-22.7 Transfer planning.** Warehouse-level coverage with `transfer` order type creates demand at the from-warehouse at delivery date − transfer lead time.
- **BR-22.8 Calendar edge cases.** Lead-time offsets skip closed days; if a date falls on a closed day it moves backward (supply) or forward (demand) to the nearest open day.
- **BR-22.9 Firming concurrency.** Firming is a saga: create target order → mark planned order firmed → on failure compensate by deleting the target; a planned order already firmed returns 409.
- **BR-22.10 Quantity precision.** Requirements carry the item's inventory UoM to 6 decimals; forecast lines in forecast UoM converted at load.
- **BR-22.11 DDMRP net-flow equation.** Net-flow position NFP = on-hand + open supply (inside DLT) − qualified demand, where qualified demand = past-due demand + demand due today + spikes (single-day demand > spike threshold × red zone inside the spike horizon). If NFP ≤ top of yellow, a planned order is created for TOG − NFP (rounded up to MOQ/multiple), dated today + DLT. Planning priority = NFP / TOG; zone shown red ≤ TOR, yellow ≤ TOY, else green. Worked example: ADU 10, DLT 10 days, LTF 0.5, VF 0.5, MOQ 0, order cycle 7 days → red base 50, red safety 25, TOR 75; yellow 100, TOY 175; green max(50, 0, 70) = 70, TOG 245. On-hand 90, open supply 60, qualified demand 20 → NFP 130 ≤ 175 → order 115, priority 53.06% (yellow).
- **BR-22.12 Quota rating.** rating_s = (allocated_s + base_s) / quota_s; ties broken by priority then source id. Example: vendor A quota 60, allocated 300; vendor B quota 40, allocated 100; requirement 100 → ratings A 5.00, B 2.50 → B is chosen; after allocation B = 200 → 5.00. With split on, the requirement is divided so that post-allocation ratings are as equal as possible, subject to min_split_qty.
- **BR-22.13 Strategy consumption.** For `assemble-to-order-final`, a sales line on date d consumes forecast in [d − backward_days, d] (latest first), then in (d, d + forward_days] (earliest first); unconsumed sales qty adds to gross requirements; forecast never goes negative.
- **BR-22.14 Simulation isolation.** A simulation version reads the operational snapshot by reference (copy-on-write) and writes only its own partitions; operational `plan.completed` consumers ignore events with `simulation = true`; firming, action messages to purchasing and delay messages to sales are suppressed.
- **BR-22.15 Shelf-life usability.** A batch with expiry e is usable for demand on date d only if e − d ≥ required remaining shelf life r; planned production supply is assumed to have expiry = delivery date + shelf_life_days.

4.22.8 Acceptance criteria

- **AC-22.1** (FR-22.3, 22.4, 22.5) Given the BR-22.2 fixture, when the static plan runs, then the planned orders PP-1…PP-4, PO-1, PO-2 exist with the stated quantities and dates and a 7-day delay message on the sales line.
- **AC-22.2** (FR-22.6) Given any planned order in the 40-fixture set, when I read its pegging, then Σ pegged qty = planned qty.
- **AC-22.3** (FR-22.14) Given the 50k-item dataset, when the plan runs three times, then all three plan hashes are equal and each run completes in < 10 min.
- **AC-22.4** (FR-22.2) Given a Min/Max item with min 100, max 400 and PAB 90, when planned, then one planned order of 310 is created.
- **AC-22.5** (FR-22.7) Given a PO due 10 working days before its earliest pegged demand and positive_days = 5, when planned, then a Postpone message suggests the demand date − receipt margin.
- **AC-22.6** (FR-22.8) Given forecast 100 and sales orders 30 in a period with `transactions` reduction, then gross requirement from forecast is 70.
- **AC-22.7** (FR-22.9) Given 24 months of intermittent history (ADI 2.1), when the statistical forecast runs, then the Croston/TSB model is chosen and MAPE is reported.
- **AC-22.8** (FR-22.10) Given 3 approved planned purchase orders for one vendor on one date with grouping on, when firmed, then one PO with 3 lines is created and the planned orders show status firmed with target id.
- **AC-22.9** (FR-22.1) Given a running static version, when a second run is requested, then 409 `PLAN_ALREADY_RUNNING`.
- **AC-22.10** (FR-22.12) Given item RM-300 in the example, when I open net requirements, then the PAB column shows 400, 190, 110 for weeks 2–4.
- **AC-22.11** (FR-22.15, BR-22.14) Given scenario "Demand +20% on group FG", when the simulation runs, then its planned production quantities for FG items are ≥ the operational plan's, no ActionMessage reaches purchasing, and `:firm` on any of its planned orders returns 409 `PLAN_SIMULATION_NOT_FIRMABLE`.
- **AC-22.12** (FR-22.16, BR-22.11) Given the BR-22.11 buffer, when planned, then one planned order of 115 is created with priority 0.5306 and zone yellow.
- **AC-22.13** (FR-22.17, BR-22.12) Given the BR-22.12 quota arrangement, when a requirement of 100 is planned, then the planned purchase order is for vendor B and B's allocated qty becomes 200 after firming.
- **AC-22.14** (FR-22.19, BR-22.13) Given strategy `assemble-to-order-final` with backward 30 days, forecast 100 on 2026-11-30 and a sales line of 40 on 2026-11-20, when planned, then gross requirement is 100 (forecast 60 + sales 40), not 140.
- **AC-22.15** (FR-22.20, BR-22.15) Given a batch expiring in 20 days and a demand 5 days out requiring 30 days remaining shelf life, when planned, then the batch is not pegged and a planned order covers the demand.
- **AC-22.16** (FR-22.18) Given item coverage at warehouse WH-2 in its own scope, when the plan runs, then WH-2 demand is covered by a planned transfer from the site warehouse and WH-2 on-hand is not netted against site demand.

4.22.9 Test plan and fixtures

Fixtures: `mrp-textbook-40` (APICS CPIM-style scenarios: lot-for-lot, FOQ, POQ, safety stock, scrap, phantom, multi-source, delays; expected planned orders as JSON); `mrp-3level` (BR-22.2); `mrp-50k` (generated: 50k items, 5-level BOMs, 12 monthly periods, 200k sales lines); `forecast-history-2k` (24 months, mix of

smooth/intermittent); `frepple-compare` (shared dataset exported to frePPLe, expected within ±1 day and ±1% qty).

Test	Type	Target	Pass criterion
Textbook scenarios	unit	engine	40/40 match
PAB identity	property	engine	PAB recurrence holds for every item/day
Pegging completeness	property	pegging	Σ peg = planned qty for all planned orders
Determinism	integration	plan run	3 identical hashes
Scale	performance	50k plan	< 10 min, memory < 8 GB
Firming saga	contract	planning ↔ production/purchasing	compensations on injected failure
Forecast models	unit (Python)	statsforecast wrapper	model selection matches fixture
frePPLe comparison	integration (optional)	shared dataset	within tolerance

Reconciliation: Σ supply – Σ demand per item/site over the horizon = closing PAB; planned production orders firmed = ProductionOrders with origin PlannedOrder.

4.22.10 Deployment and operations

Helm charts `planning` (Go, 2 replicas API + 1 planner worker pool with 8 CPU / 16 GB request, Temporal worker), `planning-db` (CloudNativePG, 200 GB), `forecast` (Python, 2 replicas). Config: `PLANNING_MAX_VERSIONS` (20), `PLANNING_DYNAMIC_DEBOUNCE_SECONDS` (60), `PLANNING_HORIZON_DAYS` (365). Flags: `planning.dynamic`, `planning.statistical-forecast`, `planning.mps-scope`. Scheduled: nightly static plan 01:00 site-local; weekly forecast generation Sunday 22:00; version purge daily. SLOs: 50k-item plan < 10 min; planned-order list p95 < 300 ms; pegging read p95 < 500 ms. Alerts: plan failed or > 20 min (page); dynamic-plan lag > 5 min (warn). Runbook: "plan stuck" — inspect Temporal workflow, cancel, re-run; "hash mismatch" — compare sorted outputs between versions. Rollback: prior digest; versions are immutable so rollback never corrupts plans.

4.22.11 Migration from AX / SAP

Source	Passage entity	Notes
ReqGroup (AX) / T438A + MRP type config (SAP)	CoverageGroup	ReqPOType→coverage_code; SAP lot-size EX→Requirement, TB→Period, FX→fixed qty via ItemCoverage standard qty, WB→Period(7)
ReqItemTable / MARC MRP views	ItemCoverage	DISMM PD/VB/ND; EISBE→safety; DISLS; PLIFZ→lead time
ForecastSales / PIR (PBIM/PBED)	ForecastLine	periods normalized to weeks or months
ReqPO (firmed only) / PLAF (firmed, PLAF-AUFFX = X)	PlannedOrder status firmed	unfirmed planned orders are not migrated — regenerated by the first run
ReqTrans, ReqTransCov / MDTB	not migrated	regenerated

Validation: first Passage plan vs last legacy plan on identical snapshot: planned order count within ±5% and 95% of planned orders match item/date ±1 day. Not migrated: planning history, AX ReqCalcTask logs, SAP MRP lists.

Migration addendum (FR-22.16–22.19). SAP T461P strategy groups map to PlanningStrategy (10 → mts-net, 11 → mts-gross, 20 → mto, 40 → assemble-to-order-final, 50/52 → plan-without-final-assembly, 70 → subassembly; 25/26/30/59/60/63/74/81/82 → nearest strategy with a migration warning). SAP EQUK/EQUP → QuotaArrangement/QuotaSource with allocated qty carried over. SAP MDLV/MDLG/MDLW → ItemCoverage planning_scope. S/4 DDMRP buffer data (PPH_DD_*) → BufferProfile/DdmrpBuffer with recalculation on first run. AX ReqPlan static plans used as simulations → ScenarioOverride with demand_source = forecast-version. AX ReqReduceKey + ReqPOType combinations → strategy per mapping table delivered with the w10 migration templates.

4.22.12 Dependencies, risks, open questions

Depends on: w21 production orders (firming target, route-based lead times); w17 PO creation; w16 on-hand and inventory transactions; w9 BOM/LLC; w6 calendars; w27 sales orders (demand arrives later — until then sales demand is loaded via fixtures and the `sales-line` ref type).

Risk	Likelihood	Impact	Mitigation
Without sales orders until w27 the plan is demo-only	Certain	Low	Forecast + fixture demand; re-proof at w27
Memory blow-up on 50k × 365 daily buckets	Medium	Medium	Sparse buckets (only dates with movement); weekly buckets beyond 90 days

Open questions: (1) Daily vs weekly buckets beyond 90 days — default weekly; decide at w22 kickoff. (2) Whether to expose `positive_days = 0` as "unlimited" (AX) or 0 — default AX semantics; decide by w22 mid-point.

4.22.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Current static plan	legal entity	plan ref	`MAIN`	any static plan	Plan whose current version drives firming and messages	Master planning parameters → General → Current static master plan	Planning file / MRP run scope per plant (OMDU, MD01 plant)
Current dynamic plan	legal entity	plan ref	`DYN`	any dynamic plan	Net-change plan for order events	Master planning parameters → Current dynamic master plan	Net change planning NETCH (MD02 processing key)
Planning horizon days	plan	int	365	30–1,095	Requirements beyond are ignored	ReqPlan coverage time fence / Master plans → time fences	Planning horizon OMDV/OPPQ (MRP planning horizon in workdays)
Daily-bucket window days	plan	int	90	0–365	Daily buckets inside, weekly beyond	n/a	Period split in MD04 (period totals)
Dynamic debounce seconds	tenant	int	60	10–3,600	Coalesces order events before net change	n/a (D365 auto-planning)	Planning file entry processing (MDAB/MDRE)
Max retained versions	plan	int	20	2–100	Purge threshold	ReqPlanVersion cleanup	n/a (single MRP list per run)
Default coverage group	legal entity	ref	`REQ`	any CoverageGroup	Used when item has none	Master planning parameters → Standard coverage → Coverage group	MRP type default per plant (OPPQ), MRP group OPPR
Planning priority on forecast vs sales	tenant	enum	sales > forecast > safety	permutations	Pegging priority	Master plans → Sequence of requirements	Requirements class priority (OVZG)

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Reduction principle default	legal entity	enum	`transactions`	none / percent / transactions / transactions-dynamic	Forecast reduction	Reduction keys (ReqReduceKey) on coverage groups	Consumption mode MARC.VRMOD, OMPA
Statistical forecast model selection	tenant	enum	`auto`	auto / ets / croston / tsb / naive	Model per item	Demand forecasting parameters → Forecast algorithm	MP30/MP31 forecast models (MPOP)
Forecast history months	tenant	int	36	12–120	History window	Demand forecasting parameters → Historical horizon	Forecast profile periods (MP38)
Firming grouping	legal entity	enum	`vendor-date`	none / vendor / vendor-date / vendor-week	PO grouping on firm	Master planning parameters → Planned orders → Firming → Grouping	MD14/MD15 collective conversion, ME59N grouping
Auto-firm in firming fence	coverage group	bool	false	true/false	Firm planned orders automatically	Coverage group → Firming time fence	Firming type in MRP type (T438A.FXHOR)
Simulation scenarios per plan	tenant	int	10	0–50	FR-22.15 limit	Static plan copies	LTP planning scenarios (MS31)
DDMRP enabled	legal entity	bool	false	true/false	FR-22.16	Priority-based planning feature	DDMRP activation (PPH_DD settings)
ADU window past/future days	buffer profile	int/int	90 / 0	7–365	ADU computation	n/a	DDMRP buffer level calculation app parameters
Spike threshold %	buffer profile	numeric	50	1–500	Qualified demand	n/a	DDMRP spike horizon/threshold
Quota arrangements enabled	legal entity	bool	false	true/false	FR-22.17	n/a	Quota usage indicator MARC.USEQU
Default planning strategy	item group	enum	`mts-net`	FR-22.19 codes	Strategy when coverage has none	Coverage group + reduction key	Strategy group OPPS / MRP group OPPR
Consumption backward/forward days	strategy	int/int	30 / 0	0–365	BR-22.13	Reduction key periods	MARC.VINT1/VINT2, OPPR
Shelf-life aware planning	item	bool	false	true/false	FR-22.20	Coverage group "Use shelf life" (PdsShelfLife)	PP/DS shelf-life planning
Planning scope per item coverage	item coverage	enum	`site`	site / warehouse / subcontractor	FR-22.18	Coverage by warehouse dimension	MRP area type 01/02/03 (OMIZ)

4.22.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Net requirements (stock/requirements list)	drill-down	item, site/warehouse, plan/version or scenario	date, ref type, qty, PAB, futures date, messages	Requirement	Net requirements form (ReqTransDisplay)	MD04 / MD05 (MRP list)
Collective net requirements	list inquiry	planner, item group, exception group, site	item, shortage date, messages count, coverage days	Requirement × ActionMessage	Net requirements list page	MD06 / MD07 collective display
Planned orders	list inquiry	plan, type, status, vendor, date range	number, item, qty, dates, status, priority, quota source	PlannedOrder	Planned orders (ReqPO) list page	MD16 / MD13 planned order list
Action messages	list inquiry	plan, action type, buyer/planner	action, ref order, qty, date, suggested date	ActionMessage	Actions (ReqTransAction) form	MD06 exception groups 1–8
Delay (futures) messages	list inquiry	plan, delay days ≥ n	demand ref, requested, expected date, delay	ActionMessage (delay)	Futures messages form	MD06 exception 10/15/30
Pegging / explain tree	drill-down	planned order id, depth	tree of supply → demand with qty	Pegging	Explosion form (ReqTransExplosion)	MD09 pegged requirements, MD4C order report
Supply schedule	operational report	items, site, bucket	per period demand, supply, PAB, safety	Requirement buckets	Supply schedule (ReqSupplySchedule), SSRS supply schedule report	MD04 period totals, MS04
Scenario comparison	operational report	scenario vs current version	planned qty/value deltas by item group, capacity load, purchase volume by vendor	PlanVersion diff	Static plan comparison (manual)	MS70 / MCEC (LTP purchasing evaluation), MS44 capacity
DDMRP buffer status	list inquiry	site, zone, priority ≤ x	item, NFP, TOR/TOY/TOG, priority %, zone	DdmrpBuffer	Priority-based planning workspace (D365)	DDMRP Replenishment planning Fiori app
Quota arrangement analysis	list inquiry	item, period	source, quota %, planned qty, actual qty, deviation	QuotaSource × orders	n/a	MEQ6/MEQ8
Forecast accuracy	KPI	item group, horizon lag	MAPE, wMAPE, bias, forecast vs actual	ForecastLine × sales history	Demand forecast accuracy (D365 Power BI)	MP33/MP32 forecast analysis
Inventory forecast	operational report	items, horizon	projected on-hand, value, days of supply	InventoryForecast	Inventory forecast (ForecastInvent)	MD04 projected stock
Plan run history	list inquiry	plan, date range	version, kind, duration, counts, hash, status	PlanVersion	Master planning log (ReqLog)	MRP run log (MDKP), MD01 list

4.22.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Supply schedule report	On demand	Header: item selection, site, bucket; per item: periods × demand/supply/PAB; footer: plan version and hash	PDF, XLSX	none	None
Supplier forecast (non-binding)	Scenario or plan publish to w20 portal	Header: vendor, horizon, version; lines: item, period, qty, unit	PDF, portal view, EDI DELFOR via w35	`supplier-forecast`	Non-binding disclaimer text per legal entity
Plan run summary	`plan.completed`	Version, duration, counts by type, top delays, hash	email, in-app	none	None
Firming result report	Firming batch	Planned orders → created POs/production/transfer orders	PDF, email	none	None
Electronic plan events	Run lifecycle	CloudEvents (4.22.5)	JSON	n/a	1 year in Iceberg

4.22.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Static plan run	daily 01:00 site-local	plan × LLC level	New PlanVersion per run; pointer swap only on success	50k items < 10 min	Version marked failed; previous stays current	Failed or > 20 min pages
Dynamic net change	event-driven, 60 s debounce	item reach set	Version per batch; dedupe by event id	< 60 s after event	Retry 3× then fall back to next static run	Lag > 5 min warns
Statistical forecast generation	weekly Sunday 22:00	item batches of 1,000	Forecast version per run; resumable per batch	50k items < 15 min	Failed batches retried; partial version not published	Failure warns
DDMRP buffer recalculation	daily 00:15	site	Upsert per (item, site) with calculated_at	20k buffers < 2 min	Keep previous zones	Zone change > 30% reported
Quota allocation recompute	daily 00:45	legal entity	Recomputes allocated_qty from orders; idempotent	< 1 min	Non-blocking	Drift vs incremental > 0 warns
Version purge	daily 03:00	plan	Drops partitions beyond max versions	< 1 min	Retry next day	n/a
Simulation runs	on demand	scenario	Version per run	50k items < 10 min (< 2 min rough-cut)	Version failed	none
Forecast consumption snapshot	daily 23:00	legal entity	Snapshot of reduced forecast for accuracy KPI	< 2 min	Non-blocking	Missing warns

4.22.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
View planning	GET /planning/*, master planning workspace, net requirements, pegging, reports	none	Planner, Buyer, Production supervisor, Sales (read delays)
Run master planning	POST /plans/{id}:run (static/dynamic)	none	Planner (lead), Batch service account
Maintain planned orders	PATCH planned orders, approve, delete, supply schedule edits	none	Planner
Firm planned orders	POST planned-orders:firm	with *Approve purchase orders* (w17) for purchase-type firming above approval limit	Planner, Buyer
Maintain forecasts	forecasts generate/lines, consensus submit	with *Approve forecast consensus*	Demand planner
Approve forecast consensus	Temporal approval step	with *Maintain forecasts* (same version)	Supply chain manager
Maintain planning setup	coverage groups, item coverage, strategies, buffer profiles, quota arrangements, parameters	with *Firm planned orders* flagged	Planning administrator
Run simulations	scenario create/run/compare, publish supplier forecast	none	Planner, S&OP manager

4.22.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
PLAN-0001	error	A run of plan {plan} is already in progress.	Concurrent run	Wait or cancel the running version	FR-22.1 (`PLAN_ALREADY_RUNNING`)
PLAN-0002	error	Planned order {po} is already firmed.	Duplicate firm	Refresh	BR-22.9
PLAN-0003	error	Planned orders of a simulation cannot be firmed.	Firm on simulation	Copy quantities to the operational plan	FR-22.15 (`PLAN_SIMULATION_NOT_FIRMABLE`)
PLAN-0004	warning	BOM cycle detected for {item}; low-level codes not updated.	Cyclic BOM	Fix BOM (w9)	BR-22.1 step 2
PLAN-0005	warning	Requirement for {item} splits into {n} orders (max {max}).	Lot size max	Review maximum qty	FR-22.4
PLAN-0006	info	Supply {ref} can be postponed to {date}.	Early supply	Apply action	FR-22.7
PLAN-0007	warning	Demand {ref} will be delayed {days} days (expected {date}).	Futures date propagation	Expedite supply or accept delay	FR-22.7
PLAN-0008	warning	No vendor for planned purchase of {item}; using none.	Missing default vendor / trade agreement	Set vendor on item coverage or quota	FR-22.3
PLAN-0009	error	Quota arrangement {qa} has no usable source in {period}.	All quotas 0 or blocked	Maintain sources	FR-22.17
PLAN-0010	warning	Buffer for {item} is in red zone (priority {pct}%).	NFP ≤ TOR	Expedite replenishment	FR-22.16
PLAN-0011	warning	Batch {batch} of {item} expires unpegged on {date}.	Shelf-life aware planning	Use in other demand or scrap	FR-22.20
PLAN-0012	error	Forecast history for {item} is shorter than {months} months; naive model used.	Short history	Accept or set manual forecast	FR-22.9
PLAN-0013	error	Plan hash mismatch with previous identical-input run.	Non-determinism	Open incident; compare sorted outputs	FR-22.14
PLAN-0014	warning	Subcontractor scope for vendor {vendor} has no vendor-location warehouse.	FR-22.18 precondition	Create vendor location (w16)	FR-22.18
PLAN-0015	info	Sales demand {ref} exceeds consumable forecast by {qty}; added on top.	BR-22.13	none	FR-22.19

4.22.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation under 3× load
Static plan run	50k items × 5 levels, 200k sales lines; 1 run/day + 5 scenarios	run < 6 / 10 / 15 min	50k items per run	20 versions × ~15M rows ≈ 30 GB steady (dropped partitions)	partition per PlanVersion	Scenarios queue behind operational run; operational < 20 min
Dynamic net change	5,000 events/day, peak 200/min	20 / 60 / 120 s after event	200 events/min coalesced	included above	reach-set computation in memory	Debounce extends to 300 s
Net requirements read	30,000/day	100 / 300 / 800 ms	200/s	n/a	(plan_version, item, site, date)	Serve from previous version
Pegging explain (depth 20)	10,000/day	150 / 500 / 1,200 ms	100/s	n/a	(demand), (supply) indexes	Lazy depth beyond 5
Firm 1,000 planned orders	50 batches/day	batch < 2 / 5 / 8 min	10 orders/s	n/a	saga per order	Batches serialized per plan
Forecast generation	50k items weekly	< 15 min	60 items/s per worker	ForecastLine 50k × 104 periods ≈ 5M rows/yr ≈ 1 GB	(model, item, site, period)	Additional Python replicas via HPA
DDMRP recalculation	20k buffers daily	< 2 min	200 buffers/s	20k × 365 history ≈ 7M rows ≈ 1 GB	(item, site, calculated_at)	Runs after plan; skip on overload with alert

4.22.20 Localization and compliance hooks

- **Calendars**: lead-time offsets use the site work calendar (w6) including country public holidays imported per country pack (wave 48 supplies holiday calendars for the 20 launch countries).
- **Units and decimals**: quantities in inventory UoM to 6 decimals; no currency rounding in planning except value projections (supplier forecast value, inventory forecast) which use w12 rounding.
- **Supplier forecasts**: EDI DELFOR (EDIFACT) / 830 (X12) formats via w35; German VDA 4905/4984 delivery schedules for automotive suppliers are a wave 48 DE pack item through the `planning.supplier-forecast.format` extension point.
- **Export control / restricted parties**: planned purchase orders to vendors flagged by w48 sanctions screening are created but blocked from firming with the w48 screening message.
- **Privacy**: none (no personal data in planning entities except planner ids).
- **Retention**: plan versions are operational data, not records; only the last 20 versions are kept; plan run summaries retained 1 year.
- **Extension points**: `planning.demand-source` (country- or industry-specific demand feeds such as call-off schedules), `planning.supplier-forecast.format`, `planning.lead-time.adjust` (OPA hook, e.g., customs lead time per country of origin from w34).

4.22.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.21.1 Master planning — Supply schedule (ReqSupplySchedule) [AX #188]	PARTIAL-22	ADDED as FR-22.21	Period-bucketed workbench with inline planned-order creation
AX 12.21.1 Master planning — Inventory forecast (ForecastInvent) [AX #189]	PARTIAL-22	ADDED as FR-22.22	Projected on-hand, value and days of supply per period
AX 12.21.2 Master planning — SSRS: Supply schedule report [AX #191]	PARTIAL-22	ADDED as FR-22.21	Report listed in 4.22.14/4.22.15
AX 12.21.4 Master planning — Priority-based planning (D365 planning priority) [AX #194]	PARTIAL-22	ADDED as FR-22.16	Planning priority = NFP / TOG; `priorityBased` coverage groups
SAP 13.12 MM-PUR — MEQ1 / MEQ3 / MEQ4 / MEQ6 / MEQ8 quota arrangement [SAP #125]	PARTIAL-22	ADDED as FR-22.17	Quota rating BR-22.12; analysis report in 4.22.14
SAP 13.12 MM-PUR Customizing — Source determination, quota arrangement rules (T001W, TMQ1) [#137]	PARTIAL-22	ADDED as FR-22.17	Quota usage parameter in 4.22.13; source-list requirement stays with w17
SAP 13.21 PP — MM02 MRP areas (MD61 per MRP area) [#237]	PARTIAL-22	ADDED as FR-22.18	Planning scope site / warehouse / subcontractor
SAP 13.21 PP — MS01–MS08 long-term planning, simulative MRP [#239]	PARTIAL-22	ADDED as FR-22.15	Plan kind `simulation` with ScenarioOverride
SAP 13.21 PP — MS31/MS32/MS33/MS44/MS47/MS64/MS65/MS70 LTP scenarios and evaluations [#240]	PARTIAL-22	ADDED as FR-22.15	Capacity and purchase-volume evaluations; supplier forecast transfer
SAP 13.21 PP — DDMRP (buffer positioning, levels, replenishment planning) [#241]	GAP (propose 22)	ADDED as FR-22.16	Coverage code `ddmrp`, BR-22.11 net-flow equation
SAP 13.21 PP — Predictive MRP (pMRP) [#242]	PARTIAL-22	ADDED as FR-22.15	Rough-cut scenario with weekly buckets and BOM depth limit
SAP 13.21 PP — Planning strategies 20/40/50/70 [#244]	PARTIAL-22	ADDED as FR-22.19	Sales-order stock segregation depends on the w16 inventory dimension (register row #145, wave 16)
SAP 13.21 PP Customizing — OPPS / OPPT strategy groups (T461S, T461P) [#247]	PARTIAL-22	ADDED as FR-22.19	Strategy mapping in migration addendum
SAP 13.21 PP Customizing — OMIZ MRP areas (MDLV, MDLG, MDLW) [#248]	PARTIAL-22	ADDED as FR-22.18	ItemCoverage.planning_scope
SAP 13.21 PP Customizing — DDMRP buffer profiles, classification runs (PPH_DD_*) [#251]	GAP (propose 22)	ADDED as FR-22.16	BufferProfile entity; ABC/XYZ classification for buffer positioning uses w16 ABC classification
SAP 13.23 PP/DS — Shelf-life planning (SAP_PP_SL001) [#278]	PARTIAL-22	ADDED as FR-22.20	BR-22.15 usability rule
SAP 13.23 PP/DS — /SAPAPO/MC01, /SAPAPO/MVM planning versions [#279]	PARTIAL-22	ADDED as FR-22.15	Scenario versions isolated per BR-22.14
SAP 13.23 PP/DS — /SAPAPO/SDP94 (DP), /SAPAPO/SNP94 (SNP), GATP [#280]	PARTIAL-22	SUPERSEDED by FR-22.9 (DP) and FR-22.17 + BR-22.7 (cross-site sourcing); GATP DEFERRED to wave 27; SNP optimizer OUT-OF-SCOPE	SNP-style network optimization (transport-lane capacity LP) is not in v1: quota and transfer planning cover sourcing across sites, and the w23 CP-SAT service is the extension path if demand justifies it

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP 13.26 CS — Service parts planning (SPP) [#306]	PARTIAL-22	DEFERRED to wave 45	Single-echelon MRP with warehouse scopes (FR-22.18) plans field stock; installed-base demand and multi-echelon service parts logic arrive with service management

4.23 Wave 23: Capacity Planning & Finite Scheduling

4.23.1 Scope and business context

Wave 23 replaces the infinite-capacity dating of waves 21–22 with a scheduler that places operations and jobs on resources with calendars and finite capacity, forward or backward, honoring precedence, setup, queue and transit times, primary/secondary resources and capabilities. Planners see a resource × time Gantt, drag jobs, see capacity load and bottlenecks, and optionally run a CP-SAT optimization that minimizes weighted lateness and makespan. Schedule dates flow back to production orders (start/end, job dates) and to master planning (planned production orders use scheduled lead times). Out of scope: shop-floor dispatch lists on the kiosk (wave 25 reads this wave's jobs), tool/fixture management, multi-plant capacity.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX job scheduling on a 2,000-job plant takes 20–40 min (WrkCtrCapRes rebuilt per order in X++); SAP CM21 capacity leveling is manual dispatch per work center and PP/DS needs a separate APO/embedded install with CIF integration	Scheduling runs per order inside the status-update dialog; capacity reservations stored row-by-row; finite scheduling is an add-on product	Scheduler service with materialized UTC capacity intervals, GiST exclusion constraint, site-level run with checkpoints	5,000 operations 30 min → < 60 s; no separate product
Gantt is read-mostly: D365 Gantt drag does not validate secondary resources or calendars until save; SAP CM21 planning board shows overloads but cannot propagate to dependent orders; Epicor Kinetic Multi-Resource Scheduling requires the APS module and locks the board during save	Validation happens server-side after commit; no dependency graph in the board	`:move` with `dryRun` preview showing violations and dependent impact before commit; canvas Gantt	Reschedule-and-verify 15 min → < 30 s; zero committed double-bookings
Load reports lag reality: AX "Capacity load" form (WrkCtrCapRes) and SAP CM01 are as-of the last scheduling run; Infor LN "Capacity Utilization" is a batch report	Load is not maintained incrementally	CapacityLoad recomputed per run and per move; bottleneck events	Load view staleness hours → < 5 s after any move
Optimization is unavailable or opaque: no optimizer in AX; SAP PP/DS optimizer configuration is a consulting project; Plex scheduling is rule-based only	Optimization tied to proprietary engines	CP-SAT via OR-Tools with a stated objective, time limit and reported solver status; heuristic kept if no improvement	Known-optimum gap ≤ 5% on small instances; optimization run self-service in 1 click
Scheduled dates do not reach planning: AX master planning re-derives lead times from the route rather than from the schedule; SAP MRP ignores PP/DS results unless CIF is configured	Planning and scheduling keep separate date models	`order.dates-changed` events feed planning; `planning.finite-capacity` flag uses scheduled dates	Date mismatch between plan and schedule 100% of orders → 0

4.23.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Operations scheduling	ProdTable scheduling (operations), WrkCtrCapRes (capacity reservations), scheduling direction	Lead-time scheduling (CA97), capacity requirements KBED/KBEZ	ScheduleRun mode `operations`, ResourceCapacity	full
Job scheduling	Job scheduling (ProdRouteJob), WrkCtrCapRes by job	Detailed scheduling (PP/DS), SAP APO	mode `jobs` (minute level)	full
Finite/infinite capacity	ReqGroup/parameters finite capacity, finite capacity on resource	Capacity leveling CM21/CM22, finite scheduling flag	per-resource `finite` flag; run option	full
Capacity load view	WrkCtrCapRes load inquiry, Capacity load graph	CM01 (load), CM05, CM07	Capacity load screen	full
Gantt	Gantt chart (D365)	Planning board CM21, PP/DS detailed scheduling board	Scheduling workspace Gantt	improved (drag with constraint check)
Calendars/efficiency	WorkCalendar, efficiency % on resource, capacity per calendar day	Factory calendar, capacity header (CR02), utilization %	WorkCalendar (w9), ResourceCapacity.efficiency_pct	full
Primary/secondary resources	Route operation resource requirements (capability/group/resource/type), priority	Alternative sequences, work center hierarchy, split (AFVV)	ProductionOperation.priority; capability matching	full
Optimization	none native	PP/DS optimizer (genetic), S/4 pMRP	CP-SAT mode	improved
Reschedule dependents	Reschedule with reference orders	Rescheduling from MRP, CM21 dispatch	`:reschedule` with dependency propagation	full
Bottleneck resource	Bottleneck resource flag (D365 scheduling)	Bottleneck work center (CM21)	utilization > threshold detection	full
Resource networks	n/a (route operation resource requirements only)	Resource network CRC5/CRC1–3 (allowed resource sequences in PP-PI)	ResourceNetwork with allowed transitions (FR-23.13)	full
Campaign planning	D365 batch order sequencing by "batch order sequence" / planning priority; no campaign object	PP-PI campaigns (COCP, CMP process order campaigns), PP/DS campaign planning	Campaign entity and campaign scheduling (FR-23.14)	full
Changeover / cleanout	Route setup time only; D365 sequencing value	Setup matrix (/SAPAPO/CDPS setup groups), cleanout orders, recipe-dependent cleaning	SetupMatrix, changeover jobs, cleanout orders (FR-23.15)	full
Campaign cost settlement	n/a	Campaign settlement / cleanout cost allocation to campaign orders	Cleanout cost allocation (FR-23.16)	full
Block planning	n/a	PP/DS block planning (/SAPAPO/CDPSC12, characteristic blocks on resources)	CapacityBlock with attribute constraints (FR-23.17)	full

4.23.3 Functional requirements

- **FR-23.1 Capacity model** — Must. The system shall materialize ResourceCapacity as open intervals (UTC minutes) per resource per day from its WorkCalendar (shifts), efficiency % and capacity-per-day override, for a rolling horizon of 365 days.
  - Notes: Rebuilt incrementally when a calendar changes (w9 event). Finite resources cannot be loaded beyond open interval minutes; infinite resources ignore the limit but still only schedule in open intervals.
- **FR-23.2 Operations scheduling** — Must. The system shall schedule operations in daily buckets, forward from a start date or backward from a delivery date, selecting a resource per operation by priority (named resource > resource group > capability) and finite availability.

- **FR-23.3 Job scheduling** — Must. The system shall schedule jobs (queue-before, setup, process, overlap, transport, queue-after) at minute granularity with the same direction and resource choice rules, splitting process jobs across shifts but not across resources.
- **FR-23.4 Constraints** — Must. The scheduler shall honor: precedence (next_oper_number), max one process job per finite resource at any minute, secondary resources booked concurrently with the primary, setup before process, overlap transfer batch (operation n+1 may start after `overlap_qty` units), transit time between resources, calendar open intervals, order start ≥ today (forward) and end ≤ delivery date (backward, else convert to forward and flag late), operation `finished` jobs frozen, firmed jobs frozen.
- **FR-23.5 Dispatch heuristic** — Must. The default algorithm is priority-list scheduling: orders sorted by (delivery date, priority, order number); each operation placed at the earliest feasible interval on the best resource (least finish time, tie → least load); complexity O(n log n) per resource.
- **FR-23.6 CP-SAT optimization** — Should. The system shall offer an optimization run via OR-Tools CP-SAT (Python `scheduler-opt` service) with objective minimize Σ weight × lateness + λ × makespan, time limit configurable (default 60 s), horizon ≤ 2,000 operations per run (larger sets are windowed).
- **FR-23.7 Gantt** — Must. The system shall render resources × time with jobs as bars colored by order, allow drag to move/reassign a job within its capability set, and show constraint violations (red outline + tooltip) before commit.
- **FR-23.8 Reschedule with propagation** — Must. Moving a job shall reschedule all successors of the same order forward and, optionally, the dependents (orders pegged to the moved order's output) with a preview of affected orders before commit.
- **FR-23.9 Capacity load** — Must. The system shall show load % = Σ booked minutes / Σ open minutes per resource/group/period (day/week), with bottleneck flag when load > 90% in ≥ 3 consecutive weeks (threshold configurable).
- **FR-23.10 Feedback to planning** — Must. The system shall publish scheduled dates so that production orders update start/end, and master planning uses scheduled lead times for firmed orders and route-based estimates for planned orders (optionally finite in the plan run via flag `planning.finite-capacity`).
- **FR-23.11 Schedule runs** — Must. Scheduling can run per order (on `:schedule`), per selection, or for the whole site nightly, as Temporal jobs with checkpoints every 500 operations.
- **FR-23.12 Capability matching** — Should. Resource selection shall honor capability requirements with minimum level and expiry dates (w9 capabilities).
- **FR-23.13 Resource networks** — Should. The system shall define a ResourceNetwork per site as a directed graph of resources (nodes) with allowed transitions (edges, with optional transfer time and maximum wait minutes), assign a network to a route operation chain, and constrain the scheduler (heuristic and CP-SAT) so that when operation n runs on resource r, operation n+1 is scheduled only on a resource r' with an edge r → r'.
    - Notes: Typical use: mixing vessel V1 feeds fillers F1/F2 only; V2 feeds F3 only. Maximum wait (e.g., product holds 120 min in the vessel) becomes a hard constraint end_n + wait_max ≥ start_{n+1}; violation in heuristic mode moves the predecessor later. A network edge may require a secondary resource (e.g., transfer line). CP-SAT encodes allowed pairs as `AddAllowedAssignments` on the resource choice variables of consecutive operations.
- **FR-23.14 Campaign planning** — Should. The system shall group production/batch orders into a Campaign (resource or resource group, campaign key such as formula family or setup group, planned start, maximum size in orders/quantity/duration, sequence rule) and schedule them contiguously on the campaign resource, inserting a leading setup and a trailing cleanout, with orders inside the campaign sequenced by the setup matrix to minimize Σ changeover minutes.
    - Notes: Campaigns are created manually from a selection, or proposed by the `campaign-proposal` job from planned/firmed orders sharing a campaign key within a horizon window (default 14 days). Orders join or leave a campaign until the first order is Started. A campaign is frozen once Started; added orders go to the next campaign. Sequence rule: setup-matrix nearest neighbor (heuristic) or CP-SAT circuit (`AddCircuit`) with time limit 30 s for ≤ 50 orders.
- **FR-23.15 Setup matrix and changeover/cleanout operations** — Should. The system shall maintain a SetupMatrix per resource or resource group with changeover minutes and cost per (from setup group → to setup group) and a cleanout definition per setup group (cleaning standard operation set, duration, mandatory after N batches or M hours), and shall generate changeover jobs between consecutive jobs with different setup groups and cleanout orders (lightweight production orders of type `cleanout`, no BOM output) at campaign end or when the mandatory condition is reached.
    - Notes: Setup group comes from the product (w8) or formula (w24) and may be overridden on the order. Changeover jobs replace the route's static setup time when a matrix entry exists. Recipe-dependent cleaning: a matrix entry may require a specific cleaning standard operation set (e.g., allergen cleanout from "contains peanut" to "peanut-free" = wet clean 240 min). Cleanout orders consume cleaning materials and labor through w21 journals.
- **FR-23.16 Campaign cost settlement** — Should. The system shall allocate the actual cost of a campaign's cleanout and leading-setup orders to the campaign's production orders by RAF quantity (default), equally, or by run minutes, posting the allocation as an overhead cost line of type `campaign-changeover` before those orders are ended.
    - Notes: Orders cannot be ended while the campaign's cleanout order is unended unless `campaign.allowEarlyEnd` is set, in which case the allocation estimates from the cleanout's planned cost and the difference posts to the last order ended.
- **FR-23.17 Block planning** — Could. The system shall define CapacityBlocks on a resource (recurring or dated time windows, e.g., Monday 06:00–14:00) each restricted to an attribute condition over product or order attributes (variant dimension values such as color = white, or batch attribute targets such as grade ≥ 2), and the scheduler shall place a job inside a block only if the job's attributes satisfy the block condition; jobs outside any matching block are placed in unblocked capacity or flagged late.

4.23.4 Data model

ResourceCapacity: id, tenant_id, resource_id, day (date), interval_start (timestamptz), interval_end, open_minutes (int), efficiency_pct (numeric 5,2), finite (bool), source (calendar/override).

ScheduledJob: id, job_id (FK ProductionJob), order_id, oper_number, resource_id, start (timestamptz), end, minutes, run_id, firmed (bool), late_minutes (int), overlap_qty (numeric), secondary_resource_ids (uuid[]).

ScheduleRun: id, scope (order/selection/site), mode (operations/jobs), direction (forward/backward), finite (bool), algorithm (heuristic/cp-sat), time_limit_s, status, started_at, finished_at, operations_count, makespan_minutes, total_lateness_minutes, objective_value, solver_status (OPTIMAL/FEASIBLE/UNKNOWN).

CapacityLoad (materialized): resource_id, period_start, period_type, open_minutes, booked_minutes, load_pct, bottleneck (bool).

Invariants: no two ScheduledJob rows on the same finite resource overlap in [start, end); every ScheduledJob lies within ResourceCapacity open intervals of its resource; for jobs j, k in the same operation chain with k successor of j: k.start ≥ j.end + transit; a ScheduledJob with firmed = true is unchanged by any run without `force`.

Indexing: ScheduledJob GiST exclusion constraint `EXCLUDE USING gist (resource_id WITH =, tstzrange(start,end) WITH &&) WHERE (finite)`; ResourceCapacity (resource_id, day); CapacityLoad refreshed per run.

ResourceNetwork / NetworkEdge (FR-23.13): ResourceNetwork(id, site_id, code, name, active_from, active_to); NetworkEdge(id, network_id, from_resource_id, to_resource_id, transfer_minutes int, max_wait_minutes int nullable, secondary_resource_id nullable).

Campaign / CampaignOrder (FR-23.14): Campaign(id, site_id, resource_id or resource_group_id, campaign_key text, status (proposed/planned/started/completed/cancelled), planned_start, planned_end, max_orders int, max_qty numeric, max_minutes int, sequence_rule (enum nearest-neighbor/cp-sat/manual), cleanout_order_id, setup_order_id, total_changeover_minutes); CampaignOrder(campaign_id, order_id, sequence int).

SetupMatrix / SetupMatrixEntry / CleanoutRule (FR-23.15): SetupMatrix(id, resource_id or resource_group_id, version, active_from); SetupMatrixEntry(matrix_id, from_setup_group, to_setup_group, changeover_minutes int, changeover_cost numeric(18,4), cleaning_operation_set_id nullable); CleanoutRule(setup_group, duration_minutes, operation_set_id, after_batches int nullable, after_hours int nullable, at_campaign_end bool).

CapacityBlock (FR-23.17): id, resource_id, recurrence (RRULE text) or start/end, condition (jsonb attribute predicate), priority int, active.

Invariants (additions): consecutive scheduled jobs of an order chain using a network have an edge between their resources; jobs of one campaign on the campaign resource are contiguous except for changeover and cleanout jobs; Σ allocated campaign changeover cost = Σ actual cost of the campaign's setup and cleanout orders.

4.23.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/scheduling/runs	Start run (scope, mode, direction, finite, algorithm)	Idempotency-Key	Run scheduling
GET	/api/v1/scheduling/runs/{id}	Run status/metrics	n/a	View scheduling
GET	/api/v1/scheduling/gantt?siteId&from&to&resourceIds	Jobs for Gantt window (≤ 5,000 bars)	n/a	View scheduling
POST	/api/v1/scheduling/jobs/{id}:move	Move/reassign with `dryRun` preview	Idempotency-Key	Maintain schedule
POST	/api/v1/scheduling/jobs/{id}:firm	Firm/unfirm	Idempotency-Key	Maintain schedule
GET	/api/v1/scheduling/capacity-load?resourceIds&period=week	Load and bottlenecks	n/a	View scheduling
GET	/api/v1/scheduling/resources/{id}/capacity?from&to	Open intervals	n/a	View scheduling
POST	/api/v1/scheduling/resources/{id}/capacity-overrides	Add downtime/extra shift	Idempotency-Key	Maintain capacity

Event	Trigger	Payload fields	Consumers
`dev.passage.scheduling.run.completed.v1`	run done	runId, scope, operations, makespanMinutes, totalLatenessMinutes, solverStatus	production, planning
`dev.passage.scheduling.job.scheduled.v1`	job placed/moved	jobId, orderId, operNumber, resourceId, start, end, runId, firmed	production, shopfloor
`passage.scheduling.order.dates-changed.v1`	order start/end changed	orderId, start, end, lateMinutes	production, planning, sales (w27)
`dev.passage.scheduling.bottleneck.detected.v1`	load > threshold	resourceId, periods[], loadPct	notifications

4.23.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Scheduling workspace (Gantt)	Zoom day/week, drag job, reassign, firm, run heuristic/optimize, preview propagation	Resource rows, job bars (order, item, qty, start/end), late markers, open-interval shading	Maintain schedule
Capacity load	Filter resource/group/period, open bottleneck, drill to jobs	Resource, period, open h, booked h, load %, bottleneck	View scheduling
Schedule runs	Start, monitor, compare runs	Scope, mode, algorithm, status, makespan, lateness, duration	Run scheduling
Resource capacity	Overrides, view calendar materialization	Day, intervals, efficiency, finite	Maintain capacity

Interaction rules: drag snaps to 5-minute grid; `Shift+drag` moves the whole order; violations block commit unless the user holds *Override scheduling constraints*; Gantt virtualizes rows and uses canvas rendering for > 1,000 bars; undo stack of 20 moves per session.

4.23.7 Business rules and edge cases

- **BR-23.1 Duration.** Process minutes = (run_time_h_per × qty / process_qty) × 60 / (efficiency_pct/100); setup minutes = setup_time_h × 60 / efficiency; both rounded up to whole minutes.
- **BR-23.2 Backward fallback.** If backward scheduling places any job before now + 1 h, the run re-schedules the order forward from now and sets late_minutes = end – delivery date end-of-day.
- **BR-23.3 Shift splitting.** A process job crossing a closed interval is split into segments on the same resource; setup is never split (if it does not fit before close, it moves to the next open interval).
- **BR-23.4 Secondary resources.** Booked for the same interval as the primary; if no secondary is free, the primary slot is skipped.
- **BR-23.5 Overlap.** With overlap_qty q > 0, operation n+1 earliest start = start_n + setup_n + process minutes for q units + transit.
- **BR-23.6 Firmed and reported jobs.** Jobs with reported hours or `finished` status keep their actual start; remaining minutes = planned – reported, floor 0.
- **BR-23.7 DST and time zones.** All intervals stored in UTC; calendars defined in site time zone; a 23-hour DST day yields fewer open minutes — property test covers 2026-03-29 (EU) and 2026-11-01 (US).
- **BR-23.8 Resource selection tie-break.** Equal finish time → lower load % → lower resource code (deterministic).
- **BR-23.9 CP-SAT model.** Interval variables per job, `NoOverlap` per finite resource, `AddNoOverlap2D` not used; precedence as start ≥ end + transit; objective Σ w_i × max(0, end_i – due_i) + λ × makespan with λ = 0.1 default; `num_workers` = 8; result accepted if FEASIBLE; heuristic result kept if solver returns UNKNOWN.
- **BR-23.10 Propagation limit.** Dependent rescheduling stops after 500 orders or 5 pegging levels and reports the truncation.
- **BR-23.11 Network wait limit.** For edge r → r' with max wait w: start_{n+1} – end_n ≤ w; the heuristic first tries r' candidates in transfer-time order; if none satisfies w, it delays operation n (and its predecessors) forward until a feasible pair exists, up to 7 days, then flags `NETWORK_INFEASIBLE`.
- **BR-23.12 Changeover sequencing.** Nearest-neighbor: start from the setup group with the lowest changeover from the resource's current state; repeatedly pick the unscheduled order with minimal changeover (tie → earliest due date → order number). Example: groups A→B 30, A→C 90, B→C 20, C→B 60; current A; orders B, C → sequence A→B→C = 50 min vs A→C→B = 150 min.
- **BR-23.13 Mandatory cleanout.** A cleanout is inserted when batches since last cleanout ≥ after_batches or run hours ≥ after_hours, whichever first, and always at campaign end if at_campaign_end; a cleanout is never split across shifts (moves to next open interval like setup, BR-23.3).
- **BR-23.14 Campaign allocation rounding.** Allocation by quantity: share_i = raf_i / Σ raf; amount_i = round(cost × share_i, 2); residual to the largest order.
- **BR-23.15 Block matching.** A job matches a block if every predicate in the block condition is true for the job's product variant dimensions and order attributes; overlapping blocks resolve by block priority; an unmatched job may use unblocked capacity only.

4.23.8 Acceptance criteria

- **AC-23.1** (FR-23.2, 23.4) Given OR-Library instance `ft06`, when scheduled with the heuristic, then the schedule is feasible (no overlap, precedence held); when optimized with CP-SAT 60 s, makespan = 55 (known optimum).

- **AC-23.2** (FR-23.4) Given 10k random jobs across 50 finite resources, then the exclusion constraint never fires and the property test "every job inside open intervals" passes.
- **AC-23.3** (FR-23.8) Given an order whose output is pegged to a sales line, when I move its last job 2 days later, then the preview lists the sales line delay and after commit the order's end date and `order.dates-changed` event carry the new date.
- **AC-23.4** (FR-23.11) Given 5,000 operations on the demo plant, when the site run executes, then it completes in < 60 s.
- **AC-23.5** (FR-23.3) Given a 300-minute process job on a resource with a shift ending in 120 minutes, then two ScheduledJob segments of 120 and 180 minutes exist on the same resource.
- **AC-23.6** (FR-23.9) Given booked 150 h against 160 h open for 3 consecutive weeks, then the resource is flagged bottleneck and an event is emitted once (not per run).
- **AC-23.7** (FR-23.1) Given a calendar change closing next Friday, then the ResourceCapacity rows for that day are removed within 30 s and affected jobs show violations.
- **AC-23.8** (FR-23.10) Given a scheduled order, when planning runs with `planning.finite-capacity` on, then its supply date equals the scheduled end date.
- **AC-23.9** (FR-23.6) Given an optimization run that times out with UNKNOWN, then the heuristic schedule remains and the run reports solver_status UNKNOWN.
- **AC-23.10** (FR-23.13, BR-23.11) Given network V1 → {F1, F2} with max wait 120 min and F1, F2 busy for the next 4 hours, when an order with mixing on V1 is scheduled, then mixing is delayed so that filling starts ≤ 120 min after mixing ends, and no filling job is placed on F3.
- **AC-23.11** (FR-23.14, FR-23.15, BR-23.12) Given the BR-23.12 matrix and two orders B and C in one campaign, when the campaign is scheduled, then the sequence is B then C with 50 changeover minutes and a cleanout job after C.
- **AC-23.12** (FR-23.16, BR-23.14) Given a cleanout costing USD 900.00 and campaign orders with RAF 100, 200 and 300 units, when allocated, then the orders receive 150.00, 300.00 and 450.00.
- **AC-23.13** (FR-23.17, BR-23.15) Given a Monday block restricted to color = white, when a red-variant job is scheduled for Monday, then it is placed outside the block or flagged late; a white job is placed inside it.

4.23.9 Test plan and fixtures

Fixtures: `orlib-jobshop-30` (ft06, ft10, ft20, la01–la20, abz5–abz9 with known optima); `sched-calendar-edge` (DST days, 24/7, zero-capacity holidays); `sched-5k-ops` (demo plant scaled); `sched-propagation` (pegged chains 5 deep).

Test	Type	Target	Pass criterion
Job-shop feasibility	unit	heuristic	30/30 feasible
CP-SAT optimality	integration	optimizer	within 5% on instances ≤ 10×10
No double-booking	property	ScheduledJob	0 violations over 10k jobs
Calendar edges	unit	capacity materialization	open minutes match fixture
Gantt move	Playwright	move + propagate	persisted and dependents updated
Scale	performance	5k ops	< 60 s, p95 Gantt load < 1 s

Reconciliation: Σ booked minutes per resource (CapacityLoad) = Σ ScheduledJob minutes; production order start/end = min/max of its scheduled jobs.

4.23.10 Deployment and operations

Helm: `scheduling` (Go, 2 API replicas + 2 worker replicas, 4 CPU / 8 GB), `scheduler-opt` (Python OR-Tools, 1 replica, 8 CPU / 16 GB, scale-to-zero via KEDA on queue depth), `scheduling-db` (CloudNativePG 50 GB). Config: `SCHED_BOTTLENECK_PCT` (90), `SCHED_OPT_TIME_LIMIT_S` (60), `SCHED_HORIZON_DAYS` (365). Flags: `scheduling.cp-sat`, `scheduling.propagate-dependents`, `planning.finite-capacity`. Jobs: nightly site run 00:30 (before planning); capacity materialization daily. SLOs: 5k-op run < 60 s; Gantt window p95 < 1 s; move p95 < 400 ms. Alerts: run failure (page), optimizer OOM (warn). Rollback: prior digest; ScheduledJob rows versioned by run_id so a bad run is reverted by re-pointing to the previous run.

4.23.11 Migration from AX / SAP

Source	Passage entity	Notes
WrkCtrCapRes (AX) / KBED+KBEZ (SAP)	ScheduledJob	only for open orders; re-scheduled on first run, legacy dates kept as `firmed` if the order is Started
WrkCtrTable capacity/efficiency, WorkCalendar / CR02 capacity header, factory calendar	ResourceCapacity (materialized)	from w9 resources and calendars
ProdRouteJob dates / AFVV dates	ProductionJob planned dates	preserved for reference

Not migrated: historical capacity reservations, PP/DS optimizer profiles, AX scheduling parameters (re-set in Passage).

4.23.12 Dependencies, risks, open questions

Depends on: w21 jobs/operations; w22 planned orders and pegging; w9 resources, capabilities, calendars.

Risk	Likelihood	Impact	Mitigation
Heuristic quality far from optimum on large instances	High	Low	CP-SAT windowing; planners can firm; measure lateness trend
Gantt performance in the browser	Medium	Medium	Canvas rendering, windowed fetch ≤ 5,000 bars

Open questions: (1) Should optimization run automatically nightly or only on demand? Default on demand; decide by w23 mid-point. (2) Tool/fixture resources as secondary resources in w23 or w44 (EAM)? Default w44.

4.23.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Default scheduling method	site	enum	`jobs`	operations / jobs	Mode on `:schedule`	Production control parameters → General → Scheduling method	Scheduling type in order type-dependent params (OPU3: forward/backward)
Default scheduling direction	site	enum	`backward`	forward / backward	Direction	Scheduling parameters → Direction	OPU3 scheduling type, adjust dates (OPJH)

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Finite capacity default	site	bool	false	true/false	Run option default	Production control parameters → Scheduling → Finite capacity	Finite scheduling in capacity planning profile (OPD0 / CM21 profile)
Capacity horizon days	site	int	365	30–730	ResourceCapacity materialization	Capacity time fence (ReqGroup / scheduling parameters)	Capacity planning horizon (OPA6 / OPDB profiles)
Bottleneck threshold %	site	numeric	90	50–200	FR-23.9 flag	Bottleneck resource (D365 scheduling)	Overload % in capacity profile (CM01 evaluation profile OPA2)
Bottleneck consecutive weeks	site	int	3	1–12	FR-23.9	n/a	n/a
Optimization time limit s	tenant	int	60	5–1,800	CP-SAT stop	n/a	PP/DS optimization profile (/SAPAPO/CDPSC11) runtime
Lateness weight λ	tenant	numeric	0.1	0–100	CP-SAT objective	n/a	PP/DS optimizer weighting (makespan/lateness/setup)
Drag snap minutes	user	int	5	1–60	Gantt grid	Gantt chart setup	Planning board profile (CM21 OPD0) time scale
Propagation limit	tenant	int pair	500 orders / 5 levels	1–5,000 / 1–20	BR-23.10	Reschedule references	Rescheduling options
Primary resource selection rule	site	enum	`least-finish`	least-finish / least-load / priority	FR-23.5	Resource selection (ProdParmSchedule)	Work center selection (alternate sequences)
Schedule run nightly time	site	time	00:30	any	Site run	Batch job ProdMultiSchedule	Collective rescheduling CO05N / CM40 background leveling
Safety time / queue defaults	resource group	minutes	0	0–10,080	Added to queue before	Resource group queue time	Queue time in work center (CR02), float before/after production (OPU3 sched margin key OPJN)
Setup matrix active	resource	bool	false	true/false	FR-23.15	n/a	Setup matrix in PP/DS resource
Campaign proposal window days	site	int	14	1–90	FR-23.14	n/a	Campaign creation settings
Allow early end in campaign	site	bool	false	true/false	FR-23.16	n/a	Campaign settlement settings
Block planning active	resource	bool	false	true/false	FR-23.17	n/a	Block planning in resource (/SAPAPO/CDPSC12)
Network max delay days	site	int	7	1–30	BR-23.11	n/a	n/a

4.23.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Capacity load by resource	list inquiry	resource/group, period type, date range	open h, booked h, load %, bottleneck	CapacityLoad	Capacity load (WrkCtrCapRes) form, Capacity load graphically	CM01 / CM02 / CM07
Capacity load detail	drill-down	resource, period	jobs, orders, minutes	ScheduledJob	Capacity reservations (WrkCtrCapRes)	CM03 / CM05 pegged requirements
Gantt	operational view	site, resources, window	job bars, late markers, violations	ScheduledJob	Gantt chart	CM21 / PP/DS detailed scheduling board
Late orders	list inquiry	site, lateness ≥ n min	order, delivery date, scheduled end, late minutes, pegged demand	ScheduledJob × order	Production orders "delayed" filter	COOIS with late date filter, CM25
Schedule run comparison	operational report	two runs	makespan, total lateness, changeover minutes, utilization	ScheduleRun	n/a	PP/DS optimization log
Resource utilization KPI	KPI	site, period	utilization %, idle %, setup %	CapacityLoad + ScheduledJob	SSRS "Resource utilization" (WrkCtrCapResUtil)	MCP7 / work center analysis (PP information system)
Campaign overview	list inquiry	site, resource, status	campaign, key, orders, planned start/end, changeover min, cleanout	Campaign	n/a	COCP campaign worklist
Changeover analysis	operational report	resource, period	changeovers count, minutes, cost by from/to group	ScheduledJob (changeover)	n/a	PP/DS setup analysis
Network feasibility	list inquiry	network, date range	orders flagged NETWORK_INFEASIBLE, wait-time violations	ScheduleRun messages	n/a	n/a (PP-PI scheduling log)
Block utilization	list inquiry	resource, period	block, matched minutes, unused block minutes	CapacityBlock × ScheduledJob	n/a	Block planning evaluation

4.23.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Dispatch list (per resource / shift)	On demand, shift start	Header: resource, shift, date; lines: sequence, order, operation, item, qty, start/end, setup group	PDF, kiosk view (w25)	none	None
Gantt print	On demand	Resource rows, time scale, bars, legend	PDF (landscape A3/Tabloid)	none	None
Campaign sheet	Campaign planned	Campaign key, sequence of orders, changeovers, cleanout instructions	PDF	`campaign`	Allergen cleanout sheets kept 2 years where food safety plans (FSMA/HACCP) require
Cleanout record	Cleanout order completed	Cleaning steps, materials, verifier, time	PDF; signature via FR-24.22 where GxP	`prod-doc`	Kept with batch records (w24 EBR)
Electronic schedule events	Job placed/moved	CloudEvents (4.23.5)	JSON	n/a	1 year

4.23.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Site schedule run	daily 00:30 site-local	site	New run_id; checkpoint every 500 operations; pointer to current run	5k operations < 60 s	Previous run stays current	Failure pages

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Capacity materialization	daily 00:05 + on calendar event	resource	Upsert per (resource, day)	1,000 resources × 365 days < 2 min	Retry; stale capacity flagged	Lag > 30 min warns
CapacityLoad refresh	after each run/move	resource × period	Recompute deltas	< 5 s per move	Full refresh nightly	Staleness > 60 s warns
Campaign proposal	daily 23:00	site × campaign key	Proposed campaigns replaced per key	< 1 min	Non-blocking	Count reported
Bottleneck detection	after nightly run	resource	One event per bottleneck episode	< 30 s	Non-blocking	Event to notifications
Optimization (on demand)	user	selection ≤ 2,000 ops (windowed)	Solver result stored; heuristic kept on UNKNOWN	≤ time limit	OOM → heuristic retained	OOM warns

4.23.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
View scheduling	GET /scheduling/*, Gantt (read), capacity load, reports	none	Planner, Production supervisor, Shop floor supervisor
Run scheduling	POST /scheduling/runs (heuristic, optimization)	none	Scheduler, Planner
Maintain schedule	`:move`, `:firm`, Gantt drag, campaign sequencing	none	Scheduler
Override scheduling constraints	commit moves with violations	with *Maintain capacity* flagged	Scheduling lead
Maintain capacity	capacity overrides, calendars on resources, capacity blocks	none	Production engineer
Maintain scheduling setup	parameters, setup matrices, resource networks, cleanout rules	none	Production engineer, ERP administrator
Maintain campaigns	create/modify/release campaigns, allocate campaign cost	with *End production orders* (w21) flagged for allocation	Scheduler, Cost accountant (allocation)

4.23.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
SCHED-0001	error	Job {job} overlaps {other} on finite resource {resource}.	Double booking on move	Choose another slot	FR-23.4
SCHED-0002	warning	Job {job} falls outside open capacity on {date}.	Calendar change	Reschedule or add capacity override	FR-23.1
SCHED-0003	warning	Order {order} is late by {minutes} min (backward scheduling fell back to forward).	BR-23.2	Expedite or accept	BR-23.2
SCHED-0004	error	No resource with capability {cap} level ≥ {level} is available.	Capability matching	Add resource or capability	FR-23.12
SCHED-0005	error	Secondary resource {resource} is not free for the selected slot.	BR-23.4	Pick another slot	BR-23.4
SCHED-0006	info	Optimization ended with status {status}; {kept} schedule kept.	Solver result	none	BR-23.9
SCHED-0007	warning	Propagation truncated after {n} orders / {levels} levels.	BR-23.10	Run site schedule	BR-23.10
SCHED-0008	error	Firmed job {job} cannot be moved without force.	Firmed	Unfirm first	FR-23.4
SCHED-0009	error	No allowed successor resource in network {network} for operation {oper} within {days} days.	Network infeasible	Add network edge or capacity	BR-23.11 (`NETWORK_INFEASIBLE`)
SCHED-0010	warning	Wait time {minutes} min exceeds network maximum {max} min.	Max wait	Move predecessor later	BR-23.11
SCHED-0011	error	Order {order} cannot join campaign {campaign}: campaign already started.	Frozen campaign	Use next campaign	FR-23.14
SCHED-0012	warning	Setup matrix has no entry {from} → {to}; route setup time used.	Missing matrix entry	Maintain matrix	FR-23.15
SCHED-0013	error	Cannot end order {order}: campaign cleanout {cleanout} is not ended.	FR-23.16	End the cleanout or allow early end	FR-23.16
SCHED-0014	warning	Job {job} does not match any block on {resource}; placed in unblocked capacity.	Block planning	Review blocks	BR-23.15
SCHED-0015	warning	Resource {resource} is a bottleneck: load {pct}% for {weeks} weeks.	FR-23.9	Add capacity or offload	FR-23.9

4.23.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation under 3× load
Site schedule run (heuristic)	5,000 operations, 200 resources, 1 nightly + 20 ad hoc/day	25 / 60 / 90 s	100 ops/s	ScheduledJob ~ 5k × 6 jobs × 365 runs retained 30 days ≈ 1M rows	GiST exclusion; (run_id)	Ad hoc runs queued; nightly priority
Gantt window load	5,000 bars; 2,000 loads/day	300 ms / 1 s / 2 s	50/s	n/a	(resource_id, start)	Window narrowed to 2,000 bars
Move with dryRun	3,000/day; peak 30/min	120 / 400 / 800 ms	20/s	ScheduleRun move log 1M rows ≈ 0.5 GB	exclusion constraint check	Propagation preview limited to 100 orders
Capacity materialization	1,000 resources × 365 days	< 2 min full; < 30 s per calendar change	n/a	365k rows/yr	(resource_id, day)	Horizon reduced to 90 days for incremental
CP-SAT optimization	≤ 2,000 ops per window, 5 runs/day	≤ time limit (60 s)	n/a	negligible	n/a	KEDA scale-out; queued beyond 3 concurrent
Campaign sequencing	≤ 50 orders per campaign	1 / 5 / 30 s (CP-SAT circuit)	n/a	negligible	n/a	Fall back to nearest neighbor

4.23.20 Localization and compliance hooks

- **Calendars and time zones**: capacity is materialized from site time zone calendars into UTC (BR-23.7); country holiday calendars come from wave 48 packs; working-time regulations (e.g., EU Working Time Directive 11-hour rest) are not enforced on resources but on workers in w25/w42.
- **Food safety**: allergen changeover rules (FR-23.15) support FSMA (US 21 CFR 117) and EU Regulation 852/2004 HACCP plans by forcing cleanout entries; cleanout records retained 2 years (configurable).
- **GxP**: for regulated plants, cleanout records require e-signature (w24 FR-24.22) and become part of the EBR.
- **Formats**: Gantt time labels follow the user locale (24h/12h), ISO week numbers in load views (configurable to US week numbering).
- **Privacy**: none (no personal data in scheduling entities).
- **Extension points**: `scheduling.constraint` (pluggable constraint provider evaluated in heuristic and CP-SAT, e.g., country-specific noise-curfew windows for resources), `scheduling.objective` (custom weights).

4.23.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP 13.22 PP-PI — CRC5 / CRC resource network [SAP #253]	PARTIAL-23	ADDED as FR-23.13	Allowed transitions and max wait; CP-SAT allowed assignments
SAP 13.22 PP-PI — Campaign planning (COCP / CMP) [#259]	PARTIAL-23	ADDED as FR-23.14	Campaign entity, proposal job, setup-matrix sequencing
SAP 13.22 PP-PI — Cleanout / changeover orders, recipe-dependent cleaning [#260]	PARTIAL-23	ADDED as FR-23.15	SetupMatrix, CleanoutRule, cleanout orders
SAP 13.22 PP-PI Periodic — Campaign settlement / cleanout cost allocation [#270]	PARTIAL-23	ADDED as FR-23.16	BR-23.14 allocation by quantity/equal/run minutes
SAP 13.22 PP-PI Customizing — Resource networks [#272]	PARTIAL-23	ADDED as FR-23.13	Network setup under *Maintain scheduling setup*
SAP 13.22 PP-PI Customizing — Campaign planning settings [#273]	PARTIAL-23	ADDED as FR-23.14	Parameters in 4.23.13
SAP 13.23 PP/DS — Block planning (/SAPAPO/CDPSC12) [#276]	GAP (propose 23)	ADDED as FR-23.17	CapacityBlock with attribute predicate
SAP 13.23 PP/DS Customizing — Block planning settings, CDP settings [#281]	GAP (propose 23)	ADDED as FR-23.17 (block planning); CDP settings ADDED as FR-24.23	CDP is planning against batch characteristics and lives with batch attributes in wave 24

4.24 Wave 24: Process Manufacturing & Lean (Formulas, Batches, Kanban)

4.24.1 Scope and business context

Wave 24 extends production to two further modes. Process manufacturers (food, chemicals, pharma) create batch orders from formulas whose lines scale with batch size, consume ingredients by potency, produce co- and by-products with cost allocation, track batch attributes, shelf life and disposition, and handle catch-weight items with a dual quantity. Lean plants define value streams and production flows with activities, replenish through kanban rules (manufacturing and withdrawal; fixed, scheduled and event replenishment), work kanban cards on a board, and cost production flows by periodic backflush. Formulas and recipes are version-controlled through engineering change (w9). After this wave a second demo plant (food/chem line) runs a batch order and a kanban-driven cell. Out of scope: PI sheets with electronic signatures (w33 quality covers signatures), process control system integration beyond the IoT connector of w25, recipe scale-up optimization.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Rescaling a batch order in AX (PmfProdTable "Rescale batch") recomputes lines but silently rounds to the inventory unit precision and does not preserve percent totals; SAP COR2 quantity change re-reads the recipe and drops manual line edits	Scaling implemented as re-explosion; rounding residual not tracked	BR-24.1 scaling with residual assigned to the largest scalable line, side-by-side preview, manual edits preserved	Percent-controlled Σ = 100.0000 on 100% of batches; rescale 12 clicks → 2
Potency adjustment is manual: AX requires the batch potency on InventBatch to be keyed before picking and the picking list does not recompute compensating lines automatically; SAP active-ingredient management needs batch characteristics plus custom proportion units	Potency not part of the consumption proposal	BR-24.2 proposal computes active and compensating quantities from recorded batch attributes	Potency re-keying 0 (attributes recorded once); consumption error < 0.1%
Co-product costing errors at End: AX TCA allocation misallocates when a co-product has zero standard cost; SAP apportionment structures (TKKAA) must be maintained per material and settlement KO88 fails silently with status messages	Allocation data in separate setup objects, no invariant check	Allocation normalized with residual rule and property test Σ = batch cost; preview before End	Allocation mismatches 0; End failures surfaced synchronously with reason
Kanban cards and boards are disconnected: AX kanban board (KanbanBoardScheduleView) refreshes manually and cannot be used on a tablet; SAP PK13N is a GUI-only status screen; Plex and Epicor Kinetic kanban rely on printed cards and scanner re-keying	Board is a list page over KanbanJob with no push updates	Touch board with SSE refresh, GS1 DataMatrix cards, event kanbans within 5 s	Card-to-board latency 5 min refresh → < 2 s; re-keying 0
Shelf-life and disposition rules are enforced only at picking in AX (PdsShelfLife) and by batch status in SAP (MSC2N), so expired or on-hold batches appear in proposals	Rules applied post hoc rather than in the proposal query	Proposals exclude expired/on-hold batches; override requires duty + reason	Expired-batch consumption incidents → 0; audit reason on 100% of overrides

4.24.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Formula	PmfFormula, BOMVersion with PmfFormulaSize, PmfFormulaLine (scalable, percent-controlled)	Master recipe (C201), material list in recipe, base quantity	Formula (BomVersion subtype), FormulaLine	full
Batch order	ProdTable with ProdType=Batch (PmfProdTable), batch order status = ProdStatus	Process order (COR1/COR2), AUFK with order category 40	BatchOrder (ProductionOrder subtype)	full
Co-/by-products	PmfCoByProd, cost allocation (TCA, manual, none), by-product cost type	Co-product settlement rule (apportionment structure), by-product with negative qty	CoProductLine, allocation_method	full
Potency	PdsBatchAttrib, PdsBatchAttribItem, base attribute, potency unit, active-ingredient line type	Active ingredient management (batch characteristic, proportion unit)	BatchAttribute with `potency` role; consumption rule	full
Batch attributes / disposition	PdsBatchAttrib, PdsBatchAttributes on InventBatch, PdsDispositionCode	Batch classification (MSC1N characteristics), batch status, usage decision	BatchAttribute, InventoryBatch.disposition (w16 batch)	full
Shelf life	InventBatch expiry, best-before, shelf-advice, PdsShelfLife	Shelf-life expiration date (SLED), minimum remaining shelf life	shelf_life_days, best_before, expiry on batch	full
Catch weight	PdsCW, CW unit, nominal/min/max qty	Catch-weight management (S/4 CWM, parallel UoM)	CatchWeightQty on product and transactions	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
PI sheet / process instructions	n/a (route card notes)	PI sheets (CORA/CO60), control recipes, process messages	operation instructions on kiosk (w25); messages via events	partial — no XSteps
Production flow / value stream	LeanProductionFlow, LeanProductionFlowActivity, ValueStream (OMOperatingUnit)	PP-REM production version, product cost collector, MF50 planning table	ProductionFlow, FlowActivity, ValueStream (w6 OperatingUnit)	full
Kanban rules and kanbans	KanbanRule (fixed/scheduled/event), Kanban, KanbanJob, KanbanJobSchedule, kanban board	Control cycle PK01, kanban board PK13N, PK05 demand source, PK17	KanbanRule, Kanban, KanbanJob	full
Backflush costing	Lean backflush costing, production flow costing (periodic)	Product cost collector (KKF6N), REM backflush MFBF, periodic settlement	FlowCostingPeriod	full
Repetitive planning table	n/a	MF50	kanban board takt view	partial
Batch attribute / shelf-life inheritance (batch derivation)	PdsInheritance (Lot inheritance setup), PdsBatchAttribInherit on formula lines, end-item / co-product inheritance	Batch derivation DVMO / DVSP / DVMAN (sender/receiver rules)	InheritanceRule, derivation on RAF and manual (FR-24.15)	full
Batch merge	PdsBatchMerge (Batch merge form)	Transfer posting batch to batch (MIGO 309/311) with manual characteristic update	`:merge` batches with attribute recomputation (FR-24.16)	improved
Consolidated batch orders (bulk → packed)	PdsConsolidatedOrder, bulk/pack relations (PdsBulkItemConv)	Process order with co-ordinated packaging orders (manual)	BulkPackRelation, consolidation job (FR-24.17)	full
Rework batch orders	PdsRework (Rework batch order), rework parameters	Rework process order (COR1 order without material), trigger points	Rework BatchOrder keeping batch identity (FR-24.18)	full
Same-batch reservation; customer sellable days; customer batch attribute requirements	PdsSameBatch, PdsCustSellableDays, PdsCustBatchAttrib	Batch search strategy for sales (VCH1, SD batch determination), minimum remaining shelf life on customer-material info	Same-batch flag, CustomerBatchRequirement (FR-24.19)	full
Lean subcontracting	LeanSubcontracting: service activity, vendor work cell, purchase agreement for service	Subcontracting in REM / kanban with external procurement (PK01 replenishment strategy external)	Service FlowActivity with vendor work cell (FR-24.20)	full
Electronic batch record	n/a (D365 batch order documents only)	COEBR batch record (creation, approval, archive)	ElectronicBatchRecord with review by exception (FR-24.21)	improved
Electronic signatures (21 CFR Part 11)	D365 electronic signature (SIGSignatureLog) on configured events	Digital signature DSIG, signature strategies (O2, DMS signature strategies)	Signature service with step-up re-authentication (FR-24.22)	full
Characteristic-dependent planning	n/a (batch attributes on sales only)	PP/DS characteristic-dependent planning (CDP)	Attribute-constrained pegging (FR-24.23)	full
Documentary batches	Tracking dimension group with batch active but not physical inventory	S/4 documentary batches	Tracking policy `batch-record-only` (FR-24.24)	full

4.24.3 Functional requirements

- **FR-24.1 Formulas** — Must. The system shall define a Formula as a BomVersion with formula_size and FormulaLine fields: scalable (bool), percent_controlled (bool, lines sum to 100%), line_type (ingredient, active-ingredient, phantom, co-product, by-product), per_series/quantity.
  - Notes: Percent-controlled formulas validate Σ scalable line % = 100.00 ± 0.01. Formula version changes go through ECO (w9); a batch order snapshots the formula at creation.
- **FR-24.2 Batch orders** — Must. The system shall create BatchOrder from a formula, scaling scalable lines proportionally to batch size (non-scalable lines fixed), with the same state machine and journals as w21 plus co-/by-product RAF lines.
- **FR-24.3 Potency** — Must. The system shall consume active-ingredient lines by potency: consumed qty = planned qty × (standard potency / actual batch potency), with compensating lines (fillers) adjusted so the batch size is kept.
  - Notes: Potency attribute defined per item (base attribute, target value, min/max); batch potency recorded at receipt or by quality (w33) on the InventoryBatch. Batches outside min/max are not pickable for the line.
- **FR-24.4 Co-/by-product costing** — Must. The system shall allocate batch cost to co-products by method: none (all cost to main item), manual (fixed %), total cost allocation (TCA: Σ co-product standard cost shares), by quantity; by-products reduce cost by burden: per series, per quantity, percent, or recycled (negative issue).
  - Notes: Allocation percentages are normalized to sum 100.0000; rounding residual goes to the main product; property test Σ allocated = batch cost.
- **FR-24.5 Batch attributes and disposition** — Must. The system shall record BatchAttribute values per inventory batch (numeric/text/enum/date with min/max/target), block batches by disposition code (available, on-hold, rejected, quarantine) and restrict picking by customer-specific attribute ranges (sales linkage in w27).
- **FR-24.6 Shelf life** — Must. The system shall compute best_before = manufacturing date + best_before_days and expiry = manufacturing date + shelf_life_days at RAF, support shelf-advice (advice days before expiry) and FEFO picking on batch orders, and block consumption of expired batches unless duty *Consume expired batch* is held.
- **FR-24.7 Catch weight** — Must. The system shall support catch-weight items with inventory quantity in the CW unit (e.g. cases) and a nominal weight with min/max, carrying both quantities on every inventory and production transaction; cost is per inventory (weight) quantity.
  - Notes: RAF requires actual weight per CW unit; variance outside min/max rejected; sales/purchase price is per weight (w26 pricing uses nominal until actual known).
- **FR-24.8 Production flows** — Must. The system shall model ValueStream → ProductionFlow (versioned, with takt time, activation dates) → FlowActivity (process or transfer; cycle time, resource/work cell, input/output locations, picking activities) and validate connectivity (every process activity has an output).
- **FR-24.9 Kanban rules** — Must. The system shall define KanbanRule with type (manufacturing/withdrawal), replenishment strategy (fixed: number of kanbans and product quantity; scheduled: via planning as planned kanbans; event: generated from sales line, BOM line, kanban line, minimum stock), lead time, fixed quantity, activity sequence, and automatic planning/firming.
- **FR-24.10 Kanbans and board** — Must. The system shall create Kanban (card) with KanbanJobs per activity, states Not planned → Planned → Prepared → In progress → Completed → Emptied (card returns), print cards with barcodes (GS1 DataMatrix), and provide a board per work cell (columns by job status) with drag completion and quantity/scrap entry.
- **FR-24.11 Kanban events** — Must. Event kanbans shall be generated within 5 s of a triggering sales line (w27 or fixture) or BOM line of a production order, and withdrawal kanbans create transfer jobs that post inventory transfers on completion.
- **FR-24.12 Lean backflush costing** — Should. The system shall cost production flows periodically: per costing period, issue materials at standard/weighted-average on kanban completion, accumulate conversion cost by activity (cycle time × cost rate), and settle at period end with variances to the flow's posting profile.
- **FR-24.13 Kanban manufacturing orders** — Must. Manufacturing kanbans shall create a lightweight ProductionOrder of type `kanban` (no routing jobs; single process activity) that reports as finished on kanban completion.
- **FR-24.14 Takt and board metrics** — Could. The board shall show takt vs cycle, WIP count per column and kanbans overdue (planned < now).

- **FR-24.15 Batch attribute and shelf-life inheritance (batch derivation)** — Must. The system shall apply InheritanceRules at RAF of batch orders (and on manual derivation) that copy or compute attribute values and shelf-life dates from sender batches (consumed ingredient batches flagged `inherit` on the formula line, or the main product batch for co-products) to receiver batches (product, co-products), with aggregation per attribute: `copy` (single sender), `min`, `max`, `weighted-average` (by consumed quantity), `earliest-date`, `latest-date`, or `fixed`.
  - Notes: Rules are defined per receiving item (AX "Lot inheritance setup") with sender item(s) or formula line; `inherit-shortest-shelf-life` (BR-24.4) is the built-in shelf-life rule. If a sender attribute is missing, the receiver attribute is left empty and the derivation log records `missing-sender-value`. Manual derivation (SAP DVMAN) re-runs rules for a batch after attribute changes upstream and propagates forward to batches already derived from it (max depth 10). Each derivation writes a DerivationLog row (sender batches, rule, values before/after) viewable in the derivation monitor (SAP DVMO).
- **FR-24.16 Batch merge** — Should. The system shall merge two or more batches of the same item and warehouse into a target batch (new or one of the sources) by posting a transfer from each source batch to the target, recomputing target attributes (numeric → weighted average by quantity; text/enum → must be equal or merge is refused unless the attribute is `merge-override`; dates → earliest expiry/best-before) and recording genealogy (all sources as parents).
  - Notes: Sources must have disposition `available` (or the same disposition) and the same catch-weight unit; merge of batches with different potency recomputes potency as weighted average; merge cost per unit = weighted average of source costs (standard-cost items unaffected). Merge requires duty *Merge batches*.
- **FR-24.17 Consolidated batch orders (bulk and packed items)** — Should. The system shall maintain BulkPackRelations (bulk item → packed item, packing quantity per packed unit in bulk UoM, pack formula/BOM) and a consolidation function that groups planned or firmed packed-item orders (planned production orders from w22 or firmed orders) within a date window into one bulk BatchOrder sized to Σ packed demand in bulk UoM (rounded up to formula multiples), creating packing orders that consume the bulk batch and are pegged to it.
  - Notes: Consolidation runs on demand or as a batch job (4.24.16); the bulk order's RAF batch is reserved for its packing orders (same-batch); packing orders inherit bulk batch attributes (FR-24.15) by default. Window default 7 days; maximum bulk batch size = formula maximum.
- **FR-24.18 Rework batch orders** — Should. The system shall create a rework BatchOrder for an existing batch: the batch (or part of it) is the first ingredient line, additional ingredients and rework operations come from a rework formula/route or are entered manually, and the output is received to the same batch number (default) or a new batch per rework parameters, with attributes re-recorded and the original batch genealogy preserved.
  - Notes: Rework parameters (AX "Rework" parameters): `keep-batch-number` (default true), `require-quality-order` (w33, default true), disposition of output (default `quarantine` until quality release). Cost: rework cost posts as variance type `rework` (w21 FR-21.20) unless `rework.capitalize`.
- **FR-24.19 Same-batch reservation and customer batch requirements** — Should. The system shall support (a) a same-batch flag on item (and overridable on sales lines) requiring a reservation or pick of the full line quantity from one batch, (b) customer sellable days (minimum remaining shelf life in days at shipment) per customer × item or item group, and (c) customer batch attribute requirements (attribute ranges per customer × item), enforced in batch reservation proposals, picking (w28) and shelf-life aware planning (w22 FR-22.20).
  - Notes: Batches not meeting the customer's requirements are excluded from proposals; override requires duty *Override customer batch requirements* with reason. Same-batch on a line with quantity larger than any single batch returns 409 `SAME_BATCH_UNAVAILABLE` with the largest available batch quantity. Sales lines arrive in w27; until then the rules apply to transfer and production picking and to fixtures.
- **FR-24.20 Lean subcontracting** — Should. The system shall model a FlowActivity of kind `service` on a vendor work cell (a Vendor resource from w9 linked to a vendor account), priced by a purchase agreement for a service item (w17), and on kanban job completion (or per period, configurable) generate or update purchase order lines for the service quantity, receive them automatically at completion, and post the service cost as conversion cost of the production flow.
  - Notes: Modes: `po-per-kanban` (one PO line per kanban job), `periodic-po` (one PO per vendor per period with released quantities accumulated, receipts per job). Material sent to the vendor moves to the vendor work cell's location via withdrawal kanbans. Vendor invoice matching uses the w18 three-way match against auto-receipts.
- **FR-24.21 Electronic batch record (EBR)** — Should. The system shall compile, for each batch order of an item whose BatchRecordProfile is active, an ElectronicBatchRecord containing: header (product, batch, formula/route version with ECO ids, batch size, dates), every posted journal line with batch/serial, attribute values with recorded-by, executed PI sheet steps and process data (w25 FR-25.16), process messages (w25 FR-25.17), equipment and cleanout records (w23), deviations (w33 nonconformances linked to the order), exceptions (any value outside limits, any override, any signature-required step), and signatures (FR-24.22); and shall route it through review (production review, QA review) with review by exception — reviewers must disposition every exception — before QA approval, after which the record is sealed (hash) and archived as PDF/A-2b plus JSON.
  - Notes: BatchRecordProfile: required sections, reviewer roles, number of signatures per stage, whether batch disposition `available` requires EBR approval (default true for GxP items). A sealed record is immutable; corrections create an addendum record linked to the original. EBR generation is incremental (sections update on events) so the review screen is current while the batch runs.
- **FR-24.22 Electronic signature service** — Must. The system shall provide a signature service usable by any service through `POST /api/v1/signatures` that records a signature only after re-authentication of the signer (Keycloak step-up: password plus a second factor; first signing in a session requires both components, subsequent signings in the same continuous session require at least one per 21 CFR 11.200(a)(1)), binds it to the signed record by a SHA-256 hash of the record's canonical JSON, and stores signer id, printed name, date/time (UTC and signer's local time zone), meaning (authored, reviewed, approved, verified, released, rejected) and reason/comment.
  - Notes: Signature strategies define for an object type and event: number of signers, sequence (parallel/serial), distinct-user rule (signers ≠ record author), required roles/duties, and the meaning per signer. Signature manifestation (11.50) is rendered on every printed or exported signed record. Signatures cannot be copied or transferred (11.70): any change to the record after signing invalidates the hash and the record shows `signature-invalidated` until re-signed. Signature log is append-only (outbox to Iceberg, 15-year retention). Three consecutive failed re-authentications lock signing for the user for 15 minutes and raise a security event (w2). Wave 48 extends this service to configured tables/fields across all modules.
- **FR-24.23 Characteristic-dependent planning** — Could. The planning engine (w22) shall carry required attribute ranges on demand (from customer batch requirements FR-24.19 or entered on the demand line) and target attribute values on supply (batch attribute values for on-hand, formula target values for planned production), and peg supply to demand only when supply attributes satisfy the demand's ranges; unmatched demand yields planned orders with the demand's attribute targets.
- **FR-24.24 Documentary batches** — Could. The system shall support a tracking policy `batch-record-only` for items that are not batch-managed in inventory: batch numbers are captured on receipts, issues and RAF and stored on InventoryTransaction for traceability and genealogy, but on-hand is not split by batch and FEFO/disposition rules do not apply.

4.24.4 Data model

FormulaLine: id, bom_version_id, line_number, item_id, line_type (enum), qty (numeric 18,6), percent (numeric 9,4), scalable (bool), percent_controlled (bool), potency_base_item (bool), unit_id, scrap_pct, is_co_product (bool), allocation_pct (numeric 7,4), by_product_cost_type (enum: per-series, per-quantity, percent, recycled).

BatchOrder (extends ProductionOrder): formula_size (numeric), batch_size (numeric), scale_factor (numeric 18,9, computed), planning_item_id (for co-product planning), potency_mode (enum), manufacturing_date, batch_number (text, generated at RAF via `batch` sequence).

CoProductLine: id, order_id, item_id, planned_qty, raf_qty, allocation_method (enum: none, manual, tca, quantity), allocation_pct, allocated_cost (numeric 18,4), is_by_product (bool), by_product_cost_type.

BatchAttribute: id, item_id, attribute_id, data_type (numeric/text/enum/date), target, min, max, role (enum: potency, generic), unit_id; **BatchAttributeValue**: batch_id (w16 InventoryBatch), attribute_id, value_numeric, value_text, value_date, recorded_at, recorded_by.

CatchWeightQty (on product): cw_unit_id, nominal_qty (numeric 18,6), min_qty, max_qty; on InventoryTransaction and journal lines: cw_qty (numeric 18,6) alongside qty.

ProductionFlow: id, value_stream_id (OperatingUnit), version, name, site_id, takt_time_s (int), status (draft/active/expired), active_from, active_to; **FlowActivity**: id, flow_id, name, kind (process/transfer), cycle_time_s, work_cell_resource_id, input_location_id, output_location_id, predecessor_ids (uuid[]), picking_required (bool).

KanbanRule: id, item_id, site_id, warehouse_id, location_id, rule_type (manufacturing/withdrawal), replenishment (fixed/scheduled/event), fixed_qty, number_of_kanbans (int), min_kanbans, max_kanbans, lead_time_h, event_source (sales/bom/kanban/min-stock), auto_plan (bool), auto_firm (bool), flow_activity_sequence (uuid[]), effective_from, effective_to.

Kanban: id, rule_id, card_number, qty, cw_qty, state (enum), handling_units (int), planned_start, due, production_order_id / transfer_id, event_ref_type, event_ref_id, printed_at; **KanbanJob**: id, kanban_id, activity_id, sequence, state (not-planned/planned/prepared/in-progress/completed), start, end, reported_qty, scrap_qty.

FlowCostingPeriod: id, flow_id, period_start, period_end, status (open/settled), material_cost, conversion_cost, output_value, variance, journal_entry_id.

Invariants: Σ FormulaLine.percent for percent-controlled scalable lines = 100.0000; Σ CoProductLine.allocated_cost + main-item cost = batch actual cost (after by-product burden); Kanban state transitions only forward except `Emptied → Not planned` on card reuse; fixed-rule active kanbans count ≤ number_of_kanbans; cw_qty present iff the item is catch-weight.

Indexing: Kanban (rule_id, state); KanbanJob (activity_id, state, planned_start) for the board; BatchAttributeValue (batch_id); FormulaLine (bom_version_id, line_number).

InheritanceRule / DerivationLog (FR-24.15): InheritanceRule(id, receiver_item_id, sender_item_id or formula_line_ref, attribute_id or `shelf-life`, aggregation (enum copy/min/max/weighted-average/earliest-date/latest-date/fixed), fixed_value, applies_to (enum product/co-product/all), active); DerivationLog(id, receiver_batch_id, sender_batch_ids uuid[], rule_id, attribute_id, value_before, value_after, trigger (raf/manual/merge), derived_at, derived_by).

BulkPackRelation (FR-24.17): id, bulk_item_id, packed_item_id, pack_qty_per_unit numeric(18,6) (bulk UoM per packed unit), pack_bom_version_id, consolidation_window_days int default 7, active.

CustomerBatchRequirement (FR-24.19): id, customer_id or customer_group_id, item_id or item_group_id, sellable_days int, attribute_id nullable, min, max, same_batch (bool), valid_from, valid_to.

ElectronicBatchRecord / EbrException / BatchRecordProfile (FR-24.21): ElectronicBatchRecord(id, order_id, batch_id, profile_id, status (compiling/in-review/approved/sealed/addendum), content_hash bytea, pdf_document_id (w5), json_document_id, sealed_at, parent_record_id (addendum)); EbrException(id, record_id, source_type, source_id, description, severity, disposition (accepted/deviation-raised/rejected), dispositioned_by, signature_id); BatchRecordProfile(id, code, item_group_id, required_sections text[], review_stages jsonb, approval_required_for_release bool).

Signature / SignatureStrategy (FR-24.22): Signature(id, tenant_id, object_type, object_id, record_hash bytea, signer_subject_id, printed_name, meaning (enum), reason text, signed_at_utc timestamptz, signer_time_zone text, auth_method (enum password+otp/otp/webauthn), session_id, strategy_id, sequence int, valid (bool)); SignatureStrategy(id, object_type, event, signers int, order (parallel/serial), distinct_from_author bool, required_duties text[], meanings text[]).

Invariants (additions): a sealed ElectronicBatchRecord's content_hash equals SHA-256 of its JSON document; every EbrException of an approved record has a disposition and a signature; a Signature with valid = true has record_hash equal to the current hash of the object; DerivationLog rows are append-only.

4.24.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/production/batch-orders	Create batch order from formula with batch_size	Idempotency-Key	Maintain batch orders
POST	/api/v1/production/batch-orders/{id}:rescale	Change batch size before Started	Idempotency-Key	Maintain batch orders
POST	/api/v1/production/batch-orders/{id}/co-products:allocate	Preview/apply allocation	Idempotency-Key	Post production journals
PUT	/api/v1/inventory/batches/{id}/attributes	Record attribute values	full replace	Maintain batch attributes
POST	/api/v1/inventory/batches/{id}:disposition	Change disposition	Idempotency-Key	Maintain batch disposition
POST	/api/v1/lean/flows	Create/version production flow	Idempotency-Key	Maintain production flows
POST	/api/v1/lean/flows/{id}:activate	Validate and activate	Idempotency-Key	Maintain production flows
POST	/api/v1/lean/kanban-rules	Create rule	Idempotency-Key	Maintain kanban rules
POST	/api/v1/lean/kanbans:generate	Generate kanbans for a rule (fixed/event)	Idempotency-Key	Maintain kanbans
POST	/api/v1/lean/kanban-jobs/{id}:complete	Complete job with qty/scrap/handling units	Idempotency-Key	Work kanban board
GET	/api/v1/lean/board?workCellId	Board view (jobs by column)	n/a	Work kanban board
POST	/api/v1/lean/flows/{id}/costing-periods/{p}:settle	Settle period	Idempotency-Key	Settle production flows

Event	Trigger	Payload fields	Consumers
`passage.production.batch-order.created.v1`	batch order created	orderId, formulaVersionId, batchSize, scaleFactor, coProducts[]	planning, analytics
`passage.production.batch.attributes-recorded.v1`	attribute values saved	batchId, itemId, values[{attributeId, value}], potency	quality (w33), sales (w27)
`passage.production.co-product.allocated.v1`	End of batch order	orderId, lines[{itemId, method, pct, amount}], byProductBurden	ledger, costing
`dev.passage.lean.kanban.created.v1`	kanban generated	kanbanId, ruleId, qty, eventRefType, eventRefId, due	production, shopfloor
`passage.lean.kanban.state-changed.v1`	job/kanban transition	kanbanId, jobId, from, to, qty, scrap, handlingUnits	production, inventory, shopfloor
`passage.lean.flow-period.settled.v1`	period settled	flowId, period, materialCost, conversionCost, outputValue, variance, journalEntryId	ledger, analytics

4.24.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Formula designer	Add lines, toggle scalable/percent, define co/by-products, validate 100%, submit ECO	Line, item, type, qty/%, scalable, potency base, allocation %	Maintain formulas
Batch order detail	Rescale, estimate, release, RAF with co-products, attributes tab	Batch size, scale factor, co-product lines, batch number, dates, attributes	Maintain batch orders
Batch attributes / disposition	Record values, change disposition, view history	Attribute, value, target/min/max, pass flag, disposition	Maintain batch attributes
Production flow designer	Canvas of activities, connect, set cycle times, activate	Activity, kind, cycle time, work cell, locations	Maintain production flows
Kanban rules	Create fixed/scheduled/event rules, compute number of kanbans	Item, type, replenishment, qty, kanbans, lead time	Maintain kanban rules
Kanban board (work cell)	Drag job between columns, complete with qty, print cards, filter	Columns Not planned/Planned/Prepared/In progress/Completed; card: item, qty, due, order	Work kanban board
Flow costing	Open/settle period, variance drill	Period, material, conversion, output, variance	Settle production flows

Interaction rules: formula percent cells show a running total with red when ≠ 100; board is touch-friendly (min 44 px targets) and refreshes via SSE; card print uses the w5 document service with GS1 DataMatrix encoding `(240)kanbanId`; rescale shows before/after line quantities side-by-side.

4.24.7 Business rules and edge cases

- **BR-24.1 Scaling.** scale_factor = batch_size / formula_size; scalable line qty = formula qty × scale_factor; non-scalable lines unchanged; percent-controlled lines qty = batch_size × percent/100; rounding to UoM precision with residual added to the largest scalable line so Σ = batch_size.
- **BR-24.2 Potency consumption.** For an active-ingredient line with standard potency P_s and picked batch potency P_a: qty = planned × P_s / P_a; the compensating line qty = planned_comp – (qty – planned); negative compensating qty is rejected with 422 `POTENCY_COMPENSATION_NEGATIVE`. Example: 100 kg at 50% standard, batch at 40% → 125 kg; filler reduced by 25 kg.
- **BR-24.3 Co-product allocation (TCA).** share_i = std_cost_i × raf_qty_i / Σ_j std_cost_j × raf_qty_j; allocated_i = round(batch_cost_after_by_products × share_i, 4); residual to the main product. Example: batch cost USD 10,000.00; main A (std 40, 200 units = 8,000), co-product B (std 20, 100 = 2,000): A gets 8,000.00, B 2,000.00. By-product C per-quantity burden 1.50 × 50 = 75.00 reduces batch cost first (9,925.00 allocated 7,940.00 / 1,985.00; C received at 75.00).
- **BR-24.4 Shelf life.** best_before/expiry computed at RAF from manufacturing_date (defaults to RAF date); if the formula has `inherit-shortest-shelf-life`, the output expiry = min(ingredient batch expiries) when earlier.
- **BR-24.5 Catch-weight RAF.** cw_qty integer ≥ 1; qty (weight) must satisfy min_qty × cw_qty ≤ qty ≤ max_qty × cw_qty; cost per weight unit.
- **BR-24.6 Kanban state machine.** Not planned → Planned (date set) → Prepared (picking complete) → In progress (first job started) → Completed (last job completed; manufacturing: RAF posted; withdrawal: transfer posted) → Emptied (consumed; card returns to Not planned for fixed rules, or is deleted for event rules). Backward moves only Planned → Not planned. Illegal moves return 409.
- **BR-24.7 Fixed rule kanban count.** number_of_kanbans = ⌈(daily demand × lead_time_days × (1 + safety%)) / fixed_qty⌉ as a suggestion; rule cannot have more active (non-Emptied) kanbans than number_of_kanbans.
- **BR-24.8 Event kanban quantity.** Event kanbans for demand D create ⌈D / fixed_qty⌉ kanbans; last kanban is full-size (overproduction allowed) unless `exact_qty` flag.
- **BR-24.9 Flow period settlement.** variance = material_cost + conversion_cost – output_value; settled at period end only if all kanbans with jobs in the period are Completed or Emptied, else 409 `FLOW_PERIOD_OPEN_KANBANS` listing them.
- **BR-24.10 Expired batch.** A picking line selecting a batch with expiry < posting date fails unless duty held, in which case the audit log records the override reason (required, ≥ 10 chars).
- **BR-24.11 Weighted-average inheritance and merge.** value = Σ (v_i × q_i) / Σ q_i over senders with a value, rounded to the attribute's decimals (default 4); example: batches 200 kg at 40.0% and 300 kg at 45.0% potency → 43.0000%. Earliest-date aggregation for expiry: min(e_i).
- **BR-24.12 Consolidation sizing.** Bulk batch qty = ⌈Σ_j packed_qty_j × pack_qty_per_unit / formula_multiple⌉ × formula_multiple; if it exceeds formula maximum, split into ⌈qty / max⌉ bulk orders, allocating packing orders by due date.
- **BR-24.13 Customer sellable days.** A batch qualifies for a customer line shipping on date d if expiry – d ≥ sellable_days; if both item default and customer value exist, the customer value wins; if both customer-group and customer exist, the customer wins.
- **BR-24.14 Signature re-authentication.** A continuous session is one without idle > 10 minutes or logout; the first signature in a session requires password + second factor; subsequent signatures require password; the signer must match the session subject; a signing attempt with a mismatched subject is rejected and logged.
- **BR-24.15 EBR review by exception.** An exception is created for: a value outside limits, any override of a block (expired batch, disposition, constraint), an unsigned required step, a deviation linked to the order, a manual journal edit after posting, or a missed process message; approval is refused while any exception is undispositioned.
- **BR-24.16 Lean subcontract receipt.** Service receipt qty = completed kanban job qty; periodic POs close at period end with quantity = Σ receipts; price from the purchase agreement valid on the job completion date.

4.24.8 Acceptance criteria

- **AC-24.1** (FR-24.2, BR-24.1) Given a 1,000 kg formula with a 50 kg non-scalable line, when I create a 2,500 kg batch, then scalable lines are ×2.5 and the non-scalable line stays 50 kg.
- **AC-24.2** (FR-24.3) Given the BR-24.2 example, when the picking list proposes consumption, then the active line is 125 kg and the filler is reduced by 25 kg.
- **AC-24.3** (FR-24.4) Given the BR-24.3 fixture for each allocation method, when the order ends, then Σ allocated cost + by-product burden = batch actual cost exactly.
- **AC-24.4** (FR-24.6) Given an expired ingredient batch, when picked without the duty, then 422 `BATCH_EXPIRED`.
- **AC-24.5** (FR-24.7) Given a catch-weight item (nominal 10 kg, min 9, max 11), when I RAF 10 cases at 95 kg, then the post fails; at 102 kg it succeeds with qty 102 and cw_qty 10.
- **AC-24.6** (FR-24.11) Given an event rule with fixed_qty 20, when a sales line for 50 arrives, then 3 kanbans exist within 5 s and completing them through the board posts RAF of 60.
- **AC-24.7** (FR-24.10) Given a kanban in Planned, when I drag it to Completed, then 409 and the board reverts.
- **AC-24.8** (FR-24.12) Given a flow period with 10 completed kanbans, when settled, then the journal entry balances and the variance equals material + conversion – output.
- **AC-24.9** (FR-24.1) Given a percent-controlled formula summing to 99.5%, when I activate it, then validation fails.
- **AC-24.10** (FR-24.5) Given a batch with disposition on-hold, when a batch order picks it, then the batch is excluded from proposals.
- **AC-24.11** (FR-24.15, BR-24.11) Given an inheritance rule `weighted-average` on potency and two ingredient batches (200 kg at 40%, 300 kg at 45%), when the batch order reports as finished, then the product batch potency is 43.0000% and the derivation monitor shows both senders.

- **AC-24.12** (FR-24.16) Given two available batches of 100 kg (expiry 2027-03-01) and 50 kg (expiry 2027-02-01), when merged into a new batch, then the target holds 150 kg with expiry 2027-02-01 and genealogy lists both sources.
- **AC-24.13** (FR-24.17, BR-24.12) Given packed orders for 400 and 600 bottles at 0.5 kg each and formula multiple 100 kg, when consolidated, then one bulk batch order of 500 kg exists and both packing orders are pegged to it.
- **AC-24.14** (FR-24.19, BR-24.13) Given customer sellable days 60 and batches expiring in 45 and 90 days, when proposing reservation for that customer, then only the 90-day batch is proposed.
- **AC-24.15** (FR-24.22, BR-24.14) Given a QA approver signing an EBR, when the second factor is wrong, then no signature is stored; after three failures signing is locked for 15 minutes and a security event exists.
- **AC-24.16** (FR-24.21, BR-24.15) Given a batch with one expired-batch override, when QA attempts approval without dispositioning the exception, then approval is refused; after disposition and signatures the record is sealed and its PDF shows signer name, meaning and UTC time.
- **AC-24.17** (FR-24.20, BR-24.16) Given a service activity on vendor V with mode `po-per-kanban`, when a kanban job of 20 completes, then a PO line for 20 service units at the agreement price is created and received.
- **AC-24.18** (FR-24.18) Given rework of batch B-77 with `keep-batch-number`, when the rework order reports as finished, then batch B-77 has the new quantity, disposition `quarantine`, and the genealogy shows the rework order.

4.24.9 Test plan and fixtures

Fixtures: `pmf-potency-20` (standard/actual potency pairs with expected consumption); `pmf-coby-12` (allocation methods × by-product cost types); `pmf-catchweight-10`; `lean-flow-demo` (3-activity flow, 2 fixed rules, 1 event rule); `lean-kanban-sim` (sales lines → event kanbans → board completion).

Test	Type	Target	Pass criterion
Potency	unit	consumption proposal	20/20
Allocation sums	property	allocation	Σ = cost for random costs/quantities
Catch weight	unit	RAF validation	min/max enforced
Kanban state machine	property	transitions	illegal rejected
Event kanban latency	integration	event→kanban	p95 < 5 s
Board E2E	Playwright	board	kanbans complete, RAF posted
Flow settlement	contract	ledger	balanced entry

Reconciliation: Σ co-product allocated cost = batch cost; Σ kanban RAF = inventory receipts from kanban orders; flow period variance = GL flow variance account.

4.24.10 Deployment and operations

Helm: `production` extended (flag `prod.process`), `lean` (Go, 2 replicas), `lean-db` (CloudNativePG 50 GB). Config: `LEAN_EVENT_KANBAN_SLA_S` (5), `PMF_POTENCY_TOLERANCE_PCT` (0.5). Flags: `prod.process`, `prod.catch-weight`, `lean.kanban`, `lean.flow-costing`. Jobs: shelf-advice notification daily 06:00; flow period auto-settle (optional) monthly. SLOs: board load p95 < 500 ms; kanban generation p95 < 5 s. Alerts: event-kanban backlog > 100 (warn); flow settlement failure (page). Rollback: prior digest; flags off disables the second plant's workspaces without data loss.

4.24.11 Migration from AX / SAP

Source	Passage entity	Notes
PmfFormula/BOMVersion+BOM lines (AX) / PLKO+PLPO+STPO recipe (SAP)	Formula, FormulaLine	formula size → base quantity; percent lines derived from quantities when SAP recipe is in base units
PmfCoByProd / co-product apportionment (TKKAA)	CoProductLine on formula	TCA percentages from apportionment structure
PdsBatchAttrib, PdsBatchAttribItem, PdsBatchAttributes / batch classification (AUSP, CABN)	BatchAttribute, BatchAttributeValue	characteristic → attribute; numeric/char/date types
InventBatch expiry/best-before / MCH1 VFDAT, HSDAT	InventoryBatch (w16) fields	already migrated in w16; this wave adds attributes
PdsCW / CWM units	CatchWeightQty	nominal/min/max
LeanProductionFlow, LeanProductionFlowActivity / production versions, REM profiles	ProductionFlow, FlowActivity	active versions only
KanbanRule, Kanban, KanbanJob / PKHD, PKPS	KanbanRule, Kanban (active cards)	completed/emptied cards not migrated

Validation: formula explosion of 100 sampled batches matches legacy consumption to 0.01%; active kanban count per rule equals legacy. Not migrated: process-order history, PI sheet archives, SAP XSteps.

Migration addendum (FR-24.15–24.24). AX PdsInheritance setup → InheritanceRule; AX PdsCustSellableDays/PdsCustBatchAttrib → CustomerBatchRequirement; AX PdsBulkItemConv → BulkPackRelation; AX LeanSubcontracting service activities → FlowActivity kind `service`; SAP DVSP derivation strategies → InheritanceRule (sender/receiver fields → attributes); SAP customer-material minimum remaining shelf life (KNMT) → sellable_days; SAP signature strategies (DSIG / O2) → SignatureStrategy. Historical EBRs and signature logs are not re-signed: legacy COEBR archives and AX SIGSignatureLog are loaded as read-only archived documents (w5) with their legacy hashes, linked to the batch.

4.24.12 Dependencies, risks, open questions

Depends on: w21 order engine; w16 batches and FEFO; w9 ECO; w6 operating units (value streams); w22 scheduled kanbans via planning.

Risk	Likelihood	Impact	Mitigation
Potency/attribute semantics vary by industry	High	Medium	Configurable role model; consultant review
Lean costing conflicts with order-based WIP reconciliation	Medium	High	Kanban orders are ProductionOrders — same WIP accounts; flow periods settle through the same End logic

Open questions: (1) Should by-product `recycled` (negative issue) be supported in v1? Default yes. (2) Board per work cell vs per flow? Default work cell; decide by w24 kickoff.

4.24.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Batch order number sequence	legal entity	sequence ref	`batch-order`	any	Batch order numbers	Production control parameters → Number sequences (batch order)	Order type PI01 number range (CO82/OPJH)
Batch number sequence	item group	sequence ref	`batch`	any	Batch numbers at RAF	Batch number groups (InventNumGroup)	Batch number assignment (OMCZ / internal number range BATCH_CLT)
Potency tolerance %	legal entity	numeric	0.5	0–10	Accepted potency deviation	Process manufacturing parameters → Potency	Active ingredient tolerance (proportion unit settings)
Default co-product allocation	item group	enum	`tca`	none/manual/tca/quantity	FR-24.4 default	Formula → Cost allocation	Apportionment structure default (OKG8 / TKKAA)
Shelf-advice days default	item group	int	30	0–3,650	Shelf-advice notification	Item → Shelf advice period	Minimum remaining shelf life (MARA.MHDRZ)
Expired batch consumption allowed	site	bool	false	true/false (duty still required)	FR-24.6	Batch disposition / expiry blocking	Batch status management (OMCT)
Catch-weight variance handling	legal entity	enum	`reject`	reject / warn	BR-24.5	CW min/max validation	CWM tolerance group
Inheritance active	legal entity	bool	true	true/false	FR-24.15	Lot inheritance (PdsInheritance)	Batch derivation activation (DVMO customizing)
Merge override attributes	attribute	bool	false	true/false	FR-24.16	Batch merge setup	n/a
Consolidation window days	bulk/pack relation	int	7	1–60	FR-24.17	Consolidated batch orders setup	n/a
Rework keep batch number	legal entity	bool	true	true/false	FR-24.18	Rework parameters	Rework order settings
Rework requires quality order	legal entity	bool	true	true/false	FR-24.18	Rework parameters	QM inspection type 04 on rework
Customer sellable days default	item group	int	0	0–3,650	BR-24.13	Customer sellable days	KNMT minimum remaining shelf life
Lean subcontract PO mode	vendor work cell	enum	`po-per-kanban`	po-per-kanban / periodic-po	FR-24.20	Lean subcontracting purchase settings	Kanban control cycle external procurement (PK01)
Batch record profile	item group	ref	none	any BatchRecordProfile	FR-24.21	n/a	EBR batch record settings (COEBR customizing)
EBR approval required for release	profile	bool	true	true/false	Disposition `available` gated	n/a	Batch record approval before usage decision
Signature strategy per event	object type × event	ref	none	any SignatureStrategy	FR-24.22	Electronic signature setup (SIGSetup)	Signature strategies (DSIG / O2)
Signing lockout attempts / minutes	tenant	int / int	3 / 15	1–10 / 1–1,440	FR-24.22	Security parameters	DSIG settings
Kanban card barcode symbology	site	enum	`gs1-datamatrix`	gs1-datamatrix / code128 / qr	Card print	Kanban card setup	PK card print (PK17)
Flow costing period length	production flow	enum	`month`	week / month	FR-24.12	Lean costing periods	Product cost collector period (KKF6N)

4.24.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Batch orders	list inquiry	status, formula, site, dates	order, formula version, batch size, scale factor, status, batch number	BatchOrder	Batch orders list page (PmfProdTable)	COOISPI process order info system
Batch attribute values	drill-down	item, batch, attribute	value, target, min, max, pass, recorded by	BatchAttributeValue	Batch attributes form (PdsBatchAttributes)	MSC3N batch classification
Batch genealogy (top-down / bottom-up)	drill-down	batch	tree of parent/child batches, orders, qty	InventoryTransaction genealogy + DerivationLog	Batch tracing (InventTrackingRegisterTrans), Item tracing	MB56 batch where-used list
Derivation monitor	list inquiry	receiver batch, date, status	rule, senders, values before/after, missing values	DerivationLog	Lot inheritance inquiry	DVMO
Co-product allocation	operational report	order, period	method, %, allocated cost by item	CoProductLine	Co-product cost allocation inquiry	KKF6N / CO88 settlement log
Shelf-life / expiry report	operational report	site, expiry within n days	batch, item, qty, expiry, best before, shelf advice	InventoryBatch	SSRS "Batch expiry" / shelf-life inquiry	MB5M BBD/SLED list
Kanban board / kanban overview	list inquiry	work cell, rule, state	card, item, qty, due, state, order	Kanban, KanbanJob	Kanban board (KanbanBoardScheduleView), Kanbans list	PK13N, PK18
Kanban rule analysis	KPI	rule, period	cycles, average lead time, overdue %, kanbans active vs max	Kanban	Kanban rule inquiry	PK13N control cycle evaluation
Flow costing period	operational report	flow, period	material, conversion, output, variance	FlowCostingPeriod	Lean costing (backflush costing) inquiry	KKF6N, MF41
Electronic batch record list	list inquiry	status, product, date, reviewer	record, batch, open exceptions, stage, sealed at	ElectronicBatchRecord	n/a	COEBR batch record list
Signature log	list inquiry	object type, signer, date	object, meaning, signer, UTC time, reason, valid	Signature	Electronic signature log inquiry (SIGSignatureLog)	DSIG signature log (DSAL)
Customer batch requirement check	list inquiry	customer, item	requirement, qualifying batches, qty available	CustomerBatchRequirement × InventoryBatch	Customer sellable days inquiry	Batch determination log (VCH1/VB(J))

4.24.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Batch ticket (picking list for batch order)	Release	Header: batch order, formula version, batch size, scale factor; lines: ingredient, qty, potency-adjusted qty, batch, weigh station; signatures block (weighed by / verified by)	PDF, kiosk (w25)	`prod-doc`	Part of EBR for GxP items
Kanban card	Kanban creation, reprint	Item, qty, CW qty, source/destination location, rule, GS1 DataMatrix `(240)kanbanId`, due	PDF (card stock 4×6 in / A6)	Kanban card number per rule	None
Batch label	RAF	Item, batch, manufacturing date, expiry/best-before, catch weight, GS1-128 `(01)(10)(17)(310n)`	ZPL via w32 printers, PDF	batch sequence	Food/pharma labeling rules per country (w48)
Certificate of analysis (batch attributes)	Batch release, on demand	Batch, attribute, specification, result, release signature	PDF, email, EDI (w35)	`coa`	Customer contract; signature manifestation if signed (FR-24.22)
Electronic batch record	QA approval (sealed)	Full EBR sections, exceptions with dispositions, signature manifestation (printed name, meaning, date/time)	PDF/A-2b + JSON, SHA-256 hash	`ebr` per legal entity	21 CFR 211.180: at least 1 year after batch expiry (default expiry + 1 year, min 5 years); EU GMP Annex 11 / EudraLex Vol 4 Ch 4: 1 year after expiry or 5 years after certification, whichever longer
Signature manifestation	Any signed record print/export	Signer printed name, date/time, meaning	Embedded in PDF	n/a	21 CFR 11.50
Lean service purchase order	FR-24.20	Vendor, service item, qty, agreement reference	PDF, EDI ORDERS (w35)	PO sequence (w17)	Purchasing retention (w17)

4.24.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Shelf-advice notification	daily 06:00 site-local	site	One notification per batch per advice date	< 1 min	Retry next run	Delivery failures warn
Consolidate batch orders	daily 21:00 (opt-in) + on demand	site × bulk item	Proposal per window replaced; firm on confirm	< 2 min	Per-bulk errors listed	Result report
Inheritance re-derivation	on attribute change event + nightly 02:30	receiver item	DerivationLog per (batch, rule, trigger id)	< 5 min for 10k batches	Max depth 10; cycles stop	Missing-sender count reported
Flow period settlement	monthly (optional auto)	production flow	One settlement per (flow, period); blocked with open kanbans	< 5 min	409 with open kanbans list	Failure pages
EBR compilation	event-driven + hourly sweep	batch order	Section upserts by source event id	< 10 s per event	Sweep repairs missed events	Missed events > 0 warns
EBR archive and seal	on approval	record	Seal once; hash stored; archive retry	< 30 s	Retry 5×; record stays approved-unsealed	Unsealed > 1 h pages
Lean periodic PO close	period end	vendor	Close once per (vendor, period)	< 1 min	Retry	Failure warns
Fixed-kanban count suggestion	weekly Monday 05:00	rule	Suggestion per rule; not applied automatically	< 1 min	Non-blocking	Deviations > 25% reported

4.24.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
Maintain formulas	Formula designer, formula ECO submit	with *Approve engineering change* (w9)	Formulator, Production engineer
Maintain batch orders	batch-orders create, `:rescale`, consolidation, rework orders	none	Production supervisor, Planner
Maintain batch attributes	PUT batch attributes, manual derivation	with *Maintain batch disposition* for the same batch flagged	Quality technician
Maintain batch disposition	`:disposition`	with *Maintain batch attributes* (same batch)	QA manager
Merge batches	batch `:merge`	none	Inventory supervisor
Consume expired batch	picking override with reason	with *Maintain batch disposition* flagged	Production supervisor
Override customer batch requirements	FR-24.19 override	none	Sales supervisor (w27)
Maintain production flows / kanban rules	flows, activities, rules, service activities	none	Lean engineer
Work kanban board	board, job complete, card print	none	Operator, Shop floor supervisor
Settle production flows	flow period settle	with *Maintain production flows*	Cost accountant
Review batch records	EBR review stage, exception disposition	with *Approve batch records* (same record)	Production reviewer
Approve batch records	EBR QA approval, seal	with *Review batch records* and with *Maintain batch orders* (same order)	QA manager
Maintain signature strategies	SignatureStrategy setup	with *Approve batch records*	Compliance administrator

4.24.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
PMF-0001	error	Percent-controlled lines sum to {sum}%, not 100%.	Formula validation	Adjust line percentages	FR-24.1
PMF-0002	error	Compensating line {line} would become negative ({qty}).	Potency adjustment	Choose another batch or change formula	BR-24.2 (`POTENCY_COMPENSATION_NEGATIVE`)
PMF-0003	error	Batch {batch} expired on {date}.	Expired batch picked	Pick another batch or use override duty	BR-24.10 (`BATCH_EXPIRED`)
PMF-0004	error	Catch weight {qty} is outside {min}–{max} for {cw} units.	BR-24.5	Re-weigh	BR-24.5
PMF-0005	warning	Batch {batch} potency {p}% is outside {min}–{max}%; not pickable.	FR-24.3	Choose batch in range	FR-24.3
PMF-0006	error	Merge refused: attribute {attr} differs ({a} vs {b}).	FR-24.16	Align values or mark merge-override	FR-24.16

Code	Severity	User text	Cause	Remediation	Raised by
PMF-0007	warning	Inheritance rule {rule}: sender value missing for {attr}.	FR-24.15	Record sender attribute and re-derive	FR-24.15
PMF-0008	error	No single batch can supply {qty} (largest {max}).	Same-batch requirement	Split line or wait for supply	FR-24.19 (`SAME_BATCH_UNAVAILABLE`)
PMF-0009	warning	Batch {batch} does not meet customer {customer} requirements ({rule}).	FR-24.19	Choose qualifying batch or override	FR-24.19
PMF-0010	error	Signature rejected: re-authentication failed.	BR-24.14	Retry with correct credentials	FR-24.22
PMF-0011	error	Signing locked for {minutes} minutes after {n} failed attempts.	Lockout	Wait or contact security	FR-24.22
PMF-0012	error	Signer must differ from the record author.	Distinct-user rule	Another user signs	FR-24.22
PMF-0013	warning	Record changed after signing; signature invalidated.	Hash mismatch	Re-sign	FR-24.22
PMF-0014	error	Batch record cannot be approved: {n} exceptions without disposition.	BR-24.15	Disposition exceptions	FR-24.21
PMF-0015	error	Batch {batch} cannot be released: batch record not approved.	Profile gating	Complete EBR approval	FR-24.21
LEAN-0001	error	Kanban move {from} → {to} is not allowed.	BR-24.6	Use allowed move	BR-24.6
LEAN-0002	error	Rule {rule} already has {n} active kanbans (max {max}).	BR-24.7	Empty a kanban or raise count	BR-24.7
LEAN-0003	error	Period cannot be settled: {n} kanbans open.	BR-24.9	Complete kanbans	BR-24.9 (`FLOW_PERIOD_OPEN_KANBANS`)
LEAN-0004	error	Flow activity {activity} has no output location.	FR-24.8 connectivity	Set output	FR-24.8
LEAN-0005	error	No valid purchase agreement for service {item} on {date}.	FR-24.20	Create or extend agreement	FR-24.20

4.24.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation under 3× load
Batch order create + scale	500/day; peak 20/min	150 / 400 / 900 ms	50/s	~5 GB (orders + lines)	as w21	Queue behind HPA
Potency proposal	2,000/day	100 / 300 / 700 ms	50/s	n/a	BatchAttributeValue (batch_id)	Proposal without potency sort (warning)
Event kanban generation	10,000/day; peak 200/min	1 / 5 / 8 s	50 kanbans/s	Kanban + jobs ~ 10M rows ≈ 4 GB	Kanban (rule_id, state); KanbanJob (activity, state, planned_start)	SLA alert; backlog processed FIFO
Board load	20,000 loads/day	150 / 500 / 1,000 ms	200/s	n/a	as above	SSE falls back to 30 s polling
Inheritance at RAF	500 RAFs/day × 20 attributes	50 / 200 / 500 ms added to RAF	n/a	DerivationLog ~ 4M rows ≈ 1 GB	(receiver_batch_id)	Deferred to async with flag `pending-derivation`
EBR compile + seal	500 batches/day, 2,000 events each	event apply p95 < 2 s; seal p95 < 30 s	1M events/day	EBR JSON+PDF ~ 5 MB each → 0.9 TB/yr in object storage	record by order; documents in object storage	Compilation lag allowed to 5 min; review screen shows lag
Signature	5,000/day	300 ms / 1 s / 2 s (excl. user input)	20/s	Signature 2M rows ≈ 1 GB/yr	(object_type, object_id)	Step-up auth latency dominated by Keycloak; no degradation of hash check

4.24.20 Localization and compliance hooks

- **21 CFR Part 11 (US) and EU GMP Annex 11**: FR-24.22 implements 11.10(e) audit trail via the signature log and w5 audit, 11.50 manifestation, 11.70 record linkage, 11.100 uniqueness (Keycloak subject), 11.200 two-component re-authentication, 11.300 controls (password policy and lockout from w2). Validation package (IQ/OQ scripts, traceability matrix to these clauses) is a wave 48 deliverable; this wave provides the OQ test cases from 4.24.9 and AC-24.15/AC-24.16.
- **Retention**: EBR retention per country/regime (US 21 CFR 211.180: expiry + 1 year; EU: max(expiry + 1 year, 5 years after certification); some countries require 10 years for blood/plasma products) set by the w48 retention policy; legal hold blocks purge.
- **Food labeling**: batch label content (allergen declaration, lot code formats, best-before vs use-by wording, e.g., EU Regulation 1169/2011) is rendered by country-specific templates in wave 48 packs via the `production.label.render` extension point.
- **Catch weight**: legal-for-trade weighing (OIML R76 / NIST Handbook 44) is a device concern; Passage stores the scale id and certification reference on catch-weight RAF lines when provided.
- **Excise / alcohol**: bonded-production reporting (US TTB Form 5110.40, EU EMCS) is a wave 48 country-pack item fed from batch order consumption and RAF events through `production.statutory-extract`.
- **Language**: batch tickets, CoA and EBR render in the site language; signature meanings are stored as codes and rendered per locale.
- **Extension points**: `production.label.render`, `production.statutory-extract`, `signature.strategy.resolve` (OPA), `ebr.section.provider` (additional EBR sections such as environmental monitoring).

4.24.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.14.1 Inventory — Batch merge (PI) (PdsBatchMerge) [AX #122]	PARTIAL-24	ADDED as FR-24.16	Attribute recomputation BR-24.11
AX 12.23.1 PI — Consolidated batch orders (PdsConsolidatedOrder) [AX #196]	PARTIAL-24	ADDED as FR-24.17	Bulk → packed with pegged packing orders
AX 12.23.1 PI — Rework batch order (PdsRework) [AX #197]	PARTIAL-24	ADDED as FR-24.18	Keeps batch identity by default
AX 12.23.1 PI — Batch attribute inheritance (PdsInheritance) [AX #198]	PARTIAL-24	ADDED as FR-24.15	Product-to-product and co-product inheritance
AX 12.23.1 PI — Batch merge (PdsBatchMerge) [AX #201]	PARTIAL-24	ADDED as FR-24.16	Same capability as #122
AX 12.23.1 PI — Same-batch reservation (PdsSameBatch) [AX #202]	PARTIAL-24	ADDED as FR-24.19	Sales-line enforcement active from wave 27
AX 12.23.1 PI — Customer-specific batch attribute requirements (PdsCustSellableDays, PdsCustBatchAttrib) [AX #203]	PARTIAL-24	ADDED as FR-24.19	BR-24.13

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.23.3 PI Periodic — Consolidate batch orders (batch) [AX #205]	PARTIAL-24	ADDED as FR-24.17	Job in 4.24.16
AX 12.23.3 PI Periodic — Attribute inheritance processing on RAF [AX #206]	PARTIAL-24	ADDED as FR-24.15	Re-derivation job in 4.24.16
AX 12.23.4 PI Setup — Lot inheritance setup [AX #208]	PARTIAL-24	ADDED as FR-24.15	InheritanceRule entity
AX 12.23.4 PI Setup — Customer sellable days / customer batch attribute requirements [AX #211]	PARTIAL-24	ADDED as FR-24.19	CustomerBatchRequirement entity
AX 12.23.4 PI Setup — Rework parameters [AX #212]	PARTIAL-24	ADDED as FR-24.18	Parameters in 4.24.13
AX 12.23.4 PI Setup — Bulk/pack item relations [AX #213]	PARTIAL-24	ADDED as FR-24.17	BulkPackRelation entity
AX 12.24.1 Lean — Subcontracting in lean (LeanSubcontracting) [AX #214]	PARTIAL-24	ADDED as FR-24.20	Service activity on vendor work cell
AX 12.24.3 Lean Periodic — Generate purchase orders for service activities [AX #215]	PARTIAL-24	ADDED as FR-24.20	Lean periodic PO close job
AX 12.24.4 Lean Setup — Subcontracting setup (service items, purchase agreements) [AX #216]	PARTIAL-24	ADDED as FR-24.20	Agreement pricing per BR-24.16
SAP 13.22 PP-PI — COEBR electronic batch record [SAP #257]	GAP (propose 24)	ADDED as FR-24.21	Review by exception, sealed PDF/A + JSON
SAP 13.22 PP-PI — Digital signature in PI sheets / batch records (DSIG) [#261]	GAP (propose 24)	ADDED as FR-24.22	Signature service reused by w25 PI sheets and generalized in w48
SAP 13.22 PP-PI — Batch derivation (DVMO / DVMAN) [#262]	PARTIAL-24	ADDED as FR-24.15	Manual derivation included
SAP 13.22 PP-PI Reports — COEBR batch record list [#264]	GAP (propose 24)	ADDED as FR-24.21	Report in 4.24.14
SAP 13.22 PP-PI Periodic — COEBR batch record generation/archive [#269]	GAP (propose 24)	ADDED as FR-24.21	EBR compilation and seal jobs in 4.24.16
SAP 13.22 PP-PI Customizing — EBR batch record profiles, archiving, approval workflow [#274]	GAP (propose 24)	ADDED as FR-24.21	BatchRecordProfile
SAP 13.22 PP-PI Customizing — Batch management in PP-PI (COB1 batch determination, class 023, derivation) [#275]	PARTIAL-24	ADDED as FR-24.15 (derivation); batch determination SUPERSEDED by w16 FEFO batch selection + FR-24.19	Class 023 characteristics = BatchAttribute (FR-24.5)
SAP 13.23 PP/DS — Characteristic-dependent planning (CDP) [#277]	PARTIAL-24	ADDED as FR-24.23	Engine change lives in the w22 planning service
SAP 13.33 Batch/Serial — DVMO / DVSP / DVMAN derivation [#365]	PARTIAL-24	ADDED as FR-24.15	—
SAP 13.33 Batch/Serial — Documentary batches (S/4) [#367]	PARTIAL-24	ADDED as FR-24.24	Tracking policy `batch-record-only` (w16 tracking dimension group)
SAP 13.33 Batch/Serial Reports — DVMO derivation monitor [#369]	PARTIAL-24	ADDED as FR-24.15	Derivation monitor in 4.24.14
SAP 13.33 Batch/Serial Periodic — Batch derivation runs (DVMO) [#370]	PARTIAL-24	ADDED as FR-24.15	Re-derivation job
SAP 13.33 Batch/Serial Customizing — Batch derivation activation, sender/receiver fields, rules [#371]	PARTIAL-24	ADDED as FR-24.15	Parameter "Inheritance active"
SAP 13.33 Batch/Serial Customizing — Documentary batch activation (S/4) [#373]	PARTIAL-24	ADDED as FR-24.24	Activation via tracking policy
SAP 13.35 DMS Customizing — Digital signature for documents (signature strategies) [#397]	GAP (propose 24)	ADDED as FR-24.22	SignatureStrategy covers any object type, including w5 documents

Scope note: the 4.24.1 out-of-scope line "PI sheets with electronic signatures (w33 quality covers signatures)" is narrowed by this closure. Electronic signatures are delivered here (FR-24.22) and PI sheets in wave 25 (FR-25.16). Wave 33 uses the same signature service for quality orders, and wave 48 extends it to configured tables and fields (AX register rows #22–#25, wave 48).

4.25 Wave 25: Shop Floor Execution, Time Registration & IoT

4.25.1 Scope and business context

Wave 25 puts production on the floor. Operators use a touch kiosk (PWA) to clock in and out, start and stop jobs (including bundling several jobs), report good and scrap quantities with reason codes, run quality checks and view instructions and drawings, and keep working for up to 30 minutes offline. Registrations are turned into pay lines (overtime, shift premiums, flex) through profiles, approved by supervisors and exported to payroll (w43). Machines publish counts, states and downtime over OPC-UA/MQTT through an edge connector into Kafka; the shopfloor service computes OEE per resource in 15-minute buckets and drives andon alerts. Labor and machine time flows into the production order's route-card postings. This is the phase-exit wave: the demo plant manufactures a configured product from planned order to finished goods with full costing. Out of scope: payroll calculation itself (w43), quality orders and usage decisions (w33; kiosk checks are simple pass/fail capture), maintenance work orders from machine events (w44 consumes the events), label printing beyond kanban cards (w32).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Job-card re-keying: AX JmgTermReg registrations are "calculated" and "transferred" by a supervisor before ProdJournalRoute lines exist, so route cards are re-keyed or posted a day late; SAP CO11N confirmations are typed by a clerk from paper tickets; Infor LN and Epicor Kinetic MES data collection posts labor in a nightly batch	Registration and production posting are separate modules joined by a transfer job	Registration stop generates and posts the job-card journal (option immediate); reversal on edit	Labor posting latency 24 h → < 5 s; re-keyed tickets 0
Kiosk needs the AOS: AX terminal registration (JmgTermReg form) is a thick client; D365 production floor execution requires connectivity; Plex is cloud-only with no offline mode	No client-side queue or idempotent replay	PWA with IndexedDB queue, client UUIDs, conflict flags	Lost registrations during outages → 0 over a 30-min disconnect
Machine data lives outside the ERP: AX has no native machine integration; SAP MII/PCo is a separate licensed stack with custom BLS transactions; Plex offers machine connectivity only through its own gateway	Proprietary connectors and point-to-point integration	`iot-edge` (OPC-UA/MQTT) → Kafka CloudEvents; any system subscribes to `shopfloor.machine-events`	Time to connect a new machine 2 weeks → 1 day; integration middleware 0
OEE is computed in spreadsheets or a BI tool from exported confirmations; SAP DM OEE requires its own master data; Epicor Advanced MES OEE updates per shift	No event-level counts in the ERP; ratios averaged incorrectly across shifts	OeeBucket per 15 min from events, ISO 22400 formula, roll-up by summing minutes/counts	OEE visibility per shift → per 15 min with p95 lag < 5 s

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Pay calculation surprises: AX JmgPayTable overtime depends on profile day types supervisors do not see, so corrections happen after payroll export; SAP time evaluation (RPTIME00) is a batch run with cryptic log	Rules engine hidden from the people approving	PayLine per rule with source registrations shown on the approval screen; recalculation on edit	Post-export corrections 5%/period → < 0.5%; approval time per team 2 h → 15 min

4.25.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Job registration terminal	JmgTermReg (terminal registration form), JmgJobTable, JmgRegistration, D365 production floor execution	SAP MII / Digital Manufacturing shop floor UI, PP-PDC terminals, CO11N	Kiosk PWA, `shopfloor-bff`, TimeRegistration	full
Clock in/out, absence	JmgStampJournalTable (raw), JmgTmpRegistration, JmgAbsenceCode, JmgProfileTable	Time events (CATS/PT), HR time types	Registration kinds clock-in/out, absence; ProfileDay	full
Job bundling	Jmg bundle jobs (JmgJobBundle)	n/a (multi-order confirmation)	bundle_id on registrations, cost split by quantity/hours	full
Feedback (qty/scrap)	JmgJobTable feedback, ProdJournalRoute from Jmg, error cause codes	CO11N yield/scrap/rework, reason for variance (AFRU.GRUND)	job-card journals auto-generated; ScrapReason	full
Pay calculation	JmgPayTable, pay agreements (JmgPayAgreement), overtime/flex, pay-item transfer	HR time evaluation (RPTIME00), wage types	PayLine, PayRule engine	full
Approval and transfer	Jmg calculate → approve → transfer to payroll (JmgPayTransfer)	CATS approval, transfer to HR	Approve + export (w43 payroll)	full
Machine integration	n/a native (IoT Intelligence for D365, Sensor Data Intelligence)	SAP MII PCo/OPC, Plant Connectivity, SAP DM	`iot-edge` connector (OPC-UA/MQTT) → Kafka	improved
OEE	D365 Sensor Data Intelligence (partial)	SAP DM OEE module	OeeBucket per resource × 15 min; ISO 22400 KPIs	full
Downtime reasons	Jmg indirect activities	PM notification/downtime reasons	DowntimeRecord with reason code	full
Andon	n/a	SAP DM andon	AndonAlert via event + kiosk banner	improved
Instructions/drawings	Route notes, document attachments (DocuRef)	PI sheets, DMS documents on operation	Operation documents (w5 attachments) on kiosk	full
Offline kiosk	n/a (requires AOS)	n/a	PWA with IndexedDB queue	improved
Control recipes / process instructions	n/a (route notes, D365 production floor execution instructions)	Control recipe CO53/CO53XT, process instruction categories (O10C/O12C/O13C), control recipe destinations	ControlRecipe generated at batch-order release, sent to kiosk or PCS (FR-25.15)	full
PI sheets / XSteps	n/a	PI sheet CO60/CO60XT/CO60_PI, worklist, standard XSteps repository (CMX, SXS)	PiSheetTemplate + StepLibrary, kiosk PI sheet execution (FR-25.16)	full
Process messages	n/a	Process messages CO54/CO54XT/CO57/CO58, message categories and destinations (O05C/O06C/O07C)	ProcessMessage with categories, destinations and monitor (FR-25.17)	full
PI sheet archive	n/a	CO64 / CO67 archive and print	PDF/A + JSON archive feeding EBR (FR-25.18)	full
Edge connector deployment	D365 IoT Intelligence (Azure IoT Hub based)	SAP Plant Connectivity (PCo) installation, agents, notification processes	`iot-edge` deployable with enrollment, mTLS, ringed rollout (FR-25.19)	improved

4.25.3 Functional requirements

- **FR-25.1 Kiosk login** — Must. The system shall let a worker log in on a shared kiosk by badge scan (GS1 barcode/NFC) or PIN mapped to a Keycloak user, with session timeout 10 min idle and switch-user in ≤ 2 taps.
- **FR-25.2 Clock in/out and absence** — Must. The system shall record clock-in/out and absence registrations with the device id and server timestamp, applying profile rounding (BR-25.2).
- **FR-25.3 Job start/stop and bundling** — Must. The system shall list dispatchable jobs for the worker's resource/work cell (from w23 ScheduledJob, ordered by start), allow start/stop of one job or a bundle, and split bundled time by quantity reported (default) or equally.
  - Notes: A worker may have one open job/bundle at a time; starting another auto-stops the previous. Jobs in status `finished` are hidden unless the supervisor reopens.
- **FR-25.4 Feedback** — Must. The system shall capture good qty, scrap qty with reason code, and "finished" flag per job; on stop it generates a job-card journal (w21) with hours and quantities and posts it (configurable: immediately or at approval).
- **FR-25.5 Quality checks** — Should. The system shall present check items attached to the operation (text/numeric/pass-fail with limits) and record results; a failed mandatory check blocks `finished` until a supervisor override.
- **FR-25.6 Instructions and documents** — Must. The kiosk shall show operation instructions (markdown) and attachments (PDF, images, video ≤ 50 MB) cached for offline use.
- **FR-25.7 Pay calculation** — Must. The system shall calculate PayLines per worker-day from registrations using the worker's profile (day types, shift windows, breaks) and pay agreement rules: base, overtime after daily/weekly thresholds, shift premium, flex balance.
  - Notes: Defaults: overtime > 8.00 h/day at 1.5×, > 40.00 h/week at 1.5×, Sunday 2.0×; rounding per BR-25.2; max 24 h/day validation.
- **FR-25.8 Approval and export** — Must. Supervisors shall approve per worker-day (bulk by team), with edits audited; approved days are locked and exported as `dev.passage.shopfloor.payday.approved.v1` for w43 and as CSV.
- **FR-25.9 IoT ingestion** — Must. The `iot-edge` connector shall subscribe to OPC-UA nodes (via `gopcua`) and MQTT topics (Mosquitto/EMQX, QoS 1), normalize to the MachineEvent schema (BR-25.4), buffer locally (SQLite ring, 24 h), and publish to Kafka topic `shopfloor.machine-events` with idempotent producer and per-machine monotonic sequence.
- **FR-25.10 OEE** — Must. The system shall compute OEE per resource per 15-minute bucket and shift (BR-25.5) from machine events and planned time from the w23 schedule/calendar, with dashboard lag p95 < 5 s.
- **FR-25.11 Downtime and andon** — Must. The system shall open a DowntimeRecord when a machine reports DOWN or no count for > `idle_threshold_s` (default 300), require a reason code within the kiosk, and raise an AndonAlert (kiosk banner + notification) on downtime > 10 min, scrap rate > 5% in a bucket, or quality fail.
- **FR-25.12 Offline kiosk** — Must. The PWA shall queue registrations and feedback in IndexedDB while disconnected (≥ 30 min, ≤ 5,000 actions), replay in order on reconnect with client-generated UUIDs for idempotency, and resolve conflicts per BR-25.7.
- **FR-25.13 Machine-to-job linkage** — Should. Machine counts shall be attributed to the job currently running on that resource (from registrations or the schedule), and may auto-fill good/scrap quantities on the kiosk for operator confirmation.
- **FR-25.14 Supervisor view** — Must. A supervisor workspace shall show who is clocked in, current jobs, open downtimes, OEE tiles, and pending approvals.

- **FR-25.15 Control recipes** — Should. On release of a batch order (w24) or production order whose route operations carry process instructions, the system shall generate a ControlRecipe per destination (kiosk PI sheet, or a process control system through the edge connector) containing the ordered process instructions of each phase with their category (instruction text, process parameter setpoint, process data request, calculation, inspection characteristic, material list, signature request, sequence definition), and shall track its status (created, sent, received, in-execution, completed, error) in a control recipe monitor with resend.
    - Notes: Destination types: `kiosk` (rendered as a PI sheet), `pcs-opcua` (setpoints written to allowlisted OPC-UA nodes through FR-25.19; process data requests become subscriptions), `pcs-mqtt` (JSON payload to a configured topic, ISA-88 recipe structure: procedure → unit procedure → operation → phase). Process instruction categories are tenant master data with characteristic definitions (data type, unit, limits). Control recipes are generated with values scaled to the batch (BR-24.1) and with potency-adjusted quantities (BR-24.2).
- **FR-25.16 PI sheets and step library** — Should. The system shall provide a PI sheet designer and a versioned StepLibrary (reusable standard steps, SAP XSteps equivalent) with step types: instruction (markdown with images), numeric input (unit, limits, mandatory), text/enum input, checklist, calculation (expression over prior inputs and order data), material consumption (posts picking-list lines), yield/RAF report, machine readout (reads an OPC-UA/MQTT tag value via the edge), timer/hold time, signature (FR-24.22 with meaning), and conditional branch; operators execute PI sheets on the kiosk with a worklist of PI sheets per work cell.
    - Notes: Step libraries are versioned and released through ECO (w9); a PI sheet template references step versions, so later library changes do not alter running batches. Out-of-limit inputs require a comment and create an EBR exception (BR-24.15); configured steps can block continuation until a supervisor signs. Steps that post (consumption, yield) generate w21 journals immediately. PI sheets work offline under FR-25.12 except machine readout and signature steps, which queue until reconnect and block completion of the sheet.
- **FR-25.17 Process messages** — Should. The system shall accept ProcessMessages from kiosk PI sheets, from the edge connector (PCS events) and by manual entry, each with a category (phase-start, phase-end, consumption, yield, process-data, quality-result, deviation, order-status, custom) and characteristic values, validate them against the category definition, and deliver them to the category's destinations: production journal posting (w21), quality inspection results (w33), EBR section (w24), Kafka topic `shopfloor.process-messages` for external consumers, or an outbound webhook (w35).
    - Notes: Process message monitor lists messages by status (received, delivered, partially-delivered, error) with per-destination results; failed destinations retry with exponential backoff (1, 5, 15, 60 min) then stay in error for manual reprocess or deletion (with reason). Messages are idempotent by (source, sourceMessageId). Manual creation requires duty *Create process messages*.
- **FR-25.18 PI sheet archive** — Should. On completion of a PI sheet the system shall render it to PDF/A-2b and canonical JSON (inputs, calculated values, timestamps, signatures with manifestation), compute a SHA-256 hash, store both in the w5 document store with the batch order's retention policy, link them to the EBR, and allow print and retrieval by order, batch or date.
- **FR-25.19 Edge connector as a deployable** — Must. The `iot-edge` connector shall ship as a signed OCI image (amd64/arm64) with a Helm chart for the plant k3s node and a single static binary for Windows/Linux industrial PCs, and shall support: enrollment with a one-time token that obtains an mTLS client certificate (issued by the cluster CA via cert-manager, 90-day validity, auto-renew at 30 days), configuration pulled from the shopfloor service (machines, OPC-UA endpoints and node ids, MQTT topics, tag mappings, write allowlist) as versioned documents, heartbeat every 30 s with buffer fill and per-machine status, ringed rollout of connector versions (canary → ring 1 → all, per plant), local admin page on localhost for diagnostics, and write-back (control recipe setpoints) restricted to allowlisted nodes with per-write audit.
    - Notes: OPC-UA security modes None, Sign and SignAndEncrypt with policies Basic256Sha256 and Aes128_Sha256_RsaOaep; server certificate trust list managed from the central UI. MQTT 3.1.1/5.0 with TLS and per-edge credentials. Store-and-forward per FR-25.9 (SQLite ring, 24 h) survives connector restart and OS reboot. Upgrades are atomic (A/B binary swap) with automatic rollback when heartbeat fails within 5 minutes. Supported deployments are documented as certified configurations (k3s 1.30+, Windows Server 2019+/IoT Enterprise, Ubuntu 22.04+). Runs without inbound ports: all traffic is outbound to Kafka (via the shopfloor ingest gateway over HTTPS/2 or Kafka TLS) and the config endpoint.

4.25.4 Data model

Worker (shopfloor projection of Party/Employee w7): id, party_id, badge_id (unique), pin_hash, default_resource_id, work_cell_id, profile_id, pay_agreement_id, cost_rate (numeric 18,4), active.

TimeRegistration: id (client UUID), tenant_id, worker_id, kind (enum: clock-in, clock-out, job-start, job-stop, absence, indirect), job_id (nullable), bundle_id (nullable), qty_good, qty_scrap, scrap_reason_id, started_at, stopped_at (timestamptz), server_received_at, device_id, offline (bool), hours (numeric 10,4, computed), status (enum: raw, calculated, approved, exported), journal_id (w21 job card), note.

ProfileDay: id, profile_id, day_type, shift_start, shift_end, break_windows (jsonb), rounding_minutes (int), flex_allowed (bool), tolerance_before_min, tolerance_after_min.

PayLine: id, worker_id, work_date, pay_type (enum: base, overtime-1.5, overtime-2.0, shift-premium, flex-in, flex-out, absence-paid, absence-unpaid), hours, rate_multiplier, amount (numeric 18,4), currency, source_registration_ids (uuid[]), approved_by, approved_at, export_batch_id.

MachineEvent: id, machine_id, resource_id, seq (bigint), kind (enum: count, state, downtime, alarm, measurement), ts (timestamptz, device), received_at, good_count, reject_count, state (RUNNING/IDLE/DOWN/SETUP/OFF), reason_code, measurement_name, measurement_value, unit, job_id (attributed), payload (jsonb).

OeeBucket: resource_id, bucket_start (15 min), shift_id, planned_minutes, run_minutes, down_minutes, ideal_cycle_s, total_count, good_count, availability, performance, quality, oee (numeric 7,4 each), source_events (int).

DowntimeRecord: id, resource_id, started_at, ended_at, reason_code (nullable until coded), category (planned/unplanned), job_id, worker_id, andon_raised (bool).

Invariants: for each worker at most one open job-start without job-stop; PayLine hours per worker-day ≤ 24.00; OeeBucket.run_minutes + down_minutes ≤ planned_minutes; MachineEvent (machine_id, seq) unique; approved PayLines are immutable (new lines + reversal on correction).

Indexing: TimeRegistration (worker_id, started_at), (job_id), unique id (client UUID); MachineEvent partitioned by month (TimescaleDB-free: native range partitions), index (resource_id, ts); OeeBucket PK (resource_id, bucket_start); retention: raw MachineEvent 90 days in Postgres, forever in Iceberg (w46); OeeBucket 3 years.

ControlRecipe / ProcessInstruction (FR-25.15): ControlRecipe(id, order_id, destination_id, status (created/sent/received/in-execution/completed/error), generated_at, sent_at, payload_hash, error_text, version); ProcessInstruction(id, control_recipe_id, phase_path text (ISA-88 path), sequence int, category_id, characteristic values jsonb, target_node (nullable OPC-UA node id/topic)); ProcessInstructionCategory(id, code, kind, characteristics jsonb [{name, type, unit, min, max}]); RecipeDestination(id, site_id, type (kiosk/pcs-opcua/pcs-mqtt), edge_id, address, active).

PiSheetTemplate / LibraryStep / PiSheetRun / PiStepResult (FR-25.16): LibraryStep(id, code, version, type (enum per FR-25.16), definition jsonb, status (draft/released/obsolete), eco_id); PiSheetTemplate(id, operation_ref (route version + oper), version, steps jsonb [{libraryStepId, version, overrides}]); PiSheetRun(id, order_id, oper_number, template_id, status (open/in-progress/held/completed), started_at, completed_at, archive_pdf_id, archive_json_id, content_hash); PiStepResult(id, run_id, step_seq, value jsonb, out_of_limit bool, comment, recorded_by, recorded_at, signature_id nullable, offline bool).

ProcessMessage / ProcessMessageDelivery (FR-25.17): ProcessMessage(id, source (kiosk/edge/manual), source_message_id, category_code, order_id, phase_path, values jsonb, occurred_at, received_at, status); ProcessMessageDelivery(message_id, destination, status, attempts int, last_error, delivered_at); unique (tenant_id, source, source_message_id).

EdgeNode / EdgeConfigVersion (FR-25.19): EdgeNode(id, site_id, name, platform (k3s/windows/linux), connector_version, ring (canary/ring1/all), cert_serial, cert_expires_at, last_heartbeat_at, buffer_fill_pct, status); EdgeConfigVersion(id, edge_id, version int, document jsonb, published_at, published_by, applied_at); EdgeWriteAudit(id, edge_id, node_id, value, control_recipe_id, actor, written_at, result).

Invariants (additions): every completed PiSheetRun has an archive_json_id whose hash equals content_hash; a ProcessMessage with status delivered has all destinations delivered; an EdgeNode accepts only one applied EdgeConfigVersion at a time and writes only to nodes in that version's allowlist.

4.25.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/shopfloor/sessions:badge	Kiosk login by badge/PIN	n/a	Kiosk login
POST	/api/v1/shopfloor/registrations	Create registration(s), batch of ≤ 500 (offline replay)	client UUID per item	Register time
GET	/api/v1/shopfloor/jobs?workerId&resourceId&status	Dispatch list	n/a	Register time
POST	/api/v1/shopfloor/jobs/{id}:feedback	Qty good/scrap/finished, checks	client UUID	Register time
GET	/api/v1/shopfloor/operations/{id}/instructions	Instructions + attachment URLs (signed, cacheable)	n/a	Register time
POST	/api/v1/shopfloor/pay:calculate	Calculate pay lines for worker-days	Idempotency-Key	Calculate time registrations
POST	/api/v1/shopfloor/pay:approve	Approve worker-days (bulk)	Idempotency-Key	Approve time registrations
POST	/api/v1/shopfloor/pay:export	Export batch (CSV + event)	Idempotency-Key	Export time to payroll
GET	/api/v1/shopfloor/oee?resourceIds&from&to&bucket=15m	OEE series	n/a	View shop floor
POST	/api/v1/shopfloor/downtimes/{id}:code	Assign reason	Idempotency-Key	Register time
GET	/api/v1/shopfloor/andon?siteId	Open alerts	n/a	View shop floor
POST	/api/v1/shopfloor/machines	Register machine → resource mapping, tags	Idempotency-Key	Maintain shop floor setup

Event	Trigger	Payload fields	Consumers
`dev.passage.shopfloor.registration.posted.v1`	registration stored	registrationId, workerId, kind, jobId, hours, qtyGood, qtyScrap, offline	production (job card), analytics
`dev.passage.shopfloor.machine.event.v1`	edge publish	see BR-25.4	shopfloor OEE, production, maintenance (w44)
`passage.shopfloor.oee.bucket-updated.v1`	bucket recomputed	resourceId, bucketStart, availability, performance, quality, oee	dashboards (w46)
`dev.passage.shopfloor.andon.raised.v1`	alert condition	alertId, resourceId, kind, severity, jobId, startedAt	notifications, kiosk
`dev.passage.shopfloor.payday.approved.v1`	approval	workerId, workDate, payLines[{type, hours, multiplier, amount}], approvedBy	payroll (w43)

4.25.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Kiosk home	Badge in/out, my jobs, start/stop, bundle, report, downtime reason, andon	Worker, clock status, current job, elapsed, qty good/scrap, offline indicator	Kiosk login / Register time
Job detail (kiosk)	View instructions/drawings, checks, report	Operation, order, item, qty planned/reported, checks	Register time
Supervisor board	Who is in, current jobs, downtimes, OEE tiles, approve days	Worker, job, elapsed, resource state, OEE A/P/Q, pending	Approve time registrations
Time approval	Calculate, edit with reason, approve, export	Worker, date, registrations, pay lines, flags (missing clock-out)	Approve time registrations
OEE dashboard	Resource/period filter, drill to downtime and buckets	OEE, A, P, Q, top losses (Pareto)	View shop floor
Machine setup	Map machine → resource, tags, ideal cycle time, thresholds	Machine, resource, protocol, node ids/topics, cycle s	Maintain shop floor setup

Interaction rules: kiosk targets ≥ 48 px, no hover-only affordances, numeric keypad for quantities, high-contrast theme, every action confirmed in ≤ 300 ms (optimistic UI); offline banner shows queued count; supervisor edits require a reason; OEE dashboard uses the w46 dashboard type when available, otherwise the Grafana panel embedded via the shell.

4.25.7 Business rules and edge cases

- **BR-25.1 Time attribution.** Hours for a job registration = (stopped_at – started_at) minus breaks that fall inside per the profile; bundle hours split by Σ qty_good (fallback equal split); hours rounded to 0.01 h on the job-card journal.
- **BR-25.2 Rounding and tolerances.** Clock-in within tolerance_before_min (default 10) of shift start rounds to shift start; clock-out within tolerance_after_min (default 5) rounds to shift end; otherwise round to nearest rounding_minutes (default 1, max 15).
- **BR-25.3 Missing clock-out.** At shift end + 4 h a missing clock-out auto-closes at shift end with flag `missing-clock-out`; the day cannot be approved until a supervisor confirms or edits.
- **BR-25.4 Machine event schema.** CloudEvents envelope `type: dev.passage.shopfloor.machine.event.v1`, `source: urn:passage:iot-edge:<site>:<edgeId>`, `subject: <machineId>`, `id: <machineId>-<seq>`, `time: <device ts RFC 3339>`, `datacontenttype: application/json`, `data: { "machineId": "M-101", "resourceId": "uuid", "seq": 184233, "kind": "count|state|downtime|alarm|measurement", "ts": "2026-10-21T08:15:03.250Z", "goodCount": 12, "rejectCount": 1, "state": "RUNNING", "reasonCode": null, "measurement": {"name": "temp_c", "value": 71.4, "unit": "Cel"}, "jobId": null, "tags": {"line": "L1"} }`. Counts are deltas since the previous count event; absolute counters are converted at the edge with rollover detection (32-bit). Events older than 7 days or with seq ≤ last seen are dropped and counted in a metric.
- **BR-25.5 OEE formula (ISO 22400 aligned).** Availability = run_minutes / planned_minutes; Performance = (ideal_cycle_s × total_count / 60) / run_minutes, capped at 1.0 with an over-rate flag; Quality = good_count / total_count (1.0 if total = 0); OEE = A × P × Q. Planned minutes = scheduled open minutes minus planned downtime (category planned). Example shift: planned 480 min, unplanned down 60 → A = 0.875; ideal cycle 60 s, total 380 → P = 380 / 420 = 0.9048; good 361 → Q = 0.95; OEE = 0.7522. Buckets roll up to shift/day by summing minutes and counts, not by averaging ratios.
- **BR-25.6 Idle detection.** No count event for idle_threshold_s while state = RUNNING flips the bucket to IDLE from the last count; a later count closes the idle gap.
- **BR-25.7 Offline conflict resolution.** Replayed registrations are accepted in client timestamp order; a job-start for a job that was finished server-side while offline is accepted but flagged `job-finished-conflict` for supervisor review; a duplicate client UUID is a no-op; clock skew > 5 min between device and server is corrected by the offset measured at reconnect and recorded.

- **BR-25.8 Pay thresholds.** Weekly overtime counts hours after 40.00 across the pay week (profile-defined start day); daily and weekly overtime do not double-count: hours already paid at 1.5× daily are excluded from the weekly threshold calculation.
- **BR-25.9 Posting to production.** Job-card journals from registrations post at cost_rate of the worker (if set) else the resource's cost category; a registration edited after posting creates a reversing journal and a new one.
- **BR-25.10 Andon de-duplication.** One open AndonAlert per (resource, kind); re-raising extends it; auto-closes when the condition clears for 5 min.
- **BR-25.11 Control recipe resend.** A control recipe in error can be resent up to 5 times; resending after any process message for the order was received requires `force` and a reason, because the PCS may have partially executed it.
- **BR-25.12 PI sheet step completion.** A sheet completes only when every mandatory step has a value, every out-of-limit value has a comment, every signature step has a valid signature, and no queued offline step remains; a held sheet (supervisor hold) cannot accept inputs except a release signature.
- **BR-25.13 Process message validation.** Values are validated against the category's characteristic types and limits; out-of-limit values are accepted with flag and an EBR exception unless the category is `reject-out-of-limit`; unknown order or phase → status error without delivery.
- **BR-25.14 Edge certificate rotation.** Renewal starts when remaining validity < 30 days; if renewal fails for 7 consecutive days an alert pages; an expired certificate stops publishing but buffering continues (24 h ring).
- **BR-25.15 Edge write safety.** A setpoint write requires the node to be on the allowlist, a value within the instruction's limits, and an active control recipe in-execution for the order on that machine; every write and its readback value are audited; a readback mismatch raises a process message of category deviation.

4.25.8 Acceptance criteria

- **AC-25.1** (FR-25.3, 25.4, BR-25.1) Given 6 operators on a shift with the fixture schedule, when they clock jobs on the kiosk (Playwright) and the day is approved, then each production order's labor cost = Σ hours × worker cost rate to the cent.
- **AC-25.2** (FR-25.7) Given a worker registering 09:00–19:30 with a 30-min break on a Tuesday, then pay lines are base 8.00 h and overtime-1.5 2.00 h.
- **AC-25.3** (FR-25.9) Given a simulated machine publishing 10,000 events/min for 60 min with a Kafka broker restart at minute 30, then every seq is present in MachineEvent and no duplicates exist.
- **AC-25.4** (FR-25.10, BR-25.5) Given the BR-25.5 shift example, then the shift OEE reads 0.7522 and the tile updates within 5 s of the last event.
- **AC-25.5** (FR-25.12) Given a kiosk disconnected for 30 min with 40 actions queued, when it reconnects, then all 40 are stored once in order and conflicts are flagged, not lost.
- **AC-25.6** (FR-25.11) Given a machine DOWN for 11 min without a reason, then an andon alert is open, the kiosk shows a banner, and coding the reason closes the downtime but keeps the alert until cleared.
- **AC-25.7** (FR-25.2, BR-25.2) Given shift start 07:00 and clock-in 06:53, then the registration time is 07:00.
- **AC-25.8** (FR-25.8) Given an approved day, when a supervisor edits a registration, then a reversal and a new PayLine are created and the export batch includes both.
- **AC-25.9** (FR-25.5) Given a mandatory numeric check with limits 10–12 and a value 13, then `finished` is blocked until override with reason.
- **AC-25.10** (phase exit) Given the demo plant, when the configured product (w8 configurator) is planned (w22), scheduled (w23), produced through kiosk registrations and machine counts (w25) and ended (w21), then the order's actual cost, variances and WIP reconcile to the journal and the recorded walkthrough is signed off.
- **AC-25.11** (FR-25.15, BR-25.15) Given a released batch order with a phase setpoint of 72.0 °C for node `ns=2;s=Vessel1.TempSP` on the allowlist, when the control recipe is sent to `pcs-opcua`, then the edge writes 72.0, the readback is audited, and the control recipe status is in-execution.
- **AC-25.12** (FR-25.16, BR-25.12) Given a PI sheet with a mandatory numeric step limits 6.5–7.5 and a signature step, when the operator enters 7.8 without a comment, then completion is refused; with a comment and a valid signature it completes and an EBR exception exists.
- **AC-25.13** (FR-25.17) Given a process message of category quality-result with destinations EBR and w33, when the w33 destination is unavailable, then the message is partially-delivered, retried at 1, 5, 15 and 60 min, and finally delivered without duplicating the EBR entry.
- **AC-25.14** (FR-25.18) Given a completed PI sheet, when archived, then a PDF/A-2b and JSON exist in the document store with equal content hash recorded on the run, and the EBR links to them.
- **AC-25.15** (FR-25.19, BR-25.14) Given a new edge node enrolled with a one-time token, when it starts, then it obtains a certificate, pulls config version 1, and heartbeats within 30 s; a canary upgrade whose heartbeat fails rolls back automatically within 5 minutes.

4.25.9 Test plan and fixtures

Fixtures: `shift-6-operators` (schedule, badge ids, profiles, expected pay lines and order costs); `pay-rules-30` (daily/weekly overtime, Sunday, flex, missing clock-out); `iot-sim-10k` (Go simulator producing counts/states with controllable gaps and rollover); `oee-shift-cases` (10 shifts with hand-computed A/P/Q); `offline-replay-40`.

Test	Type	Target	Pass criterion
Pay rules	unit	PayRule engine	30/30
OEE	unit	bucket calc	matches hand-computed to 4 decimals
Event ordering/loss	integration	iot-edge → Kafka → DB	0 lost, 0 dup over 600k events with broker restart
Kiosk shift E2E	Playwright	kiosk + production	labor cost ties
Offline sync	Playwright (offline mode)	PWA queue	40/40 once, in order
Throughput	performance	ingest	10k events/min sustained, consumer lag < 5 s
Accessibility	axe	kiosk screens	WCAG 2.1 AA

Reconciliation: Σ approved hours per order = Σ job-card hours posted; Σ PayLine hours = Σ registration hours per worker-day; Σ MachineEvent good counts attributed to a job = kiosk-confirmed good qty (variance report).

4.25.10 Deployment and operations

Helm: `shopfloor` (Go, 3 replicas; Kafka consumer group `shopfloor-oee`), `shopfloor-bff` (TS, 2 replicas, SSE), `kiosk` (static PWA served by the shell CDN path `/kiosk`), `iot-edge` (Go, deployed per plant; DaemonSet or single-node k3s at the edge; Mosquitto bundled optionally), `shopfloor-db` (CloudNativePG 200 GB, monthly partitions). Config: `SF_IDLE_THRESHOLD_S` (300), `SF_ANDON_DOWNTIME_MIN` (10), `SF_OFFLINE_MAX_ACTIONS` (5000), `IOT_BUFFER_HOURS` (24), `IOT_OPCUA_ENDPOINTS`, `IOT_MQTT_BROKER`. Flags: `shopfloor.iot`, `shopfloor.oee`, `shopfloor.pay-export`. Jobs: auto clock-out (hourly), OEE rollup (every 15 min), MachineEvent partition drop (daily, > 90 days). SLOs: kiosk action p95 < 200 ms server-side; ingest 10k events/min with consumer lag p95 < 5 s; availability 99.9%; edge buffer never exceeds 80% of 24 h. Alerts: consumer lag > 30 s (page), edge offline > 5 min (warn), seq gap detected (warn), andon storm > 50 open (warn). Runbook: "edge offline" — check MQTT/OPC-UA connectivity, buffer fill, restart connector; "seq gap" — replay from edge buffer by seq range. Rollback: prior digest; edge connector version pinned per plant with staged rollout.

4.25.11 Migration from AX / SAP

Source	Passage entity	Notes
JmgEmployee / HR mini-master (PA0001)	Worker	badge ids, default resource, profile
JmgProfileTable, JmgProfileDay / work schedule rules (T508A)	ProfileDay	shifts, breaks, rounding
JmgPayAgreement, JmgPayTable / wage types, time evaluation rules	PayRule config	overtime thresholds and multipliers
JmgAbsenceCode, Jmg error cause codes / absence types, AFRU.GRUND	reason codes	mapped to Passage code lists
JmgStampJournalTable, JmgRegistration (last 90 days) / CATSDB, AFRU	TimeRegistration (history, status exported)	read-only history for balances
Flex balances (JmgFlexTrans) / time accounts (PT_BAL00)	opening flex balance per worker	one opening line

Validation: flex balances per worker equal legacy at cutover; pay lines for the last closed week recomputed in Passage match legacy within 0.01 h per worker. Not migrated: raw terminal logs older than 90 days, machine data (none in AX/SAP core), MII configurations.

4.25.12 Dependencies, risks, open questions

Depends on: w21 job-card journals; w23 scheduled jobs (dispatch list); w24 kanban jobs on the kiosk; w7 employee party; w5 attachments and document service; w4 Kafka/Temporal; w43 payroll (consumer, later).

Risk	Likelihood	Impact	Mitigation
OPC-UA server diversity (security policies, node models)	High	Medium	Support None/Basic256Sha256 and companion-spec-agnostic tag mapping; certify against open62541 and a PLC simulator
Offline conflicts corrupt labor cost	Medium	High	Client UUID idempotency, flagged conflicts, supervisor review before posting option
Phase-exit demo depends on four prior waves' stability	High	High	Freeze w21–24 APIs two weeks before w25 close; rehearsal run weekly

Open questions: (1) Post job cards immediately or at approval? Default immediately with reversal on edit; decide by w25 kickoff. (2) Edge connector as DaemonSet in the plant cluster vs standalone binary? Default standalone k3s node per plant; decide by w25 mid-point. (3) Which OEE loss taxonomy (six big losses vs ISO 22400 element set)? Default six big losses mapped to ISO 22400 KPIs; decide by w25 kickoff.

4.25.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Kiosk idle timeout min	site	int	10	1–60	Session logout	Manufacturing execution parameters → Terminal → Log off after	PP-PDC terminal settings
Badge login mode	site	enum	`badge-or-pin`	badge / pin / badge-or-pin	FR-25.1	Manufacturing execution parameters → Terminal → Badge ID	DM/MII user login
Post job card on stop	site	enum	`immediate`	immediate / at-approval	FR-25.4	Manufacturing execution parameters → Production orders → Transfer (route card journal creation)	Confirmation parameters OPK4 (post immediately vs PDC upload)
Bundle split basis	site	enum	`quantity`	quantity / equal	BR-25.1	Job bundle setup (JmgParameters bundle)	n/a
Auto-stop previous job	site	bool	true	true/false	FR-25.3	Manufacturing execution parameters → Auto stop	n/a
Clock-in tolerance before / after min	profile	int / int	10 / 5	0–60	BR-25.2	Profile (JmgProfileTable) flex/tolerance	Work schedule rule tolerances (T550A)
Rounding minutes	profile	int	1	1–15	BR-25.2	Profile rounding	Time evaluation rounding rules
Missing clock-out close after h	site	int	4	1–24	BR-25.3	Calculation → missing clock-out	Time evaluation error message (RPTIME00)
Overtime daily / weekly thresholds	pay agreement	numeric	8.00 / 40.00	0–24 / 0–168	FR-25.7	Pay agreement (JmgPayAgreement)	Wage types / time evaluation schema
Idle threshold s	machine	int	300	30–3,600	BR-25.6	n/a	PCo notification settings
Andon downtime min / scrap %	site	int / numeric	10 / 5	1–240 / 0–100	FR-25.11	n/a	DM andon rules
OEE bucket minutes	tenant	int	15	5 / 15 / 60	FR-25.10	n/a	DM OEE time granularity
Offline max actions	site	int	5,000	100–20,000	FR-25.12	n/a	n/a
Edge buffer hours	edge node	int	24	1–168	FR-25.9	n/a	PCo storage and forward settings
Edge rollout ring	edge node	enum	`all`	canary / ring1 / all	FR-25.19	n/a	PCo agent versioning (manual)
Edge write allowlist	edge node	list	empty	OPC-UA node ids / topics	BR-25.15	n/a	PCo destination system write permissions
Control recipe destination per resource	resource	ref	`kiosk`	RecipeDestination	FR-25.15	n/a	Control recipe destination (O10C)
Process message category destinations	category	list	per category	destinations	FR-25.17	n/a	Message category destinations (O05C/O06C/O07C)
PI sheet hold requires signature	template	bool	true	true/false	BR-25.12	n/a	PI sheet XStep signature settings
Process message retry schedule	tenant	list	1,5,15,60 min	any	FR-25.17	n/a	CO54 send interval

4.25.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Who is clocked in	list inquiry	site, work cell	worker, clock-in time, current job, elapsed	TimeRegistration	Manufacturing execution → Clocked-in workers (JmgTermRegList)	PP-PDC attendance, CATS
Registrations for approval	list inquiry	team, date, flags	worker, date, registrations, pay lines, flags	TimeRegistration × PayLine	Approve (JmgCalcApprove form)	CATS approval (CATS_APPR_LITE)

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Pay lines / transfer log	list inquiry	export batch, worker	pay type, hours, amount, batch	PayLine	Pay specification (JmgPayEvents), Transfer log	PT_BAL00 / CATS transfer
Job registration history	drill-down	order, operation, worker	start/stop, hours, qty, scrap reason, journal	TimeRegistration	Registrations (JmgStampJournalTrans)	CO14 confirmations (AFRU)
OEE by resource	KPI	resource, period, shift	OEE, A, P, Q, top losses	OeeBucket	n/a (Sensor Data Intelligence)	DM OEE dashboard
Downtime Pareto	operational report	resource, period, category	reason, minutes, count, %	DowntimeRecord	Indirect activity report	PM breakdown analysis (MCI7)
Scrap by reason	operational report	order, resource, period	reason, qty, cost	TimeRegistration scrap	Error cause report (JmgErrorSpecification)	CO11N variance reasons, MCP5
Machine event / sequence gap report	operational report	machine, period	events, gaps, dropped (old/dup), lag	MachineEvent metrics	n/a	PCo logs
Control recipe monitor	list inquiry	status, destination, order	control recipe, status, sent/received, error	ControlRecipe	n/a	CO53 / CO53XT
PI sheet worklist	list inquiry	work cell, status	sheet, order, phase, status, open steps	PiSheetRun	n/a	CO60 / CMX worklist
Process message monitor	list inquiry	status, category, destination	message, category, order, per-destination status, attempts	ProcessMessage	n/a	CO54 / CO54XT
Edge fleet status	list inquiry	site, status	edge, version, ring, last heartbeat, buffer fill, cert expiry	EdgeNode	n/a	PCo management console

4.25.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Badge	Worker setup	Name, photo (optional), badge id barcode (Code 128 / NFC encode)	PDF (CR80 card)	none	Personal data — GDPR minimization (photo optional)
Time card / registration summary	Day/week approval	Worker, days, clock-in/out, jobs, hours, pay types, approver	PDF, email	none	Wage-and-hour records: US FLSA 29 CFR 516 — 3 years payroll / 2 years time cards; EU Working Time Directive national rules (e.g., DE ArbZG §16 — 2 years)
Payroll export file	Export	Worker id, date, pay type, hours, amount	CSV, event (w43)	export batch number	Retained with payroll (w43)
PI sheet (executed)	PI sheet completion	Steps, values, limits, comments, signatures with manifestation	PDF/A-2b + JSON	`pi-sheet` per legal entity	Part of EBR (w24 retention)
Control recipe	Release	ISA-88 structure, setpoints, destination	JSON (B2MML-style), PDF	`control-recipe`	Part of EBR for GxP items
OEE shift report	Shift end	Resource, A/P/Q/OEE, downtimes, scrap	PDF, email	none	None
Machine events	Edge publish	CloudEvents (BR-25.4)	JSON	n/a	Postgres 90 days; Iceberg indefinitely

4.25.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Auto clock-out	hourly	site	Closes open clock-ins per BR-25.3 once	< 30 s	Retry next hour	Count of auto-closed reported
OEE rollup	every 15 min	resource	Upsert OeeBucket by (resource, bucket_start)	< 60 s	Recompute next run	Lag > 5 min warns
Pay calculation	nightly 02:00 + on edit	worker-day	Recompute non-approved days; approved immutable	1,000 workers < 2 min	Per-worker errors listed	Errors > 0 warn
Payroll export	per pay period (configurable)	legal entity	Export batch id; re-export reuses batch	< 1 min	Retry; batch stays open	Failure pages before payroll cutoff
MachineEvent partition drop	daily 03:30	month partition	Drops partitions > 90 days after Iceberg copy confirmed	< 1 min	Skip if copy unconfirmed	Skip warns
Process message retry	every 1 min	destination	Per-delivery attempts per schedule	< 10 s	Error status after last attempt	Errors > 20 open warn
PI sheet archive sweep	hourly	site	Archives completed runs lacking archive	< 2 min	Retry 5×	Unarchived > 1 h pages
Edge certificate renewal check	daily 04:00	edge node	Renew when < 30 days	< 1 min	Alert after 7 days failing	BR-25.14
Edge rollout promotion	on schedule (ring windows)	site × ring	Promote ring after 24 h healthy	n/a	Halt rollout on failed heartbeat	Rollout halted warns

4.25.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
Kiosk login	sessions:badge, kiosk shell	none	Operator
Register time	registrations, jobs, feedback, instructions, downtime coding, PI sheet execution	with *Approve time registrations* (own registrations)	Operator
Calculate time registrations	pay:calculate	none	Shop floor supervisor
Approve time registrations	pay:approve, edits with reason	with *Register time* for own records — approval of own day blocked	Shop floor supervisor
Export time to payroll	pay:export	with *Approve time registrations* flagged	Payroll clerk
View shop floor	OEE, andon, supervisor board, reports	none	Supervisor, Plant manager
Maintain shop floor setup	machines, thresholds, profiles, pay agreements, reason codes, recipe destinations, message categories	with *Approve time registrations* (pay agreements) flagged	Manufacturing engineer, ERP administrator

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
Maintain PI sheet templates	step library, templates (ECO-released)	with *Approve batch records* (w24)	Process engineer
Create process messages	manual process message, reprocess/delete with reason	none	Shop floor supervisor
Manage edge fleet	enrollment tokens, config publish, write allowlist, rollout rings	with *Maintain PI sheet templates* (allowlist and recipe content by different people)	OT administrator

4.25.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
SF-0001	error	Badge {badge} is not recognized.	Unknown badge	Contact supervisor	FR-25.1
SF-0002	warning	Previous job {job} was stopped automatically.	One open job rule	none	FR-25.3
SF-0003	warning	Missing clock-out on {date}; closed at shift end.	BR-25.3	Supervisor confirms or edits	BR-25.3
SF-0004	warning	Job {job} was finished on the server while you were offline.	BR-25.7	Supervisor review	BR-25.7 (`job-finished-conflict`)
SF-0005	error	Mandatory check {check} failed; operation cannot be finished.	FR-25.5	Supervisor override with reason	FR-25.5
SF-0006	error	Day cannot be approved: {n} flags open.	Approval precondition	Resolve flags	FR-25.8
SF-0007	error	Hours exceed 24.00 for {worker} on {date}.	PayLine invariant	Correct registrations	FR-25.7
SF-0008	warning	Machine {machine} sequence gap {from}–{to}.	Lost or delayed events	Replay from edge buffer	FR-25.9
SF-0009	info	Andon raised on {resource}: {kind}.	FR-25.11	Respond	FR-25.11
SF-0010	error	Offline queue is full ({n} actions).	FR-25.12 limit	Reconnect device	FR-25.12
SF-0011	error	Control recipe {cr} could not be sent: {reason}.	Destination error	Fix destination; resend	FR-25.15
SF-0012	error	Resend requires force: process messages already received for order {order}.	BR-25.11	Resend with force and reason	BR-25.11
SF-0013	error	PI sheet cannot be completed: {n} mandatory steps open.	BR-25.12	Complete steps	BR-25.12
SF-0014	warning	Value {value} is outside {min}–{max}; comment required.	Out-of-limit input	Enter comment	FR-25.16
SF-0015	error	Process message rejected: unknown order or phase {ref}.	BR-25.13	Correct source mapping	BR-25.13
SF-0016	warning	Process message {id} delivery to {destination} failed (attempt {n}).	Destination unavailable	Automatic retry; reprocess later	FR-25.17
SF-0017	error	Write to node {node} refused: not on allowlist or out of limits.	BR-25.15	Update allowlist via OT admin	BR-25.15
SF-0018	warning	Edge {edge} certificate expires in {days} days; renewal failing.	BR-25.14	Check connectivity to CA	BR-25.14
SF-0019	warning	Edge {edge} rolled back to {version}: heartbeat failed after upgrade.	FR-25.19	Inspect connector logs	FR-25.19
SF-0020	error	Enrollment token is invalid or expired.	FR-25.19	Issue a new token	FR-25.19

4.25.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation under 3× load
Kiosk action (start/stop/feedback)	500 workers × 40 actions/day; peak 300/min at shift change	60 / 200 / 400 ms server-side	50/s	TimeRegistration 7M rows ≈ 4 GB	(worker_id, started_at), (job_id)	Optimistic UI; queue in PWA
Machine event ingest	200 machines × 50 events/min = 10k/min; peak 30k/min	consumer lag p95 < 5 s	500 events/s sustained	5.3B events/yr → Postgres 90 days ≈ 300 GB rolling; Iceberg ≈ 1 TB/yr compressed	monthly range partitions; (resource_id, ts)	Lag allowed to 60 s; OEE marked stale
OEE bucket recompute	200 resources × 96 buckets/day	< 5 s after last event	n/a	OeeBucket 7M rows/yr ≈ 2 GB	PK (resource_id, bucket_start)	Rollup every 30 min
Offline replay	≤ 5,000 actions per device	40 actions < 5 s; 5,000 < 2 min	100 actions/s per device	n/a	client UUID unique	Batches of 500 throttled
Process message delivery	50,000/day	200 ms / 1 s / 3 s per destination	50/s	ProcessMessage 18M rows ≈ 10 GB	(status, received_at); unique (source, source_message_id)	Retry backlog drained FIFO
PI sheet step save	100,000/day	80 / 250 / 500 ms	50/s	PiStepResult 36M rows ≈ 15 GB; archives ≈ 200 GB object storage	(run_id, step_seq)	Offline queue absorbs
Edge heartbeat / config pull	50 edges × 2/min	50 / 150 / 300 ms	10/s	EdgeNode status history 50M rows ≈ 5 GB (90-day retention)	(edge_id, ts)	Heartbeat interval auto-extends to 120 s

4.25.20 Localization and compliance hooks

- **Working-time law**: pay rules (overtime thresholds, Sunday/holiday premiums, night work) differ by country and collective agreement; launch defaults are US FLSA (40 h/week) — wave 48 country packs ship pay agreement templates (e.g., DE ArbZG 10 h/day maximum, FR 35 h/week, MX LFT) through the `shopfloor.pay-rule` extension point; holiday calendars come from w48.
- **Time-recording obligations**: EU CJEU C-55/18 (CCOO) requires objective time recording; clock registrations are kept as raw records (immutable, corrections as reversals) to satisfy it; retention per country via w48 retention policy (US 2–3 years, DE 2 years, ES 4 years).
- **Privacy and works councils**: badge, PIN hash and registration data are personal data; machine-to-worker attribution (FR-25.13) and OEE per operator are disabled per site with `shopfloor.operator-analytics = false` (default false in DE/AT/NL packs, where works-council agreements typically require it); GDPR erasure pseudonymizes workers after retention.
- **GxP**: PI sheets and process messages become EBR content under 21 CFR Part 11 / EU Annex 11 (w24 FR-24.21/24.22).
- **OT security**: edge connector deployment follows IEC 62443-3-3 zone/conduit guidance (outbound-only, mTLS, allowlisted writes); NIS2 (EU) incident reporting hooks use the w2 security event stream.
- **Language**: kiosk UI and PI sheet content are translatable per site; instruction attachments per language with fallback.
- **Extension points**: `shopfloor.pay-rule`, `shopfloor.process-message.destination`, `iot-edge.protocol-adapter` (adds protocols such as Modbus TCP or MTConnect without core changes).

4.25.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP 13.22 PP-PI — CO53 / CO53XT control recipe monitor [SAP #254]	PARTIAL-25	ADDED as FR-25.15	Destinations kiosk / pcs-opcua / pcs-mqtt; monitor in 4.25.14
SAP 13.22 PP-PI — CO54 / CO54XT / CO57 / CO58 process messages [#255]	PARTIAL-25	ADDED as FR-25.17	Categories, destinations, retry
SAP 13.22 PP-PI — CO60 / CO60XT / CO60_PI / CO64 / CO67 PI sheet [#256]	PARTIAL-25	ADDED as FR-25.16, FR-25.18	Designer, execution, archive
SAP 13.22 PP-PI — CMX / Standard XSteps (CMXSV) / SXS repository [#258]	PARTIAL-25	ADDED as FR-25.16	Versioned StepLibrary via ECO
SAP 13.22 PP-PI Reports — CO54 process message log [#263]	PARTIAL-25	ADDED as FR-25.17	Process message monitor
SAP 13.22 PP-PI Reports — CMX/CO60 PI sheet worklist [#265]	PARTIAL-25	ADDED as FR-25.16	PI sheet worklist per work cell
SAP 13.22 PP-PI Periodic — CO53 send control recipes (batch) [#266]	PARTIAL-25	ADDED as FR-25.15	Sent at release; resend per BR-25.11
SAP 13.22 PP-PI Periodic — CO54 process message processing (batch) [#267]	PARTIAL-25	ADDED as FR-25.17	Retry job in 4.25.16
SAP 13.22 PP-PI Periodic — CO64 / CO67 PI sheet archive [#268]	PARTIAL-25	ADDED as FR-25.18	Archive sweep job
SAP 13.22 PP-PI Customizing — Process management: PI categories (O10C/O12C/O13C), message categories and destinations (O05C/O06C/O07C), PI sheet layouts, XStep settings, signature strategies [#271]	PARTIAL-25	ADDED as FR-25.15, FR-25.16, FR-25.17	Signature strategies from w24 FR-24.22
SAP 13.28 HCM — PP61/PP62/PP63/PP64/PP6I/PP6A shift planning [#325]	PARTIAL-25	DEFERRED to wave 42	Rostering against staffing requirements belongs with HR core; wave 25 keeps profiles and shift windows (ProfileDay)
Coordinator-flagged — OPC-UA/MQTT edge connector as a deployable (AX 12.x Production → IoT, COVERED-25)	COVERED-25 (deployment not specified)	ADDED as FR-25.19	Enrollment, mTLS, config versions, ringed rollout, write allowlist

Related row owned by another wave: SAP #193 EWM Material Flow System (propose 32) reuses the FR-25.19 edge connector and the `iot-edge.protocol-adapter` extension point.

Part 5 — Phase 5: Order-to-Cash (waves 26–30)

5.0 Phase overview

Business outcome at phase exit. At the end of wave 30 a manufacturer runs the complete buy → make → sell → collect loop on Passage. A sales rep quotes and takes an order at a deterministically resolved price; the order is promised against cumulative ATP and checked against credit; the warehouse picks, packs and ships it with a posted packing slip that moves inventory cost to the universal journal; billing posts the invoice with tax and an ASC 606 / IFRS 15 recognition schedule; cash is applied from lockbox, remittance or bank statement; collections runs dunning and disputes; returns flow through RMA to credit note; and every subledger reconciles to the journal by automated check. Wave 30 is the first go-live candidate.

Phase-exit sign-off criteria (each produces an artifact in `reconciliation/phase-5-exit/`):

- One simulated 31-day month (fixture period 2027-03) runs end-to-end on `stage` for demo entities DMF-US (USD) and DMF-DE (EUR): ≥ 200 purchase orders received (Phase 3), ≥ 150 production orders reported as finished (Phase 4), 1,000 sales orders, ≥ 900 shipments, ≥ 950 invoices including ≥ 40 bundles with revenue schedules, 600 customer payments across manual / BAI2 lockbox / camt.053, one dunning run, 30 RMAs, one rebate settlement, inventory close and period close for 2027-03.
- Daily reconciliation passes for all 31 days with zero unexplained differences: AR open items = GL 1200; deferred revenue roll-forward = GL 2400; delivered-not-invoiced = GL 1405; inventory subledger = GL 1400; WIP = GL 1350; rebate accrual = GL 2150; tax payable = tax engine settlement = GL 2300; unapplied cash = GL 1205.
- Trial balance for 2027-03 balances; GL 4000 = Σ posted invoice revenue lines – credit notes – deferrals; GL 5000 = Σ packing-slip cost – return receipts.
- Pricing: 500 scenario fixtures and 10k-request determinism property test green; p95 < 20 ms for a 10-line order.
- Order intake 500 orders/min at p99 < 500 ms; ATP never over-promises (property test over 100 fixtures).
- Cash application ≥ 90% auto-match on the 500-payment fixture with zero false matches; dunning run produces letters for exactly the expected customer set.
- Pick-pack-ship E2E for 1,000 orders with zero inventory mismatch; packing-slip reversal contract tests green.
- Recorded walkthrough (≤ 45 min) and a reconciliation pack signed by a practicing accountant, both linked from the wave 30 issue.

Cross-wave data model summary

Entity	Owning service (Postgres db)	Key relationships
condition_record, pricing_procedure, rebate_agreement, price_journal	pricing (pricing_db, Go)	→ customer/group (w7), product (w8); explain refs consumed by sales/billing
sales_quotation, sales_order(_line), sales_agreement, order_hold, credit_profile, atp_bucket	sales (sales_db, Go)	→ condition_record; supply/demand from w16/w17/w21/w22; production order link (w21)
shipment, container, packing_slip, carrier_shipment	fulfillment (fulfillment_db, Go) + carrier-gateway (TS)	→ sales_order_line; inventory_transaction (w16); journal posting (w11)
invoice(_line), free_text_invoice, credit_note, subscription	billing (billing_db, Go)	→ packing_slip; tax (w13); journal (w11)
revenue_contract, performance_obligation, revenue_schedule	revrec (revrec_db, Go)	→ invoice_line; journal (w11)
open_item, payment, settlement, lockbox_batch, dunning_run, dispute, interest_note, return_order	receivables (receivables_db, Go) + collections-workflows (TS, Temporal)	→ invoice, customer, camt.053 statement, journal
portal_user, payment_intent	customer-portal (portal_db, TS BFF) + payments-gateway (TS)	→ party contact (w7), open_item, Stripe/Adyen adapter

Cross-wave event catalog (CloudEvents on Kafka topics `passage.<domain>`, Avro in schema registry, Debezium outbox)

Event	Producer	Consumers	Payload summary
passage.pricing.condition-record.activated.v1	pricing	sales cache, analytics	id, type, keys, validity, version
passage.pricing.rebate-accrual.posted.v1	pricing	ledger, receivables	agreement, period, amount, journal ref
dev.passage.sales.order.confirmed.v1	sales	planning, fulfillment, credit, analytics	order, customer, lines with confirmed dates, totals
passage.sales.order-hold.applied.v1 / released.v1	sales	fulfillment, collections, portal	order, hold type, reason, actor
passage.fulfillment.packing-slip.posted.v1	fulfillment	billing, inventory, ledger, sales, portal	slip, lines, qty, cost snapshot, journal ref
dev.passage.billing.invoice.posted.v1	billing	receivables, revrec, ledger, pricing, portal	invoice, lines, tax, totals, due date
dev.passage.revrec.schedule.recognized.v1	revrec	ledger, analytics	schedule line, obligation, amount, period
dev.passage.receivables.payment.applied.v1	receivables	ledger, sales (credit), portal	payment, settlements[], residual
passage.receivables.customer-balance.changed.v1	receivables	sales credit, portal	customer, balance, overdue, oldest item days, dunning level
passage.receivables.return-order.dispositioned.v1	receivables	fulfillment, billing, inventory	RMA line, disposition, qty, restocking

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
AX "find next" vs. SAP access-sequence semantics confuse migrated users	High	Medium	Two reference procedures shipped; both in the 500-scenario fixtures	Product
ATP over-promises because a supply source is missed	Medium	High	Projection rebuilt nightly from events and diffed; property test gates release	Engineering
Delivered-not-invoiced drifts from GL 1405	Medium	High	Packing slip and invoice share one cost_snapshot_id; daily check from wave 28 day 1	Engineering
ASC 606 allocation disputed by auditors	Medium	High	40 fixtures reviewed by a practicing accountant; residual-cent rule documented	Product + accountant
False auto-match in cash application	Low	High	Single candidate ≥ 85 with 20-point margin; every auto-match reversible	Engineering
Go-live simulation exposes Phase 3/4 defects late	High	Medium	Dry-run at wave 28 exit with sell/collect stubs	Both owners

5.26 Wave 26: Pricing, Discounts & Trade Agreements

5.26.1 Scope and business context

Every order, quotation, invoice and rebate in Phase 5 depends on a price that is explainable and repeatable. Wave 26 ships the `pricing` service: condition records with validity and quantity breaks, access sequences resolving from the most specific key to the most general, line/multiline/total discounts, surcharges, exclusion groups, rounding rules, campaigns, rebate agreements with accrual, atomic price journals, a simulation/explain API and commission hooks. At wave end a pricing administrator maintains agreements in the Pricing workspace and any caller receives, for a customer/product/date/quantity/currency tuple, a net price plus the ordered list of conditions that produced it. Out of scope: rebate settlement (wave 30), vendor pricing (wave 17), configurator price components beyond attribute-keyed records, storefront promotions. Added from the coverage registers (§5.26.21): vendor rebate, broker and royalty agreements with accrual (claims paid through wave 19 AP), commission transactions and posting, trade promotion funds and promotion planning, commodity pricing, free goods and retroactive accrual.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Nobody can say why a line priced at 93.40: AX "Find next" results are invisible on SalesLine; SAP needs the VA01 condition "Analysis" screen per line	Resolution is a side effect inside the order form (PriceDisc class / RV61A formulas), not a queryable function	`POST /pricing/resolve?explain=true` returns the access-by-access trace, stored with the line as `price_explain_ref`	"Why this price": 20–40 min consultant trace → < 10 s self-service; 100% of lines traced
Mass price change is a multi-hour, non-atomic job: PriceDiscAdmTrans posts row by row and leaves half-applied prices on failure; VK12 mass change has no undo	Row loop in X++/ABAP without a transaction boundary across the set	price_journal validates fully, posts in one transaction and reverses with one action	100k prices: 3–6 h plus cleanup → < 120 s; rollback 1 action
Overlapping agreements silently win by insertion order (first PriceDiscTable row found; A-table collisions blocked only per access)	No overlap validation at save	Activation rejects overlapping validity for the same key with 409 naming the conflict	Overlap-caused price disputes: ~10 per 1,000 agreements/month → 0
Rounding differs between order, packing slip and invoice (SalesLine.LineAmount re-rounded per form letter; KPEIN pricing-unit rounding)	Each consuming document re-rounds	Engine rounds once (BR-26.3); consumers reuse the stored net; invoice totals property-tested	Cent-level order/invoice differences: ~0.5% → 0
CPQ tools and storefronts re-implement price lists in middleware because pricing is unreachable outside the form call stack	No stateless pricing API	Resolve API p95 < 20 ms, condition-record events, MCP exposure via wave 46	External lookups: nightly export → live; zero duplicated price lists

5.26.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Sales price agreements	PriceDiscTable (Relation=PriceSales), PriceDiscAdmTable/Trans journal	VK11/VK12/VK13, KONH/KONP, A004/A005/A305, PR00	condition_record type PRICE	full
Line discount	PriceDiscTable (LineDiscSales), PriceDiscGroup	K004/K005/K007, KONP-KBETR	condition_record LINE_DISC	full
Multiline discount	PriceDiscTable (MultiLineDiscSales)	Group condition T685A-KZBZG, K020	MULTILINE_DISC with group key	full
Total discount	PriceDiscTable (EndDiscSales)	Header conditions HI01/RA01	TOTAL_DISC	full
Price/discount groups	PriceDiscGroup; CustTable.PriceGroup/LineDisc/MultiLineDisc/EndDisc; InventTableModule.PriceGroup	KNVV-KONDA/KDGRP, MVKE-KONDM	customer_pricing_group, product_pricing_group	full
Access order / exclusion	PriceDiscParameters, "Find next", "Search again"	V/07 access sequences (T682), V/08 procedure (T683S), exclusion T684	pricing_procedure, access_sequence, exclusion_group	improved (explain)
Smart rounding	PriceDiscRounding, journal rounding	DIFF condition, T001R	rounding_rule	full
Campaigns / promotions	Trade allowance agreements (TAMAgreement)	Sales deals VB21/VB22, promotions VB31 (KONA)	campaign	full
Rebates	PriceDiscRebate / TAMRebateAgreement, D365 Rebate management	VBO1/VBO2 (KONA, BO01–BO06, VBOX); S/4 condition contract WCOCO	rebate_agreement, rebate_accrual	partial (settlement w30)
Commissions	CommissionCalc, CommissionSalesRep, CommissionTrans	VBPA partner function; ICM	commission_rule + event hook	partial
Mass update	Price/discount agreement journals	VK12 mass change, MASS	price_journal (atomic, reversible)	improved
Vendor rebates	PdsRebate* (PI vendor rebate agreements, rebate groups, program types), D365 Rebate management vendor deals, claim journals	Supplier condition contracts (WCOCO purchasing side), settlement documents	rebate_agreement with party_role VENDOR, vendor_rebate_claim → w19 AP	full (claim payment w19)
Broker contracts and royalties	PdsBrokerContractTable, PdsRoyaltyAgreement, broker/royalty claim journals	Settlement management royalty and broker condition contracts	broker_contract, royalty_agreement, claim journal → w19 AP	full
Commission posting	CommissionCalc, CommissionSalesGroup, CommissionCustomerGroup, CommissionItemGroup, CommissionPosting, CommissionTrans	VBPA partner functions VE/PE; settlement management commission contracts; ICM	commission_rule, sales_group, commission_transaction (posted)	full (payout w19/w43)
Trade promotion funds and planning	TAMFund, TAMFundUsage, TAMMerchandisingEvent, TAMPromotionPlanning	CRM TPM; condition contracts with budget	fund, promotion_plan (baseline, lift, actuals)	partial (lift analysis basic)
Free goods	Supplementary items (PriceDiscSuppItem, order-side w27)	VBN1/VBN2/VBN3 NA00, OVKF, KOTN	condition type FREE_GOODS	full
Commodity pricing (PI)	PdsCommodityCostBasis, PdsPricingCalc, PdsPriceMarginTemplate, price calculation run	Condition contracts with index-based conditions	commodity_index, price_calculation → price_journal	full
Retroactive agreements	Rebate management recalculation	VBOF, delta accrual / delta settlement	retroactive recompute with delta posting	full

5.26.3 Functional requirements

- **FR-26.1 Condition types and procedure** — Must. The system shall define condition types (PRICE, LINE_DISC, MULTILINE_DISC, TOTAL_DISC, SURCHARGE, FREIGHT, REBATE_ACCRUAL, ROUNDING) ordered in a versioned pricing procedure per legal entity and sales channel. Notes: reference procedures `AX_STANDARD` and `SAP_RVAA01`; steps 10–990 with optional CEL requirement (≤ 512 chars); activated versions are immutable.
- **FR-26.2 Condition records** — Must. The system shall store records keyed by customer | customer group | all × variant | product | product group | all, with optional site/warehouse/channel/campaign, validity, currency, unit, ≤ 20 strictly increasing quantity breaks, amount and/or two percentages. Notes: overlapping ACTIVE records for the same key and currency are rejected with 409.
- **FR-26.3 Access sequence** — Must. The system shall resolve each condition type by trying accesses in configured order (BR-26.1) and stop at the first access yielding a candidate unless `search_next` is set, in which case it continues and keeps the lower net.

- **FR-26.4 Deterministic resolution with explain** — Must. The system shall return identical outputs for identical (procedure version, records as of pricing_date, request) and an explain trace (≤ 64 KB, content-addressed) listing every access tried, candidates, winner and reason. No wall-clock reads.
- **FR-26.5 Line, multiline and total discounts** — Must. The system shall apply line discounts per line, multiline discounts over summed quantity of lines sharing a group, and total discounts on order net after line discounts; two percentages apply sequentially (1–p1)(1–p2); amount before percentage.
- **FR-26.6 Surcharges** — Should. Fixed or percentage surcharges as separate conditions with their own revenue account; small-order threshold default 250 USD / 200 EUR.
- **FR-26.7 Rounding** — Must. Unit prices at NUMERIC(19,6); line amounts half-even to ISO 4217 minor units (wave 12); smart rounding bands for journals (0–10 → .99, 10–100 → .95, ≥ 100 → whole).
- **FR-26.8 Exclusion groups** — Must. BEST (lowest net) or FIRST (lowest step) among members; an `exclusive` winner suppresses later discount steps.
- **FR-26.9 Price journals** — Must. Mass create/update/delete through a journal that validates fully, posts atomically (100k lines < 120 s) and reverses by one action into a new journal referencing the original.
- **FR-26.10 Campaigns** — Should. Records grouped under a campaign with validity, target groups, budget and kill switch.
- **FR-26.11 Rebate agreements and accrual** — Must. Customer rebate agreements (tiered % or amount per unit; monthly/quarterly/annual; basis net invoiced or quantity) accrue on each posted invoice as Dr 4150 / Cr 2150 via the journal API; retroactive tier crossing recomputes the period and posts the delta.
- **FR-26.12 Commission hooks** — Could. Rules by sales rep × customer group × product group publish `commission.calculated` for Phase 8.
- **FR-26.13 Simulation** — Should. Simulate any tuple without a document and compare two procedure versions.
- **FR-26.14 Performance** — Must. p95 < 20 ms, p99 < 60 ms for a 10-line resolve; in-memory record cache per entity invalidated by `condition-record.activated`.
- **FR-26.15 Vendor rebate agreements** — Should. The system shall support `rebate_agreement.party_role = VENDOR`: tiered percentage or amount per unit on vendor-invoiced net value or received quantity per period, accruing Dr 2160 Vendor rebate receivable / Cr 5100 Purchase rebate income on each `vendor-invoice.posted` (w19). Notes: rebate groups (vendor) and program types BILL_BACK, FREIGHT, MARKETING; lifecycle Cumulate → Approve → Process claim as in AX PdsRebate; a claim per (agreement, period) settles by a w19 AP credit note or deduction from the next vendor payment, journal sequence `RC-`; Σ accruals – Σ claims = GL 2160 daily.
- **FR-26.16 Broker contracts** — Could. The system shall hold broker contracts (broker vendor, customer scope, product scope, rate as % of net or amount per unit, validity) and on `invoice.posted` accrue a broker claim Dr 6210 Broker commission / Cr 2175 Broker payable; claims are processed per period to w19 AP. Notes: credit notes reverse proportionally; claim statement document `BC-`.
- **FR-26.17 Royalty agreements** — Could. The system shall hold royalty agreements (licensor vendor, product or product group, basis net sales or quantity, tiered rates, minimum royalty per period) accruing Dr 6220 Royalty expense / Cr 2176 Royalty payable on `invoice.posted`, with a period claim to w19 AP. Notes: minimum royalty shortfall posts at period close; D365 price-based deals map to the same entity.
- **FR-26.18 Commission transactions and posting** — Should. The system shall define sales groups (reps with share percentages), commission customer groups, commission item groups and commission rules (basis REVENUE or MARGIN, rate) resolved specific → general, and on `invoice.posted` create `commission_transaction` rows posted Dr 6200 Commission expense / Cr 2170 Commission payable per rep. Notes: sales group and rep assignment on the order header (AX SalesTable.CommissionGroup; SAP VBPA VE/PE); credit notes create negative transactions referencing the original; a settlement export per rep and period feeds payroll (w43) or AP (w19); AX Commission report and SAP commission contract settlement list are reproduced by the Commission inquiry (§5.26.14).
- **FR-26.19 Trade promotion funds** — Could. The system shall define funds (type, budget per period, currency, customer scope, owner) whose usage is the sum of campaign accruals, rebate accruals, claims and deductions (w30) attributed to the fund; usage is visible per period and a fund close freezes it. Notes: overspend blocked unless `allow_overspend`; fund budget recalculation nightly.
- **FR-26.20 Promotion planning** — Could. The system shall plan a promotion (merchandising event category and type, campaign, fund, customers, products, dates) with a baseline (average weekly units over the prior 13 weeks), forecast lift % and planned spend; actuals are updated nightly from posted invoices and effectiveness = (actual – baseline) / baseline per product.
- **FR-26.21 Commodity pricing** — Could. The system shall store commodity cost-basis types with dated index values, pricing calculations (index × conversion factor + margin from a price margin template) and run a price calculation that produces a `price_journal` updating the affected trade agreements. Notes: run is previewable; potency pricing (w24) reuses the calculation; index values are versioned.
- **FR-26.22 Free goods** — Should. The system shall support condition type FREE_GOODS with inclusive (1 free within every 10 ordered) and exclusive (1 extra per 10 ordered) variants, calculation rule PROPORTIONAL, UNIT_REFERENCE or WHOLE_UNITS and a minimum quantity; the resolve response returns a zero-priced free-goods line proposal that w27 adds as a supplementary line (FR-27.14).
- **FR-26.23 Retroactive rebate accrual and delta settlement** — Must. The system shall, when an agreement is activated or changed with `valid_from` earlier than today, recompute accruals for invoices already posted in the scope (SAP VBOF) and post the delta per period with `retroactive=true`; settlement after a retroactive change posts only the delta against prior settlements.

5.26.4 Data model

condition_record

Field	Type	Required	Notes
id, tenant_id, legal_entity_id	uuid	Y	PK (v7); RLS
condition_type, key_combination	enum, smallint	Y	FK access_sequence_step
customer_id / customer_group_id	uuid	N	exactly one or null (all)
product_id / product_group_id / variant_dims	uuid / jsonb	N	same rule; dims {color,size,config}
site_id, warehouse_id, sales_channel, campaign_id	uuid/text	N	optional dims
currency, unit	char(3), text	Y	ISO 4217; FK uom (w6)
from_quantity, amount	numeric(19,6)	Y/N	default 0
disc_pct_1, disc_pct_2	numeric(7,4)	N	0–100
valid_from, valid_to	date	Y	inclusive
search_next, exclusive	bool	Y	default false
status, version, journal_id	enum, int, uuid	Y	DRAFT/ACTIVE/EXPIRED/WITHDRAWN

pricing_procedure: id, legal_entity_id, name, version, status, steps[] (step_no, condition_type, requirement_cel, exclusion_group_id, statistical, account_key). **access_sequence_step**: condition_type, seq_no, key_combination, required_fields[].

rebate_agreement: id, customer_id | customer_group_id, product scope, basis (NET_AMOUNT | QUANTITY), period (MONTH | QUARTER | YEAR), tiers[] (from_value, rate_pct | amount_per_unit), currency, accrual_account 2150, expense_account 4150, valid_from/to, status, settlement_method (CREDIT_NOTE | PAYMENT). **rebate_accrual**: id, agreement_id, period_id, invoice_line_id, basis_value, accrued_amount, journal_posting_id, status (ACCRUED | ADJUSTED | SETTLED | REVERSED).

price_journal: id, name, lines[] (op, before/after snapshot), status (OPEN | VALIDATED | POSTED | REVERSED), posted_at, reversed_by_journal_id.

Invariants: no two ACTIVE records share (type, keys, currency, unit, from_quantity) with overlapping validity; every ACTIVE record has a journal_id; resolve is reproducible from (procedure version, pricing_date, request); Σ rebate_accrual per period = GL 2150 movement. Indexing: composite btree (tenant, entity, type, key_combination, customer_id, product_id, currency, valid_from) plus partial indexes per key combination; partition by legal_entity; cache snapshot per (entity, pricing_date), 60 s TTL plus event invalidation.

5.26.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/pricing/resolve	Resolve header + ≤ 500 lines; `?explain=true`	stateless	PricingResolve
POST	/api/v1/pricing/condition-records, …/{id}:activate, :withdraw	Record lifecycle with overlap check	Idempotency-Key	PricingMaintain
GET	/api/v1/pricing/condition-records?customer=&product=&asOf=	Query	—	PricingView
POST	/api/v1/pricing/journals, …/{id}:post, :reverse	Mass update	Idempotency-Key	PricingJournalPost
POST	/api/v1/pricing/procedures, …/versions/{v}:activate	Procedure maintenance	Idempotency-Key	PricingAdmin
POST/GET	/api/v1/pricing/rebates/agreements; /rebates/accruals?period=	Rebates	Idempotency-Key	RebateMaintain
POST	/api/v1/pricing/simulate	Simulation, version comparison	stateless	PricingView

Validation: resolve requires legal_entity_id, customer_id, currency, pricing_date, lines[].product_id/quantity/unit; quantity ≤ 0 or unknown unit → 422. Journal post requires status VALIDATED with zero errors.

Event	Trigger	Payload fields	Consumers
passage.pricing.condition-record.activated.v1	activate / journal post	id, type, keys, currency, validity, version, journal_id, retroactive	sales cache, analytics
passage.pricing.condition-record.withdrawn.v1	withdraw / reverse	id, version, reason	sales cache
dev.passage.pricing.journal.posted.v1	post / reverse	journal_id, line_count, reversed_journal_id	audit
passage.pricing.rebate-accrual.posted.v1	invoice.posted consumed	agreement_id, period_id, invoice_line_id, amount, currency, journal_posting_id	ledger, receivables
dev.passage.pricing.commission.calculated.v1	invoice.posted consumed	invoice_line_id, sales_rep_id, basis, rate, amount	Phase 8

5.26.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Pricing workspace	tiles: expiring in 30 d, draft journals, campaigns, accrual this period	counts	PricingView
Trade agreements grid	create, copy, activate, withdraw, export	type, customer/group, product/group, currency, from qty, amount, %, validity, status	PricingMaintain
Price journal	import CSV (≤ 100k rows), validate, post, reverse	op, before/after, errors	PricingJournalPost
Procedure designer	add step, requirement, exclusion, activate version	step, type, requirement, exclusive	PricingAdmin
Simulation / explain	enter tuple, run, compare versions	per step: access, candidate, winner, amount	PricingView
Rebate agreements	tiers, accruals, close period	customer, basis, tiers, accrued	RebateMaintain

Interaction rules: keyboard row navigation and Ctrl+D duplicate; bulk edit validity and percentage on multi-select; validation on blur and at activate; explain tree navigable by arrow keys; column layouts personalizable (wave 5 shell).

5.26.7 Business rules and edge cases

- **BR-26.1 Resolution algorithm.** For each procedure step in order: (1) evaluate requirement; (2) for the step's condition type iterate accesses by seq_no — PRICE default: 1 customer × variant, 2 customer × product, 3 customer × product group, 4 customer group × product, 5 customer group × product group, 6 customer × all products, 7 all customers × product, 8 all customers × product group, 9 all × all; (3) at each access select ACTIVE records matching keys, currency = document currency (no cross-currency fallback), unit = line unit or convertible via wave 6, valid_from ≤ pricing_date ≤ valid_to, from_quantity ≤ quantity; (4) pick greatest from_quantity, then most dims matched, then lowest net unit price; (5) stop unless winner has search_next, then continue and keep the minimum; (6) no candidate at any access → nothing for discounts, `PRICE_NOT_FOUND` for PRICE unless a manual price exists; (7) exclusion groups evaluated after member steps (BEST lowest net, FIRST lowest step); (8) `exclusive` winner suppresses later discount steps; (9) round per BR-26.3.
- **BR-26.2 Pricing date** = requested ship date by default (AX), configurable to order or delivery date (SAP KOMK-PRSDT).
- **BR-26.3 Rounding.** Net unit = amount × (1−p1/100) × (1−p2/100) at 6 decimals; line net = round_half_even(qty × net unit, minor units); totals are sums of rounded lines; a percentage total discount rounds once on the total.
- **BR-26.4 Multiline grouping** sums quantities of lines with the same group and unit on one document; adding a line re-resolves the group.
- **BR-26.5 Manual override** requires duty PricingOverride, keeps the trace with a reason code, and is excluded from rebate basis when the agreement says `exclude_manual`.
- **BR-26.6 Concurrency.** Activation locks the key hash (`SELECT … FOR UPDATE`); overlap → 409 with the conflicting id.
- **BR-26.7 Retroactive journals** never re-price posted invoices; the event carries `retroactive=true`.
- **BR-26.8 Rebate tiers** evaluate on cumulative period basis; crossing a tier recomputes the whole period and posts the delta on the next invoice or at period end.
- **BR-26.9 State machines.** Record: DRAFT → ACTIVE → EXPIRED (nightly) | WITHDRAWN. Journal: OPEN → VALIDATED → POSTED → REVERSED. No return to DRAFT.
- **BR-26.10 Negative prices** rejected; 100% discount allowed and yields 0.00 with trace.
- **BR-26.11 Vendor rebate basis** = vendor invoice net (default) or received quantity; tier crossing recomputes the period as in BR-26.8; claim amount = Σ accruals − Σ prior claims for the period; a negative claim (over-accrual) creates a w19 vendor invoice instead of a credit note.
- **BR-26.12 Commission resolution** tries (sales group, customer commission group, item commission group) → (sales group, customer group, all) → (sales group, all, item group) → (sales group, all, all); basis REVENUE = line net, MARGIN = line net − cost_snapshot (w20); amount split by rep share %, rounded half-even per rep, residual cent to the first rep; a credit note creates negative transactions referencing the original.
- **BR-26.13 Free goods** inclusive reduces the priced quantity (10 ordered → 9 priced + 1 free); exclusive adds a free line; WHOLE_UNITS truncates; free goods never combine with a 100% discount on the same line and are excluded from rebate and commission basis unless the agreement sets `include_free_goods`.
- **BR-26.14 Retroactive recompute** replays `invoice.posted` for the agreement scope from `valid_from`; delta = recomputed − posted per period, posted as one journal per period; periods closed in w14 post to the first open period with the original period in the reference.
- **BR-26.15 Fund usage** = Σ attributed accruals + claims + deductions; it never exceeds budget unless `allow_overspend`; fund close freezes usage and blocks new attribution with `FUND_CLOSED`.

5.26.8 Acceptance criteria

- **AC-26.1** (FR-26.3) Given C1 in group PG-A with access-2 record 95.00 USD and access-4 record 90.00, when resolved without search_next, then 95.00 and access 4 is shown as not evaluated.
- **AC-26.2** (FR-26.3) Given search_next on the access-2 record, then 90.00 and both accesses appear in the trace.
- **AC-26.3** (FR-26.2) Given breaks 0→10.00, 100→9.50, 500→9.00 EUR, when qty = 500 then 9.00; when 499 then 9.50.
- **AC-26.4** (FR-26.4) Given 10,000 random requests resolved twice on two replicas, then outputs are byte-identical.
- **AC-26.5** (FR-26.5) Given discounts 10%/5% on 100.00 USD × 3, then line net = 256.50.
- **AC-26.6** (FR-26.8) Given exclusion group BEST with LINE_DISC 8% and campaign 12%, then only 12% applies.
- **AC-26.7** (FR-26.9) Given a 100k-line journal with one invalid line, then nothing is written and the error names the line; after correction and post, reversal restores all prior versions.
- **AC-26.8** (FR-26.11) Given a 2% monthly rebate and a 10,000.00 USD invoice, then 200.00 posts Dr 4150 / Cr 2150.
- **AC-26.9** (FR-26.14) Given 50 concurrent callers over the 500 scenarios, then p95 < 20 ms.
- **AC-26.10** (FR-26.7) Given smart rounding ".99 below 10" and a journal raising 7.41 by 5%, then 7.99 is stored.
- **AC-26.11** (FR-26.15) Given a 3% vendor rebate agreement and a 20,000.00 USD vendor invoice posted, then 600.00 posts Dr 2160 / Cr 5100 and the period claim preview shows 600.00.
- **AC-26.12** (FR-26.18) Given reps R1 60% / R2 40% in sales group SG-E and a 5% revenue rule, when a 10,000.00 USD invoice posts, then commission transactions 300.00 and 200.00 post Dr 6200 / Cr 2170 and the credit note of that invoice posts –300.00 / –200.00.
- **AC-26.13** (FR-26.22) Given inclusive free goods 1 per 10 with WHOLE_UNITS and an ordered quantity of 25, then 23 units are priced and 2 are free.
- **AC-26.14** (FR-26.23) Given a 2% agreement activated 2027-03-20 with valid_from 2027-03-01 and 50,000.00 USD invoiced in between, then a 1,000.00 delta posts Dr 4150 / Cr 2150 with retroactive=true.

5.26.9 Test plan and fixtures

Fixtures: `pricing-500` (hand-built from AX trade-agreement and SAP RVAA01 examples, with expected price and trace); `pricing-random-10k` (seeded); `price-journal-100k`; `rebate-tiers-12` (accountant-reviewed).

Test	Type	Target	Pass criterion
Resolver scenarios	unit	resolve	500/500 price and trace
Determinism	property	resolve	identical across 10k runs and replicas
Overlap rejection	unit	activate	100% rejected with 409
Journal atomicity	integration	journals	failure leaves zero rows changed
Rebate accrual = GL	reconciliation	accrual vs ledger	Σ per period = GL 2150 to the cent
Latency	performance	10-line resolve, 50 callers	p95 < 20 ms, p99 < 60 ms
Sales ↔ pricing	contract (Pact)	resolve	green

Reconciliation: Σ rebate_accrual per (agreement, period) = journal postings tagged `source=rebate_accrual`; journal snapshots replay to current record versions.

5.26.10 Deployment and operations

Helm `passage-pricing` (Go, 3 replicas, HPA CPU 70%); CloudNativePG `pricing-db` (3 instances); Atlas migrations from metadata, additive-only. Config: `PRICING_CACHE_TTL=60s`, `PRICING_MAX_LINES=500`, `PRICING_EXPLAIN_MAX_BYTES=65536`. Flags: `pricing.search_next`, `pricing.sap_procedure`, `pricing.commission_hooks`. Temporal jobs: nightly expiry 00:10 entity-local; rebate recompute at period end. SLOs: 99.95% availability, p95 20 ms. Alerts: invalidation lag > 5 s, error rate > 0.5%, journal post > 300 s. Runbook: cache rebuild, journal reversal, procedure version rollback. Rollback: previous image digest via Argo CD.

5.26.11 Migration from AX / SAP

Source	Passage entity	Notes
PriceDiscTable (Relation, AccountCode/Relation, ItemCode/Relation, QuantityAmount, Amount, Percent1/2, Currency, FromDate/ToDate, SearchAgain)	condition_record	Table/GroupId/All → customer/group/all; Relation → type; SearchAgain → search_next
PriceDiscGroup; CustTable price/discount groups	customer_pricing_group, product_pricing_group	one group per AX type
KONH/KONP + A004/A005/A305 (KSCHL, DATAB/DATBI, KBETR, KONWA, KPEIN, KMEIN), KONM scales	condition_record	A-table → key_combination; KONM → breaks; KPEIN normalized to per-unit at 6 decimals
T683S / T682	pricing_procedure, access_sequence_step	map to nearest reference procedure; manual review
KONA (BO01–BO06), VBOX; TAMRebateAgreement	rebate_agreement, opening accrual to 2150	

Validation: 1,000 sampled customer/product pairs resolve to the legacy price (AX PriceDisc::findPrice export, SAP VK13 export) to the cent. Not migrated: records expired > 24 months (archived), commission history, pricing logs, posted PriceDiscAdmTrans (lakehouse).

5.26.12 Dependencies, risks, open questions

- Depends on: w6 units, w7 customers/groups, w8 products/variants/attributes, w11 journal, w12 currencies, w25 (phase order).
- Risks: KPEIN/per-unit mismatch on import → normalize at 6 decimals; cache stampede after a 100k journal → warm cache inside the journal transaction.
- Open questions: (1) Cross-currency fallback? Default no; decide by wave 27. (2) Margin-based commissions before Phase 8? Default no; decide by wave 29.

5.26.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
price_search_mode	legal entity	enum	FIRST_MATCH	FIRST_MATCH, SEARCH_NEXT	Default for `search_next` on new records (BR-26.1 step 5)	AR parameters → Prices → "Find next" / PriceDiscParameters	V/07 access sequence "Exclusive" flag

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
pricing_date_basis	legal entity	enum	SHIP_DATE	SHIP_DATE, ORDER_DATE, DELIVERY_DATE	Pricing date per BR-26.2	Fixed to requested ship date	VOV8 pricing date default (KOMK-PRSDT)
total_discount_basis	legal entity	enum	AFTER_LINE_DISC	AFTER_LINE_DISC, GROSS	Base for TOTAL_DISC	AR parameters → Prices → "Total discount calculation"	RVAA01 header condition HI01/RA01 step base
smart_rounding_bands	legal entity	table	0–10 → .99; 10–100 → .95; ≥ 100 → whole	ordered bands	FR-26.7 journal rounding	PriceDiscRounding / AR parameters → Prices → Smart rounding	DIFF condition, T001R rounding rules
reference_procedure	legal entity × sales channel	enum	AX_STANDARD	AX_STANDARD, SAP_RVAA01, custom	Procedure determination (FR-26.1)	n/a (PriceDisc class order fixed)	OVKK procedure determination (T683V)
cross_currency_fallback	tenant	bool	false	true/false	Allow records in another currency converted at w12 rate (open question 1)	Trade agreement "Currency" strict	KONP-KONWA conversion in pricing
cache_ttl_seconds	tenant	int	60	10–600	Record cache snapshot lifetime	n/a	n/a (buffering)
max_lines_per_resolve	tenant	int	500	1–2,000	Request size limit	n/a	n/a
explain_max_bytes	tenant	int	65,536	8,192–262,144	Trace size (FR-26.4)	n/a	Condition analysis screen size
small_order_threshold	legal entity	money	250.00 USD / 200.00 EUR	≥ 0	SURCHARGE below threshold (FR-26.6)	Charges codes (MarkupAutoTable)	AMIW / AMIZ minimum value surcharge
rebate_basis_default	legal entity	enum	NET_AMOUNT	NET_AMOUNT, QUANTITY	Default on new rebate agreements	Rebate management "Calculation basis"	KONA rebate basis (subtotal 7)
rebate_accrual_accounts	legal entity	account pair	4150 / 2150	chart (w11)	FR-26.11 postings	Customer rebate posting (TAM/rebate posting profile)	VKOA ERU / ERB account keys
rebate_exclude_manual_prices	legal entity	bool	false	true/false	BR-26.5	n/a	n/a
retroactive_mode	legal entity	enum	DELTA	DELTA, FULL_REVERSAL	FR-26.23 posting shape	Rebate recalculation	VBOF update billing documents
commission_basis	legal entity	enum	REVENUE	REVENUE, MARGIN	FR-26.18	Commission calculation → Basis (Revenue/Margin)	Condition contract basis
commission_accounts	legal entity	account pair	6200 / 2170	chart (w11)	FR-26.18	Commission posting form	Condition contract account determination
commission_settlement_target	legal entity	enum	AP	AP, PAYROLL, NONE	Settlement export destination	Commission posting (vendor/employee)	Settlement to HR/AP
vendor_rebate_accounts	legal entity	account pair	2160 / 5100	chart (w11)	FR-26.15	Rebate program types → posting	Supplier condition contract account keys
vendor_rebate_claim_mode	legal entity	enum	CREDIT_NOTE	CREDIT_NOTE, PAYMENT_DEDUCTION	FR-26.15 settlement	Rebate claim journal names	Settlement document type
free_goods_calc_rule	legal entity	enum	WHOLE_UNITS	PROPORTIONAL, UNIT_REFERENCE, WHOLE_UNITS	BR-26.13	n/a	OVKF calculation rule 1/2/3 (NA00)
fund_allow_overspend	tenant	bool	false	true/false	BR-26.15	TAM fund "Allow overspend"	n/a
commodity_index_rounding	legal entity	int	6	2–6	Decimals for index × factor	PdsPricingCalc rounding	n/a
campaign_kill_switch_effect	tenant	enum	NEXT_RESOLVE	NEXT_RESOLVE, IMMEDIATE_CACHE_FLUSH	FR-26.10	n/a	VB22 deactivate

5.26.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Trade agreements	list inquiry	type, customer/group, product/group, currency, validity as-of, status	key fields, from qty, amount, %, validity, version, journal	condition_record	Trade agreements (PriceDiscTable list)	VK13 / VK33 condition display
Price list by customer	operational report	customer, product group, as-of date, currency	product, unit, net price per break	resolve (batch)	SSRS Price list (customer) / by item	V/LD price list report
Price explain	drill-down	document line or tuple	step, access, candidates, winner, reason	explain trace store	n/a (PriceDisc::findPrice has none)	VA01 item condition "Analysis"
Price journal log	list inquiry	journal, date, user, status	op, before, after, reversed_by	price_journal lines	PriceDiscAdmTrans posted journals	VK12 change documents (CDHDR)
Rebate accruals	drill-down	agreement, customer, period, status	basis, accrued, adjusted, settled, journal ref	rebate_accrual	Rebate management → Transactions	VBO3 + VBOX rebate index
Commission transactions	list inquiry	rep, sales group, period, customer, invoice	basis, rate, amount, status, journal ref	commission_transaction	Commission transactions (CommissionTrans) / SSRS Commission report	Settlement management document list
Vendor rebate claims	list inquiry	vendor, agreement, period, status	accrued, claimed, approved, settled	vendor_rebate_claim	Vendor rebate inquiries / SSRS Vendor rebate claim	Supplier condition contract settlement list
Broker and royalty claims	list inquiry	contract, vendor, period	basis, rate, accrued, claimed	broker/royalty claims	Broker/royalty claim inquiries	Settlement documents (royalty)
Fund usage	drill-down	fund, period, customer	budget, usage by source, remaining	fund_usage	Fund usage inquiry / SSRS Fund statement	TPM fund report
Promotion effectiveness	KPI	promotion, product, period	baseline, actual, lift %, spend, ROI	promotion_plan actuals	Promotion effectiveness analysis	n/a

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Commodity prices	list inquiry	cost basis type, date range	index value, source, version, affected agreements	commodity_index	Commodity price inquiries	n/a
Expiring agreements	KPI	days ahead (30)	count, value of affected open lines	condition_record	Pricing workspace tile	VK13 validity filter

5.26.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Customer price list	on demand / price list job	header: customer, currency, as-of date; lines: product, unit, break qty, net price; footer: validity note	PDF, CSV, email	none (report)	none; keep 24 months for dispute support
Rebate agreement statement	period close, on demand	header: agreement, customer, period; lines: invoice, basis, tier, accrued; footer: Σ accrued, Σ settled, open	PDF, email, portal (w30)	none	retain 7 years with accounting records
Commission statement	period close per rep	header: rep, sales group, period; lines: invoice, customer, basis, rate, amount; footer: total, settlement reference	PDF, email, CSV export to payroll/AP	none	7 years (payroll support)
Vendor rebate claim	claim generation	header: vendor, agreement, period, claim number; lines: accruals; footer: claim amount, settlement mode	PDF, email, EDI hook (w35)	`RC–` per legal entity	7 years; forms the basis of the AP credit note
Broker / royalty claim statement	claim generation	header: contract, vendor, period; lines: invoice, basis, rate, amount	PDF, email	`BC–` / `RY–`	7 years
Price journal audit extract	post / reverse	journal header, lines before/after	CSV, JSON	none	audit log retention 10 years (w48)

5.26.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Record expiry	00:10 entity-local daily	legal_entity	status transition by (id, version)	< 60 s per 1M records	retry 3×; skipped entity alerted	expiry lag > 1 day
Rebate period recompute	period end + 1 day 02:00; on tier crossing	agreement	journal Idempotency-Key = (agreement, period, run_seq)	< 10 min per 10k agreements	compensate partial journals; rerun	run > 30 min; Σ accrual ≠ GL 2150
Retroactive accrual recompute	event-driven on activation; nightly catch-up 02:30	agreement	checkpoint per invoice id processed	< 5 min per 100k invoices	resume from checkpoint	pending retro > 24 h
Commission calculation	on `invoice.posted`; nightly reconcile 03:00	legal_entity	key = invoice_line_id	< 50 ms per invoice	DLQ with replay	missing commission for posted invoice > 1 h
Vendor rebate cumulate / claim generation	nightly 03:30; claims at period end	agreement	key = (agreement, period)	< 5 min	rerun idempotent	claims not generated 2 days after period end
Promotion actuals update	nightly 04:00	promotion	watermark on invoice posting time	< 10 min	resume from watermark	lag > 1 day
Commodity price calculation run	per calculation schedule (weekly default)	cost basis type	run id; journal post idempotent	< 2 min per 10k agreements	journal stays VALIDATED for review	index older than schedule
Fund budget recalculation	nightly 04:30; fund close manual	fund	key = (fund, period)	< 1 min	rerun	usage > budget when overspend disabled
Cache warm after journal post	inside journal transaction	legal_entity	n/a	< 30 s	journal still POSTED; cache rebuilt lazily	invalidation lag > 5 s

5.26.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
PricingView	GET condition-records, simulate, explain; Pricing workspace, Simulation screen	none	Sales rep, Sales manager, Pricing analyst, Auditor
PricingMaintain	POST condition-records, :activate, :withdraw; Trade agreements grid; campaigns	with PricingJournalPost on the same journal (maker/checker when `journal_four_eyes`)	Pricing analyst
PricingJournalPost	POST journals, :post, :reverse; Price journal screen	with PricingMaintain under four-eyes; with CommodityPriceRun	Pricing manager
PricingAdmin	POST procedures, versions:activate; Procedure designer	with PricingJournalPost	Pricing manager, Application admin
PricingOverride	manual price on order lines (w27) with reason	with CreditRelease (w27)	Sales manager
RebateMaintain	POST rebates/agreements; Rebate agreements screen; accrual inquiry	with RebateSettle (w30)	Pricing analyst, AR manager
RebateView	GET rebates/accruals, claims	none	Controller, Auditor
VendorRebateClaimApprove	approve vendor rebate claims; claim job	with APPaymentPost (w19), with RebateMaintain	Purchasing manager
CommissionMaintain	commission rules, sales groups, rep assignment	with payroll settlement (w43) and APInvoicePost (w19)	Sales operations
CommissionView	commission inquiry, statements	none	Sales rep (own), Sales manager
FundMaintain	funds, promotion plans, fund close	with RebateSettle (w30)	Trade marketing manager
CommodityPriceRun	commodity indexes, calculation run	with PricingJournalPost	Pricing manager

5.26.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
PRC-0001	error	Price not found for product {product} on {date}	No PRICE candidate at any access and no manual price	Create an agreement or enter a manual price (PricingOverride)	BR-26.1 step 6
PRC-0002	error	Overlapping active record {id} for the same key, currency and unit	Validity overlap on activation	Adjust validity or withdraw the conflicting record	FR-26.2, BR-26.6
PRC-0003	error	Negative net price is not allowed	Amount or discounts produce < 0	Correct the amount or percentages	BR-26.10

Code	Severity	User text	Cause	Remediation	Raised by
PRC-0004	error	Quantity breaks must be strictly increasing	Duplicate or descending from_quantity	Reorder breaks	FR-26.2
PRC-0005	error	Journal has {n} validation errors; nothing was posted	Validation failed on post	Fix listed lines and re-validate	FR-26.9
PRC-0006	warning	Retroactive activation: {n} posted invoices will be re-accrued	valid_from before today on activation	Confirm or move valid_from	FR-26.23
PRC-0007	error	Unit {unit} is not convertible to the record unit {unit2}	No w6 conversion factor	Add conversion or record in line unit	BR-26.1 step 3
PRC-0008	error	Procedure version {v} is active and immutable	Edit on an activated version	Create a new version	FR-26.1
PRC-0009	error	Request exceeds {max} lines	max_lines_per_resolve	Split the request	FR-26.14
PRC-0010	info	search_next continued to access {n}; lower net kept	search_next in effect	none	FR-26.3
PRC-0011	error	Fund {fund} is closed	Attribution after close	Reopen (FundMaintain) or choose another fund	BR-26.15
PRC-0012	error	Fund {fund} budget exceeded by {amount}	Usage > budget and overspend disabled	Increase budget or enable overspend	BR-26.15
PRC-0013	error	Commission rule not found for sales group {sg}	No rule at any level	Add a rule or set a default	BR-26.12
PRC-0014	warning	Free goods cannot combine with a 100% discount on line {n}	Both apply	Remove one	BR-26.13
PRC-0015	error	Vendor rebate claim for {agreement} {period} already processed	Duplicate claim	View the existing claim	FR-26.15
PRC-0016	warning	Commodity index for {basis} is older than {days} days	Stale index	Load the index before the run	FR-26.21
PRC-0017	error	Minimum royalty shortfall {amount} posted at period close	Royalty below minimum	none (informational posting)	FR-26.17

5.26.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
resolve (10 lines)	2M per tenant per day; peak 5,000/min	6 / 20 / 60 ms	500 req/s per replica	none (stateless; traces 64 KB × sampled 10%)	in-memory snapshot per (entity, date)	HPA to 9 replicas; explain sampling drops to 1%; p95 ≤ 40 ms
Condition record activate	20k per day	20 / 80 / 200 ms	100/s	1M records × 1 KB = 1 GB	composite btree + partial per key combination; partition by entity	queue activations; overlap check still serialized
Price journal post (100k lines)	5 per day	— / 120 s / 180 s	1k lines/s	50 GB journals (snapshots)	partition journal lines by journal id	single journal at a time per entity; others wait VALIDATED
Rebate accrual on invoice	50k invoices per day	10 / 40 / 100 ms	200/s	20M accrual rows × 300 B = 6 GB	(agreement, period) btree; partition by period	accrual lag alert at 5 min; invoices unaffected
Commission calculation	50k invoices per day	10 / 40 / 100 ms	200/s	5 GB	(rep, period)	same as accrual
Retroactive recompute	1 per day, 100k invoices	— / 5 min / 10 min	500 invoices/s	none	checkpointed	throttled to 50% of a replica
Commodity calculation run	weekly, 10k agreements	— / 2 min / 5 min	—	100 MB	n/a	run queued

5.26.20 Localization and compliance hooks

- Currency rounding to ISO 4217 minor units per currency (w12); JPY and HUF zero-decimal handling in BR-26.3; EUR/USD examples in fixtures.
- Country packs (w48) may inject statutory surcharge conditions (for example packaging or environmental fees surfaced from w28 FR-28.14) as SURCHARGE steps through the procedure extension point `procedure.steps[].country_pack_id`; packs never alter resolution order.
- Rebate, commission, broker and royalty accruals are posted through the w11 journal with tax treatment deferred to settlement (w30 / w19); withholding on royalties (for example DE §50a, IN TDS) is applied by the w19 AP claim, not here.
- Price lists and statements render per customer language (w5 docgen locale) with date and number formats from the entity locale; explain traces are English-only engineering artifacts.
- Privacy: commission transactions reference worker ids (w42) and are subject to the w48 personal-data retention policy (7 years then pseudonymized); rep names never appear in events.
- Retention: condition records and journals 10 years (audit), promotion plans 3 years.
- Extension point: condition type registry accepts country-pack types with a CEL requirement; access sequences per pack are additive and versioned with the procedure.

5.26.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.16.1 #54 — Vendor rebates (PI vendor rebate agreements, R3)	GAP	ADDED as FR-26.15	party_role VENDOR on rebate_agreement; accrual 2160/5100
AX §12.16.1 #55 — Broker contracts (PI brokers, R3)	GAP	ADDED as FR-26.16	broker_contract, claim to w19 AP
AX §12.16.1 #56 — Royalty agreements (PI, R3)	GAP	ADDED as FR-26.17	royalty_agreement with minimum royalty
AX §12.16.2 #58 — Vendor rebate inquiries	GAP	ADDED as FR-26.15	§5.26.14 Vendor rebate claims
AX §12.16.2 #59 — Broker/royalty claim inquiries	GAP	ADDED as FR-26.16 / FR-26.17	§5.26.14 Broker and royalty claims
AX §12.16.2 #60 — SSRS Vendor rebate reports	GAP	ADDED as FR-26.15	§5.26.15 Vendor rebate claim document
AX §12.16.3 #61 — Vendor rebates → Cumulate / Approve / Process claims	GAP	ADDED as FR-26.15	§5.26.16 cumulate and claim jobs; duty VendorRebateClaimApprove

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.16.3 #62 — Process broker claims / Process royalty claims	GAP	ADDED as FR-26.16 / FR-26.17	claim jobs at period end
AX §12.16.4 #63 — Vendor rebate setup (agreements, program types, rebate groups, claim journal names)	GAP	ADDED as FR-26.15	vendor_rebate_accounts, vendor_rebate_claim_mode (§5.26.13)
AX §12.16.4 #64 — Broker contract setup / Royalty agreement setup / Claim journal names	GAP	ADDED as FR-26.16 / FR-26.17	`BC–` / `RY–` sequences
AX §12.23.1 #65 — Vendor rebates / Brokers / Royalties (PI links)	GAP	ADDED as FR-26.15–26.17	same entities
AX §12.23.3 #69 — Vendor rebates → Process (PI)	GAP	ADDED as FR-26.15	claim job
AX §12.23.4 #71 — Vendor rebates / Brokers / Royalties setup (PI)	GAP	ADDED as FR-26.15–26.17	§5.26.13
AX §12.36.1 #102 — Deal → Vendor deals (purchase rebates)	GAP	ADDED as FR-26.15	D365 vendor deal = VENDOR agreement
AX §12.36.1 #103 — Deal → Price-based deals (royalty-type)	GAP	ADDED as FR-26.17	
AX §12.36.2 #104 — Vendor deal inquiries	GAP	ADDED as FR-26.15	§5.26.14
AX §12.36.2 #105 — SSRS Vendor rebate claim	GAP	ADDED as FR-26.15	§5.26.15
AX §12.36.3 #106 — Vendor rebate claims processing	GAP	ADDED as FR-26.15	§5.26.16
AX §12.36.4 #107 — Vendor deal setup (claim via AP)	GAP	ADDED as FR-26.15	settlement through w19
AX §12.3.1 #20 — Commission transactions (CommissionTrans)	PARTIAL-26	ADDED as FR-26.18	commission_transaction posted 6200/2170
AX §12.3.2 #21 — Commission inquiry	PARTIAL-26	ADDED as FR-26.18	§5.26.14 Commission transactions
AX §12.3.2 #23 — SSRS Commission report / Commission transactions	PARTIAL-26	ADDED as FR-26.18	§5.26.15 Commission statement
AX §12.3.3 #25 — Commission → Calculate/post commissions (via invoice)	PARTIAL-26	ADDED as FR-26.18	event-driven on invoice.posted
AX §12.3.4 #29 — Commission calculation / sales groups / customer groups / item groups / posting	PARTIAL-26	ADDED as FR-26.18	commission_basis, commission_accounts (§5.26.13)
AX §12.3.4 #33 — Sales districts / Sales groups (sales reps)	PARTIAL-26	ADDED as FR-26.18	sales_group; sales district stays a w7 customer attribute
AX §12.15.1 #142 — Released product → Commodity pricing (PdsCommodity*)	PARTIAL-26	ADDED as FR-26.21	
AX §12.15.4 #150 — Commodity pricing setup (cost basis types, pricing calculations, price margin templates, commodity codes)	PARTIAL-26	ADDED as FR-26.21	commodity codes reuse w8 product attributes
AX §12.17.1 #169 — Sales → Commissions (link)	PARTIAL-26	ADDED as FR-26.18	
AX §12.17.2 #170 — SSRS Commission report (link)	PARTIAL-26	ADDED as FR-26.18	
AX §12.17.4 #172 — Commissions setup (calculation, sales groups, customer groups, item groups, posting)	PARTIAL-26	ADDED as FR-26.18	
AX §12.18.1 #174 — Merchandising events (promotion planning)	PARTIAL-26	ADDED as FR-26.20	baseline 13 weeks, lift %
AX §12.18.1 #175 — Funds (TAMFund)	PARTIAL-26	ADDED as FR-26.19	
AX §12.18.2 #177 — Fund usage inquiry	PARTIAL-26	ADDED as FR-26.19	§5.26.14 Fund usage
AX §12.18.2 #179 — Promotion effectiveness analysis (lift)	PARTIAL-26	ADDED as FR-26.20	§5.26.14 Promotion effectiveness
AX §12.18.2 #180 — SSRS Fund statement / Fund usage	PARTIAL-26	ADDED as FR-26.19	Fund usage inquiry export (PDF/CSV)
AX §12.18.3 #183 — Funds → Fund budget recalculation / Fund close	PARTIAL-26	ADDED as FR-26.19	§5.26.16 job
AX §12.18.3 #184 — Promotion → Update promotion planning actuals	PARTIAL-26	ADDED as FR-26.20	nightly actuals job
AX §12.18.4 #185 — Funds setup (fund types, budget periods)	PARTIAL-26	ADDED as FR-26.19	fund_allow_overspend
AX §12.18.4 #186 — Merchandising event categories / Event types	PARTIAL-26	ADDED as FR-26.20	
AX §12.23.1 #199 — Commodity pricing (PI link)	PARTIAL-26	ADDED as FR-26.21	
AX §12.23.2 #204 — Commodity price inquiries	PARTIAL-26	ADDED as FR-26.21	§5.26.14 Commodity prices
AX §12.23.3 #207 — Commodity pricing → Price calculation run	PARTIAL-26	ADDED as FR-26.21	§5.26.16 job → price_journal
AX §12.23.4 #209 — Commodity pricing setup (cost basis types, pricing calculations, price margin templates)	PARTIAL-26	ADDED as FR-26.21	
SAP §13.19 #217 — VBN1/VBN2/VBN3 free goods condition records	PARTIAL-26	ADDED as FR-26.22	inclusive/exclusive
SAP §13.19 #227 — OVKF free goods (NA00, NA0001, KOTN)	PARTIAL-26	ADDED as FR-26.22	free_goods_calc_rule
SAP §13.20 #228 — VBOF update billing documents (retroactive accrual)	PARTIAL-26	ADDED as FR-26.23	
SAP §13.20 #230 — Commission contracts (settlement management)	PARTIAL-26	ADDED as FR-26.18	
SAP §13.20 #232 — Royalty / licensing payments (settlement management)	GAP (propose 26)	ADDED as FR-26.17	
SAP §13.20 #233 — VBOF batch retroactive accrual update	PARTIAL-26	ADDED as FR-26.23	§5.26.16 retroactive recompute job
SAP §13.20 #234 — Delta accruals / delta settlement	PARTIAL-26	ADDED as FR-26.23	delta against prior settlements (w30 BR-30.8 reads the delta)
SAP §13.20 #236 — Commission settlement integration (HR/AP)	PARTIAL-26	ADDED as FR-26.18	commission_settlement_target; payout executes in w19 / w43
SAP §13.20 #229 — Supplier condition contracts (purchasing rebates, chargebacks)	PARTIAL-19 (note proposes w26 engine)	ADDED as FR-26.15 (accrual); chargeback claims DEFERRED to wave 30 (FR-30.22)	AP settlement remains wave 19

5.27 Wave 27: Sales Orders, Quotations, ATP & Credit

5.27.1 Scope and business context

Wave 27 ships the `sales` service: quotations with conversion, sales agreements, order types (journal, order, subscription, returned — completed in wave 30), delivery date control against cumulative ATP/CTP, credit management with limits, aging rules, holds and a Temporal release workflow, versioned change management, line-level status and a multi-channel intake API. A sales rep can quote, convert, promise a date and see holds; a credit manager releases orders. Out of scope: picking and shipping (wave 28), invoicing (wave 29), portal UI (wave 30), EDI intake (wave 35). Added from the coverage registers (§5.27.21): supplementary items,

listing/exclusion, product allocation quotas, backorder processing strategies, customer consignment, sales-order special stock, scheduling-agreement releases with cumulative quantities, product proposals, one-time customers and a cost-estimate hook.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Promise dates go stale after a planning run: AX ATP is recomputed only when the "Available ship and receipt dates" dialog opens; SAP VBEP dates change only on VA02 re-check or the nightly V_V2 rescheduling batch	ATP computed on demand inside the form; supply changes do not re-promise open lines	atp_bucket projection from inventory/planning events (lag < 30 s); re-promise on `order.changed` and supply cancellation; backward propagation (BR-27.1)	Promise accuracy ~70% → ≥ 95%; re-promise latency nightly → < 30 s
Credit holds are opaque: SalesOrderHold shows a code; OVA8 writes only VBAK-CMGST and the VKM1 list	Rules evaluated inside posting classes; only the final status persists	`credit-check.evaluated` stores every rule, value and threshold; release queue shows the failing rule	Analyst lookup per release 5–8 min (FD32/CustTable) → < 30 s; 2-business-day SLA by timer
Order entry takes 12+ clicks across SalesTable → SalesLine → InventDim → ATP dialog (VA01 overview → item → schedule lines)	Price, ATP and dims live on separate tabs	One grid with inline price and ATP within 300 ms of product + quantity	Clicks per priced, promised line 12 → 2
Channel orders are re-keyed or loaded by nightly AIF/IDoc batches with no idempotency (duplicate ORDERS05 → duplicate VBAK)	Batch integration without a business-key dedup	Intake API with (channel, external_ref) idempotency at 500 orders/min, p99 < 500 ms; `order.confirmed` events for any consumer	Duplicates per 10k channel orders ~5 → 0; latency hours → < 500 ms
Changing a confirmed order loses history (SalesLine overwritten; CDHDR/CDPOS unreadable)	No versioned document	Versions with field-level diff and selective re-pricing/ATP/credit	"What changed and who": 15 min → < 5 s

5.27.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Quotations	SalesQuotationTable/Line, SalesQuotationStatus	VA21/VA22, VBAK VBTYP B; VA11 inquiries	sales_quotation, …:convert	full
Sales orders	SalesTable/SalesLine, SalesStatus, SalesLineStatus, SalesType	VA01/VA02, VBAK/VBAP/VBEP, item categories TAN/TAS	sales_order, sales_order_line, order_schedule_line	full
Sales agreements	SalesAgreementHeader/AgreementLine, release orders	VA31 scheduling agreements, VA41 contracts	sales_agreement with release tracking	full
ATP / delivery date control	InventDim ATP (DlvDateControlType ATP/CTP), ATP time fence, SalesLine.ConfirmedDlv	OVZ9 checking rule, VBEP confirmed lines, CO09	promising module, atp_bucket	full (CTP partial)
Credit limit	CustTable.CreditMax/MandatoryCreditLimit, CustParameters.CreditLimitType, D365 credit management	FD32/UKM_BP (KNKK), OVA8, VKM1 release	credit_profile, CREDIT hold, Temporal release	full
Order holds	SalesOrderHold / hold codes	VBAK-LIFSK delivery block, FAKSK billing block	order_hold (CREDIT, FRAUD, MANUAL, PRICE_REVIEW)	full
Change management	D365 order change approval	VA02 + CDHDR/CDPOS	sales_order.version + change_request	improved
Configured products	PCConfiguration on SalesLine	Variant configuration, KMAT	configuration_id (w8) + production link (w21)	partial
Backorders / alternative items	SalesBackorder; InventTable.AltItemId	V_RA; VB11 substitution	backorder workspace; alternative in ATP fallback	full
Multi-channel intake	AIF SalesSalesOrderService	IDoc ORDERS05, BAPI_SALESORDER_CREATEFROMDAT2	POST /sales/orders with idempotency	improved
Supplementary items	PriceDiscSuppItem, supplementary items prompt on SalesLine	Free goods NA00 (w26), item proposals	supplementary_item_rule + free-goods lines	full
Listing / exclusion	— (customer-specific item blocking via InventTable stopped flags)	VB01/VB02/VB03, OV04 listing/exclusion procedures (T684)	product_listing (ALLOW / DENY per customer, group)	full
Product allocation	—	OVZJ/OVZI product allocation (T190, KOTA), aATP product allocation	allocation_quota per (product, customer/region, period)	full
Backorder processing strategies	SalesBackorder, backorder lines by item	V_V2 rescheduling; aATP BOP strategies win/gain/redistribute/fill/lose, release for delivery	backorder_run with strategy and priority	full
Customer consignment	— (consignment is vendor-side in D365)	VA01 KB/KE/KA/KR, MB58, special stock W	consignment_fill_up / issue / pick_up / return orders with owner dimension (w16)	partial (ownership dimension w16)
Sales-order special stock	Reservation against order (InventTrans marking)	Special stock E (sales order stock), MIE1, OVZG requirements class	sales_order_stock reservation segment; valuation in w21/w39	partial
Scheduling agreement releases	SalesAgreement + release orders; AIF DELFOR (w35)	VA31/VA32 with JIT/forecast delivery schedules, cumulative quantities, DELFOR/DELJIT	sales_agreement.schedule_release with cumulative reconciliation	full (EDI mapping w35)
Cross-selling / product proposal	—	VB41 cross-selling, dynamic product proposal	product_proposal rule	full
One-time customer	One-time customer (CustTable.OneTimeCustomer)	CPD account group, document-level address (OBD2)	one_time party role (w7) + order-level name/address	full

5.27.3 Functional requirements

- **FR-27.1 Quotations** — Must. Priced quotations with expiry (default +30 d) and probability, convertible whole or by line to one order preserving explain refs.
- **FR-27.2 Sales orders** — Must. Header (customer, invoice account, currency, delivery address, terms, procedure) and lines (product, variant dims, qty, unit, site/warehouse, requested dates, price fields). Notes: types JOURNAL, ORDER, SUBSCRIPTION, RETURNED; ≤ 2,000 lines; sequence `SO-`.
- **FR-27.3 Delivery date control** — Must. Confirmed ship/receipt dates per line by NONE, SALES_LEAD_TIME, ATP or CTP, applying transport days and the customer receiving calendar.
- **FR-27.4 Cumulative ATP** — Must. Per (product, dims, site, warehouse): on-hand available + scheduled receipts – scheduled issues within the time fence, with backward propagation of negative periods (BR-27.1). Notes: supply = confirmed POs (w17), released/started production orders (w21), transfers, planned orders optional (default excluded); demand = confirmed sales lines, production BOM lines, transfer issues, optional safety stock.
- **FR-27.5 CTP** — Should. When ATP fails, request a capable-to-promise date from scheduling (w23) using BOM/route lead time and finite capacity, creating a planned order link.
- **FR-27.6 Backorders and partials** — Must. Split lines into schedule lines when partial delivery is allowed; backorder list by promised date.

- **FR-27.7 Credit profile and checks** — Must. Per customer per entity: limit, currency, limit type, tolerance, review date, max dunning level; rules in BR-27.3 evaluated at entry, confirmation, release and invoicing per policy.
- **FR-27.8 Holds and release** — Must. Holds block confirmation/release; Temporal release workflow with approvers by duty and a 2-business-day SLA timer.
- **FR-27.9 Change management** — Must. Version on every post-confirmation change with field-level diff; re-run pricing/ATP/credit for affected lines; optional approval per customer.
- **FR-27.10 Sales agreements** — Should. Blanket commitment by quantity or amount; released orders inherit price and decrement commitment atomically.
- **FR-27.11 Configured products** — Should. Accept a configuration id; on confirmation create the production/planned order linkage for make-to-order items.
- **FR-27.12 Intake API** — Must. External orders with idempotency keys at 500 orders/min, p99 < 500 ms, returning id and holds.
- **FR-27.13 Alternative items** — Could. Offer the configured alternative with its ATP when the ordered product fails inside the requested date.
- **FR-27.14 Supplementary items** — Should. The system shall, at line commit, evaluate supplementary item rules (product or product group × customer or group; MANDATORY or OPTIONAL; quantity per base quantity; free or priced) and w26 FREE_GOODS proposals, add mandatory lines automatically and prompt for optional ones; supplementary lines reference the parent line and are canceled with it. Notes: AX PriceDiscSuppItem semantics; free lines carry unit_price 0.00 and `supplementary_of`.
- **FR-27.15 Listing and exclusion** — Should. The system shall validate every order line against product listings (ALLOW lists) and exclusions (DENY lists) per customer or customer group with validity, rejecting a denied line with 422 `PRODUCT_NOT_LISTED` and an override only with duty SalesOrderApprove. Notes: SAP VB01 listing/exclusion and OV04 procedures map to one `product_listing` entity.
- **FR-27.16 Product allocation** — Could. The system shall hold allocation quotas per (product or group, customer or region, period) consumed by ATP confirmation; when the quota is exhausted the line is confirmed at the next period with source `ALLOCATION`. Notes: SAP OVZJ/OVZI; quota consumption is reversible on cancellation; quota inquiry per period.
- **FR-27.17 Backorder processing strategies** — Should. The system shall run a backorder processing run per (product, site) that re-confirms open lines by strategy WIN (protect confirmed), GAIN (improve), REDISTRIBUTE (by priority), FILL (first-come) or LOSE (deconfirm low priority) with a priority from customer priority class, requested date and order value, and a preview before commit. Notes: replaces SAP V_V2 and aATP BOP; AX backorder workspace is the UI; run results emit `atp.promised` per changed line.
- **FR-27.18 Customer consignment** — Could. The system shall support consignment fill-up (ship to customer-owned location, inventory stays owned by the entity with owner dimension CUSTOMER_LOCATION from w16), consignment issue (customer consumes; invoice created), pick-up (return to own warehouse) and consignment return orders, with a stock-at-customer inquiry (SAP MB58). Notes: depends on the w16 ownership/location dimension; packing slips for fill-up post no COGS (BR-28.1 extension).
- **FR-27.19 Sales-order special stock** — Should. The system shall, for make-to-order lines, segregate supply and demand per order (`sales_order_stock = true` on the w16 reservation segment and on w21 production orders) so that ATP for that line consults only its own supply and the stock cannot be consumed by other orders; valuation of the segregated stock (valuated/non-valuated) remains with w21/w39. Notes: SAP special stock E, requirements class OVZG; AX "marking" of InventTrans.
- **FR-27.20 Scheduling agreement releases** — Should. The system shall accept forecast and JIT delivery schedules against a sales agreement (release number, cumulative received quantity, schedule lines by date, horizon), reconcile cumulative quantities against shipped quantities (BR-27.12), replace the previous release atomically and generate or adjust order schedule lines. Notes: SAP VA31/VA32 with DELFOR/DELJIT; inbound EDI 830/862 mapping arrives in w35 through this API; cumulative-quantity reset at fiscal year change per customer.
- **FR-27.21 Cross-selling and product proposals** — Could. The system shall hold product proposal rules (ordered product → proposed products with reason CROSS_SELL, UP_SELL, ACCESSORY, from w8 related products) and show proposals at line commit without creating lines unless accepted. Notes: SAP VB41, dynamic product proposal; acceptance rate recorded for analytics.
- **FR-27.22 One-time customers** — Should. The system shall allow an order and its invoice for a one-time party (w7 role ONE_TIME, one shared account per entity) with document-level name, address and tax id captured on the order header and carried to the invoice and open item. Notes: SAP CPD account group; AX one-time customer flag; credit check uses `one_time_order_limit` (default 5,000 USD / 4,500 EUR).
- **FR-27.23 Sales order cost estimate hook** — Could. The system shall expose `POST /sales/orders/{id}/lines/{l}:estimate-cost` that calls the w20 cost snapshot (stocked) or the w39 simulation API (configured/MTO) and stores `estimated_cost` and margin on the line for quotation margin display. Notes: SAP CK51N; full sales-order costing is delivered by wave 39.

5.27.4 Data model

sales_order (SalesTable / VBAK)

Field	Type	Required	Notes
id, order_number	uuid, text	Y	SO- sequence, unique per entity
tenant_id, legal_entity_id	uuid	Y	RLS
customer_id, invoice_account_id	uuid	Y	FK party (w7)
order_type, status	enum	Y	see BR-27.5
currency, pricing_procedure_version_id	char(3), uuid	Y	
delivery_address_id, delivery_terms, mode_of_delivery	uuid, text, text	Y	
payment_terms_id, cash_discount_id	uuid	Y/N	w7 terms
requested_ship_date, requested_receipt_date	date	Y	
credit_check_result	jsonb	N	last evaluation
version, etag	int, text	Y	optimistic concurrency
net/discount/tax/gross totals	numeric(19,4)	Y	derived, stored
channel, external_ref, idempotency_key	text	N	unique (tenant, channel, external_ref)

sales_order_line (SalesLine / VBAP): id, order_id, line_no, product_id, variant_dims, quantity, unit, site_id, warehouse_id, unit_price, disc_pct_1/2, line_net, tax_group, delivery_date_control, requested/confirmed ship and receipt dates, line_status (OPEN | RESERVED | PICKED | DELIVERED | INVOICED | CANCELLED), delivered_qty, invoiced_qty, remaining_qty, configuration_id, production_order_id, agreement_line_id, price_explain_ref, multiline_group, reservation_id.
order_schedule_line (VBEP): line_id, seq, qty, confirmed_ship_date, source (atp_bucket id | planned order id).

credit_profile: customer_id, legal_entity_id, credit_limit numeric(19,4), currency, limit_type (NONE | BALANCE | BALANCE_PACKING_SLIP | BALANCE_ALL), tolerance_pct (5), tolerance_amount (1,000.00), oldest_item_days_max (30), max_dunning_level (3), review_date, risk_group, mandatory_check, exposure jsonb (open_orders, delivered_not_invoiced, open_ar, unapplied_cash, updated_at).

order_hold: id, order_id, hold_type, reason_code, status (ACTIVE | RELEASED | CLEARED), applied_by, released_by, workflow_run_id, applied_at, released_at.
atp_bucket (projection): product_id, dims hash, site_id, warehouse_id, bucket_date, supply_qty, demand_qty, period_atp, cumulative_atp, source_refs, computed_at.

Invariants: Σ schedule_line.qty = line.quantity; delivered_qty ≤ quantity; remaining_qty = quantity − delivered_qty (0 after remainder cancellation); order status derives from line statuses; an order with an ACTIVE hold is never released; cumulative_atp[n] = min over k ≥ n of Σ period_atp[0..k]; every ATP-confirmed line references an atp_bucket as of confirmation. Indexing: lines (order_id, line_no); partial index on open line statuses for demand; atp_bucket PK (product, dims, site, warehouse, bucket_date); partition orders by legal_entity.

5.27.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/sales/quotations, …/{id}:send, :convert	Quotation lifecycle	Idempotency-Key	SalesQuotationMaintain
POST	/api/v1/sales/orders	Create (UI or channel)	Idempotency-Key + external_ref	SalesOrderMaintain / ChannelIntake
PATCH	/api/v1/sales/orders/{id}	Change, creates version	If-Match etag	SalesOrderMaintain
POST	/api/v1/sales/orders/{id}:confirm, :cancel; /lines/{l}:cancel-remainder	Price, ATP, credit, confirm; cancellation	Idempotency-Key	SalesOrderConfirm
POST/GET	/api/v1/sales/atp:check; /atp/buckets?product=&site=	Simulate ATP; bucket view (CO09)	stateless	SalesOrderView
POST	/api/v1/sales/orders/{id}/holds, …/holds/{h}:release	Apply / release hold	Idempotency-Key	SalesOrderHold / CreditRelease
GET/PUT	/api/v1/sales/credit-profiles/{customerId}	Credit profile	etag	CreditMaintain
POST	/api/v1/sales/agreements, …/{id}:release-order	Agreements	Idempotency-Key	SalesAgreementMaintain

Validation: confirm rejects unpriced lines (unless manual), quantity ≤ 0, customer status BLOCKED (w7); closed period on order date is a warning only (period control applies at posting).

Event	Trigger	Payload fields	Consumers
dev.passage.sales.order.confirmed.v1	confirm	order_id, customer_id, currency, lines[{line_id, product, dims, qty, site, warehouse, confirmed_ship_date}], totals, version	fulfillment, planning (w22), credit, analytics
dev.passage.sales.order.changed.v1 / cancelled.v1	PATCH / cancel	order_id, version, diff[] or line_ids, reason	fulfillment, planning, portal
dev.passage.sales.atp.promised.v1	confirm with ATP/CTP	line_id, requested, confirmed, bucket_date, source	planning, portal
passage.sales.order-hold.applied.v1 / released.v1	hold lifecycle	order_id, hold_type, reason, actor, workflow_run_id	fulfillment, collections, portal
passage.sales.credit-check.evaluated.v1	any check	customer_id, order_id, rules[{id, result, value, threshold}], outcome	audit, collections
dev.passage.sales.quotation.converted.v1	convert	quotation_id, order_id, lines	analytics, CRM

5.27.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Sales workspace	tiles: on hold, backorders, quotes expiring 7 d, intake errors	counts	SalesOrderView
Quotation detail	add line, price, send PDF, convert, mark lost	customer, lines, expiry, probability, totals	SalesQuotationMaintain
Order entry (header + grid)	add line with ATP badge, price, confirm, hold, cancel, copy	product, qty, unit, site/wh, requested/confirmed dates, price, disc, net, status	SalesOrderMaintain
ATP explorer	pick product/site, view buckets, simulate	bucket date, supply, demand, period ATP, cumulative ATP, sources	SalesOrderView
Credit release queue	view exposure, release/reject with comment	customer, order, amount, exposure vs limit, rule failed, age	CreditRelease
Backorders / change requests	re-promise, split, cancel remainder; approve diff	line, promised date, qty; field, old, new	SalesOrderMaintain / SalesOrderApprove

Interaction rules: Tab/Enter across the grid; product + quantity triggers inline price and ATP within 300 ms; validation at line commit and confirm; bulk re-promise on multi-select; default site/warehouse and column sets personalizable; changes to a confirmed order show the diff before save.

5.27.7 Business rules and edge cases

- **BR-27.1 Cumulative ATP worked example.** FG-100, site S1, warehouse W11, time fence 30 d, on-hand 100 on 2027-03-01. Supply: PO-2001 50 on 03-04; PRD-3001 80 on 03-08. Existing demand: SO-1001 60 on 03-02, SO-1002 40 on 03-05, SO-1003 70 on 03-09. Buckets open at each receipt: bucket 0 (03-01) supply 100, demand 60, period ATP 40; bucket 1 (03-04) 50 − 40 = 10; bucket 2 (03-08) 80 − 70 = 10. Cumulative ATP: 40, 50, 60. A new line of 45 requested 03-01: bucket 0 cumulative 40 < 45; first bucket with cumulative ≥ 45 is bucket 1 → confirmed ship date 2027-03-04 for complete delivery, or schedule lines 40 on 03-01 and 5 on 03-04 if partial delivery is allowed. After confirming 45 on 03-04, bucket 1 period ATP = −35; negative periods propagate backward, so cumulative becomes 5, 5, 15. Had SO-1002 been 60, bucket 1 period ATP −10 would already have reduced bucket 0 cumulative to 30.
- **BR-27.2 Dates.** confirmed_receipt = confirmed_ship + transport days (mode of delivery) rolled to the customer's next receiving day; past requested dates become today with a warning; dates beyond the fence are promised at fence end with source `TIME_FENCE`.
- **BR-27.3 Credit check rules** (any FAIL → CREDIT hold; WARN displayed only): CR-1 exposure + order amount > limit × (1 + tolerance_pct/100) and > limit + tolerance_amount → FAIL, where exposure by limit_type is BALANCE = open AR + unapplied; BALANCE_PACKING_SLIP adds delivered-not-invoiced; BALANCE_ALL adds confirmed open orders; CR-2 oldest open item overdue > oldest_item_days_max → FAIL; CR-3 dunning level ≥ max_dunning_level → FAIL; CR-4 review_date passed or profile missing while mandatory_check → FAIL; CR-5 overdue > 25% of open AR → WARN; CR-6 order > 50,000 USD / 45,000 EUR for a customer younger than 90 days → FAIL. Amounts convert to the profile currency at the wave 12 spot rate on the check date.
- **BR-27.4 Hold precedence.** CREDIT and FRAUD block confirmation and release; MANUAL and PRICE_REVIEW block release only; a CREDIT release is valid 7 days; an amount increase > 5% re-triggers CR-1.
- **BR-27.5 State machines.** Order: DRAFT → OPEN → CONFIRMED → PARTIALLY_SHIPPED → SHIPPED → INVOICED → CLOSED; CANCELLED from DRAFT/OPEN/CONFIRMED. Line: OPEN → RESERVED → PICKED → DELIVERED → INVOICED; CANCELLED from OPEN/RESERVED. Quotation: DRAFT → SENT → CONVERTED | LOST | EXPIRED | CANCELLED.
- **BR-27.6 Concurrency.** PATCH requires If-Match; concurrent confirm and change → later call 409; ATP consumption serialized per (product, dims, site, warehouse) by Postgres advisory lock.
- **BR-27.7 Reservation.** ATP confirmation creates a wave 16 reservation when `reserve_on_confirm` (default true for ORDER); JOURNAL orders never reserve or consume ATP.
- **BR-27.8 Quantities** at 6 decimals; `whole_units` products reject fractions; unit conversion via wave 6 factors, half-even.

- **BR-27.9 Agreements** decrement commitment atomically and may not exceed it beyond overdelivery_pct (default 0); subscription/agreement lines re-price only when the agreement price changes.
- **BR-27.10 Intake duplicates.** Same (channel, external_ref) within 30 days returns the original id with 200; same key with a different payload → 409.
- **BR-27.11 Backorder run.** Lines are ordered by (priority class asc, requested date asc, order value desc); WIN never reduces a confirmed quantity; LOSE may deconfirm only lines with priority class ≥ the run's `lose_threshold`; a run is a preview until committed and commits serialize per (product, dims, site, warehouse) like confirmation.
- **BR-27.12 Cumulative quantities.** A release's cumulative received quantity (customer view) is compared with Σ shipped since the cumulative reset date; a difference beyond `cum_tolerance` (default 0) raises `CUM_QTY_MISMATCH` and the release is held for review; open schedule lines = release cumulative required – Σ shipped; a new release replaces all open lines of the same agreement item in one transaction.
- **BR-27.13 Listing precedence.** DENY at customer level beats ALLOW at group level; an empty ALLOW list means no restriction; validity evaluated on the order date.
- **BR-27.14 Supplementary lines** inherit site, warehouse, dates and delivery address from the parent, cannot be re-promised independently, and are removed when the parent quantity drops below the rule's base quantity.
- **BR-27.15 Allocation consumption** happens at confirmation in requested-period order; cancellation returns quota to the period it consumed; a quota change below consumed quantity is rejected with 409.

5.27.8 Acceptance criteria

- **AC-27.1** (FR-27.4) Given the BR-27.1 fixture, when a 45-unit line requests 2027-03-01 with complete delivery, then confirmed ship date = 2027-03-04 and the promised event references bucket 1.
- **AC-27.2** (FR-27.4) Given any `atp-100` fixture, when all orders confirm, then no cumulative ATP is negative after propagation and total promised ≤ on-hand + included supply (property).
- **AC-27.3** (FR-27.7, FR-27.8) Given limit 10,000 USD, open AR 8,500, tolerance 5% / 1,000, when a 2,000 USD order confirms, then no hold (10,500 within both tolerances) and a WARN; when a 3,000 USD order confirms, then a CREDIT hold applies (11,500 > 10,500 and > 11,000), and release by a CreditRelease user confirms the order and emits `order-hold.released`.
- **AC-27.4** (FR-27.11) Given a make-to-order product with configuration CFG-77, when confirmed, then a wave 21 production order references the line and the confirmed date = scheduled end + transport days.
- **AC-27.5** (FR-27.12) Given 500 orders/min for 10 min through intake, then p99 < 500 ms and all 5,000 orders exist exactly once.
- **AC-27.6** (FR-27.1) Given a sent 3-line quotation, when lines 1 and 3 convert, then one 2-line order exists with matching prices and explain refs and the quotation is CONVERTED.
- **AC-27.7** (FR-27.9) Given a confirmed order, when line 10 changes from 5 to 8, then version increments, the diff is recorded, only line 10 is re-promised and `order.changed` is emitted.
- **AC-27.8** (FR-27.6) Given partial delivery and ATP 40 of 45, then two schedule lines exist and the backorder list shows 5 units on 2027-03-04.
- **AC-27.9** (FR-27.10) Given an agreement of 1,000 units fully released, then the next release is rejected with 422 `AGREEMENT_EXHAUSTED`.
- **AC-27.10** (FR-27.14) Given a mandatory supplementary rule "1 × MANUAL-EN free per 10 × FG-100" and a line of 25 FG-100, then a zero-priced line of 2 MANUAL-EN with `supplementary_of` exists and is canceled when the parent is canceled.
- **AC-27.11** (FR-27.15) Given customer C9 with a DENY listing for FG-120, when a line for FG-120 is committed, then 422 `PRODUCT_NOT_LISTED`; with duty SalesOrderApprove and a reason the line is accepted and the override logged.
- **AC-27.12** (FR-27.17) Given 5 backorder lines for FG-100 with 40 units of new supply and strategy REDISTRIBUTE, when the run commits, then priority-class-1 lines are confirmed first, no cumulative ATP is negative, and `atp.promised` is emitted only for changed lines.
- **AC-27.13** (FR-27.20) Given an agreement item with cumulative shipped 1,200 and a new release stating cumulative received 1,150, then the release is held with `CUM_QTY_MISMATCH` and no schedule line changes; a release stating 1,200 with required 1,500 by 2027-04-15 creates open schedule lines totaling 300.
- **AC-27.14** (FR-27.22) Given a one-time order for "Walk-in Buyer GmbH" with a document address, when invoiced (w29), then the invoice and open item carry that name and address and the one-time account's exposure is checked against `one_time_order_limit`.

5.27.9 Test plan and fixtures

Fixtures: `atp-100` (from SAP OVZ9 and AX ATP examples with expected dates); `credit-40` (profiles plus migrated AR open items); `orders-5k-intake` (generated channel payloads); `quotes-50`; `demo-plant` (wave 25 plant with FG-100…FG-120 stock and open production orders).

Test	Type	Target	Pass criterion
ATP scenarios	unit	promising	100/100 expected dates
Never over-promise	property	promising	Σ promised ≤ supply in every bucket prefix
Credit rules	unit	credit	40/40 outcomes incl. tolerance edges
Hold release	E2E (Temporal)	workflow	released; SLA timer fires at 2 business days
Intake throughput	performance	POST /orders	500/min, p99 < 500 ms, 0 duplicates
Sales ↔ pricing/inventory/production	contract (Pact)	APIs	green
Order entry UI	E2E (Playwright + axe)	shell	flows pass, 0 serious a11y violations

Reconciliation: Σ open confirmed order amounts per customer = credit_profile.exposure.open_orders; wave 16 reservations = Σ RESERVED line quantities; nightly rebuilt ATP projection = incremental projection (diff 0).

5.27.10 Deployment and operations

Helm `passage-sales` (Go, 3 replicas, HPA on RPS); `sales-db`; Temporal namespace `sales`. Config: `ATP_TIME_FENCE_DEFAULT_DAYS=30`, `ATP_INCLUDE_PLANNED_ORDERS=false`, `CREDIT_CHECK_POINTS=confirm,release`, `INTAKE_RATE_LIMIT_PER_KEY=1000/min`. Flags: `sales.ctp`, `sales.alternative_items`, `sales.change_approval`. Jobs: ATP rebuild 01:00 entity-local; quotation expiry hourly; credit review warnings daily. SLOs: create p99 500 ms, 99.95% availability, projection lag < 30 s. Alerts: lag > 120 s, hold queue age > 2 business days, intake 5xx > 0.1%. Runbook: projection rebuild, stuck release workflow, re-promise after supply cancellation. Rollback: previous digest; additive migrations.

5.27.11 Migration from AX / SAP

Source	Passage entity	Notes
SalesTable (SalesId, CustAccount, InvoiceAccount, CurrencyCode, SalesStatus) + SalesLine (ItemId, InventDimId, SalesQty, SalesPrice, LinePercent, RemainSalesPhysical/Financial)	sales_order, lines	open orders only (Backorder, or Delivered with financial remainder); InventDim → dims + site/warehouse

Source	Passage entity	Notes
SalesQuotationTable/Line (Sent); SalesAgreementHeader/Line	sales_quotation; sales_agreement	remaining commitment from released qty
CustTable.CreditMax, MandatoryCreditLimit, CustParameters.CreditLimitType	credit_profile	1:1 type mapping
VBAK/VBAP/VBEP (VBUK-GBSTK ≠ C), VBKD, VBPA	sales_order, lines, schedule lines	TAS → drop-ship flag; VBEP confirmed qty → schedule lines
KNKK (KLIMK, CTLPC), UKM_BP	credit_profile	risk class → risk_group
VBAK VBTYP B / E / G	sales_quotation / sales_agreement	

Validation: count and Σ open value per customer equal the legacy extract; every migrated line re-promises to ≤ legacy confirmed date + 1 day or is flagged. Not migrated: closed orders (lakehouse), CDHDR logs, SAP credit horizon settings (replaced by BR-27.3).

5.27.12 Dependencies, risks, open questions

- Depends on: w26 pricing; w16 on-hand/reservations; w17 PO receipts; w21 production orders; w22 planned orders and demand; w23 scheduling (CTP); w7 customers/terms; w8 configurator; w4 Temporal.
- Risks: projection staleness during planning runs (planning emits a batch-end marker; buckets flagged stale); credit exposure incomplete before waves 29/30 (migrated open AR loaded into `receivables_db.open_item`, schema created this wave).
- Open questions: (1) Default date control for migrated customers, ATP vs. lead time? Default ATP; decide by wave 28. (2) External fraud scoring? Default rule-only in v1; decide by wave 30.

5.27.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
default_order_type	legal entity	enum	ORDER	JOURNAL, ORDER, SUBSCRIPTION	Type on new orders	AR parameters → General → Sales order type default (SalesType)	VOV8 default sales document type (OR)
quotation_expiry_days	legal entity	int	30	1–365	FR-27.1	Sales and marketing parameters → Quotations → expiry	VOV8 QT validity period
quotation_probability_default	legal entity	int %	50	0–100	Pipeline weighting	Sales and marketing parameters → Quotations → probability	VBAK-AWAHR default
delivery_date_control_default	legal entity	enum	ATP	NONE, SALES_LEAD_TIME, ATP, CTP	FR-27.3	AR parameters → Updates → Delivery date control (DlvDateControlType)	OVZ9 checking rule / OVZG requirements class
atp_time_fence_days	legal entity (override per product coverage group w22)	int	30	1–365	FR-27.4 window	ATP time fence (InventModelGroup / coverage group)	OVZ9 "Check horizon"
atp_include_planned_orders	legal entity	bool	false	true/false	FR-27.4 supply set	ATP incl. planned orders	OVZ9 "Include planned orders"
atp_include_safety_stock	legal entity	bool	false	true/false	Demand set	Coverage group safety stock in ATP (ReqGroup)	OVZ9 "Check without safety stock" (inverse)
reserve_on_confirm	legal entity	bool	true	true/false	BR-27.7	AR parameters → General → Reservation (Automatic/Manual)	Schedule line category reservation (VOV6)
partial_delivery_default	customer (w7) with entity default	enum	ALLOWED	ALLOWED, COMPLETE_ONLY, ONE_SHIPMENT	FR-27.6	SalesTable / CustTable "Complete delivery" (CompleteDelivery) flag	KNVV-KZTLF partial delivery indicator
overdelivery_pct / underdelivery_pct	product × customer	numeric %	0 / 0	0–100	BR-27.9, BR-28.7	InventTable Over/underdelivery	VOV8 / KNVV over/underdelivery tolerance
credit_limit_type_default	legal entity	enum	BALANCE	NONE, BALANCE, BALANCE_PACKING_SLIP, BALANCE_ALL	FR-27.7	AR parameters → Credit rating → Credit limit type (CustParameters.CreditLimitType)	OVA8 "Open orders / deliveries / billing" update group (000012/000015/000018)
mandatory_credit_check	legal entity	bool	false	true/false	CR-4	AR parameters → "Mandatory credit limit"	OVA8 "No credit check" (inverse)
credit_check_points	legal entity	set	confirm, release	entry, confirm, release, packing_slip, invoice	FR-27.7	AR parameters → Credit management → check points	OVA8 check at order/delivery/GI
credit_tolerance_pct / amount	legal entity	numeric	5 / 1,000.00	≥ 0	CR-1	D365 credit management "Tolerance"	OVA8 "Deviation in %" / "Number of days"
credit_release_validity_days	legal entity	int	7	1–30	BR-27.4	D365 credit management release validity	OVA8 "Released documents still unchecked" days
credit_hold_sla_business_days	legal entity	int	2	1–10	FR-27.8 timer	n/a	n/a (FSCM workflow)
change_approval_required	customer (w7)	bool	false	true/false	FR-27.9	D365 Sales order change approval	n/a (VA02 unrestricted)
intake_rate_limit_per_key	tenant	text	1000/min	rate expression	FR-27.12	n/a	n/a
intake_dedup_window_days	tenant	int	30	1–90	BR-27.10	n/a	n/a
supplementary_prompt_optional	legal entity	bool	true	true/false	FR-27.14 prompt behavior	Supplementary items prompt (AR parameters → Updates)	NA00 free goods automatic
listing_override_duty	legal entity	duty	SalesOrderApprove	duties	FR-27.15	n/a	OV04 "listing check" switch
allocation_enabled	legal entity	bool	false	true/false	FR-27.16	n/a	OVZJ activation
backorder_strategy_default / lose_threshold	legal entity	enum / int	REDISTRIBUTE / 3	strategies / 1–9	FR-27.17, BR-27.11	n/a	aATP BOP segment strategy

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
cum_tolerance / cum_reset_rule	legal entity, customer	numeric / enum	0 / FISCAL_YEAR	≥ 0; FISCAL_YEAR, CALENDAR_YEAR, NEVER	BR-27.12	n/a	VA32 cumulative quantity reset / year change
one_time_order_limit	legal entity	money	5,000.00 USD / 4,500.00 EUR	≥ 0	FR-27.22	One-time customer limit (AR parameters)	OBD2 CPD account group
sales_order_stock_default	product (w8)	bool	false	true/false	FR-27.19	n/a (marking manual)	OVZG requirements class special stock E

5.27.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Sales orders list	list inquiry	customer, status, date range, site, hold, channel	order, customer, requested/confirmed dates, net, status, holds	sales_order	All sales orders (SalesTableListPage)	VA05 / VA05N
Open order lines / backorders	list inquiry	product, customer, promised date, site	line, qty, remaining, confirmed date, source	sales_order_line (open)	Backorder lines by item / by customer; SSRS Open sales orders	V.15 / VA14L (blocked) / CO06
ATP overview	drill-down	product, dims, site, warehouse, horizon	bucket date, supply, demand, period ATP, cumulative ATP, sources	atp_bucket	Available ship and receipt dates dialog	CO09 availability overview
Credit release queue	list inquiry	customer, rule failed, age, amount	customer, order, exposure vs limit, rule, age	order_hold + credit evaluations	Sales orders on hold (SalesOrderHold) / D365 Credit holds	VKM1 / VKM4
Credit exposure by customer	operational report	customer, as-of	limit, open AR, unapplied, DNI, open orders, utilization %	credit_profile.exposure	SSRS Customer balance list with credit limit	FD33 / S_ALR_87012215 (credit overview)
Quotations pipeline	KPI	status, expiry window, probability	count, weighted value, conversion rate	sales_quotation	SSRS Sales quotation / quotation follow-up	VA25 quotation list
Sales agreements and releases	drill-down	agreement, customer, item	commitment, released, remaining, release history, cum received/shipped	sales_agreement, schedule_release	Sales agreements (fulfillment tab)	VA35 scheduling agreement list, VA33 cum quantities
Order change history	drill-down	order, version range	version, field, old, new, actor, reason	sales_order versions	n/a (no versioned history)	CDHDR/CDPOS change documents
Intake errors	list inquiry	channel, error code, date	external_ref, payload hash, error, retries	intake log	AIF exceptions	WE02 / WE05 IDoc errors
Listing / allocation inquiry	list inquiry	customer, product, period	listing entries; quota, consumed, remaining	product_listing, allocation_quota	n/a	VB03 / product allocation reports
Stock at customer (consignment)	list inquiry	customer, product	qty at customer, value, age	w16 on-hand by owner dimension	n/a	MB58
Backorder run results	drill-down	run id	line, before, after, strategy reason	backorder_run	Backorder workspace	V_V2 log / aATP BOP results

5.27.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Sales quotation	:send	header: quotation number, customer, contact, validity, currency, terms; lines: product, qty, unit, price, discount, net, promised date; footer: totals, tax estimate, probability hidden, T&C block	PDF, email, portal (w30), UBL quotation JSON for w35	`SQ–` per legal entity	none; retain 3 years
Order confirmation	:confirm and on each version with `send_confirmation`	header: order, customer, invoice account, addresses, terms, mode of delivery; lines: product, qty, confirmed ship/receipt date, price, net; footer: totals, holds not shown	PDF, email, EDI ORDRSP/855 hook (w35), portal	`OC–` per legal entity (document copy number per version)	commercial record; retain 7 years with invoice
Sales agreement confirmation	agreement activation	header: agreement, customer, validity, commitment type; lines: product/group, committed qty or amount, price; footer: release terms	PDF, email	`SA–`	7 years
Scheduling agreement release acknowledgment	release accepted	header: agreement item, release number, cum received/shipped; lines: schedule dates and quantities	PDF, EDI (w35)	release number from customer	7 years (automotive audit)
Credit hold notice (internal)	hold applied	customer, order, failing rule, exposure, approver	email, in-app	none	none
Quotation lost / expired notice	status change	quotation, reason	email (internal)	none	none

5.27.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
ATP projection rebuild	01:00 entity-local daily	(product, site)	full rebuild into shadow table, swap	< 15 min per 1M buckets	keep previous projection; alert	diff incremental vs rebuilt ≠ 0; lag > 120 s
Backorder processing run	05:00 daily per site (after planning w22)	(product, site)	run id; preview persisted; commit idempotent per line version	< 10 min per 100k open lines	partial commit impossible (per-product transaction)	run > 30 min; deconfirmed lines > 5%
Quotation expiry	hourly	legal_entity	status transition by (id, version)	< 30 s	retry	none expired for 2 days while candidates exist
Credit review warnings / profile refresh	06:00 daily	customer	key = (customer, date)	< 5 min	retry	profiles with review_date passed > 100
Credit exposure recompute	event-driven; nightly reconcile 01:30	customer	exposure.updated_at	< 10 min	rerun	Σ open orders ≠ exposure.open_orders
Scheduling release reconciliation	on release receipt; nightly cum check 23:00	agreement item	release number	< 1 min	hold release	CUM_QTY_MISMATCH count > 0
Allocation period roll	first day of period 00:05	product	key = (quota, period)	< 1 min	retry	missing quota period
Intake retry / DLQ replay	every 5 min	channel	(channel, external_ref)	< 1 min	DLQ after 10 attempts	DLQ depth > 50

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Order auto-close	nightly 02:00	legal_entity	(order, version)	< 5 min	retry	none

5.27.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
SalesOrderView	GET orders, quotations, atp buckets; Sales workspace, ATP explorer	none	Sales rep, Customer service, Warehouse, Finance
SalesQuotationMaintain	POST quotations, :send, :convert; Quotation detail	none	Sales rep
SalesOrderMaintain	POST/PATCH orders, lines, :cancel-remainder; Order entry; backorders	with CreditRelease; with PricingOverride when `four_eyes_override`	Sales rep, Customer service
SalesOrderConfirm	:confirm	with CreditRelease	Sales rep (within limits), Sales manager
SalesOrderApprove	approve change requests; listing override	with SalesOrderMaintain on own orders	Sales manager
SalesOrderHold	POST holds (MANUAL, PRICE_REVIEW, FRAUD)	none	Sales manager, Fraud analyst
CreditRelease	holds:release (CREDIT); Credit release queue	with SalesOrderMaintain, SalesOrderConfirm, CreditMaintain on the same customer	Credit manager
CreditMaintain	PUT credit-profiles	with CreditRelease	Credit analyst
SalesAgreementMaintain	agreements, :release-order, schedule releases	none	Key account manager
ChannelIntake	POST orders via channel client credentials	none	Integration service accounts (w35)
BackorderRun	run and commit backorder processing	with SalesOrderMaintain on affected orders (logged, not blocked)	Order management lead
AllocationMaintain	allocation quotas	with BackorderRun	Demand planner
ConsignmentMaintain	consignment fill-up/issue/pick-up/return	with WarehouseReceive (w28)	Key account manager

5.27.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
SLS-0001	error	Line {n} has no price; enter a manual price or add an agreement	PRICE_NOT_FOUND at confirm	Add agreement or override	FR-27.2 / BR-26.1
SLS-0002	error	Customer {customer} is blocked for sales	w7 status BLOCKED	Unblock in party governance	§5.27.5 validation
SLS-0003	warning	Requested date {date} is in the past; moved to today	BR-27.2	none	BR-27.2
SLS-0004	info	Requested date beyond the ATP fence; promised at fence end {date}	BR-27.2	Shorten lead time or extend fence	BR-27.2
SLS-0005	error	Credit hold: rule {rule} failed ({value} vs {threshold})	BR-27.3	Request release in the credit queue	FR-27.7
SLS-0006	error	Order was changed by another user; reload	If-Match mismatch	Reload and reapply	BR-27.6
SLS-0007	error	Duplicate order: {channel}/{external_ref} differs from the stored payload	BR-27.10	Use a new external_ref	FR-27.12
SLS-0008	error	Agreement {id} commitment exhausted	BR-27.9	Increase commitment or create a new agreement	FR-27.10
SLS-0009	error	Product {product} rejects fractional quantities	whole_units	Enter whole units	BR-27.8
SLS-0010	warning	Overdue > 25% of open AR for {customer}	CR-5	Inform collections	BR-27.3
SLS-0011	error	Product {product} is not listed for {customer}	FR-27.15	Override with SalesOrderApprove or change product	BR-27.13
SLS-0012	warning	Mandatory supplementary item {product} added ({qty})	FR-27.14	none	BR-27.14
SLS-0013	error	Allocation for {product} in {period} exhausted; promised {date}	FR-27.16	Request quota change	BR-27.15
SLS-0014	error	Cumulative quantity mismatch: customer {cum_received}, shipped {cum_shipped}	BR-27.12	Review the release; adjust tolerance or correct shipments	FR-27.20
SLS-0015	warning	Backorder run would deconfirm {n} lines	BR-27.11 preview	Review before commit	FR-27.17
SLS-0016	error	One-time order exceeds limit {limit}	FR-27.22	Create a regular customer	FR-27.22
SLS-0017	error	Consignment issue exceeds stock at customer ({qty} available)	FR-27.18	Fill up first or correct quantity	FR-27.18
SLS-0018	info	Cost estimate {amount}, margin {pct}%	FR-27.23	none	FR-27.23

5.27.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
POST /sales/orders (intake, 10 lines)	50k per day; peak 500/min	120 / 300 / 500 ms	20 req/s per replica	20M lines × 1.5 KB = 30 GB	orders partitioned by legal entity; lines (order_id, line_no)	429 with Retry-After per key; HPA; holds evaluated asynchronously if > 1,500/min
Inline price + ATP at line commit	500k per day	80 / 300 / 600 ms	100/s	none	atp_bucket PK lookup	ATP badge shows "pending" after 300 ms; confirm still synchronous
:confirm (10 lines)	50k per day	200 / 600 / 1,200 ms	30/s	events 10 KB each	advisory lock per product	confirm queue; p95 ≤ 2 s

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
ATP projection update	1M events per day	2 / 10 / 30 s lag	1,000 events/s	50M buckets × 200 B = 10 GB	PK (product, dims, site, warehouse, bucket_date)	buckets flagged stale; promise falls back to lead time with warning
Credit check	100k per day	20 / 80 / 200 ms	100/s	evaluations 30 GB (jsonb, 2-year retention)	(customer, evaluated_at)	cached exposure (≤ 60 s) under load
Backorder run (100k lines)	1 per site per day	— / 10 min / 20 min	200 lines/s	run logs 5 GB	per-product batches	run split by product group; preview only
Scheduling release import	5k releases per day	50 / 200 / 500 ms	20/s	2 GB	(agreement_item, release_no)	queue; acknowledgment delayed

5.27.20 Localization and compliance hooks

- Tax ids and VAT validation on one-time orders use the w13 validator per country (VIES for EU, EIN format for US); the document-level address feeds tax determination exactly as a master address.
- Calendars: customer receiving calendars and transport days (BR-27.2) come from the w6 calendar service with country holiday packs (w48); cumulative-quantity reset follows the customer's fiscal year (w14).
- Currency and rounding: credit exposure converts at the w12 spot rate on the check date; quantities at 6 decimals; thresholds in BR-27.3 hold USD/EUR defaults overridable per entity.
- Documents: quotation and confirmation templates per language and entity (w5 docgen); country packs may add mandatory legal footer text (for example EU distance-selling withdrawal notice) through the template `legal_footer` slot.
- Export control and sanctioned-party screening are out of scope here (w48 compliance hooks `order.confirmed` consumer may apply a FRAUD/COMPLIANCE hold through the holds API).
- Privacy: portal and one-time contacts are personal data (w48 retention); order history pseudonymizes contact names after 7 years.
- Automotive packs (DE/US) extend FR-27.20 with EDIFACT DELFOR/DELJIT and VDA 4905/4915 mappings through w35; the cumulative model here is the extension point.

5.27.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.3.4 #31 — Trade agreements → Supplementary items (PriceDiscSuppItem)	PARTIAL-26 (proposed 27)	ADDED as FR-27.14	free-goods proposals from w26 FR-26.22
AX §12.17.1 #168 — Sales order → Supplementary items (prompt)	PARTIAL-26 (proposed 27)	ADDED as FR-27.14	supplementary_prompt_optional
AX §12.17.4 #171 — Sales orders → Supplementary items setup	PARTIAL-26 (proposed 27)	ADDED as FR-27.14	rule entity supplementary_item_rule
SAP §13.8 #90 — CK51N / CK53N sales order cost estimate	PARTIAL-27	ADDED as FR-27.23 (hook); full costing DEFERRED to wave 39	estimate stored on line
SAP §13.8 #98 — OVZG sales order costing / valuation of sales-order stock	PARTIAL-27	DEFERRED to wave 21 / 39	segregation itself ADDED as FR-27.19; valuation stays with costing waves
SAP §13.13 #141 — MB58 consignment and returnable packaging stock at customer	PARTIAL-27	ADDED as FR-27.18 (consignment); returnable packaging DEFERRED to wave 28 (FR-28.16)	needs w16 owner dimension
SAP §13.13 #143 — MIE1 / MIW1 / MIV1 physical inventory for sales-order stock and consignment	PARTIAL-27	ADDED as FR-27.18 / FR-27.19 (dimensions); counting itself SUPERSEDED by w16 counting journals on the owner / sales-order segment	
SAP §13.13 #145 — Special stock E / Q (PARTIAL-16, note names wave 27)	PARTIAL-16	ADDED as FR-27.19 for sales-order stock; project stock Q DEFERRED to wave 41	cross-reference
SAP §13.19 #216 — VB01/VB02/VB03 listing/exclusion	PARTIAL-27	ADDED as FR-27.15	
SAP §13.19 #218 — VA01 KB/KE/KA/KR consignment processes	PARTIAL-27	ADDED as FR-27.18	Could; depends on w16
SAP §13.19 #219 — VA31/VA32/VA33/VA35 scheduling agreements with JIT/forecast schedules, DELFOR/DELJIT	PARTIAL-27	ADDED as FR-27.20	EDI mapping in wave 35
SAP §13.19 #222 — VB41 cross-selling, dynamic product proposal	PARTIAL-27	ADDED as FR-27.21	
SAP §13.19 #223 — OVZJ / OVZI product allocation	GAP (propose 27)	ADDED as FR-27.16	allocation_enabled
SAP §13.19 #224 — OVZK / aATP backorder processing strategies, release for delivery	PARTIAL-27	ADDED as FR-27.17	release for delivery = w28 release rules
SAP §13.19 #225 — OVZG/OVZH/OVZI requirements classes / strategy groups	PARTIAL-27	ADDED as FR-27.19 (special stock part); planning strategies remain wave 22	
SAP §13.19 #226 — OV04 listing/exclusion procedures	PARTIAL-27	ADDED as FR-27.15	one entity, precedence BR-27.13
SAP §13.2 #23 / #37 — One-time account (CPD, OBD3) and SAP §13.3 head-office/branch #50 (PARTIAL-7, flagged for order-to-cash)	PARTIAL-7	ADDED as FR-27.22 (one-time orders); head-office/branch SUPERSEDED by sales_order.invoice_account_id (FR-27.2) and open_item.invoice_account_id (FR-30.1)	party roles themselves remain wave 7

5.28 Wave 28: Fulfillment: Picking, Packing & Shipping

5.28.1 Scope and business context

Wave 28 ships the `fulfillment` service and the `carrier-gateway` adapter. Confirmed orders release to the warehouse by rule, are picked (basic wave/batch picking; location directives and RF arrive in wave 31), packed into containers with SSCC labels, rated and labelled through a carrier abstraction, and shipped with a posted packing slip that deducts inventory and posts cost to the universal journal. Drop-ship and partial shipments are supported; tracking is published for the portal. Out of scope: location directives, license plates and mobile RF flows (wave 31), transportation planning and freight reconciliation (wave 34), hazmat and export compliance. Added from the coverage registers (§5.28.21): picking workbench, packaging materials with packaging fees, packing instructions (auto-pack) and returnable transport packaging balances.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Delivered-not-invoiced never ties: AX "Cost of units delivered" via InventPosting drifts from CustPackingSlipJour after invoice cancellations; SAP PGI posts COGS at once so revenue and COGS land in different periods	Physical and financial postings decoupled across SalesFormLetter classes; reconciliation is a month-end report	Packing slip and invoice share one cost_snapshot_id; DNI projection reconciled to GL 1405 daily; AX-style and PGI-style matrices by policy	Month-end DNI reconciliation 4–8 h → 0 (daily, diff 0.00)
Packing-slip posting blocks the user 5–20 s and fails opaquely ("Update has been cancelled"; VL02N "posting period closed" after picking)	Synchronous multi-table update with late validation	Pre-validation at PACKED (period, holds, dims, negative stock) and a Temporal saga with compensation	Failures discovered after pick ~3% → < 0.1%; post p95 < 2 s
Each carrier is a project (AX ShipCarrier via desktop add-on; SAP XSI/TM)	No provider abstraction	carrier-gateway rate/label/track contract, reference adapters, `tracking-updated` events	Adding a carrier 4–8 weeks → < 2 days; tracking visible on 100% of shipments
Short picks and partials need manual SalesLine splits or VBEP edits	Schedule lines are not first-class in fulfillment	Schedule lines drive shipments; short pick auto-creates the remainder and releases its reservation	Manual splits per 1,000 orders ~40 → 0
SSCC errors at customer receiving (bad check digit, duplicate on re-print)	SSCC generated by label software, not validated by the ERP	Gap-free SSCC per GS1 prefix with mod-10 check, uniqueness and 12-month no-reuse in the container entity	Label rejections at customer DC ~1% of pallets → 0

5.28.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Release to warehouse	Release to warehouse, SalesLine.ReleaseStatus, InventPickingList	VL01N/VL10, LIKP/LIPS	shipment (release rules)	full
Picking	WMSPickingRoute, WMSOrderTrans, picking list registration	LT03 transfer order (LTAP); EWM warehouse task	pick_work	partial (advanced w31)
Packing / HU	WHSContainerTable, packing station	VEKP/VEPO handling units, HUMO	container (nested)	full
SSCC / labels	WHSContainerTable.LicensePlate	HU external id, GS1-128	container.sscc, PDF/ZPL label	full
Packing slip posting	CustPackingSlipJour/Trans, SalesFormLetter_PackingSlip, InventTrans StatusIssue=Deducted	VL02N post goods issue, MKPF/MSEG 601, LIKP-WBSTK	packing_slip saga	full
COGS posting	InventPosting: cost of units delivered / cost of goods delivered	OBYC GBB-VAX, BSX	matrix BR-28.1	full
Carrier interface	ShipCarrierTable, ShipCarrierShippingRequest	TM / Express Ship Interface	carrier-gateway	improved
Drop-ship	Direct delivery (PurchTable.DirectDelivery)	Third-party item category TAS, statistical GR	drop_ship flow	full
Delivery schedules / ASN	SalesDeliverySchedule; AIF DESADV	VBEP; DESADV IDoc	schedule lines → shipments; ASN event (EDI w35)	partial
Picking workbench	Sales picking workbench (R3), picking routes setup, grouped picking lists	VL10 delivery due list + LT03 grouped transfer orders	pick_workbench (filtered grouped waves)	full
Packaging materials and fees	Packaging materials, packing material fees (EU), PI container types / packaging groups	Packaging material types VHAR, packaging materials (MARA MTART VERP)	packaging_material, packaging_fee, container_type	full
Packing instructions / auto-pack	— (WHS packing policies w31)	POP1/POP2/POP3 packing instructions, POF1 determination, EWM packaging specifications	packing_rule (product → container type, qty per level)	full (EWM-level rules w31)
Returnable transport packaging	—	RTP special stock M/V, MB58	returnable_packaging_balance per customer	partial (balances; charging w30 credit/debit)

5.28.3 Functional requirements

- **FR-28.1 Release rules** — Must. Release confirmed, unheld lines by rule (confirmed ship date ≤ today + horizon, default 2 d; full ATP; priority) into one shipment per (order, warehouse, ship date) unless consolidation is enabled.
- **FR-28.2 Picking** — Must. Pick work by wave (≤ 200 lines or 50 orders) or batch; physical reservation (w16); confirm quantity, batch/serial and location; short picks create a remainder and release the unpicked reservation.
- **FR-28.3 Packing** — Must. Containers (type, tare, dims), nesting pallet → carton, SSCC from the GS1 company prefix with mod-10 check digit.
- **FR-28.4 Packing slip posting** — Must. Record delivered quantities, create inventory issues, post the journal per BR-28.1 at the current cost snapshot, atomically via a Temporal saga; sequence `PS-`.
- **FR-28.5 Cancellation** — Must. Reverse an uninvoiced packing slip with mirror inventory and journal entries referencing the original.
- **FR-28.6 Carrier abstraction** — Must. rate / label / manifest / track through a provider-agnostic contract with EasyPost-style, Shippo-style and `file` adapters; labels as PDF and ZPL; tracking by 30-min poll or webhook.
- **FR-28.7 Drop-ship** — Must. Linked PO (w17) for drop-ship lines; the vendor's ship confirmation triggers the packing slip; COGS at vendor cost without inventory.
- **FR-28.8 Partial shipments** — Must. Ship each schedule line separately; order stays PARTIALLY_SHIPPED until remaining_qty = 0.
- **FR-28.9 ASN** — Should. Emit an ASN with containers, SSCCs and quantities on post for the wave 35 EDI mapper.
- **FR-28.10 Tracking visibility** — Should. Publish tracking changes to portal and order detail.
- **FR-28.11 Mobile pick (basic PWA)** — Should. Pick confirmation with GS1-128 parsing (AI 00, 01, 10, 17, 21, 30, 37) and 5-minute offline tolerance.
- **FR-28.12 DNI reconciliation** — Must. delivered_not_invoiced projection = GL 1405, checked daily.
- **FR-28.13 Picking workbench** — Should. The system shall provide a picking workbench that filters released shipment lines (customer, carrier, ship date range, product group, zone, order priority), groups them by a selectable key (customer, carrier, zone, product) into waves of ≤ `WAVE_MAX_LINES`, previews the grouping and creates pick work in one action. Notes: AX sales picking workbench (R3) semantics; saved filters per user; a line belongs to at most one open wave.
- **FR-28.14 Packaging materials and packaging fees** — Should. The system shall define packaging materials (material code, type, weight, recyclable flag, unit) assigned to products per unit (w8 packaging units) and to container types, compute packaged weight per shipment and post packaging fees (EU packaging duty: fee per kg per material and country) as a statistical line on the packing slip and a FREIGHT/SURCHARGE condition for w29 when `packaging_fee_billable`. Notes: packaging report per period per country (§5.28.14); fee tables are country-pack data (w48).
- **FR-28.15 Packing instructions** — Should. The system shall hold packing rules (product or product group × customer optional → container type, quantity per container, nesting level, max weight) and auto-pack picked lines into containers at the packing station with a one-click accept; manual repack remains available. Notes: SAP POP1/POF1 and EWM packaging specifications; rule determination specific → general like BR-26.1.

- **FR-28.16 Returnable transport packaging** — Could. The system shall track returnable packaging (pallets, crates) shipped to and returned from customers as a balance per (customer, packaging material) from packing slips and return receipts, with an aging inquiry and a hook for w30 to charge unreturned items after `rtp_charge_days`. Notes: SAP special stock M/V; no inventory valuation movement; balances reconciled to Σ container counts.

5.28.4 Data model

shipment

Field	Type	Required	Notes
id, shipment_number	uuid, text	Y	SH- sequence
legal_entity_id, warehouse_id	uuid	Y	one warehouse per shipment
customer_id, delivery_address_id	uuid	Y	
status	enum	Y	BR-28.6
planned_ship_date, actual_ship_date	date, timestamptz	Y/N	
carrier_code, service_level, mode_of_delivery	text	N	
lines[]	shipment_line	Y	order_line_id, schedule_line_id, qty_planned/picked/packed/shipped, dims
consolidation_key, packing_slip_id, carrier_shipment_id, version	text, uuid, uuid, int	N	

container: id, shipment_id, parent_container_id, container_type, sscc char(18), tare_weight, gross_weight, dimensions, uom, lines[] (shipment_line_id, qty, batch, serials[]), label_ref, status (OPEN | CLOSED | SHIPPED).

packing_slip (CustPackingSlipJour/Trans): id, packing_slip_number, shipment_id, order_id, posting_date, posted_at, status (PENDING | POSTED | CANCELLED | INVOICED), lines[] (order_line_id, qty, unit, cost_amount, cost_snapshot_id, inventory_transaction_ids[]), total_cost, journal_posting_id, reversal_of_id, document_ref.

carrier_shipment: id, shipment_id, provider, provider_ref, rate_quote, tracking_number, label_ref, status (QUOTED | LABELLED | MANIFESTED | IN_TRANSIT | DELIVERED | EXCEPTION), events[], cost_amount, currency.

Invariants: Σ container qty per line = qty_packed ≤ qty_picked ≤ qty_planned; packing slip qty = qty_shipped; one journal posting per slip plus one reversal if cancelled; SSCC check digit valid and unique per tenant; Σ cost of uninvoiced slips = GL 1405. Indexing: shipments (warehouse, status, planned_ship_date); packing slips (order_id), (posting_date), partitioned by month; containers unique sscc.

5.28.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/fulfillment/releases:run	Run release rules for a warehouse or orders	Idempotency-Key	WarehouseRelease
POST	/api/v1/fulfillment/waves, …/{id}:confirm-pick	Create wave; confirm picks	Idempotency-Key	WarehousePick
POST	/api/v1/fulfillment/shipments/{id}/containers, …/{cid}:close	Pack; close container (SSCC)	Idempotency-Key	WarehousePack
POST	/api/v1/fulfillment/shipments/{id}:rate, :label, :manifest	Carrier operations	Idempotency-Key	WarehouseShip
POST	/api/v1/fulfillment/shipments/{id}:post-packing-slip; /packing-slips/{id}:cancel	Post / reverse (saga)	Idempotency-Key	PackingSlipPost
GET	/api/v1/fulfillment/shipments/{id}/tracking	Tracking events	—	ShipmentView / Portal
POST	/api/v1/fulfillment/drop-ship/{orderLineId}:vendor-confirm	Vendor confirmation → packing slip	Idempotency-Key	PurchasingConfirm
POST	/webhooks/carriers/{provider}	Tracking webhook (HMAC)	provider event id	system

Validation: post requires status PACKED, no ACTIVE hold, posting date in an open inventory period (w14), and no negative on-hand unless the product policy allows (w16).

Event	Trigger	Payload fields	Consumers
dev.passage.fulfillment.shipment.released.v1	release	shipment_id, order_id, warehouse, lines	sales, analytics
dev.passage.fulfillment.pick.confirmed.v1	pick confirm	wave_id, lines[{order_line_id, qty, dims, location}]	inventory, sales
passage.fulfillment.packing-slip.posted.v1 / cancelled.v1	saga complete / cancel	packing_slip_id, order_id, lines[{order_line_id, qty, cost_amount, cost_snapshot_id}], journal_posting_id, posting_date	billing, sales, inventory, ledger, portal, credit
dev.passage.fulfillment.asn.issued.v1	post	shipment_id, containers[{sscc, lines}], carrier, tracking	EDI (w35)
passage.fulfillment.shipment.tracking-updated.v1	poll / webhook	shipment_id, tracking_number, status, event	portal, sales

5.28.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Fulfillment workspace	tiles: to release, picking, packed not shipped, carrier exceptions, DNI balance	counts	WarehouseView
Release board	run rules, release selected, hold	order, customer, ship date, ATP status, priority	WarehouseRelease
Wave / pick list	print, confirm, short pick, reassign	line, product, location, qty, dims, picker	WarehousePick
Packing station	scan line, add/close container, print label	container, type, weights, lines, SSCC	WarehousePack
Shipment detail	rate, label, manifest, post packing slip, cancel	carrier, service, cost, tracking, journal preview	WarehouseShip / PackingSlipPost
Mobile pick (PWA) / carrier exceptions	scan, confirm, skip; retry, switch provider, manual tracking	location, product, qty, batch/serial; provider error, age	WarehousePick / WarehouseShip

Interaction rules: scan field keeps focus; Enter confirms a line; GS1-128 auto-fills batch/serial; bulk release on multi-select; posting shows the journal preview before confirm; layouts per warehouse personalizable.

5.28.7 Business rules and edge cases

- **BR-28.1 Packing-slip COGS posting matrix** (accounts from the wave 16 posting profile; dimensions from the order line):

Scenario	At packing slip post	At invoice post (wave 29)
Stocked item, physical posting enabled (default, AX-style)	Dr 1405 Delivered-not-invoiced / Cr 1400 Inventory (physical) at cost snapshot	Reverse 1405/1400; Dr 5000 COGS / Cr 1400 Inventory (financial) at settled cost
Stocked item, policy `cogs_at_shipment` (SAP PGI-style)	Dr 5000 COGS / Cr 1400 Inventory	Revenue and AR only
Standard-cost item	Same, at standard; no variance at shipment	Same
Drop-ship line	No posting	Dr 5000 COGS / Cr 2110 GR/IR at PO cost when vendor confirmation posts
Non-stocked / service line	No posting	Revenue only
Catch-weight item	Cost on inventory quantity, not catch-weight quantity	Same
Cancellation	Mirror entries, negative amounts, same dims, reversal_of set	—

- **BR-28.2 Cost snapshot** = running average or FIFO estimate (w20) at posting, stored as cost_snapshot_id; inventory close adjusts 5000/1400, never 1405 after invoicing.
- **BR-28.3 Posting date** = actual ship timestamp in the warehouse time zone; closed inventory period → `PERIOD_CLOSED`, shipment stays PACKED.
- **BR-28.4 Saga.** (1) packing slip PENDING; (2) inventory issues; (3) journal posting with Idempotency-Key = packing_slip_id; (4) POSTED + event. Failure at 3 compensates 2; a retried 3 is a ledger no-op. PENDING > 10 min alerts.
- **BR-28.5 SSCC** = extension digit + GS1 company prefix + serial reference padded to 17 digits + mod-10 check (weights 3,1 from the right); serial from a gap-free sequence per prefix; no reuse for 12 months.
- **BR-28.6 Shipment state machine.** OPEN → RELEASED → PICKING → PICKED → PACKED → SHIPPED → DELIVERED; CANCELLED before SHIPPED (releases reservations); SHIPPED → PACKED on packing-slip cancellation.
- **BR-28.7 Over-pick** rejected; over-ship rejected unless the customer's overdelivery_pct allows.
- **BR-28.8 Consolidation** requires matching customer, address, warehouse, planned date and carrier and no `ship_separately` order.
- **BR-28.9 Carrier failures** retry 3× (2 s, 8 s, 32 s) then land in the exceptions queue; the slip may still post with a manual tracking number.
- **BR-28.10 Tracked items** must carry batch/serial at pick; the slip rejects lines missing tracking dims (w8 tracking dimension group).
- **BR-28.11 Auto-pack** fills containers in rule order (most specific rule first), closes a container when quantity per container or max weight is reached, and never splits a serial-tracked unit; lines without a rule stay unpacked and are flagged.
- **BR-28.12 Packaging fee** = Σ (material weight per unit × shipped units) per material × country rate as of the packing slip date, rounded half-even to minor units per material; fees post statistically (no journal) unless `packaging_fee_billable`, in which case w29 bills them as a SURCHARGE line with its own tax code.
- **BR-28.13 Returnable packaging balance** increases on packing-slip post (containers flagged returnable) and decreases on return receipt (w30 RMA line or standalone return); a negative balance is rejected with `RTP_NEGATIVE`.
- **BR-28.14 Workbench grouping** keeps lines of one shipment in one wave; a wave exceeding `WAVE_MAX_LINES` is split by the group key; preview shows lines per wave before creation.

5.28.8 Acceptance criteria

- **AC-28.1** (FR-28.4) Given 10 units at cost 12.50 USD with physical posting, when the slip posts, then on-hand −10, journal Dr 1405 125.00 / Cr 1400 125.00, and the event carries cost_amount 125.00.
- **AC-28.2** (FR-28.5) Given that slip cancelled before invoicing, then on-hand +10, a mirror posting exists and GL 1405 returns to its prior balance.
- **AC-28.3** (FR-28.3) Given prefix 0614141 and serial 0123456789, then SSCC = 006141410123456789 1 and the GS1-128 label scans back to it.
- **AC-28.4** (FR-28.2) Given a 100-unit line with 90 picked, then qty_picked = 90, 10 remain on backorder and their reservation is released.
- **AC-28.5** (FR-28.7) Given a drop-ship line, when the vendor confirms, then a slip posts with no inventory transaction and Dr 5000 / Cr 2110 at PO cost.
- **AC-28.6** (FR-28.6) Given recorded fixtures for both adapters, when rate/label/track run, then responses match the contract and labels are valid PDF and ZPL.
- **AC-28.7** (FR-28.1) Given 1,000 confirmed orders, when release → pick → pack → post runs, then all are SHIPPED and Σ inventory issues = Σ slip quantities with zero mismatch.
- **AC-28.8** (FR-28.12) Given 50 posted slips of which 20 invoiced, then DNI = cost of the 30 uninvoiced = GL 1405.
- **AC-28.9** (FR-28.11) Given scan `(01)00614141000012(10)LOT7(37)5`, then GTIN, batch LOT7 and qty 5 are filled.
- **AC-28.10** (FR-28.13) Given 60 released lines for carrier UPS across 12 customers, when grouped by customer with WAVE_MAX_LINES 20, then the preview shows waves of ≤ 20 lines, no shipment is split across waves, and one action creates all pick work.
- **AC-28.11** (FR-28.15) Given a rule "FG-100 → carton C12, 12 per carton, max 15 kg" and 30 picked units, when auto-pack runs, then containers of 12, 12 and 6 exist with valid SSCCs and gross weights.
- **AC-28.12** (FR-28.14) Given DMF-DE ships 100 units with 0.2 kg cardboard each at 0.50 EUR/kg, then the packing slip carries a statistical packaging fee of 10.00 EUR and the period packaging report shows 20 kg cardboard for DE.
- **AC-28.13** (FR-28.16) Given 10 returnable pallets shipped to C1 and 4 received back, then the C1 pallet balance is 6 and a return of 7 is rejected with `RTP_NEGATIVE`.

5.28.9 Test plan and fixtures

Fixtures: `warehouse-sim-1k` (1,000 orders, 120 products, 3 warehouses, batch/serial mix); `sscc-200` (GS1 examples, valid/invalid); `carrier-recorded` (HTTP recordings); `dropship-30`; `cogs-matrix-14` (per cell × costing method, accountant-reviewed).

Test	Type	Target	Pass criterion
Posting matrix	contract (ledger)	post / cancel	14/14 balanced; cancel mirrors exactly
Saga compensation	integration (Temporal)	injected ledger failure	no inventory movement remains; slip not POSTED
SSCC / GS1-128	unit, property	check digit, parser	200/200; all AI combinations round-trip
Pick-pack-ship 1k	E2E	simulated warehouse	0 mismatch, all SHIPPED < 60 min
Carrier adapters	contract	recordings	green
DNI = GL 1405	reconciliation	daily job	0.00

Reconciliation: Σ inventory issue cost = Σ journal Cr 1400 from fulfillment; DNI projection = GL 1405; line delivered_qty = Σ slip qty.

5.28.10 Deployment and operations

Helm `passage-fulfillment` (Go), `passage-carrier-gateway` (TS; egress via Envoy, provider secrets in Vault), `fulfillment-db`; PWA served by the shell. Config: `RELEASE_HORIZON_DAYS=2`, `WAVE_MAX_LINES=200`, `GS1_COMPANY_PREFIX` per entity, `COGS_AT_SHIPMENT=false`. Flags: `fulfillment.consolidation`, `fulfillment.asn`,

`fulfillment.mobile_pick`. Jobs: release every 15 min per warehouse; tracking poll 30 min; DNI check 02:00. SLOs: post p95 < 2 s, saga completion 99.9% within 60 s. Alerts: PENDING > 10 min, carrier error rate > 5%, DNI diff ≠ 0. Runbook: resume stuck saga, re-print label, manual tracking. Rollback: previous digest; Temporal workflow versioning lets in-flight sagas finish on old code.

5.28.11 Migration from AX / SAP

Source	Passage entity	Notes
CustPackingSlipJour/Trans with uninvoiced remainder	packing_slip, DNI opening balance	cost from InventTrans.CostAmountPhysical
WMSShipment, WMSPickingRoute (open); WHSContainerTable	shipment, pick_work, container	open picks reset to RELEASED; SSCC retained
LIKP/LIPS not goods-issued (WBSTK ≠ C)	shipment	A/B → RELEASED/PICKING
LIKP/LIPS goods-issued, not billed (FKSTK ≠ C)	packing_slip	cost from MSEG 601 DMBTR
VEKP/VEPO	container	EXIDV SSCC retained

Validation: DNI opening balance = legacy delivered-not-invoiced GL balance to the cent; open shipment quantities = legacy open deliveries. Not migrated: completed shipments, tracking history, carrier configuration (manual).

5.28.12 Dependencies, risks, open questions

- Depends on: w27 orders and holds; w16 inventory transactions/reservations; w20 cost snapshots; w17 POs; w11 journal; w14 period control; w8 tracking dimensions and GS1 ids; w5 PWA shell and document rendering.
- Risks: physical posting unfamiliar to SAP users (policy flag plus documented matrix); carrier sandbox instability (recorded fixtures gate CI; live sandbox nightly only).
- Open questions: (1) Default `cogs_at_shipment` for migrated SAP entities? Default true; decide by wave 29. (2) ASN EDI mapping owned by wave 28 or 35? Default 35; decide at wave 28 close.

5.28.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
release_horizon_days	warehouse	int	2	0–30	FR-28.1 release window	AR parameters → Updates → "Release to warehouse" / release rules	VL10 delivery creation date horizon (VL10 profile)
release_requires_full_atp	warehouse	bool	true	true/false	Partial release allowed when false	Release to warehouse "Partial release"	OVLK delivery split / partial delivery
consolidation_enabled	legal entity	bool	false	true/false	BR-28.8	Consolidated shipments (WHS)	OVLK combination rules (VL04 collective)
wave_max_lines / wave_max_orders	warehouse	int	200 / 50	1–2,000 / 1–500	FR-28.2, FR-28.13	WHS wave template thresholds	LT03 group limits / EWM wave template
short_pick_action	warehouse	enum	REMAINDER_BACKORDER	REMAINDER_BACKORDER, CANCEL_REMAINDER	FR-28.2	Picking list registration "Reduce quantity"	VL02N picking quantity < delivery qty handling
gs1_company_prefix / sscc_extension_digit	legal entity	text / int	— / 0	6–12 digits / 0–9	BR-28.5	WHS license plate number sequence (GS1)	HU number range (OVLS SSCC)
sscc_no_reuse_months	tenant	int	12	12–60	BR-28.5	n/a	n/a
cogs_at_shipment	legal entity	bool	false (true recommended for migrated SAP entities)	true/false	BR-28.1 matrix	AR parameters → Updates → "Post packing slip in ledger" (physical posting)	OBYC GBB-VAX at PGI (always)
posting_date_source	warehouse	enum	SHIP_TIMESTAMP	SHIP_TIMESTAMP, TODAY, MANUAL	BR-28.3	Packing slip posting date default	VL02N actual GI date
allow_negative_onhand_on_post	product policy (w16)	bool	false	true/false	§5.28.5 validation	InventModelGroup "Physical negative inventory"	MM negative stock allowed (OMJ1)
carrier_default / service_level_default	customer (w7), warehouse	text	—	configured carriers	FR-28.6	CustTable DlvMode / ShipCarrier defaults	KNVV-VSBED shipping conditions → route
carrier_retry_schedule	tenant	text	2s,8s,32s	list	BR-28.9	n/a	n/a
tracking_poll_minutes	tenant	int	30	5–240	FR-28.6	n/a	n/a
asn_enabled	legal entity	bool	false	true/false	FR-28.9	AIF DESADV outbound	DESADV output type LAVA
mobile_offline_tolerance_minutes	tenant	int	5	0–30	FR-28.11	n/a	n/a
packaging_fee_billable	legal entity	bool	false	true/false	BR-28.12	Packing material fees "Charge customer"	Packaging material pricing condition (KP00)
packaging_fee_rates	country (w48 pack)	table	—	material × rate per kg	FR-28.14	Packing material fees setup (EU)	n/a (country add-on)
auto_pack_enabled	warehouse	bool	true	true/false	FR-28.15	n/a	OVLS packing control per item category (auto-pack)
rtp_tracking_enabled / rtp_charge_days	legal entity	bool / int	false / 90	— / 30–365	FR-28.16	n/a	RTP special stock M/V
workbench_group_key_default	warehouse	enum	CUSTOMER	CUSTOMER, CARRIER, ZONE, PRODUCT	FR-28.13	Picking workbench grouping	VL10 list variant

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
dni_check_time	legal entity	time	02:00	—	FR-28.12 job	n/a (manual InventPosting report)	n/a (F.03)

5.28.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Shipments list / delivery monitor	list inquiry	warehouse, status, planned date, carrier, customer	shipment, order, customer, status, planned/actual ship, carrier, tracking	shipment	Shipments list page (WMSShipment)	VL06O / VL06F outbound delivery monitor
To release	list inquiry	warehouse, horizon, ATP status, hold	order, line, confirmed date, ATP, priority, hold	sales lines (confirmed, unheld)	Release to warehouse form	VL10 / VL10A delivery due list
Pick work / waves	list inquiry	wave, picker, zone, status	wave, lines, picked/short, picker, age	pick_work	Picking list registration (WMSPickingRoute)	LT22 / LT23 transfer order lists; VL06P
Packed not shipped	list inquiry	warehouse, age	shipment, containers, weight, carrier, label status	shipment (PACKED)	Shipments awaiting ship confirm	VL06G (for goods issue)
Packing slip journal	list inquiry / drill-down	order, customer, date range, status	slip, order, lines, qty, cost, journal ref, reversal	packing_slip	Packing slip journal (CustPackingSlipJour)	VL03N document flow; MB51 movement 601
Delivered not invoiced	operational report	legal entity, as-of, customer	slip, line, qty, cost, days since shipment, GL 1405 diff	DNI projection	Cost of units delivered (InventPosting) / Physical inventory by inventory dimension	VF04 due list value; F.03 recon
Carrier exceptions and cost	list inquiry / KPI	provider, error, age; cost per shipment	shipment, error, retries, cost quoted vs billed	carrier_shipment	Shipping carrier interface log	XSI / TM logs (VT02N)
SSCC register	list inquiry	prefix, date range, customer	sscc, container, shipment, printed_at, reused flag	container	License plate list	HU monitor HUMO
Short picks and backorders created	operational report	warehouse, period	line, planned, picked, remainder, reason	pick_work	Picking list registration log	VL06P picking variance
Packaging weights per material	statutory report (country pack)	legal entity, country, period	material, kg shipped, fee	packaging_fee lines	Packing material fees report	n/a (DE LUCID / FR CITEO exports via w48)
Returnable packaging balances	list inquiry	customer, material, age	balance, last shipped, last returned, days outstanding	returnable_packaging_balance	n/a	MB58
Picking workbench preview	drill-down	filter set, group key	wave, lines, orders, weight	pick_workbench	Sales picking workbench	VL10 grouped selection

5.28.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Pick list	wave creation / print	header: wave, warehouse, picker, zone; lines: product, dims, location, qty, barcode; footer: counts	PDF, ZPL (label printer), PWA	wave number `WV–`	none; keep 1 year
Packing slip / delivery note	packing-slip post	header: slip number, order, customer, ship-to, carrier, tracking, date; lines: product, description, qty, unit, batch/serial; footer: container summary, SSCCs, signature block	PDF, email, EDI DESADV/856 (w35), portal	`PS–` per legal entity, gap-free	commercial document; retain 7 years (10 in DE) with the invoice
GS1-128 container label	container close	SSCC (AI 00), GTIN (01), batch (10), qty (37), ship-to postal (420), order ref	ZPL, PDF	SSCC serial per prefix	GS1 General Specifications; no retention beyond SSCC register
Carrier label	:label	carrier-rendered label with tracking barcode	PDF, ZPL (provider payload)	provider tracking number	retain label_ref 1 year
Manifest / end-of-day close	:manifest	carrier, pickup date, package count, weights, tracking list	PDF, provider API	provider manifest id	retain 1 year
ASN	packing-slip post (asn_enabled)	shipment, containers with SSCC hierarchy, lines, quantities, carrier, tracking	JSON event; EDI 856 / DESADV through w35	shipment number	trading-partner agreement; retain 7 years
Bill of lading	shipment with `bol_required`	shipper, consignee, carrier, freight terms, commodity lines, weights	PDF (basic; full BOL w34)	`BOL–`	US carriage record; retain 3 years
Packaging fee statement	period close per country	material, kg, rate, fee	PDF, CSV	none	per country pack (w48)

5.28.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Release to warehouse	every 15 min per warehouse	warehouse	key = (warehouse, line, confirmed version)	< 1 min per 10k lines	retry; lines stay confirmed	lines eligible but unreleased > 1 h
Tracking poll	every 30 min	provider	last event timestamp per shipment	< 5 min	provider error → exceptions queue	poll failures > 5%
DNI reconciliation	02:00 daily	legal entity	key = (entity, date)	< 5 min	rerun; diff report	diff ≠ 0.00
Saga watchdog	every 5 min	—	PENDING slips	< 10 s	resume or compensate	PENDING > 10 min
Manifest / end-of-day close	per warehouse cutoff (default 17:00 local)	warehouse × carrier	manifest id per (warehouse, carrier, date)	< 2 min	retry; manual manifest	shipments LABELLED but unmanifested after cutoff
SSCC sequence check	nightly 00:30	prefix	—	< 1 min	alert only	duplicates or check-digit failures > 0
Packaging fee accrual report	first day of month	country	key = (country, period)	< 2 min	rerun	missing rate for material
Returnable packaging aging	nightly 03:00	customer	key = (customer, date)	< 1 min	rerun	balances outstanding > rtp_charge_days

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Label and document cleanup	weekly	—	—	< 10 min	skip	storage growth > plan

5.28.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
WarehouseView	GET shipments, tracking, DNI; Fulfillment workspace	none	Warehouse worker, Customer service, Finance
WarehouseRelease	releases:run; Release board; Picking workbench	none	Warehouse supervisor
WarehousePick	waves, :confirm-pick; Pick list, Mobile pick	with PackingSlipPost when `pick_post_separation`	Picker
WarehousePack	containers, :close; Packing station; auto-pack	none	Packer
WarehouseShip	:rate, :label, :manifest; Carrier exceptions	none	Shipping clerk
PackingSlipPost	:post-packing-slip, /packing-slips/{id}:cancel	with SalesOrderMaintain on the same order; with InvoicePost (w29) when four-eyes	Shipping clerk (post), Warehouse supervisor (cancel)
PackingRuleMaintain	packing rules, container types, packaging materials	none	Warehouse admin
CarrierAdmin	provider configuration, secrets rotation (Vault reference only)	with WarehouseShip (logged)	Logistics admin
PackagingFeeMaintain	packaging fee rates (country), statements	with InvoiceMaintain (w29)	Compliance / Finance
ReturnablePackagingMaintain	RTP balances adjustments	with WarehouseReceive	Logistics admin

5.28.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
FUL-0001	error	Shipment must be PACKED with no active hold to post	status or hold	Pack lines; release hold	§5.28.5 validation
FUL-0002	error	Inventory period {period} is closed for posting date {date}	BR-28.3	Use an open period or reopen (w14)	BR-28.3
FUL-0003	error	Negative on-hand for {product} at {warehouse}/{location}	w16 policy	Adjust stock or allow negative	§5.28.5 validation
FUL-0004	error	Over-pick: {qty} exceeds planned {planned}	BR-28.7	Reduce quantity	BR-28.7
FUL-0005	error	Over-ship exceeds customer tolerance {pct}%	BR-28.7	Reduce or raise tolerance	BR-28.7
FUL-0006	error	Batch or serial required for {product}	BR-28.10	Scan tracking dims	BR-28.10
FUL-0007	error	SSCC {sscc} failed check digit or already used	BR-28.5	Regenerate label	BR-28.5
FUL-0008	warning	Carrier {provider} failed after 3 retries; shipment moved to exceptions	BR-28.9	Retry, switch provider or enter manual tracking	BR-28.9
FUL-0009	error	Packing slip PENDING for {minutes} min; saga stalled	BR-28.4	Runbook: resume stuck saga	BR-28.4
FUL-0010	error	Cannot cancel: packing slip {id} is invoiced	FR-28.5	Credit note (w29) then cancel	FR-28.5
FUL-0011	error	Consolidation mismatch: {field} differs	BR-28.8	Ship separately	BR-28.8
FUL-0012	warning	No packing rule for {product}; line left unpacked	BR-28.11	Pack manually or add a rule	FR-28.15
FUL-0013	error	Container exceeds max weight {max} kg	BR-28.11	Close container and start a new one	BR-28.11
FUL-0014	error	Packaging fee rate missing for material {material} in {country}	BR-28.12	Load the country rate table	FR-28.14
FUL-0015	error	Returnable packaging balance would become negative for {customer}	BR-28.13	Correct the return quantity	FR-28.16
FUL-0016	info	Workbench created {n} waves from {lines} lines	FR-28.13	none	BR-28.14
FUL-0017	warning	GS1-128 scan could not be parsed: {raw}	FR-28.11	Rescan or enter manually	FR-28.11

5.28.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Release run (10k lines)	96 runs per warehouse per day	— / 60 s / 120 s	200 lines/s	shipments 5M × 2 KB = 10 GB	shipments (warehouse, status, planned_ship_date)	runs queue per warehouse; next run skipped if previous running
Pick confirm (line)	500k per day; peak 50/s	50 / 150 / 400 ms	100/s	pick_work 3 GB	(wave, status)	PWA queues offline up to 5 min
Container close (SSCC + label)	200k per day	100 / 300 / 800 ms	50/s	containers 2 GB; labels object store 20 GB	unique sscc	label rendering async; SSCC issued synchronously
Packing-slip post (saga)	100k per day; peak 20/s	600 ms / 2 s / 4 s	20/s per replica	slips 10M × 1 KB + journal refs	partition by month; (order_id)	saga workers scale; slips remain PENDING ≤ 60 s p99 under 3×
Carrier rate / label	100k per day	400 ms / 2 s / 5 s (provider-bound)	provider limit	carrier_shipment 3 GB	(provider, status)	circuit breaker per provider; exceptions queue
Tracking events	1M per day	5 / 30 / 60 s lag	50 events/s	10 GB (events jsonb, 1-year retention)	(shipment_id, event_time)	poll interval doubles; webhooks unaffected
DNI reconciliation	1 per entity per day	— / 5 min / 10 min	—	none	DNI projection (slip, status)	run moved to low-traffic window
Auto-pack (100 lines)	50k per day	100 / 400 / 1,000 ms	20/s	none	packing_rule (product, customer)	manual pack fallback

5.28.20 Localization and compliance hooks

- Packaging fees and extended-producer-responsibility reporting (DE VerpackG/LUCID, FR CITEO, ES RAP) are country-pack data and exports (w48) on top of FR-28.14; the rate table and the period statement are the extension points.
- Delivery note legal content: DE and IT require sequential delivery-note numbering and specific fields (IT DDT reason for transport); the `PS-` sequence is gap-free per entity and the docgen template exposes a `country_fields` slot.
- GS1 prefixes are issued per country member organization; the entity holds one or more prefixes, each with its own gap-free serial.
- Dangerous goods, export declarations (EU AES / US AES) and sanctioned-party screening are explicitly out of scope and arrive through w33/w34/w48 consumers of `shipment.released`.

- Weights and dimensions store SI units; US entities display lb/in through the w6 unit conversion with half-even rounding at 2 decimals.
- Time zones: posting date derives from the warehouse time zone (BR-28.3); carrier cutoffs are warehouse-local.
- Privacy: ship-to contact names and phone numbers on labels are personal data; label artifacts purge after 1 year (w48 retention policy).

5.28.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.14.4 #140 — Picking routes setup / Sales picking workbench (R3)	PARTIAL-28	ADDED as FR-28.13	grouped wave creation with preview
AX §12.15.1 #145 — Product → Packaging (packaging materials, PI container packaging)	PARTIAL-28	ADDED as FR-28.14	packaging_material on product units
AX §12.15.4 #149 — Packaging units / Packing material fees setup (EU)	PARTIAL-28	ADDED as FR-28.14	rates per country pack
AX §12.23.1 #200 — Container packaging (PI container types)	PARTIAL-28	ADDED as FR-28.14 / FR-28.15	container_type with packaging material
AX §12.23.4 #210 — Container types, Packaging groups	PARTIAL-28	ADDED as FR-28.15	packaging group = rule scope
SAP §13.17 #207 — OVLS/OVLV packing instructions (POP1/POP2/POP3/POF1), packaging material types, HU number ranges	PARTIAL-28	ADDED as FR-28.15 (packing instructions); SSCC ranges already FR-28.3	
SAP §13.16 #188 — EWM packaging specifications (/SCWM/PACKSPEC) (PARTIAL-31, note proposes 28/31)	PARTIAL-31	ADDED as FR-28.15 (basic rules); multi-level EWM specifications DEFERRED to wave 31	
SAP §13.13 #147 — Returnable transport packaging (special stock M/V)	GAP (propose 28)	ADDED as FR-28.16	balances only; charging via wave 30

5.29 Wave 29: Billing, Invoicing & Revenue Recognition

5.29.1 Scope and business context

Wave 29 ships the `billing` and `revrec` services. Posted packing slips and eligible order lines become invoices (one-off, consolidated, subscription, milestone); free-text invoices and credit notes post directly; layouts render to PDF with an e-invoice document for wave 35. Every posted invoice creates AR open items, tax postings (w13) and journal entries and, where the contract has multiple performance obligations or deferred terms, a revenue contract with standalone selling price allocation and recognition schedules. Out of scope: cash application (wave 30), intercompany invoicing (hook only; wave 40), project billing (wave 41), Peppol transmission (wave 35), metering sources beyond API-posted usage. Added from the coverage registers (§5.29.21): subscription groups, fee categories and indexation; expense deferrals; customer self-billing; bundle (solution) lines; installment terms at post; document delivery routing.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Bundle allocation lives in spreadsheets: D365 RevRecSchedule needs revenue prices per item and cannot explain allocation; SAP RAR (FARR_D_*) is a separate engine with its own posting run and FARR_RECON reconciliation	Rev rec bolted on after billing; allocation inputs not on the order	Revenue contract created at post from order lines with SSP order BR-29.2; allocation and schedules visible on the contract and in the invoice trace	Close-time rev rec effort 2–3 days → < 2 h review; 40/40 fixtures reproducible
Invoice totals differ from the order by cents (SalesLine rounding per form letter; MWST vs. BSEG rounding)	Each document re-rounds	Engine net reused; per-line tax; totals property-tested	Cent discrepancies ~0.5% of invoices → 0
Billing is batch: VF04 due list and AX summary update run nightly	Scheduled jobs over document-flow tables	`packing-slip.posted` creates the proposal at once; post p95 < 3 s; consolidation is policy, not batch	Shipment-to-invoice 12–24 h → < 5 min for non-consolidated accounts
Deferred revenue roll-forward rebuilt by hand from GL 2400 and exports	No system roll-forward that ties to GL	`GET /revrec/roll-forward` reconciled nightly to GL 2400 with drill-down	Preparation 1 day → 0; nightly diff 0.00
Credit note on a partially recognized invoice needs manual journals (manual schedule edits; RAR contract change)	Credit notes ignore recognition state	BR-29.6 reverses recognized and deferred portions automatically	Manual rev rec journals per month 10–30 → 0

5.29.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Invoice from shipment	SalesFormLetter_Invoice, CustInvoiceJour/CustInvoiceTrans, summary update	VF01/VF04, VBRK/VBRP, copy control VTFL	invoice (SHIPMENT, CONSOLIDATED)	full
Free-text invoice	CustInvoiceTable/CustInvoiceLine	FB70; order-related billing (DR)	invoice type FREE_TEXT	full
Credit note	Credit note (negative CustInvoiceJour), free-text credit	Credit memo request G2 → VF01, VBRK VBTYP O	credit_note	full
Billing plans / subscriptions	D365 Subscription billing; milestone billing	FPLA/FPLT billing plans; BRIM	subscription, milestone_schedule	partial (usage basic)
AR open item creation	CustTrans / CustTransOpen	BSID / BSAD, BKPF/BSEG	receivables.open_item via invoice.posted	full
Tax	TaxTrans, TaxGroup/TaxItemGroup	FTXP, MWST, BSET	tax engine (w13) at post	full
Revenue recognition	RevRecSchedule, RevRecContract, revenue prices (D365)	RAR: FARR_D_CONTRACT / POBS / DEFITEM	revenue_contract, performance_obligation, revenue_schedule	full
Deferred revenue postings	Deferred revenue posting on item; schedule processing	RAR posting run, contract liability	recognition run to 2400/4000	full
Layout / e-invoice	SSRS SalesInvoice; Electronic reporting	Smart Forms/Adobe Forms; DRC	docgen template + UBL 2.1 document	partial (transmission w35)
Audit trace	CustInvoiceTrans.InventTransId	Document flow VBFA	GET /invoices/{id}/trace	improved
Service subscriptions (AX)	SMASubscriptionTable, subscription groups, fee categories, index rules, subscription periods, fee transactions	Contract billing plans (VF04 periodic), price adjustment via conditions	subscription groups, fee categories, index rules, period templates on `subscription`	full
Expense deferrals	D365 Subscription billing deferral schedules (expense), RED DeferralSchedule	Prepaid expense accruals (ACCR / FBS1 recurring)	expense_deferral schedule from w19 AP invoice lines	full
Customer self-billing	—	VSB1 / VKA self-billing (GSVERF, EDI 820)	self_billing_statement reconciliation	full (EDI mapping w35)

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Solution / bundle orders	Bundle product (kit) on SalesLine	Solution order (bundles), item category group for bundles	bundle line explosion with mixed obligations (FR-29.17)	full
Installment terms	Payment schedule (CustPaymSched) on invoice	OBB9 installment terms (T052R)	installment split at post → N open items	full (terms definition w7)
Document routing	Print management (sales documents), form notes	NAST output determination (VV31), medium per partner	document_delivery_preference per customer × document	partial (conditional routing w46)

5.29.3 Functional requirements

- **FR-29.1 Invoice from shipment** — Must. Proposals from posted packing slips (and order lines for non-stocked items), one invoice per order by default, lines carrying qty, price, discounts, charges and tax codes. Notes: gap-free `INV-` per entity and fiscal year; invoice date ≥ packing slip date.
- **FR-29.2 Consolidated invoices** — Must. Combine slips for the same invoice account, currency, terms and cycle (daily/weekly/monthly).
- **FR-29.3 Posting** — Must. Atomic: journal (Dr 1200 gross; Cr 4000 per line account; Cr 2300 tax; Cr 2400 deferred obligations; COGS/DNI per BR-28.1), open item creation, `invoice.posted`. Totals = Σ line nets + charges + tax after per-currency rounding (property-tested).
- **FR-29.4 Free-text invoices** — Must. Account, dimensions, amount, tax code; recurring templates.
- **FR-29.5 Credit notes** — Must. Reference an original (whole or lines) or standalone; negative open item; reverse revenue, tax and recognition.
- **FR-29.6 Tax** — Must. Call the tax engine at proposal and post; reject post if the result moved > 0.01 without re-review; no silent zero tax when the engine is unavailable.
- **FR-29.7 Revenue contracts** — Must. Performance obligations per order (one per line or bundle templates per product), SSP (explicit, list, % of list, residual), proportional allocation, schedules.
- **FR-29.8 Recognition methods** — Must. POINT_IN_TIME (packing slip or invoice), RATABLE (daily or monthly), MILESTONE (manual/event), USAGE (posted usage); period-end run or event posts Dr 2400 / Cr 4000.
- **FR-29.9 Roll-forward** — Must. Opening, additions, recognized, adjustments, closing per period and contract, tied to GL 2400.
- **FR-29.10 Subscriptions** — Should. Monthly/quarterly/annual, in advance or arrears, daily proration on mid-period start, cancel and plan change, usage lines, daily billing job.
- **FR-29.11 Documents** — Must. PDF per entity and language; UBL 2.1 JSON for wave 35.
- **FR-29.12 Audit trace** — Should. invoice → packing slip → order line → price explain → schedules in one response.
- **FR-29.13 Intercompany hook** — Could. `invoice.posted` carries `intercompany=true` for intercompany parties (wave 40).
- **FR-29.14 Subscription groups, fee categories and indexation** — Should. The system shall extend `subscription` with subscription groups (defaults for cycle, terms, revenue account), fee categories per line (recurring, one-time, usage, index-linked), period templates (calendar month, 4-4-5, anniversary) and index rules (percentage or external index with effective date) that re-price subscription lines on the next bill with an indexation note on the invoice. Notes: AX SMA subscription semantics; an index run previews affected subscriptions; indexation history is kept per line; accrual of subscription revenue uses FR-29.8 RATABLE.
- **FR-29.15 Expense deferrals** — Should. The system shall create deferral schedules for prepaid expenses from w19 vendor invoice lines (or manual), with method STRAIGHT_LINE (daily or monthly) over a term, short-term / long-term split (within 12 months → 1300, beyond → 1310), and a monthly recognition run posting Dr expense account / Cr 1300; schedules are visible on the same roll-forward as revenue (FR-29.9) with `kind = EXPENSE`. Notes: D365 deferral schedule parity; a credit from the vendor reverses remaining schedule lines; roll-forward ties to GL 1300 + 1310.
- **FR-29.16 Customer self-billing** — Could. The system shall import a customer self-billing statement (lines: delivery or packing slip reference, quantity, unit price, amount, tax; EDI 820 / GSVERF via w35 or CSV) and reconcile it line by line against posted or proposed invoices: matching lines mark the invoice as self-billed; price differences within `self_billing_tolerance` post an adjustment credit/debit; other differences are listed for review. Notes: SAP VSB1 / VKA; the customer's document number is stored as `external_invoice_ref`; open items settle against the customer's remittance (w30) by that reference.
- **FR-29.17 Bundle (solution) lines** — Should. The system shall accept a bundle product on an order line (w8 kit/bundle definition) and at invoice proposal explode it into components with mixed revenue types (goods POINT_IN_TIME, service MILESTONE, subscription RATABLE) under one revenue contract (FR-29.7), printing the bundle as one line with components itemized beneath it when `print_components`. Notes: SAP solution order; allocation per BR-29.1; a component shipped later is invoiced when shipped while the contract already exists.
- **FR-29.18 Installment terms** — Should. The system shall, when the invoice's payment terms (w7) carry an installment plan (n installments with percentage and offset days, or fixed dates), create one open item per installment at post with the same invoice number and an installment suffix, each with its own due date and cash-discount date; dunning and aging operate per installment. Notes: SAP OBB9 / T052R and AX payment schedules; Σ installments = gross exactly with residual cents on the last installment; a credit note settles installments latest-first unless specified.
- **FR-29.19 Document delivery routing** — Should. The system shall hold a delivery preference per customer (and optionally per invoice account and document type: invoice, credit note, statement, confirmation, packing slip) with channel EMAIL, PORTAL, PRINT, EDI, NONE, recipients, copies and language, and route generated documents accordingly from the post saga with a delivery log and resend action. Notes: AX print management "per account" destinations and form notes; SAP NAST medium per partner; conditional query-based routing and designer-level overrides arrive with wave 46 and read the same preference entity.

5.29.4 Data model

invoice (CustInvoiceJour / VBRK)

Field	Type	Required	Notes
id, invoice_number	uuid, text	Y	gap-free per entity + fiscal year
legal_entity_id, invoice_account_id, customer_id	uuid	Y	
invoice_type	enum	Y	SHIPMENT, CONSOLIDATED, FREE_TEXT, SUBSCRIPTION, MILESTONE, CREDIT_NOTE
status	enum	Y	PROPOSAL, POSTED, PARTIALLY_SETTLED, SETTLED, REVERSED
invoice_date, posting_date, due_date	date	Y	due from terms
currency, exchange_rate_id	char(3), uuid	Y	three-currency amounts (w12)
net_amount, charge_amount, tax_amount, gross_amount	numeric(19,4)	Y	gross = net + charges + tax
cash_discount_date, cash_discount_amount	date, numeric	N	
journal_posting_id, tax_document_id, original_invoice_id	uuid	Y/N	original for credit notes
document_ref, einvoice_ref	text	N	PDF, UBL

invoice_line (CustInvoiceTrans / VBRP): id, invoice_id, line_no, order_line_id, packing_slip_line_id, product_id, description, qty, unit, unit_price, disc_pct_1/2, line_net, tax_code, tax_amount, revenue_account, dimensions, cost_amount, performance_obligation_id, price_explain_ref.

revenue_contract: id, order_id, customer_id, currency, transaction_price, allocation_method, status (OPEN | COMPLETE | CANCELLED). **performance_obligation**: id, contract_id, product_id | bundle_component, ssp_amount, ssp_source (EXPLICIT | LIST | PCT_OF_LIST | RESIDUAL), allocated_amount, method, term_start, term_end, satisfied_at, deferred_account 2400, revenue_account 4000. **revenue_schedule**: id, obligation_id, period_id, scheduled_date, amount, status (SCHEDULED | RECOGNIZED | REVERSED), journal_posting_id.

subscription: id, customer_id, order_id, plan, billing_cycle, bill_in_advance, start_date, end_date, next_bill_date, proration (DAILY), status (ACTIVE | PAUSED | CANCELLED), lines[] (product, qty, unit_price, usage_based, usage_unit).

Invariants: gross = Σ line_net + charges + tax exactly after per-line rounding; Σ allocated = transaction_price (residual cents to the largest obligation); Σ schedules per obligation = allocated_amount; POSTED invoices are immutable; per period opening + additions − recognized ± adjustments = closing = GL 2400. Indexing: invoices (invoice_account_id, status, due_date), unique number per (entity, fiscal_year), partitioned by posting month; schedules (period_id, status).

5.29.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/billing/proposals	Build from slips / orders / subscriptions with consolidation	Idempotency-Key	InvoiceMaintain
POST	/api/v1/billing/invoices/{id}:post	Post (tax, journal, AR)	key = invoice id	InvoicePost
POST	/api/v1/billing/free-text-invoices, /credit-notes, …/{id}:post	Free-text and credit notes	Idempotency-Key	InvoiceMaintain / CreditNotePost
GET	/api/v1/billing/invoices/{id}/document?format=pdf|ubl; …/trace	Documents; audit trace	—	InvoiceView / Portal
POST	/api/v1/revrec/contracts/{id}:allocate; /milestones/{ob}:satisfy	Allocation; milestones	Idempotency-Key	RevRecMaintain
POST/GET	/api/v1/revrec/recognition-runs; /roll-forward?period=	Period run; roll-forward	key = period	RevRecPost / RevRecView
POST	/api/v1/billing/subscriptions, …/{id}:change, :cancel; /usage	Subscriptions and usage	Idempotency-Key	SubscriptionMaintain

Validation: post rejects closed AR/GL period, lines without revenue account, tax engine unavailable (503 `TAX_UNAVAILABLE`), credit beyond the original's remaining creditable amount.

Event	Trigger	Payload fields	Consumers
dev.passage.billing.invoice.posted.v1	post	invoice_id, number, type, customer, invoice_account, currency, dates, lines[{order_line_id, product, qty, net, tax, revenue_account, obligation_id}], totals, journal_posting_id, intercompany	receivables, revrec, ledger, pricing, portal, sales
passage.billing.credit-note.posted.v1	post	credit_note_id, original_invoice_id, lines, totals, journal_posting_id	receivables, revrec, ledger, portal
dev.passage.revrec.contract.allocated.v1	allocate	contract_id, obligations[{id, ssp, allocated, method}]	analytics, audit
dev.passage.revrec.schedule.recognized.v1	run / event	schedule_id, obligation_id, period_id, amount, journal_posting_id	ledger, analytics
dev.passage.billing.subscription.billed.v1	daily job	subscription_id, invoice_id, period, proration	portal, analytics

5.29.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Billing workspace	tiles: ready to invoice, proposals, failed posts, deferred balance, subscriptions due	counts, amounts	InvoiceView
Billing due list	filter, select, propose, consolidate	customer, slips, amount, cycle	InvoiceMaintain
Invoice proposal / detail	edit charges, review tax, post, print, credit	lines, tax breakdown, totals, obligations	InvoicePost
Free-text invoice	add lines, template, post	account, dims, amount, tax code	InvoiceMaintain
Revenue contract / roll-forward	view allocation, adjust SSP, satisfy milestone; run period, drill	obligation, SSP, allocated, recognized, deferred; opening…closing, GL 2400	RevRecMaintain / RevRecView
Subscriptions	create, change plan, pause, cancel	plan, cycle, next bill, proration preview	SubscriptionMaintain

Interaction rules: proposals show tax and journal preview before post; multi-select post ≤ 500 invoices with per-row results; roll-forward drill-down opens schedules by period; column sets personalizable.

5.29.7 Business rules and edge cases

- **BR-29.1 ASC 606 allocation worked example.** Bundle: hardware (SSP 8,000.00 USD, POINT_IN_TIME at packing slip), installation (SSP 1,500.00, MILESTONE), 12-month support (SSP 2,500.00, RATABLE). Σ SSP 12,000.00; contract price 10,800.00 (10% bundle discount). Factor 10,800 / 12,000 = 0.9 → hardware 7,200.00, installation 1,350.00, support 2,250.00. Invoice at shipment 2027-03-10: Dr 1200 10,800.00 (plus tax), Cr 4000 7,200.00, Cr 2400 3,600.00. Installation satisfied 2027-03-25: Dr 2400 / Cr 4000 1,350.00. Support 2027-03-10 to 2028-03-09 at 2,250.00 / 12 = 187.50 per month; first partial month 22/31 × 187.50 = 133.06; the last period absorbs the residual so Σ = 2,250.00. With a contract price of 10,799.99 the allocation 7,199.99 / 1,349.99 / 2,249.99 sums to 10,799.97; the 0.02 residual goes to the largest obligation (hardware 7,200.01).
- **BR-29.2 SSP determination order.** Explicit SSP per product × customer group → list price (PRICE access 7) → % of list per product group → residual (at most one obligation per contract).
- **BR-29.3 Rounding.** Line net half-even to minor units; tax per line (w13) then summed unless the engine returns document-level rounding; allocation at 2 decimals with residual rule; ratable schedules at 2 decimals, last period absorbs the residual.
- **BR-29.4 Dates.** invoice_date ≥ slip posting date; posting_date in an open AR period; due_date = invoice_date + terms (incl. end-of-month variants, w7); cash discount date per terms.
- **BR-29.5 Consolidation** groups by (invoice_account, currency, terms, cycle, address if `split_by_address`); ≤ 2,000 lines; a slip belongs to at most one invoice.
- **BR-29.6 Credit note on partially recognized revenue** reverses recognized revenue proportionally (Dr 4000) and unrecognized deferred (Dr 2400) in the credit period; later schedules become REVERSED.
- **BR-29.7 State machines.** Invoice: PROPOSAL → POSTED → PARTIALLY_SETTLED → SETTLED; POSTED → REVERSED only same posting date with no settlements. Schedule: SCHEDULED → RECOGNIZED → REVERSED (credit note only).
- **BR-29.8 Proration** is daily: amount × service days / period days, including 29 February; a mid-period plan change credits unused old-plan days and bills new-plan remaining days on one invoice; cancellation effective date ≥ today.
- **BR-29.9 Gap-free numbering.** Two posts of one proposal share the Idempotency-Key and return the same invoice; the number is drawn from a serialized sequence per (entity, fiscal_year) in the saga's final step, after tax and journal succeed, so a failed post consumes no number.
- **BR-29.10 Usage records** accepted up to 45 days after period end; later ones bill in the next period with `late=true`.

- **BR-29.11 Installment split.** Installment i amount = round_half_even(gross × pct_i, minor units); the last installment = gross − Σ previous; due_date_i = invoice_date + offset_i (or the fixed date) rolled per terms rules; tax is reported once on the invoice, not per installment; the installments share one journal posting.
- **BR-29.12 Indexation** applies the index rule on the first bill date ≥ effective date; new unit price = round_half_even(old × (1 + pct/100) or old × index_new / index_base, 6 decimals); the invoice line text shows the old price, factor and new price; indexation never re-prices already-billed periods.
- **BR-29.13 Expense deferral** schedule amount per period = total × days in period / term days (daily) or total / periods (monthly) at 2 decimals with the residual on the last period; the short-term portion is recomputed at each period close as the next 12 months of unrecognized amounts and reclassified 1310 → 1300.
- **BR-29.14 Self-billing reconciliation.** Match key = packing slip or delivery number + product; quantity must equal exactly; price difference ≤ `self_billing_tolerance` (default 0.5% or 5.00) posts an adjustment credit note (customer lower) or debit note (customer higher) referencing the invoice; unmatched statement lines create review tasks; an invoice may be self-billed once.
- **BR-29.15 Document routing** evaluates (customer, invoice account, document type) → (customer, document type) → (customer) → entity default; EDI channel requires an active w35 trading-partner agreement else falls back to EMAIL with a warning; PRINT queues to the w5 print service; delivery failures retry 3× then appear in the Billing workspace tile "undelivered documents".

5.29.8 Acceptance criteria

- **AC-29.1** (FR-29.3) Given 1,000 random invoices in USD, EUR and JPY, then gross = Σ line_net + charges + tax for every invoice (property).
- **AC-29.2** (FR-29.7) Given the BR-29.1 bundle, then obligations are 7,200.00 / 1,350.00 / 2,250.00 and the first postings match the example to the cent.
- **AC-29.3** (FR-29.7, FR-29.8) Given the 40 ASC 606 fixtures, then allocation and recognition per period equal the accountant-reviewed workbook for 40/40.
- **AC-29.4** (FR-29.9) Given 2027-03 additions 3,600.00 and recognized 1,483.06, then closing 2,116.94 = GL 2400.
- **AC-29.5** (FR-29.2) Given 3 slips for one account on a weekly cycle, then one invoice is proposed and each slip becomes INVOICED on post.
- **AC-29.6** (FR-29.5) Given the posted 10,800.00 invoice with installation recognized, when a full credit note posts, then AR −10,800.00, recognized revenue reversed and remaining deferred reversed, with schedules REVERSED.
- **AC-29.7** (FR-29.10) Given a monthly 300.00 EUR subscription starting 2028-02-10 billed in advance, then the first invoice is 300 × 20/29 = 206.90 EUR.
- **AC-29.8** (FR-29.6) Given the tax engine is down, when posting, then 503 `TAX_UNAVAILABLE` and no journal or open item exists.
- **AC-29.9** (FR-29.4) Given a DMF-DE free-text line to 4900 with 19% VAT of 100.00 EUR, then Dr 1200 119.00 / Cr 4900 100.00 / Cr 2300 19.00.
- **AC-29.10** (FR-29.18) Given terms "3 installments 30/30/40 at +0/+30/+60 days" and a gross of 1,000.01 USD invoiced 2027-03-10, then open items of 300.00 (due 03-10), 300.00 (04-09) and 400.01 (05-09) exist with one invoice number and one journal posting.
- **AC-29.11** (FR-29.14) Given a monthly subscription line at 100.00 EUR with a 3% index rule effective 2027-04-01, when the April bill runs, then the line bills 103.00 with the indexation note and March remains 100.00.
- **AC-29.12** (FR-29.15) Given a 12,000.00 USD annual insurance invoice (w19) dated 2027-03-01 deferred monthly, then 1,000.00 recognizes each month Dr 6300 / Cr 1300, and at 2027-03 close the long-term portion is 0.00 and short-term 11,000.00.
- **AC-29.13** (FR-29.16) Given a self-billing statement with 10 lines of which 9 match and one prices 0.3% lower, then 9 invoices are marked self-billed, one adjustment credit note posts for the difference, and no review task exists.
- **AC-29.14** (FR-29.19) Given customer C3 with preference EMAIL for invoices and PORTAL for statements, when an invoice posts, then the PDF is emailed to the two configured recipients within 60 s and the delivery log shows status DELIVERED.
- **AC-29.15** (FR-29.17) Given a bundle product B-1 (hardware, installation, 12-month support) ordered at 10,800.00 USD, when invoiced, then one revenue contract with three obligations exists per BR-29.1 and the PDF prints one bundle line with three itemized components.

5.29.9 Test plan and fixtures

Fixtures: `asc606-40` (bundles with SSPs, discounts, milestones, modifications; workbook `fixtures/revrec/asc606-40.xlsx`); `invoices-random-1k`; `subscription-edges-25` (mid-period, leap day, plan change, cancel); `consolidation-12`; `credit-notes-15`.

Test	Type	Target	Pass criterion
Totals	property	post	gross = Σ components, 3 currencies
ASC 606	unit	revrec	40/40 match workbook
Roll-forward = GL 2400	reconciliation	nightly	0.00
Proration edges	unit	subscription	25/25
Posting saga	integration	billing ↔ tax ↔ ledger ↔ receivables	no partial state on injected failures
Documents	snapshot	docgen templates	PDF text and UBL schema valid
Billing ↔ fulfillment/receivables	contract (Pact)	APIs	green

Reconciliation: Σ POSTED gross − credit notes = AR additions = GL 1200 debits from billing; recognized revenue = GL 4000 credits from billing + revrec; DNI reversals = slip costs invoiced.

5.29.10 Deployment and operations

Helm `passage-billing`, `passage-revrec` (Go); `billing-db`, `revrec-db`; docgen templates in a ConfigMap per entity. Config: `BILLING_MAX_LINES=2000`, `BILLING_BATCH_MAX=500`, `REVREC_RESIDUAL_RULE=largest`, `USAGE_LATE_DAYS=45`. Flags: `billing.consolidation`, `billing.subscriptions`, `revrec.usage`, `billing.einvoice`. Jobs: subscription billing 03:00; recognition run on period-close trigger (w14) plus nightly preview; roll-forward check 02:30. SLOs: post p95 < 3 s, 99.95% availability. Alerts: failed posts > 1% in 15 min, roll-forward diff ≠ 0, run > 30 min. Runbook: re-run proposal, same-day reversal, regenerate document. Rollback: previous digest; recognition run idempotent per period.

5.29.11 Migration from AX / SAP

Source	Passage entity	Notes
CustInvoiceJour/Trans and CustInvoiceTable/Line with open CustTransOpen	invoice (POSTED), lines	header amounts and due dates; lines for trace
RevRecSchedule / RevRecContract (D365)	revenue_contract, obligations, unrecognized schedules	unrecognized = opening 2400
VBRK/VBRP with open BSID	invoice, lines	VBTYP O → CREDIT_NOTE
FARR_D_CONTRACT / POBS / DEFITEM (RAR)	revenue_contract, obligations, schedules	deferred items per POB → schedules
FPLA/FPLT open billing plans	subscription / milestone_schedule	

Validation: Σ migrated open balances = legacy AR trial balance per customer; Σ unrecognized schedules = legacy deferred revenue balance. Not migrated: settled invoices before the current fiscal year (lakehouse), SSRS/Smart Forms layouts (re-templated), RAR posting history.

5.29.12 Dependencies, risks, open questions

- Depends on: w28 packing slips; w27 lines and terms; w13 tax; w11/w12 journal and currencies; w14 period control; w26 explain refs and rebate basis; w5 document rendering and gap-free sequences.
- Risks: gap-free numbering under saga failure (BR-29.9); missing SSP for migrated products (residual method with warning report).
- Open questions: (1) Document- vs. line-level tax rounding per country? Default line level unless the country pack says otherwise; decide by wave 30. (2) Subscription invoices gap-free per entity or per subscription? Default per entity; decide at wave 29 mid-point.

5.29.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
summary_update_default	customer (w7) with entity default	enum	ORDER	ORDER, INVOICE_ACCOUNT, AUTOMATIC, NONE	FR-29.2 consolidation	AR parameters → Summary update tab (Summary update for invoices)	VOFA billing type collective invoice / VTFL copy control split criteria
consolidation_cycle_default	customer	enum	NONE	NONE, DAILY, WEEKLY, MONTHLY	FR-29.2	Summary update parameters	Billing schedule (KNVV-PERFK calendar)
split_by_address	legal entity	bool	false	true/false	BR-29.5	Summary update parameters "Split by delivery address"	VTFL split criteria (ZUKRI)
invoice_date_source	legal entity	enum	SHIP_DATE	SHIP_DATE, TODAY, MANUAL	BR-29.4	Invoice posting "Invoice date" default	VOFA / VTFL billing date determination (FKDAT)
invoice_number_scope	legal entity	enum	ENTITY_FISCAL_YEAR	ENTITY_FISCAL_YEAR, ENTITY, ENTITY_DOC_TYPE	BR-29.9	Number sequences (CustInvoiceId)	VOFA number range per billing type (VN01)
max_invoice_lines / batch_post_max	tenant	int	2,000 / 500	1–10,000 / 1–2,000	FR-29.3	n/a	n/a
tax_recheck_tolerance	legal entity	money	0.01	≥ 0	FR-29.6	n/a	n/a
tax_rounding_level	country pack	enum	LINE	LINE, DOCUMENT	BR-29.3	Sales tax parameters "Rounding by"	OVK1 / T001 tax rounding (TXKRS)
revrec_residual_rule	legal entity	enum	LARGEST	LARGEST, FIRST, LAST	BR-29.1	n/a (D365 allocation rounding)	RAR rounding of allocation
ssp_source_order	legal entity	list	EXPLICIT, LIST, PCT_OF_LIST, RESIDUAL	ordered subset	BR-29.2	D365 revenue prices	RAR SSP condition types
ratable_basis	legal entity	enum	DAILY	DAILY, MONTHLY	FR-29.8	Revenue schedule (RevRecSchedule) "Monthly / daily"	RAR fulfillment type time-based
deferred_revenue_account / revenue_account defaults	legal entity	account	2400 / 4000	chart (w11)	FR-29.3	Item posting profile deferred revenue	VKOA account keys ERL / ERF
usage_late_days	tenant	int	45	0–180	BR-29.10	n/a	n/a
subscription_billing_time	legal entity	time	03:00	—	FR-29.10 job	D365 subscription billing batch	VF04 periodic billing schedule
index_rule_default	subscription group	ref	none	index rules	FR-29.14	SMA index rules	Condition price adjustment (VK12 with ref)
expense_deferral_accounts	legal entity	account pair	1300 / 1310	chart	FR-29.15	D365 deferral schedule accounts	ACCR accrual object accounts
self_billing_tolerance	customer	numeric	0.5% or 5.00	≥ 0	BR-29.14	n/a	VSB1 tolerance per self-biller
print_components	product (bundle)	bool	true	true/false	FR-29.17	Kit print options	Item category "Print BOM" (VOV7)
installment_residual_to	legal entity	enum	LAST	FIRST, LAST	BR-29.11	Payment schedule rounding	OBB9 installment rounding
document_delivery_default	legal entity	enum	EMAIL	EMAIL, PORTAL, PRINT, EDI, NONE	FR-29.19	Print management (original destination)	NAST output medium default (VV31)
document_resend_retries	tenant	int	3	0–10	BR-29.15	n/a	NAST repeat processing
einvoice_enabled	legal entity	bool	false	true/false	FR-29.11 hook	Electronic reporting format setup	DRC / eDocument activation

5.29.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Billing due list	list inquiry	customer, cycle, date range, warehouse	slip, order, amount, cycle, consolidation key	packing_slip (POSTED not INVOICED)	Invoice "Select" / summary update proposal	VF04 billing due list
Invoices list	list inquiry	customer, type, status, date, currency	number, type, dates, gross, open, status	invoice	Invoice journal (CustInvoiceJour)	VF05 / VF05N
Invoice trace	drill-down	invoice	invoice → slip → order line → price explain → schedules	trace API	CustInvoiceTrans → InventTransId (no single view)	VBFA document flow
Deferred revenue roll-forward	statutory / operational report	entity, period, contract	opening, additions, recognized, adjustments, closing, GL 2400 diff	revenue_schedule	D365 revenue recognition reconciliation	FARR_RECON / RAR disaggregation reports
Revenue contracts	list inquiry / drill-down	customer, status, obligation method	price, SSP, allocated, recognized, remaining	revenue_contract	RevRecContract (D365)	FARR_CONTRACT_MAIN
Subscriptions and fee transactions	list inquiry	group, status, next bill date, fee category	subscription, lines, next bill, last index, MRR	subscription	SMA subscription inquiries / SSRS Subscription fee transactions	VA45 contract list / FPLA billing plan
Expense deferrals	list inquiry	vendor, account, period	schedule, total, recognized, short-term, long-term	expense_deferral	D365 deferral schedules	Accrual engine (ACACTREE) reports
Self-billing reconciliation	drill-down	customer, statement	line, match status, price difference, adjustment	self_billing_statement	n/a	VSB1 log / self-billing monitor

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Failed posts / undelivered documents	KPI / list	date, error	proposal, error code, retries; document, channel, failure	proposal + delivery log	Posting log / print management log	VF04 log, NAST error (V.21)
Invoice register (sales journal)	statutory report	entity, period, tax code	number, date, customer, tax id, net, tax, gross	invoice	SSRS Invoice journal / sales tax transactions	RFBELJ00 / S_ALR_87012357
Credit notes analysis	operational report	reason, customer, period	count, amount, % of revenue, recognized reversed	credit_note	Credit note journal	VF05 filter VBTYP O

5.29.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Sales invoice	post	header: entity legal name, address, tax ids, invoice number, date, due date, customer, invoice account, ship-to, order, terms, currency; lines: product, description, qty, unit, unit price, discount, net, tax code; footer: net, charges, tax by rate, gross, installments, cash discount, bank details, legal text	PDF, UBL 2.1 (w35 Peppol), email, portal, print	`INV–` gap-free per entity and fiscal year	statutory: EU VAT Directive art. 226 content; retain 10 years (DE GoBD), 7 years (US)
Free-text invoice	post	as invoice; lines: description, account-driven, tax code	PDF, UBL, email	same `INV–` series (or `FTI–` if `invoice_number_scope = ENTITY_DOC_TYPE`)	as invoice
Credit note	post	as invoice plus original invoice reference and reason	PDF, UBL CreditNote, email	`CN–` gap-free	as invoice; must reference original (art. 219)
Subscription invoice	daily billing job	as invoice plus service period, proration, indexation note	PDF, UBL, email	`INV–`	as invoice
Bundle invoice	post	bundle line with itemized components	PDF, UBL	`INV–`	as invoice
Self-billing adjustment note	reconciliation	statement reference, invoice, difference	PDF, email	`CN–` / `DN–`	as invoice; self-billing agreement on file (EU art. 224)
Revenue contract summary	on demand	obligations, SSP, allocation, schedule	PDF, CSV	none	audit support 7 years
Deferred revenue roll-forward	period close	opening → closing per contract	PDF, XLSX	none	audit workpaper 7 years

5.29.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Consolidated billing run	per cycle (daily 22:00, weekly Sunday 22:00, monthly last day 22:00)	invoice account	key = (account, cycle, period)	< 20 min per 50k slips	per-invoice saga; failures listed	failed > 1%
Subscription billing	03:00 daily	subscription	key = (subscription, period)	< 30 min per 100k subscriptions	retry; skip and report	subscriptions due not billed by 06:00
Indexation run	per index effective date, preview 7 days ahead	subscription group	key = (rule, effective date)	< 5 min	preview stays open	index missing
Revenue recognition run	period-close trigger (w14) + nightly preview 01:00	legal entity × period	key = period; per schedule line status	< 30 min per 1M schedule lines	resumable from last schedule id	run > 30 min; diff ≠ 0
Expense deferral recognition	period close + nightly preview	legal entity	key = period	< 10 min	resumable	unrecognized past periods > 0
Roll-forward check	02:30 daily	legal entity	key = (entity, date)	< 5 min	report only	diff ≠ 0.00
Document delivery retry	every 10 min	channel	delivery id	< 1 min	3 retries then workspace tile	undelivered > 50
Self-billing statement import	on file arrival (SFTP poll 15 min)	customer	SHA-256 of file	< 2 min	quarantine malformed	review tasks > 20

5.29.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
InvoiceView	GET invoices, documents, trace; Billing workspace	none	Billing clerk, AR clerk, Sales, Auditor
InvoiceMaintain	proposals, free-text invoices, due list	with ARPaymentPost (w30)	Billing clerk
InvoicePost	:post (single and batch)	with PackingSlipPost (w28) under four-eyes; with CustomerMaintain (w7)	Billing clerk
CreditNotePost	credit notes :post	with ARSettle and with InvoicePost on the same invoice when `credit_note_four_eyes`	Billing supervisor
RevRecMaintain	:allocate, SSP adjustments, milestones	with RevRecPost	Revenue accountant
RevRecPost	recognition runs, expense deferral runs	with RevRecMaintain	Controller
RevRecView	roll-forward, contracts	none	Controller, Auditor
SubscriptionMaintain	subscriptions, :change, :cancel, usage, index rules	with InvoicePost when subscription invoices auto-post	Subscription manager
SelfBillingReconcile	statement import, match, adjustments	with CreditNotePost	Billing supervisor
DocumentDeliveryMaintain	delivery preferences, resend	with CustomerMaintain (bank detail fraud risk on recipients)	AR manager

5.29.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
BIL-0001	error	Tax engine unavailable; invoice not posted	w13 outage	Retry later	FR-29.6 / AC-29.8
BIL-0002	error	Tax changed by {amount} since proposal; review required	tax recheck > tolerance	Re-review and post	FR-29.6
BIL-0003	error	Posting period {period} is closed	BR-29.4	Choose an open period	BR-29.4
BIL-0004	error	Line {n} has no revenue account	posting profile gap	Set account in w16/w11 profile	§5.29.5 validation
BIL-0005	error	Credit exceeds remaining creditable amount {amount}	FR-29.5	Reduce credit	FR-29.5
BIL-0006	error	Invoice date {date} precedes packing slip date {slip_date}	BR-29.4	Correct date	BR-29.4
BIL-0007	error	Consolidation exceeds {max} lines; split into {n} invoices	BR-29.5	none (auto split) or reduce	BR-29.5
BIL-0008	error	Same-day reversal not allowed: invoice has settlements	BR-29.7	Issue a credit note	BR-29.7
BIL-0009	warning	SSP missing for {product}; residual method used	BR-29.2	Maintain SSP	FR-29.7
BIL-0010	warning	Usage record is late; billed next period	BR-29.10	none	BR-29.10
BIL-0011	error	Installment percentages total {pct}%, expected 100%	FR-29.18	Fix terms (w7)	BR-29.11
BIL-0012	warning	Index value for {index} missing for {date}; indexation skipped	FR-29.14	Load index	BR-29.12
BIL-0013	error	Self-billing line {n} does not match any invoice	BR-29.14	Review task	FR-29.16
BIL-0014	error	Invoice {number} already self-billed	BR-29.14	none	FR-29.16
BIL-0015	warning	EDI delivery not configured for {customer}; sent by email	BR-29.15	Activate trading partner (w35)	FR-29.19
BIL-0016	error	Document delivery failed after {n} retries	BR-29.15	Resend or change preference	FR-29.19
BIL-0017	error	Expense deferral term must be ≥ 2 periods	FR-29.15	Post as expense	FR-29.15

5.29.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Invoice post (saga, 10 lines)	50k per day; peak 20/s	900 ms / 3 s / 6 s	20/s per replica	12M invoices × 4 KB + lines 60 GB	partition by posting month; (invoice_account_id, status, due_date)	saga workers scale; batch posts throttle to 50%
Proposal from packing slip	100k per day	100 / 400 / 1,000 ms	50/s	proposals purged after 90 days	(slip_id)	event lag allowed to 5 min
PDF render + delivery	60k per day	1 / 4 / 10 s	30/s	object store 300 GB (10-year retention, cold after 1 year)	content hash	rendering queued; post never waits for PDF
Recognition run	1M schedule lines per period	— / 30 min / 45 min	1k lines/s	schedules 200M rows × 150 B = 30 GB	partition by period	run split by contract range
Subscription billing	100k subscriptions per day	— / 30 min / 60 min	60 subs/s	included above	(next_bill_date, status)	finish by 06:00 or alert
Roll-forward query	200 per day	300 ms / 2 s / 5 s	—	none	materialized per (entity, period)	serve last materialization
Trace API	20k per day	80 / 300 / 800 ms	50/s	none	FK lookups	trace depth limited to explain summary

5.29.20 Localization and compliance hooks

- Invoice content and numbering: EU VAT Directive arts. 220–240 (mandatory fields, gap-free sequences, credit-note references, self-billing agreements art. 224); country packs (w48) add fields (IT codice destinatario, ES SII, FR SIREN, DE Leistungsdatum) through the docgen `country_fields` slot and the UBL extension.
- E-invoicing mandates (IT SDI, FR PPF/PDP 2026–2027, DE XRechnung/ZUGFeRD B2B from 2025, PL KSeF, MX CFDI) are transmitted in w35; this wave produces the UBL 2.1 payload and the `einvoice_ref` hook.
- Tax rounding line vs document per country (BR-29.3, open question 1); US sales tax by jurisdiction via the w13 engine; JPY zero-decimal handling.
- Revenue recognition: ASC 606 and IFRS 15 share one model; country GAAP variants (for example local GAAP point-in-time only) are a second ledger posting in w11 parallel ledgers, not a second engine.
- Down-payment invoices and tax on down payments (DE §13 UStG) are generated by w30 FR-30.21 using the invoice document type DOWN_PAYMENT_REQUEST.
- Retention: invoices and credit notes 10 years immutable (GoBD, w48 archive); PDF/A-3 for DE when `einvoice_enabled`.
- Privacy: invoices for one-time customers and individuals contain personal data; erasure requests are refused within the statutory retention period and pseudonymized after it.

5.29.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.28.1 #227 — Service subscriptions (SMASubscriptionTable, fee transactions, index)	PARTIAL-29	ADDED as FR-29.14	service agreements link from wave 45
AX §12.28.2 #228 — Subscription inquiries	PARTIAL-29	ADDED as FR-29.14	§5.29.14 Subscriptions and fee transactions
AX §12.28.2 #229 — SSRS Subscription report / Subscription fee transactions	PARTIAL-29	ADDED as FR-29.14	inquiry export (PDF/CSV)
AX §12.28.3 #230 — Create subscription fee transactions / Accrue subscription revenue / Index subscriptions	PARTIAL-29	ADDED as FR-29.14 (fees, index) and FR-29.8 (accrual)	§5.29.16 indexation run
AX §12.28.4 #231 — Subscription groups, fee categories, index rules, subscription periods	PARTIAL-29	ADDED as FR-29.14	
AX §12.37.1 #314 — Deferral schedules (revenue and expense, short/long-term split)	PARTIAL-29	ADDED as FR-29.15	1300 / 1310 split
AX §12.37.3 #315 — Expense deferral recognition	PARTIAL-29	ADDED as FR-29.15	§5.29.16 job
SAP §13.19 #220 — VKA / VSB1 customer self-billing	GAP (propose 29)	ADDED as FR-29.16	EDI GSVERF / 820 mapping in wave 35
SAP §13.19 #221 — Solution orders / bundled sales	PARTIAL-29	ADDED as FR-29.17	
SAP §13.1 #17 — OBB8 / OBB9 terms of payment / installment terms (PARTIAL-7)	PARTIAL-7	ADDED as FR-29.18 (application at invoice post); terms definition remains wave 7	flagged for order-to-cash

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP §13.3 #47 — OBB9 installment terms (customers) (PARTIAL-7)	PARTIAL-7	ADDED as FR-29.18	
SAP §13.2 #35 — OBB9 installment terms (vendors) (PARTIAL-7)	PARTIAL-7	DEFERRED to wave 19	AP invoice installments reuse the w7 terms model
AX §12.3.4 #30 — Form setup (sales documents) / Form notes / Print management	PARTIAL-46	ADDED as FR-29.19 (per-account destinations); query-based conditional routing DEFERRED to wave 46	
AX §12.17.4 #173 — Form setup, form notes, print management (sales), form sorting	PARTIAL-46	ADDED as FR-29.19; form sorting SUPERSEDED by wave 46 document designer	

5.30 Wave 30: Accounts Receivable, Collections, Returns & Customer Portal

5.30.1 Scope and business context

Wave 30 closes the loop. The `receivables` service applies cash (manual, BAI2 lockbox, remittance advice, camt.053 with ML-assisted suggestions) and settles open items; `collections-workflows` (TS on Temporal) runs aging, dunning, interest notes, promises to pay, disputes and risk scoring; return orders flow through RMA dispositions to inventory and credit notes; wave 26 accruals settle; statements are produced; and the customer portal at `customer.<env>.passage.dev` exposes orders, invoices, payments (Stripe/Adyen sandbox), returns and documents. The wave ends with the Phase 5 exit: the demo manufacturer's one-month simulation with reconciled subledgers and stakeholder sign-off. Out of scope: non-customer bank reconciliation and cash forecasting (wave 36), collections-agency integration, portal self-registration beyond invited contacts, bills of exchange (wave 36), generic case management (wave 45). Added from the coverage registers (§5.30.21): SEPA direct debit mandates and pain.008, deductions workbench, minimal collection cases, correspondence framework with email templates, balance confirmations, down payments and security deposits, distributor chargebacks, vendor dunning, GL/intercompany interest, portal guided configuration and returnable packaging charges.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Cash application is manual: AX settle open transactions (CustOpenTrans) makes the clerk find invoices per check; SAP F-28 plus FLB2 post-processing lands 30–50% of payments on account	Exact-reference matching only; no scoring, subset-sum or remittance parsing	BR-30.1 scored rules R1–R7, auto-apply ≥ 85, ML-ranked suggestions across lockbox, camt.053, remittance and portal	Auto-match 40–60% → ≥ 90%; clerk time per payment 3 min → < 20 s; false matches 0
Dunning produces wrong letters: F150 MHNK proposals ignore disputes without FSCM; CustCollectionLetterJour cannot skip promised items	Exclusion state not part of selection	Exclusions from open_item.dispute_id/promise_id at issue time; grace, interval and minimum rules (BR-30.2)	Retracted letters ~5% → 0; run preparation 2 h → < 10 min
Returns need three hand-keyed documents (ReturnTable → arrival journal → credit note; RE order → LR delivery → G2 request → VF01)	Separate forms with manual references	One RMA with dispositions (BR-30.3) driving receipt, scrap/repair/replace and credit note in a saga	Clicks per return ~45 → ≤ 8; cycle time 3–5 days → same day after receipt
Customers phone for statements, invoice copies and status; clerks re-send PDFs	No customer-facing surface on live open items	Portal on the same `open-items`, `invoices`, `shipments` APIs with card/ACH payment and return requests	"Copy of invoice/status" requests –80%; DSO –5 days via portal payments
AR-to-GL reconciliation is a month-end exercise (CustTrans vs. LedgerTrans; BSID vs. recon account via F.03)	Subledger and GL updated by different code paths	Every settlement posts through the journal API; Σ open_amount = GL 1200 daily, in the reconciliation pack	Effort 1–2 days/month → 0; diff 0.00 for 31/31 days

5.30.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Payments and settlement	CustPaymJournal (LedgerJournalTrans), CustTrans/CustSettlement, CustPaymModeTable	F-28, F-32, BSID → BSAD, OB41	payment, settlement, open_item	full
Lockbox	CustLockbox (BAI2 import)	FLB2 / FLB1 (BAI2)	lockbox_batch	full
Bank statement auto-match	Advanced bank reconciliation (camt.053)	FF_5 / EBS camt.053 auto-clearing	camt.053 import + matcher	improved (ML suggest)
Aging / collections	CustAgingLegalEntity, Collections workspace, CustCollectionsPool	FBL5N; FSCM Collections (UDM_SPECIALIST)	aging snapshot, worklist, risk score	full
Collection letters / dunning	CustCollectionLetterJour/Line, sequence, grace days	F150 (MHNK/MHND), FBMP procedure	dunning_run, dunning_letter	full
Interest notes	CustInterestJour/Trans	F.24 / F.2B	interest_note	full
Promise to pay / disputes	Collections promise; cases	FSCM dispute (UDM_DISPUTE)	promise_to_pay, dispute (Temporal)	full
Return orders / RMA	ReturnTable (SalesType=ReturnItem), ReturnDispositionCode, ReturnReasonCode	VA01 type RE, LR delivery, G2 credit memo request; S/4 advanced returns	return_order + return receipt + credit_note	full
Rebate settlement	Rebate settlement to credit note / AP	VBO1 settlement, B1/B2 credit memo	rebate_settlement → credit_note or AP payment	full
Statements / portal / card payments	CustAccountStatementExt; D365 Customer portal; Adyen connector	F.27; SAP Commerce; digital payments add-on	statement job; customer-portal; payments-gateway	improved
SEPA direct debit	CustDirectDebitMandate, CustPaymModeTable (SEPA DD), generate payments (pain.008)	FSEPA_M1..M4 mandates, F110 with payment method DD, SEPA_* customizing	dd_mandate, direct_debit_run → pain.008	full
Deductions	TAMDeductionWorkbench, deduction types, denial reasons	FSCM dispute case with residual items (F-32 residual, UDM_DISPUTE)	deduction (from short pay) + deduction workbench	full
Collection cases	CaseDetail (collections category)	FSCM UDM_DISPUTE / collection worklist case	collection_case (minimal case entity; generic cases w45)	partial
Correspondence	Customer payment advice, collection letters, CustAccountStatementExt	FB12 / F.61 / F.64, OB77 correspondence types (T048)	correspondence_type + correspondence_request via docgen	full
Balance confirmations	— (custom SSRS)	F.17 / F.18 (classic), F.99 / F.97	balance_confirmation run with reply tracking	full
Special G/L customer items	Customer prepayments (CustPrepayment), deposit journals	OBXR special G/L indicators A (down payment), H (security deposit), W (BoE)	open_item.special_gl_type DOWN_PAYMENT, SECURITY_DEPOSIT	partial (BoE w36)
Distributor chargebacks	Rebate management deals (D365)	Settlement management chargeback claims, EDI 844/849	chargeback_claim validation against w26 agreements	full (EDI mapping w35)
Vendor dunning	—	F150 for vendors with debit balances	dunning engine over AP debit open items (w19)	full

5.30.3 Functional requirements

- **FR-30.1 Open items and settlement** — Must. Open items from invoices, credit notes, payments and adjustments; full or partial settlement with cash discount, write-off within tolerance and FX gain/loss (w12) posted at settlement.
- **FR-30.2 Manual payments** — Must. Payment journal by mode (CHECK, ACH, WIRE, CARD, CASH): Dr 1000/1010 bank, Cr 1200; unapplied residual to 1205.
- **FR-30.3 Lockbox BAI2** — Must. Import record types 1–9, one payment per check with remittance detail, then match.
- **FR-30.4 Remittance advice** — Should. CSV/JSON/EDI 820 hook attached to payments for matching.
- **FR-30.5 camt.053 auto-match** — Must. Import statements, identify customer credits (CdtDbtInd=CRDT with RmtInf and debtor data), match per BR-30.1; a Python ranker reorders suggestions but never auto-applies on its own.
- **FR-30.6 Aging and worklist** — Must. Nightly aging (current, 1–30, 31–60, 61–90, 91–180, > 180 by due or invoice date) and a worklist ranked by risk score.
- **FR-30.7 Dunning** — Must. Levels, grace days, minimum amounts, blocks and intervals (BR-30.2); letters via docgen; fees posted; item level raised only at issue.
- **FR-30.8 Interest notes** — Should. Annual rate, ACT/365, minimum 5.00 USD / 5.00 EUR, posted as open items.
- **FR-30.9 Promises and disputes** — Must. Promises (amount, date) and disputes (reason, amount, owner, SLA) exclude items from dunning and interest while open; Temporal workflow to resolution (credit, paid, write-off, rejected).
- **FR-30.10 Risk scoring** — Should. 0–100 from overdue ratio, average days to pay, dunning level, disputes and broken promises; published to credit.
- **FR-30.11 Return orders (RMA)** — Must. RMA referencing invoice lines (or blind), 30-day validity, receipt to quarantine/inspection, dispositions (BR-30.3), inventory receipt at return cost, credit note with restocking charges.
- **FR-30.12 Rebate settlement** — Must. Settle wave 26 accruals per agreement period into a credit note (default) or AP payment (w19 hook), reversing 2150.
- **FR-30.13 Statements** — Should. Monthly and on-demand PDF statements and portal view.
- **FR-30.14 Customer portal** — Must. Invited contacts place orders, view invoices/documents, pay by card/ACH via payments-gateway, request returns and track shipments; Keycloak realm `portal`, OPA scoped to the contact's accounts; WCAG 2.1 AA.
- **FR-30.15 Phase-exit simulation** — Must. Harness `sim/phase5-month` drives Phases 3–5 end-to-end and produces the §5.0 reconciliation pack.
- **FR-30.16 SEPA direct debit mandates and collection** — Must. The system shall store SEPA mandates per customer bank account (unique mandate reference ≤ 35 chars, creditor identifier per legal entity, scheme CORE or B2B, signature date, sequence state FRST/RCUR/FNAL/OOFF, status PENDING → ACTIVE → SUSPENDED / CANCELLED / EXPIRED), run a direct-debit proposal over due open items of customers with payment mode SEPA_DD and an active mandate, and generate pain.008.001.08 files through the w19 ISO 20022 generator, creating DD payments that settle the items on collection date. Notes: pre-notification sent ≥ 14 calendar days before collection (configurable down to 2 by agreement); mandates unused for 36 months expire automatically; returns and rejects (pain.002 / camt.054 R-transactions) reopen the items and post the return fee as FEE; AX CustDirectDebitMandate and SAP FSEPA_M1..M4 map 1:1.
- **FR-30.17 Deductions workbench** — Should. The system shall convert the unexplained part of a short payment (R7 short-pay with reason, or a remittance line marked deduction) into a `deduction` open item (source DEDUCTION) with type and reason code, and provide a workbench to match deductions to w26 claims (rebate accruals, promotion fund usage, chargebacks), approve (credit note against the claim), deny (re-bill to the customer as a debit open item with a denial reason), split, or write off within `deduction_write_off_limit`. Notes: AX TAMDeductionWorkbench parity; matched deductions reduce the fund or accrual they consume; aging of deductions is separate from invoice aging; deductions never raise a dunning level while open.
- **FR-30.18 Collection cases** — Should. The system shall group collection work into a `collection_case` (customer, category COLLECTION or DISPUTE or DEDUCTION, owner, priority, status, SLA, linked open items, disputes, promises, activities and notes) with a Temporal timer per SLA. Notes: this is the minimal case entity that the AX/D365 generic case framework (categories, processes, knowledge articles) will replace in wave 45 through a migration of `collection_case` rows into generic cases; the API path `/receivables/cases` stays as a façade.
- **FR-30.19 Correspondence framework** — Should. The system shall define correspondence types (ACCOUNT_STATEMENT, PAYMENT_NOTICE, OPEN_ITEM_LIST, INDIVIDUAL_LETTER, DUNNING_LETTER, INTEREST_NOTE, BALANCE_CONFIRMATION, DD_PRE_NOTIFICATION) each with a docgen template per language and an email template (subject, body with merge fields), and accept correspondence requests on demand (single customer, from any open-item or payment screen) or periodic (job), delivered by the FR-29.19 routing with a delivery log. Notes: SAP FB12 / F.61 / F.64 and OB77; D365 collections email templates; the payment notice is generated automatically after each applied payment when the customer preference says so.
- **FR-30.20 Customer balance confirmations** — Should. The system shall run balance confirmations as of a key date for a selected customer set (balance ≥ threshold, or random sample of n), generate a confirmation letter with the open-item list and a reply form (portal link with a one-time token, or email reply), record replies (AGREED, DISAGREED with customer balance and comment, NO_REPLY) and produce a difference list for audit. Notes: SAP F.99 / F.17 parity; replies are immutable once recorded; a second reminder is sent after `confirmation_reminder_days` (default 14).
- **FR-30.21 Down payments and security deposits** — Should. The system shall record customer down-payment requests (noted item, no GL posting), down payments received (Dr bank / Cr 2190 Customer down payments, tagged to an order), clearing of down payments against the final invoice (Dr 2190 / Cr 1200, with tax on down payment where the country pack requires), and security deposits (Dr bank / Cr 2195, released by refund or offset). Notes: SAP special G/L indicators A, F, H (OBXR); AX customer prepayments; open items carry `special_gl_type` and are excluded from dunning and from credit exposure as a reduction only when `down_payment_reduces_exposure`; bills of exchange (indicator W) are out of scope here (wave 36).
- **FR-30.22 Distributor chargeback claims** — Could. The system shall import chargeback claims from distributors (CSV now; EDI 844 in, 849 response via w35) with lines (end customer, product, quantity, contract price, invoice reference), validate each line against the w26 agreement for the end customer (price, validity, eligibility), accept or reject per line with reason codes, and settle accepted amounts by credit note to the distributor or as deduction match (FR-30.17). Notes: chargeback claim types configurable; accepted amount per line = (distributor acquisition price – contract price) × quantity; duplicates detected by (distributor, claim number, line).
- **FR-30.23 Dunning of vendors with debit balances** — Could. The system shall let a dunning run include vendors (w19) whose net open AP balance is a debit beyond `vendor_dunning_min` (default 100.00 USD / 90.00 EUR) using the same levels and templates. Notes: SAP F150 vendor dunning; no fees and no interest on vendors by default.
- **FR-30.24 Interest on GL and intercompany balances** — Could. The system shall extend the interest engine (FR-30.8) to selected GL accounts and intercompany due-to/due-from balances with balance-based interest (daily balance × rate × days / basis ACT/360 or ACT/365), producing an interest calculation journal for review and post. Notes: SAP F.50 / F.52 and OB46 interest indicators; intercompany pairing completes in wave 40.
- **FR-30.25 Portal guided configuration** — Could. The system shall let portal contacts configure a configurable product through the w8 configurator API (attribute questions, constraint validation, price preview from w26) and add the configuration to a portal order. Notes: SAP CPQ / variant configuration web UI parity; uses the same configuration_id as FR-27.11; constraint errors shown inline.
- **FR-30.26 Returnable packaging charges** — Could. The system shall, for returnable packaging balances (FR-28.16) older than `rtp_charge_days`, propose a free-text invoice charging the deposit price per unit, and credit it when the packaging returns within 365 days.

5.30.4 Data model

open_item (CustTrans/CustTransOpen, BSID)

Field	Type	Required	Notes
id, legal_entity_id, customer_id, invoice_account_id	uuid	Y	RLS

Field	Type	Required	Notes
source_type, source_id	enum, uuid	Y	INVOICE, CREDIT_NOTE, PAYMENT, INTEREST, FEE, ADJUSTMENT, REBATE
document_number, document_date, due_date	text, date, date	Y	
currency, amount, amount_entity_currency	char(3), numeric(19,4) ×2	Y	debit +, credit –
open_amount	numeric(19,4)	Y	= amount – Σ settlements
cash_discount_date, cash_discount_amount	date, numeric	N	
dunning_level, last_dunned_at, dunning_block	smallint, date, bool	Y	defaults 0 / null / false
dispute_id, promise_id	uuid	N	exclusion flags
status	enum	Y	OPEN, PARTIALLY_SETTLED, SETTLED, REVERSED

payment: id, payment_number, customer_id (nullable until identified), payment_mode, amount, currency, received_date, value_date, bank_account_id, source (MANUAL | LOCKBOX | CAMT053 | PORTAL | REMITTANCE), source_ref (unique with source), payer_name, payer_account_hash, remittance[] (reference, amount), status (RECEIVED | SUGGESTED | APPLIED | ON_ACCOUNT | REVERSED), match_score, journal_posting_id. **settlement** (CustSettlement): id, payment_id | credit_item_id, open_item_id, settled_amount, cash_discount_taken, write_off_amount, fx_gain_loss, settled_at, journal_posting_id, reversal_of_id.

dunning_run: id, legal_entity_id, run_date, parameters, status (DRAFT | REVIEWED | ISSUED | CANCELLED). **dunning_letter**: id, run_id, customer_id, level, items[] (open_item_id, open_amount, days_overdue), fee_amount, document_ref, issued_at. **dispute**: id, customer_id, open_item_id, reason_code, disputed_amount, owner, status, sla_due, workflow_run_id.

return_order (ReturnTable): id, rma_number, customer_id, original_invoice_id, return_reason_code, status (BR-30.6), expires_at, lines[] (original_invoice_line_id, product, qty_expected, qty_received, disposition_code, return_cost_method (ORIGINAL_COST | CURRENT_COST | ZERO), restocking_pct, restocking_amount, inventory_receipt_ids[], credit_note_line_id).

Invariants: Σ settlements per item ≤ |amount|; open_amount = 0 ⇔ SETTLED; Σ OPEN open_amount per entity = GL 1200; Σ ON_ACCOUNT = GL 1205; a payment is APPLIED only when Σ settlements + residual = amount; dunning_level never decreases while open; disputed or promised items are excluded from dunning and interest; credited qty ≤ received qty × unit credit. Indexing: open items (customer_id, status, due_date), (document_number); payments (status, received_date), unique (source, source_ref); settlements partitioned by month.

5.30.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	/api/v1/receivables/open-items?customer=&status=&asOf=	Open items (FBL5N)	—	ARView / Portal (own)
POST	/api/v1/receivables/payments, …/{id}:match, :apply, :on-account, :reverse	Payment lifecycle; scored candidates	Idempotency-Key	ARPaymentPost
POST	/api/v1/receivables/lockbox/imports; /bank-statements/imports	BAI2 and camt.053 files	SHA-256 of content	ARImport
POST	/api/v1/receivables/settlements:batch	Settle with credits/payments, discount, write-off	Idempotency-Key	ARSettle
GET	/api/v1/receivables/aging?asOf=&bucketBy=	Aging snapshot	—	ARView
POST	/api/v1/receivables/dunning-runs, …/{id}:review, :issue, :cancel	Dunning	Idempotency-Key	DunningRun
POST	/api/v1/receivables/interest-notes:calculate, :post; /promises; /disputes, …/{id}:resolve	Interest, promises, disputes	Idempotency-Key	InterestPost / CollectionsMaintain
POST	/api/v1/sales/return-orders, …/{id}:authorize; /lines/{l}:receive, :disposition	RMA saga (sales + fulfillment + billing)	Idempotency-Key	ReturnMaintain / WarehouseReceive
POST	/api/v1/pricing/rebates/settlements	Settle accruals for a period	key = (agreement, period)	RebateSettle
GET/POST	/portal/api/v1/{orders,invoices,payments,returns,shipments,documents}	Portal BFF (scoped)	Idempotency-Key	PortalCustomer
POST	/api/v1/payments-gateway/intents; /webhooks/{provider}	Card/ACH intents; signed provider webhooks	provider id	system

Validation: apply requires Σ settlements + on-account residual = payment amount; cross-currency settlement only with FX settlement enabled; dunning issue requires REVIEWED; disposition requires qty_received > 0; portal payments require Σ selected open ≥ amount.

Event	Trigger	Payload fields	Consumers
dev.passage.receivables.payment.received.v1	import / manual / portal	payment_id, source, amount, currency, payer, remittance[]	matcher, analytics
dev.passage.receivables.payment.applied.v1	apply	payment_id, settlements[{open_item_id, amount, discount, write_off, fx}], residual, journal_posting_id	ledger, sales credit, portal, collections
passage.receivables.customer-balance.changed.v1	any open item change	customer_id, balance, overdue, oldest_item_days, dunning_level, risk_score	sales credit, portal
passage.receivables.dunning-letter.issued.v1	issue	run_id, customer_id, level, items[], fee, document_ref	docgen, portal, sales (hold at level 4)
dev.passage.receivables.dispute.opened.v1 / resolved.v1	dispute lifecycle	dispute_id, open_item_id, amount, reason, resolution	collections, billing, analytics
passage.receivables.return-order.authorized.v1 / dispositioned.v1	RMA	rma_id, line, disposition, qty, restocking, credit_note_id	fulfillment, inventory, billing, portal
dev.passage.pricing.rebate.settled.v1	settlement	agreement_id, period_id, amount, credit_note_id	ledger, receivables, analytics

5.30.6 User experience

Screen	Primary actions	Key fields/columns	Duty
AR workspace	tiles: unapplied cash, overdue by bucket, disputes, dunning due, RMAs awaiting inspection	amounts, counts	ARView
Cash application	import, accept/adjust suggestions, apply, on-account, split	payment, payer, amount, candidates with score and rule ids, residual	ARPaymentPost
Open items / settlement	select, settle, discount, write-off, reverse	document, due, open amount, discount available, level	ARSettle
Collections worklist / dispute detail	log contact, promise, dispute, hold order; assign, evidence, resolve	customer, risk score, overdue, oldest item, promises; amount, reason, SLA, timeline	CollectionsMaintain

Screen	Primary actions	Key fields/columns	Duty
Dunning run	create, preview letters, exclude, issue	customer, level, items, fee, blocks	DunningRun
Return order (RMA)	authorize, receive, inspect, disposition, credit	lines, disposition, restocking, credit preview	ReturnMaintain
Customer portal	place order, view invoices/documents, pay, request return, track	orders, invoices, balance, payments, RMAs, tracking	PortalCustomer

Interaction rules: cash application keyboard flow (J/K candidates, Enter accept, A apply) and bulk accept of all ≥ 85 suggestions; worklist saved filters; dunning preview opens the rendered letter; portal is mobile-first; provider hosted fields so no card data touches Passage.

5.30.7 Business rules and edge cases

- **BR-30.1 Cash application matching rules and scores.** Candidates are open items of the identified customer (all customers when unidentified). Score per candidate set: R1 exact document number in remittance/RmtInf/check memo +60; R2 amount = open amount of one item +30, or = Σ open of a subset of ≤ 5 items (0.00 tolerance) +25; R3 amount = open − available cash discount with discount date ≥ value_date +28; R4 payer bank account hash (MICR/IBAN) matches the customer's bank account +15; R5 payer name Jaro-Winkler ≥ 0.92 to customer or invoice account +10; R6 value_date within 7 days of due date +5; R7 short by < 1% or < 50.00 USD / 45.00 EUR +20 with a mandatory short-pay reason. Decision: auto-apply when exactly one set scores ≥ 85 and the runner-up ≤ winner − 20; 60–84 → SUGGESTED for one-click; < 60 → ON_ACCOUNT (1205) with top-3 suggestions kept. The ML ranker may reorder sets below 85 but cannot raise one above it. Every auto-application is reversible within the open period.
- **BR-30.2 Dunning levels and grace.** Level 1 Reminder: due date + grace days (default 5 per customer group) passed, no fee; Level 2: ≥ 15 days overdue, no fee; Level 3: ≥ 30 days, fee 25.00 USD / 20.00 EUR as open item FEE; Level 4: ≥ 45 days, fee 50.00 USD / 40.00 EUR, legal text, hold recommendation (sales applies CREDIT via CR-3). Rules: minimum letter amount 50.00 USD / 45.00 EUR per customer; ≥ 10 days since the last letter at any level; an item rises at most one level per run; letter level = max item level; items < 1.00 ignored; customers or items with dunning_block, open disputes or unbroken promises (promise date + 2 days) are excluded; a run is DRAFT until reviewed and issued, and only issue posts fees and levels; full payment resets the item, not the customer's history.
- **BR-30.3 RMA dispositions** (ReturnDispositionCode equivalents): CREDIT_ONLY — no physical return, credit at original price less restocking (default 0%); RETURN_TO_STOCK — receipt to the return warehouse at return cost (ORIGINAL_COST default), credit less restocking (default 15%, minimum 25.00 USD / 20.00 EUR); SCRAP — receipt then scrap issue to 5900, credit less restocking; REPAIR — receipt to quarantine, repair order hook (Phase 8), return to customer, no credit; REPLACE — receipt as RETURN_TO_STOCK or SCRAP plus a zero-price replacement order line referencing the RMA, credit offset by the replacement; REJECT_RETURN_TO_CUSTOMER — receipt to quarantine, ship back, no credit. Disposition is per line and immutable after credit; credit = qty received × original net unit price − restocking.
- **BR-30.4 Settlement accounting.** Cash discount within terms → Dr 4100; write-off within tolerance (≤ 5.00 or 0.5% of the item) → Dr 6100; FX difference on foreign-currency items → 7100/7200 at the settlement-date rate; partial settlement leaves PARTIALLY_SETTLED with reduced open_amount.
- **BR-30.5 Interest** = open_amount × annual_rate × days_overdue / 365 from due date + grace to the calculation date, once per item per period, excluded below the minimum and for disputed/promised items.
- **BR-30.6 State machines.** Payment: RECEIVED → SUGGESTED → APPLIED | ON_ACCOUNT → REVERSED; ON_ACCOUNT may later be applied. Dispute: OPEN → UNDER_REVIEW → RESOLVED_CREDIT | RESOLVED_PAID | RESOLVED_WRITE_OFF → CLOSED; REJECTED from OPEN/UNDER_REVIEW. RMA: REQUESTED → AUTHORIZED → RECEIVED → INSPECTED → DISPOSITIONED → CREDITED → CLOSED; REJECTED from REQUESTED; EXPIRED from AUTHORIZED after 30 days.
- **BR-30.7 Concurrency.** Settlement takes a row lock; a second clerk gets 409; imports deduplicated by SHA-256; camt.053 entries unique per (statement id, entry reference).
- **BR-30.8 Rebate settlement** runs after period close: amount = Σ accruals − prior settlements; a REBATE credit note posts Dr 2150 / Cr 1200; over-accrual posts a free-text invoice.
- **BR-30.9 Portal payments** create an intent for selected items; the signed webhook creates a PORTAL payment that auto-applies by R1 (score 100); refunds flow only through credit notes and the provider refund API with duty ARRefund.
- **BR-30.10 Statements** list items as of the statement date and never change balances; the portal reads the same `open-items` endpoint.
- **BR-30.11 SEPA collection date** = max(due date, today + pre-notification days, today + 1 TARGET2 business day); FRST and RCUR share D-1 submission since the 2016 rulebook; a mandate amended to a new debtor bank resets to FRST only if `smndr` indicator applies; an R-transaction within 8 weeks (CORE, no reason) or 13 months (unauthorized) reverses the DD payment, reopens the items and posts the bank return fee (default 3.50 EUR) as FEE.
- **BR-30.12 Deduction lifecycle.** OPEN → MATCHED (to a claim) → APPROVED (credit note posted) | DENIED (debit item re-billed) | SPLIT (children OPEN) | WRITTEN_OFF (Dr 6110 Deductions written off, ≤ `deduction_write_off_limit`, default 50.00 USD / 45.00 EUR); approval cannot exceed the claim's remaining amount; Σ children of a split = parent.
- **BR-30.13 Balance confirmation selection** uses the as-of balance in entity currency; customers with zero balance are included only when `include_zero_balances`; a disagreement creates a collection case of category DISPUTE with the customer's balance attached.
- **BR-30.14 Down payment clearing.** A final invoice for an order with down payments proposes clearing up to min(down payment remaining, invoice gross); tax on down payments posts at receipt only when the country pack sets `tax_on_down_payment` (for example DE, AT) and reverses at clearing.
- **BR-30.15 Chargeback validation.** A claim line is ACCEPTED when an ACTIVE w26 agreement exists for (end customer, product) at the claim's invoice date, contract price ≤ claimed price, and quantity ≤ the distributor's purchased quantity in the prior 12 months; otherwise REJECTED with codes CB-PRICE, CB-NOAGR, CB-QTY, CB-DUP.
- **BR-30.16 Correspondence** is rendered from a snapshot of the data at request time; re-sending reuses the snapshot unless `refresh=true`; every request is logged with its delivery result for 10 years.

5.30.8 Acceptance criteria

- **AC-30.1** (FR-30.3, FR-30.5) Given `cash-app-500`, when matching runs, then ≥ 450 payments auto-apply and zero apply to the wrong items versus the answer key.
- **AC-30.2** (FR-30.1) Given a 1,000.00 USD invoice at 2% 10 net 30 and 980.00 paid on day 8, then the item settles fully with 20.00 to 4100.
- **AC-30.3** (FR-30.7) Given 60 customers (5 blocked, 4 disputed, 3 with promises), when the run issues, then letters exist for exactly the 31 expected customers at expected levels, level-3 fees are 25.00, and disputed items appear on no letter.
- **AC-30.4** (FR-30.7) Given level 2 issued 2027-03-05, when a run executes 2027-03-12, then no letter (10-day interval).
- **AC-30.5** (FR-30.11) Given an RMA of 4 units at 100.00 USD with RETURN_TO_STOCK and 15% restocking, when received and dispositioned, then an inventory receipt of 4 at original cost exists and a credit note posts Dr 4000 400.00 / Cr 4200 Restocking 60.00 / Cr 1200 340.00 plus tax lines.
- **AC-30.6** (FR-30.11) Given SCRAP, then the receipt is followed by Dr 5900 / Cr 1400 at return cost.
- **AC-30.7** (FR-30.12) Given 10,000.00 accrued for 2027-03 on a 2% agreement, when settled, then a REBATE credit note of 10,000.00 posts and GL 2150 for the agreement is 0.00.
- **AC-30.8** (FR-30.14) Given an invited contact paying two invoices totalling 1,250.00 EUR by card in the Stripe sandbox, then a PORTAL payment is created from the webhook and both items are SETTLED within 60 s.
- **AC-30.9** (FR-30.9) Given a dispute on a 500.00 item, then dunning excludes it; when resolved as credit, a credit note posts and the item settles.

- **AC-30.10** (FR-30.15) Given `sim/phase5-month` on stage, then every §5.0 check passes and the reconciliation pack is produced and signed.
- **AC-30.11** (FR-30.16) Given DMF-DE customer C20 with an ACTIVE CORE mandate (FRST) and two items due 2027-03-20 totaling 2,380.00 EUR, when the DD run executes on 2027-03-05, then a pain.008 file with one FRST transaction of 2,380.00 and collection date 2027-03-20 validates against the XSD, a pre-notification is sent, and the mandate moves to RCUR after collection.
- **AC-30.12** (FR-30.16) Given that collection is returned with reason AC04, then the payment reverses, both items reopen and a 3.50 EUR FEE item exists.
- **AC-30.13** (FR-30.17) Given a payment short by 300.00 USD with reason "promo", when the deduction is matched to a promotion fund accrual of 300.00 and approved, then a credit note settles the deduction and the fund usage increases by 300.00.
- **AC-30.14** (FR-30.20) Given 20 customers selected for balance confirmation as of 2027-03-31, when 15 reply AGREED, 2 DISAGREED and 3 do not reply, then the difference list shows 2 rows with both balances and 2 DISPUTE cases exist.
- **AC-30.15** (FR-30.21) Given a 2,000.00 USD down payment on order SO-4000 and a final invoice of 5,000.00, then clearing posts Dr 2190 2,000.00 / Cr 1200 2,000.00 and the open item shows 3,000.00.
- **AC-30.16** (FR-30.22) Given a 10-line chargeback claim where 1 line has no agreement and 1 line duplicates a prior claim, then 8 lines are ACCEPTED, 2 REJECTED (CB-NOAGR, CB-DUP) and one credit note for the accepted total posts.
- **AC-30.17** (FR-30.19) Given a customer preference PAYMENT_NOTICE = EMAIL, when a payment applies, then a payment notice listing the settled items is emailed within 60 s and logged.

5.30.9 Test plan and fixtures

Fixtures: `cash-app-500` (300 BAI2 checks, 150 camt.053 entries, 50 remittances; short pays, discounts, multi-invoice, unidentified payers; answer key); `dunning-60`; `rma-30` (per disposition × costing method); `disputes-20`; `rebate-settle-12`; `portal-e2e` (Stripe and Adyen sandbox recordings); `sim/phase5-month` (generated month for DMF-US and DMF-DE: 200 POs, 150 production orders, 1,000 orders, 600 payments, 30 RMAs).

Test	Type	Target	Pass criterion
Matching	unit	BR-30.1	≥ 90% auto-match, 0 false matches
Subset-sum bound	property	matcher	≤ 5 items, < 50 ms per payment
Dunning	unit / E2E	run	exact letter set and levels
RMA E2E	E2E	sales + fulfillment + billing + inventory + ledger	30/30 balanced
Settlement accounting	contract (ledger)	discount, write-off, FX	balanced
BAI2 / camt.053 parsers	unit / property	importers	spec samples parse; malformed rejected with line numbers
Portal	E2E (Playwright + axe)	order, pay, return	pass; 0 serious a11y violations
Month simulation	E2E + reconciliation	sim/phase5-month	all §5.0 checks; 31/31 daily reconciliations clean

Reconciliation: Σ open_amount = GL 1200; Σ ON_ACCOUNT = GL 1205; daily settlements = GL 1000/1010 debits from receivables; fees = GL 4300; interest = GL 4310; RMA receipts = wave 16 receipts = GL 1400 debits from returns; rebate settlements = GL 2150 debits; deferred revenue, DNI, inventory, WIP, AP and tax checks re-run on simulation data; trial balance balanced.

5.30.10 Deployment and operations

Helm `passage-receivables` (Go), `passage-collections-workflows` (TS, Temporal namespace `collections`), `passage-customer-portal` (React + TS BFF; ingress `customer.<env>.passage.dev`; Keycloak realm `portal`), `passage-payments-gateway` (TS; provider keys in Vault; PCI scope limited to tokens), matcher ranker (Python, CPU). Databases `receivables-db`, `portal-db`. Config: `MATCH_AUTO_THRESHOLD=85`, `MATCH_MARGIN=20`, `DUNNING_INTERVAL_DAYS=10`, `RMA_EXPIRY_DAYS=30`, `WRITE_OFF_TOLERANCE=5.00/0.5%`. Flags: `ar.ml_ranker`, `ar.camt053`, `collections.interest`, `portal.payments`, `portal.returns`. Jobs: aging and risk score 01:30; lockbox/camt.053 SFTP poll every 15 min; dunning proposal weekly (issue manual); statements on the 1st; rebate settlement after period close; interest monthly. SLOs: open-items read p95 < 100 ms, apply p95 < 1 s, portal 99.9%, webhook processing < 60 s. Alerts: unapplied cash > 2% of monthly receipts for 3 days, webhook failures, reconciliation diff ≠ 0, run stuck in DRAFT > 7 days. Runbook: reverse a mis-application, re-import a file, re-issue a letter, rotate provider keys. Rollback: previous digest per deployable; Temporal workflows versioned.

5.30.11 Migration from AX / SAP

Source	Passage entity	Notes
CustTrans + CustTransOpen (open); CustSettlement (current year)	open_item; settlement history	AmountCur/AmountMST → amounts; CollectionLetterCode → dunning_level
CustCollectionLetterJour (last), CustInterestJour (open)	dunning history; open_item INTEREST	
ReturnTable / SalesLine (ReturnStatus not Closed)	return_order	disposition codes → BR-30.3
BSID (open), BSAD (current year)	open_item; settlement history	MANSP → dunning_block; MSCHL/MANST → level
MHNK/MHND (last run); UDM_DISPUTE (open); VBAK RE/LR (open)	dunning history; dispute; return_order	
KONA/VBOX open accruals	rebate_accrual opening	settled in Passage

Validation: Σ open items per customer = legacy AR aging to the cent; open dispute and RMA counts equal; dunning levels per item equal. Not migrated: cleared items before the current fiscal year, collection notes (lakehouse), legacy statements, CustPaymModeTable/lockbox setup (manual).

5.30.12 Dependencies, risks, open questions

- Depends on: w29 invoices and credit notes; w28 receipts for RMAs; w27 credit profile and holds; w26 accruals; w19 AP payments (rebate payout); w16 inventory; w12 FX; w11 journal; w5 documents and Keycloak realms; Phases 3–4 for the simulation.

Risk	Likelihood	Impact	Mitigation
False auto-match in production	Low	High	threshold + margin, reversible, weekly audit sample of 50
Dunning sent to disputed customers	Low	High	exclusions evaluated at issue time
RMA credit without receipt	Medium	Medium	CREDIT_ONLY requires duty ReturnCreditOnly and reason code
PCI scope creep in the portal	Low	High	hosted fields only; tokens stored; annual SAQ-A
Sign-off slips because accountant review is late	Medium	High	accountant engaged from wave 29 fixtures onward
Interim collection_case diverges from the wave 45 generic case model	Medium	Medium	collection_case fields are a subset of the w45 case schema; façade path kept; migration script owned by wave 45

Risk	Likelihood	Impact	Mitigation
SEPA R-transactions reopen items after dunning or statements already issued	Medium	Low	reopened items restart dunning at their prior level; statement shows the reversal on the next cycle

- Open questions: (1) Primary demo payment provider, Stripe or Adyen? Default Stripe sandbox; decide by wave 30 start. (2) Are dunning fees taxable in DMF-DE? Default untaxed in v1 pending the country pack (wave 48); decide at wave 30 mid-point. (3) Does the simulation include DMF-DE → DMF-US intercompany sales? Default no (wave 40); decide at wave 29 close.

5.30.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
automatic_settlement	legal entity	bool	true	true/false	Apply credits to oldest items automatically on post	AR parameters → Settlement → Automatic settlement	F.13 automatic clearing / OB74 clearing rules
settlement_priority	legal entity	list	DUE_DATE, CASH_DISCOUNT_DATE	ordered fields	Order of items for automatic settlement	AR parameters → Settlement → Settlement priority	OB74 clearing criteria
cash_discount_after_due	legal entity	bool	false	true/false	BR-30.4 discount eligibility	AR parameters → Settlement → Cash discount administration	OBA3 tolerance for cash discount
write_off_tolerance	legal entity / customer group	money + %	5.00 / 0.5%	≥ 0	BR-30.4	AR parameters → Settlement → Max. penny difference	OBA3 customer tolerance groups (T043G)
fx_settlement_enabled	legal entity	bool	true	true/false	cross-currency settlement	AR parameters → Settlement → Exchange rate at settlement	OB09 / OBA1 exchange-rate difference accounts
match_auto_threshold / match_margin	legal entity	int	85 / 20	0–100	BR-30.1	Advanced bank reconciliation matching rules	FF_5 / FEBAN search strings, interpretation algorithm
aging_period_definition	legal entity	list	0, 30, 60, 90, 180	ascending days	FR-30.6 buckets	AR parameters → Collections → Aging period definition	S_ALR_87012168 day intervals
aging_basis	legal entity	enum	DUE_DATE	DUE_DATE, INVOICE_DATE	FR-30.6	Aging "Aged by"	Net due date vs baseline
dunning_procedure	customer group	ref	STD-4	dunning procedures	BR-30.2 levels, grace, fees	Collection letter sequence (CustCollectionLetterTable)	FBMP dunning procedure (T047)
dunning_interval_days / dunning_min_amount	dunning procedure	int / money	10 / 50.00 USD / 45.00 EUR	≥ 1 / ≥ 0	BR-30.2	Collection letter "Days" / minimum	FBMP dunning interval / minimum amounts
collection_letter_basis	legal entity	enum	TRANSACTION	TRANSACTION, CUSTOMER	Level per item vs per customer	AR parameters → Collections → "Collection letter from"	FBMP "Dunning by dunning area / level per item"
interest_rate / interest_basis / interest_min	legal entity, customer group	numeric / enum / money	0 / ACT_365 / 5.00	ACT_365, ACT_360	BR-30.5	Interest codes (CustInterest)	OB46 interest indicators, OB82 terms, OBAA
sepa_creditor_identifier	legal entity	text	—	ISO creditor id (e.g. DE98ZZZ09999999999)	FR-30.16	Direct debit mandate setup → Creditor identifier	SEPA_CUST creditor id (FSEPA_M1 settings)
sepa_prenotification_days	legal entity	int	14	2–14	BR-30.11	Mandate pre-notification	SEPA pre-notification (F110 / FSEPA)
sepa_mandate_expiry_months	tenant	int	36	36	FR-30.16	Mandate expiry	SEPA_MANDATE validity
sepa_return_fee	legal entity	money	3.50 EUR	≥ 0	BR-30.11	Payment fees (CustPaymFee)	Return reason fee (OBWQ)
deduction_write_off_limit	legal entity	money	50.00 USD / 45.00 EUR	≥ 0	BR-30.12	Deduction workbench write-off limit	FSCM dispute auto-write-off (UDM_AUTO_WRITEOFF)
deduction_types / denial_reasons	legal entity	code list	PROMO, PRICE, SHORT_SHIP, DAMAGE, UNKNOWN / NOT_AUTHORIZED, EXPIRED, DUPLICATE	configurable	FR-30.17	Deduction types, denial reasons (TAM setup)	Dispute reason codes (UDM attributes)
case_sla_hours	case category	int	COLLECTION 72 / DISPUTE 120 / DEDUCTION 240	≥ 1	FR-30.18	Case category SLA	FSCM dispute planned close date
correspondence_types	legal entity	table	8 types (FR-30.19)	templates per language	FR-30.19	Collections email templates; form setup	OB77 correspondence types (T048), OB78 programs
confirmation_threshold / reminder_days	legal entity	money / int	10,000.00 / 14	≥ 0 / 1–60	FR-30.20	n/a	F.17 selection (balance limit)
down_payment_account / security_deposit_account	legal entity	account	2190 / 2195	chart	FR-30.21	Customer prepayment posting profile	OBXR special G/L A / H reconciliation accounts
down_payment_reduces_exposure	legal entity	bool	true	true/false	BR-27.3 exposure	n/a	Credit management update (special G/L in credit exposure)
rma_expiry_days / restocking defaults	legal entity	int / %	30 / 15%	1–365 / 0–100	FR-30.11, BR-30.3	Return order parameters (return item number, days)	VOV8 RE validity; KP00 restocking condition
statement_cycle	customer	enum	MONTHLY	NONE, WEEKLY, MONTHLY, QUARTERLY	FR-30.13	Customer account statement settings	KNB1-XAUSZ statement indicator (F.27)
portal_payment_providers	legal entity	set	STRIPE	STRIPE, ADYEN	FR-30.14	Payment services (credit card processor)	Digital payments add-on provider
vendor_dunning_min	legal entity	money	100.00 USD / 90.00 EUR	≥ 0	FR-30.23	n/a	F150 vendor dunning minimum

5.30.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Customer open items / line items	list inquiry	customer, status, due as-of, document type, special G/L	document, dates, amount, open, discount, level, dispute/promise flags	open_item	Customer transactions / open transactions (CustTransOpen)	FBL5N / FBL5H
Aged balances	operational report	entity, as-of, basis, buckets, customer group	current, 1–30 … > 180, total, % overdue	aging snapshot	SSRS Customer aging (Aged balances)	S_ALR_87012168 / S_ALR_87012178
Customer balance list with credit limit	operational report	entity, as-of	balance, limit, utilization, overdue	open_item + credit_profile	SSRS Customer balance list with credit limit	S_ALR_87012172 / FD33
Cash application worklist	list inquiry	source, status, score band, date	payment, payer, amount, candidates, score, residual	payment	Payment journal / CustOpenTrans settlement	FEBAN / FLB2 post-processing
Unapplied cash	KPI / list	entity, age	on-account amount, count, oldest	payment (ON_ACCOUNT)	Customer transactions not settled (payments)	FBL5N with payment on account
Dunning proposal and history	drill-down	run, customer, level	letters, items, fees, exclusions with reason	dunning_run, dunning_letter	Collection letter journal (CustCollectionLetterJour)	F150 log / MHNK / MHND
Interest notes	list inquiry	period, customer	item, days, rate, amount, status	interest_note	Interest notes (CustInterestJour)	F.24 / F.2B output
Disputes, promises and cases	list inquiry / KPI	owner, category, SLA status, reason	case, items, amount, age, SLA due, resolution	collection_case, dispute, promise	Collections workspace / Cases	UDM_SPECIALIST worklist, UDM_DISPUTE
Deductions	list inquiry / operational report	type, status, customer, age, claim match	deduction, amount, type, matched claim, decision	deduction	Deduction workbench / SSRS Deductions report	FSCM dispute case list (residual items)
Direct debit mandates and runs	list inquiry	status, sequence, scheme, last used	mandate ref, IBAN masked, scheme, sequence, last collection, expiry	dd_mandate, direct_debit_run	Direct debit mandate inquiry	FSEPA_M4 mandate display / F110 DD log
Customer account statement	operational report	customer, period, cycle	opening, transactions, closing, aging footer	open_item	SSRS Customer account statement	F.27
Balance confirmations	operational report	key date, run	customer, our balance, reply, their balance, difference	balance_confirmation	n/a	F.17 / F.18 / F.99 reply list
Return orders	list inquiry	status, reason, disposition, customer	RMA, lines, received, disposition, credit	return_order	Return orders list (ReturnTable) / SSRS Return acknowledgment	VA05 type RE / advanced returns monitor (MSR_TRAC)
Chargeback claims	list inquiry	distributor, status, period	claim, lines accepted/rejected, amount	chargeback_claim	Rebate management claim inquiry	Settlement management claim list
DSO and collection effectiveness	KPI	entity, period	DSO, best possible DSO, CEI %, average days to pay	open_item history	Collections KPIs (D365 workspace)	FSCM collection KPIs
Lockbox / statement import log	list inquiry	source, date, status	file, hash, items, matched, errors	lockbox_batch	SSRS Lockbox report	FLB2 log / FEBKO / FEBEP

5.30.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Customer account statement	monthly job (1st) / on demand / portal	header: customer, statement date, currency; lines: date, document, description, debit, credit, running balance; footer: aging buckets, remittance instructions	PDF, email, portal	none (statement date + customer)	informational; retain 7 years
Dunning / collection letter	dunning run :issue	header: level, letter date, customer, contact; lines: invoice, due date, days overdue, open amount, fee; footer: total due, payment instructions, legal text (level 4)	PDF, email, print, portal	`DL–` per entity	retain 7 years; level-4 text per country pack
Interest note	interest :post	lines: item, period, days, rate, interest	PDF, email	`IN–` gap-free (it is an open item document)	as invoice (10 years DE)
Payment notice / payment advice	payment applied (preference)	payment, date, settled items, discounts, residual	PDF, email	none	7 years
Open item list / individual letter	correspondence request	template-driven	PDF, email	none	7 years
Balance confirmation letter	confirmation run	key date, balance, open items, reply form / portal link	PDF, email	`BCF–` per run	audit evidence; retain 10 years
SEPA pre-notification	DD proposal	mandate ref, creditor id, amount, collection date	email, PDF	none	SEPA rulebook; retain 14 months minimum (with mandate)
SEPA direct debit file	DD run	GrpHdr, PmtInf per sequence type, DrctDbtTxInf with mandate data	pain.008.001.08 XML	message id per run	bank file; retain 10 years
SEPA mandate form	mandate creation	creditor, debtor, IBAN, BIC, mandate ref, scheme, signature	PDF (signed upload)	mandate ref	retain 14 months after last collection (rulebook), longer per w48
RMA authorization / return acknowledgment	:authorize / receipt	RMA number, lines, reasons, ship-to return address, expiry, label	PDF, email, portal	`RMA–`	7 years
Deduction decision (denial / approval)	workbench decision	deduction, claim, decision, reason	PDF, email	none (the re-bill is an invoice)	7 years
Chargeback response	claim decision	lines with ACCEPT/REJECT and reason codes	CSV, EDI 849 (w35)	claim number from distributor	7 years

5.30.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Aging snapshot and risk score	01:30 daily	customer	key = (customer, date)	< 10 min per 100k customers	rerun	snapshot missing by 03:00
Lockbox / camt.053 SFTP poll	every 15 min	bank account	SHA-256 of file; entry (statement id, ref)	< 2 min per 1,000 entries	quarantine malformed file with line numbers	import failure; unapplied > 2% for 3 days

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Dunning proposal	weekly Monday 06:00 (issue manual)	legal entity	run id	< 10 min per 50k items	DRAFT retained	DRAFT > 7 days
Interest calculation	monthly last day 20:00	legal entity	key = (item, period)	< 10 min	rerun idempotent	none calculated while overdue items exist
Statements	1st of month 04:00	customer	key = (customer, period)	< 30 min per 50k statements	retry delivery	undelivered > 1%
SEPA DD proposal and file	daily 07:00 (business days)	creditor id	run id; transaction end-to-end id	< 5 min per 20k items	file not released until reviewed	DD items due without active mandate
SEPA mandate expiry / R-transaction import	nightly 00:45; camt.054 with statements	mandate	(mandate, date); R-transaction ref	< 2 min	reverse idempotent	returns > 2% per run
Rebate settlement	after period close (w14 event)	agreement	key = (agreement, period)	< 10 min	per-agreement retry	accruals unsettled 5 days after close
Balance confirmation reminders	daily 08:00	run	key = (run, customer, reminder n)	< 1 min	retry	runs with no replies after 30 days
Case SLA timers	Temporal timers per case	case	workflow id	n/a	escalation to manager	SLA breach count > 0
AR to GL reconciliation	02:15 daily	legal entity	key = (entity, date)	< 5 min	report	Σ open ≠ GL 1200; ON_ACCOUNT ≠ GL 1205
RMA expiry	hourly	legal entity	status transition	< 1 min	retry	none

5.30.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
ARView	GET open-items, aging, statements; AR workspace	none	AR clerk, Collector, Controller, Auditor
ARPaymentPost	payments, :match, :apply, :on-account, :reverse	with InvoicePost (w29); with CustomerMaintain bank details (w7)	Cash application clerk
ARImport	lockbox / bank-statement imports	with ARPaymentPost when `import_apply_separation`	Treasury clerk
ARSettle	settlements:batch, write-off, discount	with CreditNotePost; write-off above tolerance needs ARWriteOffApprove	AR clerk
ARWriteOffApprove	write-offs above tolerance, deduction write-offs above limit	with ARSettle on the same item	AR manager
ARRefund	provider refunds, customer refunds via w19	with ARPaymentPost; with VendorMaintain (w7)	AR manager
DunningRun	dunning-runs, :review, :issue, :cancel	with ARSettle (letter suppression via dunning_block is logged and reviewed)	Collections manager
InterestPost	interest-notes :calculate, :post	with DunningRun under four-eyes	AR manager
CollectionsMaintain	promises, disputes, cases, contact log	none	Collector
DeductionMaintain	deduction workbench match, split, deny	with RebateSettle (w26/w30) and FundMaintain for the same fund	Deductions analyst
DirectDebitRun	mandates, DD proposal, file release	with CustomerMaintain bank details; with ARPaymentPost	Treasury clerk
ReturnMaintain / ReturnCreditOnly	return-orders, :authorize, :disposition; CREDIT_ONLY	ReturnCreditOnly with WarehouseReceive	Customer service; Returns supervisor
RebateSettle	rebates/settlements	with RebateMaintain (w26)	AR manager
BalanceConfirmationRun	confirmation runs, reply recording	with ARSettle (auditor independence)	Controller / external auditor read-only
PortalCustomer	/portal/api/v1/* scoped to own accounts	n/a (external)	Invited customer contacts

5.30.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
AR-0001	error	Settlements plus residual must equal the payment amount	apply validation	Adjust settlements or residual	§5.30.5 validation
AR-0002	error	Item {doc} is locked by another user	BR-30.7 row lock	Retry later	BR-30.7
AR-0003	error	File already imported ({hash})	duplicate import	none	BR-30.7
AR-0004	error	Line {n}: malformed BAI2 record type {type}	parser	Fix file and re-import	FR-30.3
AR-0005	warning	Payment placed on account; top suggestion score {score}	BR-30.1 < 60	Apply manually	BR-30.1
AR-0006	error	Write-off {amount} exceeds tolerance {tol}	BR-30.4	Request ARWriteOffApprove	BR-30.4
AR-0007	error	Dunning run must be REVIEWED before issue	FR-30.7	Review the run	BR-30.2
AR-0008	info	{n} customers excluded: dispute, promise or block	BR-30.2	none	BR-30.2
AR-0009	error	Disposition requires received quantity > 0	§5.30.5 validation	Receive first	BR-30.3
AR-0010	error	Credit-only return requires reason code and ReturnCreditOnly	risk control	Add reason / duty	BR-30.3
AR-0011	error	No active SEPA mandate for {customer} / {iban_masked}	FR-30.16	Activate mandate or change payment mode	BR-30.11
AR-0012	error	Mandate {ref} expired after 36 months without collection	FR-30.16	Obtain a new mandate	FR-30.16
AR-0013	warning	Collection returned ({reason}); items reopened and fee posted	R-transaction	Contact customer	BR-30.11
AR-0014	error	Deduction approval {amount} exceeds claim remaining {remaining}	BR-30.12	Split or deny remainder	FR-30.17
AR-0015	error	Chargeback line rejected: {code}	BR-30.15	Correct claim and resubmit	FR-30.22
AR-0016	warning	Balance confirmation disagreed by {customer}: difference {amount}	BR-30.13	Investigate case	FR-30.20
AR-0017	error	Down payment clearing exceeds remaining down payment {amount}	BR-30.14	Reduce clearing	FR-30.21
AR-0018	error	Case SLA breached for {case}	FR-30.18 timer	Escalate	FR-30.18
AR-0019	error	Portal payment amount exceeds selected open items	BR-30.9	Adjust selection	FR-30.14
AR-0020	error	Provider webhook signature invalid	payments-gateway	Check provider secret rotation	BR-30.9

5.30.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Open-items read (FBL5N-style, 200 rows)	500k per day; portal peak 100/s	30 / 100 / 250 ms	300/s	open items 30M rows × 500 B = 15 GB	(customer_id, status, due_date); partition by legal entity	portal served from read replica; staleness ≤ 5 s
Payment match (scored)	30k per day	20 / 50 / 120 ms	100/s	match logs 3 GB	candidates via (customer, open_amount)	ML ranker bypassed; rule score only
Payment apply	30k per day	200 ms / 1 s / 2 s	30/s	settlements partitioned by month, 5 GB	(open_item_id)	queue per customer; 409 on conflict
camt.053 / BAI2 import (5,000 entries)	50 files per day	— / 60 s / 120 s	100 entries/s	raw files object store 20 GB	(source, source_ref) unique	files processed sequentially per bank account
Dunning run (50k items)	weekly	— / 10 min / 20 min	100 items/s	letters 1 GB + PDFs	(customer, dunning_level)	proposal only; issue throttled
SEPA DD file (20k transactions)	daily	— / 3 min / 6 min	200 tx/s	2 GB XML	(mandate, collection_date)	split into multiple PmtInf / files of 10k
Portal page (dashboard)	50k sessions per day	150 / 400 / 800 ms	200 req/s	portal_db 2 GB	BFF cache 30 s	degrade to list views; payments remain available
Webhook to settled	5k per day	5 / 30 / 60 s	20/s	—	provider event id unique	queue; SLO 60 s at p99 under 3×

5.30.20 Localization and compliance hooks

- SEPA (EU/EEA): SDD Core and B2B rulebooks, pain.008.001.08 and pain.002 / camt.054 R-transactions; creditor identifier per country format; mandate retention 14 months after last collection; UK Bacs Direct Debit and US ACH debit (NACHA PPD/CCD with authorization records) are country-pack variants on the same mandate entity (w48).
- Dunning and interest: statutory late-payment interest (EU Directive 2011/7/EU: reference rate + 8 pp B2B; DE §288 BGB 40.00 EUR flat fee; FR 40.00 EUR indemnity and mandatory penalties text) set by country pack interest rules; dunning fee taxability (open question 2).
- Statements and confirmations: language, date and number formats per customer locale; balance confirmations support audit standards (ISA 505 external confirmations) with immutable replies.
- Bills of exchange (IT RiBa, ES pagaré, FR LCR, JP tegata) are not supported in v1; wave 36 adds treasury instruments and country packs enable them.
- Down payments: tax on advance payments (EU art. 65 VAT Directive) enabled per pack; DE requires a down-payment invoice with tax.
- Privacy and PCI: card data never touches Passage (SAQ-A); IBANs are stored encrypted and displayed masked; collection notes and case activities are personal data subject to w48 retention (7 years, then pseudonymized); portal contacts can export their data.
- Retention: open-item history and settlements 10 years (GoBD); dunning letters 7 years; R-transaction evidence 13 months minimum.
- Extension points: correspondence template slots, dunning procedure legal-text slot, interest rule provider, payment-file format registry (w19) and the chargeback EDI mapper (w35).

5.30.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX §12.3.1 #18 — Customer → Direct debit mandates (SEPA, CustDirectDebitMandate)	PARTIAL-30	ADDED as FR-30.16	
AX §12.3.1 #19 — Collections → Cases (collection cases)	PARTIAL-30	ADDED as FR-30.18 (minimal collection_case); generic case framework DEFERRED to wave 45	migration of collection_case to generic cases in 45
AX §12.3.2 #22 — Direct debit mandate inquiry	PARTIAL-30	ADDED as FR-30.16	§5.30.14 Direct debit mandates and runs
AX §12.3.3 #24 — Payments → Generate direct debit (pain.008)	PARTIAL-30	ADDED as FR-30.16	w19 ISO 20022 generator
AX §12.3.4 #27 — Direct debit mandate setup (SEPA creditor identifier)	PARTIAL-30	ADDED as FR-30.16	sepa_creditor_identifier
AX §12.3.4 #28 — Collections email templates (D365)	PARTIAL-30	ADDED as FR-30.19	email templates per correspondence type; SMTP transport is wave 4
AX §12.18.1 #176 — Deduction workbench (TAMDeductionWorkbench)	PARTIAL-30	ADDED as FR-30.17	
AX §12.18.2 #178 — Deduction inquiry (open, matched)	PARTIAL-30	ADDED as FR-30.17	§5.30.14 Deductions
AX §12.18.2 #181 — SSRS Deductions report	PARTIAL-30	ADDED as FR-30.17	Deductions report export
AX §12.18.3 #182 — Match deductions to claims / Write off small balances	PARTIAL-30	ADDED as FR-30.17	BR-30.12
AX §12.18.4 #187 — Deduction types / reason codes / denial reasons	PARTIAL-30	ADDED as FR-30.17	deduction_types, denial_reasons
AX §12.29.1 #233 — Cases list page / Case details (generic case management, needed by w30)	PARTIAL-45	DEFERRED to wave 45	FR-30.18 provides the interim entity; risk logged in §5.30.12
AX §12.3.1 #8, §12.3.2 #10 / #12, §12.3.3 #13, §12.3.4 #14 — Bill of exchange journals, inquiry, documents, remittance/protest, setup	GAP (propose 36)	DEFERRED to wave 36	treasury instruments theme
SAP §13.2 #29 — F150 dunning vendors with debit balances	PARTIAL-30	ADDED as FR-30.23	
SAP §13.3 #40 — FB12 / F.61 / F.64 request / print correspondence	PARTIAL-30	ADDED as FR-30.19	
SAP §13.3 #44 — SEPA mandate management (FSEPA_M1..M4)	PARTIAL-30	ADDED as FR-30.16	
SAP §13.3 #46 — OBXR special G/L customers (down payments, security deposits, BoE)	PARTIAL-30	ADDED as FR-30.21 (down payments A/F, security deposits H); BoE (W) DEFERRED to wave 36	
SAP §13.3 #48 — OB77 correspondence types / periodic account statements	PARTIAL-30	ADDED as FR-30.19	statements remain FR-30.13
SAP §13.3 #49 — SEPA mandate customizing	PARTIAL-30	ADDED as FR-30.16	§5.30.13 SEPA parameters
SAP §13.32 #358 — Customer-facing configurator (web / SAP CPQ)	PARTIAL-30	ADDED as FR-30.25	
SAP §13.1 #9 — F.50 / F.52 G/L interest calculation	GAP (propose 30)	ADDED as FR-30.24	intercompany pairing wave 40
SAP §13.1 #19 — OB46 / OB82 / OBAA interest indicators	GAP (propose 30)	ADDED as FR-30.24	interest_rate / interest_basis parameters
SAP §13.3 #43 — F.99 customer balance confirmation	GAP (propose 30)	ADDED as FR-30.20	
SAP §13.1 #11 — F.97 / F.98 / F.99 balance confirmations (GL / AP / AR; propose 19, names w30)	GAP (propose 19)	ADDED as FR-30.20 for customers; vendor (F.98) and GL (F.97) DEFERRED to wave 19	same run engine reused

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP §13.20 #231 — Chargeback / ship-and-debit claims from distributors (EDI 844/849)	GAP (propose 30)	ADDED as FR-30.22	EDI mapping in wave 35
SAP §13.20 #235 — Chargeback / claim types	GAP (propose 30)	ADDED as FR-30.22	configurable claim types
SAP §13.3 #39 — F-34 / F-36 / F-33 bill of exchange payment / discount (AR)	GAP (propose 36)	DEFERRED to wave 36	
SAP §13.3 #41 — S_ALR_87012199 customer bill of exchange list	GAP (propose 36)	DEFERRED to wave 36	
SAP §13.3 #45 — FBWD / FBWE AR bill of exchange presentation	GAP (propose 36)	DEFERRED to wave 36	

Part 6 — Phase 6: Warehouse, Quality & Logistics (waves 31–35)

6.0 Phase overview

Business outcome at phase exit. At the end of wave 35 the demo manufacturer (live since wave 30) runs its warehouses on location-directed work executed from handheld scanners, inspects inbound, in-process and outbound inventory under rule-generated quality orders, plans and rates outbound freight against real carrier rate cards, reconciles carrier invoices and lands freight into inventory cost, and exchanges purchase orders, ASNs and invoices with a demo distributor over X12 EDI and Peppol. Every one of those flows posts into the wave-16 inventory ledger and the wave-11 universal journal through the existing posting APIs; Phase 6 adds no new financial truth, only new operational sources of it. The Phase 6 services are `wms` (Go, `wms_db`), `wms-mobile` BFF (TypeScript), `quality` (Go, `quality_db`), `tms` (Go, `tms_db`) with a Python `tms-optimizer` sidecar (OR-Tools), and `integration-hub` plus `edi` (TypeScript, `integration_db`). All run on the ADR-0001 stack; nothing in this phase introduces new infrastructure.

Phase-exit sign-off criteria

- Demo distributor ↔ demo manufacturer exchange PO (850/ORDERS) → ASN (856/DESADV) → invoice (810/INVOIC and Peppol UBL) end-to-end with zero manual re-keying; the invoice lands as a wave-19 vendor invoice at the distributor and matches 3-way.
- 100 location-directive fixtures resolve to the expected location; 500-order wave release creates work and replenishment with no deadlocks and p95 task assignment under 1 s.
- Mobile E2E inbound-to-outbound run on an emulated Zebra device using scanned GS1-128 labels passes in CI; 200 queued offline transactions sync without loss or duplicate.
- Quality property test: no pick, ship or consume succeeds against a quant with `quality_status = BLOCKED` across 10,000 randomized attempts; AQL sample sizes equal ANSI/ASQ Z1.4 Table I/II-A for every lot-size class.
- VRP solutions on Solomon C1/R1/RC1 instances are feasible and within 10% of best-known distance; 200 rating fixtures produce expected charges to the cent; landed cost allocation sums to the carrier invoice exactly.
- Daily reconciliation: `wms` on-hand by location sums to the wave-16 inventory ledger on-hand per item/warehouse; freight accruals in `tms` tie to journal account 2150 (accrued freight) to the cent.

Cross-wave data model summary

Entity	Owning service	Key relationships
Location, Zone, LocationProfile	wms	Warehouse (wave 6); capacity used by Work and Quant
LicensePlate (nested)	wms	Parent LP; Quant; Load; ShipmentContainer (wave 28)
Quant (on-hand by item × dims × location × LP)	wms	Projection of inventory ledger (wave 16); carries `quality_status`
LocationDirective, WorkTemplate, WaveTemplate, ReplenishmentTemplate	wms	Metadata; evaluated by the directive engine
Work, WorkLine, Wave, Load, DockAppointment	wms	Sales shipment (wave 28), purchase receipt (wave 18), production order (wave 21), TMS Load
MobileFlow, MobileMenu, DeviceSession, OfflineTransaction	wms-mobile	Work; WorkUser (Keycloak subject)
Test, TestGroup, SamplingPlan, QualityAssociation	quality	Item/product (wave 8); Party (wave 7)
QualityOrder, QualityResult, UsageDecision	quality	Quant (blocking), PurchaseReceipt, ProductionOrder, Shipment
Nonconformance, Correction (CAPA), Instrument	quality	QualityOrder; Vendor; Temporal workflow id
Carrier, CarrierService, RateCard, Accessorial, FuelIndex	tms	Party (wave 7)
Shipment, Route, Load (TMS view), Tender, TrackingEvent	tms	wms Load; Sales shipment (wave 28)
FreightBill, CarrierInvoice, LandedCostVoyage	tms	Vendor invoice (wave 19); inventory cost (wave 20); journal (wave 11)
TradingPartner, EDIProfile, Map, Interchange, Message, Acknowledgment	edi	Party; PurchaseOrder, SalesOrder, Shipment, Invoice
ApiClient, Subscription (webhook/Kafka), DeliveryAttempt, Quota	integration-hub	Keycloak client; Kafka topics
CrossDockTemplate, CrossDockLink, SortTemplate, SortPosition, VasOrder, PackagingSpec	wms	Work; Shipment (wave 28); kit BOM (wave 9); added by FR-31.13–FR-31.17
LaborStandard, LaborLine, PerformanceDocument	wms	Work, WorkUser; payroll export (wave 42); FR-31.18
HazmatProfile, SegregationMatrix, FireContainmentSection	wms	Item (wave 8); LocationProfile; read by tms DG checks (FR-34.15); FR-31.19
AutomationChannel, TelegramProfile, CommunicationPoint, TransportOrder	wms	Location; Work; edge gateway (wave 25); FR-32.12
Printer, DeviceType, PrintQueue, RoutingRule	print service	PrintJob; FR-32.13
ComplianceList, ItemComposition, ComplianceResult, SafetyDataSheet, SdsDelivery	quality	Item, BOM (wave 9), Shipment (wave 28); FR-33.13–FR-33.14
IncomingCertificate, PhysicalSample, SpecificationSet, InspectionPoint	quality	QualityOrder; PO line (wave 17); production operation (wave 21); FR-33.15, FR-33.19–FR-33.22
StabilityStudy, PullPoint, Audit, QuestionList, Finding, Fmea, ControlPlan	quality	Batch (wave 16); Nonconformance/CAPA; TestGroup; FR-33.16–FR-33.18
VoyageLeg, ShippingContainer, Folio, GoodsInTransitOrder	tms	LandedCostVoyage; PO lines (wave 17); in-transit warehouse (wave 16); FR-34.12
YardLocation, TransportUnit, YardMove, DriverVisit, GateEvent	tms	DockAppointment (wave 31); Load; Carrier; FR-34.13–FR-34.14
DgCheckResult, Vehicle, Driver, FreightRfq, LoadPlan, ScreeningResult	tms	HazmatProfile (wave 31); RFQ (wave 17); Party (wave 7); FR-34.15–FR-34.19
PunchoutCatalog, PunchoutSession, SelfBillingDocument, ConnectorConfig	integration-hub / edi	Requisition (wave 17); customer invoice (wave 29); ECR (wave 9); FR-35.13–FR-35.16

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
dev.passage.wms.work.created.v1	wms	wms-mobile, analytics	work_id, work_type, warehouse, priority, lines[]
dev.passage.wms.work.completed.v1	wms	inventory, sales-fulfillment, production	work_id, lines with from/to location, LP, qty
dev.passage.wms.load.shipped.v1	wms	tms, sales-fulfillment, edi (856)	load_id, shipments[], LPs/SSCCs, carrier, ship_date
dev.passage.wms.receipt.registered.v1	wms	purchasing (wave 18), quality	receipt_id, po lines, LPs, qty
dev.passage.quality.order.created.v1	quality	wms (block quant), analytics	quality_order_id, trigger, item, quant_ref, sample_size

Event (CloudEvents type)	Producer	Consumers	Payload summary
dev.passage.quality.usagedecision.posted.v1	quality	wms, inventory, purchasing, production	quality_order_id, decision, accept_qty, reject_qty
dev.passage.quality.nonconformance.created.v1	quality	workflow, vendor-scorecard	ncr_id, source, severity, item, vendor
dev.passage.tms.shipment.rated.v1	tms	sales (freight charge), analytics	shipment_id, carrier, service, charges[], total
dev.passage.tms.tender.accepted.v1	tms	wms (dock), edi (204/990)	load_id, carrier, pickup_window
dev.passage.tms.freightbill.matched.v1	tms	accounts-payable (wave 19), costing (wave 20)	freight_bill_id, invoice_id, variance, landed_cost_lines[]
dev.passage.edi.message.received.v1	edi	purchasing, sales, ap	partner, standard, transaction_set, control numbers, document ref
dev.passage.edi.message.sent.v1	edi	analytics, integration-hub	partner, transaction_set, ack status
dev.passage.hub.webhook.delivered.v1	integration-hub	analytics	subscription_id, event_id, attempt, status
dev.passage.wms.crossdock.allocated.v1	wms	sales-fulfillment, analytics	receipt_id, demand_line_ref, qty, staging_location
dev.passage.wms.automation.fault.v1	wms	wms-mobile, alerting	channel_id, communication_point, lp_id, fault_code
dev.passage.quality.compliance.blocked.v1	quality	sales, purchasing	item_id, region, list_id, action (BLOCK/WARN), source_ref
dev.passage.quality.sds.distributed.v1	quality	sales-fulfillment, customer portal	item_id, sds_version, customer_id, shipment_id
dev.passage.tms.goodsintransit.shipped.v1	tms	inventory, costing	voyage_id, container_ids, lines[], estimated_costs[]
dev.passage.tms.gate.checkedin.v1 / checkedout.v1	tms	wms (door work), analytics	visit_id, appointment_id, tu_id, seal_no, door_or_slot
dev.passage.tms.dgcheck.failed.v1	tms	planners, wms	load_id, rule_ids[], classes[]
dev.passage.tms.screening.held.v1	tms	sales, compliance tasks	load_id, party_id, status, list_version
dev.passage.edi.selfbilling.matched.v1	edi	accounts receivable (wave 29), cash application (wave 30)	document_id, customer_id, matched_lines, differences[]
dev.passage.hub.punchout.returned.v1	integration-hub	purchasing (wave 17)	session_id, supplier_id, lines[], rejected[]

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Directive engine performance collapses on large location sets (50k bins)	Medium	High	Indexed candidate pre-filter per zone/profile; benchmark at 50k locations in wave 31 CI; p95 < 200 ms per line	wms lead
Scanner hardware variance (Zebra DataWedge vs Honeywell) breaks scan capture	Medium	Medium	Keyboard-wedge fallback plus intent API adapters; emulator E2E for both profiles	mobile lead
Quality blocking is bypassed by a path not covered by the property test	Low	High	Blocking enforced in `inventory` ledger reservation, not only in `wms`; property test enumerates all transaction types	quality lead
VRP solver time exceeds planning window on 1,000-stop problems	Medium	Medium	Time-limited local search (60 s) with feasible first solution guaranteed by sweep heuristic	tms lead
EDI partner specifics (implementation guides) diverge from base maps	High	Medium	Partner-level map overrides; simulator fixtures per partner; 997/CONTRL negative tests	integration lead
Peppol access point certification delays	Medium	Medium	Certified third-party AP (SMP/AS4) adapter first; own AS4 endpoint deferred to v1.x	integration lead

6.31 Wave 31: Advanced Warehouse Management

6.31.1 Scope and business context

Wave 28 ships sales orders from simple warehouses (item × location on-hand, manual pick lists). Wave 31 replaces that with the D365 WHS / SAP EWM model: a location hierarchy with capacity and profiles, license plates (LP) nested to any depth, and work that the system creates and directs by rule. After this wave a warehouse manager configures put-away and pick location directives, work templates and wave templates in metadata; releasing sales, transfer and production orders to the warehouse creates work with explicit from/to locations; replenishment runs on min/max, demand or wave triggers; loads are built and docks scheduled; receiving is ASN-driven with LPs. Out of scope: mobile execution (wave 32 — in wave 31 work is confirmed from the web workspace), quality blocking (wave 33), carrier rating (wave 34), cross-docking and catch-weight work (v1.x), labor standards (v1.x).

Scope amendment (coverage closure). The AX and SAP gap registers (coverage-ax §12.99, coverage-sap §13.99) move four items into this wave that the paragraph above lists as v1.x or does not name: planned and opportunistic cross-docking (FR-31.13), outbound sorting (FR-31.14), value-added services and kitting (FR-31.15), and engineered labor standards with planned-vs-actual reporting (FR-31.18). Transfer-order warehouse flows (FR-31.16), packaging specifications (FR-31.17) and hazardous-material storage compatibility (FR-31.19) are now explicit requirements. Catch-weight work and incentive pay calculation stay out of scope. Yard management moves to wave 34 (FR-34.13). §6.31.21 lists every register row and its disposition.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
D365 location directives (WHSLocationDirective form) fail silently: work is created with blank put location or "no location found" only visible in the work details after wave processing; debugging requires the "Location directive failures" log (10.0.x) or SQL on WHSWorkLine	Directive evaluation is a batch-side X++ class with no trace; query ranges are opaque	`/location-directives:simulate` returns a rule-by-rule trace; blocked work has status `BLOCKED_NO_LOCATION` with the failing step on the wave	Time to diagnose a directive miss: 30–60 min (consultant with SQL) → < 1 min from the trace; 0 silent blanks
EWM put-away bin determination (/SCWM/T333 storage type search sequence + bin type + storage section) spans 6 IMG nodes; a change needs a transport and a test cycle	Configuration split across customizing tables with cross-dependencies	One LocationDirective entity with ordered lines/actions, versioned metadata, simulate-before-save	Setup clicks for a new put-away rule: ~40 across 6 transactions → ≤ 8 in one designer; change lead time days → minutes
D365 wave processing (WHSWaveProcessing batch) runs every 5–15 min; pickers wait; a failed wave leaves partial work and the wave in "Held" with a generic infolog	Batch framework polling; non-atomic processing per order	Temporal `WaveReleaseWorkflow` starts on release, per-order activities with per-line failure reasons	Order release to assignable work: 5–15 min batch → < 5 s p95 for a 50-order wave; failures itemized 100%
EWM /SCWM/MON and D365 "Work details" cannot show capacity or why a location is full; over-capacity bins happen because capacity check is advisory (D365 `Location stocking limits`)	Capacity is a soft check in some paths	Capacity is an invariant: creation and confirmation both enforce it; property-tested	Over-capacity incidents: tolerated today → 0 (test-enforced)
On-hand by LP in AX (WHSInventOnhand vs InventSum) drifts and is reconciled manually by inventory journals	Two on-hand tables maintained by separate code paths	Quants are a projection of the single inventory ledger; daily automated reconciliation with alert	Monthly reconciliation effort: 4–8 h → 0 h; drift detected within 24 h

6.31.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Location master	WMSLocation, WHSLocationProfile, WHSZone, WHSLocationFormat; form WHSLocation	/SCWM/LAGP storage bins, /SCWM/T301 storage types, sections /SCWM/T302, bin types /SCWM/T303	wms.Location, Zone, LocationProfile	full
License plates / HUs	WHSLicensePlate, WHSContainerTable	/SCWM/HUHDR, /SCWM/HUITM, packaging material	wms.LicensePlate (nested), Quant	full
Put-away/pick strategy	WHSLocationDirective, WHSLocationDirectiveLine, WHSLocationDirectiveAction; form WHSLocationDirective	Put-away strategies (/SCWM/T333 storage type search, bin determination), stock removal strategies (FIFO/LIFO/partial)	wms.LocationDirective with ordered rules	full
Work templates	WHSWorkTemplateTable, WHSWorkTemplateLine, WHSWorkTransType	Warehouse process types /SCWM/TPROCTY, warehouse order creation rules /SCWM/WOCR	wms.WorkTemplate	full
Work / tasks	WHSWorkTable, WHSWorkLine; form WHSWorkTable	/SCWM/ORDIM_O warehouse tasks, /SCWM/WHO warehouse orders	wms.Work, WorkLine	full
Wave management	WHSWaveTemplateTable, WHSWaveTable, WHSWaveLine; class WHSWaveProcessing	/SCWM/WAVE, /SCWM/WAVEHDR, /SCWM/WAVEITM; wave templates /SCWM/WAVETMP	wms.WaveTemplate, Wave; Temporal `WaveReleaseWorkflow`	full
Replenishment	WHSReplenishmentTemplate, WHSReplenishmentTemplateLine; class WHSReplenishmentProcess	Replenishment control (/SCWM/REPL, planned/order-related/automatic)	wms.ReplenishmentTemplate	full
Loads / dock	WHSLoadTable, WHSLoadLine, WHSDockAppointment (appointment scheduling)	Yard management, door/transportation units /SCWM/TU, /SCWM/DOOR	wms.Load, DockAppointment	full
ASN receiving	WHSASNItem, WHSUOMStructure, packing structure import	Inbound delivery /SCWM/INB (EWM delivery), expected goods receipt	wms.AsnPackingStructure → LP receiving	full
Cycle counting	WHSCycleCountPlan, WHSCycleCountThreshold, WHSCountingWork	Physical inventory /SCWM/PI_DOCUMENT, cycle counting indicator	wms.CountPlan, counting Work (web in 31, mobile in 32)	full
Slotting	WHSSlottingTemplate (D365 10.0.x)	EWM slotting (/SCWM/SLOT)	wms.SlottingRun (ABC by picks, recommendations)	partial — recommendations only, no auto-move
Labor management	n/a (ISV)	EWM labor management	—	dropped: v1.x
Cross-docking	WHSCrossDocking, cross-docking templates, planned cross docking (D365 10.0.x), opportunistic cross docking on work policies	LE-WM LXDCA/LXDCK cross-docking monitor; EWM planned and opportunistic cross-docking (push deployment, pick from goods receipt)	wms.CrossDockTemplate, CrossDockLink; work type CROSS_DOCK	full
Outbound sorting	Outbound sorting templates and sort positions (D365 10.0.x)	EWM consolidation groups, sorter integration, /SCWM/PACK staging	wms.SortTemplate, SortPosition	full
VAS and kitting	Kit orders (D365 10.0.x), BOM journals for assembly	/SCWM/VASEXEC VAS orders, kit-to-order / kit-to-stock	wms.VasOrder; work type VAS; kit BOM (wave 9)	full
Packaging specifications	Unit sequence groups, WHSPhysDimUOM, container types, container build templates (WHSContainerBuildTemplate)	/SCWM/PACKSPEC packaging specifications with condition-based determination	wms.PackagingSpec	full
Labor management (revised)	n/a (ISV); work audit and workload visualization only	EWM labor management /SCWM/LM: engineered labor standards, planned vs actual, performance documents	wms.LaborStandard, LaborLine, PerformanceDocument	partial — standards and reports; incentive pay calculation out of scope (replaces the "dropped: v1.x" row above)
Hazardous-material storage	Hazardous material item flags; product compliance (D365 PI)	LE-WM hazardous material storage (LX06 fire-containment sections, LX24 hazmat list); EWM hazardous substance storage checks (storage class, water hazard class)	wms.HazmatProfile, SegregationMatrix, FireContainmentSection	full
Transfer order warehouse flow	WHS transfer order shipping and receiving with LP; release to warehouse	Stock transport order with EWM outbound/inbound delivery (/SCWM/PRDO, /SCWM/PRDI)	Work types TRANSFER_ISSUE/TRANSFER_RECEIPT with LP identity across sites	full

6.31.3 Functional requirements

- **FR-31.1 Location hierarchy with profiles** — Must. The system shall model locations as `warehouse / zone / aisle / rack / shelf / bin` with a LocationProfile that defines capacity (weight, volume, LP count, item count), mixing rules and allowed LP behavior.
  - Notes: capacity limits are checked when work is created and again at confirmation; a location with `allow_mixed_items = false` rejects a second item; location name is generated from a LocationFormat (`A-01-02-03` → aisle 1, rack 2, shelf 3); maximum 1,000,000 locations per warehouse.
- **FR-31.2 Nested license plates** — Must. The system shall track inventory on license plates that nest (pallet → case → inner) to depth 5, with a single move of a parent moving all children atomically.
  - Notes: LP id is an SSCC-18 when generated by Passage (extension digit 0, GS1 company prefix from wave 8 config); external LPs are accepted up to 40 characters; an LP can be at exactly one location or in transit on one Work.
- **FR-31.3 Location directives** — Must. The system shall resolve a destination (put-away) or source (pick) location for each work line by evaluating ordered directives, lines and actions as specified in BR-31.1.
  - Notes: directive scope = work order type × warehouse × optional query on item/customer/vendor; strategies: `consolidate`, `fifo`, `lifo`, `fefo`, `empty_location`, `match_packing_quantity`, `round_up_to_lp`; evaluation p95 < 200 ms per line at 50,000 locations.
- **FR-31.4 Work templates** — Must. The system shall create Work from templates whose lines define pick/put pairs, work class, stop-after-line and max lines/LPs per work header.
  - Notes: templates have a query (e.g., `shipping_method = LTL`); `work_break` rules split by customer, shipment or load; work class restricts which users may execute.
- **FR-31.5 Wave templates and release** — Must. The system shall group released order lines into waves by template (query, max lines 5,000, max weight, scheduled or manual release), create work and replenishment on release, and lock or allow late additions.
  - Notes: release runs as Temporal `WaveReleaseWorkflow` with per-order activities; automatic wave release on schedule (cron) or on order release; failed lines are listed on the wave with reason codes.
- **FR-31.6 Replenishment** — Must. The system shall create replenishment work by min/max (per location × item), by demand (wave shortage) and by load demand, from configured source zones.

    - Notes: min/max runs every 15 minutes by default; demand replenishment is created inside the wave before pick work; replenishment work carries priority 10 (higher than picks at 50) and must complete before the dependent pick is assignable.
- **FR-31.7 Work execution and exceptions** — Must. The system shall assign work by priority, work class and user filter; support short pick with reason, skip location, over-pick within tolerance and cancel-with-reversal.
    - Notes: short pick re-queues remaining qty as new work after a configurable retry; assignment is pessimistic (`SELECT … FOR UPDATE SKIP LOCKED`) so two users never receive the same work; p95 assignment < 1 s under 10k lines/day simulation.
- **FR-31.8 Loads and dock appointments** — Must. The system shall build loads from shipments (manual, rule, or by route), confirm loads, and schedule dock appointments with 15-minute slot granularity per door.
    - Notes: load confirmation posts packing slips via the wave-28 API; dock doors have calendars (wave 6); overlapping appointments on one door are rejected.
- **FR-31.9 ASN-driven receiving** — Must. The system shall import an ASN packing structure (from EDI 856 in wave 35, or JSON API now) and receive by scanning/selecting the pallet LP, creating quants per nested LP.
    - Notes: receipt without ASN remains supported (item + qty + LP); variance to ASN produces a receipt discrepancy with tolerance ±2% default.
- **FR-31.10 Cycle counting** — Should. The system shall generate count work by plan (location set, frequency), threshold (on-hand ≤ n) and ad hoc; count differences post via the wave-16 counting journal after supervisor approval above a tolerance.
    - Notes: tolerance default 1% of value or 5 units; blind counts hide expected qty.
- **FR-31.11 Slotting analysis** — Could. The system shall compute ABC velocity by picks/30 days and recommend item-to-location moves to forward pick zones.
    - Notes: output is a recommendation list; applying generates movement Work.
- **FR-31.12 Warehouse KPIs** — Should. The system shall expose lines picked per hour, work aging, dock utilization, replenishment lead time, short-pick rate via the Cube semantic layer.
    - Notes: materialized hourly; lag ≤ 60 s per PRD §8.
- **FR-31.13 Cross-docking (planned and opportunistic)** — Must. The system shall move inbound quantity from a receiving location to an outbound staging or shipping location without put-away to storage, either planned (a CrossDockTemplate links an expected supply — PO line, ASN or transfer receipt — to a released demand line before arrival) or opportunistic (at receipt registration, released demand with an open shortage for the item is found and CROSS_DOCK work is created instead of put-away).
    - Notes: allocation per BR-31.11; demand priority `SHIP_DATE`, `ORDER_PRIORITY` or `FIFO_RELEASE`; supply window default 48 h before the demand ship date; partial cross-dock splits the receipt (remainder follows put-away directives); CROSS_DOCK work uses PUT directives with the new strategy `outbound_staging`; the reservation moves from supply to demand in the wave-16 ledger so ATP never counts the quantity twice.
- **FR-31.14 Outbound sorting** — Should. The system shall sort picked LPs or cartons at a sortation area into sort positions defined by a SortTemplate (criteria: shipment, load, customer, route or carrier) and close a position into an outbound LP when it is full, when the last expected LP arrives, or on demand.
    - Notes: position capacity by LP count, weight and volume; scanning at the sort station returns the target position (p95 < 300 ms); closing prints the shipping label through the wave-32 print service and builds the LP hierarchy used by the 856 (BR-35.2).
- **FR-31.15 Value-added services and kitting** — Should. The system shall create VAS orders from sales-order line requirements or customer-item VAS profiles (labeling, tagging, gift wrap, repack, kit assembly) and run them as VAS work steps between pick and pack; kit-to-order assembles a kit BOM (wave 9 BOM type `KIT`) in the warehouse and kit-to-stock receives finished kits into inventory.
    - Notes: VAS work targets a VAS station location; assembly posts per BR-31.12; disassembly reverses the posting; VAS step time feeds labor standards (FR-31.18) and customer VAS charges (wave 29 charge code `VAS`).
- **FR-31.16 Transfer order warehouse flow** — Must. The system shall release transfer orders (wave 16) to the warehouse, create TRANSFER_ISSUE work, ship LPs to the in-transit warehouse, and receive the same LP ids at the destination with TRANSFER_RECEIPT work, applying the transfer order's under- and over-delivery tolerances.
    - Notes: LP identity is preserved across sites; an LP not on the transfer shipment is rejected (BR-31.16); in-transit quants belong to the in-transit warehouse in the ledger; a line within under-delivery tolerance (default 0%) closes on receipt.
- **FR-31.17 Packaging specifications** — Should. The system shall define packaging specifications per item or item group, optionally per customer: levels (each, inner, case, pallet) with quantity per level, packaging material item, LP type, label template and stacking limit. Work creation, the `round_up_to_lp` strategy, cartonization at pack and VAS repack use the specification found by most-specific match.
    - Notes: determination order item+customer → item → item group+customer → item group → none; packaging material consumption posts through wave 16 when `consume_material = true`; replaces D365 unit sequence groups plus container build templates and SAP /SCWM/PACKSPEC.
- **FR-31.18 Labor standards and performance** — Should. The system shall compute an engineered labor standard (planned seconds) per work line from formulas over travel distance, quantity, LP count, weight and fixed setup time, capture actual time from work confirmations and wave-32 step telemetry, and report performance (Σ planned ÷ Σ actual × 100) per user, work class, shift and warehouse.
    - Notes: formula per BR-31.14, written in the wave-3 expression engine and effective-dated; travel distance from location coordinates (x, y, z), Manhattan by default; indirect activities (wave 32 indirect activity codes) are included or excluded per setting; PerformanceDocuments are immutable once the shift closes; incentive pay is not calculated here — closed documents export to payroll (wave 42) as a time type; named per-user views require WMS_LABOR_VIEW and `labor_personal_visibility = NAMED` (§6.31.20).
- **FR-31.19 Hazardous-material storage compatibility** — Should. The system shall hold a HazmatProfile per item (UN number, proper shipping name, class/division, packing group, storage class per TRGS 510 or NFPA 30 hazard category, water hazard class, flash point in °C, net explosive mass) and enforce storage compatibility: location profiles declare allowed storage classes and a fire-containment section, and the directive engine excludes locations where the line would sit next to an incompatible class or exceed a section quantity limit.
    - Notes: segregation per BR-31.15 against a reference-data matrix (allowed / separate / prohibited per class pair); section limits in kg or L; hazmat inventory per section is an inquiry (LX06/LX24 equivalent); wave 34 reuses the same profile for transport DG checks (FR-34.15), and wave 33 links it to SDS versions (FR-33.14).

6.31.4 Data model

Location

Field	Type	Required	Notes
location_id	uuid	yes	PK
warehouse_id	uuid	yes	FK warehouse (wave 6)
name	text	yes	unique per warehouse, generated from LocationFormat
zone_id	uuid	yes	FK Zone
profile_id	uuid	yes	FK LocationProfile
aisle, rack, shelf, bin	text	no	parsed components; indexed for sort order
max_weight_kg, max_volume_m3, max_lps, max_items	numeric(18,6)/int	no	null = unlimited
pick_sequence	int	yes	travel-path ordering within zone

Field	Type	Required	Notes
status	enum	yes	ACTIVE, BLOCKED_IN, BLOCKED_OUT, BLOCKED, INACTIVE

LicensePlate

Field	Type	Required	Notes
lp_id	text(40)	yes	PK; SSCC-18 when generated
warehouse_id	uuid	yes	FK
parent_lp_id	text(40)	no	FK LicensePlate; depth ≤ 5 (trigger-enforced)
location_id	uuid	no	null only when on an in-progress Work
in_transit_work_id	uuid	no	FK Work; mutually exclusive with location_id
lp_type	enum	yes	PALLET, CASE, TOTE, INNER, CONTAINER
gross_weight_kg, volume_m3	numeric(18,6)	no	computed from children when nested
created_at	timestamptz	yes	

Work / WorkLine

Field	Type	Required	Notes
work_id	uuid	yes	PK; number sequence WRK-* (wave 5)
warehouse_id	uuid	yes	FK
work_type	enum	yes	PURCHASE_PUTAWAY, SALES_PICK, TRANSFER_ISSUE, TRANSFER_RECEIPT, PRODUCTION_PICK, RAF_PUTAWAY, REPLENISHMENT, MOVEMENT, COUNT, RETURN
status	enum	yes	see BR-31.3 state machine
priority	smallint	yes	1 (highest)–99; default by template
work_class_id	uuid	yes	FK; filters eligible users
wave_id, load_id, order_ref	uuid/text	no	FK Wave, Load; polymorphic order ref
assigned_user_id, locked_at	text/timestamptz	no	pessimistic lock
line_no, line_type (PICK/PUT), item_id, dims jsonb, qty, uom, from_location_id, to_location_id, lp_id, target_lp_id, line_status	—	yes	WorkLine; line_status OPEN/IN_PROGRESS/COMPLETED/SKIPPED/CANCELLED

LocationDirective

Field	Type	Required	Notes
directive_id	uuid	yes	PK
sequence	int	yes	evaluation order; unique per (work_type, warehouse, direction)
work_type, direction	enum	yes	direction PUT or PICK
warehouse_id	uuid	yes	
query	jsonb	yes	metadata query (wave 3 query model) against order line
lines	jsonb[]	yes	each: seq, from_qty, to_qty, uom, locate_packing_qty bool, restrict_by_uom
actions	jsonb[]	yes	per line: seq, strategy, fixed_location bool, location_query, allow_negative
multiple_sku	bool	yes	true = directive used for multi-SKU LPs

Invariants: Σ quant qty per (item, dims, warehouse) in `wms_db` equals inventory ledger physical on-hand for that warehouse at every ledger commit (eventual, reconciled daily and after each work completion); an LP is at exactly one location or on exactly one Work; location capacity ≥ Σ (quants + inbound in-progress work) at all times (property test); a WorkLine PUT always follows its PICK in `line_no` order within the same work. Indexing: `quant(warehouse_id, item_id, dims_hash, location_id)`; partial index `work(warehouse_id, priority, created_at) WHERE status = 'OPEN'`; `location(zone_id, pick_sequence)`; `license_plate(parent_lp_id)`. `wms_db` is partitioned by `tenant_id` list partitions over RLS.

6.31.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/wms/v1/locations:bulk	Create/update locations from format	Idempotency-Key header, 24 h	WMS_SETUP_MAINTAIN
POST	/wms/v1/location-directives	Create directive	key	WMS_SETUP_MAINTAIN
POST	/wms/v1/location-directives:simulate	Evaluate directives for a hypothetical line; returns trace	n/a (read)	WMS_SETUP_VIEW
POST	/wms/v1/waves/{id}:release	Release wave (starts Temporal workflow)	key; second call returns workflow id	WMS_WAVE_RELEASE
POST	/wms/v1/work:assign	Assign next work for user by filter	key per user+device	WMS_WORK_EXECUTE
POST	/wms/v1/work/{id}/lines/{n}:confirm	Confirm pick/put with actual qty, LP, location	key	WMS_WORK_EXECUTE
POST	/wms/v1/work/{id}:exception	Short pick, skip, cancel with reason	key	WMS_WORK_EXECUTE
POST	/wms/v1/receipts:from-asn	Register receipt from ASN packing structure	key = ASN id	WMS_RECEIVE
POST	/wms/v1/loads/{id}:confirm-shipment	Confirm load, post packing slips	key	WMS_SHIP_CONFIRM
GET	/wms/v1/quants?item=&location=&lp=	On-hand inquiry	—	WMS_INQUIRY
POST	/wms/v1/replenishment:run	Run min/max for a warehouse	key = warehouse+minute	WMS_REPLENISH
POST	/wms/v1/cross-dock-templates	Create or version a cross-dock template	key	WMS_SETUP_MAINTAIN
POST	/wms/v1/sort-stations/{id}:scan	Return the sort position for a scanned LP	—	WMS_WORK_EXECUTE
POST	/wms/v1/vas-orders	Create a VAS or kit order	key	WMS_VAS_MAINTAIN
POST	/wms/v1/labor/standards	Create or version a labor standard formula	key	WMS_LABOR_ADMIN
GET	/wms/v1/labor/performance?user=&shift=	Performance by user, work class or shift	—	WMS_LABOR_VIEW
POST	/wms/v1/hazmat-profiles	Maintain a hazmat profile	key	WMS_HAZMAT_MAINTAIN

Commands (CQRS): `CreateWorkFromOrderLines` validates all lines belong to one warehouse and are released; `ConfirmWorkLine` validates qty ≤ open qty + over-pick tolerance, LP exists or is new at a PUT, location status allows direction, capacity; `ReleaseWave` validates template limits before starting the workflow.

Event	Trigger	Payload fields	Consumers
dev.passage.wms.work.created.v1	Work header committed	work_id, work_type, warehouse_id, priority, wave_id, lines[]	wms-mobile, analytics
dev.passage.wms.work.completed.v1	last line confirmed	work_id, lines[{item, dims, qty, from, to, lp}], user, duration_s	inventory (post transfer), sales-fulfillment, production
dev.passage.wms.wave.released.v1	workflow finished	wave_id, orders, work_count, replenishment_count, failures[]	analytics, sales-fulfillment
dev.passage.wms.load.shipped.v1	load confirmed	load_id, shipment_ids, lp_ids, carrier_id, ship_datetime	tms, edi, sales-fulfillment
dev.passage.wms.receipt.registered.v1	receipt committed	receipt_id, po_line_refs, lp_ids, qty by item	purchasing, quality

6.31.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Warehouse setup workspace	Locations grid, profiles, zones, formats; generate locations	name, zone, profile, capacity, status	WMS_SETUP_MAINTAIN
Location directive designer	Order directives, add lines/actions, simulate	sequence, work type, query, strategy, trace	WMS_SETUP_MAINTAIN
Wave workbench	Create/release/hold waves, view failures	wave id, template, lines, weight, status, failures	WMS_WAVE_RELEASE
Work list	Filter, reassign, change priority, cancel	work id, type, status, user, age, lines	WMS_WORK_SUPERVISE
Load planning workbench	Build loads from shipments, assign door, confirm	load, shipments, weight/volume, door, appointment	WMS_SHIP_CONFIRM
Receiving workspace	Receive from ASN/PO, create LPs, print	PO, ASN, LP, item, qty, discrepancy	WMS_RECEIVE
On-hand by location/LP	Inquiry, movement, LP tree	item, dims, location, LP, qty, status	WMS_INQUIRY

Interaction rules: grids support keyboard range selection and bulk priority change (max 500 rows per action); directive designer validates on save and shows a simulation trace (rule hit/miss per step); personalization (column sets, saved filters) per user from wave 5 shell; destructive actions (cancel work with inventory in transit) require a reason code and confirmation.

6.31.7 Business rules and edge cases

- **BR-31.1 Location directive evaluation algorithm.** For a work line (item, dims, qty, uom, order attributes), evaluate: (1) select directives where `work_type`, `direction`, `warehouse` match and `query` is true, ordered by `sequence`; (2) within a directive, select the first line where `from_qty ≤ qty_in_line_uom < to_qty` (uom converted via wave 6 conversions; `locate_packing_qty` uses the item's packing hierarchy); (3) evaluate actions of that line in `seq` order: each action produces candidate locations = locations matching `location_query` ∩ status allows direction ∩ (for PUT) capacity remaining ≥ line qty/weight/volume ∩ profile mixing rules; the strategy orders candidates (`consolidate`: locations already holding item+dims, then by pick_sequence; `empty_location`: empty only; `fifo`/`fefo` for PICK: quants by receipt date / expiry ascending, tie-break pick_sequence; `round_up_to_lp`: full-LP quantities first); (4) the first action yielding ≥ 1 candidate wins and the top candidate is returned; (5) if no action of the matched line yields a candidate, continue with the next *directive* (lines within a directive are mutually exclusive by quantity band, so no other line of the same directive is tried); (6) if no directive yields a location, the line is created with `to_location = null` and work status `BLOCKED_NO_LOCATION`, and the wave lists it as a failure. Worked example: PUT line, item `BOLT-M8`, 1,200 EA, packing 100/case, 12 cases/pallet. Directives for PURCHASE_PUTAWAY in `WH1`: seq 10 query `item_group = FASTENERS`; seq 20 catch-all. Directive 10 lines: L1 0–99 EA (actions: A1 consolidate in zone `FWD`), L2 100–1199 EA (A1 consolidate zone `CASE`, A2 empty_location zone `CASE`), L3 1200–∞ EA (A1 empty_location zone `RESERVE` fixed pallet locations, A2 empty_location zone `OVERFLOW`). Qty 1,200 falls in L3. A1 candidates: RESERVE locations with status ACTIVE, 0 LPs, max_weight ≥ 1,200 × 0.012 kg = 14.4 kg → `R-05-02-01` (pick_sequence 5021) wins. If RESERVE were full, A2 would choose the first empty OVERFLOW location; if none, directive 20 is tried; if it also fails, the line is BLOCKED_NO_LOCATION.
- **BR-31.2 Capacity arithmetic.** Remaining capacity = max − Σ(quant) − Σ(in-progress PUT lines targeting the location). Rounding: weight and volume to 6 decimals; LP count and item count are integers. A location with remaining weight 0.000001 kg does not accept a 0.000002 kg line.
- **BR-31.3 Work state machine.** `OPEN → ASSIGNED → IN_PROGRESS → COMPLETED`; `OPEN|ASSIGNED → CANCELLED` (only if no line confirmed); `IN_PROGRESS → PARTIALLY_COMPLETED` (short pick; remainder re-queued as new OPEN work with `parent_work_id`); `ASSIGNED → OPEN` on unassign or lock timeout (default 30 min idle); `BLOCKED_NO_LOCATION → OPEN` after a re-evaluation succeeds. No transition out of COMPLETED; reversal creates compensating MOVEMENT work.
- **BR-31.4 Wave creation state machine.** `CREATED → RELEASING → RELEASED` or `→ RELEASE_FAILED` (any line failed and template `fail_whole_wave = true`) or `→ RELEASED_WITH_ERRORS`; `RELEASED → HELD → RELEASED`; `CREATED|RELEASED_WITH_ERRORS → CANCELLED` reverses all unstarted work.
- **BR-31.5 Replenishment precedence.** Pick work whose source location has insufficient quant after replenishment demand is created carries `depends_on_work_id`; it is not assignable until the dependency is COMPLETED. Dependency cycles are impossible because replenishment sources are restricted to zones not used as pick zones in the same directive.
- **BR-31.6 FEFO with equal expiry** breaks ties by receipt timestamp then pick_sequence; expiry is compared as a date in the warehouse's time zone.
- **BR-31.7 Over-pick tolerance** default 0%; max configurable 10%; over-pick never applies to serial-tracked items or to PICK lines sourcing from a quant with `quality_status ≠ ACCEPTED` (wave 33).
- **BR-31.8 Concurrency.** Work assignment uses `FOR UPDATE SKIP LOCKED` on the open-work partial index; quant updates use optimistic `version` with retry 3×; two confirmations of the same line with the same idempotency key return the first result; with different keys the second fails `409 LINE_ALREADY_CONFIRMED`.
- **BR-31.9 Nested LP moves** are a single transaction updating `location_id` on the root only; children derive location from the root (`location_id` null on children with a parent). Un-nesting a child sets its own location to the parent's current location.
- **BR-31.10 ASN variance.** Received qty outside ±2% (configurable) of ASN qty creates a `ReceiptDiscrepancy` with reason; receipt still posts actual qty.
- **BR-31.11 Cross-dock allocation.** At receipt registration of quantity R for item i: candidate demand D = released outbound lines for i in the same warehouse with `ship_date ≤ receipt_date + window` and open unpicked quantity > 0, ordered by template priority, then ship date, then order line id; allocate greedily; cross-dock quantity X = min(R, Σ D_open); R − X follows put-away directives. Planned links are consumed before opportunistic allocation. Supply with `quality_status ≠ ACCEPTED` (wave 33) is never cross-docked; its CROSS_DOCK work is created after the accepting usage decision if the demand is still open.
- **BR-31.12 Kit assembly posting.** One ledger transaction issues every component at BOM quantity × kit quantity (BOM scrap % applied, rounded up to the component UoM decimals) and receives the kit; failure of any component issue rolls back the whole posting; kit cost = Σ component cost + VAS labor at standard rate (wave 20).
- **BR-31.13 Packaging specification freeze.** The specification is determined at work creation and stored on the work line (`packaging_spec_id`, `version`); changing a specification does not alter open work.
- **BR-31.14 Labor standard formula.** planned_s = setup_s + travel_m × s_per_m + qty_base_uom × s_per_unit + lps × s_per_lp + weight_kg × s_per_kg, rounded to 0.1 s; actual_s = confirmed_at − started_at minus pauses longer than `labor_indirect_threshold_s`, which are logged as indirect time; a line with actual_s < 0.2 × planned_s or > 5 × planned_s is an outlier and excluded from performance.
- **BR-31.15 Hazmat segregation.** For a PUT candidate location L in section S: reject if any quant in S has class c with matrix(c, c_new) = prohibited; reject if matrix = separate and L is in the same location group (rack) as such a quant; reject if Σ hazmat quantity in S + line quantity > section limit. Each rejection appears in the simulation trace, e.g. `hazmat: class 3 vs 5.1 prohibited in FCS-02`.

- **BR-31.16 Transfer LP integrity.** TRANSFER_RECEIPT confirms only LPs on the shipped transfer; quantity variance per LP beyond tolerance creates a ReceiptDiscrepancy under BR-31.10; the in-transit quant is relieved exactly once.

6.31.8 Acceptance criteria

- **AC-31.1** (FR-31.3) Given the 100-scenario directive fixture, when each line is simulated via `/location-directives:simulate`, then every result equals the expected location and the trace shows the expected winning directive/line/action.
- **AC-31.2** (FR-31.1, BR-31.2) Given a location with max_lps = 1 holding one LP, when a PUT line targets it, then the action skips it and the simulation trace states `capacity: lps 1/1`.
- **AC-31.3** (FR-31.5, FR-31.6) Given 500 released sales orders (2,300 lines) and the demo directives, when the wave is released, then 2,300 pick lines and the expected 41 replenishment works are created, no work is BLOCKED_NO_LOCATION, and the wave is RELEASED within 120 s.
- **AC-31.4** (FR-31.7, BR-31.8) Given 50 concurrent users calling `work:assign`, when 1,000 open works exist, then no work id is returned to two users and p95 latency < 1 s.
- **AC-31.5** (FR-31.7, BR-31.3) Given an IN_PROGRESS pick of 10 EA, when the user short-picks 6 with reason `EMPTY`, then the work becomes PARTIALLY_COMPLETED, a new OPEN work for 4 EA exists with parent reference, and the inventory ledger shows a 6 EA transfer only.
- **AC-31.6** (FR-31.2, BR-31.9) Given a pallet LP with 12 case LPs, when the pallet is moved, then all 13 LPs report the new location and one `work.completed` event is emitted.
- **AC-31.7** (FR-31.9) Given an ASN with 3 pallets, when the receiver confirms pallet 2 with qty 95 of 100, then a discrepancy of –5 (–5%) is recorded and quants exist for pallet 2 only.
- **AC-31.8** (FR-31.8) Given door D1 with an appointment 09:00–09:45, when a second appointment 09:30–10:00 is requested, then it is rejected with `DOOR_SLOT_OVERLAP`.
- **AC-31.9** (property) For 10,000 generated work and confirmation sequences, location capacity is never exceeded and LP location/transit exclusivity holds.
- **AC-31.10** (FR-31.13, BR-31.11) Given a released sales line of 40 EA shipping tomorrow and an ASN pallet of 100 EA arriving today, when the pallet is received, then one CROSS_DOCK work of 40 EA to outbound staging and one put-away work of 60 EA exist and ATP counts the 40 EA once.
- **AC-31.11** (FR-31.14) Given a sort template by shipment with 12 positions, when 30 cartons for 8 shipments are scanned, then each carton is directed to its shipment's position and closing a position creates one outbound LP containing exactly those cartons.
- **AC-31.12** (FR-31.15, BR-31.12) Given a kit BOM of 3 components and a kit-to-order line of 10, when the VAS work is confirmed, then one ledger transaction issues 30 component units and receives 10 kits; with one component short, nothing posts and WMS-0114 is shown.
- **AC-31.13** (FR-31.18, BR-31.14) Given the labor formula fixture and 1,000 confirmed lines, when the shift closes, then planned seconds per line equal the fixture to 0.1 s and performance per user equals Σ planned ÷ Σ actual over non-outlier lines.
- **AC-31.14** (FR-31.19, BR-31.15) Given section FCS-02 holding class 5.1 oxidizers, when a class 3 flammable-liquid PUT line is simulated, then FCS-02 locations are excluded and the trace states the segregation reason.
- **AC-31.15** (FR-31.16, BR-31.16) Given a transfer of 3 LPs shipped from WH1, when WH2 receives 2 of them and scans an LP that is not on the transfer, then the foreign LP is rejected with WMS-0117, the 2 LPs are received with unchanged ids, and the third remains in transit.

6.31.9 Test plan and fixtures

Fixtures: `wms-demo-warehouse` (1 warehouse, 4 zones, 8,000 locations, generated from format), `directive-scenarios-100.json` (hand-authored, reviewed against D365 demo data USMF behavior), `wave-500-orders` (generated from wave-28 order fixtures), `asn-3-pallet.json`, `sim-10k-lines-day` (synthetic order stream for the simulation harness).

Test	Type	Target	Pass criterion
Directive resolution	unit/table-driven	100 scenarios	100% expected location
Capacity invariant	property (rapid/Go)	10k sequences	zero violations
Work state machine	unit	all transitions	illegal transitions rejected
Wave release	E2E (Playwright + API)	500 orders	work counts, < 120 s
Assignment contention	performance (k6)	50 VUs	p95 < 1 s, no duplicates
Simulation	performance	10k lines/day × 3 days	no deadlocked work, no BLOCKED_NO_LOCATION
Directive perf	performance	50k locations	p95 < 200 ms/line
Nested LP	contract	API	depth 6 rejected

Reconciliation: Σ quant per item/warehouse = inventory ledger on-hand; Σ in-transit on Work = ledger "in transit" status; load shipped quantities = packing slip quantities (wave 28).

6.31.10 Deployment and operations

Helm chart `passage-wms` (deployments `wms-api`, `wms-directive-engine`, `wms-worker` for Temporal activities; HPA 2–10 on CPU 70%); CloudNativePG cluster `wms-db` (3 instances, 100 GB); Atlas migrations generated from entity metadata; config: `WMS_LOCK_TIMEOUT_MIN=30`, `WMS_REPLENISH_INTERVAL_MIN=15`, `WMS_MAX_WAVE_LINES=5000`. Feature flags: `wms.advanced` per warehouse (off = wave-28 simple mode), `wms.slotting`. Scheduled jobs (Temporal schedules): min/max replenishment, automatic wave release, count plan generation (daily 02:00 warehouse-local), KPI materialization hourly. SLOs: assignment p95 < 1 s; directive simulate p95 < 200 ms; wave release of 5,000 lines < 10 min; availability 99.9%. Alerts: `WmsWorkBlockedNoLocation > 50 in 15 min`, `WmsAssignmentP95 > 2 s`, `WmsQuantLedgerDrift ≠ 0 after daily reconciliation`, `WmsWaveWorkflowFailed`. Runbook: re-evaluate blocked work; release stuck locks; reconcile drift (compare `quant` vs ledger projection, apply correction journal with approval). Rollback: Argo CD to previous digest; migrations are additive within a wave; flag `wms.advanced=false` returns a warehouse to simple mode only if no OPEN work exists.

6.31.11 Migration from AX / SAP

Source	Passage entity	Transformation
WMSLocation, WHSLocationProfile, WHSZone, WHSLocationFormat	Location, LocationProfile, Zone, LocationFormat	parse name by format; capacity from profile; status from `WHSLocationStatus`
/SCWM/LAGP, /SCWM/T301, /SCWM/T302, /SCWM/T303	Location, Zone (storage type/section), LocationProfile (bin type)	LGTYP → zone, LGBER → sub-zone attribute; capacity from LAGP `MAX_WEIGHT/MAX_CAPA`
WHSLicensePlate, WHSInventOnhand by LP	LicensePlate, Quant	parent LP via `WHSLicensePlate.ParentLicensePlateId`
/SCWM/HUHDR, /SCWM/HUITM, /SCWM/AQUA	LicensePlate, Quant	HU nesting via `HUHDR.TOP/HIGHER_LEVEL`; quants from AQUA by LGPLA
WHSLocationDirective/Line/Action	LocationDirective	strategy enum mapped 1:1; AX queries translated to metadata query

Source	Passage entity	Transformation
WHSWorkTemplateTable/Line, WHSWaveTemplateTable, WHSReplenishmentTemplate	WorkTemplate, WaveTemplate, ReplenishmentTemplate	direct
EWM put-away/removal strategies (/SCWM/T333, /SCWM/T334)	LocationDirective	storage type search sequence → ordered actions
WHSWorkTable/Line (status Open/In process), /SCWM/ORDIM_O	Work, WorkLine	only open work migrated at cutover; completed work is history (not migrated)
WHSLoadTable/Line (not shipped)	Load	open loads only

Validation: location count and names match; on-hand by location/LP equals source on-hand snapshot (wave-16 reconciliation extended by location); open work count matches. Not migrated: closed work history (WHSWorkTable confirmed, /SCWM/ORDIM_C) — kept in the wave-46 lakehouse as archived source data; labor standards; EWM yard data (v1.x).

6.31.12 Dependencies, risks, open questions

Depends on: wave 30 (go-live baseline), wave 28 (shipments, packing slips), wave 18 (purchase receipts), wave 21 (production picking/RAF), wave 16 (inventory ledger, reservations, counting journal), wave 8 (GS1 company prefix, packing hierarchy), wave 6 (warehouses, calendars, UoM), wave 4 (Temporal, outbox).

Risk	Likelihood	Impact	Mitigation
Directive semantics differ subtly from D365 and confuse migrating customers	Medium	Medium	Simulation trace; parity fixtures from D365 USMF; documented differences
Quant/ledger drift under high concurrency	Low	High	Single posting path through inventory ledger; daily reconciliation alert

Open questions: (1) Should quants be owned by `wms` or remain a projection of `inventory`? Proposed: projection with `wms` as the only writer of location/LP dims; decide by wave 31 design review. (2) SSCC extension digit per warehouse or per tenant? Proposed: per warehouse; decide by wave 32.

6.31.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
wms.advanced_enabled	warehouse	bool	false	true / false	Switches the warehouse from wave-28 simple mode to directed work	Warehouse form: "Use warehouse management processes" (InventLocation.WHSEnabled)	SPRO EWM › Master Data › Assign Warehouse Numbers (/SCWM/T300_MD)
lp_number_sequence	warehouse	sequence ref	SSCC-18 per warehouse	SSCC / custom pattern	Format of generated LP ids	Warehouse management parameters › Number sequences › License plate	SPRO EWM › Master Data › Define Number Ranges for Handling Units
sscc_extension_digit	warehouse	int	0	0–9	First digit of generated SSCCs	Warehouse management parameters › General › SSCC / GS1 setup	GS1 company prefix and SSCC number range (HU number range object)
lock_timeout_min	warehouse	int	30	5–240	Idle ASSIGNED work returns to OPEN (BR-31.3)	Work user session expiry	Resource type logon timeout (/SCWM/TRSRC_TYP)
max_wave_lines	wave template	int	5,000	1–20,000	Wave size cap (FR-31.5)	Wave template › Wave threshold	Wave template capacity profile (/SCWM/WAVETMP)
fail_whole_wave	wave template	bool	false	true / false	RELEASE_FAILED vs RELEASED_WITH_ERRORS (BR-31.4)	Wave template "Process wave at release" plus held wave status	Wave release method in wave template
over_pick_tolerance_pct	warehouse, item	numeric(5,2)	0	0–10	Over-pick limit (BR-31.7)	Mobile device menu item "Allow over-picking" with product over-pick %	Warehouse process type (/SCWM/T333) over-pick indicator
replenishment_interval_min	warehouse	int	15	5–1,440	Min/max run frequency (FR-31.6)	Replenishment batch job recurrence (WHSReplenishmentSchedule)	/SCWM/REPL background job variant
asn_variance_tolerance_pct	warehouse, vendor	numeric(5,2)	2	0–20	ReceiptDiscrepancy threshold (BR-31.10)	PurchLine over/under delivery %	Inbound delivery tolerance (MM tolerance keys T169G)
count_approval_tolerance	warehouse	value % / units	1% or 5 units	≥ 0	Count differences above it need approval (FR-31.10)	Cycle count thresholds and approval parameters (WHSCycleCountThreshold)	SPRO EWM › Physical Inventory › Define Tolerance Groups for Difference Analyzer
blind_count	warehouse	bool	true	true / false	Hides expected quantity on count work	Mobile device menu item "Display expected quantity"	Physical inventory procedure without book quantity (/SCWM/PI)
capacity_dimensions	location profile	set	{weight, volume, lps, items}	any subset	Which dimensions BR-31.2 enforces	Location profile "Use volumetrics" and location stocking limits	Storage type capacity check method (/SCWM/T331)
directive_trace_retention_days	tenant	int	30	1–365	Retention of simulate/evaluation traces	Warehouse management parameters › Location directive failures logging	Application log retention (SLG1 object /SCWM/)
crossdock_opportunistic	warehouse	bool	false	true / false	Enables opportunistic cross-dock at receipt (FR-31.13)	Work policy "Opportunistic cross docking"	SPRO EWM › Cross-Process Settings › Cross-Docking › Opportunistic Cross-Docking › Activate
crossdock_supply_window_h	cross-dock template	int	48	0–336	Supply-to-demand window (BR-31.11)	Cross docking template "Re-plan frequency" and demand window	Opportunistic cross-docking settings for inbound (time horizon)
crossdock_demand_priority	cross-dock template	enum	SHIP_DATE	SHIP_DATE / ORDER_PRIORITY / FIFO_RELEASE	Demand ordering (BR-31.11)	Cross docking template "Demand requirement prioritization"	Push deployment strategy

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
sort_position_close_rule	sort template	enum	MANUAL	MANUAL / FULL / LAST_LP	When a sort position closes (FR-31.14)	Outbound sorting template "Close position"	Consolidation group completion rule
vas_station_location	warehouse	location ref	none	locations of type VAS	Default target of VAS work (FR-31.15)	Kit order location	SPRO EWM › Value-Added Services › Define Work Center for VAS (/SCWM/TWORKST)
packaging_spec_required	warehouse	bool	false	true / false	Blocks work creation when no specification matches (FR-31.17)	Unit sequence group mandatory on WHS items	Packaging specification determination procedure (condition technique)
labor_travel_metric	warehouse	enum	MANHATTAN	MANHATTAN / EUCLIDEAN / GRAPH	Travel distance method (BR-31.14)	n/a	SPRO EWM › Labor Management › Travel Distance Calculation settings
labor_indirect_threshold_s	warehouse	int	120	30–900	Pause length counted as indirect (BR-31.14)	n/a	Indirect labor task settings (/SCWM/LM)
labor_personal_visibility	legal entity	enum	AGGREGATE_ONLY	AGGREGATE_ONLY / NAMED	Whether performance is shown per named user (FR-31.18)	n/a	Authorization for employee performance documents (/SCWM/LM)
hazmat_segregation_matrix	tenant	reference data version	TRGS510-2023	TRGS510 / NFPA30 / custom	Matrix used by BR-31.15	Product compliance / hazardous material setup	SPRO EWM › Master Data › Hazardous Substances › Define Storage Classes and storage type checks

6.31.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
On-hand by location / LP	list inquiry	warehouse, item, dims, location, LP, status	qty, reserved, available, LP tree, quality status	`quant`	On-hand by location, WHSInventOnhand	LS24/LS26, LX02, /SCWM/MON stock node
Work list and work audit	list inquiry	status, work type, user, age, wave	work id, lines, from/to, user, duration	`work`, `work_line`	All work list page, work audit templates	LT22–LT28, /SCWM/MON warehouse orders/tasks
Wave status and failures	drill-down	wave, template, status, date	lines, work count, replenishment count, failures by reason	`wave`	Wave details (WHSWaveTable)	/SCWM/WAVE, LT41/LT42
Location capacity and empty bins	operational report	zone, profile, utilization %	used/max per dimension, empty flag	`location` + `quant`	Location load, location stocking limits inquiry	LX01, LX04, /SCWM/MON storage bins
Directive failure trace	drill-down	work, wave, date range	directive, line, action, rejection reason	directive trace store	Location directive failures (D365)	Application log SLG1 (no dedicated report)
Cycle count variances	operational report	warehouse, period, counter	expected, counted, difference qty and value, approver	count work	Cycle count work approval, SSRS count sheets	LX17, LX18, LI20
Quant–ledger reconciliation	operational report	warehouse, date	Σ quant vs ledger on-hand, drift per item	reconciliation job output	Manual InventSum vs WHS comparison	LX23 stock comparison IM–WM, EWM–ERP stock comparison
Cross-docking report	operational report	warehouse, date, template	supply, demand, qty cross-docked, dwell minutes, missed opportunities	`crossdock_link`	SSRS Cross docking report	LXDCA cross-docking monitor, /SCWM/MON cross-docking node
Sort position status	list inquiry	sort station, status	position, LPs, fill %, open/closed, target shipment	`sort_position`	Outbound sorting position inquiry	/SCWM/MON consolidation groups
VAS order backlog	list inquiry	station, status, due date	order, steps, estimated vs actual time	`vas_order`	Kit orders list	/SCWM/VASEXEC worklist
Labor performance	KPI	warehouse, shift, work class, user (duty-gated)	planned s, actual s, performance %, indirect %, outliers	`labor_line`	Warehouse workload visualization (no standards)	/SCWM/MON labor management nodes, performance documents
Hazmat inventory by fire-containment section	operational / statutory report	warehouse, section, class	qty kg/L vs limit, classes present, emergency contact	`quant` + `hazmat_profile`	Product compliance regulated products inquiry	LX06, LX24
Warehouse KPIs	KPI	warehouse, hour/day	lines/h, dock-to-stock min, pick accuracy %, short-pick rate, replenishment lead time	Cube	Warehouse management workspace tiles	Fiori Warehouse KPIs, LX47

6.31.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Pick list (work document)	Work created or printed on demand	Header: work id barcode, wave, priority, warehouse; lines: sequence, from location, item, lot, qty, to location; footer: line count, total weight	PDF; ZPL variant	WRK-* (wave 5)	Operational, 90 days
Load manifest	Load confirmed	Shipper, consignee, carrier SCAC, LPs/SSCCs, weights, hazmat lines with proper shipping names (FR-31.19)	PDF	Load id	Kept with the shipment record (wave 28 retention)
Count sheet	Count work created	Location, item, lot, expected qty (blank for blind counts), counted qty field	PDF	COUNT-*	Count adjustments are ledger records (wave 16 retention, 10 years)
Cross-dock tag	CROSS_DOCK work created	Outbound order, ship date, door, SSCC	ZPL	none	None
Kit assembly ticket	VAS order released	Kit BOM, component locations, VAS steps, kit LP	PDF	VAS-*	Operational, 90 days
Hazmat section inventory list	On demand and daily	Section, classes, qty vs limit, emergency contact	PDF / CSV	none	Supplied to fire authority on request; kept 1 year
Labor performance document	Shift close	User, planned, actual, performance, indirect time	PDF / CSV export to payroll	LPD-*	Employee data retention per wave 42, default 2 years

6.31.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Min/max replenishment	Every 15 min	warehouse	Key warehouse + minute	< 30 s at 50,000 locations	Retry 3×, then skip to next interval	WmsReplenishRunFailed
Automatic wave release	Per wave template cron	wave template	Temporal workflow id per wave	500 orders < 120 s	Per-line failures; RELEASED_WITH_ERRORS	WmsWaveWorkflowFailed
Count plan generation	Daily 02:00 warehouse-local	warehouse × plan	plan id + date	< 5 min	Resume from last generated location	Zero works generated for an active plan
Quant–ledger reconciliation	Daily 03:00	warehouse × 16 item-hash partitions	snapshot timestamp	< 15 min for 2 M quants	Report only; corrections need approval	WmsQuantLedgerDrift ≠ 0
Planned cross-dock re-plan	Every 30 min	warehouse	link version	< 60 s	Keep previous links on failure	WmsCrossDockReplanFailed
Labor shift close	Shift end + 30 min	warehouse × shift	shift id	< 2 min	Idempotent rerun	Shift not closed after 2 h
KPI materialization	Hourly	warehouse	hour bucket	< 2 min	Recompute bucket	KPI lag > 60 min
Empty LP and work history cleanup	Weekly Sunday 01:00	warehouse	LP id / work id	< 30 min	Resume by id	none (log)
Directive trace purge	Daily 04:00	tenant	trace date	< 5 min	Retry next day	none (log)

6.31.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
WMS_SETUP_MAINTAIN	/wms/v1/locations:bulk, /location-directives, /cross-dock-templates; setup workspace; directive designer	WMS_COUNT_APPROVE (could change locations, then approve the resulting differences)	Warehouse administrator
WMS_SETUP_VIEW	/location-directives:simulate; setup screens read-only	none	Warehouse manager, supervisor
WMS_WAVE_RELEASE	/waves/{id}:release; wave workbench	none	Warehouse supervisor, warehouse manager
WMS_WORK_EXECUTE	/work:assign, /lines/{n}:confirm, /work/{id}:exception, /sort-stations/{id}:scan	WMS_COUNT_APPROVE on the same count document (approver ≠ counter, enforced)	Warehouse worker
WMS_WORK_SUPERVISE	Work list reassign, priority change, cancel, blocked-work re-evaluation	none	Warehouse supervisor
WMS_RECEIVE	/receipts:from-asn; receiving workspace	AP_INVOICE_APPROVE (wave 19): receiving and approving the same vendor's invoice	Receiving clerk
WMS_SHIP_CONFIRM	/loads/{id}:confirm-shipment; load planning workbench	none in wave; flagged with CUST_CREDIT_RELEASE (wave 26)	Shipping clerk
WMS_COUNT_APPROVE	Count approval screen; counting journal post	WMS_WORK_EXECUTE on same count; WMS_SETUP_MAINTAIN	Inventory controller
WMS_REPLENISH	/replenishment:run	none	Warehouse supervisor
WMS_VAS_MAINTAIN	/vas-orders; VAS backlog	none	VAS lead
WMS_LABOR_ADMIN	/labor/standards; formula designer	WMS_WORK_EXECUTE (could tune the standard for own work)	Labor analyst
WMS_LABOR_VIEW	/labor/performance with named users	none; available only where `labor_personal_visibility = NAMED`	Warehouse manager
WMS_HAZMAT_MAINTAIN	/hazmat-profiles; segregation matrix; section limits	none	EHS officer
WMS_INQUIRY	GET /wms/v1/quants and all inquiries above except labor	none	All warehouse roles

6.31.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
WMS-0101	error	No location found for {qty} {uom} of {item}; work {work} is blocked.	No directive action yielded a candidate (BLOCKED_NO_LOCATION)	Run simulate, adjust directive or free capacity, then re-evaluate	BR-31.1
WMS-0102	error	Location {loc} capacity exceeded: {dimension} {used}/{max}.	PUT would exceed remaining capacity	Choose another location or split the LP	BR-31.2
WMS-0103	error	Location {loc} does not allow mixed items.	Profile `allow_mixed_items = false`	Use an empty or same-item location	FR-31.1
WMS-0104	error	License plate nesting would exceed 5 levels.	Nest depth 6 attempted	Un-nest an intermediate level	FR-31.2
WMS-0105	error	Line {n} of work {work} is already confirmed.	Second confirmation with a different key (409 LINE_ALREADY_CONFIRMED)	Refresh the work	BR-31.8
WMS-0106	error	Quantity {qty} exceeds open quantity plus over-pick tolerance {tol}%.	Over-pick beyond tolerance or serial item	Confirm the open quantity	BR-31.7
WMS-0107	warning	Received {qty} differs from ASN {asn_qty} by {pct}%; discrepancy recorded.	Variance beyond ASN tolerance	Review the discrepancy with purchasing	BR-31.10
WMS-0108	error	Door {door} is already booked {from}–{to}.	Overlapping appointment (DOOR_SLOT_OVERLAP)	Pick another slot or door	FR-31.8
WMS-0109	info	Pick work {work} waits for replenishment {repl}.	Dependency on open replenishment	None; work becomes assignable automatically	BR-31.5
WMS-0110	error	Transition {from} → {to} is not allowed for work {work}.	Illegal state change	Use cancel or compensating movement	BR-31.3
WMS-0111	warning	Count difference {diff} exceeds tolerance; supervisor approval required.	Above count tolerance	Approve or recount	FR-31.10
WMS-0112	info	{qty} cross-docked to {loc} for order {order}; {rest} to put-away.	Cross-dock allocation	None	BR-31.11
WMS-0113	warning	Supply is under quality hold; cross-dock deferred to usage decision.	Supply not ACCEPTED	Wait for usage decision	BR-31.11
WMS-0114	error	Kit component {item} short by {qty}; kit assembly not posted.	Component issue failed	Replenish component or reduce kit qty	BR-31.12

Code	Severity	User text	Cause	Remediation	Raised by
WMS-0115	error	Storage class {c1} cannot be stored with {c2} in section {fcs}.	Segregation matrix prohibits	Use another fire-containment section	BR-31.15
WMS-0116	error	Section {fcs} limit {limit} {uom} would be exceeded.	Section quantity limit	Move stock or use another section	BR-31.15
WMS-0117	error	License plate {lp} is not on transfer shipment {ship}.	Foreign LP scanned at transfer receipt	Receive against the correct transfer	BR-31.16
WMS-0118	warning	Labor line {line} excluded as outlier (actual {a} s vs planned {p} s).	Outside 0.2×–5× planned	Review standard or confirmation	BR-31.14
WMS-0119	error	Daily reconciliation found drift for {n} item/warehouse pairs.	Quant ≠ ledger on-hand	Run drift runbook; approve correction journal	§6.31.10

6.31.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Directive evaluation	10,000 lines/day; peak 500 lines/min	40 / 200 / 350 ms per line	50 lines/s per directive-engine pod	Traces ≈ 3 GB (30-day rolling)	Candidate pre-filter by zone and profile; `location(zone_id, pick_sequence)`	Wave workflow queues lines; release time grows; no line dropped
Work assignment	50 concurrent users; 3,000 assigns/h	80 / 400 / 1,000 ms	200 assigns/s	—	Partial index on OPEN work	SKIP LOCKED keeps correctness; alert at p95 > 2 s
Line confirmation	30,000/day; peak 1,200/min	60 / 250 / 600 ms	100/s per pod	Work + lines ≈ 15 GB	Monthly range partitions on `created_at` inside tenant list partitions	HPA to 10 pods; beyond, 429 with Retry-After (mobile queues offline)
Wave release	50 waves/day, ≤ 5,000 lines each	2,300 lines < 120 s; 5,000 lines < 10 min	40 lines/s	—	Per-order Temporal activities	Task-queue backlog; scheduled waves start late; alert at 15 min delay
Cross-dock allocation at receipt	2,000 receipts/day	+50 / 150 / 300 ms on receipt	30/s	Links ≈ 0.5 GB	`demand(warehouse_id, item_id, ship_date) WHERE open`	Falls back to put-away and records a missed opportunity
Sort scan	20,000 cartons/day; 30/min per station	40 / 150 / 300 ms	20/s per station	—	In-memory position map per station	Correct but slower; latency shown on the station screen
Labor shift close	3 shifts × 200 users	< 2 min per shift	—	Performance documents ≈ 1 GB	`labor_line` monthly partitions	Close delayed; no data loss
Quant reconciliation	2 M quants	< 15 min	16 parallel partitions	Reports < 100 MB	Hash on item id	Runtime grows linearly; alert only on drift

6.31.20 Localization and compliance hooks

- **Hazardous-material storage.** Segregation and quantity limits differ by jurisdiction: Germany TRGS 510 storage classes (LGK 1–13) with water hazard classes under AwSV; United States NFPA 30, International Fire Code chapter 57 and OSHA 29 CFR 1910.106 for flammable liquids; United Kingdom HSG71 and DSEAR. Country packs (wave 48) supply the segregation matrix and section limits through the `WarehouseCompliancePack` extension point (versioned reference data; BR-31.15 is regulation-neutral).
- **Labor monitoring.** Per-person performance data is personal data under GDPR (Art. 88; DPIA under Art. 35); Germany requires works-council co-determination (BetrVG §87(1) no. 6); France requires CSE consultation. EU legal entities default to `labor_personal_visibility = AGGREGATE_ONLY`; the country pack sets defaults and an OPA policy bundle (`LaborPrivacyPolicy`) gates the named view.
- **Physical inventory.** Statutory stock-take rules (Germany HGB §241(2) permanent inventory requires each item counted at least once per fiscal year) are checked by a count-plan coverage report supplied by the country pack (`InventoryCountPolicy`).
- **GS1 and units.** SSCC company prefixes come from the GS1 member organization of the legal entity; weights and volumes are stored in SI and displayed in kg/lb and m³/ft³ by user locale (wave 5).
- **Retention.** Work, count and transfer records that support inventory valuation follow the wave-46 archive policy (10 years default; Germany GoBD/HGB §257, United States 7 years typical).
- **Calendars and time zones.** FEFO date comparison (BR-31.6), shift close and count plans use the warehouse time zone and wave-6 calendars.

6.31.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX PARTIAL #217 — §12.26.1 Outbound → Cross docking (WHSCrossDocking)	PARTIAL-31	ADDED as FR-31.13	Planned and opportunistic; BR-31.11, AC-31.10
AX PARTIAL #218 — §12.26.1 WHS → Transfer order WHS flow (ship/receive with LP)	PARTIAL-31	ADDED as FR-31.16	Transfer order document itself is wave 16; BR-31.16, AC-31.15
AX PARTIAL #219 — §12.26.2 SSRS: Cross docking report	PARTIAL-31	ADDED as FR-31.13	"Cross-docking report" in §6.31.14
AX PARTIAL #220 — §12.26.3 Cross docking → Planned cross docking processing	PARTIAL-31	ADDED as FR-31.13	"Planned cross-dock re-plan" job in §6.31.16
AX PARTIAL #221 — §12.26.4 Outbound → Cross docking templates	PARTIAL-31	ADDED as FR-31.13	`crossdock_*` parameters in §6.31.13
AX PARTIAL #222 — §12.26.4 Outbound → Outbound sorting (sort positions)	PARTIAL-31	ADDED as FR-31.14	AC-31.11
SAP #187 — §13.16 LXDCA / LXDCK planned cross-docking	GAP (propose 31)	ADDED as FR-31.13	Directive strategy `outbound_staging` as the register proposed
SAP #188 — §13.16 /SCWM/PACK, /SCWM/PACKSPEC, /SCWM/PSSTAGE packaging specifications	PARTIAL-31	ADDED as FR-31.17	Packing station flow stays FR-32.4 with wave-28 containers
SAP #189 — §13.16 /SCWM/VASEXEC VAS orders	GAP (propose 31)	ADDED as FR-31.15	Work type VAS; charge code `VAS` to wave 29
SAP #190 — §13.16 /SCWM/LM labor management	PARTIAL-31	ADDED as FR-31.18	Incentive pay calculation OUT-OF-SCOPE: performance documents export to payroll (wave 42)
SAP #191 — §13.16 /SCWM/YM_* yard management	PARTIAL-31	DEFERRED to wave 34	ADDED there as FR-34.13; AX register also proposes wave 34
SAP #194 — §13.16 EWM kit-to-stock / kit-to-order	GAP (propose 31)	ADDED as FR-31.15	BR-31.12, AC-31.12

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP #196 — §13.16 LX06 / LX07 fire-containment sections / check storage	PARTIAL-31	ADDED as FR-31.19	LX07 consistency check is covered by quant–ledger reconciliation (§6.31.16)
SAP #197 — §13.16 /SCWM/LM labor reports (planned vs actual)	PARTIAL-31	ADDED as FR-31.18	"Labor performance" in §6.31.14
SAP #198 — §13.16 Labor management period close (/SCWM/LM)	PARTIAL-31	ADDED as FR-31.18	"Labor shift close" job in §6.31.16
SAP #200 — §13.16 EWM yard management customizing	PARTIAL-31	DEFERRED to wave 34	Yard parameters in §6.34.13
SAP #202 — §13.16 Labor management: formulas, ELS, incentive	PARTIAL-31	ADDED as FR-31.18	Formulas and ELS added (BR-31.14); incentive OUT-OF-SCOPE as for #190

6.32 Wave 32: Mobile Warehouse Execution (Scanners, RF, Voice)

6.32.1 Scope and business context

Wave 31 creates work; wave 32 lets warehouse workers execute it on handheld scanners. This replaces the D365 Warehouse Management mobile app (mobile device menus, menu items and steps) and the SAP EWM RF framework (/SCWM/RFUI, ITSmobile). After this wave a worker signs in on a Zebra or Honeywell Android device (or any PWA-capable browser), picks from a configured menu, scans GS1-128 labels that the system parses into LP/GTIN/lot/expiry/serial/weight, and completes receive, put-away, pick, pack, move, count and inquiry flows with offline tolerance. An administrator changes flows without an app release. Label printing (ZPL) ships through a print service. Out of scope: full voice-directed picking (adapter interface only; a voice vendor integration is v1.x), RFID portal reads beyond the EPC ingest endpoint, iOS native shell (PWA covers iOS).

Scope amendment (coverage closure). The gap registers add two items to this wave: an automation adapter that exchanges telegrams with material flow systems (AS/RS, conveyors, sorters) through the wave-25 edge gateway (FR-32.12), and management of printers and output queues, covering device types, queue failover and browser printing beyond labels (FR-32.13). §6.32.21 lists the dispositions.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
D365 mobile menu items (WHSRFMenuItemTable) expose ~60 option fields per item; a flow change (e.g., add a lot confirmation step) needs a consultant, a config move across environments and a device re-login; D365 "mobile device steps configuration" cannot add a new step, only hide/rename	Flows are hard-coded X++ step classes parameterized by options	Flow designer with versioned step graphs, simulate, publish; devices pick up the version in ≤ 60 s	Flow change lead time: 1–3 days → < 30 min; 0 app releases per flow change
Both D365 mobile app and EWM RF (/SCWM/RFUI, ITSmobile) stop dead without Wi-Fi; workers in freezers, yards and trailers lose scans	Request/response screens with no client state	Offline queue with ordered, idempotent replay and a supervisor conflict queue	Lost transactions on disconnect: every one → 0; worker wait on reconnect: re-key → 0 keystrokes
GS1-128 labels must be scanned AI by AI in older D365 and ECC RF screens (one scan for GTIN, one for lot, one for expiry); mis-typed expiry dates reach the batch master	No composite-barcode parsing in the RF screen	GS1 parser (AI 00/01/10/17/21/310n…) fills all expected fields from one scan with check-digit verification	Scans per inbound case: 3–4 → 1; expiry/lot keying errors: ~1% of receipts → 0 (no keying)
Label printing via D365 document routing (WHSDocumentRoutingLayout) depends on the Document Routing Agent Windows service; failures are invisible on the device	Printing routed through a client-side agent with no status feedback	In-cluster print service with job status, retries and device-visible failure	Print failure detection: minutes to hours (someone notices) → < 2 s on device
EWM RF screen layouts (/SCWM/TPRES) and ITSmobile templates need ABAP/HTML work for scanner model differences	Screen rendering coupled to device generation	Responsive PWA from 320 px, DataWedge/Honeywell intent bridge, keyboard-wedge fallback	New scanner model onboarding: weeks of template work → same day (profile only)

6.32.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Mobile menu	WHSMobileAppMenu, WHSRFMenuTable, WHSRFMenuLine	/SCWM/TMENU, /SCWM/T_RF_MENU (RF menu manager /SCWM/RFUI)	wms-mobile.MobileMenu	full
Menu items / flows	WHSRFMenuItemTable (WHSRFMenuItem), work creation vs existing work modes	Logical transactions /SCWM/TLOGTR, steps /SCWM/T_RF_STEP	MobileFlow (step graph)	improved — visual designer, versioned
Detours / field config	Mobile device steps configuration (D365 10.0.x), WHSRFDetour	Presentation profile /SCWM/TPRES, personalization	FlowStep config, detours	full
Scanner integration	DataWedge via app; GS1 parsing (D365 10.0.21+)	RF barcode profiles, GS1 parsing in /SCWM/RFUI	GS1 parser service, DataWedge/Honeywell intents	full
Work user / session	WHSWorkUser, WHSWorkUserSessionState	Resource /SCWM/RSRC, /SCWM/TRSRC queue	DeviceSession, WorkUser (Keycloak)	full
Offline	n/a (online only)	n/a (online only)	OfflineTransaction queue	improved
Label printing	Document routing layouts (WHSDocumentRoutingLayout) to ZPL printers	Smart Forms / Adobe Forms via /SCWM/PRINT condition technique	PrintJob, ZPL templates	full
Voice	ISV	SAP EWM voice (ISV)	VoiceAdapter interface	partial — adapter only
RFID	n/a (ISV)	SAP AII / EWM RFID	EPC read endpoint	partial
Packing station	WHSPackingStation, container packing	Packing work center /SCWM/PACK	Pack flow + ShipmentContainer (wave 28)	full
Material flow system (automation)	Warehouse automation integration via business events and ISV warehouse control systems	EWM MFS /SCWM/MFS_*: PLC communication channels, telegram structures, communication points, conveyor segments, MFS monitor	wms.AutomationChannel, TelegramProfile, CommunicationPoint, TransportOrder	full
Output devices and queues	Network printers, Document Routing Agent, print management setup (PrintMgmtDocInstance)	SPAD output devices, device types (e.g. LB_ZEB for ZPL), access methods, spool SP01/SP02, front-end printing	Printer, DeviceType, PrintQueue, RoutingRule; browser print	full

6.32.3 Functional requirements

- **FR-32.1 Mobile app shell** — Must. The system shall provide a PWA and an Android native shell (WebView + scanner intent bridge) that run the same flow runtime, with login via Keycloak device flow and PIN re-auth after 8 h.

    - Notes: supports screen widths from 320 px; large-target mode; app updates are flow metadata, not binaries; shell release only for scanner bridge changes.
- **FR-32.2 Flow designer** — Must. The system shall let an administrator compose flows from step types (`scan`, `input`, `select`, `confirm`, `display`, `branch`, `call`), with validations, scan pattern expectations and error handling, versioned and published per warehouse.
    - Notes: a step declares expected scan classes (LP, GTIN, LOCATION, ANY); publishing validates the graph (no unreachable steps, every branch terminates); devices fetch new versions on next menu load (≤ 60 s).
- **FR-32.3 GS1 parsing** — Must. The system shall parse GS1-128 and GS1 DataMatrix element strings into application identifiers per BR-32.1, and route values to the expected step fields.
    - Notes: symbology identifiers `]C1`, `]d2`, `]Q3` accepted; FNC1 as GS (0x1D); passes the GS1 AI test suite in CI.
- **FR-32.4 Core flows** — Must. The system shall ship flows: Receive (PO/ASN/transfer), Put-away (system-directed and user-directed), Pick (work-directed, cluster to 8 orders, zone), Pack, Move (LP/qty), Count (planned/spot), Inquiry (item/location/LP).
    - Notes: all flows call the wave-31 work APIs; cluster picking assigns up to 8 works to positions on a cart with per-position LP.
- **FR-32.5 Offline queue** — Must. The system shall queue confirmations locally when offline (max 500 transactions or 24 h), replay them in order on reconnect with per-transaction idempotency keys, and surface conflicts to the user.
    - Notes: queue persisted in IndexedDB encrypted with a device key; replay uses `Idempotency-Key = device_id:seq`; conflict policy per BR-32.4.
- **FR-32.6 Device and session management** — Must. The system shall register devices, show active sessions, force sign-out, and bind a device to warehouses and menus.
    - Notes: device heartbeat every 30 s; session idle timeout 30 min aligned to the work lock.
- **FR-32.7 Label printing** — Must. The system shall render ZPL from templates (LP, location, shipping, item labels) and print via a print service supporting IPP/raw TCP 9100 printers and printer selection by step/user/location.
    - Notes: templates are Handlebars-style with GS1-128 (`^BC`) and DataMatrix (`^BX`) helpers; print service retries 3× and reports status.
- **FR-32.8 Exceptions on device** — Must. The system shall let users short pick, skip, report damaged, and request supervisor override by PIN on device.
    - Notes: override reasons configurable; all exceptions logged with user, device, location, timestamp.
- **FR-32.9 Voice and RFID integration points** — Could. The system shall expose a `VoiceAdapter` interface (prompt/response per step) and an EPC ingest endpoint mapping EPC SGTIN/SSCC to GTIN/LP.
    - Notes: EPC URI parsing for `urn:epc:id:sgtin` and `urn:epc:id:sscc`; reads deduplicated in a 2 s window.
- **FR-32.10 Scanner hardware integration** — Must. The system shall receive scans from Zebra DataWedge intents and Honeywell Data Collection intents in the native shell and from keyboard wedge in the PWA.
    - Notes: a scan is accepted only on the focused scan step; symbology prefix enabled in DataWedge profile provisioned by the shell.
- **FR-32.11 Telemetry** — Should. The system shall record per-step timing and error counts per device and user for the KPI layer.
    - Notes: OpenTelemetry spans per step; retention 90 days.
- **FR-32.12 Automation adapter (AS/RS, conveyors, sorters)** — Should. The system shall exchange work with material flow systems: send transport orders (LP, source point, destination point, priority) as telegrams to a PLC or warehouse control system, receive handshake, arrival, scan-result and fault telegrams, and confirm or raise exceptions on wms work lines accordingly.
    - Notes: transports through the wave-25 edge gateway on site: OPC UA method calls, MQTT 5 (Sparkplug B payloads optional) or raw TCP fixed-length telegrams with an EWM-style header (sender, receiver, sequence, telegram type, handshake flag); telegram layouts are versioned metadata (field, offset, length, type, padding); communication points map to wms locations; LIFE telegram every 10 s, channel DOWN after 3 missed; an LP failing scan or contour check at an identification point is routed to the reject point and raises MOB-0116.
- **FR-32.13 Printer and output-queue management** — Should. The system shall register output devices per site with a device type (ZPL II, PDF over IPP, PCL 6; 203/300/600 dpi; media sizes), group them into queues with failover, route every document type (labels and office documents such as pick lists, CoAs and bills of lading) by rule, and offer browser printing of PDF documents to users without a routed printer.
    - Notes: routing precedence per BR-32.12; a queue fails over to the next device after 2 failed attempts; job log kept 30 days; reprints carry `reprint_of` (BR-32.8); printers on site networks are reached by the site print-service pod only (no inbound connections to the cluster).

6.32.4 Data model

MobileFlow (versioned)

Field	Type	Required	Notes
flow_id	uuid	yes	PK
code	text	yes	unique per tenant (e.g., `PICK_CLUSTER`)
version	int	yes	increments on publish
status	enum	yes	DRAFT, PUBLISHED, RETIRED
steps	jsonb	yes	array of {id, type, prompt, expected_scan[], validations[], next{}}
entry_step_id	text	yes	
warehouses	uuid[]	yes	scope
published_at, published_by	timestamptz/text	no	

DeviceSession

Field	Type	Required	Notes
session_id	uuid	yes	PK
device_id	text	yes	FK Device (serial or generated)
user_id	text	yes	Keycloak subject
warehouse_id	uuid	yes	
menu_id	uuid	yes	
current_work_id	uuid	no	FK wms.Work
last_seen_at	timestamptz	yes	heartbeat
state	jsonb	yes	step pointer and field values

OfflineTransaction

Field	Type	Required	Notes
idempotency_key	text	yes	PK `device_id:seq`

Field	Type	Required	Notes
device_id, session_id	text/uuid	yes	
seq	bigint	yes	monotonic per device
command	text	yes	e.g., `ConfirmWorkLine`
payload	jsonb	yes	
captured_at	timestamptz	yes	device clock + server skew
replay_status	enum	yes	PENDING, APPLIED, CONFLICT, REJECTED
conflict_reason	text	no	

PrintJob: job_id, template_id, printer_id, zpl text, status (QUEUED/SENT/PRINTED/FAILED), attempts, created_by.

Invariants: a device has at most one active session; `seq` is strictly increasing per device and gaps are rejected; a PUBLISHED flow version is immutable. Indexing: `offline_transaction(device_id, seq)`; `device_session(user_id) WHERE state_active`.

6.32.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/mobile/v1/sessions	Start session (device, warehouse, menu)	key = device_id	WMS_WORK_EXECUTE
GET	/mobile/v1/menus/{id}	Menu with published flow versions	—	WMS_WORK_EXECUTE
POST	/mobile/v1/flows/{code}/steps/{id}:submit	Submit step input (scan or value)	key	WMS_WORK_EXECUTE
POST	/mobile/v1/scan:parse	Parse a barcode into AI map	—	WMS_WORK_EXECUTE
POST	/mobile/v1/offline:replay	Replay a batch of queued transactions	per-item keys	WMS_WORK_EXECUTE
POST	/mobile/v1/flows	Create/update draft flow	key	WMS_MOBILE_ADMIN
POST	/mobile/v1/flows/{id}:publish	Validate and publish	key	WMS_MOBILE_ADMIN
POST	/print/v1/jobs	Render template and print	key	WMS_PRINT
POST	/rfid/v1/reads	Ingest EPC reads	key = reader+epc+ts	WMS_RFID_INGEST
POST	/wms/v1/automation/channels	Define an MFS channel, telegram profile and communication points	key	WMS_AUTOMATION_ADMIN
POST	/wms/v1/automation/telegrams	Inbound telegram relayed by the edge gateway	key = channel + sequence	edge service account
POST	/wms/v1/automation/transport-orders/{id}:resend	Resend a pending transport order	key	WMS_AUTOMATION_OPERATE
POST	/print/v1/printers	Register a printer, device type and queue	key	WMS_PRINT_ADMIN
POST	/print/v1/routing-rules	Create a document routing rule	key	WMS_PRINT_ADMIN

Event	Trigger	Payload fields	Consumers
dev.passage.mobile.flow.published.v1	publish	flow_id, code, version, warehouses	devices (push via SSE), analytics
dev.passage.mobile.session.started.v1 / ended.v1	session lifecycle	session_id, device_id, user_id, warehouse	analytics
dev.passage.mobile.offline.conflict.v1	replay conflict	idempotency_key, command, reason	supervisor queue
dev.passage.print.job.completed.v1	printer ack	job_id, printer, status	wms

6.32.6 User experience

Screen	Primary actions	Key fields	Duty
Device login / menu	Sign in, pick warehouse/menu	user, warehouse, menu items	WMS_WORK_EXECUTE
Flow runner (device)	Scan, enter, confirm, exception	prompt, expected scan class, context (work, location, LP)	WMS_WORK_EXECUTE
Flow designer (web)	Build graph, validate, simulate, publish	steps, branches, validations, versions	WMS_MOBILE_ADMIN
Device management	Register, assign, force sign-out	device id, model, last seen, session	WMS_MOBILE_ADMIN
Offline conflicts	Resolve, re-apply, reject	key, command, reason	WMS_WORK_SUPERVISE
Print management	Printers, templates, job log	printer, template, status	WMS_PRINT

Interaction rules: device UI uses one input per screen; Enter/scan advances; hardware back returns one step; audio and haptic feedback on success/error; font ≥ 16 px; no mouse-only actions; designer simulation runs a flow in a browser with a virtual scanner.

6.32.7 Business rules and edge cases

- **BR-32.1 GS1 AI parsing rules.** Input begins with an optional symbology identifier (`]C1` GS1-128, `]d2` DataMatrix, `]Q3` QR), then element strings. Parse left to right: read the AI (2–4 digits per the GS1 AI table), then its data per the table; fixed-length AIs have no separator, variable-length AIs end at FNC1 (GS, 0x1D) or end of string. Supported in this wave: **AI 00** SSCC, fixed 18 numeric, check digit (mod-10, weights 3/1) verified; **AI 01** GTIN, fixed 14 numeric, check digit verified, GTIN-12/13 zero-padded on lookup; **AI 10** batch/lot, variable ≤ 20 characters (GS1 AI encodable set 82); **AI 11** production date and **AI 17** expiry, fixed 6 numeric `YYMMDD` where `DD = 00` means last day of month, century by GS1 rule (year within –49/+50 of current year); **AI 21** serial, variable ≤ 20; **AI 310n** net weight kg, fixed 6 numeric with `n` implied decimal places (`3102000125` → 1.25 kg); **AI 37** count, variable ≤ 8 numeric; **AI 400** PO number, variable ≤ 30; **AI 410/412** GLN, fixed 13; **AI 91–99** company-internal, variable ≤ 90. Unknown AI → parse error `GS1_UNKNOWN_AI` with position. Example: `]C1010950123456789310123ABC` + GS + `17261231` + `21SN0007` → {01: 09501234567893, 10: 123ABC, 17: 2026-12-31, 21: SN0007}. A barcode without symbology identifier is parsed as GS1 only if it starts with a valid AI and the step expects a GS1 class; otherwise it is a plain value.
- **BR-32.2 Scan routing.** A `scan` step expecting `LP` accepts AI 00 or a plain value matching the LP format; expecting `GTIN` accepts AI 01 or an item barcode (wave 8 item barcodes); a mismatch shows "Expected location, scanned LP" and does not advance.
- **BR-32.3 Offline ordering.** Replay applies transactions in `seq` order; the first failure stops replay for that device and marks the rest PENDING until resolved, because later steps may depend on it.
- **BR-32.4 Conflict policy.** If a replayed `ConfirmWorkLine` targets a line that was cancelled or completed by another user, the transaction is CONFLICT; a supervisor may "apply as movement" (inventory moves as scanned, order link dropped) or reject. Timestamps use the captured device time corrected by the measured server skew at the last heartbeat.
- **BR-32.5 Cluster picking** assigns at most 8 works; a cart position is bound to one order's LP; scanning the wrong position is rejected; a short pick on one position does not release the others.

- **BR-32.6 Session lock.** A device session holds the work lock; heartbeat loss > 30 min releases the lock and sets the session to STALE; resuming re-locks if the work is still OPEN.
- **BR-32.7 PIN override** authorizes exactly one action within 60 s and is logged with the supervisor's subject.
- **BR-32.8 Print idempotency.** A print job with the same key within 10 min is not re-rendered; "print again" creates a new job with `reprint_of`.
- **BR-32.9 Flow publish** requires the simulation of each core path to succeed (designer runs a smoke script attached to the flow); devices mid-flow on the previous version finish on it.
- **BR-32.10 Expiry date** `DD=00` resolves to the last calendar day of the month; `YYMMDD` with an invalid month is a parse error.
- **BR-32.11 Telegram sequencing and handshake.** Each channel keeps send and receive sequences 1–9,999 that wrap; a telegram without handshake acknowledgment after `mfs_resend_timeout_s` (default 2 s) is resent up to 3 times, after which the channel is DOWN and open transport orders show `MFS_PENDING`; a duplicate receive sequence is acknowledged and ignored; a transport order confirms the wms work line only on the arrival telegram at its final destination, never on send.
- **BR-32.12 Print routing precedence.** Rules match most specific first: user + document type → zone + document type → warehouse + document type → document-type default; ties go to the lowest rule sequence; with no match, PDF documents fall back to browser printing and ZPL jobs fail with MOB-0112.

6.32.8 Acceptance criteria

- **AC-32.1** (FR-32.3, BR-32.1) Given the GS1 AI test suite vectors (fixed and variable, with and without symbology identifiers), when parsed, then 100% match expected AI maps and check-digit failures are reported.
- **AC-32.2** (FR-32.4) Given the emulated-device Playwright suite, when a receive → put-away → pick → pack → ship flow is run with scanned GS1-128 labels from the ASN fixture, then the load ships and the inventory ledger reflects each move.
- **AC-32.3** (FR-32.5, BR-32.3) Given a device offline with 200 queued confirmations, when it reconnects, then all 200 are APPLIED exactly once, in order, and replaying the same batch again changes nothing.
- **AC-32.4** (FR-32.2) Given a published flow change adding a lot-confirmation step, when a device loads the menu, then the new version runs within 60 s with no app update.
- **AC-32.5** (FR-32.7) Given an LP label template, when a receive step completes, then a ZPL job with `^BC` SSCC is sent to the step's printer and status PRINTED is recorded.
- **AC-32.6** (BR-32.4) Given a queued confirmation for a line cancelled meanwhile, when replayed, then it is CONFLICT and appears in the supervisor queue.
- **AC-32.7** (FR-32.10) Given a Zebra DataWedge intent with `]C1` prefix, when received on a non-scan step, then it is ignored and logged.
- **AC-32.8** (FR-32.6, BR-32.6) Given a session with no heartbeat for 31 min, when another user requests work, then the stale session's work is assignable.
- **AC-32.9** (FR-32.12, BR-32.11) Given a simulated conveyor PLC on the edge gateway, when 100 LPs are routed to an AS/RS induction point and the PLC drops 5 acknowledgments, then each affected transport order is resent, no LP is confirmed before its arrival telegram, and all 100 work lines complete.
- **AC-32.10** (FR-32.13, BR-32.12) Given a queue of two printers with the first offline, when 20 LP labels print, then all 20 complete on the second printer and the job log records the failover.

6.32.9 Test plan and fixtures

Fixtures: `gs1-ai-vectors.json` (GS1 test suite plus 60 Passage vectors for edge cases), `labels-asn-3-pallet.zpl` rendered to PNG for the emulator's camera-free scan injection, `offline-200.json`, `flows-core-v1` (seven flows), device profiles `zebra-tc52`, `honeywell-ct45`.

Test	Type	Target	Pass criterion
GS1 parser	unit	parser	100% vectors
Flow graph validation	unit	designer	unreachable/non-terminating graphs rejected
Offline replay	property	queue	exactly-once under random disconnects
Device E2E	E2E (Playwright, Android emulator)	7 flows	inbound-to-outbound passes on both device profiles
Replay throughput	performance	1,000 devices × 50 tx	all applied < 5 min
ZPL rendering	snapshot	templates	byte-identical to golden ZPL
Scan latency	performance	step submit	p95 < 300 ms

Reconciliation: Σ mobile confirmations per day = Σ work line confirmations in `wms`; print jobs PRINTED = labels confirmed on device.

6.32.10 Deployment and operations

Helm `passage-wms-mobile` (BFF, SSE push, 2–8 replicas), `passage-print` (print service with printer registry; one per site network, outbound TCP 9100/IPP), PWA assets served by the shell CDN path; Android shell signed APK distributed via MDM (SOTI/Workspace ONE) or sideload. Config: `MOBILE_OFFLINE_MAX_TX=500`, `MOBILE_SESSION_IDLE_MIN=30`, `PRINT_RETRY=3`. Flags: `mobile.offline`, `mobile.voice_adapter`. SLOs: step submit p95 < 300 ms; menu load < 1 s; print job dispatched < 2 s; offline replay 200 tx < 30 s. Alerts: offline conflicts > 10/h per warehouse; print failures > 5/15 min per printer; stale sessions > 20. Runbook: printer unreachable (check print service pod on site, re-queue jobs); device clock skew > 5 min (block replay, instruct re-sync). Rollback: flow versions are retained — republish the previous version; BFF rollback via Argo CD.

6.32.11 Migration from AX / SAP

Source	Passage entity	Transformation
WHSMobileAppMenu, WHSRFMenuTable, WHSRFMenuLine	MobileMenu	hierarchy preserved
WHSRFMenuItemTable (mode, work class, directed/user-directed, detours)	MobileFlow draft	generated from the menu item's options into the equivalent core flow with configured branches; requires manual review
WHSWorkUser	WorkUser attributes on Keycloak user	user id mapping from wave 2
WHSDocumentRoutingLayout, printers	Print templates, printers	ZPL layouts imported as-is where ZPL; others re-authored
/SCWM/TMENU, /SCWM/T_RF_MENU, /SCWM/TLOGTR	MobileMenu, MobileFlow draft	logical transaction → core flow mapping table
/SCWM/RSRC, /SCWM/TRSRC	WorkUser, device bindings	resource type → work class

Not migrated: RF screen layouts (/SCWM/TPRES) and D365 step field layouts — Passage flows are redesigned; mobile session state; print spool history.

6.32.12 Dependencies, risks, open questions

Depends on: wave 31 (work APIs, LPs), wave 25 (shop-floor PWA offline pattern and service-worker library), wave 8 (item barcodes, GS1 prefix), wave 2 (Keycloak device flow), wave 28 (containers/packing).

Risk	Likelihood	Impact	Mitigation
Offline replay creates subtle inventory errors	Medium	High	Strict ordering, conflict queue, property tests; offline off by default for count flows
Android WebView versions on older scanners	Medium	Medium	Minimum WebView 100; PWA fallback

Open questions: (1) Native iOS shell — proposed: no, PWA only; decide by wave 33. (2) Voice vendor for v1.x reference adapter (Honeywell Voice vs Lydia) — proposed: defer; decide by wave 35.

6.32.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
session_idle_timeout_min	warehouse	int	30	5–240	Session goes STALE and releases the work lock (BR-32.6)	Work user session expiry (Warehouse app settings)	RF presentation device logon timeout (/SCWM/RFUI)
pin_reauth_hours	tenant	int	8	1–24	PIN re-authentication interval (FR-32.1)	Warehouse app authentication (device code, user PIN)	ITS session timeout (SICF service settings)
offline_enabled	warehouse × flow	bool	true for pick/put, false for count	true / false	Allows offline queueing for the flow (FR-32.5)	n/a (online only)	n/a (online only)
offline_max_tx	device	int	500	50–2,000	Queue cap (FR-32.5)	n/a	n/a
offline_max_age_h	device	int	24	1–72	Oldest transaction allowed in queue	n/a	n/a
flow_publish_four_eyes	tenant	bool	false	true / false	Production publish needs a second approver	Mobile device menu item change via data management (no approval)	Transport of logical transactions (SE01/STMS)
gs1_require_symbology_id	warehouse	bool	false	true / false	Unprefixed strings parsed as plain values when true (BR-32.1)	Mobile device barcode setup › GS1 symbology prefix	RF barcode verification profile
gs1_fnc1_substitute	warehouse	char	GS (0x1D)	GS / `~` / `	`	Separator accepted for FNC1	Mobile device barcode setup › FNC1 character
scanner_profile	device model	enum	ZEBRA_DATAWEDGE	ZEBRA_DATAWEDGE / HONEYWELL_DC / KEYBOARD_WEDGE	Scan capture bridge (FR-32.10)	Warehouse app device settings	ITSmobile device template
cluster_max_positions	menu / flow	int	8	1–24	Cluster pick positions (BR-32.5)	Cluster profile "Number of positions"	Multiple-order picking positions in RF
heartbeat_interval_s	tenant	int	30	10–120	Device heartbeat (FR-32.6)	n/a	n/a
print_retry_count	site	int	3	0–10	Retries before failover (FR-32.7)	Document routing agent retry	Spool retry settings (SPAD)
print_job_dedupe_min	tenant	int	10	0–60	Duplicate-key window (BR-32.8)	n/a	n/a
label_default_dpi	printer	enum	203	203 / 300 / 600	Template rendering resolution	Document routing layout printer DPI	SPAD device type (LB_ZEB 203 dpi Zebra)
browser_print_fallback	warehouse	bool	true	true / false	PDF fallback when no device routed (BR-32.12)	Print to screen / user default printer	Front-end printing (access method F/G)
mfs_heartbeat_s	automation channel	int	10	5–60	LIFE telegram interval (FR-32.12)	n/a	SPRO EWM › Material Flow System › Define PLC communication channel (life telegram)
mfs_resend_timeout_s	automation channel	int	2	1–30	Handshake timeout (BR-32.11)	n/a	MFS channel resend settings
mfs_reject_point	communication point	location ref	none	locations of type REJECT	Target for failed identification	n/a	MFS communication point "NIO" exception point
telemetry_retention_days	tenant	int	90	30–365	Step telemetry retention (FR-32.11)	n/a	n/a

6.32.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Active sessions and devices	list inquiry	warehouse, model, last seen	device, user, menu, current work, app version	`device_session`	Work user sessions	LRF1 RF monitor, /SCWM/MON resources
Offline conflict queue	list inquiry	warehouse, reason, age	key, command, captured_at, conflict reason	`offline_transaction`	n/a	n/a
Flow version history	drill-down	flow code	version, published by/at, step diff	`mobile_flow`	Mobile device menu item change history (database log)	Transport history of /SCWM/TLOGTR (SE03)
Step timing by user and flow	KPI	flow, step, user (duty-gated), date	median s, p95 s, error count	telemetry rollup	Work user activity inquiry	LRF2 queue/resource statistics
Scan error log	operational report	error type, device, date	masked raw value, AI, error position	parser log	n/a	Application log SLG1 (/SCWM/RFUI)
Print job log	list inquiry	printer, status, template, user	job, attempts, reprint_of, duration	`print_job`	Document routing job history	SP01 spool requests
Printer status	operational report	site, queue	online, queue depth, last error	printer registry	Document routing agent status	SPAD/SP01 output device status

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
MFS channel monitor	operational report	channel, communication point	state, last telegram, pending transport orders, faults	`automation_channel`	n/a	MFS monitor nodes in /SCWM/MON
Device fleet health	KPI	warehouse	step submit p95, replay backlog, stale sessions	telemetry	n/a	n/a

6.32.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
LP label	Receive or LP build	SSCC barcode (`^BC` , AI 00), human-readable SSCC, item, qty, lot, expiry, ship-from	ZPL; PDF preview	SSCC from LP sequence	GS1 logistic label guideline
Location label	On demand	Location barcode, check characters, direction arrow	ZPL	none	None
Item / GTIN label	On demand or VAS	GTIN-14 (AI 01), lot (AI 10), expiry (AI 17), serial (AI 21)	ZPL / PDF	none	Regulated serialization labels (EU FMD, US DSCSA) when item is flagged
GS1 carton shipping label	Pack or sort position close	Ship-from/to, carrier, SSCC, PO (AI 400), ship-to postal code (AI 420)	ZPL 4 × 6 in	SSCC	Partner label specifications (wave 35 partner profile)
Routed office documents	Rule match (BR-32.12)	Pick list, count sheet, CoA, BOL from their owning waves	PDF via IPP or browser	per document	per owning wave
Offline conflict report	Daily	Conflicts by reason, resolution, supervisor	PDF / CSV	none	Audit evidence, 1 year

6.32.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Stale session sweep	Every 1 min	warehouse	session `last_seen_at`	< 5 s	Releases locks; idempotent	Stale sessions > 20
Offline replay drain	Event-driven on reconnect; backlog sweep every 30 s	device_id	`device_id:seq`	200 tx < 30 s per device	Stops at first failure per device (BR-32.3)	Conflicts > 10/h per warehouse
Print queue retry and failover	Every 15 s	printer	job id + attempt	< 2 s per job	Fail over after 2 attempts	Print failures > 5/15 min per printer
MFS channel heartbeat check	Every 10 s	channel	channel id	< 100 ms	DOWN status, pending transport orders listed	MfsChannelDown
Telemetry rollup	Hourly	warehouse	hour bucket	< 1 min	Recompute bucket	Rollup lag > 2 h
Telemetry purge	Daily 01:00	tenant	day partition	< 5 min	Retry next day	none (log)
Device app-version audit	Daily 06:00	warehouse	device id	< 1 min	Report only	Devices below minimum shell version

6.32.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
WMS_WORK_EXECUTE (mobile)	/mobile/v1/sessions, /menus/{id}, /steps/{id}:submit, /scan:parse, /offline:replay	WMS_COUNT_APPROVE on the same count	Warehouse worker
WMS_MOBILE_ADMIN	/mobile/v1/flows, /flows/{id}:publish; device management	Second approver required when `flow_publish_four_eyes = true`	WMS administrator
WMS_WORK_SUPERVISE	Offline conflict resolution, PIN override, force sign-out	Overriding own work is blocked (BR-32.7)	Warehouse supervisor
WMS_PRINT	/print/v1/jobs	none	Warehouse worker, receiving and shipping clerks
WMS_PRINT_ADMIN	/print/v1/printers, /print/v1/routing-rules; print management	none	WMS administrator, site IT
WMS_RFID_INGEST	/rfid/v1/reads	Service accounts only; never assigned to people	Reader service account
WMS_AUTOMATION_ADMIN	/wms/v1/automation/channels; telegram profile designer	WMS_AUTOMATION_OPERATE in production (change and operate the same channel)	Automation engineer
WMS_AUTOMATION_OPERATE	MFS monitor: resend, block communication point, clear fault	none	Control-room operator

6.32.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
MOB-0101	error	Expected {expected}, scanned {actual}.	Scan class mismatch	Scan the requested object	BR-32.2
MOB-0102	error	Check digit invalid for AI {ai}.	SSCC/GTIN check digit failure	Rescan or relabel	BR-32.1
MOB-0103	error	Unknown GS1 application identifier at position {pos}.	GS1_UNKNOWN_AI	Check label against GS1 AI table	BR-32.1
MOB-0104	error	Invalid date {value} for AI {ai}.	Bad YYMMDD (month 13 etc.)	Relabel	BR-32.10
MOB-0105	warning	Offline: {n} transactions queued; {remaining} left before the limit.	Device offline	Move to coverage when possible	FR-32.5
MOB-0106	error	Offline queue is full; reconnect to continue.	500 transactions or 24 h reached	Reconnect	FR-32.5
MOB-0107	warning	Transaction {key} conflicts: line was {status} by {user}.	Replay CONFLICT	Supervisor resolves	BR-32.4
MOB-0108	error	Sequence gap on device {device}: expected {n}, received {m}.	Missing offline transaction	Device resync; support ticket	BR-32.3
MOB-0109	error	Wrong cart position: scanned {pos}, expected {exp}.	Cluster pick mismatch	Place item in the correct position	BR-32.5
MOB-0110	error	Supervisor PIN expired or already used.	Override outside 60 s or reused	Request a new override	BR-32.7
MOB-0111	error	Printer {printer} unreachable after {n} attempts.	Network or device failure	Check printer; job fails over	FR-32.7
MOB-0112	error	No printer routed for {document} at {location}.	No routing rule match for ZPL	Add routing rule	BR-32.12
MOB-0113	error	Flow {code} v{n} failed validation: step {id} is unreachable.	Graph validation	Fix the flow graph	FR-32.2
MOB-0114	warning	Device clock skew {s} s exceeds 300 s; replay blocked.	Device clock drift	Resync device time	§6.32.10
MOB-0115	error	Automation channel {ch} is down; transport order {to} pending.	3 missed handshakes or LIFE telegrams	Check PLC link; resend after recovery	BR-32.11

Code	Severity	User text	Cause	Remediation	Raised by
MOB-0116	warning	LP {lp} rejected at identification point {cp}: {reason}.	Scan or contour check failed	Correct LP at reject point and re-induct	FR-32.12
MOB-0117	info	Session resumed on flow version {n}.	Device was mid-flow during publish	None	BR-32.9

6.32.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Step submit	1,000 devices; 300,000 steps/day; peak 6,000/min	80 / 300 / 600 ms	100/s per BFF pod	Telemetry ≈ 40 GB (90-day rolling)	Telemetry daily partitions	Offline-capable flows queue locally; others show "busy, retry"
Scan parse	400,000/day	2 / 10 / 20 ms (in-process)	2,000/s per pod	—	—	CPU-bound; HPA adds pods
Offline replay	1,000 devices × 50 tx after an outage	200 tx < 30 s per device; fleet < 5 min	300 tx/s	Offline log 90 days ≈ 2 GB	`offline_transaction(device_id, seq)`	Replay throttled per device; ordering preserved
Menu load	20,000/day	150 / 600 / 1,000 ms	50/s	—	Published flows cached per warehouse	Cached version served; publish propagation may exceed 60 s
Print dispatch	50,000 labels/day; peak 300/min	300 ms / 1.2 s / 2 s to printer acknowledgment	20 jobs/s per site service	Job log 30 days ≈ 1 GB	`print_job(printer_id, created_at)`	FIFO backlog per printer; failover to queue peers
MFS telegram round-trip	100,000 telegrams/day; peak 50/s per site	20 / 80 / 200 ms gateway ↔ wms	200/s per site	Telegram log 30 days ≈ 3 GB	`transport_order(channel_id, status)`	Dedicated worker pool keeps automation ahead of manual traffic

6.32.20 Localization and compliance hooks

- **Language and layout.** Device UI strings come from wave-5 i18n bundles per user locale; the PWA supports right-to-left layouts (Arabic, Hebrew); voice prompts (FR-32.9) use the adapter's language model per locale.
- **Regulated serialization.** EU Falsified Medicines Directive (Delegated Regulation 2016/161) requires 2D DataMatrix with AI 01, 21, 10, 17 and national reimbursement numbers (AI 710–715); US DSCSA requires interoperable serialized traceability (EPCIS). Country packs (wave 48) add AI validation and label templates through `LabelTemplatePack`; the EPC ingest (FR-32.9) carries SGTIN data.
- **Character sets.** ZPL templates use `^CI28` (UTF-8) for non-Latin scripts; human-readable dates on labels follow locale while barcode data stays `YYMMDD`.
- **Privacy.** Per-user step telemetry is personal data (GDPR); retention 90 days; named per-user timing reports follow the wave-31 `LaborPrivacyPolicy`; works-council rules as §6.31.20.
- **Device and radio rules.** Scanner hardware certification is the device vendor's responsibility; Passage stores no location data from devices beyond the warehouse and session.

6.32.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP #193 — §13.16 EWM /SCWM/MFS* Material Flow System (PLC interface, conveyors, AS/RS)	GAP (propose 32)	ADDED as FR-32.12	On the wave-25 edge gateway as the register proposed; BR-32.11, AC-32.9
SAP #201 — §13.16 EWM MFS customizing (PLC communication channels, telegrams)	GAP (propose 32)	ADDED as FR-32.12	`mfs_*` parameters in §6.32.13; telegram profiles are versioned metadata
SAP #346 — §13.30 Printing: device types, output devices, access methods (SPAD), front-end printing, print queues (TSP03*)	PARTIAL-32	ADDED as FR-32.13	Device types, queues with failover, routing, browser printing; BR-32.12, AC-32.10

6.33 Wave 33: Quality Management & Nonconformance

6.33.1 Scope and business context

Wave 33 replaces AX quality management (test groups, quality associations, quality orders, nonconformances, corrections) and SAP QM (inspection plans, inspection lots, results recording, usage decisions, quality notifications, vendor evaluation). After this wave, quality orders are generated by association rules at purchase receipt, production report-as-finished and before shipment; samples are sized by AQL plans; results are recorded (variable and attribute); a usage decision releases or rejects inventory that was blocked from the moment the order was created; nonconformances drive corrections/CAPA through a Temporal 8D workflow; certificates of analysis print on shipment; supplier scorecards and SPC charts are computed from results; instruments have calibration schedules. Out of scope: stability studies, audit management (ISO 19011), FMEA, and the SAP QM-PM calibration order integration (Passage tracks calibration due dates and blocks results from overdue instruments).

Scope amendment (coverage closure). The gap registers move into this wave items that the paragraph above excludes or does not name: product compliance with regulated and restricted lists (FR-33.13), safety data sheets (FR-33.14), incoming vendor certificates (FR-33.15), stability studies (FR-33.16), audit management (FR-33.17), FMEA and control plans (FR-33.18), source inspection (FR-33.19), physical sample management (FR-33.20), multiple specifications (FR-33.21) and in-process inspection points (FR-33.22). The SAP QM-PM calibration-order integration stays out of scope; Instrument due-date tracking (BR-33.8) remains the Passage mechanism. §6.33.21 lists the dispositions.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX quality order blocking (InventQualityOrderTable → InventBlockingTable) blocks only the quantity registered on the receipt; production and sales paths can still pick inventory registered under a different dimension set, and "Inventory blocking" is bypassed by movement journals	Blocking is a reservation-like record checked by some, not all, transaction types	Hold lives in the inventory ledger reservation path and is checked by every transaction type; property test enumerates all	Movements of held stock: occasional (audit findings) → 0 across 10,000 randomized attempts
SAP usage decision (QA11) is a separate transaction from results recording (QE01), each 8–12 clicks with stock posting on a third tab; partial accept/reject needs manual quantity split and MIGO 321/322 follow-up	Inspection lot, results and stock posting are separate objects with separate screens	Results and decision on one order; quantity split posts both ledger legs automatically	Clicks per decision: ~25 across QE01/QA11/MIGO → ≤ 5; follow-up postings keyed manually: 2 → 0
AQL sampling in AX is fixed qty or % (InventTestItemSampling); Z1.4 tables are kept in spreadsheets and sample sizes keyed by hand; SAP QDP1 sampling procedures require master data per AQL/level combination	No standard sampling tables in the product	Z1.4 Tables I and II-A/B/C as versioned reference data with switching rules	Sample-size keying errors: ~2% of lots (spreadsheet) → 0; sampling setup per item: 15 min → 0 (plan reuse)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
SAP quality notification 8D (QM01/QM02 tasks) has no timers; overdue corrective actions are found by report ZQM or not at all; effectiveness checks are forgotten	Notification tasks are static records without workflow timers	Temporal 8D workflow with due-date timers, escalations and an automatic 90-day effectiveness-check task	Overdue CAPA actions discovered: weeks later via report → escalated within 1 business day; effectiveness checks executed: ~50% → 100%
CoA generation (AX CoA report / SAP QC21) prints whatever results exist; amended results after shipment produce inconsistent certificates	Certificate is a report over mutable results, not a versioned document	CoA generated from the accepting decision, revisioned with `supersedes`	Inconsistent certificates reported by customers: tolerated → 0 (immutable revision chain)

6.33.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Tests and groups	InventTestTable, InventTestGroup, InventTestGroupTest (lines), InventTestVariable/Outcome	Master inspection characteristics QPMK (QS21), inspection plans QP01 (PLKO/PLPO/PLMK), catalogs QPCT	quality.Test, TestGroup, Characteristic catalog	full
Sampling	InventTestItemSampling (fixed qty/%)	Sampling procedures QDP1 (QDPS), sampling schemes QDS1, dynamic modification	SamplingPlan (fixed, %, AQL Z1.4)	improved — AQL built in
Triggers	InventTestAssociationTable (quality associations)	QMAT inspection setup per material/plant; lot origins 01/04/10/89	QualityAssociation	full
Quality orders	InventQualityOrderTable, InventQualityOrderLine, InventQualityOrderLineResults; form InventQualityOrderTable	Inspection lot QALS (QA01/QA02), results QAMR/QASR (QE01), QA11 usage decision (QAVE)	QualityOrder, QualityResult, UsageDecision	full
Inventory blocking	InventBlockingTable, InventQualityOrderTable blocking	Stock type Q (quality inspection), QM stock posting at UD	Quant `quality_status`; ledger status QUALITY_HOLD	full
Nonconformance	InventNonConformanceTable, problem/diagnostic types, InventNonConformanceCorrection, InventNonConformanceOperation	Quality notifications QMEL/QMFE/QMMA/QMSM (QM01), 8D report	Nonconformance, Correction (CAPA)	full
Certificates	Certificate of analysis (InventTestCertificateOfAnalysis report)	QC21/QC22 certificate profiles (QCVK)	CoA template + generation	full
Vendor evaluation	Vendor quality on nonconformance (reports)	ME61–ME65 vendor evaluation (ELBK/ELBM/ELBP)	SupplierScorecard	full
Calibration	InventTestInstrument (instrument master only)	QM-PM calibration (inspection lot type 14, equipment)	Instrument with calibration schedule	partial
SPC	n/a	QM control charts (QGC1)	SPC (X-bar/R, p-chart)	full
Cost of quality	n/a	QM-CO appraisal costs	CostOfQuality projection	improved
Product compliance	Product compliance (D365 process industries): regulated products, restricted lists, reporting lists	S/4 Product Compliance (product marketability checks), EHS Product Safety and Stewardship	quality.ComplianceList, ItemComposition, ComplianceResult	full
Safety data sheets	Safety data sheets with validity and send-on-sales (D365 PI)	EHS report shipping (CVD1), SDS distribution	quality.SafetyDataSheet, SdsDelivery	full
Incoming certificates	n/a (attachments on product receipts)	QC51/QC52/QC53 certificate receipt, QC55 overview; certificate type in QI01	quality.IncomingCertificate	full
Stability study	n/a	QST01–QST07 stability study (storage conditions, testing schedule)	quality.StabilityStudy, PullPoint	full
Audit management	n/a	PLMD_AUDIT audit plans, question lists, findings, corrective actions	quality.Audit, QuestionList, Finding	full
FMEA / control plan	n/a	S/4 QM FMEA (FMEA cockpit), control plans	quality.Fmea, ControlPlan	full
Source inspection	n/a (quality orders only after receipt)	Source inspection flag in QI01, QI07 trigger, inspection type 0101	QualityOrder trigger SOURCE_INSPECTION	full
Sample management	Item sampling quantity only (InventTestItemSampling)	QPR1–QPR7 physical samples, sample-drawing procedures, sample types	quality.PhysicalSample	full
Multiple specifications	One test group per item/association	S/4 Multiple Specifications (customer, country, pharmacopeia objects)	quality.SpecificationSet	full
In-process inspection points	Quality associations on route operations (before/after)	Inspection points (TQ79), lot origin 03, QE11 results per point	InspectionPoint on production operations	full

6.33.3 Functional requirements

- **FR-33.1 Tests, instruments and test groups** — Must. The system shall define tests (variable with UoM, target, min/max; attribute with outcome catalog), instruments, and test groups with sequence, sampling, and acceptance per test.
  - Notes: variable results compare to tolerances with 6-decimal precision; test group versions are effective-dated; a test marked `destructive` reduces sample qty on completion.
- **FR-33.2 Sampling plans** — Must. The system shall size samples by fixed quantity, percentage, or ANSI/ASQ Z1.4 single sampling (normal, tightened, reduced) at general inspection level II by default.
  - Notes: table excerpt and switching rules in BR-33.1/BR-33.2; sample qty is capped at lot qty; percentage rounding up.
- **FR-33.3 Quality associations** — Must. The system shall generate quality orders when a trigger matches: purchase receipt (registration or product receipt), production RAF (before/after put-away), sales order (at pick/before packing slip), transfer receipt, quarantine, and manual.
  - Notes: association scope = item / item group / all × vendor/customer / group / all × site; most specific wins (BR-33.3); one order per source line per event.
- **FR-33.4 Quality orders and results** — Must. The system shall record results per test per sample, compute pass/fail per test and order, and support retest and partial sampling.
  - Notes: results recorded with instrument id and user; a result from an instrument overdue for calibration is rejected; order status per BR-33.4.
- **FR-33.5 Inventory blocking** — Must. The system shall block the inspected quantity from pick, ship, consume and transfer-out when the quality order is created and release it only on an accepting usage decision; rejection moves qty to a quarantine location or blocked status.
  - Notes: enforcement lives in the inventory ledger reservation path (wave 16), not only in `wms`; blocked qty is visible as "quality hold" in ATP (wave 27) as unavailable.
- **FR-33.6 Usage decision** — Must. The system shall record a usage decision (accept, reject, accept-with-deviation, rework, return-to-vendor, scrap) with partial quantities, and post consequential transactions.
  - Notes: reject → nonconformance auto-created; return-to-vendor → wave-18 return order draft; scrap → wave-16 scrap journal with reason.
- **FR-33.7 Nonconformance** — Must. The system shall create nonconformances (sources: customer, vendor, internal, production, service request) with problem type, severity, quarantine action, related quality order and quantity.
  - Notes: statuses NEW → APPROVED → IN_CORRECTION → CLOSED, or REFUSED; attachments and photos from the mobile app.

- **FR-33.8 Correction / CAPA workflow** — Must. The system shall run an 8D correction workflow (D1 team … D8 close) on Temporal with root cause, containment, corrective action, due dates, approvals and an effectiveness check after a configurable period (default 90 days).
  - Notes: overdue actions escalate per wave-4 workflow escalation; effectiveness check task auto-created by a Temporal timer.
- **FR-33.9 Certificates of analysis** — Must. The system shall generate CoA documents per shipment/batch from accepted results using a template per customer/item, with PDF output and e-mail/portal delivery.
  - Notes: CoA shows result, specification, method, instrument; signature block from the usage-decision user.
- **FR-33.10 Supplier quality scorecard** — Should. The system shall compute per vendor: lots inspected, lot acceptance rate, PPM defective, NCR count by severity, on-time delivery (from wave 18), and a weighted score 0–100.
  - Notes: weights configurable; monthly materialization; exposed to the wave-20 supplier portal.
- **FR-33.11 SPC charts** — Should. The system shall compute X-bar/R (variable) and p/np (attribute) charts with Western Electric rules 1–4 flagged.
  - Notes: subgroup size from sampling plan; control limits from the last 25 subgroups.
- **FR-33.12 Cost of quality** — Could. The system shall tag inspection labor (wave 24 time), scrap, rework and returns to prevention/appraisal/internal/external failure categories for reporting.
  - Notes: amounts sourced from journal entries by dimension `coq_category`.
- **FR-33.13 Regulated and restricted product lists** — Should. The system shall maintain compliance lists (regulated regions; restricted-substance lists such as REACH SVHC, RoHS Annex II, California Proposition 65, TSCA; reporting lists with thresholds) and item compositions (substance, CAS number, concentration % w/w), evaluate every item against every applicable list per BR-33.11, and block or warn on sales order lines, shipments and purchase receipts to or from regions where the item is restricted.
  - Notes: composition rolls up through the wave-9 BOM by mass fraction; results are versioned per (item, list version); block or warn per list (`compliance_block_mode`); a reporting extract per period lists quantities sold per region for reporting lists (e.g. REACH Art. 33 communication, SCIP notification data).
- **FR-33.14 Safety data sheets** — Should. The system shall store SDS versions per item × language × region (GHS 16-section reference, PDF, version, issue date, validity), record which version each customer received, and send the current SDS with the first shipment and after every new version per BR-33.12.
  - Notes: shipment confirmation (wave 28) checks `sds_required`; missing SDS holds or warns per `sds_missing_action`; validity default 1,095 days; an expiry job creates review tasks 60 days ahead; the delivery log is the distribution evidence.
- **FR-33.15 Incoming vendor certificates** — Should. The system shall require certificates (certificate of analysis, certificate of conformance, EN 10204 type 3.1 inspection certificate) per item × vendor through the quality association, record their receipt (upload, EDI attachment or supplier portal), check certificate values against the specification, and keep the quality order from acceptance until the certificate is received and accepted (BR-33.13).
  - Notes: certificate types are required or optional; a receipt without a required certificate sets `certificate_status = MISSING`; waivers need QM_USAGE_DECISION and a reason; the supplier portal (wave 20) offers upload against the ASN or PO line.
- **FR-33.16 Stability studies** — Could. The system shall plan stability studies (item, batch, protocol), storage conditions (e.g. 25 °C/60 % RH long-term, 40 °C/75 % RH accelerated per ICH Q1A(R2)), pull points (0, 3, 6, 9, 12, 18, 24, 36 months) with a test group per pull point, and create stability quality orders on schedule from retained samples (FR-33.20).
  - Notes: one Temporal workflow per study schedules pulls (BR-33.14); status PLANNED → ACTIVE → COMPLETED or TERMINATED; out-of-specification results raise an NCR and, after approval, can shorten the batch shelf life (wave 16 batch expiry).
- **FR-33.17 Audit management** — Could. The system shall plan audits (internal, supplier, customer; ISO 9001 clause references), maintain weighted question lists, record answers and findings, and convert findings into NCRs and CAPA (FR-33.8); supplier audit scores feed the scorecard (FR-33.10).
  - Notes: status PLANNED → IN_PROGRESS → REPORTED → CLOSED; score = Σ weighted answer points ÷ maximum × 100; finding severity MAJOR, MINOR or OBSERVATION; closure requires a CAPA for every major finding; an auditor cannot audit their own org unit.
- **FR-33.18 FMEA and control plans** — Could. The system shall maintain design and process FMEAs in the AIAG-VDA (2019) structure — structure, function, failure analysis, and risk analysis with severity, occurrence and detection 1–10 and action priority H/M/L — and control plans per item × routing operation whose characteristics generate test group lines and in-process inspection points (FR-33.22).
  - Notes: action priority from the AIAG-VDA AP table held as reference data (BR-33.15); RPN shown for legacy comparison only; control plan phases PROTOTYPE, PRE_LAUNCH, PRODUCTION; a control plan revision versions the linked test group; a PPAP element checklist is stored as a document set on the item.
- **FR-33.19 Source inspection** — Should. The system shall create a quality order with trigger SOURCE_INSPECTION for a PO line or ASN a configurable lead time before the confirmed delivery date, let an inspector or the vendor (portal) record results, and on acceptance mark the inspected batches so that their receipt skips receiving inspection (BR-33.16).
  - Notes: lead time default 5 days; rejection notifies purchasing and blocks the ASN; non-matching batches at receipt get the normal receipt association.
- **FR-33.20 Physical sample management** — Should. The system shall record physical samples drawn for a quality order — primary, pooled and reserve/retention samples — with sample id, quantity, container, storage location and condition, label (ZPL through wave 32) and retention expiry, under a sample-drawing procedure.
  - Notes: retention default 1 year past batch expiry (21 CFR 211.170); destructive consumption posts per BR-33.5; a daily job marks expired retention samples DUE_FOR_DISPOSAL and creates a disposal task; pooled samples reference their primary samples.
- **FR-33.21 Multiple specifications** — Should. The system shall allow specification sets per test group (default, customer, country/region, pharmacopeia) with their own limits per test, evaluate results against every applicable set, and record per set whether the lot conforms; the CoA and usage decision use the set that applies to the customer or destination (BR-33.17).
  - Notes: a lot conforms to set S if and only if every required test passes under S's limits; the lot carries `spec_conformance[]`, which wave-27 ATP uses to allocate the lot only to customers whose set it meets.
- **FR-33.22 In-process inspection points** — Should. The system shall define inspection points on production route operations by time interval (e.g. every 2 h), quantity interval (e.g. every 500 units) or free definition, generate result slots per point during production from wave-21/25 operation feedback, and halt the operation when a critical characteristic fails if configured.
  - Notes: point id = (order, operation, point number, time or quantity stamp); results are recorded on the shop-floor PWA (wave 25); a critical failure sets the operation to HALTED_QUALITY and notifies the supervisor; output since the last good point can be dispositioned as a partial lot; missed points per BR-33.18.

6.33.4 Data model

QualityOrder

Field	Type	Required	Notes
quality_order_id	uuid	yes	PK; number sequence QO-*
trigger	enum	yes	PURCHASE_RECEIPT, PRODUCTION_RAF, SALES_PICK, TRANSFER_RECEIPT, QUARANTINE, MANUAL
source_ref	jsonb	yes	{type, id, line} polymorphic
item_id, dims	uuid/jsonb	yes	batch/serial dims

Field	Type	Required	Notes
site_id, warehouse_id	uuid	yes	
lot_qty, sample_qty, uom	numeric(18,6)/text	yes	
test_group_id, test_group_version	uuid/int	yes	
sampling_plan_id, inspection_severity	uuid/enum	yes	NORMAL/TIGHTENED/REDUCED
status	enum	yes	BR-33.4
block_ref	uuid	no	inventory ledger hold id
created_at, closed_at	timestamptz	—	

QualityResult

Field	Type	Required	Notes
result_id	uuid	yes	PK
quality_order_id, test_id, sample_no	uuid/uuid/int	yes	unique together with `retest_no`
value_numeric	numeric(18,6)	no	variable tests
outcome_id	uuid	no	attribute tests
passed	bool	yes	computed
instrument_id	uuid	no	must be in calibration
recorded_by, recorded_at	text/timestamptz	yes	
retest_no	int	yes	default 0

Nonconformance

Field	Type	Required	Notes
ncr_id	uuid	yes	PK; NCR-*
source	enum	yes	CUSTOMER, VENDOR, INTERNAL, PRODUCTION, SERVICE
item_id, dims, qty	—	yes	
problem_type_id, severity	uuid/enum	yes	severity MINOR/MAJOR/CRITICAL
quality_order_id, vendor_id, customer_id, production_order_id	uuid	no	
status	enum	yes	NEW, APPROVED, REFUSED, IN_CORRECTION, CLOSED
quarantine_order_id	uuid	no	
correction_workflow_id	text	no	Temporal workflow id

SamplingPlan: plan_id, method (FIXED/PERCENT/AQL), fixed_qty, percent, aql (numeric 3,2), inspection_level (S1–S4, I–III), severity default, switching_enabled.
Instrument: instrument_id, calibration_interval_days, last_calibrated_at, due_at, status.

Invariants: a quant with an open QualityOrder referencing it has `quality_status ∈ {PENDING, BLOCKED}`; `Σ accept_qty + reject_qty + deviation_qty` of a usage decision = lot_qty; a QualityOrder is CLOSED only with a usage decision; a result cannot reference an instrument whose `due_at < recorded_at`. Indexing: `quality_order(status, site_id)`, `quality_result(quality_order_id, test_id, sample_no, retest_no)` unique, `nonconformance(vendor_id, created_at)`.

6.33.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/quality/v1/test-groups	Create/version test group	key	QM_SETUP_MAINTAIN
POST	/quality/v1/associations	Create association	key	QM_SETUP_MAINTAIN
POST	/quality/v1/sampling-plans:size	Compute sample size for lot qty/plan	—	QM_VIEW
POST	/quality/v1/quality-orders	Create manual order	key	QM_ORDER_MAINTAIN
POST	/quality/v1/quality-orders/{id}/results	Record results (batch)	key per batch	QM_RESULTS_RECORD
POST	/quality/v1/quality-orders/{id}:usage-decision	Post decision with quantities	key	QM_USAGE_DECISION
POST	/quality/v1/nonconformances	Create NCR	key	QM_NCR_MAINTAIN
POST	/quality/v1/nonconformances/{id}/corrections	Start CAPA workflow	key	QM_CAPA_MAINTAIN
POST	/quality/v1/coa:generate	Generate CoA for shipment/batch	key = shipment+batch	QM_COA
GET	/quality/v1/scorecards/vendors/{id}?period=	Scorecard	—	QM_VIEW
POST	/quality/v1/compliance-lists	Create or version a compliance list	key	QM_COMPLIANCE_MAINTAIN
POST	/quality/v1/items/{id}/sds	Upload an SDS version	key	QM_COMPLIANCE_MAINTAIN
POST	/quality/v1/quality-orders/{id}/certificates	Record an incoming certificate	key	QM_RESULTS_RECORD
POST	/quality/v1/stability-studies	Create a study and generate its schedule	key	QM_STABILITY_MAINTAIN
POST	/quality/v1/audits	Plan an audit	key	QM_AUDIT_MAINTAIN
POST	/quality/v1/fmeas	Create or version an FMEA or control plan	key	QM_FMEA_MAINTAIN
POST	/quality/v1/samples	Register a physical sample	key	QM_RESULTS_RECORD

Commands: `CreateQualityOrderFromTrigger` validates association match and that the source line is in the required state; `PostUsageDecision` validates all required tests have results (or `allow_incomplete` with deviation reason) and quantity split sums to lot qty.

Event	Trigger	Payload fields	Consumers
dev.passage.quality.order.created.v1	order created	quality_order_id, trigger, source_ref, item, dims, lot_qty, sample_qty	inventory (hold), wms, analytics
dev.passage.quality.result.recorded.v1	result batch	quality_order_id, results[]	SPC projection
dev.passage.quality.usagedecision.posted.v1	decision	quality_order_id, decision, accept_qty, reject_qty, deviation_qty, user	inventory (release/hold), wms, purchasing, production
dev.passage.quality.nonconformance.created.v1	NCR	ncr_id, source, severity, item, vendor_id	vendor scorecard, workflow
dev.passage.quality.capa.closed.v1	8D closed	ncr_id, effectiveness_result	analytics

6.33.6 User experience

Screen	Primary actions	Key fields	Duty
Quality setup	Tests, groups, sampling plans, associations, instruments	test, type, tolerance, AQL, trigger scope	QM_SETUP_MAINTAIN
Quality order workbench	List by status/site, open, record, decide	order, item, trigger, sample, due, status	QM_ORDER_MAINTAIN
Results recording (web + mobile step)	Enter values per sample/test, attach	test, sample, value, outcome, instrument	QM_RESULTS_RECORD
Usage decision	Split quantities, choose decision, reason	accept/reject/deviation qty, disposition	QM_USAGE_DECISION
Nonconformance / 8D	Create, approve, run 8D, close	NCR, severity, D1–D8 tasks, due dates	QM_NCR_MAINTAIN
Supplier scorecard	Period filter, drill to NCRs	vendor, score, PPM, acceptance rate	QM_VIEW
SPC dashboard	Chart per item/test, rule violations	subgroups, limits, rule hits	QM_VIEW

Interaction rules: results grid is keyboard-first (Tab/Enter down a sample column); out-of-tolerance values highlight immediately on blur; usage decision disabled until required results exist; CoA preview before release.

6.33.7 Business rules and edge cases

- **BR-33.1 AQL table (ANSI/ASQ Z1.4, general inspection level II, single sampling, normal inspection) — excerpt used by the sizing function.**

Lot size	Code	Sample n	AQL 1.0 Ac/Re	AQL 2.5 Ac/Re	AQL 4.0 Ac/Re
2–8	A	2	use E (13): 0/1	use D (8): 0/1	use C (5): 0/1
9–15	B	3	use E: 0/1	use D: 0/1	use C: 0/1
16–25	C	5	use E: 0/1	use D: 0/1	0/1
26–50	D	8	use E: 0/1	0/1	use E (13): 1/2
51–90	E	13	0/1	use F (20): 1/2	1/2
91–150	F	20	use G (32): 1/2	1/2	2/3
151–280	G	32	1/2	2/3	3/4
281–500	H	50	2/3	3/4	5/6
501–1,200	J	80	3/4	5/6	7/8
1,201–3,200	K	125	5/6	7/8	10/11
3,201–10,000	L	200	7/8	10/11	14/15
10,001–35,000	M	315	10/11	14/15	21/22
35,001–150,000	N	500	14/15	21/22	21/22
150,001–500,000	P	800	21/22	21/22	21/22
500,001+	Q	1,250	21/22	21/22	21/22

Arrow cells ("use X") follow the Z1.4 rule: when the table points to another sample size, use that sample size and its Ac/Re; if it exceeds lot size, inspect 100%. The full Table I (code letters for levels S-1…S-4, I–III) and Tables II-A/B/C (normal/tightened/reduced) are stored as versioned reference data `z14_table` and tested cell by cell.

- **BR-33.2 Switching rules.** Normal → tightened when 2 of 5 consecutive lots are rejected; tightened → normal after 5 consecutive accepted; normal → reduced after 10 consecutive accepted (and switching enabled); reduced → normal on a rejection. Severity is tracked per (item, vendor, test group).
- **BR-33.3 Association specificity.** Score = 4·(item match) + 2·(party match) + 1·(site match); highest score wins; ties resolved by lowest `sequence`; `item group` counts as item match with weight 3. Exactly one order per source line per trigger event; a re-triggered event (e.g., second receipt line) creates a separate order.
- **BR-33.4 Quality order state machine.** `CREATED → OPEN (sample drawn) → RESULTS_IN_PROGRESS → PENDING_DECISION → CLOSED`; `CREATED|OPEN → CANCELLED` (only if source reversed; hold released); `PENDING_DECISION → RESULTS_IN_PROGRESS` on retest request; CLOSED is final; a decision can be *amended* only by a new NCR referencing the order.
- **BR-33.5 Blocking semantics.** The hold is placed synchronously in the same transaction as source line registration (receipt registration / RAF) through the inventory ledger `PlaceHold` API; if the hold cannot be placed the source transaction fails. Hold qty = lot qty (not sample qty). Destructive test consumption posts a scrap journal for the destroyed sample qty at decision time.
- **BR-33.6 Variable test evaluation** passes if `min ≤ value ≤ max` inclusive after rounding to the test's declared decimals (half-up); attribute passes if the outcome is flagged `pass`. Order passes if every test marked `required` passes on every sample within Ac/Re for AQL plans.
- **BR-33.7 Partial decisions.** Accept 90, reject 10: accepted qty released; rejected qty moved to the quarantine location with `quality_status = REJECTED`; both legs are ledger transactions with the quality order as reference.
- **BR-33.8 Instrument calibration.** `due_at = last_calibrated_at + interval`; results are rejected if `due_at < now` with error `INSTRUMENT_OVERDUE`; a 14-day warning is raised.
- **BR-33.9 CAPA timers.** D3 containment due in 2 business days, D5 corrective action in 30 days, effectiveness check at 90 days after D7 close; business days from the site calendar.
- **BR-33.10 CoA** includes only results from the accepting decision; amended results require a new CoA revision with `supersedes`.
- **BR-33.11 Compliance evaluation.** For item i and list L, the concentration c_s of substance s = Σ over BOM components (mass fraction of the component × c_s in the component), applied recursively; i is restricted under L in region r if any s on L has c_s > threshold_s (strictly greater) and L applies to r; evaluation reruns on BOM, composition or list-version change and stores a new result version.
- **BR-33.12 SDS dispatch.** On shipment confirmation for an item with `sds_required`: select the SDS with region = ship-to country (or its region group), language = customer language else region default, status VALID; if the customer has not received that version, attach it and log the delivery; if none is valid, hold (default) or warn per `sds_missing_action`.
- **BR-33.13 Certificate gating.** A usage decision ACCEPT requires `certificate_status ∈ {RECEIVED_OK, WAIVED}` when the association demands a certificate; certificate values outside the specification set `certificate_status = RECEIVED_NOK` and create an NCR with source VENDOR.
- **BR-33.14 Stability pull scheduling.** Pull date = study start + pull-point months (same day of month; clamped to month end); the quality order is created `stability_lead_days` (default 7) before the pull date; a pull recorded more than `stability_pull_window_days` (default 3) late is flagged as a deviation.
- **BR-33.15 FMEA action priority.** Action priority = AP(S, O, D) from the AIAG-VDA reference table; an FMEA cannot be released while any H item lacks an action with owner and due date.
- **BR-33.16 Source-inspection skip.** A receipt skips receiving inspection if and only if (PO line, batch) has an accepted source inspection whose accepted quantity ≥ received quantity; excess quantity gets a normal quality order.
- **BR-33.17 Specification set precedence.** customer > country > pharmacopeia > default; conformance is computed for all sets; the CoA prints the limits of the set used for the customer.

- **BR-33.18 Inspection point frequency.** Time-based points fall due every interval from operation start in the plant time zone; quantity-based points fall due when cumulative good quantity crosses k × interval; a point without a result after a grace of `inspection_point_grace_pct` (default 25 %) of the interval is missed, and when `block_on_missed_point = true` RAF of output since the last recorded point is blocked.

6.33.8 Acceptance criteria

- **AC-33.1** (FR-33.3, BR-33.3) Given the trigger fixture of 60 receipts and 40 RAFs with overlapping associations, when posted, then exactly the expected 73 quality orders exist with the expected test groups.
- **AC-33.2** (FR-33.2, BR-33.1) Given lot sizes {5, 50, 51, 500, 501, 3200, 10001, 600000} at AQL 2.5 normal level II, when sized, then sample sizes are {5, 8, 20, 50, 80, 125, 315, 1250} (lot 51 → code E, whose AQL 2.5 cell points down to F, n = 20) and Ac/Re are {0/1, 0/1, 1/2, 3/4, 5/6, 7/8, 14/15, 21/22}, matching the Z1.4 reference cells.
- **AC-33.3** (FR-33.5) Given 10,000 random attempts to pick, ship, consume or transfer quants with open quality orders, when executed, then every attempt fails with `QUALITY_HOLD` and no inventory transaction is written.
- **AC-33.4** (FR-33.6, BR-33.7) Given a lot of 100 with results, when the decision is accept 90 / reject 10, then 90 are available, 10 sit in quarantine as REJECTED, and an NCR exists for 10.
- **AC-33.5** (FR-33.8, BR-33.9) Given an NCR with a started 8D workflow, when D7 closes and 90 days elapse (Temporal time-skipping test), then an effectiveness-check task is created and the workflow completes on its approval.
- **AC-33.6** (FR-33.4, BR-33.8) Given an instrument overdue since yesterday, when a result references it, then the request is rejected with `INSTRUMENT_OVERDUE`.
- **AC-33.7** (FR-33.9) Given an accepted batch shipped to a customer with a CoA template, when the shipment is confirmed, then a PDF CoA with every required test appears on the shipment and the customer portal.
- **AC-33.8** (BR-33.2) Given 5 consecutive lots of which 2 are rejected, when the 6th order is created, then its severity is TIGHTENED.
- **AC-33.9** (FR-33.10) Given 12 months of decisions and NCRs, when the scorecard is computed, then PPM = rejected units / inspected units × 10^6 and score matches the weighted formula fixture.
- **AC-33.10** (FR-33.13, BR-33.11) Given an item whose BOM yields 0.12 % w/w of an SVHC, when a sales order line ships to an EU customer, then the REACH Art. 33 warning QM-0121 appears and the item is on the period reporting extract; at 0.09 % w/w neither happens.
- **AC-33.11** (FR-33.14, BR-33.12) Given an item with an English SDS only and a German customer with `sds_missing_action = HOLD`, when the shipment is confirmed, then it is held with QM-0124; after a valid German SDS is uploaded the shipment releases with the SDS attached and logged.
- **AC-33.12** (FR-33.15, BR-33.13) Given an association requiring an EN 10204 3.1 certificate, when the receipt posts without one, then ACCEPT is rejected with QM-0125; after an in-specification certificate is recorded, ACCEPT succeeds.
- **AC-33.13** (FR-33.16, BR-33.14) Given a 36-month study starting 2026-01-31 with pull points 0/3/6/…/36, when the schedule is generated, then pull dates are 2026-01-31, 2026-04-30, 2026-07-31 and so on, and quality orders appear 7 days before each (Temporal time-skipping test).
- **AC-33.14** (FR-33.19, BR-33.16) Given an accepted source inspection for batch B of 1,000 units, when 1,000 units of B are received, then no receiving quality order is created; when 1,000 units of batch C arrive, a normal quality order is created.
- **AC-33.15** (FR-33.21, BR-33.17) Given a lot that meets default limits but fails customer X's tighter limit, when ATP allocates, then the lot is not allocated to customer X and remains allocatable to other customers.
- **AC-33.16** (FR-33.22, BR-33.18) Given a 2-hour inspection point and `block_on_missed_point = true`, when no result exists 2.5 h after operation start, then RAF of output since the last point is blocked and the supervisor receives QM-0130.
- **AC-33.17** (FR-33.18, BR-33.15) Given an FMEA line with S = 10, O = 8, D = 7 and no action, when release is attempted, then action priority is H and release fails with QM-0127.
- **AC-33.18** (FR-33.20) Given a retention sample whose retention expiry is today, when the daily job runs, then the sample is DUE_FOR_DISPOSAL and a disposal task exists.
- **AC-33.19** (FR-33.17) Given an audit with 2 major findings and one CAPA, when closure is attempted, then it is rejected with QM-0131.

6.33.9 Test plan and fixtures

Fixtures: `z14-tables.json` (full Z1.4 Tables I, II-A/B/C transcribed and double-checked), `quality-triggers-100.json`, `results-variable-attribute.json`, `instruments-12.json`, `ncr-8d-sample`, `vendor-scorecard-12m`.

Test	Type	Target	Pass criterion
AQL sizing	unit	every cell of II-A/B/C	equals transcribed table
Association precedence	unit	100 triggers	expected orders
Blocking property	property	inventory ledger + wms	0 successful movements on held quants
8D workflow	Temporal replay/time-skip	CAPA	completes with effectiveness check
Usage decision quantities	unit	splits	sum invariant
CoA rendering	snapshot	PDF	matches golden
Results throughput	performance	1,000 results batch	< 2 s

Reconciliation: Σ quality-held qty in ledger = Σ lot_qty of open orders; rejected qty in quarantine = Σ reject_qty of decisions – dispositions.

6.33.10 Deployment and operations

Helm `passage-quality` (api, worker, 2–6 replicas), CloudNativePG `quality-db` (3 instances, 50 GB). Config: `QM_DEFAULT_INSPECTION_LEVEL=II`, `QM_EFFECTIVENESS_DAYS=90`, `QM_INSTRUMENT_WARN_DAYS=14`. Flags: `quality.enabled` per site, `quality.spc`. Scheduled: scorecard materialization monthly (1st, 03:00), instrument due scan daily, SPC recompute hourly. SLOs: order creation added to receipt latency < 100 ms p95; results batch p95 < 2 s; decision posting p95 < 500 ms. Alerts: orders PENDING_DECISION > 5 days; held qty vs open orders drift; instrument overdue with results attempted. Runbook: release orphan holds (order cancelled but hold remains); reprocess association for a trigger event. Rollback: `quality.enabled=false` stops new orders; existing holds remain until decided (never auto-release on rollback).

6.33.11 Migration from AX / SAP

Source	Passage entity	Transformation
InventTestTable, InventTestVariable, InventTestVariableOutcome	Test, outcome catalog	type Integer/Fraction → variable; Option → attribute
InventTestGroup + lines, InventTestItemSampling	TestGroup, SamplingPlan	fixed/percent plans direct; AQL plans created new
InventTestAssociationTable	QualityAssociation	event type mapping (Registration/Product receipt/RAF/Pick)
InventQualityOrderTable/Line/LineResults (open)	QualityOrder, QualityResult	open orders only; holds recreated via ledger API

Source	Passage entity	Transformation
InventNonConformanceTable, InventNonConformanceCorrection (open)	Nonconformance, Correction	statuses mapped; closed NCRs loaded as read-only history (last 24 months)
QPMK, PLKO/PLPO/PLMK	Test, TestGroup	one group per plan usage; characteristics by `MERKNR`
QDPS/QDS1, QMAT	SamplingPlan, QualityAssociation	SAP sampling procedure type → method; `QMAT` lot origin → trigger
QALS (open, no UD), QAMR/QASR	QualityOrder, QualityResult	lot origin 01→PURCHASE_RECEIPT, 04→PRODUCTION_RAF, 10→SALES_PICK, 89→MANUAL
QMEL/QMFE/QMMA/QMSM (open)	Nonconformance, Correction tasks	notification type Q1/Q2/Q3 → source
ELBK/ELBM/ELBP	SupplierScorecard history	last 12 months of scores only

Not migrated: closed inspection lots with UD (history to lakehouse), SAP control chart history (recomputed), certificate print history.

6.33.12 Dependencies, risks, open questions

Depends on: wave 32 (mobile results step), wave 31 (quarantine locations, work), wave 28 (shipment hooks for sales trigger and CoA), wave 21 (RAF hook), wave 18 (receipt hooks, return orders), wave 16 (ledger hold API — extended in this wave), wave 4 (Temporal).

Risk	Likelihood	Impact	Mitigation
Z1.4 transcription error	Low	High	Two independent transcriptions diffed; spot-check against the ASQ tables
Hold placement adds latency to receipts	Medium	Low	Synchronous but local DB call via ledger API; measured in CI

Open questions: (1) Should sales-pick quality orders block the *order line* or the *quant*? Proposed: quant; decide by wave 33 design review. (2) Cost of quality category as a journal dimension or a tag? Proposed: dimension `coq_category`; decide by wave 37 (cost accounting).

6.33.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
quality.enabled	site	bool	false	true / false	Activates quality orders for the site	Inventory and warehouse management parameters › Quality management › "Use quality management"	QM activation per plant; material master QM view (QCC0)
default_inspection_level	tenant	enum	II	S-1…S-4, I–III	Default Z1.4 inspection level (FR-33.2)	Item sampling (no AQL levels)	Sampling scheme (QDP1)
switching_enabled	sampling plan	bool	true	true / false	Normal/tightened/reduced switching (BR-33.2)	n/a	Dynamic modification rule (QDB1)
block_on_order_create	site	bool	true	true / false	Hold placed at order creation (BR-33.5)	Quality order blocking parameters / association "Event blocking"	QM control key "post to inspection stock" (TQ08)
auto_ncr_on_reject	site	bool	true	true / false	NCR created on reject decision (FR-33.6)	Nonconformance defaults	Usage decision follow-up action "create notification"
effectiveness_check_days	site	int	90	30–365	CAPA effectiveness timer (BR-33.9)	n/a	Notification task follow-up (effectiveness check)
capa_calendar	site	calendar ref	site calendar	wave-6 calendars	Business days for CAPA due dates	Quality management parameters "Hours per day"	Notification priority response times (T356)
instrument_warn_days	site	int	14	0–90	Calibration warning horizon (BR-33.8)	Test instruments	Maintenance plan call horizon (IP10)
results_decimals	test	int	per test	0–6	Rounding before evaluation (BR-33.6)	Test "Precision"	MIC decimal places (QS21)
coa_auto_on_shipment	customer	bool	false	true / false	CoA generated at ship confirmation (FR-33.9)	Certificate of analysis "Print with packing slip"	Quality info record sales (QV51) + output determination application QC
scorecard_weights	tenant	jsonb	quality 40 / delivery 30 / NCR 20 / audit 10	Σ = 100	Scorecard formula (FR-33.10)	Vendor evaluation criteria (VendEvaluationCriterion)	Weighting keys and criteria (T147*)
compliance_block_mode	compliance list	enum	BLOCK	BLOCK / WARN	Line behavior on restriction (FR-33.13)	Restricted list "Restricted" vs "Warning"	Product marketability check rule result
sds_validity_days	item × region	int	1,095	30–3,650	SDS expiry (FR-33.14)	Product compliance › SDS parameters › "Product safety data sheet validity" / MSDS validity days	Product compliance SDS validity
sds_missing_action	legal entity	enum	HOLD	HOLD / WARN / NONE	Shipment behavior without valid SDS (BR-33.12)	SDS "Send on sales" check	EHS report shipping check in delivery (CVD1)
certificate_required_default	quality association	enum	NONE	NONE / COA / COC / EN10204_31	Certificate demand (FR-33.15)	n/a	QI01 certificate type; QM control key "certificate required"
source_inspection_lead_days	vendor × item	int	5	0–60	Source inspection trigger (FR-33.19)	n/a	QI01 source inspection indicator with QI07 lead time
retention_sample_years_after_expiry	site	int	1	0–10	Retention sample expiry (FR-33.20)	n/a	Sample type retention period (sample management customizing)
stability_lead_days	site	int	7	0–30	Order creation ahead of pull (BR-33.14)	n/a	Stability study scheduling parameters (QST)

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
stability_pull_window_days	study	int	3	0–14	Late pull deviation threshold (BR-33.14)	n/a	Stability study customizing
inspection_point_grace_pct	operation	int	25	0–100	Missed-point grace (BR-33.18)	n/a	Inspection point identification (TQ79)
block_on_missed_point	operation	bool	false	true / false	RAF block on missed point (BR-33.18)	n/a	Inspection point completion requirement
fmea_ap_table_version	tenant	reference data	AIAG-VDA-2019	versioned	Action priority lookup (BR-33.15)	n/a	FMEA customizing: risk evaluation catalog
four_eyes_decision	site	bool	false	true / false	Decision user ≠ results user	n/a	Digital signature / UD authorization (QM)

6.33.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Quality order worklist	list inquiry	status, site, trigger, age, item	order, lot qty, sample, due, status	`quality_order`	Quality orders list page	QA32 / QA33
Results worklist	list inquiry	characteristic, work center, status	open result slots, inspection point, due	`quality_result`	Quality order line results	QE51N
Results history by item and test	drill-down	item, test, vendor, period	value, limits, pass/fail, instrument	`quality_result`	Test result statistics	QGA3, QE03
NCR and CAPA status	list inquiry	source, severity, status, overdue	NCR, current D-step, due date, owner	`nonconformance` + workflow	Nonconformances list, corrections inquiry	QM10–QM12
Supplier quality scorecard	KPI	vendor, period	lots, acceptance rate, PPM, NCRs, audit score, total	scorecard	Vendor evaluation, SSRS vendor quality performance	ME6H, QGA2
SPC control chart	KPI	item, test, plant	X̄/R or p/np, limits, rule violations	SPC projection	n/a	QGC1–QGC3
Cost of quality	KPI	period, category, site	prevention, appraisal, internal and external failure	journal by `coq_category`	SSRS cost of quality	QM order cost reports (KOB1)
Quality hold reconciliation	operational report	site	held qty in ledger vs open orders	reconciliation job	n/a	QA32 with MMBE inspection stock
Certificate receipt overview	list inquiry	vendor, item, status	required, received, waived, NOK	`incoming_certificate`	n/a	QC55
Regulated and restricted product hits	list inquiry	list, region, item	substance, % w/w, threshold, action	`compliance_result`	Product compliance inquiries (regulated items)	Product marketability workbench
SDS due and distribution log	operational report	item, region, expiry window	version, valid_to, customers sent, pending	`safety_data_sheet`, `sds_delivery`	SDS due inquiry, SSRS Safety data sheet	CVD1 report shipping orders
Stability study plan	drill-down	study, condition	pull points, status, result trend	`stability_study`	n/a	QST03
Audit findings	list inquiry	audit, severity, status	finding, clause, CAPA link	`audit`	n/a	PLMD_AUDIT
Instrument calibration due	list inquiry	due within n days, site	instrument, due_at, status	`instrument`	Test instruments	IP19 / IP24 scheduling overview

6.33.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Certificate of analysis	Ship confirmation or accepting decision	Header: customer, PO, item, batch, manufacturing and expiry dates; lines: test, method, limits of the applicable specification set (FR-33.21), result, unit; footer: signer, revision, supersedes	PDF/A-2; email; customer portal; EDI attachment	COA-*	Retained ≥ 1 year past batch expiry (21 CFR 211.180) or 10 years, configurable
Quality order and sampling instruction	Order created	Lot, sample qty, tests, sample-drawing procedure	PDF	QO-*	Operational, per quality record retention
Sample label	Sample registered	Sample id barcode, order, storage condition, retention expiry	ZPL	SMP-*	None beyond sample record
Nonconformance report / NC tag	NCR created	NCR id, source, severity, qty, quarantine location	PDF; ZPL tag	NCR-*	Quality record, 10 years default
8D report	CAPA closed	D1–D8 contents, effectiveness check result	PDF	NCR id	Customer deliverable; IATF 16949 retention per customer
Safety data sheet	First shipment and each new version (BR-33.12)	GHS 16 sections	PDF attached to shipment email or portal	SDS version	REACH Art. 36: information kept 10 years
Stability study report	Pull point and study completion	Conditions, pull points, results, trend, conclusion	PDF	STB-*	Product registration file retention
Audit report	Audit REPORTED	Scope, criteria, findings, score	PDF	AUD-*	ISO 9001 §7.5 documented information
FMEA and control plan	Release	AIAG-VDA form, control plan columns	PDF / XLSX	Version	PPAP submission
Supplier scorecard	Monthly	Score components, trend, NCRs	PDF via supplier portal	none	None

6.33.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Scorecard materialization	Monthly, 1st 03:00	vendor	period	< 10 min	Rerun period	Scorecard missing on 2nd
Instrument due scan	Daily 05:00	site	date	< 1 min	Rerun	Overdue instrument with results attempted
SPC recompute	Hourly	item × test	last result id	< 5 min	Resume from last id	Lag > 3 h
Compliance re-evaluation	On BOM/list change; full run nightly 01:00	item hash range	(item, list version)	50,000 items < 30 min	Resume by hash range	Failed items > 0

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
SDS expiry and redistribution	Daily 04:00	item × region	SDS version	< 5 min	Rerun	SDS expiring in < 60 days without review task
Stability pull scheduling	Temporal timer per study	study	workflow id	n/a	Workflow retry	Missed pull deviation
Retention sample disposal	Daily 06:00	site	sample id	< 1 min	Rerun	none (tasks)
Source inspection trigger	Daily 06:00	vendor	PO line + due date	< 2 min	Rerun	Orders not created for due lines
Quality hold reconciliation	Daily 02:30	site	snapshot	< 5 min	Report only	Held qty ≠ open lot qty

6.33.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
QM_SETUP_MAINTAIN	/quality/v1/test-groups, /associations, specification sets, sampling plans	QM_USAGE_DECISION (change limits, then accept against them)	Quality administrator
QM_ORDER_MAINTAIN	/quality-orders (manual); workbench	none	Quality inspector
QM_RESULTS_RECORD	/quality-orders/{id}/results, /certificates, /samples	QM_USAGE_DECISION on the same order when `four_eyes_decision = true`	Lab technician, inspector
QM_USAGE_DECISION	/quality-orders/{id}:usage-decision; certificate waiver	QM_SETUP_MAINTAIN	QA release officer
QM_NCR_MAINTAIN	/nonconformances	none	Quality engineer
QM_CAPA_MAINTAIN	/nonconformances/{id}/corrections	Effectiveness check must be approved by another user	Quality engineer
QM_COA	/coa:generate	none	QA clerk
QM_COMPLIANCE_MAINTAIN	/compliance-lists, /items/{id}/sds	SALES_ORDER_MAINTAIN (wave 26): clearing a block on own order	Product steward / EHS
QM_STABILITY_MAINTAIN	/stability-studies	none	Stability coordinator
QM_AUDIT_MAINTAIN	/audits	Auditor may not audit own org unit (checked)	Internal auditor
QM_FMEA_MAINTAIN	/fmeas, control plans	none	Quality planner
QM_VIEW	/sampling-plans:size, /scorecards, SPC, all inquiries	none	All quality roles, purchasing

6.33.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
QM-0101	error	{item} is on quality hold (order {qo}); {action} not allowed.	QUALITY_HOLD on quant	Wait for usage decision	FR-33.5
QM-0102	error	Instrument {inst} is overdue for calibration since {date}.	INSTRUMENT_OVERDUE	Calibrate or use another instrument	BR-33.8
QM-0103	warning	Instrument {inst} calibration is due in {n} days.	Warning horizon reached	Schedule calibration	BR-33.8
QM-0104	error	Decision quantities {sum} do not equal lot quantity {lot}.	Split mismatch	Correct the split	FR-33.6
QM-0105	error	Required test {test} has no result.	Decision before results	Record result or deviate with reason	§6.33.5 commands
QM-0106	error	Inventory hold could not be placed; source transaction rolled back.	Ledger PlaceHold failed	Retry; check ledger health	BR-33.5
QM-0107	info	Inspection severity changed to {sev} for {item}/{vendor}.	Switching rule fired	None	BR-33.2
QM-0108	warning	Value {v} is outside {min}–{max} for {test}.	Out of tolerance	Retest or record failure	BR-33.6
QM-0109	error	Order {qo} is closed; amend through a nonconformance.	Change on CLOSED order	Create NCR	BR-33.4
QM-0110	warning	CAPA step {d} is {n} business days overdue; escalated.	Timer expired	Complete step	BR-33.9
QM-0121	warning	{item} contains {substance} at {c} % (threshold {t} %); {list} duty applies in {region}.	Reporting threshold exceeded	Issue required communication	BR-33.11
QM-0122	error	{item} is restricted in {region} under {list}; line blocked.	Restricted substance above limit	Change source or destination	BR-33.11
QM-0123	info	SDS {version} attached to shipment {ship}.	New version for customer	None	BR-33.12
QM-0124	error	No valid SDS for {item} in {language}/{region}; shipment held.	Missing SDS	Upload SDS	BR-33.12
QM-0125	error	Certificate {type} is required before acceptance.	Missing or NOK certificate	Obtain certificate or waive with reason	BR-33.13
QM-0126	warning	Stability pull {n} for study {s} is {d} days late.	Late pull	Record deviation	BR-33.14
QM-0127	error	FMEA {f} has high-priority items without actions.	H items without action	Add actions	BR-33.15
QM-0128	info	Batch {b} was source-inspected; receiving inspection skipped.	Accepted source inspection	None	BR-33.16
QM-0129	warning	Lot does not conform to specification set {set}.	Fails set limits	Allocate elsewhere or rework	BR-33.17
QM-0130	error	Inspection point {p} missed on operation {op}; RAF blocked.	No result within grace	Record result; supervisor review	BR-33.18
QM-0131	error	Audit cannot close: {n} major findings without CAPA.	Missing CAPA	Create CAPA	FR-33.17

6.33.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Quality order creation in receipt path	2,000 receipts/day; peak 200/min	+20 / 60 / 100 ms on receipt	40/s	Orders ≈ 1 GB	`quality_order(status, site_id)`	Hold stays synchronous; order body completes via outbox
Results batch	50,000 results/day; batches of 1,000	400 ms / 2 s / 3 s per 1,000	2,000 results/s	Results ≈ 8 GB	`quality_result` monthly partitions	Batch accepted asynchronously with status polling
Usage decision posting	1,500/day	120 / 500 / 900 ms	20/s	—	—	Queued per order; no partial postings
Compliance evaluation	50,000 items × 40 lists	20 / 80 / 200 ms per item; full run < 30 min	50 items/s	Results ≈ 2 GB	`compliance_result(item_id, list_version)`	Incremental runs continue; full run deferred to night
SDS check at shipment	5,000 shipments/day	5 / 20 / 40 ms (cached)	100/s	SDS PDFs ≈ 5 GB in object storage	`safety_data_sheet(item_id, region, language, status)`	Cache serves; holds still applied
SPC recompute	2,000 item × test series	< 5 min per hour	—	Chart points ≈ 1 GB	Series key	Recompute interval stretches to 2 h

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Inspection point due evaluation	500 active operations	1-min resolution	—	Points ≈ 0.5 GB	`inspection_point(operation_id, due_at)`	Evaluation lag ≤ 3 min; RAF block decision waits for evaluation

6.33.20 Localization and compliance hooks

- **Regulated quality.** US 21 CFR Part 11 (electronic records and signatures: usage decisions, CoA signatures and CAPA approvals re-authenticate with the meaning of signature and keep an audit trail), 21 CFR 211.166 (stability), 211.170 (reserve samples), 211.180 (record retention); EU GMP Annex 11 and Annex 16 (Qualified Person certification); medical devices 21 CFR 820 and ISO 13485; automotive IATF 16949 with AIAG-VDA FMEA and PPAP. The `RegulatedQualityPack` extension point (wave 48) sets signature policies, four-eyes defaults and retention.
- **Stability conditions.** ICH climatic zones I, II, III, IVa (30 °C/65 % RH) and IVb (30 °C/75 % RH) are reference data selected by country pack for the target market.
- **Product compliance lists.** EU REACH and CLP, RoHS; UK REACH; US TSCA and California Proposition 65; China RoHS. Lists arrive through `ComplianceListProvider` (country pack or subscription feed). SDS language follows the destination member state (REACH Art. 31) and the GHS revision adopted by each jurisdiction.
- **Certificates.** EN 10204 types 2.1, 2.2, 3.1 and 3.2 are reference data; country packs may add national forms.
- **Retention.** Batch records and CoAs: US 1 year past expiry; EU GMP 1 year past expiry or 5 years after QP certification, whichever is longer; configured per pack through the wave-46 archive.
- **Units and formats.** Results are stored with declared decimals; decimal separator and unit display follow user locale.
- **Privacy.** Audit participants, CAPA owners and inspectors are personal data under GDPR; quality records keep names under the legal-obligation basis for the retention period.

6.33.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX GAP #48 — §12.15.1 Released product → Product compliance (regulated products, restricted lists, reporting details, SDS with validity and send on sales)	GAP (propose 33)	ADDED as FR-33.13 and FR-33.14	BR-33.11, BR-33.12
AX GAP #49 — §12.15.2 Product compliance inquiries (regulated items, SDS due)	GAP (propose 33)	ADDED as FR-33.13 and FR-33.14	Inquiries in §6.33.14
AX GAP #50 — §12.15.2 SSRS: Safety data sheet	GAP (propose 33)	ADDED as FR-33.14	Document in §6.33.15
AX GAP #51 — §12.15.3 Product compliance → Expire SDS / Reporting	GAP (propose 33)	ADDED as FR-33.14 and FR-33.13	Jobs in §6.33.16
AX GAP #52 — §12.15.4 Product compliance setup (regulated regions, restricted lists, reporting lists, SDS parameters, MSDS validity days)	GAP (propose 33)	ADDED as FR-33.13 and FR-33.14	Parameters in §6.33.13
AX GAP #66 — §12.23.1 Process manufacturing: Product compliance / SDS (link)	GAP (propose 33)	ADDED as FR-33.14	Same mechanism as #48
AX GAP #67 — §12.23.2 SSRS: Safety data sheet (link)	GAP (propose 33)	ADDED as FR-33.14	Same document as #50
AX GAP #68 — §12.23.3 Compliance → SDS expiry / regulated product reporting	GAP (propose 33)	ADDED as FR-33.14 and FR-33.13	Same jobs as #51
AX GAP #70 — §12.23.4 Product compliance → regulated/restricted lists, reporting lists, SDS setup	GAP (propose 33)	ADDED as FR-33.13 and FR-33.14	Same parameters as #52
AX GAP #72 — §12.25.4 Quality management → Product compliance (link)	GAP (propose 33)	ADDED as FR-33.13	Owned by `quality` service
AX GAP #80 — §12.29.1 Organization administration: Product safety data sheets (link)	GAP (propose 33)	ADDED as FR-33.14	Same mechanism as #48
AX GAP #87 — §12.29.4 Product safety data sheets setup (link)	GAP (propose 33)	ADDED as FR-33.14	Same parameters as #52
SAP #282 — §13.24 QC51 / QC52 / QC53 / QC55 certificate receipt	PARTIAL-33	ADDED as FR-33.15	BR-33.13, AC-33.12
SAP #283 — §13.24 QST01–QST07 stability study	GAP (propose 33)	ADDED as FR-33.16	BR-33.14, AC-33.13
SAP #284 — §13.24 PLMD_AUDIT audit management	PARTIAL-33	ADDED as FR-33.17	Was out of scope in §6.33.1; scope amended
SAP #285 — §13.24 FMEA (QM-FMEA, S/4), control plan	GAP (propose 33)	ADDED as FR-33.18	BR-33.15, AC-33.17
SAP #286 — §13.24 Source inspection (QI01 flag, QI07)	PARTIAL-33	ADDED as FR-33.19	BR-33.16, AC-33.14
SAP #287 — §13.24 Sample management (QPR1–QPR7)	PARTIAL-33	ADDED as FR-33.20	AC-33.18
SAP #288 — §13.24 Multiple specifications	PARTIAL-33	ADDED as FR-33.21	BR-33.17, AC-33.15
SAP #289 — §13.24 QC55 certificate receipt overview	PARTIAL-33	ADDED as FR-33.15	"Certificate receipt overview" in §6.33.14
SAP #290 — §13.24 QST stability schedule generation	GAP (propose 33)	ADDED as FR-33.16	"Stability pull scheduling" job
SAP #291 — §13.24 TQ79 inspection points (time/quantity/free), partial lots	PARTIAL-33	ADDED as FR-33.22	BR-33.18, AC-33.16
SAP #292 — §13.24 Stability study customizing	GAP (propose 33)	ADDED as FR-33.16	`stability_*` parameters
SAP #293 — §13.24 FMEA / control plan customizing (S/4)	GAP (propose 33)	ADDED as FR-33.18	`fmea_ap_table_version`
SAP #294 — §13.24 Audit management customizing (audit types, question lists)	PARTIAL-33	ADDED as FR-33.17	Question lists are versioned metadata
SAP #295 — §13.24 Multiple specifications customizing	PARTIAL-33	ADDED as FR-33.21	Specification set types reference data
SAP #296 — §13.24 Sample management customizing (sample types, sample-drawing procedures)	PARTIAL-33	ADDED as FR-33.20	`retention_sample_years_after_expiry`

6.34 Wave 34: Transportation Management & Freight

6.34.1 Scope and business context

Wave 28 integrated parcel carriers for labels; wave 34 adds full transportation management replacing D365 TMS and SAP TM: shipments consolidated into loads and routes, rated against carrier rate cards with accessorials and fuel surcharges, rate-shopped, tendered to carriers with timeouts, tracked by EDI 214 or carrier APIs, and reconciled against carrier freight invoices with variance disputes and accruals. Freight and duties are allocated to inventory as landed cost feeding wave-20 costing. VRP route optimization runs on OR-Tools in the Python `tms-optimizer`. Out of scope: own-fleet driver apps and ELD, ocean booking, customs filing (hooks only), multi-leg international routing guides beyond two legs.

Scope amendment (coverage closure). The gap registers add the following to this wave: landed-cost voyages with shipping containers, folios and goods-in-transit orders (FR-34.12, extending FR-34.9); yard management with transportation units (FR-34.13); driver check-in/out with a gate log (FR-34.14); dangerous-goods checks for transport (FR-34.15); own-fleet resources as planning constraints (FR-34.16); lane RFQs for strategic freight procurement (FR-34.17); layer-based load stacking (FR-34.18); and foreign-trade document and screening hooks (FR-34.19). Driver mobile apps and ELD/tachograph integration stay out of scope. Customs filing, preference determination and the denied-party screening provider are wave 48 country and compliance pack content behind the hooks in FR-34.19. §6.34.21 lists the dispositions.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
D365 rate route workbench (TMSRateRouteWorkbench) shows a total per carrier with no breakdown; a wrong rate (missing rate base assignment, fuel index not loaded) is found when the carrier invoice arrives weeks later	Rate engines (TMSRateEngineBase subclasses) return totals; no explain	Rating returns an explain trace per charge step; `NO_VALID_RATE_CARD` is explicit	Rate errors detected: at invoice (2–6 weeks) → at rating (< 300 ms); rating disputes caused by setup: ~5% of invoices → < 0.5%
SAP TM carrier selection and tendering (freight order subcontracting) require TM configuration plus PI/PO for EDI 204/990; tender timeouts are monitored manually in the transportation cockpit	Tendering is a status field, not a durable process	Temporal `TenderWorkflow` with cascade, timeouts and carrier portal/EDI responses	Planner interventions per 100 tenders: ~30 (watching expiry) → < 3; tender-to-acceptance: hours of chasing → automatic cascade within the timeout
D365 freight reconciliation (TMSFreightBill ↔ TMSCarrierInvoice) is matched by hand per detail line; D365 lacks EDI 210 intake; accrual reversal needs a separate periodic job and often double-counts	Matching is a form, not a rule set; accrual lifecycle disconnected from the match	Rule-based line matching with tolerances, state machine to AP, accrual/reversal/invoice share one journal reference	Reconciliation effort: ~10 min per invoice → < 30 s for in-tolerance invoices (≥ 80% auto); account 2150 ties to zero per bill, 100%
Landed cost allocation (D365 Landed cost module voyages; SAP delivery-cost conditions FRA1/FRB1 on PO lines) rounds per line and leaves cent residuals in a variance account; multi-PO containers need manual apportionment	Per-line rounding without residual assignment; cost types not linked to receipts across POs	Allocation with residual to the highest-value line; voyage spans POs; estimated/actual variance rules by on-hand	Allocation residual: cents per voyage in variance → 0; time to land a 3-PO container: 1–2 h → < 5 min
Route planning in D365 is manual load building; SAP TM VSR optimizer requires PP/DS-style setup and licensing	No general VRP in D365; SAP optimizer is a separate engine with heavy configuration	OR-Tools VRP with explicit constraints, 60 s limit, planner override	Multi-stop planning time: 30–60 min per day of loads → < 1 min; route distance vs manual: −10% or better

6.34.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Carriers / services	TMSCarrier, TMSCarrierService, TMSCarrierServiceGroup	Business partner carrier role, /SCMTMS/ means of transport	tms.Carrier, CarrierService	full
Rate cards	TMSRateMaster, TMSRateBase, TMSRateBaseAssignment, TMSRateEngine classes (TMSRateEngineBase)	Freight agreements, rate tables, calculation sheets (/SCMTMS/TCM, /SCMTMS/TCCS)	RateCard, RateTable, CalcSheet	full
Accessorials / fuel	TMSAccessorialMaster, TMSAccessorialAssignment, TMSFuelIndex, TMSFuelIndexRegion	Charge types, fuel surcharge index tables	Accessorial, FuelIndex	full
Rate shopping / routing	Rate route workbench (TMSRateRouteWorkbench), TMSRoutePlan, TMSRouteGuide, TMSRouteSegment	Carrier selection (/SCMTMS/CAR_SEL), transportation cockpit	RateShop, RouteGuide	full
Loads / freight orders	WHSLoadTable (TMS fields), load building strategies	Freight units, freight orders /SCMTMS/TOR (TOR_ROOT/TOR_ITEM/TOR_STOP), freight bookings	Shipment, Load, Route	full
Tendering	Load tendering (manual/EDI 204, TMSLoadTender status)	Tendering (freight order subcontracting, EDI 204/990 via SAP TM)	Tender workflow (Temporal)	full
Tracking	n/a (ISV, driver check-in)	Event management (EM) / TM execution tracking	TrackingEvent (EDI 214/API)	improved
Freight reconciliation	TMSFreightBill, TMSFreightBillDetail, TMSCarrierInvoice, TMSCarrierInvoiceDetail, freight bill matching	Freight settlement document (/SCMTMS/SFIR), ERS/MIRO	FreightBill, CarrierInvoice, Match	full
Landed cost	Landed cost module (voyage, cost type, allocation)	Delivery costs on PO (FRA1/FRB1 conditions), actual costing	LandedCostVoyage, allocation to inventory	full
Route optimization	n/a (ISV)	PP/DS-based VSR optimizer	VRP (OR-Tools)	improved
Dock scheduling	WHSDockAppointment (wave 31)	Yard	wms.DockAppointment	full (wave 31)
Landed cost voyages, containers, folios, goods in transit	Landed cost module: voyages (ITMVoyage*), shipping containers, folios, cost types and auto costs, goods-in-transit orders, voyage tracking	Delivery cost conditions FRA1/FRB1, MRDC settlement, inbound delivery with transportation units, stock in transit (S/4 SIT)	LandedCostVoyage + VoyageLeg, ShippingContainer, Folio, GoodsInTransitOrder	full
Yard management	D365 yard management (yard locations, trailers, yard moves)	EWM yard management /SCWM/YM_* (yard bins, checkpoints, TU/vehicle), /SCWM/TU, /SCWM/VEH	tms.YardLocation, TransportUnit, YardMove; wms work type YARD_MOVE	full
Driver check-in / gate	Driver check-in and check-out (appointment scheduling)	EWM check-in/check-out at checkpoints (/SCWM/CICO)	DriverVisit, GateEvent	full
Dangerous goods	Hazardous material flags (limited); product compliance	EHS-DG: DG master, DG check methods and profiles in SD/LE/TM, DG documents	HazmatProfile (wave 31) + DgCheckResult at rating, load building, tender, ship	full
Fleet resources	n/a (ISV)	TM resources: vehicles, trailers, drivers, resource calendars	tms.Vehicle, Trailer, Driver as VRP constraints	partial — planning only; no driver app or ELD
Strategic freight procurement	n/a	TM strategic freight procurement (freight RFQ, bid analysis, award to freight agreements)	FreightRfq on the wave-17 RFQ engine → RateCard	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Load plan stacking	Load building templates (weight/volume only)	TM load planning (package building, 3D load plan)	LoadPlan in tms-optimizer (layer heuristic)	partial — layer stacking, no free-form 3D carton packing
Foreign trade documents and screening	Foreign trade parameters (Intrastat only)	SAP GTS: customs declarations, export control, sanctioned-party list screening, preference	Customs data on shipments, commercial invoice, `ComplianceScreeningProvider` and `CustomsFilingProvider` hooks	partial — hooks; filing and preference in wave 48

6.34.3 Functional requirements

- **FR-34.1 Shipments and loads** — Must. The system shall create TMS shipments from sales/transfer/purchase (inbound freight) shipments and consolidate them into loads by rule (same ship-from, ship-to region, mode, date window) or manually.
  - Notes: load capacity by weight, volume, pallet positions; load types TL, LTL, PARCEL, INTERMODAL; inbound loads for vendor-collect freight.
- **FR-34.2 Route planning with VRP** — Must. The system shall plan multi-stop routes with OR-Tools under the constraints in BR-34.1 and return feasible routes within 60 s for ≤ 500 stops.
  - Notes: distance/time matrix from OSRM (self-hosted) or stored lanes; first solution by parallel cheapest insertion; guided local search until the time limit.
- **FR-34.3 Rating engine** — Must. The system shall compute charges per BR-34.2 from rate cards (lane-based, zone-based, distance-based, weight-break, per pallet, per container), accessorials and fuel surcharges, with an explain trace.
  - Notes: rate cards effective-dated with currency; minimum charge; rounding to cents half-up; deterministic given the same inputs.
- **FR-34.4 Rate shopping and route guides** — Must. The system shall rate a shipment across eligible carriers/services and rank by cost, transit time or a weighted score; route guides pre-select carriers by lane with fallbacks.
  - Notes: eligibility by service area, mode, hazmat flag, max weight; results cached 10 min per input hash.
- **FR-34.5 Tendering** — Must. The system shall tender loads to carriers (e-mail/portal, EDI 204 via wave 35, API) with accept/decline/timeout, cascading to the next carrier by route guide rank.
  - Notes: Temporal `TenderWorkflow` per load; default timeout 2 h; broadcast tendering option (first accept wins).
- **FR-34.6 Tracking and POD** — Must. The system shall ingest status events (EDI 214 codes AF/X1/X3/D1 etc., carrier APIs, driver check-in) and proof of delivery (signature/photo), and expose expected vs actual per stop.
  - Notes: late-delivery alerts when ETA > window end; POD attaches to the sales shipment for wave-29 invoicing holds if required.
- **FR-34.7 Freight bills and reconciliation** — Must. The system shall create a freight bill (expected charges) at ship confirmation, match carrier invoices (EDI 210 or manual/API) line by line with tolerances, and route variances to dispute.
  - Notes: tolerance default $5 / 2% per charge; match states per BR-34.4; matched invoices create a vendor invoice in wave 19 AP.
- **FR-34.8 Freight accrual** — Must. The system shall post freight accruals (Dr freight expense or inventory-in-transit, Cr accrued freight 2150) at ship confirmation and reverse on invoice match through the universal journal.
  - Notes: journal postings via the wave-11 posting API with idempotency key = freight bill id; customer-billed freight creates a wave-29 invoice charge line.
- **FR-34.9 Landed cost** — Must. The system shall allocate freight, duty, brokerage and insurance to purchase receipt lines by weight, volume, value, quantity or equal split, and post the allocation to inventory cost (wave 20) as a cost adjustment.
  - Notes: voyage groups multiple POs/containers; estimated cost at receipt, actual on invoice with variance posting; allocation residual cents go to the highest-value line.
- **FR-34.10 Carrier portal** — Should. The system shall let carriers accept tenders, update status and upload POD via the wave-20 supplier portal framework.
  - Notes: carrier users are Keycloak external identities scoped to their loads.
- **FR-34.11 Transportation KPIs** — Should. The system shall report cost per shipment/cwt/mile, on-time pickup/delivery, tender acceptance rate, invoice variance rate.
  - Notes: Cube metrics; daily materialization.
- **FR-34.12 Landed cost voyages, containers, folios and goods in transit** — Must. The system shall track inbound international shipments as voyages with legs (origin port → destination port → warehouse), shipping containers (ISO 6346 number with check digit per BR-34.12, type 20GP/40GP/40HC/45HC/reefer, seal number), folios grouping containers per vendor or forwarder, and goods-in-transit orders that move ownership to an in-transit warehouse at the Incoterm transfer event (BR-34.11) and receive at the destination warehouse.
  - Notes: voyage status ESTIMATED → IN_TRANSIT → ARRIVED → RECEIVED → COSTED; estimated costs per cost type come from auto-cost templates (freight % of value or per container, duty from HS-code rate, insurance %) and post at goods-in-transit; actuals post at invoice with BR-34.6 variance rules; cost type codes map to accrual and clearing accounts (wave 11); goods-in-transit value sits on the wave-20 in-transit inventory account; voyage tracking stages (shipped, departed, arrived port, customs cleared, arrived warehouse) come from TrackingEvent; allocation per FR-34.9 can run per container first, then per line.
- **FR-34.13 Yard management** — Should. The system shall model yard locations (parking slots, door waiting lanes, checkpoints), transportation units (trailer, container, swap body) and vehicles, track TU position and status (EMPTY, LOADED, SEALED; reefer setpoint), and create yard-move work (wms work type YARD_MOVE) between slot and dock door, executed on the wave-32 mobile app.
  - Notes: TU ↔ load link from tender acceptance; door assignment respects DockAppointment (wave 31); slot capacity is one TU; dwell time per TU drives detention (BR-34.13, accessorial DETENTION after free time, default 2 h); a yard check is a count work over yard slots.
- **FR-34.14 Driver check-in/out and gate log** — Should. The system shall check drivers in at the gate or a kiosk against the dock appointment (appointment id or carrier SCAC + trailer/container number), capture seal number, driver name and license id, assign a yard slot or door, and check out with seal verification, recording each event in an append-only gate log.
  - Notes: matching and tolerance per BR-34.14 (default ±60 min); walk-ins allowed per site setting; check-out blocked if the load is not confirmed or the seal does not match; driver PII kept 90 days, then pseudonymized; check-in emits `dev.passage.tms.gate.checkedin.v1`, which releases the door work in wms.
- **FR-34.15 Dangerous goods checks in transport** — Should. The system shall use the wave-31 HazmatProfile to check DG compliance at rating, load building, tendering and ship confirmation: mode-specific regulation (49 CFR, ADR, RID, IMDG, IATA DGR), carrier-service DG eligibility, mixed-load segregation (49 CFR 177.848, ADR 7.5.2), ADR 1.1.3.6 quantity thresholds and limited-quantity rules, and produce DG shipping-paper lines and placard requirements.
  - Notes: result PASS/WARN/FAIL with rule ids (BR-34.15); FAIL blocks tender and ship confirmation when `dg_check_mode = BLOCK`; a 24-hour emergency response phone from legal-entity setup is mandatory before a DG shipment; DG lines print on the BOL with UN number, proper shipping name, class, packing group and quantity.
- **FR-34.16 Fleet resources** — Could. The system shall define own-fleet vehicles (weight/volume/pallet capacity, equipment type, depot, cost per km and per hour, availability calendar), trailers and drivers (qualifications such as hazmat endorsement or ADR certificate with expiry, availability calendar, hours-of-service rule set) and pass them to the VRP as heterogeneous fleet and driver-hours constraints (BR-34.16).
  - Notes: dispatch assigns vehicle + driver to a route; telematics positions ingest as TrackingEvent with source TELEMATICS; driver apps, ELD and tachograph download remain out of scope.
- **FR-34.17 Strategic freight procurement (lane RFQ)** — Could. The system shall run a freight RFQ on the wave-17 RFQ engine with lanes (origin and destination zones, mode, equipment, annual volume, weight bands) as lines, collect carrier bids through the carrier portal (FR-34.10) or spreadsheet upload,

compare bids by scenario, and award lanes into effective-dated RateCards and route-guide ranks (BR-34.18).
    - Notes: bid cost = Σ lane annual volume × bid rate + fuel at the current index; scenarios: lowest cost per lane, maximum carriers per lane, minimum incumbent share; awarded cards are drafts that need TMS_RATE_APPROVE.
- **FR-34.18 Load plan stacking** — Could. The system shall compute a load plan for a trailer or container with layer-based stacking (pallet footprint and height, maximum stack weight, stackable flag, orientation lock, load sequence in reverse stop order) and show a top view with layers, floor positions used and an axle-weight estimate.
    - Notes: first-fit decreasing by footprint with layer building in `tms-optimizer`; non-stackable pallets and DG segregation respected; < 5 s for 60 pallets; free-form 3D carton packing is not in scope.
- **FR-34.19 Foreign-trade document and screening hooks** — Should. The system shall carry customs data on shipments and loads (commodity code per item and country such as HS/HTS/Schedule B, country of origin, export classification such as ECCN, Incoterms 2020 rule with named place, customs value and currency), generate commercial invoice and customs packing list documents, call a `ComplianceScreeningProvider` for denied-party and sanctions screening at load release, tender and ship confirmation (BR-34.17), and call a `CustomsFilingProvider` (e.g. US AES/EEI, EU export declaration) at ship confirmation.
    - Notes: screening result CLEAR / POTENTIAL_MATCH / BLOCKED / SCREENING_PENDING; with no provider configured, cross-border shipments carry `NOT_SCREENED` and warning TMS-0134; provider implementations, preference/origin determination (USMCA, EU origin) and filing rules ship in wave 48.

6.34.4 Data model

Load (TMS)

Field	Type	Required	Notes
load_id	uuid	yes	PK; shared id with wms.Load
mode, equipment	enum/text	yes	TL/LTL/PARCEL/INTERMODAL; `53FT_DRY`, `REEFER`
ship_from_site_id	uuid	yes	
carrier_id, service_id	uuid	no	set on tender accept
status	enum	yes	PLANNED, RATED, TENDERED, ACCEPTED, DISPATCHED, IN_TRANSIT, DELIVERED, CLOSED, CANCELLED
total_weight_kg, total_volume_m3, pallet_positions	numeric/int	yes	
planned_pickup_start/end	timestamptz	yes	window
route_id	uuid	no	FK Route
shipments	uuid[]	yes	via LoadShipment

RateCard / RateTable

Field	Type	Required	Notes
rate_card_id	uuid	yes	PK
carrier_id, service_id	uuid	yes	
basis	enum	yes	LANE, ZONE, DISTANCE, WEIGHT_BREAK, PER_PALLET, PER_CONTAINER, FLAT
currency	char(3)	yes	
valid_from, valid_to	date	yes	no overlap per carrier/service/basis
minimum_charge	numeric(18,2)	no	
rows	jsonb	yes	[{origin, destination, break_from, break_to, rate, unit}]
fuel_index_id	uuid	no	FK FuelIndex
accessorial_ids	uuid[]	no	

FreightBill / CarrierInvoice

Field	Type	Required	Notes
freight_bill_id	uuid	yes	PK
load_id, carrier_id	uuid	yes	
expected_lines	jsonb	yes	[{charge_code, amount, currency, basis_trace}]
expected_total	numeric(18,2)	yes	
accrual_journal_id	uuid	no	
match_status	enum	yes	UNMATCHED, MATCHED, VARIANCE, DISPUTED, APPROVED, SETTLED
carrier_invoice_id	uuid	no	FK CarrierInvoice (number, date, lines, total, source EDI210/MANUAL)
variance_total	numeric(18,2)	no	
ap_invoice_id	uuid	no	wave-19 vendor invoice

Tender: tender_id, load_id, carrier_id, rank, sent_at, expires_at, response (PENDING/ACCEPTED/DECLINED/EXPIRED), rate_quoted. **LandedCostVoyage**: voyage_id, cost lines (type, amount, currency, allocation_method), receipt lines, status (ESTIMATED/ACTUAL/POSTED). **TrackingEvent**: load_id, stop_no, code (EDI 214 status), occurred_at, location, source.

Invariants: a load has at most one ACCEPTED tender; Σ allocated landed cost per voyage cost line = cost line amount (to the cent); a freight bill's accrual is reversed exactly once; shipments on a load share ship-from site and mode. Indexing: `load(status, planned_pickup_start)`, `rate_card(carrier_id, service_id, valid_from, valid_to)`, `freight_bill(match_status)`, `tracking_event(load_id, occurred_at)`.

6.34.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/tms/v1/loads:build	Build loads from shipments by rule	key	TMS_PLAN
POST	/tms/v1/routes:optimize	VRP over loads/stops; returns routes	key = request hash	TMS_PLAN
POST	/tms/v1/shipments/{id}:rate	Rate across carriers, return ranked with explain	— (cached)	TMS_RATE
POST	/tms/v1/loads/{id}:tender	Start tender workflow	key	TMS_TENDER
POST	/tms/v1/tenders/{id}:respond	Carrier accept/decline	key	TMS_CARRIER_RESPOND

Method	Path	Purpose	Idempotency	Auth duty
POST	/tms/v1/tracking-events	Ingest events (214/API)	key = source+ref	TMS_TRACK_INGEST
POST	/tms/v1/carrier-invoices	Register carrier invoice; triggers match	key = carrier+invoice no	TMS_RECONCILE
POST	/tms/v1/freight-bills/{id}:approve	Approve variance / release to AP	key	TMS_RECONCILE_APPROVE
POST	/tms/v1/landed-cost/voyages/{id}:allocate	Allocate and post	key	TMS_LANDED_COST
GET	/tms/v1/loads/{id}/tracking	Timeline	—	TMS_VIEW
POST	/tms/v1/voyages	Create a voyage with legs, containers and folios	key	TMS_LANDED_COST
POST	/tms/v1/goods-in-transit/{id}:ship	Post goods in transit at the transfer event	key	TMS_LANDED_COST
POST	/tms/v1/gate/check-in	Driver and TU check-in	key = appointment + timestamp	TMS_GATE
POST	/tms/v1/gate/check-out	Check-out with seal verification	key	TMS_GATE
POST	/tms/v1/yard-moves	Create yard-move work	key	TMS_YARD
POST	/tms/v1/loads/{id}:dg-check	Run DG checks for a load	—	TMS_PLAN
POST	/tms/v1/loads/{id}:screen	Run trade-compliance screening	—	TMS_PLAN
POST	/tms/v1/freight-rfqs/{id}:award	Award lanes into rate cards	key	TMS_RATE_APPROVE

Event	Trigger	Payload fields	Consumers
dev.passage.tms.shipment.rated.v1	rating	shipment_id, options[{carrier, service, total, transit_days}], chosen	sales (freight charge), analytics
dev.passage.tms.tender.accepted.v1 / declined.v1 / expired.v1	tender response	load_id, carrier_id, rank, rate	wms, edi (204/990), analytics
dev.passage.tms.tracking.event.v1	ingest	load_id, stop_no, code, occurred_at	sales-fulfillment, customer portal
dev.passage.tms.freightbill.matched.v1	match	freight_bill_id, carrier_invoice_id, variance, lines[]	AP, costing
dev.passage.tms.landedcost.posted.v1	allocation posted	voyage_id, receipt_lines[{line, amount}]	costing (wave 20)

6.34.6 User experience

Screen	Primary actions	Key fields	Duty
Transportation workspace	Shipments to plan, loads, tenders pending, exceptions	shipment, weight, window, status	TMS_VIEW
Load planning workbench	Drag shipments to loads, optimize routes, map view	load, stops, capacity bars, route distance/time	TMS_PLAN
Rate shop	Compare carriers, explain, select	carrier, service, charges, transit, score	TMS_RATE
Tender board	Tender, cascade, override	load, carrier rank, expires, response	TMS_TENDER
Tracking	Timeline, map, POD	event code, time, location, ETA	TMS_VIEW
Freight reconciliation	Match, dispute, approve, release to AP	expected vs invoiced per charge, variance	TMS_RECONCILE
Landed cost voyages	Add costs, allocate, post	cost type, amount, method, allocation	TMS_LANDED_COST

Interaction rules: planning workbench supports keyboard move (Alt+arrow) and multi-select; capacity bars turn red above 100%; route map uses self-hosted tiles; reconciliation grid highlights lines outside tolerance.

6.34.7 Business rules and edge cases

- **BR-34.1 VRP constraints.** Model as capacitated VRP with time windows and pickup/delivery (OR-Tools `RoutingModel`): (1) vehicle capacity dimensions weight_kg, volume_m3, pallets (hard); (2) stop time windows [earliest, latest] with soft lateness penalty $50/min up to 60 min, hard beyond; (3) max route duration 11 h driving + service times, max 14 h on-duty; (4) depot start/end per vehicle; (5) pickup-before-delivery precedence for milk runs; (6) heterogeneous fleet (equipment type must match shipment requirement, e.g., reefer); (7) vehicle-location compatibility (dock height, hazmat ban); (8) max stops per route 25; (9) objective = Σ distance cost + Σ time cost + fixed vehicle cost + lateness penalties; (10) time limit 60 s, first solution strategy `PARALLEL_CHEAPEST_INSERTION`, metaheuristic `GUIDED_LOCAL_SEARCH`. Infeasible stops are dropped with penalty 10,000 and reported, never silently.
- **BR-34.2 Rating algorithm.** For each eligible (carrier, service): select the rate card valid on ship date; compute base charge by basis (weight-break: find row with `break_from ≤ chargeable_weight < break_to`, charge = max(rate × weight, next-break deficit rule: compare with next break's minimum and take the lower); zone: origin/destination zone lookup; distance: miles × rate); chargeable weight = max(actual, dimensional weight = volume_in³ / DIM factor 139 for parcel, class-based for LTL); apply minimum charge; add accessorials (flat or % of base); add fuel surcharge = base × index % effective on ship date (weekly index); convert to shipment currency at the wave-12 rate type `FREIGHT`; round each charge to cents half-up; total = Σ charges. Explain trace lists each step with inputs.
- **BR-34.3 Tender cascade.** Rank from route guide; send to rank 1; on DECLINED or EXPIRED (timeout) send to next; after the last, status `TENDER_FAILED` and a planner task is created; a late acceptance after expiry is rejected with `TENDER_EXPIRED`.
- **BR-34.4 Match state machine.** `UNMATCHED → MATCHED` (all lines within tolerance) `→ APPROVED → SETTLED` (AP invoice posted); `UNMATCHED → VARIANCE → DISPUTED → (MATCHED | APPROVED with adjustment)`; a carrier invoice without a freight bill creates an `UNEXPECTED_INVOICE` exception and is never auto-approved. Duplicate carrier invoice numbers per carrier are rejected.
- **BR-34.5 Accrual and reversal.** Accrual posts on `load.shipped` with amount = expected total; on APPROVED, post reversal of the accrual and the AP invoice for the invoiced amount; variance to freight variance account 5250. All three postings share the freight bill id in the journal reference.
- **BR-34.6 Landed cost allocation.** Allocate each cost line by chosen method; compute each receipt line's share = round(cost × weight_i / Σ weight, 2); residual = cost – Σ shares assigned to the line with the highest value (ties: lowest line number). Estimated vs actual differences post to inventory if the item is still on hand (proportional to remaining qty) else to a purchase price variance account 5260 (wave 20 rules).
- **BR-34.7 Time windows** are stored in UTC with the stop's time zone; VRP uses the stop's local calendar; a window spanning midnight is valid.
- **BR-34.8 Load capacity exceeded** on manual planning is a warning (planner override logged); on rule-based building it is a hard split into a second load.
- **BR-34.9 Tracking event ordering**: events are applied by `occurred_at`; an out-of-order delivery event (D1) after pickup (AF) missing sets stop status DELIVERED and backfills PICKED_UP as inferred.
- **BR-34.10 Rate card overlap** on save is rejected; a card with `valid_to < ship_date` is ignored and the explain trace states `NO_VALID_RATE_CARD`.
- **BR-34.11 Goods-in-transit transfer event.** The event that moves goods into the in-transit warehouse is set per vendor × Incoterm rule; defaults: EXW, FCA, FAS, FOB, CFR, CIF, CPT, CIP → origin shipment event (risk passes at origin); DAP, DPU, DDP → destination receipt (no goods-in-transit order). Estimated landed cost posts at the transfer event; Incoterms define risk, so title transfer can be overridden per contract.
- **BR-34.12 ISO 6346 check digit.** Letters map to values A = 10, B = 12 … skipping multiples of 11 (K = 21, L = 23, U = 32, V = 34 …); each of the first 10 characters is multiplied by 2^position (position 0–9); check digit = (Σ mod 11) mod 10. Example: CSQU305438 → check digit 3, so CSQU3054383 is valid. Invalid numbers are rejected.
- **BR-34.13 Yard moves and detention.** Dwell = check-out – check-in; detention minutes = max(0, dwell – free time); expected DETENTION charge = ceil(detention minutes ÷ 60) × the carrier's hourly accessorial rate; a yard move cannot target an occupied slot or a door holding another TU's active

appointment.

- **BR-34.14 Gate check-in matching.** Match by appointment id, else by carrier SCAC + trailer/container number + date; arrival within `gate_tolerance_min` → ON_TIME, otherwise EARLY or LATE with reason; at check-out the declared seal must equal the seal recorded at load confirmation (outbound) or at check-in (inbound sealed); mismatch blocks check-out with TMS-0129.
- **BR-34.15 DG checks.** Mixed-load segregation uses the regulation's table as reference data: FAIL if two classes marked prohibited share a load; ADR 1.1.3.6 points = Σ quantity × multiplier by transport category (category 1: × 50, 2: × 3, 3: × 1, 4: × 0; category 0 always requires full ADR); points > 1,000 → WARN with full ADR marking, documentation and driver training required; limited-quantity packages within the per-inner limit → WARN for marking only; a carrier service without DG eligibility is excluded from rate shopping (FR-34.4).
- **BR-34.16 Driver hours in VRP.** Per driver rule set: US property-carrying — driving ≤ 11 h within a 14-h on-duty window, 30-min break after 8 h cumulative driving; EU 561/2006 — daily driving ≤ 9 h (10 h when `extended_day_available`), 45-min break after 4.5 h driving; violations make the route infeasible for that driver (hard constraint).
- **BR-34.17 Screening gating.** Load release, tender and ship confirmation screen every party on the shipment (sold-to, ship-to, consignee, forwarder, carrier); results are cached `screening_cache_h` (default 24 h) per party + list version; BLOCKED stays until the list changes; POTENTIAL_MATCH and SCREENING_PENDING (provider timeout) hold the load until released by TRADE_COMPLIANCE_RELEASE with a logged reason; the hook never auto-clears.
- **BR-34.18 Lane RFQ award.** Each lane is awarded to the bid with the lowest normalized annual cost subject to maximum carriers per lane and minimum incumbent share; ties go to the higher on-time %; awarded RateCards start on the award effective date and overlapping existing cards are end-dated the day before (BR-34.10 still holds).

6.34.8 Acceptance criteria

- **AC-34.1** (FR-34.2, BR-34.1) Given Solomon C101, R101 and RC101 (100 customers), when optimized with a 60 s limit, then every route satisfies capacity and time windows and total distance is within 10% of best-known.
- **AC-34.2** (FR-34.3, BR-34.2) Given the 200-shipment rating fixture, when rated, then totals equal the expected charges to the cent and the explain trace lists weight-break, minimum, accessorial and fuel steps.
- **AC-34.3** (FR-34.5, BR-34.3) Given a route guide with 3 carriers and 10-minute timeouts (test clock), when carrier 1 ignores and carrier 2 declines, then carrier 3 receives the tender and, on acceptance, the load is ACCEPTED with carrier 3.
- **AC-34.4** (FR-34.7, BR-34.4) Given a freight bill of $412.50 and a carrier invoice of $418.00, when matched with tolerance $5/2%, then the bill is VARIANCE with $5.50 and no AP invoice exists until approval.
- **AC-34.5** (FR-34.8, BR-34.5) Given a load shipped and later matched, when postings are inspected, then accrual, reversal and AP invoice tie to zero on account 2150 for that freight bill.
- **AC-34.6** (FR-34.9, BR-34.6) Given a voyage with freight $1,000.00 allocated by weight over 3 receipt lines (weights 1, 1, 1 kg), when allocated, then shares are 333.33 / 333.33 / 333.34 and Σ = 1,000.00.
- **AC-34.7** (FR-34.6) Given a 214 with status D1, when ingested, then the stop is DELIVERED and the sales shipment shows delivered timestamp.
- **AC-34.8** (FR-34.1, BR-34.8) Given 30 shipments totaling 48,000 kg for a 20,000 kg TL rule, when loads are built, then 3 loads exist, none over capacity.
- **AC-34.9** (FR-34.12, BR-34.11) Given a voyage with 2 containers for 3 POs on FOB Shanghai, when the forwarder's departure event posts, then goods-in-transit orders move ownership to the in-transit warehouse with estimated freight and duty, and after the actual invoices the variance posts per BR-34.6 with Σ allocated = Σ actual costs to the cent.
- **AC-34.10** (BR-34.12) Given container number CSQU3054383, when entered, then it is accepted; CSQU3054384 is rejected with TMS-0124.
- **AC-34.11** (FR-34.13, BR-34.13) Given a trailer checked in at 08:00 with free time 2 h and checked out at 11:20, when the freight bill is built, then a DETENTION charge of 2 h × the carrier rate is expected.
- **AC-34.12** (FR-34.14, BR-34.14) Given an outbound load sealed with 778812, when the driver declares 778813 at check-out, then check-out is blocked with TMS-0129 and the gate log records the attempt.
- **AC-34.13** (FR-34.15, BR-34.15) Given a road load with a class 1.1D explosive and a class 3 flammable liquid under 49 CFR, when the load is tendered, then the DG check FAILs with the segregation rule id and no tender is sent.
- **AC-34.14** (FR-34.15, BR-34.15) Given a fixture load of 400 L of a transport-category-2 substance, when checked under ADR, then points = 1,200 and TMS-0131 is raised; at 300 L points = 900 and no warning is raised.
- **AC-34.15** (FR-34.16, BR-34.16) Given an EU driver without an extended day available and a route needing 10.5 h of driving, when optimized, then the route is infeasible for that driver and is reassigned or split.
- **AC-34.16** (FR-34.19, BR-34.17) Given a ship-to party returning POTENTIAL_MATCH from the screening provider stub, when the load is released, then it is held with TMS-0133; after release by a TRADE_COMPLIANCE_RELEASE user with a reason, tendering proceeds and the log shows user and reason.
- **AC-34.17** (FR-34.17, BR-34.18) Given a lane RFQ with 10 lanes and 4 carrier bids, when awarded with at most 2 carriers per lane, then draft RateCards exist per awarded lane starting on the award date and prior overlapping cards end the day before.
- **AC-34.18** (FR-34.18) Given 24 pallets of which 6 are non-stackable for a 53-ft trailer, when planned, then no pallet is stacked on a non-stackable pallet and the plan reports floor positions used.

6.34.9 Test plan and fixtures

Fixtures: Solomon instances (C101, R101, RC101, C201, R201), `rating-200.json` (lanes, weights, accessorials, expected totals verified by hand in a spreadsheet committed alongside), `carriers-demo` (5 carriers, 12 services, 3 rate bases), `carrier-invoices-50.json` with 10 variance cases, `voyage-3.json`.

Test	Type	Target	Pass criterion
VRP benchmark	performance/quality	optimizer	feasible, ≤ 110% best-known
Rating determinism	unit/property	engine	same input → same output; totals exact
Tender cascade	Temporal time-skip	workflow	cascades per BR-34.3
Match tolerance	unit	50 invoices	expected statuses
Accrual tie-out	contract	journal API	zero balance per bill
Landed cost sums	property	allocation	Σ shares = cost for random inputs
214 ingest	contract	API	ordering per BR-34.9

Reconciliation: Σ accrued freight by open freight bill = journal account 2150 balance; Σ landed cost posted = Σ voyage actual costs; loads shipped in wms = loads DISPATCHED+ in tms.

6.34.10 Deployment and operations

Helm `passage-tms` (api, worker), `passage-tms-optimizer` (Python, OR-Tools, 1–4 replicas, 2 CPU/4 GB each), `passage-osrm` (optional self-hosted routing with regional extracts). CloudNativePG `tms-db` (3 instances, 50 GB). Config: `TMS_VRP_TIME_LIMIT_S=60`, `TMS_TENDER_TIMEOUT_MIN=120`, `TMS_MATCH_TOL_ABS=5`, `TMS_MATCH_TOL_PCT=2`, `TMS_DIM_FACTOR_PARCEL=139`. Flags: `tms.enabled`, `tms.landed_cost`, `tms.vrp`. Scheduled: fuel index import weekly (Monday 06:00), tender expiry sweep (Temporal timers), KPI daily. SLOs: rating p95 < 300 ms per shipment per carrier; VRP 500 stops < 60 s; match p95 < 1 s; availability 99.9%. Alerts: tender failures > 3/day; unexpected invoices > 10/day; accrual drift ≠ 0; optimizer queue depth > 20. Runbook: optimizer OOM (reduce batch or raise limits); OSRM down (fallback to stored lane distances with warning). Rollback: `tms.enabled=false` returns to wave-28 carrier labels; open tenders are cancelled with notification.

6.34.11 Migration from AX / SAP

Source	Passage entity	Transformation
TMSCarrier, TMSCarrierService, TMSCarrierServiceGroup	Carrier, CarrierService	linked to Party by vendor account
TMSRateMaster, TMSRateBase, TMSRateBaseAssignment, rate engine metadata	RateCard, RateTable	basis per rate engine type; effective dates from assignment
TMSAccessorialMaster/Assignment, TMSFuelIndex/Region	Accessorial, FuelIndex	direct
TMSRoutePlan, TMSRouteGuide, TMSRouteSegment	RouteGuide	segments → lanes
TMSFreightBill/Detail, TMSCarrierInvoice/Detail (open)	FreightBill, CarrierInvoice	unmatched/unapproved only
Landed cost voyages (open)	LandedCostVoyage	estimated costs only
/SCMTMS/TOR_ROOT/ITEM/STOP (open freight orders)	Load, Shipment, Route	TOR category FO → Load, FU → Shipment
Freight agreements, /SCMTMS/TCM rate tables, calc sheets	RateCard, CalcSheet	scales → weight breaks
/SCMTMS/SFIR (open settlements)	FreightBill	status mapping
PO delivery cost conditions (FRA1/FRB1)	LandedCostVoyage estimated	per PO line

Not migrated: historical freight invoices and tracking events (lakehouse); SAP EM event handlers; ISV route optimization data.

6.34.12 Dependencies, risks, open questions

Depends on: wave 33 (phase ordering), wave 31 (loads, docks), wave 28 (shipments, carrier labels), wave 29 (customer freight charges), wave 20 (costing adjustments, supplier portal), wave 19 (AP invoices), wave 18 (purchase receipts), wave 12 (FX rate type), wave 11 (journal), wave 7 (carrier parties).

Risk	Likelihood	Impact	Mitigation
OR-Tools solution quality on real (non-benchmark) data	Medium	Medium	Planner override; parameter presets; report gap vs heuristic baseline
Carrier invoice formats vary (EDI 210 vs PDF)	High	Low	Manual/API entry path; OCR is v1.x

Open questions: (1) OSRM self-hosted vs stored lane matrix only? Proposed: OSRM optional, lanes default; decide by wave 34 design review. (2) Should customer-billed freight use rated or actual cost? Proposed: rated at ship confirmation; decide by wave 34.

6.34.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
tender_timeout_min	carrier / route guide	int	120	5–1,440	Tender expiry (BR-34.3)	Transportation management parameters › tendering; tender response deadline	SPRO TM › Freight Order Management › Tendering › Define Tendering Profiles
broadcast_tendering	route guide	bool	false	true / false	First acceptance wins (FR-34.5)	Load tender broadcast	Tendering process type "broadcast"
vrp_time_limit_s	tenant	int	60	5–600	Optimizer time budget (BR-34.1)	n/a	Planning profile › optimizer settings (runtime)
lateness_penalty_per_min	planning profile	money	50.00 USD	≥ 0	Soft time-window penalty (BR-34.1)	n/a	Optimizer cost function settings
dim_factor_parcel	carrier service	int	139 (in³/lb)	139 / 166 / 5,000 (cm³/kg)	Dimensional weight (BR-34.2)	Rate engine parameters (dimensional weight)	Charge calculation scale base "chargeable weight"
match_tolerance_abs / match_tolerance_pct	carrier	money / pct	5.00 USD / 2 %	≥ 0	Freight invoice tolerance (BR-34.4)	Freight reconciliation › Audit masters	Freight settlement invoice verification tolerance
fuel_index_source	carrier	enum	DOE_WEEKLY	DOE_WEEKLY / EU_WEEKLY_OIL_BULLETIN / MANUAL	Fuel surcharge index (BR-34.2)	Fuel indexes (TMSFuelIndex)	Charge type FUEL with index table
freight_accrual_account	legal entity	account	2150	GL accounts	Accrual posting (BR-34.5)	Transportation management parameters › freight reconciliation posting accounts	Settlement accrual account (T030 key FR1)
freight_variance_account	legal entity	account	5250	GL accounts	Variance posting (BR-34.5)	Freight bill type posting setup	Price difference account (PRD)
landed_cost_auto_costs	vendor / item group	template ref	none	auto-cost templates	Estimated costs at goods in transit (FR-34.12)	Landed cost › Auto costs	Pricing procedure with FRA1/FRB1 conditions
goods_in_transit_transfer_event	vendor × Incoterm	enum	per BR-34.11	ORIGIN_SHIPMENT / DESTINATION_RECEIPT	Ownership timing (BR-34.11)	Landed cost parameters › goods-in-transit ownership change	Stock in transit settings (movement types 107/109)
in_transit_warehouse	site	warehouse ref	TRANSIT-{site}	in-transit warehouses	Holds goods in transit	Landed cost parameters › goods-in-transit warehouse	SIT plant/storage location
container_check_digit_required	tenant	bool	true	true / false	Enforces BR-34.12	Shipping container validation	Transportation unit identification check
detention_free_time_min	carrier	int	120	0–1,440	Detention start (BR-34.13)	Accessorial (detention)	Charge type detention/demurrage

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
gate_tolerance_min	site	int	60	0–480	ON_TIME window (BR-34.14)	Appointment scheduling early/late tolerance	SPRO EWM › Yard Management › Define Checkpoints (tolerance)
walk_in_allowed	site	bool	true	true / false	Check-in without appointment (FR-34.14)	Driver check-in setup	Check-in without planned TU
gate_pii_retention_days	legal entity	int	90	30–3,650	Driver PII pseudonymization (FR-34.14)	n/a	n/a (ILM policy)
dg_regulation_set	mode × country	enum	ADR (EU road), 49CFR (US)	ADR / RID / 49CFR / IMDG / IATA	Rules applied (FR-34.15)	n/a	EHS-DG: specify DG regulations per mode-transport category
dg_check_mode	legal entity	enum	BLOCK	BLOCK / WARN	FAIL behavior (FR-34.15)	n/a	DG check schema and check methods (SD/LE/TM)
emergency_response_phone	legal entity	text	none (required for DG)	E.164 number	Printed on DG papers	n/a	DG master emergency telephone
driver_hours_rule_set	depot	enum	FMCSA_PROPERTY	FMCSA_PROPERTY / EU_561 / NONE	Driver-hours constraint (BR-34.16)	n/a	TM resource driving-time rules
screening_provider	legal entity	adapter id	none	registered providers	Denied-party screening (FR-34.19)	n/a	SAP GTS / Watch List Screening configuration
screening_cache_h	tenant	int	24	0–168	Result cache (BR-34.17)	n/a	GTS screening buffer settings
customs_filing_provider	legal entity × country	adapter id	none	registered providers	Export filing call (FR-34.19)	n/a	GTS customs management communication
rating_cache_min	tenant	int	10	0–60	Rate shop cache (FR-34.4)	n/a	n/a

6.34.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Loads by status	list inquiry	status, pickup window, carrier, mode	load, weight, stops, tender state	`load`	Transportation management workspace; loads list	Fiori Freight Order Worklist; transportation cockpit
Rate shop log	drill-down	shipment, date	carriers, charges, explain trace	rating log	Rating results inquiry; rate route workbench	Carrier selection log
Tender history	list inquiry	carrier, outcome, period	sent, response, expiry, rank	`tender`	Tender status inquiry	Tendering overview (TM)
Late shipments / tracking timeline	operational report	late > n min, carrier	ETA vs window per stop	`tracking_event`	Load tracking inquiry	Track Freight Orders (event management)
Freight reconciliation status	operational report	carrier, match status, age	expected, invoiced, variance, dispute	`freight_bill`	Freight bill invoice reconciliation	Manage Freight Settlement Documents; VI12
Freight accrual tie-out	operational report	legal entity, period	open bills, accrual total, GL 2150 balance, difference	reconciliation	SSRS freight accrual report	VI12 with FBL3N on accrual account
Voyage cost and variance	drill-down	voyage, container, cost type	estimated, actual, variance, allocation per receipt line	`landed_cost_voyage`	Voyage cost inquiry; SSRS landed cost reports	MRDC log; MB5S GR/IR balances
Goods in transit	list inquiry	voyage, status, vendor	qty and value in transit, ETA	in-transit quants	Goods-in-transit orders	MB5T stock in transit
Yard status board	operational report	site	slots, TUs, status, dwell, reefer setpoints	`transport_unit`	Yard management workspace	/SCWM/MON yard nodes (TU, vehicle, yard bins)
Gate log	list inquiry	site, date, carrier	check-in/out times, trailer, seal, on-time flag	`gate_event`	Driver check-in log	Check-in/check-out log (/SCWM/CICO)
DG load report	operational report	load, mode, date	UN numbers, classes, ADR points, result, rule ids	`dg_check_result`	n/a	DG check log in delivery
Carrier KPIs	KPI	carrier, lane, period	cost/cwt, cost/mile, OTD %, tender acceptance %, invoice variance rate	Cube	Transportation KPIs	TM analytics / freight cost KPIs
Lane RFQ bid comparison	drill-down	RFQ, lane	bids, scenario totals, award	`freight_rfq`	n/a	Freight RFQ bid analysis
Screening holds	list inquiry	status, party, age	load, party, list version, released by	`screening_result`	n/a	GTS blocked documents

6.34.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Bill of lading (VICS straight BOL)	Load confirmed	Shipper, consignee, third-party bill-to, SCAC, PRO, handling units, NMFC class, weight, DG lines (UN, proper shipping name, class, PG, qty, emergency phone), seal	PDF	BOL-* gap-free per site	DG shipping papers: 2 years (49 CFR 172.201(e))
CMR consignment note / e-CMR	EU road load confirmed	CMR boxes 1–24	PDF; e-CMR (UN/CEFACT)	CMR-*	Per CMR convention and national law (country pack)
DG transport document	Load with DG lines	ADR 5.4.1 / 49 CFR 172.202 fields	PDF; on BOL	BOL number	ADR: 3 months minimum (5.4.4)
Load tender	Tender sent	Load, stops, windows, equipment, rate, DG flag	EDI 204; email PDF	Tender id	Contract evidence with freight bill
Freight bill reconciliation report	Match result	Expected vs invoiced per charge, variance, decision	PDF	FB-*	AP support, 10 years (wave 19)
Commercial invoice and customs packing list	Cross-border shipment confirmed	Exporter, consignee, Incoterm, commodity codes, origin, values, weights, export classification	PDF; data to `CustomsFilingProvider`	CI-*	US export records 5 years (15 CFR 762.6); country pack may extend

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Gate pass / check-in receipt	Check-in	Appointment, door or slot, trailer, instructions	Kiosk print / PDF	GATE-*	Gate log PII 90 days
Voyage cost statement	Voyage COSTED	Containers, cost types, allocation per receipt line	PDF / XLSX	Voyage id	Inventory valuation support, 10 years
Load plan sheet	Load plan computed	Top view, layers, load sequence, axle estimate	PDF	Load id	None

6.34.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Fuel index import	Weekly, Monday 06:00	index	index + week	< 1 min	Retry hourly; warn on stale index	Index older than 8 days
Tender expiry	Temporal timers	load	workflow id	n/a	Cascade (BR-34.3)	Tender failures > 3/day
Carrier invoice auto-match	Every 15 min	carrier	carrier invoice id	1,000 invoices < 5 min	Unmatched stay in queue	Unexpected invoices > 10/day
Freight accrual tie-out	Daily 02:00	legal entity	date	< 5 min	Report only	Accrual drift ≠ 0
Voyage estimate refresh	Daily 01:00	voyage	voyage version	< 5 min	Rerun	Voyages without estimates at transfer
Detention computation	On check-out + hourly sweep	site	TU visit id	< 1 min	Rerun	none
Gate PII pseudonymization	Daily 03:00	legal entity	visit id	< 2 min	Rerun	Visits past retention not pseudonymized
Screening re-check on list update	On provider list-version change	party hash range	(party, list version)	100,000 parties < 30 min	Resume by range	New BLOCKED on open loads
DG reference data update	On new regulation edition (ADR biennial, IMDG amendment)	tenant	edition	< 10 min	Keep previous edition active	Edition not applied by effective date
KPI materialization	Daily 04:00	carrier	day	< 5 min	Rerun day	KPI lag > 24 h

6.34.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
TMS_VIEW	/loads/{id}/tracking; transportation workspace	none	All transportation roles
TMS_PLAN	/loads:build, /routes:optimize, /loads/{id}:dg-check, /loads/{id}:screen	TMS_RECONCILE_APPROVE	Transportation planner
TMS_RATE	/shipments/{id}:rate	none	Planner, customer service
TMS_RATE_APPROVE	Rate card publish; /freight-rfqs/{id}:award	TMS_RECONCILE_APPROVE (sets rates and approves invoices against them)	Transportation manager
TMS_TENDER	/loads/{id}:tender; tender board	none	Transportation planner
TMS_CARRIER_RESPOND	/tenders/{id}:respond; POD upload	External identities only; never combined with internal duties	Carrier user
TMS_RECONCILE	/carrier-invoices; reconciliation grid	none	Freight auditor
TMS_RECONCILE_APPROVE	/freight-bills/{id}:approve	AP_PAYMENT_RELEASE (wave 19); TMS_RATE_APPROVE	Freight audit lead
TMS_LANDED_COST	/voyages, /goods-in-transit/{id}:ship, /landed-cost/voyages/{id}:allocate	COST_PERIOD_CLOSE (wave 20)	Import coordinator
TMS_YARD	/yard-moves; yard board	none	Yard clerk, yard driver
TMS_GATE	/gate/check-in, /gate/check-out	none	Gate guard
TMS_DG_MAINTAIN	DG reference data; WARN overrides	TMS_PLAN (overriding own load)	Dangerous goods safety adviser
TRADE_COMPLIANCE_RELEASE	Release POTENTIAL_MATCH / SCREENING_PENDING holds	SALES_ORDER_MAINTAIN (wave 26)	Trade compliance officer
TMS_FLEET_ADMIN	Vehicles, trailers, drivers, calendars	none	Fleet manager

6.34.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
TMS-0101	error	No valid rate card for {carrier}/{service} on {date}.	NO_VALID_RATE_CARD	Maintain rate card	BR-34.10
TMS-0102	error	Rate card overlaps {card} ({from}–{to}).	Overlap on save	Adjust validity	BR-34.10
TMS-0103	warning	Load {load} exceeds {dimension} capacity ({used}/{max}); override logged.	Manual over-capacity	Split load or confirm override	BR-34.8
TMS-0104	warning	{n} stops dropped as infeasible; see route report.	VRP dropped stops	Adjust windows or add vehicles	BR-34.1
TMS-0105	error	Tender {t} expired at {time}; acceptance rejected.	TENDER_EXPIRED	Re-tender	BR-34.3
TMS-0106	error	All carriers declined or timed out for load {load}; planner task created.	TENDER_FAILED	Spot quote or new route guide	BR-34.3
TMS-0107	warning	Charge {charge} differs by {amount} ({pct} %), outside tolerance.	Invoice variance	Dispute or approve	BR-34.4
TMS-0108	error	Carrier invoice {no} is already registered for {carrier}.	Duplicate invoice	Reject duplicate	BR-34.4
TMS-0109	error	Invoice has no freight bill; manual review required.	UNEXPECTED_INVOICE	Link to load or reject	BR-34.4
TMS-0110	info	Allocation residual {amount} assigned to line {line}.	Rounding residual	None	BR-34.6
TMS-0121	info	Ownership moved to in-transit warehouse at {event} ({incoterm}).	Goods-in-transit posted	None	BR-34.11
TMS-0124	error	Container number {no} fails the ISO 6346 check digit.	Invalid container number	Correct the number	BR-34.12
TMS-0125	error	Yard slot {slot} is occupied by {tu}.	Slot conflict	Choose another slot	BR-34.13
TMS-0126	info	Detention of {min} min accrued for {tu}.	Dwell beyond free time	None; charge expected	BR-34.13
TMS-0127	warning	Driver arrived {n} min {direction} for appointment {appt}.	EARLY or LATE	Reschedule door if needed	BR-34.14

Code	Severity	User text	Cause	Remediation	Raised by
TMS-0128	error	No appointment found; walk-ins are not allowed at {site}.	Unmatched check-in	Book appointment	FR-34.14
TMS-0129	error	Seal number {declared} does not match {recorded}.	Seal mismatch	Inspect trailer; supervisor decision	BR-34.14
TMS-0130	error	Dangerous goods check failed: {rule} ({class1} with {class2}).	Prohibited mixed load	Split load	BR-34.15
TMS-0131	warning	ADR 1.1.3.6 points {p} exceed 1,000; full ADR requirements apply.	Threshold exceeded	Mark vehicle; ADR-trained driver	BR-34.15
TMS-0132	error	Driver {d} would exceed {rule} ({hours} h).	Hours-of-service limit	Reassign driver or split route	BR-34.16
TMS-0133	error	Party {party} screening result {status}; load held.	Screening hit or timeout	Trade compliance review	BR-34.17
TMS-0134	warning	No screening provider configured; export shipment not screened.	NOT_SCREENED	Configure provider (wave 48)	FR-34.19
TMS-0135	error	Commodity code missing for {item} in {country}.	Customs data incomplete	Maintain commodity code	FR-34.19

6.34.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Rating	5,000 shipments/day × 8 carriers; peak 400 rate calls/min	60 / 300 / 500 ms per carrier; full rate shop p95 < 800 ms (parallel)	50 rate calls/s	Rating log 90 days ≈ 6 GB	GiST on `rate_card` validity daterange per carrier/service	Cached results served; rate shop narrows to route-guide top 3
VRP optimization	20 runs/day, ≤ 500 stops	≤ 60 s hard limit	4 concurrent runs per 2 optimizer replicas	—	—	Queue (alert at depth 20); first feasible solution returned at 50 % of budget
Tracking ingest	50,000 events/day; peak 2,000/min	20 / 80 / 200 ms	100/s	≈ 10 GB	`tracking_event` monthly partitions; `(load_id, occurred_at)`	Kafka buffering; per-load ordering preserved
Freight invoice match	1,000 invoices/day	200 ms / 1 s / 2 s	20/s	Invoices ≈ 1 GB	`freight_bill(match_status)`	Auto-match interval stretches; no auto-approval changes
Landed cost allocation	200 voyages/month × 3,000 receipt lines	< 10 s per voyage	—	Voyages ≈ 1 GB	`voyage(status)` ; receipt line FK	Allocations queued per voyage
Gate check-in	300 visits/day per site; peak 40/h	100 / 300 / 600 ms	5/s per site	Gate log ≈ 0.2 GB	`gate_event(site_id, occurred_at)`	Kiosk retries; paper fallback entered later with original time
DG check	Every load at plan, tender and ship	20 / 80 / 150 ms	100/s	Results ≈ 0.5 GB	Reference data in memory per edition	Unchanged (CPU-bound, small)
Screening call	3,000 parties/day at release	Provider budget p95 1.5 s; timeout 5 s	20/s	Results ≈ 0.3 GB	Cache keyed (party, list version)	Timeout → SCREENING_PENDING hold; never auto-clear
Load plan stacking	200 plans/day	< 5 s for 60 pallets	2/s per optimizer replica	—	—	Queued behind VRP with lower priority

6.34.20 Localization and compliance hooks

- **Dangerous goods.** US 49 CFR parts 171–180 (PHMSA), EU ADR (road), RID (rail), ADN (inland waterways), IMDG Code (sea), IATA DGR (air), Canada TDG. New editions (ADR biennial, IMDG amendments) arrive as reference data through `DgRegulationPack` (wave 48); BR-34.15 is edition-neutral.
- **Driver hours.** US FMCSA 49 CFR part 395; EU Regulation 561/2006 with tachograph Regulation 165/2014; UK domestic rules. Rule sets plug in through `DriverHoursRuleSet`.
- **Transport documents.** CMR convention and e-CMR protocol for EU road; EU eFTI Regulation 2020/1056 for electronic freight information to authorities; Mexico CFDI with Carta Porte complement for goods in transit (country pack MX, submitted through the wave-35 `MandateProvider`); Brazil CT-e/MDF-e and India e-way bill are country-pack documents on the same hook.
- **Customs, export control and sanctions.** US AES/EEI (15 CFR part 30) and EAR classification (15 CFR part 774); EU Union Customs Code export declarations and ICS2 for imports; UK CDS; denied-party lists such as the US Consolidated Screening List, EU consolidated financial sanctions list and UK OFSI list. Providers plug in through `ComplianceScreeningProvider`, `CustomsFilingProvider` and `PreferenceDeterminationProvider` (wave 48).
- **Landed cost.** Duty rates by commodity code and origin come from country-pack tariff tables; import VAT (including EU postponed accounting) posts through the wave-13 tax engine.
- **Currency and units.** Freight is rated in the carrier's currency and converted at the wave-12 rate type FREIGHT; miles/km, lb/kg and cwt (US 100 lb) vs per 100 kg follow carrier and locale.
- **Privacy and retention.** Driver names and license ids are personal data (GDPR, CCPA): 90-day retention, then pseudonymization; DG shipping papers 2 years (US) / 3 months (ADR); export records 5 years (US).

6.34.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX PARTIAL #47 — §12.8.1 Landed cost (voyages, shipping containers, folios, cost estimates, goods in transit; ITMVoyage*)	PARTIAL-34	ADDED as FR-34.12	BR-34.11, BR-34.12, AC-34.9, AC-34.10
AX PARTIAL #49 — §12.8.2 Landed cost reports (voyage cost, cost variance by voyage)	PARTIAL-34	ADDED as FR-34.12	"Voyage cost and variance" in §6.34.14
AX PARTIAL #51 — §12.8.3 Landed cost → Voyage cost allocation / Goods in transit receipts	PARTIAL-34	ADDED as FR-34.12	"Voyage estimate refresh" job; goods-in-transit receipt
AX PARTIAL #53 — §12.8.4 Landed cost setup (parameters, cost types, auto cost, container types, voyage templates, legs, tracking)	PARTIAL-34	ADDED as FR-34.12	`landed_cost_auto_costs`, `goods_in_transit_transfer_event`, `in_transit_warehouse`
AX PARTIAL #223 — §12.27.1 Landed cost (voyages, link)	PARTIAL-34	ADDED as FR-34.12	Same mechanism as #47
AX PARTIAL #224 — §12.27.2 SSRS: Landed cost reports (link)	PARTIAL-34	ADDED as FR-34.12	"Voyage cost statement" in §6.34.15
AX PARTIAL #225 — §12.27.3 Landed cost → Voyage cost actualization (link)	PARTIAL-34	ADDED as FR-34.12	Actuals at invoice with BR-34.6 variance
AX PARTIAL #226 — §12.27.4 Landed cost parameters / cost types / voyage templates (link)	PARTIAL-34	ADDED as FR-34.12	Same parameters as #53

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX GAP #73 — §12.26.4 Warehouse → Yard management (yard, trailers, yard moves)	GAP (propose 34)	ADDED as FR-34.13	BR-34.13, AC-34.11
AX GAP #74 — §12.27.1 Driver check-in / check-out	GAP (propose 34)	ADDED as FR-34.14	BR-34.14, AC-34.12
AX GAP #75 — §12.27.2 Driver check-in log	GAP (propose 34)	ADDED as FR-34.14	"Gate log" in §6.34.14
AX GAP #76 — §12.27.3 Driver check-in → Expire appointments / Gate log archive	GAP (propose 34)	ADDED as FR-34.14	"Gate PII pseudonymization" job; appointment expiry via DockAppointment status
AX GAP #77 — §12.27.4 Driver check-in setup / Gate configuration	GAP (propose 34)	ADDED as FR-34.14	`gate_tolerance_min`, `walk_in_allowed`, `gate_pii_retention_days`
AX GAP #78 — §12.27.4 Transportation → Yard management setup	GAP (propose 34)	ADDED as FR-34.13	Yard locations and detention parameters
SAP #192 — §13.16 EWM /SCWM/TU, /SCWM/VEH transportation units / vehicles	PARTIAL-34	ADDED as FR-34.13	TransportUnit linked to load and yard
SAP #195 — §13.16 EWM dangerous goods / hazmat checks in warehouse (EHS)	GAP (propose 34)	ADDED as FR-34.15	Warehouse storage side ADDED as FR-31.19
SAP #199 — §13.16 Hazardous material: regions, water pollution classes, storage classes, fire-containment sections (MGEF)	GAP (propose 34)	ADDED as FR-31.19	Master data introduced in wave 31 (first consumer); transport use FR-34.15
SAP #204 — §13.17 Dangerous goods check in delivery (EHS-DG), DG documents	GAP (propose 34)	ADDED as FR-34.15	DG transport document in §6.34.15
SAP #209 — §13.17 Dangerous goods: DG profiles, DG check methods in SD/LE (DGT*)	GAP (propose 34)	ADDED as FR-34.15	`dg_regulation_set`, `dg_check_mode`
SAP #210 — §13.18 TM transportation units / resources (vehicles, trailers, drivers), calendars	PARTIAL-34	ADDED as FR-34.16	Driver app / ELD OUT-OF-SCOPE (carriers' and telematics vendors' systems)
SAP #211 — §13.18 Dangerous goods and DG checks in TM	GAP (propose 34)	ADDED as FR-34.15	Checks at rate, build, tender, ship
SAP #213 — §13.18 Strategic freight procurement (lane RFQ, bid analysis)	GAP (propose 34)	ADDED as FR-34.17	Reuses wave-17 RFQ as the register proposed
SAP #214 — §13.18 Load consolidation / container packing optimization (3D load plan)	PARTIAL-34	ADDED as FR-34.18	Layer-based stacking; free-form 3D carton packing OUT-OF-SCOPE (low demand, high solver cost)
SAP #215 — §13.18 Dangerous goods in TM customizing	GAP (propose 34)	ADDED as FR-34.15	DG reference data job in §6.34.16
SAP #205 — §13.17 Foreign trade / export documents in delivery (customs, commercial invoice, EUR.1; GTS) — flagged for this wave by coordinator	PARTIAL-13	ADDED as FR-34.19	Commercial invoice and customs data added; EUR.1/origin certificates and customs declarations DEFERRED to wave 48
SAP #206 — §13.17 Preference determination (SAP GTS / foreign trade) — flagged	GAP (propose 48)	DEFERRED to wave 48	`PreferenceDeterminationProvider` hook; origin and commodity data captured by FR-34.19
SAP #208 — §13.17 Foreign trade: export control, preference, Intrastat settings (T604, T605) — flagged	PARTIAL-13	ADDED as FR-34.19	Export-control data and provider settings here; Intrastat settings stay wave 13; preference DEFERRED to wave 48
SAP #212 — §13.18 Customs declarations via GTS from TM — flagged	PARTIAL-13	DEFERRED to wave 48	`CustomsFilingProvider` call defined in FR-34.19; filing implementations are country-pack content
SAP #350 — §13.31 Business Partner Screening / Watch List Screening (GTS) — flagged	GAP (propose 48)	ADDED as FR-34.19	Screening gate and holds (BR-34.17) added; screening list provider DEFERRED to wave 48

6.35 Wave 35: B2B Integration: EDI, e-Invoicing & Integration Hub

6.35.1 Scope and business context

Wave 35 closes Phase 6 by replacing AX AIF / D365 data entities, Dual-write and Electronic Reporting (ER) for e-invoicing, and SAP IDoc/ALE, PI/PO (Integration Suite) and Document and Reporting Compliance (DRC). It delivers an EDI engine (X12 850/855/856/810/820/214/204/990/997 and EDIFACT ORDERS/ORDRSP/DESADV/INVOIC/CONTRL) with AS2 and SFTP transport, partner profiles and acknowledgments; Peppol BIS Billing 3.0 UBL 2.1 e-invoicing through a certified access point, Factur-X/ZUGFeRD, and a country mandate framework with hooks for IT SDI and MX CFDI; and an integration hub with public REST/GraphQL APIs, webhooks, Kafka topic subscriptions, a Backstage API catalog and developer portal with keys and quotas, plus reference connectors for Salesforce, Shopify and Stripe. Out of scope: full implementation of country e-invoice mandates beyond the framework and hooks (wave 48 country packs), EDIFACT subsets beyond the listed messages, a self-operated Peppol AS4 gateway (v1.x).

Scope amendment (coverage closure). The gap registers add four items to this wave: catalog punchout over cXML and SAP OCI that returns requisition lines to wave 17 (FR-35.13), cXML document exchange with cXML networks such as SAP Business Network (Ariba) and Coupa (FR-35.14), a PLM reference connector (FR-35.15), and intake and reconciliation of customer self-billing documents (FR-35.16). §6.35.21 lists the dispositions.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
SAP IDoc failures sit in status 51 (WE02/BD87); a 'MATERIAL not found' error on one E1EDP01 segment requires an ABAP-literate operator to read EDIDD SDATA, fix master data, then reprocess; D365 AIF (AifExceptionLog) and the data entity message processor are similar	Monitor shows raw segment data, no canonical view; reprocess is a separate transaction	Message monitor shows raw, canonical and the failing rule side by side; bulk reprocess with dry run	Mean time to resolve a failed inbound order: 30–90 min → < 5 min; operator needs ABAP/X++: yes → no
Outbound documents are produced by batch (SAP output determination NAST + RSNAST00 jobs; D365 batch jobs for outbound ports), so an ASN reaches the customer 15–60 min after shipment and 810s go out in nightly runs	Polling batch framework	Outbound documents generated from `load.shipped` / `invoice.posted` events via the outbox	Event-to-document latency: 15–60 min (batch) → < 60 s; nightly invoice runs → continuous
Every EDI partner requires a middleware project (PI/PO ICO + mapping, or an ISV EDI add-on for D365) with Java/graphical mappings nobody in the business can read	Mapping logic lives outside the ERP in a separate tool and skill set	Declarative maps in the product with partner overrides, test harness and simulator	New partner onboarding: 4–8 weeks → 1–3 days; middleware licenses for EDI: 1 → 0

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
External systems (MES, PLM, BI, agents) integrate through OData with no event feed (D365 business events cover few entities) or through BAPIs/RFCs needing SAP Gateway projects; nobody can see which APIs exist	No catalog, no subscription model, no quotas	Backstage catalog with OpenAPI/AsyncAPI for every entity, webhook/Kafka subscriptions with replay, keys and quotas, MCP (wave 47) over the same APIs	Time for an external team to first successful call: weeks (access request, gateway project) → < 1 h self-service; entities with event feeds: ~10% → 100%
ER-based e-invoicing (ERFormatMappingTable) and SAP DRC need a configuration import per country/version and validate only on submission; schematron failures come back from the network hours later	Validation happens at the authority or AP, not at creation	Schematron (CEN + Peppol) runs at validate time inside Passage; arithmetic rules checked first	Rejections from the Peppol network: ~3–5% of invoices → < 0.1%; feedback latency: hours → < 2 s

6.35.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Integration ports / endpoints	AIF: AifPort, AifInboundPort, AifOutboundPort, AifService, AifAction; D365 data entities (DMFEntity), OData	IDoc ports WE21, partner profiles WE20 (EDPP1, EDP12/EDP13/EDP21), distribution model BD64 (TBD05)	edi.TradingPartner, EDIProfile, Transport	full
Message log / monitor	AifDocumentLog, AifMessageLog, AifExceptionLog, AifGatewayQueue; D365 message processor	IDoc status EDIDS (WE02/WE05), BD87 reprocess, PI/PO message monitor	Message monitor with replay	improved — replay with state reproduction
EDI documents	ISV (partner EDI), data entity import/export	IDoc types ORDERS05, DELVRY03 (DESADV), INVOIC02, control record EDI_DC40 (EDIDC), data EDIDD (EDID4)	Map (canonical ↔ X12/EDIFACT)	full
e-Invoicing	Electronic reporting ER configurations (ERFormatMappingTable, ERModelMappingTable), Invoicing service	DRC eDocument framework (EDOCUMENT, EDOCUMENTFILE), Peppol exchange	Peppol UBL 2.1, Factur-X, mandate framework	full (Peppol), partial (country mandates: hooks)
CRM/commerce sync	Dual-write (entity maps to Dataverse)	SAP CPI prepackaged content	Reference connectors (Salesforce, Shopify, Stripe)	partial — reference connectors only
Business events / webhooks	BusinessEventsCatalog, BusinessEventsEndpoint	Event Mesh / enterprise event enablement	Webhook subscriptions, Kafka subscriptions	full
Public API catalog	OData $metadata, Open API via custom service	API Business Hub, Gateway (SEGW)	Backstage catalog, developer portal, OpenAPI/AsyncAPI	improved
API keys / quotas	Azure AD app registration only	API management (Integration Suite)	ApiClient, Quota	improved
Acknowledgments	n/a	STATUS/ALEAUD IDoc, CONTRL	997/999, CONTRL, MDN (AS2)	full
Catalog punchout	External catalogs with cXML punchout (CatVendorExternalCatalog)	OCI web-based catalogs (SPRO MM › Purchasing › Web-Based Catalogs › Define Catalogs); Ariba catalogs	integration-hub PunchoutCatalog, PunchoutSession → wave-17 requisition lines	full
cXML documents	n/a (ISV)	Ariba Network integration via Cloud Integration Gateway (cXML OrderRequest, ConfirmationRequest, ShipNoticeRequest, InvoiceDetailRequest)	cXML maps on the canonical document model	full
PLM integration	Engineering change management PLM connectors (Teamcenter, Windchill via Dataverse)	PLM system integration, Engineering Control Center	Reference connector `passage-connector-plm`	partial — reference connector
Customer self-billing	n/a (ISV for automotive)	Self-billing procedure VKA/VSB1 (IDoc GSVERF), SPRO SD › EDI › Self-Billing Procedure	SelfBillingDocument with line matching to wave-29 invoices	full

6.35.3 Functional requirements

- **FR-35.1 Trading partner profiles** — Must. The system shall define partners with identifiers (ISA sender/receiver qualifier+id, GS id, EDIFACT UNB id, Peppol participant id `0088:GLN/0192:orgnr`), transports (AS2, SFTP, HTTPS), maps per transaction set and direction, envelope options, and SLA (ack within n minutes).
  - Notes: a partner may have map overrides (per-segment rules) layered on the base map; test and production profiles; certificate rotation with overlap period.
- **FR-35.2 X12 and EDIFACT parsing/serialization** — Must. The system shall parse and generate X12 (versions 004010 and 005010) and EDIFACT (D96A, D01B) with configurable delimiters, validate against transaction-set schemas, and generate 997/999 and CONTRL acknowledgments.
  - Notes: control numbers (ISA13, GS06, ST02; UNB, UNH) are gap-free per partner per direction; duplicate inbound ISA13 within 30 days is rejected as a duplicate interchange and reported in the 997.
- **FR-35.3 Mapping engine** — Must. The system shall map between a canonical document model (order, order response, shipment notice, invoice, remittance, status, tender) and EDI segments using declarative maps (JSONata/TypeScript expressions), with lookups (partner item cross-reference, location GLN, UoM, qualifier tables).
  - Notes: base maps per BR-35.1–35.3 ship as fixtures; unit-tested; map versioning with effective dates.
- **FR-35.4 Document integration** — Must. The system shall convert inbound 850/ORDERS into wave-26 sales orders (with partner item cross-reference), outbound 855, 856/DESADV from `wms.load.shipped` with SSCC hierarchy, 810/INVOIC from wave-29 invoices, 820 remittance into wave-30 cash application, 214 into/out of TMS, 204/990 for tenders.
  - Notes: inbound order failures create a task with the raw and canonical payload; outbound documents are created from events via the outbox (no polling).
- **FR-35.5 Transport** — Must. The system shall send and receive via AS2 (RFC 4130: signed, encrypted, synchronous/asynchronous MDN) and SFTP (poll every 60 s, archive after pickup), with retries (5× exponential, max 1 h) and alerts on MDN timeout.
  - Notes: AS2 certificates stored in Vault; MDN required by default; SFTP file naming pattern per partner.
- **FR-35.6 Peppol e-invoicing** — Must. The system shall produce UBL 2.1 invoices and credit notes conforming to Peppol BIS Billing 3.0 (BR-35.4), validate against the CEN EN 16931 and Peppol schematron, and send/receive through a certified access point adapter (AS4 via the AP's API), handling the Peppol MLR/transport acks.
  - Notes: participant lookup via SML/SMP through the AP; invoices received create wave-19 vendor invoices; validation errors block sending and are shown on the invoice.
- **FR-35.7 Factur-X / ZUGFeRD and mandate framework** — Should. The system shall embed EN 16931 CII XML in PDF/A-3 (Factur-X profile EN 16931) and provide a country mandate framework (`MandateProvider` interface: validate, submit, poll status, store clearance id) with reference hooks for IT SDI (FatturaPA) and MX CFDI 4.0 (PAC stamping).
  - Notes: full IT/MX packs ship in wave 48; the framework stores clearance ids and statuses on the invoice.
- **FR-35.8 Public APIs and developer portal** — Must. The system shall publish versioned REST (OpenAPI 3.1) and GraphQL APIs for all entities exposed since wave 3, with API clients (Keycloak client credentials), scopes mapped to duties, per-client quotas (default 600 requests/min, burst 100) and a developer portal at `developers.<env>.passage.dev` generated from the Backstage catalog.
  - Notes: additive-change guarantee per PRD §8; deprecation ≥ 12 months with `Sunset` header; sandbox tenant per client.
- **FR-35.9 Webhooks and Kafka subscriptions** — Must. The system shall let clients subscribe to CloudEvents types by filter and receive them by webhook (HTTPS POST, HMAC-SHA256 signature, at-least-once, 8 retries over 24 h, dead-letter with replay) or by a dedicated Kafka topic with ACLs.

    - Notes: ordering per aggregate id; replay from a timestamp reproduces state when consumed idempotently; payload schema from the schema registry.
- **FR-35.10 Message monitor and replay** — Must. The system shall show every interchange/message with status, acks, errors and payloads; allow reprocessing inbound from raw, resending outbound, and SLA dashboards per partner.
    - Notes: payload retention 13 months (configurable); PII fields masked for non-admin duties.
- **FR-35.11 Reference connectors** — Should. The system shall ship connectors: Salesforce (accounts/opportunities ↔ parties/quotes), Shopify (orders/fulfillments/products), Stripe (payments/payouts into cash application), each built only on the public APIs and webhooks.
    - Notes: connectors are separate Helm charts; mapping configurable; they prove the hub, not full products.
- **FR-35.12 Partner simulator** — Must. The system shall include a partner simulator (AS2/SFTP endpoint plus document generator) used in CI and the demo distributor scenario.
    - Notes: generates 850/856/810 and ORDERS/DESADV/INVOIC with randomized but valid content.
- **FR-35.13 Catalog punchout (cXML and OCI)** — Should. The system shall let a requisitioner open a supplier's external catalog from a wave-17 requisition via cXML PunchOutSetupRequest or SAP OCI 4.0/5.0 (HOOK_URL form post), receive the returned cart (cXML PunchOutOrderMessage or OCI NEW_ITEM-* fields), and create requisition lines with supplier part id, description, price, currency, UoM, UNSPSC code and lead time.
    - Notes: supplier configuration holds URL, identity and shared secret (Vault), extrinsics (user email, cost center) and UoM/category mapping; validation per BR-35.13; session timeout 30 min; item-level (level-2) punchout via PunchOutSetupRequest with `ItemOut`; lines unknown to the item master become non-stock lines with a procurement category; catalog prices are flagged as catalog-sourced for wave-17 approval rules.
- **FR-35.14 cXML document exchange** — Should. The system shall exchange cXML 1.2 documents with cXML networks: outbound OrderRequest from wave-17 POs, inbound ConfirmationRequest (→ PO confirmation), ShipNoticeRequest (→ ASN for FR-31.9) and InvoiceDetailRequest (→ wave-19 vendor invoice), over HTTPS POST with shared-secret or certificate authentication and synchronous cXML Response status codes.
    - Notes: cXML maps sit on the same canonical documents as X12/EDIFACT (FR-35.3), so posting, monitoring and replay are shared; `payloadID` is the idempotency key (BR-35.14); responses 200/201 accepted, 4xx rejected with status text shown in the message monitor.
- **FR-35.15 PLM reference connector** — Could. The system shall ship a reference connector that imports released items, engineering BOMs and engineering change orders from a PLM system (Siemens Teamcenter REST services, PTC Windchill REST/OData services, or a neutral JSON/PLMXML file drop) into wave-9 engineering change requests, and writes ERP release status and effectivity back to the PLM system.
    - Notes: built only on public APIs and webhooks, like FR-35.11 connectors; PLM part number + revision map to item + BOM version; scheduled sync (default 15 min) plus event-driven updates; an ECO that targets a BOM version already released in Passage creates a task instead of overwriting; precedence per BR-35.16.
- **FR-35.16 Customer self-billing** — Could. The system shall receive customer self-billing documents (EDIFACT INVOIC with BGM 389 self-billed invoice, VDA 4908 credit notes, X12 810 flagged as self-billed by partner profile), match them line by line to Passage shipments and invoices (wave 29) by delivery note/ASN number, PO line, item and quantity, post matched documents as the legal invoice, and route differences to a dispute worklist (BR-35.15).
    - Notes: enabled per customer with a self-billing agreement reference and validity dates (EU VAT Directive Art. 224); tolerances per customer (default quantity 0, price 0.01 per unit); the customer's document number is stored as the legal invoice number; wave-29 billing rule `SELF_BILLED` suppresses Passage's own invoice for those shipments; a shipment not self-billed after `self_billing_wait_days` (default 30) creates a task; remittance follows through 820/cash application (wave 30).

6.35.4 Data model

TradingPartner / EDIProfile

Field	Type	Required	Notes
partner_id	uuid	yes	PK
party_id	uuid	yes	FK Party (wave 7)
isa_qualifier, isa_id, gs_id	text	no	X12 identifiers
unb_qualifier, unb_id	text	no	EDIFACT
peppol_participant_id	text	no	`scheme:id`
transports	jsonb	yes	[{type AS2/SFTP/HTTPS, url/host, cert_ref, mdn_mode, poll_interval}]
maps	jsonb	yes	[{transaction_set, direction, map_id, version, overrides}]
ack_sla_minutes	int	yes	default 60
environment	enum	yes	TEST, PROD

Interchange / Message

Field	Type	Required	Notes
interchange_id	uuid	yes	PK
partner_id, direction	uuid/enum	yes	IN/OUT
standard, version	enum/text	yes	X12 004010 / EDIFACT D96A / UBL 2.1
control_number	text	yes	unique per partner/direction/standard
raw_payload_ref	text	yes	object storage key (MinIO/S3-compatible, in-cluster)
received_at / sent_at	timestamptz	yes	
ack_status	enum	yes	PENDING, ACCEPTED, ACCEPTED_WITH_ERRORS, REJECTED, TIMEOUT
messages	—	—	Message rows: message_id, transaction_set, st_control, canonical_ref (document id), status (RECEIVED, MAPPED, POSTED, FAILED, SENT, ACKED), error

Subscription / DeliveryAttempt

Field	Type	Required	Notes
subscription_id	uuid	yes	PK
client_id	text	yes	Keycloak client
kind	enum	yes	WEBHOOK, KAFKA
event_types	text[]	yes	CloudEvents type patterns
filter	jsonb	no	CESQL-style expression
endpoint_url, secret_ref	text	no	webhook

Field	Type	Required	Notes
topic	text	no	Kafka `hub.<tenant>.<client>`
status	enum	yes	ACTIVE, PAUSED, DEAD_LETTERED
attempt: attempt_id, event_id, attempt_no, status_code, latency_ms, next_retry_at	—	—	DeliveryAttempt

ApiClient: client_id, scopes[], quota_per_min, burst, sandbox_tenant_id, owner contact, status. **EInvoice**: einvoice_id, invoice_id (wave 29/19), format (PEPPOL_UBL/FACTURX/CII/MANDATE), xml_ref, validation_report, transmission_id, clearance_id, status (DRAFT/VALIDATED/SENT/ACCEPTED/REJECTED).

Invariants: control numbers are strictly increasing without gaps per partner/direction; a Message in POSTED references exactly one Passage document; every outbound webhook event has ≥ 1 DeliveryAttempt or is DEAD_LETTERED; an EInvoice SENT has a passing validation report. Indexing: `interchange(partner_id, direction, control_number)` unique; `message(status, received_at)`; `delivery_attempt(subscription_id, next_retry_at) WHERE status = 'RETRY'`.

6.35.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/edi/v1/partners	Create partner profile	key	EDI_ADMIN
POST	/edi/v1/inbound/as2	AS2 receive endpoint (RFC 4130)	AS2 Message-ID	partner cert
POST	/edi/v1/interchanges/{id}:reprocess	Re-run mapping/posting from raw	key	EDI_OPERATE
POST	/edi/v1/messages/{id}:resend	Re-transmit outbound	key	EDI_OPERATE
POST	/edi/v1/maps:test	Run a map against a sample payload	—	EDI_ADMIN
POST	/einvoice/v1/invoices/{id}:validate	Build UBL/CII and run schematron	—	EINV_OPERATE
POST	/einvoice/v1/invoices/{id}:send	Transmit via AP or mandate provider	key = invoice id + format	EINV_SEND
POST	/hub/v1/clients	Register API client, scopes, quota	key	HUB_ADMIN
POST	/hub/v1/subscriptions	Create webhook/Kafka subscription	key	HUB_CLIENT
POST	/hub/v1/subscriptions/{id}:replay?from=	Replay events from timestamp	key	HUB_CLIENT
GET	/hub/v1/catalog	API/AsyncAPI catalog (Backstage-backed)	—	public (authenticated)
POST	/graphql	GraphQL endpoint	per mutation key	scopes
POST	/hub/v1/punchout/sessions	Start a punchout session (cXML or OCI)	key	PUR_REQUISITION_CREATE
POST	/hub/v1/punchout/return/{session}	Receive cart (PunchOutOrderMessage or OCI form post)	session token	session-bound
POST	/edi/v1/inbound/cxml	Receive cXML documents	payloadID	partner shared secret or certificate
POST	/edi/v1/self-billing/{id}:match	Re-run self-billing matching	key	SELF_BILLING_RECONCILE
POST	/hub/v1/connectors/{id}:sync	Trigger a connector sync (PLM, Salesforce, Shopify, Stripe)	key = connector + change token	CONNECTOR_ADMIN

Event	Trigger	Payload fields	Consumers
dev.passage.edi.message.received.v1	inbound parsed	partner_id, standard, transaction_set, control numbers, canonical_ref	purchasing, sales, AP, tms
dev.passage.edi.message.posted.v1 / failed.v1	posting outcome	message_id, document_ref, error	monitor, tasks
dev.passage.edi.message.sent.v1 / acked.v1	transmit / ack	message_id, partner_id, ack_status	analytics
dev.passage.einvoice.status.changed.v1	validation/transmission	einvoice_id, invoice_id, status, clearance_id	AR/AP, customer portal
dev.passage.hub.webhook.delivered.v1 / deadlettered.v1	delivery	subscription_id, event_id, attempts	analytics, alerting

6.35.6 User experience

Screen	Primary actions	Key fields	Duty
Partner profiles	Create, transports, certificates, maps, test	identifiers, transport, map versions, SLA	EDI_ADMIN
Map designer	Edit expressions, test with samples, diff versions	segment, element, expression, lookup	EDI_ADMIN
Message monitor	Filter, view raw/canonical, reprocess, resend	partner, set, control no, status, ack, age	EDI_OPERATE
Partner SLA dashboard	Ack latency, failure rate, volume	partner, p95 ack, failures	EDI_OPERATE
e-Invoice workspace	Validate, send, view clearance, errors	invoice, format, status, schematron errors	EINV_OPERATE
Developer portal (external)	Browse APIs/events, create keys, test console, subscriptions, usage	OpenAPI, AsyncAPI, quotas, attempts	HUB_CLIENT
Hub admin	Clients, quotas, subscriptions, dead letters	client, scopes, quota, DLQ count	HUB_ADMIN

Interaction rules: monitor supports bulk reprocess (≤ 200) with a dry run; raw payloads open in a segment-tree viewer with element hover showing the X12 dictionary name; developer portal "try it" executes against the client's sandbox tenant only.

6.35.7 Business rules and edge cases

- **BR-35.1 X12 850 inbound mapping (004010) → canonical Order.**

Segment/element	Canonical field	Rule
ISA06/ISA08, GS02/GS03	partner resolution	must match profile; else reject interchange (TA1/997 R)
BEG01	purpose	00 original, 01 cancel, 04 change → create/cancel/change order
BEG02	order_type	SA standalone, BK blanket, RL release
BEG03	customer_po	required; duplicate per customer → reject with 997 AK5 R
BEG05	po_date	CCYYMMDD
CUR02	currency	ISO 4217; default partner currency
REF*DP / REF*A	department / vendor id	optional
DTM*002 / DTM*010	requested_delivery / requested_ship	DTM02 CCYYMMDD
N1*ST, N3, N4	ship_to (GLN from N104 qualifier 92/UL)	resolve to party address; unknown → task
N1*BT	bill_to	
PO101	line_no	

Segment/element	Canonical field	Rule
PO102, PO103	qty, uom (EA, CA, LB…)	UoM via qualifier table
PO104, PO105	unit_price, basis (PE per each, TP per thousand)	price verified against wave-26 pricing; variance tolerance 1% else hold
PO106/07 pairs	item cross-reference	BP buyer part → partner item xref; VP vendor part → item; UP/EN → GTIN; first match wins in order BP, VP, UP, EN
PID*F	line description	informational
SAC (loop)	charges/allowances	C=charge, A=allowance; SAC05 amount N2 (implied 2 decimals)
CTT01	line count	must equal PO1 count else reject

- **BR-35.2 X12 856 outbound mapping (004010, structure S/O/P/I) from `wms.load.shipped`.**

Segment/element	Source	Rule
BSN01/02/03/04/05	00, shipment id (load+shipment), ship date, time, 0001	BSN05 hierarchy S-O-P-I
HL*1**S	shipment	
TD1	packaging code (PLT94), lading qty, weight (G, LB/KG)	from load totals
TD5	SCAC (TD503), mode (TD504 M motor)	from tms carrier
REF*BM, REF*CN	bill of lading, PRO number	from tms load
DTM*011	ship date	
N1*ST / N1*SF	ship-to / ship-from with GLN	qualifier 92 or UL
HL**1*O	order	PRF01 customer PO, PRF04 PO date
HL**2*P	pack (pallet LP)	MAN*GM SSCC-18 (AI 00 without AI)
HL**3*I	item	LIN (VP item, UP GTIN, BP buyer part), SN101 line, SN102 qty shipped, SN103 uom
CTT01	number of HL segments	

- **BR-35.3 X12 810 outbound mapping (004010) from the wave-29 customer invoice.**

Segment/element	Source	Rule
BIG01, BIG02, BIG03, BIG04	invoice date, invoice no, PO date, customer PO	BIG07 type DI (debit) / CR (credit note)
REF*BM, REF*CO	BOL, customer order	optional
N1*RE / N1*BT / N1*ST	remit-to / bill-to / ship-to	
ITD	terms (ITD01 code, ITD03 discount %, ITD05 discount days, ITD07 net days)	from wave-29 terms
DTM*011	ship date	
IT1	IT101 line, IT102 qty, IT103 uom, IT104 unit price, IT106/07 BP/VP	
TXI	tax type (TXI01 ST/VA/GS), amount, rate	per tax line (wave 13)
SAC	freight (SAC02 D240), allowances	charge lines
TDS01	total invoice amount, N2	equals invoice total × 100
CAD	carrier, PRO	optional
CTT01	line count	

EDIFACT ORDERS/DESADV/INVOIC maps follow the same canonical model (BGM/DTM/NAD/LIN/QTY/PRI/MOA/CPS/PAC/GIN SSCC) and are shipped as fixtures.

- **BR-35.4 Peppol BIS Billing 3.0 required elements (UBL 2.1 Invoice).** `CustomizationID = urn:cen.eu:en16931:2017#compliant#urn:fdc:peppol.eu:2017:poacc:billing:3.0`; `ProfileID = urn:fdc:peppol.eu:2017:poacc:billing:01:1.0`; `ID`; `IssueDate`; `DueDate` or `PaymentTerms`; `InvoiceTypeCode` (380 invoice, 381 credit note as `CreditNote` document); `DocumentCurrencyCode`; `BuyerReference` or `OrderReference/ID` (one required); `AccountingSupplierParty/Party`: `EndpointID @schemeID` (e.g., 0088), `PartyIdentification` (optional), `PartyName`, `PostalAddress/Country/IdentificationCode`, `PartyTaxScheme/CompanyID` with `TaxScheme/ID = VAT` when VAT applies, `PartyLegalEntity/RegistrationName`; `AccountingCustomerParty/Party` (same set, `EndpointID` required); `PaymentMeans/PaymentMeansCode` (30 credit transfer, 58 SEPA) with `PayeeFinancialAccount/ID` (IBAN); `TaxTotal/TaxAmount @currencyID`; `TaxTotal/TaxSubtotal` per category: `TaxableAmount`, `TaxAmount`, `TaxCategory/ID` (S, Z, E, AE, K, G, O), `Percent`, `TaxScheme/ID = VAT`; `LegalMonetaryTotal`: `LineExtensionAmount`, `TaxExclusiveAmount`, `TaxInclusiveAmount`, `PayableAmount` (and `AllowanceTotalAmount`/`ChargeTotalAmount` when present); ≥ 1 `InvoiceLine`: `ID`, `InvoicedQuantity @unitCode` (UN/ECE Rec 20, e.g., C62, KGM), `LineExtensionAmount`, `Item/Name`, `Item/ClassifiedTaxCategory/ID` + `Percent` + `TaxScheme`, `Price/PriceAmount`. Arithmetic rules enforced before schematron: Σ line `LineExtensionAmount` = `LineExtensionAmount`; `TaxExclusiveAmount` = that – allowances + charges; `TaxInclusiveAmount` = exclusive + `TaxAmount`; `PayableAmount` = inclusive – prepaid + rounding; amounts to 2 decimals; VAT breakdown per category/rate equals Σ lines. Validation runs CEN-EN16931-UBL then PEPPOL-EN16931-UBL schematron; any `fatal` blocks sending.
- **BR-35.5 Control numbers and duplicates.** Outbound ISA13/GS06/ST02 per partner from a gap-free number sequence (wave 5); inbound duplicate ISA13 within 30 days → TA1 code 025 / reject; a 997 is generated within 60 s for every inbound functional group; missing partner 997 after `ack_sla_minutes` → alert and resend once.
- **BR-35.6 Idempotent posting.** Inbound 850 posting uses idempotency key `partner:BEG03:BEG01`; a re-sent original (BEG01 00) with the same PO number returns the existing order and is acknowledged accepted; a change (BEG01 04) applies as a wave-26 order change with version.
- **BR-35.7 Webhook delivery.** Retries at 10 s, 1 m, 5 m, 30 m, 2 h, 6 h, 12 h, 24 h; HTTP 2xx = success; 410 Gone pauses the subscription; signature header `Passage-Signature: t=<unix>,v1=<hmac>` over `t.body`; receiver must verify within 5 min of `t`. Events per aggregate are delivered in order; a failing event blocks later events for the same aggregate only (per-aggregate queue).
- **BR-35.8 Quotas.** Token bucket per client (rate/min, burst); exceeding returns 429 with `Retry-After`; quota consumption is counted per request and per 100 GraphQL complexity points.
- **BR-35.9 Replay.** Replay from timestamp T re-delivers events with `ce-passagereplay: true`; consumers must treat them idempotently; replay is capped at 1,000,000 events per request and runs as a Temporal workflow with progress.
- **BR-35.10 e-Invoice status.** `DRAFT → VALIDATED → SENT → ACCEPTED|REJECTED`; REJECTED allows correction and a new transmission with the same invoice id and a new `transmission_id`; an ACCEPTED invoice cannot be re-sent, only credited.
- **BR-35.11 AS2 MDN.** Async MDN awaited 1 h, then resend up to 5×, then FAILED with alert; MDN signature mismatch is FAILED immediately.
- **BR-35.12 PII masking** in the monitor hides N1/N3/N4 address lines and party contact elements for users without `EDI_PII_VIEW`.
- **BR-35.13 Punchout return validation.** Each returned line needs a supplier part id, quantity > 0, unit price ≥ 0 in a currency allowed for the supplier, and a UoM mapped through the UN/ECE Recommendation 20 table; failing lines are dropped with EDI-0133 and the rest are created; the cart total is recomputed as Σ quantity × price and compared with the message total; a difference > 0.01 rejects the whole cart with EDI-0134; a price above catalog price × (1 + `punchout_price_tolerance_pct`) flags the line for approval.

- **BR-35.14 cXML idempotency.** Inbound `payloadID` is the idempotency key per partner; a repeat with identical content returns the original response; a repeat with different content is rejected with EDI-0135 and logged.
- **BR-35.15 Self-billing matching.** Key = (customer, delivery note/ASN number, PO line, item). Quantity difference > tolerance or price difference > tolerance → DIFFERENCE with reason; a self-billed line without a matching shipment → UNMATCHED; matched documents post with the customer's number as the external (legal) invoice number and Passage's internal number as reference; VAT is recomputed by the wave-13 tax engine and a difference > 0.01 is a DIFFERENCE.
- **BR-35.16 PLM import precedence.** An imported ECO with effectivity earlier than today is moved to today with EDI-0139; revisions are ordered by the PLM revision scheme; importing the same part + revision twice is idempotent.

6.35.8 Acceptance criteria

- **AC-35.1** (FR-35.2, FR-35.3, FR-35.4, FR-35.12) Given the partner simulator, when it sends 850, 855 is returned, 856 and 810 are produced after shipment and invoicing, and the simulator's 997s are matched, then every supported transaction set round-trips with no manual step and the message monitor shows ACKED for each.
- **AC-35.2** (FR-35.6, BR-35.4) Given 50 demo invoices (standard, reverse-charge, zero-rated, with allowances), when UBL is built, then all pass CEN and Peppol schematron with zero fatal errors and the amounts rules hold.
- **AC-35.3** (FR-35.9, BR-35.7) Given a webhook endpoint that fails 3 times then succeeds, when an event is published, then exactly one successful delivery is recorded after 3 retries and ordering per aggregate is preserved.
- **AC-35.4** (FR-35.9, BR-35.9) Given a consumer that rebuilds a projection from scratch, when events are replayed from the tenant's start, then the projection equals the current Passage state for the sales order aggregate fixture.
- **AC-35.5** (BR-35.5) Given an inbound interchange with an ISA13 already seen 10 days ago, when received, then it is rejected with TA1 025 and no order is created.
- **AC-35.6** (FR-35.8, BR-35.8) Given a client with quota 600/min, when 700 requests are sent in a minute, then ≥ 100 receive 429 with `Retry-After` and the portal usage chart shows the burst.
- **AC-35.7** (BR-35.1) Given an 850 with PO104 differing from the wave-26 price by 3%, when posted, then the sales order is created on price hold with a task.
- **AC-35.8** (BR-35.2) Given a shipped load with 2 pallets × 3 cases for 2 orders, when the 856 is generated, then HL levels are S(1) O(2) P(2) I(n) with MAN*GM SSCCs equal to the wms LP ids and CTT01 equals the HL count.
- **AC-35.9** (phase exit) Given the demo distributor and manufacturer tenants, when the distributor sends an 850, then the manufacturer ships and invoices and the distributor's AP shows a matched vendor invoice from the 810 and the Peppol UBL, both within 15 minutes of each event.
- **AC-35.10** (FR-35.13, BR-35.13) Given a cXML punchout simulator returning 3 items with one unmapped UoM, when the cart returns, then 2 requisition lines are created with the catalog flag and the third is reported with EDI-0133.
- **AC-35.11** (FR-35.13) Given an OCI 5.0 catalog, when the HOOK_URL form post returns NEW_ITEM-DESCRIPTION[1] and [2] with prices and units, then 2 requisition lines carry those values.
- **AC-35.12** (FR-35.14, BR-35.14) Given a PO sent as cXML OrderRequest to the network simulator, when ConfirmationRequest, ShipNoticeRequest and InvoiceDetailRequest return, then the PO is confirmed, an ASN is available for receiving, and a vendor invoice is created and 3-way matched; replaying the invoice payloadID creates nothing new.
- **AC-35.13** (FR-35.16, BR-35.15) Given 10 shipments to a self-billing customer and an INVOIC BGM 389 covering 9 of them with one price difference of 0.05 per unit, when processed, then 8 lines settle, 1 goes to the dispute worklist, and after 30 days the unbilled shipment raises EDI-0138.
- **AC-35.14** (FR-35.15, BR-35.16) Given a PLM simulator releasing part P revision B with a 3-line BOM, when the connector syncs, then a wave-9 ECR with that BOM version exists and the release status is written back; a second sync of the same revision changes nothing.

6.35.9 Test plan and fixtures

Fixtures: `x12-samples` (004010 850/855/856/810/820/214/204/990/997, 20 variants each from partner implementation guides), `edifact-samples` (ORDERS/ORDRSP/DESADV/INVOIC D96A), `peppol-invoices-50` (UBL built from wave-29 invoices, including all tax category codes), `factur-x-10.pdf`, `partner-simulator` config for the demo distributor, `webhook-chaos` (failing endpoint profiles), `replay-salesorder-projection`.

Test	Type	Target	Pass criterion
Parser/serializer	unit/property	X12/EDIFACT	round-trip byte-equal after normalization
Map tests	unit	each map	canonical equals expected JSON
Round-trip	E2E (simulator)	every transaction set	ACKED, documents posted
Schematron	contract	Peppol UBL	0 fatal on 50 invoices; known-bad fixtures produce expected rule ids
Webhook delivery	property	retries/ordering	at-least-once, ordered per aggregate
Replay	E2E	projection rebuild	equals current state
Quota	performance (k6)	gateway	429 behavior, p95 overhead < 5 ms
AS2 interop	contract	mendelson/OpenAS2	MDN exchanged signed+encrypted

Reconciliation: inbound 850 count = sales orders created + rejected + held; outbound 810 count = invoices posted with EDI partners; Peppol SENT = invoices flagged e-invoice for Peppol customers; webhook events published = delivered + dead-lettered.

6.35.10 Deployment and operations

Helm `passage-edi` (api, as2 receiver behind Envoy with client-cert passthrough, sftp poller, mapper workers), `passage-einvoice`, `passage-integration-hub` (gateway policies on Envoy Gateway for keys/quotas, webhook dispatcher, Kafka subscription provisioner using Strimzi `KafkaUser`/`KafkaTopic` CRs), `passage-devportal` (Backstage with the catalog plugin, OpenAPI/AsyncAPI docs rendered from the service registry), `passage-partner-simulator` (non-prod only), connector charts `passage-connector-{salesforce,shopify,stripe}`. CloudNativePG `integration-db` (3 instances, 200 GB; payloads in in-cluster object storage, 13-month retention). Config: `EDI_ACK_SLA_MIN=60`, `EDI_DUP_WINDOW_DAYS=30`, `HUB_DEFAULT_QUOTA=600`, `HUB_WEBHOOK_RETRIES=8`, `EINV_AP_PROVIDER=<adapter>`. Flags: `edi.enabled` per partner, `einvoice.peppol`, `einvoice.facturx`, `hub.graphql`, `hub.replay`. Scheduled: SFTP poll 60 s, ack SLA sweep 5 min, DLQ report daily, certificate expiry check daily (alert 30 days before). SLOs: inbound 850 to sales order p95 < 30 s; outbound document generated within 60 s of the source event; 997 within 60 s; webhook first attempt within 5 s of event p95; gateway overhead p95 < 5 ms; availability 99.9%. Alerts: ack SLA breaches; AS2 MDN failures; DLQ depth > 100; certificate expiring; schematron failure rate > 5%. Runbook: partner certificate rotation; reprocess failed interchanges after a map fix; drain and replay DLQ; rotate client secrets. Rollback: per-partner flag off (queue inbound, do not process); map version pin to the previous version; hub gateway policies are versioned in Git with Argo CD rollback.

6.35.11 Migration from AX / SAP

Source	Passage entity	Transformation
AifPort/AifInboundPort/AifOutboundPort, AifService/AifAction	TradingPartner transports, map assignments	port → transport; service/action → transaction set/direction
AifDocumentLog, AifMessageLog (last 13 months)	Interchange/Message history (read-only)	payloads copied to object storage
ERFormatMappingTable/ERModelMappingTable (e-invoice configurations)	EInvoice format config	ER formats are not executable in Passage; only the country/format selection and customer flags migrate
Dual-write entity maps	Connector mappings (Salesforce reference)	field map table only
BusinessEventsEndpoint/Catalog	Subscription	endpoint type → WEBHOOK
EDPP1, EDP12/EDP13/EDP21 (WE20)	TradingPartner, maps	partner type LI/KU → vendor/customer party; message type ORDERS/DESADV/INVOIC → maps
EDIPOA/port definitions (WE21), TBD05 (BD64)	Transports, subscriptions	file ports → SFTP; tRFC ports → HTTPS webhook
EDIDC/EDID4/EDIDS (last 13 months, statuses 03/53/51)	Interchange/Message history	EDI_DC40 control fields (DOCNUM, IDOCTYP, MESTYP, SNDPRN, RCVPRN, STATUS) → message columns; EDIDD SDATA kept raw
EDOCUMENT/EDOCUMENTFILE (DRC)	EInvoice with clearance ids	status mapping; XML kept as reference
PI/PO integration directory (ICO)	Partner/transport inventory	exported as a spreadsheet for manual re-creation; routing logic not migrated

Not migrated: AIF transforms (XSLT/.NET), ABAP user exits in IDoc processing, PI/PO mappings (Java/graphical) — mapping logic is re-authored in Passage maps; IDoc history older than 13 months (lakehouse archive).

6.35.12 Dependencies, risks, open questions

Depends on: wave 34 (204/990/214 and freight invoices), wave 31 (SSCC/LPs for 856), wave 30 (cash application for 820), wave 29 (customer invoices), wave 26 (sales order intake and change), wave 19 (vendor invoices), wave 17 (purchase orders for outbound 850), wave 13 (tax categories for UBL), wave 7 (parties, GLNs), wave 5 (gap-free sequences), wave 4 (outbox, schema registry), wave 3 (entity APIs).

Risk	Likelihood	Impact	Mitigation
Peppol access point certification/onboarding timeline	Medium	Medium	Certified third-party AP adapter; two providers evaluated
Partner implementation guides require per-partner map work at every customer	High	Medium	Override layer, simulator, map test harness; estimate 1–3 days per partner
Public API surface freezes the data model early	Medium	High	Additive-only versioning; v1 scope limited to entities stable since wave 30
Licensing decision (PRD §12 Q1) affects developer portal terms	Medium	Low	Portal terms page templated; decision required before wave 35 close

Open questions: (1) Which certified Peppol AP provider for the reference adapter? Proposed: evaluate two, pick by wave 35 start. (2) GraphQL as first-class or REST-only for v1 public APIs? Proposed: both, GraphQL read-only at first; decide by wave 35 design review. (3) Where do connector products live long-term (core repo vs marketplace)? Proposed: separate repos under the hub; decide by wave 46.

6.35.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
edi.enabled	trading partner	bool	false	true / false	Processes the partner's traffic (rollback queues inbound)	AIF inbound/outbound port "Enabled"	WE20 partner profile, WE21 port
ack_sla_minutes	trading partner	int	60	5–1,440	997/CONTRL expectation (BR-35.5)	n/a	ALEAUD/STATUS message settings
dup_window_days	tenant	int	30	7–365	Duplicate interchange window (BR-35.5)	n/a	EDI subsystem duplicate check
x12_version	partner × transaction set	enum	004010	004010 / 005010	Parser/serializer version (FR-35.2)	n/a (ISV)	Message type version in WE20
delimiters	trading partner	jsonb	element `*`, segment `~`, component `>`, repetition `^`	any non-alphanumeric	Envelope syntax	n/a	EDI subsystem settings
as2_mdn_mode	trading partner	enum	SYNC	SYNC / ASYNC / NONE	MDN handling (FR-35.5)	n/a	AS2 adapter channel (B2B add-on)
as2_mdn_timeout_min	trading partner	int	60	5–1,440	Resend trigger (BR-35.11)	n/a	AS2 channel timeout
sftp_poll_s	trading partner	int	60	15–3,600	Inbound poll (FR-35.5)	Recurring integration recurrence	File port / file adapter poll interval
payload_retention_months	tenant	int	13	3–120	Payload retention (FR-35.10)	AIF message log cleanup	IDoc archiving (SARA object IDOC)
pii_masking	tenant	bool	true	true / false	Masks N1/N3/N4 (BR-35.12)	n/a	n/a
peppol_ap_provider	legal entity	adapter id	none	certified AP adapters	Peppol transport (FR-35.6)	Electronic invoicing parameters (Invoicing service)	DRC interface connector (Peppol exchange)
einvoice_formats	legal entity × customer	set	{PEPPOL_UBL}	PEPPOL_UBL / FACTURX / CII / MANDATE	Formats generated (FR-35.6, FR-35.7)	Customer e-invoice settings + ER format configuration	DRC eDocument types per country
schematron_version	tenant	reference data	current Peppol BIS 3.0 release	versioned	Validation rules (BR-35.4)	ER configuration version	DRC note-based updates
hub_default_quota_per_min / hub_burst	API client	int	600 / 100	10–100,000	Token bucket (BR-35.8)	n/a	API Management rate-limit policy
webhook_retries	subscription	int	8	0–12	Retry schedule length (BR-35.7)	Business events endpoint retries	Event Mesh retry settings
replay_max_events	tenant	int	1,000,000	1–10,000,000	Replay cap (BR-35.9)	n/a	n/a

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
api_deprecation_months	tenant	int	12	≥ 12	Sunset period (FR-35.8)	n/a	n/a
punchout_session_timeout_min	supplier	int	30	5–120	Cart return window (FR-35.13)	External catalog session	OCI catalog call settings
punchout_price_tolerance_pct	supplier	numeric(5,2)	0	0–20	Approval flag threshold (BR-35.13)	n/a	Requisition price check
cxml_auth	trading partner	enum	SHARED_SECRET	SHARED_SECRET / CERTIFICATE	cXML sender credential (FR-35.14)	n/a	Ariba Network / CIG authentication
self_billing_enabled	customer	bool	false	true / false	Suppresses own invoice; expects self-billing (FR-35.16)	n/a	SPRO SD › EDI › Self-Billing Procedure › Control Self-Billing Procedure
self_billing_tolerance	customer	qty / price	0 / 0.01 per unit	≥ 0	DIFFERENCE threshold (BR-35.15)	n/a	Self-billing tolerance settings
self_billing_wait_days	customer	int	30	1–120	Unbilled shipment task (FR-35.16)	n/a	n/a
plm_sync_interval_min	connector	int	15	1–1,440	Scheduled PLM sync (FR-35.15)	Engineering PLM sync batch recurrence	PLM integration schedule

6.35.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Message monitor	list inquiry / drill-down	partner, transaction set, status, age, control number	raw, canonical, failing rule, acks	`interchange`, `message`	AIF exception log; data management job history	WE02 / WE05 / BD87; SXMB_MONI
Partner SLA dashboard	KPI	partner, period	p95 ack latency, failure %, volume	message metrics	n/a	AIF interface monitor (/AIF/IFMON)
Control number continuity	operational report	partner, direction	expected vs used, gaps	`interchange`	n/a	IDoc number analysis (WE05)
e-Invoice status	list inquiry	legal entity, country, status	invoice, format, transmission id, clearance id, errors	`einvoice`	Electronic invoicing submission log	EDOC_COCKPIT
Webhook delivery and dead letters	list inquiry	client, subscription, status	attempts, latency, last status code	`delivery_attempt`	Business events endpoint status	Event Mesh monitor
API usage by client	KPI	client, period	requests, 429 count, p95 latency	gateway metrics	n/a	API Management analytics
Punchout session log	list inquiry	supplier, user, date	session, lines returned, rejected, total	`punchout_session`	n/a	OCI shopping cart log
cXML document log	list inquiry	network, document type, status	payloadID, status code, linked document	`message`	n/a	CIG / Ariba Network message monitor
Self-billing reconciliation	operational report	customer, period	matched, DIFFERENCE, UNMATCHED, unbilled shipments	`self_billing_document`	n/a	WE02 filtered on message type GSVERF; self-billing analysis
Connector sync log	list inquiry	connector, status	objects synced, errors, change token	`connector_config` runs	Dual-write / PLM sync log	Integration Suite message processing log

6.35.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
X12 / EDIFACT interchanges (850, 855, 856, 810, 820, 214, 204, 990, 997; ORDERS, ORDRSP, DESADV, INVOIC, CONTRL)	Source events (outbox) and inbound receipt	ISA/GS/ST or UNB/UNH envelopes; maps per BR-35.1–BR-35.3	EDI	ISA13 / GS06 / ST02 gap-free per partner and direction	Payloads 13 months (configurable); invoice interchanges archived as tax records (wave 46)
Peppol UBL invoice and credit note	Invoice posted for a Peppol customer	BR-35.4 elements	UBL 2.1 XML via access point	Wave-29 invoice number	Original XML archived: DE 8 years (AO §147 as amended 2025), FR and IT 10 years
Factur-X / ZUGFeRD invoice	Invoice posted for a flagged customer	PDF/A-3 with embedded EN 16931 CII XML	PDF	Wave-29 invoice number	As Peppol invoices
Mandate submissions (FatturaPA, CFDI 4.0)	Invoice posted in mandate country	Country schema	XML via `MandateProvider`	Clearance id from authority	Country pack (wave 48)
AS2 MDN receipts	Every AS2 message	Message-ID, MIC, disposition	S/MIME	none	Kept with the payload as non-repudiation evidence
cXML documents	PO, confirmation, ship notice, invoice	cXML 1.2 envelope and payload	XML	payloadID	As EDI payloads
Self-billed invoice copy	Self-billing document posted	Customer's invoice number, "self-billing" mention, lines, VAT breakdown	PDF render + original EDI	Customer's number (legal); internal reference	Supplier keeps as its sales invoice: 10 years default

6.35.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
SFTP poll	Every 60 s	trading partner	file name + SHA-256	< 10 s	Retry 5× exponential (max 1 h)	Poll failures > 3 consecutive
Ack SLA sweep	Every 5 min	trading partner	interchange id	< 30 s	Resend once (BR-35.5)	Ack SLA breaches
Webhook retry dispatcher	Continuous on `next_retry_at` index	subscription	event id + attempt number	First attempt < 5 s p95	Dead-letter after 8 attempts	DLQ depth > 100
DLQ daily report	Daily 07:00	API client	date	< 1 min	Rerun	none
Certificate expiry check	Daily 05:00	partner certificate	certificate id	< 1 min	Rerun	Expiry within 30 days
e-Invoice status poll	Every 5 min	provider	transmission id	< 1 min	Back-off on provider errors	Invoices SENT > 24 h without status

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Payload retention purge	Daily 01:00	tenant × month partition	partition id	< 10 min	Retry next day	Partitions past retention
PLM sync	Every 15 min	connector	PLM change token	< 2 min	Resume from token	Sync failures > 3 consecutive
Self-billing unbilled check	Daily 06:00	customer	shipment id	< 2 min	Rerun	Unbilled shipments past wait days
Punchout session cleanup	Hourly	supplier	session id	< 1 min	Rerun	none

6.35.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
EDI_ADMIN	/edi/v1/partners, /edi/v1/maps:test; map designer	EDI_OPERATE in production (change map and reprocess the same traffic)	Integration administrator
EDI_OPERATE	/interchanges/{id}:reprocess, /messages/{id}:resend; message monitor	none	Integration operator
EDI_PII_VIEW	Unmasked payload view (BR-35.12)	none	Integration lead
EINV_OPERATE	/einvoice/v1/invoices/{id}:validate; e-invoice workspace	none	Accounts receivable clerk
EINV_SEND	/einvoice/v1/invoices/{id}:send	none	Accounts receivable supervisor
HUB_ADMIN	/hub/v1/clients, quotas, dead-letter handling	HUB_CLIENT on the same client (self-granting scopes)	Platform administrator
HUB_CLIENT	/hub/v1/subscriptions, /subscriptions/{id}:replay; developer portal	none	External developer
PUR_REQUISITION_CREATE (wave 17)	/hub/v1/punchout/sessions	PUR_PO_APPROVE (wave 17)	Requisitioner
SELF_BILLING_RECONCILE	/edi/v1/self-billing/{id}:match; dispute worklist	AR_CREDIT_NOTE_APPROVE (wave 29)	Accounts receivable specialist
CONNECTOR_ADMIN	/hub/v1/connectors/{id}:sync; connector configuration	none	Integration administrator

6.35.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
EDI-0101	error	Interchange sender {isa06} is not configured for any partner.	Unknown sender	Create or correct the partner profile	BR-35.1
EDI-0102	error	Duplicate interchange control number {isa13} (seen {date}); TA1 025 sent.	Duplicate ISA13 within window	Partner resends with a new number	BR-35.5
EDI-0103	error	CTT01 {n} does not match PO1 count {m}.	Truncated or malformed 850	Partner resends	BR-35.1
EDI-0104	warning	Price {p} differs from list {l} by {pct} %; order on price hold.	Variance > 1 %	Sales reviews the hold	BR-35.1
EDI-0105	error	No item cross-reference for {qualifier} {value}.	Unknown buyer/vendor part or GTIN	Add cross-reference; reprocess	BR-35.1
EDI-0106	error	Ship-to GLN {gln} not found.	Unknown location	Add GLN to party address; reprocess	BR-35.1
EDI-0107	warning	No 997 from {partner} after {n} min; resent once.	Ack SLA breach	Contact partner	BR-35.5
EDI-0108	error	AS2 MDN signature mismatch for message {id}.	Certificate or tampering	Verify partner certificate	BR-35.11
EDI-0109	error	Control number gap for {partner}: expected {e}, generated {g}.	Sequence defect	Incident; do not resend until fixed	§6.35.4 invariants
EINV-0110	error	Schematron rule {rule} failed: {text}.	Invalid UBL/CII	Correct invoice data	BR-35.4
EINV-0111	error	Arithmetic check failed: TaxInclusiveAmount ≠ TaxExclusiveAmount + TaxAmount.	Rounding or line errors	Correct invoice	BR-35.4
EINV-0112	error	Invoice {no} is already accepted; issue a credit note.	Resend of ACCEPTED invoice	Credit and reissue	BR-35.10
HUB-0113	error	Quota exceeded; retry after {s} s.	Token bucket empty (429)	Back off	BR-35.8
HUB-0114	warning	Subscription {id} paused: endpoint returned 410 Gone.	Receiver gone	Update endpoint and resume	BR-35.7
HUB-0115	error	Event {event} dead-lettered after 8 attempts.	Receiver failing	Fix receiver; replay from DLQ	BR-35.7
HUB-0116	info	Replay {id} delivered {n} of {total} events.	Replay progress	None	BR-35.9
EDI-0133	warning	Punchout line {n} dropped: {reason}.	Missing part id, bad UoM or currency	Correct supplier mapping	BR-35.13
EDI-0134	error	Punchout cart total {a} differs from line sum {b}; cart rejected.	Inconsistent cart	Repeat punchout; contact supplier	BR-35.13
EDI-0135	error	cXML payloadID {id} reused with different content.	Partner defect or replay attack	Partner sends a new payloadID	BR-35.14
EDI-0136	warning	Self-billed line {line} differs: {field} {theirs} vs {ours}.	DIFFERENCE	Dispute or accept	BR-35.15
EDI-0137	error	Self-billed line {line} has no matching shipment.	UNMATCHED	Investigate with customer	BR-35.15
EDI-0138	warning	Shipment {s} not self-billed after {n} days.	Customer has not billed	Follow up with customer	FR-35.16
EDI-0139	warning	PLM ECO {eco} effectivity moved to today.	Backdated effectivity	Review with engineering	BR-35.16

6.35.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Inbound 850 → sales order	5,000 orders/day; peak 300/min (Monday batches)	2 / 30 / 60 s end to end	10 interchanges/s	Payloads ≈ 50 GB (13-month rolling)	`interchange(partner_id, direction, control_number)` unique; `message` monthly partitions	Kafka backlog; SLA stretches to ≤ 2 min; no loss
Outbound document generation	20,000 documents/day	5 / 30 / 60 s from source event	20/s	Included above	Outbox per source service	Backlog alert; order of documents per partner preserved

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
997 / CONTRL generation	One per inbound functional group	< 60 s	20/s	—	—	Acks queued with priority over outbound documents
Peppol validation	10,000 invoices/day	300 ms / 1.5 s / 2 s	20/s	UBL ≈ 73 GB at 20 KB each	`einvoice(status, created_at)`	Schematron workers scale; sending delayed, never skipped
Webhook first attempt	2 M events/day; peak 5,000/s	1 / 5 / 10 s	5,000/s	Attempts 30 days ≈ 100 GB	`delivery_attempt` daily partitions	Per-aggregate queues; slow endpoints isolated per subscription
Gateway overhead	20 M requests/day; peak 2,000 req/s	1 / 5 / 10 ms	2,000 req/s	Metrics only	—	Quotas return 429; lowest-tier clients shed first
Replay	1 M events per request	< 10 min	2,000 events/s	—	Kafka offsets by timestamp	Throttled to protect live delivery
Punchout return	500 sessions/day	300 ms / 1 s / 2 s	5/s	Sessions ≈ 0.2 GB	`punchout_session(supplier_id, created_at)`	Return queued; session timeout still applies
Self-billing match	2,000 lines/day	< 1 s per 200-line document	10 documents/s	≈ 0.5 GB	(customer, delivery note) index	Matching queued per customer

6.35.20 Localization and compliance hooks

- **e-Invoicing mandates.** Belgium: B2B Peppol mandatory from 2026-01-01. Germany: receipt of EN 16931 e-invoices required since 2025-01-01, issuance from 2027-01-01 (turnover > EUR 800,000) and 2028-01-01 for all, as XRechnung (UBL/CII) or ZUGFeRD ≥ 2.0.1 with EN 16931 profile. France: receipt from 2026-09-01 through certified platforms (PDP), issuance for large and mid-size companies from 2026-09-01 and SMEs from 2027-09-01. Poland KSeF: 2026-02-01 for large taxpayers, 2026-04-01 for others. Italy SDI (FatturaPA) and Mexico CFDI 4.0 (PAC stamping) are clearance models. EU ViDA (Directive (EU) 2025/516) introduces digital reporting for intra-EU B2B from 2030. Each lands as a country pack (wave 48) implementing `MandateProvider`; FR-35.7 stores clearance ids and statuses.
- **Self-billing.** EU VAT Directive 2006/112/EC Art. 224 requires a prior agreement and an acceptance procedure, and Art. 226(18) requires the mention "Self-billing"; Germany UStG §14(2) "Gutschrift"; automotive formats VDA 4908 and Odette INVOIC. Country packs validate the mention and agreement fields through `SelfBillingPolicy`.
- **Retention.** e-Invoices are archived in their original XML format through wave 46 (WORM storage): Germany 8 years (AO §147, amended 2025), France and Italy 10 years (Italy with conservazione sostitutiva).
- **Privacy.** Payloads contain personal data (names, addresses, contacts): masking (BR-35.12), 13-month payload retention, and GDPR erasure requests are refused for documents under legal retention (Art. 17(3)(b)); API clients outside the EEA require a transfer assessment recorded on the ApiClient.
- **Character sets and time.** X12 basic/extended character sets; EDIFACT UNOA/UNOB/UNOC (ISO 8859-1) and UNOW (UTF-8); UBL and cXML UTF-8; ISA09/ISA10 in sender local time; UBL IssueDate in the supplier's legal date.
- **Punchout.** Currency and UoM per supplier; taxes on punchout lines are determined later by the wave-13 tax engine, never taken from the cart.

6.35.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX GAP #53 — §12.16.1 Procurement: External catalogs (punchout cXML/OCI)	GAP (propose 35)	ADDED as FR-35.13	Lines return to the wave-17 requisition; BR-35.13, AC-35.10, AC-35.11
AX PARTIAL #285 — §12.34.1 Engineering → PLM integration (Teamcenter/Windchill connectors)	PARTIAL-35	ADDED as FR-35.15	Reference connector on public APIs; AC-35.14
AX PARTIAL #291 — §12.34.3 Engineering → Sync with PLM (batch)	PARTIAL-35	ADDED as FR-35.15	"PLM sync" job in §6.35.16
AX PARTIAL #298 — §12.34.4 Engineering → PLM connector configuration	PARTIAL-35	ADDED as FR-35.15	`plm_sync_interval_min`; ConnectorConfig
SAP #139 — §13.12 MM-PUR Ariba Network integration (cXML, Cloud Integration Gateway)	PARTIAL-35	ADDED as FR-35.14	cXML catalogs also covered by FR-35.13; BR-35.14, AC-35.12
SAP #129 — §13.12 Fiori Create Purchase Requisition with catalogs (OCI punch-out) — flagged for this wave	PARTIAL-17	ADDED as FR-35.13	Requisition UI stays wave 17; punchout transport and cart intake here
SAP #220 — §13.19 VKA / VSB1 self-billing procedure (customer self-billing) — flagged	GAP (propose 29)	ADDED as FR-35.16	Adds wave-29 billing rule `SELF_BILLED`; BR-35.15, AC-35.13

Part 7 — Phase 7: Treasury, Assets & Controlling (waves 36–40)

7.0 Phase overview

Business outcome at phase exit. At the end of wave 40 the finance function runs entirely on Passage: treasury sees every bank balance intraday and a 13-week cash forecast built from the subledgers; fixed assets and leases depreciate under multiple books and post to the universal journal; controllers plan budgets, enforce budget control on commitments, and run a cost accounting ledger with allocations; product costs roll up from BOMs and routings with a cost component split, and production variances are decomposed by cause; intercompany trade posts mirror documents across legal entities and the group consolidates with eliminations and IAS 21 translation. This is the "finance parity" milestone in PRD §10 (wave 40): a practicing controller signs off that nothing they did in AX/SAP finance is missing.

Phase-exit sign-off criteria

- 1,000-line bank statement fixture auto-matches ≥ 95% of lines correctly with zero false positive matches (wave 36 proof).
- Cash forecast total per bucket equals the sum of source contributions to the cent, verified by a property test over 10,000 generated open items (wave 36).
- 100 asset × method × convention fixtures match IRS Publication 946 tables and the straight-line edge cases to the cent; the asset subledger ties to GL asset cost and accumulated depreciation accounts with zero difference (wave 37).
- IFRS 16 lease schedules match the accountant-reviewed workbook to the cent for 25 fixture leases (wave 37).
- Budget check blocks or warns every over-budget PO per policy; encumbrance liquidation on invoice is exact to the cent (wave 38).
- Every allocation cycle converges and `total cost out = total cost in` to the cent; cost accounting ledger primary costs reconcile to GL (wave 38).
- 50 multi-level cost roll-ups match the reference workbook across all cost components; variance decomposition sums to total variance for 200 ended production orders; mass roll of 50,000 items completes in < 15 minutes (wave 39).
- Intercompany mirror documents balance across entities for 1,000 random IC orders; IC reconciliation flags 100% of planted mismatches; the 3-entity, 2-currency demo group consolidates to accountant-reviewed statements (wave 40).
- Daily subledger-to-GL reconciliation jobs for bank, assets, cost accounting, costing and intercompany run green for 10 consecutive days in `stage`.

Cross-wave data model summary

Entity	Owning service (domain DB)	Key relationships
BankAccount, BankStatement, StatementLine, ReconciliationMatch, MatchingRule	`treasury-svc` (Postgres `treasury`)	BankAccount → LegalEntity, MainAccount (wave 11); StatementLine → payment (wave 19), receipt (wave 30)
CashForecastEntry, CashPosition	`cashforecast-svc` (Python, Postgres `treasury`)	Reads AP/AR open items, bank balances, recurring entries
BankConnection, OutboundFile, InboundFile	`bank-connect-svc` (TS, Postgres `treasury`)	Keys referenced by Vault path only
Asset, AssetBook, DepreciationProfile, AssetTransaction, Lease, LeaseSchedule	`asset-svc` (Postgres `assets`)	Asset → Project (wave 41 later), PurchaseLine (wave 17), InventTrans (wave 16); posts via journal API (wave 11)
BudgetRegisterEntry, BudgetPlan, BudgetControlRule, Encumbrance, BudgetCheckResult	`budget-svc` (Postgres `controlling`)	Encumbrance → Requisition/PO (wave 17), VendorInvoice (wave 18)
CostObject, CostElement, CostEntry, AllocationPolicy, AllocationRun, StatisticalMeasure	`costacct-svc` (Postgres `controlling`)	CostEntry mirrors journal lines by dimension (wave 11)
CostingVersion, CostingSheet, CostGroup, CostCalculation, CostComponentSplit, ItemCostRecord, VarianceEntry	`costing-svc` (Postgres `costing`, extends wave 20)	Reads BOM/Route versions (wave 9), production orders (wave 21), inventory cost trans (wave 20)
IntercompanyPartner, IcDocumentLink, IcJournal, IcReconciliationItem	`interco-svc` (TS, Postgres `group`)	Links SalesOrder (wave 27) ↔ PurchaseOrder (wave 17), invoices, payments
ConsolidationGroup, ConsolidationUnit, Ownership, EliminationRule, ConsolidationRun, ConsolidationEntry	`consol-svc` (Postgres `group`)	Reads balances per ledger (wave 11), rates (wave 12), statements (wave 15)

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
`dev.passage.treasury.statement.reconciled.v1`	treasury-svc	journal (wave 11), cashforecast-svc, analytics (46)	statementId, bankAccountId, matchedCount, unmatchedCount, closingBalance
`dev.passage.treasury.cashposition.updated.v1`	cashforecast-svc	dashboards, interco-svc (netting)	legalEntityId, asOf, balancesByAccount[], forecastBuckets[]
`dev.passage.assets.depreciation.posted.v1`	asset-svc	costacct-svc, analytics	runId, bookId, period, assetCount, totalAmount, journalEntryId
`dev.passage.assets.asset.disposed.v1`	asset-svc	costacct-svc, tax (wave 13)	assetId, bookId, proceeds, nbv, gainLoss
`dev.passage.budget.check.completed.v1`	budget-svc	procurement (17), AP (18), workflow	sourceDocumentId, result (pass/warn/fail), availableBefore, amount, availableAfter
`dev.passage.controlling.allocation.completed.v1`	costacct-svc	budget-svc, analytics	runId, policyId, iterations, totalIn, totalOut, residual
`dev.passage.costing.standardcost.activated.v1`	costing-svc	inventory costing (20), production (21), budget-svc	costingVersionId, itemCount, revaluationJournalId, totalRevaluation
`dev.passage.costing.variance.posted.v1`	costing-svc	costacct-svc, analytics	productionOrderId, varianceByType{lotSize,quantity,price,substitution}, byComponent[]
`dev.passage.interco.mirror.created.v1`	interco-svc	sales (27), procurement (17), consol-svc	sourceDocId, sourceEntity, mirrorDocId, mirrorEntity, type
`dev.passage.consol.run.completed.v1`	consol-svc	reporting (15), analytics	runId, groupId, period, unitCount, eliminationCount, ctaAmount

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Bank connectivity (EBICS/Open Banking) certification differs per bank and takes weeks	High	Medium	Ship SFTP first; EBICS and PSD2 against public sandboxes; keep adapters behind one `BankChannel` interface	Engineering
Depreciation or lease math diverges from auditor expectation on edge cases (short years, mid-quarter, modifications)	Medium	High	Fixture library reviewed by the accountant; property tests that total depreciation always equals depreciable basis	Product

Risk	Likelihood	Impact	Mitigation	Owner
Budget control latency slows PO entry	Medium	Medium	p95 budget check < 150 ms SLO; per-dimension-combination balance cache rebuilt from events	Engineering
Reciprocal allocations fail to converge with circular self-references	Low	Medium	Exact linear solve for ≤ 2,000 senders; capped iteration with residual sweep rule (BR-38.6)	Engineering
Cost roll-up on cyclic or 20+ level BOMs	Medium	Medium	Cycle detection at BOM approval (wave 9) and at roll time; hard depth limit 20	Engineering
Two-person team across five accounting-heavy waves	High	High	Accountant review gate per wave; AI-assisted generation per PRD §11	Owners

7.36 Wave 36: Cash & Bank Management, Reconciliation & Treasury

7.36.1 Scope and business context

Wave 36 gives treasury a system of record for bank accounts and the daily mechanics around them. Before this wave, AP payments (wave 19) and AR receipts (wave 30) post to GL bank accounts, but nobody can import a bank statement, reconcile it, know the intraday position, or forecast cash. After it, a treasury analyst imports camt.053/BAI2/MT940 statements (or receives them automatically through SFTP/EBICS/Open Banking), the matching rule engine reconciles ≥ 95% of lines without touching them, exceptions are worked in a reconciliation workspace, and a cash position and 13-week forecast refresh from the subledgers every hour. Checks can be issued, voided and reprinted; letters of credit and bank guarantees are tracked with facility limits; intercompany netting and notional cash pooling basics are available for groups.

Out of scope: hedge accounting and derivative valuation (not in v1), in-house bank virtual accounts with interest calculation, lockbox OCR, treasury trading (money market, FX deals). Payroll (wave 43) and project (wave 41) forecast sources are consumed through the forecast source interface once those waves exist; until then recurring forecast entries stand in.

Scope amendment (gap closure). The gap registers (coverage-ax §12.99 rows 1–19, coverage-sap §13.99 rows 2–84 proposed or mapped to wave 36) assign the following to this wave, and they are added below: customer bills of exchange and vendor promissory notes (FR-36.13, FR-36.14), postdated checks (FR-36.15), cash journal / petty cash (FR-36.16), bank fee analysis from camt.086 (FR-36.17), treasury deals limited to FX spot, forward and swap plus fixed-term deposits (FR-36.18), loan contracts with interest schedules including intercompany loans (FR-36.19), FX exposure (FR-36.20), treasury payment requests and bank-to-bank transfers (FR-36.21), bank chains (FR-36.22), payment hub routing (FR-36.23) and in-house cash current accounts (FR-36.24, Could, flag-gated; this supersedes the out-of-scope note above for virtual accounts with interest). Hedge accounting (IFRS 9 / ASC 815), derivatives other than FX forwards and swaps, securities and money-market trading beyond fixed-term deposits remain out of scope for v1; their disposition is recorded in 7.36.21.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
D365 advanced bank reconciliation: statement import via DMF (BankStatementFormat + XSLT transform) fails silently on bank camt dialects; the user re-imports after editing XSLT	Transform-first design; no schema validation surfaced to the user	Native camt/BAI2/MT940 parsers with per-bank profile overrides and line-level validation errors	Failed imports needing IT help: 1 in 5 → < 1 in 200
SAP FEBAN post-processing: each unmatched line is opened, the clearing document found in FBL5N, keyed and posted (≈ 14 clicks per line)	Posting rules (OT83) only test search strings; no candidate ranking	Scored candidates shown inline; `m` to match; N:1 batch matching	Clicks per exception 14 → 2; auto-match ≥ 95%
BankReconciliationMatchingRule in D365 is pass/fail; a 1-day value-date difference leaves the line unmatched	Boolean rules, no partial credit	Weighted scoring with tolerance and date-window decay	Unmatched lines after rules: 30% → < 5%
Cash position in AX = BankAccountTable.Balance updated at posting; SAP FF7A refreshes in a batch run; both ignore intraday camt.052	Batch-oriented, no intraday feed	camt.052 ingestion on arrival; position recomputed per event	Position latency 4 h batch → < 60 s after file arrival
EBICS keys held on a Windows share for a bank client tool; audit cannot prove who sent what	Connectivity outside the ERP	`bank-connect-svc` signs in Vault; every file hashed and acknowledged	100% of files with signer, hash and bank ack in audit

7.36.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Bank account master, signatories, approval	BankAccountTable, BankAccountTrans, BankGroup, form BankAccountTable; D365 bank account change workflow	FI12 house banks T012/T012K, BAM (FCLM_BAM) with signatories and workflow	`treasury-svc` BankAccount entity, workspace Bank accounts	full
Electronic statement import	BankStmtISOAccountStatement, BankStmtISOReportEntry, BankStatementFormat, DIXF ISO 20022 transform	FF_5 (FEBKO/FEBEP), FF67 manual statement, OT83 posting rules, external transaction codes	Statement import API, formats camt.053/BAI2/MT940/CSV	improved (native XSD validation, duplicate detection)
Advanced bank reconciliation	BankReconciliationHeader, BankReconciliationMatchingRule, BankReconciliationMatchingRuleSet, BankReconciliationWorkspace	FEBAN post-processing, FEB_BSPROC, search strings (OTPM), interpretation algorithms	Matching rule engine with scoring, reconciliation workspace	improved
Check management	BankChequeTable, BankChequeReprint, check void/reverse, positive pay BankPositivePayFormat	PAYR check register, FCH5/FCH8/FCH9 (void/reissue), FCHN, FCHX	Check entity with void/reprint/stop, positive pay (wave 19 export)	full
Letters of credit / guarantees	BankLCImport, BankLCExport, BankLGTable (letters of guarantee), BankFacility, BankFacilityAgreement	No native EBS equivalent; TRM/bank guarantee via FI-TRM or custom; S/4 Trade Finance	Facility, LetterOfCredit, Guarantee entities with limit tracking	full
Cash position (intraday)	Bank balance per account only (BankAccountTable.Balance), no intraday	FF7A cash position, FQM_FLOW (One Exposure), camt.052 intraday	camt.052 ingestion, CashPosition view by entity/currency/account	improved
Cash flow forecast	LedgerCov, LedgerCovDocument, Cash flow forecasting setup, workspace Cash overview	FF7B liquidity forecast, FQM planning levels/groups, Cash Management liquidity planner	`cashforecast-svc` buckets daily/weekly/monthly from AP/AR/recurring sources	full
Intercompany netting, cash pooling	Not native (centralized payments only)	IHC In-House Cash, cash pooling (concentration) in S/4 Cash Management	Netting run creating IC settlement journals; notional pool structures	partial (no IHC virtual accounts)
Bank connectivity	None native (file-based via DMF)	Bank Communication Management (BCM), Multi-Bank Connectivity (MBC)	`bank-connect-svc` adapters SFTP/EBICS 3.0/PSD2 XS2A, Vault keys	improved

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Bills of exchange (customer)	CustBillOfExchangeJour, CustBillOfExchangeTrans, LedgerJournalTable types Customer bill of exchange draw / redraw / remittance / settle / protest, CustPaymModeTable (bill of exchange), remittance types Collection / Discount, Cash and bank management parameters → Bills of exchange	F-36 (BoE payment), F-33 (discount), F-34 (collection), FBWD / FBWE presentation, special G/L indicator W (BSID/BSED), S_ALR_87012199 BoE list	BillOfExchange entity with draw / remit / protest / settle lifecycle (FR-36.13)	full
Promissory notes (vendor)	VendPromissoryNoteJour, VendPromissoryNoteTrans, journal types Vendor promissory note draw / redraw / remittance / settle, vendor posting profile per note status	F-40 (AP bill of exchange payment), FBW6 reversal, FBWE presentation, special G/L W (BSIK), S_ALR_87012116	PromissoryNote entity (FR-36.14)	full
Postdated checks	CustPostdatedChecks / VendPostdatedChecks (R3 APAC), Cash and bank management parameters → Postdated checks tab (clearing accounts, withholding tax at settlement)	No standard object; noted item (special G/L S) or BoE used in practice	PostdatedCheck entity with maturity, deposit, bounce (FR-36.15)	improved
Cash journal / petty cash	Not native (LedgerJournal against a cash main account)	FBCJ, TCJ_DOCUMENTS / TCJ_POSITIONS / TCJ_C_JOURNALS, FBCJC0–FBCJC3 setup, S_ALR_87012309 cash book	CashDesk and CashJournalDocument with daily close (FR-36.16)	full
Bank fee analysis	None	Fiori Bank Fee Analysis / Bank Relationship Overview, camt.086 import (BAM)	BankFeeStatement import and variance to fee agreement (FR-36.17)	full
Treasury deals (FX, deposits)	None	FTR_CREATE / FTR_EDIT / FTR_00 (TRM transaction manager), VTBFHA / VTBFHAPO, TBB1 flow posting, TPM1 key-date valuation, TPM10, TPM12 / TPM13 position and flow lists, TPM20 posting journal	TreasuryDeal with flows, settlement and key-date valuation (FR-36.18)	partial (FX spot / forward / swap and fixed-term deposits only; no derivatives beyond these, no securities)
Loans management	None (loans kept as journals)	CML / FN* loan contracts, FNM1 flows, F.52 balance interest used for IC loans	LoanContract with schedule and interest accrual (FR-36.19)	partial (annuity / linear / bullet, fixed or index-linked rate; no structured loans)
FX exposure	None	Treasury Exposure Management (raw exposures, exposure positions)	FxExposure derived from AP / AR / orders / forecast / loans, offset by deals (FR-36.20)	partial
Payment requests and bank-to-bank transfers	Bank account transfer via LedgerJournal (account type Bank → Bank)	FIBLFFP payment request, F111 payment program for payment requests, FRFT / FRFT_B repetitive codes	TreasuryPaymentRequest with repetitive templates and four-eyes approval (FR-36.21)	full
Bank chains	Intermediary bank fields on VendBankAccount only	FIBC bank chains (TBCHAIN11 / TBCHAIN21 / TBCHAIN31)	BankChain per currency / receiving country / receiving BIC used in pain.001 IntrmyAgt (FR-36.22)	full
Payment hub routing	Centralized payments (wave 40, FR-40.12)	Advanced Payment Management (APM) routing, batching, reformatting; BCM batch approval	PaymentRoutingRule and central payment monitor in `bank-connect-svc` (FR-36.23)	partial (no cross-tenant hub)
In-house cash	None	IHC0 / IHC1IP in-house cash center, IHC accounts, internal payment orders, interest conditions, subsidiary account statements	IhcAccount with internal payment orders, interest and generated camt.053 (FR-36.24)	partial (no external payments routed through IHC accounts in v1)
Hedge accounting	None	THMEX hedge management, hedge effectiveness test run, hedge plan reports, THM* customizing	none	dropped-with-reason: IFRS 9 / ASC 815 hedge accounting is deferred to v2; FX deals are valued through P&L (7.36.21)

7.36.3 Functional requirements

- **FR-36.1 Bank account master** — Must. The system shall maintain bank accounts per legal entity with IBAN/BIC or routing/account numbers, currency, GL main account, bank transaction types, signatories and a change-approval workflow.
  - Notes: IBAN mod-97 and ABA routing checksum validated on save; account numbers stored encrypted at rest (pgcrypto, key in Vault) and shown masked except to the `TreasuryBankAccountMaintain` duty; status lifecycle Draft → Active → Inactive → Closed; changes to number/currency/GL account require 2-step approval (Temporal workflow); maximum 500 bank accounts per legal entity.
- **FR-36.2 Statement import** — Must. The system shall import bank statements in camt.053.001.02/08, BAI2, MT940 and configurable CSV, validate structure, detect duplicates and persist header, balances (OPBD/CLBD/CLAV) and entries.
  - Notes: files up to 50 MB and 200,000 entries; duplicate detection on (bankAccount, statementId or sequence, closingBalance); gap detection on statement sequence numbers raises a warning; each entry retains the raw source fragment for audit; import of 10,000 lines completes in < 30 s.
- **FR-36.3 Matching rule engine** — Must. The system shall match statement lines to Passage bank transactions using ordered rule sets, producing a 0–100 score per candidate and auto-matching above a configurable threshold.
  - Notes: criteria: amount (exact / tolerance absolute or %), value date window (±N days), reference regex capture (end-to-end id, check number, invoice number), counterparty name trigram similarity, bank transaction code map (BTC/domain-family-subfamily); weights sum to 100; default thresholds auto-match ≥ 90, suggest 60–89, unmatched < 60; 1:1, 1:N and N:1 matches with up to 50 lines per side; rules tested against historical statements with a precision/recall report before activation.
- **FR-36.4 ML match suggestions** — Should. The system shall rank unmatched lines' candidates with a model trained on the tenant's confirmed matches and present the top 3 suggestions with reasons.
  - Notes: Python gradient-boosted model in `cashforecast-svc`; retrained nightly if ≥ 200 new confirmed matches; suggestions never auto-post; model inputs and version logged per suggestion.
- **FR-36.5 Reconciliation workspace** — Must. The system shall let users confirm, split, create new transactions (fees, interest) from unmatched lines, mark lines as in-transit, and close the reconciliation when book and bank balances agree.
  - Notes: closing requires `|bank closing − (book balance + outstanding items)| = 0.00`; new transactions created from lines post through the wave-11 journal API with the statement line as source document; reversed matches keep history; reconciliation of 10,000 lines closes in < 60 s.
- **FR-36.6 Check management** — Must. The system shall issue, void, reprint and stop checks with gap-free check numbers per bank account.
  - Notes: void keeps the number consumed; reprint issues a new number and voids the old; status Created → Printed → Sent → Cleared / Voided / Stopped; positive pay file (wave 19 format) regenerated on void.

- **FR-36.7 Letters of credit and guarantees** — Must. The system shall record import/export letters of credit and bank guarantees against bank facilities with limits, expiry and amendments, and link them to purchase or sales orders.
  - Notes: facility utilization = Σ open instruments; new instrument blocked when utilization + amount > limit unless duty `TreasuryFacilityOverride`; expiry alerts 30/7/1 days; margin deposits post via the journal.
- **FR-36.8 Cash position** — Must. The system shall present intraday position per legal entity, currency and bank account from the latest camt.052 or statement balance plus same-day unposted payments and receipts.
  - Notes: refreshed on every inbound file and at least hourly; converted to entity reporting currency at the wave-12 spot rate; age of each balance displayed; stale (> 24 h) balances flagged.
- **FR-36.9 Cash flow forecast** — Must. The system shall compute forecast buckets (daily 30 days, weekly 13 weeks, monthly 12 months) from registered sources.
  - Notes: sources: AP open items (due date adjusted by vendor average days-late), AR open items (due date + customer DSO delta), open POs/SOs not yet invoiced (expected invoice date + terms), recurring entries, manual adjustments, later payroll (43) and projects (41); each source implements `ForecastSource.Contribute(asOf) []Contribution`; every bucket total equals the sum of its contributions (property test); 1M open items forecast in < 5 min.
- **FR-36.10 Intercompany netting** — Should. The system shall run a netting cycle across participating entities, net IC receivables/payables per pair and currency, and create settlement journals.
  - Notes: cycle statuses Proposed → Approved → Settled; netting rate = wave-12 rate on settlement date; residual below 1.00 in currency written off to IC rounding account.
- **FR-36.11 Cash pooling** — Could. The system shall model notional and physical (zero-balancing) pool structures and generate sweep transfers.
  - Notes: target balance per sub-account; sweep creates payment (wave 19) with pool header as counterparty.
- **FR-36.12 Bank connectivity** — Must. The system shall send payment files and receive statements through SFTP, EBICS 3.0 (H005) and PSD2/Berlin Group XS2A adapters with keys and certificates in Vault.
  - Notes: signing via Vault transit engine; private keys never leave Vault; every outbound file hashed (SHA-256) and logged with user, time, channel and bank acknowledgment; retries with exponential backoff up to 24 h; sandbox profiles per adapter.
- **FR-36.13 Bills of exchange (customer)** — Should. The system shall draw bills of exchange against customer open items, redraw unpaid bills, remit bills to a bank for collection or discount, record protest, and settle bills at maturity, posting each status through a customer posting profile per status.
  - Notes: lifecycle Drawn → Remitted (Collection or Discount) → Honored | Protested; a Redraw creates a new bill that settles the protested one; posting: Draw Dr Bills receivable / Cr Customer (open items settled, wave 30 settlement engine); Remit for collection Dr Bills remitted for collection / Cr Bills receivable; Remit for discount Dr Bank (net), Dr Discount expense / Cr Bills remitted; Honored on bank statement match (FR-36.3 rule criterion `billNumber`) Dr Bank / Cr Bills remitted; Protest reverses to the customer open item plus protest fee line; remittance files (CFONB 160, RiBa CBI, AEB 58 / 19) come from the wave-48 country packs through the wave-19 format engine; feature flag `treasury.bills-of-exchange`; bill numbers from a per-entity number sequence; maximum 5,000 bills per remittance.
- **FR-36.14 Promissory notes (vendor)** — Should. The system shall draw promissory notes against vendor open items, redraw, remit notes to the bank, and settle them at maturity through vendor posting profiles per status.
  - Notes: lifecycle Drawn → Remitted → Settled, with Redraw replacing an unpaid note; Draw Dr Vendor (open items settled, wave 19 settlement) / Cr Notes payable; Settle on bank statement match Dr Notes payable / Cr Bank; notes due within the forecast horizon contribute to FR-36.9 on maturity date with confidence 1.0; the note document prints with the vendor remittance (7.36.15); same flag as FR-36.13.
- **FR-36.15 Postdated checks** — Should. The system shall register customer postdated checks received and vendor postdated checks issued with a check date later than the registration date, hold them in PDC clearing accounts, deposit or present them at maturity, and handle bounced checks.
  - Notes: customer PDC lifecycle Registered → Deposited (maturity deposit journal Dr Bank / Cr PDC clearing) → Cleared | Bounced (reverse, restore customer open item, bounce fee line, customer credit hold event to wave 30); vendor PDC Issued → Presented → Cleared, with the vendor open item settled at registration (parameter `settle-vendor-on-registration`) or at clearing; withholding tax (wave 13) computed at check date when `withholding-at-pdc-settlement` is on (AX R3 India / Thailand behavior); check date drives the FR-36.9 forecast bucket; the `maturity-deposit` job (7.36.16) proposes deposits per bank account daily.
- **FR-36.16 Cash journal (petty cash)** — Must. The system shall maintain cash desks per legal entity and site with a currency, GL cash account and number sequence; record receipts and payments by business transaction type; and close each day with a counted balance.
  - Notes: business transaction types Receipt, Payment, CustomerReceipt (settles wave-30 open items), VendorPayment (settles wave-19 open items), BankDeposit, BankWithdrawal, each with default main account, tax code and dimension defaults; documents are Saved (editable) then Posted (journal API, source document = cash journal document); daily close requires `counted = computed closing` where computed = opening + Σ receipts – Σ payments, otherwise a difference line posts to the cash difference account up to a parameterized limit (default USD 10.00 / EUR 10.00) and above it the close is blocked; negative desk balance is blocked at posting; receipt printout per document; maximum 50 desks per legal entity.
- **FR-36.17 Bank fee analysis (camt.086)** — Should. The system shall import camt.086 bank services billing statements, store fee lines by AFP service code with volume, unit price and tax, compare them to the bank fee agreement, and report variances.
  - Notes: BankFeeAgreement holds expected unit price per (bank, service code, currency) with effective dates; variance = (billed unit price – agreed) × volume plus unexpected service codes; tolerance default 1% per line; monthly fee accrual posts optionally (Dr Bank fees / Cr Accrued bank fees) and is cleared by the statement charge line (BTC CHRG) through FR-36.3; 12 months of statements retained online, 7 years archived.
- **FR-36.18 Treasury deals (FX spot / forward / swap, fixed-term deposits)** — Should. The system shall capture deals with counterparty bank, trade date, value dates, buy and sell amounts, rate and forward points, confirm them against the bank confirmation, generate settlement flows, value open deals at key dates and post the results.
  - Notes: deal types FxSpot, FxForward, FxSwap (near and far legs), FixedTermDeposit; status Draft → Confirmed (manual or MT300 / camt.054 match) → Settled → Closed, with Canceled from Draft or Confirmed; settlement flows become treasury payment requests (FR-36.21) on value date; key-date valuation of a forward: mark-to-market = (market forward rate for the remaining tenor – contract rate) × notional, converted at the spot rate, discounted with the wave-12 yield curve when `discount-mtm` is on; unrealized gain / loss posts to P&L (no hedge accounting) and reverses on the first day of the next period; deposits accrue interest daily with the deal's day-count convention (ACT/360 default, ACT/365, 30/360); accounting codes map (deal type, flow type) to accounts; worked example: forward sell EUR 1,000,000 against USD at 1.1000; at the key date the market forward for the remaining tenor is 1.0850, so the deal value to the EUR seller = (1.1000 – 1.0850) × 1,000,000 = USD 15,000.00 unrealized gain (undiscounted); maximum 20,000 open deals per tenant.
- **FR-36.19 Loan contracts (bank and intercompany)** — Should. The system shall record loan contracts as lender or borrower with principal, schedule type, interest terms and day-count convention, generate the repayment and interest schedule, accrue interest per period and create the settlement flows.
  - Notes: schedule types Annuity, Linear (equal principal), Bullet; rate fixed or floating (reference index + spread with fixings imported from the wave-12 rate service); day-count ACT/360, ACT/365, 30/360; accrual posts Dr Interest expense / Cr Accrued interest (borrower) or Dr Accrued interest / Cr Interest income (lender); intercompany loans create both entities' contracts from one record, post interest as an IC journal (wave 40 due-to / due-from matrix) each period, which covers F.52-style balance interest for IC loans; worked example: EUR 1,000,000 bullet, 3.00% ACT/360, 30-day period → interest = 1,000,000 × 0.03 × 30 / 360 = 2,500.00; early repayment recomputes the remaining schedule prospectively; schedules contribute to FR-36.9 and FR-36.20.

- **FR-36.20 FX exposure** — Could. The system shall derive FX exposures per currency pair and bucket from AP and AR open items, open purchase and sales orders, forecast entries, loan flows and deals, and present the net open exposure after deals.
  - Notes: exposure = Σ inflows – Σ outflows per currency and bucket (same buckets as FR-36.9) – Σ deal legs maturing in the bucket; hedge ratio = hedged / gross exposure; export CSV for the treasury committee; no automatic deal proposals.
- **FR-36.21 Treasury payment requests and bank-to-bank transfers** — Must. The system shall create payment requests that are not vendor open items (transfers between own bank accounts, third-party treasury payments, deal and loan settlements), approve them, and hand them to the wave-19 payment engine with category `TreasuryTransfer`.
  - Notes: repetitive templates (SAP FRFT equivalent) store counterparty, accounts, amount and purpose for one-click requests; four-eyes approval required above a per-entity threshold (default USD 50,000 / EUR 50,000), with the approver different from the creator (SoD check in OPA); intercompany transfers post due-to / due-from through the wave-40 matrix; a request for a Draft or Inactive bank account is rejected; cutoff times per bank from FR-36.23 determine the execution date.
- **FR-36.22 Bank chains** — Should. The system shall maintain correspondent and intermediary bank chains per currency, receiving bank country and receiving BIC, and apply them when generating payment files.
  - Notes: up to 3 hops per chain; written to pain.001 `IntrmyAgt1–3` and MT103 fields 56 / 57 by the wave-19 format engine; BIC validated (ISO 9362); chains effective-dated; a payment to a currency / country without a chain is not blocked.
- **FR-36.23 Payment hub routing and monitor** — Should. The system shall route approved payment batches from every legal entity to a channel and format profile by rule, batch them per bank cutoff, and show one monitor of all batches with bank status.
  - Notes: rule key (legal entity, bank account, payment method, currency, amount band, urgency) → (connection, format profile); batching windows per connection (for example every 30 minutes and a final run 15 minutes before the bank cutoff); status from pain.002 and camt.054 updates the batch and the underlying payments (wave 19 / wave 30); central batch approval optional per entity (BCM-style); rules are versioned and tested against the last 30 days of payments before activation (dry-run report of routing changes).
- **FR-36.24 In-house cash current accounts** — Could. The system shall run an in-house cash center at one legal entity with current accounts per participating entity, internal payment orders between them, interest conditions, and monthly statements that the participants import like bank statements.
  - Notes: flag `treasury.ihc`; IhcAccount per (participant entity, currency) with credit and debit interest conditions (fixed or index + spread), overdraft limit; internal payment order debits the payer account and credits the payee account same day and posts due-to / due-from through the wave-40 matrix in both entities; monthly interest capitalizes to the account; a camt.053 is generated per account at month end and on demand and imported by the participant through FR-36.2 so that FR-36.3 reconciles it; external payments on behalf of participants go through FR-36.23 routing in v1 without passing through the IHC account.
- **FR-36.25 Balance interest on GL and intercompany accounts** — Could. The system shall calculate interest on the daily balances of selected main accounts (typically intercompany current accounts and shareholder accounts) using interest indicators with debit and credit rates, and post the interest with an interest advice.
  - Notes: interest indicator = (rate table or reference index + spread, day-count, calculation period, minimum amount 1.00); interest = Σ_days balance_d × rate_d / basis, separately for debit and credit days; balance per value date from the wave-11 balance projection; postings Dr Interest receivable / Cr Interest income (or the reverse) per account and dimension; for intercompany accounts the counterpart posts in the partner entity through the wave-40 IC journal (FR-40.5); reuses the day-count library of FR-36.19; run is per legal entity and period and is rerunnable after reversal only.

7.36.4 Data model

BankAccount

Field	Type	Required	Notes
id	uuid	yes	PK
legal_entity_id	uuid	yes	FK org (wave 6)
account_code	text(20)	yes	unique per entity
iban / account_number	text, encrypted	yes	one of the two; masked in API
bic / routing_number	text	no	validated
currency	char(3)	yes	ISO 4217
main_account_id	uuid	yes	FK ledger account (wave 11)
status	enum	yes	Draft/Active/Inactive/Closed
next_check_number	bigint	no	gap-free sequence
signatories	jsonb	no	[{partyId, limit, currency}]

BankStatementLine

Field	Type	Required	Notes
id	uuid	yes	PK
statement_id	uuid	yes	FK BankStatement
line_no	int	yes	unique in statement
booking_date / value_date	date	yes	
amount	numeric(19,4)	yes	signed, statement currency
btc_code	text	no	ISO 20022 domain/family/subfamily
references	jsonb	no	EndToEndId, InstrId, Ustrd, chequeNumber
counterparty_name	text	no	normalized for trigram index
status	enum	yes	Unmatched/Suggested/Matched/Created/InTransit
raw_fragment	text	yes	source XML/text slice

ReconciliationMatch

Field	Type	Required	Notes
id	uuid	yes	PK
reconciliation_id	uuid	yes	FK ReconciliationHeader
statement_line_ids	uuid[]	yes	1..50
bank_transaction_ids	uuid[]	yes	1..50

Field	Type	Required	Notes
score	smallint	yes	0–100
rule_id	uuid	no	null when manual
matched_by	text	yes	user or `rule-engine`
matched_at	timestamptz	yes	
reversed_at	timestamptz	no	

CashForecastEntry: id, legal_entity_id, source_type (AP/AR/PO/SO/Recurring/Manual/Payroll/Project), source_id, expected_date, amount, currency, confidence (0–1), bucket_key, computed_at.

Entities added by FR-36.13–FR-36.25 (summaries; tables follow the same conventions as above):

- **TreasuryInstrument**: id, legal_entity_id, kind (BillOfExchange/PromissoryNote/PostdatedCheck), direction (Receivable/Payable), number (per-entity sequence), party_id, bank_account_id, amount numeric(19,4), currency, issue_date, maturity_date (check date for PDC), status (Drawn/Registered/Remitted/Deposited/Presented/Honored/Settled/Cleared/Protested/Bounced/Redrawn), remittance_id, settled_open_item_ids uuid[], protest_reason_code, replaced_by_id.
- **CashDesk**: id, legal_entity_id, site_id, code, currency, main_account_id, difference_account_id, difference_limit numeric(19,4), number_sequence_id, status. **CashJournalDocument**: id, desk_id, document_no, business_date, transaction_type, amount, tax_code, party_id, open_item_ids, dimensions, status (Saved/Posted/Reversed), journal_entry_id. **CashDeskDay**: desk_id, business_date, opening, receipts, payments, computed_closing, counted, difference, status (Open/Closed/Reopened), closed_by.
- **BankFeeStatement / BankFeeLine**: statement_id, bank_party_id, billing_period, line: service_code (AFP), description, volume numeric(19,4), unit_price numeric(19,6), amount, tax, agreed_unit_price, variance, status (Within/Over/Under/Unexpected).
- **TreasuryDeal**: id, legal_entity_id, deal_no, type (FxSpot/FxForward/FxSwap/FixedTermDeposit), counterparty_bank_id, trade_date, near_value_date, far_value_date, buy_currency, buy_amount, sell_currency, sell_amount, contract_rate numeric(19,8), forward_points numeric(19,8), interest_rate numeric(9,6), day_count, status, confirmation_ref, flows jsonb [{flowType, valueDate, amount, currency, paymentRequestId, journalEntryId}].
- **LoanContract**: id, legal_entity_id, role (Borrower/Lender), counterparty_party_id, ic_partner_id, principal, currency, start_date, maturity_date, schedule_type (Annuity/Linear/Bullet), rate_type (Fixed/Floating), fixed_rate, index_code, spread_bp, day_count, status (Draft/Active/Repaid/Terminated); **LoanScheduleLine**: period_no, start, end, opening_principal, interest, principal_repayment, closing_principal, fixing_rate, accrual_journal_id, settlement_request_id.
- **TreasuryPaymentRequest**: id, legal_entity_id, category (TreasuryTransfer/DealSettlement/LoanSettlement/ThirdParty/IhcExternal), from_bank_account_id, to_bank_account_id or beneficiary jsonb, amount, currency, requested_date, execution_date, template_id, status (Draft/Submitted/Approved/Routed/Sent/Executed/Rejected), approver_id. **BankChain**: currency, receiving_country, receiving_bic, hops[1..3] (BIC, account), effective_from. **PaymentRoutingRule**: version, match keys, connection_id, format_profile_id, batch_window_min, cutoff_local_time.
- **IhcAccount**: id, center_entity_id, participant_entity_id, currency, credit_condition, debit_condition, overdraft_limit, balance (projection), status; **IhcPaymentOrder**: id, payer_account_id, payee_account_id, amount, value_date, status, journal_pair_id.
- **InterestIndicator / InterestRun**: indicator code, rate table, day_count, period; run id, legal_entity_id, period, accounts[], total_debit_interest, total_credit_interest, journal_entry_ids.

Invariants: Σ amounts of matched statement lines = Σ amounts of matched bank transactions ± configured tolerance; a bank transaction belongs to at most one non-reversed match; statement closing balance = opening balance + Σ lines; a closed reconciliation is immutable. Indexes: statement lines by (bank_account_id, value_date), GIN trigram on counterparty_name, partial index on status = Unmatched; CashForecastEntry partitioned monthly by expected_date, retention 24 months.

7.36.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/treasury/bank-accounts	Create account	Idempotency-Key	TreasuryBankAccountMaintain
POST	/api/v1/treasury/statements/import	Upload statement file (multipart)	file SHA-256	TreasuryStatementImport
POST	/api/v1/treasury/reconciliations/{id}/run-rules	Execute rule set	runId	TreasuryReconcile
POST	/api/v1/treasury/reconciliations/{id}/matches	Manual match	Idempotency-Key	TreasuryReconcile
POST	/api/v1/treasury/reconciliations/{id}/close	Close	state check	TreasuryReconcileApprove
POST	/api/v1/treasury/checks/{id}/void	Void check	state check	TreasuryCheckMaintain
GET	/api/v1/treasury/cash-position?entity=&asOf=	Position	n/a	TreasuryView
GET	/api/v1/treasury/cash-forecast?entity=&horizon=	Forecast buckets	n/a	TreasuryView
POST	/api/v1/treasury/netting-cycles	Propose netting	Idempotency-Key	TreasuryNetting
POST	/api/v1/treasury/connections/{id}/send	Send file via channel	file hash	TreasuryBankSend
POST	/api/v1/treasury/instruments	Draw bill / note, register PDC	Idempotency-Key	TreasuryInstrumentMaintain
POST	/api/v1/treasury/instruments/{id}/remit	Remit (collection/discount) or deposit	state check	TreasuryInstrumentMaintain
POST	/api/v1/treasury/instruments/{id}/protest	Protest bill / bounce PDC	state check	TreasuryInstrumentSettle
POST	/api/v1/treasury/cash-desks/{id}/documents	Post cash journal document	Idempotency-Key	TreasuryCashDeskPost
POST	/api/v1/treasury/cash-desks/{id}/days/{date}/close	Close desk day with count	state check	TreasuryCashDeskPost
POST	/api/v1/treasury/fee-statements/import	Import camt.086	file SHA-256	TreasuryStatementImport
POST	/api/v1/treasury/deals	Capture deal	Idempotency-Key	TreasuryDealCapture
POST	/api/v1/treasury/deals/{id}/confirm	Confirm against bank confirmation	state check	TreasuryDealConfirm
POST	/api/v1/treasury/valuation-runs	Key-date valuation (dryRun option)	(entity, keyDate)	TreasuryDealValuate
POST	/api/v1/treasury/loans	Create loan and schedule	Idempotency-Key	TreasuryLoanMaintain
POST	/api/v1/treasury/payment-requests	Create request or from template	Idempotency-Key	TreasuryPaymentRequestCreate
POST	/api/v1/treasury/payment-requests/{id}/approve	Approve (four-eyes)	state check	TreasuryPaymentRequestApprove
PUT	/api/v1/treasury/routing-rules/{version}	Publish routing rules after dry run	version	TreasuryRoutingConfigure
POST	/api/v1/treasury/ihc/payment-orders	Internal payment order	Idempotency-Key	TreasuryIhcOperate
POST	/api/v1/treasury/interest-runs	GL balance interest run	(entity, period)	TreasuryInterestRun

Commands: `ImportStatement` (validates schema, rejects duplicates), `RunMatching` (rule set must be Active), `ConfirmMatch` (amount tolerance ≤ rule tolerance), `CloseReconciliation` (difference = 0). Queries served from read models.

Event	Trigger	Payload fields	Consumers
`dev.passage.treasury.statement.imported.v1`	import committed	statementId, bankAccountId, lineCount, openingBalance, closingBalance	cashforecast-svc, treasury-svc matcher
`dev.passage.treasury.statement.reconciled.v1`	reconciliation closed	statementId, matchedCount, createdCount, closingBalance	journal, analytics
`dev.passage.treasury.check.voided.v1`	void	checkId, bankAccountId, number, reason	AP (19) positive pay
`dev.passage.treasury.cashposition.updated.v1`	position refresh	legalEntityId, asOf, balances[]	dashboards, interco-svc
`dev.passage.treasury.file.sent.v1`	outbound file acknowledged	connectionId, fileHash, channel, ackStatus	audit
`dev.passage.treasury.instrument.statuschanged.v1`	bill / note / PDC status change	instrumentId, kind, fromStatus, toStatus, amount, maturityDate	AR (30), AP (19), cashforecast-svc
`dev.passage.treasury.pdc.bounced.v1`	PDC bounced	instrumentId, customerId, amount, fee, reasonCode	AR (30) credit hold
`dev.passage.treasury.cashdesk.closed.v1`	desk day closed	deskId, businessDate, closing, difference	journal audit, analytics
`dev.passage.treasury.deal.valued.v1`	valuation posted	runId, keyDate, dealCount, unrealizedTotal, journalEntryId	analytics, consol-svc
`dev.passage.treasury.paymentrequest.approved.v1`	request approved	requestId, category, amount, currency, executionDate	bank-connect-svc routing
`dev.passage.treasury.ihc.statement.generated.v1`	IHC month-end statement	ihcAccountId, participantEntityId, period, closingBalance, fileRef	treasury-svc import (participant)

7.36.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Bank accounts	Create, edit (workflow), close, view balances	code, IBAN (masked), currency, GL account, status, last statement	TreasuryBankAccountMaintain
Reconciliation workspace	Run rules, confirm, split, create transaction, close	line date/amount/reference, candidate score, difference, unmatched count	TreasuryReconcile
Checks	Print, void, reprint, stop	number, payee, amount, status	TreasuryCheckMaintain
Facilities and instruments	Create LC/guarantee, amend, close	facility, limit, utilization, expiry	TreasuryFacilityMaintain
Cash position	Filter entity/currency, drill to statement	account, balance, age, same-day pending	TreasuryView
Cash forecast	Change horizon, add manual entry, drill to source	bucket, inflow, outflow, net, cumulative	TreasuryView
Bank connections	Test, send, view log	channel, status, last file, ack	TreasuryBankSend

Interaction rules: the reconciliation grid supports keyboard `m` (match selected), `u` (unmatch), `n` (new transaction); multi-select up to 50 lines each side with running difference shown; rule changes require re-run before close; column layouts and filters persist per user via the wave-5 shell personalization.

7.36.7 Business rules and edge cases

- **BR-36.1** Score = Σ (criterion weight × criterion result ∈ [0,1]); amount tolerance criterion result = 1 − |Δ|/tolerance clipped at 0; date window result = 1 − |daysl/window.
- **BR-36.2** Ties above the auto threshold with more than one candidate are downgraded to Suggested; auto-match requires a unique top candidate at least 5 points above the next.
- **BR-36.3** Reconciliation state machine: Open → InProgress → Closed; Closed → Reopened only by `TreasuryReconcileApprove` within the open fiscal period; reopening reverses nothing, it only allows new matches.
- **BR-36.4** Statement currency ≠ bank account currency is rejected; FX differences between book and bank on foreign receipts post to realized gain/loss (wave 12) with the statement line as source.
- **BR-36.5** A void after Sent and before Cleared triggers a positive pay void record; void after Cleared is rejected.
- **BR-36.6** Forecast bucket boundaries use the legal entity's calendar; a contribution whose expected date falls on a non-banking day moves to the next banking day.
- **BR-36.7** Concurrency: two users confirming matches on the same line receive 409 on the second write (optimistic version on StatementLine).
- **BR-36.8** Duplicate statement file with identical hash returns 200 with the existing statement id; same statement id with different closing balance returns 409.
- **BR-36.9** Facility utilization uses the instrument's original currency converted at the daily rate; limit breaches recomputed nightly and on rate change > 2%.
- **BR-36.10** Bill of exchange state machine: Drawn → Remitted → Honored | Protested; Protested → Redrawn (new bill, old bill closed) or → ReturnedToCustomer (open item restored with protest fee); a bill cannot be remitted twice; remittance for discount books the discount expense on remittance date, not at maturity.
- **BR-36.11** Promissory note state machine: Drawn → Remitted → Settled; Drawn → Redrawn; settlement requires a matched bank statement line or a manual settlement with duty `TreasuryInstrumentSettle`; the vendor open item is settled at draw and never reopened except through a redraw reversal.
- **BR-36.12** Postdated check: a deposit before the check date is rejected unless the parameter `allow-early-deposit` is on; a bounce after Cleared is a new customer transaction (fee + reversal) with reason code; bounced customer checks raise `dev.passage.treasury.pdc.bounced.v1` consumed by wave-30 credit hold.
- **BR-36.13** Cash desk day close: closing balance = opening + Σ receipts − Σ payments in desk currency; |counted − closing| ≤ cash difference limit posts a difference line, above the limit the close is blocked; no document can post to a closed day; reopening a closed day requires `TreasuryCashDeskApprove` and the day's reopen is audited.
- **BR-36.14** Deal valuation entries are per deal and key date, reversed on the first calendar day of the next period before the new valuation; a deal settled between two key dates realizes gain / loss = settlement rate difference × notional and the last unrealized entry is reversed in the same journal.
- **BR-36.15** Interest accrual uses the contract day-count; ACT/360 for a leap-year February period counts 29 days; accrual rounding half-up to the minor unit per period, and the final period of a schedule absorbs the rounding residual so Σ interest = Σ scheduled interest.
- **BR-36.16** Treasury payment requests above the four-eyes threshold require an approver different from the creator and from the last editor; OPA denies self-approval; an edit after approval returns the request to Draft.
- **BR-36.17** In-house cash internal payment orders value the same day; an order exceeding the payer's overdraft limit is rejected unless the `TreasuryIhcOverride` duty approves; interest capitalizes on the last calendar day of the month at the conditions in force per day (daily balance method).
- **BR-36.18** Bank fee variance: a fee line is Within tolerance when |billed − agreed| / agreed ≤ 1%; a service code with no agreement line is Unexpected; a Within line never raises a message, an Over line raises TRS-0418 and an Unexpected line TRS-0419.

7.36.8 Acceptance criteria

- **AC-36.1** (FR-36.2) Given a camt.053 file with 1,000 entries, when imported, then 1,000 lines persist, opening + Σ lines = closing, and a re-upload returns the same statement id.
- **AC-36.2** (FR-36.3) Given the 1,000-line reconciliation fixture, when rules run, then ≥ 950 lines are auto-matched correctly and 0 are matched to the wrong transaction.

- **AC-36.3** (FR-36.5) Given a line for a USD 12.50 bank fee, when the user creates a transaction, then a journal entry Dr Bank fees 12.50 / Cr Bank 12.50 posts with the line as source document.
- **AC-36.4** (FR-36.5) Given a difference of 0.01, when closing, then the close is rejected with the difference displayed.
- **AC-36.5** (FR-36.6) Given check 1001 Printed, when voided and reprinted, then 1001 is Voided and 1002 is Printed with the same payee and amount.
- **AC-36.6** (FR-36.7) Given a facility of EUR 1,000,000 with EUR 950,000 utilized, when an LC of EUR 100,000 is created, then it is blocked without the override duty.
- **AC-36.7** (FR-36.9) Given 10,000 generated open items, when the forecast computes, then every bucket equals the sum of its contributions to the cent.
- **AC-36.8** (FR-36.12) Given an EBICS sandbox, when a pain.001 is sent, then the file hash, signature reference and bank acknowledgment are in the audit log and no private key bytes appear in any log or DB.
- **AC-36.9** (FR-36.13) Given a customer bill of EUR 10,000.00 drawn against two invoices and remitted for discount at 2% fee, when remitted, then the journal shows Dr Bank 9,800.00, Dr Discount expense 200.00 / Cr Bills remitted 10,000.00, and the two invoices are settled.
- **AC-36.10** (FR-36.13) Given a remitted bill, when the bank reports non-payment and the user protests it with a EUR 25.00 fee, then the customer open item of 10,025.00 is restored and the bill status is Protested.
- **AC-36.11** (FR-36.15) Given a customer PDC of USD 5,000.00 dated 2026-11-15 registered on 2026-10-20, when the forecast computes, then 5,000.00 sits in the 2026-11-15 bucket; when deposited on 2026-11-15 and bounced on 2026-11-18, then the open item of 5,000.00 plus a 15.00 fee is restored.
- **AC-36.12** (FR-36.16) Given a desk opening at EUR 500.00 with receipts 120.00 and payments 80.00, when the cashier counts 539.50 and closes, then a difference line of –0.50 posts and the day is Closed; when the cashier counts 525.00, then the close is rejected with TRS-0431.
- **AC-36.13** (FR-36.17) Given a camt.086 with service code `010100` billed at 0.26 for 1,000 items and an agreement of 0.25, when imported, then the variance line shows 10.00 Over (4%) and raises TRS-0418.
- **AC-36.14** (FR-36.18) Given the forward sell EUR 1,000,000 at 1.1000 example and a market forward of 1.0850 at the key date, when valued, then an unrealized gain of USD 15,000.00 posts and reverses on the first day of the next period.
- **AC-36.15** (FR-36.19) Given the EUR 1,000,000 bullet IC loan at 3.00% ACT/360, when the 30-day period accrues, then the lender posts Dr Due from borrower 2,500.00 / Cr Interest income 2,500.00 and the borrower the mirror entry with one ic_pair_id.
- **AC-36.16** (FR-36.21) Given a transfer request of USD 75,000.00 created by user A, when A attempts to approve, then OPA denies it; when user B approves, then a pain.001 with category `TreasuryTransfer` is routed by FR-36.23 to the bank's connection.
- **AC-36.17** (FR-36.25) Given an IC current account with a debit balance of EUR 200,000.00 for 31 days at 4.00% ACT/360, when the interest run executes for the month, then interest = 200,000 × 0.04 × 31 / 360 = 688.89 posts in the lender and the mirror entry posts in the borrower.

7.36.9 Test plan and fixtures

Fixtures: `recon-1000` (1,000 statement lines across camt.053/BAI2/MT940 with 60% exact, 25% reference-only, 10% N:1 batch receipts, 5% fees, generated from wave 19/30 payment fixtures); `forecast-10k` (generated open items with due dates across 12 months); `ebics-sandbox` (vendor sandbox recordings for contract tests); `facility-20` (LCs and guarantees with amendments).

Test	Type	Target	Pass criterion
Scoring function	Unit	rule engine	deterministic scores per criterion table
Statement parsers	Contract	camt.053 v02/v08, BAI2, MT940	100% of fixture files parse; schema errors reported with line
Match uniqueness	Property	reconciliation	no transaction in two active matches
Forecast additivity	Property	cashforecast-svc	bucket = Σ contributions
Reconcile-and-close	E2E	workspace	recon-1000 closes with difference 0
Import throughput	Performance	import API	10,000 lines < 30 s; 200,000 lines < 10 min
Adapter contracts	Contract	SFTP/EBICS/XS2A	sandbox round-trip with acknowledgment

Reconciliation checks: Σ BankAccount book balance = GL bank main account balance per entity nightly; Σ outstanding checks = positive pay open items; cash position = latest statement closing + pending.

7.36.10 Deployment and operations

Helm charts `treasury-svc`, `bank-connect-svc`, `cashforecast-svc`; config: `TREASURY_AUTO_MATCH_THRESHOLD=90`, `TREASURY_MAX_FILE_MB=50`, Vault role `bank-connect`. Atlas migrations generate tables from entity metadata. Feature flags: `treasury.ml-suggestions`, `treasury.cash-pooling`. Scheduled jobs (Temporal): `statement-poll` every 15 min per connection, `cash-position-refresh` hourly, `forecast-rebuild` hourly and on demand, `facility-expiry-check` daily 06:00. SLOs: import p95 < 30 s per 10k lines, rule run p95 < 20 s per 10k lines, forecast API p95 < 300 ms. Alerts: unreconciled lines > 7 days, connection failure > 3 retries, key rotation due. Runbook: resend file, rebuild forecast, rotate EBICS keys. Rollback: Helm rollback; matches are append-only with reversals, so no data rollback.

7.36.11 Migration from AX / SAP

Source	Passage entity	Notes
BankAccountTable, BankAccountTrans (open items)	BankAccount, BankTransaction	opening balance as of cutover; unreconciled items migrated as InTransit
BankChequeTable	Check	status map; next number continues
BankLCImport/Export, BankLGTable, BankFacility	LetterOfCredit, Guarantee, Facility	open instruments only
T012/T012K, BAM accounts	BankAccount	house bank + account id → account code
FEBKO/FEBEP (last 90 days)	BankStatement, StatementLine	for matching history and ML training
PAYR	Check	
LedgerCov / FQM_FLOW	not migrated	forecast is recomputed from sources

Validation: Σ migrated bank balances = GL bank accounts; IBAN checksums pass. Not migrated: closed reconciliations older than 90 days, matching rule configurations (re-authored, since scoring semantics differ).

7.36.12 Dependencies, risks, open questions

Depends on: wave 11 journal posting, wave 12 rates, wave 19 payment files and positive pay, wave 30 cash application, wave 35 integration hub for SFTP transport, wave 5 Vault integration.

Risk	Likelihood	Impact	Mitigation
Bank-specific camt dialects	High	Medium	Per-bank profile overrides; fixture per bank
ML suggestions mislead users	Medium	Low	Suggestions only, reasons shown, precision tracked

Open questions: (1) Default auto-match threshold 90 or 95? Proposed 90; decide by wave 36 build start. (2) EBICS versions to support (2.5 and 3.0)? Proposed 3.0 only; decide by wave 36. (3) Cash pooling in v1? Proposed Could; decide at phase 7 exit.

7.36.13 Parameters and setup

Parameters live in the `Treasury parameters` page (one tab per area, mirroring AX `BankParameters` tabs) and are versioned with effective dates; changes are audited and take effect for documents created after the change.

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`recon.autoMatchThreshold`	legal entity	int	90	60–100	score at or above which the engine auto-matches (FR-36.3)	BankReconciliationMatchingRuleSet, Cash and bank management parameters → Bank reconciliation	FEBAN interpretation algorithm + OT83 posting rule assignment
`recon.suggestThreshold`	legal entity	int	60	1–(auto – 1)	lower bound of Suggested status	matching rule "Mark as matched" vs manual	OTPM search string relevance (no score)
`recon.tieMargin`	legal entity	int	5	0–20	points the top candidate must lead by (BR-36.2)	none	none
`statement.maxFileMb`	tenant	int	50	1–200	rejects larger uploads (FR-36.2)	DMF file size limit	FF_5 file size (no limit setting)
`statement.sequenceGapPolicy`	bank account	enum	Warn	Ignore / Warn / Block	handling of missing statement numbers	none	FF_5 "check statement number"
`statement.btcMap`	bank account	table	ISO 20022 defaults	domain/family/subfamily → transaction type	maps bank codes to bank transaction types and default offset accounts	BankTransType, Bank statement transaction code mapping	OT83 external transaction types → posting rules (T028G)
`position.staleHours`	legal entity	int	24	1–168	flags old balances in cash position (FR-36.8)	none	FF7A display (no flag)
`forecast.horizon`	legal entity	struct	daily 30 / weekly 13 / monthly 12	daily 1–90, weekly 1–52, monthly 1–36	bucket layout (FR-36.9)	Cash flow forecasting setup (LedgerCovParameters)	OT14 planning levels, OT13 planning groups, FF7B horizon
`forecast.arDelayMode`	legal entity	enum	CustomerDsoDelta	None / CustomerDsoDelta / CustomerAverageDaysLate	adjusts AR expected dates	Cash flow forecasting setup → receivables payment history	FF7B planning level "customer payment history"
`check.voidReasonRequired`	bank account	bool	true	true / false	void without reason rejected	Bank account → Checks → "Require void reason"	FCH8 reason for void (T012V void reason codes)
`check.regeneratePositivePay`	bank account	bool	true	true / false	positive pay record on void (BR-36.5)	BankPositivePayFormat	FCHX positive pay extract
`facility.expiryAlertDays`	legal entity	int[]	30, 7, 1	1–365 each	LC / guarantee expiry alerts (FR-36.7)	Letters of credit → expiry notification	none native
`netting.residualWriteOff`	legal entity	money	1.00	0.00–100.00	residual to IC rounding account (FR-36.10)	none	IHC netting tolerance
`paymentRequest.fourEyesThreshold`	legal entity	money	USD 50,000 / EUR 50,000	≥ 0	second approver required above (FR-36.21)	Bank → Bank transfer workflow	FIBLFFP payment request approval (F8BT / BCM release strategy)
`instrument.billPostingProfile`	legal entity	table	none	status (Drawn/Remitted-Collection/Remitted-Discount/Protested) → account	GL accounts per bill status (FR-36.13)	Accounts receivable parameters → Bill of exchange tab, customer posting profile per bill status	OBXR special G/L indicator W accounts, OBYN bill of exchange charges

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`instrument.notePostingProfile`	legal entity	table	none	status (Drawn/Remitted/Settled) → account	GL accounts per promissory-note status (FR-36.14)	Accounts payable parameters → Promissory note tab, vendor posting profile	OBYR special G/L W (vendor)
`instrument.remittanceTypes`	bank account	enum[]	Collection	Collection / Discount	which remittance modes the bank accepts	Bill of exchange remittance types	F-33 / F-34 discount vs collection
`pdc.clearingAccounts`	legal entity	struct	none	receivable PDC account, payable PDC account	holding accounts until clearing (FR-36.15)	Cash and bank management parameters → Postdated checks tab	special G/L indicator S (noted items)
`pdc.settleVendorOnRegistration`	legal entity	bool	true	true / false	vendor open item settled at issue or at clearing	Postdated checks → "Settle postdated check"	none
`pdc.allowEarlyDeposit`	legal entity	bool	false	true / false	BR-36.12 deposit before check date	none	none
`pdc.withholdingAtSettlement`	legal entity	bool	false	true / false	withholding tax at check date (wave 13)	R3 India/Thailand "Calculate withholding tax at settlement"	extended withholding tax at payment (OBWW
`cashDesk.differenceLimit`	cash desk	money	10.00 desk currency	0.00–1,000.00	close blocked above limit (BR-36.13)	none	FBCJC0 cash journal definition (no limit)
`cashDesk.businessTransactions`	legal entity	table	6 seeded types	type, main account, tax code, dimensions	defaults per cash journal line (FR-36.16)	none	FBCJC2 business transactions (TCJ_TRANSACTIONS
`cashDesk.numberSequence`	cash desk	sequence	per desk, gap-free	number sequence id	document numbering	none	FBCJC1 number ranges
`fees.tolerancePct`	legal entity	decimal	1.00%	0.00–10.00%	Within vs Over (BR-36.18)	none	BAM bank fee analysis tolerance
`deal.dayCountDefault`	legal entity	enum	ACT/360	ACT/360 / ACT/365 / 30/360 / ACT/ACT	interest on deposits and loans	none	TRM product type interest calculation method (FTR customizing JBD*)
`deal.accountingCodes`	legal entity	table	seeded per deal type	(deal type, flow type, update type) → accounts	posting of flows and valuations (FR-36.18)	none	TRM accounting codes, update types, valuation areas (TPM customizing)
`deal.discountMtm`	legal entity	bool	false	true / false	discount MTM with the wave-12 curve	none	TPM valuation class / NPV method
`loan.defaultSchedule`	legal entity	enum	Annuity	Annuity / Linear / Bullet	schedule template (FR-36.19)	none	CML product types / condition types
`interest.indicators`	main account	table	none	indicator code, rates, day count, period, min amount	GL balance interest (FR-36.25)	none	OB46 interest indicators, OB82 interest terms, OB83 reference rates
`bankChain.searchMode`	legal entity	enum	ByCurrencyAndCountry	ByCurrency / ByCurrencyAndCountry / ByReceivingBic	chain lookup order (FR-36.22)	none	FIBC bank chain scenario (TBCHAIN11/21/31)
`routing.batchWindowMin`	connection	int	30	5–1,440	batching window (FR-36.23)	none	BCM batch rules (Bank Communication Management → merge criteria)
`routing.cutoffLocalTime`	connection	time	15:00 bank local	00:00–23:59	requests after cutoff move to next banking day	Bank → Bank holidays / payment calendar (wave 19)	BCM payment medium cutoff, FBZP value days
`ihc.conditions`	IHC account	struct	none	credit / debit rate, index + spread, overdraft limit	IHC interest and limits (FR-36.24)	none	IHC0 conditions, IHC account limits

7.36.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Reconciliation status	list inquiry	entity, bank account, period, status	statement, lines, matched, suggested, unmatched, difference, age	`recon_status_rm`	Bank reconciliation list (BankReconciliationHeader)	FEBAN, FEB_BSPROC
Unmatched lines aging	operational report	entity, account, age bands 0–7 / 8–30 / > 30 days	count, amount per band, top counterparties	`statement_line` partial index Unmatched	Bank statement lines (unreconciled)	FEBAN worklist, FF_6
Cash position	KPI / drill-down	entity, currency, as-of	balance, age, same-day pending in/out, converted total	`cash_position_rm`	Bank account balance FactBox	FF7A
Cash flow forecast	drill-down	entity, horizon, source types, currency	inflow, outflow, net, cumulative, by source	`cash_forecast_entry`	Cash flow forecast workspace (LedgerCov)	FF7B, FQM liquidity forecast
Check register / outstanding checks	list inquiry	bank account, status, date range	number, payee, amount, status, cleared date	`check`	Check (BankChequeTable), Outstanding checks report	FCHN, FCH6 (PAYR)
Facility utilization	KPI	facility, bank, currency	limit, utilization, headroom, expiring in 30 days	`facility_rm`	Bank facilities, LC inquiry	none native (TRM / Trade Finance)
Bill of exchange list (customer)	list inquiry	status, maturity range, remittance, customer	number, customer, amount, maturity, status, bank	`treasury_instrument` kind = Bill	Bill of exchange inquiry (CustBillOfExchangeJour)	S_ALR_87012199
Promissory note list (vendor)	list inquiry	status, maturity range, vendor	number, vendor, amount, maturity, status	`treasury_instrument` kind = Note	Promissory note inquiry (VendPromissoryNoteJour)	S_ALR_87012116
Remittance list	operational report	remittance id, bank, type	bills, totals, discount fee, net proceeds	`instrument_remittance`	SSRS Bill of exchange remittance list	FBWE output list
Postdated check inquiry	list inquiry	direction, check date range, status, party	check no, bank, check date, amount, status	`treasury_instrument` kind = PDC	Postdated checks (CustPostdatedChecks / VendPostdatedChecks)	none
Cash book	statutory report	cash desk, date range	opening, documents (no, type, text, receipt, payment), running balance, closing, difference	`cash_journal_document`, `cash_desk_day`	none	S_ALR_87012309, FBCJ print
Bank fee analysis	operational report	bank, period, service code	volume, billed, agreed, variance, status, trend 12 months	`bank_fee_line`	none	Fiori Bank Fee Analysis / Bank Relationship Overview
Treasury position list	list inquiry	entity, deal type, counterparty, key date	deal, nominal, rate, MTM, accrued interest, maturity	`deal_position_rm`	none	TPM12
Treasury flow list	list inquiry	entity, value date range, flow type	deal / loan, flow type, amount, currency, posted, payment request	`deal_flow`, `loan_schedule_line`	none	TPM13
Treasury posting journal	operational report	entity, period, deal type	flow / valuation postings with accounts and journal ids	`deal_posting_log`	none	TPM20
FX exposure	KPI / drill-down	entity, currency pair, horizon	gross exposure, hedged, net open, hedge ratio	`fx_exposure_rm`	none	Exposure management 2.0 reports
Loan schedule and interest	drill-down	loan, role, period	opening, interest, repayment, closing, fixing	`loan_schedule_line`	none	FNO5 / CML flow overview
Payment batch monitor	list inquiry	entity, connection, status, date	batch, payments, amount, channel, ack, pain.002 status	`payment_batch_rm` (bank-connect-svc)	none	BNK_MONI, Fiori Monitor Payments
IHC account balances	list inquiry	center, participant, currency	balance, overdraft limit, accrued interest, last statement	`ihc_account`	none	IHC account balance (IHC1IP / F9K* reports)
Interest run result	operational report	entity, period, indicator	account, debit days interest, credit days interest, posted amount	`interest_run`	none	F.52 log

7.36.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Check	payment with method Check (wave 19) or reprint	header: payer entity, bank, check no, date; body: payee, amount in words and figures; footer: MICR E-13B line (routing, account, check no), signature image per signatory limit	PDF to check stock (laser), print queue	gap-free per bank account (`next_check_number`)	voided checks retained as records 7 years (US), 10 years (DE)
Positive pay file	check print / void	header: account, file date; lines: check no, amount, issue date, payee, void flag	bank CSV / BAI-like per profile	per bank account per day	7 years
Bill of exchange	draw (FR-36.13)	drawer, drawee, amount, maturity, place of payment, domiciliation bank, acceptance box	PDF; LCR (CFONB 160), RiBa (CBI), AEB 19 / 58 files via wave-48 pack	per-entity bill sequence	statutory form per country (FR Code de commerce L511-1); 10 years
Remittance list and file	remittance (FR-36.13)	remittance id, bank, type, bills (no, drawee, maturity, amount), totals, discount fee	PDF + bank file	per remittance	10 years
Promissory note	draw (FR-36.14)	issuer, beneficiary, amount, maturity, domiciliation	PDF; pagaré layout (ES) via pack	per-entity note sequence	10 years
Cash journal receipt	cash document post (FR-36.16)	desk, document no, date, type, party, amount, tax, cashier, signature line	PDF (A6 / thermal 80 mm)	per desk gap-free	GoBD (DE) single-record immutability; 10 years
Cash book (daily / monthly)	day close, on demand	desk, period, opening, document lines, closing, count, difference, closer	PDF, CSV	n/a	DE §146 AO daily recording; 10 years
Deal confirmation (outbound)	deal capture (FR-36.18)	deal no, trade date, value dates, amounts, rate, settlement instructions	PDF, MT300 via SWIFT service bureau (wave 35 hub)	deal sequence	EMIR record keeping 5 years after termination
Loan interest advice / statement	accrual or settlement (FR-36.19), interest run (FR-36.25)	contract, period, principal, rate, days, interest, withholding	PDF, email to counterparty; IC copy to partner entity	per contract period	transfer-pricing file 10 years
IHC account statement	month end / on demand (FR-36.24)	account, opening, internal orders, interest, closing	camt.053.001.08, PDF	per account per period	10 years
Treasury payment order	approval (FR-36.21)	debtor account, creditor, amount, purpose, approvers	pain.001.001.09	payment request number	10 years

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Bank fee variance report	camt.086 import	bank, period, service lines, variance	PDF, CSV	none	none (management report)

7.36.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`statement-poll`	every 15 min	bank connection	file SHA-256 dedupe	< 30 s per connection	retry 5× exponential; channel marked degraded	connection failure > 3 retries
`cash-position-refresh`	hourly + on file arrival	legal entity	as-of timestamp	< 20 s per entity	keep last good position, flag stale	position age > 2 h
`forecast-rebuild`	hourly + on demand	legal entity	(entity, computedAt) snapshot swap	< 5 min for 1M items	previous snapshot stays live	rebuild failure
`facility-expiry-check`	daily 06:00	legal entity	(instrument, alert day) sent flag	< 1 min	retry next run	none
`ml-match-retrain`	nightly 01:00 if ≥ 200 new matches	tenant	model version	< 20 min	keep prior model	precision drop > 5 pts
`bank-gl-reconcile`	nightly 02:30	bank account	(account, date)	< 5 min	report differences	difference ≠ 0
`pdc-maturity-deposit`	daily 07:00	bank account	(instrument, check date)	< 2 min	proposals only; no posting	proposals older than 2 days
`instrument-maturity`	daily 07:00	legal entity	(instrument, maturity)	< 2 min	remittance / settle proposals	bills past maturity unhonored > 5 days
`deal-valuation`	period end (close checklist task, wave 14) and on demand	legal entity	(entity, key date) run id; reversal first	< 5 min for 20,000 deals	run fails whole; nothing posts	missing market rates
`valuation-reversal`	first calendar day of period 00:30	legal entity	(run id) reversal flag	< 2 min	retry; blocks next valuation	reversal missing
`loan-interest-accrual`	period end	legal entity	(contract, period)	< 3 min for 5,000 loans	per-contract isolation; failures listed	missing fixings
`gl-interest-run`	period end	legal entity	(entity, period)	< 5 min for 10,000 accounts	rerun only after reversal	difference vs prior period > 50%
`payment-batch-window`	every `routing.batchWindowMin`	connection	batch id	< 60 s per 10,000 payments	batch held; payments stay Approved	batch past cutoff unsent
`fee-statement-poll`	monthly day 5	bank connection	file hash	< 1 min	retry daily for 5 days	statement not received
`ihc-interest-and-statements`	month end 23:00	IHC center	(account, period)	< 5 min for 500 accounts	per-account checkpoint	statement generation failure
`cash-desk-open-day-check`	daily 23:00 site local	cash desk	(desk, date)	< 30 s	alert only	days left open > 1

7.36.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
TreasuryBankAccountMaintain	POST/PUT /treasury/bank-accounts; Bank accounts screen	TreasuryBankAccountApprove; AccountsPayablePaymentPost (wave 19)	Treasury analyst
TreasuryBankAccountApprove	bank-account change workflow approve	TreasuryBankAccountMaintain	Treasurer
TreasuryStatementImport	POST /statements/import, /fee-statements/import	none	Treasury analyst, integration service account
TreasuryReconcile	/reconciliations/*/run-rules, /matches; Reconciliation workspace	AccountsPayablePaymentPost, AccountsReceivableCashPost on the same bank account	Treasury analyst
TreasuryReconcileApprove	/reconciliations/*/close, reopen	TreasuryReconcile (same reconciliation)	Treasurer, Controller
TreasuryCheckMaintain	/checks/*/void, reprint, stop	AccountsPayablePaymentApprove	AP payment clerk
TreasuryFacilityMaintain / TreasuryFacilityOverride	Facilities and instruments screen; override limit	override vs maintain	Treasury analyst / Treasurer
TreasuryInstrumentMaintain	/instruments (draw, register, remit, deposit)	TreasuryInstrumentSettle	AR clerk, AP clerk
TreasuryInstrumentSettle	/instruments/*/protest, settle	TreasuryInstrumentMaintain	Treasury analyst
TreasuryCashDeskPost	/cash-desks/*/documents, day close	TreasuryCashDeskApprove; AccountsReceivableCashPost on same desk	Cashier
TreasuryCashDeskApprove	reopen closed day, raise difference limit	TreasuryCashDeskPost	Site controller
TreasuryDealCapture	POST /deals	TreasuryDealConfirm (front office vs back office)	Treasury dealer
TreasuryDealConfirm	/deals/*/confirm, settlement release	TreasuryDealCapture	Treasury back office
TreasuryDealValuate	/valuation-runs	TreasuryDealCapture	Treasury accountant
TreasuryLoanMaintain	/loans	TreasuryPaymentRequestApprove on the same loan	Treasury analyst
TreasuryPaymentRequestCreate	/payment-requests, templates	TreasuryPaymentRequestApprove (BR-36.16)	Treasury analyst
TreasuryPaymentRequestApprove	/payment-requests/*/approve	TreasuryPaymentRequestCreate, TreasuryBankAccountMaintain	Treasurer
TreasuryRoutingConfigure	/routing-rules, bank chains, connections	TreasuryBankSend	Treasury system admin
TreasuryBankSend	/connections/*/send	TreasuryRoutingConfigure	Treasury back office, payment service account
TreasuryIhcOperate / TreasuryIhcOverride	/ihc/payment-orders; overdraft override	operate vs override	IHC operator / Treasurer
TreasuryInterestRun	/interest-runs	GeneralLedgerJournalApprove (wave 11)	Treasury accountant
TreasuryView	read-only GETs, cash position, forecast, all treasury inquiries	none	Treasurer, CFO, Controller

SoD conflicts are enforced as OPA rules (wave 2) at role assignment (warning, requires mitigation record) and at execution for the same object (deny).

7.36.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
TRS-0401	error	IBAN `{iban}` fails the mod-97 check	checksum invalid	correct the IBAN	FR-36.1
TRS-0402	info	Change to bank account `{code}` awaits approval	number / currency / GL change	approver acts in workflow	FR-36.1
TRS-0405	info	Statement already imported as `{statementId}`	identical file hash	none; existing statement returned	BR-36.8
TRS-0406	error	Statement `{id}` exists with a different closing balance	same statement id, different content	contact bank; import corrected file with new id	BR-36.8
TRS-0407	warning	Statement sequence gap: expected `{n}`, received `{m}`	missing statement	request missing statement	FR-36.2
TRS-0408	error	Statement currency `{c1}` differs from account currency `{c2}`	wrong account mapping	fix bank profile account mapping	BR-36.4
TRS-0410	error	Reconciliation cannot close: difference `{amount}`	book ≠ bank + outstanding	match or create remaining items	FR-36.5
TRS-0411	error	Line `{line}` was matched by `{user}`	concurrent match	refresh grid	BR-36.7
TRS-0414	error	Check `{no}` is cleared and cannot be voided	void after Cleared	post a reversal through AP instead	BR-36.5
TRS-0415	error	Facility `{facility}` limit exceeded by `{amount}`	utilization + amount > limit	request override or reduce amount	FR-36.7
TRS-0416	warning	Balance for `{account}` is `{hours}` h old	no statement within stale window	check connection / poll	FR-36.8
TRS-0418	warning	Bank fee `{service}` billed `{billed}` above agreed `{agreed}`	variance beyond tolerance	dispute with bank or update agreement	BR-36.18
TRS-0419	warning	Unexpected bank service code `{service}`	no agreement line	add service to agreement	BR-36.18
TRS-0421	error	Bill `{no}` is already remitted on `{remittanceId}`	second remittance	none	BR-36.10
TRS-0422	error	Protest reason code required	protest without reason	choose a reason code	BR-36.10
TRS-0424	error	Postdated check `{no}` cannot be deposited before `{checkDate}`	early deposit	wait or enable `pdc.allowEarlyDeposit`	BR-36.12
TRS-0425	info	Postdated check `{no}` bounced; open item restored	bank return	follow up with customer	FR-36.15
TRS-0431	error	Cash count difference `{diff}` exceeds limit `{limit}`	count vs computed closing	recount or request approver	BR-36.13
TRS-0432	error	Cash desk `{desk}` day `{date}` is closed	posting to closed day	post on the open day or reopen with approval	BR-36.13
TRS-0433	error	Cash desk balance would become negative	payment > balance	record a bank withdrawal first	FR-36.16
TRS-0441	error	Deal `{no}` confirmation differs: `{field}`	MT300 / manual confirmation mismatch	correct deal or contact counterparty	FR-36.18
TRS-0442	error	No market rate for `{pair}` tenor `{tenor}` on `{date}`	missing wave-12 rate / points	load rates and rerun	FR-36.18
TRS-0445	error	Missing fixing for index `{index}` on `{date}`	floating loan period without fixing	import fixing	FR-36.19
TRS-0451	error	You cannot approve a request you created or edited	self-approval	another approver acts	BR-36.16
TRS-0452	error	No routing rule matches payment batch `{batch}`	rule gap	add rule or default route	FR-36.23
TRS-0453	warning	Bank cutoff passed; execution date moved to `{date}`	request after cutoff	none or mark urgent channel	FR-36.21
TRS-0461	error	IHC account `{account}` overdraft limit exceeded	internal order > limit	override duty or reduce	BR-36.17
TRS-0471	error	EBICS signature failed: key `{keyId}` expired	key rotation due	rotate keys in Vault, re-send	FR-36.12

7.36.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Statement import	30 files/day, 50,000 lines/day; peak 20 files/min at 06:00	8 s / 30 s / 60 s per 10k lines	200,000 lines in < 10 min	18M lines × 1.2 KB ≈ 22 GB	statement_line partitioned monthly by booking_date; GIN trigram on counterparty	queue by connection; imports serialize per bank account; p95 doubles, no rejections
Rule run	50,000 lines/day	5 s / 20 s / 40 s per 10k lines	4 parallel accounts	n/a	partial index status = Unmatched	runs queue in Temporal; UI shows queued
Manual match API	2,000/day; peak 60/min	40 / 150 / 300 ms	50 req/s	1M matches ≈ 0.5 GB	match by (reconciliation_id)	p95 < 400 ms
Cash position query	5,000/day; peak 100/min	50 / 300 / 600 ms	30 req/s	n/a (read model)	read model keyed (entity, currency)	serve cached snapshot with age
Forecast rebuild	1M open items, hourly	n/a	1M items < 5 min	24 months × 12M rows ≈ 6 GB	monthly partitions on expected_date	rebuild cadence drops to every 3 h; snapshot age shown
Check print batch	2,000 checks/run	n/a	2,000 checks < 2 min	50k PDFs ≈ 5 GB (object store)	check by (bank_account_id, number) unique	batch split by 500
Cash journal posting	500 desks × 40 docs/day	60 / 300 / 600 ms	20 docs/s	7M docs ≈ 4 GB	by (desk_id, business_date)	posting queue; receipts print after commit
Instrument status changes	5,000 bills/day	80 / 300 / 600 ms	5,000 per remittance in < 60 s	2M instruments ≈ 2 GB	by (legal_entity_id, kind, status, maturity_date)	remittance batches queue
Deal valuation run	20,000 open deals	n/a	< 5 min	240k valuation rows ≈ 0.3 GB	deal by (entity, status); valuation partitioned by key date	run takes < 15 min; close checklist task stays InProgress
Loan accrual	5,000 contracts	n/a	< 3 min	60k schedule lines ≈ 0.1 GB	schedule by (loan_id, period_no)	runtime scales linearly
Payment routing	100,000 payments/day; peak 10,000 per window	20 ms per payment	10,000 per batch < 60 s	36M routing log rows ≈ 10 GB	log partitioned monthly	batches fill next window; cutoff warnings TRS-0453
GL interest run	10,000 accounts × 31 days	n/a	< 5 min	small	balance projection by (account, value_date)	< 15 min

7.36.20 Localization and compliance hooks

- **Bills of exchange and promissory notes** vary by country: FR LCR / BOR (CFONB 160 remittance), IT RiBa (CBI), ES letra de cambio / pagaré (AEB 19 / 58, Norma 32), PT letras, JP tegata and densai electronic records, BR duplicata / boleto (CNAB 240 / 400). Country packs (wave 48) provide the `InstrumentRemittanceFormat` implementation, the document layout, stamp-duty lines (IT, JP stamp tax by amount band) and protest deadlines; the core supplies the lifecycle and postings.
- **Postdated checks** are common in IN, AE, SA, PH, MY, TH; the IN pack sets withholding at settlement, the AE pack records bounce reason codes required for legal action. Extension point `PdcPolicy` (deposit window, bounce fee, credit hold).

- **Cash journal**: DE GoBD / §146 AO require daily recording, immutability after post, and a cash book export (DSFinV-K is out of scope because desks are not retail POS); PL requires KP / KW cash receipt documents; FR and IT require a sequential cash book. Extension point `CashJournalPolicy` (numbering, mandatory fields, max cash payment amounts such as IT EUR 5,000 / FR EUR 1,000 cash payment limits, warning TRS-0433 extended by pack rules).
- **Bank statement and payment formats**: BAI2 (US), CODA (BE), Norma 43 (ES), CFONB 120 (FR), MT940 / camt.053 (DE, NL), CAMT for SEPA; positive pay (US), CPA-005 (CA); bank chain usage for non-SEPA cross-border payments. Extension point `BankFormatProfile` in the wave-19 format engine.
- **FX deal regulatory reporting**: EU EMIR (FX forwards and swaps are reportable; spot is not), UK EMIR, US Dodd-Frank for swap counterparties; Passage records the LEI on every counterparty and the UTI on deals and exports the trade report through the wave-48 `RegulatoryTradeReport` hook; delegated reporting by the bank is the default.
- **Intercompany loans and GL interest**: thin-capitalization and interest-limitation data (DE §4h EStG, US §163(j), EU ATAD art. 4) and transfer-pricing safe-harbor rates feed the wave-40 transfer-pricing evidence export; withholding tax on cross-border interest uses wave-13 tax codes.
- **Retention**: statements, checks, cash books and deal records 10 years (DE HGB §257, FR), 7 years (US), 5 years after termination for EMIR records; enforced through the wave-5 retention policies.
- **Calendars and rounding**: banking-day calendars per country and currency (TARGET2 for EUR); day-count conventions per currency market (GBP ACT/365, EUR / USD money market ACT/360).

7.36.21 Coverage closure

Register rows assigned to wave 36 in coverage-ax.md §12.99 (16 rows) and coverage-sap.md §13.99 (39 rows), plus two SAP rows proposed for wave 30 that this wave closes (GL balance interest).

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.2.1 Payments → Promissory note journals (Draw / Redraw / Remittance / Settle)	GAP	ADDED as FR-36.14	lifecycle BR-36.11
AX 12.2.2 SSRS Promissory note (vendor) documents	GAP	ADDED as FR-36.14	document in 7.36.15
AX 12.2.3 Promissory notes → Draw / Redraw / Remit / Settle periodic batches	GAP	ADDED as FR-36.14	`instrument-maturity` job 7.36.16
AX 12.2.4 Payment → Promissory note setup (journal names, posting profiles, protest)	GAP	ADDED as FR-36.14	`instrument.notePostingProfile` 7.36.13
AX 12.3.1 Payments → Bill of exchange journals (Draw / Redraw / Protest / Remittance / Settle)	GAP	ADDED as FR-36.13	BR-36.10
AX 12.3.1 Payments → Postdated checks (R3 APAC)	GAP	ADDED as FR-36.15	customer and vendor
AX 12.3.2 Bill of exchange inquiry	GAP	ADDED as FR-36.13	7.36.14 Bill of exchange list
AX 12.3.2 Postdated check inquiry	GAP	ADDED as FR-36.15	7.36.14
AX 12.3.2 SSRS Bill of exchange documents / Remittance list	GAP	ADDED as FR-36.13	7.36.15
AX 12.3.3 Payments → Bill of exchange remittance / protest batches	GAP	ADDED as FR-36.13	`instrument-maturity` job
AX 12.3.4 Payment → Bill of exchange setup (journal names, posting profiles, protest reasons, remittance types)	GAP	ADDED as FR-36.13	`instrument.billPostingProfile`, `instrument.remittanceTypes`
AX 12.3.4 Payment → Postdated checks setup (R3 APAC)	GAP	ADDED as FR-36.15	`pdc.*` parameters
AX 12.5.1 Bank → Bills of exchange / Promissory notes (links)	GAP	ADDED as FR-36.13 / FR-36.14	bank-side remittance and honor via FR-36.3 `billNumber` criterion
AX 12.5.1 Bank → Postdated checks (links)	GAP	ADDED as FR-36.15	deposit proposals per bank account
AX 12.5.4 Bills of exchange / promissory notes setup (bank side)	GAP	ADDED as FR-36.13 / FR-36.14	remittance types per bank account
AX 12.5.4 Postdated checks parameters (R3 APAC)	GAP	ADDED as FR-36.15	7.36.13
SAP 13.1 FBCJ Cash journal	GAP	ADDED as FR-36.16	
SAP 13.1 FBCJC0 / FBCJC1 / FBCJC2 / FBCJC3 cash journal setup, number ranges, business transactions, print	GAP	ADDED as FR-36.16	`cashDesk.*` parameters, receipt and cash book documents
SAP 13.1 S_ALR_87012309 Cash book	GAP	ADDED as FR-36.16	7.36.14 Cash book
SAP 13.1 F.50 / F.52 G/L interest calculation	GAP (proposed 30)	ADDED as FR-36.25	treasury owns IC current-account interest; wave 30 interest notes unchanged
SAP 13.1 OB46 / OB82 / OBAA interest indicators	GAP (proposed 30)	ADDED as FR-36.25	`interest.indicators`
SAP 13.2 F-40 / F-36 (FBW6) Bill of exchange payment (AP)	GAP	ADDED as FR-36.14	AP bills modeled as promissory notes payable
SAP 13.2 FIBLFFP / F111 Payment request	PARTIAL-36	ADDED as FR-36.21	
SAP 13.2 S_ALR_87012116 Bill of exchange list (AP)	GAP	ADDED as FR-36.14	7.36.14
SAP 13.2 F111 Payment program for payment requests	PARTIAL-36	ADDED as FR-36.21	executed by wave-19 payment engine, routed by FR-36.23
SAP 13.2 FBWE AP bill of exchange presentation	GAP	ADDED as FR-36.14	remit / settle
SAP 13.2 Bank-to-bank transfers (payment request types, FIBL)	PARTIAL-36	ADDED as FR-36.21	category `TreasuryTransfer`

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP 13.3 F-34 / F-36 / F-33 Bill of exchange payment / discount (AR)	GAP	ADDED as FR-36.13	collection and discount
SAP 13.3 S_ALR_87012199 Customer bill of exchange list	GAP	ADDED as FR-36.13	
SAP 13.3 FBWD / FBWE AR bill presentation	GAP	ADDED as FR-36.13	
SAP 13.5 FBCJ Cash journal	GAP	ADDED as FR-36.16	duplicate of FI-GL row
SAP 13.5 FRFT / FRFT_B Repetitive payment codes / bank-to-bank transfer	PARTIAL-36	ADDED as FR-36.21	repetitive templates
SAP 13.5 IHC0 / IHC1IP In-House Cash	PARTIAL-36	ADDED as FR-36.24	external payments via IHC accounts out of v1
SAP 13.5 Advanced Payment Management (Approve / Monitor Payments)	PARTIAL-36	ADDED as FR-36.23	central monitor, routing, batching
SAP 13.5 FBWE / FBWD (bills) presentation	GAP	ADDED as FR-36.13	
SAP 13.5 S_ALR_87012309 Cash journal report	GAP	ADDED as FR-36.16	
SAP 13.5 Bank Fee Analysis (camt.086)	GAP	ADDED as FR-36.17	
SAP 13.5 Cash pooling / sweeping run (IHC)	PARTIAL-36	ADDED as FR-36.24	FR-36.11 sweeps + IHC accounts and interest
SAP 13.5 F111 (treasury payment requests) periodic	PARTIAL-36	ADDED as FR-36.21	
SAP 13.5 Bank fee statement import (camt.086)	GAP	ADDED as FR-36.17	`fee-statement-poll` job
SAP 13.5 Bank chains (FIBC)	PARTIAL-36	ADDED as FR-36.22	
SAP 13.5 IHC customizing (centers, accounts, conditions, payment orders)	PARTIAL-36	ADDED as FR-36.24	`ihc.conditions`
SAP 13.5 Cash journal customizing (FBCJC0..3)	GAP	ADDED as FR-36.16	
SAP 13.6 FTR_CREATE / FTR_EDIT / FTR_00 Treasury transaction manager	GAP	ADDED as FR-36.18	FX spot / forward / swap, fixed-term deposits; securities, money-market paper and options OUT-OF-SCOPE for v1 (no customer demand in the manufacturing segment; v2 backlog)
SAP 13.6 TBB1 / TPM10 Post flows	GAP	ADDED as FR-36.18	flows post on value date
SAP 13.6 TPM1 / TPM60 Valuation, amortization, position management	GAP	ADDED as FR-36.18	key-date MTM; amortization of deposits by accrual
SAP 13.6 THMEX Hedge management and accounting (IFRS 9)	GAP	OUT-OF-SCOPE hedge accounting (IFRS 9 / ASC 815 designation, OCI cash-flow hedge reserve) requires effectiveness documentation and auditor-reviewed models; v1 values FX deals through P&L	revisit at v2 planning
SAP 13.6 Exposure management (TEM)	GAP	ADDED as FR-36.20	
SAP 13.6 Loans management (FN*, CML)	GAP	ADDED as FR-36.19	bank and IC loans
SAP 13.6 TPM12 / TPM13 Position / flow list	GAP	ADDED as FR-36.18	7.36.14
SAP 13.6 TPM20 Posting journal	GAP	ADDED as FR-36.18	7.36.14
SAP 13.6 Hedge plan reports	GAP	OUT-OF-SCOPE hedge accounting deferred to v2	FX exposure hedge ratio in FR-36.20 covers the planning view
SAP 13.6 Treasury information system	GAP	ADDED as FR-36.18 / FR-36.20	position, flow, exposure inquiries
SAP 13.6 TBB1 periodic / TPM1 valuation run / accruals	GAP	ADDED as FR-36.18 / FR-36.19	`deal-valuation`, `loan-interest-accrual` jobs
SAP 13.6 Hedge effectiveness test run	GAP	OUT-OF-SCOPE hedge accounting deferred to v2	
SAP 13.6 TRM customizing: product, transaction, flow, update types, valuation areas, accounting codes	GAP	ADDED as FR-36.18	`deal.accountingCodes`, `deal.dayCountDefault`
SAP 13.6 Hedge management customizing (THM*)	GAP	OUT-OF-SCOPE hedge accounting deferred to v2	

7.37 Wave 37: Fixed Assets & Leases

7.37.1 Scope and business context

Wave 37 adds the asset subledger. Capital spend currently lands on expense or clearing accounts; after this wave an accountant acquires assets from AP invoices, projects (CIP) or inventory, depreciates them monthly under several books (local GAAP, IFRS, tax) with different methods and conventions, transfers, splits, revalues, impairs and disposes of them with correct gain/loss, and accounts for leases under IFRS 16 / ASC 842 with right-of-use assets and lease liabilities. A physical inventory with barcode scanning reconciles the register to what exists.

Out of scope: enterprise asset maintenance (wave 44), tax depreciation reports beyond MACRS/ADS tables for the US and straight-line for other v1 countries (country packs in wave 48), component accounting for infrastructure-grade asset hierarchies deeper than 3 levels.

Scope amendment (gap closure). The registers (coverage-ax §12.99 rows 40–43, coverage-sap §13.99 rows 51–55 and 315–321) add asset additions / component records (FR-37.14), insurance records with insured-value indexing (FR-37.15), group assets for pooled tax depreciation (FR-37.16) and capital appropriation requests (FR-37.17) to this wave. Investment program hierarchies with budget distribution and carry-forward land in wave 38 (FR-38.21) because they need budget control; the IRS Form 4562 layout ships with the US country pack (wave 48).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX depreciation proposal → LedgerJournalTrans → post: three forms, and one bad asset aborts the whole journal with an infolog	Journal-centric batch without per-asset isolation	Temporal run with per-asset checkpoints; failing assets reported, others post	Failed runs per month 2 → 0; run restart time 1 h → < 1 min
SAP AFAB runs in background only, errors in SM37 spool; test run then real run, 2–4 h for 100k assets	Batch job architecture	Online proposal preview with formula inputs; 1M asset-periods/h	Run time for 100k assets 2 h → < 6 min
AssetBook convention changes in D365 silently recalculate history; auditors cannot see why a prior charge changed	No immutable transaction log per book	Append-only AssetTransaction; prospective changes only, with explanation	Unexplained changes in audit sample 100% traceable
Mid-quarter test performed in Excel because neither AX nor SAP computes the 40% Q4 test	Tax rules outside the engine	Automatic test per tax book per year with delta posting	Manual tax workbooks: 1 per entity per year → 0
Acquisition from AP needs the invoice line fixed-asset fields, then an asset journal; SAP needs F-90 with asset account assignment keyed again	Two-step with re-keying	AP invoice line with asset flag posts directly to the book	Clicks 18 → 3; re-keyed acquisitions 0

7.37.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Asset master and groups	AssetTable, AssetGroup, AssetGroupBookSetup, form AssetTable	ANLA, ANLZ (time-dependent), asset classes ANKA/ANKT (OAOA), AS01/AS02/AS03	Asset, AssetClass entities	full
Books / depreciation areas	AssetBook (D365 books; AX 2012 value models + AssetDepBook), posting layer per book	ANLB per depreciation area (OADB), T093 chart of depreciation, ledger assignment	AssetBook per asset × book definition, posts to ledger layers (wave 12)	full
Depreciation profiles	AssetDepreciationProfile (StraightLine, ReducingBalance, Manual, Factor, Consumption, StraightLineLifeRemaining, 200%/175%/150%/125% DB, RB/SL), conventions	Depreciation keys T090NA/T090NAZ (AFAMA), multilevel methods T090NP, period controls T090NR	DepreciationProfile with method, convention, factor, tables	full
Depreciation run	AssetDepreciationProposal → LedgerJournalTrans (Asset journal), AssetTrans	AFAB / FAA_DEPRECIATION_POST, ANLP, ANLC	Depreciation run (Temporal) → journal entry per book/period	improved (restartable, per-asset diagnostics)
Acquisition from AP / project / inventory	PurchLine fixed asset fields, ProjTable capitalize, InventJournal to fixed asset	F-90, ABZON, MIGO with asset account assignment (A), AIAB/AIBU AuC settlement	AcquisitionSource links from wave 18 invoice, wave 41 project, wave 16 inventory journal	full
Transfer / split	AssetTransfer, "Split fixed asset" (D365)	ABUMN (intra/inter-company), partial transfer	Transfer and Split transactions preserving history	full
Disposal, revaluation, impairment	AssetTrans types DisposalSale, DisposalScrap, WriteUpAdj, WriteDownAdj, Revaluation	ABAON, ABAVN, F-92, ABAW revaluation, impairment via write-down	Disposal (sale/scrap, partial), Revaluation, Impairment	full
Leases	D365 Asset leasing (AssetLeaseLease, AssetLeaseBook, AssetLeasePaymentSchedule, AssetLeaseLiabilityAmortization)	RE-FX Contract and Lease Management (RECN, RECEEP valuation), IFRS 16 valuation rules	Lease, LeaseSchedule, ROU asset auto-created	full
Physical inventory	Not native (AX 2012 lacked); D365 none; partner add-ons	ANLA-INVNR only; third-party scanning	Count sessions with PWA barcode scanning	improved
Asset additions / components	AssetAddition (Fixed asset → Additions), AssetMajorType, AssetTable.MajorType	Sub-numbers (AS11 asset sub-number), ANLA-ANLN2	AssetAddition records under an asset (FR-37.14)	full
Insurance	AssetInsurance, AssetInsuranceType, insurance companies on AssetTable (Insurance tab), SSRS Fixed asset insurance report	ANLA insurance fields, OAVI insurance types, OAVF / T094 index series, S_ALR_87011951	AssetInsurancePolicy with index series (FR-37.15)	full
Group assets	none (AX uses consolidated books per group manually)	AS21 / AS22 / AS23 group asset, OAYM / OAAX group asset areas (T093A)	GroupAsset per tax book (FR-37.16)	full
Appropriation requests	none native (Budget planning + project quotations)	IMA11 / IMA2 / IMA3 appropriation requests, variants, approval status, IMCCP1	CapexRequest with variants and Temporal approval (FR-37.17)	full

7.37.3 Functional requirements

- **FR-37.1 Asset master** — Must. The system shall maintain assets with class, location, responsible worker, serial/barcode, parent/child (≤ 3 levels), and default dimensions.
  - Notes: asset number from wave-5 number sequences; status NotYetAcquired → Open → Sold / Scrapped / Transferred; barcode unique per legal entity.
- **FR-37.2 Books and profiles** — Must. The system shall attach N books per asset (default from class), each with a depreciation profile, service life, salvage value, convention, posting layer and ledger.
  - Notes: methods: straight-line (service life and life-remaining variants), declining balance (factor 125/150/175/200%, with or without switch to SL), units of production, manual, MACRS tables (GDS 3/5/7/10/15/20-year, ADS); conventions: none, half-year, mid-month (1st or 15th), mid-quarter, full-month, half-year next year; up to 10 books per asset.
- **FR-37.3 Straight-line formula** — Must. The system shall compute period depreciation as `(cost − salvage − accumulated adjustments) / remaining periods` with the convention applied to the first and last periods.
  - Notes: worked mid-month example: cost USD 12,000, salvage 0, 5-year life, in service 2026-03-10, mid-month (15th) convention. Monthly charge = 12,000 / 60 = 200.00. March 2026 takes half a month: 100.00. FY2026 = 100 + 9 × 200 = 1,900.00; FY2027–FY2030 = 2,400.00 each (9,600.00); FY2031 = Jan 200 + Feb 200 + Mar 100 = 500.00. Total 12,000.00. The last period absorbs any rounding residual so Σ depreciation = depreciable basis exactly.
- **FR-37.4 MACRS** — Must. The system shall compute MACRS using the declining-balance-with-switch formula and verify it against the IRS Publication 946 percentage tables.
  - Notes: worked 5-year GDS, half-year convention, cost USD 25,000: rate = 200%/5 = 40%. Y1 = 25,000 × 40% × ½ = 5,000.00 (20.00%). Y2 = 20,000 × 40% = 8,000.00 (32.00%). Y3 = 12,000 × 40% = 4,800.00 (19.20%). Y4: DB = 7,200 × 40% = 2,880.00; SL over remaining 2.5 years = 7,200 / 2.5 = 2,880.00 → switch (11.52%). Y5 = 2,880.00 (11.52%). Y6 = 1,440.00 (5.76%). Σ = 25,000.00. Mid-quarter applies when > 40% of the year's basis is placed in Q4; the system computes the test per tax book per year and alerts. Bonus depreciation percentage and Section 179 amounts recorded as first-year adjustments.
- **FR-37.5 Depreciation run** — Must. The system shall propose and post depreciation per book, period and legal entity as one journal entry per book with lines per asset (or summarized per account/dimension by option).
  - Notes: Temporal workflow with per-asset checkpoints; rerun for the same book/period is rejected unless the prior run is reversed; 1,000,000 asset-book-periods per hour; proposal preview shows per-asset amount and formula inputs.
- **FR-37.6 Acquisition sources** — Must. The system shall create acquisitions from AP invoice lines flagged as asset, project CIP capitalization, inventory journal issue to asset, and manual journal.
  - Notes: AP invoice with asset flag posts Dr Asset cost / Cr AP (instead of expense); CIP collects to the asset's under-construction book value and capitalizes on a date with transfer from CIP account; inventory issue uses the item's current cost (wave 20).

- **FR-37.7 Transfer and split** — Must. The system shall transfer assets between dimensions, locations or legal entities and split by percentage or amount into new assets with proportional cost and accumulated depreciation.
  - Notes: intercompany transfer posts disposal in the source and acquisition in the target at NBV or agreed price (wave 40 IC due-to/due-from); split percentages must sum to 100.00%.
- **FR-37.8 Disposal** — Must. The system shall dispose assets by sale (to a customer invoice, wave 29, or free text) or scrap, including partial disposal, posting cost reversal, accumulated depreciation reversal and gain/loss.
  - Notes: gain/loss = proceeds − NBV of the disposed portion; partial disposal uses the disposed share of cost and accumulated depreciation; depreciation up to the disposal date is posted first when `depreciate-on-disposal` is set per book.
- **FR-37.9 Revaluation and impairment** — Must. The system shall post revaluation (IAS 16 revaluation model) to OCI reserve or P&L per policy and impairment (IAS 36) as write-down with reversal limits.
  - Notes: impairment reversal cannot exceed the carrying amount that would have existed without impairment.
- **FR-37.10 Leases (IFRS 16 / ASC 842)** — Must. The system shall record leases with payment schedules, incremental borrowing rate, options, and generate the liability amortization and ROU depreciation schedules, posting initial recognition, monthly interest, payments and depreciation.
  - Notes: worked example, annual payments in arrears USD 10,000 for 5 years, IBR 5%: liability = 10,000 × (1 − 1.05^{-5})/0.05 = 43,294.77. Schedule: Y1 interest 2,164.74, closing 35,459.51; Y2 1,772.98 → 27,232.49; Y3 1,361.62 → 18,594.11; Y4 929.71 → 9,523.82; Y5 476.18 (plugged from 476.19 to clear the 0.01 residual) → 0.00. Total interest 6,705.23 = 50,000 − 43,294.77. ROU asset 43,294.77 depreciated straight-line 8,658.95 for Y1–Y4 and 8,658.97 in Y5. Monthly schedules use periodic rate = nominal/12 by default (option: effective (1+r)^(1/12) − 1). Modifications remeasure the liability with the ROU adjusted by the same amount; short-term (≤ 12 months) and low-value (< USD 5,000 / EUR 5,000) leases are expensed under election.
- **FR-37.11 Physical inventory** — Should. The system shall run count sessions with the PWA scanner, record found/missing/moved assets and create transfer or scrap proposals.
- **FR-37.12 Asset budgeting** — Should. The system shall forecast depreciation for existing assets and planned acquisitions and publish it as a budget source to wave 38.
- **FR-37.13 Subledger reconciliation** — Must. The system shall reconcile Σ asset cost and Σ accumulated depreciation per book to the GL accounts nightly and list differences by asset.
- **FR-37.14 Asset additions and component records** — Should. The system shall record additions and components under an asset (improvement, major part, accessory) with description, quantity, serial number, date, cost reference and optional link to an acquisition adjustment, without creating a separate depreciable asset unless the user promotes the record to a sub-asset.
  - Notes: an addition with `financial = true` creates an AcquisitionAdj transaction on the selected books in the same request; non-financial additions are descriptive only; promote-to-sub-asset creates a child asset (≤ 3 levels, FR-37.1) and moves the referenced cost by split (FR-37.7); maximum 500 additions per asset.
- **FR-37.15 Insurance records and insured-value indexing** — Should. The system shall record insurance policies per asset (insurer, policy number, insurance type, coverage start / end, premium, insured value basis) and compute the indexed insured value from replacement value and an index series.
  - Notes: insurance types (property, machinery breakdown, transit, fleet) and insurance companies are master data; insured value_t = base replacement value × index_t ÷ index_base; index series per asset class and country, yearly values loaded by the `insurance-index-update` job; policies expiring within 30 days raise alerts; the insurance report (7.37.14) lists insured vs NBV vs replacement value and under-insured assets (insured < 90% of indexed replacement value).
- **FR-37.16 Group assets** — Should. The system shall group assets of the same class and tax year into a group asset for a designated tax book and compute depreciation on the group's aggregate basis, while other books keep per-asset depreciation.
  - Notes: covers US general asset accounts (IRC §168(i)(4)), DE Sammelposten (EUR 250.01–1,000.00 pool depreciated over 5 years), UK capital allowance pools and IN block of assets via profiles from the wave-48 packs; member assets carry the book with `depreciatedAtGroup = true` and no own charge; disposal of a member under the default rule credits proceeds to the group (no per-asset gain / loss) and under `terminate-on-disposal` computes gain / loss per member; the group's depreciation run line references every member for drill.
- **FR-37.17 Capital appropriation requests** — Should. The system shall let requesters create capital expenditure requests with one or more variants (scope, amount per year, expected in-service date, depreciation profile assumption), route them through a value-based approval workflow, and on approval create planned acquisitions and, when the request is executed, the asset under construction or project.
  - Notes: status Draft → Submitted → InApproval → Approved → Released → Closed, with Rejected and Postponed; approval levels by amount (defaults ≤ USD 50,000 plant manager, ≤ 500,000 CFO, above board) through the wave-4 workflow; the approved variant's yearly amounts publish as planned acquisitions to FR-37.12 depreciation forecasting and as a budget source to wave 38; assignment to an investment program position (FR-38.21) is optional; actual spend from linked CIP / project (wave 41) is tracked against the approved amount with an overrun warning at 10%.

7.37.4 Data model

Asset: id, legal_entity_id, asset_number (unique per entity), class_id (FK AssetClass), name, status, location_id, responsible_worker_id, barcode, parent_asset_id, default_dimensions jsonb, acquired_date, quantity numeric(19,4).

AssetBook

Field	Type	Required	Notes
id	uuid	yes	PK
asset_id	uuid	yes	FK Asset
book_definition_id	uuid	yes	FK BookDefinition (ledger, posting layer, currency)
profile_id	uuid	yes	FK DepreciationProfile
service_life_periods	int	yes	1..1200
salvage_value	numeric(19,4)	yes	default 0
convention	enum	yes	None/HalfYear/MidMonth1/MidMonth15/MidQuarter/FullMonth/HalfYearNextYear
in_service_date	date	yes	
cost, accumulated_depreciation, nbv	numeric(19,4)	yes	projections from AssetTransaction
last_depreciated_period	text	no	fiscal period key
status	enum	yes	NotYetAcquired/Open/Closed/Sold/Scrapped

AssetTransaction

Field	Type	Required	Notes
id	uuid	yes	PK, append-only
asset_book_id	uuid	yes	FK

Field	Type	Required	Notes
type	enum	yes	Acquisition/AcquisitionAdj/Depreciation/DepreciationAdj/Transfer/Split/Revaluation/Impairment/DisposalSale/DisposalScrap/CIPCapitalization
transaction_date	date	yes	
amount	numeric(19,4)	yes	signed
journal_entry_id	uuid	yes	FK wave-11 journal
source_type, source_id	text, uuid	no	AP invoice, project, inventory journal, lease
reversal_of_id	uuid	no	
run_id	uuid	no	depreciation run

Lease: id, legal_entity_id, lease_number, lessor_party_id, commencement_date, term_months, payment_frequency, payment_amount, payment_timing (arrears/advance), ibr_rate numeric(9,6), classification (Finance/Operating for ASC 842), rou_asset_id, status (Draft/Active/Modified/Terminated), election flags.

LeaseScheduleLine: lease_id, period_no, opening_liability, interest, payment, closing_liability, rou_depreciation, posted_journal_id.

Entities added by FR-37.14–FR-37.17: **AssetAddition** (id, asset_id, type Improvement/Component/Accessory, description, quantity, serial_number, added_date, financial bool, asset_transaction_id, promoted_asset_id); **AssetInsurancePolicy** (id, asset_id, insurer_party_id, policy_number, insurance_type_id, coverage_from, coverage_to, premium, basis Replacement/NBV/Agreed, base_value, index_series_id, index_base_year); **IndexSeries / IndexValue** (series code, country, class, year, value numeric(9,4)); **GroupAsset** (id, legal_entity_id, book_definition_id, class_id, tax_year, profile_id, basis, accumulated_depreciation, member_count, status); **CapexRequest** (id, legal_entity_id, number, title, requester_id, status, selected_variant_id, program_position_id, approved_amount, currency, linked_asset_ids[], linked_project_id) and **CapexRequestVariant** (id, request_id, description, amounts_by_year jsonb, in_service_date, profile_assumption_id, npv, irr).

Invariants: Σ AssetTransaction amounts by type = projected cost / accumulated depreciation; NBV ≥ 0 unless revaluation policy allows; Σ depreciation over life = cost – salvage; LeaseScheduleLine closing of period n = opening of n+1 and final closing = 0.00. Indexes: AssetTransaction by (asset_book_id, transaction_date), by run_id; AssetBook by (book_definition_id, status).

7.37.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/assets	Create asset	Idempotency-Key	AssetMaintain
POST	/api/v1/assets/{id}/books	Add book	Idempotency-Key	AssetMaintain
POST	/api/v1/assets/depreciation-runs	Propose run (book, period)	(book, period) unique	AssetDepreciate
POST	/api/v1/assets/depreciation-runs/{id}/post	Post proposal	state check	AssetPostJournal
POST	/api/v1/assets/{id}/transactions	Acquisition/adjustment/revaluation/impairment	Idempotency-Key	AssetPostJournal
POST	/api/v1/assets/{id}/dispose	Sale/scrap, partial %	Idempotency-Key	AssetDispose
POST	/api/v1/assets/{id}/transfer	Transfer / split	Idempotency-Key	AssetMaintain
GET	/api/v1/assets/{id}/schedule?book=	Projected depreciation	n/a	AssetView
POST	/api/v1/leases	Create lease and compute schedule	Idempotency-Key	LeaseMaintain
POST	/api/v1/leases/{id}/modify	Remeasure	Idempotency-Key	LeaseMaintain
POST	/api/v1/assets/count-sessions	Physical inventory	Idempotency-Key	AssetCount
POST	/api/v1/assets/{id}/additions	Record addition / component	Idempotency-Key	AssetMaintain
POST	/api/v1/assets/{id}/insurance-policies	Add insurance policy	Idempotency-Key	AssetInsuranceMaintain
POST	/api/v1/assets/group-assets	Create group asset and assign members	Idempotency-Key	AssetConfigure
POST	/api/v1/assets/capex-requests	Create appropriation request with variants	Idempotency-Key	CapexRequestCreate
POST	/api/v1/assets/capex-requests/{id}/submit	Start approval workflow	state check	CapexRequestCreate

Validation: book must be Open and period open per ledger × module (wave 14); disposal percentage in (0,100]; lease IBR in (0, 50%].

Event	Trigger	Payload fields	Consumers
`dev.passage.assets.asset.acquired.v1`	acquisition posted	assetId, bookIds[], cost, sourceType, sourceId	budget-svc, analytics
`dev.passage.assets.depreciation.posted.v1`	run posted	runId, bookDefinitionId, period, assetCount, totalAmount, journalEntryId	costacct-svc, analytics
`dev.passage.assets.asset.disposed.v1`	disposal posted	assetId, bookId, percentage, proceeds, nbv, gainLoss	tax, analytics
`dev.passage.assets.lease.recognized.v1`	lease activated	leaseId, liability, rouAsset, termMonths, rate	analytics
`dev.passage.assets.capexrequest.approved.v1`	request approved	requestId, variantId, amountsByYear, programPositionId	budget-svc (38), asset forecasting
`dev.passage.assets.insurance.expiring.v1`	policy within 30 days of expiry	policyId, assetId, coverageTo, insuredValue	workspace, notifications

7.37.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Fixed assets workspace	New asset, run depreciation, exceptions, reconciliation tile	assets by status, pending acquisitions, unreconciled difference	AssetView
Asset detail	Books tab, transactions tab, schedule, dispose, transfer	number, class, NBV per book, location	AssetMaintain
Depreciation run	Propose, review, post, reverse	book, period, asset count, total, exceptions	AssetDepreciate
Lease detail	Schedule, modify, terminate	liability, ROU, interest, next payment	LeaseMaintain
Count session (PWA)	Scan, mark found/missing, submit	barcode, location, found at	AssetCount

Interaction rules: schedule preview recalculates on any book field change before save; bulk class change via grid multi-select with validation at submit; depreciation run proposals paginate 500 assets per page with per-asset formula tooltip.

7.37.7 Business rules and edge cases

- **BR-37.1** Period depreciation rounds half-up to currency minor unit; the final period of the life is a plug so Σ = depreciable basis.
- **BR-37.2** Asset state machine: NotYetAcquired → Open (first acquisition) → Sold | Scrapped | Transferred (full IC transfer); Open → Open on partial disposal; no transaction on Sold/Scrapped except reversal of the disposal.

- **BR-37.3** Depreciation for a period cannot post if a prior period for the same book is unposted (sequential), except when the book's `catch-up` option is set, which posts the arrears into the current period.
- **BR-37.4** Change of service life or salvage after acquisition applies prospectively: new charge = (NBV – salvage) / remaining periods.
- **BR-37.5** Mid-quarter test: if Q4 in-service basis / full-year basis > 40% for a tax book, all assets of that year switch to mid-quarter; assets already depreciated are recomputed and the delta posts in the current period.
- **BR-37.6** Disposal date before last depreciated period requires reversal of later depreciation first; the system offers the reversal set.
- **BR-37.7** Partial disposal percentage p: disposed cost = cost × p, disposed accumulated depreciation = acc × p, rounded; residual cents stay on the remaining asset.
- **BR-37.8** Lease liability remeasurement on modification: new PV at revised rate (if scope change) or original rate (if only payment change under ASC 842 for operating leases); ROU adjusts by the same amount; if ROU would go below zero the excess hits P&L.
- **BR-37.9** Concurrency: the depreciation run holds a per-(book, period) Postgres advisory lock; a second run request returns 409.
- **BR-37.10** Indexed insured value rounds half-up to the currency minor unit; a missing index value for the current year uses the latest available year and raises AST-0531.
- **BR-37.11** A member asset can belong to one group asset per book; adding a member after the group's first depreciation period posts a catch-up in the current period on the group; removing a member is allowed only through disposal or transfer.
- **BR-37.12** Capex request state machine: Draft → Submitted → InApproval → Approved → Released → Closed; InApproval → Rejected | Postponed → Draft; an amount change after Approved returns the request to Submitted when the increase exceeds 10% or USD 10,000 / EUR 10,000, whichever is lower.
- **BR-37.13** Promoting a financial addition to a sub-asset moves cost and accumulated depreciation of the referenced transaction pro rata; the parent's remaining life is unchanged.

7.37.8 Acceptance criteria

- **AC-37.1** (FR-37.3) Given the USD 12,000 mid-month example, when the schedule is generated, then FY2026 = 1,900.00, FY2027–2030 = 2,400.00 each, FY2031 = 500.00.
- **AC-37.2** (FR-37.4) Given the USD 25,000 MACRS 5-year example, when computed, then the yearly amounts are 5,000.00 / 8,000.00 / 4,800.00 / 2,880.00 / 2,880.00 / 1,440.00.
- **AC-37.3** (FR-37.4) Given 100 fixture assets across methods and conventions, when depreciated for full lives, then every amount matches the reference table to the cent.
- **AC-37.4** (FR-37.5) Given 10,000 assets, when the monthly run posts, then one journal entry per book exists, Σ lines = Σ proposal, and a second run for the same period is rejected.
- **AC-37.5** (FR-37.8) Given cost 10,000, accumulated 4,000, proceeds 7,000, 50% disposal, when disposed, then cost –5,000, accumulated +2,000, gain 4,000 posts and the remaining NBV is 3,000.
- **AC-37.6** (FR-37.10) Given the 5-year USD 10,000 lease at 5%, when activated, then liability = 43,294.77, Y5 interest = 476.18 and closing = 0.00.
- **AC-37.7** (FR-37.10) Given 25 fixture leases, when schedules compute, then every line matches the accountant workbook to the cent.
- **AC-37.8** (FR-37.13) Given the demo entity after a month, when reconciliation runs, then Σ asset cost and Σ accumulated depreciation equal the GL accounts with 0.00 difference.
- **AC-37.9** (FR-37.15) Given a replacement value of USD 100,000.00 at index 112.0 (base year 2024) and an index of 118.5 for 2026, when the index update runs, then the insured value is 105,803.57 and assets insured below 95,223.21 (90%) are listed as under-insured.
- **AC-37.10** (FR-37.16) Given 40 DE assets of EUR 600.00 each acquired in 2026 in a Sammelposten group, when the tax book depreciates FY2026, then one group line of 24,000.00 / 5 = 4,800.00 posts and each member shows 0.00 own depreciation with a drill to the group.
- **AC-37.11** (FR-37.17) Given a capex request of USD 320,000.00, when submitted, then the workflow routes to the CFO level; when approved, then planned acquisitions of the selected variant appear in the depreciation forecast and `capexrequest.approved` reaches budget-svc.
- **AC-37.12** (FR-37.14) Given an asset with a financial addition of USD 2,000.00, when saved, then an AcquisitionAdj of 2,000.00 posts on every Open book and the addition links to the transaction.

7.37.9 Test plan and fixtures

Fixtures: `dep-100` (100 assets × SL, DB 150/200%, UoP, MACRS 3/5/7-year, conventions, short first years, salvage > 0), `lease-25` (accountant workbook: arrears/advance, monthly/annual, modifications, terminations), `disposal-30` (full/partial, before and after period end), `migration-nbv` (opening NBVs for the demo entity).

Test	Type	Target	Pass criterion
Method formulas	Unit	depreciation engine	match tables to the cent
Life totals	Property	engine	Σ depreciation = cost – salvage for random inputs
Lease PV and schedule	Unit/Property	lease engine	closing = 0.00; Σ interest = Σ payments – PV
Run posting	Contract	journal API	one entry per book; idempotent
Disposal flows	E2E	workspace	gain/loss per fixture
Run throughput	Performance	run	1M asset-book-periods/hour

Reconciliation checks: asset subledger to GL per book nightly; lease liability schedule to GL lease liability; ROU asset to GL ROU accounts.

7.37.10 Deployment and operations

Helm chart `asset-svc`; env `ASSET_RUN_BATCH_SIZE=500`, `ASSET_MAX_BOOKS=10`; Atlas migrations; flags `assets.physical-inventory`, `assets.asc842-operating`. Jobs: `asset-gl-reconcile` nightly 02:00, `lease-schedule-post` monthly on period open, `mid-quarter-test` on tax year end. SLOs: schedule API p95 < 200 ms; run throughput as above; reconciliation job < 10 min for 1M assets. Alerts: reconciliation difference ≠ 0, run failure, lease schedule unposted after period close. Rollback: Helm rollback; posted runs reversed through the reversal API, never deleted.

7.37.11 Migration from AX / SAP

Source	Passage entity	Notes
AssetTable, AssetGroup	Asset, AssetClass	
AssetBook / AssetDepBook, AssetDepreciationProfile	AssetBook, DepreciationProfile	method/convention mapped; opening NBV as acquisition + accumulated depreciation transactions dated cutover
AssetTrans	AssetTransaction (summarized)	history summarized to opening balances; detail retained in archive
AssetLease*	Lease, LeaseScheduleLine	remaining schedule recomputed; variance to legacy shown

Source	Passage entity	Notes
ANLA, ANLZ, ANKA	Asset, AssetClass	
ANLB, ANLC, T090NA	AssetBook, DepreciationProfile	depreciation key → profile map table; ANLC year values → opening balances
ANLP, ANEP	AssetTransaction (summarized)	
RE-FX contracts	Lease	

Validation: Σ opening NBV per book = GL; count of active assets matches; recomputed next-period depreciation within 0.5% of legacy (differences listed). Not migrated: fully depreciated and retired assets older than 7 years (archive), AX value model posting profiles (re-authored).

7.37.12 Dependencies, risks, open questions

Depends on: wave 11 journal and dimensions, wave 12 posting layers, wave 14 period control, wave 18 AP invoice, wave 16 inventory journals, wave 36 for IC transfer settlement; forward-references wave 41 projects (CIP) and wave 44 EAM.

Risk	Likelihood	Impact	Mitigation
Legacy NBV differs from recomputed schedules	High	Medium	Load NBV as balances, not recomputed; variance report
ASC 842 operating lease single-cost rule complexity	Medium	Medium	Flag-gated; IFRS 16 first

Open questions: (1) Posting summarization default per account or per asset? Proposed per asset; decide by wave 37 start. (2) Support 175%/125% DB in v1? Proposed yes (cheap); decide by wave 37. (3) Bonus depreciation handled as profile or adjustment? Proposed adjustment; decide by wave 37.

7.37.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`assets.numberSequence`	legal entity	sequence	FA-######	number sequence id	asset numbers (FR-37.1)	Fixed assets parameters → Number sequences	AS08 number ranges
`assets.capitalizationThreshold`	legal entity	money	USD 2,500 / EUR 800	≥ 0	AP lines below threshold do not propose asset creation	Fixed assets parameters → "Capitalization threshold" (D365)	OAYK low-value asset amount
`assets.createFromPurchasing`	legal entity	enum	ProductReceipt	Off / ProductReceipt / VendorInvoice	when AP-flagged lines create assets (FR-37.6)	Fixed assets parameters → "Create asset during product receipt or vendor invoice posting"	asset account assignment category A (OME9), ANLA creation in ME21N
`assets.defaultBooksByClass`	asset class	table	none	book definitions + profiles	books attached on create (FR-37.2)	Fixed asset groups → Books setup (AssetGroupBookSetup)	OAYZ depreciation areas per asset class
`assets.bookDefinition`	legal entity	table	Local GAAP, IFRS, Tax	ledger, posting layer, currency, posting allowed	posting target per book	Books (AssetBookTable), posting layer	OADB depreciation areas, OAOB chart of depreciation
`assets.postingProfile`	legal entity	table	none	(class, transaction type, book) → accounts	account determination	Fixed asset posting profiles (AssetLedgerAccounts)	AO90 account determination
`assets.depreciationSummarization`	book	enum	PerAsset	PerAsset / PerAccountDimension	journal line granularity (FR-37.5)	Fixed assets parameters → "Summarize depreciation"	OAYR posting rules (period and method per area)
`assets.depreciationPeriodicity`	book	enum	Monthly	Monthly / Quarterly / Yearly	run frequency	Depreciation profile → Depreciation year / frequency	OAYR period interval
`assets.catchUp`	book	bool	false	true / false	BR-37.3 arrears into current period	"Catch up depreciation" option	AFAB "unplanned/repeat run"
`assets.depreciateOnDisposal`	book	bool	true	true / false	depreciation to disposal date first (FR-37.8)	Fixed assets parameters → "Depreciate asset when disposing"	depreciation key period control for retirement (T090NR)
`assets.roundingMinorUnit`	book	enum	currency minor unit	minor unit / whole unit	BR-37.1 rounding	Depreciation profile → Rounding	OAYO rounding specifications
`assets.maxBooks`	tenant	int	10	1–20	books per asset (env `ASSET_MAX_BOOKS`)	none	max 99 depreciation areas
`assets.disposalApproval`	legal entity	enum	AboveNbv10k	None / Always / AboveNbv10k	workflow on disposal	Fixed assets parameters → "Approval required for disposals" (workflow)	none native (ABAVN no workflow)
`assets.midQuarterTest`	tax book	bool	true for US tax books	true / false	BR-37.5	none	US-specific depreciation keys with mid-quarter (T090NA)
`assets.bonusPercent`	tax book × year	decimal	per wave-48 US pack	0–100%	first-year bonus adjustment (FR-37.4)	Special depreciation allowance profile	special depreciation area (bonus)
`lease.periodicRateMethod`	legal entity	enum	Nominal/12	Nominal/12 / Effective	monthly rate (FR-37.10)	Asset leasing parameters → interest calculation	RE-FX valuation rule (RECEEP)
`lease.lowValueThreshold`	legal entity	money	USD 5,000 / EUR 5,000	≥ 0	low-value election	Asset leasing parameters → "Low value threshold"	RE-FX contract type exemption
`lease.shortTermMonths`	legal entity	int	12	1–12	short-term election	Asset leasing parameters	RE-FX valuation rule term
`insurance.types` / `insurance.companies`	tenant	table	seeded 4 types	code, description	FR-37.15 master data	Insurance types / Insurance companies	OAVI insurance types, insurance companies (T086)
`insurance.indexSeries`	tenant	table	none	series code, country, class, yearly values	FR-37.15 indexing	none	OAVF index series (T094)
`assets.groupAssetBooks`	tax book	bool	false	true / false	book allows group assets (FR-37.16)	none	OAYM / OAAX group asset areas (T093A)

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`capex.approvalLevels`	legal entity	table	50,000 / 500,000 / above	amount bands → approver role	FR-37.17 routing	Budget planning workflow (stand-in)	IMA approval status profile, appropriation request types (TAPRT)
`capex.overrunWarnPct`	legal entity	decimal	10%	0–100%	warning on spend vs approved	none	IM budget availability control tolerance
`count.sessionTolerance`	site	int	0	0–10	missing items allowed before escalation (FR-37.11)	none	none

7.37.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Asset register	list inquiry	entity, class, book, status, location	number, name, in-service, cost, accumulated, NBV, life remaining	`asset_book`	Fixed assets list page, Fixed asset balances	AR01 asset balances
Asset history sheet (roll-forward)	statutory report	entity, book, fiscal year, class	opening cost, acquisitions, retirements, transfers, revaluation, closing cost; opening acc., depreciation, impairment, closing acc.; NBV	`asset_transaction` aggregated	SSRS Fixed asset statement / Fixed asset note	S_ALR_87011990 asset history sheet
Depreciation forecast	drill-down	entity, book, horizon months, include planned (FR-37.17)	period, existing assets, planned acquisitions, total	forecast projection (FR-37.12)	SSRS Fixed asset projection / Depreciation expense projection	S_ALR_87012936 depreciation simulation
Acquisitions / retirements list	operational report	entity, book, date range, source type	asset, date, amount, source document, gain/loss	`asset_transaction`	SSRS Fixed asset acquisitions / disposals	S_ALR_87012050 acquisitions, S_ALR_87012052 retirements
Asset transactions	list inquiry	asset, book, type, date	type, date, amount, journal, run	`asset_transaction`	Fixed asset transactions (AssetTrans)	AW01N asset explorer, S_ALR_87012048
Tax depreciation (US 4562 data)	statutory report	entity, tax book, year	Part I §179, Part II bonus, Part III MACRS by recovery class and convention, mid-quarter flag	tax book projection	none	S_ALR_87010127 / 87010129
Lease roll-forward and maturity analysis	statutory report	entity, period, classification	opening liability, additions, interest, payments, remeasurements, closing; undiscounted maturities < 1 / 1–5 / > 5 years (IFRS 16.58)	`lease_schedule_line`	Asset leasing → Lease summary / liability amortization report	RE-FX RECEEP evaluation, IFRS 16 disclosure reports
Insurance report	operational report	entity, insurer, type, expiry window	asset, policy, insured value, indexed replacement value, NBV, under-insured flag	`asset_insurance_policy`	SSRS Fixed asset insurance report	S_ALR_87011951
Physical inventory list	operational report	site, location, session	expected, found, missing, moved	`count_session`	none	S_ALR_87011979 asset inventory list
Subledger to GL reconciliation	KPI / drill-down	entity, book, period	subledger cost / acc., GL balance, difference per account	reconciliation job output (FR-37.13)	Fixed asset reconciliation (D365 report)	ABST2 reconciliation analysis
Group asset members	drill-down	group asset, year	members, basis, group depreciation share	`group_asset`	none	AS23 group asset display
Capex request pipeline	KPI	entity, status, program, year	count, requested, approved, spent, overrun %	`capex_request`	none	IM appropriation request information system, IMCCV1

7.37.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Asset tag label	asset create / reprint	asset number, barcode (Code 128 or QR), entity, location	PDF label (50 × 25 mm), ZPL for thermal printers	asset number	none
Asset transfer slip	transfer / IC transfer (FR-37.7)	asset, from / to dimension, location, entity, NBV, approver	PDF	transfer document number	10 years (IC transfer evidence)
Disposal / scrap certificate	scrap disposal (FR-37.8)	asset, date, method, witness, NBV written off	PDF	disposal number	10 years; environmental proof for regulated waste where required by pack
Asset history sheet	period / year end	as 7.37.14	PDF, XLSX, XBRL tags via wave 15	none	statutory notes to accounts; 10 years
Lease schedule	lease activation / modification	lease, payments, interest, liability, ROU	PDF, XLSX	lease number	IFRS 16 / ASC 842 audit file; 10 years
IRS Form 4562	tax year end	US pack layout	PDF (wave 48)	none	7 years (US)
Count sheet	count session start	location, expected assets, barcode, found box	PDF, PWA list	session number	none
Capex request approval record	approval	request, variant, amounts, NPV/IRR, approvers with timestamps	PDF	request number	10 years (capex governance)

7.37.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`depreciation-run`	monthly on period open + on demand	(book, period, legal entity)	per-asset checkpoint; (book, period) advisory lock	< 6 min for 100k assets	failing assets listed; others post	run failure, exceptions > 0
`lease-schedule-post`	monthly on period open	legal entity	(lease, period) posted flag	< 2 min for 5,000 leases	per-lease isolation	unposted after close
`asset-gl-reconcile`	nightly 02:00	(entity, book)	(entity, book, date)	< 10 min for 1M assets	report only	difference ≠ 0
`mid-quarter-test`	tax year end + on demand	(entity, tax book, year)	(entity, book, year) result	< 2 min	alert; no auto-post without approval	Q4 share > 40%

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`depreciation-forecast-publish`	nightly 03:00	legal entity	snapshot version	< 5 min	previous snapshot kept	budget source stale > 2 days
`insurance-index-update`	yearly 01-15 + on load	index series	(series, year)	< 1 min	missing year → AST-0531	under-insured count change
`insurance-expiry-check`	daily 06:00	legal entity	(policy, alert day)	< 1 min	retry	expiring policies
`capex-overrun-check`	daily 05:00	legal entity	(request, date)	< 1 min	warning only	overrun > `capex.overrunWarnPct`

7.37.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
AssetView	GET /assets/*, workspace, all 7.37.14 inquiries	none	Fixed asset accountant, Controller, Auditor
AssetMaintain	POST /assets, /books, /additions, /transfer	AssetDispose on same asset when disposal approval is on	Fixed asset accountant
AssetConfigure	classes, book definitions, profiles, posting profiles, group assets	AssetPostJournal	Finance system admin
AssetDepreciate	/depreciation-runs (propose)	none	Fixed asset accountant
AssetPostJournal	/depreciation-runs/*/post, /transactions	AssetConfigure; AccountsPayableInvoiceApprove for AP-sourced acquisitions	Fixed asset accountant
AssetDispose	/dispose	AccountsReceivableInvoicePost for the sale invoice	Fixed asset accountant
AssetDisposeApprove	disposal workflow approve	AssetDispose	Controller
LeaseMaintain	/leases, /modify	LeasePostApprove	Lease accountant
AssetInsuranceMaintain	/insurance-policies, index series	none	Risk manager
AssetCount	/count-sessions, PWA scanner	AssetMaintain (counter cannot adjust register)	Site asset custodian
CapexRequestCreate	/capex-requests, submit	CapexRequestApprove on own request	Plant manager, Engineering lead
CapexRequestApprove	workflow approve	CapexRequestCreate (same request)	CFO, Controller

7.37.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
AST-0501	error	Book `{book}` period `{period}` is closed for fixed assets	period control (wave 14)	open period or post in current	FR-37.5
AST-0502	error	Depreciation for `{book}` `{period}` is already posted	rerun without reversal	reverse the prior run first	FR-37.5
AST-0503	error	Previous period `{period}` is not depreciated	sequential rule	run prior period or enable catch-up	BR-37.3
AST-0504	error	Another depreciation run holds `{book}` `{period}`	advisory lock	wait for completion	BR-37.9
AST-0507	error	Split percentages total `{pct}`; they must total 100.00%	split input	correct percentages	FR-37.7
AST-0508	error	Disposal date precedes last depreciated period `{period}`	backdated disposal	accept proposed reversal set	BR-37.6
AST-0509	error	Asset `{asset}` is `{status}`; only the disposal can be reversed	posting on Sold / Scrapped	reverse disposal	BR-37.2
AST-0511	warning	Q4 placed-in-service share `{pct}` exceeds 40%; mid-quarter applies	mid-quarter test	review and approve delta posting	BR-37.5
AST-0512	warning	Impairment reversal limited to `{amount}`	carrying amount cap	none	FR-37.9
AST-0515	error	Lease IBR `{rate}` outside (0, 50%]	invalid input	correct rate	FR-37.10
AST-0516	info	Lease remeasured; ROU adjusted by `{amount}`	modification	none	BR-37.8
AST-0520	error	Subledger differs from GL by `{amount}` on `{account}`	reconciliation	investigate listed assets	FR-37.13
AST-0525	warning	Barcode `{code}` not in register	count scan	create asset or mark found-unknown	FR-37.11
AST-0531	warning	Index `{series}` has no value for `{year}`; using `{lastYear}`	missing index	load index value	BR-37.10
AST-0532	warning	Asset `{asset}` is under-insured: `{insured}` vs `{indexed}`	insured < 90% indexed	review policy	FR-37.15
AST-0535	error	Asset `{asset}` already belongs to group `{group}` for book `{book}`	duplicate membership	remove via disposal / transfer	BR-37.11
AST-0541	warning	Capex request `{no}` spend `{spent}` exceeds approval by `{pct}`	overrun	request supplement	FR-37.17
AST-0542	info	Capex request `{no}` returned to Submitted after amount increase	BR-37.12	approvers re-approve	BR-37.12

7.37.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Depreciation run	100k assets × 3 books monthly	n/a	1M asset-book-periods/h	3.6M transactions ≈ 2 GB	AssetTransaction partitioned yearly by transaction_date; index (asset_book_id, transaction_date)	workers scale to 8; run < 20 min
Schedule API	3,000/day; peak 60/min	40 / 200 / 400 ms	30 req/s	n/a	computed on read, cached per (book, version)	cache hit rate rises; p95 < 500 ms
Asset create from AP	2,000/day; peak 200/min at invoice batch	80 / 250 / 500 ms	20/s	0.7M assets ≈ 1 GB	Asset unique (entity, number), barcode unique	queued through outbox
Lease schedule compute	50/day	100 / 400 / 800 ms for 360 periods	10/s	5,000 leases × 120 lines ≈ 0.1 GB	by (lease_id, period_no)	none needed
GL reconciliation	1M assets	n/a	< 10 min	small	aggregates by (book, account)	< 30 min; alerts delayed
Count scan (PWA)	20 scanners × 1 scan/5 s	50 / 150 / 300 ms	10 scans/s	0.5M scans ≈ 0.1 GB	by (session_id, barcode)	offline queue on device, sync later
Asset history sheet	50 runs/month	2 / 8 / 15 s for 100k assets	n/a	n/a	materialized yearly roll-forward	served from previous night snapshot with age

7.37.20 Localization and compliance hooks

- **Tax depreciation methods and tables** vary by country: US MACRS / ADS, §179, bonus, Form 4562 (US pack); DE AfA tables, GWG (≤ EUR 800 immediate expense), Sammelposten pool (FR-37.16); FR dégressif coefficients 1.25 / 1.75 / 2.25 and amortissement dérogatoire (excess tax over book posted to a regulated provision); UK capital allowances (AIA, WDA 18% / 6% pools, full expensing) as group assets; IN Companies Act Schedule II lives and income-tax block of assets WDV; MX Ley del ISR maximum rates; CA CCA classes with half-year rule. Packs provide DepreciationProfile templates, conventions and statutory layouts through the `DepreciationMethodProvider` and `StatutoryAssetReport` extension points.
- **Statutory asset registers**: FR tableau des immobilisations (liasse 2054 / 2055), DE Anlagenspiegel (HGB §284), IT libro dei cespiti; layouts via the wave-48 electronic reporting designer.
- **Leases**: IFRS 16 (default), ASC 842 operating lease single-cost (flag), local GAAP packs where leases remain off-balance (DE HGB, FR PCG) use a book with `leaseCapitalization = false`.
- **Retention**: asset records for the life of the asset plus 10 years (DE, FR), 7 years after disposal (US).
- **Currency and rounding**: books in non-functional currencies (e.g., hyperinflationary IAS 29 restatement) are out of v1; rounding per book currency minor unit.
- **Privacy**: responsible-worker field is personal data; masked for users without HR visibility (wave 42 policy).

7.37.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.6.1 Fixed asset → Fixed asset additions (AssetAddition)	PARTIAL-37	ADDED as FR-37.14	
AX 12.6.1 Fixed asset → Insurance records (AssetInsurance)	PARTIAL-37	ADDED as FR-37.15	
AX 12.6.2 SSRS Fixed asset insurance report	PARTIAL-37	ADDED as FR-37.15	Insurance report in 7.37.14
AX 12.6.4 Fixed asset → Insurance types / Insurance companies	PARTIAL-37	ADDED as FR-37.15	`insurance.types` / `insurance.companies`
SAP 13.4 AS21 / AS22 / AS23 Group asset	PARTIAL-37	ADDED as FR-37.16	
SAP 13.4 Insurance values / indexing (S_ALR_87011951)	PARTIAL-37	ADDED as FR-37.15	index series formula BR-37.10
SAP 13.4 S_ALR_87010127 / 87010129 Tax depreciation reports (Form 4562 / mid-quarter)	PARTIAL-37	DEFERRED to wave 48	data (Part I–III amounts, mid-quarter flag) is in 7.37.14; the IRS form layout ships with the US pack
SAP 13.4 OAYM / OAAX Group assets per area (T093A)	PARTIAL-37	ADDED as FR-37.16	`assets.groupAssetBooks`
SAP 13.4 OAVI / OAVF Insurance types, companies, index series (T086, T094)	PARTIAL-37	ADDED as FR-37.15	`insurance.indexSeries`
SAP 13.27 IM01–IM36 / IMA11 / IMA2 / IMA3 / IMCCP1 Investment programs and appropriation requests	PARTIAL-37	ADDED as FR-37.17 (appropriation requests) and FR-38.21 (programs)	program budget distribution needs wave-38 budget control
SAP 13.27 IM reports (S_ALR_87012…) Investment program reports	PARTIAL-37	ADDED as FR-37.17 / FR-38.21	Capex request pipeline (7.37.14), Investment program report (7.38.14)
SAP 13.27 IM27 / IM32 IM budget rollup / carry-forward	PARTIAL-37	DEFERRED to wave 38	ADDED there as FR-38.21
SAP 13.27 IM customizing: program types, appropriation request types, status, budget categories, plan / budget profiles (TAIF, TAPRT)	PARTIAL-37	ADDED as FR-37.17 / FR-38.21	`capex.approvalLevels` here; program types in 7.38.13

7.38 Wave 38: Cost Accounting, Budgeting & Budget Control

7.38.1 Scope and business context

Wave 38 gives controllers a planning and control layer separate from statutory accounting. Before it, the universal journal records actuals by dimension but there is no budget to compare against, no way to stop a purchase order that exceeds a department's funds, and no management view that reclassifies and allocates costs the way the business is run. After it, budget register entries and planning scenarios flow through workflow into budget models; budget control checks requisitions, POs and invoices against available funds, recording encumbrances and liquidating them exactly; a cost accounting ledger mirrors primary costs from the journal into cost objects and cost elements, runs assessment/distribution allocations that converge and reconcile, supports activity-based costing and internal orders, and feeds the management reporting cubes consumed by wave 46.

Out of scope: public-sector fund accounting (appropriations, apportionments, year-end encumbrance carry-forward beyond the simple option), profitability analysis segment reporting (CO-PA style, wave 46 semantic layer), project budgets (wave 41).

Scope amendment (gap closure). The registers (coverage-ax §12.99 rows 34–37, 45–46; coverage-sap §13.99 rows 86–123 mapped to wave 38, plus the investment-program rows handed over from wave 37) add Excel worksheet templates (FR-38.13), plan generation from sources (FR-38.14), position forecasting (FR-38.15), cost behavior / overhead rate / cost rollup policies (FR-38.16), template (driver-formula) allocation (FR-38.17), cost object hierarchies (FR-38.18), top-down distribution (FR-38.19), budget reservation documents (earmarked funds, FR-38.20) and investment programs (FR-38.21). Fund and funded-program accounting uses financial dimensions (7.38.21); appropriations and apportionments remain out of scope.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
D365 budget check on PO happens in a batch-like synchronous call that takes 3–8 s per line on large BudgetSourceTracking tables; buyers disable it	Available-funds query aggregates BudgetSourceTrackingDetail at check time	Per dimension-combination balance cache rebuilt from events; check is one indexed read	Check latency 5 s → p95 < 150 ms
SAP FM availability control (FMAVCR01) rebuilds the AVC ledger (FMAVCT) when budget structure changes; FMAVCREINIT takes hours	Derived ledger with offline rebuild	Balance cache with incremental event replay	Rebuild for 1M documents 3 h → < 10 min
Encumbrance liquidation on invoice in AX leaves orphan amounts when the PO was partially received at a different price (BudgetSourceTracking residual)	Relief computed from invoice amount, not from PO reservation	Relief = min(invoice-matched share of PO reservation, remaining reservation); property-tested to the cent	Orphan encumbrances per month 20–50 → 0
KSU5/KSV5 cycles in SAP: a cycle with a circular segment fails with "iteration not converging" and the controller must restructure segments	Fixed 1,000-iteration loop with no exact solve	Exact linear solve ≤ 2,000 senders; residual sweep rule	Failed cycles per close 2–3 → 0

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
CAM cost accounting ledger in D365 requires "Process data" batch and loads GL into CAMCostEntry hours after posting	Separate ledger filled by batch import	Cost entries mirrored from journal events within seconds	Lag 4–24 h → < 5 s

7.38.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Budget register entries	BudgetTransactionHeader, BudgetTransactionLine, BudgetTransactionCode, BudgetModel, BudgetCycle	FMBB budget entry documents, FMBH; KP06/KP26 plan versions	BudgetRegisterEntry, BudgetModel	full
Budget planning	BudgetPlanHeader/Line, BudgetPlanningProcess, BudgetPlanStage, BudgetPlanScenario, BudgetPlanWorkflow, BudgetPlanTemplate (Excel)	KP06 plan versions, BPC/SAC planning	BudgetPlan with scenarios, stages, Temporal workflow	full
Budget control configuration	BudgetControlConfiguration, BudgetControlRule, BudgetGroup, BudgetControlUserGroup, budget threshold, funds available calculation	FM: FMDERIVE, funds centers FMFCTR, commitment items FMCI, tolerance profiles (OFUP/OF11), AVC ledger FMAVCT	BudgetControlRule with dimension subset, time span, tolerances, overrides	full
Encumbrances / commitments	BudgetSourceTracking, BudgetSourceTrackingDetail, BudgetCheckResult, LedgerSourceDocument encumbrance posting (purchase order encumbrance posting type)	COOI commitments, FMIOI open items, KSB2	Encumbrance entity with reservation/relief lines	full
Cost accounting ledger	CAMCostAccountingLedger, CAMCostObject, CAMCostElement, CAMCostElementDimension, CAMDimensionHierarchy, CAMCostEntry	CO area, KS01/KS02 CSKS, KSH1 hierarchy, CSKA/CSKB cost elements (KA01/KA06), COEP/COBK line items, COSP/COSS totals	CostObject, CostElement, CostEntry, DimensionHierarchy	full
Allocations	CAMCostAllocationPolicy, CAMCostDistributionPolicy, CAMAllocationBase, CAMCostBehaviorPolicy	KSV5 distribution, KSU5 assessment, cycles T811C/T811S/T811K, KSC6 indirect activity allocation	AllocationPolicy (assessment/distribution/reciprocal), AllocationBase	improved (exact solve)
Statistical measures / activity types	CAMStatisticalDimension, statistical measure provider templates	KL01 CSLA activity types, KP26 prices, KB21N	StatisticalMeasure, ActivityType with rates	full
Internal orders	Not native (projects used)	KO01 AUFK internal orders, KO88 settlement	InternalOrder cost object type with settlement rules	full
Cost control workspace	CAMCostControlWorkspace, overhead calculation, variance by cost object	S_ALR_87013611 (cost centers actual/plan/variance), KSB1	Cost control workspace with plan/actual/variance drill	full
Budget plan worksheets	BudgetPlanTemplate, Excel budget plan worksheet (Office add-in), BudgetPlanLayout (D365 layouts)	KP06 with planner profiles (KP34), SAC / BPC input forms	WorksheetTemplate with xlsx round-trip (FR-38.13)	full
Generate plan from source	Budget plan generation: from GL actuals, from budget plan, from fixed assets, from project forecasts, from demand forecast, from position forecasts	KP97 copy planning, KP98 copy actual to plan	PlanGenerationRun with sources (FR-38.14)	full
Position forecasting	HcmPositionForecast, HcmPositionForecastBudgetAcctLine, budget cost elements	HCM personnel cost planning (PCP, HRHCP00*)	PositionForecast with cost elements (FR-38.15)	partial (HR-driven source when wave 42 ships)
Cost behavior / overhead / rollup policies	CAMCostBehaviorPolicy, CAMOverheadRatePolicy, CAMCostRollupPolicy	Cost center categories (OKA2), overhead costing sheets on cost centers (KZS2 / KGI2), cost center group rollups (KSH1)	CostBehaviorPolicy, OverheadRatePolicy, CostRollupPolicy (FR-38.16)	full
Template allocation	none	CPT1 templates, CPTA / CPTB template allocation (actual / plan)	AllocationTemplate with driver formulas (FR-38.17)	full
Cost object hierarchy	CAMDimensionHierarchy (reporting only)	KKPHIE / KKP2 cost object hierarchy, KKP4 collective entry, KKPX distribution	CostObjectHierarchy with node collection and distribution (FR-38.18)	full
Top-down distribution	none	KE28 top-down distribution (CO-PA)	Allocation type TopDown (FR-38.19)	full
Earmarked funds	Budget reservations (D365 public sector, BudgetReservation_PSN)	FMX1 / FMX2 / FMX3 funds reservation, FMY1 precommitment, FMZ1 commitment, FMW1 revenue forecast, FMV1 funds blocking, FMEF document types	BudgetReservation with types (FR-38.20)	full
Investment programs	none native	IM01–IM36 programs and positions, IM52 budget distribution, IM32 budget carry-forward, IM27 rollup	InvestmentProgram, ProgramPosition (FR-38.21)	full

7.38.3 Functional requirements

- **FR-38.1 Budget models and register entries** — Must. The system shall record budget register entries (Original, Revision, Transfer, Carry-forward, Pre-encumbrance, Encumbrance reservation) per budget model, cycle, dimension combination and period, with workflow approval.
  - Notes: a transfer must net to zero across its lines; entries post to a budget ledger separate from the universal journal; budget model hierarchies (sub-models) roll up; 2M register lines per model supported.
- **FR-38.2 Budget planning** — Must. The system shall run planning processes with stages, scenarios (up to 10 per plan), top-down and bottom-up aggregation, distribution keys, and Excel import/export, ending in a register entry.
  - Notes: scenarios compared side-by-side; comments per cell; stage transitions by Temporal workflow; budget plan line count up to 500,000 per plan.
- **FR-38.3 Budget control configuration** — Must. The system shall define control rules per legal entity: budget control dimensions (subset of ledger dimensions), budget cycle time span (year / quarter / period / custom), funds-available formula, documents under control (requisition, PO, vendor invoice, GL journal, expense, project), thresholds (warn %, block %), and user-group overrides.
  - Notes: rule activation versioned with effective date; disabling control does not delete history.
- **FR-38.4 Budget check order of operations** — Must. The system shall evaluate a budget check in this fixed order: (1) determine whether the document type and main account range are under control; (2) derive the budget control dimension combination from the line's ledger dimension; (3) resolve the budget cycle and time span containing the line's accounting date; (4) relieve the document's own prior reservation (revision of a PO, invoice matching a PO) so only the net change is checked; (5) compute available = original + revisions + transfers in − transfers out + carry-forward − actuals − encumbrances − pre-encumbrances (− unposted documents if enabled); (6) compare the net change against available with the rule's thresholds: ≤ warn% → pass, between warn% and block% → warn, > block% → fail; (7) apply user-group override (fail → warn for groups with `BudgetOverride`); (8) record BudgetCheckResult and, on pass/warn, reserve the amount atomically.
  - Notes: evaluation uses a per-combination balance row with `SELECT … FOR UPDATE` so two concurrent POs cannot both pass against the last dollar; p95 < 150 ms; results retained 7 years.

- **FR-38.5 Encumbrance lifecycle** — Must. The system shall create pre-encumbrances for approved requisitions, convert to encumbrances on PO confirmation, adjust on PO change, relieve on invoice posting or PO cancellation, and optionally carry forward at year end.
  - Notes: relief = min(invoice quantity × PO unit price, remaining reservation) per PO line; price differences on invoice create a new check for the excess; closed POs release residuals.
- **FR-38.6 Cost accounting ledger** — Must. The system shall maintain cost objects (cost centers, internal orders, products, projects as references), cost elements (primary mapped from main accounts, secondary for allocations), dimension hierarchies, and cost entries mirrored from journal events within 5 s.
  - Notes: cost behavior (fixed/variable) classified by policy; one ledger per legal entity and currency; versioned hierarchies effective-dated.
- **FR-38.7 Allocation policies** — Must. The system shall run distribution (primary elements retained) and assessment (secondary element) allocations by fixed percentage, statistical measure, or actual cost basis, in ordered steps with reciprocal support.
  - Notes: iteration convergence rule: for reciprocal steps with n senders, when n ≤ 2,000 solve `(I − A) x = b` exactly (LU with partial pivoting); otherwise iterate Gauss–Seidel until max absolute unallocated residual on any sender ≤ 0.01 currency units or 100 iterations; after loop, any residual is swept to that sender's largest receiver; invariant `Σ cost in = Σ cost out` to the cent; rounding residual per step posted to the largest receiver.
- **FR-38.8 Activity-based costing** — Should. The system shall define activity types with planned rates, record activity quantities from production (wave 21) and manual entries, and allocate at rate × quantity with rate variances.
- **FR-38.9 Internal orders** — Should. The system shall create internal orders with budgets, collect costs, and settle to cost centers, assets (wave 37) or P&L by settlement rule.
- **FR-38.10 Cost control workspace** — Must. The system shall show plan vs actual vs committed per cost object and element, with variance %, drill to cost entries and to journal lines, and period comparison.
- **FR-38.11 Reconciliation** — Must. The system shall reconcile Σ primary cost entries per element and period to the journal balances and list differences.
- **FR-38.12 Management cubes** — Should. The system shall publish cost ledger facts and hierarchies to the lakehouse via CDC with a documented star schema for wave 46.
- **FR-38.13 Budget plan worksheet templates** — Should. The system shall define worksheet templates per planning process (row dimensions, column sets per scenario / period / prior-year actual, editable and protected ranges, formulas) and round-trip them as xlsx files whose upload validates and writes back only editable cells.
  - Notes: each exported workbook carries the plan id, stage, template version and a SHA-256 per protected range in a hidden sheet; upload is rejected when the stage changed, the template version differs or a protected range checksum fails; cell-level changes are applied as a diff with an audit row per cell; maximum 200,000 editable cells per upload; formulas evaluate in Passage (not trusted from the file) so totals always recompute server-side.
- **FR-38.14 Generate budget plan from sources** — Must. The system shall create or refresh plan lines from: GL actuals of a period range (with % uplift), another plan or scenario, depreciation forecast (FR-37.12) and approved capex requests (FR-37.17), demand forecast (wave 22) × standard cost (wave 39) or price, project forecasts (wave 41, when available), and position forecasts (FR-38.15).
  - Notes: generation modes Replace, Add, Update-if-unchanged (keeps manually edited cells); every generated line keeps its source reference; the run reports lines created / updated / skipped; 500,000 lines generate in < 3 min.
- **FR-38.15 Position forecasting** — Could. The system shall maintain forecast positions with start / end dates, FTE, compensation group and budget cost elements (salary, benefits %, payroll tax %), compute monthly cost per position and publish it as a plan source.
  - Notes: positions are entered manually or imported from CSV in wave 38; when wave 42 ships, the `HrPositionSource` populates them from effective-dated positions and compensation plans; cost element formulas: salary / 12 × FTE, benefits = salary component × %, tax capped by an annual ceiling per cost element; vacancy factor % per department.
- **FR-38.16 Cost behavior, overhead rate and cost rollup policies** — Should. The system shall classify cost entries as fixed or variable by policy rules (cost element range × cost object type, or regression result with a manual override), apply overhead rates per cost object and element to produce overhead cost entries, and define rollup policies that aggregate cost objects into hierarchy nodes for reporting.
  - Notes: policy evaluation order: explicit rule → element default → Unclassified; regression option fits cost against a statistical measure over ≥ 12 periods and classifies the intercept as fixed; overhead rate entries carry origin `Overhead` and a secondary element; rollup policies do not post, they define reporting aggregation; policies effective-dated and versioned.
- **FR-38.17 Template (driver-formula) allocation** — Should. The system shall allocate costs from a sender to receivers using templates whose lines evaluate formulas over receiver attributes and statistical measures (for example 25.00 × inspection lots + 0.5% × material cost) for actual and plan versions.
  - Notes: formula language is a restricted expression grammar (numbers, + − × ÷, min / max, if, references to measures, attributes and cost element balances); evaluation is per receiver per period; a template run produces one allocation run (FR-38.7 conservation invariant applies); formulas are validated at save and dry-run against the last closed period.
- **FR-38.18 Cost object hierarchies** — Should. The system shall define cost object hierarchies (for example product group → product → production order), collect costs at a node that cannot be assigned to a lower object, and distribute them to the lower objects by a basis (actual output, standard cost of output, fixed %) at period end.
  - Notes: a node may have collected costs only when its hierarchy type allows it; distribution posts origin `HierarchyDistribution` with the receiving order and feeds wave-39 variance (the distributed cost counts as actual input of category `Distributed`); undistributable residual remains on the node with a warning.
- **FR-38.19 Top-down distribution** — Should. The system shall distribute an amount booked at an aggregated level (for example freight per legal entity) to detailed segments (customer, product, region) in proportion to a reference basis from a reference period.
  - Notes: reference basis: revenue, quantity, COGS or any measure in the cost ledger or the wave-46 semantic layer; zero-basis segments are excluded; result posts as statistical (reporting) or actual cost entries per option; worked example: freight USD 12,000.00 with revenue shares 300,000 / 150,000 / 50,000 → 7,200.00 / 3,600.00 / 1,200.00.
- **FR-38.20 Budget reservation documents (earmarked funds)** — Should. The system shall create budget reservation documents of types FundsReservation, Precommitment, FundsBlocking and RevenueForecast that reserve or block available funds without a requisition, and let requisitions, POs, invoices and journals consume them by reference.
  - Notes: a reservation runs the FR-38.4 check on creation; consumption by a follow-on document relieves the reservation and creates the follow-on encumbrance in one transaction; FundsBlocking cannot be consumed; RevenueForecast increases available funds only when the rule sets `revenue-increases-budget`; residuals release on close or on expiry date; per-type number sequences.
- **FR-38.21 Investment programs** — Should. The system shall maintain investment programs per approval year with a hierarchy of program positions, distribute program budget top-down to positions, assign capex requests (FR-37.17), assets under construction and projects to positions, roll up plan / budget / actual, and carry forward unused budget to the next approval year.
  - Notes: a child position's budget cannot exceed the parent's undistributed budget; budget control (FR-38.4) for capex documents uses the program position as a control dimension; carry-forward moves (budget − actual − commitments) per position into the new program year as a CarryForward register entry; program types define allowed position levels (≤ 6) and budget categories (Capex, Opex-related, Contingency).

7.38.4 Data model

BudgetRegisterEntry: id, legal_entity_id, budget_model_id, entry_type (Original/Revision/Transfer/CarryForward/PreEncumbrance/Encumbrance), budget_cycle_id, status (Draft/Submitted/Approved/Posted/Rejected), lines[] (dimension_combination_id, period_key, amount numeric(19,4), currency), workflow_id, posted_at.

BudgetControlBalance

Field	Type	Required	Notes
id	uuid	yes	PK
rule_version_id	uuid	yes	FK BudgetControlRule version
control_dimension_key	text	yes	hashed dimension subset
time_span_key	text	yes	e.g. FY2026, 2026-Q3, 2026-09
budget_total	numeric(19,4)	yes	original + revisions + transfers + carry-forward
actuals	numeric(19,4)	yes	from journal events
encumbrances	numeric(19,4)	yes	
pre_encumbrances	numeric(19,4)	yes	
version	bigint	yes	optimistic + row lock on check

Encumbrance

Field	Type	Required	Notes
id	uuid	yes	PK
source_type	enum	yes	Requisition/PurchaseOrder/Invoice/Journal
source_line_id	uuid	yes	
balance_id	uuid	yes	FK BudgetControlBalance
reserved	numeric(19,4)	yes	
relieved	numeric(19,4)	yes	Σ relief lines
status	enum	yes	Open/PartiallyRelieved/Relieved/Released
lines	jsonb	yes	[{type: Reserve/Adjust/Relieve/Release, amount, documentId, at}]

CostEntry

Field	Type	Required	Notes
id	uuid	yes	PK, append-only
ledger_id	uuid	yes	cost accounting ledger
cost_object_id	uuid	yes	FK CostObject
cost_element_id	uuid	yes	FK CostElement (primary/secondary)
period_key	text	yes	
amount	numeric(19,4)	yes	
behavior	enum	yes	Fixed/Variable/Unclassified
origin	enum	yes	Journal/Allocation/Activity/Statistical/Manual
source_id	uuid	yes	journal line, allocation run, etc.
allocation_run_id	uuid	no	

AllocationPolicy: id, ledger_id, name, type (Distribution/Assessment/Reciprocal), steps[] (order, sender filter, receiver filter, basis {Fixed%/StatisticalMeasure/ActualCost}, secondary_element_id), schedule, version.

Entities added by FR-38.13–FR-38.21: **WorksheetTemplate** (id, planning_process_id, version, row_dimensions[], column_sets jsonb, protected_ranges jsonb, formulas jsonb); **PlanGenerationRun** (id, plan_id, scenario_id, source_type, parameters jsonb, mode, created, updated, skipped); **PositionForecast** (id, legal_entity_id, position_code, department_dim, start_date, end_date, fte numeric(5,2), compensation_group_id, cost_elements jsonb, source Manual/HR); **CostBehaviorPolicy / OverheadRatePolicy / CostRollupPolicy** (id, ledger_id, version, effective_from, rules jsonb); **AllocationTemplate** (id, ledger_id, sender_filter, lines [{receiverFilter, formula, secondaryElementId}], version); **CostObjectHierarchy / HierarchyNode** (id, type, parent_id, cost_object_id, collects bool, distribution_basis); **BudgetReservation** (id, legal_entity_id, type FundsReservation/Precommitment/FundsBlocking/RevenueForecast, number, lines [{dimension_combination_id, period_key, amount, consumed}], expiry_date, status Open/PartiallyConsumed/Consumed/Closed/Expired); **InvestmentProgram / ProgramPosition** (program: id, code, approval_year, program_type_id, currency, status; position: id, program_id, parent_id, code, level, budget, distributed, plan, actual, commitment, assigned_object_refs[]).

Invariants: a BudgetControlBalance's available = budget_total – actuals – encumbrances – pre_encumbrances; Σ Encumbrance.reserved – relieved for a balance = balance.encumbrances; Σ CostEntry(origin=Allocation) per run = 0 (senders negative, receivers positive). Indexes: BudgetControlBalance unique (rule_version_id, control_dimension_key, time_span_key); CostEntry partitioned by period_key, index (cost_object_id, cost_element_id, period_key).

7.38.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/budget/register-entries	Create entry	Idempotency-Key	BudgetMaintain
POST	/api/v1/budget/register-entries/{id}/submit	Start workflow	state check	BudgetMaintain
POST	/api/v1/budget/plans/{id}/scenarios	Add scenario	Idempotency-Key	BudgetPlan
POST	/api/v1/budget/checks	Perform check (document lines)	documentId+version	BudgetCheck (system)
GET	/api/v1/budget/available?rule=&dims=&span=	Available funds	n/a	BudgetView
POST	/api/v1/budget/encumbrances/{id}/relieve	Relief from invoice	documentId	BudgetCheck (system)
POST	/api/v1/controlling/allocation-runs	Run policy for period	(policy, period, version)	CostAccountingRun
GET	/api/v1/controlling/cost-objects/{id}/entries?period=	Entries	n/a	CostAccountingView
POST	/api/v1/controlling/statistical-measures/{id}/values	Load measure values	Idempotency-Key	CostAccountingMaintain
POST	/api/v1/controlling/internal-orders/{id}/settle	Settle	(order, period)	CostAccountingRun
POST	/api/v1/budget/plans/{id}/worksheets/export	Export xlsx for template and stage	n/a	BudgetPlan
POST	/api/v1/budget/plans/{id}/worksheets/import	Upload xlsx diff	file SHA-256	BudgetPlan

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/budget/plans/{id}/generate	Generate from source	(plan, scenario, source, version)	BudgetPlan
POST	/api/v1/budget/reservations	Create earmarked funds document	Idempotency-Key	BudgetReserve
POST	/api/v1/budget/investment-programs/{id}/distribute	Distribute budget to positions	Idempotency-Key	InvestmentProgramMaintain
POST	/api/v1/controlling/template-allocations/runs	Run template allocation	(template, period, version)	CostAccountingRun
POST	/api/v1/controlling/top-down-runs	Run top-down distribution	(policy, period, version)	CostAccountingRun

Validation: check requests must carry accounting date, ledger dimension and amount per line; allocation run rejects when period is closed in the cost ledger or a prior step's run for the period is missing.

Event	Trigger	Payload fields	Consumers
`dev.passage.budget.register.posted.v1`	entry posted	entryId, modelId, type, lines summary	balance projector, analytics
`dev.passage.budget.check.completed.v1`	check	sourceDocumentId, lineResults[{result, available, amount}]	procurement, AP, workflow
`dev.passage.budget.encumbrance.relieved.v1`	relief	encumbranceId, amount, invoiceId, remaining	analytics
`dev.passage.controlling.allocation.completed.v1`	run complete	runId, policyId, iterations, totalIn, totalOut, residual	budget-svc, analytics
`dev.passage.controlling.costentry.created.v1`	mirror or allocation	entryId, costObjectId, elementId, amount, origin	lakehouse
`dev.passage.budget.reservation.consumed.v1`	follow-on document consumes reservation	reservationId, documentId, amount, remaining	analytics
`dev.passage.budget.program.carriedforward.v1`	year-end carry-forward	programId, fromYear, toYear, positions[], amount	budget-svc, asset-svc
`dev.passage.controlling.hierarchy.distributed.v1`	node distribution	runId, nodeId, period, distributed, residual	costing-svc (39), analytics

7.38.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Budgeting workspace	New entry, approvals, available funds by rule	model, cycle, approved/pending, over-budget alerts	BudgetView
Budget plan grid	Edit cells, distribute, compare scenarios, submit stage	dimension rows × period columns, scenario columns, comments	BudgetPlan
Budget control rules	Configure dimensions, thresholds, user groups, activate	rule version, effective date, documents in control	BudgetConfigure
Budget check result (inline in PO)	View reason, request override	available, amount, result, threshold	Buyer
Cost accounting workspace	Run allocations, process status, reconciliation	ledger, period, steps status, in/out totals	CostAccountingRun
Cost control	Filter object/hierarchy node, drill	plan, actual, committed, variance, variance %	CostAccountingView

Interaction rules: plan grid supports copy/paste from Excel, keyboard fill-down, and distribute-by-key; budget check result renders inside the PO line without leaving the form; allocation runs show step progress and residual live via server-sent events.

7.38.7 Business rules and edge cases

- **BR-38.1** Budget check amounts convert to the ledger's accounting currency at the document's rate; a later rate change does not re-check.
- **BR-38.2** Register entry state machine: Draft → Submitted → Approved → Posted; Submitted → Rejected → Draft; Posted entries are immutable and corrected by a Revision.
- **BR-38.3** A PO revision that lowers the amount never fails a check; it releases the difference.
- **BR-38.4** Invoice exceeding the PO reservation: the matched part relieves; the excess is checked as a new amount under the invoice's document rule.
- **BR-38.5** Time span resolution for a date on a period boundary uses the legal entity's fiscal calendar; dates outside any open cycle fail the check with reason `NoBudgetCycle`.
- **BR-38.6** Allocation convergence: iteration stops at residual ≤ 0.01 or 100 iterations; residual sweep goes to the sender's largest receiver by basis share; step totals must satisfy `Σ out = Σ in` to the cent or the run fails and rolls back.
- **BR-38.7** Allocation runs for a period are versioned; rerun reverses the prior run's entries (origin Allocation) and creates new ones; actuals-based runs after new journal postings require rerun, flagged by the reconciliation job.
- **BR-38.8** A sender with zero basis in all receivers keeps its cost and is reported as unallocated; the run does not fail.
- **BR-38.9** Internal order settlement cannot exceed collected cost; over-settlement is rejected.
- **BR-38.10** Year-end carry-forward of open encumbrances creates a CarryForward register entry in the new cycle and releases the old reservation atomically.
- **BR-38.11** Worksheet import applies only to cells whose server value equals the value at export time or that were not changed on the server since; conflicting cells are rejected and listed (CO-0627); the rest apply.
- **BR-38.12** Template allocation formulas returning a negative receiver amount fail the run unless the template sets `allow-negative`; division by a zero measure evaluates to 0 for that receiver and raises a warning.
- **BR-38.13** Top-down distribution: residual cents from rounding go to the segment with the largest basis; Σ distributed = source amount to the cent.
- **BR-38.14** Budget reservation state machine: Open → PartiallyConsumed → Consumed → Closed; Open | PartiallyConsumed → Expired at expiry date (residual released); FundsBlocking goes Open → Closed only.
- **BR-38.15** Investment program distribution: Σ child budgets ≤ parent budget at all times; reducing a parent below Σ children is rejected; carry-forward is allowed once per position and year.
- **BR-38.16** Cost behavior reclassification after period close applies from the next open period only; closed-period entries keep their classification.

7.38.8 Acceptance criteria

- **AC-38.1** (FR-38.4) Given budget 100,000.00 with actuals 60,000.00 and encumbrances 30,000.00, when a PO line of 12,000.00 is checked with warn 0% and block 5%, then the result is `fail` (available 10,000.00 < 12,000.00, over by 20%).
- **AC-38.2** (FR-38.4) Given the same balance, when a PO of 10,400.00 is checked, then the result is `warn` (over by 4%) and 10,400.00 is reserved.
- **AC-38.3** (FR-38.5) Given a PO line of 10 × 50.00 encumbered, when an invoice for 10 × 52.00 posts, then relief = 500.00 and a new check runs for 20.00.
- **AC-38.4** (FR-38.4) Given two concurrent POs of 8,000.00 against 10,000.00 available, when checked simultaneously, then exactly one passes.
- **AC-38.5** (FR-38.7) Given a reciprocal step with 3 service centers serving each other, when run, then Σ out = Σ in to the cent and the solution matches the analytic solve.
- **AC-38.6** (FR-38.11) Given a month of journal postings, when reconciliation runs, then Σ primary cost entries per element equal the journal balances with 0.00 difference.
- **AC-38.7** (FR-38.2) Given a plan with 3 scenarios and 2 stages, when approvers act, then the workflow completes and a register entry posts from the selected scenario.

- **AC-38.8** (FR-38.6) Given a journal entry posted, when 5 s elapse, then matching cost entries exist for every line with a primary element mapping.
- **AC-38.9** (FR-38.13) Given an exported worksheet whose protected range was edited offline, when uploaded, then the upload is rejected with CO-0626 and no cell is written.
- **AC-38.10** (FR-38.14) Given GL actuals of FY2025 of USD 1,200,000.00 for a department and a 3% uplift, when the plan generates for FY2026 in Replace mode, then plan lines total 1,236,000.00 with source references to the actuals.
- **AC-38.11** (FR-38.19) Given freight of USD 12,000.00 and revenue shares 300,000 / 150,000 / 50,000, when top-down distribution runs, then the segments receive 7,200.00 / 3,600.00 / 1,200.00 and Σ = 12,000.00.
- **AC-38.12** (FR-38.20) Given a funds reservation of EUR 20,000.00, when a PO of EUR 8,000.00 references it, then the reservation shows 12,000.00 remaining and an encumbrance of 8,000.00 exists, with available funds unchanged by the conversion.
- **AC-38.13** (FR-38.21) Given a program with budget USD 5,000,000.00 and children distributed 4,800,000.00, when a user distributes 300,000.00 more, then the distribution is rejected with CO-0641.
- **AC-38.14** (FR-38.17) Given a template line `25.00 × inspectionLots + 0.005 × materialCost` and an order with 4 lots and material cost 20,000.00, when run, then the receiver gets 200.00 and the sender is credited 200.00.

7.38.9 Test plan and fixtures

Fixtures: `budget-check-200` (200 PO/invoice scenarios across thresholds, overrides, revisions, cycles); `alloc-reciprocal-10` (service centers with circular references and a 2,500-sender case for the iterative path); `plan-3-scenarios` (demo entity planning process); `costledger-month` (one month of journal postings mapped to elements).

Test	Type	Target	Pass criterion
Check order of operations	Unit	budget-svc	each step observable in result trace
Relief exactness	Property	encumbrance	Σ reserved − relieved ≥ 0, never negative; relief ≤ matched share
Concurrent checks	E2E	API	no double pass
Allocation conservation	Property	costacct-svc	Σ in = Σ out for random matrices
Convergence	Unit	iterative solver	residual ≤ 0.01 within 100 iterations for diagonally dominant cases
Check latency	Performance	API	p95 < 150 ms at 50 checks/s
Allocation throughput	Performance	run	1M cost entries, 5 steps < 10 min

Reconciliation checks: cost ledger primary costs = journal by element/period; Σ encumbrances = Σ open PO lines under control; budget balances = Σ posted register entries.

7.38.10 Deployment and operations

Helm charts `budget-svc`, `costacct-svc` (shared Postgres `controlling`); env `BUDGET_CHECK_TIMEOUT_MS=500`, `ALLOC_MAX_ITER=100`, `ALLOC_EXACT_SOLVE_MAX_N=2000`; flags `budget.unposted-in-available`, `controlling.abc`; jobs: `cost-mirror-lag-check` every 5 min, `cost-gl-reconcile` nightly, `allocation-schedule` per policy, `budget-balance-rebuild` on demand. SLOs: check p95 < 150 ms, mirror lag p95 < 5 s, allocation 1M entries < 10 min. Alerts: mirror lag > 60 s, reconciliation difference ≠ 0, check timeouts > 0.1%. Rollback: Helm; balance cache rebuilt from events if corrupted.

7.38.11 Migration from AX / SAP

Source	Passage entity	Notes
BudgetTransactionHeader/Line (current and next cycle)	BudgetRegisterEntry	posted entries summarized per combination/period
BudgetControlConfiguration, BudgetControlRule, BudgetGroup	BudgetControlRule	re-validated; user groups mapped to Keycloak groups
BudgetSourceTracking (open)	Encumbrance	open POs only, tied to migrated POs (wave 17)
CAMCostObject, CAMCostElement, CAMDimensionHierarchy	CostObject, CostElement, DimensionHierarchy	
CSKS, CSKT, SETHEADER/SETNODE (KSH1)	CostObject, DimensionHierarchy	
CSKA/CSKB	CostElement	primary mapped by main account
T811C/T811S/T811K cycles	AllocationPolicy	segment → step; tracing factors → basis
FMAVCT, COOI, FMIOI	Encumbrance	open commitments only
COSP/COSS plan versions, KP06	BudgetRegisterEntry (Original)	current year plan
COEP history	not migrated	cost entries mirrored from migrated journal balances

Validation: Σ migrated budget = legacy budget report per combination; open commitments count and total match. Not migrated: closed-cycle budget history older than 2 years (lakehouse archive), AX budget plan Excel templates.

7.38.12 Dependencies, risks, open questions

Depends on: wave 11 dimensions and balances, wave 14 fiscal calendars and allocation engine (reused for GL), wave 17 requisitions/POs, wave 18 invoices, wave 21 activity quantities, wave 37 depreciation forecasts.

Risk	Likelihood	Impact	Mitigation
Balance cache drift vs journal	Medium	High	Nightly reconcile; rebuild job; alert
Procurement UX slowed by checks	Medium	Medium	SLO; async pre-check on requisition

Open questions: (1) Include unposted documents in available funds by default? Proposed off; decide by wave 38 start. (2) Reciprocal exact solve threshold 2,000? Proposed yes; revisit at phase exit. (3) Internal orders as cost object type or separate module? Proposed cost object type; decide by wave 38.

7.38.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`budget.registerNumberSequence`	legal entity	sequence	BR-######	sequence id	register entry numbers	Budgeting parameters → Number sequences	FMBB document number ranges (FM area)
`budget.transferRules`	legal entity	table	net zero only	dimension constraints for transfers	FR-38.1 transfer validation	Budgeting parameters → "Use budget transfer rules", Budget transfer rules	FMBB transfer document types, budget structure (FMMD)
`budget.cycleTimeSpan`	legal entity	enum	FiscalYear	FiscalYear / Quarter / Period / Custom	time span for control (FR-38.3)	Budget cycle time spans	FM budget profile (annual / overall)
`budget.controlDimensions`	rule version	dimension[]	MainAccount + Department	subset of ledger dimensions	control combination (FR-38.4)	Budget control configuration → Define parameters → budget control dimensions	FMDERIVE derivation, FM account assignment elements (fund, funds center, commitment item)
`budget.fundsAvailableFormula`	rule version	flags	budget – actuals – encumbrances – pre-encumbrances	include unposted / include revenue	available formula	Budget funds available calculation	AVC control ledger FMAVCT, AVC profiles
`budget.thresholds`	rule version	struct	warn 0% / block 0%	0–100% each	FR-38.4 step 6	Budget control rules → thresholds	FM availability control tolerance profiles
`budget.overrideGroups`	rule version	group[]	none	Keycloak groups	fail → warn (FR-38.4 step 7)	BudgetControlUserGroup	FM tolerance profile per user (activity group)
`budget.documentsUnderControl`	rule version	enum[]	Requisition, PO, VendorInvoice	Requisition / PO / VendorInvoice / Journal / Expense / Project / Reservation	FR-38.3	Budget control configuration → Select source documents	FM update profile, commitment-relevant documents
`budget.carryForward`	legal entity	enum	Off	Off / OpenEncumbrances / OpenEncumbrancesAndUnusedBudget	year-end FR-38.5 / BR-38.10	Budgeting → Year-end encumbrance process (D365)	FMJ2 / FMJ1 commitment carry-forward, FMCB budget carry-forward
`plan.worksheetMaxCells`	tenant	int	200,000	1,000–1,000,000	FR-38.13 upload limit	Excel budget plan layout limits	KP06 planner profile (KP34) layout
`plan.generationMode`	planning process	enum	UpdateIfUnchanged	Replace / Add / UpdateIfUnchanged	FR-38.14	Generate budget plan from … → "Replace / Add"	KP97 copy planning (overwrite / reset)
`position.costElements`	legal entity	table	Salary, Benefits 20%, Payroll tax 7.65% capped	element, formula, ceiling	FR-38.15	Budget cost elements (HcmBudgetPurposeType)	PCP cost items / cost planning variants
`reservation.documentTypes`	legal entity	table	4 types seeded	type, number sequence, consumable	FR-38.20	Budget reservation types (PSN)	FMEF earmarked funds document types, number ranges
`program.types`	tenant	table	Capex (6 levels)	levels, budget categories	FR-38.21	none	IM program types (TAIF), budget categories, plan / budget profiles
`ledger.costAccountingLedger`	legal entity	struct	one per entity	currency, fiscal calendar, chart of cost elements	FR-38.6	Cost accounting ledger (CAMCostAccountingLedger)	OKKP controlling area
`ledger.versions`	ledger	table	Actual (0), Plan (1)	version code, type, locked periods	plan / actual separation	Cost accounting versions (CAMVersion)	OKEQ CO versions
`ledger.periodLock`	ledger	table	none	period × activity (allocation, entry)	blocks runs in closed periods	Cost accounting → Ledger period status	OKP1 period lock
`ledger.elementMapping`	ledger	rules	main-account range → primary element	ranges	journal mirroring (FR-38.6)	Cost element dimension members from main accounts (CAMCostElementDimension)	KA01 primary / automatic creation of cost elements (S/4 GL account type "primary costs")
`ledger.defaultCostObject`	ledger	rules	none	main account → cost object	default assignment for journal lines without object	Cost accounting → default cost objects	OKB9 default account assignment
`alloc.exactSolveMaxN`	ledger	int	2,000	100–10,000	FR-38.7 solver choice	none	KSU5 iteration (no exact solve)

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`alloc.maxIterations`	ledger	int	100	10–1,000	FR-38.7	none	cycle iteration flag (KSU1 / KSV1)
`alloc.cycles`	ledger	policy	none	steps, sender / receiver filters, basis	FR-38.7	Cost allocation / distribution policies (CAMCostAllocationPolicy)	KSU1 assessment cycles, KSV1 distribution cycles (T811C)
`alloc.templateEnvironment`	ledger	enum	Controlling	Controlling / CostObject / Segment	which attributes formulas may reference (FR-38.17)	none	CPT1 template environments
`policy.costBehavior`	ledger	policy	element default	rules / regression	FR-38.16	Cost behavior policies	cost center categories (OKA2), fixed / variable plan split (KP06 layouts)
`policy.overheadRate`	ledger	policy	none	rate tables	FR-38.16	Overhead rate policies	KZS2 costing sheet on cost centers / orders, KZZ2 overhead rates
`abc.activityTypes`	ledger	table	none	activity type, unit, planned rate	FR-38.8	Statistical dimension / cost behavior templates	KL01 activity types, KP26 activity prices
`fund.dimensions`	legal entity	bool	false	true / false	seeds `Fund` and `FundedProgram` dimensions and adds them to control dimensions	Financial dimensions (Fund, PSN)	FM5I funds, FMMEASURE funded programs

7.38.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Budget control statistics	drill-down	rule, combination, time span	budget, actuals, encumbrances, pre-encumbrances, reservations, available, % used	`budget_control_balance`	Budget control statistics (BudgetControlStatistics)	FMAVCR01 / FMAVCR02
Budget vs actual vs committed	operational report	entity, hierarchy node, period range, version	plan, actual, committed, variance, variance %	cost ledger + budget balance	Budget control statistics by period, Actual vs budget (D365 report)	S_ALR_87013611, FMRP_RFFMEP1AX
Budget register entries	list inquiry	model, type, status, date	entry, type, lines total, workflow status	`budget_register_entry`	Budget register entries (BudgetTransactionHeader)	FMBB document display, FMEDDW
Open encumbrances	list inquiry	entity, source type, age	document, reserved, relieved, remaining, status	`encumbrance`	Encumbrance inquiry / BudgetSourceTracking	KSB2 commitments, FMRP_RFFMEP1OX
Earmarked funds list	list inquiry	type, status, expiry	document, amount, consumed, remaining, expiry	`budget_reservation`	Budget reservation list (PSN)	S_P99_41000099 (earmarked funds journal)
Cost object line items	drill-down	cost object, element, period, origin	entry, amount, origin, source document	`cost_entry`	Cost object statement (CAMCostObjectStatement)	KSB1, KOB1
Allocation run log	operational report	policy / template, period, version	step, senders, receivers, in, out, residual, iterations	`allocation_run`	Overhead calculation results	KSU5 / KSV5 / CPTA logs
Plan scenario comparison	drill-down	plan, scenarios (≤ 3), dimension rows	scenario values, deltas, %	`budget_plan_line`	Budget plan comparison	KP06 version comparison, S_ALR_87013612
Activity price and quantity	operational report	activity type, period	planned rate, actual rate, quantity, rate variance	`activity_type`	none	KSBT activity prices, KSII actual price calculation
Investment program overview	KPI / drill-down	program, year, position level	budget, distributed, plan, actual, commitments, available	`program_position`	none	IM information system program reports
Position forecast cost	operational report	department, scenario, year	positions, FTE, salary, benefits, tax, total by month	`position_forecast`	Position forecasting reports	PCP cost plan reports
Cost behavior classification	list inquiry	ledger, element, object type	fixed, variable, unclassified amounts, policy rule applied	`cost_entry`	Cost behavior inquiry	KSB1 by cost element category

7.38.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Budget plan worksheet	export (FR-38.13)	header sheet (plan, stage, template version), data sheet, hidden checksum sheet	xlsx	n/a	none
Budget register entry approval record	workflow completion	entry, lines, approvers, timestamps	PDF	register entry number	7 years (internal control evidence, SOX 404 where applicable)
Budget transfer advice	transfer posted	from / to combinations, amount, reason	PDF, email to budget owners	register entry number	7 years
Earmarked funds document	create / change	type, lines, expiry, approver	PDF	per-type sequence	7 years (public-sector packs 10 years)
Cost center report	period close	plan / actual / variance per element	PDF, XLSX	none	none (management)
Allocation run statement	run completion	sender → receiver matrix, totals, residual	PDF, CSV	run id	7 years (cost allocation audit trail; government contract cost accounting standards where applicable)
Investment program budget letter	distribution	position, budget, conditions	PDF	program number	7 years

7.38.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`cost-mirror-lag-check`	every 5 min	ledger	offsets per Kafka partition	< 10 s	alert only	lag > 60 s
`cost-gl-reconcile`	nightly 02:15	(ledger, period)	(ledger, period, date)	< 5 min	report differences	difference ≠ 0
`allocation-schedule`	per policy (default period close day 2)	(policy, period)	(policy, period, version); rerun reverses	< 10 min for 1M entries	rollback run; retry	non-convergence / residual > 0.01
`template-allocation`	per template	(template, period)	as above	< 5 min	as above	formula errors
`hierarchy-distribution`	period end after allocations	(hierarchy, period)	(node, period)	< 5 min	residual left on node	residual > 0
`budget-balance-rebuild`	on demand	rule version	event replay offset	< 10 min for 1M documents	old cache serves until swap	drift detected
`encumbrance-carry-forward`	fiscal year end (close checklist)	legal entity	(entity, year)	< 10 min for 100k encumbrances	atomic per combination	items failed
`reservation-expiry`	daily 01:00	legal entity	(reservation, date)	< 1 min	retry	expiring in 7 days
`program-carry-forward`	approval year end	program	(position, year)	< 2 min	per-position isolation	skipped positions
`position-forecast-refresh`	nightly 03:30 (when wave 42 active)	legal entity	snapshot version	< 5 min	keep previous	source failure

7.38.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
BudgetView	GET /budget/*, budgeting workspace, 7.38.14 budget reports	none	Budget owner, Controller
BudgetMaintain	/register-entries, submit	BudgetApprove (same entry)	Budget analyst
BudgetApprove	register entry / transfer workflow approve	BudgetMaintain	Budget manager, CFO
BudgetPlan	/plans/*, worksheets, generate	BudgetApprove at the same stage	Budget preparer
BudgetConfigure	control rules, thresholds, override groups, worksheet templates	BudgetOverride, BudgetMaintain	Finance system admin
BudgetOverride	fail → warn on check	BudgetConfigure; ProcurementPurchaseOrderApprove for own POs	Budget manager
BudgetReserve	/reservations	BudgetApprove of the same reservation	Budget analyst
InvestmentProgramMaintain	/investment-programs, distribute, carry-forward	CapexRequestApprove (37) on assigned requests	Capex controller
CostAccountingView	GET /controlling/*, cost control workspace	none	Controller, Department head
CostAccountingMaintain	cost objects, elements, hierarchies, measures, policies	CostAccountingRun	Cost accountant
CostAccountingRun	allocation, template, top-down, hierarchy, settlement runs	CostAccountingMaintain (policy author cannot run the period)	Cost accountant lead

7.38.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
CO-0601	error	Budget check failed: available `{available}`, requested `{amount}`	over block threshold	reduce amount or request override	FR-38.4
CO-0602	warning	Budget exceeded by `{pct}` within tolerance	between warn and block	none or request override	FR-38.4
CO-0603	error	No budget cycle contains `{date}`	NoBudgetCycle	open cycle / change date	BR-38.5
CO-0604	error	Transfer lines do not net to zero (difference `{amount}`)	unbalanced transfer	correct lines	FR-38.1
CO-0605	info	Budget check overridden by `{user}`	override group	none	FR-38.4
CO-0608	error	Budget check timed out	check > 500 ms	retry; SRE alert	FR-38.4
CO-0611	error	Allocation did not converge: residual `{residual}` after `{n}` iterations	iterative solve	review reciprocal structure	BR-38.6
CO-0612	error	Allocation totals differ: in `{in}`, out `{out}`	conservation failure	run rolled back; report defect	BR-38.6
CO-0613	warning	Sender `{sender}` has zero basis; cost not allocated	no basis	load statistical measures	BR-38.8
CO-0614	error	Period `{period}` is locked for allocations	ledger period lock	unlock or run next period	7.38.13 `ledger.periodLock`
CO-0615	warning	Journal postings after the last allocation run for `{period}`	late postings	rerun allocation	BR-38.7
CO-0618	error	Settlement `{amount}` exceeds collected cost `{collected}`	over-settlement	reduce settlement	BR-38.9
CO-0621	error	Formula error in template `{template}` line `{n}`: `{detail}`	invalid expression	fix formula	FR-38.17
CO-0622	warning	Measure `{measure}` is zero for receiver `{receiver}`	division by zero	load measure or accept	BR-38.12
CO-0626	error	Worksheet protected range `{range}` was modified	checksum mismatch	export again	FR-38.13
CO-0627	warning	`{n}` cells conflicted with server changes and were not applied	concurrent edit	review conflict list	BR-38.11
CO-0631	error	Funds blocking `{doc}` cannot be consumed	FundsBlocking consumption	use a reservation type	BR-38.14
CO-0632	info	Reservation `{doc}` expired; `{amount}` released	expiry	none	BR-38.14
CO-0641	error	Distribution exceeds parent position budget by `{amount}`	BR-38.15	reduce distribution or raise parent	BR-38.15
CO-0645	warning	Hierarchy node `{node}` keeps `{amount}` undistributed	no basis on children	review basis	FR-38.18

7.38.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Budget check	20,000 checks/day; peak 50/s at month-end PO runs	30 / 150 / 300 ms	50 checks/s sustained	7M check results ≈ 4 GB (7-year retention ≈ 28 GB)	BudgetControlBalance unique (rule_version, key, span); results partitioned monthly	row locks serialize hot combinations; p95 < 400 ms; timeouts fail open to `warn` only if rule sets `fail-open` (default fail closed)
Available funds query	10,000/day	20 / 80 / 150 ms	100 req/s	n/a	same	cached 5 s

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Cost entry mirroring	2M journal lines/day	lag 1 / 5 / 15 s	500 lines/s	730M entries ≈ 180 GB	CostEntry partitioned by period_key; index (cost_object_id, cost_element_id, period_key)	lag up to 60 s; alert
Allocation run	1M entries, 5 steps	n/a	< 10 min	allocation entries ≈ 20% of mirrored volume	per-run batch insert, COPY	< 30 min
Template / top-down run	200k receivers	n/a	< 5 min	small	receivers streamed by partition	< 15 min
Plan grid load	500,000 lines per plan; 50 concurrent planners	300 ms / 1.2 s / 2.5 s per 10k-row page	50 concurrent users	5M plan lines ≈ 3 GB	plan lines by (plan_id, scenario_id, combination)	page size drops to 2k rows
Worksheet import	200 uploads/day near deadline	5 / 20 / 40 s for 200k cells	10 concurrent	xlsx stored 1 year ≈ 20 GB	n/a	uploads queue, status polled
Budget balance rebuild	1M documents	n/a	< 10 min	n/a	replay by rule version	< 30 min, old cache serves

7.38.20 Localization and compliance hooks

- **Public-sector budgeting**: US federal (USSGL, appropriation / apportionment / allotment, Treasury Account Symbol) and state / local GASB 34 fund accounting are out of v1; the `fund.dimensions` parameter and earmarked funds (FR-38.20) cover nonprofit restricted funds (ASC 958) and EU grant tracking. Public-sector packs (post-v1) extend through `BudgetStructureProvider`.
- **Government contract cost accounting**: US CAS (FAR Part 30) and DCAA require documented, consistently applied allocation bases with an audit trail; allocation run statements (7.38.15) and versioned policies satisfy the trail; indirect rate (fringe, overhead, G&A) pools are configured as assessment steps.
- **Statutory cost accounting**: FR comptabilité analytique is not statutory; DE Betriebsabrechnungsbogen (BAB) is a report layout over the cost ledger (wave-48 DE pack template).
- **Personnel cost planning**: payroll tax and social security ceilings in FR-38.15 cost elements are country data from wave-43 payroll packs once available; privacy: position forecasts with named workers are personal data and use wave-42 HR visibility rules.
- **Retention**: budget check results and approval records 7 years; plan worksheets 1 year online then archive.
- **Fiscal calendars**: 4-4-5 and 13-period calendars from wave 14 drive time spans and allocation periods.

7.38.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.4.1 Budget plan → Budget plan templates / worksheets (BudgetPlanTemplate)	PARTIAL-38	ADDED as FR-38.13	
AX 12.4.3 Budget planning → Generate budget plan from source (GL actuals, previous plan, FA, projects, demand forecast, HR positions)	PARTIAL-38	ADDED as FR-38.14	project source activates with wave 41
AX 12.4.4 Budget planning → Budget planning templates (Excel)	PARTIAL-38	ADDED as FR-38.13	
AX 12.4.4 Budget planning → Position forecasting (D365 HR-driven)	PARTIAL-38	ADDED as FR-38.15	manual / CSV positions in wave 38; HR-driven source when wave 42 ships
AX 12.7.1 Cost behavior / Cost classification (CAMCostBehavior)	PARTIAL-38	ADDED as FR-38.16	
AX 12.7.4 Policies — cost behavior, distribution, overhead rate, allocation, rollup	PARTIAL-38	ADDED as FR-38.16	distribution and allocation policies already FR-38.7
SAP 13.7 CPTA / CPTB ABC template allocation	PARTIAL-38	ADDED as FR-38.17	
SAP 13.7 ABC template allocation environments, functions (CPT1)	PARTIAL-38	ADDED as FR-38.17	`alloc.templateEnvironment`
SAP 13.8 KKPHIE / KKP2 Cost object hierarchy	PARTIAL-38	ADDED as FR-38.18	
SAP 13.9 KE28 Top-down distribution (actual)	PARTIAL-38	ADDED as FR-38.19	
SAP 13.11 FM5I / FM5S Fund	PARTIAL-38	SUPERSEDED by wave-11 financial dimension `Fund` seeded by `fund.dimensions` and usable as a budget control dimension (FR-38.3)	appropriation-based fund accounting out of v1
SAP 13.11 FMMEASURE Funded program	PARTIAL-38	SUPERSEDED by financial dimension `FundedProgram` (same mechanism)	
SAP 13.11 FMX1 / FMX2 / FMX3 Earmarked funds: funds reservation	PARTIAL-38	ADDED as FR-38.20	
SAP 13.11 FMY1 / FMZ1 / FMW1 / FMV1 Precommitment / commitment / revenue forecast / funds blocking	PARTIAL-38	ADDED as FR-38.20	FMZ1 funds commitment maps to FundsReservation with consumption
SAP 13.11 Earmarked funds document types, number ranges (FMEF*)	PARTIAL-38	ADDED as FR-38.20	`reservation.documentTypes`
SAP 13.27 IM27 / IM32 / IM52 Investment program budget distribution, rollup, carry-forward (handed over from wave 37)	PARTIAL-37	ADDED as FR-38.21	see 7.37.21

7.39 Wave 39: Product Costing, Cost Roll-Up & Variance Analysis

7.39.1 Scope and business context

Wave 20 delivered the inventory costing engine with a standard cost method but no way to derive the standard: standards were keyed. Wave 39 computes them. Cost accountants define cost groups, costing sheets with indirect cost nodes, and costing versions; roll up costs from BOMs and routings per site and quantity with explosion modes and fallback rules; compare estimates; activate pending costs, which revalues on-hand inventory and posts the revaluation; and analyze purchase price and production variances decomposed into lot size, quantity, price and substitution by cost component. At the end of the wave the demo plant runs on standard cost with a monthly variance report.

Out of scope: actual costing / material ledger periodic unit price (wave 20 covers FIFO/weighted average), mixed costing across procurement alternatives, co-product cost allocation for process manufacturing beyond fixed percentage (wave 24 rules reused), sales-order costing for configured items.

Scope amendment (gap closure). The registers (coverage-ax §12.99 rows 50 and 54; coverage-sap §13.99 rows 89–101 mapped to wave 39; plus SAP row 90 mapped to wave 27 and AX rows 20–21 proposed for wave 20, whose notes point here) add standard cost conversion (FR-39.12), ad-hoc cost estimates with costing models (FR-39.13), mixed costing over procurement alternatives (FR-39.14), COGS and price-difference posting by cost component (FR-39.15), sales order cost estimates for configured and make-to-order lines (FR-39.16), parallel valuation views (FR-39.17) and parallel inventory valuation ledgers (FR-39.18, flag-gated). These supersede the out-of-scope notes above for mixed costing and sales-order costing. Material-ledger actual costing remains with wave 20.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX BOMCalcTable "Calculate items" for a costing version runs single-threaded; 50k items take 6–10 h and lock InventItemPriceSim	Sequential explosion per item re-exploding shared subassemblies	Topological roll-up computes each node once; parallel by level	50k items 8 h → < 15 min
CK40N costing run in SAP: selection, structure explosion, costing, analysis, marking, release as separate steps with logs in CKAPP; errors found at release	Multi-step batch with late validation	One run with upfront validation (missing prices, cycles, inactive routes) and per-item diagnostics	Rework runs per cycle 3 → 1
Cost component split lost in AX when a subassembly rolls into its parent (only cost group breakdown if "cost breakdown" enabled, which doubles InventCostTrans volume)	Breakdown as optional duplicate storage	Split is the primary representation; parent carries child split natively	Items with full split 40% → 100%
Production variance report in AX shows lot size/quantity/price/substitution totals that do not sum to the posted variance when rounding occurs across InventCostTrans	Independent rounding per variance type	Decomposition with residual assignment rule; property-tested	Unexplained variance per order 0.00
Standard cost activation (InventItemPrice activate / CK24 release) revalues inventory in a batch without preview; controllers discover the P&L hit after posting	No simulation of revaluation	Activation preview shows revaluation by item/site before commit	Unplanned revaluation surprises 0; preview < 60 s

7.39.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Costing versions	CostingVersion (type Standard cost / Planned cost), blocking, fallback principle, InventItemPrice, InventItemPriceSim	Costing variant OKKN, valuation variant, costing version (OKKN), KEKO header	CostingVersion with status, fallback, site scope	full
Cost groups and categories	CostGroup (direct material / direct manufacturing / indirect / undefined; fixed/variable), CostCategory with cost category prices, BOMCalcGroup	Cost component structure OKTZ (TCKH1–3), cost elements per component, activity types/prices KP26	CostGroup, CostComponent, CostCategory rate tables	full
Costing sheet	CostSheetNodeTable, CostSheetCalculationBasis, CostSheetCalculationFactor (surcharge/rate nodes, absorption basis)	Costing sheet (KZS2): base rows, overhead rows (percentage/quantity), credit rows; T683	CostingSheet with Basis/Surcharge/Rate/Total nodes	full
BOM calculation	BOMCalcTable, BOMCalcTrans, BOMCalcItem, explosion mode (multilevel / single level / make-to-order / according to BOM line type), profit settings	CK11N single item, CK40N run, CKIS items, KEPH component split, CK13N display	CostCalculation with explosion mode, split retained	full
Mass roll and activation	Costing version "Calculate items", "Activate pending prices", InventSettlement cost change variance	CK40N marking (CK24 mark/release), MBEW STPRS update, MR21	Mass roll (Temporal), activation with revaluation preview	improved
Purchase price variance	InventCostTrans variance type PurchPrice posted on receipt for standard cost items	Price difference account PRD on GR/IR (OBYC PRD)	VarianceEntry type PurchasePrice from wave 18 receipts/invoices	full
Production variances	ProdCalcTrans estimate, InventCostTransVariance (ProdLotSize, ProdQuantity, ProdPrice, ProdSubstitution) at ending	KKS1/KKS2 variance calculation categories (input price, input quantity, resource usage, remaining input, lot size, output price, remaining), CO88 settlement	Variance decomposition on production order end	full
Inventory cost revaluation	Cost change variance (InventSettlement), inventory value report	Price change document (MR21/CKMPRPUNMARK), ML revaluation	Revaluation journal posted on activation	full
Standard cost conversion	Standard cost conversion checklist (D365 InventStdCostConv, InventStdCostConvItem), "Convert to standard cost"	Price control change S → V / V → S on MBEW-VPRSV (MM02) with MR21 price change	ConversionRun with checklist (FR-39.12)	improved (preview, validation)
Ad-hoc cost estimate	none (Excel)	CKECP / CKECP1 easy cost planning, costing models (CKF*), CKUC unit costing	CostingModel and AdHocEstimate (FR-39.13)	full
Mixed costing	none (single source per costing version)	CK91N procurement alternatives, CK94 mixing ratios, OMXL quantity structure types	ProcurementAlternative with mixing ratios (FR-39.14)	full
COGS and PPV by component	Cost breakdown posting (InventCostTrans cost group detail), limited	S/4 COGS splitting (FINSC_COGS*), price-difference splitting	Component-split posting option (FR-39.15)	full
Sales order cost estimate	Item price for configured items (PCConfiguration → BOMCalcTable for order)	CK51N / CK53N sales order cost estimate, KKE1 base planning	CostCalculation type SalesOrder (FR-39.16)	full
Parallel valuation views	none	Legal / group / profit-center valuation (OKKP, TKA01 valuation views, transfer price variants)	Costing version valuation view (FR-39.17)	partial (no ML actual costing per view)
Global inventory accounting	Global inventory accounting add-in (D365 10.0.3x: inventory accounting ledgers, accounting conventions, costing policies)	Material ledger multiple currencies / valuations (CKMLCP per valuation view)	InventoryAccountingLedger per convention (FR-39.18)	partial (flag-gated; periodic settlement per ledger only)

7.39.3 Functional requirements

- **FR-39.1 Cost groups and components** — Must. The system shall classify item costs, cost categories (labor/machine) and indirect nodes into cost groups with type and behavior, and map cost groups to cost components for the split.
  - Notes: ≤ 50 cost groups, ≤ 20 components per tenant; an item has exactly one cost group per site override.
- **FR-39.2 Costing sheet** — Must. The system shall define a costing sheet tree with Basis nodes (cost groups selected), Surcharge nodes (% on a basis), Rate nodes (amount per routing hour or per unit), and Total nodes, per site and costing version.

    - Notes: surcharge basis evaluates only the current level's direct lines unless the node is flagged `include-rolled-subassemblies`; absorption posts to indirect accounts via posting profiles at production end.
- **FR-39.3 Costing versions** — Must. The system shall maintain costing versions (Standard, Planned, Simulation) with blocking of pending prices, fallback principle (None / Active version / Specified version), site scope, and effective date.
- **FR-39.4 Cost roll-up algorithm** — Must. The system shall compute item costs by: (1) building the BOM graph for the selection and detecting cycles (reject with path); (2) ordering nodes topologically, leaves first, ≤ 20 levels; (3) for each node computing direct material = Σ child quantity × (1 + scrap%) × child cost (purchased: price from version, else fallback; manufactured: rolled value with its split); direct manufacturing = Σ operations (setup time / costing lot size + run time per unit) × cost category rate; indirect = costing sheet nodes evaluated on this level's basis; (4) retaining the cost component split of every child inside the parent (not collapsed); (5) storing the result as a pending price in the version with explanation lines.
    - Notes: worked 2-level example (USD). Subassembly SA-200: 3 × RM-301 at 2.50 = 7.50; 1 × RM-302 at 1.00 with 10% scrap = 1.10; machining 0.25 h × 60.00 = 15.00; sheet: material overhead 10% on direct material = 0.86; labor overhead 50% on direct labor = 7.50. SA-200 = 31.96 (DM 8.60, DL 15.00, IND-MAT 0.86, IND-LAB 7.50). Finished item FG-100: 2 × SA-200 = 63.92 carried as split (DM 17.20, DL 30.00, IND-MAT 1.72, IND-LAB 15.00); 1 × RM-300 at 4.00 → DM 4.00; assembly 0.5 h × 40.00 = 20.00 plus setup 1 h × 40.00 / costing lot size 10 = 4.00 → DL 24.00; material overhead 10% on own direct material purchased lines = 0.40; labor overhead 50% × 24.00 = 12.00. FG-100 = 104.32 with split DM 21.20, DL 54.00, IND-MAT 2.12, IND-LAB 27.00 (sums to 104.32). Explosion modes: multilevel (as above), single level (children at their active cost), make-to-order (explode only phantom and MTO lines), by BOM line type.
- **FR-39.5 Mass roll** — Must. The system shall roll all items in a version/site selection in parallel by topological level, with per-item diagnostics (missing price, inactive BOM/route, cycle), completing 50,000 items in < 15 minutes.
- **FR-39.6 Activation and revaluation** — Must. The system shall activate pending prices as of a date, compute on-hand revaluation = Σ on-hand quantity × (new – old standard) per item/site/dimension, show a preview, and post the revaluation journal (Dr/Cr Inventory vs cost change variance account).
    - Notes: activation is atomic per version/site; wave-20 standard cost engine switches at the activation timestamp; open production orders estimated at the old standard keep their estimate but variance at end uses the new standard for output (configurable).
- **FR-39.7 Purchase price variance** — Must. The system shall post PPV = (receipt unit cost – standard) × quantity at product receipt and adjust at invoice for standard-cost items, by cost component DM.
- **FR-39.8 Production variance decomposition** — Must. The system shall, when a production order ends, compute total variance = actual cost – standard cost of reported-as-finished (RAF) output, decomposed per cost component as: lot size variance LSV = Σ over fixed-per-lot elements (setup, constant scrap) of (standard per unit × Q_raf – estimated per lot ÷ Q_order × Q_raf); quantity variance QV = Σ_i (actual qty_i – standard qty_i for Q_raf) × standard unit cost_i for planned inputs i (inputs planned but not consumed give negative QV); price variance PV = Σ_i actual qty_i × (actual unit cost_i – standard unit cost_i) for inputs whose cost differs from standard (labor rate differences, components not on standard cost at that site, subcontract invoices); substitution variance SV = Σ_j actual qty_j × actual unit cost_j for inputs j not in the estimate. Identity: LSV + QV + PV + SV = total variance to the cent; residual from rounding assigned to QV of the largest planned line.
    - Notes: unfavorable is positive; output quantity variance for co-products handled by wave 24 percentages; variances post to GL by posting profile per type and component.
- **FR-39.9 Variance analysis workspace** — Must. The system shall present variances by order, item, site, period, type and component with drill to consumption and route transactions, and the monthly variance report.
- **FR-39.10 Cost estimate comparison and what-if** — Should. The system shall compare two versions or dates item-by-item by component, and allow a simulation version with overridden prices, rates and routes.
- **FR-39.11 Standard cost reconciliation** — Must. The system shall reconcile Σ on-hand quantity × active standard to GL inventory accounts per site nightly (extends the wave-20 inventory value report).
- **FR-39.12 Standard cost conversion** — Should. The system shall convert items from an actual costing method (FIFO, LIFO, weighted average) to standard cost at a conversion date through a checklist: select items and sites, roll pending standards into a conversion version, validate, preview the conversion revaluation, run inventory close up to the conversion date, and activate.
    - Notes: validation blocks conversion for items with negative physical on-hand, open in-transit transfers, or production orders started but not ended (option: allow, with variance at the new standard); conversion revaluation = Σ on-hand × standard – on-hand value at actual cost after the close, posted Dr / Cr Inventory vs cost change variance; after conversion the wave-20 engine stops settling issues for converted items; the checklist records each step's user and timestamp; reverse conversion (standard → actual) uses the same checklist with value carried at the last standard.
- **FR-39.13 Ad-hoc cost estimates (easy cost planning)** — Should. The system shall define costing models with characteristics (for example cavities, machine hours, material weight) whose formulas produce item, activity, subcontract and overhead lines, and compute estimates for non-material objects (quotes, internal orders, project tasks) with the costing sheet applied.
    - Notes: formula grammar shared with FR-38.17; prices resolved from the selected costing version with fallback; an estimate can be copied as plan to an internal order (wave 38), as a project estimate (wave 41), or as quote cost on a sales quotation (wave 27); estimates are versioned and comparable through FR-39.10.
- **FR-39.14 Mixed costing** — Should. The system shall maintain procurement alternatives per item and site (purchase from vendor, production version or BOM / route alternative, transfer from another site) with mixing ratios per costing version and compute the item's cost as the ratio-weighted cost of the alternatives with a weighted component split.
    - Notes: ratios must total 100.00% per item, site and version; each alternative is rolled with its own quantity structure; worked example: purchase from vendor A at 10.00 (60%) and in-house production at 12.50 (40%) → mixed cost = 0.6 × 10.00 + 0.4 × 12.50 = 11.00, and the split is weighted the same way (DM = 0.6 × 10.00 + 0.4 × production DM, other components 0.4 × production amounts); alternatives without a valid cost fail the item unless fallback applies; ratios can be derived from the last 12 months of receipts with `derive-ratios`.
- **FR-39.15 COGS and price-difference posting by cost component** — Should. The system shall, per site option, post cost of goods sold on sales issues (wave 28) to separate accounts per cost component from the item's active split, and split purchase price variances proportionally to the purchased item's own components.
    - Notes: worked example: 10 × FG-100 at 104.32 issued → Dr COGS-material 212.00, Dr COGS-labor 540.00, Dr COGS-material overhead 21.20, Dr COGS-labor overhead 270.00 / Cr Inventory 1,043.20; PPV split uses the ratio of each component in the standard of the purchased item (for example material 92%, freight surcharge 5%, duty 3%); rounding residual to the largest component; option is off by default so wave-28 postings are unchanged unless enabled.
- **FR-39.16 Sales order cost estimates** — Should. The system shall cost a configured or make-to-order sales order line using its order-specific BOM and route (wave 27 configurator output, wave 9 versions) at the order quantity, store the result as a calculation of type SalesOrder linked to the line, and show the margin.
    - Notes: margin = net line amount – estimated cost; the estimate recomputes on configuration change before confirmation and freezes at confirmation; option `use-order-estimate-as-standard` makes the MTO production order's variance compare against the order estimate instead of the item standard; estimates for lines with ≤ 6 levels return in < 2 s.
- **FR-39.17 Parallel valuation views** — Should. The system shall maintain costing versions per valuation view (Legal, Group, ProfitCenter): the Group view rolls intercompany-purchased components at the supplying entity's Legal-view cost (excluding intercompany markup), and the ProfitCenter view uses internal transfer prices between profit centers within an entity.
    - Notes: the Group view standard is stored alongside the Legal view per item and site; inventory receipts from IC partners (wave 40) store the group cost per receipt layer so that unrealized profit = Legal-view value – Group-view value of remaining stock; ProfitCenter view transfer prices come from a

transfer-price table or cost-plus % per profit-center pair; only the Legal view posts to the statutory ledger, other views post to dedicated ledger layers (wave 12) when the flag `costing.valuation-views-post` is on.

- **FR-39.18 Parallel inventory valuation ledgers (global inventory accounting)** — Could. The system shall maintain inventory accounting ledgers per legal entity, each with an accounting convention (costing method per item group, currency, target ledger layer), and run the wave-20 costing engine per ledger on the same inventory transactions to produce per-ledger inventory values, COGS and settlements.
  - Notes: flag `costing.global-inventory-accounting`; example: IFRS layer FIFO, local GAAP layer weighted average, group layer standard (FR-39.17 Group view); each ledger has its own inventory close, value report and GL reconciliation; the primary ledger is the one wave 20 already posts, additional ledgers post adjustments into their layer only; capacity budget: 3 ledgers per entity in v1.

7.39.4 Data model

CostingVersion: id, legal_entity_id, code, type (Standard/Planned/Simulation), status (Open/Blocked/Active/Closed), fallback_principle, fallback_version_id, site_ids[], effective_from, activated_at.

CostCalculation

Field	Type	Required	Notes
id	uuid	yes	PK
costing_version_id	uuid	yes	FK
item_id, site_id	uuid	yes	
calc_quantity	numeric(19,4)	yes	costing lot size
explosion_mode	enum	yes	Multilevel/SingleLevel/MakeToOrder/ByLineType
bom_version_id, route_version_id	uuid	yes	wave 9
unit_cost	numeric(19,6)	yes	6-decimal unit cost
status	enum	yes	Pending/Active/Superseded/Error
diagnostics	jsonb	no	[{code, message, ref}]
run_id	uuid	no	mass roll

CostComponentSplit

Field	Type	Required	Notes
calculation_id	uuid	yes	FK, part of PK
component_id	uuid	yes	FK CostComponent, part of PK
level	enum	yes	Own/Rolled (child-carried)
amount	numeric(19,6)	yes	per calc_quantity
fixed_amount, variable_amount	numeric(19,6)	yes	behavior split

CostingSheetNode: id, sheet_id, parent_id, type (Basis/Surcharge/Rate/Total), cost_group_ids[], percentage numeric(9,4), rate numeric(19,6), rate_unit (Hour/Unit), include_rolled bool, absorption_account_id, order.

VarianceEntry

Field	Type	Required	Notes
id	uuid	yes	PK
source_type, source_id	enum, uuid	yes	ProductionOrder/Receipt/Invoice/Activation
type	enum	yes	LotSize/Quantity/Price/Substitution/PurchasePrice/CostChange
component_id	uuid	yes	
item_id, site_id	uuid	yes	
amount	numeric(19,4)	yes	unfavorable positive
journal_entry_id	uuid	yes	
detail	jsonb	no	per input line actual/standard qty and cost

Entities added by FR-39.12–FR-39.18: **ConversionRun** (id, legal_entity_id, site_ids[], item_ids[], conversion_date, costing_version_id, checklist jsonb [{step, status, user, at}], revaluation_journal_id, status Draft/Validated/Closed/Activated/Reversed); **CostingModel** (id, code, characteristics [{name, unit, default}], lines [{type Item/Activity/Subcontract/Overhead, ref, quantityFormula, priceFormula}], version) and **AdHocEstimate** (id, model_id, object_type Quote/InternalOrder/ProjectTask, object_id, characteristic_values jsonb, costing_version_id, total, split jsonb); **ProcurementAlternative** (id, item_id, site_id, costing_version_id, type Purchase/Production/Transfer, vendor_id or production_version_id or source_site_id, ratio numeric(7,4)); **CostCalculation** gains `calc_type` (Item/SalesOrder/AdHoc/Mixed) and `valuation_view` (Legal/Group/ProfitCenter), with the unique-Active partial index extended to (version, item, site, valuation_view); **InventoryAccountingLedger** (id, legal_entity_id, code, currency, ledger_layer, costing_policies [{itemGroupId, method}], primary bool).

Invariants: Σ CostComponentSplit.amount per calculation = unit_cost × calc_quantity (6 decimals); exactly one Active calculation per (version, item, site); Σ VarianceEntry per production order = actual − standard of RAF to the cent. Indexes: CostCalculation unique partial (version, item, site) where status = Active; VarianceEntry by (site_id, period, type).

7.39.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/costing/versions	Create version	Idempotency-Key	CostingMaintain
POST	/api/v1/costing/sheets	Create/version sheet	Idempotency-Key	CostingMaintain
POST	/api/v1/costing/calculations	Roll one item (explosion mode, qty)	(version, item, site, bom, route, qty)	CostingCalculate
POST	/api/v1/costing/versions/{id}/roll	Mass roll selection	runId	CostingCalculate
GET	/api/v1/costing/calculations/{id}/explain	Explanation tree with split	n/a	CostingView
POST	/api/v1/costing/versions/{id}/activate	Preview (dryRun=true) or activate	(version, date)	CostingActivate

Method	Path	Purpose	Idempotency	Auth duty
GET	/api/v1/costing/variances?site=&period=&type=	Variances	n/a	CostingView
POST	/api/v1/costing/variances/compute	Compute for ended order (system)	productionOrderId+version	CostingSystem
GET	/api/v1/costing/compare?a=&b=	Version comparison	n/a	CostingView
POST	/api/v1/costing/conversions	Start standard cost conversion checklist	Idempotency-Key	CostingActivate
POST	/api/v1/costing/conversions/{id}/steps/{step}	Execute checklist step	(conversion, step)	CostingActivate
POST	/api/v1/costing/estimates	Ad-hoc estimate from costing model	Idempotency-Key	CostingCalculate
PUT	/api/v1/costing/procurement-alternatives/{itemId}	Set alternatives and ratios	version	CostingMaintain
POST	/api/v1/costing/sales-order-estimates	Cost a sales order line	(lineId, configVersion)	CostingCalculate (system for sales)

Event	Trigger	Payload fields	Consumers
`dev.passage.costing.calculation.completed.v1`	item rolled	calculationId, itemId, siteId, unitCost, split[]	analytics
`dev.passage.costing.rollrun.completed.v1`	mass roll finished	runId, itemCount, errorCount, durationMs	workspace
`dev.passage.costing.standardcost.activated.v1`	activation	costingVersionId, itemCount, revaluationJournalId, totalRevaluation	inventory costing, production, budget
`dev.passage.costing.variance.posted.v1`	variance computed	sourceId, type totals, byComponent[], journalEntryId	costacct-svc, analytics
`dev.passage.costing.conversion.activated.v1`	conversion activated	conversionId, itemCount, revaluationJournalId, conversionDate	inventory costing (20), production (21)
`dev.passage.costing.salesorderestimate.completed.v1`	order line costed	lineId, calculationId, cost, margin, marginPct	sales (27), analytics

7.39.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Costing workspace	Roll, activate, compare, diagnostics tile	version, pending count, errors, last activation	CostingView
Costing sheet designer	Add node, set basis/%/rate, validate	tree, cost groups, percentage, rate, absorption account	CostingMaintain
Calculation explain	Expand levels, view split, export	level, line, qty, unit cost, component amounts	CostingView
Activation preview	Filter item/site, approve	item, old, new, on-hand, revaluation	CostingActivate
Variance analysis	Pivot by type/component/order, drill	order, item, standard, actual, LSV, QV, PV, SV	CostingView

Interaction rules: the explain tree lazy-loads levels; keyboard `e` expands all; activation preview requires an explicit confirmation of the total revaluation amount; variance pivot persists per user.

7.39.7 Business rules and edge cases

- **BR-39.1** Unit costs are stored with 6 decimals; postings round to the currency minor unit at the transaction level; split rounding residual goes to the DM component.
- **BR-39.2** Version state machine: Open → Blocked (no edits of pending) → Active (per site activation) → Closed; Simulation versions cannot activate.
- **BR-39.3** A cyclic BOM (item appears in its own explosion path) fails the calculation with the path listed; phantoms are exploded in place.
- **BR-39.4** Fallback: missing price with fallback Active uses the active standard; with None, the calculation errors; fallback use is recorded in diagnostics.
- **BR-39.5** Setup cost per unit = setup time × rate ÷ costing lot size; costing lot size defaults from the item's default order settings (wave 8) and can be overridden per calculation.
- **BR-39.6** Scrap: BOM line scrap % multiplies quantity; route operation scrap % compounds backward through preceding operations.
- **BR-39.7** Activation while a mass roll is running for the same version/site is rejected (advisory lock).
- **BR-39.8** Revaluation uses on-hand at the activation timestamp from the wave-16 projection; inventory transactions timestamped after activation use the new standard.
- **BR-39.9** Variance identity: if ILSV + QV + PV + SV − totall > 0.00 after residual assignment, the computation fails and the order end is blocked with an error.
- **BR-39.10** Ended orders reopened (wave 21) reverse their variance entries; re-ending recomputes.
- **BR-39.11** Standard cost conversion requires a completed wave-20 inventory close with end date = conversion date − 1 day; activation without it is rejected (CST-0731).
- **BR-39.12** Mixing ratios are validated per (item, site, version) to total 100.00%; a ratio of 0.00% keeps the alternative for reporting but excludes it from the roll.
- **BR-39.13** COGS component posting: Σ component lines = total COGS of the issue to the cent; residual from rounding goes to the material component.
- **BR-39.14** A sales order estimate freezes at line confirmation; later configuration changes create a new estimate version and mark the old one Superseded; production order variance uses the frozen version when `use-order-estimate-as-standard` is on.
- **BR-39.15** Group-view unit cost of an IC-purchased component equals the supplier entity's Legal-view cost at the receipt date converted at the receipt's rate; when the supplier has no Legal-view standard, the IC purchase price applies and the receipt is flagged `groupCostUnknown` for wave 40 elimination review.

7.39.8 Acceptance criteria

- **AC-39.1** (FR-39.4) Given the worked SA-200/FG-100 fixture, when rolled multilevel, then FG-100 = 104.32 with split DM 21.20 / DL 54.00 / IND-MAT 2.12 / IND-LAB 27.00.
- **AC-39.2** (FR-39.4) Given 50 multi-level fixture products, when rolled, then every component amount matches the reference workbook to the cent.
- **AC-39.3** (FR-39.5) Given 50,000 items across 8 levels, when mass rolled, then the run completes in < 15 min with diagnostics for every error.
- **AC-39.4** (FR-39.6) Given 1,000 units on hand at 10.00 and a new standard 10.50, when activated, then a journal Dr Inventory 500.00 / Cr Cost change variance 500.00 posts and the inventory value report ties.
- **AC-39.5** (FR-39.8) Given an order for 100 planned, 90 RAF, consuming 290 of a component standard 3/unit at 2.00 plus 5 units of an unplanned part at 1.50, with setup 40.00 per lot and costing lot size 100, then QV = (290 − 270) × 2.00 = 40.00, SV = 7.50, LSV = 90 × (0.40 − 0.40) = 0.00 and total = 47.50 (plus any labor PV), and the identity holds.
- **AC-39.6** (FR-39.8) Given 200 ended fixture orders, when variances compute, then Σ types = total for every order.
- **AC-39.7** (FR-39.7) Given a receipt of 10 at 5.20 for an item standard 5.00, when posted, then PPV = 2.00 posts to DM.
- **AC-39.8** (FR-39.3) Given a cyclic BOM, when rolled, then the calculation errors with the cycle path and the run continues for other items.
- **AC-39.9** (FR-39.11) Given month end, when reconciliation runs, then Σ on-hand × standard equals GL inventory per site.
- **AC-39.10** (FR-39.12) Given 500 units of an item at FIFO value 4,925.00 and a standard of 10.00, when conversion activates on 2026-11-01 after the October close, then a journal Dr Inventory 75.00 / Cr Cost change variance 75.00 posts and later issues use 10.00.
- **AC-39.11** (FR-39.14) Given alternatives vendor A 10.00 at 60% and production 12.50 at 40%, when rolled, then the mixed cost is 11.00 and the split totals 11.00.

- **AC-39.12** (FR-39.15) Given the component posting option and an issue of 10 × FG-100, when posted, then four COGS lines 212.00 / 540.00 / 21.20 / 270.00 post and total 1,043.20.
- **AC-39.13** (FR-39.16) Given a configured line priced at 2,400.00 whose configuration rolls to 1,860.00, when costed, then the estimate shows margin 540.00 (22.50%) and freezes on confirmation.
- **AC-39.14** (FR-39.17) Given a component bought from IC partner B at 120.00 whose Legal-view cost in B is 100.00, when the Group view rolls, then the parent's Group-view material component uses 100.00 and the Legal view uses 120.00.

7.39.9 Test plan and fixtures

Fixtures: `rollup-50` (50 products, up to 6 levels, phantoms, scrap, setup, reference workbook); `rollup-50k` (generated 50,000 items, 8 levels, for performance); `variance-200` (ended orders with each variance type, reference workbook); `activation-demo` (demo plant on-hand with old/new standards).

Test	Type	Target	Pass criterion
Node cost formulas	Unit	roll-up	worked example exact
Split conservation	Property	roll-up	Σ split = unit cost × qty for random BOMs
Cycle detection	Unit	graph	every planted cycle reported
Mass roll time	Performance	run	50k items < 15 min on 4 workers
Activation revaluation	E2E	activation + journal	journal = preview; value report ties
Variance identity	Property	decomposition	LSV + QV + PV + SV = total for random orders

Reconciliation checks: inventory at standard = GL inventory per site; Σ variance journals per period = Σ VarianceEntry; activation revaluation journal = Σ preview rows.

7.39.10 Deployment and operations

Helm chart `costing-svc` (extends wave 20 chart, Postgres `costing`); env `COSTING_MAX_LEVELS=20`, `COSTING_ROLL_WORKERS=4`; flags `costing.whatif`, `costing.variance-new-standard-for-open-orders`; jobs: `variance-compute` on production order end events, `standard-cost-reconcile` nightly, `monthly-variance-report` on period close. SLOs: single calculation p95 < 500 ms for ≤ 6 levels; explain API p95 < 300 ms; mass roll as above. Alerts: roll error rate > 2%, variance identity failures > 0, reconciliation difference ≠ 0. Rollback: Helm; activation reversal is a new activation of the prior version (reversing journal), never a delete.

7.39.11 Migration from AX / SAP

Source	Passage entity	Notes
CostingVersion, InventItemPrice (active standard)	CostingVersion, CostCalculation (Active, no explanation)	active standards loaded as calculations with status Active and diagnostics `Migrated`
CostGroup, CostCategory, CostSheetNodeTable/CalculationBasis/Factor	CostGroup, CostCategory, CostingSheet	
BOMCalcTable/BOMCalcTrans	not migrated	re-rolled from migrated BOMs; comparison report vs legacy
InventCostTransVariance history	lakehouse archive	
MBEW (STPRS, PEINH), KEKO/KEPH released estimates	CostCalculation, CostComponentSplit	KEPH components mapped to CostComponent
OKTZ cost component structure, KZS2 sheets, KP26 rates	CostComponent, CostingSheet, CostCategory rates	
COSS/CO88 settled variance history	lakehouse archive	

Validation: re-rolled standard within 1% of legacy active standard for 95% of items (exceptions listed with explanation); Σ on-hand × migrated standard = migrated GL inventory. Not migrated: simulation versions, legacy calculation logs.

7.39.12 Dependencies, risks, open questions

Depends on: wave 9 BOM/route versions, wave 16 on-hand, wave 20 standard cost engine and inventory value report, wave 21 production order end and consumption, wave 24 co-product rules, wave 38 cost elements for absorption.

Risk	Likelihood	Impact	Mitigation
Legacy standards unreproducible	High	Medium	Load active standards as-is; re-roll comparison
Deep BOMs exceed memory in parallel roll	Low	Medium	Level batching; worker memory limit 2 GiB

Open questions: (1) Surcharge basis includes rolled subassemblies by default? Proposed no (AX behavior); decide by wave 39 start. (2) Variance for open orders at new standard? Proposed flag off; decide by wave 39. (3) Store split at 6 or 4 decimals? Proposed 6; decide by wave 39.

7.39.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`costing.versionType`	costing version	enum	Standard	Standard / Planned / Simulation / Conversion	allowed operations (FR-39.3)	Costing versions → Costing type (Standard cost / Planned cost)	OKKN costing variant + OKKI costing type
`costing.valuationVariant`	costing version	struct	price strategy: version → active → purchase price	ordered price sources per item type	price lookup order for purchased items	Costing versions → Fallback principle, "Use purchase price"	OKK4 valuation variant (material price strategy)
`costing.fallback`	costing version	enum	Active	None / Active / SpecifiedVersion	BR-39.4	Fallback principle	valuation variant strategy sequence
`costing.blockPending`	costing version	bool	false	true / false	blocks edits of pending prices (BR-39.2)	Costing version → Block pending prices	costing run marking allowance (CK24 "allow marking")
`costing.componentStructure`	tenant	table	DM, DL, IND-MAT, IND-LAB, SUB	≤ 20 components, element / cost group assignment	split layout (FR-39.1)	Cost groups (CostGroup) and cost group types	OKTZ cost component structure (TCKH1–3)
`costing.sheet`	site × version	tree	none	Basis / Surcharge / Rate / Total nodes	FR-39.2	Costing sheets (CostSheetNodeTable)	KZS2 costing sheet

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`costing.overheadRates`	site × version × period	table	none	% or rate per node	surcharge / rate values	Costing sheet → Calculation factors	KZZ2 overhead rates
`costing.includeRolledSubassemblies`	site	bool	false	true / false	surcharge basis scope (FR-39.2)	Costing sheet node "Include in calculation basis"	costing sheet base row definition
`costing.maxLevels`	tenant	int	20	1–40	BR / env `COSTING_MAX_LEVELS`	BOM calculation "Maximum level"	costing variant explosion limit
`costing.lotSizeSource`	item	enum	DefaultOrderSettings	DefaultOrderSettings / Override	BR-39.5 costing lot size	Item → Default order settings → Standard order quantity	MARC-LOSGR costing lot size
`costing.postingProfile`	site	table	per variance type × component	accounts	variance and revaluation accounts	Posting profiles (Inventory posting → Production variances)	OBYC PRD / UMB / GBB account determination
`costing.varianceCategories`	tenant	enum[]	LotSize, Quantity, Price, Substitution	as listed + Distributed (FR-38.18)	categories computed	Production variances (production control parameters)	OKV1 variance keys, variance variants (input price / quantity / resource-usage / remaining / lot size / output price)
`costing.newStandardForOpenOrders`	site	bool	false	true / false	FR-39.6 open-order behavior	none	KKS1 target cost version 0 vs 1
`costing.conversionBlockers`	legal entity	enum[]	NegativeOnHand, InTransit, StartedOrders	as listed	FR-39.12 validation	Standard cost conversion setup / checklist validations	MM02 price control change checks (open orders)
`costing.mixedCosting`	site	bool	false	true / false	FR-39.14 enabled	none	OMXL quantity structure types, procurement alternatives
`costing.mixingRatioSource`	site	enum	Manual	Manual / DeriveLast12Months	FR-39.14 ratio source	none	CK94 mixing ratios
`costing.cogsComponentPosting`	site	bool	false	true / false	FR-39.15	Cost breakdown posting on inventory model group	FINSC_COGS splitting profile
`costing.ppvSplit`	site	bool	false	true / false	FR-39.15 PPV	none	price-difference splitting (S/4)
`costing.orderEstimateAsStandard`	site	bool	false	true / false	BR-39.14	Item price for configured items	KKE1 / CK51N with sales-order controlling (strategy 20)
`costing.valuationViews`	legal entity	enum[]	Legal	Legal / Group / ProfitCenter	FR-39.17	none	OKKP valuation views, transfer price variants (8KEM)
`costing.globalInventoryAccounting`	legal entity	struct	off	ledgers ≤ 3 with conventions	FR-39.18	Global inventory accounting → Ledgers, Accounting conventions, Costing policies	material ledger valuation types / currency types (OMX2, OMX3)
`costing.rollWorkers`	tenant	int	4	1–32	parallelism (env `COSTING_ROLL_WORKERS`)	batch task count	CK40N parallel processing (server group)

7.39.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Cost calculation explain	drill-down	version, item, site, view	level, line, qty, unit cost, component amounts, sheet nodes	`cost_calculation`, `cost_component_split`	BOM calculation results (BOMCalcTrans)	CK13N, KEKO / KEPH display
Mass roll diagnostics	operational report	run, error code	item, site, code, message, reference	`cost_calculation.diagnostics`	Calculation infolog	CK40N log, CKAPP
Pending vs active standard	list inquiry	version, site, threshold %	item, active, pending, delta, delta %	`cost_calculation`	Item price (InventItemPrice) pending vs active	S_P99_41000111 cost estimate comparison, CK40N analysis
Activation revaluation preview	operational report	version, site, date	item, on-hand, old, new, revaluation	activation preview	Activate pending prices (no preview)	CK24 / MR21 (no preview)
Production variance by order	drill-down	site, period, item, type	order, standard, actual, LSV, QV, PV, SV, by component	`variance_entry`	Production variances (InventCostTransVariance)	KKS1 / KKBC_ORD
Purchase price variance	list inquiry	site, vendor, item, period	receipt, standard, actual, PPV, split	`variance_entry` type PurchasePrice	PPV inquiry (InventCostTrans)	MR11 GR/IR clearing log, FBL3N on the PRD price-difference account
Monthly variance report	KPI	site, period	variance by type / component, % of standard production cost	`variance_entry` aggregated	Production variance report (SSRS)	KKS1 summary log, CO-PC information system summarized order analysis
Inventory value at standard	statutory report	site, date, view, ledger	item, qty, standard, value, GL balance, difference	wave-20 value report + `cost_calculation`	Inventory value report	MB5L, MB52 with value
Standard cost conversion status	operational report	conversion run	step, status, user, timestamp, revaluation	`conversion_run`	Standard cost conversion checklist	MM02 change documents
Mixed cost composition	drill-down	item, site, version	alternative, ratio, cost, contribution, split	`procurement_alternative`	none	CK91N / CK94 display

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Sales order margin	list inquiry	order, customer, margin % band	line, price, estimate, margin, frozen	`cost_calculation` type SalesOrder	Configured item price	CK53N, KKE3
Valuation view comparison	drill-down	item, site, views	Legal, Group, ProfitCenter cost and component deltas	`cost_calculation` by view	none	CKM3 price analysis per valuation view

7.39.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Cost estimate sheet	explain export	item, version, quantity, levels, split, sheet nodes	PDF, XLSX	calculation id	supporting evidence for inventory valuation; 10 years
Activation revaluation report	activation	item, site, old / new standard, on-hand, revaluation, journal id	PDF, CSV	activation id	10 years (inventory valuation audit)
Standard cost conversion record	conversion activated	checklist steps with users and timestamps, revaluation	PDF	conversion run id	10 years
Monthly variance report	period close	variance by type / component / order	PDF, XLSX	none	management; 7 years
Sales order cost estimate	line confirmation	configuration summary, cost by component, margin	PDF (internal)	calculation id	7 years (pricing evidence)
Transfer-price evidence (cost basis)	period end	Group vs Legal view cost per IC item	CSV feeding FR-40.4 export	none	transfer-pricing documentation 10 years

7.39.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`mass-roll`	on demand; typical yearly + quarterly	(version, site, level)	per-level checkpoint; item results upserted	< 15 min for 50k items	per-item error, run continues	error rate > 2%
`variance-compute`	on production order end event	production order	(order, version)	< 2 s per order	retries 3×; order end blocked on identity failure	identity failures > 0
`standard-cost-reconcile`	nightly 02:45	site	(site, date)	< 5 min	report	difference ≠ 0
`monthly-variance-report`	period close	site	(site, period)	< 2 min	rerun	not generated by close day 3
`mixing-ratio-derive`	quarterly	site	(site, quarter)	< 3 min	keep previous ratios	ratio change > 20 pts
`valuation-view-roll`	after each Legal mass roll	(version, site, view)	per-level checkpoint	< 15 min	view marked stale	stale views at activation
`gia-ledger-settlement`	per inventory close	(entity, ledger)	(ledger, close date)	< 30 min per ledger for 1M transactions	per-ledger isolation	ledger close failed
`sales-order-estimate-refresh`	on configuration change event	sales line	(line, config version)	< 2 s	warning on line	estimate missing at confirmation

7.39.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
CostingView	GET /costing/*, explain, variances, compare	none	Cost accountant, Controller, Plant manager
CostingMaintain	versions, sheets, cost groups, components, alternatives, models	CostingActivate	Cost accountant
CostingCalculate	/calculations, /roll, /estimates, /sales-order-estimates	none	Cost accountant, sales service account
CostingActivate	/activate, /conversions	CostingMaintain (author of the pending prices cannot activate them)	Controller
CostingSystem	/variances/compute	n/a (service account only)	costing-svc worker
CostingConfigureLedgers	valuation views, inventory accounting ledgers, posting profiles	CostingActivate	Finance system admin

7.39.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
CST-0701	error	Cycle detected: `{path}`	cyclic BOM	fix BOM	BR-39.3
CST-0702	error	No price for `{item}` in version `{version}` and fallback None	missing price	add price or set fallback	BR-39.4
CST-0703	warning	Fallback price used for `{item}` from `{source}`	fallback	review	BR-39.4
CST-0704	error	BOM depth `{n}` exceeds limit `{max}`	> 20 levels	restructure or raise limit	FR-39.4
CST-0705	error	Route `{route}` inactive for `{site}` at `{date}`	invalid route	approve route version	FR-39.5
CST-0710	error	Version `{version}` is being rolled; activation blocked	advisory lock	wait	BR-39.7
CST-0711	error	Simulation versions cannot be activated	BR-39.2	copy to Standard	BR-39.2
CST-0712	info	Activation will revalue inventory by `{amount}`	preview	confirm amount	FR-39.6
CST-0720	error	Variance identity failed for order `{order}`: residual `{r}`	rounding / data defect	order end blocked; contact support	BR-39.9
CST-0721	warning	Order `{order}` reopened; variances reversed	BR-39.10	re-end order	BR-39.10
CST-0731	error	Inventory close through `{date}` required before conversion	BR-39.11	run inventory close	BR-39.11
CST-0732	error	`{n}` items have negative on-hand; conversion blocked	conversion blocker	correct on-hand	FR-39.12
CST-0741	error	Mixing ratios for `{item}` total `{pct}`	≠ 100.00%	correct ratios	BR-39.12
CST-0751	warning	Sales order estimate superseded by configuration change	BR-39.14	none	BR-39.14
CST-0761	warning	Group cost unknown for IC receipt `{receipt}`; IC price used	BR-39.15	roll supplier Legal view	BR-39.15

Code	Severity	User text	Cause	Remediation	Raised by
CST-0771	error	Inventory accounting ledger `{ledger}` has no costing policy for item group `{group}`	GIA setup gap	add policy	FR-39.18

7.39.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Single calculation	2,000/day interactive	150 / 500 / 1,000 ms (≤ 6 levels)	20/s	≈ 1 KB per calculation + split	unique partial (version, item, site, view) where Active	p95 < 1.5 s
Mass roll	50k items × 2 views, 4×/year	n/a	50k items < 15 min per view	400k calculations × 3 KB ≈ 1.2 GB	level batches; worker memory 2 GiB	workers scale to 8; < 30 min
Explain API	500/day	80 / 300 / 600 ms	10/s	n/a	lazy load per level	page levels
Variance compute	3,000 orders ended/day; peak 300/h	400 ms / 2 s / 4 s per order	5 orders/s	1M variance entries ≈ 0.6 GB	VarianceEntry by (site_id, period, type)	queue; order end confirmed async
Sales order estimate	5,000 configured lines/day	400 ms / 2 s / 4 s	10/s	5,000 × 365 × 3 KB ≈ 5.5 GB	by (sales_line_id)	estimates queue; confirmation waits ≤ 10 s
COGS component posting	200k issues/day	adds ≤ 10 ms per issue	300 issues/s	4× COGS lines in journal ≈ +40 GB	journal partitions (wave 11)	posting batched
GIA ledger settlement	1M inventory transactions/month × 3 ledgers	n/a	< 30 min per ledger	3× settlement rows ≈ 10 GB	per-ledger partitions	ledgers run sequentially

7.39.20 Localization and compliance hooks

- **Inventory valuation rules** differ: IFRS (IAS 2) prohibits LIFO; US GAAP allows LIFO (LIFO conformity rule ties tax and book); DE HGB allows Festwert and group valuation; FR PCG allows CUMP / FIFO. Global inventory accounting ledgers (FR-39.18) let one entity keep an IFRS FIFO ledger and a US LIFO tax ledger; country packs (wave 48) seed accounting conventions.
- **Standard cost capitalization**: IAS 2.13 requires fixed production overhead absorption at normal capacity; the costing sheet rate nodes carry `normalCapacityHours` and the variance report shows under-absorption that must be expensed; US §263A (UNICAP) additional costs are a tax-ledger costing sheet variant supplied by the US pack.
- **Transfer pricing**: Group and ProfitCenter views feed OECD Guidelines evidence (cost-plus margins) exported in FR-40.4; local file requirements (DE §90 AO, MX, IN Form 3CEB) consume it through wave 48.
- **Retention**: cost estimates and activation revaluation evidence 10 years (DE HGB §257, FR), 7 years (US).
- **Rounding**: 6-decimal unit costs, posting in currency minor unit; JPY and other zero-decimal currencies round to whole units at posting.

7.39.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.8.3 Standard cost → Revaluation on activation / Conversion to standard cost (Standard cost conversion checklist)	PARTIAL-39	ADDED as FR-39.12	revaluation on activation already FR-39.6
AX 12.8.4 Standard cost conversion setup	PARTIAL-39	ADDED as FR-39.12	`costing.conversionBlockers`
AX 12.8.1 Global inventory accounting (multiple cost ledgers / accounting conventions)	GAP (proposed 20)	ADDED as FR-39.18	costing-svc extends the wave-20 engine; flag-gated
AX 12.8.4 Global inventory accounting setup (ledgers, accounting policies, conventions)	GAP (proposed 20)	ADDED as FR-39.18	`costing.globalInventoryAccounting`
SAP 13.8 CKECP / CKECP1 Easy cost planning / ad-hoc cost estimate	PARTIAL-39	ADDED as FR-39.13	project use in wave 41 consumes the estimate
SAP 13.8 CK51N / CK53N Sales order cost estimate	PARTIAL-27 (note proposes 39)	ADDED as FR-39.16	
SAP 13.8 CK91N / CK92 Procurement alternatives / mixing ratios	PARTIAL-39	ADDED as FR-39.14	
SAP 13.8 Mixed costing customizing: quantity structure types, procurement alternatives (OMXL)	PARTIAL-39	ADDED as FR-39.14	`costing.mixedCosting`, `costing.mixingRatioSource`
SAP 13.8 Easy cost planning customizing: costing models, characteristics (CKF*)	PARTIAL-39	ADDED as FR-39.13	CostingModel entity
SAP 13.8 COGS split by cost component (FINSC_COGS*)	PARTIAL-39	ADDED as FR-39.15	
SAP 13.8 Price-difference splitting (S/4)	PARTIAL-39	ADDED as FR-39.15	

7.40 Wave 40: Intercompany & Financial Consolidation

7.40.1 Scope and business context

Wave 40 closes phase 7 and delivers finance parity. Groups with several legal entities trade with each other, share costs and must report as one. Before this wave each entity is an island: an intercompany sale is keyed twice, due-to/due-from balances drift, and consolidated statements are assembled in spreadsheets. After it, pairing a customer in entity A with a vendor in entity B makes every sales order, packing slip, invoice and payment create its mirror in the partner automatically (including direct-delivery chains); intercompany journal lines post due-to/due-from entries in both entities; a reconciliation workspace matches both sides and flags mismatches; and the consolidation service builds group structures with ownership percentages, collects balances per unit, translates under IAS 21, eliminates intercompany balances, transactions, unrealized profit in inventory and investments, computes non-controlling interest, and produces consolidated statements through the wave-15 reporting designer with drill-down to the entity journal.

Out of scope: equity-method accounting for associates (v1.x), complex multi-tier ownership changes mid-year (step acquisitions), statutory group tax, transfer-pricing calculation itself (hooks only: documented IC prices and margin evidence export).

Scope amendment (gap closure). The registers (coverage-ax §12.99 rows 11, 73, 82, 97, 103, 126, 135, 190, 192, 193; coverage-sap §13.99 rows 87–116 mapped to wave 40; plus the regrouping rows proposed for wave 14) add centralized payments (FR-40.12), intercompany charges for expenses, time and vendor-invoice lines (FR-40.13), intercompany on-hand visibility (FR-40.14), intercompany planning groups (FR-40.15), group valuation for unrealized profit (FR-40.16), reclassification and regrouping rules (FR-40.17), consolidation of investments including the equity method and divestitures (FR-40.18), top-side group journals (FR-40.19) and the consolidation monitor with validation rules (FR-40.20). This supersedes the out-of-scope note above for equity-method accounting; step acquisitions (remeasuring a previously held interest when control is obtained) and proportional consolidation remain out of scope, with dispositions in 7.40.21.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
D365 intercompany: the mirror PO/SO is created by a synchronous X++ call; if the partner entity's number sequence or dimension validation fails, the user sees an infolog and the pair is left half-created (SalesTable without InterCompanyPurchId)	No saga; no compensation	Temporal saga with compensating cancel; both documents or neither	Orphan IC documents per month 5–20 → 0
SAP cross-company FB01 posts via OBYA clearing accounts; FBICR3 (ICR) reconciles weeks later by downloading both ledgers	Reconciliation is a separate batch extract	Every IC posting carries a pair id; reconciliation is continuous per posting	Mismatch detection latency 30 days → < 1 min
Intercompany billing in SAP SD (RD04 output → INVOIC IDoc → FB01 vendor invoice) fails silently in WE02 when a tax code is missing; AP sees nothing	Message-based integration inside one system	Mirror vendor invoice created in the same saga with validation before commit	IDoc-style failures to triage per month 10–30 → 0
AX consolidation (LedgerConsolidate online) copies balances into a consolidation company; eliminations (LedgerEliminationRule) are posted journals with no drill to source entity; reruns duplicate unless reversed manually	Copy-based consolidation into a ledger	Consolidation run is a derived, versioned dataset with elimination entries linked to source balances; rerun supersedes	Rerun effort 2 h → < 2 min; drill to entity 100%
Currency translation differences computed in Excel because AX "Consolidate online" translates all accounts at one rate type unless per-account overrides are maintained	Rate type per account instead of per statement category	IAS 21 methods by account category with CTA derived automatically	Manual CTA journals per period 1–3 → 0

7.40.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Trading partner setup	InterCompanyTradingPartner, InterCompanyEndpoint, InterCompanyEndpointActionPolicy, InterCompanyTradingValueMap, CustTable/VendTable intercompany flags	Customer/vendor with company code assignment, VBAK-VKORG IC sales org, OBYA clearing	IntercompanyPartner (customer in A ↔ vendor in B) with action policies and value maps	full
Mirror sales/purchase orders	SalesTable.InterCompanyOriginalPurchId, PurchTable.InterCompanyOriginalSalesId, InterCompanySalesOrder/PurchaseOrder classes, chains	Cross-company stock transport order (ME21N NB) with delivery and IC billing (VF01 IV)	Mirror document saga	full
Direct delivery chains	InterCompany direct delivery, SalesLine.InterCompanyInventDim, chain of orders	Third-party order processing, cross-company drop-ship	DirectDeliveryChain entity (3 orders linked)	full
Intercompany invoices	InterCompany invoice posting (sales invoice → vendor invoice), InterCompanyPurchInvoice	SD IV intercompany billing, RD04 → INVOIC → FB01	Mirror invoice creation in same saga	improved
Intercompany accounting (journals)	LedgerJournalTrans.OffsetCompany, LedgerIntercompany (due-to/due-from accounts), intercompany journal	FB01 cross-company (BVORG), OBYA, FAGL cross-company postings	IcJournal with due-to/due-from derivation	full
Intercompany reconciliation	Intercompany reconciliation report (D365 10.0.x)	ICR FBICR3 / FBIC010 (GL open items), FBICS3	IcReconciliationItem continuous matching	improved
Group structure and ownership	Consolidation legal entity, LedgerConsolidate, LedgerConsolidateLegalEntity (share %)	Consolidation units/groups (CXCUNIT), ownership in COI, ECCS CX5x	ConsolidationGroup, ConsolidationUnit, Ownership	full
Eliminations	LedgerEliminationRule, LedgerEliminationRuleLine, elimination journal (LedgerJournalType Elimination), ConsolidationAccount	Group Reporting reclassification/elimination tasks (ACDOCU document types 2x), IC profit in inventory, COI	EliminationRule engine, ConsolidationEntry	full
Currency translation	Consolidate online rate types per account, LedgerConsolidateExchangeRate	Translation methods (closing, average, historical), CTA	IAS 21 per account category, CTA derived	full
Consolidated statements	Management Reporter consolidated reporting tree	Group Reporting reports, ACDOCU-based analytics	Wave 15 designer over consolidated dataset with drill	full
Centralized payments	Centralized payments (Organization administration → Centralized payments, VendPaymentJournal paying for other legal entities), LedgerInterCompany	FBZP paying company code / sending company code, F110 cross-company payment	CentralizedPaymentSetup with due-to / due-from settlement (FR-40.12)	full
Intercompany charges (expenses, timesheets)	TrvExpTrans intercompany legal entity, ProjIntercompany parameters (lending / borrowing), intercompany timesheets and vendor invoices (R3)	CO cross-company allocations, PS intercompany billing (DP90 with cross-company resource-related billing)	IcChargeRequest contract (FR-40.13)	full
Intercompany on-hand	Intercompany on-hand (R3 Inventory → On-hand across legal entities)	MMBE across plants of different company codes	IC on-hand read model (FR-40.14)	full
Intercompany master planning	Intercompany planning groups (ReqIntercompanyPlanningGroup), intercompany master scheduling	MRP with special procurement key (cross-company stock transfer), MD01 sequence by plant	IcPlanningGroup run sequence (FR-40.15)	full
Group valuation of IC profit	none (profit in inventory computed manually)	Group / profit-center valuation views (OKKP, TKA01), transfer prices 8KEM / 8KEN	Group view cost per receipt layer (FR-40.16, with FR-39.17)	partial (no ML actual group costing)
Reclassification and regrouping	Financial reports reclass rows; no posting rule	FAGLF101 / FAGL_CL_REGROUP regrouping, Group Reporting reclassification methods and runs	ReclassificationRule engine (FR-40.17)	full
Consolidation of investments	LedgerConsolidate share % only	CoI activities (first consolidation, subsequent, increase / decrease, divestiture), equity method (ECCS CoI, Group Reporting "Manage Investments / Equity data")	ConsolidationOfInvestments activities (FR-40.18)	partial (no step acquisition)
Top-side group journals	Consolidation company journals (LedgerJournal in consolidation company)	Group Reporting "Post Group Journal Entries" (document types 0C / 1x / 2x), ECCS manual postings	GroupJournal per run with document types (FR-40.19)	full
Consolidation monitor and validation	Consolidate online process log	CXCD data monitor, consolidation monitor tasks, validation rules (Group Reporting "Define Validation Rules")	ConsolidationMonitor tasks and ValidationRule (FR-40.20)	full

7.40.3 Functional requirements

- **FR-40.1 Trading partners** — Must. The system shall pair a customer in entity A with a vendor in entity B (and vice versa) with action policies (auto-create, auto-confirm, auto-post packing slip, auto-post invoice), value maps (item numbers, units, dimensions, charges) and an IC pricing policy (agreed price list, cost plus %, transfer price table).
  - Notes: one active pairing per entity pair and direction; policies versioned; mirror creation requires both entities on the same tenant.
- **FR-40.2 Mirror-document matrix** — Must. The system shall create mirror documents per the matrix below in a Temporal saga with compensation.
  - Notes: matrix (source in selling/holding entity → mirror in buying/receiving entity): sales order → purchase order; sales order change (qty/date/price) → PO change with re-approval if beyond policy tolerance; packing slip → product receipt; sales invoice → vendor invoice (matched 3-way to the mirror PO and receipt); customer payment received → vendor payment settled (optional, when netting is not used); sales return (RMA) → purchase return; free-text invoice → vendor invoice journal; IC journal line with offset entity → due-to/due-from entries both sides; fixed asset IC transfer (wave 37) → disposal + acquisition; reverse direction (PO created in buyer first) → sales order in seller. Saga steps: validate partner and policy → reserve number sequences in both entities → create source → create mirror → link (`IcDocumentLink`) → publish; on any failure compensate by canceling created documents.
- **FR-40.3 Direct delivery chains** — Must. The system shall support customer order in A, IC PO from A to B, IC SO in B shipping directly to A's customer, with B's packing slip posting A's product receipt and A's customer packing slip in one saga.
- **FR-40.4 Intercompany pricing and transfer-pricing hooks** — Should. The system shall price IC lines from the policy, record the pricing basis on the link, and export a transfer-pricing evidence file (item, qty, price, basis, cost, margin) per period.
- **FR-40.5 Intercompany accounting** — Must. The system shall derive due-to/due-from accounts per entity pair from the IC accounting setup and post balanced entries in both entities for IC journal lines, allocations (wave 14/38) and expense re-charges, each carrying a shared `ic_pair_id`.
  - Notes: both entries post atomically through the journal API per entity; if the second post fails the first is reversed; amounts in the counterparty entity convert at the document rate with FX difference booked to IC FX accounts.
- **FR-40.6 Intercompany reconciliation** — Must. The system shall match IC balances and transactions per entity pair, currency and period, classifying each as Matched (difference ≤ 0.01 or FX tolerance), Timing (counterpart exists in the next period), or Mismatch, with a workspace to resolve.
  - Notes: FX tolerance default 0.5% of amount; period-end summary required to be free of Mismatch before consolidation unless overridden by `ConsolidationApprove`.
- **FR-40.7 Group structure** — Must. The system shall define consolidation groups and units (legal entities or sub-groups) with effective-dated ownership %, consolidation method (full / proportional — v1 full only), group currency and chart mapping (entity main account → consolidation account).
- **FR-40.8 Data collection and translation** — Must. The system shall collect trial balances per unit and period from the wave-11 balance projection, translate under IAS 21: assets/liabilities at closing rate, income/expense at average rate (or transaction-date rates by option), equity at historical rates, and derive CTA = translated closing net assets − (translated opening net assets + translated net income); CTA posts to the OCI consolidation account.
- **FR-40.9 Eliminations** — Must. The system shall apply elimination rules in order: (E1) IC revenue/COGS and other P&L; (E2) IC receivables/payables and loans with translation differences to CTA; (E3) unrealized profit in inventory (upstream shared with NCI, downstream parent only); (E4) investment in subsidiary against equity at acquisition with goodwill; (E5) NCI share of current-year profit and equity; (E6) dividends within group. Each produces ConsolidationEntry lines linked to the source balances.
  - Notes: worked 2-entity example (group currency USD; P owns 80% of S, S in EUR; closing rate 1.10, average 1.08, historical at acquisition 1.05). S sold goods to P for EUR 100,000 (cost EUR 70,000); 40% remain in P's inventory; S's receivable EUR 100,000 open; P's payable recorded at USD 108,500. E1: Dr Revenue 108,000.00 / Cr COGS 108,000.00 (100,000 × 1.08). E2: Dr AP 108,500.00, Dr CTA 1,500.00 / Cr AR 110,000.00 (100,000 × 1.10). E3: unrealized profit = 30,000 × 40% = EUR 12,000 × 1.08 = 12,960.00: Dr COGS 12,960.00 / Cr Inventory 12,960.00; upstream so NCI shares 20%: Dr NCI (equity) 2,592.00 / Cr NCI share of profit 2,592.00. E4: S share capital EUR 500,000 and pre-acquisition retained earnings EUR 200,000 at 1.05 = 735,000.00; P investment USD 620,000.00; Dr Share capital 525,000.00, Dr Pre-acquisition RE 210,000.00, Dr Goodwill 32,000.00 / Cr Investment in S 620,000.00, Cr NCI 147,000.00 (20% × 735,000). E5: S net income EUR 150,000 × 1.08 = 162,000.00; Dr Profit attributable to NCI 32,400.00 / Cr NCI (equity) 32,400.00. Each entry balances; the run reports Σ debits = Σ credits = 1,032,952.00 across eliminations E1–E5.
- **FR-40.10 Consolidation run** — Must. The system shall execute collection → translation → eliminations → NCI → validation as one versioned run per group/period with partial recompute for changed units, producing a consolidated trial balance and statements.
  - Notes: run of 50 units × 10,000 accounts in < 5 min; validation: consolidated balance sheet balances; Σ eliminations per rule balanced; no unresolved Mismatch items.
- **FR-40.11 Consolidated reporting and drill** — Must. The system shall expose the consolidated dataset to the wave-15 designer with drill from a consolidated line to unit balances, elimination entries and entity journal lines.
- **FR-40.12 Centralized payments** — Must. The system shall let a paying entity settle vendor invoices of other entities (pay on behalf of) and receive customer payments for other entities, posting due-to / due-from in both entities and settling the open item in the owning entity.
  - Notes: setup per (paying entity, owning entity, payment method) with the due-to / due-from accounts from the BR-40.4 matrix; worked example: P pays S's vendor invoice EUR 10,000.00 → P: Dr Due from S 10,000.00 / Cr Bank 10,000.00; S: Dr Vendor 10,000.00 / Cr Due to P 10,000.00, open item settled in S; cash discount belongs to the owning entity; FX difference in the owning entity when currencies differ; customer receipts work in reverse; a payment proposal in the paying entity (wave 19) selects open items across the owning entities it is set up for; both postings carry one `ic_pair_id` and run in the mirror saga (FR-40.2 compensation rules).
- **FR-40.13 Intercompany charges** — Must. The system shall accept intercompany charge requests from expense lines (wave 43), timesheet and project transactions (wave 41) and vendor invoice lines (wave 18) whose cost belongs to another entity, price them by the IC charge pricing policy, and post the cost in the borrowing entity with a due-to and the recovery in the lending entity with a due-from or an IC invoice.
  - Notes: `IcChargeRequest` fields: source document, lending entity, borrowing entity, category (Expense / Hour / Item / Fee), quantity, cost amount, dimensions in the borrowing entity; pricing: cost, cost-plus % per category, or transfer price table per resource role and category (AX intercompany transfer prices); settlement mode Journal (due-to / due-from) or Invoice (IC customer invoice in lender → vendor invoice in borrower through FR-40.2); tax on cross-border recharges from wave 13; the consuming waves (41, 43) call the same API, so no IC logic lives in those services.
- **FR-40.14 Intercompany on-hand** — Should. The system shall show on-hand and available quantities of trading partners' sites for items mapped by the partner value map when entering IC orders, without exposing cost.
  - Notes: read model built from wave-16 on-hand events of partner entities in the same tenant, filtered by the IC partner's allowed sites; refresh lag p95 < 60 s; visibility requires the partner policy `share-on-hand = true` in both directions.
- **FR-40.15 Intercompany planning groups** — Should. The system shall define planning groups of entities with a run sequence so that planned IC purchase orders created by MRP in a buying entity become IC demand in the supplying entity before that entity is planned, and firming creates the mirror documents through FR-40.2.
  - Notes: sequence derived from the IC partner graph (buyers before suppliers) with cycle detection; each entity is planned by the wave-22 engine in sequence inside one Temporal workflow; IC demand carries the requested date minus the IC transport lead time; replanning a buyer after its supplier marks the supplier plan stale (IC-0815).
- **FR-40.16 Group valuation for unrealized profit** — Must. The system shall compute unrealized profit in inventory (E3) per IC receipt layer as remaining quantity × (Legal-view unit cost in the holder − Group-view unit cost from the supplier, FR-39.17), and fall back to the seller's average margin (BR-40.7) when a layer carries `groupCostUnknown`.

    - Notes: layers follow the holder's costing method (FIFO for remaining layers by default); supplier group cost converts at the receipt-date rate and the elimination translates at the average rate of the period; the elimination detail lists every layer for drill.
- **FR-40.17 Reclassification and regrouping rules** — Must. The system shall run rule-based reclassifications at entity level (posting through the journal with automatic reversal on the first day of the next period) and at group level (consolidation entries), covering sign-based moves, remaining-term splits and two-sided intercompany elimination methods with difference accounts and thresholds.
    - Notes: entity-level rule types: DebitBalanceVendors → other receivables, CreditBalanceCustomers → other payables, RemainingTerm (receivables / payables / loans due > 12 months → non-current, per open item due date), invoked as a wave-14 close checklist task (SAP FAGLF101 / FAGL_CL_REGROUP equivalent); group-level types: SignBased (for example negative consolidated cash → bank overdrafts), FS item remap, and two-sided IC elimination with difference posted to an IC difference account when |difference| ≤ threshold (default 1,000.00 group currency) and blocking the run above it; rules are ordered, versioned and dry-runnable.
- **FR-40.18 Consolidation of investments and equity method** — Should. The system shall record investments and equity holdings per investor and investee with effective-dated ownership and cost, and run CoI activities: first consolidation (goodwill per E4), subsequent consolidation, ownership increase or decrease without change of control (equity transaction, no goodwill change), full divestiture with deconsolidation gain or loss, and the equity method for associates and joint ventures.
    - Notes: divestiture gain = proceeds – (group carrying amount of net assets + goodwill – NCI) + CTA recycled from OCI; equity method: share of profit = ownership % × associate net income, dividends received reduce the investment, worked example: 30% associate, investment USD 300,000.00, associate net income 200,000.00, dividends paid to the investor 20,000.00 → Dr Investment in associate 60,000.00 / Cr Share of profit of associates 60,000.00; Dr Dividend income 20,000.00 / Cr Investment in associate 20,000.00; closing carrying amount 340,000.00; associates are not line-consolidated and need only net income, equity and dividends, collected through FR-40.8 or imported (`/consol/imports`).
- **FR-40.19 Top-side group journals** — Must. The system shall let consolidation accountants post manual group journals per run and document type (Adjustment, Reclassification, Elimination, CoI) at group or unit level, with optional carry-forward into later periods and approval.
    - Notes: document types decide whether entries are period-only or cumulative (carry forward in the same fiscal year) and whether they need approval (default: amount ≥ 100,000.00 group currency); top-side entries survive reruns (they attach to the group and period, not to a run version) and are re-applied to each new version; every entry needs a reason text and attachment.
- **FR-40.20 Consolidation monitor and validation rules** — Must. The system shall present a monitor per group and period with tasks per unit (collect, release, validate, translate) and per group (IC eliminations, profit in inventory, CoI, reclassification, top-side, group validation), enforce task dependencies, lock released unit data, and evaluate validation rules at unit and group level.
    - Notes: task status NotStarted / Running / Completed / CompletedWithWarnings / Error / Blocked; unit data release locks the unit's collected balances for the period (re-collection needs `ConsolidationApprove`); validation rules are expressions over FS items (for example total assets = total liabilities + equity; IC partner present on IC FS items; change in retained earnings = net income – dividends ± tolerance 1.00) with severity Error (blocks) or Warning; the monitor feeds the wave-14 close checklist.

7.40.4 Data model

IcDocumentLink

Field	Type	Required	Notes
id	uuid	yes	PK
ic_pair_id	uuid	yes	shared by all documents in a chain
partner_id	uuid	yes	FK IntercompanyPartner
source_entity_id, source_doc_type, source_doc_id	uuid, enum, uuid	yes	
mirror_entity_id, mirror_doc_type, mirror_doc_id	uuid, enum, uuid	yes	
chain_id	uuid	no	direct delivery chain
saga_id	text	yes	Temporal workflow id
status	enum	yes	Creating/Linked/Compensated/Failed
pricing_basis	jsonb	no	policy, price list, markup

IcReconciliationItem: id, entity_a_id, entity_b_id, period_key, currency, ic_pair_id, amount_a, amount_b, difference, fx_tolerance_applied, status (Matched/Timing/Mismatch/Resolved), resolution_note, resolved_by.

ConsolidationUnit: id, group_id, legal_entity_id (or child_group_id), functional_currency, ownership[] ({from, to, percent numeric(7,4)}), method (Full), acquisition_date, historical_rate numeric(19,8), chart_mapping_id.

ConsolidationRun

Field	Type	Required	Notes
id	uuid	yes	PK
group_id	uuid	yes	
period_key	text	yes	
version	int	yes	increments per rerun
status	enum	yes	Collecting/Translating/Eliminating/Validating/Completed/Failed/Superseded
rates	jsonb	yes	closing, average, historical by currency
cta_amount	numeric(19,4)	no	
validation	jsonb	no	checks and results
started_at, completed_at	timestamptz	yes/no	

ConsolidationEntry: id, run_id, rule_code (E1–E6/Translation/Collection), unit_id, consolidation_account_id, debit, credit, group_currency, source_refs jsonb ({unitId, accountId, amount, rate}), nci_share numeric(19,4).

Entities added by FR-40.12–FR-40.20: **CentralizedPaymentSetup** (paying_entity_id, owning_entity_id, payment_method, direction Pay/Receive, due_to_account, due_from_account, effective_from); **IcChargeRequest** (id, source_type Expense/Timesheet/ProjectTransaction/VendorInvoiceLine, source_id, lending_entity_id, borrowing_entity_id, category, quantity, cost_amount, price_amount, pricing_basis jsonb, settlement_mode Journal/Invoice, ic_pair_id, status Received/Priced/Posted/Failed); **IcTransferPrice** (lending_entity_id, borrowing_entity_id, category, resource_role, method Cost/CostPlus/Table, markup_pct, price, currency, effective_from); **IcPlanningGroup** (id, members [{entityId, sequence}], lead_time_days per pair); **ReclassificationRule** (id, level Entity/Group, type, selection jsonb, target_account_or_fs_item, threshold, order, version, auto_reverse bool); **Investment** (investor_unit_id, investee_unit_id, method Full/Equity,

ownership[] {from, percent}, cost, acquisition_date, goodwill, fair_value_adjustments jsonb); **GroupJournal** (id, group_id, period_key, document_type, unit_id, lines [{fsItem, debit, credit, partnerUnit}], carry_forward bool, approval_status, reason); **ConsolidationTask** (group_id, period_key, unit_id, task_type, status, started_at, completed_at, messages jsonb); **ValidationRule** (id, level, expression, severity, tolerance, version).

Invariants: every IcDocumentLink with status Linked has both documents existing and amounts equal at document rate; Σ ConsolidationEntry debit = Σ credit per run and per rule_code; ownership percentages per unit never exceed 100.00% at any date; a Completed run is immutable (rerun creates a new version and marks the old Superseded). Indexes: IcDocumentLink by ic_pair_id, by (mirror_entity_id, mirror_doc_id); ConsolidationEntry by (run_id, consolidation_account_id); IcReconciliationItem by (entity_a_id, entity_b_id, period_key, status).

7.40.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/interco/partners	Create pairing and policy	Idempotency-Key	IntercompanyConfigure
POST	/api/v1/interco/mirrors	Start mirror saga for a source document (system)	sourceDocId+version	IntercompanySystem
GET	/api/v1/interco/links?pairId=	Chain view	n/a	IntercompanyView
POST	/api/v1/interco/journals	IC journal with offset entity	Idempotency-Key	IntercompanyPost
POST	/api/v1/interco/reconciliations/run?period=	Run matching	(period, version)	IntercompanyReconcile
POST	/api/v1/interco/reconciliations/items/{id}/resolve	Resolve	state check	IntercompanyReconcile
POST	/api/v1/consol/groups	Create group/units/ownership	Idempotency-Key	ConsolidationConfigure
POST	/api/v1/consol/runs	Start run (group, period)	(group, period) new version	ConsolidationRun
GET	/api/v1/consol/runs/{id}/trial-balance	Consolidated TB	n/a	ConsolidationView
GET	/api/v1/consol/runs/{id}/entries?rule=	Elimination entries with source refs	n/a	ConsolidationView
POST	/api/v1/consol/imports	External unit TB import	file hash	ConsolidationConfigure
POST	/api/v1/interco/centralized-payments/setup	Configure pay / receive on behalf	Idempotency-Key	IntercompanyConfigure
POST	/api/v1/interco/charges	Submit IC charge request (system)	sourceId+version	IntercompanySystem
GET	/api/v1/interco/onhand?partner=&item=	Partner on-hand	n/a	IntercompanyView
POST	/api/v1/interco/planning-groups/{id}/run	Sequenced IC planning run	runId	IntercompanyPlan
POST	/api/v1/consol/reclassification-runs	Entity- or group-level reclass (dryRun option)	(scope, period, version)	ConsolidationRun
POST	/api/v1/consol/group-journals	Post top-side journal	Idempotency-Key	ConsolidationJournal
POST	/api/v1/consol/investments	Record investment / ownership change	Idempotency-Key	ConsolidationConfigure
GET	/api/v1/consol/monitor?group=&period=	Monitor tasks and statuses	n/a	ConsolidationView
POST	/api/v1/consol/units/{id}/release?period=	Release and lock unit data	state check	ConsolidationUnitRelease

Validation: mirror saga rejects when the partner policy is inactive or the mirror entity's period is closed; consolidation run rejects unresolved Mismatch items unless override; rates must exist for every unit currency.

Event	Trigger	Payload fields	Consumers
`dev.passage.interco.mirror.created.v1`	saga linked	icPairId, sourceDocId, sourceEntity, mirrorDocId, mirrorEntity, type	sales, procurement, analytics
`dev.passage.interco.mirror.compensated.v1`	saga failed	icPairId, failedStep, reason	workflow, alerts
`dev.passage.interco.journal.posted.v1`	IC journal both sides posted	icPairId, entityA, entityB, amount, currency, journalIds[]	reconciliation, consol-svc
`dev.passage.interco.reconciliation.flagged.v1`	mismatch	itemId, entities, difference	workspace, workflow
`dev.passage.consol.run.completed.v1`	run completed	runId, groupId, period, version, unitCount, eliminationCount, ctaAmount	reporting, analytics
`dev.passage.interco.charge.posted.v1`	IC charge posted in both entities	chargeId, icPairId, lendingEntity, borrowingEntity, amount, settlementMode	wave 41 projects, wave 43 expenses, reconciliation
`dev.passage.interco.planning.completed.v1`	planning group run done	runId, groupId, entitiesPlanned[], icDemandCount	master planning (22), workspace
`dev.passage.consol.unit.released.v1`	unit data released	groupId, unitId, period, balancesHash	consol-svc monitor
`dev.passage.consol.validation.failed.v1`	validation rule error	runId, ruleId, unitId, difference	workspace, close checklist (14)

7.40.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Intercompany workspace	Pending sagas, failed mirrors, mismatches, netting link	pair, source, mirror, status, age	IntercompanyView
Partner setup	Pair, policies, value maps, pricing	entity A/B, customer/vendor, actions, tolerances	IntercompanyConfigure
Chain view	Open any document in chain, retry/compensate	SO, PO, receipts, invoices, payments with status	IntercompanyView
IC reconciliation	Run, filter status, resolve, drill both sides	pair, currency, amount A, amount B, difference, status	IntercompanyReconcile
Consolidation workspace	Define group, start run, view validation, open statements	group, period, version, status, CTA, checks	ConsolidationRun
Elimination entries	Filter by rule/unit, drill to source	rule, unit, account, debit, credit, NCI share	ConsolidationView

Interaction rules: chain view renders as a graph with keyboard navigation between nodes; reconciliation grid pairs rows side by side with difference highlighted; consolidation run progress streams per step; statements open in the wave-15 viewer with a "consolidated" badge and drill menu.

7.40.7 Business rules and edge cases

- **BR-40.1** Mirror saga state machine: Creating → Linked; Creating → Compensated (all created documents canceled) → Failed when compensation itself fails (alert, manual); Linked documents cannot be deleted, only canceled through the chain.
- **BR-40.2** Source and mirror amounts are equal in document currency; each entity's accounting currency conversion uses its own rate at the document date; differences surface in reconciliation as FX-tolerated.
- **BR-40.3** A change to a linked sales order beyond policy tolerance (default qty ±0, price ±0.5%, date ±2 days) puts the mirror PO into re-approval (wave 17 workflow); within tolerance it updates silently.

- **BR-40.4** Due-to/due-from derivation: entity A line with offset entity B posts A: Dr/Cr account, Cr/Dr "Due from B"; B: Dr/Cr "Due to A", Cr/Dr the offset account; accounts resolved from the IC accounting matrix (A,B); missing matrix entry fails the post.
- **BR-40.5** Reconciliation Timing status requires the counterpart within the next period; after that it becomes Mismatch.
- **BR-40.6** Translation: historical rate per equity movement date; CTA computed per unit, then NCI share of CTA (ownership %) reclassified to NCI.
- **BR-40.7** Unrealized profit in inventory uses the seller's margin on the remaining quantity (FIFO assumption on IC receipts) and the average rate of the sale period.
- **BR-40.8** Ownership changes mid-period apply from the effective date; income is split pro rata by days unless a measured sub-period balance is provided.
- **BR-40.9** Consolidation rounding: entries computed at 6 decimals, rounded to the group currency minor unit per line; a per-rule rounding line keeps each rule balanced.
- **BR-40.10** Rerun with no changed units and same rates returns the existing version (idempotent); any change creates version +1 and supersedes.
- **BR-40.11** Centralized payment: the owning entity's open item is settled only when the paying entity's payment posts; a failed bank payment (wave 19 return) reverses both entities' postings in one saga.
- **BR-40.12** IC charge pricing resolves the transfer price effective on the source transaction date; a missing price fails the charge with IC-0821 and the source document stays posted in the lending entity on a suspense IC account until repriced.
- **BR-40.13** IC planning sequence: an entity appears once per run; cycles in the partner graph are broken by the group's explicit sequence, else the run fails with IC-0816.
- **BR-40.14** Entity-level reclassification entries reverse on the first calendar day of the next period with the same journal source; a rerun in the same period reverses the previous run first.
- **BR-40.15** Two-sided IC elimination: difference = Σ receivable side − Σ payable side per partner pair and FS item group after translation; |difference| ≤ threshold posts to the IC difference item; above threshold the task ends in Error.
- **BR-40.16** Ownership change without loss of control posts the difference between consideration and the NCI carrying amount to group equity; goodwill is unchanged.
- **BR-40.17** Unit data released for a period cannot be re-collected without `ConsolidationApprove`; re-collection invalidates every downstream task of that unit and period.
- **BR-40.18** Carry-forward top-side journals are re-applied to every later period of the same fiscal year and stop at year end unless the document type is `Cumulative`.

7.40.8 Acceptance criteria

- **AC-40.1** (FR-40.2) Given a sales order of 100 × 25.00 EUR in A to the IC customer, when confirmed, then a PO 100 × 25.00 EUR exists in B linked by ic_pair_id, and both documents balance.
- **AC-40.2** (FR-40.2) Given 1,000 random IC orders including planted number-sequence failures, when sagas run, then every pair is either Linked with equal amounts or Compensated with no orphan document.
- **AC-40.3** (FR-40.3) Given a direct-delivery chain, when B posts the packing slip, then A's product receipt and A's customer packing slip post in the same saga with matching quantities.
- **AC-40.4** (FR-40.5) Given an IC journal line of USD 1,000.00 expense in A offset to B, when posted, then A shows Dr Expense 1,000.00 / Cr Due from B 1,000.00 and B shows Dr Due to A 1,000.00 / Cr Cash 1,000.00 with one ic_pair_id.
- **AC-40.5** (FR-40.6) Given 50 planted mismatches (missing mirror, wrong amount, wrong period), when reconciliation runs, then all 50 are flagged and no matched pair is flagged.
- **AC-40.6** (FR-40.8) Given unit S with opening net assets EUR 700,000 at 1.05, net income EUR 150,000 at 1.08, closing net assets EUR 850,000 at 1.10, when translated, then CTA = 935,000 − (735,000 + 162,000) = 38,000.00 USD.
- **AC-40.7** (FR-40.9) Given the worked P/S example, when the run executes, then entries E1–E5 post with the stated amounts and each rule balances.
- **AC-40.8** (FR-40.9) Given the 3-entity, 2-currency demo group fixture, when consolidated, then the consolidated balance sheet and P&L match the accountant-reviewed eliminations to the cent.
- **AC-40.9** (FR-40.11) Given a consolidated revenue line, when drilled, then unit balances, E1 entries and entity journal lines are reachable within 3 clicks.
- **AC-40.10** (FR-40.12) Given P pays S's vendor invoice of EUR 10,000.00, when the payment posts, then P shows Dr Due from S / Cr Bank 10,000.00, S shows Dr Vendor / Cr Due to P 10,000.00 with the invoice settled, and both lines share one ic_pair_id.
- **AC-40.11** (FR-40.13) Given 8 hours by a lending-entity consultant at cost 60.00 with cost-plus 10%, when the charge posts in Journal mode, then the borrower books 528.00 project cost with Due to lender, and the lender books Due from borrower 528.00 against cost recovery 480.00 and IC margin 48.00.
- **AC-40.12** (FR-40.16) Given 40 units remaining from an IC receipt at Legal cost 120.00 and Group cost 100.00, when E3 runs, then unrealized profit of 800.00 in the receipt currency is eliminated and the detail lists the layer.
- **AC-40.13** (FR-40.17) Given a vendor with a debit balance of USD 5,000.00 at period end, when the regrouping rule runs, then Dr Other receivables 5,000.00 / Cr Vendor reclass 5,000.00 posts and reverses on the first day of the next period.
- **AC-40.14** (FR-40.18) Given the 30% associate example, when the equity method runs, then the investment carrying amount is 340,000.00 and the share of profit is 60,000.00.
- **AC-40.15** (FR-40.20) Given a unit whose balance sheet does not balance by 12.00, when validation runs, then the unit task ends in Error and the group translation task is Blocked.
- **AC-40.16** (FR-40.15) Given buyer A with a planned IC PO of 100 units for 2026-12-10 and transport lead time 3 days, when the planning group runs, then supplier B is planned after A with IC demand of 100 units on 2026-12-07.

7.40.9 Test plan and fixtures

Fixtures: `ic-orders-1000` (generated across 3 entities, 2 currencies, with planted failures and direct-delivery chains); `ic-recon-planted-50` (mismatch types); `consol-group-3` (P in USD, S1 in EUR 80%, S2 in USD 100%, IC sales, inventory, loans, dividends; accountant workbook); `translation-cases` (rate scenarios, mid-year acquisition).

Test	Type	Target	Pass criterion
Saga compensation	Property	interco-svc	no orphan documents under random failures
Mirror amount equality	Property	links	source = mirror in document currency
Due-to/due-from derivation	Unit	IC accounting	matrix cases
Reconciliation classifier	Unit	matcher	Matched/Timing/Mismatch per table
Translation and CTA	Unit	consol-svc	worked example exact
Elimination rules	Unit/E2E	run	P/S example; demo group workbook
Run performance	Performance	run	50 units × 10k accounts < 5 min
Drill-down	E2E	reporting	3 clicks to journal

Reconciliation checks: Σ due-from in A = Σ due-to in B per pair and currency (after FX tolerance); consolidated TB balances; Σ eliminations per rule = 0; consolidated equity = parent equity + NCI + CTA.

7.40.10 Deployment and operations

Helm charts `interco-svc` (TS, Temporal workers), `consol-svc` (Go), Postgres `group`; env `IC_SAGA_TIMEOUT_S=120`, `IC_FX_TOLERANCE_PCT=0.5`, `CONSOL_MAX_UNITS=500`; flags `interco.payment-mirroring`, `consol.external-import`; jobs: `ic-reconcile` hourly and at period close, `ic-saga-sweeper` every 5 min for stuck sagas, `consol-run-scheduled` on close checklist task (wave 14). SLOs: mirror saga p95 < 5 s, reconciliation run < 2 min for 100k items, consolidation as above. Alerts: compensated sagas > 0 per hour, stuck sagas > 10 min, run failures. Runbook: retry saga, manual compensation, rerun consolidation version. Rollback: Helm; runs are versioned datasets, links are append-only.

7.40.11 Migration from AX / SAP

Source	Passage entity	Notes
InterCompanyTradingPartner, InterCompanyEndpoint*, value maps	IntercompanyPartner, policies	
SalesTable/PurchTable IC links (open orders)	IcDocumentLink	open chains only, status Linked
LedgerIntercompany (due-to/due-from setup)	IC accounting matrix	
LedgerConsolidate*, LedgerEliminationRule/Line, ConsolidationAccount mapping	ConsolidationGroup, Unit, EliminationRule, chart mapping	
OBYA clearing accounts, customer/vendor IC assignments	IC accounting matrix, IntercompanyPartner	
Open cross-company STOs and IC billing documents	IcDocumentLink	
Group Reporting units/groups/FS item mapping, ECCS CX hierarchy	ConsolidationGroup, Unit, chart mapping	
ACDOCU / ECMCT history	lakehouse archive	prior consolidated results kept for comparison, not re-run

Validation: IC balances per pair equal legacy after migration; first Passage consolidation for the last closed legacy period matches legacy consolidated statements within 0.01% per line (differences explained). Not migrated: closed IC chains, legacy elimination journals (replaced by derived runs).

7.40.12 Dependencies, risks, open questions

Depends on: waves 11–15 (journal, rates, period control, reporting), wave 17 PO and approval, wave 18 vendor invoice matching, wave 27–29 sales orders, shipping and invoicing, wave 30 payments, wave 36 netting, wave 37 IC asset transfers, wave 39 cost for unrealized profit.

Risk	Likelihood	Impact	Mitigation
Cross-domain saga partial failure leaves stuck state	Medium	High	Sweeper job, compensation tests, manual runbook
Accountant disagreement on NCI/CTA presentation	Medium	Medium	Workbook-driven fixtures; sign-off gate
Group chart mapping effort per customer	High	Medium	Mapping import template; validation report

Open questions: (1) Mirror customer payments by default or rely on netting? Proposed netting; decide by wave 40 start. (2) Proportional consolidation in v1? Proposed no; decide at phase exit. (3) Transfer-pricing evidence format (CSV vs OECD XML)? Proposed CSV; decide by wave 40.

7.40.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`ic.accountingMatrix`	entity pair	table	none	(A, B) → due-to, due-from, IC FX accounts	BR-40.4 derivation	Intercompany accounting setup (LedgerInterCompany)	OBYA cross-company clearing accounts
`ic.partnerPolicy`	entity pair × direction	struct	auto-create on, auto-post off	create / confirm / post packing slip / post invoice flags	FR-40.1 action policies	InterCompanyEndpointActionPolicy	IC billing output (VF01 IV, RD04 output type), customer / vendor per company code
`ic.valueMaps`	entity pair	table	none	item, unit, dimension, charge maps	FR-40.1 mapping	InterCompanyTradingValueMap	customer-material info records (VD51), plant assignment
`ic.pricingPolicy`	entity pair	enum	PriceList	PriceList / CostPlus / TransferPriceTable	IC line price (FR-40.4)	Intercompany trade → Price / InterCompanyTradingRelation pricing	pricing procedure with PI01 / IV01 conditions, 8KEM / 8KEN transfer price variants
`ic.changeTolerance`	entity pair	struct	qty ±0, price ±0.5%, date ±2 days	ranges	BR-40.3	none	none
`ic.fxTolerancePct`	tenant	decimal	0.5%	0–5%	FR-40.6 Matched	none	FBICR3 tolerance settings (ICR)
`ic.sagaTimeoutS`	tenant	int	120	30–600	saga timeout (env `IC_SAGA_TIMEOUT_S`)	none	none
`ic.centralizedPayments`	entity pair	struct	none	paying entity, payment methods, direction	FR-40.12	Centralized payments setup	FBZP paying company code / sending company code
`ic.chargeSettlementMode`	entity pair	enum	Journal	Journal / Invoice	FR-40.13 settlement	Intercompany project parameters (lending / borrowing, invoice vs journal)	PS intercompany billing (DP90 cross-company)
`ic.transferPrices`	entity pair × category	table	cost	Cost / CostPlus % / Table	FR-40.13 pricing	Project → Transfer prices (intercompany)	CO transfer price variants (TKA01 valuation)

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`ic.shareOnHand`	entity pair	bool	false	true / false	FR-40.14 visibility	Intercompany on-hand (R3) visibility	plant assignment / ATP across company codes
`ic.planningGroups`	tenant	table	none	members, sequence, lead times	FR-40.15	Intercompany planning groups / members	MRP special procurement key (cross-company), plant sequencing
`consol.groupCurrency`	consolidation group	char(3)	USD	ISO 4217	translation target	Consolidation legal entity currency	consolidation group currency (Group Reporting)
`consol.chartMapping`	unit	table	none	main account → consolidation account / FS item	FR-40.7	Consolidation accounts (MainAccount.ConsolidationMainAccount), consolidation account groups	FS item mapping (Group Reporting), ECCS FS item hierarchy
`consol.translationMethods`	group	table	closing / average / historical by account category	per category method	FR-40.8	Consolidate online → exchange rate type per account	translation methods (closing, average, historical)
`consol.eliminationRules`	group	ordered rules	E1–E6	rule sets	FR-40.9	Elimination rules (LedgerEliminationRule)	elimination / reclassification methods (Group Reporting), ECCS elimination of IC payables / receivables
`consol.icThreshold`	group	money	1,000.00 group currency	≥ 0	BR-40.15	none	reclassification method difference threshold
`consol.reclassRules`	entity / group	ordered rules	DebitBalanceVendors, CreditBalanceCustomers, RemainingTerm > 12 months	rule types (FR-40.17)	FR-40.17	none (financial report rows only)	FAGL_CL_RGRP sort methods and adjustment accounts, FAGLF101
`consol.coiMethods`	investment	enum	Full	Full / Equity	FR-40.18	LedgerConsolidateLegalEntity share %	CoI methods (purchase, equity)
`consol.documentTypes`	group	table	Adjustment, Reclassification, Elimination, CoI	period / cumulative, approval threshold	FR-40.19	Consolidation journal names	Group Reporting document types (0C, 1x, 2x, 3x)
`consol.validationRules`	unit / group	expressions	3 seeded	expression, severity, tolerance	FR-40.20	none	validation rules (Define Validation Rules)
`consol.maxUnits`	tenant	int	500	1–2,000	env `CONSOL_MAX_UNITS`	none	none

7.40.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
IC chain view	drill-down	pair id, document, status	documents in chain, status, amounts	`ic_document_link`	Intercompany sales order / purchase order references	VF03 / ME23N document flow
Failed / compensated sagas	operational report	date, partner, failed step	saga, source, step, reason, age	`ic_document_link` status Compensated / Failed	none	WE02 IDoc errors (INVOIC)
IC reconciliation	list inquiry	pair, period, currency, status	amount A, amount B, difference, status, resolution	`ic_reconciliation_item`	Intercompany reconciliation report (D365)	FBICR3, FBIC010, FBICS3
Due-to / due-from balances	statutory report	entity pair, period	due-to, due-from, difference, ageing	journal balances by IC accounts	Intercompany accounting balances	FAGLB03 on clearing accounts, ICR reports
Centralized payments log	list inquiry	paying entity, owning entity, date	payment, invoices settled, amounts, ic pair	`ic_document_link` type CentralPayment	Centralized payments journal lines	F110 payment list by paying company code
IC charges	list inquiry	lender, borrower, category, period	charge, source, cost, price, margin, mode	`ic_charge_request`	Intercompany expense / timesheet lines	KSB1 cross-company allocations
Transfer-pricing evidence	statutory report	entity pair, period	item / category, qty, price, basis, cost, margin %	`ic_document_link.pricing_basis` + FR-39.17	none	TP documentation from 8KEM conditions
Consolidated trial balance	statutory report	group, period, version	FS item, unit amounts, eliminations, consolidated	`consolidation_entry`	SSRS Consolidation (consolidated balances)	ECCS CX5x reports, Group Reporting "Group Data Analysis"
Elimination detail	drill-down	run, rule, unit	entry, FS item, debit, credit, NCI, source refs	`consolidation_entry`	Elimination journal lines	CoI / elimination document display
Consolidation monitor	KPI	group, period	tasks by unit / group, status, duration	`consolidation_task`	none	Consolidation monitor (Group Reporting), CXCD data monitor
CTA and NCI roll-forward	statutory report	group, period	opening, translation movement, NCI share, closing	`consolidation_entry` (Translation, E5)	none	Group Reporting statement of changes in equity
Investment / equity holdings	list inquiry	group, investor, method	investee, ownership %, cost, goodwill, carrying amount	`investment`	Consolidation legal entity share %	Manage Investments / Equity data

7.40.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
IC invoice (sales) and mirror vendor invoice	FR-40.2 invoice step, FR-40.13 Invoice mode	standard invoice layout (wave 29) plus IC partner, pair id, transfer-pricing basis	PDF, e-invoice per country pack (wave 48) where IC invoices are in mandate scope	seller's invoice sequence	VAT invoice rules apply to cross-border IC supplies; 10 years
IC netting / settlement statement	centralized payment or netting (FR-36.10)	pair, documents settled, amounts, FX	PDF, email to both entity controllers	settlement number	10 years
IC balance confirmation	period end	pair, balances by account and currency, difference, confirm box	PDF, portal acknowledgment	per pair per period	audit evidence 10 years
Transfer-pricing evidence file	period / year end	per FR-40.4 and FR-39.17	CSV (OECD XML optional per open question 3)	none	local file / master file support 10 years
Consolidated financial statements	run completed	balance sheet, P&L, OCI, equity changes, cash flow (wave 15 designer)	PDF, XLSX, iXBRL / ESEF via wave 15	run id + version	statutory group accounts; 10 years
Elimination and top-side journal listing	run completed	rule, entries, reasons, approvers	PDF, CSV	run id	10 years

7.40.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`ic-reconcile`	hourly + period close	(pair, period)	(pair, period, version)	< 2 min for 100k items	rerun	mismatches > 0 at close
`ic-saga-sweeper`	every 5 min	saga id	Temporal workflow id	< 30 s	compensate or alert	stuck > 10 min
`ic-onhand-projection`	continuous (events)	partner pair	event offset	lag < 60 s	replay from offset	lag > 5 min
`ic-planning-run`	nightly 01:00 before entity MRP	planning group	(group, run date)	< 60 min for 5 entities	entity-level retry; plan marked stale	stale supplier plan
`regrouping-run`	period end (close checklist task)	legal entity	(entity, period, version) with reversal	< 5 min for 200k open items	rollback and retry	not run by close day 2
`regrouping-reversal`	first calendar day of period	legal entity	(run id)	< 2 min	retry	reversal missing
`consol-run-scheduled`	on close checklist task	(group, period)	(group, period, version)	< 5 min for 50 units × 10k accounts	task-level error, run Failed	run failure
`equity-method-run`	inside consolidation run	investment	(investment, period)	< 1 min	task Error	missing associate data
`ic-balance-confirmation`	period close day 3	pair	(pair, period)	< 2 min	resend	unconfirmed after 5 days

7.40.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles
IntercompanyView	GET /interco/*, chain view, IC on-hand, reconciliation read	none	IC accountant, Controllers of both entities
IntercompanyConfigure	/partners, matrix, value maps, pricing, centralized payment setup, planning groups	IntercompanyPost, IntercompanyReconcile	Group finance system admin
IntercompanyPost	/interco/journals	IntercompanyReconcile (same pair and period)	IC accountant
IntercompanyReconcile	/reconciliations/*, resolve	IntercompanyPost	Group accountant
IntercompanySystem	/mirrors, /charges (service accounts only)	n/a	interco-svc, project-svc, expense-svc
IntercompanyPlan	/planning-groups/*/run	none	Supply chain planner (group)
ConsolidationConfigure	/consol/groups, investments, chart mapping, rules, validation rules	ConsolidationRun, ConsolidationApprove	Consolidation lead
ConsolidationUnitRelease	/units/*/release	ConsolidationApprove for the same unit	Entity controller
ConsolidationRun	/consol/runs, reclassification runs	ConsolidationConfigure	Consolidation accountant
ConsolidationJournal	/group-journals	ConsolidationApprove for own journals	Consolidation accountant
ConsolidationApprove	override Mismatch gate, approve top-side ≥ threshold, re-collect released unit	ConsolidationJournal, ConsolidationRun	Group CFO
ConsolidationView	GET /consol/*, monitor, statements	none	Group controller, Auditor

7.40.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
IC-0801	error	No active IC partner policy between `{a}` and `{b}`	missing / inactive pairing	create or activate partner	FR-40.1
IC-0802	error	Mirror creation failed at `{step}`; source document canceled	saga compensation	fix cause (number sequence, dimension) and recreate	BR-40.1
IC-0803	error	Compensation failed for pair `{pairId}`; manual action required	Failed state	runbook manual compensation	BR-40.1
IC-0804	warning	Change exceeds partner tolerance; mirror PO sent for re-approval	BR-40.3	approver acts	BR-40.3
IC-0805	error	No IC accounting matrix entry for `{a}` → `{b}`	missing due-to / due-from	add matrix row	BR-40.4
IC-0806	error	Mirror entity `{b}` period `{period}` is closed	period control	open period or date	FR-40.2
IC-0810	warning	Reconciliation item `{id}` moved from Timing to Mismatch	counterpart missing after next period	resolve	BR-40.5
IC-0811	info	Centralized payment settled `{n}` invoices of `{entity}`	FR-40.12	none	FR-40.12
IC-0815	warning	Supplier plan for `{entity}` is stale after buyer replanning	FR-40.15	rerun group	FR-40.15
IC-0816	error	Planning group has a cycle without explicit sequence	BR-40.13	define sequence	BR-40.13
IC-0821	error	No transfer price for `{category}` / `{role}` on `{date}`	BR-40.12	add price	BR-40.12
IC-0825	warning	Partner on-hand for `{entity}` is `{s}` s old	projection lag	none	FR-40.14

Code	Severity	User text	Cause	Remediation	Raised by
CNS-0901	error	Consolidation blocked: `{n}` unresolved IC mismatches	FR-40.6 gate	resolve or override	FR-40.6
CNS-0902	error	Missing `{rateType}` rate for `{currency}` on `{date}`	rates missing	load rates (wave 12)	FR-40.8
CNS-0903	error	Rule `{rule}` unbalanced by `{amount}`	elimination defect	run fails; report defect	BR-40.9
CNS-0904	error	Ownership for `{unit}` exceeds 100.00% on `{date}`	invalid ownership	correct ownership	FR-40.7
CNS-0905	error	IC difference `{diff}` exceeds threshold `{t}` for `{pair}`	BR-40.15	resolve in IC reconciliation	BR-40.15
CNS-0906	error	Validation `{rule}` failed for `{unit}`: `{difference}`	FR-40.20	correct unit data and re-release	FR-40.20
CNS-0907	error	Unit `{unit}` data is released; re-collection needs approval	BR-40.17	request approval	BR-40.17
CNS-0908	warning	Group cost unknown for `{n}` layers; margin fallback used	FR-40.16	roll Group view	FR-40.16
CNS-0909	error	Top-side journal ≥ `{threshold}` requires approval	FR-40.19	approver acts	FR-40.19
CNS-0910	error	Change of control requires step-acquisition accounting, not supported	ownership crosses 50% from an existing interest	post manual top-side entries (FR-40.19)	FR-40.18

7.40.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Mirror saga	5,000 IC documents/day; peak 300/min at invoicing	1.5 / 5 / 10 s	10 sagas/s	1.8M links ≈ 1 GB	IcDocumentLink by ic_pair_id and (mirror_entity_id, mirror_doc_id)	Temporal task queue backlog; p95 < 15 s; no lost sagas
IC charge posting	50,000 charges/day (time + expense)	200 ms / 800 ms / 2 s	50/s	18M charges ≈ 9 GB	partitioned monthly by created_at	batched by pair per minute
IC reconciliation run	100k items per pair-period	n/a	< 2 min	1.2M items ≈ 0.6 GB	(entity_a, entity_b, period, status)	< 6 min
IC on-hand query	3,000/day	30 / 120 / 250 ms	50 req/s	read model ≈ size of partner on-hand	keyed (partner, item, site)	stale data with age shown
Consolidation run	50 units × 10k accounts, 5 versions per close	n/a	< 5 min	50 × 10k × 12 × 5 ≈ 30M entries ≈ 12 GB	ConsolidationEntry partitioned by period; index (run_id, account)	partial recompute for changed units; < 15 min
Drill-down consolidated → journal	2,000/day during close	200 ms / 800 ms / 1.5 s	20 req/s	n/a	source_refs GIN index	paginated drill
Regrouping run	200k open items per entity	n/a	< 5 min	2.4M reclass lines ≈ 1 GB	open items by due date (wave 19 / 30 indexes)	< 15 min

7.40.20 Localization and compliance hooks

- **Consolidation frameworks**: IFRS 10 / IFRS 3 / IAS 21 / IAS 28 (equity method) default; US GAAP ASC 810 / 830 / 323 differences (for example, NCI measured at fair value vs proportionate share, a choice under IFRS 3) are set per group; local GAAP group accounts (DE HGB §§290–315, FR règlement ANC 2020-01) through wave-48 pack chart mappings and statement layouts; ESEF iXBRL tagging for EU listed groups through the wave-15 designer.
- **Proportional consolidation** is dropped because IFRS 11 removed it for joint ventures (equity method applies); HGB §310 still allows it, so a DE pack may request it post-v1.
- **Transfer pricing**: OECD Guidelines master / local file and country-by-country reporting (CbCR, OECD XML schema) for groups with revenue ≥ EUR 750 million; Passage supplies the data export (FR-40.4, FR-40.13, FR-39.17); filing formats are wave-48 pack outputs (`TransferPricingReport` extension point).
- **IC invoicing tax**: cross-border IC supplies follow wave-13 VAT rules (EU intra-community supply, reverse charge); IC invoices fall under e-invoicing mandates where applicable (MX CFDI, IN IRP, FR / DE B2B schedules) through wave-48 packs.
- **Regrouping**: balance-sheet classification of current / non-current (IAS 1.66–1.76) and HGB §268 (residual terms > 1 year disclosed); the RemainingTerm rule threshold is a pack parameter (12 months default).
- **Retention**: consolidation runs, top-side journals and IC confirmations 10 years.
- **Currency**: translation of hyperinflationary units (IAS 29) out of v1; rounding per group currency minor unit (BR-40.9).

7.40.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.2.1 Payments → Centralized payments (cross-company)	PARTIAL-40	ADDED as FR-40.12	
AX 12.10.1 Expense reports → Intercompany expenses (R3)	PARTIAL-40	ADDED as FR-40.13	expense UI arrives with wave 43 and calls the IC charge API
AX 12.10.3 Expense → Intercompany expense posting (due-to / due-from)	PARTIAL-40	ADDED as FR-40.13	Journal settlement mode
AX 12.11.3 Project → Intercompany project invoicing (intercompany timesheets / expenses)	PARTIAL-40	ADDED as FR-40.13	posting and invoicing contract here; project resource lending screens DEFERRED to wave 41
AX 12.11.4 Intercompany → Intercompany project parameters (lending / borrowing, timesheet / expense / vendor invoice transfer)	PARTIAL-40	ADDED as FR-40.13	`ic.chargeSettlementMode`, `ic.transferPrices`
AX 12.11 Prices → Transfer prices (intercompany) (noted PARTIAL-40 in the COVERED-41 row)	PARTIAL-40	ADDED as FR-40.13	IcTransferPrice
AX 12.14.2 Inventory → Intercompany on-hand (R3)	PARTIAL-40	ADDED as FR-40.14	
AX 12.14.3 Inventory → Intercompany: on-hand sync	PARTIAL-40	ADDED as FR-40.14	event-driven projection, no batch sync
AX 12.21.1 Intercompany planned orders (R3 intercompany master scheduling)	PARTIAL-40	ADDED as FR-40.15	

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.21.3 Intercompany master scheduling (run across entities)	PARTIAL-40	ADDED as FR-40.15	`ic-planning-run`
AX 12.21.4 Intercompany planning groups / members (sequence, plan)	PARTIAL-40	ADDED as FR-40.15	`ic.planningGroups`
SAP 13.7 CO versions for parallel valuation (TKA07, TKA09, TKA01)	PARTIAL-40	ADDED as FR-40.16 (with FR-39.17)	
SAP 13.8 Transfer prices / multiple valuation views (OKKP, OKEQ group)	PARTIAL-40	ADDED as FR-40.16 (with FR-39.17)	
SAP 13.9 8KEM / 8KEN Transfer prices: pricing procedures, conditions, variants	PARTIAL-40	ADDED as FR-40.13 / FR-39.17	profit-center valuation view in FR-39.17; IC pricing in FR-40.4 / FR-40.13
SAP 13.10 Manage Reclassifications / Define Reclassification Rules	PARTIAL-40	ADDED as FR-40.17	
SAP 13.10 Consolidation of Investments (first, subsequent, step acquisition, divestiture, equity method)	PARTIAL-40	ADDED as FR-40.18	step acquisition OUT-OF-SCOPE for v1 (IFRS 3.42 remeasurement of the previously held interest is rare in the target segment); CNS-0910 directs users to top-side journals (FR-40.19)
SAP 13.10 Manage Investments / Equity data	PARTIAL-40	ADDED as FR-40.18	Investment entity
SAP 13.10 Consolidation monitor tasks (IC elimination incl. dividends, profit in inventory, CoI, group validation)	PARTIAL-40	ADDED as FR-40.20	dividends are E6 (FR-40.9); CoI FR-40.18
SAP 13.10 Reclassification runs	PARTIAL-40	ADDED as FR-40.17	
SAP 13.10 Reclassification / elimination methods (two-sided, differences to FS item, thresholds, dividends)	PARTIAL-40	ADDED as FR-40.17	BR-40.15
SAP 13.10 CoI methods (purchase, equity, proportional), activities	PARTIAL-40	ADDED as FR-40.18	proportional OUT-OF-SCOPE: IFRS 11 removed proportional consolidation for joint ventures (7.40.20)
SAP 13.1 FAGLF101 / F101 Regroup receivables / payables	GAP (proposed 14)	ADDED as FR-40.17	runs as a wave-14 close checklist task; engine lives in consol-svc rules
SAP 13.1 FAGL_CL_REGROUP / FAGL_CL_MIGRATION Closing regrouping (S/4)	GAP (proposed 14)	ADDED as FR-40.17	
SAP 13.1 FAGL_CL_RGRP sort method and adjustment accounts for regrouping	GAP (proposed 14)	ADDED as FR-40.17	`consol.reclassRules`
SAP 13.2 FAGLF101 (AP) Regroup vendors with debit balances	GAP (proposed 14)	ADDED as FR-40.17	DebitBalanceVendors rule
SAP 13.3 FAGLF101 (AR) Regroup customers with credit balances	GAP (proposed 14)	ADDED as FR-40.17	CreditBalanceCustomers rule

Part 8 — Phase 8: Projects, People & Service (waves 41–45)

8.0 Phase overview

Business outcome at phase exit. At the end of wave 45 the demo group (manufacturer plus the engineering-services entity added in wave 41) runs project operations, HR core, payroll, expense management, plant maintenance and field service on Passage, with every financial effect posted through the universal journal (wave 11) and reconciled daily. A project manager plans a WBS, books resources, captures time and expense, bills by T&M, fixed-price milestone or retainer, and recognizes revenue by percentage-of-completion. An HR administrator manages workers, positions and jobs with effective dating; employees self-serve absence and benefits. A payroll administrator runs a US gross-to-net cycle that matches IRS Publication 15-T to the cent and posts to the GL. A maintenance planner keeps plant machines on preventive and counter-based plans whose work orders cost to the asset and GL. A service coordinator dispatches field technicians under SLA with OR-Tools, and technicians execute offline on a PWA. This phase is where Passage reaches parity with AX Project management and accounting, HR, Payroll, Expense management, D365 Asset Management and Service management, and with SAP PS, HCM PA/OM/PT, PY, Concur, PM and CS/FSM.

Phase-exit sign-off criteria

- Percentage-of-completion estimate fixtures (12 projects, 4 periods each) match the accountant-reviewed workbook to 0.01 USD/EUR per period.
- Property test over 10,000 generated transaction sets: every invoice proposal contains exactly the billable transactions selected by its billing rule, no duplicates, no omissions.
- Effective-dated position/manager as-of queries return the correct answer for 100% of 50,000 randomized dates across the demo workforce (1,200 workers).
- 200 US payroll fixture scenarios match Pub 15-T percentage-method computations to the cent; payroll posting reconciles to GL accounts 6100–6199 and 2300–2349 with zero variance.
- Expense policy engine flags violations exactly per rule set on a 500-report fixture; reimbursements post to AP and settle in a wave-19 payment run.
- Maintenance plan fixtures (40 plans, time and counter) generate work orders on the expected dates/readings; work order cost (parts + labor) reconciles to asset cost and GL maintenance accounts.
- IoT threshold breach creates a maintenance request in p95 < 5 s; recorded downtime reduces wave-23 scheduling capacity.
- Dispatch property test: 1,000 generated schedules are feasible (skills, time windows, shifts) with a solver time p95 < 30 s for 200 work orders × 25 technicians.
- Recorded phase-exit walkthrough: projects, HR, payroll, maintenance and field service end-to-end on `stage`, signed off by both owners.

Cross-wave data model summary

Entity	Owning service	Key relationships
Project, WbsNode, ProjectTransaction, InvoiceProposal, ProjectEstimate	`projects` (Go, Postgres `projects`)	Customer (wave 7 party), Worker (wave 42), Journal (wave 11), Fixed asset CIP (wave 37)
Worker, Employment, Position, Job, Department, AbsencePlan, BenefitPlan	`hr` (TS, Postgres `hr`)	Legal entity/org hierarchy (wave 6), Time registrations (wave 25), User (Keycloak)
PayCycle, PayRun, PayStatement, EarningCode, DeductionCode, TaxRuleSet	`payroll` (Go, Postgres `payroll`)	Worker/Employment (wave 42), Journal (wave 11), Bank/payment file (wave 36)
ExpenseReport, ExpenseLine, ExpensePolicyRule, CardTransaction	`expense` (TS, Postgres `expense`)	Worker, Project (wave 41), AP invoice (wave 18), Tax (wave 13)
FunctionalLocation, Asset, MaintenancePlan, MaintenanceRequest, MaintWorkOrder, Counter	`eam` (Go, Postgres `eam`)	Resource (wave 22), Inventory item (wave 16), Fixed asset (wave 37), IoT reading (wave 25)
ServiceObject (installed base), ServiceAgreement, Case, ServiceWorkOrder, DispatchAssignment	`service` (TS) + `dispatch` (Python/OR-Tools), Postgres `service`	Customer, Sales shipment serial (wave 28), Worker (technician), Truck stock location (wave 16), Invoice (wave 29)

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
`dev.passage.projects.transaction.posted.v1`	projects	journal, analytics, expense (link-back)	projectId, wbsId, txType, qty, cost, salesPrice, currency, journalId
`dev.passage.hr.employment.changed.v1`	hr	payroll, projects (resource validity), service (technician), Keycloak sync	workerId, employmentId, action, effectiveFrom, effectiveTo
`dev.passage.hr.absence.approved.v1`	hr	time (wave 25), projects (availability), payroll (paid-leave earnings)	workerId, planId, from, to, hours
`dev.passage.payroll.run.posted.v1`	payroll	journal, bank (wave 36), analytics	payRunId, payCycleId, grossTotal, netTotal, employerTaxTotal, journalId
`dev.passage.expense.report.approved.v1`	expense	ap (wave 18), projects	reportId, workerId, totalAmount, currency, lines[]
`dev.passage.eam.request.created.v1`	eam (or IoT bridge)	service desk UI, notifications	requestId, assetId, source, severity, readingValue
`dev.passage.eam.workorder.completed.v1`	eam	journal (cost settlement), fixed assets, scheduling (wave 23) capacity release	workOrderId, assetId, partsCost, laborCost, downtimeMinutes
`dev.passage.service.sla.breached.v1`	service	notifications, analytics	caseId or workOrderId, slaType, dueAt, breachedAt
`dev.passage.service.dispatch.published.v1`	dispatch	technician PWA, service	scheduleDate, assignments[] (workOrderId, technicianId, start, end)

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Payroll tax legal exposure (wrong withholding, late forms)	Medium	High	Table-driven rule sets versioned per tax year; 200-scenario Pub 15-T fixture gate; ADP/Gusto-style provider adapter as alternative path	Engineering/security
PII breach or over-exposure of HR/payroll data	Medium	High	PII fields in a separate vaulted schema with column-level encryption, OPA duty `HcmPIIRead`, masked by default, every read audited	Engineering/security
POC revenue disputes with auditors	Low	High	Accountant-reviewed workbook fixtures; estimate reversal/re-run is idempotent; loss-provision rule explicit	Product
Dispatch optimizer infeasible or slow at scale	Medium	Medium	CP-SAT with 30 s cap and greedy fallback; property test over generated instances	Engineering
Offline technician PWA conflicts	Medium	Medium	Per-work-order append-only action log with server-side merge; last-writer-wins only for free-text	Engineering

Risk	Likelihood	Impact	Mitigation	Owner
Scope of five large modules in five waves	High	High	Each wave ships the thin vertical slice in its deliverable list; talent/recruitment, subscriptions and predictive maintenance are explicitly basic	Product

8.41 Wave 41: Project Operations: WBS, Time, Expense, Billing & Accounting

8.41.1 Scope and business context

Project-centric businesses (engineering services, contract manufacturing, construction-adjacent, professional services) need a project ledger that AX delivers through Project management and accounting (ProjTable, ProjGroup, ProjWBS, hour/expense/item journals, ProjProposalJour, ProjEstimate) and SAP through PS (PROJ/PRPS WBS, AUFK networks, CJ20N, DP90 resource-related billing, KKA2 results analysis, CJ88 settlement). At the end of this wave a user can create a project hierarchy and WBS with scheduled tasks and resource bookings, record hour/expense/item/fee transactions that cost the project and post to the GL, generate invoice proposals by T&M, fixed-price milestone or retainer, run percentage-of-completion estimates that post WIP and accrued revenue, and read committed cost, EVM and profitability. A demo engineering-services legal entity (`ES-US`) is added alongside the manufacturer.

Out of scope: grant management beyond customer funding sources with limits; project quotations (use wave 26 quotes with a project link); multi-currency funding-rule splitting beyond percentage splits; resource scheduling optimization (wave 23 engine is reused for capacity only); D365 Project Operations "Project for the web" scheduling parity.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX invoice proposal (ProjProposalJour via "Create invoice proposal" dialog) takes 9–12 clicks per project and silently skips transactions whose line property changed after posting; SAP DP90 requires a DIP profile (ODP1) that most sites never get right, so billing runs produce missing lines nobody notices until the customer disputes	Selection is a batch class with no explainable predicate; eligibility state lives on the transaction and is mutated by other processes	Declarative billing rules with a stated predicate (BR-41.3), a "why not billed" explain endpoint per transaction, and proposals built in one action from the project workspace	Clicks 12 → 2 per proposal; omitted-billable-transaction rate 0 (property test); dispute rate attributable to billing errors –90%
Fixed-price estimates (ProjEstimate, "Create estimate" periodic job; SAP KKA2 + CJ88) run as month-end batch taking 1–4 h on 2,000 projects; reversal (ProjEstimate "Reverse") and re-run is manual and frequently leaves orphan WIP	Batch framework; estimate and settlement are two separate steps with separate postings	Estimate run as a Temporal workflow per project with idempotent reverse-and-rerun in one call; WIP projection reconciled nightly to GL	Estimate run 4 h → < 10 min for 2,000 projects; orphan WIP incidents 0; re-run effort 30 min → 1 click
Committed cost (ProjCommittedCost) refreshed by batch ("Update committed costs") so budget checks are hours stale; SAP availability control fires only at PO save and cannot be queried live	Polling batch over PurchLine; no event stream	Committed cost projection updated from PO/receipt events within 10 s and queryable per WBS node	Staleness 4 h batch → < 10 s; PO-over-budget surprises at month-end –95%
WBS scheduling in ProjWBS form recalculates the whole plan synchronously (30–90 s on 1,000 activities) and ignores worker calendars unless "Scheduling with resources" is enabled; SAP CJ20N networks require AFVC maintenance per activity	Form-bound X++ scheduler; calendars optional	CPM scheduler over role calendars, < 500 ms for 2,000 nodes, baseline versioning	Reschedule 60 s → < 0.5 s; calendar-violating task dates 0
Hour journal → ledger reconciliation (ProjEmplTrans vs LedgerTrans) done in Excel each close; SAP CATS → CO → PS needs three transfer steps (CAT5/CAT7)	Separate posting paths with no shared id	Every transaction carries journal_id; subledger-to-GL reconciliation is an automated nightly check with zero tolerance	Reconciliation effort 2 person-days per close → 0; variance 0.00
External systems (PLM, MES, BI) read project cost via nightly SQL extracts of ProjTable/ProjCostTrans	No supported API/event surface	REST/GraphQL endpoints plus CloudEvents on Kafka and MCP exposure (wave 47)	Integration lead time weeks → same day; data latency 24 h → < 60 s

8.41.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Project master and hierarchy	ProjTable, ProjGroup, ProjType (TimeMaterial, FixedPrice, Internal, Cost, Investment, Time), form ProjTable	PROJ (project definition), PRPS (WBS elements), CJ20N Project Builder	`projects` service, `Project` entity, screen Projects › Project	full
WBS and scheduling	ProjWBS, ProjPlanVersion, ProjWBSActivityEstimate	PRPS, AUFK networks, AFVC activities, CJ20N	`WbsNode`, `WbsDependency`, CPM scheduler	full
Hour/expense/item journals	ProjJournalTable/ProjJournalTrans → ProjEmplTrans, ProjCostTrans, ProjItemTrans, ProjRevenueTrans	CATS (CATSDB) → CO postings; MM goods issue to WBS; KB21N activity allocation	`ProjectJournal`, `ProjectTransaction` (type HOUR/COST/ITEM/FEE)	full
Invoice proposals	ProjProposalJour, ProjProposalEmpl/Cost/Item/OnAcc, ProjInvoiceJour	DP90/DP91 resource-related billing, billing request → VF01	`InvoiceProposal` → wave-29 `Invoice`	full
Fixed-price estimates	ProjEstimate, ProjPeriodTable, completed-contract / percentage-of-completion, cost template	KKA2 results analysis (RA keys, RA version), CJ88 settlement to CO-PA/GL	`ProjectEstimate` with method COMPLETED_CONTRACT / POC_COST / POC_MILESTONE	full
Ledger posting profiles	ProjPosting (ProjLedgerPostingSetup), ProjLinePropertySetup, ProjCategory	Cost elements, OKB9 default account assignment, settlement profile	`ProjectPostingProfile` keyed by (projectGroup, category, lineProperty)	full
Customer funding / grants	ProjFundingSource, ProjFundingRule, ProjFundingLimit, ProjGrant	Not native (grants management GM module)	`FundingSource`, `FundingRule` (percentage split, limits)	partial — grant compliance reporting dropped; v1.x
Resource roles and bookings	ResResource, ResRole, ResResourceRole, PSAProjectTeam; D365 PO bookable resources	HR mini-master in PS, capacity planning CMP2	`ResourceRole`, `ResourceBooking` (soft/hard)	full
Committed cost	ProjCommittedCost, ProjCommitmentTmp	Commitment items, CJI5	`CommittedCost` projection from PO/requisition events	full
EVM and reporting	Project statements, cash flow, invoice control	CJI3, S_ALR_87013532 cost/revenue reports	`/reports/projects/evm` (PV/EV/AC, CPI, SPI)	improved — live rather than batch
Project quotations	SalesQuotationTable (type Project), "Transfer to project" (ProjQuotationTransfer)	VA21 quotation with WBS account assignment, DP80 sales pricing on project	`ProjectQuotation` = wave-27 quotation kind PROJECT with estimate lines, convert-to-project (FR-41.12)	full
Weekly timesheets	TSTimesheetTable/TSTimesheetLine, TSTimesheetPeriod, TSFavorites, TSTimesheetDelegate, timesheet workflow	CATS: CAT2 (data entry profile CAC1/CAC2), CATS_APPR_LITE approval, CAT5 transfer to PS, CADO/CATC inquiries	`Timesheet`, `TimesheetPeriod`, `TimesheetFavorite`, `TimesheetDelegate` (FR-41.13)	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Retention (retainage) terms	ProjRetentionTerms (customer), vendor retention terms on project PO, "Retention release"	Retention in payment terms (OBB9 installments), S/4 PO retention (ME21N "Retention" tab), special G/L indicator for retention	`RetentionTerm`, `RetentionLedger` (FR-41.14)	full
Indirect cost components	ProjIndirectCostComponentGroup, ProjIndirectCostComponent, ProjIndirectCostRule (R3 PSA)	Costing sheet KZS2, overhead calculation CJ44/CJ45 (KGI2) on WBS	`IndirectCostComponent`, `IndirectCostRule` (FR-41.15)	full
Project interest	not native	CJZ1/CJZ2/CJZ3/CJZ5 interest calculation on projects; interest profiles OPIB/OPIC/OPID	`InterestProfile`, interest run (FR-41.16)	full
Simulation versions	ProjPlanVersion copy (no transfer)	CJV1–CJV6 simulation versions, OPTS/OPT version profiles	`PlanVersion.kind = SIMULATION` with transfer and diff (FR-41.17)	full
Change orders / claims	not native (PSA change requests via workflow)	CLM1/CLM2/CLM3/CLM10/CLM11 claims, TQ80 claim notification type CL	`ProjectChangeOrder` (FR-41.18)	full
Project stock and deliveries	Item requirements (ProjItemRequirement), project inventory journals	Special stock Q (project stock), OPS valuated/non-valuated, MD51 project MRP, CN33 BOM transfer, CNS0 delivery from project	wave-16 ownership dimension `project_id`, wave-22 planning scope PROJECT, wave-28 shipment from project stock (FR-41.19)	full
Forecasts, risks and issues	ProjForecastEmpl/Cost/Item/Revenue/OnAcc forecast models	CPM (Commercial Project Management) financial planning, risk/issue management	`ProjectForecast` models, `ProjectRiskIssue` (FR-41.20)	partial — CPM multi-version financial plans reduced to forecast models

8.41.3 Functional requirements

- **FR-41.1 Project master and hierarchy** — Must. The system shall maintain projects with type (TIME_MATERIAL, FIXED_PRICE, INTERNAL, COST, INVESTMENT, TIME), group, customer, contract, currency, dates, stage and an unlimited parent/child hierarchy.
  - Stages: CREATED → IN_PROCESS → FINISHED (and ON_HOLD from IN_PROCESS); transitions gated by duty `ProjManage`.
  - Sub-projects inherit group, customer and contract unless overridden; a sub-project cannot change customer.
  - Project ID from wave-5 number sequence `PROJ`, 20 chars max.
- **FR-41.2 WBS with scheduling** — Must. The system shall maintain WBS nodes per project plan version with effort, duration, dependencies (FS/SS/FF/SF with lag), constraints and a CPM scheduler respecting resource calendars.
  - Plan versions: one BASELINE (immutable after approval), one WORKING; copying creates a new version.
  - Scheduler recomputes early/late start/finish, slack and critical path in < 500 ms for 2,000 nodes.
  - Calendars come from wave-8 resource calendars; a task cannot be scheduled on a non-working day of its assigned role calendar.
- **FR-41.3 Resource roles and bookings** — Must. The system shall define roles with cost and sales prices and allow soft/hard bookings of workers (wave 42) or generic resources to WBS nodes.
  - Hard booking over 100% utilization on a day is rejected; soft booking is warned.
  - Price precedence for hours: worker-specific → role × project → role × customer → role.
- **FR-41.4 Project transactions** — Must. The system shall record HOUR, COST (expense), ITEM, FEE and ON_ACCOUNT transactions through journals or integrated documents (timesheets wave 25, expense wave 43, PO/receipt wave 17, inventory journal wave 16).
  - Each transaction carries cost price, sales price, line property (chargeable/non-chargeable), category, WBS node, quantity, three-currency amounts.
  - Posting is atomic with the universal journal through the wave-11 posting API with idempotency key `projects:txn:{journalId}:{lineNo}`.
- **FR-41.5 Posting profiles** — Must. The system shall resolve GL accounts for cost, WIP, accrued revenue, invoiced revenue and on-account by (projectGroup, category, lineProperty) with fallbacks to (group, *, *) and (*, *, *).
  - Resolution failure blocks posting with error `PROJ-POST-001` naming the missing key.
- **FR-41.6 Invoice proposals** — Must. The system shall build invoice proposals selecting billable transactions per billing rule (T&M, fixed-price milestone, retainer/on-account, unit-of-delivery) and hand them to wave-29 invoicing.
  - Selection filters: project set, date range, transaction types, line property CHARGEABLE, not previously proposed, funding-rule allocation.
  - A proposal is editable until POSTED; removing a line returns it to the pool.
- **FR-41.7 Fixed-price estimates** — Must. The system shall run periodic estimates with methods COMPLETED_CONTRACT, POC_COST (cost-to-cost) and POC_MILESTONE, posting WIP, accrued revenue and loss provisions, with reversal and re-run.
  - Period source: wave-14 fiscal periods; one estimate per project per period; re-run reverses the prior posting first.
- **FR-41.8 Committed cost** — Should. The system shall project committed costs from requisitions, POs and item requirements in near-real time (< 10 s lag) and surface remaining budget.
- **FR-41.9 Customer funding sources** — Should. The system shall split project costs/revenue among funding sources by percentage with per-source limits and a limit-exceeded hold.
- **FR-41.10 EVM and profitability reporting** — Must. The system shall compute PV, EV (POC × BAC), AC, CPI, SPI, EAC and gross margin per project and WBS node as of any date.
- **FR-41.11 Project purchase orders and subcontracts** — Must. The system shall allow wave-17 POs to reference project/WBS so that receipts create ITEM/COST transactions and open POs appear as committed cost.
- **FR-41.12 Project quotations** — Should. The system shall create project quotations as wave-27 quotations of kind PROJECT with estimate lines (hours by role, expenses, items, fees, on-account milestones) and convert a WON quotation into a project with a BASELINE plan version built from the estimate lines.
  - Conversion creates the project in stage CREATED, copies customer, contract, currency and dimensions, and keeps `quotation_id` for quote-to-actual comparison.
  - A quotation converts once; a second attempt raises `PROJ-QUO-001`.
  - Hour estimate lines are priced by role sales price (FR-41.3 precedence); margin preview shows cost, sales and margin % before sending.
- **FR-41.13 Weekly timesheets** — Must. The system shall provide per-worker timesheets per timesheet period (period types WEEK, SEMI_MONTH, MONTH per legal entity) with lines by project, WBS node, category and line property, hours per day, internal/external comments, favorites, delegates, workflow approval routed per project and posting to HOUR transactions.
  - Favorites are saved (project, WBS, category) combinations; "copy previous period" copies lines without hours.
  - A delegate (TimesheetDelegate with validity) may enter and submit on behalf of a worker; the audit log records both identities.
  - Periodic creation generates DRAFT timesheets from favorites at period start; the missing-timesheets inquiry lists workers without a SUBMITTED sheet after period end.
  - Wave-25 time registrations on project jobs appear as lines with source TIME_REG and cannot be edited in the timesheet.
  - Validation per BR-41.9; statuses DRAFT → SUBMITTED → APPROVED (per line) → POSTED; RETURNED → DRAFT.
- **FR-41.14 Retention terms** — Should. The system shall apply customer retention terms (percent held per invoice until milestone or project completion, release schedule) on invoice proposals and vendor retention terms on project purchase orders, keep a retention ledger per project and party, and release retention through release proposals (customer) or released vendor due lines (wave 18).

- **FR-41.15 Indirect cost components** — Should. The system shall define indirect cost component groups, components (basis HOUR_COST, HOUR_QTY or EXPENSE_COST; rate or amount per unit; cost and optional sales side) and rules by (legal entity, project group, category, worker group, effective date), and post INDIRECT transactions at hour posting time per BR-41.11.
- **FR-41.16 Project interest** — Could. The system shall define interest profiles (annual rate source, basis WIP or cost balance, day count ACT/365 or ACT/360, treatment CAPITALIZE or EXPENSE) assigned per project group or project, and run a monthly interest calculation that posts capitalized interest to WIP/CIP for INVESTMENT and FIXED_PRICE projects per BR-41.12.
- **FR-41.17 Simulation plan versions** — Should. The system shall allow plan versions of kind SIMULATION (copied from WORKING or BASELINE), scheduled and costed independently, compared side by side (dates, effort, cost, critical path) and transferred into WORKING with a diff and conflict resolution per BR-41.14.
- **FR-41.18 Change orders and claims** — Must. The system shall record project change orders of type SCOPE_CHANGE, CLAIM_CUSTOMER, CLAIM_VENDOR or INTERNAL with cost and revenue impact lines per WBS node, attachments, customer approval evidence, and a lifecycle DRAFT → SUBMITTED → APPROVED | REJECTED → APPLIED that adjusts contract value, budget and baseline per BR-41.13.
- **FR-41.19 Project stock and deliveries** — Should. The system shall segregate project-owned inventory through the wave-16 ownership dimension `project_id` (VALUATED, NON_VALUATED or NONE per project group), plan project demand per project in wave 22, and ship from project stock through wave-28 shipments that create ITEM transactions on the WBS node.
- **FR-41.20 Forecasts, risks and issues** — Should. The system shall maintain forecast models (hour, expense, item, fee, on-account lines by period and WBS node) with copy-to-budget and forecast-vs-actual, and a risk and issue register per project (probability 1–5, impact 1–5, owner, mitigation, due date, status).

8.41.4 Data model

Project

Field	Type	Required	Notes
id	text(20)	yes	PK; number sequence PROJ
tenant_id	uuid	yes	RLS
legal_entity_id	uuid	yes	FK wave 6
parent_project_id	text(20)	no	FK self
project_group_id	text(10)	yes	FK ProjectGroup (type, posting profile)
project_type	enum	yes	TIME_MATERIAL, FIXED_PRICE, INTERNAL, COST, INVESTMENT, TIME
customer_party_id	uuid	cond	required for external types
contract_id	uuid	cond	FK ProjectContract (billing rules, funding)
currency	char(3)	yes	project currency
stage	enum	yes	CREATED, IN_PROCESS, ON_HOLD, FINISHED
start_date, end_date	date	yes/no	end_date ≥ start_date
dimension_set_id	uuid	yes	default financial dimensions
version	int	yes	optimistic concurrency

WbsNode

Field	Type	Required	Notes
id	uuid	yes	PK
project_id, plan_version_id	text, uuid	yes	FK; plan version BASELINE/WORKING
wbs_code	text(40)	yes	unique per plan version, e.g. `1.2.3`
parent_id	uuid	no	FK self
name	text(120)	yes	
effort_hours	numeric(12,2)	no	≥ 0
duration_days	numeric(8,2)	no	working days
early_start, early_finish, late_start, late_finish	date	computed	scheduler output
constraint_type, constraint_date	enum, date	no	ASAP, SNET, FNLT, MSO
role_id	uuid	no	default resource role
is_billable_milestone	bool	yes	default false

ProjectTransaction

Field	Type	Required	Notes
id	uuid	yes	PK
project_id, wbs_node_id	text, uuid	yes/no	
txn_type	enum	yes	HOUR, COST, ITEM, FEE, ON_ACCOUNT
category_id	text(20)	yes	FK ProjectCategory (type-bound)
line_property	enum	yes	CHARGEABLE, NON_CHARGEABLE
txn_date	date	yes	
quantity	numeric(14,3)	yes	hours, units
cost_amount, sales_amount	numeric(18,4)	yes	in project currency; plus accounting and reporting currency columns
currency	char(3)	yes	
worker_id, item_id, vendor_party_id	uuid	cond	by txn_type
source_doc_type, source_doc_id	text, uuid	yes	TIMESHEET, EXPENSE, PO_RECEIPT, INV_JOURNAL, MANUAL
journal_id	uuid	yes	universal journal reference
proposal_id, invoice_id	uuid	no	set when billed
estimate_id	uuid	no	set when captured in POC

InvoiceProposal: id, project_id (or contract_id for multi-project), billing_rule_id, customer_party_id, currency, status (DRAFT, APPROVED, POSTED, CANCELED), from_date, to_date, total_cost, total_sales, tax_amount, lines[] (transaction_id, qty, unit_price, amount, funding_source_id).

ProjectEstimate: id, project_id, period_id, method, cost_to_date, estimated_cost_to_complete, estimated_total_cost, contract_value, poc_percent numeric(7,4), revenue_to_date, revenue_this_period, billed_to_date, accrued_revenue, deferred_revenue, loss_provision, status (DRAFT, POSTED, REVERSED), journal_id.

Timesheet

Field	Type	Required	Notes
id	uuid	yes	PK; number sequence PROJ-TS
worker_id, legal_entity_id	uuid	yes	
period_id	uuid	yes	FK TimesheetPeriod (period_type WEEK/SEMI_MONTH/MONTH, from, to, status OPEN/CLOSED)
status	enum	yes	DRAFT, SUBMITTED, APPROVED, RETURNED, POSTED
entered_by_worker_id	uuid	yes	delegate when ≠ worker_id
lines[]	TimesheetLine	yes	(project_id, wbs_node_id, category_id, line_property, hours[7..31] by day, internal_comment, external_comment, source MANUAL/FAVORITE/TIME_REG, line_status, approver_id)
total_hours	numeric(8,2)	computed	
submitted_at, posted_at	timestamptz	no	
version	int	yes	optimistic concurrency

ProjectChangeOrder: id (sequence PROJ-CO), project_id, co_type (SCOPE_CHANGE, CLAIM_CUSTOMER, CLAIM_VENDOR, INTERNAL), title, description, status (DRAFT, SUBMITTED, APPROVED, REJECTED, APPLIED), lines[] (wbs_node_id, cost_impact, revenue_impact, effort_hours_delta, schedule_days_delta), customer_approval_ref, vendor_party_id (claims), applied_baseline_version_id, applied_at. **RetentionTerm**: id, party_role (CUSTOMER/VENDOR), retention_pct numeric(5,2), release_trigger (MILESTONE, PROJECT_FINISHED, DATE_OFFSET), release_schedule[] (milestone_wbs_id or offset_days, release_pct). **RetentionLedger**: project_id, party_id, party_role, source_doc_id, retained_amount, released_amount, currency. **IndirectCostComponent**: group_id, code, basis (HOUR_COST, HOUR_QTY, EXPENSE_COST), rate numeric(9,4), amount_per_unit, cost_category_id, sales_applicable bool. **IndirectCostRule**: component_id, legal_entity_id, project_group_id, category_id, worker_group_id, validity. **InterestProfile**: code, rate_source (FIXED, wave-12 rate table), annual_rate numeric(7,4), basis (WIP, COST), day_count (ACT_365, ACT_360), treatment (CAPITALIZE, EXPENSE). **PlanVersion**: id, project_id, kind (BASELINE, WORKING, SIMULATION), sequence, copied_from_version_id, locked bool. **ProjectForecast**: model_id, project_id, wbs_node_id, period_id, txn_type, qty, cost_amount, sales_amount. **ProjectRiskIssue**: id, project_id, kind (RISK, ISSUE), probability smallint, impact smallint, owner_worker_id, mitigation, due_date, status (OPEN, MITIGATED, CLOSED). **ProjectQuotation**: wave-27 quotation id, project_id (after conversion), estimate_lines[] (txn_type, role_id/category_id/item_id, qty, cost_price, sales_price).

Invariants: Σ ProjectTransaction.cost_amount per project equals the project cost subledger balance and the GL WIP+cost accounts for the posting profile; a transaction appears on at most one POSTED proposal; one POSTED estimate per (project, period); WBS codes unique per plan version. Indexing: `ProjectTransaction (tenant_id, project_id, txn_date)`, partial index on `proposal_id IS NULL AND line_property = 'CHARGEABLE'` for proposal selection; `WbsNode (plan_version_id, parent_id)`; table partitioned by tenant hash (16).

8.41.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/projects`	Create project	`Idempotency-Key` header	ProjMaintain
PUT	`/api/v1/projects/{id}/plans/{ver}/wbs`	Replace WBS (bulk)	version ETag	ProjPlan
POST	`/api/v1/projects/{id}/plans/{ver}/schedule`	Run CPM scheduler	pure	ProjPlan
POST	`/api/v1/projects/journals`	Create hour/cost/item/fee journal	key	ProjJournalMaintain
POST	`/api/v1/projects/journals/{id}/post`	Post journal to project and GL	key	ProjJournalPost
POST	`/api/v1/projects/invoice-proposals`	Build proposal from selection criteria	key	ProjBillingMaintain
POST	`/api/v1/projects/invoice-proposals/{id}/post`	Hand to invoicing	key	ProjBillingPost
POST	`/api/v1/projects/estimates/run`	Run estimates for period	key per (project, period)	ProjEstimatePost
POST	`/api/v1/projects/estimates/{id}/reverse`	Reverse estimate	key	ProjEstimatePost
GET	`/api/v1/projects/{id}/evm?asOf=`	EVM metrics	n/a	ProjInquire
POST	`/api/v1/projects/timesheets`	Create/copy timesheet for (worker, period)	key per (worker, period)	ProjTimesheetEnter
POST	`/api/v1/projects/timesheets/{id}/submit`	Submit for approval	key + version	ProjTimesheetEnter
POST	`/api/v1/projects/timesheets/{id}/lines/{lineId}/decide`	Approve/return a line	key	ProjTimesheetApprove
POST	`/api/v1/projects/timesheets/{id}/post`	Post approved lines as HOUR transactions	key	ProjJournalPost
GET	`/api/v1/projects/timesheets/missing?periodId=`	Missing timesheets	n/a	ProjInquire
POST	`/api/v1/projects/quotations/{id}/convert`	Convert WON project quotation to project	key	ProjMaintain
POST	`/api/v1/projects/change-orders`	Create change order / claim	key	ProjMaintain
POST	`/api/v1/projects/change-orders/{id}/apply`	Apply APPROVED change order (new baseline)	key	ProjChangeOrderApprove
POST	`/api/v1/projects/{id}/plans/{ver}/transfer`	Transfer SIMULATION into WORKING	key + ETag	ProjPlan
POST	`/api/v1/projects/interest/run`	Monthly interest calculation	key per (project, period)	ProjEstimatePost
POST	`/api/v1/projects/retention/release`	Build retention release proposal	key	ProjBillingMaintain
GET	`/api/v1/projects/{id}/forecasts/{modelId}`	Forecast vs actual	n/a	ProjInquire

Commands validate: journal lines must reference a project in IN_PROCESS; category type must match txn_type; posting date within open period for module PROJECTS (wave 14); proposal selection requires a billing rule whose currency equals project currency.

Event	Trigger	Payload fields	Consumers
`dev.passage.projects.transaction.posted.v1`	journal/document posted	see 8.0	journal, analytics, expense
`dev.passage.projects.invoice-proposal.posted.v1`	proposal → POSTED	proposalId, invoiceId, lines[]	invoicing (wave 29), analytics
`dev.passage.projects.estimate.posted.v1`	estimate posted or reversed	estimateId, projectId, periodId, pocPercent, revenueThisPeriod, journalId, reversed	journal, analytics

Event	Trigger	Payload fields	Consumers
`dev.passage.projects.wbs.scheduled.v1`	scheduler run	projectId, planVersionId, criticalPath[], finishDate	resource planning, notifications
`dev.passage.projects.timesheet.submitted.v1`	timesheet SUBMITTED	timesheetId, workerId, periodId, totalHours, projectIds[]	workflow (wave 4), notifications
`dev.passage.projects.change-order.applied.v1`	change order APPLIED	changeOrderId, projectId, contractValueDelta, budgetDelta, newBaselineVersionId	budget control (wave 38), analytics
`dev.passage.projects.retention.released.v1`	release proposal POSTED	projectId, partyId, releasedAmount, invoiceId	invoicing, AP

8.41.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Projects workspace	tiles: proposals to approve, estimates due, over-budget, unposted journals	project, stage, budget remaining, CPI/SPI	ProjInquire
Project detail	stage, copy, open WBS, forecasts, statements	header fields, funding, posting profile	ProjMaintain
WBS Gantt	add/indent/outdent, link dependencies, schedule, baseline	code, name, effort, dates, slack, role, % complete	ProjPlan
Resource board	book, split, resolve over-allocation	worker/role × week heatmap	ProjPlan
Journals (hour/cost/item/fee)	add line, validate, post	project, WBS, category, qty, cost, sales, line property	ProjJournalMaintain
Invoice proposals	build, review, remove line, approve, post	transaction rows, totals, funding split	ProjBillingMaintain
Estimates	run, review, reverse, post	cost to date, ETC, POC, revenue, WIP	ProjEstimatePost
Project statements / EVM	drill to transaction and journal	PV/EV/AC, CPI, SPI, EAC, margin	ProjInquire
My timesheet (ESS)	add favorite, copy previous period, enter hours per day, comment, submit; delegate switch	project, WBS, category, Mon–Sun hours, period total, status per line	ProjTimesheetEnter
Timesheet approval (MSS / project manager)	approve/return per line or whole sheet, view comments	worker, project, hours, line property	ProjTimesheetApprove
Change orders	create, add impact lines, attach approval, submit, approve, apply	type, WBS, cost/revenue impact, status	ProjMaintain / ProjChangeOrderApprove
Plan version compare	pick two versions, diff dates/effort/cost, transfer simulation	node, Δ start, Δ finish, Δ effort, Δ cost	ProjPlan
Forecasts, risks and issues	add forecast lines by period, copy to budget, add risk/issue	model, period, qty, cost, sales; probability, impact, owner	ProjPlan
Retention ledger	view retained/released by party, build release	party, source invoice/PO, retained, released, open	ProjBillingMaintain

Interaction rules: Gantt supports keyboard indent (Tab), outdent (Shift+Tab), link (Ctrl+L); bulk edit of category/line property on journal lines; validation on line commit with inline errors; proposal grid personalization (column set saved per user).

8.41.7 Business rules and edge cases

- **BR-41.1 Cost-to-cost POC** — poc = cost_to_date ÷ estimated_total_cost, rounded to 4 decimals; revenue_to_date = contract_value × poc, rounded to currency minor unit; revenue_this_period = revenue_to_date − prior revenue_to_date. Worked example (USD, fixed-price, contract 500,000.00, original estimate 400,000.00): period 1 cost to date 120,000.00, ETC 280,000.00 → EAC 400,000.00, poc 0.3000, revenue to date 150,000.00, billed to date 100,000.00 → accrued revenue 50,000.00 (debit accrued revenue, credit revenue). Period 2 cost to date 260,000.00, ETC revised to 190,000.00 → EAC 450,000.00, poc 0.5778 (260,000/450,000 = 0.57777…), revenue to date 288,888.89, period revenue 138,888.89, billed to date 300,000.00 → deferred (billed in excess) 11,111.11 as liability; gross profit to date 28,888.89. A EUR project uses the same arithmetic in EUR with accounting-currency amounts at the period-end rate.
- **BR-41.2 Loss provision** — if EAC > contract_value, the full expected loss (EAC − contract_value) is provisioned in the period discovered; later improvements release the provision, never below zero.
- **BR-41.3 Invoice proposal selection** — a transaction is selected when: line_property = CHARGEABLE, proposal_id IS NULL, txn_date ∈ [from, to], project stage ∈ {IN_PROCESS, FINISHED}, txn_type ∈ rule.types, and for FIXED_PRICE rules only ON_ACCOUNT milestones with is_billable_milestone WBS nodes whose % complete = 100 or whose date ≤ to. Transactions under a funding source whose limit is reached are selected on the next source by priority; if none, the line is HELD with reason `FUND-LIMIT`.
- **BR-41.4 Retainers** — on-account (retainer) invoices create ON_ACCOUNT transactions with negative sales; T&M proposals apply open retainer balance before invoicing net, oldest first.
- **BR-41.5 Price precedence and rounding** — hour sales price resolved per FR-41.3; amounts = qty × price rounded half-even to minor unit per line, never per total.
- **BR-41.6 Period control** — a transaction with txn_date in a closed PROJECTS period is rejected (`PROJ-PERIOD-001`); estimates run only after the period's projects sub-ledger is closed for transactions.
- **BR-41.7 Stage FINISHED** — allowed only when no open journals, no DRAFT/APPROVED proposals, WIP balance zero or an estimate with method COMPLETED_CONTRACT posted for the final period.
- **BR-41.8 Concurrency** — proposal building takes a `SELECT … FOR UPDATE SKIP LOCKED` on candidate transactions; two concurrent builds for overlapping criteria never select the same transaction.
- **BR-41.9 Timesheet validation** — one timesheet per (worker, period); the period must be OPEN and every line date inside it; Σ hours per calendar day across lines ≤ 24.00 (`PROJ-TS-003`), hours in 0.25 increments by default (parameter); the project must be IN_PROCESS (`PROJ-TS-002`); a delegate must hold a valid TimesheetDelegate row (`PROJ-TS-004`); approval routes each line to the project manager of its project (parameter "workflow by project", else to the worker's manager as-of period end per BR-42.1); a sheet is APPROVED when all lines are approved; posting creates one HOUR transaction per line per day with source TIMESHEET, idempotency key `projects:ts:{timesheetId}:{lineId}:{date}`; returned lines reopen the sheet as RETURNED; lines with source TIME_REG are read-only and pre-approved.
- **BR-41.10 Retention arithmetic** — retained amount per invoice line = round_half_even(line amount × retention_pct); retention is excluded from the invoice due amount and recorded in RetentionLedger; release is triggered by the release schedule (milestone WBS node reaches 100 % complete, project stage FINISHED, or offset days after invoice) and billed as RETENTION_RELEASE lines on a release proposal; vendor retention is withheld at wave-18 invoice posting as a separate due line whose due date is the release trigger date; a release before its trigger raises `PROJ-RET-001` unless duty ProjBillingPost overrides with a reason.
- **BR-41.11 Indirect cost application** — at HOUR posting, for each component of the project's component group the most specific matching rule (worker group > category > project group > legal entity) applies: amount = basis × rate (HOUR_COST: cost amount; HOUR_QTY: hours × amount_per_unit) rounded half-even per line; posted as INDIRECT transactions with the same WBS node and date, line property NON_CHARGEABLE unless component.sales_applicable, and with `source_doc_id` = the hour transaction; reversing the hour transaction reverses its indirect lines.

- **BR-41.12 Project interest** — monthly interest = average daily balance of the basis (WIP or cost) × annual_rate × days_in_month ÷ 365 (ACT/365) or ÷ 360 (ACT/360), rounded half-even to the minor unit; CAPITALIZE debits WIP (or CIP via wave 37 for INVESTMENT projects) and credits capitalized-interest contra; EXPENSE debits project interest expense; capitalization stops when the project reaches FINISHED or the asset is placed in service; reversal follows the estimate reversal of the same period. Example: average WIP 1,000,000.00 USD, 6.00 %, October (31 days), ACT/365 → 5,095.89.
- **BR-41.13 Change order application** — only APPROVED change orders apply; application is atomic: contract_value += Σ revenue_impact, project budget (wave 38) += Σ cost_impact, a new BASELINE plan version is created from the current baseline plus effort/schedule deltas, and the change order becomes APPLIED and immutable (`PROJ-CO-001` on a second apply); CLAIM_VENDOR creates a wave-18 vendor debit note draft for Σ cost_impact; CLAIM_CUSTOMER requires `customer_approval_ref` when the parameter "customer approval required" is on.
- **BR-41.14 Simulation transfer** — transfer computes a three-way diff (common ancestor = copied_from version, SIMULATION, current WORKING); nodes changed only in the simulation are applied; nodes changed in both since the copy are listed as conflicts (`PROJ-SIM-001`) and must be resolved node by node (keep working / take simulation); after transfer the simulation is locked; at most N simulation versions per project (parameter, default 20).

Estimate state machine: DRAFT → POSTED → REVERSED; REVERSED → (new DRAFT for same period). Proposal: DRAFT → APPROVED → POSTED; DRAFT/APPROVED → CANCELED.

8.41.8 Acceptance criteria

- **AC-41.1** (FR-41.7, BR-41.1) Given the fixture project in BR-41.1, when estimates run for periods 1 and 2, then revenue_this_period is 150,000.00 and 138,888.89 and the posted journal balances per the posting profile.
- **AC-41.2** (FR-41.7, BR-41.2) Given EAC revised to 520,000.00 in period 3, when the estimate runs, then loss_provision = 20,000.00 and the GL loss-provision account carries 20,000.00.
- **AC-41.3** (FR-41.6, BR-41.3) Given 10,000 generated transactions and a T&M rule for 2026-09, when a proposal builds, then every selected line satisfies the predicate and no qualifying line is omitted (property test).
- **AC-41.4** (FR-41.2) Given a WBS with FS dependency and lag 2 days on a role calendar closed on weekends, when scheduled, then successor early start is the next working day after lag.
- **AC-41.5** (FR-41.4, FR-41.5) Given an hour journal line with category LABOR and line property CHARGEABLE, when posted, then the journal debits WIP-cost and credits payroll allocation per profile and the transaction's journal_id is set.
- **AC-41.6** (FR-41.11) Given a PO line referencing WBS 1.2, when the receipt posts, then an ITEM transaction exists and committed cost decreases by the received amount within 10 s.
- **AC-41.7** (FR-41.9) Given two funding sources 70/30 with a 10,000.00 limit on source B, when proposals exceed the limit, then the excess line is HELD with reason FUND-LIMIT.
- **AC-41.8** (FR-41.10) Given BAC 400,000, poc 0.30, AC 120,000 and PV 100,000, when EVM is queried, then EV = 120,000, CPI = 1.00, SPI = 1.20.
- **AC-41.9** (FR-41.13, BR-41.9) Given a worker with two favorites and WEEK period 2026-10-19 to 2026-10-25, when periodic creation runs, then a DRAFT timesheet with two zero-hour lines exists; when 25.00 h are entered on 2026-10-21, then submit is rejected with `PROJ-TS-003`; when 8.00 h per weekday are approved and posted, then 10 HOUR transactions exist with source TIMESHEET.
- **AC-41.10** (FR-41.14, BR-41.10) Given a customer retention term of 10 % released at PROJECT_FINISHED and a T&M proposal of 20,000.00 USD, when posted, then the invoice due amount is 18,000.00 and the retention ledger shows 2,000.00 retained; when the project reaches FINISHED and the release proposal posts, then a RETENTION_RELEASE invoice of 2,000.00 exists.
- **AC-41.11** (FR-41.15, BR-41.11) Given component OVH-LABOR at 12 % of HOUR_COST for category LABOR, when an hour line of 8.00 h × 50.00 posts, then an INDIRECT transaction of 48.00 exists on the same WBS node, and reversing the hour line reverses it.
- **AC-41.12** (FR-41.18, BR-41.13) Given an APPROVED SCOPE_CHANGE with +50,000.00 revenue and +30,000.00 cost on WBS 1.3, when applied, then contract_value increases by 50,000.00, the wave-38 budget by 30,000.00, a new BASELINE version exists and a second apply returns `PROJ-CO-001`.
- **AC-41.13** (FR-41.16, BR-41.12) Given average WIP 1,000,000.00 USD and profile 6.00 % ACT/365 CAPITALIZE, when the October 2026 interest run posts, then 5,095.89 debits WIP and the estimate for October includes it in cost_to_date.
- **AC-41.14** (FR-41.12) Given a WON project quotation with 3 estimate lines, when converted, then a project in CREATED with a BASELINE plan version of 3 nodes exists and a second conversion returns `PROJ-QUO-001`.
- **AC-41.15** (FR-41.17, BR-41.14) Given a SIMULATION copied from WORKING where node 1.2 was later moved in WORKING and node 1.4 only in the simulation, when transferred, then 1.4 is applied, 1.2 is reported as a conflict, and the simulation is locked after resolution.

8.41.9 Test plan and fixtures

Fixtures: `proj-estimates-12x4` (12 projects × 4 periods, accountant workbook `fixtures/projects/poc-workbook.xlsx`); `proj-txn-gen` (Hypothesis generator, 10,000 transactions, seeds committed); `proj-wbs-2000` (2,000-node WBS with 3,500 dependencies); `es-us-demo` (demo entity with 25 live projects derived from wave-29 customers).

Test	Type	Target	Pass criterion
POC arithmetic	unit	estimate engine	matches workbook to 0.01
Proposal selection	property	selection predicate	0 false positives/negatives over 10k
CPM scheduling	unit/perf	scheduler	correct dates; 2,000 nodes < 500 ms
Posting profile resolution	contract	posting API	every (group, category, property) in fixture resolves
Journal → GL	E2E	journals, journal service	balanced entries; event emitted
EVM	unit	report service	CPI/SPI per formulas
Proposal build concurrency	integration	two parallel builds	disjoint selections

Reconciliation: Σ transactions cost by project = project cost subledger = GL cost/WIP accounts; Σ POSTED proposal sales = invoiced revenue per wave-29; accrued – deferred revenue = GL accrued/deferred accounts.

8.41.10 Deployment and operations

Helm chart `passage-projects` (API, scheduler worker, estimate worker), Postgres cluster `projects` (CloudNativePG, 3 replicas), Atlas migration `projects/0001_initial`. Config: `PROJ_SCHEDULER_MAX_NODES=5000`, `PROJ_PROPOSAL_BATCH=500`. Feature flags: `projects.intercompany-hours`, `projects.funding-limits`. Temporal schedules: nightly `projects-reconcile` 02:00 tenant-local, `committed-cost-refresh` every 60 s. SLOs: journal post p95 < 400 ms for 100 lines; proposal build p95 < 5 s for 5,000 candidates; availability 99.9%. Alerts: reconciliation variance ≠ 0; estimate run failure; outbox lag > 60 s. Runbook: re-run estimate (reverse then run), rebuild committed cost projection from events. Rollback: Argo digest rollback; migrations are additive within the wave.

8.41.11 Migration from AX / SAP

Source	Passage entity	Notes
ProjTable, ProjGroup, ProjType	Project, ProjectGroup	Status map: Created→CREATED, InProcess→IN_PROCESS, Finished→FINISHED
ProjWBS, ProjPlanVersion	WbsNode, PlanVersion	AX activity numbers → wbs_code
ProjEmplTrans, ProjCostTrans, ProjItemTrans, ProjRevenueTrans, ProjOnAccTrans	ProjectTransaction	open (unbilled) only; billed history as read-only archive
ProjPosting, ProjCategory, ProjLinePropertySetup	ProjectPostingProfile, ProjectCategory, line property	
ProjEstimate, ProjPeriodTable	ProjectEstimate	last posted estimate per project as opening balance
PROJ, PRPS, AUFK/AFVC	Project, WbsNode	PRPS.POSID → wbs_code; networks flattened to nodes
CATSDB (released, unbilled)	ProjectTransaction HOUR	
KKA2 RA results (COSB)	ProjectEstimate opening	RA version 0 only

Validation: per-project cost to date, billed to date and WIP must equal source within 0.01; count of unbilled chargeable transactions equal. Not migrated: billed history detail (archived in lakehouse, wave 46), ProjForecast models older than current year, SAP settlement rules (CJ88) — replaced by posting profiles.

8.41.12 Dependencies, risks, open questions

Depends on: wave 11 posting API, wave 14 periods, wave 17 PO project references, wave 25 timesheets, wave 29 invoicing, wave 37 CIP, wave 38 budget control, wave 40 intercompany. Coverage-closure additions also depend on: wave 27 quotations (FR-41.12), wave-16 ownership dimension `project_id` and wave-22 planning scope PROJECT (FR-41.19), wave-28 shipments (FR-41.19), wave 18 vendor invoices for vendor retention and claims (FR-41.14, FR-41.18), wave 12 rate tables (FR-41.16) and wave-4 workflow for timesheet routing (FR-41.13).

Risk	Likelihood	Impact	Mitigation
POC method disputes	Low	High	Workbook sign-off before wave close
Proposal performance on large backlogs	Medium	Medium	Partial index + batch 500

Open questions: (1) Support POC by effort (hours) in v1? Default no; decide by wave 42. (2) Grant compliance reporting (AX ProjGrant)? Default v1.x; decide by wave 46. (3) Intercompany resource hours at transfer price (wave 40 mirror documents)? Default v1.x; decide by wave 42.

8.41.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Default project group per project type	legal entity	lookup	—	ProjectGroup	Pre-fills group and posting profile on project create	Project management and accounting parameters › General › Default project group (ProjParameters)	OPSA project profile (TCJ41) default values
Journal approval required (HOUR / COST / ITEM / FEE)	legal entity	bool per type	false		Routes journal post through wave-4 workflow before FR-41.4 posting	PM&A parameters › Journals › Approval required	CATS approval profile (CAC2) for hours; n/a for KB21N
Timesheet period type	legal entity	enum	WEEK	WEEK, SEMI_MONTH, MONTH	Generates TimesheetPeriod rows (FR-41.13)	PM&A parameters › Timesheet › Period type; Timesheet periods (ProjPeriodTable)	CATS data entry profile period (CAC1)
Timesheet workflow by project	legal entity	bool	true		Line approval routed to project manager (BR-41.9)	PM&A parameters › Timesheet › Enable workflow by project	CATS_APPR_LITE approver determination by WBS
Timesheet hour increment	legal entity	numeric	0.25	0.01, 0.10, 0.25, 0.50, 1.00	Validation on line hours	PM&A parameters › Timesheet › Hour increments	CAC1 "rounding of hours"
Timesheet max hours per day	legal entity	numeric	24.00	1.00–24.00	BR-41.9 `PROJ-TS-003`	PM&A parameters › Timesheet › Max hours per day	CAC1 "max. hours per day"
Allow timesheet delegates	tenant	bool	true		Enables TimesheetDelegate (FR-41.13)	PM&A parameters › Timesheet › Allow delegates (TSTimesheetDelegate)	CATS substitutes (CATS_DELEGATE / HR substitution)
Default line property	legal entity	lookup	CHARGEABLE	CHARGEABLE, NON_CHARGEABLE	Pre-fills line property by (project group, category)	Line property setup (ProjLinePropertySetup)	DIP profile ODP1 material determination / billing relevance
Proposal selection includes FINISHED projects	legal entity	bool	true		Stage set in BR-41.3	PM&A parameters › Invoice › Include finished projects	DP90 selection variant
Estimate method default	project group	enum	POC_COST	COMPLETED_CONTRACT, POC_COST, POC_MILESTONE	FR-41.7 default on new projects	Project groups › Estimate › Completion method (ProjGroup)	OKG1 RA method per RA key; OKG2 RA version
Estimate cost template	legal entity	lookup	—	ProjectCostTemplate	Which categories count toward cost_to_date and ETC	Cost templates (ProjCostTemplate)	OKG4 line IDs / OKG5 assignment of cost elements
Committed cost sources	legal entity	multi-enum	PO, REQUISITION, ITEM_REQUIREMENT	+ VENDOR_INVOICE_UNPOSTED	FR-41.8 projection inputs	PM&A parameters › Cost control › Committed cost sources	OPSA "commitment management" indicator; OKKP activate commitments
Committed cost refresh fallback interval	tenant	int (s)	60	10–600	Fallback poll when the event stream lags	"Update committed costs" batch recurrence	CJI5 batch variant
Customer retention default term	legal entity	lookup	—	RetentionTerm	FR-41.14 default on project contracts	Project contracts › Retention terms (ProjRetentionTerms)	OBB9 installment terms / S/4 PO retention settings

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Vendor retention default term	legal entity	lookup	—	RetentionTerm	Applied to project POs (wave 17)	Vendor retention terms (ProjVendRetentionTerms)	S/4 MM retention (ME21N Retention tab)
Indirect cost component group	legal entity / project group	lookup	—	IndirectCostComponentGroup	FR-41.15 / BR-41.11	PM&A parameters › Indirect costs › Default component group	KZS2 costing sheet in OPSA project profile
Interest profile	project group / project	lookup	—	InterestProfile	FR-41.16	n/a	OPIB/OPIC/OPID interest profile in project profile
Interest day count	interest profile	enum	ACT_365	ACT_365, ACT_360	BR-41.12	n/a	OPIC interest indicator calendar type
Scheduling with resource calendars	legal entity	bool	true		FR-41.2 calendar enforcement	PM&A parameters › Scheduling › Scheduling with resources	OPUU network profile scheduling parameters
Max simulation versions per project	tenant	int	20	1–100	BR-41.14	n/a	OPTS simulation profile
Change order customer approval required	legal entity	bool	true		BR-41.13 for CLAIM_CUSTOMER	n/a	TQ80 claim type (notification type CL) settings
Project stock valuation	project group	enum	VALUATED	VALUATED, NON_VALUATED, NONE	FR-41.19	Item requirements consumption setting (ProjParameters › Item requirements)	OPS project stock (TCJ41 valuated project stock)
Funding limit hold	legal entity	bool	true		BR-41.3 `FUND-LIMIT`	Project contracts › Funding limits	n/a (GM budget)
Number sequences	legal entity	lookup	PROJ, PROJ-TS, PROJ-PROP, PROJ-EST, PROJ-CO	wave-5 sequences	Identifiers	PM&A parameters › Number sequences	OPSJ project number ranges; OPSK project coding mask

8.41.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Projects list	list inquiry	stage, type, group, customer, project manager, legal entity	id, name, stage, budget, actual, committed, remaining, CPI, SPI	`proj_summary`	Projects list page (ProjTable)	CN41N structure overview
Project statement	drill-down	project (set), date range, cost/revenue/hours/WIP tabs	cost by category, revenue invoiced/accrued, hours, WIP, gross margin	`proj_statement`	Project statements (ProjStatement); SSRS Project statements	S_ALR_87013531 / 87013532 / 87013533 cost-revenue-commitment; CJI3 line items
Posted transactions	list inquiry	project, WBS, type, category, date, worker, source document	transaction fields, journal id, proposal id, estimate id	`proj_txn`	Posted transactions (ProjTransPosted), Hour/Cost/Item/Revenue transactions forms	CJI3 actual line items; CJI5 commitment line items
Invoice control	drill-down	project, customer, period	proposed, invoiced, retainer applied, retention held, open to bill	`proj_billing`	Invoice control (ProjInvoiceControl); Project invoice journals (ProjInvoiceJour)	DP90 billing request list; VF05 billing documents by WBS
WIP and estimates	drill-down	project group, period, method	cost to date, ETC, EAC, POC, revenue to date, accrued/deferred, loss provision	`proj_estimate`	Estimates (ProjEstimate); SSRS Project WIP	KKA2 / KKAJ results analysis; S_ALR_87013537 WIP
Project cash flow	operational report	project, horizon, forecast model	expected receipts (proposals, retention release), payments (POs, timesheets), net by period	`proj_cashflow`	Cash flow (ProjCashFlow); SSRS Project cash flow	S_ALR_87013567 project cash flow
Budget vs actual vs committed	drill-down	project, WBS, period, dimension	budget, actual, committed, remaining, % consumed	`proj_budget` with wave-38	Project budget balances; Committed costs (ProjCommittedCost)	S_ALR_87013558 budget/actual/commitment; CJ31 budget display
EVM	KPI	project, as-of date, WBS level	PV, EV, AC, CPI, SPI, EAC, VAC	`proj_evm`	none native (BI cube)	CNE5 progress analysis; CNE1 progress determination
Missing timesheets	list inquiry	period, legal entity, department	worker, period, status, days missing, manager	`ts_missing`	Missing timesheets (TSMissingTimesheets); SSRS Timesheet report	CATC time leveling; CADO display working times
Timesheet hours	list inquiry	worker, project, period, line status	hours by day, line property, approver, posted	`ts_hours`	Timesheet hours inquiry (TSTimesheetTable)	CATS_DA display CATS data
Utilization	KPI	role, worker, department, period	billable hours, non-billable, available, utilization %	`proj_utilization`	Hour utilization (ProjUtilization); SSRS Utilization	CMP9 workforce planning evaluation
Retention ledger	list inquiry	project, party, role, open only	retained, released, open, next release trigger	`proj_retention`	Retention transactions (ProjRetentionTrans)	FBL5N / FBL1N with retention special G/L
Change order log	list inquiry	project, type, status	change order, type, cost impact, revenue impact, applied baseline	`proj_change_order`	n/a	CLM10 claim list
Forecast vs actual	drill-down	project, model, period	forecast qty/cost/sales, actual, variance	`proj_forecast`	Project forecasts (ProjForecastEmpl/Cost/Item)	CJ40/CJ42 cost planning vs actual (S_ALR_87013543)
Risks and issues	list inquiry	project, kind, status, owner	probability, impact, score, owner, due	`proj_risk_issue`	n/a	CPM risk/issue list

8.41.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Project invoice	invoice proposal POSTED → wave-29 invoice	header: customer, project, contract, period, billing rule; lines: transactions (detail or summary per rule), retainer applied, retention held; footer: tax by code, totals, payment terms, bank details	PDF, e-invoice (wave-29 country formats), email	wave-29 `INV`	Fiscal invoice; retention per wave-29 country rule (US 7 years, DE 10 years)
Invoice proposal (pre-bill review)	proposal DRAFT/APPROVED	same as invoice plus "why not billed" exclusions list	PDF, XLSX, email	`PROJ-PROP`	none (internal)
Retention release invoice	release proposal POSTED	header: project, original invoices; lines: RETENTION_RELEASE per source invoice	PDF, e-invoice, email	wave-29 `INV`	fiscal invoice
Project quotation	wave-27 quotation SENT	header: customer, validity; lines: estimate lines by type, totals, margin hidden	PDF, email	wave-27 `QUO`	none
Timesheet	print from sheet or approval	header: worker, period, status; lines: project/WBS/category × days; footer: totals, approvals with timestamps	PDF	`PROJ-TS`	labor record: FLSA 3 years (US), ArbZG §16 2 years (DE)
Project statement	on demand	cost/revenue/hours/WIP sections per filter	PDF, XLSX	none	none
Change order (customer-facing)	change order SUBMITTED	header: project, contract, change order no; lines: scope, cost/revenue impact, schedule impact; footer: signature blocks	PDF, email, e-signature link (wave 35 hub)	`PROJ-CO`	contract amendment; keep with contract (10 years DE HGB §257 for commercial letters)
Estimate voucher	estimate POSTED	wave-11 journal voucher with estimate id, POC, revenue, WIP	PDF	wave-11 voucher	accounting record
Project purchase order	wave-17 PO with project reference	wave-17 PO layout plus project/WBS and vendor retention clause	PDF, EDI, email	wave-17 `PO`	per wave 17

8.41.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`projects-reconcile`	02:00 tenant-local daily	tenant, legal entity	run id per (tenant, date); re-run replaces results	< 10 min for 2,000 projects	retry 3×; variance report written on failure	variance ≠ 0.00 → page project accountant; run missing by 03:00
`committed-cost-refresh`	event-driven; fallback every 60 s	tenant, project	Kafka consumer offset; projection rebuild from events is idempotent	< 10 s lag	consumer restart; rebuild on checksum mismatch	lag > 60 s
`projects-estimate-run`	on demand at period close (Temporal workflow per project)	project, period	key (project, period); reverse-and-rerun in one workflow	< 10 min for 2,000 projects	workflow retry per project; failed projects listed, others continue	any project failed; run > 30 min
`timesheet-periodic-create`	period start 00:00 tenant-local (Monday for WEEK)	legal entity, worker	key (worker, period)	< 2 min for 10,000 workers	per-worker retry	created count < active workers – exemptions
`timesheet-missing-reminder`	day after period end 09:00; repeat daily until submitted	legal entity	idempotent per (worker, period, day)	< 1 min	notification retry via wave 35	reminders undelivered
`projects-interest-run`	monthly after period close	project	key (project, period)	< 5 min for 2,000 projects	per-project retry	missing interest for projects with a profile
`retention-release-check`	nightly 03:00	project	evaluates triggers; creates release proposals once per trigger	< 2 min	retry	proposals not created for FINISHED projects with open retention
`projects-schedule-recalc`	nightly 01:00 for projects whose calendars changed; on demand	project	pure recompute	< 500 ms per 2,000 nodes	retry	runtime > 5 s per project
`projects-forecast-rollover`	first day of fiscal period	project, model	key (model, period)	< 1 min	retry	none rolled

8.41.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
ProjMaintain	POST/PUT `/api/v1/projects`, change orders create, quotation convert; screens Project detail, Change orders	with ProjChangeOrderApprove on the same change order; with ProjSetupAdmin	Project manager, Project administrator
ProjPlan	WBS replace, schedule, plan version transfer, forecasts, risks; screens WBS Gantt, Resource board, Plan version compare	none	Project manager, Resource manager
ProjJournalMaintain	journals create/edit; screens Journals	with ProjJournalPost	Project accountant, Project coordinator
ProjJournalPost	journals post, timesheet post	with ProjJournalMaintain; with ProjTimesheetEnter on own sheet	Project accountant
ProjBillingMaintain	proposals build/edit, retention release build; screens Invoice proposals, Retention ledger	with ProjBillingPost	Project accountant
ProjBillingPost	proposals post, release post	with ProjBillingMaintain; with ProjMaintain (contract value)	Project controller
ProjEstimatePost	estimates run/reverse/post, interest run	with ProjSetupAdmin (posting profiles)	Project controller
ProjTimesheetEnter	timesheets create/submit for self or delegated worker; screen My timesheet	with ProjTimesheetApprove on own sheet	Employee (ESS), Contractor
ProjTimesheetApprove	timesheet line decide; screen Timesheet approval	with ProjTimesheetEnter (own)	Project manager, Line manager (MSS)
ProjChangeOrderApprove	change order approve/apply	with ProjMaintain on the same change order (creator ≠ approver)	Project controller, Account executive
ProjInquire	all GET inquiries and reports	none	Project manager, Project accountant, Executive
ProjSetupAdmin	parameters (8.41.13), posting profiles, categories, retention terms, indirect cost components, interest profiles	with ProjEstimatePost, ProjJournalPost	Project administrator

8.41.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
PROJ-POST-001	error	No posting profile for project group {group}, category {category}, line property {property}	Resolution failed at all fallback levels	Add a profile row or a wildcard fallback	FR-41.5
PROJ-PERIOD-001	error	Transaction date {date} is in a closed PROJECTS period	Period closed in wave 14	Change the date or reopen the period	BR-41.6
PROJ-STAGE-001	error	Project {id} cannot move from {from} to {to}	Transition not allowed	Use an allowed transition	FR-41.1
PROJ-STAGE-002	error	Project {id} cannot be FINISHED: {reasons}	Open journals, proposals or non-zero WIP	Post or cancel the listed items; post a final estimate	BR-41.7
PROJ-WBS-001	error	Dependency {a} → {b} creates a cycle	Circular dependency	Remove one dependency	FR-41.2
PROJ-WBS-002	error	WBS code {code} already exists in plan version {ver}	Duplicate code	Change the code	FR-41.2
PROJ-BOOK-001	error	Hard booking exceeds 100 % on {date} for {resource}	Over-allocation	Reduce hours or use a soft booking	FR-41.3
PROJ-BOOK-002	warning	Soft booking exceeds 100 % on {date} for {resource}	Over-allocation	Review on the resource board	FR-41.3
PROJ-PROP-001	error	Billing rule currency {a} differs from project currency {b}	Mismatch	Pick a rule in the project currency	FR-41.6
PROJ-PROP-002	info	No billable transactions matched the selection	Empty candidate set	Widen the date range or check line properties with "why not billed"	FR-41.6
FUND-LIMIT	warning	Line held: funding source {source} limit {limit} reached	Funding limit	Raise the limit or add a source	BR-41.3
PROJ-EST-001	error	A POSTED estimate already exists for {project} in period {period}	Duplicate run	Reverse it first (or use reverse-and-rerun)	FR-41.7
PROJ-EST-002	error	Projects sub-ledger for period {period} is not closed for transactions	Sequence	Close the sub-ledger period	BR-41.6
PROJ-TS-001	error	Timesheet period {period} is closed	Period CLOSED	Ask the administrator to reopen	BR-41.9
PROJ-TS-002	error	Project {id} is not IN_PROCESS	Stage check	Pick another project or change stage	BR-41.9
PROJ-TS-003	error	Hours on {date} total {hours}, more than {max}	Daily cap	Correct the hours	BR-41.9
PROJ-TS-004	error	You are not a delegate for {worker} on {date}	Missing or expired delegation	Ask the worker or administrator to add a delegate row	BR-41.9
PROJ-TS-005	warning	Hours on line {line} are not a multiple of {increment}	Increment parameter	Round the entry	8.41.13
PROJ-QUO-001	error	Quotation {id} has already been converted to project {project}	Second conversion	Open the existing project	FR-41.12
PROJ-CO-001	error	Change order {id} is already APPLIED	Second apply	None required	BR-41.13
PROJ-CO-002	error	Customer approval reference required for a customer claim	Parameter on	Attach the approval evidence	BR-41.13
PROJ-RET-001	error	Retention for {invoice} cannot be released before its trigger ({trigger})	Early release	Wait for the trigger or override with ProjBillingPost and a reason	BR-41.10
PROJ-INT-001	warning	Project {id} has no interest profile; skipped	No profile	Assign a profile on the project or group	FR-41.16
PROJ-SIM-001	warning	{n} nodes changed in both WORKING and simulation {ver}; resolve conflicts	Three-way diff conflicts	Resolve node by node	BR-41.14
PROJ-STOCK-001	error	Insufficient project stock for item {item} on project {project}	Project-owned on-hand short	Transfer to project stock or change ownership	FR-41.19

8.41.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Journal post (100 lines)	2,000 journals/day per tenant; peak 50/min	150 / 400 / 800 ms	200 lines/s sustained	ProjectTransaction ≈ 1.2 KB × 5 M rows = 6 GB	partitioned by tenant hash (16); (tenant, project, txn_date)	queue journals; posting API back-pressure; no partial posts
Timesheet submit + line approval	10,000 workers × 1 sheet/week; peak Monday 09:00 2,000/min	100 / 300 / 600 ms	100 submits/s	Timesheet + lines ≈ 3 GB	(worker_id, period_id) unique; lines by timesheet	submit accepted, workflow routing lags; UI shows "pending routing"
Invoice proposal build (5,000 candidates)	500 proposals/month	1 / 5 / 10 s	20 builds in parallel	small	partial index proposal_id IS NULL AND CHARGEABLE	batch of 500 per transaction; longer builds, same result
Estimate run	2,000 projects per period	per project 50 / 300 / 800 ms; run < 10 min	10 workflows in parallel	ProjectEstimate ≈ 50 MB	(project_id, period_id) unique	Temporal rate-limits; run completes later
CPM scheduler	2,000 nodes, 3,500 dependencies	120 / 500 / 900 ms	5 runs/s	PlanVersion copies ≈ 200 MB	(plan_version_id, parent_id)	reject > 5,000 nodes (`PROJ_SCHEDULER_MAX_NODES`)
EVM / statement as-of query	5,000 queries/day	80 / 250 / 500 ms	50 q/s	projections ≈ 1 GB	materialized daily rollup + delta	serve rollup without delta, flagged stale
Committed cost projection update	20,000 PO events/day	1 / 5 / 10 s lag	100 events/s	≈ 300 MB	(project_id, wbs_node_id)	consumer lag alert; projection remains consistent
Interest run	2,000 projects monthly	run < 5 min	—	negligible	—	later completion
Change order apply	200/month	200 / 600 / 1,200 ms	—	negligible	—	serialized per project

8.41.20 Localization and compliance hooks

- Revenue recognition: POC_COST and POC_MILESTONE map to the input and output methods of ASC 606 / IFRS 15; COMPLETED_CONTRACT remains for tax books under local GAAP (e.g. DE HGB §252 realization principle). The method is a per-project-group parameter (8.41.13) and a ledger-specific override is exposed through the wave-11 multi-ledger posting (statutory vs IFRS ledger).
- Capitalized interest: IAS 23 (mandatory capitalization on qualifying assets) vs ASC 835-20; the interest profile treatment and the stop rule ("placed in service") are per legal entity; wave-48 country packs set the default profile.
- Retention: statutory retainage caps and release deadlines vary — US state prompt-payment acts (5–10 % caps, release within 30–45 days of substantial completion), DE VOB/B §17 Sicherheitseinbehalt (5 %, bank guarantee substitution), FR retenue de garantie (5 %, 1 year); the RetentionTerm model carries

`max_pct` and `release_deadline_days` that wave-48 packs populate; extension point `projects.retention.validate.<cc>`.

- Project invoices follow wave-29 e-invoicing per country (DE XRechnung/ZUGFeRD, IT SDI FatturaPA, FR Factur-X, US none); wave 41 only supplies line content and the project reference tags (buyer reference / contract number per EN 16931).
- Timesheet records retention: FLSA 29 CFR 516 (3 years payroll, 2 years time cards) in the US; EU Working Time Directive implementations (DE ArbZG §16: 2 years; FR Code du travail: 1 year for durations); retention class `LABOR_TIME` in the wave-5 retention policy with per-country durations from wave 48.
- Calendars and holidays for scheduling come from wave-8 calendars per site/country; timesheet period boundaries follow the legal entity's calendar week (ISO 8601 Monday start by default, Sunday start where the pack says so, e.g. US).
- Currency rounding: per-line rounding to the minor unit (JPY 0 decimals, CHF cash rounding to 0.05 only on printed totals per wave 29); accounting-currency amounts at the period-end rate for estimates (wave 12 rate types `CLOSING` for WIP, `AVERAGE` for revenue when the pack prescribes it).
- Language: category names, line properties and billing-rule descriptions are translatable through wave-3 metadata registry labels; invoice line texts print in the customer's language (wave 7 party language).
- Privacy: timesheet comments and worker names on external invoice detail are redacted when the customer contract flag `redact_worker_names` is set (GDPR data minimization); worker ids never print.

8.41.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX #88 — 12.11.1 Project quotations (SalesQuotationTable type Project, transfer to project)	PARTIAL	ADDED as FR-41.12	Quotation document stays in wave 27; conversion, estimate lines and link live here
AX #89 — 12.11.1 Timesheets list page (TSTimesheetTableListPage)	PARTIAL	ADDED as FR-41.13	Screen "My timesheet" / "Timesheet approval" in 8.41.6
AX #90 — 12.11.1 Timesheet details (TSTimesheetTable)	PARTIAL	ADDED as FR-41.13	Lines, comments, submit; BR-41.9
AX #93 — 12.11.1 Project → Retention terms (ProjRetentionTerms)	PARTIAL	ADDED as FR-41.14	Customer and vendor terms; BR-41.10
AX #94 — 12.11.2 Timesheet inquiries (missing timesheets, timesheet hours)	PARTIAL	ADDED as FR-41.13	Rows "Missing timesheets", "Timesheet hours" in 8.41.14
AX #95 — 12.11.2 SSRS Timesheet report / Missing timesheets	PARTIAL	ADDED as FR-41.13	Timesheet document in 8.41.15; missing-timesheets inquiry
AX #96 — 12.11.3 Timesheets → Timesheet workflow / Post timesheets / Create periodic timesheets	PARTIAL	ADDED as FR-41.13	Jobs `timesheet-periodic-create`, `timesheet-missing-reminder`; post endpoint
AX #98 — 12.11.3 Project → Retention release (vendor/customer)	PARTIAL	ADDED as FR-41.14	Job `retention-release-check`; endpoint `/retention/release`
AX #99 — 12.11.4 Billing → Billing rule templates / Retention terms	PARTIAL	ADDED as FR-41.14	Billing rule templates are the wave-41 billing rules (FR-41.6) stored as reusable templates per contract; retention terms setup in 8.41.13
AX #100 — 12.11.4 Timesheets → Timesheet periods / Period types / Timesheet parameters (favorites, delegates, workflow by project)	PARTIAL	ADDED as FR-41.13	Parameters rows in 8.41.13
AX #101 — 12.11.4 Indirect costs (component groups, components, rules — R3 PSA)	PARTIAL	ADDED as FR-41.15	BR-41.11; parameter "Indirect cost component group"
AX #272 — 12.31.1 Timesheets (Home → Timesheets link)	PARTIAL	ADDED as FR-41.13	ESS entry "My timesheet"
AX #91 — 12.11.1 Beginning balance journals (ProjJournalTable beginning balance)	PARTIAL (wave 49)	DEFERRED to wave 49	8.41.11 loads opening WIP/estimates through the migration pack; a user-facing beginning-balance journal type is delivered with the wave-49 migration tooling
SAP #310 — 13.27 CJZ1/CJZ2/CJZ3/CJZ5 Interest calculation on projects	GAP	ADDED as FR-41.16	BR-41.12; capitalized interest IAS 23 / ASC 835-20
SAP #311 — 13.27 CJV1–CJV6 Simulation versions	PARTIAL	ADDED as FR-41.17	BR-41.14 transfer with three-way diff
SAP #312 — 13.27 CLM1/CLM2/CLM3/CLM10/CLM11 Claims	PARTIAL	ADDED as FR-41.18	Change orders and claims; BR-41.13
SAP #313 — 13.27 CNS0 Project deliveries	PARTIAL	ADDED as FR-41.19	Shipment from project stock via wave 28
SAP #314 — 13.27 Project stock (Q), PS material components, MD51, CN33	PARTIAL	ADDED as FR-41.19	Ownership dimension `project_id` in wave 16; planning scope PROJECT in wave 22; CN33 BOM transfer = copy wave-9 BOM lines as item estimate lines
SAP #316 — 13.27 Commercial Project Management (risks/issues, financial planning)	PARTIAL	ADDED as FR-41.20	Forecast models and risk/issue register; CPM multi-version financial plans remain partial (parity map row)
SAP #318 — 13.27 CJZ2 Interest calculation run	GAP	ADDED as FR-41.16	Job `projects-interest-run`
SAP #320 — 13.27 OPS project stock settings (valuated/non-valuated), OPUV, MRP for project	PARTIAL	ADDED as FR-41.19	Parameter "Project stock valuation"
SAP #322 — 13.27 Claim management customizing (TQ80)	PARTIAL	ADDED as FR-41.18	Parameter "Change order customer approval required"; co_type enum replaces claim types
SAP #323 — 13.27 Simulation / project version profiles (OPTS/OPT)	PARTIAL	ADDED as FR-41.17	Parameter "Max simulation versions per project"
SAP #324 — 13.27 Interest profiles (OPIB/OPIC/OPID)	GAP	ADDED as FR-41.16	InterestProfile entity; parameters "Interest profile", "Interest day count"

8.42 Wave 42: Human Resources Core: Workers, Positions, Absence & Talent

8.42.1 Scope and business context

HR core replaces AX Human resources (HcmWorker, HcmEmployment, HcmPosition, HcmJob, HcmPositionWorkerAssignment, LeaveRequest/LeavePlan, HcmBenefit) and SAP HCM PA/OM (PA20/PA30 infotypes 0000 Actions, 0001 Org Assignment, 0002 Personal Data, 0008 Basic Pay, 2001 Absences; HRP1000/HRP1001 objects and relationships; PT time types; benefits PA-BN). After this wave, an HR administrator manages workers, employments, positions, jobs and departments with effective dating; employees request absence and enroll in benefits; managers approve and run simple reviews. The wave also establishes the PII vault used by payroll.

Out of scope: full recruitment (ATS), learning management, compensation cycles beyond fixed/variable plans, SuccessFactors-style succession, country-specific HR legal fields beyond the framework.

Coverage closure (8.42.21) narrows those exclusions as follows: an internal course catalog with sessions, registrations and completion-to-skills is in scope (FR-42.14) while external LMS content players (SCORM/xAPI) are not; a questionnaire engine (FR-42.15), injury and illness register (FR-42.16), loaned items (FR-42.17), skill mapping (FR-42.18), onboarding/offboarding checklists (FR-42.19), HR reminders (FR-42.20), requisitions and offers (FR-42.21), compensation planning cycles with budgets and manager worksheets (FR-42.22) and concurrent/global assignments (FR-42.23) are in scope; career sites, candidate sourcing and succession/talent pools stay out.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX HcmPositionWorkerAssignment edits in the Position form require "Maintain versions" dialogs; as-of reporting needs SQL over ValidFrom/ValidTo and routinely returns two managers for a day because ranges are inclusive-inclusive; SAP PA30 infotype splits (BEGDA/ENDDA) produce the same overlap at delimit time	Date-effective framework with inclusive end dates and no exclusion constraints	`[from, to)` ranges with Postgres exclusion constraints; one `as-of` endpoint; org chart with a date picker	Duplicate/overlapping manager answers 0 (property test over 50,000 dates); as-of query p95 < 50 ms vs 2–5 s SSRS reports
Hire in AX takes 25+ clicks across Worker, Employment, Position assignment, Compensation and User relation forms; SAP PA40 hire action walks 8–12 infotype screens and aborts midway on validation, leaving a partial employee	Action not modeled as a unit; each screen commits separately	One `PersonnelAction` with a diff, workflow approval, and atomic apply (future-dated via Temporal); SCIM provisions the user automatically	Clicks 25 → 4; partial-hire records 0; user provisioning 1–2 days manual → < 60 s
Leave accruals (LeaveBankTransaction) in AX and SAP time evaluation (PT60, schema TM04) run as batch and silently truncate proration to whole days; carry-over expiry needs custom PCRs	Accrual arithmetic hidden in batch/schema code; no stated formula	Formula stated in BR-42.3 with proration, caps, carry-over buckets and expiry; balance change events to ESS in real time	Accrual disputes –90%; balance visible to employee at T+0 instead of next payroll run
Benefits eligibility in HcmBenefit relies on static "Eligibility rules" with a query form that administrators cannot test; SAP PA-BN eligibility groupings need feature trees (PESTA)	Rule engine not simulatable	CEL eligibility expressions with a simulate endpoint over the whole workforce	Eligibility errors at open enrollment 0 on 100-employee fixture; rule authoring 1 day consultant → 10 min admin
PII (national ID, bank) is readable by any user with HR tables access; DB log auditing (AX) off by default, SAP PA20 display not logged	PII stored inline with no masking layer	Vaulted `hr_pii` schema, column encryption, masked default, every unmasked read audited with reason	Unaudited PII reads 0; fields exposed to non-HR roles 0

8.42.2 AX / SAP parity map

Capability	AX 2012 / D365	SAP ECC / S/4	Passage equivalent	Parity
Worker/person	HcmWorker, DirPerson, HcmPersonPrivateDetails	PA0002 Personal Data, PA0006 Address	`Worker` (links wave-7 party person), `WorkerPII` vault	full
Employment	HcmEmployment (ValidFrom/To), HcmEmploymentDetail	PA0000 Actions, PA0001 Org Assignment	`Employment` (effective-dated)	full
Positions and assignments	HcmPosition, HcmPositionDetail, HcmPositionWorkerAssignment, HcmPositionHierarchy	HRP1000 (S objects), HRP1001 A/B 008 holder, A 002 reports-to	`Position`, `PositionAssignment`, `PositionHierarchy`	full
Jobs	HcmJob, HcmJobDetail, HcmJobType, HcmJobFunction	HRP1000 C (job)	`Job`	full
Departments/org	OMOperatingUnit (Department), HcmPositionHierarchyType	HRP1000 O (org unit), PPOME	wave-6 org hierarchy, type DEPARTMENT	full
Compensation	HcmCompensationFixedPlan, HcmCompensationVariablePlan, HcmEmploymentFixedCompensation	PA0008 Basic Pay, wage types	`CompensationPlan`, `FixedCompensation` (effective-dated)	partial — no grade/step matrices in v1
Leave and absence	LeavePlan, LeaveBankTransaction, LeaveRequest; D365 HR leave types	PA2001 Absences, PT quota types (PA2006), accrual via time evaluation	`AbsencePlan`, `AbsenceBalance`, `AbsenceRequest`	full
Benefits	HcmBenefit, HcmBenefitPlan, HcmBenefitOption, HcmWorkerEnrolledBenefit	PA0167/0168/0169 benefit infotypes, PA-BN plans	`BenefitPlan`, `BenefitEnrollment`, `LifeEvent`	full
Skills/certifications	HcmSkill, HcmPersonSkill, HcmPersonCertificate	PD qualifications (Q objects), HRP1001 A 032	`Skill`, `WorkerSkill`, `Certification`	full
Performance	HcmDiscussion, HcmGoal, HcmRatingModel	Performance management (PMP), appraisals	`Review`, `Goal`	partial — basics only
Personnel actions	HcmPersonnelAction, HcmPersonnelActionType	PA40 actions, PA0000	`PersonnelAction` with workflow	full
Courses / learning	HRMCourseTable, HRMCourseSession, HRMCourseAttendee, HRMCourseInstructor, HRMCourseLocation/Classroom, agenda templates, course confirmation, "Transfer course completion to competencies"	PSV1/PSV2/PSV3 business events (training & event management), PSVP, LSO_PSV; SF Learning	`Course`, `CourseSession`, `CourseRegistration` (FR-42.14)	partial — no SCORM/xAPI content player
Questionnaires	KMQuestionnaire, KMQuestion, KMAnswerGroup, KMQuestionnaireSchedule, KMResultGroup, results analysis (Org admin › Questionnaires; Home › My questionnaires)	Appraisal templates (PHAP_CATALOG), HR surveys; SAP Qualtrics	`Questionnaire`, `QuestionnaireSchedule`, `AnswerSession` (FR-42.15)	full
Injury and illness	HRMInjuryIncident, HRMInjuryTreatment, HRMInjuryBodyPart, OSHA 300/300A/301	EHS Incident Management (EHSM) incident recording	`IncidentReport` (FR-42.16); OSHA layouts via wave 48	full for register; forms via US pack
Loaned items	HRMLoanTable, HRMLoanItem, loan types	PA0040 Objects on loan	`LoanedItem` (FR-42.17) linked to wave-44 assets	full
Skill mapping	HcmSkillMapping (skill mapping, skill gap analysis)	PEPM profile matchup, PPPM qualification search	skill search and gap analysis (FR-42.18)	full
Worker tasks / onboarding checklists	HcmWorkerTask, checklist templates	SF Onboarding/Offboarding tasks	`ChecklistTemplate`, `WorkerTask` (FR-42.19)	full
Reminders	Worker reminders (certificate expiry, probation end) via alerts	PA0019 Monitoring of tasks (date monitoring)	`HrReminderRule` on wave-5 alert rules (FR-42.20)	full
Recruitment requisitions and offers	HRMRecruitingTable, HRMApplication, HRMApplicantTable	E-Recruiting (HRRCF_*), SF Recruiting	`Requisition`, `Candidate`, `Offer` (FR-42.21)	partial — no career site
Compensation planning cycles	HcmCompensationProcessEvent, HcmCompensationBudget, merit guideline matrix	PECM_* enterprise compensation management (T71*); SF Compensation	`CompensationCycle`, `CompensationWorksheet` (FR-42.22)	full
Concurrent / global employment	several HcmEmployment per worker (per legal entity only)	HRUS_D2 / HRPAD concurrent employment, global employment (host/home)	`Employment.assignment_kind` PRIMARY/CONCURRENT/GLOBAL_HOST (FR-42.23)	full

Capability	AX 2012 / D365	SAP ECC / S/4	Passage equivalent	Parity
COBRA (US)	n/a	HRBEN00COBRA / HRBEN00ICOBRA	qualifying-event record on enrollment end; notices in wave-48 US pack	partial — deferred

8.42.3 Functional requirements

- **FR-42.1 Worker records** — Must. The system shall maintain workers (EMPLOYEE, CONTRACTOR) linked to one wave-7 person party, with personnel number from sequence `WRK`, and PII in a vaulted schema.
  - PII fields (national ID, date of birth, bank details, home address) encrypted at column level (AES-256-GCM, tenant key in Vault); masked in all reads unless duty `HcmPIIRead`.
- **FR-42.2 Effective-dated employment** — Must. The system shall maintain employments per legal entity with validity ranges and derived status (ACTIVE, ON_LEAVE, TERMINATED).
  - Multiple employments allowed across entities; at most one ACTIVE per entity at any instant.
- **FR-42.3 Positions, jobs, departments** — Must. The system shall maintain positions (FTE, job, department, reports-to position, cost dimensions) and jobs (title, type, function) with effective dating.
  - Position FTE default 1.00, range 0.05–2.00 (job-share); position assignments cannot exceed FTE in total unless `allowOverfill`.
- **FR-42.4 Position hierarchy as-of** — Must. The system shall answer manager, chain-of-command and direct reports for any worker as-of any date within p95 < 50 ms.
- **FR-42.5 Personnel actions** — Must. The system shall model hire, rehire, transfer, promotion, compensation change, leave of absence and termination as personnel actions with workflow approval and future-dated effect.
- **FR-42.6 Absence plans and accruals** — Must. The system shall accrue balances per plan (frequency MONTHLY/BIWEEKLY/ANNUAL/PER_HOUR_WORKED), with proration, caps, carry-over limits and expiry, and reconcile balances to approved requests and wave-25 time registrations.
- **FR-42.7 Absence requests** — Must. The system shall let employees request absence in hours or days, validate against balance and blackout periods, route to manager, and publish to time and payroll.
- **FR-42.8 Benefits** — Must. The system shall define plans (MEDICAL, DENTAL, VISION, LIFE, RETIREMENT_401K, HSA, FSA, OTHER) with options, eligibility rules, employee/employer rates, open enrollment windows and life events.
- **FR-42.9 Employee and manager self-service** — Must. The system shall provide ESS (personal data, absence, benefits, pay statements from wave 43) and MSS (approvals, team calendar, team positions).
- **FR-42.10 Skills and certifications** — Should. The system shall maintain skill profiles with rating scales and certifications with expiry, feeding service technician skills (wave 45).
- **FR-42.11 Performance basics** — Should. The system shall support goals, review templates and a review cycle with manager/employee sign-off.
- **FR-42.12 Compliance framework** — Must. The system shall support consent records, retention policies per data class and a country-specific field framework (extension slots per country code).
- **FR-42.13 Identity sync** — Must. The system shall provision/deprovision Keycloak users via SCIM on hire/termination actions.
- **FR-42.14 Courses and learning** — Should. The system shall maintain a course catalog (course types, groups, agenda templates), sessions (dates, location/classroom or virtual link, instructor, capacity, cost), registrations with waitlist, attendance and completion, and transfer completion into skills and certifications.
  - Registration statuses: REQUESTED → CONFIRMED | WAITLISTED → ATTENDED | NO_SHOW | CANCELED; manager approval optional per course.
  - Completion with result PASS writes `WorkerSkill` rows (rating from the course's skill outcomes) and/or a `Certification` whose expiry = completion date + validity months.
  - Course confirmation letters and attendance lists are documents (8.42.15); registering over capacity goes to the waitlist, and a full waitlist raises `HCM-CRS-001`.
- **FR-42.15 Questionnaire engine** — Should. The system shall design questionnaires (question types SINGLE, MULTI, SCALE 1–n, NUMERIC, TEXT, DATE; answer groups; result groups with weights and scoring; media attachments), schedule them to a population (CEL over worker/department/position/course session, or external respondents by wave-35 link), collect answers in ESS "My questionnaires", and analyze results by question, result group and respondent group for reuse by reviews (FR-42.11), course evaluation (FR-42.14) and wave-32 quality audits.
  - Anonymous mode stores no respondent id and refuses breakdowns with fewer than 5 respondents (`HCM-QST-002`).
- **FR-42.16 Injury and illness register** — Should. The system shall record workplace injury and illness incidents (incident type, occurred/reported timestamps, site and location, body parts, severity, treatments, days away and restricted, outcome, witnesses, root cause, corrective action, privacy-case flag, OSHA-recordable flag) and provide OSHA 300/300A/301 data extracts that the wave-48 US pack renders and submits (ITA).
- **FR-42.17 Loaned items** — Could. The system shall track items loaned to workers (loan type, free-text item or wave-44 asset, serial, issued/due/returned dates, condition out/in) with due reminders and an offboarding task that blocks the termination checklist while items are OUT.
- **FR-42.18 Skill mapping and gap analysis** — Should. The system shall search workers (and FR-42.21 candidates) by required skills with minimum ratings, certifications (unexpired) and availability, and compute the gap between a job's or position's skill profile and a worker's profile as a list of (skill, required, actual, delta).
- **FR-42.19 Worker tasks and checklists** — Should. The system shall define checklist templates (ONBOARDING, OFFBOARDING, TRANSFER) with tasks (owner role, due offset from the action's effective date, instructions, evidence required) and instantiate them as worker tasks when a personnel action applies; completion is tracked on the action, in MSS and in ESS.
- **FR-42.20 HR reminders** — Should. The system shall provide reminder rules (certificate expiry, probation end, contract end, visa/permit expiry, loaned item due, review due) on the wave-5 alert rule engine with HR selectors, recipients (worker, manager, HR partner) and lead times (default 60/30/7 days), replacing BR-42.8's fixed certification notification.
- **FR-42.21 Requisitions, candidates and offers** — Could. The system shall manage job requisitions (position, headcount, hiring manager, approval workflow, budget check against the wave-38 position forecast), a candidate pipeline (APPLIED → SCREEN → INTERVIEW → OFFER → HIRED | REJECTED), offer letters generated from templates with compensation and start date, and hire-from-offer that creates the hire personnel action and the onboarding checklist.
- **FR-42.22 Compensation planning cycles** — Should. The system shall run merit and bonus planning cycles with a budget per department, a guideline matrix (performance rating × compa-ratio band → recommended % range), manager worksheets with proposed amounts and justification, roll-up approval through the position hierarchy, and bulk application as compensation-change personnel actions on the cycle effective date.
- **FR-42.23 Concurrent and global assignments** — Should. The system shall allow a worker to hold several concurrent employments (within one legal entity or across entities) with `assignment_kind` PRIMARY, CONCURRENT or GLOBAL_HOST (home entity, assignment type, planned end date), each with its own position, compensation and pay group, while absence balances, PII and the user identity remain per worker.

8.42.4 Data model

Worker

Field	Type	Required	Notes
id	uuid	yes	PK
personnel_number	text(20)	yes	unique per tenant; sequence WRK
party_person_id	uuid	yes	FK wave-7 person
worker_type	enum	yes	EMPLOYEE, CONTRACTOR
preferred_name, legal_name	text	yes	legal_name from party
user_id	uuid	no	Keycloak subject
pii_ref	uuid	cond	FK WorkerPII (vault schema)
status	enum	computed	from employments

Employment

Field	Type	Required	Notes
id	uuid	yes	PK
worker_id	uuid	yes	FK
legal_entity_id	uuid	yes	FK
validity	tstzrange	yes	`[start, end)`; exclusion constraint on (worker_id, legal_entity_id) && validity
employment_type	enum	yes	FULL_TIME, PART_TIME, TEMP, CONTRACTOR
hire_date, seniority_date, termination_date	date	yes/no/no	
termination_reason	text(20)	cond	code table
pay_group_id	uuid	no	FK payroll (wave 43)
work_calendar_id	uuid	yes	FK wave 8

Position

Field	Type	Required	Notes
id	uuid	yes	PK
position_number	text(20)	yes	unique; sequence POS
job_id	uuid	yes	FK Job
department_id	uuid	yes	FK org unit type DEPARTMENT
legal_entity_id	uuid	yes	
fte	numeric(4,2)	yes	0.05–2.00
validity	tstzrange	yes	position existence
reports_to_position_id	uuid	no	effective-dated via PositionHierarchy
dimension_set_id	uuid	no	default dims for payroll/projects
allow_overfill	bool	yes	default false

PositionAssignment: id, position_id, worker_id, employment_id, validity tstzrange, fte_share numeric(4,2), is_primary bool. Exclusion constraint: (position_id && validity) unless allow_overfill; (worker_id, is_primary = true && validity).

AbsencePlan: id, code, unit (HOURS/DAYS), accrual_frequency, accrual_rate numeric(8,4), accrual_day (LAST_DAY/FIRST_DAY/PAYDATE), proration (DAYS_IN_PERIOD/NONE), max_balance, carry_over_max, carry_over_expiry_months, plan_year_start (MM-DD), negative_balance_allowed, min_service_days. **AbsenceBalance**: id, worker_id, plan_id, bucket (CURRENT/CARRY_OVER), period_start, accrued, used, adjusted, expired, balance. **AbsenceRequest**: id, worker_id, plan_id, from, to, quantity, status (DRAFT, SUBMITTED, APPROVED, REJECTED, CANCELED), approver_id, reason.

BenefitPlan: id, code, type, provider_party_id, options[] (code, coverage level, employee_rate, employer_rate, rate_basis PER_PAY_PERIOD/MONTHLY), eligibility_rule (CEL expression), enrollment_window, waiting_days, pre_tax bool, payroll_deduction_code. **BenefitEnrollment**: worker_id, plan_id, option_code, validity, dependents[], life_event_id, end_reason (TERMINATION, LIFE_EVENT, PLAN_END), cobra_qualifying_event bool (US; notices via wave 48).

Course / CourseSession / CourseRegistration

Field	Type	Required	Notes
course.id, code, name	uuid, text(20), text(120)	yes	course_type_id, course_group_id, agenda_template_id
course.skill_outcomes[]	jsonb	no	(skill_id, rating) written on completion
course.certification_id, validity_months	uuid, int	no	certification issued on PASS
course.requires_manager_approval	bool	yes	default false
session.id, course_id	uuid	yes	
session.starts_at, ends_at	timestamptz	yes	agenda lines for multi-day sessions
session.location_id, classroom_id, virtual_url	uuid, uuid, text	cond	one of location or virtual
session.instructor_worker_id / instructor_party_id	uuid	no	internal or external instructor
session.capacity, waitlist_capacity	int	yes	≥ 1
session.cost_per_attendee, currency	numeric(12,2), char(3)	no	charged to the attendee's department dimension
registration.worker_id, session_id, status	uuid, uuid, enum	yes	REQUESTED, CONFIRMED, WAITLISTED, ATTENDED, NO_SHOW, CANCELED
registration.result, score	enum, numeric(5,2)	no	PASS, FAIL

Questionnaire: id, code, type_id, questions[] (id, text, question_type, answer_group_id, weight, mandatory, media_ref), result_groups[] (code, question_ids[], scoring), anonymous bool, status (DRAFT, ACTIVE, RETIRED). **QuestionnaireSchedule**: id, questionnaire_id, population (CEL), opens_at, closes_at, reminder_days[], status (PLANNED, OPEN, CLOSED). **AnswerSession**: id, schedule_id, respondent_worker_id (NULL when anonymous), answers[] (question_id, value), started_at, completed_at, score_by_result_group jsonb.

IncidentReport: id (sequence HCM-INC), worker_id, legal_entity_id, site_id, incident_type_id, occurred_at, reported_at, location_text, body_parts[], severity_id, treatment_ids[], days_away, days_restricted, outcome_id, privacy_case bool, description, witnesses[], root_cause, corrective_action, osha_recordable bool, asset_id

(wave 44, optional), status (REPORTED, UNDER_REVIEW, CLOSED). **LoanedItem**: id, worker_id, loan_type_id, item_description, asset_id, serial, issued_on, due_on, returned_on, condition_out, condition_in, status (OUT, RETURNED, LOST). **ChecklistTemplate**: id, kind (ONBOARDING, OFFBOARDING, TRANSFER), tasks[] (code, title, owner_role, due_offset_days, instructions, evidence_required). **WorkerTask**: id, worker_id, action_id, template_task_code, owner_worker_id, due_on, status (OPEN, DONE, SKIPPED), evidence_ref. **HrReminderRule**: id, selector (CERT_EXPIRY, PROBATION_END, CONTRACT_END, VISA_EXPIRY, LOAN_DUE, REVIEW_DUE), lead_days[], recipients[], alert_rule_id (wave 5). **Requisition**: id (sequence HCM-REQ), position_id, headcount, hiring_manager_worker_id, status (DRAFT, APPROVED, OPEN, FILLED, CANCELED), budget_check_ref. **Candidate**: id, party_person_id, requisition_id, stage, source, skills[]. **Offer**: id, candidate_id, compensation jsonb, start_date, template_id, status (DRAFT, SENT, ACCEPTED, DECLINED, WITHDRAWN), document_ref. **CompensationCycle**: id, name, effective_date, kind (MERIT, BONUS, BOTH), budget_lines[] (department_id, budget_amount, currency), guideline_matrix jsonb (rating × compa_ratio_band → pct_min, pct_max), status (PLANNING, MANAGER_INPUT, APPROVAL, APPLIED, CANCELED). **CompensationWorksheet**: cycle_id, manager_worker_id, rows[] (worker_id, current_pay, rating, compa_ratio, guideline_pct_min, guideline_pct_max, proposed_pct, proposed_amount, justification), status (DRAFT, SUBMITTED, APPROVED). **Employment** adds: assignment_kind (PRIMARY, CONCURRENT, GLOBAL_HOST; default PRIMARY), home_legal_entity_id, planned_end_date, global_assignment_type.

Invariants: every ACTIVE employment has exactly one primary position assignment or an explicit `unassigned` flag; balances = accrued – used + adjusted – expired per bucket; hierarchy is acyclic at every instant (checked on write with a recursive CTE as-of the new validity range). Indexing: GiST on validity ranges; `PositionHierarchy (tenant_id, child_position_id, validity)`; PII schema `hr_pii` with separate role and no RLS bypass.

8.42.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/hr/workers`	Create worker (+ party)	key	HcmWorkerMaintain
POST	`/api/v1/hr/actions`	Submit personnel action (hire/transfer/terminate…)	key	HcmActionMaintain
GET	`/api/v1/hr/workers/{id}/as-of?date=`	Employment, position, manager as-of	n/a	HcmWorkerInquire
GET	`/api/v1/hr/positions/{id}/reports?asOf=&depth=`	Direct/indirect reports	n/a	HcmOrgInquire
POST	`/api/v1/hr/absence/requests`	Create request	key	EssAbsence
POST	`/api/v1/hr/absence/requests/{id}/decide`	Approve/reject	key	MssApprove
POST	`/api/v1/hr/absence/accrue`	Run accrual for plan/period	key per (plan, period)	HcmAbsenceAdmin
POST	`/api/v1/hr/benefits/enrollments`	Enroll (open enrollment or life event)	key	EssBenefits
GET	`/api/v1/hr/workers/{id}/pii`	Unmasked PII (audited)	n/a	HcmPIIRead
POST	`/api/v1/hr/courses/sessions/{id}/registrations`	Register worker (self or manager)	key per (worker, session)	EssLearning / HcmLearningAdmin
POST	`/api/v1/hr/courses/sessions/{id}/complete`	Record attendance and results; transfer to skills	key	HcmLearningAdmin
POST	`/api/v1/hr/questionnaires/schedules`	Schedule questionnaire to a population	key	HcmQuestionnaireAdmin
POST	`/api/v1/hr/questionnaires/answer-sessions/{id}/submit`	Submit answers	key	EssBase
GET	`/api/v1/hr/questionnaires/schedules/{id}/results`	Results analysis	n/a	HcmQuestionnaireAdmin
POST	`/api/v1/hr/incidents`	Report injury or illness incident	key	HcmIncidentReport
GET	`/api/v1/hr/incidents/osha-extract?year=&siteId=`	OSHA 300/300A/301 data extract	n/a	HcmIncidentAdmin
POST	`/api/v1/hr/loaned-items`	Issue loaned item	key	HcmWorkerMaintain
POST	`/api/v1/hr/loaned-items/{id}/return`	Return item	key	HcmWorkerMaintain
GET	`/api/v1/hr/skills/search?skills=&minRating=&certs=&availableFrom=`	Skill mapping search	n/a	HcmOrgInquire
GET	`/api/v1/hr/workers/{id}/skill-gap?positionId=`	Gap analysis	n/a	HcmOrgInquire
POST	`/api/v1/hr/worker-tasks/{id}/complete`	Complete checklist task	key	MssBase / EssBase
POST	`/api/v1/hr/requisitions`	Create requisition	key	HcmRecruitMaintain
POST	`/api/v1/hr/offers/{id}/hire`	Hire from accepted offer (creates hire action)	key	HcmActionMaintain
PUT	`/api/v1/hr/compensation/cycles/{id}/worksheets/{managerId}`	Save manager worksheet	ETag	MssBase
POST	`/api/v1/hr/compensation/cycles/{id}/apply`	Apply cycle as compensation-change actions	key	HcmCompAdmin

Validation: action effective date may be future-dated up to 365 days; termination date ≥ hire date; absence request quantity ≤ balance + negative allowance; benefit enrollment only inside window or within 30 days of a life event.

Event	Trigger	Payload	Consumers
`dev.passage.hr.employment.changed.v1`	action applied	see 8.0	payroll, projects, service, SCIM
`dev.passage.hr.position.assignment.changed.v1`	assignment start/end	workerId, positionId, validity, managerPositionId	authorization (manager roles), workflow routing
`dev.passage.hr.absence.approved.v1`	request approved	workerId, planId, from, to, hours	time, projects, payroll
`dev.passage.hr.benefit.enrollment.changed.v1`	enrollment start/end	workerId, planId, option, validity, employeeRate	payroll deductions
`dev.passage.hr.course.completed.v1`	registration ATTENDED with PASS	workerId, courseId, sessionId, skills[], certificationId	wave-45 dispatch skills, notifications
`dev.passage.hr.incident.reported.v1`	incident created	incidentId, siteId, severity, oshaRecordable, assetId	EHS notifications, wave-44 asset history
`dev.passage.hr.compensation.cycle.applied.v1`	cycle APPLIED	cycleId, effectiveDate, actionIds[]	payroll (wave 43), budget (wave 38)

8.42.6 User experience

Screen	Primary actions	Key fields/columns	Duty
HR workspace	tiles: pending actions, expiring certifications, open enrollment		HcmWorkerInquire
Worker detail	tabs: employment, positions, compensation, absence, benefits, skills, documents	masked PII with reveal (audited)	HcmWorkerMaintain
Org chart	navigate as-of date, drag worker to position (creates action)	position, incumbent, FTE, vacancies	HcmOrgInquire
Personnel actions	create, approve, apply	type, effective date, changes diff	HcmActionMaintain

Screen	Primary actions	Key fields/columns	Duty
Absence admin	run accrual, adjust, view balances	plan, bucket, accrued/used/balance	HcmAbsenceAdmin
ESS portal	request absence, view balances, update address, enroll benefits, view pay statements		EssBase
MSS portal	approve requests, team calendar, team positions		MssBase
Benefits enrollment wizard	select options, add dependents, confirm	per-pay-period cost preview	EssBenefits
Course catalog and sessions	browse, register, waitlist, print confirmation; admin: record attendance/results	course, session dates, location, instructor, seats left, cost	EssLearning / HcmLearningAdmin
Questionnaire designer and results	design questions, schedule, view statistics	question, type, answer group, result group, response rate	HcmQuestionnaireAdmin
My questionnaires (ESS)	answer, resume, submit	open questionnaires, due date	EssBase
Incident register	report, review, close; OSHA extract	incident, site, severity, days away/restricted, recordable	HcmIncidentReport / HcmIncidentAdmin
Loaned items	issue, return, mark lost	worker, item/asset, due, condition	HcmWorkerMaintain
Skill search and gap	search by skills/certs/availability; compare to position profile	worker, skill, rating, required, delta	HcmOrgInquire
Checklist tasks (MSS / ESS)	complete task, attach evidence, skip with reason	task, owner, due, status	MssBase / EssBase
Requisitions and candidates	create requisition, move candidate stage, generate offer, hire	requisition, position, stage, offer status	HcmRecruitMaintain
Compensation worksheet (MSS)	propose %, justify, submit; cycle admin: budgets, guidelines, apply	worker, current pay, rating, compa-ratio, guideline, proposed, budget used	MssBase / HcmCompAdmin

Rules: as-of date picker persists per session in Org chart; PII reveal requires reason text (≥ 10 chars) and logs to audit; all ESS screens pass axe WCAG 2.1 AA and work at 360 px width.

8.42.7 Business rules and edge cases

- **BR-42.1 Effective-dated position logic** — assignments, hierarchy edges and position attributes are stored as `[valid_from, valid_to)` ranges; a change on date D closes the current row at D and opens a new row from D; as-of queries use `validity @> asOf::timestamptz`. Manager of worker W as-of D = incumbent (primary assignment) of `reports_to_position` of W's primary position as-of D; if that position is vacant as-of D, walk up until an incumbent is found, max 10 levels, else NULL (AX "vacant position" behavior).
- **BR-42.2 Future-dated actions** — applied by a Temporal timer at 00:00 tenant-local on the effective date; the action can be withdrawn until then; applied rows are immutable (corrections create new actions).
- **BR-42.3 Accrual formula** — per period: `accrual = rate × proration_factor`, proration_factor = days_employed_in_period ÷ days_in_period when proration = DAYS_IN_PERIOD, rounded half-up to 2 decimals; balance capped at max_balance (excess discarded, logged). Example plan `US-PTO-15` (DAYS, MONTHLY, rate 1.25 on LAST_DAY, max 22.50, carry_over_max 5.00, expiry 3 months, plan year 01-01): hire 2026-03-16 → March accrual 1.25 × 16/31 = 0.65; April–December 9 × 1.25 = 11.25; total accrued 11.90; used 4.00; balance 7.90 at 2026-12-31. Year-end: carry-over bucket = min(7.90, 5.00) = 5.00; 2.90 expired. 2027-01-31: CURRENT accrues 1.25, total 6.25; consumption takes CARRY_OVER first; unused carry-over expires 2027-03-31 23:59:59.
- **BR-42.4 Negative balances** — allowed only if plan.negative_balance_allowed, down to –(rate × 2); termination with negative balance raises a payroll deduction suggestion.
- **BR-42.5 Overlapping absence** — a request overlapping an APPROVED request for the same worker is rejected; overlapping a SUBMITTED one warns.
- **BR-42.6 Benefit eligibility** — evaluated as-of enrollment effective date with CEL expression over {employment_type, fte, service_days, job.type, legal_entity, age}; waiting_days counted from hire_date; a terminated employment ends enrollments at the end of the month of termination (COBRA handling is v1.x).
- **BR-42.7 Hierarchy cycles** — rejected on write (`HCM-ORG-001`).
- **BR-42.8 Certifications** — expiry generates a notification at 60/30/7 days; expired certification removes skill match from wave-45 dispatch.
- **BR-42.9 Course registration and completion** — seats are allocated in registration order; when confirmed = capacity, new registrations are WAITLISTED up to waitlist_capacity, else `HCM-CRS-001`; a cancellation promotes the oldest waitlisted registration; ATTENDED with PASS writes skill outcomes (rating = max(existing, course rating)) and a certification valid from the session end date for validity_months; session cost per attendee posts to the attendee's department dimension through the wave-11 posting API when the parameter "Post course costs" is on.
- **BR-42.10 Questionnaire scoring and anonymity** — result group score = Σ(answer value × question weight) ÷ Σ weights, rounded to 2 decimals; a schedule closes at closes_at and late submissions are rejected (`HCM-QST-001`); anonymous schedules store no respondent id and the population list is kept only as a count; any breakdown with fewer than 5 respondents returns `HCM-QST-002`.
- **BR-42.11 Incident recordability** — osha_recordable defaults to true when the incident involves death, days away, restricted work or transfer, medical treatment beyond first aid, or loss of consciousness (29 CFR 1904.7); privacy_case = true hides the worker name in extracts and lists; a CLOSED incident is immutable (corrections are new rows linked by `supersedes_id`).
- **BR-42.12 Checklist instantiation** — when a personnel action applies, tasks are created with due_on = effective date + due_offset_days (negative offsets allowed for pre-boarding) and owner = the worker holding the owner role as-of the effective date (manager via BR-42.1, HR partner from the department, IT from the role mapping); a termination action's OFFBOARDING checklist includes an automatic task "Return loaned items" that stays OPEN while any LoanedItem is OUT.
- **BR-42.13 Compensation cycle arithmetic** — compa_ratio = current annual pay ÷ job pay range midpoint, rounded to 2 decimals; guideline band from the matrix row (rating) and column (compa_ratio band); proposed_pct outside [pct_min, pct_max] requires justification (`HCM-CMP-001` warning); Σ proposed amounts per department ≤ budget_amount, else submit is blocked (`HCM-CMP-002`) unless the cycle flag `allow_over_budget` is on; apply creates one COMPENSATION_CHANGE action per row with effective date = cycle effective date and new pay = current × (1 + proposed_pct), rounded to the minor unit.
- **BR-42.14 Concurrent assignments** — a worker may have at most one PRIMARY employment per legal entity at an instant (existing exclusion constraint) plus any number of CONCURRENT or GLOBAL_HOST employments; each employment carries its own pay group and position assignment; a GLOBAL_HOST employment requires home_legal_entity_id ≠ legal_entity_id and a planned_end_date; absence balances accrue on the PRIMARY employment only unless the plan is marked per-employment.
- **BR-42.15 Requisition and offer** — a requisition requires an approved position with vacancy as-of the planned start date or `allow_overfill`; the budget check calls the wave-38 position forecast and blocks OPEN when the forecast is exceeded (`HCM-REQ-001`); an ACCEPTED offer converts to a hire action once (`HCM-OFR-001` on repeat); DECLINED or WITHDRAWN offers reopen the requisition headcount.

Absence request state machine: DRAFT → SUBMITTED → APPROVED | REJECTED; SUBMITTED/APPROVED → CANCELED (approved cancellation reverses usage if start date is in the future or within 30 days past with manager approval).

8.42.8 Acceptance criteria

- **AC-42.1** (FR-42.4, BR-42.1) Given 1,200 workers with 50,000 randomized as-of dates, when manager is queried, then results match the reference model 100%, p95 < 50 ms.
- **AC-42.2** (FR-42.3) Given position POS-100 with FTE 1.00 and an active assignment, when a second overlapping assignment is created, then it is rejected unless allow_overfill.
- **AC-42.3** (FR-42.6, BR-42.3) Given plan US-PTO-15 and hire 2026-03-16, when accruals run through 2027-01-31, then balances equal 0.65, 11.90, 7.90, 5.00 carry-over, 2.90 expired, 6.25 as computed.
- **AC-42.4** (FR-42.7) Given a balance of 2.00 days and a request for 3.00 on a plan that forbids negatives, when submitted, then it is rejected with `HCM-ABS-002`.
- **AC-42.5** (FR-42.8, BR-42.6) Given 100 fixture employees and eligibility rule `employment_type in ['FULL_TIME'] && service_days >= 30`, when evaluated, then eligible set matches the fixture expectation exactly.
- **AC-42.6** (FR-42.5, BR-42.2) Given a transfer action effective 2026-12-01, when the date arrives, then the old assignment closes at 2026-12-01T00:00 tenant-local and the new one opens.
- **AC-42.7** (FR-42.1) Given a user without HcmPIIRead, when reading a worker, then national ID is `***-**-1234` and the audit log records the masked read.
- **AC-42.8** (FR-42.13) Given a hire action applied, when SCIM sync runs, then a Keycloak user exists within 60 s; on termination it is disabled within 60 s.
- **AC-42.9** (FR-42.14, BR-42.9) Given a session with capacity 10 and waitlist 2, when 13 workers register, then 10 are CONFIRMED, 2 WAITLISTED and the 13th receives `HCM-CRS-001`; when one confirmed registration cancels, then the oldest waitlisted becomes CONFIRMED; when attendance is recorded with PASS for a course mapping skill FORKLIFT rating 3 and a 24-month certification, then the worker's skill rating is 3 and the certification expires 24 months after the session end.
- **AC-42.10** (FR-42.15, BR-42.10) Given an anonymous questionnaire scheduled to 12 workers with 4 responses, when results are requested by department, then the overall statistics return and the department breakdown returns `HCM-QST-002`.
- **AC-42.11** (FR-42.16, BR-42.11) Given an incident with 3 days away, when saved, then osha_recordable = true and the OSHA 300 extract for the year lists it with days away 3 and the worker name masked when privacy_case is set.
- **AC-42.12** (FR-42.19, BR-42.12) Given an ONBOARDING template with a task at offset –5 days and a hire action effective 2026-11-02, when the action is approved, then a task due 2026-10-28 is assigned to the manager as-of 2026-11-02.
- **AC-42.13** (FR-42.22, BR-42.13) Given a MERIT cycle with department budget 50,000.00 USD and a worksheet whose proposals sum to 52,000.00, when submitted, then it is blocked with `HCM-CMP-002`; at 49,500.00 it submits, and apply creates one COMPENSATION_CHANGE action per row on the cycle effective date.
- **AC-42.14** (FR-42.23, BR-42.14) Given a worker with a PRIMARY employment in US-01 and a GLOBAL_HOST employment in DE-01 with planned end 2027-06-30, when the as-of query runs for 2027-01-15, then both employments and positions are returned, and payroll (wave 43) receives separate pay-group memberships.
- **AC-42.15** (FR-42.17, BR-42.12) Given a worker with a laptop loaned item OUT, when a termination action applies, then the offboarding task "Return loaned items" stays OPEN until the item is returned, and the checklist cannot complete before that.
- **AC-42.16** (FR-42.18) Given position POS-200 requiring skills WELDING ≥ 3 and PLC ≥ 2, when the gap analysis runs for a worker rated WELDING 4, PLC 1, then the result lists (PLC, required 2, actual 1, delta –1) only.

8.42.9 Test plan and fixtures

Fixtures: `hr-workforce-1200` (generated workers, employments, positions across 3 entities with 5 years of actions); `hr-accrual-cases` (36 hire-date/plan combinations with expected balances); `hr-benefit-100` (100 employees, 6 plans, expected eligibility); `hr-pii-contract` (role × field masking matrix).

Test	Type	Target	Pass criterion
As-of manager/position	property	hierarchy queries	100% match reference, p95 < 50 ms
Accrual engine	unit	accrual formula	all 36 cases exact
Eligibility CEL	unit	rule evaluator	100/100
PII masking	security contract	OPA + API	matrix holds; every unmasked read audited
Action workflow	E2E	Temporal + API	future-dated transfer applies on date
SCIM	integration	Keycloak	user lifecycle within 60 s

Reconciliation: Σ absence usage = approved request hours = wave-25 absence time registrations; headcount by department as-of = Σ primary assignments.

8.42.10 Deployment and operations

Chart `passage-hr` (API, workflow worker, SCIM worker, ESS BFF), Postgres `hr` with schema `hr_pii` under a separate role and Vault-managed key. Flags: `hr.performance`. Schedules: `hr-apply-actions` 00:00 tenant-local, `hr-accrue` per plan frequency, `hr-cert-expiry` daily. SLOs: as-of query p95 < 50 ms; ESS page p95 < 300 ms; availability 99.9%. Alerts: action application failures; PII decrypt errors; SCIM lag > 5 min. Runbook: replay actions from event log; rotate tenant PII key (re-encrypt job). Rollback by digest; `hr_pii` migrations never drop columns.

8.42.11 Migration from AX / SAP

Source	Passage entity	Notes
HcmWorker, DirPerson, HcmPersonPrivateDetails	Worker, WorkerPII	PII to vault at load time
HcmEmployment, HcmEmploymentDetail	Employment	ValidFrom/ValidTo → validity range
HcmPosition, HcmPositionDetail, HcmPositionDuration	Position	
HcmPositionWorkerAssignment	PositionAssignment	
HcmPositionHierarchy	PositionHierarchy	type Line only
HcmJob, HcmJobDetail	Job	
LeavePlan, LeaveBankTransaction, LeaveRequest	AbsencePlan, AbsenceBalance (opening), AbsenceRequest (future only)	
HcmBenefit*, HcmWorkerEnrolledBenefit	BenefitPlan, BenefitEnrollment	
PA0000/0001/0002/0006/0008	Employment, Position assignment, Worker, PII, FixedCompensation	infotype BEGDA/ENDDA → ranges
HRP1000/HRP1001 (O, S, C; A/B 002, 003, 007, 008)	org units (wave 6), Position, Job, hierarchy, assignments	
PA2006 quotas, PA2001	AbsenceBalance opening, historical absences (archive)	

Validation: active headcount per entity equal; per-worker current position/manager equal; quota balances within 0.01. Not migrated: closed historical actions older than 5 years (archive), AX HRM recruiting history, SAP PD development plans.

8.42.12 Dependencies, risks, open questions

Depends on: wave 6 org hierarchy, wave 7 person party, wave 8 calendars, wave 2 Keycloak/OPA, wave 4 workflow, wave 25 time registration.

Risk	Likelihood	Impact	Mitigation
Effective-dating bugs	Medium	High	Range types + exclusion constraints + property tests
Country HR field explosion	High	Medium	Extension slot framework; first countries via wave 48

Open questions: (1) Should contractors live in HR or only in the party model? Default HR with worker_type CONTRACTOR; decide by wave 43. (2) ESS as separate PWA or shell module? Default shell module; decide at wave 42 start.

8.42.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Personnel number sequence	legal entity	lookup	WRK	wave-5 sequences	Worker ids (FR-42.1)	Human resources parameters › Number sequences (HcmSharedParameters)	Feature NUMKR, PA04 number ranges
Position / requisition / incident sequences	legal entity	lookup	POS, HCM-REQ, HCM-INC	wave-5 sequences	Identifiers	HR parameters › Number sequences	PA04 / OM number ranges (OONR)
Effective-date granularity	tenant	enum	DAY	DAY, TIMESTAMP	Range boundaries at 00:00 tenant-local or exact instant (BR-42.1)	HR shared parameters › Date-effective (ValidFrom/ValidTo)	T77S0 PLOGI/TIMCO time constraints (T777Z)
Max future-dating of actions (days)	legal entity	int	365	0–730	Validation in 8.42.5	Personnel action parameters	T529A action type rules
Personnel action approval required per action type	legal entity	bool per type	true for HIRE, TERMINATE, COMP_CHANGE		Workflow routing (FR-42.5)	HR parameters › Personnel actions › Enable personnel actions + workflow	T588C / T588D infogroups; dynamic actions T588Z
Position overfill allowed by default	legal entity	bool	false		Default for `allow_overfill` (FR-42.3)	HR parameters › Positions	T77S0 PPVAC vacancy switch
Vacant-manager walk depth	tenant	int	10	1–20	BR-42.1	n/a	Evaluation path SAP_MANG / chief determination
PII masking pattern per field	tenant	text per field	national ID `***-**-####`; bank `******####`	regex template	FR-42.1	n/a (no masking)	n/a (authorization P_PERNR only)
PII reveal reason minimum length	tenant	int	10	0–200	8.42.6 rule	n/a	n/a
Absence negative balance floor multiplier	absence plan	numeric	2	0–12	BR-42.4	Leave plan › Allow negative balance (LeavePlan)	T559L quota deduction beyond entitlement
Absence request unit rounding	absence plan	numeric	0.25 h / 0.5 day	0.01–1	Request quantity rounding	Leave and absence parameters › Rounding	T554S absence type counting rules (T556C)
Approved-absence cancellation window (days past)	legal entity	int	30	0–365	Absence state machine	Leave and absence parameters	T554S changes to past absences
Benefit life-event window (days)	legal entity	int	30	0–90	Validation in 8.42.5	Benefits parameters › Life event window	T5UB* (US benefits) adjustment reasons
Course waitlist default size	tenant	int	5	0–100	BR-42.9	Course parameters (HRMParameters › Course)	Business event type capacity (PV10 / T77* LSO settings)
Post course costs to GL	legal entity	bool	false		BR-42.9	Course session cost posting (not native)	PSV activity allocation / internal billing LSO
Questionnaire anonymity threshold	tenant	int	5	3–50	BR-42.10	Questionnaire parameters (KMParameters)	n/a (Qualtrics)
Questionnaire reminder days	schedule	int[]	[3, 1]	0–30	Notifications before close	Questionnaire schedules	n/a
OSHA establishment id per site	site	text	—	9-digit/alnum	FR-42.16 extracts	Injury and illness parameters (US)	EHS location master
Checklist template per action type	legal entity	lookup	—	ChecklistTemplate	FR-42.19	Worker tasks / Checklist templates (HcmWorkerTask)	SF Onboarding process configuration
HR reminder lead days	rule	int[]	[60, 30, 7]	1–365	FR-42.20	Worker reminders / alert rules	PA0019 task types (T531) reminder dates
Compensation guideline rounding	cycle	numeric	0.1 %	0.01–1	BR-42.13	Compensation parameters (HcmCompensationParameters)	T71ADM* rounding rules (ECM)
Allow over-budget compensation submit	cycle	bool	false		BR-42.13	Compensation budget control	PECM budget check
Concurrent employment enabled	tenant	bool	true		FR-42.23	n/a (multiple employments per entity)	T77S0 CCURE MAINC (concurrent employment switch)
Requisition budget check	legal entity	enum	BLOCK	NONE, WARN, BLOCK	BR-42.15	Recruitment parameters	E-Recruiting requisition approval
Retention policy per data class	tenant	duration per class	per wave 48 pack	1–30 years	FR-42.12	Data retention (not native)	ILM objects HRPA_* / IRF

8.42.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Headcount and FTE as-of	KPI	as-of date, legal entity, department, job, worker type	headcount, FTE, vacancies, open requisitions	`hr_headcount`	Headcount analysis (HcmWorkerHeadcount); SSRS Employee statistics	S_AHR_61016362 headcount; S_PH9_46000223 headcount changes

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Org chart / reporting lines	drill-down	as-of date, root position, depth	position, incumbent, FTE, manager	`hr_hierarchy`	Organization chart (HcmPositionHierarchy)	PPOSE / PPOME; S_AHR_61016497 staff assignments
Personnel actions log	list inquiry	type, status, effective range, department	action, worker, effective date, approver, diff	`hr_actions`	Personnel actions (HcmPersonnelAction)	S_AHR_61016373 / S_AHR_61016380 entries/leavings
Absence balances	list inquiry	plan, bucket, as-of, department	accrued, used, adjusted, expired, balance	`hr_absence_bal`	Leave balances (LeaveBankTransaction inquiry)	PT_QTA10 quota overview; PT50 quota overview
Absence calendar	operational report	department, period, plan	worker × day grid of absences	`hr_absence_cal`	Leave calendar; SSRS Absence report	PT90 / PT_BAL00 absence data overview; S_AHR_61016369
Benefits enrollment	list inquiry	plan, option, as-of	worker, option, dependents, employee/employer cost	`hr_benefits`	Enrolled benefits (HcmWorkerEnrolledBenefit)	HRBEN0001 participation; HRBEN0005 costs
Certifications expiring	list inquiry	days ahead, type, department	worker, certification, expiry	`hr_certs`	Certificates expiring (HcmPersonCertificate)	S_AHR_61015532 qualifications expiring
Skill search / gap	drill-down	skills, min rating, position	worker, rating, required, delta	`hr_skills`	Skill mapping / skill gap (HcmSkillMapping)	PEPM profile matchup
Course attendance	operational report	course, session, date range	registrations by status, attendance %, pass %, cost	`hr_courses`	Course inquiries; SSRS Attendance list, Course confirmation	S_PH9_46000424 attendance list; S_AHR_61016216 attendance statistics
Questionnaire results	drill-down	schedule, result group, respondent group	response rate, score by question and result group, distribution	`hr_qst_results`	Questionnaire results analysis; SSRS Questionnaire statistics	Appraisal results (APPCREATE reports)
Injury and illness log	statutory report	year, site, recordable only	OSHA 300 columns (case no, name or "privacy case", job, date, location, description, classification, days away/restricted, injury type)	`hr_incidents`	SSRS OSHA 300/300A/301	EHS incident reports (EHSM)
Loaned items outstanding	list inquiry	loan type, overdue only	worker, item, issued, due, days overdue	`hr_loans`	Loaned items inquiry (HRMLoanItem)	S_AHR_61016399 objects on loan (IT0040)
Onboarding progress	KPI	action type, department, period	open tasks, overdue tasks, % complete per action	`hr_tasks`	Worker tasks	SF Onboarding dashboard
Compensation cycle status	KPI	cycle, department	budget, proposed, % used, worksheets submitted/approved	`hr_comp_cycle`	Compensation process results	PECM_EVALUATE_CRV / PECM_DISPLAY_BUDGETS
PII access audit	operational report	worker, user, period	read time, user, fields, reason	`hr_pii_audit`	Database log (not default)	S_AHR_61016380 n/a; SM20 security audit log

8.42.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Personnel action letter (hire, transfer, promotion, termination)	action APPLIED (template per action type)	header: worker, entity, effective date; body: position, job, compensation, reason; footer: signature blocks	PDF, email, e-signature (wave 35)	action id	employment record; retention per data class (US 7 years after termination common practice; DE 3 years after end of employment per §195 BGB limitation)
Offer letter	offer SENT	candidate, position, start date, compensation, conditions, acceptance block	PDF, email, e-signature	`HCM-OFR`	candidate record (EU GDPR: delete 6 months after rejection unless consent)
Course confirmation letter	registration CONFIRMED	course, session dates, location/link, agenda, instructor	PDF, email, iCalendar	none	none
Course attendance list	session start / end	session, attendee list with signature column, result	PDF	session id	safety training evidence (OSHA 29 CFR 1910 training records per standard)
Course certificate	registration ATTENDED with PASS	worker, course, date, validity, issuer	PDF	certificate id	per certification body
OSHA 301 incident report data / 300 log / 300A summary	incident saved; year end (300A posting Feb 1–Apr 30)	OSHA form fields	data extract (CSV/JSON) to wave-48 US pack for PDF and ITA CSV	HCM-INC	29 CFR 1904.33: 5 years after year end
Loaned item receipt	item issued / returned	worker, item, serial, condition, due date, signatures	PDF	loan id	until return + 1 year
Compensation statement (total rewards)	cycle APPLIED	current/new pay, bonus, benefits employer cost	PDF, ESS	none	employment record
Questionnaire	schedule OPEN (external respondents)	questions and answer controls	web form, PDF	none	per questionnaire data class

8.42.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`hr-apply-actions`	00:00 tenant-local daily	tenant, legal entity	action id; applied actions skipped	< 2 min for 1,000 actions	per-action retry 3×; failures listed in HR workspace	any failed action; run not complete by 00:30
`hr-accrue`	per plan frequency (month end, pay date)	plan, period	key (plan, period, worker)	< 5 min for 10,000 workers	per-worker retry; partial results kept	accrual missing for active workers
`hr-carryover-expiry`	plan-year start and carry-over expiry date	plan	key (plan, plan year)	< 5 min	retry	balances not rolled
`hr-cert-expiry` / `hr-reminders`	daily 06:00	tenant	key (rule, subject, lead day)	< 1 min	wave-5 alert retry	undelivered reminders
`hr-questionnaire-distribute`	schedule opens_at; reminders per reminder_days	schedule	key (schedule, respondent, day)	< 2 min for 5,000 respondents	retry	distribution count ≠ population
`hr-course-completion-transfer`	on attendance save; nightly sweep 02:00	session	key (registration)	< 1 min	retry	ATTENDED+PASS without skill rows

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`hr-scim-sync`	event-driven; reconcile hourly	tenant	Keycloak subject id	< 60 s per event	retry with backoff; reconcile job fixes drift	SCIM lag > 5 min
`hr-retention-purge`	weekly Sunday 03:00	data class	key (record, policy version)	< 30 min	stop on error; legal holds respected	purge errors; records past retention > 0
`hr-pii-rekey`	on demand (key rotation)	tenant	per-row key version	< 1 h for 100,000 workers	resumable from checkpoint	decrypt errors

8.42.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
HcmWorkerMaintain	`/hr/workers`, loaned items; Worker detail	with PayrollWorkerMaintain (bank details) for the same user	HR administrator
HcmActionMaintain	`/hr/actions`, hire from offer; Personnel actions	with approving own action; with PayrollRunFinalize	HR administrator, HR partner
HcmPIIRead	`/hr/workers/{id}/pii`; reveal control	none (audited)	HR administrator (restricted), Payroll administrator
HcmOrgInquire / HcmWorkerInquire	as-of, reports, skill search/gap; Org chart	none	HR partner, Manager (own subtree via OPA)
HcmAbsenceAdmin	accrue, adjust balances; Absence admin	with EssAbsence approval of own requests	Absence administrator
HcmLearningAdmin	course catalog, sessions, attendance, completion	none	Training coordinator
HcmQuestionnaireAdmin	questionnaire design, schedule, results	with answering an anonymous schedule's breakdown below threshold (blocked by rule)	HR partner, Quality auditor
HcmIncidentReport / HcmIncidentAdmin	report incidents / review, close, OSHA extract	admin cannot close an incident they reported	Supervisor (report), EHS manager (admin)
HcmRecruitMaintain	requisitions, candidates, offers	with HcmCompAdmin approving own offer compensation above range	Recruiter
HcmCompAdmin	compensation cycles, budgets, guidelines, apply	with MssBase worksheet approval of own pay	Compensation manager
EssBase / EssAbsence / EssBenefits / EssLearning	own data, requests, enrollments, registrations, questionnaires, tasks	none	Employee
MssBase / MssApprove	team approvals, worksheets, team tasks	approve own request blocked	Manager
HcmSetupAdmin	parameters 8.42.13, plans, templates, reminder rules	with HcmAbsenceAdmin adjustments	HR system administrator

8.42.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
HCM-ORG-001	error	Position {a} cannot report to {b}: creates a cycle on {date}	Hierarchy cycle	Choose another reports-to position	BR-42.7
HCM-EMP-001	error	Worker {w} already has an ACTIVE PRIMARY employment in {entity} on {date}	Exclusion constraint	Use CONCURRENT or end the current employment	FR-42.2, BR-42.14
HCM-EMP-002	error	Termination date {t} is before hire date {h}	Date order	Correct the date	8.42.5 validation
HCM-POS-001	error	Position {p} FTE {fte} would be exceeded	Overfill	Reduce FTE share or enable overfill	FR-42.3
HCM-ACT-001	error	Effective date is more than {n} days in the future	Future-dating limit	Use an earlier date	8.42.13
HCM-ACT-002	info	Action {id} scheduled for {date}	Future-dated action	None	BR-42.2
HCM-ABS-001	error	Request overlaps an approved absence {id}	Overlap	Change dates	BR-42.5
HCM-ABS-002	error	Requested {q} exceeds balance {b}	Insufficient balance	Reduce quantity or choose another plan	FR-42.7
HCM-ABS-003	warning	Request overlaps submitted request {id}	Overlap with pending	Check before approving	BR-42.5
HCM-BEN-001	error	Enrollment outside the open window or life-event period	Window	Wait for the window or record a life event	8.42.5 validation
HCM-BEN-002	error	Worker not eligible for plan {plan}: {rule result}	Eligibility rule false	Review the CEL result	BR-42.6
HCM-PII-001	error	Reason of at least {n} characters is required to reveal PII	Missing reason	Enter a reason	8.42.6
HCM-CRS-001	error	Session {s} and its waitlist are full	Capacity	Pick another session	BR-42.9
HCM-CRS-002	warning	Manager approval required for course {c}	Course setting	Await approval	FR-42.14
HCM-QST-001	error	Questionnaire closed on {date}	Late submission	Contact the owner	BR-42.10
HCM-QST-002	error	Breakdown hidden: fewer than {n} respondents	Anonymity threshold	Use a wider group	BR-42.10
HCM-INC-001	warning	Incident appears OSHA-recordable; confirm classification	BR-42.11 default	Confirm or justify override	BR-42.11
HCM-LOAN-001	warning	{n} loaned items still out for {worker}	Offboarding block	Record returns or mark lost	BR-42.12
HCM-CMP-001	warning	Proposed {pct} % is outside guideline {min}–{max} %; justification required	Guideline	Enter justification	BR-42.13
HCM-CMP-002	error	Proposals {sum} exceed budget {budget} for {department}	Budget	Reduce proposals	BR-42.13
HCM-REQ-001	error	Requisition exceeds position forecast for {department}	Budget check	Adjust forecast (wave 38) or headcount	BR-42.15
HCM-OFR-001	error	Offer {id} has already been converted to hire action {a}	Repeat conversion	Open the existing action	BR-42.15

8.42.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
As-of manager/position query	50,000/day; peak 500/min	10 / 50 / 120 ms	200 q/s	n/a	GiST on validity; PositionHierarchy (tenant, child, validity)	cache by (worker, date) 5 min; never stale across an applied action
Personnel action submit/apply	1,000/day	100 / 300 / 600 ms	20/s	≈ 200 MB	(worker_id, effective_date)	queue apply; ordering per worker preserved
Absence request + balance check	2,000/day; peak 200/min	60 / 200 / 400 ms	50/s	≈ 300 MB	(worker_id, plan_id, period_start)	balance read from projection; request still validated on write

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Accrual run	10,000 workers per period	run < 5 min	50 workers/s	≈ 120k rows/year	partition AbsenceBalance by plan year	longer run; ESS shows "accrual in progress"
ESS page load	10,000 workers, peak 2,000/min at open enrollment	120 / 300 / 600 ms	100 pages/s	n/a	BFF caching per user	shed non-critical tiles; enrollment path preserved
Skill search	500/day	100 / 400 / 800 ms	10 q/s	n/a	GIN on WorkerSkill (skill_id, rating)	limit 500 results
Questionnaire submit	5,000 respondents in 48 h	80 / 250 / 500 ms	50/s	≈ 1 GB answers	AnswerSession by schedule	accept and score asynchronously
PII reveal (audited)	200/day	50 / 150 / 300 ms	10/s	audit ≈ 50 MB	audit log append-only, monthly partitions	no degradation allowed: reveal fails closed if audit write fails

8.42.20 Localization and compliance hooks

- Country fields: the extension-slot framework (FR-42.12) holds per-country worker fields (US SSN, I-9 status, EEO ethnicity/veteran codes; DE Steuer-ID, Sozialversicherungsnummer, Konfession for Kirchensteuer; UK NI number); wave-48 packs register the slot schemas, validation and masking.
- US statutory: OSHA 300/300A/301 layouts and the ITA electronic submission, EEO-1 Component 1, VETS-4212, ACA 1095-C, I-9 document tracking, FMLA eligibility (12 months / 1,250 hours / 50 employees within 75 miles) and COBRA notices are produced by the wave-48 US pack from wave-42 data (incident register, worker slots, benefit end reasons, absence plans).
- EU: GDPR Art. 9 special-category data (health data in incidents, religion for Kirchensteuer) stored only in `hr_pii` with explicit purpose; works-council (DE BetrVG §87) approval flag on questionnaires and monitoring reminders; candidate data deletion 6 months after rejection by default (retention class `CANDIDATE`).
- Absence: statutory minimum leave (DE BUrlG 20 days for a 5-day week; EU WTD 4 weeks), sick-leave certification rules and public holidays come from wave-8 calendars and country absence plan templates; extension point `hr.absence.validate.<cc>`.
- Language and formats: names follow wave-7 party name formats (family-name-first locales); dates ISO 8601 in APIs, locale formats in UI; documents print in the worker's preferred language.
- Retention: per data class (EMPLOYMENT, PAYROLL_LINK, INCIDENT 5 years after year end, CANDIDATE, TRAINING) in the wave-5 retention policy, durations from the country pack; legal holds suspend purge.
- Compensation: pay transparency (EU Directive 2023/970 pay ranges in offers and reports from 2026-06-07) supported by job pay ranges on `Job` and gender pay gap report hooks in the wave-48 pack.

8.42.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX #27 — 12.12.1 Courses (HRMCourseTable, sessions, registrations, instructors, locations)	GAP	ADDED as FR-42.14	BR-42.9
AX #31 — 12.12.2 Course inquiries	GAP	ADDED as FR-42.14	"Course attendance" in 8.42.14
AX #35 — 12.12.2 SSRS Course reports (Attendance, Course confirmation letters)	GAP	ADDED as FR-42.14	Documents in 8.42.15
AX #38 — 12.12.3 Course confirmations / Transfer course completion to competencies	GAP	ADDED as FR-42.14	Job `hr-course-completion-transfer`
AX #40 — 12.12.4 Course types, groups, locations, classrooms, instructors, agenda templates	GAP	ADDED as FR-42.14	Course entities in 8.42.4
AX #28 — 12.12.1 Questionnaires (design, schedules, answer sessions, results)	GAP	ADDED as FR-42.15	BR-42.10
AX #32 — 12.12.2 Questionnaire results analysis	GAP	ADDED as FR-42.15	"Questionnaire results" in 8.42.14
AX #36 — 12.12.2 SSRS Questionnaire reports	GAP	ADDED as FR-42.15	Results export PDF/XLSX
AX #39 — 12.12.3 Distribute questionnaires / Collect results	GAP	ADDED as FR-42.15	Job `hr-questionnaire-distribute`
AX #41 — 12.12.4 Questionnaire types, question types, answer groups, result groups, media types, parameters	GAP	ADDED as FR-42.15	Parameters in 8.42.13
AX #83 — 12.29.1 Questionnaires (Org admin link)	GAP	ADDED as FR-42.15	One engine for HR, quality (wave 32) and org admin
AX #89 — 12.29.4 Questionnaires setup (link)	GAP	ADDED as FR-42.15	Same setup
AX #93 — 12.31.1 My questionnaires / Answer questionnaire / Results	GAP	ADDED as FR-42.15	ESS "My questionnaires"
AX #96 — 12.31.2 Questionnaire statistics / Results analysis	GAP	ADDED as FR-42.15	8.42.14
AX #97 — 12.31.2 SSRS Questionnaire reports (link)	GAP	ADDED as FR-42.15	8.42.14
AX #99 — 12.31.3 Schedule questionnaires (batch)	GAP	ADDED as FR-42.15	8.42.16
AX #101 — 12.31.4 Questionnaire setup (link)	GAP	ADDED as FR-42.15	8.42.13
AX #29 — 12.12.1 Injury and illness (incidents, treatments, body parts, OSHA logs)	GAP	ADDED as FR-42.16	BR-42.11
AX #33 — 12.12.2 Injury/illness inquiries	GAP	ADDED as FR-42.16	"Injury and illness log"
AX #37 — 12.12.2 SSRS OSHA 300/300A/301	GAP	ADDED as FR-42.16	Data extract here; form layouts and ITA submission DEFERRED to wave 48 US pack
AX #42 — 12.12.4 Incident types, body parts, treatments, severity levels, outcomes	GAP	ADDED as FR-42.16	Code tables in wave-3 registry
AX #30 — 12.12.1 Loaned items (HRMLoanTable, HRMLoanItem)	GAP	ADDED as FR-42.17	Linked to wave-44 assets
AX #34 — 12.12.2 Loaned items inquiry	GAP	ADDED as FR-42.17	"Loaned items outstanding"
AX #43 — 12.12.4 Loan types	GAP	ADDED as FR-42.17	Code table
AX #104 — 12.12.1 Skill mapping / skill gap analysis (HcmSkillMapping)	PARTIAL	ADDED as FR-42.18	AC-42.16
AX #106 — 12.12.1 Worker tasks / reminders / checklists (HcmWorkerTask)	PARTIAL	ADDED as FR-42.19	BR-42.12
AX #113 — 12.12.4 Worker tasks / checklist templates	PARTIAL	ADDED as FR-42.19	ChecklistTemplate
AX #110 — 12.12.3 Worker reminders (certificate expiry, probation end)	PARTIAL	ADDED as FR-42.20	On wave-5 alert rule engine
AX #105 — 12.12.1 Worker compliance (US) I-9, ADA, immigration/visa, union agreements	PARTIAL (wave 48)	DEFERRED to wave 48	Extension slots and visa-expiry reminder (FR-42.20) exist here; US records in the US pack
AX #112 — 12.12.4 I-9 document types, ADA requirements, visa types, union agreements	PARTIAL (wave 48)	DEFERRED to wave 48	Same

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX #109 — 12.12.2 SSRS EEO-1 / VETS	PARTIAL (wave 48)	DEFERRED to wave 48	Source fields in US extension slots
AX #111 — 12.12.3 HR year-end US ACA 1095-C	PARTIAL (wave 48)	DEFERRED to wave 48	Uses benefit enrollment and hours data
AX #118 — 12.13.4 Leave of absence / FMLA tracking (US)	PARTIAL (wave 48)	DEFERRED to wave 48	Absence plan type FMLA with eligibility CEL is configurable here; entitlement tracking rules in US pack
SAP #328 — 13.28 HRBEN00COBRA / HRBEN00ICOBRA COBRA	PARTIAL-42	DEFERRED to wave 48	`cobra_qualifying_event` captured on BenefitEnrollment; notices and premium billing in US pack
SAP #329 — 13.28 E-Recruiting (HRRCF_*) / SF Recruiting & Onboarding	PARTIAL-42	ADDED as FR-42.21	Requisitions, candidates, offers; career site and sourcing OUT-OF-SCOPE (non-goal)
SAP #330 — 13.28 PPCP / PPSS succession planning / SF Succession	PARTIAL-42	OUT-OF-SCOPE talent-suite non-goal for v1	Skill gap (FR-42.18) and reviews (FR-42.11) cover the data; succession pools via wave-35 connector to a talent suite
SAP #331 — 13.28 PECM_* enterprise compensation management	PARTIAL-42	ADDED as FR-42.22	BR-42.13; long-term incentives OUT-OF-SCOPE
SAP #337 — 13.28 SPRO compensation areas, plans, guidelines, budgets (T71*)	PARTIAL-42	ADDED as FR-42.22	Parameters in 8.42.13
SAP #332 — 13.28 PSV1/PSV2/PSV3/PSVP/LSO_PSV training & event management	PARTIAL-42	ADDED as FR-42.14	No SCORM/xAPI player
SAP #338 — 13.28 SPRO training & event management / LSO settings (T77*)	PARTIAL-42	ADDED as FR-42.14	8.42.13
SAP #333 — 13.28 HRUS_D2 / HRPAD concurrent / global employment	PARTIAL-42	ADDED as FR-42.23	BR-42.14
SAP #340 — 13.28 SPRO concurrent / global employment configuration	PARTIAL-42	ADDED as FR-42.23	Parameter "Concurrent employment enabled"
SAP #334 — 13.28 SF Onboarding / Offboarding	PARTIAL-42	ADDED as FR-42.19	BR-42.12
AX #108 — 12.12.1 Human resources → Cases (HR cases)	PARTIAL (wave 45)	DEFERRED to wave 45	Closed by FR-45.12 generic case management with category type HR; listed in 8.45.21

8.43 Wave 43: Payroll Engine & Expense Management

8.43.1 Scope and business context

This wave delivers a country-pluggable gross-to-net payroll engine replacing AX 2012 Payroll (PayrollPayStatement, PayrollEarningCode, PayrollTaxCode, PayrollBenefit, PayrollPayCycle) and SAP PY (schemas, PCRs, wage types, RT results table), and expense management replacing AX Travel and expense (TrvExpTable, TrvExpTrans, TrvPolicy, TrvPerDiem, TrvCreditCardTrans) and SAP Concur. After the wave, a payroll administrator runs a pay cycle from HR and time data, reviews gross-to-net, posts to the GL and produces a payment file; employees submit expense reports with receipts, policy checks, per diems and card transactions, and get reimbursed through AP.

Out of scope: country packs beyond US and the EU reference (DE assumed, see open question 3 in the master PRD); year-end form filing (structures only); garnishment court-order interfaces; Concur-style travel booking.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX 2012 payroll tax updates arrive as hotfix-delivered PayrollTaxCode data (Ceridian engine) with no diff view; SAP PY tax changes ship as HR support packs (BSI TUBs) that require a Basis transport cycle and regression of schemas/PCRs; a wrong table version is discovered only when net pay looks wrong	Rules embedded in code/transports, not versioned data	Rule sets as signed, versioned JSON packs with diff, publish approval and a frozen binding on each run (PayStatement.rule_version per line)	Tax-table deployment 1–2 weeks transport cycle → < 1 h; mis-versioned runs 0 (binding recorded on every line)
Gross-to-net results are opaque: AX pay statement lines do not show basis or bracket; SAP RT (PC_PAYRESULT) shows wage types only decipherable by a payroll consultant	No explain path from result to rule	Each `PayResultLine` stores basis_amount, rule_version and bracket; statement viewer drills to the rule	"Why is my tax X?" tickets resolved by admin in < 2 min vs consultant escalation; 200/200 fixtures explainable
Retro pay: AX PayrollRetroactiveEarning only handles earning-rate changes; SAP retro recalculation (`/551`, `/552` wage types) reruns whole periods and produces deltas nobody can reconcile	Recalculation designed around the batch, not around the delta	Retro engine replays original rule-set versions and books explicit RETRO_* delta lines	Retro reconciliation effort 4 h per run → 0; delta fixtures 25/25
Payroll GL posting (AX "Generate ledger journal" → LedgerJournalTrans; SAP PC00_M99_CIPE) is a separate step, often weeks late, and rounds per-worker amounts so the journal differs from statements by cents	Separate batch with its own rounding	Posting in the same workflow as finalize, per-line rounded amounts summed (no re-rounding) through the wave-11 posting API	Statement-to-GL variance 0.00; posting lag days → same run
Expense policy in TrvPolicy uses X++ expressions a consultant must change; violations appear only at submit; Concur audit rules are a separate SaaS admin console with no API to test	Policy not data, evaluated late	CEL rule schema (BR-43.5) evaluated at line commit and submit with a simulate-against-history endpoint	Rule change lead time 1–2 weeks → 10 min; employee rework (returned reports) –80%; 500/500 fixture exactness
Card transactions (TrvCreditCardTrans) imported by batch and matched by hand; receipts re-keyed	No matching logic, no OCR	Hourly feed import, tolerance matching, OCR draft lines	Manual line keying –90%; auto-match ≥ 95%

8.43.2 AX / SAP parity map

Capability	AX 2012 / D365	SAP ECC / S/4	Passage equivalent	Parity
Pay cycles and periods	PayrollPayCycle, PayrollPayPeriod	Payroll area, period parameters (T549A/T549Q)	`PayCycle`, `PayPeriod`	full
Earnings	PayrollEarningCode, PayrollEarningCodeRate, earning statements	Wage types (T512W), PCR rules	`EarningCode` (rule set versioned)	full
Deductions/benefits	PayrollBenefit, PayrollDeductionCode (AX via benefits)	Deduction wage types, PA0014/0015	`DeductionCode` (pre/post-tax, limits)	full
Taxes	PayrollTaxCode, PayrollTaxEngine (Ceridian), PayrollWorkerTaxCode	Tax schemas (US: UTAX via BSI TaxFactory)	`TaxRuleSet` per country/year; US federal/state tables	full for US; EU reference
Gross-to-net	PayrollPayStatement, PayrollPayStatementLine	Schema U000/D000, RT results table, CRT	`PayStatement` with `PayResultLine`	full
Retro pay	PayrollRetroactiveEarning	Retroactive accounting (recalculation)	`RetroCalculation`	full
GL posting	PayrollPayStatement distribution → LedgerJournal	PC00_M99_CIPE posting run	journal via wave-11 posting API	full

Capability	AX 2012 / D365	SAP ECC / S/4	Passage equivalent	Parity
Pay statement/payment	PayrollPayStatement, direct deposit	Pay slip (PE51), DME (PC00_M99_FPAYM)	`PayStatement` PDF; NACHA/SEPA via wave 36	full
Year-end	W-2 data (PayrollW2)	Tax reporter (PU19)	`YearEndForm` structures W-2, 1099, DE Lohnsteuerbescheinigung	partial — structures, not filing
Expense reports	TrvExpTable, TrvExpTrans, TrvCategory	Concur expense reports; PR05 (classic)	`ExpenseReport`, `ExpenseLine`	full
Policies and per diems	TrvPolicy, TrvPolicyRule, TrvPerDiem	Concur audit rules, per diem tables (T706*)	`ExpensePolicyRule`, `PerDiemRate`	full
Card feeds	TrvCreditCardTrans import	Concur card feeds	`CardTransaction` (CSV/OFX/Visa VCF)	full
Cash advances	TrvCashAdvance (request, approve, pay via AP, settle against reports), cash advance parameters	PR02/PR05 advances, advance wage types, settlement on trip posting (PRRW)	`CashAdvance` (FR-43.15)	full
Travel requisitions (pre-approval)	TrvRequisitionTable/Line, requisition policies	TP01/TP02/TP03/PR01 travel request, T702 travel plan settings, approval workflow	`TravelRequisition` (FR-43.16)	full — booking stays out
Deduction arrears	Arrears recovery processing (PayrollArrears), deduction priority	ARRRS / DDNTK tables, deduction priority T51P6	arrears ledger per deduction code (FR-43.17)	full
Off-cycle runs	Manual pay statements, PayrollPayStatement (off-cycle)	PUOC_10 off-cycle workbench (bonus, correction, manual check, check replacement)	`PayRun.run_type` OFF_CYCLE subtypes (FR-43.18)	full
Overpayment claims	Negative net carried as advance (manual)	PC00_M10_CLAIMS claims processing	`PayrollClaim` (FR-43.19)	full
Quarterly/annual statutory data	Payroll tax reports (941, 940, state UI)	PC00_M10_* tax reporter, PU19	`StatutoryWageAggregate` (FR-43.20); filing via wave 48	partial — filing in US pack
Workers' compensation	Workers' compensation report (US)	US workers' comp (IT0001 WC code via feature)	class code on job/position by state (FR-43.21)	full for data; state reports via wave 48

8.43.3 Functional requirements

- **FR-43.1 Pay groups, cycles, periods** — Must. The system shall define pay groups per legal entity with frequency (WEEKLY, BIWEEKLY, SEMIMONTHLY, MONTHLY), generated pay periods and pay dates, and a cutoff for time/absence.
- **FR-43.2 Versioned rule sets** — Must. The system shall express earnings, deductions, taxes and employer contributions as rule sets versioned by country and effective date; a pay run binds to the rule-set versions active on its period end.
- **FR-43.3 Earnings** — Must. The system shall compute REGULAR, OVERTIME (1.5×, 2.0×), PTO, HOLIDAY, BONUS, COMMISSION and SHIFT_DIFF from salary (FixedCompensation ÷ periods) and wave-25 time registrations.
- **FR-43.4 Deductions and contributions** — Must. The system shall compute pre-tax (401(k), Section 125 medical, HSA/FSA) and post-tax deductions with per-period, annual and percent-of-gross limits, and employer contributions (match with cap).
- **FR-43.5 US tax engine** — Must. The system shall compute federal income tax withholding per Pub 15-T percentage method (automated payroll systems), Social Security (6.2% to the annual wage base), Medicare (1.45% + 0.9% additional over 200,000), FUTA, and state income tax/SUI for the states covered by the year's table pack.
- **FR-43.6 EU reference pack (DE)** — Should. The system shall compute Lohnsteuer (via the BMF programmablauf tables), Solidaritätszuschlag, Kirchensteuer and social insurance contributions (KV/PV/RV/AV with Beitragsbemessungsgrenzen) for a demo employee set.
- **FR-43.7 Gross-to-net and pay statements** — Must. The system shall produce a pay statement per worker per period with earnings, pre-tax, taxes, post-tax, net, employer costs and YTD; statements are immutable after the run is FINALIZED.
- **FR-43.8 Retro pay** — Must. The system shall recalculate prior periods when backdated changes arrive (rate, hours, absence) and post deltas in the current period with an explanatory line.
- **FR-43.9 GL posting and payments** — Must. The system shall post the run to the GL by (earning/deduction/tax code × dimension from position) and generate NACHA/SEPA payment batches through wave 36.
- **FR-43.10 Provider adapter** — Should. The system shall export a provider payroll file (ADP/Gusto-style CSV/JSON) and import results for entities that outsource payroll, posting the same GL structure.
- **FR-43.11 Expense reports** — Must. The system shall let workers create reports with lines (category, date, amount, currency, merchant, project/WBS, attendees, receipt), split lines, itemize hotels and submit for approval.
- **FR-43.12 Policy engine** — Must. The system shall evaluate rule sets at line commit and submit, producing WARN, BLOCK, REQUIRE_JUSTIFICATION or REQUIRE_RECEIPT outcomes, with per-diem computation by location/date.
- **FR-43.13 Card feeds and receipts** — Must. The system shall import card transactions (CSV, OFX, Visa VCF 4.0) into a worker's unassigned pool, match receipts by amount/date/merchant (tolerance ±1% or 2 days), and OCR receipts into draft lines.
- **FR-43.14 Reimbursement** — Must. The system shall post approved reports as wave-18 vendor invoices against the worker's employee-vendor account (company-paid card lines against the card issuer vendor), with tax recovery lines where VAT is recoverable.
- **FR-43.15 Cash advances** — Should. The system shall let workers request cash advances (amount, currency, purpose, optional travel requisition and project), route them for approval, pay them through wave-18/19 as a prepayment to the employee vendor, and settle them against expense reports per BR-43.9 with return or payroll recovery of unused amounts.
  - Statuses: DRAFT → SUBMITTED → APPROVED → PAID → PARTIALLY_SETTLED → SETTLED; SUBMITTED → REJECTED.
  - Aging: unsettled advances older than the parameter limit (default 30 days after trip end) raise reminders and block new advances.
- **FR-43.16 Travel requisitions** — Should. The system shall capture pre-trip requisitions (purpose, destinations, dates, estimated lines by category, project/WBS, attendees) with policy checks, approval workflow, an optional linked cash advance, and comparison of the resulting expense report against the approved estimate.
- **FR-43.17 Deduction arrears recovery** — Should. The system shall, when net pay is insufficient, take deductions in priority order, record untaken amounts in an arrears ledger per deduction code (policy ARREARS, NO_ARREARS or PARTIAL), and recover arrears in later periods within a per-period recovery limit per BR-43.10.
- **FR-43.18 Off-cycle runs** — Must. The system shall run OFF_CYCLE pay runs of subtype BONUS, FINAL_PAY, MANUAL_CHECK, CHECK_REPLACEMENT and REVERSAL for single or selected workers, with supplemental-wage tax methods per BR-43.11 and the same finalize/post/pay workflow as regular runs.
  - FINAL_PAY includes accrued unused paid leave per the absence plan's payout flag (wave 42) and state final-pay deadlines from the US pack.
  - CHECK_REPLACEMENT voids the original payment (wave 36) and reissues without recalculation.
- **FR-43.19 Overpayment claims** — Should. The system shall create a payroll claim when a recalculation (retro, correction, reversal) yields negative net for a worker, with recovery as a payroll deduction schedule, direct repayment or write-off, and gross-vs-net treatment by tax year per BR-43.12.
- **FR-43.20 Statutory wage aggregates** — Should. The system shall aggregate wages and taxes per legal entity, jurisdiction and quarter/year (federal 941/940 lines, state UI wages and excess wages, local tax) from finalized statements, reconcile them to GL liability accounts, and expose them to the wave-48 country pack that renders and files the returns.
- **FR-43.21 Workers' compensation class codes** — Could. The system shall assign workers' compensation class codes per state to jobs (override per position), compute the premium base (gross minus overtime premium portion, minus excluded earnings flagged on earning codes), and report premium base by class, state and period.

8.43.4 Data model

PayRun

Field	Type	Required	Notes
id	uuid	yes	PK
pay_group_id, pay_period_id	uuid	yes	FK
run_type	enum	yes	REGULAR, OFF_CYCLE, CORRECTION
status	enum	yes	DRAFT, CALCULATED, REVIEWED, FINALIZED, POSTED, PAID, CANCELED
rule_set_binding	jsonb	yes	{country, taxYear, versions{}} frozen at calculate
gross_total, net_total, employer_cost_total	numeric(18,2)	computed	
journal_id, payment_batch_id	uuid	no	
calculated_at, finalized_by	timestamptz, uuid	no	

PayStatement

Field	Type	Required	Notes
id	uuid	yes	PK
pay_run_id, worker_id, employment_id	uuid	yes	
gross, taxable_wages_fed, fica_wages, net	numeric(14,2)	yes	
lines[]	PayResultLine	yes	(code, kind EARNING/PRETAX/TAX/POSTTAX/EMPLOYER, hours, rate, amount, ytd_amount, basis_amount, rule_version)
w4	jsonb	cond	{filingStatus, multipleJobs, dependentsCredit, otherIncome, deductions, extraWithholding}
pdf_ref	text	no	object store key
hash	text	yes	SHA-256 of lines; immutability check

TaxRuleSet: id, country, jurisdiction (FED, state code, locality), tax_year, version, effective_from, brackets[] (filing_status, over, not_over, base_tax, rate), wage_base, rates{}, status (DRAFT, PUBLISHED). **EarningCode**: code, name, kind (HOURLY/SALARY/FLAT), multiplier, taxable flags (fed, fica, futa, state), gl_account_rule. **DeductionCode**: code, pre_tax bool, basis (FLAT/PERCENT_GROSS), limits {per_period, annual}, employer_match {percent, cap_percent}, gl_rule, priority.

ExpenseReport: id, worker_id, legal_entity_id, purpose, status (DRAFT, SUBMITTED, APPROVED, REJECTED, POSTED, PAID), total_amount, reimbursable_amount, company_paid_amount, currency, approver_chain[], ap_invoice_id. **ExpenseLine**: id, report_id, category_id, txn_date, amount, currency, exchange_rate, amount_accounting, merchant, city/country, project_id, wbs_node_id, card_txn_id, receipt_id, attendees[], itemizations[], policy_results[] (rule_id, outcome, justification), tax_code, personal bool. **ExpensePolicyRule** (schema in BR-43.5). **CardTransaction**: id, worker_id, card_last4, txn_date, post_date, merchant, mcc, amount, currency, status (UNASSIGNED, ASSIGNED, DISPUTED, PERSONAL).

CashAdvance: id (sequence EXP-ADV), worker_id, legal_entity_id, amount, currency, purpose, travel_requisition_id, project_id, status (DRAFT, SUBMITTED, APPROVED, REJECTED, PAID, PARTIALLY_SETTLED, SETTLED), paid_amount, settled_amount, returned_amount, recovered_via_payroll_amount, ap_prepayment_id, due_settlement_date. **TravelRequisition**: id (sequence EXP-TR), worker_id, purpose, destinations[] (country, city, from, to), estimate_lines[] (category_id, amount, currency), estimated_total, project_id, wbs_node_id, status (DRAFT, SUBMITTED, APPROVED, REJECTED, CLOSED), linked_report_ids[]. **ArrearsLedger**: worker_id, deduction_code, pay_run_id, amount_due, amount_taken, arrears_created, arrears_recovered, balance. **PayrollClaim**: id, worker_id, origin_pay_run_id, tax_year, gross_amount, net_amount, recovery_method (PAYROLL_DEDUCTION, DIRECT_REPAYMENT, WRITE_OFF), per_period_amount, balance, status (OPEN, RECOVERING, SETTLED, WRITTEN_OFF). **StatutoryWageAggregate**: legal_entity_id, jurisdiction, form_code (941, 940, SUI_, …), period (YYYY-Qn or YYYY), line_code, amount, statement_count, reconciled_at. **WorkersCompClass**: state, class_code, description, rate_per_100, job_ids[]. PayRun adds `off_cycle_subtype` (BONUS, FINAL_PAY, MANUAL_CHECK, CHECK_REPLACEMENT, REVERSAL) and `original_pay_run_id`; DeductionCode adds `arrears_policy` and `recovery_limit_pct`; EarningCode adds `supplemental` bool and `wc_excluded` bool.

Invariants: Σ PayStatement.net = PayRun.net_total = payment batch total; Σ GL debits = gross + employer cost; a FINALIZED statement's hash never changes; a card transaction belongs to at most one line; report total = Σ lines (reimbursable + company-paid). Indexing: `PayStatement (worker_id, pay_run_id)` unique; `CardTransaction (worker_id, status)`; payroll tables partitioned by tax_year.

8.43.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/payroll/runs`	Create run for pay period	key	PayrollRunMaintain
POST	`/api/v1/payroll/runs/{id}/calculate`	Gross-to-net calculation (Temporal workflow)	key + status guard	PayrollRunMaintain
POST	`/api/v1/payroll/runs/{id}/finalize`	Freeze statements	key	PayrollRunFinalize
POST	`/api/v1/payroll/runs/{id}/post`	GL post + payment batch	key	PayrollRunPost
PUT	`/api/v1/payroll/rule-sets/{id}/publish`	Publish rule-set version	ETag	PayrollRuleAdmin
POST	`/api/v1/expense/reports`	Create report	key	EssExpense
POST	`/api/v1/expense/reports/{id}/lines`	Add line (policy evaluated)	key	EssExpense
POST	`/api/v1/expense/reports/{id}/submit`	Submit	key	EssExpense
POST	`/api/v1/expense/card-transactions/import`	Import feed	key per file hash	ExpenseAdmin
POST	`/api/v1/expense/receipts`	Upload + OCR	key	EssExpense
POST	`/api/v1/expense/cash-advances`	Request cash advance	key	EssExpense
POST	`/api/v1/expense/cash-advances/{id}/decide`	Approve/reject advance	key	MssApprove
POST	`/api/v1/expense/cash-advances/{id}/return`	Record returned cash	key	ExpenseAdmin
POST	`/api/v1/expense/travel-requisitions`	Create requisition	key	EssExpense
POST	`/api/v1/expense/travel-requisitions/{id}/submit`	Submit requisition	key	EssExpense
POST	`/api/v1/payroll/runs/off-cycle`	Create off-cycle run (subtype, workers)	key	PayrollRunMaintain
POST	`/api/v1/payroll/claims/{id}/recovery`	Set claim recovery method/schedule	key	PayrollRunMaintain
GET	`/api/v1/payroll/statutory-aggregates?entity=&form=&period=`	Quarterly/annual wage aggregates	n/a	PayrollInquire

Validation: calculate requires HR cutoff passed and all time registrations for the period approved (or policy allows unapproved as 0 hours); finalize requires zero BLOCK validation results; expense submit requires every REQUIRE_* satisfied.

Event	Trigger	Payload	Consumers
`dev.passage.payroll.run.posted.v1`	GL post	see 8.0	journal, bank, analytics
`dev.passage.payroll.statement.finalized.v1`	finalize	statementId, workerId, net, payDate	ESS, provider adapter
`dev.passage.expense.report.approved.v1`	approval	see 8.0	ap, projects
`dev.passage.expense.policy.violated.v1`	BLOCK outcome at submit	reportId, lineId, ruleId, outcome	analytics, audit
`dev.passage.expense.cash-advance.paid.v1`	AP prepayment paid	advanceId, workerId, amount, currency, prepaymentId	expense settlement, analytics
`dev.passage.payroll.claim.created.v1`	negative net on recalculation	claimId, workerId, taxYear, gross, net	payroll admin notifications, AR (direct repayment)
`dev.passage.payroll.statutory-aggregate.closed.v1`	quarter/year aggregate reconciled	legalEntityId, formCode, period, totals	wave-48 country pack filing

8.43.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Payroll workspace	tiles: runs by status, retro pending, exceptions		PayrollInquire
Pay run detail	calculate, review exceptions, compare to prior run (variance %), finalize, post	worker, gross, taxes, net, Δ vs prior	PayrollRunMaintain
Statement viewer	drill line → rule version and basis	earnings, pre-tax, taxes, post-tax, net, YTD	PayrollInquire / EssPayStatement
Rule-set editor	edit brackets/rates, diff versions, publish	jurisdiction, tax year, brackets	PayrollRuleAdmin
Worker payroll profile	W-4 (2020+ fields), state forms, direct deposit (vaulted), garnishments		PayrollWorkerMaintain
Expense report (ESS, mobile)	add line, attach receipt (camera), match card, itemize, submit	category, amount, policy badges	EssExpense
Expense approval (MSS)	approve/reject/return, view receipt side-by-side		MssApprove
Policy admin	rule list, simulate against past reports		ExpenseAdmin
Cash advances (ESS / admin)	request, view settlement, record return; admin aging list	amount, paid, settled, open, due date	EssExpense / ExpenseAdmin
Travel requisition (ESS / MSS)	create with estimates, submit, approve, link report	destinations, dates, estimate by category, actual vs estimate	EssExpense / MssApprove
Off-cycle workbench	pick workers and subtype, calculate, finalize, void/reissue	worker, subtype, gross, supplemental tax, net	PayrollRunMaintain
Claims and arrears	view balances, set recovery schedule, write off	worker, code, arrears balance, claim balance	PayrollRunMaintain
Statutory aggregates	review quarter totals, reconcile, release to pack	form, line, amount, GL balance, variance	PayrollInquire

Rules: pay run variance view highlights workers whose net changed > 20% vs prior period; rule-set publish requires a second approver (SoD); expense mobile works offline for draft lines and receipt capture.

8.43.7 Business rules and edge cases

- **BR-43.1 US gross-to-net worked example** (rule set `US-FED-2025.1`, biweekly, 26 periods; Colorado flat 4.40% on federal taxable wages; W-4 2020+, Single, no Step 2/3/4 entries; 401(k) 6% pre-tax; employer match 50% of first 6%). Gross REGULAR 4,000.00. Pre-tax 401(k) 240.00. Federal taxable wages 3,760.00; FICA wages 4,000.00 (401(k) does not reduce FICA). Pub 15-T Step 1: adjusted annual wage = 3,760.00 × 26 = 97,760.00. Step 2 (standard withholding schedule, Single, 2025 table): bracket 54,875–115,125 → 5,578.50 + 22% × (97,760.00 – 54,875.00) = 5,578.50 + 9,434.70 = 15,013.20; ÷ 26 = 577.43. Step 3 credits 0; Step 4 extra 0 → federal withholding 577.43. Social Security 4,000.00 × 6.2% = 248.00 (YTD below wage base 176,100.00). Medicare 4,000.00 × 1.45% = 58.00. Colorado 3,760.00 × 4.40% = 165.44. Net = 4,000.00 – 240.00 – 577.43 – 248.00 – 58.00 – 165.44 = 2,711.13. Employer: match 120.00, SS 248.00, Medicare 58.00, FUTA 0.6% × 4,000 = 24.00 until YTD 7,000. GL: Dr salary expense 4,000.00, Dr employer tax expense 330.00, Dr 401(k) match expense 120.00; Cr net pay clearing 2,711.13, Cr 401(k) payable 360.00, Cr federal WH payable 577.43, Cr FICA payable 612.00, Cr FUTA payable 24.00, Cr CO WH payable 165.44.
- **BR-43.2 Rounding** — each tax computed on unrounded basis and rounded half-up to 0.01 per line; annualized federal amount divided by periods then rounded; YTD caps applied on the rounded cumulative.
- **BR-43.3 Wage base caps** — SS stops when YTD FICA wages reach the wage base, applying partial tax on the crossing period; additional Medicare starts on the crossing period for wages over 200,000.00 regardless of filing status.
- **BR-43.4 Retro** — a retro trigger is any change with effective date < current period start affecting a FINALIZED statement; recalculation re-runs the original rule-set versions, computes delta per line, and books deltas as RETRO_* lines in the current period; taxes on retro use current-period method (no amended returns).
- **BR-43.5 Expense policy rule schema** — JSON: `{ "id": "EXP-MEAL-075", "version": 3, "scope": {"legalEntities": ["ES-US"], "categories": ["MEALS"], "workerGroups": ["*"]}, "basis": "PER_DAY" | "PER_LINE" | "PER_REPORT" | "PER_TRIP", "condition": "line.amountAccounting > 75.00 && !line.attendees.size() > 0" (CEL over line, report, worker, perDiem), "threshold": {"amount": 75.00, "currency": "USD"}, "outcome": "WARN" | "BLOCK" | "REQUIRE_JUSTIFICATION" | "REQUIRE_RECEIPT", "message": "Meals over 75 USD/day need attendees", "effectiveFrom": "2026-01-01" }`. Example set: receipts required for any line ≥ 25.00 USD/EUR; lodging per night ≤ GSA/per-diem table × 1.5 else justification; mileage at IRS rate from table; alcohol category BLOCK; submission later than 60 days after txn_date BLOCK; duplicate (same merchant, amount, date within worker) BLOCK.
- **BR-43.6 Per diem** — rate by (country, city tier, date); first/last day 75%; meals provided reduce by 20/30/50% (breakfast/lunch/dinner); computed in rate currency then converted at line date.
- **BR-43.7 Finalization** — FINALIZED statements are immutable; corrections are OFF_CYCLE or CORRECTION runs that reference the original.
- **BR-43.8 Separation of duties** — the user who calculates cannot finalize; the user who publishes a rule set cannot be its author (enforced by wave-48 SoD engine, interim OPA check).
- **BR-43.9 Cash advance settlement** — at expense report approval, open PAID advances of the worker in the report currency are applied oldest first to reimbursable lines: applied = min(advance open, reimbursable total); the AP invoice for the report is reduced by the applied amount and the prepayment is settled against it (wave 18); if the report total < advance, the open remainder stays on the advance until (a) the worker records a cash return, (b) the settlement due date passes and the parameter "recover through payroll" creates a post-tax deduction in the next regular run, or (c) it is written off with ExpenseAdmin approval. Advances in another currency are converted at the report approval date rate (wave 12) and the FX difference posts to the realized FX account. Example: advance 1,000.00 USD; report 820.00 → applied 820.00, open 180.00; worker returns 180.00 → SETTLED. Report 1,150.00 → applied 1,000.00, AP invoice 150.00.

- **BR-43.10 Arrears** — deductions are taken in ascending priority (taxes and court-ordered garnishments first, then pre-tax benefits, then post-tax voluntary); when net would go below zero (or below the state minimum-net rule from the pack), the remaining deduction is: NO_ARREARS → skipped; ARREARS → full untaken amount added to the ledger; PARTIAL → taken up to net, rest to ledger. Recovery in later periods adds up to recovery_limit_pct (default 100 %) of the normal per-period deduction; pre-tax arrears recovered in a later tax year are recovered post-tax. Example: net before medical 150.00, medical 200.00 (PARTIAL) → take 150.00, arrears 50.00; next period normal 200.00 + recovery 50.00 = 250.00.
- **BR-43.11 Supplemental wage tax (US)** — earning codes flagged supplemental, paid in an OFF_CYCLE BONUS run or identified separately, are withheld at the flat supplemental rate (22 % for 2025; 37 % on the portion of YTD supplemental wages above 1,000,000.00) unless the worker's W-4 or the parameter selects the aggregate method; FICA applies as for regular wages. Example: bonus 5,000.00 USD → federal 1,100.00, SS 310.00, Medicare 72.50, state per pack (CO 4.40 % → 220.00), net 3,297.50.
- **BR-43.12 Overpayment claims** — a claim is created when the recalculated net for a worker < 0; in the same tax year the claim is recovered net (taxes reversed through the recalculation); across tax years the claim amount is gross (W-2c correction data flagged for the pack) per IRS rules; recovery as payroll deduction is limited by the pack's state rule (e.g. written consent and maximum per period); write-off posts to payroll overpayment expense.
- **BR-43.13 Travel requisition variance** — an expense report linked to an approved requisition is compared by category; when actual total > estimated total × (1 + tolerance) (parameter, default 10 %) the report requires justification (REQUIRE_JUSTIFICATION outcome); travel on dates outside the requisition's range is flagged WARN; a requisition closes when its linked report is POSTED or 60 days after the last travel date.

Pay run state machine: DRAFT → CALCULATED → REVIEWED → FINALIZED → POSTED → PAID; CALCULATED/REVIEWED → DRAFT (recalculate); DRAFT… REVIEWED → CANCELED. Expense report: DRAFT → SUBMITTED → APPROVED → POSTED → PAID; SUBMITTED → REJECTED ! RETURNED (→ DRAFT).

8.43.8 Acceptance criteria

- **AC-43.1** (FR-43.5, BR-43.1) Given the worked-example worker, when the run calculates, then federal WH = 577.43, SS = 248.00, Medicare = 58.00, CO = 165.44, net = 2,711.13.
- **AC-43.2** (FR-43.5, BR-43.3) Given YTD FICA wages 174,000.00 and gross 4,000.00, when calculated, then SS = 130.20 (6.2% × 2,100.00).
- **AC-43.3** (FR-43.5) Given 200 fixture scenarios across filing statuses, Step 2 checkbox, Step 3/4 values and 6 states, when calculated, then all match Pub 15-T expectations to 0.01.
- **AC-43.4** (FR-43.8, BR-43.4) Given a rate increase effective two periods back, when retro evaluates and the next run calculates, then RETRO_REGULAR equals the summed delta and taxes recompute on the new gross.
- **AC-43.5** (FR-43.9) Given a posted run, when reconciliation runs, then GL payroll expense and liability accounts tie to Σ statements with 0.00 variance.
- **AC-43.6** (FR-43.12, BR-43.5) Given the 500-report fixture, when submitted, then policy outcomes equal the expected matrix exactly (0 false flags).
- **AC-43.7** (FR-43.13) Given a card feed with 1,000 transactions and 800 receipts, when matched, then ≥ 95% auto-match within tolerance and no cross-worker matches.
- **AC-43.8** (FR-43.14) Given an approved report with 300.00 USD reimbursable and 200.00 company-paid, when posted, then an AP invoice of 300.00 exists for the employee vendor and 200.00 for the issuer vendor, and the next payment run pays 300.00.
- **AC-43.9** (FR-43.15, BR-43.9) Given a PAID advance of 1,000.00 USD and an approved report of 820.00, when the report posts, then 820.00 settles against the prepayment, the advance shows open 180.00 and status PARTIALLY_SETTLED; after a recorded return of 180.00 it is SETTLED.
- **AC-43.10** (FR-43.16, BR-43.13) Given an approved requisition estimating 2,000.00 and a linked report totaling 2,300.00 with tolerance 10 %, when submitted, then the report requires justification.
- **AC-43.11** (FR-43.17, BR-43.10) Given net before medical 150.00 and a PARTIAL medical deduction of 200.00, when calculated, then 150.00 is taken, the arrears ledger holds 50.00, and the next period takes 250.00.
- **AC-43.12** (FR-43.18, BR-43.11) Given a BONUS off-cycle run of 5,000.00 for the BR-43.1 worker, when calculated, then federal 1,100.00, SS 310.00, Medicare 72.50, CO 220.00 and net 3,297.50.
- **AC-43.13** (FR-43.19, BR-43.12) Given a correction that reduces a finalized gross by 1,000.00 after net was paid, when recalculated in the same tax year, then a claim with net amount equal to the net difference is created and a recovery schedule of 4 equal deductions can be set.
- **AC-43.14** (FR-43.20) Given a quarter with 13 finalized biweekly runs, when the 941 aggregate closes, then lines 2 (wages), 3 (federal withholding), 5a/5c (SS/Medicare wages) equal Σ statements and tie to GL liability accounts with 0.00 variance.

8.43.9 Test plan and fixtures

Fixtures: `payroll-us-200` (200 scenarios, expected values in `fixtures/payroll/us-2025-expected.csv` derived from Pub 15-T worksheets); `payroll-de-20` (20 DE employees against BMF test cases); `payroll-retro-25`; `expense-policy-500` (reports × rules with expected outcomes); `card-feed-1000` (VCF and OFX samples).

Test	Type	Target	Pass criterion
Federal/state withholding	unit	tax engine	200/200 to 0.01
Caps and crossing periods	property	FICA/FUTA	never over-withholds; crossing period correct
Retro deltas	unit	retro engine	25/25
GL posting	contract	posting API	balanced; account per code rule
Policy engine	unit	CEL evaluator	500/500
Receipt OCR	integration	OCR service	amount/date extracted ≥ 90% on sample set
Pay run perf	performance	calculate workflow	10,000 workers < 10 min

Reconciliation: Σ statement net = payment batch; Σ tax lines by jurisdiction = liability accounts; Σ reimbursable = AP employee-vendor invoices.

8.43.10 Deployment and operations

Charts `passage-payroll` (Go API, Temporal calc workers, statement renderer) and `passage-expense` (TS API, OCR Python worker, feed importer); Postgres `payroll` (encrypted bank columns via `hr_pii` key pattern) and `expense`. Rule-set packs delivered as signed JSON bundles (`rulesets/us/2025.1.json`) loaded by migration job. Flags: `payroll.de-pack`, `payroll.provider-adapter`, `expense.ocr`. Schedules: `payroll-cutoff-reminder`, `expense-card-import` hourly (SFTP pull), `expense-aging` daily. SLOs: calculate 10k workers < 10 min; statement fetch p95 < 200 ms; policy evaluation p95 < 100 ms per line. Alerts: calculation failures; tax table pack missing for period end; card import failures. Runbook: re-run calculation from DRAFT; reissue payment batch; rotate SFTP keys. Rollback by digest; FINALIZED data is never altered by migrations.

8.43.11 Migration from AX / SAP

Source	Passage entity	Notes
PayrollPayCycle, PayrollPayPeriod	PayCycle, PayPeriod	
PayrollEarningCode, PayrollTaxCode, PayrollBenefit	EarningCode, TaxRuleSet mapping, DeductionCode	AX tax engine codes mapped to jurisdictions
PayrollWorkerTaxCode, PayrollWorkerTaxRegion, W-4 data	worker payroll profile	
PayrollPayStatement (current year)	PayStatement (YTD opening balances only)	historical statements archived
TrvExpTable/TrvExpTrans (open)	ExpenseReport/ExpenseLine	DRAFT/SUBMITTED only
TrvPolicy/TrvPolicyRule, TrvPerDiem	ExpensePolicyRule, PerDiemRate	X++ expression rules rewritten to CEL (manual)
T512W wage types, T549*	EarningCode/DeductionCode, PayCycle	
RT (PCL2 cluster) current year	YTD opening balances	
PA0210 (W-4), PA0207/0208/0209	worker payroll profile	
Concur exports	ExpenseReport (open)	via CSV

Validation: YTD gross/tax per worker equal to source within 0.01; count of open expense reports equal. Not migrated: prior-year statements (lakehouse archive), SAP schemas/PCRs (rebuilt as rule sets), closed expense reports.

8.43.12 Dependencies, risks, open questions

Depends on: wave 42 HR, wave 25 time, wave 18 AP, wave 19 payments, wave 36 bank files, wave 13 tax codes, wave 12 FX.

Risk	Likelihood	Impact	Mitigation
Tax table maintenance each year	High	High	Signed rule-set packs; provider adapter path
OCR accuracy	Medium	Low	Draft lines always user-confirmed

Open questions: (1) EU reference DE vs UK (master PRD Q3). Default DE; decide before wave 43 start. (2) Store direct-deposit details in payroll or wave-36 bank domain? Default payroll with vault; decide by wave 43.

8.43.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Pay group frequency and pay-date offset	pay group	enum + int	BIWEEKLY, +5 days after period end	WEEKLY, BIWEEKLY, SEMIMONTHLY, MONTHLY; 0–15	FR-43.1 period generation	Payroll parameters › Pay cycles (PayrollPayCycle)	T549A payroll area; T549Q periods; PA03 control record
Time/absence cutoff (hours before pay date)	pay group	int	72	0–240	Calculate precondition	Payroll parameters › Cutoff	PA03 control record "released for payroll"; T549S date modifiers
Unapproved time policy	pay group	enum	ZERO_HOURS	ZERO_HOURS, INCLUDE, BLOCK	8.43.5 validation	Payroll parameters › Pay statements	CATS transfer status / PT time evaluation
Rule-set pack per country/year	legal entity	lookup	latest PUBLISHED	TaxRuleSet versions	FR-43.2 binding	Tax engine update (PayrollTaxEngine)	T511K constants, T5UTZ US tax authorities, BSI TaxFactory version
Supplemental tax method	legal entity	enum	FLAT	FLAT, AGGREGATE	BR-43.11	Payroll parameters › Supplemental tax	T5UTN tax models / BSI method
Variance highlight threshold	tenant	pct	20 %	1–100 %	Pay run variance view	n/a	PC00_M99_PA03_CORR / payroll log thresholds n/a
Rounding mode	country pack	enum	HALF_UP per line	HALF_UP, HALF_EVEN	BR-43.2	Payroll parameters › Rounding	T511K rounding constants
Arrears policy per deduction code	deduction code	enum	ARREARS	ARREARS, NO_ARREARS, PARTIAL	BR-43.10	Deduction code › Arrears (PayrollDeductionCode)	T51P6 processing class / deduction priority and arrears
Arrears recovery limit	deduction code	pct	100 %	0–500 %	BR-43.10	Arrears recovery settings	T51P6 recovery rules
Payment method default	pay group	enum	DIRECT_DEPOSIT	DIRECT_DEPOSIT, CHECK, PAYCARD	FR-43.9	Payroll parameters › Payment	T042Z payment methods; IT0009 bank details
Payroll GL posting granularity	legal entity	enum	CODE_X_DIMENSION	SUMMARY, CODE_X_DIMENSION, WORKER	FR-43.9	Payroll parameters › Ledger › Summarize	T52EL / T52EK symbolic accounts; PC00_M99_CIPE variant
Expense report number sequence and approval chain	legal entity	lookup	EXP-REP; manager → cost-center owner > 5,000.00	wave-4 workflow	FR-43.11	Travel and expense parameters › Number sequences, workflow	T702N trip number range; travel workflow WS20000050
Receipt required threshold	legal entity	amount	25.00 USD/EUR	≥ 0	BR-43.5	Expense management parameters › Receipt required	T706B1 receipt types "receipt required"
Submission deadline (days after txn)	legal entity	int	60	1–365	BR-43.5	Expense policy (TrvPolicy)	T706D trip provision variant deadlines
Mileage rate table	legal entity	table	IRS / BMF tables	per vehicle type, date	BR-43.5	Mileage rates (TrvMileage)	T706K / T706F mileage rates
Per diem tables and meal reduction	legal entity	table	GSA / BMF	country, city tier, date	BR-43.6	Per diems (TrvPerDiem)	T706V/T706U per diem tables
Card match tolerance	tenant	pct + days	±1 %, 2 days	0–10 %, 0–10 days	FR-43.13	Credit card parameters	Concur card matching
Cash advance max per request	legal entity	amount	2,000.00	≥ 0	FR-43.15	Expense management parameters › Cash advance limit	T706* advance limits
Cash advance settlement due (days after trip end)	legal entity	int	30	1–180	BR-43.9 aging	Cash advance parameters › Settlement	PR05 advance settlement
Recover unsettled advances through payroll	legal entity	bool	false		BR-43.9	Cash advance parameters › Recover via payroll	Trip transfer to payroll (PRPY)
Travel requisition required	legal entity / category	bool	false		FR-43.16	Travel requisition parameters	T702 travel plan settings: request required

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Requisition variance tolerance	legal entity	pct	10 %	0–100 %	BR-43.13	Requisition policies	Travel request estimated cost check
Workers' comp class code source	legal entity	enum	JOB	JOB, POSITION	FR-43.21	Workers compensation setup	Feature for WC code (IT0001 assignment)

8.43.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Payroll register	operational report	pay run, pay group, department	worker, hours, earnings, pre-tax, taxes, post-tax, net, employer cost	`pay_register`	Pay statement report (PayrollPayStatementReport)	PC00_M10_CEDT remuneration statement; PC00_M10_REC payroll reconciliation
Wage type / code reporter	drill-down	code, period range, entity	amount, hours, worker count by code	`pay_codes`	Earning statements inquiry	H99CWTR0 / PC00_M99_CWTR wage type reporter
Tax liability reconciliation	operational report	jurisdiction, period	withheld, employer tax, GL liability, variance	`pay_tax_recon`	Payroll tax reports	PC00_M10_RPCTRCU0 tax reconciliation
Variance vs prior run	drill-down	pay run, threshold	worker, Δ gross, Δ net, Δ %	`pay_variance`	n/a	PC00_M99_PA03_CHECK / payroll journal comparison
Arrears and deductions not taken	list inquiry	code, worker, open only	due, taken, arrears balance	`pay_arrears`	Arrears inquiry	ARRRS / DDNTK cluster tables via H99CWTR0
Claims	list inquiry	status, tax year	worker, gross, net, balance, method	`pay_claims`	n/a	PC00_M10_CLAIMS
Statutory aggregates (941/940/SUI)	statutory report	entity, form, period	form lines, amounts, variance to GL	`pay_statutory`	Payroll tax forms (941, 940)	PU19 tax reporter; PC00_M10_*
YTD by worker	list inquiry	worker, year	YTD per code, wage bases	`pay_ytd`	Pay statement YTD	PC00_M10_CLJN payroll journal
Workers' comp premium base	operational report	state, class, period	premium base, overtime premium excluded, estimated premium	`pay_wc`	SSRS Workers compensation report	US workers' comp report (RPCWCOU0)
Expense reports	list inquiry	status, worker, period, project	report, total, reimbursable, company-paid, policy flags	`exp_reports`	Expense reports list (TrvExpTableListPage); SSRS Expense report	PRTE trip data; PR10 trip overview
Policy violations	KPI	rule, department, period	violation count, amount, justified %	`exp_violations`	Policy violation inquiry	Concur audit rule reports
Unassigned card transactions	list inquiry	worker, age	txn, merchant, amount, days unassigned	`exp_cards`	Unattached credit card transactions (TrvPBSMaindata)	Concur card feed exceptions
Cash advance aging	operational report	status, days overdue, worker	advance, paid, settled, open, days past due	`exp_advances`	Cash advance inquiry; SSRS Cash advance report	PR05 advances list
Requisition vs actual	drill-down	requisition, department	estimate by category, actual, variance	`exp_req_variance`	Travel requisition inquiry	Travel request vs trip (TP03 / PRTE)

8.43.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Pay statement	statement FINALIZED	header: employer, worker, pay period, pay date; lines: earnings (hours, rate, amount, YTD), pre-tax, taxes, post-tax, employer contributions, arrears/claims; footer: net, payment split, accrued leave balances	PDF, ESS	pay run id + worker	US state pay-stub content laws (e.g. CA Labor Code §226, 3 years); DE GewO §108 Entgeltabrechnung; retain 7 years (payroll records)
Direct deposit advice	payment batch released	net, bank last 4, pay date	PDF, email	statement id	as pay statement
Payroll check	MANUAL_CHECK / CHECK payment / CHECK_REPLACEMENT	MICR line, check number, payee, amount in words, stub	PDF (check stock), positive-pay file to bank (wave 36)	check number per bank account	bank and payroll record retention
NACHA / SEPA payment file	run POSTED	wave-36 layouts (PPD entries, pain.001 salary category SALA)	NACHA, ISO 20022 pain.001	wave-36 batch id	bank record
W-2 / 1099 data, DE Lohnsteuerbescheinigung data	year end	form box structures	data to wave-48 pack (PDF, EFW2, ELSTER)	year + worker	4 years (IRS), 6 years (DE AO §147)
Payroll GL voucher	run POSTED	wave-11 voucher with code × dimension lines	PDF	voucher	accounting record
Expense report	report SUBMITTED / POSTED	header: worker, purpose, period; lines: date, category, merchant, amount, currency, rate, project, policy outcome; receipts appendix	PDF, ESS	EXP-REP	tax receipt retention (US IRS 3 years; DE 10 years for Belege under AO §147)
Cash advance request	advance SUBMITTED	worker, amount, purpose, approval, settlement	PDF	EXP-ADV	accounting record
Travel requisition	SUBMITTED	destinations, dates, estimates, approval	PDF	EXP-TR	none beyond 2 years

8.43.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`payroll-calculate` (Temporal)	on demand per run	pay run, worker shard of 500	key per (run, worker); status guard	10,000 workers < 10 min	per-shard retry; failed workers listed; run stays CALCULATED-with-errors	failure count > 0; runtime > 15 min
`payroll-cutoff-reminder`	cutoff – 24 h	pay group	key (group, period)	< 1 min	retry	reminders undelivered

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`payroll-retro-detect`	on every effective-dated change event; nightly sweep	worker	key (worker, change id)	< 1 s per event	retry	retro pending > 2 periods
`payroll-statutory-aggregate`	quarter end + 5 days; year end	legal entity, form	key (entity, form, period); re-run replaces	< 5 min	retry; variance report	variance ≠ 0.00
`payroll-ruleset-load`	on pack release (migration job)	country, year	signature + version unique	< 1 min	reject unsigned	pack missing for next period end (14 days ahead)
`expense-card-import`	hourly (SFTP pull)	issuer feed	key per file hash	< 2 min per 10,000 txns	quarantine bad file	import failure; no file for 24 h
`expense-aging`	daily 07:00	legal entity	key (report, day)	< 1 min	retry	reports older than submission deadline
`cash-advance-aging`	daily 07:00	legal entity	key (advance, day)	< 1 min	retry	advances past due; payroll recovery created
`travel-requisition-close`	daily 02:00	legal entity	key (requisition)	< 1 min	retry	none closed for 7 days

8.43.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
PayrollRunMaintain	runs create/calculate, off-cycle, claims recovery; Pay run detail, Off-cycle workbench	with PayrollRunFinalize (same run); with HcmActionMaintain	Payroll administrator
PayrollRunFinalize	`/runs/{id}/finalize`	with PayrollRunMaintain for the same run (BR-43.8)	Payroll manager
PayrollRunPost	`/runs/{id}/post`, payment batch release	with PayrollWorkerMaintain (bank details)	Payroll manager, Treasury
PayrollRuleAdmin	rule-set edit/publish	author ≠ publisher (BR-43.8)	Payroll configuration specialist
PayrollWorkerMaintain	W-4, state forms, direct deposit, garnishments	with PayrollRunPost	Payroll administrator
PayrollInquire / EssPayStatement	registers, reports, statutory aggregates; own statements	none	Payroll administrator, Controller; Employee
EssExpense	reports, receipts, cash advances, requisitions (own)	with MssApprove on own items	Employee
MssApprove	approve/reject/return reports, advances, requisitions	own items blocked	Manager
ExpenseAdmin	card import, policy rules, advance returns, write-offs	with EssExpense for own items	Expense administrator
ExpenseAuditor	read all reports, receipts, violations; return to worker	with ExpenseAdmin rule publish	Internal audit

8.43.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
PAY-RUN-001	error	Cutoff for {period} has not passed	Calculate too early	Wait for cutoff or change parameter	8.43.5 validation
PAY-RUN-002	error	{n} time registrations for {period} are not approved	Unapproved time with BLOCK policy	Approve time or change policy	8.43.5
PAY-RUN-003	error	Run has {n} blocking validation results	Finalize precondition	Resolve exceptions	8.43.5
PAY-RUN-004	error	You calculated this run; another user must finalize	SoD	Ask a payroll manager	BR-43.8
PAY-RULE-001	error	No published rule set for {country} {year} on {date}	Missing pack	Load and publish the pack	FR-43.2
PAY-RULE-002	error	Publisher must differ from author	SoD	Second approver	BR-43.8
PAY-STMT-001	error	Statement {id} is FINALIZED and cannot change	Immutability	Use a CORRECTION or OFF_CYCLE run	BR-43.7
PAY-ARR-001	warning	Deduction {code} short by {amount}; added to arrears	Insufficient net	Review arrears ledger	BR-43.10
PAY-CLM-001	warning	Negative net {amount} for {worker}; claim {id} created	Overpayment	Set recovery method	BR-43.12
PAY-OFF-001	error	Original payment {id} not found or already voided	Check replacement	Select the original payment	FR-43.18
PAY-STAT-001	error	Aggregate {form} {period} differs from GL by {variance}	Reconciliation	Investigate postings before release	FR-43.20
EXP-POL-001	error	{rule message}	BLOCK outcome	Correct the line	BR-43.5
EXP-POL-002	warning	{rule message}	WARN outcome	Optional justification	BR-43.5
EXP-REC-001	error	Receipt required for lines ≥ {threshold}	REQUIRE_RECEIPT	Attach receipt	BR-43.5
EXP-CARD-001	error	Card transaction {id} already linked to line {line}	Double use	Unlink first	8.43.4 invariant
EXP-CARD-002	error	Feed file {hash} was already imported	Duplicate import	None	FR-43.13
EXP-ADV-001	error	Advance exceeds limit {limit}	Parameter	Reduce amount	FR-43.15
EXP-ADV-002	error	You have advances past settlement due; new advances blocked	Aging	Settle open advances	FR-43.15
EXP-ADV-003	info	{amount} of advance {id} applied to report {report}	Settlement	None	BR-43.9
EXP-TR-001	warning	Report exceeds approved requisition by {pct} %; justification required	Variance	Justify	BR-43.13
EXP-TR-002	error	Requisition required for category {category}	Parameter	Create a requisition	FR-43.16

8.43.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Pay run calculate	10,000 workers per run; 4 runs/month	run < 10 min; per worker 20 / 60 / 150 ms	20 workers/s per worker process, 8 processes	PayResultLine ≈ 40 lines × 260k statements = 10 M rows/year ≈ 4 GB	partitioned by tax_year; (worker_id, pay_run_id) unique	add Temporal workers; run takes longer, never partial-finalizes
Statement fetch (ESS)	10,000 views on pay date; peak 2,000/min	50 / 200 / 400 ms	100/s	PDFs ≈ 150 KB × 260k = 39 GB object store	object store key per statement	serve PDF from cache; drill-down delayed
Policy evaluation per line	50,000 lines/month; peak 300/min	20 / 100 / 200 ms	50/s	policy_results inline	rules cached per (entity, category)	evaluate at submit only, line-commit warnings deferred
Card import	100,000 txns/month	per file 10,000 rows < 2 min	100 rows/s	≈ 200 MB	(worker_id, status); file hash unique	queue files; matching delayed

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Receipt OCR	40,000/month	2 / 6 / 12 s	5/s	images ≈ 400 KB each = 16 GB	object store	draft lines created later; user can key manually
Statutory aggregate	4 quarters × 20 forms per entity	< 5 min	—	negligible	(entity, form, period)	runs later
Cash advance settlement at approval	1,000/month	100 / 300 / 600 ms	10/s	negligible	(worker_id, status)	settlement queued; AP invoice waits

8.43.20 Localization and compliance hooks

- Payroll is country-pluggable through rule-set packs (FR-43.2); wave 43 ships US (federal, 6 reference states) and DE as the EU reference; wave-48 packs add jurisdictions, statutory forms (941, 940, state UI, W-2/W-3 EFW2, 1099-NEC, DE Lohnsteueranmeldung, DEÜV/SV-Meldungen, Beitragsnachweis), e-filing (IRS FIRE/EFW2, ELSTER, sv.net/dakota) and local taxes.
- Supplemental tax rates, wage bases (SS 176,100.00 for 2025), FUTA credit-reduction states, state minimum-net and final-pay deadlines, and garnishment limits (CCPA 25 % of disposable earnings) are pack data, never code.
- New-hire reporting (state directories), workers' comp state reports and ACA hours data are rendered by the US pack from wave 42/43 data; extension point `payroll.statutory.<cc>.<form>`.
- Expense: VAT recovery per country (wave 13 tax codes; DE Vorsteuer on hotel 7 %/19 % split, UK VAT receipts), per diem tables (US GSA/DoD, DE BMF Verpflegungsmehraufwand 14/28 EUR, 2025 values), mileage (IRS 0.70 USD/mile 2025; DE 0.30 EUR/km) and fringe-benefit treatment of non-reimbursable items are pack data.
- Currency: pay in the pay group currency only; expense lines in any currency converted at the transaction-date rate type configured by the pack (card issuer rate when present).
- Privacy: pay statements and bank details use the `hr_pii` key pattern; receipt images may contain card data — OCR masks PANs to last 4 before storage (PCI DSS); EU retention limits on receipts follow the country pack.
- Calendars: pay dates shift for bank holidays per wave-8 bank calendar (US Federal Reserve, TARGET2); semi-monthly periods follow the 15th/last-day convention.

8.43.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX #71 — 12.10.1 Cash advances (TrvCashAdvance)	PARTIAL	ADDED as FR-43.15	BR-43.9
AX #74 — 12.10.2 Cash advance inquiry	PARTIAL	ADDED as FR-43.15	"Cash advance aging" in 8.43.14
AX #76 — 12.10.2 SSRS Cash advance report	PARTIAL	ADDED as FR-43.15	Document in 8.43.15
AX #79 — 12.10.3 Reconcile/settle cash advances	PARTIAL	ADDED as FR-43.15	Settlement at report approval + `cash-advance-aging`
AX #83 — 12.10.4 Expense management parameters — Intercompany / Cash advance	PARTIAL	ADDED as FR-43.15	Cash advance parameters in 8.43.13; intercompany part stays with wave 40 (register #73/#82)
AX #84 — 12.10.4 Cash advance parameters, posting	PARTIAL	ADDED as FR-43.15	Prepayment posting via wave 18
AX #72 — 12.10.1 Travel requisitions (TrvRequisitionTable)	PARTIAL	ADDED as FR-43.16	BR-43.13
AX #75 — 12.10.2 Travel requisition inquiry	PARTIAL	ADDED as FR-43.16	"Requisition vs actual"
AX #85 — 12.10.4 Requisition policies / types	PARTIAL	ADDED as FR-43.16	Policy engine (BR-43.5) evaluates requisition lines
AX #117 — 12.13.3 Arrears recovery processing	PARTIAL	ADDED as FR-43.17	BR-43.10
AX #115 — 12.13.2 SSRS Workers compensation report	PARTIAL (wave 48)	ADDED as FR-43.21	Class codes and premium base here; state-specific report layouts DEFERRED to wave 48
AX #116 — 12.13.2 SSRS New hire reporting (US state)	PARTIAL (wave 48)	DEFERRED to wave 48	Source: `hr.employment.changed.v1` HIRE events; state layouts in US pack
SAP #326 — 13.28 PC00_M10_OSUP / PUOC_10 off-cycle workbench	PARTIAL-43	ADDED as FR-43.18	BR-43.11
SAP #327 — 13.28 PC00_M10_CLAIMS claims processing	PARTIAL-43	ADDED as FR-43.19	BR-43.12
SAP #335 — 13.28 PC00_M10_* US reports (941, 940, W-2, SUI, new hire, EEO-1, workers' comp)	PARTIAL-43	ADDED as FR-43.20	Aggregates and reconciliation here; rendering/filing DEFERRED to wave 48; EEO-1 is wave-42 data via wave 48
SAP #336 — 13.28 PU19 tax reporter runs	PARTIAL-43	ADDED as FR-43.20	Job `payroll-statutory-aggregate`
SAP #341 — 13.29 TP01/TP02/TP03/PR01 travel request	PARTIAL-43	ADDED as FR-43.16	Booking (TP01 planning) OUT-OF-SCOPE per 8.43.1
SAP #342 — 13.29 PR02/PR05 cash advances	PARTIAL-43	ADDED as FR-43.15	
SAP #343 — 13.29 Travel request / plan settings, approval workflows, substitutes (T702)	PARTIAL-43	ADDED as FR-43.16	Substitutes via wave-4 workflow delegation

8.44 Wave 44: Enterprise Asset Management (Maintenance)

8.44.1 Scope and business context

EAM replaces D365 Asset Management (EntAssetObject, EntAssetFunctionalLocation, EntAssetMaintenancePlan, EntAssetWorkOrder, EntAssetCounter, EntAssetMaintenanceRequest, EntAssetBOM) and SAP PM (IFLOT functional locations, EQUI equipment, MPLA/MPOS plans with IP10 scheduling, QMEL notifications IW21, AUFK orders IW31/IW32, IMPTT measuring points, AFIH order header). After this wave a maintenance planner models the plant as functional locations and assets, sets time/counter/condition-based plans that generate work orders, dispatches technicians against wave-23 resource capacity, consumes spares from wave-16 inventory, records labor and downtime, and sees costs settle to the asset and GL. IoT readings from wave 25 trigger requests.

Out of scope: predictive ML models (hooks only), calibration management, fleet management, linear assets.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
SAP IP10/IP30 deadline monitoring is a batch job (RISTRA20) run daily or weekly; calls are generated late or in bursts and counter-based plans need IK11 readings entered by hand; D365 "Schedule maintenance plans" is likewise a batch with a fixed horizon	Scheduling is periodic recalculation, not event-driven	Scheduler runs on every completion/reading event plus nightly; forecast from rolling usage; algorithm stated in BR-44.1	Call latency 1–7 days batch → < 60 s after triggering event; 40/40 plan fixtures exact
Creating a corrective work order in D365 Asset Management from a request takes 8–10 clicks across Maintenance request → Work order → Job → Lines; SAP IW21 → IW31 → components → IW32 release is 4 transactions	Request and order are separate forms with re-keyed fields	Request triage converts to a WO in one action carrying asset, location, symptom, parts and checklist	Clicks 10 → 2; re-keyed fields 0
IoT condition data lives in a separate product (D365 IoT Intelligence, SAP ASPM/IoT) and raises alerts in another UI; maintenance requests are created by people copying values	No in-ERP condition engine	Condition rules on the wave-25 stream create deduplicated requests in < 5 s	Reading-to-request 1 h–1 day manual → p95 < 5 s; duplicate requests 0
Maintenance downtime never reaches production scheduling: SAP PM order has no capacity link to PP work-center scheduling except manual CM01 entries; D365 AM downtime is reporting-only	Separate modules without a shared resource model	WOs reference wave-22 resources; downtime emits a capacity exception consumed by the wave-23 scheduler	Planned-downtime scheduling conflicts –90%; capacity reflects downtime within 10 s
WO cost reconciliation: SAP KO88 settlement to asset/cost center with settlement rules per order; D365 AM costs post via project and need "project invoice/estimate" steps	Order cost is a separate settlement object	Cost rolls to the asset and GL on completion through the posting API; nightly reconciliation	Settlement step eliminated; WO-cost-to-GL variance 0.00

8.44.2 AX / SAP parity map

Capability	AX / D365	SAP ECC / S/4	Passage equivalent	Parity
Functional locations	EntAssetFunctionalLocation, types, structure	IFLOT, IFLOTX, IL01/IL02	`FunctionalLocation` (ltree path)	full
Assets/equipment	EntAssetObject, EntAssetObjectType, asset lifecycle states	EQUI, EQKT, IE01/IE02, equipment categories	`Asset` with `AssetType` and lifecycle	full
Counters/measuring points	EntAssetCounter, EntAssetObjectCounter, counter registration	IMPTT measuring points, IMRG measurement documents (IK11)	`Counter`, `CounterReading`	full
Maintenance plans	EntAssetMaintenancePlan, plan lines (time/counter), plan job types	MPLA/MPOS, IP10 scheduling, IP30 deadline monitoring, strategy plans (IP11)	`MaintenancePlan`, `PlanLine`, scheduler	full
Requests/notifications	EntAssetMaintenanceRequest, request lifecycle	QMEL/QMIH notifications M1/M2/M3, IW21/IW22	`MaintenanceRequest`	full
Work orders	EntAssetWorkOrder, EntAssetWorkOrderLine, job types, lifecycle states, checklists	AUFK/AFIH orders PM01/PM02/PM03, IW31/IW32, operations AFVC	`MaintWorkOrder`, `WorkOrderJob`, `Checklist`	full
Scheduling	Work order scheduling against resources (production resources)	Capacity planning CM01, work centers	wave-23 scheduler with `MAINTENANCE` job class	full
Parts and labor	Work order item requirements → inventory journal; hour journal	Component reservation (RESB), goods issue MIGO 261, confirmation IW41 (AFRU)	`WorkOrderPart` → wave-16 reservation/issue; `WorkOrderLabor`	full
Costing	Work order costs to asset (project-based in D365 AM)	Order settlement KO88 to cost center/asset, cost analysis	cost rollup to asset and GL via posting API; optional wave-37 capitalization	full
Condition monitoring	Counter thresholds; IoT Intelligence	PM-PRM, IoT via SAP IoT/ASPM	`ConditionRule` on IoT readings (wave 25)	improved
KPIs	Asset KPIs (D365 AM)	MCI7 MTTR/MTBF	`/reports/eam/kpis` MTBF, MTTR, availability, cost/asset	full
Maintenance rounds	EntAssetMaintenanceRound (route of assets inspected together)	Object list on order (OBJK), task list for multiple objects, IP41 single-cycle plan with object list	`MaintenanceRound`, plan line type ROUND (FR-44.14)	full
Maintenance sequences	EntAssetMaintenanceSequence (ordered job types)	Strategy plans IP11/IP42 with packages and hierarchy (T351P)	`MaintenanceSequence` with step hierarchy (FR-44.15)	full
Fault registration and designer	EntAssetFault (symptoms, areas, types, causes, remedies), fault designer per asset type	Catalog profiles (QS41 code groups/codes, OIS2 catalog profile), notification items damage/cause/activity (QMFE/QMUR/QMMA)	`FaultCatalog` per asset type (FR-44.16)	full
Asset warranties	EntAssetWarranty, warranty types, vendor warranty setup, warranty check on work order	Warranty master BGMK (BGM1), equipment warranty (IE02 warranty tab), warranty check on IW31	`AssetWarranty` (FR-44.17)	full
Maintenance budgets	EntAssetMaintenanceBudget, budget calculation from plans	IP30 cost estimates / plan cost simulation (IP31 / IP-costing), investment programs	`MaintenanceBudget` (FR-44.18) → wave 38	full
Permits and LOTO	not native	Permits (T357G, IPMD), Work Clearance Management WCM (WCA* objects: work approval, application, lockout/tagout)	`Permit`, `IsolationPoint`, `LotoLock` (FR-44.19)	full
Refurbishment	not native	IW81 refurbishment order (PM04), IW8W goods receipt, OMWC split valuation by condition (T025)	refurbishment WO type + wave-16 condition valuation (FR-44.20)	full
Fleet objects	Asset type attributes	IE31/IE36/IE37 fleet objects (FLEET table: license plate, VIN, fuel), MCIZ fuel analysis	asset type FLEET_VEHICLE attributes + fuel counters (FR-44.21)	full
Criticality / RCM	Asset criticality field	ASPM (Asset Strategy and Performance Management) criticality assessment, RCM; Predictive Asset Insights	`CriticalityAssessment` (FR-44.22); predictive models as hooks	partial — no ML models in v1

8.44.3 Functional requirements

- **FR-44.1 Functional locations** — Must. The system shall maintain a functional-location hierarchy per site (max depth 12) with types, with assets installable at locations and location history.
- **FR-44.2 Assets** — Must. The system shall maintain assets (type, manufacturer/model, serial, criticality 1–5, lifecycle state, parent asset, production resource link, fixed asset link, warranty) with install/remove history.
    - Lifecycle: CREATED → ACTIVE → INACTIVE → DISPOSED; only ACTIVE assets can receive work orders (override duty).

- **FR-44.3 Counters** — Must. The system shall maintain counters (unit, cumulative/reset, source MANUAL/IOT) with readings validated monotonic for cumulative counters and an average daily usage rate (rolling 30 days).
- **FR-44.4 Maintenance plans** — Must. The system shall define plans with lines of type TIME (interval + unit, calendar), COUNTER (interval + counter), or TIME_OR_COUNTER (whichever first) / TIME_AND_COUNTER, with job type, checklist, estimated hours and parts, early/late tolerances and generation horizon.
- **FR-44.5 Plan scheduling and generation** — Must. The system shall compute next-due dates/readings per BR-44.1 and generate work orders within the horizon, merging lines due within a merge window into one work order per asset.
- **FR-44.6 Maintenance requests** — Must. The system shall capture requests from users, mobile, IoT condition rules and wave-32 quality, with triage to work order, reject or defer.
- **FR-44.7 Work orders** — Must. The system shall manage work orders with jobs (asset, job type, trade, variant), lifecycle, checklists, planned/actual parts and hours, downtime, failure codes (symptom/cause/remedy) and completion notes.
- **FR-44.8 Scheduling** — Must. The system shall schedule work order jobs against wave-23 resources (maintenance teams, technicians via wave-42 workers) with finite capacity and record planned downtime on the production resource.
- **FR-44.9 Parts** — Must. The system shall reserve, pick and issue parts through wave-16 inventory journals with cost flowing to the work order.
- **FR-44.10 Labor** — Must. The system shall record labor via hour registrations (wave 25 kiosk or mobile) at trade cost rates.
- **FR-44.11 Costing and GL** — Must. The system shall accumulate parts, labor, external services (wave-17 PO) and overhead per work order, post to maintenance expense accounts by (asset type, job type, cost dimension of the functional location) and roll up to the asset; CAPITAL job types create wave-37 CIP postings.
- **FR-44.12 Condition rules** — Must. The system shall evaluate IoT readings against thresholds (above/below/rate-of-change over window) and create requests within 5 s of breach, with deduplication.
- **FR-44.13 KPIs** — Should. The system shall compute MTBF, MTTR, availability, cost per asset and backlog hours by asset, location and type.
- **FR-44.14 Maintenance rounds** — Should. The system shall define rounds (ordered stops of assets or functional locations, each with a job type and checklist, optional walking/driving minutes between stops) scheduled by plan lines of type ROUND that generate one work order per round occurrence with one job per stop.
    - A stop that cannot be performed is marked SKIPPED with a reason; skipped stops generate a follow-up request when the job type is mandatory.
- **FR-44.15 Maintenance sequences** — Should. The system shall define sequences of job types on an asset or asset type with interval multiples and a hierarchy (e.g. step A every 1×, B every 4×, C every 12× of a base interval of 250 h), generating for each due occurrence only the highest-hierarchy step plus steps not included in it, per BR-44.10.
- **FR-44.16 Fault catalog and analysis** — Must. The system shall maintain fault catalogs per asset type (symptoms, areas, types, causes, remedies, with allowed combinations), require fault registration on CORRECTIVE and EMERGENCY work orders at completion, and analyze faults by asset, type, symptom, cause and period (count, downtime, cost, repeat rate within 30 days).
    - Replaces the free-text failure_symptom/cause/remedy codes on `MaintWorkOrder` with FK references; existing columns stay for migration (populated from the catalog code).
- **FR-44.17 Asset warranties** — Should. The system shall record vendor/manufacturer warranties per asset (warranty type, vendor party, start from purchase/installation, duration in months or counter units, parts/labor coverage, exclusions), check coverage when a work order is created or parts are issued, flag covered lines and create a vendor warranty claim (wave-45 FR-45.15) on completion.
- **FR-44.18 Maintenance budgets** — Should. The system shall calculate maintenance budgets per asset, functional location and period from active plans over a horizon (forecast occurrences × est_hours × trade cost rate + planned parts at standard cost + external services), allow manual budget lines, transfer the result to a wave-38 budget plan, and report budget vs actual vs committed.
- **FR-44.19 Permits and lockout/tagout** — Must. The system shall define permit types (HOT_WORK, CONFINED_SPACE, ELECTRICAL_ISOLATION, WORKING_AT_HEIGHT, LINE_BREAKING, custom) required by job type, asset type or functional location; maintain isolation points per asset (energy type, location, isolation method, verification step); and manage permits with lifecycle REQUESTED → APPROVED → ISSUED → SUSPENDED → CLOSED and personal LOTO locks per technician per BR-44.11.
- **FR-44.20 Refurbishment orders** — Should. The system shall support rotable spares valued by condition (NEW, REFURBISHED, DAMAGED as a wave-16 inventory condition dimension with cost per condition) and refurbishment work orders that issue DAMAGED units, collect labor, parts and external repair cost, and receive the same serials as REFURBISHED at the condition's valuation per BR-44.12.
- **FR-44.21 Fleet attributes** — Could. The system shall provide asset type FLEET_VEHICLE with license plate, VIN, fuel type, tank capacity, assigned driver (wave-42 worker), registration and inspection expiry dates, and fuel log entries (date, quantity, cost, odometer) that update odometer and fuel-consumption counters.
- **FR-44.22 Criticality assessment** — Should. The system shall assess asset criticality by a matrix of consequence categories (SAFETY, ENVIRONMENT, PRODUCTION, QUALITY, COST, each 1–5) and failure likelihood (1–5), derive the asset criticality (1–5) per BR-44.15, version assessments, and use criticality to default work order priority and plan review frequency; predictive models remain hooks (`eam.predictive-hooks`).

8.44.4 Data model

Asset

Field	Type	Required	Notes
id	uuid	yes	PK
asset_number	text(30)	yes	unique; sequence AST
asset_type_id	uuid	yes	FK
name	text(120)	yes	
functional_location_id	uuid	no	current location
parent_asset_id	uuid	no	
site_id	uuid	yes	wave 6
serial_number, manufacturer, model	text	no	
criticality	smallint	yes	1–5
lifecycle_state	enum	yes	
production_resource_id	uuid	no	wave 22
fixed_asset_id	uuid	no	wave 37
warranty_until	date	no	
cost_dimension_set_id	uuid	no	

MaintenancePlan / PlanLine

Field	Type	Required	Notes
id	uuid	yes	PK (line)
plan_id	uuid	yes	FK plan (name, asset or asset type/location scope, active)
line_type	enum	yes	TIME, COUNTER, TIME_OR_COUNTER, TIME_AND_COUNTER
interval_value, interval_unit	int, enum	cond	DAYS, WEEKS, MONTHS, YEARS
calendar_id	uuid	no	wave 8; default 7-day
counter_id, counter_interval	uuid, numeric(14,2)	cond	
job_type_id, trade_id, variant	uuid, uuid, text	yes/no/no	
checklist_id	uuid	no	
est_hours, parts[]	numeric, jsonb	no	
early_tolerance_pct, late_tolerance_pct	smallint	yes	default 10/10 (SAP tolerance analog)
shift_factor_early, shift_factor_late	smallint	yes	0–100%, default 0/100
horizon_days, merge_window_days	int	yes	default 90 / 7
last_completed_at, last_completed_reading	timestamptz, numeric	no	updated on completion
next_due_at, next_due_reading, next_due_forecast_at	computed		

MaintWorkOrder

Field	Type	Required	Notes
id	uuid	yes	PK; number WO-{seq}
asset_id, functional_location_id	uuid	yes/no	
work_order_type	enum	yes	PREVENTIVE, CORRECTIVE, INSPECTION, CAPITAL, EMERGENCY
status	enum	yes	CREATED, SCHEDULED, RELEASED, IN_PROGRESS, ON_HOLD, COMPLETED, CLOSED, CANCELED
priority	smallint	yes	1 (emergency)–5
request_id, plan_line_id	uuid	no	provenance
due_at, planned_start, planned_end, actual_start, actual_end	timestamptz		
jobs[]	WorkOrderJob	yes	(job_type, trade, est_hours, actual_hours, checklist results)
parts[]	WorkOrderPart	no	(item, qty_planned, qty_issued, reservation_id, cost)
downtime_minutes	int	no	
failure_symptom, failure_cause, remedy	text(20)	no	code tables
cost_parts, cost_labor, cost_external, cost_overhead	numeric(14,2)	computed	
journal_ids[]	uuid[]		

Counter: id, asset_id, name, unit, kind (CUMULATIVE/RESETTABLE), source, last_reading, last_reading_at, avg_daily_usage numeric(12,4). **CounterReading**: id, counter_id, reading_at, value, source, iot_message_id. **MaintenanceRequest**: id, asset_id, source (USER, MOBILE, IOT, QUALITY), severity, description, reading_value, status (NEW, TRIAGED, CONVERTED, REJECTED, DEFERRED), work_order_id, dedupe_key.

MaintenanceRound: id, code, site_id, stops[] (sequence, asset_id or functional_location_id, job_type_id, checklist_id, travel_minutes_from_previous), est_total_minutes, active. **MaintenanceSequence**: id, scope (asset_id or asset_type_id), base_interval_value, base_interval_unit or counter_id, steps[] (code, job_type_id, multiple int, hierarchy smallint, includes_step_codes[]), last_occurrence_number. **FaultCatalog**: asset_type_id, symptoms[], areas[], types[], causes[], remedies[] (code, text), allowed_combinations[] (symptom, area, type). **WorkOrderFault**: work_order_id, symptom_code, area_code, type_code, cause_code, remedy_code, notes. **AssetWarranty**: id, asset_id, warranty_type_id, vendor_party_id, start_basis (PURCHASE, INSTALL), starts_on, months, counter_id, counter_limit, parts_covered, labor_covered, exclusions[], status. **MaintenanceBudget**: id, version, scope (asset/location), period_id, source (PLAN_CALC, MANUAL), labor_amount, parts_amount, external_amount, total, wave38_plan_id. **PermitType**: code, name, checklist_id, approver_role, max_validity_hours, requires_gas_test bool. **Permit**: id (sequence EAM-PMT), permit_type_code, work_order_id, asset_id, valid_from, valid_to, status (REQUESTED, APPROVED, ISSUED, SUSPENDED, CLOSED, CANCELED), approved_by, issued_by, checklist_results[]. **IsolationPoint**: id, asset_id, energy_type (ELECTRICAL, PNEUMATIC, HYDRAULIC, MECHANICAL, THERMAL, CHEMICAL), location_text, method, verification_step, tag_code. **LotoLock**: id, permit_id, isolation_point_id, technician_worker_id, lock_number, applied_at, verified_at, removed_at. **CriticalityAssessment**: id, asset_id, version, consequence {SAFETY, ENVIRONMENT, PRODUCTION, QUALITY, COST}, likelihood, risk_score, criticality, assessed_by, assessed_at. **Asset** adds: fleet attributes (license_plate, vin, fuel_type, tank_capacity, driver_worker_id, registration_expiry, inspection_expiry) for type FLEET_VEHICLE; `rotable` bool and `condition` for serialized rotables. MaintWorkOrder adds work_order_type REFURBISHMENT and ROUND, `fault_id`, `permit_ids[]`, `warranty_covered` bool.

Invariants: cumulative counter readings are non-decreasing in time; a COMPLETED work order has actual_start ≤ actual_end; Σ work order costs = Σ linked journal lines to maintenance accounts; one open (NEW/TRIAGED) IoT request per (asset, rule) dedupe_key. Indexing: `FunctionalLocation.path ltree` GiST; `PlanLine (next_due_at)` and `(next_due_forecast_at)`; `MaintWorkOrder (site_id, status, due_at)`; `CounterReading` partitioned monthly.

8.44.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/eam/assets`	Create asset	key	EamAssetMaintain
POST	`/api/v1/eam/counters/{id}/readings`	Register reading	key (source + message id)	EamCounterRegister / IoT bridge
POST	`/api/v1/eam/plans/schedule`	Recompute next due and generate WOs	key per (plan, run date)	EamPlanAdmin
POST	`/api/v1/eam/requests`	Create request	key (dedupe_key)	EamRequestCreate
POST	`/api/v1/eam/requests/{id}/convert`	Convert to work order	key	EamPlanner
POST	`/api/v1/eam/work-orders`	Create WO	key	EamPlanner
POST	`/api/v1/eam/work-orders/{id}/status`	Transition status	key + version	EamPlanner / EamTechnician
POST	`/api/v1/eam/work-orders/{id}/parts/issue`	Issue parts (inventory journal)	key	EamTechnician
POST	`/api/v1/eam/work-orders/{id}/labor`	Record labor	key	EamTechnician
GET	`/api/v1/eam/kpis?scope=&from=&to=`	MTBF/MTTR/cost	n/a	EamInquire
POST	`/api/v1/eam/rounds`	Create maintenance round	key	EamPlanAdmin
POST	`/api/v1/eam/sequences`	Create maintenance sequence	key	EamPlanAdmin

Method	Path	Purpose	Idempotency	Auth duty
PUT	`/api/v1/eam/fault-catalogs/{assetTypeId}`	Maintain fault catalog	ETag	EamPlanAdmin
POST	`/api/v1/eam/work-orders/{id}/faults`	Register fault	key	EamTechnician
POST	`/api/v1/eam/budgets/calculate`	Calculate budget from plans (scope, horizon, version)	key per (scope, version)	EamBudgetAdmin
POST	`/api/v1/eam/permits`	Request permit for a work order	key	EamPlanner
POST	`/api/v1/eam/permits/{id}/transition`	Approve / issue / suspend / close permit	key + version	EamPermitIssuer
POST	`/api/v1/eam/permits/{id}/locks`	Apply / verify / remove LOTO lock	key per lock action	EamTechnician
POST	`/api/v1/eam/work-orders/refurbishment`	Create refurbishment WO for rotable serials	key	EamPlanner
POST	`/api/v1/eam/assets/{id}/criticality`	Record criticality assessment	key	EamAssetMaintain
POST	`/api/v1/eam/assets/{id}/fuel-log`	Record fuel entry (fleet)	key	EamCounterRegister

Validation: readings for cumulative counters must be ≥ last_reading else `EAM-CTR-001` (reset requires explicit reset flag); COMPLETED requires all checklist mandatory items answered and all reserved parts issued or released; CAPITAL work orders require a fixed_asset_id or CIP project.

Event	Trigger	Payload	Consumers
`dev.passage.eam.request.created.v1`	request created	see 8.0	planner UI, notifications
`dev.passage.eam.workorder.status.changed.v1`	transition	workOrderId, from, to, at, actor	mobile, analytics
`dev.passage.eam.workorder.completed.v1`	COMPLETED	see 8.0	journal, fixed assets, scheduling
`dev.passage.eam.counter.reading.registered.v1`	reading	counterId, assetId, value, at, source	plan scheduler, condition rules
`dev.passage.eam.downtime.recorded.v1`	downtime on resource	resourceId, from, to, workOrderId	scheduling (wave 23), OEE (wave 25)
`dev.passage.eam.permit.status.changed.v1`	permit transition	permitId, workOrderId, from, to, at, actor	mobile, safety notifications
`dev.passage.eam.warranty.claimable.v1`	covered WO completed	workOrderId, assetId, warrantyId, vendorPartyId, coveredParts, coveredLabor	service (FR-45.15 supplier claims)
`dev.passage.eam.budget.calculated.v1`	budget calculation done	budgetVersion, scope, periods[], total	budgeting (wave 38)

8.44.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Asset management workspace	tiles: requests to triage, WOs due in 7 days, overdue, backlog hours		EamInquire
Plant structure tree	expand locations, install/move assets, open asset	location path, asset, state, criticality	EamStructureMaintain
Asset detail	counters, plans, WO history, costs, KPIs, documents		EamAssetMaintain
Maintenance plan editor	add line, simulate schedule 12 months, activate	type, interval, counter, next due, forecast	EamPlanAdmin
Request triage	convert, reject, defer, merge	severity, source, asset, age	EamPlanner
Work order board (kanban + Gantt)	schedule, release, assign, print	status, priority, due, assigned team	EamPlanner
Technician mobile (PWA)	start/stop, checklist, parts scan, labor, photos, complete		EamTechnician
KPI dashboard	filter by site/type/period	MTBF, MTTR, availability, cost	EamInquire
Round and sequence editor	order stops, set jobs and checklists; define steps and hierarchy; simulate 12 months	stop, asset, job type, minutes; step, multiple, hierarchy	EamPlanAdmin
Fault catalog designer	add codes, allowed combinations per asset type	symptom, area, type, cause, remedy	EamPlanAdmin
Fault analysis	filter, Pareto by symptom/cause	count, downtime, cost, repeat %	EamInquire
Permit board	request, approve, issue, suspend, close; LOTO lock status	permit, type, WO, valid to, locks applied/verified/removed	EamPermitIssuer
Maintenance budget	calculate, adjust, transfer to wave 38, budget vs actual	asset/location, period, labor, parts, external, actual, variance	EamBudgetAdmin
Rotables and refurbishment	view serials by condition, create refurbishment WO	item, serial, condition, location, value	EamPlanner

Rules: board drag changes planned_start with conflict warning; checklist items support numeric with tolerance (auto-flag out of range); mobile offline for up to 72 h with action replay.

8.44.7 Business rules and edge cases

- **BR-44.1 Plan scheduling algorithm** — run nightly and on each completion/reading event for the affected plan lines. For each active line: (a) base = last_completed_at (or plan start if none); time due `T = calendar.add(base, interval)` — if shift factors apply and completion was late/early by Δ versus the previous due, shift base by `Δ × shift_factor_late/early` (SAP IP10 analog); (b) counter due `R = last_completed_reading + counter_interval`; forecast date `F = now + (R − counter.last_reading) ÷ avg_daily_usage` (if avg_daily_usage = 0, F = NULL); (c) next_due_at = T for TIME; = F for COUNTER (next_due_reading = R); = min(T, F) for TIME_OR_COUNTER; = max(T, F) for TIME_AND_COUNTER; (d) generate a work order when `next_due_at − early_tolerance_pct × interval ≤ now + horizon_days` and no open WO exists for the line; (e) lines for the same asset due within merge_window_days are merged into one WO with multiple jobs; (f) a counter-based WO is released only when the actual reading ≥ R – tolerance or the forecast date arrives. Example: line TIME_OR_COUNTER, 90 days or 500 hours; last completed 2026-08-01 at 12,000 h; counter 12,300 h on 2026-10-21 with avg usage 6.5 h/day → T = 2026-10-30, R = 12,500, F = 2026-10-21 + 30.8 days = 2026-11-21 → next_due_at = 2026-10-30; generated on 2026-10-21 (horizon 90) with due 2026-10-30.
- **BR-44.2 Completion resets** — completion sets last_completed_at = actual_end and last_completed_reading = reading at completion (prompted if counter-based); the line reschedules immediately.
- **BR-44.3 Status transitions** — CREATED → SCHEDULED → RELEASED → IN_PROGRESS → COMPLETED → CLOSED; IN_PROGRESS ↔ ON_HOLD; any pre-COMPLETED → CANCELED (releases reservations). CLOSED requires all costs posted (no pending inventory or hour journals) and is final; cost corrections after CLOSED use a CORRECTION WO.
- **BR-44.4 Condition rules** — rule {counter or tag, operator >, <, RATE_OVER(window), value, severity, cooldown_minutes}; breach creates one request with dedupe_key `(asset, rule)` while a request is NEW/TRIAGED or within cooldown (default 60 min); latency target from reading ingest to request event p95 < 5 s.
- **BR-44.5 Downtime** — recording downtime on an asset linked to a production resource creates a capacity exception on wave-23 for [from, to]; planned downtime from SCHEDULED work orders is tentative and becomes firm at RELEASED.

- **BR-44.6 Parts costing** — issued at wave-16 running cost at issue time; returns credit at the same cost; partial issues allowed.
- **BR-44.7 Emergency WOs** — priority 1 skips SCHEDULED and may be IN_PROGRESS before RELEASED; must be regularized within 24 h (alert).
- **BR-44.8 KPIs** — MTBF = Σ uptime ÷ number of failures (CORRECTIVE/EMERGENCY WOs with failure codes) over the window; MTTR = Σ (actual_end – actual_start) of those WOs ÷ count; availability = 1 – downtime ÷ scheduled time.
- **BR-44.9 Rounds** — a ROUND plan line schedules per BR-44.1 (TIME only); one work order per occurrence with jobs in stop sequence; the work order completes when every stop is DONE or SKIPPED with a reason; completion resets the line once for the whole round; a skipped mandatory stop creates a MaintenanceRequest (source ROUND) for that asset.
- **BR-44.10 Sequence step selection** — for occurrence number k (1-based, k = last_occurrence_number + 1), candidate steps = {s : k mod s.multiple = 0}; the selected step is the candidate with the highest hierarchy; steps listed in its `includes_step_codes` are suppressed; remaining candidates not included are added as extra jobs. Example: A×1 (h1), B×4 (h2, includes A), C×12 (h3, includes A, B): k = 4 → B; k = 12 → C; k = 6 → A.
- **BR-44.11 Permits and LOTO** — a job requiring a permit type cannot move its work order to IN_PROGRESS until a permit of that type is ISSUED and valid now (`EAM-PMT-001`); approver ≠ requester; ELECTRICAL_ISOLATION and other isolation permits are ISSUED only when every isolation point of the asset has a LotoLock applied and verified (zero-energy check recorded); each technician working the job applies a personal lock (one lock per technician per point); the permit can CLOSE only when all locks are removed and the work order is COMPLETED or the work is suspended; permits past valid_to move to SUSPENDED automatically and IN_PROGRESS jobs are flagged; all transitions are append-only audit records retained 5 years.
- **BR-44.12 Refurbishment valuation** — issue of a DAMAGED serial to a refurbishment WO credits inventory at the DAMAGED condition value; WO costs (labor, parts, external) accumulate; receipt as REFURBISHED debits inventory at the REFURBISHED condition standard (or moving average) value; the difference between (damaged value + WO costs) and the refurbished value posts to the refurbishment variance account; serials keep their identity and history across conditions.
- **BR-44.13 Budget calculation** — for each active plan line, forecast occurrences in the horizon per BR-44.1 (counter lines use avg_daily_usage; usage 0 → no occurrences); amount per occurrence = est_hours × trade cost rate (wave-42/23 rate effective at the occurrence date) + Σ planned parts × standard cost + planned external amount; amounts roll up per (asset, period); sequence lines use BR-44.10 step selection; recalculation creates a new budget version and never overwrites a version transferred to wave 38.
- **BR-44.14 Warranty check** — at WO creation and at each part issue, a warranty covers the line when the asset's warranty is ACTIVE on the WO date (and counter ≤ counter_limit when set), the coverage flag matches the line kind, and the job type/part is not in exclusions; covered lines still cost the WO, and a warranty claimable event is emitted at completion with covered amounts.
- **BR-44.15 Criticality** — risk_score = max(consequence categories) × likelihood (1–25); criticality = 5 when risk ≥ 20 or SAFETY = 5, 4 when ≥ 15, 3 when ≥ 10, 2 when ≥ 5, else 1; a new assessment version replaces asset.criticality; work orders default priority = 6 – criticality for CORRECTIVE types; assets with criticality ≥ 4 require plan review every 12 months (reminder).

8.44.8 Acceptance criteria

- **AC-44.1** (FR-44.5, BR-44.1) Given the BR-44.1 example, when the scheduler runs on 2026-10-21, then a WO exists with due 2026-10-30 and next_due_reading 12,500.
- **AC-44.2** (FR-44.5) Given 40 plan fixtures, when scheduled across a simulated year, then generated WO dates/readings match expected within 0 days / 0 units.
- **AC-44.3** (FR-44.11) Given a completed WO with 350.00 USD parts and 8 h labor at 45.00/h, when posted, then the GL maintenance expense is 710.00 and the asset cost history shows 710.00.
- **AC-44.4** (FR-44.12, BR-44.4) Given a bearing temperature rule > 85 °C, when a reading of 87 °C arrives, then a request exists within 5 s and a second reading of 88 °C within 60 min creates none.
- **AC-44.5** (FR-44.8, BR-44.5) Given a RELEASED WO with planned downtime 4 h on resource M-12, when wave-23 schedules, then M-12 capacity for that window is zero.
- **AC-44.6** (FR-44.3) Given a cumulative counter at 12,300, when a reading of 12,250 is posted without reset flag, then it is rejected with EAM-CTR-001.
- **AC-44.7** (FR-44.7, BR-44.3) Given a WO with an unanswered mandatory checklist item, when COMPLETED is requested, then it is rejected with `EAM-WO-004`.
- **AC-44.8** (FR-44.13, BR-44.8) Given 3 failures and 2,700 h uptime in a window, when KPIs are queried, then MTBF = 900 h.
- **AC-44.9** (FR-44.14, BR-44.9) Given a weekly round of 6 stops, when the occurrence WO is completed with stop 4 SKIPPED on a mandatory job, then the round line resets once and a MaintenanceRequest with source ROUND exists for stop 4's asset.
- **AC-44.10** (FR-44.15, BR-44.10) Given sequence A×1, B×4 (includes A), C×12 (includes A, B), when occurrences 1–12 generate, then jobs are A, A, A, B, A, A, A, B, A, A, A, C.
- **AC-44.11** (FR-44.19, BR-44.11) Given a WO whose job requires ELECTRICAL_ISOLATION on an asset with 3 isolation points, when IN_PROGRESS is requested with 2 verified locks, then it is rejected with `EAM-PMT-002`; with 3 verified locks and an ISSUED permit it succeeds; closing the permit with one lock still applied returns `EAM-PMT-003`.
- **AC-44.12** (FR-44.20, BR-44.12) Given a DAMAGED pump serial valued 400.00 USD, WO costs 650.00 and REFURBISHED standard 1,000.00, when the serial is received REFURBISHED, then inventory shows the serial at 1,000.00 and the refurbishment variance account carries 50.00 debit.
- **AC-44.13** (FR-44.17, BR-44.14) Given an asset with a 24-month parts-only vendor warranty started 2025-11-01, when a part is issued to a WO on 2026-10-21, then the part line is warranty-covered, labor is not, and completion emits `eam.warranty.claimable.v1` with the part amount.
- **AC-44.14** (FR-44.18, BR-44.13) Given a TIME plan line every 90 days with 6 h at 45.00/h and parts 120.00, when a 12-month budget calculates from 2027-01-01 with last completion 2026-12-15, then 4 occurrences budget 390.00 each (1,560.00 total) in the periods of 2027-03-15, 2027-06-13, 2027-09-11 and 2027-12-10.
- **AC-44.15** (FR-44.16) Given a CORRECTIVE WO with no fault registered, when COMPLETED is requested, then it is rejected with `EAM-FLT-001`; a symptom/area/type combination not allowed for the asset type returns `EAM-FLT-002`.
- **AC-44.16** (FR-44.22, BR-44.15) Given consequences SAFETY 3, PRODUCTION 5 and likelihood 4, when assessed, then risk_score = 20, criticality = 5 and a new CORRECTIVE WO defaults to priority 1.

8.44.9 Test plan and fixtures

Fixtures: `eam-plant-demo` (3 sites, 120 locations, 400 assets, 60 counters from wave-22 resources); `eam-plans-40` with a 365-day expected generation calendar; `eam-iot-stream` (10 readings/s replay from wave-25 fixture); `eam-wo-cost-50`.

Test	Type	Target	Pass criterion
Scheduler	unit/property	next-due computation	matches calendar; monotonic under completions
WO generation	E2E	nightly job	40/40 plans
IoT → request	performance	condition engine	p95 < 5 s at 10 readings/s
Cost posting	contract	posting API	balanced; asset rollup equal
Downtime → capacity	integration	scheduling API	capacity reflects downtime
Mobile offline	E2E (Playwright, offline mode)	PWA	72 h replay without loss

Reconciliation: Σ WO costs by account = GL maintenance accounts; Σ parts issued = inventory transactions with reference WO.

8.44.10 Deployment and operations

Chart `passage-eam` (Go API, scheduler worker, condition engine consumer, mobile BFF); Postgres `eam` with ltree and monthly partitions on readings. Flags: `eam.capital-wo`, `eam.predictive-hooks`. Schedules: `eam-plan-schedule` nightly 01:00 site-local plus event-driven; `eam-kpi-rollup` hourly. SLOs: condition engine p95 < 5 s; WO status API p95 < 200 ms; availability 99.9%. Alerts: scheduler run failure; condition consumer lag > 30 s; emergency WO not regularized in 24 h. Runbook: replay readings from Kafka by offset; re-run scheduler for a plan. Rollback by digest.

8.44.11 Migration from AX / SAP

Source	Passage entity	Notes
EntAssetFunctionalLocation	FunctionalLocation	
EntAssetObject, EntAssetObjectType	Asset, AssetType	lifecycle state map
EntAssetCounter, EntAssetObjectCounter	Counter, last reading	
EntAssetMaintenancePlan (+ lines)	MaintenancePlan, PlanLine	last completed from WO history
EntAssetWorkOrder (open)	MaintWorkOrder	closed as history
EntAssetMaintenanceRequest (open)	MaintenanceRequest	
IFLOT/IFLOTX	FunctionalLocation	TPLNR structure indicator → path
EQUI/EQKT/EQUZ	Asset	EQUZ for location/time slices
MPLA/MPOS/MHIS	MaintenancePlan, PlanLine, last call	
IMPTT/IMRG	Counter, CounterReading (last 12 months)	
QMEL (open), AUFK/AFIH PM orders (open)	MaintenanceRequest, MaintWorkOrder	

Validation: asset count per site; open WO count and planned cost equal; next due dates within tolerance of source IP30 output. Not migrated: closed orders older than 3 years (archive), SAP strategy packages (converted to explicit lines).

8.44.12 Dependencies, risks, open questions

Depends on: wave 16 inventory, wave 17 PO, wave 22 resources, wave 23 scheduler, wave 25 IoT, wave 37 fixed assets, wave 42 workers. Coverage-closure additions also depend on: wave-16 inventory condition dimension with cost per condition (FR-44.20 refurbishment; if wave 16 ships without it, refurbishment runs as a valuation-adjustment journal until wave 16 is extended), wave 38 budget plans (FR-44.18), wave 42 workers for permit roles and fleet drivers (FR-44.19, FR-44.21), and the wave-45 supplier warranty claim (FR-45.15) consuming `eam.warranty.claimable.v1`.

Risk	Likelihood	Impact	Mitigation
Counter forecast inaccuracy	Medium	Low	Rolling usage; manual override
IoT volume	Medium	Medium	Partitioned readings; downsampled retention 13 months

Open questions: (1) Capitalize CAPITAL WOs directly to fixed assets or always via CIP project? Default CIP project; decide by wave 44. (2) Calibration in v1? Default no.

8.44.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Asset / WO / request / permit number sequences	site	lookup	AST, WO, MR, EAM-PMT	wave-5 sequences	Identifiers	Asset management parameters › Number sequences (EntAssetParameters)	IW order number ranges (OION / T003O), notification ranges (OIN1/TQ80), equipment ranges (OIEN)
Work order types enabled	site	multi-enum	PREVENTIVE, CORRECTIVE, INSPECTION, EMERGENCY	+ CAPITAL, ROUND, REFURBISHMENT	FR-44.7	Asset management parameters › Work orders › Work order types	Order types PM01–PM04 (T003O, T350, T399X)
Request types and auto-convert priority	site	lookup + int	auto-convert priority 1	1–5	FR-44.6	Maintenance request types (EntAssetRequestType)	Notification types M1/M2/M3 (TQ80); notification-order type link (T399A)
Plan generation horizon / merge window	plan line	int	90 / 7 days	1–730 / 0–60	BR-44.1 (d), (e)	Asset management parameters › Maintenance plans › Schedule horizon	IP10 call horizon, scheduling period (MPLA, T399W plan category)
Early/late tolerance and shift factors	plan line	pct	10/10; 0/100	0–100	BR-44.1 (a)	Maintenance plan line tolerances	Scheduling parameters: shift factor late/early, tolerance (IP42 / MPLA)
Counter avg-usage window	counter	int days	30	7–365	FR-44.3 forecast	Counter setup (EntAssetCounterType)	IK11 estimated annual performance (IMPTT)
Counter reading tolerance for release	plan line	pct	5 %	0–50	BR-44.1 (f)	n/a	Call horizon % for performance-based plans
Default criticality → priority map	site	table	6 – criticality	1–5	BR-44.15	Criticality setup	Priority type PM (T356)
Fault registration required on completion	work order type	bool	true for CORRECTIVE, EMERGENCY		FR-44.16	Asset management parameters › Fault registration mandatory	Catalog profile mandatory on notification (OIS2, T352B)
Permit required by job type / asset type	job type / asset type	multi-lookup	—	PermitType	BR-44.11	n/a	Permit categories (T357G) assigned to equipment/FL; WCM work clearance usage (WCA*)
Permit max validity (hours)	permit type	int	12	1–168	BR-44.11 auto-suspend	n/a	WCM approval validity

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Emergency WO regularization window	site	int hours	24	1–72	BR-44.7	n/a	Order type "immediate order" settings
Mobile offline limit	tenant	int hours	72	8–168	8.44.6 rule	Mobile app (D365 Field) settings	SAP Asset Manager offline settings
Parts issue at release vs on demand	site	enum	ON_DEMAND	AT_RELEASE, ON_DEMAND	FR-44.9	Work order item requirements	Reservation relevance / goods movement on release (T399X)
Maintenance expense posting key	site	lookup	(asset type, job type, FL dimension)	posting profiles	FR-44.11	Asset management parameters › Journals / project link	Settlement profile (OKO7) and default settlement rule (KO88 replacement)
Condition valuation for rotables	item group	bool + table	false	conditions NEW/REFURBISHED/DAMAGED with value basis	BR-44.12	n/a	Split valuation by condition (OMWC, T025 valuation types)
Budget calculation horizon and version	site	int months + text	12, "BUD-{year}"	1–60	BR-44.13	Maintenance budget parameters	IP30/IP31 cost simulation horizon
Warranty check on WO create	site	bool	true		BR-44.14	Warranty parameters › Check on work order	Warranty check on order (IW31 warranty indicator, BGMK)
Condition rule cooldown default	rule	int minutes	60	1–1,440	BR-44.4	n/a	n/a (ASPM / IoT rules)

8.44.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Work order list	list inquiry	site, status, type, priority, due range, team, asset/location	WO, asset, status, due, planned/actual dates, est/actual hours, cost	`eam_wo`	All work orders (EntAssetWorkOrderListPage)	IW38 change / IW39 display orders
Request list	list inquiry	status, source, severity, age	request, asset, source, severity, age, WO	`eam_req`	Maintenance requests (EntAssetRequestTable)	IW28 / IW29 notifications
Scheduling overview	operational report	plan, date range	plan line, next due, forecast, generated WO, status	`eam_schedule`	Maintenance schedule (EntAssetMaintenanceSchedule)	IP24 scheduling overview list; IP19 scheduling overview graphic
Asset structure and history	drill-down	site, location path, type	asset, location history, WO history, counters	`eam_structure`	Functional location / asset views	IH01 structural display; IH06 FL list; IH08 equipment list
Measurement documents	list inquiry	counter, period, source	readings, deltas, source	`eam_readings`	Counter registrations	IK17 measurement documents list
Cost per asset	drill-down	asset, type, period	parts, labor, external, overhead, total vs budget	`eam_cost`	Asset cost (EntAssetObjectCost)	MCI3 cost analysis (PMIS); KOB1 order line items
MTBF / MTTR / availability	KPI	site, type, criticality, window	MTBF, MTTR, availability, failures	`eam_kpi`	Asset KPIs	MCI7 breakdown statistics; MCI1 object statistics
Fault analysis	KPI	asset type, symptom, cause, period	count, downtime, cost, repeat rate 30 days, Pareto	`eam_faults`	Fault analysis; SSRS Fault report	IW69 notification items; MCI5 damage analysis
Backlog	KPI	site, trade, priority	open WOs, backlog hours, weeks of backlog	`eam_backlog`	Work order backlog	IW38 with capacity; CM01 capacity load
Budget vs actual	drill-down	scope, version, period	budget, actual, committed, variance	`eam_budget`	Budget vs actual (maintenance)	IP31 cost simulation vs KOB1 actuals; S_ALR_87013611
Warranty inquiry	list inquiry	vendor, expiring within, asset type	asset, warranty, vendor, ends, covered cost YTD	`eam_warranty`	Warranty inquiry (EntAssetWarranty)	BGM3 display warranty; IE03 warranty tab
Permit register	list inquiry	status, type, site, valid range	permit, WO, asset, valid from/to, locks open	`eam_permits`	n/a	WCM work clearance document lists (WCM_WCD)
Round compliance	KPI	round, period	occurrences, stops done, skipped %, on-time %	`eam_rounds`	Maintenance rounds results	IW38 for object-list orders
Rotables by condition	list inquiry	item, condition, location	serial, condition, value, refurbishment count	`eam_rotables`	n/a	MMBE / MB52 split valuation stock; IW8W history
Fuel consumption (fleet)	KPI	vehicle, driver, period	liters/gallons, cost, distance, consumption per 100 km / mpg	`eam_fuel`	n/a	MCIZ fleet consumption analysis

8.44.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Work order package	WO RELEASED (print from board)	header: WO, asset, location, priority, due, permits required; jobs: instructions, checklist, safety notes; parts pick list; labor sign-off; footer: QR code to mobile	PDF, mobile	WO	maintenance record: life of asset + 3 years; regulated assets per country (e.g. pressure vessels DE BetrSichV records)
Permit to work	permit ISSUED	permit type, WO, asset, hazards, isolation points with lock numbers, gas test results, validity, signatures (requester, approver, issuer, each technician)	PDF (posted at worksite), mobile	EAM-PMT	OSHA 29 CFR 1910.147 periodic inspection certification records; retain 5 years
LOTO lock tag	lock applied	technician name, lock number, point, date/time, contact	label PDF (thermal printer)	lock number	per 29 CFR 1910.147(c)(5)
Round checklist	round WO released	ordered stops, checklist items, readings fields	PDF, mobile	WO	maintenance record
Maintenance request acknowledgment	request created (user/mobile)	request no, asset, description, status link	email, notification	MR	none
Refurbishment shipment to repair vendor	external refurbishment step	serials, condition, vendor PO reference	PDF with wave-28 shipment / wave-17 PO	wave-17 PO	per wave 17/28
Fault report	on demand	fault Pareto per filter	PDF, XLSX	none	none

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Maintenance budget	version transferred	scope × period budget lines	XLSX, PDF	budget version	budget record

8.44.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`eam-plan-schedule`	nightly 01:00 site-local + on completion/reading events	site, plan line	key per (plan line, run date); open-WO check prevents duplicates	< 5 min for 10,000 lines	per-line retry; failures listed	run failure; lines with next_due in past and no WO
`eam-condition-engine`	streaming (Kafka consumer)	asset hash	dedupe_key + cooldown	p95 < 5 s per reading	consumer restart from offset	consumer lag > 30 s
`eam-kpi-rollup`	hourly	site	rollup per (site, hour) replace	< 2 min	retry	stale > 3 h
`eam-permit-expiry`	every 5 min	site	key (permit, transition)	< 10 s	retry	permits ISSUED past valid_to
`eam-emergency-regularize-check`	hourly	site	key (WO, hour)	< 10 s	retry	emergency WO not regularized within window
`eam-budget-calculate`	on demand; annual budget cycle	site, version	key (scope, version)	< 10 min for 10,000 lines × 12 months	resumable per site	calculation failed
`eam-warranty-expiry`	daily 06:00	site	key (warranty, lead day)	< 1 min	retry	none
`eam-readings-retention`	monthly	counter partition	drop/downsample partitions older than 13 months	< 30 min	retry next day	partition count over budget

8.44.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
EamStructureMaintain / EamAssetMaintain	functional locations, assets, warranties, criticality; Plant structure tree, Asset detail	none	Maintenance engineer
EamPlanAdmin	plans, rounds, sequences, fault catalogs, schedule run	none	Maintenance planner (senior)
EamPlanner	requests convert, WO create/schedule/release, permits request, refurbishment WOs	with EamPermitIssuer on the same permit	Maintenance planner, Supervisor
EamTechnician	WO status (IN_PROGRESS/COMPLETED), parts issue, labor, faults, LOTO lock actions; Technician mobile	with EamPermitIssuer approving a permit for own job	Technician
EamPermitIssuer	permit approve/issue/suspend/close	requester ≠ approver; issuer cannot be the only lock holder	Area authority, Safety officer
EamCounterRegister	readings, fuel log	none	Operator, IoT bridge service account
EamBudgetAdmin	budget calculate, adjust, transfer	with BudgetApprove (wave 38) on the same plan	Maintenance manager
EamInquire	all GET reports and KPIs	none	Maintenance manager, Plant manager
EamSetupAdmin	parameters 8.44.13, permit types, isolation points, posting keys	with EamPermitIssuer	EAM system administrator

8.44.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
EAM-CTR-001	error	Reading {v} is below last reading {last} for cumulative counter {c}	Non-monotonic	Set the reset flag or correct the value	FR-44.3
EAM-AST-001	error	Asset {a} is {state}; work orders need ACTIVE	Lifecycle	Activate or use override duty	FR-44.2
EAM-FL-001	error	Functional location depth exceeds 12	Hierarchy limit	Flatten the structure	FR-44.1
EAM-PLN-001	warning	Counter {c} has zero average usage; forecast unavailable	No usage	Register readings or set manual usage	BR-44.1
EAM-PLN-002	info	{n} work orders generated, {m} merged	Scheduler result	None	FR-44.5
EAM-WO-001	error	Transition {from} → {to} not allowed	State machine	Use an allowed transition	BR-44.3
EAM-WO-002	error	Work order cannot CLOSE: {n} pending journals	Costs not posted	Post or cancel journals	BR-44.3
EAM-WO-003	warning	Emergency work order {wo} not regularized within {h} h	BR-44.7	Complete planning fields	BR-44.7
EAM-WO-004	error	Mandatory checklist item {i} unanswered	Completion check	Answer the item	8.44.5 validation
EAM-WO-005	error	CAPITAL work order needs a fixed asset or CIP project	Validation	Link a fixed asset or project	FR-44.11
EAM-FLT-001	error	Register a fault before completing {type} work orders	Fault mandatory	Add symptom/cause/remedy	FR-44.16
EAM-FLT-002	error	Combination {symptom}/{area}/{type} not allowed for {assetType}	Catalog	Choose an allowed combination	FR-44.16
EAM-PMT-001	error	Job requires an ISSUED {permitType} permit	No valid permit	Request and issue the permit	BR-44.11
EAM-PMT-002	error	{n} isolation points lack a verified lock	LOTO incomplete	Apply and verify locks	BR-44.11
EAM-PMT-003	error	Permit cannot close: {n} locks still applied	Locks open	Remove locks	BR-44.11
EAM-PMT-004	warning	Permit {p} expired and was suspended	Validity passed	Revalidate or reissue	BR-44.11
EAM-PMT-005	error	Approver must differ from requester	SoD	Another approver	BR-44.11
EAM-REF-001	error	Serial {s} is not in condition DAMAGED	Refurbishment input	Select a damaged unit	BR-44.12
EAM-WAR-001	info	Line covered by warranty {w} ({vendor})	Warranty check	Claim created at completion	BR-44.14
EAM-BUD-001	error	Budget version {v} already transferred; create a new version	Immutability	Recalculate as new version	BR-44.13
EAM-IOT-001	info	Duplicate breach suppressed for {asset}/{rule}	Dedupe/cooldown	None	BR-44.4

8.44.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
IoT reading → request	10 readings/s per site, 30 sites; peak 1,000/s	1 / 5 / 8 s	1,000 readings/s	CounterReading ≈ 9.5 B rows raw → downsampled to 1/min after 30 days ≈ 200 GB	monthly partitions; (counter_id, reading_at)	evaluate thresholds on every reading, rate-of-change rules sampled at 1/s; lag alert
Plan scheduling (nightly)	10,000 plan lines per tenant	run < 5 min; per line 5 / 20 / 50 ms	100 lines/s	negligible	PlanLine (next_due_at), (next_due_forecast_at)	event-driven part continues; nightly sweep extends
WO status transition (mobile)	20,000 transitions/day; peak 300/min	50 / 200 / 400 ms	50/s	WO ≈ 2 KB × 150k/year = 300 MB	(site_id, status, due_at)	mobile action log buffers; server applies in order
Work order board load	200 planners, 2,000 open WOs per site	300 / 800 / 1,500 ms	20 loads/s	n/a	projection per site	Gantt lazy-loads by week
Permit transition	1,000/day	80 / 250 / 500 ms	10/s	permits + locks ≈ 100 MB	(work_order_id), (status, valid_to)	none permitted: permit checks fail closed
Budget calculation	10,000 lines × 12 months	< 10 min	20 lines/s per worker	MaintenanceBudget ≈ 120k rows/version	(version, scope, period)	longer run
KPI query	2,000/day	100 / 400 / 900 ms	20 q/s	rollups ≈ 2 GB	hourly rollup tables	serve last rollup

8.44.20 Localization and compliance hooks

- Lockout/tagout: US OSHA 29 CFR 1910.147 (energy control procedures, periodic inspection at least annually, lock per authorized employee), EU Directive 2009/104/EC and DE BetrSichV / TRBS 1112 (Instandhaltung), UK PUWER 1998; permit types and checklists are pack templates per country; extension point `eam.permit.template.<cc>`.
- Statutory inspections: pressure equipment (EU PED, DE BetrSichV §§15–16 intervals), lifting equipment (UK LOLER 6/12 months), fire systems (NFPA 25), vehicle inspection (DE HU §29 StVZO, US DOT 49 CFR 396.17 annual) delivered as plan templates with mandatory checklists by the wave-48 packs.
- Environmental: refrigerant leak checks (EU F-gas Regulation 2024/573 intervals by CO2-equivalent; US EPA Section 608 for ≥ 50 lb appliances) as plan templates using counter "refrigerant charge".
- Records retention: maintenance and inspection records retained for the asset life plus country-specific periods (US OSHA 1910.147 certification records; DE BetrSichV records until next inspection; aviation/rail regulated sectors out of scope).
- Units: counters support metric and US customary units with wave-8 UoM conversion; temperature thresholds stored in SI with display conversion.
- Fleet: fuel tax and mileage logs for company vehicles (DE Fahrtenbuch per §6 EStG; US IRS Pub 463 substantiation) use fuel log and odometer counters; wave-48 packs format the log.
- Language: checklists, permit texts and LOTO procedures are translatable (wave-3 labels); permits print in the site language plus English when the site parameter requests a second language.

8.44.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX #299 — 12.35.1 Maintenance rounds (EntAssetMaintenanceRound)	PARTIAL	ADDED as FR-44.14	BR-44.9
AX #300 — 12.35.1 Maintenance sequences (EntAssetMaintenanceSequence)	PARTIAL	ADDED as FR-44.15	BR-44.10
AX #301 — 12.35.1 Work order fault registration (EntAssetFault)	PARTIAL	ADDED as FR-44.16	WorkOrderFault
AX #302 — 12.35.1 Warranties (EntAssetWarranty, warranty check on WO)	PARTIAL	ADDED as FR-44.17	BR-44.14; vendor claim in FR-45.15
AX #303 — 12.35.1 Maintenance budgets (EntAssetMaintenanceBudget)	PARTIAL	ADDED as FR-44.18	BR-44.13
AX #304 — 12.35.2 Fault analysis	PARTIAL	ADDED as FR-44.16	"Fault analysis" in 8.44.14
AX #305 — 12.35.2 Warranty inquiry	PARTIAL	ADDED as FR-44.17	"Warranty inquiry"
AX #306 — 12.35.2 Budget vs actual (maintenance)	PARTIAL	ADDED as FR-44.18	"Budget vs actual"
AX #307 — 12.35.2 SSRS Fault report	PARTIAL	ADDED as FR-44.16	Document in 8.44.15
AX #308 — 12.35.3 Schedule maintenance rounds / sequences	PARTIAL	ADDED as FR-44.14	Covered by `eam-plan-schedule` with ROUND and sequence lines (also FR-44.15)
AX #309 — 12.35.3 Maintenance budget calculation	PARTIAL	ADDED as FR-44.18	Job `eam-budget-calculate`
AX #310 — 12.35.4 Plan types, plan line defaults, rounds setup, sequences setup	PARTIAL	ADDED as FR-44.14	Parameters in 8.44.13; sequences setup via FR-44.15
AX #311 — 12.35.4 Fault symptoms, areas, types, causes, remedies, fault designer	PARTIAL	ADDED as FR-44.16	Fault catalog designer screen
AX #312 — 12.35.4 Warranty types, vendor warranty setup	PARTIAL	ADDED as FR-44.17	AssetWarranty.warranty_type_id
AX #313 — 12.35.4 Maintenance budget parameters	PARTIAL	ADDED as FR-44.18	Parameter "Budget calculation horizon and version"
SAP #297 — 13.25 IE31/IE36/IE37 fleet objects	PARTIAL-44	ADDED as FR-44.21	Fuel log and odometer counters
SAP #298 — 13.25 IW81/IW8W refurbishment order / GR for refurbishment	PARTIAL-44	ADDED as FR-44.20	BR-44.12; needs wave-16 condition dimension (flagged in 8.44.12)
SAP #299 — 13.25 IW32 permits (IPMD) / WCM lockout/tagout	PARTIAL-44	ADDED as FR-44.19	BR-44.11

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP #300 — 13.25 Asset Intelligence Network / Predictive Asset Insights / ASPM	PARTIAL-44	ADDED as FR-44.22	Criticality/RCM-lite here; predictive ML OUT-OF-SCOPE for v1 (hooks only; models candidate for wave 47); AIN manufacturer network OUT-OF-SCOPE
SAP #301 — 13.25 SPRO refurbishment valuation types (OMWC), order type PM04	PARTIAL-44	ADDED as FR-44.20	Parameter "Condition valuation for rotables"; WO type REFURBISHMENT
SAP #302 — 13.25 SPRO permits (T357G), WCM objects/applications/approvals	PARTIAL-44	ADDED as FR-44.19	PermitType, IsolationPoint setup

8.45 Wave 45: Service Management & Field Service

8.45.1 Scope and business context

Service management replaces AX Service management (SMAServiceAgreementTable, SMAServiceOrderTable, SMAServiceObjectTable, SMADispatchBoard, SMARepair, service subscriptions), D365 Field Service (work orders, bookings, agreements, IoT alerts, mobile) and SAP CS/FSM (service notifications, service orders, IBase, warranties, SAP Field Service Management). After this wave: an installed base is built from shipped serials; agreements define entitlements, SLAs and pricing; cases become service work orders; a dispatcher schedules technicians with OR-Tools respecting skills, territories, time windows and travel; technicians execute offline on a PWA with photos, signatures and truck-stock parts; billing posts to the customer or warranty. This is the phase-exit wave.

Out of scope: customer-facing self-scheduling portal (v1.x), subscription billing beyond wave-29 recurring invoices, remote assist/AR, depot repair RMA beyond wave-30 RMA linkage.

Coverage closure (8.45.21) adds the cross-module case framework that earlier waves deferred to this wave: generic case management with categories, processes, activities and associations to any record (FR-45.12), a knowledge base of symptom/solution articles (FR-45.13), activities on cases, projects and other records (FR-45.14), supplier warranty claims (FR-45.15) and email-to-case intake (FR-45.16). Telephony/CTI and omnichannel contact-center routing remain out of scope (CRM non-goal).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX SMADispatchBoard is a manual Gantt: a dispatcher drags each SMAServiceOrderTable row by hand, with no skill or travel check; SAP FSM/RSO optimization is a separate licensed product with its own master data sync	No optimizer in the ERP; duplicated technician master	OR-Tools CP-SAT dispatch inside the service domain using wave-42 skills and wave-16 truck stock (BR-45.3)	Dispatch time per territory-day 2–3 h → < 30 s solve + 10 min review; skill/window violations 0
Entitlement: AX service orders pick the agreement the user chooses (SMAServiceOrderTable.ServiceAgreementId) with no SLA clock; SAP CS warranty check (IW31 "warranty" pop-up) is informational and SLA profiles live in CRM only	No deterministic entitlement or SLA engine in ERP	Resolver (BR-45.1) and business-hour SLA clocks with pause states and breach events (BR-45.2)	Wrong-agreement billing 0 on fixtures; SLA breaches discovered after the fact → warning at 75%
Installed base built by hand: AX service objects created manually after shipment; SAP IBase needs IB51/IB52 maintenance or a custom user exit from delivery	No event from shipment to service	Service objects created from wave-28 serialized shipment events with warranty start	Manual IBase upkeep 100% → 0; serial-to-object lag days → < 60 s
Technician execution on paper or a disconnected app; parts used from vans re-keyed into inventory journals days later, causing negative stock and wrong COGS	No mobile execution or truck-location model	Offline PWA with append-only action log; TRUCK locations with consumption at completion	Parts re-keying 0; inventory accuracy on trucks ≥ 99%; completion-to-invoice 5–10 days → same day
Service billing in AX goes through project invoice proposals (service orders post to ProjEmplTrans/ProjItemTrans), a second module to learn; SAP DP90 again	Service reuses project billing plumbing	Work order lines carry coverage (BILLABLE/WARRANTY/AGREEMENT) and post directly to wave-29 invoicing	Billing steps 6 → 1; warranty-vs-billable misclassification 0 on 30 fixtures

8.45.2 AX / SAP parity map

Capability	AX / D365	SAP ECC / S/4	Passage equivalent	Parity
Installed base / service objects	SMAServiceObjectTable, SMAServiceObjectRelationTable; D365 FS customer assets	IBase (IBIN/IBIB), equipment with customer, serial numbers (SERI)	`ServiceObject` from wave-28 shipment serials	full
Warranties	Limited (SMAServiceObject + agreement) ; D365 FS warranties	Warranty master (BGMK), warranty check on order	`Warranty` (parts/labor/travel, start trigger)	improved
Service agreements	SMAServiceAgreementTable/Line, SMAAgreementLine, service intervals	Service contracts (VA41, VBAK type WV/SC), SLA profiles	`ServiceAgreement`, `Entitlement`, `SlaProfile`	full
Cases/notifications	smmCase (CRM case), D365 FS cases	QMEL service notifications (IW51), CRM cases	`Case`	full
Service orders	SMAServiceOrderTable/Line, stages	Service orders (IW31 type SM01/SM02), CS order	`ServiceWorkOrder`	full
Dispatch board	SMADispatchBoard (Gantt), SMAServiceTechnician	SAP FSM planning board / Resource Scheduling Optimizer; CRM MRS	`dispatch` service (OR-Tools CP-SAT) + board	improved
Technician mobile	D365 FS mobile (Power Apps)	SAP FSM mobile	technician PWA	full
Parts from trucks	D365 FS warehouse per resource	Van stock (storage location per technician)	wave-16 location type TRUCK per technician	full
Billing	SMAServiceOrder → project invoice (service orders post via projects)	DP90 to billing, warranty claim	`ServiceWorkOrder` → wave-29 invoice; WARRANTY lines no-charge	full
KPIs	Service management reports	CS reports	SLA attainment, first-time fix, utilization	improved
Generic case management	CaseDetail, CaseCategory (category types: HR, customer service, collections, product change, audit, service, project…), CaseProcess (stages), case associations, case category security	QMEL notifications by type, S/4 Service cases, CRM case management (CRMD_ORDER case), dispute cases (UDM)	`Case` with `CaseCategory`, `CaseProcess`, `CaseAssociation` (FR-45.12)	full
Knowledge base	CaseKnowledgeArticle (knowledge articles on cases)	Solution database IS01/IS02 (symptoms/solutions), CRM knowledge articles	`KnowledgeArticle` (FR-45.13)	full
Activities	smmActivities (tasks, appointments) on cases and projects	Notification tasks/activities (QMSM/QMMA), CRM activities	`Activity` (FR-45.14)	full
Supplier warranty claims	n/a	WTY warranty claim processing (WTY1, claim types, vendor recovery)	`SupplierWarrantyClaim` (FR-45.15)	full

Capability	AX / D365	SAP ECC / S/4	Passage equivalent	Parity
Front office intake	Case via email (manual)	CIC0 customer interaction center, CRM IC email	email-to-case via wave-35 hub (FR-45.16)	partial — no CTI/telephony
Service subscriptions	SMASubscriptionTable, fee transactions, indexation	Service contract billing plans (VA42 billing plan), periodic billing	wave-29 subscription billing with agreement billing_model SUBSCRIPTION	improved — single engine

8.45.3 Functional requirements

- **FR-45.1 Installed base** — Must. The system shall create service objects automatically from wave-28 shipments of serialized items (and manually), with customer, site address, parent/child structure, install date and current status.
- **FR-45.2 Warranties** — Must. The system shall attach warranty terms (parts/labor/travel coverage flags, duration from SHIP/INSTALL/REGISTRATION, exclusions) to service objects and evaluate coverage per work order line.
- **FR-45.3 Service agreements** — Must. The system shall define agreements per customer with lines covering service objects or categories, entitlements (included visits/hours, discounts, price lists), SLA profile, validity and billing (prepaid, per-visit, subscription via wave 29).
- **FR-45.4 SLA profiles** — Must. The system shall define response and resolution targets by priority with business-hour calendars and pause states per BR-45.2.
- **FR-45.5 Cases** — Must. The system shall capture cases (phone, email, portal, IoT from wave 44/25) and resolve entitlement to the best agreement per BR-45.1, starting the SLA clock.
- **FR-45.6 Service work orders** — Must. The system shall create work orders from cases or agreements (preventive visits) with incident type, required skills, estimated duration, time window, service object, address, and lines (labor, parts, travel, fees) each billable, covered-by-warranty or covered-by-agreement.
- **FR-45.7 Dispatch optimization** — Must. The system shall produce daily schedules per territory with OR-Tools CP-SAT satisfying BR-45.3 constraints, with manual override on the board and incremental re-optimization on changes.
- **FR-45.8 Technician mobile PWA** — Must. The system shall provide an offline-capable PWA: today's route, navigate, start travel/work, checklists, photos, parts from truck stock, signature capture, customer-facing summary.
- **FR-45.9 Truck stock** — Must. The system shall treat each technician's vehicle as a wave-16 location (type TRUCK) with replenishment transfer orders and consumption on work order completion.
- **FR-45.10 Billing** — Must. The system shall post completed work orders to wave-29 invoicing: billable lines at agreement/standard prices, warranty lines at zero with a warranty cost record, agreement-covered lines consuming entitlement.
- **FR-45.11 Service KPIs** — Should. The system shall compute SLA attainment, first-time-fix rate, mean time to respond/resolve, technician utilization and travel ratio.
- **FR-45.12 Generic case management** — Must. The system shall generalize `Case` to any business domain through case categories (hierarchical, with category type SERVICE, CUSTOMER_SERVICE, HR, COLLECTIONS, PRODUCT_CHANGE, AUDIT, PROJECT, VENDOR, INTERNAL), case processes (ordered stages with entry/exit criteria and required activities), associations to any record (entity type + id from the wave-3 metadata registry: customer, vendor, worker, invoice, item, project, asset, work order, audit finding), a case log, resolution codes, per-category SLA profile and per-category OPA security.
  - SERVICE and CUSTOMER_SERVICE categories keep the entitlement and SLA behavior of FR-45.5 (BR-45.1, BR-45.2); other categories use the category's SLA profile (or none).
  - customer_party_id becomes optional; required for SERVICE/CUSTOMER_SERVICE/COLLECTIONS categories.
  - Minimal case entities used by earlier waves (wave-30 disputes, wave-9 product change requests) migrate into categories COLLECTIONS and PRODUCT_CHANGE with their ids kept as associations (migration job `case-consolidate`).
- **FR-45.13 Knowledge base** — Should. The system shall maintain knowledge articles (title, symptom, cause, solution steps, applicability to item, item group, incident type, asset type and case category, language, attachments) with versioning and lifecycle DRAFT → IN_REVIEW → PUBLISHED → RETIRED, full-text and structured search, suggestions on case creation, linking of articles to cases, and usage and helpfulness statistics.
- **FR-45.14 Activities** — Should. The system shall record activities (TASK, APPOINTMENT, CALL_NOTE, EMAIL_NOTE) with owner, due/start/end, status (OPEN, DONE, CANCELED) and reminders on cases, projects (wave 41), service work orders and any associable record, with a personal "My activities" list and calendar export (iCalendar).
- **FR-45.15 Supplier warranty claims** — Should. The system shall create supplier warranty claims from WARRANTY-covered service work order lines and from `eam.warranty.claimable.v1` events, with claim lines (part, serial, failure code, labor hours at the vendor's agreed rate), submission to the vendor (PDF/email or wave-35 EDI), vendor decision (ACCEPTED, PARTIALLY_ACCEPTED, REJECTED), failed-part return through a wave-30/17 vendor return, and settlement by matching the vendor credit note (wave 18) per BR-45.12.
- **FR-45.16 Email-to-case intake** — Should. The system shall create or update cases from inbound email to category mailboxes through the wave-35 integration hub, threading replies by a case token in the subject (`[CS-{seq}]`) or the Message-ID chain, storing attachments, matching the sender to a customer contact, sending an acknowledgment, and logging outbound replies on the case.

8.45.4 Data model

ServiceAgreement

Field	Type	Required	Notes
id	uuid	yes	PK; number SA-{seq}
customer_party_id	uuid	yes	
legal_entity_id	uuid	yes	
validity	daterange	yes	
sla_profile_id	uuid	yes	
billing_model	enum	yes	PREPAID, PER_VISIT, SUBSCRIPTION
price_list_id, discount_pct	uuid, numeric(5,2)	no	
priority_rank	smallint	yes	entitlement tie-break
lines[]	AgreementLine	yes	(service_object_id or category, included_visits, included_hours, parts_covered bool, labor_covered bool, travel_covered bool)
status	enum	yes	DRAFT, ACTIVE, SUSPENDED, EXPIRED, CANCELED

Case

Field	Type	Required	Notes
id	uuid	yes	PK; number CS-{seq}
customer_party_id, service_object_id	uuid	yes/no	
channel	enum	yes	PHONE, EMAIL, PORTAL, IOT, INTERNAL

Field	Type	Required	Notes
priority	smallint	yes	1–4
agreement_id, entitlement_result	uuid, jsonb	no	resolution trace
sla_response_due, sla_resolution_due	timestamptz	computed	
sla_state	enum	yes	RUNNING, PAUSED, MET, BREACHED
sla_paused_minutes	int	yes	
status	enum	yes	NEW, IN_PROGRESS, WAITING_CUSTOMER, WAITING_PARTS, RESOLVED, CLOSED, CANCELED
opened_at, responded_at, resolved_at	timestamptz		

ServiceWorkOrder

Field	Type	Required	Notes
id	uuid	yes	PK; number SWO-{seq}
case_id, agreement_id, service_object_id	uuid	no/no/no	
customer_party_id, address_id, geo (lat, lon)		yes	
territory_id	uuid	yes	
incident_type_id, required_skills[]	uuid, uuid[]	yes	
est_duration_minutes	int	yes	
time_window	tstzrange	no	customer-promised
priority, sla_due	smallint, timestamptz		
status	enum	yes	UNSCHEDULED, SCHEDULED, DISPATCHED, TRAVELING, IN_PROGRESS, ON_HOLD, COMPLETED, POSTED, CANCELED
assignment (technician_id, start, end, sequence)		no	from dispatch
lines[]	WorkOrderLine		(kind LABOR/PART/TRAVEL/FEE, qty, unit_price, coverage BILLABLE/WARRANTY/AGREEMENT, amount)
signature_ref, photos[]		no	object store
first_time_fix	bool	no	set at completion

CaseCategory: id, code, parent_id, category_type, process_id, sla_profile_id, default_priority, opa_policy_ref, resolution_codes[]. **CaseProcess**: id, stages[] (code, sequence, entry_criteria CEL, exit_criteria CEL, required_activity_templates[]). Case adds: category_id, stage_code, resolution_code, resolution_text, case_log[] (at, actor, kind, text), and customer_party_id becomes conditional. **CaseAssociation**: case_id, entity_type (registry name), entity_id, role (SUBJECT, RELATED). **KnowledgeArticle**: id (sequence KB-{seq}), version, title, symptom, cause, solution, applicability {item_ids[], item_group_ids[], incident_type_ids[], asset_type_ids[], category_ids[]}, language, status (DRAFT, IN_REVIEW, PUBLISHED, RETIRED), author_id, reviewer_id, published_at, tsv tsvector, views, helpful_yes, helpful_no. **CaseArticleLink**: case_id, article_id, article_version, outcome (SOLVED, NOT_HELPFUL). **Activity**: id, kind, subject, owner_worker_id, due_at, start_at, end_at, status, reminder_minutes, associations[] (entity_type, entity_id). **SupplierWarrantyClaim**: id (sequence SWC-{seq}), vendor_party_id, source (SERVICE_WO, EAM_WO), source_ids[], lines[] (item_id, serial, failure_code, part_amount, labor_hours, labor_rate, claimed_amount, accepted_amount), status (DRAFT, SUBMITTED, ACCEPTED, PARTIALLY_ACCEPTED, REJECTED, SETTLED, CLOSED), submitted_at, vendor_reference, return_order_id, credit_note_id, recovered_amount. **InboundMessage**: id, mailbox, message_id, in_reply_to, from_address, subject, received_at, case_id, attachments[].

ServiceObject: id, item_id, serial, customer_party_id, site_address_id, parent_id, installed_at, status (ACTIVE, INACTIVE, RETURNED), shipment_line_id, warranty_id. **Warranty**: id, template, starts_on (SHIP/INSTALL/REGISTRATION), months, parts_covered, labor_covered, travel_covered, exclusions[]. **Technician**: worker_id, territory_ids[], skills[] (from wave 42), shift calendar, home geo, truck_location_id, max_daily_minutes, overtime_allowed. **DispatchRun**: id, territory_id, date, status, solver_seconds, objective, assignments[].

Invariants: a case has at most one resolved agreement; SLA due timestamps are recomputed whenever paused minutes change; a work order has at most one active assignment; parts lines must reference the technician's truck location or a wave-16 reservation; invoiced amount = Σ BILLABLE lines. Indexing: `ServiceWorkOrder (territory_id, status, sla_due)`; geo via PostGIS point index; `Case (sla_state, sla_resolution_due)` for the breach monitor.

8.45.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/service/agreements`	Create agreement	key	SvcAgreementMaintain
POST	`/api/v1/service/cases`	Create case (entitlement + SLA start)	key	SvcCaseCreate
POST	`/api/v1/service/cases/{id}/status`	Transition (pauses/resumes SLA)	key + version	SvcCaseMaintain
POST	`/api/v1/service/work-orders`	Create WO	key	SvcWoMaintain
POST	`/api/v1/service/dispatch/runs`	Optimize territory/date	key per (territory, date, version)	SvcDispatch
POST	`/api/v1/service/dispatch/runs/{id}/publish`	Publish schedule	key	SvcDispatch
POST	`/api/v1/service/work-orders/{id}/actions`	Mobile action log (travel start, work start, part, photo, signature, complete)	key per action id	SvcTechnician
POST	`/api/v1/service/work-orders/{id}/post`	Bill / warranty post	key	SvcBillingPost
GET	`/api/v1/service/kpis?territory=&from=&to=`	KPIs	n/a	SvcInquire
POST	`/api/v1/cases`	Create case in any category (generic)	key	CaseCreate (category-scoped by OPA)
POST	`/api/v1/cases/{id}/stage`	Advance case process stage	key + version	CaseMaintain
POST	`/api/v1/cases/{id}/associations`	Associate a record	key	CaseMaintain
GET	`/api/v1/knowledge/articles/search?q=&itemId=&incidentTypeId=&lang=`	Search / suggest articles	n/a	KbRead
POST	`/api/v1/knowledge/articles/{id}/publish`	Publish article version	key + ETag	KbPublish
POST	`/api/v1/activities`	Create activity on any record	key	ActivityMaintain
POST	`/api/v1/service/supplier-claims`	Create supplier warranty claim	key	SvcClaimMaintain

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/service/supplier-claims/{id}/decision`	Record vendor decision	key	SvcClaimMaintain

Validation: case creation requires a customer; dispatch run requires all WOs to have geo and skills; complete action requires signature unless incident type exempts; post requires COMPLETED and all parts lines consumed.

Event	Trigger	Payload	Consumers
`dev.passage.service.case.opened.v1`	case created	caseId, customerId, priority, agreementId, responseDue, resolutionDue	notifications, analytics
`dev.passage.service.sla.breached.v1`	breach monitor	see 8.0	notifications, analytics
`dev.passage.service.dispatch.published.v1`	publish	see 8.0	PWA, customer notifications
`dev.passage.service.workorder.completed.v1`	complete action	workOrderId, technicianId, lines[], firstTimeFix, durationMinutes	billing, inventory (truck consumption), KPIs
`dev.passage.service.workorder.posted.v1`	billing post	workOrderId, invoiceId, billableTotal, warrantyCost, entitlementConsumed	invoicing, analytics
`dev.passage.case.stage.changed.v1`	stage transition (any category)	caseId, categoryId, from, to, at, actor	workflow, notifications, domain owners (HR, collections, audit)
`dev.passage.knowledge.article.published.v1`	article PUBLISHED	articleId, version, language, applicability	search index, copilot retrieval (wave 47)
`dev.passage.service.supplier-claim.settled.v1`	credit note matched	claimId, vendorId, claimedAmount, recoveredAmount	analytics, GL warranty recovery

8.45.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Service workspace	tiles: SLA at risk (> 75% elapsed), unscheduled WOs, today's completion rate		SvcInquire
Case detail	entitle, create WO, pause/resume, resolve	SLA timers, agreement, object, timeline	SvcCaseMaintain
Agreement detail	lines, entitlement usage, SLA profile, billing		SvcAgreementMaintain
Dispatch board (Gantt + map)	optimize, drag assign, lock, publish, re-optimize	technician rows, WO blocks, travel segments, SLA due markers	SvcDispatch
Work order detail	lines, coverage, parts, photos, signature, post		SvcWoMaintain
Technician PWA	route, navigate, start/stop, checklist, parts, photos, signature, complete	offline indicator, sync queue	SvcTechnician
Installed base explorer	search by customer/serial, hierarchy, history		SvcInquire
KPI dashboard	filter by territory/technician	SLA %, FTF %, utilization	SvcInquire
My cases / case list (any category)	create, filter by category/stage/SLA, bulk assign	case, category, stage, owner, SLA due, associations	CaseMaintain
Case category and process setup	define categories, stages, criteria, SLA, security	category tree, process stages	CaseSetupAdmin
Knowledge base	search, view, rate helpful, author, review, publish	title, applicability, language, version, status, helpful %	KbRead / KbAuthor / KbPublish
My activities	list, calendar, complete, reschedule	activity, related record, due, status	ActivityMaintain
Supplier warranty claims	create from WO lines, submit, record decision, match credit note	claim, vendor, lines, claimed, accepted, recovered	SvcClaimMaintain
Case mailbox	view unmatched inbound email, attach to case, create case	sender, subject, received, match suggestion	CaseMaintain

Rules: board locks (pinned assignments) survive re-optimization; dragging a WO outside skill/time window shows a hard-block explanation; PWA photo max 5 per action at 1,600 px long edge; signature stored as PNG with timestamp and geo.

8.45.7 Business rules and edge cases

- **BR-45.1 Entitlement resolution** — candidates = ACTIVE agreements of the customer (or parent party) whose validity covers opened_at and whose lines match the service object (exact object > object category > any) ; rank by match specificity, then priority_rank, then latest start; the chosen agreement sets the SLA profile and coverage; if none, the default SLA profile of the legal entity applies and all lines are BILLABLE unless warranty covers them.
- **BR-45.2 SLA clock rules** — response target measured from opened_at to first of (responded_at, assignment DISPATCHED); resolution target from opened_at to resolved_at. Clocks count only business minutes of the SLA profile calendar (e.g. Mon–Fri 08:00–18:00 in the service site's time zone, holidays excluded), unless the profile is 24×7. Pause states: WAITING_CUSTOMER pauses both clocks; WAITING_PARTS pauses resolution only if profile.pause_on_parts = true. Due timestamps are stored as absolute times and recomputed on each pause/resume: due = add_business_minutes(opened_at, target_minutes + paused_minutes). Warning event at 75% elapsed; breach event when now > due while RUNNING; a breached SLA stays BREACHED even if resolved later (attainment reporting counts it as missed). Example: priority 2 resolution target 480 business minutes, opened Fri 2026-10-23 16:00 (America/Chicago), paused 120 minutes Mon → due Tue 2026-10-27 12:00.
- **BR-45.3 Dispatch optimization constraints** — decision: assign each WO to at most one technician and a start time. Hard constraints: required_skills ⊆ technician.skills (certifications unexpired per BR-42.8); WO.territory ∈ technician.territory_ids; start ≥ time_window.lower and end ≤ time_window.upper if set; start/end within technician shift; sequence feasibility with travel time from an OSRM/Valhalla matrix (self-hosted) plus 5-minute buffer; Σ (work + travel) ≤ max_daily_minutes (+ overtime up to 120 min if overtime_allowed); pinned assignments fixed; parts required must be on the truck or a reservation picked. Objective (lexicographic weights): minimize SLA lateness minutes × 100, unassigned priority-weighted WOs × 1,000, travel minutes × 1, overtime minutes × 5, technician count × 50. Solver: CP-SAT with 30 s time limit, 8 workers; fallback greedy insertion when no feasible solution in time; incremental re-optimization only moves WOs not yet DISPATCHED or with the dispatcher's consent. Scale target: 200 WOs × 25 technicians per territory-day.
- **BR-45.4 Coverage per line** — PART: WARRANTY if warranty.parts_covered and within term and not excluded; else AGREEMENT if line.parts_covered; else BILLABLE. LABOR/TRAVEL analogous. Agreement included_hours consumed FIFO; when exhausted, remaining hours BILLABLE at agreement price list.
- **BR-45.5 Work order status machine** — UNSCHEDULED → SCHEDULED → DISPATCHED → TRAVELING → IN_PROGRESS → COMPLETED → POSTED; IN_PROGRESS ↔ ON_HOLD; any pre-COMPLETED → CANCELED; a COMPLETED WO with follow-up creates a new WO linked by `follow_up_of` and sets first_time_fix = false.
- **BR-45.6 Offline merge** — mobile actions are append-only with client-generated UUIDs and device timestamps; server orders by (device timestamp, receipt order); conflicting status transitions resolve to the furthest valid state; free text last-writer-wins; parts consumption is idempotent per action id.

- **BR-45.7 Truck stock** — consumption posts an inventory issue from the TRUCK location at completion; negative truck stock blocked unless policy `service.allow-negative-truck`; replenishment transfer orders generated when below min per item.
- **BR-45.8 Pricing** — BILLABLE labor at agreement price list → customer price group → standard rate; parts via wave-27 pricing engine with agreement discount; rounding per line half-even.
- **BR-45.9 Case process stages** — a case starts at the first stage of its category's process; advancing requires the current stage's exit criteria (CEL over case, associations, activities) to be true and all required activities of the stage DONE (`CASE-STG-001`); moving back one stage is allowed with a reason; RESOLVED requires a resolution code; CLOSED is final, and reopening creates a new case linked by `reopened_from` unless the category allows reopen within N days (parameter, default 14).
- **BR-45.10 Case category security** — read and write access to a case is the union of: owner, members of the category's owning team, and users whose OPA policy for the category type grants access (e.g. HR cases only to HR partners and the case subject's manager chain; COLLECTIONS to the customer's collector); associations to records the user cannot read are shown as "restricted record"; HR category cases store free text in the `hr_pii`-protected column set.
- **BR-45.11 Knowledge article lifecycle and suggestion** — only one PUBLISHED version per (article, language); publishing a new version retires the previous one; reviewer ≠ author; suggestions on case creation rank PUBLISHED articles by ts_rank of the case description against the article tsvector × 0.6 + applicability match (item 0.25, incident type 0.10, asset type 0.05) and return the top 5; helpfulness % = helpful_yes ÷ (helpful_yes + helpful_no) shown when ≥ 10 votes.
- **BR-45.12 Supplier warranty claims** — claimable amount per line = part cost at issue (wave 16) + labor_hours × vendor labor_rate (from the vendor warranty agreement; default 0 when labor not covered); claims must be submitted within the vendor's claim window (default 30 days after WO completion, `SVC-SWC-001` after); accepted_amount ≤ claimed_amount; settlement matches the wave-18 vendor credit note by vendor_reference, and the recovered amount credits the warranty expense account; rejected claims close with reason and no posting; a failed-part return is required when the vendor flag `return_required` is set before SUBMITTED.
- **BR-45.13 Email threading** — an inbound message is attached to case X when the subject contains `[CS-X]` and the case is not CLOSED, or when In-Reply-To matches a message already on case X; otherwise a new case is created in the mailbox's default category with the sender matched to a party contact by email (exact, case-insensitive); messages to a CLOSED case create a new case linked by `reopened_from`; auto-acknowledgment is sent once per new case and suppressed for auto-replies (headers `Auto-Submitted`, `X-Autoreply`).

8.45.8 Acceptance criteria

- **AC-45.1** (FR-45.5, BR-45.1) Given 50 entitlement fixtures (overlapping agreements, categories, parent parties), when cases open, then each resolves to the expected agreement and SLA profile.
- **AC-45.2** (FR-45.4, BR-45.2) Given the BR-45.2 example, when the case is paused 120 minutes on Monday, then resolution due = 2026-10-27T12:00-05:00.
- **AC-45.3** (BR-45.2) Given a RUNNING case at 75% elapsed, when the monitor runs, then a warning notification fires; at 100% a `sla.breached` event fires once.
- **AC-45.4** (FR-45.7, BR-45.3) Given 1,000 generated territory-days (≤ 200 WOs, ≤ 25 technicians), when optimized, then every assignment satisfies skills, windows, shifts and travel feasibility, and solver time p95 < 30 s (property test).
- **AC-45.5** (FR-45.7) Given a pinned assignment, when re-optimization runs, then it is unchanged.
- **AC-45.6** (FR-45.10, BR-45.4) Given a WO with a part covered by warranty and 3 labor hours under an agreement with 1 included hour, when posted, then the invoice contains 2 billable labor hours, the part is zero-priced with warranty cost recorded, and entitlement usage shows 1 hour consumed.
- **AC-45.7** (FR-45.8, BR-45.6) Given a technician offline for 8 h performing start/part/complete actions, when reconnected, then the server state equals the device state and inventory shows one issue per part action.
- **AC-45.8** (FR-45.1) Given a wave-28 shipment of 10 serialized units, when confirmed, then 10 ACTIVE service objects exist with warranty starting on ship date.
- **AC-45.9** (phase exit) Given the demo group, when the recorded walkthrough runs projects → HR → payroll → maintenance → field service, then every step completes on `stage` with reconciliations at zero variance.
- **AC-45.10** (FR-45.12, BR-45.9) Given an HR category with stages INTAKE → INVESTIGATION → RESOLUTION and a required "Interview" activity in INVESTIGATION, when the case advances to RESOLUTION with the activity OPEN, then it is rejected with `CASE-STG-001`; after the activity is DONE it advances.
- **AC-45.11** (FR-45.12, BR-45.10) Given an HR case on worker W, when a collector without HR policy queries it, then 403 is returned; the subject's manager (as-of today per BR-42.1) can read it.
- **AC-45.12** (FR-45.13, BR-45.11) Given 3 PUBLISHED articles of which one applies to the case item, when a case is created with matching description text, then that article ranks first in the suggestions; a reviewer equal to the author cannot publish (`KB-PUB-001`).
- **AC-45.13** (FR-45.15, BR-45.12) Given a WO with a WARRANTY part line of 180.00 USD and 1.5 labor hours at a vendor rate of 60.00, when a claim is created and the vendor accepts 250.00 with credit note matched, then recovered_amount = 250.00, the warranty expense account is credited 250.00 and the claim is SETTLED.
- **AC-45.14** (FR-45.16, BR-45.13) Given an open case CS-1042, when an email with subject "Re: [CS-1042] pump noise" arrives, then it is logged on CS-1042 and no new case or acknowledgment is created; an email with no token from an unknown sender creates a new case in the mailbox's default category.
- **AC-45.15** (FR-45.14) Given a project (wave 41) and a case each with an open TASK for the same owner, when "My activities" loads, then both appear with their related records, and completing one sets status DONE with timestamp.

8.45.9 Test plan and fixtures

Fixtures: `svc-installed-base` (derived from wave-28 shipped serials, 2,000 objects); `svc-entitlement-50`; `svc-dispatch-gen` (Hypothesis generator for WOs/technicians with seeded travel matrix); `svc-mobile-replay` (offline action logs); `svc-billing-30`.

Test	Type	Target	Pass criterion
Entitlement	unit	resolver	50/50
SLA calendar arithmetic	property	business-minute functions	add/subtract inverse; DST-safe
Dispatch feasibility	property	CP-SAT model	1,000/1,000 feasible; p95 < 30 s
Offline merge	unit/E2E	action log merge	deterministic result
Billing coverage	contract	invoicing API	30/30
PWA	E2E (Playwright, offline)	technician flow	complete with signature offline
Breach monitor	performance	100k open cases	scan < 60 s

Reconciliation: Σ POSTED billable = wave-29 invoices with source SWO; Σ warranty cost = GL warranty expense; truck stock issues = Σ part actions.

8.45.10 Deployment and operations

Charts `passage-service` (TS API, breach monitor, mobile BFF) and `passage-dispatch` (Python OR-Tools worker, OSRM sidecar with regional extracts); Postgres `service` with PostGIS. Flags: `service.allow-negative-truck`. Schedules: `service-sla-monitor` every 60 s; `service-dispatch-auto` 18:00 territory-local for next day; `service-truck-replenish` nightly. SLOs: case create p95 < 300 ms; dispatch run ≤ 30 s; PWA sync p95 < 2 s per 100 actions; availability 99.9%. Alerts: SLA monitor lag > 2 min; solver fallback rate > 10%; OSRM unavailable. Runbook: re-run dispatch with greedy; replay mobile actions; refresh OSRM extract. Rollback by digest; dispatch runs are versioned and republishable.

8.45.11 Migration from AX / SAP

Source	Passage entity	Notes
SMAServiceObjectTable, SMAServiceObjectRelationTable	ServiceObject	link to inventory serials where found
SMAServiceAgreementTable/Line, SMAAgreementLine	ServiceAgreement, lines	interval lines → preventive visit WOs
SMAServiceOrderTable/Line (open)	ServiceWorkOrder	stage map to status
SMAServiceTechnician	Technician	links to wave-42 workers
D365 FS msdyn_workorder, msdyn_agreement, msdyn_customerasset	ServiceWorkOrder, ServiceAgreement, ServiceObject	via Dataverse export
IBase (IBIN/IBIB/IBST), EQUI with customer	ServiceObject	
BGMK warranties	Warranty	
VBAK/VBAP contracts type WV/SC, SLA profiles	ServiceAgreement, SlaProfile	
QMEL service notifications (open), CS orders (open)	Case, ServiceWorkOrder	
SAP FSM activities (open)	ServiceWorkOrder assignments	via FSM API

Validation: installed-base count per customer equal; open WO count and value equal; active agreement count equal. Not migrated: closed service orders (archive), AX subscriptions (recreated as wave-29 recurring invoices), FSM historical GPS traces.

8.45.12 Dependencies, risks, open questions

Depends on: wave 16 locations (TRUCK), wave 27 pricing, wave 28 serial shipments, wave 29 invoicing, wave 35 notification hub, wave 42 workers/skills, wave 44 condition rules, wave 23 scheduler patterns.

Risk	Likelihood	Impact	Mitigation
Travel matrix accuracy/availability	Medium	Medium	Self-hosted OSRM; cached matrix; straight-line fallback × 1.3
Offline conflict edge cases	Medium	Medium	Append-only actions; replay fixtures
Phase-exit demo breadth	High	High	Walkthrough script frozen at wave start; dry run at week 4

Open questions: (1) Customer self-scheduling portal in v1? Default v1.x; decide by wave 46. (2) Should IoT-originated cases bypass the case stage and create WOs directly? Default create case with auto-conversion for priority 1; decide by wave 45 start.

8.45.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Agreement / case / WO / claim number sequences	legal entity	lookup	SA, CS, SWO, SWC, KB	wave-5 sequences	Identifiers	Service management parameters › Number sequences (SMAParameters)	Service order number ranges (T003O SM01/SM02), notification ranges (TQ80 S1/S2/S3)
Default SLA profile	legal entity	lookup	—	SlaProfile	BR-45.1 fallback	Service management parameters › Service level agreements	SLA profiles (CRM service profile / response profile)
SLA warning threshold	tenant	pct	75 %	50–95 %	BR-45.2 warning event	n/a	Response profile escalation
Pause resolution on WAITING_PARTS	SLA profile	bool	false		BR-45.2	n/a	Status-dependent SLA pause (CRM date profile)
Entitlement match includes parent party	legal entity	bool	true		BR-45.1	Service agreement › Customer hierarchy	Contract determination (VA41 partner functions)
Warranty start trigger default	item group	enum	SHIP	SHIP, INSTALL, REGISTRATION	FR-45.2	Service object › Warranty dates	Warranty start (BGMK master warranty; IE01 warranty tab)
Installed base auto-creation from shipments	legal entity	bool	true		FR-45.1	Service management parameters › Create service objects	IB creation from delivery (IBase user exit / VL02N serial)
Dispatch auto-run time	territory	time	18:00	00:00–23:59	`service-dispatch-auto`	Dispatch board parameters	FSM planning settings
Solver time limit / workers	tenant	int s / int	30 / 8	5–300 / 1–32	BR-45.3	n/a	RSO optimization run settings
Travel buffer minutes	territory	int	5	0–60	BR-45.3	Service management parameters › Travel time	FSM travel time setup
Overtime allowance	technician	int minutes	120	0–240	BR-45.3	Technician setup	FSM workforce rules
Signature required by incident type	incident type	bool	true		8.45.5 validation	Service order › Sign-off (SMASignOff)	Service confirmation completion
Allow negative truck stock	tenant	bool	false		BR-45.7	n/a	Van stock storage location settings (T001L negative stock)
Truck replenishment min/max per item	truck location	numeric	from item	≥ 0	BR-45.7	Warehouse per resource (D365 FS)	MRP area per storage location (OMIZ)
Case category default and reopen window	category	lookup + int days	14	0–365	BR-45.9	Case management › Case categories, Case parameters (Organization administration › Setup › Cases)	Notification type settings (TQ80), CRM case types
Case process per category	category	lookup	—	CaseProcess	BR-45.9	Case processes (CaseProcess)	Notification status profile (BS02) / task lists (QMSM)
Knowledge article review required	tenant	bool	true		BR-45.11	Knowledge article parameters	Solution database settings (IS01 symptom/solution types)

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
Suggestion count and weights	tenant	int + weights	5; 0.6/0.25/0.10/0.05	1–20	BR-45.11	n/a	n/a
Supplier claim window (days)	vendor	int	30	1–365	BR-45.12	n/a	WTY claim types and deadlines (WTY customizing)
Case mailboxes and default categories	legal entity	list	—	email addresses	BR-45.13	Email to case (not native)	CIC email routing (CRM ERMS)

8.45.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Service work order list	list inquiry	territory, status, priority, SLA due, technician, agreement	WO, customer, object, status, window, SLA due, assigned	`svc_wo`	Service orders (SMAServiceOrderTable list)	IW72 change / IW73 display service orders
Case list (any category)	list inquiry	category, stage, owner, SLA state, association	case, category, stage, priority, SLA due, age	`case_list`	Cases list page (CaseDetailListPage); Case inquiries	IW58 / IW59 notifications; CRM case search
Case report / case activities	operational report	category, period	opened, resolved, backlog, mean time to resolve, activities per case	`case_stats`	SSRS Case report / Case activities	IW65 notification activities; IW69 notification items
SLA attainment	KPI	territory, category, priority, period	response %, resolution %, breaches, mean minutes	`svc_sla`	n/a (SSRS service level reports)	CRM SLA analysis
First-time fix and utilization	KPI	technician, territory, period	FTF %, utilization %, travel ratio	`svc_kpi`	Service management reports	FSM analytics
Installed base	drill-down	customer, item, serial, warranty status	object, structure, install date, warranty end, history	`svc_ib`	Service objects (SMAServiceObjectTable)	IB53 IBase display; IH08 equipment with customer
Agreement entitlement usage	list inquiry	agreement, period	included vs consumed visits/hours, revenue	`svc_entitlement`	Service agreements › Lines	IW73 by contract; VA45 contract list
Warranty cost and recovery	KPI	item, vendor, period	warranty cost, claimed, recovered, recovery %	`svc_warranty`	n/a	WTY claim worklist and claim analysis
Supplier claims	list inquiry	vendor, status, age	claim, lines, claimed, accepted, recovered, days open	`svc_claims`	n/a	WTY claim worklist (WTY)
Knowledge article usage	KPI	category, language, period	views, links to cases, solved %, helpful %	`kb_stats`	Knowledge articles (CaseKnowledgeArticle)	IS02 solution usage
My activities	list inquiry	owner, due range, status	activity, related record, due	`activities`	Activities (smmActivities)	CRM activity search
Truck stock	list inquiry	technician, item, below min	on hand, min, max, open transfers	`svc_truck`	On-hand by resource warehouse	MMBE by van storage location

8.45.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Service report (customer-facing summary)	WO COMPLETED	header: customer, site, object, WO; work performed, parts used, labor and travel, coverage per line, photos (optional), signature with timestamp/geo	PDF, email, PWA	SWO	service record; consumer-repair disclosure laws where applicable (e.g. EU right-to-repair directive 2024/1799 information)
Service invoice	WO POSTED	wave-29 invoice with WO reference, BILLABLE lines only	PDF, e-invoice, email	wave-29 `INV`	fiscal invoice
Service agreement	agreement ACTIVE	parties, validity, covered objects, entitlements, SLA targets, pricing, billing model, terms	PDF, e-signature	SA	contract record, 10 years (DE HGB §257)
Case acknowledgment / resolution notice	case created / RESOLVED	case no, category, summary, SLA targets or resolution	email, portal	CS	none
Dispatch schedule (technician day sheet)	schedule published	route order, windows, addresses, required parts	PWA, PDF	dispatch run id	none
Supplier warranty claim	claim SUBMITTED	vendor, claim no, lines (item, serial, failure code, part amount, labor), return reference	PDF, email, EDI (wave 35)	SWC	supporting document for credit note; accounting retention
Knowledge article	on demand	title, applicability, solution steps, version	PDF, portal	KB-{seq} version	none

8.45.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
`service-sla-monitor`	every 60 s	tenant, category	event once per (case, threshold)	100k open cases < 60 s	retry; skip-locked batches	lag > 2 min
`service-dispatch-auto`	18:00 territory-local	territory, date	key per (territory, date, version)	≤ 30 s solve per territory	greedy fallback	fallback rate > 10 %
`service-truck-replenish`	nightly 02:00	truck location	key (location, date)	< 5 min	retry	below-min items without transfer
`service-ib-from-shipments`	event-driven	shipment	key (shipment line, serial)	< 60 s per shipment	retry from offset	serial without object after 5 min
`service-agreement-visits`	daily 03:00	agreement	key (agreement line, visit date)	< 5 min	retry	missed preventive visit WOs
`case-email-ingest`	every 1 min (wave-35 mailbox poll or push)	mailbox	Message-ID unique	< 30 s per batch	quarantine parse failures	mailbox backlog > 100
`case-consolidate` (one-off per tenant at cut-in)	on demand	source entity	key (source id)	< 30 min for 100k	resumable	unmapped source cases
`supplier-claim-deadline`	daily 07:00	vendor	key (claim, day)	< 1 min	retry	drafts within 5 days of window end
`kb-reindex`	on publish; nightly verify	language	article version	< 10 s per article	retry	index lag > 5 min

8.45.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
SvcAgreementMaintain	agreements create/change; Agreement detail	with SvcBillingPost for the same customer	Service contract manager
SvcCaseCreate / SvcCaseMaintain	service cases create, transitions	none	Service agent
CaseCreate / CaseMaintain (category-scoped)	`/api/v1/cases`, stage, associations; case list, mailbox	per-category OPA policy (BR-45.10)	Agent of the owning team (HR partner, collector, auditor)
CaseSetupAdmin	categories, processes, mailboxes, SLA profiles	with CaseMaintain on HR categories (setup admin cannot read HR cases by default)	Case administrator
SvcWoMaintain / SvcDispatch	WOs, dispatch runs, publish	none	Service coordinator, Dispatcher
SvcTechnician	mobile action log, parts, signatures	none	Field technician
SvcBillingPost	WO post to invoicing	with SvcAgreementMaintain	Service billing clerk
SvcClaimMaintain	supplier claims create, submit, decision	with AP credit note posting (wave 18) for the same claim	Warranty administrator
KbRead / KbAuthor / KbPublish	search, author, publish	author ≠ publisher (BR-45.11)	All users / Senior agent / Knowledge manager
ActivityMaintain	activities on readable records	none	All internal users
SvcInquire	KPIs, lists, reports	none	Service manager

8.45.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
SVC-ENT-001	info	No agreement matched; default SLA {profile} applied, lines billable	Entitlement	Check agreements	BR-45.1
SVC-SLA-001	warning	SLA {type} at {pct} % for case {case}	Warning threshold	Prioritize	BR-45.2
SVC-SLA-002	error	SLA {type} breached for {case}	Breach	Escalate	BR-45.2
SVC-DSP-001	error	Work order {wo} lacks geo or skills; cannot optimize	Precondition	Complete address/skills	8.45.5 validation
SVC-DSP-002	warning	Solver fell back to greedy for {territory} {date}	Time limit	Review assignments	BR-45.3
SVC-DSP-003	error	Technician {t} lacks skill {s} or certification expired	Hard constraint on drag	Choose another technician	BR-45.3
SVC-WO-001	error	Signature required to complete	Incident type rule	Capture signature	8.45.5 validation
SVC-WO-002	error	Parts lines not consumed; cannot post	Post precondition	Consume or remove parts	8.45.5 validation
SVC-TRK-001	error	Truck stock for {item} would go negative	BR-45.7	Transfer stock or use reservation	BR-45.7
SVC-SYNC-001	warning	{n} offline actions merged; {m} status conflicts resolved to {state}	Offline merge	Review timeline	BR-45.6
CASE-STG-001	error	Stage {stage} exit criteria not met: {reasons}	Process gate	Complete required activities	BR-45.9
CASE-SEC-001	error	You do not have access to cases in category {category}	Category security	Request access	BR-45.10
CASE-RES-001	error	Resolution code required	RESOLVED transition	Pick a code	BR-45.9
CASE-MAIL-001	info	Email attached to case {case}	Threading	None	BR-45.13
KB-PUB-001	error	Reviewer must differ from author	SoD	Assign another reviewer	BR-45.11
KB-PUB-002	warning	Publishing retires version {v}	Versioning	None	BR-45.11
SVC-SWC-001	error	Claim window for vendor {v} ended on {date}	Deadline	Request vendor exception	BR-45.12
SVC-SWC-002	error	Failed-part return required before submission	Vendor flag	Create return	BR-45.12
SVC-SWC-003	error	Accepted amount exceeds claimed amount	Validation	Correct decision	BR-45.12

8.45.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation behavior under 3× load
Case create (entitlement + SLA)	5,000/day; peak 100/min	100 / 300 / 600 ms	20/s	Case + log ≈ 5 KB × 1.8 M = 9 GB	(category_id, status), (sla_state, sla_resolution_due)	entitlement result cached per (customer, object) 5 min
SLA monitor scan	100k open cases	< 60 s per cycle	—	n/a	partial index on RUNNING	cycle extends; warning events late by ≤ 1 cycle
Dispatch optimization	200 WOs × 25 technicians per territory-day; 50 territories	≤ 30 s per territory	8 territories in parallel	DispatchRun ≈ 1 GB	(territory_id, date)	greedy fallback; staggered start times
Mobile sync	1,000 technicians × 200 actions/day	2 s p95 per 100 actions	500 actions/s	action log ≈ 40 GB incl. photos refs	(work_order_id, action_id)	queue; device retries with backoff
Knowledge search	20,000/day; peak 50/s	50 / 150 / 300 ms	50 q/s	articles ≈ 50k × 20 KB = 1 GB	GIN on tsvector per language	return top 5 from title-only index
Email ingest	10,000 messages/day	< 30 s to case	20 msg/s	attachments ≈ 200 GB object store	Message-ID unique	backlog processed in order; ack delayed
Supplier claim lifecycle	2,000/month	100 / 300 / 600 ms	5/s	≈ 100 MB	(vendor_party_id, status)	none required

8.45.20 Localization and compliance hooks

- Consumer warranty law: EU Directive 2019/771 (2-year legal guarantee, 1-year burden-of-proof reversal), DE BGB §§437–439, UK Consumer Rights Act 2015, US Magnuson-Moss Warranty Act; legal minimum coverage is a pack-provided warranty template that `Warranty` cannot undercut for consumer customers; extension point `service.warranty.minimum.<cc>`.
- Right to repair: EU Directive 2024/1799 repair information obligations reflected in the service report fields (repair price, parts availability) when the pack enables them.
- SLA calendars: business hours and public holidays per service site from wave-8 calendars; time zone of the service site, DST-safe arithmetic (8.45.9 property test).
- Privacy: technician location (geo on signatures, travel traces) is personal data; retention 90 days for traces by default, works-council approval flag (DE BetrVG §87(1) 6) before enabling live tracking; HR-category cases inherit HR retention and `hr_pii` protection; email intake honors data-subject requests by purging

InboundMessage bodies with the case.

- Language: case acknowledgments, service reports and knowledge articles per customer language (articles are per-language versions; fallback to English).
- E-invoicing for service invoices follows wave 29 per country; supplier claims to EU vendors can be sent as EN 16931 credit requests when the vendor supports it (pack data).
- Tax: warranty replacements free of charge are non-taxable supplies in most jurisdictions, but recovered supplier claim amounts follow the vendor's tax treatment (wave 13 tax codes on the credit note).

8.45.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX #233 — 12.29.1 Cases list page / case details (CaseDetailListPage / CaseDetail)	PARTIAL	ADDED as FR-45.12	Categories, processes, associations, log, resolution
AX #234 — 12.29.1 Cases → Knowledge articles (CaseKnowledgeArticle)	PARTIAL	ADDED as FR-45.13	CaseArticleLink
AX #241 — 12.29.1 Case processes / case categories (CaseProcess, CaseCategory)	PARTIAL	ADDED as FR-45.12	BR-45.9
AX #242 — 12.29.2 Case inquiries (by category/status, SLA)	PARTIAL	ADDED as FR-45.12	"Case list (any category)", "SLA attainment"
AX #244 — 12.29.2 SSRS Case report / Case activities	PARTIAL	ADDED as FR-45.12	"Case report / case activities"
AX #248 — 12.29.3 Close cases / case SLA breach notifications	PARTIAL	ADDED as FR-45.12	`service-sla-monitor` extended to all categories; BR-45.9 close rules
AX #254 — 12.29.4 Case categories / category types / processes / category security	PARTIAL	ADDED as FR-45.12	BR-45.10 OPA per category
AX #271 — 12.31.1 Cases (My cases)	PARTIAL	ADDED as FR-45.12	Screen "My cases / case list"
AX #275 — 12.31.4 Cases → Case setup (link)	PARTIAL	ADDED as FR-45.12	Screen "Case category and process setup"
AX #108 — 12.12.1 Human resources → Cases (HR cases)	PARTIAL	ADDED as FR-45.12	Category type HR with `hr_pii` protection (BR-45.10)
AX #92 — 12.11.1 Project → Activities (smmActivities)	PARTIAL	ADDED as FR-45.14	Activities associate to projects (wave 41)
AX #227 — 12.28.1 Service subscriptions (SMASubscriptionTable, fee transactions, index)	PARTIAL (wave 29)	SUPERSEDED by wave-29 subscription billing	Agreement billing_model SUBSCRIPTION drives wave-29 recurring invoices; fee indexation uses wave-29 price escalation
AX #228 — 12.28.2 Subscription inquiries	PARTIAL (wave 29)	SUPERSEDED by wave-29 subscription billing	wave-29 subscription inquiries filtered by agreement
AX #229 — 12.28.2 SSRS Subscription report / fee transactions	PARTIAL (wave 29)	SUPERSEDED by wave-29 subscription billing	
AX #230 — 12.28.3 Create subscription fee transactions / accrue subscription revenue / index subscriptions	PARTIAL (wave 29)	SUPERSEDED by wave-29 subscription billing	Revenue accrual by wave-29 revenue recognition schedules
AX #231 — 12.28.4 Subscription groups, fee categories, index rules, periods	PARTIAL (wave 29)	SUPERSEDED by wave-29 subscription billing	
SAP #303 — 13.26 Warranty claim processing (WTY1, vendor recovery)	PARTIAL-45	ADDED as FR-45.15	BR-45.12; dealer-to-OEM claims (automotive WTY inbound) OUT-OF-SCOPE
SAP #307 — 13.26 SPRO warranty claim processing (WTY)	PARTIAL-45	ADDED as FR-45.15	Parameter "Supplier claim window"
SAP #304 — 13.26 Solution database IS01/IS02 (knowledge base)	GAP	ADDED as FR-45.13	BR-45.11; wave-47 copilot consumes published articles
SAP #308 — 13.26 Solution database / knowledge base settings	GAP	ADDED as FR-45.13	Parameters "Knowledge article review required", "Suggestion count and weights"
SAP #305 — 13.26 CIC0 / CRM Service / S/4 Service front office	PARTIAL-45	ADDED as FR-45.16	Email-to-case; telephony/CTI OUT-OF-SCOPE (CRM non-goal); service quotations via wave-27 quotations linked to cases
SAP #309 — 13.26 CIC / contact center framework	PARTIAL-45	OUT-OF-SCOPE CRM/contact-center non-goal	Mailbox routing covered by FR-45.16; CTI, agent desktop and channel routing are external (integrate via wave 35)

Part 9 — Phase 9: Intelligence, Compliance & Go-Live (waves 46–50)

9.0 Phase overview

Business outcome at phase exit. A manufacturer runs its entire business on Passage in production, off AX/SAP, with the vendor dependencies of the legacy systems removed: analytics no longer needs Entity store, BYOD or BW extractors because every domain streams into one governed lakehouse; users ask questions and delegate work to agents that act only within their own permissions; the tenant runs legally in seven countries with statutory outputs validated against regulator schemas; segregation of duties, SOC 2 evidence and privacy requests are system functions rather than spreadsheets; the customer's open items, balances and history were moved by a migration factory that reconciled every control total; and the platform has demonstrated 10k concurrent users, a region failover and a reconciled production month-end. v1.0 is tagged, signed and published.

Phase-exit sign-off criteria (all measured in the production-candidate environment unless stated)

- Lakehouse freshness: p95 OLTP-commit-to-dashboard-visible lag < 60 s while the wave-50 load profile runs.
- Semantic-layer `revenue` equals GL revenue (universal journal, account category REVENUE) to the cent for 30 consecutive days; `inventory_value` equals the inventory subledger within USD 0.01 / EUR 0.01.
- Agent evals in CI: AP coding ≥ 92% top-1 accuracy, cash application ≥ 90% precision; permission-inheritance contract tests 100% pass; per-tenant token budget enforced with an alert fired in a test.
- Seven country packs (US, UK, DE, FR, IN, MX, CA) each ship a statutory-report fixture that validates against the regulator's XSD/JSON schema in CI.
- 50 planted SoD conflicts detected (0 misses); mitigation workflow records approver, reason and expiry.
- Backup/restore drill: tenant restored to a point in time with RPO ≤ 5 min and RTO ≤ 1 h; external penetration test shows zero open high/critical findings.
- Synthetic AX and SAP datasets (full year, 3 entities each) migrate with 100% match on all control totals; a parallel-run month shows zero unexplained variances; cutover rehearsal < 48 h; rollback rehearsal restores legacy coexistence.
- Load test: 10k concurrent users, 1M journal lines/h, 100k order lines/h with p99 within the SLO table in 9.50.10.
- DR game day: region failover RTO ≤ 1 h, RPO ≤ 5 min, zero committed-transaction loss (ledger hash chain verified).
- First customer closes a full production month-end with all subledgers reconciled to the journal; hypercare exit criteria (9.50.1) met.
- v1.0 tag signed (cosign), SBOMs attached, release notes and documentation published, v2 roadmap opened.

Cross-wave data model summary

Entity	Owning service	Key relationships
`lake_table` / `cdc_pipeline_state`	analytics-ingest	1 per (domain, table); references schema-registry subject
`semantic_metric`, `semantic_model`	semantic (Cube)	metric → model → lake tables; metric → `reconciliation_check`
`dashboard`, `dashboard_tile`, `report_schedule`	analytics-ui	tile → metric; schedule → dashboard or export
`document_template`, `document_render`	docgen	template → source entity type (SalesInvoice, PurchOrder, PackingSlip, Check)
`tool_definition`, `agent`, `agent_run`, `agent_action`, `approval_task`	mcp-gateway, agent-runtime	run → agent version; action → tool_definition; approval → Temporal task
`eval_dataset`, `eval_run`, `prompt_version`, `token_budget`	agent-evals	eval_run → agent version; budget → tenant
`country_pack`, `er_format`, `er_mapping`, `mandate`	localization, ereporting	format → mandate → country; pack → legal entity
`sod_rule`, `sod_conflict`, `mitigation`, `certification_campaign`	sod	rule → (duty_a, duty_b); conflict → user, rule; mitigation → conflict
`control`, `control_evidence`, `subject_request`, `retention_policy`, `legal_hold`	compliance, privacy	evidence → control; hold overrides policy
`migration_pack`, `pack_run`, `control_total`, `legacy_xref`, `parallel_run`, `variance`	migration-factory	run → pack; variance → parallel_run; xref → every migrated entity
`cutover_plan`, `cutover_step`, `go_nogo_item`	cutover	step → plan; item → step, owner
`region`, `failover_event`, `capacity_model`, `tenant_cost`	platform-ops	failover → region pair; cost → tenant, month
`print_rule`, `text_block`, `correspondence_type`, `dq_rule`, `archive_partition`	docgen, catalog, analytics-ingest	rule → document type, account; archive partition → retention policy (wave 48)
`esign_requirement`, `esign_record`, `audit_policy`, `audit_violation`, `audit_case`, `e_message`	compliance, ereporting	esign record → record version hash; violation → document, case; message → er_format run
`screening_list`, `screening_hit`, `trade_classification`, `export_license`, `elevated_session`, `security_incident`	trade-compliance, security	hit → party/document (blocks release); incident → breach regimes

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
`dev.passage.analytics.snapshot.committed.v1`	analytics-ingest	semantic, catalog	table, snapshot_id, row_delta, max_source_lsn, watermark
`dev.passage.analytics.metric.reconciled.v1`	semantic	compliance, alerting	metric_id, date, semantic_value, source_value, delta, status
`dev.passage.agent.run.completed.v1`	agent-runtime	audit, copilot-ui, evals	run_id, agent, version, tool_calls[], tokens, outcome
`dev.passage.agent.approval.requested.v1`	agent-runtime	workflow inbox	run_id, action_id, proposed_change, risk_score
`dev.passage.ereporting.document.submitted.v1`	ereporting	AR/AP, compliance	mandate, format_version, document_ref, regulator_ack
`dev.passage.sod.conflict.detected.v1`	sod	compliance, workflow	user_id, rule_id, roles[], severity
`dev.passage.privacy.request.completed.v1`	privacy	audit	request_id, type, subject_ref, affected_domains[]
`dev.passage.migration.pack.reconciled.v1`	migration-factory	cutover	run_id, pack, control_totals[], status
`dev.passage.migration.parallelrun.diffed.v1`	migration-factory	cutover	run_id, period, variances_unexplained, variances_explained
`dev.passage.ops.failover.completed.v1`	platform-ops	all services (fencing)	from_region, to_region, promoted_lsn, rpo_seconds, rto_seconds
`dev.passage.documents.print.routed.v1`	docgen	print agent, audit	render_id, rule_id, destinations[]
`dev.passage.compliance.esign.signed.v1`	compliance	quality, production, audit	esign_id, record, meaning, signer
`dev.passage.trade.screening.hit.v1`	trade-compliance	sales, procurement, payments	subject, list_code, score
`dev.passage.security.incident.breach_confirmed.v1`	security	privacy, notifications	incident_id, regimes[], deadlines[]

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Lakehouse metrics drift from GL (duplicate CDC, late data)	Medium	High	Daily reconciliation job with hard fail; equality-delete upserts keyed on LSN	Analytics lead
Agent takes an action outside user permission	Low	Critical	Tool calls carry the user's token; OPA evaluated at the API, never in the agent; contract tests per tool	Security lead
Country-pack legal error (wrong tax box, wrong XML)	Medium	High	Regulator schema fixtures in CI; provider adapters (Avalara, Sovos) as fallback; packs behind flags	Localization lead
Migration reconciliation hides netting errors	Medium	High	Control totals at line grain, not just sums (counts, hashes, by-dimension)	Migration lead
Cutover window overrun	Medium	High	Two full rehearsals; runbook timed per step; abort criteria at T+36 h	Cutover manager
Region failover loses committed journal entries	Low	Critical	Synchronous quorum replication for ledger Postgres; hash-chain verification post-failover	SRE lead
Two-person team cannot staff hypercare	High	High	Implementation partner on-call rota; agents in suggest-mode triage support tickets	Product

9.46 Wave 46: Data Platform, Analytics & Embedded BI

9.46.1 Scope and business context

AX customers build analytics on Entity store aggregate measurements refreshed by batch, BYOD exports to Azure SQL and Power BI embedded workspaces; SSRS runs document layouts through `SrsReportDataProvider` classes. SAP customers extract into BW/4HANA InfoProviders via ODP, expose `@Analytics.dataCategory: #CUBE` CDS views and embed SAC stories. Both put analytical load on OLTP or on a copy that is hours stale. This wave gives every Passage domain a CDC path into an Apache Iceberg lakehouse, a Cube semantic layer with governed metrics whose row-level security mirrors OPA, embedded dashboards in each workspace, a self-service explorer, scheduled reports, an operational document designer that replaces SSRS layouts, and OpenMetadata for catalog and lineage. At wave exit a controller sees revenue on a dashboard within 60 s of posting and knows it equals the GL. Out of scope: natural-language querying (wave 47), historical legacy data (wave 49), customer-built ML models.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Entity store refresh (`AggregateMeasurement`, batch job "Refresh entity store") runs hourly at best; Power BI tiles show numbers up to 4 h old and controllers re-query the GL to confirm	Batch copy from OLTP into a separate star schema; no change feed	Debezium CDC → Iceberg → Cube; freshness badge on every tile	Latency 4 h batch → p95 < 60 s
Revenue on a dashboard and revenue on the trial balance disagree; nobody knows which is right (BYOD export gaps, BW extractor delta failures in `RSA7`)	Export jobs drop deltas silently; no reconciliation step	Daily automated metric ↔ GL reconciliation with hard fail and banner	Reconciliation effort 2 person-days/month → 0; delta tolerance 0.005
Changing an SSRS invoice layout needs a developer: edit `.rdl`, extend `SrsReportDataProvider`, build, deploy through LCS; SAP needs `SE71`/`SFP` ABAP skills	Layout bound to code artifacts in a deployable package	Metadata document designer with fixture preview and visual regression	Turnaround 2 weeks → same day; 0 code deploys
Building one BW query or SSAS cube perspective for a new KPI takes an analyst through `RSA1`, transformations, DTPs and authorizations (`RSECADMIN`)	Separate security and modeling stack from the application	Metric registry; RLS compiled from the same OPA policies as OLTP	New governed metric 10 days → 1 h; 0 separate security objects
Exporting to Excel from a data entity (`Open in Office`) caps at 10k rows or times out; users build shadow spreadsheets	OLTP OData serving analytical loads	Explorer exports via Trino as Temporal jobs	1M rows in < 10 min; OLTP load from reporting 0

9.46.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Analytical store	Entity store, `AggregateMeasurement` (e.g. `LedgerActivityMeasure`), batch refresh	BW/4HANA ADSO, CompositeProvider, ODP extractors (`RSA1`)	Debezium CDC → Iceberg `raw.*`, dbt `core.*`/`mart.*`	Improved (streaming, no batch refresh)
Data export for BI	BYOD (`DMFDataEntity` export to Azure SQL), Export to Data Lake	SLT, Datasphere replication flows	Iceberg tables queryable via Trino; CDC is the export	Improved
Embedded dashboards	Power BI embedded workspaces (`PBIWorkspace`, `SysPBIReportController`)	SAC stories in Fiori launchpad, Overview pages	`<DashboardHost>` web component per workspace	Full
Operational documents	SSRS `SalesInvoice.Report`, `PurchPurchaseOrder.Report`, `SrsReportDataProvider`, Print management	SAPscript (`SE71`), Smart Forms, Adobe Forms (`SFP`), `NACE` output	Document designer, `document_template` with print-management rules	Full
OLAP cubes	SSAS cubes (`SalesCube`, `GeneralLedgerCube`), perspectives	BEx queries (`RSRT`), BW queries	Cube semantic models and measures	Improved
Embedded analytical views	Data entities with `IsAggregateDataEntity`	CDS `@Analytics.dataCategory #CUBE`, `@Analytics.query`	Cube `cubes/views` generated from entity metadata	Full
Self-service / Excel	Open in Excel (`DataEntity` + OData), Power BI Desktop over BYOD	Analysis for Office, Query Browser (`/UI2/QUERY_BROWSER`)	Explorer; export to XLSX, Power BI (ODBC via Trino), Tableau	Full
Row-level security	XDS policies, Power BI RLS roles	Analysis authorizations (`RSECADMIN`)	OPA policy compiled to Cube `queryRewrite` filters	Full
Financial statements	Management Reporter	FSV (`FSE2`), Fiori financial statement apps	Wave-15 statement designer; marts feed trend tiles	Full
Catalog and lineage	None (AOT only)	BW metadata repository, Datasphere catalog	OpenMetadata with Debezium/dbt/Cube lineage ingestion	Improved
Print management routing	Print management setup (`PrintMgmtSettings`, `PrintMgmtDocInstance`, conditional settings by query, original/copy destinations), Form setup, Form sorting	`NACE` output determination (condition tables, access sequences, `NAST`), S/4 Output Parameter Determination (BRF+ `OPD`), `SPAD` printers	`print_rule` with condition expressions, original/copy destinations and print agent (FR-46.14)	Full
Form notes and standard texts	Form notes (`FormLetterRemarks`), document notes (`DocuRef`) printed per document type	`SO10` standard texts, text IDs (`SE75`), text determination (`VOTXN`)	`text_block` with placeholders and language versions (FR-46.15)	Full
Correspondence	Customer/vendor account statements (`CustAccountStatementExt`), payment advice	`F.27` periodic statements, `F.64` correspondence (`OB77` types, `T048`), `SAP01`–`SAP19`	`correspondence_type` with trigger and template (FR-46.15)	Full
Data quality monitoring	Data validation checklist (DMF), none for production data	Information Steward rules and scorecards	`dq_rule` on catalog assets with scorecard tile (FR-46.16)	Full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Process mining	None	Signavio Process Intelligence, Process Observer (`POC`), Process AI	`mart.process_event` event log, variant explorer, OCEL/XES export (FR-46.17)	Partial (no conformance engine in v1)
Data archiving	D365 Archive (long-term retention via Dataverse), `SysDatabaseLog` cleanup	`SARA` archiving objects, `SARI` archive info system, ILM retention warehouse	Archive tier `archive.<domain>.<table>` in Iceberg driven by wave-48 residence times (FR-46.18)	Full

9.46.3 Functional requirements

- **FR-46.1 CDC from every domain** — Must. The system shall run one Debezium Postgres connector per domain database using `pgoutput`, publishing to topics `cdc.<domain>.<table>` keyed on `(tenant_id, primary key)`, Avro-encoded with schema-registry compatibility `BACKWARD_TRANSITIVE`.
    - Notes: every table listed in entity metadata with `analytics: true` is captured; `REPLICA IDENTITY FULL` enforced by the Atlas migration generator; snapshot mode `initial` on first run then `never`; heartbeat every 10 s to advance slots on idle domains; slot lag > 1 GB alerts.
- **FR-46.2 Iceberg landing contract** — Must. The system shall write each CDC topic into Iceberg table `raw.<domain>.<table>` (format v2, merge-on-read, equality deletes on the key) partitioned by `days(_source_ts)` and `bucket(16, tenant_id)`.
    - Notes: every row carries `_op` (c/u/d), `_source_lsn`, `_source_ts`, `_ingested_at`; commits every 30 s or 50 MB, whichever first; snapshot expiry 7 days, orphan-file cleanup daily; compaction rewrites files < 64 MB nightly.
- **FR-46.3 Schema evolution** — Must. The system shall apply additive source changes automatically and quarantine incompatible ones.
    - Notes: add column → add nullable Iceberg column; type widening (`int→bigint`, `decimal(18,4)→decimal(28,6)`) applied; dropped column retained and tagged `deprecated_since`; rename → new column plus `renamed_from` property in catalog; narrowing or NOT NULL additions pause the sink, route to `cdc.dlq.<domain>` and alert within 60 s.
- **FR-46.4 Late-arriving data and exactly-once effect** — Must. The system shall produce exactly one logical state per key regardless of Kafka redelivery.
    - Notes: sink deduplicates on `(tenant_id, pk, _source_lsn)`; Iceberg commit and Kafka offset are written atomically by the sink coordinator (control topic `iceberg.control`), so replay after crash is idempotent; watermark = max(`_source_ts`) − 5 min; rows with `_source_ts` older than the watermark are merged into closed partitions; dbt incremental models use a 3-day lookback; a row arriving more than 30 days late is written and flagged `late=true` for reconciliation.
- **FR-46.5 dbt transformation layers** — Must. The system shall maintain `core.*` (deduplicated current-state and history tables, SCD2 on master data) and `mart.*` (finance, sales, procurement, inventory, production, projects, HR) dbt models with tests on every primary key, every FK and every reconciled measure.
    - Notes: `mart.fin_journal_line` is the only source for monetary metrics; models run every 60 s incremental, nightly full for SCD2 rebuild; dbt test failure blocks downstream models and raises `metric.reconciled` with status `BLOCKED`.
- **FR-46.6 Governed metrics** — Must. The system shall define metrics in the semantic layer with owner, version, grain, allowed dimensions and formula; the initial set is:
    - `revenue` = `-SUM(amount_reporting) FROM mart.fin_journal_line WHERE account_category='REVENUE' AND status='POSTED'` (credit shown positive).
    - `cogs` = `SUM(amount_reporting) WHERE account_category='COGS'`; `gross_margin` = `revenue − cogs`; `gross_margin_pct` = `gross_margin / NULLIF(revenue,0)`.
    - `dso_days` = `ar_open_balance_at(end) / NULLIF(revenue_trailing(90d),0) * 90`; countback variant selectable per tenant.
    - `dpo_days` = `ap_open_balance_at(end) / NULLIF(cogs_trailing(90d),0) * 90`.
    - `inventory_turns` = `cogs_trailing(365d) / NULLIF(AVG(inventory_value_eom over last 13 month-ends),0)`.
    - `otif_pct` = `COUNT(sales_line WHERE shipped_ts <= requested_ship_date AND qty_shipped = qty_ordered) / NULLIF(COUNT(sales_line shipped),0)`.
    - `oee_pct` = `availability * performance * quality` where `availability = run_time / planned_time`, `performance = (ideal_cycle_time * total_count) / run_time`, `quality = good_count / total_count`, sourced from `mart.prod_resource_shift`.
- **FR-46.7 Row-level security mirrored from OPA** — Must. The system shall compile the user's OPA data-scope decision (legal entities, sites, dimension ranges) into Cube `queryRewrite` filters on every query, using the same Keycloak token as the OLTP APIs.
    - Notes: cache keyed by `(user_id, policy_bundle_version)` for 5 min; a query touching a model with no scope mapping is rejected (`403 SCOPE_UNMAPPED`); aggregations never bypass filters (pre-aggregations are built per security context).
- **FR-46.8 Embedded dashboards** — Must. The system shall render a dashboard host in every workspace with tiles bound to semantic metrics, auto-refresh ≤ 60 s, and drill-through to the OLTP record.
- **FR-46.9 Self-service explorer and export** — Should. The system shall let users compose metric × dimension × filter queries, save them, and export to XLSX (≤ 1M rows), CSV, Power BI (Trino ODBC/JDBC connection string) and Tableau (same).
    - Notes: exports > 100k rows run as Temporal jobs and deliver via signed URL valid 24 h; every export is audit-logged with row count and filter.
- **FR-46.10 Scheduled reports** — Should. The system shall schedule dashboards and explorer queries (cron, time-zone aware) with delivery to email, SFTP or in-app inbox, PDF/XLSX, retry 3× with backoff.
- **FR-46.11 Operational document designer** — Must. The system shall provide a layout designer (bands, repeaters, barcodes, conditional sections, locale formats) for SalesInvoice, PurchOrder, PackingSlip, Check, Statement, PickList and Remittance with versioned templates and print-management rules (per entity, per customer/vendor, per destination).
    - Notes: render via headless Chromium to PDF/A-3; data source is the document's REST read model, not the lakehouse; p95 render < 800 ms for a 50-line invoice.
- **FR-46.12 Catalog and lineage** — Should. The system shall register every Postgres table, Iceberg table, dbt model, Cube metric and dashboard in OpenMetadata with column-level lineage and owner, refreshed hourly.

- **FR-46.13 Daily metric reconciliation** — Must. The system shall compare `revenue`, `cogs`, `ar_open_balance`, `ap_open_balance`, `inventory_value` against the universal journal and subledgers at 02:00 tenant local time and fail the dashboard with a banner if ldeltal > 0.005.
- **FR-46.14 Print management routing** — Must. The system shall resolve, for every document render, the destinations of the ORIGINAL and each COPY from `print_rule` sets maintained per (document type, legal entity), with account-level overrides (customer, vendor, project, bank) and conditional rules evaluated by priority.
    - Notes: destinations are PRINTER (IPP queue reached through the site print agent), EMAIL (template, recipients from party contact roles such as `INVOICE_TO`, `REMIT_TO`), E_INVOICE (wave-35 channel), ARCHIVE_ONLY, PORTAL, NONE; resolution order is account override → first matching conditional rule → legal-entity default → tenant default; conditions use the wave-3 expression subset (FR-3.3) over the document read model, or reference a FR-3.13 decision table by code (`rules.eval`) (e.g. `amount_total >= 10000 AND customer.group = 'KEY'`); a rule may override the template version, copy count (0–9) and the caption printed per copy (`ORIGINAL`, `COPY 1`, `FILE COPY`); the resolved destination is shown on the posting dialog and recorded on `document_render.destinations_json`; the print agent is a Go sidecar per site polling `/api/v1/documents/print-queue`, buffering 24 h offline and acknowledging each job with the printer's job id.
- **FR-46.15 Standard text blocks and correspondence** — Should. The system shall maintain `text_block` (id, language versions, placeholders such as `{{customer.name}}`, `{{document.due_date|date}}`), form notes per (document type, legal entity, position header/footer), and `correspondence_type` definitions (periodic account statement, payment notice, balance confirmation, dunning cover letter, remittance cover) with trigger (schedule, event, manual), selection (party group, open-item age) and template.
    - Notes: placeholders are resolved against the document read model; a missing placeholder renders empty and raises `DOC-0407`; periodic statements run on the wave-46 schedule engine (default last calendar day, 20:00 local); correspondence output follows FR-46.14 routing; text blocks are exported with country packs for statutory wording (wave 48).
- **FR-46.16 Data-quality rules** — Should. The system shall evaluate `dq_rule` definitions (completeness: null share per column; validity: regex or domain list; uniqueness; referential; freshness: max age of `_ingested_at`) against catalog assets every 60 min, compute a score per entity (0–100), and show a scorecard tile with drill-down to failing rows.
    - Notes: rules compile to dbt tests or Trino queries; a rule has an owner, a threshold (default: fail below 98% pass rate) and a severity; failures create a task in the owner's wave-4 inbox once per rule per day; scores are stored in `dq_score` (asset, rule, date, pass_pct, failed_rows) and are exposed as Cube measures.
- **FR-46.17 Process event log and mining views** — Could. The system shall build `mart.process_event` (case_id, case_type, activity, event_ts, actor, resource, amount, source_event_type) from the CloudEvents stream, chaining documents into cases (P2P: requisition → PO → receipt → invoice → payment; O2C: quote → order → shipment → invoice → cash; production: order → release → RAF → close), and provide a variant explorer tile (top 20 variants, median and p90 cycle time per step, rework count, automation share) with export to OCEL 2.0 JSON and XES.
    - Notes: case stitching uses document reference fields, not heuristics; events older than 24 months are rolled up; conformance checking against a reference model is not in v1 (external tools consume the export).
- **FR-46.18 Archive tier** — Should. The system shall move closed OLTP partitions (journal lines, inventory transactions, audit, workflow history, agent runs) whose age exceeds the wave-48 `retention_policy.residence_months` (default 24) to Iceberg `archive.<domain>.<table>` in a Temporal job, verify row count and content hash per partition, then detach and drop the OLTP partition.
    - Notes: records under legal hold or in an open fiscal period are never moved; archived records stay readable from the record detail (badge `ARCHIVED`) and in explorer with `source=archive`; a partition can be restored into a side table for 30 days on `ArchiveAdmin` request; archive moves are evidence for wave-48 control CC6.5 (data disposal).

9.46.4 Data model

`semantic_metric` (semantic service, Postgres `analytics`)

Field	Type	Required	Notes
metric_id	text	Y	PK, e.g. `revenue`
version	int	Y	monotonically increasing; (metric_id, version) unique
formula	text	Y	Cube measure SQL
grain	text	Y	`journal_line`, `sales_line`, `resource_shift`
allowed_dimensions	text[]	Y	from `semantic_model.dimensions`
owner_user_id	uuid	Y	FK users
status	enum	Y	DRAFT, PUBLISHED, DEPRECATED
reconciliation_source	text	N	e.g. `ledger.balance_by_account`
tolerance	numeric(18,6)	N	default 0.005

`cdc_pipeline_state` (analytics-ingest)

Field	Type	Required	Notes
domain	text	Y	PK part
table_name	text	Y	PK part
schema_version	int	Y	registry version last applied
last_lsn	bigint	Y	last committed source LSN
last_snapshot_id	bigint	Y	Iceberg snapshot
watermark_ts	timestamptz	Y	
status	enum	Y	RUNNING, PAUSED_SCHEMA, PAUSED_ERROR
dlq_count_24h	int	Y	default 0

`document_template` (docgen)

Field	Type	Required	Notes
template_id	uuid	Y	PK
tenant_id	uuid	Y	RLS
document_type	enum	Y	SalesInvoice, PurchOrder, PackingSlip, Check, Statement, PickList, Remittance
version	int	Y	unique with template_id
locale	text	Y	BCP 47, e.g. `de-DE`
layout_json	jsonb	Y	band tree; max 512 KB
status	enum	Y	DRAFT, PUBLISHED, RETIRED
visual_hash	text	N	perceptual hash of fixture render
published_at	timestamptz	N	

`report_schedule`: schedule_id, tenant_id, target_type (DASHBOARD/QUERY), target_id, cron, timezone, format, destination_json, last_run_at, next_run_at, failure_count.

Invariants: a PUBLISHED metric is immutable (changes create a new version); every mart measure has exactly one reconciliation source or is tagged `non_financial`; a template can be PUBLISHED only if its visual regression delta ≤ 0.1%; `last_lsn` never decreases. Indexing: `cdc_pipeline_state` PK (domain, table_name); Iceberg partition spec as FR-46.2 plus sort order `(tenant_id, _source_lsn)`; `document_template` index (tenant_id, document_type, status).

Additional entities (FR-46.14–46.18). `print_rule`: rule_id, tenant_id, legal_entity_id, document_type, scope (DEFAULT, ACCOUNT, CONDITIONAL), account_ref (party or project), priority, condition_expr (wave-3 expression subset or `rules.eval` reference, ≤ 2 KB), destinations_json (per ORIGINAL/COPY n: type, target, template_version, caption), copies int default 1, status (ACTIVE, RETIRED), version. `text_block`: block_id, tenant_id, code (unique per tenant), language (BCP 47), body (≤ 16 KB), placeholders_json, status. `correspondence_type`: type_id, tenant_id, code, trigger (SCHEDULE, EVENT, MANUAL), cron, selection_json, template_id, status. `dq_rule`: rule_id, asset_fqn, rule_type (COMPLETENESS, VALIDITY, UNIQUENESS, REFERENTIAL, FRESHNESS), expression, threshold_pct numeric(5,2) default 98, severity, owner_user_id, status. `archive_partition`: partition_id, domain, table_name, period, rows_oltp, rows_archive, content_hash, moved_at, status (PENDING, VERIFIED, DROPPED, RESTORED), legal_hold_checked_at.

9.46.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	`/api/v1/analytics/metrics`	list published metrics with formulas	n/a	AnalyticsView
POST	`/api/v1/analytics/query`	Cube query (measures, dimensions, filters, timeDimensions)	n/a (read)	AnalyticsView + data scope
GET	`/api/v1/analytics/dashboards/{id}`	dashboard definition and tiles	n/a	AnalyticsView
PUT	`/api/v1/analytics/dashboards/{id}`	create/update dashboard	ETag	AnalyticsAuthor
POST	`/api/v1/analytics/exports`	start export job	`Idempotency-Key`	AnalyticsExport
POST	`/api/v1/analytics/schedules`	create schedule	`Idempotency-Key`	AnalyticsAuthor
GET	`/api/v1/analytics/reconciliation?date=`	reconciliation results per metric	n/a	FinanceView
POST	`/api/v1/documents/templates/{id}/versions`	publish template version	`Idempotency-Key`	DocumentDesign
POST	`/api/v1/documents/render`	render document (type, record_id, template override)	by (record, version)	per-document duty
GET	`/api/v1/catalog/lineage/{asset_fqn}`	upstream/downstream lineage	n/a	CatalogView
GET/PUT	`/api/v1/documents/print-rules`	list/maintain print rules per document type and entity	ETag	PrintManagementSetup
POST	`/api/v1/documents/print-rules/resolve`	resolve destinations for a record without rendering	n/a (read)	per-document duty
GET	`/api/v1/documents/print-queue?site=`	print agent pulls pending jobs; `PATCH` acknowledges	by job id	PrintAgent (service account)
GET/PUT	`/api/v1/documents/text-blocks/{code}`	maintain text block language versions	ETag	DocumentDesign
POST	`/api/v1/documents/correspondence/{type}/run`	run a correspondence type (statements, notices)	`Idempotency-Key`	per-module duty
GET/PUT	`/api/v1/catalog/dq-rules`	maintain data-quality rules; `GET /scores?asset=` returns history	ETag	CatalogSteward
POST	`/api/v1/analytics/process-events/export`	export OCEL/XES for a case type and period	`Idempotency-Key`	AnalyticsExport
POST	`/api/v1/archive/partitions/{id}/restore`	restore an archived partition for 30 days	by partition id	ArchiveAdmin

Query validation: ≤ 10 measures, ≤ 8 dimensions, time range ≤ 5 years at day grain, result cap 50k rows interactive; otherwise redirected to export.

Event	Trigger	Payload fields	Consumers
`dev.passage.analytics.snapshot.committed.v1`	Iceberg commit	domain, table, snapshot_id, rows_added, rows_deleted, max_lsn, watermark_ts	semantic (cache invalidation), catalog
`dev.passage.analytics.metric.reconciled.v1`	daily job	metric_id, tenant_id, date, semantic_value, source_value, delta, status	compliance evidence, alerting
`dev.passage.analytics.report.delivered.v1`	schedule run	schedule_id, run_id, format, destination, row_count, bytes	audit
`dev.passage.documents.template.published.v1`	version publish	template_id, version, document_type, visual_hash	print management, catalog
`dev.passage.documents.print.routed.v1`	destination resolved at render	render_id, document_type, record_id, rule_id, destinations[]	print agent, audit
`dev.passage.catalog.dq.scored.v1`	hourly evaluation	asset_fqn, rule_id, pass_pct, failed_rows, severity	data owner inbox, dashboard
`dev.passage.archive.partition.moved.v1`	archive job verified a partition	domain, table, period, rows, content_hash	compliance evidence (CC6.5), catalog

9.46.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Workspace dashboard host	refresh, drill-through, change period, pin tile	metric tiles, sparklines, freshness badge	AnalyticsView
Explorer	add measure/dimension, filter, pivot, save, export	metric picker with formula tooltip, dimension tree	AnalyticsView
Metric registry	publish version, set owner, view reconciliation history	metric_id, formula, grain, tolerance, last delta	AnalyticsAuthor
Pipeline health	pause/resume, replay from LSN, view DLQ	domain, table, lag_s, status, dlq_count	AnalyticsAdmin
Document designer	drag bands, bind fields, preview with fixture, publish	band tree, locale, print-management rules	DocumentDesign
Schedules	create, run now, view deliveries	cron, timezone, destination, last status	AnalyticsAuthor

Screen	Primary actions	Key fields/columns	Duty
Print management setup	add rule, set condition, order by priority, preview resolution for a record	document type, scope, account, condition, destinations, copies	PrintManagementSetup
Text blocks and correspondence	edit language versions, insert placeholder, run statement job	code, language, body, placeholders, correspondence trigger	DocumentDesign
Data-quality scorecard	drill failing rows, assign owner, mute rule for 7 days	asset, rule, pass %, failed rows, trend	CatalogSteward
Process variant explorer	pick case type and period, compare variants, export OCEL	variant, count, median cycle time, rework	AnalyticsView
Archive partitions	view moved partitions, request restore	domain, table, period, rows, hash, status	ArchiveAdmin

Interaction rules: explorer queries run on 500 ms debounce; `Ctrl+E` exports current grid; freshness badge turns amber at lag > 60 s, red at > 300 s; designer preview reruns on each property change with a 300 ms debounce; personalization (pinned tiles, saved explorer views) stored per user and exported with user profile.

9.46.7 Business rules and edge cases

- **BR-46.1** Monetary metrics use `amount_reporting` in the ledger's reporting currency; rounding at display only, half-even, to the currency's minor units.
- **BR-46.2** A deleted OLTP row (`_op='d'`) is tombstoned in `raw.*`, removed from `core.*` current tables and kept in history with `valid_to`.
- **BR-46.3** Reversal journals net to zero in `revenue` on the reversal's posting date, not the original date.
- **BR-46.4** DSO/DPO use calendar days; trailing windows end at the selected period end inclusive; if the trailing window has zero revenue, the metric is NULL, never infinity.
- **BR-46.5** OEE `planned_time` excludes shifts flagged `no_production`; `performance` is capped at 100%.
- **BR-46.6** Pipeline state machine: RUNNING → PAUSED_SCHEMA (incompatible change) → RUNNING after operator approves mapping; RUNNING → PAUSED_ERROR after 5 consecutive sink failures → RUNNING on replay from `last_lsn`.
- **BR-46.7** Export files are deleted 24 h after creation; a second identical request within 1 h returns the existing file.
- **BR-46.8** Template version state: DRAFT → PUBLISHED (requires visual regression pass) → RETIRED; a RETIRED version remains renderable for re-prints of documents posted under it.
- **BR-46.9** Dashboard queries for a user with no legal-entity scope return an empty result with reason `NO_SCOPE`, not an error.
- **BR-46.10** A reconciliation delta inside tolerance for 30 days auto-marks the metric `TRUSTED`; one failure resets the counter.
- **BR-46.11** Print rule resolution is deterministic: account override beats conditional rules; among conditional rules the lowest `priority` number wins and ties are rejected at save; a rule referencing a RETIRED template version cannot be ACTIVE.
- **BR-46.12** Copies are numbered from 1 and carry the caption of their rule; a re-print of a posted document always renders with the template version recorded on `document_render`, never the current one.
- **BR-46.13** A print job unacknowledged by the print agent for 30 min is marked `STALE`, re-queued once, then `FAILED` with `DOC-0412` to the requesting user.
- **BR-46.14** An archive partition is dropped from OLTP only after `rows_oltp = rows_archive` and the content hash matches; a mismatch leaves both copies and opens an incident of severity Sev-3.
- **BR-46.15** DQ scores below threshold for 3 consecutive evaluations escalate to the catalog steward; a muted rule still scores but does not notify.
- **BR-46.16** Process event case ids are the originating document id; a document that cannot be chained lands in case type `UNLINKED` and is excluded from cycle-time measures.

9.46.8 Acceptance criteria

- **AC-46.1** (FR-46.1, 46.2) Given a journal posted in `ledger` with 1,000 lines, when 60 s elapse, then `mart.fin_journal_line` contains all 1,000 rows with matching `_source_lsn` order.
- **AC-46.2** (FR-46.3) Given a migration adds nullable column `memo2` to `sales_line`, when CDC resumes, then the Iceberg table has the column and no DLQ rows.
- **AC-46.3** (FR-46.3) Given a migration changes `qty` from `decimal(28,6)` to `decimal(18,4)`, when CDC resumes, then the pipeline is PAUSED_SCHEMA within 60 s and an alert fires.
- **AC-46.4** (FR-46.4) Given Kafka redelivers 10,000 already-applied change events, when the sink processes them, then row counts and sums in `core.*` are unchanged.
- **AC-46.5** (FR-46.4) Given a row with `_source_ts` 10 days in the past arrives, when dbt runs, then the row is present in the correct day partition and `late=true`.
- **AC-46.6** (FR-46.6, 46.13) Given a day's postings, when reconciliation runs, then semantic `revenue` equals ledger REVENUE balance movement to USD 0.00 and EUR 0.00.
- **AC-46.7** (FR-46.7) Given user A scoped to legal entity USMF and user B to DEMF, when both open the sales dashboard, then A sees only USMF rows and B only DEMF rows, including in totals.
- **AC-46.8** (FR-46.11) Given the 7 standard document templates and their fixtures, when rendered in CI, then perceptual-hash delta vs baseline ≤ 0.1% for each.
- **AC-46.9** (FR-46.9) Given an explorer query returning 250k rows, when the user exports XLSX, then a Temporal job produces the file and a signed URL within 5 min.
- **AC-46.10** (FR-46.10) Given a schedule `0 6 * * 1` Europe/Berlin, when Monday 06:00 CET arrives, then the PDF is delivered and `report.delivered` is emitted.
- **AC-46.11** (FR-46.14) Given a customer with account override EMAIL and a conditional rule `amount_total >= 10000 → PRINTER + COPY 1 ARCHIVE_ONLY`, when a USD 12,000 invoice is posted, then the original goes to the customer's e-mail (override wins) and `document_render.destinations_json` records the rule id.

- **AC-46.12** (FR-46.14) Given the site print agent offline for 2 h, when 40 packing slips are posted, then all 40 print within 5 min of the agent reconnecting and none duplicates.
- **AC-46.13** (FR-46.15) Given correspondence type `ACCOUNT_STATEMENT` scheduled monthly for customer group `KEY`, when the schedule fires, then one statement per customer with open items is rendered in the customer's language and routed per FR-46.14.
- **AC-46.14** (FR-46.16) Given a completeness rule on `vendor.tax_id` with threshold 98%, when 3% of vendors lack a tax id, then the scorecard shows the rule failed, the owner receives one inbox task, and the failing vendor list is downloadable.
- **AC-46.15** (FR-46.18) Given journal-line partitions older than 24 months and no legal hold, when the archive job runs, then the partitions exist in `archive.ledger.journal_line`, hashes match, the OLTP partitions are dropped, and the records still open from the journal inquiry with badge `ARCHIVED`.
- **AC-46.16** (FR-46.17) Given one month of P2P documents, when `mart.process_event` is built, then every paid invoice traces to its PO case and the variant explorer lists the straight-through variant with its share.

9.46.9 Test plan and fixtures

Fixtures: `fx-lake-usmf-year` (one fiscal year, 3 entities, 2.4M journal lines, 180k sales lines, from the wave-30 demo dataset replayed); `fx-schema-evolution` (12 migration scripts: add, widen, drop, rename, narrow, not-null); `fx-late-data` (5k events with `_source_ts` skewed 1 h–45 d); `fx-docs-7` (7 document types × 3 locales × fixture records); `fx-oee-shifts` (90 days, 12 resources).

Test	Type	Target	Pass criterion
Sink idempotency	property	Iceberg sink	any permutation/duplication of events yields identical table state
Schema evolution matrix	contract	connector + sink	12 scripts classified correctly (8 auto, 4 pause)
Metric formula unit tests	unit	dbt/Cube	each metric matches hand-computed value on 20-row fixture
OPA→Cube filter equivalence	contract	semantic	for 200 users, Cube result set ⊆ OLTP API result set
Dashboard lag	performance	end-to-end	p95 < 60 s at 1M journal lines/h
Visual regression	E2E	docgen	21 renders ≤ 0.1% delta
Export at scale	performance	explorer	1M-row XLSX in < 10 min
Lineage completeness	contract	OpenMetadata	100% of dbt models and metrics have upstream edges

Reconciliation: `revenue`, `cogs` ↔ ledger balance movements by account category; `ar_open_balance`, `ap_open_balance` ↔ open-item subledgers; `inventory_value` ↔ inventory close valuation; row counts `raw.*` ↔ `core.*` current ↔ Postgres `COUNT(*)` per tenant.

9.46.10 Deployment and operations

Helm charts: `passage-cdc` (Kafka Connect cluster with Debezium and Iceberg sink, 3 workers), `passage-lakehouse` (Iceberg REST catalog, Nessie-compatible, object storage via S3 API), `passage-trino` (coordinator + 4 workers, autoscaling to 16), `passage-clickhouse` (optional accelerator for `mart.*` pre-aggregations), `passage-cube` (API + refresh workers), `passage-dbt-runner` (Temporal worker), `passage-docgen`, `passage-openmetadata`. Config: `CDC_DOMAINS` list, `ICEBERG_COMMIT_INTERVAL=30s`, `LATE_DATA_WATERMARK=5m`, `RECON_TOLERANCE=0.005`, object-store credentials from Vault. Migrations: `analytics` database (Atlas); Iceberg schemas managed by the sink. Feature flags: `analytics.dashboards`, `analytics.explorer`, `docgen.designer` per tenant. Jobs: dbt incremental every 60 s, SCD2 nightly 01:00, reconciliation 02:00, compaction 03:00, lineage sync hourly. SLOs: dashboard lag p95 < 60 s; query p95 < 2 s for ≤ 50k rows; document render p95 < 800 ms; pipeline availability 99.9%. Alerts: slot lag > 1 GB, DLQ > 0 for 5 min, reconciliation delta > tolerance, Trino queue > 50. Runbook: replay from LSN, rebuild `core.*` from `raw.*`, rotate object-store keys. Rollback: Helm rollback of services; Iceberg `rollback_to_snapshot` per table; connectors restart from stored offsets.

9.46.11 Migration from AX / SAP

Source	Passage	Notes
Entity store `AggregateMeasurement` definitions	`semantic_metric`	mapped by hand; measures become Cube measures, dimensions become joins
Power BI `.pbix` over BYOD	Dashboards	not converted; rebuilt on Cube; Power BI Desktop can connect to Trino
SSRS `.rdl` layouts via `SrsReportDataProvider`	`document_template`	semi-automatic: bands, fields and expressions imported; SSRS expressions in VB are flagged for manual rewrite
Print management settings (`PrintMgmtSettings`)	print-management rules	full
BW InfoProviders, BEx queries	`mart.*` + metrics	metadata import creates draft metrics; data not migrated here (wave 49)
SAC stories	Dashboards	rebuilt
Smart Forms / Adobe Forms	`document_template`	layout imported from XDP where possible; scripting rewritten

Not migrated: SSAS cube partitions, BYOD Azure SQL contents, Power BI datasets — all derivable from the lakehouse.

9.46.12 Dependencies, risks, open questions

Depends on: wave 3 (outbox/CDC infrastructure), wave 5 (audit from CDC), wave 11–15 (universal journal, statements), wave 30 (sales and AR marts), wave 25 (OEE source data), wave 45 (all domains emitting).

Risk	Likelihood	Impact	Mitigation
Iceberg small-file explosion at 30 s commits	High	Medium	nightly compaction; adaptive commit size
Cube pre-aggregations leak across security contexts	Low	Critical	pre-aggregations partitioned by legal entity; contract test
Trino cost at 10k users	Medium	Medium	ClickHouse accelerator for top-20 dashboards

Open questions: (1) Trino-only vs Trino+ClickHouse by default — proposed default: Trino only, ClickHouse behind flag; decide by wave 47. (2) Document rendering engine fallback (Chromium vs Typst) — default Chromium; decide by wave 48.

9.46.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`analytics.cdc.commit_interval_s`	tenant (platform default)	int	30	10–300	Iceberg commit cadence per sink	Entity store refresh batch recurrence (`AggregateMeasurement`)	ODP delta queue schedule (`ODQMON`), DTP frequency
`analytics.cdc.late_watermark_min`	tenant	int	5	1–60	watermark lag before partitions close	none	BW late-arriving delta handling
`analytics.recon.tolerance`	legal entity	numeric	0.005	0–1.00	max	delta	before dashboard banner
`analytics.recon.run_time_local`	tenant	time	02:00	any	daily reconciliation start	batch job recurrence	`SM36` job start
`analytics.dashboard.refresh_s`	user	int	60	15–600	tile auto-refresh	Power BI tile refresh	SAC story auto-refresh
`analytics.freshness.amber_s` / `red_s`	tenant	int	60 / 300	≥ 10	badge thresholds	none	none
`analytics.query.max_rows_interactive`	tenant	int	50,000	1,000–200,000	redirect to export above cap	OData `maxpagesize`	BICS result limit (`RSADMIN MAX_NUMBER_OF_CELLS`)
`analytics.export.max_rows`	tenant	int	1,000,000	≤ 5,000,000	export job cap	Open in Excel row limit	ALV/AfO export limits
`analytics.export.link_ttl_h`	tenant	int	24	1–168	signed URL validity	none	none
`analytics.schedule.retry`	tenant	int/backoff	3 / 1,4,16 min	0–5	delivery retry	batch retry count	`SM37` restart
`docgen.engine`	tenant	enum	CHROMIUM	CHROMIUM, TYPST (flag)	render engine	SSRS	ADS (`SFP`)
`docgen.pdf_profile`	legal entity	enum	PDF/A-3	PDF/A-3, PDF 1.7	archival format	none	ADS PDF/A setting
`docgen.visual_regression_max_pct`	tenant	numeric	0.1	0–5	publish gate	none	none
`print.default_destination.<document_type>`	legal entity	enum	ARCHIVE_ONLY	PRINTER, EMAIL, E_INVOICE, ARCHIVE_ONLY, PORTAL, NONE	fallback when no rule matches	Print management default	`NACE` default condition record
`print.copies_default`	legal entity	int	1	0–9	copies when rule silent	Print management copies	`NAST` number of messages
`print.agent_offline_buffer_h`	site	int	24	1–72	job retention while agent offline	Document routing agent	`SPAD` spool retention
`print.stale_job_min`	site	int	30	5–240	BR-46.13 threshold	none	spool request timeout
`text.default_language`	legal entity	BCP 47	entity language	installed languages	fallback for text blocks	Form notes language	`SO10` logon language
`correspondence.statement_schedule`	legal entity	cron	`0 20 L * *`	cron	periodic statements	Account statement batch	`F.27` variant
`dq.eval_interval_min`	tenant	int	60	15–1440	rule evaluation cadence	none	Information Steward task schedule
`dq.default_threshold_pct`	tenant	numeric	98	50–100	default rule threshold	none	IS rule threshold
`process.event_retention_months`	tenant	int	24	6–120	event-log roll-up age	none	Process Observer retention
`archive.residence_months.<domain>`	tenant	int	24	6–120 (≥ wave-48 statutory minimum)	FR-46.18 move age	none	`AOBJ` residence time
`archive.restore_ttl_days`	tenant	int	30	1–90	restored partition lifetime	none	`SARI` retrieval
`catalog.sync_interval_min`	tenant	int	60	15–1440	lineage refresh	none	BW metadata repository refresh

Setup sequence per legal entity: install document templates → map party contact roles for e-mail routing → define print rules (defaults, then overrides) → load text blocks and form notes → enable dashboards per workspace → register reconciliation sources for financial metrics → assign DQ rule owners → set archive residence per domain (must not undercut the wave-48 retention minimum for the entity's country).

9.46.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Workspace dashboards	KPI	period, legal entity, site, dimension filters	governed metrics, sparklines, freshness	Cube over `mart.*`	Workspace tiles, Power BI embedded, role center KPIs	Fiori Overview Pages, SAC stories, Smart Business tiles
Explorer	drill-down	measures × dimensions × filters, time grain	any published metric and dimension	Cube/Trino	Open in Excel, Power BI Desktop over BYOD	Analysis for Office, Query Browser, Multidimensional apps
Metric reconciliation results	operational report	metric, date range, status	semantic value, source value, delta, status, days trusted	`metric_reconciliation`	none	none
Pipeline health	list inquiry	domain, status	lag seconds, last LSN, DLQ count, snapshot id	`cdc_pipeline_state`	Entity store refresh status	`RSA7` delta queue, `ODQMON`
Export log	list inquiry	user, date, format	rows, bytes, filter hash, destination	`export_job`	none	download audit (`SM20` partial)
Schedule deliveries	list inquiry	schedule, status	run time, destination, format, retries, error	`report_delivery`	Batch job history	`SM37`, `SOST`, `RSRD_LOG`

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Document output log	list inquiry	document type, record, destination, status	render id, template version, destinations, agent ack, error	`document_render`	Print management journal / document history	`VV13` / `VV23` NAST status, Manage Output Items
Print rule resolution trace	drill-down	record	matched rule, evaluated conditions, final destinations	`print_rule` + read model	none	`NACE` determination analysis
Data-quality scorecard	KPI	asset, rule type, owner	pass %, failed rows, trend, score	`dq_score`	none	Information Steward scorecard
Process variant explorer	drill-down	case type, period, entity	variants, cycle times, rework, automation share	`mart.process_event`	none	Signavio Process Intelligence
Archive partition inquiry	list inquiry	domain, period, status	rows, hash, moved_at, restore state	`archive_partition`	none	`SARI`, `SARA` job log
Lineage view	drill-down	asset	upstream/downstream assets, owners, last refresh	OpenMetadata	none	BW metadata repository, Datasphere catalog
Usage statistics	KPI	period	dashboard opens, explorer queries, exports per user	`analytics_usage`	none	`RSRT` statistics, SAC usage

9.46.15 Printed and electronic documents

Document	Trigger	Layout elements (header/line/footer)	Formats	Numbering sequence	Legal / retention requirement
Sales invoice	invoice posting (wave 30)	entity branding, bill-to/ship-to, tax ids, invoice number/date, payment terms, IBAN; lines with item, qty, unit price, discount, tax code; totals per tax rate, footer text blocks	PDF/A-3 with embedded e-invoice XML (wave 35), e-mail, EDI, portal	invoice sequence per legal entity (gap-free where the country pack requires)	country pack statutory content; retain 7–10 years per wave-48 policy
Purchase order	PO confirmation (wave 17)	vendor, delivery address, terms, buyer; lines with item, qty, price, requested date; footer terms text block	PDF, e-mail, cXML/EDI (wave 35), portal	PO number	retain with PO record
Packing slip	shipment confirmation (wave 28)	ship-to, carrier, tracking; lines with item, qty, batch/serial, weight	PDF, label printer (wave 32)	packing slip sequence	customs copy per wave-48 export rules
Check	payment run (wave 19)	MICR line, payee, amount in words, remittance stub	PDF to check printer; positive-pay file	check number per bank account	bank retention; voided checks kept
Customer/vendor statement	correspondence schedule or manual	balance forward, open items by age, totals	PDF, e-mail, portal	statement run id	none
Pick list	wave release (wave 31)	wave, locations in pick sequence, item, qty, license plate	PDF, mobile	pick list id	none
Remittance advice	payment run	payee, invoices paid, discounts, net	PDF, e-mail, ISO 20022 `remt.001`	payment id	none
Payment notice / balance confirmation	correspondence type	party, period, items, confirmation text block, signature line	PDF, e-mail	correspondence run id	audit support
Scheduled report	schedule	dashboard snapshot or explorer grid, parameters, generated-at, freshness	PDF, XLSX, CSV	run id	none; deleted after 30 days
Export file	explorer export	grid as requested, filter manifest sheet	XLSX, CSV	export job id	audit log kept 1 year; file deleted after 24 h
Process event export	FR-46.17	cases, events, objects	OCEL 2.0 JSON, XES	export job id	none

9.46.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
dbt incremental run	every 60 s	domain	`_source_lsn` high-water mark per model	< 45 s for 1M lines/h	skip cycle, alert after 5 consecutive failures	`dbt_run_duration_s`, lag p95
SCD2 rebuild	nightly 01:00	master-data entity	full rebuild into staging then swap	< 20 min	keep previous tables; alert	job status
Metric reconciliation	daily 02:00 local	tenant, metric	(tenant, metric, date) unique	< 5 min	banner on dashboard; `metric.reconciled` BLOCKED	delta > tolerance
Iceberg compaction and snapshot expiry	nightly 03:00	table	Iceberg commit	< 60 min	retry next night; small-file count alert	files < 64 MB count
Lineage sync	hourly	asset type	OpenMetadata ingestion checkpoint	< 10 min	retry 3×	missing-edge count
Schedule dispatcher	every minute	schedule	`(schedule_id, next_run_at)`	< 1 s dispatch	BR: retry 3× with backoff, then `report_delivery` FAILED	failed deliveries
Export cleanup	hourly	export job	delete files > 24 h	< 1 min	none	none
Periodic correspondence (statements)	per `correspondence_type.cron`	legal entity, party group	run id	< 15 min per 10k statements	partial run resumable per party	failed renders
Print queue sweeper	every 5 min	site	job state	< 1 s	BR-46.13	stale jobs > 0
DQ evaluation	every 60 min	asset	`(rule, evaluated_at)`	< 10 min	skip rule after timeout 5 min	rules failing
Process event build	every 15 min	case type	event offset	< 5 min	resume from offset	build lag
Archive mover	weekly Sunday 04:00	domain, partition	`archive_partition` state machine	< 2 h per 100M rows	BR-46.14	hash mismatch, partitions pending > 7 days

9.46.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
AnalyticsView	`GET /analytics/metrics`, `POST /analytics/query`, dashboard host, explorer (read)	none	every business role

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
AnalyticsAuthor	`PUT /analytics/dashboards`, `POST /analytics/schedules`, metric registry publish	with FinanceClose when publishing financial metrics (MEDIUM, `RPT-001`)	Controller, Analytics author
AnalyticsExport	`POST /analytics/exports`, process-event export	subject to wave-48 export governance	Controller, Analyst
AnalyticsAdmin	pipeline health, replay from LSN, DLQ release, Cube cache flush	with AnalyticsAuthor (LOW)	Platform operator
DocumentDesign	template designer, `POST /documents/templates/{id}/versions`, text blocks	with PrintManagementSetup when the same user routes payments (`P2P-007`, MEDIUM)	Application administrator
PrintManagementSetup	`PUT /documents/print-rules`, resolution trace	with ProcessVendorPayments (HIGH, check and remittance redirection)	Application administrator
PrintAgent (service)	`GET/PATCH /documents/print-queue`	n/a	site print agent identity
CatalogView	lineage, catalog browse	none	every business role
CatalogSteward	DQ rules, owners, asset metadata	none	Data steward
ArchiveAdmin	archive partitions, restore	with LegalHoldAdmin (MEDIUM)	Platform operator, Compliance

9.46.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
ANA-0101	error	Query exceeds the interactive limit of {max_rows} rows. Use Export.	result cap	run as export job	FR-46.9
ANA-0102	error	You have no data scope for this model (NO_SCOPE).	OPA scope returns empty	request legal-entity access	BR-46.9
ANA-0103	error	Model {model} has no security mapping (SCOPE_UNMAPPED).	missing scope mapping	AnalyticsAdmin maps the model	FR-46.7
ANA-0104	warning	Metric {metric} failed reconciliation on {date}: delta {delta}.		delta	> tolerance
ANA-0105	error	CDC for {domain}.{table} paused: incompatible schema change ({detail}).	narrowing or NOT NULL	operator approves mapping, resume	FR-46.3
ANA-0106	warning	Data is {lag} s old.	freshness above amber threshold	wait or check pipeline health	FR-46.8
ANA-0107	info	Export ready: {rows} rows, link valid until {expires_at}.	export completed	download	FR-46.9
ANA-0108	error	Schedule {name} failed after 3 attempts: {error}.	delivery failure	fix destination, run now	FR-46.10
ANA-0109	warning	Data-quality rule {rule} on {asset} at {pass_pct}% (threshold {threshold}%).	rule below threshold	owner reviews failing rows	FR-46.16
ANA-0110	error	Archive verification failed for {domain}.{table} {period}: counts {a} vs {b}.	mismatch	incident opened; partition kept	BR-46.14
DOC-0401	error	Template version {v} is RETIRED and cannot be assigned to a print rule.	rule references retired version	pick a PUBLISHED version	BR-46.11
DOC-0402	error	Two print rules have priority {p} for {document_type} in {entity}.	tie	change priority	BR-46.11
DOC-0403	error	Visual regression delta {pct}% exceeds {max}%; version not published.	render drift	review diff, re-baseline with reason	BR-46.8
DOC-0404	error	Render failed for {document_type} {id}: {reason}.	data or engine error	retry; check read model	FR-46.11
DOC-0405	warning	No print rule matched; using entity default {destination}.	fallback	add rule if unwanted	FR-46.14
DOC-0406	error	Recipient role {role} has no e-mail on party {party}.	missing contact	maintain contact	FR-46.14
DOC-0407	warning	Placeholder {name} is empty in text block {code}.	unresolved placeholder	fix binding or text	FR-46.15
DOC-0412	error	Print job {id} failed: no acknowledgment from print agent at {site}.	agent offline > 2 × stale window	check agent, reprint	BR-46.13

9.46.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation under 3× load
CDC commit to `raw.*`	1M journal lines/h peak, 20M change events/day	lag 10 s / 30 s / 60 s	6k events/s per domain	400 GB raw (Parquet, zstd)	`days(_source_ts)`, `bucket(16, tenant_id)`	commit size adapts to 100 MB; lag badge amber
dbt incremental cycle	60 s cadence	20 s / 45 s / 60 s	n/a	`core.*` 250 GB, `mart.*` 120 GB	lookback 3 days	cycle stretches to 120 s; no data loss
Cube query (≤ 50k rows)	10k users, 5% querying, peak 500 queries/min	300 ms / 2 s / 4 s	50 queries/s	pre-aggregation 50 GB	pre-aggregations per legal entity and day	queue; p95 4 s; explorer debounce raised to 1 s
Dashboard load (8 tiles)	10k opens/day	800 ms / 2 s / 3 s	100 loads/s	n/a	cached per (user scope, tile, minute)	cache TTL 2 min
Export job	200/day, up to 1M rows	1 min / 8 min / 10 min	4 concurrent per tenant	24 h retention only	Trino spill to disk	queued; ETA shown
Document render (50 lines)	50k documents/day, peak 2k/min	300 ms / 800 ms / 1.5 s	40 renders/s per docgen pod	1.5 TB PDF/A per 10M docs (object store)	by tenant prefix and month	autoscale to 12 pods; print queue absorbs
Print rule resolution	per render	5 ms / 20 ms / 50 ms	n/a	negligible	index (tenant, document_type, scope, priority)	n/a
DQ evaluation	500 rules	2 min / 8 min / 10 min	n/a	`dq_score` 2 GB	by (asset, date)	interval doubles
Archive move	100M rows/week	< 2 h	15k rows/s	OLTP shrinks ~40%	partition by month	job spans two windows

9.46.20 Localization and compliance hooks

- Document layouts vary by country: legal mentions (DE §14 UStG invoice contents, FR mention of late-payment penalties, IN GSTIN and HSN codes, MX CFDI UUID and QR, UK VAT number), address formats and number/date formats come from the wave-48 `country_pack.document_layouts[]` which installs template variants keyed by `locale`; the designer refuses to publish a `SalesInvoice` variant lacking the pack's mandatory elements.
- Currency rounding at display uses the currency's minor units (JPY 0, KWD 3) and the entity's rounding rule from wave 11; amounts in words on checks use the pack's language rules.
- Fiscal calendars (wave 11) drive the Cube `timeDimension` for period comparisons; 4-4-5 and IN April–March years are first-class.
- Retention of rendered documents, exports and archive partitions follows wave-48 `retention_policy` per record class and country (DE 10 years GoBD, US 7 years, FR 10 years, IN 8 years); the archive mover reads the minimum and never undercuts it.
- Privacy: explorer and exports are subject to wave-48 export governance (FR-48.24) and the PII classification from wave 42; the process event log stores actor ids, not names, and is pseudonymized on erasure.

- Language: dashboards, metric names and text blocks are translated through the wave-2 label framework; text blocks keep per-language bodies and the document locale picks them.
- Extension points: `country_pack.document_layouts[]`, `country_pack.text_blocks[]`, `print_rule.condition_expr` (wave-3 expression subset / FR-3.13 decision tables with a `country` input column), Cube `queryRewrite` for country-specific data scopes, `retention_policy.residence_months` per domain.

9.46.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.99.4 row 17 — 12.2.4 Forms → Print management (AP)	PARTIAL-46	ADDED as FR-46.14	conditional routing per vendor and query, original/copy destinations
AX 12.99.4 row 30 — 12.3.4 Form setup (sales documents) / Form notes / Print management	PARTIAL-46	ADDED as FR-46.14, FR-46.15	form notes via text blocks; routing via print rules
AX 12.99.4 row 102 — 12.11.4 Form setup (project invoices) / Print management	PARTIAL-46	ADDED as FR-46.14	project account override scope
AX 12.99.4 row 167 — 12.16.4 Form setup (purchase documents) / Form notes / Print management (purchase)	PARTIAL-46	ADDED as FR-46.14, FR-46.15	
AX 12.99.4 row 173 — 12.17.4 Form setup (sales), Form notes, Print management (sales), Form sorting	PARTIAL-46	ADDED as FR-46.14, FR-46.15	form sorting already in designer (line sort band property)
AX 12.99.4 row 240 — 12.29.1 Print management (org-wide) (`PrintMgmtDocType`)	PARTIAL-46	ADDED as FR-46.14	tenant and legal-entity defaults
AX 12.99.4 row 259 — 12.29.4 Print management setup (per module/document: original/copy destinations, conditional settings by query)	PARTIAL-46	ADDED as FR-46.14	query conditions via wave-3 expression subset or FR-3.13 decision table
AX 12.30 inventory rows "Data archiving / Long-term retention", "Archive/purge jobs (D365)"	COVERED-46 (by name only)	ADDED as FR-46.18	archive tier now specified; retention minimums from wave 48
SAP 13.99.2 row 33 — 13.2 `F.64` AP correspondence (PARTIAL-19, "define on wave 46 designer")	PARTIAL-19	ADDED as FR-46.15	correspondence types with triggers; wave 19/30 bind their events
SAP 13.99.2 row 349 — 13.31 `S010`/`S072`/`SE75` standard texts (PARTIAL-5, "propose wave 46 designer text blocks")	PARTIAL-5	ADDED as FR-46.15	text blocks with placeholders and language versions
SAP 13.99.2 row 353 — 13.31 SAP Process Mining (Signavio) / Process Observer (`POC`)	PARTIAL-46	ADDED as FR-46.17 (partial: event log, variants, OCEL/XES export; no conformance engine)	conformance checking remains OUT-OF-SCOPE for v1 — external tools consume the export
SAP 13.99.2 row 399 — 13.36 Data quality dashboards (Information Steward)	PARTIAL-46	ADDED as FR-46.16	DQ rules on catalog assets, scorecard tile
SAP 13.99.2 row 401 — 13.37 Signavio Process AI / Process Insights	PARTIAL-46	SUPERSEDED by FR-46.17 + wave-47 copilot (`ask_data` over `mart.process_event`)	recommendations are copilot answers over the event log, not a separate product
SAP 13.31 rows "S/4 Output Management (BRF+ OPD)", "output control: output types, determination"	SUPERSEDED-46	ADDED as FR-46.14	determination rules are print rules; channels map 1:1
SAP 13.99.2 rows 348/352/354 — BRF+ / decision tables (PARTIAL-3, not this wave)	PARTIAL-3	SUPERSEDED by FR-3.13 (wave 3 `RuleSet`/`DecisionTable`); output determination here consumes it via `print_rule.condition_expr`, no private rules engine	listed for traceability only
AX 12.99.1 theme "Alerts and notification rules" (wave 5, not this wave)	GAP-5	DEFERRED to wave 5 (owner); reconciliation banners, DQ failures and schedule failures deliver through the wave-5 notification center	listed for traceability only

9.47 Wave 47: AI Copilots, Agents & Automation

9.47.1 Scope and business context

D365 ships Copilot for finance and operations (summaries, draft e-mails, Copilot in workflow history) and Power Automate flows triggered from business events; SAP ships Joule and SAP Build Process Automation. Neither lets a governed agent execute a multi-step ERP process under the invoking user's exact permissions with an eval gate. This wave exposes every Passage API, entity and report as MCP tools through a gateway that forwards the user's token to OPA-protected APIs; adds a copilot in the shell (ask-the-data over the wave-46 semantic layer, explain-this-record, draft documents, summarize workflows); runs agents on Temporal with approval gates; gates deployment on an eval harness; and enforces responsible-AI controls with a provider-agnostic model layer (Claude default; Bedrock, Azure OpenAI, open-weights via vLLM selectable). Out of scope: training custom models, autonomous posting without approval in v1, voice.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Coding a vendor invoice in `VendInvoiceJour`/`MIRO` means opening the PO, the vendor history and the chart of accounts in three forms, then keying account, dimensions and tax code per line	No suggestion engine; copilots cannot read the posting context	AP coding agent proposes account, dimensions, tax code, PO match per line; approver confirms	Clicks per invoice 18 → 2; coding time 4 min → 20 s; top-1 accuracy ≥ 92%
Cash application: `CustPaymMatch` rules and `F-28` match exact amounts only; partials and multi-invoice remittances go to a manual queue	Rule-based matching without reading remittance text	Cash application agent ranks settlements from remittance text and tolerances	Auto-matched share 60% → 90%+; unapplied cash aged > 7 d reduced 10×
Power Automate / SAP Build flows need a connector project, a service account with broad rights and a separate audit trail	Automation runs outside the ERP's authorization model	MCP tools with user-token inheritance; Temporal agent runtime; `agent_action` audit	Integration project 6 weeks → 0 (tools generated); 0 actions beyond user permission
"Why is this number what it is?" requires reading workflow history, `SysDatabaseLog` and voucher transactions across forms	Context scattered across tables with no narrative	Explain-this-record over audit trail and related documents with citations	Investigation 15 min → 30 s
D365 Copilot / Joule answer from a vendor model with no accuracy gate, no cost control and no kill switch	Vendor-managed black box	Eval harness in CI, token budgets, kill switch, provider choice	100% of agent releases gated by evals; kill in ≤ 10 s

9.47.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Conversational assistant	D365 Copilot sidecar (`CopilotFeatureManagement`), Copilot summaries on workflow history	Joule in S/4HANA Cloud, Fiori launchpad	Copilot panel in shell	Improved (acts, not only answers)
Ask the data	Copilot "ask about data" over Dataverse virtual entities	Joule analytics via SAC	`ask_data` tool → Cube query with OPA scope	Full
Process automation	Power Automate flows on business events (`BusinessEventsCatalog`), Flow connectors	SAP Build Process Automation, workflow agents	Agent runtime on Temporal, triggered by CloudEvents	Full
Invoice coding suggestions	Vendor invoice automation, `VendInvoiceAutomation`	Cash Application ML, Invoice assistant (Business AI)	AP coding agent (suggest → approve)	Full
Cash application	Customer payment advanced matching rules (`CustPaymMatch`)	Cash Application ML (`FINS_CASH_ML`)	Cash application agent	Full
Planning exceptions	Planning Optimization action messages	MRP Cockpit (`MD04`, `PPH_MRP_COCKPIT`), Situation Handling	MRP exception triage agent	Improved
Collections outreach	Collections workspace, `CustCollectionLetter`	FSCM Collections worklist (`UDM_SUPERVISOR`)	Collections agent drafts outreach	Full
Anomaly detection	Finance insights (cash-flow forecast, late-payment prediction)	Business AI journal anomaly detection	Journal anomaly agent	Full
Audit of AI actions	None dedicated	None dedicated	`agent_action` audit with inputs, tool calls, outputs, approver	Improved
Payment-date prediction	Customer payment predictions (Finance insights, Dataverse ML)	Cash Application ML, FSCM collections scoring	`payment_prediction` model and scores on open invoices (FR-47.15)	Full
Knowledge retrieval	Copilot over Microsoft Learn docs	Joule answers from SAP Help, solution database (`IS01`/`IS02`)	`search_knowledge` tool over wave-45 knowledge articles and the wave-50 docs site (FR-47.14)	Full
AI scenario activation and thresholds	Feature management flags, `VendInvoiceAutomation` parameters	`ISLM` intelligent scenarios, Business AI activation per company code	`agent.<id>.mode` per legal entity with thresholds (FR-47.5, 9.47.13)	Full

9.47.3 Functional requirements

- **FR-47.1 MCP gateway** — Must. The system shall expose Passage capabilities as MCP tools from a single gateway (`mcp.passage.<tenant>`), generated from OpenAPI/AsyncAPI and entity metadata.
    - Notes: tool names `passage.<domain>.<entity>.<verb>` (e.g. `passage.ap.invoice.list`, `passage.ap.invoice.propose_coding`, `passage.ledger.journal.get`); `inputSchema` is the endpoint's JSON Schema with tenant_id removed (injected from token); each tool declares `annotations.readOnlyHint`, `destructiveHint`, `idempotentHint`, and a Passage extension `x-passage-duty` (required duty) and `x-passage-approval` (`none`, `suggest`, `approve_before_commit`); read tools return ≤ 200 rows per call with cursor; a tool call carries `traceparent` and `x-agent-run-id`.
- **FR-47.2 Permission inheritance** — Must. The system shall execute every tool call with the invoking user's Keycloak token (on-behalf-of exchange, `act` claim = agent id), so OPA evaluates the user's duties and data scope at the API; agents hold no credentials of their own.
    - Notes: token lifetime for agent runs 15 min, refreshed by Temporal activity; a tool whose `x-passage-duty` the user lacks is not listed to the model; contract test: for each of 400 tools, a user without the duty receives 403 from the gateway and the API.
- **FR-47.3 Copilot UI** — Must. The system shall provide a shell panel with modes ask-the-data, explain-this-record, draft (e-mail, PO text, dunning letter), summarize (workflow history, case) with citations to records and metrics used.
    - Notes: streaming responses; every answer lists tool calls made; "explain" uses the record's audit trail and related documents; drafts are inserted only on user click.
- **FR-47.4 Agent runtime** — Must. The system shall run agents as Temporal workflows (`AgentRunWorkflow`) with steps plan → tool calls → proposal → approval gate → commit, each tool call an activity with retry policy and 60 s timeout, run timeout 24 h including approval wait.
- **FR-47.5 AP invoice coding agent** — Must. The system shall propose account, dimensions, tax code and PO match for an unposted vendor invoice using vendor history, PO lines and text, in suggest mode (approver confirms) or auto mode when confidence ≥ 0.95 and amount < tenant threshold (default USD 5,000 / EUR 4,500).
- **FR-47.6 Cash application agent** — Must. The system shall propose settlements for unapplied customer payments (remittance text, amounts, tolerances), ranked by confidence; applies only after approval in v1.
- **FR-47.7 MRP exception triage, collections outreach, journal anomaly agents** — Should. The system shall ship these three in suggest mode: triage groups action messages by cause and proposes expedite/postpone; collections drafts outreach per dunning policy; anomaly scores journals on amount, account pairing, time of posting and user, flagging score > 0.8.
- **FR-47.8 Approval gates** — Must. The system shall route any `approve_before_commit` action to the wave-4 workflow inbox with proposed change, diff, confidence and rationale; approver decisions are recorded on `agent_action`.
- **FR-47.9 Eval harness** — Must. The system shall maintain golden datasets per agent and run evals in CI on every prompt, model or tool change, blocking deploy below thresholds.
    - Metrics and thresholds: AP coding top-1 accuracy ≥ 0.92, top-3 ≥ 0.98; cash application precision ≥ 0.90, recall ≥ 0.75; MRP triage macro-F1 ≥ 0.80; anomaly AUROC ≥ 0.85, false-positive rate ≤ 0.05; ask-the-data metric-selection accuracy ≥ 0.95 and SQL-equivalence ≥ 0.90; refusal correctness ≥ 0.99 on an adversarial set of 500 prompts; p95 latency per agent run < 30 s excluding approval; cost per run ≤ budgeted USD; regression gate = no metric drops > 1 point vs last release.
- **FR-47.10 Prompt and version registry** — Must. The system shall version system prompts, tool lists, model id and parameters as `agent_version`; production pins a version; rollback is a pointer change.
- **FR-47.11 Token budgets** — Must. The system shall enforce per-tenant monthly token budgets (default 50M input / 10M output), soft alert at 80%, hard stop at 100% with copilot degraded to deterministic features; per-run cap 400k tokens.
- **FR-47.12 Responsible-AI controls** — Must. The system shall: redact PII (names, IBAN, tax ids, emails) from prompts to external providers using the wave-42 PII classification, replacing with stable pseudonyms resolved after response; log every action (prompt hash, tool calls, outputs, approver); provide a tenant and

global kill switch that halts all runs within 10 s; prohibit model-initiated posting without `approve_before_commit` or auto-mode policy; label all generated text in UI; keep a model card per agent version; allow provider selection per tenant with data-residency constraint checked against wave-48 tenant residency.

- **FR-47.13 Provider layer** — Must. The system shall abstract chat completion, tool use and embeddings behind one interface with adapters for Anthropic API, Amazon Bedrock, Azure, and OpenAI-compatible open-weights endpoints; feature parity test per adapter.
- **FR-47.14 Knowledge retrieval tool** — Should. The system shall expose `passage.knowledge.search` (hybrid BM25 + embedding search over wave-45 knowledge articles, wave-50 documentation, and tenant policy documents uploaded by an administrator) so copilot answers to "how do I" questions cite the article or document section; the tool respects article visibility (internal, portal) through the user's token.
    - Notes: embeddings are computed by the provider layer (FR-47.13) and stored in Postgres `pgvector` in the `knowledge` database; index refresh within 5 min of article publish; answers without a citation above similarity 0.75 say so instead of guessing.
- **FR-47.15 Payment-date prediction** — Should. The system shall train a per-tenant model (Python, gradient-boosted trees) weekly on paid invoices (customer, terms, amount, invoice day-of-month, dunning history, disputes, seasonality) and score every open customer invoice with `predicted_payment_date` and `p_late_30d`, feeding the collections agent (FR-47.7), the wave-30 collections workspace and the `dso_days` forecast tile.
    - Notes: minimum 500 paid invoices per tenant before scoring is enabled, else a terms-plus-average-delay heuristic is used and labeled; model card per version; MAE on held-out set ≤ 6 days or the previous model stays active.

9.47.4 Data model

`tool_definition` (mcp-gateway)

Field	Type	Required	Notes
tool_name	text	Y	PK, `passage.<domain>.<entity>.<verb>`
version	int	Y	from OpenAPI version
input_schema	jsonb	Y	JSON Schema draft 2020-12
output_schema	jsonb	Y	
required_duty	text	Y	OPA duty id
approval_mode	enum	Y	none, suggest, approve_before_commit
read_only	bool	Y	
source_ref	text	Y	OpenAPI operationId or report id

`agent_run` (agent-runtime)

Field	Type	Required	Notes
run_id	uuid	Y	PK = Temporal workflow id
tenant_id	uuid	Y	RLS
agent_id	text	Y	e.g. `ap_coding`
agent_version	int	Y	FK agent_version
invoked_by_user_id	uuid	Y	whose permissions apply
trigger	enum	Y	USER, EVENT, SCHEDULE
status	enum	Y	PLANNING, ACTING, AWAITING_APPROVAL, COMMITTED, REJECTED, FAILED, KILLED
tokens_in / tokens_out	bigint	Y	
cost_usd	numeric(12,6)	Y	
started_at / ended_at	timestamptz	Y/N	

`agent_action`: action_id, run_id, seq, tool_name, input_json (redacted), output_hash, output_json (redacted, ≤ 64 KB), approval_task_id, approver_user_id, decision (APPROVED/REJECTED/AUTO), decided_at, committed_entity_ref.

`eval_run`: eval_run_id, agent_id, agent_version, dataset_id, dataset_version, metrics_json, passed bool, ci_ref, created_at. **`token_budget`**: tenant_id, month, limit_in, limit_out, used_in, used_out, state (OK/SOFT/HARD).

Invariants: `agent_action.input_json` never contains raw PII (checked by a redaction scanner at write); an action with approval_mode `approve_before_commit` has `committed_entity_ref` only if decision ∈ {APPROVED}; a run's `invoked_by_user_id` is immutable; a production `agent_version` has a passing `eval_run`. Indexing: `agent_run` (tenant_id, status, started_at), partition by month; `agent_action` (run_id, seq).

Additional entities (FR-47.14, 47.15). `knowledge_chunk`: chunk_id, tenant_id, source_type (ARTICLE, DOC, POLICY), source_ref, section, text (≤ 2 KB), embedding vector(1024), visibility, updated_at. `payment_prediction`: invoice_id, tenant_id, model_version, predicted_payment_date, p_late_30d numeric(5,4), features_hash, scored_at; `prediction_model`: model_version, tenant_id, trained_at, rows, mae_days, status (CANDIDATE, ACTIVE, RETIRED).

9.47.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	`/mcp` (SSE/streamable HTTP)	MCP server endpoint; `tools/list`, `tools/call`	per JSON-RPC id	user token
POST	`/api/v1/copilot/chat`	copilot conversation turn (stream)	n/a	CopilotUse
POST	`/api/v1/agents/{agent_id}/runs`	start a run	`Idempotency-Key`	AgentInvoke + agent's duty
GET	`/api/v1/agents/runs/{run_id}`	run status, actions, costs	n/a	AgentView
POST	`/api/v1/agents/runs/{run_id}/actions/{id}/decision`	approve/reject	by action id	approver duty
POST	`/api/v1/agents/kill-switch`	halt tenant or global	by scope	AgentAdmin / PlatformAdmin
GET	`/api/v1/agents/budgets/{tenant}`	budget status	n/a	AgentAdmin
POST	`/api/v1/agents/evals/runs`	trigger eval (CI)	by commit sha	CI service account
PUT	`/api/v1/agents/{agent_id}/versions/{v}/promote`	pin production version	ETag	AgentAdmin

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/knowledge/search`	hybrid search with citations (also exposed as MCP tool)	n/a (read)	CopilotUse
POST	`/api/v1/knowledge/documents`	upload tenant policy document for retrieval	`Idempotency-Key`	KnowledgeAdmin
GET	`/api/v1/ar/invoices/{id}/prediction`	payment prediction for an open invoice	n/a	CollectionsView
POST	`/api/v1/predictions/payment/retrain`	trigger retrain (also weekly job)	by week	AgentAdmin

Validation: run start rejects when tenant budget state HARD, kill switch active, or agent version lacks passing eval; decision endpoint rejects if the approver equals `invoked_by_user_id` and the action's amount exceeds the SoD self-approval limit (wave 48).

Event	Trigger	Payload fields	Consumers
`dev.passage.agent.run.completed.v1`	run end	run_id, agent_id, version, status, tool_calls[], tokens, cost_usd	audit, evals, copilot
`dev.passage.agent.approval.requested.v1`	gate reached	run_id, action_id, tool_name, proposal, confidence, risk_score	workflow inbox
`dev.passage.agent.budget.threshold.v1`	80%/100%	tenant_id, month, pct, state	alerting, admin UI
`dev.passage.agent.killswitch.changed.v1`	toggle	scope, active, by_user	all agent workers
`dev.passage.prediction.payment.scored.v1`	weekly scoring or invoice change	invoice_id, model_version, predicted_payment_date, p_late_30d	collections workspace, analytics
`dev.passage.agent.version.promoted.v1`	production pointer change	agent_id, from_version, to_version, eval_run_id, by_user	audit, workers

9.47.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Copilot panel (shell)	ask, explain record, draft, summarize, view tool calls	conversation, citations, "insert draft"	CopilotUse
Agent inbox (in workflow inbox)	approve, reject, edit proposal, bulk approve ≤ 50	proposal diff, confidence, rationale, amount	approver duty
Agent runs	filter, inspect timeline, replay in sandbox	run_id, agent, status, tokens, cost	AgentView
Agent administration	enable/disable per entity, set mode, thresholds, kill switch	agent, version, mode, auto threshold	AgentAdmin
Eval dashboard	compare versions, open failing cases	metric per version, trend	AgentAdmin

Rules: `Ctrl+/` opens copilot with current record context; approvals support keyboard `A/R`; generated text is marked with a badge and never auto-saved; copilot remembers mode per workspace per user.

9.47.7 Business rules and edge cases

- **BR-47.1** Run state machine: PLANNING → ACTING → (AWAITING_APPROVAL → COMMITTED | REJECTED) | COMMITTED (auto) | FAILED | KILLED; no transition out of COMMITTED/REJECTED/KILLED.
- **BR-47.2** Auto mode commits only when confidence ≥ threshold AND amount < limit AND the vendor/customer is not flagged `high_risk` AND no SoD conflict exists for the invoking user.
- **BR-47.3** Approval requests expire after 72 h → REJECTED with reason `EXPIRED`; the source document stays unposted.
- **BR-47.4** Token accounting uses provider-reported usage; if absent, estimated at 4 chars/token and flagged `estimated`.
- **BR-47.5** Kill switch active → running activities finish their current API call, no new tool calls; runs move to KILLED within 10 s.
- **BR-47.6** A tool error 4xx is not retried; 5xx/timeouts retry 3× with exponential backoff (1 s, 4 s, 16 s).
- **BR-47.7** Redaction pseudonyms are stable per (tenant, value) for 24 h so the model can refer back; the mapping is stored only in memory of the run worker and in Vault transit-encrypted form on `agent_run`.
- **BR-47.8** Eval datasets are versioned; a dataset change requires re-baselining all agent versions before promotion.
- **BR-47.9** Provider failover: if the default provider returns 5xx for 60 s, runs switch to the tenant's secondary provider only if it satisfies residency; otherwise runs pause.
- **BR-47.10** Cash application proposals never split a payment across more than 20 invoices, and never create write-offs above the wave-30 tolerance.
- **BR-47.11** A copilot answer that would require a tool the user lacks is answered with the reason (`AGT-0203`) and never with partial data from another user's cache.
- **BR-47.12** Payment predictions are recomputed for an invoice on any dunning, dispute or partial-payment event; a prediction older than 7 days is shown with a stale badge.
- **BR-47.13** Generated drafts inserted into a dunning letter or e-mail keep `origin=agent` and the run id on the resulting document; editing by the user clears neither.

9.47.8 Acceptance criteria

- **AC-47.1** (FR-47.1) Given an authenticated user, when `tools/list` is called, then only tools whose `required_duty` the user holds are returned.
- **AC-47.2** (FR-47.2) Given user without duty `APInvoicePost`, when the agent calls `passage.ap.invoice.post`, then gateway returns 403 and the action is logged `DENIED`.
- **AC-47.3** (FR-47.5, 47.9) Given the `eval-ap-coding-v3` dataset (2,000 invoices), when evals run, then top-1 accuracy ≥ 0.92 and the CI job passes.
- **AC-47.4** (FR-47.6, 47.9) Given `eval-cash-app-v2` (1,500 payments), when evals run, then precision ≥ 0.90.
- **AC-47.5** (FR-47.8) Given a coding proposal with confidence 0.80, when the run reaches the gate, then an approval task appears in the inbox and nothing is posted.
- **AC-47.6** (FR-47.12) Given a tenant kill switch toggled, when 10 s elapse, then all runs for that tenant are KILLED and no provider calls occur.

- **AC-47.7** (FR-47.11) Given a tenant at 100% budget, when a run is started, then it is rejected with `BUDGET_EXHAUSTED` and an alert was raised at 80%.
- **AC-47.8** (FR-47.12) Given an invoice containing a vendor IBAN, when the prompt is sent, then the provider request contains a pseudonym and the response shows the real IBAN only in the UI.
- **AC-47.9** (FR-47.3) Given "what was USMF revenue last month vs the month before", when asked, then the answer cites metric `revenue`, the two periods, and matches the explorer to the cent.
- **AC-47.10** (FR-47.13) Given provider switched to Bedrock, when the AP coding eval runs, then thresholds are still met.
- **AC-47.11** (FR-47.14) Given a published knowledge article "How to reverse a posted journal", when a user asks the copilot how to reverse a journal, then the answer cites the article section and no tool call touches ledger data.
- **AC-47.12** (FR-47.15) Given a tenant with 2,000 paid invoices, when the weekly retrain runs, then a model with MAE ≤ 6 days becomes ACTIVE and every open invoice has a prediction within 10 min.

9.47.9 Test plan and fixtures

Fixtures: `eval-ap-coding-v3` (2,000 labeled invoices, 40 vendors, 3 entities, from wave-19 fixtures plus hand labels); `eval-cash-app-v2` (1,500 payments with remittances, 12% partials, 5% multi-invoice); `eval-mrp-triage-v1` (800 action messages); `eval-anomaly-v1` (50k journals, 400 injected anomalies); `eval-ask-data-v1` (300 questions with expected Cube queries); `adv-prompts-500` (injection, exfiltration, policy bypass).

Test	Type	Target	Pass criterion
Permission inheritance	contract	gateway + 400 tools	0 cases where agent exceeds user
Tool schema conformance	contract	gateway	every tool validates against MCP spec and `x-passage-*`
Agent evals	eval	5 agents	thresholds in FR-47.9
Adversarial refusal	eval	copilot	≥ 0.99 correct refusals
Redaction scanner	unit/property	redaction	0 PII patterns in stored prompts across 100k generated samples
Kill switch latency	E2E	runtime	≤ 10 s
Run throughput	performance	runtime	500 concurrent runs, p95 < 30 s
Provider parity	contract	4 adapters	identical tool-use round trips

Reconciliation: tokens billed by provider ↔ `agent_run.tokens_*` within 1%; `agent_action` COMMITTED count ↔ posted documents tagged `origin=agent`.

9.47.10 Deployment and operations

Helm: `passage-mcp-gateway` (TS, 3 replicas), `passage-agent-runtime` (TS workers on Temporal namespace `agents`), `passage-copilot-bff`, `passage-agent-evals` (Python job runner). Config: `MODEL_PROVIDER_DEFAULT=anthropic`, `MODEL_ID` pinned per agent version, provider keys in Vault, `TOKEN_BUDGET_DEFAULT_IN=50000000`. Flags: `copilot.enabled`, `agent.<id>.mode` (off/suggest/auto) per tenant and legal entity. Jobs: budget rollover monthly, eval nightly on production datasets, model-card export weekly. SLOs: copilot first-token p95 < 2 s; run success rate ≥ 99%; gateway availability 99.9%. Alerts: eval regression, budget 80/100%, provider error rate > 5%, kill switch toggled. Runbook: rotate provider keys, replay a run in sandbox, demote version. Rollback: `promote` previous `agent_version`; Helm rollback.

9.47.11 Migration from AX / SAP

Source	Passage	Notes
Power Automate flows on business events	Agent or Temporal workflow definitions	catalogued; rebuilt by hand; the business-event catalog maps to CloudEvents types
`VendInvoiceAutomation` parameters	AP coding agent thresholds	tolerance and auto-post limits imported
`CustPaymMatch` rules	Cash application agent rules	matching rules become deterministic pre-filters
SAP Build Process Automation projects	Agent definitions	not convertible; documented
Joule / D365 Copilot settings	Copilot flags	n/a

Not migrated: conversation histories, Copilot usage data (no business value).

9.47.12 Dependencies, risks, open questions

Depends on: wave 46 (semantic layer for ask-the-data), wave 4 (Temporal, workflow inbox), wave 2 (OPA duties, token exchange), wave 19/30 (AP invoices, cash application APIs), wave 42 (PII classification).

Risk	Likelihood	Impact	Mitigation
Prompt injection via invoice text	High	High	tool allow-list per agent; adversarial eval; `approve_before_commit`
Eval datasets overfit to demo data	Medium	Medium	customer-labeled holdout from wave 49 parallel run
Provider cost spikes	Medium	Medium	budgets; prompt caching; smaller models for classification

Open questions: (1) Auto mode for AP coding in v1 for amounts below threshold — proposed default on, decide by wave 49 with first customer. (2) Open-weights model as residency fallback for DE tenants — default: vLLM adapter shipped, not certified; decide by wave 50.

9.47.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivale
`ai.provider.default`	tenant	enum	ANTHROPIC	ANTHROPIC, BEDROCK, AZURE, OPENAI_COMPATIBLE	provider for new runs	Copilot (Azure OpenAI, Microsoft-managed)	Joule (SAP-managed), AI Core deploym
`ai.provider.secondary`	tenant	enum	none	same, residency-checked (BR-47.9)	failover target	none	none
`ai.residency_region`	tenant	enum	from wave-48 residency	EU, US, IN, CA, UK, MX	restricts provider endpoints	LCS region	RISE region / Core region

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivale
`ai.model.<agent_id>`	tenant	text	per `agent_version`	provider model ids	model pinned per agent version	none	`ISLM` model version
`copilot.enabled`	tenant / legal entity	bool	true		shell panel on/off	Feature management "Copilot"	Joule activatic
`copilot.modes`	tenant	set	ask, explain, draft, summarize	subset	enabled modes	`CopilotFeatureManagement`	Joule capabili
`agent.<id>.mode`	legal entity	enum	SUGGEST	OFF, SUGGEST, AUTO	FR-47.5 behavior	`VendInvoiceAutomation` automatic posting switch	Business AI "propose vs a apply"
`agent.ap_coding.auto_threshold`	legal entity	numeric	0.95	0.80–1.00	min confidence for AUTO	none	Cash App ML threshold (`FINS_CASH_`
`agent.ap_coding.auto_amount_limit`	legal entity	money	USD 5,000 / EUR 4,500	≥ 0	AUTO cap	invoice automation tolerance	none
`agent.cash_app.max_invoices_per_payment`	tenant	int	20	1–50	BR-47.10	`CustPaymMatch` settings	Cash App ML parameters
`agent.approval_expiry_h`	tenant	int	72	1–240	BR-47.3	workflow escalation timeout	`SWWA` deadlin
`agent.run_timeout_h`	tenant	int	24	1–72	run including approval wait	none	none
`agent.tool_timeout_s`	tenant	int	60	10–300	per activity	none	none
`ai.budget.monthly_in` / `monthly_out`	tenant	bigint	50M / 10M	≥ 0	FR-47.11	none	AI Core quota
`ai.budget.soft_pct`	tenant	int	80	50–99	alert threshold	none	none
`ai.run_token_cap`	tenant	int	400,000	≤ 2M	per run	none	none
`ai.redaction.external`	tenant	bool	true (locked for external providers)	true	FR-47.12	none	Joule "no trair on customer c
`ai.pseudonym_ttl_h`	tenant	int	24	1–72	BR-47.7	none	none
`ai.killswitch`	tenant / global	bool	false		halt runs	none	`ISLM` deactiv scenario
`ai.run_log_retention_months`	tenant	int	24	6–120	`agent_run` / `agent_action` retention (archived per FR-46.18)	none	none
`ai.eval.regression_points`	global	numeric	1.0	0–5	FR-47.9 gate	none	none
`knowledge.sources`	tenant	set	ARTICLES, DOCS	ARTICLES, DOCS, POLICY	FR-47.14	none	none
`prediction.payment.min_rows`	tenant	int	500	100–10,000	FR-47.15 enablement	none	none
`prediction.payment.mae_max_days`	tenant	numeric	6	1–30	promotion gate	none	none

Setup sequence: confirm tenant residency (wave 48) → select provider and store keys in Vault → enable copilot per legal entity → set each agent to OFF/SUGGEST/AUTO with thresholds → assign approver duties → set budgets → run the eval suite against the tenant's labeled holdout before switching any agent to AUTO.

9.47.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Agent runs	list inquiry	agent, status, user, date, entity	run id, version, trigger, tokens, cost, duration, outcome	`agent_run`	Vendor invoice automation workspace, Workflow history	`ISLM` scenario monitoring, Cash App ML log
Agent action audit	drill-down	run	seq, tool, redacted input/output hash, decision, approver, committed ref	`agent_action`	`SysDatabaseLog` (partial)	change documents (`CDHDR`) partial
Approval queue (agent)	list inquiry	agent, amount, confidence	proposal, diff, rationale, age	workflow inbox view	Workflow work items	`SBWP` inbox
Eval dashboard	KPI	agent, version, dataset	accuracy, precision, recall, F1, AUROC, latency, cost per run, trend	`eval_run`	none	none
Token budget status	KPI	tenant, month	used/limit in and out, state, forecast exhaustion date	`token_budget`	none	AI Core quota usage
Copilot usage	KPI	period, workspace	sessions, questions per mode, citation rate, refusal rate	`copilot_session`	none	Joule usage analytics
Tool catalog	list inquiry	domain, duty, approval mode	tool name, version, read-only, required duty	`tool_definition`	none	none
Kill-switch and promotion log	list inquiry	scope, date	action, by user, reason, timestamp	audit events	none	none
Prompt and version registry	drill-down	agent	versions, model id, prompt hash, eval status, production pointer	`agent_version`	none	`ISLM` versions
Payment prediction accuracy	operational report	period, customer group	MAE days, p_late calibration, predicted vs actual	`payment_prediction`	Customer payment predictions (Finance insights)	FSCM collections scoring reports
Provider error and latency	operational report	provider, period	5xx rate, p95 latency, failovers	OpenTelemetry metrics	none	none

9.47.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Model card	agent version promoted	agent, version, model id, training/eval data description, metrics, limitations, residency, approver	Markdown + PDF	agent version	retained with version (EU AI Act transparency support); evidence for CC7.2
Eval report	CI eval run	dataset version, per-metric results vs thresholds, regressions, failing cases sample	HTML/PDF attached to CI	eval run id	24 months
Agent run audit extract	request by auditor	run timeline, actions, approvals, redaction markers	JSON + PDF	run id	per wave-48 retention (24 months then archive)
Generated drafts (dunning letter, e-mail, PO text)	user clicks "insert draft"	the wave-30/17 document carries the text; badge `origin=agent`	as the host document	host document's sequence	host document's rules

No document is posted or sent by this wave on its own; every outbound artifact is a host document from another wave with the agent origin recorded.

9.47.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Budget rollover	monthly, 1st 00:00 UTC	tenant	(tenant, month) unique	< 1 min	retry; previous month frozen	missing rollover
Token reconciliation	daily 03:00	tenant, provider	provider usage report id	< 5 min	flag `estimated` rows	variance > 1%
Nightly production eval	daily 01:00	agent	(agent_version, dataset_version, date)	< 30 min	alert on regression	metric drop > 1 pt
Approval expiry sweeper	hourly	tenant	action state	< 1 min	BR-47.3	expired count
Model-card export	weekly Monday	agent	version	< 5 min	retry	none
Tool catalog regeneration	on deploy	domain	OpenAPI hash	< 2 min	deploy blocked if schema invalid	catalog drift
Knowledge index refresh	every 5 min / on publish event	source	source updated_at	< 1 min	retry 3×	index lag > 15 min
Payment model retrain and scoring	weekly Sunday 02:00	tenant	model_version	< 20 min per tenant	keep previous ACTIVE	MAE > max
Run-log archive	weekly	tenant, month	FR-46.18 mover	< 30 min	as FR-46.18	pending > 7 days
Provider health probe	every 30 s	provider	n/a	< 1 s	BR-47.9 failover	error rate > 5%

9.47.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
CopilotUse	`POST /copilot/chat`, `/mcp` tools/list and read tools, `POST /knowledge/search`	none (tools are gated by the user's own duties)	every business role when `copilot.enabled`
AgentInvoke	`POST /agents/{id}/runs`	with the agent's commit duty when in AUTO mode (`AI-001`, HIGH: a user cannot both invoke AUTO and approve)	AP clerk (ap_coding), Cash clerk (cash_app), Planner (mrp_triage), Collector (collections)
AgentApprove (per agent)	`POST /agents/runs/{id}/actions/{a}/decision`	with AgentInvoke for the same run (BR: approver ≠ invoker) and SoD rules of the underlying posting duty	AP supervisor, AR supervisor, Planning lead, Credit manager
AgentView	`GET /agents/runs`, timeline, costs	none	Auditor, Finance manager
AgentAdmin	mode and thresholds, `PUT /versions/{v}/promote`, tenant kill switch, budgets	with AgentApprove (MEDIUM, `AI-002`)	Application administrator
PlatformAdmin (global kill switch)	`POST /agents/kill-switch` scope=global	n/a	Platform operator (break-glass, wave 48 FR-48.23)
PromptAuthor	create `agent_version` drafts, edit prompts and tool lists	with AgentAdmin promote (MEDIUM, `AI-003`: author cannot promote own version)	AI engineer
EvalRun (service)	`POST /agents/evals/runs`	n/a	CI service account
KnowledgeAdmin	`POST /knowledge/documents`, source settings	none	Application administrator

9.47.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
AGT-0201	error	Agent runs are halted by the {scope} kill switch.	kill switch active	AgentAdmin/PlatformAdmin clears it	FR-47.12
AGT-0202	error	Monthly token budget exhausted (BUDGET_EXHAUSTED); copilot runs in deterministic mode until {reset_date}.	budget state HARD	raise budget or wait	FR-47.11
AGT-0203	info	I cannot do that: you do not hold duty {duty} required by {tool}.	missing duty	request access	FR-47.2 / BR-47.11
AGT-0204	error	Agent version {v} has no passing evaluation and cannot be promoted.	eval missing or failed	run evals	FR-47.9
AGT-0205	warning	Token budget at {pct}% for {month}.	soft threshold	review usage	FR-47.11
AGT-0206	error	Provider {provider} is unavailable and no residency-compliant secondary is configured; run paused.	provider outage	configure secondary or wait	BR-47.9
AGT-0207	error	Provider endpoint {endpoint} is outside tenant residency {region}.	residency violation	choose compliant endpoint	FR-47.12 / BR-48.7
AGT-0208	warning	Approval request expired after {hours} h; nothing was posted.	BR-47.3	re-run agent	BR-47.3
AGT-0209	error	You cannot approve an action from a run you started above the self-approval limit.	approver = invoker	route to another approver	9.47.5 validation
AGT-0210	error	Prompt rejected: unredacted PII detected before provider call.	redaction scanner	report; run is FAILED	FR-47.12

Code	Severity	User text	Cause	Remediation	Raised by
AGT-0211	info	Token usage estimated (provider reported no usage).	BR-47.4	none	BR-47.4
AGT-0212	warning	No knowledge source matches your question with sufficient confidence.	similarity < 0.75	rephrase or ask an administrator	FR-47.14
AGT-0213	warning	Payment prediction not available: fewer than {min_rows} paid invoices.	enablement threshold	heuristic shown	FR-47.15
AGT-0214	error	Tool {tool} failed with {status}; not retried.	4xx	fix input or permissions	BR-47.6

9.47.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation under 3× load
Copilot first token	10k users, 3 questions/user/day, peak 200 questions/min	800 ms / 2 s / 4 s	50 concurrent streams per BFF pod	`copilot_session` 20 GB	by (tenant, month)	queue with position shown; deterministic answers for ask-data when provider saturated
`tools/list`	per session	20 ms / 80 ms / 150 ms	500/s	n/a	cached per (user duties hash, catalog version) 5 min	n/a
Tool call via gateway	5M/month	+5 ms / +15 ms / +30 ms over API	1k/s	n/a	n/a	gateway autoscale
Agent run (excl. approval)	20k runs/day, peak 500 concurrent	8 s / 30 s / 60 s	500 concurrent per Temporal namespace	`agent_run` 15 GB, `agent_action` 80 GB (redacted)	partition by month	runs queue; SLA warning at 2×
Eval suite (5 agents)	per commit	< 30 min	n/a	`eval_run` 5 GB	n/a	nightly only
Redaction pass	per prompt ≤ 100 KB	20 ms / 60 ms / 120 ms	n/a	n/a	n/a	n/a
Knowledge search	50k/day	100 ms / 300 ms / 600 ms	50/s	`knowledge_chunk` 10 GB per 100k articles	`pgvector` HNSW (m=16, ef=64) per tenant	ef lowered to 32
Payment scoring	1M open invoices	< 10 min batch	2k invoices/s	`payment_prediction` 3 GB	(tenant, invoice)	spreads over 2 windows

9.47.20 Localization and compliance hooks

- Response language follows the user's locale (wave 2); glossary terms (duty names, document types) come from the label framework so German users see "Buchungsbeleg", not a translation of the English.
- PII categories redacted before external providers follow the wave-42 classification extended by country packs (DE: religion, works-council membership; IN: Aadhaar; US: SSN; MX: CURP/RFC; FR: NIR); pseudonymization is mandatory for all tenants with residency EU or UK.
- Provider residency: tenants with residency EU/UK/IN may only use endpoints in-region (BR-48.7); the open-weights adapter is the fallback where no in-region hosted provider exists; the admin console shows the compliant provider list per region.
- Transparency: generated text is labeled in the UI in the user's language; the model card and action audit are the evidence objects for AI-governance questions (EU AI Act Article 50 transparency support; NIST AI RMF map/measure functions).
- Extension points: `country_pack.pii_categories[]`, `ai.residency_region` from wave-48 tenant residency, `agent.<id>.mode` per legal entity, prompt glossaries per language in `agent_version`.

9.47.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
(no GAP or PARTIAL register row carries current or proposed wave 47)	—	—	rows below are secondary references and inventory COVERED-47 rows verified against this wave
SAP 13.99.2 row 304 — 13.26 Solution database `IS01` / `IS02` (GAP, propose 45 "or wave 47 copilot answers")	GAP-45	ADDED as FR-47.14 (retrieval over wave-45 articles); article authoring itself DEFERRED to wave 45 (owner)	copilot cites knowledge articles; symptom/solution structure lives in wave 45
SAP 13.99.2 row 401 — 13.37 Signavio Process AI / Process Insights (PARTIAL-46)	PARTIAL-46	SUPERSEDED by FR-46.17 + copilot `ask_data` over `mart.process_event`	closed in 9.46.21
AX 12.3 inventory row "Customer payment predictions (D365 Finance insights)"	COVERED-47 (by name only)	ADDED as FR-47.15	prediction model now specified with gates
AX 12.30 inventory row "AI: model provider configuration, token budgets, kill switch"	COVERED-47	confirmed by FR-47.11–47.13 and 9.47.13	
AX 12.31 inventory rows "Copilot / Ask-the-data", "Copilot → Agent runs", "Copilot settings"	COVERED-47	confirmed by FR-47.3, 47.4, 9.47.13	
AX 12.21 inventory row "Planning → MRP exception triage agent"	COVERED-47	confirmed by FR-47.7	
SAP 13.37 inventory rows "Responsible AI", "Agent action audit", "Token/cost budget", "AI governance policies", "Business AI scenario activation"	COVERED-47	confirmed by FR-47.5, 47.11, 47.12 and 9.47.13 (`agent.<id>.mode`)	
SAP 13.99.2 rows 348/352/354 — BRF+ / decision tables (PARTIAL-3, not this wave)	PARTIAL-3	SUPERSEDED by FR-3.13 (wave 3); `POST /v1/rules/{code}/evaluate` is exposed as MCP tool `passage.rules.ruleset.evaluate`, agents never embed rules in prompts	listed for traceability only

9.48 Wave 48: Globalization, Localization & Compliance Controls

9.48.1 Scope and business context

AX delivers country/region localizations, Electronic Reporting (`ERModel`, `ERFormat`, `ERModelMapping` configurations from the Regulatory Configuration Service), segregation of duties (`SysSecSegregationOfDutiesRule`, conflict table `SysSecSegregationOfDutiesConflict`) and `SysDatabaseLog`. SAP delivers country versions, Document and Reporting Compliance (e-documents, statutory reports), GRC Access Control rulesets with risk ids, and the Audit Information System. This wave adds a localization framework with seven country packs, an electronic reporting designer with a mandates registry, an SoD engine with mitigation and access certification, security hardening with SOC 2 control mapping and evidence automation, and GDPR/CCPA subject requests with retention and legal holds. Out of scope: additional countries (v1.x), payroll localizations beyond wave 43, ISO 27001 certification itself (evidence only).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
A regulator changes an XML schema; the ER configuration must be re-imported from RCS/LCS, tested in a sandbox and deployed, or an SAP note applied via `SNOTE` with a transport	Formats bundled with the application release cycle	Mandates registry with effective-dated `er_format` versions, schema validation in CI	Schema update 3 weeks → 1 day; 0 regulator rejections for schema errors
SoD conflicts in `SysSecSegregationOfDutiesConflict` are found after assignment; GRC AC risk analysis runs as a batch and mitigations live in a separate system	Detection decoupled from role assignment	`sod/simulate` before save; nightly evaluation; mitigation workflow in the same workspace	Detection latency 1 day batch → < 2 s at assignment; unmitigated HIGH conflicts 0
SOC 2 evidence is screenshots and exports gathered by hand each audit (realm settings, backup logs, scan reports)	No system of record for controls	Control library with scheduled collectors and hashed artifacts	Evidence effort 20 person-days/audit → 1; 100% evidence machine-collected
GDPR requests require searching every table by hand (person search report in AX, IRF in SAP) and erasure risks breaking postings	PII not classified; no fan-out workflow	Classification-driven discovery, Temporal fan-out per domain, pseudonymization	Request fulfillment 25 days → 2; 0 broken financial references
`SysDatabaseLog` slows posting and fills the database; `CDHDR/CDPOS` is incomplete for custom tables	Application-side audit writes	CDC-derived audit, configuration only chooses what to surface	Posting overhead from audit 15% → 0%; 100% field coverage

9.48.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Country localization	Country/region-specific features gated by `CountryRegionCodes`, Localization feature keys	Country versions, `T005` country settings, localization BC sets	`country_pack` with tax, reports, layouts, bank formats, calendars	Full (7 countries)
Electronic reporting	ER `ERModel`, `ERFormat`, `ERModelMapping`, RCS, `ERFormatMappingRun`	DRC e-documents (`EDOC_COCKPIT`), statutory reports (`SRF`), Advanced Compliance Reporting	ER designer: `er_model`, `er_mapping`, `er_format` with mandates registry	Full
E-invoicing mandates	Electronic invoicing service (Italy SDI, Mexico CFDI)	DRC eDocument Framework (SDI, CFDI, Peppol, India IRN)	Mandate adapters: MX CFDI 4.0, IN IRP e-invoice, Peppol BIS 3 (wave 35), UK MTD VAT	Full for first set
Statutory tax reports	Sales tax reporting, EU sales list, Intrastat (`Intrastat` form)	`RFUMSV00` VAT return, ESL (`RFASLM00`), Intrastat (`VE01`)	ER formats: US sales tax, UK MTD VAT return, DE UStVA/ZM, FR CA3/DEB, IN GSTR-1/3B, MX DIOT, CA GST/HST	Full
Segregation of duties	`SysSecSegregationOfDutiesRule`, conflict detection, `SysSecSegregationOfDutiesConflict` resolution	GRC Access Control rulesets, risk ids (e.g. P001, F002), mitigating controls, UAR	`sod_rule`, `sod_conflict`, `mitigation`, `certification_campaign`	Improved (campaigns included)
Audit log	`SysDatabaseLog`, `DatabaseLog` form	`CDHDR/CDPOS`, AIS (`SECR`), `SM20` security audit log	wave-5 CDC audit + compliance evidence views	Full
Security compliance	Microsoft-managed SOC reports	SAP-managed for RISE; customer for on-prem	SOC 2 control mapping with automated evidence	Improved (customer-visible)
Data privacy	Person search report, GDPR request form (`GDPR request`)	Information Retrieval Framework (IRF), ILM retention (`IRM_CUST`), blocking	`subject_request`, `retention_policy`, `legal_hold`	Full
Data residency	LCS region selection	RISE region	tenant residency policy enforced on storage, lakehouse, AI provider	Full
Electronic signatures (21 CFR Part 11)	Electronic signature parameters, Signature requirements (`SIGSignatureSetup`), Signature log (`SIGSignatureLog`), Reason codes, user certificates	Digital signature (`DSIG`) in PI sheets/EBR, `ELSIG*` signature strategies, SSF	`esign_requirement`, `esign_record` with re-authentication and meaning (FR-48.13)	Full
Audit policies and audit cases	Audit policies (`AuditPolicyRule`, `AuditPolicyViolation`), rule types duplicate/keyword/list search/sampling, Audit cases (`CaseDetail`)	GRC Process Control continuous monitoring, Fraud Management detection strategies, AIS	`audit_policy` rules on FR-3.13 decision tables, `audit_violation`, `audit_case` (FR-48.14, 48.15)	Full
Process, risk and control library	Internal controls (`SysComplianceProcess`, `SysComplianceRisk`, `SysComplianceFiles`)	GRC Process Control (`GRFN_*`: processes, subprocesses, controls, risks, test plans)	`process_node`, `risk`, `control` linked to SOC 2 controls and evidence (FR-48.16)	Full
Electronic messaging	Electronic messages (`EMElectronicMessage`, `EMMessageProcessing`, processing actions, web service settings)	DRC eDocument cockpit (`EDOC_COCKPIT`), statutory reporting submission and response	`e_message` with type, status machine, processing actions, response tracking (FR-48.17)	Full
XBRL / iXBRL disclosure	Financial reporting XBRL export (limited), ER XBRL formats	Disclosure Management, Group Reporting XBRL, SAC for financial statements	`er_format` output type IXBRL with taxonomy registry (FR-48.18)	Full
Sanctions / denied-party screening	Third-party add-ons only	GTS Sanctioned Party List screening, Business Partner Screening (`/SAPSLL/SPL*`)	`screening_list`, `screening_hit` on parties and documents (FR-48.19)	Full
Export control and customs documents	Foreign trade parameters, Intrastat, commodity codes; export documents via third parties	GTS export management, license determination, customs declarations (`/SAPSLL/*`), `VX11N`	`trade_classification`, `export_license`, export document set (FR-48.20)	Partial (declarations through customs broker adapters)
Preference / origin	none	GTS preference processing (vendor declarations, long-term vendor declaration `/SAPSLL/PRE*`)	`origin_determination` from BOM roll-up (FR-48.21)	Full for USMCA and EU PEM
Emergency access	none (temporary security role assignment by admin)	GRC EAM firefighter IDs, FF log review	`elevated_session` with reason, time box and reviewer sign-off (FR-48.23)	Full
Read access logging	none	Read Access Logging (`SRAL`)	`read_log_policy` on classified fields (FR-48.25)	Full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Masked tenant copy	LCS database refresh (no masking)	TDMS, client copy with scrambling (`SCC4`/`SCCL`)	masked tenant clone (FR-48.26)	Improved
US workforce compliance	Worker compliance (US): I-9, ADA, visa, union agreements, EEO, VETS, ACA, FMLA	US HR infotypes (0077, 0094, 0442, 0495), ACA reporting (`PU_ACA`), new-hire reporting	US pack workforce data and outputs (FR-48.22, 48.31)	Full
Security incident and breach	none	none (customer ITSM)	`security_incident` with breach clock and notification tasks (FR-48.29)	Improved

9.48.3 Functional requirements

- **FR-48.1 Country pack framework** — Must. The system shall package per-country tax rules, statutory report formats, document layouts, bank formats (NACHA, BACS, SEPA CT/DD, NEFT/RTGS, SPEI, EFT), holiday calendars and address/tax-id validators as a versioned `country_pack`, installable per legal entity.
    - Notes: packs are metadata plus ER formats, never code forks; a legal entity has exactly one primary pack; pack upgrade is a versioned migration with preview.
- **FR-48.2 First seven packs** — Must. US (sales/use tax, 1099-NEC/MISC, ACH), UK (MTD VAT, BACS, Companies House XBRL tags), DE (USt‑VA, ZM, SEPA, GoBD export, DATEV), FR (CA3, DEB, FEC file, SEPA), IN (GST returns GSTR-1/3B, e-invoice IRP, TDS), MX (CFDI 4.0 with PAC, DIOT, SPEI), CA (GST/HST/PST, EFT, T4A).
- **FR-48.3 Electronic reporting designer** — Must. The system shall provide `er_model` (logical data model), `er_mapping` (Passage entities → model), `er_format` (XML/JSON/CSV/fixed-width output with XSD/JSON-schema validation), versioned with effective dates and a mandates registry (`mandate`: regulator, jurisdiction, schema version, effective_from, sunset).
    - Notes: format execution is a Temporal job; every output is stored with hash and regulator acknowledgment; schema validation failure blocks submission.
- **FR-48.4 SoD rules engine** — Must. The system shall evaluate rules of the form `duty_a CONFLICTS duty_b` (and privilege-level) against every user's effective roles on each assignment change and nightly, raising conflicts with severity.
    - Rule schema: `rule_id`, `name`, `duty_a`, `duty_b`, `severity` (HIGH/MEDIUM/LOW), `risk_id` (GRC-style, e.g. `P2P-001`), `description`, `mitigation_allowed` bool, `max_mitigation_days` (default 365), `status`.
    - Seed rules (10 of 50): `P2P-001` MaintainVendors ↔ ProcessVendorPayments (HIGH); `P2P-002` CreatePurchaseOrders ↔ ApprovePurchaseOrders (HIGH); `P2P-003` PostVendorInvoices ↔ ApproveVendorInvoices (HIGH); `P2P-004` ReceiveGoods ↔ PostVendorInvoices (MEDIUM); `O2C-001` MaintainCustomers ↔ PostCustomerPayments (HIGH); `O2C-002` CreateSalesOrders ↔ ApproveCreditLimits (HIGH); `O2C-003` IssueCreditNotes ↔ PostCustomerPayments (HIGH); `GL-001` PostGeneralJournals ↔ ApproveGeneralJournals (HIGH); `GL-002` MaintainChartOfAccounts ↔ PostGeneralJournals (MEDIUM); `SEC-001` MaintainUsersAndRoles ↔ any posting duty (HIGH); `HR-001` MaintainWorkerBankDetails ↔ RunPayroll (HIGH).
- **FR-48.5 Mitigation workflow** — Must. The system shall allow a conflict to be mitigated with a compensating control, approver (not the conflicted user), reason, and expiry ≤ `max_mitigation_days`; unmitigated HIGH conflicts block role assignment when tenant policy `sod.block_high=true`.
- **FR-48.6 Access certification campaigns** — Should. The system shall run campaigns where managers certify or revoke each direct report's roles within a window (default 14 days), auto-revoking uncertified roles at close when policy says so, with evidence export.
- **FR-48.7 SOC 2 control mapping and evidence** — Must. The system shall ship a control library mapped to TSC and automate evidence collection on a schedule.
    - Excerpt: CC6.1 logical access — evidence: Keycloak realm export, OPA bundle hash, RLS policy list (daily); CC6.2 provisioning — SCIM join/leave log, certification results (weekly); CC6.3 role changes — `sod_conflict` report, role-assignment audit (daily); CC6.6 boundary protection — Envoy Gateway TLS config, NetworkPolicy manifests (daily); CC6.7 data in transit/at rest — cert-manager inventory, CNPG encryption settings (weekly); CC6.8 malware/unauthorized software — image signature verification logs, SBOM diff (per deploy); CC7.1 vulnerability detection — Trivy scan results, pen-test tracker (daily/quarterly); CC7.2 anomaly monitoring — alert rule inventory, journal-anomaly agent summary (daily); CC7.3 incident evaluation — incident records with severity and timeline (per incident); CC7.4 incident response — post-incident review, runbook link (per incident); CC7.5 recovery — backup/restore drill results, RPO/RTO measured (monthly).
- **FR-48.8 Security hardening** — Must. The system shall rotate all secrets ≤ 90 days via Vault, enforce signed images, run Trivy and dependency scanning per build, remediate external pen-test findings (high/critical = 0 at exit), and enforce tenant data residency (region for Postgres, object store, lakehouse, AI provider).
- **FR-48.9 Backup and restore** — Must. The system shall take continuous WAL archives and daily base backups per domain cluster (CloudNativePG), support point-in-time restore of one tenant into an isolated namespace, and run a monthly drill measuring RPO ≤ 5 min and RTO ≤ 1 h.
- **FR-48.10 Subject requests** — Must. The system shall process GDPR/CCPA requests of type ACCESS, PORTABILITY, ERASURE, RECTIFICATION, RESTRICTION with the flow: intake (portal or admin) → identity verification (two attributes + e-mail link) → scope discovery (search party, worker, user, contact across all domains via the PII classification) → legal-hold and retention check → domain fan-out (Temporal child workflow per domain) → compile (ACCESS/PORTABILITY: JSON + PDF bundle) or apply (ERASURE: pseudonymize PII fields, keep financial records with `erased=true`) → approval by privacy officer → deliver via signed link → close within 30 days (GDPR) / 45 days (CCPA), with the deadline tracked.
- **FR-48.11 Retention and legal hold** — Must. The system shall apply retention policies per record class (e.g. financial documents 10 years DE, 7 years US; HR 6 years post-termination; logs 1 year) and legal holds that suspend erasure and retention purge for named scopes.
- **FR-48.12 Database log equivalent** — Should. The system shall offer `SysDatabaseLog`-style configuration: pick entities/fields to surface in the compliance audit view, backed by the wave-5 CDC audit, with no extra write path.
- **FR-48.13 Electronic signatures** — Must. The system shall require an electronic signature, configured in `esign_requirement` per (entity type, action or field, legal entity), before the guarded action commits: the signer re-authenticates (password plus TOTP or WebAuthn through Keycloak step-up, `acr=esign`), selects a reason code, sees the meaning of the signature (authored, reviewed, approved, released, verified) and the record summary, and the system stores an `esign_record` bound to the record version hash.
    - Notes: satisfies 21 CFR Part 11 §11.50 (name, date/time, meaning on the signed record and its printouts), §11.70 (signature-to-record linking by hash), §11.200 (two components at the first signing of a session, one at subsequent signings within 15 min); optional X.509 user certificate (PAdES/XAdES) for

documents leaving the system; three failed re-authentications lock signing for 30 min and raise `ESG-0603`; consumers: quality release (wave 33), batch record and PI-sheet steps (wave 24), master-recipe approval, ECO release (wave 9), payment approval, period close; signatures are never editable — a "reset" creates a revocation record with reason by `ESignAdmin` and invalidates downstream approvals; the signature log is a read projection of `esign_record`.

- **FR-48.14 Audit policies** — Must. The system shall run post-hoc audit policies over expense reports (wave 43), vendor invoices (wave 19), purchase orders (wave 17) and general journals (wave 11), with rule types DUPLICATE (same vendor + amount ± tolerance + date window), KEYWORD (list match in text fields), LIST_SEARCH (party in a watch list, e.g. employees as vendors), SAMPLING (random n% or every k-th, seeded), THRESHOLD (amount above a value per category, premium/mileage limits), WEEKEND_HOLIDAY (posted outside business calendar) and AGGREGATE (sum per party per period above limit).
    - Notes: rule conditions and thresholds are FR-3.13 decision tables (`rules.eval('audit.<rule_type>', …)`), never a private engine; policies are scoped by organization node (legal entity, department) and effective-dated; a run evaluates documents in a date window, creates `audit_violation` rows (document, rule, score, evidence fields) and is idempotent on (policy version, document id); SAMPLING uses a stored seed so a rerun selects the same documents.
- **FR-48.15 Audit cases** — Must. The system shall group violations into `audit_case` records by case category (e.g. duplicate payment, policy breach, sampled review) with assignee, due date, status (OPEN, IN_REVIEW, RESOLVED_NO_ISSUE, RESOLVED_ISSUE, CLOSED, ARCHIVED), findings, recovery amount and linked documents; closing requires a resolution code; closed cases older than `audit.case_archive_days` move to the archive tier (FR-46.18).
- **FR-48.16 Process, risk and control library** — Should. The system shall maintain a hierarchical process library (process → subprocess → activity, up to 5 levels), a risk library (risk id, description, likelihood, impact, owner) and controls (control id, type preventive/detective, automation manual/automated/IT-dependent, frequency, owner, test procedure) linked many-to-many; controls map to SOC 2 TSC ids, SoD rules, audit policies and evidence collectors; process documents are files in the wave-5 document store with templates per document type and versions.
    - Notes: an automated control links to its evidence collector (FR-48.7) so the library shows last evidence and status; control tests (design and operating effectiveness) are recorded with sample, tester, result and exceptions.
- **FR-48.17 Electronic messaging lifecycle** — Must. The system shall track statutory submissions as `e_message` records of a message type (UK MTD VAT return, DE ELSTER USt VA, FR EDI-TVA, IN GSTR-1, MX DIOT, CA GST34 NETFILE, US 1099 FIRE) through statuses defined per type (e.g. CREATED → GENERATED → VALIDATED → SENT → ACKNOWLEDGED | REJECTED → CORRECTED), with processing actions per transition (run ER format, validate schema, call web service, import response, await user input) and additional fields (period key, receipt id, obligation id).
    - Notes: web service settings (endpoint, OAuth client or certificate from Vault, fraud-prevention headers for HMRC MTD) are per message type and legal entity; responses are parsed into structured status and stored with the raw payload; a rejected message keeps its history and a correction creates a new item with `supersedes` (BR-48.3); e-invoice clearance (CFDI, IRP, Peppol) stays in wave 35 and reports its status here for one inbox.
- **FR-48.18 XBRL and iXBRL disclosure** — Should. The system shall add output type IXBRL and XBRL to `er_format`, with a taxonomy registry (UK FRS 102/IFRS taxonomies for Companies House and HMRC CT600, ESEF ESMA taxonomy, US GAAP taxonomy reference) and a tagging map from wave-15 financial statement lines to taxonomy concepts, contexts (entity, period, dimensions) and units.
    - Notes: output validates against the taxonomy (XBRL 2.1 and Inline XBRL 1.1 conformance via Arelle in a Python job); ESEF packages are zipped per the reporting-package spec with the block-tagged notes; filing itself is through electronic messaging (FR-48.17) where the regulator offers an API, else a download.
- **FR-48.19 Denied-party and sanctions screening** — Must. The system shall screen parties (customers, vendors, contacts, banks, workers on request) and documents (sales orders, deliveries, purchase orders, payments) against `screening_list` sources (OFAC SDN and Consolidated, EU Consolidated Financial Sanctions, UK HMT/OFSI, UN Security Council, US BIS Entity List, Denied Persons List, Unverified List) refreshed daily, using normalized fuzzy matching (token-set ratio plus transliteration; default hit threshold 0.88) on name, address and country.
    - Notes: a hit sets the party to `SCREENING_BLOCK` and blocks order release, shipment, PO send and payment until a `TradeComplianceOfficer` clears it as false positive (with reason; whitelist entry per party and list entry) or confirms it; rescreening runs on any list delta and on party change; a provider adapter (e.g. Descartes, LexisNexis Bridger, Dow Jones) can replace the internal matcher per tenant; every decision is retained 5 years.
- **FR-48.20 Export control and foreign trade documents** — Should. The system shall classify products by export classification (US ECCN/EAR99, EU dual-use Annex I number, HS/HTS tariff codes from wave 13 Intrastat setup), determine license requirements per (classification, destination country, end use) from a FR-3.13 decision table shipped by the pack, track `export_license` quantities and values, block deliveries that need a missing or exhausted license, and generate the export document set: commercial invoice, packing list for customs, certificate of origin, EUR.1 / origin statement, and US EEI data (AES) or EU export declaration data for the broker.
    - Notes: shipment execution and carrier documents are wave 34; this FR adds the compliance decision, documents and broker data file (JSON/EDIFACT CUSDEC via wave-35 hub); direct submission to customs systems (ACE, ATLAS, Delta G) is through broker adapters only in v1.
- **FR-48.21 Origin and preference determination** — Could. The system shall compute preferential origin for manufactured items per trade agreement (USMCA, EU–UK TCA, EU PEM) by rolling up the wave-21 BOM with component origins from supplier declarations (`supplier_origin_declaration`, long-term, validity ≤ 24 months), applying tariff-shift and regional-value-content rules (USMCA RVC transaction-value and net-cost methods; EU max non-originating content %) and storing the result per (item, agreement, period) for the commercial invoice and origin statement.
- **FR-48.22 US workforce compliance data** — Should. The system shall add to the US pack the worker compliance fields and workflows: Form I-9 (sections 1–3, document types List A/B/C, expiry tickler 90/30 days, E-Verify case number), ADA accommodation requests, visa/immigration records (type, expiry, sponsor), union agreements (local, seniority date, dues), FMLA leave tracking (12-week entitlement on a rolling 12-month look-back, intermittent leave in hours, eligibility check 1,250 hours/12 months) and workers' compensation class codes per position and state.
    - Notes: fields use the wave-42 country-specific fields framework; I-9 documents are stored with access limited to `HRComplianceUS` and are covered by read access logging (FR-48.25).
- **FR-48.23 Emergency (break-glass) access** — Must. The system shall grant elevated roles through an `elevated_session` request with reason code, ticket reference, requested roles and duration (≤ 4 h, default 1 h), approved by a designated approver (or self-activated with mandatory post-review when `breakglass.self_activate=true`), issued as a short-lived Keycloak token with claim `elev=<session_id>`, and every API call in the session logged; the session log is assigned to a reviewer who signs off within 3 business days.
    - Notes: SoD evaluation still runs and records conflicts for the session; sessions end automatically; unreviewed sessions older than 3 business days raise `SEC-0708` and appear on the CC6.3 evidence report.

- **FR-48.24 Export governance** — Should. The system shall apply download policies per data classification (PII, FINANCIAL, CONFIDENTIAL from wave 42 and entity metadata): allow, allow with watermark and audit, require reason, require approval, deny; per role and per channel (grid export, explorer export, report download, API bulk read > 10k rows, MCP tool bulk read).
    - Notes: every export records user, purpose, row count, columns, classification and file hash; XLSX/PDF carry a visible watermark with user and timestamp when the policy says so; the export log is evidence for CC6.1/CC6.7.
- **FR-48.25 Read access logging** — Should. The system shall log reads of fields marked in `read_log_policy` (default: national ids, bank accounts, salary, health, I-9 documents, date of birth) at the API layer — who, when, which record and field set, channel, purpose — without logging the values, retained per policy (default 2 years) and searchable by subject for subject-access requests.
- **FR-48.26 Masked tenant clone** — Should. The system shall clone a tenant (all domain schemas, lakehouse namespace, documents optional) into a non-production tenant with masking rules applied in-flight per field classification: deterministic tokenization for keys that must stay joinable (party names, e-mails), format-preserving randomization (IBAN with valid checksum, tax ids), date shifting (± 0–30 days per subject, consistent), nulling (free text notes), keeping amounts and quantities.
    - Notes: masking runs in the clone pipeline, never on the source; clones are tagged `masked=true` and refuse outbound integrations (bank, e-invoice, e-mail) by default; used by wave-49 rehearsals and wave-50 load tests (BR-50.5).
- **FR-48.27 India pack depth** — Must. The IN pack shall cover GST components (CGST, SGST/UTGST, IGST, cess) with place-of-supply rules, reverse charge, ITC eligibility and GSTR-2B reconciliation; TDS sections (194C, 194J, 194I, 194Q, 194H) with thresholds, PAN-based rates (206AA), lower-deduction certificates, challan ITNS 281 generation and CIN capture, quarterly 26Q/27Q returns (FVU file) and Form 16A certificates; TCS (206C(1H)) with collection and 27EQ return; 26AS/AIS reconciliation import.
- **FR-48.28 HR legal content per country pack** — Should. Each pack shall ship HR-legal fields, rules and outputs: DE (works-council co-determination flags on HR actions, contract types befristet/unbefristet, statutory leave minimums, Mutterschutz/Elternzeit), FR (CDI/CDD/intérim contract types with CDD maximum durations, DPAE pre-hire declaration data, mutuelle), UK (right-to-work checks, statutory sick/maternity pay eligibility, pension auto-enrolment flags), IN (PF/ESI numbers, gratuity eligibility after 5 years), MX (IMSS registration, aguinaldo and PTU entitlements), CA (Record of Employment data, provincial statutory holidays).
- **FR-48.29 Security incident and breach notification** — Must. The system shall maintain `security_incident` records (source: alert, user report, vendor, pen test; severity Sev-1–4; affected tenants, data categories and approximate subject count; timeline entries; containment, eradication, recovery steps; root cause; post-incident review) and, when a personal-data breach is confirmed, start breach clocks per applicable regime: GDPR Art. 33 supervisory authority within 72 h of awareness, Art. 34 subjects without undue delay when high risk, UK ICO 72 h, US state laws (per state table, e.g. 30–60 days), CA PIPEDA "as soon as feasible", IN CERT-In 6 h for reportable incidents, with notification tasks, templates and a record of what was sent when.
    - Notes: incidents are evidence for CC7.3/CC7.4 (FR-48.7); a Sev-1 incident automatically links its on-call page (wave 50 FR-50.10); customer-facing notifications are generated per affected tenant with the tenant's DPA contact.
- **FR-48.30 Evidence store and auditor access** — Must. The system shall store control evidence artifacts in an object-store bucket with object lock (compliance mode, retention = audit period + 1 year, default 2 years), hash each artifact (SHA-256) into a per-control hash chain, and give external auditors a time-boxed read-only `Auditor` role scoped to an audit period, with sample requests (auditor selects population items, owners attach evidence) tracked to completion.
    - Notes: the audit package export (ZIP with manifest, hashes and chain heads) can be verified offline with a published script; evidence cannot be deleted before lock expiry by any role.
- **FR-48.31 US pack statutory outputs (workforce and assets)** — Should. The US pack shall produce EEO-1 Component 1 (job category × race/ethnicity × sex), VETS-4212, ACA 1094-C/1095-C (IRS AIR XML), state new-hire reports (state formats, 20-day deadline tracking), workers' compensation payroll report by class code and state, OSHA Forms 300/300A/301 and the ITA CSV upload from the wave-42 injury and illness register, and IRS Form 4562 depreciation detail from the wave-37 tax book.

9.48.4 Data model

`sod_rule` (sod)

Field	Type	Required	Notes
rule_id	text	Y	PK, e.g. `P2P-001`
duty_a / duty_b	text	Y	OPA duty ids; `*POST*` wildcard allowed
severity	enum	Y	HIGH, MEDIUM, LOW
risk_id	text	Y	GRC-style id
mitigation_allowed	bool	Y	default true
max_mitigation_days	int	Y	default 365
status	enum	Y	ACTIVE, RETIRED
version	int	Y	

`sod_conflict`: conflict_id, tenant_id, user_id, rule_id, roles_json (which roles grant each duty), severity, detected_at, state (OPEN, MITIGATED, RESOLVED, ACCEPTED_RISK), mitigation_id, resolved_at.

`er_format` (ereporting)

Field	Type	Required	Notes
format_id	uuid	Y	PK
mandate_id	uuid	Y	FK mandate
version	int	Y	
output_type	enum	Y	XML, JSON, CSV, FIXED, PDF
schema_ref	text	N	XSD/JSON schema location, hash
template_json	jsonb	Y	element tree with bindings
effective_from / effective_to	date	Y/N	

Field	Type	Required	Notes
status	enum	Y	DRAFT, PUBLISHED, SUPERSEDED

`subject_request`: request_id, tenant_id, type, subject_ref (party/worker/user), verified_at, deadline_at, state (RECEIVED, VERIFIED, DISCOVERED, HELD, APPROVED, DELIVERED, CLOSED, REJECTED), domains_json, bundle_ref, privacy_officer_id.

`control_evidence`: evidence_id, control_id (e.g. `CC6.1`), collected_at, collector, artifact_ref (object store), hash, period_start, period_end, status (COLLECTED, REVIEWED, EXCEPTION).

Invariants: a user cannot approve their own mitigation; a PUBLISHED `er_format` with a schema_ref has a validated fixture; an ERASURE cannot complete while a legal hold covers the subject; `control_evidence.hash` matches the stored artifact. Indexing: `sod_conflict` (tenant_id, state, severity); `subject_request` (tenant_id, deadline_at); `control_evidence` (control_id, collected_at).

Additional entities (FR-48.13–48.31).

`esign_record` (compliance)

Field	Type	Required	Notes
esign_id	uuid	Y	PK
tenant_id	uuid	Y	RLS
requirement_id	uuid	Y	FK `esign_requirement` (entity_type, action, legal_entity, meaning, reason_codes[], certificate_required)
entity_type / record_id	text / uuid	Y	signed record
record_version_hash	text	Y	SHA-256 of the canonical record version
signer_user_id	uuid	Y	
meaning	enum	Y	AUTHORED, REVIEWED, APPROVED, RELEASED, VERIFIED
reason_code	text	Y	from `esign_reason`
auth_method	enum	Y	PASSWORD_TOTP, WEBAUTHN, CERTIFICATE
signed_at	timestamptz	Y	server time, UTC
certificate_fingerprint	text	N	when certificate signing is used
revoked_by_id	uuid	N	FK to revocation record

`audit_policy` / `audit_violation` / `audit_case`: `audit_policy` (policy_id, tenant_id, document_type EXPENSE/VENDOR_INVOICE/PURCHASE_ORDER/GENERAL_JOURNAL, rule_type, ruleset_code (FR-3.13), org_scope_json, effective_from/to, schedule, version, status); `audit_violation` (violation_id, policy_id, policy_version, document_type, document_id, score, evidence_json, run_id, case_id, state OPEN/IN_CASE/DISMISSED); `audit_case` (case_id, tenant_id, category, assignee_id, due_at, status, resolution_code, recovery_amount, currency, findings, closed_at, archived_at).

`e_message`: message_id, tenant_id, legal_entity_id, message_type, period_key, status, format_run_id, payload_ref, payload_hash, response_ref, receipt_id, supersedes_id, additional_fields_json, last_action, last_action_at.

`screening_hit`: hit_id, tenant_id, subject_type (PARTY/DOCUMENT), subject_id, list_code, list_entry_id, score numeric(4,3), matched_fields, state (OPEN, FALSE_POSITIVE, CONFIRMED), decided_by, reason, decided_at; `screening_list`: list_code, source_url, version, loaded_at, entry_count.

`trade_classification` / `export_license` / `origin_determination`: classification (item_id, regime US_EAR/EU_DUALUSE/HS, code, valid_from); license (license_id, regime, number, destination_countries[], items[], qty_limit, value_limit, used_qty, used_value, expires_at); origin (item_id, agreement, period, originating bool, method, rvc_pct, non_originating_pct, computed_at, declaration_refs[]).

`elevated_session`: session_id, user_id, roles[], reason_code, ticket_ref, requested_minutes, approved_by, started_at, ended_at, reviewer_id, reviewed_at, review_result.

`security_incident`: incident_id, severity, source, detected_at, aware_at, affected_tenants[], data_categories[], subject_estimate, personal_data_breach bool, regimes_json (regime, deadline_at, notified_at), status (TRIAGE, CONTAINED, ERADICATED, RECOVERED, CLOSED), root_cause, pir_ref.

Other entities: `read_log_policy` (field_fqn, classification, retention_days), `read_log` (partitioned by day), `export_policy`, `export_log`, `process_node`, `risk`, `control`, `control_test`, `supplier_origin_declaration`, `taxonomy`, `xbrl_tag_map`, `clone_job`.

Invariants (additional): an `esign_record` is append-only and its `record_version_hash` equals the hash of the version committed by the guarded action; a document blocked by an OPEN `screening_hit` cannot transition to released/shipped/paid; `elevated_session.ended_at − started_at ≤ 240 min`; an `audit_violation` belongs to at most one case; evidence objects are under object lock until `retain_until`. Indexing: `esign_record` (entity_type, record_id); `audit_violation` (policy_id, state); `screening_hit` (subject_type, subject_id, state); `read_log` partition by day, index (subject_ref, read_at); `e_message` (legal_entity_id, message_type, period_key).

9.48.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/localization/packs/{code}/install`	install pack on legal entity	by (entity, version)	LocalizationAdmin
POST	`/api/v1/ereporting/formats/{id}/run`	execute format for period	`Idempotency-Key`	per report duty
POST	`/api/v1/ereporting/submissions/{id}/submit`	submit to regulator adapter	by submission id	per report duty
GET	`/api/v1/sod/conflicts?state=OPEN`	list conflicts	n/a	SoDReview
POST	`/api/v1/sod/conflicts/{id}/mitigations`	create mitigation	`Idempotency-Key`	SoDApprove
POST	`/api/v1/sod/simulate`	simulate role assignment	n/a	SecurityAdmin
POST	`/api/v1/sod/campaigns`	start certification campaign	`Idempotency-Key`	SoDAdmin
GET	`/api/v1/compliance/controls/{id}/evidence?period=`	evidence listing	n/a	ComplianceView
POST	`/api/v1/privacy/requests`	intake subject request	`Idempotency-Key`	PrivacyIntake or portal
POST	`/api/v1/privacy/holds`	create legal hold	`Idempotency-Key`	LegalHoldAdmin
POST	`/api/v1/ops/backups/restore-drill`	run PITR drill into namespace	by drill id	PlatformAdmin

Method	Path	Purpose	Idempotency	Auth duty
GET/PUT	`/api/v1/compliance/esign/requirements`	maintain signature requirements and reason codes	ETag	ESignAdmin
POST	`/api/v1/compliance/esign/sign`	sign a pending action (step-up token required)	by (requirement, record_version_hash)	signer's action duty
POST	`/api/v1/compliance/esign/{id}/revoke`	revoke with reason	by id	ESignAdmin
POST	`/api/v1/compliance/audit-policies/{id}/runs`	run audit policy for a window	by (policy version, window)	AuditPolicyRun
POST	`/api/v1/compliance/audit-cases`	create case from violations	`Idempotency-Key`	AuditCaseManage
POST	`/api/v1/ereporting/messages/{id}/actions/{action}`	execute processing action on an electronic message	by (message, action, status)	per report duty
POST	`/api/v1/trade/screening/check`	screen a party or document on demand	n/a (read)	TradeComplianceView
POST	`/api/v1/trade/screening/hits/{id}/decision`	clear or confirm a hit	by hit	TradeComplianceOfficer
POST	`/api/v1/trade/export-checks`	license determination for a delivery	n/a (read)	TradeComplianceView
POST	`/api/v1/security/elevated-sessions`	request break-glass session	`Idempotency-Key`	any user with an eligible role mapping
POST	`/api/v1/security/incidents`	open security incident	`Idempotency-Key`	SecurityIncidentManage
POST	`/api/v1/ops/tenants/{id}/masked-clone`	clone tenant with masking	by clone job id	PlatformAdmin + second approver

Validation: pack install rejects if the entity's fiscal calendar is open-period incompatible; `sod/simulate` returns conflicts before assignment is saved; a `run` of a format outside `effective_from..to` is rejected.

Event	Trigger	Payload fields	Consumers
`dev.passage.ereporting.document.submitted.v1`	regulator ack	mandate_id, format_version, document_ref, ack_id, status	AR/AP (mark invoice sent), compliance
`dev.passage.sod.conflict.detected.v1`	evaluation	user_id, rule_id, severity, roles[]	workflow, compliance
`dev.passage.sod.mitigation.approved.v1`	approval	conflict_id, approver, expires_at	audit
`dev.passage.privacy.request.completed.v1`	close	request_id, type, affected_domains[], delivered_at	audit
`dev.passage.compliance.evidence.collected.v1`	collector	control_id, evidence_id, hash	compliance UI
`dev.passage.compliance.esign.signed.v1`	signature stored	esign_id, entity_type, record_id, meaning, signer, signed_at	quality, production, audit
`dev.passage.compliance.audit.violation.created.v1`	audit policy run	policy_id, document_type, document_id, score	audit case UI, workflow
`dev.passage.trade.screening.hit.v1`	screening match	subject_type, subject_id, list_code, score	sales, procurement, payments (block)
`dev.passage.security.elevated.started.v1` / `.ended.v1`	break-glass	session_id, user_id, roles[], reason	SOC, audit, reviewer inbox
`dev.passage.security.incident.breach_confirmed.v1`	breach flag set	incident_id, regimes[], deadlines[]	privacy, notifications
`dev.passage.ereporting.message.status_changed.v1`	e-message transition	message_id, type, from, to, receipt_id	compliance UI, tax

9.48.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Country packs	install, upgrade preview, view contents	code, version, entity, components	LocalizationAdmin
ER designer	edit model/mapping/format, validate against schema, run with fixture	element tree, binding, schema status	ERDesign
Mandates registry	add mandate, set effective dates, link formats	regulator, jurisdiction, schema version	ERDesign
SoD conflicts	mitigate, resolve, accept risk, export	user, rule, severity, roles, age	SoDReview/SoDApprove
Role assignment (wave 2)	simulate conflicts before save	conflicts preview	SecurityAdmin
Certification campaign	certify/revoke per user, progress	manager, user, roles, decision	Manager
Compliance workspace	evidence calendar, exceptions, export audit package	control, last evidence, status	ComplianceView
Privacy requests	verify, review scope, approve, deliver	type, subject, deadline, state	PrivacyOfficer
Signature requirements	add requirement per entity/action, set meaning and reason codes	entity, action, entity scope, meaning, certificate flag	ESignAdmin
Signature dialog (shell component)	re-authenticate, pick reason, sign	record summary, meaning, reason, signer	signer
Signature log	filter by record, user, meaning; export	signed_at, signer, meaning, reason, record, hash	ComplianceView
Audit policies	define rule, link decision table, schedule, run now	policy, rule type, scope, schedule, last run	AuditPolicyAdmin
Audit cases	assign, review evidence, resolve, close	case, category, violations, assignee, due, status	AuditCaseManage
Process, risk and control library	tree edit, link risks/controls/evidence, record tests	process, risk, control, owner, last test	ComplianceAdmin
Electronic messages	run next action, view response, correct	type, period, status, receipt, last action	per report duty
Screening workbench	review hits, compare fields, clear/confirm	subject, list, entry, score, matched fields	TradeComplianceOfficer
Export compliance	classify items, maintain licenses, view blocked deliveries	item, ECCN/HS, license, usage, blocks	TradeComplianceOfficer
Break-glass sessions	request, approve, review log	user, roles, reason, duration, review state	SecurityAdmin / reviewer
Security incidents	triage, timeline, breach clocks, notifications	severity, status, deadlines, notified	SecurityIncidentManage

Rules: SoD grid supports bulk mitigation ≤ 25 with one approver; evidence exports as a ZIP with manifest hash; privacy request deadline shows days remaining with red at ≤ 5.

9.48.7 Business rules and edge cases

- **BR-48.1** Conflict state machine: OPEN → MITIGATED (expires → OPEN) | RESOLVED (role removed) | ACCEPTED_RISK (LOW/MEDIUM only, by SoDAdmin).
- **BR-48.2** Conflicts are evaluated on effective duties including inherited roles and temporary delegations; a delegation that creates a HIGH conflict is refused.
- **BR-48.3** ER format outputs are immutable once submitted; corrections create a new submission with `supersedes`.
- **BR-48.4** Tax-period statutory runs lock the source tax settlement; re-running requires reversal of the settlement (wave 13).

- **BR-48.5** Erasure pseudonymizes `name`, `address`, `email`, `phone`, `tax_id`, `bank_account`, `date_of_birth` fields; monetary documents keep amounts, dates and dimension values.
- **BR-48.6** Retention purge runs monthly and skips any record under hold, in an open fiscal period, or referenced by an open document.
- **BR-48.7** Residency: a tenant with `residency=EU` cannot be assigned an AI provider endpoint outside the EU; the API rejects the configuration.
- **BR-48.8** Backup drill failing RPO or RTO opens a Sev-2 incident automatically.
- **BR-48.9** Certification campaign close: uncertified roles are revoked only when `campaign.auto_revoke=true`; otherwise flagged.
- **BR-48.10** Country-pack upgrade with a changed tax rate effective-dates the new rate; it never rewrites posted tax.
- **BR-48.11** E-signature: the guarded action and the `esign_record` commit in one transaction; if the record changed between dialog open and sign (hash mismatch), signing fails with `ESG-0602` and the user re-reviews.
- **BR-48.12** Within a continuous session a second signature needs only the password component if the first signing was ≤ 15 min ago; after 15 min of inactivity both components are required again.
- **BR-48.13** A signed record's printouts and PDF renders show signer name, date/time in the legal entity's time zone with UTC offset, and meaning (wave-46 template element `esign_block`).
- **BR-48.14** Audit DUPLICATE rule: two vendor invoices are duplicates when vendor equal, amount within max(0.01, 0.5%), invoice date within 30 days and normalized invoice numbers (strip non-alphanumerics, leading zeros) equal or Levenshtein ≤ 1; posted credit notes are excluded.
- **BR-48.15** Screening: a party cleared as false positive is not re-flagged for the same list entry unless the party's name, address or country changes or the list entry changes.
- **BR-48.16** A delivery to a destination with a comprehensive embargo (pack table, e.g. CU, IR, KP, SY, Crimea/DNR/LNR regions) is blocked regardless of classification; only `TradeComplianceOfficer` with a license reference can release.
- **BR-48.17** Electronic message transitions follow the type's status machine; an action not allowed from the current status is rejected; a message in SENT for longer than the type's response timeout (default 72 h) raises `EMS-0904`.
- **BR-48.18** Breach clock starts at `aware_at`, not `detected_at`; deadlines are computed per regime in UTC and displayed in the tenant time zone; a missed deadline cannot be closed without a late-notification justification.
- **BR-48.19** Break-glass sessions cannot grant `SecurityAdmin` or `PlatformAdmin` to the requester's own user; those require a second PlatformAdmin.
- **BR-48.20** FMLA entitlement uses the rolling 12-month period measured backward from the first day of leave; intermittent leave is deducted in hours at the worker's scheduled weekly hours.

9.48.8 Acceptance criteria

- **AC-48.1** (FR-48.2, 48.3) Given each of the 7 packs' statutory fixture, when the format runs, then the output validates against the regulator schema with 0 errors.
- **AC-48.2** (FR-48.4) Given 50 users with planted conflicting role combinations, when evaluation runs, then 50 conflicts are detected with the correct rule ids and 0 false negatives.
- **AC-48.3** (FR-48.5) Given an OPEN HIGH conflict and `sod.block_high=true`, when the role assignment is saved, then it is rejected with the rule id.
- **AC-48.4** (FR-48.5) Given a mitigation, when approved by a non-conflicted approver with reason and 180-day expiry, then the conflict is MITIGATED and `mitigation.approved` is emitted.
- **AC-48.5** (FR-48.9) Given a tenant with 10 GB across domains, when a PITR drill to T–20 min runs, then the restored namespace is queryable within 1 h and the lost-data window ≤ 5 min.
- **AC-48.6** (FR-48.10) Given an ACCESS request for a customer contact, when processed, then the bundle contains data from every domain holding the subject and the request closes within 30 days.
- **AC-48.7** (FR-48.10, 48.11) Given an ERASURE request for a party under legal hold, when processed, then state is HELD and no field is changed.
- **AC-48.8** (FR-48.7) Given control CC7.1, when the daily collector runs, then a Trivy report evidence record exists with a matching hash.
- **AC-48.9** (FR-48.8) Given the external pen-test report, when wave exit is reviewed, then open high/critical findings = 0.
- **AC-48.10** (FR-48.1) Given the DE pack installed on DEMF, when an invoice is printed, then the layout uses `de-DE` formats and the UStVA mapping includes the invoice's tax lines.
- **AC-48.11** (FR-48.13) Given a signature requirement on quality-order release with meaning RELEASED, when an inspector releases a lot, then the dialog requires password plus TOTP, the `esign_record` hash matches the released version, and the certificate of analysis PDF shows name, time and meaning.
- **AC-48.12** (FR-48.13) Given three failed re-authentications, when the user tries a fourth time within 30 min, then signing is locked with `ESG-0603` and a security event is logged.
- **AC-48.13** (FR-48.14, 48.15) Given 1,000 vendor invoices with 12 planted duplicates, when the DUPLICATE policy runs, then 12 violations are created, a rerun creates none, and grouping them yields one case per vendor.
- **AC-48.14** (FR-48.17) Given a UK MTD VAT return for period 26A1, when the user runs GENERATE → VALIDATE → SEND, then HMRC sandbox returns a receipt id stored on the message and status ACKNOWLEDGED.
- **AC-48.15** (FR-48.18) Given the UK entity's annual statements, when the iXBRL format runs, then the file passes Arelle validation against the FRS 102 taxonomy with 0 errors.

- **AC-48.16** (FR-48.19) Given a customer named like an OFAC SDN entry at score 0.92, when a sales order is released, then release is blocked, a hit is OPEN, and after clearance as false positive the order releases and no hit recurs on rescreen.
- **AC-48.17** (FR-48.20) Given an item with ECCN 5A002 shipped to a destination requiring a license and no license on file, when the delivery is confirmed, then it is blocked with `TRD-1003`.
- **AC-48.18** (FR-48.23) Given a break-glass session of 60 min, when it ends, then the token is invalid, every call is in the session log, and the reviewer inbox has a review task due in 3 business days.
- **AC-48.19** (FR-48.29) Given a confirmed personal-data breach affecting EU subjects with `aware_at` 2026-11-02T09:00Z, when the incident is saved, then a GDPR authority deadline of 2026-11-05T09:00Z and a notification task exist.
- **AC-48.20** (FR-48.26) Given a masked clone of the demo tenant, when compared to source, then 0 original e-mails, IBANs or tax ids appear, IBAN checksums are valid, and trial balances are identical.
- **AC-48.21** (FR-48.27) Given TDS section 194J at 10% on a vendor without PAN, when an invoice is posted, then 20% is withheld (206AA) and the next 26Q FVU file includes the deduction.

9.48.9 Test plan and fixtures

Fixtures: `fx-statutory-7` (one period of postings per country with expected outputs and regulator schemas: UK HMRC MTD VAT JSON, DE ELSTER USt VA XML, FR FEC, IN GSTR JSON, MX CFDI 4.0 XSD, US 1099 IRS FIRE, CA GST34 fields); `fx-sod-50` (50 planted users); `fx-privacy-subjects` (20 subjects spanning 9 domains); `fx-residency` (3 tenants EU/US/IN).

Test	Type	Target	Pass criterion
Schema validation	contract	7 formats	0 schema errors
SoD detection	unit/property	engine	50/50; random role sets never miss a seeded conflict
Mitigation workflow	E2E	sod	approver ≠ user enforced; expiry reopens
Privacy fan-out	E2E	privacy	all domains respond; bundle complete
PITR drill	performance	CNPG	RPO ≤ 5 min, RTO ≤ 1 h
Secrets rotation	contract	Vault	all leases ≤ 90 days
Residency enforcement	contract	config API	cross-region assignment rejected
Evidence collectors	unit	11 controls	artifact + hash per schedule

Reconciliation: statutory VAT return boxes ↔ tax settlement totals (wave 13) to the cent; conflicts count ↔ independent recomputation from role assignments; evidence count ↔ schedule expectations.

9.48.10 Deployment and operations

Helm: `passage-localization`, `passage-ereporting` (TS + Temporal), `passage-sod` (Go), `passage-compliance`, `passage-privacy`. Config: `COUNTRY_PACKS_ENABLED`, regulator adapter credentials (HMRC, PAC, IRP) in Vault, `SOD_BLOCK_HIGH`, `PRIVACY_DEADLINE_DAYS`. Migrations: `compliance` and `localization` databases. Flags per tenant: `pack.<code>`, `sod.enforce`, `privacy.portal`. Jobs: SoD nightly 00:30, evidence collectors per control schedule, retention purge monthly, backup drill monthly, secrets rotation. SLOs: format run p95 < 5 min for 100k lines; SoD evaluation < 2 s per assignment. Alerts: schema validation failure, regulator rejection, HIGH conflict unmitigated > 7 days, drill failure. Rollback: pack version downgrade with preview; ER format supersede; Helm rollback.

9.48.11 Migration from AX / SAP

Source	Passage	Notes
`SysSecSegregationOfDutiesRule`	`sod_rule`	duties mapped via wave-2 role mapping; severity imported
`SysSecSegregationOfDutiesConflict` with resolution	`sod_conflict` + `mitigation`	open conflicts re-evaluated; mitigations imported with original approver and expiry
GRC AC rulesets (risk ids, functions, actions)	`sod_rule`	function→duty mapping table; mitigating controls → `mitigation`
ER configurations (`ERModel`, `ERFormat` XML exports)	`er_model`, `er_format`	importer parses ER XML; formulas in ER expression language are translated where possible, else flagged
`SysDatabaseLog` configuration	FR-48.12 configuration	table/field list
`CDHDR/CDPOS` history	wave-49 lakehouse history	read-only
GDPR request logs	`subject_request` history	import closed requests for evidence

Not migrated: Microsoft/SAP SOC reports (vendor-owned), regulator credentials (re-issued).

9.48.12 Dependencies, risks, open questions

Depends on: wave 2 (roles/duties, OPA), wave 5 (CDC audit), wave 13 (tax), wave 35 (Peppol, e-invoicing transport), wave 42 (PII classification), wave 46 (evidence from reconciliation).

Risk	Likelihood	Impact	Mitigation
Regulator schema changes mid-wave	High	Medium	mandates registry with effective dates; schema fixtures pinned
SoD rule set incomplete for customer	Medium	High	50-rule seed plus GRC import
Erasure breaks referential integrity	Medium	High	pseudonymize, never delete rows

Open questions: (1) PAC provider for MX (default: Finkok sandbox); decide by wave 49. (2) Whether to pursue SOC 2 Type I before go-live — proposed yes, readiness assessment by wave 50.

9.48.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`pack.primary`	legal entity	code	none	US, UK, DE, FR, IN, MX, CA	installed country pack	Legal entity country/region, `CountryRegionCodes` features	Company code country (`OBY6`), country version
`pack.version_pin`	legal entity	semver	latest	published versions	blocks auto-upgrade	Feature management	Support package level
`er.schema_validation`	tenant	enum	BLOCK	BLOCK, WARN	FR-48.3 gate	ER format validation	DRC validation
`er.output_retention_years`	legal entity	int	from pack (DE 10, FR 10, US 7, UK 6, IN 8, MX 5, CA 6)	≥ pack minimum	output archive	ER job archive document type	DRC archive, ILM
`emsg.response_timeout_h.<type>`	legal entity	int	72	1–720	BR-48.17	Electronic message processing settings	eDocument cockpit timeout
`sod.block_high`	tenant	bool	true		FR-48.5 enforcement	SoD rule "block" behavior	GRC AC mitigation required before provisioning
`sod.max_mitigation_days`	tenant	int	365	1–730	mitigation expiry cap	none	Mitigating control validity
`sod.nightly_time`	tenant	time	00:30		batch evaluation	Verify compliance batch	GRC batch risk analysis (`GRAC_BATCH_RISK_ANALYSIS`)
`certification.window_days`	tenant	int	14	3–60	campaign window	none	UAR review period
`certification.auto_revoke`	tenant	bool	false		BR-48.9	none	UAR auto-remove
`esign.second_component`	tenant	enum	TOTP	TOTP, WEBAUTHN	Part 11 two-component	Electronic signature parameters (certificate/password)	`ELSIG` strategy (system signature with authorization)
`esign.session_reuse_min`	tenant	int	15	0–30	BR-48.12	none	none
`esign.lockout_attempts` / `lockout_min`	tenant	int	3 / 30	1–10 / 5–1440	lock after failures	none	SU01 lock
`audit.case_archive_days`	tenant	int	365	30–3650	FR-48.15 archiving	Close/archive audit cases	none
`audit.sampling_seed_policy`	tenant	enum	PER_RUN_STORED	PER_RUN_STORED, FIXED	reproducible samples	Audit sampling parameters	none
`screening.threshold`	tenant	numeric	0.88	0.70–0.99	hit score	none	SPL comparison threshold (`/SAPSLL/SPL_CUS`)
`screening.lists`	tenant	set	OFAC_SDN, OFAC_CONS, EU_CFSP, UK_OFSI, UN_SC, BIS_EL, BIS_DPL, BIS_UVL		sources	none	SPL list types
`screening.provider`	tenant	enum	INTERNAL	INTERNAL, DESCARTES, BRIDGER, DOWJONES	adapter	none	GTS or third party
`trade.license_check`	legal entity	enum	BLOCK	BLOCK, WARN, OFF	FR-48.20	none	GTS legal control (`/SAPSLL/LCLIC*`)
`breakglass.max_minutes`	tenant	int	60	15–240	session length	none	Firefighter validity
`breakglass.self_activate`	tenant	bool	false		FR-48.23	none	EAM decentralized firefighting
`breakglass.review_days`	tenant	int	3	1–10	review SLA	none	FF log review workflow
`readlog.retention_days`	tenant	int	730	90–3650	FR-48.25	none	`SRAL` log retention
`export.policy.<classification>`	tenant / role	enum	PII: REASON+WATERMARK; FINANCIAL: AUDIT	ALLOW, AUDIT, WATERMARK, REASON, APPROVAL, DENY	FR-48.24	none	none (custom)
`privacy.deadline_days`	tenant	int	30 (GDPR) / 45 (CCPA)	regulation table	FR-48.10	GDPR request settings	DPW deadlines
`retention.<record_class>`	legal entity	int years	from pack	≥ statutory	FR-48.11	D365 data retention (archive)	ILM policies (`IRMPOL`), residence rules
`residency.region`	tenant	enum	at provisioning	deployed regions	BR-48.7	LCS region	RISE region
`secrets.max_age_days`	platform	int	90	1–90	FR-48.8	none	none
`backup.drill_cron`	platform	cron	monthly 1st Sun 03:00	cron	FR-48.9	LCS DR	HANA backup test
`evidence.lock_years`	tenant	int	2	1–10	FR-48.30	none	none
`incident.breach_regimes`	tenant	set	from packs + residency	GDPR, UK_GDPR, US_STATE, PIPEDA, CERT_IN, LFPDPPP	FR-48.29	none	none
`masking.profile`	tenant	code	`default-v1`	profiles	FR-48.26	none	TDMS scrambling rules

Setup sequence per legal entity: assign primary pack → review pack upgrade preview → set retention per record class (cannot go below pack minimum) → map ER formats to mandates and message types → configure regulator credentials in Vault → load SoD seed rules and GRC import → define signature requirements (regulated tenants) → configure audit policies → enable screening lists → set export policies → schedule evidence collectors.

9.48.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Statutory reports catalog	list inquiry	country, mandate, period	report, format version, due date, last run, status	`mandate`, `er_format`	Electronic reporting workspace	DRC Run Statutory Reports
Electronic messages	list inquiry	type, entity, period, status	message, status, receipt id, last action, response	`e_message`	Electronic messages	`EDOC_COCKPIT`
VAT return (per pack)	statutory report	entity, period	boxes per pack (UK 1–9, DE Kz 81/86/66…, FR CA3 lines)	tax settlement (wave 13)	Sales tax payment report, ER VAT declaration	`RFUMSV00`, DRC VAT return

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
SoD conflicts	list inquiry	state, severity, rule, user	user, rule, roles, age, mitigation expiry	`sod_conflict`	SoD conflicts / unresolved	GRC risk analysis (user level)
Mitigation report	operational report	expiry window, approver	conflict, control, approver, expires_at	`mitigation`	none	GRC mitigation reports
Certification campaign status	KPI	campaign	% certified, revoked, overdue by manager	`certification_campaign`	none	UAR status
Signature log	list inquiry	record, signer, meaning, date	signed_at, signer, meaning, reason, hash	`esign_record`	Electronic signature log / report	`ELSIG` log, DSIG log
Audit policy violations	list inquiry	policy, document type, state	document, rule, score, case	`audit_violation`	Audit policy violations	GRC PC exceptions
Audit cases	list inquiry / operational report	category, status, assignee	case, violations, recovery, age	`audit_case`	Audit case inquiry / report	Fraud Management alerts
Control matrix	operational report	process, TSC, owner	control, risk, last test result, last evidence, exceptions	library + evidence	Internal control documents	GRC PC control matrix
Screening hits	list inquiry	state, list, subject type	subject, list entry, score, age, decision	`screening_hit`	none	`/SAPSLL/SPL_BL` blocked documents
Export-blocked deliveries	list inquiry	reason, destination	delivery, item, classification, license needed	`export_check`	none	GTS blocked documents (`/SAPSLL/BLRGSD`)
Read access log	drill-down	subject, user, field, date	read_at, user, record, fields, channel, purpose	`read_log`	none	`SRALMANAGER` log
Export log	list inquiry	user, classification, date	rows, columns, purpose, hash, policy	`export_log`	none	none
Break-glass sessions	list inquiry	user, review state	roles, reason, duration, calls, reviewer, result	`elevated_session`	none	GRC firefighter log report
Subject requests	list inquiry	type, state, deadline	days remaining, domains, officer	`subject_request`	GDPR request list	DPW, IRF results
Security incidents and breach clocks	KPI	severity, status	open incidents, MTTR, deadlines, notified on time %	`security_incident`	none	none
Evidence calendar	KPI	control, period	expected vs collected, exceptions	`control_evidence`	none	AIS (`SECR`)

9.48.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
VAT / GST / sales-tax returns (7 packs)	tax period close	boxes per regulator schema, entity tax id, period, declarant	UK MTD JSON, ELSTER XML, FR EDI-TVA, GSTR JSON, DIOT TXT, GST34 fields, US state files	`e_message` id; regulator receipt	statutory; 6–10 years per pack
FEC (FR), GoBD export (DE), DATEV export	on demand / year-end	journal lines with prescribed columns	FEC TXT (18 columns), GoBD index.xml + CSV, DATEV CSV	run id	10 years
SAF-T / audit file (where pack requires)	on demand	header, master files, GL entries, source documents	XML per national schema	run id	statutory
iXBRL accounts / ESEF package	annual	tagged statements and notes	XHTML iXBRL, ESEF ZIP	filing id	10 years
1099-NEC/MISC, 1096 (US)	year-end	payer, recipient, boxes	PDF copy B, IRS FIRE file	form sequence	7 years
ACA 1094-C/1095-C, EEO-1, VETS-4212, OSHA 300/300A/301, Form 4562 (US)	annual	per form	PDF, IRS AIR XML, EEOC/OSHA ITA CSV	form year	5–7 years (OSHA 5 years)
TDS 26Q/27Q, Form 16A, TCS 27EQ (IN)	quarterly	deductor TAN, deductees, sections, challans	FVU text file, PDF certificate	quarter + certificate number	8 years
CFDI 4.0 (MX)	invoice (wave 35 clearance)	per SAT schema with UUID and QR	XML + PDF	series/folio	5 years
Commercial invoice, packing list, certificate of origin, EUR.1/origin statement	export delivery	exporter, consignee, HS codes, origin, Incoterms, values, weights, license refs	PDF, broker JSON/EDIFACT	export document number per delivery	5 years (customs)
E-signature manifest	signed record print	signer, date/time with offset, meaning, reason	PDF block (wave-46 element), PAdES when certificate	n/a	with the signed record
Subject access bundle	privacy request	index, data per domain, processing purposes	JSON + PDF	request id	delivery receipt kept 3 years
Breach notification letters	breach confirmed	controller, nature, categories, counts, contact, measures	PDF, e-mail, regulator portal forms	incident id	5 years
SOC 2 audit package	on request	manifest, hashes, evidence index	ZIP	package id	evidence lock period
Certification campaign evidence	campaign close	manager decisions per user and role	XLSX, PDF	campaign id	3 years

9.48.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
SoD nightly evaluation	00:30 tenant time	tenant, user hash	(tenant, date)	< 10 min for 10k users	retry; last good result kept	new HIGH conflicts
Mitigation expiry sweep	daily 01:00	tenant	conflict state	< 1 min	reopen per BR-48.1	expiring in 30 days
Evidence collectors	per control (daily/weekly/per deploy)	control	(control, period)	< 5 min each	EXCEPTION status, retry 3×	missing evidence
Statutory format runs	per mandate calendar	entity, period	`Idempotency-Key` per (format, period)	< 5 min per 100k lines	message stays GENERATED with error	schema failure

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Electronic message response poller	every 15 min	message type	receipt id	< 1 min	BR-48.17 timeout	SENT > timeout
Audit policy runs	per policy schedule (default nightly 02:30)	policy, document type	(policy version, document id)	< 15 min per 1M documents	partial resume by window slice	violations spike > 3× baseline
Screening list refresh and delta rescreen	daily 05:00 UTC	list	list version	< 20 min for 1M parties	keep previous list; alert	list age > 48 h
Retention purge	monthly 2nd Sunday	record class, domain	purge manifest	< 4 h	BR-48.6 skips; manifest diff	purge errors
Read-log partition roll	daily	day	partition	< 5 min	retry	none
Secrets rotation	per secret ≤ 90 days	secret	Vault lease	< 1 min	rollback to previous version	lease age > 80 days
Backup PITR drill	monthly	tenant sample	drill id	< 1 h	Sev-2 (BR-48.8)	RPO/RTO breach
Breach deadline monitor	every 15 min	incident	n/a	< 1 s	page at T−12 h	deadline at risk
Certification campaign close	at window end	campaign	campaign state	< 10 min	BR-48.9	overdue managers
Masked clone	on request	tenant, domain	clone job checkpoint per table	< 2 h for 100 GB	resume per table	masking rule miss

9.48.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
LocalizationAdmin	pack install/upgrade, `retention.*` minimum checks	with ERDesign on the same pack (LOW)	Application administrator
ERDesign	ER designer, mandates registry, message types	with statutory submission duty (MEDIUM, `TAX-002`: design and submit)	Tax technologist
TaxReportSubmit	`ereporting/formats/{id}/run`, `submissions/{id}/submit`, message actions	with ERDesign (MEDIUM)	Tax accountant
SoDReview / SoDApprove / SoDAdmin	conflicts, mitigations, campaigns, rule maintenance	SoDApprove with SecurityAdmin (HIGH, `SEC-002`)	Internal control, Compliance officer
ESignAdmin	signature requirements, reason codes, revoke	with any signing duty on the same requirement (HIGH, `QA-001`)	QA systems administrator
AuditPolicyAdmin / AuditPolicyRun	audit policies, decision-table links, runs	with AP invoice posting (MEDIUM, `AUD-001`: audit own postings)	Internal audit
AuditCaseManage	cases, findings, recovery	with the audited document's posting duty (HIGH)	Internal audit
ComplianceAdmin / ComplianceView	control library, tests, evidence, audit package	ComplianceAdmin with PlatformAdmin (MEDIUM)	Compliance officer; Auditor (view, time-boxed)
TradeComplianceOfficer	screening decisions, licenses, classifications, embargo release	with SalesOrderRelease (HIGH, `TRD-001`)	Trade compliance
PrivacyOfficer / PrivacyIntake / LegalHoldAdmin	subject requests, holds	LegalHoldAdmin with ArchiveAdmin (MEDIUM)	DPO, Legal
SecurityIncidentManage	incidents, breach clocks, notifications	none	Security lead
BreakGlassApprove	approve and review elevated sessions	with requesting the same session (HIGH, BR-48.19)	Security lead
HRComplianceUS	I-9, visa, ADA, FMLA records	with Payroll run (MEDIUM, `HR-002`)	HR compliance

9.48.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
LOC-0501	error	Pack {code} {version} cannot be installed: fiscal calendar {cal} has open periods incompatible with the pack.	calendar mismatch	close or align periods	9.48.5 validation
LOC-0502	error	Retention for {class} cannot be below the statutory minimum of {years} years for {country}.	parameter under minimum	raise value	FR-48.11
ERP-0801	error	Output fails schema {schema}: {n} errors (first: {xpath} {msg}).	schema validation	fix mapping or data	FR-48.3
ERP-0802	error	Format {format} is not effective for period {period}.	effective dates	choose correct version	9.48.5 validation
EMS-0901	error	Action {action} is not allowed in status {status} for message type {type}.	status machine	run allowed action	BR-48.17
EMS-0904	warning	No response from {regulator} for {hours} h on message {id}.	timeout	poll manually; contact regulator	BR-48.17
EMS-0905	error	Regulator rejected message {id}: {code} {text}.	rejection	correct and resubmit	FR-48.17
SOD-1101	error	Assignment blocked: conflict {rule_id} ({severity}) between {duty_a} and {duty_b}.	HIGH conflict with block policy	remove role or mitigate	FR-48.5
SOD-1102	error	You cannot approve a mitigation for your own conflict.	self-approval	route to another approver	9.48.4 invariant
ESG-0601	info	Signature required: {meaning} for {entity} {id}.	requirement matched	sign or cancel	FR-48.13
ESG-0602	error	The record changed since you opened it; review the current version before signing.	hash mismatch	reopen	BR-48.11
ESG-0603	error	Signing locked for {minutes} min after repeated failed authentication.	lockout	wait or contact ESignAdmin	FR-48.13
AUD-0701	warning	{n} documents violate policy {policy}.	audit run	review in audit cases	FR-48.14
TRD-1001	error	{subject} matches {list} entry {entry} (score {score}); processing is blocked.	screening hit	TradeComplianceOfficer decides	FR-48.19
TRD-1002	error	Destination {country} is under embargo; delivery blocked.	embargo	release with license reference	BR-48.16

Code	Severity	User text	Cause	Remediation	Raised by
TRD-1003	error	Export license required for {item} ({classification}) to {country}; none valid.	license determination	obtain or assign license	FR-48.20
SEC-0707	error	Break-glass cannot grant {role} to yourself.	BR-48.19	request second PlatformAdmin	BR-48.19
SEC-0708	warning	Break-glass session {id} not reviewed after {days} business days.	review overdue	reviewer signs off	FR-48.23
SEC-0709	error	Download denied by export policy for {classification} data.	export policy DENY	request approval	FR-48.24
PRV-1201	error	Erasure blocked: subject is under legal hold {hold}.	hold	wait for hold release	FR-48.11
INC-1301	warning	Breach notification to {authority} due in {hours} h.	breach clock	send notification	FR-48.29
OPS-1401	error	Restore drill missed target: RPO {rpo} s / RTO {rto} min.	drill failure	Sev-2 opened	BR-48.8

9.48.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation under 3× load
SoD simulate at assignment	500 assignments/day	200 ms / 2 s / 3 s	20/s	`sod_conflict` < 1 GB	(tenant, user)	5 s; never skipped
SoD nightly	10k users × 300 roles	< 10 min	n/a	n/a	in-memory duty bitsets	30 min
ER format run	100k lines per return	1 min / 5 min / 8 min	10 concurrent per tenant	outputs 50 GB (10 years)	by (entity, period)	queued
E-signature commit	5k signatures/day (regulated tenant)	300 ms / 800 ms / 1.5 s (excl. user step-up)	20/s	`esign_record` 2 GB	(entity_type, record_id)	n/a
Audit policy run	1M documents/night	< 15 min	1.5k docs/s	`audit_violation` 1 GB	(policy, state), by month	window split
Screening on document release	50k checks/day, peak 200/min	30 ms / 120 ms / 250 ms	50/s	`screening_hit` < 1 GB; lists 2 GB	trigram + phonetic index on list names	fall back to async check that holds release
Delta rescreen	1M parties	< 20 min	1k parties/s	n/a	name token index	completes in next window
Read access logging	2M sensitive reads/day	+2 ms / +5 ms / +10 ms	async write via Kafka	300 GB/year	partition by day	buffer in Kafka, never drop
Subject request discovery	50 requests/month	< 10 min per request	n/a	bundles 5 GB	PII classification index	queued
Evidence collection	11 controls, ~400 artifacts/month	< 5 min each	n/a	50 GB/year (locked)	by control/period	delayed, flagged
Masked clone	100 GB tenant	< 2 h	15 MB/s per table stream	n/a	per-table parallelism 8	4 h

9.48.20 Localization and compliance hooks

This wave is the localization layer for every other wave. What varies and where it plugs in:

Area	Varies by	Country packs touching it	Extension point
Tax determination and returns	country, state/province, item and party tax groups	all 7	FR-3.13 decision tables with `country` input (sys layer), `er_format` per mandate, `e_message` types
Withholding	country, section	US (backup withholding, 1099), IN (TDS/TCS), MX (ISR/IVA retention)	wave-13 withholding engine rule sets; FR-48.27
Statutory files and audit exports	country	DE (GoBD, DATEV), FR (FEC), IN, MX (contabilidad electrónica), UK (MTD)	`er_format`
Document layouts and legal mentions	country, language	all 7	`country_pack.document_layouts[]`, wave-46 text blocks
Bank formats	country, bank	NACHA (US), BACS (UK), SEPA (DE, FR), NEFT/RTGS (IN), SPEI (MX), CPA 005 EFT (CA)	wave-19 payment format registry fed by pack
Calendars and fiscal years	country	IN April–March; holidays all	wave-6 calendars from pack
Number formats, address formats, tax-id validation	country	all	pack validators (GSTIN, RFC, USt-IdNr, SIREN/SIRET, VAT GB, BN)
Currency rounding	currency	JPY 0, KWD 3, CHF 0.05 cash rounding	wave-11 rounding rules
Retention	country, record class	DE/FR 10 y, IN 8 y, US 7 y, UK/CA 6 y, MX 5 y	`retention_policy`, archive mover (FR-46.18)
Privacy and breach regimes	residency, subject country	GDPR (DE, FR), UK GDPR, CCPA/US state, PIPEDA (CA), DPDP Act 2023 (IN), LFPDPPP (MX)	`incident.breach_regimes`, `privacy.deadline_days`
E-signature regimes	industry, country	US 21 CFR Part 11; EU eIDAS advanced signatures via certificate option	`esign_requirement.certificate_required`
Export control	exporter country, destination	US EAR, EU dual-use Regulation 2021/821, UK export control order, CA ECL	screening lists, classification regimes, license decision tables
HR legal	country	all 7 (FR-48.28, 48.22)	wave-42 country-specific fields framework
Language	user and document locale	en-US, en-GB, de-DE, fr-FR, hi-IN/en-IN, es-MX, fr-CA/en-CA	label framework import/export (wave 2 / AX "Translations")

9.48.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.99.3 row 22 — 12.9.1 Compliance → Electronic signatures (link)	GAP	ADDED as FR-48.13	
AX 12.99.3 row 23 — 12.9.2 Electronic signature log inquiry	GAP	ADDED as FR-48.13	signature log inquiry in 9.48.14

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.99.3 row 24 — 12.9.2 SSRS: Electronic signature report	GAP	ADDED as FR-48.13	signature log export / manifest in 9.48.15
AX 12.99.3 row 25 — 12.9.3 Electronic signature → Reset/revoke signatures	GAP	ADDED as FR-48.13	revocation record by ESignAdmin
AX 12.99.3 row 26 — 12.9.4 Electronic signature parameters, Signature requirements, Reason codes	GAP	ADDED as FR-48.13	parameters in 9.48.13
AX 12.99.3 row 79 — 12.29.1 Electronic signatures → Signature log / Reset signature	GAP	ADDED as FR-48.13	
AX 12.99.3 row 81 — 12.29.1 Electronic signatures → Certificates (user signing certificates)	GAP	ADDED as FR-48.13	optional certificate signing (PAdES/XAdES)
AX 12.99.3 row 84 — 12.29.2 Electronic signature inquiry	GAP	ADDED as FR-48.13	
AX 12.99.3 row 85 — 12.29.3 Electronic signatures → Signature verification / Reset	GAP	ADDED as FR-48.13	verification = hash recompute against record version
AX 12.99.3 row 86 — 12.29.4 Electronic signature parameters / requirements / reason codes / certificates	GAP	ADDED as FR-48.13	
AX 12.99.3 row 90 — 12.30.1 Sys admin → Electronic signature (link)	GAP	ADDED as FR-48.13	
AX 12.99.3 row 92 — 12.30.4 Sys admin → Electronic signature parameters (link)	GAP	ADDED as FR-48.13	
AX 12.99.3 row 37 — 12.12.2 SSRS: OSHA 300/300A/301 (US) (wave 42, US pack output 48)	GAP-42	ADDED as FR-48.31 (output); incident register DEFERRED to wave 42 (owner)	
AX 12.99.4 row 8 — 12.1.4 Tax (IN) → India-specific setup (tax components, TDS/TCS)	PARTIAL-48	ADDED as FR-48.27	
AX 12.99.4 row 12 — 12.2.3 Vendor invoice policy violations (R3 audit)	PARTIAL-48	ADDED as FR-48.14	
AX 12.99.4 row 16 — 12.2.4 Vendor invoice policies (R3 audit policy)	PARTIAL-48	ADDED as FR-48.14	
AX 12.99.4 row 55 — 12.9.1 Internal controls → Files (`SysComplianceFiles`)	PARTIAL-48	ADDED as FR-48.16	process documents in wave-5 document store
AX 12.99.4 row 56 — 12.9.1 Internal controls → Process library (`SysComplianceProcess`)	PARTIAL-48	ADDED as FR-48.16	
AX 12.99.4 row 57 — 12.9.1 Internal controls → Risk library (`SysComplianceRisk`)	PARTIAL-48	ADDED as FR-48.16	
AX 12.99.4 row 58 — 12.9.1 Audit → Audit cases (`CaseDetail`)	PARTIAL-48	ADDED as FR-48.15	audit cases here; generic case management stays wave 45
AX 12.99.4 row 59 — 12.9.1 Audit → Audit policy violations list (`AuditPolicyViolation`)	PARTIAL-48	ADDED as FR-48.14	
AX 12.99.4 row 60 — 12.9.2 Audit policy violations inquiry	PARTIAL-48	ADDED as FR-48.14	
AX 12.99.4 row 61 — 12.9.2 Audit case inquiry	PARTIAL-48	ADDED as FR-48.15	
AX 12.99.4 row 62 — 12.9.2 SSRS: Audit policy report / Audit case report	PARTIAL-48	ADDED as FR-48.14, FR-48.15	
AX 12.99.4 row 63 — 12.9.2 SSRS: Internal control documents	PARTIAL-48	ADDED as FR-48.16	control matrix report
AX 12.99.4 row 64 — 12.9.3 Run audit policies (batch)	PARTIAL-48	ADDED as FR-48.14	job in 9.48.16
AX 12.99.4 row 65 — 12.9.3 Close/archive audit cases	PARTIAL-48	ADDED as FR-48.15	archive via FR-46.18
AX 12.99.4 row 66 — 12.9.4 Audit policies (rule types duplicate, keyword, list search, sampling, premium/mileage)	PARTIAL-48	ADDED as FR-48.14	rules on FR-3.13 decision tables
AX 12.99.4 row 67 — 12.9.4 Audit policy rule types	PARTIAL-48	ADDED as FR-48.14	
AX 12.99.4 row 68 — 12.9.4 Audit case categories	PARTIAL-48	ADDED as FR-48.15	
AX 12.99.4 row 69 — 12.9.4 Audit keyword lists / Audit sampling parameters	PARTIAL-48	ADDED as FR-48.14	`audit.sampling_seed_policy`
AX 12.99.4 row 70 — 12.9.4 Risk/control types, Process document templates	PARTIAL-48	ADDED as FR-48.16	
AX 12.99.4 row 77 — 12.10.2 SSRS: Expense audit (audit policy results)	PARTIAL-48	ADDED as FR-48.14	
AX 12.99.4 row 80 — 12.10.3 Run audit policies on expense reports	PARTIAL-48	ADDED as FR-48.14	
AX 12.99.4 row 87 — 12.10.4 Expense → Audit policies (link)	PARTIAL-48	ADDED as FR-48.14	
AX 12.99.4 row 105 — 12.12.1 Worker compliance (US) → I-9, ADA, Immigration/visa, Union agreements	PARTIAL-48	ADDED as FR-48.22	
AX 12.99.4 row 109 — 12.12.2 SSRS: EEO-1 / VETS (US)	PARTIAL-48	ADDED as FR-48.31	
AX 12.99.4 row 111 — 12.12.3 HR → Year-end (US ACA 1095-C)	PARTIAL-48	ADDED as FR-48.31	
AX 12.99.4 row 112 — 12.12.4 Worker compliance (US) setup: I-9 document types, ADA, visa types, union agreements	PARTIAL-48	ADDED as FR-48.22	
AX 12.99.4 row 115 — 12.13.2 SSRS: Workers compensation report	PARTIAL-48	ADDED as FR-48.22 (class codes), FR-48.31 (report)	
AX 12.99.4 row 116 — 12.13.2 SSRS: New hire reporting (US state)	PARTIAL-48	ADDED as FR-48.31	
AX 12.99.4 row 118 — 12.13.4 Leave of absence / FMLA tracking (US)	PARTIAL-48	ADDED as FR-48.22	BR-48.20
AX 12.99.4 row 260 — 12.29.4 Organization → Audit policies (link)	PARTIAL-48	ADDED as FR-48.14	
AX 12.99.4 row 276 — 12.33.1 Electronic messages (`EMElectronicMessage*`)	PARTIAL-48	ADDED as FR-48.17	
AX 12.99.4 row 277 — 12.33.2 Electronic messages inquiry	PARTIAL-48	ADDED as FR-48.17	
AX 12.99.4 row 278 — 12.33.3 Electronic messages → Generate / Send / Receive response	PARTIAL-48	ADDED as FR-48.17	poller in 9.48.16
AX 12.99.4 row 279 — 12.33.4 Electronic messages setup (message types, item types, statuses, processing actions, web service settings)	PARTIAL-48	ADDED as FR-48.17	

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP 13.99.2 row 112 — 13.10 Disclosure management / XBRL (SAP DM, SAC)	GAP	ADDED as FR-48.18	
SAP 13.99.2 row 206 — 13.17 Preference determination (GTS)	GAP	ADDED as FR-48.21	Could priority; USMCA and EU PEM first
SAP 13.99.2 row 205 — 13.17 Foreign trade / export documents in delivery (GTS) (PARTIAL-13, propose 48)	PARTIAL-13	ADDED as FR-48.20	shipment execution stays wave 34; customs submission via broker adapters
SAP 13.99.2 row 339 — 13.28 Country-specific HR infotypes and legal changes	PARTIAL-48	ADDED as FR-48.28 (non-US), FR-48.22 (US)	
SAP 13.99.2 row 344 — 13.30 GRC EAM firefighter	PARTIAL-48	ADDED as FR-48.23	
SAP 13.99.2 row 345 — 13.30 Anonymized production-to-test refresh (TDMS / client copy)	PARTIAL-48	ADDED as FR-48.26	used by waves 49 and 50
SAP 13.99.2 row 347 — 13.30 Read Access Logging (`SRAL`)	PARTIAL-48	ADDED as FR-48.25	
SAP 13.99.2 row 350 — 13.31 Business Partner Screening / Watch List Screening (GTS)	GAP	ADDED as FR-48.19	
SAP 13.99.2 row 355 — 13.31 Watch list screening / GTS compliance configuration	GAP	ADDED as FR-48.19	parameters in 9.48.13
SAP 13.99.2 row 400 — 13.36 Export governance (Excel export restrictions, download audit)	PARTIAL-48	ADDED as FR-48.24	
SAP 13.99.2 row 25 — 13.2 `J1INCHLN`/`J1INBANK`/`J1INCERT` (PARTIAL-13, IN pack)	PARTIAL-13	ADDED as FR-48.27	challans, CIN, Form 16A
SAP 13.99.2 row 53 — 13.4 `S_ALR_87010127`/`87010129` US Form 4562 (PARTIAL-37, US pack)	PARTIAL-37	ADDED as FR-48.31	from wave-37 tax book
SAP 13.99.2 row 257 — 13.22 `COEBR` electronic batch record (GAP-24, +48)	GAP-24	ADDED as FR-48.13 (signature service); EBR itself DEFERRED to wave 24 (owner)	
SAP 13.99.2 row 261 — 13.22 Digital signature in PI sheets / batch records (`DSIG`) (GAP-24, cross-cutting 24/48)	GAP-24	ADDED as FR-48.13	wave 24 binds PI-sheet steps to `esign_requirement`
SAP 13.99.2 row 24 — 13.2 `F-40`/`F-36`/`FBW6` bills of exchange (GAP-36, or v1.x pack under 48)	GAP-36	DEFERRED to wave 36 (owner of instruments); FR LCR and MX pagaré formats ship as pack ER formats once wave 36 provides the instrument lifecycle	none of the first 7 packs is blocked without it
SAP 13.99.2 row 383 — 13.35 PLM labeling (GLM) / specification management (PARTIAL-9, propose 48 labeling)	PARTIAL-9	DEFERRED to wave 33 (product compliance owner); packs supply GHS/CLP phrase sets as text blocks (FR-46.15)	
AX 12.99.1 theme "Compliance extras" — SOC 2 evidence store (task brief)	PARTIAL (FR-48.7 collection only)	ADDED as FR-48.30	object lock, hash chain, auditor role
AX 12.99.1 theme "Compliance extras" — incident/breach process (CC7.3/7.4 evidence without an incident record)	PARTIAL	ADDED as FR-48.29	
AX 12.99.1 theme "Compliance extras" — license usage reporting (3)	PARTIAL-3	SUPERSEDED by FR-3.14 (wave 3 usage snapshot)	listed for traceability
SAP 13.99.2 rows 348/352/354 — BRF+ in GRC (PARTIAL-3)	PARTIAL-3	SUPERSEDED by FR-3.13; audit policies, license determination and embargo rules are FR-3.13 rule sets	
AX 12.99.1 theme "Alerts and notification rules" (wave 5)	GAP-5	DEFERRED to wave 5 (owner); SoD, breach, screening and evidence alerts deliver through it	

9.49 Wave 49: Migration Factory: Full AX/SAP Transactional Cutover Tooling

9.49.1 Scope and business context

AX 2012 → D365 uses the data upgrade scripts (`ReleaseUpdateDB` classes, LCS data-upgrade deployable package) and DIXF; SAP uses SUM with DMO for conversion, the Migration Cockpit (`LTMC`/`LTMOM`, staging tables) and Selective Data Transition. None of them moves a customer to a different ERP. Building on the wave-10 framework (connectors, declarative mapping, idempotent runs), this wave adds transactional packs with control totals, historical loads into the lakehouse with legacy cross-references, a parallel-run harness, a cutover runbook generator and coexistence adapters. Out of scope: code migration (X++/ABAP), custom-table migration beyond the mapping DSL, SAP archive (ADK) files.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Open items loaded through DIXF (`DMFDefinitionGroup`) or Migration Cockpit staging tables (`/1LT/DS*`) are reconciled in Excel against `VendTransOpen`/`BSIK` totals; mismatches found days later	No built-in control totals at sub-total grain	`control_total` per pack at (entity, account, currency) grain with hashes, computed in-run	Reconciliation 5 person-days per entity → 10 min automated; mismatches found in-run
Re-running a failed load creates duplicates (`InventSum` doubled, duplicate `BKPF` opening documents)	Loaders without natural-key idempotency	Idempotent packs keyed on `legacy_xref`	Duplicate rate after re-run 0
Parallel runs compare printed trial balances by eye; variances are explained in e-mail threads	No diff tooling	Normalized tuple diff with EXPECTED/EXPLAINED/UNEXPLAINED classification and sign-off	Variance review 3 weeks → 2 days; 0 unexplained at sign-off
Cutover plans are spreadsheets; step durations are guesses; rollback is untested	No runbook tooling tied to the data loads	Generated `cutover_plan` with timed steps from rehearsals, go/no-go gates, rehearsed rollback	Cutover < 48 h; 2 rehearsals; rollback rehearsed
Users lose the legacy voucher/document id after migration and keep the old system online for lookups	No cross-reference model	`legacy_xref` on every record and in the lakehouse history	Legacy lookups needing the old system 100% → 0

9.49.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Open AP/AR items	`VendTrans`/`VendTransOpen`, `CustTrans`/`CustTransOpen`, `VendSettlement`	`BSIK`/`BSID` (ECC), `ACDOCA` open items (S/4), `BSEG`	pack `open_items` → AP/AR open items with `legacy_xref`	Full
GL balances	`GeneralJournalAccountEntry`, `DimensionAttributeValueCombination`, `LedgerBalance*`	`GLT0`/`FAGLFLEXT` (ECC), `ACDOCA` (S/4)	pack `gl_balances` → opening journal per dimension	Full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
On-hand inventory	`InventSum`, `InventTrans`, `InventDim`, `InventTransOrigin`	`MARD`, `MCHB` (batch), `MSKA`, `MBEW` valuation	pack `onhand` → inventory opening transactions by dimension	Full
Open POs / SOs	`PurchTable`/`PurchLine`, `SalesTable`/`SalesLine` with remaining qty	`EKKO`/`EKPO`/`EKET`, `VBAK`/`VBAP`/`VBEP`	pack `open_orders` → orders with delivered/invoiced quantities	Full
In-process production	`ProdTable`, `ProdBOM`, `ProdRoute`, `ProdJournalProd`	`AFKO`/`AFPO`/`AFVC`, `AUFK`	pack `wip_prod` → orders at status with consumed/reported quantities and WIP value	Full
Fixed assets	`AssetTable`, `AssetBook`, `AssetTrans`	`ANLA`/`ANLB`/`ANLC`/`ANEP`	pack `fixed_assets` → NBV per book with depreciation profile	Full
Project WIP	`ProjTable`, `ProjTrans*`, `ProjWIP*`	`PRPS`, `COSP`/`COSS`, RA results	pack `project_wip`	Full
Employee/payroll YTD	`HcmWorker`, `PayrollEarningStatement*`	`PA0001`/`PA0008`, payroll results cluster (`PCL2`)	pack `payroll_ytd`	Partial (US + DE only)
Upgrade tooling	AX 2012→D365 data upgrade (LCS), DIXF	SUM/DMO, Migration Cockpit, SDT	migration factory packs + runbook generator	Improved (cross-vendor)
History	Legacy database kept online	Archive / SDT history	lakehouse `history.*` with `legacy_xref`	Full (read-only)
Project beginning balances	Project beginning balance journals (`ProjJournalTable` type Beginning balance, `ProjBegBalJournalTrans_CostSales`, `ProjBegBalJournalTrans_OnAcc`)	Migration Cockpit objects for WBS, settlement rules; RA results transfer (`KKA2`), legacy cost transfer (`KB11N`/`KB15N`)	pack `project_wip` posting `project_opening_line` by category (FR-49.12)	Full

9.49.3 Functional requirements

- **FR-49.1 Transactional packs** — Must. The system shall provide packs `open_items`, `gl_balances`, `onhand`, `open_orders`, `wip_prod`, `fixed_assets`, `project_wip`, `payroll_ytd`, `bank_balances`, each with AX and SAP extractors, mapping, validation (Great Expectations suites), load and reconciliation steps.
- **FR-49.2 Control totals** — Must. The system shall compute and compare per pack: `open_items`: count, sum amount in transaction currency per currency, sum in accounting currency, per (entity, account, currency) sub-totals, hash of sorted (legacy_id, amount); `gl_balances`: per (entity, account, dimension combination, currency) balance, trial balance debit = credit, count of non-zero combinations; `onhand`: per (item, site, warehouse, location, batch, serial) quantity and per (item, site) value, total value per entity; `open_orders`: order count, line count, remaining quantity, remaining amount; `wip_prod`: order count, WIP value per order; `fixed_assets`: count, acquisition cost, accumulated depreciation, NBV per book; `project_wip`: WIP value per project; `payroll_ytd`: gross, tax, net per worker; `bank_balances`: per account.
  - Notes: every total must match exactly (0.00) unless a tolerance rule is explicitly configured with reason; totals are stored on `control_total` with source and target values and timestamp.
- **FR-49.3 Legacy cross-reference** — Must. The system shall store `legacy_xref` (system, company, table, legacy_id → passage entity, id) for every migrated record and expose it on every record detail and in the lakehouse.
- **FR-49.4 Historical load** — Must. The system shall load closed legacy transactions (journals, invoices, inventory transactions, orders) for N years (default 7) into Iceberg `history.<system>.*` with the legacy schema preserved, mapped views `history.unified.*` on the Passage model, queryable in explorer with `source=legacy` filter.
- **FR-49.5 Parallel-run harness** — Must. The system shall replay a legacy period's source documents (vendor invoices, customer invoices, inventory movements, journals, production postings) into a Passage rehearsal tenant through the public posting APIs in original sequence and compare results.
  - Diff algorithm: (1) normalize both sides to canonical tuples `(entity, account, dimension set, currency, period, amount)` for trial balance, `(item, dims, qty, value)` for inventory, `(subledger, open item id, amount)` for subledgers; (2) outer-join on key; (3) classify each difference: EXPECTED (rule-based: known rounding ≤ 0.01, policy differences registered in `variance_rule`, e.g. different FX rounding step), EXPLAINED (operator attaches a reason), UNEXPLAINED; (4) aggregate by rule; (5) report unexplained count and amount; (6) exit criterion unexplained = 0 and explained amount ≤ 0.1% of period revenue.
- **FR-49.6 Cutover runbook generator** — Must. The system shall generate `cutover_plan` with timed steps from T−14d to T+30d from the selected packs, entities and legacy systems, with owners, durations from the last rehearsal, go/no-go items and rollback steps.
  - Timeline baseline: T−14d freeze master-data changes in legacy, final pack dry run; T−7d second rehearsal, sign-off on variance report; T−3d freeze configuration in Passage; T−1d 18:00 legacy transaction freeze, final extracts of master data and static reference data; T−0 00:00 extract transactional packs; T+4h load and reconcile master and balances; T+12h load open items, orders, on-hand, WIP; T+24h reconciliation complete, go/no-go meeting; T+30h user access opened; T+36h abort deadline if go/no-go fails; T+1d first postings, hypercare starts; T+7d first weekly reconciliation; T+30d first month-end close, hypercare review.
- **FR-49.7 Coexistence adapters** — Should. The system shall support entity-by-entity cutover with intercompany documents and consolidation data exchanged with the legacy system through wave-35 integration hub adapters (AX OData/`DMF` entities; SAP IDoc/BAPI) for `INVOIC`, `ORDERS`, GL postings, and master-data sync one way (legacy → Passage) until the last entity cuts over.
- **FR-49.8 Rehearsal environment** — Must. The system shall provision a rehearsal tenant from pipeline (Terraform + Helm + seed) in ≤ 2 h, with masked extracts, and tear it down.
- **FR-49.9 Rollback plan** — Must. The system shall generate a rollback plan per cutover: stop Passage access, restore legacy write access, export any Passage postings made after go-live as journal/document files for re-keying, with a decision point at T+36h.
- **FR-49.10 Idempotent re-runs** — Must. The system shall re-run any pack without duplicates (wave-10 natural keys plus `legacy_xref` uniqueness) and record every run with inputs' hashes.
- **FR-49.11 Validation suites** — Must. The system shall run Great Expectations suites per pack before load: referential (every vendor on open items exists), domain (currency codes ISO 4217), balance (debit = credit), uniqueness; failures above 0 block load unless the record is quarantined to `pack_quarantine` with reason.
- **FR-49.12 Project beginning balance journals** — Must. The system shall load, per open project, opening transactions by category (hours, expense, item, fee, on-account billing) with quantity, cost, sales value, transaction date, worker/resource, activity and dimensions as `project_opening_line`, flagged

`beginning_balance=true`, together with WIP already recognized (WIP cost, WIP sales value, accrued revenue, invoiced on-account) and the remaining estimate, so that wave-41 estimates, billing rules and revenue recognition continue from the cutover state.

- Notes: opening lines never post to the GL (the GL WIP and revenue balances arrive in pack `gl_balances`); control totals per project and category (cost, sales value, WIP, on-account) must equal the legacy `ProjTrans*/COSP-COSS` and RA results to 0.00; lines marked "chargeable, not invoiced" are invoiceable in Passage with the legacy transaction date; fixed-price projects carry percent complete and recognized revenue to date; time-and-material projects carry unbilled transactions at legacy sales price.

9.49.4 Data model

`pack_run` (migration-factory)

Field	Type	Required	Notes
run_id	uuid	Y	PK = Temporal workflow id
tenant_id	uuid	Y	
pack	enum	Y	the 9 packs
source_system	enum	Y	AX2012, D365FO, SAPECC, S4HANA
source_company	text	Y	`DataAreaId` / `BUKRS`
cutoff_ts	timestamptz	Y	extraction cutoff
input_hash	text	Y	sha256 of extract
status	enum	Y	EXTRACTED, VALIDATED, LOADED, RECONCILED, FAILED, ROLLED_BACK
records_in / records_loaded / records_quarantined	int	Y	

`control_total`: total_id, run_id, name, key_json (e.g. {entity, account, currency}), source_value numeric(28,6), target_value numeric(28,6), delta, tolerance, status (MATCH, WITHIN_TOLERANCE, MISMATCH), computed_at.

`legacy_xref`: xref_id, tenant_id, source_system, source_company, source_table, source_id, passage_entity, passage_id, run_id; unique (source_system, source_company, source_table, source_id).

`parallel_run`: parallel_run_id, tenant_id, period, source_system, documents_replayed, state (REPLAYING, DIFFING, REVIEW, SIGNED_OFF), unexplained_count, unexplained_amount, explained_amount, signed_off_by. **`variance`**: variance_id, parallel_run_id, area (TB/INV/AP/AR/FA), key_json, legacy_value, passage_value, delta, class (EXPECTED/EXPLAINED/UNEXPLAINED), rule_id, reason.

`cutover_step`: step_id, plan_id, seq, offset_minutes (from T–0), name, owner, planned_duration_min, actual_start, actual_end, status, rollback_step_id, go_nogo bool.

Invariants: a pack run reaches RECONCILED only if every `control_total.status != MISMATCH`; `legacy_xref` is append-only; a `parallel_run` can be SIGNED_OFF only when unexplained_count = 0; cutover steps with `go_nogo=true` must all be complete before the "open user access" step can start. Indexing: `legacy_xref` unique index as above plus (passage_entity, passage_id); `control_total` (run_id, status); `variance` (parallel_run_id, class).

Additional entity (FR-49.12). `project_opening_line`: line_id, tenant_id, project_id, category (HOUR, EXPENSE, ITEM, FEE, ON_ACCOUNT), transaction_date, qty numeric(28,6), cost_amount, sales_amount, currency, worker_id, activity_id, dimensions_json, billing_status (CHARGEABLE_UNBILLED, BILLED, NON_CHARGEABLE), wip_cost, wip_sales, legacy_xref_id, run_id; unique (tenant_id, legacy_xref_id).

9.49.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/migration/packs/{pack}/runs`	start extract/validate/load	by input_hash	MigrationRun
GET	`/api/v1/migration/runs/{id}/control-totals`	totals and status	n/a	MigrationView
POST	`/api/v1/migration/runs/{id}/quarantine/{rec}/release`	fix and release quarantined record	by record	MigrationRun
POST	`/api/v1/migration/history/loads`	start historical load	by (system, years)	MigrationRun
GET	`/api/v1/migration/xref?system=&table=&id=`	resolve legacy id	n/a	any authenticated
POST	`/api/v1/migration/parallel-runs`	start replay for period	by (period, system)	MigrationRun
POST	`/api/v1/migration/parallel-runs/{id}/variances/{v}/explain`	attach reason	by variance	MigrationReview
POST	`/api/v1/cutover/plans/generate`	generate plan from packs	by inputs hash	CutoverManager
POST	`/api/v1/cutover/plans/{id}/steps/{s}/complete`	mark step done	by step	step owner
POST	`/api/v1/cutover/plans/{id}/rollback`	execute rollback plan	by plan	CutoverManager + second approver
GET	`/api/v1/migration/runs/{id}/projects/{project}/opening`	opening lines and totals per project and category	n/a	MigrationView

Validation: pack run rejects extracts with `cutoff_ts` later than the legacy freeze; `complete` rejects out-of-sequence go/no-go steps; `rollback` requires two distinct approvers.

Event	Trigger	Payload fields	Consumers
`dev.passage.migration.pack.reconciled.v1`	reconciliation done	run_id, pack, totals_match, totals_mismatch, quarantined	cutover, notifications
`dev.passage.migration.parallelrun.diffed.v1`	diff done	parallel_run_id, period, unexplained_count, unexplained_amount	cutover
`dev.passage.cutover.step.completed.v1`	step done	plan_id, step_id, actual_duration_min, status	cutover dashboard
`dev.passage.cutover.decision.recorded.v1`	go/no-go	plan_id, decision, approvers[], timestamp	audit, platform-ops

9.49.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Migration factory workspace	run pack, view totals, open quarantine, re-run	pack, source, status, records, mismatches	MigrationRun
Control-total board	drill mismatch to records, export	name, key, source, target, delta	MigrationView
Quarantine	edit field, release, reject	record, rule failed, reason	MigrationRun
Parallel-run review	classify variances, attach reason, sign off	area, key, legacy, passage, delta, class	MigrationReview

Screen	Primary actions	Key fields/columns	Duty
Cutover runbook	start/complete step, timer, go/no-go checklist, rollback	seq, T-offset, owner, planned vs actual, status	CutoverManager
Legacy reference (on any record)	open xref	system, company, table, id	all

Rules: control-total board auto-refreshes every 10 s during a run; variance grid supports bulk classify by rule; runbook shows elapsed vs planned with amber at 110%, red at 125%.

9.49.7 Business rules and edge cases

- **BR-49.1** Pack run state machine: EXTRACTED → VALIDATED → LOADED → RECONCILED; any step → FAILED; LOADED/RECONCILED → ROLLED_BACK (compensating journals and deletion of unposted documents).
- **BR-49.2** Open items are loaded against an opening-balance clearing account; the sum of open items per control account must equal the GL balance migrated for that account to 0.00 before RECONCILED.
- **BR-49.3** On-hand is loaded at legacy cost per dimension; the inventory GL account opening balance is migrated as a single line and must equal the loaded value per entity.
- **BR-49.4** Open orders are loaded with `qty_delivered` and `qty_invoiced` history so remaining quantities match; prior deliveries are not re-posted.
- **BR-49.5** WIP production orders carry consumed material and reported hours as a WIP opening entry; remaining operations are rebuilt from the Passage route.
- **BR-49.6** Fixed assets load acquisition date and accumulated depreciation per book; the next depreciation run must equal legacy's projected amount within 0.01 or the asset is flagged.
- **BR-49.7** Amount rounding during replay follows Passage rules; a difference ≤ 0.01 per document is classified EXPECTED by rule `RND-001`.
- **BR-49.8** Historical loads are read-only; a legacy document with no mapped Passage type lands in `history.<system>.unmapped`.
- **BR-49.9** Cutover abort at T+36h: if any go/no-go item is open, the rollback plan executes automatically after CutoverManager confirmation.
- **BR-49.10** Coexistence: a document received from legacy that already exists by `legacy_xref` is acknowledged and ignored.
- **BR-49.11** Project opening lines are excluded from GL posting and from the wave-41 cost-to-GL reconciliation for the cutover period; they are included in project estimates, billing and revenue-recognition calculations.
- **BR-49.12** A project whose opening totals do not reconcile to the legacy project subledger cannot be invoiced or estimated in Passage until the mismatch is resolved (status `OPENING_MISMATCH`).

9.49.8 Acceptance criteria

- **AC-49.1** (FR-49.1, 49.2) Given the synthetic AX dataset (3 entities, full year), when all packs run, then every `control_total.status = MATCH`.
- **AC-49.2** (FR-49.1, 49.2) Given the synthetic SAP dataset, when all packs run, then every `control_total.status = MATCH`.
- **AC-49.3** (FR-49.11) Given an open item for a vendor missing in master data, when validation runs, then the record is quarantined and load proceeds for the rest.
- **AC-49.4** (FR-49.10) Given a completed `open_items` run, when the same extract is run again, then records_loaded = 0 and totals are unchanged.
- **AC-49.5** (FR-49.5) Given one month of legacy documents replayed, when the diff runs, then unexplained_count = 0 and explained amount ≤ 0.1% of revenue.
- **AC-49.6** (FR-49.6) Given the generated runbook, when the rehearsal is executed, then all steps from T−0 to "open user access" complete in < 48 h.
- **AC-49.7** (FR-49.9) Given a rehearsal cutover, when rollback executes, then the legacy coexistence adapters resume and Passage postings since T−0 are exported as re-keying files.
- **AC-49.8** (FR-49.3) Given migrated customer invoice legacy id `CIV-00123` (AX `CustInvoiceJour.InvoiceId`), when a user opens the Passage invoice, then the legacy reference panel shows system, company and id.
- **AC-49.9** (FR-49.4) Given 7 years of legacy journals loaded, when the explorer queries `revenue` with `source=legacy` for a closed year, then it equals the legacy trial balance for that year.
- **AC-49.10** (FR-49.8) Given a pipeline trigger, when a rehearsal tenant is requested, then it is ready in ≤ 2 h with masked data.
- **AC-49.11** (FR-49.12) Given an AX T&M project with 120 chargeable unbilled hour transactions, when `project_wip` loads, then 120 opening lines exist, totals per category match legacy, no GL posting was created, and the first Passage invoice proposal lists the 120 lines at legacy sales price.

9.49.9 Test plan and fixtures

Fixtures: `fx-ax-synth-year` (AX 2012 R3 schema: `CustTrans`, `VendTrans`, `InventSum`, `GeneralJournalAccountEntry`, `PurchLine`, `SalesLine`, `ProdTable`, `AssetBook`, 3 `DataAreaId`, 1.1M rows); `fx-sap-synth-year` (ECC 6.0 and S/4 2023 variants: `BSID/BSIK/ACDOCA`, `MARD/MBEW`, `EKPO`, `VBAP`, `AFKO`, `ANLC`, 3 `BUKRS`, 1.3M rows); `fx-replay-month` (one month, 42k documents); `fx-cutover-plan` (plan with 84 steps).

Test	Type	Target	Pass criterion
Pack reconciliation	E2E	9 packs × 2 systems	100% MATCH
Idempotent re-run	property	loader	re-run of any subset loads 0 new rows
GE suites	unit	validation	all expectations defined per pack with ≥ 95% coverage of mapped fields
Parallel-run diff	E2E	harness	0 unexplained
Diff classifier	unit	classifier	100 seeded variances classified per rule
Cutover timing	performance	runbook	< 48 h; each step within 125% of plan

Test	Type	Target	Pass criterion
Rollback	E2E	rollback	coexistence resumes; exported files complete
History load throughput	performance	Iceberg	≥ 200k rows/min

Reconciliation: trial balance (Passage opening) ↔ legacy trial balance per entity/currency; open items ↔ control accounts; on-hand value ↔ inventory account; NBV ↔ asset accounts; parallel-run TB ↔ legacy period TB.

9.49.10 Deployment and operations

Helm: `passage-migration-factory` (Go extractors, TS orchestration on Temporal namespace `migration`), `passage-legacy-adapters` (AX OData/SQL, SAP RFC/ODP via wave-35 hub), `passage-rehearsal-provisioner` (pipeline job). Config: legacy connection strings in Vault, `HISTORY_YEARS=7`, `MASK_PII=true` for rehearsal. Flags: `migration.coexistence` per entity. Jobs: nightly delta extract during coexistence; weekly reconciliation post go-live for 30 days. SLOs: pack load ≥ 50k records/min; control totals computed within 10 min of load. Alerts: MISMATCH, quarantine > 1%, step overrun > 125%. Runbook entries: resume a failed pack from VALIDATED; rebuild xref; execute rollback. Rollback: per BR-49.1 and FR-49.9.

9.49.11 Migration from AX / SAP

This wave is the migration. Summary mapping:

Source	Passage entity	Transformation
`CustTransOpen/VendTransOpen`, `BSID/BSIK`	AR/AP open items	amount in transaction and accounting currency; due date; settlement history collapsed to remaining amount
`GeneralJournalAccountEntry` + `DimensionAttributeValueCombination`, `ACDOCA/FAGLFLEXT`	opening journal	balances per account × dimension × currency at cutoff; P&L closed to retained earnings if cutoff is year-end
`InventSum/InventDim/MBEW/MARD/MCHB`	inventory opening transactions	qty and cost per dimension; batch/serial attributes
`PurchLine/SalesLine`, `EKPO/VBAP`	open order lines	remaining = ordered − delivered; price and tax re-derived and compared
`ProdTable/AFKO`	production orders	status mapping (Started → RELEASED+WIP); WIP value
`AssetBook/ANLC`	asset books	NBV, accumulated depreciation, remaining life
`ProjTrans*/COSP`	project WIP	WIP per project, by category

Not migrated: legacy number sequences (new sequences, old ids kept in xref); AX `SysDatabaseLog` rows and SAP `CDHDR` (loaded to history only); settled items and closed orders (history only); X++/ABAP customizations.

9.49.12 Dependencies, risks, open questions

Depends on: wave 10 (framework, connectors), wave 35 (integration hub for coexistence), wave 46 (lakehouse for history), wave 48 (PII masking rules), all transactional waves for posting APIs.

Risk	Likelihood	Impact	Mitigation
Legacy dimension model does not map 1:1	High	High	mapping DSL with default dimension and exception report; sign-off before load
Extract window exceeds freeze	Medium	High	delta extracts; parallelized extractors; rehearsal timing
Coexistence creates duplicate intercompany documents	Medium	Medium	xref dedupe (BR-49.10)

Open questions: (1) History years default 7 vs 10 for DE GoBD — proposed 10 when DE pack installed; decide by wave 50. (2) Payroll YTD for countries without a Passage payroll engine — proposed: load as balances only; decide by wave 50.

9.49.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`migration.source_system`	legal entity	enum	none	AX2012, D365FO, SAPECC, S4HANA	extractor selection	DIXF source data format	Migration Cockpit project source (`LTMC` direct transfer / staging)
`migration.cutoff_ts`	legal entity	timestamptz	none	≤ freeze	extract cutoff	Data upgrade cutoff	Legacy data transfer date (`OAYE`)
`migration.opening_clearing_account`	legal entity	account	per pack	balance-sheet accounts	BR-49.2 offset	Opening balance journal offset	Data transfer clearing accounts (`OASV` offset)
`migration.tolerance.<pack>`	legal entity	money + reason	0.00	≥ 0 with mandatory reason	FR-49.2	none	none
`migration.quarantine_max_pct`	tenant	numeric	1.0	0–10	load blocks above share	DIXF error threshold	Migration Cockpit error handling
`migration.history_years`	tenant	int	7 (10 when DE pack installed)	1–20	FR-49.4	none	SDT / archive
`migration.parallel.unexplained_max`	tenant	int	0	0	FR-49.5 exit	none	none
`migration.parallel.explained_max_pct_revenue`	tenant	numeric	0.1	0–1	FR-49.5 exit	none	none
`migration.variance_rules`	tenant	rule set	`RND-001`	FR-3.13 decision table	EXPECTED classification	none	none
`migration.extract_parallelism`	tenant	int	8	1–32	extractor workers	DIXF threads (bundle size)	Migration Cockpit background jobs
`migration.load_batch_size`	tenant	int	5,000	500–50,000	API batch	DIXF bundle size	Migration Cockpit portion size
`migration.mask_pii`	rehearsal tenant	bool	true		FR-48.26 profile	none	TDMS scrambling
`cutover.t0`	plan	timestamptz	none		runbook anchor	none	none

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`cutover.abort_offset_h`	plan	int	36	12–72	BR-49.9	none	none
`cutover.step_overrun_amber_pct` / `red_pct`	plan	int	110 / 125	≥ 100	runbook colors	none	none
`coexistence.entities`	tenant	list	none	legal entities	FR-49.7 scope	none	none
`coexistence.sync_cron`	tenant	cron	`*/15 * * * *`	cron	master-data sync	none	ALE distribution model schedule (`BD64`)
`migration.project_opening.categories`	legal entity	set	HOUR, EXPENSE, ITEM, FEE, ON_ACCOUNT	subset	FR-49.12	Beginning balance journal line types	Migration object project cost categories

9.49.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Pack runs	list inquiry	pack, system, entity, status	records in/loaded/quarantined, duration, input hash	`pack_run`	DIXF execution history (`DMFExecution`)	Migration Cockpit monitoring (`LTMC` → Monitoring)
Control-total board	operational report	run, status	name, key, source, target, delta, tolerance	`control_total`	none (Excel)	Migration Cockpit "Migration results"
Quarantine	list inquiry	run, rule	record, rule failed, reason, age	`pack_quarantine`	DIXF staging errors	Staging table messages
Opening trial balance vs legacy	statutory/operational report	entity, currency	account × dimension, legacy, Passage, delta	`control_total` + ledger	Trial balance	`S_ALR_87012277`, `FAGLB03`
Open items reconciliation	operational report	entity, control account	open items sum, GL balance, delta	AP/AR + ledger	Vendor/customer balance list with aging	`S_ALR_87012082`/`87012172`
Inventory opening valuation	operational report	entity, site	qty, value by item/dims vs legacy	inventory + `control_total`	Inventory value report	`MB5L`, `MC.9`
Project opening balances	operational report	project, category	cost, sales, WIP, on-account vs legacy	`project_opening_line`	Project beginning balance journal list	`CJI3`/RA results (`KKA3`)
Parallel-run variance report	drill-down	parallel run, area, class	key, legacy, Passage, delta, rule, reason	`variance`	none	none
Cutover runbook status	KPI	plan	steps done/total, elapsed vs plan, overruns, go/no-go items	`cutover_step`	none	none
Legacy cross-reference lookup	list inquiry	system, table, legacy id	Passage entity and id	`legacy_xref`	none	none
History queries	drill-down	`source=legacy`, period	any metric	`history.unified.*`	AX legacy database kept online	SARI/archive, SDT history

9.49.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Reconciliation certificate per pack	pack RECONCILED	run, entity, cutoff, each control total with source/target/delta, signer	PDF with e-signature (FR-48.13)	run id	audit evidence; 10 years with opening balances
Opening trial balance sign-off	`gl_balances` RECONCILED	entity, currency, account × dimension balances, legacy reference, controller signature	PDF, XLSX	entity + cutoff	statutory link between legacy and Passage books (GoBD/FEC continuity)
Parallel-run sign-off report	parallel run SIGNED_OFF	period, unexplained = 0, explained by rule, approvers	PDF	parallel run id	audit evidence
Cutover runbook and go/no-go record	plan generated / decision	steps, owners, timings, decision, approvers	PDF, CSV	plan id	7 years
Rollback re-keying files	rollback executed	postings since T–0 as journals and documents	CSV, legacy import formats (AX DIXF package, SAP `LTMC` template)	rollback id	until re-keyed and reconciled
Legacy cross-reference export	on demand	system, table, legacy id, Passage id	CSV, Parquet	n/a	retained with history (FR-49.4)

9.49.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Pack run (extract → validate → load → reconcile)	on demand / runbook step	pack, entity	`input_hash`, `legacy_xref` uniqueness	≥ 50k records/min	resume from last completed stage (BR-49.1)	MISMATCH, quarantine > 1%
Historical load	on demand	system, table, year	Iceberg snapshot per (table, year)	≥ 200k rows/min	resume per year	rows vs source count
Parallel-run replay	on demand per period	document type, day	replay cursor (legacy doc id order)	42k documents in < 6 h	pause on API 5xx; resume	replay errors
Variance diff	after replay	area	`parallel_run_id`	< 30 min	rerun idempotent	unexplained > 0
Coexistence delta sync	every 15 min	entity, object	legacy change pointer / timestamp	< 2 min	BR-49.10 dedupe; retry	backlog > 1 h
Post-go-live weekly reconciliation	weekly for 30 days	entity	(entity, week)	< 20 min	ticket	delta ≠ 0
Rehearsal tenant provision / teardown	on pipeline trigger	tenant	clone job (FR-48.26)	≤ 2 h	retry stage	> 2 h
Cutover step timer	continuous during cutover	plan	step state	n/a	amber/red per parameters	overrun > 125%

9.49.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
MigrationRun	start pack runs, history loads, release quarantine, parallel runs	with MigrationReview (HIGH, `MIG-001`: load and sign off own variances)	Migration engineer
MigrationView	control totals, runs, xref	none	Controller, Auditor, Migration team
MigrationReview	classify and explain variances, sign off parallel run	with MigrationRun (HIGH)	Controller
OpeningBalanceSign	sign reconciliation certificates and opening trial balance (FR-48.13)	with MigrationRun (HIGH)	Controller, CFO
CutoverManager	generate plan, record go/no-go, initiate rollback (with second approver)	with step owner of go/no-go items they approve (MEDIUM)	Cutover manager
CutoverStepOwner	complete assigned steps	none	named per step
LegacyConnectionAdmin	legacy connection strings, coexistence adapter settings	with MigrationRun in production (MEDIUM)	Platform operator

9.49.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
MIG-1501	error	Extract cutoff {cutoff} is after the legacy freeze {freeze}.	late cutoff	re-extract	9.49.5 validation
MIG-1502	error	Control total {name}{key} mismatch: source {s}, target {t}, delta {d}.	load difference	drill to records	FR-49.2
MIG-1503	warning	{n} records quarantined ({pct}%) in run {run}.	validation failures	fix and release	FR-49.11
MIG-1504	error	Quarantine share {pct}% exceeds {max}%; load stopped.	too many failures	fix mapping	FR-49.11
MIG-1505	info	Re-run loaded 0 new records; all {n} already present by cross-reference.	idempotent re-run	none	FR-49.10
MIG-1506	error	Open items for control account {acct} ({sum}) differ from migrated GL balance ({bal}).	BR-49.2	reconcile both packs	BR-49.2
MIG-1507	warning	Asset {asset}: next depreciation {p} differs from legacy projection {l}.	BR-49.6	review depreciation profile	BR-49.6
MIG-1508	error	Project {project} opening totals do not reconcile ({category} delta {d}).	FR-49.12 mismatch	fix extract or mapping	BR-49.12
MIG-1509	error	Parallel run cannot be signed off: {n} unexplained variances ({amount}).	open variances	explain or fix	FR-49.5
CUT-1601	error	Step {step} cannot start: go/no-go items {items} are open.	sequence rule	complete items	9.49.4 invariant
CUT-1602	warning	Step {step} at {pct}% of planned duration.	overrun	escalate	9.49.6 rules
CUT-1603	error	Rollback requires two distinct approvers.	single approver	add second	9.49.5 validation
CUT-1604	warning	Abort deadline T+{h} h in {minutes} min with open go/no-go items.	BR-49.9	decide	BR-49.9

9.49.19 Performance budgets and capacity

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation under 3× load
Pack load	1.3M rows per synthetic year, up to 50M for large customers	n/a (batch)	≥ 50k records/min	`legacy_xref` ~1 KB per migrated record (50 GB per 50M)	`legacy_xref` unique (system, company, table, id) + (passage_entity, passage_id)	parallelism capped; window extends, rehearsal timing updated
Control-total computation	per run	< 10 min after load	n/a	`control_total` < 1 GB	(run_id, status)	30 min
Historical load	7 years, 300M rows	n/a	≥ 200k rows/min	1–3 TB Parquet	`history.<system>.*` partitioned by (year, company)	longer load before cutover, not during
Parallel-run replay	42k–500k documents/month	per document p95 < 1 s via public APIs	50 docs/s	rehearsal tenant only	n/a	replay over two nights
Xref lookup	per record detail open	5 ms / 15 ms / 30 ms	500/s	n/a	as above	n/a
Coexistence sync	20k changes/day	delivery lag p95 < 15 min	100 msgs/s	n/a	n/a	lag up to 1 h; alert

9.49.20 Localization and compliance hooks

- Opening balances must preserve statutory continuity: DE GoBD and FR FEC require that the closing balances of the legacy books equal the opening balances in Passage and that the link is documented (opening trial balance sign-off, 9.49.15); the FR FEC for the cutover year is produced from legacy history plus Passage postings with `history.unified.*`.
- History years default to 10 when the DE or FR pack is installed (retention minimum), 8 for IN, 7 for US, 6 for UK/CA, 5 for MX (`migration.history_years` cannot go below the wave-48 retention minimum).
- Gap-free document numbering: legacy invoice numbers are kept in `legacy_xref` and shown on reprints; new fiscal sequences start in Passage per pack rules (MX CFDI series/folio, IN invoice series per GSTIN and fiscal year, FR chronological numbering).
- Tax: open AR/AP items keep their original tax amounts and tax codes mapped through the pack tax mapping; IN GST ITC balances and TDS certificates pending issue are migrated as balances with references; MX CFDI UUIDs of open invoices are retained for payment complements (REP).
- Payroll YTD (pack `payroll_ytd`) is limited to US and DE engines in v1; other countries load balances only (open question 2).
- Privacy: rehearsal extracts are masked per FR-48.26; production extracts are encrypted in transit and at rest, retained 90 days after go-live then deleted with a destruction record.
- Extension points: pack-specific mapping tables (tax code, payment method, chart of accounts) as FR-3.13 decision tables; `variance_rule` set per tenant; extractor plug-ins per source schema version.

9.49.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
AX 12.99.4 row 91 — 12.11.1 Project management → Beginning balance journals	PARTIAL-49	ADDED as FR-49.12	opening lines by category with WIP and estimate state
AX 12.32 inventory row "Parallel-run replay (Passage migration factory)"	COVERED-49	confirmed by FR-49.5	

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
SAP 13.4 inventory rows `AS91`–`AS94` legacy asset transfer, `OASV` balance transfer, `OAYE`/`OAYF` legacy transfer date	COVERED-49	confirmed by FR-49.1 (`fixed_assets`, `gl_balances`) and `migration.cutoff_ts`	
SAP 13.99.2 row 345 — TDMS masked refresh (PARTIAL-48)	PARTIAL-48	ADDED as FR-48.26; consumed here by FR-49.8 rehearsal tenants	closed in 9.48.21
AX 12.99.1 theme "Project depth — beginning balance journals (49)"	PARTIAL	ADDED as FR-49.12	same row as above, theme view
SAP 13.99.2 row 24 — bills of exchange (GAP-36)	GAP-36	DEFERRED to wave 36 (owner); pack `open_items` carries a `instrument_type` column mapped once wave 36 ships	migration of open bills of exchange depends on the instrument lifecycle

9.50 Wave 50: Scale Hardening, Multi-Region & Production Go-Live

9.50.1 Scope and business context

AX customers rely on LCS-managed production environments and Microsoft SLAs; SAP customers on Basis operations, HA/DR setups (HANA system replication, cluster failover) or RISE. Passage must prove the same in a self-operated, cloud-agnostic stack: a performance campaign to 10k concurrent users, 1M journal lines/h and 100k order lines/h; active/passive multi-region with cross-region Postgres replication, Kafka MirrorMaker 2 and DNS failover; SRE runbooks, capacity and cost models, zero-downtime upgrades, tenant provisioning automation; complete documentation; the first customer's production go-live and the public v1.0 release. Out of scope: active/active multi-region, per-tenant dedicated clusters, marketplace listings.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
D365 One Version updates and SAP SUM upgrades require a downtime window (hours) and a sandbox cycle; customers defer and accumulate risk	Monolithic application server; schema changes applied in place	Rolling deploys with expand/contract migrations and N/N–1 API contract tests	Upgrade downtime 4–8 h → 0; release cadence quarterly → weekly
DR is a service request to LCS or a Basis project with HANA system replication; failover is rarely rehearsed and RPO/RTO are unknown	DR owned by vendor or built per customer	One-command failover, quarterly game day with measured RTO/RPO	RTO ≤ 1 h, RPO ≤ 5 min, measured every quarter
Performance regressions appear in production (`SysOperation` batch queues, `ST05` traces after the fact)	No standing load test tied to releases	k6 scenarios S1–S7 and SLO burn-rate gates in the release pipeline	Regressions caught pre-release 100%; p99 SLOs published
Environment sizing is a vendor estimator (LCS subscription estimator, Quick Sizer) with opaque cost	No per-tenant metering in the product	Capacity model and `tenant_cost` meter per month	Cost estimate error ±50% → ±5%
Provisioning a new company environment takes days of service requests	Manual environment lifecycle	One-call tenant provisioning with residency and license	Provisioning 3 days → ≤ 15 min

Cutover go/no-go checklist (phase exit, executed at T+24h of the production cutover): (1) all wave-49 packs RECONCILED, 0 MISMATCH; (2) parallel run for the prior month SIGNED_OFF with 0 unexplained; (3) opening trial balance signed by the customer controller; (4) SoD evaluation shows 0 unmitigated HIGH conflicts for go-live users; (5) country packs installed and one statutory fixture executed per entity; (6) load test on the production cluster at 50% of target passed within SLOs; (7) DR game day passed within 30 days before cutover; (8) backup verified: PITR drill on production data within 7 days; (9) security: pen-test high/critical = 0, secrets rotated, images signed; (10) integration endpoints (bank, EDI, e-invoicing, carriers) smoke-tested in production; (11) user acceptance sign-off for 12 core processes; (12) support rota, escalation and rollback decision owner named; (13) monitoring dashboards and alerts green for 24 h; (14) legacy freeze confirmed and rollback plan rehearsed. Any open item = NO-GO; the decision is recorded via `cutover.decision.recorded`.

Hypercare (T+1d to T+30d): 24×5 on-call with 15-minute response for Sev-1, 1 h for Sev-2; daily 09:00 customer stand-up; daily reconciliation of all subledgers to the journal and of lakehouse metrics; weekly reconciliation against legacy coexistence feeds; agents in suggest mode only; change freeze except Sev-1 fixes; exit criteria: first month-end closed and reconciled, Sev-1 count = 0 for 14 consecutive days, Sev-2 backlog ≤ 5, SLOs met for 30 days, customer sign-off.

v1.0 release criteria: all 50 waves' proofs green in CI on the release commit; phase-exit criteria of 9.0 met; CVE scan 0 critical, 0 high older than 30 days; API reference generated from OpenAPI/AsyncAPI with 0 undocumented operations; administrator, implementer and per-module functional guides published; upgrade path from the last stage build tested; license and pricing published; tag `v1.0.0` signed with cosign, SBOM (SPDX) attached, Helm charts and images published by digest.

9.50.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Production environment management	LCS production, sandbox tiers (Tier-2–5), service requests	Basis: `SM51` servers, `RZ10` profiles, RISE managed	Tenant provisioning automation, environment tiers via Helm values	Full
HA/DR	Microsoft-managed geo-secondary, LCS disaster recovery	HANA system replication, Pacemaker clusters, DR site	CNPG replica clusters cross-region, MirrorMaker 2, DNS failover	Full
Capacity and sizing	LCS subscription estimator, usage profile	Quick Sizer, SAPS	Capacity model per tenant (users, lines/h → nodes, storage)	Full
Performance tooling	Performance SDK, Trace parser, `SysOperation` batch tuning	`ST05`, `ST12`, `SAT`, `SQLM`, HANA PlanViz	k6 scenarios, OpenTelemetry traces, pg_stat_statements dashboards	Full
Monitoring	LCS Environment monitoring, Application Insights	`RZ20` CCMS, Solution Manager, Focused Run	Prometheus/Grafana SLO dashboards, Loki, Tempo	Full
Upgrades	One Version monthly updates, LCS deployable packages, downtime	SUM, maintenance planner, zero-downtime option (ZDO)	Expand/contract Atlas migrations, rolling deploys, no downtime	Improved
Batch and job scheduling	Batch framework (`SysOperation`), batch groups	`SM36`/`SM37`, Redwood	Temporal schedules (from wave 4) with multi-region continuity	Full
Licensing	Dynamics 365 per-user licenses	SAP named user licenses, FUE	Passage license file per tenant, user-type metering	Full
Documentation	Microsoft Learn	SAP Help Portal	Docs site: admin, implementer, API, functional guides, training	Full

9.50.3 Functional requirements

- **FR-50.1 Load-test scenarios** — Must. The system shall meet the SLOs below under the k6 scenario set run for 2 h sustained with 10k virtual users on the production-sized cluster:

- S1 journal posting: 1M lines/h via `/api/v1/ledger/journals` (batches of 500); S2 order intake: 100k order lines/h via sales order API plus pricing and ATP; S3 interactive mix: 10k users with 60% list/detail reads, 25% form saves, 10% dashboard loads, 5% explorer queries (think time 10 s); S4 warehouse mobile: 2k scanners at 1 scan/5 s; S5 MRP: 50k-item regeneration concurrent with S3; S6 month-end: inventory close, FX revaluation, allocations concurrent with S1 at 25%; S7 CDC lag: lakehouse freshness measured during S1+S2.

Operation	SLO (p99 unless noted)	Error budget
Entity read (generated)	p95 < 50 ms, p99 < 150 ms	0.1%
Form save	p99 < 400 ms	0.1%
Order intake (line)	p99 < 500 ms	0.1%
Pricing	p95 < 20 ms	0.1%
Journal post (500-line batch)	p99 < 2 s; sustained 5k lines/s	0.05%
Mobile scan round trip	p99 < 300 ms	0.1%
Dashboard load	p95 < 2 s	0.5%
Lakehouse lag	p95 < 60 s	n/a
MRP 50k items	< 10 min	n/a
Availability per tenant	99.9% (99.95% with multi-region)	monthly

- **FR-50.2 Postgres tuning** — Must. The system shall partition the journal line table by month (range) and hash by tenant (16), inventory transactions by month, audit by week; CNPG clusters with 1 primary + 2 synchronous-quorum replicas (`synchronous_standby_names ANY 1`); PgBouncer transaction pooling; `pg_stat_statements` top-50 reviewed with indexes added before exit.
- **FR-50.3 Kafka and cache tuning** — Must. The system shall run Strimzi with 6 brokers, `min.insync.replicas=2`, `acks=all`, topic partitions sized at ≥ 2× consumer parallelism, and Redis-compatible caches for pricing and policy bundles with ≤ 5 s invalidation.
- **FR-50.4 Multi-region active/passive** — Must. The system shall run a passive region with CNPG replica clusters streaming from the primary (async, lag SLO < 30 s; ledger cluster quorum-synchronous across regions when the customer accepts the latency), MirrorMaker 2 for all topics with offset translation, object-store replication for lakehouse and documents, Temporal multi-cluster replication, and DNS failover (health-checked, TTL 60 s).
- **FR-50.5 Failover procedure** — Must. The system shall promote the passive region by one command (`passage-ops failover --to eu-west`), which fences the old primary (revoke writes, scale to zero), promotes CNPG replicas, switches MirrorMaker direction, activates Temporal, flips DNS, and verifies the ledger hash chain; RTO ≤ 1 h, RPO ≤ 5 min.
- **FR-50.6 DR game day** — Must. The system shall run a quarterly game day: (1) announce window, freeze deploys; (2) take baseline hashes (ledger chain head per tenant, row counts per domain, Kafka end offsets); (3) inject failure (primary region network partition via NetworkPolicy, then node pool scale-down); (4) on-call executes failover runbook with timer; (5) verify: hash chain head matches or exceeds baseline, no gap in gap-free sequences, in-flight Temporal workflows resumed, reconciliation jobs green; (6) run smoke suite of 12 processes; (7) measure RTO/RPO; (8) fail back during a planned window; (9) post-game review with actions. Pass: RTO ≤ 1 h, RPO ≤ 5 min, zero committed-transaction loss.
- **FR-50.7 Zero-downtime upgrades** — Must. The system shall deploy services with rolling updates and expand/contract schema migrations (add → backfill → switch → drop in separate releases), with API compatibility tests between N and N−1 and a 30-minute automatic rollback window on SLO breach.
- **FR-50.8 Tenant provisioning** — Must. The system shall create a tenant (Keycloak realm, databases/schemas with RLS, Kafka ACLs, lakehouse namespace, object-store prefixes, residency region, license) from one API call in ≤ 15 min, and decommission with retention export.
- **FR-50.9 Capacity and cost model** — Must. The system shall publish a model mapping (concurrent users, journal lines/h, order lines/h, storage GB) to node counts, Postgres sizes, Kafka brokers and monthly cost per provider, and meter per-tenant resource usage monthly into `tenant_cost`.
- **FR-50.10 SRE runbooks and on-call** — Must. The system shall ship runbooks for the 30 alert rules shipped by the platform, incident severity definitions, escalation, and a status page; every alert links to its runbook.
- **FR-50.11 Documentation** — Must. The system shall publish the administrator guide, implementer guide (migration, localization, cutover), API reference (OpenAPI/AsyncAPI generated), functional guides per module (one per phase 1–8 module), and a training curriculum (role-based, 6 tracks).
- **FR-50.12 Licensing and release** — Must. The system shall enforce a signed license file per tenant (user types: full, operational, self-service; expiry; grace 30 days), and the release pipeline shall tag, sign, attach SBOMs and publish charts and images by digest.

9.50.4 Data model

`region` (platform-ops)

Field	Type	Required	Notes
region_id	text	Y	PK, e.g. `us-east`, `eu-west`
role	enum	Y	PRIMARY, PASSIVE, FENCED
provider	text	Y	eks/gke/aks/bare
cnpg_lag_seconds	int	Y	observed
mm2_lag_messages	bigint	Y	observed
last_failover_event_id	uuid	N	
dns_weight	int	Y	0 or 100

`failover_event`: event_id, from_region, to_region, initiated_by, started_at, dns_flipped_at, promoted_at, completed_at, rpo_seconds, rto_seconds, ledger_hash_ok bool, gapfree_ok bool, notes.

`tenant_license`: tenant_id, license_id, issued_at, expires_at, full_users, operational_users, selfservice_users, signature, state (VALID, GRACE, EXPIRED).

`tenant_cost`: tenant_id, month, cpu_core_hours, memory_gb_hours, storage_gb, kafka_gb, lakehouse_gb, ai_tokens, cost_usd, cost_eur, model_version.

`capacity_model`: model_version, inputs_json (users, lines/h, storage), outputs_json (nodes, pg size, brokers), provider, cost_usd_month.

Invariants: exactly one region is PRIMARY; a FENCED region accepts no writes (enforced by revoking the application role); license `full_users` ≥ active full users else state GRACE; `tenant_cost` is computed from Prometheus data with a stored query version. Indexing: `failover_event` (started_at); `tenant_cost` PK (tenant_id, month).

9.50.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/ops/tenants`	provision tenant	`Idempotency-Key`	PlatformAdmin
DELETE	`/api/v1/ops/tenants/{id}`	decommission with export	by id	PlatformAdmin + second approver
POST	`/api/v1/ops/failover`	initiate failover	by event id	PlatformAdmin (break-glass)
GET	`/api/v1/ops/regions`	region status and lag	n/a	PlatformView
POST	`/api/v1/ops/gameday`	start game day workflow	by id	PlatformAdmin
GET	`/api/v1/ops/capacity/estimate`	capacity estimate for inputs	n/a	PlatformView
GET	`/api/v1/ops/tenants/{id}/cost?month=`	metered cost	n/a	TenantAdmin
PUT	`/api/v1/ops/tenants/{id}/license`	install license	ETag	PlatformAdmin
GET	`/api/v1/ops/slo`	SLO status and error budgets	n/a	PlatformView
GET	`/healthz/region`	region health for DNS checks	n/a	public

Validation: failover rejects if target region replication lag > 300 s unless `--accept-data-loss` with second approver; provisioning rejects a residency region not deployed.

Event	Trigger	Payload fields	Consumers
`dev.passage.ops.failover.completed.v1`	promotion done	from_region, to_region, promoted_lsn, rpo_seconds, rto_seconds	all services (reconnect), audit
`dev.passage.ops.tenant.provisioned.v1`	provisioning done	tenant_id, region, realm, databases[]	billing, catalog
`dev.passage.ops.slo.breached.v1`	SLO burn rate	slo_id, window, burn_rate, tenant_id	alerting, release gate
`dev.passage.ops.license.state_changed.v1`	license check	tenant_id, state, expires_at	shell banner, admin

9.50.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Platform operations console	view regions, initiate failover (two-person), view game-day timeline	region, role, lag, last failover	PlatformAdmin
Tenant management	provision, license, residency, decommission	tenant, region, license state, users	PlatformAdmin
SLO dashboard (Grafana embedded)	filter by tenant/service, view burn rate	SLO, target, current, budget remaining	PlatformView
Cost and capacity	estimate, view monthly meter, export	inputs, outputs, cost USD/EUR	TenantAdmin
Status page (public)	subscribe	component, status, incidents	public
Release notes / docs site	search, version switch	version, module	public

Rules: failover button requires two distinct PlatformAdmin confirmations within 5 min; SLO dashboard default window 30 days; license banner appears in the shell at GRACE.

9.50.7 Business rules and edge cases

- **BR-50.1** Region state machine: PASSIVE → PRIMARY (promotion) with the former PRIMARY → FENCED; FENCED → PASSIVE only after re-sync verified; never two PRIMARY.
- **BR-50.2** During failover, any request to the fenced region returns 503 with `Retry-After: 30`; clients retry idempotent requests with the same key, so no double posting.
- **BR-50.3** Ledger hash chain: each journal carries `prev_hash`; after promotion the chain head per tenant must be verifiable; a break is a Sev-1 and blocks user access.
- **BR-50.4** Gap-free sequences (fiscal invoice numbers) are allocated in the ledger Postgres (synchronous replication), never in caches, so failover cannot duplicate numbers.
- **BR-50.5** Load tests run against a copy of the first customer's masked data, never production.
- **BR-50.6** An SLO burn rate > 14.4× over 1 h (fast burn) pages; > 6× over 6 h (slow burn) tickets; a release with any fast-burn alert in the 30-minute window auto-rolls back.
- **BR-50.7** Expand/contract: a migration that drops a column cannot ship in the same release as the code that stops using it.
- **BR-50.8** License GRACE allows logins for 30 days with a banner; EXPIRED blocks new logins but never data export or admin access.
- **BR-50.9** Tenant decommission retains an encrypted export for the retention period defined by the tenant's country packs before deletion.
- **BR-50.10** Cost model rounds to USD 0.01 / EUR 0.01; FX for EUR at the month-average ECB rate.

9.50.8 Acceptance criteria

- **AC-50.1** (FR-50.1) Given scenarios S1–S7 run concurrently for 2 h, when results are published, then every SLO row is met and error rates are within budget.
- **AC-50.2** (FR-50.2) Given 1M journal lines/h, when posting runs, then ledger Postgres replica lag p95 < 5 s and no query exceeds 1 s in pg_stat_statements.
- **AC-50.3** (FR-50.5, 50.6) Given the game day partition, when on-call runs the failover command, then RTO ≤ 1 h, RPO ≤ 5 min, ledger hash chain verified, gap-free sequences continuous.
- **AC-50.4** (FR-50.6) Given failover completed, when the 12-process smoke suite runs in the new primary, then all pass and CDC lag recovers to < 60 s within 15 min.
- **AC-50.5** (FR-50.7) Given release N+1 with an expand migration, when rolled out, then zero failed requests and N clients continue to work.
- **AC-50.6** (FR-50.8) Given a provisioning request with residency EU, when it runs, then the tenant exists in `eu-west` with RLS, realm, ACLs and license in ≤ 15 min.
- **AC-50.7** (FR-50.9) Given one month of production, when the cost meter runs, then `tenant_cost` is within 5% of provider billing for the cluster.
- **AC-50.8** (FR-50.12) Given a tenant whose license expired 31 days ago, when a user logs in, then login is blocked and admin export works.
- **AC-50.9** (9.50.1 checklist) Given the production cutover, when T+24h arrives, then all 14 go/no-go items are complete and the decision is recorded.
- **AC-50.10** (9.50.1 hypercare) Given the first production month, when month-end closes, then all subledgers reconcile to the journal, lakehouse revenue equals GL, and the customer signs off.
- **AC-50.11** (FR-50.11, 50.12) Given the release commit, when the pipeline runs, then `v1.0.0` is tagged and signed, SBOMs attached, docs published with 0 broken links and 0 undocumented API operations.

9.50.9 Test plan and fixtures

Fixtures: `fx-load-10k` (masked copy of the first customer plus synthetic growth to 10k users, 3 years of transactions, 40M journal lines); `k6-scenarios` (S1–S7 scripts with data pools); `fx-gameday` (baseline hash manifest generator); `fx-smoke-12` (12 core processes: PO→receipt→invoice→payment, SO→ship→invoice→cash, production order→RAF, journal post, bank reconciliation, asset depreciation, project invoice, payroll run, statutory report, dashboard, copilot query, mobile pick).

Test	Type	Target	Pass criterion
Load campaign	performance	whole platform	SLO table met, 2 h
Soak	performance	whole platform	24 h at 50% load, no memory growth > 10%
Failover	E2E	multi-region	RTO/RPO met, hash chain ok
Fail-back	E2E	multi-region	re-sync and promotion without loss
Upgrade compatibility	contract	APIs N vs N–1	100% of contract tests pass
Expand/contract linter	unit	migrations	no drop in same release as last use
Provisioning	E2E	tenant API	≤ 15 min, all components
Chaos	E2E	Kafka broker loss, Postgres primary kill	no failed postings, recovery < 5 min
Docs	contract	docs site	0 broken links, OpenAPI coverage 100%
Smoke 12	E2E	production	all pass post-cutover and post-failover

Reconciliation: post-failover row counts and ledger hash heads ↔ baseline; production daily subledger ↔ journal; lakehouse metrics ↔ GL; `tenant_cost` ↔ provider billing; license active users ↔ Keycloak sessions.

9.50.10 Deployment and operations

Helm: `passage-platform` umbrella with per-region values; `passage-mm2` (MirrorMaker 2), `passage-ops` (Go control plane), `passage-status-page`, `passage-docs`. CNPG `Cluster` with `replica.enabled` in the passive region; Temporal multi-cluster. Config: `REGION_ROLE`, `DNS_PROVIDER` (thin Terraform module per provider), `FAILOVER_MAX_LAG_S=300`, `LICENSE_PUBLIC_KEY`. Migrations: `ops` database. Flags: `ops.failover.enabled`, `license.enforce`. Jobs: cost meter monthly, capacity model nightly, game day quarterly (scheduled Temporal workflow with human task), backup drill monthly, SLO report weekly. SLOs: as FR-50.1; control-plane API p99 < 500 ms. Alerts: replication lag > 60 s, MM2 lag > 100k messages, SLO fast/slow burn, license expiry 30/7/1 days, failover in progress. Runbooks: failover, fail-back, broker replacement, Postgres primary loss, slot bloat, cache stampede, Temporal namespace replication lag, certificate expiry, object-store replication lag, release rollback. Rollback: Argo CD rollback by digest; database contract step deferred to the next release; DNS weight restore.

9.50.11 Migration from AX / SAP

Source	Passage	Notes
LCS environment inventory (production, sandboxes)	Tenant and environment tiers	documented; sandboxes re-created from masked production via provisioning
LCS deployable packages / Basis transports (`STMS`)	GitOps releases	not migrated; release notes map customer changes to metadata extensions
Batch groups and job schedules (`BatchJob`, `SM37`)	Temporal schedules	imported in wave 4 and verified for production cadence here
Monitoring thresholds (CCMS `RZ20`, LCS alerts)	Prometheus alert rules	mapped per alert category
D365 / SAP user licenses	`tenant_license` user types	full ↔ Dynamics full user / SAP professional; operational ↔ activity / functional; self-service ↔ team member / productivity

Not migrated: Microsoft/SAP SLAs and support contracts (replaced by Passage support terms), proprietary performance traces.

9.50.12 Dependencies, risks, open questions

Depends on: wave 1 (pipeline, Argo CD), wave 3 (Kafka, outbox), wave 4 (Temporal), wave 11 (ledger hash chain, gap-free sequences from wave 5), wave 46 (lakehouse lag measurement), wave 48 (backup drills, residency, pen test), wave 49 (customer migration and cutover runbook).

Risk	Likelihood	Impact	Mitigation
Cross-region synchronous ledger replication adds latency beyond SLO	Medium	High	quorum within region plus async cross-region by default; customer opt-in to cross-region sync
First customer month-end reveals unmapped edge cases	High	High	hypercare change process; parallel-run month before cutover; agents in suggest mode
Load test environment differs from production	Medium	Medium	same Helm values and node classes; 50% load re-run on production before cutover
DNS TTL caching delays failover for some clients	Medium	Medium	TTL 60 s; client retry with alternate endpoint list

Open questions: (1) Active/active for read-only analytics in the passive region — proposed yes for Trino only; decide before v1.1. (2) Public pricing unit (per full user per month vs per tenant tier) — proposed per user type; decide at v1.0 release. (3) Licensing model (open-core vs source-available) carried from PRD §12 — must be decided before the v1.0 tag.

9.50.13 Parameters and setup

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`REGION_ROLE`	region	enum	PRIMARY / PASSIVE	PRIMARY, PASSIVE, FENCED	write enablement	LCS environment type	HANA system replication role (`hdbnsutil -sr_state`)
`FAILOVER_MAX_LAG_S`	platform	int	300	30–3600	failover rejection threshold	LCS DR configuration	HSR log retention / `logshipping_timeout`
`failover.two_person_window_min`	platform	int	5	1–30	second confirmation window	none	none
`dns.ttl_s`	platform	int	60	30–300	failover propagation	Azure Traffic Manager TTL	DNS/virtual hostname switch
`cnpg.sync_quorum`	cluster	text	`ANY 1`	`ANY n`, `FIRST n`	FR-50.2 durability	Azure SQL geo-replication	HSR `SYNC`/`SYNCMEM` mode
`cnpg.cross_region_sync.ledger`	tenant (opt-in)	bool	false		FR-50.4 cross-region sync	none	HSR multi-target sync

Parameter	Scope	Type	Default	Allowed values	Effect	AX equivalent	SAP equivalent
`kafka.min_insync_replicas`	platform	int	2	2–3	FR-50.3	none	none
`mm2.lag_alert_messages`	platform	int	100,000	≥ 1,000	alert	none	none
`slo.<id>.target`	platform / tenant	numeric	per FR-50.1 table		SLO and error budget	LCS SLA	RISE SLA
`slo.burn.fast` / `slow`	platform	ratio	14.4×/1 h, 6×/6 h		BR-50.6	none	Focused Run alert thresholds
`release.auto_rollback_window_min`	platform	int	30	10–120	FR-50.7	One Version rollback (none automatic)	SUM ZDO fallback
`release.cadence`	tenant	enum	WEEKLY	WEEKLY, BIWEEKLY, MONTHLY (pinned ring)	update ring	One Version update cadence, pause updates	Maintenance planner, FPS schedule
`provisioning.default_tier`	platform	enum	PROD	DEV, TEST, UAT, PROD	resource profile	LCS Tier-1 to Tier-5	system landscape DEV/QAS/PRD
`provisioning.timeout_min`	platform	int	15	5–60	FR-50.8	LCS deployment	client copy duration
`license.grace_days`	tenant	int	30	0–60	BR-50.8	license enforcement grace	SAP license key temporary (28 days)
`license.enforce`	tenant	bool	true		FR-50.12	none	`SLICENSE`
`cost.fx_source`	platform	enum	ECB_MONTH_AVG	ECB_MONTH_AVG, FIXED	BR-50.10	none	none
`capacity.model_version`	platform	text	latest	published	estimates	LCS subscription estimator	Quick Sizer version
`gameday.cron`	platform	cron	quarterly first Tuesday 14:00 UTC	cron	FR-50.6	none	DR test schedule
`hypercare.days`	tenant	int	30	14–90	hypercare window	none	hypercare
`hypercare.change_freeze`	tenant	bool	true during hypercare		Sev-1 fixes only	none	transport freeze (`STMS` lock)
`status_page.components`	platform	list	API, UI, Analytics, Integrations, Mobile		public status	Service health (LCS)	SAP for Me availability
`docs.version_retention`	platform	int	6 releases	3–24	docs site versions	Microsoft Learn versions	Help Portal versions

9.50.14 Inquiries and reports

Name	Type	Filters & parameters	Columns / measures	Source projection	AX equivalent	SAP equivalent
Region status	list inquiry	region	role, CNPG lag, MM2 lag, Temporal replication, DNS weight	`region` + Prometheus	LCS environment monitoring	`HDBSettings` replication status, `RZ20`
Failover history	list inquiry	date	from/to, RPO, RTO, hash ok, gap-free ok	`failover_event`	LCS DR history	DR test log
SLO and error budget	KPI	service, tenant, window	target, current, budget remaining, burn rate	Prometheus recording rules	LCS health metrics	Focused Run / Solution Manager SLR
Load-test campaign report	operational report	campaign, scenario	p50/p95/p99, error rate, throughput per S1–S7	k6 results store	Performance SDK results	`ST03N` workload, LoadRunner reports
Top SQL	drill-down	cluster, period	query, calls, mean/p99 time, rows	`pg_stat_statements`	Trace parser, LCS SQL insights	`ST04` / `DBACOCKPIT`, `SQLM`
Capacity estimate	operational report	users, lines/h, storage	nodes, PG sizes, brokers, cost	`capacity_model`	LCS subscription estimator	Quick Sizer
Tenant cost	KPI	tenant, month	CPU, memory, storage, Kafka, lakehouse, AI tokens, USD/EUR	`tenant_cost`	none	none
License status	list inquiry	tenant	user types used vs licensed, state, expiry	`tenant_license` + wave-3 `UsageSnapshot`	License usage report	`USMM`, `SLAW`
Tenant inventory	list inquiry	region, tier	tenant, region, tier, residency, version ring	`tenant`	LCS environment list	`SM51`, landscape directory
Release history	list inquiry	version	digest, SBOM, migrations, rollbacks	release pipeline	LCS update history	`SPAM` / `SAINT` history
Hypercare dashboard	KPI	tenant	Sev-1/2 counts, backlog, daily reconciliation status, SLO	incidents + recon	none	hypercare reports

9.50.15 Printed and electronic documents

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Release notes	each release	version, digest, features by module, migrations, breaking changes (none allowed for N–1), CVE fixes	HTML, Markdown, PDF	semver	kept for all versions
SBOM	each release	packages, versions, licenses, hashes	SPDX 2.3 JSON, CycloneDX	per image digest	5 years (supply-chain evidence, CC8.1)
Signature attestation	release tag	cosign signature, Rekor entry	Sigstore bundle	per digest	permanent
DR game-day report	after game day	timeline, RTO/RPO measured, hash verification, actions	PDF	game day id	SOC 2 CC7.5 evidence (wave 48 lock)
Load-test certificate	campaign pass	scenarios, SLO table, environment, commit	PDF	campaign id	release evidence
Tenant license file	license install	tenant, user-type counts, expiry, signature	signed JSON (Ed25519)	license id	contract term
Monthly tenant cost statement	month close	resource lines, cost USD/EUR, model version	PDF, CSV	(tenant, month)	billing records 7 years
Go/no-go decision record	T+24 h	14 checklist items, evidence links, approvers	PDF with e-signature (FR-48.13)	plan id	audit evidence
Hypercare exit report	hypercare end	exit criteria, incidents, reconciliation results, customer sign-off	PDF	tenant + date	project record

Document	Trigger	Layout elements	Formats	Numbering sequence	Legal / retention requirement
Status-page incident notice	incident	component, impact, timeline, resolution	HTML, e-mail, RSS	incident id	1 year public history

9.50.16 Periodic and batch jobs

Job	Schedule default	Partitioning key	Idempotency / checkpoint	Typical runtime target	Failure handling	Monitor / alert
Cost meter	monthly, 2nd 02:00 UTC	tenant	(tenant, month)	< 30 min	rerun idempotent	variance vs billing > 5%
Capacity model refresh	nightly	provider	model_version	< 10 min	keep previous	none
DR game day	quarterly	region pair	Temporal workflow with human tasks	< 4 h including fail-back	abort and fail back	RTO/RPO breach
Backup verification on production	weekly	cluster	drill id	< 1 h	Sev-2	failure
SLO report	weekly Monday	service	week	< 5 min	retry	budget < 25%
License check	hourly	tenant	state transition	< 1 min	keep last state	expiry 30/7/1 days
Soak test	before each minor release	scenario	campaign id	24 h	release blocked	memory growth > 10%
Load campaign S1–S7	before each minor release and pre-cutover	scenario	campaign id	2 h	release blocked	any SLO miss
Certificate expiry scan	daily	namespace	n/a	< 1 min	auto-renew via cert-manager; page at 7 days	expiry < 14 days
Replication slot and bloat check	every 15 min	cluster	n/a	< 10 s	runbook	slot lag > 1 GB
Docs build and link check	per release	site	commit	< 10 min	release blocked	broken links > 0

9.50.17 Security duties and privileges

Duty	Privileges (entry points)	Segregation-of-duties conflicts	Default roles that include it
PlatformAdmin	`POST /ops/tenants`, `DELETE /ops/tenants/{id}` (with second approver), `POST /ops/failover`, `PUT /ops/tenants/{id}/license`, game day	with any tenant business duty (HIGH, `OPS-001`: operators do not post)	Platform operator (break-glass for failover)
PlatformView	`GET /ops/regions`, `/ops/slo`, `/ops/capacity/estimate`	none	SRE, Support
ReleaseManager	promote release by digest, pause ring, trigger rollback	with code approval of the same change (MEDIUM, `CHG-001`: four-eyes)	Release manager
OnCallResponder	runbook actions (restart, scale, replay), incident updates	none	SRE on-call
TenantAdmin	`GET /ops/tenants/{id}/cost`, user-type assignment within license	with PlatformAdmin (MEDIUM)	Customer administrator
LicenseIssuer (offline)	sign license files with the Ed25519 private key (HSM/Vault transit)	with PlatformAdmin (HIGH)	Passage commercial operations
StatusPagePublisher	publish incident notices	none	Support lead
CutoverDecisionOwner	record go/no-go at production cutover	with CutoverStepOwner of checklist items (MEDIUM)	Customer CFO / project sponsor

9.50.18 Message catalog

Code	Severity	User text	Cause	Remediation	Raised by
OPS-1701	error	Failover rejected: target region lag {lag} s exceeds {max} s. Use --accept-data-loss with a second approver.	lag above limit	wait or accept with approval	9.50.5 validation
OPS-1702	error	Failover needs a second PlatformAdmin confirmation within {min} min.	two-person rule	second confirmation	9.50.6 rules
OPS-1703	info	This region is read-only during failover; retry in {s} s.	fenced region (503)	client retry	BR-50.2
OPS-1704	error	Ledger hash chain broken for tenant {tenant} at journal {id}; user access blocked.	BR-50.3	Sev-1 runbook	BR-50.3
OPS-1705	error	Residency region {region} is not deployed; tenant not provisioned.	provisioning validation	deploy region or change residency	9.50.5 validation
OPS-1706	warning	Release {v} rolled back automatically: fast burn on {slo}.	BR-50.6	investigate before re-release	BR-50.6
OPS-1707	error	Migration drops column {col} still referenced in this release.	expand/contract linter	split releases	BR-50.7
OPS-1708	warning	Replication lag {lag} s on {cluster} exceeds 60 s.	lag	check network and WAL volume	9.50.10 alerts
LIC-1801	warning	License expires in {days} days.	expiry approaching	renew	FR-50.12
LIC-1802	warning	License in grace period: {used} full users active, {licensed} licensed.	over-use or expiry	true-up or reassign users	BR-50.8
LIC-1803	error	License expired; sign-in blocked. Administrators can still sign in and export data.	EXPIRED	renew	BR-50.8
LIC-1804	error	License signature invalid.	tampered or wrong key	reinstall issued file	FR-50.12
COST-1901	warning	Metered cost for {month} differs from provider billing by {pct}%.	meter drift	review query version	AC-50.7

9.50.19 Performance budgets and capacity

The SLO table in FR-50.1 is the binding budget. Capacity assumptions per reference tenant (10k concurrent users) follow.

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation under 3× load
Journal posting	1M lines/h peak, 8M lines/day	500-line batch 600 ms / 1.2 s / 2 s	5k lines/s sustained	1.2 TB ledger (lines + audit)	range by month, hash by tenant (16)	queue in outbox, 429 with `Retry-After` for batch APIs; interactive saves prioritized

Operation	Volume assumption	Latency target (p50 / p95 / p99)	Throughput target	Storage growth per year	Index / partition strategy	Degradation under 3× load
Order intake	100k lines/h	150 ms / 300 ms / 500 ms	30 lines/s per pod	200 GB	by month	ATP degrades to cached availability (flagged)
Interactive mix	10k users, think time 10 s	read 20 / 50 / 150 ms; save 120 / 250 / 400 ms	1k req/s	n/a	PgBouncer transaction pooling	p99 doubles; no errors
Mobile scan	2k scanners, 1 scan/5 s	80 / 180 / 300 ms	400 scans/s	100 GB	by month	offline queue on device (wave 32)
Kafka	60k msgs/s peak	produce p99 < 50 ms	6 brokers	30 TB retained 7 days (tiered)	partitions ≥ 2× consumers	consumer lag grows; lakehouse badge amber
Cross-region replication	WAL 40 MB/s peak	lag p95 < 30 s	n/a	n/a	n/a	lag alert > 60 s; failover window widens
Tenant provisioning	20 tenants/day	< 15 min	5 concurrent	~2 GB per empty tenant	per-tenant schemas	queued
Failover	per event	RTO ≤ 1 h, RPO ≤ 5 min	n/a	n/a	n/a	runbook manual steps

9.50.20 Localization and compliance hooks

- Region placement follows wave-48 tenant residency: EU tenants run primary and passive in EU regions only (e.g. `eu-west` ↔ `eu-central`); IN tenants in-country; failover never crosses a residency boundary (`POST /ops/failover` validates the pair).
- Retention on decommission follows the tenant's country packs (BR-50.9): encrypted export kept for the longest applicable record-class minimum (DE/FR 10 years) unless the customer takes custody and signs a release.
- Status page and release notes are published in English with German and French translations for customer-facing incident notices (contractual notice language per tenant).
- Time zones: maintenance windows, game days and hypercare stand-ups are scheduled in the tenant's time zone; SLO reports use UTC.
- Licensing and cost: cost statements show USD and EUR with ECB month-average FX (BR-50.10); taxes on Passage's own invoices are outside the product.
- Compliance evidence produced here (game-day reports, SBOMs, signatures, backup verification, change records) flows into the wave-48 evidence store for CC7.5, CC8.1 and A1.2/A1.3 (availability criteria).
- Extension points: per-provider Terraform modules, Helm values per region, `slo.<id>.target` per contract tier.

9.50.21 Coverage closure

Register row (inventory section + AX form or SAP T-code)	Status before	Disposition	Note
(no GAP or PARTIAL register row carries current or proposed wave 50)	—	—	rows below are inventory COVERED-50 rows and cross-references verified against this wave
AX 12.30 inventory rows "Tenant provisioning / Environment management", "Tenant provisioning templates / Environment catalog"	COVERED-50	confirmed by FR-50.8 and 9.50.13 `provisioning.*`	
AX 12.30 inventory row "DR → Game days / Failover drills"	COVERED-50	confirmed by FR-50.5, FR-50.6	
AX 12.30 inventory row "Database → Backup / Restore / Refresh from production (LCS)"	COVERED-48 (DR in 50)	confirmed by FR-48.9, FR-48.26 (masked refresh) and FR-50.4	
SAP 13.99.2 row 345 — TDMS masked refresh (PARTIAL-48, "propose wave 48/50")	PARTIAL-48	ADDED as FR-48.26; consumed here by BR-50.5 load tests	closed in 9.48.21
AX 12.30 rows "License inquiries", "License report / Named user counts", "Named user license count refresh" (PARTIAL-3)	PARTIAL-3	SUPERSEDED by FR-3.14 (wave 3 `UsageSnapshot`); FR-50.12 enforcement reads it	
AX 12.99.1 theme "Alerts and notification rules" (wave 5)	GAP-5	DEFERRED to wave 5 (owner); platform alerting is FR-5.14 + 9.50.10	

Part 10 — Appendices

These appendices are normative for every phase file (`docs/prd/phase-N-*.md`). Where a phase file and an appendix disagree, the appendix wins for conventions (naming, envelopes, error codes, security, testing) and the phase file wins for business behavior. The stack is fixed by ADR-0001; nothing here proposes infrastructure outside it.

A. Glossary

Terms are drawn from AX 2012 / D365 F&SCM, SAP ECC / S/4HANA, the standards Passage implements, and Passage itself. The last column is the name used in Passage metadata, APIs and UI.

Term	Origin	Definition	Passage term
AOT	AX	Application Object Tree: the AX metadata repository for tables, classes, forms, menus.	Metadata registry
DDIC	SAP	ABAP Data Dictionary: domains, data elements, tables, append structures.	Metadata registry
EDT	AX	Extended Data Type: a typed, reusable field definition (e.g. `CustAccount`).	Field type
Base enum	AX	Fixed enumeration stored as an integer; extensible via enum extensions.	Enum (metadata)
Extension (vs overlayering)	AX	D365 model that adds to standard objects instead of editing them.	Entity extension / hook
DataAreaId	AX	The company (legal entity) column present on every company-specific table.	`legal_entity_id`
MANDT	SAP	Client key present on every client-dependent table.	`tenant_id`
Tenant	Passage	Top-level isolation boundary: one customer organization, one RLS scope, one Keycloak realm.	Tenant
Legal entity	AX / Passage	A company that produces its own financial statements. SAP: company code (BUKRS).	LegalEntity
Operating unit	AX	Department, cost center, business unit, value stream: non-legal organization units.	OperatingUnit
Company code (BUKRS)	SAP	Smallest organizational unit with a complete self-contained set of accounts.	LegalEntity
Plant (WERKS)	SAP	Physical production/logistics location. AX: site.	Site
Storage location (LGORT)	SAP	Sub-location of a plant where stock is held. AX: warehouse.	Warehouse
Role / duty / privilege	AX	Three-layer security model: privileges grant entry points, duties group privileges, roles group duties.	Role / Duty / Privilege
Authorization object	SAP	Named set of fields (e.g. `F_BKPF_BUK`) checked at runtime.	Privilege
SoD	AX / SAP GRC	Segregation of duties: rules preventing one user from holding conflicting duties.	SoD rule
Number sequence	AX	Generator of document identifiers per scope; continuous (gap-free) or not. SAP: number range object (SNRO).	NumberSequence
Database log	AX	Table/field change log. SAP: change documents (CDHDR/CDPOS).	Audit log (CDC)
Workflow	AX / SAP	Approval and process orchestration (AX workflow framework, SAP Business Workflow).	Temporal workflow
DIXF / DMF	AX	Data Import Export Framework / Data Management Framework. SAP: LSMW, Migration Cockpit.	Migration pipeline
LCS	AX	Lifecycle Services: Microsoft's hosted deployment/upgrade portal. SAP: STMS/CTS+.	Argo CD GitOps pipeline
DirParty / Global Address Book	AX	Shared party model with customer/vendor/contact roles.	Party
Business Partner (BP)	SAP	S/4 single party record with roles; CVI links to KNA1/LFA1.	Party
CustTable / KNA1	AX / SAP	Customer master table.	Customer (party role)
VendTable / LFA1	AX / SAP	Vendor master table.	Vendor (party role)
Posting profile	AX	Mapping from a customer/vendor/item group to GL control accounts. SAP: reconciliation account.	PostingProfile
Product master / variant	AX	A product with dimensions (size, color, style, configuration) that explode into distinct products. SAP: configurable material.	ProductMaster / Product
Released product	AX	A product made available to one legal entity with entity-specific defaults. SAP: MARC plant view.	ReleasedProduct
MARA / MARC / MARD	SAP	Material master: general, plant and storage-location views.	Product / ReleasedProduct / on-hand
Dimension group	AX	Product, storage, and tracking dimension groups controlling which dimensions apply to an item.	DimensionGroup
Item model group	AX	Costing method, negative inventory policy, physical/financial posting flags. SAP: valuation class + MRP/accounting views.	ItemModelGroup
GTIN / SSCC	GS1	Global Trade Item Number; Serial Shipping Container Code (18-digit pallet/case id).	GTIN / SSCC
BOM	AX / SAP	Bill of materials: component list by version with effective dates. SAP: STKO/STPO.	BOM
Formula	AX	Process-industry BOM with potency, co-/by-products, scalable quantities. SAP: master recipe.	Formula
Route / routing	AX / SAP	Operation sequence with resources and times. SAP: PLKO/PLPO.	Routing
ECM / ECO	AX / SAP	Engineering change management; engineering change order.	ChangeOrder
Universal journal	SAP	ACDOCA: single line-item table for FI, CO, AA and ML. AX: GeneralJournalAccountEntry.	JournalEntry / JournalLine
Financial dimension	AX	Analytical attribute on a ledger line (department, cost center, project). SAP: CO object / profit center / segment.	Dimension
Account structure	AX	Rules for which dimension combinations are valid per main account.	AccountStructure
Posting layer	AX	Parallel accounting layer (Current, Operations, Tax). SAP: ledger group / non-leading ledger.	Ledger (parallel)
Currency types	SAP	10 company code, 30 group, 40 hard currency amounts per line. AX: accounting/reporting currency.	Transaction / accounting / reporting currency
OB52	SAP	Posting period control per account type and ledger. AX: ledger calendar period status per module.	PeriodControl
Allocation rule	AX	Basis/fixed distribution of balances across dimensions. SAP: assessment/distribution (KSU5/KSV5).	AllocationRule
FSV	SAP	Financial statement version: hierarchical statement definition (OB58). AX: Management Reporter row/column definitions.	StatementDefinition
InventTrans	AX	Immutable inventory transaction with issue/receipt status. SAP: MSEG material document line.	InventoryTransaction
InventSum	AX	On-hand aggregate per dimension combination. SAP: MARD/MCHB stock tables.	OnHand (projection)
Physical / financial status	AX	Transaction lifecycle: ordered → reserved → picked → deducted / received → registered → posted (financially).	InventoryTransaction status
Reservation	AX / SAP	Allocation of on-hand or expected receipts to a demand.	Reservation

Term	Origin	Definition	Passage term
Inventory close	AX	Periodic settlement of issues against receipts to finalize cost. SAP: material ledger closing (CKMLCP).	Cost close
Standard cost / costing version	AX	Frozen unit cost by version and site, activated per date. SAP: standard price, costing run (CK40N).	CostingVersion
Costing sheet	AX / SAP	Indirect cost (overhead) rule tree applied during cost roll-up.	CostingSheet
Product receipt	AX	Posting that records goods received against a PO. SAP: goods receipt (MIGO, movement type 101).	ProductReceipt
GR/IR	SAP	Goods-receipt / invoice-receipt clearing account; accrues liability between receipt and invoice. AX: accrued purchases.	GR/IR accrual account
Three-way match	AX / SAP	Invoice compared to PO and receipt within price/quantity tolerances (MIRO, tolerance keys).	Matching
ERS	SAP	Evaluated receipt settlement: self-billing from receipts without a vendor invoice.	ERS
Open item	SAP	Unsettled AP/AR line (FBL1N/FBL5N). AX: customer/vendor transaction.	OpenItem
Settlement	AX	Application of payment or credit against open items. SAP: clearing.	Settlement
F110	SAP	Automatic payment program. AX: payment proposal.	PaymentProposal
pain.001 / camt.053	ISO 20022	Credit-transfer initiation message / end-of-day bank statement message.	Payment file / BankStatement
NACHA	Standard	US ACH file format.	Payment file (US)
Production order	AX / SAP	Order to make a quantity of an item via BOM and route (CO01).	ProductionOrder
RAF	AX	Report as finished: posts finished goods receipt from a production order. SAP: order confirmation (CO11N) + GR 101.	ReportAsFinished
Coverage group	AX	Planning parameters per item: requirement/period/min-max coverage, time fences, safety stock. SAP: MRP type + lot-sizing procedure.	CoverageGroup
MRP / MPS	Standard	Material requirements planning; master production schedule.	Planning run
Planned order	AX / SAP	MRP proposal for purchase, production or transfer, to be firmed.	PlannedOrder
Action / delay message	AX	Planner advice: advance, postpone, cancel; or dates that cannot be met. SAP: exception messages.	PlanningMessage
Finite / infinite capacity	AX / SAP	Scheduling that respects or ignores resource load (CM01, PP/DS).	Scheduling mode
Catch weight	AX	Dual-UoM inventory (nominal units plus actual weight).	CatchWeight
Potency	AX	Active-ingredient concentration used to scale formula consumption.	Potency
OEE	ISO 22400	Overall equipment effectiveness = availability × performance × quality.	OEE
Trade agreement	AX	Price/discount list by account/group/all × item/group/all with effective dates. SAP: condition record.	PriceCondition
Condition technique	SAP	Pricing procedure of condition types resolved via access sequences (VK11).	Pricing procedure / access sequence
ATP / CTP	AX / SAP	Available-to-promise from on-hand and supply; capable-to-promise considers capacity.	ATP
Packing slip	AX	Delivery document posting inventory deduction and COGS. SAP: outbound delivery + post goods issue.	PackingSlip (Shipment posting)
Handling unit / license plate	SAP / AX	Physical container (pallet, case) with its own id holding nested items.	LicensePlate
Location directive	AX	WMS rule choosing put-away/pick locations. SAP: storage type search strategy.	LocationDirective
Quality order	AX	Inspection with tests and results leading to a disposition. SAP: inspection lot + usage decision.	QualityOrder
NCR / CAPA	Quality	Nonconformance report; corrective and preventive action. SAP: quality notification.	Nonconformance / CAPA
AQL	ANSI/ASQ Z1.4	Acceptable quality level sampling plan.	SamplingPlan
Landed cost	Standard	Freight, duty and handling allocated to inventory cost.	LandedCost
EDI 850 / 856 / 810	X12	Purchase order / advance ship notice / invoice transaction sets. EDIFACT: ORDERS / DESADV / INVOIC.	EDI message
Peppol / UBL	Standard	Pan-European e-invoicing network; Universal Business Language XML.	E-invoice
Dunning	SAP	Reminder letters by escalation level (F150). AX: collection letters.	CollectionLetter
RMA	Standard	Return merchandise authorization. AX: return order; SAP: RE order.	ReturnOrder
SSP	ASC 606	Standalone selling price used to allocate transaction price across performance obligations.	SSP
Performance obligation	ASC 606 / IFRS 15	Distinct promise in a contract to which revenue is allocated.	PerformanceObligation
ROU asset	IFRS 16 / ASC 842	Right-of-use asset recognized for a lease.	Lease
Budget control / encumbrance	AX / SAP	Reservation of budget by requisitions and POs (commitments).	Encumbrance
Elimination	Consolidation	Removal of intercompany balances and unrealized profit in group statements.	EliminationRule
POC	Accounting	Percentage-of-completion revenue estimate. SAP: results analysis.	ProjectEstimate
Functional location	SAP PM	Hierarchical place where equipment is installed. D365: functional location.	FunctionalLocation
Installed base	SAP CS	Serialized equipment at customer sites covered by service.	InstalledBase
Entity store	AX	D365 aggregate measurement store for Power BI. SAP: BW/CDS views.	Lakehouse (Iceberg)
MCP	Standard	Model Context Protocol: tool interface for AI agents.	MCP server
Copilot / agent	Passage	Copilot answers and drafts; an agent executes multi-step work under approval gates.	Copilot / Agent
Outbox	Pattern	Table written in the same transaction as the state change and relayed to Kafka by Debezium.	Outbox
CloudEvent	CNCF	Standard event envelope (specversion, id, source, type, subject, time, data).	Event
RLS	Postgres	Row-level security policy evaluated on every statement.	Tenant isolation
Wave (delivery)	Passage	A 4–8 week increment with deliverable, proof, deploy and references. Not to be confused with a WMS pick wave.	Wave

B. Cross-module data model

B.1 Bounded contexts

Passage has 12 core bounded contexts. Each is a deployable set of services with exactly one Postgres database (CloudNativePG cluster) whose schema is generated from entity metadata. Three supporting contexts (`integration`, `analytics`, `agents`) own only operational tables (partner profiles, message logs, prompt

registry, eval runs) and never master or transactional business data.

Context	Database	Owns	Language	Waves
`platform`	`passage_platform`	Tenants, users, roles/duties/privileges, metadata registry, number sequences, audit log, workflow/batch definitions, legal entities, operating units, sites, warehouses, hierarchies, fiscal calendars, ISO reference data, UoM	Go (sequences, audit), TS (admin, workflow)	1–6
`party`	`passage_party`	Party, roles (customer, vendor, contact, employee link), addresses, tax registrations, groups, payment and delivery terms, credit limits	TS	7
`product`	`passage_product`	Product, product master, released product, dimension groups, attributes, BOM, formula, routing, resources, calendars, change orders	Go (explosion), TS	8–9
`ledger`	`passage_ledger`	Accounts, account structures, dimensions, journal entries/lines, balances, ledgers, exchange rates, tax codes and transactions, period control, allocations, accruals, statements, budgets, cost accounting ledger, consolidation groups	Go	11–15, 38, 40
`inventory`	`passage_inventory`	Inventory transactions, on-hand projection, reservations, inventory journals, counting, cost layers, costing versions, costing sheets, quality orders, nonconformances	Go	16, 20, 33, 39
`procurement`	`passage_procurement`	Requisitions, RFQs, purchase agreements, purchase orders, product receipts, vendor invoices, match results, AP open items, payment proposals, payments	Go (matching, settlement), TS	17–20
`manufacturing`	`passage_manufacturing`	Production/batch orders, picking lists, route cards, planning runs, planned orders, planning messages, schedules, kanbans, time registrations, machine events	Go; Python (forecasting)	21–25
`sales`	`passage_sales`	Price conditions, quotations, sales orders, holds, shipments/packing slips, customer invoices, revenue schedules, AR open items, cash applications, collection cases, return orders, rebate agreements	Go (pricing, ATP, settlement), TS	26–30
`warehouse`	`passage_warehouse`	Locations, license plates, location directives, work templates, work, waves, replenishment, loads, routes, carrier rates, freight invoices, mobile flow definitions	Go (directives), TS	28, 31–34
`treasury`	`passage_treasury`	Bank accounts, statements, reconciliations, cash forecasts, checks, letters of credit, fixed assets, asset books, asset transactions, leases	Go	36–37
`workforce`	`passage_workforce`	Workers, positions, jobs, employment, absence, benefits, payroll runs, pay rules, expense reports	TS; Go (payroll calc)	25 (time), 42–43
`projects`	`passage_projects`	Projects, WBS, project transactions, estimates, invoice proposals, functional locations, maintenance assets, maintenance plans, maintenance work orders, installed base, service agreements, cases, service work orders	TS	41, 44–45

B.2 Cross-context reference rule

A row in one database may reference an entity in another context only by its identifier (UUIDv7 or, for user-facing documents, the number-sequence id) in a column named `<entity>_id` plus, when needed for display, a denormalized `<entity>_number` and `<entity>_name` snapshot. There are no foreign keys, views or `dblink` connections across databases. The owning context publishes `created`/`changed`/`deleted` events; consumers maintain a local read model (`<context>_ref.<entity>` tables) refreshed from those events and never query another context's database. Referential existence is validated at write time by calling the owning context's API (`HEAD /v1/<context>/<collection>/{id}`) or against the local read model when staleness of up to 5 s is acceptable (documents, not postings). The universal journal is the only synchronous cross-context write: subledgers call `ledger.Post` and store the returned `journal_entry_id`.

Identity rules: every table has `id uuid` (UUIDv7, time-ordered), `tenant_id uuid NOT NULL`, `legal_entity_id uuid` where the entity is company-specific, `version int` for optimistic concurrency, `created_at/created_by/updated_at/updated_by`. Fiscal documents additionally have `document_number text` (from a number sequence) and `posted_at timestamptz`.

B.3 Core entities

Entity	Context	Database	Key identifiers	Main relationships
Tenant	platform	passage_platform	`id`, `slug`	has many LegalEntity, User, Role
User	platform	passage_platform	`id`, `keycloak_sub`, `email`	many-to-many Role scoped by LegalEntity with `valid_from/to`
Role / Duty / Privilege	platform	passage_platform	`code` (e.g. `AP.Clerk`, `MaintainVendors`, `vendor.write`)	Role ⊃ Duty ⊃ Privilege; compiled to OPA data
EntityDefinition	platform	passage_platform	`name`, `version`	has Fields, Enums, Relations, Extensions
NumberSequence	platform	passage_platform	`code`, `legal_entity_id`, `scope`	referenced by every document type (Appendix G)
AuditEntry	platform	passage_platform	`id`, `entity`, `record_id`, `trace_id`, `hash`	hash-chained to previous entry; immutable
LegalEntity	platform	passage_platform	`id`, `code` (e.g. `USMF`), `country`	has Sites, FiscalCalendar, ledger set
OperatingUnit	platform	passage_platform	`id`, `type`, `code`	effective-dated Hierarchy nodes by purpose
Site / Warehouse	platform	passage_platform	`id`, `code`, `legal_entity_id`	Warehouse belongs to Site; referenced by inventory and warehouse contexts
FiscalCalendar / Period	platform	passage_platform	`id`, `fiscal_year`, `period_no`	PeriodControl rows per Ledger × module
Party	party	passage_party	`id`, `party_number`	has Roles, Addresses, TaxRegistrations, Contacts
Customer	party	passage_party	`id`, `party_id`, `customer_account`	belongs to CustomerGroup, PostingProfile; credit limit
Vendor	party	passage_party	`id`, `party_id`, `vendor_account`	belongs to VendorGroup, PostingProfile, payment terms
Product / ProductMaster	product	passage_product	`id`, `product_number`, `gtin`	Master has Variants by dimension values; DimensionGroups
ReleasedProduct	product	passage_product	`product_id`, `legal_entity_id`	ItemModelGroup, defaults per Site/Warehouse
BOM / BOMLine	product	passage_product	`id`, `bom_number`, `version`	lines reference Product ids; effective dates; phantom, scrap
Routing / Operation	product	passage_product	`id`, `routing_number`, `version`	Operations require Resource capabilities
Account / AccountStructure	ledger	passage_ledger	`main_account`, `structure_id`	Dimensions allowed per structure
JournalEntry / JournalLine	ledger	passage_ledger	`id`, `document_number`, `idempotency_key`	lines carry 3 currency amounts, dimension set, `source_document_id`; `reverses_id`
Balance	ledger	passage_ledger	(`ledger_id`, `account`, `dimension_hash`, `period`)	projection rebuilt from JournalLine
Ledger / RateType / ExchangeRate	ledger	passage_ledger	`ledger_code`, `rate_type`, `from/to`, `valid_from`	LegalEntity has leading + parallel Ledgers
TaxCode / TaxTransaction	ledger	passage_ledger	`tax_code`, `jurisdiction_id`	TaxTransaction references source line id and JournalLine

Entity	Context	Database	Key identifiers	Main relationships
InventoryTransaction	inventory	passage_inventory	`id`, `lot_id`, `reference_type/id`	dimensions (site, warehouse, location, batch, serial, LP); status pair
OnHand	inventory	passage_inventory	(`product_id`, dimension hash)	projection = Σ InventoryTransaction
CostLayer / CostingVersion	inventory	passage_inventory	`id`, `receipt_transaction_id`, `version_code`	settlements link issue to receipt layers
QualityOrder	inventory	passage_inventory	`id`, `qo_number`	triggered by receipt/production/shipment reference; blocks OnHand
PurchaseRequisition / RFQ / PurchaseAgreement	procurement	passage_procurement	`document_number`	Requisition lines → PO lines; Agreement releases
PurchaseOrder / Line	procurement	passage_procurement	`document_number`, `version`	`vendor_id`, `product_id`, `site/warehouse_id`; delivery schedules
ProductReceipt	procurement	passage_procurement	`document_number`	references PO lines, creates InventoryTransaction ids and JournalEntry id
VendorInvoice / MatchResult	procurement	passage_procurement	`document_number`, `vendor_invoice_number`	matches PO + Receipt lines; posts JournalEntry; creates APOpenItem
APOpenItem / Payment	procurement	passage_procurement	`id`, `payment_number`	Settlement links Payment to OpenItems; payment file id
ProductionOrder	manufacturing	passage_manufacturing	`document_number`, `status`	BOM/Routing snapshot; PickingList, RouteCard, JournalEntry ids for WIP
PlannedOrder / PlanningRun	manufacturing	passage_manufacturing	`id`, `run_id`	pegged to demand ids (SalesOrderLine, ProductionOrder, Forecast)
TimeRegistration / MachineEvent	manufacturing	passage_manufacturing	`id`, `worker_id`, `resource_id`	pay types exported to PayrollRun; OEE aggregates
PriceCondition	sales	passage_sales	`id`, `condition_type`, access key	resolved by access sequence; effective dates
SalesOrder / Line	sales	passage_sales	`document_number`, `status`	`customer_id`, `product_id`; Holds; ATP results; linked ProductionOrder/PlannedOrder ids
Shipment / PackingSlip	sales	passage_sales	`document_number`, `sscc`	references SalesOrder lines, Load id, InventoryTransaction ids, COGS JournalEntry id
CustomerInvoice / RevenueSchedule	sales	passage_sales	`document_number`	invoice lines → PerformanceObligations → schedule lines → JournalEntry ids
AROpenItem / CashApplication	sales	passage_sales	`id`, `receipt_number`	settlement to invoices; BankStatement line id
CollectionCase / ReturnOrder	sales	passage_sales	`case_number`, `rma_number`	customer, disputed invoice ids; RMA disposition → InventoryTransaction
Location / LicensePlate	warehouse	passage_warehouse	`location_code`, `lp_number`	Zone ⊃ Aisle ⊃ Rack ⊃ Bin; LP nests LPs
Work / PickWave	warehouse	passage_warehouse	`work_id`, `wave_id`	work lines reference order lines; directives; assigned user
BankAccount / BankStatement	treasury	passage_treasury	`id`, `iban/account_number` (P3)	statement lines reconcile to JournalLines, Payments, CashApplications
FixedAsset / AssetBook / AssetTransaction	treasury	passage_treasury	`asset_number`, `book_code`	acquisition from VendorInvoice/Project ids; depreciation JournalEntry ids
Worker / Position / Job	workforce	passage_workforce	`personnel_number`, `position_id`	effective-dated Employment; Position reports to Position; `party_id`
PayrollRun / ExpenseReport	workforce	passage_workforce	`run_id`, `report_number`	pay statements; JournalEntry ids; reimbursement → VendorInvoice id
Project / WBS / ProjectTransaction	projects	passage_projects	`project_number`, `wbs_code`	transactions reference TimeRegistration, ExpenseReport, PO, inventory ids; JournalEntry ids
MaintenanceAsset / MaintenanceWorkOrder	projects	passage_projects	`asset_number`, `work_order_number`	FunctionalLocation tree; parts → InventoryTransaction; labor → TimeRegistration
ServiceAgreement / Case / ServiceWorkOrder	projects	passage_projects	`agreement_number`, `case_number`	InstalledBase serials from Shipment; billing → CustomerInvoice id

C. Event catalog conventions

C.1 Envelope

Every event is a CloudEvents 1.0 message in structured JSON mode on Kafka with the following attributes. Payload schemas are JSON Schema 2020-12 registered in the schema registry and described in AsyncAPI 3.0 documents generated from entity metadata.

Attribute	Rule
`specversion`	`1.0`
`id`	UUIDv7; equals the outbox row id, so redelivery carries the same id
`source`	`/passage/<context>/<service>` e.g. `/passage/ledger/posting`
`type`	`dev.passage.<context>.<entity>.<verb>.v<n>` — lowercase, no hyphens; verb is past tense (`posted`, `confirmed`, `changed`)
`subject`	the aggregate id (UUIDv7)
`time`	commit time of the producing transaction, RFC 3339 UTC
`datacontenttype`	`application/json`
`dataschema`	registry URL `https://schemas.passage.dev/<type>.json`
`tenantid` (ext)	tenant UUID; always present
`legalentityid` (ext)	legal entity UUID when company-scoped
`correlationid` (ext)	id of the originating request or workflow; propagated unchanged
`causationid` (ext)	id of the event or command that directly caused this event
`traceparent` (ext)	W3C trace context from the producing span
`actor` (ext)	`user:<sub>`, `agent:<agent-id>@user:<sub>`, or `system:<job>`
`sequence` (ext)	monotonically increasing per aggregate (`version` after the change)

C.2 Topics and partitions

- Topic: `passage.<context>.<entity>` (one topic per entity, all verbs and versions share it); 12 partitions default, 48 for `ledger.journalentry`, `inventory.transaction`, `manufacturing.machineevent`.
- Partition key: `tenantid + ":" + subject` so events for one aggregate are ordered; `machineevent` uses `tenantid:resource_id`.
- Retention: 30 days for transactional topics, 7 days for `machineevent`, compacted `passage.<context>.<entity>.snapshot` topics for reference-data read models.
- Dead letter: `passage.<context>.<entity>.dlq.<consumer>` with the original event, error and attempt count in headers.
- Access: ACLs per consumer service principal (SCRAM-SHA-512); cross-tenant consumers (analytics) run under a dedicated principal and are the only ones allowed to read all tenants.

C.3 Schema evolution

1. Changes within a version are additive only: new optional fields, new enum values flagged `x-passage-added-in`. Removing, renaming or retyping a field, or making an optional field required, requires a new `v<n+1>` type.
2. Schema registry compatibility mode is `BACKWARD_TRANSITIVE`; CI rejects a non-additive change (wave 4 proof).
3. When `v<n+1>` ships, the producer emits both `v<n>` and `v<n+1>` for a deprecation window of two minor releases of the producing service, then drops `v<n>`. Consumers must migrate inside the window; a consumer still subscribed to a dropped type fails its startup health check.
4. Enum consumers must treat unknown values as `other` and log at warn, never fail.

C.4 Outbox table DDL

Every service database contains exactly this table; Debezium's outbox event router relays rows to Kafka and deletes them after acknowledgment.

```
CREATE TABLE outbox (
  id             uuid        PRIMARY KEY,                    -- UUIDv7, becomes CloudEvents id
  tenant_id      uuid        NOT NULL,
  aggregate_type text        NOT NULL,                       -- e.g. 'journalentry'
  aggregate_id   uuid        NOT NULL,                       -- CloudEvents subject
  event_type     text        NOT NULL,                       -- full CloudEvents type
  partition_key  text        NOT NULL,                       -- tenant_id:aggregate_id
  sequence       bigint      NOT NULL,
  headers        jsonb       NOT NULL DEFAULT '{}'::jsonb,   -- correlationid, causationid, traceparent, actor
  payload        jsonb       NOT NULL,
  created_at     timestamptz NOT NULL DEFAULT clock_timestamp()
);
CREATE INDEX outbox_created_at_idx ON outbox (created_at);
ALTER TABLE outbox REPLICA IDENTITY FULL;
```

Rules: the outbox insert happens in the same transaction as the state change; the payload is the event `data` only (envelope attributes come from columns); payload size limit 256 KiB (larger content is stored and referenced by URL); the `outbox` table is excluded from RLS because only the service role writes it.

C.5 Consumer idempotency

Every consumer keeps an inbox table and processes each event inside one database transaction:

```
CREATE TABLE consumer_inbox (
  consumer   text        NOT NULL,
  event_id   uuid        NOT NULL,
  processed_at timestamptz NOT NULL DEFAULT now(),
  PRIMARY KEY (consumer, event_id)
);
```

1. `INSERT INTO consumer_inbox ... ON CONFLICT DO NOTHING`; if zero rows were inserted, skip the event (duplicate delivery).
2. Apply the effect and write any outgoing outbox rows in the same transaction; commit; then acknowledge the Kafka offset.
3. Out-of-order protection: consumers compare `sequence` with the stored aggregate version and park events that arrive early for up to 60 s before sending to the DLQ.
4. Inbox rows are pruned after 30 days (the topic retention) by the batch framework.
5. Retries: 5 attempts with exponential backoff (1 s → 60 s), then DLQ; DLQ depth per consumer is an alert at 100 and a page at 1,000.

C.6 Replay procedure

1. Open a change ticket naming consumer, topic, time window and reason; replay of a fiscal consumer (ledger, AP, AR) requires controller approval via the platform workflow.
2. Pause the consumer group (`passage-ctl consumer pause <group>`), snapshot its inbox table.
3. Reset the offset to the timestamp (`kafka-consumer-groups --reset-offsets --to-datetime`) or to a DLQ range; for a rebuild, truncate the read model and reset to earliest on the compacted snapshot topic.
4. Resume; the inbox guarantees already-processed events are skipped, so replay is safe by construction. A replay that must reprocess (bug fix) deletes inbox rows for the window first, inside the ticket.
5. Verify with the reconciliation checks in Appendix I.6 and close the ticket with the before/after totals.

C.7 Principal event types

Event type	Producer	Wave	Key consumers
`dev.passage.platform.entitydefinition.changed.v1`	platform/metadata	3	generators, shell, analytics
`dev.passage.platform.user.provisioned.v1`	platform/identity	2	workforce, audit
`dev.passage.platform.workflowtask.completed.v1`	platform/workflow	4	originating context, notifications
`dev.passage.platform.legalentity.created.v1`	platform/org	6	ledger, party, product, number sequences
`dev.passage.platform.periodcontrol.changed.v1`	platform/org	14	all posting services (cache invalidation)
`dev.passage.party.party.created.v1`	party	7	sales, procurement, workforce read models

Event type	Producer	Wave	Key consumers
`dev.passage.party.party.merged.v1`	party	7	every context holding `party_id` (rewrite references)
`dev.passage.party.customer.creditchanged.v1`	party	7	sales (credit checks)
`dev.passage.product.product.released.v1`	product	8	inventory, sales, procurement, manufacturing
`dev.passage.product.bom.versionactivated.v1`	product	9	manufacturing, inventory (cost roll)
`dev.passage.product.changeorder.released.v1`	product	9	manufacturing, warehouse, projects
`dev.passage.ledger.journalentry.posted.v1`	ledger	11	balances, analytics, audit, budgets
`dev.passage.ledger.journalentry.reversed.v1`	ledger	11	subledgers, analytics
`dev.passage.ledger.exchangerate.published.v1`	ledger	12	sales, procurement, treasury
`dev.passage.ledger.period.closed.v1`	ledger	14	all posting services, analytics
`dev.passage.ledger.taxsettlement.posted.v1`	ledger	13	statutory reporting, analytics
`dev.passage.inventory.transaction.posted.v1`	inventory	16	on-hand, sales (ATP), planning, analytics
`dev.passage.inventory.reservation.changed.v1`	inventory	16	sales, warehouse, manufacturing
`dev.passage.inventory.costclose.completed.v1`	inventory	20	ledger reconciliation, analytics
`dev.passage.inventory.qualityorder.decided.v1`	inventory	33	warehouse, procurement (scorecards), sales
`dev.passage.procurement.purchaseorder.confirmed.v1`	procurement	17	planning, budgets (encumbrance), integration (EDI 850)
`dev.passage.procurement.productreceipt.posted.v1`	procurement	18	inventory, ledger (GR/IR), quality, warehouse
`dev.passage.procurement.vendorinvoice.posted.v1`	procurement	18	ledger, treasury (forecast), budgets, analytics
`dev.passage.procurement.payment.posted.v1`	procurement	19	ledger, treasury, integration (payment file)
`dev.passage.manufacturing.productionorder.released.v1`	manufacturing	21	inventory (reservations), warehouse, shop floor
`dev.passage.manufacturing.productionorder.reportedasfinished.v1`	manufacturing	21	inventory, quality, sales (ATP)
`dev.passage.manufacturing.productionorder.ended.v1`	manufacturing	21	ledger (variances), inventory (cost), analytics
`dev.passage.manufacturing.plannedorder.firmed.v1`	manufacturing	22	procurement (PO creation), manufacturing orders
`dev.passage.manufacturing.schedule.changed.v1`	manufacturing	23	production orders, planning, shop floor
`dev.passage.manufacturing.machineevent.received.v1`	manufacturing/iot	25	OEE, projects (condition monitoring)
`dev.passage.sales.salesorder.confirmed.v1`	sales	27	inventory (reservation), planning, warehouse, integration (EDI 855)
`dev.passage.sales.shipment.posted.v1`	sales	28	inventory, ledger (COGS), billing, integration (ASN 856)
`dev.passage.sales.invoice.posted.v1`	sales	29	ledger, AR, revenue schedules, integration (810/Peppol), analytics
`dev.passage.sales.payment.applied.v1`	sales	30	ledger, collections, treasury
`dev.passage.sales.returnorder.received.v1`	sales	30	inventory, quality, billing (credit note)
`dev.passage.warehouse.work.completed.v1`	warehouse	31	inventory (status change), sales, procurement
`dev.passage.warehouse.load.shipped.v1`	warehouse	34	sales (shipment posting), integration (EDI 214), treasury (freight accrual)
`dev.passage.treasury.bankstatement.reconciled.v1`	treasury	36	ledger, AR, AP
`dev.passage.treasury.asset.depreciated.v1`	treasury	37	ledger, analytics
`dev.passage.workforce.payrollrun.posted.v1`	workforce	43	ledger, treasury (payments), projects
`dev.passage.projects.transaction.posted.v1`	projects	41	ledger (WIP), billing, analytics
`dev.passage.agents.action.executed.v1`	agents	47	audit, cost guard, analytics

D. API conventions

D.1 Resources and naming

- Base path `https://api.<env>.passage.dev/v1/<context>/<collection>`; collections are plural kebab-case (`/v1/ledger/journal-entries`, `/v1/sales/sales-orders`, `/v1/product/released-products`). Ids in paths are UUIDv7; document numbers are reachable through `?filter=documentNumber eq 'INV-USMF-2026-000123'`.
- Sub-resources only for composition (`/sales-orders/{id}/lines`). Lifecycle actions are custom methods with a colon: `POST /v1/ledger/journal-entries/{id}:post`, `:reverse`, `:confirm`, `:release`, `:cancel`. Actions are the only way to change status; `PATCH` on a `status` field returns 422 `state_transition_invalid`.
- JSON field names are camelCase in the wire format and snake_case in Postgres; the generator maintains the mapping. Money is `{ "amount": "1234.56", "currency": "USD" }` with a decimal string; dates are ISO 8601 (`2026-10-21`, `2026-10-21T14:03:00Z`); quantities are decimal strings with `unit`.
- Every entity declared in metadata gets `GET` list, `GET` by id, `POST`, `PUT`, `PATCH` (JSON Merge Patch), `DELETE` (soft, unless metadata says hard), plus `:batch` and `:export`, generated. Hand-written endpoints are limited to engines (post, price, plan, schedule, match, settle).

D.2 Versioning

`/v1` is the only version in v1.0. Within a major version changes are additive only (new optional fields, new endpoints, new enum values). Breaking changes ship as `/v2` for the affected context with both versions served for at least 12 months; the `Sunset` and `Deprecation` headers (RFC 8594) announce retirement. The `X-Passage-Api-Build` response header carries the service build for support.

D.3 Pagination, filtering, sorting, shape

- Cursor pagination: `?limit=50&cursor=<opaque>`; `limit` max 500. Response: `{ "items": [...], "nextCursor": "...", "estimatedTotal": 1234 }`. Cursors encode the sort key and are valid for 24 h; a cursor from a different filter returns 400 `invalid_cursor`.
- Filter grammar (subset of OData): `field op value` with `eq ne gt ge lt le in like isnull`, combined with `and or not` and parentheses; nested fields by dot (`customer.group eq 'WHOLESALE'`); dimension filters via `dimensions.costCenter eq 'CC-100'`. Max 20 predicates; filters on non-indexed fields over 100k rows are rejected with 422 `filter_not_indexed`.
- `?sort=-postedAt,documentNumber`; `?fields=id,documentNumber,total` for projection; `?expand=lines,customer` for one level of embedded reads (never across contexts; cross-context fields come from the local read model and are marked `x-passage-snapshot` in OpenAPI).
- `?asOf=2026-01-01` on effective-dated entities (hierarchies, BOMs, positions, prices).

D.4 Error envelope

Errors are RFC 9457 problem details (`Content-Type: application/problem+json`):

```
{
  "type": "https://passage.dev/problems/period_closed",
  "title": "Period is closed",
  "status": 422,
  "detail": "Ledger USMF-IFRS period 2026-09 is closed for module AP since 2026-10-05T18:00:00Z.",
  "instance": "/v1/procurement/vendor-invoices/0192f3a1-...:post",
  "code": "period_closed",
  "traceId": "4bf92f3577b34da6a3ce929d0e0e4736",
  "errors": [ { "field": "postingDate", "code": "period_closed", "message": "..." } ]
}
```

Code	HTTP	Meaning
`validation_failed`	400	Payload fails JSON Schema; `errors[]` lists fields
`required_field_missing`	400	A metadata-required field is absent
`invalid_enum_value`	400	Value not in the enum (including extensions)
`invalid_cursor`	400	Cursor expired or from another query
`filter_not_indexed`	422	Filter would scan an unindexed column on a large table
`unauthenticated`	401	Missing or invalid token
`forbidden_duty`	403	OPA denied: principal lacks the duty/privilege
`forbidden_tenant`	403	Resource belongs to another tenant or legal entity outside scope
`sod_conflict`	403	Action would violate a segregation-of-duties rule
`approval_required`	403	Action requires an approval workflow; a task was created
`not_found`	404	No such resource in the caller's scope
`conflict_version`	412	`If-Match` does not equal the current ETag
`precondition_required`	428	Mutating request without `If-Match`
`idempotency_key_reuse`	422	Same `Idempotency-Key` with a different payload
`idempotency_in_progress`	409	Original request with this key is still executing
`unique_violation`	409	Unique constraint (e.g. vendor invoice number per vendor)
`state_transition_invalid`	422	Action not allowed from the current status
`period_closed`	422	Posting date in a closed/on-hold period for this module
`unbalanced_entry`	422	Debits ≠ credits in any currency
`dimension_invalid`	422	Dimension combination violates the account structure
`inventory_insufficient`	422	On-hand would go negative and the item model group forbids it
`credit_hold`	422	Customer on hold or limit exceeded; hold created
`number_sequence_exhausted`	422	Sequence reached its upper bound or is stopped
`rate_limited`	429	Tenant tier quota exceeded; `Retry-After` set
`payload_too_large`	413	Body over 10 MiB (bulk) or 1 MiB (single)
`upstream_unavailable`	503	Tax/rate/carrier/bank provider failed and no fallback applied
`internal_error`	500	Unexpected; `traceId` for support

D.5 Idempotency

`Idempotency-Key` (UUID or ≤ 128-char string) is required on every `POST` that creates or posts a document and on every custom method; it is optional on `PUT/PATCH` (which rely on ETags). The key is scoped to tenant + principal + method + path. The server stores the request body hash and the final response for 24 h: a repeat with the same hash returns the stored response with `Idempotent-Replayed: true`; a repeat with a different hash returns 422 `idempotency_key_reuse`; a repeat while the first is in flight returns 409 `idempotency_in_progress`. Posting engines additionally persist the key on the document (`idempotency_key` column, unique per tenant) so the guarantee survives the 24 h cache.

D.6 Concurrency

Every readable resource returns a strong `ETag` derived from `version` (e.g. `"v17"`). `PUT`, `PATCH`, `DELETE` and custom methods require `If-Match`; a missing header is 428, a mismatch is 412 `conflict_version` with the current representation in `detail`. List responses include per-item `etag`.

D.7 Rate limits

Tier	Sustained (requests/min per tenant)	Burst	Bulk rows/day	Webhook deliveries/min	Concurrent exports
Starter	600	100	200k	60	2
Professional	3,000	500	2M	300	5
Enterprise	12,000 (negotiable)	2,000	20M	1,200	20

Enforced at Envoy Gateway per tenant and per API key; headers `RateLimit-Limit`, `RateLimit-Remaining`, `RateLimit-Reset` (IETF draft) on every response. Agents share the tenant budget and have a separate per-agent cap (Appendix E). Shop-floor and mobile apps use a dedicated limit class (`device`) of 120 requests/min per device.

D.8 Bulk endpoints

`POST /v1/<context>/<collection>:batch` accepts up to 1,000 items with `"atomic": true|false`; atomic batches roll back on the first error, non-atomic batches return per-item results with the same problem-details shape. Larger loads use NDJSON upload to `POST /v1/platform/jobs` (type `import`), which runs in the batch framework

with checkpoints and returns a job id, progress and an error file. `:export` streams NDJSON or CSV with the same filter grammar. Postings via bulk are still one journal entry per document; the batch framework never bypasses posting rules.

D.9 OpenAPI and SDKs

The metadata generator emits one OpenAPI 3.1 document per context (`openapi/<context>.v1.yaml`) with schemas, examples, error codes and `x-passage-duty` on every operation; hand-written engine endpoints are merged from annotated Go/TS handlers. CI runs Spectral (style) and oasdiff (breaking-change detection) against the last released spec; the developer portal (wave 35) publishes specs and the AsyncAPI documents. SDKs generated per release: TypeScript (first-party, used by the shell), Go, Python in v1.0; C# and Java in v1.x for AX and SAP shops respectively. SDKs expose idempotency keys, ETags and pagination as first-class helpers and never hide problem details.

E. Security and authorization model

E.1 Tenancy

A tenant is one customer organization: one Keycloak realm, one `tenant_id`, one row-level security scope across all 12 databases, one Kafka ACL principal set, one Vault namespace, one cost budget. Legal entities are scopes inside the tenant; users are assigned roles per legal entity (or `*`). Data residency is per tenant (wave 48): a tenant is pinned to a region's cluster set and its Iceberg warehouse. Cross-tenant access does not exist in the application; only the analytics and platform-operations principals can read across tenants and only through audited service accounts.

The request path sets the scope: Envoy validates the JWT, the service opens a Postgres connection from a pool bound to the non-superuser application role and executes `SET LOCAL passage.tenant_id = '<uuid>'; SET LOCAL passage.legal_entity_ids = '<uuid>,<uuid>'; SET LOCAL passage.actor = '<sub>'` inside the transaction. The application role has no `BYPASSRLS`.

E.2 RLS policy template

Generated for every tenant-scoped table; the generator adds the `legal_entity_scope` policy only when the entity declares `legal_entity_id`.

```
ALTER TABLE ledger.journal_entry ENABLE ROW LEVEL SECURITY;
ALTER TABLE ledger.journal_entry FORCE ROW LEVEL SECURITY;

CREATE POLICY tenant_isolation ON ledger.journal_entry
  AS RESTRICTIVE FOR ALL TO passage_app
  USING      (tenant_id = current_setting('passage.tenant_id', true)::uuid)
  WITH CHECK (tenant_id = current_setting('passage.tenant_id', true)::uuid);

CREATE POLICY legal_entity_scope ON ledger.journal_entry
  AS RESTRICTIVE FOR ALL TO passage_app
  USING (
    current_setting('passage.legal_entity_ids', true) = '*'
    OR legal_entity_id = ANY (string_to_array(current_setting('passage.legal_entity_ids', true), ',')::uuid[])
  )
  WITH CHECK (
    legal_entity_id = ANY (string_to_array(current_setting('passage.legal_entity_ids', true), ',')::uuid[])
  );

-- Batch and CDC roles: separate role with the same policies; Debezium reads the WAL and is outside RLS by design,
-- so the outbox payload carries tenant_id and the relay sets the CloudEvents tenantid from it.
```

A missing `passage.tenant_id` makes `current_setting(..., true)` return NULL and the policy returns no rows: the fail-safe is empty, never cross-tenant.

E.3 Role / duty / privilege schema

```
CREATE TABLE platform.privilege (
  code text PRIMARY KEY,                  -- 'vendor.write', 'journalentry.post', 'payment.approve'
  resource text NOT NULL,                 -- entity or endpoint family
  action text NOT NULL,                   -- read | write | post | approve | delete | export | admin
  description text NOT NULL
);
CREATE TABLE platform.duty (
  code text PRIMARY KEY,                  -- 'MaintainVendors', 'PostVendorInvoices', 'ApprovePayments'
  tenant_id uuid,                         -- NULL = shipped by Passage, else customer-defined
  description text NOT NULL
);
CREATE TABLE platform.duty_privilege (duty_code text REFERENCES platform.duty, privilege_code text REFERENCES platform.privilege, PRIMARY KEY
(duty_code, privilege_code));
CREATE TABLE platform.role (code text PRIMARY KEY, tenant_id uuid, description text NOT NULL);
CREATE TABLE platform.role_duty (role_code text REFERENCES platform.role, duty_code text REFERENCES platform.duty, PRIMARY KEY (role_code,
duty_code));
CREATE TABLE platform.user_role (
  user_id uuid NOT NULL, role_code text NOT NULL, tenant_id uuid NOT NULL,
  legal_entity_id uuid,                   -- NULL = all legal entities in the tenant
  valid_from date NOT NULL DEFAULT current_date, valid_to date,
  granted_by uuid NOT NULL, PRIMARY KEY (user_id, role_code, legal_entity_id, valid_from)
);
CREATE TABLE platform.sod_rule (
  code text PRIMARY KEY, duty_a text NOT NULL, duty_b text NOT NULL,
  severity text NOT NULL CHECK (severity IN ('high','medium','low')), description text NOT NULL
);
```

Passage ships ~60 roles, ~400 duties and ~2,500 privileges, named after AX (e.g. role `AccountsPayableClerk`, duty `MaintainVendorInvoices`). Customers add duties and roles; they cannot add privileges, which are generated from metadata (`<entity>.<action>`) and engine endpoints.

E.4 OPA policy structure

The compiled policy bundle contains `data.passage.grants[tenant][user] = {duty: [legal_entity_ids]}` and `data.passage.duty_privileges[duty] = [privileges]`, rebuilt by the platform service on every role change (push to the OPA sidecars via bundle server, ≤ 10 s propagation). The input is `{ "principal": {sub, tenant, agent?, on_behalf_of?}, "action": "<privilege>", "resource": {context, entity, id?, legal_entity_id?, amount?, owner?} }`. Every service calls `POST /v1/data/passage/authz/allow` on its sidecar; the gateway performs a coarse check (token, tenant, route duty) and the service performs the fine-grained one (record scope).

Example 1 — duty-based allow with legal-entity scope:

```
package passage.authz

default allow := false

allow if {
  some duty
  priv_set := data.passage.duty_privileges[duty]
  priv_set[_] == input.action
  scopes := data.passage.grants[input.principal.tenant][input.principal.sub][duty]
  scope_ok(scopes, input.resource.legal_entity_id)
}

scope_ok(scopes, le) if scopes[_] == "*"
scope_ok(scopes, le) if scopes[_] == le
```

Example 2 — agents never exceed the user they act for, and are additionally limited by their own allow-list:

```
allow if {
  input.principal.agent != null
  user_input := object.union(input, {"principal": {"sub": input.principal.on_behalf_of, "tenant": input.principal.tenant}})
  data.passage.authz.allow with input as user_input
  data.passage.agents[input.principal.agent].allowed_actions[_] == input.action
  not data.passage.agents[input.principal.agent].kill_switch
}
```

Example 3 — amount-banded approval (payment release above a threshold needs `ApprovePaymentsHighValue`, and the approver may not be the preparer):

```
deny contains msg if {
  input.action == "payment.approve"
  input.resource.amount_usd > 50000
  not has_duty("ApprovePaymentsHighValue")
  msg := "payments over USD 50,000 require ApprovePaymentsHighValue"
}

deny contains msg if {
  input.action == "payment.approve"
  input.resource.prepared_by == input.principal.sub
  msg := "sod: preparer cannot approve own payment"
}

has_duty(d) if data.passage.grants[input.principal.tenant][input.principal.sub][d]
```

E.5 Segregation-of-duties enforcement points

Point	Behavior
Role design (wave 48 SoD engine)	Creating or editing a role whose duties conflict is blocked (high) or warned (medium/low)
User assignment	Assigning a role that creates a conflict with the user's existing roles opens a mitigation workflow; the assignment is inactive until a mitigating control (compensating review, time-boxed access) is approved
Runtime (OPA `deny`)	Same-person checks on document pairs: prepare/approve journals, create vendor/approve vendor invoice, create PO/post receipt over threshold, enter payment/release payment, maintain bank account/run payments
Agents	An agent's action set is checked against SoD rules at registration; an agent cannot hold both sides of any high-severity rule
Certification	Quarterly access-certification campaigns; uncertified roles expire after 30 days
Evidence	Every conflict, mitigation and certification decision is an audit entry exported to the SOC 2 evidence store

E.6 PII classification and handling

Level	Examples	Storage	Logging / tracing	Display	Export / analytics	Retention
P0 Public	Product catalog, public prices, legal entity name	Plain	Allowed	Full	Allowed	Business need
P1 Internal	Order numbers, amounts, dimension values, internal notes	Plain, RLS	Allowed (ids only, no free text)	Full with duty	Tenant-scoped	Business need
P2 Personal	Person names, business emails, phone, postal address, personnel number	Plain, RLS, `pii: true` in metadata	Never in logs; hashed in traces	Full with duty; masked in list exports by default	Lakehouse with column tags; erasure propagates by `party_id`	Per GDPR/CCPA policy; erasure request honored within 30 days except fiscal documents (pseudonymized)
P3 Sensitive	National IDs, tax IDs of persons, bank account/IBAN, card tokens, salary, health/absence reasons, biometrics, dates of birth	Column-level encryption (pgcrypto, key per tenant in Vault transit), separate `*_secure` tables, no full-text index	Never; access itself is audited	Masked (`****1234`) unless a `ViewSensitive*` duty is held; reveal is an audited action	Excluded from lakehouse except aggregated; excluded from exports except statutory formats	Minimum legal; vaulted in `workforce` with per-field retention

Metadata flags (`pii: P2|P3`) drive generation: masking in generated APIs and forms, exclusion from the Debezium CDC allow-list for P3, and the subject-request tooling (wave 48) that enumerates every P2/P3 field per `party_id`.

E.7 Secrets management

HashiCorp Vault is the only secret store: Kubernetes auth per service account, one namespace per tenant for customer-specific credentials (bank, tax provider, carrier). External Secrets Operator materializes only the minimum into Kubernetes secrets with 1 h TTL. Database credentials are dynamic (Vault database engine, 24 h leases). Bank signing keys and EBICS keys live in Vault transit and never leave it; files are signed by calling transit. Rotation: 90 days for static secrets, automatic on lease expiry for dynamic ones, immediate on offboarding. Git is scanned by gitleaks in CI; a leak fails the build. Keycloak client secrets rotate per release.

E.8 Audit and retention

Audit is derived from Debezium CDC into an append-only `platform.audit_entry` table with a per-tenant SHA-256 hash chain (`hash = sha256(prev_hash || canonical_json(entry))`) verified nightly; the chain head is written to the lakehouse and to object storage with object lock. Retention: audit entries for fiscal documents (journal entries, invoices, payments, receipts, packing slips, production orders, asset transactions, payroll) 7 years from fiscal-year end, extended by country pack (DE and FR 10 years, IN 8 years); other entities 3 years; agent actions 7 years. Audit entries cannot be deleted; GDPR erasure replaces P2/P3 values with pseudonyms and records the erasure itself.

E.9 Encryption

In transit: TLS 1.3 at the gateway (cert-manager, ACME or customer CA), mTLS between services and to Postgres and Kafka with cert-manager-issued certificates rotated every 30 days; Kafka SCRAM-SHA-512 plus TLS; Debezium and Temporal over mTLS. At rest: encrypted storage class on every persistent volume (provider KMS or LUKS on bare metal), CloudNativePG backups and WAL archives encrypted with a per-tenant key via Vault transit (envelope), Kafka tiered storage and Iceberg objects encrypted server-side; P3 columns encrypted additionally at the application level. Key rotation re-wraps envelopes without rewriting data; a tenant's crypto-shred (key deletion) is the final step of tenant offboarding.

F. Non-functional requirements detail

ID	Area	Metric	Target	Measurement	First enforced
NFR-A1	Availability	Monthly availability per tenant (API + shell)	≥ 99.9% (v1), ≥ 99.95% with multi-region	Synthetic probes every 30 s from two regions; SLO in Grafana	Wave 5 (measured), wave 50 (99.95%)
NFR-A2	Availability	RPO	≤ 5 min	Backup/restore drill; WAL archive lag metric	Wave 48
NFR-A3	Availability	RTO	≤ 1 h	DR game day, timed	Wave 48, region failover wave 50
NFR-A4	Availability	Zero-downtime deploys and migrations	0 failed requests during rollout	Argo rollout with k6 traffic; expand/contract migrations only	Wave 3
NFR-A5	Availability	Rollback time	≤ 15 min to prior digest	Timed in Argo CD history	Wave 1
NFR-P1	Performance	Journal posting throughput	≥ 5,000 lines/s sustained, p99 < 200 ms	k6 against ledger on 3-node stage	Wave 11
NFR-P2	Performance	Generated entity simple read	p95 < 50 ms at 1,000 rps	k6	Wave 3
NFR-P3	Performance	Price resolution (10-line order)	p95 < 20 ms	Go benchmark + k6	Wave 26
NFR-P4	Performance	Order intake API	500 orders/min, p99 < 500 ms	k6	Wave 27
NFR-P5	Performance	MRP run	50k items × 12 periods × 5 levels < 10 min	Scheduled benchmark in CI nightly	Wave 22
NFR-P6	Performance	Scheduling	5k operations < 60 s	Benchmark fixtures	Wave 23
NFR-P7	Performance	BOM explosion (7 levels, 2,000 components)	< 500 ms	Go benchmark	Wave 9
NFR-P8	Performance	Inventory close (1M transactions)	< 20 min	Batch job timing	Wave 20
NFR-P9	Performance	Cost roll (50k items)	< 15 min	Batch job timing	Wave 39
NFR-P10	Performance	Dashboard freshness (OLTP → lakehouse)	p95 < 60 s	Debezium/Iceberg lag metric	Wave 46
NFR-P11	Performance	Shell first contentful paint (warm)	< 1.5 s on a mid-range laptop	Lighthouse in CI	Wave 5
NFR-P12	Performance	Warehouse task assignment	p95 < 1 s	Simulation	Wave 31
NFR-S1	Scale	Concurrent users	10,000	k6 load campaign	Wave 50
NFR-S2	Scale	Journal lines per hour	1M	Load campaign	Wave 50
NFR-S3	Scale	Order lines per hour	100k	Load campaign	Wave 50
NFR-S4	Scale	IoT ingest	10k events/min per plant, no loss on broker restart	Simulator + chaos	Wave 25
NFR-S5	Scale	Migration load	50k customers, 20k vendors, 100k items, 10k BOMs < 30 min	Migration run ledger	Wave 10
NFR-S6	Scale	Tenants per cluster set	100 with noisy-neighbor limits	Capacity model, rate limits	Wave 50
NFR-SEC1	Security	Authorization on every request	100% of routes carry `x-passage-duty`; 0 unprotected	OpenAPI lint + gateway audit	Wave 2
NFR-SEC2	Security	Tenant isolation	0 cross-tenant rows in RLS test suite	Automated RLS test per table	Wave 2
NFR-SEC3	Security	Supply chain	100% images signed, SBOM attached, 0 critical CVEs at deploy	cosign verify, Grype gate	Wave 1
NFR-SEC4	Security	Penetration test	0 high/critical open at wave 48 exit	External test report	Wave 48
NFR-SEC5	Security	Secrets in source	0	gitleaks	Wave 1
NFR-SEC6	Security	Agent permission ceiling	0 actions exceeding the user's OPA decision	Contract tests	Wave 47
NFR-C1	Compliance	SOC 2 TSC mapping	100% of in-scope controls with automated evidence	Compliance workspace	Wave 48
NFR-C2	Compliance	Subject access/erasure request	Fulfilled ≤ 30 days; export ≤ 24 h	Workflow SLA timer	Wave 48

ID	Area	Metric	Target	Measurement	First enforced
NFR-C3	Compliance	Statutory outputs validate	SAF-T, Peppol, pain.001, X12 pass schema/schematron	CI fixture validation	Waves 13, 19, 35
NFR-C4	Compliance	Fiscal document immutability	0 updates to posted documents (DB trigger)	Trigger + CDC audit test	Wave 11
NFR-ACC1	Accessibility	WCAG 2.1 AA on generated forms	100% pass axe, 0 serious/critical	Playwright + axe in CI	Wave 5
NFR-ACC2	Accessibility	Keyboard operability	100% of workspaces navigable without a mouse	Keyboard test suite	Wave 5
NFR-ACC3	Accessibility	Mobile/kiosk apps	WCAG 2.1 AA, touch targets ≥ 44 px	axe on emulated devices	Waves 25, 32
NFR-O1	Observability	Trace coverage	100% of services emit OTel traces; one trace id per request end-to-end	Tempo query in CI smoke	Wave 5
NFR-O2	Observability	SLO dashboards and alerts	Every service has latency/error/saturation SLOs with burn-rate alerts	Grafana provisioning review	Wave 5
NFR-O3	Observability	Audit latency	Field change visible in audit viewer ≤ 2 s	E2E test	Wave 5
NFR-PT1	Portability	Provider-managed services in the critical path	0	Architecture review per wave	Wave 1
NFR-PT2	Portability	Local parity	`make deploy-local` on k3d < 10 min	Timed in CI	Wave 1
NFR-PT3	Portability	Provider modules	EKS, GKE, AKS, bare-metal Terraform modules with identical interface	Module tests	Wave 1 (k3d + one cloud), wave 50 (all)
NFR-D1	Data integrity	Subledger-to-GL reconciliation	0 unexplained variance, daily	CI reconciliation job (Appendix I.6)	Wave 16
NFR-D2	Data integrity	Balance projection = journal re-aggregation	Exact	Nightly rebuild compare	Wave 11
NFR-D3	Data integrity	Gap-free sequences	0 gaps, 0 duplicates under 10k concurrent requests	Jepsen-style test	Wave 5
NFR-D4	Data integrity	Negative inventory	0 unless item model group allows	Concurrency test	Wave 16
NFR-D5	Data integrity	Exactly-once event effect	0 duplicate effects after chaos kill	Outbox/inbox chaos test	Wave 4
NFR-D6	Data integrity	Migration reconciliation	100% control totals match	Migration run ledger	Waves 10, 49

G. Numbering and document conventions

G.1 Number sequence service

Sequences are defined per tenant with a scope of `tenant`, `legal_entity`, `legal_entity+fiscal_year` or `site`. Format strings use `{LE}` (legal entity code), `{FY}` (fiscal year), `{YY}`/`{YYYY}`, `{SITE}`, and `{######}` (zero-padded counter). Two modes:

- **Continuous (gap-free)**: the number is allocated inside the posting transaction from a per-sequence row locked with `SELECT ... FOR UPDATE`; if the transaction aborts the number is never consumed. Throughput is bounded (≈ 2,000/s per sequence on stage); used only where the law or audit requires it.
- **Non-continuous**: numbers are pre-allocated in blocks of 100 per service instance (cached); gaps are possible on restart. Used for everything that is not a fiscal document.

A sequence can be `stopped` (posting returns `number_sequence_exhausted`), has `lowest/highest` bounds, an optional yearly reset (only with `{FY}` in the format) and an alert at 90% consumption.

G.2 Formats per document type

Document	Sequence code	Format (default)	Example	Mode	Scope / reset
General journal entry (posted)	`GJ`	`{LE}-GJ-{FY}-{#######}`	`USMF-GJ-2026-0001234`	Gap-free	Legal entity × fiscal year
Journal batch (unposted)	`JB`	`JB-{########}`	`JB-00004521`	Non-continuous	Tenant
Customer invoice / credit note	`INV` / `CN`	`{LE}-INV-{FY}-{######}`	`DEMF-INV-2026-000087`	Gap-free	Legal entity × fiscal year (legal requirement in DE, FR, IN, MX)
Vendor invoice (internal number)	`VINV`	`{LE}-VINV-{FY}-{######}`	`USMF-VINV-2026-003310`	Gap-free	Legal entity × fiscal year
Purchase requisition	`REQ`	`REQ-{######}`	`REQ-001204`	Non-continuous	Tenant
RFQ	`RFQ`	`RFQ-{######}`	`RFQ-000051`	Non-continuous	Tenant
Purchase order	`PO`	`{LE}-PO-{######}`	`USMF-PO-000912`	Non-continuous	Legal entity
Product receipt	`PR`	`{LE}-PR-{FY}-{######}`	`USMF-PR-2026-000455`	Gap-free	Legal entity × fiscal year
Quotation	`QUO`	`QUO-{######}`	`QUO-000210`	Non-continuous	Tenant
Sales order	`SO`	`{LE}-SO-{######}`	`USMF-SO-004417`	Non-continuous	Legal entity
Packing slip / delivery note	`PS`	`{LE}-PS-{FY}-{######}`	`USMF-PS-2026-002031`	Gap-free	Legal entity × fiscal year
Return order (RMA)	`RMA`	`RMA-{######}`	`RMA-000078`	Non-continuous	Tenant
Vendor payment	`PAY`	`{LE}-PAY-{FY}-{######}`	`USMF-PAY-2026-000660`	Gap-free	Legal entity × fiscal year
Customer receipt / cash application	`RCPT`	`{LE}-RCPT-{FY}-{######}`	`USMF-RCPT-2026-001102`	Gap-free	Legal entity × fiscal year
Check number	`CHK-<bank>`	`{######}`	`100245`	Gap-free	Bank account (void keeps the number)
Production / batch order	`PRD`	`{SITE}-PRD-{######}`	`S1-PRD-000318`	Non-continuous	Site
Planned order	`PLN`	`PLN-{########}`	`PLN-00019902`	Non-continuous	Tenant (reset per planning run)
Warehouse work	`WRK`	`WRK-{########}`	`WRK-00031177`	Non-continuous	Tenant
Load	`LD`	`LD-{######}`	`LD-000504`	Non-continuous	Tenant
Quality order	`QO`	`QO-{######}`	`QO-000912`	Non-continuous	Tenant
Fixed asset	`FA`	`{LE}-FA-{#####}`	`USMF-FA-00317`	Non-continuous	Legal entity

Document	Sequence code	Format (default)	Example	Mode	Scope / reset
Asset transaction	`FAT`	`{LE}-FAT-{FY}-{######}`	`USMF-FAT-2026-000045`	Gap-free	Legal entity × fiscal year
Maintenance work order	`MWO`	`MWO-{######}`	`MWO-001560`	Non-continuous	Tenant
Service case / service work order	`CASE` / `SWO`	`CASE-{######}`	`CASE-000731`	Non-continuous	Tenant
Project	`PRJ`	`PRJ-{######}`	`PRJ-000142`	Non-continuous	Tenant
Payroll run	`PAYR`	`{LE}-PAYR-{FY}-{###}`	`USMF-PAYR-2026-021`	Gap-free	Legal entity × fiscal year
Expense report	`EXP`	`EXP-{######}`	`EXP-004012`	Non-continuous	Tenant
Bank statement	`BST`	`{bank}-{YYYY}-{####}`	`CHASE01-2026-0210`	Non-continuous	Bank account

Country packs may override formats and modes (Appendix H), for example a per-series gap-free invoice number for MX CFDI (`serie` + `folio`) or the FR requirement that credit notes use a separate chronological sequence.

G.3 Fiscal document immutability

A document is fiscal when it posts to the universal journal or is legally required to be retained (invoices, credit notes, receipts, packing slips, payments, journal entries, asset transactions, payroll statements, production order settlements). Rules:

1. After `posted_at` is set, the header and lines are immutable: a database trigger rejects `UPDATE`/`DELETE` on posted rows except the status columns `settled_amount`, `settlement_status` and `printed_at`, and the metadata generator marks all other fields read-only in the API.
2. Corrections are new documents: reversal (an exact negating journal entry linked by `reverses_id`), credit note, correcting invoice (country packs that require it), or an adjustment journal. The original stays and shows its correction links.
3. The rendered PDF/XML of a fiscal document is stored at posting time in object storage with object lock for the retention period and its SHA-256 is written to the document; re-rendering produces a copy marked `Duplicate`.
4. The posting date determines the fiscal period; the document date may differ. Both are immutable after posting.
5. Document numbers are unique per tenant forever; a sequence reset per fiscal year includes the year in the number.

G.4 Attachments

Attachments are stored in S3-compatible object storage (bucket per tenant, server-side encrypted) and referenced by `platform.attachment` rows (`entity`, `record_id`, `kind`, `content_sha256`, `size`, `retention_class`). Limits: 25 MiB per file, 100 files per record, file types allow-listed (PDF, images, Office, XML, CSV, TXT, ZIP ≤ 25 MiB); antivirus scan (ClamAV) before the attachment becomes visible. Attachments on fiscal documents inherit the document's retention and object lock; attachments on other records follow the entity's retention. Deleting a record soft-deletes attachments; hard purge runs after the retention window. Inbound documents (OCR invoices, EDI, e-invoices) keep the original payload as a `source` attachment.

H. Localization scope

Country packs are metadata and code bundles installed per legal entity (wave 48). Every pack ships tax determination data, statutory report formats for the electronic reporting designer, document layouts, bank formats, calendar/number defaults and a regulator-schema fixture set. The matrix below is the v1.0 scope; items marked v1.x ship after go-live.

Capability	US	UK	DE	FR	IN	MX	CA
Tax model	Sales & use tax by jurisdiction (state/county/city/district), nexus, exemption certificates, use-tax accrual; Avalara/Vertex adapter	VAT 20/5/0%, reverse charge, postponed import VAT, CIS withholding, Northern Ireland protocol	USt 19/7%, §13b reverse charge, intra-EU supplies, withholding for construction (Bauabzugsteuer)	TVA 20/10/5.5/2.1%, autoliquidation, intra-EU, DAS2 withholding hooks	GST (CGST/SGST/IGST/UTGST), HSN/SAC codes, reverse charge, TDS/TCS withholding sections	IVA 16/8/0%, ISR and IVA retentions, IEPS, SAT product/service catalogs	GST/HST 5–15%, provincial PST/QST, input tax credits, non-resident rules
Statutory reports (v1.0)	1099-MISC/NEC/INT, sales tax returns by state (data extract), W-2 structure	MTD VAT return (HMRC API), VAT100 boxes, EC sales list for NI	UStVA (ELSTER XML), ZM (EC sales list), Intrastat, GoBD-compliant export (GDPdU), E-Bilanz v1.x	CA3 VAT return, DEB/Intrastat, FEC (fichier des écritures comptables), liasse fiscale v1.x	GSTR-1, GSTR-3B, e-way bill payload, TDS Form 26Q structure, 26AS reconciliation hook	DIOT, Contabilidad Electrónica (catálogo, balanza, pólizas XML), ISR provisional data	GST/HST return (GST34 boxes), T4/T4A/T5018 structures, QST return
E-invoicing mandate	None; Peppol/DBNAlliance optional	None in v1.0 (B2B mandate consultation); Peppol BIS optional	B2G XRechnung mandatory; B2B receive (2025) and issue (2027) via XRechnung/ZUGFeRD 2.x	B2B mandate via PDP: receive 2026, issue phased 2026–2027; Factur-X profiles	IRP e-invoice (IRN + signed QR) mandatory above turnover threshold; GSP adapter	CFDI 4.0 via PAC mandatory for every invoice, payment complement, carta porte v1.x; cancellation workflow	None; Peppol optional
Bank formats	NACHA ACH (PPD/CCD/CTX), check printing (MICR), positive pay, BAI2 statements, lockbox BAI2, ISO 20022 for wires	BACS Standard 18, Faster Payments via ISO 20022 pain.001, CHAPS, MT940/camt.053 statements	SEPA SCT/SDD (pain.001/pain.008), camt.052/053/054, EBICS 3.0	SEPA SCT/SDD, camt.053, EBICS TS (signed)	NEFT/RTGS/IMPS bank-specific layouts via adapter, ISO 20022 pain.001 for H2H banks, MT940	SPEI bank layouts via adapter, ISO 20022 for multinational banks, BAI2/camt.053	CPA Standard 005 EFT, Lynx wires via ISO 20022, cheque printing, camt.053
Payroll	Full engine: federal/state/local withholding (Pub. 15-T), FICA, FUTA/SUTA, garnishments, W-2/941 structures	PAYE RTI (FPS/EPS), NI, auto-enrolment — engine only if chosen as EU/UK reference (open question 3), else provider adapter	Lohnsteuer, Sozialversicherung (DEÜV), ELStAM — reference EU engine candidate (open question 3), else provider adapter	DSN monthly declaration — v1.x, provider adapter in v1.0	PF, ESI, professional tax, Form 16 — v1.x, provider adapter in v1.0	CFDI nómina, IMSS, ISR tables — v1.x, provider adapter in v1.0	CRA/RQ source deductions, CPP/EI, ROE — v1.x, provider adapter in v1.0
Calendar and formats	Fiscal year any; MM/DD/YYYY display; USD 2 decimals; Letter paper	Tax year 6 April (payroll), company FY any; DD/MM/YYYY; GBP; A4	Calendar FY default; DD.MM.YYYY; EUR with `1.234,56`; A4; DIN 5008 letters	Calendar FY default; DD/MM/YYYY; EUR with `1 234,56`; A4	FY 1 April–31 March; DD/MM/YYYY; INR lakh/crore grouping (`12,34,567.00`)	Calendar FY; DD/MM/YYYY; MXN; Letter	FY any; YYYY-MM-DD; CAD; Letter; bilingual EN/FR documents (Quebec)

Capability	US	UK	DE	FR	IN	MX	CA
Document layouts	Invoice, PO, packing slip, check, 1099	Invoice with VAT breakdown, credit note, remittance	Rechnung with §14 UStG mandatory fields, Gutschrift, Mahnung levels	Facture with mentions obligatoires, avoir, relance	Tax invoice with GSTIN/HSN/IRN QR, debit/credit note, e-way bill	CFDI PDF representation with UUID/QR, complemento de pago	Invoice with GST/HST registration, bilingual variants
Fiscal retention	7 years	6 years	10 years (GoBD)	10 years	8 years	5 years	6 years

Common to every pack: ISO 3166/4217/639 reference data, address formats and postal validation rules, tax-registration validators (EIN, UTR/VAT, USt-IdNr, SIREN/SIRET/TVA, GSTIN/PAN, RFC, BN/GST), number-sequence overrides, holiday calendars, and a regulator-schema fixture that must validate in CI before the pack can be released.

I. Testing standards

I.1 Test pyramid per service type

Service type	Unit	Property	Contract	Integration	E2E	Performance
Go engine (ledger, inventory, pricing, MRP, scheduling, costing, matching, settlement)	`go test`, table-driven, ≥ 85% line coverage on domain packages	`rapid` on invariants (balance, conservation, determinism, state machines)	OpenAPI contract via `oapi-codegen` validators; Pact for consumer contracts between contexts	Testcontainers Postgres + Kafka; outbox/inbox chaos	Through the shell for the wave demo path	Go benchmarks in CI (`benchstat` regression gate ±10%); k6 for throughput targets
TypeScript service (workflow, integration, BFF, admin)	Vitest, ≥ 80% coverage	`fast-check` for mappers, parsers (EDI, GS1, bank files), policy routing	OpenAPI + AsyncAPI schema validation; Pact	Testcontainers + Temporal test server	Playwright	k6
Python service (forecasting, ML, agent evals)	pytest, ≥ 80%	`hypothesis` for numeric routines and parsers	JSON Schema on inputs/outputs	Docker compose fixtures	n/a	Benchmark notebooks exported to CI metrics
React shell and module-federated workspaces	Vitest + Testing Library	`fast-check` for form state reducers	Storybook interaction tests; generated-form snapshot per entity	MSW-backed integration	Playwright + axe across every generated form and workspace	Lighthouse CI
PWA (shop floor, warehouse, field)	Vitest	`fast-check` for the offline queue (commutativity, idempotence)	Same as shell	Device emulation	Playwright on emulated Android and tablet, offline toggling	Scan-to-ack latency budget 300 ms

The ratio target per service is roughly 70% unit/property, 20% contract/integration, 10% E2E/performance by count; every wave's proof items map to at least one row.

I.2 Fixture naming

```
fixtures/<context>/w<NN>-<slug>/<nnn>-<scenario>.<json|csv|ndjson>
fixtures/<context>/w<NN>-<slug>/<nnn>-<scenario>.expected.<json|csv>
fixtures/<context>/w<NN>-<slug>/README.md               -- what the set covers, source, size
fixtures/<context>/w<NN>-<slug>/REVIEW.yaml              -- accountant review record (I.5)
```

Example: `fixtures/ledger/w12-currency-parallel-ledgers/017-revaluation-eur-usd-unrealized.json`. Scenario numbers are stable once published (a retired scenario keeps its number with `retired: true`). Shared master-data seeds live in `fixtures/seed/<demo-company>/` (`USMF` manufacturer, `DEMF` process plant, `GBSI` services entity, `MXDS` distributor) and are loaded by the migration framework, so fixtures and demos use the same data.

I.3 Property-based testing

Libraries are fixed: Go `pgregory.net/rapid`, TypeScript `fast-check`, Python `hypothesis`. Every engine declares its invariants in a `PROPERTIES.md` next to the code, and each invariant has one property test with a minimum of 1,000 generated cases in CI (10,000 nightly). Required invariant families: conservation (Σ debits = Σ credits; Σ inventory transactions = on-hand; Σ allocations = source), determinism (same input → same output across runs and goroutines), idempotence (apply twice = apply once), monotonicity (sequences, versions), state machines (every illegal transition rejected), and round-trips (serialize/parse, UoM conversions, currency rounding). Shrunken counterexamples are committed as regression fixtures with the originating seed.

I.4 Golden files

Golden outputs (generated migrations, OpenAPI specs, rendered documents, statement packs, bank files, EDI messages, Rego bundles) live in `testdata/golden/` next to the test. A golden test fails on any byte difference. Updates happen only through `make update-golden PKG=<path>`, which rewrites files and records the reason in the commit; a PR that touches `testdata/golden/` must carry the label `golden-update` and a reviewer other than the author must approve (enforced by CODEOWNERS). Rendered PDFs are compared as rasterized pages with a 0.1% pixel tolerance; XML and JSON are canonicalized before comparison; timestamps and ids are normalized by the test harness.

I.5 Accountant-reviewed fixtures

Finance waves (11–15, 18–20, 29, 36–41, 43) require fixture sets reviewed by a practicing accountant before the wave can close. Process: (1) the engineer drafts scenarios and expected results in a workbook (`.xlsx`) exported to `.expected.csv`; (2) the reviewer works the scenarios independently and signs `REVIEW.yaml` (`reviewer`, `credential`, `reviewed_on`, `standard` such as ASC 606, IAS 21, IRS Pub. 946, `scope`, `exceptions[]`); (3) CI fails if a finance fixture set lacks a `REVIEW.yaml` dated after its last change; (4) any later change to the set re-opens review. The reviewer owner is tracked in the PRD validation checklist.

I.6 Reconciliation checks in CI

A scheduled job (`reconcile-daily`) runs at 02:00 UTC on stage against the demo companies and on every merge to `main` against the fixture databases; any non-zero variance fails the job and pages the wave owner.

Check	Ties	Tolerance	From wave
Trial balance	Σ debits − Σ credits per ledger per period	0.00	11

Check	Ties	Tolerance	From wave
Balance projection	`Balance` table vs re-aggregated `JournalLine`	0.00	11
Parallel ledgers	Each ledger's trial balance sums to zero	0.00	12
Tax	Σ `TaxTransaction` per tax code vs GL tax accounts	0.00	13
Statements	Balance sheet balances; P&L net = retained earnings movement; cash flow = Δ cash	0.00	15
Inventory quantity	Σ `InventoryTransaction` per dimension vs `OnHand`	0	16
Inventory value	Inventory value report vs GL inventory accounts by item group	0.00	20
GR/IR	Σ receipts – Σ invoices per PO line vs GR/IR account	0.00	18
AP	Σ open vendor items vs AP control accounts by posting profile	0.00	19
WIP	Open production order cost vs GL WIP accounts	0.00	21
AR and deferred revenue	Σ open customer items vs AR control; revenue schedule balance vs deferred revenue	0.00	29–30
Bank	Reconciled statement balance vs GL bank account	0.00	36
Fixed assets	Σ NBV per book vs GL asset and accumulated depreciation	0.00	37
Cost accounting	Primary costs in cost ledger vs GL	0.00	38
Intercompany	Due-to/due-from pairs net to zero per pair	0.00	40
Projects	Project WIP/accrued revenue vs GL	0.00	41
Payroll	Payroll register vs GL payroll accounts	0.00	43
Semantic layer	Revenue/COGS metrics vs GL	0.00	46
Audit chain	Hash chain verifies end-to-end	0 breaks	5

I.7 Performance and E2E harness

k6 scripts live in `perf/<context>/<scenario>.js` with thresholds encoded in the script (`http_req_duration{scenario:post}: p(99)<200`); CI runs smoke profiles on every merge and full profiles nightly on stage, publishing results to Grafana with a trend alert at +10% over 7 days. Playwright suites live in `e2e/<workspace>/` with page objects per generated form type and run against the stage URL after Argo reports healthy; every suite calls `axe` on each page it visits and fails on serious/critical violations. Flaky tests are quarantined by label for at most one wave before deletion or repair.

J. Traceability matrix

Each master-PRD functional requirement maps to its primary wave (FR-p.n → wave 5p + n), the later waves that extend it, the phase-exit criterion that proves it, and the test types that are primary evidence. Phase-exit criteria are the proof items of waves 5, 10, 15, 20, 25, 30, 35, 40, 45 and 50.

FR	Requirement (short)	Primary wave	Extended by	Phase-exit criterion	Primary test types
FR-0.1	GitOps deploy, signed images, SBOM, digest rollback	1	48, 50	Wave 5 platform demo: deploy → login → extend → approve → trace	CI negative tests, timed rollback, cosign/Grype gates
FR-0.2	Federated identity; roles/duties/privileges → OPA; RLS	2	47, 48	Wave 5 demo (login + permission viewer)	Contract (20 duties), RLS per table, federation in CI
FR-0.3	Metadata-declared entities generate schema/API/UI; no-code extensions	3	5, 46	Wave 5 demo (extend entity live)	Golden generation, backward-compat, load 1,000 rps
FR-0.4	CloudEvents via outbox; Temporal workflows and batch	4	47	Wave 5 demo (approval + trace)	Chaos outbox, Temporal timer tests, schema-registry gate
FR-0.5	Shell (WCAG AA), gap-free sequences, immutable audit, tracing	5	46	Wave 5 demo signed off	Playwright + axe, Jepsen-style sequence test, audit E2E
FR-1.1	Enterprise structure, hierarchies, calendars, ISO data, UoM	6	14, 40	Wave 10: demo master data fully migrated	Property (as-of hierarchy), UoM round-trip, calendar fixtures
FR-1.2	Single party model, dedup/merge, tax-id validation, governance	7	30, 42	Wave 10	Dedup precision/recall, merge integrity audit, Temporal approval
FR-1.3	Products, dimension groups, released products, GS1, configurator	8	24, 27	Wave 10	Variant explosion perf, configurator fixtures, GTIN vectors
FR-1.4	BOMs, formulas, routings, resources, ECM	9	21, 24, 39	Wave 10	Explosion golden + perf, effective-date tests, ECO workflow
FR-1.5	Migration framework with AX/SAP connectors and reconciliation	10	49	Wave 10: synthetic AX extract loads at 100% reconciliation	Load timing, idempotent re-run, mapping DSL units
FR-2.1	Append-only universal journal, dimensions, idempotent posting, balances	11	14, 15, 40	Wave 15: closed period with statements tying to TB	Property (1M entries), negative posting tests, 5k lines/s k6
FR-2.2	Three currencies, rate providers, parallel ledgers, revaluation	12	40	Wave 15	Rounding property, parallel-ledger fixtures, revaluation workbook
FR-2.3	Tax determination/calculation/posting, providers, SAF-T	13	35, 48	Wave 15	500-scenario matrix, settlement reconciliation, XSD validation
FR-2.4	Period control, close checklist, allocations, accruals, year-end	14	38	Wave 15: recorded close walkthrough	Contract (closed period on every path), allocation fixtures
FR-2.5	Statement designer, standard statements, drill-down, audit report	15	40, 46	Wave 15 sign-off by finance stakeholders	Golden statements, drill-down property, designer round-trip
FR-3.1	Immutable inventory transactions, journals, reservations, counting	16	31, 33	Wave 20: buy → receive → pay → value walkthrough	Concurrency (no negative), rebuild compare, FEFO test
FR-3.2	Requisition → RFQ → agreement → PO with policies and versions	17	35, 38	Wave 20	Approval matrix, versioning tests, price-lookup property
FR-3.3	Receiving with GR/IR, invoice capture, 2/3-way match, ERS	18	35	Wave 20	200-scenario matching fixtures, GR/IR nets to zero
FR-3.4	AP settlement, payment proposals, ISO 20022/NACHA/check, positive pay	19	36	Wave 20	300 settlement fixtures, pain.001 XSD, idempotent run

FR	Requirement (short)	Primary wave	Extended by	Phase-exit criterion	Primary test types
FR-3.5	FIFO/LIFO/average/standard costing, close, GL reconciliation, portal	20	39	Wave 20	Costing workbooks, close timing, value report ties to GL
FR-4.1	Production order lifecycle, WIP, variances, subcontracting	21	24, 25, 39	Wave 25: planned order → finished goods with full costing	State-machine property, costing fixtures, backflush test
FR-4.2	MRP/MPS, coverage groups, messages, pegging, forecasting	22	23, 27	Wave 25	40 APICS scenarios, scale 50k items, pegging completeness
FR-4.3	Finite/infinite scheduling, Gantt, CP-SAT	23	44, 45	Wave 25	OR-Library fixtures, no-double-booking property, 5k ops perf
FR-4.4	Process manufacturing and lean	24	33	Wave 25	Potency/co-product fixtures, kanban simulation, catch-weight tests
FR-4.5	Shop-floor kiosk, time registration, IoT, OEE, offline	25	43, 44	Wave 25	Kiosk E2E, 10k events/min ingest, 30-min offline test
FR-5.1	Deterministic pricing engine with explain	26	30, 40	Wave 30: one simulated month buy → make → sell → collect, all subledgers reconcile	500 pricing fixtures, determinism property, p95 < 20 ms
FR-5.2	Quotations, orders, ATP/CTP, credit, holds, intake API	27	35	Wave 30	ATP fixtures + over-promise property, credit workflow, 500 orders/min
FR-5.3	Release, pick/pack/ship, packing slips with COGS, carriers, drop-ship	28	31, 32, 34	Wave 30	Posting contract tests, 1,000-order E2E, SSCC validation
FR-5.4	Invoicing variants and ASC 606/IFRS 15	29	40, 41	Wave 30	Totals property, 40 ASC 606 fixtures, roll-forward reconciliation
FR-5.5	Cash application, collections, RMA, rebates, customer portal	30	36, 47	Wave 30 GO-LIVE CANDIDATE sign-off	Cash-application fixtures, dunning tests, RMA E2E
FR-6.1	Location directives, work, waves, license plates, replenishment, loads	31	34	Wave 35: EDI PO → ASN → invoice between demo partners	100 directive fixtures, 10k-line simulation, capacity property
FR-6.2	Mobile scanner app, flow designer, GS1, offline, labels	32	45	Wave 35	Device E2E, GS1 parser suite, 200-transaction offline sync
FR-6.3	Quality orders, sampling, usage decisions, NCR/CAPA, CoA	33	44	Wave 35	Trigger fixtures, blocking property, Z1.4 tables
FR-6.4	Load/route planning, rating, tendering, freight reconciliation, landed cost	34	40	Wave 35	Solomon VRP fixtures, 200 rating fixtures, tender timeouts
FR-6.5	EDI X12/EDIFACT, Peppol, integration hub, developer portal	35	48	Wave 35	Partner-simulator round trips, schematron, webhook at-least-once
FR-7.1	Bank accounts, statements, reconciliation, cash forecast, connectivity	36	48	Wave 40: group consolidated statements, finance parity sign-off	1,000-line reconciliation, forecast property, EBICS contract tests
FR-7.2	Fixed assets, multi-book depreciation, CIP, IFRS 16	37	41	Wave 40	100 asset fixtures vs MACRS tables, lease workbook, subledger tie
FR-7.3	Budgeting, budget control, cost accounting, ABC	38	41	Wave 40	Budget-check fixtures, allocation convergence, cost ledger tie
FR-7.4	Costing sheets, roll-up, standard cost activation, variances	39	—	Wave 40	50-product workbook, activation revaluation, variance sums
FR-7.5	Intercompany trade/accounting, consolidation with eliminations	40	—	Wave 40 sign-off by a practicing controller	IC balance property, 3-entity eliminations, mismatch detection
FR-8.1	WBS, resources, journals, billing modes, WIP/POC, EVM	41	45	Wave 45: projects, HR, payroll, maintenance, field service walkthrough	POC workbook, invoice-proposal property, WBS scheduling
FR-8.2	Workers/positions/jobs, absence, benefits, self-service, PII	42	43	Wave 45	Effective-date property, accrual fixtures, PII contract tests
FR-8.3	Country-pluggable payroll, expense management	43	48	Wave 45	200 Pub. 15-T scenarios, retro fixtures, payroll GL tie
FR-8.4	Functional locations, maintenance plans, work orders, IoT triggers	44	—	Wave 45	Plan fixtures, cost posting tie, 5 s IoT trigger
FR-8.5	Installed base, agreements/SLAs, cases, dispatch, technician app	45	—	Wave 45	Entitlement fixtures, dispatch property, warranty billing
FR-9.1	Lakehouse, semantic layer, dashboards, document designer, catalog	46	—	Wave 50: first customer month-end in production; load and DR targets met	Lag < 60 s, metric = GL check, RLS in dashboards, visual regression
FR-9.2	MCP APIs, copilot, governed agents, evals, cost budgets	47	—	Wave 50	Eval thresholds in CI, permission-ceiling contracts, audit of actions
FR-9.3	Localization framework, country packs, ER designer, SoD, SOC 2, privacy	48	—	Wave 50	Regulator-schema fixtures, 50 SoD conflicts, restore drill, pen test
FR-9.4	Transactional migration packs, parallel-run diff, cutover, coexistence	49	—	Wave 50	100% control totals, zero unexplained variance, 48 h rehearsal
FR-9.5	Scale targets, multi-region DR, operations, first customer, v1.0	50	—	Wave 50	k6 campaign, DR game day, production close, signed release

K. Reference index

Consolidated from the `references` arrays in `roadmap/waves.json` (183 distinct URLs, deduplicated by URL). Wave numbers in parentheses. SAP Help URLs are the identifiers recorded in `waves.json`; verify against the current SAP Help portal before citing in ADRs.

K.1 AX 2012 / D365 documentation (Microsoft Learn)

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- AX 2012 role-based security — https://learn.microsoft.com/en-us/dynamicsax-2012/developer/role-based-security-in-microsoft-dynamics-ax (2)
- AX 2012 workflow overview — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/workflow-overview (4)
- AX 2012 number sequences — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/about-number-sequences (5)

- AX 2012 organization model — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/organizations-and-organizational-hierarchies (6)
- AX 2012 global address book — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/global-address-book-overview (7)
- AX 2012 product information management — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/about-products (8)
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- AX 2012 sales tax overview — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/about-sales-tax (13)
- AX 2012 allocation rules — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/about-ledger-allocation-rules (14)
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- AX 2012 procurement and sourcing — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/procurement-and-sourcing (17)
- AX 2012 invoice matching — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/about-accounts-payable-invoice-matching (18)
- AX 2012 accounts payable — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/accounts-payable (19)
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- AX 2012 master planning — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/master-planning (22)
- AX 2012 scheduling — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/about-scheduling (23)
- AX 2012 process manufacturing — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/process-manufacturing-production-and-logistics (24)
- AX 2012 lean manufacturing — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/lean-manufacturing (24)
- AX 2012 trade agreements — https://learn.microsoft.com/en-us/dynamicsax-2012/appuser-itpro/about-trade-agreements (26)
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L. Competitive benchmark — top-tier manufacturing ERPs

L.1 Capability matrix

The matrix compares the six systems a mid-to-large manufacturer shortlists against the Passage v1.0 target. Cells name the actual feature or module and give typical numbers where they are public or widely reported by customers and analysts; "n/a" means the capability is absent or only available through a partner product. Figures marked "typical" are customer-reported ranges, not vendor commitments, and should be re-validated per deal.

Capability	SAP S/4HANA	D365 F&SCM	Infor LN / CloudSuite Industrial	Epicor Kinetic	IFS Cloud	Plex	Passage target
MRP full regeneration, 50k items	MRP Live on HANA: minutes (typically < 10 min; MD01N)	Planning Optimization (Azure service): typically 5–30 min; legacy master planning: hours	LN: batch overnight typical; CSI (SyteLine) APS: minutes to hours	MRP regeneration batch, typically overnight; net-change intraday	MRP/MS batch, typically hours; incremental supported	Net-change MRP continuous, minutes	< 10 min for 50k items × 12 periods × 5 BOM levels; deterministic; net-change continuous
Live pegging / explain plan	MD04 stock/requirements list; PP/DS pegging	Planning Optimization pegging view	LN order pegging (project pegging); CSI APS pegging	Time Phase inquiry pegging	Supply/demand pegging in Supply Chain Planning	Limited (order-level links)	Pegging tree API explaining 100% of any planned order's quantity
Finite capacity scheduling	Embedded PP/DS with optimizer (extra license)	Finite capacity in master planning + Gantt; no optimizer	Infor Scheduling / CSI APS finite scheduling	APS module (add-on), finite with rules	Constraint-Based Scheduling (CBS) module	Production Scheduling (finite, drag-drop)	Finite/infinite + CP-SAT optimizer built in; 5k operations < 60 s
Shop-floor execution UX	SAP Digital Manufacturing (separate cloud, extra license) or MII	Production floor execution (touch, job cards)	Infor Factory Track (separate)	Kinetic MES (Advanced MES, ex-Mattec)	Shop Floor Workbench; IFS MES	Native MES, machine-integrated; strongest of the group	Touch PWA kiosk, 30-min offline, time → costing and payroll in one flow
Machine / IoT connectivity	SAP DM Edge, SAP IoT	Sensor Data Intelligence (Azure IoT Hub)	Infor OS IoT + Factory Track	Advanced MES direct PLC monitoring	IoT Business Connector	Native PLC data collection	OPC-UA/MQTT edge connector → Kafka; 10k events/min per plant, no loss on broker restart
OEE out of the box	In SAP DM	Via Sensor Data Intelligence / Power BI; limited native	Factory Track	Advanced MES real-time OEE	Via MES; limited native	Native real-time OEE	Native, ISO 22400 definitions, 5 s refresh

Capability	SAP S/4HANA	D365 F&SCM	Infor LN / CloudSuite Industrial	Epicor Kinetic	IFS Cloud	Plex	Passage target
Product configurator (CTO)	Variant Configuration (LO-VC / AVC)	Product configuration models (constraint-based)	Infor CPQ (separate); LN configurator	Kinetic Configurator (rules-based)	Sales Configurator	Basic option selection	Constraint-based (OR-Tools CP-SAT), 50-constraint fixtures, drives BOM/route
Process manufacturing: formula, potency, co/by-products, catch weight	PP-PI recipes, batch management, active ingredient	Full: formulas, potency, co/by-products, catch weight, batch attributes	LN: limited; CloudSuite Food & Beverage (M3) strong	Limited in Kinetic	Recipe/formula, batch; limited potency	Lot, catch weight; formulas for food/bev	Wave 24: formulas, potency, co/by allocation, catch weight, shelf life, batch disposition
Lean / kanban	KANBAN (PK*), REM	Lean manufacturing: production flows, kanban rules/boards	CSI kanban basic	Kanban basic	Kanban/repetitive basic	Kanban native	Production flows, event/fixed/withdrawal kanban, backflush costing
Quality management	QM: inspection lots, usage decisions, notifications; CAPA via QIM	Quality orders, nonconformances, CoA, test groups	Infor Quality / LN quality basic	Advanced Quality (AQM) with NCR/CAPA	Quality Management: control plans, nonconformance	Native: control plans, SPC, gage R&R, PPAP	Rule-triggered quality orders, ANSI Z1.4 AQL, 8D CAPA, SPC, CoA, supplier scorecards, calibration
Advanced WMS	EWM (basic embedded or advanced, extra license)	Advanced warehouse: location directives, waves, LPs	Infor WMS (separate); LN warehouse basic	Kinetic warehouse basic + Advanced Mobile Warehouse	Warehouse data collection (WADACO) basic	Native basic/mid WMS	Location directives, work templates, waves, LPs, replenishment, slotting, dock scheduling
WMS mobile	EWM RF framework / Fiori	Warehouse Management mobile app (Android, configurable flows)	Factory Track mobile	Mobile Warehouse	Aurena mobile scanning	Native mobile	PWA + Android shell, flow designer, offline queue, GS1 AI parsing, ZPL printing
Transportation management	SAP TM (embedded)	TMS native (rating, routing, load building)	Infor Nexus (separate)	n/a (partner)	Limited	n/a	VRP load/route planning, rate shopping, tendering, freight reconciliation, landed cost
Inventory costing methods	Standard, moving average, actual (material ledger)	FIFO, LIFO, weighted average, standard, moving average	Standard, actual, FIFO/LIFO (lot)	Standard, average, last, FIFO, lot FIFO	Standard, weighted average, FIFO (lot), actual	Standard, actual (average)	FIFO, LIFO, weighted average (running/periodic), standard, moving average; close ties to GL
Universal journal (single line-item truth)	ACDOCA: one table for FI/CO/AA/ML	GeneralJournalAccountEntry + separate subledger tables	GL with integration transactions	GL with posting (booking) rules	GL vouchers	GL	Append-only journal; balances are rebuilt projections; every subledger reconciles daily by automated check
Parallel ledgers / multi-GAAP	Leading + non-leading ledgers, currency types 10/30/40	Posting layers (current/operations/tax + custom)	Limited (parallel companies / dimensions)	Multiple GL books	Parallel accounting (IFRS/local)	n/a	Ledger groups per legal entity; 3 currency amounts per line
Month-end close, typical	3–5 days with Advanced Financial Closing	5–8 days	5–10 days	5–10 days	5–10 days	5–10 days	≤ 3 days; continuous close via daily reconciliation (Appendix I.6)
ASC 606 / IFRS 15	Revenue Accounting and Reporting (RAR)	Revenue recognition module (being replaced by subscription billing)	n/a (partner)	n/a	Project-based rev rec	n/a	Native: performance obligations, SSP allocation, ratable/milestone/usage schedules
Intercompany & consolidation	Cross-company postings; Group Reporting	Intercompany trade; consolidation	CloudSuite consolidation	Multi-company, consolidation	Group consolidation	Multi-plant; limited consolidation	Mirror documents, due-to/due-from matching, eliminations, IAS 21 translation
EAM and field service	PM + Field Service Management	Asset Management + Field Service (separate app)	Infor EAM (now HxGN EAM, separate)	Maintenance basic	Strongest EAM/FSM of the group	Basic maintenance	Native EAM (plans, work orders, IoT triggers) + field service (dispatch optimization, technician app)
Localization coverage	60+ country versions	40+ countries	40+ (LN)	~20 country packs	50+ countries	US-centric, ~10	7 packs in v1.0 (US, UK, DE, FR, IN, MX, CA); framework for v1.x
Extensibility model	ABAP (key-user, developer, BTP side-by-side)	X++ extensions (no overlayering), Power Platform	ION, Mongoose (CSI), LN Extensions	Application Studio (low-code), BPM, C#	Aurena low-code configuration + PL/SQL	Configuration; limited custom code	Metadata extensions (entities, enums, fields, workflows, forms, reports) + sanctioned hooks; no proprietary language
API openness	OData/SOAP on SAP Business Accelerator Hub (thousands, uneven coverage)	OData data entities (2,000+), custom services	ION APIs (REST), BODs	REST API v2 (OData) across the product	REST projections (OData) across the product	REST/SOAP via developer portal	OpenAPI 3.1 generated for every entity and engine; GraphQL read layer; additive-only within /v1
Event streaming	Event Mesh / Advanced Event Mesh (CloudEvents)	Business events → Event Grid / Service Bus / Dataverse	ION BOD publishing (XML)	Limited (Epicor Functions, webhooks)	IFS Connect events	Limited webhooks	Every state change is a replayable CloudEvent on Kafka via outbox; AsyncAPI published
Analytics / data access	Embedded CDS analytics; Datasphere	Entity store; Synapse/Fabric link	Birst; Data Lake	Epicor Data Analytics (Grow)	Analysis models; IFS Analytics	Plex Intelligence	Debezium CDC → Iceberg lakehouse, Cube semantic layer, p95 lag < 60 s, open to any BI tool
AI agents	Joule agents; Business AI	Copilot; autonomous agents (reconciliation, supplier comms)	Infor GenAI / Industry AI agents	Prism agents	IFS.ai agents	Copilot features	MCP-exposed APIs; agents inherit user OPA permissions; evals gate deploys; model-agnostic
Audit trail	Change documents (per table, configurable); AIS	Database log (opt-in per table, performance cost)	Configurable audit trail	Change log per table (opt-in)	History logging (opt-in)	Change logs	Field-level from CDC on every table, hash-chained, 7-year fiscal retention
Segregation of duties	GRC Access Control (separate)	Native SoD rules	Infor GRC (separate)	n/a	n/a	n/a	Native SoD engine with mitigation workflow and certification campaigns

Capability	SAP S/4HANA	D365 F&SCM	Infor LN / CloudSuite Industrial	Epicor Kinetic	IFS Cloud	Plex	Passage target
Deployment model	Public cloud, private cloud (RISE), on-premise	Microsoft-managed Azure only (on-premise LBD retired)	Multi-tenant AWS (CloudSuite) or on-premise LN	Epicor-managed Azure or on-premise	IFS-managed Azure or customer-managed Kubernetes (remote deployment)	SaaS only, multi-tenant	Any conformant Kubernetes (EKS/GKE/AKS/bare-metal/k3d); self-hosted or managed; no provider-managed service in the critical path
Upgrade cadence and effort	Annual (on-prem) / biannual (public cloud) mandatory; on-prem upgrades are multi-month projects	4 mandatory service updates per year (One Version); regression testing each	Monthly cloud updates; LN on-prem by project	Biannual releases; cloud updates mandatory	Biannual releases + monthly service updates (evergreen)	Continuous (weekly)	Continuous; zero-downtime expand/contract migrations; digest rollback < 15 min; customer-controlled cadence
Licensing	Named users (Professional/Functional), FUE under RISE; list prices not public	Per user per month (Finance / SCM list ≈ USD 180–210 per full user; Activity/Team tiers lower)	Subscription per user; not public	Subscription per user; not public	Subscription per user; not public	Subscription per plant and user; not public	Open question 1 (decide by wave 35): open-core or source-available; no per-seat charge for shop-floor, portal or agent users
Implementation, mid-size manufacturer (typical)	12–24 months	9–18 months	6–12 months (CSI)	6–12 months	9–18 months	6–12 months	3–6 months using the migration factory (wave 49) and parallel-run diff
Accessibility	Fiori: WCAG 2.1 AA for most apps	WCAG 2.1 AA	Partial	Partial	Partial	Partial	WCAG 2.1 AA on every generated form and workspace, axe in CI

Reading the matrix: the incumbents win today on breadth of localization, installed-base references and depth in a few modules (SAP finance, IFS EAM, Plex MES). Passage does not try to out-feature them row by row in v1.0; it competes on the rows where every incumbent is structurally weak — one journal with automated reconciliation, metadata extensibility without a proprietary language, event streaming and open APIs as the default integration surface, cloud-agnostic deployment, continuous zero-downtime upgrades, and governed AI agents that act under the same authorization as users.

L.2 Interoperability contract

Principle: any system on the customer's network — MES, PLM, SCADA/OPC-UA, a third-party WMS, EDI trading partners, BI tools, AI agents — consumes Passage through the same APIs, events and MCP tools that Passage's own modules use. There is no separate "integration layer" product, no licensed middleware, and no integration project to stand up before the first interface works. What SAP sells as Integration Suite, Microsoft as dual-write plus Logic Apps, and Infor as ION is, in Passage, the integration hub shipped in wave 35: a catalog, keys and quotas, a message monitor and replay over surfaces that already exist from wave 3 (REST), wave 4 (Kafka) and wave 47 (MCP).

The contract every surface honors:

1. **One authorization model.** Every call, event subscription, webhook and tool invocation is authorized by the same OPA bundle against the same roles/duties/privileges (Appendix E). Partner credentials are users or service accounts in the tenant's Keycloak realm; there is no integration back door.
2. **One schema source.** OpenAPI, AsyncAPI, GraphQL SDL, MCP tool definitions and EDI mappings are generated from the same entity metadata, so a field added by a customer extension appears on every surface in the same release.
3. **Additive evolution.** `/v1` and `v<n>` event types change additively only; breaking changes ship as a new version with a 12-month (API) or two-minor-release (event) overlap, announced with `Deprecation` / `Sunset` headers and registry flags.
4. **Correlation and idempotency everywhere.** Every inbound request carries or receives a `correlationid`; every outbound event and webhook carries it; every mutating call is idempotent by key (Appendix D.5); every consumer is expected to be idempotent by event id (Appendix C.5).
5. **Replayability.** Events are retained 30 days and replayable per consumer; webhooks can be re-driven from the message monitor; file and EDI inbound payloads are kept as source attachments.
6. **Sandbox and conformance.** Every tenant gets a sandbox tenant with the demo companies; a conformance kit (Postman/k6/Pact suites) lets a partner prove their integration before production keys are issued.
7. **Observability shared with the customer.** Per-key traces, delivery logs, DLQ contents and SLA dashboards are visible in the integration hub, not only to Passage operators.
8. **No data hostage.** Full export (NDJSON per entity, Iceberg tables, event archive) is a standard API, not a service engagement.

Surface	Typical consumers	Protocol / format	Authentication and authorization	SLA (per tenant, measured)	Notes
REST	MES, PLM, custom apps, iPaaS, mobile	HTTPS/JSON, OpenAPI 3.1, RFC 9457 errors	OAuth 2.0 (Keycloak): authorization code for users, client credentials for services; JWT validated at Envoy, OPA per request; tenant API keys for partner quotas	99.9% availability; p95 < 50 ms simple reads, p99 < 500 ms document posts; rate limits per tier (D.7)	Idempotency keys, ETags, cursor pagination, bulk `:batch` / `:export`
GraphQL	BI tools, portals, composite screens, low-code builders	HTTPS, GraphQL over the generated SDL (read-only in v1.0; mutations via REST)	Same JWT and OPA; field-level masking for P2/P3 per duty	99.9%; p95 < 200 ms for queries ≤ 1,000 cost points; depth ≤ 8, cost-limited	Persisted queries for portals; mirrors REST filter grammar
Kafka topics	MES, third-party WMS, data platform, CRM sync, customer event consumers	Kafka protocol, CloudEvents JSON, AsyncAPI 3.0, schema registry	mTLS + SCRAM-SHA-512 per principal; topic ACLs per tenant; cross-tenant principals only for the customer's own analytics	99.9% broker availability; p95 commit-to-consumer < 5 s; 30-day retention; replay on request	External consumers connect through the tenant's dedicated consumer group; DLQ visible in hub
Webhooks	SaaS partners, iPaaS (Workato, Make), Salesforce/Shopify connectors	HTTPS POST of the CloudEvent, JSON	HMAC-SHA256 signature in `X-Passage-Signature` with per-endpoint secret (rotation supported), optional mTLS, IP allow-list	At-least-once; p95 first delivery < 10 s; retries with exponential backoff for 24 h, then DLQ; per-tier delivery quotas	Subscriptions filtered by event type and legal entity; re-drive from monitor
MCP	AI agents (Passage copilot, Claude, customer-built agents, partner agents)	MCP over HTTPS (streamable), tool/resource definitions generated from OpenAPI and the semantic layer	OAuth 2.1 with user-delegated token; agent identity registered in the tenant; OPA evaluates `agent@user` so the agent never exceeds the user (E.4 example 2); kill switch	99.9%; p95 tool call < 1 s excluding downstream posting time; per-tenant token and cost budgets	Every tool call audited with inputs/outputs; approval gates for posting tools

Surface	Typical consumers	Protocol / format	Authentication and authorization	SLA (per tenant, measured)	Notes
EDI	Trading partners, 3PLs, carriers, retailers	X12 (850/855/856/810/820/214) and EDIFACT (ORDERS/DESADV/INVOIC) over AS2 or SFTP; Peppol UBL 2.1 via access point	AS2 certificates per partner, SFTP key pairs, Peppol access-point credentials; partner profile binds to a service account with scoped duties	997/CONTRL functional acknowledgment within 15 min; inbound processing p95 < 5 min; partner SLA monitor with alerting	Mappings versioned per partner; replay from message monitor; partner simulator in sandbox
OPC-UA / MQTT	SCADA, PLCs, machines, sensors, historians, Andon displays	OPC-UA client (Sign & Encrypt) and MQTT 3.1.1/5 over TLS via the edge connector; events normalized to CloudEvents on Kafka	X.509 device certificates issued per plant; per-device MQTT credentials; connector runs under a plant service account	10k events/min per plant; store-and-forward on edge for 24 h; end-to-end p95 < 5 s to OEE/condition monitoring; no loss on broker restart	Tag/topic mapping is metadata; thresholds raise maintenance requests (wave 44)
File / SFTP	Banks, payroll providers, tax providers, legacy systems, lockbox	SFTP (ISO 20022 pain/camt, NACHA, BAI2, MT940, CSV/NDJSON), optional PGP	SFTP key authentication per partner, PGP keys and bank signing keys in Vault (never exported), checksum manifests	Polling every 5 min; inbound processing p95 < 15 min; outbound files signed and audited (E.7); bank acknowledgments (pain.002) ingested	Idempotent by file hash; duplicates rejected and logged; retention per fiscal rules

A customer therefore adds an MES by subscribing it to `passage.manufacturing.productionorder` and `passage.inventory.transaction`, pointing it at `/v1/manufacturing/production-orders/{id}:report` for confirmations, and issuing it a service account with the `ReportProduction` duty — the same three artifacts Passage's own shop-floor kiosk uses. Replacing that MES later means revoking one principal and one subscription, not unwinding a middleware project.

Part 12 — Coverage Inventory: Dynamics AX 2012 R3 / D365 F&SCM → Passage

Status	Draft v0.1 — completeness oracle for the 50-wave roadmap
Date	2026-10-21
Source of truth for waves	roadmap/waves.json (wave numbers 1–50); roadmap/phases.json
Scope rules	docs/PRD.md §2 goals and non-goals, §5 scope map
Purpose	The team's fear is "missing something". This part walks the AX 2012 R3 navigation pane module by module (plus the D365 F&SCM additions that matter to a manufacturer) and assigns every form, inquiry, report, periodic job and setup form to a Passage wave — or says, honestly, that nothing covers it yet.

12.0 How to read this inventory

Each module section mirrors the AX 2012 R3 menu structure and has four tables:

1. **Common / Journals** — every list page, details form and journal the user opens from *Common* or *Journals*. Columns: Form name | AOT form or menu item (best effort from memory of the AX 2012 R3 AOT; treat as a search key, not gospel) | Purpose | Passage status | Wave | Notes.
2. **Inquiries & Reports** — every *Inquiries* entry and every SSRS report under *Reports*. Columns: Name | Purpose | Passage status | Wave | Notes.
3. **Periodic** — every periodic job or process. Same columns as 2.
4. **Setup** — every setup form, with the module's *parameters* form listed first and its tabs spelled out as separate rows where the tab changes behavior that a wave must implement. Columns: Name | Key parameters / options | Passage status | Wave | Notes.

Status legend (the *Passage status* column always starts with one of these tokens so the document can be counted mechanically):

- `COVERED-<wave>` — a deliverable in waves.json for that wave clearly implements the AX capability, at parity or better.
- `PARTIAL-<wave> (what's missing)` — the wave covers the core but a named piece of the AX feature is not in any wave deliverable. The missing piece is listed in the gap register (§12.99) with a proposed wave.
- `GAP` — no wave in waves.json clearly covers the item. The Notes column always begins with `Propose wave NN:` so the gap register can be generated from the rows. Wave column shows `-`.
- `OUT-OF-SCOPE (reason)` — excluded by a PRD non-goal (CRM/marketing automation, retail POS/e-commerce, country packs beyond US/UK/DE/FR/IN/MX/CA, X++/ABAP code migration) or because AX itself deprecated it before R3. Wave column shows `-`.
- `SUPERSEDED (how)` — Passage deliberately does it a different way; the replacing mechanism is named. Wave column shows the wave that ships the replacement.

Conventions

- Wave numbers refer to `n` in waves.json. Phase boundaries: 0 = 1–5, 1 = 6–10, 2 = 11–15, 3 = 16–20, 4 = 21–25, 5 = 26–30, 6 = 31–35, 7 = 36–40, 8 = 41–45, 9 = 46–50.
- "SSRS report" rows are assessed against two things: the *data* (does a Passage wave produce it?) and the *rendering* (wave 46 replaces every SSRS layout with the operational report designer and embedded dashboards). A report is `COVERED-<n>` when the data wave exists and wave 46 renders it; it is `PARTIAL` only when the data itself is missing.
- Forms that are shared across modules (Terms of payment, Methods of payment, Charges codes, Print management, Form setup, Number sequences) are listed in every module where AX shows them, so each module section is self-contained for its reviewer. They map to the same wave each time.
- Where AX 2012 R3 and D365 F&SCM differ (e.g. Collections, Credit management, Cost accounting ledger, Landed cost), the D365 name is given in Notes and the row is assessed against the D365 behavior because that is what a manufacturer migrating today expects.
- Country-specific forms are listed only where they belong to one of the seven v1 country packs; everything else is `OUT-OF-SCOPE (country pack beyond v1 set)`.

12.1 General ledger

The general ledger is the heart of AX. In Passage it is the universal journal (wave 11) plus currencies/parallel ledgers (12), tax (13), period close (14) and financial reporting (15). Consolidation moves to wave 40; budgeting to wave 38.

12.1.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Main accounts (list page)	MainAccountListPage	Browse chart of accounts with balances FactBox	COVERED-11	11	Metadata-driven list page with balance projection FactBox
Main account details	MainAccount	Account type, category, DB/CR default, suspended, validation (currency/user/posting type), allocation terms, default dimensions, translations	COVERED-11	11	Account structure validation rules at posting API
Chart of accounts	LedgerChartOfAccounts	Shared COA header; main accounts belong to a COA shared by ledgers	COVERED-11	11	Shared COA across legal entities (tenant-level)
Ledger	Ledger	Per-legal-entity ledger: COA, fiscal calendar, accounting/reporting currency, exchange rate types, account structures, balance control	COVERED-11	11	Ledger service entity; parallel ledgers in 12
Financial dimensions	DimensionAttribute	Define dimensions backed by entities (dept, cost center, customer…) or custom lists	COVERED-11	11	Up to 20 dims per ledger
Financial dimension values	DimensionAttributeValue (DimensionValueDetails)	Values with active dates, suspended, owner, group dimension	COVERED-11	11	Effective-dated values
Financial dimension sets	DimensionFocus	Balance views by dimension set, rebuild balances	COVERED-11	11	Balance projection by account × dimension × period (event-sourced)
General journal (list)	LedgerJournalTable (menu LedgerJournalTable)	List of journal batches by name with status/approval	COVERED-11	11	Journal workspace

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
General journal lines	LedgerJournalTransDaily	Voucher lines, account/offset, dimensions, tax, invoice, due date, payment info	COVERED-11	11	
Journal voucher (lines) — Fixed assets journal	LedgerJournalTransAsset	FA transactions from GL journal	COVERED-37	37	FA transaction types posted through journal
Periodic journals	LedgerJournalPeriodic	Recurring journal templates with period unit, last date, end date	COVERED-14	14	Recurring journal scheduling
Allocation journal	LedgerJournalTable (type Allocation)	Journal created by allocation rule processing, review before post	COVERED-14	14	Allocation engine output
Elimination journal	LedgerJournalTable (type Elimination)	Journal created by elimination proposals in consolidation company	COVERED-40	40	
Intercompany journal	LedgerJournalTable (intercompany lines)	Lines posting due-to/due-from across legal entities	COVERED-40	40	Due-to/due-from entries wave 40; posting API wave 11
Global general journal (D365)	LedgerJournalTable (global)	Enter journals for any legal entity without company switch	COVERED-11	11	Shell re-scopes per legal entity; cross-entity journal entry in workspace
Journal approval (workflow)	LedgerJournalTable + workflow	Submit journal for approval	COVERED-11	11	Temporal approval template (wave 4) applied to journals in 11
Posted journals (list)	LedgerJournalTable (posted)	Review posted batches, reverse	COVERED-11	11	Reversal creates linked negating entry
Voucher attachments / document handling	DocuView	Attach documents to vouchers	PARTIAL-11 (no document types, archiving policy or external store yet)	11	Attachments named in wave 11; general document management is a gap under Org admin §12.29
Advanced ledger entries (public sector)	LedgerAdvancedEntry (PSA)	Multi-line entries with posting definitions	OUT-OF-SCOPE (public sector feature set is not in the PRD scope map)	—	Public sector is not a target vertical
Journal names list	LedgerJournalName	Also reachable from Common in some role centers	COVERED-11	11	See Setup
Reverse transaction (from voucher)	LedgerTransReverse	Reverse one voucher / trace reversal	COVERED-11	11	
Reverse entire journal	LedgerJournalTableReverse	Reverse all vouchers of a batch	COVERED-11	11	
Fiscal calendars (Common link in R3)	FiscalCalendars	Shared calendars, years, periods	COVERED-6	6	Owned by Org admin in Passage

12.1.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Trial balance (inquiry, dimension set)	Opening/debit/credit/closing by account and dimension set, date range	COVERED-15	15	Drill-down to lines
Ledger transactions (Voucher transactions)	Lines by voucher with origin trace	COVERED-11	11	Journal lines inquiry with source-document link
Audit trail	Who posted which voucher when, journal/source	COVERED-15	15	Audit trail report (wave 15) over CDC audit (wave 5)
Transaction origin	Drill from a GL line to originating subledger document	COVERED-15	15	Statement → balance → lines → source document
Posted sales tax (Sales tax transactions inquiry)	Tax amounts by code/period	COVERED-13	13	
Settled sales tax (sales tax payments inquiry)	Settlement history	COVERED-13	13	
Ledger budget transactions inquiry	Budget register entries by account	COVERED-38	38	
Currency exchange rate inquiry	Rates by type/pair/date	COVERED-12	12	Rate audit
Dimension statement (Dimension balance inquiry)	Balances for a single dimension value	COVERED-11	11	Balance projection query
Closing sheet (periodic, country-dependent)	Closing adjustments worksheet (used in ES/IT/EE)	OUT-OF-SCOPE (country pack beyond v1 set; closing adjustments in Passage are allocation/accrual journals)	—	
Ledger reconciliation inquiry (Inventory/ledger, AR/AP)	Compare subledger to GL	COVERED-20	20	Daily automated subledger-to-GL reconciliation from wave 16; inventory value report wave 20
SSRS: Trial balance	Printed trial balance	COVERED-15	15	Rendered via wave 46 designer
SSRS: Trial balance summary	Summary by account	COVERED-15	15	
SSRS: Ledger transaction list	Lines by date range	COVERED-11	11	
SSRS: Ledger account balance	Balance per account/period	COVERED-15	15	
SSRS: Dimension statement	Dimension-based P&L/balance	COVERED-15	15	
SSRS: Financial statement (old LedgerBalanceSheet row/column definitions)	Legacy AX statement engine	SUPERSEDED (report definitions in wave 15 designer, Management Reporter model)	15	AX 2012 had both; Passage keeps only the MR-style model
SSRS: Chart of accounts	List of accounts with attributes	COVERED-11	11	
SSRS: Journal (LedgerJournal report)	Unposted journal printout	COVERED-11	11	
SSRS: Posted journals	Posted batch printout	COVERED-11	11	
SSRS: Voucher (LedgerTransVoucher)	Single voucher with all lines	COVERED-11	11	
SSRS: Audit trail	Printed audit trail	COVERED-15	15	
SSRS: Ledger reconciliation (Ledger/Inventory, Ledger/AR, Ledger/AP, Ledger/Project, Ledger/Bank)	Reconciliation listings	COVERED-20	20	Reconciliation checks per subledger, wave 20 report; bank in 36; project in 41
SSRS: Sales tax payments	Settlement report	COVERED-13	13	
SSRS: Sales tax specification by ledger transaction	Tax by voucher	COVERED-13	13	
SSRS: Sales tax transactions — details	Tax detail listing	COVERED-13	13	
SSRS: Sales tax list / Sales tax codes	Setup listing	COVERED-13	13	
SSRS: EU sales list	Statutory EU sales listing	COVERED-13	13	

Name	Purpose	Passage status	Wave	Notes
SSRS: Exchange rate (Currency exchange rates)	Rate table printout	COVERED-12	12	
SSRS: Fiscal year close report / Opening balances	Closing results	COVERED-14	14	
SSRS: Consolidation (Consolidated balances)	Consolidated trial balance	COVERED-40	40	
SSRS: Eliminations	Elimination detail	COVERED-40	40	
SSRS: Allocation rule report	Allocation definitions	COVERED-14	14	
SSRS: Journal names, Posting setup listings	Setup listings	COVERED-11	11	Any entity can be exported/printed from the shell (wave 5)
SSRS: Cash flow forecast (GL cash flow)	Cash flow forecast from open items	COVERED-36	36	
SSRS: Balance control (balance sheet with balance control accounts)	Control accounts	COVERED-15	15	
SSRS: Fiscal LIFO reports (Italy)	Italian fiscal LIFO valuation	OUT-OF-SCOPE (country pack beyond v1 set)	—	
SSRS: Withholding tax reports	Withholding by vendor/period	COVERED-13	13	1099 in AP
Management Reporter / Financial reporting (D365)	Row/column/tree report designer and viewer	COVERED-15	15	

12.1.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Journals → Periodic journals (retrieve)	Generate journal lines from periodic journal template	COVERED-14	14	
Journals → Allocation journal (Process allocation request)	Run allocation rules, create allocation journal	COVERED-14	14	Fixed, basis, iterative
Journals → Accrual schemes (Ledger accruals)	Spread an expense/revenue across periods	COVERED-14	14	
Journals → Ledger accrual (from journal line)	Apply scheme on a journal line	COVERED-14	14	
Fiscal year close → Fiscal year close (year-end close)	Transfer P&L to retained earnings, opening balances	COVERED-14	14	Per-dimension carry-forward
Fiscal year close → Opening transactions	Create opening transactions per account/dimension	COVERED-14	14	
Fiscal year close → Transfer opening balances (D365 year-end close templates)	Templates for retained earnings by dimension	COVERED-14	14	
Consolidate → Consolidate [Online]	Pull subsidiary balances into consolidation company	COVERED-40	40	
Consolidate → Consolidate [Export] / [Import]	File-based consolidation	COVERED-40	40	Import path for non-Passage subsidiaries
Consolidate → Elimination proposal	Generate elimination journal	COVERED-40	40	
Currency revaluation (Foreign currency revaluation, GL)	Unrealized gain/loss on monetary accounts	COVERED-12	12	Reversal next period
Currency revaluation → Reporting currency adjustment (D365)	Restate reporting currency balances	COVERED-12	12	Currency translation
Transaction reversal (Reverse transaction / Trace reversal)	Reverse posted vouchers	COVERED-11	11	
Sales tax payment (Settle and post sales tax)	Settle tax period to authority	COVERED-13	13	
Sales tax → Sales tax report / EU sales list / VAT return	Statutory extracts	COVERED-13	13	VAT return, EU sales list, Intrastat, SAF-T
Conditional sales tax settlement	Settle tax when payment made (cash basis)	COVERED-13	13	Rules engine supports cash-basis trigger; verify fixture
Ledger balances rebuild (Financial dimension sets balance update)	Rebuild balance table	COVERED-11	11	Balance projection rebuild from journal
Close period (Ledger calendar period status change)	Open/on hold/closed per module	COVERED-14	14	Period control per ledger × module × user group
Financial period close workspace (D365)	Close checklist, tasks, dependencies	COVERED-14	14	
Delete / Compress transactions (Ledger transaction compression)	Compress old GL transactions	SUPERSEDED (append-only journal; history archived to lakehouse wave 46, never compressed in place)	46	
Reason code assignment (reversal reasons)	Require reasons on reversal	PARTIAL-11 (reason codes as a shared framework are not named in any wave)	11	See Org admin §12.29 reason codes
Intercompany journal post (periodic)	Post pending intercompany lines	COVERED-40	40	
Advanced ledger entries (public sector posting definitions)		OUT-OF-SCOPE (public sector)	—	
Fiscal LIFO calculation (IT)		OUT-OF-SCOPE (country pack beyond v1 set)	—	
Ledger budget: transfer budget to ledger (old budget model)		SUPERSEDED (budget register entries, wave 38)	38	
Financial reports generation (MR schedule)	Scheduled report packs	COVERED-15	15	Scheduled distribution PDF/XLSX/email
Export ledger (SAF-T, audit file)	Statutory audit file	COVERED-13	13	SAF-T extract; other country audit files in wave 48

12.1.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
General ledger parameters — Ledger tab	Check for voucher used, correction (storno), journal-based accrual, transaction reversal, batch transfer rules, use document date for voucher, balance control	COVERED-11	11	Storno (negative debit) handled as posting option
General ledger parameters — Sales tax tab	Apply sales tax taxation rules, cash discount on invoice incl. tax, tax on prepayment, rounding, conditional sales tax, sales tax settlement account, reverse charge	COVERED-13	13	
General ledger parameters — Inventory dimensions tab	Dimensions shown on ledger lines	COVERED-16	16	
General ledger parameters — Number sequences tab	Voucher sequences for journals, tax settlement	COVERED-5	5	Number sequence service
General ledger parameters — Batch transfer rules (D365 summarize subledger)	Asynchronous subledger transfer, summarize by account	SUPERSEDED (subledgers post synchronously via the universal journal API; summarization is a projection)	11	
Ledger	COA, calendar, currencies, account structures, balance control	COVERED-11	11	
Chart of accounts → Chart of accounts	Shared COA	COVERED-11	11	
Chart of accounts → Main accounts	Accounts	COVERED-11	11	
Chart of accounts → Main account categories	Categories for reporting/cubes	COVERED-15	15	Used by statement designer and wave 46 semantic layer
Chart of accounts → Account structures (Configure account structures)	Segments, allowed values, constraints, activation	COVERED-11	11	
Chart of accounts → Advanced rule structures / Advanced rules	Conditional additional segments	PARTIAL-11 (advanced rules (conditional extra segments) are not explicitly listed; account structures are)	11	Validation rules may need conditional branch
Chart of accounts → Account structure activation (periodic)	Activate draft structures	COVERED-11	11	
Chart of accounts → Financial dimensions		COVERED-11	11	
Chart of accounts → Financial dimension sets		COVERED-11	11	
Chart of accounts → Derived financial hierarchies (R3)	Category hierarchies used to derive dimension groupings for reporting	PARTIAL-15 (derived hierarchies for reporting are not named)	15	Report tree designer likely covers; confirm
Chart of accounts → Default dimension templates (D365)	Templates for default dimensions on master data	COVERED-11	11	Default dimensions on master data
Ledger calendar (Fiscal calendars + Ledger calendar)	Period statuses per module (open/on hold/closed), per user group access	COVERED-14	14	Calendar definition in 6; period control in 14
Fiscal calendars (Org admin shared)	Years, periods, closing periods	COVERED-6	6	
Journals → Journal names	Type, voucher series, approval workflow, posting restrictions, lines limit, fixed offset account, default currency/exchange rate type, private for user group	COVERED-11	11	
Journals → Journal texts	Reusable transaction text codes	COVERED-11	11	Journal templates
Journals → Journal controls (posting restrictions by account type/segment)	Restrict accounts per journal name	COVERED-11	11	
Journals → Default descriptions (Transaction text)	Auto text by transaction type with %1 placeholders	PARTIAL-11 (default description templates per posting type not named)	11	Small; put in wave 11 journal workspace
Journals → Accrual schemes	Ledger accrual schemes with debit/credit accounts, period key	COVERED-14	14	
Journals → Periodic journal names		COVERED-14	14	
Posting → Accounts for automatic transactions (System accounts)	Penny difference, exchange rate gain/loss, year-end result, error account, cash discount etc.	COVERED-11	11	
Posting → Posting definitions (public sector)		OUT-OF-SCOPE (public sector)	—	
Posting → Transaction posting definitions (PSA)		OUT-OF-SCOPE (public sector)	—	
Posting → Intercompany accounting	Due-to/due-from account pairs between legal entities, journal names	COVERED-40	40	
Posting → Reason codes (financial reasons)	Reasons for reversals, FA, customer/vendor transactions	PARTIAL-11 (shared reason code framework not named)	11	Gap register
Posting → Ledger posting groups (sales tax)	Tax payable/receivable/settlement accounts	COVERED-13	13	
Posting → Inventory posting (link)	Inventory posting profiles	COVERED-16	16	
Allocation → Ledger allocation rules	Fixed/basis/spread, source filters, destination, offset, reversal	COVERED-14	14	
Allocation → Allocation terms (on main account)	Fixed percentage allocation on account	COVERED-14	14	
Currency → Currencies	ISO code, rounding (price/unit/sales/purchase/general rounding rules), symbol	COVERED-6	6	ISO 4217 reference data
Currency → Currency exchange rates	Rates by type/pair/date, triangulation	COVERED-12	12	
Currency → Exchange rate types	Rate types (default, budget, closing)	COVERED-12	12	
Currency → Currency revaluation accounts	Realized/unrealized gain/loss by currency	COVERED-12	12	
Currency → Import currency exchange rates (provider)	Configure providers (ECB, OANDA) and schedule	COVERED-12	12	ECB/OpenExchangeRates adapters
Currency → Euro triangulation / Denomination currency	Legacy euro conversion	OUT-OF-SCOPE (obsolete; triangulation itself is in wave 12)	—	

Name	Key parameters / options	Passage status	Wave	Notes
Sales tax → Sales tax codes	Rate, limits, unit, calculation method (whole amount/interval), period, ledger posting group, use tax, exempt	COVERED-13	13	
Sales tax → Sales tax groups	Grouping by customer/vendor with exempt/reverse charge flags	COVERED-13	13	
Sales tax → Item sales tax groups	Grouping by item	COVERED-13	13	
Sales tax → Sales tax authorities	Authority vendor account, rounding	COVERED-13	13	
Sales tax → Sales tax settlement periods	Period intervals per authority, terms	COVERED-13	13	
Sales tax → Ledger posting groups		COVERED-13	13	
Sales tax → Sales tax exempt codes	Reason codes for exemption	COVERED-13	13	Exemption certificates
Sales tax → Sales tax jurisdictions (US)	Jurisdiction hierarchy, codes	COVERED-13	13	
Sales tax → Sales tax reporting codes	Codes for statutory return boxes	COVERED-13	13	VAT return box mapping
Sales tax → Sales tax (country-specific tab e.g. VAT registration numbers by country)		COVERED-7	7	Tax registration numbers on party
Sales tax → Withholding tax codes / groups	Withholding rates, item withholding groups (D365)	COVERED-13	13	
Sales tax → Tax branch (IN/other)		OUT-OF-SCOPE (country pack beyond v1 set)	—	India is in v1 pack set but tax branch is GST-era D365; handled in wave 48 India pack as needed
Sales tax → External (Sales tax, Avalara-type)	External provider connection	COVERED-13	13	
Sales tax → Conditional sales tax	Tax on payment	COVERED-13	13	
Sales tax → Sales tax on prepayment		COVERED-13	13	
Foreign trade → Intrastat parameters, commodity codes, transaction codes, transport methods, ports, statistics procedure	Intrastat master data	PARTIAL-13 (Intrastat extract is named; the setup forms (commodity codes, transaction codes, transport methods, ports, country properties) are not)	13	Gap register proposes wave 13
Foreign trade → EU sales list parameters / reporting codes		COVERED-13	13	
Consolidation → Consolidation accounts, Elimination rules, Consolidation groups	Account mapping, elimination rules, legal entity groups	COVERED-40	40	
Fiscal LIFO (IT)		OUT-OF-SCOPE (country pack beyond v1 set)	—	
Financial reporting → Report definitions (MR)	Rows, columns, trees, report	COVERED-15	15	
Financial reporting → Financial statements (legacy)	Row/column definitions	SUPERSEDED (wave 15 designer)	15	
Financial reporting → Financial reporting parameters (D365)	Integration settings	COVERED-15	15	
Budgeting (links)		COVERED-38	38	See §12.4
Ledger → Balance control accounts	Accounts for balance control on journal name	COVERED-11	11	
Ledger → Posting layers	Current/operations/tax layers	COVERED-12	12	Posting layers = ledger groups
Cash flow forecasts (GL setup: main accounts cash flow, dependent accounts)	Forecast settings per account	COVERED-36	36	
Organization → Legal entities (link)		COVERED-6	6	
Number sequences (GL)	Voucher series	COVERED-5	5	
Print management (GL)		COVERED-46	46	Document designer
Form setup (GL letters)		COVERED-46	46	
Workflow → General ledger workflows (journal approval, advanced ledger entry)	Workflow configuration per journal	COVERED-4	4	Workflow designer
Ledger → Closing sheets setup (country)		OUT-OF-SCOPE (country pack beyond v1 set)	—	
Tax (IN) → India-specific setup (tax components, TDS/TCS)		PARTIAL-48 (India pack listed for wave 48 but TDS/TCS/GST depth unspecified)	48	

12.2 Accounts payable

Accounts payable in AX spans vendor master (Passage wave 7), purchase orders (17), receiving and invoice matching (18), payments and settlement (19) and 1099/positive pay (19, 36). Vendor collaboration goes to wave 20.

12.2.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Vendors (list page)	VendTableListPage	All vendors with FactBoxes (balance, contacts, recent POs)	COVERED-7	7	Party model with Vendor role
Vendor details	VendTable	General, addresses, contacts, misc details, vendor profile, purchasing demographics, invoice and delivery, purchase order defaults, payment, tax 1099, financial dimensions	COVERED-7	7	Role-specific vendor data entity

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Vendors on hold	VendTable (Blocked = Invoice/All/Payment/Requisition/Never)	Hold statuses	COVERED-7	7	Hold types in vendor role
Vendor requests (R3)	VendRequest (VendRequestListPage)	New vendor / change request workflow (employee-initiated, unsolicited via EP)	PARTIAL-20 (vendor request workflow and prospective-vendor self-registration are not named; supplier portal is)	20	Gap register
Vendor bank accounts	VendBankAccount	Bank details per vendor, SEPA, prenote status	COVERED-19	19	
Purchase orders (list page)	PurchTableListPage	All POs	COVERED-17	17	
Purchase order details	PurchTable	Header/lines, charges, confirmation, change management	COVERED-17	17	
Purchase orders → Received but not invoiced	PurchTableListPage (filter)	POs awaiting invoice	COVERED-18	18	GR/IR reconciliation
Purchase orders → Open purchase order lines	PurchLine list	Open line view	COVERED-17	17	
Purchase orders → Backordered purchase lines	PurchLineBackOrder	Overdue deliveries	COVERED-17	17	
Purchase orders → Purchase order confirmations (list of journals)	PurchConfirmJournal (VendPurchOrderJour)	Confirmation history	COVERED-17	17	
Purchase orders → Product receipts (Packing slip journal)	VendPackingSlipJour	Posted receipts	COVERED-18	18	
Purchase orders → Prepayment invoices	PurchPrepayTable	Prepayment on PO	COVERED-19	19	Prepayments
Vendor invoices → Open vendor invoices	VendInvoiceJournal (CustVendInvoiceJour)	Posted invoices not yet paid	COVERED-19	19	Open-item AP
Vendor invoices → Pending vendor invoices	VendInvoiceInfoListPage (VendInvoiceInfoTable)	Unposted header/lines invoices with matching	COVERED-18	18	
Vendor invoices → Vendor invoice register (journal)	LedgerJournalTable (VendInvoiceRegister)	Register invoice to arrival account before approval	COVERED-18	18	Invoice capture (two-step register/approve)
Vendor invoices → Invoice approval journal	LedgerJournalTable (VendInvoiceApprove)	Approve registered invoices, assign ledger accounts	COVERED-18	18	
Vendor invoices → Invoice journal	LedgerJournalTable (VendInvoiceRecord)	Direct expense invoices (no PO)	COVERED-18	18	
Vendor invoices → Invoice pool excl. posting	VendInvoicePool (LedgerJournalPool)	Pending pool of invoices	COVERED-18	18	
Vendor invoices → Vendor invoice workflow (D365)	VendInvoiceInfoTable + workflow	Header/line approval	COVERED-18	18	Exception workflow
Vendor invoices → Invoice matching details	VendInvoiceMatchingDetails	Price/qty/total match status per line	COVERED-18	18	
Vendor invoices → Vendor invoice automation (D365 OCR / "Invoice capture")	VendInvoiceCapture	Capture, classify, auto-apply prepayment, submit	COVERED-18	18	OCR/AP-automation adapter; AI coding agent wave 47
Payments → Payment journal	LedgerJournalTable (VendPaymentJournal)	Payment lines, proposals, generate payments, settle	COVERED-19	19	
Payments → Promissory note journals (Draw / Redraw / Remittance / Settle)	LedgerJournalTable (VendPromissoryNote*)	Vendor bills of exchange	GAP	—	Propose wave 36: promissory notes / bills of exchange (vendor side) with remittance and settlement journals; not in any wave
Payments → Payment proposal	VendPaymProposal	Select by due date/discount date	COVERED-19	19	
Payments → Vendor payment advice (print)	VendPaymAdvice	Advice document	COVERED-19	19	
Payments → Positive pay (link)	BankPositivePayTable	Export issued checks	COVERED-19	19	
Payments → Prenotes (ACH)	VendPaymPrenote	Prenote generation and confirmation	PARTIAL-19 (ACH prenote (zero-dollar validation) flow is not named)	19	Gap register
Payments → Centralized payments (cross-company)	VendPaymentJournal (centralized)	Pay on behalf of other legal entities	PARTIAL-40 (centralized payments across entities are not named in 19 or 40)	40	Gap register
Vendor transactions	VendTrans	Posted vendor ledger	COVERED-19	19	
Vendor open transactions (Settle open transactions)	VendOpenTrans	Settlement, mark, cross-currency, cash discount	COVERED-19	19	
Closed transaction editing	VendTransClosed? (VendSettledTrans)	Reverse settlements	COVERED-19	19	Reversal handling
Vendor invoice (from PO) — Invoice posting form	PurchEditLines (VendInvoiceInfoTable)	Post PO invoice	COVERED-18	18	
Purchase order → Receipts list (Arrival overview)	WMSArrivalOverview	Inbound arrival planning	COVERED-18	18	Arrival/registration

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Purchase order → Registration	InventTransRegister	Register receipt by dimension/serial	COVERED-18	18	
Vendor → Consignment (D365 consignment replenishment orders)	VendConsignmentReplenishmentOrder	Vendor-owned stock, ownership change	GAP	—	Propose wave 16: consignment inventory (vendor-owned on-hand, ownership-change journal, consignment replenishment orders)
1099 → Vendor 1099 transactions	Vend1099Trans (Tax1099Summary)	Tax 1099 amounts per vendor	COVERED-13	13	1099 reporting hooks
Vendor → Vendor collaboration invoices (D365)	VendCollabInvoice	Vendor-submitted invoices	COVERED-20	20	Supplier portal invoice submission
Vendor → Vendor evaluation (R3 procurement)	VendEvaluation	Rate vendors on criteria	COVERED-33	33	Supplier quality scorecards
Purchase agreements (link)	PurchAgreement	Blanket agreements	COVERED-17	17	
Vendor → Charges transactions	MarkupTrans	Posted charges	COVERED-17	17	
Vendor → Deliveries (Vendor delivery schedule)	PurchLineDeliverySchedule	Delivery schedules on PO lines	COVERED-17	17	

12.2.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Vendor transactions inquiry	Posted transactions by vendor	COVERED-19	19	
Vendor balance (Vendor balance list)	Balances by vendor	COVERED-19	19	
Vendor aging (Aged balances inquiry)	Aging by period definition	COVERED-19	19	Vendor aging
Invoice history and matching details	Match status per invoice	COVERED-18	18	
Invoice journal inquiry / Invoice (Vendor invoice journal)	Posted invoice headers	COVERED-18	18	
Payment history / Payment inquiry	Payments by vendor	COVERED-19	19	
Settlement inquiry (Settled transactions)	Settlement pairs, cash discounts	COVERED-19	19	
Purchase order lines / Purchase order history	PO status history	COVERED-17	17	Versioned change management
Product receipt journal inquiry (Packing slip journal)	Receipts	COVERED-18	18	
Cash discount inquiry	Discounts taken/lost	COVERED-19	19	
Vendor invoice transactions (line-level)	Invoice lines	COVERED-18	18	
Journals → Confirmations / Receipts lists / Product receipts / Invoices (journal inquiries)	Posted document journals	COVERED-17	17	
Accrued purchases (Received not invoiced)	GR/IR accrual balance	COVERED-18	18	
Charges inquiry	Charges by PO	COVERED-17	17	
Withholding tax inquiry	Withholding by vendor	COVERED-13	13	
1099 → Tax 1099 summary / detail / state	1099 inquiries	COVERED-13	13	US country pack wave 48 produces IRS file formats
Vendor hold history		COVERED-7	7	Audit log
Positive pay inquiry	Exported checks	COVERED-19	19	
SSRS: Vendor aging (Vendor aging report)	Aged AP	COVERED-19	19	
SSRS: Vendor balance list with credit limit		COVERED-19	19	
SSRS: Vendor invoice (printed invoice copy)		COVERED-18	18	
SSRS: Vendor transactions report		COVERED-19	19	
SSRS: Vendor payment advice / Payment summary		COVERED-19	19	
SSRS: Check (BankChequePrint) / Check register	Check layout printing	COVERED-19	19	Check printing wave 19; check management wave 36
SSRS: Vendor 1099 (1099-MISC/INT/DIV/S/G forms and electronic file)	IRS forms and e-file	COVERED-48	48	US country pack; hooks in 13
SSRS: Vendor invoice specification / Invoice register journal		COVERED-18	18	
SSRS: Purchase order (document), Purchase order confirmation, Purchase inquiry (price inquiry)	PO documents	COVERED-17	17	Rendered wave 46
SSRS: Product receipt (packing slip) document		COVERED-18	18	
SSRS: Receipts list		COVERED-18	18	
SSRS: Vendor statement (Vendor account statement)		COVERED-19	19	
SSRS: Vendor base data / Vendor list		COVERED-7	7	
SSRS: Purchase order status / Purchase order overview / Backorder lines		COVERED-17	17	
SSRS: Purchase analysis (Purchase by vendor/item/period)	Statistics	COVERED-46	46	Semantic layer
SSRS: Vendor payment journal / Payment history		COVERED-19	19	
SSRS: Cash discount list / Terms of payment list	Setup listings	COVERED-7	7	
SSRS: Expected cash flow (vendor)		COVERED-36	36	
SSRS: Vendor posting profile / Vendor groups listing		COVERED-7	7	
SSRS: Withholding tax (US/other)		COVERED-13	13	
SSRS: Purchase order lines — Vendor delivery performance (On-time delivery)	Vendor OTD	COVERED-33	33	Supplier scorecards

Name	Purpose	Passage status	Wave	Notes
SSRS: Promissory note (vendor) documents		GAP	—	Propose wave 36: with promissory notes
SSRS: Vendor due date list (Open invoices by due date)		COVERED-19	19	
SSRS: Prepayment statement		COVERED-19	19	
SSRS: Vendor invoice totals (country)		OUT-OF-SCOPE (country pack beyond v1 set)	—	

12.2.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Payment proposal (batch)	Generate payment lines	COVERED-19	19	
Generate payments (payment media)	Produce ACH/SEPA/check/wire files	COVERED-19	19	pain.001, NACHA, check print
Payment file return import (pain.002)	Bank status ingestion	COVERED-19	19	
Positive pay export		COVERED-19	19	
Foreign currency revaluation (AP)	Revalue open vendor items	COVERED-12	12	
Settlement → Settle open transactions (batch) / Automatic settlement	Auto-settle by rules	COVERED-19	19	
Reverse settlement		COVERED-19	19	
Cash discount → Recalculate cash discount dates		COVERED-19	19	
Invoice → Post pending vendor invoices (batch)		COVERED-18	18	
Invoice → Vendor invoice policy violations (R3 audit)	Policy check batch	PARTIAL-48 (audit policies are not named; SoD engine is)	48	Gap register
Purchase order → Confirm / Receive / Invoice (batch posting)	Mass posting	COVERED-17	17	Batch job framework wave 4
Purchase order → Update → Delivery date / Finalize PO (close)		COVERED-17	17	
Purchase order → Purchase order confirmation requests (vendor collaboration)		COVERED-20	20	
Purchase order → Evaluated receipt settlement (ERS)	Self-billing invoices from receipts	COVERED-18	18	
Purchase order → Delete purchase orders / lines (clean up)	Housekeeping	COVERED-17	17	
Purchase order → Vendor delivery schedule update		COVERED-17	17	
1099 → Update 1099 (year-end) / Print 1099 / electronic filing		COVERED-48	48	US pack
Vendor → Vendor hold release (mass)		COVERED-7	7	
Vendor → Update vendor bank accounts status (prenote results)		PARTIAL-19 (prenote)	19	
Promissory notes → Draw / Redraw / Remit / Settle (periodic batches)		GAP	—	Propose wave 36: promissory notes
Clean up → Delete settled transactions / Compress vendor transactions		SUPERSEDED (append-only ledger; archive to lakehouse wave 46)	46	
Consignment → Consignment replenishment order processing		GAP	—	Propose wave 16: consignment
Invoice capture → Process OCR queue (D365)		COVERED-18	18	
Vendor collaboration → Provision users / invite (D365)		COVERED-20	20	
Purchase requisition → Release approved requisitions / Consolidate (procurement, link)		COVERED-17	17	

12.2.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Accounts payable parameters — General tab	Posting profile default, settlement priority, cash discount administration, credit note as correction, delivery address, intercompany	COVERED-19	19	
Accounts payable parameters — Updates tab	PO defaults (confirmation/receipt/invoice posting rules), safety levels on PO, enable change management, cancel/reverse rules, product receipt posting	COVERED-17	17	
Accounts payable parameters — Invoice tab	Invoice matching (2-way/3-way/none), post product receipt in ledger, post invoice automatically, approval workflow, invoice totals matching, charges matching, allow price tolerance, use invoice register, posting of invoice: tax, prepayment	COVERED-18	18	
Accounts payable parameters — Settlement tab	Automatic settlement, cash discount mark, exchange rate at settlement, reverse settlement options	COVERED-19	19	
Accounts payable parameters — Ledger and sales tax tab	Sales tax, posting type, financial dimension defaults	COVERED-13	13	
Accounts payable parameters — Credit rating / Blocking	Block types	COVERED-7	7	
Accounts payable parameters — Number sequences	Invoice, payment, PO sequences	COVERED-5	5	
Accounts payable parameters — Summary update	Consolidation of POs at invoicing by invoice account/order/automatic summary	COVERED-18	18	Summary invoice posting across POs
Accounts payable parameters — Prices (Price tolerance)	Default price tolerance percentage, charges tolerance	COVERED-18	18	
Accounts payable parameters — Vendor invoice workflow (D365 Invoice capture)		COVERED-18	18	
Vendors → Vendor groups	Group, default terms, default tax group	COVERED-7	7	
Vendors → Vendor posting profiles	Summary account, settle account, sales tax prepayment, arrival account per group/vendor/table	COVERED-7	7	Posting profile hooks in 7; accounts resolved in 11

Name	Key parameters / options	Passage status	Wave	Notes
Vendors → Vendor holds reasons (D365)		COVERED-7	7	
Vendors → Vendor request types (R3)	Request workflow types	PARTIAL-20 (vendor request workflow)	20	
Vendors → Vendor evaluation criteria / criterion groups (R3)	Scoring criteria	COVERED-33	33	
Vendors → Vendor profiles / Vendor user requests (EP)		SUPERSEDED (supplier portal wave 20 with Keycloak identities)	20	
Payment → Terms of payment	Net days, cash on delivery, current month, COD, payment schedule, grace	COVERED-7	7	
Payment → Cash discounts	Discount %, days, next discount code, ledger account	COVERED-7	7	Also in 19 settlement
Payment → Methods of payment (vendor)	Period, account type, file format (export), bridging posting, payment specification, remittance, bank transaction type, check/EFT/SEPA, positive pay format, validation	COVERED-19	19	
Payment → Payment specifications	Per method of payment code (e.g. SEPA purpose)	COVERED-19	19	
Payment → Payment fees	Fee per bank/method with ledger account	COVERED-19	19	
Payment → Payment days	Fixed days of week/month for payment date	COVERED-19	19	
Payment → Payment schedules	Installments (fixed amount/quantity/percentage)	COVERED-7	7	
Payment → Payment calendar (D365)	Bank holidays by country for due dates	PARTIAL-19 (payment calendar / bank holiday due-date shifting is not named)	19	Gap register
Payment → Promissory note setup (journal names, posting profiles, protest)		GAP	—	Propose wave 36: promissory notes
Payment → Positive pay formats	XSLT-defined bank formats	COVERED-19	19	Positive pay export
Payment → Payment purpose codes (country)		COVERED-19	19	SEPA purpose codes
Payment → Direct debit (vendor)		OUT-OF-SCOPE (vendor-side direct debit is country-specific; SEPA creditor side covered via customer direct debit in AR)	—	
Charges → Charges codes (purchase)	Debit/credit type (item/ledger/customer/vendor), tax group, posting	COVERED-17	17	
Charges → Auto charges	Automatic charges by account/item code, mode of delivery	COVERED-17	17	
Charges → Vendor charges groups / Item charges groups / Delivery charges groups		COVERED-17	17	
Charges → Charges allocation (to lines)	Allocate header charges to lines by net amount/quantity	COVERED-17	17	Landed cost in 34
Invoice matching → Price tolerances (item/vendor/group)	Tolerance % per item/vendor combination	COVERED-18	18	
Invoice matching → Charges tolerances		COVERED-18	18	
Invoice matching → Invoice totals tolerances (D365: Invoice totals matching)		COVERED-18	18	
Invoice matching → Matching policy by item/vendor	2-way/3-way per item/vendor	COVERED-18	18	
Invoice → Vendor invoice policies (R3 audit policy)	Policy rules for invoice audit (duplicates, missing receipts)	PARTIAL-48 (audit policies)	48	
Invoice → Invoice register setup (arrival accounts)	Via posting profile	COVERED-18	18	
Invoice → Vendor invoice automation parameters (D365)		COVERED-18	18	
Forms → Form setup (PO, receipt, invoice print options)	Document notes, print settings, number of copies	COVERED-46	46	
Forms → Form notes	Notes per document type	COVERED-46	46	
Forms → Print management	Conditional print destinations per document/vendor	PARTIAL-46 (print management conditional routing per account is not named; document designer is)	46	Gap register
Tax 1099 → 1099 fields / 1099 boxes / State 1099 / 1099 software vendor info	IRS box mapping	COVERED-48	48	US pack; hooks in 13
Distribution → Financial dimension defaults on vendor		COVERED-11	11	
Vendor bank → Bank transaction types, Bank parameters (link)		COVERED-36	36	
Trade agreements → Price/discount agreement journal names		COVERED-26	26	Pricing engine applies to purchase prices too (wave 17 uses it)
Trade agreements → Activate price/discount (Trade agreement activation)		COVERED-26	26	
Trade agreements → Price/discount groups (vendor)		COVERED-26	26	
Trade agreements → Purchase price tolerance		COVERED-18	18	
Purchase → Purchase order defaults (order type, charges)		COVERED-17	17	
Purchase → Purchase order change management (activation per vendor)		COVERED-17	17	
Purchase → Prepayment setup (prepayment posting profile)		COVERED-19	19	

Name	Key parameters / options	Passage status	Wave	Notes
External codes (vendor)	Partner codes for EDI	COVERED-35	35	Partner profiles
Vendor → Vendor item catalog (Vendor external item numbers)	Vendor item numbers, descriptions	COVERED-8	8	External item numbers on released product
Withholding tax (vendor group assignment)		COVERED-13	13	
Vendor → Purchase order approval policies (link to Procurement)		COVERED-17	17	
Vendor → Vendor consignment setup (D365 consignment parameters, ownership change journal names)		GAP	—	Propose wave 16: consignment
Workflow → Accounts payable workflows (vendor invoice, PO, vendor change)		COVERED-4	4	
Number sequences (AP)		COVERED-5	5	
Vendor → Vendor collaboration parameters (D365)		COVERED-20	20	
Payment → Electronic payment file formats (AIF-based/ER)		COVERED-19	19	ISO 20022 generator; ER designer in 48 for exotic formats
Payment → Check layout (bank account)	Check printing layout per bank account	COVERED-19	19	Check layout in document designer; bank check management in 36

12.3 Accounts receivable

Accounts receivable covers customer master (wave 7), free-text and sales invoices (29), payments, settlement, collections, interest and lockbox (30), credit management (27) and commissions (26). AX 2012 R3 also places the sales order list here; it is assessed in §12.17.

12.3.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Customers (list page)	CustTableListPage	All customers with balance/aging FactBoxes	COVERED-7	7	Party model with Customer role
Customer details	CustTable	General, addresses, contacts, misc details, sales demographics, credit and collections, invoice and delivery, sales order defaults, payment defaults, financial dimensions, retail	COVERED-7	7	
Customer → Credit limit / credit rating / mandatory credit limit	CustTable (credit tab)	Limit, rating, mandatory check	COVERED-27	27	Credit management
Customer bank accounts	CustBankAccount	Bank details for direct debit/refunds	COVERED-30	30	
Customer → Direct debit mandates (SEPA, R3)	CustDirectDebitMandate	Mandate lifecycle, sequence type	PARTIAL-30 (SEPA direct debit mandates (pain.008) are not named; collections and cash application are)	30	Gap register
Sales orders (list page, also in AR)	SalesTableListPage		COVERED-27	27	See §12.17
Free text invoices (list page)	CustFreeInvoiceListPage	Non-item invoices with ledger lines	COVERED-29	29	
Free text invoice details	CustFreeInvoice	Lines with ledger account, tax, dimensions, charges	COVERED-29	29	
Recurring free text invoices (R3 templates)	CustInvoiceTemplate / CustRecurrenceInvoice	Template-based recurring invoices	COVERED-29	29	Subscription/recurring schedules
Customer invoices → Open customer invoices	CustInvoiceJournal	Posted invoices open	COVERED-30	30	
Customer invoices → Invoice journal (customer)	LedgerJournalTable (CustInvoiceJournal)	Ledger-based customer invoices	COVERED-29	29	
Customer invoices → Sales invoice journal (posted)	CustInvoiceJour	Posted sales invoices	COVERED-29	29	
Payments → Payment journal	LedgerJournalTable (CustPaymentJournal)	Customer receipts, settlement, deposit slips	COVERED-30	30	Cash application
Payments → Lockbox transactions (R3 US)	CustLockbox / CustPaymLockbox	Import BAI2 lockbox files, match, create payment journal	COVERED-30	30	
Payments → Bill of exchange journals (Draw / Redraw / Protest / Remittance / Settle)	LedgerJournalTable (CustBillOfExchange*)	Customer bills of exchange	GAP	—	Propose wave 36: bills of exchange (customer) with remittance for collection/discount, protest, settle; not in any wave
Payments → Deposit slips	BankDeposit	Group receipts into bank deposit	COVERED-36	36	
Payments → Payment advice (customer)	CustPaymAdvice		COVERED-30	30	
Payments → Credit card payments (Payment services)	CreditCardAuthTrans	Authorize/capture via payment service	COVERED-30	30	Stripe/Adyen adapter
Payments → Postdated checks (R3 APAC)	CustPostDatedCheck	Register, mature, settle PDCs	GAP	—	Propose wave 36: postdated checks (customer and vendor) as a treasury instrument
Collections → Collections (list page, R3 collections workspace)	CustCollectionsListPage	Aging, activities, promises, cases per customer	COVERED-30	30	Collections workspace
Collections → Collections (customer detail page)	CustCollections	Transactions, activities, promise to pay, dispute, write-off, NSF	COVERED-30	30	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Collections → Collection letter notes (journal)	CustCollectionLetterJour	Generated collection letters	COVERED-30	30	Dunning levels
Collections → Interest notes (journal)	CustInterestJour	Interest invoices	COVERED-30	30	
Collections → Customer pools	CustCollectionsPool	Query-based customer groupings for collectors	COVERED-30	30	
Collections → Collection agents	CustCollectionsAgent	Agent assignment	COVERED-30	30	
Collections → Aging snapshot	CustAgingLegalEntity (CustAgingSnapshot)	Snapshot of aging for workspace	COVERED-30	30	Aging computed live from open items; snapshot for KPI history in 46
Collections → Cases (collection cases)	CaseDetail	Case per customer dispute	PARTIAL-30 (disputes workflow named; generic case management not)	30	See §12.29 case management gap
Collections → Promise to pay / NSF / Write-off actions	CustCollections (actions)		COVERED-30	30	
Customer transactions	CustTrans	Posted customer ledger	COVERED-30	30	
Customer open transactions (Settle open transactions)	CustOpenTrans	Settlement with marking, cross-currency, discounts	COVERED-30	30	
Closed transaction editing (customer)	CustSettledTrans	Reverse settlement	COVERED-30	30	
Customer → Interest adjustments	CustInterestAdj	Adjust interest notes	COVERED-30	30	
Customer → Trade agreements (price/discount agreement journals)	PriceDiscAdmTable	Customer pricing journals	COVERED-26	26	
Customer → Charges transactions	MarkupTrans		COVERED-26	26	
Customer → Prepayments (sales order prepayment)	SalesPrepayment (via payment journal)	Customer prepayment and application	COVERED-30	30	
Customer → Credit management (D365 add-in): Credit holds list, Credit limit adjustments journal, Blocking rules	CustCreditManagement*	Credit holds, temporary limits, rules	COVERED-27	27	Credit holds, limits, release workflow
Customer → Customer self-service (D365 customer portal)			COVERED-30	30	Customer portal
Returns → Return orders (link)	ReturnTable	RMA	COVERED-30	30	
Commission transactions	CommissionTrans	Posted commission	PARTIAL-26 (commission calculation hooks are named, but commission posting to vendor/employee and transactions are not)	26	Gap register
Customer → Customer invoice (packing slip based) posting — Summary update	SalesEditLines	Consolidated invoicing	COVERED-29	29	
Customer → Global customer (cross-company)	CustTable (global)		COVERED-7	7	Party shared across entities
Customer → Rebates (Customer rebate agreements — TAM link)	TAMRebateAgreement		COVERED-26	26	Rebate accrual 26, settlement 30

12.3.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Customer transactions inquiry		COVERED-30	30	
Customer balance list / Customer balance		COVERED-30	30	
Aged balances (customer aging inquiry)		COVERED-30	30	
Invoice journal / Invoice lines inquiry	Posted invoices	COVERED-29	29	
Payment history / Payment inquiry		COVERED-30	30	
Settlement inquiry (Settled transactions)		COVERED-30	30	
Interest note inquiry		COVERED-30	30	
Collection letter inquiry		COVERED-30	30	
Credit card transactions inquiry		COVERED-30	30	
Lockbox import history		COVERED-30	30	
Sales order inquiries (Sales orders, Backorders, Open order lines, Delivered not invoiced)		COVERED-27	27	
Journals → Confirmations / Picking lists / Packing slips / Invoices	Posted document journals	COVERED-28	28	Invoices in 29
Commission inquiry		PARTIAL-26 (commission posting)	26	
Credit management inquiries (D365: credit holds history, credit limit history)		COVERED-27	27	
Customer hold / on-hold history		COVERED-7	7	
Direct debit mandate inquiry		PARTIAL-30 (direct debit)	30	
Bill of exchange inquiry		GAP	—	Propose wave 36: bills of exchange
Postdated check inquiry		GAP	—	Propose wave 36: postdated checks
Free text invoice inquiry		COVERED-29	29	

Name	Purpose	Passage status	Wave	Notes
Customer self-service inquiries (portal)		COVERED-30	30	
SSRS: Customer aging (Aged balances report)		COVERED-30	30	
SSRS: Customer balance list / with credit limit		COVERED-30	30	
SSRS: Customer account statement (external / internal)	Statements	COVERED-30	30	Customer statements
SSRS: Customer transactions		COVERED-30	30	
SSRS: Invoice (sales invoice document) / Free text invoice document / Invoice specification	Customer-facing documents	COVERED-29	29	Rendered wave 46; e-invoice 35
SSRS: Collection letter note document		COVERED-30	30	
SSRS: Interest note document		COVERED-30	30	
SSRS: Customer payment advice / Payment summary		COVERED-30	30	
SSRS: Deposit slip		COVERED-36	36	
SSRS: Customer due date list (Open invoices by due date)		COVERED-30	30	
SSRS: Expected cash flow (customer)		COVERED-36	36	
SSRS: Customer base data / Customer list		COVERED-7	7	
SSRS: Sales order confirmation / Picking list / Packing slip (documents)		COVERED-28	28	Confirmation in 27
SSRS: Sales analysis (Sales by customer/item/period, Top customers)		COVERED-46	46	Semantic layer
SSRS: Commission report / Commission transactions		PARTIAL-26 (commission posting)	26	
SSRS: Cash discount list / Terms of payment / Posting profiles listings		COVERED-7	7	
SSRS: Sales tax reporting (customer)		COVERED-13	13	
SSRS: Bill of exchange documents / Remittance list		GAP	—	Propose wave 36: bills of exchange
SSRS: Credit card payment report		COVERED-30	30	
SSRS: Customer invoice totals (country-specific)		OUT-OF-SCOPE (country pack beyond v1 set)	—	
SSRS: Lockbox report		COVERED-30	30	
SSRS: Collections (Collections activities, Promises)		COVERED-30	30	
SSRS: Customer/customer group statistics		COVERED-46	46	
SSRS: Prepayment statement (customer)		COVERED-30	30	
SSRS: External codes listing		COVERED-35	35	

12.3.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Collections → Create collection letters	Generate letters by sequence/level	COVERED-30	30	Dunning run
Collections → Print/post collection letters		COVERED-30	30	
Collections → Interest calculation (Create interest notes)	Calculate interest on overdue	COVERED-30	30	
Collections → Print/post interest notes		COVERED-30	30	
Collections → Age customer balances (aging snapshot)		COVERED-30	30	
Collections → Customer pool refresh		COVERED-30	30	
Settlement → Settle open transactions (batch) / Automatic settlement		COVERED-30	30	
Foreign currency revaluation (AR)		COVERED-12	12	
Payments → Lockbox import (batch)		COVERED-30	30	
Payments → Payment file import (customer payments / camt.054)		COVERED-30	30	camt.053/054 auto-match
Payments → Generate direct debit (pain.008)	SEPA direct debit collection	PARTIAL-30 (direct debit)	30	
Payments → Bill of exchange remittance/protest batches		GAP	—	Propose wave 36: bills of exchange
Payments → Credit card settlement/capture (payment services)	Capture authorized cards on invoice	COVERED-30	30	
Sales update → Confirmation / Picking list / Packing slip / Invoice (batch)	Mass posting	COVERED-28	28	Invoice 29
Sales update → Summary update (consolidated invoicing)		COVERED-29	29	
Sales update → Recurring free text invoices (generate)		COVERED-29	29	
Sales update → Reservation (batch) / Explode BOM		COVERED-16	16	
Sales update → Delete orders (clean up)		COVERED-27	27	
Sales update → Update credit holds / Credit management periodic (D365)		COVERED-27	27	
Customer statements (periodic generate)		COVERED-30	30	
Clean up → Delete settled transactions / Compress customer transactions		SUPERSEDED (append-only; archive to lakehouse)	46	
Commission → Calculate/post commissions (via invoice)		PARTIAL-26 (commission posting)	26	
Credit card → Authorization expiry cleanup		COVERED-30	30	
Rebates → Process rebates / Cumulate / Approve / Pay (TAM link)		COVERED-30	30	Rebate settlement
Customer → Customer payment predictions (D365 Finance insights)	ML prediction of payment date	COVERED-47	47	Collections outreach agent; prediction model under wave 47
Customer → Update customer bank account status		COVERED-30	30	

Name	Purpose	Passage status	Wave	Notes
Customer → Send e-invoices (ER)		COVERED-35	35	Peppol
Interest → Interest adjustment batch		COVERED-30	30	

12.3.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Accounts receivable parameters — General tab	Posting profile, settlement priority, delivery address, credit rating, mandatory credit limit, credit limit type (none/balance/balance+packing slip/all), message when exceeding, intercompany	COVERED-27	27	
Accounts receivable parameters — Updates tab	Posting defaults for confirmation/picking/packing slip/invoice, safety levels, cancel options, picking list posting, packing slip in ledger, delivery date control defaults	COVERED-28	28	
Accounts receivable parameters — Settlement tab	Automatic settlement, cash discount rules, exchange rate at settlement, settlement of overpayments	COVERED-30	30	
Accounts receivable parameters — Collections tab	Collection letter from (per customer/transaction), interest calculation basis, collection agents, aging period definition	COVERED-30	30	
Accounts receivable parameters — Credit rating / Credit management (D365)	Credit management automation	COVERED-27	27	
Accounts receivable parameters — Ledger and sales tax tab		COVERED-13	13	
Accounts receivable parameters — Prices tab	Price/discount search defaults, total discount calculation, smart rounding	COVERED-26	26	
Accounts receivable parameters — Summary update tab	Summary update by invoice account/order/automatic/none, summary update parameters	COVERED-29	29	
Accounts receivable parameters — Shipments tab (AR delivery)	Output order, shipment defaults	SUPERSEDED (wave 28 fulfillment / wave 31 WHS model replaces WMS-I shipments)	28	
Accounts receivable parameters — Number sequences		COVERED-5	5	
Accounts receivable parameters — Payment services (credit card)	Processor connection, authorization expiry	COVERED-30	30	
Accounts receivable parameters — Sales order holds	Default hold behaviors	COVERED-27	27	
Customers → Customer groups		COVERED-7	7	
Customers → Customer posting profiles	Summary account, settle account, collection letter sequence, interest code, write-off account	COVERED-7	7	
Customers → Customer hold reasons / On-hold statuses (Invoice/All/Payment/Requisition)		COVERED-7	7	
Customers → Customer self-service parameters (EP)		SUPERSEDED (customer portal wave 30)	30	
Payment → Terms of payment		COVERED-7	7	
Payment → Cash discounts		COVERED-7	7	
Payment → Methods of payment (customer)	Period, account type, file format (import), bridging, lockbox, credit card, direct debit, payment specification	COVERED-30	30	
Payment → Payment specifications / Payment fees / Payment days / Payment schedules		COVERED-30	30	Schedules in 7
Payment → Payment calendar (D365)		PARTIAL-19 (payment calendar)	19	
Payment → Lockbox setup (lockbox, lockbox file format, customer MICR numbers)	BAI2 format mapping	COVERED-30	30	
Payment → Direct debit mandate setup (SEPA creditor identifier)		PARTIAL-30 (direct debit)	30	
Payment → Bill of exchange setup (journal names, posting profiles, protest reasons, remittance types)		GAP	—	Propose wave 36: bills of exchange
Payment → Postdated checks setup (R3 APAC)		GAP	—	Propose wave 36: postdated checks
Payment → Credit card → Payment services / Credit card types / Credit card processors	Authorize.net-style connector	COVERED-30	30	
Payment → Prepayment posting profile		COVERED-30	30	
Collections → Collection letter sequences	Levels, days, fees, letter text, block/hold actions	COVERED-30	30	
Collections → Interest codes	Rate by period, grace, min/max, ranges, calculation by date/days	COVERED-30	30	
Collections → Aging period definitions	Buckets by day/month/period	COVERED-30	30	
Collections → Customer pools		COVERED-30	30	
Collections → Collection agents / Collections team (R3 team)		COVERED-30	30	
Collections → Collections email templates (D365)		PARTIAL-30 (email templates for dunning/collections communication not named; letters are)	30	Gap register (email framework, Org admin)
Collections → Write-off reason codes / Write-off account		COVERED-30	30	
Collections → NSF information (bank → NSF fees)		COVERED-36	36	
Credit management (D365) → Credit management parameters, Blocking rules, Scoring groups, Risk groups, Credit limit adjustment journal names, Guarantees/insurance, Credit management groups	Rule-based holds and scoring	COVERED-27	27	Aging-based rules, risk scoring
Charges → Charges codes (sales) / Auto charges / Customer charges groups / Item charges groups / Delivery charges groups		COVERED-26	26	Charges are a pricing condition
Commission → Commission calculation / Commission sales groups / Commission customer groups / Commission item groups / Commission posting		PARTIAL-26 (commission posting and sales group/rep assignment)	26	

Name	Key parameters / options	Passage status	Wave	Notes
Forms → Form setup (sales documents) / Form notes / Print management		PARTIAL-46 (print management routing)	46	
Forms → Form sorting (sort lines on documents)		COVERED-46	46	Document designer
Trade agreements → Price/discount agreement journal names / Activate / Price/discount groups (customer, line, multiline, total)		COVERED-26	26	
Trade agreements → Smart rounding (Round-off) rules	Psychological price rounding	COVERED-26	26	
Trade agreements → Supplementary items	Free/mandatory add-on items	PARTIAL-26 (supplementary items are not named in pricing or order waves)	27	Gap register
Sales → Order types / Order hold codes / Order event setup		COVERED-27	27	
Sales → Delivery (Modes of delivery, Terms of delivery, Delivery schedule defaults, Destination codes)		COVERED-28	28	Terms of delivery in 7
Sales → Sales tax exemption numbers on customer		COVERED-13	13	
Sales → Customer item numbers (External item descriptions)		COVERED-8	8	
Sales → Sales reasons / Reason codes (customer)		PARTIAL-11 (reason codes)	11	
Sales → Sales statistics / Statistics groups		COVERED-46	46	
Sales → Customer classification groups / Line of business / Segments / Subsegments		COVERED-7	7	Customer attributes
Sales → Sales districts / Sales groups (sales reps)		PARTIAL-26 (sales groups/reps for commission)	26	
Sales → Campaign/prospect setup (link to Sales & marketing)		OUT-OF-SCOPE (CRM non-goal)	—	
External codes (customer)		COVERED-35	35	
Workflow → Accounts receivable workflows (free text invoice, credit limit, customer change)		COVERED-4	4	
Statement → Customer statement setup (statement layout, statement intervals)		COVERED-30	30	
Interest → Interest adjustment reasons		COVERED-30	30	
Number sequences (AR)		COVERED-5	5	
Rebates → Rebate agreements (TAM) / Rebate program types / Rebate groups		COVERED-26	26	See §12.18
Returns → Return reason codes / Return reason code groups / Disposition codes / Disposition actions (link to Sales & marketing)		COVERED-30	30	
E-invoice setup (Electronic invoice parameters, EDI)		COVERED-35	35	
Customer → Customer tax registration / VAT numbers		COVERED-7	7	

12.4 Budgeting

AX 2012 R2/R3 budgeting = budget register entries, budget control, budget planning (R2+). All of it lands in wave 38.

12.4.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Budget register entries (list page)	BudgetTransactionListPage	All register entries by code/model	COVERED-38	38	
Budget register entry details	BudgetTransaction	Header/lines, budget type (original, transfer, revision, encumbrance, pre-encumbrance, carry-forward), allocation, workflow	COVERED-38	38	
Budget plans (list page, R2+)	BudgetPlanListPage	Budget planning documents by process/scenario/org	COVERED-38	38	
Budget plan details	BudgetPlan	Lines per scenario, worksheets (Excel), justification, workflow stages	COVERED-38	38	Top-down/bottom-up
Budget control statistics	BudgetControlStatistics	Budget vs actual vs encumbrance by dimension	COVERED-38	38	
Budget plan → Budget plan templates / worksheets	BudgetPlanTemplate	Excel-based planning worksheets	PARTIAL-38 (Excel worksheet round-trip for budget plans is not named; planning workflow and scenarios are)	38	Gap register
Budget account entries (fixed assets budgets link)	AssetBudget	Capex budgets from FA	COVERED-37	37	Asset budgeting
Budget → Reservations for encumbrances (inquiry)	BudgetSourceTracking	Encumbrance tracking per source document	COVERED-38	38	Commitments

12.4.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Budget control statistics inquiry	Budget funds available	COVERED-38	38	
Actual vs budget inquiry		COVERED-38	38	
Budget register entry inquiry / Budget balances		COVERED-38	38	
Budget plan inquiries (Budget plan lines, Budget plan amounts by scenario)		COVERED-38	38	
Budget check results (Budget check errors/warnings)	Per source document	COVERED-38	38	
Budget control activity (source document encumbrance)		COVERED-38	38	
SSRS: Budget funds available		COVERED-38	38	
SSRS: Actual versus budget		COVERED-38	38	

Name	Purpose	Passage status	Wave	Notes
SSRS: Budget control activity		COVERED-38	38	
SSRS: Budget register entries / Budget details		COVERED-38	38	
SSRS: Budget plan (printed)		COVERED-38	38	
SSRS: Budget transactions by model		COVERED-38	38	
SSRS: Ledger budget (legacy LedgerBudget)		SUPERSEDED (register entries)	38	

12.4.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Budget transfers (process transfer register entries)		COVERED-38	38	
Budget → Recalculate budget control balances	Rebuild available balances	COVERED-38	38	
Budget → Carry forward budget / Year-end budget processing		COVERED-38	38	
Budget planning → Generate budget plan from source (GL actuals, previous plan, FA, projects, demand forecast, HR positions)		PARTIAL-38 (generating plans from position/HR forecasts and demand forecasts is not named)	38	Gap register
Budget planning → Budget plan workflow stage advance (batch)		COVERED-38	38	
Budget planning → Generate budget register entry from budget plan		COVERED-38	38	
Budget → Allocate budget (allocation terms / period allocation keys / dimension allocation)		COVERED-38	38	
Budget → Import/export budget register entries (Excel/DIXF)		COVERED-5	5	Excel import/export
Budget → Delete budget balances / purge		COVERED-38	38	
Budget control → Activate/deactivate configuration		COVERED-38	38	
Budget → Encumbrance year-end process (public sector)		OUT-OF-SCOPE (public sector)	—	

12.4.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Budgeting parameters	Default budget model, budget transfer journal, number sequences, budget balances update	COVERED-38	38	
Basic budgeting → Budget models	Models with submodels, stop/blocked	COVERED-38	38	Budget versions
Basic budgeting → Budget codes	Codes per budget type with workflow	COVERED-38	38	
Basic budgeting → Budget cycles / Budget cycle time spans	Fiscal cycle mapping	COVERED-38	38	
Basic budgeting → Budget transfer rules	Allowed transfers between dimensions	COVERED-38	38	
Basic budgeting → Budget allocation terms	Allocation by dimension percent	COVERED-38	38	
Basic budgeting → Period allocation keys	Spread across periods	COVERED-38	38	
Basic budgeting → Dimensions for budgeting	Which financial dimensions are budgeted	COVERED-38	38	
Basic budgeting → Budget register entry workflow		COVERED-38	38	
Budget control → Budget control configuration (Define parameters, Budget funds available calculation, Documents and journals, Assign budget models, Define budget control rules, Budget groups, Over budget permissions, Activate)	Full budget control configuration	COVERED-38	38	Budget checks with tolerances and overrides
Budget control → Budget control rules / Budget groups		COVERED-38	38	
Budget control → User groups with override permissions		COVERED-38	38	
Budget planning → Budget planning configuration (Scenarios, Stages, Workflows, Stage rules, Budget plan priorities, Templates, Columns, Layouts, Parameters)		COVERED-38	38	
Budget planning → Budget planning processes	Process per cycle × ledger × org hierarchy	COVERED-38	38	
Budget planning → Budget planning templates (Excel)		PARTIAL-38 (Excel worksheets)	38	
Budget planning → Budget plan document layouts (D365)		COVERED-38	38	
Budget planning → Position forecasting (D365 HR-driven)	Forecast positions/compensation into plan	PARTIAL-38 (position forecasting)	38	Depends on HR wave 42; sequencing note in gap register
Budget planning → Budget planning allocation schedules		COVERED-38	38	
Fixed asset budgets (link)		COVERED-37	37	
Project budgets (link)		COVERED-41	41	
Workflow → Budgeting workflows		COVERED-4	4	
Number sequences (budgeting)		COVERED-5	5	

12.5 Cash and bank management

Everything lands in wave 36 except payment generation (19) and customer receipts (30). Bills of exchange / promissory notes and postdated checks are gaps (proposed wave 36).

12.5.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Bank accounts (list page)	BankAccountTableListPage	All bank accounts	COVERED-36	36	
Bank account details	BankAccountTable	Account, currency, main account, check setup (layout, numbering), positive pay, SEPA/IBAN, bank groups, reconciliation settings, signatories	COVERED-36	36	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Bank transactions (per account)	BankAccountTrans	Posted bank transactions	COVERED-36	36	
Bank statements (list page, Advanced bank reconciliation R2+)	BankStatementListPage	Imported statements (camt.053, BAI2, MT940)	COVERED-36	36	
Bank statement details	BankStatement	Lines, status, validation	COVERED-36	36	
Bank reconciliation worksheets (Advanced)	BankReconciliationWorksheet	Match statement to book, rules, reconcile	COVERED-36	36	
Account reconciliation (classic bank reconciliation)	BankReconciliation	Manual tick-off reconciliation	SUPERSEDED (advanced reconciliation with rules and ML suggestions, wave 36)	36	Classic mode retained as manual matching within the worksheet
Checks (list)	BankChequeTable	Issued checks, void, reprint, status	COVERED-36	36	Check management
Deposit slips	BankDeposit	Deposit of customer receipts	COVERED-36	36	
Letters of credit (Import/Export letters of credit, R2+)	BankLC* (BankLCImport, BankLCExport)	LC lifecycle, shipments, margins, charges	COVERED-36	36	
Letters of guarantee	BankLG*	Guarantee issue, extend, cancel, liquidate	COVERED-36	36	
Bank facilities (Facility agreements)	BankFacilityAgreement	Facility limits per bank, types	COVERED-36	36	
Bank → Positive pay transactions	BankPositivePayTable	Export status	COVERED-19	19	
Bank → Cash flow forecast (D365 Cash flow forecasting workspace)	CashFlowForecast*	Forecast from AP/AR/budget/projects	COVERED-36	36	
Cash management (D365 Cash positioning)		Intraday positions	COVERED-36	36	camt.052
Bank → Bank document (LC/LG) transactions	BankDocument*		COVERED-36	36	
Bank → Bills of exchange / Promissory notes (links)			GAP	—	Propose wave 36: bills of exchange and promissory notes (see AR/AP)
Bank → Postdated checks (links)			GAP	—	Propose wave 36: postdated checks
Bank → Payment journal (bank-originated)	LedgerJournalTable (bank)		COVERED-36	36	
Bank → Bank account signatories (D365)	BankAccountSignatory	Signers and approvals	COVERED-36	36	
Intercompany netting / cash pooling (in-house banking)	— (not in AX; SAP IHC)		COVERED-36	36	In-house banking basics

12.5.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Bank transactions inquiry		COVERED-36	36	
Bank balance inquiry / Bank account balance by date		COVERED-36	36	
Check inquiry / Check register		COVERED-36	36	
Reconciliation inquiry / Reconciliation history		COVERED-36	36	
Bank statement inquiry (imported statements)		COVERED-36	36	
Deposit slip inquiry		COVERED-36	36	
Letter of credit / guarantee inquiries		COVERED-36	36	
Cash flow forecast inquiry (Cash flow forecasts by period)		COVERED-36	36	
Bank facility usage inquiry		COVERED-36	36	
Positive pay inquiry		COVERED-19	19	
SSRS: Bank transactions / Bank account statement (internal)		COVERED-36	36	
SSRS: Check register / Checks report / Void checks		COVERED-36	36	
SSRS: Reconciliation summary / Bank reconciliation report		COVERED-36	36	
SSRS: Deposit slip (document)		COVERED-36	36	
SSRS: Cash flow forecast report		COVERED-36	36	
SSRS: Bank account list / Bank groups		COVERED-36	36	
SSRS: Payment reconciliation (bank)		COVERED-36	36	
SSRS: Letter of credit / guarantee documents		COVERED-36	36	
SSRS: Unreconciled transactions		COVERED-36	36	
SSRS: Bank transaction types list		COVERED-36	36	

12.5.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Import bank statements (batch)	camt.053/BAI2/MT940 import	COVERED-36	36	
Reconcile bank statements (auto-match batch)		COVERED-36	36	
Cash flow forecast → Calculate / Update forecast		COVERED-36	36	
Positive pay → Generate positive pay file		COVERED-19	19	

Name	Purpose	Passage status	Wave	Notes
Checks → Void / Reprint / Delete checks		COVERED-36	36	
Bank → Foreign currency revaluation (bank accounts)		COVERED-12	12	
Bank → Transfer bank accounts (bank transfer journal)	Account-to-account transfer	COVERED-36	36	
Letters of credit → Expiry / margin update batches		COVERED-36	36	
Bank → Delete reconciliation history / Compress		SUPERSEDED (append-only)	46	
Bank → Post bank fees / NSF		COVERED-36	36	
Bank connectivity → Send/receive files (SFTP/EBICS/host-to-host)		COVERED-36	36	Vault-managed keys
Cash positioning → Intraday statement import (camt.052)		COVERED-36	36	

12.5.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Cash and bank management parameters — General	Allow checks for bank accounts, check number method, prenotes, bank statement reconciliation defaults, advanced reconciliation enabled, letter of credit/guarantee settings	COVERED-36	36	
Cash and bank management parameters — Number sequences	Check, deposit, statement sequences	COVERED-5	5	
Bank groups	Bank master (routing, SWIFT, address)	COVERED-36	36	
Bank transaction types	Classify transactions (fee, interest, check) with GL accounts	COVERED-36	36	
Bank transaction groups	Grouping for reconciliation	COVERED-36	36	
Check layout (per bank account)	Layout, MICR, signature, copies, check form	COVERED-19	19	Check printing; management 36
Bank statement format (Advanced bank reconciliation: Bank statement formats, Transformation (XSLT), Reconciliation matching rules, Matching rule sets)	Import formats and rules	COVERED-36	36	Standard ISO 20022/BAI2/MT940 parsers; custom via ER wave 48
Bank reconciliation matching rules / rule sets	Auto-match criteria, tolerance, grouping	COVERED-36	36	
Positive pay formats		COVERED-19	19	
Letters of credit → Bank documents facility types, LC types (import/export), LC charges, Document types		COVERED-36	36	
Letters of guarantee → Guarantee types, Letters of guarantee parameters		COVERED-36	36	
Bank facilities → Facility agreements, Facility groups, Facility types		COVERED-36	36	
Cash flow forecasting setup (Cash flow forecast setup D365: AP/AR/budget/ledger/inventory/project/sales/purchase/ external forecasts; dependent accounts; liquidity accounts)		COVERED-36	36	
Payment purpose codes / Bank codes (country)		COVERED-36	36	
Bank account → Prenote setup		PARTIAL-19 (prenote)	19	
Bank → Bank holidays / Payment calendar		PARTIAL-19 (payment calendar)	19	
Bills of exchange / promissory notes setup (bank side)		GAP	—	Propose wave 36: bills of exchange
Postdated checks parameters (R3 APAC)		GAP	—	Propose wave 36: postdated checks
Bank connectivity (D365 Bank connectivity, Electronic payment setup)	Host-to-host, SWIFT, EBICS	COVERED-36	36	
Workflow → Bank workflows (bank account change approval, signatory)		COVERED-36	36	
Number sequences (bank)		COVERED-5	5	
Credit card → Payment services (link)		COVERED-30	30	
NSF information (bank fees for returned payments)		COVERED-36	36	

12.6 Fixed assets

The AX fixed asset module (asset groups, books/value models, depreciation profiles, journals, mass update, physical inventory) maps to wave 37; leases (D365 Asset leasing) are also wave 37.

12.6.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Fixed assets (list page)	AssetTableListPage	All assets	COVERED-37	37	
Fixed asset details	AssetTable	General, technical info (serial, make/model), location, responsible worker, insurance, lease, sale/scrap, value models/books, financial dimensions	COVERED-37	37	
Value models (AX 2012) / Books (D365) per asset	AssetBook (AssetBookTable)	Depreciation profile, service life, posting layer, status, acquisition info	COVERED-37	37	Multi-book
Depreciation books (tax, AX 2012)	AssetDepBookTable	Non-ledger tax books	COVERED-37	37	Tax books merged into "books" per D365
Fixed asset journal (GL journal, type fixed asset)	LedgerJournalTable (AssetJournal)	Acquisition, acquisition adjustment, depreciation, depreciation adjustment, revaluation, write-up/down, disposal sale/scrap, extraordinary depreciation, bonus depreciation	COVERED-37	37	Full transaction set
Fixed asset budget journal	AssetBudgetJournal	Budget entries for assets	COVERED-37	37	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Depreciation book journal (tax)	AssetDepBookJournal	Non-posting tax depreciation	COVERED-37	37	
Fixed asset transactions	AssetTrans	Posted asset ledger	COVERED-37	37	
Fixed asset → Physical inventory (D365 asset count)	AssetPhysicalInventory	Barcode count and reconciliation	COVERED-37	37	
Fixed asset → Lease (D365 Asset leasing: Lease, Lease book, Payment schedule, Lease classification)	AssetLease*	Lease records with classification tests	COVERED-37	37	IFRS 16 / ASC 842
Fixed asset → Asset leasing journals (lease payment, interest, depreciation)	AssetLeaseJournal		COVERED-37	37	
Fixed asset → Asset acquisition from PO/AP invoice (Fixed asset on purchase line)	PurchLine (asset)	Create/acquire asset from purchase	COVERED-37	37	Acquisition from AP invoice
Fixed asset → Acquisition from project (Estimate posting to FA)	ProjTable (asset)	CIP to asset	COVERED-37	37	CIP from projects
Fixed asset → Acquisition from inventory (Inventory to fixed asset journal)	InventJournalTable (asset)	Issue inventory to asset	COVERED-37	37	
Fixed asset → Split / Transfer (Mass transfer, Transfer fixed asset)	AssetTransfer	Transfer between dimensions/locations, split	COVERED-37	37	
Fixed asset → Reclassify (change group)	AssetReclassify		COVERED-37	37	
Fixed asset → Fixed asset additions	AssetAddition	Sub-components/additions tracking	PARTIAL-37 (asset additions (non-financial component records) not named)	37	Gap register
Fixed asset → Maintenance / Service (asset maintenance records)	AssetMaintenance		COVERED-44	44	EAM
Fixed asset → Insurance records	AssetInsurance		PARTIAL-37 (insurance records not named)	37	Gap register
Fixed asset → Asset statement (D365 Fixed asset roll forward)	AssetRollForward		COVERED-37	37	

12.6.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Fixed asset transactions inquiry		COVERED-37	37	
Fixed asset balances inquiry (Fixed asset balances by book)	NBV per book/period	COVERED-37	37	
Depreciation profile preview (Profile inquiry)	Projected depreciation	COVERED-37	37	
Fixed asset budget inquiry		COVERED-37	37	
Value model / book inquiry		COVERED-37	37	
Lease inquiries (lease liability, ROU schedules)		COVERED-37	37	
SSRS: Fixed asset list / Fixed asset base data		COVERED-37	37	
SSRS: Fixed asset transactions report		COVERED-37	37	
SSRS: Fixed asset balances / Fixed asset note (statement)		COVERED-37	37	
SSRS: Fixed asset roll forward		COVERED-37	37	
SSRS: Fixed asset depreciation forecast		COVERED-37	37	
SSRS: Fixed assets due for replacement		COVERED-37	37	
SSRS: Fixed asset insurance report		PARTIAL-37 (insurance)	37	
SSRS: Fixed asset acquisitions / disposals / Scrapped assets		COVERED-37	37	
SSRS: Fixed asset listing by location / responsible		COVERED-37	37	
SSRS: Lease reports (liability amortization, ROU, maturity analysis, disclosure)		COVERED-37	37	
SSRS: Bonus depreciation report (US)		COVERED-37	37	
SSRS: Fixed asset physical inventory report		COVERED-37	37	
SSRS: Fixed asset budget report		COVERED-37	37	
SSRS: Reconciliation Fixed assets/Ledger		COVERED-37	37	Subledger reconciles to GL

12.6.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Depreciation proposal (create depreciation journal)	Depreciation run per book/period	COVERED-37	37	
Depreciation book → Depreciation proposal (tax book)		COVERED-37	37	
Mass update (change depreciation profile/service life for many assets)		COVERED-37	37	
Mass transfer		COVERED-37	37	
Revaluation of fixed asset groups (Consumption/Revaluation batch)		COVERED-37	37	
Copy fixed asset / Copy value model		COVERED-37	37	
Fixed asset budget → Transfer to budget register		COVERED-38	38	
Fixed asset → Periodic depreciation (batch)		COVERED-37	37	
Consumption depreciation → Consumption proposal (units of production)		COVERED-37	37	
Asset leasing → Batch journal creation (payments, interest, amortization) / Lease impairment / Lease modification		COVERED-37	37	
Fixed asset → Year-end / Close value models		COVERED-37	37	
Fixed asset → Delete/compress transactions		SUPERSEDED (append-only)	46	
Fixed asset → Physical inventory count import (barcode)		COVERED-37	37	

Name	Purpose	Passage status	Wave	Notes
Fixed asset → Import assets (DIXF/Excel)		COVERED-10	10	

12.6.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Fixed assets parameters — General	Automatically create depreciation adjustments, minimum depreciation, consider prior-year depreciation, bonus depreciation, consumption depreciation, capitalization threshold, delete asset on journal post, allow multiple acquisitions, low-value pool (AU), default value models	COVERED-37	37	AU low-value pool is OOS country
Fixed assets parameters — Purchase orders	Allow asset acquisition from PO, create asset during receipt/invoice, check capitalization threshold	COVERED-37	37	
Fixed assets parameters — Number sequences		COVERED-5	5	
Fixed asset groups	Group defaults: value models, number sequence, depreciation profiles, service life, posting profile defaults	COVERED-37	37	
Fixed asset posting profiles	Ledger accounts per transaction type × group/asset × book	COVERED-37	37	
Value models (AX) / Books (D365)	Posting layer, calendar, depreciation profile, alternative profile, round-off, derived value models/books (auto-post to multiple)	COVERED-37	37	
Depreciation books (tax)	Non-posting books	COVERED-37	37	
Depreciation profiles	Methods: straight line service life, straight line life remaining, reducing balance, manual, factor, consumption, 200%/175%/150%/125% reducing, manual, RB/SL (US), Lo-Value pool; conventions (half year, mid month, mid quarter, full month, half-year start of life); period frequency	COVERED-37	37	MACRS tables
Depreciation profiles → Manual depreciation schedules (percentages per period)		COVERED-37	37	
Consumption factors / Consumption units	Units-of-production setup	COVERED-37	37	
Depreciation conventions		COVERED-37	37	
Fixed asset → Special depreciation allowance (bonus depreciation, US)		COVERED-37	37	
Fixed asset → Journal names (fixed asset journal, budget journal, depreciation book journal)		COVERED-37	37	
Fixed asset → Location / Major types / Condition / Asset attributes	Reference lists	COVERED-37	37	
Fixed asset → Insurance types / Insurance companies		PARTIAL-37 (insurance)	37	
Fixed asset → Reason codes (fixed assets)		PARTIAL-11 (reason codes)	11	
Fixed asset → Transaction reversal setup		COVERED-37	37	
Fixed asset → Fixed asset activity codes		COVERED-37	37	
Fixed asset → Sorting / Fixed asset groups → Value model defaults		COVERED-37	37	
Fixed asset → Derived value models / derived books	Post to multiple books from one journal	COVERED-37	37	
Asset leasing → Asset leasing parameters, Lease books, Lease groups, Payment frequencies, Classification criteria (ownership transfer, purchase option, lease term %, PV %), Journal names, Posting accounts		COVERED-37	37	
Fixed asset → Physical inventory parameters (barcode, worker)		COVERED-37	37	
Workflow → Fixed asset workflows (journal approval)		COVERED-4	4	
Number sequences (fixed assets)		COVERED-5	5	
Fixed asset → Country-specific (RU/PL/CZ depreciation, DE "AfA tables")		OUT-OF-SCOPE (country pack beyond v1 set; DE AfA handled in DE pack wave 48 if required)	—	

12.7 Cost accounting

AX 2012 R3 ships the legacy *Cost accounting* module (COS: dimensions, cost categories, cost lines, service categories, allocations, budget cost). D365 replaced it with the *Cost accounting ledger* (cost objects, cost elements, allocation policies, statistical measures, cost control workspace). Passage builds the D365 model in wave 38; every legacy COS form is therefore `SUPERSEDED` by its wave-38 equivalent.

12.7.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Cost accounting dimensions (legacy)	COSDimension	Cost center/cost unit dimensions	SUPERSEDED (cost object dimensions and hierarchies, wave 38)	38	
Cost categories (legacy COS)	COSCostCategory	Cost types mapped from ledger accounts	SUPERSEDED (cost elements, wave 38)	38	
Cost lines (legacy, manual cost postings)	COSLine	Manual cost entries	SUPERSEDED (cost accounting ledger journals, wave 38)	38	
Cost journal (legacy COS journal)	COSJournalTable	Manual cost accounting journal	SUPERSEDED (cost entries journal, wave 38)	38	
Service categories / Service quantities (legacy)	COSServiceCategory	Activity quantities for allocation	SUPERSEDED (statistical measures, wave 38)	38	
Cost distribution (legacy)	COSAllocation (distribution)	Distribute costs by key	SUPERSEDED (allocation policies: distribution, wave 38)	38	
Cost allocation (legacy)	COSAllocation	Allocate costs via allocation key	SUPERSEDED (allocation policies: assessment, wave 38)	38	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Cost budget (legacy COS budget)	COSBudget	Budget by cost category/dimension	SUPERSEDED (budget data in cost accounting ledger from budget register entries, wave 38)	38	
Cost accounting ledger (D365)	CAMCostAccountingLedger	Ledger per fiscal calendar/currency with GL/budget source	COVERED-38	38	
Cost objects / Cost object dimensions (D365)	CAMDimension	Cost center, product, project cost objects	COVERED-38	38	
Cost elements / Cost element dimensions (D365)	CAMCostElement	Primary (from GL) and secondary cost elements	COVERED-38	38	
Dimension hierarchies (D365 cost accounting)	CAMDimensionHierarchy	Hierarchies for allocation/reporting	COVERED-38	38	
Cost entries (D365)	CAMCostEntry	Processed source entries	COVERED-38	38	
Cost control workspace (D365)	CAMCostControlWorkspace	Manager cost reporting with variance and drill	COVERED-38	38	
Statistical measures / Statistical entries (D365)	CAMStatisticalMeasure	Quantities (headcount, kWh) used as allocation bases	COVERED-38	38	
Cost behavior / Cost classification (fixed/variable) (D365)	CAMCostBehavior	Classify fixed vs variable	PARTIAL-38 (cost behavior (fixed/variable split) policies not named)	38	Gap register
Internal orders (SAP concept; AX via projects)	—		COVERED-38	38	Internal orders named in wave 38
Activity-based costing	— (not in AX)		COVERED-38	38	

12.7.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Cost transactions inquiry (legacy)		SUPERSEDED (cost entries, wave 38)	38	
Cost balances by dimension (legacy)		SUPERSEDED (cost control workspace)	38	
Cost accounting ledger → Cost entries inquiry (D365)		COVERED-38	38	
Cost control → Cost by cost object/element with budget variance (D365)		COVERED-38	38	
Allocation results inquiry (D365: Allocation journal)		COVERED-38	38	
SSRS: Cost accounting (legacy) — Cost distribution sheet, Cost unit statement, Cost category list, Cost allocation list		SUPERSEDED (cost control reports, wave 38/46)	38	
D365: Cost control workspace reports / Power BI cost accounting analysis		COVERED-46	46	Management cubes
SSRS: Service category quantities		SUPERSEDED (statistical measures)	38	

12.7.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Transfer ledger transactions to cost accounting (legacy)		SUPERSEDED (source data processing from GL, wave 38)	38	
Cost distribution / Cost allocation processing (legacy)		SUPERSEDED (overhead calculation, wave 38)	38	
Cost budget transfer (legacy)		SUPERSEDED	38	
Source data processing (D365: GL, budget, statistical)	Pull entries into cost ledger	COVERED-38	38	
Overhead calculation (D365: cost behavior, cost distribution, overhead rate, cost allocation)	Run allocation policies in sequence	COVERED-38	38	Allocation cycles converge
Cost accounting ledger → Reverse overhead calculation		COVERED-38	38	
Statistical measure data import		COVERED-38	38	
Cost ledger period close		COVERED-38	38	

12.7.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Cost accounting parameters (legacy)	Dimension mapping, journal names	SUPERSEDED (cost accounting ledger setup)	38	
Cost accounting dimensions / Cost category groups / Allocation keys / Calculation versions (legacy)		SUPERSEDED	38	
D365: Cost accounting ledgers	Calendar, currency, exchange rate type	COVERED-38	38	
D365: Dimensions (cost element, cost object, statistical) and Dimension member data providers (GL main accounts, financial dimensions, budget)		COVERED-38	38	
D365: Dimension hierarchies		COVERED-38	38	
D365: Data source providers / Source data (GL actual, budget, budget register)		COVERED-38	38	
D365: Policies — Cost behavior policies, Cost distribution policies, Overhead rate policies, Cost allocation policies, Cost rollup policies (D365 2020+)		PARTIAL-38 (cost behavior and cost rollup policies not named; allocation policies are)	38	
D365: Policy assignments / Policy versions		COVERED-38	38	
D365: Cost control workspace configurations		COVERED-38	38	
D365: Cost accounting ledger security (dimension access)		COVERED-2	2	OPA policies
D365: Statistical measure providers		COVERED-38	38	
D365: Cost accounting parameters		COVERED-38	38	
Number sequences (cost accounting)		COVERED-5	5	

12.8 Cost management (D365)

D365's *Cost management* module consolidates inventory accounting (AX Inventory management → Setup → Costing), costing versions, costing sheets and the inventory value reports. Passage splits it: inventory costing and inventory close in wave 20, product costing/cost roll-up/costing sheets/variances in wave 39, landed cost in wave 34.

12.8.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Costing versions (Costing version setup)	CostingVersion	Standard/planned cost versions, blocking, fallback, site	COVERED-20	20	Costing versions in 20; BOM calc in 39
Item prices (Costing version maintenance: pending/active prices)	InventItemPrice (CostingVersionMaintenance)	Pending cost, activation, cost breakdown by cost group	COVERED-39	39	
Cost object (D365 inventory accounting cost objects)	CostObject	Item × dimension cost object with running average	COVERED-20	20	
Inventory value / Cost explorer	CostExplorer (InventCostTransExplorer)	Trace cost from issue to receipts	COVERED-20	20	Settlement trace
Closing and adjustment (Inventory close list)	InventClosing	Close runs, recalculation, adjustments (on-hand, transactions)	COVERED-20	20	
Inventory posting (link)	InventPosting		COVERED-16	16	
Production → Standard cost transactions (variances)	InventCostTrans	Purchase/production variances	COVERED-39	39	
Cost management → Inventory accounting journals (Moving average revaluation, Standard cost revaluation, Inventory cost revaluation)	InventJournalTable (revaluation)	Revalue on-hand	COVERED-39	39	Standard cost revaluation in 39; moving average settlement in 20
Landed cost (D365 Landed cost module: Voyages, Shipping containers, Folios, Cost estimates, Goods in transit)	ITMVoyage*	Multi-leg inbound costing with estimates and actuals	PARTIAL-34 (landed cost allocation is named; D365 voyage/container/folio tracking and goods-in-transit orders are not)	34	Gap register
Global inventory accounting (D365 add-in: multiple cost ledgers / accounting conventions)	GIA ledgers	Parallel inventory valuation (IFRS vs local)	GAP	—	Propose wave 20: parallel inventory valuation ledgers (one costing method per ledger group) aligned with wave 12 parallel ledgers

12.8.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Inventory value report (configurable: Inventory value report by resource/resource group, by dimension)	On-hand value vs GL by date	COVERED-20	20	
Inventory value report storage / configuration (D365)	Persisted report runs	COVERED-20	20	
Inventory aging report (D365)	Aging of on-hand by receipt date	PARTIAL-20 (inventory aging buckets not named)	20	Gap register
Cost rollup inquiry / Calculated cost breakdown (BOM calculation results)		COVERED-39	39	
Costing version comparison / Compare item prices		COVERED-39	39	Cost estimate comparison
Inventory close inquiry / Settlements / Adjustment history		COVERED-20	20	
Standard cost variance inquiry (purchase price variance, production variances)		COVERED-39	39	
SSRS: Inventory value (Physical inventory by inventory dimension, Inventory value by dimension)		COVERED-20	20	
SSRS: Inventory cost/value reconciliation (Inventory/Ledger reconciliation)		COVERED-20	20	
SSRS: Cost estimates and costing (BOM calculation report)		COVERED-39	39	
SSRS: Indirect cost transactions / Costing sheet listing		COVERED-39	39	
SSRS: Variance analysis statement (production)		COVERED-39	39	
SSRS: Settled transactions / Inventory close report		COVERED-20	20	
SSRS: Items without cost		COVERED-39	39	
Landed cost reports (voyage cost, cost variance by voyage)		PARTIAL-34 (landed cost voyages)	34	

12.8.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Closing and adjustment → Close inventory	Settle issues vs receipts per model	COVERED-20	20	
Closing and adjustment → Recalculation	Interim recalculation	COVERED-20	20	
Closing and adjustment → Adjustment (on-hand / transactions)	Manual value adjustments	COVERED-20	20	
Closing and adjustment → Cancel close / Reopen		COVERED-20	20	
Costing version → Calculation (BOM calculation, mass)	Roll up costs	COVERED-39	39	Mass roll 50k items < 15 min
Costing version → Activate pending prices (mass activation)	Activate standard costs; revalue inventory	COVERED-39	39	
Costing version → Copy item prices between versions		COVERED-39	39	
Inventory accounting → Post inventory to ledger (transfer)		SUPERSEDED (synchronous posting via universal journal)	16	

Name	Purpose	Passage status	Wave	Notes
Moving average → Revaluation for moving average		COVERED-20	20	
Standard cost → Revaluation on activation / Conversion to standard cost (Standard cost conversion checklist)	Convert items from actual to standard	PARTIAL-39 (standard cost conversion checklist (switching costing model mid-life) not named)	39	Gap register
Landed cost → Voyage cost allocation / Goods in transit receipts		PARTIAL-34 (landed cost voyages)	34	
Cost management → Inventory value report generation (batch)		COVERED-20	20	
Costing → Delete pending prices / Clean up costing versions		COVERED-39	39	
Inventory cost (D365) → Update running average / Refresh on-hand cost		COVERED-20	20	

12.8.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Inventory and warehouse management parameters — Costing/ Inventory accounting tab (D365 Cost management parameters)	Post physical inventory, post financial inventory, break-even cost, use latest cost price, variance by cost group	COVERED-20	20	
Costing versions	Type (standard/planned), block activation/changes, fallback (active/none/version), site/date filters, explosion mode	COVERED-39	39	
Cost groups	Direct material, direct manufacturing, indirect, undefined; cost group type and behavior (fixed/variable)	COVERED-39	39	
Costing sheets (Costing sheet setup)	Nodes: totals, price nodes (cost groups), surcharge/rate indirect costs with absorption basis	COVERED-39	39	
Indirect cost (surcharge / rate) definitions and versions	Rates per costing version/site	COVERED-39	39	
Cost categories (routing costs; shared with Production)	Cost category price per version, cost group	COVERED-39	39	
BOM calculation groups	Cost price model, sales price model, warnings, stop explosion	COVERED-39	39	
Cost basis types (D365 commodity pricing)		COVERED-26	26	Commodity price based on index; see PI §12.23
Inventory posting profiles (link)		COVERED-16	16	
Item model groups → Costing method (FIFO/LIFO/LIFO date/weighted avg/weighted avg date/standard/moving average), include physical value, fixed receipt price, latest cost price		COVERED-20	20	
Inventory value report configuration (D365: Inventory value reports setup — columns, dimensions, resource type)		COVERED-20	20	
Inventory aging report setup (D365)		PARTIAL-20 (inventory aging)	20	
Landed cost setup (D365: Landed cost parameters, Cost types, Cost type codes, Auto cost, Apportionment, Shipping container types, Vessels, Voyage templates, Legs, Journeys, Tracking)		PARTIAL-34 (landed cost voyages)	34	
Global inventory accounting setup (ledgers, accounting policies, conventions)		GAP	—	Propose wave 20: parallel inventory valuation ledgers
Standard cost conversion setup		PARTIAL-39 (standard cost conversion)	39	
Number sequences (cost management)		COVERED-5	5	

12.9 Compliance and internal controls

AX 2012's *Compliance and internal controls* module holds internal control documentation (process library/files), the risk library, audit policies (R3 audit workbench) and the database log link. Environmental sustainability was removed before R3. Passage: SoD and SOC 2 evidence in wave 48; audit policies are a gap proposed for wave 48; database log is wave 5.

12.9.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Internal controls → Files (process documents)	SysComplianceFiles (EP)	Document library for control documentation	PARTIAL-48 (SOC 2 evidence automation is named; a controls/process document library is not)	48	Gap register
Internal controls → Process library / Process documents	SysComplianceProcess	Hierarchical process documentation with risk/control links	PARTIAL-48 (process library)	48	
Internal controls → Risk library	SysComplianceRisk	Risks with controls	PARTIAL-48 (risk library)	48	
Audit → Audit cases	AuditCase (CaseDetail audit)	Cases from audit policy violations	PARTIAL-48 (audit policies)	48	
Audit → Audit policy violations list	AuditPolicyViolation	Violations by policy rule	PARTIAL-48 (audit policies)	48	
Database log (link to Sys admin)	SysDatabaseLog		COVERED-5	5	Audit log from CDC
Environmental sustainability (substance flows, process maps)	ENV*		OUT-OF-SCOPE (removed from AX 2012 R3; v2 ESG module candidate)	—	
Compliance → Segregation of duties conflicts (link to Sys admin)	SysSecSegregationOfDutiesConflict		COVERED-48	48	SoD engine

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Compliance → Electronic signatures (link)	SysSignature		GAP	—	Propose wave 48: electronic signatures (21 CFR Part 11 style) on configured tables/fields with reason codes and signature log

12.9.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Audit policy violations inquiry		PARTIAL-48 (audit policies)	48	
Audit case inquiry		PARTIAL-48 (audit policies)	48	
Database log inquiry		COVERED-5	5	
Electronic signature log inquiry		GAP	—	Propose wave 48: electronic signatures
Segregation of duties conflicts / unresolved conflicts		COVERED-48	48	
SSRS: Audit policy report / Audit case report		PARTIAL-48 (audit policies)	48	
SSRS: Database log report		COVERED-5	5	
SSRS: Segregation of duties rules / conflicts		COVERED-48	48	
SSRS: Electronic signature report		GAP	—	Propose wave 48: electronic signatures
SSRS: Internal control documents		PARTIAL-48 (process library)	48	
SOC 2 control evidence reports (Passage)		COVERED-48	48	

12.9.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Audit → Run audit policies (batch: expense reports, vendor invoices, purchase orders, GL journals)	Rule-based post-hoc audit (duplicates, keywords, sampling, list search)	PARTIAL-48 (audit policies)	48	
Audit → Close/archive audit cases		PARTIAL-48 (audit policies)	48	
Database log → Clean up database log		SUPERSEDED (immutable audit store with retention policy wave 48)	48	
SoD → Verify compliance of user-role assignments (batch)		COVERED-48	48	
Access certification campaigns (Passage)		COVERED-48	48	
Electronic signature → Reset/revoke signatures		GAP	—	Propose wave 48: electronic signatures

12.9.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Compliance parameters	Portal site, file locations	SUPERSEDED (no Enterprise Portal; compliance workspace wave 48)	48	
Audit → Audit policies (policy framework: policy rule types — expense/vendor invoice/PO/GL journal; rule types: duplicate, keyword, list search, sampling, premium/mileage)		PARTIAL-48 (audit policies)	48	
Audit → Audit policy rule types		PARTIAL-48 (audit policies)	48	
Audit → Audit case categories		PARTIAL-48 (audit policies)	48	
Audit → Audit keyword lists / Audit sampling parameters		PARTIAL-48 (audit policies)	48	
Database log setup (tables/fields to log)		COVERED-5	5	Field-level audit flag in metadata
Electronic signature → Electronic signature parameters, Signature requirements (table/field), Reason codes for signatures		GAP	—	Propose wave 48: electronic signatures
Segregation of duties rules (link)		COVERED-48	48	
Risk/control types, Process document templates		PARTIAL-48 (process library)	48	
Workflow → Audit case workflow		COVERED-4	4	

12.10 Travel and expense

Expense management (expense reports, cash advances, travel requisitions, policies, per diems, credit card import, project integration) lands in wave 43; approval infrastructure in wave 4; VAT recovery relies on wave 13.

12.10.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Expense reports (list page)	TrvExpTableListPage	My/all expense reports by status	COVERED-43	43	
Expense report details	TrvExpTable / TrvExpTrans	Lines by category, itemization, receipts, guests, project/dimension, policy violations	COVERED-43	43	
Expense reports → Unattached credit card transactions (Credit card transactions pool)	TrvPBSMaindata (TrvUnreconciledExpense)	Imported card transactions awaiting assignment	COVERED-43	43	Credit-card feed import
Cash advances (Cash advance requests)	TrvCashAdvance	Request, approve, pay via AP, settle against reports	PARTIAL-43 (cash advances are not named)	43	Gap register
Travel requisitions	TrvRequisitionTable	Pre-trip approval with estimates	PARTIAL-43 (travel requisitions / pre-approval are not named)	43	Gap register
Expense reports → Itemization / Guests / Per diem lines	TrvExpTransItemization	Hotel itemization, meals per diem	COVERED-43	43	
Expense reports → Receipts (attach, OCR)	TrvDocuView		COVERED-43	43	OCR receipts

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Expense reports → Intercompany expenses (R3)	TrvExpTrans (intercompany)	Charge another legal entity	PARTIAL-40 (intercompany expenses not named in 40 or 43)	40	Gap register
Expense reports → Expense report workflow / Approval	Workflow		COVERED-43	43	
Delegates (expense delegates)	TrvDelegates	Enter on behalf	COVERED-43	43	
Expense reports → Posted expense reports / Expense transactions	TrvExpTransPosted		COVERED-43	43	
Expense → Mobile receipt capture (D365 mobile workspace)			COVERED-43	43	PWA

12.10.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Expense report inquiry / Expense transactions		COVERED-43	43	
Credit card transactions inquiry (imported / unreconciled)		COVERED-43	43	
Cash advance inquiry		PARTIAL-43 (cash advances)	43	
Travel requisition inquiry		PARTIAL-43 (travel requisitions)	43	
Policy violations inquiry		COVERED-43	43	
Expense reports pending approval (Work items)		COVERED-4	4	
SSRS: Expense report (printed) / Expense report details		COVERED-43	43	
SSRS: Expense transactions by category / employee / project		COVERED-46	46	
SSRS: Cash advance report		PARTIAL-43 (cash advances)	43	
SSRS: Unreconciled credit card transactions		COVERED-43	43	
SSRS: Policy violations report		COVERED-43	43	
SSRS: Per diem rates list		COVERED-43	43	
SSRS: Expense audit (audit policy results)		PARTIAL-48 (audit policies)	48	
SSRS: VAT recovery report (expense tax)		PARTIAL-13 (VAT recovery on expenses (partial recoverability per country) not named)	13	Gap register

12.10.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Import credit card transactions (card feed: AMEX GL1025/Visa VCF/MasterCard CDF)		COVERED-43	43	
Post expense reports (batch)	Posts to AP (reimbursement) and GL/project	COVERED-43	43	Reimbursement via AP
Reconcile/settle cash advances		PARTIAL-43 (cash advances)	43	
Expense → Mileage rate update / Per diem rate import		COVERED-43	43	
Expense → Run audit policies on expense reports		PARTIAL-48 (audit policies)	48	
Expense → Delete unposted / Clean up card pool		COVERED-43	43	
Expense → Recover VAT (ER-based claim)		PARTIAL-13 (VAT recovery)	13	
Expense → Intercompany expense posting (due-to/due-from)		PARTIAL-40 (intercompany expenses)	40	

12.10.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Expense management parameters — General	Default currency, posting to AP (vendor per worker), mileage/per diem defaults, require receipts threshold, allow itemization, project posting	COVERED-43	43	
Expense management parameters — Policies / Workflow / Credit card	Policy violation behaviors, approval, card import settings	COVERED-43	43	
Expense management parameters — Intercompany / Cash advance		PARTIAL-43 (cash advances)	43	
Expense management parameters — Number sequences		COVERED-5	5	
Expense categories (shared categories flagged for expense; expense type, default payment method, tax group, ledger account)		COVERED-43	43	
Shared categories (link to Org admin category hierarchy)		COVERED-8	8	Category hierarchies
Payment methods (expense: company card, personal, cash advance)	Who pays / who is reimbursed	COVERED-43	43	
Per diem locations / Per diem rates / Per diem rate tiers (meals, hotel, first/last day %)		COVERED-43	43	
Mileage rates / Mileage tiers		COVERED-43	43	
Expense policies (Policy framework: expense policy rules by category: amount limits, receipts, per diem, mileage, hotel; actions: warn/justify/block)		COVERED-43	43	
Expense policy violation types / justifications		COVERED-43	43	
Credit card → Credit card types, Credit card category codes (MCC mapping), Credit card transaction import formats, Card issuer connection		COVERED-43	43	
Delegates setup		COVERED-43	43	
Cash advance → Cash advance parameters, posting		PARTIAL-43 (cash advances)	43	
Travel requisition → Requisition policies / types		PARTIAL-43 (travel requisitions)	43	

Name	Key parameters / options	Passage status	Wave	Notes
Expense → Tax configuration (recoverable VAT per country, tax groups on expenses)		PARTIAL-13 (VAT recovery)	13	
Expense → Project integration (billable categories, line properties)		COVERED-41	41	
Expense → Audit policies (link)		PARTIAL-48 (audit policies)	48	
Expense → Display fields / Expense report fields configuration (optional fields)		COVERED-3	3	Metadata extensions
Expense → Ledger posting (expense posting profiles / posting by category)		COVERED-43	43	
Workflow → Expense workflows (report, line-level, cash advance, requisition)		COVERED-4	4	
Number sequences (expense)		COVERED-5	5	
Expense → Mobile app configuration (D365)		COVERED-43	43	

12.11 Project management and accounting

The project module is wave 41 (WBS, resources, journals, billing, estimates, grants), with time registration shared with wave 25, expenses with wave 43 and fixed-asset capitalization with wave 37.

12.11.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Projects (list page)	ProjTableListPage	All projects, hierarchy	COVERED-41	41	
Project details	ProjTable	Type (T&M, fixed-price, investment, cost, internal, time), stage, group, contract, funding, dimensions, WBS, invoice/billing, budget, resource	COVERED-41	41	
Project contracts	ProjInvoiceTable	Contract with funding sources, funding rules, limits, billing rules	COVERED-41	41	
Project quotations	SalesQuotationTable (project)	Quote with WBS/estimates, convert to project	PARTIAL-41 (project quotations with convert-to-project are not named; sales quotations are in 27)	41	Gap register
Work breakdown structure	ProjWBS (HierarchyTreeLookup)	Tasks, dependencies, effort, resources, estimates	COVERED-41	41	
Project hierarchy / Project tree	ProjTable (parent)	Parent/sub-projects	COVERED-41	41	
Project stage change (Create/In process/Finished…)	ProjTable (stage)	Stage gates with rules	COVERED-41	41	
Project forecasts (Hour, Expense, Item, Fee, On-account forecasts)	ProjForecast*	Forecast models	COVERED-41	41	
Project budgets / Budget revisions (Project budgeting)	ProjBudget	Original budget, revisions, budget control per project	COVERED-41	41	
Timesheets (list page)	TSTimesheetTableListPage	Weekly timesheets with workflow	PARTIAL-41 (hour journals are named; weekly timesheet UI with favorites/workflow is not explicit)	41	Gap register (note wave 25 time registration covers shop floor, not project timesheets)
Timesheet details	TSTimesheetTable	Lines, comments, submit	PARTIAL-41 (timesheets)	41	
Hour journals	ProjJournalTable (hour)	Post hours	COVERED-41	41	
Expense journals (project)	LedgerJournalTable (project expense)	Post expenses	COVERED-41	41	
Item journals (project)	InventJournalTable (project)	Consume inventory on project	COVERED-41	41	
Fee journals	ProjJournalTable (fee)	Post fees/revenue	COVERED-41	41	
Beginning balance journals	ProjJournalTable (beginning balance)	Opening WIP/transactions on migration	PARTIAL-41 (beginning balance journals for project migration not named; wave 49 covers project WIP migration)	49	Gap register
Project purchase orders / Purchase requisitions (project)	PurchTable (project)	PO tied to project	COVERED-41	41	
Project item requirements	SalesTable (item requirement)	Item requirements consumed from inventory	COVERED-41	41	
Project sales orders	SalesTable (project)		COVERED-41	41	
Project invoice proposals	ProjProposalJour	Build invoice from transactions	COVERED-41	41	
Project invoices (posted)	ProjInvoiceJour		COVERED-41	41	
On-account transactions (Milestones)	ProjOnAccTrans	Fixed-price billing milestones	COVERED-41	41	
Estimates (Estimate projects, Completed contract / Percentage completion)	ProjControlPeriodTable (ProjEstimate)	Revenue recognition for fixed-price	COVERED-41	41	
Grants (Grant management, R2 PSA)	ProjGrant	Grant records, grant types, funding	COVERED-41	41	Customer funding/grants
Project → Resources (Resource list, Resource assignments, Resource availability, Resource requests)	ProjResource*, ResRoleAssignment	Role-based resource booking	COVERED-41	41	
Project → Resource scheduling (Resource scheduling form, R3)	ResResourceScheduling	Book resources to WBS tasks	COVERED-41	41	
Project → Activities (Activity list for projects)	smmActivities	Activities on projects	PARTIAL-45 (generic activities/case management framework)	45	
Project → Committed costs	ProjCommittedCost	Commitments from PO/requisition/timesheet	COVERED-41	41	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Project → Funding limits / Funding rules	ProjFundingLimit		COVERED-41	41	
Project → Billing rules (Unit of delivery, Progress, Milestone, Fee, Time and material, Retainage)	ProjBillingRule		COVERED-41	41	
Project → Retention terms (customer/vendor retention)	ProjRetentionTerms	Hold back % until milestone	PARTIAL-41 (customer and vendor retention (retainage) terms are not named)	41	Gap register
Project → Project transactions (posted)	ProjTrans*		COVERED-41	41	
Project → Adjustments (Project transaction adjustments)	ProjAdjust	Reallocate posted transactions	COVERED-41	41	
Project → Subcontracts / Vendor invoices for project			COVERED-41	41	
Project → Project workspace (D365 Project management workspace)			COVERED-41	41	
Microsoft Project integration (MPP sync)	ProjMSProject*		SUPERSEDED (WBS scheduling native; MS Project import/export via wave 5 Excel/CSV and wave 35 APIs)	41	

12.11.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Project transactions inquiry (posted transactions, by type)		COVERED-41	41	
Project statements (Project statement inquiry: cost, revenue, WIP, profit)		COVERED-41	41	
Cash flow (project)		COVERED-36	36	
Committed cost inquiry		COVERED-41	41	
Invoice proposals / Invoices inquiry		COVERED-41	41	
Estimate inquiry (Estimate details)		COVERED-41	41	
Budget vs actual (project budget balances)		COVERED-41	41	
Resource utilization / availability inquiry		COVERED-41	41	
Timesheet inquiries (missing timesheets, timesheet hours)		PARTIAL-41 (timesheets)	41	
Hour utilization		COVERED-41	41	
Project ledger reconciliation (Project/Ledger)		COVERED-41	41	
Funding source / contract inquiries		COVERED-41	41	
Grant inquiries		COVERED-41	41	
SSRS: Project statement / Project statement by category		COVERED-41	41	
SSRS: Project invoice (document) / Invoice proposal / Credit note		COVERED-41	41	
SSRS: Hour utilization / Employee utilization		COVERED-41	41	
SSRS: Project transactions (hours, expenses, items, fees, on-account)		COVERED-41	41	
SSRS: WIP / Accrued revenue / Cost-to-complete / Percentage complete		COVERED-41	41	
SSRS: Project list / Project base data		COVERED-41	41	
SSRS: Cash flow forecast (project)		COVERED-36	36	
SSRS: Committed costs report		COVERED-41	41	
SSRS: Timesheet report / Missing timesheets		PARTIAL-41 (timesheets)	41	
SSRS: Estimate report / Estimate journal		COVERED-41	41	
SSRS: Project budget / Budget balances		COVERED-41	41	
SSRS: Resource scheduling / Resource assignment		COVERED-41	41	
SSRS: Project invoice journal / Customer invoice register		COVERED-41	41	
SSRS: Earned value (CPI/SPI)		COVERED-41	41	EVM
SSRS: Project profitability		COVERED-46	46	
SSRS: Grant report		COVERED-41	41	

12.11.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Journals → Post hour/expense/item/fee journals (batch)		COVERED-41	41	
Timesheets → Timesheet workflow / Post timesheets / Create periodic timesheets		PARTIAL-41 (timesheets)	41	
Estimates → Create / Post / Reverse / Eliminate estimates	Fixed-price revenue recognition cycle	COVERED-41	41	
Invoice proposals → Create invoice proposals (batch) / Post		COVERED-41	41	
Project → Transfer to fixed assets (investment project estimate elimination)		COVERED-37	37	
Project → Accrue revenue (T&M WIP accrual)		COVERED-41	41	
Project → Reverse accrued revenue		COVERED-41	41	
Project → Cost control / Budget control recalc		COVERED-41	41	
Project → Project stage update (mass)		COVERED-41	41	
Project → Line property update / Adjust transactions (mass)		COVERED-41	41	
Project → Intercompany project invoicing (R3 intercompany timesheets/expenses)	Cross-entity resource lending and invoicing	PARTIAL-40 (intercompany project resourcing/invoicing not named)	40	Gap register

Name	Purpose	Passage status	Wave	Notes
Project → Project purchase requisition release		COVERED-17	17	
Project → Copy project / Copy WBS from template		COVERED-41	41	
Project → Delete / Close projects		COVERED-41	41	
Project → Project forecast → Copy/transfer forecast to budget		COVERED-41	41	
Project → Project sales tax / VAT on invoice proposals		COVERED-13	13	
Project → Retention release (vendor/customer)		PARTIAL-41 (retention)	41	
Project → Resource availability recalculation		COVERED-41	41	
Project → Grant periodic (grant renewals, closeouts)		COVERED-41	41	
Project → Project invoice e-invoicing (ER)		COVERED-35	35	

12.11.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Project management and accounting parameters — General	Default project types, stages, line property, WBS defaults, ledger posting (balance/P&L), hour/expense/item/fee posting, cost price defaults, sales price defaults, invoice proposal rules	COVERED-41	41	
Project parameters — Financial (ledger posting rules by project type: cost, revenue, WIP, accrued)		COVERED-41	41	
Project parameters — Cost control / Budget / Estimate / Invoice / Timesheet / Scheduling / Resources / Journals / Number sequences		COVERED-41	41	Timesheet tab PARTIAL per 12.11.1
Project groups	Posting rules per group (P&L/balance), WIP accounts, estimate method, cost template	COVERED-41	41	
Project types / Project stages / Project stage rules		COVERED-41	41	
Categories → Project categories / Category groups / Shared categories	Hour, expense, item, fee categories with posting	COVERED-41	41	
Line properties	Chargeable / capitalize / efficiency %	COVERED-41	41	
Posting → Ledger posting setup (Cost accounts, Revenue accounts by category/project/customer)		COVERED-41	41	
Posting → Project posting profile (customer invoice)		COVERED-41	41	
Prices → Cost prices (hour, expense), Sales prices (hour, expense, fee, subscription), Transfer prices (intercompany)	Price lists by resource/project/category	COVERED-41	41	Transfer prices PARTIAL-40 for intercompany
Estimates → Cost templates / Period types / Estimate periods	Completion methods	COVERED-41	41	
Project contracts → Funding types / Funding rule templates		COVERED-41	41	
Billing → Billing rule templates / Retention terms		PARTIAL-41 (retention)	41	
Timesheets → Timesheet periods / Period types / Timesheet parameters (favorites, delegates, workflow by project)		PARTIAL-41 (timesheets)	41	
Resources → Resource roles, Competencies, Calendars, Resource groups, Availability parameters		COVERED-41	41	
Scheduling → Scheduling parameters (WBS: default calendar, dependencies)		COVERED-41	41	
WBS → WBS templates		COVERED-41	41	
Project → Project templates		COVERED-41	41	
Project → Validation (Project/category, Project/worker, Category/worker validation groups)	Who may post what to which project	COVERED-41	41	
Budget → Project budget parameters / Budget control configuration (project)		COVERED-41	41	
Grants → Grant types, Grant statuses, CFDA numbers (US)		COVERED-41	41	
Indirect costs (Indirect cost component groups, Indirect cost components, Indirect cost rules — R3 PSA)	Overhead on project hours	PARTIAL-41 (project indirect cost components/rules not named; cost accounting allocations in 38 do not apply to project transactions)	41	Gap register
Journals → Journal names (project journals)		COVERED-41	41	
Forms → Form setup (project invoices) / Print management		PARTIAL-46 (print management routing)	46	
Workflow → Project workflows (timesheet, expense, hour journal, project stage)		COVERED-4	4	
Number sequences (project)		COVERED-5	5	
Microsoft Project → MS Project integration parameters		SUPERSEDED (native WBS)	41	
Intercompany → Intercompany project parameters (lending/borrowing entity, timesheet/expense/vendor invoice transfer)		PARTIAL-40 (intercompany project)	40	

12.12 Human resources

AX 2012 R3 Human resources (workers, positions, jobs, departments, benefits, leave, compensation, performance, skills/competencies, recruitment, courses, questionnaires, injury/illness, loaned items, worker compliance) maps to wave 42, with absence calendars shared with wave 25 and payroll in 43. Courses,

questionnaires, injury/illness and loaned items are not in any wave and are listed as gaps.

12.12.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Workers (list page)	HcmWorkerListPage	All workers (employees + contractors)	COVERED-42	42	Party with Employee role
Worker details	HcmWorker	Profile, employment, position assignments, compensation, leave, benefits, identification (SSN/passport), bank accounts, contacts, personal contacts, financial dimensions, competencies	COVERED-42	42	PII vaulting
Employees / Contractors (list pages)	HcmEmployeeListPage / HcmContractorListPage	Filtered worker lists	COVERED-42	42	
Employment (worker employment history, effective-dated)	HcmEmployment	Hire, terminate, rehire with dates	COVERED-42	42	
Positions (list page)	HcmPositionListPage	Position hierarchy, vacancies	COVERED-42	42	
Position details	HcmPosition	Job, department, reports-to, position type, duration, FTE, worker assignment, payroll, budget	COVERED-42	42	
Jobs	HcmJob	Job templates: title, type, function, tasks, skills, education, compensation level	COVERED-42	42	
Departments (Org admin link)	OMDepartment		COVERED-6	6	
Personnel actions (D365: hire, transfer, terminate, position change with workflow)	HcmPersonnelAction	Action-based changes with approval	COVERED-42	42	Effective-dated with workflow
Leave and absence → Absence registrations / Absence journals (AX 2012)	HRMAbsenceTable	Register absences, approve, transfer	COVERED-42	42	
Leave and absence → Leave requests / Leave balances (D365)	HcmLeaveRequest	ESS requests with balances	COVERED-42	42	
Compensation → Fixed compensation / Variable compensation (plans, enrollments, events)	HcmFixedCompensationEvent / HcmVariableComp*	Salary structures, merit, bonus plans	COVERED-42	42	Compensation plans
Benefits → Benefit enrollment (worker benefits)	HcmWorkerBenefit	Enroll in plans, dependents, beneficiaries	COVERED-42	42	
Benefits → Open enrollment / Benefits management (D365 benefits management with life events)	BenefitsManagement*	Enrollment periods, eligibility processing	COVERED-42	42	
Performance → Performance journals / Discussions / Goals / Reviews (D365 performance management)	HcmPerformance*	Goals, reviews, ratings	COVERED-42	42	
Competencies → Skills / Certificates / Education / Courses / Tests / Positions of trust / Professional experience (worker competencies)	HcmPersonSkill, HcmPersonCertificate, …	Worker competency records	COVERED-42	42	Skills, certifications
Competencies → Skill mapping / Skill gap analysis	HcmSkillMapping	Search workers by skills	PARTIAL-42 (skill mapping/gap analysis not named)	42	Gap register
Recruitment → Recruitment projects / Applicants / Applications / Application routing	HRMRecruitingTable, HRMApplicant, HRMApplication	Applicant tracking	COVERED-42	42	Recruitment pipeline basics
Courses (training) → Courses, Course sessions, Course registrations, Instructors, Course locations	HRMCourseTable	Training administration	GAP	—	Propose wave 42: learning/training management (courses, sessions, registrations, instructors, completion to competencies)
Questionnaires → Questionnaires, Questionnaire schedules, Answer sessions, Results analysis	KMQuestionnaire, KMKnowledgeCollectorSchedule	Surveys/tests (360 feedback, course evaluations)	GAP	—	Propose wave 42: questionnaire engine (design, schedule, answer, analyze) reused by performance reviews and quality audits
Injury and illness → Injury/illness incidents, Treatments, Body parts, OSHA logs	HRMInjuryIncident	Workplace incident tracking	GAP	—	Propose wave 42: injury and illness incident register with OSHA 300 extracts (US pack)
Loaned items → Loaned items, Loan items	HRMLoanTable	Equipment lent to workers	GAP	—	Propose wave 42: loaned items (assets issued to workers) tied to EAM assets wave 44
Worker compliance (US) → I-9, ADA requirements, Immigration/visa, Union agreements	HcmWorkerI9Document, HcmPersonVisa	Compliance records	PARTIAL-42 (country-specific compliance fields framework is named; I-9/visa/union specifics come with US pack)	48	
Worker → Worker tasks / reminders / Checklists (onboarding)	HcmWorkerTask	Onboarding/offboarding checklists	PARTIAL-42 (onboarding checklists not named)	42	Gap register
Employee self-service (Role center, EP)	EP ESS pages	View/edit personal info, requests	COVERED-42	42	ESS/MSS portal
Manager self-service	EP MSS pages	Team view, approvals	COVERED-42	42	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Worker → Worker action workflow (position changes)	HcmWorkerActionWorkflow		COVERED-42	42	
Teams (Org admin: Teams, Team types)	OMTeam	Cross-org teams for collaboration	PARTIAL-6 (teams (ad hoc cross-org groups) not named; hierarchies are)	6	Gap register
Workforce → Worker calendars / Working time (link to calendars)	WorkCalendarTable		COVERED-9	9	
Human resources → Cases (HR cases)	CaseDetail		PARTIAL-45 (generic case management)	45	

12.12.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Worker inquiries (Workers by department/position, Seniority, Birthdays, Anniversaries)		COVERED-42	42	
Position inquiries (Vacant positions, Position hierarchy, Headcount)		COVERED-42	42	
Absence inquiries (Absence statistics, Absence per worker)		COVERED-42	42	
Benefit inquiries (Enrollments, Eligibility)		COVERED-42	42	
Compensation inquiries (Pay history, Compensation events)		COVERED-42	42	
Competency inquiries (Skills by worker, Certificate expiry)		COVERED-42	42	
Recruitment inquiries (Applicants by project)		COVERED-42	42	
Course inquiries		GAP	—	Propose wave 42: learning management
Questionnaire results analysis		GAP	—	Propose wave 42: questionnaire engine
Injury/illness inquiries		GAP	—	Propose wave 42: injury and illness
Loaned items inquiry		GAP	—	Propose wave 42: loaned items
SSRS: Worker list / Worker base data / Worker contact list		COVERED-42	42	
SSRS: Position hierarchy / Position list / Vacancies		COVERED-42	42	
SSRS: Absence report / Absence statistics		COVERED-42	42	
SSRS: Benefit enrollment report / Benefit cost		COVERED-42	42	
SSRS: Compensation report / Pay rate history		COVERED-42	42	
SSRS: Skills / Certificates / Education reports		COVERED-42	42	
SSRS: Recruitment reports (Applicant list, Application status)		COVERED-42	42	
SSRS: Course reports (Attendance, Course confirmation letters)		GAP	—	Propose wave 42: learning management
SSRS: Questionnaire reports		GAP	—	Propose wave 42: questionnaire engine
SSRS: OSHA 300/300A/301 (US)		GAP	—	Propose wave 42: injury and illness (US pack output wave 48)
SSRS: EEO-1 / VETS (US compliance)		PARTIAL-42 (US compliance reporting with wave 48 US pack)	48	
SSRS: Headcount / Turnover analytics		COVERED-46	46	
SSRS: Organization chart		COVERED-6	6	Hierarchy designer

12.12.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Absence → Transfer absence registrations (to payroll/projects)		COVERED-42	42	
Leave → Accrue leave balances (D365 accrual processing)		COVERED-42	42	Accrual fixtures
Leave → Carry-forward / expire balances		COVERED-42	42	
Benefits → Mass benefit enrollment / Update benefit rates / Expire benefits / Process life events		COVERED-42	42	
Compensation → Process compensation events (merit cycle)		COVERED-42	42	
Positions → Update position assignments (effective dating batch) / Position budget sync		COVERED-42	42	
Recruitment → Close recruitment projects / Mass applicant correspondence		COVERED-42	42	
Courses → Course confirmations / Transfer course completion to competencies		GAP	—	Propose wave 42: learning management
Questionnaires → Distribute questionnaires (schedule) / Collect results		GAP	—	Propose wave 42: questionnaire engine
Worker → Delete worker history / Anonymize (GDPR)		COVERED-48	48	Subject requests
Worker → Reminders (certificate expiry, probation end)		PARTIAL-42 (reminder/alert engine; see alerts gap in §12.31)	42	
HR → Import workers (DIXF)		COVERED-10	10	
HR → Year-end (US ACA 1095-C)		PARTIAL-42 (ACA reporting with US pack)	48	

12.12.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Human resources parameters — General / Recruitment / Workers / Absence / Benefits / Compensation / Number sequences / Shared (HR shared parameters: identification, worker numbering)		COVERED-42	42	

Name	Key parameters / options	Passage status	Wave	Notes
Organization → Departments, Jobs, Positions setup (Position types, Position hierarchy types, Job functions, Job types, Job tasks, Areas of responsibility, Titles)		COVERED-42	42	
Workers → Employment types, Reasons for termination, Employment categories, Seniority, Identification types, Ethnic origins, Veteran statuses, Marital status, Honorifics	Reference lists	COVERED-42	42	
Workers → Personal contact relationship types / Dependents / Beneficiaries		COVERED-42	42	
Absence → Absence codes / Absence code groups / Absence setups / Absence statuses / Leave types (D365: Leave and absence plans, Accrual schedules, Leave calendars, Holiday corrections)		COVERED-42	42	
Benefits → Benefit types, Benefit plans, Benefit options, Benefit eligibility policies, Benefit rates, Contribution calculation rates, Benefit periods, Life events (D365), Flex credits (D365)		COVERED-42	42	
Compensation → Compensation grids, Compensation levels, Reference point setups, Fixed compensation plans, Variable compensation plans, Compensation actions, Pay frequencies, Compensation regions, Surveys		COVERED-42	42	
Performance → Rating models, Review types, Goal types, Discussion types		COVERED-42	42	
Competencies → Skill types, Skills, Rating models, Certificate types, Education disciplines, Education levels, Institutions, Course types		COVERED-42	42	
Recruitment → Recruitment project types, Application statuses, Applicant sources, Media, Correspondence actions, Application routing templates		COVERED-42	42	
Courses → Course types, Course groups, Course locations, Classrooms, Instructors, Agenda templates		GAP	—	Propose wave 42: learning management
Questionnaires → Questionnaire types, Question types, Answer groups, Result groups, Media types, Questionnaire parameters		GAP	—	Propose wave 42: questionnaire engine
Injury and illness → Incident types, Body parts, Injury/illness treatments, Severity levels, Outcomes		GAP	—	Propose wave 42: injury and illness
Loaned items → Loan types		GAP	—	Propose wave 42: loaned items
Worker compliance (US) → I-9 document types, ADA requirements, Visa types, Union agreements		PARTIAL-42 (US compliance)	48	
Worker → Worker tasks / Checklist templates (onboarding)		PARTIAL-42 (onboarding checklists)	42	
Teams → Team types, Teams		PARTIAL-6 (teams)	6	
Employee self-service parameters (EP) / Manager self-service parameters		SUPERSEDED (ESS/MSS portal wave 42)	42	
HR → Workflow (worker actions, leave, compensation, benefits)		COVERED-4	4	
HR → Number sequences		COVERED-5	5	
HR → Privacy / PII masking policies (Passage)		COVERED-42	42	
HR → Calendars (working time)		COVERED-9	9	
HR → Country-specific fields (localization framework)		COVERED-42	42	
HR → Payroll information on worker (link)		COVERED-43	43	

12.13 Payroll

AX 2012 R2/R3 US Payroll (earnings, benefits/deductions, taxes via Symmetry tax engine, pay statements, GL posting, W-2) maps to wave 43 (country-pluggable gross-to-net with US and one EU pack, plus provider adapter).

12.13.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Earnings statements (list page)	PayrollEarningStatementListPage	Generated earnings per worker per period	COVERED-43	43	
Earnings statement details	PayrollEarningStatement	Earning lines (salary, hourly, overtime, retro), source (timesheet, schedule, manual)	COVERED-43	43	
Pay statements (list page)	PayrollPayStatementListPage	Gross-to-net statements	COVERED-43	43	
Pay statement details	PayrollPayStatement	Earnings, benefits, deductions, taxes, net pay, payment method	COVERED-43	43	
Worker → Payroll information (Worker payroll setup: tax regions, tax codes, earnings, benefits, garnishments, pay cycle, payment method/direct deposit)	PayrollWorkerDetails (HcmWorker → Payroll tab)	Per-worker payroll setup	COVERED-43	43	
Worker → Benefit (payroll) enrollment / Deduction enrollment	PayrollWorkerBenefit	Payroll deduction/contribution enrollment	COVERED-43	43	
Worker → Garnishments and tax levies	PayrollWorkerGarnishment	Garnishment rules, limits, disposable income	COVERED-43	43	
Worker → Tax codes / Tax regions (W-4 info)	PayrollWorkerTaxCode	Federal/state withholding settings	COVERED-43	43	
Worker → Earning codes / Default earnings (recurring earnings)	PayrollWorkerEarningCode	Standing earnings	COVERED-43	43	
Worker → Payroll calendar assignment / Pay cycles			COVERED-43	43	
Worker → Direct deposit / Payment methods	PayrollWorkerBankAccountDisbursement	Split deposits	COVERED-43	43	
Payroll → Payroll transactions / Posted payroll journal (GL journal from payroll)	LedgerJournalTable (payroll)		COVERED-43	43	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Payroll → Vendor invoices for payroll liabilities (taxes, benefit providers)	VendInvoice (payroll)		COVERED-43	43	Liabilities via AP
Payroll → Tax (Symmetry tax engine integration)	PayrollTaxEngine*	External tax calculation	COVERED-43	43	Versioned rule sets; provider adapter alternative
Payroll → Retroactive earnings	PayrollEarningStatement (retro)	Retro pay lines	COVERED-43	43	
Payroll → Pay statement reversal / Void			COVERED-43	43	
Payroll → Time registration import (from T&A / timesheets)			COVERED-25	25	Export to payroll from wave 25

12.13.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Earnings statement inquiry / Pay statement inquiry		COVERED-43	43	
Worker payroll history (earnings, taxes, deductions YTD)		COVERED-43	43	
Tax inquiries (Tax summary by region, Tax liabilities)		COVERED-43	43	
Benefit/deduction register inquiry		COVERED-43	43	
Garnishment inquiry		COVERED-43	43	
Payroll posting inquiry (journal)		COVERED-43	43	
SSRS: Pay statement (document) / Check stub		COVERED-43	43	
SSRS: Payroll register / Payroll summary		COVERED-43	43	
SSRS: Tax register / Tax liability / Form 941 / State unemployment (SUTA)	US statutory	COVERED-48	48	US pack
SSRS: W-2 / W-3 / Electronic W-2		COVERED-43	43	Year-end forms framework
SSRS: Benefit register / Deduction register / Garnishment register		COVERED-43	43	
SSRS: Earnings register		COVERED-43	43	
SSRS: Payroll journal (GL)		COVERED-43	43	
SSRS: Workers compensation report		PARTIAL-43 (workers' compensation class codes/reporting not named)	48	Gap register
SSRS: Payroll audit / reconciliation reports		COVERED-43	43	
SSRS: Positive pay for payroll checks		COVERED-19	19	
SSRS: New hire reporting (US state)		PARTIAL-43 (new hire reporting with US pack)	48	

12.13.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Generate earnings statements	Create earnings per pay period from schedules/time	COVERED-43	43	
Release earnings statements		COVERED-43	43	
Generate pay statements (calculate gross-to-net)		COVERED-43	43	
Submit pay statements for payment (payment journal, direct deposit file, checks)		COVERED-43	43	NACHA via wave 19
Post payroll (GL journal)		COVERED-43	43	
Issue payments / Print payroll checks		COVERED-19	19	
Pay statement reversal		COVERED-43	43	
Retroactive pay processing		COVERED-43	43	
Generate vendor invoices for payroll liabilities		COVERED-43	43	
Tax updates (apply tax engine updates)		COVERED-43	43	Versioned rule sets
Year-end → Generate W-2 / Electronic filing / Archive		COVERED-43	43	
Benefit accrual (payroll-based leave accrual: PTO)		COVERED-42	42	
Payroll → Arrears recovery processing		PARTIAL-43 (deduction arrears recovery not named)	43	Gap register
Payroll → Pay period roll / Close pay period		COVERED-43	43	
Payroll → Provider export (ADP/Gusto adapter)		COVERED-43	43	
Payroll → Import time from T&A		COVERED-25	25	

12.13.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Payroll parameters — General / Earnings / Posting / Payment / Tax / Number sequences	Default earning codes, accrual, posting defaults, payment journal names, tax engine	COVERED-43	43	
Pay cycles and pay periods	Weekly/biweekly/semimonthly/monthly, period generation	COVERED-43	43	
Payroll calendars (D365: work cycles)		COVERED-43	43	
Earning codes	Type (hourly/salary/overtime/retro), rate basis, multiplier, tax treatment, productive, accounting	COVERED-43	43	
Earning code groups / Earning rate groups		COVERED-43	43	
Benefits (payroll) → Benefit elements: Benefit types, Benefit plans, Benefit options; Benefit accrual plans (PTO accrual with tiers)	Deduction/contribution plans with tax rules	COVERED-43	43	

Name	Key parameters / options	Passage status	Wave	Notes
Taxes → Tax regions, Tax codes (federal/state/local), Tax groups, Tax engine configuration (Symmetry), Tax update schedule, Workers' compensation codes		COVERED-43	43	Workers' comp PARTIAL
Garnishments → Garnishment types, Disposable income definitions, Garnishment rules, Administrative fees		COVERED-43	43	
Schedules → Work schedules (schedule templates for salaried)		COVERED-43	43	
Posting → Payroll posting definitions (earnings/benefits/taxes to accounts), Payroll posting rules by position/department		COVERED-43	43	
Payment → Payment methods (check, direct deposit), Pay statement layouts, Positive pay		COVERED-43	43	
Vendors for payroll liabilities (tax authorities, benefit providers)		COVERED-43	43	
Pay statement → Pay statement check layout / Direct deposit advice		COVERED-46	46	
Year-end → W-2 box mapping, Electronic filing setup		COVERED-43	43	
Country packs → US tables (federal/state) / EU reference country (DE or UK)		COVERED-43	43	Open question 3 in PRD
Third-party payroll provider adapter configuration		COVERED-43	43	
Workflow → Payroll workflows		COVERED-4	4	
Number sequences (payroll)		COVERED-5	5	
Payroll → Leave of absence / FMLA tracking (US)		PARTIAL-42 (FMLA tracking not named)	48	Gap register

12.14 Inventory management

AX 2012 R3 Inventory management (minus Quality, treated in §12.25, and minus WMS I/II which Passage supersedes with wave 31) maps to wave 16 (ledger, journals, counting, reservations, blocking, quarantine), wave 20 (costing, close), wave 28 (picking/shipping of orders) and wave 8 (item master). Transfer orders are only partially covered and are listed in the gap register.

12.14.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Released products (list page, link)	EcoResProductDetailsExtended (InventTable)	Item master per legal entity	COVERED-8	8	See §12.15
On-hand inventory (inquiry entry)	InventOnhandItem / InventDimOnHand	On-hand by dimension with physical/financial split	COVERED-16	16	
Inventory transactions	InventTrans	All transactions with statuses	COVERED-16	16	
Transfer orders (list page)	InventTransferOrderListPage	Ship/receive between warehouses with in-transit	PARTIAL-16 (transfer *journal* is named; transfer *orders* with shipment, in-transit warehouse, receive, underdelivery and transfer-order picking are not named in 16, 22 or 28)	16	Gap register
Transfer order details	InventTransferTable / InventTransferLine	Lines, shipment/receipt, dates, charges	PARTIAL-16 (transfer orders)	16	
Journals → Movement journal	InventJournalTable (Movement)	Qty in/out with offset account	COVERED-16	16	
Journals → Inventory adjustment journal (Profit/loss)	InventJournalTable (ProfitLoss)	Adjustments to item posting accounts	COVERED-16	16	
Journals → Transfer journal	InventJournalTable (Transfer)	Dimension-to-dimension transfer	COVERED-16	16	
Journals → Bills of materials journal	InventJournalTable (BOM)	Report BOM as finished without production	COVERED-16	16	
Journals → Counting journal	InventJournalTable (Count)	Count sheets, variances, approval	COVERED-16	16	
Journals → Tag counting journal	InventJournalTable (TagCount)	Tag-based count	PARTIAL-16 (tag counting (numbered tags, two-step) not named; cycle counting is)	16	Gap register
Journals → Item arrival journal / Production input journal	WMSJournalTable (Arrival / ProdInput)	Register arrivals; production input	COVERED-18	18	Arrival; production input in 21
Journals → Inventory ownership change journal (D365 consignment)	InventJournalTable (OwnershipChange)		GAP	—	Propose wave 16: consignment
Quarantine orders	InventQuarantineOrder	Quarantine warehouse, start/end/report as finished	COVERED-16	16	Quarantine
Inventory blocking	InventBlocking	Block qty (manual/quality)	COVERED-16	16	
Quality orders / Nonconformances (links)	InventQualityOrderTable		COVERED-33	33	§12.25
Batches (Inventory batches)	InventBatch	Batch attributes, expiry, shelf-life dates, disposition, vendor batch, country of origin	COVERED-24	24	Batch core in 16; attributes/shelf life in 24
Serial numbers	InventSerial	Serial records	COVERED-16	16	
Batch disposition codes (status on batch)	PdsDispositionCode	Available/unavailable with blocking	COVERED-24	24	
Batch merge (PI)	PdsBatchMerge	Merge batches with attribute recalculation	PARTIAL-24 (batch merge/split with attribute recalculation not named)	24	Gap register
Pallets (WMS II)	WMSPallet	Pallet master	SUPERSEDED (license plates wave 31)	31	
Locations (WMS II) / Store areas / Store zones	WMSLocation, WMSStoreArea, WMSStoreZone	Legacy location model	SUPERSEDED (location model wave 31)	31	
Pallet transports / Output orders / Shipments (WMS I)	WMSTransport, WMSOrder, WMSShipment	Legacy warehouse execution	SUPERSEDED (work/waves/loads wave 31; fulfillment wave 28)	31	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Picking routes / Picking list registration (sales/inventory)	WMSPickingRoute, WMSOrderTrans	Legacy picking	SUPERSEDED (pick confirmation wave 28; work wave 31)	28	
Arrival overview	WMSArrivalOverview	Expected receipts planning	COVERED-18	18	
Item arrival registration / Registration form	InventTransRegister	Register by dimension/serial/batch	COVERED-18	18	
Picking → Pick (Picking form on orders) / Unpick	InventTransPick	Pick/unpick quantities	COVERED-28	28	
Reservation form (manual reservation on order lines)	InventTransReserve	Reserve by dimension/batch	COVERED-16	16	
Inventory → Bar codes (item bar codes)	InventItemBarcode		COVERED-8	8	
Inventory → Dimensions display / Dimension defaults			COVERED-16	16	
Inventory → Counting → Cycle counting work (WHS)	WHSWorkTable (count)		COVERED-31	31	
Inventory → ABC classification (on item)	InventTable (ABC)	ABC codes by revenue/value/margin/carrying cost	PARTIAL-16 (ABC classification fields/calculation not named)	16	Gap register
Inventory → Forecast (Inventory forecast: sales/purchase forecasts, link)	ForecastSales / ForecastPurch		COVERED-22	22	
Inventory → Item coverage (link)	ReqItemTable		COVERED-22	22	
Inventory → Inventory statuses (WHS)	WHSInventoryStatus	Available/blocked statuses	COVERED-16	16	Inventory status
Inventory → Catch weight quantities on transactions	PdsCatchWeight	Dual UoM	COVERED-24	24	
Inventory → Commodity codes (Intrastat on item)			PARTIAL-13 (Intrastat setup)	13	
Inventory → Inventory visibility add-in (D365)		Real-time cross-system on-hand	COVERED-16	16	On-hand projection; API wave 35

12.14.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
On-hand inquiry (by dimension, with cost)		COVERED-16	16	
On-hand → Physical inventory by dimension / Available physical		COVERED-16	16	
Transactions inquiry (Inventory transactions by item/dimension/reference)		COVERED-16	16	
Transfer order inquiries (Transfer order lines, shipped/received)		PARTIAL-16 (transfer orders)	16	
Journal inquiries (posted journals by type)		COVERED-16	16	
Batch inquiries (Batch attributes, Batch disposition, Expired batches, Shelf life advice)		COVERED-24	24	
Serial number inquiry / Serial trace (Item tracing R3)	Trace batch/serial forward/backward	COVERED-24	24	Genealogy; also wave 33 quality
Quarantine order inquiry		COVERED-16	16	
Reservations inquiry (Reserved by/for)		COVERED-16	16	
Inventory blocking inquiry		COVERED-16	16	
Settlements / Cost explorer (link)		COVERED-20	20	
Inventory statistics (Item statistics, Inventory turnover)		COVERED-46	46	Inventory turns metric
Item arrival inquiry (Arrival overview)		COVERED-18	18	
Picking list inquiry (Picking list registration journal)		COVERED-28	28	
Shipments / Output orders inquiries (WMS I)		SUPERSEDED (wave 28/31)	31	
Pallet / Location inquiries (WMS II)		SUPERSEDED (wave 31)	31	
Counting history / Count variance inquiry		COVERED-16	16	
Inventory dimension inquiry (dimension combinations)		COVERED-16	16	
Inventory → Intercompany on-hand (R3)	On-hand across legal entities	PARTIAL-40 (intercompany on-hand visibility not named)	40	Gap register
SSRS: Physical inventory by inventory dimension / On-hand		COVERED-16	16	
SSRS: Inventory value (link)		COVERED-20	20	
SSRS: Inventory transactions report		COVERED-16	16	
SSRS: Journal reports (Movement, Counting list, Counting journal, Transfer, BOM, Tag counting list)		COVERED-16	16	Tag counting PARTIAL
SSRS: Transfer order (shipping document, receiving document, Transfer order overview)		PARTIAL-16 (transfer orders)	16	
SSRS: Inventory aging (link)		PARTIAL-20 (inventory aging)	20	
SSRS: ABC classification report		PARTIAL-16 (ABC classification)	16	
SSRS: Inventory turnover / Slow-moving items / Items without transactions		COVERED-46	46	
SSRS: Item base data / Item list / Item price list		COVERED-8	8	
SSRS: Batch / Serial number listings (Batch attributes, Expired batches)		COVERED-24	24	
SSRS: Quarantine orders report		COVERED-16	16	
SSRS: Bar code label (item)		COVERED-32	32	ZPL labels
SSRS: Item arrival / Receipts list		COVERED-18	18	

Name	Purpose	Passage status	Wave	Notes
SSRS: Picking list / Picking list registration		COVERED-28	28	
SSRS: Packing slip (inventory transfer)		PARTIAL-16 (transfer orders)	16	
SSRS: Inventory reconciliation (Inventory/Ledger)		COVERED-20	20	
SSRS: Item model group / Dimension group listings		COVERED-8	8	
SSRS: Consignment inventory reports (D365)		GAP	—	Propose wave 16: consignment
SSRS: Catch weight on-hand / transactions		COVERED-24	24	
SSRS: Country of origin / Shelf life reports (PI)		COVERED-24	24	

12.14.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Closing and adjustment (link)		COVERED-20	20	
Counting → Create counting journal (by item/warehouse/location, counting groups, periodic)		COVERED-16	16	Cycle counting plans
Counting → Post counting / Approve variances		COVERED-16	16	
Tag counting → Create tag counting / Transfer to counting journal		PARTIAL-16 (tag counting)	16	
Transfer orders → Ship / Receive (batch) / Transfer order picking release		PARTIAL-16 (transfer orders)	16	
Transfer orders → Delete transfer orders		PARTIAL-16 (transfer orders)	16	
Quarantine orders → Start / End / Report as finished (batch)		COVERED-16	16	
Reservation → Automatic reservation (batch) / Cancel reservations		COVERED-16	16	
Batches → Expire batches / Update shelf life dates / Batch disposition update		COVERED-24	24	
Batches → Shelf life advice (batch rotation)		COVERED-24	24	
ABC classification (periodic calculation)		PARTIAL-16 (ABC classification)	16	
Forecast → Transfer forecast (link)		COVERED-22	22	
Bar codes → Generate bar codes (batch)		COVERED-8	8	
Inventory → Recalculate on-hand (InventSum rebuild) / Check inventory consistency		COVERED-16	16	Projection rebuild
Inventory → Clean up (Delete inventory journals, Delete InventSum zero records, Delete warehouse journals)		COVERED-16	16	Housekeeping job
Inventory → Inventory close (link) / Transfer inventory to ledger		COVERED-20	20	
Inventory → Warehouse management: Pallet transports / Shipments / Output orders processing (WMS I/II)		SUPERSEDED (wave 31)	31	
Inventory → Update unit conversions (mass)		COVERED-6	6	
Inventory → Release products to legal entities (mass)		COVERED-8	8	
Inventory → Consignment: ownership change batch		GAP	—	Propose wave 16: consignment
Inventory → Intercompany: on-hand sync		PARTIAL-40 (intercompany on-hand)	40	
Inventory → Item tracing (run trace)		COVERED-24	24	
Inventory → Catch weight: update nominal weights		COVERED-24	24	

12.14.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Inventory and warehouse management parameters — General	Default dimensions, pallet/location usage (WMS II), date-controlled reservations, allow negative physical on quarantine, bill of materials: explode, counting reason codes, reservation by batch/date	COVERED-16	16	WMS II flags superseded
Inventory and warehouse management parameters — Journals	Journal names defaults, delete lines after posting, journal size	COVERED-16	16	
Inventory and warehouse management parameters — Inventory dimensions / Inventory accounting (Costing)		COVERED-20	20	
Inventory and warehouse management parameters — Quality management	Hours per day, nonconformance defaults	COVERED-33	33	
Inventory and warehouse management parameters — Transport (WMS I) / Locations (WMS II) / Bar codes / Pallets		SUPERSEDED (wave 31)	31	
Inventory and warehouse management parameters — Transfer orders	Default in-transit warehouse, underdelivery, picking	PARTIAL-16 (transfer orders)	16	
Inventory and warehouse management parameters — Number sequences		COVERED-5	5	
Inventory and warehouse management parameters — Quarantine / Blocking		COVERED-16	16	
Inventory → Item model groups	Costing method, include physical value, post physical/financial inventory, negative physical/financial, quarantine management, registration/picking requirements, receiving/issue requirements, stocked product, FIFO date-controlled, backward from ship date	COVERED-8	8	Costing behavior wave 20
Inventory → Item groups	Posting defaults per group	COVERED-8	8	
Inventory → Dimension groups: Product dimension groups, Storage dimension groups, Tracking dimension groups (active, primary stocking, physical/financial inventory, for purchase prices/sales prices, blank receipt/issue allowed, coverage plan by dimension)		COVERED-8	8	
Inventory → Inventory dimension display settings (Dimensions display)		COVERED-5	5	Saved views
Inventory → Reservation hierarchies (WHS)		COVERED-31	31	
Inventory → Unit sequence groups (WHS)		COVERED-31	31	
Inventory → Inventory statuses / Inventory status transitions (D365)		COVERED-16	16	
Inventory → Posting → Inventory posting profiles (Purchase order, Sales order, Inventory, Production, Standard cost variance tabs; by item code/account code)	Full inventory posting matrix	COVERED-16	16	Standard cost variance accounts in 39

Name	Key parameters / options	Passage status	Wave	Notes
Inventory → Posting → Item posting groups (link)		COVERED-16	16	
Inventory → Journals → Journal names, inventory (type, voucher, posting restrictions, reserve on post)		COVERED-16	16	
Inventory → Journals → Warehouse journal names (arrival, production input)		COVERED-18	18	
Inventory → Counting → Counting groups (periodic/zero/positive/negative/manual counting codes, interval)		COVERED-16	16	Cycle counting plans
Inventory → Counting → Counting reason code policies		PARTIAL-11 (reason codes)	11	
Inventory → Batch → Batch attributes, Batch attribute groups, Product-specific batch attributes, Customer-specific batch attributes		COVERED-24	24	
Inventory → Batch → Batch disposition master		COVERED-24	24	
Inventory → Batch → Shelf life setup on item (shelf advice period, best before, expiry), Batch number groups (manual/auto, by date)		COVERED-24	24	
Inventory → Serial → Serial number groups (auto numbering rules)		COVERED-16	16	
Inventory → Warehouses (Warehouse details: type, site, quarantine warehouse, transit warehouse, WHS flags, default locations, location format, reservation)		COVERED-6	6	WHS flags wave 31
Inventory → Sites		COVERED-6	6	
Inventory → Locations (WMS II) / Location names / Location types / Store areas / Zones / Aisles	Legacy	SUPERSEDED (location model wave 31)	31	
Inventory → Pallets → Pallet types / Pallet IDs / Pallet transports setup (WMS II)		SUPERSEDED (license plates wave 31)	31	
Inventory → Bar codes → Bar code setup (types: EAN13, Code 39, Code 128, Interleaved 2 of 5, UPC-A…), Bar code mask characters, Bar code masks		COVERED-8	8	GS1 identifiers
Inventory → Quarantine → Quarantine zones (link to Quality)		COVERED-33	33	
Inventory → Inventory blocking reasons		COVERED-16	16	
Inventory → Distribution → Item allocation keys (link to planning)		COVERED-22	22	
Inventory → Forecast models / Forecast plans (link)		COVERED-22	22	
Inventory → ABC classification setup (ABC calculation parameters)		PARTIAL-16 (ABC classification)	16	
Inventory → Catch weight → Catch weight item setup, CW units, nominal/min/max weights		COVERED-24	24	
Inventory → Transfer orders → Transfer order parameters, In-transit warehouses		PARTIAL-16 (transfer orders)	16	
Inventory → Consignment → Consignment parameters (D365)		GAP	—	Propose wave 16: consignment
Inventory → Country of origin / Product origin setup (PI)		COVERED-24	24	
Inventory → Dimension-based cost price (Cost by site/warehouse)		COVERED-20	20	
Inventory → Unit of measure (link) / Unit conversions (standard, inter-class, intra-class, product-specific)		COVERED-6	6	
Inventory → Charges on inventory (Misc. charges on journals)		COVERED-16	16	
Inventory → Inventory visibility add-in setup (D365)		SUPERSEDED (native on-hand projection + public API wave 35)	35	
Workflow → Inventory workflows (inventory journal approval)		COVERED-16	16	
Number sequences (inventory)		COVERED-5	5	
Shipping carriers (Shipping carrier interface, AX 2012 carrier integration: UPS/FedEx via shipping software)		COVERED-28	28	Carrier integration abstraction
Delivery → Modes of delivery / Terms of delivery / Transport (link)		COVERED-28	28	
Inventory → Picking → Picking routes setup / Picking workbench (R3 sales picking workbench)	Batch create picking lists by filters	PARTIAL-28 (picking workbench (grouped picking list generation across sales orders, non-WHS) is not named; wave/batch picking is)	28	Gap register

12.15 Product information management

Products, product masters, variants, released products, dimension groups, attributes, categories, BOMs/formulas (also §12.22/§12.23), configuration — wave 8 and wave 9 — with product compliance (PI) a gap.

12.15.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Products (all products and product masters, shared)	EcoResProductListPage / EcoResProductDetails	Tenant-level product definitions	COVERED-8	8	
Product masters	EcoResProductMasterListPage	Masters with dimension groups and configuration technology (predefined variant / dimension-based / constraint-based)	COVERED-8	8	
Product variants	EcoResProductVariantListPage (EcoResDistinctProductVariant)	Variants per master	COVERED-8	8	1,000 variants in < 5 s
Released products (list page)	EcoResProductDetailsExtended (InventTable)	Per legal entity: general, purchase, sell, manage inventory, engineer, plan, manage projects, manage costs, financial dimensions, retail, warehouse, transportation, product variants	COVERED-8	8	Site/warehouse defaults

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Released product variants	InventDimCombination	Variant per entity with price/cost	COVERED-8	8	
Released product → Default order settings / Site specific order settings	InventItemSetup (InventItemPurchSetup/SalesSetup/InventSetup)	Default site/warehouse, order qty (min/max/multiple/standard), lead time, stopped	COVERED-8	8	
Released product → Item coverage (link)	ReqItemTable		COVERED-22	22	
Released product → BOM versions / Formula versions / Route versions	BOMVersion, RouteVersion	Approved/active versions per item	COVERED-9	9	
Released product → External item descriptions (customer/vendor item numbers)	CustVendExternalItem		COVERED-8	8	
Released product → Bar codes / GTIN	InventItemBarcode / InventItemGTIN		COVERED-8	8	
Released product → Unit conversions (product-specific)	UnitOfMeasureConversion		COVERED-6	6	
Released product → Translations (product name/description per language)	EcoResProductTranslation		COVERED-8	8	Translations via metadata
Released product → Product attributes / Attribute values	EcoResProductAttributeValue	Attribute values by category	COVERED-8	8	
Released product → Product categories assignment	EcoResProductCategory		COVERED-8	8	
Released product → Related products	EcoResProductRelation	Up-sell/cross-sell/accessory	PARTIAL-8 (related products (cross/up-sell relations) not named)	8	Gap register
Released product → Product images / documents	DocuRef		COVERED-8	8	
Released product → Potency / Active ingredient (PI)	PdsIngredient	Base attribute, potency item	COVERED-24	24	
Released product → Catch weight (PI)	PdsCatchWeightItem		COVERED-24	24	
Released product → Planning items / Co-products / By-products (PI)	PdsCoProduct		COVERED-24	24	
Released product → Commodity pricing (PI: Pricing calculation, Price margins, Commodity codes)	PdsCommodity*	Cost basis types, price calculation, margins	PARTIAL-26 (commodity price index-based pricing for process industries (PI commodity pricing) not named)	26	Gap register
Released product → Product compliance (PI: Regulated products, Restricted lists, Reporting details, Safety data sheets (MSDS/SDS) with validity and send on sales)	PdsProductComplianceTable, PdsMSDS	Hazardous material compliance	GAP	—	Propose wave 33: product compliance (regulated/restricted lists by country/state, SDS versions with shipment delivery rules, reporting thresholds)
Released product → Sequencing (PI sequence groups)	PdsSequence	Sequence attributes for scheduling	COVERED-23	23	Setup optimization
Released product → Pricing (Item price, Price simulation link)	InventItemPrice		COVERED-26	26	Cost in 39
Released product → Charges (Item charges groups)			COVERED-17	17	
Released product → Product lifecycle state (D365)	EcoResProductLifecycleState	Active/obsolete states controlling transactions	PARTIAL-8 (product lifecycle states with per-process blocking not named; stopped flags are)	8	Gap register
Product configuration → Product configuration models (R2+: constraint-based)	PCProductConfigurationModel	Components, attributes, constraints, calculations, BOM/route lines, UI	COVERED-8	8	Configurator foundation (OR-Tools/MiniZinc)
Product configuration → Configure line (configure from order)	PCRuntimeConfigurator	Interactive configuration	COVERED-27	27	Order with configured product
Product configuration → Product models (legacy Product builder, deprecated R2)	PBATable		OUT-OF-SCOPE (deprecated in AX 2012 R2; constraint-based configurator replaces)	—	
Product configuration → Configuration variants / Dimension-based configuration (configurations)	ConfigTable	Config dimension values	COVERED-8	8	
Categories → Category hierarchies (Procurement, Sales, Commodity code, Retail, Project)	EcoResCategoryHierarchy	Hierarchies with attributes	COVERED-8	8	
Categories → Category hierarchy roles	EcoResCategoryHierarchyRole		COVERED-8	8	
Bills of materials (list page, BOM designer)	BOMTable, BOMDesigner		COVERED-9	9	
Formulas (PI)	BOMTable (formula)		COVERED-9	9	Formula BOMs
Routes (link)	RouteTable		COVERED-9	9	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Product change cases (case management for products)	CaseDetail (product change)	Link cases to product/BOM changes	COVERED-9	9	ECR/ECO
Engineering change (D365 ECM link)			COVERED-9	9	§12.34
Product → Product readiness checks (D365)	EcoResProductReadiness	Pre-release checklists per legal entity	PARTIAL-9 (product readiness policies/checks not named)	9	Gap register
Item sales tax groups / Tax on product			COVERED-13	13	
Product → Packaging (D365 packaging materials, PI container packaging)		Packaging material for fees	PARTIAL-28 (packaging materials and packaging fees (EU packaging duty) not named)	28	Gap register

12.15.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Released products inquiry / Products by category		COVERED-8	8	
Product variants inquiry		COVERED-8	8	
Where-used (BOM where-used)		COVERED-9	9	
BOM lines / BOM versions inquiry		COVERED-9	9	
Configuration inquiries (Configured products, Configuration history)		COVERED-8	8	
External item descriptions inquiry		COVERED-8	8	
Product attributes inquiry / Attribute search		COVERED-8	8	
Product compliance inquiries (Regulated items, SDS due)		GAP	—	Propose wave 33: product compliance
SSRS: Item base data / Released product list		COVERED-8	8	
SSRS: BOM list / BOM lines / BOM where-used / BOM versions		COVERED-9	9	
SSRS: Formula report (PI)		COVERED-9	9	
SSRS: Item price list / Price list by customer		COVERED-26	26	
SSRS: Bar code labels		COVERED-32	32	
SSRS: Product configuration model / Configuration report		COVERED-8	8	
SSRS: Category hierarchy report		COVERED-8	8	
SSRS: Safety data sheet (SDS document, PI)		GAP	—	Propose wave 33: product compliance
SSRS: Product images catalog		COVERED-46	46	
SSRS: Unit conversions list		COVERED-6	6	

12.15.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Release products (to legal entities, mass release)		COVERED-8	8	
Product → Change product dimension groups / Change item model group (periodic conversions)		PARTIAL-8 (changing dimension/model groups on items with transactions (conversion jobs) not named)	8	Gap register
Product → Generate variants (from dimension values)		COVERED-8	8	
Product → Update default order settings (mass)		COVERED-8	8	
BOM → Copy BOM / Change BOM item (mass replace component)		COVERED-9	9	
BOM → Approve/activate versions (batch)		COVERED-9	9	
BOM → Calculate (BOM calculation, link)		COVERED-39	39	
Product configuration → Model versioning / Reuse configurations / Clean up configurations		COVERED-8	8	
Product → Import products / variants / attributes (DIXF)		COVERED-10	10	
Product → Update product translations (batch)		COVERED-8	8	
Product compliance → Expire SDS / Reporting (periodic)		GAP	—	Propose wave 33: product compliance
Product → Delete released products / Mark obsolete		PARTIAL-8 (lifecycle states)	8	
Categories → Update category attributes / Reassign products		COVERED-8	8	
Product → Price update (Item price copy / Trade agreement bulk update)		COVERED-26	26	

12.15.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Product information management parameters	Product dimension defaults, variant number nomenclature (D365), released product defaults, number sequences	COVERED-8	8	
Product → Product dimension groups / Storage dimension groups / Tracking dimension groups		COVERED-8	8	
Product → Product dimensions: Sizes, Colors, Styles, Configurations (shared value lists)		COVERED-8	8	
Product → Product nomenclature / Variant nomenclature (D365)	Naming rules for variants	COVERED-8	8	
Product → Item model groups (link) / Item groups (link)		COVERED-8	8	

Name	Key parameters / options	Passage status	Wave	Notes
Product → Product lifecycle states (D365)		PARTIAL-8 (lifecycle states)	8	
Product → Product types / Product subtypes (item vs service; product vs master)		COVERED-8	8	
Product → Units (Units, Unit classes, Unit conversions)		COVERED-6	6	
Product → Attributes: Attribute types, Attributes, Attribute groups (D365 product attributes), Category attributes		COVERED-8	8	
Product → Category hierarchies / Category hierarchy roles		COVERED-8	8	
Product → Bar code setup / Bar code masks / GTIN setup		COVERED-8	8	
Product → Product configuration models setup (Attribute types, Components, Subcomponents, Tables constraints (system-defined/user-defined), Template constraints, BOM/route line templates, Calculation templates, Pricing rules, UI setup, Model versions, Product configuration parameters)		COVERED-8	8	
Product → Configuration groups (dimension-based config) / Configuration rules (BOM config rules, legacy)		COVERED-8	8	
Product → Default order settings templates		COVERED-8	8	
Product → External item description setup		COVERED-8	8	
Product → Item sales tax groups default / Charges groups default		COVERED-13	13	
Product → Packaging units / Packing material fees setup (EU)		PARTIAL-28 (packaging materials)	28	
Product → Product compliance setup (PI: Regulated regions, Restricted lists, Reporting lists, SDS parameters, Product safety data sheet validity, Item-specific: MSDS validity days)		GAP	—	Propose wave 33: product compliance
Product → Country of origin setup / Product origin rules (PI)		COVERED-24	24	
Product → Commodity pricing setup (Cost basis types, Pricing calculations, Price margin templates, Commodity codes)		PARTIAL-26 (commodity pricing)	26	
Product → Potency / Base attribute setup / Batch attributes (link)		COVERED-24	24	
Product → Catch weight setup (link)		COVERED-24	24	
Product → Sequencing setup (Sequences, Sequence groups)		COVERED-23	23	
BOM → BOM versions parameters (approval required, activation rules), BOM calculation groups (link), BOM designer parameters		COVERED-9	9	
BOM → Scrap / Rounding-up (BOM line types: item, phantom, pegged supply, vendor)		COVERED-9	9	Phantom, subcontract (vendor line type) in 21
Engineering → Engineering change management parameters (D365, link)		COVERED-9	9	
Product → Product readiness policies (D365)		PARTIAL-9 (product readiness)	9	
Product → Product master → Product search / Product image settings		COVERED-8	8	
Product → Released product templates (record templates)		COVERED-5	5	Record templates in shell
Product → Workflow (product release/change approval)		COVERED-4	4	
Number sequences (products, variants, BOMs)		COVERED-5	5	
Product → Retail product attributes / Retail categories		OUT-OF-SCOPE (retail non-goal)	—	

12.16 Procurement and sourcing

Requisitions, RFQs, agreements, POs, purchasing policies, categories and vendor evaluation map to wave 17; receiving to 18; catalogs are partially covered; external catalogs (punchout), vendor rebates and broker contracts are gaps.

12.16.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Purchase requisitions (list page: all, prepared by me, assigned to me)	PurchReqTableListPage	Requisition lifecycle (draft, in review, approved, closed)	COVERED-17	17	
Purchase requisition details	PurchReqTable / PurchReqLine	Lines from catalog/non-catalog, requester, accounting distribution, approval	COVERED-17	17	
Purchase requisitions → Release approved purchase requisitions (workbench)	PurchReqReleaseApproved	Consolidate and create POs	COVERED-17	17	Consolidation into POs
Purchase requisitions → Consolidation opportunities	PurchReqConsolidation	Group lines for better pricing	COVERED-17	17	
Requests for quotations (list page)	PurchRFQCaseTableListPage	RFQ cases	COVERED-17	17	
RFQ details / RFQ replies (Vendor replies) / Compare replies	PurchRFQCaseTable, PurchRFQReplyTable, PurchRFQCompare	Send, receive, compare, accept/reject	COVERED-17	17	Supplier reply portal
RFQ → Bidding (Vendor portal RFQ reply, sealed bids, scoring criteria R3)	PurchRFQVendorPortal	Scoring, sealed bidding, amendments	PARTIAL-17 (sealed bids, scoring criteria and amendments/addenda not named)	17	Gap register

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Purchase agreements (list page)	PurchAgreementListPage	Blanket with commitment types (product qty/value, category value, total value), release lines	COVERED-17	17	
Purchase agreement classifications	AgreementClassification		COVERED-17	17	
Purchase orders (list page: all, received not invoiced, etc.)	PurchTableListPage		COVERED-17	17	
Purchase order details	PurchTable	Header/lines, delivery schedule, pricing, charges, dimensions, change management, intercompany	COVERED-17	17	Intercompany 40
Purchase orders → Purchase order change requests / Versions (change management)	PurchTableVersion	Versioned, re-approval	COVERED-17	17	
Purchase orders → Purchase order confirmation (vendor confirmation via collaboration)	PurchConfirmation		COVERED-20	20	
Purchase orders → Delivery schedule on line	PurchLineDeliverySchedule		COVERED-17	17	
Purchase orders → Direct delivery (drop ship from sales)	PurchTable (direct delivery)		COVERED-28	28	Drop-ship
Purchase orders → Subcontracting (purchase line from production)	PurchLine (subcontract)		COVERED-21	21	
Vendors (list page, link)	VendTableListPage		COVERED-7	7	
Procurement catalogs	CatCatalog (ProcurementCatalog)	Internal catalog of products per category	PARTIAL-17 (requisitions and category policies are named; an internal procurement catalog with product images, category navigation, catalog publication by org unit is not)	17	Gap register
External catalogs (punchout cXML/OCI)	CatExternalCatalog	Vendor punchout	GAP	—	Propose wave 35: external catalog punchout (cXML/OCI) via integration hub, requisition lines returned to wave 17
Vendor catalogs (vendor catalog import, CMR files)	CatVendorCatalog (CatCMR)	Vendor-maintained product catalog import	PARTIAL-20 (vendor catalog upload via supplier portal not named)	20	Gap register
Procurement categories (category hierarchy for procurement)	EcoResCategory (procurement role)		COVERED-8	8	
Vendor requests (link)	VendRequest		PARTIAL-20 (vendor request workflow)	20	
Vendor evaluation (R3 vendor rating / criteria)	VendEvaluation	Rate vendors per criteria at receipt/invoice	COVERED-33	33	Supplier scorecards
Vendor rebates (PI vendor rebate agreements, R3)	PdsRebateAgreement (vendor)	Rebate agreements with vendors, claims, accruals	GAP	—	Propose wave 26: vendor rebates (agreements, accrual on purchase, claim/credit note settlement via AP) mirroring customer rebates
Broker contracts (PI brokers and royalties, R3)	PdsBrokerContract	Broker commissions on sales, claim via AP	GAP	—	Propose wave 26: broker contracts (commission to broker vendor on sales invoices, claims via AP)
Royalty agreements (PI, R3)	PdsRoyaltyAgreement	Royalties payable on sales to licensors	GAP	—	Propose wave 26: royalty agreements (accrual on sales, claim to vendor)
Consignment (link)			GAP	—	Propose wave 16: consignment
Purchase orders → Product receipt / Receipts list (links)			COVERED-18	18	
Purchase orders → Invoice (link)			COVERED-18	18	
Purchase orders → Charges / Charges allocation	MarkupTrans		COVERED-17	17	
Purchase orders → Prepayment (link)			COVERED-19	19	
Purchase inquiries (Purchase inquiry documents, R3 "purchase inquiry" journal)	PurchInquiry (VendPurchOrderJour inquiry type)	Price inquiry documents to vendors	COVERED-17	17	RFQ covers
Procurement → Purchasing policies (link)	PurchPolicy		COVERED-17	17	
Procurement → Signing limits (link to Org admin)	SigningLimit		PARTIAL-17 (signing limits (self-service limit requests per job/compensation) not named; amount-band policies are)	17	Gap register
Procurement → Workflow work items (link)			COVERED-4	4	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Procurement → Vendor collaboration (D365 PO confirmation, invoices, consignment)			COVERED-20	20	
Procurement → Vendor onboarding (D365 prospective vendor registration)	VendProspectiveVendorRegistration	External registration → request → vendor	PARTIAL-20 (vendor onboarding)	20	
Procurement → Vendor certifications / Vendor diversity (D365)	VendCertification	Certifications with expiry, diversity classifications	PARTIAL-7 (vendor certifications/diversity classifications not named)	7	Gap register
Procurement → Vendor contacts / Vendor users (portal)			COVERED-20	20	

12.16.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Purchase order inquiries (POs, lines, history, versions, backorders, open lines)		COVERED-17	17	
Purchase requisition inquiries (by requester, status, consolidation)		COVERED-17	17	
RFQ inquiries (RFQ status, replies, award history)		COVERED-17	17	
Purchase agreement inquiries (Fulfillment / released quantities)		COVERED-17	17	
Vendor evaluation inquiry / Vendor performance		COVERED-33	33	
Journals → Confirmations / Receipts lists / Product receipts / Invoices		COVERED-18	18	
Charges inquiry		COVERED-17	17	
Procurement category spend inquiry (Spend analysis cube)		COVERED-46	46	
Vendor rebate inquiries		GAP	—	Propose wave 26: vendor rebates
Broker/royalty claim inquiries		GAP	—	Propose wave 26: broker contracts
SSRS: Purchase order (document), Purchase order confirmation, PO status, PO lines		COVERED-17	17	
SSRS: Purchase requisition (document) / Requisition status		COVERED-17	17	
SSRS: Request for quotation (document) / RFQ comparison / RFQ reply		COVERED-17	17	
SSRS: Purchase agreement (document) / Agreement fulfillment		COVERED-17	17	
SSRS: Vendor evaluation report / Vendor performance (on-time, quality)		COVERED-33	33	
SSRS: Purchase analysis (Spend by vendor/category/period, Purchase statistics)		COVERED-46	46	
SSRS: Backorder lines / Overdue deliveries		COVERED-17	17	
SSRS: Procurement catalog report		PARTIAL-17 (procurement catalog)	17	
SSRS: Vendor rebate reports		GAP	—	Propose wave 26: vendor rebates
SSRS: Purchase price variance (link)		COVERED-39	39	
SSRS: Policy violations (purchasing policy)		COVERED-17	17	

12.16.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Purchase requisitions → Release approved requisitions (batch auto-create POs)		COVERED-17	17	
Purchase requisitions → Expire / Clean up requisitions		COVERED-17	17	
Purchase orders → Confirm / Receipts list / Product receipt / Invoice (batch posting)		COVERED-17	17	
Purchase orders → Update delivery dates / Finalize / Cancel remaining		COVERED-17	17	
Purchase orders → Firm planned purchase orders (link)		COVERED-22	22	
Purchase agreements → Update agreement release status / Expire		COVERED-17	17	
RFQ → Send RFQ (batch), Close bidding, Purchase order from accepted reply		COVERED-17	17	
Catalog → Import vendor catalogs (CMR) / Publish procurement catalog		PARTIAL-17 (procurement catalog)	17	
Vendor evaluation → Calculate vendor ratings		COVERED-33	33	
Vendor rebates → Cumulate / Approve / Process claims		GAP	—	Propose wave 26: vendor rebates
Broker → Process broker claims / Royalty → Process royalty claims		GAP	—	Propose wave 26: broker contracts
Procurement → Vendor collaboration: provision users, send PO (batch)		COVERED-20	20	
Procurement → Policy evaluation (re-evaluate policies on open requisitions)		COVERED-17	17	
Procurement → Delete purchase orders / Clean up update history		COVERED-17	17	
Procurement → EDI 850 outbound (batch)		COVERED-35	35	

12.16.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Procurement and sourcing parameters — General	Default order type, create direct delivery, safety levels, intercompany PO defaults, delivery date control	COVERED-17	17	
Procurement and sourcing parameters — Purchase requisitions / Purchase agreements / RFQ / Catalog / Vendor collaboration / Number sequences		COVERED-17	17	Catalog tab PARTIAL
Policies → Purchasing policies (policy rules: catalog policy, category access policy, category policy, purchase agreement policy, purchase order creation and demand consolidation, purchase requisition control, RFQ policy, re-approval rule for POs)	Policy framework with org hierarchy scope	COVERED-17	17	Amount band/category policies

Name	Key parameters / options	Passage status	Wave	Notes
Policies → Purchase requisition approval amount thresholds / Signing limit policies		PARTIAL-17 (signing limits)	17	
Policies → Purchasing policy organization hierarchy purpose		COVERED-6	6	Hierarchy purposes
Categories → Procurement categories, Category policy rules, Category managers, Category vendor assignment, Category questionnaires		PARTIAL-8 (category-level vendor assignment and category questionnaires not named)	17	Gap register
Catalogs → Procurement catalogs, Catalog policies, External catalogs (cXML config), Vendor catalog import parameters, Catalog images		PARTIAL-17 (procurement catalog)	17	External catalogs GAP (wave 35)
Vendors → Vendor evaluation criterion groups / criteria		COVERED-33	33	
Vendors → Vendor request types / Vendor onboarding questionnaires (D365)		PARTIAL-20 (vendor request workflow)	20	
Vendors → Vendor certification types / Diversity classifications		PARTIAL-7 (vendor certifications)	7	
RFQ → RFQ types, RFQ scoring criteria/groups, Reply templates, Solicitation types (D365), Bidding parameters		PARTIAL-17 (sealed bids/scoring)	17	
Purchase agreements → Agreement classifications, Commitment types		COVERED-17	17	
Purchase orders → Order types, Purchase order change management parameters per vendor, PO approval workflow		COVERED-17	17	
Charges → Charges codes / Auto charges / Charges groups (link)		COVERED-17	17	
Prices and discounts → Trade agreement journal names, Price/discount groups (vendor), Activate price/discount, Purchase price tolerance, Vendor price search parameters (vendor > group > all)		COVERED-26	26	Wave 17 uses pricing engine for vendor prices
Vendor rebates → Rebate agreements, Rebate program types, Rebate groups (vendor), Rebate claim journal names		GAP	—	Propose wave 26: vendor rebates
Brokers and royalties → Broker contract setup, Royalty agreement setup, Claim journal names		GAP	—	Propose wave 26: broker contracts
Delivery → Modes of delivery, Terms of delivery, Delivery schedule parameters		COVERED-28	28	
Forms → Form setup (purchase documents) / Form notes / Print management (purchase)		PARTIAL-46 (print management routing)	46	
Vendor collaboration → Vendor collaboration parameters, Consignment parameters		COVERED-20	20	Consignment GAP wave 16
Workflow → Procurement workflows (requisition line/header, PO, agreement, RFQ)		COVERED-4	4	
Number sequences (procurement)		COVERED-5	5	
Intercompany → Intercompany purchase order setup (link)		COVERED-40	40	
Procurement → Unit of measure defaults per category		COVERED-6	6	
Procurement → Receiving: Registration/receipt requirements on item model group (link)		COVERED-18	18	
Procurement → EDI partner setup (link)		COVERED-35	35	

12.17 Sales and marketing

Sales quotations, orders, agreements, delivery and returns map to waves 27–30 (orders/ATP/credit 27, fulfillment 28, invoicing 29, returns 30) with pricing in 26. The CRM half (leads, opportunities, campaigns, telemarketing, mailings, activities, responsibilities) is out of scope per the PRD non-goal; Passage integrates with Salesforce/HubSpot via wave 35 connectors.

12.17.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Sales orders (list page: all, open, backorders, delivered not invoiced, recent)	SalesTableListPage		COVERED-27	27	
Sales order details	SalesTable / SalesLine	Header/lines, delivery, price/discount, charges, dimensions, holds, intercompany, project	COVERED-27	27	
Sales order → Order types (journal, subscription, sales order, returned order, item requirements)	SalesTable.SalesType		COVERED-27	27	
Sales order → Delivery date control (ATP, CTP, sales lead time, none)	SalesLine (dlv date control)		COVERED-27	27	
Sales order → Credit limit check / Order holds	SalesTableHold		COVERED-27	27	
Sales order → Order events / Order history	SalesOrderEvent		COVERED-27	27	Audit
Sales order → Configure line / Product configuration	PCRuntimeConfigurator		COVERED-27	27	
Sales order → Direct delivery / Drop ship to vendor	SalesLine (direct delivery) → PurchTable		COVERED-28	28	
Sales order → Alternative items (substitutes)	InventItemAlternative	Alternate when out of stock	COVERED-27	27	
Sales order → Supplementary items (prompt)	PriceDiscSuppItem		PARTIAL-26 (supplementary items)	27	
Sales order → Charges / Charges allocation	MarkupTrans		COVERED-26	26	
Sales order → Prepayment			COVERED-30	30	
Sales order → Confirmation / Picking list / Packing slip / Invoice posting	SalesEditLines		COVERED-28	28	Invoice 29; confirmation 27
Sales order → Reservation	InventTransReserve		COVERED-16	16	
Sales order → Delivery schedule (split line into schedule)	SalesLineDeliverySchedule		COVERED-28	28	
Sales order → Blanket / Sales agreements	SalesAgreementListPage / SalesAgreement	Commitment types, release orders	COVERED-27	27	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Sales order → Release orders from agreement	SalesAgreementRelease		COVERED-27	27	
Sales order → Intercompany chain (direct delivery chains)	InterCompany*		COVERED-40	40	
Sales order → Backorder lines / Backorder overview	SalesBackorderLines		COVERED-27	27	
Sales order → Item requirements (project)	SalesTable (item req)		COVERED-41	41	
Sales order → Packing (packing form, HU)			COVERED-28	28	
Sales order → Enter shipping details / Carrier			COVERED-28	28	
Sales quotations (list page)	SalesQuotationListPage	Quotes with prospects/customers, win/lose, convert	COVERED-27	27	
Sales quotation details	SalesQuotationTable	Lines, expiry, follow-up, templates, alternative quotes, price simulation	COVERED-27	27	Price simulation in 26
Sales quotation → Quotation templates / Quotation journals (confirm, send, lose, cancel)	SalesQuotationTemplate, SalesQuotationJour		COVERED-27	27	
Sales quotation → Mass create quotations (campaign)	SalesQuotationMassCreate		OUT-OF-SCOPE (CRM non-goal: campaign-driven mass quotes)	—	
Return orders (list page)	ReturnTableListPage	RMA with reason, disposition, replacement	COVERED-30	30	
Return order details	ReturnTable (SalesTable returned)	Return lines, find sales order, disposition code, inspection, replacement order, credit	COVERED-30	30	
Return order → Arrival / Registration / Quarantine / Disposition	WMSArrivalOverview (return)		COVERED-30	30	
Customers (link) / Prospects	CustTableListPage / smmBusRelTable	Customers and prospects	COVERED-7	7	Prospect role in party model
Contacts	smmContactPersonListPage	Contact persons	COVERED-7	7	
Leads (R2+)	smmLeadTableListPage	Lead management	OUT-OF-SCOPE (CRM non-goal; integrate Salesforce/HubSpot wave 35)	—	
Opportunities (R2+)	smmOpportunityTableListPage	Opportunity pipeline	OUT-OF-SCOPE (CRM non-goal)	—	
Campaigns	smmCampaignTable	Campaign targets, responses	OUT-OF-SCOPE (CRM non-goal)	—	
Campaign targets / Mailings / Direct marketing	smmCampaignSelection, smmMailings		OUT-OF-SCOPE (CRM non-goal)	—	
Telemarketing / Call lists	smmTMCallListTable		OUT-OF-SCOPE (CRM non-goal)	—	
Activities (sales activities: tasks, appointments, actions, events)	smmActivities		OUT-OF-SCOPE (CRM non-goal; project/service activities handled by case framework)	—	
Responsibilities (sales responsibilities on customers/opportunities)	smmResponsibilities		OUT-OF-SCOPE (CRM non-goal)	—	
Sales management (sales targets, sales units, sales management statistics)	smmSalesUnit, smmSalesTarget		OUT-OF-SCOPE (CRM non-goal; targets via wave 46 analytics)	—	
Business relations (legacy prospects)	smmBusRelTable		COVERED-7	7	Prospect role
Competitors (on opportunities)	smmCompetitorTable		OUT-OF-SCOPE (CRM non-goal)	—	
Outlook synchronization (contacts/appointments)	smmOutlookSync		OUT-OF-SCOPE (CRM non-goal)	—	
Customer self-service (D365 customer portal link)			COVERED-30	30	
Trade allowance (link)			COVERED-26	26	§12.18
Sales → Price simulation (quotation / order)	PriceSimulation	What-if price and margin	COVERED-26	26	Price simulation and explain
Sales → Commissions (link)			PARTIAL-26 (commission posting)	26	
Sales → Sales order → Shipping carrier interface (R3)	ShipCarrier*	Rate/label via carrier software	COVERED-28	28	
Sales → Credit card authorization on order	CreditCardAuthTrans		COVERED-30	30	
Sales → Order intake API / Multi-channel (EDI 850 inbound, e-commerce)			COVERED-27	27	EDI in 35
Sales → Call center (link)			OUT-OF-SCOPE (call center)	—	§12.19

12.17.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Sales order inquiries (orders, lines, history, open lines, backorders)		COVERED-27	27	
Sales quotation inquiries (quotation lines, follow-up, win/loss)		COVERED-27	27	
Sales agreement fulfillment inquiry		COVERED-27	27	
Return order inquiries (return status, disposition history)		COVERED-30	30	
Journals → Confirmations / Picking lists / Packing slips / Invoices		COVERED-28	28	
Delivery → Backorders / Delivery performance (OTIF)		COVERED-46	46	OTIF metric
ATP inquiry (Available-to-promise per item/date)		COVERED-27	27	
CRM inquiries (Leads/Opportunities pipeline, Campaign results, Activities)		OUT-OF-SCOPE (CRM non-goal)	—	
SSRS: Sales order confirmation / Picking list / Packing slip / Invoice (documents)		COVERED-28	28	
SSRS: Sales quotation (document) / Quotation confirmation		COVERED-27	27	
SSRS: Sales agreement (document)		COVERED-27	27	
SSRS: Return order / Return acknowledgment (RMA document)		COVERED-30	30	
SSRS: Sales order status / Open sales orders / Backorder lines by item/customer		COVERED-27	27	

Name	Purpose	Passage status	Wave	Notes
SSRS: Sales analysis (Sales by customer/item/sales group, Top items, Gross margin)		COVERED-46	46	
SSRS: Price list (customer) / Price list by item		COVERED-26	26	
SSRS: Delivery performance / On-time delivery		COVERED-46	46	
SSRS: Campaign / Lead / Opportunity / Activity / Telemarketing reports		OUT-OF-SCOPE (CRM non-goal)	—	
SSRS: Mailings / Labels (customer address labels)		OUT-OF-SCOPE (CRM non-goal)	—	
SSRS: Sales statistics / Customer group statistics		COVERED-46	46	
SSRS: Sales tax on sales (link)		COVERED-13	13	
SSRS: Certificate of analysis on shipment (link)		COVERED-33	33	
SSRS: Commission report (link)		PARTIAL-26 (commission posting)	26	
SSRS: Shipping labels / Bill of lading		COVERED-28	28	BOL in 34

12.17.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Sales update → Confirmation / Picking list / Packing slip / Invoice (batch)		COVERED-28	28	
Sales update → Release sales order picking (Release to warehouse)		COVERED-28	28	
Sales update → Summary update		COVERED-29	29	
Sales update → Delivery date update / Backorder processing		COVERED-27	27	
Sales update → Delete orders / Clean up update history / Delete quotations		COVERED-27	27	
Sales quotations → Expire quotations / Mass update		COVERED-27	27	
Sales agreements → Update status / Expire		COVERED-27	27	
Return orders → Clean up expired RMA		COVERED-30	30	
Credit management → Update holds / Auto release (D365)		COVERED-27	27	
Campaigns → Campaign processing / Mailings		OUT-OF-SCOPE (CRM non-goal)	—	
Activities → Synchronize with Outlook		OUT-OF-SCOPE (CRM non-goal)	—	
Sales → EDI 855/856/810 outbound (batch)		COVERED-35	35	
Sales → Intercompany sales order creation		COVERED-40	40	
Sales → Order holds → Clear holds (batch)		COVERED-27	27	
Sales → Shipping carrier → Export shipments / Import tracking		COVERED-28	28	
Sales → Trade allowance processing (link)		COVERED-26	26	

12.17.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Sales and marketing parameters — General	Default quotation settings, prospect conversion, sales unit	COVERED-27	27	
Sales and marketing parameters — Quotations (expiry, follow-up, templates, probability)		COVERED-27	27	
Sales and marketing parameters — Leads / Opportunities / Campaigns / Activities / Telemarketing / Mailings / Outlook		OUT-OF-SCOPE (CRM non-goal)	—	
Sales and marketing parameters — Number sequences		COVERED-5	5	
Sales orders → Order types / Order hold codes / Order event setup / Sales order pools		COVERED-27	27	
Sales orders → Delivery date control (Sales lead time, Order deadlines, Transport days, Coverage calendars, Delivery date control defaults)		COVERED-27	27	
Sales orders → Order deadlines (cut-off times)		COVERED-27	27	
Sales orders → Transport days (shipping calendar between sites/warehouses/customers)		COVERED-27	27	
Sales orders → Customer item numbers / External item descriptions		COVERED-8	8	
Sales orders → Alternative items setup		COVERED-27	27	
Sales orders → Supplementary items		PARTIAL-26 (supplementary items)	27	
Sales agreements → Agreement classifications		COVERED-27	27	
Quotations → Quotation templates / Quotation types / Reasons (win/loss/cancel) / Quotation journal names / Probability		COVERED-27	27	
Returns → Return reason codes / Reason code groups / Disposition codes / Disposition actions (credit only, replace and credit, replace and scrap, return to customer, scrap)		COVERED-30	30	
Returns → RMA parameters (expiry, inspection warehouse)		COVERED-30	30	
Prices and discounts → Trade agreement journal names, Price/discount groups (customer, line discount, multiline discount, total discount, price group), Activate price/discount, Smart rounding, Discount combination rules (price/discount parameters), Price search (customer > group > all)		COVERED-26	26	
Prices and discounts → Price adjustment / Price list creation (customer price lists)		COVERED-26	26	Mass price updates
Charges → Charges codes (sales), Auto charges, Charges groups		COVERED-26	26	
Commissions → Commission calculation, Sales groups (sales reps), Commission customer groups, Commission item groups, Commission posting		PARTIAL-26 (commission posting)	26	
Delivery → Modes of delivery, Terms of delivery, Delivery charges groups, Shipping carriers, Carrier services (R3 shipping carrier interface: Carrier company, Carrier interface config)		COVERED-28	28	
Delivery → Destination codes (R3 Transportation link)		COVERED-34	34	
Customers → Customer groups / Posting profiles / Classification groups / Line of business / Segments / Subsegments / Sales districts (link)		COVERED-7	7	

Name	Key parameters / options	Passage status	Wave	Notes
Customers → Prospects setup (Business relation types, statuses, Conversion to customer templates)		COVERED-7	7	Prospect role
Contacts → Contact titles, Loyalty, Decision maker, Character, Contact responsibilities		COVERED-7	7	Contact role attributes
Leads → Lead types, Lead sources, Lead ratings, Lead statuses, Lead qualification stages		OUT-OF-SCOPE (CRM non-goal)	—	
Opportunities → Opportunity types, Sales phases, Probabilities, Reasons, Processes		OUT-OF-SCOPE (CRM non-goal)	—	
Campaigns → Campaign types, Campaign groups, Campaign media, Campaign processes, Response types, Mailing types, Target groups		OUT-OF-SCOPE (CRM non-goal)	—	
Activities → Activity types, Activity categories, Activity phases, Activity plans		OUT-OF-SCOPE (CRM non-goal; service activities in 45)	—	
Telemarketing → Call list types, Call statuses, Scripts		OUT-OF-SCOPE (CRM non-goal)	—	
Responsibilities → Responsibility types / Sales management: Sales units, Sales targets, Sales management periods		OUT-OF-SCOPE (CRM non-goal)	—	
Management statistics (sales statistics setup, Sales/management statistics cubes)		COVERED-46	46	
Document management → Mail merge / Document templates (sales letters)		OUT-OF-SCOPE (CRM non-goal; document templates for business documents handled in wave 46)	—	
Credit management (D365) setup (link)		COVERED-27	27	
ATP → ATP parameters (time fence, backward/forward consumption, include planned orders), CTP via master planning		COVERED-27	27	
Forms → Form setup (sales documents), Form notes, Print management (sales), Form sorting		PARTIAL-46 (print management routing)	46	
Workflow → Sales workflows (sales order approval D365, sales agreement)		COVERED-4	4	
Number sequences (sales)		COVERED-5	5	
Intercompany → Intercompany sales setup (link)		COVERED-40	40	
Sales → Customer self-service parameters (D365 portal)		COVERED-30	30	
Sales → E-commerce connector (Shopify/Medusa)		COVERED-35	35	Non-goal says integrate, not build

12.18 Trade allowance management

AX 2012 R3 TAM: trade allowance agreements (bill-back, lump sum, off-invoice), funds, customer categories, merchandising events, deductions workbench, claims via AR. Passage: promotions and rebate accruals in wave 26, settlement and deductions in wave 30. The deduction workbench and funds are only partially covered.

12.18.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Trade allowance agreements (list page)	TAMTradeAllowanceAgreementListPage	Agreements with merchandising events: bill-back, lump sum, off-invoice	COVERED-26	26	Promotions + rebate agreements
Trade allowance agreement details	TAMTradeAllowanceAgreement	Customers, items, period, funds, approval, lines (bill-back rebate lines, lump sum, off-invoice discounts)	COVERED-26	26	
Merchandising events (Promotion planning)	TAMMerchandisingEvent	Events with analysis (lift, baseline)	PARTIAL-26 (promotion planning with baseline/lift analysis not named; promotions are)	26	Gap register
Funds	TAMFund	Budget funds per customer/period with usage tracking	PARTIAL-26 (trade promotion funds with budget/usage tracking not named)	26	Gap register
Bill-back workbench (Rebate workbench)	TAMBillBackWorkbench (PdsRebate*)	Cumulate, approve, process, pay rebates	COVERED-30	30	Rebate settlement
Deduction workbench	TAMDeductionWorkbench	Match short-pays (deductions) to claims, write-off, deny, split	PARTIAL-30 (deductions workbench — matching customer short-pays to promotions, denial/write-off/split — not named; disputes workflow is)	30	Gap register
Lump sum payments (via AP claim / credit note)	TAMLumpSum		COVERED-30	30	
Customer rebate transactions (PdsRebateTable)	PdsRebateTable	Accrued rebate lines	COVERED-26	26	
Customer categories (Customer category hierarchy for TAM)	TAMCustomerCategory	Customer groupings for agreements	COVERED-7	7	Customer groups/attributes
Trade allowance → Sales order promotion application (promotion on order lines)	SalesLine (TAM)	Off-invoice applied at order	COVERED-26	26	

12.18.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Trade allowance agreement inquiry / Agreement status		COVERED-26	26	
Fund usage inquiry		PARTIAL-26 (funds)	26	
Rebate accrual inquiry / Bill-back transactions		COVERED-26	26	
Deduction inquiry (open deductions, matched)		PARTIAL-30 (deductions workbench)	30	
Promotion effectiveness analysis (lift)		PARTIAL-26 (promotion planning)	26	
SSRS: Trade allowance agreement (document)		COVERED-26	26	
SSRS: Fund statement / Fund usage		PARTIAL-26 (funds)	26	
SSRS: Bill-back / Rebate report		COVERED-30	30	

Name	Purpose	Passage status	Wave	Notes
SSRS: Deductions report		PARTIAL-30 (deductions workbench)	30	
SSRS: Customer promotions summary		COVERED-46	46	

12.18.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Cumulate rebates / Approve rebates / Process rebates (create credit notes or AP claims)		COVERED-30	30	
Deductions → Match deductions to claims / Write off small balances		PARTIAL-30 (deductions workbench)	30	
Funds → Fund budget recalculation / Fund close		PARTIAL-26 (funds)	26	
Agreements → Approve / Expire / Mass copy agreements		COVERED-26	26	
Lump sum → Generate lump sum claims		COVERED-30	30	
Promotion → Update promotion planning actuals		PARTIAL-26 (promotion planning)	26	

12.18.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Trade allowance management parameters	Default fund, rebate journal names, deduction journal, approval workflow, claim via AR/AP	COVERED-26	26	
Trade allowance → Customer categories / Customer category hierarchy		COVERED-7	7	
Trade allowance → Funds setup (fund types, budget periods)		PARTIAL-26 (funds)	26	
Trade allowance → Merchandising event categories / Event types		PARTIAL-26 (promotion planning)	26	
Trade allowance → Deduction types / Deduction reason codes / Denial reasons		PARTIAL-30 (deductions workbench)	30	
Trade allowance → Rebate program types / Rebate groups / Rebate journal names / Rebate posting accounts		COVERED-26	26	
Trade allowance → Lump sum setup		COVERED-30	30	
Trade allowance → Bill-back calculation basis (invoice/payment)		COVERED-26	26	
Trade allowance → Workflow (agreement approval)		COVERED-4	4	
Trade allowance → Number sequences		COVERED-5	5	

12.19 Call center

AX 2012 R3 Call center (part of Retail) is a catalog-and-script driven order-taking channel. Per the PRD non-goal (retail POS/e-commerce), it is out of scope except for the pieces that are generic order-desk capabilities already built in wave 27 (order entry, holds, payments).

12.19.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Call center sales orders (order entry with scripts, prices from catalog, payment capture)	SalesTable (call center)	Operator order entry	OUT-OF-SCOPE (retail/call-center channel non-goal; generic order desk in wave 27)	—	Wave 27 order entry covers manufacturer order desk
Customer service form (CustomerService)	MCRCustomerService	360° customer view for operators	SUPERSEDED (customer 360 workspace in wave 7/30 shell with FactBoxes)	30	
Catalogs (call center catalogs, catalog products, source codes)	MCRCatalogTable	Mail-order catalog with source codes	OUT-OF-SCOPE (retail non-goal)	—	
Scripts	MCRScript	Operator scripts	OUT-OF-SCOPE (retail non-goal)	—	
Continuity programs / Continuity orders	MCRContinuity*	Subscription shipments (book-of-the-month)	OUT-OF-SCOPE (retail non-goal; subscription billing wave 29 covers recurring invoicing)	—	
Installment plans / Payment schedules (call center)	MCRInstallment		OUT-OF-SCOPE (retail non-goal; payment schedules wave 7)	—	
Fraud / Order holds (fraud rules)	MCRFraud*	Fraud hold rules	SUPERSEDED (order holds (credit/fraud/manual) wave 27)	27	
Returns (call center RMA with return reason, disposition)			COVERED-30	30	
Direct mail / Direct marketing	MCRDirectMail		OUT-OF-SCOPE (CRM/retail non-goal)	—	
Price match / Price override journals	MCRPriceMatch		OUT-OF-SCOPE (retail non-goal)	—	
Coupons (call center coupons)	MCRCoupon		OUT-OF-SCOPE (retail non-goal; promotions wave 26 for B2B)	—	
Credit card payments / Tender on order			COVERED-30	30	
Customer → Call center customer details (RFM, source)	MCRCustTable		OUT-OF-SCOPE (retail non-goal)	—	

12.19.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Call center sales statistics (orders per operator, catalog performance)		OUT-OF-SCOPE (retail non-goal)	—	
Continuity / Installment inquiries		OUT-OF-SCOPE (retail non-goal)	—	
Fraud hold inquiry		SUPERSEDED (order holds inquiry wave 27)	27	
SSRS: Catalog / Source code analysis		OUT-OF-SCOPE (retail non-goal)	—	

Name	Purpose	Passage status	Wave	Notes
SSRS: Call center order status		COVERED-27	27	

12.19.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Continuity → Process continuity orders		OUT-OF-SCOPE (retail non-goal)	—	
Installments → Generate installment invoices		OUT-OF-SCOPE (retail non-goal)	—	
Fraud → Run fraud checks		SUPERSEDED (order holds wave 27)	27	
Direct mail → Generate mailing		OUT-OF-SCOPE (CRM non-goal)	—	
Call center → Clean up expired catalogs / coupons		OUT-OF-SCOPE (retail non-goal)	—	

12.19.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Call center parameters	Channel defaults, scripts, fraud, order holds, catalogs, payment	OUT-OF-SCOPE (retail non-goal)	—	
Call centers (channel setup, operators, payment methods)		OUT-OF-SCOPE (retail non-goal)	—	
Scripts / Script categories		OUT-OF-SCOPE (retail non-goal)	—	
Fraud → Fraud rules, Fraud scoring, Fraud statuses		SUPERSEDED (order hold codes/rules wave 27)	27	
Catalogs → Catalog source codes, Catalog categories		OUT-OF-SCOPE (retail non-goal)	—	
Continuity → Continuity programs, Schedules		OUT-OF-SCOPE (retail non-goal)	—	
Installment plans setup		OUT-OF-SCOPE (retail non-goal)	—	
Coupons setup		OUT-OF-SCOPE (retail non-goal)	—	
Price match setup / Reason codes		OUT-OF-SCOPE (retail non-goal)	—	
Payment → Call center payment methods / Credit card types		COVERED-30	30	
Direct marketing setup		OUT-OF-SCOPE (CRM non-goal)	—	

12.20 Retail

AX 2012 R3 Retail (Retail headquarters, POS, Commerce Data Exchange, channels, assortments, loyalty, retail scheduler) is out of scope per the PRD non-goal; Passage integrates with Shopify/Medusa via wave 35. Rows are listed so reviewers can see the exclusion is deliberate.

12.20.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Retail stores / Retail channels (store, online store, call center)	RetailChannelTable	Channel master	OUT-OF-SCOPE (retail non-goal)	—	
Registers / Devices / POS terminals	RetailTerminalTable		OUT-OF-SCOPE (retail non-goal)	—	
Retail product catalogs / Assortments	RetailCatalog, RetailAssortment		OUT-OF-SCOPE (retail non-goal)	—	
Retail statements (POS transaction posting)	RetailStatementTable	Post POS sales to AR/inventory	OUT-OF-SCOPE (retail non-goal; e-commerce orders arrive as sales orders via wave 35 connector)	—	
Retail transactions (POS transactions)	RetailTransactionTable		OUT-OF-SCOPE (retail non-goal)	—	
Loyalty (programs, cards, schemes, points)	RetailLoyalty*		OUT-OF-SCOPE (retail non-goal)	—	
Retail discounts / Price groups / Promotions (Retail pricing)	RetailPeriodicDiscount		OUT-OF-SCOPE (retail non-goal; B2B promotions wave 26)	—	
Gift cards	RetailGiftCard		OUT-OF-SCOPE (retail non-goal)	—	
Replenishment (Retail replenishment: buyer's push, cross docking, replenishment rules)	RetailReplenishment*	Store replenishment	OUT-OF-SCOPE (retail non-goal; warehouse replenishment wave 31)	—	
Retail workers / Commissions			OUT-OF-SCOPE (retail non-goal)	—	
Online store / Sharepoint storefront			OUT-OF-SCOPE (e-commerce storefront non-goal)	—	

12.20.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Retail sales reports (sales by store/register/hour, Top items)		OUT-OF-SCOPE (retail non-goal)	—	
Statement inquiries / Transaction inquiries		OUT-OF-SCOPE (retail non-goal)	—	
Loyalty inquiries		OUT-OF-SCOPE (retail non-goal)	—	
Commerce Data Exchange status (sync inquiry)		OUT-OF-SCOPE (retail non-goal)	—	
Retail cubes (Retail BI)		OUT-OF-SCOPE (retail non-goal)	—	

12.20.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Retail scheduler (A/N jobs: distribution schedule, upload/download sessions)	Commerce Data Exchange sync jobs	OUT-OF-SCOPE (retail non-goal)	—	
Calculate / Post statements		OUT-OF-SCOPE (retail non-goal)	—	
Replenishment → Buyer's push / Cross docking		OUT-OF-SCOPE (retail non-goal)	—	
Loyalty → Process loyalty points / Expire		OUT-OF-SCOPE (retail non-goal)	—	
Retail → Publish catalog / Online store sync		OUT-OF-SCOPE (retail non-goal; Shopify connector wave 35)	—	

12.20.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Retail parameters / Retail shared parameters		OUT-OF-SCOPE (retail non-goal)	—	
Retail scheduler parameters / Channel database / Commerce Data Exchange		OUT-OF-SCOPE (retail non-goal)	—	
POS → POS profiles (functionality, receipt, hardware, visual, offline), Screen layouts, Operations, Permission groups		OUT-OF-SCOPE (retail non-goal)	—	
Store → Store setup, Registers, Hardware stations, Payment methods per store, Tender types, Info codes, Reason codes (retail)		OUT-OF-SCOPE (retail non-goal)	—	
Pricing → Retail price groups, Discount setup, Affiliations		OUT-OF-SCOPE (retail non-goal)	—	
Loyalty setup		OUT-OF-SCOPE (retail non-goal)	—	
Sales tax → Retail tax setup (destination-based, tax override)		COVERED-13	13	Tax engine handles destination-based
Retail hierarchy / Organization hierarchy for retail		COVERED-6	6	Hierarchy purposes
Online store → Storefront / Sharepoint site setup		OUT-OF-SCOPE (e-commerce storefront non-goal)	—	

12.21 Master planning

Static/dynamic plans, coverage, planned orders, action/futures messages, forecasting, safety stock and intercompany planning map to wave 22 (MRP/MPS/forecasting) with capacity in 23 and intercompany planning in 40. D365 Planning Optimization and priority-based planning are superseded by the Passage planning engine.

12.21.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Planned orders (list page: all, planned purchase, planned production, planned transfer, planned kanban)	ReqPOListPage	Review/firm/change planned orders	COVERED-22	22	
Planned order details	ReqPO	Qty, dates, pegging (requirement profile), action/futures messages, approve, firm, group	COVERED-22	22	
Planned orders → Firm (to PO/production/transfer/kanban)	ReqTransFirm	Firming with grouping (by vendor, period, buyer group)	COVERED-22	22	
Planned orders → Change planned orders (mass change)	ReqPOChange	Bulk date/qty	COVERED-22	22	
Net requirements (per item)	ReqTransOverview	Requirements/supply with pegging per plan	COVERED-22	22	Explain-this-plan pegging tree
Explosion (per order/line)	ReqTransExplosion	Multi-level explosion tree with pegging and futures	COVERED-22	22	
Action messages (list)	ReqTransAction	Advance/postpone/decrease/increase/cancel/derived actions	COVERED-22	22	
Futures messages / Delay messages (list)	ReqTransFutures	Delays and calculated dates	COVERED-22	22	
Supply schedule (R3)	ReqSupplySchedule	Period-bucketed supply/demand view with planned order creation	PARTIAL-22 (period-bucketed supply schedule workbench not named; planning engine and planned orders are)	22	Gap register
Demand forecast (Forecast → Demand forecast / Sales forecast lines)	ForecastSales	Forecast lines by item/model/allocation key	COVERED-22	22	
Supply forecast (Purchase forecast)	ForecastPurch		COVERED-22	22	
Inventory forecast	ForecastInvent? (InventForecast)	On-hand level forecast	PARTIAL-22 (inventory (stock level) forecast type not named; demand/supply forecasts are)	22	Gap register
Demand forecasting (R3: Generate statistical baseline forecast, Adjusted demand forecast workbook, Forecast accuracy)	ReqDemPlan*	Azure ML/SSAS baseline forecasts with Excel adjustment	COVERED-22	22	ETS/Croston service with override and consensus
Master plans (list)	ReqPlanSched	Plans: static, dynamic, others	COVERED-22	22	
Item coverage (per item/dimension)	ReqItemTable	Coverage group override, min/max, lead times, planned order type, vendor	COVERED-22	22	
Safety stock journal	ReqSafetyStockJournal	Proposed min levels from history	COVERED-22	22	Safety stock
Planned production orders → Scheduling (Gantt from planned orders)			COVERED-23	23	
Intercompany planned orders (R3 intercompany master scheduling)	ReqPO (intercompany)	Planned orders across legal entities with intercompany planning groups	PARTIAL-40 (intercompany master planning across legal entities not named in 22 or 40)	40	Gap register
Planned orders → Sequencing (PI)	PdsSequence	Sequence planned batch orders	COVERED-23	23	
Planned orders → Capacity reservations (link)	WrkCtrCapRes		COVERED-23	23	
Demand forecasting → Forecast models / Submodels	ForecastModel		COVERED-22	22	
Planning → Planning Optimization (D365 service), Priority-based planning (D365)	ReqPlanningOptimization	External planning service	SUPERSEDED (native Go planning engine wave 22 with priority as an attribute)	22	Priority-based planning (D365 2021+): PARTIAL — see gap register entry under setup

12.21.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Net requirements inquiry / Requirement profile		COVERED-22	22	
Explosion inquiry / Trace (why was this planned)		COVERED-22	22	
Planned order inquiries (by item, vendor, period)		COVERED-22	22	
Action message inquiry / Futures inquiry		COVERED-22	22	
Capacity → Capacity load / Capacity reservations (link)		COVERED-23	23	
Master plan run history / Plan log (ReqLog)	Run statistics, errors	COVERED-22	22	
Forecast inquiries (Forecast lines, Forecast consumption / reduction)		COVERED-22	22	
Demand forecast accuracy (MAPE)		COVERED-22	22	
Item coverage inquiry (Coverage settings by item)		COVERED-22	22	
Planned order pegging (bidirectional)		COVERED-22	22	
SSRS: Planned orders report / Planned purchase orders by vendor		COVERED-22	22	
SSRS: Net requirements report		COVERED-22	22	
SSRS: Action messages / Futures messages		COVERED-22	22	
SSRS: Capacity load (planned)		COVERED-23	23	
SSRS: Forecast (Sales forecast, Purchase forecast, Inventory forecast)		COVERED-22	22	Inventory forecast PARTIAL
SSRS: Coverage groups listing / Item coverage listing		COVERED-22	22	
SSRS: Safety stock journal report		COVERED-22	22	
SSRS: Master plan run log		COVERED-22	22	
SSRS: Supply schedule report		PARTIAL-22 (supply schedule)	22	

12.21.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Master scheduling (run plan: regeneration / net change / net change minimized)		COVERED-22	22	Incremental < 2 min target
Forecast scheduling (run forecast plan)		COVERED-22	22	
Planned order → Firm planned orders (batch by filter)		COVERED-22	22	
Planned order → Approve planned orders (batch)		COVERED-22	22	
Planned order → Delete planned orders		COVERED-22	22	
Planned order → Scheduling of planned orders (Gantt scheduling batch)		COVERED-23	23	
Plan → Copy plan / Delete plan		COVERED-22	22	
Safety stock → Calculate proposal (from history) / Update minimum keys		COVERED-22	22	
Forecast → Copy / Transfer forecast / Create forecast from history (Forecast allocation)		COVERED-22	22	
Forecast → Forecast reduction (consumption by actual orders)	Reduction keys and principles	COVERED-22	22	Forecast with reduction
Demand forecasting → Generate statistical baseline forecast / Import adjusted forecast / Remove outliers		COVERED-22	22	
Intercompany master scheduling (run across entities)		PARTIAL-40 (intercompany planning)	40	
Capacity → Delete capacity reservations (link)		COVERED-23	23	
Clean up → Delete futures/action messages / Clean plan history		COVERED-22	22	
Planning → Planning Optimization sync (D365)		SUPERSEDED (native engine)	22	
Planning → Kanban quantity calculation (link to Lean)		COVERED-24	24	
Planning → Item allocation key refresh		COVERED-22	22	
Planning → Generate planned kanbans		COVERED-24	24	
Planning → MRP exception triage agent (Passage)		COVERED-47	47	

12.21.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Master planning parameters — General	Current static/dynamic plan, automatic copy static→dynamic, use dynamic negative days, planned order defaults	COVERED-22	22	
Master planning parameters — Planned orders	Group by vendor/buyer/period on firming, planned purchase order type, firm journal	COVERED-22	22	
Master planning parameters — Standard update / Number sequences / Performance (threads, cache)		COVERED-22	22	
Master planning parameters — Planning Optimization (D365 connector, parameters)		SUPERSEDED (native engine)	22	
Plans → Master plans (Plan: include on-hand/transactions/forecast/safety stock/inquiries/intercompany; time fences: coverage, freeze, firming, explosion, capacity, forecast plan, action, futures, BOM explosion; scheduling method: operations/job; finite capacity/material/property; action messages on/off; futures on/off)	Full plan parameters	COVERED-22	22	Capacity flags 23
Plans → Forecast plans (include forecast models, reduction)		COVERED-22	22	
Coverage → Coverage groups (requirement/period/min-max/manual, negative/positive days, coverage time fence, firming/freeze fences, action rules, futures, safety margins (receipt/issue/reorder), period lengths, action message tolerances, BOM/formula explosion fence, use dynamic negative days, coverage planned order type, planning time fence)		COVERED-22	22	
Coverage → Item coverage (link) / Minimum/maximum keys	Seasonal min/max profiles	COVERED-22	22	
Coverage → Period templates (period coverage)		COVERED-22	22	

Name	Key parameters / options	Passage status	Wave	Notes
Coverage → Planned order defaults (Planned purchase/production/transfer defaults)		COVERED-22	22	
Forecast → Forecast models / Submodels		COVERED-22	22	
Forecast → Reduction keys / Reduction principles (none, percent-reduction key, transactions-dynamic period, transactions-reduction key)		COVERED-22	22	
Forecast → Item allocation keys / Period allocation keys (Period allocation categories)	Distribute forecast across items/periods	COVERED-22	22	
Forecast → Customer/Vendor forecast groups (forecast at group level)		COVERED-22	22	
Demand forecasting → Demand forecasting parameters (algorithm, horizon, aggregation, history start, item allocation keys, Azure ML service)		COVERED-22	22	statsforecast service
Demand forecasting → Forecast algorithm parameters (seasonality, confidence, outlier removal)		COVERED-22	22	
Action messages → Action message settings (per coverage group)		COVERED-22	22	
Futures → Futures settings (per coverage group)		COVERED-22	22	
Safety stock → Safety stock journal names		COVERED-22	22	
Intercompany → Intercompany planning groups / Intercompany planning group members (sequence, plan)		PARTIAL-40 (intercompany planning)	40	
Scheduling → Scheduling parameters (link to Production: operations/job scheduling defaults)		COVERED-23	23	
Sequencing → Sequencing setup (PI)		COVERED-23	23	
Planning → Planning item groups (process planning item, link)		COVERED-24	24	
Planning → Delivery date control / Transport days / Coverage calendars (shared with Sales)		COVERED-27	27	
Planning → Planned order grouping rules (vendor, buyer group, period)		COVERED-22	22	
Planning → Buyer groups	Grouping for planned purchase orders	COVERED-22	22	
Planning → Planner / Planner groups (D365: planner on item)		COVERED-22	22	
Planning → Priority-based planning (D365: planning priority models, priority rules on coverage group)	Priority-driven reorder instead of dates (DDMRP-style)	PARTIAL-22 (priority-based / DDMRP-style buffer planning not named)	22	Gap register
Planning → Subcontracting planning (vendor line type / planned purchase for subcontract)		COVERED-21	21	
Planning → Capability-based scheduling (link)		COVERED-23	23	
Master planning → Number sequences		COVERED-5	5	
Master planning → Workflow (planned order approval D365)		COVERED-4	4	

12.22 Production control

Production orders, journals, routes/operations, resources, scheduling, Gantt, manufacturing execution (job registration, time and attendance) and production posting map to waves 21 (orders/WIP/costing), 23 (capacity/scheduling), 25 (shop floor, T&A), 9 (routes/resources master data) and 39 (variances/costing sheets).

12.22.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Production orders (list page: all, created, estimated, scheduled, released, started, reported as finished, ended; "my" pools)	ProdTableListPage	Order lifecycle	COVERED-21	21	
Production order details	ProdTable	Header: item, qty, dates, BOM/route, status, pool, group, reservation, site/warehouse, priority, properties, financial dimensions, costing	COVERED-21	21	
Production order → BOM (production BOM lines)	ProdBOM	Order-specific BOM with line type (item/phantom/pegged supply/vendor), scrap, consumption, resource consumption	COVERED-21	21	
Production order → Route (production route operations)	ProdRoute	Order-specific operations, times, resources, queue/setup/process/transport	COVERED-21	21	
Production order → Jobs (route jobs)	ProdRouteJob	Job-level (setup, process, queue after/before, transport, overlap)	COVERED-23	23	Job scheduling
Production order → Estimate / Schedule / Release / Start / Report as finished / End (update dialogs)	ProdUpdEstimation, ProdUpdScheduling, ProdUpdRelease, ProdUpdStartUp, ProdUpdReportFinished, ProdUpdCostCalc (End)	Status updates with options (auto-post journals, reference-level update, BOM/route consumption)	COVERED-21	21	
Production order → Reset status	ProdUpdStatusReset	Back-step status	COVERED-21	21	Lifecycle state machine
Production order → Split / Group (ProdSplit)	ProdSplit	Split order quantity	PARTIAL-21 (production order split/merge not named)	21	Gap register
Production order → Price calculation / Cost estimation view	ProdCalcTrans	Estimated cost by cost group	COVERED-21	21	
Production order → Production transactions (view cost, consumption)	ProdTransView		COVERED-21	21	
Production order → Planned production orders (link)	ReqPOListPage		COVERED-22	22	
Production order → Production pools	ProdPool	Grouping/statuses for processing	COVERED-21	21	
Production order → Reservation / Pick list release			COVERED-21	21	
Production order → Subcontracting (vendor lines → PO)	ProdBOM (vendor)		COVERED-21	21	
Production order → Attachments / Instructions / Drawings	DocuRef		COVERED-25	25	Shop-floor instructions

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Journals → Picking list	ProdJournalTable (Picking list) / ProdJournalBOM	Material consumption, backflush, negative consumption	COVERED-21	21	
Journals → Route card	ProdJournalTable (Route card) / ProdJournalRoute	Hours/qty per operation (non-job level)	COVERED-21	21	
Journals → Job card	ProdJournalTable (Job card) / ProdJournalRoute (job)	Per job registration	COVERED-21	21	
Journals → Report as finished	ProdJournalTable (RAF) / ProdJournalProd	RAF qty, error qty, accept error, end job, by/co-products	COVERED-21	21	
Journals → Production journals (all) / Posted journals	ProdJournalTable		COVERED-21	21	
Journals → Production input journal (WMS arrival for production)	WMSJournalTable (ProdInput)		COVERED-21	21	
Resources (list page)	WrkCtrResourceListPage (WrkCtrTable)	Resources: type (vendor/human/machine/tool/location), capacity, efficiency, calendar, capabilities, costs	COVERED-9	9	
Resource groups	WrkCtrResourceGroup	Group with site, warehouse, input/output locations, calendar, ledger posting	COVERED-9	9	
Resource capabilities (link)	WrkCtrCapability		COVERED-9	9	
Routes (list page, Route details, Route versions)	RouteTableListPage, RouteTable, RouteVersion	Routes with operations, relations	COVERED-9	9	
Operations	RouteOprTable	Operation master with relations (per item/route/site)	COVERED-9	9	
Operation relations (route operation details: setup/run time, resources requirements, cost categories, scrap)	RouteOpr		COVERED-9	9	
Gantt chart (Resource/Order views)	GanttControl (ProdGantt)	Interactive scheduling	COVERED-23	23	
Capacity reservations	WrkCtrCapRes	Reserved capacity per job/resource	COVERED-23	23	
Job list (all jobs, ready jobs, started)	ProdRouteJobListPage	Jobs across orders	COVERED-23	23	
Manufacturing execution → Job registration (terminal: clock-in, start job, bundle jobs, report feedback, clock-out)	JmgRegistration (JmgRegistrationTouch)	Shop-floor terminal	COVERED-25	25	Kiosk UI
Manufacturing execution → Job list (electronic job list / dispatch list)	JmgJobList	Prioritized job queue per resource	COVERED-25	25	
Manufacturing execution → Registrations (raw registrations)	JmgTermReg	Clock/job registrations awaiting calculation	COVERED-25	25	
Manufacturing execution → Current activities / Absences	JmgCurrentActivities	Who is on which job	COVERED-25	25	
Manufacturing execution → Calculate / Approve (registrations per worker/day)	JmgCalculate, JmgApprove	Pay-time calculation and supervisor approval	COVERED-25	25	
Manufacturing execution → Transfer (to payroll/projects/production)	JmgTransfer		COVERED-25	25	Export to payroll wave 43
Manufacturing execution → Time and attendance (clock registrations, flex balance, absence registration)	JmgRegistration (T&A)		COVERED-25	25	
Manufacturing execution → Production floor execution (D365 PFE)	JmgProductionFloorExecution	Touch UI for jobs, materials, quality, maintenance requests	COVERED-25	25	
Production control → Lean manufacturing (links)			COVERED-24	24	§12.24
Production control → Batch orders (PI)	ProdTable (batch)		COVERED-24	24	§12.23
Production control → Production order cost analysis (Cost estimates and costing)	ProdCalc	Planned vs actual by cost group/type	COVERED-21	21	Variance decomposition 39
Production control → Delays (R3 Production orders with delays)	ProdTable (delays)		COVERED-23	23	
Production control → Scrap / Error cause (job card error)	ProdErrorCause		COVERED-21	21	
Production control → Production order properties (property for scheduling)	ProdProperty		COVERED-23	23	
Production control → Production groups	ProdGroup	Ledger posting grouping	COVERED-21	21	

12.22.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Production order inquiries (orders, BOM, route, jobs, transactions, status log)		COVERED-21	21	
Production transactions (Route transactions, BOM transactions)		COVERED-21	21	
Cost inquiry (Production cost: estimated vs realized, WIP)		COVERED-21	21	
Capacity → Capacity load (per resource/group/period, graphical)		COVERED-23	23	
Capacity → Capacity reservations		COVERED-23	23	
Capacity → Resource load (Gantt)		COVERED-23	23	
Jobs → Job list / Job status / Ready jobs		COVERED-23	23	
Journals → Picking list / Route card / Job card / RAF journal inquiries		COVERED-21	21	
Manufacturing execution → Registration inquiries (Raw registrations, Pay time, Flex balance, Absence, Job registrations, Overtime)		COVERED-25	25	
Manufacturing execution → Production feedback / Operator efficiency		COVERED-25	25	OEE
Lean → Kanban inquiries (link)		COVERED-24	24	
Scrap / Error cause inquiry		COVERED-21	21	
Production → Delayed orders / Production performance (OTD)		COVERED-46	46	

Name	Purpose	Passage status	Wave	Notes
SSRS: Production orders (status list) / Production order overview		COVERED-21	21	
SSRS: Production papers (Job card, Route card, Picking list, Route card with jobs, Production order document bundle)		COVERED-21	21	Rendered 46
SSRS: Production order — BOM lines / Route lines		COVERED-21	21	
SSRS: Capacity load / Capacity reservations / Resource load		COVERED-23	23	
SSRS: Production cost per period / Completed production order cost analysis / Production cost by cost group		COVERED-39	39	
SSRS: Route consumption / BOM consumption (planned vs actual)		COVERED-21	21	
SSRS: Job transactions / Route transactions		COVERED-21	21	
SSRS: Resources / Resource groups / Capabilities listings		COVERED-9	9	
SSRS: Calendars / Working time		COVERED-9	9	
SSRS: Routes / Operations listings		COVERED-9	9	
SSRS: Cost categories / Cost category prices		COVERED-39	39	
SSRS: Manufacturing execution reports (Registrations, Pay statistics, Absence, Flex, Job efficiency, Timecard)		COVERED-25	25	
SSRS: Production variance reports (link)		COVERED-39	39	
SSRS: Scrap report / Error cause statistics		COVERED-21	21	
SSRS: Work-in-progress report		COVERED-21	21	
SSRS: Production order label (barcode)		COVERED-32	32	

12.22.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Production orders → Estimate / Schedule / Release / Start / Report as finished / End (batch)		COVERED-21	21	
Production orders → Update costing (Price calculation batch)		COVERED-21	21	
Production orders → Delete production orders / Clean up ended orders		COVERED-21	21	
Production orders → Reset status (batch)		COVERED-21	21	
Journals → Post journals (batch) / Delete posted journals		COVERED-21	21	
Scheduling → Operations scheduling / Job scheduling (batch, by filter)		COVERED-23	23	
Scheduling → Reschedule all / Delete capacity reservations		COVERED-23	23	
Scheduling → Compose working times (generate calendar days from templates)		COVERED-9	9	
Scheduling → Update calendars (calendar expiry extension)		COVERED-9	9	
Manufacturing execution → Calculate registrations (batch) / Approve (batch) / Transfer (batch)		COVERED-25	25	
Manufacturing execution → Generate flex balances / Reset flex / Reset statistics balances		COVERED-25	25	
Manufacturing execution → Update job list / Clean up registrations		COVERED-25	25	
Manufacturing execution → Auto-clock-out		COVERED-25	25	
Lean → Backflush costing (periodic)		COVERED-24	24	
Lean → Kanban periodic (update kanban quantities, process scheduled kanbans)		COVERED-24	24	
Production → Production order status refresh for WHS (release to warehouse)		COVERED-31	31	
Production → IoT: machine event ingestion / OEE calc		COVERED-25	25	
Production → Update route/BOM on open orders (mass re-explosion)		COVERED-21	21	
Production → Production order cost roll-up refresh (planned cost re-estimate)		COVERED-21	21	

12.22.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Production control parameters — General	Production order defaults: BOM/route version selection, reservation (manual/estimation/scheduling/release/start), automatic BOM consumption (flushing principle: start/finish/manual), route consumption, posting of journals, scheduling method default, lock route operations, remove reservations, status defaults	COVERED-21	21	
Production control parameters — Standard update tab	Default update parameters per status update (estimation, scheduling, release, start, RAF, end)	COVERED-21	21	
Production control parameters — Start tab	Picking list posting at start (quantity: start qty/remaining), route card at start, end-mark, reference status update	COVERED-21	21	
Production control parameters — Operations tab	Route card/job card defaults, hours/qty posting, scrap handling	COVERED-21	21	
Production control parameters — Journals tab	Default journal names per type	COVERED-21	21	
Production control parameters — Automatic update tab	Automatic report as finished, end-mark, auto-start, reference level update	COVERED-21	21	
Production control parameters — General ledger tab	Ledger posting: item+category, item+resource, none; post costs during production (WIP); profit setting; posting of RAF by cost group	COVERED-21	21	
Production control parameters — Number sequences tab		COVERED-5	5	
Production control parameters — Lean tab / Process tab (PI)		COVERED-24	24	
Production parameters by site (Site-specific production parameters)	Overrides per site	COVERED-21	21	
Production → Production groups	Ledger posting profile per group (WIP accounts, issue/receipt accounts)	COVERED-21	21	
Production → Production pools		COVERED-21	21	

Name	Key parameters / options	Passage status	Wave	Notes
Production → Production journal names (picking list, route card, job card, RAF)		COVERED-21	21	
Production → Production posting profiles (in Inventory posting: Production tab: estimated cost consumed, WIP issue, WIP account, cost of materials consumed, issue, receipt, RAF)		COVERED-21	21	
Production → Error causes (scrap reason codes)		COVERED-21	21	
Production → Properties (scheduling property per resource/period)		COVERED-23	23	
Production → Priority (production priority)		COVERED-23	23	
Routes → Route groups (setup time/run time/queue time estimation flags, job management, capacity, reporting, scheduling of jobs, working time, cost categories)		COVERED-9	9	
Routes → Operations (link)		COVERED-9	9	
Routes → Cost categories (hourly/piece rates, cost group, ledger accounts)		COVERED-39	39	
Routes → Shared categories (project/production category flags)		COVERED-9	9	
Routes → Cost category price per costing version		COVERED-39	39	
Routes → Route version parameters (approval, activation)		COVERED-9	9	
Routes → Route types (simple/network: sequential/parallel)		COVERED-9	9	
Routes → Route operation relations: Primary/secondary resources, Link types (hard/soft), Overlap quantity, Transfer batch		COVERED-9	9	
Resources → Resource groups		COVERED-9	9	
Resources → Resources (types: vendor, human resources, machine, tool, location)		COVERED-9	9	
Resources → Capabilities / Resource capabilities (level, priority, expiry)		COVERED-9	9	
Resources → Resource types / Competency-based (skills, courses, certificates for human resources)		COVERED-9	9	HR competency in 42
Resources → Resource bottleneck / Finite capacity flag / Efficiency % / Capacity unit		COVERED-23	23	
Resources → Resource ledger posting (operations resource, WIP, issue accounts)		COVERED-21	21	
Resources → Vendor as resource (subcontracting)		COVERED-21	21	
Calendars → Calendars (working time calendars), Working time templates (shifts, efficiency %, properties), Compose working times		COVERED-9	9	
Calendars → Calendar properties / Working time intervals		COVERED-9	9	
Scheduling → Scheduling parameters (Scheduling direction, Scheduling method default, Capacity: finite/infinite, Finite material, Finite property, Capability-based scheduling, Schedule references, Sync references, Cancel capacity reservations, Primary resource selection (duration/first available/ etc.), Scheduling time fence)		COVERED-23	23	
Scheduling → Operations scheduling percentage / Scheduling of jobs: Job types, Job management		COVERED-23	23	
Scheduling → Load (capacity load calculation: setup/process/queue/transport included)		COVERED-23	23	
Sequencing → Sequence groups / Sequences (PI)		COVERED-23	23	
Manufacturing execution → Manufacturing execution parameters (General: registration validation, job bundling, feedback, Pay: pay agreement, flex, overtime; Production: picking list/RAF posting from registrations; Absence; Terminal)		COVERED-25	25	
Manufacturing execution → Registration configuration / Terminal configuration (touch screen layout, buttons, barcode)		COVERED-25	25	Kiosk configuration
Manufacturing execution → Workers (time registration workers: profile group, calculation group, approval group, pay agreement, flex allowed, terminal)		COVERED-25	25	
Manufacturing execution → Profiles (clock profiles with profile types: standard time, flex+, flex-, overtime, break, pay)		COVERED-25	25	Pay-type calculation
Manufacturing execution → Profile groups / Profile calendar / Profile relations		COVERED-25	25	
Manufacturing execution → Pay: Pay agreements, Pay agreement lines (pay types by time/overtime/shift), Pay types, Pay items, Statistics balances, Counters		COVERED-25	25	
Manufacturing execution → Indirect activities (categories, activities: break, meeting, maintenance)		COVERED-25	25	Downtime reason codes

Name	Key parameters / options	Passage status	Wave	Notes
Manufacturing execution → Absence (codes, groups, statistics)		COVERED-42	42	Shared with HR
Manufacturing execution → Groups (Calculation groups, Approval groups, Supervisors)		COVERED-25	25	
Manufacturing execution → Flex (flex profiles, flex groups, flex balance limits)		COVERED-25	25	
Manufacturing execution → Switch codes / Job bundling / Job priority		COVERED-25	25	
Manufacturing execution → Production feedback (feedback parameters, auto-report)		COVERED-25	25	
Manufacturing execution → Production floor execution: Configure PFE interface (tabs, buttons, kiosk), Device setup (D365)		COVERED-25	25	
Manufacturing execution → Time and attendance parameters (shifts, rounding, grace)		COVERED-25	25	
Lean manufacturing setup (link)		COVERED-24	24	§12.24
Process manufacturing setup (link)		COVERED-24	24	§12.23
Production → IoT (D365 Sensor Data Intelligence: scenarios, sensors, notifications)		COVERED-25	25	OPC-UA/MQTT
Production → Subcontracting setup (vendor resource, line type vendor, purchase on release)		COVERED-21	21	
Production → Costing sheet / Cost groups (link)		COVERED-39	39	
Production → BOM/Route version activation rules		COVERED-9	9	
Production → Quality associations for production (link)		COVERED-33	33	
Production → Warehouse: Production input/output locations, License plate RAF (WHS)		COVERED-31	31	
Production → Workflow (production order release approval D365)		COVERED-4	4	
Production → Number sequences		COVERED-5	5	
Production → Gantt chart configuration (colors, views, user settings)		COVERED-23	23	
Production → Andon / alerts (Passage)		COVERED-25	25	

12.23 Process manufacturing (PI)

AX 2012 R3 folds process industries into PIM, Inventory, Production control, Master planning and Sales. Formulas, batch orders, co/by-products, potency, catch weight, batch attributes/shelf life, batch balancing, consolidated batch orders, rework and sequencing map to waves 24 (process manufacturing), 9 (formula versions) and 23 (sequencing). Commodity pricing, vendor rebates, brokers/royalties and product compliance are gaps (see §12.15, §12.16).

12.23.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Formulas (list page) / Formula designer	PdsFormulaListPage (BOMTable formula), BOMDesigner	Formula with co/by-products, scalable lines, percentage lines, step consumption	COVERED-9	9	Formula BOMs
Formula versions (per item, with formula size, co-product/by-product lines, approval)	BOMVersion (formula)		COVERED-9	9	
Formula lines (line types: scalable, step consumption, percent, variable/fixed, catch weight)	BOMLine (formula)		COVERED-9	9	
Co-products / By-products on formula (cost allocation %, burden: by-product cost allocation: none, percent of cost, per quantity, per series, recycled)	PdsCoByProduct		COVERED-24	24	Co/by-product allocation methods
Batch orders (list page)	ProdTableListPage (batch)	Batch orders from formulas	COVERED-24	24	
Batch order details	ProdTable (PdsBatch)	Formula explosion, co/by-products, batch attributes, potency, batch balancing, rework	COVERED-24	24	
Batch order → Batch balancing	PdsBatchBalancing	Adjust ingredient quantities by actual potency of selected batches	COVERED-24	24	Potency fixtures
Batch order → Consolidated batch orders (batch order consolidation / bulk item → packed items)	PdsConsolidatedOrder	Bulk and packed order relation	PARTIAL-24 (consolidated batch orders (bulk/pack relationship) not named)	24	Gap register
Batch order → Rework batch order	PdsRework	Rework a finished batch	PARTIAL-24 (rework batch orders not named)	24	Gap register
Batch order → Report as finished with co/by-products	ProdJournalProd (coby)		COVERED-24	24	
Planning items (formula for planning multiple co-products)	InventTable (planning item)		COVERED-24	24	
Batch attributes (Inventory batches, link)	PdsBatchAttrib	Attributes with min/max/target/tolerance	COVERED-24	24	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Batch attribute inheritance (lot inheritance: product-to-product, co-product inheritance)	PdsInheritance	Inherit attributes/shelf life from ingredients	PARTIAL-24 (batch attribute and shelf-life inheritance rules not named)	24	Gap register
Catch weight (CW) transactions on orders (sales/purchase/production/inventory)	PdsCatchWeight		COVERED-24	24	
Potency (active ingredient) on batches / Potency pricing (price per unit of active ingredient)	PdsIngredient / PdsPotencyPricing		COVERED-24	24	Potency pricing PARTIAL-26 (commodity pricing)
Shelf life (best before, expiry, shelf advice) / Batch disposition codes	InventBatch (shelf life)		COVERED-24	24	
Sequencing → Sequenced planned orders / Sequencing of batch orders	PdsSequence		COVERED-23	23	
Commodity pricing (Pricing calculation, link)	PdsCommodity*		PARTIAL-26 (commodity pricing)	26	
Vendor rebates / Brokers / Royalties (links)			GAP	—	Propose wave 26: vendor rebates (see §12.16)
Product compliance / Safety data sheets (link)			GAP	—	Propose wave 33: product compliance (see §12.15)
Container packaging (PI packaging / Container types)	PdsContainer	Packaging hierarchy	PARTIAL-28 (packaging materials)	28	
Country of origin (batch-level)	PdsCountryOfOrigin		COVERED-24	24	
Item tracing (batch/serial genealogy)	InventTrack		COVERED-24	24	
Batch merge	PdsBatchMerge		PARTIAL-24 (batch merge)	24	
Same-batch reservation (batch reservation rules: consolidated/same batch on sales line)	PdsSameBatch		PARTIAL-24 (same-batch reservation requirement on sales lines not named)	24	Gap register
Customer-specific batch attribute requirements (customer sellable days / batch attribute requirements on sales)	PdsCustSellableDays, PdsCustBatchAttrib	Shelf life minimum per customer, attribute ranges per customer	PARTIAL-24 (customer sellable days / customer batch attribute requirements on sales lines not named)	24	Gap register

12.23.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Batch order inquiries (co-product cost, batch balancing results)		COVERED-24	24	
Batch attribute inquiries / Attribute search for batches		COVERED-24	24	
Item tracing / Genealogy inquiry		COVERED-24	24	
Catch weight on-hand / transactions		COVERED-24	24	
Shelf life inquiries (Expired, Expiring, Shelf advice)		COVERED-24	24	
Potency inquiries		COVERED-24	24	
Sequencing inquiry		COVERED-23	23	
Commodity price inquiries		PARTIAL-26 (commodity pricing)	26	
SSRS: Formula report / Formula with co-products		COVERED-9	9	
SSRS: Batch order papers (Batch card, Picking list with potency)		COVERED-24	24	
SSRS: Batch attribute certificate (link to CoA)		COVERED-33	33	
SSRS: Lot/batch traceability report		COVERED-24	24	
SSRS: Catch weight reports		COVERED-24	24	
SSRS: Shelf life / Expiry reports		COVERED-24	24	
SSRS: Country of origin report		COVERED-24	24	
SSRS: Safety data sheet (link)		GAP	—	Propose wave 33: product compliance
SSRS: Co-product cost allocation report		COVERED-24	24	

12.23.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Batch orders → Estimate / Schedule / Release / Start / RAF / End (batch)		COVERED-24	24	
Batch orders → Consolidate batch orders (batch)		PARTIAL-24 (consolidated batch orders)	24	
Batches → Update shelf life / Expire batches / Reset disposition		COVERED-24	24	
Batches → Attribute inheritance processing on RAF		PARTIAL-24 (inheritance)	24	
Commodity pricing → Price calculation run (update trade agreements from commodity index)		PARTIAL-26 (commodity pricing)	26	
Sequencing → Run sequencing on planned orders		COVERED-23	23	
Catch weight → Update nominal weights		COVERED-24	24	
Compliance → SDS expiry / regulated product reporting		GAP	—	Propose wave 33: product compliance
Vendor rebates → Process		GAP	—	Propose wave 26: vendor rebates

12.23.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Production control parameters — Process tab (formula defaults, co-product posting, batch balancing, rework)		COVERED-24	24	
Formula → Formula parameters (scalable, formula size, yield %, step consumption tables)		COVERED-9	9	
Formula → Co-product cost allocation methods / By-product burden types		COVERED-24	24	
Batch → Batch attributes, Attribute groups, Product-specific attributes, Customer-specific attributes		COVERED-24	24	
Batch → Batch disposition master / Disposition codes		COVERED-24	24	
Batch → Shelf life setup, Batch number groups, Expiry calculation (from production date/receipt date)		COVERED-24	24	
Batch → Lot inheritance setup (inheritance of attributes/shelf life by product)		PARTIAL-24 (inheritance)	24	
Batch → Same-batch reservation / Batch reservation parameters (FEFO date-controlled)		COVERED-16	16	Same-batch PARTIAL-24
Potency → Base attribute, Potency item setup, Potency pricing (per active ingredient)		COVERED-24	24	
Catch weight → CW item setup, CW units, Nominal/min/max weight		COVERED-24	24	
Sequencing → Sequences, Sequence groups, Sequence priorities (link)		COVERED-23	23	
Commodity pricing → Cost basis types, Pricing calculations, Price margin templates		PARTIAL-26 (commodity pricing)	26	
Product compliance → Regulated/restricted lists, Reporting lists, SDS setup		GAP	—	Propose wave 33: product compliance
Vendor rebates / Brokers / Royalties setup		GAP	—	Propose wave 26: vendor rebates
Container packaging → Container types, Packaging groups		PARTIAL-28 (packaging materials)	28	
Country of origin setup		COVERED-24	24	
Planning item setup (planning formula)		COVERED-24	24	
Customer sellable days / Customer batch attribute requirements		PARTIAL-24 (customer sellable days)	24	
Rework → Rework parameters		PARTIAL-24 (rework)	24	
Consolidated batch orders → Bulk/pack item relations		PARTIAL-24 (consolidated batch orders)	24	
Number sequences (PI)		COVERED-5	5	

12.24 Lean manufacturing

Value streams, production flows, activities, kanban rules, kanban boards and backflush costing map to wave 24. Value streams as operating units are in wave 6.

12.24.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Production flows (list page)	LeanProductionFlowListPage	Flow with versions and activities	COVERED-24	24	
Production flow details / Versions (activate, expire, takt, cycle time)	LeanProductionFlow / LeanProductionFlowVersion		COVERED-24	24	
Activities (process/transfer activities with resource, locations, run time, takt)	LeanProductionFlowActivity (PlanActivity)		COVERED-24	24	
Activity relations (successor/predecessor, cycle time ratio)	PlanActivityRelation		COVERED-24	24	
Kanban rules (list page)	KanbanRuleListPage	Manufacturing/withdrawal; fixed/scheduled/event; quantities, lead time, circulating cards	COVERED-24	24	
Kanbans (list page: all kanbans, by rule)	KanbanListPage / Kanban	Kanban cards with jobs and status	COVERED-24	24	
Kanban board for process jobs	KanbanJobSchedulingBoard	Drag jobs between planned/in-progress/completed; board per cell	COVERED-24	24	
Kanban board for transfer jobs	KanbanJobTransferBoard	Transfer execution board	COVERED-24	24	
Kanban schedule board (scheduling of kanban jobs)	KanbanJobSchedulingPeriod	Period/resource schedule	COVERED-24	24	
Kanban quantity calculation (Kanban quantity calculation policies, calculation runs)	KanbanQuantityCalculation	Recalculate kanban quantities from demand	COVERED-24	24	
Event kanbans (sales/production/kanban line/stock replenishment events)	Kanban (event)		COVERED-24	24	Event kanban simulation
Kanban cards (print, card circulation)	KanbanCard		COVERED-24	24	
Lean → Backflush costing (cost accounting for production flows: periodic conversion costs, variance)	LeanCostingPeriodicProcess	Period-based costing of production flows	COVERED-24	24	
Lean → Production flow costing (cost group breakdown, WIP by flow)	LeanCostFlow		COVERED-24	24	
Lean → Subcontracting in lean (service activities with vendor, purchase agreements)	LeanSubcontracting	Service-based subcontracting on activities	PARTIAL-24 (lean subcontracting (activity-based service purchase from vendor, purchase agreement linkage) not named; discrete subcontracting is in 21)	24	Gap register
Lean → Picking list (kanban material pick) / Kanban job pick list	KanbanJobPickingList		COVERED-24	24	
Lean → Scheduled kanban jobs (from master planning planned kanbans)			COVERED-22	22	Planned kanbans
Value streams (Org admin operating unit)	OMValueStream		COVERED-6	6	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Lean → Kanban supermarket / Kanban location setup (WHS)			COVERED-31	31	

12.24.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Kanban job inquiries (jobs by status, late jobs, cycle time)		COVERED-24	24	
Kanban rule inquiries / Kanban quantities history		COVERED-24	24	
Production flow inquiries (activity load, takt vs actual)		COVERED-24	24	
Lean costing inquiries (period costs, variances per flow)		COVERED-24	24	
Event kanban inquiry (source events)		COVERED-24	24	
SSRS: Kanban card (print) / Kanban card circulation report		COVERED-24	24	
SSRS: Kanban rule report / Kanban overview		COVERED-24	24	
SSRS: Production flow report / Activity report		COVERED-24	24	
SSRS: Kanban quantity calculation report		COVERED-24	24	
SSRS: Backflush costing report / Lean variances		COVERED-24	24	
SSRS: Kanban job pick list		COVERED-24	24	

12.24.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Kanban → Process event kanbans (generate from events: sales, BOM line, min stock)		COVERED-24	24	
Kanban → Kanban quantity calculation run / Update kanban rules from calculation		COVERED-24	24	
Kanban → Print kanban cards (batch) / Reprint		COVERED-24	24	
Kanban → Schedule kanban jobs (automatic planning)		COVERED-24	24	
Kanban → Expire fixed kanbans / Clean up completed kanbans		COVERED-24	24	
Lean → Backflush costing period processing (calculate conversion costs, post, close period)		COVERED-24	24	
Lean → Production flow version activation/expiry (batch)		COVERED-24	24	
Lean → Transfer job auto-complete (on receive)		COVERED-24	24	
Lean → Lean subcontracting: generate purchase orders for service activities		PARTIAL-24 (lean subcontracting)	24	
Lean → Kanban rule effectivity update from master planning		COVERED-22	22	

12.24.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Production control parameters — Lean tab (kanban defaults, costing, picking list posting, transfer defaults)		COVERED-24	24	
Lean → Production flow models (costing method per flow: backflush costing)		COVERED-24	24	
Lean → Kanban rule policies (replenishment strategy: fixed, scheduled, event; kanban quantity, min/max, lead time, circulating/one-time; automatic planning)		COVERED-24	24	
Lean → Kanban quantity calculation policies (demand period, forecast/orders, safety stock, rounding)		COVERED-24	24	
Lean → Kanban card configuration (card layout, card number sequence, barcode)		COVERED-24	24	
Lean → Kanban board configuration (board per work cell, columns, colors, job filters)		COVERED-24	24	
Lean → Work cells (resource groups of type work cell, capacity units, calendars)		COVERED-9	9	
Lean → Lean schedule groups		COVERED-24	24	
Lean → Kanban handling units (bins/containers)		COVERED-24	24	
Lean → Activity resource/location defaults (input/output locations, supermarket)		COVERED-24	24	
Lean → Lean costing: Cost groups for lean, Cost category for activities, Conversion cost posting, WIP accounts per production flow		COVERED-24	24	
Lean → Subcontracting setup (service items, purchase agreements for activities)		PARTIAL-24 (lean subcontracting)	24	
Lean → Kanban job registration (manufacturing execution integration)		COVERED-25	25	
Lean → Value streams (link)		COVERED-6	6	
Lean → Number sequences (kanban, production flow)		COVERED-5	5	
Lean → Mobile device kanban flows (WHS mobile: receive kanban, empty kanban)		COVERED-32	32	

12.25 Quality management

In AX 2012 R3 quality lives under Inventory management (quality orders, nonconformances, corrections, test groups, quality associations, certificates of analysis). It maps to wave 33, with inventory blocking from wave 16 and calibration/SPC additions.

12.25.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Quality orders (list page)	InventQualityOrderTableListPage	Orders generated by associations or manually; reference (purchase, sales, production, quarantine, inventory, RMA)	COVERED-33	33	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Quality order details	InventQualityOrderTable / InventQualityOrderLine	Tests, sampling, results (variable/attribute), pass/fail, validate	COVERED-33	33	Results recording
Quality order → Results (test results per sample)	InventQualityOrderLineResults		COVERED-33	33	
Quality order → Validate / Close	InventQualityOrderValidate	Usage decision	COVERED-33	33	Usage decision
Nonconformances (list page)	InventNonConformanceTableListPage	NCR with problem type, source (customer, service request, internal, vendor, production), quarantine zone	COVERED-33	33	NCR/CAPA
Nonconformance details	InventNonConformanceTable	Approve/refuse, related operations, corrections, quarantine link	COVERED-33	33	
Nonconformance → Operations (nonconformance operations with cost items, timesheets, charges)	InventNonConformanceOperation	Work to resolve NC with cost	COVERED-33	33	Cost of quality
Corrections	InventNonConformanceCorrection	Corrective actions with diagnostic types, planned/actual dates	COVERED-33	33	CAPA with 8D
Certificates of analysis (print from sales packing slip / quality order)	InventTestCertificateOfAnalysis	CoA document	COVERED-33	33	
Quality → Test instruments / Device calibration (Passage addition)			COVERED-33	33	Calibration tracking
Quality → Supplier quality scorecards (Passage)			COVERED-33	33	
Quality → SPC charts (Passage)			COVERED-33	33	
Quality → Inventory blocking (quality)	InventBlocking (quality)	Auto-block on quality order creation	COVERED-33	33	Automatic blocking/release
Quality → Quarantine orders (link)	InventQuarantineOrder		COVERED-16	16	
Quality → Batch disposition on failed quality (link)			COVERED-24	24	
Quality → D365 quality management mobile flows (quality order on mobile device)	WHSQuality*	Quality on handheld	COVERED-32	32	
Quality → Vendor quality (quality by vendor, test by vendor)	InventTestAssociationTable (vendor)		COVERED-33	33	

12.25.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Quality order inquiries (by status, item, vendor, test results)		COVERED-33	33	
Nonconformance inquiries (by problem type, source, status)		COVERED-33	33	
Correction inquiries		COVERED-33	33	
Test result statistics (pass rate by item/vendor)		COVERED-33	33	
SSRS: Certificate of analysis		COVERED-33	33	
SSRS: Quality order (document) / Quality order test results		COVERED-33	33	
SSRS: Nonconformance (document) / Nonconformance tag / Nonconformance report		COVERED-33	33	
SSRS: Correction report		COVERED-33	33	
SSRS: Nonconformance operations / Cost of quality		COVERED-33	33	
SSRS: Test groups / Tests listings		COVERED-33	33	
SSRS: Quality associations listing		COVERED-33	33	
SSRS: Vendor quality performance		COVERED-33	33	
SSRS: Item sampling listing		COVERED-33	33	

12.25.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Quality → Create quality orders automatically (on events per association)		COVERED-33	33	Trigger fixtures
Quality → Validate quality orders (batch)		COVERED-33	33	
Quality → Delete / Close nonconformances		COVERED-33	33	
Quality → Calibration due processing (Passage)		COVERED-33	33	
Quality → Supplier scorecard recalculation		COVERED-33	33	
Quality → Nonconformance cost posting (operations charges)		COVERED-33	33	

12.25.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Inventory and warehouse management parameters — Quality management tab	Use quality management, hours per day, report setup (CoA, NC tag), nonconformance defaults	COVERED-33	33	
Quality management → Quality associations (reference type: purchase/sales/production/quarantine/inventory; item code; account code; event type: before/after; execution: before/after; test group; document type; dimensions)		COVERED-33	33	

Name	Key parameters / options	Passage status	Wave	Notes
Quality management → Test groups (tests, sequence, sampling, acceptable quality level, instrument, destructive, test variable/outcome)		COVERED-33	33	
Quality management → Tests (type: integer/fraction/option, unit, instrument)		COVERED-33	33	
Quality management → Test variables / Test variable outcomes (pass/fail outcomes)		COVERED-33	33	
Quality management → Test instruments		COVERED-33	33	
Quality management → Item sampling (fixed quantity / percent / license plate / full blocking)		COVERED-33	33	AQL sampling plans
Quality management → Quality groups / Item quality groups (D365)		COVERED-33	33	
Quality management → Nonconformance: Problem types, Quarantine zones, Diagnostic types, Operations (NC operation types), Charges on NC operations		COVERED-33	33	
Quality management → Nonconformance types (customer, service request, internal, vendor, production)		COVERED-33	33	
Quality management → Certificate of analysis setup (print per customer / item)		COVERED-33	33	
Quality management → Quality orders number sequences		COVERED-5	5	
Quality management → Quality order blocking parameters (block on create, release on validate)		COVERED-33	33	
Quality management → Calibration schedules (Passage)		COVERED-33	33	
Quality management → CAPA templates (8D) (Passage)		COVERED-33	33	
Quality management → Workflow (nonconformance approval)		COVERED-4	4	
Quality management → Mobile device quality menu items (D365)		COVERED-32	32	
Quality management → Product compliance (PI, link)		GAP	—	Propose wave 33: product compliance

12.26 Warehouse management (WHS)

The AX 2012 R3 advanced warehouse module (Warehouse management) — location directives, work templates, waves, loads, license plates, mobile device menu items, replenishment, cycle counting, packing, labels — maps to wave 31 (rules, work, waves, loads, replenishment, slotting) and wave 32 (mobile app, labels, RFID/voice). Dock scheduling is wave 31; TMS pieces (§12.27) wave 34.

12.26.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
All loads (list page) / Load details	WHSLoadTableListPage / WHSLoadTable	Loads with lines, shipments, status, carrier	COVERED-31	31	Load planning
Load planning workbench	WHSLoadPlanningWorkbench	Build loads from order lines, templates, release	COVERED-31	31	
Shipments (list page) / Shipment details	WHSShipmentTableListPage / WHSShipmentTable	Shipment per order/load with containers	COVERED-28	28	WHS shipments in 31
Waves (list page) / Wave details	WHSWaveTableListPage / WHSWaveTable	Wave lines, process, release, status	COVERED-31	31	
All work (list page) / Work details	WHSWorkTableListPage / WHSWorkTable	Work headers/lines (pick/put), status, user, LP	COVERED-31	31	
Work → Open work / Work by user / Work audit	WHSWorkListPage (filters)		COVERED-31	31	
Work → Cancel work / Reassign work / Unblock	WHSWorkCancel, WHSWorkReassign	Exceptions	COVERED-31	31	
License plates (list) / License plate details	WHSLicensePlate	LP with nested parents, on-hand	COVERED-31	31	
Locations (WHS)	WHSLocation (WMSLocation with profile)	Bins with profile, zone, stocking limits	COVERED-31	31	
Containers (list) / Containerization	WHSContainerTable	Packing containers, SSCC	COVERED-28	28	Containers/HU
Pack (packing station form)	WHSPackContainer (Pack)	Pack items into containers at packing station	COVERED-28	28	
Replenishment → Replenishment work / Min-max	WHSReplenishment		COVERED-31	31	
Cycle counting → Cycle count work / Cycle count plans runs	WHSCycleCountPlan (run)		COVERED-31	31	
Cycle counting → Count approvals (variance review)	WHSCycleCountWorkApproval		COVERED-31	31	
Inventory status (on-hand by status)	WHSInventoryStatus		COVERED-16	16	
Dock appointment scheduling → Appointments / Dock management	TMSAppointment* (TMSDockMgmt)	Carrier appointments per dock	COVERED-31	31	Dock scheduling
Inbound → ASN (Advanced shipping notice import) / Inbound loads	WHSUOMStructure / WHSASN	ASN with packing structure to LP	COVERED-31	31	ASN-driven receiving

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Inbound → Receiving (mobile) / Receipt lines	WHSRFReceive		COVERED-32	32	
Outbound → Release to warehouse (sales / transfer / production)	WHSReleaseToWarehouse		COVERED-28	28	
Outbound → Manual packing / Ship confirm loads / Bill of lading	WHSLoadShipConfirm, WHSBillOfLading		COVERED-28	28	BOL doc
Outbound → Cross docking (D365 planned cross docking, opportunistic)	WHSCrossDocking	Cross-dock inbound to outbound	PARTIAL-31 (cross docking (planned/opportunistic) not named)	31	Gap register
Outbound → Wave label printing / Document routing	WHSDocumentRouting		COVERED-32	32	
Mobile device → Work queue / Mobile device portal (AX 2012 R3 WHS mobile portal, IIS)	WHSWorkExecute (mobile)	Handheld execution	SUPERSEDED (PWA + native shell mobile app wave 32)	32	
Mobile device → System-directed / user-directed picking, cluster picking, work with LP, print	WHSWorkExecute (menu items)		COVERED-32	32	
Work users / Work user sessions	WHSWorkUser / WHSWorkUserSession		COVERED-32	32	Session/device management
Product → Warehouse tab on released product (unit sequence group, reservation hierarchy, filter codes, physical dimensions)	InventTable (WHS)		COVERED-31	31	
Inventory → Movement by template (ad hoc moves)	WHSWorkTemplate (movement)		COVERED-31	31	
Inventory → Inventory adjustments (WHS)	WHSRFAdjustment		COVERED-32	32	
Inventory → License plate build / Consolidate / Split			COVERED-31	31	
Inventory → Inventory status change	WHSInventoryStatusChange		COVERED-16	16	
Production → Raw material picking / RAF with LP / Put-away from production (WHS)	WHSProdPickingList		COVERED-31	31	
Kanban → Kanban supermarket (WHS lean)			COVERED-31	31	
Quality → Quality orders on mobile / Quality item sampling LP (link)			COVERED-32	32	
WHS → Warehouse slotting (D365 slotting templates, slotting demand)	WHSSlotting		COVERED-31	31	Slotting analysis
WHS → Warehouse workload visualization (D365 workspace)			COVERED-31	31	Warehouse KPIs
WHS → Catch weight on mobile (D365)			COVERED-24	24	
WHS → Warehouse app task validator / Detours (D365)	WHSMobileDeviceDetour		COVERED-32	32	Flow designer
WHS → Sales order picking with packing structure (D365 packing structure)			COVERED-28	28	
WHS → Transfer order WHS flow (ship/receive with LP)			PARTIAL-16 (transfer orders)	31	WHS execution of transfer orders depends on transfer orders gap

12.26.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
On-hand by location / On-hand by license plate		COVERED-31	31	
Work inquiries (Work line history, Work audit, Work exceptions log)		COVERED-31	31	
Location load / Location stocking limits inquiry		COVERED-31	31	Capacity never exceeded
Wave inquiries (Wave history, Wave lines)		COVERED-31	31	
Load inquiries (Load lines, Load history, Shipments per load)		COVERED-31	31	
License plate inquiry / LP history		COVERED-31	31	
Cycle count inquiries (Count history, Variances)		COVERED-31	31	
Replenishment inquiry		COVERED-31	31	
Mobile device inquiries (User sessions, Work user activity)		COVERED-32	32	
Inventory status inquiry		COVERED-16	16	
Dock appointment inquiries		COVERED-31	31	
Warehouse KPIs (throughput, pick rates, accuracy)		COVERED-46	46	
SSRS: Pick list (WHS work document) / Work pick list		COVERED-31	31	
SSRS: Packing slip (WHS) / Bill of lading / Container label / Shipping label (SSCC)		COVERED-28	28	Labels 32
SSRS: License plate label / Location label / Wave label		COVERED-32	32	ZPL
SSRS: Load report / Load manifest		COVERED-31	31	
SSRS: Cycle count report / Count sheets		COVERED-31	31	
SSRS: Location report / Zone listing / Location profile listing		COVERED-31	31	
SSRS: Work template / Location directive listings		COVERED-31	31	
SSRS: Replenishment report		COVERED-31	31	
SSRS: Inventory status report		COVERED-16	16	
SSRS: Dock schedule		COVERED-31	31	
SSRS: Work user productivity		COVERED-46	46	
SSRS: Cross docking report		PARTIAL-31 (cross docking)	31	

12.26.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Wave processing (batch wave release / scheduled waves)		COVERED-31	31	
Replenishment → Min-max replenishment / Demand replenishment / Wave demand replenishment (batch)		COVERED-31	31	
Replenishment → Load demand replenishment (D365)		COVERED-31	31	
Cycle counting → Cycle count plan processing (scheduled) / Threshold counting		COVERED-31	31	
Cycle counting → Approve/reject count variances (batch)		COVERED-31	31	
Work → Delete work / Cancel expired work / Clean up work history		COVERED-31	31	
Loads → Ship confirm (batch) / Load posting (packing slip from load)		COVERED-28	28	
Loads → Outbound load planning (batch build by template)		COVERED-31	31	
Inbound → ASN import (EDI 856) batch		COVERED-35	35	
Mobile → Label reprint batch / Document routing cleanup		COVERED-32	32	
Inventory status → Mass status change		COVERED-16	16	
Slotting → Run slotting / Generate slotting replenishment		COVERED-31	31	
License plates → Clean up empty license plates		COVERED-31	31	
Work users → Expire sessions		COVERED-32	32	
Cross docking → Planned cross docking processing		PARTIAL-31 (cross docking)	31	
WHS → Release to warehouse (batch, automatic release by rule)		COVERED-28	28	Order release rules
WHS → Warehouse workload refresh		COVERED-31	31	
WHS → Delete expired mobile device user sessions / Clean up WHS history (work line history retention)		COVERED-31	31	

12.26.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Warehouse management parameters — General	Default location profiles, LP numbering, user/system-directed defaults, work audit, pack defaults, cycle count defaults, replenishment defaults, dock management, number sequences	COVERED-31	31	
Warehouse management parameters — Mobile device / Print / Barcode	Mobile device settings, display, printers	COVERED-32	32	
Warehouse (Warehouse details with "Use warehouse management processes", default receipt/shipping/production locations, license plate receiving)		COVERED-31	31	
Location setup → Location profiles (allow mixed items/batches/status, LP tracking, cycle counting, stocking limits, location format)		COVERED-31	31	
Location setup → Location formats (segments, separators)		COVERED-31	31	
Location setup → Location types (quarantine, shipping, receiving, staging, final shipping)		COVERED-31	31	
Location setup → Zones / Zone groups		COVERED-31	31	
Location setup → Locations (bins) with coordinates, sort codes		COVERED-31	31	
Location setup → Fixed locations (item fixed locations)		COVERED-31	31	
Location setup → Location stocking limits (by item/UoM/container type/physical dimensions)		COVERED-31	31	
Location setup → Location directives (work order type, direction, work type; lines with query, strategy (consolidate, FEFO batch, LP empty, round up, lowest qty…), location directive actions with query, fixed location usage, allow negative, restrict by UoM)		COVERED-31	31	Directive fixtures
Location setup → Location directive failures (D365 troubleshooting)		COVERED-31	31	
Location setup → Directive codes		COVERED-31	31	
Work → Work templates (work order type, work pools, work classes, work template lines: pick/put/print/inventory status change/custom; work header breaks; max pick quantity; directive codes)		COVERED-31	31	
Work → Work pools		COVERED-31	31	
Work → Work classes (valid location types per class)		COVERED-31	31	
Work → Work users (user id, default warehouse, menu, allowed work classes, process cycle counting, max LP)		COVERED-32	32	
Work → Work exceptions (short pick exceptions, pick overrides)		COVERED-31	31	Short picks
Work → Work priorities / Work blocking reasons (D365)		COVERED-31	31	
Work → Work policies (D365: work creation rules for production/transfer)		COVERED-31	31	
Work → Work cancel reasons (D365)		COVERED-31	31	
Waves → Wave templates (shipping, manufacturing, kanban; wave methods: allocate wave, create loads, create work, containerize, replenish, release to warehouse; automatic processing, wave thresholds, attributes)		COVERED-31	31	
Waves → Wave process methods (register custom methods)		COVERED-31	31	
Waves → Wave attributes / Wave step codes (D365)		COVERED-31	31	
Waves → Wave label templates (D365)		COVERED-32	32	
Load → Load templates (capacity: weight/volume/floor space, load template group)		COVERED-31	31	
Load → Load building templates / Load building strategies (D365: volume-based)		COVERED-31	31	
Load → Load posting methods (packing slip, sales invoice, transfer shipment)		COVERED-28	28	
Load → Load line type / Load status (customizable)		COVERED-31	31	

Name	Key parameters / options	Passage status	Wave	Notes
Mobile device → Mobile device menu items (work: pick/put, user-directed, system-directed, cluster picking, cycle counting, inquiry, adjustments, movement, LP build, print, production RAF, kanban, quality; indirect activities; options per item: generate LP, override, validation)		COVERED-32	32	Flow designer
Mobile device → Mobile device menus (hierarchy of menu items)		COVERED-32	32	
Mobile device → Mobile device display settings / Work user display / Mobile device steps configuration (D365 step configuration, field names)		COVERED-32	32	
Mobile device → Mobile device overrides / Detours (D365)		COVERED-32	32	
Mobile device → Mobile device barcode setup / Scan data parsing (GS1) (D365)		COVERED-32	32	GS1 AI parser
Mobile device → Work user sessions / Device management (D365 Warehouse app)		COVERED-32	32	
Mobile device → Cluster profiles (cluster picking)		COVERED-32	32	
Mobile device → Work confirmation / Validation (location, LP, item confirmation per menu item)		COVERED-32	32	
Mobile device → Indirect activity codes (warehouse)		COVERED-32	32	
Mobile device → Mobile device user authentication (D365 device token)		COVERED-2	2	Keycloak device auth
Inventory → Inventory statuses / Inventory status transitions		COVERED-16	16	
Inventory → Disposition codes (WHS: receiving disposition → inventory status + work template)		COVERED-31	31	
Inventory → Unit sequence groups (pallet/case/each with LP grouping and packing)		COVERED-31	31	
Inventory → Reservation hierarchies		COVERED-31	31	
Inventory → Filters (Filter codes, Filter groups, Product filter assignment)		COVERED-31	31	
Inventory → Physical dimensions (per item/UoM)		COVERED-31	31	
Inventory → Warehouse-specific item setup (fixed locations, min/max, replenishment)		COVERED-31	31	
Packing → Container types / Container groups / Container build templates (containerization during wave) / Container packing policies / Packing profiles / Packing station setup		COVERED-28	28	
Packing → Packing policies (D365 manual packing: packing slip/label on close)		COVERED-28	28	
Replenishment → Replenishment templates (min/max, demand, wave demand, load demand, slotting; replenishment units, work template)		COVERED-31	31	
Replenishment → Min/max replenishment rules per item/location		COVERED-31	31	
Replenishment → Fixed location replenishment / Immediate replenishment (D365)		COVERED-31	31	
Cycle counting → Cycle count plans (query, days between counts, work pool, max locations)		COVERED-31	31	
Cycle counting → Cycle count thresholds (quantity/percent threshold triggers)		COVERED-31	31	
Cycle counting → Cycle count approval parameters / Counting reason codes (link)		COVERED-31	31	
Cycle counting → Spot cycle counting / Supervisor approval		COVERED-31	31	
Label → Document routing (printer per work/user/warehouse), Document routing layouts (ZPL templates), Printers (print service)		COVERED-32	32	ZPL print service
Label → License plate label layouts / Container label layouts (D365 label layouts)		COVERED-32	32	
Label → Wave label printing setup (D365)		COVERED-32	32	
Dock scheduling → Appointment rules, Dock profiles, Dock management profiles, Appointment schedules, Appointment durations		COVERED-31	31	
Inbound → ASN setup (packing structure import, EDI 856 mapping)		COVERED-35	35	
Inbound → Inbound load statuses / Receiving policies (LP receiving, mixed LP receiving)		COVERED-31	31	
Outbound → Release to warehouse rules (automatic release, fulfillment rate %, partial release)		COVERED-28	28	
Outbound → Ship confirm setup / Bill of lading setup		COVERED-28	28	
Outbound → Shipping carrier / Mode of delivery mapping to WHS		COVERED-28	28	
Outbound → Cross docking templates (D365)		PARTIAL-31 (cross docking)	31	
Outbound → Outbound sorting (D365 outbound sorting positions)		PARTIAL-31 (outbound sorting (sort pick-to-shipment at a sortation area) not named)	31	Gap register
Slotting → Slotting templates / Slotting specifications (D365)		COVERED-31	31	
Warehouse → Production: Production input/output location setup, Raw material picking policies, RAF put-away policies, Material staging		COVERED-31	31	
Warehouse → Kanban: Kanban board for WHS, Supermarket replenishment		COVERED-31	31	
Warehouse → Catch weight on WHS (D365 CW tags)		COVERED-24	24	
Warehouse → Warehouse management workload visualization setup		COVERED-31	31	
Warehouse → Warehouse app task validator / Process guide steps (D365)		COVERED-32	32	
Warehouse → RFID / Voice adapter configuration (Passage)		COVERED-32	32	

Name	Key parameters / options	Passage status	Wave	Notes
Warehouse → Transportation (link to TMS)		COVERED-34	34	
Warehouse → Number sequences (work, wave, load, LP, container)		COVERED-5	5	
Warehouse → Work audit templates		COVERED-31	31	
Warehouse → Warehouse management (WHS) workflows (D365 none)		COVERED-4	4	
Warehouse → Yard management (D365 2024: yard, trailers, yard moves)		GAP	—	Propose wave 34: yard management (trailer/yard locations, check-in/out, yard moves, dock door assignment) alongside TMS driver check-in

12.27 Transportation management (TMS)

AX 2012 R3 TMS: rating engines, rate masters, carriers, route plans/guides, load building, tendering, freight bills and reconciliation, driver check-in, transportation tenders, dock appointment scheduling. Passage wave 34 covers rating, routing, load building, tendering, tracking, freight reconciliation and landed cost; yard/driver check-in is a gap.

12.27.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Loads (planning workbench, TMS view)	WHSLoadPlanningWorkbench (TMS)	Rate loads, assign carriers	COVERED-34	34	
Rate route workbench	TMSRateRouteWorkbench	Rate shop across carriers/routes for a load	COVERED-34	34	Rate shopping
Transportation tenders / Carrier tendering	TMSLoadTender	Tender to carrier, accept/reject, timeouts	COVERED-34	34	Tendering workflow
Freight bills	TMSFreightBill	Expected freight bill per load leg with charges	COVERED-34	34	
Freight bill invoices / Freight invoice matching	TMSFreightBillInvoice, TMSFreightBillMatching	Carrier invoice vs freight bill, reconciliation	COVERED-34	34	Freight reconciliation
Driver check-in / check-out	TMSDriverCheckIn	Driver/trailer arrival at dock, appointment link	GAP	—	Propose wave 34: driver check-in/out with appointment link, trailer/seal capture, gate log (with yard management)
Routes (transportation routes: multi-leg, hubs)	TMSRoute	Route with segments and hubs	COVERED-34	34	Route planning
Route plans / Route guides	TMSRoutePlan, TMSRouteGuide	Planned legs and automatic carrier selection rules	COVERED-34	34	
Shipments (TMS view) / Shipment tracking	WHSShipmentTable (TMS)	Tracking events, POD	COVERED-34	34	EDI 214/API
Load → Rate loads / Rate shipments	TMSRating	Rating per load/shipment	COVERED-34	34	
Load → Hub consolidation / Load building (manual)	TMSHub	Consolidate at hubs	COVERED-34	34	
Load → Freight charge allocation to orders (customer charges from freight)	TMSFreightChargeAllocation	Pass freight to sales orders as charges	COVERED-34	34	
Load → Transportation status / Carrier invoice reasons	TMSTransportationStatus		COVERED-34	34	
Landed cost (D365 module; voyages, link)			PARTIAL-34 (landed cost voyages)	34	
Carrier → Carrier rate cards (Rate masters, Rate base)	TMSRateMaster		COVERED-34	34	
Dock appointment scheduling (link)			COVERED-31	31	
Inbound → Inbound load rating / Transportation on purchase orders			COVERED-34	34	
Transportation → Transportation management workspace (D365 outbound/inbound)			COVERED-34	34	

12.27.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Freight bill inquiries / Freight invoice reconciliation status		COVERED-34	34	
Tender history / Tender status inquiry		COVERED-34	34	
Load tracking inquiry (tracking events)		COVERED-34	34	
Rating results inquiry (rate shopping log)		COVERED-34	34	
Carrier performance (on-time, cost per mile)		COVERED-46	46	
Driver check-in log		GAP	—	Propose wave 34: driver check-in
SSRS: Bill of lading (master/underlying)		COVERED-28	28	
SSRS: Load manifest / Load tender document		COVERED-34	34	
SSRS: Freight bill / Freight invoice reconciliation report		COVERED-34	34	
SSRS: Carrier rate listing / Rate master listing		COVERED-34	34	
SSRS: Route guide listing / Route plan listing		COVERED-34	34	
SSRS: Transportation tender summary		COVERED-34	34	
SSRS: Freight accrual report		COVERED-34	34	
SSRS: Dock schedule (link)		COVERED-31	31	

Name	Purpose	Passage status	Wave	Notes
SSRS: Landed cost reports (link)		PARTIAL-34 (landed cost voyages)	34	

12.27.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Rate loads (batch rating) / Auto-rate on release		COVERED-34	34	
Tendering → Send tenders (batch) / Tender timeout processing / Cascade to next carrier		COVERED-34	34	
Freight reconciliation → Match freight invoices (auto) / Post freight accruals / Generate AP invoice for matched freight		COVERED-34	34	
Tracking → Import tracking events (EDI 214 / carrier API polling)		COVERED-34	34	
Load building → Automatic load building (strategy)		COVERED-34	34	
Route → Route guide evaluation (batch assign carriers)		COVERED-34	34	
Fuel index → Update fuel surcharges from index		COVERED-34	34	
Landed cost → Voyage cost actualization (link)		PARTIAL-34 (landed cost voyages)	34	
Freight charges → Allocate freight to sales/purchase orders (batch)		COVERED-34	34	
Transportation → Clean up tenders / rating logs		COVERED-34	34	
Driver check-in → Expire appointments / Gate log archive		GAP	—	Propose wave 34: driver check-in

12.27.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Transportation management parameters — General	Default rating profile, tender settings, freight bill types, dock management, number sequences, appointment settings	COVERED-34	34	
Carriers → Shipping carriers (carrier vendor, services, rating profile, tender method, tracking)		COVERED-34	34	
Carriers → Carrier services / Carrier groups / Carrier rating profiles		COVERED-34	34	
Carriers → Transportation methods / Transportation modes		COVERED-34	34	
Carriers → Carrier invoice reasons / Reconciliation reasons		COVERED-34	34	
Rating → Rating profiles (rating metadata: rate base, rate master, engine)		COVERED-34	34	
Rating → Rate masters / Rate base / Rate base type / Rate base assignments (effective-dated rate tables: per mile, per weight break, per zone)		COVERED-34	34	
Rating → Rating metadata (rate structure definitions)		COVERED-34	34	
Rating → Accessorial masters / Accessorial assignments / Accessorial charges (fuel surcharge, liftgate, residential)		COVERED-34	34	
Rating → Fuel indexes / Fuel surcharge tables		COVERED-34	34	
Rating → Zone masters / Zones / Zone engines (postal code zones)		COVERED-34	34	
Rating → Engines: Rating engines, Generic engines, Transit time engines, Mileage engines (PC*MILER/Google Maps adapter), Apportionment engines, Rate engines (custom)		COVERED-34	34	Adapter interface; OR-Tools for VRP
Rating → Transit time masters		COVERED-34	34	
Routing → Route plans / Route guides / Route schedules / Hubs (hub masters) / Transportation constraints (carrier/item/customer constraints)		COVERED-34	34	
Routing → Route segments / Legs		COVERED-34	34	
Load building → Load building strategies / Load templates (link)		COVERED-34	34	
Freight billing → Freight bill types / Freight bill type assignments (which charges belong to which party) / Audit masters (tolerances for matching)		COVERED-34	34	
Freight billing → Freight accrual posting (ledger accounts) / Freight charge allocation rules (to orders)		COVERED-34	34	
Tendering → Tender types / Tender timeout rules / Tender broadcast (D365)		COVERED-34	34	
Tracking → Transportation status master / Status codes (EDI 214 mapping) / Proof of delivery setup		COVERED-34	34	
Dock appointment → Appointment rules / Dock profiles / Appointment scheduling calendar		COVERED-31	31	
Driver check-in → Check-in setup / Gate configuration		GAP	—	Propose wave 34: driver check-in
Landed cost → Landed cost parameters / Cost types / Voyage templates (link)		PARTIAL-34 (landed cost voyages)	34	
Transportation → Container / Trailer types		COVERED-34	34	
Transportation → Transportation document types / Bill of lading setup		COVERED-28	28	
Transportation → Destination codes / Postal code masters		COVERED-34	34	
Transportation → Number sequences (load tender, freight bill)		COVERED-5	5	
Transportation → Workflow (freight invoice approval)		COVERED-4	4	
Transportation → Carrier EDI/API connections (204/990/214 and carrier APIs)		COVERED-35	35	
Transportation → Yard management setup (D365 2024)		GAP	—	Propose wave 34: yard management

12.28 Service management

Service agreements, service orders, dispatch board, service objects/tasks, repair, subscriptions and service BOM in AX 2012 R3 map to wave 45 (installed base, agreements/SLA, cases, work orders, dispatch, mobile). Service subscriptions billing shares wave 29.

12.28.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Service agreements (list page)	SMAAgreementTableListPage	Agreements with lines (service objects/tasks, intervals, prices)	COVERED-45	45	Service agreements with entitlements/SLAs
Service agreement details	SMAAgreementTable / SMAAgreementLine	Lines, service level agreement, project link, templates	COVERED-45	45	
Service orders (list page)	SMAServiceOrderTableListPage	Service orders from agreements/manual; lines (hour, item, expense, fee)	COVERED-45	45	Service work orders
Service order details	SMAServiceOrderTable / SMAServiceOrderLine	Technician, service object, task, repair, stage, SLA timer, posting to project	COVERED-45	45	
Dispatch board	SMADispatchBoard	Gantt of technicians × service orders, drag/drop	COVERED-45	45	Dispatch optimization (OR-Tools)
Service objects (installed base)	SMAServiceObjectTable	Customer equipment, serial, template BOM	COVERED-45	45	Installed base from shipped serials
Service object relations (object ↔ agreement/customer)	SMAServiceObjectRelation		COVERED-45	45	
Service tasks / Service task relations	SMAServiceTaskTable		COVERED-45	45	
Repair → Repair lines (symptom, diagnosis, resolution, repair stage)	SMARepairLine	Repair tracking	COVERED-45	45	Repair orders
Service subscriptions (subscription fee transactions, index)	SMASubscriptionTable	Recurring subscription fees via project	PARTIAL-45 (AX service subscriptions (fee indexation, accrual via project) not explicitly named; subscription billing is wave 29)	29	Gap register
Service → Service stages / Stage transitions	SMAServiceStage		COVERED-45	45	
Service → Service level agreements (time recording, SLA compliance)	SMAServiceLevelAgreement		COVERED-45	45	SLA attainment
Service → Service order sign-off / Customer signature (mobile)			COVERED-45	45	Mobile app signatures
Service → Cases (customer service cases, link to Org admin case management)	CaseDetail (service)		COVERED-45	45	Cases → service work orders
Service → Technician mobile (D365 Field Service mobile)			COVERED-45	45	
Service → Warranty tracking (D365 warranty / AX via service object)			COVERED-45	45	Warranty terms
Service → Service BOM (service object BOM / template BOM)	SMAServiceObjectBOM		COVERED-45	45	
Service → Parts consumption from technician van stock			COVERED-45	45	Mobile inventory locations
Service → Service order project posting (hours/items/expenses to project)			COVERED-41	41	
Service → Service order invoicing (via project invoice proposal)			COVERED-41	41	Billing billable vs warranty

12.28.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Service order inquiries (by stage, technician, customer, SLA status)		COVERED-45	45	
Service agreement inquiries (expiring agreements, lines due)		COVERED-45	45	
Service object inquiries (history per object)		COVERED-45	45	
Repair statistics (symptom/diagnosis/resolution frequency)		COVERED-45	45	
SLA compliance inquiry		COVERED-45	45	
Subscription inquiries		PARTIAL-45 (service subscriptions)	29	
Service KPIs (first-time fix, SLA attainment, technician utilization)		COVERED-45	45	
SSRS: Service order (document) / Service order lines / Service order status		COVERED-45	45	
SSRS: Service agreement (document) / Agreement lines		COVERED-45	45	
SSRS: Service object list / Service object history		COVERED-45	45	
SSRS: Repair report / Repair statistics		COVERED-45	45	
SSRS: Dispatch list / Technician schedule		COVERED-45	45	
SSRS: SLA report		COVERED-45	45	
SSRS: Subscription report / Subscription fee transactions		PARTIAL-45 (service subscriptions)	29	
SSRS: Service invoice (via project)		COVERED-41	41	

12.28.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Create service orders from agreements (periodic generation by interval)		COVERED-45	45	
Service agreements → Copy / Renew / Expire agreements		COVERED-45	45	

Name	Purpose	Passage status	Wave	Notes
Subscriptions → Create subscription fee transactions / Accrue subscription revenue / Index subscriptions		PARTIAL-45 (service subscriptions)	29	
Service orders → Post service orders (to project) / Invoice (via project)		COVERED-41	41	
Service orders → Update stages (batch) / Close		COVERED-45	45	
Dispatch → Automatic dispatch optimization run		COVERED-45	45	
Service → SLA breach notifications		COVERED-45	45	
Service → Synchronize installed base from shipments (serials)		COVERED-45	45	
Service → Mobile sync / Offline reconciliation		COVERED-45	45	

12.28.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Service management parameters — General / Service orders / Agreements / Subscriptions / Dispatching / Number sequences	Default stages, project posting, time recording, SLA settings, subscription defaults	COVERED-45	45	Subscriptions PARTIAL
Service → Service stages / Stage reasons / Stage transitions		COVERED-45	45	
Service → Service order types / Service priorities		COVERED-45	45	
Service → Service level agreements (SLA definitions: time to respond/resolve, calendars)		COVERED-45	45	
Service → Service objects: Service object groups, Service object templates		COVERED-45	45	
Service → Service tasks / Service task groups		COVERED-45	45	
Service → Service agreement groups / Agreement templates / Service intervals		COVERED-45	45	
Service → Repair: Conditions, Symptom areas, Symptom codes, Diagnosis areas, Diagnosis codes, Resolutions, Repair stages		COVERED-45	45	
Service → Dispatching: Dispatch teams, Technicians (workers), Skills, Territories, Dispatch board settings		COVERED-45	45	
Service → Subscription: Subscription groups, Subscription fee categories, Index rules, Subscription periods		PARTIAL-45 (service subscriptions)	29	
Service → Project integration (project group for service, categories, line properties)		COVERED-41	41	
Service → Warranty types / Entitlement types (Passage)		COVERED-45	45	
Service → Mobile app configuration (offline, photos, signatures)		COVERED-45	45	
Service → Van stock / Technician warehouses		COVERED-45	45	
Service → Workflow (service order approval)		COVERED-4	4	
Service → Number sequences		COVERED-5	5	
Service → Case categories for service (link)		COVERED-45	45	
Service → Customer portal: case submission / service request (link)		COVERED-30	30	

12.29 Organization administration

Legal entities, operating units, hierarchies, address book, number sequences, document management, case management, workflow configuration, calendars, units, electronic signatures, foreign trade, policies and signing limits. Passage: waves 6 (structure, calendars, UoM, address book), 7 (global address book), 5 (number sequences, audit), 4 (workflow). Case management, document management depth, reason codes, teams and electronic signatures are partial or gaps.

12.29.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Legal entities (list) / Legal entity details	OMLegalEntity (CompanyInfo)	Company, addresses, tax registration, bank, statutory reporting, foreign trade, ER tabs	COVERED-6	6	
Operating units (departments, cost centers, business units, value streams, retail channels)	OMOperatingUnit		COVERED-6	6	
Organization hierarchies / Hierarchy designer (with purposes, publish with effective date, view hierarchy)	OMHierarchyDesigner, OMHierarchyPurpose	Multi-purpose effective-dated hierarchies	COVERED-6	6	
Global address book (list) / Party details	DirPartyTableListPage / DirPartyTable	Parties with roles, addresses, contact info, relationships	COVERED-7	7	
Global address book → Duplicate check / Merge parties	DirDuplicateCheck, DirPartyMerge		COVERED-7	7	
Address books (address book groupings with team access)	DirAddressBook		COVERED-7	7	
Teams / Team types	OMTeam		PARTIAL-6 (teams)	6	
Cases (list page) / Case details	CaseDetailListPage / CaseDetail	Case with category, process, activities, log, knowledge articles, associations, resolution	PARTIAL-45 (generic case management (categories, processes, activities, knowledge articles, associations to any record) is not named in any wave; service cases are in 45, disputes in 30, product change in 9)	45	Gap register
Cases → Knowledge articles	CaseKnowledgeArticle		PARTIAL-45 (generic case management)	45	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Document management → Document handling (attachments on any record), Document types, File types	DocuView, DocuType		PARTIAL-5 (attachments exist in 5/11; document types, storage locations (SharePoint/S3), archive directories, retention are not named)	5	Gap register
Document management → Document templates (Word/Excel mail merge)	DocuTemplate		OUT-OF-SCOPE (CRM mail merge non-goal; business documents via wave 46)	—	
Number sequences (list) / Number sequence details / Number sequence wizard	NumberSequenceTableListPage, NumberSequenceTable, NumberSeqWizard	Scopes (shared, company, legal entity, operating unit, fiscal calendar period), continuous, pre-allocation, format segments	COVERED-5	5	
Number sequences → References (per data type/scope)	NumberSequenceReference		COVERED-5	5	
Workflow → Workflow configurations (per type: PO, requisition, journal, etc.) / Workflow editor	WorkflowTable, Workflow editor (WF4)	Graphical designer, versions, activation	COVERED-4	4	Workflow designer
Workflow → Workflow history / Work items	WorkflowStatus, WorkflowWorkItem		COVERED-4	4	
Calendars (Org admin: working time calendars shared with production/HR)	WorkCalendarTable		COVERED-9	9	
Units / Unit conversions / Unit classes / Unit translations	UnitOfMeasure, UnitOfMeasureConversion		COVERED-6	6	
Foreign trade → Intrastat (journal) / EU sales list (link)	Intrastat, TaxEUSalesList		PARTIAL-13 (Intrastat setup)	13	Intrastat extract named; journal maintenance/corrections not
Electronic signatures → Signature log / Reset signature	SysSignatureLog		GAP	—	Propose wave 48: electronic signatures
Policies → Signing limits (Signing limit policies, Worker signing limit requests, Default signing limits by job/compensation level)	SigningLimit*		PARTIAL-17 (signing limits)	17	
Policies → Policy framework forms (Purchasing policies, Expense policies, Audit policies, Vendor invoice policies, Signing limit policies)	SysPolicy*	Policy framework with org hierarchy scoping	PARTIAL-17 (a generic policy framework scoped by org hierarchy is implied by 17/43 but not named as reusable)	17	Gap register
Product safety data sheets (link, PI)			GAP	—	Propose wave 33: product compliance
Reason codes (shared financial reasons across modules)	ReasonTable	Fixed assets, customer, vendor, bank, ledger reasons	PARTIAL-11 (reason codes)	11	
Address → Address setup: Address formats, Address purposes, Name sequences, Name affixes, ZIP/postal codes, Counties, States/provinces, Countries/regions, Districts	LogisticsAddress*		COVERED-6	6	Address book with validated postal addresses
Address → Address books parameters / Global address book parameters (security by address book, name sequence defaults, duplicate check on)	DirParameters		COVERED-7	7	
Global address book → Party relationship types / Role security	DirRelationshipType		COVERED-7	7	
Organization → Translations (legal entity/operating unit name translations)			COVERED-6	6	
Organization → Legal entity logos / Document branding	CompanyImage		COVERED-46	46	Document designer branding
Electronic signatures → Certificates (user signing certificates)	SysSignatureCertificate		GAP	—	Propose wave 48: electronic signatures
Print management (org-wide)	PrintMgmtDocType		PARTIAL-46 (print management routing)	46	
Alerts (org-level alert rules)	EventRule		GAP	—	Propose wave 5: alerts and notification rules (change-based and due-date alerts on any entity/field, delivered in-app/email/webhook) built on the wave 4 event backbone
Questionnaires (link, Org admin > Questionnaires in R3)	KMQuestionnaire		GAP	—	Propose wave 42: questionnaire engine
Case management → Case processes / Case categories (link)	CaseProcess, CaseCategory		PARTIAL-45 (generic case management)	45	
Organization → Vendor collaboration user provisioning (link)			COVERED-20	20	

12.29.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Organization hierarchy inquiry (as-of date view)		COVERED-6	6	
Number sequence inquiries (status list, pre-allocation, gaps)		COVERED-5	5	
Workflow inquiries (Workflow history, Workflow performance, Pending work items)		COVERED-4	4	
Case inquiries (cases by category/status, SLA)		PARTIAL-45 (generic case management)	45	

Name	Purpose	Passage status	Wave	Notes
Global address book inquiries (parties by role, duplicate candidates)		COVERED-7	7	
Document management inquiries (documents by type/record)		PARTIAL-5 (document management)	5	
Electronic signature inquiry		GAP	—	Propose wave 48: electronic signatures
SSRS: Legal entities list / Operating units list / Organization hierarchy report		COVERED-6	6	
SSRS: Number sequence list / Number sequence references		COVERED-5	5	
SSRS: Workflow configurations / Workflow history / Workflow tracking		COVERED-4	4	
SSRS: Address book (parties, contacts, addresses) / Mailing labels		COVERED-7	7	Labels OOS (CRM)
SSRS: Case report / Case activities		PARTIAL-45 (generic case management)	45	
SSRS: Units / Unit conversions list		COVERED-6	6	
SSRS: Calendars report		COVERED-9	9	
SSRS: Intrastat report / EU sales list (link)		COVERED-13	13	
SSRS: Signing limits report		PARTIAL-17 (signing limits)	17	
SSRS: Policies report (policy rules per hierarchy)		PARTIAL-17 (policy framework)	17	

12.29.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Organization hierarchy → Publish hierarchy version (effective date)		COVERED-6	6	
Number sequences → Clean up pre-allocated / Status list cleanup / Generate pre-allocation		COVERED-5	5	
Workflow → Workflow message processing / Workflow due date processing / Workflow cleanup (batch jobs)		COVERED-4	4	Temporal timers
Workflow → Line-item workflow batch activation		COVERED-4	4	
Global address book → Run duplicate detection (batch) / Party merge batch		COVERED-7	7	
Global address book → Address validation / Geocoding refresh		COVERED-6	6	
Foreign trade → Intrastat transfer (from invoices) / Intrastat validation / Intrastat export (file)		PARTIAL-13 (Intrastat setup)	13	
Foreign trade → EU sales list transfer / Export		COVERED-13	13	
Cases → Close cases / Case SLA breach notifications		PARTIAL-45 (generic case management)	45	
Document management → Archive documents / Purge by retention		PARTIAL-5 (document management)	5	
Electronic signatures → Signature verification / Reset		GAP	—	Propose wave 48: electronic signatures
Calendars → Compose working times / Update calendar expiry		COVERED-9	9	
Signing limits → Signing limit request processing (annual recertification)		PARTIAL-17 (signing limits)	17	
Translations → Import/export system translations		COVERED-48	48	Localization framework
Organization → Delete legal entity / Legal entity copy (template)		COVERED-6	6	Tenant provisioning automation in 50

12.29.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Organization → Legal entities	Name, addresses, registration numbers, tax, bank, statutory, foreign trade, ER	COVERED-6	6	
Organization → Operating units / Departments / Cost centers / Business units / Value streams		COVERED-6	6	
Organization → Organization hierarchies / Organization hierarchy purposes (security, procurement, expenditure, budget planning, benefits, centralized payments, signing limits, audit, retail, etc.)		COVERED-6	6	
Organization → Teams / Team types		PARTIAL-6 (teams)	6	
Global address book → Global address book parameters / Address books / Address parameters (formats, purposes, name sequences, ZIP, counties, states, countries, districts, address setup)		COVERED-7	7	
Global address book → Party relationship types / Name affixes / Contact information types / Electronic address purposes		COVERED-7	7	
Global address book → Address validation (external verification service connector)		COVERED-6	6	Geocoding hook
Number sequences → Number sequences / References / Number sequence parameters (continuous, cleanup, pre-allocation) / Segment configuration (scopes)		COVERED-5	5	
Document management → Document management parameters (archive directory, SharePoint, file types, max size), Document types (file, note, URL, image, template; group; location: archive/database/SharePoint), Active document tables, File types		PARTIAL-5 (document management)	5	
Document management → Document management: OneDrive/SharePoint integration (D365), Attachment migration		PARTIAL-5 (document management)	5	Object storage (S3-compatible) is the Passage equivalent
Case management → Case categories / Case category types (HR, customer service, collections, product change, audit, service, project, knowledge, etc.) / Case processes (stages) / Case category security		PARTIAL-45 (generic case management)	45	
Workflow → Workflow parameters (notifications, email, Enterprise Portal links, due date escalation) / Workflow version management		COVERED-4	4	
Workflow → Workflow types (registered workflow types per module) / Workflow infrastructure configuration (message/due date/line-item batch)		COVERED-4	4	Temporal
Workflow → Workflow delegation (user options) / Workflow participants (hierarchy, user group, expenditure participants, queue)		COVERED-4	4	
Calendars → Calendars / Working time templates		COVERED-9	9	

Name	Key parameters / options	Passage status	Wave	Notes
Units → Units / Unit classes / Unit conversions / Unit translations / Unit of measure parameters (decimal precision)		COVERED-6	6	UCUM
Electronic signatures → Electronic signature parameters / Signature requirements / Reason codes / Certificates		GAP	—	Propose wave 48: electronic signatures
Foreign trade → Foreign trade parameters, Commodity codes, Transaction codes, Transport methods, Ports, Statistics procedures, Country/region properties (EU), Intrastat compression, Intrastat check setup, EU sales list reporting		PARTIAL-13 (Intrastat setup)	13	
Policies → Signing limit policies / Signing limit parameters / Default signing limits (per job/compensation) / Signing limit request workflow		PARTIAL-17 (signing limits)	17	
Policies → Policy parameters (per policy type: organization hierarchy purpose)		PARTIAL-17 (policy framework)	17	
Reason codes (Financial reasons: ledger, bank, customer, vendor, fixed asset)		PARTIAL-11 (reason codes)	11	
Product safety data sheets setup (link, PI)		GAP	—	Propose wave 33: product compliance
Print management → Print management setup (per module/document: original/copy destinations, conditional settings by query)		PARTIAL-46 (print management routing)	46	
Alerts → Alert rules (Change-based, Due-date; event rule setup, alert email templates)		GAP	—	Propose wave 5: alerts and notification rules
Questionnaires setup (link)		GAP	—	Propose wave 42: questionnaire engine
Organization → Fiscal calendars (shared)		COVERED-6	6	
Organization → Vendor collaboration / Customer self-service user provisioning (link)		COVERED-20	20	
Organization → Translations (system labels, custom text translations)		COVERED-48	48	
Organization → Shared categories (category hierarchies, link)		COVERED-8	8	
Organization → Record templates (per table)		COVERED-5	5	
Organization → Document branding / Report headers per legal entity		COVERED-46	46	
Organization → Data partitions (link to Sys admin)		SUPERSEDED (tenant model wave 2)	2	
Organization → Audit policies (link)		PARTIAL-48 (audit policies)	48	
Organization → Workflow approval email templates		PARTIAL-4 (email templates / SMTP outbound not explicitly named in 4 or 5)	4	Gap register

12.30 System administration

Users, security roles/duties/privileges, SoD, licensing, batch jobs, AOS/cluster, AIF services, database log, data partitions, virtual companies, Enterprise Portal, reporting/analysis services, help, alerts/email, model store. Passage supersedes most of this with waves 1 (pipeline), 2 (identity/authz), 3 (metadata), 4 (batch/workflow/events), 5 (audit/observability) and 35 (integration hub).

12.30.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Users (list) / User details (relations, roles, user options, profiles)	SysUserInfoPage / UserInfo	User accounts, domain/claims, roles per organization	COVERED-2	2	Keycloak + SCIM; admin UI
User groups	UserGroupInfo	Groups for period access/journal approval	COVERED-2	2	
User relations (worker ↔ user)	DirPersonUser		COVERED-7	7	Employee role ↔ identity
Security → Security roles / Duties / Privileges / Entry points (Security configuration, Security development tool)	SecurityRole, SecurityDuty, SecurityPrivilege	Role-based security model	COVERED-2	2	Compiled to OPA
Security → Assign users to roles (rules: automatic role assignment by query)	SysSecRoleAssignmentRule		COVERED-2	2	
Security → Segregation of duties rules / Verify compliance / Conflicts	SysSecSegregationOfDuties*		COVERED-48	48	
Security → Extensible data security policies (XDS)	SysDataSecurityPolicy	Row-level security via policy queries	SUPERSEDED (Postgres RLS + OPA data policies wave 2)	2	
Security → Table permissions framework / Record-level security (legacy)	SysRecordLevelSecurity		SUPERSEDED (RLS/OPA)	2	
Security → Effective permissions viewer (Passage)			COVERED-2	2	
Batch jobs (list) / Batch job details / Batch tasks	BatchJob, BatchTask	Jobs, recurrence, dependencies, alerts, history	COVERED-4	4	Batch job framework + monitor UI
Batch groups / Server configuration (batch servers, threads)	BatchGroup, SysServerConfig		SUPERSEDED (Kubernetes workers, Temporal task queues)	4	
AOS → Cluster configuration / Load balancing	SysClusterConfig		SUPERSEDED (Kubernetes + Envoy, wave 1)	1	
Online users	SysUsersOnline	Sessions, kill session	COVERED-2	2	Keycloak sessions
Database → Database information / SQL administration (indexes, synchronize) / Database log setup	SysSqlAdmin, SysDatabaseLogSetup		SUPERSEDED (Atlas migrations from metadata, wave 3; audit from CDC, wave 5)	3	
Data partitions	SysDataPartition	Partition isolation	SUPERSEDED (tenant model with RLS wave 2)	2	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Virtual company accounts / Table collections	VirtualDataAreaList	Share tables across companies	SUPERSEDED (tenant-level shared master data (party, product, COA) by design, waves 7/8/11)	7	
Services and AIF → Inbound/Outbound ports, Adapters (file, MSMQ, HTTP, NetTcp), Document data policies, Exceptions, Queue manager, History, Document services	AifInboundPort, AifOutboundPort, AifDocumentLog	Integration framework	SUPERSEDED (integration hub: REST/GraphQL/webhooks/Kafka/EDI, wave 35; message monitor with replay)	35	
Licensing → License information / License configuration (configuration keys)	SysLicenseCodeReadFile, SysConfig	Feature keys	SUPERSEDED (feature flags/tenant entitlements in metadata registry wave 3; licensing model decided before wave 35)	3	
Setup checklists (Initialization checklist, Partition checklist, Upgrade checklist)	SysCheckList*		SUPERSEDED (tenant provisioning automation wave 50; migration runbooks wave 49)	50	
Enterprise Portal → Sites / EP parameters / Web parts	EP*		SUPERSEDED (React shell + portals, waves 5/20/30/42)	5	
Reporting Services / Analysis Services → Report servers, Deploy reports, Cubes (SQL Server Analysis Services projects), Configure analysis extensions	SRSServers, BIAnalysisServer		SUPERSEDED (document designer and semantic layer/lakehouse, wave 46)	46	
Help system (Help server, Help content)			SUPERSEDED (documentation site, in-app help wave 50)	50	
Alerts → Alert parameters / Email templates / Email parameters (SMTP) / Outbound email status (E-mail sending status)	EventParameters, SysEmailTable, SysEmailParameters		PARTIAL-4 (email templates/SMTP outbound)	4	Alerts GAP wave 5
Email → Email distributor batch	SysEmailBatch		PARTIAL-4 (email templates/SMTP outbound)	4	
Document handling (link)			PARTIAL-5 (document management)	5	
Global search configuration / Search configuration (Enterprise search, full-text)	SysSearchConfiguration		COVERED-5	5	Global search in shell
Client performance options / Usage data / Personalization reset	SysUserSetup		COVERED-5	5	Personalization
System parameters (System parameters form: language, timezone defaults, security, alerts, help, Unicode)	SystemParameters		COVERED-2	2	Tenant settings
Model store / Layers / Models (AOT) / Code upgrade / Compile / CIL	SysModelStore		SUPERSEDED (metadata registry + GitOps, waves 1/3; no X++ per PRD non-goal)	3	
Trace parser / SQL statement trace log / Tracing cockpit	SysTraceParser		SUPERSEDED (OpenTelemetry traces/metrics/logs wave 5)	5	
Sys admin → Periodic → Data import/export (link to DIXF)			COVERED-10	10	
Sys admin → Office add-ins (Excel add-in configuration, Document data sources)	DocuDataSource		COVERED-5	5	Excel/CSV import-export
Sys admin → Workflow infrastructure configuration (link)			COVERED-4	4	
Sys admin → Business events catalog (D365)	BusinessEventsCatalog	Event subscriptions to endpoints	COVERED-4	4	Kafka/webhooks
Sys admin → Feature management (D365)	FeatureManagement	Feature flags	COVERED-3	3	
Sys admin → Data entities / OData (D365 link)			COVERED-3	3	Generated OpenAPI
Sys admin → Dual-write (D365, link)			SUPERSEDED (wave 35)	35	§12.40
Sys admin → LCS environment monitoring / Telemetry (D365)			SUPERSEDED (Grafana/Tempo/Loki, wave 5)	5	
Sys admin → Mobile workspaces (D365 mobile app designer)			SUPERSEDED (PWA shell, metadata-driven forms)	5	
Sys admin → Inquiries → Database log / System log / Batch job history / SQL statements			COVERED-5	5	
Sys admin → Users → User log (login history)	SysUserLog		COVERED-2	2	Keycloak events
Sys admin → Security diagnostics for task recordings (D365)			SUPERSEDED (effective-permission viewer wave 2)	2	
Sys admin → Task recorder / Business process modeler (LCS)	SysTaskRecorder		SUPERSEDED (Playwright E2E suites; training curriculum wave 50)	50	
Sys admin → Electronic signature (link)			GAP	—	Propose wave 48: electronic signatures
Sys admin → Printers / Document routing agent (D365 network printers)			COVERED-32	32	Print service
Sys admin → Personalization management (D365: import/export user personalizations, saved views publish)			COVERED-5	5	Saved views
Sys admin → Data archiving / Long-term retention (D365 archive)			COVERED-46	46	Lakehouse archive
Sys admin → Secrets (key vault parameters, D365)			COVERED-36	36	Vault; secrets rotation 48
Sys admin → Tenant provisioning / Environment management (Passage)			COVERED-50	50	

12.30.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Online users / User sessions		COVERED-2	2	
Batch job history / Batch job errors		COVERED-4	4	

Name	Purpose	Passage status	Wave	Notes
Database log inquiry		COVERED-5	5	
System log / Event log		COVERED-5	5	Loki
SQL statement trace / Long-running queries		SUPERSEDED (OpenTelemetry/Grafana)	5	
Security inquiries (Users by role, Role permissions, Security roles report, Duties by role, Privileges by duty, Users with SoD conflicts, Effective permissions)		COVERED-2	2	
AIF inquiries (Document history, Exceptions, Queue manager)		SUPERSEDED (message monitor wave 35)	35	
License inquiries (Named user license counts, License usage by role)		PARTIAL-3 (license usage reporting is not named; depends on licensing decision)	3	Gap register
Workflow infrastructure inquiries		COVERED-4	4	
User log / Login history		COVERED-2	2	
SSRS: Security roles / Users / User permissions (Security role report)		COVERED-2	2	
SSRS: Database log report		COVERED-5	5	
SSRS: Batch job report		COVERED-4	4	
SSRS: License report / Named user counts		PARTIAL-3 (license usage reporting)	3	
SSRS: Number sequence status (link)		COVERED-5	5	
SSRS: Table information / AOT object listings		SUPERSEDED (metadata registry diff/export wave 3)	3	
Observability dashboards (SLOs, traces) (Passage)		COVERED-5	5	

12.30.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Batch → Batch job scheduling / Recurrence / Batch alerts / Clean up batch history		COVERED-4	4	
Database log → Clean up database log		SUPERSEDED (retention policy)	48	
Database → Synchronize database / Rebuild indexes / Compile / Generate CIL / Deploy		SUPERSEDED (CI/CD pipeline wave 1; Atlas migrations wave 3)	1	
Database → Backup / Restore / Refresh from production (LCS)		COVERED-48	48	Backup/restore drills; DR in 50
AIF → Process inbound/outbound queues / Send e-mails / Clean up document log		SUPERSEDED (integration hub wave 35)	35	
Email → Send email distributor batch / Retry failed		PARTIAL-4 (email templates/SMTP outbound)	4	
Alerts → Change-based alerts / Due date alerts (batch processors)		GAP	—	Propose wave 5: alerts and notification rules
Notifications → Clean up notifications		COVERED-5	5	
Security → Verify SoD compliance (batch) / Automatic role assignment batch		COVERED-48	48	
Security → User provisioning sync (SCIM/AD import wizard)		COVERED-2	2	
Licensing → Named user license count refresh		PARTIAL-3 (license usage reporting)	3	
Data → Data import/export (DIXF batch, link)		COVERED-10	10	
Data → Delete company data / Copy company (SysDataExport/Import)		COVERED-10	10	Migration framework; tenant provisioning 50
Data → Data partition management (create/initialize)		SUPERSEDED (tenant provisioning)	50	
Enterprise Portal → Deploy EP / Refresh AOT elements		SUPERSEDED (shell deploys via GitOps)	1	
Reporting → Deploy reports / Process cubes (SSAS)		SUPERSEDED (wave 46 lakehouse/dbt)	46	
Help → Update help content		SUPERSEDED (docs site wave 50)	50	
Upgrade → Code upgrade / Data upgrade (AX 2009 → 2012, 2012 → D365)		SUPERSEDED (migration factory wave 49, data only per non-goal)	49	
Model store → Import/export model / Install hotfix (axutil)		SUPERSEDED (GitOps, signed images, wave 1)	1	
Monitoring → Performance counters / Trace capture		SUPERSEDED (OpenTelemetry wave 5)	5	
Personalization → Reset usage data (batch)		COVERED-5	5	
Business events → Process business events queue / Retry failed endpoints (D365)		COVERED-4	4	Outbox + Kafka
Data archiving → Archive/purge jobs (D365)		COVERED-46	46	
Secrets → Rotate secrets (Passage)		COVERED-48	48	
DR → Game days / Failover drills (Passage)		COVERED-50	50	

12.30.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
System parameters	Default language, timezone, alerts, help, Unicode, security defaults	COVERED-2	2	
Users → User options defaults (language, timezone, number formats, workflow notifications, delegation)		COVERED-5	5	My user options
Users → Active Directory import wizard / Claims users / Identity providers		COVERED-2	2	OIDC/SAML federation, SCIM
Users → User profiles (role center profiles)		SUPERSEDED (workspaces per role, wave 5)	5	
Security → Security roles / Duties / Privileges / Process cycles (security configuration)		COVERED-2	2	
Security → Segregation of duties rules		COVERED-48	48	
Security → Automatic role assignment rules (query-based)		COVERED-2	2	

Name	Key parameters / options	Passage status	Wave	Notes
Security → Extensible data security (XDS) policies / Policy contexts		SUPERSEDED (RLS/OPA data policies wave 2)	2	
Security → Security development tool / Entry point permissions		SUPERSEDED (policy compiler from metadata wave 2/3)	2	
Security → Password/MFA policies (Passage via Keycloak)		COVERED-2	2	
Security → API keys / Service accounts / Rate limits (Passage)		COVERED-35	35	Developer portal keys and quotas
Licensing → License information / Configuration keys / Named user licensing types		SUPERSEDED (feature flags/entitlements wave 3; licensing model open question)	3	License usage PARTIAL
Batch → Batch groups / Batch servers / Batch thread configuration / Batch job alerts		SUPERSEDED (Temporal task queues, K8s worker pools; monitor UI wave 4)	4	
AOS → Server configuration / Cluster configuration / Load balancer / Services (AIF on AOS)		SUPERSEDED (Kubernetes, Envoy, wave 1)	1	
Database → Database log setup / Database information / SQL administration		SUPERSEDED (metadata audit flags wave 3/5)	5	
Data partitions / Partition checklist		SUPERSEDED (tenant model wave 2)	2	
Virtual company accounts / Table collections		SUPERSEDED (shared tenant-level master data)	7	
Services and AIF → Inbound ports / Outbound ports / Adapters / Pipelines (XSLT transforms) / Document data policies / Value mapping / Services configuration		SUPERSEDED (integration hub wave 35: REST/GraphQL/webhooks/Kafka; EDI mapping engine)	35	
Enterprise Portal → Web sites / EP parameters / Web menus / Web parts / Role centers		SUPERSEDED (React shell + portals)	5	
Reporting Services → Report servers / Reporting parameters / Report deployment / Print destinations (file, email, printer)		SUPERSEDED (document designer wave 46; print service wave 32)	46	
Analysis Services → Analysis servers / Cubes / Perspectives / Analysis extensions		SUPERSEDED (semantic layer wave 46)	46	
Alerts → Alert parameters (polling intervals, retention, drill-down links) / Email templates / Email parameters (SMTP, Exchange, mail sending mode: batch/client) / Email distributor		PARTIAL-4 (email templates/SMTP outbound)	4	Alerts GAP wave 5
Help → Help system parameters / Help server / Help content languages		SUPERSEDED (docs site wave 50)	50	
Global search / Enterprise search parameters / Search configuration		COVERED-5	5	
Client performance → Client performance options / Usage data policies		COVERED-5	5	
Document management (link)		PARTIAL-5 (document management)	5	
Office add-ins → Document data sources (Excel add-in entities), Office integration parameters (D365)		COVERED-5	5	
Workflow infrastructure configuration (message processing, due date, line-item batch jobs)		COVERED-4	4	
Model store → Models / Layers / Configuration (Development: AOT, X++ compile, hotfixes)		SUPERSEDED (no X++; metadata + GitOps)	3	PRD non-goal
Setup checklists → Initialization checklist (steps: license, config keys, partitions, company, users)		SUPERSEDED (tenant provisioning automation wave 50)	50	
System → Timezones / Languages / Date-time settings		COVERED-6	6	Reference data
Number sequences (system-level)		COVERED-5	5	
Sys admin → Feature management configuration (D365)		COVERED-3	3	
Sys admin → Business events configuration (endpoints: Azure Service Bus, Event Grid, HTTPS, Power Automate) (D365)		COVERED-4	4	Webhooks/Kafka; replay wave 35
Sys admin → Data entities / OData / Entity store refresh (D365)		COVERED-3	3	Lakehouse 46
Sys admin → Integration: Recurring integrations / Data management framework scheduling (D365)		COVERED-35	35	
Sys admin → Electronic signature parameters (link)		GAP	—	Propose wave 48: electronic signatures
Sys admin → Personalization management / Saved views publish (D365)		COVERED-5	5	
Sys admin → Network printers / Document routing agent (D365)		COVERED-32	32	
Sys admin → Key vault parameters / Certificates (D365)		COVERED-36	36	Vault
Sys admin → Data residency / Backup policies / Retention (Passage)		COVERED-48	48	
Sys admin → Tenant provisioning templates / Environment catalog (Passage)		COVERED-50	50	
Sys admin → Observability: SLOs, alert routes, PagerDuty integration (Passage)		COVERED-5	5	
Sys admin → AI: model provider configuration, token budgets, kill switch (Passage)		COVERED-47	47	

12.31 Home / role centers / workflow

AX 2012 R3 Home module: role centers (Enterprise Portal pages with cues, KPIs, work lists), work items (workflow), activities, cases, questionnaires, alerts and user options. Passage: workspaces and personalization in wave 5, workflow in wave 4, dashboards/KPIs in wave 46, copilot in wave 47.

12.31.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Role center (per profile: Accountant, Buyer, Production manager, Sales manager, CFO, … 30+ role centers)	EP role center pages (SysUserProfile)	Cues, KPIs (SSAS), work list, quick links, reports, Power BI parts	SUPERSEDED (role-based workspaces with tiles/lists wave 5 and embedded dashboards/KPIs wave 46)	5	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Work items assigned to me / Work items queue / Work item queue assignments	WorkflowWorkItemListPage, WorkflowWorkItemQueue	Approvals and tasks	COVERED-4	4	Human tasks on Temporal
Work item → Approve/Reject/Delegate/Request change/Complete	WorkflowWorkItemActionDialog		COVERED-4	4	
My activities (Activities: tasks, appointments, events, actions)	smmActivities (Home)	Personal activity list	OUT-OF-SCOPE (CRM activity model non-goal; tasks for cases/projects handled by case framework and workflow)	—	
Cases (My cases)	CaseDetailListPage		PARTIAL-45 (generic case management)	45	
Questionnaires → My questionnaires / Answer questionnaire / Questionnaire results	KMQuestionnaire (Home)		GAP	—	Propose wave 42: questionnaire engine
Alerts (My alerts / Notification list)	EventInbox	Alert inbox	GAP	—	Propose wave 5: alerts and notification rules (notification center exists in shell; rule-based alerts do not)
Notification center (Passage shell)		In-app notifications from workflow/batch	COVERED-5	5	
Cues (saved queries with counts)	SysCue		COVERED-5	5	Saved views with counts in workspaces
My user options (General, Status bar, Preferences: language/timezone/number format/date format, Workflow: notifications/delegation, Usage data, Account)	SysUserSetup (UserOptions)		COVERED-5	5	
Workflow delegation (My delegations: all/by workflow/by module, date ranges)	WorkflowDelegation		COVERED-4	4	
Timesheets (Home → Timesheets, link)	TSTimesheetTable		PARTIAL-41 (timesheets)	41	
Expense reports (Home → Expense reports, link)	TrvExpTable		COVERED-43	43	
Purchase requisitions (Home → My requisitions, link)	PurchReqTable		COVERED-17	17	
Employee self-service pages (My information, My team)	EP ESS		COVERED-42	42	
Reports (Home → Common reports / My reports)			COVERED-46	46	
Global search / Record search	SysSearch		COVERED-5	5	
Favorites / Navigation personalization	SysFavorites		COVERED-5	5	Personalization
Document handling on any record	DocuView		PARTIAL-5 (document management)	5	
Copilot / Ask-the-data (Passage)			COVERED-47	47	
Job monitor / My batch jobs (Passage)			COVERED-4	4	

12.31.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Workflow history (per document) / Workflow tracking details		COVERED-4	4	
Work item inquiries (Work items by user/queue/status, overdue work items)		COVERED-4	4	
Delegation inquiry		COVERED-4	4	
Role center KPIs (SSAS cube KPIs per role)		COVERED-46	46	Governed metrics
Cue counts inquiry		COVERED-5	5	
Alert inquiry (alerts by rule/user)		GAP	—	Propose wave 5: alerts and notification rules
Activity inquiries		OUT-OF-SCOPE (CRM non-goal)	—	
Questionnaire statistics / Results analysis		GAP	—	Propose wave 42: questionnaire engine
SSRS: Workflow history report / Workflow performance		COVERED-4	4	
SSRS: Work items report		COVERED-4	4	
SSRS: My reports (user-specific report list)		COVERED-46	46	
SSRS: Questionnaire reports (link)		GAP	—	Propose wave 42: questionnaire engine

12.31.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Workflow → Due date escalation / Reminders (batch)		COVERED-4	4	SLA timers
Workflow → Work item queue reassignment (batch)		COVERED-4	4	
Alerts → Process change-based alerts / Due-date alerts		GAP	—	Propose wave 5: alerts and notification rules
Role center → Refresh cues / Refresh KPIs (cube processing)		COVERED-46	46	
Questionnaires → Schedule questionnaires (batch)		GAP	—	Propose wave 42: questionnaire engine
Notifications → Clean up / Expire notifications		COVERED-5	5	
Personalization → Publish saved views to roles (D365)		COVERED-5	5	
Copilot → Agent runs (suggest mode) (Passage)		COVERED-47	47	

12.31.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Role center setup → User profiles / Profile defaults / Role center page assignment (EP)		SUPERSEDED (workspaces per role, wave 5)	5	

Name	Key parameters / options	Passage status	Wave	Notes
Role center → Cues setup (create cue from filter, cue groups) / KPI setup (KPI parts, cube perspectives)		COVERED-5	5	KPIs 46
Workflow → Workflow configurations (link) / Workflow parameters / Workflow participants / Delegation settings		COVERED-4	4	
Workflow → Work item queues / Queue groups / Queue assignment rules		COVERED-4	4	
Workflow → Notification settings (email, in-app, digest) / Workflow email templates		PARTIAL-4 (email templates/SMTP outbound)	4	
Alerts → Alert rules (create alert rule on field/record: changes to, due date; subject/message; recipients; organization-wide) / Alert rule templates		GAP	—	Propose wave 5: alerts and notification rules
My user options → Language, Timezone, Number/date formats, Default company, Status bar, Workflow notifications, Delegation, Email, Usage data, Accessibility (D365 enhanced tab sequence)		COVERED-5	5	WCAG 2.1 AA
Questionnaires → Questionnaire setup (link)		GAP	—	Propose wave 42: questionnaire engine
Activities → Activity setup (link, CRM)		OUT-OF-SCOPE (CRM non-goal)	—	
Cases → Case setup (link)		PARTIAL-45 (generic case management)	45	
Favorites / Navigation → Navigation pane customization, Personalization (form personalization, saved views, workspace tiles)		COVERED-5	5	
Global search → Search configuration (searchable tables/fields)		COVERED-5	5	
Home → Keyboard shortcuts / Accessibility settings (Passage)		COVERED-5	5	Keyboard-first
Home → Copilot settings (opt-in, data scope) (Passage)		COVERED-47	47	
Home → Mobile workspace configuration (D365 mobile app)		SUPERSEDED (PWA shell)	5	
Home → Dashboard/tile configuration per workspace (D365 workspace tiles, Power BI embed)		COVERED-46	46	

12.32 Data import/export framework

DIXF (AX 2012 R3) / Data management framework (D365): source data formats, processing groups, entities (target entities, custom entity wizard), staging, execution history, compare, copy between companies, recurring integrations. Maps to wave 10 (migration framework) and wave 35 (recurring integrations), with Excel in wave 5.

12.32.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Processing groups (list) / Processing group details	DMFDefinitionGroup	Group of entities with source format, mapping, sequence	COVERED-10	10	Migration pipeline: extract/stage/map/validate/load
Processing group → Entities (select entities, source format, sample file, generate mapping, modify mapping, validate)	DMFDefinitionGroupEntity		COVERED-10	10	Declarative mapping DSL
Processing group → Get staging data (source → staging)	DMFStagingWriter		COVERED-10	10	Postgres staging schema
Processing group → Copy data to target (staging → target)	DMFStagingTarget		COVERED-10	10	
Processing group → Preview source data / Validate	DMFPreview		COVERED-10	10	
Staging data (view/edit staging records, errors)	DMFStaging* (per entity)	Review and correct before load	COVERED-10	10	Exception queue with re-run
Execution history (jobs, status, errors, logs)	DMFExecution		COVERED-10	10	Migration run ledger with lineage
Target entities (entity list: Customers, Vendors, Products, BOM, Routes, Open transactions, Fixed assets, GL balances, Budget, Projects, HR, etc. ~150 entities in R3)	DMFEntity		COVERED-10	10	AX connectors for master data in 10; transactional packs in 49
Custom entity wizard (create entity from table)	DMFEntityWizard		COVERED-3	3	Any metadata entity is importable (generated)
Entity structure / Entity attributes (field metadata)	DMFEntityAttribute		COVERED-3	3	
Compare data (source vs target)	DMFCompareData		COVERED-10	10	Reconciliation: counts, hashes, sampled compare
Copy entity data between companies	DMFCopyEntity		COVERED-10	10	
Data import/export → Export (to file by processing group)	DMFExport		COVERED-5	5	Excel/CSV export of any entity
D365 Data management workspace (Import/Export projects, Data packages, Templates, Job history, Recurring data jobs, Entity list, Configure entity export to database (BYOD))	DMFDataManagement		COVERED-10	10	BYOD superseded by lakehouse 46
D365 → Recurring data jobs (enqueue/dequeue APIs)	DMFRecurringJob		COVERED-35	35	Integration hub; Kafka
D365 → Data package (zip with manifest) / LCS asset library			COVERED-10	10	Migration packs
D365 → Entity dependencies / Sequencing / Data templates (default templates)			COVERED-10	10	
D365 → Configuration data projects (copy config across environments)			COVERED-10	10	Also GitOps config
DIXF → Source → Staging via SSIS (service on AOS)	DMFService		SUPERSEDED (pipeline service in Go/Python, no SSIS)	10	

12.32.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Execution history / Job status inquiry		COVERED-10	10	
Staging data inquiry (errors by entity)		COVERED-10	10	

Name	Purpose	Passage status	Wave	Notes
Compare data results		COVERED-10	10	
Entity mapping inquiry (field mapping per entity)		COVERED-10	10	
Recurring job status / Message status (D365)		COVERED-35	35	Message monitor
SSRS: Processing group report / Entity list		COVERED-10	10	
SSRS: Execution log / Error log report		COVERED-10	10	
Reconciliation report (row counts, hash totals, sampled field compare) (Passage)		COVERED-10	10	

12.32.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Run processing group (batch: get staging, copy to target)		COVERED-10	10	
Clean up staging tables / Execution history		COVERED-10	10	
Recurring import/export jobs (D365 scheduling)		COVERED-35	35	
Entity store refresh (D365 BYOD/entity store)		SUPERSEDED (CDC → lakehouse wave 46)	46	
Dry-run / Idempotent re-run (Passage)		COVERED-10	10	
Parallel-run replay (Passage migration factory)		COVERED-49	49	

12.32.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Data import/export framework parameters (shared working directory, SSIS service, entity settings, source formats defaults)		COVERED-10	10	
Source data formats (file: CSV/XML/Excel/fixed width; ODBC; AX) with code page, delimiters, headers		COVERED-10	10	Extract: ODBC from AX SQL, SAP RFC/OData, flat files
Target entities setup (entity structure, staging table, query, validation class)		COVERED-3	3	Metadata-generated
Entity mapping defaults / Default values / Conversion (enum mapping, number sequence generation)		COVERED-10	10	Mapping DSL: coercion, enum mapping, default fill
Processing group sequencing / Entity dependencies		COVERED-10	10	
DIXF service configuration (AOS service)		SUPERSEDED (K8s pipeline service)	10	
D365 → Data management parameters (entity settings, performance: parallel threads, set-based processing, skip staging)		COVERED-10	10	
D365 → Data management: Entity export to database (BYOD) configuration		SUPERSEDED (lakehouse 46)	46	
D365 → Recurring integration endpoints / Integration keys		COVERED-35	35	
D365 → Data templates / Default templates per module		COVERED-10	10	
Passage → Legacy connectors (AX 2012/D365 F&O, SAP ECC/S4) configuration		COVERED-10	10	
Passage → Great Expectations validation suites		COVERED-10	10	
Passage → Migration run ledger / Lineage		COVERED-10	10	

12.33 Electronic reporting / Regulatory Configuration Service (D365)

ER (D365; backported to AX 2012 R3 CU13 partially): configurations (models, model mappings, formats), configuration providers, repositories (LCS, RCS, Dataverse, file system, global repo), destinations, ER parameters, business document management, electronic messages, Globalization Studio (tax calculation service, e-invoicing service). Passage: wave 48 (ER designer, localization framework), wave 35 (e-invoicing, Peppol), wave 13 (tax calculation), wave 46 (business documents).

12.33.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Electronic reporting workspace / Configurations (tree: data model, model mapping, format, metadata)	ERWorkspace, ERSolutionTable	Design and manage configurations	COVERED-48	48	ER designer: configurable XML/JSON/CSV with versioning
Configuration → Model designer (data model)	ERModelDesigner		COVERED-48	48	
Configuration → Model mapping designer (data sources: tables, entities, calculated fields, parameters)	ERModelMappingDesigner		COVERED-48	48	Over metadata entities
Configuration → Format designer (XML/JSON/text/Excel/Word output, bindings, formulas)	ERFormatDesigner		COVERED-48	48	Excel/Word output via wave 46 designer
Configuration → Versions (draft/completed/shared/discontinued), Import/Export (XML), Rebase	ERSolutionVersionTable		COVERED-48	48	Versioning
Configuration providers / Active provider	ERVendorTable		COVERED-48	48	
Repositories (LCS asset library, RCS, Dataverse, File system, Global repository, Operations resources)	ERRepository*	Fetch Microsoft-published configurations	SUPERSEDED (Passage country packs ship with Helm/packs; registry of regulator mandates wave 48)	48	
ER jobs (Electronic reporting jobs: run history, output files)	ERFormatMappingRunJobTable		COVERED-48	48	
ER destinations (per format: file, email, print, archive (document management), screen, Power BI, SharePoint)	ERFormatDestinationTable		COVERED-48	48	Archive depends on document management PARTIAL-5

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Business document management (Word/Excel templates for business documents, editable by users; template versions)	ERBusinessDocumentManagement		COVERED-46	46	Operational report designer
Electronic messages (Electronic messaging framework: message types, message items, processing actions, additional fields; used for VAT returns, SAF-T, UK MTD, IT e-invoice SDI, etc.)	EMElectronicMessage*, EMMessageProcessing	Statutory message lifecycle (generate, send, receive response)	PARTIAL-48 (electronic messaging lifecycle (statutory submission with response tracking, e.g. UK MTD, DE/FR mandates) not named; e-invoicing in 35 covers Peppol/SDI/CFDI submissions)	48	Gap register
Globalization Studio → Tax calculation service (D365 Tax Calculation: tax configurations, tax feature setup, tax applicability rules, tax codes designer)	TaxCalculationService	Configurable tax determination	COVERED-13	13	Internal rules engine default + provider adapters
Globalization Studio → Electronic invoicing service (Invoicing service: submissions, processing pipelines, customization)	EInvoiceService*	Cloud e-invoicing	COVERED-35	35	Peppol access point, SDI/CFDI hooks
Globalization Studio → Globalization features / Feature setup (per country)			COVERED-48	48	Country packs
Regulatory Configuration Service (RCS: standalone config editing, Globalization feature management)	RCS portal		SUPERSEDED (ER designer in-product wave 48; configs in Git-backed packs)	48	
ER → Data model/format parameters (application-specific parameters: lookup tables maintained by users, e.g. VAT box mapping)	ERFormatMappingParameters		COVERED-48	48	
ER → Format mapping run (run report manually with parameters)	ERFormatMappingRun		COVERED-48	48	
ER → Configuration dependencies / Model-to-format links			COVERED-48	48	

12.33.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
ER jobs history / Output archive inquiry		COVERED-48	48	
Configuration version history / Dependency inquiry		COVERED-48	48	
Electronic messages inquiry (status, responses, errors)		PARTIAL-48 (electronic messaging)	48	
E-invoicing submission log (D365 invoicing service)		COVERED-35	35	
Tax calculation service logs		COVERED-13	13	
Business document template usage inquiry		COVERED-46	46	
ER → Configuration comparison (diff versions)		COVERED-48	48	
ER → Format validation report (schema validation)		COVERED-48	48	Regulator schema fixtures
Statutory reports catalog per country (Passage)		COVERED-48	48	

12.33.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Run ER format in batch (scheduled statutory reports)		COVERED-48	48	
Import configurations from repository (update to latest versions)		SUPERSEDED (pack updates via Helm/GitOps wave 48)	48	
Electronic messages → Generate / Send / Receive response (batch processing actions)		PARTIAL-48 (electronic messaging)	48	
E-invoicing → Submit documents / Poll status (batch)		COVERED-35	35	
ER → Clean up job history / Archived outputs		COVERED-48	48	
Business document management → Publish templates		COVERED-46	46	
Tax calculation service → Sync configurations		COVERED-13	13	
Regulatory mandates registry refresh (Passage)		COVERED-48	48	

12.33.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Electronic reporting parameters (attachments document types for job archive/baseline/others, LCS/RCS connection, Dataverse)		COVERED-48	48	
Configuration providers (own provider for custom configs)		COVERED-48	48	
Repositories (LCS, RCS, Global repo, file system, Dataverse)		SUPERSEDED (Git-backed packs)	48	
ER destinations per format component (file/email/print/archive/screen/Power BI) + Electronic reporting destination parameters		COVERED-48	48	
ER → Application-specific parameters setup (per format version)		COVERED-48	48	
ER → Format mapping assignment to business processes (e.g. payment formats on method of payment, invoice formats in print management, customer/vendor statements)		COVERED-48	48	Payment formats 19; print 46
Business document management → Parameters (SharePoint/OneDrive for template editing), Template access by role		COVERED-46	46	
Electronic messages → Message types, Message item types, Message statuses, Processing actions (ER export/import, web service call, user input), Additional fields, Executable classes, Web service settings (endpoint, certificates), Message processing		PARTIAL-48 (electronic messaging)	48	
Globalization Studio → Tax calculation parameters (connection, feature per LE), Tax features (versions), Tax applicability (customer/vendor/item tax groups by rules)		COVERED-13	13	
Globalization Studio → Electronic invoicing parameters (invoicing service connection, feature per LE, processing pipelines, submission channels), E-invoicing document types		COVERED-35	35	
Globalization Studio → Globalization feature setup per legal entity		COVERED-48	48	
RCS → Globalization feature management (standalone)		SUPERSEDED (in-product)	48	

Name	Key parameters / options	Passage status	Wave	Notes
ER → Key vault certificates for signing (XAdES)		COVERED-36	36	Vault; signing in 35/48
ER → Configuration localization (labels per language)		COVERED-48	48	
Passage → Country pack registry (US, UK, DE, FR, IN, MX, CA) with regulator schema fixtures		COVERED-48	48	
Passage → Regulator mandate registry (effective dates, versions)		COVERED-48	48	

12.34 Engineering change management (D365)

D365 ECM: engineering products, engineering categories/attributes, engineering versions, product readiness, engineering change requests/orders with impact analysis, release to legal entities, product lifecycle states. Passage wave 9 covers ECR/ECO workflow, impact analysis (where-used), versions and release to production; engineering organization/release policies and readiness checks are partial.

12.34.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Engineering products (list) / Engineering product details	EngChgEngineeringProduct	Version-tracked products owned by engineering org	COVERED-9	9	
Engineering versions (version per product with BOM/route/attributes, effective dates, active)	EngChgEcoResProductVersion		COVERED-9	9	Effective-dated versions
Engineering change requests (list) / details	EngChgEngineeringChangeRequest	Request with impacted products, severity, workflow	COVERED-9	9	
Engineering change orders (list) / details	EngChgEngineeringChangeOrder	Order with product changes (BOM, route, attributes, new version), impact analysis, approval, processing	COVERED-9	9	ECO workflow with impact analysis
Engineering change order → Impact analysis (where-used, open orders, on-hand, planned orders)	EngChgImpactAnalysis		COVERED-9	9	Where-used; open order impact should be verified against 21/22
Engineering change order → Process changes (apply to products, create version)	EngChgProcessChange		COVERED-9	9	
Release product structure (release engineering product + BOM/route to legal entities)	EngChgReleaseProductStructure	Release to operational LEs with policies	PARTIAL-9 (engineering organization vs operational legal entity split, and release product structure to legal entities with per-LE release policies, not named)	9	Gap register
Product readiness checks (per LE checklist before product is usable)	EngChgProductReadiness		PARTIAL-9 (product readiness)	9	
Product acceptance (accept released product in receiving LE)	EngChgProductAcceptance		PARTIAL-9 (release product structure)	9	
Product lifecycle states (per version, controlling process enablement)	EcoResProductLifecycleState		PARTIAL-8 (lifecycle states)	8	
Engineering attributes / Attribute search (engineering attribute values per version)	EngChgEngineeringAttribute		COVERED-8	8	Attributes & classification
Engineering product categories (category assigns version nomenclature, attributes, readiness policy)	EngChgEngineeringProductCategory		PARTIAL-9 (engineering product categories with version nomenclature not named; category hierarchies are)	9	Gap register
Engineering change → Open order impact (sales/purchase/production orders referencing superseded version)			COVERED-9	9	
Engineering → Document/drawing attachments per version	DocuRef (version)		COVERED-9	9	
Engineering → Product change cases (AX 2012 R3 case management for product changes)	CaseDetail (product change)		COVERED-9	9	
Engineering → PLM integration (D365 Teamcenter/Windchill connectors via Dataverse)			PARTIAL-35 (PLM connector not named among iPaaS reference connectors)	35	Gap register

12.34.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Engineering version inquiry / Version comparison (BOM diff between versions)		COVERED-9	9	
Where-used inquiry (across versions)		COVERED-9	9	
Change request/order inquiries (by status, product, severity)		COVERED-9	9	
Impact analysis report		COVERED-9	9	
Release status inquiry (product × legal entity)		PARTIAL-9 (release product structure)	9	
Readiness check status inquiry		PARTIAL-9 (product readiness)	9	
SSRS: Engineering change order (document) / Change request		COVERED-9	9	
SSRS: Engineering version list / BOM comparison		COVERED-9	9	
SSRS: Product release report		PARTIAL-9 (release product structure)	9	

12.34.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Process engineering change orders (batch apply)		COVERED-9	9	
Release product structures (batch release to LEs)		PARTIAL-9 (release product structure)	9	
Expire old versions / Activate versions by effective date		COVERED-9	9	
Readiness check evaluation (batch)		PARTIAL-9 (product readiness)	9	
Engineering → Sync with PLM (batch)		PARTIAL-35 (PLM connector)	35	
Engineering → Update open orders to new version (mass)		COVERED-9	9	

12.34.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Engineering change management parameters (engineering organization, default severity, change order defaults, version activation, number sequences)		COVERED-9	9	
Engineering organizations (which LE owns engineering data)		PARTIAL-9 (release product structure)	9	
Engineering product categories (version nomenclature, attributes, readiness policy, release policy)		PARTIAL-9 (engineering product categories)	9	
Version nomenclature (number/letter/date segments)		PARTIAL-9 (engineering product categories)	9	
Product readiness policies / Readiness checks (per-LE checklists: tax, prices, coverage, warehouse setup)		PARTIAL-9 (product readiness)	9	
Product release policies (what to copy to receiving LE: BOM, route, prices, default order settings)		PARTIAL-9 (release product structure)	9	
Engineering attributes / Attribute types / Attribute search setup		COVERED-8	8	
Product lifecycle states (enabled business processes per state, default states)		PARTIAL-8 (lifecycle states)	8	
Change severity / Change priority / Change categories / Change reasons		COVERED-9	9	
Impact analysis configuration (which documents to scan)		COVERED-9	9	
ECM workflows (change request, change order approval)		COVERED-4	4	
Engineering → Number sequences (versions, change orders)		COVERED-5	5	
Engineering → PLM connector configuration		PARTIAL-35 (PLM connector)	35	

12.35 Asset management (D365)

D365 Asset management (EAM): functional locations, assets, asset types/lifecycle, maintenance plans/rounds/sequences, job types, checklists, maintenance requests, work orders with scheduling and journals, counters, fault management, condition assessment, warranties, costing and KPIs. Maps to wave 44.

12.35.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Assets (list) / Asset details (type, lifecycle state, functional location, model/manufacturer, attributes, BOM/spare parts, counters, warranties, documents)	EntAssetObjectTable		COVERED-44	44	
Functional locations (hierarchy, types, lifecycle)	EntAssetFunctionalLocation		COVERED-44	44	
Asset lifecycle state changes / Asset history (installation, transfers)	EntAssetObjectLifecycleState		COVERED-44	44	
Maintenance plans (time-based, counter-based, condition) / Maintenance plan lines	EntAssetMaintenancePlan		COVERED-44	44	
Maintenance rounds (route of assets inspected together)	EntAssetMaintenanceRound		PARTIAL-44 (maintenance rounds (multi-asset inspection routes) not named)	44	Gap register
Maintenance sequences (ordered job types on an asset)	EntAssetMaintenanceSequence		PARTIAL-44 (maintenance sequences not named)	44	Gap register
Maintenance requests (list) / details (request types, convert to work order)	EntAssetMaintenanceRequest		COVERED-44	44	IoT-triggered requests
Work orders (list) / details (lines per asset/job type, lifecycle state, scheduled, workers, checklist, tools, items, hours, expenses, project)	EntAssetWorkOrderTable		COVERED-44	44	
Work order → Schedule (scheduling of workers/capacity; work order pools; dispatch)	EntAssetWorkOrderSchedule		COVERED-44	44	Scheduling against resources wave 23
Work order → Journals (hours, items, expenses) and posting to project/GL	EntAssetWorkOrderJournal		COVERED-44	44	Costs → asset and GL
Work order → Checklists / Checklist results	EntAssetWorkOrderChecklist		COVERED-44	44	
Work order → Fault registration (symptoms, causes, remedies)	EntAssetFault		PARTIAL-44 (fault designer: symptom/area/type/cause/remedy taxonomy and fault analysis not named)	44	Gap register
Work order → Downtime registration (maintenance downtime affecting production resource)	EntAssetMaintenanceDowntime		COVERED-44	44	Downtime → scheduling capacity
Work order → Tools (resources booked)	EntAssetWorkOrderTool		COVERED-44	44	
Work order → Project / Cost control (work order cost, budget)	EntAssetWorkOrderProject		COVERED-44	44	
Counters (asset counters, counter registrations) / Measurements	EntAssetCounter		COVERED-44	44	Counters and measurement points
Condition assessment (records per asset)	EntAssetConditionAssessment		COVERED-44	44	Condition monitoring
Asset BOM / Spare parts (parts per asset with where-used)	EntAssetObjectBOM		COVERED-44	44	
Warranties (vendor warranty per asset, warranty check on work orders)	EntAssetWarranty		PARTIAL-44 (asset warranties (vendor warranty terms, warranty check on work order) not named)	44	Gap register

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Maintenance budgets (asset/functional location budget lines, budget calculation from plans)	EntAssetMaintenanceBudget		PARTIAL-44 (maintenance budgeting from plans not named)	44	Gap register
Asset insurance / Asset attributes			COVERED-44	44	Attributes via wave 8 framework
Maintenance workers / Worker groups / Worker skills (technicians)	EntAssetWorker		COVERED-44	44	
Maintenance planning workspace / Asset management workspace			COVERED-44	44	
Mobile maintenance (D365 Asset management mobile workspace)			COVERED-44	44	PWA
Asset → IoT sensors / Sensor Data Intelligence (asset condition from IoT)			COVERED-25	25	IoT ingestion; thresholds in 44
Asset → Link to fixed asset (financial asset)	EntAssetObjectFixedAsset		COVERED-37	37	
Asset → Link to production resource (machine)	EntAssetObjectResource		COVERED-44	44	

12.35.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Work order inquiries (by state, asset, worker, overdue)		COVERED-44	44	
Maintenance request inquiries		COVERED-44	44	
Asset cost inquiry (cost per asset/functional location: hours, items, expenses, by period)		COVERED-44	44	
Asset KPIs (MTBF, MTTR, availability, cost per asset, downtime)		COVERED-44	44	
Fault analysis (fault statistics by symptom/cause)		PARTIAL-44 (fault designer)	44	
Counter history / Condition trends		COVERED-44	44	
Maintenance plan forecast (upcoming work by plan)		COVERED-44	44	
Warranty inquiry		PARTIAL-44 (warranties)	44	
Budget vs actual (maintenance)		PARTIAL-44 (maintenance budgets)	44	
SSRS: Work order (document) / Work order report / Work order pick list / Checklist		COVERED-44	44	
SSRS: Maintenance plan report / Forecast		COVERED-44	44	
SSRS: Asset list / Functional location list / Asset hierarchy		COVERED-44	44	
SSRS: Asset cost report / Cost control		COVERED-44	44	
SSRS: Fault report		PARTIAL-44 (fault designer)	44	
SSRS: Downtime report		COVERED-44	44	
SSRS: Asset KPI report		COVERED-44	44	
SSRS: Spare parts where-used		COVERED-44	44	

12.35.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Schedule maintenance plans (generate work orders from time/counter/condition plans)		COVERED-44	44	Plan fixtures
Schedule maintenance rounds / Maintenance sequences		PARTIAL-44 (maintenance rounds)	44	
Work order scheduling (auto-schedule, capacity)		COVERED-44	44	
Counter updates (import from IoT/production)		COVERED-44	44	
Condition threshold evaluation (IoT)		COVERED-44	44	
Post work order journals (batch)		COVERED-44	44	
Maintenance budget calculation		PARTIAL-44 (maintenance budgets)	44	
Asset KPI calculation (batch)		COVERED-44	44	
Close work orders / Update lifecycle states (batch)		COVERED-44	44	
Mobile sync		COVERED-44	44	

12.35.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Asset management parameters (work order defaults, journal names, project integration, scheduling, number sequences, mobile)		COVERED-44	44	
Assets → Asset types, Asset type defaults (job types, counters, attributes, checklists), Asset lifecycle models/states, Asset models, Manufacturers, Asset attribute types, Asset criticality, Asset service levels		COVERED-44	44	
Functional locations → Functional location types, Lifecycle models/states, Attributes		COVERED-44	44	
Maintenance jobs → Maintenance job types, Job type categories, Job type variants, Trades, Job type defaults (hours, items, checklists, tools), Checklist templates/variables		COVERED-44	44	
Maintenance plans → Plan types, Plan line defaults, Maintenance rounds setup, Maintenance sequences setup		PARTIAL-44 (maintenance rounds)	44	
Work orders → Work order types, Work order lifecycle models/states, Work order pools, Work order priorities, Service levels, Work order projects (project groups/categories), Scheduling parameters (workers, capacity, preferred workers)		COVERED-44	44	
Maintenance requests → Request types, Request lifecycle states, Request-to-work-order defaults		COVERED-44	44	
Workers → Maintenance workers, Worker groups, Skills, Worker calendars		COVERED-44	44	
Counters → Counter types, Counter units		COVERED-44	44	
Fault → Fault symptoms, Fault areas, Fault types, Fault causes, Fault remedies, Fault designer (per asset type)		PARTIAL-44 (fault designer)	44	

Name	Key parameters / options	Passage status	Wave	Notes
Condition assessment → Condition assessment templates		COVERED-44	44	
Warranties → Warranty types, Vendor warranty setup		PARTIAL-44 (warranties)	44	
Budget → Maintenance budget parameters		PARTIAL-44 (maintenance budgets)	44	
Cost control → Cost types, Asset cost posting (project categories/GL)		COVERED-44	44	
Downtime → Downtime reasons, Capacity impact on production resources		COVERED-44	44	
IoT → Sensor scenarios, Thresholds, Notification rules		COVERED-25	25	
Mobile → Mobile workspace configuration		COVERED-44	44	
Workflow → Asset management workflows (request approval, work order approval)		COVERED-4	4	
Number sequences (assets, work orders, requests)		COVERED-5	5	

12.36 Rebate management (D365)

D365 Rebate management (2020+): deals (customer or vendor), deal lines with tiers, transaction accruals, deal statuses, provisions, rebate posting via journals, claims. Passage: customer rebates in waves 26 (agreements/accrual) and 30 (settlement); vendor-side deals are a gap (proposed wave 26).

12.36.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Rebate management deals (list) / Deal details (customer/vendor, products, tiers, dates, calculation basis, posting)	RebateManagementDeal		COVERED-26	26	Customer deals; vendor deals GAP
Deal → Vendor deals (purchase rebates)	RebateManagementDeal (vendor)		GAP	—	Propose wave 26: vendor rebates
Deal lines / Tiers (amount/quantity break, percentage/amount/price)	RebateManagementDealLine		COVERED-26	26	
Rebate management transactions (accrued per invoice line)	RebateManagementTransaction		COVERED-26	26	
Provisions (accrual journal) / Reversal of provisions	RebateManagementProvision		COVERED-26	26	Accrual postings
Process rebates (create claims: credit note, AP invoice, customer payment journal line)	RebateManagementProcess		COVERED-30	30	Rebate settlement
Deal status (draft, active, closed, cancelled) / Deal approval workflow			COVERED-26	26	
Deal → Price-based deals (royalty-type)			GAP	—	Propose wave 26: royalty agreements
Rebate management workspace			COVERED-26	26	

12.36.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Rebate transactions inquiry (by deal/customer/period)		COVERED-26	26	
Provision balance inquiry (accrued vs settled)		COVERED-30	30	
Deal performance (achieved tier, projected rebate)		COVERED-26	26	
Vendor deal inquiries		GAP	—	Propose wave 26: vendor rebates
SSRS: Rebate deal (document) / Rebate statement to customer		COVERED-30	30	
SSRS: Rebate accrual report		COVERED-26	26	
SSRS: Vendor rebate claim		GAP	—	Propose wave 26: vendor rebates

12.36.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Generate rebate transactions (from posted invoices)		COVERED-26	26	
Post provisions (accruals) / Reverse provisions		COVERED-26	26	
Process rebates (claims) / Settle		COVERED-30	30	
Close/expire deals		COVERED-26	26	
Vendor rebate claims processing		GAP	—	Propose wave 26: vendor rebates

12.36.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Rebate management parameters (journal names, posting profiles, default calculation, workflow, number sequences)		COVERED-26	26	
Rebate management posting profiles (accrual/expense accounts)		COVERED-26	26	
Deal types / Rebate reduction principles / Calculation methods (per unit, percentage, fixed)		COVERED-26	26	
Rebate groups (customer rebate groups, item rebate groups)		COVERED-26	26	
Vendor deal setup (claim via AP)		GAP	—	Propose wave 26: vendor rebates
Workflow → Deal approval		COVERED-4	4	
Number sequences (deals)		COVERED-5	5	

12.37 Subscription billing (D365)

D365 Subscription billing (2021+): Recurring contract billing (billing schedules, groups, escalation, milestones, unbilled revenue), Revenue and expense deferrals (deferral schedules, templates, recognition), Multiple element revenue allocation (MEA templates, standalone selling prices). Maps to wave 29.

12.37.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Billing schedules (list) / Billing schedule details (lines, frequency, start/end, escalation, milestones, pricing, deferral)	RCB BillingSchedule	Recurring contract billing	COVERED-29	29	Subscription billing: recurring schedules, proration, usage
Billing schedule → Generate invoices (free text / sales order)	RCB GenerateInvoice		COVERED-29	29	
Billing schedule → Escalation/de-escalation (price changes by period)	RCB Escalation		COVERED-29	29	
Billing schedule → Milestone billing (milestone templates)	RCB Milestone		COVERED-29	29	
Billing schedule → Usage-based billing (consumption import)	RCB Usage		COVERED-29	29	
Billing schedule → Renewal / Termination / Put on hold / Credit (prorated)			COVERED-29	29	
Unbilled revenue (unbilled revenue schedules, invoiced-in-advance)	RCB UnbilledRevenue		COVERED-29	29	Deferred revenue postings
Deferral schedules (list) / details (revenue and expense deferrals, short-term/long-term split)	RED DeferralSchedule		PARTIAL-29 (expense deferrals (prepaid expense amortization from AP invoices) not named; revenue deferrals are)	29	Gap register
Deferral → Recognition processing (recognize period amounts, generate journal)	RED Recognition		COVERED-29	29	
Deferral → Deferral templates / Straight-line vs event-based	RED Template		COVERED-29	29	
Multiple element revenue allocation (MEA) → MEA templates, SSP allocation, allocation on sales order/billing schedule	MEA*		COVERED-29	29	SSP allocation, ASC 606 bundles
Contract renewal workspace / Subscription billing workspace			COVERED-29	29	
Billing schedule → Customer portal self-service (view subscriptions)			COVERED-30	30	

12.37.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Billing schedule inquiries (upcoming invoices, renewals due, MRR/ARR)		COVERED-29	29	
Unbilled revenue inquiry / Deferred revenue roll-forward		COVERED-29	29	Reconciles to GL
Deferral schedule inquiry (balance by period, short/long-term)		COVERED-29	29	Expense deferral PARTIAL
MEA allocation inquiry		COVERED-29	29	
SSRS: Billing schedule (document) / Invoice preview		COVERED-29	29	
SSRS: Deferred revenue report / Revenue waterfall		COVERED-29	29	
SSRS: Unbilled revenue report		COVERED-29	29	
SSRS: Subscription metrics (MRR, churn)		COVERED-46	46	

12.37.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Generate invoices from billing schedules (batch by date)		COVERED-29	29	
Recognition processing (deferral recognition batch)		COVERED-29	29	
Short-term/long-term reclassification (batch)		COVERED-29	29	
Escalation processing (price updates)		COVERED-29	29	
Usage import (batch)		COVERED-29	29	
Renewal processing / Expire schedules		COVERED-29	29	
Unbilled revenue posting (batch)		COVERED-29	29	
Expense deferral recognition		PARTIAL-29 (expense deferrals)	29	

12.37.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Recurring contract billing parameters (invoice type, proration method, number sequences, posting)		COVERED-29	29	
Billing schedule groups / Billing frequencies / Item billing types / Escalation types / Milestone templates / Termination reasons / Renewal settings		COVERED-29	29	
Revenue and expense deferrals parameters (posting, short/long-term accounts, recognition method)		COVERED-29	29	Expense PARTIAL
Deferral templates / Deferral defaults (per item/customer/account) / Deferral item groups / Event-based deferral events		COVERED-29	29	
MEA parameters / MEA templates / Standalone selling prices (by item/customer group, date-effective) / Allocation methods (relative SSP, residual)		COVERED-29	29	

Name	Key parameters / options	Passage status	Wave	Notes
Workflow → Billing schedule approval		COVERED-4	4	
Number sequences		COVERED-5	5	

12.38 Revenue recognition (D365)

D365 Revenue recognition (2019–2023, now deprecated in favor of Subscription billing): revenue schedules, revenue prices (SSP), bundles, revenue recognition journals, reallocation, expected revenue. Maps to wave 29.

12.38.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Revenue schedules (per sales order line: schedule lines by period, status, hold)	RevRecSchedule		COVERED-29	29	Schedules (ratable, milestone, usage)
Revenue recognition journal (recognize scheduled revenue, post)	RevRecJournal (LedgerJournalTable)		COVERED-29	29	
Revenue price allocation (on sales order: allocate per SSP across lines/bundles)	RevRecAllocation		COVERED-29	29	
Expected revenue / Expected revenue schedule (before invoicing)	RevRecExpectedRevenue		COVERED-29	29	
Bundles (bundle items with components for allocation)	RevRecBundle		COVERED-29	29	Bundles
Reallocation (after order change, re-run allocation)	RevRecReallocation		COVERED-29	29	
Revenue recognition → Contract (contract terms per order)	RevRecContract		COVERED-29	29	
Revenue recognition workspace			COVERED-29	29	

12.38.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Revenue schedule inquiry (deferred/recognized by period)		COVERED-29	29	
Allocation details inquiry		COVERED-29	29	
Deferred revenue roll-forward		COVERED-29	29	
SSRS: Revenue recognition report / Revenue schedule report		COVERED-29	29	
SSRS: Deferred revenue by customer		COVERED-29	29	

12.38.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Create revenue recognition journal (batch)		COVERED-29	29	
Post revenue recognition journal		COVERED-29	29	
Reallocation processing		COVERED-29	29	
Update expected revenue		COVERED-29	29	

12.38.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Revenue recognition parameters (journal names, deferred revenue accounts, recognition basis, bundle handling)		COVERED-29	29	
Revenue schedules (schedule templates: occurrences, frequency, recognition basis: contract start/invoice date, milestone)		COVERED-29	29	
Revenue prices (SSP by item/customer/currency, allocation %)		COVERED-29	29	
Item groups / Released product → Revenue type (post-contract support, essential, non-essential), Revenue schedule default		COVERED-29	29	
Bundle setup (bundle components)		COVERED-29	29	
Revenue recognition posting (ledger accounts for deferred revenue, contract liability)		COVERED-29	29	
Number sequences		COVERED-5	5	

12.39 Vendor collaboration / customer self-service (D365)

D365 Vendor collaboration (PO confirmation, consignment, invoices, RFQ reply, onboarding/registration), vendor collaboration user provisioning, and customer self-service (Power Pages/portal). Passage: supplier portal (wave 20), customer portal (wave 30).

12.39.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Vendor collaboration workspace (vendor view: POs to confirm, confirmed, rejected)	VendVendorPortalPurchaseOrder		COVERED-20	20	Supplier portal: PO acknowledgement
Vendor → Confirm/reject PO with changes (substitute, split, dates)	VendVendorPortalPurchaseOrderConfirm		COVERED-20	20	
Vendor → Confirm with notes / attachments			COVERED-20	20	
Vendor → Consignment inventory view (on-hand, replenishment orders)	VendVendorPortalConsignment		GAP	—	Propose wave 16: consignment
Vendor → Invoices (submit invoice against PO/receipt; view invoice status)	VendVendorPortalInvoice		COVERED-20	20	
Vendor → RFQ replies (bid via portal)	VendVendorPortalRFQ		COVERED-17	17	
Vendor → ASN submission (D365 vendor shipping notice)			COVERED-20	20	ASN
Vendor → Document exchange / Contacts / Company information maintenance (bank account change requests)			COVERED-20	20	Document exchange; change approval workflow 7

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Vendor → Prospective vendor registration (onboarding wizard, questionnaires)	VendProspectiveVendorRegistration		PARTIAL-20 (vendor onboarding)	20	
Vendor collaboration user provisioning (invite users, roles: vendor admin, vendor user; security per vendor)	VendVendorPortalUserRequest		COVERED-20	20	Keycloak external identities
Vendor → Vendor collaboration: Product receipt visibility / Payment status			COVERED-20	20	Payment advice via portal (wave 19)
Customer self-service (D365 customer portal template: orders, invoices, cases, account)	Customer portal (Power Pages)		COVERED-30	30	Customer portal
Customer portal → Order entry / Order status / Shipment tracking			COVERED-27	27	Portal completed in 30
Customer portal → Invoices and payments (card/ACH)			COVERED-30	30	
Customer portal → Returns request			COVERED-30	30	
Customer portal → Cases / Service requests			COVERED-45	45	
Customer portal → Documents (statements, CoA)			COVERED-30	30	

12.39.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Vendor collaboration inquiries (confirmations pending, response time)		COVERED-20	20	
Portal user activity / provisioning status		COVERED-20	20	
Customer portal usage inquiry		COVERED-30	30	
SSRS: Vendor confirmation performance		COVERED-33	33	Supplier scorecards

12.39.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Send POs to portal / Notify vendors (batch)		COVERED-20	20	
Expire pending confirmations / Escalate		COVERED-20	20	
Provision portal users (batch invite) / Deactivate		COVERED-20	20	
Sync customer portal data		SUPERSEDED (portal reads APIs directly; no sync)	30	

12.39.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Vendor collaboration parameters (PO confirmation required, auto-confirm rules, invoice submission, consignment, onboarding questionnaires)		COVERED-20	20	
Vendor collaboration → Security roles for external users / Vendor collaboration user provisioning workflow		COVERED-2	2	
Vendor collaboration → Procurement: Purchase order change settings for vendor-proposed changes		COVERED-17	17	
Vendor collaboration → Onboarding: Registration wizard configuration, Questionnaires		PARTIAL-20 (vendor onboarding)	20	
Customer self-service → Portal configuration (branding, features, payment adapter)		COVERED-30	30	
Customer self-service → External identity (customer users, B2B invitations)		COVERED-2	2	
Customer self-service → Dual-write/Dataverse dependency (D365 portal uses Dataverse)		SUPERSEDED (portal on Passage APIs)	30	

12.40 Dual-write / Dataverse integration (D365)

D365 Dual-write synchronizes F&SCM tables with Dataverse for Power Platform and Dynamics 365 CE apps (Sales, Field Service, Customer Service). Passage supersedes the entire mechanism with the integration hub: generated APIs, Kafka topics, webhooks, MCP and reference connectors (Salesforce, Shopify, Stripe). Every row is `SUPERSEDED` and points at the replacing wave.

12.40.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Dual-write administration (Data integration → Dual-write: table maps, running/paused status)	DualWriteAdmin	Manage entity maps	SUPERSEDED (integration hub: Kafka topics per entity + webhooks, wave 35)	35	
Table maps (standard maps: customers, vendors, products, orders, invoices, etc.) / Map versions	DualWriteTableMap		SUPERSEDED (AsyncAPI-described topics from entity metadata, wave 4)	4	
Initial sync (initial write) / Catch-up sync	DualWriteInitialSync		SUPERSEDED (replay from Kafka/outbox; migration framework wave 10)	35	
Dual-write → Error log / Alerts	DualWriteErrorLog		SUPERSEDED (message monitor with replay and DLQ, wave 35)	35	
Dual-write → Company (legal entity) mapping to Dataverse business units			SUPERSEDED (tenant/legal entity in API scope)	2	
Dataverse → Virtual entities (F&O virtual tables in Dataverse)			SUPERSEDED (GraphQL/REST APIs, wave 35)	35	
Power Platform integration (Power Apps embed, Power Automate flows, Power BI)			SUPERSEDED (embedded dashboards wave 46; Temporal workflows/agents wave 47; webhooks 35)	46	
Dynamics 365 Sales / Customer Service / Field Service integration via Dual-write			SUPERSEDED (Salesforce/HubSpot connectors wave 35; native service wave 45)	35	CRM non-goal
Business events to Power Automate / Logic Apps			SUPERSEDED (CloudEvents on Kafka + webhooks, wave 4/35)	4	
Data integrator (legacy CDS data integration projects)			SUPERSEDED (integration hub)	35	
Finance and Operations virtual entities / OData endpoints			SUPERSEDED (generated OpenAPI/GraphQL, wave 3/35)	3	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Export to Data Lake (D365 Export to Azure Data Lake / Synapse Link)			SUPERSEDED (CDC → Iceberg lakehouse, wave 46)	46	

12.40.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Dual-write sync status inquiry		SUPERSEDED (message monitor wave 35)	35	
Dual-write error inquiry		SUPERSEDED (message monitor/DLQ wave 35)	35	
Dataverse → Entity sync health (Power Platform admin center)		SUPERSEDED (observability wave 5; integration SLAs wave 35)	35	
Synapse Link status		SUPERSEDED (lakehouse freshness metrics wave 46)	46	

12.40.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Initial sync / Resync table maps		SUPERSEDED (replay)	35	
Pause/resume maps (maintenance windows)		SUPERSEDED (consumer pause/resume on Kafka)	35	
Error retry batch		SUPERSEDED (retry with backoff + DLQ)	35	
Export to Data Lake incremental refresh		SUPERSEDED (CDC streaming)	46	

12.40.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Dual-write setup (link environment, Dataverse connection, solution install)		SUPERSEDED (integration hub setup: API keys, webhook endpoints, Kafka consumer groups, wave 35)	35	
Table map configuration (field mapping, transforms, filters, direction)		SUPERSEDED (event schemas from metadata; mapping in connectors)	4	
Dual-write → Integration key configuration (alternate keys)		SUPERSEDED (entity natural keys in metadata)	3	
Power Platform environment linkage (LCS)		SUPERSEDED (developer portal wave 35)	35	
Export to Data Lake configuration		SUPERSEDED (lakehouse wave 46)	46	
Dataverse security roles for virtual entities		SUPERSEDED (OPA policies on APIs wave 2)	2	

12.99 Gap register

Every row in §12.1–§12.40 whose status is `GAP` or `PARTIAL` appears here with a proposed wave assignment. `GAP` rows carry their proposal in the Notes column of the source table ("Propose wave NN: …"); `PARTIAL` rows propose the wave shown in their Wave column unless the Notes say otherwise. The register is generated from the tables above by `scripts`-style tooling so it cannot drift from the inventory; re-run the assembler after editing any module section.

12.99.1 Gap themes and proposed wave assignments

The detailed registers below are generated from every `GAP` and `PARTIAL` row in §12.1–§12.40. This table groups them into the themes a wave owner would actually plan. Proposed waves respect the dependency chain in waves.json (a theme is never proposed before the wave whose data model it needs). Items marked **(wave-splitting candidate)** are large enough that the wave's estimate should be revisited at the next phase-exit re-estimation (PRD §11).

Theme	AX / D365 feature(s)	Proposed wave	Why this wave	Size
Transfer orders	Transfer orders with in-transit warehouse, ship/receive, underdelivery, transfer picking, WHS transfer flows	16	Inventory ledger wave owns movements; MRP (22) and WHS (31) consume transfer orders as supply/work	**(wave-splitting candidate)**
Consignment inventory	Vendor-owned on-hand, ownership-change journal, consignment replenishment orders, vendor portal consignment view	16	Needs inventory ledger ownership dimension; surfaces in AP (19) and supplier portal (20)	Medium
Inventory extras	Tag counting, ABC classification, inventory aging, picking workbench	16 / 20 / 28	Natural extensions of counting, costing and fulfillment waves	Small each
Alerts and notification rules	Change-based and due-date alert rules on any entity/field, alert inbox, email delivery	5	Shell + event backbone exist by wave 5; every later module expects alerts	Medium
Email and templates	SMTP/provider outbound, templated email for workflow, collections, dunning, portals	4	Workflow notifications in 4 need it; collections (30) reuses	Small
Document management depth	Document types, storage backends (S3-compatible), archive/retention, document routing to records	5	Attachments exist in 5/11; policy layer missing	Medium
Reason codes framework	Shared financial reason codes (ledger, bank, customer, vendor, FA, counting)	11	First consumer is journal reversal in 11	Small
Policy framework and signing limits	Reusable org-hierarchy-scoped policy engine (purchasing, expense, audit, signing limits) and signing-limit requests	17	First consumer is purchasing policy in 17; expense (43) and audit (48) reuse	Medium
Procurement catalogs and punchout	Internal procurement catalog, vendor catalog import (CMR), external catalogs via cXML/OCI punchout	17 / 20 / 35	Catalog in 17, vendor upload via portal 20, punchout through integration hub 35	Medium
RFQ depth	Sealed bids, scoring criteria, amendments	17	Extends RFQ cycle	Small
Vendor master depth	Vendor requests/onboarding workflow, certifications and diversity classifications, vendor catalog	7 / 20	Party governance (7) and supplier portal (20)	Small
Payments depth	ACH prenotes, payment calendars (bank holidays), centralized cross-entity payments	19 / 40	Payment media in 19; centralized payments need intercompany due-to/due-from (40)	Small / Medium

Theme	AX / D365 feature(s)	Proposed wave	Why this wave	Size
Treasury instruments	Customer bills of exchange, vendor promissory notes, postdated checks (draw/remit/protest/settle journals)	36	Cash and bank wave owns instruments; AR/AP journals already exist	**(wave-splitting candidate)**
Direct debit mandates	SEPA mandates (pain.008), mandate lifecycle	30	Cash application wave; ISO 20022 generator from 19	Small
Rebates, brokers, royalties, commissions	Vendor rebates, broker contracts, royalty agreements, commission posting, TAM funds, promotion planning, deductions workbench, supplementary items, commodity pricing	26 / 30	Pricing engine (26) owns agreements/accruals; settlement and deductions in 30	**(wave-splitting candidate)**
Process manufacturing depth	Consolidated batch orders, rework batch orders, batch merge, attribute/shelf-life inheritance, same-batch reservation, customer sellable days, lean subcontracting	24	Process/lean wave	Medium
Planning depth	Supply schedule workbench, inventory forecast, priority-based (DDMRP-style) planning, intercompany master planning	22 / 40	Planning engine in 22; intercompany in 40	Medium
Production order split/merge	Split a production order quantity, merge	21	Lifecycle state machine owner	Small
Product master depth	Product lifecycle states, related products, dimension/model-group conversion jobs, packaging materials/fees	8 / 28	Product master and fulfillment	Small
Engineering change depth	Engineering organization and release product structure to LEs, readiness checks/policies, engineering categories and version nomenclature, PLM connector	9 / 35	ECM wave; PLM connector via integration hub	Medium
Product compliance	Regulated/restricted lists, safety data sheets with delivery rules, reporting thresholds	33	Quality wave is the natural home (documents on shipment, blocking)	Medium
Intrastat and VAT recovery	Intrastat setup (commodity/transaction codes, transport methods, ports), Intrastat journal corrections, VAT recovery on expenses	13	Tax engine wave	Small
Financial reporting hierarchies	Derived financial hierarchies, advanced rule structures, default descriptions	11 / 15	Journal and reporting waves	Small
Cost accounting policies	Cost behavior (fixed/variable) and cost rollup policies	38	Cost accounting ledger wave	Small
Budget planning depth	Excel worksheet round-trip, position forecasting from HR, plan generation from demand forecast/FA	38 (+42 dependency)	Position forecasting needs HR (42): deliver stub in 38, complete in 42	Medium
Costing extras	Standard cost conversion checklist, parallel inventory valuation ledgers (Global inventory accounting)	39 / 20	Costing waves	Medium
Landed cost (D365 module)	Voyages, shipping containers, folios, goods-in-transit orders, cost estimates vs actuals	34	Transportation wave owns landed cost allocation	**(wave-splitting candidate)**
Yard and gate	Driver check-in/out, yard management (trailers, yard moves, dock assignment)	34	TMS wave with dock scheduling from 31	Medium
WHS extras	Cross docking (planned/opportunistic), outbound sorting	31	Advanced warehouse wave	Small
Fixed asset extras	Asset additions (components), insurance records	37	Fixed assets wave	Small
Intercompany depth	Intercompany expenses, intercompany project resourcing/invoicing, intercompany on-hand visibility, centralized payments, intercompany planning	40	Intercompany wave	Medium
Project depth	Project quotations, weekly timesheets with workflow, retention (retainage) terms, project indirect cost components, beginning balance journals (49)	41 / 49	Project wave; migration pack for opening WIP	Medium
HR extras	Courses/learning, questionnaire engine, injury and illness (OSHA), loaned items, skill mapping, onboarding checklists, teams (6), US compliance specifics (48)	42 (+6, +48)	HR wave; questionnaire engine reused by performance reviews and quality	**(wave-splitting candidate)**
Expense extras	Cash advances, travel requisitions, deduction arrears recovery (payroll), workers' compensation and new-hire reporting (48)	43 (+48)	Payroll/expense wave; statutory outputs via US pack	Medium
EAM extras	Maintenance rounds and sequences, fault designer, asset warranties, maintenance budgets	44	Asset management wave	Medium
Service subscriptions	AX service subscription fees with indexation and accrual	29 (+45)	Subscription billing engine in 29 is the implementation; service wave links agreements	Small
Generic case management	Case categories/processes/activities/knowledge articles on any record (HR, collections, product change, audit, service)	45	Service wave builds cases; earlier consumers (30, 9) use a minimal case entity until then — flag as risk	Medium
Compliance extras	Audit policies (rule-based post-hoc audit of expenses/invoices/POs/journals), process/risk library, electronic signatures (21 CFR Part 11), electronic messaging lifecycle, India TDS/TCS depth, license usage reporting (3)	48 (+3)	Compliance wave; electronic signatures are a prerequisite for pharma/food manufacturers and should be pulled earlier if the first customer is regulated	**(wave-splitting candidate)**
Print management routing	Conditional print destinations per document/account/query	46	Document designer wave	Small

12.99.2 Proposed load by wave

Proposed wave	GAP rows	PARTIAL rows	Total
3	0	3	3
4	0	6	6
5	7	8	15
6	0	4	4
7	0	2	2
8	0	7	7
9	0	16	16
11	0	10	10
13	0	8	8
15	0	1	1
16	9	15	24
17	0	16	16

Proposed wave	GAP rows	PARTIAL rows	Total
19	0	6	6
20	2	11	13
21	0	1	1
22	0	4	4
24	0	16	16
26	19	24	43
27	0	3	3
28	0	5	5
29	0	7	7
30	0	11	11
31	0	6	6
33	12	0	12
34	6	8	14
35	1	3	4
36	16	0	16
37	0	4	4
38	0	6	6
39	0	2	2
40	0	10	10
41	0	12	12
42	24	4	28
43	0	10	10
44	0	15	15
45	0	11	11
46	0	7	7
48	12	34	46
49	0	1	1
Total	**108**	**317**	**425**

12.99.3 GAP rows (no wave covers the item)

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
1	12.2 Accounts payable	12.2.1 Common / Journals	Payments → Promissory note journals (Draw / Redraw / Remittance / Settle)	GAP	Propose wave 36: promissory notes / bills of exchange (vendor side) with remittance and settlement journals; not in any wave	36
2	12.2 Accounts payable	12.2.1 Common / Journals	Vendor → Consignment (D365 consignment replenishment orders)	GAP	Propose wave 16: consignment inventory (vendor-owned on-hand, ownership-change journal, consignment replenishment orders)	16
3	12.2 Accounts payable	12.2.2 Inquiries & Reports	SSRS: Promissory note (vendor) documents	GAP	Propose wave 36: with promissory notes	36
4	12.2 Accounts payable	12.2.3 Periodic	Promissory notes → Draw / Redraw / Remit / Settle (periodic batches)	GAP	Propose wave 36: promissory notes	36
5	12.2 Accounts payable	12.2.3 Periodic	Consignment → Consignment replenishment order processing	GAP	Propose wave 16: consignment	16
6	12.2 Accounts payable	12.2.4 Setup	Payment → Promissory note setup (journal names, posting profiles, protest)	GAP	Propose wave 36: promissory notes	36
7	12.2 Accounts payable	12.2.4 Setup	Vendor → Vendor consignment setup (D365 consignment parameters, ownership change journal names)	GAP	Propose wave 16: consignment	16
8	12.3 Accounts receivable	12.3.1 Common / Journals	Payments → Bill of exchange journals (Draw / Redraw / Protest / Remittance / Settle)	GAP	Propose wave 36: bills of exchange (customer) with remittance for collection/discount, protest, settle; not in any wave	36
9	12.3 Accounts receivable	12.3.1 Common / Journals	Payments → Postdated checks (R3 APAC)	GAP	Propose wave 36: postdated checks (customer and vendor) as a treasury instrument	36
10	12.3 Accounts receivable	12.3.2 Inquiries & Reports	Bill of exchange inquiry	GAP	Propose wave 36: bills of exchange	36
11	12.3 Accounts receivable	12.3.2 Inquiries & Reports	Postdated check inquiry	GAP	Propose wave 36: postdated checks	36
12	12.3 Accounts receivable	12.3.2 Inquiries & Reports	SSRS: Bill of exchange documents / Remittance list	GAP	Propose wave 36: bills of exchange	36
13	12.3 Accounts receivable	12.3.3 Periodic	Payments → Bill of exchange remittance/protest batches	GAP	Propose wave 36: bills of exchange	36
14	12.3 Accounts receivable	12.3.4 Setup	Payment → Bill of exchange setup (journal names, posting profiles, protest reasons, remittance types)	GAP	Propose wave 36: bills of exchange	36
15	12.3 Accounts receivable	12.3.4 Setup	Payment → Postdated checks setup (R3 APAC)	GAP	Propose wave 36: postdated checks	36

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
16	12.5 Cash and bank management	12.5.1 Common / Journals	Bank → Bills of exchange / Promissory notes (links)	GAP	Propose wave 36: bills of exchange and promissory notes (see AR/AP)	36
17	12.5 Cash and bank management	12.5.1 Common / Journals	Bank → Postdated checks (links)	GAP	Propose wave 36: postdated checks	36
18	12.5 Cash and bank management	12.5.4 Setup	Bills of exchange / promissory notes setup (bank side)	GAP	Propose wave 36: bills of exchange	36
19	12.5 Cash and bank management	12.5.4 Setup	Postdated checks parameters (R3 APAC)	GAP	Propose wave 36: postdated checks	36
20	12.8 Cost management (D365)	12.8.1 Common / Journals	Global inventory accounting (D365 add-in: multiple cost ledgers / accounting conventions)	GAP	Propose wave 20: parallel inventory valuation ledgers (one costing method per ledger group) aligned with wave 12 parallel ledgers	20
21	12.8 Cost management (D365)	12.8.4 Setup	Global inventory accounting setup (ledgers, accounting policies, conventions)	GAP	Propose wave 20: parallel inventory valuation ledgers	20
22	12.9 Compliance and internal controls	12.9.1 Common / Journals	Compliance → Electronic signatures (link)	GAP	Propose wave 48: electronic signatures (21 CFR Part 11 style) on configured tables/fields with reason codes and signature log	48
23	12.9 Compliance and internal controls	12.9.2 Inquiries & Reports	Electronic signature log inquiry	GAP	Propose wave 48: electronic signatures	48
24	12.9 Compliance and internal controls	12.9.2 Inquiries & Reports	SSRS: Electronic signature report	GAP	Propose wave 48: electronic signatures	48
25	12.9 Compliance and internal controls	12.9.3 Periodic	Electronic signature → Reset/revoke signatures	GAP	Propose wave 48: electronic signatures	48
26	12.9 Compliance and internal controls	12.9.4 Setup	Electronic signature → Electronic signature parameters, Signature requirements (table/field), Reason codes for signatures	GAP	Propose wave 48: electronic signatures	48
27	12.12 Human resources	12.12.1 Common / Journals	Courses (training) → Courses, Course sessions, Course registrations, Instructors, Course locations	GAP	Propose wave 42: learning/training management (courses, sessions, registrations, instructors, completion to competencies)	42
28	12.12 Human resources	12.12.1 Common / Journals	Questionnaires → Questionnaires, Questionnaire schedules, Answer sessions, Results analysis	GAP	Propose wave 42: questionnaire engine (design, schedule, answer, analyze) reused by performance reviews and quality audits	42
29	12.12 Human resources	12.12.1 Common / Journals	Injury and illness → Injury/illness incidents, Treatments, Body parts, OSHA logs	GAP	Propose wave 42: injury and illness incident register with OSHA 300 extracts (US pack)	42
30	12.12 Human resources	12.12.1 Common / Journals	Loaned items → Loaned items, Loan items	GAP	Propose wave 42: loaned items (assets issued to workers) tied to EAM assets wave 44	42
31	12.12 Human resources	12.12.2 Inquiries & Reports	Course inquiries	GAP	Propose wave 42: learning management	42
32	12.12 Human resources	12.12.2 Inquiries & Reports	Questionnaire results analysis	GAP	Propose wave 42: questionnaire engine	42
33	12.12 Human resources	12.12.2 Inquiries & Reports	Injury/illness inquiries	GAP	Propose wave 42: injury and illness	42
34	12.12 Human resources	12.12.2 Inquiries & Reports	Loaned items inquiry	GAP	Propose wave 42: loaned items	42
35	12.12 Human resources	12.12.2 Inquiries & Reports	SSRS: Course reports (Attendance, Course confirmation letters)	GAP	Propose wave 42: learning management	42
36	12.12 Human resources	12.12.2 Inquiries & Reports	SSRS: Questionnaire reports	GAP	Propose wave 42: questionnaire engine	42
37	12.12 Human resources	12.12.2 Inquiries & Reports	SSRS: OSHA 300/300A/301 (US)	GAP	Propose wave 42: injury and illness (US pack output wave 48)	42
38	12.12 Human resources	12.12.3 Periodic	Courses → Course confirmations / Transfer course completion to competencies	GAP	Propose wave 42: learning management	42
39	12.12 Human resources	12.12.3 Periodic	Questionnaires → Distribute questionnaires (schedule) / Collect results	GAP	Propose wave 42: questionnaire engine	42
40	12.12 Human resources	12.12.4 Setup	Courses → Course types, Course groups, Course locations, Classrooms, Instructors, Agenda templates	GAP	Propose wave 42: learning management	42
41	12.12 Human resources	12.12.4 Setup	Questionnaires → Questionnaire types, Question types, Answer groups, Result groups, Media types, Questionnaire parameters	GAP	Propose wave 42: questionnaire engine	42
42	12.12 Human resources	12.12.4 Setup	Injury and illness → Incident types, Body parts, Injury/illness treatments, Severity levels, Outcomes	GAP	Propose wave 42: injury and illness	42
43	12.12 Human resources	12.12.4 Setup	Loaned items → Loan types	GAP	Propose wave 42: loaned items	42
44	12.14 Inventory management	12.14.1 Common / Journals	Journals → Inventory ownership change journal (D365 consignment)	GAP	Propose wave 16: consignment	16
45	12.14 Inventory management	12.14.2 Inquiries & Reports	SSRS: Consignment inventory reports (D365)	GAP	Propose wave 16: consignment	16

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
46	12.14 Inventory management	12.14.3 Periodic	Inventory → Consignment: ownership change batch	GAP	Propose wave 16: consignment	16
47	12.14 Inventory management	12.14.4 Setup	Inventory → Consignment → Consignment parameters (D365)	GAP	Propose wave 16: consignment	16
48	12.15 Product information management	12.15.1 Common / Journals	Released product → Product compliance (PI: Regulated products, Restricted lists, Reporting details, Safety data sheets (MSDS/SDS) with validity and send on sales)	GAP	Propose wave 33: product compliance (regulated/restricted lists by country/state, SDS versions with shipment delivery rules, reporting thresholds)	33
49	12.15 Product information management	12.15.2 Inquiries & Reports	Product compliance inquiries (Regulated items, SDS due)	GAP	Propose wave 33: product compliance	33
50	12.15 Product information management	12.15.2 Inquiries & Reports	SSRS: Safety data sheet (SDS document, PI)	GAP	Propose wave 33: product compliance	33
51	12.15 Product information management	12.15.3 Periodic	Product compliance → Expire SDS / Reporting (periodic)	GAP	Propose wave 33: product compliance	33
52	12.15 Product information management	12.15.4 Setup	Product → Product compliance setup (PI: Regulated regions, Restricted lists, Reporting lists, SDS parameters, Product safety data sheet validity, Item-specific: MSDS validity days)	GAP	Propose wave 33: product compliance	33
53	12.16 Procurement and sourcing	12.16.1 Common / Journals	External catalogs (punchout cXML/OCI)	GAP	Propose wave 35: external catalog punchout (cXML/OCI) via integration hub, requisition lines returned to wave 17	35
54	12.16 Procurement and sourcing	12.16.1 Common / Journals	Vendor rebates (PI vendor rebate agreements, R3)	GAP	Propose wave 26: vendor rebates (agreements, accrual on purchase, claim/credit note settlement via AP) mirroring customer rebates	26
55	12.16 Procurement and sourcing	12.16.1 Common / Journals	Broker contracts (PI brokers and royalties, R3)	GAP	Propose wave 26: broker contracts (commission to broker vendor on sales invoices, claims via AP)	26
56	12.16 Procurement and sourcing	12.16.1 Common / Journals	Royalty agreements (PI, R3)	GAP	Propose wave 26: royalty agreements (accrual on sales, claim to vendor)	26
57	12.16 Procurement and sourcing	12.16.1 Common / Journals	Consignment (link)	GAP	Propose wave 16: consignment	16
58	12.16 Procurement and sourcing	12.16.2 Inquiries & Reports	Vendor rebate inquiries	GAP	Propose wave 26: vendor rebates	26
59	12.16 Procurement and sourcing	12.16.2 Inquiries & Reports	Broker/royalty claim inquiries	GAP	Propose wave 26: broker contracts	26
60	12.16 Procurement and sourcing	12.16.2 Inquiries & Reports	SSRS: Vendor rebate reports	GAP	Propose wave 26: vendor rebates	26
61	12.16 Procurement and sourcing	12.16.3 Periodic	Vendor rebates → Cumulate / Approve / Process claims	GAP	Propose wave 26: vendor rebates	26
62	12.16 Procurement and sourcing	12.16.3 Periodic	Broker → Process broker claims / Royalty → Process royalty claims	GAP	Propose wave 26: broker contracts	26
63	12.16 Procurement and sourcing	12.16.4 Setup	Vendor rebates → Rebate agreements, Rebate program types, Rebate groups (vendor), Rebate claim journal names	GAP	Propose wave 26: vendor rebates	26
64	12.16 Procurement and sourcing	12.16.4 Setup	Brokers and royalties → Broker contract setup, Royalty agreement setup, Claim journal names	GAP	Propose wave 26: broker contracts	26
65	12.23 Process manufacturing (PI)	12.23.1 Common / Journals	Vendor rebates / Brokers / Royalties (links)	GAP	Propose wave 26: vendor rebates (see §12.16)	26
66	12.23 Process manufacturing (PI)	12.23.1 Common / Journals	Product compliance / Safety data sheets (link)	GAP	Propose wave 33: product compliance (see §12.15)	33
67	12.23 Process manufacturing (PI)	12.23.2 Inquiries & Reports	SSRS: Safety data sheet (link)	GAP	Propose wave 33: product compliance	33
68	12.23 Process manufacturing (PI)	12.23.3 Periodic	Compliance → SDS expiry / regulated product reporting	GAP	Propose wave 33: product compliance	33
69	12.23 Process manufacturing (PI)	12.23.3 Periodic	Vendor rebates → Process	GAP	Propose wave 26: vendor rebates	26
70	12.23 Process manufacturing (PI)	12.23.4 Setup	Product compliance → Regulated/restricted lists, Reporting lists, SDS setup	GAP	Propose wave 33: product compliance	33
71	12.23 Process manufacturing (PI)	12.23.4 Setup	Vendor rebates / Brokers / Royalties setup	GAP	Propose wave 26: vendor rebates	26
72	12.25 Quality management	12.25.4 Setup	Quality management → Product compliance (PI, link)	GAP	Propose wave 33: product compliance	33
73	12.26 Warehouse management (WHS)	12.26.4 Setup	Warehouse → Yard management (D365 2024: yard, trailers, yard moves)	GAP	Propose wave 34: yard management (trailer/yard locations, check-in/out, yard moves, dock door assignment) alongside TMS driver check-in	34
74	12.27 Transportation management (TMS)	12.27.1 Common / Journals	Driver check-in / check-out	GAP	Propose wave 34: driver check-in/out with appointment link, trailer/seal capture, gate log (with yard management)	34
75	12.27 Transportation management (TMS)	12.27.2 Inquiries & Reports	Driver check-in log	GAP	Propose wave 34: driver check-in	34

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
76	12.27 Transportation management (TMS)	12.27.3 Periodic	Driver check-in → Expire appointments / Gate log archive	GAP	Propose wave 34: driver check-in	34
77	12.27 Transportation management (TMS)	12.27.4 Setup	Driver check-in → Check-in setup / Gate configuration	GAP	Propose wave 34: driver check-in	34
78	12.27 Transportation management (TMS)	12.27.4 Setup	Transportation → Yard management setup (D365 2024)	GAP	Propose wave 34: yard management	34
79	12.29 Organization administration	12.29.1 Common / Journals	Electronic signatures → Signature log / Reset signature	GAP	Propose wave 48: electronic signatures	48
80	12.29 Organization administration	12.29.1 Common / Journals	Product safety data sheets (link, PI)	GAP	Propose wave 33: product compliance	33
81	12.29 Organization administration	12.29.1 Common / Journals	Electronic signatures → Certificates (user signing certificates)	GAP	Propose wave 48: electronic signatures	48
82	12.29 Organization administration	12.29.1 Common / Journals	Alerts (org-level alert rules)	GAP	Propose wave 5: alerts and notification rules (change-based and due-date alerts on any entity/field, delivered in-app/email/webhook) built on the wave 4 event backbone	5
83	12.29 Organization administration	12.29.1 Common / Journals	Questionnaires (link, Org admin > Questionnaires in R3)	GAP	Propose wave 42: questionnaire engine	42
84	12.29 Organization administration	12.29.2 Inquiries & Reports	Electronic signature inquiry	GAP	Propose wave 48: electronic signatures	48
85	12.29 Organization administration	12.29.3 Periodic	Electronic signatures → Signature verification / Reset	GAP	Propose wave 48: electronic signatures	48
86	12.29 Organization administration	12.29.4 Setup	Electronic signatures → Electronic signature parameters / Signature requirements / Reason codes / Certificates	GAP	Propose wave 48: electronic signatures	48
87	12.29 Organization administration	12.29.4 Setup	Product safety data sheets setup (link, PI)	GAP	Propose wave 33: product compliance	33
88	12.29 Organization administration	12.29.4 Setup	Alerts → Alert rules (Change-based, Due-date; event rule setup, alert email templates)	GAP	Propose wave 5: alerts and notification rules	5
89	12.29 Organization administration	12.29.4 Setup	Questionnaires setup (link)	GAP	Propose wave 42: questionnaire engine	42
90	12.30 System administration	12.30.1 Common / Journals	Sys admin → Electronic signature (link)	GAP	Propose wave 48: electronic signatures	48
91	12.30 System administration	12.30.3 Periodic	Alerts → Change-based alerts / Due date alerts (batch processors)	GAP	Propose wave 5: alerts and notification rules	5
92	12.30 System administration	12.30.4 Setup	Sys admin → Electronic signature parameters (link)	GAP	Propose wave 48: electronic signatures	48
93	12.31 Home / role centers / workflow	12.31.1 Common / Journals	Questionnaires → My questionnaires / Answer questionnaire / Questionnaire results	GAP	Propose wave 42: questionnaire engine	42
94	12.31 Home / role centers / workflow	12.31.1 Common / Journals	Alerts (My alerts / Notification list)	GAP	Propose wave 5: alerts and notification rules (notification center exists in shell; rule-based alerts do not)	5
95	12.31 Home / role centers / workflow	12.31.2 Inquiries & Reports	Alert inquiry (alerts by rule/user)	GAP	Propose wave 5: alerts and notification rules	5
96	12.31 Home / role centers / workflow	12.31.2 Inquiries & Reports	Questionnaire statistics / Results analysis	GAP	Propose wave 42: questionnaire engine	42
97	12.31 Home / role centers / workflow	12.31.2 Inquiries & Reports	SSRS: Questionnaire reports (link)	GAP	Propose wave 42: questionnaire engine	42
98	12.31 Home / role centers / workflow	12.31.3 Periodic	Alerts → Process change-based alerts / Due-date alerts	GAP	Propose wave 5: alerts and notification rules	5
99	12.31 Home / role centers / workflow	12.31.3 Periodic	Questionnaires → Schedule questionnaires (batch)	GAP	Propose wave 42: questionnaire engine	42
100	12.31 Home / role centers / workflow	12.31.4 Setup	Alerts → Alert rules (create alert rule on field/record: changes to, due date; subject/message; recipients; organization-wide) / Alert rule templates	GAP	Propose wave 5: alerts and notification rules	5
101	12.31 Home / role centers / workflow	12.31.4 Setup	Questionnaires → Questionnaire setup (link)	GAP	Propose wave 42: questionnaire engine	42
102	12.36 Rebate management (D365)	12.36.1 Common / Journals	Deal → Vendor deals (purchase rebates)	GAP	Propose wave 26: vendor rebates	26
103	12.36 Rebate management (D365)	12.36.1 Common / Journals	Deal → Price-based deals (royalty-type)	GAP	Propose wave 26: royalty agreements	26
104	12.36 Rebate management (D365)	12.36.2 Inquiries & Reports	Vendor deal inquiries	GAP	Propose wave 26: vendor rebates	26
105	12.36 Rebate management (D365)	12.36.2 Inquiries & Reports	SSRS: Vendor rebate claim	GAP	Propose wave 26: vendor rebates	26

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
106	12.36 Rebate management (D365)	12.36.3 Periodic	Vendor rebate claims processing	GAP	Propose wave 26: vendor rebates	26
107	12.36 Rebate management (D365)	12.36.4 Setup	Vendor deal setup (claim via AP)	GAP	Propose wave 26: vendor rebates	26
108	12.39 Vendor collaboration / customer self-service (D365)	12.39.1 Common / Journals	Vendor → Consignment inventory view (on-hand, replenishment orders)	GAP	Propose wave 16: consignment	16

12.99.4 PARTIAL rows (wave covers the core; named piece missing)

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
1	12.1 General ledger	12.1.1 Common / Journals	Voucher attachments / document handling	PARTIAL	no document types, archiving policy or external store yet	11
2	12.1 General ledger	12.1.3 Periodic	Reason code assignment (reversal reasons)	PARTIAL	reason codes as a shared framework are not named in any wave	11
3	12.1 General ledger	12.1.4 Setup	Chart of accounts → Advanced rule structures / Advanced rules	PARTIAL	advanced rules (conditional extra segments) are not explicitly listed; account structures are	11
4	12.1 General ledger	12.1.4 Setup	Chart of accounts → Derived financial hierarchies (R3)	PARTIAL	derived hierarchies for reporting are not named	15
5	12.1 General ledger	12.1.4 Setup	Journals → Default descriptions (Transaction text)	PARTIAL	default description templates per posting type not named	11
6	12.1 General ledger	12.1.4 Setup	Posting → Reason codes (financial reasons)	PARTIAL	shared reason code framework not named	11
7	12.1 General ledger	12.1.4 Setup	Foreign trade → Intrastat parameters, commodity codes, transaction codes, transport methods, ports, statistics procedure	PARTIAL	Intrastat extract is named; the setup forms (commodity codes, transaction codes, transport methods, ports, country properties) are not	13
8	12.1 General ledger	12.1.4 Setup	Tax (IN) → India-specific setup (tax components, TDS/TCS)	PARTIAL	India pack listed for wave 48 but TDS/TCS/GST depth unspecified	48
9	12.2 Accounts payable	12.2.1 Common / Journals	Vendor requests (R3)	PARTIAL	vendor request workflow and prospective-vendor self-registration are not named; supplier portal is	20
10	12.2 Accounts payable	12.2.1 Common / Journals	Payments → Prenotes (ACH)	PARTIAL	ACH prenote (zero-dollar validation) flow is not named	19
11	12.2 Accounts payable	12.2.1 Common / Journals	Payments → Centralized payments (cross-company)	PARTIAL	centralized payments across entities are not named in 19 or 40	40
12	12.2 Accounts payable	12.2.3 Periodic	Invoice → Vendor invoice policy violations (R3 audit)	PARTIAL	audit policies are not named; SoD engine is	48
13	12.2 Accounts payable	12.2.3 Periodic	Vendor → Update vendor bank accounts status (prenote results)	PARTIAL	prenote	19
14	12.2 Accounts payable	12.2.4 Setup	Vendors → Vendor request types (R3)	PARTIAL	vendor request workflow	20
15	12.2 Accounts payable	12.2.4 Setup	Payment → Payment calendar (D365)	PARTIAL	payment calendar / bank holiday due-date shifting is not named	19
16	12.2 Accounts payable	12.2.4 Setup	Invoice → Vendor invoice policies (R3 audit policy)	PARTIAL	audit policies	48
17	12.2 Accounts payable	12.2.4 Setup	Forms → Print management	PARTIAL	print management conditional routing per account is not named; document designer is	46
18	12.3 Accounts receivable	12.3.1 Common / Journals	Customer → Direct debit mandates (SEPA, R3)	PARTIAL	SEPA direct debit mandates (pain.008) are not named; collections and cash application are	30
19	12.3 Accounts receivable	12.3.1 Common / Journals	Collections → Cases (collection cases)	PARTIAL	disputes workflow named; generic case management not	30
20	12.3 Accounts receivable	12.3.1 Common / Journals	Commission transactions	PARTIAL	commission calculation hooks are named, but commission posting to vendor/employee and transactions are not	26
21	12.3 Accounts receivable	12.3.2 Inquiries & Reports	Commission inquiry	PARTIAL	commission posting	26
22	12.3 Accounts receivable	12.3.2 Inquiries & Reports	Direct debit mandate inquiry	PARTIAL	direct debit	30
23	12.3 Accounts receivable	12.3.2 Inquiries & Reports	SSRS: Commission report / Commission transactions	PARTIAL	commission posting	26
24	12.3 Accounts receivable	12.3.3 Periodic	Payments → Generate direct debit (pain.008)	PARTIAL	direct debit	30
25	12.3 Accounts receivable	12.3.3 Periodic	Commission → Calculate/post commissions (via invoice)	PARTIAL	commission posting	26
26	12.3 Accounts receivable	12.3.4 Setup	Payment → Payment calendar (D365)	PARTIAL	payment calendar	19
27	12.3 Accounts receivable	12.3.4 Setup	Payment → Direct debit mandate setup (SEPA creditor identifier)	PARTIAL	direct debit	30

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
28	12.3 Accounts receivable	12.3.4 Setup	Collections → Collections email templates (D365)	PARTIAL	email templates for dunning/collections communication not named; letters are	30
29	12.3 Accounts receivable	12.3.4 Setup	Commission → Commission calculation / Commission sales groups / Commission customer groups / Commission item groups / Commission posting	PARTIAL	commission posting and sales group/rep assignment	26
30	12.3 Accounts receivable	12.3.4 Setup	Forms → Form setup (sales documents) / Form notes / Print management	PARTIAL	print management routing	46
31	12.3 Accounts receivable	12.3.4 Setup	Trade agreements → Supplementary items	PARTIAL	supplementary items are not named in pricing or order waves	27
32	12.3 Accounts receivable	12.3.4 Setup	Sales → Sales reasons / Reason codes (customer)	PARTIAL	reason codes	11
33	12.3 Accounts receivable	12.3.4 Setup	Sales → Sales districts / Sales groups (sales reps)	PARTIAL	sales groups/reps for commission	26
34	12.4 Budgeting	12.4.1 Common / Journals	Budget plan → Budget plan templates / worksheets	PARTIAL	Excel worksheet round-trip for budget plans is not named; planning workflow and scenarios are	38
35	12.4 Budgeting	12.4.3 Periodic	Budget planning → Generate budget plan from source (GL actuals, previous plan, FA, projects, demand forecast, HR positions)	PARTIAL	generating plans from position/HR forecasts and demand forecasts is not named	38
36	12.4 Budgeting	12.4.4 Setup	Budget planning → Budget planning templates (Excel)	PARTIAL	Excel worksheets	38
37	12.4 Budgeting	12.4.4 Setup	Budget planning → Position forecasting (D365 HR-driven)	PARTIAL	position forecasting	38
38	12.5 Cash and bank management	12.5.4 Setup	Bank account → Prenote setup	PARTIAL	prenote	19
39	12.5 Cash and bank management	12.5.4 Setup	Bank → Bank holidays / Payment calendar	PARTIAL	payment calendar	19
40	12.6 Fixed assets	12.6.1 Common / Journals	Fixed asset → Fixed asset additions	PARTIAL	asset additions (non-financial component records) not named	37
41	12.6 Fixed assets	12.6.1 Common / Journals	Fixed asset → Insurance records	PARTIAL	insurance records not named	37
42	12.6 Fixed assets	12.6.2 Inquiries & Reports	SSRS: Fixed asset insurance report	PARTIAL	insurance	37
43	12.6 Fixed assets	12.6.4 Setup	Fixed asset → Insurance types / Insurance companies	PARTIAL	insurance	37
44	12.6 Fixed assets	12.6.4 Setup	Fixed asset → Reason codes (fixed assets)	PARTIAL	reason codes	11
45	12.7 Cost accounting	12.7.1 Common / Journals	Cost behavior / Cost classification (fixed/variable) (D365)	PARTIAL	cost behavior (fixed/variable split) policies not named	38
46	12.7 Cost accounting	12.7.4 Setup	D365: Policies — Cost behavior policies, Cost distribution policies, Overhead rate policies, Cost allocation policies, Cost rollup policies (D365 2020+)	PARTIAL	cost behavior and cost rollup policies not named; allocation policies are	38
47	12.8 Cost management (D365)	12.8.1 Common / Journals	Landed cost (D365 Landed cost module: Voyages, Shipping containers, Folios, Cost estimates, Goods in transit)	PARTIAL	landed cost allocation is named; D365 voyage/container/folio tracking and goods-in-transit orders are not	34
48	12.8 Cost management (D365)	12.8.2 Inquiries & Reports	Inventory aging report (D365)	PARTIAL	inventory aging buckets not named	20
49	12.8 Cost management (D365)	12.8.2 Inquiries & Reports	Landed cost reports (voyage cost, cost variance by voyage)	PARTIAL	landed cost voyages	34
50	12.8 Cost management (D365)	12.8.3 Periodic	Standard cost → Revaluation on activation / Conversion to standard cost (Standard cost conversion checklist)	PARTIAL	standard cost conversion checklist (switching costing model mid-life) not named	39
51	12.8 Cost management (D365)	12.8.3 Periodic	Landed cost → Voyage cost allocation / Goods in transit receipts	PARTIAL	landed cost voyages	34
52	12.8 Cost management (D365)	12.8.4 Setup	Inventory aging report setup (D365)	PARTIAL	inventory aging	20
53	12.8 Cost management (D365)	12.8.4 Setup	Landed cost setup (D365: Landed cost parameters, Cost types, Cost type codes, Auto cost, Apportionment, Shipping container types, Vessels, Voyage templates, Legs, Journeys, Tracking)	PARTIAL	landed cost voyages	34
54	12.8 Cost management (D365)	12.8.4 Setup	Standard cost conversion setup	PARTIAL	standard cost conversion	39
55	12.9 Compliance and internal controls	12.9.1 Common / Journals	Internal controls → Files (process documents)	PARTIAL	SOC 2 evidence automation is named; a controls/process document library is not	48
56	12.9 Compliance and internal controls	12.9.1 Common / Journals	Internal controls → Process library / Process documents	PARTIAL	process library	48
57	12.9 Compliance and internal controls	12.9.1 Common / Journals	Internal controls → Risk library	PARTIAL	risk library	48
58	12.9 Compliance and internal controls	12.9.1 Common / Journals	Audit → Audit cases	PARTIAL	audit policies	48

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
59	12.9 Compliance and internal controls	12.9.1 Common / Journals	Audit → Audit policy violations list	PARTIAL	audit policies	48
60	12.9 Compliance and internal controls	12.9.2 Inquiries & Reports	Audit policy violations inquiry	PARTIAL	audit policies	48
61	12.9 Compliance and internal controls	12.9.2 Inquiries & Reports	Audit case inquiry	PARTIAL	audit policies	48
62	12.9 Compliance and internal controls	12.9.2 Inquiries & Reports	SSRS: Audit policy report / Audit case report	PARTIAL	audit policies	48
63	12.9 Compliance and internal controls	12.9.2 Inquiries & Reports	SSRS: Internal control documents	PARTIAL	process library	48
64	12.9 Compliance and internal controls	12.9.3 Periodic	Audit → Run audit policies (batch: expense reports, vendor invoices, purchase orders, GL journals)	PARTIAL	audit policies	48
65	12.9 Compliance and internal controls	12.9.3 Periodic	Audit → Close/archive audit cases	PARTIAL	audit policies	48
66	12.9 Compliance and internal controls	12.9.4 Setup	Audit → Audit policies (policy framework: policy rule types — expense/vendor invoice/PO/GL journal; rule types: duplicate, keyword, list search, sampling, premium/mileage)	PARTIAL	audit policies	48
67	12.9 Compliance and internal controls	12.9.4 Setup	Audit → Audit policy rule types	PARTIAL	audit policies	48
68	12.9 Compliance and internal controls	12.9.4 Setup	Audit → Audit case categories	PARTIAL	audit policies	48
69	12.9 Compliance and internal controls	12.9.4 Setup	Audit → Audit keyword lists / Audit sampling parameters	PARTIAL	audit policies	48
70	12.9 Compliance and internal controls	12.9.4 Setup	Risk/control types, Process document templates	PARTIAL	process library	48
71	12.10 Travel and expense	12.10.1 Common / Journals	Cash advances (Cash advance requests)	PARTIAL	cash advances are not named	43
72	12.10 Travel and expense	12.10.1 Common / Journals	Travel requisitions	PARTIAL	travel requisitions / pre-approval are not named	43
73	12.10 Travel and expense	12.10.1 Common / Journals	Expense reports → Intercompany expenses (R3)	PARTIAL	intercompany expenses not named in 40 or 43	40
74	12.10 Travel and expense	12.10.2 Inquiries & Reports	Cash advance inquiry	PARTIAL	cash advances	43
75	12.10 Travel and expense	12.10.2 Inquiries & Reports	Travel requisition inquiry	PARTIAL	travel requisitions	43
76	12.10 Travel and expense	12.10.2 Inquiries & Reports	SSRS: Cash advance report	PARTIAL	cash advances	43
77	12.10 Travel and expense	12.10.2 Inquiries & Reports	SSRS: Expense audit (audit policy results)	PARTIAL	audit policies	48
78	12.10 Travel and expense	12.10.2 Inquiries & Reports	SSRS: VAT recovery report (expense tax)	PARTIAL	VAT recovery on expenses (partial recoverability per country) not named	13
79	12.10 Travel and expense	12.10.3 Periodic	Reconcile/settle cash advances	PARTIAL	cash advances	43
80	12.10 Travel and expense	12.10.3 Periodic	Expense → Run audit policies on expense reports	PARTIAL	audit policies	48
81	12.10 Travel and expense	12.10.3 Periodic	Expense → Recover VAT (ER-based claim)	PARTIAL	VAT recovery	13
82	12.10 Travel and expense	12.10.3 Periodic	Expense → Intercompany expense posting (due-to/due-from)	PARTIAL	intercompany expenses	40
83	12.10 Travel and expense	12.10.4 Setup	Expense management parameters — Intercompany / Cash advance	PARTIAL	cash advances	43
84	12.10 Travel and expense	12.10.4 Setup	Cash advance → Cash advance parameters, posting	PARTIAL	cash advances	43
85	12.10 Travel and expense	12.10.4 Setup	Travel requisition → Requisition policies / types	PARTIAL	travel requisitions	43
86	12.10 Travel and expense	12.10.4 Setup	Expense → Tax configuration (recoverable VAT per country, tax groups on expenses)	PARTIAL	VAT recovery	13
87	12.10 Travel and expense	12.10.4 Setup	Expense → Audit policies (link)	PARTIAL	audit policies	48
88	12.11 Project management and accounting	12.11.1 Common / Journals	Project quotations	PARTIAL	project quotations with convert-to-project are not named; sales quotations are in 27	41
89	12.11 Project management and accounting	12.11.1 Common / Journals	Timesheets (list page)	PARTIAL	hour journals are named; weekly timesheet UI with favorites/workflow is not explicit	41

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
90	12.11 Project management and accounting	12.11.1 Common / Journals	Timesheet details	PARTIAL	timesheets	41
91	12.11 Project management and accounting	12.11.1 Common / Journals	Beginning balance journals	PARTIAL	beginning balance journals for project migration not named; wave 49 covers project WIP migration	49
92	12.11 Project management and accounting	12.11.1 Common / Journals	Project → Activities (Activity list for projects)	PARTIAL	generic activities/case management framework	45
93	12.11 Project management and accounting	12.11.1 Common / Journals	Project → Retention terms (customer/vendor retention)	PARTIAL	customer and vendor retention (retainage) terms are not named	41
94	12.11 Project management and accounting	12.11.2 Inquiries & Reports	Timesheet inquiries (missing timesheets, timesheet hours)	PARTIAL	timesheets	41
95	12.11 Project management and accounting	12.11.2 Inquiries & Reports	SSRS: Timesheet report / Missing timesheets	PARTIAL	timesheets	41
96	12.11 Project management and accounting	12.11.3 Periodic	Timesheets → Timesheet workflow / Post timesheets / Create periodic timesheets	PARTIAL	timesheets	41
97	12.11 Project management and accounting	12.11.3 Periodic	Project → Intercompany project invoicing (R3 intercompany timesheets/expenses)	PARTIAL	intercompany project resourcing/invoicing not named	40
98	12.11 Project management and accounting	12.11.3 Periodic	Project → Retention release (vendor/customer)	PARTIAL	retention	41
99	12.11 Project management and accounting	12.11.4 Setup	Billing → Billing rule templates / Retention terms	PARTIAL	retention	41
100	12.11 Project management and accounting	12.11.4 Setup	Timesheets → Timesheet periods / Period types / Timesheet parameters (favorites, delegates, workflow by project)	PARTIAL	timesheets	41
101	12.11 Project management and accounting	12.11.4 Setup	Indirect costs (Indirect cost component groups, Indirect cost components, Indirect cost rules — R3 PSA)	PARTIAL	project indirect cost components/rules not named; cost accounting allocations in 38 do not apply to project transactions	41
102	12.11 Project management and accounting	12.11.4 Setup	Forms → Form setup (project invoices) / Print management	PARTIAL	print management routing	46
103	12.11 Project management and accounting	12.11.4 Setup	Intercompany → Intercompany project parameters (lending/borrowing entity, timesheet/expense/vendor invoice transfer)	PARTIAL	intercompany project	40
104	12.12 Human resources	12.12.1 Common / Journals	Competencies → Skill mapping / Skill gap analysis	PARTIAL	skill mapping/gap analysis not named	42
105	12.12 Human resources	12.12.1 Common / Journals	Worker compliance (US) → I-9, ADA requirements, Immigration/visa, Union agreements	PARTIAL	country-specific compliance fields framework is named; I-9/visa/union specifics come with US pack	48
106	12.12 Human resources	12.12.1 Common / Journals	Worker → Worker tasks / reminders / Checklists (onboarding)	PARTIAL	onboarding checklists not named	42
107	12.12 Human resources	12.12.1 Common / Journals	Teams (Org admin: Teams, Team types)	PARTIAL	teams (ad hoc cross-org groups) not named; hierarchies are	6
108	12.12 Human resources	12.12.1 Common / Journals	Human resources → Cases (HR cases)	PARTIAL	generic case management	45
109	12.12 Human resources	12.12.2 Inquiries & Reports	SSRS: EEO-1 / VETS (US compliance)	PARTIAL	US compliance reporting with wave 48 US pack	48
110	12.12 Human resources	12.12.3 Periodic	Worker → Reminders (certificate expiry, probation end)	PARTIAL	reminder/alert engine; see alerts gap in §12.31	42
111	12.12 Human resources	12.12.3 Periodic	HR → Year-end (US ACA 1095-C)	PARTIAL	ACA reporting with US pack	48
112	12.12 Human resources	12.12.4 Setup	Worker compliance (US) → I-9 document types, ADA requirements, Visa types, Union agreements	PARTIAL	US compliance	48
113	12.12 Human resources	12.12.4 Setup	Worker → Worker tasks / Checklist templates (onboarding)	PARTIAL	onboarding checklists	42
114	12.12 Human resources	12.12.4 Setup	Teams → Team types, Teams	PARTIAL	teams	6
115	12.13 Payroll	12.13.2 Inquiries & Reports	SSRS: Workers compensation report	PARTIAL	workers' compensation class codes/reporting not named	48
116	12.13 Payroll	12.13.2 Inquiries & Reports	SSRS: New hire reporting (US state)	PARTIAL	new hire reporting with US pack	48
117	12.13 Payroll	12.13.3 Periodic	Payroll → Arrears recovery processing	PARTIAL	deduction arrears recovery not named	43

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
118	12.13 Payroll	12.13.4 Setup	Payroll → Leave of absence / FMLA tracking (US)	PARTIAL	FMLA tracking not named	48
119	12.14 Inventory management	12.14.1 Common / Journals	Transfer orders (list page)	PARTIAL	transfer *journal* is named; transfer *orders* with shipment, in-transit warehouse, receive, underdelivery and transfer-order picking are not named in 16, 22 or 28	16
120	12.14 Inventory management	12.14.1 Common / Journals	Transfer order details	PARTIAL	transfer orders	16
121	12.14 Inventory management	12.14.1 Common / Journals	Journals → Tag counting journal	PARTIAL	tag counting (numbered tags, two-step) not named; cycle counting is	16
122	12.14 Inventory management	12.14.1 Common / Journals	Batch merge (PI)	PARTIAL	batch merge/split with attribute recalculation not named	24
123	12.14 Inventory management	12.14.1 Common / Journals	Inventory → ABC classification (on item)	PARTIAL	ABC classification fields/calculation not named	16
124	12.14 Inventory management	12.14.1 Common / Journals	Inventory → Commodity codes (Intrastat on item)	PARTIAL	Intrastat setup	13
125	12.14 Inventory management	12.14.2 Inquiries & Reports	Transfer order inquiries (Transfer order lines, shipped/received)	PARTIAL	transfer orders	16
126	12.14 Inventory management	12.14.2 Inquiries & Reports	Inventory → Intercompany on-hand (R3)	PARTIAL	intercompany on-hand visibility not named	40
127	12.14 Inventory management	12.14.2 Inquiries & Reports	SSRS: Transfer order (shipping document, receiving document, Transfer order overview)	PARTIAL	transfer orders	16
128	12.14 Inventory management	12.14.2 Inquiries & Reports	SSRS: Inventory aging (link)	PARTIAL	inventory aging	20
129	12.14 Inventory management	12.14.2 Inquiries & Reports	SSRS: ABC classification report	PARTIAL	ABC classification	16
130	12.14 Inventory management	12.14.2 Inquiries & Reports	SSRS: Packing slip (inventory transfer)	PARTIAL	transfer orders	16
131	12.14 Inventory management	12.14.3 Periodic	Tag counting → Create tag counting / Transfer to counting journal	PARTIAL	tag counting	16
132	12.14 Inventory management	12.14.3 Periodic	Transfer orders → Ship / Receive (batch) / Transfer order picking release	PARTIAL	transfer orders	16
133	12.14 Inventory management	12.14.3 Periodic	Transfer orders → Delete transfer orders	PARTIAL	transfer orders	16
134	12.14 Inventory management	12.14.3 Periodic	ABC classification (periodic calculation)	PARTIAL	ABC classification	16
135	12.14 Inventory management	12.14.3 Periodic	Inventory → Intercompany: on-hand sync	PARTIAL	intercompany on-hand	40
136	12.14 Inventory management	12.14.4 Setup	Inventory and warehouse management parameters — Transfer orders	PARTIAL	transfer orders	16
137	12.14 Inventory management	12.14.4 Setup	Inventory → Counting → Counting reason code policies	PARTIAL	reason codes	11
138	12.14 Inventory management	12.14.4 Setup	Inventory → ABC classification setup (ABC calculation parameters)	PARTIAL	ABC classification	16
139	12.14 Inventory management	12.14.4 Setup	Inventory → Transfer orders → Transfer order parameters, In-transit warehouses	PARTIAL	transfer orders	16
140	12.14 Inventory management	12.14.4 Setup	Inventory → Picking → Picking routes setup / Picking workbench (R3 sales picking workbench)	PARTIAL	picking workbench (grouped picking list generation across sales orders, non-WHS) is not named; wave/batch picking is	28
141	12.15 Product information management	12.15.1 Common / Journals	Released product → Related products	PARTIAL	related products (cross/up-sell relations) not named	8
142	12.15 Product information management	12.15.1 Common / Journals	Released product → Commodity pricing (PI: Pricing calculation, Price margins, Commodity codes)	PARTIAL	commodity price index-based pricing for process industries (PI commodity pricing) not named	26
143	12.15 Product information management	12.15.1 Common / Journals	Released product → Product lifecycle state (D365)	PARTIAL	product lifecycle states with per-process blocking not named; stopped flags are	8
144	12.15 Product information management	12.15.1 Common / Journals	Product → Product readiness checks (D365)	PARTIAL	product readiness policies/checks not named	9
145	12.15 Product information management	12.15.1 Common / Journals	Product → Packaging (D365 packaging materials, PI container packaging)	PARTIAL	packaging materials and packaging fees (EU packaging duty) not named	28
146	12.15 Product information management	12.15.3 Periodic	Product → Change product dimension groups / Change item model group (periodic conversions)	PARTIAL	changing dimension/model groups on items with transactions (conversion jobs) not named	8
147	12.15 Product information management	12.15.3 Periodic	Product → Delete released products / Mark obsolete	PARTIAL	lifecycle states	8

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
148	12.15 Product information management	12.15.4 Setup	Product → Product lifecycle states (D365)	PARTIAL	lifecycle states	8
149	12.15 Product information management	12.15.4 Setup	Product → Packaging units / Packing material fees setup (EU)	PARTIAL	packaging materials	28
150	12.15 Product information management	12.15.4 Setup	Product → Commodity pricing setup (Cost basis types, Pricing calculations, Price margin templates, Commodity codes)	PARTIAL	commodity pricing	26
151	12.15 Product information management	12.15.4 Setup	Product → Product readiness policies (D365)	PARTIAL	product readiness	9
152	12.16 Procurement and sourcing	12.16.1 Common / Journals	RFQ → Bidding (Vendor portal RFQ reply, sealed bids, scoring criteria R3)	PARTIAL	sealed bids, scoring criteria and amendments/addenda not named	17
153	12.16 Procurement and sourcing	12.16.1 Common / Journals	Procurement catalogs	PARTIAL	requisitions and category policies are named; an internal procurement catalog with product images, category navigation, catalog publication by org unit is not	17
154	12.16 Procurement and sourcing	12.16.1 Common / Journals	Vendor catalogs (vendor catalog import, CMR files)	PARTIAL	vendor catalog upload via supplier portal not named	20
155	12.16 Procurement and sourcing	12.16.1 Common / Journals	Vendor requests (link)	PARTIAL	vendor request workflow	20
156	12.16 Procurement and sourcing	12.16.1 Common / Journals	Procurement → Signing limits (link to Org admin)	PARTIAL	signing limits (self-service limit requests per job/compensation) not named; amount-band policies are	17
157	12.16 Procurement and sourcing	12.16.1 Common / Journals	Procurement → Vendor onboarding (D365 prospective vendor registration)	PARTIAL	vendor onboarding	20
158	12.16 Procurement and sourcing	12.16.1 Common / Journals	Procurement → Vendor certifications / Vendor diversity (D365)	PARTIAL	vendor certifications/diversity classifications not named	7
159	12.16 Procurement and sourcing	12.16.2 Inquiries & Reports	SSRS: Procurement catalog report	PARTIAL	procurement catalog	17
160	12.16 Procurement and sourcing	12.16.3 Periodic	Catalog → Import vendor catalogs (CMR) / Publish procurement catalog	PARTIAL	procurement catalog	17
161	12.16 Procurement and sourcing	12.16.4 Setup	Policies → Purchase requisition approval amount thresholds / Signing limit policies	PARTIAL	signing limits	17
162	12.16 Procurement and sourcing	12.16.4 Setup	Categories → Procurement categories, Category policy rules, Category managers, Category vendor assignment, Category questionnaires	PARTIAL	category-level vendor assignment and category questionnaires not named	17
163	12.16 Procurement and sourcing	12.16.4 Setup	Catalogs → Procurement catalogs, Catalog policies, External catalogs (cXML config), Vendor catalog import parameters, Catalog images	PARTIAL	procurement catalog	17
164	12.16 Procurement and sourcing	12.16.4 Setup	Vendors → Vendor request types / Vendor onboarding questionnaires (D365)	PARTIAL	vendor request workflow	20
165	12.16 Procurement and sourcing	12.16.4 Setup	Vendors → Vendor certification types / Diversity classifications	PARTIAL	vendor certifications	7
166	12.16 Procurement and sourcing	12.16.4 Setup	RFQ → RFQ types, RFQ scoring criteria/groups, Reply templates, Solicitation types (D365), Bidding parameters	PARTIAL	sealed bids/scoring	17
167	12.16 Procurement and sourcing	12.16.4 Setup	Forms → Form setup (purchase documents) / Form notes / Print management (purchase)	PARTIAL	print management routing	46
168	12.17 Sales and marketing	12.17.1 Common / Journals	Sales order → Supplementary items (prompt)	PARTIAL	supplementary items	27
169	12.17 Sales and marketing	12.17.1 Common / Journals	Sales → Commissions (link)	PARTIAL	commission posting	26
170	12.17 Sales and marketing	12.17.2 Inquiries & Reports	SSRS: Commission report (link)	PARTIAL	commission posting	26
171	12.17 Sales and marketing	12.17.4 Setup	Sales orders → Supplementary items	PARTIAL	supplementary items	27
172	12.17 Sales and marketing	12.17.4 Setup	Commissions → Commission calculation, Sales groups (sales reps), Commission customer groups, Commission item groups, Commission posting	PARTIAL	commission posting	26
173	12.17 Sales and marketing	12.17.4 Setup	Forms → Form setup (sales documents), Form notes, Print management (sales), Form sorting	PARTIAL	print management routing	46
174	12.18 Trade allowance management	12.18.1 Common / Journals	Merchandising events (Promotion planning)	PARTIAL	promotion planning with baseline/lift analysis not named; promotions are	26
175	12.18 Trade allowance management	12.18.1 Common / Journals	Funds	PARTIAL	trade promotion funds with budget/usage tracking not named	26
176	12.18 Trade allowance management	12.18.1 Common / Journals	Deduction workbench	PARTIAL	deductions workbench — matching customer short-pays to promotions, denial/write-off/split — not named; disputes workflow is	30

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
177	12.18 Trade allowance management	12.18.2 Inquiries & Reports	Fund usage inquiry	PARTIAL	funds	26
178	12.18 Trade allowance management	12.18.2 Inquiries & Reports	Deduction inquiry (open deductions, matched)	PARTIAL	deductions workbench	30
179	12.18 Trade allowance management	12.18.2 Inquiries & Reports	Promotion effectiveness analysis (lift)	PARTIAL	promotion planning	26
180	12.18 Trade allowance management	12.18.2 Inquiries & Reports	SSRS: Fund statement / Fund usage	PARTIAL	funds	26
181	12.18 Trade allowance management	12.18.2 Inquiries & Reports	SSRS: Deductions report	PARTIAL	deductions workbench	30
182	12.18 Trade allowance management	12.18.3 Periodic	Deductions → Match deductions to claims / Write off small balances	PARTIAL	deductions workbench	30
183	12.18 Trade allowance management	12.18.3 Periodic	Funds → Fund budget recalculation / Fund close	PARTIAL	funds	26
184	12.18 Trade allowance management	12.18.3 Periodic	Promotion → Update promotion planning actuals	PARTIAL	promotion planning	26
185	12.18 Trade allowance management	12.18.4 Setup	Trade allowance → Funds setup (fund types, budget periods)	PARTIAL	funds	26
186	12.18 Trade allowance management	12.18.4 Setup	Trade allowance → Merchandising event categories / Event types	PARTIAL	promotion planning	26
187	12.18 Trade allowance management	12.18.4 Setup	Trade allowance → Deduction types / Deduction reason codes / Denial reasons	PARTIAL	deductions workbench	30
188	12.21 Master planning	12.21.1 Common / Journals	Supply schedule (R3)	PARTIAL	period-bucketed supply schedule workbench not named; planning engine and planned orders are	22
189	12.21 Master planning	12.21.1 Common / Journals	Inventory forecast	PARTIAL	inventory (stock level) forecast type not named; demand/supply forecasts are	22
190	12.21 Master planning	12.21.1 Common / Journals	Intercompany planned orders (R3 intercompany master scheduling)	PARTIAL	intercompany master planning across legal entities not named in 22 or 40	40
191	12.21 Master planning	12.21.2 Inquiries & Reports	SSRS: Supply schedule report	PARTIAL	supply schedule	22
192	12.21 Master planning	12.21.3 Periodic	Intercompany master scheduling (run across entities)	PARTIAL	intercompany planning	40
193	12.21 Master planning	12.21.4 Setup	Intercompany → Intercompany planning groups / Intercompany planning group members (sequence, plan)	PARTIAL	intercompany planning	40
194	12.21 Master planning	12.21.4 Setup	Planning → Priority-based planning (D365: planning priority models, priority rules on coverage group)	PARTIAL	priority-based / DDMRP-style buffer planning not named	22
195	12.22 Production control	12.22.1 Common / Journals	Production order → Split / Group (ProdSplit)	PARTIAL	production order split/merge not named	21
196	12.23 Process manufacturing (PI)	12.23.1 Common / Journals	Batch order → Consolidated batch orders (batch order consolidation / bulk item → packed items)	PARTIAL	consolidated batch orders (bulk/pack relationship) not named	24
197	12.23 Process manufacturing (PI)	12.23.1 Common / Journals	Batch order → Rework batch order	PARTIAL	rework batch orders not named	24
198	12.23 Process manufacturing (PI)	12.23.1 Common / Journals	Batch attribute inheritance (lot inheritance: product-to-product, co-product inheritance)	PARTIAL	batch attribute and shelf-life inheritance rules not named	24
199	12.23 Process manufacturing (PI)	12.23.1 Common / Journals	Commodity pricing (Pricing calculation, link)	PARTIAL	commodity pricing	26
200	12.23 Process manufacturing (PI)	12.23.1 Common / Journals	Container packaging (PI packaging / Container types)	PARTIAL	packaging materials	28
201	12.23 Process manufacturing (PI)	12.23.1 Common / Journals	Batch merge	PARTIAL	batch merge	24
202	12.23 Process manufacturing (PI)	12.23.1 Common / Journals	Same-batch reservation (batch reservation rules: consolidated/same batch on sales line)	PARTIAL	same-batch reservation requirement on sales lines not named	24
203	12.23 Process manufacturing (PI)	12.23.1 Common / Journals	Customer-specific batch attribute requirements (customer sellable days / batch attribute requirements on sales)	PARTIAL	customer sellable days / customer batch attribute requirements on sales lines not named	24
204	12.23 Process manufacturing (PI)	12.23.2 Inquiries & Reports	Commodity price inquiries	PARTIAL	commodity pricing	26
205	12.23 Process manufacturing (PI)	12.23.3 Periodic	Batch orders → Consolidate batch orders (batch)	PARTIAL	consolidated batch orders	24
206	12.23 Process manufacturing (PI)	12.23.3 Periodic	Batches → Attribute inheritance processing on RAF	PARTIAL	inheritance	24

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
207	12.23 Process manufacturing (PI)	12.23.3 Periodic	Commodity pricing → Price calculation run (update trade agreements from commodity index)	PARTIAL	commodity pricing	26
208	12.23 Process manufacturing (PI)	12.23.4 Setup	Batch → Lot inheritance setup (inheritance of attributes/shelf life by product)	PARTIAL	inheritance	24
209	12.23 Process manufacturing (PI)	12.23.4 Setup	Commodity pricing → Cost basis types, Pricing calculations, Price margin templates	PARTIAL	commodity pricing	26
210	12.23 Process manufacturing (PI)	12.23.4 Setup	Container packaging → Container types, Packaging groups	PARTIAL	packaging materials	28
211	12.23 Process manufacturing (PI)	12.23.4 Setup	Customer sellable days / Customer batch attribute requirements	PARTIAL	customer sellable days	24
212	12.23 Process manufacturing (PI)	12.23.4 Setup	Rework → Rework parameters	PARTIAL	rework	24
213	12.23 Process manufacturing (PI)	12.23.4 Setup	Consolidated batch orders → Bulk/pack item relations	PARTIAL	consolidated batch orders	24
214	12.24 Lean manufacturing	12.24.1 Common / Journals	Lean → Subcontracting in lean (service activities with vendor, purchase agreements)	PARTIAL	lean subcontracting (activity-based service purchase from vendor, purchase agreement linkage) not named; discrete subcontracting is in 21	24
215	12.24 Lean manufacturing	12.24.3 Periodic	Lean → Lean subcontracting: generate purchase orders for service activities	PARTIAL	lean subcontracting	24
216	12.24 Lean manufacturing	12.24.4 Setup	Lean → Subcontracting setup (service items, purchase agreements for activities)	PARTIAL	lean subcontracting	24
217	12.26 Warehouse management (WHS)	12.26.1 Common / Journals	Outbound → Cross docking (D365 planned cross docking, opportunistic)	PARTIAL	cross docking (planned/opportunistic) not named	31
218	12.26 Warehouse management (WHS)	12.26.1 Common / Journals	WHS → Transfer order WHS flow (ship/receive with LP)	PARTIAL	transfer orders	31
219	12.26 Warehouse management (WHS)	12.26.2 Inquiries & Reports	SSRS: Cross docking report	PARTIAL	cross docking	31
220	12.26 Warehouse management (WHS)	12.26.3 Periodic	Cross docking → Planned cross docking processing	PARTIAL	cross docking	31
221	12.26 Warehouse management (WHS)	12.26.4 Setup	Outbound → Cross docking templates (D365)	PARTIAL	cross docking	31
222	12.26 Warehouse management (WHS)	12.26.4 Setup	Outbound → Outbound sorting (D365 outbound sorting positions)	PARTIAL	outbound sorting (sort pick-to-shipment at a sortation area) not named	31
223	12.27 Transportation management (TMS)	12.27.1 Common / Journals	Landed cost (D365 module; voyages, link)	PARTIAL	landed cost voyages	34
224	12.27 Transportation management (TMS)	12.27.2 Inquiries & Reports	SSRS: Landed cost reports (link)	PARTIAL	landed cost voyages	34
225	12.27 Transportation management (TMS)	12.27.3 Periodic	Landed cost → Voyage cost actualization (link)	PARTIAL	landed cost voyages	34
226	12.27 Transportation management (TMS)	12.27.4 Setup	Landed cost → Landed cost parameters / Cost types / Voyage templates (link)	PARTIAL	landed cost voyages	34
227	12.28 Service management	12.28.1 Common / Journals	Service subscriptions (subscription fee transactions, index)	PARTIAL	AX service subscriptions (fee indexation, accrual via project) not explicitly named; subscription billing is wave 29	29
228	12.28 Service management	12.28.2 Inquiries & Reports	Subscription inquiries	PARTIAL	service subscriptions	29
229	12.28 Service management	12.28.2 Inquiries & Reports	SSRS: Subscription report / Subscription fee transactions	PARTIAL	service subscriptions	29
230	12.28 Service management	12.28.3 Periodic	Subscriptions → Create subscription fee transactions / Accrue subscription revenue / Index subscriptions	PARTIAL	service subscriptions	29
231	12.28 Service management	12.28.4 Setup	Service → Subscription: Subscription groups, Subscription fee categories, Index rules, Subscription periods	PARTIAL	service subscriptions	29
232	12.29 Organization administration	12.29.1 Common / Journals	Teams / Team types	PARTIAL	teams	6
233	12.29 Organization administration	12.29.1 Common / Journals	Cases (list page) / Case details	PARTIAL	generic case management (categories, processes, activities, knowledge articles, associations to any record) is not named in any wave; service cases are in 45, disputes in 30, product change in 9	45
234	12.29 Organization administration	12.29.1 Common / Journals	Cases → Knowledge articles	PARTIAL	generic case management	45
235	12.29 Organization administration	12.29.1 Common / Journals	Document management → Document handling (attachments on any record), Document types, File types	PARTIAL	attachments exist in 5/11; document types, storage locations (SharePoint/S3), archive directories, retention are not named	5
236	12.29 Organization administration	12.29.1 Common / Journals	Foreign trade → Intrastat (journal) / EU sales list (link)	PARTIAL	Intrastat setup	13
237	12.29 Organization administration	12.29.1 Common / Journals	Policies → Signing limits (Signing limit policies, Worker signing limit requests, Default signing limits by job/compensation level)	PARTIAL	signing limits	17

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
238	12.29 Organization administration	12.29.1 Common / Journals	Policies → Policy framework forms (Purchasing policies, Expense policies, Audit policies, Vendor invoice policies, Signing limit policies)	PARTIAL	a generic policy framework scoped by org hierarchy is implied by 17/43 but not named as reusable	17
239	12.29 Organization administration	12.29.1 Common / Journals	Reason codes (shared financial reasons across modules)	PARTIAL	reason codes	11
240	12.29 Organization administration	12.29.1 Common / Journals	Print management (org-wide)	PARTIAL	print management routing	46
241	12.29 Organization administration	12.29.1 Common / Journals	Case management → Case processes / Case categories (link)	PARTIAL	generic case management	45
242	12.29 Organization administration	12.29.2 Inquiries & Reports	Case inquiries (cases by category/status, SLA)	PARTIAL	generic case management	45
243	12.29 Organization administration	12.29.2 Inquiries & Reports	Document management inquiries (documents by type/record)	PARTIAL	document management	5
244	12.29 Organization administration	12.29.2 Inquiries & Reports	SSRS: Case report / Case activities	PARTIAL	generic case management	45
245	12.29 Organization administration	12.29.2 Inquiries & Reports	SSRS: Signing limits report	PARTIAL	signing limits	17
246	12.29 Organization administration	12.29.2 Inquiries & Reports	SSRS: Policies report (policy rules per hierarchy)	PARTIAL	policy framework	17
247	12.29 Organization administration	12.29.3 Periodic	Foreign trade → Intrastat transfer (from invoices) / Intrastat validation / Intrastat export (file)	PARTIAL	Intrastat setup	13
248	12.29 Organization administration	12.29.3 Periodic	Cases → Close cases / Case SLA breach notifications	PARTIAL	generic case management	45
249	12.29 Organization administration	12.29.3 Periodic	Document management → Archive documents / Purge by retention	PARTIAL	document management	5
250	12.29 Organization administration	12.29.3 Periodic	Signing limits → Signing limit request processing (annual recertification)	PARTIAL	signing limits	17
251	12.29 Organization administration	12.29.4 Setup	Organization → Teams / Team types	PARTIAL	teams	6
252	12.29 Organization administration	12.29.4 Setup	Document management → Document management parameters (archive directory, SharePoint, file types, max size), Document types (file, note, URL, image, template; group; location: archive/database/SharePoint), Active document tables, File types	PARTIAL	document management	5
253	12.29 Organization administration	12.29.4 Setup	Document management → Document management: OneDrive/SharePoint integration (D365), Attachment migration	PARTIAL	document management	5
254	12.29 Organization administration	12.29.4 Setup	Case management → Case categories / Case category types (HR, customer service, collections, product change, audit, service, project, knowledge, etc.) / Case processes (stages) / Case category security	PARTIAL	generic case management	45
255	12.29 Organization administration	12.29.4 Setup	Foreign trade → Foreign trade parameters, Commodity codes, Transaction codes, Transport methods, Ports, Statistics procedures, Country/region properties (EU), Intrastat compression, Intrastat check setup, EU sales list reporting	PARTIAL	Intrastat setup	13
256	12.29 Organization administration	12.29.4 Setup	Policies → Signing limit policies / Signing limit parameters / Default signing limits (per job/compensation) / Signing limit request workflow	PARTIAL	signing limits	17
257	12.29 Organization administration	12.29.4 Setup	Policies → Policy parameters (per policy type: organization hierarchy purpose)	PARTIAL	policy framework	17
258	12.29 Organization administration	12.29.4 Setup	Reason codes (Financial reasons: ledger, bank, customer, vendor, fixed asset)	PARTIAL	reason codes	11
259	12.29 Organization administration	12.29.4 Setup	Print management → Print management setup (per module/document: original/copy destinations, conditional settings by query)	PARTIAL	print management routing	46
260	12.29 Organization administration	12.29.4 Setup	Organization → Audit policies (link)	PARTIAL	audit policies	48
261	12.29 Organization administration	12.29.4 Setup	Organization → Workflow approval email templates	PARTIAL	email templates / SMTP outbound not explicitly named in 4 or 5	4
262	12.30 System administration	12.30.1 Common / Journals	Alerts → Alert parameters / Email templates / Email parameters (SMTP) / Outbound email status (E-mail sending status)	PARTIAL	email templates/SMTP outbound	4
263	12.30 System administration	12.30.1 Common / Journals	Email → Email distributor batch	PARTIAL	email templates/SMTP outbound	4
264	12.30 System administration	12.30.1 Common / Journals	Document handling (link)	PARTIAL	document management	5
265	12.30 System administration	12.30.2 Inquiries & Reports	License inquiries (Named user license counts, License usage by role)	PARTIAL	license usage reporting is not named; depends on licensing decision	3
266	12.30 System administration	12.30.2 Inquiries & Reports	SSRS: License report / Named user counts	PARTIAL	license usage reporting	3

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
267	12.30 System administration	12.30.3 Periodic	Email → Send email distributor batch / Retry failed	PARTIAL	email templates/SMTP outbound	4
268	12.30 System administration	12.30.3 Periodic	Licensing → Named user license count refresh	PARTIAL	license usage reporting	3
269	12.30 System administration	12.30.4 Setup	Alerts → Alert parameters (polling intervals, retention, drill-down links) / Email templates / Email parameters (SMTP, Exchange, mail sending mode: batch/client) / Email distributor	PARTIAL	email templates/SMTP outbound	4
270	12.30 System administration	12.30.4 Setup	Document management (link)	PARTIAL	document management	5
271	12.31 Home / role centers / workflow	12.31.1 Common / Journals	Cases (My cases)	PARTIAL	generic case management	45
272	12.31 Home / role centers / workflow	12.31.1 Common / Journals	Timesheets (Home → Timesheets, link)	PARTIAL	timesheets	41
273	12.31 Home / role centers / workflow	12.31.1 Common / Journals	Document handling on any record	PARTIAL	document management	5
274	12.31 Home / role centers / workflow	12.31.4 Setup	Workflow → Notification settings (email, in-app, digest) / Workflow email templates	PARTIAL	email templates/SMTP outbound	4
275	12.31 Home / role centers / workflow	12.31.4 Setup	Cases → Case setup (link)	PARTIAL	generic case management	45
276	12.33 Electronic reporting / Regulatory Configuration Service (D365)	12.33.1 Common / Journals	Electronic messages (Electronic messaging framework: message types, message items, processing actions, additional fields; used for VAT returns, SAF-T, UK MTD, IT e-invoice SDI, etc.)	PARTIAL	electronic messaging lifecycle (statutory submission with response tracking, e.g. UK MTD, DE/FR mandates) not named; e-invoicing in 35 covers Peppol/SDI/CFDI submissions	48
277	12.33 Electronic reporting / Regulatory Configuration Service (D365)	12.33.2 Inquiries & Reports	Electronic messages inquiry (status, responses, errors)	PARTIAL	electronic messaging	48
278	12.33 Electronic reporting / Regulatory Configuration Service (D365)	12.33.3 Periodic	Electronic messages → Generate / Send / Receive response (batch processing actions)	PARTIAL	electronic messaging	48
279	12.33 Electronic reporting / Regulatory Configuration Service (D365)	12.33.4 Setup	Electronic messages → Message types, Message item types, Message statuses, Processing actions (ER export/import, web service call, user input), Additional fields, Executable classes, Web service settings (endpoint, certificates), Message processing	PARTIAL	electronic messaging	48
280	12.34 Engineering change management (D365)	12.34.1 Common / Journals	Release product structure (release engineering product + BOM/route to legal entities)	PARTIAL	engineering organization vs operational legal entity split, and release product structure to legal entities with per-LE release policies, not named	9
281	12.34 Engineering change management (D365)	12.34.1 Common / Journals	Product readiness checks (per LE checklist before product is usable)	PARTIAL	product readiness	9
282	12.34 Engineering change management (D365)	12.34.1 Common / Journals	Product acceptance (accept released product in receiving LE)	PARTIAL	release product structure	9
283	12.34 Engineering change management (D365)	12.34.1 Common / Journals	Product lifecycle states (per version, controlling process enablement)	PARTIAL	lifecycle states	8
284	12.34 Engineering change management (D365)	12.34.1 Common / Journals	Engineering product categories (category assigns version nomenclature, attributes, readiness policy)	PARTIAL	engineering product categories with version nomenclature not named; category hierarchies are	9
285	12.34 Engineering change management (D365)	12.34.1 Common / Journals	Engineering → PLM integration (D365 Teamcenter/Windchill connectors via Dataverse)	PARTIAL	PLM connector not named among iPaaS reference connectors	35
286	12.34 Engineering change management (D365)	12.34.2 Inquiries & Reports	Release status inquiry (product × legal entity)	PARTIAL	release product structure	9
287	12.34 Engineering change management (D365)	12.34.2 Inquiries & Reports	Readiness check status inquiry	PARTIAL	product readiness	9
288	12.34 Engineering change management (D365)	12.34.2 Inquiries & Reports	SSRS: Product release report	PARTIAL	release product structure	9
289	12.34 Engineering change management (D365)	12.34.3 Periodic	Release product structures (batch release to LEs)	PARTIAL	release product structure	9
290	12.34 Engineering change management (D365)	12.34.3 Periodic	Readiness check evaluation (batch)	PARTIAL	product readiness	9
291	12.34 Engineering change management (D365)	12.34.3 Periodic	Engineering → Sync with PLM (batch)	PARTIAL	PLM connector	35
292	12.34 Engineering change management (D365)	12.34.4 Setup	Engineering organizations (which LE owns engineering data)	PARTIAL	release product structure	9
293	12.34 Engineering change management (D365)	12.34.4 Setup	Engineering product categories (version nomenclature, attributes, readiness policy, release policy)	PARTIAL	engineering product categories	9

#	Module	Table	Item	Status	Missing / proposal	Proposed wave
294	12.34 Engineering change management (D365)	12.34.4 Setup	Version nomenclature (number/letter/date segments)	PARTIAL	engineering product categories	9
295	12.34 Engineering change management (D365)	12.34.4 Setup	Product readiness policies / Readiness checks (per-LE checklists: tax, prices, coverage, warehouse setup)	PARTIAL	product readiness	9
296	12.34 Engineering change management (D365)	12.34.4 Setup	Product release policies (what to copy to receiving LE: BOM, route, prices, default order settings)	PARTIAL	release product structure	9
297	12.34 Engineering change management (D365)	12.34.4 Setup	Product lifecycle states (enabled business processes per state, default states)	PARTIAL	lifecycle states	8
298	12.34 Engineering change management (D365)	12.34.4 Setup	Engineering → PLM connector configuration	PARTIAL	PLM connector	35
299	12.35 Asset management (D365)	12.35.1 Common / Journals	Maintenance rounds (route of assets inspected together)	PARTIAL	maintenance rounds (multi-asset inspection routes) not named	44
300	12.35 Asset management (D365)	12.35.1 Common / Journals	Maintenance sequences (ordered job types on an asset)	PARTIAL	maintenance sequences not named	44
301	12.35 Asset management (D365)	12.35.1 Common / Journals	Work order → Fault registration (symptoms, causes, remedies)	PARTIAL	fault designer: symptom/area/type/cause/remedy taxonomy and fault analysis not named	44
302	12.35 Asset management (D365)	12.35.1 Common / Journals	Warranties (vendor warranty per asset, warranty check on work orders)	PARTIAL	asset warranties (vendor warranty terms, warranty check on work order) not named	44
303	12.35 Asset management (D365)	12.35.1 Common / Journals	Maintenance budgets (asset/functional location budget lines, budget calculation from plans)	PARTIAL	maintenance budgeting from plans not named	44
304	12.35 Asset management (D365)	12.35.2 Inquiries & Reports	Fault analysis (fault statistics by symptom/cause)	PARTIAL	fault designer	44
305	12.35 Asset management (D365)	12.35.2 Inquiries & Reports	Warranty inquiry	PARTIAL	warranties	44
306	12.35 Asset management (D365)	12.35.2 Inquiries & Reports	Budget vs actual (maintenance)	PARTIAL	maintenance budgets	44
307	12.35 Asset management (D365)	12.35.2 Inquiries & Reports	SSRS: Fault report	PARTIAL	fault designer	44
308	12.35 Asset management (D365)	12.35.3 Periodic	Schedule maintenance rounds / Maintenance sequences	PARTIAL	maintenance rounds	44
309	12.35 Asset management (D365)	12.35.3 Periodic	Maintenance budget calculation	PARTIAL	maintenance budgets	44
310	12.35 Asset management (D365)	12.35.4 Setup	Maintenance plans → Plan types, Plan line defaults, Maintenance rounds setup, Maintenance sequences setup	PARTIAL	maintenance rounds	44
311	12.35 Asset management (D365)	12.35.4 Setup	Fault → Fault symptoms, Fault areas, Fault types, Fault causes, Fault remedies, Fault designer (per asset type)	PARTIAL	fault designer	44
312	12.35 Asset management (D365)	12.35.4 Setup	Warranties → Warranty types, Vendor warranty setup	PARTIAL	warranties	44
313	12.35 Asset management (D365)	12.35.4 Setup	Budget → Maintenance budget parameters	PARTIAL	maintenance budgets	44
314	12.37 Subscription billing (D365)	12.37.1 Common / Journals	Deferral schedules (list) / details (revenue and expense deferrals, short-term/long-term split)	PARTIAL	expense deferrals (prepaid expense amortization from AP invoices) not named; revenue deferrals are	29
315	12.37 Subscription billing (D365)	12.37.3 Periodic	Expense deferral recognition	PARTIAL	expense deferrals	29
316	12.39 Vendor collaboration / customer self-service (D365)	12.39.1 Common / Journals	Vendor → Prospective vendor registration (onboarding wizard, questionnaires)	PARTIAL	vendor onboarding	20
317	12.39 Vendor collaboration / customer self-service (D365)	12.39.4 Setup	Vendor collaboration → Onboarding: Registration wizard configuration, Questionnaires	PARTIAL	vendor onboarding	20

12.99.5 Inventory totals

Status	Rows
COVERED	2472
PARTIAL	317
GAP	108
OUT-OF-SCOPE	102
SUPERSEDED	147
Total assessed rows	**3146**

Part 13 — Coverage Inventory: SAP ECC 6.0 / S/4HANA → Passage

This part is the completeness oracle for the SAP side of the parity goal (PRD §2 goal 1, §5 scope map). It inventories SAP ECC 6.0 and S/4HANA functionality module by module — by transaction code and by IMG (SPRO) node — and maps every row to one of the 50 Passage waves in roadmap/waves.json. Its purpose is to make "we missed something" a question that can be answered by grep rather than by fear. The companion inventory for Dynamics AX / D365 F&SCM is Part 12.

13.0 How to read this inventory

Each SAP application component gets one section with four tables:

1. **Master data & documents** — the create/change/display transactions and the documents they produce (`T-code | Name | Purpose | Passage status | Wave | Notes`).
2. **Reports & inquiries** — standard list, line-item and analysis reports (`T-code | Name | Passage status | Wave | Notes`).
3. **Periodic processing** — closing, batch and mass-processing transactions (`T-code | Name | Passage status | Wave | Notes`).
4. **Customizing (SPRO)** — the IMG nodes, their customizing tables and the key options Passage must offer an equivalent for (`IMG node / T-code | Table(s) | Key options | Passage status | Wave | Notes`).

Status legend (exactly one token per row, always in the "Passage status" column):

- `COVERED-<wave>` — the wave's `deliverable/proof` in waves.json clearly includes an equivalent. The wave column repeats the number.
- `PARTIAL-<wave>` — the wave covers the core but the Notes column names what is missing. Every PARTIAL row is repeated in the gap register (§13.99).
- `GAP` — no wave clearly covers the item. The wave column holds a *proposed* assignment (`propose <n>`); the gap register collects them for a roadmap decision.
- `OUT-OF-SCOPE` — excluded by a PRD non-goal (§2) or by the manufacturer focus; the reason is in Notes.
- `SUPERSEDED-<wave>` — Passage deliberately does it a different way; the Notes column names the Passage mechanism. A SUPERSEDED row is **not** a gap: the SAP transaction has no one-to-one twin because the underlying mechanism (e.g. the condition technique's separate rebate module, SAP GUI, ABAP enhancement spots) is replaced by something structurally different.

Conventions:

- T-codes are given in their ECC 6.0 form; where S/4HANA replaced a transaction with a Fiori app or a new transaction, both are named (e.g. `FD32 → UKM_BP`, `VBO1 → WCOCO`). Fiori apps are cited by name, not by app number.
- "Wave" refers to the `n` in waves.json. Phase-exit waves (5, 10, 15, 20, 25, 30, 35, 40, 45, 50) are the sign-off points at which a module's rows must be demonstrably closed.
- IMG rows name the customizing table where it is well known (T001, T001B, T030, T156, T161, T685, TVAK…). Passage stores all customizing as metadata-registry configuration (wave 3) scoped per tenant / legal entity (wave 6); the "Key options" column lists the switches Passage must expose, not every SAP field.
- Rows are intentionally granular (a create/change/display triple is one row; a report family such as `S_ALR_87012*` is one row per distinct report, not one row per variant).
- Where SAP offers both an ECC transaction and a New GL / S/4 successor, the successor is inventoried and the predecessor noted, so classic GL (`GLT0`-based) transactions appear once under the FI-GL section.

A per-section roll-up of the same counts is in §13.98 and the gap register (every GAP and PARTIAL row with its current or proposed wave) is in §13.99. Status totals across all four tables of every section (computed from the rows, excluding §13.98–13.99):

Status	Rows	Share
COVERED	2049	74.6%
PARTIAL	310	11.3%
GAP	91	3.3%
OUT-OF-SCOPE	22	0.8%
SUPERSEDED	274	10.0%
Total rows	**2746**	100%

13.1 FI-GL — General Ledger (classic GL, New GL, ledgers, ACDOCA)

SAP context: FI-GL in ECC 6.0 runs either classic GL (`GLT0` totals, special-purpose ledgers) or New GL (`FAGLFLEXT`, document splitting, parallel ledgers). S/4HANA collapses FI and CO totals into the universal journal `ACDOCA` (one line item table; CO objects and ledgers as columns). Passage's wave 11 universal journal is built on the S/4 model: one append-only line-item store, balances as projections, dimensions as columns, parallel ledgers via ledger groups (wave 12).

13.1.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
FS00	G/L account master (centrally)	Create/change/display chart-of-accounts and company-code segment of a G/L account	COVERED-11	11	Account entity with per-legal-entity attributes (currency, open-item flag, posting block, tax category, field status)
FSP0	G/L account in chart of accounts	Chart segment only	COVERED-11	11	Same entity; chart-level vs entity-level attributes split in metadata
FSS0	G/L account in company code	Company-code segment only	COVERED-11	11	Per-legal-entity release of an account
FS15 / FS16	Copy G/L accounts (send/receive)	Copy company-code segments between company codes	COVERED-11	11	"Release account to legal entity" bulk action
FSM1 / FSM2 / FSM3	Sample accounts	Templates for company-code segments	SUPERSEDED-11	11	Account templates in the journal workspace; no separate sample-account object
KDH1 / KDH2 / KDH3	Account groups (G/L account groups hierarchy)	Hierarchies for FSV/reporting	COVERED-15	15	Account structures and report trees

T-code	Name	Purpose	Passage status	Wave	Notes
FB50	Enter G/L account document	Single-screen G/L posting (Enjoy)	COVERED-11	11	Manual journal; idempotent Post API
FB50L	Enter G/L account document for ledger group	Ledger-specific manual posting	COVERED-12	12	Journal with ledger-group scope
F-02	Enter G/L account posting (classic)	Posting-key based general posting	SUPERSEDED-11	11	Posting keys replaced by debit/credit amount sign on lines
FB01	Post document (general)	Any document type, any account type	COVERED-11	11	Single Post API for all subledger-aware lines
FV50 / FBV0 / FBV2 / FBV3	Park / post / change / display parked G/L document	Draft documents for later posting or approval	COVERED-11	11	Draft status + approval workflow (Temporal)
FBV4 / FBV5 / FBV6	Change parked header / display changes / reject	Parked document administration	COVERED-11	11	Draft audit from CDC (wave 5)
FB02 / FB03 / FB04	Change / display / display changes to document	Document maintenance (header text, reference, assignment only)	COVERED-11	11	Post-posting edit limited to non-financial fields; audit log
FB08 / F.80	Reverse document / mass reversal	Negating entry linked to the original	COVERED-11	11	Reversal API creates linked negating entry; mass reversal via batch job (wave 4)
FBRA	Reset cleared items	Undo clearing before reversal	COVERED-19	19	Settlement reversal in AP/AR; GL open-item accounts see §13.1 note on F-03
FBD1 / FBD2 / FBD3 / FBD4	Recurring entry document create/change/display/changes	Recurring journals with run schedule	COVERED-11	11	Recurring journals with schedule; execution via wave 4 batch
F.14 / F.15	Execute recurring entries / list recurring entries	Batch creation from recurring templates	COVERED-11	11	Scheduled recurring-journal job
FBS1	Enter accrual/deferral document	Reversing document with reversal date	COVERED-14	14	Accrual schemes with automatic reversal
F.81	Reverse accrual/deferral documents	Mass reversal of FBS1 documents	COVERED-14	14	Automatic reversal on schedule
FBM1 / FBM2 / FBM3	Sample documents	Non-posting template documents	SUPERSEDED-11	11	Journal templates
F-03 / F-04	Clear G/L account / post with clearing	Open-item clearing on OI-managed G/L accounts	PARTIAL-11	11	Open-item management specified only for AP/AR (waves 19/30) and GR/IR (wave 18); generic OI-managed G/L accounts (clearing accounts, bank sub-accounts) need the same clearing engine exposed on GL
F-05	Post foreign currency valuation (manual)	Manual FX adjustment posting	COVERED-12	12	Manual journal in reporting currency
F-65	Preliminary posting (park)	Classic parking	COVERED-11	11	Draft journals
FB41	Post tax payable	Transfer of tax balances	COVERED-13	13	Tax settlement job posts to tax payable
OB08	Enter exchange rates (also master data)	Rates by rate type, date, pair	COVERED-12	12	Exchange rate service with provider adapters and audit
FBCJ	Cash journal	Petty cash with receipts/expenses and GL posting	GAP	propose 36	No wave mentions a petty-cash/cash-journal document; small but common on plant floors — add to wave 36 (cash & bank) as "cash journal with receipts, business transactions and daily closing"
FBCJC0 / FBCJC1 / FBCJC2 / FBCJC3	Cash journal setup, number ranges, business transactions, print	Cash journal customizing	GAP	propose 36	Follows FBCJ
FAGLL03 (document)	Ledger-specific line item entry context	See reports	COVERED-12	12	—
GB01 / GB02	Enter/post G/L documents with document splitting characteristics	Classic FI-SL documents	SUPERSEDED-12	12	Special-purpose ledger replaced by parallel ledgers + dimensions
OB_GLACC11 / OB_GLACC12 / OB_GLACC13	Mass maintenance of G/L accounts (chart / company code / names)	Mass change	COVERED-5	5	Grid mass edit + Excel import/export on any entity
FAGL_ACTIVATE_OP	Activate open-item management on G/L accounts	Conversion utility	PARTIAL-11	11	Depends on the GL OI gap above
FAGL_SWITCH_TO_OPEN_ITEM	Switch to OI management (S/4)	Same	PARTIAL-11	11	Same
GCAC	Ledger comparison	Compare two ledgers	COVERED-12	12	Per-ledger trial balances with diff
FAGLB03 (document view)	Ledger balance display → line items	See reports	COVERED-15	15	—
FINS_ACDOC_CUST	Universal journal customizing overview (S/4)	ACDOCA extensibility	SUPERSEDED-3	3	Journal-line extensions via the metadata kernel (customer fields on JournalLine)
FAGL_MIG_*	New GL migration cockpit	Classic → New GL migration	SUPERSEDED-49	49	Migration factory loads GL balances by dimension directly
ACC_DOCUMENT (BAPI_ACC_DOCUMENT_POST)	Accounting document BAPI	Programmatic posting	COVERED-11	11	Post API (REST/gRPC) with idempotency keys

13.1.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
FBL3N	G/L account line items (classic)	COVERED-11	11	Journal line inquiry with saved views, grid, export
FAGLL03	G/L account line items (New GL, ledger-aware)	COVERED-12	12	Ledger filter on the same inquiry
FAGLL03H	G/L line item browser (HANA)	COVERED-46	46	Lakehouse-backed line-item browser for history; OLTP inquiry for recent
FS10N / FAGLB03	G/L account balance display (classic / New GL)	COVERED-11	11	Balance projection by account × dimension × period with drill to lines
FS10NA	Balance display with drill-down	COVERED-15	15	Statement → balance → lines drill
F.01	Financial statements (RFBILA00)	COVERED-15	15	Balance sheet / P&L from financial statement version equivalent
S_ALR_87012284	Financial statements (FSV, with comparison)	COVERED-15	15	Comparative columns in report designer
S_ALR_87012277	G/L account balances	COVERED-15	15	Trial balance
S_ALR_87012301	Totals and balances	COVERED-15	15	Trial balance by period
S_ALR_87012282	G/L line items, list for printing	COVERED-11	11	Printable line-item list

T-code	Name	Passage status	Wave	Notes
S_ALR_87012287	Document journal	COVERED-15	15	Audit trail / document journal
S_ALR_87012289	Compact document journal	COVERED-15	15	Same, compact layout
S_ALR_87012291	Line item journal	COVERED-15	15	Same
S_ALR_87012293	Display of changed documents	COVERED-5	5	Audit log viewer
S_ALR_87012308	Display changes to G/L accounts	COVERED-5	5	Audit log on Account entity
S_ALR_87012326	Chart of accounts list	COVERED-11	11	Account list export
S_ALR_87012328	G/L account list	COVERED-11	11	Same
S_ALR_87012332	G/L account statements	COVERED-15	15	Account statement report
S_ALR_87012341 / 87012342	Invoice numbers allocated twice / gaps in document number assignment	COVERED-5	5	Number-sequence gap report (gap-free mode audit)
S_ALR_87012357 / 87012356	Advance return for tax on sales/purchases / sales tax report	COVERED-13	13	VAT return extracts
S_ALR_87012382	Cash flow (direct method, from FSV)	COVERED-15	15	Indirect cash flow standard; direct via report designer
S_PL0_86000028 / S_PL0_86000029	Financial statement: actual/actual comparison, plan/actual	COVERED-15	15	Comparative statements; plan column after wave 38
S_PL0_86000030	G/L account balances (New GL)	COVERED-12	12	Ledger-aware trial balance
S_PL0_86000031	Transaction figures: account balance	COVERED-11	11	—
S_E38_98000088	Profit center group: plan/actual/variance	COVERED-38	38	Dimension = profit center; plan from budgeting
S_AC0_52000887 / 52000888	Receivables/payables by profit center (New GL)	COVERED-15	15	Dimension-level subledger report
FAGLF03	Reconciliation ledger vs. documents (New GL)	COVERED-16	16	Daily automated subledger-to-journal reconciliation check
F.03	Financial accounting comparative analysis (SAPF190)	COVERED-16	16	Same reconciliation job
GR55 / GRR1 / GRR2 / GRR3	Report Painter / Report Writer report groups and reports	SUPERSEDED-15	15	Row/column report designer (Management-Reporter model) replaces Report Painter; wave 46 semantic layer for ad-hoc
FGI0 / FGI1 / FGI3	Drill-down reporting (New GL): execute / create report / display	SUPERSEDED-15	15	Report designer + drill-down
FGI4 / FGI5 / FGI6	Drill-down report forms	SUPERSEDED-15	15	Same
FAGL_SCHEDULE_LOG (FAGLCOFITRACEADMIN)	Trace for CO-FI real-time integration	SUPERSEDED-11	11	No CO→FI reconciliation needed: single journal
S_ALR_87012309	Cash book / cash journal report	GAP	propose 36	Follows FBCJ gap
FAGLGVTR (log)	Balance carry-forward log	COVERED-14	14	Year-end job log
Fiori: Trial Balance	Trial balance (ledger, dimensions, comparison)	COVERED-15	15	—
Fiori: Display Line Items in General Ledger	S/4 line-item app	COVERED-11	11	—
Fiori: Display Financial Statement	S/4 statements app	COVERED-15	15	—
Fiori: Balance Sheet/Income Statement — Multidimensional	Dimension-pivot statements	COVERED-46	46	Semantic layer pivot
Fiori: Manage Journal Entries	Search/display/reverse journal entries	COVERED-11	11	—
Fiori: Verify General Journal Entries (approval)	Workflow-based journal approval	COVERED-11	11	Approval workflow on journals
Fiori: Audit Journal	Chronological audit journal	COVERED-15	15	Audit trail report
Fiori: Journal Entry Analyzer	Analyze journal entries by dimension	COVERED-46	46	—

13.1.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
OB52 (operational use)	Open/close posting periods	COVERED-14	14	Period control per ledger × module × user group
FAGL_FC_VAL / F.05 (classic)	Foreign currency valuation (GL, AP, AR open items)	COVERED-12	12	Revaluation job with next-period reversal
FAGL_FC_TRANS	Currency translation (balances into group currency)	COVERED-12	12	Reporting-currency translation
FAGLGVTR / F.16 (classic)	Balance carry-forward (New GL / classic)	COVERED-14	14	Year-end close: P&L to retained earnings, BS carry-forward per dimension
F.07	Balance carry-forward AP/AR	COVERED-14	14	Subledger carry-forward in the same job
F.13 / F13E	Automatic clearing (with/without specification of clearing currency)	PARTIAL-11	11	Automatic clearing specified for GR/IR (wave 18) and AP/AR settlement; generic rule-based GL auto-clearing is part of the GL OI gap
FAGLF101 / F101	Regroup receivables/payables (sorted list, reclassification by remaining term)	GAP	propose 14	Balance-sheet reclassification (short/long-term, debit-balance vendors, credit-balance customers) is a standard close step not named in wave 14's checklist/allocations — add as a close task type
F.19	GR/IR analysis / regrouping	COVERED-18	18	GR/IR reconciliation report and reclass
F.50 / F.52	G/L interest calculation (balance / account interest)	GAP	propose 30	Interest on GL balances (e.g. intercompany loans) is not named; wave 30 covers AR interest notes only — extend the interest engine to GL/intercompany accounts
ACACACT / ACACTREE01 / ACACPSITEMS	Accrual engine: periodic run / accrual objects / display postings	COVERED-14	14	Accrual schemes; recurring accrual objects
ACACADJ	Accrual engine: adjust accrual	COVERED-14	14	—
FAGL_CL_REGROUP / FAGL_CL_MIGRATION	Closing: regrouping (S/4)	GAP	propose 14	Same as FAGLF101
F.97 / F.98 / F.99	Balance confirmations for G/L / AP / AR	GAP	propose 19	Balance confirmation letters (vendor/customer/GL) not named in waves 19 or 30 — add to wave 30 customer statements + wave 19 vendor confirmations
FB1S / FB1D / FB1K	Clear G/L / customer / vendor (classic clearing)	COVERED-19	19	Settlement; GL part in GL OI gap

T-code	Name	Passage status	Wave	Notes
GCAC	Ledger comparison (closing check)	COVERED-12	12	—
FINS_CO_RECON (FINS_RECON_RC0..RC3)	S/4 reconciliation reports (ACDOCA vs. BSEG/COEP)	SUPERSEDED-11	11	No dual tables to reconcile; balance projection vs journal check is the equivalent
FINSC_LEDGER (admin run)	Ledger settings activation	COVERED-12	12	Ledger group configuration
CLOCO / CLOCOC / CLOCOS	Financial Closing Cockpit: task list / templates / monitor	COVERED-14	14	Close checklist with templates, dependencies, assignees, status board
SCMA	Schedule Manager (classic)	SUPERSEDED-14	14	Close checklist + wave 4 batch scheduler
Fiori: Manage Closing Tasks / Advanced Financial Closing	S/4 close orchestration	COVERED-14	14	Close checklist across entities
SAFT / Fiori: Run Statutory Reports (ACR)	Advanced Compliance Reporting (SAF-T, VAT returns)	COVERED-13	13	SAF-T extract validated to OECD XSD; VAT returns; statutory packs in wave 48
FAGL_SCHEDULE (FAGL_PLAN_*)	Plan data entry in New GL (GP12N etc.)	COVERED-38	38	Budget register against ledger dimensions
GP12N / GP12NA	Plan data entry / display (New GL)	COVERED-38	38	—
FAGL_CO_PLAN	Transfer CO plan data to New GL	SUPERSEDED-38	38	One planning store
S_ALR_87099918 (RFUMSV00 run)	Advance tax return run	COVERED-13	13	Tax settlement job

13.1.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OX02 — Company code	T001	Country, currency, language, fiscal year variant, chart of accounts	COVERED-6	6	LegalEntity entity
OBY6 — Company code global parameters	T001	FYV, posting period variant, field status variant, negative postings, business area FS, tax base net/gross, cash management active	COVERED-11	11	LegalEntity accounting settings
OB13 / OB62 — Chart of accounts / assign to company code	T004, T001	Operational / group / country chart; length; integration with CO	COVERED-11	11	Account structure per ledger; alternative account numbers via dimension mapping
OBD4 — Account groups	T077S	Number ranges, field status per group	COVERED-11	11	Account groups with field visibility rules (metadata field attributes)
OB53 — Retained earnings account	T030 (BIL)	P&L statement account type	COVERED-14	14	Year-end close settings
OB29 — Fiscal year variants	T009, T009B	Calendar/non-calendar, special periods, shortened FY	COVERED-6	6	Fiscal calendars: 4-4-5, 13-period, calendar; special periods needed — covered by period types
OB37 — Assign fiscal year variant to company code	T001	—	COVERED-6	6	Calendar per legal entity / ledger
OB52 — Posting period variants: open/close periods	T001B	Account type ranges, from/to period, authorization group (2nd interval)	COVERED-14	14	Period control per ledger × module × user group
OBBO / OBBP — Posting period variant define / assign	T010O, T001	—	COVERED-14	14	—
OBA7 — Document types	T003	Number range, account types allowed, reversal doc type, net posting, authorization group, reference/doc header text required	COVERED-11	11	Journal/document types with number sequence and allowed line types
FBN1 / OBH2 — Number ranges for accounting documents / copy	NRIV	Year-dependent, external, interval	COVERED-5	5	Number sequence service, gap-free mode for fiscal docs
OB41 — Posting keys	TBSL	D/C, account type, field status, sales-related, special G/L	SUPERSEDED-11	11	No posting keys: line = account + signed amount + line type
OBC4 — Field status variants / groups	T004F, T004V	Required/optional/suppressed per field group	COVERED-3	3	Metadata field attributes per account group / line type
OBC5 — Assign company code to field status variant	T001	—	COVERED-11	11	—
OB28 / GGB0 — Validations (FI)	GB90*, T001Q	Prerequisite/check/message at header, line, document level	PARTIAL-11	11	Wave 11 has dimension validation rules; a general user-defined validation rule language (Boolean rules over header/line fields with message classes) is only implied by wave 3 "validation" — scope it explicitly
OBBH / GGB1 — Substitutions (FI)	GB91*, T001R	Field substitution with exits at line/document	PARTIAL-11	11	Default dimensions on master data exist; rule-based substitution (derive dimension from another field) needs an explicit rule engine on the Post API
OBBZ / OBBW — Substitution/validation for cost of sales accounting	—	Functional area derivation	PARTIAL-11	11	Functional area as a dimension with derivation rules (see above)
OKKP (FI side) — Controlling area assignment	TKA01, TKA02	Company code ↔ controlling area, currency type	SUPERSEDED-11	11	No separate controlling area: dimensions on the journal; wave 38 management reporting
OB22 / FINSC_LEDGER — Parallel currencies per company code / ledger settings (S/4)	T001A, FINSC_LEDGER	Currency types 10/30/40, valuation view, ledger type (standard/extension), accounting principle, FYV per ledger	COVERED-12	12	Three-currency lines; ledger groups; extension ledgers (adjustment-only) need confirming — treat as covered by ledger groups
FAGL_ACTIVATION / FAGL_ACTIVEC — Activate New GL	FAGL_ACTIVEC	—	SUPERSEDED-11	11	Single journal always on
Define ledgers for GL accounting (FAGL_LEDGER)	T881, T882G	Leading ledger, representative ledger	COVERED-12	12	—
Define ledger groups (FAGL_LEDGER_GRP)	T8G*, FAGL_TLDGRP	Group of ledgers for postings	COVERED-12	12	—
Define accounting principles / assign to ledger groups	TACC_PRINCIPLE, TACC_TRGT_LDGR	IFRS/local GAAP/tax	COVERED-12	12	—

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Document splitting: classify G/L accounts, document types, splitting method, zero-balance, constants (FAGL_SPLIT*)	T8G01…T8G40	Active/passive split, characteristics (PC, segment, business area), zero-balance clearing account, inheritance	PARTIAL-11	11	Wave 11 requires every line to carry dimensions, so dimension-complete documents are native; but *inheritance to balance-sheet lines* (AP/AR/tax lines split by the expense lines' profit centers) and zero-balance clearing per dimension are not named — needed for balance sheet by profit center/segment
Define segment / derive segment from profit center	FAGL_SEGM, CEPC	—	COVERED-11	11	Segment as a dimension; derivation default from profit center dimension
OB08 / OBBS / OBBT — Exchange rate types, ratios, rate maintenance	TCURR, TCURV, TCURF, TCURX	Rate types (M, EURX, P), inversion, triangulation via reference currency, decimals per currency	COVERED-12	12	Rate types, triangulation, decimals from ISO 4217
OB07 — Exchange rate types	TCURV	—	COVERED-12	12	—
OBA1 — Accounts for exchange rate differences / FX valuation	T030H	Realized/unrealized gain-loss, valuation and translation accounts per G/L account	COVERED-12	12	Posting profile for realized/unrealized
OB59 / OBBD — Valuation methods and valuation areas (FC valuation)	T044A, T044G, FAGL_VAL_AREA	Lowest value principle, always valuate, reset, exchange rate type per method; valuation area ↔ accounting principle	COVERED-12	12	Revaluation method per ledger / accounting principle
OBXR / OBYR — Special G/L indicators (customers / vendors)	T074	Down payments, bills of exchange, guarantees, noted items	PARTIAL-19	19	Prepayments/down payments covered (waves 19/30); bills of exchange and noted items not named — see FI-AP/AR sections
OBY7 / OBY8 / OBY9 — Copy chart of accounts / company code / delete	—	—	COVERED-6	6	Entity templates ("copy legal entity") in enterprise structure
OBYA — Clearing accounts for cross-company-code transactions	T001U	Due-to/due-from per company-code pair	COVERED-40	40	Intercompany accounting due-to/due-from
OBYC (FI-side view) — Automatic postings (MM/SD integration)	T030	Transaction keys BSX, GBB, WRX, PRD, KDM, UMB, FRL, BSV, AUM, VAX/VAY	COVERED-16	16	Inventory posting profiles (item model group × site × account)
OB40 — Tax accounts	T030K	Per tax transaction key (MWS, VST, NAV…)	COVERED-13	13	Tax posting profile
OBB8 / OBB9 — Terms of payment / installment terms	T052, T052R	Day limits, baseline date, discount days/percent, installment split	PARTIAL-7	7	Payment terms covered; installment terms (split one invoice into several due dates) not named — add to wave 7 terms model
OBA3 / OBA4 — Tolerance groups (customers/vendors) / (employees, posting amount limits)	T043G, T043T	Max amount per document, per line, cash discount, payment difference tolerances	COVERED-19	19	Amount limits via OPA policy + settlement tolerances (wave 19)
OBBE / OBXL — Reason codes for payment differences / accounts	T053R, T030	Charge off, residual, disputes	COVERED-30	30	Reason codes on cash application
OBD2 — Customer account groups / OBD3 — Vendor account groups	T077D, T077K	Number range, field status, one-time account	COVERED-7	7	Party role groups, field attributes; one-time vendors/customers — see §13.2
OBXU / OBXI — Cash discount received/granted accounts	T030	Net/gross method	COVERED-19	19	—
OBXK / OBXB — Bank charges / bank sub-accounts	T030	—	COVERED-36	36	Bank posting profiles
OBYP — GR/IR clearing accounts	T030	WRX	COVERED-18	18	—
OBXM / OBXO / OBXW — Retained earnings-related: acquisition/retirement, rounding, over/under payments	T030	—	COVERED-19	19	Rounding and payment difference accounts
OBA0 — Tolerance groups for G/L accounts (clearing)	T043S	—	PARTIAL-11	11	GL OI gap
OB32 — Document change rules	TBAER	Which fields editable after posting	COVERED-11	11	Field-level edit rules post-posting
OB46 — Interest indicators / OB82 / OBAA	T056*	Balance interest, item interest, calendar type	GAP	propose 30	Interest engine (see F.52)
OBD5 / Define sort keys (allocation field)	TZUN	Assignment field fill rule	SUPERSEDED-11	11	Sort/assignment via configurable reference fields and saved views
OBR1 / OBR2 / OBR3 — Reset transaction data / delete master data / delete customer-vendor	—	Test-client tooling	SUPERSEDED-1	1	Environment reset through pipeline/tenant provisioning
FINSC_FIELDSTATUS / Define field status for ledger (S/4)	—	—	COVERED-3	3	Metadata
OB74 — Additional rules for automatic clearing	TF123	Criteria fields per account type/range	PARTIAL-11	11	GL OI gap (AP/AR criteria covered by settlement rules)
OBB1 / OBB2 — Define accounts for bank interest / tax on cash discount	T030	—	COVERED-13	13	—
SNUM / SNRO (FI number range objects)	NRIV	RF_BELEG, etc.	COVERED-5	5	Number sequences
OBBG — Assign country to tax calculation procedure	T005	TAXUSJ, TAXUSX, TAXEUR…	COVERED-13	13	Jurisdiction hierarchy per country
OBH1 — Copy number range intervals to fiscal year	NRIV	—	COVERED-5	5	Year-dependent sequences
Define Reasons for Reversal (OB_REVERSE / T041C)	T041C	Negative posting allowed, alternative posting date	COVERED-11	11	Reversal reason codes
FAGL_CL_RGRP / Define sort method & adjustment accounts for regrouping	T043/ T030	Remaining term intervals	GAP	propose 14	FAGLF101 gap
Define Posting Keys for Clearing (OBXH etc.)	T030	—	SUPERSEDED-11	11	No posting keys
OBD4 → "Define Screen Layout per Company Code"	T001	—	COVERED-3	3	—

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Cost of sales accounting: functional areas (OKBD), activate (FAGL_FUNC_AREA)	TFKB, T001	—	PARTIAL-11	11	Functional area dimension + derivation rules (substitution gap)
Define Business Areas (OX03) / Consolidation business areas	TGSB, TGSBK	—	COVERED-6	6	Dimension values
Profit center: standard hierarchy, dummy PC (KCH1/KE59 — see CO-PA)	CEPC, SETHEADER	—	COVERED-6	6	Profit center as dimension with hierarchy purposes

13.2 FI-AP — Accounts Payable

SAP context: vendor master (LFA1/LFB1/LFM1, replaced by Business Partner with FLVN00/FLVN01 roles in S/4), invoices and credit memos (FB60/FB65 or MIRO for PO-based), open items (BSIK/BSAK), payment program F110, dunning (rare for AP), down payments, withholding tax. Passage maps vendor master to wave 7 (party model), non-PO invoices to wave 18 (capture) and open items/payments to wave 19.

13.2.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
XK01 / XK02 / XK03 (→ BP)	Vendor master create/change/display centrally	General, company-code and purchasing data	COVERED-7	7	Party + Vendor role + per-legal-entity and per-purchasing-org data
FK01 / FK02 / FK03	Vendor master (accounting)	Company-code segment only	COVERED-7	7	Vendor role, legal-entity scope
MK01 / MK02 / MK03	Vendor master (purchasing)	Purchasing-org segment	COVERED-7	7	Vendor role, purchasing scope; see MM-PUR
XK05 / XK06 / FK05 / FK06	Block / mark for deletion	Posting/payment/purchasing blocks	COVERED-7	7	Role status and block flags with approval
XK07	Change account group	Reassign vendor group	COVERED-7	7	Group change under governance workflow
XK99 / MASS (LFA1)	Mass maintenance vendor	Mass change	COVERED-5	5	Grid mass edit
BP (role FLVN00/FLVN01)	Business partner — vendor roles (S/4)	Unified BP	COVERED-7	7	Native model
CVI_FS_CHECK_CUSTOMIZING / MDS_LOAD_COCKPIT	Customer-vendor integration synchronization	ECC → S/4 BP sync	SUPERSEDED-7	7	One party model, no sync
One-time vendor (account group CPD)	One-time accounts with address on document	Ad-hoc payee	PARTIAL-7	7	Not named in wave 7; add "one-time party" role with document-level address capture
FK08 / FK09	Confirm vendor master changes (dual control)	Sensitive-field dual control	COVERED-7	7	Change approval workflow on master data (bank details, payment terms)
FB60	Enter vendor invoice (non-PO)	Single-screen AP invoice	COVERED-18	18	Manual invoice capture without PO
FB65	Enter vendor credit memo	—	COVERED-19	19	Vendor credit notes
F-43 / F-41	Enter vendor invoice / credit memo (classic)	Posting-key based	SUPERSEDED-18	18	Single invoice form
FV60 / FV65	Park vendor invoice / credit memo	Draft for approval	COVERED-18	18	Invoice draft + approval workflow
FBV0	Post parked document	—	COVERED-18	18	—
F-48	Vendor down payment	Special G/L down payment posting	COVERED-19	19	Prepayments
F-47	Down payment request	Noted item for F110	COVERED-19	19	Prepayment request on PO/invoice workflow
F-54	Clear vendor down payment	Transfer down payment against invoice	COVERED-19	19	Prepayment application on settlement
F-53 / F-58	Post outgoing payment / with printed form	Manual payment with clearing	COVERED-19	19	Manual payment + settlement; check print
F-44	Clear vendor	Account clearing without payment	COVERED-19	19	Settlement without payment (credit vs invoice)
F-51	Post with clearing (transfer)	Transfer with clearing	COVERED-19	19	—
F-40 / F-36 (AP side: FBW6)	Bill of exchange payment (AP)	Bills of exchange payable	GAP	propose 36	Bills of exchange (AP and AR) are not in any wave; relevant in IT/ES/FR/JP/BR; propose as a wave 36 (bank) option or defer to a localization v1.x pack under wave 48 — needs a decision
FB10	Invoice/credit memo fast entry	Multi-document fast entry	COVERED-18	18	Grid entry / import
FBL1N (as document finder)	—	—	COVERED-19	19	—
FCHN / FCH5 / FCH6	Check register / create manual check / change check info	Check management	COVERED-36	36	Check management (void/reprint), check register
FCH7 / FCH8 / FCH9	Reprint / cancel payment + check / void check	—	COVERED-36	36	—
FCHR / FCHT	Online cashed checks / assign checks to payments	—	COVERED-36	36	Bank statement cashed-check matching
FCHU / FCHX	Create reference for check / check extract (positive pay)	Positive pay file	COVERED-19	19	Positive pay export
F110 (document side: payment documents)	—	—	COVERED-19	19	—
FBZ5	Print check for payment document	—	COVERED-19	19	Check printing
FB03 (vendor doc)	Display document	—	COVERED-19	19	—
FBD1 (vendor recurring)	Recurring vendor invoice	—	COVERED-11	11	Recurring journal with vendor lines

T-code	Name	Purpose	Passage status	Wave	Notes
J1INCHLN / J1INBANK / J1INCERT (IN WHT)	Withholding tax challan / bank challan / certificate (India)	Country-specific WHT remittance	PARTIAL-13	13	Withholding engine + 26AS hooks named; Indian TDS challan/certificate flows belong to the IN country pack (wave 48) — partial until pack ships
S_P00_07000134 / RFIDYYWT	Generic withholding tax reporting (1099, DME)	Statutory WHT reporting	COVERED-13	13	1099 reporting hooks (year-end forms in wave 43 share structures)
FIBLFFP / F111	Payment request / payment program for payment requests	Treasury payment requests outside AP	PARTIAL-36	36	Payment requests from treasury (not vendor items) not named; add to wave 36 in-house banking
VIM (OpenText) / Fiori: Manage Supplier Invoices / Upload Supplier Invoices	AP automation	OCR capture, approval	COVERED-18	18	OCR/AP automation adapter
Ariba Network invoice	Supplier-submitted invoice	—	COVERED-20	20	Supplier portal invoice submission

13.2.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
FBL1N / FBL1H	Vendor line items / line item browser	COVERED-19	19	Vendor open/cleared item inquiry
FK10N / FK10NA	Vendor balance display	COVERED-19	19	Vendor balance with drill
S_ALR_87012077	Vendor information system	COVERED-46	46	Vendor analytics
S_ALR_87012078	Due date analysis for open items	COVERED-19	19	Aging
S_ALR_87012082	Vendor balances in local currency	COVERED-19	19	—
S_ALR_87012083	List of vendor open items for printing	COVERED-19	19	—
S_ALR_87012084	Open items — vendor due date forecast	COVERED-36	36	Cash forecast input
S_ALR_87012085	Vendor payment history	COVERED-19	19	—
S_ALR_87012086	Vendor list	COVERED-7	7	—
S_ALR_87012087	Address list	COVERED-7	7	—
S_ALR_87012089	Display changes to vendors	COVERED-5	5	Audit viewer
S_ALR_87012090	Display/confirm critical vendor changes	COVERED-7	7	Dual-control queue
S_ALR_87012093	Vendor business (purchasing volume)	COVERED-46	46	—
S_ALR_87012103	List of vendor line items	COVERED-19	19	—
S_ALR_87012104	Vendor open items (OI list)	COVERED-19	19	—
S_ALR_87012105	List of down payments open at key date	COVERED-19	19	Prepayment report
S_P99_41000099	Payment list (F110)	COVERED-19	19	Payment run output
S_ALR_87012119	Cashed checks per bank account	COVERED-36	36	—
S_P99_41000101	Check register	COVERED-36	36	—
S_ALR_87012176 (AP aging via RFKOPR00)	Vendor open item aging	COVERED-19	19	—
S_ALR_87012116	Bill of exchange list	GAP	propose 36	Follows bills gap
FBL1N variant: withholding tax	Withholding tax line items	COVERED-13	13	—
S_ALR_87012111 / S_P00_07000134	Withholding tax report / 1099-MISC	COVERED-13	13	—
Fiori: Supplier Balances / Display Supplier List / Manage Supplier Line Items	S/4 apps	COVERED-19	19	—
Fiori: Accounts Payable Overview (analytical)	AP KPI page	COVERED-46	46	Embedded dashboards
Fiori: Cash Discount Utilization / Days Payable Outstanding	AP KPIs	COVERED-46	46	Governed metrics

13.2.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
F110	Automatic payment program (proposal, edit, schedule, print)	COVERED-19	19	Payment proposal → approval → run → media; idempotent
F110S	Payment program scheduled (batch variant)	COVERED-19	19	Scheduled via wave 4 jobs
F111	Payment program for payment requests	PARTIAL-36	36	See payment requests
FBPM / FBPM1	Payment medium workbench (PMW) run / DME engine	COVERED-19	19	ISO 20022 pain.001, NACHA, check files
DMEE / DMEE1	DME engine format tree maintenance	SUPERSEDED-19	19	Format adapters (ISO 20022, NACHA) + wave 48 electronic reporting designer for new formats
FDTA	DME administration (download files)	COVERED-19	19	Payment file vault with audit (keys never leave Vault — wave 36)
SAPF110V (F110 proposal log)	Payment proposal log	COVERED-19	19	—
FBZ0	Edit payment proposal (display/change)	COVERED-19	19	—
F150 (AP dunning)	Dunning vendors (debit balances)	PARTIAL-30	30	Dunning covers customers; dunning vendors with debit balances is an edge case — reuse engine
FAGL_FC_VAL (AP)	FX valuation of vendor open items	COVERED-12	12	—
FAGLF101 (AP)	Regroup vendors with debit balances	GAP	propose 14	Reclassification gap
F.98	Vendor balance confirmation	GAP	propose 19	Balance confirmations gap
MRRL (ERS — see MM-IV)	Evaluated receipt settlement	COVERED-18	18	—
J1INCHLN run / withholding tax remittance	WHT payment to authority	COVERED-13	13	Tax settlement for withholding
S_P00_07000134 (year-end 1099 run)	1099 generation	COVERED-13	13	—

T-code	Name	Passage status	Wave	Notes
F.07 (AP)	Carry forward vendor balances	COVERED-14	14	—
FBWE (AP bill of exchange presentation)	Bill of exchange presentation	GAP	propose 36	Bills gap
Fiori: Schedule Accounts Payable Jobs	Job templates (payment proposal, correspondence)	COVERED-4	4	Job framework
Fiori: Manage Automatic Payments	S/4 payment run app	COVERED-19	19	—
FPAYM / FBPM payment advice	Payment advice generation (REMADV)	COVERED-19	19	Vendor payment advice email/portal
F.64 (AP correspondence)	Correspondence: maintain / print	PARTIAL-19	19	Correspondence types (account statement, payment notice) only partially named (payment advice) — define correspondence framework (templates + triggers) on wave 46 document designer

13.2.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OBD3 — Vendor account groups / field status	T077K	Number range, one-time, field status	COVERED-7	7	—
XKN1 / OBAS — Number ranges for vendor accounts / assign	NRIV, T077K	Internal/external	COVERED-5	5	—
OBB8 — Terms of payment (vendors)	T052	—	COVERED-7	7	—
FBZP — Payment program configuration: all company codes, paying company codes, payment methods per country, per company code, bank determination (ranking, bank accounts, available amounts, value date), house banks	T042, T042A, T042B, T042D, T042E, T042I, T042Z, T042V	Payment method classification (check/transfer/bill), min/max amounts, foreign payments allowed, form/DME program, optimization by bank group/postal code, bank selection ranking	COVERED-19	19	Payment methods, bank ranking and limits per legal entity; house banks in wave 36
FI12 / FI13 — House banks and bank accounts	T012, T012K	Bank key, account id, G/L account, currency	COVERED-36	36	Bank account master (BAM)
OBPM1 / OBPM2 / OBPM3 / OBPM4 — Payment medium formats / note-to-payee / payment method / selection variants	TFPM*, T042Z*	—	COVERED-19	19	Format adapters
OBYR — Special G/L indicators vendors (down payments, guarantees, BoE)	T074	Reconciliation accounts per SGL	PARTIAL-19	19	Down payments yes; BoE no
OBXT / OBA3 — Tolerances for vendors / payment differences	T043G	Residual, charge-off limits	COVERED-19	19	Settlement tolerances
OBA4 — Tolerance groups for employees (posting limits)	T043T	Max amount per doc/line/discount	COVERED-2	2	OPA policy on amounts
Define withholding tax types / codes / assign to company code (extended WHT)	T059P, T059Z, T001WT	Invoice/payment time, base, rates, exemptions, certificates, accumulation	COVERED-13	13	Withholding engine
Classic withholding tax (OBCS / T059Q)	T059Q	—	SUPERSEDED-13	13	Extended model only
OBZT — Tax codes allowed in FB60/FB70	T007V	Per transaction	COVERED-13	13	Allowed tax codes per document type
OBCL — Default tax code for non-taxable transactions	T001	—	COVERED-13	13	—
OB71 — Define reason codes for payment block / OB27	T008	Block for payment, manual/automatic	COVERED-19	19	Payment holds
OBB9 — Installment terms	T052R	—	PARTIAL-7	7	Installment gap
Define alternative payee / permitted payee (BP role)	LFA1-LNRZA, LFZA	—	COVERED-7	7	Alternate payee on vendor role
Define correspondence types / forms (OB77, OB78, OB79)	T048, T048B	—	PARTIAL-19	19	Correspondence framework gap
Define payment advice forms (SAPscript/PDF)	—	—	COVERED-19	19	Payment advice templates; wave 46 designer
Check lots (FCHI) / check number ranges	PCEC	Lot per house bank/account	COVERED-36	36	Check lots
OBYP / OBA7 (vendor doc types KR, KG, KZ, KA)	T003	—	COVERED-11	11	Document types
Define sensitive fields for dual control (S_ALR_87003378 / T055F)	T055F	Fields requiring confirmation	COVERED-7	7	Change approval on sensitive fields
OBXS — Down payment request accounts	T030	—	COVERED-19	19	—
Define vendor activity groups (no) — Define payment block reasons	T008	—	COVERED-19	19	—
Evaluated receipt settlement defaults (vendor master ERS flag, OMRR)	LFM1	—	COVERED-18	18	—
OBXF — Posting keys for AP (OBXK group)	TBSL	—	SUPERSEDED-11	11	—
OBD3 "one-time account"	T077K	—	PARTIAL-7	7	One-time party gap
Define intercompany vendor/customer clearing (OBYA)	T001U	—	COVERED-40	40	—
Bank-to-bank transfers (payment request types, FIBL)	—	—	PARTIAL-36	36	Payment requests gap

13.3 FI-AR — Accounts Receivable

SAP context: customer master (KNA1/KNB1/KNVV → BP roles FLCU00/FLCU01), customer invoices via SD billing (VF01 → RV document) or FB70, incoming payments (F-28), lockbox (FLB2), dunning (F150), correspondence, interest calculation, credit management (classic FD32 or FSCM-CR). Passage: customer master wave 7, billing wave 29, AR/cash application/collections wave 30, credit wave 27.

13.3.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
XD01 / XD02 / XD03 (→ BP)	Customer master centrally	General, company-code, sales-area data	COVERED-7	7	Party + Customer role + sales-scope data
FD01 / FD02 / FD03	Customer master (accounting)	Company-code segment	COVERED-7	7	—

T-code	Name	Purpose	Passage status	Wave	Notes
VD01 / VD02 / VD03	Customer master (sales)	Sales-area segment	COVERED-7	7	Sales-scope role data (see SD)
XD05 / XD06 / FD05 / FD06 / VD05 / VD06	Block / flag for deletion	—	COVERED-7	7	—
XD07	Change customer account group	—	COVERED-7	7	—
XD99 / MASS (KNA1)	Mass maintenance customers	—	COVERED-5	5	—
FD08 / FD09	Confirm critical customer changes	Dual control	COVERED-7	7	—
FD32 → UKM_BP / UKM_MY_DCDS	Customer credit management master	Credit limit, risk category, blocks	COVERED-27	27	Credit limits, aging-based rules, risk scoring (wave 30)
FD24 / FD33	Credit management changes / display	—	COVERED-27	27	—
FB70	Enter customer invoice (FI, non-SD)	Direct AR invoice	COVERED-29	29	Free-text invoice
FB75	Enter customer credit memo	—	COVERED-29	29	Credit notes
F-22 / F-27	Customer invoice / credit memo (classic)	—	SUPERSEDED-29	29	Single invoice form
FV70 / FV75	Park customer invoice / credit memo	—	COVERED-29	29	Draft + approval
F-28	Post incoming payment	Manual cash application with clearing	COVERED-30	30	Manual cash application
F-26	Incoming payment fast entry	—	COVERED-30	30	—
F-29	Customer down payment	—	COVERED-30	30	Customer prepayments (deposit on order)
F-37	Customer down payment request	—	COVERED-27	27	Deposit request on order
F-39	Clear customer down payment	—	COVERED-30	30	—
F-30 / F-32	Post with clearing / clear customer	—	COVERED-30	30	Settlement without payment
F-34 / F-36 / F-33	Bill of exchange payment / discount (AR)	—	GAP	propose 36	Bills of exchange gap
FBE1 / FBE2 / FBE3	Payment advice create/change/display	Remittance advices for matching	COVERED-30	30	Remittance advice ingestion and matching
FBE6	Delete payment advices	—	COVERED-30	30	—
FLB1 / FLB2	Postprocess lockbox / import lockbox file (BAI2)	Lockbox	COVERED-30	30	Lockbox BAI2
FEBA_LOCKBOX	Postprocess lockbox (New)	—	COVERED-30	30	—
F-21 / F-30	Transfer posting (customer)	—	COVERED-30	30	—
FDM1 / UDM_DISPUTE	Create dispute case (FSCM-DM)	—	COVERED-30	30	Disputes workflow
F.27 (document side)	Account statement (periodic correspondence)	—	COVERED-30	30	Customer statements
FBL5N (document finder)	—	—	COVERED-30	30	—
FB12 / F.61 / F.64	Request / print correspondence	Correspondence requests	PARTIAL-30	30	Statements and dunning named; general correspondence framework (payment notices, individual letters, open item lists) only partial
F.2A / F.2B	Interest calculation (customer items) run	—	COVERED-30	30	Interest notes
VKM1 / VKM4 / VKM3	Release blocked SD documents (credit)	Credit release	COVERED-27	27	Credit hold release workflow
BP (FLCU00/FLCU01)	Customer roles in BP (S/4)	—	COVERED-7	7	—
Fiori: Manage Customer Line Items / Post Incoming Payments / Reprocess Bank Statement Items	S/4 AR apps	COVERED-30	30	—	
Biller Direct (FSCM-BD)	Customer self-service for invoices/payments	—	SUPERSEDED-30	30	Customer portal with payment adapter

13.3.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
FBL5N / FBL5H	Customer line items / browser	COVERED-30	30	—
FD10N / FD10NA	Customer balance display	COVERED-30	30	—
FD11	Customer account analysis	COVERED-30	30	—
S_ALR_87012167	Accounts receivable information system	COVERED-46	46	AR analytics
S_ALR_87012168	Due date analysis for open items	COVERED-30	30	Aging
S_ALR_87012169 / 87012170	Transaction figures: account balance / special sales	COVERED-30	30	—
S_ALR_87012172	Customer balances in local currency	COVERED-30	30	—
S_ALR_87012173	List of customer open items for printing	COVERED-30	30	—
S_ALR_87012174	List of customer open items	COVERED-30	30	—
S_ALR_87012175	Open items — customer due date forecast	COVERED-36	36	Cash forecast
S_ALR_87012176	Customer evaluation with OI sorted list (aging)	COVERED-30	30	—

T-code	Name	Passage status	Wave	Notes
S_ALR_87012177	Customer payment history	COVERED-30	30	Payment behaviour feeds risk scoring
S_ALR_87012178	Customer open item analysis by balance of overdue items	COVERED-30	30	—
S_ALR_87012179	Customer list	COVERED-7	7	—
S_ALR_87012180	Address list	COVERED-7	7	—
S_ALR_87012182	Display changes to customers	COVERED-5	5	—
S_ALR_87012183	Display/confirm critical customer changes	COVERED-7	7	—
S_ALR_87012186	Customer sales	COVERED-46	46	—
S_ALR_87012197	List of customer line items	COVERED-30	30	—
S_ALR_87012198	List of down payments open on key date	COVERED-30	30	—
S_ALR_87012199	Customer: bill of exchange list	GAP	propose 36	Bills gap
S_ALR_87012215	Customer sales per period (RFDUML00)	COVERED-46	46	—
F.21 / S_ALR_87012218	Interest scale / interest calculation list	COVERED-30	30	—
F.31	Credit management overview	COVERED-27	27	Credit exposure
F.32 / FDK43	Credit management: missing data / master data list	COVERED-27	27	—
FCV3 / S_ALR_87012215 (credit)	Credit overview (SD)	COVERED-27	27	—
UKM_COMMITMENTS / UKM_LOGS_DISPLAY	FSCM credit exposure / logs	COVERED-27	27	—
UDM_SUPERVISOR / UDM_SPECIALIST	Collections worklists (FSCM-COL)	COVERED-30	30	Collections workspace
UDM_DISPUTE (list)	Dispute case list	COVERED-30	30	—
S_ALR_87012085 (customer) / F.23	Dunning history	COVERED-30	30	—
FBMP (display)	Dunning procedure list	COVERED-30	30	—
Fiori: Days Sales Outstanding / Overdue Receivables / Future Receivables / Cash Collection Tracker	AR KPIs	COVERED-46	46	—
Fiori: Process Receivables	Collections work center	COVERED-30	30	—

13.3.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
F150	Dunning run (parameters, schedule, print)	COVERED-30	30	Dunning levels, grace days, blocks; letters only for expected customers
F.24 / F.26 / F.2A / F.2B	Interest calculation (items / balances)	COVERED-30	30	Interest notes
F.27	Periodic account statements	COVERED-30	30	Customer statements
FLB2 / FEBA_LOCKBOX	Lockbox import and postprocessing	COVERED-30	30	—
FF_5 / FEBAN (AR side)	Bank statement → customer clearing	COVERED-30	30	camt.053 auto-match for AR (wave 36 for the bank side)
F.13 (AR)	Automatic clearing customers	COVERED-30	30	Rule-based auto-match
FAGL_FC_VAL (AR)	FX valuation of customer open items	COVERED-12	12	—
FAGLF101 (AR)	Regroup customers with credit balances	GAP	propose 14	Reclassification gap
F.99	Customer balance confirmation	GAP	propose 30	Balance confirmation gap
F.07 (AR)	Carry forward	COVERED-14	14	—
UKM_MALUS_DSP / UKM_MASS / UKM_CREDIT_LIMIT_MASS	Mass credit limit recalculation / scoring run	COVERED-30	30	Risk scoring batch
F.28 / FD24 mass	Reset credit limit data / recalculate	COVERED-27	27	—
UDM_BP_PROF / UDM_GENWL	Generate collections worklist (FSCM-COL)	COVERED-30	30	Worklist generation
F110 (customer direct debits)	Direct debit run (SEPA SDD)	COVERED-30	30	Customer direct debit via payment adapter / pain.008 — confirm pain.008 is in wave 19 formats (wave 19 lists pain.001 only)
FDTA (DD)	DME admin for direct debits	COVERED-19	19	—
SEPA mandate management (FSEPA_M1..M4)	Mandates	PARTIAL-30	30	SEPA direct-debit mandates not named; required for EU customers paying by direct debit — add mandate entity to wave 30
VF44 / VF45 (see SD)	Revenue recognition run (classic)	COVERED-29	29	—
FBWD / FBWE (AR BoE)	Bill of exchange presentation	GAP	propose 36	Bills gap
Fiori: Schedule Accounts Receivable Jobs	Job templates (dunning, statements, interest)	COVERED-4	4	—

13.3.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OBD2 — Customer account groups / field status	T077D	—	COVERED-7	7	—

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
XDN1 / OBAR — Number ranges customers / assign	NRIV, T077D	—	COVERED-5	5	—
OBB8 — Terms of payment (customers)	T052	—	COVERED-7	7	—
OBXR — Special G/L customers (down payments, security deposits, BoE)	T074	—	PARTIAL-30	30	BoE gap
OBA3 — Customer tolerances	T043G	—	COVERED-30	30	—
FBMP — Dunning procedures (levels, intervals, charges, min amounts, grace days, forms)	T047*, T047A..T047M	Dunning levels 1–9, dunning charges by amount, interest indicator, print parameters, dunning by dunning area	COVERED-30	30	Dunning levels + letters
OBL6 / OB61 — Dunning areas / block reasons	T047M, T040S	—	COVERED-30	30	—
OB46 / OB82 / OB83 — Interest indicators, calculation types, reference rates	T056*, T056P, T056R	Item/balance interest, calendar, rounding	COVERED-30	30	AR interest; GL interest is a gap
OB45 — Credit control areas (classic)	T014	Credit limit currency, update group, risk categories	SUPERSEDED-27	27	FSCM-style credit segments modeled as credit profiles per legal entity/group
OVA8 — Automatic credit control (SD)	T691F	Static/dynamic check, horizon, open orders/deliveries, reactions	COVERED-27	27	Credit rules (limit, aging, dynamic horizon)
OB01 — Risk categories / OVFL credit groups	T691A, T691B	—	COVERED-27	27	—
UKM_* FSCM credit customizing (credit segments, rules, scoring formulas, BRF+ rules)	UKMCRED_SGM*, UKM_*	Score formulas, limit rules, check rules per segment	COVERED-27	27	Credit profiles + scoring (wave 30)
UDM_* collections customizing (collection strategies, segments, profiles, priorities, worklist rules)	UDM_*	Strategy rules (BRF), priorities, groups	COVERED-30	30	Collection strategies
UDM_* dispute case customizing (case types, attributes, status, reasons, RMS)	SCMG_*	—	COVERED-30	30	Disputes
OBB9 — Installment terms (customers)	T052R	—	PARTIAL-7	7	Installment gap
OB10 / Lockbox control (OBAY, OBAX)	T049A, T049B, T049L	Lockbox destination, BAI format, posting rules	COVERED-30	30	—
OBXL / OBBE — Payment difference reason codes and accounts	T053R, T030	—	COVERED-30	30	—
Define payment advice types / notes to payee (OBBE)	T053	—	COVERED-30	30	—
Define correspondence types (OB77) / periodic account statements	T048	—	PARTIAL-30	30	Correspondence framework gap
SEPA mandate customizing	SEPA_*	—	PARTIAL-30	30	Mandate gap
Define alternate payer / head office–branch	KNA1-KNRZE, KNB1	Head office/branch accounting	PARTIAL-7	7	Head-office/branch (invoice to branch, pay from HQ) not named in wave 7 party roles — add "bill-to/pay-from parent" relationship
OBYR (customer side) sales/FI account determination — see VKOA	T030, C001..	—	COVERED-29	29	Revenue account determination
Define reason codes for credit holds (OVA8/OVAK)	T691*	—	COVERED-27	27	—
OB38 — Assign credit control area to company code	T001	—	SUPERSEDED-27	27	—
Define cash discount base, tax on discount (OBY6)	T001	—	COVERED-13	13	—
Define dunning forms (SAPscript F150_DUNN_01)	—	—	COVERED-30	30	Letter templates; wave 46 designer

13.4 FI-AA — Asset Accounting

SAP context: asset classes, charts of depreciation with depreciation areas (book, tax, group, IFRS), asset master (ANLA/ANLB/ANLC), transactions (acquisition ABZON/F-90, retirement ABAVN/F-92, transfer ABUMN, post-capitalization ABNAN, write-up ABZU, manual depreciation ABMA, unplanned ABAA), depreciation run AFAB (S/4: FAA_DEPRECIATION_POST), year-end AJAB/AJRW (S/4: FAA_CMP), AuC settlement AIAB/AIBU, asset history sheet. Passage: wave 37 with books ≈ depreciation areas and posting to the universal journal per ledger group.

13.4.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
AS01 / AS02 / AS03	Asset master create/change/display	Asset with depreciation areas, time-dependent data, origin	COVERED-37	37	Asset master with books
AS11 / AS12 / AS13	Asset sub-number create/change/display	Component sub-assets	COVERED-37	37	Sub-assets / components
AS21 / AS22 / AS23	Group asset create/change/display	Group depreciation (US tax)	PARTIAL-37	37	Group assets (depreciate at group level, e.g. ADR) not named in wave 37 — add to tax book features
AS05 / AS06 / AS25 / AS26	Block / delete asset / block group asset	—	COVERED-37	37	—
AS91 / AS92 / AS93 / AS94	Legacy asset data transfer (create/change/display old asset, sub-number)	Opening NBVs	COVERED-49	49	Fixed-asset migration pack; wave 37 demo migration

T-code	Name	Purpose	Passage status	Wave	Notes
AS100	Legacy data transfer via Excel	—	COVERED-10	10	Migration framework + Excel import
ABZON / F-90	Acquisition with automatic offsetting entry / from vendor (AP)	Acquisition	COVERED-37	37	Acquisition from AP invoice
ABZP / ABZE	Acquisition from affiliated company / in-house production	—	COVERED-37	37	From intercompany (wave 40) and from production/project
ABSO	Miscellaneous transactions (transaction type driven)	Generic asset posting	COVERED-37	37	Transaction type model
ABUMN / ABUML	Transfer within company code / to another asset class	Intra-company transfer	COVERED-37	37	Transfer
ABT1N	Intercompany asset transfer	Cross-entity transfer with NBV/gross methods	COVERED-40	40	Intercompany asset transfer via wave 40 IC + wave 37
ABAVN / F-92 / ABAON	Retirement by scrapping / with customer (sale) / sale without customer	Disposals	COVERED-37	37	Disposal (sale/scrap) with gain/loss
ABNAN	Post-capitalization	Capitalize in a later year	COVERED-37	37	Adjustment transactions
ABZU	Write-up	Reverse excess depreciation	COVERED-37	37	—
ABAA	Unplanned depreciation	Impairment-like manual posting	COVERED-37	37	Impairment
ABMA	Manual depreciation	—	COVERED-37	37	—
ABMR	Manual transfer of reserves	Special reserves (German HGB)	OUT-OF-SCOPE	—	Special-reserve accounting (Sonderposten) is a DE localization topic; revisit with the DE country pack
ABAW	Balance sheet revaluation	Revaluation (IAS 16)	COVERED-37	37	Revaluation transaction
ABGF / ABGL	Credit memo in invoice year / in following year	Acquisition credit	COVERED-37	37	—
AB08	Reverse asset document	—	COVERED-37	37	Reversal
AIAB / AIBU / AIST	AuC settlement rule / settlement / reverse	Assets under construction	COVERED-37	37	CIP / under-construction from projects (wave 41 feeds)
OASV	Post balances to asset accounts (data transfer)	Data transfer reconciliation	COVERED-49	49	Migration reconciliation
AW01N	Asset Explorer	Values, transactions, parameters per area	COVERED-37	37	Asset inquiry with per-book values
ASKB / ASKBN	Periodic APC posting (non-leading areas)	Post values of areas that don't post real-time	COVERED-37	37	Multi-book posting to ledger groups
ABST / ABST2	Reconciliation FI-AA vs. G/L	—	COVERED-37	37	Asset subledger reconciles to GL automatically
OAMK	Change reconciliation accounts (asset)	Setup	COVERED-37	37	—
AS08	Asset number ranges	—	COVERED-5	5	—
Fiori: Manage Fixed Assets / Post Acquisition / Post Retirement / Transfer Asset	S/4 apps	COVERED-37	37	—	
RE-FX: RECN / RECECN (lease contracts, IFRS 16 valuation)	Contract and lease management	Lease accounting	COVERED-37	37	IFRS 16 / ASC 842 schedules, ROU asset & liability
RE-FX: REISCN / RERAPP / RERAPL	Lease contract lists / periodic posting / planned-record list	—	COVERED-37	37	Lease schedules and postings
RE-FX real estate objects (business entity, building, rental object, rental contracts)	Real estate management	Property management	OUT-OF-SCOPE	—	Real-estate management (tenancy, service charge settlement) is beyond the manufacturer scope; only the lease-accounting part is in wave 37
Asset barcode inventory (AIS / third-party)	Physical asset inventory	—	COVERED-37	37	Barcode physical inventory with reconciliation
Insurance values (asset master tab, S_ALR_87011951)	Insurance values / indexing	—	PARTIAL-37	37	Insurance values and index series not named — add to asset master

13.4.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
S_ALR_87011990	Asset history sheet	COVERED-37	37	Asset roll-forward by class/book
S_ALR_87011963 / 87011964 / 87011965 / 87011966	Asset balances by asset number / class / business area / cost center	COVERED-37	37	—
S_ALR_87011967 / 87011968	Asset balances by plant / location	COVERED-37	37	—
S_ALR_87011979 / 87011981	Physical inventory list / by location	COVERED-37	37	—
S_ALR_87011994	Asset balances (summary)	COVERED-37	37	—
S_ALR_87012004 / 87012005	Total depreciation / depreciation by class	COVERED-37	37	—
S_ALR_87012006	Ordinary depreciation	COVERED-37	37	—
S_ALR_87012007 / 87012008	Special / unplanned depreciation	COVERED-37	37	—
S_ALR_87012013 / 87012018	Depreciation comparison / depreciation simulation	COVERED-37	37	Simulation on planned assets
S_ALR_87012026	Depreciation current year	COVERED-37	37	—

T-code	Name	Passage status	Wave	Notes
S_ALR_87012028	Asset transactions	COVERED-37	37	—
S_ALR_87012033	Asset acquisitions	COVERED-37	37	—
S_ALR_87012037	Changes to asset master records	COVERED-5	5	—
S_ALR_87012039	Asset transactions (intracompany transfers)	COVERED-37	37	—
S_ALR_87012041 / 87012043	Asset portfolio / asset list	COVERED-37	37	—
S_ALR_87012048	Asset retirements	COVERED-37	37	—
S_ALR_87012050 / 87012052	Asset acquisitions / retirements (gain-loss)	COVERED-37	37	—
S_ALR_87012054	Intracompany transfers	COVERED-37	37	—
S_ALR_87012056	Directory of unposted assets	COVERED-37	37	—
S_ALR_87012058	List of origins of asset debits	COVERED-37	37	Acquisition source trace (AP, project, inventory)
S_ALR_87012060	List of origins by cost elements	COVERED-37	37	—
S_ALR_87012063	Asset list (inventory)	COVERED-37	37	—
S_ALR_87012075	Leased assets (classic leasing)	COVERED-37	37	Lease schedules
S_ALR_87012936 / 87012937	Depreciation on capitalized assets (simulation) / planned depreciation	COVERED-37	37	—
S_P99_41000192	Posted depreciation by cost center	COVERED-37	37	Dimension-level depreciation
S_ALR_87010127 / 87010129 (US: form 4562 / mid-quarter)	Tax depreciation reports	PARTIAL-37	37	MACRS tables yes; IRS Form 4562 layout belongs to the US pack (wave 48)
AR01 / AR02 / AR03 / AR11	Asset balances / history sheet / depreciation list / mid-year listing (ad hoc)	COVERED-37	37	—
AR31 / AR32	Mass change of assets (worklist execute/display)	COVERED-5	5	Mass edit
Fiori: Asset Balances / Asset History Sheet / Depreciation Lists / Asset Transactions	S/4 apps	COVERED-37	37	—

13.4.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
AFAB / AFABN / FAA_DEPRECIATION_POST (S/4)	Depreciation posting run (planned, repeat, restart, unplanned)	COVERED-37	37	Depreciation run per book with restart safety (Temporal)
AFAR	Recalculate depreciation	COVERED-37	37	Recalc after parameter change
AFBP	Display depreciation posting log	COVERED-37	37	—
AJRW	Fiscal year change (open new year)	SUPERSEDED-37	37	No explicit year open; values projected per period; year-end close via wave 14 checklist
AJAB / FAA_CMP (S/4)	Year-end closing asset accounting / close fiscal year	COVERED-14	14	Close checklist task with asset-period lock
OAAQ / OAAR	Reverse year-end closing / per area	COVERED-14	14	Controlled reopen with audit
ASKB / ASKBN	Periodic posting of non-leading depreciation areas	COVERED-37	37	—
AIAB / AIBU (periodic)	AuC settlement run	COVERED-37	37	—
ABST2	FI-AA / G/L reconciliation	COVERED-37	37	Automated daily
AR29 / AR29N	Revaluation / index-based revaluation and new valuation	COVERED-37	37	Revaluation
J1AI / country inflation accounting (AJ)	Inflation accounting	OUT-OF-SCOPE	—	Hyperinflation (IAS 29) country functions deferred beyond first country packs
RASIMU02 (depreciation simulation)	Simulation / forecast	COVERED-37	37	—
RERAPP (RE-FX)	Periodic lease posting run	COVERED-37	37	—
Fiori: Schedule Asset Accounting Jobs	Job templates	COVERED-4	4	—

13.4.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
EC08 / OAOB — Copy reference chart of depreciation / assign company code	T093C, T093	—	COVERED-37	37	Books template per country
OADB / OABC / OABD — Depreciation areas: define, area type, transfer of APC/depreciation terms	T093, T093A	Real/derived area, posting to G/L (0/1/2/3/4), ledger group, currency, value takeover rules	COVERED-37	37	Books (tax/IFRS/local) posting to ledger groups; derived books (delta) — treat as covered
OAYZ / OAYR — Asset class default depreciation / posting rules (periodic/monthly)	T093B	Depreciation key, useful life per class and area; posting cycle	COVERED-37	37	Asset class defaults, posting cycle
OAOA — Asset classes	ANKA, ANKB	Account determination, number range, screen layout, AuC flag, low-value asset flag, history	COVERED-37	37	Asset groups/classes
AO90 — Account determination (APC, acc. depreciation, expense, gain/loss, revaluation, AuC)	T095, T095B	Per area and account determination key	COVERED-37	37	Asset posting profiles
AO21 / AO25 / AO71 — Depreciation keys, methods (base, declining, multi-level, period control), maximum amount, cut-off value	T090*, T090NA, T090NAZ, T090NP, T090NR	Base methods (SL, DB, units of production), multi-level, period control (fiscal year start, mid-month, half-year), changeover methods, shutdown, scrap value	COVERED-37	37	Depreciation profiles + conventions (half-year, mid-month, mid-quarter) incl. MACRS tables
OAVH / OAVS — Period control / calendar assignment	T090NP, T090NV	—	COVERED-37	37	Conventions
AO73 / AO74 / AO76 — Transaction types, transaction type groups	TABW, TABWG	Debit/credit, retirement w/ revenue, capitalization, history sheet group	COVERED-37	37	Transaction type model
OAYK — Low-value asset limits per area	T093B	Maximum LVA amount, quantity LVA	COVERED-37	37	LVA threshold rules

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OAYE / OAYF — Legacy data transfer date / last period posted	T093C	—	COVERED-49	49	Migration parameters
OA79 / OA80 — Asset history sheet versions / groups	T082A, T082B, T082C	—	COVERED-37	37	Roll-forward report definition
AO11 — Number range assets	NRIV	—	COVERED-5	5	—
AO21 / OA77 — Screen layout rules for asset master / depreciation areas	T082G, T082H	Field status, maintenance level (class/main/sub)	COVERED-3	3	Metadata field attributes
OAYR intervals — Specify intervals and posting rules (AFAB period)	T093D	Smoothing vs catch-up	COVERED-37	37	Catch-up / smoothing option
OABN / OABW — Area for revaluation / unplanned depreciation allowed	T093A	Which transaction types allowed per area	COVERED-37	37	—
OAV8 / OAVA — Asset evaluation groups / environmental protection indicators / asset locations	T087, T499S	Reporting attributes	COVERED-37	37	Classification attributes on asset; site/location dimensions (wave 6)
OAYM / OAAX — Group assets per area	T093A	Group asset areas	PARTIAL-37	37	Group asset gap
OAVI / OAVF — Insurance types, companies, index series	T086, T094	—	PARTIAL-37	37	Insurance gap
OAOA AuC settings — Asset under construction: settlement profile, line-item settlement	T093	—	COVERED-37	37	—
RE-FX IMG: contract types, valuation rules (IFRS 16/ASC 842), accounting principles, condition types, posting rules	TIVRA*, VICN*	Lease classification, discount rate, IBR	COVERED-37	37	Lease parameters
Parallel valuation: ledger approach vs accounts approach (FAA_CMP settings)	T093, FINSC_LEDGER	—	COVERED-37	37	Ledger-group approach only
Depreciation keys for tax (MACRS, ADS) tables	T090*	—	COVERED-37	37	MACRS tables

13.5 FI-BL / FSCM-CM / BAM — Bank Accounting, Cash Management & Bank Account Management

SAP context: house banks (FI12 → Bank Account Management in S/4 with workflow), electronic bank statement (FF_5/FEBA/FEBAN; S/4: Manage Bank Statements), manual statement (FF67), check deposit (FF68), cash position/liquidity forecast (FF7A/FF7B; S/4 Cash Management "One Exposure" FQM), memo records (FF63), bank communication management (BCM), in-house cash (IHC), cash journal. Passage: wave 36, with bank-side AR/AP matching in waves 19/30.

13.5.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
FI01 / FI02 / FI03 / FI06	Bank master (bank key) create/change/display/flag	Bank directory	COVERED-7	7	Bank as party role; bank directory import (SWIFT/ABA)
BAUP	Bank directory upload (file)	Mass bank key import	COVERED-7	7	Import adapter
FI12 / FI12_HBANK / Fiori: Manage Bank Accounts (BAM)	House banks and bank accounts	Bank account master with workflow, signatories	COVERED-36	36	Bank account master with signatories and approval
BAM: bank account review, signatory, overdraft limit	Bank account governance	—	COVERED-36	36	—
FF67	Manual bank statement entry	—	COVERED-36	36	Manual statement entry
FF_5 / FF.5	Import electronic bank statement (MT940, BAI2, camt.053)	EBS import	COVERED-36	36	camt.053/BAI2/MT940 import
FEBA / FEBAN / FEBA_BANK_STATEMENT	Postprocess bank statement	Fix unmatched items	COVERED-36	36	Reconciliation exception queue
FEBP	Post bank statement (batch)	—	COVERED-36	36	—
Fiori: Reprocess Bank Statement Items	S/4	—	COVERED-36	36	—
FF68	Check deposit (manual entry)	Deposit list	COVERED-36	36	Check deposit
FF63 / FF65	Memo records create / list (planned items)	Manual cash plan items	COVERED-36	36	Forecast manual items
FF7A / FF7B → Fiori: Cash Position / Liquidity Forecast	Cash position / liquidity forecast	Treasury daily view	COVERED-36	36	Cash positioning (intraday camt.052) and forecast from AP/AR/payroll/projects
FF/1 / FF/4	Payment advice / planning type management	—	COVERED-36	36	—
FCH* (check management — see FI-AP)	—	—	COVERED-36	36	—
FLB1 / FLB2 (see FI-AR)	Lockbox	—	COVERED-30	30	—
FBCJ	Cash journal	Petty cash	GAP	propose 36	Cash journal gap (see FI-GL)
FRFT / FRFT_B	Repetitive payment codes / bank-to-bank transfer	Treasury transfers	PARTIAL-36	36	Bank-to-bank transfer / payment request gap
BNK_MONI / BNK_APP / BNK_MONIP (BCM)	Bank Communication Management: batch monitor, approve payment batches, payment status	Payment batch approval and status (pain.002)	COVERED-36	36	Bank file audit and approval; pain.002 ingestion (wave 19)

T-code	Name	Purpose	Passage status	Wave	Notes
IHC0 / IHC1IP (In-House Cash)	In-house cash center, IHC accounts, internal payment orders	Intercompany banking	PARTIAL-36	36	"In-house banking basics: netting, cash pooling" named; full IHC (internal accounts, interest, statements for subsidiaries) is partial
Fiori: Bank Statement Monitor	Bank statement monitor (S/4)	—	COVERED-36	36	—
Fiori: Manage Bank Statements / Cash Flow Analyzer	S/4 Cash	—	COVERED-36	36	—
Fiori: Approve Bank Payments / Monitor Payments (Advanced Payment Management)	APM	Payment hub routing	PARTIAL-36	36	Payment approval covered; APM-style central payment hub (route, batch, reformat across entities) only partially (adapters)
Letters of credit / bank guarantees (TRM FTR_CREATE or BL)	Guarantees	—	COVERED-36	36	LC and guarantees named
FBWE / FBWD (bills)	Bill of exchange presentation	—	GAP	propose 36	Bills gap

13.5.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
FF7A	Cash position	COVERED-36	36	—
FF7B	Liquidity forecast	COVERED-36	36	—
FF_6	Display bank statement (electronic)	COVERED-36	36	—
FF.6	Display statement overview	COVERED-36	36	—
FEBA (list mode)	Statement items overview	COVERED-36	36	—
S_ALR_87012309 (cash journal)	Cash journal report	GAP	propose 36	—
S_P99_41000101 / FCHN	Check register	COVERED-36	36	—
FF.8	Payment advice journal	COVERED-36	36	—
FLQAB / FLQAD (liquidity planner)	Liquidity analysis / actual assignment	COVERED-36	36	Cash flow forecast buckets
FQM_* (One Exposure) / Fiori: Cash Flow Analyzer	Cash flow analysis by liquidity item	COVERED-36	36	—
Fiori: Bank Relationship Overview / Bank Fee Analysis	Bank fees (camt.086)	GAP	propose 36	Bank fee analysis (camt.086) not named; propose as wave 36 report
Fiori: Cash Position Details / Check Cash Flow Items	—	COVERED-36	36	—
BNK_MONI (report mode)	Payment batch status	COVERED-36	36	—

13.5.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
FF_5 (scheduled)	Nightly statement import	COVERED-36	36	Scheduled job
FEBC / FEBP	Batch posting of statements	COVERED-36	36	—
FF67 → FEBA (manual cycle)	Manual statement close	COVERED-36	36	—
FF.D	Delete / archive memo records	COVERED-36	36	—
FLQAC / FLQAD	Liquidity planner assignment run	COVERED-36	36	—
FQM_FLOW_BUILDER (One Exposure rebuild)	Rebuild cash flows	SUPERSEDED-36	36	Forecast projected from events; rebuild = projection rebuild
BNK_BATCH_* (BCM batching job)	Payment batching	COVERED-36	36	—
Bank reconciliation statement (S/4 bank reconciliation)	—	COVERED-36	36	Auto-reconciliation with matching rules + ML
Cash pooling / sweeping run (IHC)	—	PARTIAL-36	36	Basics named
Intercompany netting run (IHC netting)	—	COVERED-36	36	Netting named
F111 (treasury payment requests)	—	PARTIAL-36	36	—
Bank fee statement import (camt.086)	—	GAP	propose 36	—

13.5.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OBXK / FI12 — House banks, accounts, G/L accounts, bank sub-accounts (clearing per payment method)	T012, T012K, T012D	Sub-accounts per payment method (outgoing checks, transfers)	COVERED-36	36	Bank sub-account posting profile
OT83 / OT51 — Electronic bank statement: transaction types, external transaction codes, posting rules, account symbols, assign bank to transaction type	T028B, T028D, T028G, T033*	Posting rule (bank side / subledger side), interpretation algorithms (document number, reference, note-to-payee), BAI/MT940/camt format	COVERED-36	36	Matching rules engine + posting rules
OTPM — Search strings for EBS	T028SS	Regex-like note-to-payee parsing	COVERED-36	36	Matching rules with pattern extraction (ML suggestions)
OT41 / FF/1 — Planning levels / planning groups / cash management account names	T036, T035, T037	Level per G/L and subledger group; forecast groups	COVERED-36	36	Forecast categories/buckets
Liquidity items and derivation rules (FQM / FLQ)	FLQITEM*, FQM_*	Query sequences, derivation from G/L, BP, doc type	COVERED-36	36	Forecast source rules
Define check lots / void reason codes (FCHI / FCHV)	PCEC, TVOID	—	COVERED-36	36	—
Bank chains (FIBC)	TBCHAIN*	Intermediary banks per currency/country	PARTIAL-36	36	Correspondent/intermediary bank chains not named — add to bank connectivity
BCM: rules for batching, approval (release) rules, payment status mapping	BNK_*	Approval patterns (n-eyes), batch rules	COVERED-36	36	Approval workflow for bank files
IHC customizing: IHC centers, accounts, conditions, payment orders	IHC_*	—	PARTIAL-36	36	IHC partial

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
BAM: account types, review process, signatory groups, workflow	FCLM_*	—	COVERED-36	36	—
Cash journal customizing (FBCJC0..3)	TCJ_*	—	GAP	propose 36	—
Bank connectivity (EBICS / SWIFT / Multi-Bank Connectivity settings)	—	—	COVERED-36	36	SFTP/EBICS/Open-Banking adapters; keys in Vault
Lockbox customizing (OBAY/OBAX) — see FI-AR	—	—	COVERED-30	30	—

13.6 FSCM — Credit, Collections, Dispute Management & Treasury (TRM)

SAP context: FIN-FSCM-CR (credit management with scoring/BRF+), FIN-FSCM-COL (collections worklists, strategies), FIN-FSCM-DM (dispute cases on SAP Case Management), FIN-FSCM-BD (Biller Direct), FIN-FSCM-TRM (Treasury & Risk: transaction manager for money market/FX/derivatives/securities, market risk analyzer, hedge management, exposure management). Passage: credit wave 27, collections/disputes wave 30, cash/bank wave 36; TRM largely not covered.

13.6.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
UKM_BP	Credit management business partner (credit profile, segments, limits)	Credit master	COVERED-27	27	Credit profile per customer/credit group
UKM_CASE	Credit limit request / case	Workflow for limit increase	COVERED-7	7	Credit limit change approval (wave 7 governance + wave 27)
UKM_MY_DCDS	Documented credit decisions	Credit decision log	COVERED-27	27	Credit hold release workflow with reasons
UKM_COMMITMENTS	Credit exposure (liability) display	—	COVERED-27	27	—
UDM_DISPUTE	Create/process dispute case	Dispute	COVERED-30	30	Disputes workflow
UDM_SUPERVISOR / UDM_SPECIALIST	Collections supervisor / specialist worklist	Collections	COVERED-30	30	Collections workspace
Promise to pay / customer contacts / notes (collections)	—	—	COVERED-30	30	Promises to pay
Resubmissions (collections)	Follow-ups	—	COVERED-30	30	Follow-up tasks
Create dispute from FBL5N line item	Dispute from line item	—	COVERED-30	30	—
Biller Direct web app	Customer invoice portal	—	SUPERSEDED-30	30	Customer portal
FTR_CREATE / FTR_EDIT / FTR_00	Treasury transaction manager: create financial transaction (money market, FX forward/spot/swap, derivatives, securities, loans)	Deal capture	GAP	propose 36	TRM transaction manager is not in any wave. Manufacturers with FX exposure need at least FX spot/forward deal capture, settlement and valuation; propose a wave 36 "treasury deals (FX, loans, deposits)" extension, with derivatives/securities deferred to v2
TBB1 / TPM10	Post flows / treasury posting	Deal accounting	GAP	propose 36	Follows TRM gap
TPM1 / TPM60	Valuation (key date), amortization, position management	Mark-to-market	GAP	propose 36	—
THMEX (Hedge management & accounting, IFRS 9)	Hedge relationships, effectiveness, hedge accounting postings	Hedge accounting	GAP	propose 36	Hedge accounting per IFRS 9/ASC 815 — v2 candidate; register as gap
Exposure management (raw exposures, TEM)	Exposure capture from AP/AR/forecast	—	GAP	propose 36	FX exposure from open items is derivable from waves 12/36 data; deal side missing
Market Risk Analyzer / Credit Risk Analyzer	VaR, limits	—	OUT-OF-SCOPE	—	Market/credit risk analyzers are bank-grade treasury; beyond manufacturer v1
TBDM market data import (rates, prices)	Market data management	—	PARTIAL-12	12	Exchange-rate providers covered; interest curves/security prices not
Loans management (FN*, CML)	Loan contracts	—	GAP	propose 36	Intercompany and bank loans: schedules + interest; propose minimal loan contract in wave 36
Commodity management (CPE, commodity pricing engine)	Commodity pricing in MM/SD	—	OUT-OF-SCOPE	—	Commodity derivatives/pricing engine not in v1; price-index-based condition could be done via pricing engine (wave 26) later

13.6.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
UKM_LOGS_DISPLAY	Credit check logs	COVERED-27	27	—
UKM_MALUS_DSP	Score/limit history	COVERED-30	30	—
F.31	Credit overview	COVERED-27	27	—
UDM_SUPERVISOR (reports)	Collections KPI (promises kept, contacts)	COVERED-46	46	—
UDM_DISPUTE (list/analytics)	Dispute analytics	COVERED-46	46	—
Fiori: Credit Limit Utilization / Manage Credit Accounts / Manage Collection Worklist / Manage Dispute Cases	S/4 apps	COVERED-30	30	—
TPM12 / TPM13	Position list / flow list (TRM)	GAP	propose 36	TRM gap
TPM20	Posting journal (TRM)	GAP	propose 36	TRM gap
Hedge plan reports	—	GAP	propose 36	—
Treasury information system (TRM)	—	GAP	propose 36	—

13.6.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
UKM mass scoring / limit recalculation	Mass credit scoring	COVERED-30	30	—
UKM_TRANSFER_ITEMS	Transfer FI/SD liabilities to FSCM credit	SUPERSEDED-27	27	Exposure computed from the journal/orders directly
UDM_GENWL	Generate collections worklist	COVERED-30	30	—
F150 (triggered strategies)	Dunning	COVERED-30	30	—
Auto-create disputes from residuals / close promises	—	COVERED-30	30	—
TBB1 (periodic) / TPM1 valuation run / accruals	TRM period-end	GAP	propose 36	TRM gap
Hedge effectiveness test run	—	GAP	propose 36	—
TBDM import run	Market data import	PARTIAL-12	12	—

13.6.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
UKM: credit segments, risk classes, rules, formulas (BRF+), check rules, score ranges, blocking reasons	UKMCRED_SGM*, UKM_*	Static/dynamic checks, oldest open item, max dunning level, scoring formula, follow-on process	COVERED-27	27	Credit rules + scoring
UKM: integration (SD credit group, FI-AR liability update)	T691*, UKM_*	—	COVERED-27	27	—
UDM collections: segments, groups, strategies (rules, priorities), promise-to-pay, resubmission reasons, customer contact types	UDM_*	—	COVERED-30	30	—
Dispute case management: RMS, case types, attributes profile, status profile, reason codes, escalation, action profile	SCMG*	—	COVERED-30	30	—
Biller Direct configuration	—	—	SUPERSEDED-30	30	—
TRM: product types, transaction types, flow types, update types, valuation areas/classes, accounting codes, position management procedures	TRL*, TRD*, TPM*	—	GAP	propose 36	TRM gap
Hedge management: hedging profiles, exposure categories	THM*	—	GAP	propose 36	—
Market data: rate types, yield curves, volatility	VTB*, TCURR	—	PARTIAL-12	12	—

13.7 CO-OM — Overhead Cost Controlling (cost centers, internal orders, activity types, allocations)

SAP context: cost center accounting (CO-OM-CCA), internal orders (CO-OM-OPA), activity-based costing (CO-OM-ABC), cost element accounting (CO-OM-CEL; S/4 merges cost elements into G/L accounts FS00 type P/S). Planning (KP06/KP26), actual postings (KB11N/KB21N/KB31N), period-end allocations (KSV5/KSU5/KSC6), settlement (KO88), commitments. Passage: wave 38 cost accounting ledger with allocation policies; primary postings come straight from the universal journal (wave 11 dimension = cost center).

13.7.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
KS01 / KS02 / KS03	Cost center create/change/display	Cost center master (validity, hierarchy, category, lock flags)	COVERED-6	6	OperatingUnit (cost center) with effective dating; CO attributes in wave 38
KS04 / KS05 / KS13	Delete / display changes / list	—	COVERED-6	6	—
KS12 / KS14	Collective change / cost center list	Mass	COVERED-5	5	—
KSH1 / KSH2 / KSH3	Cost center group create/change/display	Groups (sets)	COVERED-6	6	Hierarchy purposes; groups = hierarchy nodes
OKEON / OKENN	Standard hierarchy change/display	—	COVERED-6	6	Hierarchy designer
KA01 / KA02 / KA03 / KA06 (→ FS00 in S/4)	Cost element primary / secondary create/change/display	Cost elements	SUPERSEDED-11	11	S/4 model: accounts of type primary/secondary cost in the single chart
KA04 / KA05 / KA23	Delete / changes / list cost elements	—	COVERED-11	11	—
KAH1 / KAH2 / KAH3	Cost element group	—	COVERED-11	11	Account structures/groups
KL01 / KL02 / KL03 / KL04 / KL05	Activity type create/change/display/delete/changes	Activity types (labor hour, machine hour)	COVERED-38	38	Activity types with rates; used by routing (wave 9) and production costing (wave 21/39)
KLH1 / KLH2 / KLH3	Activity type group	—	COVERED-38	38	—
KK01 / KK02 / KK03	Statistical key figure create/change/display	SKFs for allocation bases	COVERED-38	38	Statistical measures
KBH1 / KBH2 / KBH3	SKF groups	—	COVERED-38	38	—
KO01 / KO02 / KO03	Internal order create/change/display	Orders (overhead, investment, accrual, revenue)	COVERED-38	38	Internal orders (cost objects)
KO04 / KOK2 / KOK3 / KOK4	Order manager / collective processing / display / automatic collective	Mass order processing	COVERED-38	38	—
KOH1 / KOH2 / KOH3	Order group	—	COVERED-38	38	—
KO22 / KO23 / KO24 / KO25 / KO26 / KO27	Order budget: change/display, supplement, return, release	Budget on internal orders with availability control	COVERED-38	38	Budget control on cost objects
KO30 / KO32	Activate / deactivate availability control	—	COVERED-38	38	—
KOB1 / KOB2 / KOB3 / KOB4 / KOB5	Order line items: actual / commitment / variance / budget / settlement	Inquiry	COVERED-38	38	—
KB11N / KB14N / KB13N	Manual reposting of costs (enter/reverse/display)	Reposting primary costs	COVERED-38	38	Cost reposting journal (dimension-only move)
KB15N / KB16N / KB17N	Manual cost allocation (enter/display/reverse)	—	COVERED-38	38	Manual allocation
KB21N / KB23N / KB24N	Activity allocation (enter/display/reverse)	Direct activity allocation (secondary)	COVERED-38	38	Activity allocation entries; from shop floor (wave 25)

T-code	Name	Purpose	Passage status	Wave	Notes
KB31N / KB33N / KB34N	Statistical key figures: enter/display/reverse	—	COVERED-38	38	—
KB41N / KB43N / KB44N	Reposting revenues	—	COVERED-38	38	—
KB51N / KB53N / KB54N	Activity posting (sender activities)	—	COVERED-38	38	—
KB61	Reposting of line items (document-based)	Repost with reference	COVERED-38	38	—
KP06 / KP07	Cost center plan: primary costs (change/display)	Planning	COVERED-38	38	Budget planning with versions/scenarios
KP26 / KP27	Activity type plan: quantities and prices	—	COVERED-38	38	Activity rate planning
KP46 / KP47	SKF planning	—	COVERED-38	38	—
KPF6 / KPF7	Order plan: primary costs	—	COVERED-38	38	—
KP04 / KP65 / KP66 / KP67	Planner profiles / planning layouts	Planning UX	SUPERSEDED-38	38	Budget planning forms via metadata grids
KP90 / KP91 / KP98	Delete plan data / copy plan versions	—	COVERED-38	38	Scenario copy
KPSI	Plan reconciliation / iterative plan cost calculation	—	COVERED-38	38	—
CP01 / CP02 / CP03	Business process (ABC) create/change/display	ABC processes	COVERED-38	38	Activity-based costing
CPH1 / CPH2 / CPH3	Business process group	—	COVERED-38	38	—
KSAZ (also customizing)	Overhead structure (costing sheet)	—	COVERED-39	39	Costing sheets
Commitments from ME21N/ME51N with account assignment K/F	Commitments	—	COVERED-38	38	Encumbrances/commitments from PR/PO
Fiori: Manage Cost Centers / Manage Activity Types / Manage Internal Orders	S/4 apps	—	COVERED-38	38	—
Fiori: Reassign Costs and Revenues / Enter Activity Allocation / Enter Statistical Key Figures	S/4 apps	—	COVERED-38	38	—
Fiori: Import Financial Plan Data (SAC/Excel)	Plan upload	—	COVERED-38	38	Excel import / SAC-like planning via wave 46

13.7.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
S_ALR_87013611	Cost centers: actual/plan/variance	COVERED-38	38	Cost control workspace
S_ALR_87013612	Cost centers: range: actual/plan/variance	COVERED-38	38	—
S_ALR_87013613 / 87013614	Actual/plan/variance by cost element / cost element groups	COVERED-38	38	—
S_ALR_87013617 / 87013618 / 87013619	Actual/target/variance; range cost centers; cost center groups	COVERED-38	38	Target costs need activity-dependent plan — covered in wave 38/39
S_ALR_87013620	Actual comparison across fiscal years / periods	COVERED-38	38	—
S_ALR_87013621	Range: actual/plan/commitments	COVERED-38	38	—
S_ALR_87013623 / 87013624 / 87013625	Quarterly / year comparison / plan-plan versions	COVERED-38	38	—
S_ALR_87013626 / 87013627 / 87013628	Activity types: plan/actual; prices; comparison	COVERED-38	38	—
S_ALR_87013631 / 87013632	SKF actual/plan; activity quantities	COVERED-38	38	—
S_ALR_87013633 / 87013635 / 87013636	Cost component split; statistical key figures; area comparison	COVERED-38	38	—
S_ALR_87013640	Cost centers: planning overview	COVERED-38	38	—
S_ALR_87013645 / 87013646 / 87013647	Cost center: variance by cost element groups	COVERED-38	38	—
KSB1 / KSB2 / KSB5 / KSBP	Cost center actual line items / commitments / CO documents / plan line items	COVERED-38	38	—
KSBT	Activity prices report	COVERED-38	38	—
KSBL	Cost center planning overview	COVERED-38	38	—
KOB1 … KOB8	Order line items (actual, commitment, variance, budget, settlement, results analysis)	COVERED-38	38	—
S_ALR_87012993 / 87012994 / 87012995	Orders: actual/plan/variance; actual/plan/commitment; list	COVERED-38	38	—
S_ALR_87012996 / 87012997 / 87012998	Orders: budget/actual/commitment; actual/target; cumulative	COVERED-38	38	—
S_ALR_87012999 / 87013000 / 87013001	Orders: current period / year comparison	COVERED-38	38	—
S_ALR_87013002 … 87013010	Order group reports	COVERED-38	38	—
S_ALR_87013019	Orders: list of orders with budget	COVERED-38	38	—
KOC4	Order selection (order manager)	COVERED-38	38	—
ABC business process reports (S_ALR_87013 range)	Business process reports	COVERED-38	38	—
KS13	Cost center master data report	COVERED-6	6	—
KSB9	Plan/actual comparison	COVERED-38	38	—
Fiori: Cost Centers – Plan/Actual / Internal Orders – Plan/Actual / Cost Center Line Items	S/4 analytical apps	COVERED-46	46	—
Report Painter CO reports (GR55 library 1VK, 6O1)	—	SUPERSEDED-15	15	Report designer + semantic layer

13.7.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
KSV1 / KSV2 / KSV3 / KSV5 / KSV6	Distribution cycle create/change/display/execute/overview (actual)	COVERED-38	38	Allocation policies (distribution keeps primary cost element)
KSU1 / KSU2 / KSU3 / KSU5 / KSU6	Assessment cycle (actual)	COVERED-38	38	Assessment (secondary cost element)
KSW1 … KSW5	Periodic reposting cycles	COVERED-38	38	—

T-code	Name	Passage status	Wave	Notes
KSC1 … KSC6	Indirect activity allocation cycles	COVERED-38	38	—
KSVB / KSUB / KSCB	Plan distribution / plan assessment / plan indirect activity allocation	COVERED-38	38	Plan allocations
KSII / KSPI	Actual price calculation / plan price calculation	COVERED-38	38	Activity rate calculation (plan and actual)
KSS1 / KSS2	Actual cost splitting / variance calculation for cost centers	COVERED-38	38	Variance analysis for cost centers (target vs actual)
KSA3	Accrual calculation (percentage method)	COVERED-14	14	Accrual schemes
KO88 / KO8G	Settlement of orders (individual / collective)	COVERED-38	38	Settlement rules to cost objects/assets/GL
KO8B	Display settlement documents	COVERED-38	38	—
KGI2 / KGI4	Overhead calculation on orders (individual / collective)	COVERED-39	39	Costing-sheet overhead application
KSI4	Overhead calculation on cost centers	COVERED-39	39	—
CPTA / CPTB	ABC: template allocation (actual/plan)	PARTIAL-38	38	ABC named; template (driver-formula) allocation is partial — specify driver formulas
KO9E	Order planning: plan settlement	COVERED-38	38	—
OKP1 / OKP2	Period lock change/display (CO)	COVERED-14	14	Period control per module (CO)
KALA / KALB	Reconciliation ledger (classic CO-FI)	SUPERSEDED-11	11	Single journal
CO-FI real-time integration (FAGLCOFIC)	—	SUPERSEDED-11	11	—
KEU5 (CO-PA assessment — see CO-PA)	—	COVERED-38	38	—
KSV5/KSU5 run via Schedule Manager (SCMA)	—	COVERED-14	14	Close checklist
Fiori: Run Allocations / Manage Allocations (S/4 universal allocation)	S/4 allocation apps	COVERED-38	38	—
Commitment carry-forward (KOCF / CJCF)	Carry forward open commitments	COVERED-38	38	Encumbrance year-end handling
KO02 status: TECO/close orders (mass via KOK4)	Order status management	COVERED-38	38	—

13.7.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OKKP — Maintain controlling area (activation of components, currency, fiscal year variant, CoA)	TKA00, TKA01	Cost centers, order management, commitments, ProfitAnalysis, PCA active; variance calculation	SUPERSEDED-6	6	No controlling area; features switched per tenant/legal entity
OX19 — Assign company codes to controlling area	TKA02	1:1 or cross-company	SUPERSEDED-6	6	—
KANK — Number ranges for CO documents	NRIV	Business transactions grouped	COVERED-5	5	—
OKEQ — Maintain versions (plan versions, valuation settings)	TKA07, TKA09	Plan/actual, exchange rate type, price calculation settings	COVERED-38	38	Budget versions / scenarios
OKA2 — Cost center categories	TKT05	Activity/actual revenue allowed, quantity	COVERED-6	6	Cost center type attributes
OKEN / OKEON — Standard hierarchy	SETHEADER, SETNODE	—	COVERED-6	6	—
OKB2 / OKB3 / OKB9 — Cost element default (batch input), automatic account assignment	TKA3A	Default cost center per account/company code	COVERED-11	11	Default dimensions per account
Sender/receiver rules for cycles, allocation structures (OKO6), PA transfer structures (KEI1)	T811*, T811C, T811K	Segments, tracing factors (fixed %, portions, SKF, actual costs), iterative, cumulative	COVERED-38	38	Allocation policies: fixed/basis/iterative
OKO7 — Settlement profiles, allocation structures, source structures, PA transfer structure	TKB1, T811*	Receivers allowed (CTR, ORD, FXA, G/L, WBS, PSG), settlement cost elements, amount/percent/equivalence	COVERED-38	38	Settlement rules
KOT2_OPA / KOT2 — Order types (internal orders)	T003O	Order category, number range, settlement profile, planning profile, status profile, release immediately	COVERED-38	38	Cost object types
BS02 — Status profiles (user statuses, allowed transactions)	TJ30	—	COVERED-4	4	Workflow state machines per entity
OKOB — Budget profiles, availability control tolerances	TBP*	Tolerance limits, actions (warning, mail, error), usage %	COVERED-38	38	Budget check tolerances/overrides
KP65 / KP75 / KP85 — Planner profiles, planning layouts	TKA50, T240	—	SUPERSEDED-38	38	Metadata grids
KL01 customizing — Activity type categories (manual, indirect, as needed)	TKA*	Price indicator (plan/actual/manual), allocation cost element	COVERED-38	38	—
Define periodic reposting / assessment cost elements	—	—	COVERED-38	38	—
OKV1 / OKVW — Target cost versions / variance variants for cost centers	TKV01	Fixed/variable variance categories	COVERED-38	38	—
OKTZ (cost component structure) — see CO-PC	—	—	COVERED-39	39	—
Commitments: activate commitment management (OKKP), commitment item mapping	TKA00	—	COVERED-38	38	—
CO versions for parallel valuation (group/profit-center valuation, transfer prices)	TKA07, TKA09, TKA01	Legal/group/PC valuation	PARTIAL-40	40	Transfer-pricing documentation hooks named; parallel group/profit-center valuation views (three valuation views in CO/ML) are not — register as partial for consolidation of profit in inventory

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
ABC: template allocation environments, functions (CPT1)	—	—	PARTIAL-38	38	—
Statistical key figure categories (fixed/totals)	—	—	COVERED-38	38	—
Activity types: allocation structure assignment	—	—	COVERED-38	38	—

13.8 CO-PC — Product Cost Controlling (product cost planning, cost object controlling, material ledger)

SAP context: CO-PC-PCP (cost estimates CK11N, costing runs CK40N, marking/releasing CK24, cost component split OKTZ, costing variants OKKN, costing sheets KZS2), CO-PC-OBJ (order-related: WIP KKAX, variances KKS1/KKS2, settlement CO88; product cost collectors KKF6N for REM; sales-order costing; results analysis KKA1-3), CO-PC-ACT (material ledger/actual costing CKMLCP, price change MR21, CKM3). Passage: wave 20 (valuation, material ledger, MR21), wave 21 (order costing, WIP, settlement), wave 39 (standard cost planning, roll-up, costing sheets, variances), wave 41 (results analysis for projects).

13.8.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
CK11N / CK13N	Create / display material cost estimate with quantity structure	Standard cost estimate from BOM + routing	COVERED-39	39	BOM calculation by costing version/site/qty
CKUC	Multilevel unit costing (costing without quantity structure)	Manual cost estimate	COVERED-39	39	Manual cost estimate lines (unit costing)
CKECP / CKECP1	Easy cost planning / ad-hoc cost estimate	Templates for cost planning	PARTIAL-39	39	What-if simulations named; template-based "easy" cost planning for non-material objects (projects/orders) — partial, map to wave 41 project estimates
CK24	Price update: mark and release standard cost	Activate standard cost → MBEW	COVERED-39	39	Standard cost activation with inventory revaluation posting
CK40N	Costing run (selection, structure explosion, costing, analysis, marking, release)	Mass costing	COVERED-39	39	Mass roll of 50k items < 15 min
CK51N / CK53N	Sales order cost estimate create / display	Costing per sales order item (MTO)	PARTIAL-27	27	Order-specific costing for configured/MTO products not explicitly named in waves 27/39 — add "sales order cost estimate" to wave 39 simulations
CK74N / CK75N	Additive costs create / change	Manual add-ons to cost estimate	COVERED-39	39	Additive cost lines
CK91N / CK92	Procurement alternatives / mixing ratios (mixed costing)	Multiple sources blended	PARTIAL-39	39	Mixed costing (weighted sources: make vs buy, multiple vendors/production versions) not named — add to wave 39 fallback rules
KKF6N / KKF6M / KKF1	Product cost collector create/change (REM) / collective	Cost collector per material/production version	COVERED-24	24	Backflush costing for lean (periodic cost collector)
KKF2 / KKF3	Cost object (general) create/change	Cost object nodes	COVERED-38	38	Cost objects
KKPHIE / KKP2	Cost object hierarchy	—	PARTIAL-38	38	Cost object hierarchies for process cost distribution — partial
KZS2 (master)	Costing sheet maintain	Overhead rates	COVERED-39	39	Costing sheets
KZZ2 / KZB2 / KZM2	Overhead rates (percentage / quantity) / credits	—	COVERED-39	39	Surcharge/rate nodes
KEKO / KEPH (cost estimate tables)	—	Data	COVERED-39	39	Cost estimate entity with component split
MR21	Price change (material)	Standard/MAP price change with revaluation	COVERED-20	20	Cost changes, revaluation
MR22	Debit/credit material (value only)	Inventory value adjustment	COVERED-20	20	Value adjustment journal
CKMPRPUN	Price determination / periodic unit price	—	COVERED-20	20	Periodic weighted average
CKM3 / CKM3N	Material price analysis (ML)	Per-period price history, categories (receipts, consumption, price diffs)	COVERED-20	20	Costing inquiry per item/period
CKMLCP (document view)	Costing cockpit	—	COVERED-20	20	—
CKMM	Change price determination (2 → 3)	ML activation per material	SUPERSEDED-20	20	Costing method is an item model group attribute
CKMLPC / CKMLDM	ML price change / debit-credit	—	COVERED-20	20	—
CKMSTART	ML startup	—	SUPERSEDED-20	20	Always-on inventory ledger
KKAX / KKAO (document)	WIP calculation individual / collective	WIP per order	COVERED-21	21	WIP postings
KKA1 / KKA2 / KKA3	Results analysis (sales order / WBS / order)	POC / revenue-based RA	COVERED-41	41	POC estimates for projects; sales-order RA via wave 29 rev rec
KKS1 / KKS2 (document)	Variance calculation (collective / individual)	Variance categories	COVERED-39	39	Variance decomposition
CKR1	Reorganization of cost estimates	—	SUPERSEDED-39	39	Versioned estimates retained
Fiori: Analyze Costs by Work Center / Production Cost Analysis / Material Price Analysis	S/4	—	COVERED-39	39	—

13.8.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
CK84_99	Itemization of cost estimates	COVERED-39	39	Itemization view
CK80_99 / CK86_99 / CK88_99	Cost components / costed multilevel BOM / partner cost component split	COVERED-39	39	Cost component split by level
CK85_99	Cost estimate comparison	COVERED-39	39	Cost estimate comparison
S_P99_41000111	Analyze/compare material cost estimates	COVERED-39	39	—
S_ALR_87099930 / S_P99_41000112	Material cost estimate list / price analysis	COVERED-39	39	—
KKBC_ORD / KKBC_HOE / KKBC_PKO	Cost analysis: order / hierarchy / product cost collector	COVERED-21	21	Production order cost inquiry (plan vs actual, by cost component)
KKBC_KTR / KKBC_KUN	Cost object / sales order cost analysis	COVERED-29	29	Sales-order profitability
KOB1 (production orders)	Order actual line items	COVERED-21	21	—
S_ALR_87013127 / 87013128 / 87013129	Order selection: target/actual/variance; WIP; variance categories	COVERED-39	39	—
S_ALR_87013139 / 87013140	Cumulative / period-based analysis	COVERED-39	39	—
S_ALR_87013146	Order summarization hierarchy	COVERED-46	46	—
KKR0 / KKRC	Summarization hierarchy (define / run)	SUPERSEDED-46	46	Semantic layer aggregates
CKM3N / CKMVFM	Material price analysis / value flow monitor	COVERED-20	20	—
S_P99_41000062	ML reports: price history, consumption	COVERED-20	20	—
MB5L / MC.9 (value)	Inventory value list (see MM-IM)	COVERED-20	20	—
KKAQ / KKAV	Results analysis reports / WIP reports	COVERED-21	21	—
Variance category reports (lot size / price / quantity / resource usage)	—	COVERED-39	39	—
Fiori: Production Cost Analysis / Analyze Work in Process / Analyze Production Variances	S/4	COVERED-39	39	—

13.8.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
CK40N (periodic costing run)	Monthly/annual standard costing run	COVERED-39	39	—
CK24 (mass release)	Release standard prices	COVERED-39	39	—
KKAX / KKAO	WIP calculation (order / collective)	COVERED-21	21	—
KKS1 / KKS2 / KKS5 / KKS6	Variance calculation (collective / individual; cost collectors)	COVERED-39	39	—
CO88 / KO88 / KK87	Settlement production orders (collective / individual / cost collectors)	COVERED-21	21	Order settlement to inventory/COGS/variance
KGI2 / CO43	Overhead calculation on production orders (individual / collective)	COVERED-39	39	—
KKA1/KKA2/KKA3 (runs), KKAJ / KKAK	Results analysis (collective)	COVERED-41	41	—
CKMLCP	Material ledger costing cockpit (single-level, multi-level, revaluation of consumption, WIP revaluation, post closing)	COVERED-20	20	Periodic actual costing with settlement of issues against receipts
CKMLCPAVR (Alternative valuation run)	Alternative valuation run (cumulative periods)	PARTIAL-20	20	Multi-period cumulative actual costing (AVR) not named — partial
CKMDUVACT / CKMDUVMAT	Distribution of usage variances	PARTIAL-20	20	Usage-variance distribution to consumers — partial
MMPV / MMRV	Close period / allow posting to previous period	COVERED-14	14	—
WIP at actual cost (ML)	—	PARTIAL-20	20	WIP revaluation at actual cost — partial
CKMLCP "post closing" → FI postings	ML closing entries	COVERED-20	20	GL adjustments
KKAO/KKAX scheduling via SCMA	—	COVERED-14	14	Close checklist
Fiori: Schedule Production Accounting Jobs	S/4 jobs	COVERED-4	4	—
Product cost collector periodic settlement (KK87 / CO88 with PKO)	—	COVERED-24	24	—

13.8.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OKKN — Costing variants (costing type, valuation variant, date control, quantity structure control, transfer control)	TCK03, TCK05, TCK06, TCK07	Valuation strategy for materials (plan price 1/2/3, std, MAP, PO price), activities (plan/actual rate), subcontracting, external processing; BOM/routing selection id; transfer existing estimates	COVERED-39	39	Costing versions with fallback rules and explosion modes
OKK4 — Valuation variants	TCK05	Strategy sequences	COVERED-39	39	—
OKKM — Costing types	TCK03	Price update allowed (std/tax/commercial)	COVERED-39	39	—
OKTZ — Cost component structure (components, assignment of cost elements, primary cost split, rollup)	TCKH1..TCKH8	Fixed/variable split, roll-up flag, COGS/inventory relevant	COVERED-39	39	Cost component split / cost groups
OKYD — Cost component views / groups	TCKH*	—	COVERED-39	39	—
KZS2 / KZM2 / KZZ2 / KZB2 / KZE2 — Costing sheets: calculation bases, percentage overheads, quantity overheads, credits, dependencies	TKZS*, T685*	Base cost element groups, overhead keys/groups, dependency (plant, order type), credit cost element and cost center	COVERED-39	39	Costing sheet nodes: surcharge/rate with dependency keys
OKKI / OKOZ — Overhead keys / overhead groups	TKZU*	Per material	COVERED-39	39	—
OPJI / OPJF — Selection ID for BOM/routing in costing	TCK19	Alternatives selection	COVERED-39	39	—

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OKK3 — Date control, quantity structure control	TCK06	—	COVERED-39	39	—
Period lock for ML (OKP1)	—	—	COVERED-14	14	—
OMX1 / OMX2 / OMX3 — Activate material ledger per valuation area; ML types; assign currency types	T001K, TCKM*	Price determination 2/3; currencies (10/30)	SUPERSEDED-20	20	Costing method per item model group; multi-currency valuation covered by wave 12/20
CKMLCP settings: activate actual costing, revaluation of consumption, WIP at actual	TCKM*, MLAUF	—	COVERED-20	20	—
OKG1..OKG9 — Results analysis: RA keys, versions, methods (POC, revenue-based, cost-based), valuation methods, line IDs, assignment, update, posting rules	TKKA*	15 RA methods, WIP/reserves categories	COVERED-41	41	POC methods (project) + rev rec (wave 29)
OKGC / OKGD — WIP calculation: RA keys for production orders	TKKA*	WIP at target vs actual	COVERED-21	21	—
OKV1 / OKVG / OKVW — Variance keys, variance variants, target cost versions	TKV01, TKV02, TKV03	Variance categories (input price, input quantity, resource usage, remaining input, mixed price, output price, lot size, remaining); target cost version basis (current std, plan, alternative)	COVERED-39	39	Variance decomposition categories (purchase price, production lot size/quantity/price/substitution)
OKO7 — Settlement profiles / allocation structures for production orders	TKB1, T811*	Settlement to material (MAT), variance to PA (PSG)	COVERED-21	21	—
OPL8 (order-type parameters: costing variants plan/actual, RA key) — see PP	T399X	—	COVERED-21	21	—
KKF6N settings: REM profile cost collector	—	—	COVERED-24	24	—
Mixed costing: quantity structure types, procurement alternatives (OMXL)	CKMLMV*	—	PARTIAL-39	39	Mixed costing gap
Easy cost planning: costing models, characteristics	CKF*	—	PARTIAL-39	39	—
Sales order costing: requirements class settings (OVZG) valuation/costing	T459K	Valuated/non-valuated sales order stock, costing id, RA key	PARTIAL-27	27	Sales-order stock valuation (valuated MTO stock) belongs to wave 16/21 — not named; partial
Transfer prices / multiple valuation views (OKKP, OKEQ group)	TKA01	—	PARTIAL-40	40	See CO-OM
Cost of goods sold split by cost component (S/4 COGS splitting)	FINSC_COGS*	Split COGS posting into components	PARTIAL-39	39	COGS posting by cost component (material/labor/overhead accounts) not named in wave 28/39 — add posting profile option
Price-difference splitting (S/4)	—	Split PPV by cost component	PARTIAL-39	39	Same

13.9 CO-PA — Profitability Analysis (costing-based, account-based) and EC-PCA Profit Center Accounting

SAP context: CO-PA with operating concern (KEA0), characteristics/value fields, derivation (KEDR), valuation (KE4I), actual postings from SD billing/FI, assessments (KEU5), top-down distribution (KE28), planning (KEPM), reports (KE30). S/4HANA favours account-based ("margin analysis") in ACDOCA with attributed profitability segments. EC-PCA: profit centers (KE51), PC hierarchy, transfer prices, PCA reports (S_ALR_87013326). Passage: profitability = dimensions on the universal journal (wave 11) + semantic layer (wave 46); PCA = profit center dimension + hierarchy (wave 6); allocations wave 38.

13.9.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
KEA0 / KEA5 / KEA6	Maintain operating concern / characteristics / value fields	Define profitability dimensions	SUPERSEDED-11	11	Profitability characteristics are financial dimensions (up to 20) on every journal line; value fields = accounts (account-based model)
KEDR / KEDE	Derivation strategy / rules (derive characteristics)	Characteristic derivation	PARTIAL-11	11	Default dimensions on master data exist; rule-based derivation tables (lookup A×B→C with validity) = substitution gap (see FI-GL)
KE4I / KE4J / KE4IM	Assign SD condition types / MM conditions to value fields	Valuation mapping	SUPERSEDED-29	29	Pricing-condition amounts post to revenue/discount accounts with dimensions; no separate value field mapping
KE4U / KE40	Valuation strategies / costing key (COGS split from cost estimate)	Valuation with cost component split	COVERED-39	39	COGS by cost component (see CO-PC partial for posting)
KE21N / KE23N / KE24	Create line item / display line item / line item list	Manual PA postings & inquiry	COVERED-11	11	Journal with profitability dimensions
KE51 / KE52 / KE53 / KE54	Profit center create/change/display/delete	PC master	COVERED-6	6	Profit center as OperatingUnit type / dimension
KE55 / KE56	Profit center collective / mass assignment to company codes	—	COVERED-5	5	—
KCH1 / KCH2 / KCH3 / KCH5N / KCH6N	Profit center groups / standard hierarchy	—	COVERED-6	6	Hierarchy designer
KE59	Dummy profit center	Fallback PC	SUPERSEDED-11	11	Dimension required-ness rules; no dummy
1KE4 / 1KEK	PCA: transfer postings of receivables/payables	PCA periodic transfer	SUPERSEDED-11	11	Real-time: dimensions on every line (document splitting partial noted)
1KE1 / 1KEE	PC assignment monitor	Check master data assignment to PC	COVERED-6	6	Unassigned-dimension report
9KE0	PCA: manual (statistical) postings	Statistical PC postings	COVERED-11	11	Statistical/memo dimension journals (non-balancing) — treat covered via statistical ledger of wave 38
KEPM	CO-PA planning framework	Sales/profit planning	COVERED-38	38	Planning on dimensions (budget register)
7KE1 / 7KE3	PC planning: costs/revenues, balance sheet items	—	COVERED-38	38	—
KES1 / KES3	Characteristic values maintenance / hierarchies	—	COVERED-6	6	Dimension values & hierarchies

T-code	Name	Purpose	Passage status	Wave	Notes
KEQ3 / KEN1	Segment-level characteristics / number ranges (PA documents)	—	SUPERSEDED-11	11	—
KEDB / KEDU	Delete test data / rebuild summarization levels	—	SUPERSEDED-46	46	Projections/lakehouse
Fiori: Market Segments – Actuals / Product Profitability / Customer Profitability	S/4 margin analysis apps	—	COVERED-46	46	Governed metrics (revenue, margin) by dimension
Fiori: Manage Profit Centers / Manage Profit Center Groups	S/4	—	COVERED-6	6	—

13.9.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
KE30 / KE31 / KE32 / KE33	Execute / create / change / display profitability report (drill-down)	SUPERSEDED-46	46	Semantic-layer explorer and report designer replace KE3x drill-down reports
KE24 / KE25	Line item display actual / plan	COVERED-11	11	—
KE3I / KE35	Report transport / forms	SUPERSEDED-46	46	—
KE4L / KE41 (pricing reports)	—	COVERED-26	26	—
S_ALR_87013326	PCA: plan/actual comparison	COVERED-38	38	—
S_ALR_87013340 / 87013336 / 87013337	PC group: plan/actual; balance sheet by PC; receivables/payables by PC	COVERED-15	15	Dimension statements (document splitting partial noted)
KE5Z / KE5Y	PCA actual / plan line items	COVERED-11	11	—
S_ALR_87013342 / S_ALR_87009712	PC: return on investment / key figures	COVERED-46	46	—
2KEE / 2KET	PC totals records / balance carry-forward display	COVERED-14	14	—
KE3L	Report trees	SUPERSEDED-5	5	Workspace navigation
Fiori: Profit Center – Plan/Actual / Margin Analysis apps	—	COVERED-46	46	—

13.9.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
KEU1 / KEU2 / KEU3 / KEU5	CO-PA assessment cycles (cost centers → segments)	COVERED-38	38	Allocation to profitability dimensions
KE28	Top-down distribution (actual)	PARTIAL-38	38	Top-down distribution of aggregated values (e.g. freight by revenue share) to detailed segments not named — add as allocation type "top-down"
KE27	Periodic valuation (revaluate with actual costs from ML)	PARTIAL-20	20	Revaluation of COGS in profitability with actual cost — tied to ML revaluation of consumption partial
KE2B / KE2C	Subsequent posting of billing documents / FI docs to PA	SUPERSEDED-11	11	Real-time single journal
KE4ST / KE4SMM	Simulate/transfer order settlements to PA	SUPERSEDED-21	21	Settlement posts with dimensions
KEND / KEDR realignment run	Realignment (re-derive characteristics on history)	PARTIAL-11	11	Mass re-derivation of dimensions on posted lines is a controlled restatement; propose explicit "dimension realignment" tool (audit-logged) in wave 11/15
1KEK / 1KE8 / F.5D / F.5E	PCA periodic transfers (payables/receivables, assets, materials)	SUPERSEDED-11	11	—
2KES	PCA balance carry-forward	COVERED-14	14	—
4KE5 / 4KE1-4KE3	PCA distribution/assessment cycles	COVERED-38	38	—
KEPM batch / plan copy (KE1Z)	Plan copy/versions	COVERED-38	38	—
KEDV / KEDB	Summarization levels build (CO-PA)	SUPERSEDED-46	46	—

13.9.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
KEA0 — Operating concern: attributes (currency, fiscal year), data structure, environment	TKEB, CE1xxxx..CE4xxxx	Costing-based vs account-based; company-code currency	SUPERSEDED-11	11	Account-based margin analysis = universal journal dimensions
KEKK — Assign controlling area to operating concern	TKA01	—	SUPERSEDED-6	6	—
KEDR — Derivation rules (table lookup, derivation rule, move, clear, enhancement)	TKEDR*, K9R*	—	PARTIAL-11	11	Derivation gap
KE4I / KE4J / KE4IM / KE4W — Value field assignment, reset of value fields, quantity fields	TKEZU, TKEVA	—	SUPERSEDED-29	29	—
KE4U / KE40 / KE4H — Valuation strategy & costing keys, assignment	TKEVA*, TKEPS	Costing key by material type/plant/point of valuation	COVERED-39	39	COGS split
KEI1 / KEI2 — PA transfer structures (settlement/assessment to PA)	T811*, TKEPC	Cost element groups → value fields	SUPERSEDED-38	38	Allocations target dimensions directly
KEPM planning: planning levels, packages, methods, layouts	TKEPL*	—	SUPERSEDED-38	38	Budget planning
KE3 report customizing: forms, key figure schemes (KER1)	—	—	SUPERSEDED-46	46	—
0KE5 — PCA: controlling area settings (PCA active, dummy PC, elimination of internal business volume)	TKA01, TKA00	—	SUPERSEDED-6	6	—

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
0KEM / 0KEP — PCA substitution rules / derivation of PC from SD (sales org, material), MM	T8A*	—	PARTIAL-11	11	Derivation gap
3KEH / 3KEI — Additional balance sheet/P&L accounts in PCA (classic)	T8A3*	—	SUPERSEDED-11	11	—
8KEM / 8KEN — Transfer prices: pricing procedures, conditions, transfer price variants	T8PT*	PC valuation view	PARTIAL-40	40	Transfer pricing documentation hooks only; PC valuation view not covered
GB02 — Number ranges for PCA documents	NRIV	—	COVERED-5	5	—
S/4: Define attributed profitability segment settings	FINSC_*	Attribution for cost center/order postings	SUPERSEDED-11	11	—
KES3 — Characteristic hierarchies	—	—	COVERED-6	6	Dimension hierarchies

13.10 EC-CS / SEM-BCS / S/4HANA Group Reporting — Consolidation

SAP context: EC-CS (ECC), SEM-BCS (BW-based), S/4HANA Finance for Group Reporting (consolidation units, data monitor, consolidation monitor, reclassifications, IC elimination, consolidation of investments, currency translation, group reports via Fiori/SAC). Passage: wave 40 (group structures, ownership %, eliminations, translation, minority interest, consolidated statements via wave 15).

13.10.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
CX1N (EC-CS) / Fiori: Manage Consolidation Units	Consolidation units, groups, versions	Group structure	COVERED-40	40	Group structures, ownership %
CX1X / Fiori: Manage Group Structure	Consolidation group hierarchy, periods of inclusion/divestiture	—	COVERED-40	40	Ownership with effective dates
CX5S / Fiori: Define FS Items / FS item hierarchies	Financial statement items, breakdown categories, sub-items	Group chart of accounts	COVERED-40	40	Group account structure (wave 15 account structures)
CXCD / Fiori: Data Monitor	Data collection: flexible upload, read from universal journal, manual journal	Data collection	COVERED-40	40	Entity data from the journal; upload for non-Passage entities (wave 10 import)
CX20 / Fiori: Post Group Journal Entries	Manual consolidation entries (document types, posting levels 00/01/10/20/30)	Adjustments	COVERED-40	40	Group adjustment journals at posting levels (entity, group)
CXCE / Fiori: Consolidation Monitor	Run tasks: reclass, IC elim, CoI, translation, validation	Consolidation run	COVERED-40	40	—
Fiori: Manage Reclassifications / Define Reclassification Rules	Rule-based reclassifications	—	PARTIAL-40	40	Elimination rules named; general reclassification rules (e.g. move FS item based on sign) partial
Fiori: Consolidation of Investments (activity-based: first consolidation, subsequent, step acquisition, divestiture; equity method)	CoI	Goodwill, minority interest	PARTIAL-40	40	Minority interest named; purchase-method CoI with goodwill, step acquisitions, divestitures and equity-method accounting are not — partial
Fiori: Manage Investments / Equity data	Investment and equity holdings	—	PARTIAL-40	40	—
Fiori: Define Validation Rules / Run Validations	Data validation before consolidation	—	COVERED-40	40	Validation rules (wave 3 validation + reconciliation)
Fiori: Define Currency Translation Methods	Translation methods, rates per FS item type (closing, average, historical), CTA	—	COVERED-40	40	IAS 21 translation
Fiori: Intercompany Matching & Reconciliation (ICMR)	IC matching at document level, auto-match rules, communication	IC reconciliation	COVERED-40	40	IC matching and reconciliation workspace
Fiori: Manage Substitution/Validation Rules (Group Reporting)	—	—	COVERED-40	40	—
Fiori: Group Data Analysis / Consolidated Balance Sheet / P&L / Cash flow	Group reports with drill to entity	—	COVERED-40	40	Consolidated statements via wave 15 with drill
SEM-BCS: UCWB (workbench), UCMON (monitor)	BW-based consolidation	—	SUPERSEDED-40	40	Native consolidation on the journal/lakehouse
EC-CS: data entry layouts, report writer reports	—	—	SUPERSEDED-40	40	—
Disclosure management / XBRL (SAP DM, SAC)	Statutory filing packages	—	GAP	propose 48	XBRL/ESEF filing output not named; propose electronic reporting designer (wave 48) output type
Fiori: Manage Ledgers for Group Reporting (Y1/Y2), versions, extension versions	Consolidation ledgers/versions	—	COVERED-40	40	Consolidation versions (actual, budget, forecast)

13.10.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
Fiori: Group Data Analysis (with reporting rules, hierarchies)	Consolidated analysis	COVERED-40	40	—
Fiori: Consolidated Balance Sheet / Income Statement / Cash Flow / Changes in Equity	Group statements	COVERED-40	40	Changes in equity statement — add to wave 15 standard statements (minor)
Fiori: Data Analysis – Journal Entries (group)	Group line items	COVERED-40	40	—
Fiori: Group Journal Entry display / task logs	Audit of consolidation run	COVERED-40	40	—
Fiori: Intercompany Reconciliation reports / ICMR dashboards	—	COVERED-40	40	—
EC-CS report writer (database listing)	—	SUPERSEDED-15	15	Report designer
S_ALR_87012 consolidation reports (EC-CS)	—	SUPERSEDED-15	15	—

13.10.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
Data monitor tasks: release universal journal, data validation, currency translation, balance carry-forward	Entity-level close for consolidation	COVERED-40	40	—
Consolidation monitor tasks: IC elimination (AR/AP, revenue/expense, dividends), elimination of IC profit in inventory, CoI, group validation	Group-level tasks	PARTIAL-40	40	IC balances and inventory profit named; dividend elimination and CoI partial
Period lock / task lock (period control for consolidation)	—	COVERED-14	14	Period control per ledger incl. consolidation ledger
Balance carry-forward (group)	—	COVERED-14	14	—
Reclassification runs	—	PARTIAL-40	40	—
Matching run (ICMR) scheduled	—	COVERED-40	40	—

13.10.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Global settings: consolidation chart of accounts, versions, ledgers, fiscal year variant	TF*, FINCS_*	—	COVERED-40	40	—
FS item attributes, breakdown categories, selections	FINCS_FSI*, TF*	Sub-assignments (partner unit, movement type/transaction type, functional area)	COVERED-40	40	Group dimensions incl. partner and movement type (movement types needed for cash flow/equity — confirm wave 40 includes "transaction/movement type")
Document types (posting levels, auto-reversal, deferred tax)	FINCS_DOCTYPE	—	COVERED-40	40	—
Reclassification methods / elimination methods (IC elimination of AR/AP, revenue/expense, dividends)	TF*	Two-sided elimination, differences to FS item, thresholds	PARTIAL-40	40	—
CoI methods (purchase, equity, proportional), activities	TF*	Goodwill treatment, minority	PARTIAL-40	40	—
Currency translation methods, exchange rate indicators, differences	TF*	—	COVERED-40	40	—
Validation rules / reporting rules	FINCS_*	—	COVERED-40	40	—
Data release/collection settings: mapping of operational CoA to group CoA, substitution	FINCS_MAP*	—	COVERED-40	40	Account mapping entity ↔ group
ICMR: matching methods, reconciliation cases, auto-adjustment rules, communication templates	ICA_*	—	COVERED-40	40	—
Group reporting data mapping for external (non-SAP) units (flexible upload formats)	—	—	COVERED-10	10	Import DSL

13.11 FM / PSM — Funds Management & Public Sector Management (budgeting)

SAP context: FM (budget control system BCS: FMBB budgeting workbench, FMAVCR availability control, funds centers FMSA/FMSB, commitment items FMCIA, funds FM5I, FMDERIVE account assignment derivation, earmarked funds FMX1/FMY1, FM reports FMRP), PSM (grants management GM, funded programs, PSCD). Manufacturers rarely activate FM; D365 F&O's budget control is the model Passage follows (wave 38). FM rows map to wave 38 where they are budget control; genuinely public-sector functions are OUT-OF-SCOPE.

13.11.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
FMSA / FMSB / FMSC	Funds center create/change/display	Budget responsibility unit	COVERED-38	38	Budget dimensions (cost center/department)
FMCIA / FMCIC	Commitment item create/display	Budget line category	COVERED-38	38	Budget account categories (main accounts/groups)
FM5I / FM5S	Fund create/display	Funding source	PARTIAL-38	38	"Fund" dimension (restricted/unrestricted funding) not named; grants/projects usually carry it — partial; usable via a financial dimension
FMMEASURE (funded program)	Funded program	—	PARTIAL-38	38	Same
FMBB	Budgeting workbench (BCS): enter, supplement, return, transfer, release, carry-over	Budget documents	COVERED-38	38	Budget register entries with types (original, transfer, supplement)
FMBBC / FMMPDA	Budget mass copy / plan to budget	—	COVERED-38	38	—
FMX1 / FMX2 / FMX3	Earmarked funds: funds reservation create/change/display	Pre-commitment reservation	PARTIAL-38	38	Pre-encumbrance from requisition is named; manual "funds reservation" without a requisition is not — add budget reservation document
FMY1 / FMZ1 / FMW1 / FMV1	Funds precommitment / commitment / forecast of revenue / funds blocking	—	PARTIAL-38	38	Same
FMAVCR01 / FMAVCR02	Availability control: display annual values / overall	Budget consumption	COVERED-38	38	Budget check and consumption inquiry
FMDERIVE	Derivation of FM account assignment	—	PARTIAL-11	11	Derivation gap
FMRP_RFFMEP1AX / FMRP_RFFMEP1GX	FM line items (actuals/commitments)	—	COVERED-38	38	—
GM: GMGRANT, grants budgeting	Grant master, grant budgeting	Grants management	OUT-OF-SCOPE	—	Public-sector/university grants; PRD scope is manufacturers (wave 41 has "customer funding/grants" for projects which covers project funding sources)

T-code	Name	Purpose	Passage status	Wave	Notes
PSCD / FI-CA (public sector collection)	Contract accounts	—	OUT-OF-SCOPE	—	Utilities/public-sector contract accounting
Fiori: Budget Control Overview / Manage Budget Documents	S/4 FM apps	—	COVERED-38	38	—

13.11.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
FMRP_RW_BUDGET / FMRP_RW_BUDCON	Budget overview / budget consumption	COVERED-38	38	—
FMRP_RFFMAV01X / FMAVCR01	Availability control overview	COVERED-38	38	—
FMEDDW	Budget entry documents, totals	COVERED-38	38	—
FM commitments/actuals reports (FMRP_*)	—	COVERED-38	38	—
GM reports (S_ALN_*)	Grants reports	OUT-OF-SCOPE	—	—

13.11.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
FMJ2 / FMJ_SPLIT	Commitment carry-forward (open PO/PR to next year)	COVERED-38	38	Encumbrance year-end handling
FMMC / FMMA	Budget carry-forward (residual)	COVERED-38	38	—
FMAVCREINIT	Reinitialize AVC ledger	SUPERSEDED-38	38	Projection rebuild
FMCG / FMCJ	Close/open budget period (BCS period control)	COVERED-14	14	Period control (budget module)
FMBB release run / mass release	—	COVERED-38	38	Budget workflow
FM reconstruction (FMN3N / FMN4N / FMN5N)	Reconstruct commitments/actuals	SUPERSEDED-38	38	—

13.11.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
FM area definition, assignment to company codes, fiscal year variant	FM01, T001	—	SUPERSEDED-6	6	Budget scope = legal entity + dimensions
Activate FM / BCS	FMAVC*	—	SUPERSEDED-38	38	Budget control activation per entity
Budget types, budget categories, versions (FMBB types)	FMBCS*	Payment/commitment budget	COVERED-38	38	Budget types
Availability control ledgers, tolerance profiles, activation per budget address, filter settings	FMAVC*, FMAVCT	Warning %, error %, derivation of control objects	COVERED-38	38	Budget check tolerances and overrides
Commitment item categories (expense/revenue/balance), financial transactions	FMCI*	—	COVERED-38	38	—
FMDERIVE strategy	FMDERIVE*	—	PARTIAL-11	11	—
Update profile (payment budget vs commitment budget, update dates)	—	—	SUPERSEDED-38	38	—
Earmarked funds document types, number ranges	FMEF*	—	PARTIAL-38	38	—
Grants management customizing	GM*	—	OUT-OF-SCOPE	—	—

13.12 MM-PUR — Purchasing

SAP context: purchase requisitions (ME51N), RFQ/quotations (ME41–ME49), purchase orders (ME21N), outline agreements — contracts (ME31K) and scheduling agreements (ME31L/ME38), info records (ME11), source lists (ME01), quota arrangements (MEQ1), release strategies (classification-based), purchasing conditions (MEK1), vendor evaluation (ME61), output (ME9F), stock transport orders, subcontracting, consignment, third-party. S/4: Fiori Manage Purchase Orders/Requisitions, central procurement, Ariba. Passage: wave 17 (requisitions, RFQ, agreements, POs, pricing, approvals), wave 7 (vendor), wave 20 (supplier portal), wave 33 (vendor evaluation), wave 16 (transfers), wave 40 (intercompany).

13.12.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
ME51N / ME52N / ME53N	Purchase requisition create/change/display	Internal demand	COVERED-17	17	Requisition with policy approvals
ME54N / ME55	Release PR individual / collective	Approval	COVERED-17	17	Approval workflow (Temporal)
ME56 / ME57	Assign source of supply / assign and process (convert to PO)	Sourcing	COVERED-17	17	Consolidation into POs; source determination from agreements/price lists
ME58 / ME59N	Create PO from assigned PRs / automatic PO creation	—	COVERED-17	17	Auto-PO from approved requisitions
ME5A / ME5J / ME5K	PR list display / by project / by account assignment	—	COVERED-17	17	Saved views
ME5F	Release reminder (PR)	—	COVERED-4	4	SLA timers/escalation
ME41 / ME42 / ME43	RFQ create/change/display	Sourcing event	COVERED-17	17	RFQ cycle
ME44 / ME45	Maintain RFQ supplement / release RFQ	—	COVERED-17	17	—
ME47 / ME48	Maintain / display quotation	Supplier reply	COVERED-17	17	Supplier reply portal (wave 20) + manual entry
ME49	Price comparison list	Compare quotes	COVERED-17	17	Compare & award
ME4L / ME4M / ME4N / ME4S	RFQ lists by vendor / material / number / collective number	—	COVERED-17	17	—
ME21N / ME22N / ME23N	Purchase order create/change/display	PO	COVERED-17	17	PO lifecycle with versions
ME21 / ME22 / ME23 (classic)	—	—	SUPERSEDED-17	17	Single PO form

T-code	Name	Purpose	Passage status	Wave	Notes
ME24 / ME25 / ME27	Maintain PO supplement / create PO with vendor unknown / stock transport order (classic)	—	COVERED-17	17	Vendor-unknown PO = requisition
ME28 / ME29N	Release PO collective / individual	Approval	COVERED-17	17	—
ME2O	Subcontracting stock per vendor (SC stock monitor)	Components at subcontractor	COVERED-21	21	Subcontracting with vendor PO linkage; vendor-held inventory location
ME2ON	SC stock monitoring (S/4)	—	COVERED-21	21	—
ME31K / ME32K / ME33K	Contract (quantity/value) create/change/display	Blanket agreement	COVERED-17	17	Purchase agreements with release tracking
ME34K / ME35K	Maintain contract supplement / release contract	—	COVERED-17	17	—
ME31L / ME32L / ME33L	Scheduling agreement create/change/display	SA with delivery schedules	COVERED-17	17	PO delivery schedules + agreements; JIT/forecast schedule lines (ME84/ME38) — see ME84 row
ME34L / ME35L	SA supplement / release	—	COVERED-17	17	—
ME38 / ME39	Maintain / display SA delivery schedule	Schedule lines	COVERED-17	17	Delivery schedules on agreement lines
ME84 / ME85 / ME86	Create SA releases (FRC/JIT) / renumber / display release documentation	Forecast / JIT delivery schedules to supplier (EDI DELFOR/DELJIT)	PARTIAL-17	17	Vendor-facing forecast/JIT release generation (cumulative quantities, release documentation) is a supplier-schedule capability not explicitly named in wave 17; EDI DELFOR/DELJIT in wave 35 lists 850/855/856/810/820/214 only — partial
ME9F / ME9K / ME9L / ME9E	Message output PO / contracts / SA / SA releases	Print/EDI	COVERED-17	17	Supplier output (PDF/email), EDI in wave 35
ME11 / ME12 / ME13 / ME14 / ME15	Purchasing info record create/change/display/changes/flag delete	Vendor–material price & lead time	COVERED-17	17	Trade agreements / vendor price lists per vendor×product×site with lead time; info-record "last PO" updates
ME1L / ME1M / ME1W / ME1P	Info records by vendor / material / material group / order price history	—	COVERED-17	17	—
ME01 / ME03 / ME04 / ME05 / ME06 / ME07	Source list maintain/display/changes/generate/analyze/reorganize	Allowed/fixed sources, MRP-relevant	COVERED-17	17	Approved vendor list per item/site with MRP relevance flag (wave 22 uses it — confirm "fixed source for planned purchase orders")
MEQ1 / MEQ3 / MEQ4 / MEQ6 / MEQ8	Quota arrangement maintain/display/changes/analyze	Split demand across sources	PARTIAL-22	22	Quota arrangements (percentage split across vendors/own production for MRP) not named in wave 17/22 — partial
MEK1 / MEK2 / MEK3 / MEK4	Create/change/display conditions (purchasing) / create with reference	Condition records (PB00, RA01, FRA1)	COVERED-17	17	Vendor pricing conditions (price, discounts, freight) via pricing engine shared with wave 26 — confirm purchasing uses the wave 26 engine (wave 17 says "trade agreements/price lists")
MEKA / MEKB / MEKC / MEKD / MEKE	Condition lists by vendor/material group/material/plant	—	COVERED-17	17	—
MEKP / MEKR / MEKH	Price change conditions mass / by vendor	—	COVERED-26	26	Mass price updates
ME61 / ME62 / ME63 / ME64 / ME65	Vendor evaluation maintain/display/calculate scores/evaluation comparison/ranking list	Vendor scorecards	COVERED-33	33	Supplier quality scorecards (quality, delivery, price, service criteria)
ME6A / ME6B / ME6C / ME6D / ME6E / ME6F / ME6G / ME6H	Vendor evaluation lists, changes, without weighting, log	—	COVERED-33	33	—
MK01 / XK01 (purchasing views)	Vendor purchasing data	—	COVERED-7	7	—
MKVZ / MKVG	Vendor list purchasing / vendor-material	—	COVERED-7	7	—
ME21N item category L (subcontracting)	Subcontract PO with components (BOM explosion)	—	COVERED-21	21	—
ME21N item category K (consignment)	Vendor consignment PO	—	PARTIAL-16	16	Consignment stock (vendor-owned at our site, settle on consumption MRKO) not named in waves 16/18 — partial; needs ownership attribute on inventory + settlement run
ME21N item category S (third-party)	Third-party / drop-ship PO from sales order	—	COVERED-28	28	Drop-ship / direct delivery
ME21N item category U (stock transfer) / UB STO / NB cross-company STO	Stock transport orders (intra/intercompany) with delivery	—	PARTIAL-16	16	Wave 16 "transfer journals" and wave 40 intercompany trade cover the postings; an explicit *transfer order document* with delivery, in-transit stock and receiving (plant-to-plant with shipping) is not named — partial; propose "transfer orders with in-transit status" in wave 16/28
ME21N pipeline (P)	Pipeline materials (utilities)	—	OUT-OF-SCOPE	—	Pipeline handling (no PO, settlement on withdrawal) — niche; can be modeled as consignment later
ME21N service item (D)	Services PO	—	PARTIAL-18	18	See MM-SRV
ME21N limit items / blanket PO (FO)	Framework orders with value limit	—	COVERED-17	17	Blanket agreements; limit PO as order type — confirm
ME2V / ME2W	Goods receipt forecast / vendor delivery forecast	—	COVERED-18	18	Expected receipts
MEMASSPO / MEMASSRQ / MEMASSIN / MEMASSSA	Mass change PO / PR / info record / SA	—	COVERED-5	5	Mass edit
ME80FN / ME80RN / ME80AN	General analyses: PO / RFQ / agreements	—	COVERED-17	17	—
ME91F / ME91A / ME91E	Reminders/expediters (urging) for PO / RFQ / SA	Expediting letters	COVERED-17	17	Expediting output and dunning of overdue confirmations — confirm in wave 17 ("confirmation" named)
ME92F / ME92K / ME92L	Monitor order acknowledgment	Confirmation tracking	COVERED-17	17	PO confirmation; supplier portal acknowledgement (wave 20)
ME97 / ME98	Archive PRs / POs	—	COVERED-48	48	Retention policies
ME2A	Monitor confirmations	—	COVERED-17	17	—
ME23N → "Purchase order history"	PO history (GR, IR, DP)	—	COVERED-18	18	PO line document flow

T-code	Name	Purpose	Passage status	Wave	Notes
Fiori: Manage Purchase Requisitions / Orders / Contracts / Scheduling Agreements / Supplier Quotations / Purchasing Info Records / Source Lists	S/4 apps	—	COVERED-17	17	—
Fiori: Create Purchase Requisition (self-service, catalog, free text)	Self-service requisitioning with catalogs (OCI punch-out)	—	PARTIAL-17	17	Requisition yes; catalog/punch-out (OCI/cXML) item selection not named — partial; propose wave 20 (supplier collaboration) or wave 35 (integration)
Fiori: Monitor Purchase Order Items / Purchase Order Items by Account Assignment	—	—	COVERED-17	17	—
Central procurement (hub-and-spoke, central PR/PO/contract)	Multi-system procurement	—	SUPERSEDED-17	17	Single multi-entity tenant; cross-entity contracts via wave 40
SAP Ariba Buying / Sourcing / Contracts / SLP (supplier lifecycle)	Cloud procurement suite	—	PARTIAL-20	20	Supplier portal (ack, ASN, invoice, docs) covers Ariba Network basics; strategic sourcing events (auctions, weighted RFP scoring), supplier onboarding/qualification questionnaires and contract authoring are not — partial; propose v1.x

13.12.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
ME2L / ME2M / ME2N / ME2K / ME2C / ME2J / ME2B / ME2W	Purchasing documents by vendor / material / number / account assignment / material group / project / requirement tracking / supplying plant	COVERED-17	17	List pages with saved views
ME3L / ME3M / ME3N / ME3K / ME3C / ME3B / ME3J	Outline agreements by vendor / material / number / account assignment / material group / tracking / project	COVERED-17	17	—
ME5A / ME5K / ME5J	Requisitions lists	COVERED-17	17	—
ME80FN / ME80RN / ME80AN / ME81N	General analyses; analysis of order values	COVERED-46	46	—
ME1P	Order price history	COVERED-17	17	—
ME2V	Goods receipt forecast (expected deliveries)	COVERED-18	18	—
ME6H	Vendor evaluation: standard analysis	COVERED-33	33	—
MC$G / MC$I / MC$4 / MC$6 / MC$O / MC$<	PURCHIS: vendor analysis (purchasing values, quantities), material, material group, delivery reliability, PO value	COVERED-46	46	Purchasing analytics in semantic layer
MCE1 / MCE3 / MCE5 / MCE7 / MCE8	PURCHIS: purchasing group / vendor / material group / material / long-term planning analyses	COVERED-46	46	—
MELB	Purchasing transactions per requirement tracking number	COVERED-17	17	—
MSRV1 / MSRV3 (services)	—	PARTIAL-18	18	See MM-SRV
Open POs by vendor / open PO value reports	—	COVERED-17	17	—
ME2L with selection parameter WE101/RECHNUNG	Open GR / open invoices by PO	COVERED-18	18	—
Fiori: Purchase Order Items by Supplier (analytical), Supplier Evaluation by Time/Quantity/Price/Quality, Spend Analysis apps	S/4 analytical apps	COVERED-46	46	—
Fiori: Monitor Supplier Confirmations / Monitor Scheduling Agreement Releases	—	PARTIAL-17	17	SA release documentation partial
RFQ analytics / quotation comparison	—	COVERED-17	17	—

13.12.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
ME59N (batch)	Automatic PO generation from PRs	COVERED-17	17	Scheduled auto-PO
ME84 (batch)	SA release creation run	PARTIAL-17	17	SA release partial
ME9F (batch) / RSNAST00	Output processing (print/EDI)	COVERED-17	17	Output jobs; wave 35 EDI
ME91F (batch)	Expediting run	COVERED-17	17	—
ME63 / ME6G (batch)	Vendor evaluation scoring run	COVERED-33	33	—
MEKP / MEKPE (batch)	Price change mass run	COVERED-26	26	—
ME07	Source list reorganization	SUPERSEDED-17	17	Effective dating
ME97 / SARA (MM_EKKO)	Archiving POs	COVERED-48	48	Retention
ME2K (commitments)	Commitment update	COVERED-38	38	Encumbrance from PO
MRKO (consignment settlement)	—	PARTIAL-16	16	Consignment gap
Contract release value update / expiry monitoring (ME3M with validity)	Contract expiry alerts	COVERED-17	17	Agreement validity alerts
MEI1 / MEI2 / MEI4	Subsequent price change (automatic document adjustment after condition change)	PARTIAL-17	17	Retroactive re-pricing of open POs after condition change — not named; partial

T-code	Name	Passage status	Wave	Notes
Fiori: Schedule Purchasing Jobs	Job templates	COVERED-4	4	—

13.12.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OX08 / OX01 / OX17 — Purchasing organizations; assign to company code / plant; reference purch. org	T024E, T024W, T024Z	Plant-specific, cross-plant, cross-company, reference purch org	COVERED-6	6	Procurement hierarchy purpose in enterprise structure
OME4 — Purchasing groups	T024	—	COVERED-6	6	Buyer groups
OMEC / OMEB / OMEE / OMED — Document types PO / RFQ / contract / SA; number ranges; allowed item categories; link PR↔PO	T161, T161A, T161P	Field selection, control indicators (GR/IR required), time-dependent conditions, release groups	COVERED-17	17	Order types with number sequences and allowed line types
OMH7 / OMH6 — Number ranges for purchasing documents	NRIV	—	COVERED-5	5	—
OMF4 — Screen layout at document level / field selection keys	T162, T162K	Required/optional per field selection key (e.g. ME21N, NBF, AKTH)	COVERED-3	3	Metadata field attributes by order type
OMET — Function authorizations for buyers (value limits, order types)	T160B	—	COVERED-2	2	OPA policy on amounts/order types
Item categories / account assignment categories (OME9)	T163, T163K, T163B	Account assignment (K, F, P, A, C, Q), GR/IR indicators, consumption posting	COVERED-17	17	Account assignment line types; project (wave 41), asset (wave 37)
OMGQ / OMGS / OMGSCK — Release procedure for POs (with classification) / PRs (with/without classification): release groups, codes, indicators, strategies, prerequisites, workflow	T16FG, T16FC, T16FS, T16FK, T16FW, T16FV	Classification characteristics (value, plant, group), release simulation, workflow assignment	SUPERSEDED-17	17	Policy-driven approval chains (amount bands, category, site) configured in the wave 4 workflow designer; no classification system needed
OMKK — Price variance tolerance limits for PO vs. PR / info record (PE, SE keys)	T169G	Percentage/absolute, warning/error	COVERED-17	17	Price tolerance on PO vs agreement
OMFI / OMFJ — Version management for PR / PO (reasons, fields)	T160VS, T160VSK	Version-relevant fields, reasons	COVERED-17	17	Versioned change management
M/03 … M/08 — Purchasing condition tables / access sequences / condition types / calculation schema (RM0000, RM1000), schema groups for vendor/purchasing org, schema determination	T681*, T682*, T685, T683	Condition types PB00/PBXX/RA00/FRA1/SKTO, header/item, group conditions, time-dependent	SUPERSEDED-26	26	Shared pricing engine (condition tables/access sequences) — wave 26 engine used by purchasing; wave 17 price lists
Price determination process / condition control at plant level	T001W	—	COVERED-17	17	—
Default values for buyers (user parameters EVO/EFB)	T160, TMGU	—	COVERED-5	5	User defaults / personalization
Texts for purchasing documents, text types, copying rules	T165*, TTXID	—	COVERED-17	17	Document text blocks (header/line notes)
Partner roles in purchasing (OA ordering address, PI invoicing party, GS goods supplier), partner schemas	TPAR, TPAER	—	COVERED-7	7	Party roles & document partners
Reasons for ordering, reasons for rejection (RFQ)	T161*	—	COVERED-17	17	Reason codes
Confirmation control (confirmation control keys, categories AB/LA)	T163D, T163G	Confirmation sequences, GR-relevant, MRP-relevant	COVERED-17	17	Confirmation + ASN (wave 20/28)
OMGN — Set up stock transport order (delivery type, checking rule per supplying plant)	T161V, T161W	STO with/without delivery, one-step/two-step	PARTIAL-16	16	Transfer order gap
Subcontracting settings (SC vendor stock types)	T156*	—	COVERED-21	21	—
Vendor evaluation — weighting keys, criteria, scoring methods, points scores	T147*, T147A..T147M	Main/sub criteria, automatic (price, GR deviation, quality, QM) vs manual	COVERED-33	33	Scorecard criteria
Source determination — source list requirement per plant, quota arrangement rules	T001W, TMQ1	—	PARTIAL-22	22	Quota partial
Message determination for purchasing (NACE EF/EV/EL/EA), condition tables, output types NEU, print programs, EDI partner profile	NAST, TNAPR, T685B	Medium (print, fax, EDI, email), timing	COVERED-17	17	Output rules + wave 35 EDI; see cross-application
Default document types in requisition creation from MRP (OPPI / OMD)	—	—	COVERED-22	22	Planned purchase order → requisition/PO conversion settings
Account assignment — automatic account determination (OBYC KBS)	T030	—	COVERED-16	16	—
OMEE — Contract document types, target value, release order documentation	T161	—	COVERED-17	17	—
Scheduling agreement — release creation profile, firm zone/trade-off zone, JIT indicator	T163M	—	PARTIAL-17	17	SA release partial
Taxes in purchasing (tax code default per vendor/plant, MEK1 NAVS)	T007A	—	COVERED-13	13	Tax determination on PO
Incoterms / terms of delivery (OVIM)	TINC	—	COVERED-7	7	Delivery terms
Purchasing value keys (reminder days, tolerances, order acknowledgment)	T405	Under/over-delivery tolerances, reminders 1/2/3	COVERED-17	17	Item defaults for expediting and delivery tolerances
Material groups (OMSF) — see MM master data	T023	—	COVERED-8	8	Product categories
Authorization groups / field selection per activity (classic)	—	—	COVERED-2	2	—
Flexible workflow for PR/PO/contract approval (S/4 SWDD_SCENARIO / Manage Workflows app)	SWF_*	Step conditions, agent rules	COVERED-4	4	Workflow designer
Ariba Network integration (cXML, Cloud Integration Gateway)	—	—	PARTIAL-35	35	cXML not listed among EDI formats (X12/EDIFACT/Peppol); partial

13.13 MM-IM — Inventory Management & Physical Inventory

SAP context: material documents (MKPF/MSEG; S/4 MATDOC), goods movements via MIGO (receipts, issues, transfers, reversals, 100+ movement types), reservations (MB21), stock overview (MMBE), stock types (unrestricted, quality, blocked, in transit, consignment, subcontractor, sales order stock, project stock), physical inventory (MI01–MI20, cycle counting MICN), period close (MMPV). Passage: wave 16 inventory ledger, wave 18 receiving, wave 21 production movements, wave 28 shipping movements, wave 31 warehouse detail.

13.13.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
MM01 / MM02 / MM03 (IM views: general plant, storage, accounting)	Material master — plant/storage/valuation views	Item per site/warehouse	COVERED-8	8	Released product per legal entity with site/warehouse defaults; item model group (costing, negative stock)
MIGO	Goods movement (GR for PO/order/other, GI, transfer posting, place in storage, remove from storage, return delivery, subsequent delivery/adjustment, cancel, display)	Universal movement UI	COVERED-16	16	Inventory journals (movement/transfer/adjustment) + receiving (wave 18) + production issues/receipts (wave 21)
MB1A / MB1B / MB1C (classic)	Goods issue / transfer posting / other receipt	—	SUPERSEDED-16	16	Journals by type
MB01 / MB0A / MB31 (classic)	GR for PO / GR for PO unknown / GR for production order	—	SUPERSEDED-18	18	Receiving (wave 18) / RAF (wave 21)
MB11 (classic)	Goods movement (general)	—	SUPERSEDED-16	16	—
MB02 / MB03	Change (texts) / display material document	—	COVERED-16	16	Inventory transaction inquiry
MBST / MBSU	Cancel material document / place in storage for mat doc	Reversal	COVERED-16	16	Reversal of inventory transaction with linked negating movement
MB21 / MB22 / MB23 / MB24 / MB25	Reservation create/change/display/list by material/by account assignment	Manual reservations	COVERED-16	16	Manual/auto reservations
MBVR	Manage reservations (delete/close)	—	COVERED-16	16	—
MB90	Output of material documents (GR slip, labels)	—	COVERED-32	32	Label printing (ZPL) via print service; GR slips via document designer (wave 46)
MMBE	Stock overview (by plant/SLoc/batch/special stock)	On-hand inquiry	COVERED-16	16	On-hand by dimension
MB52	List warehouse stocks on hand	—	COVERED-16	16	—
MB53 / MB54	Plant availability / consignment stock	—	PARTIAL-16	16	Vendor consignment gap
MB58	Consignment and returnable packaging stock at customer	Customer consignment	PARTIAL-27	27	Customer consignment (fill-up/issue/pick-up/return orders) not named in SD/inventory waves — partial; needs "stock at customer" ownership dimension
MBLB	Stocks at subcontractor	—	COVERED-21	21	—
MMSC	Collective entry of storage locations (extend material)	—	COVERED-8	8	Release to warehouse mass
MI01 / MI02 / MI03	Physical inventory document create/change/display	Count documents	COVERED-16	16	Count sheets
MI04 / MI05 / MI06	Enter count / change count / display count	—	COVERED-16	16	Count entry (mobile in wave 32)
MI07 / MI08 / MI10	Post differences / post with count / post document without reference	Variance posting	COVERED-16	16	Variance posting to GL
MI09	Enter count without reference document	—	COVERED-16	16	—
MI11	Recount	—	COVERED-16	16	—
MI20	Difference list	—	COVERED-16	16	—
MI21 / MI22 / MI23 / MI24	Print PI documents / display PI docs for material / display PI data for material / difference list	—	COVERED-16	16	—
MI31 / MI32 / MI33 / MI34 / MI35 / MI37 / MI38 / MI39 / MI40	Batch create PI documents (all stocks / special stocks / enter count via batch input / post differences via batch)	Mass PI	COVERED-16	16	Cycle counting plans + mass count generation
MICN	Create PI documents for cycle counting	ABC-based cycle counting	COVERED-16	16	Cycle counting plans (ABC classification MIBC)
MIBC	ABC analysis for cycle counting (set CC indicator)	—	COVERED-16	16	—
MIK1 / MIQ1 / MIM1	PI for vendor consignment / project stock / returnable transport packaging	Special stock PI	PARTIAL-16	16	Consignment/RTP gaps
MIE1 / MIW1 / MIV1	PI for sales order stock / customer consignment / returnable packaging at customer	—	PARTIAL-27	27	Sales-order stock & customer consignment partial
MIS1 / MIS2 (inventory sampling)	Inventory sampling (statistical)	—	GAP	propose 16	Statistical inventory sampling (count a sample, extrapolate) — rarely used but a recognized GAAP-compatible method; low priority
OMJJ (movement types) — see customizing	—	—	COVERED-16	16	—
Special stock: sales order stock (E), project stock (Q)	Stock segregated by SO/WBS	—	PARTIAL-16	16	Sales-order/project-specific stock (valuated/non-valuated) not explicitly named as an inventory dimension — partial; needed for MTO/ETO manufacturers (waves 21/27/41)
Stock in transit (MB5T; S/4 stock in transit for STO)	In-transit stock	—	PARTIAL-16	16	Transfer order gap
Returnable transport packaging (M, V)	RTP at vendor/customer	—	GAP	propose 28	Returnable packaging (pallets, crates) tracking at customers/vendors not named; propose wave 28 (packaging/HU) option
Goods receipt blocked stock (103/105)	GR to blocked, release	—	COVERED-18	18	Arrival/registration vs product receipt posting (two-step receiving)
Fiori: Post Goods Movement / Post Goods Receipt for PO / Transfer Stock – In Plant / Cross-Plant / Stock – Single Material / Multiple Materials / Manage Stock	S/4 apps	—	COVERED-16	16	—

T-code	Name	Purpose	Passage status	Wave	Notes
Fiori: Manage Physical Inventory Documents / Process Physical Inventory Count / Overview of Count Results	—	—	COVERED-16	16	—
Fiori: Manage Material Documents / Material Documents Overview	—	—	COVERED-16	16	—

13.13.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
MB51	Material document list (by material/plant/movement type/date)	COVERED-16	16	Inventory transactions inquiry
MB52	Warehouse stocks on hand (with values)	COVERED-16	16	—
MB5B	Stocks on posting date (period opening/closing, movements)	COVERED-16	16	On-hand as-of date from immutable transactions
MB5L	List of stock values: balances (vs GL)	COVERED-20	20	Inventory value report reconciling to GL
MB5T	Stock in transit (cross-company/plant)	PARTIAL-16	16	Transfer order gap
MB5M	Expiration date list (shelf life)	COVERED-24	24	Shelf life / expiry (batch attributes)
MB5W	List of stock values (period)	COVERED-20	20	—
MB5S	GR/IR balances per PO	COVERED-18	18	—
MB5C	Pick-up list for batch where-used	COVERED-24	24	Batch traceability
MB54 / MB58 / MBLB	Consignment / customer consignment / subcontractor stock	PARTIAL-16	16	Consignment gaps
MBBS	Valuated special stock (sales order/project)	PARTIAL-16	16	Special stock gap
MBSM	Cancelled material documents	COVERED-16	16	—
MMBE	Stock overview	COVERED-16	16	—
MMBE_OLD	Legacy stock overview	SUPERSEDED-16	16	—
MC.1 / MC.2 / MC.3 / MC.4 / MC.5 / MC.6 / MC.7 / MC.8 / MC.9 / MC.A / MC.B / MC.C / MC.D / MC.E	INVCO stock analyses: plant / storage location / material / MRP controller / business area / by receipts/issues/stock value/ranges of coverage/inventory turnover/slow-moving	COVERED-46	46	Inventory analytics (turns, coverage, slow-moving) in semantic layer; MC.9 current stock/values also in wave 20 value report
MC50 / MC46 / MC44 / MC45	Dead stock / slow-moving items / inventory turnover / usage values	COVERED-46	46	—
MC48 / MC49	Stock value: current / mean stock value	COVERED-46	46	—
MMPV (log) / MMRV	Period close status	COVERED-14	14	—
MI20 / MI24	PI difference list	COVERED-16	16	—
S_P00_07000139 / S_P00_07000140	Stock card and inventory list (country-specific)	COVERED-16	16	—
MB5OA / MB5U	Stock in open PO / consumption	COVERED-17	17	—
MBGR	Material documents by reason for movement	COVERED-16	16	Reason codes
Fiori: Stock – Multiple Materials / Inventory Turnover Analysis / Slow or Non-Moving Materials / Dead Stock Analysis / Goods Movement Analysis / Inventory Aging	Analytical apps	COVERED-46	46	—
Fiori: Material Documents Overview / Reservations overview	—	COVERED-16	16	—

13.13.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
MMPV	Close period for material master records	COVERED-14	14	Period control (inventory)
MMRV	Allow posting to previous period	COVERED-14	14	Period status on-hold/open per module
MI31 → MI21 → MI04 → MI07 (PI cycle)	Annual physical inventory	COVERED-16	16	—
MICN (scheduled)	Cycle count document generation	COVERED-16	16	—
MIBC (periodic)	ABC re-classification	COVERED-16	16	—
MB5K	Stock consistency check (MARD vs MSEG)	SUPERSEDED-16	16	On-hand projection rebuild-and-compare proof
MBVR	Reservation cleanup	COVERED-16	16	—
MRKO	Consignment & pipeline settlement	PARTIAL-16	16	Consignment gap
MB5L (month-end)	Inventory to GL reconciliation	COVERED-20	20	Daily automated
CKMLCP / MR21 (see CO-PC)	—	COVERED-20	20	—
SARA MM_MATBEL	Archiving material documents	COVERED-48	48	Retention + lakehouse history (wave 46)
Subcontracting component provision (541) via ME2O	—	COVERED-21	21	—
Fiori: Schedule Physical Inventory Document creation	—	COVERED-4	4	—

13.13.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OX10 / OX09 — Plants / storage locations	T001W, T001L	Address, factory calendar, purchasing org assignment	COVERED-6	6	Sites, warehouses
OX18 / OX14 — Assign plant to company code / valuation area setting (plant vs company code)	T001K	Valuation level	COVERED-6	6	Valuation at site level (default) — confirm valuation level choice

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OMJJ — Movement types: copy, define, allowed fields, update control (quantity/value), reversal movement, reason codes, account grouping (GBB modifiers), screen layout, allowed transactions	T156, T156B, T156S, T156X, T156M, T156W, T158B	100+ movement types; auto-created vs custom	SUPERSEDED-16	16	Inventory transaction types are a fixed typed set (receipt, issue, transfer, adjustment, count, RAF, consumption …) with posting profiles per item model group; custom "movement types" become journal types with configurable posting profiles
OMBS — Reason for movement	T157D	Reason codes per movement type	COVERED-16	16	Reason codes on journals
OMBT — Number ranges for material documents / PI documents / reservations	NRIV	—	COVERED-5	5	—
OMJ1 — Allow negative stocks (per valuation area / storage location)	T001K, T001L	—	COVERED-16	16	Negative-inventory policy per item model group/warehouse
OMJX / OMJR — Field selection for MIGO / print control	T158, T159P	—	COVERED-3	3	Metadata
OMJ5 — Shelf life expiration date check (per plant / movement type)	T159L, T156	Minimum remaining shelf life check at GR	COVERED-24	24	Shelf life checks on receipt
OMC0 — Tolerance limits for GR (under/over delivery, B1/B2 keys)	T169G	B1/B2 keys	COVERED-18	18	Receipt tolerances
OMBW — Screen layout per movement type (field selection)	T156B	—	COVERED-3	3	—
OMCJ / OMCH — Default values for goods receipt (stock type default: quality/blocked), GR slip print	T158, T159	—	COVERED-18	18	Receive-to-quarantine by item/vendor (wave 33 QM triggers)
OMJD — Automatic storage location creation on movements	T001L	—	COVERED-16	16	—
Stock types / GR blocked stock settings	—	—	COVERED-16	16	Inventory status (available, quarantine, blocked, in-transit)
Batch / serial checks in IM (see batch section)	—	—	COVERED-16	16	—
OMWB / OBYC — Account determination (valuation grouping code, valuation class, account modifier, transaction key)	T030, T025, T001K, T030W	BSX inventory, GBB offsets (VBR consumption, AUF order, VAX/VAY COGS, ZOB other, INV PI differences), PRD price differences, WRX GR/IR, UMB revaluation, FRL external processing, BSV subcontracting, KDM FX	COVERED-16	16	Inventory posting profiles by item model group × site × transaction type
OMWD / OMWC / OMSK — Valuation grouping code / split valuation (valuation categories/types) / valuation classes & account category reference	T001K, T025, T025K, T149*	Split valuation by origin/batch (valuation type)	PARTIAL-20	20	Split valuation (different values per valuation type, e.g. new vs refurbished, domestic vs import) not named — partial; batch-level valuation needed for process industries
OMW0 / OMW1 — Price control (standard/moving average) per material type / valuation area	T134, T001K	V/S, price control mandatory	COVERED-20	20	Costing method on item model group
OMS2 — Material types (quantity/value update per valuation area)	T134, T134M	Quantity/value updating, allowed views, number range	COVERED-8	8	Product types (stocked/non-stocked/service) + item model group flags (physical/financial posting)
OMSL / OMSF — Material number format / material groups	TMCNM, T023	Length, leading zeros, lexicographic	COVERED-8	8	Product number sequences; categories
OMSR / OMS9 — Field selection material master (field references, screen sequences)	T130F, T133*	—	COVERED-3	3	Metadata
OMT3B / OMT3E — Material master screen sequence / data screens	T133K	—	SUPERSEDED-5	5	Metadata-driven forms
OMSY — Fiscal year/period per company code (material periods)	MARV	Current period, allow previous	COVERED-14	14	—
OMBG / OMJN — Physical inventory settings: default values, tolerances per user group (posting differences max), cycle counting indicators (OMCO) and intervals, inventory sampling	T159*, T043I, T159C	ABC thresholds, count intervals, float days	COVERED-16	16	Cycle counting plans; variance posting limits via OPA
OMJ2 / OMJ3 — Print settings for GR/GI slips, labels	T159*, T159P	—	COVERED-46	46	Document designer / label printing (wave 32)
OMCQ — System messages (warning/error) for inventory management	T100C	Make messages errors/warnings	SUPERSEDED-3	3	Validation rules with severity in metadata
OMJA — Movement type statistics relevance (LIS)	T156S	—	SUPERSEDED-46	46	Lakehouse
Special stock settings (consignment price determination, pipeline)	—	—	PARTIAL-16	16	Consignment gap
OMCT — batch level (see batch)	—	—	COVERED-8	8	—
OMCP — Dynamic availability check per movement type	T156, T441V	Check rule per movement	COVERED-16	16	ATP/availability policy on issues (no negative unless allowed)
Stock determination (OSPX / OSPW) strategies for issues (consignment vs own, SLoc priority)	—	—	PARTIAL-16	16	Picking policies FEFO/FIFO named; ownership-based stock determination partial (consignment)

13.14 MM-IV — Logistics Invoice Verification

SAP context: MIRO (PO-based, delivery-note based, GR-based IV, planned/unplanned delivery costs, subsequent debit/credit), parking (MIR7), blocked invoices (MRBR; price/quantity/date/stochastic blocks), tolerance keys (OMR6), GR/IR clearing maintenance (MR11), ERS (MRRL), consignment settlement (MRKO), invoicing plans (MRIS), revaluation (MRNB), duplicate invoice check, OBYC WRX/PRD/KDM. Passage: wave 18 (capture, 2/3-way match, tolerances, exceptions, GR/IR, ERS), wave 19 (AP open items), wave 34 (freight invoices/landed cost).

13.14.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
MIRO	Enter incoming invoice (invoice, credit memo, subsequent debit/credit; PO reference, delivery note, bill of lading, service entry sheet, vendor)	PO-based invoice	COVERED-18	18	Vendor invoice capture with PO/receipt reference; subsequent debit/credit = price adjustment invoice
MIR7 / MIR4	Park invoice / display invoice document	Draft	COVERED-18	18	Draft + approval
MIR6	Invoice overview (parked, held, blocked, posted)	Worklist	COVERED-18	18	AP invoice workspace
MR8M	Cancel invoice document	Reversal	COVERED-18	18	Reversal with inventory/GR-IR re-open
MRBR	Release blocked invoices (price/qty/date/manual block; automatic release)	Exception resolution	COVERED-18	18	Match exception workflow and release
MR11 / MR11SHOW	GR/IR clearing account maintenance (write off small balances) / display	GR/IR cleanup	COVERED-18	18	GR/IR reconciliation report + write-off
MRRL	Evaluated receipt settlement	Self-billing	COVERED-18	18	ERS
MRKO	Consignment and pipeline settlement	—	PARTIAL-16	16	Consignment gap
MRIS	Invoicing plan settlement (periodic/partial invoicing plans on PO)	Recurring PO invoicing (rent, leases)	PARTIAL-17	17	Invoicing plans on POs (periodic settlement without GR) not named — partial; relevant to leases/services; propose wave 17 PO line "invoicing plan"
MRNB	Revaluation (retroactive price change on invoiced items)	Retro-billing by vendor	PARTIAL-17	17	Retro re-pricing partial (MEI1 row)
MRDC	Automatic delivery cost settlement (freight, customs via planned delivery costs)	Planned delivery costs settlement	COVERED-34	34	Freight reconciliation + landed cost
MIRA	Enter invoice for background verification	Batch invoice entry	COVERED-18	18	Bulk capture/OCR queue
MR01 / MR02 / MR03 / MR41 (classic)	Classic invoice verification	—	SUPERSEDED-18	18	—
MR90	Output messages for invoices (ERS printouts)	—	COVERED-18	18	ERS self-billing document output
MIR5	Display list of invoice documents	—	COVERED-18	18	—
Invoice reduction (complaint document)	Invoice reduction with vendor complaint letter	—	COVERED-18	18	Short-pay / invoice reduction with dispute to vendor — treat covered under exception workflow
GR-based IV flag (PO item)	Match per GR document	—	COVERED-18	18	3-way at receipt-document level
Unplanned delivery costs	Distribute across items or post to separate account	—	COVERED-34	34	Landed cost allocation
Duplicate invoice check (vendor flag + OMRDC)	Prevent duplicates (ref no, date, amount)	—	COVERED-18	18	Duplicate detection rules
Withholding tax on MIRO	—	—	COVERED-13	13	—
Fiori: Create Supplier Invoice / Manage Supplier Invoices / Release Blocked Invoices / Clear GR/IR Clearing Account / Supplier Invoices List	S/4 apps	—	COVERED-18	18	—
Fiori: Upload Supplier Invoices / Supplier Invoice from OCR (Central Invoice Management, Business Network)	—	—	COVERED-18	18	OCR adapter; e-invoice (wave 35)

13.14.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
MIR5 / MIR6	Invoice document list / overview	COVERED-18	18	—
MB5S	GR/IR balances per PO	COVERED-18	18	—
MR11SHOW	GR/IR maintenance documents	COVERED-18	18	—
MRBR (list)	Blocked invoices list	COVERED-18	18	—
Invoice register (AP)	Invoice register	COVERED-18	18	—
ME2L (RECHNUNG)	Open invoice quantities by PO	COVERED-18	18	—
F.19	GR/IR analysis	COVERED-18	18	—
PURCHIS invoice values	Invoice value analysis	COVERED-46	46	—
Fiori: Invoice Cycle Time / Touchless invoice rate KPIs	—	COVERED-46	46	—

13.14.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
MRBR (automatic release in batch)	Release blocked invoices automatically when block reasons no longer apply	COVERED-18	18	Re-evaluate match status on new receipts
MRRL (scheduled)	ERS run	COVERED-18	18	—
MRKO (scheduled)	Consignment settlement	PARTIAL-16	16	—
MRIS (scheduled)	Invoicing plan run	PARTIAL-17	17	—
MRNB (scheduled)	Revaluation run	PARTIAL-17	17	—
MRDC (scheduled)	Delivery cost settlement	COVERED-34	34	—
MR11 (periodic)	GR/IR clearing	COVERED-18	18	—
F.19 (periodic regroup)	GR/IR balance sheet regrouping	COVERED-18	18	—
SARA MM_REBEL	Archive invoice documents	COVERED-48	48	—
Fiori: Schedule Supplier Invoice Jobs	—	COVERED-4	4	—

13.14.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OMR6 — Tolerance limits (keys AN, AP, BD, BR, BW, DQ, DW, KW, LA, LD, PP, PS, ST, VP) per company code	T169G	Absolute/percent upper/lower, check/no check; price variance (PP), quantity (DQ), date (ST), blanket PO (LA/LD), order price unit (PS), moving average (VP)	COVERED-18	18	Price/quantity/total/date tolerances per legal entity and vendor group
OMR4 / OMRJ — Document types for invoice verification (RE, KG) / number ranges	T003, T169F, NRIV	—	COVERED-11	11	—
OMRDC — Duplicate invoice check settings	T169P	Check company code, reference, date	COVERED-18	18	—
OMRM — Activate direct posting to G/L and material accounts in MIRO	T169P	—	COVERED-18	18	Non-PO lines on PO invoices
Item amount check (manual block by amount)	T169P	—	COVERED-18	18	Approval threshold
OMRX — Stochastic block (random sampling thresholds)	T169P	—	GAP	propose 18	Random audit sampling of invoices — simple to add to wave 18 exception rules; register as gap
OMRN — Maintain default values for tax codes (MIRO)	T169V	—	COVERED-13	13	—
Vendor-specific tolerances (tolerance groups for vendors: small differences, automatic acceptance)	T169*, T043*	—	COVERED-18	18	Vendor-group tolerances
OBYC — WRX, PRD, KDM, DIF for IV	T030	—	COVERED-18	18	—
ERS settings: vendor flag, document type, tax code default, message type ERS	LFM1, T169*	—	COVERED-18	18	—
Invoice reduction: document types, reason codes	T169*	—	COVERED-18	18	—
Define payment block reasons via IV	T008	—	COVERED-19	19	—
Invoicing plan types (periodic/partial)	T163*	—	PARTIAL-17	17	Invoicing plan gap
Delivery costs: condition types FRA1/FRB1/ZOTH, provision accounts (FR1/FR3 keys)	T030, T685	—	COVERED-34	34	Landed cost charge codes with accrual accounts
Flexible workflow for supplier invoice approval (S/4)	SWF_*	—	COVERED-4	4	—
Default values for IV (unplanned delivery costs distribution)	T169P	—	COVERED-34	34	—
GR/IR regrouping accounts	T030	—	COVERED-18	18	—

13.15 MM-SRV — External Services Management (incl. S/4 Lean Services)

SAP context: service master (AC03), model service specifications (ML10), standard service catalog (SSC), service POs (item category D with service lines and value limits), service entry sheets (ML81N) with acceptance → GR, service conditions (ML33/ML39), invoice via MIRO referencing SES. S/4 "lean services" simplifies: service items as product type Service, no SES (Fiori Manage Service Entry Sheets – Lean). Passage: non-stock service products (wave 8), PO (wave 17), service confirmation/receipt (wave 18) — **no explicit service entry sheet**, so mostly PARTIAL.

13.15.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
AC01 / AC02 / AC03 / AC04 / AC06	Service master create/change/display/list/changes	Service master (activity numbers, UoM, valuation class)	PARTIAL-8	8	Product of type "service" is covered; a dedicated service master with service categories, purchasing text, standard service catalog is partial — add "service product type" attributes to wave 8
ML10 / ML11 / ML12	Model service specifications create/change/display	Reusable service spec templates	GAP	propose 17	Service specification templates (hierarchical outlines of services with quantities) not named; propose "PO line templates" in wave 17
Standard service catalog (SSC)	Standard texts for services	—	GAP	propose 17	Same
ME21N (item cat D) with service lines, value limits, contract limits	Service PO with hierarchical service outline and limits	—	PARTIAL-17	17	Limit/blanket POs named; service outlines (hierarchy of service lines under one PO item) and contract-limit references are partial
ML81N / ML82 / ML83 / ML84 / ML85	Service entry sheet create-maintain / display / list / accept-revoke / collective release	Service performance confirmation → GR	PARTIAL-18	18	Receiving covers "product receipt"; a *service entry sheet* (lines of performed services with quantities/prices, acceptance workflow, partial entries, unplanned services within limits) is not named — partial; add "service receipt/entry sheet with acceptance" to wave 18
ML33 / ML34 / ML35 / ML36 / ML39 / ML40 / ML41 / ML45 / ML46 / ML47 / ML48	Service conditions: vendor-level / service-level / vendor+plant	Service pricing	PARTIAL-17	17	Vendor price lists for service products via pricing engine — partial until service products confirmed
MSRV1 … MSRV6	Service lists: by service / by PO / by SES / by vendor / by contract / entry sheets	—	PARTIAL-18	18	—
MIRO with SES reference	Invoice for services (SES-based)	—	PARTIAL-18	18	Match to service receipt
ML91 / ML92 / ML93	Service entry sheet archiving / lists	—	PARTIAL-18	18	—

T-code	Name	Purpose	Passage status	Wave	Notes
Fiori: Manage Service Entry Sheets – Lean Services / Create Service Entry Sheet	S/4 lean services	—	PARTIAL-18	18	Same
Fiori: Manage Purchase Orders (service items), Manage Supplier Quotations for services	—	—	PARTIAL-17	17	—
Supplier self-service SES (SAP Business Network / Fieldglass)	Supplier enters SES / timesheets	—	PARTIAL-20	20	Supplier portal: add service confirmation/timesheet entry (contingent labor) — partial
Fieldglass (contingent workforce, SOW)	External workforce management	—	OUT-OF-SCOPE	—	Contingent workforce VMS beyond v1; contractor records in HR (wave 42) + service POs suffice

13.15.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
MSRV1 … MSRV6	Service lists	PARTIAL-18	18	—
ME2S	Services per purchase order	PARTIAL-18	18	—
MSRV3 (entry sheets)	SES list	PARTIAL-18	18	—
ML83 / ML84	SES acceptance list	PARTIAL-18	18	—
PURCHIS services analysis	Service purchasing analysis	COVERED-46	46	—

13.15.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
ML85 (collective release of SES)	Mass acceptance	PARTIAL-18	18	—
MRRL for services	ERS for SES	PARTIAL-18	18	—
SES archiving (SARA MM_SRVMAS)	—	COVERED-48	48	—

13.15.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Service master: number ranges, service categories, organizational status, field selection	ASMD	—	PARTIAL-8	8	—
Service specifications: screen layout, number ranges for SES, release procedure for SES (classification)	ESSR	—	PARTIAL-18	18	SES approval via workflow designer
Service conditions: calculation schema MS0000, condition types PRS	T685	—	PARTIAL-17	17	—
Default values for SES acceptance (immediate GR, account assignment)	—	—	PARTIAL-18	18	—
Source determination for services (contracts)	—	—	PARTIAL-17	17	—
Lean services: activate, product type service, service entry sheet (lean) settings	—	—	PARTIAL-18	18	—

13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)

SAP context: classic WM — warehouse number, storage types/sections/bins (LS01N), transfer requirements (LB01), transfer orders (LT01/LT03/LT06/LT10), confirmation (LT12), put-away/picking strategies, replenishment (LP21/LX16?), WM physical inventory (LI01N–LI21), RF (LM00–LM77), HU management (HU02), storage unit management. EWM (decentral or embedded in S/4) — warehouse product, storage types/bins, warehouse tasks/orders, waves, POSC/LOSC process steps, packing specifications, labor management, yard, slotting, MFS, RF framework (/SCWM/RFUI), warehouse monitor (/SCWM/MON). Passage: wave 31 (locations, LPs/HUs, directives, work templates, waves, replenishment, load planning, dock scheduling, slotting, KPIs), wave 32 (mobile/RF, voice, RFID, labels), wave 16 (counting), wave 28 (basic pick/pack/ship).

13.16.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
LS01N / LS02N / LS03N	Storage bin create/change/display	Bins with type, section, capacity, coordinates	COVERED-31	31	Location model: zones/aisles/racks/bins with capacity/profile
LS05 / LS10	Generate storage bins automatically / define bin structure	Mass bin creation	COVERED-31	31	Bin generator
LS06 / LS07 / LS08	Block/unblock bins (by bin / by storage type / list of blocked bins)	Bin blocking	COVERED-31	31	Location status
LS11 / LS12	Change several bins / block storage type	—	COVERED-31	31	—
LS24 / LS25 / LS26	Stock per material (bin stock) / per bin / stock overview (warehouse)	Bin-level inquiry	COVERED-31	31	On-hand by location/LP
LS22 / LS23 / LS27 / LS28	Change / display quants; quants by storage unit	Quant maintenance	COVERED-31	31	Inventory by location/LP/batch
LS32 / LS33	Change / display storage unit	SU (handling unit)	COVERED-31	31	License plates
LB01 / LB02 / LB03 / LB10 / LB11 / LB12 / LB13	Transfer requirement create/change/display/lists (by storage type, material, movement)	TR = open work demand	SUPERSEDED-31	31	Work demand generated directly from source documents (ASN, order release, replenishment)
LT01 / LT02 / LT03 / LT04 / LT05 / LT06 / LT07 / LT08 / LT09 / LT10	Transfer order: create (manual) / from TR / for delivery / from TR / for posting change / for material document / for mixed SU / by storage type / stock transfer	Work (tasks) creation	COVERED-31	31	Work templates generate tasks (put-away, pick, move, replenish)
LT11 / LT12 / LT13 / LT14 / LT15 / LT16	Confirm TO item / TO / SU / by picking list / cancel TO / cancel TO item	Work confirmation	COVERED-31	31	Work execution (confirm, exceptions, short picks)

T-code	Name	Purpose	Passage status	Wave	Notes
LT21 / LT22 / LT23 / LT24 / LT25 / LT26 / LT27 / LT28	Display TO / list by storage type / by number / by material / by SU / by bin / by delivery / archive	Work inquiry	COVERED-31	31	Work list pages
LT31 / LT32	Print TO / TO group	Pick lists	COVERED-32	32	Printing + mobile
LT41 / LT42 / LT43 / LT44 / LT45	Create / display / confirm TO group (wave) for delivery groups	Group picking (waves, 2-step picking)	COVERED-31	31	Wave templates; cluster/zone picking
LT0A / LT0B / LT0C / LT0D / LT0E / LT0F / LT0G / LT0H / LT0I / LT0J / LT0R / LT0S	TOs for: inbound delivery / outbound delivery / multiple deliveries / return to stock / stock pickup / 2-step picking	—	COVERED-31	31	—
LP10 / LP11 / LP12 / LP21 / LP22	Replenishment: fixed bins (planned) / for storage type / for production / replenishment of fixed bins / replenishment for storage types	Replenishment	COVERED-31	31	Replenishment (min/max, demand, wave)
LX LD10 / LD11 / LD12 / LD13	Picking for delivery monitor / collective TO creation for deliveries	—	COVERED-31	31	—
LQ01 / LQ02	Transfer posting in WM (stock category change: Q→unrestricted, blocked)	Status change	COVERED-31	31	Inventory status changes per location
LU01 / LU02 / LU03 / LU04	Posting change notices	—	SUPERSEDED-31	31	Status changes are direct
LI01N / LI02N / LI03N	WM physical inventory document create/change/display	Bin counts	COVERED-16	16	Cycle counting at location level (wave 31 adds bins)
LI04 / LI05 / LI06	Print / list PI documents	—	COVERED-16	16	—
LI11N / LI12N / LI13N / LI14	Enter count / change count / display / start recount	—	COVERED-32	32	Mobile count
LI20 / LI21	Clear differences WM / clear differences in IM	Variance posting (two-step: WM then IM)	SUPERSEDED-16	16	Single inventory ledger: one variance posting
LX15 / LX16 / LX17 / LX18	Continuous inventory / zero-stock check / annual inventory evaluation / stock placement counting	PI variants	COVERED-16	16	Cycle counting policies incl. zero-count trigger
LICC / LX26	Cycle counting per quant / continuous inventory	—	COVERED-16	16	—
HU02 / HU03 / HUMO	Create/change HU / display HU / HU monitor	Handling units (nested)	COVERED-28	28	Containers/HU with SSCC; nested LPs (wave 31)
HUPAST / HU04 / HUIBD	Pack in HU monitor; HU for inbound delivery	—	COVERED-31	31	—
VL01N/VL02N packing (see LE-SHP)	Packing in delivery	—	COVERED-28	28	—
LM00 → LM01 … LM77 (RF menus: putaway, picking, inquiry, counting, load/unload, pack)	RF transactions (ITSmobile)	RF execution	COVERED-32	32	Mobile app flows (receive, put-away, pick, pack, move, count, inquiry)
LRF1 / LRF2	RF monitor / RF queue assignment	RF supervision	COVERED-32	32	Device/session management; task queues (wave 31 priorities)
LRFMD	RF device master	—	COVERED-32	32	Device registry
LX01 … LX47 (lists — see reports)	—	—	COVERED-31	31	—
LXDCA / cross-docking (LXDCK)	Planned cross-docking	—	GAP	propose 31	Cross-docking (inbound → outbound without put-away, planned/opportunistic) not named in wave 31 — propose directive type "cross-dock"
Dynamic cycle counting / task interleaving	—	—	COVERED-31	31	Work assignment by priority; interleaving — confirm
EWM: /SCWM/MON	Warehouse management monitor (documents, stock, resources, exceptions)	Supervisor cockpit	COVERED-31	31	Warehouse workspace & KPIs
EWM: /SCWM/PRDI / /SCWM/PRDO	Maintain inbound / outbound delivery (warehouse request)	Delivery documents in EWM	SUPERSEDED-31	31	Single delivery/receipt documents (no ERP↔EWM replication)
EWM: /SCWM/TODLV_I / /SCWM/TODLV_O / /SCWM/TO_CONF	Create warehouse tasks for inbound/outbound delivery / confirm	—	COVERED-31	31	—
EWM: /SCWM/ADPROD / /SCWM/ADGI	Goods issue/receipt for production (staging)	Production staging	COVERED-31	31	Staging for production orders (wave 21 picking lists + wave 31 directives)
EWM: /SCWM/WAVE / /SCWM/WAVETMP	Wave management: create/release waves, wave templates	Wave picking	COVERED-31	31	Wave templates
EWM: /SCWM/PACK / /SCWM/PACKSPEC / /SCWM/PSSTAGE	Packing work center / packaging specifications	Packing specs (how to pack product into HU)	PARTIAL-31	31	HU/packing covered; *packaging specifications* (rules: product → packaging material, quantities per level) only implied — partial; propose wave 28/31 "packaging rules"
EWM: /SCWM/VASEXEC / VAS orders	Value-added services (kitting, labeling)	—	GAP	propose 31	VAS/kit-to-order in warehouse not named; propose wave 31 work type "VAS" and wave 24 kitting via lean BOM
EWM: /SCWM/RSRC / /SCWM/USER / /SCWM/QUEUE	Resources, users, queues	Resource management	COVERED-31	31	Users/resources/queues
EWM: /SCWM/LM (labor management: engineered labor standards, planned vs actual, performance docs)	Labor management	—	PARTIAL-31	31	Warehouse KPIs named; engineered labor standards, operator performance/incentive documents — partial
EWM: /SCWM/SLOT slotting & rearrangement	Slotting	—	COVERED-31	31	Slotting analysis
EWM: /SCWM/YM_* yard management (yard bins, vehicles, TUs, checkpoints)	Yard management	—	PARTIAL-31	31	Dock scheduling named; yard (trailer positions, moves, check-in/out) partial
EWM: /SCWM/TU, /SCWM/VEH	Transportation units / vehicles	—	PARTIAL-34	34	Load/TU in wave 34 — partial for yard link
EWM: /SCWM/PI_PROCESS / /SCWM/PI_CREATE / /SCWM/PI_COUNT / /SCWM/DIFF_ANALYZER	Physical inventory in EWM + difference analyzer	—	COVERED-16	16	—
EWM: /SCWM/QINSP Quality inspection engine (QIE)	Inspection in warehouse	—	COVERED-33	33	Quality orders at receipt

T-code	Name	Purpose	Passage status	Wave	Notes
EWM: /SCWM/RFUI	RF framework (menu, screens, logical transactions)	—	COVERED-32	32	Configurable flow designer
EWM: /SCWM/MFS* (Material Flow System: PLC interface, conveyors, ASRS)	Automation integration	—	GAP	propose 32	MFS/PLC integration (telegrams to conveyor/AS-RS) not named; wave 25 OPC-UA/MQTT edge is the natural base — propose wave 32 "automation adapter (ASRS/conveyor)"
EWM: /SCWM/DLVPPF / /SCWM/PPF	Post processing framework actions (print, msg)	—	COVERED-4	4	Event-triggered actions
EWM: Catch weight, serial numbers in WM, batch in WM	—	—	COVERED-24	24	Catch weight (wave 24), serial (wave 8)
EWM: Kit-to-stock / kit-to-order	—	—	GAP	propose 31	See VAS
EWM: Dangerous goods / HazMat checks in warehouse (EHS)	—	—	GAP	propose 34	Hazmat classification, storage compatibility, DG documents — not named; propose wave 34 (transport) + wave 8 attributes
Fiori: Warehouse KPIs, Warehouse Monitor apps, Outbound Delivery Order Overview, Process Warehouse Tasks	—	—	COVERED-31	31	—

13.16.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
LX01	List of empty storage bins	COVERED-31	31	—
LX02	Stock list (bin stock by material)	COVERED-31	31	—
LX03	Bin status report	COVERED-31	31	—
LX04	Capacity used (storage type)	COVERED-31	31	Capacity never exceeded (proof)
LX05	Block bins in bulk storage	COVERED-31	31	—
LX06 / LX07	Fire-containment sections / check storage (inconsistencies)	PARTIAL-31	31	Fire/hazard sections tie to hazmat gap
LX08 / LX09	Accident regulations / overview of all TRs	COVERED-31	31	—
LX10 / LX11 / LX12 / LX13	Stock movements per storage type / document overview / document overview (SU) / analysis of differences	COVERED-31	31	—
LX14 / LX15 / LX16 / LX17	Material situation (ABC/movement frequency) / selection of PI by date / zero-stock check / list of inventory differences	COVERED-31	31	ABC/velocity analysis for slotting
LX18 / LX19 / LX20 / LX21	Statistics of inventory differences / inventory data takeover / generate IM inventory from WM / pick list for several TOs	COVERED-31	31	—
LX22 / LX23 / LX24 / LX25 / LX26 / LX27 / LX29	Inventory status / stock comparison IM–WM / hazardous material / inventory status / cycle counting / stock below minimum / fixed bin supervision	COVERED-31	31	IM–WM comparison superseded (single ledger)
LX31 / LX32 / LX33 / LX34	Analysis of print control / archived TOs / archived TRs / posting change notices	COVERED-31	31	—
LX35 / LX36 / LX37 / LX38 / LX39	Archived PI docs / archived PI history / linked objects / check report Customizing strategy K / evaluation of 2-step picking	COVERED-31	31	—
LX40 / LX41 / LX42 / LX43 / LX44 / LX45 / LX46 / LX47	Material master data in WM / bin status (SU) / stock in storage units / check consistency / HU inquiry / RF monitor-related / TO analysis / analysis of delays in delivery	COVERED-31	31	—
LL01	Warehouse activity monitor (critical TOs, TRs, deliveries, negative stock, posting changes)	COVERED-31	31	Exceptions dashboard
LT22 … LT28	TO lists	COVERED-31	31	—
LS24 … LS26	Bin stock	COVERED-31	31	—
HUMO	HU monitor	COVERED-28	28	—
MCL1 / MCL5 / MCL9 / MCLD (WM LIS)	Warehouse controlling: stock placement/removal, flow of quantities, material movements	COVERED-46	46	—
EWM: /SCWM/MON node reports (stock, documents, resources, PI, labor)	—	COVERED-31	31	—
EWM: /SCWM/LM_ labor reports (planned vs actual)	—	PARTIAL-31	31	—
Fiori: Warehouse KPIs (dock-to-stock, order cycle time, pick accuracy)	—	COVERED-31	31	—

13.16.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
LP21 / LP22 (scheduled)	Replenishment runs	COVERED-31	31	—
LX16 (scheduled)	Zero-stock check	COVERED-16	16	—
LICC / LX26 (scheduled)	Cycle counting generation	COVERED-16	16	—
LT0S / LT0R	2-step picking: removal / allocation	COVERED-31	31	—

T-code	Name	Passage status	Wave	Notes
LT41/LT42 (TO groups, scheduled waves)	Wave release	COVERED-31	31	—
LX / LT15 cleanup / TO archiving (SARA LE_TRANSORD / RL_TA)	Archiving TOs	COVERED-48	48	—
LX14 (periodic ABC)	ABC analysis	COVERED-31	31	—
EWM: /SCWM/WAVE scheduled release, replenishment (/SCWM/REPL), slotting (/SCWM/SLOT), rearrangement (/SCWM/REAR), PI generation, archiving	—	COVERED-31	31	—
EWM: /SCWM/ERP_* delivery replication, queue monitor (SMQ1/SMQ2, qRFC)	ERP↔EWM integration	SUPERSEDED-31	31	Embedded single system; Kafka events instead of qRFC
EWM: Delta/initial stock load (/SCWM/ISU, /SCWM/ERP_STOCKCHECK)	Stock comparison ERP↔EWM	SUPERSEDED-16	16	Single ledger
Labor management period close (/SCWM/LM)	—	PARTIAL-31	31	—

13.16.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OMLW / Define warehouse number, assign to plant/storage location (OMLY)	T300, T320	Warehouse number ↔ plant/SLoc	COVERED-6	6	Warehouse entity (wave 6) with WMS-enabled flag (wave 31)
Define storage types (OMLX / T301), storage sections (T302), storage bin types (T303), storage unit types (T307), bin structure	T301, T302, T303, T307, T305	Put-away strategy (A/B/C/F/I/K/L/P/Q/R), removal strategy (A/F/L/M/P/S/H), mixed storage, capacity check method, SU management, addition to stock, bin blocking on movement, return/negative stock	SUPERSEDED-31	31	Storage strategies replaced by location directives (rules engine) + location profiles; capacity check methods covered
Storage section search (OMLH), storage type search (OMLY/T334T), bin type search (T334B)	T334T, T334B, T334E	Search sequences per movement type/stock category/special stock	SUPERSEDED-31	31	Directive rules
WM movement types (OMLR), reference movement types IM→WM (OMLM LE_WM_MVT)	T333, T321	TO creation automatic/immediate, confirmation required, source/dest bins fixed	SUPERSEDED-31	31	Work templates per source document type
Define control for TO creation / confirmation (OMLV print control, OMLN number ranges)	T329*, T340	Print immediately, pick list sorting, confirmation per item/HU	COVERED-31	31	Work template options
Storage unit management activation, SU number ranges	T300, NRIV	—	COVERED-31	31	LP sequences
Replenishment control for fixed bins / storage types (OMLH / T331B)	T331, T331B, T*	Min/max per bin, replenishment movement type	COVERED-31	31	—
Physical inventory in WM: activate, PI types per storage type, difference posting tolerances	T301, T*	—	COVERED-16	16	—
Hazardous material: regions, water pollution classes, storage classes, fire-containment sections	T* (MGEF)	—	GAP	propose 34	Hazmat gap
Interface to QM / PP (control cycles for WM staging PK05 / LPK1), kanban in WM	PKHD, T*	—	COVERED-31	31	Staging for production (control cycles) — confirm WM staging strategies (pick parts, crate parts, release order parts) in wave 31
RF: menu management (LRFMD LRF1), screen management, verification profiles, bar code types (OMLC)	T* , LRF_*	—	COVERED-32	32	Flow designer
Decentralized WM / EWM integration (SLG, qRFC, distribution model)	—	—	SUPERSEDED-31	31	—
EWM: warehouse number, storage types/sections/bins/activity areas, storage bin sorting, warehouse process types, POSC/LOSC (process-oriented/layout-oriented storage control), put-away/stock removal strategies & search sequences, HU types & packaging specs, work centers, queues, resource types, wave templates, labor management, exceptions codes, PPF	/SCWM/T3* tables	—	COVERED-31	31	Rules engine + work templates + location profiles; packaging specs partial (noted)
EWM: delivery document types / item types, mapping from ERP	/SCDL/*	—	SUPERSEDED-28	28	Single delivery model
EWM: QIE, catch weight, batch management, serial number profiles in EWM	—	—	COVERED-33	33	—
EWM: Yard management customizing	—	—	PARTIAL-31	31	—
EWM: MFS customizing (PLC communication channels, telegrams)	—	—	GAP	propose 32	—
EWM: Labor management: formulas, ELS, incentive	—	—	PARTIAL-31	31	—

13.17 LE-SHP / LE-TRA — Shipping & (classic) Transportation

SAP context: outbound deliveries (VL01N, VL10*), picking (LT03 / WM, or picking without WM), packing with HUs, post goods issue (VL02N), inbound deliveries (VL31N) and ASN, delivery monitor (VL06O/VL06I), shipping points, route determination, delivery scheduling, POD (VLPOD), output (VL71/VL74), shipment documents (VT01N) and shipment cost (VI01) in LE-TRA (deprecated in S/4 in favour of TM). Passage: wave 28 (release, pick/pack/ship, HU/SSCC, carrier adapters, ASN, drop-ship, partial shipments), wave 18 (inbound ASN receiving), wave 34 (shipment/freight).

13.17.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
VL01N / VL02N / VL03N	Outbound delivery create (w/ reference to order) / change / display	Delivery document	COVERED-28	28	Shipment/packing-slip document from order release
VL01NO	Delivery without order reference	—	COVERED-28	28	Ad-hoc shipment (e.g. samples) — confirm
VL10A / VL10B / VL10C / VL10D / VL10E / VL10F / VL10G / VL10H / VL10I	Delivery due list: sales orders / POs (STO) / items / items (STO) / …, collective delivery creation	Release to warehouse in batch	COVERED-28	28	Order release rules (batch/wave)

T-code	Name	Purpose	Passage status	Wave	Notes
VL04	Process delivery due list (classic)	—	SUPERSEDED-28	28	—
VL06O / VL06G / VL06P / VL06U / VL06C / VL06D / VL06L / VL06T / VL06F	Outbound delivery monitor: for GI / for picking / unchecked / confirmation / for distribution / loading / transportation planning / general	Delivery worklist by status	COVERED-28	28	Fulfillment workspace with status views
VL06I / VL06IG / VL06IP / VL06IC	Inbound delivery monitor	—	COVERED-18	18	Inbound ASN/receiving worklist
VL31N / VL32N / VL33N	Inbound delivery (ASN) create/change/display	ASN-based receiving	COVERED-18	18	ASN-driven receiving with LPs (wave 31)
VL34	Worklist inbound deliveries (collective creation from POs)	—	COVERED-18	18	—
VL09	Reverse goods movement (cancel PGI/PGR)	Reversal	COVERED-28	28	Packing slip cancellation reverses COGS
VL02N → Post goods issue	PGI	COGS/inventory posting	COVERED-28	28	—
VL22	Display delivery change documents	—	COVERED-5	5	—
VL23 / VL21	Post goods issue in background / collective PGI	—	COVERED-28	28	Batch PGI
VL35 / VL35_ST / VL36 / VL37	Create picking waves (by delivery time / shipment) / change wave / wave monitor	Wave picking (basic)	COVERED-28	28	Basic wave picking (wave 28), advanced (wave 31)
LT03 / LT0A (picking via WM)	Picking TO for delivery	—	COVERED-28	28	Pick confirmation
VL02N → Picking without WM (pick quantity) / "Lean WM"	Picking without WM	—	COVERED-28	28	Basic pick
VL02N → Pack (HU01 HU02 in delivery), VL74	Packing in delivery; HU creation; packing proposals	Packing	COVERED-28	28	Packing with containers/HU, SSCC
HU02 / HU03 / HUMO / HUPAST	HU management	—	COVERED-28	28	—
VLPOD / VLPODL / VLPODQ	Proof of delivery: change / list / quantity differences	POD-based billing	COVERED-34	34	POD events from carrier/EDI 214; billing on POD — confirm wave 29 billing trigger "on POD" option
VL71 / VL74 / VL70	Output from deliveries / HU output / message overview	Packing slip, labels, ASN (DESADV/856)	COVERED-28	28	ASN output, labels
VLSP	Subsequent outbound delivery split	Split delivery	COVERED-28	28	Partial shipments/split
VL_ VLUNIV	Delivery mass change (VL_MASS)	—	COVERED-5	5	—
VT01N / VT02N / VT03N	Shipment document create/change/display (LE-TRA)	Shipment (consolidated deliveries, legs, stages, carrier)	COVERED-34	34	Load/shipment (wave 34); basic shipment in wave 28 ("shipments")
VT04 / VT05 / VT06 / VT70	Shipment creation in collective processing / shipment list / shipment output	—	COVERED-34	34	Load building
VT11 / VT12 / VT14 / VT16 / VT22	Shipment lists: selection / status / change documents	—	COVERED-34	34	—
VI01 / VI02 / VI03 / VI04 / VI05 / VI11 / VI12	Shipment cost document create/change/display/list; settlement; transfer to FI	Freight cost calc & accrual	COVERED-34	34	Rating + freight accrual
VI07 / VI08	Shipment cost settlement with service agent (self-billing)	—	COVERED-34	34	Freight reconciliation / self-billing
VTFL / VFSC	Shipment cost → billing of freight to customer	—	COVERED-29	29	Freight charges on invoice
VL10 → STO deliveries (replenishment delivery NL/NLCC)	Deliveries for stock transport orders	—	PARTIAL-16	16	Transfer order gap
VRRE / VL0 (returns delivery)	Returns delivery (inbound) for RE orders	—	COVERED-30	30	RMA receipt
VL02N → batch split, serial numbers in delivery	Batch/serial assignment at shipment	—	COVERED-28	28	Tracking dimensions on packing slip
Delivery scheduling (route, transit, pick/pack times, loading)	Backward/forward delivery scheduling	—	COVERED-27	27	Delivery date control (transport days, pick/pack lead)
Fiori: Manage Outbound Deliveries / Create Outbound Deliveries / Pick Outbound Deliveries / Pack Outbound Deliveries / Post Goods Issue / Manage Inbound Deliveries / Change Inbound Deliveries	S/4 apps	—	COVERED-28	28	—
Fiori: Outbound Delivery Overview / Delivery Performance (analytical)	—	—	COVERED-46	46	OTIF metric
Dangerous goods check in delivery (EHS-DG), DG documents	—	—	GAP	propose 34	Hazmat gap
Foreign trade / export documents in delivery (customs, commercial invoice, EUR.1, Intrastat from delivery; GTS)	Export compliance	—	PARTIAL-13	13	Intrastat extract named (wave 13); export documentation (commercial invoice, packing list for customs, certificate of origin), export control screening (denied parties, licenses) and customs declarations (GTS) are not — partial; propose wave 48 "foreign trade/customs" pack
Preference determination (SAP GTS / foreign trade)	Origin & preference (USMCA/EU)	—	GAP	propose 48	Rules-of-origin/preference calculation from BOM — gap

13.17.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
VL06O / VL06I family	Delivery monitors	COVERED-28	28	—
VL06F	General delivery list	COVERED-28	28	—

T-code	Name	Passage status	Wave	Notes
VL_ VLSI	Delivery lists	COVERED-28	28	—
V_SA / V_UC / V_V2	Backorders by delivery / incomplete deliveries / rescheduling	COVERED-27	27	Backorder handling
VA14L / V.15	Blocked deliveries / backorders	COVERED-27	27	—
VT11 / VT16 / VT20	Shipment lists	COVERED-34	34	—
VI11 / VI12	Shipment cost lists	COVERED-34	34	—
MC(/ MCTA (SIS deliveries) / MC-O	Shipping analyses (on-time, lead time)	COVERED-46	46	OTIF/delivery performance
HUMO	HU monitor	COVERED-28	28	—
VLPODL	POD list	COVERED-34	34	—
Fiori: Outbound Delivery Log/ Delivery Performance	—	COVERED-46	46	—

13.17.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
VL10 family (batch)	Scheduled delivery creation	COVERED-28	28	Release rules scheduled
VL23 / VL21 (batch PGI)	Batch goods issue	COVERED-28	28	—
VL71 / RSNAST00	Output run (packing slips, ASN EDI)	COVERED-28	28	—
VT04 (batch)	Shipment creation run	COVERED-34	34	—
VI04 (batch)	Shipment cost creation	COVERED-34	34	—
VI05 / Shipment cost settlement run	Freight accrual settlement	COVERED-34	34	—
VLPODQ (batch)	POD quantity processing	COVERED-34	34	—
V_V2 (scheduled)	Rescheduling sales documents (backorder re-allocation)	COVERED-27	27	ATP re-allocation job
SARA RV_LIKP / SD_VTTK	Archiving deliveries/shipments	COVERED-48	48	—
Intrastat (VE01 / MEIS / VI99) declaration run	Intrastat from deliveries/receipts	COVERED-13	13	—
Fiori: Schedule Delivery Creation / Schedule Goods Issue	—	COVERED-4	4	—

13.17.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OVXD / OVX6 — Shipping points define / assign to plant; shipping point determination (OVL2)	TVST, TVSWZ, TVSTZ	Loading group × shipping condition × plant → shipping point	COVERED-28	28	Warehouse/shipping site determination rules
0VLK — Delivery types (LF, LR, NL, NLCC, EL)	TVLK	Order required, item requirement, route determination, output, text determination, number range	COVERED-28	28	Shipment document types
0VLP — Delivery item categories; 0184 determination	TVLP, T184L	Picking relevance, packing control, availability check, batch split allowed, zero-quantity allowed	COVERED-28	28	Line types
OVLK / 0VLG — Delivery split criteria, combination rules (per ship-to, route, shipping point)	TVLK	—	COVERED-28	28	Consolidation rules
OVLY / OVZ9 (delivery scheduling): shipping point/route times (pick/pack, loading, transit, transportation lead time), factory calendars	TVST, TVRO, TROLZ	Backward/forward scheduling, working times	COVERED-27	27	Delivery date control
0VTC / OVRF — Routes, route stages, route determination (departure zone × destination zone × shipping condition × transportation group)	TVRO, TVRSZ, TVTW, TROLZ	Transit time, carrier default, route schedule (VL51)	COVERED-34	34	Routing/rating in wave 34; basic route/transit in wave 27/28
VL51 / VL52 / VL53 — Route schedules	TVSR	—	COVERED-34	34	—
OVLT — Picking: lean WM, picking location determination (MALA/RETA/MARE rules), storage location rule	TVLK, TVKOL	—	COVERED-28	28	Picking location rules
OVLS / OVLV — Packing: packaging material types (VHAR), packing instructions (POP1/POP2/POP3/POF1), allowed packaging materials, HU number ranges (SSCC), packing control per item category	TVTY, TVEGR, PACKKP	Automatic packing via packing instructions	PARTIAL-28	28	SSCC/containers yes; packing instructions (auto-pack rules) partial (same as EWM packaging specs)
OVLP — POD: activate POD per item category / ship-to, reasons for quantity differences	TVLP, TVPODG	—	COVERED-34	34	—
V/30 / VV21 / VV22 / VV23 — Output determination for deliveries (LD00 delivery note, LALE ASN, HU labels), output types, access sequences, condition records	T685B, NAST, TNAPR	Medium, timing, partner	COVERED-28	28	Output rules (see cross-app)
VTLA — Copy control order → delivery; VTFL delivery → billing	TVCPL, TVCPF	Copy requirements, data transfer routines	SUPERSEDED-28	28	Document flow in domain model with explicit mapping rules; no copy-control routines
OVZ9 / OVZ2 / OVZ3 — Availability check in deliveries (checking groups, scope)	TVLP, T441V	—	COVERED-27	27	ATP
OVLK: Partner determination in delivery (VOPAN LF), text determination (VOTXN), incompletion (OVA2/OVA0 delivery incompletion)	TVLK, TPAER, TVIMF	—	COVERED-28	28	Partners/notes/completeness validations via metadata
0VLK: Serial numbers / batch split in delivery; batch determination (VCH1 for delivery)	—	—	COVERED-28	28	—
OVTZ / Transportation: shipment types (VTTK), shipment stages, legs, shipment cost types (T_55), pricing procedure for freight (TVFT), service agent condition records (TK11), shipment cost settlement settings	TVTK, TVFT, T_56, TVFTS	Shipment cost item categories, calculation basis (weight/volume/distance), settlement to vendor	SUPERSEDED-34	34	TM-style rating engine (rate cards, accessorials) replaces LE-TRA condition-technique freight pricing
Foreign trade: export control, preference, Intrastat settings (VEB, VE0)	T604, T605	—	PARTIAL-13	13	Foreign trade gap

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Dangerous goods: DG profiles, DG check methods in SD/LE	DGT*	—	GAP	propose 34	Hazmat gap
Loading groups, transportation groups (material master)	TLGR, TTGR	—	COVERED-8	8	Product attributes
Define reasons for cancellation of PGI, goods movement types for delivery (601/602/641/643/651…) via schedule line	T156, TVEP	—	COVERED-28	28	Transaction types
OVLN Define number ranges deliveries / HUs	NRIV	—	COVERED-5	5	—

13.18 TM — Transportation Management (SAP TM / embedded TM in S/4)

SAP context: freight units (FU), freight orders (FO) and bookings (FB), transportation cockpit (/SCMTMS/PLAN? or NWBC/Fiori), carrier selection and tendering, charge management (agreements, rate tables, calculation sheets), freight settlement documents (FSD) → MM-IV, forwarding orders/settlement (LSP scenario), tracking events (EM), strategic freight procurement. Passage: wave 34 (load/route planning with VRP, rating, tendering, tracking, freight reconciliation, landed cost).

13.18.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
/SCMTMS/TRQ (forwarding order) / Fiori: Create Forwarding Order	Forwarding order (LSP scenario: customer orders transport)	LSP business	OUT-OF-SCOPE	—	Logistics service provider (3PL/forwarder) scenario is beyond manufacturer scope; shipper scenario only
Freight units (FU) built from deliveries/orders (FUBR rules)	Freight unit building	—	COVERED-34	34	Load building from shipments
/SCMTMS/FRE Freight order (road), freight booking (air/ocean)	FO/FB documents	—	COVERED-34	34	Loads with carrier; ocean/air bookings — confirm multi-modal (wave 34 says loads/routes/carriers; treat covered with ocean/air as modes)
Transportation cockpit (/SCMTMS/PLAN NWBC) / Fiori: Transportation Cockpit	Manual + optimizer planning (VSR)	Planning UI	COVERED-34	34	Load/route planning with OR-Tools VRP
Carrier selection (manual/automatic, cost/priority/allocation)	—	—	COVERED-34	34	Rate shopping
Tendering (peer-to-peer, broadcast, carrier portal/collaboration portal, EDI 990)	Tendering workflow	—	COVERED-34	34	Tendering workflow with timeouts cascade
Freight agreements, rate tables, calculation sheets, charge types, scales	Charge management master data	—	COVERED-34	34	Carrier rate cards, accessorials, fuel surcharge
Freight settlement document (FSD) / Fiori: Manage Freight Settlement Documents → MM-IV (service PO)	Carrier cost settlement	—	COVERED-34	34	Freight reconciliation + accrual posting
Cost distribution (to deliveries/items → CO-PA/landed cost)	Freight cost distribution	—	COVERED-34	34	Landed cost allocation
Tracking & tracing (Event Management / Fiori: Track Freight Orders), event handlers	Visibility	—	COVERED-34	34	Tracking events (EDI 214/API), POD
Transportation units / resources (vehicles, trailers, drivers), calendars	Fleet resources	—	PARTIAL-34	34	Own fleet (vehicles/drivers/driver hours) not named — partial; carriers assumed external
Transportation network: locations, zones, lanes, transshipment locations, schedules	Network master	—	COVERED-34	34	Routes/lanes/zones
Dangerous goods & DG checks in TM	—	—	GAP	propose 34	Hazmat gap
Customs declarations via GTS from TM	—	—	PARTIAL-13	13	Foreign trade gap
Fiori: Monitor Freight Orders / Freight Order Worklist	—	—	COVERED-34	34	—
Strategic freight procurement (RFQ for lanes, bid analysis)	—	—	GAP	propose 34	Lane-based freight RFQ/bid analysis — gap; could reuse wave 17 RFQ
Load consolidation / container packing optimization (load plan 3D)	—	—	PARTIAL-34	34	Load building with capacity constraints named; 3D load plan/visual stacking partial
Parcel manifesting (multi-carrier, label)	—	—	COVERED-28	28	Carrier adapters (EasyPost/Shippo-style)

13.18.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
Fiori: Freight Order Overview / Transportation Cockpit lists	—	COVERED-34	34	—
Fiori: Analyze Freight Costs / Freight cost per lane / carrier performance	—	COVERED-46	46	—
Fiori: Track Shipments / event monitor	—	COVERED-34	34	—
Fiori: Carrier On-time performance	—	COVERED-46	46	—
TM BW content (freight cost analytics)	—	SUPERSEDED-46	46	—

13.18.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
Background planning (VSR optimizer runs)	Scheduled planning	COVERED-34	34	—
Freight unit building run	—	COVERED-34	34	—
Automatic carrier selection run / tendering jobs	—	COVERED-34	34	—
FSD creation & transfer to ERP (batch)	Freight settlement run	COVERED-34	34	—
Freight accrual posting / reversal	—	COVERED-34	34	—
Event message processing (EM)	—	COVERED-34	34	—

T-code	Name	Passage status	Wave	Notes
Archiving freight documents	—	COVERED-48	48	—

13.18.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Freight unit building rules, freight order types, freight booking types	/SCMTMS/C_*	Split rules, stages	COVERED-34	34	—
Planning profiles, optimizer settings, selection profiles, capacity selection, incompatibilities, constraints	/SCMTMS/C_*	Cost functions, time windows	COVERED-34	34	VRP constraints
Carrier selection settings, tendering profiles, tendering process types	—	—	COVERED-34	34	—
Charge management: charge types, calculation profiles, rate table templates, scale templates, agreement types	—	—	COVERED-34	34	—
Settlement: FSD types, cost distribution profiles, invoice integration (service PO types), accruals	—	—	COVERED-34	34	—
Event management configuration (expected events, event types)	—	—	COVERED-34	34	—
Transportation zones, lanes, means of transport, equipment types	—	—	COVERED-34	34	—
Output management (PPF) for TM documents (BOL, load plan)	—	—	COVERED-46	46	Document designer
Integration: ERP order/delivery-based transportation requirements, TM–EWM integration	—	—	SUPERSEDED-34	34	Single system events
Dangerous goods in TM	—	—	GAP	propose 34	—

13.19 SD — Sales & Distribution (pricing, orders, deliveries, billing, credit)

SAP context: SD-MD (customer sales data, customer-material info, conditions), SD-SLS (inquiry, quotation, order, contract, scheduling agreement, returns, credit/debit memo requests, consignment, free-of-charge, rush/cash sales, third-party, make-to-order with configuration), SD-BF (pricing condition technique, availability check, partner/text/output determination, incompletion, credit management), SD-SHP (see LE-SHP), SD-BIL (billing documents, billing plans, invoice lists, credit/debit memos, pro-forma, intercompany billing, rebates, revenue account determination, SD-FI interface). Passage: wave 26 (pricing), wave 27 (quotes/orders/ATP/credit/holds), wave 28 (fulfillment), wave 29 (billing/rev rec), wave 30 (AR/returns/portal), wave 40 (intercompany).

13.19.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
VD01 / VD02 / VD03 / XD01 (sales area data)	Customer master sales area view	Shipping, billing, partner functions, pricing groups per sales org/channel/division	COVERED-7	7	Customer role with sales-scope data (sales org/channel/division ≈ hierarchy purpose)
VD51 / VD52 / VD53 / VD59	Customer-material info record create/change/display/list	Customer part numbers, delivery agreements	COVERED-27	27	Customer item numbers/external item references — confirm in wave 27 (named in AX "external item descriptions"); treat covered
VK11 / VK12 / VK13 / VK14	Condition record create/change/display/create with reference (prices PR00, discounts K004/K005/K007, freight, taxes MWST)	Pricing conditions	COVERED-26	26	Pricing engine condition tables/access sequences
VK31 / VK32 / VK33 / VK34	Condition maintenance via area menu (mass)	—	COVERED-26	26	Mass price updates
V-41 … V-48 / V_I7	Create material/customer/price list conditions, scales	—	COVERED-26	26	Quantity breaks (scales)
VK30 / VKP	Condition maintenance / pricing report	—	COVERED-26	26	—
VB11 / VB12 / VB13 / VB14	Material determination (substitution) create/change/display/reference	Product substitution rules	COVERED-27	27	Alternative items
VB01 / VB02 / VB03	Listing/exclusion	Allowed/forbidden products per customer	PARTIAL-27	27	Product listing/exclusion per customer not named in wave 27 — partial; simple rule on order validation
VBN1 / VBN2 / VBN3	Free goods condition records (inclusive/exclusive)	Free goods	PARTIAL-26	26	Free goods (buy 10 get 1) not named; promotions with quantity breaks are — partial
VB21 / VB22 / VB23 / VB31 / VB32 / VB33	Sales deals / promotions	Promotions	COVERED-26	26	Promotions & campaigns
VA51 / VA52 / VA53	Item proposal create/change/display	Order templates	COVERED-27	27	Order templates / product proposals — confirm (AX "item proposal")
VA11 / VA12 / VA13 / VA15	Inquiry create/change/display/list	Pre-sales inquiry	COVERED-27	27	Quotation stage "inquiry" or CRM; wave 27 quotations cover
VA21 / VA22 / VA23 / VA25 / VA26	Quotation create/change/display/list/selection	Quotation	COVERED-27	27	Quotation → order conversion
VA01 / VA02 / VA03 / VA05 / VA05N	Sales order create/change/display/list	Standard order (OR), rush (SO), cash sales (BV), free-of-charge (FD/KL), returns (RE), credit/debit memo request (CR/DR), consignment (KB/KE/KA/KR), third-party, MTO/configurable	COVERED-27	27	Order types: journal/order/subscription/returned; CR/DR requests → credit note workflow (wave 29/30)
VA01 order type RE (returns) + VRRE	Returns order with returns delivery	RMA	COVERED-30	30	Return orders (RMA) with disposition
VA01 CR / DR (credit / debit memo request) with billing block	Credit/debit memo requests	—	COVERED-29	29	Credit notes with approval
VA01 KB / KE / KA / KR (consignment fill-up/issue/pick-up/returns)	Customer consignment processes	—	PARTIAL-27	27	Customer consignment gap (see MM-IM)

T-code	Name	Purpose	Passage status	Wave	Notes
VA01 BV / SO (cash sales / rush order)	Immediate delivery and invoice	—	COVERED-27	27	Order type with immediate release/invoice — confirm
VA01 FD / KL / SD (free-of-charge delivery / subsequent free-of-charge)	Free-of-charge	—	COVERED-27	27	Zero-price order lines with COGS
VA01 third-party (item cat TAS) → PR/PO → MIRO → VF01	Third-party order processing	—	COVERED-28	28	Drop-ship
VA01 individual purchase order (TAB)	Purchase-to-order (stock via PO)	—	COVERED-27	27	Direct delivery / back-to-back PO — covered by drop-ship & "order with configured product creates production/planned order linkage" (wave 27); back-to-back purchase covered via wave 22 MRP; treat covered
VA01 with variant configuration (TAC)	Configure-to-order	—	COVERED-27	27	Configurator (wave 8)
VA01 MTO (TAK) → production order	Make-to-order linkage	—	COVERED-27	27	Order-linked production order
VA31 / VA32 / VA33 / VA35	Customer scheduling agreement create/change/display/list (SA with JIT/forecast schedules VA32 → delivery schedule)	Customer SA (automotive)	PARTIAL-27	27	"Scheduling agreements" named in wave 27 replaces list; inbound EDI DELFOR/DELJIT (830/862) customer schedules with cumulative quantities and schedule reconciliation are not named — partial (automotive)
VA41 / VA42 / VA43 / VA45	Contract create/change/display/list (quantity contract, value contract, service contract, master contract)	Sales agreements	COVERED-27	27	Sales agreements (blanket); service contracts in wave 45
VA44 / VA46	Contract: actual overview / value contract release orders	—	COVERED-27	27	Release tracking on agreements
VA88 / VA08	Sales document mass processing (MASS VBAK)	Mass change	COVERED-5	5	—
VA14L	Sales documents blocked for delivery	Holds	COVERED-27	27	Order holds
VKM1 / VKM2 / VKM3 / VKM4 / VKM5	Credit: blocked SD documents / released / all / release / delivery	Credit release	COVERED-27	27	Credit hold workflow
V.01 / V.02 / V.03 / V.04 / V.05	Incomplete sales documents lists (orders, deliveries, contracts…)	Incompleteness	COVERED-27	27	Validation/incomplete status
V_R1 / V_R2 / V_RA	Rescheduling / backorder processing	ATP rebalancing	COVERED-27	27	Backorder handling
CO06 / CO09 / CO24	Backorder processing (ATP manual allocation) / availability overview / missing parts	ATP	COVERED-27	27	ATP/CTP with cumulative ATP view
V_V2	Rescheduling sales documents (batch)	—	COVERED-27	27	—
VKA / VSB1 (self-billing)	Self-billing procedure (customer self-billing, automotive)	—	GAP	propose 29	Customer self-billing (EDI 820/GSVERF reconciliation against invoices) not named; automotive; propose wave 29/35 option
VF01 / VF02 / VF03	Billing document create/change/display (invoice F2, pro-forma F5/F8, credit memo G2, debit memo L2, intercompany IV, returns credit RE, cancellation S1)	Billing	COVERED-29	29	Invoice posting; credit notes; pro-forma — confirm pro-forma (no posting) document type; treat covered
VF04	Maintain billing due list (order-related, delivery-related, collective)	Billing run	COVERED-29	29	Invoice generation from shipments/orders; consolidated invoices
VF05 / VF05N	List billing documents	—	COVERED-29	29	—
VF06	Batch billing	—	COVERED-29	29	—
VF11	Cancel billing document	Reversal	COVERED-29	29	Credit note / cancellation
VF21 / VF22 / VF23 / VF24 / VF25	Invoice list create/change/display/worklist/list	Invoice lists (periodic collective invoice with factoring discount)	COVERED-29	29	Consolidated/summary invoices
VF31	Output from billing documents	Print/EDI invoice (810/INVOIC), e-invoice	COVERED-29	29	Invoice layouts; e-invoice via wave 35
VFX3	Release billing documents to accounting (blocked)	—	COVERED-29	29	Posting errors queue
VF44 / VF45 / VF46 / VF47	Revenue recognition (classic SD rev rec): worklist / display / cancel / inconsistency check	Classic rev rec	SUPERSEDED-29	29	ASC 606/IFRS 15 rev rec engine
RAR: FARR_* (contracts, POBs, schedules), Fiori: Manage Revenue Contracts	Revenue Accounting and Reporting (IFRS 15)	—	COVERED-29	29	Performance obligations, SSP allocation, schedules
Billing plans (periodic/milestone) on order items (VA02 → billing plan)	Periodic / milestone billing	—	COVERED-29	29	Subscription (recurring) + milestone billing
Down payment processing in SD (billing plan FAZ, F-29)	Down payment request/invoice	—	COVERED-27	27	Deposit requests; wave 30 customer prepayments
VF01 IV (intercompany billing)	Intercompany invoice from cross-company delivery	—	COVERED-40	40	IC mirror documents
Resource-related billing DP90/DP91 (see CS/PS)	—	—	COVERED-41	41	T&M billing
Customer returns (S/4 Advanced Returns Management: MSR_*)	Returns with inspection, refund determination, logistics follow-up (return to vendor)	—	COVERED-30	30	RMA: inspection, disposition, credit/refund; return-to-vendor flow — confirm return-to-vendor; treat covered
VKP0 / Pricing report (V/LD)	Pricing reports (condition lists)	—	COVERED-26	26	Price simulation and explain
VOFM (routines) — see customizing	—	—	SUPERSEDED-26	26	—

T-code	Name	Purpose	Passage status	Wave	Notes
Fiori: Manage Sales Orders / Sales Quotations / Sales Contracts / Sales Scheduling Agreements / Credit Memo Requests / Debit Memo Requests / Customer Returns / Billing Documents / Create Billing Documents	S/4 apps	—	COVERED-27	27	—
Fiori: Sales Order Fulfillment Issues (monitor), Track Sales Orders	—	—	COVERED-27	27	Order holds/exceptions workspace
Fiori: Manage Prices – Sales / Pricing Elements	—	—	COVERED-26	26	—
Fiori: Manage Solution Orders / Solution Business (bundles)	Bundled sales (products, services, subscriptions)	—	PARTIAL-29	29	Bundles with mixed revenue types partially via rev rec bundles; solution-order document type not named — partial
SAP CPQ / Sales Cloud	Configure-price-quote (cloud)	—	OUT-OF-SCOPE	—	CRM/CPQ suites: PRD non-goal (integrate with Salesforce/HubSpot); internal quoting with configurator (waves 8/27) covers ERP-side CPQ
Sales activities / CAS (VC01N)	Sales support (contacts, activities, mailings)	—	OUT-OF-SCOPE	—	CRM non-goal
Cross-selling (VB41), product proposal/dynamic product proposal	Order-entry suggestions	—	PARTIAL-27	27	Product proposals partially (item proposal); cross-sell rules not named
Payment cards in SD (authorization, settlement — CCARD)	Card payments on orders	—	COVERED-30	30	Stripe/Adyen adapter on portal; card on order/invoice — confirm card authorization on order (wave 30 says portal payments); treat covered

13.19.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
VA05 / VA05N	List of sales orders (by customer, material, status)	COVERED-27	27	—
VA15 / VA25 / VA35 / VA45	Inquiries / quotations / scheduling agreements / contracts lists	COVERED-27	27	—
VA26 / VA46 (expiring)	Quotation / contract expiry lists	COVERED-27	27	—
V.01 … V.05	Incomplete documents	COVERED-27	27	—
V.14 / V.15	Sales orders by object status / backorders	COVERED-27	27	—
V.21 / V.23	Log of collective run / release orders for billing	COVERED-29	29	—
VA14L / VKM1	Blocked documents (delivery / credit)	COVERED-27	27	—
VF05 / VF05N	Billing documents list	COVERED-29	29	—
VF24 / VFX3	Invoice list worklist / blocked for accounting	COVERED-29	29	—
V/LD	Pricing report (condition lists)	COVERED-26	26	—
VK13 lists / V_NL	Condition overview	COVERED-26	26	—
CO09 / CO06	Availability overview / backorder processing	COVERED-27	27	—
MC+E / MC+A / MC+Q / MC+2 / MC(A / MC(I / MC-A / MC-E	SIS: customer analysis (incoming orders, sales, returns, credit memos), material analysis, sales org, shipping point, sales office/employee	COVERED-46	46	Sales analytics in semantic layer
MCTA / MCTC / MCTE / MCTG / MCTI / MCTK	SIS standard analyses: customer / material / sales org / sales office / sales employee / shipping point	COVERED-46	46	—
MCSI	SIS flexible analyses	SUPERSEDED-46	46	Self-service explorer
S_ALR_87012 (SD-FI: revenue by account) / VKOA simulation	Revenue account determination analysis	COVERED-29	29	—
F.31 / FDK43 (credit)	Credit overview	COVERED-27	27	—
VA88 / SDO1	Document flow	COVERED-27	27	Document flow: order → shipment → invoice → payment
Fiori: Sales Order Fulfillment Monitor / Sales Volume / Incoming Sales Orders / Sales Order Backlog / Quotation Conversion Rates / Customer Returns analytics	S/4 analytical apps	COVERED-46	46	—
Fiori: Sales Management Overview	KPI page	COVERED-46	46	—

13.19.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
VF04 / VF06 (batch)	Billing due list run (daily/periodic, collective invoices)	COVERED-29	29	—
VF21 (batch) / RV60SBAT	Invoice list creation	COVERED-29	29	—
VF31 (batch) / RSNAST00	Invoice output run	COVERED-29	29	—
VFX3 (batch)	Release to accounting	COVERED-29	29	—
V_V2 (batch)	Rescheduling / backorder re-allocation	COVERED-27	27	—
VKM / credit re-check batch (RVKRED06/08)	Credit re-check of blocked docs, credit exposure rebuild	COVERED-27	27	—
F.28 (reset credit limit data)	Credit data rebuild	SUPERSEDED-27	27	Exposure computed
VA05 → mass reject / mass close	Mass order status changes	COVERED-5	5	—
VF44 (batch)	Classic rev rec run	SUPERSEDED-29	29	—

T-code	Name	Passage status	Wave	Notes
FARR_* rev rec period-end (contract liability/asset posting, revenue posting run)	RAR period close	COVERED-29	29	Deferred revenue roll-forward postings
Billing plan date update (periodic billing dates)	—	COVERED-29	29	Subscription schedules
SARA SD_VBAK / SD_VBRK	Archiving sales/billing	COVERED-48	48	—
MCVCHECK / S014 SIS update (OMO1)	LIS statistics update	SUPERSEDED-46	46	CDC to lakehouse
Fiori: Schedule Billing Creation / Schedule Billing Output / Schedule Billing Release	S/4 job templates	COVERED-4	4	—

13.19.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OVX5 / OVXA / OVXB / OVXG / OVX3 / OVXK / OVXM — Sales organizations, distribution channels, divisions, sales areas, sales offices/groups, assignments to company code/plant	TVKO, TVTW, TSPA, TVKOV, TVKOS, TVBUR, TVKGR, TVKWZ	—	COVERED-6	6	Sales hierarchy purpose; sales scope dimensions
VOV8 — Sales document types (OR, RE, CR, DR, KB, BV, SO, QT, KA, IN, WK1, LZ…)	TVAK	Number range, screen sequence, delivery/billing type defaults, credit check, incompletion procedure, pricing procedure determination (doc pricing proc), immediate delivery, lead time, transaction group	COVERED-27	27	Order types with defaults
VOV7 — Item categories (TAN, TAS, TAB, TAK, TAC, TANN, TAQ, TAD, REN, G2N, L2N…)	TVAP	Billing relevance (A/B/F/I), pricing relevant, schedule lines allowed, returns, BOM explosion, credit active, completion rule	COVERED-27	27	Line types
VOV4 — Item category determination (doc type × item category group × usage × higher-level)	T184	—	SUPERSEDED-27	27	Line type derived from product type/order type via metadata rules
VOV6 / VOV5 — Schedule line categories and determination (CP, CN, CV, DN…)	TVEP, T184	Movement type, requirement transfer, availability check, purchase requisition creation (third-party)	SUPERSEDED-27	27	Delivery schedule lines are native; movement/requirement behaviour by line type
OVKK — Pricing procedure determination (sales area × doc pricing procedure × customer pricing procedure)	T683V	—	COVERED-26	26	Pricing procedure selection rules
V/08 — Pricing procedures (RVAA01 etc.): steps, counters, condition types, from/to, manual, required, statistical, subtotals, requirements, calculation/base formulas, account keys	T683S	Requirements/formulas (VOFM), subtotals, account keys ERL/ERS/ERF/MWS	COVERED-26	26	Pricing procedure = ordered condition steps with rounding; VOFM routines replaced by expression rules
V/06 — Condition types (PR00, K004, K005, K007, KF00, HD00, RA01, MWST, VPRS, SKTO, BO01…)	T685, T685A	Condition class (price/discount/tax/rebate), calculation type (qty/percent/fixed/scale), scale basis, group condition, header/item, manual, plus/minus, text, reference condition	COVERED-26	26	Condition type attributes
V/07 — Access sequences	T682, T682I, T682Z	Accesses with condition tables, exclusive, requirements	COVERED-26	26	Access sequences specific → general
V/03 / V/04 / V/05 — Condition tables create/change/display	T681, A001..A999	Field catalog, header/with validity	COVERED-26	26	Condition tables generated from metadata
OV31 / V/N — Condition exclusion groups (VB(1 / OV32)	T684, T684G	Best price among groups	COVERED-26	26	Exclusion/best-of rules
OVKI / OVKJ — Document pricing procedure / customer pricing procedure indicators	TVKV, TVKD	—	COVERED-26	26	—
V/76 / V/77 — Material pricing group / customer pricing group / price lists / price groups	T178, T151, T189, T188	—	COVERED-26	26	Price groups
V/LD — Pricing reports definition	—	—	COVERED-26	26	—
VOFM — Requirements, formulas (pricing, copy, output, data transfer)	TFRM	ABAP routines	SUPERSEDED-26	26	Expression/rule language in metadata; no ABAP (PRD non-goal)
OVA8 / OVAK / OVAD / OVA7 — Automatic credit control, credit check per sales doc type, delivery type, item category credit active	T691F, TVAK, TVLK, TVAP	Static/dynamic, horizon, max doc value, oldest open item, reaction, release via VKM	COVERED-27	27	Credit rules
OB45 / OB38 — Credit control areas / assignment (classic)	T014, T001	—	SUPERSEDED-27	27	—
OVZ9 / OVZ2 / OVZ3 / OVZ1 / OVZG / OVZH / OVZ0 — Availability check: checking groups, checking rules, scope of check (include POs, reservations, production orders, safety stock), requirement classes/types, strategy groups (planning), delivery date control	TMVF, T441V, T459K, T459A, TVEP	ATP scope, replenishment lead time, backorders, product allocation	COVERED-27	27	ATP configuration: supply/demand elements included, delivery date control; planning strategy in wave 22
OVZJ / OVZI — Product allocation (quota per customer/period)	T190, KOTA	—	GAP	propose 27	Product allocation (ATP quotas per customer/region in scarcity) not named — propose wave 27 ATP option
OVZK / OVZ — Availability check against product allocation; aATP (S/4): backorder processing strategies (BOP), alternative-based confirmation (ABC), release for delivery	—	—	PARTIAL-27	27	ATP/CTP covered; aATP backorder processing with strategies (win/gain/redistribute/fill/lose) partially named ("backorder handling")
VOPAN — Partner determination procedures (sales doc header/item, delivery, billing, customer master)	TPAR, TPAER, TVPG	Partner functions SP/SH/BP/PY, mandatory, unique	COVERED-7	7	Document partner roles (sold-to, ship-to, bill-to, payer)
VOTXN — Text determination (text types, procedures, access sequences)	TTXID, TTXERN	Copy from customer/material/preceding doc	COVERED-27	27	Notes/text blocks with inheritance — confirm inheritance rules
OVA2 / OVA0 / OVA — Incompletion procedures (order/delivery/billing), status groups	TVUVG, TVUV, TVUVF	Fields required for next step	COVERED-27	27	Validation/completeness rules via metadata

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
VTAA / VTLA / VTFA / VTFL / VTAF / VTFF — Copy control order→order, order→delivery, order→billing, delivery→billing, quotation→order, billing→billing	TVCPA, TVCPL, TVCPF	Copy requirements, data transfer routines, item category proposal, quantity/value update (+/-)	SUPERSEDED-27	27	Document conversion rules (quote→order→shipment→invoice) in the domain model with explicit mapping and quantity tracking; no VOFM copy routines
VOFA — Billing types (F2, F5, F8, G2, L2, RE, IV, S1, S2, FAZ, FAS)	TVFK	Number range, posting block, account determination procedure, cancellation type, invoice list type, rebate relevant, statistics	COVERED-29	29	Invoice document types
OVBI / OVBJ / OVBK / OVBL / OVBM — Billing plan types (periodic, milestone), date descriptions, date categories, date proposals, assignment to item categories	T6B0, TFPLA, TFPLB, TFPLT	Horizon, in advance/arrears, billing block until milestone, milestone from PS	COVERED-29	29	Billing schedules (ratable/milestone/usage)
VKOA — Revenue account determination (condition technique: application V, procedure KOFI/KOFK, tables 001-005: sales org/account assignment group customer/material/account key)	C001..C005, T030, TVKV	Account keys ERL, ERS, ERF, MWS, EVV, DIF	COVERED-29	29	Revenue/discount/freight posting profiles by customer group × item group × sales scope
OVK1 / OVK3 / OVK4 / OVK6 — Tax determination rules: tax categories per country (MWST, UTXJ), customer tax classifications, material tax classifications, plant/region tax codes	TSTL, TSKD, TSKM, T005, TVKWZ	Tax classification combinations → tax code via condition records (VK11 MWST)	COVERED-13	13	Tax groups × item tax groups → tax codes (tax engine); jurisdiction codes (TAXUSJ)
OVFL / OVF — Define plant/region tax jurisdiction defaults	—	—	COVERED-13	13	—
OVXD / OVX — Shipping points etc. — see LE-SHP	—	—	COVERED-28	28	—
OVLK — Delivery scheduling and transportation scheduling activation per sales doc type	TVAK	—	COVERED-27	27	—
OVKP / OVRC — Order reasons, reasons for rejection, delivery blocks, billing blocks, order blocks	TVAU, TVAG, TVLS, TVFS, TVAST	—	COVERED-27	27	Reason codes; holds (credit, manual)
OVB — Define sales document status groups, user statuses (BS02)	TJ30	—	COVERED-4	4	—
OVK — Customer groups, customer account assignment groups, material account assignment groups, item category groups, material groups 1-5	T151, TVKT, TVKM, TPTM, TVM1..TVM5	—	COVERED-7	7	Customer groups (wave 7), item groups (wave 8)
OVSH — Sales districts, sales offices, sales groups, customer hierarchy types (OVH1/OVH2/OVH3 hierarchy assignments)	T171, TVBUR, TVKGR, T179, KNVH	Hierarchy nodes for pricing/rebates	COVERED-7	7	Customer hierarchies (party hierarchy purposes)
OVZ "Define requirements classes/types, assignment to strategy group" (OVZG/OVZH/OVZI)	T459K, T459A, T459U	MTO/MTS/CTO behaviour, costing, special stock E	PARTIAL-27	27	Planning strategies (wave 22) + sales-order stock partial
Output determination SD (NACE V1/V2/V3, V/30 V/40 V/42 V/48 V/57): output types BA00/LD00/RD00, access sequences, condition records VV11/VV21/VV31, print programs/forms	NAST, TNAPR, T685B, B001..	Medium (print/EDI/email/ALE), partner, timing, repetition	COVERED-27	27	Output rules per document type; EDI 855/856/810 wave 35
OV "Define material determination types/procedures" (OV11 OV12)	T475, T475T	Substitution reasons, automatic/manual	COVERED-27	27	Alternative items rules
OV "Listing/exclusion procedures" (OV04)	T684	—	PARTIAL-27	27	Listing gap
OVKF — Free goods: condition type NA00, procedure NA0001, determination	T685, KOTN	Inclusive/exclusive, calculation rule	PARTIAL-26	26	Free goods gap
OVSD / OVSE / OVSG — Sales activities (CAS) customizing	—	—	OUT-OF-SCOPE	—	CRM non-goal
OMO1 / MC18 / MC21 / MC24 — SIS info structures, update rules, update groups	TMC*, S0xx	—	SUPERSEDED-46	46	Lakehouse + semantic layer
OVB6 / V/86 — Define rounding (unit of measure rounding, rounding profiles OVR)	T006R, TVRPL	Rounding profiles for order qty	COVERED-27	27	Order quantity rounding (UoM); smart rounding for prices (wave 26)
OVRQ / Define units of measure groups, delivery units, minimum order qty (material master)	T006, MVKE	—	COVERED-8	8	Product sales attributes
Advanced Returns Management settings: return reasons, refund types, inspection codes, follow-up activities, logistical follow-up	MSR_*	—	COVERED-30	30	Disposition codes
RAR customizing: accounting principles, POB types, SSP determination, allocation methods, contract categories, inflight checks	FARR_C_*	—	COVERED-29	29	Rev rec configuration
Payment card customizing (CCARD): card types, authorization horizon, merchant IDs, clearing house interface	TCCAA, TCCT	—	COVERED-30	30	Payment adapter
Define sales deal / promotion types (OVB / VB(2)	T6B1, T6B2	—	COVERED-26	26	—
Intercompany billing settings (IV billing type, internal customer per sales org, OVV9 / OVV8)	TVKO, T001, TVVF	—	COVERED-40	40	—
Define number ranges SD documents (VN01, VN02, VN03, VN04)	NRIV	—	COVERED-5	5	—
Define sales document field selection / screen sequence (SHD0 transaction variants)	—	—	SUPERSEDED-5	5	Metadata-driven forms + personalization
Mass maintenance of SD conditions (VK12 scales), condition index (V/I / V/I7), condition table index	—	—	COVERED-26	26	—

13.20 SD Rebates / Settlement Management (condition contracts)

SAP context: ECC SD rebate processing — rebate agreements (VBO1/VBO2/VBO3), accruals via condition types BO01–BO06 on billing, settlement (VB(7), partial settlement, retroactive rebates (VBOF), rebate index rebuild (VBOF/OVB3). S/4HANA replaces this with Settlement Management (condition contracts WCOCO,

business volume determination, settlement runs WB2R_SC, settlement documents, accruals; also supplier rebates / chargebacks / commissions). Passage: wave 26 (rebate agreements + accrual), wave 30 (rebate settlement), wave 19 (vendor side via AP).

13.20.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
VBO1 / VBO2 / VBO3	Rebate agreement create/change/display (types 0001 group, 0002 material, 0003 customer, 0004 hierarchy, 0005 independent of sales volume)	Customer rebate agreements	COVERED-26	26	Rebate agreements and accrual calculation
VB(D	Extend rebate agreement (validity)	—	COVERED-26	26	—
VB(7	Rebate settlement (partial/final; creates credit memo request B1/B3)	Settlement	COVERED-30	30	Rebate settlement from wave 26 accruals
VBOF	Update billing documents (retroactive rebate accrual)	Retroactive agreements	PARTIAL-26	26	Retroactive rebate (apply agreement to already-invoiced volume) not named — partial
VB(8 / VBOE	Rebate index: list / rebuild	—	SUPERSEDED-26	26	Accruals computed from journal/billing events
WCOCO (S/4) / Fiori: Manage Condition Contracts	Condition contract create/change (customer, supplier, commission; contract types)	Settlement management master	COVERED-26	26	Agreement model covers customer rebates; supplier rebates (vendor-side accrual + AP credit) — confirm wave 19/26 cover supplier rebates; see below
Supplier condition contracts (purchasing rebates, chargebacks)	Vendor rebates / ship-and-debit chargebacks	—	PARTIAL-19	19	Supplier rebate accrual and settlement (credit from vendor) and chargeback/ship-and-debit claims are not named in waves 17/19 — partial; propose wave 26 engine reused for purchasing with settlement via AP in wave 19
Commission contracts (sales rep commissions via settlement management)	Commissions	—	PARTIAL-26	26	"Commission calculation hooks" only — partial; settlement to payroll/AP not named
WB2R_BUSVOL / Fiori: Business Volume	Business volume determination (condition contract business volume selection)	—	COVERED-26	26	Accrual base from billed/shipped volume
WB2R_SC / WB2R_SV	Settlement run (single / mass) → settlement documents (WB2R_SD display)	—	COVERED-30	30	—
Settlement documents (credit memo / invoice) WB2R_* / Fiori: Manage Settlement Documents	Settlement documents with accounting	—	COVERED-30	30	Credit note / vendor invoice from settlement
Fiori: Condition Contract Overview / Settlement Dates / Accruals monitor	—	—	COVERED-26	26	—
Promotion pricing / trade promotion management (TPM, CRM)	Trade promotions with funds	—	OUT-OF-SCOPE	—	CRM TPM non-goal; wave 26 promotions cover pricing side
Chargeback / ship-and-debit claims from distributors	Claim validation against agreements	—	GAP	propose 30	Distributor chargeback claims (EDI 844/849) validation and settlement — gap for manufacturers selling via distribution; propose wave 30 + wave 35
Royalty / licensing payments (settlement management use)	Royalty accrual/settlement	—	GAP	propose 26	Royalty accrual by sales volume (license fees to third parties) — gap; reuse rebate engine with vendor-side settlement

13.20.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
VB(8 / VBO3 → sales volume / verification level	Rebate agreement sales volume & accrual verification	COVERED-26	26	Explain accrual
VB(5	Rebate agreement list	COVERED-26	26	—
S_ALR_87012 (rebate accruals in FI)	Accrual balances	COVERED-26	26	Accrual account reconciliation
Fiori: Condition Contract Overview / Business Volume analytics / Settlement document list	—	COVERED-26	26	—
WB2R_* reports: settlement document list, accrual list	—	COVERED-30	30	—

13.20.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
VB(7 (periodic settlement run)	Partial settlement per period	COVERED-30	30	—
VBOF (batch)	Retroactive accrual update	PARTIAL-26	26	—
WB2R_SC (batch) / settlement calendar	Settlement run by settlement date type	COVERED-30	30	—
Accrual run / accrual reversal (settlement management)	Periodic accrual posting	COVERED-26	26	Rebate accrual job
WB2R_BUSVOL (batch)	Business volume update	COVERED-26	26	—
Delta accruals / delta settlement	Delta settlement after retro changes	PARTIAL-26	26	Retroactive gap

13.20.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OVB3 / VB(2 — Rebate agreement types, condition type groups, rebate condition types (BO01-BO06), accrual account keys (ERB/ERU)	T6B1, T6B2, T685, T687	Manual accruals, settlement periods, verification levels	COVERED-26	26	Rebate agreement types
OVB1 / OVB2 — Activate rebate processing (sales org, billing type, payer)	TVKO, TVFK, KNVV	—	COVERED-26	26	—

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
V/08 rebate steps in pricing procedure (BO01 subtotal, statistical accrual)	T683S	—	COVERED-26	26	—
Settlement management: condition contract types, settlement process types, settlement document types, business volume selection criteria, settlement calendars, accrual settings, pricing procedures for settlement (WB2R)	WCB_*, WB2_*	Customer/supplier/commission contract types, settlement date types (partial/final/delta), accrual condition types	COVERED-26	26	Agreement types + settlement calendars (wave 30 settlement)
Settlement management: account determination for settlement docs, tax determination	T030, WB2_*	—	COVERED-30	30	—
Chargeback / claim types	—	—	GAP	propose 30	—
Commission settlement integration (HR/AP)	—	—	PARTIAL-26	26	—

13.21 PP — Production Planning (master data, MRP, SOP/demand management, production orders, capacity, KANBAN, REM)

SAP context: PP-BD (material MRP views, BOMs CS01, work centers CR01, routings CA01, production versions C223, ECM CC01), PP-MRP (MD01/MD02/MD03, MD04 stock/requirements, MD05/MD06 MRP lists, planned orders MD11–MD16, planning file MD20/MD21, MRP areas, MRP Live MD01N), PP-MP (SOP MC87/MC88, flexible planning MC93/MC94, demand management MD61–MD63, PIR transfer MC74/MC75, long-term planning MS01–MS70), PP-SFC (production orders CO01–CO15, COOIS, COHV, CO40/CO41 conversion, confirmations CO11N, COGI, availability check, TECO/close), PP-CRP (capacity CM01–CM07, leveling CM21–CM29, CM25 planning table), PP-KAB (control cycles PK01, kanban board PK13N), PP-REM (REM profile, planning table MF50, backflush MFBF, cost collectors), PP-PDC. Passage: wave 9 (BOM/routing/work centers/ECM), wave 21 (production orders, confirmations, WIP), wave 22 (MRP/MPS/forecasting), wave 23 (capacity/scheduling), wave 24 (lean/kanban/REM), wave 25 (shop floor/PDC), wave 39 (costing).

13.21.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
MM01 / MM02 / MM03 (MRP 1-4, work scheduling views)	Material master planning data: MRP type, lot size, procurement type, special procurement, safety stock, planning strategy, scheduling margin key, availability check, production version, backflush, batch entry	Planning parameters per item/site	COVERED-22	22	Coverage groups (requirement/period/min-max), lead times, safety stock, planning strategy on released product; wave 8 for master
MM02 → MRP areas (MD61 per MRP area)	MRP area assignment (storage location / subcontractor level planning)	—	PARTIAL-22	22	Planning at warehouse (sub-site) level and at subcontractor level not explicitly named — partial; wave 22 plans by site; add "planning scope = site or warehouse"
CS01 / CS02 / CS03	Material BOM create/change/display (usage 1 production, 2 engineering, 3 universal; alternatives; item categories L/N/R/D/T/K/M/I)	BOMs	COVERED-9	9	BOM header/version/lines with effective dates, phantoms, scrap, by-/co-products
CS05 / CS06 / CS07 / CS08	Change/display BOM group / plant assignment BOM (allocate BOM to plant)	—	COVERED-9	9	BOM versions per site
CS11 / CS12 / CS13	BOM explosion: level by level / multilevel / summarized	—	COVERED-9	9	Explosion API
CS14	BOM comparison	—	COVERED-9	9	Version diff — confirm; treat covered
CS15	Where-used list (single-level/multilevel)	—	COVERED-9	9	Where-used
CS20	Mass change BOM	—	COVERED-9	9	Mass change with ECO
CS40 / CS41 / CS42	Configurable BOM (super BOM) assignment / create / change	Super BOM for VC	COVERED-8	8	Configurator BOM with constraints
CS51 / CS52 / CS53 / CS61 / CS62 / CS63 / CS71 …	Standard BOM / document BOM / equipment BOM / functional location BOM / sales order BOM / WBS BOM	Other BOM categories	PARTIAL-9	9	Material BOM yes; equipment/functional location BOMs in wave 44; sales-order BOMs (order-specific) and WBS BOMs not named — partial (ETO)
CS80 / CS81 / CS82	BOM change documents	—	COVERED-5	5	Audit
CR01 / CR02 / CR03	Work center create/change/display (capacities, scheduling formulas, costing activity types, default values, hierarchy)	Resources	COVERED-9	9	Resources (machine/tool/labor/vendor/location) with capabilities, calendars, cost categories
CR05 / CR06 / CR07 / CR08	Work center list / assignment to cost center / capacity list / hierarchy	—	COVERED-9	9	—
CR11 / CR12 / CR13	Capacity create/change/display (pooled capacities)	Capacity master	COVERED-9	9	Resource capacity with shifts
CR21 / CR22 / CR23	Work center hierarchy create/change/display	Resource groups	COVERED-9	9	Resource groups
CRC1 / CRC2 / CRC3	Resource (PP-PI) create/change/display	PI resources	COVERED-9	9	Same resource model
CA01 / CA02 / CA03	Routing create/change/display (operations, sequences — standard/alternative/parallel, component allocation, PRTs, inspection characteristics, user fields)	Routings	COVERED-9	9	Routing versions, operations, operation relations (sequential/parallel), setup/run times, component allocation — confirm component-to-operation allocation; treat covered
CA11 / CA12 / CA13	Reference operation set create/change/display	Reusable operation sequences	COVERED-9	9	Operation templates — confirm; treat covered
CA21 / CA22 / CA23	Rate routing (REM) create/change/display	—	COVERED-24	24	Production flow / rate routing
CA31 / CA32 / CA33	Reference rate routing	—	COVERED-24	24	—
CA80 / CA81 / CA82 / CA83 / CA85 / CA85N / CA90	Where-used (work center / PRT / ref op set), mass change routings, routing change documents	—	COVERED-9	9	—
CA97 / CA97N	Update material master (lot size data from routing)	—	COVERED-9	9	Lead time roll-up from routing (CA97) — covered by routing capacity calc proof

T-code	Name	Purpose	Passage status	Wave	Notes
CA98 / CA99	Delete routings / display task list change docs	—	COVERED-9	9	—
C223	Production versions (BOM alternative + routing + lot size range + validity)	Production versions	COVERED-9	9	BOM/route version pairing per site/quantity — confirm production-version concept (AX has BOM/route versions separately); treat covered
CF01 / CF02 / CF03 / CF10 …	Production resources/tools (PRT: material/equipment/document/misc)	Tools	COVERED-9	9	Resource type "tool"
CC01 / CC02 / CC03 / CC04 / CC05 / CC06 / CC07	Engineering change master (ECR/ECO) create/change/display/info system/alternative dates/leading change master	ECM	COVERED-9	9	ECR/ECO workflow with effective dates, impact analysis
CC11 / CC12 / CC13	Change request (ECR) create/change/display	—	COVERED-9	9	—
CC31 / CC32 / CC33	Change package / change number release	—	COVERED-9	9	—
CEWB	Engineering workbench (mass BOM/routing maintenance)	—	COVERED-9	9	Engineering workspace grid
MD61 / MD62 / MD63	Planned independent requirements (PIR) create/change/display	Forecast/demand input per strategy	COVERED-22	22	Forecasts with reduction/consumption modes
MD64 / MD65 / MD66 / MD67	PIR: create/change planning table / display / split / copy	—	COVERED-22	22	—
MD70 / MD71 / MD72 / MD73 / MD74 / MD75 / MD76	Copy PIR from forecast / planning / delete old PIRs / reorganize	—	COVERED-22	22	—
MD81 / MD82 / MD83	Customer independent requirements	—	COVERED-22	22	Manual demand lines — confirm; treat covered
MP30 / MP31 / MP32 / MP33 / MP38 / MP39	Material forecast (execute / change / display parameters / total forecast)	Statistical forecast	COVERED-22	22	ETS/Croston forecast service with manual override
MC87 / MC88 / MC89	SOP standard: create/change/display plan (product group / material)	Sales & operations planning	COVERED-22	22	Forecast consensus workflow (S&OP) — rough-cut capacity planning (MC88 RCCP) partially via wave 23 capacity load; treat covered
MC84 / MC85 / MC86	Product group create/display/change (SOP aggregation)	Planning hierarchies	COVERED-22	22	Forecast at product family level with disaggregation — confirm; treat covered
MC93 / MC94 / MC95 / MC96	Flexible planning: create/change/display/copy plan versions (info structures)	Flexible planning	SUPERSEDED-22	22	Planning on lakehouse/semantic layer + forecast service (wave 46/22)
MC74 / MC75 / MC76	Transfer SOP to demand management (material / product group / disaggregate)	—	COVERED-22	22	Consensus → PIR
MC8B / MC8E / MC8V	Event, mass processing SOP	—	COVERED-22	22	—
MS01 / MS02 / MS03 / MS04 / MS05 / MS06 / MS07 / MS08	Long-term planning (LTP): planning scenario, simulative MRP, stock/requirements (simulation), evaluations	Simulation planning	PARTIAL-22	22	What-if simulation plans (separate planning versions with simulated demand, evaluating capacity/purchasing) not explicitly named — partial; propose "static plans" (AX-style) in wave 22
MS31 / MS32 / MS33 / MS44 / MS47 / MS64 / MS65 / MS70	LTP: planning scenario create/change/display, planned orders in scenario, evaluation (capacity, PURCHIS transfer)	—	PARTIAL-22	22	Same
MD01 / MD02 / MD03	MRP total planning / single-item multilevel / single-item single-level	MRP run	COVERED-22	22	Planning engine with regenerative/net-change modes
MD01N (MRP Live, S/4)	MRP Live (HANA-based total run)	—	COVERED-22	22	50k items < 10 min target
MD40 / MD41 / MD42 / MD43	MPS total / single-item multilevel / single-level / interactive	MPS	COVERED-22	22	MPS with firming (planning time fence)
MD50 / MD51	MRP for sales order / project	Order-specific MRP (MTO)	COVERED-22	22	Pegged planning for MTO — covered via pegging; sales-order stock partial noted in MM-IM
MD04	Stock/requirements list (current)	Net requirements view	COVERED-22	22	Net requirements / pegging inquiry
MD05 / MD06	MRP list (individual / collective)	Planning results at run time	COVERED-22	22	Plan run results snapshot
MD07	Stock/requirements list collective (by MRP controller, exception groups, traffic lights)	Planner worklist	COVERED-22	22	Planner workspace with action messages
MD09	Pegged requirements	Pegging	COVERED-22	22	Pegging tree
MD4C	Multilevel order report	Order network	COVERED-22	22	Explain-this-plan
MD11 / MD12 / MD13 / MD16 / MD19	Planned order create/change/display/collective display/firm	Planned orders	COVERED-22	22	Planned orders with firming, grouping
MD14 / MD15 / MD17	Convert planned order → PR (individual / collective) / conversion collective PR	Conversion	COVERED-22	22	Firm → requisition/PO
CO40 / CO41 / CO48	Convert planned order → production order (individual / collective / partial)	Conversion	COVERED-21	21	Firm → production order
MD20 / MD21	Create / display planning file entry	Net-change triggers	SUPERSEDED-22	22	Event-driven net change
MD25 / MD26 / MD27	Planning calendar create/change/display	Period lot-sizing calendars	COVERED-22	22	Period coverage calendars — confirm custom period calendars; treat covered
MDAB / MDRE	Planning file setup / consistency check	—	SUPERSEDED-22	22	—
MDBT	MRP run in background (variants)	—	COVERED-22	22	Scheduled nightly plan
MDVP	Collective availability check (planned orders)	—	COVERED-22	22	—
MDSA / MDSB	Planned orders with/without components (BOM explosion numbers)	—	COVERED-22	22	—
MF50	REM planning table (rate-based line loading)	Lean planning board	COVERED-24	24	Production flow / kanban board; rate-based planning — confirm line-loading planning table; treat covered
MFBF / MF41 / MF42N / MF70	REM backflush (assembly/component/activity) / reverse / collective / aggregate collective confirmation	REM execution	COVERED-24	24	Backflush costing for lean (periodic)

T-code	Name	Purpose	Passage status	Wave	Notes
MF60 / MF63 / MF65 / MF68	Pull list (material staging) / stock transfer / situation / printing	Material staging	COVERED-24	24	Kanban/withdrawal + wave 31 staging
MF12 / MF26 / MF27	Document log / display postprocessing errors	—	COVERED-24	24	—
MF30 / KKF6N	Create product cost collector (REM)	—	COVERED-24	24	—
MFPR	Production version check (REM)	—	COVERED-24	24	—
PK01 / PK02 / PK03 / PKMC	Control cycle create/change/display / maintenance (S/4)	Kanban rules (supply area, container qty, replenishment strategy: in-house/external/stock transfer, status sequence)	COVERED-24	24	Kanban rules (manufacturing/withdrawal/fixed/event)
PK05 / PK05S	Supply area create / display	Supply areas (PSA)	COVERED-24	24	Supply areas / kanban locations
PK13N / PK12N / PK11	Kanban board (demand source / supply source / status change)	Kanban board	COVERED-24	24	Kanban board, cards
PK21 / PK22 / PK23 / PK31 / PK41 / PK51	Kanban status change (barcode), kanban correction, event-driven kanban (PK23), kanban calculation (PK41), card print	—	COVERED-24	24	Event kanbans; card printing (wave 32 labels); kanban quantity calculation (PK41/PK51) — confirm kanban quantity calculation; treat covered
PK17 / PK18	Kanban plan/actual, kanban evaluation	—	COVERED-24	24	—
CO01 / CO02 / CO03	Production order create/change/display (with material)	Production order	COVERED-21	21	Production order lifecycle
CO07 / CO08 / CO10	Production order without material / for sales order / for project	Variants	COVERED-21	21	Rework orders (CO07), MTO orders (CO08), project orders (wave 41)
CO05N / CO05	Collective release of orders	—	COVERED-21	21	Mass release
CO11N / CO11 / CO12 / CO14 / CO15 / CO16N / CO13	Confirmations: single operation (time ticket) / fast entry / collective / display / order confirmation / postprocessing errors / cancel	Confirmations	COVERED-21	21	Report as finished + route card/job card confirmations; shop-floor kiosk (wave 25)
CO1F / CO1L / CO1P / CO1V	Progress confirmation / confirmation list / predefined confirmation processes / fast entry	—	COVERED-21	21	—
COGI	Postprocess failed goods movements (backflush errors)	Error queue	COVERED-21	21	Backflush error queue with retry — confirm; essential; treat covered
CO27 / CO28	Picking list / collective picking	Component picking	COVERED-21	21	Picking lists (backflush/manual)
MB1A 261 / MIGO GI for order, MB31 / MIGO GR for order (101), 531 by-product, 545	Goods movements for production orders	—	COVERED-21	21	Material issues, RAF with by-/co-products
CO02 → TECO / CLSD / mass via COHV	Technically complete / close orders	Status transitions	COVERED-21	21	Lifecycle state machine (end)
COHV / COHVPI / COMAC	Mass processing of production orders (release, print, schedule, costing, TECO, close, availability check)	Mass processing	COVERED-21	21	Mass operations with job framework
COOIS / COOISPI	Order information system (orders, operations, components, confirmations, documented goods movements, capacities, PRTs)	Inquiry	COVERED-21	21	Production order inquiry: cost roll, status log, transactions
CO24	Missing parts information system	Shortages	COVERED-21	21	Component shortage list (ATP on components)
CO46 / CO50	Order progress report / collective	—	COVERED-21	21	—
CO04 / CO04N	Print order papers (shop papers: job ticket, pick list, time ticket, confirmation slip)	Shop papers	COVERED-21	21	Route cards, job cards; document designer wave 46
CORS / CO88 (see CO-PC)	—	—	COVERED-21	21	—
CM01 / CM02 / CM03 / CM04 / CM05	Capacity planning: load (standard overview) / orders / pool / backlog / overload	Capacity evaluation	COVERED-23	23	Capacity load views by resource/group/period
CM07 / CM50 / CM51 / CM52	Variable capacity evaluation / work center view / order view / extended	—	COVERED-23	23	—
CM21 / CM22 / CM23 / CM25 / CM26 / CM27 / CM28 / CM29 / CM31	Capacity leveling: planning table (graphical/tabular), dispatching, work center view, individual capacity	Finite scheduling / dispatching	COVERED-23	23	Interactive Gantt with drag-reschedule; finite scheduling
CM40 / CM41	Capacity leveling in background / evaluation of background leveling	—	COVERED-23	23	Scheduler service batch
CM99 / CM38	Capacity leveling profiles	—	COVERED-23	23	—
CR15 / CR16	Where-used work center / capacity	—	COVERED-9	9	—
Production scheduling board (Fiori Manage Production Operations / Schedule Production), PP/DS board	—	—	COVERED-23	23	Gantt
Fiori: Manage Production Orders / Confirm Production Operation / Monitor Production Orders / Manage Material Coverage / Monitor Material Coverage / Display MRP Master Data Issues / Manage Change Requests (MRP) / Schedule MRP Runs / Manage Planned Orders / Manage Supply Demand / Monitor External Requirements / Monitor Internal Requirements	S/4 PP Fiori apps	—	COVERED-22	22	Planner workspace
Fiori: Manage Work Center Capacity / Schedule Production / Monitor Work Center Schedules / Monitor Capacity Utilization	—	—	COVERED-23	23	—
Fiori: Kanban Board / Manage Control Cycles	—	—	COVERED-24	24	—
Fiori: Manage Production Versions / Manage Bills of Material / Manage Routings / Manage Work Centers / Manage Change Records (ECM)	—	—	COVERED-9	9	—

T-code	Name	Purpose	Passage status	Wave	Notes
Demand-driven replenishment (DDMRP: buffer positioning, buffer levels, replenishment planning, Fiori apps)	DDMRP (S/4)	—	GAP	propose 22	DDMRP (decoupling points, buffer zones red/yellow/green, net flow equation) not named; propose as wave 22 coverage-group type "DDMRP buffer"
Predictive MRP (pMRP, S/4)	Rough simulation of future demand vs capacity	—	PARTIAL-22	22	Overlaps LTP partial
Production planning with PP/DS — see §13.23	—	—	COVERED-23	23	—
Subcontracting with MRP (special procurement 30), components provided, subcontract orders from planned orders	—	—	COVERED-21	21	Subcontracting operations with vendor PO linkage (wave 21) + MRP (wave 22)
Co-products / by-products in BOM & order settlement (apportionment structure)	—	—	COVERED-24	24	Co/by-product costing allocation
Phantom assemblies, alternative items (BOM item alternative groups), discontinuation (MM02 follow-up material)	—	—	PARTIAL-9	9	Phantoms yes; alternative item groups (priority/strategy) and discontinuation/follow-up (run-out) are not named — partial; propose wave 9 BOM line attributes + wave 22 MRP logic
Assembly processing / assemble-to-order (planning strategies 81/82)	ATO: production order created from sales order with dynamic ATP	—	COVERED-27	27	Order with configured product creates production/planned order linkage
Make-to-order with sales order stock (strategy 20), planning with final assembly (40), planning without final assembly (50), subassembly planning (70)	Planning strategies	—	PARTIAL-22	22	Coverage groups + forecast reduction named; the full strategy-group semantics (consumption modes, MTO stock segmentation, planning at assembly level) are partially covered — partial; sales-order stock noted in MM-IM
Scrap: assembly scrap, component scrap, operation scrap (routing), net ID	Scrap planning	—	COVERED-9	9	BOM scrap and route scrap — confirm operation scrap on routing; treat covered
Rework (CO07 / trigger points CO31 / CO33)	Rework orders / trigger points inserting rework operations	—	PARTIAL-21	21	Rework via order w/o material covered; trigger points (automatic rework operation insertion on confirmation) not named — partial
Serial numbers in production orders (header/components), batch determination in production (COB1)	—	—	COVERED-21	21	Tracking dimensions
Order split (CO02 → split), order change management (OCM: COCM / COCM1)	Order split / order change management after ECO	—	PARTIAL-21	21	Order change management (propagating BOM/routing changes to open orders with change costs) not named — partial; propose wave 9/21

13.21.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
MD04 / MD05 / MD06 / MD07	Stock/requirements and MRP lists	COVERED-22	22	—
MD09 / MD4C	Pegging / order report	COVERED-22	22	—
MD45 / MD46 / MD47 / MD48	Evaluation of planning result (individual/collective) / product group planning / cross-plant evaluation	COVERED-22	22	—
MDLD	Print MRP list	COVERED-22	22	—
MD44 / MD4x	Planning situation (material/product group)	COVERED-22	22	—
COOIS / COOISPI	Production order info system (all list variants)	COVERED-21	21	—
CO24	Missing parts	COVERED-21	21	—
CO26 / CO28	Order information system: planned/production orders	COVERED-21	21	—
COID	Object overview (order info system)	COVERED-21	21	—
CM01 … CM07	Capacity evaluations	COVERED-23	23	—
CS11 / CS12 / CS13 / CS15	BOM explosion / where-used	COVERED-9	9	—
CA80 / CA81 / CA82	Routing where-used	COVERED-9	9	—
CC04	Change master information system	COVERED-9	9	—
MCP1 / MCP3 / MCP5 / MCP7 / MCPB / MCPD / MCPF (PP-IS, Shop Floor Information System)	Work center analysis / operation analysis / material analysis / production order analysis (lead times, quantities, scrap)	COVERED-46	46	Production analytics (OEE in wave 25)
MCRE / MCRI / MCRM (REM analysis) / MCRP / MCRK (kanban analysis)	REM/kanban LIS	COVERED-46	46	—
MCP6 / MCPK	Work center capacity analysis (LIS)	COVERED-46	46	—
MD4 / S_ALR "Planning file entries"	—	SUPERSEDED-22	22	—
PK17 / PK18 / PK19	Kanban evaluations (plan vs actual, status history)	COVERED-24	24	—
MF26 / MF12	REM document log / postprocessing list	COVERED-24	24	—
Fiori: Production Order Overview / Production Execution analytics / Production Cost Analysis / OEE dashboards (SAP Digital Manufacturing)	—	COVERED-25	25	OEE
Fiori: MRP Cockpit (Monitor Material Coverage / Shortages), Display MRP Key Figures	—	COVERED-22	22	—

13.21.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
MD01 / MDBT / MD01N (scheduled)	Nightly MRP / MRP Live run	COVERED-22	22	—
MD40 (scheduled)	MPS run	COVERED-22	22	—
MP38 / MPBT	Mass forecast run	COVERED-22	22	Forecast batch
MC8E / MC8D / MC8G	SOP mass processing (copy, transfer)	COVERED-22	22	—
MD74 / MD75 / MD76	Reorganize PIRs (delete history, adjust)	COVERED-22	22	—

T-code	Name	Passage status	Wave	Notes
MD15 / CO41 (batch)	Mass conversion of planned orders	COVERED-22	22	—
CO05N / COHV (batch release, print, costing)	Mass release / scheduling / costing	COVERED-21	21	—
COGI (periodic)	Backflush error reprocessing	COVERED-21	21	—
CO1P	Predefined confirmation processes (background)	COVERED-21	21	—
CM40 (scheduled)	Background capacity leveling	COVERED-23	23	—
CO88 / KKAX / KKS1 (see CO-PC)	Settlement / WIP / variances	COVERED-21	21	—
MFBF / MF70 (periodic) / KK87 (REM settlement)	REM period-end	COVERED-24	24	—
PK41 / PK51 (kanban calculation run)	Kanban quantity recalculation	COVERED-24	24	—
CORUPROC / CORUAFWP	Confirmation background process (actual costs, goods movements)	SUPERSEDED-21	21	Synchronous posting via journal/inventory APIs; job framework for retries
SARA PP_ORDER / PR_ORDER	Archiving production orders	COVERED-48	48	—
CA97 (periodic)	Update lot-size dependent lead times	COVERED-9	9	—
Fiori: Schedule MRP Runs / Schedule Production Order Jobs	—	COVERED-4	4	—

13.21.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OS20 / OS21 / OS22 / OS23 / OS24 / OS25 / OS27 / OS28 / OS30 — BOM: usages, status, item categories, default values, modification parameters, BOM explosion/applications, alternative determination	T415S, T416, T418, TCS	Item category control (stock/non-stock/variable size/document/text), usage-specific item status	COVERED-9	9	BOM line types and usages
OPJM / OPJI / OPJF — Selection ID for alternative BOM/routing (application, priorities)	TCS, T407	—	COVERED-9	9	Version selection rules
OPJL — Define checking groups / checking rules for order availability (OPJJ scope, OPJK checking control per plant/order type/business event: component, PRT, capacity check; release behaviour)	T441V, T441R, T399X	Availability check at creation/release; collective conversion rules	COVERED-21	21	Component availability on release
OPJH — Production order types (PP01, PP02…)	T003O	Number range, settlement profile, residence time, status profile, CO partner update, classification	COVERED-21	21	Order types
OPL8 — Order type dependent parameters (planning: production scheduling profile, routing selection, BOM application; implementation: documentation of goods movements; cost accounting: costing variants, RA key)	T399X	—	COVERED-21	21	Order type parameters
OPKP — Production scheduling profiles (automatic release/GR/print at creation, batch management, capacity leveling flags, overall profile)	TCO43	—	COVERED-21	21	Production scheduling profile → order defaults
OPJ / OPU3 / OPU5 — Scheduling parameters production orders / planned orders (scheduling type: forward/backward/today, reduction strategies, capacity requirements generation, scheduling horizon)	T399D, TCX	Lead time scheduling vs duration, reduction levels, float before/after (OPJ scheduling margin key OMI8)	COVERED-23	23	Scheduler settings (forward/backward, finite/infinite)
OPK4 — Confirmation parameters (per plant/order type): goods movements (backflush), actual costs posting, error handling, propose dates/quantities, decoupled/synchronous, HR integration (time tickets), termination on error	TCORU, T399	—	COVERED-21	21	Confirmation behaviour
OPK0 / OPK1 / OPK2 / OPK3 — Confirmation screens, single-screen entry layout, reasons for variances	TCORV, T*	—	COVERED-21	21	Reason codes for scrap/variance
OPKF / OPK8 — Print control (shop papers): lists, forms, print programs	T496	—	COVERED-21	21	Document designer
OPJB / BS02 — Status profiles (user statuses for orders)	TJ30	—	COVERED-4	4	—
OPJ2 / OPJ — Define number ranges production orders / confirmations / reservations	NRIV	—	COVERED-5	5	—
OPJ6 / OPJ7 / OPJ8 — Production order field selection, component/operation defaults	T*	—	COVERED-3	3	—
OPPQ — Plant parameters for MRP (number ranges, planning run defaults, floats, scheduling margin keys, direct procurement, BOM explosion, planning file, external procurement defaults, rescheduling horizon, exception messages)	T399D, T436A (margin keys)	—	COVERED-22	22	Planning parameters per site
OPPR — MRP groups (strategy group, consumption mode, planning horizon, creation indicators)	T438M	—	COVERED-22	22	Coverage group / planning group
OMDQ / OMDR — MRP types (PD, VB, VM, VV, ND, M0-M4, P1-P4 with firming types, planning time fence, roll-forward)	T438A	Reorder point, forecast-based, MPS, time fence, firming types 0-4	COVERED-22	22	Coverage groups: requirement/period/min-max; firming/time fences
OMI4 — Lot-sizing procedures (EX, FX, HB, WB, MB, TB, PK, optimum: dynamic/least unit cost/part period balancing, rounding, min/max, takt)	T439A	Static/period/optimum, scheduling, rounding profiles	COVERED-22	22	Lot-sizing: requirement/period/min-max/fixed/multiples/EOQ — confirm optimum procedures (least-unit-cost, part-period) — treat covered with note
OMD9 — Special procurement types (subcontracting 30, phantom 50, stock transfer 40, withdrawal from alternative plant, direct production 52, production in other plant 70/80)	T460A	—	COVERED-22	22	Procurement type per item/site incl. transfer from other site (transfer order gap noted)
OMD0 — MRP controllers	T024D	—	COVERED-22	22	Planner assignments
OMDX / OMDU — Activate MRP / planning file per plant	T399D, MDLV	—	SUPERSEDED-22	22	—

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OMDY / OMD3 — Exception messages, exception groups	T458A, T458B	—	COVERED-22	22	Action/futures messages
OPPS / OPPT — Planning strategies / strategy groups (10, 11, 20, 25, 30, 40, 50, 52, 59, 60, 63, 70, 74, 81, 82…)	T461S, T461P	Requirements type, consumption, assembly processing	PARTIAL-22	22	Planning strategy semantics partial (noted)
OMPO / OMI — Consumption mode, consumption periods (backward/forward)	T438M	—	COVERED-22	22	Forecast reduction
OMIK / OMI8 — Scheduling margin key, floats	T436A	—	COVERED-22	22	Safety margins
OMIZ / MRP areas (OMIZ / S/4 MRP areas per storage location, subcontractor)	MDLV, MDLG, MDLW	—	PARTIAL-22	22	MRP area partial
OMIX / OMD — Define planning horizon, rescheduling check (tolerances), firm receipts handling	T399D	—	COVERED-22	22	Action message tolerances
OMDT / Safety stock availability %, dynamic safety stock (range of coverage profiles OMIA)	T438R	—	COVERED-22	22	Safety stock + period coverage — confirm range-of-coverage profile (dynamic safety stock) — treat covered
MC7F / OMPB / OMPC — SOP: info structure S076 planning parameters, planning types (MC8A/MC8B), macros, period split, aggregation/disaggregation (proportional factors MC9)	S076, T445, TMC	—	COVERED-22	22	Forecast hierarchy disaggregation
OMPN / OMPK — Demand management: requirement types, PIR versions, consumption, requirements class	T459K, T459A, T459U	—	COVERED-22	22	—
MP80 / OMPH / OMPG — Forecast profiles, forecast parameters (model selection, alpha/beta/gamma, periods)	T* , MPOP	Constant, trend, seasonal, Croston	COVERED-22	22	Forecast models
OPPB / OPPP — Planning calendars	T439	—	COVERED-22	22	—
OP40 / OP41 / OP42 / OP43 / OP44 / OP45 … — Work center: categories, field selection, default values, capacity categories, formulas, standard value keys, performance efficiency rate, shift definitions, shift sequences, move time matrix, location groups	TC26, TC27, TC30, TC31, TC37, TC24, T430, T435, T436	Formulas for setup/processing/teardown/capacity; standard value keys (SAP1, SAP2); shifts	COVERED-9	9	Resources: capabilities, calendars with shifts, time formulas — confirm formula-based durations (quantity-dependent) — treat covered
OP00 / OP01 / OP4 — Capacity planning profiles (overall profiles, selection, time, evaluation, strategy, period profiles for CM21/CM25), dispatching strategy (finite, insert, midpoint, date entry, sequence dependent setup)	TCY*	—	COVERED-23	23	Scheduler settings; setup optimization
OP4A / OP4B — Capacity reduction, available capacity versions, intervals of available capacity	TC26, KAKO, KAZY	—	COVERED-23	23	Capacity versions
OP5A / OP8B / OP8 — Routing: task list types, usages, status, control keys (OPJ8 OP67 / OP00 — control key: scheduling, capacity, costing, confirmation required, external processing, print, milestone), operation default values, number ranges, user fields, PRT control keys	TCA*, T430 (control keys), T430T	Control key flags	COVERED-9	9	Operation control attributes (schedule/cost/confirm/external)
OP67 / OPJ — Control keys (PP01, PP02, PP03 external, PP99 milestone)	T430	—	COVERED-9	9	—
OPMP / OPM — Trigger points (standard trigger points, functions)	TTRIG	—	PARTIAL-21	21	Trigger point gap
OS40 / OS41 / OS44 — ECM: change types, object types, status, ECR/ECO profiles, release keys, effectivity types (date, serial, parameter)	TCC*, TECH*	Date effectivity vs parameter/serial effectivity	PARTIAL-9	9	Date effectivity covered; serial-number/parameter effectivity not named — partial
OPZ / OSPT / OSP2 — REM profiles (reporting point backflush, cost collector, GI/GR posting, aggregate backflush, RP confirmation)	T437S	Reporting points (milestones), pull-list, cost collector	COVERED-24	24	Lean profiles
OSP / OPKA — Kanban: control cycle categories, replenishment strategies (in-house: 0001 production order, 0002 run schedule, 0003 manual KANBAN; external: PO, SA, source list; stock transfer: reservation, TO), status sequences, kanban calculation profiles, print control	TPKM, TPK*	—	COVERED-24	24	Kanban rules types
OPKW / OPKX — PDC (plant data collection) interfaces: CC2/CC4 communication parameters, upload/download, time events	TCORU, T*	—	COVERED-25	25	Shop-floor kiosk + IoT replaces PDC; HR time events in wave 25
OPJG / OPK — Batch determination in production (search procedure, strategy types), serial number profiles per order type	TLSP, T*	—	COVERED-21	21	—
OPJW / OPJ — Define default values for conversion (planned → production), order types per MRP group	T399X, T438M	—	COVERED-22	22	—
OPPE / OPPI — Default document types for PRs from MRP, purchasing group, account assignment	T399D	—	COVERED-22	22	—
OPJJ — Availability check scope (production)	T441V	—	COVERED-21	21	—
OPJK — Checking control (order type × plant × business transaction)	T399X / T441R	—	COVERED-21	21	—
OMK / OMI — Define MRP areas, MRP list layout, MD04 navigation profiles, period totals	T*	—	COVERED-22	22	—
OPPS "Assembly processing" settings (order type for ATO)	T461S	—	COVERED-27	27	—
DDMRP buffer profiles, classification runs (S/4 DD)	PPH_DD_*	—	GAP	propose 22	DDMRP gap
OCM profiles (order change management: change profiles, change cost)	COCM*	—	PARTIAL-21	21	OCM partial

13.22 PP-PI — Production Planning for Process Industries

SAP context: master recipes (C201–C203; recipe header, operations/phases, resources, materials, secondary resources, process instructions/XSteps), process orders (COR1–COR3, COR5 collective, COR6N confirmation, CORR? mass), process management (control recipes CO53, process messages CO54/CO57, PI sheets CO60), electronic batch record (COEBR), batch management and batch determination, resource networks (CMX? / RN?), campaign planning (PP-PI campaigns), process manufacturing cockpit (CO50?), material quantity calculation. Passage: wave 24 (formulas/batch orders, potency, co/by-products, catch weight, recipe version control), wave 21 (order lifecycle), wave 25 (shop floor/PI sheets), wave 33 (QM in process).

13.22.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
C201 / C202 / C203	Master recipe create/change/display (recipe group, operations, phases, material list, resources, process instructions, QM characteristics, relationships)	Recipes	COVERED-24	24	Formula + route (recipe) with version control integrated with ECM
C223 (PI)	Production versions (recipe + formula)	—	COVERED-24	24	—
C251 / C252 / C253	Print recipes / recipe lists / where-used	—	COVERED-24	24	—
C260 / C261	Change recipe with ECM / recipe change documents	—	COVERED-9	9	—
C298	Mass deletion of recipes	—	COVERED-24	24	—
CRC1 / CRC2 / CRC3	Resource (PI) create/change/display (resource category: processing unit, labor, storage, vessel)	Resources	COVERED-9	9	Resource model
CRC5 / CRC	Resource network (RN)	Resource networks for scheduling	PARTIAL-23	23	Resource networks (allowed sequences of resources, e.g. vessel → filler) not named — partial; scheduling with CP-SAT could model as constraints
CS01 (BOM for PI = formula) / material quantity calculation (C201 → MQC formulas)	Formula with material quantity calculation (active ingredient scaling, yield)	—	COVERED-24	24	Potency-based consumption, scalable quantities
COR1 / COR2 / COR3	Process order create/change/display	Batch/process orders	COVERED-24	24	Batch orders from formulas
COR5 / CORS / COR7 / COR7_PI / COR8	Collective release / cancel / create from planned order (individual / collective)	—	COVERED-24	24	—
COR6 / COR6N / CORK / CORR / CORT / CORZ	Confirmation: phase / order / collective / display / cancel	Confirmations	COVERED-24	24	Batch order reporting (RAF with co/by-products)
CORA / CORM / CORP	Process order scheduling / mass processing (COHVPI)	—	COVERED-24	24	—
CO53 / CO53XT	Control recipe monitor (send control recipes to process control / PI sheet)	Process management	PARTIAL-25	25	Shop-floor instructions/drawings on kiosk; structured *control recipes* with process instruction categories sent to a destination (PCS/PI sheet) are partial
CO54 / CO54XT / CO57 / CO58	Process message monitor / create process message manually / send	Process messages (from shop floor/PCS → ERP: consumption, yield, quality results)	PARTIAL-25	25	IoT/MQTT machine events + kiosk reporting cover inbound; configurable message categories with destinations (e.g. to QM results, to batch record) partial
CO60 / CO60XT / CO60_PI / CO64 / CO67	PI sheet: find/maintain / worklist / print / archive	Electronic work instructions (XSteps/PI sheets)	PARTIAL-25	25	Kiosk instructions; a designer for interactive PI sheets (XSteps with input fields, calculations, signatures, process data requests) is not named — partial
COEBR	Electronic batch record (EBR) creation/approval/archive (21 CFR Part 11)	Batch record	GAP	propose 24	EBR with digital signatures, deviation review and archiving (GxP) not named — propose wave 24 (+ wave 48 compliance)
CMX / Standard XSteps (CMXSV) / SXS repository	Standard XSteps repository	—	PARTIAL-25	25	Same as PI sheet
CO50 (Process manufacturing cockpit) / COPOC	Process operator cockpit	—	COVERED-25	25	Shop-floor kiosk
MSC1N / MSC2N / MSC3N / MSC4N (batch master — see §13.33)	Batch create/change/display/changes	—	COVERED-24	24	—
Campaign planning (PP-PI campaign, process order campaigns: COCP / CMP)	Group process orders into campaigns (setup/cleanout)	—	PARTIAL-23	23	Campaign/sequence planning (group orders to minimize changeover, cleanout orders) not named — partial; CP-SAT setup optimization partly covers
Cleanout / changeover orders, recipe-dependent cleaning	—	—	PARTIAL-23	23	Same
Digital signature in PI sheets / batch records (DSIG)	21 CFR Part 11 signatures	—	GAP	propose 24	E-signature framework (meaning of signature, re-authentication) not named; propose cross-cutting in wave 24/48
Shelf life / batch expiry, batch-specific UoM (potency UoM, proportion/product units)	—	—	COVERED-24	24	Batch attributes, shelf life, catch weight dual UoM, potency
Batch derivation (DVMO / DVMAN) — see §13.33	—	—	PARTIAL-24	24	Batch derivation (inherit attributes from component batches to product batch) not named — partial
Fiori: Manage Process Orders / Confirm Process Order Phase / Monitor Process Orders	S/4 apps	—	COVERED-24	24	—

13.22.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
COOISPI	Process order information system	COVERED-24	24	—
CO54 (monitor)	Process message log	PARTIAL-25	25	—
COEBR (list)	Batch record list	GAP	propose 24	—
MB56 / BMBC	Batch where-used / batch information cockpit	COVERED-24	24	Batch traceability (top-down/bottom-up)
MCP (PP-PI LIS: resource analysis, phase analysis)	—	COVERED-46	46	—
C251 recipe lists	—	COVERED-24	24	—
CMX/CO60 worklists	PI sheet worklist	PARTIAL-25	25	—

13.22.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
COR5 / COHVPI (batch)	Mass release / scheduling	COVERED-24	24	—
CO53 (send control recipes, batch)	—	PARTIAL-25	25	—
CO54 (process message processing, batch)	—	PARTIAL-25	25	—
CO64 / CO67 (PI sheet archive)	—	PARTIAL-25	25	—
COEBR (batch record generation/archive)	—	GAP	propose 24	—
CO88 / KKAX / KKS1 (process orders)	Settlement / WIP / variances	COVERED-21	21	—
Campaign settlement / cleanout cost allocation	—	PARTIAL-23	23	—
SARA PR_ORDER	Archive process orders	COVERED-48	48	—

13.22.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Master recipe: profiles, task list types (2), status, control keys (PI01, PI02…), phase/operation categories, relationship types	TCA*, T430	—	COVERED-24	24	—
Process order types (PI01…), order type dependent parameters (COR4), production scheduling profile (CORY), confirmation parameters (CORO / OPK4 PI), availability check (OPJJ/OPJK PI)	T003O, T399X, TCO43	—	COVERED-24	24	—
Material quantity calculation: formulas, functions	—	—	COVERED-24	24	Potency scaling
Process management: process instruction categories/characteristics (O10C O12C / O13C), process message categories & destinations (O05C / O06C / O07C), control recipe destinations (O10C), PI sheet layouts, XStep settings (SXS), signature strategies (O2 / DSIG)	TPM*, CPM*	Destinations: PI sheet, process control system (PCS via PI-PCS interface), ERP functions (goods movement, confirmation, QM results)	PARTIAL-25	25	PI sheet / message framework partial
Resource networks	—	—	PARTIAL-23	23	—
Campaign planning settings	—	—	PARTIAL-23	23	—
EBR: batch record profiles, archiving, approval workflow	—	—	GAP	propose 24	—
Batch management in PP-PI: batch determination procedures for process orders (COB1), classification (023), batch derivation	TLSP, T*	—	PARTIAL-24	24	Derivation partial
Co-product apportionment structures (CK / OPJ)	—	—	COVERED-24	24	Allocation methods

13.23 PP/DS — Advanced Planning & Detailed Scheduling (APO PP/DS and S/4 embedded PP/DS)

SAP context: APO PP/DS (CIF integration CFM1/CFM2, product view /SAPAPO/RRP3, detailed scheduling planning board /SAPAPO/CDPS0, heuristics, optimizer, pegging, alert monitor /SAPAPO/AMON1, product planning table /SAPAPO/PPT1, PP/DS runs /SAPAPO/CDPSB0, order views /SAPAPO/RRP4, block planning, setup matrices, characteristic-dependent planning (CDP), MRP heuristics, shelf-life planning). S/4HANA embedded PP/DS: advanced planning flag on material, PP/DS planning via MD04/Fiori, same planning board. Passage: wave 23 (scheduler service with finite/infinite, OR-Tools CP-SAT optimization, Gantt, bottlenecks) + wave 22 (pegging). PP/DS-specific supply-chain-wide planning (SNP, GATP) outside this component.

13.23.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
CFM1 / CFM2 / CFM3 / CFM4 / CFM5 / CFM6	CIF integration models: generate / activate / display / deactivate / delete / change transfer	ERP ↔ APO master & transactional data	SUPERSEDED-23	23	Embedded single system, Kafka events; no CIF
CFC9 / CFG1 / CFP1 / CFP3	CIF settings / application log / queue display	—	SUPERSEDED-23	23	—
/SAPAPO/MAT1 / /SAPAPO/LOC3 / /SAPAPO/RES01 / /SAPAPO/PDS_DISP / /SAPAPO/SCC03 (PPM)	APO product / location / resource / PDS (production data structure) / PPM master	Planning master data	SUPERSEDED-23	23	Native product/site/resource/BOM-route data (waves 6/8/9)
/SAPAPO/RRP3	Product view (receipts/requirements, pegging, order creation)	Planner view	COVERED-22	22	Net requirements view with pegging
/SAPAPO/RRP4 / /SAPAPO/RRP7	Receipts view / product overview	—	COVERED-22	22	—
/SAPAPO/CDPS0 / /SAPAPO/CDPS1 / CDPSC	Detailed scheduling planning board (variable/interactive), DS strategies	Gantt scheduling	COVERED-23	23	Interactive Gantt with drag-reschedule

T-code	Name	Purpose	Passage status	Wave	Notes
/SAPAPO/PPT1	Product planning table (REM/line loading)	—	COVERED-24	24	Lean/rate-based board
/SAPAPO/CDPSB0 / /SAPAPO/CDPSB1	Production planning run (heuristics: SAP_PP_002 planning standard lots, SAP_PP_003, SAP_MRP_001, SAP_PP_C001 bottom-up, SAP_PP_SL001 shelf life…) in background / display	Planning run	COVERED-22	22	Planning engine runs with pluggable heuristics (lot sizing); bottleneck/multi-level heuristics via CP-SAT
/SAPAPO/OPT (PP/DS optimizer profiles & runs), /SAPAPO/CDPSOPT	Optimizer (genetic algorithm / constraint propagation; objectives: makespan, lateness, setup time/cost, mode cost)	Optimization	COVERED-23	23	CP-SAT objectives: minimize lateness/makespan; setup optimization
Setup matrices / setup groups (/SAPAPO/CDPSC4 / CDPSC5)	Sequence-dependent setup	—	COVERED-23	23	Setup optimization (sequence-dependent setup matrix) — confirm matrix data model; treat covered
Block planning (resource blocks per characteristic, /SAPAPO/CDPSC12)	Reserve capacity blocks by product characteristics	—	GAP	propose 23	Block planning (time blocks on resources restricting which characteristic values can be produced) not named — propose wave 23 constraint type
Characteristic-dependent planning (CDP) / CDP configuration	Planning by batch characteristics (e.g., color, grade)	—	PARTIAL-24	24	Batch attributes exist; planning that matches demand characteristics to supply batches (CDP) — partial; propose wave 22/24
Shelf-life planning (SAP_PP_SL001)	Pegging respecting expiry (FEFO planning)	—	PARTIAL-22	22	FEFO picking (wave 16) yes; planning-time shelf-life pegging (min remaining shelf life at demand date) not named — partial
Pegging (dynamic/fixed), /SAPAPO/PEG fixed pegging	Pegging	—	COVERED-22	22	Pegging tree; fixed pegging (manual lock) — confirm "firming"; treat covered
/SAPAPO/AMON1 / /SAPAPO/AMON3	Alert monitor (alert profiles, overall profiles)	Exceptions	COVERED-22	22	Action/delay messages + wave 5 alerts
/SAPAPO/RRPCUST1 / /SAPAPO/CDPSC (strategy profiles, planning board profiles)	Planning strategies (finite/infinite, insert/append/squeeze-in, planning direction, scheduling mode)	—	COVERED-23	23	Scheduler strategies
/SAPAPO/MC01 (models/versions), /SAPAPO/MVM	Planning versions (simulation versions)	—	PARTIAL-22	22	Simulation versions = LTP partial
/SAPAPO/SDP94 (DP), /SAPAPO/SNP94 (SNP), GATP (/SAPAPO/AC03)	Demand planning / supply network planning / global ATP	—	PARTIAL-22	22	DP ≈ wave 22 forecasting (statistical + consensus); SNP (multi-site network optimization: sourcing across plants/DCs with transport lanes and capacity) and GATP (rules-based ATP, multi-site substitution) are not named — partial; register multi-site network planning as a v1.x candidate under wave 22/27
S/4 embedded PP/DS: "Advanced planning" flag (MM02), PP/DS planning via MD04 extension, Fiori: Monitor Capacity Utilization / Schedule Production / Production Scheduling Board	—	—	COVERED-23	23	—
Multi-resource planning (modes, alternative resources, secondary resources)	—	—	COVERED-23	23	Primary/secondary resources, alternative resources via capabilities
Finite scheduling with time-continuous capacity, calendar, breaks, downtime (from PM)	—	—	COVERED-23	23	Downtime from wave 44 affects capacity

13.23.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
/SAPAPO/RRP3 / RRP4 / RRP7	Product/receipt views	COVERED-22	22	—
/SAPAPO/CDPS0 (views: resource, order, operation)	DS board views	COVERED-23	23	—
/SAPAPO/AMON1	Alert monitor	COVERED-22	22	—
/SAPAPO/PEG1	Pegging overview	COVERED-22	22	—
/SAPAPO/OPT11 (optimizer logs)	Optimizer explanation	COVERED-23	23	Explain schedule (constraint warnings)
/SAPAPO/CDPSC11 (resource load / capacity utilization)	Capacity utilization	COVERED-23	23	—
Fiori: Monitor Capacity Utilization / Monitor Resource Schedule	—	COVERED-23	23	—
/SAPAPO/CCR (CIF comparison/reconciliation)	ERP vs APO consistency	SUPERSEDED-23	23	Single system

13.23.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
/SAPAPO/CDPSB0 (scheduled planning runs with heuristic steps)	Nightly PP/DS run	COVERED-22	22	—
Optimizer background runs	—	COVERED-23	23	—
/SAPAPO/RLCDEL (delete old orders), /SAPAPO/OM17 (consistency)	Housekeeping	SUPERSEDED-23	23	—
CIF delta/initial transfers, queue monitoring (SMQ1/SMQ2, /SAPAPO/CQ)	—	SUPERSEDED-23	23	—
Conversion of planned orders (PP/DS → ERP), /SAPAPO/RRP_	—	COVERED-22	22	—
Alert determination runs	—	COVERED-22	22	—

13.23.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Global settings for PP/DS (planning version, pegging, order types), planning procedures (per product), heuristics (algorithms, settings), heuristic profiles	/SAPAPO/*	Automatic planning immediately / in planning run; planning procedure 1-5	COVERED-22	22	Planning policies per coverage group
Strategy profiles (scheduling modes, planning direction, finiteness level, validity periods, propagation ranges), planning board profiles, time profiles, work areas	/SAPAPO/*	—	COVERED-23	23	—
Optimizer profiles (objectives weights, runtime, modes), setup matrices	/SAPAPO/*	—	COVERED-23	23	—
Alert profiles / alert types	/SAPAPO/*	—	COVERED-22	22	—

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
CIF: integration model settings, publication types, filter objects, error handling (CFC)	CIF*	—	SUPERSEDED-23	23	—
Block planning settings, CDP settings	—	—	GAP	propose 23	Block planning gap
Embedded PP/DS activation (S/4): advanced planning settings, PP/DS material flag defaults	—	—	SUPERSEDED-23	23	—
Resource categories (single/multi/mixed), capacity variants, breaks, shifts, downtimes	/SAPAPO/RES*	—	COVERED-9	9	—

13.24 QM — Quality Management

SAP context: QM-PT (inspection plans QP01, master inspection characteristics QS21, inspection methods QS31, catalogs QS41, sampling procedures QDV1, sampling schemes QDP1, dynamic modification rules QDB1, quality info records QI01/QV51), QM-IM (inspection lots QA01/QA32, results recording QE01/QE51N, defects recording, usage decisions QA11, stock postings), QM-QC (control charts QGC1, quality level QDL1), QM-QN (quality notifications QM01, tasks, 8D), QM-CA (certificates QC01/QC21, CoA on delivery), QM-IT (test equipment calibration via PM), vendor evaluation integration, stability studies (QST01), audit management (PLMD_AUDIT), FMEA, control plan (S/4 QM). Passage: wave 33 (tests, sampling, associations, quality orders, UD, blocking, NCR/CAPA, CoA, scorecards, SPC, calibration), wave 16 (quarantine status), wave 18 (receive to QI).

13.24.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
QS21 / QS22 / QS23 / QS24 / QS25 / QS26 / QS27 / QS28	Master inspection characteristic create/change/display (quantitative/qualitative, tolerances, catalogs, sampling) / versions / where-used / list / replace	Test definitions	COVERED-33	33	Tests (variable/attribute) with tolerances, catalogs
QS31 / QS32 / QS33 / QS34 / QS37 / QS38 / QS39	Inspection method create/change/display/versions/where-used/list	Test methods	COVERED-33	33	Test instructions/methods
QS41 / QS42 / QS43 / QS44 / QS45 / QS46 / QS47 / QS48 / QS49	Catalogs: code groups/codes (characteristic attributes, defect types, defect locations, causes, tasks, activities, usage decisions) create/change/display/list; selected sets (QS51 / QS52 / QS53)	Code catalogs	COVERED-33	33	Catalogs: attribute values, defect codes, cause codes, disposition codes
QP01 / QP02 / QP03 / QP05 / QP06 / QP07 / QP08	Inspection plan create/change/display/print/list/mass change/where-used (operations, characteristics, inspection points, test equipment)	Inspection plans	COVERED-33	33	Test groups + quality associations
QP11 / QP12 / QP13	Reference operation set (QM)	—	COVERED-33	33	Test group templates
QDV1 / QDV2 / QDV3 / QDV6 / QDV7	Sampling procedure create/change/display/uses/replace (sample size: fixed, %, sampling scheme, 100%; valuation mode: attributive, variable, manual)	Sampling	COVERED-33	33	Sampling plans (AQL per ANSI/ASQ Z1.4)
QDP1 / QDP2 / QDP3	Sampling scheme create/change/display (AQL tables, inspection severity)	—	COVERED-33	33	—
QDB1 / QDB2 / QDB3	Dynamic modification rule create/change/display (skip lots, tightened/reduced/normal)	Skip-lot / dynamic modification	COVERED-33	33	Inspection skip rules based on quality level — confirm skip-lot (switching rules) in wave 33 sampling; treat covered
QDL1 / QDL2 / QDL3	Quality level create/change/display	—	COVERED-33	33	—
QI01 / QI02 / QI03 / QI05 / QI06 / QI07 / QI08	Quality info record procurement (vendor release, QM control key, certificate requirement, inspection skip) / mass change / list / release status	Vendor-material quality agreement	COVERED-33	33	Supplier/item quality status (approved, blocked, probation)
QV51 / QV52 / QV53	Quality info record sales (CoA profile per customer)	—	COVERED-33	33	CoA per customer profile
QC01 / QC02 / QC03	Certificate profile create/change/display	CoA layout	COVERED-33	33	CoA templates
QC11 / QC12 / QC13 / QC15	Certificate profile assignment (material/customer)	—	COVERED-33	33	—
QC21 / QC22	Quality certificate for batch / for inspection lot (print)	CoA output	COVERED-33	33	CoA generation on shipments
QC51 / QC52 / QC53 / QC55	Certificate receipt (incoming certificates from vendor)	Incoming CoA tracking	PARTIAL-33	33	Incoming vendor certificate receipt/tracking (required certificate at GR, block until received) not named — partial
QA01 / QA02 / QA03	Inspection lot create/change/display (manual; automatic at GR 01, production 03/04, delivery 10, stock transfer 08, recurring 09, calibration 14)	Quality orders	COVERED-33	33	Quality orders generated by associations (receipt/production/shipment); recurring inspection (09) — confirm recurring/periodic re-inspection of batches; treat covered
QA32 / QA33	Inspection lot worklist: change / display (results, UD, stock posting)	Worklist	COVERED-33	33	Quality workspace
QA11 / QA12 / QA13 / QA14	Usage decision record/change/display/change with history	Disposition	COVERED-33	33	Usage decisions with inventory blocking/release
QA16 / QA17	Collective usage decision / job planning for UD	—	COVERED-33	33	Mass UD
QE01 / QE02 / QE03 / QE11 / QE12 / QE13 / QE17	Results recording (per lot / operation / inspection point / for test equipment)	Results	COVERED-33	33	Results recording (variable/attribute)
QE51N / QE52 / QE72	Results recording worklist / defects worklist	—	COVERED-33	33	—
QF01 / QF02 / QF03 / QF11	Defects recording (lot / operation / characteristic)	Defects	COVERED-33	33	Nonconformance capture

T-code	Name	Purpose	Passage status	Wave	Notes
QA07 / QA08	Deadline monitoring (recurring inspection, shelf life) / mass change inspection setup in material master	—	COVERED-33	33	Periodic inspection triggers
QA10 / QA05 / QA06	Trigger recurring inspection / job planning	—	COVERED-33	33	—
QE71 / QE72 / QE73	Tabular results recording (inspection points / characteristics / lots)	Mass results	COVERED-33	33	Grid results entry
QM01 / QM02 / QM03	Quality notification create/change/display (types Q1 customer complaint, Q2 vendor complaint, Q3 internal problem; items, causes, tasks, activities, 8D)	NCR / complaints	COVERED-33	33	Nonconformance → correction/CAPA (8D) workflow; customer complaint and supplier complaint types — confirm customer complaint type links to RMA (wave 30); treat covered
QM10 / QM11 / QM12 / QM13 / QM14 / QM19	Notification lists (worklist, tasks, items, activities)	—	COVERED-33	33	—
QM50 / QM (notification cost / order)	QM order for notification costs (appraisal cost)	—	COVERED-33	33	Cost of quality
QK01 / QK02 / QK04	Assign QM order to material / settlement (appraisal costs)	—	COVERED-33	33	Cost of quality
QGC1 / QGC2 / QGC3	Control charts (SPC) for inspection lots / characteristics / tabular	SPC	COVERED-33	33	SPC charts from results
QGA3 / QGA2 / QGP1 / QGP2	Results history / quality score / inspection results analysis	—	COVERED-33	33	—
QST01 / QST02 / QST03 / QST04 / QST05 / QST06 / QST07	Stability study (initial sample, storage conditions, testing schedule)	Stability	GAP	propose 33	Stability studies (pharma/food shelf-life testing schedules) not named; propose wave 33 extension for regulated process manufacturers
QE17 / IE01 (test equipment as equipment, calibration inspection type 14, IP10 calibration plans)	Calibration inspection	—	COVERED-33	33	Device/instrument calibration tracking
PLMD_AUDIT (Audit management: audits, questions, corrective actions)	Audit management	—	PARTIAL-33	33	CAPA yes; audit planning/questionnaires/audit findings (ISO 9001 internal audits, supplier audits) not named — partial
FMEA (QM-FMEA, S/4), control plan (S/4 QM)	FMEA / control plan	—	GAP	propose 33	FMEA and control plans (APQP/PPAP) — gap for automotive/aerospace suppliers; propose wave 33 extension
Quality inspection in EWM (QIE) / inspection at warehouse	—	—	COVERED-33	33	—
Source inspection (QI01 source inspection flag, QI07)	Inspection at vendor before GR	—	PARTIAL-33	33	Source inspection lots (before delivery) not named — partial
Sample management (QPR1 / QPR2 / QPR3 / QPR5 / QPR6 / QPR7)	Physical samples, sample drawing procedure, sample labels	—	PARTIAL-33	33	Sampling sizes yes; physical sample records (primary/pooled samples with labels, retention samples) not named — partial
Multiple specifications (customer/country-specific specs, S/4 "Multiple Specifications")	—	—	PARTIAL-33	33	Customer-specific test specs/limits — partial
Fiori: Manage Inspection Lots / Record Inspection Results / Record Defects / Manage Quality Tasks / Manage Quality Notifications / Process Inspection Lots / Manage Usage Decisions	S/4 apps	—	COVERED-33	33	—
Fiori: Manage Quality Levels / Manage Inspection Plans / Manage Master Inspection Characteristics / Manage Sampling Procedures / Manage Sampling Schemes	—	—	COVERED-33	33	—

13.24.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
QA32 / QA33	Inspection lot lists	COVERED-33	33	—
QM10 / QM11	Notification lists	COVERED-33	33	—
QGA3 / QGA	Results history / quality score	COVERED-33	33	—
QGP1	Results for inspection lot / characteristic (graphics)	COVERED-33	33	—
QGC1 … QGC3	Control charts	COVERED-33	33	—
MCXA / MCXB / MCXC / MCXD / MCXE / MCXF / MCXG / MCXI / MCXK / MCXM / MCXP / MCXV / MCXX (QMIS)	QMIS: inspection lots (material/vendor/customer), notifications, results, defects, quality scores, problems (vendor/customer/material)	COVERED-46	46	Quality analytics; supplier scorecards (wave 33)
MCVA / MCVB / MCVC / MCVD	QMIS problem analyses (notifications)	COVERED-46	46	—
QS28 / QS38 / QS49	Master data lists (MICs, methods, catalogs)	COVERED-33	33	—
QP06 / QP07	Inspection plan lists / where-used	COVERED-33	33	—
QI06	Quality info records list	COVERED-33	33	—

T-code	Name	Passage status	Wave	Notes
QA07 (list)	Deadline monitoring list	COVERED-33	33	—
QC55	Certificate receipt overview	PARTIAL-33	33	—
ME6H / ME64 (vendor evaluation with QM score)	Supplier quality score	COVERED-33	33	—
Fiori: Quality Technician Overview / Quality Engineer Overview / Inspection Lot Analytics / Defect Analysis / Supplier Quality	—	COVERED-46	46	—

13.24.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
QA07 (scheduled)	Deadline monitoring: recurring inspection lots, shelf-life expiry, batch status change	COVERED-33	33	—
QA16 (batch UD)	Automatic usage decisions (auto UD for lots without defects)	COVERED-33	33	Auto-UD rules — confirm; treat covered
QDL (quality level update) / dynamic modification update	Quality level update after UD	COVERED-33	33	—
QEH / QM notification follow-up (SWU3 workflow tasks)	Task monitoring / escalation	COVERED-4	4	SLA timers
QK04 / KO88 (QM orders settlement)	Settle appraisal costs	COVERED-33	33	—
QST (stability schedule generation)	Stability testing schedule	GAP	propose 33	—
Calibration plan scheduling (IP10/IP30 for test equipment)	—	COVERED-33	33	—
ME63 (vendor evaluation with QM)	—	COVERED-33	33	—
SARA QM_QMEL / QM_CONTROL	Archiving notifications/lots	COVERED-48	48	—
Quality score calculation / quality level reorganization	—	COVERED-33	33	—

13.24.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
QCC0 — QM customizing entry (inspection types 01-14; inspection lot origins; assignment of inspection types to lot origins; default values per inspection type: plan usage, auto UD, skip, inspection with task list/material spec, sample calculation, QM order)	TQ30, TQ32, TQ34	Inspection type per origin (GR, production, delivery, recurring, calibration, stock transfer, audit, customer return)	COVERED-33	33	Quality associations: when/what triggers a quality order (receipt, production, shipment, return)
Material master QM view defaults (QM control key, inspection setup QM02)	QMAT, TQ08	—	COVERED-33	33	Item quality settings
QM control keys in procurement (QM in procurement active, certificate required, vendor release, technical delivery terms)	TQ08, TQ	Blocks PO/GR if vendor not released / certificate missing	COVERED-33	33	Supplier quality status enforced at PO/GR — confirm enforcement at PO; treat covered
Define catalog types (1 characteristic attributes, 2 tasks, 3 , 8 activities, 9 defect types, E defect locations, 5 causes…), code groups	TQ15, QPGR	—	COVERED-33	33	—
Define sampling procedures/schemes defaults; valuation modes; inspection severities; dynamic modification (lot-based/characteristic-based)	TQ	—	COVERED-33	33	—
Define inspection points (time/quantity/freely defined), partial lots, inspection point identification	TQ79	—	PARTIAL-33	33	In-process inspection points (per time/quantity during production) partially covered by "production" trigger — partial
Define results recording settings: processing modes, valuation rules, characteristic weighting, defect classes (OK/critical/major/minor)	TQ	—	COVERED-33	33	—
Define usage decision catalogs, follow-up actions (e.g. auto email, batch status change, stock posting to blocked) , UD selected sets per plant	TQ , TQ07	Follow-up actions as function modules	COVERED-33	33	Disposition actions
Quality notification types, screen areas, partner determination, priorities, catalogs profile, action box (follow-up functions), tasks workflow	TQ80, TQ8*, T*	Q1/Q2/Q3/F1/F2/F3/G types	COVERED-33	33	NCR types; 8D template
Quality certificate: certificate types, profile characteristics, output determination (NACE QC)	TQ , NAST	—	COVERED-33	33	—
QM orders: order types (QN01), settlement, costing	T003O	—	COVERED-33	33	—
QM-IT: calibration inspection type 14, PM integration (IP10 plans with inspection), equipment categories for test equipment	TQ30, T370T	—	COVERED-33	33	—
Vendor evaluation QM criteria (quality score, audit score, complaints)	T147*	—	COVERED-33	33	—
Stability study customizing	—	—	GAP	propose 33	—
FMEA / control plan customizing (S/4)	—	—	GAP	propose 33	—
Audit management customizing (audit types, question lists)	—	—	PARTIAL-33	33	—
Multiple specifications customizing	—	—	PARTIAL-33	33	—
Sample management customizing (sample types, sample-drawing procedures)	—	—	PARTIAL-33	33	—
Define number ranges for lots/notifications/MICs	NRIV	—	COVERED-5	5	—

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Define status profiles for lots/notifications (BS02)	TJ30	—	COVERED-4	4	—

13.25 PM — Plant Maintenance (Enterprise Asset Management)

SAP context: technical objects (functional locations IL01, equipment IE01, equipment BOMs IB01, serial numbers IQ01, measuring points IK01, measurement documents IK11, object links, classification), maintenance planning (task lists IA01/IA05, maintenance plans IP01/IP41/IP42 single-cycle/strategy/multiple counter, scheduling IP10/IP30, strategies IP11), maintenance processing (notifications IW21 M1/M2/M3, orders IW31 PM01-PM06, permits, refurbishment orders, confirmations IW41, completion IW32 TECO, settlement KO88), PMIS (MCI1-MCI8), work clearance management (WCM), Linear asset management, MRS integration, mobile asset management (SAP Service and Asset Manager). Passage: wave 44 (functional locations, assets/equipment, BOMs/spares, counters/measurement, plans, requests, work orders with parts/labor/costs, IoT condition monitoring, KPIs) + wave 23 scheduling + wave 25 IoT + wave 37 fixed asset link.

13.25.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
IL01 / IL02 / IL03 / IL04 / IL05	Functional location create/change/display/list/changes	Location hierarchy (structure indicator)	COVERED-44	44	Functional locations
IL07 / IL08 / IL09 / IL10	FL structure list / installation / dismantle / mass change	—	COVERED-44	44	—
IL11 / IL12 / IL13 / IL14	Reference functional location	—	COVERED-44	44	FL templates — confirm; treat covered
IE01 / IE02 / IE03 / IE05 / IE06 / IE4N	Equipment create/change/display/list/changes/install-dismantle with goods movement	Equipment master (categories M/P/Q/S/F; serial-number link to material)	COVERED-44	44	Assets/equipment hierarchy; serial-tracked equipment from shipments (wave 45 installed base)
IE07 / IE08 / IE10	Equipment list (multilevel) / create equipment (standard) / mass change	—	COVERED-44	44	—
IE31 / IE36 / IE37	Fleet object create / list / change	Fleet management (vehicles)	PARTIAL-44	44	Fleet-specific attributes (fuel, odometer, driver) not named — partial; counters cover odometer
IH01 / IH02 / IH03 / IH04 / IH05 / IH06 / IH08 / IH09	Structural display: FL structure / equipment structure / where-used / list	—	COVERED-44	44	Asset hierarchy inquiry
IB01 / IB02 / IB03 / IB11 / IB12 / IB13 / IB05	Equipment BOM / functional location BOM create/change/display	Spares BOM	COVERED-44	44	BOMs/spares per asset
IQ01 / IQ02 / IQ03 / IQ04 / IQ08 / IQ09	Serial number (material serial) create/change/display/list	Serialized material = equipment	COVERED-8	8	Serial tracking dimension; equipment link (wave 44/45)
IK01 / IK02 / IK03 / IK07 / IK08	Measuring point create/change/display/list/mass change (counters/measuring points)	Counters & measurement points	COVERED-44	44	Counters and measurement points
IK11 / IK12 / IK13 / IK16 / IK17 / IK18 / IK21 / IK22 / IK31 / IK32 / IK33 / IK34	Measurement document create/change/display/list/collective entry/reverse / transfer	Readings	COVERED-44	44	Readings (manual, IoT via wave 25)
IA01 / IA02 / IA03 / IA05 / IA06 / IA07 / IA11 / IA12 / IA13 / IA08 / IA09 / IA10	Task lists: equipment / general maintenance / functional location (operations, components, PRTs, maintenance packages)	Maintenance task lists	COVERED-44	44	Checklists & job templates (operations, parts) — confirm multi-operation task lists with parts; treat covered
IP01 / IP02 / IP03 / IP41 / IP42 / IP43 / IP11 / IP12 / IP13 / IP14 / IP15 / IP16 / IP17 / IP18	Maintenance plans: create (strategy / single-cycle / multiple counter) change / display; strategies create/change/display (packages, hierarchy, cycle sets) / mass change / lists	Preventive maintenance plans	COVERED-44	44	Time/counter/condition-based plans; strategy packages (e.g. 1M/3M/1Y with hierarchy) — confirm package hierarchy; treat covered
IP10 / IP30 / IP30H / IP19 / IP24 / IP31	Schedule maintenance plan (manual) / deadline monitoring (batch) / scheduling overview (graphical) / scheduling list / cost overview	Plan scheduling	COVERED-44	44	Plan generation of work orders/requests
IW21 / IW22 / IW23 / IW24 / IW25 / IW26	Maintenance notification create (M1 request, M2 malfunction, M3 activity)/change/display, breakdown report	Maintenance requests	COVERED-44	44	Maintenance requests; downtime/breakdown reporting (affects scheduling capacity)
IW28 / IW29 / IW30 / IW64 / IW65 / IW66 / IW67 / IW68 / IW69	Notification lists / activities / tasks / items / causes	—	COVERED-44	44	—
IW31 / IW32 / IW33 / IW34 / IW36	Maintenance order create (PM01 maintenance, PM02 preventive, PM03 refurbishment, PM04 calibration, PM05 investment, PM06)/change/display/from notification/sub-order	Work orders	COVERED-44	44	Work orders with planning, parts reservation, labor, costs to asset/GL; refurbishment orders (repair spares to stock) — confirm refurbishment with valuation types; treat PARTIAL via note below
IW81 / IW8W	Refurbishment order create / goods receipt for refurbishment	Refurbish spares (valuation type change)	PARTIAL-44	44	Refurbishment of rotables (condition-based valuation: damaged → refurbished) not named — partial; split valuation gap in MM
IW37 / IW37N / IW38 / IW39 / IW40 / IW49 / IW49N / IW47 / IW48	Order lists / operations lists / multilevel list / confirmations list	—	COVERED-44	44	—
IW41 / IW42 / IW43 / IW44 / IW45 / IW46 / IW47 / IW48	Confirmations: individual / overall completion / display / collective / cancel / postprocessing errors	Labor confirmation	COVERED-44	44	Labor time to work orders (mobile technician app wave 45; kiosk wave 25)
IW3D / IW3K / IW3L / IW3M	Print order / change components / list components / material where-used in orders	—	COVERED-44	44	—
IW32 → Permits (IPMD / WCM)	Permits, work clearance management (lockout/tagout)	Safety permits	PARTIAL-44	44	Permits/LOTO work clearance (WCM) not named — partial; propose wave 44 "permit/LOTO checklist"
IW51 / IW52 / IW53 (CS notifications) — see §13.26	—	—	COVERED-45	45	—

T-code	Name	Purpose	Passage status	Wave	Notes
MB11 / MIGO 261/262 for PM order, MB21 reservations from orders	Parts issues/reservations	—	COVERED-44	44	Parts reservation & consumption
ME51N from PM order (non-stock components, external services)	External services on work orders	—	COVERED-44	44	Contractor services via PO — partial due to service entry sheet gap; treat covered via PO
CL20N (class assignment to equipment/FL), characteristics	Technical object classification	—	COVERED-8	8	Classification
IE4N	Install/dismantle equipment with goods movement	—	COVERED-44	44	—
Linear asset management (LAM: linear data on FL/equipment, network)	Pipelines, roads, rails	—	OUT-OF-SCOPE	—	Linear assets (utilities/rail) outside manufacturer scope
Asset Intelligence Network / Predictive Asset Insights / Asset Strategy & Performance Management	Cloud EAM extensions	—	PARTIAL-44	44	Condition monitoring + predictive hooks named; RCM/criticality analysis (ASPM) partial
Work orders → fixed asset capitalization (investment orders PM05 / settlement to AuC)	Capital maintenance	—	COVERED-37	37	Settlement to asset (CIP)
Fiori: Manage Maintenance Notifications / Orders / Create Maintenance Request / Perform Maintenance Jobs / Report Malfunction / Find Technical Objects / Manage Maintenance Plans / Schedule Maintenance Plans / Manage Work Orders (Phase model)	S/4 apps	—	COVERED-44	44	—
Fiori: Maintenance Planner Overview / Maintenance Scheduling Board / Resource Scheduling (RSH)	Scheduling technicians	—	COVERED-23	23	Scheduling against resources (wave 23)
SAP Service and Asset Manager (mobile)	Mobile EAM app	—	COVERED-45	45	Technician mobile app (wave 45) used for maintenance too — confirm maintenance mobile in wave 44/45; treat covered

13.25.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
IW28 / IW29 / IW38 / IW39 / IW37N / IW49N	Notification / order / operation lists	COVERED-44	44	—
IH01 … IH09	Structure displays	COVERED-44	44	—
IK17 / IK18	Measurement documents list	COVERED-44	44	—
IP19 / IP24	Scheduling overview (graphical) / list	COVERED-44	44	—
IW3M / IW13	Material where-used in orders / material where-used (equipment BOM)	COVERED-44	44	—
MCI1 / MCI2 / MCI3 / MCI4 / MCI5 / MCI6 / MCI7 / MCI8	PMIS: object class / manufacturer / location / planner group / damage analysis / object statistics / breakdown analysis (MTBF/MTTR) / cost analysis	COVERED-44	44	Asset lifecycle KPIs: MTBF/MTTR, cost per asset, downtime
MCJB / MCJC	PMIS: MTTR/MTBF analyses	COVERED-44	44	—
S_ALR_87013 (PM cost reports: order cost analysis, planned/actual)	—	COVERED-44	44	—
IW37N with costs / KOB1 (PM orders)	Order cost line items	COVERED-44	44	—
IP16 / IP17 / IP18	Maintenance plan lists / items / cost	COVERED-44	44	—
Fiori: Maintenance Backlog Overview / Asset Health KPIs / Technical Object Breakdowns / Maintenance Order Costs	—	COVERED-46	46	—

13.25.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
IP30 / IP30H (scheduled)	Deadline monitoring: generate maintenance calls (orders/notifications)	COVERED-44	44	—
IK34 / IoT measurement document upload (MDoc interface)	Counter reading upload	COVERED-25	25	IoT → readings
IW42 / IW44 (collective confirmation)	Confirmation batch	COVERED-44	44	—
KO88 / KO8G (PM order settlement)	Settlement to cost center/asset	COVERED-44	44	Costs → asset and GL
IW32 TECO/CLSD mass (IW38 → mass TECO)	Order completion	COVERED-44	44	—
MCI LIS update / PMIS rebuild (OLI)	—	SUPERSEDED-46	46	—
SARA PM_ORDER / PM_QMEL	Archiving	COVERED-48	48	—
Condition-based triggers (IoT thresholds → maintenance request)	—	COVERED-44	44	IoT threshold → request < 5 s
Capacity requirements transfer to scheduling (work center load from PM orders)	—	COVERED-23	23	Downtime affects production capacity

13.25.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Technical objects: structure indicators for FL (edit masks, hierarchy levels), equipment categories (M machine, P production resource, S customer equipment, Q test equipment, F fleet), equipment types, object types, planner groups, plant sections, ABC indicators, authorization groups	T370F, T370T, T370U, T024I, T357, T370C	—	COVERED-44	44	FL numbering masks; equipment categories
Define maintenance planning plants / maintenance work centers / planner groups	T001W, T024I	—	COVERED-6	6	Sites + maintenance resources
Notification types (M1/M2/M3), catalogs profile, priorities (OIO), screen structures, partner determination, action box, number ranges	TQ80, T352B, TQ8	Breakdown flag, malfunction start/end, damage/cause/activity catalogs	COVERED-44	44	Request types; downtime capture
Order types (PM01–PM06), order type ↔ planning plant, number ranges, default values (control key, cost center), settlement profiles, costing variants, budget profiles	T003O, T350, T399X, TKB1	—	COVERED-44	44	Work order types with costing
Control keys (PM01 internal, PM02 external, PM03 external with service), activity types	T430	—	COVERED-44	44	—
Maintenance plan categories (order/notification/service entry sheet), strategies, packages, scheduling parameters (shift factors, tolerance, call horizon, cycle modification)	T351*, T351X, T399W	Call horizon %, scheduling period, completion requirement	COVERED-44	44	Plan scheduling parameters
Measurement/counter settings: characteristics (CT04), counter overflow, reversal	T*	—	COVERED-44	44	—
Task list types (A general, E equipment, T functional location), usages, status, condition codes	TCA*	—	COVERED-44	44	—
Refurbishment: valuation types for condition (OMWC split valuation), order type PM04 settings	T025, T003O	—	PARTIAL-44	44	Refurbishment gap
Permits (define permit types, categories), WCM (work clearance objects, applications, approvals)	T357G, WCA*	—	PARTIAL-44	44	Permit/LOTO gap
PMIS info structures (S061/S062/S065/S070/S115/S116), update groups	—	—	SUPERSEDED-46	46	—
Integration: PM ↔ FI-AA (asset sync from equipment, settlement), PM ↔ MM (reservations, PRs), PM ↔ PP (capacity), PM ↔ QM (calibration), PM ↔ HR (work center persons)	—	—	COVERED-44	44	—
Phase model for maintenance orders (S/4 "Reactive Maintenance"/"Proactive" processes, phases/subphases, approvals)	—	—	COVERED-44	44	Work order state machine (wave 4)
Mobile EAM configuration (Service and Asset Manager)	—	—	COVERED-45	45	—

13.26 CS / Customer Service / SAP Field Service Management

SAP context: CS (SM): service notifications (IW51–IW59, S1 problem/S2 service request/S3 activity), service orders (IW31 SM01/SM02/SM03 with revenue; billing via DP90 resource-related billing, DIP profile ODP1), service contracts (VA41 WV with billing plans, service products, SLA via contract), warranties (BGM1/BGM2 master warranties, warranty check on notification/order), installed base (IB51/IB52/IB53; IBase components), repairs processing (RAS order with repair procedure, returns/loaner/repair/delivery), in-house repair (S/4 Fiori), solution database, call center (CIC), SAP Field Service Management (cloud dispatch/mobile), S/4 Service (service orders, contracts, confirmations in S/4 Service). Passage: wave 45 (installed base, warranties, agreements/SLAs, cases → work orders, repair orders, dispatch optimization, technician mobile app, parts from trucks, billing to customer/warranty, KPIs) + wave 41 (resource-related billing) + wave 30 (RMA).

13.26.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
IB51 / IB52 / IB53 / IB55 / IB56 / IB61 / IB62	Installed base create/change/display/list; IBase components (equipment, materials, documents, text); IBase by sales order	Installed base	COVERED-45	45	Installed base/service objects with serial tracking from shipments
IE01 (equipment category S customer equipment), IE4N	Customer equipment	—	COVERED-45	45	—
BGM1 / BGM2 / BGM3	Master warranty create/change/display (counters, services covered, validity)	Warranty terms	COVERED-45	45	Warranty terms on installed base
IW51 / IW52 / IW53 / IW54 / IW55 / IW56 / IW57	Service notification create (S1/S2/S3)/change/display/create from call center/complete	Cases	COVERED-45	45	Cases
IW58 / IW59	Service notification lists	—	COVERED-45	45	—
IW31 (SM01/SM02/SM03) / IW32 / IW33	Service order create/change/display (revenue-bearing, resource-related billing, DIP profile)	Service work orders	COVERED-45	45	Service work orders with parts, labor, billing
IW34 / IW36	Service order from notification / sub-order	—	COVERED-45	45	—
IW41 / IW42 (service confirmations)	Technician time/parts confirmation	—	COVERED-45	45	Mobile confirmations
DP90 / DP91 / DP93 / DP95 / DP96 / DP97 / DP98	Resource-related billing: billing request from service order / sales order / contract; collective; display	T&M billing	COVERED-41	41	T&M invoice proposals (wave 41 billing rules; wave 45 billable vs warranty lines)
ODP1 / ODP2 (DIP profile) — customizing	—	—	COVERED-41	41	Billing rules
VA41 / VA42 / VA43 (WV service contract, SC, QP quantity contract with service items)	Service contracts with billing plans (periodic), SLA/response profiles, contract items (service products), price agreements	Service agreements	COVERED-45	45	Service agreements (entitlements, SLAs, pricing)
VA01 RAS (repair order: repair procedure — return, loaner, repair, scrap, deliver) + IW32 repair orders	Repairs processing (in-house repair)	—	COVERED-45	45	Repair orders; RMA (wave 30)
Fiori: Manage In-House Repairs (S/4)	—	—	COVERED-45	45	—
Warranty claim processing (WTY: WTY1 warranty claims from dealers/to vendors)	Warranty claims (supplier recovery)	—	PARTIAL-45	45	Customer warranty handling yes; supplier warranty claim recovery (claims to OEM/vendor, reimbursement) not named — partial
Service products (material type DIEN), service packages (KMAT/configurable services)	Service catalog	—	COVERED-8	8	Service product type (noted partial in MM-SRV for purchasing side)

T-code	Name	Purpose	Passage status	Wave	Notes
Solution database (IS01 / IS02 symptom/solution)	Knowledge base	—	GAP	propose 45	Knowledge base/solution search for cases not named; propose wave 45 (or wave 47 copilot answers)
Customer Interaction Center (CIC0) / SAP CRM Service / S/4 Service (Service orders, service contracts, service confirmations, service quotation)	Service front office	—	PARTIAL-45	45	Case management yes; omnichannel contact center (telephony/CTI, email routing) not named — partial (CRM non-goal boundary)
SAP FSM: Planning & dispatching board, technician mobile app, smartforms/checklists, crowd service, customer self-service portal	Field service cloud	—	COVERED-45	45	Dispatch board, OR-Tools scheduling, mobile app (offline, photos, signatures, parts); checklists via wave 44
Service parts planning (SPP / advanced service parts)	Spare parts planning for service networks	—	PARTIAL-22	22	MRP covers; multi-echelon service parts planning partial
Technician van stock (storage locations per technician, mobile inventory)	Truck stock	—	COVERED-45	45	Mobile inventory locations
Subscription billing for service contracts (SAP BRIM / Subscription Billing)	Usage-based/subscription	—	COVERED-29	29	Subscription billing (recurring, proration, usage)
Entitlement management (S/4 Entitlement Management)	Entitlements	—	COVERED-45	45	Entitlements on agreements
Fiori: Manage Service Orders / Service Confirmations / Service Contracts / Service Quotations / Solution Quotations / Customer Service Overview / Manage Service Requests	S/4 Service apps	—	COVERED-45	45	—

13.26.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
IW58 / IW59 / IW38 / IW39 (service)	Notification / order lists	COVERED-45	45	—
IB56 / IH (IBase structure)	Installed base displays	COVERED-45	45	—
MCI (service CS-IS: notifications, orders by customer)	Service information system	COVERED-46	46	—
DP96 / VF04 (service billing due)	Billing requests list	COVERED-41	41	—
Contract lists (VA45 WV), expiring contracts	—	COVERED-45	45	—
Warranty check log (notification/order)	—	COVERED-45	45	—
Fiori: Service KPIs (SLA attainment, first-time fix, utilization, backlog)	—	COVERED-45	45	Service KPIs
FSM analytics / reporting	—	COVERED-46	46	—

13.26.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
DP90 (batch) / DP95	Collective billing request creation	COVERED-41	41	—
VF04 (service contracts periodic billing)	Contract billing run	COVERED-29	29	—
IP30 for service maintenance plans (contract-based preventive visits)	Contract-based plans generate service orders	COVERED-45	45	Preventive visits from agreements — confirm via wave 44 plans linked to agreements; treat covered
KO88 (service order settlement to profitability)	Settlement	COVERED-45	45	—
Contract renewal/expiry run, SLA breach monitoring	—	COVERED-45	45	SLA tracking
SARA PM_ORDER / SM notifications	Archiving	COVERED-48	48	—
Dispatch optimization run (FSM auto-scheduling)	—	COVERED-45	45	OR-Tools dispatch

13.26.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Service notification types (S1/S2/S3), catalogs, priorities, response/service profiles (SLA: response time, service window)	TQ80, T* (response profiles OIS)	—	COVERED-45	45	SLA profiles
Service order types (SM01/SM02/SM03), revenues on orders, settlement to PA/sales order, DIP profiles (ODP1: usage, characteristics, sources, material determination)	T003O, DIP*	—	COVERED-41	41	Billing rules
Contract types (WV), contract profiles, cancellation rules, renewal, billing plan types for contracts, SLA assignment, price agreements	TVAK, TVKV, T*	—	COVERED-45	45	Agreements
Repairs processing: repair procedures, actions (101 returns, 102 loaner, 103 repair, 104 scrap, 105 deliver), item categories IRRE/IRRP/IRRS/IRLA	T* (VRPP)	—	COVERED-45	45	Repair order flows
Warranty: warranty types, counters, warranty check settings (notification/order)	BGMK, T*	—	COVERED-45	45	—
IBase categories, component types	IBIN, T*	—	COVERED-45	45	—
Warranty claim processing (WTY) customizing	—	—	PARTIAL-45	45	—
FSM configuration (skills, regions, business rules, checklists, mobile settings)	—	—	COVERED-45	45	—
S/4 Service: service order types, item categories, service confirmation types, service quotation	—	—	COVERED-45	45	—
Solution database / knowledge base settings	—	—	GAP	propose 45	—
CIC / contact center framework	—	—	PARTIAL-45	45	—

13.27 PS — Project System

SAP context: project definitions (CJ01/CJ06/CJ20N project builder), WBS elements (CJ11–CJ13), networks and activities (CN21–CN23, CN01 standard networks), milestones (CN11?), standard WBS (CJ91), planning (CJ40 cost planning, CJ42 revenue, CJR2 cost element planning, CJ9? capacity), budgeting (CJ30 original, CJ36 supplement, CJ37 return, CJ32 release, availability control), dates/scheduling (CJ21? CJ2B? CN24?), execution (CN25 confirmations, CAT2 timesheet), procurement (PRs from networks, PS material components), billing (billing plans on WBS/sales order, DP90 resource-related, milestone billing), period-end (CJ44 overhead, CJ8G settlement, KKA2 results analysis, CJB1 settlement rules, interest CJZ2?), reporting (CN41/CN42/CN43/CN45/CN46/CN47/CN48 structure/info system, S_ALR_87013532-87013575 cost reports, CJI3 line items, CJE0 hierarchy reports), project versions (CN72), simulation (CJV1), claims (CLM1), progress analysis (CNE1/CNE5 EVM), investment management (IM: IM01 program, IM22? IM52 budget distribution, appropriation requests IMA11). S/4: Project Control (Fiori Project Builder, Commercial Project Management CPM), Project Financials Control. Passage: wave 41 (WBS, scheduling, resources, forecasts/budgets, hour/expense/item/fee journals, project POs, billing T&M/FP/milestones/retainers, WIP/POC/RA, EVM, grants) + wave 38 budget control + wave 37 CIP.

13.27.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
CJ20N	Project Builder (project definition, WBS, networks, activities, milestones, documents, PS texts, materials)	Central project maintenance	COVERED-41	41	Project hierarchy + WBS with scheduling
CJ01 / CJ02 / CJ03 / CJ06 / CJ07 / CJ08	Project definition create/change/display (with/without template)	—	COVERED-41	41	Project types & templates
CJ11 / CJ12 / CJ13 / CJ14	WBS element create/change/display	WBS	COVERED-41	41	—
CJ91 / CJ92 / CJ93	Standard WBS create/change/display	Templates	COVERED-41	41	Project templates — confirm; treat covered
CN01 / CN02 / CN03	Standard network create/change/display	Network templates	COVERED-41	41	—
CN21 / CN22 / CN23 / CN24 / CN25 / CN26 / CN27 / CN28 / CN29	Network create/change/display / overall scheduling / confirm activity / collective confirmation / cancel confirmation / display confirmation	Networks & activities (internal, external, cost, service), relationships (FS/SS/FF/SF), scheduling	COVERED-41	41	WBS scheduling respecting dependencies & resource calendars; activity confirmations (hours)
CN11 / CN12 / CN13	Standard milestones	Milestones	COVERED-41	41	Milestones (billing & progress)
CJ2B / CJ2C / CJ2D / CJ21 / CJ22 / CJ23 / CJ24 / CJ29	Project planning board (Gantt) change/display; WBS scheduling; project scheduling	Gantt	COVERED-41	41	Scheduling Gantt (reuse wave 23 Gantt)
CJ40 / CJ41	WBS cost planning change/display (structure/detailed planning, cost element)	Cost planning	COVERED-41	41	Forecasts/budgets
CJ42 / CJ43	Revenue planning	—	COVERED-41	41	—
CJR2 / CJR3 / CJR4 / CJR5 / CJR6 / CJR7	Cost element / activity input planning (WBS)	—	COVERED-41	41	—
CJ9B / CJ9BS / CJ9C / CJ9D / CJ9E / CJ9F / CJ9FS / CJ9G / CJ9K / CJ9L / CJ9M	Copy plan versions, plan → budget, copy actual to plan, delete	Planning utilities	COVERED-41	41	Forecast versions
CJ30 / CJ31 / CJ32 / CJ33 / CJ34 / CJ35 / CJ36 / CJ37 / CJ38 / CJ3A / CJ3B	Budget: original/display/release/display release/transfer/return/supplement/carry forward/display documents	Project budget & availability control	COVERED-38	38	Budget control on projects (wave 38) + wave 41 project budgets
CJBV / CJBW / CJBN / CJEN	Activate/deactivate availability control; reconstruct	—	COVERED-38	38	—
CJB1 / CJB2	Generate settlement rules (collective / individual)	—	COVERED-41	41	Ledger posting profiles / settlement to assets (CIP) or COGS
CJ8G / CJ88 / CJ8A / CJ9	Settlement (collective / individual / AuC settlement)	Settlement	COVERED-41	41	Project settlement; CIP to fixed assets (wave 37)
CJ44 / CJ45 / CJ46 / CJ47	Overhead calculation (individual/collective) actual/plan	Overhead	COVERED-39	39	Costing sheet overheads on projects
CJZ1 / CJZ2 / CJZ3 / CJZ5	Interest calculation on projects	Capitalized interest	GAP	propose 41	Project interest calculation (interest on WIP/capitalized interest IAS 23) not named — propose wave 41 option
KKA2 / KKAJ (RA for WBS)	Results analysis (POC, revenue-based, cost-based)	WIP / accrued revenue	COVERED-41	41	POC estimates, WIP/accrued revenue
CJI3 / CJI4 / CJI5 / CJI8 / CJI9 / CJIA / CJIB / CJIC / CJID / CJIE / CJIF / CJIG	Line items: actual / plan / commitments / budget / hierarchy / document display	Inquiry	COVERED-41	41	Project transactions inquiry
CN41 / CN41N / CN42 / CN42N / CN43 / CN43N / CN44 / CN45 / CN46 / CN47 / CN48 / CN49 / CN50 / CN51 / CN52 / CN53 / CN54 / CN55	Structure overview / project definitions / WBS / networks / activities / confirmations / components / milestones / PRTs / capacities / PS texts / documents / relationships	Project info system (structures)	COVERED-41	41	—
CN72 / CN71	Project version create (snapshots)	Versions/baselines	COVERED-41	41	Baselines — confirm; treat covered
CJV1 / CJV2 / CJV3 / CJV4 / CJV5 / CJV6	Simulation version create/change/display/transfer	Simulation	PARTIAL-41	41	Project what-if simulation versions not named — partial
CLM1 / CLM2 / CLM3 / CLM10 / CLM11	Claims (variation/change requests with cost impact)	Claim management	PARTIAL-41	41	Change orders/claims on projects (scope change with cost/revenue impact and customer approval) not named — partial; propose wave 41 "project change order"
CNE1 / CNE2 / CNE5	Progress analysis (EVM: POC methods, BCWS/BCWP/ACWP) individual/collective/display	EVM	COVERED-41	41	EVM (CPI/SPI)
CNS0 / CN (shipping from projects — delivery from project)	Project deliveries (CNS0)	—	PARTIAL-41	41	Shipping items from project stock/WBS not named — partial (ETO)
Project stock (Q), PS material components, MD51 project MRP, CN33 (BOM transfer PS-BOM)	Project procurement/stock	—	PARTIAL-41	41	Project items/POs named; project stock segregation partial (MM-IM note)

T-code	Name	Purpose	Passage status	Wave	Notes
CAT2 / CATS / CATSXT / CAPS / CAT6 / CAT7 / CAT9 / CATA	Timesheet entry / approval / transfer to PS/CO/PM/HR	Time capture	COVERED-41	41	Hour journals; timesheet with approval (shared with wave 25/42)
PR05 (expenses to WBS)	Expense to project	—	COVERED-43	43	Expense journals (wave 41 & 43)
VA01 with WBS account assignment / milestone billing plan; DP90	Project sales & billing	—	COVERED-41	41	T&M/fixed-price/milestone billing, retainers
IM01 / IM02 / IM03 / IM22 / IM23 / IM24 / IM25 / IM27 / IM28 / IM30 / IM31 / IM32 / IM33 / IM34 / IM35 / IM36 / IM52 / IMA11 / IMA2 / IMA3 / IMCCP1	Investment Management: investment programs (positions, hierarchy), budget distribution/supplement/return, program planning, appropriation requests (IMA11 create, variants, approval), rollup/carry-forward	Capital investment programs	PARTIAL-37	37	Asset budgeting named in wave 37; investment program hierarchy with appropriation requests (approval of capex before projects/orders) and program budget distribution — partial; propose wave 38 budgeting + wave 41
Commercial Project Management (CPM: project workspace, financial planning, change/risk/issue management)	Professional-services project cockpit	—	PARTIAL-41	41	Risks/issues register, cost/revenue forecasting workbook — partial
Multi-project / portfolio (SAP PPM: portfolio items, resource planning, project management)	PPM	—	OUT-OF-SCOPE	—	Portfolio & project management suite beyond ERP-project v1 (resource bookings & WBS in wave 41 suffice)
Fiori: Project Builder / Project Control – Enterprise Projects / Project Financial Controller Overview / Monitor Projects / Manage Project Procurement / Plan Project Costs / Project Briefs	S/4 apps	—	COVERED-41	41	—
Fiori: Manage Project Billing / Release Billing Proposals (S/4 Project Billing for professional services)	—	—	COVERED-41	41	Invoice proposals

13.27.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
S_ALR_87013532 / 87013533 / 87013534 / 87013535 / 87013536	Plan/actual/variance; plan/actual/commitment/remaining plan; budget/actual/commitment; budget updates; actual in CO area currency	COVERED-41	41	Project profitability, committed costs
S_ALR_87013537 … 87013575 (cost element, revenue, hierarchy, period breakdown, forecast, payments)	PS cost/revenue/payment reports	COVERED-41	41	—
S_ALR_87013558 / 87013559 / 87013560	Budget/actual/variance; budget/distributed/plan; commitment	COVERED-38	38	—
S_ALR_87013562 / 87013563 / 87013564	Actual/plan/variance by cost element; period; revenue	COVERED-41	41	—
S_ALR_87013566 / 87013567 / 87013568 / 87013569 / 87013570 / 87013571 / 87013572 / 87013573 / 87013574 / 87013575	Results analysis, WIP, payments, receipts/payments, cash flow	COVERED-41	41	—
S_ALR_87100185 / 87100186 / 87100187 / 87100188 / 87100189 / 87100190 / 87100191	Progress reports (EVM), milestone trend analysis	COVERED-41	41	—
CJE0 / CJE1 / CJE2 / CJE3 / CJE4 / CJE5 / CJE6 / CJEB	Hierarchy reports (Report Painter) execute/create/change/display/forms	SUPERSEDED-15	15	Report designer
CJI3 / CJI5 / CJI8	Line items	COVERED-41	41	—
CN41N / CN42N / CN43N / CN45N / CN46N / CN47N / CN48N	Project info system structure reports	COVERED-41	41	—
CN41 → progress, dates comparisons (CN / CNMT milestone trend analysis)	Milestone trend analysis	COVERED-41	41	—
CNE5	Progress analysis report	COVERED-41	41	—
CJ31 / CJ33	Budget display	COVERED-38	38	—
IM (IM reports: S_ALR_87012)	Investment program reports	PARTIAL-37	37	—
MCI / project capacity (CM / CMP2 / CMP3 / CMP9 workforce planning)	Capacity/workforce planning reports	COVERED-41	41	Resource bookings
Fiori: Project Cost Report / Project Profitability / Budget Consumption / Project Status Overview / Utilization Analysis	—	COVERED-46	46	—

13.27.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
CJ44 / CJ45 (overhead)	Overhead allocation	COVERED-39	39	—
KKA2 / KKAJ (results analysis)	RA run	COVERED-41	41	—
CJ8G / CJ88	Settlement run (to CO/PA, assets, cost centers)	COVERED-41	41	—
CJZ2 (interest)	Interest calculation run	GAP	propose 41	—
CJB1 (periodic rule generation)	—	COVERED-41	41	—
CJCF / CJCO	Carry forward commitments / budget	COVERED-38	38	—
CJEN / CJBN	Reconstruct project info database / availability control	SUPERSEDED-41	41	Projections
CNE2 (progress determination collective)	EVM progress run	COVERED-41	41	—
CN72 (periodic versions)	Snapshot	COVERED-41	41	—
DP90/DP95 (billing requests)	Billing	COVERED-41	41	—
CAT6 / CAT7 / CAT9 (transfer CATS → PS/CO/PM/HR)	Timesheet transfer	SUPERSEDED-41	41	Direct posting of approved time
IM27 / IM32 (IM budget rollup / carry-forward)	—	PARTIAL-37	37	—
SARA PS_PROJECT	Archiving	COVERED-48	48	—
CJ02 mass status (TECO/CLSD) / CN mass change (CNMASS)	—	COVERED-5	5	—

13.27.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OPSA — Project profile (WBS defaults, status profile, planning profile, budget profile, version profile, settlement, planning board, object class)	TCJ41	—	COVERED-41	41	Project types/templates
OPSJ — Budget profiles (availability control activation, tolerance limits, actions)	TBP*, TBPFE	—	COVERED-38	38	—
OPSB — Planning profiles (cost/revenue planning, versions, automatic revenue planning)	TCJ4	—	COVERED-41	41	—
OPSC / OPSD / OPSE — Project coding mask (CJ / OPSK special characters, edit mask), project types, priorities, applicant/responsible persons	TCJE, TCJS, TCJP	—	COVERED-41	41	Project numbering masks
OPSV / OPUV / OPUU / OPSU — Network types, network profiles, control keys (PS01/PS02/PS05), network parameters (CO integration, scheduling, number ranges)	TCN41, T003O, TCN	—	COVERED-41	41	—
OPTP / OPU6 / OPU — Scheduling parameters for WBS/networks (planning method, reduction, scheduling type, adjust basic dates)	TCX	—	COVERED-41	41	—
OPSQ / OPS — Status profiles (BS02), user status for projects	TJ30	—	COVERED-4	4	—
OKG (RA keys) / OPS — Results analysis keys per project profile, valuation methods	TKKA*	—	COVERED-41	41	—
OPSJ / OKOA / Settlement profiles & strategies (CJB1 strategy: settlement to AuC/cost center/PA)	TKB1, T811	—	COVERED-41	41	—
ODP1 — DIP profiles (resource-related billing)	DIP*	Usage (billing/quotation), characteristics, sources (actual costs, SD)	COVERED-41	41	Billing rules
OPSX / Milestone usage, milestone billing, progress analysis settings (progress version, measurement methods OPS)	T*	—	COVERED-41	41	EVM measurement methods
OPS — Project stock settings (valuated/non-valuated), PS material components (OPUV), MRP for project	TCJ41, T*	—	PARTIAL-41	41	Project stock partial
CATS: data entry profiles (CAC1), CATS fields, approval workflow, cost assignment	TCATS*, TCAT	—	COVERED-41	41	Timesheet profiles
Investment management: program types, appropriation request types, status, budget categories, plan/budget profiles	TAIF, TAPRT	—	PARTIAL-37	37	IM partial
Claim management customizing (notification type CL, claim types)	TQ80	—	PARTIAL-41	41	—
Simulation versions / project versions profiles (OPTS / OPT)	TCJ	—	PARTIAL-41	41	—
PS texts, documents (DMS link), PS info system profiles (OPSM / OPTZ / CJ)	—	—	COVERED-41	41	—
Project billing (S/4): billing element profiles, billing rules, invoice proposal settings	—	—	COVERED-41	41	—
Interest profiles (OPIB / OPIC / OPID)	T*	—	GAP	propose 41	—

13.28 HCM — PA / OM / PT / PY + SuccessFactors Employee Central

SAP context: HCM-PA (personnel administration: infotypes 0000–0999, PA30/PA40 actions, PA20, PA70 fast entry, master data), PA-OS/OM (org management: org units, positions, jobs, PPOME/PP01, evaluation paths, relationships), PT (time management: work schedules, absences/attendances IT2001/2002, quotas IT2006, time evaluation PT60/RPTIME00, CATS, time events PDC, PT50 quota overview, shift planning PP61), PY (payroll: PC00_Mxx_CALC, schemas PE01, rules PE02, wage types, payroll results PC_PAYRESULT, posting PC00_M99_CIPE, off-cycle, retro, year-end PU19 tax reporter US, garnishments IT0194/0195, remuneration statement), PA-BN (benefits: plans, enrollment HRBEN0001, HRBEN0071 eligibility), PA-RC (recruitment PB10–PB80, e-Recruiting), PA-PD (personnel development: qualifications PPPM, appraisals PHAP_*, career/succession), PE (training & event management, LSO), PA-CM (compensation management), HR ESS/MSS, SuccessFactors Employee Central (EC), EC Payroll, SF Recruiting/Onboarding/Learning/Performance & Goals/Compensation/Succession. Passage: wave 42 (workers/positions/jobs/compensation, absence/leave, benefits, ESS/MSS, talent basics, recruitment basics, PII controls), wave 43 (payroll engine, country packs US + 1 EU, provider adapter, year-end forms, expense management), wave 25 (time & attendance core).

13.28.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
PA30 / PA20	Maintain / display HR master data (infotypes: 0000 actions, 0001 org assignment, 0002 personal data, 0006 address, 0007 planned working time, 0008 basic pay, 0009 bank details, 0014 recurring payments, 0015 additional payments, 0016 contract, 0021 family, 0041 date specs, 0105 communication, 0167/0168/0169 benefits, 0207-0210 US tax, 0194/0195 garnishments, 2001 absences, 2002 attendances, 2006 quotas, 2010 remuneration info)	Employee master	COVERED-42	42	Worker/employee/contractor records with effective-dated employment, org assignments, compensation plans; bank details (wave 43 payroll); country-specific fields framework
PA40	Personnel actions (hire, org reassignment, pay change, leave of absence, termination, rehire)	Lifecycle actions	COVERED-42	42	Effective-dated employment actions with workflow
PA41 / PA42	Change entry/leaving date / fast entry for actions	—	COVERED-42	42	—
PA70 / PA71	Fast entry (infotype for many employees) / fast entry of time data	Mass maintenance	COVERED-5	5	Grid mass edit
PA10	Personnel file (overview of all infotypes)	—	COVERED-42	42	Worker 360 view
PRMD / PRMM / PRMS	Maintain/display HR master data (travel privileges)	—	COVERED-43	43	—
PPOME / PPOMA / PPOSE / PPOM_OLD	Organization and staffing: org units, positions, jobs, holders, cost center assignment (change/display)	Org management	COVERED-42	42	Positions, jobs, departments (wave 6 operating units) with effective dating

T-code	Name	Purpose	Passage status	Wave	Notes
PO03 / PO10 / PO13 / PO01 / PP01 / PP02 / PP03	Maintain job / org unit / position / work center / general object maintenance (infotypes 1000-1999: relationships 1001, description 1002, planned compensation 1005, vacancy 1007, account assignment 1008, cost distribution 1018, obsolete 1014, work schedule 1011)	OM objects	COVERED-42	42	Position/job attributes, cost distribution, vacancy
PQ01 … PQ / PPO	OM actions (create org unit with position etc.)	—	COVERED-42	42	—
PPCI / PPSC / PPSM	Create structure / evaluation paths	—	COVERED-6	6	Hierarchy purposes
PP61 / PP62 / PP63 / PP64 / PP6I / PP6A	Shift planning (target plan, requirements, assignments)	Shift planning / rostering	PARTIAL-25	25	Time registration & shifts in wave 25; shift rostering (planning employees to shifts against requirements, swaps) not named — partial; propose wave 42
PT01 / PT02 / PT03	Work schedule generation / change / display (period work schedules, daily WS, WS rules)	Work schedules	COVERED-42	42	Work calendars/shifts (wave 9 resource calendars + wave 42 employee schedules)
PA61 / PA62 / PA63 / PA64	Maintain time data / list entry / display / calendar	Absences/attendances	COVERED-42	42	Absence/leave requests; attendance (wave 25)
PT50 / PT_QTA00 / PT_QTA10	Quota overview / generate absence quotas	Leave balances	COVERED-42	42	Accruals, carry-over
PT60 / PT61 / PT62 / PT63 / PT64 / PT65 / PT66 / PT67 / PT68 / PT69	Time evaluation (RPTIME00) / time statement / attendance list / absence list / time leveling / time balances	Time evaluation	COVERED-25	25	Pay-type calculation (overtime, shifts); time statement
PT40 / PT41 / PT42 / PT43 / PT44 / PT45 / PT46 / PT47 / PT80 / PT90	Time events / PDC (upload, postprocessing), time pool	Clock in/out	COVERED-25	25	Kiosk registrations
CAT2 / CATS (see PS)	Timesheet	—	COVERED-41	41	—
PC00_Mxx_CALC (e.g. PC00_M10_CALC US, PC00_M01_CALC DE, PC00_M08_CALC GB) / PC00_M99_CALC_SIMU	Payroll run (country) / simulation	Gross-to-net	COVERED-43	43	Payroll engine (US + one EU reference; provider adapter)
PA03	Payroll control record (release/exit/corrections, period)	Payroll area status	COVERED-43	43	Pay cycle status with locking
PU03	Change payroll status (infotype 0003: retro accounting dates, locks)	—	COVERED-43	43	Retro pay triggers
PC00_M99_CIPE / PCP0 / PC00_M99_CIPC	Posting to accounting (create, release, post documents) / display posting runs	Payroll GL posting	COVERED-43	43	Payroll posting to GL (reconciled)
PC00_M99_CWTR	Wage type reporter	—	COVERED-43	43	Payroll reconciliation reports
PC_PAYRESULT / PC00_M99_DISPLAY	Display payroll results (clusters)	—	COVERED-43	43	Pay statements / results
PC00_Mxx_CEDT / PC00_M99_CEDT	Remuneration statement (pay slip)	—	COVERED-43	43	Pay statements
PC00_M99_CMLI0 / PC00_M99_FPAYM / PC00_M99_CDTA	Bank transfers (DME), pre-DME	Payroll payments	COVERED-43	43	Payment via wave 19 formats (ACH/SEPA)
PC00_M99_URMP / PC00_M99_CLJN / PC00_M99_CLGA	Payroll journal / account / cumulated	—	COVERED-43	43	—
PC00_M10_OSUP / PC00_Mxx_OC (off-cycle workbench PUOC_10)	Off-cycle payroll (bonus, correction, final pay)	—	PARTIAL-43	43	Off-cycle/ad-hoc pay runs (bonus, termination final pay, check replacement) not explicitly named — partial
PU19	Tax reporter (US: 941, W-2, state)	US statutory	COVERED-43	43	Year-end forms framework (W-2/1099); quarterly filings — partial via "framework"; treat covered
PC00_M10_CLAIMS / PC00_M10_UTXX	US claims processing / tax authorities	—	PARTIAL-43	43	Claims (overpayment recovery) not named — partial
PA30 IT0194/0195 garnishments; PC00_M10_CGTD	Garnishments (orders, documents, disbursement)	—	COVERED-43	43	Garnishments named
HRBEN0001 / HRBEN0003 / HRBEN0005 / HRBEN0006 / HRBEN0014 / HRBEN0071 / HRBEN0075 / HRBEN0079	Benefits: enrollment / participation monitor / termination / eligibility change / open enrollment / plan overview / cost summary / default plans	Benefits administration	COVERED-42	42	Benefits: plans, eligibility, open enrollment, life events; ESS enrollment
HRBEN00ICOBRA / HRBEN00COBRA	COBRA (US)	—	PARTIAL-42	42	COBRA notices/continuation (US) belongs to US pack — partial
PB10 / PB20 / PB30 / PB40 / PB50 / PB60 / PB80 / PBAA / PBAB / PBAE / PBAK	Recruitment (classic): initial entry of applicant, display, maintain, applicant actions, vacancy assignment, applicant activities, mass processing	Recruiting	COVERED-42	42	Recruitment pipeline basics
E-Recruiting (HRRCF_*) / SF Recruiting & Onboarding	Modern recruiting	—	PARTIAL-42	42	Pipeline basics; career site, requisition approval, offer letters, onboarding tasks — partial
PPPM / PPPE / PPQD / OOQA	Qualifications catalog, profile matchup, development plans	Skills	COVERED-42	42	Skills, certifications
PHAP_CATALOG / PHAP_CREATE / PHAP_CHANGE / PHAP_ADMIN / PHAP_SEARCH / PHAP_PREPARE	Appraisal templates / create / change / administration (performance management, objective setting)	Performance	COVERED-42	42	Performance reviews, goals
PPCP / PPSS (career/succession planning) / SF Succession	Succession planning	—	PARTIAL-42	42	Talent basics; succession planning/talent pools not named — partial
PECM_* (enterprise compensation management: planning, budgets, long-term incentives, pay scale OH) / SF Compensation	Compensation planning cycles	—	PARTIAL-42	42	Compensation plans yes; merit/bonus planning cycles with budgets and manager worksheets — partial
PSV1 / PSV2 / PSV3 / PSVP / LSO_PSV (training & event management / LSO) / SF Learning	Training catalog, bookings, resources, follow-up	Learning	PARTIAL-42	42	Certifications tracked; course catalog/LMS (sessions, bookings, completion) not named — partial
PP01 IT1050/1051 (job pricing, pay grades) / OH11 (pay scale)	Pay structures (pay scale groups/levels)	—	COVERED-42	42	Compensation plans / pay grades
HRUS_D2 / HRPAD (personnel data: concurrent employment, global employment)	Concurrent/global employment	—	PARTIAL-42	42	Multi-assignment and global assignments not named — partial

T-code	Name	Purpose	Passage status	Wave	Notes
ESS/MSS (Fiori My Profile, My Leave Requests, My Timesheet, My Paystubs, Team Calendar, Approve Leave Requests, Approve Timesheets)	Self-service	—	COVERED-42	42	ESS/MSS
SuccessFactors Employee Central (person/employment/job info, foundation objects, position management, time off, EC time sheet, global benefits, document generation, MDF custom objects, Intelligent Services, EC Payroll, People Analytics)	Cloud HRIS	—	COVERED-42	42	HR core (worker, positions, absence, benefits, ESS); MDF-style custom objects via metadata kernel (wave 3)
SF Onboarding / Offboarding	Onboarding tasks	—	PARTIAL-42	42	Onboarding checklists not named — partial
Employee Central integration with ERP (BIB, EC-ERP replication, cost center replication)	—	—	SUPERSEDED-42	42	Single system
HR PII: HR authorizations (structural P_ORGIN/P_PERNR/structural authorizations OOSB), data privacy (ILM blocking of employee data)	—	—	COVERED-42	42	PII vaulting, consent, retention, role-based masking (contract tests)
Fiori: Manage Employee Master Data / Employee Lookup / Org Chart	—	—	COVERED-42	42	—

13.28.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
S_AHR_61016369 / 61016362 / 61016360 / 61016380 / 61016373 … (PA standard reports: employee list, flexible employee data, headcount, entries/leavings, birthday list, service anniversaries, nationality, family members, education)	HR master data reports	COVERED-42	42	HR reporting via saved views + semantic layer
S_AHR_61016532 / 61016529 / 61016528 … (OM reports: org structure, staff assignments, position descriptions, vacancies, existing jobs/positions/org units)	OM reports	COVERED-42	42	Org chart
S_AHR_61018 (time: absences/attendances overview, quota overview, time statement, cumulated time balances, attendance check) / PT64 / PT65 / PT90	Time reports	COVERED-42	42	—
S_AHR_61016150 / PC00_M99_CWTR / PC00_M99_CLJN / PC00_M99_CLGA / PC00_M99_URME (payroll: wage type reporter, payroll journal, payroll account, reconciliation, posting documents)	Payroll reports	COVERED-43	43	Payroll audit and reconciliation reports
PC00_M10_* US reports (941, 940, W-2, state unemployment, new hire, EEO-1, workers' comp)	US statutory	PARTIAL-43	43	Year-end W-2/1099 named; quarterly 941/state UI and EEO/VETS reporting belong to US pack — partial
HRBEN reports (participation, costs, benefit statements, HRBEN0075 / 0079)	Benefits reports	COVERED-42	42	—
Ad-hoc query (S_PH0_48000510 / PQAH / SQ01 HR InfoSet query)	HR ad hoc query	SUPERSEDED-46	46	Self-service explorer with PII RLS
HIS (PPIS / S_AHR_61016 HIS reporting via org structure)	HR information system	SUPERSEDED-46	46	—
SF People Analytics / Workforce Analytics / Report Center	—	COVERED-46	46	—
Fiori: Headcount Overview / Workforce KPIs	—	COVERED-46	46	—

13.28.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
PA03 release → PC00_Mxx_CALC → corrections → exit	Payroll cycle	COVERED-43	43	Pay cycles
PC00_M99_CIPE (posting) → PCP0 release	Payroll GL posting run	COVERED-43	43	—
PC00_M99_CMLI / FPAYM / PC00_M99_FPAYM	Payroll payment run (DME)	COVERED-43	43	—
PT60 (RPTIME00) scheduled	Time evaluation nightly	COVERED-25	25	—
PT_QTA00 / RPTQTA00	Quota generation/accrual run	COVERED-42	42	Leave accruals
RPUAUD00 / PT_BAL00	Time balance / audit	COVERED-42	42	—
PU19 (quarterly/annual tax reporter runs)	Statutory filings	PARTIAL-43	43	—
PC00_M99_CRET (retro) / PC00_M99_PA03_RELEA	Retro payroll processing	COVERED-43	43	Retro pay
HRBEN0001 open enrollment period / HRBEN0071 eligibility run	Open enrollment cycle	COVERED-42	42	—
RPUDELPP / PU01 (delete payroll results)	Payroll result corrections	COVERED-43	43	—
PA40 mass actions (RPUPAA / RPMACT batch input)	Mass actions	COVERED-5	5	—
HR data archiving / ILM destruction of employee data (SARA PA_*, ILM)	—	COVERED-48	48	Retention policies, erasure
SF: scheduled jobs (EC → payroll point-to-point replication, Intelligent Services events)	—	SUPERSEDED-42	42	Events
Third-party payroll provider file exchange (ADP/Gusto adapters)	Outsourced payroll	COVERED-43	43	Provider adapter

13.28.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Enterprise structure (HR): personnel areas/subareas (T500P, T001P), employee groups/subgroups (T501, T503), payroll areas (T549A), assignment to company code	T500P, T001P, T501, T503, T549A	—	COVERED-42	42	HR org dimensions per legal entity
Infotype customizing: infotype attributes (T582A), screen control (T588M), infotype menus (T588B), dynamic actions (T588Z), features (PE03: NUMKR, ABKRS, LGMST, TARIF…)	T582A, T588M, T588B, T588Z, T549	Time constraints, retro relevance, subtypes	SUPERSEDED-3	3	Metadata-driven entities/fields with effective dating; dynamic actions → workflow/event hooks (wave 4)
PA40 personnel actions / action types / info groups / reasons (T529A, T529U, T588C)	T529A, T529U, T588D	—	COVERED-42	42	Action types with reason codes

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OM: object types, relationships, evaluation paths, plan versions, number ranges (OONR), infotype 1000-series settings	T778O, T778V, T778A, T778P, T77S0	—	COVERED-42	42	—
OM–PA integration switches (T77S0 PLOGI ORGA), position vacancy, cost distribution	T77S0	—	SUPERSEDED-42	42	Single model
Time management: work schedule rules (T508A), daily work schedules (T550A), period work schedules (T551A), holiday calendars (SCAL), absence/attendance types (T554S), quota types (T556A), counting rules (T556C), quota generation rules (T559L), time types (T555A), time evaluation schemas (TM00/TM04) and PCRs (PE02), time pairs	T508A, T550A, T551A, T554S, T556A, T556C, T559L, T555A	—	COVERED-42	42	Absence plans, accrual rules, work calendars; pay-type calculation rules (wave 25) as versioned rule sets replace schemas/PCRs
Payroll: wage types (T512W, T511, wage type catalog OH11), processing/cumulation/evaluation classes, schemas (PE01: U000 US, D000 DE), personnel calculation rules (PE02), payroll periods (T549Q/T549S), pay scale structure (T510), tax models (T5UTM/T5UTE US), BSI TaxFactory integration (US), posting characteristics (T52EK symbolic accounts, T52EZ, OBYE account assignment), garnishment customizing, off-cycle reasons	T512W, T511, T510, T549Q, T549S, T52E*, T5U*	—	COVERED-43	43	Earnings/deductions/taxes as versioned rule sets; US federal/state tables (BSI-like provider adapter option); posting profiles
Benefits: benefit areas, plan types, plans (health/insurance/savings/stock/credit/misc), cost variants, eligibility rules, termination rules, enrollment events, open enrollment periods, EDI to providers (T5UB*)	T5UB*, T74*	—	COVERED-42	42	Benefit plans/eligibility/events
Recruitment: applicant groups/ranges, activities, vacancies, applicant status	T750*, T751*	—	COVERED-42	42	—
PD: qualification scales, appraisal templates (PHAP_CATALOG), development plans, career/succession settings	T77*	—	COVERED-42	42	—
Compensation management: compensation areas, plans, guidelines, budgets, eligibility	T71*	—	PARTIAL-42	42	Planning cycles partial
Training & event management / LSO: business event types, resources, booking, follow-up, LSO settings	T77*	—	PARTIAL-42	42	LMS partial
HR authorizations: P_ORGIN, P_PERNR, structural authorizations (OOSP/OOSB), test procedures (T584A), context-sensitive	T77PR, T77UA	—	COVERED-42	42	OPA policies with structural scope (org hierarchy) + PII masking
ESS/MSS configuration (launchpad, Fiori apps, HR Renewal)	—	—	COVERED-42	42	—
SuccessFactors EC: foundation objects, MDF objects, business rules (rules engine), workflows, role-based permissions, data models (XML), picklists, event reasons	—	—	COVERED-42	42	Metadata kernel + workflow + OPA
EC Payroll / point-to-point replication configuration	—	—	SUPERSEDED-42	42	—
Country-specific HR infotypes and legal changes (country packs: US, DE, GB, FR, IN, MX, CA)	—	—	PARTIAL-48	48	Country-specific fields framework (wave 42) + packs (wave 48) cover tax/statutory; HR-legal (works council, contract types, statutory leave) per country partially
Concurrent employment / global employment configuration	—	—	PARTIAL-42	42	—

13.29 FI-TV — Travel Management / SAP Concur

SAP context: FI-TV travel requests (TRIP: TP01? request, PR01? plan), travel planning (flight/hotel/car booking via Amadeus/Sabre, PR03?), travel expenses (PR05 create/change expense report, TRIP, PR04? , receipts, per diems/flat rates, mileage, credit card clearing PRCC, approval workflow, settlement PREC, posting to FI PRFI/PRRW, payment via payroll or AP/DME PRDX/PRDH?), travel privileges IT0017, Concur Expense/Travel/Invoice/Request with SAP ICS integration. Passage: wave 43 (expense reports, receipts OCR, policies/per diems, approvals, credit-card feed, reimbursement via AP, project expense) — travel booking is not covered.

13.29.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
PA30 IT0017 travel privileges / PRMD	Employee travel privileges (reimbursement group, trip rules)	—	COVERED-43	43	Expense policy groups per employee
TRIP / PR05 / PR04	Travel manager / create-change travel expense report (receipts, per diems, mileage, advances, deductions)	Expense report	COVERED-43	43	Expense reports with receipts (OCR), per diems, mileage
TP01 / TP02 / TP03 / PR01 (travel request)	Travel request (pre-trip approval with estimated cost)	Pre-approval	PARTIAL-43	43	Pre-trip travel request/approval with estimated cost and cash advance not named — partial; propose wave 43 "pre-approval request"
TP04 / PR03 (travel plan: flight, hotel, car via GDS)	Travel booking	Booking	OUT-OF-SCOPE	—	Travel booking/GDS integration is not in scope; integrate with Concur Travel/Navan via wave 35 integration hub
PR02 / PR05 advances (PR cash advance)	Cash advances	—	PARTIAL-43	43	Advances and offset against expenses not named — partial
PRCC / PRCCD / PRCCE / PRCCF	Credit card clearing: import card transactions, buffer, assign to trips	Card feed	COVERED-43	43	Credit-card feed import
PR05 → receipts attachments (ArchiveLink)	Receipt images	—	COVERED-43	43	Receipt capture
PR10 / PR11 (per diem / flat rate maintenance — customizing)	—	—	COVERED-43	43	—
PRAP / PR05 approval (workflow WS20000040)	Expense approval	—	COVERED-43	43	Approvals
Concur Expense (reports, receipts, ExpenseIt, audit rules, allocations to cost objects, card feeds), Concur Request, Concur Invoice, Concur Travel	Cloud T&E	—	COVERED-43	43	Expense management covers Concur Expense; Request partial; Travel OOS; Invoice = AP capture (wave 18)
Concur–SAP ICS integration (financial posting, employee sync)	—	—	SUPERSEDED-43	43	Native
Fiori: My Travel Requests / My Travel and Expenses / Approve Travel Expenses	—	—	COVERED-43	43	—

13.29.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
PRTS / S_AHR_61016401 … 61016411 (trip statistics, receipts, cost assignment, general trip data, trips by period/cost center, travel expense statement)	Travel reports	COVERED-43	43	Expense reporting
PR05 overview / PRTE (trip overview)	—	COVERED-43	43	—
PRCC statistics	Card transactions not assigned	COVERED-43	43	—
S_AHR_61016 (taxable amounts per employee)	Taxable benefits	COVERED-43	43	Taxable expense items to payroll
Concur reporting (Cognos/Intelligence)	—	COVERED-46	46	—

13.29.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
PREC	Settlement of trips (calculate reimbursement, tax)	COVERED-43	43	Expense settlement
PRFI / PRRW	Create posting run / manage posting runs (post to FI/CO)	COVERED-43	43	Posting to GL/projects
PRDX / PRDH / PRDE	DME payment of trips (direct payment) / via payroll (PRPY) / data medium	COVERED-43	43	Reimbursement via AP payment run (wave 19)
PRCC (scheduled import)	Card feed import	COVERED-43	43	—
PRPY / transfer to payroll (IT0017/ wage types)	Reimbursement via payroll	COVERED-43	43	Payroll reimbursement option — confirm; treat covered
PRAA (archiving) / SARA FI_TV	Archiving	COVERED-48	48	—
Concur SAE (standard accounting extract) import	Concur → GL	SUPERSEDED-43	43	—

13.29.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Travel expense control (trip provision variants, trip schemas, trip types statutory/enterprise, reimbursement groups)	T702N, T702O, T706*	—	COVERED-43	43	Policy groups
Per diems/flat rates: meals, accommodation, mileage (country/region/time), deductions (T706H T706F T706U)	T706*	Statutory vs enterprise rates, 3-month rule (DE), IRS rates (US)	COVERED-43	43	Per-diem tables per country pack (wave 48)
Expense types (T706B1, individual receipts), max amounts, receipt requirement, VAT/tax treatment (input tax recovery), wage types for payroll	T706B*, T706K	—	COVERED-43	43	Expense categories with policy limits and tax
Travel request / travel plan settings, approval workflows, substitutes	T702, SWF	—	PARTIAL-43	43	Pre-approval partial
Credit card clearing: card companies, transaction keys mapping to expense types	T706_CCC	—	COVERED-43	43	—
Posting: symbolic accounts, account determination (T706K), cost assignment rules (cost center/WBS/order), posting runs	T706K, T030	—	COVERED-43	43	Posting profiles; project expense
Travel planning (GDS, policies, travel agency)	—	—	OUT-OF-SCOPE	—	—
Concur configuration (audit rules, policies, workflows, allocation forms)	—	—	COVERED-43	43	Policy rule engine

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13.30 Basis & Security (users, roles, GRC Access Control, change documents, transports, system administration)

SAP context: user administration (SU01, SU10, SU3, SU53, SU56, SUIM, SUGR, SU0?), CUA (SCUA/SCUM), role maintenance (PFCG, SU24, SU25, SU22, SU21 auth objects, SU20 fields), security audit (SM19/SM20/RSAU_CONFIG, SM18), password/security policies (RZ10, SECPOL), SSO/SNC/SAML (STRUST, SAML2, SNC), GRC Access Control (ARA risk analysis, ARM access requests, EAM emergency access/firefighter, BRM role management, UAR user access review), transports (SE09/SE10/STMS/SE01/SE03, CTS+, ChaRM), system admin (SM36/SM37 jobs, SM50/SM51/SM66 processes, SM12 locks, SM13 updates, SM21 syslog, ST22 dumps, ST02/ST03/ST04/ST05/ST06/ST11/ST12 performance, SMLG, RZ04, RZ20 CCMS, RZ70/SLD, SM59 RFC, SICF, SOAMANAGER, SPAD/SP01/SP02 spool, SCOT/SOST mail, SM30/SM31 table maintenance, SE16/SE16N/SE16H table display, SE11/SE80/SE38/SE37/SE24/SE18/SE19/CMOD development, SNOTE/SPAM/SAINT/SUM patching, SCC4/SCC1/SCCL client admin, SCDO/SCU3 change docs & table logging, SNRO number ranges, SE93 t-codes, SM35 batch input, SLG1 app log, SARA archiving, DB13/DB02 database, SAP Solution Manager, EarlyWatch, SAP Router, Web Dispatcher, HANA cockpit, SAP Cloud ALM). Passage: waves 1 (pipeline/transports), 2 (identity/authz), 3 (metadata/DDIC), 4 (jobs/workflow), 5 (shell, number sequences, audit, observability), 48 (SoD, SOC 2, privacy), 50 (operations/DR).

13.30.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
SU01 / SU01D / SU10	User maintenance (address, logon data, roles, profiles, parameters, defaults, groups) / display / mass changes	Users	COVERED-2	2	Keycloak users with SCIM provisioning; admin UI (users, roles, duties)
SU3 / SU0 / SU50 / SU51 / SU52	User own data (defaults, parameters, address)	—	COVERED-5	5	Personalization/preferences
SU01 → SNC / SAML2 / STRUST / SSO2	SSO configuration	—	COVERED-2	2	OIDC/SAML federation; MFA
SCUA / SCUM / SCUG / SCUL	Central User Administration	Multi-system users	SUPERSEDED-2	2	Single IdP (Keycloak)
SUGR	User groups	—	COVERED-2	2	Groups
PFCG	Role maintenance (menu, authorizations, profiles, users; single/composite/derived roles, org levels)	Roles	COVERED-2	2	Role/duty/privilege model compiled to OPA/Rego; derived roles by org level = role scoping per legal entity/site

T-code	Name	Purpose	Passage status	Wave	Notes
SU24 / SU25 / SU22	Authorization default values per transaction (check indicators) / upgrade/profile generator / SAP defaults	Auth proposals	SUPERSEDED-2	2	Privileges attached to API endpoints/entities in metadata
SU21 / SU20	Authorization objects / fields	—	SUPERSEDED-2	2	Privilege = (resource, action, scope) in policy
SU53 / SU56 / ST01 / STAUTHTRACE	Authorization check analysis / user buffer / system trace / auth trace	Troubleshooting	COVERED-2	2	Effective-permission viewer + OPA decision logs
SUIM	User information system (users by role/auth/transaction, change docs for users/roles, comparisons)	Reporting	COVERED-2	2	Effective-permission report; access reports
SM19 / SM20 / RSAU_CONFIG / RSAU_READ_LOG / SM18	Security audit log configuration / evaluation / reorganization	Security audit	COVERED-5	5	Audit log + OTel logs; wave 48 SOC 2 evidence
SECPOL / RZ10 login parameters	Password/security policies	—	COVERED-2	2	Keycloak policies
GRC AC: NWBC Access Request (ARM) / Access Risk Analysis (ARA) / Business Role Management (BRM) / Emergency Access Management (EAM firefighter IDs, logs) / User Access Review (UAR) / SoD Review	Access governance	—	COVERED-48	48	SoD rules engine, conflict detection, mitigation workflow, access certification campaigns; emergency (break-glass) access — confirm firefighter/break-glass with session log; treat PARTIAL via next row
GRC EAM firefighter (superuser privilege management with reason codes and log review)	Emergency access	—	PARTIAL-48	48	Break-glass elevated access with time-boxed sessions, reason codes and reviewer sign-off not explicitly named — partial
SE09 / SE10 / SE01 / SE03 / STMS / STMS_IMPORT / SE06 / SE07	Transport organizer (workbench/customizing requests), transport management, import queues, system change option	Transports	COVERED-1	1	GitOps: config/metadata changes as versioned artifacts promoted through environments by digest; customizing-as-code
SCC4 / SCC1 / SCCL / SCC3 / SCC5 / SCC7 / SCC8 / SCC9	Client maintenance, client copy (local/remote), client transport, delete	Clients	SUPERSEDED-2	2	Tenants + environments; tenant provisioning automation (wave 50)
SE11 / SE12 / SE13 / SE14 / SE15 / SE16 / SE16N / SE16H / SE17	ABAP Dictionary: tables/views/data elements/domains/search helps/lock objects, technical settings, DB utility, repository info system, table browser	DDIC	COVERED-3	3	Metadata registry (entities, fields, enums); generated schema; table browser = entity explorer
SE80 / SE38 / SE37 / SE24 / SE41 / SE51 / SE71 / SE91 / SE93 / SE63 / SE54 / SM30 / SM31 / SM34	Development workbench, programs, function modules, classes, menus, screens, SAPscript, messages, transaction codes, translation, table maintenance generator, table maintenance	ABAP development	OUT-OF-SCOPE	—	PRD non-goal: no proprietary extension language; metadata extensions (wave 3) and public APIs (wave 35) instead
SE18 / SE19 / SE20 / CMOD / SMOD / SPAU / SPDD / SE95	BAdIs, enhancement spots, customer exits, modification adjustment	Enhancement framework	SUPERSEDED-3	3	Lifecycle hooks/extension points in metadata; webhooks and event subscribers (waves 4/35)
SNOTE / SPAM / SAINT / SUM / SPAU / SPDD	Note assistant, support packages, add-ons, upgrade	Patching	SUPERSEDED-1	1	Continuous delivery; zero-downtime migrations (wave 50)
SM36 / SM37 / SM62 / SM64 / SM61 / SM65 / SMX	Background jobs: define, monitor, events, trigger event, job servers, check	Batch	COVERED-4	4	Batch job framework with job monitor UI
SM50 / SM51 / SM66 / SM04 / AL08 / SM12 / SM13 / SM14 / SM21 / ST22 / SM58 / SMQ1 / SMQ2 / SMQR / SMQS	Work processes, servers, global process overview, user sessions, locks, update requests, system log, dumps, tRFC, qRFC queues	System monitoring	SUPERSEDED-5	5	Kubernetes + OpenTelemetry (Grafana/Tempo/Loki/Prometheus), SLO dashboards, alerting; Kafka lag monitoring; Temporal UI
ST02 / ST03 / ST03N / ST04 / ST05 / ST06 / ST07 / ST10 / ST11 / ST12 / STAD / SQLM / SWLT / ST14	Buffers, workload analysis, DB performance, SQL trace, OS monitor, app monitor, table call stats, developer traces, single transaction analysis, statistics records, SQL monitor	Performance	SUPERSEDED-5	5	OTel traces/metrics; pgBadger/pg_stat_statements; k6 load campaigns (wave 50)
RZ20 / RZ21 / RZ03 / RZ04 / RZ10 / RZ11 / RZ12 / SMLG / SM63	CCMS monitoring, operation modes, instance profiles, parameters, RFC server groups, logon groups	—	SUPERSEDED-1	1	Helm values / K8s config; HPA; Envoy routing
SM59 / SICF / SOAMANAGER / SRT_* / WE21 / SMICM / SMMS	RFC destinations, HTTP services, web service config, ICM	Connectivity	SUPERSEDED-35	35	Integration hub (REST/GraphQL/webhooks/Kafka), API gateway, keys & quotas
STRUST / SSFA / SSO2 / SAML2 / SNCWIZARD	Trust manager (certificates), SSF, SAML	PKI	COVERED-1	1	cert-manager, Vault, Keycloak
SPAD / SP01 / SP02 / SPAD printers / SP12	Spool administration, output requests, device types	Printing	COVERED-32	32	Print service (ZPL, PDF) + document designer; cloud print queues — confirm generic print queue management for office printers (not just ZPL); treat PARTIAL via note in cross-app
SCOT / SOST / SOSG / SOSV / SBWP / SO10 / SO16	SAPconnect (email/fax), send status, business workplace, standard texts	Messaging	COVERED-5	5	Notification service (email/SMS/in-app) — confirm; treat covered
SM35 / SHDB	Batch input sessions / recorder	Data entry automation	SUPERSEDED-10	10	Import/export (wave 5) and migration pipeline (wave 10); APIs
SLG0 / SLG1 / SLG2	Application log objects / display / delete	App log	COVERED-5	5	Structured logs (Loki) with trace ids
SCDO / SCU3 / SE16 CDHDR-CDPOS / RSSCD100	Change document objects / table logging evaluation / change documents display	Change documents	COVERED-5	5	Field-level immutable audit from CDC with viewer
SNRO / SNUM / NRIV	Number range objects/intervals/buffering	Number ranges	COVERED-5	5	Number sequence service (gap-free option)
SARA / SARI / SARJ / DB15 / AOBJ / ILM (IRM_CUST, ILMARA, ILM_DESTRUCTION) — see cross-app	Archiving	—	COVERED-48	48	—
DB02 / DB12 / DB13 / DB14 / DB20 / DBACOCKPIT / HANA cockpit / HDBSQL	Database administration: space, backups, planning calendar, statistics	DBA	SUPERSEDED-50	50	CloudNativePG backups/PITR, backup/restore drills (wave 48), partitioning (wave 50)
SM02 / SM13 / SE38 RSPARAM / SU10 lock	System messages, update admin	—	SUPERSEDED-5	5	Shell banner/notifications — confirm system-wide announcement banner; treat covered
SM30 → table maintenance views (customizing data)	Customizing table maintenance	—	SUPERSEDED-3	3	Configuration entities generated from metadata with UI
SE93 transaction codes / SMEN favorites / SSM2 area menus	Navigation	—	SUPERSEDED-5	5	Workspaces, global search, saved views

T-code	Name	Purpose	Passage status	Wave	Notes
SPRO / SPRO_ADMIN / S_IMG (IMG projects, project views, activities)	Implementation guide, IMG projects, customizing activities, documentation	Customizing navigator	SUPERSEDED-3	3	Configuration workspace organized per module; implementer guide (wave 50); configuration-as-code with diff/versioning (wave 3 registry)
SOLMAN (Solution Manager: ChaRM, test suite, process documentation, monitoring, EarlyWatch, Focused Build) / SAP Cloud ALM / Signavio	ALM	—	SUPERSEDED-1	1	GitHub + Argo CD + observability; process documentation = functional guides (wave 50)
SAP Router, Web Dispatcher, SAProuter, Gateway (SMGW), Cloud Connector	Network components	—	SUPERSEDED-1	1	Envoy Gateway, ingress, cert-manager
SLD / LMDB / RZ70	System landscape directory	—	SUPERSEDED-1	1	Backstage catalog (wave 35)
SU01 → user types (dialog, system, communication, service, reference)	Technical users	—	COVERED-2	2	Service accounts/API keys (wave 35)
SAP Identity Management (IdM) / Identity Authentication Service (IAS) / Identity Provisioning (IPS)	Cloud identity	—	SUPERSEDED-2	2	Keycloak + SCIM
UI5/Fiori launchpad admin (/UI2/FLPD_CUST, /UI2/FLPCM_CUST, /UI2/SEMOBJ, /UI2/FLP, spaces & pages /UI2/FLPD_SPACE) — see cross-app	—	—	COVERED-5	5	—
SAP GUI (Windows/Java/HTML), SAP Business Client (NWBC), WebGUI, Personas	Clients	—	SUPERSEDED-5	5	React shell; Personas-style personalization = shell personalization
HANA: HDB user/role admin, SQL privileges, calculation views	DB-level security	—	SUPERSEDED-2	2	Postgres roles + RLS; no end-user DB access

13.30.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
SUIM (all reports)	Users by roles/authorizations/t-codes; roles by authorization; change documents (users/roles/profiles); comparison; critical authorizations	COVERED-2	2	Access reports; SoD (wave 48)
RSUSR002 / RSUSR003 / RSUSR008_009_NEW / RSUSR100 / RSUSR200	Users by complex criteria / standard users password check / critical combinations / change docs / logon dates & password changes	COVERED-2	2	—
SM20 / RSAU_READ_LOG	Security audit log evaluation	COVERED-5	5	—
ST03N / STAD / SM21 / ST22	Usage statistics, statistics records, syslog, dumps	SUPERSEDED-5	5	Grafana dashboards
SM37 job overview / SM36	Job logs	COVERED-4	4	Job monitor
SLG1	Application log	COVERED-5	5	—
SCU3 / RSVTPROT	Table change logs	COVERED-5	5	Audit viewer for configuration entities
SE16/SE16N/SE16H	Table display	SUPERSEDED-3	3	Entity explorer + lakehouse SQL (wave 46) with RLS
GRC AC reports: risk analysis (user/role/profile level), mitigation reports, firefighter logs, access request status, UAR status	Access governance reports	COVERED-48	48	—
STMS import history / SE10 request logs	Transport history	COVERED-1	1	Argo CD history, git log
AIS (Audit Information System: SECR / S_ALR_87100 system audit, business audit)	Audit information system	COVERED-48	48	SOC 2 control evidence; wave 15 audit trail
EarlyWatch Alert / Solution Manager dashboards	System health	SUPERSEDED-5	5	SLO dashboards
License administration (SLAW / USMM / LAW2)	License measurement	SUPERSEDED-50	50	Tenant usage/cost model (wave 50)

13.30.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
SM36 standard jobs (SAP_REORG_JOBS, SAP_REORG_SPOOL, SAP_COLLECTOR_FOR_PERFMONITOR, SAP_REORG_ABAPDUMPS, SAP_CHECK_ACTIVE_JOBS…)	Housekeeping jobs	SUPERSEDED-4	4	Platform housekeeping (log retention, vacuum) via K8s CronJobs / job framework
SM18 / RSAU_FILE_ADMIN	Security audit log reorganization	COVERED-5	5	Log retention policies
SUIM/PFCG mass user comparison (PFUD / PFCG_TIME_DEPENDENCY)	Role assignment validity cleanup	COVERED-2	2	Effective-dated role assignments
SU10 mass lock / unlock, inactive users (RSUSR200)	User lifecycle	COVERED-2	2	SCIM deprovisioning; wave 48 access reviews
GRC UAR campaigns, periodic risk analysis batch, firefighter log sync	Access certification	COVERED-48	48	Access certification campaigns
STMS import jobs / transport of copies / ChaRM cycles	Release management	COVERED-1	1	GitOps promotion
DB13 backups / statistics / reorg	DBA calendar	SUPERSEDED-50	50	CloudNativePG scheduled backups; wave 48 restore drills
SPAM/SUM/SNOTE patch cycles, kernel upgrades	Patching	SUPERSEDED-1	1	CI/CD with SBOM/vuln scans (Grype/Trivy), image signing
Client copy refresh (SCCL) for test systems	Environment refresh	SUPERSEDED-2	2	Tenant clone/anonymized refresh — confirm anonymized test-data refresh tooling (wave 48 privacy + wave 50 ops); treat PARTIAL via row below
Anonymized production-to-test data refresh (TDMS / client copy with scrambling)	Test data management	PARTIAL-48	48	Tenant clone with PII scrambling not explicitly named — partial; propose wave 48/50
SLG2 / SARA application log deletion	Log cleanup	COVERED-5	5	—

T-code	Name	Passage status	Wave	Notes
License measurement (USMM)	—	SUPERSEDED-50	50	—
SM13 update error reprocessing / SM58 tRFC retry / SMQ2 queue restart	Error reprocessing	SUPERSEDED-4	4	Outbox + Temporal retries, DLQ replay (wave 35)

13.30.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Profile parameters (RZ10/RZ11: login/, *auth/*, rdisp/, *icm/*, rsau/*)	TPFET, PAHI	Password rules, lock on failed logons, SSO settings, session timeout, audit	COVERED-2	2	Keycloak realm policies; gateway settings
PFCG role concept: org levels (PFCG_ORGFIELD_CREATE), derived roles, composite roles, menu, SU24 check indicators	AGR_*, USOBT_C, USOBX_C	—	COVERED-2	2	Role/duty/privilege metadata; org-scoped assignment
SU25 steps / SU24 maintenance (check indicators)	USOBT_C	—	SUPERSEDED-2	2	—
Security audit log filters (SM19/RSAU_CONFIG), audit classes	RSAU*	—	COVERED-5	5	—
GRC AC: rule sets (functions, risks, SoD matrix), mitigating controls, risk owners, workflow MSMP, firefighter configuration, connectors	GRAC*	—	COVERED-48	48	SoD rules + mitigation workflow
GRC AC: access request types, approval stages (MSMP), provisioning settings, UAR settings	GRAC*	—	COVERED-48	48	Access request workflow via wave 4 + Keycloak provisioning
STMS configuration: transport domain, routes, layers, QA approval procedure, import strategies; CTS+ for non-ABAP	TMS*	—	SUPERSEDED-1	1	Argo CD environments, promotion by digest, approval gates
SCC4 client settings (customizing changes allowed, cross-client, protection)	T000	—	SUPERSEDED-2	2	Environment-level write protection via GitOps
SE06 system change option, SE03 namespaces, software components	—	—	SUPERSEDED-1	1	—
Operation modes (RZ04), logon groups (SMLG), RFC server groups (RZ12)	—	—	SUPERSEDED-1	1	K8s scheduling/HPA
Background processing: job classes, server groups (SM61), event definitions (SM62)	TBTCO, BTCSEV	—	COVERED-4	4	Job priorities/partitioning; event triggers (Kafka)
Printing: device types, output devices, access methods (SPAD), front-end printing, print queues	TSP03*	—	PARTIAL-32	32	Print service for labels/documents; generic office print queue/printer management partial
SAPconnect nodes (SCOT), SMTP, fax, domains, send jobs	SX*	—	COVERED-5	5	Email provider adapter
Table logging activation (rec/client, SE13 log data changes), change document objects (SCDO)	—	—	SUPERSEDED-5	5	CDC audit for everything flagged audit=true in metadata
Number range buffering (SNRO: no buffering, main memory, parallel)	TNRO	—	COVERED-5	5	Gap-free vs buffered sequences
Archiving customizing: archiving objects, residence times, archive infostructures (SARI/SARJ), content repository (OAC0), ILM policies	AOBJ, ARCH_*, ILM*	—	COVERED-48	48	Retention policies, legal holds
ArchiveLink / KPro: content repositories (OAC0/OACT/OAC2/OAC3), document types, links to business objects (GOS)	TOA*, TOAOM	—	COVERED-5	5	Attachments object store (S3-compatible) — confirm object storage for attachments; treat covered
SAML/OIDC/SNC configuration, trusted IdPs, user mapping	SAML2*	—	COVERED-2	2	—
HTTP whitelist, CSRF, UCON (unified connectivity) RFC allowlist	UCON*	—	COVERED-2	2	API gateway policies
Read Access Logging (SRAL) configuration	SRAL*	—	PARTIAL-48	48	PII access logged (wave 42) for HR; configurable read-access logging for arbitrary sensitive fields partial — propose wave 48
UI data protection masking (UI masking/logging add-on)	—	—	COVERED-42	42	PII masking per role
Data destruction / blocking (ILM, BP blocking BUPA_PRE_EOP, end of purpose)	ILM*	—	COVERED-48	48	Retention + erasure; end-of-purpose checks per party — confirm EoP checks across modules; treat covered
SAP Cloud ALM / Solution Manager process management, test management (ECATT SECATT, CBTA)	—	—	SUPERSEDED-1	1	CI tests, Playwright, fixtures

13.31 Cross-Application (workflow, IDoc/ALE, PI/PO, DRC e-documents, output management NAST/BRF+, Fiori launchpad, archiving/ILM, GDPR/IRF, attachments, texts, enterprise search)

13.31.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
SWDD / SWDD_SCENARIO / SWDM	Workflow builder (classic) / flexible workflow scenario editor / business workflow explorer	Workflow design	COVERED-4	4	Workflow designer (configuration-driven approval chains) + Temporal
PFTC / PFTC_INS / PFTC_CHG / PFTC_DIS	Task maintenance (standard tasks, workflow templates)	—	SUPERSEDED-4	4	Workflow templates in designer
PFAC / PFAC_INS / PFAC_CHG	Rules (agent determination: responsibilities, org, function module)	Agent rules	COVERED-4	4	Approver resolution (role, hierarchy, amount band, delegation)
SWO1 / SWO2 / SWO3 / SWO6 / SE24 (ABAP OO)	Business object builder / repository / delegation	BOR	SUPERSEDED-3	3	Entities + events from metadata
SWE2 / SWETYPV / SWEC / SWEL / SWELS / SWE3 / SWEINST	Event type linkages, change-document events, event trace, instance linkages	Event linkage	COVERED-4	4	Entity-changed CloudEvents trigger workflows; AsyncAPI

T-code	Name	Purpose	Passage status	Wave	Notes
SBWP / Fiori My Inbox / SWNCONFIG / SWNADMIN / SOST	Business workplace inbox, My Inbox app, extended notifications (email)	Task inbox	COVERED-4	4	Human tasks UI in shell + notifications; delegation/substitution (HRUS_D2 / SBWP substitutes) — confirm substitution rules; treat covered
SWIA / SWI1 / SWI2_FREQ / SWI2_DEAD / SWI2_DURA / SWI2_ADM1 / SWI5 / SWI6 / SWI14 / SWPR / SWPC / SWU_OBUF / SWWL / SWWH	Work item administration (execute as admin, selection, frequency, deadlines, durations, without agents), workload, workflows for object, restart after error, continue after system crash, buffer sync, delete work items	Workflow admin	COVERED-4	4	Temporal UI + workflow admin (retry, restart, reassign, SLA/deadline monitor)
SWU3 / SWU0 / SWUS / SWUD / SWU2 / SWU_EWBTE	Automatic workflow customizing / event simulation / test workflow / diagnosis / RFC monitor	—	COVERED-4	4	Workflow test harness (Temporal test server in CI)
SWF_ADM_ / SWFVISU / SWF_PROCESS_	Flexible workflow admin, task visualization (My Inbox UI), process monitor	—	COVERED-4	4	—
WE02 / WE05 / WE09 / WE19 / WE07 / WE10	IDoc display / list / search by content / test tool / statistics	IDoc monitoring	COVERED-35	35	Message monitor with replay
WE20 / WE21 / WE30 / WE31 / WE60 / WE81 / WE82 / WE41 / WE42 / WE57 / WE47 / WE46 / WE40 / WE44 / BD51 / BD55 / BD56 / BD62 / BD64 / BD87 / BD20 / BD21 / BD50 / BD52 / BD53 / BD61 / BD10 / BD11 / BD12 / BD13 / BD14 / BD15 / BD16 / BD17 / BD18	Partner profiles, ports, IDoc types, segments, documentation, message types, assignment, process codes (outbound/inbound), status processing, distribution model, reprocess, inbound/outbound processing, change pointers (activate, per message type, reduce), send material/customer/vendor/cost center…	ALE/IDoc configuration & distribution	SUPERSEDED-35	35	EDI engine (X12/EDIFACT) + Kafka topic subscriptions + webhooks + REST/GraphQL; partner profiles, acknowledgements, message monitor; master-data distribution = events (no change pointers)
SALE / BDLS / BD54 / BD97 / SM59	ALE customizing, logical system conversion, define logical systems, assign RFC	—	SUPERSEDED-35	35	—
PI/PO: ESR (Enterprise Services Repository), Integration Directory (ID), SXMB_MONI / SXI_MONITOR / SXMB_ADM / SXMB_IFR / SRT_MONI / SRT_UTIL / IDX1 / IDX2 / IDX5 / SLDCHECK / RWB / PIMON	Process integration design/configuration/monitoring, IDoc adapter metadata, web service monitoring	Middleware	SUPERSEDED-35	35	Integration hub (API catalog via Backstage, developer portal, webhooks, Kafka, iPaaS connectors Salesforce/Shopify/Stripe); message monitor with replay and partner SLAs
SAP Integration Suite (Cloud Integration/CPI iFlows, API Management, Open Connectors, Trading Partner Management, Event Mesh, Integration Advisor)	Cloud middleware	—	SUPERSEDED-35	35	Same; Event Mesh = Kafka topics; TPM = EDI partner profiles
EDOC_COCKPIT / EDOC_RESUBMIT / EDOC_BACKGROUND / Fiori: eDocument Cockpit (DRC) / Manage Electronic Documents	Document and Reporting Compliance: e-invoices/e-documents per country (IT SDI, MX CFDI, ES SII, HU RTIR, IN e-invoice/e-way bill, PL KSeF, Peppol, FR, BR NF-e…)	e-documents	COVERED-35	35	Peppol UBL 2.1, Factur-X/ZUGFeRD, country mandates framework (IT SDI, MX CFDI hooks); further mandates per country pack (wave 48) — note: only IT/MX hooks named; first packs US/UK/DE/FR/IN/MX/CA — IN e-invoice & FR 2026 mandate expected in packs
DRC Statutory Reporting (Run Statutory Reports / Manage Compliance Reports: VAT returns, EC sales list, SAF-T, withholding, Intrastat, e-filing)	Statutory reporting framework	—	COVERED-48	48	Electronic reporting designer + regulator mandates registry; wave 13 extracts
NACE / NACZ / NACO / VV11-VV13 / VV21-23 / VV31-33 / MN01-MN27 / VOK2 / V/30 / V/40 / V/57 / V/59 / V/G6 / VN	Output determination (condition technique per application: V1 sales, V2 shipping, V3 billing, EF purchasing, EA RFQ, EV contracts, EL SA, ME inventory, Q QM, PM , MR invoice verif, KM) — output types, access sequences, procedures, condition records, print programs/forms, partners, media, dispatch time	Output management (NAST)	SUPERSEDED-46	46	Output rules: per document type/event → channel (print/email/EDI/portal) with templates from document designer; events (wave 4) for timing; wave 35 EDI; no condition technique for output
S/4 Output Management (BRF+ output determination: OPD, output parameter determination, Manage Output Items , Adobe forms via ADS, email templates, printer queues)	S/4 output control	—	SUPERSEDED-46	46	Same; BRF+ decision tables ↔ rules engine
BRF+ (BRFPLUS / FDT_WORKBENCH) / Decision Service Management (DSM)	Business rules framework (decision tables, formulas, rulesets)	Rules engine	PARTIAL-3	3	Validation rules and OPA policies exist; a general-purpose, user-maintainable decision-table/rules engine (used by output, tax, credit, dispute, GRC, substitution) is implied across waves but not named as a shared service — partial; propose explicit "rules engine (decision tables)" in wave 3/4
SE71 (SAPscript) / SMARTFORMS / SFP (Adobe forms) / ADS / Fiori form templates (Maintain Form Templates)	Form design	Forms	SUPERSEDED-46	46	Operational report/document designer (invoices, POs, packing slips, checks)
SO10 / SO72 / SE75 (text objects/IDs)	Standard texts, text management	Text blocks	PARTIAL-5	5	Document/line notes exist; a reusable standard-text library with placeholders and language versions for documents is not named — partial; propose wave 46 designer text blocks
GOS (Generic Object Services): attachments, notes, links, relationships, workflow overview, ArchiveLink (OAAD / OAWD / OAOR)	Attachments & notes on any object	—	COVERED-5	5	Attachments & notes on any entity (journal attachments wave 11; DMS §13.35)
Document Relationship Browser (DRB) / document flow	Cross-document flow	—	COVERED-27	27	Document flow (order→shipment→invoice→payment), trace ids
/UI2/FLP / /UI2/FLPD_CUST / /UI2/FLPCM_CUST / /UI2/FLPD_CONF / /UI2/SEMOBJ / /UI2/NWBC_CFG_CUST / /UI2/FLPCM_ (catalogs, groups, tiles, target mappings, spaces & pages, launchpad content manager) / /UI2/FLP_SYS_CONF / /IWFND/MAINT_SERVICE (OData services) / /IWFND/ERROR_LOG / /IWFND/GW_CLIENT / /UI5/THEME_DESIGNER / /UI5/UI5_REPOSITORY_LOAD	Fiori launchpad configuration, OData gateway, UI theme designer	Launchpad & gateway	COVERED-5	5	React shell: workspaces by role, navigation, global search, personalization, theming; API gateway
SAP Screen Personas / GuiXT	Screen simplification	—	SUPERSEDED-5	5	Metadata-driven forms + personalization

T-code	Name	Purpose	Passage status	Wave	Notes
SAP Enterprise Search (ESH_COCKPIT / ESH_ADMIN) / Fiori search	Enterprise search	—	COVERED-5	5	Global search
SARA / SARI / SARJ / SARE / AOBJ / SAR / DB15 / ILM: IRM_CUST / ILMARA / ILM_DESTRUCTION / ILM_STOR / ILM_SNAPSHOT / LHM (legal hold) / ILM retention warehouse	Data archiving (write/delete/store/retrieve), archive info system, ILM retention rules, legal holds, data destruction, retention warehouse (system decommissioning)	Archiving & ILM	COVERED-48	48	Retention policies, legal holds, erasure; history in lakehouse (wave 46) — archive retrieval = lakehouse query; system decommissioning (retention warehouse) partially covered by migration-factory history load (wave 49)
GDPR: IRF (Information Retrieval Framework: data model, purposes, retrieval runs), blocking/deletion (BUPA_PRE_EOP, EoP checks), consent management (ECM), read access logging (SRAL), UI masking, data privacy workbench (DPW)	Data protection	—	COVERED-48	48	GDPR/CCPA subject requests (export/erasure), retention; consent in HR (wave 42); SRAL partial (noted)
SAP Information Steward / MDG (Master Data Governance: change requests, data models, validation rules, replication)	Master data governance	—	COVERED-7	7	Data governance: ownership, stewardship queue, change approval workflow (party); product/BOM governance via ECO (wave 9) — note MDG-style central governance for all master domains confirm; treat covered
MDG consolidation & mass processing (duplicate check, best record)	—	—	COVERED-7	7	Dedup/merge with survivorship
SAP Business Partner Screening / Watch List Screening (GTS)	Sanctions/denied-party screening	—	GAP	propose 48	Denied-party/sanctions screening of parties and documents not named; propose wave 48 (compliance) with provider adapter
Legal Content Management / contract management (SAP CLM, Ariba Contracts)	Legal contracts	—	PARTIAL-17	17	Purchase/sales agreements exist; legal contract authoring, clause library, obligations — partial (noted in Ariba row)
SAP Jam / SAP Build Work Zone / Collaboration	Collaboration	—	OUT-OF-SCOPE	—	Integrate with Slack/Teams via webhooks (wave 35); no native collaboration suite
SAP Mobile Platform / SAP Mobile Services / Mobile Start / Fiori Client	Mobile platform	—	COVERED-5	5	PWA shell; mobile apps (waves 25/32/45)
Business Application Studio / Web IDE / Eclipse ADT (SE80 successors) / ABAP Cloud / Steampunk / BTP extensions (side-by-side)	Extension development	—	SUPERSEDED-3	3	Metadata extensions + public APIs/webhooks/MCP (waves 3/35/47)
Key User Extensibility (Custom Fields and Logic, Custom Business Objects, Custom CDS Views, Custom Analytical Queries, Adapt UI)	In-app extensibility (S/4)	—	COVERED-3	3	Metadata kernel: entity extensions (fields, enums, hooks), custom entities; custom views (wave 46); adapt UI (wave 5)
SAP Business Technology Platform (BTP) services: Workflow Management, Business Rules, Process Automation, Event Mesh, API Management, Integration Suite, Document Management, Launchpad service	Platform services	—	SUPERSEDED-4	4	Temporal, rules (partial), Kafka, hub, object storage, shell
SAP Cloud Appliance Library / S/4 activation (Best Practices content, SAP Activate)	Pre-configured content	—	SUPERSEDED-6	6	Seed data / demo manufacturer; tenant templates (wave 50 provisioning)
Data Quality Management (DQM: address validation, duplicate check via SAP DQM microservices)	Address validation	—	COVERED-6	6	Validated postal addresses, geocoding hook
SAP Geographical Enablement Framework (GEF) / maps	Geo	—	COVERED-6	6	Geocoding; maps in TM/FSM (waves 34/45)
Business Event Handling (BTE: FIBF/BF / open FI events)	Event hooks in FI	—	SUPERSEDED-4	4	Lifecycle hooks + events
SAP Business Rules (BRF+ in GRC, tax, dispute, output) — see above	—	—	PARTIAL-3	3	—
Application Interface Framework (AIF: /AIF/CUST, /AIF/ERR, interface monitor)	Interface monitoring & error handling	—	COVERED-35	35	Message monitor with replay, dead-letter, partner SLAs
SAP Process Mining (Signavio) / Process Observer (POC)	Process mining	—	PARTIAL-46	46	Event stream + lakehouse enables mining; no native process-mining UI — partial

13.31.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
SWI1 / SWI2_* / SWI5 / SWI6	Work item selection, frequency, deadlines, durations, workload	COVERED-4	4	Workflow dashboards (SLA, throughput)
SWEL	Event trace	COVERED-4	4	Kafka topic browser / event log
WE02 / WE05 / WE07 / WE09	IDoc lists & statistics	COVERED-35	35	Message monitor
BD87 (status overview)	IDoc status overview	COVERED-35	35	—
SXMB_MONI / SRT_MONI / /AIF/ERR	Integration message monitors	COVERED-35	35	—
EDOC_COCKPIT (status by country)	e-document status	COVERED-35	35	—
NAST analysis (VV13 / WE output status) / Fiori: Manage Output Items	Output status lists	COVERED-46	46	Output log per document
SOST	Sent emails/fax status	COVERED-5	5	Notification log
SARI / ILM reports	Archive info system queries	COVERED-46	46	Lakehouse history queries
IRF retrieval run results	Subject access report	COVERED-48	48	—
/IWFND/ERROR_LOG / /IWFND/APPS_LOG	OData gateway logs	SUPERSEDED-5	5	OTel traces
/UI2/FLP usage analytics (Fiori usage)	Usage analytics	COVERED-5	5	Product analytics on shell (OTel)
ESH search analytics	—	COVERED-5	5	—

13.31.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
SWWDHEX / SWWERRE / SWWCOND / SWWCLEAR (deadline monitoring, error monitoring, condition evaluation, clear work items)	Workflow background jobs	SUPERSEDED-4	4	Temporal timers/retries
RSWWWIDE / SWWL (delete work items) / SWW_SARA archiving	Workflow housekeeping	COVERED-48	48	Workflow history retention
RSEOUT00 / RBDAPP01 / RBDMANI2 / RBDMIDOC (change pointers → IDocs) / RSEIDOC2 / RBDAGAIN / RBDAGAIE	IDoc outbound processing, inbound posting, reprocess, change pointer processing	SUPERSEDED-35	35	Event-driven outbox; replay
RSNAST00 / RSNASTED	Output processing (print/EDI) batch	COVERED-46	46	Output jobs
EDOC_BACKGROUND / DRC submission jobs	e-document submission/polling	COVERED-35	35	—
SARA write/delete/store jobs; ILM destruction runs	Archiving runs	COVERED-48	48	Retention jobs
IRF retrieval runs; blocking runs (BUPA_PRE_EOP/EoP)	GDPR runs	COVERED-48	48	—
/IWFND/ cache cleanup, /UI2/ cache invalidation	Gateway maintenance	SUPERSEDED-5	5	—
ESH index updates (ESH_COCKPIT)	Search indexing	COVERED-5	5	Search index from CDC
SXMS_ / integration archiving & retention	Message retention	COVERED-35	35	—
MDG replication / data load jobs (DRFOUT)	Master data replication	SUPERSEDED-7	7	Events

13.31.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
SWU3 basic workflow settings (RFC destination WORKFLOW_LOCAL, WF-BATCH user, prefix numbers SWU9 , monitoring jobs)	SWF_*	—	SUPERSEDED-4	4	Temporal cluster config
Flexible workflow scenarios (per document type: PR, PO, contract, journal entry, supplier invoice, credit memo…), step conditions, agent determination rules, deadlines, email notifications	SWF_FLEX*	—	COVERED-4	4	Workflow designer with templates per entity
Substitution/delegation rules (SBWP, HRUS_D2), vacation handling	HRUS_D2	—	COVERED-4	4	Delegation/escalation with SLA timers
ALE: logical systems, distribution model, partner profiles, ports, serialization groups, filters, change pointers activation per message type	TBDLS, TBD*, EDP*, EDIPO*	—	SUPERSEDED-35	35	Partner profiles for EDI; topic subscriptions
EDI: converter/subsystem settings, status codes, message control	EDI*	—	COVERED-35	35	EDI engine (AS2/SFTP, acknowledgments 997/CONTRL)
PI/PO / Integration Suite configuration (communication channels, adapters, mappings, SLAs)	—	—	SUPERSEDED-35	35	Hub
DRC: e-document types per country, activation, interface connectors (PEPPOL access point, SDI, CFDI PAC), source document types, status mapping	EDOC*	—	COVERED-35	35	Country mandates framework; packs wave 48
DRC statutory reporting: report definitions, categories, parameters, document and data sources, correspondence	—	—	COVERED-48	48	ER designer
Output determination customizing (NACE per application: output types, procedures, access sequences, condition tables, print programs, forms, partner functions, media, dispatch times, storage mode)	T685B, TNAPR, NAST, T681*	—	SUPERSEDED-46	46	Output rules/channels/templates
S/4 output control: output types, determination (BRF+), form templates, email templates, printer settings, channels	APOC_*	—	SUPERSEDED-46	46	—
BRF+ applications, functions, rulesets, decision tables, expressions	FDT_*	—	PARTIAL-3	3	Rules engine gap
Text determination (text objects, IDs, procedures) across SD/MM/FI	TTXID, TTXERN	—	COVERED-27	27	Notes types per entity
GOS/ArchiveLink: document types, object types, links (OAC3), content repositories (OAC0), bar code entry (OAM1) , storage scenarios	TOA*	—	COVERED-5	5	Attachment types, object storage
Fiori launchpad: catalogs, groups, spaces, pages, target mappings, roles (PFCG menu), theming, user defaults (default values per parameter)	/UI2/*	—	COVERED-5	5	Workspaces, navigation by role, personalization
OData/Gateway activation (/IWFND/MAINT_SERVICE), service groups, system aliases	/IWFND/*	—	SUPERSEDED-35	35	Generated OpenAPI/GraphQL; API catalog
Archiving objects: residence/retention, variants, archive routing, info structures, ILM rules (audit areas, policies: retention/destruction), legal hold management	AOBJ, ILM*	—	COVERED-48	48	Retention policies per entity
Data protection: purposes, EoP (end-of-purpose) checks per application, blocking, IRF data model/purposes, SRAL, consent	ILM*, IRF*, SRAL*	—	COVERED-48	48	Privacy requests; SRAL partial
Enterprise search: connectors, models, authorization	ESH*	—	COVERED-5	5	—
Key user extensibility: custom fields (business contexts), custom logic (BAdI implementations in-app), custom business objects, custom CDS views, UI adaptation	—	—	COVERED-3	3	Metadata extensions
Application log objects (SLG0), log retention	BALOBJ	—	COVERED-5	5	—
AIF interfaces, error handling recipients, alerts	/AIF/*	—	COVERED-35	35	—
MDG data models, change request types, workflows, validation (BRF+), replication (DRF)	MDG*	—	COVERED-7	7	Governance workflow
Watch list screening / GTS compliance configuration	—	—	GAP	propose 48	Screening gap

13.32 LO-VC — Variant Configuration & AVC (Advanced Variant Configuration)

SAP context: classification-based configuration: characteristics (CT04), classes type 300 (CL02), configurable material KMAT (MM01), configuration profile (CU41), object dependencies (CU01: preconditions, selection conditions, procedures, actions, constraints/constraint nets), variant tables (CU60/CU61), variant functions (CU65), super BOM (CS40) and super routing with dependencies, configuration simulation (CU50), material variants (MM01 with configured values), order BOM (CU51), variant pricing (VA00/VA01 condition types with $SET_PRICING_FACTOR), VC modeling environment (PMEVC), AVC (S/4 Advanced Variant Configuration: new engine, Fiori Manage/Simulate Configuration Models, Manage Classes/Characteristics, product modeling environment), Product Structure Management (PSM/iPPE). Passage: wave 8 (constraint-based configurator on MiniZinc/OR-Tools; attributes & classification), wave 27 (configured order → production), wave 26 (variant pricing via engine), wave 9 (BOM/routing).

13.32.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
CT04 / CT06	Characteristics maintain / where-used (values, intervals, multi-value, required, restrictable, reference to table field)	Configuration attributes	COVERED-8	8	Attributes with value domains; reference attributes — confirm "reference characteristics" (bind to document field e.g. order qty); treat covered
CL02 / CL01 / CL03 / CL6B / CL6C	Class (type 300 variant) create/change/display, class hierarchy	Configuration classes	COVERED-8	8	Classes
MM01 (KMAT, configurable flag)	Configurable material	—	COVERED-8	8	Product master (configurable)
CU41 / CU42 / CU43	Configuration profile create/change/display (class assignment, BOM explosion, interface design, process: planned/production order BOM, order BOM, sales order)	Config profile	COVERED-8	8	Configurator model binding to product + explosion mode
CU01 / CU02 / CU03 / CU04 / CU05	Object dependencies (global) create/change/display/list/where-used (preconditions, selection conditions, procedures, actions, constraints)	Dependencies	COVERED-8	8	Constraint model (MiniZinc/OR-Tools): constraints, selection rules, defaults; procedure-style value inference — confirm procedural "procedures" semantics (set defaults, $SELF.x = ...) map to constraint model; treat covered
CU21 / CU22 / CU23	Dependency nets (constraint nets) create/change/display	Constraint nets	COVERED-8	8	—
CU60 / CU61 / CU62 / CU65 / CU66 / CU67	Variant tables: structure / content / display; variant functions (ABAP)	Lookup tables / functions	PARTIAL-8	8	Variant tables (decision tables) map to constraint tables — covered; *variant functions* (ABAP-coded) → SUPERSEDED by expression functions; table maintenance UI not named — partial
CU50	Configuration simulation (for material with profile)	Test configuration	COVERED-8	8	Configurator simulation/test (fixture set proof)
CU51 / CU52 / CU55 / CU56	Order BOM maintain (sales order BOM — result-oriented/knowledge-based) / display / instantiation	Order BOM (ETO adjustments)	PARTIAL-21	21	Order-specific BOM edits after configuration (ETO) not named — partial; propose wave 21/27 "order BOM"
CU70 / CU71 / CU72	Configurable material: variant matching / material variant matching	Material variants	COVERED-8	8	Variants: pre-defined variants (dimension combinations) and configured variants — matching an entered configuration to an existing variant (type matching) — confirm; treat covered
MM01 material variants (configured values in MRP 3 / Basic data)	Material variant (stockable configured product)	—	COVERED-8	8	Released variant per configuration
CS40 / CS41 / CS42	Super BOM allocation to configurable material / dependencies on BOM items (CS02 → dependencies)	Super BOM	COVERED-8	8	Configurator BOM selection rules
CA01 super routing with dependencies on operations	Super routing	—	COVERED-9	9	Operation selection rules — confirm; treat covered
CU80 / CUMODEL / PMEVC	Product modeling environment for VC (model tree, dependencies, variant tables)	Modeling IDE	COVERED-8	8	Configurator model editor
VA01 → configuration (CUCFG screen, characteristic value assignment, consistency check)	Configure in sales order / quote	—	COVERED-27	27	Configured products on quotes/orders; portal configurator (wave 30) — confirm customer-portal configurator; treat PARTIAL via row below
Customer-facing configurator (web, Fiori / SAP CPQ)	Guided selling	—	PARTIAL-30	30	Customer portal order entry named; guided configurator UI for portal/e-commerce not explicitly named — partial
VK11 VA00 / VA01 (variant conditions, $SET_PRICING_FACTOR)	Variant pricing (surcharges by characteristic values)	—	COVERED-26	26	Pricing engine with attribute-based conditions — confirm condition on configuration attribute; treat covered
CU44 / CU45 (configuration overview, config display in documents)	Display configuration (orders, deliveries, production orders)	—	COVERED-27	27	Configuration on documents
CU7 / CUCPT (configuration copy/ templates)	Configuration templates	—	PARTIAL-8	8	Saved configurations/templates for reuse — partial
AVC: Fiori Simulate Configuration Models / Manage Configuration Models / Manage Characteristics / Manage Classes / Configuration Model Overview / Object Dependencies (AVC syntax: constraints only, no procedures), AVC engine (variant configuration engine with HANA), "Classic vs Advanced"	S/4 AVC	—	COVERED-8	8	Constraint-only engine matches AVC direction
PSM / iPPE (Product Structure Management: product variant structures, CMP nodes, PVS)	Product structure for complex configurable products (automotive)	—	PARTIAL-9	9	Multi-level configurable structures covered by super BOM; iPPE-style product variant structures with alternatives/variants at node level — partial; propose wave 9 v1.x
Configuration in production (planned/production order configuration, CO02 → configuration), configured BOM explosion at order	—	—	COVERED-21	21	Configured order explosion (phase 4 exit: configured product manufactured)
Configuration in purchasing (configurable PO items for configurable vendor products), configuration in service (IBase configuration)	—	—	PARTIAL-17	17	Configured PO items not named — partial
VC with batch/serial effectivity, parameter effectivity (ECM)	Effectivity	—	PARTIAL-9	9	Noted in PP (serial/parameter effectivity partial)
Fiori: Configuration Reporting / Where-used of characteristics in configurations	—	—	COVERED-8	8	—

13.32.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
CU05 (where-used dependencies) / CT06 (char where-used) / CL6C	Where-used lists	COVERED-8	8	—
CU50 (simulation trace / explanation)	Configuration explanation (why value set/excluded)	COVERED-8	8	Explain configuration (constraint explanation) — confirm explanation output; treat covered
CU44 / CUTRACE (configuration trace)	Trace dependencies	COVERED-8	8	—
Configured documents lists (VA05 with config, COOIS with config)	—	COVERED-27	27	—
AVC: Configuration Model Overview / usage analytics (most used values)	—	COVERED-46	46	—

13.32.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
CU mass release of dependencies / knowledge base (CU31 / CU32 / CU33 / CU34 / CU36 runtime versions for SCE/IPC)	Knowledge-base objects & runtime versions (for external configurators, IPC)	SUPERSEDED-8	8	Single configurator service with versioned models
Model versioning / release (AVC model release)	Model release	COVERED-8	8	Versioned configurator models with ECO (wave 9)
Re-explosion of configured orders after model change (OCM)	—	PARTIAL-21	21	OCM partial (noted)
Material variant generation (mass create variants)	—	COVERED-8	8	Variant explosion (1,000 variants < 5 s)

13.32.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Class types (300 variant, 200 material, 001…), class status, characteristics groups, organizational areas	TCLA, TCLAT, TCLO	—	COVERED-8	8	—
Configuration profile settings: BOM explosion (none/single/multi-level), configuration process (planned/production order, sales order, order BOM), result-oriented vs knowledge-based order BOM, interface design (UI groups)	CUCO	—	COVERED-8	8	Explosion mode per model; UI groups → configurator form layout
Dependency editor settings, variant table DB-tables, variant functions (function modules)	—	—	SUPERSEDED-8	8	Expression language
Pricing: variant condition types (VA00), $SET_PRICING_FACTOR, pricing procedure steps	T685	—	COVERED-26	26	—
Material variant settings (MRP 3: configured material, variant/configuration type matching activation)	—	—	COVERED-8	8	—
AVC activation (per configuration profile: Advanced), AVC engine settings, Fiori apps	—	—	SUPERSEDED-8	8	—
Configurable BOM/routing settings (item/operation dependencies allowed), ECM for VC objects	—	—	COVERED-9	9	—
Order BOM settings (CU51 processing)	—	—	PARTIAL-21	21	—
Integration with CPQ / IPC / SCE knowledge-base runtime	—	—	SUPERSEDED-8	8	Configurator API for portals

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13.33 Batch Management & Serial Numbers

SAP context: batch master (MSC1N/MSC2N/MSC3N/MSC4N; batch level plant/material/client OMCT; batch status management OMCS?; batch classification class type 022/023 with characteristics e.g. expiry, potency, grade), batch determination (search strategies by application: MM MBC1, SD VCH1, PP COB1, WM LS?; condition technique with selection classes and sort rules CU70?), batch where-used (MB56, BMBC batch information cockpit), batch derivation (DVMO monitor, DVSP? setup, sender/receiver), shelf life (SLED, min remaining shelf life, OMJ5 checks), batch-specific units of measure (proportion/product units), electronic batch record, batch input/output in all movements, Batch Management in EWM/QM (quality inspection per batch, batch status), serial numbers (serial number profiles OIS2, serial number master IQ01-IQ09 = equipment, serial procedures MMSL/SDRE/SDLS/PPAU/PPRL/QMSL/PPSF/MMSL, serial in deliveries/orders/inventory, Fiori Manage Serial Numbers?), S/4 "Batch Management: Manage Batches, Batch Genealogy? Fiori Batch Information Cockpit". Passage: wave 8 (batch/serial tracking dimensions, profiles), wave 16 (batch/serial on movements, FEFO), wave 18 (serial capture at receipt, duplicate block), wave 24 (batch attributes, shelf life, disposition, potency, catch weight), wave 28 (batch/serial on shipments), wave 33 (QM per batch), wave 45 (serial → installed base).

13.33.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
MSC1N / MSC2N / MSC3N / MSC4N	Batch create/change/display/changes (basic data: production/expiry dates, status, vendor batch, classification values)	Batch master	COVERED-24	24	Batch entity with attributes, shelf life, status; wave 8 tracking dimension
MSC5N / MSC (mass change batches)	Mass batch change	—	COVERED-5	5	—
MCH1 / MCHA / MCHB (tables)	Batch tables (client/plant/stock level)	Data	COVERED-16	16	Batch on-hand per warehouse/location
MBC1 / MBC2 / MBC3	Batch search strategy (MM) create/change/display	Batch determination rules (MM)	COVERED-16	16	Picking policies FEFO/FIFO + attribute-based selection — confirm attribute-based batch selection (e.g. grade ≥ X, customer-specific restrictions); treat covered
VCH1 / VCH2 / VCH3	Batch search strategy (SD)	—	COVERED-28	28	Customer-specific batch rules (e.g. min remaining shelf life per customer) — confirm; treat covered
COB1 / COB2 / COB3	Batch search strategy (PP/PI)	—	COVERED-21	21	Batch determination in production issues

T-code	Name	Purpose	Passage status	Wave	Notes
LS (WM batch strategy) / /SCWM/ batch determination	—	—	COVERED-31	31	—
MB56	Batch where-used list (top-down / bottom-up)	Genealogy	COVERED-24	24	Batch traceability; batch where-used (recall) — confirm "batch genealogy" explicitly for recall reporting; treat covered
BMBC	Batch information cockpit (search by attributes, stock, where-used, documents, QM results)	Batch cockpit	COVERED-24	24	Batch workspace
DVMO / DVSP / DVMAN	Batch derivation: monitor / set up derivation (sender/receiver, rules) / manual derivation	Derivation	PARTIAL-24	24	Batch derivation (inherit attribute values from component batches) partial (noted in PP-PI)
BMSM / batch status change (restricted/unrestricted) via MSC2N / QA11 UD	Batch status management	—	COVERED-24	24	Batch disposition (QM UD, manual)
Batch split in deliveries / orders; batch entry in confirmations	—	—	COVERED-28	28	—
Shelf life: SLED/BBD, remaining shelf life checks (OMJ5), batch expiry re-inspection (QA07), expired batch stock reports (MB5M)	Shelf life	—	COVERED-24	24	Best-before/expiry, FEFO
Batch-specific UoM (CUNI with batch-specific conversion, potency conversion, active-ingredient management)	Proportion/product units	—	COVERED-24	24	Potency-based consumption & catch weight dual UoM
Batch classification (CL20N class 023, characteristics CT04: e.g. expiry date, potency, country of origin, vendor batch)	Batch attributes	—	COVERED-24	24	Batch attributes
Batch valuation (split valuation per batch, OMWC valuation type = batch)	Batch-level valuation	—	PARTIAL-20	20	Split valuation gap
Batch numbering (internal/external, OMAD number ranges, batch number templates, customer-specific numbering via exit)	Batch numbering	—	COVERED-5	5	Number sequences (per item/site templates) — confirm template-based batch numbers (e.g. YYMMDD-xxx); treat covered
Documentary batches (S/4 "documentary batch" for non-batch-managed materials)	Documentary batches	—	PARTIAL-24	24	Recording batch numbers for info only (no stock segregation) not named — partial
Batch in EWM (batch-managed stock in bins), QIE	—	—	COVERED-31	31	—
OIS2 — Serial number profiles (customizing; see below)	—	—	COVERED-8	8	—
IQ01 / IQ02 / IQ03 / IQ04 / IQ08 / IQ09	Serial number master (material serial = equipment category) create/change/display/list/display list	Serial master	COVERED-8	8	Serial entity (tracking dimension) with status and history; equipment link (wave 44/45)
Serial numbers in: GR/GI (MIGO), deliveries (VL02N), sales orders, production orders (CO02 header/components), inspection lots, PM orders, physical inventory	Serial tracking	—	COVERED-16	16	Serial on all movements; duplicate block (wave 18 proof)
Serial number history (IQ03 → history: stock, location, customer)	Serial genealogy	—	COVERED-45	45	Installed base serial tracking from shipments; serial history — confirm full movement history per serial; treat covered
Fiori: Manage Batches / Batch Information Cockpit / Display Batch Genealogy / Manage Serial Numbers	—	—	COVERED-24	24	—
Serial numbers at component level in finished-goods serial (as-built genealogy)	As-built records	—	PARTIAL-21	21	Serial-to-serial component genealogy (as-built) not explicitly named — partial; propose wave 21 RAF capture

13.33.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
MB56 / BMBC	Batch where-used / cockpit	COVERED-24	24	—
MB5M	Expiration date list	COVERED-24	24	—
MMBE / MB52 (batch level)	Stock by batch	COVERED-16	16	—
MB5C	Batch pick-up list	COVERED-24	24	—
CL30N / CL6BN (find objects/batches by characteristic values)	Batch search by attributes	COVERED-24	24	—
IQ08 / IQ09	Serial number lists	COVERED-8	8	—
QA33 (per batch) / QGA3	Batch quality results	COVERED-33	33	—
DVMO (derivation monitor)	—	PARTIAL-24	24	—
Fiori: Batch Genealogy / Expiring Batches / Serial Number History	—	COVERED-24	24	—

13.33.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
QA07 (recurring inspection / expiry monitoring, batch status change)	Expiry handling	COVERED-33	33	—
MB5M-based expiry alerts	—	COVERED-24	24	—
Batch archiving (SARA MM_SPSTOCK / batch deletion flag MSC2N, MM71)	—	COVERED-48	48	—
Batch derivation runs (DVMO)	—	PARTIAL-24	24	—
Serial number archiving / equipment	—	COVERED-48	48	—
Batch level conversion (plant → material level, RVBLEVEL)	—	SUPERSEDED-8	8	Batch uniqueness scope is a product attribute

13.33.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OMCT — Batch level (plant / material / client)	TCUCH	Uniqueness scope	COVERED-8	8	Batch uniqueness scope per product
OMCS — Batch status management activation (restricted/unrestricted), initial status per material type/plant (OMAB)	TCUCH, T134, T001W	—	COVERED-24	24	Batch status
OMAD / OMAE — Internal batch number assignment, number ranges, automatic creation at GR/GI	TCUCH, NRIV	—	COVERED-5	5	—
OMCV / OMCZ — Batch classification activation per movement type/transaction, class type 023 settings, characteristics for batch class (standard characteristics LOBM_*)	T156, TCLA	—	COVERED-24	24	—
OMJ5 — Shelf life expiration check per plant/movement type, SLED calculation (production date + shelf life)	T159L	—	COVERED-24	24	—
Batch determination: condition tables, access sequences, strategy types (per application MM/SD/PP/WM), search procedures, assignment to movement types/order types/sales doc types, selection classes, sort rules (CU70) , quantity proposal, batch split settings	T6 , TLSP	—	COVERED-16	16	Picking policies & batch selection rules
Batch derivation: activation, sender/receiver fields, derivation events, rules	DVMO*	—	PARTIAL-24	24	—
Batch-specific units of measure: activation, proportion/product units, characteristics (LOBM_*)	T006, TCUBSUOM	—	COVERED-24	24	—
Batch valuation (valuation type = batch) activation	T001K, T149	—	PARTIAL-20	20	—
Documentary batch activation (S/4)	—	—	PARTIAL-24	24	—
OIS2 — Serial number profiles (serialization level, serializing procedures per business transaction: MMSL goods movement, SDRE returns, SDLS deliveries, PPAU production order header, PPRL production release, PPSF serial in production, QMSL inspection, HUSL HU; equipment requirement, stock check)	T377, T377P, T377X	Optional/mandatory serial per transaction; stock validation	COVERED-8	8	Serial tracking profiles (track at receipt/issue/shipment/production)
OIS / OIEQ — Serial numbers: equipment category for serial numbers, number assignment (serial = equipment), equipment creation automatic	T377, T370T	—	COVERED-8	8	—
Serial number status management, serial number history activation	—	—	COVERED-8	8	—
EWM/WM batch & serial settings	—	—	COVERED-31	31	—

13.34 CA-CL — Classification System

SAP context: characteristics (CT04: formats CHAR/NUM/DATE/TIME/CURR, values, intervals, multi-value, required, additional values, reference to table fields, documents, long texts), classes (CL01/CL02/CL03: class types 001 material, 002 equipment, 003 functional location, 017 document, 022 batch (plant), 023 batch (material), 032 release strategy, 300 variant, 010 vendor, 011 customer, 012 BP…), class hierarchies (CL24N? CL6?), object assignment (CL20N single object/multiple classes, CL24N class/multiple objects, CL22N?), find objects (CL30N/CL6BN), where-used (CL6AN?), mass change (CLMM), class type customizing (O1CL), characteristic groups, organizational areas, status, keywords, classification data transfer (BAPI_OBJCL_*), standard characteristics (LOBM_, etc.). Passage: wave 8 ("Attributes & classification (SAP CL-style classes) and category hierarchies") used by products, batches, equipment, documents; metadata kernel typed fields (wave 3).

13.34.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
CT04 / CT06 / CT10 / CT12	Characteristics maintain / where-used / list / dependencies	Attribute definitions (typed, value sets, intervals, multi-value, units, templates)	COVERED-8	8	Attribute definitions with typed value domains and UoM
CL01 / CL02 / CL03 / CL04	Class create/change/display/delete (class type, characteristics, keywords, standards, documents, status, organizational areas, hierarchy)	Classes	COVERED-8	8	Attribute groups/classes per entity type
CL24N / CL20N / CL22N / CL25	Assign objects to class / assign class(es) to object / display assignments / class hierarchy maintenance	Object classification	COVERED-8	8	Attribute assignment to products/batches/equipment/documents/parties — confirm classification applies to parties (010/011) and documents (017); treat covered
CL6AN / CL6BN / CL30N / CL31	Where-used (class), find objects in classes (search by characteristic values), class list	Search	COVERED-8	8	Attribute search (global search + filters)
CLMM	Mass change classification data	—	COVERED-5	5	—
CL2B / CLHP	Class hierarchy / hierarchy display	—	COVERED-8	8	Category hierarchies
CLGT / CLWA / CL6 (class statistics, characteristics overview)	—	—	COVERED-8	8	—
CTBW (characteristic ↔ BW extraction)	—	—	SUPERSEDED-46	46	Attributes stream to lakehouse
CL02 class type 032 (release strategy)	Classification used for release strategies	—	SUPERSEDED-17	17	Policy-driven approvals
CL02 class type 300 (variant) — see §13.32	—	—	COVERED-8	8	—
CL02 class types 022/023 (batch) — see §13.33	—	—	COVERED-24	24	—
CL02 class type 017 (document) — see §13.35	—	—	COVERED-8	8	—
CL02 class types 002/003 (equipment / functional location)	Technical object classification	—	COVERED-44	44	—

T-code	Name	Purpose	Passage status	Wave	Notes
Engineering/technical attributes with UoM & tolerances (e.g. for parametric search of parts)	Parametric search	—	COVERED-8	8	Attribute search
Fiori: Manage Characteristics / Manage Classes / Classify Products	—	—	COVERED-8	8	—

13.34.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
CL30N / CL6BN	Find objects in class (by values)	COVERED-8	8	—
CL6AN / CT06	Where-used	COVERED-8	8	—
CL6C / CLGT	Class/characteristic lists, statistics	COVERED-8	8	—
Fiori: Classification analytics (products by attribute)	—	COVERED-46	46	—

13.34.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
CLMM (mass update)	—	COVERED-5	5	—
Classification reorganization / redundancy cleanup (RCCLUKA / CL)	—	SUPERSEDED-8	8	Metadata registry
Classification data transfer (BAPI_OBJCL_CREATE, LSMW object CLASSIFICATION)	Migration	COVERED-10	10	Migration DSL for attributes

13.34.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
O1CL — Class types per object (table, screen, key fields, multiple classification, standard class, hierarchy allowed, variant class)	TCLA, TCLAO, TCLAT	—	SUPERSEDED-3	3	Attribute groups attachable per entity in metadata
Characteristics groups (CT04), characteristic status, organizational areas, keywords	TCMG, TCLST	—	COVERED-8	8	—
Class status (released/locked), class groups	TCLST	—	COVERED-8	8	—
Characteristic templates / formats, default values, currency/UoM settings	—	—	COVERED-8	8	—
Object key settings per class type (e.g. material+plant for 022)	TCLAO	—	COVERED-8	8	—
Classification in BW/extraction settings (CTBW)	—	—	SUPERSEDED-46	46	—
Attribute inheritance in class hierarchies	—	—	COVERED-8	8	Category hierarchies with inherited attributes — confirm inheritance; treat covered

13.35 DMS — Document Management System (CA-DMS), Easy DMS, ECTR, CAD integration

SAP context: document info records (CV01N/CV02N/CV03N/CV04N; document type, part, version, status network, originals stored in KPro/content server, object links to material/BOM/equipment/orders/customers/WBS, classification 017, document hierarchies/structures CV11/CV12/CV13?, document BOM), document distribution (CVI?), status management with release workflow, change management via ECM, Easy DMS (Windows Explorer integration), SAP Engineering Control Center (ECTR) for CAD (AutoCAD/SolidWorks/NX) with BOM sync, conversion (PDF/viewer), ArchiveLink (OAAD) for scanned docs, Fiori: Manage Documents, SAP Document Management service (BTP), Product Lifecycle Management (PLM: engineering record, PLM web UI, Enterprise Product Development EPD). Passage: wave 5 (attachments on any entity), wave 8 (product images/documents), wave 9 (ECO workflow, "PLM integration" named), wave 48 (retention) — a true DMS (versioned, status-controlled, CAD-linked documents) is only PARTIAL.

13.35.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
CV01N / CV02N / CV03N	Document info record create/change/display (document type, number, part, version; status; originals; object links; classification; texts)	Document master	PARTIAL-8	8	Attachments (wave 5) and product documents (wave 8) exist; a *document info record* with type, versioning, part/version numbering, status network and release is not named — partial; propose wave 9 "engineering documents" (drawings/specs under ECO control)
CV04N	Find document (by attributes, classification, object links, full text)	Search	PARTIAL-8	8	Global search covers attachments metadata; attribute/classification search on documents partial
CV11 / CV12 / CV13 / CV15	Document structure (document BOM) create/change/display / where-used	Document hierarchies	GAP	propose 9	Document structures (assemblies of documents, e.g. drawing sets) not named — propose wave 9
CV01N → originals (check-in/check-out, KPro storage categories, viewer, conversion)	Original files (CAD, PDF, Office) with check-out locking	Files	PARTIAL-5	5	Attachments stored; check-out/check-in locking, multiple originals per version, viewer/conversion (2D/3D) not named — partial
CV01N → object links (MARA, STKO, EQUI, IFLOT, AUFK, KNA1, LFA1, PRPS, VBAK, EKPO…)	Link documents to business objects	—	COVERED-5	5	Attachments on any entity; cross-entity links — confirm one document linked to many entities; treat covered
CV01N → status network (in work → released → obsolete), status with release/approval workflow, digital signature	Document status & release	—	PARTIAL-9	9	ECO workflow covers BOM/route release; document release state machine with signature — partial
CV01N → version / part / change number (ECM)	Document versioning under ECM	—	PARTIAL-9	9	ECO-controlled document versions partial
CV01N → classification (class type 017)	Document attributes	—	COVERED-8	8	Attribute classes on documents
CV01N → document distribution (recipients, distribution orders CVI1 / CVI2 / CVI8)	Distribution of documents (e.g. to suppliers)	—	PARTIAL-20	20	Supplier portal document exchange named; controlled distribution with acknowledgement partial

T-code	Name	Purpose	Passage status	Wave	Notes
Easy DMS / SAP DMS Connector for Windows	Explorer-integrated document management	—	SUPERSEDED-5	5	Web shell drag-drop + Excel/CSV import; desktop sync not planned
ECTR (Engineering Control Center): CAD integration (SolidWorks, Inventor, NX, CATIA, AutoCAD, ECAD), BOM derivation from CAD, 3D viewer, DMS for CAD	CAD–ERP integration	—	GAP	propose 9	CAD integration (CAD file management, CAD BOM → engineering BOM sync) not named; wave 9 says "PLM integration" only as reference — propose wave 9 (v1.x) PLM/CAD connector (e.g. via a PLM system) rather than native CAD management
OAAD / OAWD / OAOR / OAC0 (ArchiveLink scanned documents, business document navigator)	Scanned document linking (incoming invoices, delivery notes)	—	COVERED-5	5	Attachments; OCR capture (wave 18)
Document BOM items (CS01 item category D)	Documents as BOM components (e.g., drawings, instructions)	—	PARTIAL-9	9	BOM line type "document" not named — partial
PLM: Engineering Record (change management cockpit), Product Structure Management, Visual Enterprise (3D visualization), Recipe Development (PLM-RDP), Specification management (EHS specs), Labeling (GLM)	PLM suite	—	PARTIAL-9	9	ECR/ECO covered; recipe development with formula trials/specs (process industries), specification management, global label management — partial; propose wave 24 (recipe dev) & wave 48 (labeling compliance)
SAP Document Management service (BTP) / Document Information Extraction	Cloud DMS	—	COVERED-5	5	Object storage; OCR (wave 18)
Fiori: Manage Documents / Manage Document Info Records	—	—	PARTIAL-8	8	—

13.35.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
CV04N	Document search	PARTIAL-8	8	—
CV13 / CV15	Document structure display / where-used	GAP	propose 9	—
CVI8 (distribution log)	Distribution status	PARTIAL-20	20	—
Document lists per object (object links)	—	COVERED-5	5	Attachment lists
ECTR reports (CAD BOM comparison)	—	GAP	propose 9	—

13.35.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
CV conversion runs (originals → PDF/neutral formats)	Conversion server runs	PARTIAL-5	5	Preview generation for attachments — partial
Document archiving / retention (SARA CV_DVS / KPro cleanup)	—	COVERED-48	48	—
Document distribution runs (CVI)	—	PARTIAL-20	20	—
ECTR sync jobs	—	GAP	propose 9	—

13.35.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
DC10 — Document types (number assignment, version/part numbering, status network, object links, screen layout, classification class, originals count, versioning)	TDWA, TDWS, TDWO, TDWP	—	PARTIAL-8	8	Attachment types exist; full document type model partial
DC20 — Status (document status definitions, release flag, content version)	TDWS	—	PARTIAL-9	9	—
DC30 — Workstation applications (file types, viewer/editor, conversion)	TDWP	—	PARTIAL-5	5	Preview/viewer mapping partial
OACT / OAC0 / KPro storage categories, content repositories, cache servers	SDOKSTCA, TOAAR	—	COVERED-5	5	Object storage buckets
Object link customizing (DC screen numbers, object types allowed per document type)	TDWO	—	COVERED-5	5	—
Document distribution customizing (recipient lists, distribution types)	—	—	PARTIAL-20	20	—
ECTR configuration (CAD applications, templates, BOM sync rules)	—	—	GAP	propose 9	—
Digital signature for documents (signature strategies)	—	—	GAP	propose 24	E-signature framework gap (noted in PP-PI)
Full-text search activation (TREX/HANA) for originals	—	—	PARTIAL-5	5	Attachment content indexing not named — partial

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13.36 Analytics — BW/4HANA, CDS views & embedded analytics, SAP Analytics Cloud, LIS, Report Painter, SAP Query

SAP context: BW (RSA1 workbench, DataSources RSA5/RSA6/RSA3, process chains RSPC, InfoProviders, transformations/DTPs, queries BEx RSRT/BEx Query Designer/Analyzer, Analysis for Office, BW/4HANA, Open ODS, HANA calculation views, BW Bridge), embedded analytics (CDS views with @Analytics annotations, analytical queries, Fiori analytical list pages/overview pages, KPI modeler & Smart Business, Query Browser, View Browser, Custom Analytical Queries, Multidimensional reports), SAP Analytics Cloud (stories, planning, predictive, Digital Boardroom, live connection to S/4), Datasphere, SAP Query (SQ01/SQ02/SQ03, SQVI QuickViewer, InfoSets), Report Painter/Writer (GRR1/GR51/GR55), LIS (SIS/PURCHIS/INVCO/PP-IS/QMIS/PMIS; MC* transactions), Drill-down reporting (FGI0, KE30, FM/PS), Fiori Overview pages. Passage: wave 46 (CDC → Iceberg lakehouse, Trino/ClickHouse, dbt, Cube semantic layer with governed metrics and RLS, embedded dashboards, self-service explorer, scheduled reports, export to Excel/Power BI/Tableau, operational document designer, OpenMetadata catalog/lineage), wave 15 (financial report designer), wave 5 (saved views, export).

13.36.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
RSA1	BW Data Warehousing Workbench (InfoObjects, DataSources, InfoProviders, transformations, DTPs, process chains)	Warehouse modeling	SUPERSEDED-46	46	CDC → Iceberg lakehouse; dbt models replace transformations/InfoProviders
RSA5 / RSA6 / RSA3 / RSA7 / LBWE / LBWQ / LBWG / SBIW	Install/maintain DataSources, extractor checker, delta queue, LO cockpit (logistics extraction), setup tables	Extraction	SUPERSEDED-46	46	Debezium CDC from every domain; no extractors/setup tables
RSPC / RSPCM / RSPC1	Process chains (maintenance / monitor)	ETL orchestration	SUPERSEDED-46	46	dbt + job framework (wave 4) with lineage
RSRT / RSRT2 / RRMX / BEx Query Designer / BEx Analyzer / Analysis for Office (AO) / BEx Web	Query monitor / BEx reporting / Excel-based analysis	Query & reporting	SUPERSEDED-46	46	Semantic layer (Cube) + self-service explorer; Excel export & Power BI/Tableau connectivity
RSD1 / RSD5 / RSDS / RSDCUBE / RSDODS / RSDMPRO	InfoObjects, DataSources, InfoCubes, DSOs, MultiProviders, CompositeProviders (BW/4: aDSO, CompositeProvider, Open ODS)	Modeling objects	SUPERSEDED-46	46	dbt models + semantic cubes
RSECADMIN	BW analysis authorizations	RLS in BW	SUPERSEDED-46	46	RLS mirrored from OPA
RSMO / RSRV / RSDDV / RSKC / RSBATCH	Monitor, analysis & repair, aggregates/BWA/HANA, special chars, batch manager	BW admin	SUPERSEDED-46	46	Lakehouse ops, OpenMetadata
HANA calculation views (HANA Studio / Web IDE), HANA Live, Smart Data Integration	HANA native modeling	—	SUPERSEDED-46	46	Trino/ClickHouse views; dbt
CDS views (ADT: @Analytics.query, @Analytics.dataCategory CUBE/DIMENSION, @ObjectModel), Custom CDS Views (Fiori), View Browser, Query Browser	Embedded analytics virtual data model	—	COVERED-46	46	Semantic layer models over lakehouse (and OLTP for operational views); custom views via self-service explorer with governed metrics
Custom Analytical Queries (Fiori), Manage KPIs and Reports (KPI modeler, Smart Business tiles), Analytical List Page, Overview Pages	Embedded analytics apps	—	COVERED-46	46	Embedded dashboards in every workspace; KPI tiles (OTIF, OEE, DSO…)
Multidimensional reports (Design Studio/Lumira/WebDynpro grid), Fiori "Multidimensional" apps	Pivot analysis	—	COVERED-46	46	Self-service explorer pivot
SAP Analytics Cloud (stories, data models, live/import connections, planning models, predictive scenarios (Smart Predict), Digital Boardroom, Analytics Designer, SAC for Group Reporting)	Cloud BI & planning	—	COVERED-46	46	Embedded dashboards + explorer; planning in wave 38; predictive via wave 22 forecasting & wave 47 agents
SAP Datasphere (spaces, data builder, semantic layer, data marketplace), BW Bridge	Cloud data warehouse	—	COVERED-46	46	Lakehouse + semantic layer + catalog
SQ01 / SQ02 / SQ03 / SQVI	SAP Query (queries, InfoSets, user groups), QuickViewer	Ad hoc query	SUPERSEDED-46	46	Self-service explorer with RLS
GRR1 / GRR2 / GRR3 / GR51 / GR52 / GR53 / GR55 / GR32 / GR37 / GRCT / GRS	Report Painter / Report Writer: reports, libraries, sets, variables, report groups	Financial/CO reporting	SUPERSEDED-15	15	Row/column report designer (Management-Reporter model)
GS01 / GS02 / GS03 / GS11	Sets (basic, single, multi) for Report Writer / allocation	Sets	SUPERSEDED-6	6	Hierarchies & dimension groups
FGI0 / FGI1 / KE30 / CJE0 / FMEN / S_ALR drill-down reports	Drill-down reporting tools	—	SUPERSEDED-46	46	Explorer drill-down
MC* (LIS: MCSI, MC21 info structures, MC18 field catalogs, MC24 update rules, MC3 planning, OMO1 update control, standard analyses MC.x, MCE*, MCT*, MCP*, MCI*, MCX*, MCL*)	Logistics Information System	—	SUPERSEDED-46	46	Lakehouse + semantic metrics; see per-module rows
S_ALR_* report trees (SAP Easy Access: Information Systems) / SARP / SERP (report trees) / SE43 area menus	Report navigation	—	SUPERSEDED-5	5	Workspace navigation + saved reports
Report variants (SE38 variants, SU3 defaults), ALV layouts (/N ALV grid, layout management), ALV export to Excel	Report personalization	—	COVERED-5	5	Saved views, grid personalization, Excel/CSV export
Scheduled report distribution (SM36 + SP01 + SOST; BW broadcasting RSRD_ADMIN; SAC schedule publications)	Report broadcasting	—	COVERED-46	46	Scheduled reports (PDF/XLSX/email)
SAP BusinessObjects BI (Web Intelligence, Crystal Reports, Lumira, Dashboards)	Enterprise BI	—	SUPERSEDED-46	46	Embedded BI + export to Power BI/Tableau
Crystal Reports / Adobe forms for operational documents (invoice, PO layouts)	Operational documents	—	COVERED-46	46	Operational report/document designer (replaces SSRS/Adobe layouts)
Data catalog / metadata (SAP Information Steward, Datasphere catalog, BW metadata repository RSOR)	Catalog & lineage	—	COVERED-46	46	OpenMetadata catalog & lineage
Data quality dashboards (Information Steward)	DQ monitoring	—	PARTIAL-46	46	Metric reconciliation checks exist; generic data-quality rule dashboards (completeness, validity per entity) partial — propose wave 46 DQ rules on catalog
SAP Predictive Analytics / Predictive Analytics Integrator (PAI) / ISLM (Intelligent Scenario Lifecycle Management) / HANA PAL/APL	Embedded ML scenarios	—	SUPERSEDED-47	47	Agent runtime + Python forecasting/ML services; eval harness (wave 47)

13.36.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
BEx queries / AO workbooks / SAC stories (all standard BW business content: FI, CO, SD, MM, PP, QM, PM, HR InfoCubes/queries)	Standard BI content	SUPERSEDED-46	46	Governed metric library (revenue, margin, DSO, DPO, OTIF, OEE, inventory turns, etc.) + standard dashboards per workspace
Fiori Overview Pages (Procurement, Sales, Finance, Inventory, Production, Quality, Maintenance, HR)	Role-based KPI pages	COVERED-46	46	Embedded dashboards in every workspace

T-code	Name	Passage status	Wave	Notes
Fiori analytical apps (hundreds: e.g. Sales Volume, Overdue Receivables, Inventory Turnover, Supplier Evaluation, Production Order Cycle Time)	—	COVERED-46	46	Covered per-module rows
Query Browser / View Browser	Discover CDS views/queries	COVERED-46	46	Catalog
S_ALR_* standard reports (all modules)	Classic reports	COVERED-15	15	Report designer + explorer; see per-module tables
SQVI / SQ01 outputs	Ad hoc lists	SUPERSEDED-46	46	—
LIS standard analyses (MC.x etc.)	—	SUPERSEDED-46	46	—
Usage/monitoring: RSRT statistics, BW statistics, SAC usage	BI usage analytics	COVERED-46	46	—

13.36.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
RSPC process chains (daily loads, delta, rollup, compression, aggregates)	ETL schedule	SUPERSEDED-46	46	Streaming CDC (p95 lag < 60 s) + dbt scheduled models
LBWE / LBWQ / setup table fills (OLI*BW: OLI7BW, OLI3BW…)	LO extraction setup	SUPERSEDED-46	46	—
OMO1 / LIS update runs, OLIX (LIS rebuild)	LIS statistics	SUPERSEDED-46	46	—
RSRD_ADMIN / broadcasting / SAC publication schedules	Report distribution	COVERED-46	46	Scheduled reports
Data archiving of BW (near-line storage NLS)	—	SUPERSEDED-46	46	Iceberg retention/partitioning
Metric reconciliation (revenue in semantic layer = GL daily)	—	COVERED-46	46	Automated check (proof)
ISLM model retraining schedules	—	SUPERSEDED-47	47	Agent/ML eval & retraining pipelines

13.36.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
BW source system connection (RSA1: source systems, ODP, SLT/RFC), BW settings (RSCUSTV*, RSADMIN)	RSADMIN, RSBASIDOC	—	SUPERSEDED-46	46	Debezium connectors per domain DB
LIS customizing (MC21 info structures, MC24 update rules, OMO1 update control per application, period units)	TMC*	—	SUPERSEDED-46	46	—
Report Painter libraries, standard layouts, report groups; Report Writer sets	T803*, T801*	—	SUPERSEDED-15	15	—
SAP Query: InfoSets (SQ02), user groups (SQ03), query areas (global/standard)	AQ*	—	SUPERSEDED-46	46	—
Embedded analytics: analytical authorizations (CDS DCL), KPI/Smart Business configuration (KPI catalog, evaluations, tiles), analytical app configuration	—	—	COVERED-46	46	RLS from OPA; KPI tiles
SAC: connections, models, teams/roles, data access control	—	—	COVERED-46	46	—
Datasphere spaces, connections, semantic layer	—	—	COVERED-46	46	—
Data catalog/lineage configuration	—	—	COVERED-46	46	OpenMetadata
Report scheduling & distribution (broadcast settings, email templates)	—	—	COVERED-46	46	—
ALV layout administration (global vs user layouts, default layouts)	LTDX	—	COVERED-5	5	Saved views (shared/personal)
Export governance (Excel export restrictions, download audit)	—	—	PARTIAL-48	48	Export of PII/financial data audit & restriction policies not named — partial; propose wave 48

13.37 AI — SAP Joule, SAP Build Process Automation, Business AI (SUPERSEDED by wave 47)

SAP context: Joule (generative AI copilot in S/4HANA Cloud, SuccessFactors, Ariba: navigational, transactional, informational, Joule agents, Joule Studio), SAP Build Process Automation (BPA: workflow + RPA bots, decisions/business rules, process visibility), SAP CoPilot (legacy), SAP Business AI scenarios (cash application ML, AP invoice matching proposals, MRP exception triage, predictive delivery, intelligent product proposals, GR/IR account reconciliation ML, situation handling (Situation Framework: situation types, situation pages), Intelligent Scenario Lifecycle Management (ISLM), Document Information Extraction, Business Document Processing, SAP AI Core/AI Launchpad, generative AI hub), SAP Signavio process AI. Passage: wave 47 (MCP server over every API/entity/report with OPA, copilot UI, Temporal-backed agent runtime with approval gates — AP coding, cash application, MRP exception triage, collections outreach, journal anomaly detection — eval harness in CI, prompt/version registry, cost budgets, responsible-AI controls, model-agnostic provider layer).

13.37.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
Joule (navigational/transactional/informational capabilities; natural-language navigation, create documents, Q&A over data/documentation)	Copilot in S/4 Fiori	Copilot	SUPERSEDED-47	47	Copilot UI in the shell: ask-the-data (semantic layer), explain-this-record, draft documents, summarize workflows; MCP server exposes every API/entity/report
Joule Agents (e.g. dispute resolution agent, cash collection agent, maintenance planner agent, sourcing agent, financial accruals agent) / Joule Studio (build custom agents/skills)	Autonomous agents	Agents	SUPERSEDED-47	47	Agent runtime (Temporal) with approval gates: AP invoice coding, cash application suggestions, MRP exception triage, collections outreach, journal anomaly detection; agent builder = MCP tools + workflow designer
SAP Build Process Automation: processes (workflow), automations (RPA bots), decisions (business rules), forms, process visibility dashboards, pre-built content (SAP Store)	Low-code process automation + RPA	—	SUPERSEDED-4	4	Workflow designer (wave 4) + rules (partial, §13.31) + agents (wave 47) for unstructured steps; RPA screen-scraping bots replaced by APIs/MCP tools
SAP CoPilot (legacy digital assistant)	—	—	SUPERSEDED-47	47	—

T-code	Name	Purpose	Passage status	Wave	Notes
Situation Handling (Situation Framework: standard/extended situation types, situation pages, My Situations)	Proactive alerts with context and actions	—	SUPERSEDED-47	47	Wave 5 alerting + wave 22 exception messages + wave 47 agents triage; "situations" = proactive tasks from events
Business AI scenarios: Cash Application (ML matching), Intelligent Invoice Matching / AP, GR/IR reconciliation ML, Predicted delivery delay (sales), Intelligent product proposal, Slow-moving inventory prediction, Demand forecasting ML (IBP/S/4), Maintenance predictive (PAI), Anomaly detection in journals (Financial Statement Insights)	Embedded ML	—	SUPERSEDED-47	47	Agents in suggest-mode (AP coding ≥ 92%, cash application ≥ 90% precision); ML forecasting (wave 22); predictive maintenance hooks (wave 44); anomaly detection on journals
Document Information Extraction / Business Document Processing (invoice OCR, PO extraction)	Document AI	—	COVERED-18	18	OCR/AP-automation adapter (wave 18); receipts OCR (wave 43)
ISLM (Intelligent Scenario Lifecycle Management: scenarios, models, training, activation) / SAP AI Core / AI Launchpad / generative AI hub (model access, prompt management, orchestration, grounding)	ML/AI lifecycle & model hub	—	SUPERSEDED-47	47	Model-agnostic provider layer (Claude default, Bedrock/Azure/open-weights), prompt/version registry, eval harness in CI, cost/token budgets
SAP Signavio Process AI / Process Insights (process mining recommendations)	Process intelligence	—	PARTIAL-46	46	Process mining partial (noted §13.31)
Joule in SuccessFactors / Ariba / Concur (HR Q&A, procurement assistant, expense assistant)	Domain copilots	—	SUPERSEDED-47	47	Copilot enabled in all workspaces
SAP Conversational AI (chatbots, legacy)	Chatbots	—	SUPERSEDED-47	47	Copilot UI
Responsible AI: SAP AI ethics policy, Joule data privacy (no training on customer data), AI governance	AI governance	—	COVERED-47	47	PII redaction, action logging, kill switch, permission inheritance (agent never exceeds user), NIST AI RMF

13.37.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
Joule informational queries (Q&A over business data and documentation)	Ask-the-data	SUPERSEDED-47	47	Ask-the-data over semantic layer
ISLM model monitoring (accuracy, drift) / AI Launchpad metrics	Model monitoring	SUPERSEDED-47	47	Eval dashboards; cost/token budget alerts
BPA process visibility dashboards	Process KPIs	SUPERSEDED-46	46	Workflow dashboards (wave 4/46)
Agent action audit (Joule agent logs)	—	COVERED-47	47	Every agent action audited with inputs, tool calls, outputs, approver
Situation analytics (situations by type, resolution time)	—	SUPERSEDED-47	47	—

13.37.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
ISLM training/retraining jobs, model activation	—	SUPERSEDED-47	47	Eval runs in CI gate deploys; retraining pipelines (Python services)
Cash application ML proposal runs (nightly)	—	SUPERSEDED-47	47	Cash application agent (suggest-mode)
Situation determination batch runs	—	SUPERSEDED-47	47	Event-driven
RPA bot schedules (BPA)	—	SUPERSEDED-4	4	Job framework + APIs
Token/cost budget resets & reports per tenant	—	COVERED-47	47	Cost guard

13.37.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Joule activation per scope item, capability assignment to roles, Joule Studio skill/agent configuration, grounding sources	—	—	SUPERSEDED-47	47	Copilot/agent enablement per workspace & role (OPA); MCP tool catalog
Situation types (standard/extended), triggers, conditions, responsibility management (teams & responsibilities), notification channels	—	—	SUPERSEDED-47	47	Alert rules + responsibility assignment via org model (wave 6)
ISLM scenario settings, AI Core resource groups, model deployment configs, generative AI hub orchestration (prompt templates, content filtering, data masking)	—	—	SUPERSEDED-47	47	Provider layer config, prompt registry, PII redaction, kill switch
BPA: process templates, decision tables (business rules), RPA agent settings, environment/destinations	—	—	SUPERSEDED-4	4	Workflow designer; rules engine (partial)
AI governance policies (usage logging, consent, data residency for AI processing)	—	—	COVERED-47	47	Responsible-AI controls; data residency per tenant (wave 48)
Business AI scenario activation (per company code / app), confidence thresholds, auto-apply vs propose	—	—	COVERED-47	47	Suggest-mode vs auto with approval gates; precision thresholds gating deploy

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13.98 Coverage summary by module

Row counts per section (all four tables). Modules with a high PARTIAL/GAP share are where the roadmap needs decisions.

Section	COVERED	PARTIAL	GAP	OUT-OF-SCOPE	SUPERSEDED	Total
13.1 FI-GL — General Ledger (classic GL, New GL, ledgers, ACDOCA)	123	13	9	0	19	164
13.2 FI-AP — Accounts Payable	91	11	5	0	5	112
13.3 FI-AR — Accounts Receivable	96	7	5	0	4	112
13.4 FI-AA — Asset Accounting	89	5	0	3	1	98
13.5 FI-BL / FSCM-CM / BAM — Bank Accounting, Cash Management & Bank Account Management	48	7	6	0	1	62
13.6 FSCM — Credit, Collections, Dispute Management & Treasury (TRM)	23	3	14	2	3	45
13.7 CO-OM — Overhead Cost Controlling (cost centers, internal orders, activity types, allocations)	99	3	0	0	8	110
13.8 CO-PC — Product Cost Controlling (product cost planning, cost object controlling, material ledger)	67	13	0	0	5	85
13.9 CO-PA — Profitability Analysis (costing-based, account-based) and EC-PCA Profit Center Accounting	27	7	0	0	22	56

Section	COVERED	PARTIAL	GAP	OUT-OF-SCOPE	SUPERSEDED	Total
13.10 EC-CS / SEM-BCS / S/4HANA Group Reporting — Consolidation	29	7	1	0	4	41
13.11 FM / PSM — Funds Management & Public Sector Management (budgeting)	18	7	0	4	5	34
13.12 MM-PUR — Purchasing	95	16	0	1	5	117
13.13 MM-IM — Inventory Management & Physical Inventory	80	13	2	0	9	104
13.14 MM-IV — Logistics Invoice Verification	49	7	1	0	1	58
13.15 MM-SRV — External Services Management (incl. S/4 Lean Services)	2	22	2	1	0	27
13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	78	9	7	0	11	105
13.17 LE-SHP / LE-TRA — Shipping & (classic) Transportation	69	4	3	0	3	79
13.18 TM — Transportation Management (SAP TM / embedded TM in S/4)	31	3	3	1	2	40
13.19 SD — Sales & Distribution (pricing, orders, deliveries, billing, credit)	117	10	2	3	13	145
13.20 SD Rebates / Settlement Management (condition contracts)	22	6	3	1	1	33
13.21 PP — Production Planning (master data, MRP, SOP/demand management, production orders, capacity, KANBAN, REM)	174	14	2	0	6	196
13.22 PP-PI — Production Planning for Process Industries	26	18	5	0	0	49
13.23 PP/DS — Advanced Planning & Detailed Scheduling (APO PP/DS and S/4 embedded PP/DS)	29	4	2	0	8	43
13.24 QM — Quality Management	70	10	5	0	0	85
13.25 PM — Plant Maintenance (Enterprise Asset Management)	58	6	0	1	2	67
13.26 CS / Customer Service / SAP Field Service Management	42	5	2	0	0	49
13.27 PS — Project System	64	12	3	1	3	83
13.28 HCM — PA / OM / PT / PY + SuccessFactors Employee Central	66	16	0	0	7	89
13.29 FI-TV — Travel Management / SAP Concur	25	3	0	2	2	32
13.30 Basis & Security (users, roles, GRC Access Control, change documents, transports, system administration)	51	4	0	1	39	95
13.31 Cross-Application (workflow, IDoc/ALE, PI/PO, DRC e-documents, output management NAST/BRF+, Fiori launchpad, archiving/ILM, GDPR/IRF, attachments, texts, enterprise search)	57	6	2	1	25	91
13.32 LO-VC — Variant Configuration & AVC (Advanced Variant Configuration)	30	9	0	0	4	43
13.33 Batch Management & Serial Numbers	44	9	0	0	1	54
13.34 CA-CL — Classification System	24	0	0	0	5	29
13.35 DMS — Document Management System (CA-DMS), Easy DMS, ECTR, CAD integration	8	18	7	0	1	34
13.36 Analytics — BW/4HANA, CDS views & embedded analytics, SAP Analytics Cloud, LIS, Report Painter, SAP Query	22	2	0	0	28	52
13.37 AI — SAP Joule, SAP Build Process Automation, Business AI (SUPERSEDED by wave 47)	6	1	0	0	21	28
All	**2049**	**310**	**91**	**22**	**274**	**2746**

13.99 Gap register

Every row above whose status is `GAP` or `PARTIAL–<wave>`, in document order, with the wave it is currently mapped to (PARTIAL) or the *proposed* wave (GAP). The "What is missing / proposal" column repeats the row's Notes so this register can be read on its own and pasted into the wave issues. Proposed waves are suggestions for the roadmap owners; a GAP stays a GAP until a wave's `deliverable` in waves.json names it.

13.99.1 Rollup by (current or proposed) wave

Wave	PARTIAL rows mapped	GAP rows proposed	Total
3	3	0	3
5	5	0	5
7	6	0	6
8	9	0	9
9	10	6	16
11	17	0	17
12	3	0	3
13	4	0	4
14	0	5	5
16	17	1	18
17	18	2	20
18	16	1	17
19	5	2	7
20	13	0	13
21	8	0	8
22	13	2	15
23	6	2	8
24	9	6	15
25	11	0	11
26	7	1	8
27	11	1	12
28	1	1	2
29	1	1	2
30	7	5	12

Wave	PARTIAL rows mapped	GAP rows proposed	Total
31	8	3	11
32	1	2	3
33	10	5	15
34	3	7	10
35	1	0	1
36	10	29	39
37	9	0	9
38	9	0	9
39	6	0	6
40	10	0	10
41	8	3	11
42	10	0	10
43	7	0	7
44	6	0	6
45	4	2	6
46	3	0	3
48	5	4	9
All	**310**	**91**	**401**

13.99.2 Register (document order)

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
1	13.1 FI-GL	Master data & documents	F-03 / F-04	Clear G/L account / post with clearing	PARTIAL–11	11	Open-item management specified only for AP/AR (waves 19/30) and GR/IR (wave 18); generic OI-managed G/L accounts (clearing accounts, bank sub-accounts) need the same clearing engine exposed on GL
2	13.1 FI-GL	Master data & documents	FBCJ	Cash journal	GAP	propose 36	No wave mentions a petty-cash/cash-journal document; small but common on plant floors — add to wave 36 (cash & bank) as "cash journal with receipts, business transactions and daily closing"
3	13.1 FI-GL	Master data & documents	FBCJC0 / FBCJC1 / FBCJC2 / FBCJC3	Cash journal setup, number ranges, business transactions, print	GAP	propose 36	Follows FBCJ
4	13.1 FI-GL	Master data & documents	FAGL_ACTIVATE_OP	Activate open-item management on G/L accounts	PARTIAL–11	11	Depends on the GL OI gap above
5	13.1 FI-GL	Master data & documents	FAGL_SWITCH_TO_OPEN_ITEM	Switch to OI management (S/4)	PARTIAL–11	11	Same
6	13.1 FI-GL	Reports & inquiries	S_ALR_87012309	Cash book / cash journal report	GAP	propose 36	Follows FBCJ gap
7	13.1 FI-GL	Periodic processing	F.13 / F13E	Automatic clearing (with/without specification of clearing currency)	PARTIAL–11	11	Automatic clearing specified for GR/IR (wave 18) and AP/AR settlement; generic rule-based GL auto-clearing is part of the GL OI gap
8	13.1 FI-GL	Periodic processing	FAGLF101 / F101	Regroup receivables/payables (sorted list, reclassification by remaining term)	GAP	propose 14	Balance-sheet reclassification (short/long-term, debit-balance vendors, credit-balance customers) is a standard close step not named in wave 14's checklist/allocations — add as a close task type
9	13.1 FI-GL	Periodic processing	F.50 / F.52	G/L interest calculation (balance / account interest)	GAP	propose 30	Interest on GL balances (e.g. intercompany loans) is not named; wave 30 covers AR interest notes only — extend the interest engine to GL/intercompany accounts
10	13.1 FI-GL	Periodic processing	FAGL_CL_REGROUP / FAGL_CL_MIGRATION	Closing: regrouping (S/4)	GAP	propose 14	Same as FAGLF101
11	13.1 FI-GL	Periodic processing	F.97 / F.98 / F.99	Balance confirmations for G/L / AP / AR	GAP	propose 19	Balance confirmation letters (vendor/customer/GL) not named in waves 19 or 30 — add to wave 30 customer statements + wave 19 vendor confirmations
12	13.1 FI-GL	Customizing (SPRO)	OB28 / GGB0 — Validations (FI)	GB90*, T001Q	PARTIAL–11	11	Wave 11 has dimension validation rules; a general user-defined validation rule language (Boolean rules over header/line fields with message classes) is only implied by wave 3 "validation" — scope it explicitly
13	13.1 FI-GL	Customizing (SPRO)	OBBH / GGB1 — Substitutions (FI)	GB91*, T001R	PARTIAL–11	11	Default dimensions on master data exist; rule-based substitution (derive dimension from another field) needs an explicit rule engine on the Post API

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
14	13.1 FI-GL	Customizing (SPRO)	OBBZ / OBBW — Substitution/validation for cost of sales accounting	—	PARTIAL–11	11	Functional area as a dimension with derivation rules (see above)
15	13.1 FI-GL	Customizing (SPRO)	Document splitting: classify G/L accounts, document types, splitting method, zero-balance, constants (FAGL_SPLIT*)	T8G01…T8G40	PARTIAL–11	11	Wave 11 requires every line to carry dimensions, so dimension-complete documents are native; but *inheritance to balance-sheet lines* (AP/AR/tax lines split by the expense lines' profit centers) and zero-balance clearing per dimension are not named — needed for balance sheet by profit center/segment
16	13.1 FI-GL	Customizing (SPRO)	OBXR / OBYR — Special G/L indicators (customers / vendors)	T074	PARTIAL–19	19	Prepayments/down payments covered (waves 19/30); bills of exchange and noted items not named — see FI-AP/AR sections
17	13.1 FI-GL	Customizing (SPRO)	OBB8 / OBB9 — Terms of payment / installment terms	T052, T052R	PARTIAL–7	7	Payment terms covered; installment terms (split one invoice into several due dates) not named — add to wave 7 terms model
18	13.1 FI-GL	Customizing (SPRO)	OBA0 — Tolerance groups for G/L accounts (clearing)	T043S	PARTIAL–11	11	GL OI gap
19	13.1 FI-GL	Customizing (SPRO)	OB46 — Interest indicators / OB82 / OBAA	T056*	GAP	propose 30	Interest engine (see F.52)
20	13.1 FI-GL	Customizing (SPRO)	OB74 — Additional rules for automatic clearing	TF123	PARTIAL–11	11	GL OI gap (AP/AR criteria covered by settlement rules)
21	13.1 FI-GL	Customizing (SPRO)	FAGL_CL_RGRP / Define sort method & adjustment accounts for regrouping	T043/ T030	GAP	propose 14	FAGLF101 gap
22	13.1 FI-GL	Customizing (SPRO)	Cost of sales accounting: functional areas (OKBD), activate (FAGL_FUNC_AREA)	TFKB, T001	PARTIAL–11	11	Functional area dimension + derivation rules (substitution gap)
23	13.2 FI-AP	Master data & documents	One-time vendor (account group CPD)	One-time accounts with address on document	PARTIAL–7	7	Not named in wave 7; add "one-time party" role with document-level address capture
24	13.2 FI-AP	Master data & documents	F-40 / F-36 (AP side: FBW6)	Bill of exchange payment (AP)	GAP	propose 36	Bills of exchange (AP and AR) are not in any wave; relevant in IT/ES/FR/JP/BR; propose as a wave 36 (bank) option or defer to a localization v1.x pack under wave 48 — needs a decision
25	13.2 FI-AP	Master data & documents	J1INCHLN / J1INBANK / J1INCERT (IN WHT)	Withholding tax challan / bank challan / certificate (India)	PARTIAL–13	13	Withholding engine + 26AS hooks named; Indian TDS challan/certificate flows belong to the IN country pack (wave 48) — partial until pack ships
26	13.2 FI-AP	Master data & documents	FIBLFFP / F111	Payment request / payment program for payment requests	PARTIAL–36	36	Payment requests from treasury (not vendor items) not named; add to wave 36 in-house banking
27	13.2 FI-AP	Reports & inquiries	S_ALR_87012116	Bill of exchange list	GAP	propose 36	Follows bills gap
28	13.2 FI-AP	Periodic processing	F111	Payment program for payment requests	PARTIAL–36	36	See payment requests
29	13.2 FI-AP	Periodic processing	F150 (AP dunning)	Dunning vendors (debit balances)	PARTIAL–30	30	Dunning covers customers; dunning vendors with debit balances is an edge case — reuse engine
30	13.2 FI-AP	Periodic processing	FAGLF101 (AP)	Regroup vendors with debit balances	GAP	propose 14	Reclassification gap
31	13.2 FI-AP	Periodic processing	F.98	Vendor balance confirmation	GAP	propose 19	Balance confirmations gap
32	13.2 FI-AP	Periodic processing	FBWE (AP bill of exchange presentation)	Bill of exchange presentation	GAP	propose 36	Bills gap
33	13.2 FI-AP	Periodic processing	F.64 (AP correspondence)	Correspondence: maintain / print	PARTIAL–19	19	Correspondence types (account statement, payment notice) only partially named (payment advice) — define correspondence framework (templates + triggers) on wave 46 document designer
34	13.2 FI-AP	Customizing (SPRO)	OBYR — Special G/L indicators vendors (down payments, guarantees, BoE)	T074	PARTIAL–19	19	Down payments yes; BoE no
35	13.2 FI-AP	Customizing (SPRO)	OBB9 — Installment terms	T052R	PARTIAL–7	7	Installment gap
36	13.2 FI-AP	Customizing (SPRO)	Define correspondence types / forms (OB77, OB78, OB79)	T048, T048B	PARTIAL–19	19	Correspondence framework gap
37	13.2 FI-AP	Customizing (SPRO)	OBD3 "one-time account"	T077K	PARTIAL–7	7	One-time party gap
38	13.2 FI-AP	Customizing (SPRO)	Bank-to-bank transfers (payment request types, FIBL)	—	PARTIAL–36	36	Payment requests gap
39	13.3 FI-AR	Master data & documents	F-34 / F-36 / F-33	Bill of exchange payment / discount (AR)	GAP	propose 36	Bills of exchange gap
40	13.3 FI-AR	Master data & documents	FB12 / F.61 / F.64	Request / print correspondence	PARTIAL–30	30	Statements and dunning named; general correspondence framework (payment notices, individual letters, open item lists) only partial

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
41	13.3 FI-AR	Reports & inquiries	S_ALR_87012199	Customer: bill of exchange list	GAP	propose 36	Bills gap
42	13.3 FI-AR	Periodic processing	FAGLF101 (AR)	Regroup customers with credit balances	GAP	propose 14	Reclassification gap
43	13.3 FI-AR	Periodic processing	F.99	Customer balance confirmation	GAP	propose 30	Balance confirmation gap
44	13.3 FI-AR	Periodic processing	SEPA mandate management (FSEPA_M1..M4)	Mandates	PARTIAL–30	30	SEPA direct-debit mandates not named; required for EU customers paying by direct debit — add mandate entity to wave 30
45	13.3 FI-AR	Periodic processing	FBWD / FBWE (AR BoE)	Bill of exchange presentation	GAP	propose 36	Bills gap
46	13.3 FI-AR	Customizing (SPRO)	OBXR — Special G/L customers (down payments, security deposits, BoE)	T074	PARTIAL–30	30	BoE gap
47	13.3 FI-AR	Customizing (SPRO)	OBB9 — Installment terms (customers)	T052R	PARTIAL–7	7	Installment gap
48	13.3 FI-AR	Customizing (SPRO)	Define correspondence types (OB77) / periodic account statements	T048	PARTIAL–30	30	Correspondence framework gap
49	13.3 FI-AR	Customizing (SPRO)	SEPA mandate customizing	SEPA_*	PARTIAL–30	30	Mandate gap
50	13.3 FI-AR	Customizing (SPRO)	Define alternate payer / head office–branch	KNA1-KNRZE, KNB1	PARTIAL–7	7	Head-office/branch (invoice to branch, pay from HQ) not named in wave 7 party roles — add "bill-to/pay-from parent" relationship
51	13.4 FI-AA	Master data & documents	AS21 / AS22 / AS23	Group asset create/change/display	PARTIAL–37	37	Group assets (depreciate at group level, e.g. ADR) not named in wave 37 — add to tax book features
52	13.4 FI-AA	Master data & documents	Insurance values (asset master tab, S_ALR_87011951)	Insurance values / indexing	PARTIAL–37	37	Insurance values and index series not named — add to asset master
53	13.4 FI-AA	Reports & inquiries	S_ALR_87010127 / 87010129 (US: form 4562 / mid-quarter)	Tax depreciation reports	PARTIAL–37	37	MACRS tables yes; IRS Form 4562 layout belongs to the US pack (wave 48)
54	13.4 FI-AA	Customizing (SPRO)	OAYM / OAAX — Group assets per area	T093A	PARTIAL–37	37	Group asset gap
55	13.4 FI-AA	Customizing (SPRO)	OAVI / OAVF — Insurance types, companies, index series	T086, T094	PARTIAL–37	37	Insurance gap
56	13.5 FI-BL / FSCM-CM / BAM	Master data & documents	FBCJ	Cash journal	GAP	propose 36	Cash journal gap (see FI-GL)
57	13.5 FI-BL / FSCM-CM / BAM	Master data & documents	FRFT / FRFT_B	Repetitive payment codes / bank-to-bank transfer	PARTIAL–36	36	Bank-to-bank transfer / payment request gap
58	13.5 FI-BL / FSCM-CM / BAM	Master data & documents	IHC0 / IHC1IP (In-House Cash)	In-house cash center, IHC accounts, internal payment orders	PARTIAL–36	36	"In-house banking basics: netting, cash pooling" named; full IHC (internal accounts, interest, statements for subsidiaries) is partial
59	13.5 FI-BL / FSCM-CM / BAM	Master data & documents	Fiori: Approve Bank Payments / Monitor Payments (Advanced Payment Management)	APM	PARTIAL–36	36	Payment approval covered; APM-style central payment hub (route, batch, reformat across entities) only partially (adapters)
60	13.5 FI-BL / FSCM-CM / BAM	Master data & documents	FBWE / FBWD (bills)	Bill of exchange presentation	GAP	propose 36	Bills gap
61	13.5 FI-BL / FSCM-CM / BAM	Reports & inquiries	S_ALR_87012309 (cash journal)	Cash journal report	GAP	propose 36	—
62	13.5 FI-BL / FSCM-CM / BAM	Reports & inquiries	Fiori: Bank Relationship Overview / Bank Fee Analysis	Bank fees (camt.086)	GAP	propose 36	Bank fee analysis (camt.086) not named; propose as wave 36 report
63	13.5 FI-BL / FSCM-CM / BAM	Periodic processing	Cash pooling / sweeping run (IHC)	—	PARTIAL–36	36	Basics named
64	13.5 FI-BL / FSCM-CM / BAM	Periodic processing	F111 (treasury payment requests)	—	PARTIAL–36	36	—
65	13.5 FI-BL / FSCM-CM / BAM	Periodic processing	Bank fee statement import (camt.086)	—	GAP	propose 36	—
66	13.5 FI-BL / FSCM-CM / BAM	Customizing (SPRO)	Bank chains (FIBC)	TBCHAIN*	PARTIAL–36	36	Correspondent/intermediary bank chains not named — add to bank connectivity
67	13.5 FI-BL / FSCM-CM / BAM	Customizing (SPRO)	IHC customizing: IHC centers, accounts, conditions, payment orders	IHC_*	PARTIAL–36	36	IHC partial
68	13.5 FI-BL / FSCM-CM / BAM	Customizing (SPRO)	Cash journal customizing (FBCJC0..3)	TCJ_*	GAP	propose 36	—
69	13.6 FSCM	Master data & documents	FTR_CREATE / FTR_EDIT / FTR_00	Treasury transaction manager: create financial transaction (money market, FX forward/spot/swap, derivatives, securities, loans)	GAP	propose 36	TRM transaction manager is not in any wave. Manufacturers with FX exposure need at least FX spot/forward deal capture, settlement and valuation; propose a wave 36 "treasury deals (FX, loans, deposits)" extension, with derivatives/securities deferred to v2

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
70	13.6 FSCM	Master data & documents	TBB1 / TPM10	Post flows / treasury posting	GAP	propose 36	Follows TRM gap
71	13.6 FSCM	Master data & documents	TPM1 / TPM60	Valuation (key date), amortization, position management	GAP	propose 36	—
72	13.6 FSCM	Master data & documents	THMEX (Hedge management & accounting, IFRS 9)	Hedge relationships, effectiveness, hedge accounting postings	GAP	propose 36	Hedge accounting per IFRS 9/ASC 815 — v2 candidate; register as gap
73	13.6 FSCM	Master data & documents	Exposure management (raw exposures, TEM)	Exposure capture from AP/AR/forecast	GAP	propose 36	FX exposure from open items is derivable from waves 12/36 data; deal side missing
74	13.6 FSCM	Master data & documents	TBDM market data import (rates, prices)	Market data management	PARTIAL–12	12	Exchange-rate providers covered; interest curves/security prices not
75	13.6 FSCM	Master data & documents	Loans management (FN*, CML)	Loan contracts	GAP	propose 36	Intercompany and bank loans: schedules + interest; propose minimal loan contract in wave 36
76	13.6 FSCM	Reports & inquiries	TPM12 / TPM13	Position list / flow list (TRM)	GAP	propose 36	TRM gap
77	13.6 FSCM	Reports & inquiries	TPM20	Posting journal (TRM)	GAP	propose 36	TRM gap
78	13.6 FSCM	Reports & inquiries	Hedge plan reports	—	GAP	propose 36	—
79	13.6 FSCM	Reports & inquiries	Treasury information system (TRM)	—	GAP	propose 36	—
80	13.6 FSCM	Periodic processing	TBB1 (periodic) / TPM1 valuation run / accruals	TRM period-end	GAP	propose 36	TRM gap
81	13.6 FSCM	Periodic processing	Hedge effectiveness test run	—	GAP	propose 36	—
82	13.6 FSCM	Periodic processing	TBDM import run	Market data import	PARTIAL–12	12	—
83	13.6 FSCM	Customizing (SPRO)	TRM: product types, transaction types, flow types, update types, valuation areas/classes, accounting codes, position management procedures	TRL*, TRD*, TPM*	GAP	propose 36	TRM gap
84	13.6 FSCM	Customizing (SPRO)	Hedge management: hedging profiles, exposure categories	THM*	GAP	propose 36	—
85	13.6 FSCM	Customizing (SPRO)	Market data: rate types, yield curves, volatility	VTB*, TCURR	PARTIAL–12	12	—
86	13.7 CO-OM	Periodic processing	CPTA / CPTB	ABC: template allocation (actual/plan)	PARTIAL–38	38	ABC named; template (driver-formula) allocation is partial — specify driver formulas
87	13.7 CO-OM	Customizing (SPRO)	CO versions for parallel valuation (group/profit-center valuation, transfer prices)	TKA07, TKA09, TKA01	PARTIAL–40	40	Transfer-pricing documentation hooks named; parallel group/profit-center valuation views (three valuation views in CO/ML) are not — register as partial for consolidation of profit in inventory
88	13.7 CO-OM	Customizing (SPRO)	ABC: template allocation environments, functions (CPT1)	—	PARTIAL–38	38	—
89	13.8 CO-PC	Master data & documents	CKECP / CKECP1	Easy cost planning / ad-hoc cost estimate	PARTIAL–39	39	What-if simulations named; template-based "easy" cost planning for non-material objects (projects/orders) — partial, map to wave 41 project estimates
90	13.8 CO-PC	Master data & documents	CK51N / CK53N	Sales order cost estimate create / display	PARTIAL–27	27	Order-specific costing for configured/MTO products not explicitly named in waves 27/39 — add "sales order cost estimate" to wave 39 simulations
91	13.8 CO-PC	Master data & documents	CK91N / CK92	Procurement alternatives / mixing ratios (mixed costing)	PARTIAL–39	39	Mixed costing (weighted sources: make vs buy, multiple vendors/production versions) not named — add to wave 39 fallback rules
92	13.8 CO-PC	Master data & documents	KKPHIE / KKP2	Cost object hierarchy	PARTIAL–38	38	Cost object hierarchies for process cost distribution — partial
93	13.8 CO-PC	Periodic processing	CKMLCPAVR (Alternative valuation run)	Alternative valuation run (cumulative periods)	PARTIAL–20	20	Multi-period cumulative actual costing (AVR) not named — partial
94	13.8 CO-PC	Periodic processing	CKMDUVACT / CKMDUVMAT	Distribution of usage variances	PARTIAL–20	20	Usage-variance distribution to consumers — partial
95	13.8 CO-PC	Periodic processing	WIP at actual cost (ML)	—	PARTIAL–20	20	WIP revaluation at actual cost — partial
96	13.8 CO-PC	Customizing (SPRO)	Mixed costing: quantity structure types, procurement alternatives (OMXL)	CKMLMV*	PARTIAL–39	39	Mixed costing gap
97	13.8 CO-PC	Customizing (SPRO)	Easy cost planning: costing models, characteristics	CKF*	PARTIAL–39	39	—

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
98	13.8 CO-PC	Customizing (SPRO)	Sales order costing: requirements class settings (OVZG) valuation/costing	T459K	PARTIAL–27	27	Sales-order stock valuation (valuated MTO stock) belongs to wave 16/21 — not named; partial
99	13.8 CO-PC	Customizing (SPRO)	Transfer prices / multiple valuation views (OKKP, OKEQ group)	TKA01	PARTIAL–40	40	See CO-OM
100	13.8 CO-PC	Customizing (SPRO)	Cost of goods sold split by cost component (S/4 COGS splitting)	FINSC_COGS*	PARTIAL–39	39	COGS posting by cost component (material/labor/overhead accounts) not named in wave 28/39 — add posting profile option
101	13.8 CO-PC	Customizing (SPRO)	Price-difference splitting (S/4)	—	PARTIAL–39	39	Same
102	13.9 CO-PA	Master data & documents	KEDR / KEDE	Derivation strategy / rules (derive characteristics)	PARTIAL–11	11	Default dimensions on master data exist; rule-based derivation tables (lookup A×B→C with validity) = substitution gap (see FI-GL)
103	13.9 CO-PA	Periodic processing	KE28	Top-down distribution (actual)	PARTIAL–38	38	Top-down distribution of aggregated values (e.g. freight by revenue share) to detailed segments not named — add as allocation type "top-down"
104	13.9 CO-PA	Periodic processing	KE27	Periodic valuation (revaluate with actual costs from ML)	PARTIAL–20	20	Revaluation of COGS in profitability with actual cost — tied to ML revaluation of consumption partial
105	13.9 CO-PA	Periodic processing	KEND / KEDR realignment run	Realignment (re-derive characteristics on history)	PARTIAL–11	11	Mass re-derivation of dimensions on posted lines is a controlled restatement; propose explicit "dimension realignment" tool (audit-logged) in wave 11/15
106	13.9 CO-PA	Customizing (SPRO)	KEDR — Derivation rules (table lookup, derivation rule, move, clear, enhancement)	TKEDR*, K9R*	PARTIAL–11	11	Derivation gap
107	13.9 CO-PA	Customizing (SPRO)	0KEM / 0KEP — PCA substitution rules / derivation of PC from SD (sales org, material), MM	T8A*	PARTIAL–11	11	Derivation gap
108	13.9 CO-PA	Customizing (SPRO)	8KEM / 8KEN — Transfer prices: pricing procedures, conditions, transfer price variants	T8PT*	PARTIAL–40	40	Transfer pricing documentation hooks only; PC valuation view not covered
109	13.10 EC-CS / SEM-BCS / S/4HANA Group Reporting	Master data & documents	Fiori: Manage Reclassifications / Define Reclassification Rules	Rule-based reclassifications	PARTIAL–40	40	Elimination rules named; general reclassification rules (e.g. move FS item based on sign) partial
110	13.10 EC-CS / SEM-BCS / S/4HANA Group Reporting	Master data & documents	Fiori: Consolidation of Investments (activity-based: first consolidation, subsequent, step acquisition, divestiture; equity method)	CoI	PARTIAL–40	40	Minority interest named; purchase-method CoI with goodwill, step acquisitions, divestitures and equity-method accounting are not — partial
111	13.10 EC-CS / SEM-BCS / S/4HANA Group Reporting	Master data & documents	Fiori: Manage Investments / Equity data	Investment and equity holdings	PARTIAL–40	40	—
112	13.10 EC-CS / SEM-BCS / S/4HANA Group Reporting	Master data & documents	Disclosure management / XBRL (SAP DM, SAC)	Statutory filing packages	GAP	propose 48	XBRL/ESEF filing output not named; propose electronic reporting designer (wave 48) output type
113	13.10 EC-CS / SEM-BCS / S/4HANA Group Reporting	Periodic processing	Consolidation monitor tasks: IC elimination (AR/AP, revenue/expense, dividends), elimination of IC profit in inventory, CoI, group validation	Group-level tasks	PARTIAL–40	40	IC balances and inventory profit named; dividend elimination and CoI partial
114	13.10 EC-CS / SEM-BCS / S/4HANA Group Reporting	Periodic processing	Reclassification runs	—	PARTIAL–40	40	—
115	13.10 EC-CS / SEM-BCS / S/4HANA Group Reporting	Customizing (SPRO)	Reclassification methods / elimination methods (IC elimination of AR/AP, revenue/expense, dividends)	TF*	PARTIAL–40	40	—
116	13.10 EC-CS / SEM-BCS / S/4HANA Group Reporting	Customizing (SPRO)	CoI methods (purchase, equity, proportional), activities	TF*	PARTIAL–40	40	—
117	13.11 FM / PSM	Master data & documents	FM5I / FM5S	Fund create/display	PARTIAL–38	38	"Fund" dimension (restricted/unrestricted funding) not named; grants/projects usually carry it — partial; usable via a financial dimension
118	13.11 FM / PSM	Master data & documents	FMMEASURE (funded program)	Funded program	PARTIAL–38	38	Same
119	13.11 FM / PSM	Master data & documents	FMX1 / FMX2 / FMX3	Earmarked funds: funds reservation create/change/display	PARTIAL–38	38	Pre-encumbrance from requisition is named; manual "funds reservation" without a requisition is not — add budget reservation document
120	13.11 FM / PSM	Master data & documents	FMY1 / FMZ1 / FMW1 / FMV1	Funds precommitment / commitment / forecast of revenue / funds blocking	PARTIAL–38	38	Same
121	13.11 FM / PSM	Master data & documents	FMDERIVE	Derivation of FM account assignment	PARTIAL–11	11	Derivation gap

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
122	13.11 FM / PSM	Customizing (SPRO)	FMDERIVE strategy	FMDERIVE*	PARTIAL–11	11	—
123	13.11 FM / PSM	Customizing (SPRO)	Earmarked funds document types, number ranges	FMEF*	PARTIAL–38	38	—
124	13.12 MM-PUR	Master data & documents	ME84 / ME85 / ME86	Create SA releases (FRC/JIT) / renumber / display release documentation	PARTIAL–17	17	Vendor-facing forecast/JIT release generation (cumulative quantities, release documentation) is a supplier-schedule capability not explicitly named in wave 17; EDI DELFOR/DELJIT in wave 35 lists 850/855/856/810/820/214 only — partial
125	13.12 MM-PUR	Master data & documents	MEQ1 / MEQ3 / MEQ4 / MEQ6 / MEQ8	Quota arrangement maintain/display/changes/analyze	PARTIAL–22	22	Quota arrangements (percentage split across vendors/own production for MRP) not named in wave 17/22 — partial
126	13.12 MM-PUR	Master data & documents	ME21N item category K (consignment)	Vendor consignment PO	PARTIAL–16	16	Consignment stock (vendor-owned at our site, settle on consumption MRKO) not named in waves 16/18 — partial; needs ownership attribute on inventory + settlement run
127	13.12 MM-PUR	Master data & documents	ME21N item category U (stock transfer) / UB STO / NB cross-company STO	Stock transport orders (intra/intercompany) with delivery	PARTIAL–16	16	Wave 16 "transfer journals" and wave 40 intercompany trade cover the postings; an explicit *transfer order document* with delivery, in-transit stock and receiving (plant-to-plant with shipping) is not named — partial; propose "transfer orders with in-transit status" in wave 16/28
128	13.12 MM-PUR	Master data & documents	ME21N service item (D)	Services PO	PARTIAL–18	18	See MM-SRV
129	13.12 MM-PUR	Master data & documents	Fiori: Create Purchase Requisition (self-service, catalog, free text)	Self-service requisitioning with catalogs (OCI punch-out)	PARTIAL–17	17	Requisition yes; catalog/punch-out (OCI/cXML) item selection not named — partial; propose wave 20 (supplier collaboration) or wave 35 (integration)
130	13.12 MM-PUR	Master data & documents	SAP Ariba Buying / Sourcing / Contracts / SLP (supplier lifecycle)	Cloud procurement suite	PARTIAL–20	20	Supplier portal (ack, ASN, invoice, docs) covers Ariba Network basics; strategic sourcing events (auctions, weighted RFP scoring), supplier onboarding/qualification questionnaires and contract authoring are not — partial; propose v1.x
131	13.12 MM-PUR	Reports & inquiries	MSRV1 / MSRV3 (services)	—	PARTIAL–18	18	See MM-SRV
132	13.12 MM-PUR	Reports & inquiries	Fiori: Monitor Supplier Confirmations / Monitor Scheduling Agreement Releases	—	PARTIAL–17	17	SA release documentation partial
133	13.12 MM-PUR	Periodic processing	ME84 (batch)	SA release creation run	PARTIAL–17	17	SA release partial
134	13.12 MM-PUR	Periodic processing	MRKO (consignment settlement)	—	PARTIAL–16	16	Consignment gap
135	13.12 MM-PUR	Periodic processing	MEI1 / MEI2 / MEI4	Subsequent price change (automatic document adjustment after condition change)	PARTIAL–17	17	Retroactive re-pricing of open POs after condition change — not named; partial
136	13.12 MM-PUR	Customizing (SPRO)	OMGN — Set up stock transport order (delivery type, checking rule per supplying plant)	T161V, T161W	PARTIAL–16	16	Transfer order gap
137	13.12 MM-PUR	Customizing (SPRO)	Source determination — source list requirement per plant, quota arrangement rules	T001W, TMQ1	PARTIAL–22	22	Quota partial
138	13.12 MM-PUR	Customizing (SPRO)	Scheduling agreement — release creation profile, firm zone/trade-off zone, JIT indicator	T163M	PARTIAL–17	17	SA release partial
139	13.12 MM-PUR	Customizing (SPRO)	Ariba Network integration (cXML, Cloud Integration Gateway)	—	PARTIAL–35	35	cXML not listed among EDI formats (X12/EDIFACT/Peppol); partial
140	13.13 MM-IM	Master data & documents	MB53 / MB54	Plant availability / consignment stock	PARTIAL–16	16	Vendor consignment gap
141	13.13 MM-IM	Master data & documents	MB58	Consignment and returnable packaging stock at customer	PARTIAL–27	27	Customer consignment (fill-up/issue/pick-up/return orders) not named in SD/inventory waves — partial; needs "stock at customer" ownership dimension
142	13.13 MM-IM	Master data & documents	MIK1 / MIQ1 / MIM1	PI for vendor consignment / project stock / returnable transport packaging	PARTIAL–16	16	Consignment/RTP gaps
143	13.13 MM-IM	Master data & documents	MIE1 / MIW1 / MIV1	PI for sales order stock / customer consignment / returnable packaging at customer	PARTIAL–27	27	Sales-order stock & customer consignment partial

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
144	13.13 MM-IM	Master data & documents	MIS1 / MIS2 (inventory sampling)	Inventory sampling (statistical)	GAP	propose 16	Statistical inventory sampling (count a sample, extrapolate) — rarely used but a recognized GAAP-compatible method; low priority
145	13.13 MM-IM	Master data & documents	Special stock: sales order stock (E), project stock (Q)	Stock segregated by SO/WBS	PARTIAL–16	16	Sales-order/project-specific stock (valuated/non-valuated) not explicitly named as an inventory dimension — partial; needed for MTO/ETO manufacturers (waves 21/27/41)
146	13.13 MM-IM	Master data & documents	Stock in transit (MB5T; S/4 stock in transit for STO)	In-transit stock	PARTIAL–16	16	Transfer order gap
147	13.13 MM-IM	Master data & documents	Returnable transport packaging (M, V)	RTP at vendor/customer	GAP	propose 28	Returnable packaging (pallets, crates) tracking at customers/vendors not named; propose wave 28 (packaging/HU) option
148	13.13 MM-IM	Reports & inquiries	MB5T	Stock in transit (cross-company/plant)	PARTIAL–16	16	Transfer order gap
149	13.13 MM-IM	Reports & inquiries	MB54 / MB58 / MBLB	Consignment / customer consignment / subcontractor stock	PARTIAL–16	16	Consignment gaps
150	13.13 MM-IM	Reports & inquiries	MBBS	Valuated special stock (sales order/project)	PARTIAL–16	16	Special stock gap
151	13.13 MM-IM	Periodic processing	MRKO	Consignment & pipeline settlement	PARTIAL–16	16	Consignment gap
152	13.13 MM-IM	Customizing (SPRO)	OMWD / OMWC / OMSK — Valuation grouping code / split valuation (valuation categories/types) / valuation classes & account category reference	T001K, T025, T025K, T149*	PARTIAL–20	20	Split valuation (different values per valuation type, e.g. new vs refurbished, domestic vs import) not named — partial; batch-level valuation needed for process industries
153	13.13 MM-IM	Customizing (SPRO)	Special stock settings (consignment price determination, pipeline)	—	PARTIAL–16	16	Consignment gap
154	13.13 MM-IM	Customizing (SPRO)	Stock determination (OSPX / OSPW) strategies for issues (consignment vs own, SLoc priority)	—	PARTIAL–16	16	Picking policies FEFO/FIFO named; ownership-based stock determination partial (consignment)
155	13.14 MM-IV	Master data & documents	MRKO	Consignment and pipeline settlement	PARTIAL–16	16	Consignment gap
156	13.14 MM-IV	Master data & documents	MRIS	Invoicing plan settlement (periodic/partial invoicing plans on PO)	PARTIAL–17	17	Invoicing plans on POs (periodic settlement without GR) not named — partial; relevant to leases/services; propose wave 17 PO line "invoicing plan"
157	13.14 MM-IV	Master data & documents	MRNB	Revaluation (retroactive price change on invoiced items)	PARTIAL–17	17	Retro re-pricing partial (MEI1 row)
158	13.14 MM-IV	Periodic processing	MRKO (scheduled)	Consignment settlement	PARTIAL–16	16	—
159	13.14 MM-IV	Periodic processing	MRIS (scheduled)	Invoicing plan run	PARTIAL–17	17	—
160	13.14 MM-IV	Periodic processing	MRNB (scheduled)	Revaluation run	PARTIAL–17	17	—
161	13.14 MM-IV	Customizing (SPRO)	OMRX — Stochastic block (random sampling thresholds)	T169P	GAP	propose 18	Random audit sampling of invoices — simple to add to wave 18 exception rules; register as gap
162	13.14 MM-IV	Customizing (SPRO)	Invoicing plan types (periodic/partial)	T163*	PARTIAL–17	17	Invoicing plan gap
163	13.15 MM-SRV	Master data & documents	AC01 / AC02 / AC03 / AC04 / AC06	Service master create/change/display/list/changes	PARTIAL–8	8	Product of type "service" is covered; a dedicated service master with service categories, purchasing text, standard service catalog is partial — add "service product type" attributes to wave 8
164	13.15 MM-SRV	Master data & documents	ML10 / ML11 / ML12	Model service specifications create/change/display	GAP	propose 17	Service specification templates (hierarchical outlines of services with quantities) not named; propose "PO line templates" in wave 17
165	13.15 MM-SRV	Master data & documents	Standard service catalog (SSC)	Standard texts for services	GAP	propose 17	Same
166	13.15 MM-SRV	Master data & documents	ME21N (item cat D) with service lines, value limits, contract limits	Service PO with hierarchical service outline and limits	PARTIAL–17	17	Limit/blanket POs named; service outlines (hierarchy of service lines under one PO item) and contract-limit references are partial
167	13.15 MM-SRV	Master data & documents	ML81N / ML82 / ML83 / ML84 / ML85	Service entry sheet create-maintain / display / list / accept-revoke / collective release	PARTIAL–18	18	Receiving covers "product receipt"; a *service entry sheet* (lines of performed services with quantities/prices, acceptance workflow, partial entries, unplanned services within limits) is not named — partial; add "service receipt/entry sheet with acceptance" to wave 18

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
168	13.15 MM-SRV	Master data & documents	ML33 / ML34 / ML35 / ML36 / ML39 / ML40 / ML41 / ML45 / ML46 / ML47 / ML48	Service conditions: vendor-level / service-level / vendor+plant	PARTIAL–17	17	Vendor price lists for service products via pricing engine — partial until service products confirmed
169	13.15 MM-SRV	Master data & documents	MSRV1 … MSRV6	Service lists: by service / by PO / by SES / by vendor / by contract / entry sheets	PARTIAL–18	18	—
170	13.15 MM-SRV	Master data & documents	MIRO with SES reference	Invoice for services (SES-based)	PARTIAL–18	18	Match to service receipt
171	13.15 MM-SRV	Master data & documents	ML91 / ML92 / ML93	Service entry sheet archiving / lists	PARTIAL–18	18	—
172	13.15 MM-SRV	Master data & documents	Fiori: Manage Service Entry Sheets – Lean Services / Create Service Entry Sheet	S/4 lean services	PARTIAL–18	18	Same
173	13.15 MM-SRV	Master data & documents	Fiori: Manage Purchase Orders (service items), Manage Supplier Quotations for services	—	PARTIAL–17	17	—
174	13.15 MM-SRV	Master data & documents	Supplier self-service SES (SAP Business Network / Fieldglass)	Supplier enters SES / timesheets	PARTIAL–20	20	Supplier portal: add service confirmation/timesheet entry (contingent labor) — partial
175	13.15 MM-SRV	Reports & inquiries	MSRV1 … MSRV6	Service lists	PARTIAL–18	18	—
176	13.15 MM-SRV	Reports & inquiries	ME2S	Services per purchase order	PARTIAL–18	18	—
177	13.15 MM-SRV	Reports & inquiries	MSRV3 (entry sheets)	SES list	PARTIAL–18	18	—
178	13.15 MM-SRV	Reports & inquiries	ML83 / ML84	SES acceptance list	PARTIAL–18	18	—
179	13.15 MM-SRV	Periodic processing	ML85 (collective release of SES)	Mass acceptance	PARTIAL–18	18	—
180	13.15 MM-SRV	Periodic processing	MRRL for services	ERS for SES	PARTIAL–18	18	—
181	13.15 MM-SRV	Customizing (SPRO)	Service master: number ranges, service categories, organizational status, field selection	ASMD	PARTIAL–8	8	—
182	13.15 MM-SRV	Customizing (SPRO)	Service specifications: screen layout, number ranges for SES, release procedure for SES (classification)	ESSR	PARTIAL–18	18	SES approval via workflow designer
183	13.15 MM-SRV	Customizing (SPRO)	Service conditions: calculation schema MS0000, condition types PRS	T685	PARTIAL–17	17	—
184	13.15 MM-SRV	Customizing (SPRO)	Default values for SES acceptance (immediate GR, account assignment)	—	PARTIAL–18	18	—
185	13.15 MM-SRV	Customizing (SPRO)	Source determination for services (contracts)	—	PARTIAL–17	17	—
186	13.15 MM-SRV	Customizing (SPRO)	Lean services: activate, product type service, service entry sheet (lean) settings	—	PARTIAL–18	18	—
187	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Master data & documents	LXDCA / cross-docking (LXDCK)	Planned cross-docking	GAP	propose 31	Cross-docking (inbound → outbound without put-away, planned/opportunistic) not named in wave 31 — propose directive type "cross-dock"
188	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Master data & documents	EWM: /SCWM/PACK / /SCWM/PACKSPEC / /SCWM/PSSTAGE	Packing work center / packaging specifications	PARTIAL–31	31	HU/packing covered; *packaging specifications* (rules: product → packaging material, quantities per level) only implied — partial; propose wave 28/31 "packaging rules"
189	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Master data & documents	EWM: /SCWM/VASEXEC / VAS orders	Value-added services (kitting, labeling)	GAP	propose 31	VAS/kit-to-order in warehouse not named; propose wave 31 work type "VAS" and wave 24 kitting via lean BOM
190	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Master data & documents	EWM: /SCWM/LM (labor management: engineered labor standards, planned vs actual, performance docs)	Labor management	PARTIAL–31	31	Warehouse KPIs named; engineered labor standards, operator performance/incentive documents — partial
191	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Master data & documents	EWM: /SCWM/YM_* yard management (yard bins, vehicles, TUs, checkpoints)	Yard management	PARTIAL–31	31	Dock scheduling named; yard (trailer positions, moves, check-in/out) partial

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
192	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Master data & documents	EWM: /SCWM/TU, /SCWM/VEH	Transportation units / vehicles	PARTIAL–34	34	Load/TU in wave 34 — partial for yard link
193	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Master data & documents	EWM: /SCWM/MFS* (Material Flow System: PLC interface, conveyors, ASRS)	Automation integration	GAP	propose 32	MFS/PLC integration (telegrams to conveyor/AS-RS) not named; wave 25 OPC-UA/MQTT edge is the natural base — propose wave 32 "automation adapter (ASRS/conveyor)"
194	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Master data & documents	EWM: Kit-to-stock / kit-to-order	—	GAP	propose 31	See VAS
195	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Master data & documents	EWM: Dangerous goods / HazMat checks in warehouse (EHS)	—	GAP	propose 34	Hazmat classification, storage compatibility, DG documents — not named; propose wave 34 (transport) + wave 8 attributes
196	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Reports & inquiries	LX06 / LX07	Fire-containment sections / check storage (inconsistencies)	PARTIAL–31	31	Fire/hazard sections tie to hazmat gap
197	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Reports & inquiries	EWM: /SCWM/LM_ labor reports (planned vs actual)	—	PARTIAL–31	31	—
198	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Periodic processing	Labor management period close (/SCWM/LM)	—	PARTIAL–31	31	—
199	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Customizing (SPRO)	Hazardous material: regions, water pollution classes, storage classes, fire-containment sections	T* (MGEF)	GAP	propose 34	Hazmat gap
200	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Customizing (SPRO)	EWM: Yard management customizing	—	PARTIAL–31	31	—
201	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Customizing (SPRO)	EWM: MFS customizing (PLC communication channels, telegrams)	—	GAP	propose 32	—
202	13.16 LE-WM (classic Warehouse Management) and EWM (Extended Warehouse Management)	Customizing (SPRO)	EWM: Labor management: formulas, ELS, incentive	—	PARTIAL–31	31	—
203	13.17 LE-SHP / LE-TRA	Master data & documents	VL10 → STO deliveries (replenishment delivery NL/NLCC)	Deliveries for stock transport orders	PARTIAL–16	16	Transfer order gap
204	13.17 LE-SHP / LE-TRA	Master data & documents	Dangerous goods check in delivery (EHS-DG), DG documents	—	GAP	propose 34	Hazmat gap
205	13.17 LE-SHP / LE-TRA	Master data & documents	Foreign trade / export documents in delivery (customs, commercial invoice, EUR.1, Intrastat from delivery; GTS)	Export compliance	PARTIAL–13	13	Intrastat extract named (wave 13); export documentation (commercial invoice, packing list for customs, certificate of origin), export control screening (denied parties, licenses) and customs declarations (GTS) are not — partial; propose wave 48 "foreign trade/customs" pack
206	13.17 LE-SHP / LE-TRA	Master data & documents	Preference determination (SAP GTS / foreign trade)	Origin & preference (USMCA/EU)	GAP	propose 48	Rules-of-origin/preference calculation from BOM — gap

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
207	13.17 LE-SHP / LE-TRA	Customizing (SPRO)	OVLS / OVLV — Packing: packaging material types (VHAR), packing instructions (POP1/POP2/POP3/POF1), allowed packaging materials, HU number ranges (SSCC), packing control per item category	TVTY, TVEGR, PACKKP	PARTIAL–28	28	SSCC/containers yes; packing instructions (auto-pack rules) partial (same as EWM packaging specs)
208	13.17 LE-SHP / LE-TRA	Customizing (SPRO)	Foreign trade: export control, preference, Intrastat settings (VEB, VE0)	T604, T605	PARTIAL–13	13	Foreign trade gap
209	13.17 LE-SHP / LE-TRA	Customizing (SPRO)	Dangerous goods: DG profiles, DG check methods in SD/LE	DGT*	GAP	propose 34	Hazmat gap
210	13.18 TM	Master data & documents	Transportation units / resources (vehicles, trailers, drivers), calendars	Fleet resources	PARTIAL–34	34	Own fleet (vehicles/drivers/driver hours) not named — partial; carriers assumed external
211	13.18 TM	Master data & documents	Dangerous goods & DG checks in TM	—	GAP	propose 34	Hazmat gap
212	13.18 TM	Master data & documents	Customs declarations via GTS from TM	—	PARTIAL–13	13	Foreign trade gap
213	13.18 TM	Master data & documents	Strategic freight procurement (RFQ for lanes, bid analysis)	—	GAP	propose 34	Lane-based freight RFQ/bid analysis — gap; could reuse wave 17 RFQ
214	13.18 TM	Master data & documents	Load consolidation / container packing optimization (load plan 3D)	—	PARTIAL–34	34	Load building with capacity constraints named; 3D load plan/visual stacking partial
215	13.18 TM	Customizing (SPRO)	Dangerous goods in TM	—	GAP	propose 34	—
216	13.19 SD	Master data & documents	VB01 / VB02 / VB03	Listing/exclusion	PARTIAL–27	27	Product listing/exclusion per customer not named in wave 27 — partial; simple rule on order validation
217	13.19 SD	Master data & documents	VBN1 / VBN2 / VBN3	Free goods condition records (inclusive/exclusive)	PARTIAL–26	26	Free goods (buy 10 get 1) not named; promotions with quantity breaks are — partial
218	13.19 SD	Master data & documents	VA01 KB / KE / KA / KR (consignment fill-up/issue/pick-up/returns)	Customer consignment processes	PARTIAL–27	27	Customer consignment gap (see MM-IM)
219	13.19 SD	Master data & documents	VA31 / VA32 / VA33 / VA35	Customer scheduling agreement create/change/display/list (SA with JIT/forecast schedules VA32 → delivery schedule)	PARTIAL–27	27	"Scheduling agreements" named in wave 27 replaces list; inbound EDI DELFOR/DELJIT (830/862) customer schedules with cumulative quantities and schedule reconciliation are not named — partial (automotive)
220	13.19 SD	Master data & documents	VKA / VSB1 (self-billing)	Self-billing procedure (customer self-billing, automotive)	GAP	propose 29	Customer self-billing (EDI 820/GSVERF reconciliation against invoices) not named; automotive; propose wave 29/35 option
221	13.19 SD	Master data & documents	Fiori: Manage Solution Orders / Solution Business (bundles)	Bundled sales (products, services, subscriptions)	PARTIAL–29	29	Bundles with mixed revenue types partially via rev rec bundles; solution-order document type not named — partial
222	13.19 SD	Master data & documents	Cross-selling (VB41), product proposal/dynamic product proposal	Order-entry suggestions	PARTIAL–27	27	Product proposals partially (item proposal); cross-sell rules not named
223	13.19 SD	Customizing (SPRO)	OVZJ / OVZI — Product allocation (quota per customer/period)	T190, KOTA	GAP	propose 27	Product allocation (ATP quotas per customer/region in scarcity) not named — propose wave 27 ATP option
224	13.19 SD	Customizing (SPRO)	OVZK / OVZ — Availability check against product allocation; aATP (S/4): backorder processing strategies (BOP), alternative-based confirmation (ABC), release for delivery	—	PARTIAL–27	27	ATP/CTP covered; aATP backorder processing with strategies (win/gain/redistribute/fill/lose) partially named ("backorder handling")
225	13.19 SD	Customizing (SPRO)	OVZ "Define requirements classes/types, assignment to strategy group" (OVZG/OVZH/OVZI)	T459K, T459A, T459U	PARTIAL–27	27	Planning strategies (wave 22) + sales-order stock partial
226	13.19 SD	Customizing (SPRO)	OV "Listing/exclusion procedures" (OV04)	T684	PARTIAL–27	27	Listing gap
227	13.19 SD	Customizing (SPRO)	OVKF — Free goods: condition type NA00, procedure NA0001, determination	T685, KOTN	PARTIAL–26	26	Free goods gap
228	13.20 SD Rebates / Settlement Management (condition contracts)	Master data & documents	VBOF	Update billing documents (retroactive rebate accrual)	PARTIAL–26	26	Retroactive rebate (apply agreement to already-invoiced volume) not named — partial
229	13.20 SD Rebates / Settlement Management (condition contracts)	Master data & documents	Supplier condition contracts (purchasing rebates, chargebacks)	Vendor rebates / ship-and-debit chargebacks	PARTIAL–19	19	Supplier rebate accrual and settlement (credit from vendor) and chargeback/ship-and-debit claims are not named in waves 17/19 — partial; propose wave 26 engine reused for purchasing with settlement via AP in wave 19

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
230	13.20 SD Rebates / Settlement Management (condition contracts)	Master data & documents	Commission contracts (sales rep commissions via settlement management)	Commissions	PARTIAL–26	26	"Commission calculation hooks" only — partial; settlement to payroll/AP not named
231	13.20 SD Rebates / Settlement Management (condition contracts)	Master data & documents	Chargeback / ship-and-debit claims from distributors	Claim validation against agreements	GAP	propose 30	Distributor chargeback claims (EDI 844/849) validation and settlement — gap for manufacturers selling via distribution; propose wave 30 + wave 35
232	13.20 SD Rebates / Settlement Management (condition contracts)	Master data & documents	Royalty / licensing payments (settlement management use)	Royalty accrual/settlement	GAP	propose 26	Royalty accrual by sales volume (license fees to third parties) — gap; reuse rebate engine with vendor-side settlement
233	13.20 SD Rebates / Settlement Management (condition contracts)	Periodic processing	VBOF (batch)	Retroactive accrual update	PARTIAL–26	26	—
234	13.20 SD Rebates / Settlement Management (condition contracts)	Periodic processing	Delta accruals / delta settlement	Delta settlement after retro changes	PARTIAL–26	26	Retroactive gap
235	13.20 SD Rebates / Settlement Management (condition contracts)	Customizing (SPRO)	Chargeback / claim types	—	GAP	propose 30	—
236	13.20 SD Rebates / Settlement Management (condition contracts)	Customizing (SPRO)	Commission settlement integration (HR/AP)	—	PARTIAL–26	26	—
237	13.21 PP	Master data & documents	MM02 → MRP areas (MD61 per MRP area)	MRP area assignment (storage location / subcontractor level planning)	PARTIAL–22	22	Planning at warehouse (sub-site) level and at subcontractor level not explicitly named — partial; wave 22 plans by site; add "planning scope = site or warehouse"
238	13.21 PP	Master data & documents	CS51 / CS52 / CS53 / CS61 / CS62 / CS63 / CS71 …	Standard BOM / document BOM / equipment BOM / functional location BOM / sales order BOM / WBS BOM	PARTIAL–9	9	Material BOM yes; equipment/functional location BOMs in wave 44; sales-order BOMs (order-specific) and WBS BOMs not named — partial (ETO)
239	13.21 PP	Master data & documents	MS01 / MS02 / MS03 / MS04 / MS05 / MS06 / MS07 / MS08	Long-term planning (LTP): planning scenario, simulative MRP, stock/requirements (simulation), evaluations	PARTIAL–22	22	What-if simulation plans (separate planning versions with simulated demand, evaluating capacity/purchasing) not explicitly named — partial; propose "static plans" (AX-style) in wave 22
240	13.21 PP	Master data & documents	MS31 / MS32 / MS33 / MS44 / MS47 / MS64 / MS65 / MS70	LTP: planning scenario create/change/display, planned orders in scenario, evaluation (capacity, PURCHIS transfer)	PARTIAL–22	22	Same
241	13.21 PP	Master data & documents	Demand-driven replenishment (DDMRP: buffer positioning, buffer levels, replenishment planning, Fiori apps)	DDMRP (S/4)	GAP	propose 22	DDMRP (decoupling points, buffer zones red/yellow/green, net flow equation) not named; propose as wave 22 coverage-group type "DDMRP buffer"
242	13.21 PP	Master data & documents	Predictive MRP (pMRP, S/4)	Rough simulation of future demand vs capacity	PARTIAL–22	22	Overlaps LTP partial
243	13.21 PP	Master data & documents	Phantom assemblies, alternative items (BOM item alternative groups), discontinuation (MM02 follow-up material)	—	PARTIAL–9	9	Phantoms yes; alternative item groups (priority/strategy) and discontinuation/follow-up (run-out) are not named — partial; propose wave 9 BOM line attributes + wave 22 MRP logic
244	13.21 PP	Master data & documents	Make-to-order with sales order stock (strategy 20), planning with final assembly (40), planning without final assembly (50), subassembly planning (70)	Planning strategies	PARTIAL–22	22	Coverage groups + forecast reduction named; the full strategy-group semantics (consumption modes, MTO stock segmentation, planning at assembly level) are partially covered — partial; sales-order stock noted in MM-IM
245	13.21 PP	Master data & documents	Rework (CO07 / trigger points CO31 / CO33)	Rework orders / trigger points inserting rework operations	PARTIAL–21	21	Rework via order w/o material covered; trigger points (automatic rework operation insertion on confirmation) not named — partial
246	13.21 PP	Master data & documents	Order split (CO02 → split), order change management (OCM: COCM / COCM1)	Order split / order change management after ECO	PARTIAL–21	21	Order change management (propagating BOM/routing changes to open orders with change costs) not named — partial; propose wave 9/21
247	13.21 PP	Customizing (SPRO)	OPPS / OPPT — Planning strategies / strategy groups (10, 11, 20, 25, 30, 40, 50, 52, 59, 60, 63, 70, 74, 81, 82…)	T461S, T461P	PARTIAL–22	22	Planning strategy semantics partial (noted)
248	13.21 PP	Customizing (SPRO)	OMIZ / MRP areas (OMIZ / S/4 MRP areas per storage location, subcontractor)	MDLV, MDLG, MDLW	PARTIAL–22	22	MRP area partial

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
249	13.21 PP	Customizing (SPRO)	OPMP / OPM — Trigger points (standard trigger points, functions)	TTRIG	PARTIAL–21	21	Trigger point gap
250	13.21 PP	Customizing (SPRO)	OS40 / OS41 / OS44 — ECM: change types, object types, status, ECR/ECO profiles, release keys, effectivity types (date, serial, parameter)	TCC*, TECH*	PARTIAL–9	9	Date effectivity covered; serial-number/parameter effectivity not named — partial
251	13.21 PP	Customizing (SPRO)	DDMRP buffer profiles, classification runs (S/4 DD)	PPH_DD_*	GAP	propose 22	DDMRP gap
252	13.21 PP	Customizing (SPRO)	OCM profiles (order change management: change profiles, change cost)	COCM*	PARTIAL–21	21	OCM partial
253	13.22 PP-PI	Master data & documents	CRC5 / CRC	Resource network (RN)	PARTIAL–23	23	Resource networks (allowed sequences of resources, e.g. vessel → filler) not named — partial; scheduling with CP-SAT could model as constraints
254	13.22 PP-PI	Master data & documents	CO53 / CO53XT	Control recipe monitor (send control recipes to process control / PI sheet)	PARTIAL–25	25	Shop-floor instructions/drawings on kiosk; structured *control recipes* with process instruction categories sent to a destination (PCS/PI sheet) are partial
255	13.22 PP-PI	Master data & documents	CO54 / CO54XT / CO57 / CO58	Process message monitor / create process message manually / send	PARTIAL–25	25	IoT/MQTT machine events + kiosk reporting cover inbound; configurable message categories with destinations (e.g. to QM results, to batch record) partial
256	13.22 PP-PI	Master data & documents	CO60 / CO60XT / CO60_PI / CO64 / CO67	PI sheet: find/maintain / worklist / print / archive	PARTIAL–25	25	Kiosk instructions; a designer for interactive PI sheets (XSteps with input fields, calculations, signatures, process data requests) is not named — partial
257	13.22 PP-PI	Master data & documents	COEBR	Electronic batch record (EBR) creation/approval/archive (21 CFR Part 11)	GAP	propose 24	EBR with digital signatures, deviation review and archiving (GxP) not named — propose wave 24 (+ wave 48 compliance)
258	13.22 PP-PI	Master data & documents	CMX / Standard XSteps (CMXSV) / SXS repository	Standard XSteps repository	PARTIAL–25	25	Same as PI sheet
259	13.22 PP-PI	Master data & documents	Campaign planning (PP-PI campaign, process order campaigns: COCP / CMP)	Group process orders into campaigns (setup/cleanout)	PARTIAL–23	23	Campaign/sequence planning (group orders to minimize changeover, cleanout orders) not named — partial; CP-SAT setup optimization partly covers
260	13.22 PP-PI	Master data & documents	Cleanout / changeover orders, recipe-dependent cleaning	—	PARTIAL–23	23	Same
261	13.22 PP-PI	Master data & documents	Digital signature in PI sheets / batch records (DSIG)	21 CFR Part 11 signatures	GAP	propose 24	E-signature framework (meaning of signature, re-authentication) not named; propose cross-cutting in wave 24/48
262	13.22 PP-PI	Master data & documents	Batch derivation (DVMO / DVMAN) — see §13.33	—	PARTIAL–24	24	Batch derivation (inherit attributes from component batches to product batch) not named — partial
263	13.22 PP-PI	Reports & inquiries	CO54 (monitor)	Process message log	PARTIAL–25	25	—
264	13.22 PP-PI	Reports & inquiries	COEBR (list)	Batch record list	GAP	propose 24	—
265	13.22 PP-PI	Reports & inquiries	CMX/CO60 worklists	PI sheet worklist	PARTIAL–25	25	—
266	13.22 PP-PI	Periodic processing	CO53 (send control recipes, batch)	—	PARTIAL–25	25	—
267	13.22 PP-PI	Periodic processing	CO54 (process message processing, batch)	—	PARTIAL–25	25	—
268	13.22 PP-PI	Periodic processing	CO64 / CO67 (PI sheet archive)	—	PARTIAL–25	25	—
269	13.22 PP-PI	Periodic processing	COEBR (batch record generation/archive)	—	GAP	propose 24	—
270	13.22 PP-PI	Periodic processing	Campaign settlement / cleanout cost allocation	—	PARTIAL–23	23	—
271	13.22 PP-PI	Customizing (SPRO)	Process management: process instruction categories/characteristics (O10C O12C / O13C), process message categories & destinations (O05C / O06C / O07C), control recipe destinations (O10C), PI sheet layouts, XStep settings (SXS), signature strategies (O2 / DSIG)	TPM*, CPM*	PARTIAL–25	25	PI sheet / message framework partial
272	13.22 PP-PI	Customizing (SPRO)	Resource networks	—	PARTIAL–23	23	—
273	13.22 PP-PI	Customizing (SPRO)	Campaign planning settings	—	PARTIAL–23	23	—
274	13.22 PP-PI	Customizing (SPRO)	EBR: batch record profiles, archiving, approval workflow	—	GAP	propose 24	—

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
275	13.22 PP-PI	Customizing (SPRO)	Batch management in PP-PI: batch determination procedures for process orders (COB1), classification (023), batch derivation	TLSP, T*	PARTIAL–24	24	Derivation partial
276	13.23 PP/DS	Master data & documents	Block planning (resource blocks per characteristic, /SAPAPO/CDPSC12)	Reserve capacity blocks by product characteristics	GAP	propose 23	Block planning (time blocks on resources restricting which characteristic values can be produced) not named — propose wave 23 constraint type
277	13.23 PP/DS	Master data & documents	Characteristic-dependent planning (CDP) / CDP configuration	Planning by batch characteristics (e.g., color, grade)	PARTIAL–24	24	Batch attributes exist; planning that matches demand characteristics to supply batches (CDP) — partial; propose wave 22/24
278	13.23 PP/DS	Master data & documents	Shelf-life planning (SAP_PP_SL001)	Pegging respecting expiry (FEFO planning)	PARTIAL–22	22	FEFO picking (wave 16) yes; planning-time shelf-life pegging (min remaining shelf life at demand date) not named — partial
279	13.23 PP/DS	Master data & documents	/SAPAPO/MC01 (models/versions), /SAPAPO/MVM	Planning versions (simulation versions)	PARTIAL–22	22	Simulation versions = LTP partial
280	13.23 PP/DS	Master data & documents	/SAPAPO/SDP94 (DP), /SAPAPO/SNP94 (SNP), GATP (/SAPAPO/AC03)	Demand planning / supply network planning / global ATP	PARTIAL–22	22	DP ≈ wave 22 forecasting (statistical + consensus); SNP (multi-site network optimization: sourcing across plants/DCs with transport lanes and capacity) and GATP (rules-based ATP, multi-site substitution) are not named — partial; register multi-site network planning as a v1.x candidate under wave 22/27
281	13.23 PP/DS	Customizing (SPRO)	Block planning settings, CDP settings	—	GAP	propose 23	Block planning gap
282	13.24 QM	Master data & documents	QC51 / QC52 / QC53 / QC55	Certificate receipt (incoming certificates from vendor)	PARTIAL–33	33	Incoming vendor certificate receipt/tracking (required certificate at GR, block until received) not named — partial
283	13.24 QM	Master data & documents	QST01 / QST02 / QST03 / QST04 / QST05 / QST06 / QST07	Stability study (initial sample, storage conditions, testing schedule)	GAP	propose 33	Stability studies (pharma/food shelf-life testing schedules) not named; propose wave 33 extension for regulated process manufacturers
284	13.24 QM	Master data & documents	PLMD_AUDIT (Audit management: audits, questions, corrective actions)	Audit management	PARTIAL–33	33	CAPA yes; audit planning/questionnaires/audit findings (ISO 9001 internal audits, supplier audits) not named — partial
285	13.24 QM	Master data & documents	FMEA (QM-FMEA, S/4), control plan (S/4 QM)	FMEA / control plan	GAP	propose 33	FMEA and control plans (APQP/PPAP) — gap for automotive/aerospace suppliers; propose wave 33 extension
286	13.24 QM	Master data & documents	Source inspection (QI01 source inspection flag, QI07)	Inspection at vendor before GR	PARTIAL–33	33	Source inspection lots (before delivery) not named — partial
287	13.24 QM	Master data & documents	Sample management (QPR1 / QPR2 / QPR3 / QPR5 / QPR6 / QPR7)	Physical samples, sample drawing procedure, sample labels	PARTIAL–33	33	Sampling sizes yes; physical sample records (primary/pooled samples with labels, retention samples) not named — partial
288	13.24 QM	Master data & documents	Multiple specifications (customer/country-specific specs, S/4 "Multiple Specifications")	—	PARTIAL–33	33	Customer-specific test specs/limits — partial
289	13.24 QM	Reports & inquiries	QC55	Certificate receipt overview	PARTIAL–33	33	—
290	13.24 QM	Periodic processing	QST (stability schedule generation)	Stability testing schedule	GAP	propose 33	—
291	13.24 QM	Customizing (SPRO)	Define inspection points (time/quantity/freely defined), partial lots, inspection point identification	TQ79	PARTIAL–33	33	In-process inspection points (per time/quantity during production) partially covered by "production" trigger — partial
292	13.24 QM	Customizing (SPRO)	Stability study customizing	—	GAP	propose 33	—
293	13.24 QM	Customizing (SPRO)	FMEA / control plan customizing (S/4)	—	GAP	propose 33	—
294	13.24 QM	Customizing (SPRO)	Audit management customizing (audit types, question lists)	—	PARTIAL–33	33	—
295	13.24 QM	Customizing (SPRO)	Multiple specifications customizing	—	PARTIAL–33	33	—
296	13.24 QM	Customizing (SPRO)	Sample management customizing (sample types, sample-drawing procedures)	—	PARTIAL–33	33	—
297	13.25 PM	Master data & documents	IE31 / IE36 / IE37	Fleet object create / list / change	PARTIAL–44	44	Fleet-specific attributes (fuel, odometer, driver) not named — partial; counters cover odometer
298	13.25 PM	Master data & documents	IW81 / IW8W	Refurbishment order create / goods receipt for refurbishment	PARTIAL–44	44	Refurbishment of rotables (condition-based valuation: damaged → refurbished) not named — partial; split valuation gap in MM

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
299	13.25 PM	Master data & documents	IW32 → Permits (IPMD / WCM)	Permits, work clearance management (lockout/tagout)	PARTIAL–44	44	Permits/LOTO work clearance (WCM) not named — partial; propose wave 44 "permit/LOTO checklist"
300	13.25 PM	Master data & documents	Asset Intelligence Network / Predictive Asset Insights / Asset Strategy & Performance Management	Cloud EAM extensions	PARTIAL–44	44	Condition monitoring + predictive hooks named; RCM/criticality analysis (ASPM) partial
301	13.25 PM	Customizing (SPRO)	Refurbishment: valuation types for condition (OMWC split valuation), order type PM04 settings	T025, T003O	PARTIAL–44	44	Refurbishment gap
302	13.25 PM	Customizing (SPRO)	Permits (define permit types, categories), WCM (work clearance objects, applications, approvals)	T357G, WCA*	PARTIAL–44	44	Permit/LOTO gap
303	13.26 CS / Customer Service / SAP Field Service Management	Master data & documents	Warranty claim processing (WTY: WTY1 warranty claims from dealers/to vendors)	Warranty claims (supplier recovery)	PARTIAL–45	45	Customer warranty handling yes; supplier warranty claim recovery (claims to OEM/vendor, reimbursement) not named — partial
304	13.26 CS / Customer Service / SAP Field Service Management	Master data & documents	Solution database (IS01 / IS02 symptom/solution)	Knowledge base	GAP	propose 45	Knowledge base/solution search for cases not named; propose wave 45 (or wave 47 copilot answers)
305	13.26 CS / Customer Service / SAP Field Service Management	Master data & documents	Customer Interaction Center (CIC0) / SAP CRM Service / S/4 Service (Service orders, service contracts, service confirmations, service quotation)	Service front office	PARTIAL–45	45	Case management yes; omnichannel contact center (telephony/CTI, email routing) not named — partial (CRM non-goal boundary)
306	13.26 CS / Customer Service / SAP Field Service Management	Master data & documents	Service parts planning (SPP / advanced service parts)	Spare parts planning for service networks	PARTIAL–22	22	MRP covers; multi-echelon service parts planning partial
307	13.26 CS / Customer Service / SAP Field Service Management	Customizing (SPRO)	Warranty claim processing (WTY) customizing	—	PARTIAL–45	45	—
308	13.26 CS / Customer Service / SAP Field Service Management	Customizing (SPRO)	Solution database / knowledge base settings	—	GAP	propose 45	—
309	13.26 CS / Customer Service / SAP Field Service Management	Customizing (SPRO)	CIC / contact center framework	—	PARTIAL–45	45	—
310	13.27 PS	Master data & documents	CJZ1 / CJZ2 / CJZ3 / CJZ5	Interest calculation on projects	GAP	propose 41	Project interest calculation (interest on WIP/capitalized interest IAS 23) not named — propose wave 41 option
311	13.27 PS	Master data & documents	CJV1 / CJV2 / CJV3 / CJV4 / CJV5 / CJV6	Simulation version create/change/display/transfer	PARTIAL–41	41	Project what-if simulation versions not named — partial
312	13.27 PS	Master data & documents	CLM1 / CLM2 / CLM3 / CLM10 / CLM11	Claims (variation/change requests with cost impact)	PARTIAL–41	41	Change orders/claims on projects (scope change with cost/revenue impact and customer approval) not named — partial; propose wave 41 "project change order"
313	13.27 PS	Master data & documents	CNS0 / CN (shipping from projects — delivery from project)	Project deliveries (CNS0)	PARTIAL–41	41	Shipping items from project stock/WBS not named — partial (ETO)
314	13.27 PS	Master data & documents	Project stock (Q), PS material components, MD51 project MRP, CN33 (BOM transfer PS-BOM)	Project procurement/stock	PARTIAL–41	41	Project items/POs named; project stock segregation partial (MM-IM note)
315	13.27 PS	Master data & documents	IM01 / IM02 / IM03 / IM22 / IM23 / IM24 / IM25 / IM27 / IM28 / IM30 / IM31 / IM32 / IM33 / IM34 / IM35 / IM36 / IM52 / IMA11 / IMA2 / IMA3 / IMCCP1	Investment Management: investment programs (positions, hierarchy), budget distribution/supplement/return, program planning, appropriation requests (IMA11 create, variants, approval), rollup/carry-forward	PARTIAL–37	37	Asset budgeting named in wave 37; investment program hierarchy with appropriation requests (approval of capex before projects/orders) and program budget distribution — partial; propose wave 38 budgeting + wave 41
316	13.27 PS	Master data & documents	Commercial Project Management (CPM: project workspace, financial planning, change/risk/issue management)	Professional-services project cockpit	PARTIAL–41	41	Risks/issues register, cost/revenue forecasting workbook — partial
317	13.27 PS	Reports & inquiries	IM (IM reports: S_ALR_87012)	Investment program reports	PARTIAL–37	37	—
318	13.27 PS	Periodic processing	CJZ2 (interest)	Interest calculation run	GAP	propose 41	—
319	13.27 PS	Periodic processing	IM27 / IM32 (IM budget rollup / carry-forward)	—	PARTIAL–37	37	—
320	13.27 PS	Customizing (SPRO)	OPS — Project stock settings (valuated/non-valuated), PS material components (OPUV), MRP for project	TCJ41, T*	PARTIAL–41	41	Project stock partial
321	13.27 PS	Customizing (SPRO)	Investment management: program types, appropriation request types, status, budget categories, plan/budget profiles	TAIF, TAPRT	PARTIAL–37	37	IM partial
322	13.27 PS	Customizing (SPRO)	Claim management customizing (notification type CL, claim types)	TQ80	PARTIAL–41	41	—
323	13.27 PS	Customizing (SPRO)	Simulation versions / project versions profiles (OPTS / OPT)	TCJ	PARTIAL–41	41	—
324	13.27 PS	Customizing (SPRO)	Interest profiles (OPIB / OPIC / OPID)	T*	GAP	propose 41	—

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
325	13.28 HCM	Master data & documents	PP61 / PP62 / PP63 / PP64 / PP6I / PP6A	Shift planning (target plan, requirements, assignments)	PARTIAL–25	25	Time registration & shifts in wave 25; shift rostering (planning employees to shifts against requirements, swaps) not named — partial; propose wave 42
326	13.28 HCM	Master data & documents	PC00_M10_OSUP / PC00_Mxx_OC (off-cycle workbench PUOC_10)	Off-cycle payroll (bonus, correction, final pay)	PARTIAL–43	43	Off-cycle/ad-hoc pay runs (bonus, termination final pay, check replacement) not explicitly named — partial
327	13.28 HCM	Master data & documents	PC00_M10_CLAIMS / PC00_M10_UTXX	US claims processing / tax authorities	PARTIAL–43	43	Claims (overpayment recovery) not named — partial
328	13.28 HCM	Master data & documents	HRBEN00ICOBRA / HRBEN00COBRA	COBRA (US)	PARTIAL–42	42	COBRA notices/continuation (US) belongs to US pack — partial
329	13.28 HCM	Master data & documents	E-Recruiting (HRRCF_*) / SF Recruiting & Onboarding	Modern recruiting	PARTIAL–42	42	Pipeline basics; career site, requisition approval, offer letters, onboarding tasks — partial
330	13.28 HCM	Master data & documents	PPCP / PPSS (career/succession planning) / SF Succession	Succession planning	PARTIAL–42	42	Talent basics; succession planning/talent pools not named — partial
331	13.28 HCM	Master data & documents	PECM_* (enterprise compensation management: planning, budgets, long-term incentives, pay scale OH) / SF Compensation	Compensation planning cycles	PARTIAL–42	42	Compensation plans yes; merit/bonus planning cycles with budgets and manager worksheets — partial
332	13.28 HCM	Master data & documents	PSV1 / PSV2 / PSV3 / PSVP / LSO_PSV (training & event management / LSO) / SF Learning	Training catalog, bookings, resources, follow-up	PARTIAL–42	42	Certifications tracked; course catalog/LMS (sessions, bookings, completion) not named — partial
333	13.28 HCM	Master data & documents	HRUS_D2 / HRPAD (personnel data: concurrent employment, global employment)	Concurrent/global employment	PARTIAL–42	42	Multi-assignment and global assignments not named — partial
334	13.28 HCM	Master data & documents	SF Onboarding / Offboarding	Onboarding tasks	PARTIAL–42	42	Onboarding checklists not named — partial
335	13.28 HCM	Reports & inquiries	PC00_M10_* US reports (941, 940, W-2, state unemployment, new hire, EEO-1, workers' comp)	US statutory	PARTIAL–43	43	Year-end W-2/1099 named; quarterly 941/state UI and EEO/VETS reporting belong to US pack — partial
336	13.28 HCM	Periodic processing	PU19 (quarterly/annual tax reporter runs)	Statutory filings	PARTIAL–43	43	—
337	13.28 HCM	Customizing (SPRO)	Compensation management: compensation areas, plans, guidelines, budgets, eligibility	T71*	PARTIAL–42	42	Planning cycles partial
338	13.28 HCM	Customizing (SPRO)	Training & event management / LSO: business event types, resources, booking, follow-up, LSO settings	T77*	PARTIAL–42	42	LMS partial
339	13.28 HCM	Customizing (SPRO)	Country-specific HR infotypes and legal changes (country packs: US, DE, GB, FR, IN, MX, CA)	—	PARTIAL–48	48	Country-specific fields framework (wave 42) + packs (wave 48) cover tax/statutory; HR-legal (works council, contract types, statutory leave) per country partially
340	13.28 HCM	Customizing (SPRO)	Concurrent employment / global employment configuration	—	PARTIAL–42	42	—
341	13.29 FI-TV	Master data & documents	TP01 / TP02 / TP03 / PR01 (travel request)	Travel request (pre-trip approval with estimated cost)	PARTIAL–43	43	Pre-trip travel request/approval with estimated cost and cash advance not named — partial; propose wave 43 "pre-approval request"
342	13.29 FI-TV	Master data & documents	PR02 / PR05 advances (PR cash advance)	Cash advances	PARTIAL–43	43	Advances and offset against expenses not named — partial
343	13.29 FI-TV	Customizing (SPRO)	Travel request / travel plan settings, approval workflows, substitutes	T702, SWF	PARTIAL–43	43	Pre-approval partial
344	13.30 Basis & Security (users, roles, GRC Access Control, change documents, transports, system administration)	Master data & documents	GRC EAM firefighter (superuser privilege management with reason codes and log review)	Emergency access	PARTIAL–48	48	Break-glass elevated access with time-boxed sessions, reason codes and reviewer sign-off not explicitly named — partial
345	13.30 Basis & Security (users, roles, GRC Access Control, change documents, transports, system administration)	Periodic processing	Anonymized production-to-test data refresh (TDMS / client copy with scrambling)	Test data management	PARTIAL–48	48	Tenant clone with PII scrambling not explicitly named — partial; propose wave 48/50
346	13.30 Basis & Security (users, roles, GRC Access Control, change documents, transports, system administration)	Customizing (SPRO)	Printing: device types, output devices, access methods (SPAD), front-end printing, print queues	TSP03*	PARTIAL–32	32	Print service for labels/documents; generic office print queue/printer management partial

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
347	13.30 Basis & Security (users, roles, GRC Access Control, change documents, transports, system administration)	Customizing (SPRO)	Read Access Logging (SRAL) configuration	SRAL*	PARTIAL–48	48	PII access logged (wave 42) for HR; configurable read-access logging for arbitrary sensitive fields partial — propose wave 48
348	13.31 Cross-Application (workflow, IDoc/ALE, PI/PO, DRC e-documents, output management NAST/BRF+, Fiori launchpad, archiving/ILM, GDPR/IRF, attachments, texts, enterprise search)	Master data & documents	BRF+ (BRFPLUS / FDT_WORKBENCH) / Decision Service Management (DSM)	Business rules framework (decision tables, formulas, rulesets)	PARTIAL–3	3	Validation rules and OPA policies exist; a general-purpose, user-maintainable decision-table/rules engine (used by output, tax, credit, dispute, GRC, substitution) is implied across waves but not named as a shared service — partial; propose explicit "rules engine (decision tables)" in wave 3/4
349	13.31 Cross-Application (workflow, IDoc/ALE, PI/PO, DRC e-documents, output management NAST/BRF+, Fiori launchpad, archiving/ILM, GDPR/IRF, attachments, texts, enterprise search)	Master data & documents	SO10 / SO72 / SE75 (text objects/IDs)	Standard texts, text management	PARTIAL–5	5	Document/line notes exist; a reusable standard-text library with placeholders and language versions for documents is not named — partial; propose wave 46 designer text blocks
350	13.31 Cross-Application (workflow, IDoc/ALE, PI/PO, DRC e-documents, output management NAST/BRF+, Fiori launchpad, archiving/ILM, GDPR/IRF, attachments, texts, enterprise search)	Master data & documents	SAP Business Partner Screening / Watch List Screening (GTS)	Sanctions/denied-party screening	GAP	propose 48	Denied-party/sanctions screening of parties and documents not named; propose wave 48 (compliance) with provider adapter
351	13.31 Cross-Application (workflow, IDoc/ALE, PI/PO, DRC e-documents, output management NAST/BRF+, Fiori launchpad, archiving/ILM, GDPR/IRF, attachments, texts, enterprise search)	Master data & documents	Legal Content Management / contract management (SAP CLM, Ariba Contracts)	Legal contracts	PARTIAL–17	17	Purchase/sales agreements exist; legal contract authoring, clause library, obligations — partial (noted in Ariba row)
352	13.31 Cross-Application (workflow, IDoc/ALE, PI/PO, DRC e-documents, output management NAST/BRF+, Fiori launchpad, archiving/ILM, GDPR/IRF, attachments, texts, enterprise search)	Master data & documents	SAP Business Rules (BRF+ in GRC, tax, dispute, output) — see above	—	PARTIAL–3	3	—
353	13.31 Cross-Application (workflow, IDoc/ALE, PI/PO, DRC e-documents, output management NAST/BRF+, Fiori launchpad, archiving/ILM, GDPR/IRF, attachments, texts, enterprise search)	Master data & documents	SAP Process Mining (Signavio) / Process Observer (POC)	Process mining	PARTIAL–46	46	Event stream + lakehouse enables mining; no native process-mining UI — partial
354	13.31 Cross-Application (workflow, IDoc/ALE, PI/PO, DRC e-documents, output management NAST/BRF+, Fiori launchpad, archiving/ILM, GDPR/IRF, attachments, texts, enterprise search)	Customizing (SPRO)	BRF+ applications, functions, rulesets, decision tables, expressions	FDT_*	PARTIAL–3	3	Rules engine gap

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
355	13.31 Cross-Application (workflow, IDoc/ALE, PI/PO, DRC e-documents, output management NAST/BRF+, Fiori launchpad, archiving/ILM, GDPR/IRF, attachments, texts, enterprise search)	Customizing (SPRO)	Watch list screening / GTS compliance configuration	—	GAP	propose 48	Screening gap
356	13.32 LO-VC	Master data & documents	CU60 / CU61 / CU62 / CU65 / CU66 / CU67	Variant tables: structure / content / display; variant functions (ABAP)	PARTIAL–8	8	Variant tables (decision tables) map to constraint tables — covered; *variant functions* (ABAP-coded) → SUPERSEDED by expression functions; table maintenance UI not named — partial
357	13.32 LO-VC	Master data & documents	CU51 / CU52 / CU55 / CU56	Order BOM maintain (sales order BOM — result-oriented/knowledge-based) / display / instantiation	PARTIAL–21	21	Order-specific BOM edits after configuration (ETO) not named — partial; propose wave 21/27 "order BOM"
358	13.32 LO-VC	Master data & documents	Customer-facing configurator (web, Fiori / SAP CPQ)	Guided selling	PARTIAL–30	30	Customer portal order entry named; guided configurator UI for portal/e-commerce not explicitly named — partial
359	13.32 LO-VC	Master data & documents	CU7 / CUCPT (configuration copy/ templates)	Configuration templates	PARTIAL–8	8	Saved configurations/templates for reuse — partial
360	13.32 LO-VC	Master data & documents	PSM / iPPE (Product Structure Management: product variant structures, CMP nodes, PVS)	Product structure for complex configurable products (automotive)	PARTIAL–9	9	Multi-level configurable structures covered by super BOM; iPPE-style product variant structures with alternatives/variants at node level — partial; propose wave 9 v1.x
361	13.32 LO-VC	Master data & documents	Configuration in purchasing (configurable PO items for configurable vendor products), configuration in service (IBase configuration)	—	PARTIAL–17	17	Configured PO items not named — partial
362	13.32 LO-VC	Master data & documents	VC with batch/serial effectivity, parameter effectivity (ECM)	Effectivity	PARTIAL–9	9	Noted in PP (serial/parameter effectivity partial)
363	13.32 LO-VC	Periodic processing	Re-explosion of configured orders after model change (OCM)	—	PARTIAL–21	21	OCM partial (noted)
364	13.32 LO-VC	Customizing (SPRO)	Order BOM settings (CU51 processing)	—	PARTIAL–21	21	—
365	13.33 Batch Management & Serial Numbers	Master data & documents	DVMO / DVSP / DVMAN	Batch derivation: monitor / set up derivation (sender/receiver, rules) / manual derivation	PARTIAL–24	24	Batch derivation (inherit attribute values from component batches) partial (noted in PP-PI)
366	13.33 Batch Management & Serial Numbers	Master data & documents	Batch valuation (split valuation per batch, OMWC valuation type = batch)	Batch-level valuation	PARTIAL–20	20	Split valuation gap
367	13.33 Batch Management & Serial Numbers	Master data & documents	Documentary batches (S/4 "documentary batch" for non-batch-managed materials)	Documentary batches	PARTIAL–24	24	Recording batch numbers for info only (no stock segregation) not named — partial
368	13.33 Batch Management & Serial Numbers	Master data & documents	Serial numbers at component level in finished-goods serial (as-built genealogy)	As-built records	PARTIAL–21	21	Serial-to-serial component genealogy (as-built) not explicitly named — partial; propose wave 21 RAF capture
369	13.33 Batch Management & Serial Numbers	Reports & inquiries	DVMO (derivation monitor)	—	PARTIAL–24	24	—
370	13.33 Batch Management & Serial Numbers	Periodic processing	Batch derivation runs (DVMO)	—	PARTIAL–24	24	—
371	13.33 Batch Management & Serial Numbers	Customizing (SPRO)	Batch derivation: activation, sender/receiver fields, derivation events, rules	DVMO*	PARTIAL–24	24	—
372	13.33 Batch Management & Serial Numbers	Customizing (SPRO)	Batch valuation (valuation type = batch) activation	T001K, T149	PARTIAL–20	20	—
373	13.33 Batch Management & Serial Numbers	Customizing (SPRO)	Documentary batch activation (S/4)	—	PARTIAL–24	24	—
374	13.35 DMS	Master data & documents	CV01N / CV02N / CV03N	Document info record create/change/display (document type, number, part, version; status; originals; object links; classification; texts)	PARTIAL–8	8	Attachments (wave 5) and product documents (wave 8) exist; a *document info record* with type, versioning, part/version numbering, status network and release is not named — partial; propose wave 9 "engineering documents" (drawings/specs under ECO control)
375	13.35 DMS	Master data & documents	CV04N	Find document (by attributes, classification, object links, full text)	PARTIAL–8	8	Global search covers attachments metadata; attribute/classification search on documents partial

#	Section	Table	T-code / IMG node	Name	Status	Wave (current / proposed)	What is missing / proposal
376	13.35 DMS	Master data & documents	CV11 / CV12 / CV13 / CV15	Document structure (document BOM) create/change/display / where-used	GAP	propose 9	Document structures (assemblies of documents, e.g. drawing sets) not named — propose wave 9
377	13.35 DMS	Master data & documents	CV01N → originals (check-in/check-out, KPro storage categories, viewer, conversion)	Original files (CAD, PDF, Office) with check-out locking	PARTIAL–5	5	Attachments stored; check-out/check-in locking, multiple originals per version, viewer/conversion (2D/3D) not named — partial
378	13.35 DMS	Master data & documents	CV01N → status network (in work → released → obsolete), status with release/approval workflow, digital signature	Document status & release	PARTIAL–9	9	ECO workflow covers BOM/route release; document release state machine with signature — partial
379	13.35 DMS	Master data & documents	CV01N → version / part / change number (ECM)	Document versioning under ECM	PARTIAL–9	9	ECO-controlled document versions partial
380	13.35 DMS	Master data & documents	CV01N → document distribution (recipients, distribution orders CVI1 / CVI2 / CVI8)	Distribution of documents (e.g. to suppliers)	PARTIAL–20	20	Supplier portal document exchange named; controlled distribution with acknowledgement partial
381	13.35 DMS	Master data & documents	ECTR (Engineering Control Center): CAD integration (SolidWorks, Inventor, NX, CATIA, AutoCAD, ECAD), BOM derivation from CAD, 3D viewer, DMS for CAD	CAD–ERP integration	GAP	propose 9	CAD integration (CAD file management, CAD BOM → engineering BOM sync) not named; wave 9 says "PLM integration" only as reference — propose wave 9 (v1.x) PLM/CAD connector (e.g. via a PLM system) rather than native CAD management
382	13.35 DMS	Master data & documents	Document BOM items (CS01 item category D)	Documents as BOM components (e.g., drawings, instructions)	PARTIAL–9	9	BOM line type "document" not named — partial
383	13.35 DMS	Master data & documents	PLM: Engineering Record (change management cockpit), Product Structure Management, Visual Enterprise (3D visualization), Recipe Development (PLM-RDP), Specification management (EHS specs), Labeling (GLM)	PLM suite	PARTIAL–9	9	ECR/ECO covered; recipe development with formula trials/specs (process industries), specification management, global label management — partial; propose wave 24 (recipe dev) & wave 48 (labeling compliance)
384	13.35 DMS	Master data & documents	Fiori: Manage Documents / Manage Document Info Records	—	PARTIAL–8	8	—
385	13.35 DMS	Reports & inquiries	CV04N	Document search	PARTIAL–8	8	—
386	13.35 DMS	Reports & inquiries	CV13 / CV15	Document structure display / where-used	GAP	propose 9	—
387	13.35 DMS	Reports & inquiries	CVI8 (distribution log)	Distribution status	PARTIAL–20	20	—
388	13.35 DMS	Reports & inquiries	ECTR reports (CAD BOM comparison)	—	GAP	propose 9	—
389	13.35 DMS	Periodic processing	CV conversion runs (originals → PDF/neutral formats)	Conversion server runs	PARTIAL–5	5	Preview generation for attachments — partial
390	13.35 DMS	Periodic processing	Document distribution runs (CVI)	—	PARTIAL–20	20	—
391	13.35 DMS	Periodic processing	ECTR sync jobs	—	GAP	propose 9	—
392	13.35 DMS	Customizing (SPRO)	DC10 — Document types (number assignment, version/part numbering, status network, object links, screen layout, classification class, originals count, versioning)	TDWA, TDWS, TDWO, TDWP	PARTIAL–8	8	Attachment types exist; full document type model partial
393	13.35 DMS	Customizing (SPRO)	DC20 — Status (document status definitions, release flag, content version)	TDWS	PARTIAL–9	9	—
394	13.35 DMS	Customizing (SPRO)	DC30 — Workstation applications (file types, viewer/editor, conversion)	TDWP	PARTIAL–5	5	Preview/viewer mapping partial
395	13.35 DMS	Customizing (SPRO)	Document distribution customizing (recipient lists, distribution types)	—	PARTIAL–20	20	—
396	13.35 DMS	Customizing (SPRO)	ECTR configuration (CAD applications, templates, BOM sync rules)	—	GAP	propose 9	—
397	13.35 DMS	Customizing (SPRO)	Digital signature for documents (signature strategies)	—	GAP	propose 24	E-signature framework gap (noted in PP-PI)
398	13.35 DMS	Customizing (SPRO)	Full-text search activation (TREX/HANA) for originals	—	PARTIAL–5	5	Attachment content indexing not named — partial
399	13.36 Analytics	Master data & documents	Data quality dashboards (Information Steward)	DQ monitoring	PARTIAL–46	46	Metric reconciliation checks exist; generic data-quality rule dashboards (completeness, validity per entity) partial — propose wave 46 DQ rules on catalog
400	13.36 Analytics	Customizing (SPRO)	Export governance (Excel export restrictions, download audit)	—	PARTIAL–48	48	Export of PII/financial data audit & restriction policies not named — partial; propose wave 48
401	13.37 AI	Master data & documents	SAP Signavio Process AI / Process Insights (process mining recommendations)	Process intelligence	PARTIAL–46	46	Process mining partial (noted §13.31)